

Tables for Evaluating Local Structural-After-Measurement (LSAM) and Traditional Approaches for the Estimation of Complex Nonlinear Effects Among Latent Variables

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**Tables for Evaluating Local Structural-After-Measurement (LSAM) and Traditional
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Performance Measures**

Table S1*Complete performance measures for evaluating parameter estimates in the simulation studies*

Performance Measure	Empirical Calculation	MCSE
Mean-Based Metrics		
Absolute Bias	$\bar{B} - \beta$	$\sqrt{S_B^2/K}$
Relative Bias	$(\frac{\bar{B}}{\beta}) - 1$	$\sqrt{S_B^2/(K\beta^2)}$
Percent Relative Bias	$\left[(\frac{\bar{B}}{\beta}) - 1 \right] \times 100\%$	—
MSE	$\frac{1}{K} \sum_{k=1}^K (B_k - \beta)^2$	$\sqrt{\frac{1}{K} [S_B^4(k_B - 1) + 4S_B^3g_B(\bar{B} - \beta) + 4S_B^2(\bar{B} - \beta)^2]}$
RMSE	$\sqrt{\frac{1}{K} \sum_{k=1}^K (B_k - \beta)^2}$	$\sqrt{\frac{K-1}{K} \sum_{j=1}^K (\text{RMSE}_{(j)} - \text{RMSE})^2}$
Relative MSE	$\frac{(\bar{B} - \beta)^2 + S_B^2}{\beta^2}$	$\sqrt{\frac{1}{K\beta^4} [S_B^4(k_B - 1) + 4S_B^3g_B(\bar{B} - \beta) + 4S_B^2(\bar{B} - \beta)^2]}$
Relative RMSE	$\sqrt{\frac{(\bar{B} - \beta)^2 + S_B^2}{\beta^2}}$	$\sqrt{\frac{K-1}{K} \sum_{j=1}^K (\text{rRMSE}_{(j)} - \text{rRMSE})^2}$
SE/SD Ratio	$\bar{S}E/S_B$	—
Rejection Rate	$r_\alpha = \frac{1}{K} \sum_{k=1}^K I(P_k < \alpha)$	$\sqrt{r_\alpha(1 - r_\alpha)/K}$
Coverage	$c_\beta = \frac{1}{K} \sum_{k=1}^K I(L_k \leq \beta \leq U_k)$	$\sqrt{c_\beta(1 - c_\beta)/K}$
Average Width	$\bar{W} = \bar{U} - \bar{L}$	$\sqrt{S_W^2/K}$
Median-Based Metrics		
Median Bias	$Mdn(B) - \beta$	—
Relative Median Bias	$\frac{Mdn(B) - \beta}{\beta}$	—
Percent Relative Median Bias	$\frac{Mdn(B) - \beta}{\beta} \times 100\%$	—
MAD	$Mdn(\ B_k - Mdn(B)\)$	—
Median RMSE	$\sqrt{\text{Bias}_{Mdn}^2 + \text{MAD}^2}$	—

Note. B = estimator for parameter β ; K = number of replications producing estimates $B_1, \dots, B_K$; $\bar{B} = (1/K) \sum_{k=1}^K B_k$ = sample mean;
 $S_B^2 = (1/(K-1)) \sum_{k=1}^K (B_k - \bar{B})^2$ = sample variance; $S_B = \sqrt{S_B^2}$ = sample standard deviation; $\overline{SE} = (1/K) \sum_{k=1}^K \widehat{SE}_k$ = mean standard error; k_B = standardized kurtosis; g_B = standardized skewness; $I(\cdot)$ = indicator function; P_k = p-value from replication k ; α = significance level; L_k, U_k = lower and upper CI endpoints; S_W^2 = variance of CI widths; r_α and c_β denote rejection rate and coverage, respectively; $Mdn(B)$ = sample median of estimates; MAD = median absolute deviation with scaling factor 1.4826. MCSE = Monte Carlo Standard Error, dashes indicate MCSEs not computed. MSE = Mean Squared Error. RMSE = Root Mean Squared Error.

Complete Results Tables — Simulation 1

Table S1

Simulation 1 - Condition 1: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.4

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LMS	beta31	0.316	0.297	-0.019 $\pm$ 0.003	-0.061 $\pm$ 0.010	0.009 $\pm$ 0.000	0.096 $\pm$ 0.002	0.010 $\pm$ 0.000	0.098 $\pm$ 0.003	0.095 $\pm$ 0.005	0.309 $\pm$ 0.009	0.291	-0.025	-8.05	0.0915	0.0950	0.997	93.414 $\pm$ 0.822	0.374 $\pm$ 0.002	90.560 $\pm$ 0.969
LMS	beta32	0.316	0.302	-0.014 $\pm$ 0.003	-0.046 $\pm$ 0.010	0.009 $\pm$ 0.000	0.093 $\pm$ 0.002	0.009 $\pm$ 0.000	0.094 $\pm$ 0.003	0.089 $\pm$ 0.004	0.298 $\pm$ 0.009	0.298	-0.018	-5.72	0.0931	0.0948	1.021	94.951 $\pm$ 0.725	0.373 $\pm$ 0.002	91.987 $\pm$ 0.900
LMS	beta33	0.139	0.111	-0.028 $\pm$ 0.004	-0.203 $\pm$ 0.032	0.018 $\pm$ 0.001	0.133 $\pm$ 0.003	0.019 $\pm$ 0.001	0.136 $\pm$ 0.004	0.961 $\pm$ 0.047	0.980 $\pm$ 0.029	0.104	-0.035	-25.51	0.1267	0.1315	1.005	93.853 $\pm$ 0.796	0.525 $\pm$ 0.006	15.038 $\pm$ 1.184
LMS	beta34	0.101	0.108	0.007 $\pm$ 0.003	0.069 $\pm$ 0.030	0.008 $\pm$ 0.000	0.091 $\pm$ 0.003	0.008 $\pm$ 0.001	0.092 $\pm$ 0.003	0.822 $\pm$ 0.050	0.907 $\pm$ 0.031	0.104	0.003	3.12	0.0861	0.0861	0.998	95.499 $\pm$ 0.687	0.357 $\pm$ 0.004	23.930 $\pm$ 1.414
LMS	beta35	0.101	0.103	0.002 $\pm$ 0.003	0.022 $\pm$ 0.029	0.008 $\pm$ 0.000	0.088 $\pm$ 0.002	0.008 $\pm$ 0.000	0.088 $\pm$ 0.003	0.761 $\pm$ 0.040	0.872 $\pm$ 0.027	0.101	0.000	0.12	0.0882	0.0882	1.004	94.292 $\pm$ 0.769	0.347 $\pm$ 0.004	23.381 $\pm$ 1.402
LSAM	beta31	0.316	0.320	0.004 $\pm$ 0.004	0.012 $\pm$ 0.012	0.013 $\pm$ 0.001	0.114 $\pm$ 0.003	0.013 $\pm$ 0.001	0.114 $\pm$ 0.004	0.131 $\pm$ 0.007	0.362 $\pm$ 0.011	0.314	-0.002	-0.59	0.1171	0.1171	1.073	96.707 $\pm$ 0.591	0.481 $\pm$ 0.011	81.778 $\pm$ 1.279
LSAM	beta32	0.316	0.327	0.011 $\pm$ 0.004	0.035 $\pm$ 0.011	0.012 $\pm$ 0.001	0.110 $\pm$ 0.003	0.012 $\pm$ 0.001	0.110 $\pm$ 0.003	0.121 $\pm$ 0.006	0.348 $\pm$ 0.011	0.324	0.008	2.44	0.1119	0.1122	1.117	98.353 $\pm$ 0.422	0.480 $\pm$ 0.009	82.547 $\pm$ 1.258
LSAM	beta33	0.139	0.151	0.012 $\pm$ 0.006	0.083 $\pm$ 0.040	0.028 $\pm$ 0.002	0.169 $\pm$ 0.005	0.029 $\pm$ 0.002	0.169 $\pm$ 0.006	1.478 $\pm$ 0.092	1.216 $\pm$ 0.043	0.142	0.003	1.97	0.1599	0.1599	1.121	97.695 $\pm$ 0.497	0.741 $\pm$ 0.025	13.941 $\pm$ 1.148
LSAM	beta34	0.101	0.106	0.005 $\pm$ 0.004	0.049 $\pm$ 0.039	0.014 $\pm$ 0.001	0.120 $\pm$ 0.004	0.014 $\pm$ 0.001	0.120 $\pm$ 0.004	1.419 $\pm$ 0.086	1.191 $\pm$ 0.041	0.103	0.002	1.83	0.1147	0.1148	1.081	97.914 $\pm$ 0.473	0.510 $\pm$ 0.014	11.196 $\pm$ 1.045
LSAM	beta35	0.101	0.105	0.004 $\pm$ 0.004	0.043 $\pm$ 0.041	0.016 $\pm$ 0.001	0.126 $\pm$ 0.004	0.016 $\pm$ 0.001	0.126 $\pm$ 0.005	1.557 $\pm$ 0.099	1.248 $\pm$ 0.045	0.096	-0.005	-4.63	0.1104	0.1105	1.055	98.024 $\pm$ 0.461	0.521 $\pm$ 0.020	12.404 $\pm$ 1.092
QML	beta31	0.316	0.297	-0.019 $\pm$ 0.003	-0.060 $\pm$ 0.010	0.009 $\pm$ 0.000	0.096 $\pm$ 0.002	0.010 $\pm$ 0.000	0.098 $\pm$ 0.003	0.095 $\pm$ 0.005	0.309 $\pm$ 0.009	0.290	-0.026	-8.23	0.0905	0.0942	1.019	93.524 $\pm$ 0.815	0.383 $\pm$ 0.009	90.340 $\pm$ 0.979
QML	beta32	0.316	0.302	-0.014 $\pm$ 0.003	-0.044 $\pm$ 0.010	0.009 $\pm$ 0.000	0.094 $\pm$ 0.002	0.009 $\pm$ 0.000	0.095 $\pm$ 0.003	0.089 $\pm$ 0.004	0.299 $\pm$ 0.009	0.298	-0.018	-5.75	0.0928	0.0946	1.030	94.841 $\pm$ 0.733	0.378 $\pm$ 0.005	91.877 $\pm$ 0.905
QML	beta33	0.139	0.112	-0.027 $\pm$ 0.004	-0.191 $\pm$ 0.032	0.018 $\pm$ 0.001	0.134 $\pm$ 0.004	0.019 $\pm$ 0.001	0.137 $\pm$ 0.004	0.970 $\pm$ 0.049	0.985 $\pm$ 0.029	0.105	-0.034	-24.64	0.1295	0.1339	1.008	94.402 $\pm$ 0.762	0.531 $\pm$ 0.006	15.807 $\pm$ 1.209
QML	beta34	0.101	0.107	0.006 $\pm$ 0.003	0.060 $\pm$ 0.030	0.008 $\pm$ 0.000	0.092 $\pm$ 0.003	0.008 $\pm$ 0.000	0.092 $\pm$ 0.003	0.826 $\pm$ 0.048	0.909 $\pm$ 0.031	0.103	0.002	1.86	0.0861	0.0861	1.001	95.719 $\pm$ 0.671	0.360 $\pm$ 0.005	22.832 $\pm$ 1.391
QML	beta35	0.101	0.103	0.002 $\pm$ 0.003	0.018 $\pm$ 0.029	0.008 $\pm$ 0.000	0.089 $\pm$ 0.002	0.008 $\pm$ 0.000	0.089 $\pm$ 0.003	0.781 $\pm$ 0.043	0.884 $\pm$ 0.028	0.101	0.000	-0.25	0.0884	0.0884	0.997	93.963 $\pm$ 0.789	0.349 $\pm$ 0.004	22.722 $\pm$ 1.388
UPI	beta31	0.316	0.319	0.003 $\pm$ 0.004	0.009 $\pm$ 0.012	0.014 $\pm$ 0.001	0.119 $\pm$ 0.003	0.014 $\pm$ 0.001	0.119 $\pm$ 0.004	0.142 $\pm$ 0.008	0.377 $\pm$ 0.012	0.315	-0.001	-0.25	0.1161	0.1161	1.129	96.597 $\pm$ 0.601	0.527 $\pm$ 0.008	74.314 $\pm$ 1.448
UPI	beta32	0.316	0.328	0.012 $\pm$ 0.004	0.039 $\pm$ 0.012	0.013 $\pm$ 0.001	0.114 $\pm$ 0.003	0.013 $\pm$ 0.001	0.115 $\pm$ 0.004	0.131 $\pm$ 0.007	0.362 $\pm$ 0.011	0.324	0.008	2.47	0.1160	0.1162	1.175	98.353 $\pm$ 0.422	0.525 $\pm$ 0.009	76.729 $\pm$ 1.400
UPI	beta33	0.139	0.167	0.028 $\pm$ 0.006	0.200 $\pm$ 0.046	0.037 $\pm$ 0.002	0.193 $\pm$ 0.006	0.038 $\pm$ 0.002	0.195 $\pm$ 0.007	1.968 $\pm$ 0.127	1.403 $\pm$ 0.051	0.146	0.007	4.81	0.1666	0.1668	1.305	98.902 $\pm$ 0.345	0.988 $\pm$ 0.053	8.562 $\pm$ 0.927
UPI	beta34	0.101	0.109	0.008 $\pm$ 0.005	0.076 $\pm$ 0.047	0.021 $\pm$ 0.001	0.144 $\pm$ 0.005	0.021 $\pm$ 0.001	0.144 $\pm$ 0.005	2.034 $\pm$ 0.129	1.426 $\pm$ 0.051	0.099	-0.002	-2.33	0.1245	0.1246	1.248	98.134 $\pm$ 0.448	0.704 $\pm$ 0.021	4.830 $\pm$ 0.710
UPI	beta35	0.101	0.099	-0.002 $\pm$ 0.005	-0.017 $\pm$ 0.048	0.021 $\pm$ 0.001	0.146 $\pm$ 0.004	0.021 $\pm$ 0.001	0.145 $\pm$ 0.005	2.075 $\pm$ 0.127	1.441 $\pm$ 0.050	0.091	-0.010	-9.44	0.1298	0.1301	1.252	97.914 $\pm$ 0.473	0.714 $\pm$ 0.026	5.928 $\pm$ 0.782

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S2

Simulation 1 - Condition 2: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.4

Method	Param.	True	Mean-based									Median-based					Inference				
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE	
LMS	beta31	0.316	0.297	-0.019 $\pm$ 0.002	-0.061 $\pm$ 0.006	0.003 $\pm$ 0.000	0.058 $\pm$ 0.001	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.037 $\pm$ 0.002	0.193 $\pm$ 0.005	0.296	-0.020	-6.22	0.0566	0.0599	1.026	93.656 $\pm$ 0.774	0.233 $\pm$ 0.002	99.899 $\pm$ 0.101	
LMS	beta32	0.316	0.295	-0.021 $\pm$ 0.002	-0.067 $\pm$ 0.006	0.003 $\pm$ 0.000	0.056 $\pm$ 0.001	0.004 $\pm$ 0.000	0.060 $\pm$ 0.001	0.036 $\pm$ 0.001	0.190 $\pm$ 0.005	0.293	-0.023	-7.32	0.0588	0.0632	1.054	94.058 $\pm$ 0.750	0.232 $\pm$ 0.001	99.899 $\pm$ 0.101	
LMS	beta33	0.139	0.106	-0.033 $\pm$ 0.003	-0.234 $\pm$ 0.019	0.007 $\pm$ 0.000	0.083 $\pm$ 0.002	0.008 $\pm$ 0.000	0.089 $\pm$ 0.002	0.411 $\pm$ 0.019	0.641 $\pm$ 0.018	0.109	-0.030	-21.38	0.0799	0.0852	0.998	92.749 $\pm$ 0.823	0.325 $\pm$ 0.002	27.392 $\pm$ 1.415	
LMS	beta34	0.101	0.109	0.008 $\pm$ 0.002	0.081 $\pm$ 0.017	0.003 $\pm$ 0.000	0.055 $\pm$ 0.001	0.003 $\pm$ 0.000	0.055 $\pm$ 0.002	0.298 $\pm$ 0.015	0.546 $\pm$ 0.016	0.108	0.007	6.64	0.0515	0.0519	1.011	95.065 $\pm$ 0.687	0.216 $\pm$ 0.002	52.367 $\pm$ 1.585	
LMS	beta35	0.101	0.108	0.007 $\pm$ 0.002	0.071 $\pm$ 0.017	0.003 $\pm$ 0.000	0.055 $\pm$ 0.001	0.003 $\pm$ 0.000	0.056 $\pm$ 0.002	0.305 $\pm$ 0.014	0.553 $\pm$ 0.016	0.107	0.006	6.37	0.0527	0.0531	0.996	95.065 $\pm$ 0.687	0.216 $\pm$ 0.002	51.964 $\pm$ 1.585	
LSAM	beta31	0.316	0.319	0.003 $\pm$ 0.002	0.009 $\pm$ 0.007	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	0.044 $\pm$ 0.002	0.210 $\pm$ 0.006	0.318	0.002	0.59	0.0633	0.0634	1.016	95.670 $\pm$ 0.646	0.265 $\pm$ 0.001	99.496 $\pm$ 0.225	
LSAM	beta32	0.316	0.316	-0.000 $\pm$ 0.002	-0.001 $\pm$ 0.006	0.004 $\pm$ 0.000	0.064 $\pm$ 0.001	0.004 $\pm$ 0.000	0.064 $\pm$ 0.002	0.041 $\pm$ 0.002	0.202 $\pm$ 0.005	0.314	-0.002	-0.68	0.0673	0.0674	1.051	96.677 $\pm$ 0.569	0.263 $\pm$ 0.001	99.799 $\pm$ 0.142	
LSAM	beta33	0.139	0.139	-0.000 $\pm$ 0.003	-0.002 $\pm$ 0.021	0.008 $\pm$ 0.000	0.092 $\pm$ 0.002	0.008 $\pm$ 0.000	0.091 $\pm$ 0.003	0.433 $\pm$ 0.022	0.658 $\pm$ 0.020	0.140	0.001	0.58	0.0877	0.0877	1.007	96.777 $\pm$ 0.560	0.361 $\pm$ 0.003	37.362 $\pm$ 1.535	
LSAM	beta34	0.101	0.106	0.005 $\pm$ 0.002	0.053 $\pm$ 0.020	0.004 $\pm$ 0.000	0.065 $\pm$ 0.002	0.004 $\pm$ 0.000	0.065 $\pm$ 0.002	0.417 $\pm$ 0.023	0.646 $\pm$ 0.020	0.106	0.005	4.97	0.0599	0.0601	1.005	95.569 $\pm$ 0.653	0.256 $\pm$ 0.002	38.973 $\pm$ 1.548	
LSAM	beta35	0.101	0.107	0.006 $\pm$ 0.002	0.061 $\pm$ 0.021	0.004 $\pm$ 0.000	0.067 $\pm$ 0.002	0.005 $\pm$ 0.000	0.067 $\pm$ 0.002	0.441 $\pm$ 0.023	0.664 $\pm$ 0.020	0.106	0.005	5.05	0.0663	0.0665	0.992	95.972 $\pm$ 0.624	0.260 $\pm$ 0.002	38.872 $\pm$ 1.547	
QML	beta31	0.316	0.297	-0.019 $\pm$ 0.002	-0.061 $\pm$ 0.006	0.003 $\pm$ 0.000	0.058 $\pm$ 0.001	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.037 $\pm$ 0.002	0.193 $\pm$ 0.005	0.297	-0.019	-6.14	0.0567	0.0599	1.018	93.555 $\pm$ 0.779	0.231 $\pm$ 0.001	100.000 $\pm$ 0.000	
QML	beta32	0.316	0.295	-0.021 $\pm$ 0.002	-0.067 $\pm$ 0.006	0.003 $\pm$ 0.000	0.056 $\pm$ 0.001	0.004 $\pm$ 0.000	0.060 $\pm$ 0.001	0.036 $\pm$ 0.001	0.190 $\pm$ 0.005	0.293	-0.023	-7.36	0.0597	0.0640	1.052	94.260 $\pm$ 0.738	0.231 $\pm$ 0.001	99.899 $\pm$ 0.101	
QML	beta33	0.139	0.108	-0.031 $\pm$ 0.003	-0.221 $\pm$ 0.019	0.007 $\pm$ 0.000	0.083 $\pm$ 0.002	0.008 $\pm$ 0.000	0.089 $\pm$ 0.002	0.406 $\pm$ 0.019	0.638 $\pm$ 0.017	0.111	-0.028	-19.98	0.0797	0.0844	0.998	93.253 $\pm$ 0.796	0.325 $\pm$ 0.002	29.104 $\pm$ 1.441	
QML	beta34	0.101	0.108	0.007 $\pm$ 0.002	0.069 $\pm$ 0.017	0.003 $\pm$ 0.000	0.055 $\pm$ 0.001	0.003 $\pm$ 0.000	0.055 $\pm$ 0.002	0.296 $\pm$ 0.015	0.544 $\pm$ 0.016	0.106	0.005	5.37	0.0515	0.0518	1.010	94.763 $\pm$ 0.707	0.216 $\pm$ 0.002	51.561 $\pm$ 1.586	
QML	beta35	0.101	0.107	0.006 $\pm$ 0.002	0.060 $\pm$ 0.017	0.003 $\pm$ 0.000	0.056 $\pm$ 0.001	0.003 $\pm$ 0.000	0.056 $\pm$ 0.002	0.305 $\pm$ 0.014	0.553 $\pm$ 0.016	0.107	0.006	6.04	0.0527	0.0531	0.993	95.065 $\pm$ 0.687	0.216 $\pm$ 0.002	51.561 $\pm$ 1.586	
UPI	beta31	0.316	0.319	0.003 $\pm$ 0.002	0.008 $\pm$ 0.007	0.005 $\pm$ 0.000	0.067 $\pm$ 0.002	0.005 $\pm$ 0.000	0.067 $\pm$ 0.002	0.045 $\pm$ 0.002	0.213 $\pm$ 0.006	0.318	0.002	0.70	0.0647	0.0647	1.023	95.770 $\pm$ 0.639	0.270 $\pm$ 0.001	99.396 $\pm$ 0.246	
UPI	beta32	0.316	0.316	0.000 $\pm$ 0.002	0.001 $\pm$ 0.006	0.004 $\pm$ 0.000	0.064 $\pm$ 0.001	0.004 $\pm$ 0.000	0.064 $\pm$ 0.002	0.041 $\pm$ 0.002	0.202 $\pm$ 0.005	0.314	-0.002	-0.63	0.0665	0.0665	1.069	97.080 $\pm$ 0.534	0.268 $\pm$ 0.001	99.899 $\pm$ 0.101	
UPI	beta33	0.139	0.142	0.003 $\pm$ 0.003	0.022 $\pm$ 0.022	0.009 $\pm$ 0.000	0.097 $\pm$ 0.003	0.009 $\pm$ 0.000	0.097 $\pm$ 0.003	0.487 $\pm$ 0.025	0.698 $\pm$ 0.021	0.140	0.001	0.82	0.0905	0.0905	1.028	97.281 $\pm$ 0.516	0.391 $\pm$ 0.004	33.132 $\pm$ 1.494	
UPI	beta34	0.101	0.108	0.007 $\pm$ 0.002	0.068 $\pm$ 0.023	0.005 $\pm$ 0.000	0.073 $\pm$ 0.002	0.005 $\pm$ 0.000	0.073 $\pm$ 0.002	0.524 $\pm$ 0.029	0.724 $\pm$ 0.023	0.103	0.002	2.25	0.0685	0.0686	1.012	97.080 $\pm$ 0.534	0.289 $\pm$ 0.003	31.420 $\pm$ 1.473	
UPI	beta35	0.101	0.106	0.005 $\pm$ 0.002	0.050 $\pm$ 0.023	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.546 $\pm$ 0.030	0.739 $\pm$ 0.023	0.102	0.001	0.97	0.0691	0.0691	0.994	95.267 $\pm$ 0.674	0.290 $\pm$ 0.004	30.211 $\pm$ 1.457	

Note. MCSE = Monte Carlo Standard Error shown as $\backslash u00b1$ value. Coverage, Type I Error, and Power expressed as percentages. $\backslash u2014$ indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S3

Simulation 1 - Condition 3: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.6

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LMS	beta31	0.316	0.307	-0.009 $\pm$ 0.002	-0.028 $\pm$ 0.007	0.004 $\pm$ 0.000	0.067 $\pm$ 0.002	0.005 $\pm$ 0.000	0.067 $\pm$ 0.002	0.045 $\pm$ 0.002	0.213 $\pm$ 0.006	0.305	-0.011	-3.36	0.0653	0.0662	1.042	95.691 $\pm$ 0.643	0.272 $\pm$ 0.001	99.599 $\pm$ 0.200
LMS	beta32	0.316	0.307	-0.009 $\pm$ 0.002	-0.028 $\pm$ 0.007	0.005 $\pm$ 0.000	0.069 $\pm$ 0.002	0.005 $\pm$ 0.000	0.070 $\pm$ 0.002	0.049 $\pm$ 0.002	0.221 $\pm$ 0.006	0.303	-0.013	-3.97	0.0700	0.0711	1.004	95.291 $\pm$ 0.671	0.272 $\pm$ 0.001	99.299 $\pm$ 0.264
LMS	beta33	0.139	0.122	-0.017 $\pm$ 0.003	-0.123 $\pm$ 0.021	0.009 $\pm$ 0.000	0.094 $\pm$ 0.002	0.009 $\pm$ 0.000	0.095 $\pm$ 0.003	0.470 $\pm$ 0.024	0.686 $\pm$ 0.020	0.120	-0.019	-13.34	0.0848	0.0868	0.914	91.784 $\pm$ 0.869	0.336 $\pm$ 0.003	33.467 $\pm$ 1.494
LMS	beta34	0.101	0.104	0.003 $\pm$ 0.002	0.034 $\pm$ 0.019	0.004 $\pm$ 0.000	0.061 $\pm$ 0.001	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.365 $\pm$ 0.017	0.604 $\pm$ 0.017	0.104	0.003	3.22	0.0622	0.0623	0.949	94.289 $\pm$ 0.735	0.227 $\pm$ 0.002	45.992 $\pm$ 1.578
LMS	beta35	0.101	0.103	0.002 $\pm$ 0.002	0.022 $\pm$ 0.019	0.004 $\pm$ 0.000	0.061 $\pm$ 0.001	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.367 $\pm$ 0.017	0.606 $\pm$ 0.017	0.102	0.001	0.95	0.0575	0.0575	0.952	94.188 $\pm$ 0.741	0.228 $\pm$ 0.002	45.391 $\pm$ 1.576
LSAM	beta31	0.316	0.320	0.004 $\pm$ 0.002	0.012 $\pm$ 0.007	0.005 $\pm$ 0.000	0.072 $\pm$ 0.002	0.005 $\pm$ 0.000	0.072 $\pm$ 0.002	0.051 $\pm$ 0.002	0.227 $\pm$ 0.006	0.319	0.003	1.09	0.0715	0.0716	1.043	95.992 $\pm$ 0.621	0.293 $\pm$ 0.001	98.998 $\pm$ 0.315
LSAM	beta32	0.316	0.319	0.003 $\pm$ 0.002	0.010 $\pm$ 0.007	0.006 $\pm$ 0.000	0.074 $\pm$ 0.002	0.006 $\pm$ 0.000	0.074 $\pm$ 0.002	0.055 $\pm$ 0.003	0.235 $\pm$ 0.007	0.315	-0.001	-0.32	0.0746	0.0746	1.006	95.691 $\pm$ 0.643	0.293 $\pm$ 0.001	98.898 $\pm$ 0.330
LSAM	beta33	0.139	0.141	0.002 $\pm$ 0.003	0.016 $\pm$ 0.023	0.010 $\pm$ 0.001	0.100 $\pm$ 0.003	0.010 $\pm$ 0.001	0.100 $\pm$ 0.003	0.513 $\pm$ 0.028	0.716 $\pm$ 0.022	0.141	0.002	1.46	0.0886	0.0886	0.906	92.585 $\pm$ 0.829	0.354 $\pm$ 0.003	38.477 $\pm$ 1.540
LSAM	beta34	0.101	0.104	0.003 $\pm$ 0.002	0.029 $\pm$ 0.021	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.448 $\pm$ 0.022	0.670 $\pm$ 0.020	0.103	0.002	2.25	0.0664	0.0664	0.942	94.990 $\pm$ 0.691	0.250 $\pm$ 0.002	40.882 $\pm$ 1.556
LSAM	beta35	0.101	0.103	0.002 $\pm$ 0.002	0.024 $\pm$ 0.021	0.004 $\pm$ 0.000	0.067 $\pm$ 0.002	0.004 $\pm$ 0.000	0.067 $\pm$ 0.002	0.434 $\pm$ 0.021	0.659 $\pm$ 0.019	0.100	-0.001	-0.54	0.0622	0.0622	0.956	94.389 $\pm$ 0.728	0.249 $\pm$ 0.002	39.679 $\pm$ 1.549
QML	beta31	0.316	0.307	-0.009 $\pm$ 0.002	-0.028 $\pm$ 0.007	0.004 $\pm$ 0.000	0.067 $\pm$ 0.002	0.005 $\pm$ 0.000	0.067 $\pm$ 0.002	0.045 $\pm$ 0.002	0.213 $\pm$ 0.006	0.305	-0.011	-3.38	0.0653	0.0661	1.042	95.691 $\pm$ 0.643	0.273 $\pm$ 0.001	99.599 $\pm$ 0.200
QML	beta32	0.316	0.307	-0.009 $\pm$ 0.002	-0.028 $\pm$ 0.007	0.005 $\pm$ 0.000	0.069 $\pm$ 0.002	0.005 $\pm$ 0.000	0.070 $\pm$ 0.002	0.049 $\pm$ 0.002	0.221 $\pm$ 0.006	0.304	-0.012	-3.78	0.0692	0.0703	1.007	94.990 $\pm$ 0.691	0.273 $\pm$ 0.001	99.299 $\pm$ 0.264
QML	beta33	0.139	0.123	-0.016 $\pm$ 0.003	-0.118 $\pm$ 0.021	0.009 $\pm$ 0.000	0.094 $\pm$ 0.002	0.009 $\pm$ 0.000	0.095 $\pm$ 0.003	0.467 $\pm$ 0.024	0.684 $\pm$ 0.020	0.121	-0.018	-13.04	0.0854	0.0873	0.916	91.884 $\pm$ 0.864	0.336 $\pm$ 0.002	33.567 $\pm$ 1.495
QML	beta34	0.101	0.104	0.003 $\pm$ 0.002	0.027 $\pm$ 0.019	0.004 $\pm$ 0.000	0.061 $\pm$ 0.001	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.363 $\pm$ 0.016	0.602 $\pm$ 0.017	0.104	0.003	2.68	0.0627	0.0628	0.951	94.188 $\pm$ 0.741	0.227 $\pm$ 0.002	45.892 $\pm$ 1.577
QML	beta35	0.101	0.103	0.002 $\pm$ 0.002	0.015 $\pm$ 0.019	0.004 $\pm$ 0.000	0.061 $\pm$ 0.001	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.367 $\pm$ 0.017	0.606 $\pm$ 0.017	0.101	0.000	0.32	0.0572	0.0572	0.951	94.188 $\pm$ 0.741	0.228 $\pm$ 0.002	45.291 $\pm$ 1.576
UPI	beta31	0.316	0.319	0.003 $\pm$ 0.002	0.010 $\pm$ 0.007	0.005 $\pm$ 0.000	0.072 $\pm$ 0.002	0.005 $\pm$ 0.000	0.072 $\pm$ 0.002	0.052 $\pm$ 0.002	0.228 $\pm$ 0.006	0.318	0.002	0.77	0.0720	0.0721	1.021	96.293 $\pm$ 0.598	0.289 $\pm$ 0.001	98.798 $\pm$ 0.345
UPI	beta32	0.316	0.319	0.003 $\pm$ 0.002	0.011 $\pm$ 0.007	0.005 $\pm$ 0.000	0.074 $\pm$ 0.002	0.006 $\pm$ 0.000	0.074 $\pm$ 0.002	0.055 $\pm$ 0.003	0.235 $\pm$ 0.007	0.316	0.000	0.08	0.0722	0.0722	0.994	95.391 $\pm$ 0.664	0.289 $\pm$ 0.001	99.098 $\pm$ 0.299
UPI	beta33	0.139	0.143	0.004 $\pm$ 0.003	0.032 $\pm$ 0.023	0.011 $\pm$ 0.001	0.103 $\pm$ 0.003	0.011 $\pm$ 0.001	0.103 $\pm$ 0.003	0.547 $\pm$ 0.030	0.740 $\pm$ 0.023	0.141	0.002	1.79	0.0952	0.0952	0.926	93.587 $\pm$ 0.775	0.373 $\pm$ 0.003	35.972 $\pm$ 1.519
UPI	beta34	0.101	0.104	0.003 $\pm$ 0.002	0.031 $\pm$ 0.022	0.005 $\pm$ 0.000	0.072 $\pm$ 0.002	0.005 $\pm$ 0.000	0.072 $\pm$ 0.002	0.503 $\pm$ 0.025	0.709 $\pm$ 0.021	0.104	0.003	2.84	0.0695	0.0696	0.957	96.092 $\pm$ 0.613	0.268 $\pm$ 0.003	34.669 $\pm$ 1.506
UPI	beta35	0.101	0.103	0.002 $\pm$ 0.002	0.017 $\pm$ 0.022	0.005 $\pm$ 0.000	0.069 $\pm$ 0.002	0.005 $\pm$ 0.000	0.069 $\pm$ 0.002	0.469 $\pm$ 0.022	0.685 $\pm$ 0.020	0.100	-0.001	-1.07	0.0633	0.0634	0.998	94.890 $\pm$ 0.697	0.271 $\pm$ 0.003	33.066 $\pm$ 1.489

Note. MCSE = Monte Carlo Standard Error shown as $\backslash 00b1$ value. Coverage, Type I Error, and Power expressed as percentages. $\backslash 2014$ indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S4

Simulation 1 - Condition 4: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.6

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LMS	beta31	0.316	0.304	-0.012 $\pm$ 0.001	-0.038 $\pm$ 0.004	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.019 $\pm$ 0.001	0.140 $\pm$ 0.004	0.304	-0.012	-3.73	0.0407	0.0423	1.031	93.900 $\pm$ 0.757	0.172 $\pm$ 0.001	100.000 $\pm$ 0.000
LMS	beta32	0.316	0.306	-0.010 $\pm$ 0.001	-0.033 $\pm$ 0.004	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.021 $\pm$ 0.001	0.144 $\pm$ 0.004	0.304	-0.012	-3.65	0.0453	0.0468	0.990	93.800 $\pm$ 0.763	0.171 $\pm$ 0.001	99.900 $\pm$ 0.100
LMS	beta33	0.139	0.120	-0.019 $\pm$ 0.002	-0.135 $\pm$ 0.012	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.003 $\pm$ 0.000	0.056 $\pm$ 0.001	0.160 $\pm$ 0.007	0.400 $\pm$ 0.010	0.118	-0.021	-14.91	0.0526	0.0566	1.015	94.000 $\pm$ 0.751	0.208 $\pm$ 0.001	61.500 $\pm$ 1.539
LMS	beta34	0.101	0.106	0.005 $\pm$ 0.001	0.046 $\pm$ 0.011	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.128 $\pm$ 0.006	0.357 $\pm$ 0.010	0.105	0.004	3.76	0.0344	0.0346	1.005	94.400 $\pm$ 0.727	0.141 $\pm$ 0.001	85.300 $\pm$ 1.120
LMS	beta35	0.101	0.105	0.004 $\pm$ 0.001	0.037 $\pm$ 0.011	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.126 $\pm$ 0.006	0.355 $\pm$ 0.010	0.106	0.005	4.48	0.0364	0.0366	1.011	95.300 $\pm$ 0.669	0.141 $\pm$ 0.001	82.500 $\pm$ 1.202
LSAM	beta31	0.316	0.316	-0.000 $\pm$ 0.001	-0.001 $\pm$ 0.004	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.020 $\pm$ 0.001	0.142 $\pm$ 0.004	0.315	-0.001	-0.19	0.0426	0.0426	1.032	95.000 $\pm$ 0.689	0.181 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta32	0.316	0.317	0.001 $\pm$ 0.001	0.003 $\pm$ 0.005	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.022 $\pm$ 0.001	0.147 $\pm$ 0.004	0.315	-0.001	-0.31	0.0486	0.0486	0.992	95.600 $\pm$ 0.649	0.181 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta33	0.139	0.137	-0.002 $\pm$ 0.002	-0.011 $\pm$ 0.012	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.144 $\pm$ 0.006	0.379 $\pm$ 0.010	0.137	-0.002	-1.69	0.0526	0.0527	1.017	95.400 $\pm$ 0.662	0.210 $\pm$ 0.001	73.000 $\pm$ 1.404
LSAM	beta34	0.101	0.103	0.002 $\pm$ 0.001	0.020 $\pm$ 0.012	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.143 $\pm$ 0.007	0.378 $\pm$ 0.011	0.101	0.000	0.15	0.0382	0.0382	0.998	94.500 $\pm$ 0.721	0.149 $\pm$ 0.001	79.000 $\pm$ 1.288
LSAM	beta35	0.101	0.102	0.001 $\pm$ 0.001	0.012 $\pm$ 0.012	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.142 $\pm$ 0.006	0.377 $\pm$ 0.010	0.102	0.001	0.84	0.0390	0.0390	1.004	95.000 $\pm$ 0.689	0.150 $\pm$ 0.001	76.300 $\pm$ 1.345
QML	beta31	0.316	0.304	-0.012 $\pm$ 0.001	-0.037 $\pm$ 0.004	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.019 $\pm$ 0.001	0.140 $\pm$ 0.004	0.304	-0.012	-3.67	0.0406	0.0422	1.027	93.800 $\pm$ 0.763	0.171 $\pm$ 0.000	100.000 $\pm$ 0.000
QML	beta32	0.316	0.306	-0.010 $\pm$ 0.001	-0.033 $\pm$ 0.004	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.021 $\pm$ 0.001	0.144 $\pm$ 0.004	0.304	-0.012	-3.72	0.0455	0.0470	0.988	93.700 $\pm$ 0.768	0.171 $\pm$ 0.000	100.000 $\pm$ 0.000
QML	beta33	0.139	0.121	-0.018 $\pm$ 0.002	-0.132 $\pm$ 0.012	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.003 $\pm$ 0.000	0.056 $\pm$ 0.001	0.160 $\pm$ 0.007	0.399 $\pm$ 0.010	0.119	-0.020	-14.33	0.0529	0.0565	1.013	94.200 $\pm$ 0.739	0.208 $\pm$ 0.001	61.900 $\pm$ 1.536
QML	beta34	0.101	0.105	0.004 $\pm$ 0.001	0.040 $\pm$ 0.011	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.127 $\pm$ 0.006	0.357 $\pm$ 0.010	0.104	0.003	3.13	0.0344	0.0345	1.003	94.400 $\pm$ 0.727	0.141 $\pm$ 0.001	85.000 $\pm$ 1.129
QML	beta35	0.101	0.104	0.003 $\pm$ 0.001	0.032 $\pm$ 0.011	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.126 $\pm$ 0.006	0.355 $\pm$ 0.010	0.105	0.004	4.07	0.0367	0.0369	1.010	95.300 $\pm$ 0.669	0.141 $\pm$ 0.001	81.900 $\pm$ 1.218
UPI	beta31	0.316	0.316	-0.000 $\pm$ 0.001	-0.001 $\pm$ 0.004	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.020 $\pm$ 0.001	0.142 $\pm$ 0.004	0.316	0.000	0.04	0.0426	0.0426	1.010	94.800 $\pm$ 0.702	0.178 $\pm$ 0.000	100.000 $\pm$ 0.000
UPI	beta32	0.316	0.317	0.001 $\pm$ 0.001	0.002 $\pm$ 0.005	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.022 $\pm$ 0.001	0.148 $\pm$ 0.004	0.315	-0.001	-0.20	0.0484	0.0484	0.971	94.700 $\pm$ 0.708	0.178 $\pm$ 0.000	100.000 $\pm$ 0.000
UPI	beta33	0.139	0.138	-0.001 $\pm$ 0.002	-0.005 $\pm$ 0.012	0.003 $\pm$ 0.000	0.055 $\pm$ 0.001	0.003 $\pm$ 0.000	0.054 $\pm$ 0.002	0.154 $\pm$ 0.007	0.392 $\pm$ 0.011	0.138	-0.001	-0.92	0.0527	0.0527	1.008	95.300 $\pm$ 0.669	0.215 $\pm$ 0.001	72.600 $\pm$ 1.410
UPI	beta34	0.101	0.102	0.001 $\pm$ 0.001	0.012 $\pm$ 0.012	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.150 $\pm$ 0.007	0.387 $\pm$ 0.011	0.101	0.000	-0.06	0.0379	0.0379	1.006	95.500 $\pm$ 0.656	0.154 $\pm$ 0.001	75.800 $\pm$ 1.354
UPI	beta35	0.101	0.102	0.001 $\pm$ 0.001	0.009 $\pm$ 0.012	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.148 $\pm$ 0.007	0.384 $\pm$ 0.011	0.102	0.001	0.76	0.0384	0.0384	1.020	95.300 $\pm$ 0.669	0.155 $\pm$ 0.001	74.000 $\pm$ 1.387

Note. MCSE = Monte Carlo Standard Error shown as $\backslash 00b1$ value. Coverage, Type I Error, and Power expressed as percentages. $\backslash 2014$ indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S5

Simulation 1 - Condition 5: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.8

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LMS	beta31	0.316	0.315	-0.001 $\pm$ 0.002	-0.004 $\pm$ 0.006	0.003 $\pm$ 0.000	0.059 $\pm$ 0.001	0.003 $\pm$ 0.000	0.059 $\pm$ 0.002	0.035 $\pm$ 0.002	0.187 $\pm$ 0.005	0.313	-0.003	-0.85	0.0569	0.0570	0.970	94.100 $\pm$ 0.745	0.225 $\pm$ 0.001	100.000 $\pm$ 0.000
LMS	beta32	0.316	0.310	-0.006 $\pm$ 0.002	-0.018 $\pm$ 0.006	0.003 $\pm$ 0.000	0.059 $\pm$ 0.001	0.003 $\pm$ 0.000	0.059 $\pm$ 0.002	0.035 $\pm$ 0.001	0.186 $\pm$ 0.005	0.311	-0.005	-1.72	0.0581	0.0584	0.982	93.400 $\pm$ 0.785	0.225 $\pm$ 0.001	100.000 $\pm$ 0.000
LMS	beta33	0.139	0.134	-0.005 $\pm$ 0.002	-0.033 $\pm$ 0.015	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.237 $\pm$ 0.010	0.487 $\pm$ 0.013	0.134	-0.005	-3.57	0.0673	0.0675	0.936	93.100 $\pm$ 0.801	0.248 $\pm$ 0.001	55.600 $\pm$ 1.571
LMS	beta34	0.101	0.102	0.001 $\pm$ 0.001	0.014 $\pm$ 0.014	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.203 $\pm$ 0.009	0.451 $\pm$ 0.012	0.101	0.000	-0.00	0.0452	0.0452	0.961	93.600 $\pm$ 0.774	0.171 $\pm$ 0.001	64.100 $\pm$ 1.517
LMS	beta35	0.101	0.101	0.000 $\pm$ 0.001	0.001 $\pm$ 0.014	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.206 $\pm$ 0.009	0.453 $\pm$ 0.012	0.101	0.000	0.49	0.0454	0.0454	0.955	93.800 $\pm$ 0.763	0.171 $\pm$ 0.001	63.600 $\pm$ 1.522
LSAM	beta31	0.316	0.320	0.004 $\pm$ 0.002	0.012 $\pm$ 0.006	0.004 $\pm$ 0.000	0.060 $\pm$ 0.001	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.037 $\pm$ 0.002	0.192 $\pm$ 0.005	0.317	0.001	0.43	0.0603	0.0603	0.972	94.300 $\pm$ 0.733	0.230 $\pm$ 0.001	100.000 $\pm$ 0.000
LSAM	beta32	0.316	0.316	-0.000 $\pm$ 0.002	-0.001 $\pm$ 0.006	0.004 $\pm$ 0.000	0.060 $\pm$ 0.001	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	0.036 $\pm$ 0.002	0.189 $\pm$ 0.005	0.316	0.000	-0.09	0.0595	0.0595	0.986	94.200 $\pm$ 0.739	0.230 $\pm$ 0.001	100.000 $\pm$ 0.000
LSAM	beta33	0.139	0.142	0.003 $\pm$ 0.002	0.020 $\pm$ 0.016	0.005 $\pm$ 0.000	0.069 $\pm$ 0.002	0.005 $\pm$ 0.000	0.069 $\pm$ 0.002	0.244 $\pm$ 0.011	0.494 $\pm$ 0.014	0.142	0.003	2.17	0.0686	0.0687	0.927	93.000 $\pm$ 0.807	0.249 $\pm$ 0.001	59.400 $\pm$ 1.553
LSAM	beta34	0.101	0.102	0.001 $\pm$ 0.001	0.012 $\pm$ 0.015	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.218 $\pm$ 0.010	0.467 $\pm$ 0.013	0.100	-0.001	-0.77	0.0475	0.0475	0.952	93.100 $\pm$ 0.801	0.176 $\pm$ 0.001	62.000 $\pm$ 1.535
LSAM	beta35	0.101	0.101	-0.000 $\pm$ 0.002	-0.002 $\pm$ 0.015	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.222 $\pm$ 0.010	0.471 $\pm$ 0.013	0.101	0.000	-0.11	0.0478	0.0478	0.943	93.700 $\pm$ 0.768	0.176 $\pm$ 0.001	61.200 $\pm$ 1.541
QML	beta31	0.316	0.314	-0.002 $\pm$ 0.002	-0.005 $\pm$ 0.006	0.003 $\pm$ 0.000	0.059 $\pm$ 0.001	0.003 $\pm$ 0.000	0.059 $\pm$ 0.002	0.035 $\pm$ 0.002	0.187 $\pm$ 0.005	0.313	-0.003	-0.86	0.0567	0.0568	0.968	94.100 $\pm$ 0.745	0.224 $\pm$ 0.001	100.000 $\pm$ 0.000
QML	beta32	0.316	0.310	-0.006 $\pm$ 0.002	-0.019 $\pm$ 0.006	0.003 $\pm$ 0.000	0.059 $\pm$ 0.001	0.003 $\pm$ 0.000	0.059 $\pm$ 0.002	0.035 $\pm$ 0.001	0.186 $\pm$ 0.005	0.311	-0.005	-1.71	0.0583	0.0585	0.979	93.300 $\pm$ 0.791	0.225 $\pm$ 0.001	100.000 $\pm$ 0.000
QML	beta33	0.139	0.134	-0.005 $\pm$ 0.002	-0.035 $\pm$ 0.015	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.237 $\pm$ 0.010	0.487 $\pm$ 0.013	0.134	-0.005	-3.60	0.0669	0.0670	0.936	93.200 $\pm$ 0.796	0.248 $\pm$ 0.001	55.600 $\pm$ 1.571
QML	beta34	0.101	0.102	0.001 $\pm$ 0.001	0.014 $\pm$ 0.014	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.203 $\pm$ 0.009	0.451 $\pm$ 0.012	0.101	0.000	-0.27	0.0454	0.0454	0.960	93.400 $\pm$ 0.785	0.171 $\pm$ 0.001	64.000 $\pm$ 1.518
QML	beta35	0.101	0.101	0.000 $\pm$ 0.001	0.000 $\pm$ 0.014	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.206 $\pm$ 0.009	0.454 $\pm$ 0.012	0.102	0.001	0.52	0.0451	0.0451	0.953	93.700 $\pm$ 0.768	0.171 $\pm$ 0.001	63.600 $\pm$ 1.522
UPI	beta31	0.316	0.320	0.004 $\pm$ 0.002	0.012 $\pm$ 0.006	0.004 $\pm$ 0.000	0.060 $\pm$ 0.001	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.037 $\pm$ 0.002	0.192 $\pm$ 0.005	0.318	0.002	0.66	0.0598	0.0598	0.939	93.500 $\pm$ 0.780	0.223 $\pm$ 0.001	100.000 $\pm$ 0.000
UPI	beta32	0.316	0.315	-0.001 $\pm$ 0.002	-0.002 $\pm$ 0.006	0.004 $\pm$ 0.000	0.060 $\pm$ 0.001	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	0.036 $\pm$ 0.002	0.189 $\pm$ 0.005	0.316	0.000	-0.00	0.0586	0.0586	0.949	93.200 $\pm$ 0.796	0.223 $\pm$ 0.001	100.000 $\pm$ 0.000
UPI	beta33	0.139	0.142	0.003 $\pm$ 0.002	0.025 $\pm$ 0.016	0.005 $\pm$ 0.000	0.069 $\pm$ 0.002	0.005 $\pm$ 0.000	0.070 $\pm$ 0.002	0.250 $\pm$ 0.011	0.500 $\pm$ 0.014	0.141	0.002	1.56	0.0704	0.0705	0.927	93.100 $\pm$ 0.801	0.252 $\pm$ 0.001	58.800 $\pm$ 1.556
UPI	beta34	0.101	0.102	0.001 $\pm$ 0.002	0.005 $\pm$ 0.015	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.221 $\pm$ 0.010	0.470 $\pm$ 0.013	0.099	-0.002	-2.21	0.0454	0.0454	0.961	93.400 $\pm$ 0.785	0.179 $\pm$ 0.001	60.300 $\pm$ 1.547
UPI	beta35	0.101	0.101	-0.000 $\pm$ 0.002	-0.002 $\pm$ 0.015	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.236 $\pm$ 0.011	0.486 $\pm$ 0.014	0.100	-0.001	-0.75	0.0484	0.0484	0.933	94.100 $\pm$ 0.745	0.180 $\pm$ 0.001	60.200 $\pm$ 1.548

Note. MCSE = Monte Carlo Standard Error shown as $\backslash u00b1$ value. Coverage, Type I Error, and Power expressed as percentages. $\backslash u2014$ indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S6

Simulation 1 - Condition 6: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.8

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LMS	beta31	0.316	0.311	-0.005 $\pm$ 0.001	-0.016 $\pm$ 0.004	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.001 $\pm$ 0.000	0.039 $\pm$ 0.001	0.015 $\pm$ 0.001	0.122 $\pm$ 0.003	0.312	-0.004	-1.19	0.0393	0.0395	0.948	93.500 $\pm$ 0.780	0.143 $\pm$ 0.000	100.000 $\pm$ 0.000
LMS	beta32	0.316	0.311	-0.005 $\pm$ 0.001	-0.016 $\pm$ 0.004	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.014 $\pm$ 0.001	0.119 $\pm$ 0.003	0.310	-0.006	-1.95	0.0372	0.0377	0.972	94.400 $\pm$ 0.727	0.143 $\pm$ 0.000	100.000 $\pm$ 0.000
LMS	beta33	0.139	0.134	-0.005 $\pm$ 0.001	-0.037 $\pm$ 0.009	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.082 $\pm$ 0.004	0.287 $\pm$ 0.008	0.131	-0.008	-5.58	0.0384	0.0392	1.015	95.600 $\pm$ 0.649	0.157 $\pm$ 0.001	92.500 $\pm$ 0.833
LMS	beta34	0.101	0.102	0.001 $\pm$ 0.001	0.008 $\pm$ 0.009	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.081 $\pm$ 0.004	0.285 $\pm$ 0.008	0.101	0.000	-0.23	0.0273	0.0273	0.967	93.200 $\pm$ 0.796	0.109 $\pm$ 0.000	94.100 $\pm$ 0.745
LMS	beta35	0.101	0.101	0.000 $\pm$ 0.001	0.003 $\pm$ 0.009	0.001 $\pm$ 0.000	0.028 $\pm$ 0.001	0.001 $\pm$ 0.000	0.028 $\pm$ 0.001	0.076 $\pm$ 0.004	0.276 $\pm$ 0.008	0.102	0.001	0.81	0.0276	0.0276	0.992	95.600 $\pm$ 0.649	0.108 $\pm$ 0.000	94.800 $\pm$ 0.702
LSAM	beta31	0.316	0.316	0.000 $\pm$ 0.001	0.000 $\pm$ 0.004	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.015 $\pm$ 0.001	0.124 $\pm$ 0.003	0.316	0.000	0.10	0.0403	0.0403	0.953	93.800 $\pm$ 0.763	0.146 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta32	0.316	0.316	-0.000 $\pm$ 0.001	-0.000 $\pm$ 0.004	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.014 $\pm$ 0.001	0.120 $\pm$ 0.003	0.315	-0.001	-0.24	0.0380	0.0380	0.978	94.500 $\pm$ 0.721	0.146 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta33	0.139	0.141	0.002 $\pm$ 0.001	0.015 $\pm$ 0.009	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.080 $\pm$ 0.004	0.282 $\pm$ 0.008	0.139	0.000	-0.20	0.0376	0.0376	1.017	95.700 $\pm$ 0.641	0.156 $\pm$ 0.001	95.000 $\pm$ 0.689
LSAM	beta34	0.101	0.101	-0.000 $\pm$ 0.001	-0.003 $\pm$ 0.009	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.086 $\pm$ 0.004	0.294 $\pm$ 0.008	0.101	0.000	0.10	0.0282	0.0282	0.955	93.300 $\pm$ 0.791	0.111 $\pm$ 0.000	93.300 $\pm$ 0.791
LSAM	beta35	0.101	0.100	-0.001 $\pm$ 0.001	-0.007 $\pm$ 0.009	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.081 $\pm$ 0.004	0.284 $\pm$ 0.008	0.100	-0.001	-0.87	0.0292	0.0293	0.983	95.300 $\pm$ 0.669	0.110 $\pm$ 0.000	93.800 $\pm$ 0.763
QML	beta31	0.316	0.311	-0.005 $\pm$ 0.001	-0.016 $\pm$ 0.004	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.001 $\pm$ 0.000	0.039 $\pm$ 0.001	0.015 $\pm$ 0.001	0.122 $\pm$ 0.003	0.312	-0.004	-1.23	0.0394	0.0395	0.949	93.500 $\pm$ 0.780	0.143 $\pm$ 0.000	100.000 $\pm$ 0.000
QML	beta32	0.316	0.311	-0.005 $\pm$ 0.001	-0.016 $\pm$ 0.004	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.014 $\pm$ 0.001	0.119 $\pm$ 0.003	0.310	-0.006	-1.99	0.0372	0.0377	0.973	94.400 $\pm$ 0.727	0.143 $\pm$ 0.000	100.000 $\pm$ 0.000
QML	beta33	0.139	0.134	-0.005 $\pm$ 0.001	-0.038 $\pm$ 0.009	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.082 $\pm$ 0.004	0.287 $\pm$ 0.008	0.131	-0.008	-5.59	0.0385	0.0393	1.014	95.600 $\pm$ 0.649	0.157 $\pm$ 0.001	92.400 $\pm$ 0.838
QML	beta34	0.101	0.102	0.001 $\pm$ 0.001	0.008 $\pm$ 0.009	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.081 $\pm$ 0.004	0.285 $\pm$ 0.008	0.101	0.000	-0.24	0.0276	0.0276	0.966	93.300 $\pm$ 0.791	0.109 $\pm$ 0.000	94.100 $\pm$ 0.745
QML	beta35	0.101	0.101	0.000 $\pm$ 0.001	0.003 $\pm$ 0.009	0.001 $\pm$ 0.000	0.028 $\pm$ 0.001	0.001 $\pm$ 0.000	0.028 $\pm$ 0.001	0.076 $\pm$ 0.004	0.276 $\pm$ 0.008	0.102	0.001	0.82	0.0276	0.0276	0.991	95.300 $\pm$ 0.669	0.108 $\pm$ 0.000	94.500 $\pm$ 0.721
UPI	beta31	0.316	0.316	-0.000 $\pm$ 0.001	-0.000 $\pm$ 0.004	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.015 $\pm$ 0.001	0.124 $\pm$ 0.003	0.316	0.000	0.13	0.0400	0.0400	0.920	92.400 $\pm$ 0.838	0.141 $\pm$ 0.000	100.000 $\pm$ 0.000
UPI	beta32	0.316	0.316	-0.000 $\pm$ 0.001	-0.000 $\pm$ 0.004	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.014 $\pm$ 0.001	0.120 $\pm$ 0.003	0.315	-0.001	-0.24	0.0384	0.0385	0.948	93.800 $\pm$ 0.763	0.141 $\pm$ 0.000	100.000 $\pm$ 0.000
UPI	beta33	0.139	0.141	0.002 $\pm$ 0.001	0.017 $\pm$ 0.009	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.082 $\pm$ 0.004	0.285 $\pm$ 0.008	0.138	-0.001	-0.41	0.0384	0.0384	1.015	94.900 $\pm$ 0.696	0.158 $\pm$ 0.001	94.600 $\pm$ 0.715
UPI	beta34	0.101	0.101	-0.000 $\pm$ 0.001	-0.002 $\pm$ 0.009	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.089 $\pm$ 0.004	0.298 $\pm$ 0.009	0.100	-0.001	-0.55	0.0291	0.0291	0.953	93.700 $\pm$ 0.768	0.112 $\pm$ 0.000	92.800 $\pm$ 0.817
UPI	beta35	0.101	0.100	-0.001 $\pm$ 0.001	-0.007 $\pm$ 0.009	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.082 $\pm$ 0.004	0.286 $\pm$ 0.008	0.100	-0.001	-0.59	0.0294	0.0294	0.988	95.100 $\pm$ 0.683	0.112 $\pm$ 0.000	93.600 $\pm$ 0.774

Note. MCSE = Monte Carlo Standard Error shown as $\backslash u00b1$ value. Coverage, Type I Error, and Power expressed as percentages. $\backslash u2014$ indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S7

Simulation 1 - Condition 7: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.4

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LMS	beta31	0.316	0.344	0.028 $\pm$ 0.004	0.090 $\pm$ 0.011	0.011 $\pm$ 0.001	0.105 $\pm$ 0.003	0.012 $\pm$ 0.001	0.109 $\pm$ 0.003	0.118 $\pm$ 0.006	0.344 $\pm$ 0.011	0.341	0.025	8.07	0.1041	0.1072	1.029	95.957 $\pm$ 0.679	0.423 $\pm$ 0.003	91.201 $\pm$ 0.977
LMS	beta32	0.316	0.340	0.024 $\pm$ 0.004	0.075 $\pm$ 0.011	0.011 $\pm$ 0.001	0.104 $\pm$ 0.003	0.011 $\pm$ 0.001	0.106 $\pm$ 0.003	0.113 $\pm$ 0.006	0.336 $\pm$ 0.010	0.335	0.019	5.89	0.1080	0.1096	1.044	96.314 $\pm$ 0.650	0.424 $\pm$ 0.003	89.417 $\pm$ 1.061
LMS	beta33	0.139	0.014	-0.125 $\pm$ 0.005	-0.899 $\pm$ 0.033	0.017 $\pm$ 0.001	0.131 $\pm$ 0.003	0.033 $\pm$ 0.001	0.181 $\pm$ 0.004	1.700 $\pm$ 0.077	1.304 $\pm$ 0.032	0.014	-0.125	-89.60	0.1354	0.1839	1.060	82.996 $\pm$ 1.295	0.545 $\pm$ 0.007	4.875 $\pm$ 0.743
LMS	beta34	0.101	0.220	0.119 $\pm$ 0.003	1.177 $\pm$ 0.030	0.008 $\pm$ 0.000	0.087 $\pm$ 0.002	0.022 $\pm$ 0.001	0.147 $\pm$ 0.003	2.129 $\pm$ 0.089	1.459 $\pm$ 0.032	0.214	0.113	111.92	0.0820	0.1396	1.041	78.240 $\pm$ 1.423	0.356 $\pm$ 0.005	76.813 $\pm$ 1.455
LMS	beta35	0.101	0.218	0.117 $\pm$ 0.003	1.160 $\pm$ 0.030	0.008 $\pm$ 0.000	0.087 $\pm$ 0.003	0.021 $\pm$ 0.001	0.146 $\pm$ 0.003	2.086 $\pm$ 0.094	1.444 $\pm$ 0.034	0.213	0.112	110.46	0.0835	0.1393	1.038	79.905 $\pm$ 1.382	0.354 $\pm$ 0.005	77.170 $\pm$ 1.447
LSAM	beta31	0.316	0.321	0.005 $\pm$ 0.008	0.016 $\pm$ 0.025	0.052 $\pm$ 0.003	0.229 $\pm$ 0.007	0.052 $\pm$ 0.003	0.228 $\pm$ 0.008	0.523 $\pm$ 0.030	0.723 $\pm$ 0.024	0.321	0.005	1.71	0.2205	0.2206	1.135	98.573 $\pm$ 0.409	1.017 $\pm$ 0.024	32.699 $\pm$ 1.618
LSAM	beta32	0.316	0.319	0.003 $\pm$ 0.008	0.009 $\pm$ 0.025	0.054 $\pm$ 0.003	0.233 $\pm$ 0.007	0.054 $\pm$ 0.003	0.233 $\pm$ 0.008	0.543 $\pm$ 0.031	0.737 $\pm$ 0.025	0.312	-0.004	-1.37	0.2298	0.2298	1.163	97.979 $\pm$ 0.485	1.062 $\pm$ 0.031	31.391 $\pm$ 1.600
LSAM	beta33	0.139	0.170	0.031 $\pm$ 0.006	0.224 $\pm$ 0.045	0.033 $\pm$ 0.002	0.181 $\pm$ 0.006	0.034 $\pm$ 0.002	0.183 $\pm$ 0.007	1.737 $\pm$ 0.109	1.318 $\pm$ 0.047	0.164	0.025	18.30	0.1695	0.1714	1.111	96.076 $\pm$ 0.670	0.786 $\pm$ 0.025	11.653 $\pm$ 1.106
LSAM	beta34	0.101	0.104	0.003 $\pm$ 0.004	0.025 $\pm$ 0.035	0.011 $\pm$ 0.001	0.103 $\pm$ 0.003	0.011 $\pm$ 0.001	0.103 $\pm$ 0.004	1.040 $\pm$ 0.063	1.020 $\pm$ 0.036	0.097	-0.004	-4.41	0.0942	0.0943	1.051	97.265 $\pm$ 0.562	0.424 $\pm$ 0.011	24.495 $\pm$ 1.483
LSAM	beta35	0.101	0.101	0.000 $\pm$ 0.004	0.003 $\pm$ 0.036	0.011 $\pm$ 0.001	0.104 $\pm$ 0.003	0.011 $\pm$ 0.001	0.104 $\pm$ 0.004	1.065 $\pm$ 0.068	1.032 $\pm$ 0.038	0.101	0.000	0.48	0.0912	0.0912	1.115	97.384 $\pm$ 0.550	0.456 $\pm$ 0.019	25.684 $\pm$ 1.507
QML	beta31	0.316	0.344	0.028 $\pm$ 0.004	0.090 $\pm$ 0.012	0.011 $\pm$ 0.001	0.105 $\pm$ 0.003	0.012 $\pm$ 0.001	0.109 $\pm$ 0.003	0.119 $\pm$ 0.006	0.345 $\pm$ 0.011	0.340	0.024	7.60	0.1053	0.1080	1.029	95.600 $\pm$ 0.707	0.425 $\pm$ 0.003	90.131 $\pm$ 1.028
QML	beta32	0.316	0.340	0.024 $\pm$ 0.004	0.075 $\pm$ 0.011	0.011 $\pm$ 0.001	0.104 $\pm$ 0.003	0.011 $\pm$ 0.001	0.107 $\pm$ 0.003	0.114 $\pm$ 0.006	0.338 $\pm$ 0.010	0.335	0.019	6.15	0.1082	0.1099	1.041	96.433 $\pm$ 0.640	0.425 $\pm$ 0.003	89.298 $\pm$ 1.066
QML	beta33	0.139	0.016	-0.123 $\pm$ 0.005	-0.887 $\pm$ 0.033	0.018 $\pm$ 0.001	0.133 $\pm$ 0.004	0.033 $\pm$ 0.002	0.182 $\pm$ 0.005	1.706 $\pm$ 0.082	1.306 $\pm$ 0.034	0.017	-0.122	-87.91	0.1355	0.1825	1.057	85.018 $\pm$ 1.231	0.553 $\pm$ 0.007	4.756 $\pm$ 0.734
QML	beta34	0.101	0.220	0.119 $\pm$ 0.003	1.180 $\pm$ 0.031	0.008 $\pm$ 0.000	0.090 $\pm$ 0.003	0.022 $\pm$ 0.001	0.149 $\pm$ 0.003	2.180 $\pm$ 0.097	1.476 $\pm$ 0.034	0.213	0.112	110.73	0.0824	0.1389	1.011	78.121 $\pm$ 1.426	0.356 $\pm$ 0.005	76.219 $\pm$ 1.468
QML	beta35	0.101	0.217	0.116 $\pm$ 0.003	1.150 $\pm$ 0.030	0.008 $\pm$ 0.000	0.087 $\pm$ 0.003	0.021 $\pm$ 0.001	0.145 $\pm$ 0.003	2.066 $\pm$ 0.094	1.437 $\pm$ 0.034	0.211	0.110	108.79	0.0834	0.1379	1.027	79.667 $\pm$ 1.388	0.350 $\pm$ 0.005	77.170 $\pm$ 1.447
UPI	beta31	0.316	0.337	0.021 $\pm$ 0.008	0.066 $\pm$ 0.025	0.052 $\pm$ 0.003	0.229 $\pm$ 0.007	0.053 $\pm$ 0.003	0.229 $\pm$ 0.008	0.527 $\pm$ 0.032	0.726 $\pm$ 0.025	0.331	0.015	4.74	0.2327	0.2331	1.214	98.692 $\pm$ 0.392	1.088 $\pm$ 0.032	33.175 $\pm$ 1.624
UPI	beta32	0.316	0.323	0.007 $\pm$ 0.008	0.022 $\pm$ 0.025	0.054 $\pm$ 0.003	0.233 $\pm$ 0.007	0.054 $\pm$ 0.003	0.233 $\pm$ 0.008	0.541 $\pm$ 0.032	0.736 $\pm$ 0.025	0.320	0.004	1.12	0.2256	0.2256	1.235	99.049 $\pm$ 0.335	1.126 $\pm$ 0.038	31.272 $\pm$ 1.599
UPI	beta33	0.139	0.147	0.008 $\pm$ 0.007	0.056 $\pm$ 0.048	0.037 $\pm$ 0.003	0.193 $\pm$ 0.007	0.037 $\pm$ 0.003	0.193 $\pm$ 0.008	1.924 $\pm$ 0.134	1.387 $\pm$ 0.054	0.133	-0.006	-4.31	0.1731	0.1732	1.469	95.957 $\pm$ 0.679	1.110 $\pm$ 0.059	6.064 $\pm$ 0.823
UPI	beta34	0.101	0.101	0.000 $\pm$ 0.004	0.002 $\pm$ 0.036	0.011 $\pm$ 0.001	0.104 $\pm$ 0.003	0.011 $\pm$ 0.001	0.104 $\pm$ 0.004	1.067 $\pm$ 0.068	1.033 $\pm$ 0.037	0.095	-0.006	-6.19	0.0959	0.0961	1.163	96.433 $\pm$ 0.640	0.476 $\pm$ 0.017	20.095 $\pm$ 1.382
UPI	beta35	0.101	0.104	0.003 $\pm$ 0.004	0.028 $\pm$ 0.036	0.011 $\pm$ 0.001	0.107 $\pm$ 0.004	0.011 $\pm$ 0.001	0.107 $\pm$ 0.004	1.119 $\pm$ 0.074	1.058 $\pm$ 0.040	0.094	-0.007	-6.93	0.0908	0.0911	1.203	97.146 $\pm$ 0.574	0.504 $\pm$ 0.022	22.354 $\pm$ 1.437

Note. MCSE = Monte Carlo Standard Error shown as $\backslash$ u00b1 value. Coverage, Type I Error, and Power expressed as percentages. $\backslash$ u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S8

Simulation 1 - Condition 8: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.4

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LMS	beta31	0.316	0.340	0.024 $\pm$ 0.002	0.075 $\pm$ 0.007	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	0.005 $\pm$ 0.000	0.070 $\pm$ 0.002	0.049 $\pm$ 0.002	0.221 $\pm$ 0.006	0.339	0.023	7.40	0.0624	0.0667	1.011	94.490 $\pm$ 0.729	0.260 $\pm$ 0.001	99.796 $\pm$ 0.144
LMS	beta32	0.316	0.339	0.023 $\pm$ 0.002	0.072 $\pm$ 0.006	0.004 $\pm$ 0.000	0.064 $\pm$ 0.001	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.046 $\pm$ 0.002	0.214 $\pm$ 0.006	0.338	0.022	6.93	0.0664	0.0700	1.044	95.816 $\pm$ 0.640	0.260 $\pm$ 0.001	99.898 $\pm$ 0.102
LMS	beta33	0.139	0.010	-0.129 $\pm$ 0.003	-0.927 $\pm$ 0.020	0.007 $\pm$ 0.000	0.085 $\pm$ 0.002	0.024 $\pm$ 0.001	0.155 $\pm$ 0.003	1.236 $\pm$ 0.041	1.112 $\pm$ 0.019	0.012	-0.127	-91.52	0.0834	0.1521	0.993	63.163 $\pm$ 1.541	0.332 $\pm$ 0.008	7.143 $\pm$ 0.823
LMS	beta34	0.101	0.223	0.122 $\pm$ 0.002	1.207 $\pm$ 0.017	0.003 $\pm$ 0.000	0.055 $\pm$ 0.001	0.018 $\pm$ 0.000	0.134 $\pm$ 0.002	1.756 $\pm$ 0.048	1.325 $\pm$ 0.019	0.218	0.117	115.53	0.0537	0.1284	0.981	33.571 $\pm$ 1.509	0.213 $\pm$ 0.003	99.592 $\pm$ 0.204
LMS	beta35	0.101	0.225	0.124 $\pm$ 0.002	1.225 $\pm$ 0.016	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.018 $\pm$ 0.000	0.134 $\pm$ 0.002	1.757 $\pm$ 0.044	1.325 $\pm$ 0.017	0.224	0.123	121.47	0.0522	0.1333	1.052	31.429 $\pm$ 1.483	0.211 $\pm$ 0.003	99.694 $\pm$ 0.176
LSAM	beta31	0.316	0.317	0.001 $\pm$ 0.004	0.003 $\pm$ 0.013	0.016 $\pm$ 0.001	0.128 $\pm$ 0.003	0.016 $\pm$ 0.001	0.128 $\pm$ 0.004	0.165 $\pm$ 0.008	0.406 $\pm$ 0.012	0.313	-0.003	-0.94	0.1198	0.1199	1.008	95.612 $\pm$ 0.654	0.507 $\pm$ 0.004	70.306 $\pm$ 1.460
LSAM	beta32	0.316	0.314	-0.002 $\pm$ 0.004	-0.006 $\pm$ 0.013	0.016 $\pm$ 0.001	0.128 $\pm$ 0.003	0.016 $\pm$ 0.001	0.128 $\pm$ 0.004	0.165 $\pm$ 0.008	0.406 $\pm$ 0.012	0.307	-0.009	-2.90	0.1211	0.1214	1.006	97.041 $\pm$ 0.541	0.506 $\pm$ 0.004	72.245 $\pm$ 1.430
LSAM	beta33	0.139	0.150	0.011 $\pm$ 0.003	0.080 $\pm$ 0.021	0.008 $\pm$ 0.000	0.092 $\pm$ 0.002	0.008 $\pm$ 0.000	0.092 $\pm$ 0.003	0.440 $\pm$ 0.024	0.663 $\pm$ 0.021	0.141	0.002	1.75	0.0827	0.0827	0.974	95.102 $\pm$ 0.689	0.349 $\pm$ 0.004	38.878 $\pm$ 1.557
LSAM	beta34	0.101	0.103	0.002 $\pm$ 0.002	0.015 $\pm$ 0.016	0.003 $\pm$ 0.000	0.050 $\pm$ 0.001	0.003 $\pm$ 0.000	0.050 $\pm$ 0.001	0.245 $\pm$ 0.012	0.495 $\pm$ 0.015	0.101	0.000	-0.43	0.0473	0.0473	0.959	94.286 $\pm$ 0.741	0.188 $\pm$ 0.002	61.224 $\pm$ 1.556
LSAM	beta35	0.101	0.104	0.003 $\pm$ 0.002	0.031 $\pm$ 0.016	0.003 $\pm$ 0.000	0.050 $\pm$ 0.001	0.003 $\pm$ 0.000	0.050 $\pm$ 0.002	0.249 $\pm$ 0.013	0.499 $\pm$ 0.015	0.102	0.001	1.36	0.0472	0.0472	0.938	94.592 $\pm$ 0.723	0.185 $\pm$ 0.002	65.306 $\pm$ 1.521
QML	beta31	0.316	0.340	0.024 $\pm$ 0.002	0.076 $\pm$ 0.007	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	0.005 $\pm$ 0.000	0.070 $\pm$ 0.002	0.049 $\pm$ 0.002	0.221 $\pm$ 0.006	0.340	0.024	7.44	0.0626	0.0669	1.007	94.388 $\pm$ 0.735	0.260 $\pm$ 0.001	99.898 $\pm$ 0.102
QML	beta32	0.316	0.339	0.023 $\pm$ 0.002	0.072 $\pm$ 0.006	0.004 $\pm$ 0.000	0.064 $\pm$ 0.001	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.046 $\pm$ 0.002	0.215 $\pm$ 0.006	0.339	0.023	7.35	0.0668	0.0708	1.037	95.408 $\pm$ 0.669	0.260 $\pm$ 0.001	99.898 $\pm$ 0.102
QML	beta33	0.139	0.013	-0.126 $\pm$ 0.003	-0.905 $\pm$ 0.020	0.007 $\pm$ 0.000	0.086 $\pm$ 0.002	0.023 $\pm$ 0.001	0.152 $\pm$ 0.003	1.198 $\pm$ 0.041	1.094 $\pm$ 0.019	0.014	-0.125	-89.82	0.0833	0.1501	0.966	64.898 $\pm$ 1.525	0.324 $\pm$ 0.002	7.449 $\pm$ 0.839
QML	beta34	0.101	0.221	0.120 $\pm$ 0.002	1.193 $\pm$ 0.018	0.003 $\pm$ 0.000	0.056 $\pm$ 0.001	0.018 $\pm$ 0.000	0.133 $\pm$ 0.002	1.726 $\pm$ 0.049	1.314 $\pm$ 0.019	0.217	0.116	114.52	0.0550	0.1281	0.947	33.163 $\pm$ 1.504	0.207 $\pm$ 0.002	99.592 $\pm$ 0.204
QML	beta35	0.101	0.223	0.122 $\pm$ 0.002	1.210 $\pm$ 0.016	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.018 $\pm$ 0.000	0.132 $\pm$ 0.002	1.720 $\pm$ 0.043	1.311 $\pm$ 0.017	0.222	0.121	119.41	0.0520	0.1313	1.027	32.041 $\pm$ 1.491	0.206 $\pm$ 0.002	99.796 $\pm$ 0.144
UPI	beta31	0.316	0.321	0.005 $\pm$ 0.004	0.015 $\pm$ 0.013	0.017 $\pm$ 0.001	0.131 $\pm$ 0.003	0.017 $\pm$ 0.001	0.131 $\pm$ 0.004	0.172 $\pm$ 0.008	0.415 $\pm$ 0.012	0.320	0.004	1.41	0.1248	0.1248	1.015	96.327 $\pm$ 0.601	0.522 $\pm$ 0.005	69.286 $\pm$ 1.474
UPI	beta32	0.316	0.319	0.003 $\pm$ 0.004	0.010 $\pm$ 0.013	0.018 $\pm$ 0.001	0.132 $\pm$ 0.003	0.018 $\pm$ 0.001	0.132 $\pm$ 0.004	0.175 $\pm$ 0.009	0.419 $\pm$ 0.012	0.315	-0.001	-0.22	0.1286	0.1286	1.007	96.735 $\pm$ 0.568	0.522 $\pm$ 0.005	70.510 $\pm$ 1.457
UPI	beta33	0.139	0.146	0.007 $\pm$ 0.003	0.053 $\pm$ 0.025	0.011 $\pm$ 0.001	0.107 $\pm$ 0.003	0.011 $\pm$ 0.001	0.107 $\pm$ 0.004	0.592 $\pm$ 0.038	0.769 $\pm$ 0.027	0.134	-0.005	-3.46	0.0907	0.0908	1.061	95.510 $\pm$ 0.661	0.444 $\pm$ 0.011	26.327 $\pm$ 1.407
UPI	beta34	0.101	0.102	0.001 $\pm$ 0.002	0.010 $\pm$ 0.017	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.003 $\pm$ 0.000	0.053 $\pm$ 0.002	0.273 $\pm$ 0.014	0.522 $\pm$ 0.016	0.099	-0.002	-1.57	0.0515	0.0515	0.959	94.592 $\pm$ 0.723	0.198 $\pm$ 0.003	58.367 $\pm$ 1.575
UPI	beta35	0.101	0.103	0.002 $\pm$ 0.002	0.021 $\pm$ 0.017	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.003 $\pm$ 0.000	0.053 $\pm$ 0.002	0.279 $\pm$ 0.015	0.528 $\pm$ 0.017	0.098	-0.003	-2.57	0.0472	0.0473	0.939	94.592 $\pm$ 0.723	0.196 $\pm$ 0.003	62.347 $\pm$ 1.548

Note. MCSE = Monte Carlo Standard Error shown as $\backslash u00b1$ value. Coverage, Type I Error, and Power expressed as percentages. $\backslash u2014$ indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S9

Simulation 1 - Condition 9: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, $N = 400$, Reliability = 0.6

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LMS	beta31	0.316	0.324	0.008 $\pm$ 0.003	0.026 $\pm$ 0.009	0.007 $\pm$ 0.000	0.086 $\pm$ 0.002	0.007 $\pm$ 0.000	0.086 $\pm$ 0.002	0.075 $\pm$ 0.003	0.273 $\pm$ 0.008	0.323	0.007	2.08	0.0841	0.0843	1.005	94.833 $\pm$ 0.705	0.339 $\pm$ 0.001	95.947 $\pm$ 0.628
LMS	beta32	0.316	0.328	0.012 $\pm$ 0.003	0.037 $\pm$ 0.009	0.007 $\pm$ 0.000	0.085 $\pm$ 0.002	0.007 $\pm$ 0.000	0.086 $\pm$ 0.002	0.074 $\pm$ 0.003	0.271 $\pm$ 0.008	0.324	0.008	2.64	0.0836	0.0840	1.019	95.339 $\pm$ 0.671	0.339 $\pm$ 0.002	96.555 $\pm$ 0.581
LMS	beta33	0.139	0.082	-0.057 $\pm$ 0.003	-0.408 $\pm$ 0.020	0.008 $\pm$ 0.000	0.090 $\pm$ 0.002	0.011 $\pm$ 0.000	0.106 $\pm$ 0.003	0.581 $\pm$ 0.025	0.762 $\pm$ 0.018	0.080	-0.059	-42.42	0.0879	0.1058	0.958	87.133 $\pm$ 1.066	0.336 $\pm$ 0.003	19.149 $\pm$ 1.252
LMS	beta34	0.101	0.155	0.054 $\pm$ 0.002	0.536 $\pm$ 0.016	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.547 $\pm$ 0.023	0.740 $\pm$ 0.016	0.153	0.052	51.08	0.0506	0.0723	0.961	81.358 $\pm$ 1.240	0.194 $\pm$ 0.002	89.159 $\pm$ 0.990
LMS	beta35	0.101	0.154	0.053 $\pm$ 0.002	0.525 $\pm$ 0.016	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.006 $\pm$ 0.000	0.074 $\pm$ 0.002	0.544 $\pm$ 0.023	0.737 $\pm$ 0.017	0.154	0.053	52.73	0.0488	0.0723	0.949	81.358 $\pm$ 1.240	0.195 $\pm$ 0.002	88.045 $\pm$ 1.033
LSAM	beta31	0.316	0.320	0.004 $\pm$ 0.004	0.012 $\pm$ 0.014	0.019 $\pm$ 0.001	0.136 $\pm$ 0.003	0.019 $\pm$ 0.001	0.136 $\pm$ 0.004	0.186 $\pm$ 0.009	0.431 $\pm$ 0.012	0.317	0.001	0.18	0.1328	0.1328	1.002	95.947 $\pm$ 0.628	0.535 $\pm$ 0.004	66.869 $\pm$ 1.498
LSAM	beta32	0.316	0.324	0.008 $\pm$ 0.004	0.026 $\pm$ 0.014	0.019 $\pm$ 0.001	0.137 $\pm$ 0.003	0.019 $\pm$ 0.001	0.137 $\pm$ 0.004	0.189 $\pm$ 0.009	0.435 $\pm$ 0.012	0.318	0.002	0.61	0.1273	0.1274	0.996	94.833 $\pm$ 0.705	0.536 $\pm$ 0.004	68.693 $\pm$ 1.476
LSAM	beta33	0.139	0.152	0.013 $\pm$ 0.003	0.092 $\pm$ 0.022	0.009 $\pm$ 0.000	0.096 $\pm$ 0.002	0.009 $\pm$ 0.000	0.097 $\pm$ 0.003	0.488 $\pm$ 0.024	0.699 $\pm$ 0.021	0.148	0.009	6.14	0.0928	0.0932	0.949	94.630 $\pm$ 0.718	0.358 $\pm$ 0.004	40.729 $\pm$ 1.564
LSAM	beta34	0.101	0.100	-0.001 $\pm$ 0.002	-0.010 $\pm$ 0.017	0.003 $\pm$ 0.000	0.055 $\pm$ 0.001	0.003 $\pm$ 0.000	0.055 $\pm$ 0.002	0.301 $\pm$ 0.016	0.549 $\pm$ 0.017	0.100	-0.001	-0.91	0.0484	0.0484	0.922	93.212 $\pm$ 0.801	0.200 $\pm$ 0.003	57.649 $\pm$ 1.573
LSAM	beta35	0.101	0.098	-0.003 $\pm$ 0.002	-0.030 $\pm$ 0.018	0.003 $\pm$ 0.000	0.058 $\pm$ 0.002	0.003 $\pm$ 0.000	0.058 $\pm$ 0.002	0.325 $\pm$ 0.018	0.570 $\pm$ 0.018	0.098	-0.003	-2.56	0.0498	0.0499	0.900	91.489 $\pm$ 0.888	0.203 $\pm$ 0.003	55.117 $\pm$ 1.583
QML	beta31	0.316	0.325	0.009 $\pm$ 0.003	0.027 $\pm$ 0.009	0.007 $\pm$ 0.000	0.086 $\pm$ 0.002	0.008 $\pm$ 0.000	0.087 $\pm$ 0.002	0.075 $\pm$ 0.003	0.274 $\pm$ 0.008	0.323	0.007	2.08	0.0840	0.0842	1.002	94.833 $\pm$ 0.705	0.339 $\pm$ 0.001	96.150 $\pm$ 0.612
QML	beta32	0.316	0.328	0.012 $\pm$ 0.003	0.038 $\pm$ 0.009	0.007 $\pm$ 0.000	0.085 $\pm$ 0.002	0.007 $\pm$ 0.000	0.086 $\pm$ 0.002	0.074 $\pm$ 0.003	0.272 $\pm$ 0.008	0.325	0.009	2.90	0.0840	0.0845	1.017	95.339 $\pm$ 0.671	0.339 $\pm$ 0.002	96.758 $\pm$ 0.564
QML	beta33	0.139	0.083	-0.056 $\pm$ 0.003	-0.401 $\pm$ 0.021	0.008 $\pm$ 0.000	0.090 $\pm$ 0.002	0.011 $\pm$ 0.000	0.106 $\pm$ 0.003	0.577 $\pm$ 0.025	0.760 $\pm$ 0.018	0.081	-0.058	-41.78	0.0872	0.1048	0.953	87.335 $\pm$ 1.059	0.335 $\pm$ 0.003	19.858 $\pm$ 1.270
QML	beta34	0.101	0.154	0.053 $\pm$ 0.002	0.527 $\pm$ 0.016	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.005 $\pm$ 0.000	0.074 $\pm$ 0.002	0.538 $\pm$ 0.023	0.734 $\pm$ 0.016	0.152	0.051	50.66	0.0510	0.0722	0.954	80.952 $\pm$ 1.250	0.193 $\pm$ 0.002	88.956 $\pm$ 0.998
QML	beta35	0.101	0.153	0.052 $\pm$ 0.002	0.516 $\pm$ 0.016	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.005 $\pm$ 0.000	0.074 $\pm$ 0.002	0.532 $\pm$ 0.022	0.730 $\pm$ 0.016	0.154	0.053	52.36	0.0483	0.0716	0.946	81.459 $\pm$ 1.237	0.193 $\pm$ 0.002	87.943 $\pm$ 1.036
UPI	beta31	0.316	0.327	0.011 $\pm$ 0.004	0.034 $\pm$ 0.014	0.018 $\pm$ 0.001	0.136 $\pm$ 0.003	0.019 $\pm$ 0.001	0.136 $\pm$ 0.004	0.185 $\pm$ 0.009	0.431 $\pm$ 0.012	0.323	0.007	2.25	0.1316	0.1318	1.002	95.745 $\pm$ 0.642	0.533 $\pm$ 0.004	69.098 $\pm$ 1.471
UPI	beta32	0.316	0.329	0.013 $\pm$ 0.004	0.042 $\pm$ 0.014	0.019 $\pm$ 0.001	0.136 $\pm$ 0.003	0.019 $\pm$ 0.001	0.137 $\pm$ 0.004	0.188 $\pm$ 0.009	0.433 $\pm$ 0.013	0.324	0.008	2.53	0.1241	0.1243	1.000	95.035 $\pm$ 0.691	0.534 $\pm$ 0.004	70.517 $\pm$ 1.451
UPI	beta33	0.139	0.148	0.009 $\pm$ 0.003	0.067 $\pm$ 0.023	0.010 $\pm$ 0.001	0.101 $\pm$ 0.003	0.010 $\pm$ 0.001	0.101 $\pm$ 0.003	0.532 $\pm$ 0.029	0.729 $\pm$ 0.023	0.142	0.003	2.02	0.0976	0.0977	1.036	95.339 $\pm$ 0.671	0.410 $\pm$ 0.007	32.016 $\pm$ 1.485
UPI	beta34	0.101	0.098	-0.003 $\pm$ 0.002	-0.030 $\pm$ 0.018	0.003 $\pm$ 0.000	0.056 $\pm$ 0.002	0.003 $\pm$ 0.000	0.056 $\pm$ 0.002	0.312 $\pm$ 0.017	0.559 $\pm$ 0.018	0.099	-0.002	-2.35	0.0505	0.0506	0.925	93.110 $\pm$ 0.806	0.204 $\pm$ 0.003	54.914 $\pm$ 1.584
UPI	beta35	0.101	0.097	-0.004 $\pm$ 0.002	-0.042 $\pm$ 0.019	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	0.350 $\pm$ 0.020	0.592 $\pm$ 0.020	0.096	-0.005	-4.64	0.0505	0.0507	0.893	91.793 $\pm$ 0.874	0.209 $\pm$ 0.003	54.002 $\pm$ 1.586

Note. MCSE = Monte Carlo Standard Error shown as $\backslash u00b1$ value. Coverage, Type I Error, and Power expressed as percentages. $\backslash u2014$ indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S10

Simulation 1 - Condition 10: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.6

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LMS	beta31	0.316	0.321	0.005 $\pm$ 0.002	0.017 $\pm$ 0.005	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.028 $\pm$ 0.001	0.166 $\pm$ 0.005	0.320	0.004	1.25	0.0501	0.0502	1.030	94.985 $\pm$ 0.691	0.211 $\pm$ 0.001	100.000 $\pm$ 0.000
LMS	beta32	0.316	0.321	0.005 $\pm$ 0.002	0.016 $\pm$ 0.006	0.003 $\pm$ 0.000	0.055 $\pm$ 0.001	0.003 $\pm$ 0.000	0.055 $\pm$ 0.001	0.030 $\pm$ 0.001	0.174 $\pm$ 0.005	0.322	0.006	1.79	0.0558	0.0560	0.979	94.985 $\pm$ 0.691	0.211 $\pm$ 0.001	100.000 $\pm$ 0.000
LMS	beta33	0.139	0.084	-0.055 $\pm$ 0.002	-0.396 $\pm$ 0.013	0.003 $\pm$ 0.000	0.056 $\pm$ 0.001	0.006 $\pm$ 0.000	0.078 $\pm$ 0.002	0.317 $\pm$ 0.013	0.563 $\pm$ 0.013	0.085	-0.054	-38.69	0.0530	0.0755	0.942	80.241 $\pm$ 1.261	0.205 $\pm$ 0.001	38.616 $\pm$ 1.542
LMS	beta34	0.101	0.158	0.057 $\pm$ 0.001	0.564 $\pm$ 0.010	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	0.004 $\pm$ 0.000	0.065 $\pm$ 0.001	0.420 $\pm$ 0.013	0.648 $\pm$ 0.011	0.157	0.056	55.37	0.0321	0.0645	0.951	53.661 $\pm$ 1.579	0.120 $\pm$ 0.001	100.000 $\pm$ 0.000
LMS	beta35	0.101	0.157	0.056 $\pm$ 0.001	0.555 $\pm$ 0.010	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	0.004 $\pm$ 0.000	0.065 $\pm$ 0.001	0.410 $\pm$ 0.013	0.640 $\pm$ 0.010	0.155	0.054	53.75	0.0308	0.0624	0.940	52.257 $\pm$ 1.582	0.119 $\pm$ 0.001	99.799 $\pm$ 0.142
LSAM	beta31	0.316	0.316	0.000 $\pm$ 0.003	0.001 $\pm$ 0.008	0.006 $\pm$ 0.000	0.081 $\pm$ 0.002	0.006 $\pm$ 0.000	0.081 $\pm$ 0.002	0.065 $\pm$ 0.003	0.255 $\pm$ 0.007	0.314	-0.002	-0.67	0.0755	0.0756	1.003	95.286 $\pm$ 0.671	0.317 $\pm$ 0.001	96.690 $\pm$ 0.567
LSAM	beta32	0.316	0.314	-0.002 $\pm$ 0.003	-0.005 $\pm$ 0.008	0.007 $\pm$ 0.000	0.081 $\pm$ 0.002	0.007 $\pm$ 0.000	0.081 $\pm$ 0.002	0.066 $\pm$ 0.003	0.257 $\pm$ 0.007	0.314	-0.002	-0.60	0.0802	0.0802	0.994	94.483 $\pm$ 0.723	0.316 $\pm$ 0.001	96.189 $\pm$ 0.606
LSAM	beta33	0.139	0.144	0.005 $\pm$ 0.002	0.037 $\pm$ 0.012	0.003 $\pm$ 0.000	0.054 $\pm$ 0.001	0.003 $\pm$ 0.000	0.054 $\pm$ 0.002	0.150 $\pm$ 0.007	0.387 $\pm$ 0.011	0.145	0.006	4.09	0.0504	0.0507	0.953	93.681 $\pm$ 0.771	0.200 $\pm$ 0.001	82.046 $\pm$ 1.216
LSAM	beta34	0.101	0.102	0.001 $\pm$ 0.001	0.007 $\pm$ 0.010	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.096 $\pm$ 0.005	0.310 $\pm$ 0.009	0.101	0.000	-0.14	0.0312	0.0312	0.921	92.578 $\pm$ 0.830	0.113 $\pm$ 0.001	91.073 $\pm$ 0.903
LSAM	beta35	0.101	0.102	0.001 $\pm$ 0.001	0.014 $\pm$ 0.010	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.097 $\pm$ 0.005	0.311 $\pm$ 0.010	0.103	0.002	1.49	0.0294	0.0294	0.914	91.876 $\pm$ 0.865	0.113 $\pm$ 0.001	91.274 $\pm$ 0.894
QML	beta31	0.316	0.321	0.005 $\pm$ 0.002	0.017 $\pm$ 0.005	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.028 $\pm$ 0.001	0.167 $\pm$ 0.005	0.321	0.005	1.44	0.0495	0.0497	1.026	94.985 $\pm$ 0.691	0.211 $\pm$ 0.001	100.000 $\pm$ 0.000
QML	beta32	0.316	0.321	0.005 $\pm$ 0.002	0.017 $\pm$ 0.006	0.003 $\pm$ 0.000	0.055 $\pm$ 0.001	0.003 $\pm$ 0.000	0.055 $\pm$ 0.001	0.031 $\pm$ 0.001	0.175 $\pm$ 0.005	0.322	0.006	1.78	0.0559	0.0562	0.976	94.784 $\pm$ 0.704	0.210 $\pm$ 0.001	100.000 $\pm$ 0.000
QML	beta33	0.139	0.084	-0.055 $\pm$ 0.002	-0.393 $\pm$ 0.013	0.003 $\pm$ 0.000	0.056 $\pm$ 0.001	0.006 $\pm$ 0.000	0.078 $\pm$ 0.002	0.316 $\pm$ 0.013	0.562 $\pm$ 0.013	0.086	-0.053	-37.98	0.0540	0.0755	0.937	80.241 $\pm$ 1.261	0.205 $\pm$ 0.001	39.318 $\pm$ 1.547
QML	beta34	0.101	0.157	0.056 $\pm$ 0.001	0.559 $\pm$ 0.010	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	0.004 $\pm$ 0.000	0.065 $\pm$ 0.001	0.415 $\pm$ 0.013	0.644 $\pm$ 0.011	0.156	0.055	54.56	0.0323	0.0639	0.943	53.561 $\pm$ 1.579	0.120 $\pm$ 0.001	100.000 $\pm$ 0.000
QML	beta35	0.101	0.157	0.056 $\pm$ 0.001	0.550 $\pm$ 0.010	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	0.004 $\pm$ 0.000	0.064 $\pm$ 0.001	0.404 $\pm$ 0.013	0.636 $\pm$ 0.010	0.155	0.054	53.59	0.0312	0.0625	0.936	52.357 $\pm$ 1.582	0.118 $\pm$ 0.001	99.799 $\pm$ 0.142
UPI	beta31	0.316	0.319	0.003 $\pm$ 0.003	0.010 $\pm$ 0.008	0.007 $\pm$ 0.000	0.081 $\pm$ 0.002	0.007 $\pm$ 0.000	0.081 $\pm$ 0.002	0.065 $\pm$ 0.003	0.255 $\pm$ 0.007	0.318	0.002	0.72	0.0762	0.0763	0.998	95.286 $\pm$ 0.671	0.315 $\pm$ 0.001	96.991 $\pm$ 0.541
UPI	beta32	0.316	0.317	0.001 $\pm$ 0.003	0.004 $\pm$ 0.008	0.007 $\pm$ 0.000	0.081 $\pm$ 0.002	0.007 $\pm$ 0.000	0.081 $\pm$ 0.002	0.066 $\pm$ 0.003	0.256 $\pm$ 0.007	0.318	0.002	0.71	0.0793	0.0793	0.990	94.183 $\pm$ 0.741	0.314 $\pm$ 0.001	96.489 $\pm$ 0.583
UPI	beta33	0.139	0.145	0.006 $\pm$ 0.002	0.041 $\pm$ 0.013	0.003 $\pm$ 0.000	0.057 $\pm$ 0.001	0.003 $\pm$ 0.000	0.057 $\pm$ 0.002	0.170 $\pm$ 0.008	0.412 $\pm$ 0.012	0.144	0.005	3.60	0.0554	0.0556	0.960	93.982 $\pm$ 0.753	0.214 $\pm$ 0.002	78.335 $\pm$ 1.305
UPI	beta34	0.101	0.101	0.000 $\pm$ 0.001	0.002 $\pm$ 0.010	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	0.100 $\pm$ 0.005	0.316 $\pm$ 0.009	0.101	0.000	-0.45	0.0319	0.0319	0.923	91.976 $\pm$ 0.860	0.116 $\pm$ 0.001	90.371 $\pm$ 0.934
UPI	beta35	0.101	0.101	0.000 $\pm$ 0.001	0.004 $\pm$ 0.010	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	0.099 $\pm$ 0.005	0.314 $\pm$ 0.009	0.101	0.000	0.18	0.0303	0.0303	0.925	92.377 $\pm$ 0.840	0.115 $\pm$ 0.001	90.672 $\pm$ 0.921

Note. MCSE = Monte Carlo Standard Error shown as $\backslash u00b1$ value. Coverage, Type I Error, and Power expressed as percentages. $\backslash u2014$ indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S11

Simulation 1 - Condition 11: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.8

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.318	0.002 ± 0.002	0.006 ± 0.008	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	0.062 ± 0.003	0.249 ± 0.007	0.321	0.005	1.67	0.0758	0.0760	0.989	94.684 ± 0.711	0.305 ± 0.001	97.392 ± 0.505
LMS	beta32	0.316	0.321	0.005 ± 0.003	0.014 ± 0.008	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.080 ± 0.002	0.063 ± 0.003	0.252 ± 0.007	0.320	0.004	1.42	0.0743	0.0744	0.980	94.584 ± 0.717	0.305 ± 0.001	97.693 ± 0.475
LMS	beta33	0.139	0.116	-0.023 ± 0.002	-0.167 ± 0.015	0.004 ± 0.000	0.064 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.238 ± 0.011	0.488 ± 0.013	0.115	-0.024	-16.92	0.0621	0.0664	0.948	90.873 ± 0.912	0.237 ± 0.002	49.850 ± 1.584
LMS	beta34	0.101	0.122	0.021 ± 0.001	0.203 ± 0.012	0.001 ± 0.000	0.037 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.176 ± 0.009	0.419 ± 0.012	0.121	0.020	19.38	0.0337	0.0390	0.916	87.964 ± 1.030	0.133 ± 0.001	90.973 ± 0.908
LMS	beta35	0.101	0.120	0.019 ± 0.001	0.188 ± 0.012	0.001 ± 0.000	0.038 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.175 ± 0.009	0.418 ± 0.012	0.119	0.018	17.42	0.0360	0.0401	0.891	87.262 ± 1.056	0.132 ± 0.001	90.471 ± 0.930
LSAM	beta31	0.316	0.317	0.001 ± 0.003	0.003 ± 0.010	0.010 ± 0.000	0.099 ± 0.002	0.010 ± 0.000	0.099 ± 0.003	0.098 ± 0.005	0.313 ± 0.009	0.323	0.007	2.28	0.0950	0.0952	0.974	94.383 ± 0.729	0.378 ± 0.002	88.465 ± 1.012
LSAM	beta32	0.316	0.321	0.005 ± 0.003	0.016 ± 0.010	0.010 ± 0.001	0.100 ± 0.003	0.010 ± 0.001	0.100 ± 0.003	0.100 ± 0.005	0.316 ± 0.010	0.321	0.005	1.54	0.0934	0.0935	0.964	93.280 ± 0.793	0.377 ± 0.002	90.070 ± 0.947
LSAM	beta33	0.139	0.139	0.000 ± 0.002	0.001 ± 0.015	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.215 ± 0.011	0.464 ± 0.014	0.136	-0.003	-1.82	0.0613	0.0614	0.930	92.076 ± 0.855	0.235 ± 0.002	64.895 ± 1.512
LSAM	beta34	0.101	0.102	0.001 ± 0.001	0.005 ± 0.012	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.154 ± 0.008	0.393 ± 0.012	0.101	0.000	-0.39	0.0369	0.0369	0.880	91.675 ± 0.875	0.137 ± 0.002	79.438 ± 1.280
LSAM	beta35	0.101	0.100	-0.001 ± 0.001	-0.012 ± 0.013	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.158 ± 0.008	0.397 ± 0.012	0.098	-0.003	-2.75	0.0375	0.0376	0.865	91.274 ± 0.894	0.136 ± 0.002	76.931 ± 1.334
QML	beta31	0.316	0.318	0.002 ± 0.002	0.007 ± 0.008	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	0.062 ± 0.003	0.250 ± 0.007	0.321	0.005	1.60	0.0769	0.0770	1.013	94.784 ± 0.704	0.313 ± 0.008	97.292 ± 0.514
QML	beta32	0.316	0.320	0.004 ± 0.003	0.014 ± 0.008	0.006 ± 0.000	0.080 ± 0.002	0.006 ± 0.000	0.080 ± 0.002	0.064 ± 0.003	0.252 ± 0.007	0.320	0.004	1.27	0.0742	0.0743	1.009	94.383 ± 0.729	0.315 ± 0.010	97.492 ± 0.495
QML	beta33	0.139	0.116	-0.023 ± 0.002	-0.166 ± 0.015	0.004 ± 0.000	0.064 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.238 ± 0.011	0.488 ± 0.013	0.116	-0.023	-16.39	0.0617	0.0658	0.948	90.973 ± 0.908	0.237 ± 0.002	50.351 ± 1.583
QML	beta34	0.101	0.121	0.020 ± 0.001	0.199 ± 0.012	0.001 ± 0.000	0.037 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.174 ± 0.009	0.417 ± 0.012	0.120	0.019	19.12	0.0340	0.0391	0.909	88.064 ± 1.027	0.132 ± 0.001	90.873 ± 0.912
QML	beta35	0.101	0.120	0.019 ± 0.001	0.188 ± 0.012	0.001 ± 0.000	0.038 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.175 ± 0.009	0.418 ± 0.012	0.118	0.017	17.29	0.0362	0.0402	0.888	86.861 ± 1.070	0.132 ± 0.001	90.572 ± 0.925
UPI	beta31	0.316	0.320	0.004 ± 0.003	0.012 ± 0.010	0.010 ± 0.000	0.099 ± 0.002	0.010 ± 0.000	0.099 ± 0.003	0.098 ± 0.005	0.314 ± 0.009	0.324	0.008	2.50	0.0958	0.0962	0.958	93.882 ± 0.759	0.372 ± 0.002	90.271 ± 0.939
UPI	beta32	0.316	0.323	0.007 ± 0.003	0.022 ± 0.010	0.010 ± 0.001	0.100 ± 0.003	0.010 ± 0.001	0.100 ± 0.003	0.100 ± 0.005	0.317 ± 0.009	0.321	0.005	1.63	0.0950	0.0952	0.951	93.380 ± 0.787	0.372 ± 0.002	90.672 ± 0.921
UPI	beta33	0.139	0.140	0.001 ± 0.002	0.005 ± 0.015	0.004 ± 0.000	0.067 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	0.229 ± 0.011	0.479 ± 0.014	0.140	0.001	0.73	0.0644	0.0644	0.937	92.678 ± 0.825	0.245 ± 0.002	62.387 ± 1.534
UPI	beta34	0.101	0.100	-0.001 ± 0.001	-0.007 ± 0.013	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.158 ± 0.008	0.397 ± 0.012	0.100	-0.001	-0.90	0.0368	0.0368	0.873	91.073 ± 0.903	0.137 ± 0.002	77.533 ± 1.322
UPI	beta35	0.101	0.099	-0.002 ± 0.001	-0.016 ± 0.013	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.161 ± 0.008	0.401 ± 0.012	0.097	-0.004	-3.79	0.0366	0.0368	0.867	89.870 ± 0.956	0.138 ± 0.002	76.329 ± 1.346

Note. MCSE = Monte Carlo Standard Error shown as $\backslash u00b1$ value. Coverage, Type I Error, and Power expressed as percentages. $\backslash u2014$ indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S12

Simulation 1 - Condition 12: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, $N = 1000$, Reliability = 0.8

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LMS	beta31	0.316	0.316	-0.000 $\pm$ 0.001	-0.000 $\pm$ 0.005	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.022 $\pm$ 0.001	0.150 $\pm$ 0.004	0.317	0.001	0.16	0.0467	0.0467	1.032	96.697 $\pm$ 0.565	0.191 $\pm$ 0.000	100.000 $\pm$ 0.000
LMS	beta32	0.316	0.314	-0.002 $\pm$ 0.002	-0.006 $\pm$ 0.005	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	0.025 $\pm$ 0.001	0.158 $\pm$ 0.004	0.314	-0.002	-0.60	0.0476	0.0476	0.980	93.894 $\pm$ 0.758	0.191 $\pm$ 0.000	100.000 $\pm$ 0.000
LMS	beta33	0.139	0.120	-0.019 $\pm$ 0.001	-0.140 $\pm$ 0.009	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.104 $\pm$ 0.005	0.322 $\pm$ 0.008	0.119	-0.020	-14.28	0.0401	0.0448	0.935	88.589 $\pm$ 1.006	0.148 $\pm$ 0.001	86.687 $\pm$ 1.075
LMS	beta34	0.101	0.122	0.021 $\pm$ 0.001	0.211 $\pm$ 0.006	0.000 $\pm$ 0.000	0.021 $\pm$ 0.000	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.086 $\pm$ 0.003	0.293 $\pm$ 0.006	0.122	0.021	20.67	0.0204	0.0292	1.012	83.083 $\pm$ 1.186	0.081 $\pm$ 0.001	100.000 $\pm$ 0.000
LMS	beta35	0.101	0.122	0.021 $\pm$ 0.001	0.208 $\pm$ 0.007	0.001 $\pm$ 0.000	0.023 $\pm$ 0.001	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.096 $\pm$ 0.004	0.311 $\pm$ 0.007	0.122	0.021	20.34	0.0228	0.0307	0.883	79.279 $\pm$ 1.282	0.081 $\pm$ 0.001	99.800 $\pm$ 0.141
LSAM	beta31	0.316	0.316	0.000 $\pm$ 0.002	0.000 $\pm$ 0.006	0.003 $\pm$ 0.000	0.058 $\pm$ 0.001	0.003 $\pm$ 0.000	0.058 $\pm$ 0.002	0.033 $\pm$ 0.002	0.182 $\pm$ 0.005	0.315	-0.001	-0.33	0.0561	0.0561	1.027	96.096 $\pm$ 0.613	0.232 $\pm$ 0.001	99.800 $\pm$ 0.141
LSAM	beta32	0.316	0.313	-0.003 $\pm$ 0.002	-0.011 $\pm$ 0.006	0.004 $\pm$ 0.000	0.061 $\pm$ 0.001	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.037 $\pm$ 0.002	0.193 $\pm$ 0.005	0.314	-0.002	-0.65	0.0571	0.0571	0.971	93.594 $\pm$ 0.775	0.232 $\pm$ 0.001	99.900 $\pm$ 0.100
LSAM	beta33	0.139	0.140	0.001 $\pm$ 0.001	0.004 $\pm$ 0.009	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.081 $\pm$ 0.004	0.285 $\pm$ 0.008	0.139	0.000	0.27	0.0400	0.0400	0.921	91.592 $\pm$ 0.878	0.143 $\pm$ 0.001	94.995 $\pm$ 0.690
LSAM	beta34	0.101	0.102	0.001 $\pm$ 0.001	0.008 $\pm$ 0.007	0.000 $\pm$ 0.000	0.021 $\pm$ 0.000	0.000 $\pm$ 0.000	0.021 $\pm$ 0.001	0.043 $\pm$ 0.002	0.206 $\pm$ 0.006	0.101	0.000	-0.12	0.0205	0.0205	0.979	93.994 $\pm$ 0.752	0.080 $\pm$ 0.001	98.899 $\pm$ 0.330
LSAM	beta35	0.101	0.102	0.001 $\pm$ 0.001	0.013 $\pm$ 0.007	0.001 $\pm$ 0.000	0.024 $\pm$ 0.001	0.001 $\pm$ 0.000	0.024 $\pm$ 0.001	0.055 $\pm$ 0.003	0.234 $\pm$ 0.007	0.102	0.001	0.80	0.0223	0.0224	0.870	90.390 $\pm$ 0.932	0.081 $\pm$ 0.001	98.799 $\pm$ 0.345
QML	beta31	0.316	0.316	0.000 $\pm$ 0.002	0.001 $\pm$ 0.005	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.023 $\pm$ 0.001	0.150 $\pm$ 0.004	0.317	0.001	0.22	0.0465	0.0465	1.035	95.996 $\pm$ 0.620	0.192 $\pm$ 0.001	99.900 $\pm$ 0.100
QML	beta32	0.316	0.314	-0.002 $\pm$ 0.002	-0.007 $\pm$ 0.005	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	0.025 $\pm$ 0.001	0.158 $\pm$ 0.004	0.314	-0.002	-0.74	0.0479	0.0480	0.983	93.894 $\pm$ 0.758	0.192 $\pm$ 0.001	99.900 $\pm$ 0.100
QML	beta33	0.139	0.120	-0.019 $\pm$ 0.001	-0.138 $\pm$ 0.009	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.104 $\pm$ 0.005	0.322 $\pm$ 0.008	0.119	-0.020	-14.05	0.0402	0.0447	0.929	88.789 $\pm$ 0.998	0.148 $\pm$ 0.001	86.887 $\pm$ 1.068
QML	beta34	0.101	0.122	0.021 $\pm$ 0.001	0.205 $\pm$ 0.006	0.000 $\pm$ 0.000	0.021 $\pm$ 0.000	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.083 $\pm$ 0.003	0.288 $\pm$ 0.006	0.121	0.020	19.93	0.0209	0.0290	0.997	82.983 $\pm$ 1.189	0.080 $\pm$ 0.001	100.000 $\pm$ 0.000
QML	beta35	0.101	0.122	0.021 $\pm$ 0.001	0.208 $\pm$ 0.007	0.001 $\pm$ 0.000	0.023 $\pm$ 0.001	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.096 $\pm$ 0.004	0.311 $\pm$ 0.007	0.122	0.021	20.41	0.0228	0.0307	0.879	79.379 $\pm$ 1.280	0.080 $\pm$ 0.001	99.800 $\pm$ 0.141
UPI	beta31	0.316	0.317	0.001 $\pm$ 0.002	0.005 $\pm$ 0.006	0.003 $\pm$ 0.000	0.058 $\pm$ 0.001	0.003 $\pm$ 0.000	0.058 $\pm$ 0.002	0.034 $\pm$ 0.002	0.183 $\pm$ 0.005	0.316	0.000	0.13	0.0561	0.0561	1.009	95.395 $\pm$ 0.663	0.229 $\pm$ 0.001	99.800 $\pm$ 0.141
UPI	beta32	0.316	0.314	-0.002 $\pm$ 0.002	-0.007 $\pm$ 0.006	0.004 $\pm$ 0.000	0.061 $\pm$ 0.001	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.037 $\pm$ 0.002	0.193 $\pm$ 0.005	0.315	-0.001	-0.33	0.0580	0.0580	0.957	93.493 $\pm$ 0.780	0.229 $\pm$ 0.001	99.800 $\pm$ 0.141
UPI	beta33	0.139	0.140	0.001 $\pm$ 0.001	0.005 $\pm$ 0.009	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.085 $\pm$ 0.004	0.291 $\pm$ 0.008	0.139	0.000	0.01	0.0400	0.0400	0.924	91.692 $\pm$ 0.873	0.147 $\pm$ 0.001	94.294 $\pm$ 0.734
UPI	beta34	0.101	0.101	0.000 $\pm$ 0.001	0.003 $\pm$ 0.007	0.000 $\pm$ 0.000	0.021 $\pm$ 0.000	0.000 $\pm$ 0.000	0.021 $\pm$ 0.001	0.044 $\pm$ 0.002	0.210 $\pm$ 0.006	0.101	0.000	-0.08	0.0206	0.0206	0.972	93.293 $\pm$ 0.791	0.081 $\pm$ 0.001	98.799 $\pm$ 0.345
UPI	beta35	0.101	0.102	0.001 $\pm$ 0.001	0.011 $\pm$ 0.007	0.001 $\pm$ 0.000	0.024 $\pm$ 0.001	0.001 $\pm$ 0.000	0.024 $\pm$ 0.001	0.056 $\pm$ 0.003	0.236 $\pm$ 0.007	0.101	0.000	-0.35	0.0226	0.0226	0.871	90.490 $\pm$ 0.928	0.081 $\pm$ 0.001	98.599 $\pm$ 0.372

Note. MCSE = Monte Carlo Standard Error shown as $\backslash u00b1$ value. Coverage, Type I Error, and Power expressed as percentages. $\backslash u2014$ indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S13

Simulation 1 - Condition 13: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.4

Method	Param.	True	Mean-based										Median-based					Inference			
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE	
LMS	beta31	0.316	0.306	-0.010 $\pm$ 0.004	-0.032 $\pm$ 0.013	0.009 $\pm$ 0.001	0.093 $\pm$ 0.003	0.009 $\pm$ 0.001	0.094 $\pm$ 0.004	0.088 $\pm$ 0.006	0.297 $\pm$ 0.012	0.307	-0.009	-2.82	0.0941	0.0945	1.008	93.818 $\pm$ 1.027	0.368 $\pm$ 0.004	92.909 $\pm$ 1.09	
LMS	beta32	0.316	0.308	-0.008 $\pm$ 0.004	-0.027 $\pm$ 0.011	0.007 $\pm$ 0.000	0.084 $\pm$ 0.003	0.007 $\pm$ 0.000	0.085 $\pm$ 0.003	0.072 $\pm$ 0.004	0.268 $\pm$ 0.010	0.302	-0.014	-4.43	0.0927	0.0938	1.109	96.364 $\pm$ 0.798	0.367 $\pm$ 0.002	95.091 $\pm$ 0.921	
LMS	beta33	0.139	0.131	-0.008 $\pm$ 0.005	-0.060 $\pm$ 0.039	0.016 $\pm$ 0.001	0.127 $\pm$ 0.005	0.016 $\pm$ 0.001	0.128 $\pm$ 0.005	0.842 $\pm$ 0.062	0.918 $\pm$ 0.039	0.125	-0.014	-10.35	0.1239	0.1247	1.067	95.636 $\pm$ 0.871	0.533 $\pm$ 0.028	17.273 $\pm$ 1.612	
LMS	beta34	0.101	0.045	-0.056 $\pm$ 0.004	-0.555 $\pm$ 0.038	0.008 $\pm$ 0.001	0.090 $\pm$ 0.003	0.011 $\pm$ 0.001	0.106 $\pm$ 0.003	1.102 $\pm$ 0.060	1.050 $\pm$ 0.033	0.047	-0.054	-53.78	0.0930	0.1077	1.125	87.273 $\pm$ 1.421	0.397 $\pm$ 0.033	8.000 $\pm$ 1.157	
LMS	beta35	0.101	0.047	-0.054 $\pm$ 0.004	-0.538 $\pm$ 0.040	0.009 $\pm$ 0.001	0.094 $\pm$ 0.003	0.012 $\pm$ 0.001	0.109 $\pm$ 0.004	1.158 $\pm$ 0.075	1.076 $\pm$ 0.039	0.042	-0.059	-58.90	0.0944	0.1116	1.006	86.909 $\pm$ 1.438	0.371 $\pm$ 0.008	8.545 $\pm$ 1.192	
LSAM	beta31	0.316	0.327	0.011 $\pm$ 0.005	0.036 $\pm$ 0.016	0.014 $\pm$ 0.001	0.119 $\pm$ 0.004	0.014 $\pm$ 0.001	0.120 $\pm$ 0.005	0.144 $\pm$ 0.010	0.379 $\pm$ 0.016	0.327	0.011	3.56	0.1190	0.1195	1.230	96.182 $\pm$ 0.817	0.576 $\pm$ 0.020	73.818 $\pm$ 1.875	
LSAM	beta32	0.316	0.325	0.009 $\pm$ 0.005	0.028 $\pm$ 0.015	0.012 $\pm$ 0.001	0.111 $\pm$ 0.004	0.012 $\pm$ 0.001	0.111 $\pm$ 0.005	0.124 $\pm$ 0.009	0.352 $\pm$ 0.015	0.322	0.006	1.98	0.1087	0.1089	1.355	98.727 $\pm$ 0.478	0.590 $\pm$ 0.024	76.182 $\pm$ 1.816	
LSAM	beta33	0.139	0.141	0.002 $\pm$ 0.008	0.018 $\pm$ 0.059	0.037 $\pm$ 0.003	0.193 $\pm$ 0.008	0.037 $\pm$ 0.003	0.193 $\pm$ 0.009	1.919 $\pm$ 0.153	1.385 $\pm$ 0.063	0.132	-0.007	-5.31	0.1723	0.1725	1.310	98.545 $\pm$ 0.511	0.989 $\pm$ 0.055	8.182 $\pm$ 1.169	
LSAM	beta34	0.101	0.105	0.004 $\pm$ 0.010	0.042 $\pm$ 0.095	0.051 $\pm$ 0.004	0.225 $\pm$ 0.009	0.051 $\pm$ 0.004	0.225 $\pm$ 0.010	4.959 $\pm$ 0.396	2.227 $\pm$ 0.101	0.110	0.009	8.42	0.2148	0.2150	1.403	98.545 $\pm$ 0.511	1.238 $\pm$ 0.076	1.455 $\pm$ 0.511	
LSAM	beta35	0.101	0.114	0.013 $\pm$ 0.011	0.127 $\pm$ 0.105	0.062 $\pm$ 0.005	0.250 $\pm$ 0.011	0.062 $\pm$ 0.005	0.250 $\pm$ 0.012	6.108 $\pm$ 0.532	2.471 $\pm$ 0.120	0.098	-0.003	-3.28	0.2219	0.2220	1.293	99.091 $\pm$ 0.405	1.264 $\pm$ 0.074	0.727 $\pm$ 0.362	
QML	beta31	0.316	0.306	-0.010 $\pm$ 0.004	-0.031 $\pm$ 0.013	0.009 $\pm$ 0.001	0.094 $\pm$ 0.003	0.009 $\pm$ 0.001	0.094 $\pm$ 0.004	0.088 $\pm$ 0.006	0.297 $\pm$ 0.012	0.306	-0.010	-3.19	0.0920	0.0925	0.999	93.455 $\pm$ 1.055	0.366 $\pm$ 0.002	93.091 $\pm$ 1.081	
QML	beta32	0.316	0.308	-0.008 $\pm$ 0.004	-0.025 $\pm$ 0.011	0.007 $\pm$ 0.000	0.085 $\pm$ 0.003	0.007 $\pm$ 0.000	0.085 $\pm$ 0.003	0.073 $\pm$ 0.004	0.269 $\pm$ 0.010	0.302	-0.014	-4.42	0.0907	0.0917	1.104	96.182 $\pm$ 0.817	0.367 $\pm$ 0.002	95.818 $\pm$ 0.854	
QML	beta33	0.139	0.132	-0.007 $\pm$ 0.006	-0.048 $\pm$ 0.040	0.017 $\pm$ 0.001	0.129 $\pm$ 0.005	0.017 $\pm$ 0.001	0.129 $\pm$ 0.005	0.864 $\pm$ 0.062	0.929 $\pm$ 0.039	0.127	-0.012	-8.77	0.1160	0.1166	1.024	95.273 $\pm$ 0.905	0.518 $\pm$ 0.009	17.091 $\pm$ 1.605	
QML	beta34	0.101	0.044	-0.057 $\pm$ 0.004	-0.561 $\pm$ 0.039	0.008 $\pm$ 0.001	0.092 $\pm$ 0.003	0.012 $\pm$ 0.001	0.108 $\pm$ 0.004	1.146 $\pm$ 0.065	1.070 $\pm$ 0.035	0.046	-0.055	-54.11	0.0908	0.1059	1.054	87.636 $\pm$ 1.404	0.381 $\pm$ 0.011	7.273 $\pm$ 1.107	
QML	beta35	0.101	0.046	-0.055 $\pm$ 0.004	-0.542 $\pm$ 0.041	0.009 $\pm$ 0.001	0.096 $\pm$ 0.004	0.012 $\pm$ 0.001	0.111 $\pm$ 0.004	1.199 $\pm$ 0.079	1.095 $\pm$ 0.040	0.042	-0.059	-58.83	0.0919	0.1095	0.977	87.818 $\pm$ 1.395	0.368 $\pm$ 0.005	8.000 $\pm$ 1.157	
UPI	beta31	0.316	0.325	0.009 $\pm$ 0.005	0.029 $\pm$ 0.017	0.016 $\pm$ 0.001	0.126 $\pm$ 0.005	0.016 $\pm$ 0.001	0.126 $\pm$ 0.005	0.159 $\pm$ 0.012	0.398 $\pm$ 0.017	0.313	-0.003	-0.85	0.1266	0.1266	1.525	96.545 $\pm$ 0.779	0.751 $\pm$ 0.037	65.091 $\pm$ 2.033	
UPI	beta32	0.316	0.324	0.008 $\pm$ 0.005	0.027 $\pm$ 0.015	0.013 $\pm$ 0.001	0.114 $\pm$ 0.004	0.013 $\pm$ 0.001	0.114 $\pm$ 0.005	0.130 $\pm$ 0.010	0.361 $\pm$ 0.016	0.325	0.009	2.88	0.1161	0.1164	1.639	98.909 $\pm$ 0.443	0.732 $\pm$ 0.030	62.727 $\pm$ 2.062	
UPI	beta33	0.139	0.163	0.024 $\pm$ 0.008	0.176 $\pm$ 0.059	0.037 $\pm$ 0.003	0.193 $\pm$ 0.008	0.038 $\pm$ 0.003	0.194 $\pm$ 0.009	1.954 $\pm$ 0.154	1.398 $\pm$ 0.063	0.152	0.013	9.20	0.1830	0.1835	1.709	99.273 $\pm$ 0.362	1.292 $\pm$ 0.065	4.182 $\pm$ 0.854	
UPI	beta34	0.101	0.100	-0.001 $\pm$ 0.012	-0.011 $\pm$ 0.116	0.075 $\pm$ 0.006	0.275 $\pm$ 0.012	0.075 $\pm$ 0.006	0.275 $\pm$ 0.013	7.387 $\pm$ 0.617	2.718 $\pm$ 0.128	0.073	-0.028	-27.61	0.2389	0.2405	2.127	97.455 $\pm$ 0.672	2.291 $\pm$ 0.161	0.182 $\pm$ 0.182	
UPI	beta35	0.101	0.102	0.001 $\pm$ 0.012	0.005 $\pm$ 0.121	0.083 $\pm$ 0.008	0.288 $\pm$ 0.014	0.083 $\pm$ 0.008	0.287 $\pm$ 0.015	8.100 $\pm$ 0.759	2.846 $\pm$ 0.147	0.063	-0.038	-38.04	0.2287	0.2319	1.935	97.273 $\pm$ 0.695	2.182 $\pm$ 0.142	0.364 $\pm$ 0.257	

Note. MCSE = Monte Carlo Standard Error shown as $\backslash$ u00b1 value. Coverage, Type I Error, and Power expressed as percentages. $\backslash$ u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S14

Simulation 1 - Condition 14: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.4

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LMS	beta31	0.316	0.307	-0.009 $\pm$ 0.002	-0.030 $\pm$ 0.006	0.003 $\pm$ 0.000	0.059 $\pm$ 0.001	0.004 $\pm$ 0.000	0.059 $\pm$ 0.002	0.035 $\pm$ 0.002	0.188 $\pm$ 0.005	0.307	-0.009	-2.77	0.0571	0.0577	0.991	93.846 $\pm$ 0.797	0.228 $\pm$ 0.001	100.000 $\pm$ 0.000
LMS	beta32	0.316	0.302	-0.014 $\pm$ 0.002	-0.044 $\pm$ 0.006	0.004 $\pm$ 0.000	0.060 $\pm$ 0.001	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.038 $\pm$ 0.002	0.195 $\pm$ 0.005	0.299	-0.017	-5.39	0.0629	0.0652	0.965	92.967 $\pm$ 0.848	0.227 $\pm$ 0.001	99.890 $\pm$ 0.110
LMS	beta33	0.139	0.138	-0.001 $\pm$ 0.003	-0.010 $\pm$ 0.019	0.006 $\pm$ 0.000	0.080 $\pm$ 0.002	0.006 $\pm$ 0.000	0.080 $\pm$ 0.002	0.330 $\pm$ 0.016	0.574 $\pm$ 0.017	0.136	-0.003	-2.20	0.0790	0.0791	1.008	94.396 $\pm$ 0.762	0.316 $\pm$ 0.002	42.418 $\pm$ 1.638
LMS	beta34	0.101	0.042	-0.059 $\pm$ 0.002	-0.587 $\pm$ 0.019	0.003 $\pm$ 0.000	0.057 $\pm$ 0.001	0.007 $\pm$ 0.000	0.082 $\pm$ 0.002	0.661 $\pm$ 0.025	0.813 $\pm$ 0.017	0.040	-0.061	-60.28	0.0599	0.0854	1.014	79.231 $\pm$ 1.345	0.226 $\pm$ 0.002	10.549 $\pm$ 1.018
LMS	beta35	0.101	0.039	-0.062 $\pm$ 0.002	-0.609 $\pm$ 0.019	0.003 $\pm$ 0.000	0.057 $\pm$ 0.001	0.007 $\pm$ 0.000	0.084 $\pm$ 0.002	0.688 $\pm$ 0.028	0.829 $\pm$ 0.018	0.042	-0.059	-58.66	0.0536	0.0799	1.002	79.011 $\pm$ 1.350	0.224 $\pm$ 0.002	10.440 $\pm$ 1.014
LSAM	beta31	0.316	0.321	0.005 $\pm$ 0.002	0.015 $\pm$ 0.007	0.004 $\pm$ 0.000	0.067 $\pm$ 0.002	0.004 $\pm$ 0.000	0.067 $\pm$ 0.002	0.045 $\pm$ 0.002	0.212 $\pm$ 0.006	0.321	0.005	1.69	0.0647	0.0649	1.044	95.824 $\pm$ 0.663	0.274 $\pm$ 0.002	99.560 $\pm$ 0.219
LSAM	beta32	0.316	0.317	0.001 $\pm$ 0.002	0.004 $\pm$ 0.007	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.046 $\pm$ 0.002	0.215 $\pm$ 0.006	0.316	0.000	-0.04	0.0683	0.0683	1.029	96.374 $\pm$ 0.620	0.274 $\pm$ 0.002	99.121 $\pm$ 0.309
LSAM	beta33	0.139	0.138	-0.001 $\pm$ 0.003	-0.006 $\pm$ 0.023	0.009 $\pm$ 0.001	0.096 $\pm$ 0.003	0.009 $\pm$ 0.001	0.096 $\pm$ 0.003	0.479 $\pm$ 0.026	0.692 $\pm$ 0.022	0.140	0.001	0.45	0.0895	0.0895	1.046	96.923 $\pm$ 0.572	0.395 $\pm$ 0.005	35.495 $\pm$ 1.586
LSAM	beta34	0.101	0.111	0.010 $\pm$ 0.004	0.097 $\pm$ 0.041	0.016 $\pm$ 0.001	0.126 $\pm$ 0.004	0.016 $\pm$ 0.001	0.126 $\pm$ 0.004	1.554 $\pm$ 0.094	1.246 $\pm$ 0.043	0.097	-0.004	-4.14	0.1192	0.1193	1.066	97.253 $\pm$ 0.542	0.525 $\pm$ 0.008	8.462 $\pm$ 0.923
LSAM	beta35	0.101	0.101	0.000 $\pm$ 0.004	0.002 $\pm$ 0.041	0.015 $\pm$ 0.001	0.123 $\pm$ 0.004	0.015 $\pm$ 0.001	0.123 $\pm$ 0.004	1.492 $\pm$ 0.092	1.221 $\pm$ 0.043	0.095	-0.006	-6.18	0.1116	0.1118	1.048	96.923 $\pm$ 0.572	0.507 $\pm$ 0.009	7.912 $\pm$ 0.895
QML	beta31	0.316	0.307	-0.009 $\pm$ 0.002	-0.029 $\pm$ 0.006	0.003 $\pm$ 0.000	0.059 $\pm$ 0.001	0.004 $\pm$ 0.000	0.059 $\pm$ 0.002	0.035 $\pm$ 0.002	0.188 $\pm$ 0.005	0.308	-0.008	-2.62	0.0563	0.0569	0.990	94.066 $\pm$ 0.783	0.228 $\pm$ 0.001	100.000 $\pm$ 0.000
QML	beta32	0.316	0.302	-0.014 $\pm$ 0.002	-0.044 $\pm$ 0.006	0.004 $\pm$ 0.000	0.060 $\pm$ 0.001	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.038 $\pm$ 0.002	0.194 $\pm$ 0.005	0.298	-0.018	-5.56	0.0631	0.0655	0.964	92.747 $\pm$ 0.860	0.227 $\pm$ 0.001	99.890 $\pm$ 0.110
QML	beta33	0.139	0.139	-0.000 $\pm$ 0.003	-0.003 $\pm$ 0.019	0.006 $\pm$ 0.000	0.080 $\pm$ 0.002	0.006 $\pm$ 0.000	0.080 $\pm$ 0.002	0.330 $\pm$ 0.016	0.574 $\pm$ 0.017	0.139	0.000	-0.35	0.0779	0.0779	1.012	94.396 $\pm$ 0.762	0.317 $\pm$ 0.002	43.187 $\pm$ 1.642
QML	beta34	0.101	0.041	-0.060 $\pm$ 0.002	-0.594 $\pm$ 0.019	0.003 $\pm$ 0.000	0.057 $\pm$ 0.001	0.007 $\pm$ 0.000	0.083 $\pm$ 0.002	0.669 $\pm$ 0.025	0.818 $\pm$ 0.017	0.039	-0.062	-61.24	0.0592	0.0856	1.025	78.681 $\pm$ 1.358	0.228 $\pm$ 0.002	10.440 $\pm$ 1.014
QML	beta35	0.101	0.039	-0.062 $\pm$ 0.002	-0.615 $\pm$ 0.019	0.003 $\pm$ 0.000	0.057 $\pm$ 0.001	0.007 $\pm$ 0.000	0.084 $\pm$ 0.002	0.698 $\pm$ 0.028	0.836 $\pm$ 0.018	0.041	-0.060	-58.99	0.0540	0.0804	0.997	78.571 $\pm$ 1.360	0.223 $\pm$ 0.002	10.549 $\pm$ 1.018
UPI	beta31	0.316	0.322	0.006 $\pm$ 0.002	0.018 $\pm$ 0.007	0.005 $\pm$ 0.000	0.070 $\pm$ 0.002	0.005 $\pm$ 0.000	0.070 $\pm$ 0.002	0.049 $\pm$ 0.002	0.221 $\pm$ 0.007	0.320	0.004	1.42	0.0703	0.0705	1.114	96.593 $\pm$ 0.601	0.304 $\pm$ 0.004	97.692 $\pm$ 0.498
UPI	beta32	0.316	0.316	-0.000 $\pm$ 0.002	-0.001 $\pm$ 0.007	0.005 $\pm$ 0.000	0.070 $\pm$ 0.002	0.005 $\pm$ 0.000	0.070 $\pm$ 0.002	0.049 $\pm$ 0.002	0.221 $\pm$ 0.006	0.315	-0.001	-0.27	0.0692	0.0692	1.114	97.033 $\pm$ 0.562	0.306 $\pm$ 0.004	97.473 $\pm$ 0.520
UPI	beta33	0.139	0.146	0.007 $\pm$ 0.003	0.053 $\pm$ 0.024	0.010 $\pm$ 0.001	0.100 $\pm$ 0.003	0.010 $\pm$ 0.001	0.100 $\pm$ 0.003	0.522 $\pm$ 0.029	0.723 $\pm$ 0.023	0.142	0.003	2.09	0.0912	0.0912	1.191	97.582 $\pm$ 0.509	0.468 $\pm$ 0.010	27.143 $\pm$ 1.474
UPI	beta34	0.101	0.116	0.015 $\pm$ 0.005	0.147 $\pm$ 0.050	0.023 $\pm$ 0.002	0.153 $\pm$ 0.005	0.024 $\pm$ 0.002	0.154 $\pm$ 0.006	2.314 $\pm$ 0.160	1.521 $\pm$ 0.058	0.098	-0.003	-2.56	0.1359	0.1360	1.296	97.473 $\pm$ 0.520	0.777 $\pm$ 0.021	2.088 $\pm$ 0.474
UPI	beta35	0.101	0.107	0.006 $\pm$ 0.005	0.060 $\pm$ 0.050	0.023 $\pm$ 0.001	0.152 $\pm$ 0.005	0.023 $\pm$ 0.001	0.152 $\pm$ 0.006	2.266 $\pm$ 0.146	1.505 $\pm$ 0.055	0.091	-0.010	-9.99	0.1343	0.1346	1.298	96.484 $\pm$ 0.611	0.774 $\pm$ 0.024	1.429 $\pm$ 0.393

Note. MCSE = Monte Carlo Standard Error shown as $\backslash 00b1$ value. Coverage, Type I Error, and Power expressed as percentages. $\backslash 2014$ indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S15

Simulation 1 - Condition 15: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.6

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LMS	beta31	0.316	0.308	-0.008 $\pm$ 0.002	-0.024 $\pm$ 0.007	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	0.004 $\pm$ 0.000	0.067 $\pm$ 0.002	0.044 $\pm$ 0.002	0.211 $\pm$ 0.006	0.307	-0.009	-2.88	0.0653	0.0659	1.030	96.165 $\pm$ 0.610	0.267 $\pm$ 0.001	99.697 $\pm$ 0.175
LMS	beta32	0.316	0.307	-0.009 $\pm$ 0.002	-0.029 $\pm$ 0.007	0.005 $\pm$ 0.000	0.069 $\pm$ 0.002	0.005 $\pm$ 0.000	0.069 $\pm$ 0.002	0.048 $\pm$ 0.002	0.219 $\pm$ 0.006	0.304	-0.012	-3.65	0.0686	0.0695	0.992	94.551 $\pm$ 0.721	0.267 $\pm$ 0.001	99.495 $\pm$ 0.225
LMS	beta33	0.139	0.143	0.004 $\pm$ 0.003	0.029 $\pm$ 0.019	0.007 $\pm$ 0.000	0.085 $\pm$ 0.002	0.007 $\pm$ 0.000	0.085 $\pm$ 0.002	0.374 $\pm$ 0.017	0.611 $\pm$ 0.017	0.141	0.002	1.43	0.0842	0.0842	0.993	94.955 $\pm$ 0.695	0.330 $\pm$ 0.002	39.354 $\pm$ 1.552
LMS	beta34	0.101	0.058	-0.043 $\pm$ 0.002	-0.424 $\pm$ 0.022	0.005 $\pm$ 0.000	0.070 $\pm$ 0.002	0.007 $\pm$ 0.000	0.082 $\pm$ 0.002	0.660 $\pm$ 0.029	0.812 $\pm$ 0.020	0.059	-0.042	-41.65	0.0662	0.0785	0.969	87.689 $\pm$ 1.044	0.266 $\pm$ 0.002	13.925 $\pm$ 1.100
LMS	beta35	0.101	0.057	-0.044 $\pm$ 0.002	-0.433 $\pm$ 0.022	0.005 $\pm$ 0.000	0.069 $\pm$ 0.002	0.007 $\pm$ 0.000	0.082 $\pm$ 0.002	0.658 $\pm$ 0.028	0.811 $\pm$ 0.019	0.056	-0.045	-44.12	0.0654	0.0792	0.993	89.001 $\pm$ 0.994	0.270 $\pm$ 0.002	13.320 $\pm$ 1.079
LSAM	beta31	0.316	0.317	0.001 $\pm$ 0.002	0.003 $\pm$ 0.007	0.005 $\pm$ 0.000	0.073 $\pm$ 0.002	0.005 $\pm$ 0.000	0.072 $\pm$ 0.002	0.053 $\pm$ 0.002	0.229 $\pm$ 0.006	0.317	0.001	0.21	0.0723	0.0723	1.039	96.266 $\pm$ 0.602	0.295 $\pm$ 0.001	98.789 $\pm$ 0.347
LSAM	beta32	0.316	0.316	0.000 $\pm$ 0.002	0.001 $\pm$ 0.007	0.005 $\pm$ 0.000	0.074 $\pm$ 0.002	0.005 $\pm$ 0.000	0.074 $\pm$ 0.002	0.054 $\pm$ 0.002	0.233 $\pm$ 0.007	0.316	0.000	-0.10	0.0748	0.0748	1.020	95.661 $\pm$ 0.647	0.295 $\pm$ 0.001	98.789 $\pm$ 0.347
LSAM	beta33	0.139	0.141	0.002 $\pm$ 0.003	0.015 $\pm$ 0.021	0.009 $\pm$ 0.000	0.093 $\pm$ 0.002	0.009 $\pm$ 0.000	0.093 $\pm$ 0.003	0.446 $\pm$ 0.023	0.668 $\pm$ 0.020	0.138	-0.001	-0.91	0.0911	0.0911	0.983	96.165 $\pm$ 0.610	0.358 $\pm$ 0.003	36.327 $\pm$ 1.528
LSAM	beta34	0.101	0.103	0.002 $\pm$ 0.004	0.020 $\pm$ 0.036	0.013 $\pm$ 0.001	0.114 $\pm$ 0.003	0.013 $\pm$ 0.001	0.114 $\pm$ 0.004	1.275 $\pm$ 0.067	1.129 $\pm$ 0.035	0.101	0.000	-0.07	0.1093	0.1093	0.990	95.156 $\pm$ 0.682	0.443 $\pm$ 0.004	13.824 $\pm$ 1.096
LSAM	beta35	0.101	0.103	0.002 $\pm$ 0.004	0.015 $\pm$ 0.036	0.013 $\pm$ 0.001	0.116 $\pm$ 0.003	0.013 $\pm$ 0.001	0.116 $\pm$ 0.004	1.314 $\pm$ 0.071	1.146 $\pm$ 0.036	0.098	-0.003	-3.14	0.1100	0.1101	0.969	95.459 $\pm$ 0.661	0.440 $\pm$ 0.004	13.017 $\pm$ 1.069
QML	beta31	0.316	0.309	-0.007 $\pm$ 0.002	-0.024 $\pm$ 0.007	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	0.004 $\pm$ 0.000	0.067 $\pm$ 0.002	0.044 $\pm$ 0.002	0.211 $\pm$ 0.006	0.307	-0.009	-2.70	0.0654	0.0659	1.029	96.165 $\pm$ 0.610	0.267 $\pm$ 0.001	99.697 $\pm$ 0.175
QML	beta32	0.316	0.307	-0.009 $\pm$ 0.002	-0.029 $\pm$ 0.007	0.005 $\pm$ 0.000	0.069 $\pm$ 0.002	0.005 $\pm$ 0.000	0.069 $\pm$ 0.002	0.048 $\pm$ 0.002	0.219 $\pm$ 0.006	0.304	-0.012	-3.67	0.0685	0.0694	0.991	94.551 $\pm$ 0.721	0.267 $\pm$ 0.001	99.495 $\pm$ 0.225
QML	beta33	0.139	0.144	0.005 $\pm$ 0.003	0.034 $\pm$ 0.019	0.007 $\pm$ 0.000	0.085 $\pm$ 0.002	0.007 $\pm$ 0.000	0.085 $\pm$ 0.002	0.376 $\pm$ 0.017	0.613 $\pm$ 0.017	0.141	0.002	1.63	0.0843	0.0843	0.994	95.358 $\pm$ 0.668	0.332 $\pm$ 0.002	40.464 $\pm$ 1.559
QML	beta34	0.101	0.057	-0.044 $\pm$ 0.002	-0.434 $\pm$ 0.022	0.005 $\pm$ 0.000	0.070 $\pm$ 0.002	0.007 $\pm$ 0.000	0.082 $\pm$ 0.002	0.667 $\pm$ 0.029	0.817 $\pm$ 0.020	0.058	-0.043	-42.37	0.0665	0.0791	0.975	87.689 $\pm$ 1.044	0.267 $\pm$ 0.002	13.925 $\pm$ 1.100
QML	beta35	0.101	0.056	-0.045 $\pm$ 0.002	-0.441 $\pm$ 0.022	0.005 $\pm$ 0.000	0.070 $\pm$ 0.002	0.007 $\pm$ 0.000	0.083 $\pm$ 0.002	0.668 $\pm$ 0.028	0.817 $\pm$ 0.019	0.055	-0.046	-45.50	0.0663	0.0806	0.990	89.001 $\pm$ 0.994	0.270 $\pm$ 0.002	13.017 $\pm$ 1.069
UPI	beta31	0.316	0.317	0.001 $\pm$ 0.002	0.004 $\pm$ 0.007	0.005 $\pm$ 0.000	0.073 $\pm$ 0.002	0.005 $\pm$ 0.000	0.073 $\pm$ 0.002	0.054 $\pm$ 0.002	0.231 $\pm$ 0.007	0.316	0.000	0.05	0.0717	0.0717	1.033	96.165 $\pm$ 0.610	0.296 $\pm$ 0.002	98.486 $\pm$ 0.388
UPI	beta32	0.316	0.317	0.001 $\pm$ 0.002	0.002 $\pm$ 0.007	0.006 $\pm$ 0.000	0.074 $\pm$ 0.002	0.006 $\pm$ 0.000	0.074 $\pm$ 0.002	0.055 $\pm$ 0.003	0.235 $\pm$ 0.007	0.318	0.002	0.54	0.0741	0.0741	1.012	94.854 $\pm$ 0.702	0.295 $\pm$ 0.001	98.688 $\pm$ 0.361
UPI	beta33	0.139	0.147	0.008 $\pm$ 0.003	0.061 $\pm$ 0.023	0.010 $\pm$ 0.001	0.099 $\pm$ 0.003	0.010 $\pm$ 0.001	0.099 $\pm$ 0.003	0.509 $\pm$ 0.029	0.714 $\pm$ 0.023	0.147	0.008	5.97	0.0956	0.0960	0.997	96.670 $\pm$ 0.570	0.386 $\pm$ 0.004	35.721 $\pm$ 1.522
UPI	beta34	0.101	0.105	0.004 $\pm$ 0.004	0.036 $\pm$ 0.041	0.017 $\pm$ 0.001	0.129 $\pm$ 0.004	0.017 $\pm$ 0.001	0.129 $\pm$ 0.004	1.630 $\pm$ 0.092	1.277 $\pm$ 0.041	0.101	0.000	-0.13	0.1202	0.1202	1.083	96.670 $\pm$ 0.570	0.547 $\pm$ 0.008	7.669 $\pm$ 0.845
UPI	beta35	0.101	0.106	0.005 $\pm$ 0.004	0.047 $\pm$ 0.041	0.017 $\pm$ 0.001	0.129 $\pm$ 0.004	0.017 $\pm$ 0.001	0.129 $\pm$ 0.004	1.626 $\pm$ 0.091	1.275 $\pm$ 0.041	0.101	0.000	-0.41	0.1136	0.1136	1.077	96.367 $\pm$ 0.594	0.544 $\pm$ 0.008	6.862 $\pm$ 0.803

Note. MCSE = Monte Carlo Standard Error shown as $\backslash u00b1$ value. Coverage, Type I Error, and Power expressed as percentages. $\backslash u2014$ indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S16

Simulation 1 - Condition 16: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.6

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LMS	beta31	0.316	0.309	-0.007 $\pm$ 0.001	-0.021 $\pm$ 0.004	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.018 $\pm$ 0.001	0.136 $\pm$ 0.004	0.310	-0.006	-1.92	0.0417	0.0421	1.009	94.383 $\pm$ 0.729	0.168 $\pm$ 0.000	100.000 $\pm$ 0.000
LMS	beta32	0.316	0.308	-0.008 $\pm$ 0.001	-0.024 $\pm$ 0.004	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.018 $\pm$ 0.001	0.134 $\pm$ 0.004	0.309	-0.007	-2.31	0.0415	0.0422	1.025	95.186 $\pm$ 0.678	0.167 $\pm$ 0.000	100.000 $\pm$ 0.000
LMS	beta33	0.139	0.142	0.003 $\pm$ 0.002	0.020 $\pm$ 0.012	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.143 $\pm$ 0.007	0.379 $\pm$ 0.011	0.140	0.001	0.93	0.0519	0.0519	1.003	94.684 $\pm$ 0.711	0.207 $\pm$ 0.001	77.332 $\pm$ 1.326
LMS	beta34	0.101	0.059	-0.042 $\pm$ 0.001	-0.416 $\pm$ 0.013	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.004 $\pm$ 0.000	0.060 $\pm$ 0.001	0.349 $\pm$ 0.013	0.591 $\pm$ 0.012	0.058	-0.043	-42.62	0.0401	0.0588	1.004	83.651 $\pm$ 1.171	0.167 $\pm$ 0.001	26.580 $\pm$ 1.399
LMS	beta35	0.101	0.060	-0.041 $\pm$ 0.001	-0.406 $\pm$ 0.013	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.003 $\pm$ 0.000	0.059 $\pm$ 0.001	0.336 $\pm$ 0.013	0.580 $\pm$ 0.012	0.060	-0.041	-40.98	0.0417	0.0588	1.014	82.648 $\pm$ 1.199	0.166 $\pm$ 0.001	28.887 $\pm$ 1.435
LSAM	beta31	0.316	0.318	0.002 $\pm$ 0.001	0.005 $\pm$ 0.004	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.020 $\pm$ 0.001	0.141 $\pm$ 0.004	0.318	0.002	0.60	0.0446	0.0446	1.024	95.286 $\pm$ 0.671	0.179 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta32	0.316	0.317	0.001 $\pm$ 0.001	0.003 $\pm$ 0.004	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.019 $\pm$ 0.001	0.139 $\pm$ 0.004	0.316	0.000	0.01	0.0436	0.0436	1.040	96.289 $\pm$ 0.599	0.179 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta33	0.139	0.140	0.001 $\pm$ 0.002	0.009 $\pm$ 0.012	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.003 $\pm$ 0.000	0.053 $\pm$ 0.002	0.148 $\pm$ 0.007	0.384 $\pm$ 0.011	0.140	0.001	0.76	0.0520	0.0520	1.011	95.186 $\pm$ 0.678	0.212 $\pm$ 0.001	74.423 $\pm$ 1.382
LSAM	beta34	0.101	0.103	0.002 $\pm$ 0.002	0.016 $\pm$ 0.020	0.004 $\pm$ 0.000	0.065 $\pm$ 0.002	0.004 $\pm$ 0.000	0.065 $\pm$ 0.002	0.415 $\pm$ 0.020	0.644 $\pm$ 0.018	0.098	-0.003	-2.96	0.0637	0.0637	1.024	95.587 $\pm$ 0.650	0.261 $\pm$ 0.001	32.798 $\pm$ 1.487
LSAM	beta35	0.101	0.103	0.002 $\pm$ 0.002	0.023 $\pm$ 0.020	0.004 $\pm$ 0.000	0.065 $\pm$ 0.001	0.004 $\pm$ 0.000	0.065 $\pm$ 0.002	0.416 $\pm$ 0.019	0.645 $\pm$ 0.018	0.101	0.000	-0.02	0.0643	0.0643	1.022	94.885 $\pm$ 0.698	0.261 $\pm$ 0.001	33.400 $\pm$ 1.494
QML	beta31	0.316	0.310	-0.006 $\pm$ 0.001	-0.020 $\pm$ 0.004	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.018 $\pm$ 0.001	0.136 $\pm$ 0.004	0.310	-0.006	-1.87	0.0417	0.0421	1.009	94.483 $\pm$ 0.723	0.168 $\pm$ 0.000	100.000 $\pm$ 0.000
QML	beta32	0.316	0.308	-0.008 $\pm$ 0.001	-0.024 $\pm$ 0.004	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.018 $\pm$ 0.001	0.134 $\pm$ 0.004	0.309	-0.007	-2.32	0.0416	0.0422	1.027	95.186 $\pm$ 0.678	0.168 $\pm$ 0.000	100.000 $\pm$ 0.000
QML	beta33	0.139	0.142	0.003 $\pm$ 0.002	0.023 $\pm$ 0.012	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.143 $\pm$ 0.007	0.379 $\pm$ 0.011	0.141	0.002	1.33	0.0519	0.0519	1.005	94.483 $\pm$ 0.723	0.207 $\pm$ 0.001	77.633 $\pm$ 1.320
QML	beta34	0.101	0.058	-0.043 $\pm$ 0.001	-0.422 $\pm$ 0.013	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.004 $\pm$ 0.000	0.060 $\pm$ 0.001	0.354 $\pm$ 0.013	0.595 $\pm$ 0.012	0.057	-0.044	-43.40	0.0401	0.0594	1.005	82.949 $\pm$ 1.191	0.167 $\pm$ 0.001	26.279 $\pm$ 1.394
QML	beta35	0.101	0.059	-0.042 $\pm$ 0.001	-0.413 $\pm$ 0.013	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.003 $\pm$ 0.000	0.059 $\pm$ 0.001	0.341 $\pm$ 0.013	0.584 $\pm$ 0.012	0.059	-0.042	-41.79	0.0415	0.0592	1.017	82.146 $\pm$ 1.213	0.167 $\pm$ 0.001	28.285 $\pm$ 1.426
UPI	beta31	0.316	0.318	0.002 $\pm$ 0.001	0.005 $\pm$ 0.004	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.020 $\pm$ 0.001	0.142 $\pm$ 0.004	0.319	0.003	0.86	0.0448	0.0449	1.004	95.587 $\pm$ 0.650	0.176 $\pm$ 0.000	100.000 $\pm$ 0.000
UPI	beta32	0.316	0.317	0.001 $\pm$ 0.001	0.003 $\pm$ 0.004	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.019 $\pm$ 0.001	0.139 $\pm$ 0.004	0.316	0.000	0.15	0.0432	0.0432	1.021	95.888 $\pm$ 0.629	0.176 $\pm$ 0.000	100.000 $\pm$ 0.000
UPI	beta33	0.139	0.142	0.003 $\pm$ 0.002	0.019 $\pm$ 0.012	0.003 $\pm$ 0.000	0.054 $\pm$ 0.001	0.003 $\pm$ 0.000	0.054 $\pm$ 0.002	0.152 $\pm$ 0.007	0.390 $\pm$ 0.011	0.142	0.003	2.18	0.0515	0.0515	1.029	95.787 $\pm$ 0.636	0.218 $\pm$ 0.001	73.621 $\pm$ 1.396
UPI	beta34	0.101	0.103	0.002 $\pm$ 0.002	0.025 $\pm$ 0.021	0.005 $\pm$ 0.000	0.069 $\pm$ 0.002	0.005 $\pm$ 0.000	0.069 $\pm$ 0.002	0.461 $\pm$ 0.022	0.679 $\pm$ 0.020	0.099	-0.002	-1.87	0.0650	0.0650	1.044	96.189 $\pm$ 0.606	0.281 $\pm$ 0.002	27.482 $\pm$ 1.414
UPI	beta35	0.101	0.105	0.004 $\pm$ 0.002	0.037 $\pm$ 0.022	0.005 $\pm$ 0.000	0.071 $\pm$ 0.002	0.005 $\pm$ 0.000	0.071 $\pm$ 0.002	0.494 $\pm$ 0.025	0.703 $\pm$ 0.021	0.100	-0.001	-0.73	0.0676	0.0676	1.016	95.587 $\pm$ 0.650	0.282 $\pm$ 0.002	29.589 $\pm$ 1.446

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S17

Simulation 1 - Condition 17: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.8

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LMS	beta31	0.316	0.311	-0.005 $\pm$ 0.002	-0.016 $\pm$ 0.006	0.003 $\pm$ 0.000	0.056 $\pm$ 0.001	0.003 $\pm$ 0.000	0.056 $\pm$ 0.002	0.031 $\pm$ 0.001	0.177 $\pm$ 0.005	0.310	-0.006	-1.87	0.0582	0.0585	1.010	95.600 $\pm$ 0.649	0.221 $\pm$ 0.001	99.900 $\pm$ 0.100
LMS	beta32	0.316	0.314	-0.002 $\pm$ 0.002	-0.006 $\pm$ 0.006	0.003 $\pm$ 0.000	0.057 $\pm$ 0.001	0.003 $\pm$ 0.000	0.057 $\pm$ 0.002	0.032 $\pm$ 0.001	0.179 $\pm$ 0.005	0.313	-0.003	-1.00	0.0585	0.0586	0.995	95.000 $\pm$ 0.689	0.221 $\pm$ 0.001	100.000 $\pm$ 0.000
LMS	beta33	0.139	0.145	0.006 $\pm$ 0.002	0.046 $\pm$ 0.014	0.004 $\pm$ 0.000	0.064 $\pm$ 0.001	0.004 $\pm$ 0.000	0.064 $\pm$ 0.002	0.211 $\pm$ 0.009	0.460 $\pm$ 0.012	0.148	0.009	6.13	0.0631	0.0637	0.987	94.400 $\pm$ 0.727	0.246 $\pm$ 0.001	63.400 $\pm$ 1.523
LMS	beta34	0.101	0.076	-0.025 $\pm$ 0.002	-0.249 $\pm$ 0.018	0.003 $\pm$ 0.000	0.057 $\pm$ 0.001	0.004 $\pm$ 0.000	0.063 $\pm$ 0.002	0.385 $\pm$ 0.017	0.621 $\pm$ 0.016	0.077	-0.024	-24.17	0.0572	0.0621	1.005	92.700 $\pm$ 0.823	0.226 $\pm$ 0.001	25.700 $\pm$ 1.382
LMS	beta35	0.101	0.077	-0.024 $\pm$ 0.002	-0.236 $\pm$ 0.018	0.003 $\pm$ 0.000	0.057 $\pm$ 0.001	0.004 $\pm$ 0.000	0.062 $\pm$ 0.002	0.374 $\pm$ 0.016	0.612 $\pm$ 0.015	0.075	-0.026	-25.66	0.0566	0.0622	1.013	93.100 $\pm$ 0.801	0.226 $\pm$ 0.001	25.900 $\pm$ 1.385
LSAM	beta31	0.316	0.315	-0.001 $\pm$ 0.002	-0.004 $\pm$ 0.006	0.003 $\pm$ 0.000	0.058 $\pm$ 0.001	0.003 $\pm$ 0.000	0.058 $\pm$ 0.002	0.033 $\pm$ 0.001	0.182 $\pm$ 0.005	0.314	-0.002	-0.59	0.0572	0.0572	1.012	96.000 $\pm$ 0.620	0.229 $\pm$ 0.001	100.000 $\pm$ 0.000
LSAM	beta32	0.316	0.318	0.002 $\pm$ 0.002	0.006 $\pm$ 0.006	0.003 $\pm$ 0.000	0.058 $\pm$ 0.001	0.003 $\pm$ 0.000	0.058 $\pm$ 0.002	0.034 $\pm$ 0.002	0.185 $\pm$ 0.005	0.317	0.001	0.24	0.0589	0.0589	0.999	94.700 $\pm$ 0.708	0.229 $\pm$ 0.001	100.000 $\pm$ 0.000
LSAM	beta33	0.139	0.143	0.004 $\pm$ 0.002	0.031 $\pm$ 0.015	0.004 $\pm$ 0.000	0.064 $\pm$ 0.001	0.004 $\pm$ 0.000	0.064 $\pm$ 0.002	0.213 $\pm$ 0.009	0.461 $\pm$ 0.012	0.144	0.005	3.70	0.0628	0.0630	0.983	94.100 $\pm$ 0.745	0.246 $\pm$ 0.001	62.600 $\pm$ 1.530
LSAM	beta34	0.101	0.098	-0.003 $\pm$ 0.002	-0.028 $\pm$ 0.022	0.005 $\pm$ 0.000	0.072 $\pm$ 0.002	0.005 $\pm$ 0.000	0.072 $\pm$ 0.002	0.505 $\pm$ 0.022	0.710 $\pm$ 0.019	0.100	-0.001	-0.54	0.0714	0.0714	1.000	94.900 $\pm$ 0.696	0.281 $\pm$ 0.001	27.900 $\pm$ 1.418
LSAM	beta35	0.101	0.099	-0.002 $\pm$ 0.002	-0.017 $\pm$ 0.022	0.005 $\pm$ 0.000	0.071 $\pm$ 0.002	0.005 $\pm$ 0.000	0.071 $\pm$ 0.002	0.492 $\pm$ 0.023	0.701 $\pm$ 0.020	0.096	-0.005	-4.64	0.0696	0.0697	1.014	95.800 $\pm$ 0.634	0.282 $\pm$ 0.001	27.700 $\pm$ 1.415
QML	beta31	0.316	0.311	-0.005 $\pm$ 0.002	-0.017 $\pm$ 0.006	0.003 $\pm$ 0.000	0.056 $\pm$ 0.001	0.003 $\pm$ 0.000	0.056 $\pm$ 0.002	0.031 $\pm$ 0.001	0.177 $\pm$ 0.005	0.310	-0.006	-1.85	0.0576	0.0579	1.007	95.600 $\pm$ 0.649	0.221 $\pm$ 0.001	100.000 $\pm$ 0.000
QML	beta32	0.316	0.314	-0.002 $\pm$ 0.002	-0.006 $\pm$ 0.006	0.003 $\pm$ 0.000	0.057 $\pm$ 0.001	0.003 $\pm$ 0.000	0.057 $\pm$ 0.002	0.032 $\pm$ 0.001	0.179 $\pm$ 0.005	0.313	-0.003	-1.07	0.0584	0.0585	0.993	95.000 $\pm$ 0.689	0.220 $\pm$ 0.000	100.000 $\pm$ 0.000
QML	beta33	0.139	0.145	0.006 $\pm$ 0.002	0.045 $\pm$ 0.014	0.004 $\pm$ 0.000	0.064 $\pm$ 0.001	0.004 $\pm$ 0.000	0.064 $\pm$ 0.002	0.211 $\pm$ 0.009	0.459 $\pm$ 0.012	0.148	0.009	6.18	0.0627	0.0633	0.987	94.300 $\pm$ 0.733	0.246 $\pm$ 0.001	63.100 $\pm$ 1.526
QML	beta34	0.101	0.076	-0.025 $\pm$ 0.002	-0.252 $\pm$ 0.018	0.003 $\pm$ 0.000	0.058 $\pm$ 0.001	0.004 $\pm$ 0.000	0.063 $\pm$ 0.002	0.388 $\pm$ 0.017	0.623 $\pm$ 0.016	0.076	-0.025	-24.70	0.0568	0.0620	1.003	92.300 $\pm$ 0.843	0.226 $\pm$ 0.001	26.100 $\pm$ 1.389
QML	beta35	0.101	0.077	-0.024 $\pm$ 0.002	-0.240 $\pm$ 0.018	0.003 $\pm$ 0.000	0.057 $\pm$ 0.001	0.004 $\pm$ 0.000	0.062 $\pm$ 0.002	0.377 $\pm$ 0.016	0.614 $\pm$ 0.015	0.075	-0.026	-25.97	0.0567	0.0625	1.013	93.500 $\pm$ 0.780	0.227 $\pm$ 0.001	25.700 $\pm$ 1.382
UPI	beta31	0.316	0.315	-0.001 $\pm$ 0.002	-0.004 $\pm$ 0.006	0.003 $\pm$ 0.000	0.058 $\pm$ 0.001	0.003 $\pm$ 0.000	0.058 $\pm$ 0.002	0.033 $\pm$ 0.002	0.183 $\pm$ 0.005	0.314	-0.002	-0.53	0.0579	0.0579	0.974	94.500 $\pm$ 0.721	0.221 $\pm$ 0.001	100.000 $\pm$ 0.000
UPI	beta32	0.316	0.318	0.002 $\pm$ 0.002	0.006 $\pm$ 0.006	0.003 $\pm$ 0.000	0.058 $\pm$ 0.001	0.003 $\pm$ 0.000	0.058 $\pm$ 0.002	0.034 $\pm$ 0.002	0.185 $\pm$ 0.005	0.316	0.000	0.13	0.0596	0.0596	0.962	93.700 $\pm$ 0.768	0.220 $\pm$ 0.001	100.000 $\pm$ 0.000
UPI	beta33	0.139	0.144	0.005 $\pm$ 0.002	0.039 $\pm$ 0.015	0.004 $\pm$ 0.000	0.065 $\pm$ 0.001	0.004 $\pm$ 0.000	0.065 $\pm$ 0.002	0.219 $\pm$ 0.010	0.468 $\pm$ 0.013	0.145	0.006	3.98	0.0640	0.0642	0.984	94.600 $\pm$ 0.715	0.250 $\pm$ 0.001	62.900 $\pm$ 1.528
UPI	beta34	0.101	0.099	-0.002 $\pm$ 0.002	-0.020 $\pm$ 0.023	0.005 $\pm$ 0.000	0.074 $\pm$ 0.002	0.005 $\pm$ 0.000	0.074 $\pm$ 0.002	0.534 $\pm$ 0.023	0.731 $\pm$ 0.020	0.099	-0.002	-1.63	0.0728	0.0728	1.011	95.300 $\pm$ 0.669	0.292 $\pm$ 0.001	25.200 $\pm$ 1.373
UPI	beta35	0.101	0.100	-0.001 $\pm$ 0.002	-0.012 $\pm$ 0.023	0.005 $\pm$ 0.000	0.073 $\pm$ 0.002	0.005 $\pm$ 0.000	0.073 $\pm$ 0.002	0.518 $\pm$ 0.024	0.720 $\pm$ 0.020	0.098	-0.003	-3.33	0.0676	0.0677	1.029	95.600 $\pm$ 0.649	0.293 $\pm$ 0.001	24.700 $\pm$ 1.364

Note. MCSE = Monte Carlo Standard Error shown as $\backslash$ u00b1 value. Coverage, Type I Error, and Power expressed as percentages. $\backslash$ u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S18*Simulation 1 - Condition 18: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.8*

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LMS	beta31	0.316	0.313	-0.003 $\pm$ 0.001	-0.010 $\pm$ 0.004	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.013 $\pm$ 0.001	0.112 $\pm$ 0.003	0.314	-0.002	-0.70	0.0356	0.0356	1.030	94.600 $\pm$ 0.715	0.142 $\pm$ 0.003	99.900 $\pm$ 0.100
LMS	beta32	0.316	0.312	-0.004 $\pm$ 0.001	-0.014 $\pm$ 0.004	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.013 $\pm$ 0.001	0.116 $\pm$ 0.003	0.311	-0.005	-1.57	0.0360	0.0363	0.985	94.600 $\pm$ 0.715	0.140 $\pm$ 0.001	99.900 $\pm$ 0.100
LMS	beta33	0.139	0.142	0.003 $\pm$ 0.001	0.019 $\pm$ 0.009	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.079 $\pm$ 0.004	0.282 $\pm$ 0.008	0.140	0.001	0.60	0.0389	0.0389	1.022	95.000 $\pm$ 0.689	0.157 $\pm$ 0.001	94.600 $\pm$ 0.715
LMS	beta34	0.101	0.076	-0.025 $\pm$ 0.001	-0.243 $\pm$ 0.011	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.189 $\pm$ 0.008	0.435 $\pm$ 0.011	0.076	-0.025	-25.11	0.0366	0.0445	1.000	90.400 $\pm$ 0.932	0.143 $\pm$ 0.000	56.700 $\pm$ 1.567
LMS	beta35	0.101	0.078	-0.023 $\pm$ 0.001	-0.226 $\pm$ 0.012	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.187 $\pm$ 0.008	0.432 $\pm$ 0.010	0.078	-0.023	-22.49	0.0369	0.0433	0.988	89.900 $\pm$ 0.953	0.144 $\pm$ 0.001	56.500 $\pm$ 1.568
LSAM	beta31	0.316	0.317	0.001 $\pm$ 0.001	0.003 $\pm$ 0.004	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.013 $\pm$ 0.001	0.114 $\pm$ 0.003	0.318	0.002	0.72	0.0361	0.0362	1.011	95.700 $\pm$ 0.641	0.143 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta32	0.316	0.315	-0.001 $\pm$ 0.001	-0.002 $\pm$ 0.004	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.014 $\pm$ 0.001	0.118 $\pm$ 0.003	0.314	-0.002	-0.50	0.0376	0.0377	0.980	95.100 $\pm$ 0.683	0.143 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta33	0.139	0.140	0.001 $\pm$ 0.001	0.004 $\pm$ 0.009	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.078 $\pm$ 0.003	0.279 $\pm$ 0.008	0.139	0.000	-0.33	0.0394	0.0394	1.022	95.400 $\pm$ 0.662	0.156 $\pm$ 0.000	94.400 $\pm$ 0.727
LSAM	beta34	0.101	0.099	-0.002 $\pm$ 0.001	-0.024 $\pm$ 0.014	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.203 $\pm$ 0.009	0.451 $\pm$ 0.013	0.099	-0.002	-2.34	0.0449	0.0450	0.992	94.600 $\pm$ 0.715	0.177 $\pm$ 0.000	59.600 $\pm$ 1.552
LSAM	beta35	0.101	0.101	-0.000 $\pm$ 0.001	-0.004 $\pm$ 0.014	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.208 $\pm$ 0.009	0.456 $\pm$ 0.012	0.100	-0.001	-0.62	0.0460	0.0460	0.976	93.900 $\pm$ 0.757	0.176 $\pm$ 0.000	60.900 $\pm$ 1.543
QML	beta31	0.316	0.313	-0.003 $\pm$ 0.001	-0.010 $\pm$ 0.004	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.013 $\pm$ 0.001	0.112 $\pm$ 0.003	0.313	-0.003	-0.84	0.0352	0.0353	1.010	94.600 $\pm$ 0.715	0.139 $\pm$ 0.001	100.000 $\pm$ 0.000
QML	beta32	0.316	0.312	-0.004 $\pm$ 0.001	-0.014 $\pm$ 0.004	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.013 $\pm$ 0.001	0.116 $\pm$ 0.003	0.311	-0.005	-1.57	0.0360	0.0363	0.981	94.700 $\pm$ 0.708	0.140 $\pm$ 0.000	100.000 $\pm$ 0.000
QML	beta33	0.139	0.141	0.002 $\pm$ 0.001	0.017 $\pm$ 0.009	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.079 $\pm$ 0.004	0.281 $\pm$ 0.008	0.140	0.001	0.43	0.0386	0.0386	1.020	95.000 $\pm$ 0.689	0.156 $\pm$ 0.000	94.700 $\pm$ 0.708
QML	beta34	0.101	0.076	-0.025 $\pm$ 0.001	-0.245 $\pm$ 0.011	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.190 $\pm$ 0.008	0.436 $\pm$ 0.011	0.076	-0.025	-25.11	0.0367	0.0446	0.999	90.300 $\pm$ 0.936	0.143 $\pm$ 0.000	56.500 $\pm$ 1.568
QML	beta35	0.101	0.078	-0.023 $\pm$ 0.001	-0.228 $\pm$ 0.012	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.188 $\pm$ 0.008	0.433 $\pm$ 0.010	0.078	-0.023	-23.07	0.0369	0.0436	0.978	89.700 $\pm$ 0.961	0.143 $\pm$ 0.000	56.200 $\pm$ 1.569
UPI	beta31	0.316	0.317	0.001 $\pm$ 0.001	0.003 $\pm$ 0.004	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.013 $\pm$ 0.001	0.115 $\pm$ 0.003	0.318	0.002	0.58	0.0366	0.0366	0.975	94.500 $\pm$ 0.721	0.138 $\pm$ 0.000	100.000 $\pm$ 0.000
UPI	beta32	0.316	0.315	-0.001 $\pm$ 0.001	-0.002 $\pm$ 0.004	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.014 $\pm$ 0.001	0.118 $\pm$ 0.003	0.314	-0.002	-0.53	0.0373	0.0373	0.947	94.400 $\pm$ 0.727	0.139 $\pm$ 0.000	100.000 $\pm$ 0.000
UPI	beta33	0.139	0.140	0.001 $\pm$ 0.001	0.005 $\pm$ 0.009	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.079 $\pm$ 0.003	0.281 $\pm$ 0.008	0.138	-0.001	-0.57	0.0394	0.0394	1.024	95.600 $\pm$ 0.649	0.157 $\pm$ 0.000	94.400 $\pm$ 0.727
UPI	beta34	0.101	0.099	-0.002 $\pm$ 0.001	-0.016 $\pm$ 0.015	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.214 $\pm$ 0.010	0.462 $\pm$ 0.013	0.099	-0.002	-2.22	0.0466	0.0467	0.986	94.600 $\pm$ 0.715	0.180 $\pm$ 0.001	59.400 $\pm$ 1.553
UPI	beta35	0.101	0.101	0.000 $\pm$ 0.001	0.001 $\pm$ 0.015	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.216 $\pm$ 0.010	0.465 $\pm$ 0.013	0.100	-0.001	-0.92	0.0462	0.0462	0.975	93.900 $\pm$ 0.757	0.180 $\pm$ 0.001	58.900 $\pm$ 1.556

Note. MCSE = Monte Carlo Standard Error shown as $\backslash$ u00b1 value. Coverage, Type I Error, and Power expressed as percentages. $\backslash$ u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S19

Simulation 1 - Condition 19: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.4

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.303	-0.013 ± 0.007	-0.042 ± 0.024	0.011 ± 0.002	0.104 ± 0.009	0.011 ± 0.002	0.105 ± 0.009	0.110 ± 0.017	0.332 ± 0.028	0.305	-0.011	-3.41	0.1129	0.1134	0.993	93.846 ± 1.721	0.406 ± 0.015	86.154 ± 2.473
LMS	beta32	0.316	0.296	-0.020 ± 0.007	-0.065 ± 0.022	0.010 ± 0.001	0.098 ± 0.006	0.010 ± 0.001	0.100 ± 0.006	0.100 ± 0.011	0.316 ± 0.020	0.290	-0.026	-8.27	0.1227	0.1255	1.092	95.897 ± 1.420	0.420 ± 0.024	85.128 ± 2.548
LMS	beta33	0.139	0.191	0.052 ± 0.010	0.374 ± 0.071	0.019 ± 0.003	0.138 ± 0.010	0.022 ± 0.003	0.147 ± 0.010	1.120 ± 0.138	1.058 ± 0.074	0.188	0.049	35.20	0.1538	0.1614	1.186	94.359 ± 1.652	0.641 ± 0.055	35.385 ± 3.424
LMS	beta34	0.101	0.056	-0.045 ± 0.007	-0.443 ± 0.068	0.009 ± 0.001	0.097 ± 0.006	0.011 ± 0.001	0.106 ± 0.006	1.105 ± 0.116	1.051 ± 0.064	0.042	-0.059	-58.76	0.1081	0.1233	1.123	84.615 ± 2.584	0.425 ± 0.044	7.692 ± 1.908
LMS	beta35	0.101	0.069	-0.032 ± 0.007	-0.314 ± 0.070	0.010 ± 0.001	0.098 ± 0.006	0.011 ± 0.001	0.103 ± 0.006	1.043 ± 0.110	1.022 ± 0.064	0.058	-0.043	-43.00	0.1081	0.1165	1.156	87.179 ± 2.394	0.446 ± 0.044	10.256 ± 2.173
LSAM	beta31	0.316	0.324	0.008 ± 0.011	0.025 ± 0.035	0.025 ± 0.003	0.157 ± 0.010	0.024 ± 0.003	0.156 ± 0.012	0.245 ± 0.033	0.495 ± 0.038	0.317	0.001	0.46	0.1467	0.1467	1.144	97.436 ± 1.132	0.702 ± 0.038	52.308 ± 3.577
LSAM	beta32	0.316	0.327	0.011 ± 0.011	0.036 ± 0.035	0.024 ± 0.003	0.156 ± 0.009	0.024 ± 0.003	0.156 ± 0.011	0.242 ± 0.028	0.492 ± 0.034	0.328	0.012	3.92	0.1583	0.1588	1.148	96.923 ± 1.237	0.700 ± 0.033	55.897 ± 3.556
LSAM	beta33	0.139	0.292	0.153 ± 0.015	1.101 ± 0.106	0.042 ± 0.006	0.205 ± 0.016	0.065 ± 0.010	0.256 ± 0.020	3.380 ± 0.504	1.838 ± 0.145	0.277	0.138	99.31	0.1852	0.2310	1.032	96.410 ± 1.332	0.830 ± 0.053	28.718 ± 3.240
LSAM	beta34	0.101	0.003	-0.098 ± 0.009	-0.970 ± 0.089	0.016 ± 0.002	0.125 ± 0.009	0.025 ± 0.003	0.159 ± 0.011	2.477 ± 0.319	1.574 ± 0.109	0.001	-0.100	-98.98	0.1152	0.1525	1.103	84.103 ± 2.618	0.542 ± 0.036	1.026 ± 0.722
LSAM	beta35	0.101	0.010	-0.091 ± 0.010	-0.899 ± 0.097	0.019 ± 0.003	0.137 ± 0.011	0.027 ± 0.003	0.164 ± 0.011	2.635 ± 0.318	1.623 ± 0.107	-0.008	-0.109	-107.85	0.1206	0.1625	1.036	85.128 ± 2.548	0.556 ± 0.035	4.615 ± 1.503
QML	beta31	0.316	0.289	-0.027 ± 0.006	-0.087 ± 0.020	0.008 ± 0.001	0.089 ± 0.004	0.009 ± 0.001	0.093 ± 0.005	0.087 ± 0.008	0.295 ± 0.017	0.295	-0.021	-6.66	0.0981	0.1004	1.063	92.821 ± 1.849	0.372 ± 0.005	86.154 ± 2.473
QML	beta32	0.316	0.288	-0.028 ± 0.007	-0.088 ± 0.021	0.008 ± 0.001	0.092 ± 0.004	0.009 ± 0.001	0.096 ± 0.005	0.092 ± 0.008	0.303 ± 0.017	0.289	-0.027	-8.40	0.1123	0.1154	1.135	96.410 ± 1.332	0.408 ± 0.038	87.692 ± 2.353
QML	beta33	0.139	0.187	0.048 ± 0.009	0.342 ± 0.067	0.017 ± 0.002	0.130 ± 0.007	0.019 ± 0.002	0.138 ± 0.009	0.991 ± 0.108	0.995 ± 0.063	0.193	0.054	39.08	0.1437	0.1536	1.054	93.846 ± 1.721	0.538 ± 0.016	32.821 ± 3.363
QML	beta34	0.101	0.056	-0.045 ± 0.006	-0.446 ± 0.062	0.008 ± 0.001	0.088 ± 0.005	0.010 ± 0.001	0.099 ± 0.006	0.953 ± 0.091	0.976 ± 0.055	0.044	-0.057	-56.86	0.0898	0.1066	0.976	85.128 ± 2.548	0.336 ± 0.011	9.231 ± 2.073
QML	beta35	0.101	0.062	-0.039 ± 0.007	-0.383 ± 0.065	0.008 ± 0.001	0.092 ± 0.005	0.010 ± 0.001	0.099 ± 0.006	0.970 ± 0.100	0.985 ± 0.059	0.051	-0.050	-49.46	0.0933	0.1058	1.000	87.692 ± 2.353	0.360 ± 0.012	13.333 ± 2.434
UPI	beta31	0.316	0.312	-0.004 ± 0.016	-0.013 ± 0.051	0.050 ± 0.007	0.223 ± 0.017	0.050 ± 0.007	0.223 ± 0.018	0.497 ± 0.073	0.705 ± 0.058	0.332	0.016	5.11	0.2458	0.2463	1.837	92.821 ± 1.849	1.607 ± 0.188	25.641 ± 3.127
UPI	beta32	0.316	0.322	0.006 ± 0.016	0.018 ± 0.052	0.052 ± 0.006	0.228 ± 0.013	0.052 ± 0.006	0.227 ± 0.016	0.517 ± 0.060	0.719 ± 0.049	0.319	0.003	0.99	0.2597	0.2597	1.709	88.205 ± 2.310	1.526 ± 0.154	28.205 ± 3.223
UPI	beta33	0.139	0.257	0.118 ± 0.017	0.852 ± 0.126	0.060 ± 0.009	0.244 ± 0.020	0.073 ± 0.012	0.271 ± 0.024	3.800 ± 0.617	1.949 ± 0.171	0.221	0.082	58.77	0.2265	0.2408	1.857	97.949 ± 1.015	1.779 ± 0.210	7.179 ± 1.849
UPI	beta34	0.101	-0.019	-0.120 ± 0.018	-1.184 ± 0.178	0.063 ± 0.009	0.250 ± 0.018	0.077 ± 0.010	0.277 ± 0.019	7.516 ± 0.946	2.742 ± 0.192	-0.018	-0.119	-118.29	0.2354	0.2640	1.811	90.256 ± 2.124	1.777 ± 0.231	1.538 ± 0.881
UPI	beta35	0.101	-0.038	-0.139 ± 0.018	-1.378 ± 0.179	0.064 ± 0.010	0.253 ± 0.019	0.083 ± 0.012	0.288 ± 0.022	8.127 ± 1.129	2.851 ± 0.215	-0.016	-0.117	-115.99	0.2121	0.2423	1.535	87.692 ± 2.353	1.521 ± 0.145	3.077 ± 1.237

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S20

Simulation 1 - Condition 20: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.4

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.295	-0.021 ± 0.004	-0.066 ± 0.012	0.003 ± 0.000	0.059 ± 0.003	0.004 ± 0.000	0.062 ± 0.003	0.039 ± 0.003	0.198 ± 0.009	0.293	-0.023	-7.35	0.0710	0.0747	1.002	94.656 ± 1.389	0.232 ± 0.002	100.000 ± 0.000
LMS	beta32	0.316	0.293	-0.023 ± 0.003	-0.072 ± 0.011	0.003 ± 0.000	0.055 ± 0.003	0.004 ± 0.000	0.059 ± 0.003	0.035 ± 0.003	0.188 ± 0.010	0.295	-0.021	-6.60	0.0660	0.0692	1.067	94.656 ± 1.389	0.230 ± 0.001	100.000 ± 0.000
LMS	beta33	0.139	0.204	0.065 ± 0.005	0.470 ± 0.036	0.006 ± 0.001	0.080 ± 0.004	0.011 ± 0.001	0.103 ± 0.004	0.554 ± 0.040	0.744 ± 0.030	0.213	0.074	52.93	0.0993	0.1236	1.053	87.405 ± 2.050	0.332 ± 0.006	70.611 ± 2.814
LMS	beta34	0.101	0.047	-0.054 ± 0.003	-0.536 ± 0.034	0.003 ± 0.000	0.055 ± 0.003	0.006 ± 0.000	0.077 ± 0.003	0.585 ± 0.046	0.765 ± 0.033	0.045	-0.056	-55.27	0.0601	0.0820	0.989	78.626 ± 2.533	0.214 ± 0.004	9.160 ± 1.782
LMS	beta35	0.101	0.047	-0.054 ± 0.003	-0.530 ± 0.034	0.003 ± 0.000	0.055 ± 0.003	0.006 ± 0.000	0.077 ± 0.003	0.579 ± 0.037	0.761 ± 0.027	0.042	-0.059	-58.01	0.0633	0.0863	0.985	74.809 ± 2.682	0.213 ± 0.004	10.687 ± 1.909
LSAM	beta31	0.316	0.328	0.012 ± 0.006	0.038 ± 0.018	0.009 ± 0.001	0.093 ± 0.005	0.009 ± 0.001	0.094 ± 0.006	0.088 ± 0.009	0.297 ± 0.018	0.319	0.003	0.91	0.0905	0.0906	1.000	96.947 ± 1.063	0.365 ± 0.005	94.656 ± 1.389
LSAM	beta32	0.316	0.322	0.006 ± 0.005	0.019 ± 0.016	0.007 ± 0.001	0.084 ± 0.004	0.007 ± 0.001	0.084 ± 0.005	0.071 ± 0.007	0.266 ± 0.016	0.317	0.001	0.33	0.0808	0.0808	1.093	96.565 ± 1.125	0.360 ± 0.005	95.420 ± 1.292
LSAM	beta33	0.139	0.274	0.135 ± 0.006	0.973 ± 0.046	0.011 ± 0.001	0.104 ± 0.005	0.029 ± 0.002	0.171 ± 0.007	1.507 ± 0.116	1.228 ± 0.049	0.263	0.124	88.88	0.0975	0.1574	0.981	78.626 ± 2.533	0.401 ± 0.008	81.679 ± 2.390
LSAM	beta34	0.101	-0.001	-0.102 ± 0.004	-1.010 ± 0.042	0.005 ± 0.000	0.069 ± 0.003	0.015 ± 0.001	0.123 ± 0.005	1.484 ± 0.105	1.218 ± 0.045	-0.001	-0.102	-101.08	0.0616	0.1192	0.975	60.305 ± 3.023	0.263 ± 0.005	2.672 ± 0.996
LSAM	beta35	0.101	0.002	-0.099 ± 0.004	-0.976 ± 0.040	0.004 ± 0.000	0.065 ± 0.004	0.014 ± 0.001	0.118 ± 0.004	1.364 ± 0.087	1.168 ± 0.039	0.003	-0.098	-97.31	0.0618	0.1161	1.009	65.649 ± 2.934	0.257 ± 0.006	4.580 ± 1.292
QML	beta31	0.316	0.293	-0.023 ± 0.004	-0.073 ± 0.011	0.003 ± 0.000	0.059 ± 0.003	0.004 ± 0.000	0.063 ± 0.003	0.040 ± 0.003	0.199 ± 0.009	0.288	-0.028	-8.95	0.0626	0.0687	1.006	94.275 ± 1.435	0.232 ± 0.002	100.000 ± 0.000
QML	beta32	0.316	0.291	-0.025 ± 0.003	-0.079 ± 0.011	0.003 ± 0.000	0.055 ± 0.003	0.004 ± 0.000	0.060 ± 0.003	0.036 ± 0.003	0.190 ± 0.010	0.293	-0.023	-7.40	0.0592	0.0636	1.068	93.511 ± 1.522	0.230 ± 0.001	100.000 ± 0.000
QML	beta33	0.139	0.202	0.063 ± 0.005	0.451 ± 0.036	0.006 ± 0.001	0.081 ± 0.004	0.010 ± 0.001	0.102 ± 0.004	0.538 ± 0.040	0.733 ± 0.031	0.208	0.069	49.68	0.0939	0.1166	1.050	89.313 ± 1.909	0.331 ± 0.005	67.557 ± 2.892
QML	beta34	0.101	0.043	-0.058 ± 0.003	-0.574 ± 0.034	0.003 ± 0.000	0.055 ± 0.003	0.006 ± 0.000	0.080 ± 0.003	0.625 ± 0.049	0.790 ± 0.033	0.040	-0.061	-60.15	0.0550	0.0819	0.986	77.863 ± 2.565	0.212 ± 0.003	11.069 ± 1.938
QML	beta35	0.101	0.044	-0.057 ± 0.003	-0.565 ± 0.033	0.003 ± 0.000	0.054 ± 0.003	0.006 ± 0.000	0.079 ± 0.003	0.606 ± 0.039	0.778 ± 0.028	0.038	-0.063	-62.64	0.0572	0.0853	0.991	74.046 ± 2.708	0.211 ± 0.003	11.832 ± 1.995
UPI	beta31	0.316	0.310	-0.006 ± 0.015	-0.018 ± 0.048	0.062 ± 0.009	0.248 ± 0.018	0.061 ± 0.009	0.248 ± 0.020	0.614 ± 0.088	0.784 ± 0.062	0.328	0.012	3.90	0.2672	0.2675	1.876	81.298 ± 2.409	1.824 ± 0.256	38.168 ± 3.001
UPI	beta32	0.316	0.297	-0.019 ± 0.014	-0.059 ± 0.044	0.051 ± 0.007	0.226 ± 0.016	0.051 ± 0.007	0.227 ± 0.017	0.514 ± 0.068	0.717 ± 0.053	0.305	-0.011	-3.43	0.2512	0.2514	1.885	83.206 ± 2.309	1.672 ± 0.225	42.748 ± 3.056
UPI	beta33	0.139	0.267	0.128 ± 0.012	0.920 ± 0.089	0.040 ± 0.005	0.201 ± 0.014	0.057 ± 0.007	0.238 ± 0.016	2.928 ± 0.366	1.711 ± 0.115	0.229	0.090	65.05	0.1652	0.1883	1.699	99.237 ± 0.538	1.338 ± 0.124	22.137 ± 2.565
UPI	beta34	0.101	-0.091	-0.192 ± 0.016	-1.903 ± 0.161	0.069 ± 0.010	0.263 ± 0.020	0.106 ± 0.013	0.325 ± 0.022	10.374 ± 1.302	3.221 ± 0.215	-0.073	-0.174	-172.64	0.2452	0.3009	1.784	81.298 ± 2.409	1.839 ± 0.208	1.908 ± 0.845
UPI	beta35	0.101	-0.054	-0.155 ± 0.013	-1.536 ± 0.130	0.045 ± 0.005	0.212 ± 0.013	0.069 ± 0.007	0.263 ± 0.015	6.762 ± 0.724	2.600 ± 0.150	-0.032	-0.133	-131.38	0.2145	0.2522	1.687	80.916 ± 2.428	1.404 ± 0.148	1.527 ± 0.758

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S21

Simulation 1 - Condition 21: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.6

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LMS	beta31	0.316	0.302	-0.014 $\pm$ 0.002	-0.044 $\pm$ 0.007	0.005 $\pm$ 0.000	0.071 $\pm$ 0.002	0.005 $\pm$ 0.000	0.072 $\pm$ 0.002	0.052 $\pm$ 0.002	0.229 $\pm$ 0.006	0.300	-0.016	-4.93	0.0707	0.0724	0.990	92.821 $\pm$ 0.821	0.276 $\pm$ 0.001	99.090 $\pm$ 0.302
LMS	beta32	0.316	0.304	-0.012 $\pm$ 0.002	-0.039 $\pm$ 0.008	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.006 $\pm$ 0.000	0.076 $\pm$ 0.002	0.057 $\pm$ 0.003	0.239 $\pm$ 0.007	0.302	-0.014	-4.50	0.0718	0.0732	0.947	92.012 $\pm$ 0.862	0.277 $\pm$ 0.001	99.191 $\pm$ 0.285
LMS	beta33	0.139	0.154	0.015 $\pm$ 0.003	0.104 $\pm$ 0.020	0.007 $\pm$ 0.000	0.085 $\pm$ 0.002	0.007 $\pm$ 0.000	0.087 $\pm$ 0.002	0.388 $\pm$ 0.017	0.623 $\pm$ 0.017	0.153	0.014	10.28	0.0860	0.0872	1.016	94.944 $\pm$ 0.697	0.340 $\pm$ 0.003	46.006 $\pm$ 1.585
LMS	beta34	0.101	0.083	-0.018 $\pm$ 0.002	-0.176 $\pm$ 0.019	0.003 $\pm$ 0.000	0.059 $\pm$ 0.001	0.004 $\pm$ 0.000	0.062 $\pm$ 0.002	0.373 $\pm$ 0.016	0.611 $\pm$ 0.016	0.081	-0.020	-19.65	0.0582	0.0615	0.980	92.518 $\pm$ 0.837	0.227 $\pm$ 0.002	31.850 $\pm$ 1.481
LMS	beta35	0.101	0.084	-0.017 $\pm$ 0.002	-0.172 $\pm$ 0.018	0.003 $\pm$ 0.000	0.058 $\pm$ 0.001	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.359 $\pm$ 0.018	0.599 $\pm$ 0.017	0.084	-0.017	-16.39	0.0558	0.0582	1.023	92.619 $\pm$ 0.831	0.232 $\pm$ 0.002	31.446 $\pm$ 1.476
LSAM	beta31	0.316	0.312	-0.004 $\pm$ 0.002	-0.012 $\pm$ 0.008	0.006 $\pm$ 0.000	0.078 $\pm$ 0.002	0.006 $\pm$ 0.000	0.078 $\pm$ 0.002	0.061 $\pm$ 0.003	0.247 $\pm$ 0.007	0.312	-0.004	-1.18	0.0778	0.0779	1.004	94.540 $\pm$ 0.722	0.307 $\pm$ 0.001	97.573 $\pm$ 0.489
LSAM	beta32	0.316	0.314	-0.002 $\pm$ 0.003	-0.006 $\pm$ 0.008	0.007 $\pm$ 0.000	0.083 $\pm$ 0.002	0.007 $\pm$ 0.000	0.083 $\pm$ 0.002	0.070 $\pm$ 0.003	0.264 $\pm$ 0.008	0.313	-0.003	-0.81	0.0782	0.0783	0.938	92.821 $\pm$ 0.821	0.307 $\pm$ 0.001	98.180 $\pm$ 0.425
LSAM	beta33	0.139	0.178	0.039 $\pm$ 0.003	0.281 $\pm$ 0.021	0.009 $\pm$ 0.000	0.093 $\pm$ 0.002	0.010 $\pm$ 0.000	0.101 $\pm$ 0.003	0.529 $\pm$ 0.025	0.727 $\pm$ 0.020	0.176	0.037	26.32	0.0919	0.0989	0.988	93.428 $\pm$ 0.788	0.361 $\pm$ 0.003	51.972 $\pm$ 1.589
LSAM	beta34	0.101	0.072	-0.029 $\pm$ 0.002	-0.283 $\pm$ 0.021	0.004 $\pm$ 0.000	0.065 $\pm$ 0.002	0.005 $\pm$ 0.000	0.071 $\pm$ 0.002	0.499 $\pm$ 0.022	0.707 $\pm$ 0.018	0.071	-0.030	-30.02	0.0653	0.0720	0.968	90.495 $\pm$ 0.933	0.248 $\pm$ 0.002	22.346 $\pm$ 1.325
LSAM	beta35	0.101	0.073	-0.028 $\pm$ 0.002	-0.273 $\pm$ 0.020	0.004 $\pm$ 0.000	0.065 $\pm$ 0.002	0.005 $\pm$ 0.000	0.070 $\pm$ 0.002	0.484 $\pm$ 0.023	0.696 $\pm$ 0.019	0.075	-0.026	-25.72	0.0611	0.0664	1.001	90.597 $\pm$ 0.928	0.254 $\pm$ 0.002	21.031 $\pm$ 1.296
QML	beta31	0.316	0.302	-0.014 $\pm$ 0.002	-0.045 $\pm$ 0.007	0.005 $\pm$ 0.000	0.071 $\pm$ 0.002	0.005 $\pm$ 0.000	0.072 $\pm$ 0.002	0.052 $\pm$ 0.002	0.229 $\pm$ 0.006	0.300	-0.016	-5.08	0.0702	0.0721	0.988	92.619 $\pm$ 0.831	0.275 $\pm$ 0.001	99.090 $\pm$ 0.302
QML	beta32	0.316	0.303	-0.013 $\pm$ 0.002	-0.040 $\pm$ 0.008	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.006 $\pm$ 0.000	0.076 $\pm$ 0.002	0.057 $\pm$ 0.003	0.239 $\pm$ 0.007	0.302	-0.014	-4.50	0.0718	0.0732	0.944	91.911 $\pm$ 0.867	0.276 $\pm$ 0.001	99.191 $\pm$ 0.285
QML	beta33	0.139	0.154	0.015 $\pm$ 0.003	0.109 $\pm$ 0.019	0.007 $\pm$ 0.000	0.085 $\pm$ 0.002	0.007 $\pm$ 0.000	0.086 $\pm$ 0.002	0.387 $\pm$ 0.017	0.622 $\pm$ 0.017	0.154	0.015	11.03	0.0845	0.0859	1.017	95.046 $\pm$ 0.690	0.339 $\pm$ 0.003	46.613 $\pm$ 1.586
QML	beta34	0.101	0.082	-0.019 $\pm$ 0.002	-0.186 $\pm$ 0.019	0.003 $\pm$ 0.000	0.059 $\pm$ 0.001	0.004 $\pm$ 0.000	0.062 $\pm$ 0.002	0.376 $\pm$ 0.016	0.613 $\pm$ 0.016	0.081	-0.020	-19.72	0.0587	0.0620	0.980	92.417 $\pm$ 0.842	0.227 $\pm$ 0.002	31.547 $\pm$ 1.478
QML	beta35	0.101	0.083	-0.018 $\pm$ 0.002	-0.183 $\pm$ 0.018	0.003 $\pm$ 0.000	0.058 $\pm$ 0.001	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.360 $\pm$ 0.017	0.600 $\pm$ 0.017	0.084	-0.017	-16.84	0.0563	0.0588	1.024	92.417 $\pm$ 0.842	0.232 $\pm$ 0.002	31.041 $\pm$ 1.471
UPI	beta31	0.316	0.321	0.005 $\pm$ 0.003	0.015 $\pm$ 0.008	0.006 $\pm$ 0.000	0.080 $\pm$ 0.002	0.006 $\pm$ 0.000	0.081 $\pm$ 0.002	0.065 $\pm$ 0.003	0.255 $\pm$ 0.008	0.320	0.004	1.19	0.0750	0.0751	1.012	94.641 $\pm$ 0.716	0.319 $\pm$ 0.003	96.562 $\pm$ 0.579
UPI	beta32	0.316	0.323	0.007 $\pm$ 0.003	0.022 $\pm$ 0.009	0.007 $\pm$ 0.000	0.086 $\pm$ 0.002	0.007 $\pm$ 0.000	0.086 $\pm$ 0.002	0.074 $\pm$ 0.004	0.272 $\pm$ 0.008	0.322	0.006	1.98	0.0815	0.0818	0.950	93.428 $\pm$ 0.788	0.319 $\pm$ 0.003	97.169 $\pm$ 0.527
UPI	beta33	0.139	0.166	0.027 $\pm$ 0.003	0.193 $\pm$ 0.023	0.010 $\pm$ 0.000	0.099 $\pm$ 0.002	0.010 $\pm$ 0.001	0.102 $\pm$ 0.003	0.541 $\pm$ 0.027	0.736 $\pm$ 0.021	0.162	0.023	16.73	0.0960	0.0988	1.016	96.461 $\pm$ 0.588	0.393 $\pm$ 0.005	41.658 $\pm$ 1.568
UPI	beta34	0.101	0.089	-0.012 $\pm$ 0.002	-0.123 $\pm$ 0.024	0.006 $\pm$ 0.000	0.077 $\pm$ 0.002	0.006 $\pm$ 0.000	0.078 $\pm$ 0.002	0.594 $\pm$ 0.029	0.771 $\pm$ 0.023	0.088	-0.013	-12.52	0.0730	0.0741	0.980	93.327 $\pm$ 0.794	0.295 $\pm$ 0.004	25.581 $\pm$ 1.387
UPI	beta35	0.101	0.088	-0.013 $\pm$ 0.002	-0.130 $\pm$ 0.023	0.005 $\pm$ 0.000	0.073 $\pm$ 0.002	0.005 $\pm$ 0.000	0.074 $\pm$ 0.002	0.534 $\pm$ 0.027	0.731 $\pm$ 0.022	0.086	-0.015	-14.42	0.0662	0.0678	1.056	95.248 $\pm$ 0.677	0.301 $\pm$ 0.004	24.267 $\pm$ 1.363

Note. MCSE = Monte Carlo Standard Error shown as $\backslash u00b1$ value. Coverage, Type I Error, and Power expressed as percentages. $\backslash u2014$ indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S22

Simulation 1 - Condition 22: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.6

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LMS	beta31	0.316	0.303	-0.013 $\pm$ 0.002	-0.041 $\pm$ 0.005	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	0.025 $\pm$ 0.001	0.157 $\pm$ 0.004	0.301	-0.015	-4.79	0.0498	0.0520	0.927	91.300 $\pm$ 0.891	0.174 $\pm$ 0.001	99.800 $\pm$ 0.141
LMS	beta32	0.316	0.304	-0.012 $\pm$ 0.001	-0.039 $\pm$ 0.004	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.021 $\pm$ 0.001	0.145 $\pm$ 0.004	0.304	-0.012	-3.93	0.0432	0.0449	1.006	94.700 $\pm$ 0.708	0.174 $\pm$ 0.001	99.900 $\pm$ 0.100
LMS	beta33	0.139	0.158	0.019 $\pm$ 0.002	0.135 $\pm$ 0.013	0.003 $\pm$ 0.000	0.056 $\pm$ 0.001	0.004 $\pm$ 0.000	0.059 $\pm$ 0.002	0.182 $\pm$ 0.008	0.427 $\pm$ 0.011	0.159	0.020	14.74	0.0545	0.0582	0.973	93.300 $\pm$ 0.791	0.215 $\pm$ 0.001	81.000 $\pm$ 1.241
LMS	beta34	0.101	0.080	-0.021 $\pm$ 0.001	-0.213 $\pm$ 0.012	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.185 $\pm$ 0.008	0.431 $\pm$ 0.010	0.080	-0.021	-20.55	0.0375	0.0428	0.979	88.800 $\pm$ 0.997	0.145 $\pm$ 0.001	57.400 $\pm$ 1.564
LMS	beta35	0.101	0.080	-0.021 $\pm$ 0.001	-0.207 $\pm$ 0.012	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.181 $\pm$ 0.008	0.426 $\pm$ 0.010	0.079	-0.022	-21.47	0.0373	0.0431	0.977	89.100 $\pm$ 0.985	0.144 $\pm$ 0.001	58.000 $\pm$ 1.561
LSAM	beta31	0.316	0.313	-0.003 $\pm$ 0.002	-0.009 $\pm$ 0.005	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.027 $\pm$ 0.001	0.166 $\pm$ 0.005	0.312	-0.004	-1.39	0.0520	0.0522	0.929	92.900 $\pm$ 0.812	0.190 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta32	0.316	0.313	-0.003 $\pm$ 0.002	-0.010 $\pm$ 0.005	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.023 $\pm$ 0.001	0.153 $\pm$ 0.004	0.313	-0.003	-1.06	0.0463	0.0464	1.001	95.500 $\pm$ 0.656	0.189 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta33	0.139	0.179	0.040 $\pm$ 0.002	0.288 $\pm$ 0.013	0.003 $\pm$ 0.000	0.058 $\pm$ 0.001	0.005 $\pm$ 0.000	0.071 $\pm$ 0.002	0.259 $\pm$ 0.011	0.508 $\pm$ 0.012	0.179	0.040	28.58	0.0576	0.0700	0.955	87.800 $\pm$ 1.035	0.218 $\pm$ 0.001	89.600 $\pm$ 0.965
LSAM	beta34	0.101	0.069	-0.032 $\pm$ 0.001	-0.318 $\pm$ 0.013	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.268 $\pm$ 0.011	0.518 $\pm$ 0.012	0.068	-0.033	-32.64	0.0416	0.0531	0.950	84.500 $\pm$ 1.144	0.154 $\pm$ 0.001	43.600 $\pm$ 1.568
LSAM	beta35	0.101	0.070	-0.031 $\pm$ 0.001	-0.310 $\pm$ 0.013	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.261 $\pm$ 0.011	0.511 $\pm$ 0.012	0.069	-0.032	-31.71	0.0404	0.0515	0.949	83.900 $\pm$ 1.162	0.153 $\pm$ 0.001	43.200 $\pm$ 1.566
QML	beta31	0.316	0.303	-0.013 $\pm$ 0.002	-0.041 $\pm$ 0.005	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	0.025 $\pm$ 0.001	0.158 $\pm$ 0.004	0.301	-0.015	-4.86	0.0497	0.0520	0.925	91.600 $\pm$ 0.877	0.174 $\pm$ 0.001	99.900 $\pm$ 0.100
QML	beta32	0.316	0.303	-0.013 $\pm$ 0.001	-0.040 $\pm$ 0.004	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.021 $\pm$ 0.001	0.145 $\pm$ 0.004	0.303	-0.013	-4.08	0.0431	0.0450	1.004	94.700 $\pm$ 0.708	0.173 $\pm$ 0.001	100.000 $\pm$ 0.000
QML	beta33	0.139	0.158	0.019 $\pm$ 0.002	0.137 $\pm$ 0.013	0.003 $\pm$ 0.000	0.056 $\pm$ 0.001	0.004 $\pm$ 0.000	0.059 $\pm$ 0.002	0.182 $\pm$ 0.008	0.426 $\pm$ 0.011	0.160	0.021	14.75	0.0542	0.0580	0.974	93.400 $\pm$ 0.785	0.214 $\pm$ 0.001	81.100 $\pm$ 1.238
QML	beta34	0.101	0.079	-0.022 $\pm$ 0.001	-0.220 $\pm$ 0.012	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.188 $\pm$ 0.008	0.434 $\pm$ 0.011	0.080	-0.021	-21.11	0.0371	0.0428	0.977	88.200 $\pm$ 1.020	0.145 $\pm$ 0.001	56.800 $\pm$ 1.566
QML	beta35	0.101	0.079	-0.022 $\pm$ 0.001	-0.214 $\pm$ 0.012	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.184 $\pm$ 0.008	0.429 $\pm$ 0.010	0.078	-0.023	-22.38	0.0370	0.0433	0.978	88.600 $\pm$ 1.005	0.144 $\pm$ 0.001	57.700 $\pm$ 1.562
UPI	beta31	0.316	0.317	0.001 $\pm$ 0.002	0.003 $\pm$ 0.005	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.003 $\pm$ 0.000	0.053 $\pm$ 0.002	0.028 $\pm$ 0.001	0.168 $\pm$ 0.005	0.316	0.000	0.13	0.0517	0.0517	0.914	91.900 $\pm$ 0.863	0.190 $\pm$ 0.001	100.000 $\pm$ 0.000
UPI	beta32	0.316	0.317	0.001 $\pm$ 0.002	0.002 $\pm$ 0.005	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.023 $\pm$ 0.001	0.153 $\pm$ 0.004	0.318	0.002	0.62	0.0467	0.0467	0.996	95.200 $\pm$ 0.676	0.189 $\pm$ 0.001	100.000 $\pm$ 0.000
UPI	beta33	0.139	0.159	0.020 $\pm$ 0.002	0.145 $\pm$ 0.014	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	0.004 $\pm$ 0.000	0.064 $\pm$ 0.002	0.210 $\pm$ 0.011	0.458 $\pm$ 0.013	0.157	0.018	12.89	0.0572	0.0599	0.963	94.100 $\pm$ 0.745	0.228 $\pm$ 0.001	79.700 $\pm$ 1.272
UPI	beta34	0.101	0.084	-0.017 $\pm$ 0.001	-0.165 $\pm$ 0.015	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.240 $\pm$ 0.011	0.490 $\pm$ 0.013	0.083	-0.018	-17.91	0.0460	0.0494	0.949	91.400 $\pm$ 0.887	0.173 $\pm$ 0.001	50.400 $\pm$ 1.581
UPI	beta35	0.101	0.085	-0.016 $\pm$ 0.001	-0.158 $\pm$ 0.014	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.223 $\pm$ 0.010	0.472 $\pm$ 0.013	0.084	-0.017	-16.40	0.0438	0.0469	0.970	91.000 $\pm$ 0.905	0.171 $\pm$ 0.001	50.300 $\pm$ 1.581

Note. MCSE = Monte Carlo Standard Error shown as $\backslash 00b1$ value. Coverage, Type I Error, and Power expressed as percentages. $\backslash 02014$ indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S23

Simulation 1 - Condition 23: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.8

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LMS	beta31	0.316	0.309	-0.007 $\pm$ 0.002	-0.021 $\pm$ 0.006	0.003 $\pm$ 0.000	0.058 $\pm$ 0.001	0.003 $\pm$ 0.000	0.058 $\pm$ 0.002	0.034 $\pm$ 0.002	0.185 $\pm$ 0.005	0.307	-0.009	-2.96	0.0593	0.0601	0.997	94.800 $\pm$ 0.702	0.227 $\pm$ 0.001	99.800 $\pm$ 0.141
LMS	beta32	0.316	0.311	-0.005 $\pm$ 0.002	-0.017 $\pm$ 0.006	0.003 $\pm$ 0.000	0.057 $\pm$ 0.001	0.003 $\pm$ 0.000	0.057 $\pm$ 0.001	0.033 $\pm$ 0.001	0.181 $\pm$ 0.005	0.312	-0.004	-1.38	0.0577	0.0579	1.019	96.100 $\pm$ 0.612	0.227 $\pm$ 0.001	99.900 $\pm$ 0.100
LMS	beta33	0.139	0.140	0.001 $\pm$ 0.002	0.010 $\pm$ 0.015	0.004 $\pm$ 0.000	0.064 $\pm$ 0.001	0.004 $\pm$ 0.000	0.064 $\pm$ 0.002	0.214 $\pm$ 0.010	0.462 $\pm$ 0.013	0.139	0.000	-0.33	0.0600	0.0600	1.003	94.700 $\pm$ 0.708	0.253 $\pm$ 0.001	59.900 $\pm$ 1.550
LMS	beta34	0.101	0.096	-0.005 $\pm$ 0.001	-0.051 $\pm$ 0.014	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.207 $\pm$ 0.009	0.455 $\pm$ 0.012	0.095	-0.006	-6.00	0.0461	0.0465	0.969	94.300 $\pm$ 0.733	0.174 $\pm$ 0.001	58.200 $\pm$ 1.560
LMS	beta35	0.101	0.095	-0.006 $\pm$ 0.001	-0.061 $\pm$ 0.014	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.195 $\pm$ 0.009	0.442 $\pm$ 0.012	0.093	-0.008	-8.10	0.0434	0.0441	1.009	94.000 $\pm$ 0.751	0.175 $\pm$ 0.001	56.800 $\pm$ 1.566
LSAM	beta31	0.316	0.314	-0.002 $\pm$ 0.002	-0.007 $\pm$ 0.006	0.004 $\pm$ 0.000	0.059 $\pm$ 0.001	0.004 $\pm$ 0.000	0.059 $\pm$ 0.002	0.035 $\pm$ 0.002	0.188 $\pm$ 0.005	0.310	-0.006	-1.77	0.0600	0.0603	1.001	95.700 $\pm$ 0.641	0.233 $\pm$ 0.001	100.000 $\pm$ 0.000
LSAM	beta32	0.316	0.315	-0.001 $\pm$ 0.002	-0.003 $\pm$ 0.006	0.003 $\pm$ 0.000	0.058 $\pm$ 0.001	0.003 $\pm$ 0.000	0.058 $\pm$ 0.002	0.034 $\pm$ 0.001	0.185 $\pm$ 0.005	0.316	0.000	0.06	0.0576	0.0576	1.018	96.100 $\pm$ 0.612	0.233 $\pm$ 0.001	100.000 $\pm$ 0.000
LSAM	beta33	0.139	0.148	0.009 $\pm$ 0.002	0.063 $\pm$ 0.015	0.004 $\pm$ 0.000	0.065 $\pm$ 0.001	0.004 $\pm$ 0.000	0.065 $\pm$ 0.002	0.219 $\pm$ 0.010	0.468 $\pm$ 0.013	0.148	0.009	6.20	0.0608	0.0614	1.006	94.000 $\pm$ 0.751	0.254 $\pm$ 0.001	63.400 $\pm$ 1.523
LSAM	beta34	0.101	0.095	-0.006 $\pm$ 0.002	-0.064 $\pm$ 0.015	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.226 $\pm$ 0.010	0.475 $\pm$ 0.013	0.092	-0.009	-8.47	0.0469	0.0476	0.960	93.900 $\pm$ 0.757	0.179 $\pm$ 0.001	55.700 $\pm$ 1.571
LSAM	beta35	0.101	0.093	-0.008 $\pm$ 0.001	-0.077 $\pm$ 0.014	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.212 $\pm$ 0.010	0.460 $\pm$ 0.013	0.091	-0.010	-9.56	0.0450	0.0460	1.002	94.000 $\pm$ 0.751	0.180 $\pm$ 0.001	52.700 $\pm$ 1.579
QML	beta31	0.316	0.309	-0.007 $\pm$ 0.002	-0.022 $\pm$ 0.006	0.003 $\pm$ 0.000	0.058 $\pm$ 0.001	0.003 $\pm$ 0.000	0.058 $\pm$ 0.002	0.034 $\pm$ 0.002	0.185 $\pm$ 0.005	0.307	-0.009	-2.97	0.0595	0.0602	0.993	94.800 $\pm$ 0.702	0.226 $\pm$ 0.001	99.900 $\pm$ 0.100
QML	beta32	0.316	0.310	-0.006 $\pm$ 0.002	-0.018 $\pm$ 0.006	0.003 $\pm$ 0.000	0.057 $\pm$ 0.001	0.003 $\pm$ 0.000	0.057 $\pm$ 0.001	0.033 $\pm$ 0.001	0.181 $\pm$ 0.005	0.311	-0.005	-1.46	0.0576	0.0578	1.015	96.100 $\pm$ 0.612	0.226 $\pm$ 0.001	100.000 $\pm$ 0.000
QML	beta33	0.139	0.140	0.001 $\pm$ 0.002	0.009 $\pm$ 0.015	0.004 $\pm$ 0.000	0.064 $\pm$ 0.001	0.004 $\pm$ 0.000	0.064 $\pm$ 0.002	0.214 $\pm$ 0.010	0.463 $\pm$ 0.013	0.139	0.000	0.01	0.0601	0.0601	1.001	94.600 $\pm$ 0.715	0.253 $\pm$ 0.001	60.000 $\pm$ 1.549
QML	beta34	0.101	0.096	-0.005 $\pm$ 0.001	-0.053 $\pm$ 0.014	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.208 $\pm$ 0.009	0.456 $\pm$ 0.012	0.094	-0.007	-6.58	0.0463	0.0468	0.967	94.200 $\pm$ 0.739	0.174 $\pm$ 0.001	58.000 $\pm$ 1.561
QML	beta35	0.101	0.095	-0.006 $\pm$ 0.001	-0.063 $\pm$ 0.014	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.196 $\pm$ 0.009	0.443 $\pm$ 0.012	0.093	-0.008	-8.12	0.0433	0.0440	1.007	93.800 $\pm$ 0.763	0.175 $\pm$ 0.001	56.500 $\pm$ 1.568
UPI	beta31	0.316	0.318	0.002 $\pm$ 0.002	0.007 $\pm$ 0.006	0.003 $\pm$ 0.000	0.059 $\pm$ 0.001	0.003 $\pm$ 0.000	0.059 $\pm$ 0.002	0.035 $\pm$ 0.002	0.187 $\pm$ 0.005	0.314	-0.002	-0.60	0.0603	0.0604	0.967	94.500 $\pm$ 0.721	0.224 $\pm$ 0.001	100.000 $\pm$ 0.000
UPI	beta32	0.316	0.320	0.004 $\pm$ 0.002	0.011 $\pm$ 0.006	0.003 $\pm$ 0.000	0.058 $\pm$ 0.001	0.003 $\pm$ 0.000	0.058 $\pm$ 0.002	0.034 $\pm$ 0.001	0.184 $\pm$ 0.005	0.320	0.004	1.42	0.0568	0.0570	0.986	95.100 $\pm$ 0.683	0.225 $\pm$ 0.001	100.000 $\pm$ 0.000
UPI	beta33	0.139	0.144	0.005 $\pm$ 0.002	0.035 $\pm$ 0.015	0.004 $\pm$ 0.000	0.066 $\pm$ 0.001	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	0.223 $\pm$ 0.010	0.473 $\pm$ 0.013	0.146	0.007	4.72	0.0636	0.0639	0.989	94.200 $\pm$ 0.739	0.254 $\pm$ 0.001	62.100 $\pm$ 1.534
UPI	beta34	0.101	0.099	-0.002 $\pm$ 0.002	-0.018 $\pm$ 0.015	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.235 $\pm$ 0.011	0.485 $\pm$ 0.013	0.096	-0.005	-4.47	0.0488	0.0491	0.947	94.000 $\pm$ 0.751	0.182 $\pm$ 0.001	58.100 $\pm$ 1.560
UPI	beta35	0.101	0.097	-0.004 $\pm$ 0.001	-0.040 $\pm$ 0.015	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.216 $\pm$ 0.010	0.465 $\pm$ 0.013	0.096	-0.005	-5.30	0.0446	0.0449	0.992	94.500 $\pm$ 0.721	0.182 $\pm$ 0.001	56.100 $\pm$ 1.569

Note. MCSE = Monte Carlo Standard Error shown as $\backslash u00b1$ value. Coverage, Type I Error, and Power expressed as percentages. $\backslash u2014$ indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S24

Simulation 1 - Condition 24: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.8

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.309	-0.007 ± 0.001	-0.023 ± 0.004	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.013 ± 0.001	0.116 ± 0.003	0.309	-0.007	-2.34	0.0360	0.0368	1.020	95.095 ± 0.683	0.143 ± 0.000	100.000 ± 0.000
LMS	beta32	0.316	0.310	-0.006 ± 0.001	-0.020 ± 0.004	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.014 ± 0.001	0.117 ± 0.003	0.310	-0.006	-1.81	0.0357	0.0361	1.000	94.294 ± 0.734	0.143 ± 0.000	100.000 ± 0.000
LMS	beta33	0.139	0.138	-0.001 ± 0.001	-0.008 ± 0.010	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.095 ± 0.004	0.308 ± 0.009	0.138	-0.001	-0.93	0.0413	0.0413	0.947	92.893 ± 0.813	0.159 ± 0.001	90.490 ± 0.928
LMS	beta34	0.101	0.099	-0.002 ± 0.001	-0.024 ± 0.009	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.077 ± 0.003	0.277 ± 0.007	0.099	-0.002	-2.43	0.0293	0.0294	1.003	95.195 ± 0.677	0.110 ± 0.000	94.394 ± 0.728
LMS	beta35	0.101	0.096	-0.005 ± 0.001	-0.050 ± 0.009	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.081 ± 0.004	0.284 ± 0.008	0.096	-0.005	-5.42	0.0279	0.0284	0.992	93.994 ± 0.752	0.110 ± 0.000	91.992 ± 0.859
LSAM	beta31	0.316	0.313	-0.003 ± 0.001	-0.009 ± 0.004	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.014 ± 0.001	0.116 ± 0.003	0.313	-0.003	-0.95	0.0376	0.0377	1.020	95.596 ± 0.649	0.147 ± 0.000	100.000 ± 0.000
LSAM	beta32	0.316	0.314	-0.002 ± 0.001	-0.006 ± 0.004	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.014 ± 0.001	0.117 ± 0.003	0.314	-0.002	-0.56	0.0368	0.0369	1.010	94.494 ± 0.722	0.147 ± 0.000	100.000 ± 0.000
LSAM	beta33	0.139	0.145	0.006 ± 0.001	0.043 ± 0.010	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.094 ± 0.004	0.307 ± 0.008	0.145	0.006	4.45	0.0421	0.0425	0.955	93.794 ± 0.763	0.158 ± 0.001	93.694 ± 0.769
LSAM	beta34	0.101	0.097	-0.004 ± 0.001	-0.043 ± 0.009	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.082 ± 0.004	0.287 ± 0.008	0.097	-0.004	-4.37	0.0304	0.0307	0.995	94.895 ± 0.696	0.112 ± 0.000	92.492 ± 0.834
LSAM	beta35	0.101	0.094	-0.007 ± 0.001	-0.071 ± 0.009	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.087 ± 0.004	0.295 ± 0.008	0.093	-0.008	-8.07	0.0296	0.0307	0.992	93.594 ± 0.775	0.112 ± 0.000	90.490 ± 0.928
QML	beta31	0.316	0.309	-0.007 ± 0.001	-0.023 ± 0.004	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.013 ± 0.001	0.116 ± 0.003	0.308	-0.008	-2.39	0.0360	0.0367	1.023	95.395 ± 0.663	0.144 ± 0.001	99.900 ± 0.100
QML	beta32	0.316	0.310	-0.006 ± 0.001	-0.020 ± 0.004	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.014 ± 0.001	0.117 ± 0.003	0.310	-0.006	-1.82	0.0356	0.0360	1.003	93.994 ± 0.752	0.143 ± 0.001	100.000 ± 0.000
QML	beta33	0.139	0.138	-0.001 ± 0.001	-0.009 ± 0.010	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.095 ± 0.004	0.308 ± 0.009	0.138	-0.001	-1.02	0.0410	0.0410	0.947	92.993 ± 0.808	0.159 ± 0.001	90.591 ± 0.924
QML	beta34	0.101	0.098	-0.003 ± 0.001	-0.025 ± 0.009	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.077 ± 0.003	0.277 ± 0.007	0.099	-0.002	-2.32	0.0292	0.0293	1.001	95.295 ± 0.670	0.110 ± 0.000	94.394 ± 0.728
QML	beta35	0.101	0.096	-0.005 ± 0.001	-0.051 ± 0.009	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.081 ± 0.004	0.285 ± 0.008	0.095	-0.006	-5.58	0.0279	0.0285	0.990	93.894 ± 0.758	0.110 ± 0.000	91.992 ± 0.859
UPI	beta31	0.316	0.315	-0.001 ± 0.001	-0.002 ± 0.004	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.014 ± 0.001	0.116 ± 0.003	0.314	-0.002	-0.56	0.0365	0.0365	0.980	94.595 ± 0.715	0.141 ± 0.000	100.000 ± 0.000
UPI	beta32	0.316	0.317	0.001 ± 0.001	0.002 ± 0.004	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.014 ± 0.001	0.117 ± 0.003	0.317	0.001	0.19	0.0377	0.0377	0.971	93.694 ± 0.769	0.141 ± 0.000	100.000 ± 0.000
UPI	beta33	0.139	0.141	0.002 ± 0.001	0.013 ± 0.010	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.094 ± 0.004	0.307 ± 0.009	0.139	0.000	0.31	0.0428	0.0428	0.949	93.093 ± 0.802	0.159 ± 0.001	91.692 ± 0.873
UPI	beta34	0.101	0.101	0.000 ± 0.001	0.001 ± 0.009	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.083 ± 0.004	0.287 ± 0.008	0.101	0.000	0.01	0.0305	0.0305	0.996	95.596 ± 0.649	0.113 ± 0.000	93.493 ± 0.780
UPI	beta35	0.101	0.098	-0.003 ± 0.001	-0.031 ± 0.009	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.084 ± 0.004	0.290 ± 0.008	0.097	-0.004	-3.61	0.0299	0.0302	0.994	94.494 ± 0.722	0.113 ± 0.000	91.992 ± 0.859

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S25

Simulation 1 - Condition 25: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.4

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.333	0.017 ± 0.005	0.053 ± 0.016	0.015 ± 0.001	0.121 ± 0.004	0.015 ± 0.001	0.122 ± 0.005	0.149 ± 0.010	0.386 ± 0.016	0.339	0.023	7.12	0.1241	0.1261	0.999	95.819 ± 0.818	0.473 ± 0.005	79.264 ± 1.658
LMS	beta32	0.316	0.336	0.020 ± 0.005	0.065 ± 0.016	0.015 ± 0.001	0.124 ± 0.004	0.016 ± 0.001	0.125 ± 0.005	0.158 ± 0.011	0.397 ± 0.016	0.341	0.025	7.87	0.1289	0.1313	0.984	95.819 ± 0.818	0.478 ± 0.005	81.271 ± 1.595
LMS	beta33	0.139	0.095	-0.044 ± 0.008	-0.314 ± 0.055	0.035 ± 0.002	0.188 ± 0.006	0.037 ± 0.003	0.192 ± 0.008	1.917 ± 0.134	1.384 ± 0.056	0.118	-0.021	-15.37	0.1918	0.1930	1.007	93.813 ± 0.985	0.740 ± 0.018	16.722 ± 1.526
LMS	beta34	0.101	0.192	0.091 ± 0.005	0.906 ± 0.045	0.013 ± 0.001	0.112 ± 0.004	0.021 ± 0.001	0.145 ± 0.005	2.054 ± 0.145	1.433 ± 0.054	0.177	0.076	75.25	0.1045	0.1293	0.993	91.137 ± 1.162	0.437 ± 0.012	49.164 ± 2.044
LMS	beta35	0.101	0.188	0.087 ± 0.004	0.858 ± 0.042	0.011 ± 0.001	0.103 ± 0.004	0.018 ± 0.001	0.134 ± 0.005	1.771 ± 0.124	1.331 ± 0.049	0.180	0.079	78.40	0.1073	0.1333	1.061	91.639 ± 1.132	0.428 ± 0.011	48.997 ± 2.044
LSAM	beta31	0.316	0.345	0.029 ± 0.053	0.091 ± 0.169	1.706 ± 0.110	1.306 ± 0.042	1.704 ± 0.110	1.305 ± 0.050	17.064 ± 1.102	4.131 ± 0.158	0.369	0.053	16.74	1.3921	1.3931	2.556	99.833 ± 0.167	13.086 ± 0.462	2.007 ± 0.573
LSAM	beta32	0.316	0.464	0.148 ± 0.054	0.469 ± 0.172	1.773 ± 0.112	1.332 ± 0.042	1.792 ± 0.115	1.339 ± 0.051	17.948 ± 1.147	4.236 ± 0.161	0.362	0.046	14.54	1.3665	1.3672	2.562	100.000 ± 0.000	13.372 ± 0.484	3.010 ± 0.699
LSAM	beta33	0.139	0.297	0.158 ± 0.015	1.135 ± 0.111	0.141 ± 0.012	0.376 ± 0.017	0.166 ± 0.015	0.408 ± 0.020	8.597 ± 0.782	2.932 ± 0.143	0.251	0.112	80.63	0.2959	0.3164	2.202	99.164 ± 0.372	3.246 ± 0.154	4.682 ± 0.864
LSAM	beta34	0.101	0.041	-0.060 ± 0.020	-0.590 ± 0.203	0.251 ± 0.021	0.501 ± 0.021	0.254 ± 0.021	0.504 ± 0.024	24.927 ± 2.094	4.993 ± 0.234	0.055	-0.046	-45.85	0.4388	0.4412	2.431	99.331 ± 0.333	4.775 ± 0.200	2.843 ± 0.680
LSAM	beta35	0.101	-0.004	-0.105 ± 0.021	-1.036 ± 0.212	0.275 ± 0.021	0.525 ± 0.020	0.286 ± 0.023	0.535 ± 0.024	28.017 ± 2.210	5.293 ± 0.234	0.050	-0.051	-50.90	0.4111	0.4143	2.415	99.666 ± 0.236	4.967 ± 0.212	3.177 ± 0.717
QML	beta31	0.316	0.335	0.019 ± 0.005	0.059 ± 0.015	0.013 ± 0.001	0.116 ± 0.004	0.014 ± 0.001	0.117 ± 0.005	0.137 ± 0.009	0.371 ± 0.015	0.336	0.020	6.27	0.1187	0.1204	0.997	95.485 ± 0.849	0.452 ± 0.003	83.278 ± 1.526
QML	beta32	0.316	0.338	0.022 ± 0.005	0.069 ± 0.015	0.014 ± 0.001	0.119 ± 0.004	0.015 ± 0.001	0.121 ± 0.005	0.146 ± 0.010	0.383 ± 0.015	0.340	0.024	7.66	0.1223	0.1246	0.980	95.819 ± 0.818	0.457 ± 0.004	83.946 ± 1.501
QML	beta33	0.139	0.103	-0.036 ± 0.007	-0.259 ± 0.050	0.028 ± 0.002	0.168 ± 0.005	0.030 ± 0.002	0.172 ± 0.006	1.533 ± 0.097	1.238 ± 0.046	0.116	-0.023	-16.86	0.1778	0.1794	0.993	94.147 ± 0.960	0.656 ± 0.013	17.057 ± 1.538
QML	beta34	0.101	0.184	0.083 ± 0.004	0.821 ± 0.041	0.010 ± 0.001	0.102 ± 0.004	0.017 ± 0.001	0.131 ± 0.005	1.687 ± 0.111	1.299 ± 0.046	0.176	0.075	73.85	0.0961	0.1217	0.958	87.793 ± 1.339	0.382 ± 0.008	54.181 ± 2.037
QML	beta35	0.101	0.179	0.078 ± 0.004	0.775 ± 0.038	0.009 ± 0.001	0.093 ± 0.003	0.015 ± 0.001	0.121 ± 0.004	1.445 ± 0.082	1.202 ± 0.037	0.177	0.076	75.37	0.0968	0.1231	1.033	88.963 ± 1.281	0.376 ± 0.007	52.843 ± 2.041
UPI	beta31	0.316	0.324	0.008 ± 0.021	0.026 ± 0.067	0.269 ± 0.021	0.519 ± 0.021	0.269 ± 0.021	0.518 ± 0.023	2.690 ± 0.215	1.640 ± 0.074	0.335	0.019	6.01	0.4722	0.4726	2.019	99.666 ± 0.236	4.105 ± 0.200	5.853 ± 0.960
UPI	beta32	0.316	0.345	0.029 ± 0.021	0.092 ± 0.067	0.268 ± 0.021	0.518 ± 0.021	0.268 ± 0.022	0.518 ± 0.023	2.687 ± 0.216	1.639 ± 0.074	0.324	0.008	2.40	0.4749	0.4750	2.001	99.164 ± 0.372	4.061 ± 0.185	5.686 ± 0.947
UPI	beta33	0.139	0.170	0.031 ± 0.011	0.219 ± 0.082	0.077 ± 0.007	0.278 ± 0.012	0.078 ± 0.007	0.280 ± 0.014	4.048 ± 0.369	2.012 ± 0.101	0.155	0.016	11.31	0.2383	0.2389	1.916	97.993 ± 0.573	2.090 ± 0.108	2.676 ± 0.660
UPI	beta34	0.101	0.076	-0.025 ± 0.008	-0.249 ± 0.083	0.042 ± 0.004	0.205 ± 0.009	0.042 ± 0.004	0.206 ± 0.010	4.158 ± 0.354	2.039 ± 0.096	0.064	-0.037	-36.53	0.1827	0.1864	2.016	98.328 ± 0.524	1.617 ± 0.082	3.177 ± 0.717
UPI	beta35	0.101	0.068	-0.033 ± 0.008	-0.323 ± 0.078	0.037 ± 0.003	0.193 ± 0.008	0.038 ± 0.003	0.196 ± 0.009	3.762 ± 0.318	1.940 ± 0.091	0.063	-0.038	-37.36	0.1809	0.1848	2.074	98.495 ± 0.498	1.572 ± 0.075	1.839 ± 0.549

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S26

Simulation 1 - Condition 26: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.4

Method	Param.	True	Mean-based									Median-based					Inference				
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE	
LMS	beta31	0.316	0.339	0.023 $\pm$ 0.003	0.072 $\pm$ 0.009	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.005 $\pm$ 0.000	0.071 $\pm$ 0.002	0.051 $\pm$ 0.003	0.226 $\pm$ 0.008	0.338	0.022	6.99	0.0730	0.0762	1.072	95.208 $\pm$ 0.854	0.284 $\pm$ 0.001	99.521 $\pm$ 0.27	
LMS	beta32	0.316	0.336	0.020 $\pm$ 0.003	0.065 $\pm$ 0.009	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.005 $\pm$ 0.000	0.071 $\pm$ 0.002	0.050 $\pm$ 0.003	0.223 $\pm$ 0.008	0.336	0.020	6.43	0.0734	0.0762	1.073	95.847 $\pm$ 0.797	0.284 $\pm$ 0.002	99.042 $\pm$ 0.389	
LMS	beta33	0.139	0.119	-0.020 $\pm$ 0.004	-0.146 $\pm$ 0.031	0.012 $\pm$ 0.001	0.109 $\pm$ 0.003	0.012 $\pm$ 0.001	0.111 $\pm$ 0.004	0.632 $\pm$ 0.039	0.795 $\pm$ 0.029	0.119	-0.020	-14.41	0.1196	0.1213	1.014	95.048 $\pm$ 0.867	0.432 $\pm$ 0.006	24.601 $\pm$ 1.721	
LMS	beta34	0.101	0.182	0.081 $\pm$ 0.002	0.807 $\pm$ 0.023	0.003 $\pm$ 0.000	0.058 $\pm$ 0.002	0.010 $\pm$ 0.000	0.100 $\pm$ 0.003	0.978 $\pm$ 0.047	0.989 $\pm$ 0.025	0.180	0.079	77.88	0.0638	0.1013	1.071	76.038 $\pm$ 1.706	0.243 $\pm$ 0.003	88.019 $\pm$ 1.298	
LMS	beta35	0.101	0.181	0.080 $\pm$ 0.002	0.791 $\pm$ 0.023	0.003 $\pm$ 0.000	0.059 $\pm$ 0.002	0.010 $\pm$ 0.000	0.099 $\pm$ 0.003	0.967 $\pm$ 0.048	0.983 $\pm$ 0.026	0.178	0.077	75.80	0.0628	0.0990	1.034	77.636 $\pm$ 1.665	0.239 $\pm$ 0.003	87.220 $\pm$ 1.334	
LSAM	beta31	0.316	0.393	0.077 $\pm$ 0.023	0.243 $\pm$ 0.074	0.339 $\pm$ 0.030	0.582 $\pm$ 0.026	0.344 $\pm$ 0.031	0.587 $\pm$ 0.029	3.445 $\pm$ 0.307	1.856 $\pm$ 0.091	0.367	0.051	16.06	0.6254	0.6275	1.935	99.681 $\pm$ 0.226	4.415 $\pm$ 0.261	9.105 $\pm$ 1.150	
LSAM	beta32	0.316	0.343	0.027 $\pm$ 0.023	0.086 $\pm$ 0.074	0.338 $\pm$ 0.029	0.581 $\pm$ 0.025	0.338 $\pm$ 0.029	0.581 $\pm$ 0.028	3.385 $\pm$ 0.293	1.840 $\pm$ 0.088	0.344	0.028	8.73	0.6458	0.6464	1.909	98.882 $\pm$ 0.420	4.350 $\pm$ 0.257	11.342 $\pm$ 1.267	
LSAM	beta33	0.139	0.233	0.094 $\pm$ 0.005	0.676 $\pm$ 0.039	0.018 $\pm$ 0.002	0.135 $\pm$ 0.006	0.027 $\pm$ 0.002	0.164 $\pm$ 0.007	1.393 $\pm$ 0.110	1.180 $\pm$ 0.050	0.239	0.100	71.86	0.1507	0.1808	1.528	96.166 $\pm$ 0.767	0.806 $\pm$ 0.039	34.665 $\pm$ 1.902	
LSAM	beta34	0.101	0.043	-0.058 $\pm$ 0.008	-0.575 $\pm$ 0.074	0.035 $\pm$ 0.003	0.188 $\pm$ 0.009	0.039 $\pm$ 0.004	0.197 $\pm$ 0.010	3.795 $\pm$ 0.347	1.948 $\pm$ 0.096	0.054	-0.047	-46.35	0.1978	0.2033	1.943	98.403 $\pm$ 0.501	1.433 $\pm$ 0.091	6.709 $\pm$ 1.000	
LSAM	beta35	0.101	0.056	-0.045 $\pm$ 0.008	-0.441 $\pm$ 0.077	0.037 $\pm$ 0.003	0.193 $\pm$ 0.009	0.039 $\pm$ 0.004	0.198 $\pm$ 0.010	3.859 $\pm$ 0.360	1.964 $\pm$ 0.100	0.066	-0.035	-34.86	0.2037	0.2067	1.862	98.083 $\pm$ 0.548	1.412 $\pm$ 0.092	7.029 $\pm$ 1.022	
QML	beta31	0.316	0.339	0.023 $\pm$ 0.003	0.072 $\pm$ 0.008	0.004 $\pm$ 0.000	0.067 $\pm$ 0.002	0.005 $\pm$ 0.000	0.070 $\pm$ 0.002	0.049 $\pm$ 0.003	0.222 $\pm$ 0.007	0.338	0.022	6.98	0.0733	0.0765	1.062	95.048 $\pm$ 0.867	0.277 $\pm$ 0.001	100.000 $\pm$ 0.000	
QML	beta32	0.316	0.336	0.020 $\pm$ 0.003	0.064 $\pm$ 0.008	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	0.005 $\pm$ 0.000	0.069 $\pm$ 0.002	0.047 $\pm$ 0.003	0.218 $\pm$ 0.007	0.337	0.021	6.63	0.0722	0.0752	1.071	95.527 $\pm$ 0.826	0.276 $\pm$ 0.001	99.681 $\pm$ 0.226	
QML	beta33	0.139	0.114	-0.025 $\pm$ 0.004	-0.179 $\pm$ 0.029	0.010 $\pm$ 0.001	0.099 $\pm$ 0.003	0.010 $\pm$ 0.001	0.102 $\pm$ 0.004	0.543 $\pm$ 0.032	0.737 $\pm$ 0.026	0.113	-0.026	-18.44	0.1071	0.1101	1.004	93.770 $\pm$ 0.966	0.391 $\pm$ 0.004	23.802 $\pm$ 1.702	
QML	beta34	0.101	0.181	0.080 $\pm$ 0.002	0.789 $\pm$ 0.021	0.003 $\pm$ 0.000	0.054 $\pm$ 0.002	0.009 $\pm$ 0.000	0.096 $\pm$ 0.002	0.906 $\pm$ 0.040	0.952 $\pm$ 0.022	0.176	0.075	74.37	0.0600	0.0961	1.054	71.406 $\pm$ 1.806	0.223 $\pm$ 0.003	89.137 $\pm$ 1.244	
QML	beta35	0.101	0.180	0.079 $\pm$ 0.002	0.778 $\pm$ 0.022	0.003 $\pm$ 0.000	0.055 $\pm$ 0.002	0.009 $\pm$ 0.000	0.096 $\pm$ 0.002	0.902 $\pm$ 0.042	0.950 $\pm$ 0.023	0.177	0.076	75.28	0.0582	0.0957	1.016	72.684 $\pm$ 1.781	0.220 $\pm$ 0.002	90.735 $\pm$ 1.159	
UPI	beta31	0.316	0.311	-0.005 $\pm$ 0.015	-0.016 $\pm$ 0.047	0.135 $\pm$ 0.010	0.368 $\pm$ 0.014	0.135 $\pm$ 0.010	0.367 $\pm$ 0.016	1.352 $\pm$ 0.101	1.163 $\pm$ 0.050	0.324	0.008	2.54	0.3324	0.3325	1.795	99.840 $\pm$ 0.160	2.588 $\pm$ 0.129	10.863 $\pm$ 1.254	
UPI	beta32	0.316	0.315	-0.001 $\pm$ 0.016	-0.003 $\pm$ 0.049	0.151 $\pm$ 0.012	0.388 $\pm$ 0.015	0.150 $\pm$ 0.012	0.388 $\pm$ 0.017	1.507 $\pm$ 0.116	1.227 $\pm$ 0.054	0.306	-0.010	-3.31	0.3313	0.3315	1.776	99.840 $\pm$ 0.160	2.703 $\pm$ 0.143	11.182 $\pm$ 1.260	
UPI	beta33	0.139	0.170	0.031 $\pm$ 0.006	0.226 $\pm$ 0.047	0.026 $\pm$ 0.002	0.162 $\pm$ 0.007	0.027 $\pm$ 0.002	0.165 $\pm$ 0.008	1.403 $\pm$ 0.121	1.185 $\pm$ 0.056	0.162	0.023	16.56	0.1497	0.1514	1.753	98.243 $\pm$ 0.525	1.112 $\pm$ 0.068	14.856 $\pm$ 1.421	
UPI	beta34	0.101	0.073	-0.028 $\pm$ 0.005	-0.276 $\pm$ 0.051	0.016 $\pm$ 0.001	0.128 $\pm$ 0.005	0.017 $\pm$ 0.001	0.131 $\pm$ 0.006	1.682 $\pm$ 0.131	1.297 $\pm$ 0.057	0.067	-0.034	-33.33	0.1200	0.1246	1.854	98.562 $\pm$ 0.476	0.931 $\pm$ 0.052	3.834 $\pm$ 0.767	
UPI	beta35	0.101	0.072	-0.029 $\pm$ 0.005	-0.291 $\pm$ 0.052	0.017 $\pm$ 0.001	0.131 $\pm$ 0.006	0.018 $\pm$ 0.001	0.134 $\pm$ 0.006	1.767 $\pm$ 0.145	1.329 $\pm$ 0.060	0.069	-0.032	-31.27	0.1187	0.1228	1.837	98.403 $\pm$ 0.501	0.944 $\pm$ 0.054	8.946 $\pm$ 1.141	

Note. MCSE = Monte Carlo Standard Error shown as $\backslash u00b1$ value. Coverage, Type I Error, and Power expressed as percentages. $\backslash u2014$ indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S27

Simulation 1 - Condition 27: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, $N = 400$, Reliability = 0.6

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LMS	beta31	0.316	0.323	0.007 $\pm$ 0.003	0.023 $\pm$ 0.009	0.008 $\pm$ 0.000	0.087 $\pm$ 0.002	0.008 $\pm$ 0.000	0.087 $\pm$ 0.003	0.076 $\pm$ 0.004	0.276 $\pm$ 0.008	0.324	0.008	2.55	0.0937	0.0940	1.055	95.706 $\pm$ 0.681	0.359 $\pm$ 0.003	95.141 $\pm$ 0.723
LMS	beta32	0.316	0.322	0.006 $\pm$ 0.003	0.018 $\pm$ 0.009	0.008 $\pm$ 0.000	0.089 $\pm$ 0.002	0.008 $\pm$ 0.000	0.089 $\pm$ 0.003	0.079 $\pm$ 0.004	0.281 $\pm$ 0.008	0.322	0.006	1.85	0.0973	0.0974	1.034	95.254 $\pm$ 0.715	0.360 $\pm$ 0.004	93.785 $\pm$ 0.812
LMS	beta33	0.139	0.108	-0.031 $\pm$ 0.003	-0.221 $\pm$ 0.022	0.009 $\pm$ 0.000	0.092 $\pm$ 0.002	0.009 $\pm$ 0.000	0.097 $\pm$ 0.003	0.489 $\pm$ 0.025	0.699 $\pm$ 0.021	0.107	-0.032	-23.35	0.0948	0.1002	1.000	93.333 $\pm$ 0.838	0.362 $\pm$ 0.003	24.181 $\pm$ 1.439
LMS	beta34	0.101	0.145	0.044 $\pm$ 0.002	0.439 $\pm$ 0.018	0.003 $\pm$ 0.000	0.055 $\pm$ 0.001	0.005 $\pm$ 0.000	0.071 $\pm$ 0.002	0.491 $\pm$ 0.023	0.701 $\pm$ 0.018	0.143	0.042	41.26	0.0556	0.0695	0.944	85.537 $\pm$ 1.182	0.204 $\pm$ 0.002	79.774 $\pm$ 1.350
LMS	beta35	0.101	0.144	0.043 $\pm$ 0.002	0.422 $\pm$ 0.018	0.003 $\pm$ 0.000	0.055 $\pm$ 0.002	0.005 $\pm$ 0.000	0.069 $\pm$ 0.002	0.471 $\pm$ 0.023	0.687 $\pm$ 0.019	0.142	0.041	40.81	0.0558	0.0694	0.947	86.441 $\pm$ 1.151	0.203 $\pm$ 0.002	78.870 $\pm$ 1.372
LSAM	beta31	0.316	0.325	0.009 $\pm$ 0.006	0.030 $\pm$ 0.018	0.029 $\pm$ 0.001	0.171 $\pm$ 0.004	0.029 $\pm$ 0.001	0.171 $\pm$ 0.005	0.292 $\pm$ 0.015	0.541 $\pm$ 0.017	0.321	0.005	1.58	0.1760	0.1761	1.066	95.706 $\pm$ 0.681	0.713 $\pm$ 0.011	48.475 $\pm$ 1.680
LSAM	beta32	0.316	0.323	0.007 $\pm$ 0.006	0.022 $\pm$ 0.019	0.031 $\pm$ 0.002	0.176 $\pm$ 0.005	0.031 $\pm$ 0.002	0.176 $\pm$ 0.006	0.311 $\pm$ 0.016	0.558 $\pm$ 0.017	0.318	0.002	0.48	0.1743	0.1743	1.040	95.932 $\pm$ 0.664	0.718 $\pm$ 0.013	49.944 $\pm$ 1.681
LSAM	beta33	0.139	0.170	0.031 $\pm$ 0.003	0.225 $\pm$ 0.024	0.010 $\pm$ 0.001	0.100 $\pm$ 0.003	0.011 $\pm$ 0.001	0.104 $\pm$ 0.003	0.564 $\pm$ 0.032	0.751 $\pm$ 0.024	0.162	0.023	16.33	0.0995	0.1021	0.991	93.785 $\pm$ 0.812	0.387 $\pm$ 0.005	40.904 $\pm$ 1.653
LSAM	beta34	0.101	0.090	-0.011 $\pm$ 0.002	-0.105 $\pm$ 0.023	0.005 $\pm$ 0.000	0.069 $\pm$ 0.002	0.005 $\pm$ 0.000	0.070 $\pm$ 0.002	0.481 $\pm$ 0.027	0.694 $\pm$ 0.023	0.089	-0.012	-11.72	0.0659	0.0670	0.961	92.994 $\pm$ 0.858	0.261 $\pm$ 0.005	38.870 $\pm$ 1.639
LSAM	beta35	0.101	0.089	-0.012 $\pm$ 0.002	-0.122 $\pm$ 0.024	0.005 $\pm$ 0.000	0.072 $\pm$ 0.002	0.005 $\pm$ 0.000	0.073 $\pm$ 0.002	0.517 $\pm$ 0.030	0.719 $\pm$ 0.024	0.088	-0.013	-12.47	0.0680	0.0692	0.955	93.672 $\pm$ 0.818	0.268 $\pm$ 0.006	37.740 $\pm$ 1.629
QML	beta31	0.316	0.324	0.008 $\pm$ 0.003	0.024 $\pm$ 0.009	0.008 $\pm$ 0.000	0.087 $\pm$ 0.002	0.008 $\pm$ 0.000	0.087 $\pm$ 0.003	0.076 $\pm$ 0.004	0.276 $\pm$ 0.008	0.324	0.008	2.54	0.0901	0.0904	1.042	95.593 $\pm$ 0.690	0.355 $\pm$ 0.002	95.367 $\pm$ 0.707
QML	beta32	0.316	0.322	0.006 $\pm$ 0.003	0.019 $\pm$ 0.009	0.008 $\pm$ 0.000	0.089 $\pm$ 0.002	0.008 $\pm$ 0.000	0.089 $\pm$ 0.003	0.080 $\pm$ 0.004	0.282 $\pm$ 0.008	0.322	0.006	2.00	0.0957	0.0959	1.015	95.028 $\pm$ 0.731	0.354 $\pm$ 0.002	93.898 $\pm$ 0.805
QML	beta33	0.139	0.109	-0.030 $\pm$ 0.003	-0.215 $\pm$ 0.022	0.008 $\pm$ 0.000	0.092 $\pm$ 0.002	0.009 $\pm$ 0.000	0.097 $\pm$ 0.003	0.484 $\pm$ 0.024	0.696 $\pm$ 0.020	0.107	-0.032	-23.01	0.0943	0.0996	0.998	92.994 $\pm$ 0.858	0.360 $\pm$ 0.003	24.181 $\pm$ 1.439
QML	beta34	0.101	0.145	0.044 $\pm$ 0.002	0.433 $\pm$ 0.018	0.003 $\pm$ 0.000	0.055 $\pm$ 0.001	0.005 $\pm$ 0.000	0.070 $\pm$ 0.002	0.483 $\pm$ 0.023	0.695 $\pm$ 0.018	0.143	0.042	41.30	0.0538	0.0681	0.939	85.311 $\pm$ 1.190	0.202 $\pm$ 0.002	79.548 $\pm$ 1.356
QML	beta35	0.101	0.143	0.042 $\pm$ 0.002	0.416 $\pm$ 0.018	0.003 $\pm$ 0.000	0.055 $\pm$ 0.002	0.005 $\pm$ 0.000	0.069 $\pm$ 0.002	0.465 $\pm$ 0.023	0.682 $\pm$ 0.019	0.140	0.039	38.89	0.0551	0.0677	0.938	86.328 $\pm$ 1.155	0.201 $\pm$ 0.002	79.435 $\pm$ 1.359
UPI	beta31	0.316	0.328	0.012 $\pm$ 0.007	0.036 $\pm$ 0.022	0.045 $\pm$ 0.003	0.211 $\pm$ 0.006	0.045 $\pm$ 0.003	0.211 $\pm$ 0.007	0.447 $\pm$ 0.027	0.668 $\pm$ 0.023	0.335	0.019	5.99	0.2083	0.2091	1.156	97.401 $\pm$ 0.535	0.956 $\pm$ 0.036	41.695 $\pm$ 1.657
UPI	beta32	0.316	0.328	0.012 $\pm$ 0.007	0.037 $\pm$ 0.023	0.046 $\pm$ 0.003	0.213 $\pm$ 0.006	0.046 $\pm$ 0.003	0.214 $\pm$ 0.007	0.457 $\pm$ 0.026	0.676 $\pm$ 0.022	0.322	0.006	1.92	0.1974	0.1975	1.138	97.175 $\pm$ 0.557	0.951 $\pm$ 0.034	41.921 $\pm$ 1.659
UPI	beta33	0.139	0.156	0.017 $\pm$ 0.004	0.124 $\pm$ 0.028	0.013 $\pm$ 0.001	0.115 $\pm$ 0.004	0.013 $\pm$ 0.001	0.116 $\pm$ 0.004	0.696 $\pm$ 0.048	0.835 $\pm$ 0.032	0.145	0.006	4.00	0.1061	0.1062	1.052	95.028 $\pm$ 0.731	0.473 $\pm$ 0.012	29.153 $\pm$ 1.528
UPI	beta34	0.101	0.091	-0.010 $\pm$ 0.003	-0.100 $\pm$ 0.028	0.007 $\pm$ 0.000	0.084 $\pm$ 0.003	0.007 $\pm$ 0.000	0.085 $\pm$ 0.003	0.701 $\pm$ 0.047	0.837 $\pm$ 0.031	0.093	-0.008	-8.23	0.0746	0.0750	1.067	93.333 $\pm$ 0.838	0.351 $\pm$ 0.015	33.672 $\pm$ 1.589
UPI	beta35	0.101	0.089	-0.012 $\pm$ 0.003	-0.123 $\pm$ 0.028	0.007 $\pm$ 0.000	0.085 $\pm$ 0.003	0.007 $\pm$ 0.000	0.086 $\pm$ 0.003	0.723 $\pm$ 0.047	0.850 $\pm$ 0.031	0.090	-0.011	-10.47	0.0724	0.0731	1.067	94.350 $\pm$ 0.776	0.355 $\pm$ 0.016	34.689 $\pm$ 1.600

Note. MCSE = Monte Carlo Standard Error shown as $\backslash u00b1$ value. Coverage, Type I Error, and Power expressed as percentages. $\backslash u2014$ indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S28

Simulation 1 - Condition 28: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.6

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LMS	beta31	0.316	0.324	0.008 $\pm$ 0.002	0.026 $\pm$ 0.006	0.003 $\pm$ 0.000	0.055 $\pm$ 0.001	0.003 $\pm$ 0.000	0.056 $\pm$ 0.002	0.031 $\pm$ 0.001	0.176 $\pm$ 0.005	0.323	0.007	2.27	0.0605	0.0609	1.061	95.464 $\pm$ 0.676	0.229 $\pm$ 0.008	99.789 $\pm$ 0.149
LMS	beta32	0.316	0.322	0.006 $\pm$ 0.002	0.019 $\pm$ 0.006	0.003 $\pm$ 0.000	0.054 $\pm$ 0.001	0.003 $\pm$ 0.000	0.055 $\pm$ 0.002	0.030 $\pm$ 0.001	0.173 $\pm$ 0.005	0.321	0.005	1.55	0.0547	0.0549	1.062	95.359 $\pm$ 0.683	0.227 $\pm$ 0.005	99.789 $\pm$ 0.149
LMS	beta33	0.139	0.115	-0.024 $\pm$ 0.002	-0.169 $\pm$ 0.014	0.003 $\pm$ 0.000	0.059 $\pm$ 0.001	0.004 $\pm$ 0.000	0.063 $\pm$ 0.002	0.207 $\pm$ 0.010	0.455 $\pm$ 0.013	0.117	-0.022	-16.01	0.0587	0.0628	0.964	90.295 $\pm$ 0.961	0.222 $\pm$ 0.001	54.641 $\pm$ 1.617
LMS	beta34	0.101	0.143	0.042 $\pm$ 0.001	0.418 $\pm$ 0.010	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.275 $\pm$ 0.010	0.525 $\pm$ 0.010	0.143	0.042	41.43	0.0314	0.0523	0.985	72.468 $\pm$ 1.451	0.124 $\pm$ 0.001	98.312 $\pm$ 0.418
LMS	beta35	0.101	0.144	0.043 $\pm$ 0.001	0.425 $\pm$ 0.010	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	0.003 $\pm$ 0.000	0.054 $\pm$ 0.001	0.282 $\pm$ 0.010	0.531 $\pm$ 0.010	0.144	0.043	42.28	0.0314	0.0530	0.989	70.992 $\pm$ 1.474	0.125 $\pm$ 0.001	98.101 $\pm$ 0.443
LSAM	beta31	0.316	0.326	0.010 $\pm$ 0.003	0.033 $\pm$ 0.010	0.010 $\pm$ 0.000	0.098 $\pm$ 0.002	0.010 $\pm$ 0.000	0.099 $\pm$ 0.003	0.098 $\pm$ 0.005	0.313 $\pm$ 0.009	0.326	0.010	3.04	0.1033	0.1037	1.020	95.570 $\pm$ 0.668	0.394 $\pm$ 0.002	89.557 $\pm$ 0.993
LSAM	beta32	0.316	0.320	0.004 $\pm$ 0.003	0.011 $\pm$ 0.010	0.009 $\pm$ 0.000	0.095 $\pm$ 0.002	0.009 $\pm$ 0.000	0.095 $\pm$ 0.003	0.090 $\pm$ 0.004	0.301 $\pm$ 0.008	0.320	0.004	1.12	0.0999	0.0999	1.060	95.675 $\pm$ 0.661	0.395 $\pm$ 0.003	88.924 $\pm$ 1.019
LSAM	beta33	0.139	0.162	0.023 $\pm$ 0.002	0.165 $\pm$ 0.013	0.003 $\pm$ 0.000	0.057 $\pm$ 0.001	0.004 $\pm$ 0.000	0.062 $\pm$ 0.002	0.198 $\pm$ 0.010	0.445 $\pm$ 0.012	0.162	0.023	16.48	0.0563	0.0608	0.954	91.983 $\pm$ 0.882	0.215 $\pm$ 0.001	82.489 $\pm$ 1.234
LSAM	beta34	0.101	0.090	-0.011 $\pm$ 0.001	-0.114 $\pm$ 0.012	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.140 $\pm$ 0.007	0.374 $\pm$ 0.011	0.090	-0.011	-10.82	0.0348	0.0365	0.976	91.772 $\pm$ 0.892	0.138 $\pm$ 0.002	71.203 $\pm$ 1.471
LSAM	beta35	0.101	0.091	-0.010 $\pm$ 0.001	-0.094 $\pm$ 0.012	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.135 $\pm$ 0.007	0.367 $\pm$ 0.011	0.091	-0.010	-9.61	0.0332	0.0346	0.997	92.932 $\pm$ 0.832	0.140 $\pm$ 0.002	71.519 $\pm$ 1.466
QML	beta31	0.316	0.325	0.009 $\pm$ 0.002	0.027 $\pm$ 0.006	0.003 $\pm$ 0.000	0.055 $\pm$ 0.001	0.003 $\pm$ 0.000	0.056 $\pm$ 0.002	0.031 $\pm$ 0.001	0.177 $\pm$ 0.005	0.325	0.009	2.74	0.0594	0.0600	1.011	95.359 $\pm$ 0.683	0.219 $\pm$ 0.001	100.000 $\pm$ 0.000
QML	beta32	0.316	0.322	0.006 $\pm$ 0.002	0.019 $\pm$ 0.006	0.003 $\pm$ 0.000	0.055 $\pm$ 0.001	0.003 $\pm$ 0.000	0.055 $\pm$ 0.002	0.030 $\pm$ 0.001	0.174 $\pm$ 0.005	0.321	0.005	1.67	0.0542	0.0545	1.024	95.464 $\pm$ 0.676	0.219 $\pm$ 0.001	100.000 $\pm$ 0.000
QML	beta33	0.139	0.116	-0.023 $\pm$ 0.002	-0.166 $\pm$ 0.014	0.003 $\pm$ 0.000	0.058 $\pm$ 0.001	0.004 $\pm$ 0.000	0.063 $\pm$ 0.002	0.204 $\pm$ 0.010	0.452 $\pm$ 0.012	0.116	-0.023	-16.60	0.0576	0.0621	0.961	90.084 $\pm$ 0.971	0.220 $\pm$ 0.001	55.169 $\pm$ 1.615
QML	beta34	0.101	0.143	0.042 $\pm$ 0.001	0.414 $\pm$ 0.010	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.272 $\pm$ 0.010	0.521 $\pm$ 0.010	0.142	0.041	40.90	0.0311	0.0517	0.978	72.152 $\pm$ 1.456	0.123 $\pm$ 0.001	98.207 $\pm$ 0.431
QML	beta35	0.101	0.144	0.043 $\pm$ 0.001	0.423 $\pm$ 0.010	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.279 $\pm$ 0.010	0.528 $\pm$ 0.010	0.144	0.043	42.09	0.0307	0.0524	0.988	70.886 $\pm$ 1.475	0.124 $\pm$ 0.001	98.418 $\pm$ 0.405
UPI	beta31	0.316	0.323	0.007 $\pm$ 0.004	0.021 $\pm$ 0.012	0.013 $\pm$ 0.001	0.114 $\pm$ 0.003	0.013 $\pm$ 0.001	0.114 $\pm$ 0.003	0.130 $\pm$ 0.006	0.361 $\pm$ 0.011	0.324	0.008	2.47	0.1160	0.1163	1.031	96.414 $\pm$ 0.604	0.461 $\pm$ 0.006	79.852 $\pm$ 1.303
UPI	beta32	0.316	0.314	-0.002 $\pm$ 0.004	-0.006 $\pm$ 0.011	0.012 $\pm$ 0.001	0.111 $\pm$ 0.003	0.012 $\pm$ 0.001	0.111 $\pm$ 0.003	0.123 $\pm$ 0.006	0.351 $\pm$ 0.010	0.311	-0.005	-1.60	0.1106	0.1107	1.076	96.519 $\pm$ 0.595	0.467 $\pm$ 0.008	78.165 $\pm$ 1.342
UPI	beta33	0.139	0.150	0.011 $\pm$ 0.002	0.077 $\pm$ 0.014	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.193 $\pm$ 0.010	0.439 $\pm$ 0.014	0.148	0.009	6.70	0.0579	0.0587	0.967	93.565 $\pm$ 0.797	0.228 $\pm$ 0.002	75.844 $\pm$ 1.390
UPI	beta34	0.101	0.092	-0.009 $\pm$ 0.001	-0.090 $\pm$ 0.013	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.172 $\pm$ 0.010	0.414 $\pm$ 0.014	0.092	-0.009	-8.70	0.0368	0.0379	0.987	92.616 $\pm$ 0.849	0.158 $\pm$ 0.002	66.350 $\pm$ 1.535
UPI	beta35	0.101	0.094	-0.007 $\pm$ 0.001	-0.069 $\pm$ 0.013	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.169 $\pm$ 0.010	0.411 $\pm$ 0.013	0.093	-0.008	-7.98	0.0379	0.0388	1.028	94.726 $\pm$ 0.726	0.165 $\pm$ 0.004	66.667 $\pm$ 1.531

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S29

Simulation 1 - Condition 29: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.8

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LMS	beta31	0.316	0.317	0.001 $\pm$ 0.003	0.002 $\pm$ 0.008	0.006 $\pm$ 0.000	0.079 $\pm$ 0.002	0.006 $\pm$ 0.000	0.079 $\pm$ 0.002	0.063 $\pm$ 0.003	0.250 $\pm$ 0.007	0.316	0.000	0.11	0.0750	0.0751	1.008	94.900 $\pm$ 0.696	0.313 $\pm$ 0.001	97.100 $\pm$ 0.531
LMS	beta32	0.316	0.318	0.002 $\pm$ 0.003	0.005 $\pm$ 0.008	0.006 $\pm$ 0.000	0.080 $\pm$ 0.002	0.006 $\pm$ 0.000	0.080 $\pm$ 0.002	0.064 $\pm$ 0.003	0.252 $\pm$ 0.007	0.317	0.001	0.32	0.0789	0.0789	0.998	94.900 $\pm$ 0.696	0.312 $\pm$ 0.001	97.400 $\pm$ 0.503
LMS	beta33	0.139	0.121	-0.018 $\pm$ 0.002	-0.126 $\pm$ 0.015	0.005 $\pm$ 0.000	0.067 $\pm$ 0.002	0.005 $\pm$ 0.000	0.070 $\pm$ 0.002	0.251 $\pm$ 0.012	0.501 $\pm$ 0.014	0.123	-0.016	-11.55	0.0654	0.0673	0.922	91.800 $\pm$ 0.868	0.244 $\pm$ 0.002	53.100 $\pm$ 1.578
LMS	beta34	0.101	0.121	0.020 $\pm$ 0.001	0.199 $\pm$ 0.012	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.193 $\pm$ 0.009	0.439 $\pm$ 0.012	0.121	0.020	19.82	0.0363	0.0414	0.887	86.000 $\pm$ 1.097	0.138 $\pm$ 0.001	88.100 $\pm$ 1.024
LMS	beta35	0.101	0.120	0.019 $\pm$ 0.001	0.189 $\pm$ 0.012	0.001 $\pm$ 0.000	0.039 $\pm$ 0.001	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.181 $\pm$ 0.009	0.426 $\pm$ 0.012	0.119	0.018	18.26	0.0370	0.0413	0.883	88.300 $\pm$ 1.016	0.134 $\pm$ 0.001	88.700 $\pm$ 1.001
LSAM	beta31	0.316	0.315	-0.001 $\pm$ 0.003	-0.003 $\pm$ 0.010	0.011 $\pm$ 0.000	0.103 $\pm$ 0.002	0.011 $\pm$ 0.000	0.103 $\pm$ 0.003	0.106 $\pm$ 0.005	0.326 $\pm$ 0.009	0.314	-0.002	-0.72	0.0992	0.0992	0.984	94.400 $\pm$ 0.727	0.398 $\pm$ 0.002	86.200 $\pm$ 1.091
LSAM	beta32	0.316	0.317	0.001 $\pm$ 0.003	0.004 $\pm$ 0.010	0.011 $\pm$ 0.000	0.103 $\pm$ 0.002	0.011 $\pm$ 0.000	0.103 $\pm$ 0.003	0.106 $\pm$ 0.005	0.326 $\pm$ 0.009	0.315	-0.001	-0.39	0.1032	0.1033	0.978	94.400 $\pm$ 0.727	0.395 $\pm$ 0.002	86.400 $\pm$ 1.084
LSAM	beta33	0.139	0.145	0.006 $\pm$ 0.002	0.040 $\pm$ 0.016	0.005 $\pm$ 0.000	0.069 $\pm$ 0.002	0.005 $\pm$ 0.000	0.069 $\pm$ 0.002	0.247 $\pm$ 0.012	0.497 $\pm$ 0.015	0.142	0.003	1.99	0.0658	0.0658	0.899	92.100 $\pm$ 0.853	0.243 $\pm$ 0.002	64.800 $\pm$ 1.510
LSAM	beta34	0.101	0.101	0.000 $\pm$ 0.001	0.004 $\pm$ 0.013	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.182 $\pm$ 0.010	0.427 $\pm$ 0.013	0.103	0.002	1.55	0.0378	0.0379	0.863	90.700 $\pm$ 0.918	0.146 $\pm$ 0.002	75.200 $\pm$ 1.366
LSAM	beta35	0.101	0.100	-0.001 $\pm$ 0.001	-0.007 $\pm$ 0.013	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.177 $\pm$ 0.010	0.421 $\pm$ 0.013	0.100	-0.001	-1.07	0.0389	0.0389	0.846	90.300 $\pm$ 0.936	0.141 $\pm$ 0.002	77.000 $\pm$ 1.331
QML	beta31	0.316	0.317	0.001 $\pm$ 0.003	0.002 $\pm$ 0.008	0.006 $\pm$ 0.000	0.079 $\pm$ 0.002	0.006 $\pm$ 0.000	0.079 $\pm$ 0.002	0.063 $\pm$ 0.003	0.251 $\pm$ 0.007	0.317	0.001	0.22	0.0755	0.0755	1.006	94.900 $\pm$ 0.696	0.313 $\pm$ 0.001	97.000 $\pm$ 0.539
QML	beta32	0.316	0.317	0.001 $\pm$ 0.003	0.005 $\pm$ 0.008	0.006 $\pm$ 0.000	0.080 $\pm$ 0.002	0.006 $\pm$ 0.000	0.080 $\pm$ 0.002	0.064 $\pm$ 0.003	0.253 $\pm$ 0.007	0.318	0.002	0.56	0.0807	0.0807	0.994	94.900 $\pm$ 0.696	0.311 $\pm$ 0.001	97.200 $\pm$ 0.522
QML	beta33	0.139	0.121	-0.018 $\pm$ 0.002	-0.126 $\pm$ 0.015	0.005 $\pm$ 0.000	0.067 $\pm$ 0.002	0.005 $\pm$ 0.000	0.070 $\pm$ 0.002	0.251 $\pm$ 0.012	0.501 $\pm$ 0.014	0.123	-0.016	-11.78	0.0656	0.0676	0.918	91.500 $\pm$ 0.882	0.243 $\pm$ 0.002	53.100 $\pm$ 1.578
QML	beta34	0.101	0.121	0.020 $\pm$ 0.001	0.195 $\pm$ 0.012	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.192 $\pm$ 0.009	0.438 $\pm$ 0.012	0.120	0.019	19.30	0.0360	0.0410	0.877	85.300 $\pm$ 1.120	0.136 $\pm$ 0.001	88.300 $\pm$ 1.016
QML	beta35	0.101	0.120	0.019 $\pm$ 0.001	0.188 $\pm$ 0.012	0.001 $\pm$ 0.000	0.039 $\pm$ 0.001	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.182 $\pm$ 0.009	0.426 $\pm$ 0.012	0.119	0.018	18.16	0.0369	0.0412	0.879	87.900 $\pm$ 1.031	0.133 $\pm$ 0.001	88.600 $\pm$ 1.005
UPI	beta31	0.316	0.319	0.003 $\pm$ 0.003	0.010 $\pm$ 0.010	0.010 $\pm$ 0.000	0.102 $\pm$ 0.002	0.010 $\pm$ 0.000	0.102 $\pm$ 0.003	0.103 $\pm$ 0.005	0.321 $\pm$ 0.009	0.317	0.001	0.27	0.0960	0.0960	0.977	93.600 $\pm$ 0.774	0.389 $\pm$ 0.002	88.300 $\pm$ 1.016
UPI	beta32	0.316	0.322	0.006 $\pm$ 0.003	0.021 $\pm$ 0.010	0.010 $\pm$ 0.000	0.102 $\pm$ 0.002	0.010 $\pm$ 0.000	0.102 $\pm$ 0.003	0.104 $\pm$ 0.005	0.323 $\pm$ 0.009	0.322	0.006	2.00	0.1004	0.1006	0.969	93.800 $\pm$ 0.763	0.387 $\pm$ 0.002	88.400 $\pm$ 1.013
UPI	beta33	0.139	0.140	0.001 $\pm$ 0.002	0.007 $\pm$ 0.016	0.005 $\pm$ 0.000	0.070 $\pm$ 0.002	0.005 $\pm$ 0.000	0.070 $\pm$ 0.002	0.251 $\pm$ 0.013	0.501 $\pm$ 0.015	0.138	-0.001	-0.95	0.0655	0.0655	0.903	91.900 $\pm$ 0.863	0.246 $\pm$ 0.002	62.500 $\pm$ 1.531
UPI	beta34	0.101	0.102	0.001 $\pm$ 0.001	0.006 $\pm$ 0.013	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.179 $\pm$ 0.010	0.423 $\pm$ 0.013	0.102	0.001	1.48	0.0381	0.0381	0.860	90.300 $\pm$ 0.936	0.144 $\pm$ 0.002	75.900 $\pm$ 1.352
UPI	beta35	0.101	0.100	-0.001 $\pm$ 0.001	-0.011 $\pm$ 0.013	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.172 $\pm$ 0.009	0.414 $\pm$ 0.013	0.100	-0.001	-0.64	0.0375	0.0375	0.853	90.100 $\pm$ 0.944	0.140 $\pm$ 0.002	75.600 $\pm$ 1.358

Note. MCSE = Monte Carlo Standard Error shown as $\backslash u00b1$ value. Coverage, Type I Error, and Power expressed as percentages. $\backslash u2014$ indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S30

Simulation 1 - Condition 30: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.8

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LMS	beta31	0.316	0.314	-0.002 $\pm$ 0.002	-0.006 $\pm$ 0.005	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.024 $\pm$ 0.001	0.155 $\pm$ 0.004	0.315	-0.001	-0.42	0.0477	0.0477	1.014	95.200 $\pm$ 0.676	0.195 $\pm$ 0.001	99.800 $\pm$ 0.141
LMS	beta32	0.316	0.317	0.001 $\pm$ 0.002	0.003 $\pm$ 0.005	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	0.025 $\pm$ 0.001	0.158 $\pm$ 0.004	0.315	-0.001	-0.17	0.0504	0.0504	0.995	95.600 $\pm$ 0.649	0.194 $\pm$ 0.001	100.000 $\pm$ 0.000
LMS	beta33	0.139	0.125	-0.014 $\pm$ 0.001	-0.103 $\pm$ 0.009	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.095 $\pm$ 0.005	0.308 $\pm$ 0.009	0.123	-0.016	-11.29	0.0399	0.0429	0.950	92.900 $\pm$ 0.812	0.151 $\pm$ 0.001	88.100 $\pm$ 1.024
LMS	beta34	0.101	0.120	0.019 $\pm$ 0.001	0.188 $\pm$ 0.007	0.000 $\pm$ 0.000	0.022 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.083 $\pm$ 0.004	0.288 $\pm$ 0.007	0.119	0.018	17.90	0.0212	0.0279	0.951	83.700 $\pm$ 1.168	0.082 $\pm$ 0.001	99.700 $\pm$ 0.173
LMS	beta35	0.101	0.119	0.018 $\pm$ 0.001	0.182 $\pm$ 0.007	0.000 $\pm$ 0.000	0.022 $\pm$ 0.000	0.001 $\pm$ 0.000	0.028 $\pm$ 0.001	0.079 $\pm$ 0.003	0.282 $\pm$ 0.006	0.119	0.018	18.26	0.0215	0.0283	0.952	82.900 $\pm$ 1.191	0.081 $\pm$ 0.001	99.900 $\pm$ 0.100
LSAM	beta31	0.316	0.314	-0.002 $\pm$ 0.002	-0.006 $\pm$ 0.006	0.004 $\pm$ 0.000	0.061 $\pm$ 0.001	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.037 $\pm$ 0.002	0.192 $\pm$ 0.006	0.315	-0.001	-0.28	0.0584	0.0584	1.006	94.600 $\pm$ 0.715	0.240 $\pm$ 0.001	99.700 $\pm$ 0.173
LSAM	beta32	0.316	0.317	0.001 $\pm$ 0.002	0.002 $\pm$ 0.006	0.004 $\pm$ 0.000	0.061 $\pm$ 0.001	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.038 $\pm$ 0.002	0.194 $\pm$ 0.005	0.316	0.000	-0.11	0.0627	0.0627	0.996	95.500 $\pm$ 0.656	0.240 $\pm$ 0.001	99.900 $\pm$ 0.100
LSAM	beta33	0.139	0.143	0.004 $\pm$ 0.001	0.032 $\pm$ 0.009	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	0.085 $\pm$ 0.004	0.292 $\pm$ 0.009	0.143	0.004	2.67	0.0395	0.0397	0.926	93.600 $\pm$ 0.774	0.147 $\pm$ 0.001	95.400 $\pm$ 0.662
LSAM	beta34	0.101	0.100	-0.001 $\pm$ 0.001	-0.010 $\pm$ 0.007	0.001 $\pm$ 0.000	0.023 $\pm$ 0.001	0.001 $\pm$ 0.000	0.023 $\pm$ 0.001	0.051 $\pm$ 0.002	0.226 $\pm$ 0.006	0.099	-0.002	-1.59	0.0221	0.0221	0.925	92.800 $\pm$ 0.817	0.083 $\pm$ 0.001	97.900 $\pm$ 0.453
LSAM	beta35	0.101	0.100	-0.001 $\pm$ 0.001	-0.012 $\pm$ 0.007	0.001 $\pm$ 0.000	0.023 $\pm$ 0.001	0.001 $\pm$ 0.000	0.023 $\pm$ 0.001	0.050 $\pm$ 0.002	0.224 $\pm$ 0.006	0.100	-0.001	-0.62	0.0206	0.0207	0.928	92.300 $\pm$ 0.843	0.082 $\pm$ 0.001	98.400 $\pm$ 0.397
QML	beta31	0.316	0.315	-0.001 $\pm$ 0.002	-0.004 $\pm$ 0.005	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.024 $\pm$ 0.001	0.155 $\pm$ 0.004	0.315	-0.001	-0.25	0.0480	0.0480	1.012	95.000 $\pm$ 0.689	0.194 $\pm$ 0.001	99.900 $\pm$ 0.100
QML	beta32	0.316	0.317	0.001 $\pm$ 0.002	0.002 $\pm$ 0.005	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	0.025 $\pm$ 0.001	0.157 $\pm$ 0.004	0.315	-0.001	-0.29	0.0504	0.0504	0.995	95.500 $\pm$ 0.656	0.194 $\pm$ 0.000	100.000 $\pm$ 0.000
QML	beta33	0.139	0.125	-0.014 $\pm$ 0.001	-0.102 $\pm$ 0.009	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.095 $\pm$ 0.005	0.308 $\pm$ 0.009	0.124	-0.015	-11.00	0.0401	0.0429	0.945	92.500 $\pm$ 0.833	0.150 $\pm$ 0.001	88.000 $\pm$ 1.028
QML	beta34	0.101	0.119	0.018 $\pm$ 0.001	0.183 $\pm$ 0.007	0.000 $\pm$ 0.000	0.022 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.081 $\pm$ 0.004	0.284 $\pm$ 0.007	0.119	0.018	17.35	0.0210	0.0274	0.943	83.800 $\pm$ 1.165	0.081 $\pm$ 0.001	99.700 $\pm$ 0.173
QML	beta35	0.101	0.119	0.018 $\pm$ 0.001	0.182 $\pm$ 0.007	0.000 $\pm$ 0.000	0.022 $\pm$ 0.000	0.001 $\pm$ 0.000	0.028 $\pm$ 0.001	0.079 $\pm$ 0.003	0.281 $\pm$ 0.006	0.119	0.018	18.09	0.0211	0.0279	0.948	82.500 $\pm$ 1.202	0.081 $\pm$ 0.001	99.900 $\pm$ 0.100
UPI	beta31	0.316	0.315	-0.001 $\pm$ 0.002	-0.003 $\pm$ 0.006	0.004 $\pm$ 0.000	0.060 $\pm$ 0.001	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	0.037 $\pm$ 0.002	0.191 $\pm$ 0.006	0.317	0.001	0.20	0.0597	0.0597	0.994	94.600 $\pm$ 0.715	0.236 $\pm$ 0.001	99.700 $\pm$ 0.173
UPI	beta32	0.316	0.318	0.002 $\pm$ 0.002	0.006 $\pm$ 0.006	0.004 $\pm$ 0.000	0.061 $\pm$ 0.001	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.037 $\pm$ 0.002	0.192 $\pm$ 0.005	0.318	0.002	0.49	0.0621	0.0621	0.988	95.700 $\pm$ 0.641	0.235 $\pm$ 0.001	99.900 $\pm$ 0.100
UPI	beta33	0.139	0.140	0.001 $\pm$ 0.001	0.009 $\pm$ 0.009	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.081 $\pm$ 0.004	0.285 $\pm$ 0.008	0.140	0.001	0.51	0.0390	0.0390	0.935	93.300 $\pm$ 0.791	0.145 $\pm$ 0.001	95.200 $\pm$ 0.676
UPI	beta34	0.101	0.101	-0.000 $\pm$ 0.001	-0.005 $\pm$ 0.007	0.001 $\pm$ 0.000	0.023 $\pm$ 0.001	0.001 $\pm$ 0.000	0.023 $\pm$ 0.001	0.051 $\pm$ 0.002	0.225 $\pm$ 0.006	0.100	-0.001	-1.00	0.0222	0.0223	0.922	92.700 $\pm$ 0.823	0.082 $\pm$ 0.001	98.200 $\pm$ 0.420
UPI	beta35	0.101	0.100	-0.001 $\pm$ 0.001	-0.006 $\pm$ 0.007	0.001 $\pm$ 0.000	0.022 $\pm$ 0.001	0.001 $\pm$ 0.000	0.022 $\pm$ 0.001	0.049 $\pm$ 0.002	0.222 $\pm$ 0.006	0.101	0.000	-0.34	0.0215	0.0215	0.928	92.700 $\pm$ 0.823	0.082 $\pm$ 0.001	98.400 $\pm$ 0.397

Note. MCSE = Monte Carlo Standard Error shown as $\backslash u00b1$ value. Coverage, Type I Error, and Power expressed as percentages. $\backslash u0021$ indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S31

Simulation 1 - Condition 31: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.4

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.308	-0.008 ± 0.012	-0.027 ± 0.039	0.008 ± 0.002	0.090 ± 0.012	0.008 ± 0.002	0.089 ± 0.013	0.080 ± 0.020	0.282 ± 0.041	0.327	0.011	3.61	0.0951	0.0958	1.019	88.889 ± 4.277	0.358 ± 0.017	94.444 ± 3.117
LMS	beta32	0.316	0.304	-0.012 ± 0.011	-0.039 ± 0.036	0.007 ± 0.001	0.084 ± 0.009	0.007 ± 0.001	0.085 ± 0.010	0.072 ± 0.014	0.267 ± 0.033	0.310	-0.006	-2.00	0.1282	0.1284	1.097	88.889 ± 4.277	0.363 ± 0.010	96.296 ± 2.570
LMS	beta33	0.139	0.207	0.068 ± 0.016	0.487 ± 0.117	0.014 ± 0.003	0.120 ± 0.012	0.019 ± 0.004	0.137 ± 0.016	0.967 ± 0.193	0.983 ± 0.114	0.220	0.081	58.10	0.1409	0.1624	1.029	87.037 ± 4.571	0.483 ± 0.061	55.556 ± 6.762
LMS	beta34	0.101	-0.037	-0.138 ± 0.009	-1.368 ± 0.088	0.004 ± 0.001	0.065 ± 0.008	0.023 ± 0.003	0.153 ± 0.009	2.284 ± 0.247	1.511 ± 0.084	-0.042	-0.143	-141.25	0.1051	0.1772	1.259	46.296 ± 6.785	0.323 ± 0.044	5.556 ± 3.117
LMS	beta35	0.101	-0.014	-0.115 ± 0.009	-1.137 ± 0.094	0.005 ± 0.001	0.070 ± 0.005	0.018 ± 0.002	0.134 ± 0.009	1.761 ± 0.230	1.327 ± 0.090	-0.023	-0.124	-122.97	0.1072	0.1640	1.339	61.111 ± 6.634	0.366 ± 0.051	5.556 ± 3.117
LSAM	beta31	0.316	0.367	0.051 ± 0.028	0.160 ± 0.087	0.041 ± 0.008	0.203 ± 0.021	0.043 ± 0.009	0.207 ± 0.026	0.431 ± 0.089	0.656 ± 0.083	0.352	0.036	11.42	0.1811	0.1847	1.167	94.444 ± 3.117	0.928 ± 0.117	53.704 ± 6.785
LSAM	beta32	0.316	0.330	0.014 ± 0.023	0.044 ± 0.074	0.029 ± 0.006	0.171 ± 0.020	0.029 ± 0.007	0.170 ± 0.024	0.289 ± 0.066	0.538 ± 0.075	0.309	-0.007	-2.35	0.1563	0.1564	1.369	100.000 ± 0.000	0.917 ± 0.098	46.296 ± 6.785
LSAM	beta33	0.139	0.301	0.162 ± 0.027	1.164 ± 0.197	0.041 ± 0.010	0.202 ± 0.028	0.066 ± 0.016	0.257 ± 0.035	3.420 ± 0.843	1.849 ± 0.251	0.262	0.123	88.66	0.1562	0.1990	1.053	98.148 ± 1.835	0.833 ± 0.111	40.741 ± 6.686
LSAM	beta34	0.101	-0.097	-0.198 ± 0.021	-1.958 ± 0.211	0.024 ± 0.006	0.156 ± 0.021	0.063 ± 0.010	0.251 ± 0.022	6.186 ± 0.989	2.487 ± 0.213	-0.083	-0.184	-182.11	0.1590	0.2432	1.170	66.667 ± 6.415	0.717 ± 0.079	0.000 ± 0.000
LSAM	beta35	0.101	-0.038	-0.139 ± 0.025	-1.379 ± 0.247	0.034 ± 0.009	0.183 ± 0.026	0.052 ± 0.009	0.229 ± 0.022	5.133 ± 0.862	2.266 ± 0.219	-0.057	-0.158	-156.80	0.1493	0.2176	1.390	83.333 ± 5.072	0.998 ± 0.191	0.000 ± 0.000
QML	beta31	0.316	0.307	-0.009 ± 0.012	-0.030 ± 0.039	0.008 ± 0.002	0.090 ± 0.012	0.008 ± 0.002	0.090 ± 0.013	0.081 ± 0.020	0.285 ± 0.042	0.317	0.001	0.38	0.0825	0.0825	0.974	88.889 ± 4.277	0.345 ± 0.008	94.444 ± 3.117
QML	beta32	0.316	0.301	-0.015 ± 0.012	-0.047 ± 0.037	0.007 ± 0.002	0.086 ± 0.010	0.007 ± 0.001	0.086 ± 0.011	0.074 ± 0.015	0.272 ± 0.034	0.296	-0.020	-6.29	0.1086	0.1104	1.065	88.889 ± 4.277	0.357 ± 0.008	96.296 ± 2.570
QML	beta33	0.139	0.211	0.072 ± 0.017	0.516 ± 0.119	0.015 ± 0.003	0.122 ± 0.011	0.020 ± 0.004	0.140 ± 0.016	1.017 ± 0.193	1.008 ± 0.112	0.224	0.085	61.10	0.1401	0.1638	0.896	90.741 ± 3.945	0.427 ± 0.014	53.704 ± 6.785
QML	beta34	0.101	-0.043	-0.144 ± 0.009	-1.422 ± 0.093	0.005 ± 0.001	0.069 ± 0.008	0.025 ± 0.003	0.159 ± 0.009	2.479 ± 0.276	1.575 ± 0.090	-0.041	-0.142	-140.50	0.0916	0.1689	1.059	44.444 ± 6.762	0.286 ± 0.012	5.556 ± 3.117
QML	beta35	0.101	-0.023	-0.124 ± 0.010	-1.223 ± 0.095	0.005 ± 0.001	0.071 ± 0.006	0.020 ± 0.003	0.142 ± 0.010	1.977 ± 0.260	1.406 ± 0.096	-0.034	-0.135	-133.45	0.0909	0.1626	1.109	59.259 ± 6.686	0.307 ± 0.016	11.111 ± 4.277
UPI	beta31	0.316	0.275	-0.041 ± 0.032	-0.131 ± 0.101	0.055 ± 0.012	0.234 ± 0.028	0.055 ± 0.012	0.235 ± 0.030	0.554 ± 0.115	0.744 ± 0.095	0.351	0.035	11.12	0.3310	0.3329	1.636	85.185 ± 4.834	1.500 ± 0.218	20.370 ± 5.481
UPI	beta32	0.316	0.440	0.124 ± 0.050	0.391 ± 0.160	0.137 ± 0.027	0.371 ± 0.039	0.150 ± 0.034	0.388 ± 0.052	1.505 ± 0.345	1.227 ± 0.165	0.284	-0.032	-9.98	0.5581	0.5590	1.491	88.889 ± 4.277	2.168 ± 0.447	25.926 ± 5.964
UPI	beta33	0.139	0.237	0.098 ± 0.032	0.705 ± 0.231	0.056 ± 0.015	0.236 ± 0.034	0.064 ± 0.019	0.253 ± 0.043	3.324 ± 0.997	1.823 ± 0.309	0.197	0.058	41.58	0.2565	0.2630	1.413	96.296 ± 2.570	1.307 ± 0.200	1.852 ± 1.835
UPI	beta34	0.101	-0.123	-0.224 ± 0.048	-2.221 ± 0.474	0.124 ± 0.037	0.352 ± 0.057	0.172 ± 0.042	0.415 ± 0.057	16.844 ± 4.131	4.104 ± 0.560	-0.138	-0.239	-237.06	0.5137	0.5667	1.829	77.778 ± 5.658	2.522 ± 0.575	0.000 ± 0.000
UPI	beta35	0.101	-0.216	-0.317 ± 0.066	-3.139 ± 0.653	0.235 ± 0.071	0.485 ± 0.081	0.331 ± 0.090	0.575 ± 0.087	32.450 ± 8.832	5.697 ± 0.860	-0.075	-0.176	-173.79	0.5604	0.5873	1.350	85.185 ± 4.834	2.565 ± 0.564	0.000 ± 0.000

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S32

Simulation 1 - Condition 32: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.4

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.298	-0.018 ± 0.007	-0.057 ± 0.022	0.002 ± 0.000	0.050 ± 0.005	0.003 ± 0.001	0.052 ± 0.006	0.027 ± 0.005	0.166 ± 0.019	0.309	-0.007	-2.27	0.0669	0.0673	1.266	94.118 ± 3.295	0.246 ± 0.029	98.039 ± 1.941
LMS	beta32	0.316	0.304	-0.012 ± 0.008	-0.038 ± 0.025	0.003 ± 0.001	0.056 ± 0.007	0.003 ± 0.001	0.057 ± 0.008	0.033 ± 0.007	0.181 ± 0.024	0.298	-0.018	-5.72	0.0659	0.0683	0.986	94.118 ± 3.295	0.218 ± 0.003	100.000 ± 0.000
LMS	beta33	0.139	0.206	0.067 ± 0.009	0.484 ± 0.063	0.004 ± 0.001	0.063 ± 0.010	0.008 ± 0.002	0.092 ± 0.011	0.434 ± 0.097	0.659 ± 0.081	0.226	0.087	62.61	0.0798	0.1180	1.245	88.235 ± 4.512	0.307 ± 0.036	86.275 ± 4.819
LMS	beta34	0.101	-0.027	-0.128 ± 0.009	-1.266 ± 0.085	0.004 ± 0.001	0.062 ± 0.007	0.020 ± 0.003	0.142 ± 0.010	1.967 ± 0.259	1.403 ± 0.096	-0.033	-0.134	-132.51	0.0721	0.1520	1.345	39.216 ± 6.837	0.325 ± 0.128	7.843 ± 3.765
LMS	beta35	0.101	-0.014	-0.115 ± 0.007	-1.142 ± 0.073	0.003 ± 0.001	0.053 ± 0.006	0.016 ± 0.002	0.127 ± 0.007	1.574 ± 0.159	1.254 ± 0.065	-0.025	-0.126	-124.91	0.0668	0.1427	0.940	35.294 ± 6.692	0.195 ± 0.007	7.843 ± 3.765
LSAM	beta31	0.316	0.338	0.022 ± 0.015	0.069 ± 0.047	0.011 ± 0.002	0.107 ± 0.011	0.012 ± 0.002	0.108 ± 0.014	0.117 ± 0.025	0.342 ± 0.045	0.335	0.019	6.09	0.1222	0.1237	1.047	100.000 ± 0.000	0.439 ± 0.023	88.235 ± 4.512
LSAM	beta32	0.316	0.314	-0.002 ± 0.015	-0.005 ± 0.047	0.011 ± 0.003	0.106 ± 0.013	0.011 ± 0.003	0.105 ± 0.015	0.111 ± 0.026	0.333 ± 0.047	0.321	0.005	1.72	0.0855	0.0857	1.035	98.039 ± 1.941	0.431 ± 0.023	86.275 ± 4.819
LSAM	beta33	0.139	0.257	0.118 ± 0.014	0.846 ± 0.099	0.010 ± 0.003	0.098 ± 0.020	0.023 ± 0.006	0.153 ± 0.021	1.207 ± 0.304	1.099 ± 0.151	0.262	0.123	88.48	0.1062	0.1625	0.958	86.275 ± 4.819	0.370 ± 0.019	84.314 ± 5.092
LSAM	beta34	0.101	-0.066	-0.167 ± 0.016	-1.653 ± 0.163	0.014 ± 0.004	0.117 ± 0.017	0.041 ± 0.008	0.203 ± 0.021	4.058 ± 0.797	2.015 ± 0.209	-0.044	-0.145	-143.40	0.0930	0.1721	0.930	54.902 ± 6.968	0.428 ± 0.025	7.843 ± 3.765
LSAM	beta35	0.101	-0.038	-0.139 ± 0.017	-1.372 ± 0.164	0.014 ± 0.004	0.118 ± 0.017	0.033 ± 0.004	0.181 ± 0.013	3.220 ± 0.400	1.794 ± 0.124	-0.057	-0.158	-156.48	0.0992	0.1866	0.912	62.745 ± 6.770	0.422 ± 0.031	5.882 ± 3.295
QML	beta31	0.316	0.298	-0.018 ± 0.007	-0.056 ± 0.022	0.003 ± 0.000	0.051 ± 0.005	0.003 ± 0.001	0.053 ± 0.006	0.028 ± 0.005	0.168 ± 0.020	0.304	-0.012	-3.77	0.0599	0.0610	1.110	94.118 ± 3.295	0.220 ± 0.003	100.000 ± 0.000
QML	beta32	0.316	0.300	-0.016 ± 0.007	-0.052 ± 0.023	0.003 ± 0.001	0.053 ± 0.006	0.003 ± 0.001	0.055 ± 0.007	0.030 ± 0.007	0.173 ± 0.024	0.297	-0.019	-6.06	0.0546	0.0578	1.061	94.118 ± 3.295	0.219 ± 0.003	100.000 ± 0.000
QML	beta33	0.139	0.209	0.070 ± 0.009	0.504 ± 0.067	0.004 ± 0.001	0.067 ± 0.011	0.009 ± 0.002	0.096 ± 0.012	0.481 ± 0.108	0.694 ± 0.085	0.226	0.087	62.65	0.0765	0.1159	1.088	90.196 ± 4.164	0.285 ± 0.007	88.235 ± 4.512
QML	beta34	0.101	-0.029	-0.130 ± 0.008	-1.286 ± 0.080	0.003 ± 0.001	0.058 ± 0.006	0.020 ± 0.002	0.142 ± 0.009	1.977 ± 0.235	1.406 ± 0.086	-0.033	-0.134	-132.89	0.0660	0.1496	0.897	37.255 ± 6.770	0.204 ± 0.006	11.765 ± 4.512
QML	beta35	0.101	-0.022	-0.123 ± 0.007	-1.223 ± 0.071	0.003 ± 0.000	0.051 ± 0.005	0.018 ± 0.002	0.134 ± 0.007	1.749 ± 0.173	1.322 ± 0.067	-0.029	-0.130	-128.99	0.0598	0.1434	0.982	37.255 ± 6.770	0.198 ± 0.006	9.804 ± 4.164
UPI	beta31	0.316	0.362	0.046 ± 0.066	0.146 ± 0.207	0.219 ± 0.059	0.468 ± 0.069	0.217 ± 0.063	0.466 ± 0.079	2.174 ± 0.629	1.474 ± 0.249	0.341	0.025	7.90	0.5911	0.5916	1.586	84.314 ± 5.092	2.911 ± 0.571	13.725 ± 4.819
UPI	beta32	0.316	0.328	0.012 ± 0.083	0.038 ± 0.262	0.350 ± 0.120	0.591 ± 0.113	0.343 ± 0.120	0.586 ± 0.119	3.433 ± 1.206	1.853 ± 0.375	0.370	0.054	17.17	0.6381	0.6404	1.189	88.235 ± 4.512	2.754 ± 0.747	19.608 ± 5.560
UPI	beta33	0.139	0.205	0.066 ± 0.081	0.478 ± 0.581	0.332 ± 0.129	0.577 ± 0.125	0.330 ± 0.134	0.575 ± 0.132	17.098 ± 6.942	4.135 ± 0.953	0.172	0.033	24.07	0.5046	0.5057	0.952	98.039 ± 1.941	2.151 ± 0.429	5.882 ± 3.295
UPI	beta34	0.101	-0.579	-0.680 ± 0.226	-6.737 ± 2.236	2.602 ± 1.073	1.613 ± 0.380	3.014 ± 1.347	1.736 ± 0.439	295.448 ± 132.075	17.189 ± 4.344	-0.180	-0.281	-277.77	0.9473	0.9880	0.902	88.235 ± 4.512	5.700 ± 1.276	1.961 ± 1.941
UPI	beta35	0.101	-0.489	-0.590 ± 0.201	-5.844 ± 1.993	2.067 ± 0.740	1.438 ± 0.288	2.375 ± 0.932	1.541 ± 0.339	232.819 ± 91.341	15.258 ± 3.359	-0.415	-0.516	-511.08	1.1532	1.2635	0.858	94.118 ± 3.295	4.833 ± 0.989	0.000 ± 0.000

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S33

Simulation 1 - Condition 33: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.6

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LMS	beta31	0.316	0.306	-0.010 $\pm$ 0.003	-0.032 $\pm$ 0.008	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.005 $\pm$ 0.000	0.069 $\pm$ 0.002	0.047 $\pm$ 0.003	0.217 $\pm$ 0.008	0.304	-0.012	-3.85	0.0659	0.0670	1.061	94.084 $\pm$ 0.896	0.282 $\pm$ 0.014	99.278 $\pm$ 0.321
LMS	beta32	0.316	0.310	-0.006 $\pm$ 0.002	-0.018 $\pm$ 0.008	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	0.043 $\pm$ 0.002	0.208 $\pm$ 0.007	0.307	-0.009	-2.72	0.0660	0.0665	1.081	96.970 $\pm$ 0.651	0.278 $\pm$ 0.010	99.856 $\pm$ 0.144
LMS	beta33	0.139	0.173	0.034 $\pm$ 0.003	0.242 $\pm$ 0.022	0.007 $\pm$ 0.000	0.082 $\pm$ 0.002	0.008 $\pm$ 0.000	0.089 $\pm$ 0.003	0.406 $\pm$ 0.023	0.637 $\pm$ 0.021	0.170	0.031	22.28	0.0770	0.0830	1.041	94.084 $\pm$ 0.896	0.334 $\pm$ 0.003	56.277 $\pm$ 1.884
LMS	beta34	0.101	0.030	-0.071 $\pm$ 0.003	-0.705 $\pm$ 0.026	0.005 $\pm$ 0.000	0.069 $\pm$ 0.002	0.010 $\pm$ 0.000	0.099 $\pm$ 0.002	0.969 $\pm$ 0.043	0.984 $\pm$ 0.024	0.028	-0.073	-72.70	0.0710	0.1021	1.006	75.325 $\pm$ 1.638	0.274 $\pm$ 0.003	6.926 $\pm$ 0.964
LMS	beta35	0.101	0.029	-0.072 $\pm$ 0.003	-0.710 $\pm$ 0.025	0.004 $\pm$ 0.000	0.067 $\pm$ 0.002	0.010 $\pm$ 0.000	0.098 $\pm$ 0.002	0.943 $\pm$ 0.041	0.971 $\pm$ 0.023	0.028	-0.073	-71.80	0.0701	0.1009	1.033	77.489 $\pm$ 1.587	0.271 $\pm$ 0.004	6.782 $\pm$ 0.955
LSAM	beta31	0.316	0.313	-0.003 $\pm$ 0.003	-0.011 $\pm$ 0.009	0.006 $\pm$ 0.000	0.077 $\pm$ 0.002	0.006 $\pm$ 0.000	0.077 $\pm$ 0.003	0.060 $\pm$ 0.004	0.245 $\pm$ 0.009	0.310	-0.006	-1.86	0.0761	0.0764	1.024	96.104 $\pm$ 0.735	0.311 $\pm$ 0.002	97.835 $\pm$ 0.553
LSAM	beta32	0.316	0.316	0.000 $\pm$ 0.003	0.001 $\pm$ 0.009	0.006 $\pm$ 0.000	0.076 $\pm$ 0.002	0.006 $\pm$ 0.000	0.076 $\pm$ 0.003	0.057 $\pm$ 0.003	0.239 $\pm$ 0.008	0.315	-0.001	-0.45	0.0780	0.0780	1.045	97.691 $\pm$ 0.570	0.310 $\pm$ 0.002	98.268 $\pm$ 0.496
LSAM	beta33	0.139	0.181	0.042 $\pm$ 0.003	0.300 $\pm$ 0.024	0.008 $\pm$ 0.000	0.089 $\pm$ 0.003	0.010 $\pm$ 0.001	0.098 $\pm$ 0.003	0.498 $\pm$ 0.028	0.706 $\pm$ 0.023	0.180	0.041	29.60	0.0799	0.0898	1.026	93.651 $\pm$ 0.926	0.357 $\pm$ 0.003	53.968 $\pm$ 1.893
LSAM	beta34	0.101	0.043	-0.058 $\pm$ 0.004	-0.573 $\pm$ 0.041	0.012 $\pm$ 0.001	0.110 $\pm$ 0.003	0.015 $\pm$ 0.001	0.124 $\pm$ 0.004	1.502 $\pm$ 0.084	1.225 $\pm$ 0.039	0.040	-0.061	-60.68	0.1048	0.1214	0.994	86.003 $\pm$ 1.318	0.427 $\pm$ 0.005	6.205 $\pm$ 0.916
LSAM	beta35	0.101	0.045	-0.056 $\pm$ 0.004	-0.557 $\pm$ 0.040	0.011 $\pm$ 0.001	0.106 $\pm$ 0.003	0.014 $\pm$ 0.001	0.120 $\pm$ 0.003	1.409 $\pm$ 0.070	1.187 $\pm$ 0.034	0.037	-0.064	-62.95	0.0984	0.1172	1.016	89.610 $\pm$ 1.159	0.422 $\pm$ 0.005	4.618 $\pm$ 0.797
QML	beta31	0.316	0.306	-0.010 $\pm$ 0.003	-0.032 $\pm$ 0.008	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.005 $\pm$ 0.000	0.069 $\pm$ 0.002	0.047 $\pm$ 0.003	0.217 $\pm$ 0.008	0.304	-0.012	-3.85	0.0652	0.0664	1.012	94.084 $\pm$ 0.896	0.270 $\pm$ 0.001	99.423 $\pm$ 0.288
QML	beta32	0.316	0.310	-0.006 $\pm$ 0.002	-0.018 $\pm$ 0.008	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	0.044 $\pm$ 0.002	0.209 $\pm$ 0.007	0.307	-0.009	-2.88	0.0654	0.0660	1.042	96.825 $\pm$ 0.666	0.269 $\pm$ 0.001	99.856 $\pm$ 0.144
QML	beta33	0.139	0.174	0.035 $\pm$ 0.003	0.249 $\pm$ 0.022	0.007 $\pm$ 0.000	0.082 $\pm$ 0.002	0.008 $\pm$ 0.000	0.089 $\pm$ 0.003	0.411 $\pm$ 0.023	0.641 $\pm$ 0.021	0.171	0.032	22.83	0.0771	0.0833	1.033	93.651 $\pm$ 0.926	0.333 $\pm$ 0.003	55.988 $\pm$ 1.886
QML	beta34	0.101	0.028	-0.073 $\pm$ 0.003	-0.722 $\pm$ 0.026	0.005 $\pm$ 0.000	0.069 $\pm$ 0.002	0.010 $\pm$ 0.000	0.100 $\pm$ 0.002	0.988 $\pm$ 0.044	0.994 $\pm$ 0.024	0.026	-0.075	-74.67	0.0701	0.1030	1.006	75.180 $\pm$ 1.641	0.272 $\pm$ 0.003	6.926 $\pm$ 0.964
QML	beta35	0.101	0.028	-0.073 $\pm$ 0.003	-0.725 $\pm$ 0.025	0.004 $\pm$ 0.000	0.067 $\pm$ 0.002	0.010 $\pm$ 0.000	0.099 $\pm$ 0.002	0.965 $\pm$ 0.042	0.983 $\pm$ 0.023	0.027	-0.074	-73.34	0.0697	0.1017	1.020	76.912 $\pm$ 1.601	0.268 $\pm$ 0.003	6.782 $\pm$ 0.955
UPI	beta31	0.316	0.324	0.008 $\pm$ 0.004	0.026 $\pm$ 0.011	0.009 $\pm$ 0.001	0.096 $\pm$ 0.003	0.009 $\pm$ 0.001	0.096 $\pm$ 0.004	0.092 $\pm$ 0.007	0.303 $\pm$ 0.012	0.324	0.008	2.47	0.0898	0.0901	1.653	97.403 $\pm$ 0.604	0.619 $\pm$ 0.042	79.942 $\pm$ 1.521
UPI	beta32	0.316	0.327	0.011 $\pm$ 0.004	0.034 $\pm$ 0.012	0.009 $\pm$ 0.001	0.097 $\pm$ 0.003	0.010 $\pm$ 0.001	0.098 $\pm$ 0.004	0.095 $\pm$ 0.007	0.309 $\pm$ 0.012	0.325	0.009	2.87	0.0923	0.0927	1.667	98.846 $\pm$ 0.406	0.634 $\pm$ 0.046	80.808 $\pm$ 1.496
UPI	beta33	0.139	0.190	0.051 $\pm$ 0.004	0.370 $\pm$ 0.027	0.010 $\pm$ 0.001	0.099 $\pm$ 0.003	0.012 $\pm$ 0.001	0.112 $\pm$ 0.004	0.645 $\pm$ 0.044	0.803 $\pm$ 0.030	0.183	0.044	31.60	0.0906	0.1007	1.323	97.547 $\pm$ 0.588	0.514 $\pm$ 0.019	39.683 $\pm$ 1.858
UPI	beta34	0.101	0.056	-0.045 $\pm$ 0.007	-0.444 $\pm$ 0.072	0.037 $\pm$ 0.003	0.192 $\pm$ 0.007	0.039 $\pm$ 0.003	0.197 $\pm$ 0.008	3.810 $\pm$ 0.278	1.952 $\pm$ 0.080	0.053	-0.048	-47.56	0.1678	0.1745	1.379	94.084 $\pm$ 0.896	1.039 $\pm$ 0.066	4.906 $\pm$ 0.821
UPI	beta35	0.101	0.059	-0.042 $\pm$ 0.007	-0.413 $\pm$ 0.067	0.031 $\pm$ 0.002	0.177 $\pm$ 0.006	0.033 $\pm$ 0.002	0.182 $\pm$ 0.007	3.235 $\pm$ 0.220	1.799 $\pm$ 0.070	0.048	-0.053	-52.13	0.1563	0.1650	1.411	93.651 $\pm$ 0.926	0.979 $\pm$ 0.058	4.473 $\pm$ 0.785

Note. MCSE = Monte Carlo Standard Error shown as $\backslash$ u00b1 value. Coverage, Type I Error, and Power expressed as percentages. $\backslash$ u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S34

Simulation 1 - Condition 34: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.6

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.305	-0.011 ± 0.002	-0.035 ± 0.005	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.021 ± 0.001	0.144 ± 0.004	0.305	-0.011	-3.41	0.0433	0.0446	0.975	93.041 ± 0.913	0.168 ± 0.000	100.000 ± 0.000
LMS	beta32	0.316	0.308	-0.008 ± 0.002	-0.024 ± 0.005	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.019 ± 0.001	0.139 ± 0.005	0.310	-0.006	-1.99	0.0406	0.0410	0.990	93.170 ± 0.906	0.168 ± 0.000	100.000 ± 0.000
LMS	beta33	0.139	0.172	0.033 ± 0.002	0.234 ± 0.013	0.003 ± 0.000	0.051 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.188 ± 0.009	0.433 ± 0.012	0.172	0.033	23.65	0.0502	0.0600	1.029	90.979 ± 1.028	0.205 ± 0.001	89.304 ± 1.109
LMS	beta34	0.101	0.032	-0.069 ± 0.002	-0.682 ± 0.015	0.002 ± 0.000	0.042 ± 0.001	0.007 ± 0.000	0.081 ± 0.001	0.639 ± 0.022	0.799 ± 0.014	0.032	-0.069	-68.08	0.0431	0.0811	1.013	63.273 ± 1.730	0.167 ± 0.001	9.665 ± 1.061
LMS	beta35	0.101	0.029	-0.072 ± 0.002	-0.710 ± 0.015	0.002 ± 0.000	0.042 ± 0.001	0.007 ± 0.000	0.083 ± 0.001	0.681 ± 0.022	0.825 ± 0.014	0.029	-0.072	-71.45	0.0420	0.0835	0.999	59.149 ± 1.765	0.166 ± 0.001	9.665 ± 1.061
LSAM	beta31	0.316	0.309	-0.007 ± 0.002	-0.022 ± 0.006	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.050 ± 0.002	0.025 ± 0.001	0.157 ± 0.005	0.308	-0.008	-2.66	0.0488	0.0495	0.985	94.588 ± 0.812	0.190 ± 0.001	100.000 ± 0.000
LSAM	beta32	0.316	0.314	-0.002 ± 0.002	-0.006 ± 0.006	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.002	0.024 ± 0.001	0.156 ± 0.005	0.314	-0.002	-0.51	0.0448	0.0448	0.985	93.814 ± 0.865	0.190 ± 0.001	100.000 ± 0.000
LSAM	beta33	0.139	0.179	0.040 ± 0.002	0.287 ± 0.013	0.003 ± 0.000	0.052 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	0.220 ± 0.010	0.469 ± 0.012	0.180	0.041	29.15	0.0520	0.0659	1.041	89.948 ± 1.079	0.211 ± 0.001	91.624 ± 0.994
LSAM	beta34	0.101	0.048	-0.053 ± 0.002	-0.528 ± 0.023	0.004 ± 0.000	0.066 ± 0.002	0.007 ± 0.000	0.085 ± 0.002	0.706 ± 0.033	0.840 ± 0.022	0.047	-0.054	-53.52	0.0673	0.0863	0.980	83.119 ± 1.345	0.254 ± 0.002	9.536 ± 1.054
LSAM	beta35	0.101	0.042	-0.059 ± 0.002	-0.582 ± 0.023	0.004 ± 0.000	0.066 ± 0.002	0.008 ± 0.000	0.088 ± 0.002	0.767 ± 0.033	0.876 ± 0.021	0.040	-0.061	-60.83	0.0603	0.0861	0.979	82.345 ± 1.369	0.254 ± 0.002	9.021 ± 1.028
QML	beta31	0.316	0.305	-0.011 ± 0.002	-0.036 ± 0.005	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.021 ± 0.001	0.144 ± 0.004	0.305	-0.011	-3.57	0.0432	0.0446	0.976	92.912 ± 0.921	0.168 ± 0.001	100.000 ± 0.000
QML	beta32	0.316	0.308	-0.008 ± 0.002	-0.024 ± 0.005	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.019 ± 0.001	0.139 ± 0.005	0.310	-0.006	-1.97	0.0406	0.0411	0.994	93.428 ± 0.890	0.169 ± 0.001	99.871 ± 0.129
QML	beta33	0.139	0.172	0.033 ± 0.002	0.237 ± 0.013	0.003 ± 0.000	0.051 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.190 ± 0.009	0.436 ± 0.012	0.172	0.033	23.88	0.0507	0.0606	1.030	90.593 ± 1.048	0.206 ± 0.001	89.433 ± 1.104
QML	beta34	0.101	0.031	-0.070 ± 0.002	-0.694 ± 0.015	0.002 ± 0.000	0.042 ± 0.001	0.007 ± 0.000	0.082 ± 0.001	0.654 ± 0.022	0.809 ± 0.014	0.031	-0.070	-69.25	0.0423	0.0817	1.019	62.113 ± 1.741	0.167 ± 0.001	9.021 ± 1.028
QML	beta35	0.101	0.028	-0.073 ± 0.002	-0.722 ± 0.015	0.002 ± 0.000	0.042 ± 0.001	0.007 ± 0.000	0.084 ± 0.001	0.697 ± 0.022	0.835 ± 0.014	0.028	-0.073	-72.19	0.0417	0.0840	1.002	57.474 ± 1.775	0.167 ± 0.001	9.923 ± 1.073
UPI	beta31	0.316	0.314	-0.002 ± 0.002	-0.007 ± 0.007	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.043 ± 0.002	0.208 ± 0.007	0.317	0.001	0.29	0.0655	0.0655	1.310	96.392 ± 0.669	0.337 ± 0.015	91.495 ± 1.001
UPI	beta32	0.316	0.317	0.001 ± 0.002	0.005 ± 0.007	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.042 ± 0.003	0.206 ± 0.007	0.318	0.002	0.56	0.0602	0.0602	1.324	96.778 ± 0.634	0.338 ± 0.015	92.268 ± 0.959
UPI	beta33	0.139	0.179	0.040 ± 0.002	0.290 ± 0.015	0.003 ± 0.000	0.059 ± 0.001	0.005 ± 0.000	0.071 ± 0.002	0.262 ± 0.013	0.512 ± 0.014	0.181	0.042	29.98	0.0595	0.0726	1.228	94.330 ± 0.830	0.282 ± 0.006	75.902 ± 1.535
UPI	beta34	0.101	0.055	-0.046 ± 0.004	-0.459 ± 0.041	0.013 ± 0.001	0.116 ± 0.003	0.016 ± 0.001	0.125 ± 0.004	1.531 ± 0.084	1.237 ± 0.039	0.052	-0.049	-48.61	0.1064	0.1172	1.200	92.397 ± 0.951	0.546 ± 0.023	9.021 ± 1.028
UPI	beta35	0.101	0.045	-0.056 ± 0.004	-0.552 ± 0.040	0.013 ± 0.001	0.112 ± 0.003	0.016 ± 0.001	0.125 ± 0.004	1.531 ± 0.087	1.237 ± 0.039	0.043	-0.058	-57.27	0.1012	0.1166	1.273	92.268 ± 0.959	0.558 ± 0.028	7.861 ± 0.966

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S35

Simulation 1 - Condition 35: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, $N = 400$, Reliability = 0.8

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LMS	beta31	0.316	0.313	-0.003 $\pm$ 0.002	-0.010 $\pm$ 0.006	0.003 $\pm$ 0.000	0.055 $\pm$ 0.001	0.003 $\pm$ 0.000	0.055 $\pm$ 0.002	0.031 $\pm$ 0.001	0.175 $\pm$ 0.005	0.313	-0.003	-0.80	0.0551	0.0552	1.023	95.000 $\pm$ 0.689	0.221 $\pm$ 0.001	99.800 $\pm$ 0.141
LMS	beta32	0.316	0.314	-0.002 $\pm$ 0.002	-0.007 $\pm$ 0.006	0.003 $\pm$ 0.000	0.057 $\pm$ 0.001	0.003 $\pm$ 0.000	0.057 $\pm$ 0.002	0.033 $\pm$ 0.001	0.182 $\pm$ 0.005	0.314	-0.002	-0.73	0.0578	0.0579	0.986	94.700 $\pm$ 0.708	0.222 $\pm$ 0.001	99.900 $\pm$ 0.100
LMS	beta33	0.139	0.150	0.011 $\pm$ 0.002	0.082 $\pm$ 0.014	0.004 $\pm$ 0.000	0.063 $\pm$ 0.001	0.004 $\pm$ 0.000	0.064 $\pm$ 0.002	0.213 $\pm$ 0.009	0.462 $\pm$ 0.012	0.151	0.012	8.45	0.0649	0.0659	1.007	94.700 $\pm$ 0.708	0.249 $\pm$ 0.001	65.400 $\pm$ 1.504
LMS	beta34	0.101	0.069	-0.032 $\pm$ 0.002	-0.321 $\pm$ 0.019	0.004 $\pm$ 0.000	0.059 $\pm$ 0.001	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.447 $\pm$ 0.020	0.669 $\pm$ 0.017	0.070	-0.031	-30.98	0.0601	0.0677	0.989	89.400 $\pm$ 0.973	0.230 $\pm$ 0.001	23.100 $\pm$ 1.333
LMS	beta35	0.101	0.070	-0.031 $\pm$ 0.002	-0.309 $\pm$ 0.018	0.003 $\pm$ 0.000	0.058 $\pm$ 0.001	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	0.423 $\pm$ 0.018	0.650 $\pm$ 0.016	0.069	-0.032	-31.21	0.0565	0.0647	1.012	90.900 $\pm$ 0.909	0.230 $\pm$ 0.001	22.400 $\pm$ 1.318
LSAM	beta31	0.316	0.316	-0.000 $\pm$ 0.002	-0.002 $\pm$ 0.006	0.003 $\pm$ 0.000	0.057 $\pm$ 0.001	0.003 $\pm$ 0.000	0.057 $\pm$ 0.002	0.032 $\pm$ 0.001	0.180 $\pm$ 0.005	0.316	0.000	-0.12	0.0559	0.0559	1.030	95.700 $\pm$ 0.641	0.230 $\pm$ 0.001	99.900 $\pm$ 0.100
LSAM	beta32	0.316	0.317	0.001 $\pm$ 0.002	0.002 $\pm$ 0.006	0.004 $\pm$ 0.000	0.059 $\pm$ 0.001	0.004 $\pm$ 0.000	0.059 $\pm$ 0.002	0.035 $\pm$ 0.001	0.188 $\pm$ 0.005	0.317	0.001	0.46	0.0599	0.0600	0.989	95.000 $\pm$ 0.689	0.230 $\pm$ 0.001	100.000 $\pm$ 0.000
LSAM	beta33	0.139	0.149	0.010 $\pm$ 0.002	0.072 $\pm$ 0.014	0.004 $\pm$ 0.000	0.064 $\pm$ 0.001	0.004 $\pm$ 0.000	0.064 $\pm$ 0.002	0.214 $\pm$ 0.009	0.463 $\pm$ 0.012	0.149	0.010	7.11	0.0641	0.0649	1.004	95.100 $\pm$ 0.683	0.250 $\pm$ 0.001	65.700 $\pm$ 1.501
LSAM	beta34	0.101	0.087	-0.014 $\pm$ 0.002	-0.140 $\pm$ 0.023	0.005 $\pm$ 0.000	0.074 $\pm$ 0.002	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.553 $\pm$ 0.025	0.744 $\pm$ 0.020	0.086	-0.015	-14.56	0.0736	0.0750	0.986	93.200 $\pm$ 0.796	0.285 $\pm$ 0.001	23.100 $\pm$ 1.333
LSAM	beta35	0.101	0.088	-0.013 $\pm$ 0.002	-0.131 $\pm$ 0.023	0.005 $\pm$ 0.000	0.072 $\pm$ 0.002	0.005 $\pm$ 0.000	0.073 $\pm$ 0.002	0.524 $\pm$ 0.023	0.724 $\pm$ 0.019	0.087	-0.014	-13.77	0.0695	0.0708	1.010	93.600 $\pm$ 0.774	0.285 $\pm$ 0.001	23.000 $\pm$ 1.331
QML	beta31	0.316	0.313	-0.003 $\pm$ 0.002	-0.011 $\pm$ 0.006	0.003 $\pm$ 0.000	0.055 $\pm$ 0.001	0.003 $\pm$ 0.000	0.055 $\pm$ 0.002	0.031 $\pm$ 0.001	0.175 $\pm$ 0.005	0.313	-0.003	-0.90	0.0552	0.0553	1.025	95.000 $\pm$ 0.689	0.222 $\pm$ 0.001	99.800 $\pm$ 0.141
QML	beta32	0.316	0.314	-0.002 $\pm$ 0.002	-0.008 $\pm$ 0.006	0.003 $\pm$ 0.000	0.057 $\pm$ 0.001	0.003 $\pm$ 0.000	0.057 $\pm$ 0.002	0.033 $\pm$ 0.001	0.182 $\pm$ 0.005	0.313	-0.003	-0.84	0.0580	0.0581	0.987	94.900 $\pm$ 0.696	0.222 $\pm$ 0.001	99.900 $\pm$ 0.100
QML	beta33	0.139	0.150	0.011 $\pm$ 0.002	0.083 $\pm$ 0.014	0.004 $\pm$ 0.000	0.063 $\pm$ 0.001	0.004 $\pm$ 0.000	0.064 $\pm$ 0.002	0.213 $\pm$ 0.009	0.462 $\pm$ 0.012	0.151	0.012	8.49	0.0653	0.0663	1.008	94.600 $\pm$ 0.715	0.250 $\pm$ 0.001	65.300 $\pm$ 1.505
QML	beta34	0.101	0.068	-0.033 $\pm$ 0.002	-0.326 $\pm$ 0.019	0.004 $\pm$ 0.000	0.059 $\pm$ 0.001	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.451 $\pm$ 0.020	0.671 $\pm$ 0.017	0.069	-0.032	-31.38	0.0597	0.0676	0.991	89.400 $\pm$ 0.973	0.230 $\pm$ 0.001	22.900 $\pm$ 1.329
QML	beta35	0.101	0.069	-0.032 $\pm$ 0.002	-0.314 $\pm$ 0.018	0.003 $\pm$ 0.000	0.058 $\pm$ 0.001	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	0.426 $\pm$ 0.018	0.653 $\pm$ 0.016	0.069	-0.032	-31.63	0.0564	0.0648	1.015	90.900 $\pm$ 0.909	0.230 $\pm$ 0.001	22.300 $\pm$ 1.316
UPI	beta31	0.316	0.320	0.004 $\pm$ 0.002	0.013 $\pm$ 0.006	0.003 $\pm$ 0.000	0.057 $\pm$ 0.001	0.003 $\pm$ 0.000	0.057 $\pm$ 0.002	0.033 $\pm$ 0.001	0.180 $\pm$ 0.005	0.322	0.006	1.77	0.0566	0.0569	0.992	94.600 $\pm$ 0.715	0.221 $\pm$ 0.001	100.000 $\pm$ 0.000
UPI	beta32	0.316	0.321	0.005 $\pm$ 0.002	0.016 $\pm$ 0.006	0.004 $\pm$ 0.000	0.059 $\pm$ 0.001	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	0.036 $\pm$ 0.002	0.189 $\pm$ 0.005	0.322	0.006	1.87	0.0603	0.0606	0.952	94.200 $\pm$ 0.739	0.222 $\pm$ 0.001	100.000 $\pm$ 0.000
UPI	beta33	0.139	0.145	0.006 $\pm$ 0.002	0.046 $\pm$ 0.015	0.004 $\pm$ 0.000	0.064 $\pm$ 0.001	0.004 $\pm$ 0.000	0.065 $\pm$ 0.002	0.216 $\pm$ 0.009	0.464 $\pm$ 0.013	0.144	0.005	3.86	0.0647	0.0649	0.998	95.300 $\pm$ 0.669	0.251 $\pm$ 0.001	63.300 $\pm$ 1.524
UPI	beta34	0.101	0.096	-0.005 $\pm$ 0.002	-0.046 $\pm$ 0.024	0.006 $\pm$ 0.000	0.077 $\pm$ 0.002	0.006 $\pm$ 0.000	0.077 $\pm$ 0.002	0.585 $\pm$ 0.028	0.765 $\pm$ 0.022	0.096	-0.005	-5.20	0.0752	0.0754	0.988	94.200 $\pm$ 0.739	0.299 $\pm$ 0.002	25.100 $\pm$ 1.371
UPI	beta35	0.101	0.096	-0.005 $\pm$ 0.002	-0.045 $\pm$ 0.024	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.558 $\pm$ 0.025	0.747 $\pm$ 0.021	0.098	-0.003	-3.15	0.0725	0.0726	1.013	94.800 $\pm$ 0.702	0.299 $\pm$ 0.002	25.000 $\pm$ 1.369

Note. MCSE = Monte Carlo Standard Error shown as $\backslash u00b1$ value. Coverage, Type I Error, and Power expressed as percentages. $\backslash u2014$ indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S36

Simulation 1 - Condition 36: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, $N = 1000$, Reliability = 0.8

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LMS	beta31	0.316	0.310	-0.006 $\pm$ 0.001	-0.020 $\pm$ 0.004	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.013 $\pm$ 0.001	0.113 $\pm$ 0.003	0.310	-0.006	-2.01	0.0358	0.0364	1.015	95.200 $\pm$ 0.676	0.140 $\pm$ 0.000	100.000 $\pm$ 0.000
LMS	beta32	0.316	0.312	-0.004 $\pm$ 0.001	-0.012 $\pm$ 0.004	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.013 $\pm$ 0.001	0.114 $\pm$ 0.003	0.312	-0.004	-1.11	0.0368	0.0370	0.994	94.900 $\pm$ 0.696	0.140 $\pm$ 0.000	100.000 $\pm$ 0.000
LMS	beta33	0.139	0.149	0.010 $\pm$ 0.001	0.074 $\pm$ 0.009	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	0.088 $\pm$ 0.004	0.297 $\pm$ 0.008	0.149	0.010	7.14	0.0398	0.0410	1.005	94.800 $\pm$ 0.702	0.157 $\pm$ 0.000	95.500 $\pm$ 0.656
LMS	beta34	0.101	0.071	-0.030 $\pm$ 0.001	-0.293 $\pm$ 0.011	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.211 $\pm$ 0.008	0.460 $\pm$ 0.010	0.071	-0.030	-29.35	0.0368	0.0473	1.027	87.000 $\pm$ 1.063	0.144 $\pm$ 0.000	48.600 $\pm$ 1.581
LMS	beta35	0.101	0.069	-0.032 $\pm$ 0.001	-0.315 $\pm$ 0.012	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.239 $\pm$ 0.010	0.489 $\pm$ 0.011	0.069	-0.032	-31.76	0.0374	0.0493	0.980	85.100 $\pm$ 1.126	0.145 $\pm$ 0.000	45.800 $\pm$ 1.576
LSAM	beta31	0.316	0.312	-0.004 $\pm$ 0.001	-0.012 $\pm$ 0.004	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.013 $\pm$ 0.001	0.115 $\pm$ 0.003	0.312	-0.004	-1.22	0.0363	0.0365	1.019	95.100 $\pm$ 0.683	0.145 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta32	0.316	0.315	-0.001 $\pm$ 0.001	-0.002 $\pm$ 0.004	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.014 $\pm$ 0.001	0.117 $\pm$ 0.003	0.316	0.000	-0.05	0.0391	0.0391	1.001	94.900 $\pm$ 0.696	0.145 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta33	0.139	0.149	0.010 $\pm$ 0.001	0.070 $\pm$ 0.009	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	0.087 $\pm$ 0.004	0.296 $\pm$ 0.008	0.148	0.009	6.51	0.0388	0.0399	1.002	94.500 $\pm$ 0.721	0.157 $\pm$ 0.000	95.400 $\pm$ 0.662
LSAM	beta34	0.101	0.090	-0.011 $\pm$ 0.001	-0.108 $\pm$ 0.014	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.204 $\pm$ 0.009	0.452 $\pm$ 0.012	0.089	-0.012	-11.75	0.0449	0.0465	1.027	94.600 $\pm$ 0.715	0.178 $\pm$ 0.000	49.900 $\pm$ 1.581
LSAM	beta35	0.101	0.087	-0.014 $\pm$ 0.001	-0.141 $\pm$ 0.015	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.232 $\pm$ 0.011	0.482 $\pm$ 0.013	0.086	-0.015	-14.75	0.0459	0.0483	0.981	93.200 $\pm$ 0.796	0.179 $\pm$ 0.001	48.100 $\pm$ 1.580
QML	beta31	0.316	0.310	-0.006 $\pm$ 0.001	-0.020 $\pm$ 0.004	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.013 $\pm$ 0.001	0.113 $\pm$ 0.003	0.310	-0.006	-2.00	0.0357	0.0363	1.014	95.200 $\pm$ 0.676	0.140 $\pm$ 0.000	100.000 $\pm$ 0.000
QML	beta32	0.316	0.312	-0.004 $\pm$ 0.001	-0.013 $\pm$ 0.004	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.013 $\pm$ 0.001	0.114 $\pm$ 0.003	0.312	-0.004	-1.16	0.0372	0.0374	0.993	94.900 $\pm$ 0.696	0.140 $\pm$ 0.000	100.000 $\pm$ 0.000
QML	beta33	0.139	0.149	0.010 $\pm$ 0.001	0.074 $\pm$ 0.009	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	0.088 $\pm$ 0.004	0.297 $\pm$ 0.008	0.149	0.010	7.23	0.0398	0.0410	1.004	94.700 $\pm$ 0.708	0.157 $\pm$ 0.000	95.600 $\pm$ 0.649
QML	beta34	0.101	0.071	-0.030 $\pm$ 0.001	-0.296 $\pm$ 0.011	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.214 $\pm$ 0.008	0.462 $\pm$ 0.010	0.071	-0.030	-29.62	0.0366	0.0472	1.025	86.700 $\pm$ 1.074	0.144 $\pm$ 0.000	48.500 $\pm$ 1.580
QML	beta35	0.101	0.069	-0.032 $\pm$ 0.001	-0.318 $\pm$ 0.012	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	0.241 $\pm$ 0.010	0.491 $\pm$ 0.011	0.068	-0.033	-32.25	0.0372	0.0494	0.977	84.600 $\pm$ 1.141	0.145 $\pm$ 0.000	45.700 $\pm$ 1.575
UPI	beta31	0.316	0.315	-0.001 $\pm$ 0.001	-0.003 $\pm$ 0.004	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.013 $\pm$ 0.001	0.115 $\pm$ 0.003	0.314	-0.002	-0.49	0.0366	0.0367	0.981	93.800 $\pm$ 0.763	0.140 $\pm$ 0.000	100.000 $\pm$ 0.000
UPI	beta32	0.316	0.318	0.002 $\pm$ 0.001	0.005 $\pm$ 0.004	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.014 $\pm$ 0.001	0.117 $\pm$ 0.003	0.318	0.002	0.66	0.0390	0.0391	0.966	94.100 $\pm$ 0.745	0.140 $\pm$ 0.000	100.000 $\pm$ 0.000
UPI	beta33	0.139	0.143	0.004 $\pm$ 0.001	0.030 $\pm$ 0.009	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	0.086 $\pm$ 0.004	0.294 $\pm$ 0.008	0.143	0.004	2.74	0.0409	0.0411	0.988	94.900 $\pm$ 0.696	0.157 $\pm$ 0.000	93.800 $\pm$ 0.763
UPI	beta34	0.101	0.099	-0.002 $\pm$ 0.001	-0.018 $\pm$ 0.014	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.206 $\pm$ 0.009	0.454 $\pm$ 0.013	0.099	-0.002	-1.86	0.0470	0.0470	1.034	95.800 $\pm$ 0.634	0.186 $\pm$ 0.001	55.900 $\pm$ 1.570
UPI	beta35	0.101	0.096	-0.005 $\pm$ 0.002	-0.048 $\pm$ 0.015	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.227 $\pm$ 0.011	0.477 $\pm$ 0.013	0.097	-0.004	-4.45	0.0452	0.0454	0.988	93.900 $\pm$ 0.757	0.186 $\pm$ 0.001	54.200 $\pm$ 1.576

Note. MCSE = Monte Carlo Standard Error shown as $\backslash u00b1$ value. Coverage, Type I Error, and Power expressed as percentages. $\backslash u2014$ indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S37

Simulation 1 - Condition 37: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.4

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LMS	beta31	0.316	0.320	0.004 $\pm$ 0.004	0.011 $\pm$ 0.012	0.012 $\pm$ 0.001	0.111 $\pm$ 0.003	0.012 $\pm$ 0.001	0.111 $\pm$ 0.004	0.123 $\pm$ 0.007	0.351 $\pm$ 0.012	0.310	-0.006	-2.00	0.1081	0.1083	1.100	97.743 $\pm$ 0.512	0.479 $\pm$ 0.009	82.542 $\pm$ 1.308
LMS	beta32	0.316	0.311	-0.005 $\pm$ 0.004	-0.015 $\pm$ 0.012	0.012 $\pm$ 0.001	0.110 $\pm$ 0.003	0.012 $\pm$ 0.001	0.110 $\pm$ 0.004	0.122 $\pm$ 0.007	0.349 $\pm$ 0.011	0.311	-0.005	-1.65	0.1121	0.1122	1.095	95.843 $\pm$ 0.688	0.473 $\pm$ 0.010	80.760 $\pm$ 1.358
LMS	beta33	0.139	0.007	-0.132 $\pm$ 0.011	-0.949 $\pm$ 0.082	0.108 $\pm$ 0.008	0.329 $\pm$ 0.012	0.125 $\pm$ 0.010	0.354 $\pm$ 0.015	6.491 $\pm$ 0.523	2.548 $\pm$ 0.110	0.090	-0.049	-35.14	0.2324	0.2375	0.921	94.299 $\pm$ 0.799	1.188 $\pm$ 0.057	20.309 $\pm$ 1.386
LMS	beta34	0.101	0.205	0.104 $\pm$ 0.007	1.033 $\pm$ 0.074	0.047 $\pm$ 0.004	0.217 $\pm$ 0.008	0.058 $\pm$ 0.005	0.241 $\pm$ 0.011	5.689 $\pm$ 0.480	2.385 $\pm$ 0.106	0.149	0.048	47.68	0.1460	0.1537	0.898	93.705 $\pm$ 0.837	0.765 $\pm$ 0.031	10.095 $\pm$ 1.038
LMS	beta35	0.101	0.193	0.092 $\pm$ 0.007	0.912 $\pm$ 0.067	0.039 $\pm$ 0.003	0.196 $\pm$ 0.008	0.047 $\pm$ 0.004	0.217 $\pm$ 0.010	4.601 $\pm$ 0.391	2.145 $\pm$ 0.096	0.142	0.041	40.69	0.1480	0.1536	1.018	94.537 $\pm$ 0.783	0.783 $\pm$ 0.038	10.926 $\pm$ 1.075
LSAM	beta31	0.316	0.326	0.010 $\pm$ 0.004	0.031 $\pm$ 0.012	0.012 $\pm$ 0.001	0.111 $\pm$ 0.003	0.012 $\pm$ 0.001	0.111 $\pm$ 0.004	0.123 $\pm$ 0.007	0.351 $\pm$ 0.012	0.322	0.006	1.92	0.1076	0.1078	1.069	97.150 $\pm$ 0.573	0.463 $\pm$ 0.005	83.017 $\pm$ 1.294
LSAM	beta32	0.316	0.322	0.006 $\pm$ 0.004	0.020 $\pm$ 0.012	0.012 $\pm$ 0.001	0.111 $\pm$ 0.003	0.012 $\pm$ 0.001	0.111 $\pm$ 0.004	0.124 $\pm$ 0.007	0.352 $\pm$ 0.011	0.320	0.004	1.36	0.1083	0.1084	1.068	96.793 $\pm$ 0.607	0.465 $\pm$ 0.007	81.354 $\pm$ 1.342
LSAM	beta33	0.139	0.153	0.014 $\pm$ 0.006	0.103 $\pm$ 0.042	0.029 $\pm$ 0.002	0.170 $\pm$ 0.005	0.029 $\pm$ 0.002	0.171 $\pm$ 0.006	1.507 $\pm$ 0.095	1.228 $\pm$ 0.044	0.147	0.008	5.49	0.1594	0.1596	1.082	98.337 $\pm$ 0.441	0.722 $\pm$ 0.024	16.033 $\pm$ 1.264
LSAM	beta34	0.101	0.107	0.006 $\pm$ 0.004	0.062 $\pm$ 0.041	0.014 $\pm$ 0.001	0.119 $\pm$ 0.004	0.014 $\pm$ 0.001	0.119 $\pm$ 0.004	1.400 $\pm$ 0.093	1.183 $\pm$ 0.044	0.099	-0.002	-1.59	0.1069	0.1069	1.069	97.862 $\pm$ 0.498	0.500 $\pm$ 0.011	10.095 $\pm$ 1.038
LSAM	beta35	0.101	0.107	0.006 $\pm$ 0.004	0.056 $\pm$ 0.040	0.014 $\pm$ 0.001	0.118 $\pm$ 0.004	0.014 $\pm$ 0.001	0.118 $\pm$ 0.004	1.364 $\pm$ 0.083	1.168 $\pm$ 0.041	0.102	0.001	1.06	0.1064	0.1064	1.092	97.387 $\pm$ 0.550	0.505 $\pm$ 0.014	9.739 $\pm$ 1.022
QML	beta31	0.316	0.303	-0.013 $\pm$ 0.003	-0.041 $\pm$ 0.010	0.009 $\pm$ 0.000	0.095 $\pm$ 0.002	0.009 $\pm$ 0.000	0.096 $\pm$ 0.003	0.092 $\pm$ 0.005	0.304 $\pm$ 0.009	0.297	-0.019	-5.88	0.0964	0.0982	1.049	95.962 $\pm$ 0.678	0.391 $\pm$ 0.003	90.261 $\pm$ 1.022
QML	beta32	0.316	0.299	-0.017 $\pm$ 0.003	-0.054 $\pm$ 0.011	0.009 $\pm$ 0.000	0.096 $\pm$ 0.002	0.010 $\pm$ 0.000	0.098 $\pm$ 0.003	0.096 $\pm$ 0.005	0.309 $\pm$ 0.009	0.298	-0.018	-5.85	0.0978	0.0996	1.028	94.656 $\pm$ 0.775	0.388 $\pm$ 0.002	88.717 $\pm$ 1.090
QML	beta33	0.139	0.107	-0.032 $\pm$ 0.005	-0.227 $\pm$ 0.035	0.019 $\pm$ 0.001	0.139 $\pm$ 0.004	0.020 $\pm$ 0.001	0.143 $\pm$ 0.005	1.057 $\pm$ 0.059	1.028 $\pm$ 0.033	0.110	-0.029	-20.57	0.1373	0.1402	1.043	96.437 $\pm$ 0.639	0.570 $\pm$ 0.008	18.171 $\pm$ 1.329
QML	beta34	0.101	0.115	0.014 $\pm$ 0.003	0.143 $\pm$ 0.032	0.009 $\pm$ 0.001	0.095 $\pm$ 0.003	0.009 $\pm$ 0.001	0.096 $\pm$ 0.003	0.896 $\pm$ 0.053	0.946 $\pm$ 0.032	0.106	0.005	5.15	0.0932	0.0933	1.043	95.131 $\pm$ 0.742	0.387 $\pm$ 0.006	14.014 $\pm$ 1.196
QML	beta35	0.101	0.113	0.012 $\pm$ 0.003	0.122 $\pm$ 0.031	0.008 $\pm$ 0.000	0.092 $\pm$ 0.002	0.009 $\pm$ 0.000	0.092 $\pm$ 0.003	0.835 $\pm$ 0.044	0.914 $\pm$ 0.028	0.108	0.007	6.44	0.0912	0.0914	1.072	95.843 $\pm$ 0.688	0.385 $\pm$ 0.006	14.371 $\pm$ 1.209
UPI	beta31	0.316	0.328	0.012 $\pm$ 0.004	0.037 $\pm$ 0.012	0.013 $\pm$ 0.001	0.113 $\pm$ 0.003	0.013 $\pm$ 0.001	0.113 $\pm$ 0.004	0.129 $\pm$ 0.007	0.359 $\pm$ 0.012	0.323	0.007	2.16	0.1159	0.1161	1.152	98.456 $\pm$ 0.425	0.510 $\pm$ 0.007	79.097 $\pm$ 1.401
UPI	beta32	0.316	0.324	0.008 $\pm$ 0.004	0.024 $\pm$ 0.012	0.013 $\pm$ 0.001	0.113 $\pm$ 0.003	0.013 $\pm$ 0.001	0.113 $\pm$ 0.004	0.127 $\pm$ 0.007	0.357 $\pm$ 0.012	0.320	0.004	1.39	0.1125	0.1126	1.176	97.981 $\pm$ 0.485	0.519 $\pm$ 0.010	76.722 $\pm$ 1.456
UPI	beta33	0.139	0.174	0.035 $\pm$ 0.007	0.248 $\pm$ 0.049	0.040 $\pm$ 0.003	0.199 $\pm$ 0.006	0.041 $\pm$ 0.003	0.202 $\pm$ 0.007	2.113 $\pm$ 0.139	1.454 $\pm$ 0.054	0.159	0.020	14.07	0.1812	0.1822	1.192	99.169 $\pm$ 0.313	0.931 $\pm$ 0.032	10.095 $\pm$ 1.038
UPI	beta34	0.101	0.105	0.004 $\pm$ 0.005	0.039 $\pm$ 0.049	0.021 $\pm$ 0.001	0.144 $\pm$ 0.005	0.021 $\pm$ 0.001	0.144 $\pm$ 0.005	2.027 $\pm$ 0.133	1.424 $\pm$ 0.053	0.091	-0.010	-9.41	0.1235	0.1239	1.241	97.150 $\pm$ 0.573	0.699 $\pm$ 0.020	4.869 $\pm$ 0.742
UPI	beta35	0.101	0.104	0.003 $\pm$ 0.005	0.028 $\pm$ 0.045	0.018 $\pm$ 0.001	0.133 $\pm$ 0.004	0.018 $\pm$ 0.001	0.133 $\pm$ 0.005	1.723 $\pm$ 0.110	1.313 $\pm$ 0.048	0.099	-0.002	-1.77	0.1190	0.1190	1.307	97.268 $\pm$ 0.562	0.680 $\pm$ 0.023	4.988 $\pm$ 0.750

Note. MCSE = Monte Carlo Standard Error shown as $\backslash$ u00b1 value. Coverage, Type I Error, and Power expressed as percentages. $\backslash$ u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S38

Simulation 1 - Condition 38: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.4

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.313	-0.003 ± 0.002	-0.010 ± 0.007	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.042 ± 0.002	0.205 ± 0.006	0.311	-0.005	-1.51	0.0671	0.0673	1.082	96.347 ± 0.634	0.274 ± 0.003	98.402 ± 0.424
LMS	beta32	0.316	0.314	-0.002 ± 0.002	-0.007 ± 0.007	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	0.042 ± 0.002	0.204 ± 0.006	0.309	-0.007	-2.23	0.0633	0.0636	1.099	96.689 ± 0.604	0.277 ± 0.005	98.630 ± 0.393
LMS	beta33	0.139	0.119	-0.020 ± 0.004	-0.141 ± 0.029	0.014 ± 0.001	0.118 ± 0.005	0.014 ± 0.001	0.119 ± 0.005	0.734 ± 0.060	0.857 ± 0.038	0.118	-0.021	-15.44	0.1158	0.1178	1.409	96.689 ± 0.604	0.649 ± 0.042	30.936 ± 1.562
LMS	beta34	0.101	0.133	0.032 ± 0.003	0.322 ± 0.026	0.006 ± 0.000	0.077 ± 0.003	0.007 ± 0.001	0.084 ± 0.004	0.686 ± 0.055	0.828 ± 0.035	0.135	0.034	33.40	0.0814	0.0881	1.440	97.603 ± 0.517	0.435 ± 0.035	27.055 ± 1.501
LMS	beta35	0.101	0.134	0.033 ± 0.003	0.328 ± 0.029	0.007 ± 0.001	0.086 ± 0.003	0.009 ± 0.001	0.092 ± 0.004	0.837 ± 0.069	0.915 ± 0.040	0.132	0.031	30.75	0.0816	0.0873	1.264	95.890 ± 0.671	0.428 ± 0.024	27.283 ± 1.505
LSAM	beta31	0.316	0.321	0.005 ± 0.002	0.016 ± 0.007	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.043 ± 0.002	0.207 ± 0.006	0.316	0.000	-0.06	0.0651	0.0651	1.026	96.119 ± 0.653	0.262 ± 0.001	99.886 ± 0.114
LSAM	beta32	0.316	0.322	0.006 ± 0.002	0.019 ± 0.007	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.043 ± 0.002	0.207 ± 0.006	0.321	0.005	1.60	0.0647	0.0649	1.030	96.575 ± 0.614	0.263 ± 0.001	100.000 ± 0.000
LSAM	beta33	0.139	0.158	0.019 ± 0.003	0.136 ± 0.021	0.008 ± 0.000	0.088 ± 0.002	0.008 ± 0.000	0.090 ± 0.003	0.419 ± 0.023	0.647 ± 0.021	0.148	0.009	6.49	0.0951	0.0955	1.044	97.717 ± 0.505	0.360 ± 0.004	44.064 ± 1.677
LSAM	beta34	0.101	0.099	-0.002 ± 0.002	-0.020 ± 0.020	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	0.358 ± 0.020	0.598 ± 0.020	0.104	0.003	3.20	0.0615	0.0616	1.072	97.032 ± 0.573	0.254 ± 0.003	33.904 ± 1.599
LSAM	beta35	0.101	0.098	-0.003 ± 0.002	-0.034 ± 0.022	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.419 ± 0.023	0.647 ± 0.021	0.100	-0.001	-1.45	0.0644	0.0645	0.996	95.091 ± 0.730	0.255 ± 0.003	35.616 ± 1.618
QML	beta31	0.316	0.298	-0.018 ± 0.002	-0.057 ± 0.006	0.004 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.038 ± 0.002	0.196 ± 0.006	0.297	-0.019	-6.10	0.0608	0.0638	1.018	94.064 ± 0.798	0.237 ± 0.001	99.886 ± 0.114
QML	beta32	0.316	0.299	-0.017 ± 0.002	-0.053 ± 0.006	0.004 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.038 ± 0.002	0.195 ± 0.005	0.295	-0.021	-6.73	0.0577	0.0615	1.022	95.091 ± 0.730	0.237 ± 0.001	99.886 ± 0.114
QML	beta33	0.139	0.120	-0.019 ± 0.003	-0.140 ± 0.020	0.006 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.083 ± 0.003	0.355 ± 0.019	0.595 ± 0.018	0.112	-0.027	-19.78	0.0852	0.0895	1.075	95.662 ± 0.688	0.339 ± 0.003	31.507 ± 1.570
QML	beta34	0.101	0.104	0.003 ± 0.002	0.033 ± 0.017	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.002	0.259 ± 0.014	0.509 ± 0.016	0.110	0.009	9.37	0.0558	0.0565	1.123	96.575 ± 0.614	0.226 ± 0.002	43.950 ± 1.677
QML	beta35	0.101	0.104	0.003 ± 0.002	0.033 ± 0.019	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.308 ± 0.016	0.555 ± 0.017	0.108	0.007	7.36	0.0554	0.0559	1.033	94.635 ± 0.761	0.227 ± 0.002	43.379 ± 1.674
UPI	beta31	0.316	0.321	0.005 ± 0.002	0.016 ± 0.007	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.043 ± 0.002	0.208 ± 0.006	0.317	0.001	0.38	0.0672	0.0672	1.036	96.347 ± 0.634	0.266 ± 0.001	99.772 ± 0.161
UPI	beta32	0.316	0.322	0.006 ± 0.002	0.018 ± 0.007	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.043 ± 0.002	0.207 ± 0.006	0.321	0.005	1.62	0.0628	0.0630	1.042	96.575 ± 0.614	0.266 ± 0.001	99.886 ± 0.114
UPI	beta33	0.139	0.163	0.024 ± 0.003	0.173 ± 0.023	0.009 ± 0.000	0.094 ± 0.003	0.009 ± 0.001	0.097 ± 0.003	0.488 ± 0.027	0.699 ± 0.022	0.149	0.010	7.36	0.0962	0.0968	1.075	98.288 ± 0.438	0.397 ± 0.005	39.612 ± 1.652
UPI	beta34	0.101	0.099	-0.002 ± 0.002	-0.020 ± 0.022	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.426 ± 0.024	0.653 ± 0.021	0.100	-0.001	-0.91	0.0667	0.0667	1.099	97.146 ± 0.563	0.284 ± 0.004	26.941 ± 1.499
UPI	beta35	0.101	0.098	-0.003 ± 0.002	-0.027 ± 0.024	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.492 ± 0.030	0.702 ± 0.024	0.098	-0.003	-2.73	0.0681	0.0681	1.036	95.091 ± 0.730	0.288 ± 0.004	27.169 ± 1.503

Note. MCSE = Monte Carlo Standard Error shown as $\backslash u00b1$ value. Coverage, Type I Error, and Power expressed as percentages. $\backslash u2014$ indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S39

Simulation 1 - Condition 39: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.6

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.311	-0.005 ± 0.002	-0.016 ± 0.007	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.074 ± 0.002	0.054 ± 0.002	0.233 ± 0.006	0.313	-0.003	-0.92	0.0754	0.0755	1.023	95.483 ± 0.665	0.294 ± 0.002	97.741 ± 0.476
LMS	beta32	0.316	0.313	-0.003 ± 0.002	-0.009 ± 0.007	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.054 ± 0.003	0.232 ± 0.007	0.314	-0.002	-0.62	0.0712	0.0712	1.027	95.483 ± 0.665	0.295 ± 0.002	98.357 ± 0.407
LMS	beta33	0.139	0.119	-0.020 ± 0.003	-0.144 ± 0.023	0.010 ± 0.001	0.100 ± 0.003	0.010 ± 0.001	0.102 ± 0.003	0.539 ± 0.033	0.734 ± 0.025	0.124	-0.015	-10.93	0.0920	0.0933	1.089	96.715 ± 0.571	0.428 ± 0.010	32.957 ± 1.506
LMS	beta34	0.101	0.120	0.019 ± 0.002	0.186 ± 0.023	0.005 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.076 ± 0.003	0.564 ± 0.035	0.751 ± 0.026	0.109	0.008	7.59	0.0648	0.0653	1.018	95.072 ± 0.694	0.293 ± 0.006	32.136 ± 1.496
LMS	beta35	0.101	0.120	0.019 ± 0.002	0.190 ± 0.023	0.005 ± 0.000	0.072 ± 0.002	0.006 ± 0.000	0.075 ± 0.003	0.545 ± 0.037	0.738 ± 0.027	0.111	0.010	10.14	0.0661	0.0669	1.060	96.715 ± 0.571	0.300 ± 0.007	31.725 ± 1.491
LSAM	beta31	0.316	0.317	0.001 ± 0.002	0.003 ± 0.007	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.053 ± 0.002	0.230 ± 0.006	0.318	0.002	0.49	0.0705	0.0705	1.021	94.764 ± 0.714	0.291 ± 0.001	99.179 ± 0.289
LSAM	beta32	0.316	0.319	0.003 ± 0.002	0.009 ± 0.007	0.005 ± 0.000	0.074 ± 0.002	0.005 ± 0.000	0.074 ± 0.002	0.054 ± 0.003	0.233 ± 0.007	0.317	0.001	0.38	0.0720	0.0720	1.009	95.585 ± 0.658	0.292 ± 0.001	99.281 ± 0.271
LSAM	beta33	0.139	0.143	0.004 ± 0.003	0.028 ± 0.019	0.007 ± 0.000	0.085 ± 0.002	0.007 ± 0.000	0.085 ± 0.002	0.370 ± 0.017	0.608 ± 0.017	0.142	0.003	1.94	0.0808	0.0809	1.040	95.893 ± 0.636	0.345 ± 0.003	40.349 ± 1.572
LSAM	beta34	0.101	0.103	0.002 ± 0.002	0.021 ± 0.020	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	0.391 ± 0.019	0.626 ± 0.018	0.098	-0.003	-2.70	0.0604	0.0605	0.987	94.456 ± 0.733	0.244 ± 0.002	37.372 ± 1.550
LSAM	beta35	0.101	0.102	0.001 ± 0.002	0.009 ± 0.019	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	0.370 ± 0.018	0.608 ± 0.018	0.099	-0.002	-1.70	0.0601	0.0602	1.020	96.304 ± 0.605	0.246 ± 0.002	37.166 ± 1.548
QML	beta31	0.316	0.306	-0.010 ± 0.002	-0.033 ± 0.007	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.052 ± 0.002	0.228 ± 0.006	0.308	-0.008	-2.64	0.0728	0.0733	1.012	94.353 ± 0.740	0.283 ± 0.001	98.871 ± 0.339
QML	beta32	0.316	0.307	-0.009 ± 0.002	-0.028 ± 0.007	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.052 ± 0.002	0.227 ± 0.006	0.308	-0.008	-2.59	0.0695	0.0700	1.013	94.969 ± 0.700	0.283 ± 0.001	98.973 ± 0.323
QML	beta33	0.139	0.119	-0.020 ± 0.003	-0.142 ± 0.020	0.007 ± 0.000	0.086 ± 0.002	0.008 ± 0.000	0.088 ± 0.003	0.403 ± 0.020	0.635 ± 0.019	0.120	-0.019	-13.57	0.0843	0.0864	1.054	95.688 ± 0.651	0.355 ± 0.004	32.957 ± 1.506
QML	beta34	0.101	0.108	0.007 ± 0.002	0.070 ± 0.020	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	0.381 ± 0.019	0.618 ± 0.019	0.101	0.000	0.48	0.0569	0.0569	1.001	93.943 ± 0.764	0.243 ± 0.003	38.604 ± 1.560
QML	beta35	0.101	0.108	0.007 ± 0.002	0.072 ± 0.019	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	0.366 ± 0.020	0.605 ± 0.019	0.104	0.003	2.72	0.0579	0.0580	1.030	95.996 ± 0.628	0.245 ± 0.003	38.501 ± 1.559
UPI	beta31	0.316	0.317	0.001 ± 0.002	0.003 ± 0.007	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.054 ± 0.002	0.232 ± 0.006	0.318	0.002	0.72	0.0725	0.0725	1.001	94.764 ± 0.714	0.288 ± 0.001	98.871 ± 0.339
UPI	beta32	0.316	0.319	0.003 ± 0.002	0.011 ± 0.008	0.006 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.074 ± 0.002	0.055 ± 0.003	0.235 ± 0.007	0.318	0.002	0.69	0.0749	0.0750	0.991	95.483 ± 0.665	0.288 ± 0.001	99.076 ± 0.307
UPI	beta33	0.139	0.146	0.007 ± 0.003	0.054 ± 0.021	0.008 ± 0.000	0.089 ± 0.002	0.008 ± 0.000	0.089 ± 0.003	0.414 ± 0.020	0.644 ± 0.019	0.141	0.002	1.49	0.0849	0.0849	1.040	96.920 ± 0.554	0.364 ± 0.003	38.193 ± 1.557
UPI	beta34	0.101	0.102	0.001 ± 0.002	0.009 ± 0.021	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.067 ± 0.002	0.444 ± 0.022	0.667 ± 0.020	0.097	-0.004	-4.19	0.0629	0.0631	0.986	94.764 ± 0.714	0.260 ± 0.002	32.444 ± 1.500
UPI	beta35	0.101	0.102	0.001 ± 0.002	0.012 ± 0.021	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.421 ± 0.022	0.649 ± 0.020	0.101	0.000	0.19	0.0608	0.0608	1.033	96.304 ± 0.605	0.265 ± 0.003	29.979 ± 1.468

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S40

Simulation 1 - Condition 40: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, $N = 1000$, Reliability = 0.6

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LMS	beta31	0.316	0.312	-0.004 $\pm$ 0.001	-0.013 $\pm$ 0.005	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.022 $\pm$ 0.001	0.149 $\pm$ 0.004	0.311	-0.005	-1.69	0.0466	0.0469	0.987	94.372 $\pm$ 0.731	0.182 $\pm$ 0.001	99.899 $\pm$ 0.100
LMS	beta32	0.316	0.311	-0.005 $\pm$ 0.001	-0.016 $\pm$ 0.005	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.022 $\pm$ 0.001	0.147 $\pm$ 0.004	0.310	-0.006	-2.03	0.0468	0.0473	0.999	94.874 $\pm$ 0.699	0.181 $\pm$ 0.001	99.899 $\pm$ 0.100
LMS	beta33	0.139	0.125	-0.014 $\pm$ 0.002	-0.099 $\pm$ 0.014	0.004 $\pm$ 0.000	0.062 $\pm$ 0.002	0.004 $\pm$ 0.000	0.064 $\pm$ 0.002	0.211 $\pm$ 0.011	0.459 $\pm$ 0.014	0.127	-0.012	-8.64	0.0604	0.0616	1.001	94.573 $\pm$ 0.718	0.245 $\pm$ 0.003	58.693 $\pm$ 1.561
LMS	beta34	0.101	0.114	0.013 $\pm$ 0.001	0.131 $\pm$ 0.014	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.002	0.218 $\pm$ 0.013	0.467 $\pm$ 0.016	0.111	0.010	9.55	0.0415	0.0426	0.970	96.281 $\pm$ 0.600	0.172 $\pm$ 0.003	81.608 $\pm$ 1.228
LMS	beta35	0.101	0.112	0.011 $\pm$ 0.001	0.108 $\pm$ 0.014	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.195 $\pm$ 0.011	0.442 $\pm$ 0.014	0.109	0.008	7.71	0.0414	0.0421	0.999	95.980 $\pm$ 0.623	0.169 $\pm$ 0.002	82.513 $\pm$ 1.204
LSAM	beta31	0.316	0.317	0.001 $\pm$ 0.001	0.003 $\pm$ 0.005	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.022 $\pm$ 0.001	0.147 $\pm$ 0.004	0.315	-0.001	-0.28	0.0456	0.0456	0.991	94.673 $\pm$ 0.712	0.180 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta32	0.316	0.317	0.001 $\pm$ 0.001	0.002 $\pm$ 0.005	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.021 $\pm$ 0.001	0.145 $\pm$ 0.004	0.316	0.000	-0.13	0.0470	0.0470	1.002	95.176 $\pm$ 0.679	0.181 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta33	0.139	0.139	-0.000 $\pm$ 0.002	-0.000 $\pm$ 0.013	0.003 $\pm$ 0.000	0.055 $\pm$ 0.001	0.003 $\pm$ 0.000	0.055 $\pm$ 0.002	0.159 $\pm$ 0.007	0.399 $\pm$ 0.011	0.138	-0.001	-0.63	0.0545	0.0545	0.959	93.769 $\pm$ 0.766	0.209 $\pm$ 0.001	73.568 $\pm$ 1.398
LSAM	beta34	0.101	0.103	0.002 $\pm$ 0.001	0.015 $\pm$ 0.013	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.159 $\pm$ 0.008	0.398 $\pm$ 0.012	0.102	0.001	0.72	0.0392	0.0392	0.945	94.372 $\pm$ 0.731	0.149 $\pm$ 0.001	77.186 $\pm$ 1.330
LSAM	beta35	0.101	0.100	-0.001 $\pm$ 0.001	-0.007 $\pm$ 0.012	0.001 $\pm$ 0.000	0.039 $\pm$ 0.001	0.001 $\pm$ 0.000	0.039 $\pm$ 0.001	0.147 $\pm$ 0.007	0.383 $\pm$ 0.011	0.100	-0.001	-1.23	0.0397	0.0397	0.982	94.874 $\pm$ 0.699	0.149 $\pm$ 0.001	76.985 $\pm$ 1.334
QML	beta31	0.316	0.306	-0.010 $\pm$ 0.001	-0.032 $\pm$ 0.005	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.022 $\pm$ 0.001	0.149 $\pm$ 0.004	0.305	-0.011	-3.61	0.0456	0.0470	0.985	92.965 $\pm$ 0.811	0.177 $\pm$ 0.001	100.000 $\pm$ 0.000
QML	beta32	0.316	0.305	-0.011 $\pm$ 0.001	-0.034 $\pm$ 0.005	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.022 $\pm$ 0.001	0.148 $\pm$ 0.004	0.304	-0.012	-3.71	0.0467	0.0481	0.995	94.573 $\pm$ 0.718	0.177 $\pm$ 0.001	100.000 $\pm$ 0.000
QML	beta33	0.139	0.120	-0.019 $\pm$ 0.002	-0.138 $\pm$ 0.013	0.003 $\pm$ 0.000	0.057 $\pm$ 0.001	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	0.187 $\pm$ 0.009	0.432 $\pm$ 0.012	0.120	-0.019	-13.94	0.0555	0.0588	0.981	93.869 $\pm$ 0.761	0.219 $\pm$ 0.001	59.497 $\pm$ 1.556
QML	beta34	0.101	0.106	0.005 $\pm$ 0.001	0.051 $\pm$ 0.013	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.159 $\pm$ 0.008	0.398 $\pm$ 0.012	0.105	0.004	3.77	0.0387	0.0389	0.962	95.075 $\pm$ 0.686	0.150 $\pm$ 0.001	82.211 $\pm$ 1.212
QML	beta35	0.101	0.105	0.004 $\pm$ 0.001	0.035 $\pm$ 0.012	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.149 $\pm$ 0.007	0.387 $\pm$ 0.011	0.103	0.002	1.88	0.0385	0.0386	0.985	94.472 $\pm$ 0.724	0.150 $\pm$ 0.001	83.719 $\pm$ 1.170
UPI	beta31	0.316	0.317	0.001 $\pm$ 0.001	0.004 $\pm$ 0.005	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.022 $\pm$ 0.001	0.147 $\pm$ 0.004	0.314	-0.002	-0.52	0.0464	0.0464	0.974	94.573 $\pm$ 0.718	0.177 $\pm$ 0.000	100.000 $\pm$ 0.000
UPI	beta32	0.316	0.317	0.001 $\pm$ 0.001	0.002 $\pm$ 0.005	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.021 $\pm$ 0.001	0.146 $\pm$ 0.004	0.316	0.000	-0.13	0.0489	0.0489	0.981	94.472 $\pm$ 0.724	0.178 $\pm$ 0.000	100.000 $\pm$ 0.000
UPI	beta33	0.139	0.140	0.001 $\pm$ 0.002	0.009 $\pm$ 0.013	0.003 $\pm$ 0.000	0.057 $\pm$ 0.001	0.003 $\pm$ 0.000	0.057 $\pm$ 0.002	0.168 $\pm$ 0.008	0.410 $\pm$ 0.012	0.138	-0.001	-0.49	0.0527	0.0527	0.960	93.769 $\pm$ 0.766	0.215 $\pm$ 0.001	72.362 $\pm$ 1.418
UPI	beta34	0.101	0.102	0.001 $\pm$ 0.001	0.011 $\pm$ 0.013	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.170 $\pm$ 0.009	0.412 $\pm$ 0.013	0.101	0.000	0.02	0.0406	0.0406	0.949	94.573 $\pm$ 0.718	0.155 $\pm$ 0.001	75.176 $\pm$ 1.370
UPI	beta35	0.101	0.100	-0.001 $\pm$ 0.001	-0.010 $\pm$ 0.013	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.158 $\pm$ 0.007	0.398 $\pm$ 0.011	0.100	-0.001	-1.41	0.0393	0.0393	0.982	94.472 $\pm$ 0.724	0.155 $\pm$ 0.001	74.975 $\pm$ 1.373

Note. MCSE = Monte Carlo Standard Error shown as $\backslash u00b1$ value. Coverage, Type I Error, and Power expressed as percentages. $\backslash u2014$ indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S41

Simulation 1 - Condition 41: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.8

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LMS	beta31	0.316	0.311	-0.005 $\pm$ 0.002	-0.017 $\pm$ 0.006	0.003 $\pm$ 0.000	0.059 $\pm$ 0.001	0.003 $\pm$ 0.000	0.059 $\pm$ 0.002	0.035 $\pm$ 0.002	0.187 $\pm$ 0.005	0.308	-0.008	-2.45	0.0543	0.0549	0.996	94.394 $\pm$ 0.728	0.229 $\pm$ 0.001	99.800 $\pm$ 0.141
LMS	beta32	0.316	0.313	-0.003 $\pm$ 0.002	-0.010 $\pm$ 0.006	0.003 $\pm$ 0.000	0.059 $\pm$ 0.001	0.003 $\pm$ 0.000	0.059 $\pm$ 0.002	0.035 $\pm$ 0.002	0.187 $\pm$ 0.005	0.313	-0.003	-0.80	0.0597	0.0597	0.994	94.595 $\pm$ 0.715	0.230 $\pm$ 0.001	99.900 $\pm$ 0.100
LMS	beta33	0.139	0.132	-0.007 $\pm$ 0.002	-0.048 $\pm$ 0.016	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.243 $\pm$ 0.011	0.493 $\pm$ 0.014	0.134	-0.005	-3.93	0.0663	0.0666	0.974	94.595 $\pm$ 0.715	0.260 $\pm$ 0.002	54.855 $\pm$ 1.574
LMS	beta34	0.101	0.106	0.005 $\pm$ 0.001	0.046 $\pm$ 0.015	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.221 $\pm$ 0.011	0.470 $\pm$ 0.014	0.103	0.002	1.63	0.0454	0.0454	0.980	93.093 $\pm$ 0.802	0.182 $\pm$ 0.002	64.464 $\pm$ 1.514
LMS	beta35	0.101	0.107	0.006 $\pm$ 0.002	0.056 $\pm$ 0.015	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.227 $\pm$ 0.011	0.476 $\pm$ 0.014	0.104	0.003	3.44	0.0452	0.0454	0.974	93.994 $\pm$ 0.752	0.182 $\pm$ 0.002	66.066 $\pm$ 1.498
LSAM	beta31	0.316	0.314	-0.002 $\pm$ 0.002	-0.005 $\pm$ 0.006	0.003 $\pm$ 0.000	0.059 $\pm$ 0.001	0.003 $\pm$ 0.000	0.059 $\pm$ 0.002	0.035 $\pm$ 0.002	0.187 $\pm$ 0.006	0.313	-0.003	-0.97	0.0565	0.0566	0.995	94.895 $\pm$ 0.696	0.230 $\pm$ 0.001	99.900 $\pm$ 0.100
LSAM	beta32	0.316	0.316	0.000 $\pm$ 0.002	0.001 $\pm$ 0.006	0.003 $\pm$ 0.000	0.059 $\pm$ 0.001	0.003 $\pm$ 0.000	0.059 $\pm$ 0.002	0.035 $\pm$ 0.002	0.186 $\pm$ 0.005	0.316	0.000	0.03	0.0583	0.0583	0.997	94.895 $\pm$ 0.696	0.230 $\pm$ 0.001	99.900 $\pm$ 0.100
LSAM	beta33	0.139	0.140	0.001 $\pm$ 0.002	0.010 $\pm$ 0.015	0.004 $\pm$ 0.000	0.065 $\pm$ 0.002	0.004 $\pm$ 0.000	0.065 $\pm$ 0.002	0.221 $\pm$ 0.010	0.470 $\pm$ 0.013	0.142	0.003	2.02	0.0636	0.0637	0.965	94.494 $\pm$ 0.722	0.247 $\pm$ 0.002	60.661 $\pm$ 1.546
LSAM	beta34	0.101	0.102	0.001 $\pm$ 0.001	0.011 $\pm$ 0.014	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.210 $\pm$ 0.010	0.458 $\pm$ 0.013	0.101	0.000	-0.24	0.0453	0.0453	0.967	92.292 $\pm$ 0.844	0.175 $\pm$ 0.001	64.565 $\pm$ 1.513
LSAM	beta35	0.101	0.102	0.001 $\pm$ 0.001	0.013 $\pm$ 0.014	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.208 $\pm$ 0.010	0.456 $\pm$ 0.013	0.102	0.001	0.54	0.0436	0.0437	0.969	93.994 $\pm$ 0.752	0.175 $\pm$ 0.001	64.464 $\pm$ 1.514
QML	beta31	0.316	0.309	-0.007 $\pm$ 0.002	-0.022 $\pm$ 0.006	0.003 $\pm$ 0.000	0.059 $\pm$ 0.001	0.003 $\pm$ 0.000	0.059 $\pm$ 0.002	0.035 $\pm$ 0.002	0.187 $\pm$ 0.005	0.307	-0.009	-2.89	0.0542	0.0549	0.994	94.294 $\pm$ 0.734	0.229 $\pm$ 0.001	99.900 $\pm$ 0.100
QML	beta32	0.316	0.311	-0.005 $\pm$ 0.002	-0.015 $\pm$ 0.006	0.003 $\pm$ 0.000	0.059 $\pm$ 0.001	0.003 $\pm$ 0.000	0.059 $\pm$ 0.002	0.035 $\pm$ 0.002	0.186 $\pm$ 0.005	0.312	-0.004	-1.15	0.0595	0.0596	0.994	94.595 $\pm$ 0.715	0.229 $\pm$ 0.001	99.900 $\pm$ 0.100
QML	beta33	0.139	0.131	-0.008 $\pm$ 0.002	-0.059 $\pm$ 0.015	0.004 $\pm$ 0.000	0.067 $\pm$ 0.002	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.236 $\pm$ 0.011	0.486 $\pm$ 0.014	0.132	-0.007	-5.08	0.0660	0.0664	0.973	94.795 $\pm$ 0.703	0.256 $\pm$ 0.002	55.255 $\pm$ 1.573
QML	beta34	0.101	0.104	0.003 $\pm$ 0.001	0.031 $\pm$ 0.015	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.212 $\pm$ 0.010	0.460 $\pm$ 0.013	0.102	0.001	0.63	0.0448	0.0448	0.978	92.693 $\pm$ 0.823	0.178 $\pm$ 0.001	64.464 $\pm$ 1.514
QML	beta35	0.101	0.105	0.004 $\pm$ 0.001	0.039 $\pm$ 0.015	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.215 $\pm$ 0.010	0.463 $\pm$ 0.013	0.103	0.002	2.19	0.0450	0.0450	0.976	93.894 $\pm$ 0.758	0.178 $\pm$ 0.001	66.667 $\pm$ 1.491
UPI	beta31	0.316	0.314	-0.002 $\pm$ 0.002	-0.005 $\pm$ 0.006	0.003 $\pm$ 0.000	0.059 $\pm$ 0.001	0.003 $\pm$ 0.000	0.059 $\pm$ 0.002	0.035 $\pm$ 0.002	0.186 $\pm$ 0.006	0.312	-0.004	-1.17	0.0550	0.0551	0.962	94.094 $\pm$ 0.746	0.222 $\pm$ 0.001	99.900 $\pm$ 0.100
UPI	beta32	0.316	0.316	0.000 $\pm$ 0.002	0.001 $\pm$ 0.006	0.003 $\pm$ 0.000	0.059 $\pm$ 0.001	0.003 $\pm$ 0.000	0.059 $\pm$ 0.002	0.035 $\pm$ 0.002	0.187 $\pm$ 0.005	0.318	0.002	0.54	0.0578	0.0579	0.964	93.794 $\pm$ 0.763	0.223 $\pm$ 0.001	99.900 $\pm$ 0.100
UPI	beta33	0.139	0.141	0.002 $\pm$ 0.002	0.016 $\pm$ 0.015	0.004 $\pm$ 0.000	0.067 $\pm$ 0.002	0.004 $\pm$ 0.000	0.067 $\pm$ 0.002	0.231 $\pm$ 0.011	0.481 $\pm$ 0.014	0.142	0.003	2.06	0.0644	0.0644	0.958	93.894 $\pm$ 0.758	0.251 $\pm$ 0.002	59.560 $\pm$ 1.553
UPI	beta34	0.101	0.102	0.001 $\pm$ 0.001	0.009 $\pm$ 0.015	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.219 $\pm$ 0.010	0.468 $\pm$ 0.013	0.102	0.001	0.82	0.0445	0.0445	0.966	91.792 $\pm$ 0.868	0.179 $\pm$ 0.001	62.763 $\pm$ 1.530
UPI	beta35	0.101	0.102	0.001 $\pm$ 0.001	0.014 $\pm$ 0.015	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.217 $\pm$ 0.010	0.466 $\pm$ 0.013	0.101	0.000	-0.28	0.0445	0.0445	0.970	93.894 $\pm$ 0.758	0.179 $\pm$ 0.001	62.763 $\pm$ 1.530

Note. MCSE = Monte Carlo Standard Error shown as $\backslash u00b1$ value. Coverage, Type I Error, and Power expressed as percentages. $\backslash u2014$ indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S42

Simulation 1 - Condition 42: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, $N = 1000$, Reliability = 0.8

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LMS	beta31	0.316	0.311	-0.005 $\pm$ 0.001	-0.015 $\pm$ 0.004	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.014 $\pm$ 0.001	0.120 $\pm$ 0.003	0.311	-0.005	-1.66	0.0385	0.0389	0.993	95.000 $\pm$ 0.689	0.146 $\pm$ 0.002	99.900 $\pm$ 0.100
LMS	beta32	0.316	0.313	-0.003 $\pm$ 0.001	-0.008 $\pm$ 0.004	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.013 $\pm$ 0.001	0.116 $\pm$ 0.003	0.313	-0.003	-1.00	0.0381	0.0382	1.018	95.200 $\pm$ 0.676	0.146 $\pm$ 0.002	99.900 $\pm$ 0.100
LMS	beta33	0.139	0.133	-0.006 $\pm$ 0.001	-0.040 $\pm$ 0.009	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.091 $\pm$ 0.004	0.302 $\pm$ 0.008	0.135	-0.004	-3.15	0.0390	0.0393	0.988	95.800 $\pm$ 0.634	0.161 $\pm$ 0.001	85.800 $\pm$ 1.104
LMS	beta34	0.101	0.104	0.003 $\pm$ 0.001	0.029 $\pm$ 0.009	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.085 $\pm$ 0.004	0.292 $\pm$ 0.008	0.103	0.002	1.88	0.0287	0.0288	0.989	95.000 $\pm$ 0.689	0.114 $\pm$ 0.001	97.900 $\pm$ 0.453
LMS	beta35	0.101	0.103	0.002 $\pm$ 0.001	0.020 $\pm$ 0.009	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.082 $\pm$ 0.004	0.286 $\pm$ 0.008	0.102	0.001	0.93	0.0291	0.0291	1.009	94.800 $\pm$ 0.702	0.114 $\pm$ 0.001	97.300 $\pm$ 0.513
LSAM	beta31	0.316	0.315	-0.001 $\pm$ 0.001	-0.003 $\pm$ 0.004	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.014 $\pm$ 0.001	0.118 $\pm$ 0.003	0.315	-0.001	-0.39	0.0393	0.0394	0.988	95.500 $\pm$ 0.656	0.145 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta32	0.316	0.317	0.001 $\pm$ 0.001	0.003 $\pm$ 0.004	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.013 $\pm$ 0.001	0.116 $\pm$ 0.003	0.316	0.000	0.15	0.0386	0.0386	1.004	95.600 $\pm$ 0.649	0.144 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta33	0.139	0.140	0.001 $\pm$ 0.001	0.004 $\pm$ 0.009	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.081 $\pm$ 0.004	0.285 $\pm$ 0.008	0.141	0.002	1.32	0.0381	0.0382	0.984	95.400 $\pm$ 0.662	0.153 $\pm$ 0.001	91.200 $\pm$ 0.896
LSAM	beta34	0.101	0.101	-0.000 $\pm$ 0.001	-0.000 $\pm$ 0.009	0.001 $\pm$ 0.000	0.028 $\pm$ 0.001	0.001 $\pm$ 0.000	0.028 $\pm$ 0.001	0.078 $\pm$ 0.004	0.279 $\pm$ 0.008	0.101	0.000	0.17	0.0271	0.0271	0.988	94.700 $\pm$ 0.708	0.109 $\pm$ 0.001	96.800 $\pm$ 0.557
LSAM	beta35	0.101	0.100	-0.001 $\pm$ 0.001	-0.007 $\pm$ 0.009	0.001 $\pm$ 0.000	0.028 $\pm$ 0.001	0.001 $\pm$ 0.000	0.028 $\pm$ 0.001	0.076 $\pm$ 0.004	0.275 $\pm$ 0.008	0.100	-0.001	-1.22	0.0274	0.0275	1.006	93.600 $\pm$ 0.774	0.110 $\pm$ 0.001	96.400 $\pm$ 0.589
QML	beta31	0.316	0.310	-0.006 $\pm$ 0.001	-0.019 $\pm$ 0.004	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.014 $\pm$ 0.001	0.120 $\pm$ 0.003	0.309	-0.007	-2.18	0.0384	0.0391	0.986	95.100 $\pm$ 0.683	0.145 $\pm$ 0.001	99.900 $\pm$ 0.100
QML	beta32	0.316	0.312	-0.004 $\pm$ 0.001	-0.013 $\pm$ 0.004	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.014 $\pm$ 0.001	0.116 $\pm$ 0.003	0.311	-0.005	-1.46	0.0380	0.0383	1.008	95.300 $\pm$ 0.669	0.144 $\pm$ 0.001	99.900 $\pm$ 0.100
QML	beta33	0.139	0.132	-0.007 $\pm$ 0.001	-0.054 $\pm$ 0.009	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.090 $\pm$ 0.004	0.300 $\pm$ 0.008	0.133	-0.006	-4.44	0.0390	0.0395	0.987	95.700 $\pm$ 0.641	0.159 $\pm$ 0.001	85.600 $\pm$ 1.110
QML	beta34	0.101	0.103	0.002 $\pm$ 0.001	0.017 $\pm$ 0.009	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.082 $\pm$ 0.004	0.286 $\pm$ 0.008	0.102	0.001	0.78	0.0282	0.0282	0.988	94.500 $\pm$ 0.721	0.112 $\pm$ 0.001	97.900 $\pm$ 0.453
QML	beta35	0.101	0.102	0.001 $\pm$ 0.001	0.008 $\pm$ 0.009	0.001 $\pm$ 0.000	0.028 $\pm$ 0.001	0.001 $\pm$ 0.000	0.028 $\pm$ 0.001	0.079 $\pm$ 0.004	0.281 $\pm$ 0.008	0.101	0.000	-0.30	0.0285	0.0285	1.007	94.300 $\pm$ 0.733	0.112 $\pm$ 0.001	97.200 $\pm$ 0.522
UPI	beta31	0.316	0.315	-0.001 $\pm$ 0.001	-0.003 $\pm$ 0.004	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.014 $\pm$ 0.001	0.118 $\pm$ 0.003	0.314	-0.002	-0.64	0.0387	0.0387	0.957	94.900 $\pm$ 0.696	0.140 $\pm$ 0.000	100.000 $\pm$ 0.000
UPI	beta32	0.316	0.317	0.001 $\pm$ 0.001	0.003 $\pm$ 0.004	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.014 $\pm$ 0.001	0.116 $\pm$ 0.003	0.317	0.001	0.18	0.0382	0.0382	0.970	94.900 $\pm$ 0.696	0.140 $\pm$ 0.000	100.000 $\pm$ 0.000
UPI	beta33	0.139	0.140	0.001 $\pm$ 0.001	0.007 $\pm$ 0.009	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.084 $\pm$ 0.004	0.289 $\pm$ 0.008	0.141	0.002	1.31	0.0397	0.0397	0.978	94.900 $\pm$ 0.696	0.154 $\pm$ 0.001	91.000 $\pm$ 0.905
UPI	beta34	0.101	0.101	0.000 $\pm$ 0.001	0.000 $\pm$ 0.009	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.080 $\pm$ 0.004	0.282 $\pm$ 0.008	0.101	0.000	-0.06	0.0280	0.0280	0.990	94.400 $\pm$ 0.727	0.111 $\pm$ 0.001	96.700 $\pm$ 0.565
UPI	beta35	0.101	0.100	-0.001 $\pm$ 0.001	-0.008 $\pm$ 0.009	0.001 $\pm$ 0.000	0.028 $\pm$ 0.001	0.001 $\pm$ 0.000	0.028 $\pm$ 0.001	0.077 $\pm$ 0.004	0.277 $\pm$ 0.008	0.100	-0.001	-1.43	0.0280	0.0280	1.013	94.300 $\pm$ 0.733	0.111 $\pm$ 0.001	96.300 $\pm$ 0.597

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S43

Simulation 1 - Condition 43: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.4

Method	Param.	True	Mean-based										Median-based					Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE		
LMS	beta31	0.316	0.336	0.020 ± 0.005	0.063 ± 0.016	0.020 ± 0.001	0.143 ± 0.004	0.021 ± 0.001	0.144 ± 0.005	0.208 ± 0.012	0.456 ± 0.016	0.345	0.029	9.14	0.1368	0.1399	1.031	95.209 ± 0.749	0.577 ± 0.012	69.165 ± 1.619		
LMS	beta32	0.316	0.334	0.018 ± 0.005	0.057 ± 0.015	0.020 ± 0.001	0.140 ± 0.004	0.020 ± 0.001	0.141 ± 0.005	0.198 ± 0.012	0.445 ± 0.015	0.341	0.025	7.81	0.1330	0.1353	1.054	96.437 ± 0.650	0.577 ± 0.010	69.042 ± 1.620		
LMS	beta33	0.139	-0.206	-0.345 ± 0.016	-2.484 ± 0.117	0.216 ± 0.012	0.465 ± 0.013	0.335 ± 0.022	0.579 ± 0.020	17.335 ± 1.114	4.163 ± 0.142	-0.054	-0.193	-139.20	0.3779	0.4245	0.824	87.346 ± 1.165	1.501 ± 0.064	11.548 ± 1.120		
LMS	beta34	0.101	0.365	0.264 ± 0.009	2.613 ± 0.093	0.071 ± 0.005	0.267 ± 0.010	0.141 ± 0.009	0.375 ± 0.013	13.806 ± 0.921	3.716 ± 0.129	0.271	0.170	168.57	0.1742	0.2436	0.905	92.015 ± 0.950	0.947 ± 0.049	47.420 ± 1.750		
LMS	beta35	0.101	0.360	0.259 ± 0.009	2.567 ± 0.090	0.067 ± 0.005	0.259 ± 0.009	0.134 ± 0.009	0.366 ± 0.012	13.149 ± 0.857	3.626 ± 0.123	0.283	0.182	179.90	0.1844	0.2589	0.960	92.015 ± 0.950	0.974 ± 0.065	45.455 ± 1.745		
LSAM	beta31	0.316	0.320	0.004 ± 0.008	0.011 ± 0.025	0.052 ± 0.003	0.229 ± 0.007	0.052 ± 0.003	0.229 ± 0.008	0.524 ± 0.033	0.724 ± 0.026	0.321	0.005	1.67	0.2228	0.2229	1.126	98.034 ± 0.487	1.011 ± 0.022	29.853 ± 1.604		
LSAM	beta32	0.316	0.322	0.006 ± 0.008	0.017 ± 0.025	0.052 ± 0.003	0.229 ± 0.007	0.052 ± 0.003	0.229 ± 0.008	0.524 ± 0.032	0.724 ± 0.026	0.310	-0.006	-1.81	0.2164	0.2164	1.118	98.771 ± 0.386	1.003 ± 0.030	33.538 ± 1.655		
LSAM	beta33	0.139	0.169	0.030 ± 0.006	0.217 ± 0.045	0.032 ± 0.002	0.180 ± 0.006	0.033 ± 0.002	0.182 ± 0.007	1.722 ± 0.118	1.312 ± 0.050	0.164	0.025	17.64	0.1703	0.1720	1.106	96.437 ± 0.650	0.781 ± 0.036	11.548 ± 1.120		
LSAM	beta34	0.101	0.100	-0.001 ± 0.003	-0.015 ± 0.032	0.009 ± 0.001	0.093 ± 0.003	0.009 ± 0.001	0.093 ± 0.003	0.842 ± 0.052	0.918 ± 0.033	0.098	-0.003	-3.10	0.0898	0.0898	1.140	97.420 ± 0.556	0.414 ± 0.014	24.324 ± 1.504		
LSAM	beta35	0.101	0.102	0.001 ± 0.003	0.015 ± 0.034	0.010 ± 0.001	0.098 ± 0.003	0.010 ± 0.001	0.098 ± 0.004	0.941 ± 0.064	0.970 ± 0.037	0.100	-0.001	-0.98	0.0892	0.0893	1.066	96.683 ± 0.628	0.410 ± 0.015	26.044 ± 1.538		
QML	beta31	0.316	0.341	0.025 ± 0.004	0.078 ± 0.012	0.013 ± 0.001	0.113 ± 0.003	0.013 ± 0.001	0.115 ± 0.004	0.133 ± 0.007	0.364 ± 0.012	0.342	0.026	8.23	0.1154	0.1183	1.026	95.332 ± 0.739	0.453 ± 0.003	83.292 ± 1.308		
QML	beta32	0.316	0.340	0.024 ± 0.004	0.077 ± 0.012	0.012 ± 0.001	0.111 ± 0.003	0.013 ± 0.001	0.114 ± 0.003	0.129 ± 0.007	0.359 ± 0.011	0.340	0.024	7.65	0.1159	0.1184	1.040	96.069 ± 0.681	0.452 ± 0.004	84.398 ± 1.272		
QML	beta33	0.139	0.008	-0.131 ± 0.005	-0.944 ± 0.038	0.023 ± 0.001	0.151 ± 0.004	0.040 ± 0.002	0.200 ± 0.005	2.067 ± 0.100	1.438 ± 0.038	0.010	-0.129	-93.05	0.1491	0.1974	1.025	88.575 ± 1.115	0.606 ± 0.009	7.002 ± 0.894		
QML	beta34	0.101	0.224	0.123 ± 0.003	1.221 ± 0.031	0.008 ± 0.000	0.090 ± 0.003	0.023 ± 0.001	0.153 ± 0.004	2.282 ± 0.104	1.511 ± 0.036	0.215	0.114	113.31	0.0900	0.1456	1.087	85.872 ± 1.221	0.383 ± 0.007	73.833 ± 1.541		
QML	beta35	0.101	0.222	0.121 ± 0.003	1.202 ± 0.032	0.008 ± 0.001	0.092 ± 0.003	0.023 ± 0.001	0.152 ± 0.004	2.277 ± 0.107	1.509 ± 0.037	0.216	0.115	113.89	0.0914	0.1469	1.036	83.538 ± 1.300	0.374 ± 0.007	73.342 ± 1.550		
UPI	beta31	0.316	0.327	0.011 ± 0.008	0.036 ± 0.026	0.054 ± 0.003	0.232 ± 0.007	0.054 ± 0.003	0.232 ± 0.008	0.537 ± 0.034	0.733 ± 0.027	0.338	0.022	7.07	0.2280	0.2291	1.191	97.912 ± 0.501	1.081 ± 0.030	32.924 ± 1.647		
UPI	beta32	0.316	0.335	0.019 ± 0.008	0.060 ± 0.025	0.053 ± 0.003	0.230 ± 0.007	0.053 ± 0.003	0.230 ± 0.008	0.532 ± 0.034	0.729 ± 0.027	0.323	0.007	2.20	0.2215	0.2216	1.149	98.157 ± 0.471	1.035 ± 0.023	34.029 ± 1.661		
UPI	beta33	0.139	0.151	0.012 ± 0.007	0.085 ± 0.050	0.040 ± 0.003	0.199 ± 0.007	0.040 ± 0.003	0.199 ± 0.008	2.055 ± 0.146	1.433 ± 0.057	0.134	-0.005	-3.30	0.1703	0.1703	1.385	95.700 ± 0.711	1.080 ± 0.049	6.143 ± 0.842		
UPI	beta34	0.101	0.101	-0.000 ± 0.003	-0.001 ± 0.034	0.009 ± 0.001	0.097 ± 0.003	0.009 ± 0.001	0.097 ± 0.004	0.924 ± 0.058	0.961 ± 0.035	0.096	-0.005	-4.50	0.0902	0.0903	1.216	96.806 ± 0.616	0.463 ± 0.017	21.622 ± 1.443		
UPI	beta35	0.101	0.099	-0.002 ± 0.003	-0.016 ± 0.034	0.009 ± 0.001	0.097 ± 0.003	0.009 ± 0.001	0.097 ± 0.004	0.928 ± 0.062	0.963 ± 0.036	0.097	-0.004	-3.66	0.0894	0.0895	1.134	96.069 ± 0.681	0.433 ± 0.014	23.464 ± 1.485		

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S44

Simulation 1 - Condition 44: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.4

Method	Param.	True	Mean-based										Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE	
LMS	beta31	0.316	0.338	0.022 ± 0.003	0.069 ± 0.010	0.009 ± 0.000	0.095 ± 0.003	0.010 ± 0.000	0.098 ± 0.003	0.096 ± 0.005	0.310 ± 0.009	0.340	0.024	7.54	0.0900	0.0931	1.016	91.675 ± 0.891	0.380 ± 0.010	88.137 ± 1.043	
LMS	beta32	0.316	0.329	0.013 ± 0.003	0.042 ± 0.009	0.009 ± 0.000	0.093 ± 0.002	0.009 ± 0.000	0.094 ± 0.003	0.088 ± 0.004	0.297 ± 0.009	0.334	0.018	5.72	0.0913	0.0931	1.048	93.757 ± 0.780	0.382 ± 0.011	87.513 ± 1.066	
LMS	beta33	0.139	-0.212	-0.351 ± 0.014	-2.523 ± 0.101	0.188 ± 0.009	0.434 ± 0.010	0.311 ± 0.018	0.558 ± 0.017	16.100 ± 0.912	4.012 ± 0.120	-0.034	-0.173	-124.13	0.2664	0.3174	0.677	81.582 ± 1.250	1.151 ± 0.045	21.644 ± 1.328	
LMS	beta34	0.101	0.355	0.254 ± 0.007	2.516 ± 0.070	0.048 ± 0.003	0.220 ± 0.007	0.113 ± 0.006	0.336 ± 0.009	11.049 ± 0.597	3.324 ± 0.093	0.267	0.166	164.49	0.1283	0.2099	0.761	68.574 ± 1.497	0.654 ± 0.022	77.003 ± 1.357	
LMS	beta35	0.101	0.367	0.266 ± 0.007	2.632 ± 0.074	0.053 ± 0.003	0.231 ± 0.007	0.124 ± 0.007	0.352 ± 0.010	12.146 ± 0.661	3.485 ± 0.098	0.278	0.177	175.69	0.1325	0.2214	0.754	67.950 ± 1.505	0.683 ± 0.027	78.356 ± 1.328	
LSAM	beta31	0.316	0.324	0.008 ± 0.004	0.026 ± 0.013	0.016 ± 0.001	0.128 ± 0.003	0.016 ± 0.001	0.128 ± 0.004	0.165 ± 0.008	0.406 ± 0.012	0.317	0.001	0.27	0.1291	0.1291	1.007	95.317 ± 0.682	0.505 ± 0.004	72.737 ± 1.436	
LSAM	beta32	0.316	0.312	-0.004 ± 0.004	-0.013 ± 0.013	0.017 ± 0.001	0.129 ± 0.003	0.017 ± 0.001	0.129 ± 0.004	0.167 ± 0.008	0.409 ± 0.012	0.310	-0.006	-2.00	0.1267	0.1269	1.008	96.566 ± 0.587	0.511 ± 0.004	70.239 ± 1.475	
LSAM	beta33	0.139	0.154	0.015 ± 0.003	0.107 ± 0.022	0.009 ± 0.000	0.094 ± 0.003	0.009 ± 0.001	0.096 ± 0.003	0.473 ± 0.027	0.688 ± 0.022	0.147	0.008	5.51	0.0851	0.0854	0.947	94.797 ± 0.716	0.351 ± 0.004	42.144 ± 1.593	
LSAM	beta34	0.101	0.100	-0.001 ± 0.002	-0.007 ± 0.015	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.002	0.224 ± 0.012	0.474 ± 0.015	0.099	-0.002	-1.52	0.0462	0.0462	0.998	95.421 ± 0.674	0.187 ± 0.002	61.811 ± 1.567	
LSAM	beta35	0.101	0.105	0.004 ± 0.002	0.037 ± 0.017	0.003 ± 0.000	0.052 ± 0.002	0.003 ± 0.000	0.052 ± 0.002	0.270 ± 0.016	0.519 ± 0.018	0.105	0.004	3.70	0.0487	0.0488	0.943	96.046 ± 0.629	0.194 ± 0.002	62.123 ± 1.565	
QML	beta31	0.316	0.344	0.028 ± 0.002	0.088 ± 0.008	0.005 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	0.062 ± 0.003	0.249 ± 0.007	0.342	0.026	8.13	0.0729	0.0773	0.969	91.883 ± 0.881	0.280 ± 0.002	98.855 ± 0.343	
QML	beta32	0.316	0.338	0.022 ± 0.002	0.068 ± 0.007	0.005 ± 0.000	0.071 ± 0.002	0.006 ± 0.000	0.074 ± 0.002	0.055 ± 0.003	0.235 ± 0.006	0.337	0.021	6.50	0.0740	0.0768	1.004	93.444 ± 0.798	0.279 ± 0.001	98.959 ± 0.327	
QML	beta33	0.139	0.011	-0.128 ± 0.003	-0.919 ± 0.024	0.010 ± 0.001	0.102 ± 0.003	0.027 ± 0.001	0.163 ± 0.003	1.383 ± 0.054	1.176 ± 0.024	0.011	-0.128	-91.95	0.0984	0.1613	0.928	73.465 ± 1.424	0.371 ± 0.004	7.908 ± 0.871	
QML	beta34	0.101	0.223	0.122 ± 0.002	1.210 ± 0.018	0.003 ± 0.000	0.057 ± 0.001	0.018 ± 0.001	0.135 ± 0.002	1.786 ± 0.052	1.336 ± 0.020	0.219	0.118	116.41	0.0551	0.1299	1.009	40.895 ± 1.586	0.227 ± 0.002	99.480 ± 0.232	
QML	beta35	0.101	0.228	0.127 ± 0.002	1.253 ± 0.019	0.004 ± 0.000	0.061 ± 0.002	0.020 ± 0.001	0.140 ± 0.002	1.932 ± 0.060	1.390 ± 0.022	0.221	0.120	118.35	0.0556	0.1318	0.971	39.334 ± 1.576	0.232 ± 0.003	99.168 ± 0.293	
UPI	beta31	0.316	0.330	0.014 ± 0.004	0.046 ± 0.013	0.017 ± 0.001	0.129 ± 0.003	0.017 ± 0.001	0.130 ± 0.004	0.170 ± 0.008	0.412 ± 0.012	0.327	0.011	3.62	0.1278	0.1283	1.018	95.525 ± 0.667	0.517 ± 0.005	74.506 ± 1.406	
UPI	beta32	0.316	0.315	-0.001 ± 0.004	-0.002 ± 0.013	0.017 ± 0.001	0.130 ± 0.003	0.017 ± 0.001	0.130 ± 0.004	0.169 ± 0.009	0.411 ± 0.012	0.311	-0.005	-1.68	0.1286	0.1287	1.025	96.566 ± 0.587	0.522 ± 0.005	70.239 ± 1.475	
UPI	beta33	0.139	0.153	0.014 ± 0.003	0.100 ± 0.025	0.012 ± 0.001	0.108 ± 0.003	0.012 ± 0.001	0.109 ± 0.004	0.616 ± 0.037	0.785 ± 0.027	0.142	0.003	2.00	0.0963	0.0963	1.026	94.797 ± 0.716	0.435 ± 0.012	28.512 ± 1.456	
UPI	beta34	0.101	0.098	-0.003 ± 0.002	-0.031 ± 0.015	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.227 ± 0.012	0.477 ± 0.015	0.096	-0.005	-5.01	0.0445	0.0448	1.034	95.317 ± 0.682	0.195 ± 0.003	57.336 ± 1.595	
UPI	beta35	0.101	0.105	0.004 ± 0.002	0.036 ± 0.017	0.003 ± 0.000	0.054 ± 0.002	0.003 ± 0.000	0.054 ± 0.002	0.286 ± 0.017	0.534 ± 0.018	0.104	0.003	2.49	0.0496	0.0497	0.966	96.254 ± 0.613	0.204 ± 0.003	57.856 ± 1.593	

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S45

Simulation 1 - Condition 45: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.6

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.321	0.005 ± 0.003	0.016 ± 0.010	0.009 ± 0.000	0.092 ± 0.002	0.009 ± 0.000	0.092 ± 0.003	0.085 ± 0.004	0.292 ± 0.009	0.321	0.005	1.67	0.0940	0.0941	1.045	95.319 ± 0.689	0.378 ± 0.006	86.809 ± 1.104
LMS	beta32	0.316	0.327	0.011 ± 0.003	0.034 ± 0.010	0.009 ± 0.000	0.094 ± 0.002	0.009 ± 0.000	0.095 ± 0.003	0.089 ± 0.004	0.299 ± 0.009	0.328	0.012	3.94	0.0936	0.0944	1.012	94.255 ± 0.759	0.373 ± 0.005	90.426 ± 0.960
LMS	beta33	0.139	0.089	-0.050 ± 0.004	-0.362 ± 0.025	0.012 ± 0.001	0.108 ± 0.003	0.014 ± 0.001	0.119 ± 0.004	0.736 ± 0.043	0.858 ± 0.028	0.092	-0.047	-33.54	0.1001	0.1104	1.064	93.936 ± 0.778	0.451 ± 0.012	21.383 ± 1.337
LMS	beta34	0.101	0.168	0.067 ± 0.002	0.662 ± 0.021	0.004 ± 0.000	0.064 ± 0.002	0.009 ± 0.000	0.093 ± 0.003	0.845 ± 0.045	0.919 ± 0.026	0.160	0.059	58.61	0.0595	0.0839	1.005	90.000 ± 0.978	0.254 ± 0.006	83.191 ± 1.220
LMS	beta35	0.101	0.163	0.062 ± 0.002	0.612 ± 0.019	0.003 ± 0.000	0.059 ± 0.002	0.007 ± 0.000	0.085 ± 0.002	0.714 ± 0.038	0.845 ± 0.023	0.158	0.057	56.58	0.0544	0.0789	1.106	89.681 ± 0.992	0.255 ± 0.011	83.617 ± 1.207
LSAM	beta31	0.316	0.315	-0.001 ± 0.005	-0.005 ± 0.015	0.020 ± 0.001	0.141 ± 0.004	0.020 ± 0.001	0.141 ± 0.004	0.198 ± 0.010	0.445 ± 0.014	0.317	0.001	0.30	0.1379	0.1379	0.973	95.957 ± 0.642	0.536 ± 0.004	66.170 ± 1.543
LSAM	beta32	0.316	0.327	0.011 ± 0.005	0.033 ± 0.015	0.021 ± 0.001	0.143 ± 0.004	0.021 ± 0.001	0.144 ± 0.004	0.207 ± 0.011	0.455 ± 0.014	0.324	0.008	2.47	0.1372	0.1375	0.948	93.723 ± 0.791	0.533 ± 0.004	67.447 ± 1.528
LSAM	beta33	0.139	0.156	0.017 ± 0.003	0.120 ± 0.022	0.009 ± 0.000	0.092 ± 0.002	0.009 ± 0.000	0.094 ± 0.003	0.455 ± 0.023	0.674 ± 0.020	0.151	0.012	8.61	0.0911	0.0919	0.975	94.043 ± 0.772	0.353 ± 0.004	45.532 ± 1.624
LSAM	beta34	0.101	0.104	0.003 ± 0.002	0.032 ± 0.019	0.003 ± 0.000	0.058 ± 0.002	0.003 ± 0.000	0.058 ± 0.002	0.327 ± 0.019	0.572 ± 0.019	0.103	0.002	1.95	0.0540	0.0540	0.902	92.340 ± 0.867	0.204 ± 0.003	60.000 ± 1.598
LSAM	beta35	0.101	0.099	-0.002 ± 0.002	-0.023 ± 0.018	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.291 ± 0.016	0.539 ± 0.017	0.101	0.000	0.47	0.0514	0.0514	0.943	92.553 ± 0.856	0.201 ± 0.003	54.574 ± 1.624
QML	beta31	0.316	0.322	0.006 ± 0.003	0.019 ± 0.009	0.008 ± 0.000	0.088 ± 0.002	0.008 ± 0.000	0.089 ± 0.003	0.079 ± 0.004	0.280 ± 0.008	0.321	0.005	1.62	0.0878	0.0880	1.008	95.106 ± 0.704	0.349 ± 0.002	92.234 ± 0.873
QML	beta32	0.316	0.328	0.012 ± 0.003	0.037 ± 0.009	0.008 ± 0.000	0.091 ± 0.002	0.008 ± 0.000	0.091 ± 0.003	0.083 ± 0.004	0.289 ± 0.008	0.327	0.011	3.43	0.0897	0.0903	0.973	94.043 ± 0.772	0.346 ± 0.002	93.936 ± 0.778
QML	beta33	0.139	0.086	-0.053 ± 0.003	-0.384 ± 0.022	0.009 ± 0.000	0.093 ± 0.002	0.011 ± 0.001	0.107 ± 0.003	0.595 ± 0.030	0.771 ± 0.022	0.086	-0.053	-37.84	0.0866	0.1013	0.980	90.000 ± 0.978	0.357 ± 0.004	21.489 ± 1.340
QML	beta34	0.101	0.161	0.060 ± 0.002	0.589 ± 0.018	0.003 ± 0.000	0.056 ± 0.001	0.007 ± 0.000	0.082 ± 0.002	0.656 ± 0.031	0.810 ± 0.020	0.155	0.054	53.52	0.0559	0.0777	0.952	85.426 ± 1.151	0.210 ± 0.003	88.936 ± 1.023
QML	beta35	0.101	0.156	0.055 ± 0.002	0.544 ± 0.016	0.003 ± 0.000	0.051 ± 0.001	0.006 ± 0.000	0.075 ± 0.002	0.549 ± 0.025	0.741 ± 0.018	0.154	0.053	52.31	0.0483	0.0716	1.021	85.213 ± 1.158	0.203 ± 0.003	88.617 ± 1.036
UPI	beta31	0.316	0.319	0.003 ± 0.005	0.010 ± 0.015	0.020 ± 0.001	0.141 ± 0.004	0.020 ± 0.001	0.141 ± 0.004	0.199 ± 0.010	0.446 ± 0.014	0.322	0.006	1.90	0.1422	0.1423	0.968	95.638 ± 0.666	0.535 ± 0.004	67.340 ± 1.530
UPI	beta32	0.316	0.332	0.016 ± 0.005	0.049 ± 0.015	0.020 ± 0.001	0.143 ± 0.004	0.021 ± 0.001	0.143 ± 0.004	0.206 ± 0.011	0.454 ± 0.014	0.328	0.012	3.93	0.1391	0.1396	0.948	93.617 ± 0.797	0.530 ± 0.004	69.362 ± 1.504
UPI	beta33	0.139	0.153	0.014 ± 0.003	0.102 ± 0.024	0.010 ± 0.001	0.101 ± 0.003	0.010 ± 0.001	0.102 ± 0.003	0.537 ± 0.032	0.733 ± 0.025	0.143	0.004	2.88	0.0935	0.0935	1.040	95.213 ± 0.696	0.412 ± 0.008	35.000 ± 1.556
UPI	beta34	0.101	0.103	0.002 ± 0.002	0.021 ± 0.019	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	0.357 ± 0.021	0.598 ± 0.020	0.101	0.000	0.45	0.0552	0.0552	0.885	90.532 ± 0.955	0.209 ± 0.003	55.638 ± 1.620
UPI	beta35	0.101	0.097	-0.004 ± 0.002	-0.035 ± 0.018	0.003 ± 0.000	0.055 ± 0.002	0.003 ± 0.000	0.055 ± 0.002	0.295 ± 0.017	0.543 ± 0.018	0.099	-0.002	-1.52	0.0496	0.0497	0.954	92.979 ± 0.833	0.205 ± 0.003	51.702 ± 1.630

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S46

Simulation 1 - Condition 46: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.6

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LMS	beta31	0.316	0.321	0.005 $\pm$ 0.002	0.016 $\pm$ 0.006	0.003 $\pm$ 0.000	0.057 $\pm$ 0.001	0.003 $\pm$ 0.000	0.058 $\pm$ 0.002	0.033 $\pm$ 0.001	0.182 $\pm$ 0.005	0.322	0.006	1.94	0.0556	0.0559	0.995	94.663 $\pm$ 0.713	0.224 $\pm$ 0.001	99.597 $\pm$ 0.201
LMS	beta32	0.316	0.323	0.007 $\pm$ 0.002	0.022 $\pm$ 0.006	0.003 $\pm$ 0.000	0.058 $\pm$ 0.001	0.003 $\pm$ 0.000	0.058 $\pm$ 0.002	0.034 $\pm$ 0.002	0.184 $\pm$ 0.005	0.325	0.009	2.82	0.0560	0.0567	0.993	93.857 $\pm$ 0.762	0.225 $\pm$ 0.001	99.295 $\pm$ 0.265
LMS	beta33	0.139	0.087	-0.052 $\pm$ 0.002	-0.372 $\pm$ 0.016	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.007 $\pm$ 0.000	0.086 $\pm$ 0.002	0.381 $\pm$ 0.020	0.617 $\pm$ 0.017	0.091	-0.048	-34.31	0.0628	0.0789	0.968	90.232 $\pm$ 0.942	0.260 $\pm$ 0.005	36.757 $\pm$ 1.530
LMS	beta34	0.101	0.161	0.060 $\pm$ 0.001	0.597 $\pm$ 0.012	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.005 $\pm$ 0.000	0.071 $\pm$ 0.001	0.497 $\pm$ 0.019	0.705 $\pm$ 0.014	0.156	0.055	54.14	0.0339	0.0643	0.948	63.948 $\pm$ 1.524	0.141 $\pm$ 0.002	99.597 $\pm$ 0.201
LMS	beta35	0.101	0.162	0.061 $\pm$ 0.001	0.604 $\pm$ 0.012	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.005 $\pm$ 0.000	0.072 $\pm$ 0.001	0.504 $\pm$ 0.019	0.710 $\pm$ 0.014	0.158	0.057	56.30	0.0341	0.0663	0.966	64.552 $\pm$ 1.518	0.143 $\pm$ 0.003	99.597 $\pm$ 0.201
LSAM	beta31	0.316	0.316	-0.000 $\pm$ 0.003	-0.001 $\pm$ 0.008	0.007 $\pm$ 0.000	0.083 $\pm$ 0.002	0.007 $\pm$ 0.000	0.083 $\pm$ 0.002	0.068 $\pm$ 0.003	0.262 $\pm$ 0.008	0.317	0.001	0.27	0.0795	0.0795	0.970	94.159 $\pm$ 0.744	0.315 $\pm$ 0.001	96.073 $\pm$ 0.616
LSAM	beta32	0.316	0.318	0.002 $\pm$ 0.003	0.006 $\pm$ 0.008	0.007 $\pm$ 0.000	0.083 $\pm$ 0.002	0.007 $\pm$ 0.000	0.083 $\pm$ 0.002	0.068 $\pm$ 0.003	0.261 $\pm$ 0.007	0.318	0.002	0.52	0.0804	0.0804	0.972	93.857 $\pm$ 0.762	0.315 $\pm$ 0.001	95.871 $\pm$ 0.631
LSAM	beta33	0.139	0.141	0.002 $\pm$ 0.002	0.014 $\pm$ 0.012	0.003 $\pm$ 0.000	0.055 $\pm$ 0.001	0.003 $\pm$ 0.000	0.055 $\pm$ 0.002	0.154 $\pm$ 0.008	0.393 $\pm$ 0.011	0.138	-0.001	-0.64	0.0507	0.0507	0.935	93.152 $\pm$ 0.801	0.200 $\pm$ 0.001	77.744 $\pm$ 1.320
LSAM	beta34	0.101	0.102	0.001 $\pm$ 0.001	0.006 $\pm$ 0.010	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	0.102 $\pm$ 0.005	0.320 $\pm$ 0.010	0.100	-0.001	-1.41	0.0294	0.0294	0.886	93.454 $\pm$ 0.785	0.112 $\pm$ 0.001	91.843 $\pm$ 0.869
LSAM	beta35	0.101	0.102	0.001 $\pm$ 0.001	0.005 $\pm$ 0.010	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.094 $\pm$ 0.005	0.306 $\pm$ 0.009	0.100	-0.001	-0.76	0.0296	0.0296	0.924	92.346 $\pm$ 0.844	0.112 $\pm$ 0.001	92.548 $\pm$ 0.833
QML	beta31	0.316	0.321	0.005 $\pm$ 0.002	0.014 $\pm$ 0.006	0.003 $\pm$ 0.000	0.056 $\pm$ 0.001	0.003 $\pm$ 0.000	0.056 $\pm$ 0.002	0.031 $\pm$ 0.001	0.177 $\pm$ 0.005	0.321	0.005	1.68	0.0546	0.0549	0.995	94.260 $\pm$ 0.738	0.217 $\pm$ 0.001	99.799 $\pm$ 0.142
QML	beta32	0.316	0.323	0.007 $\pm$ 0.002	0.021 $\pm$ 0.006	0.003 $\pm$ 0.000	0.056 $\pm$ 0.001	0.003 $\pm$ 0.000	0.056 $\pm$ 0.002	0.032 $\pm$ 0.001	0.178 $\pm$ 0.005	0.323	0.007	2.21	0.0548	0.0552	0.991	93.958 $\pm$ 0.756	0.217 $\pm$ 0.001	99.799 $\pm$ 0.142
QML	beta33	0.139	0.079	-0.060 $\pm$ 0.002	-0.432 $\pm$ 0.014	0.004 $\pm$ 0.000	0.060 $\pm$ 0.001	0.007 $\pm$ 0.000	0.085 $\pm$ 0.002	0.374 $\pm$ 0.016	0.611 $\pm$ 0.014	0.080	-0.059	-42.19	0.0569	0.0817	0.936	81.873 $\pm$ 1.223	0.221 $\pm$ 0.002	35.045 $\pm$ 1.514
QML	beta34	0.101	0.158	0.057 $\pm$ 0.001	0.563 $\pm$ 0.011	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.004 $\pm$ 0.000	0.067 $\pm$ 0.001	0.435 $\pm$ 0.016	0.659 $\pm$ 0.012	0.154	0.053	52.38	0.0328	0.0623	0.929	59.517 $\pm$ 1.558	0.126 $\pm$ 0.001	99.899 $\pm$ 0.101
QML	beta35	0.101	0.158	0.057 $\pm$ 0.001	0.566 $\pm$ 0.011	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.004 $\pm$ 0.000	0.066 $\pm$ 0.001	0.431 $\pm$ 0.014	0.657 $\pm$ 0.011	0.156	0.055	54.53	0.0326	0.0640	0.953	59.919 $\pm$ 1.555	0.126 $\pm$ 0.001	100.000 $\pm$ 0.000
UPI	beta31	0.316	0.318	0.002 $\pm$ 0.003	0.006 $\pm$ 0.008	0.007 $\pm$ 0.000	0.083 $\pm$ 0.002	0.007 $\pm$ 0.000	0.083 $\pm$ 0.002	0.068 $\pm$ 0.003	0.261 $\pm$ 0.007	0.319	0.003	0.93	0.0813	0.0814	0.969	94.663 $\pm$ 0.713	0.314 $\pm$ 0.001	95.871 $\pm$ 0.631
UPI	beta32	0.316	0.321	0.005 $\pm$ 0.003	0.016 $\pm$ 0.008	0.007 $\pm$ 0.000	0.082 $\pm$ 0.002	0.007 $\pm$ 0.000	0.082 $\pm$ 0.002	0.068 $\pm$ 0.003	0.261 $\pm$ 0.007	0.322	0.006	1.96	0.0792	0.0795	0.971	93.555 $\pm$ 0.779	0.314 $\pm$ 0.001	96.375 $\pm$ 0.593
UPI	beta33	0.139	0.141	0.002 $\pm$ 0.002	0.013 $\pm$ 0.013	0.003 $\pm$ 0.000	0.057 $\pm$ 0.001	0.003 $\pm$ 0.000	0.057 $\pm$ 0.002	0.169 $\pm$ 0.009	0.412 $\pm$ 0.012	0.137	-0.002	-1.49	0.0539	0.0539	0.967	94.058 $\pm$ 0.750	0.217 $\pm$ 0.002	73.716 $\pm$ 1.397
UPI	beta34	0.101	0.101	0.000 $\pm$ 0.001	0.000 $\pm$ 0.010	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.104 $\pm$ 0.005	0.322 $\pm$ 0.010	0.099	-0.002	-1.83	0.0301	0.0301	0.903	92.346 $\pm$ 0.844	0.115 $\pm$ 0.001	91.339 $\pm$ 0.893
UPI	beta35	0.101	0.100	-0.001 $\pm$ 0.001	-0.005 $\pm$ 0.010	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.096 $\pm$ 0.005	0.310 $\pm$ 0.009	0.099	-0.002	-1.68	0.0303	0.0303	0.928	92.649 $\pm$ 0.828	0.114 $\pm$ 0.001	92.044 $\pm$ 0.859

Note. MCSE = Monte Carlo Standard Error shown as $\backslash u00b1$ value. Coverage, Type I Error, and Power expressed as percentages. $\backslash u2014$ indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S47

Simulation 1 - Condition 47: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.8

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.315	-0.001 ± 0.002	-0.005 ± 0.008	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	0.062 ± 0.003	0.249 ± 0.007	0.315	-0.001	-0.23	0.0782	0.0782	1.005	95.477 ± 0.659	0.310 ± 0.003	95.779 ± 0.637
LMS	beta32	0.316	0.314	-0.002 ± 0.002	-0.005 ± 0.008	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	0.060 ± 0.003	0.245 ± 0.007	0.314	-0.002	-0.59	0.0759	0.0759	1.024	95.075 ± 0.686	0.311 ± 0.003	96.181 ± 0.608
LMS	beta33	0.139	0.115	-0.024 ± 0.002	-0.171 ± 0.015	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.264 ± 0.013	0.514 ± 0.015	0.118	-0.021	-15.43	0.0639	0.0674	0.942	92.261 ± 0.847	0.249 ± 0.002	49.849 ± 1.585
LMS	beta34	0.101	0.125	0.024 ± 0.001	0.237 ± 0.013	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.212 ± 0.011	0.461 ± 0.014	0.121	0.020	20.16	0.0362	0.0416	0.880	89.347 ± 0.978	0.138 ± 0.002	93.467 ± 0.783
LMS	beta35	0.101	0.124	0.023 ± 0.001	0.225 ± 0.012	0.001 ± 0.000	0.038 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.194 ± 0.010	0.441 ± 0.013	0.120	0.019	18.64	0.0361	0.0407	0.919	90.955 ± 0.909	0.138 ± 0.002	92.663 ± 0.827
LSAM	beta31	0.316	0.315	-0.001 ± 0.003	-0.004 ± 0.010	0.010 ± 0.000	0.098 ± 0.002	0.010 ± 0.000	0.098 ± 0.003	0.096 ± 0.004	0.310 ± 0.009	0.314	-0.002	-0.71	0.0977	0.0977	0.984	94.874 ± 0.699	0.378 ± 0.002	87.337 ± 1.054
LSAM	beta32	0.316	0.314	-0.002 ± 0.003	-0.006 ± 0.010	0.009 ± 0.000	0.097 ± 0.002	0.009 ± 0.000	0.097 ± 0.003	0.094 ± 0.004	0.306 ± 0.009	0.313	-0.003	-1.02	0.0926	0.0927	1.000	94.472 ± 0.724	0.379 ± 0.002	88.844 ± 0.998
LSAM	beta33	0.139	0.138	-0.001 ± 0.002	-0.004 ± 0.015	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.223 ± 0.011	0.472 ± 0.014	0.137	-0.002	-1.32	0.0625	0.0625	0.908	92.965 ± 0.811	0.234 ± 0.002	64.523 ± 1.517
LSAM	beta34	0.101	0.103	0.002 ± 0.001	0.023 ± 0.013	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.158 ± 0.008	0.397 ± 0.012	0.102	0.001	0.70	0.0361	0.0361	0.865	91.055 ± 0.905	0.136 ± 0.002	81.307 ± 1.236
LSAM	beta35	0.101	0.102	0.001 ± 0.001	0.013 ± 0.012	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.148 ± 0.008	0.385 ± 0.012	0.100	-0.001	-0.91	0.0353	0.0353	0.895	92.261 ± 0.847	0.137 ± 0.002	81.709 ± 1.226
QML	beta31	0.316	0.315	-0.001 ± 0.002	-0.004 ± 0.008	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	0.062 ± 0.003	0.249 ± 0.007	0.315	-0.001	-0.24	0.0787	0.0787	0.992	95.276 ± 0.673	0.306 ± 0.001	96.281 ± 0.600
QML	beta32	0.316	0.314	-0.002 ± 0.002	-0.006 ± 0.008	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	0.060 ± 0.003	0.245 ± 0.007	0.313	-0.003	-0.95	0.0760	0.0760	1.012	94.673 ± 0.712	0.307 ± 0.001	96.683 ± 0.568
QML	beta33	0.139	0.113	-0.026 ± 0.002	-0.187 ± 0.015	0.004 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.264 ± 0.013	0.514 ± 0.015	0.115	-0.024	-17.13	0.0628	0.0672	0.935	91.558 ± 0.881	0.244 ± 0.002	49.749 ± 1.585
QML	beta34	0.101	0.124	0.023 ± 0.001	0.231 ± 0.012	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.207 ± 0.011	0.455 ± 0.013	0.121	0.020	19.71	0.0366	0.0417	0.872	88.040 ± 1.029	0.135 ± 0.002	93.668 ± 0.772
QML	beta35	0.101	0.123	0.022 ± 0.001	0.220 ± 0.012	0.001 ± 0.000	0.038 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.190 ± 0.010	0.435 ± 0.012	0.120	0.019	18.43	0.0362	0.0407	0.914	90.553 ± 0.927	0.136 ± 0.002	92.462 ± 0.837
UPI	beta31	0.316	0.318	0.002 ± 0.003	0.006 ± 0.010	0.010 ± 0.000	0.098 ± 0.002	0.010 ± 0.000	0.098 ± 0.003	0.096 ± 0.004	0.309 ± 0.009	0.319	0.003	0.80	0.0957	0.0957	0.972	93.568 ± 0.778	0.372 ± 0.002	89.146 ± 0.986
UPI	beta32	0.316	0.316	0.000 ± 0.003	0.001 ± 0.010	0.009 ± 0.000	0.096 ± 0.002	0.009 ± 0.000	0.096 ± 0.003	0.092 ± 0.004	0.303 ± 0.008	0.314	-0.002	-0.59	0.0923	0.0923	0.994	94.271 ± 0.737	0.373 ± 0.002	89.045 ± 0.990
UPI	beta33	0.139	0.139	0.000 ± 0.002	0.003 ± 0.016	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.244 ± 0.013	0.494 ± 0.015	0.137	-0.002	-1.32	0.0644	0.0644	0.910	92.764 ± 0.821	0.245 ± 0.003	62.915 ± 1.531
UPI	beta34	0.101	0.102	0.001 ± 0.001	0.010 ± 0.013	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.162 ± 0.008	0.403 ± 0.012	0.101	0.000	-0.33	0.0377	0.0377	0.862	91.055 ± 0.905	0.137 ± 0.002	79.095 ± 1.289
UPI	beta35	0.101	0.101	0.000 ± 0.001	0.001 ± 0.012	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.144 ± 0.007	0.379 ± 0.011	0.100	-0.001	-0.73	0.0365	0.0365	0.916	91.960 ± 0.862	0.138 ± 0.002	79.799 ± 1.273

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u00b10.014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S48

Simulation 1 - Condition 48: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.8

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.317	0.001 ± 0.002	0.004 ± 0.005	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.025 ± 0.001	0.157 ± 0.004	0.318	0.002	0.50	0.0509	0.0510	0.984	95.200 ± 0.676	0.192 ± 0.001	100.000 ± 0.000
LMS	beta32	0.316	0.316	0.000 ± 0.002	0.001 ± 0.005	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.024 ± 0.001	0.153 ± 0.004	0.316	0.000	-0.14	0.0485	0.0485	1.005	95.600 ± 0.649	0.191 ± 0.001	100.000 ± 0.000
LMS	beta33	0.139	0.124	-0.015 ± 0.001	-0.104 ± 0.009	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.098 ± 0.004	0.313 ± 0.008	0.124	-0.015	-10.52	0.0413	0.0439	0.968	93.100 ± 0.801	0.155 ± 0.001	86.400 ± 1.084
LMS	beta34	0.101	0.122	0.021 ± 0.001	0.212 ± 0.007	0.000 ± 0.000	0.022 ± 0.000	0.001 ± 0.000	0.031 ± 0.001	0.093 ± 0.004	0.305 ± 0.007	0.121	0.020	19.37	0.0218	0.0293	0.973	84.900 ± 1.132	0.084 ± 0.001	99.900 ± 0.100
LMS	beta35	0.101	0.122	0.021 ± 0.001	0.206 ± 0.007	0.001 ± 0.000	0.022 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.092 ± 0.004	0.303 ± 0.008	0.121	0.020	19.36	0.0205	0.0283	0.943	87.000 ± 1.063	0.083 ± 0.001	100.000 ± 0.000
LSAM	beta31	0.316	0.318	0.002 ± 0.002	0.005 ± 0.006	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.037 ± 0.002	0.192 ± 0.005	0.317	0.001	0.30	0.0609	0.0609	0.976	94.900 ± 0.696	0.232 ± 0.001	99.900 ± 0.100
LSAM	beta32	0.316	0.315	-0.001 ± 0.002	-0.003 ± 0.006	0.004 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.035 ± 0.002	0.187 ± 0.005	0.315	-0.001	-0.25	0.0599	0.0599	0.997	95.800 ± 0.634	0.231 ± 0.001	100.000 ± 0.000
LSAM	beta33	0.139	0.143	0.004 ± 0.001	0.028 ± 0.009	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.076 ± 0.003	0.276 ± 0.008	0.142	0.003	2.13	0.0387	0.0388	0.955	94.800 ± 0.702	0.143 ± 0.001	96.100 ± 0.612
LSAM	beta34	0.101	0.101	0.000 ± 0.001	0.001 ± 0.007	0.000 ± 0.000	0.022 ± 0.000	0.000 ± 0.000	0.022 ± 0.001	0.046 ± 0.002	0.214 ± 0.006	0.100	-0.001	-1.18	0.0208	0.0208	0.945	93.100 ± 0.801	0.080 ± 0.001	99.400 ± 0.244
LSAM	beta35	0.101	0.102	0.001 ± 0.001	0.007 ± 0.007	0.000 ± 0.000	0.022 ± 0.001	0.000 ± 0.000	0.022 ± 0.001	0.048 ± 0.002	0.220 ± 0.007	0.101	0.000	-0.11	0.0192	0.0192	0.916	92.300 ± 0.843	0.080 ± 0.001	99.200 ± 0.282
QML	beta31	0.316	0.317	0.001 ± 0.002	0.004 ± 0.005	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.025 ± 0.001	0.158 ± 0.004	0.317	0.001	0.40	0.0504	0.0504	0.983	95.000 ± 0.689	0.192 ± 0.001	100.000 ± 0.000
QML	beta32	0.316	0.316	-0.000 ± 0.002	-0.001 ± 0.005	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.024 ± 0.001	0.154 ± 0.004	0.315	-0.001	-0.33	0.0485	0.0485	1.006	95.500 ± 0.656	0.192 ± 0.001	100.000 ± 0.000
QML	beta33	0.139	0.122	-0.017 ± 0.001	-0.121 ± 0.009	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.100 ± 0.004	0.316 ± 0.008	0.123	-0.016	-11.82	0.0417	0.0448	0.960	92.000 ± 0.858	0.153 ± 0.001	85.600 ± 1.110
QML	beta34	0.101	0.122	0.021 ± 0.001	0.205 ± 0.007	0.000 ± 0.000	0.022 ± 0.000	0.001 ± 0.000	0.030 ± 0.001	0.089 ± 0.004	0.298 ± 0.007	0.120	0.019	18.98	0.0217	0.0290	0.968	85.000 ± 1.129	0.083 ± 0.001	99.900 ± 0.100
QML	beta35	0.101	0.122	0.021 ± 0.001	0.204 ± 0.007	0.001 ± 0.000	0.022 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.091 ± 0.004	0.301 ± 0.008	0.120	0.019	19.15	0.0203	0.0280	0.937	86.300 ± 1.087	0.082 ± 0.001	100.000 ± 0.000
UPI	beta31	0.316	0.319	0.003 ± 0.002	0.010 ± 0.006	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.037 ± 0.002	0.191 ± 0.005	0.318	0.002	0.66	0.0616	0.0617	0.967	94.200 ± 0.739	0.229 ± 0.001	99.900 ± 0.100
UPI	beta32	0.316	0.316	-0.000 ± 0.002	-0.000 ± 0.006	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.035 ± 0.002	0.187 ± 0.005	0.317	0.001	0.25	0.0591	0.0591	0.985	95.300 ± 0.669	0.228 ± 0.001	100.000 ± 0.000
UPI	beta33	0.139	0.143	0.004 ± 0.001	0.030 ± 0.009	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.079 ± 0.004	0.280 ± 0.008	0.142	0.003	2.51	0.0381	0.0383	0.966	94.300 ± 0.733	0.147 ± 0.001	95.900 ± 0.627
UPI	beta34	0.101	0.101	-0.000 ± 0.001	-0.005 ± 0.007	0.000 ± 0.000	0.022 ± 0.000	0.000 ± 0.000	0.022 ± 0.001	0.047 ± 0.002	0.217 ± 0.006	0.100	-0.001	-1.32	0.0206	0.0206	0.945	92.300 ± 0.843	0.081 ± 0.001	99.500 ± 0.223
UPI	beta35	0.101	0.101	0.000 ± 0.001	0.005 ± 0.007	0.001 ± 0.000	0.022 ± 0.001	0.001 ± 0.000	0.022 ± 0.001	0.050 ± 0.003	0.223 ± 0.007	0.100	-0.001	-0.68	0.0196	0.0196	0.918	91.800 ± 0.868	0.081 ± 0.001	99.000 ± 0.315

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S49

Simulation 1 - Condition 49: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.4

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LMS	beta31	0.316	0.317	0.001 $\pm$ 0.005	0.004 $\pm$ 0.015	0.012 $\pm$ 0.001	0.108 $\pm$ 0.004	0.012 $\pm$ 0.001	0.108 $\pm$ 0.005	0.117 $\pm$ 0.009	0.342 $\pm$ 0.016	0.317	0.001	0.26	0.1040	0.1040	1.019	94.610 $\pm$ 0.974	0.432 $\pm$ 0.007	84.758 $\pm$ 1.550
LMS	beta32	0.316	0.315	-0.001 $\pm$ 0.005	-0.003 $\pm$ 0.014	0.011 $\pm$ 0.001	0.106 $\pm$ 0.004	0.011 $\pm$ 0.001	0.106 $\pm$ 0.004	0.112 $\pm$ 0.008	0.334 $\pm$ 0.013	0.314	-0.002	-0.51	0.1149	0.1149	1.029	95.911 $\pm$ 0.854	0.426 $\pm$ 0.006	87.918 $\pm$ 1.405
LMS	beta33	0.139	0.090	-0.049 $\pm$ 0.009	-0.355 $\pm$ 0.066	0.046 $\pm$ 0.005	0.214 $\pm$ 0.012	0.048 $\pm$ 0.006	0.219 $\pm$ 0.014	2.494 $\pm$ 0.291	1.579 $\pm$ 0.098	0.118	-0.021	-15.18	0.1865	0.1877	0.947	96.654 $\pm$ 0.775	0.795 $\pm$ 0.030	22.491 $\pm$ 1.800
LMS	beta34	0.101	0.109	0.008 $\pm$ 0.008	0.076 $\pm$ 0.076	0.032 $\pm$ 0.003	0.178 $\pm$ 0.010	0.032 $\pm$ 0.003	0.178 $\pm$ 0.011	3.091 $\pm$ 0.339	1.758 $\pm$ 0.104	0.079	-0.022	-21.54	0.1345	0.1363	0.852	85.502 $\pm$ 1.518	0.593 $\pm$ 0.024	4.647 $\pm$ 0.908
LMS	beta35	0.101	0.107	0.006 $\pm$ 0.007	0.062 $\pm$ 0.071	0.028 $\pm$ 0.003	0.166 $\pm$ 0.009	0.028 $\pm$ 0.003	0.166 $\pm$ 0.010	2.705 $\pm$ 0.299	1.645 $\pm$ 0.098	0.083	-0.018	-17.98	0.1284	0.1297	0.899	87.546 $\pm$ 1.424	0.585 $\pm$ 0.020	5.019 $\pm$ 0.941
LSAM	beta31	0.316	0.321	0.005 $\pm$ 0.005	0.016 $\pm$ 0.016	0.014 $\pm$ 0.001	0.117 $\pm$ 0.004	0.014 $\pm$ 0.001	0.117 $\pm$ 0.005	0.136 $\pm$ 0.010	0.369 $\pm$ 0.016	0.318	0.002	0.69	0.1167	0.1167	1.300	97.212 $\pm$ 0.710	0.595 $\pm$ 0.025	74.907 $\pm$ 1.869
LSAM	beta32	0.316	0.320	0.004 $\pm$ 0.005	0.011 $\pm$ 0.016	0.013 $\pm$ 0.001	0.114 $\pm$ 0.003	0.013 $\pm$ 0.001	0.114 $\pm$ 0.004	0.129 $\pm$ 0.008	0.359 $\pm$ 0.013	0.328	0.012	3.79	0.1270	0.1276	1.344	97.398 $\pm$ 0.686	0.599 $\pm$ 0.023	72.305 $\pm$ 1.929
LSAM	beta33	0.139	0.164	0.025 $\pm$ 0.008	0.177 $\pm$ 0.056	0.033 $\pm$ 0.003	0.182 $\pm$ 0.008	0.034 $\pm$ 0.003	0.184 $\pm$ 0.009	1.745 $\pm$ 0.157	1.321 $\pm$ 0.066	0.145	0.006	4.10	0.1781	0.1782	1.481	98.327 $\pm$ 0.553	1.057 $\pm$ 0.072	12.639 $\pm$ 1.433
LSAM	beta34	0.101	0.090	-0.011 $\pm$ 0.010	-0.112 $\pm$ 0.098	0.052 $\pm$ 0.005	0.229 $\pm$ 0.010	0.052 $\pm$ 0.005	0.229 $\pm$ 0.011	5.145 $\pm$ 0.454	2.268 $\pm$ 0.112	0.096	-0.005	-5.27	0.2096	0.2096	1.495	98.327 $\pm$ 0.553	1.342 $\pm$ 0.105	1.301 $\pm$ 0.489
LSAM	beta35	0.101	0.105	0.004 $\pm$ 0.010	0.044 $\pm$ 0.102	0.057 $\pm$ 0.005	0.239 $\pm$ 0.010	0.057 $\pm$ 0.005	0.239 $\pm$ 0.011	5.602 $\pm$ 0.473	2.367 $\pm$ 0.113	0.115	0.014	13.58	0.2175	0.2180	1.419	98.885 $\pm$ 0.453	1.330 $\pm$ 0.081	1.487 $\pm$ 0.522
QML	beta31	0.316	0.305	-0.011 $\pm$ 0.004	-0.035 $\pm$ 0.013	0.009 $\pm$ 0.001	0.096 $\pm$ 0.003	0.009 $\pm$ 0.001	0.096 $\pm$ 0.004	0.093 $\pm$ 0.006	0.305 $\pm$ 0.012	0.305	-0.011	-3.60	0.0966	0.0973	0.994	93.494 $\pm$ 1.063	0.373 $\pm$ 0.003	91.636 $\pm$ 1.194
QML	beta32	0.316	0.303	-0.013 $\pm$ 0.004	-0.041 $\pm$ 0.013	0.009 $\pm$ 0.001	0.095 $\pm$ 0.003	0.009 $\pm$ 0.001	0.096 $\pm$ 0.004	0.093 $\pm$ 0.006	0.304 $\pm$ 0.011	0.306	-0.010	-3.10	0.1033	0.1038	1.004	93.866 $\pm$ 1.034	0.375 $\pm$ 0.003	91.822 $\pm$ 1.181
QML	beta33	0.139	0.133	-0.006 $\pm$ 0.005	-0.044 $\pm$ 0.039	0.016 $\pm$ 0.001	0.125 $\pm$ 0.004	0.016 $\pm$ 0.001	0.125 $\pm$ 0.005	0.806 $\pm$ 0.051	0.898 $\pm$ 0.034	0.135	-0.004	-3.04	0.1331	0.1332	1.100	96.283 $\pm$ 0.816	0.538 $\pm$ 0.011	23.234 $\pm$ 1.821
QML	beta34	0.101	0.048	-0.053 $\pm$ 0.004	-0.523 $\pm$ 0.041	0.009 $\pm$ 0.001	0.096 $\pm$ 0.004	0.012 $\pm$ 0.001	0.109 $\pm$ 0.004	1.169 $\pm$ 0.068	1.081 $\pm$ 0.036	0.051	-0.050	-49.47	0.0956	0.1079	1.027	83.086 $\pm$ 1.616	0.385 $\pm$ 0.008	3.717 $\pm$ 0.816
QML	beta35	0.101	0.052	-0.049 $\pm$ 0.004	-0.485 $\pm$ 0.040	0.009 $\pm$ 0.001	0.095 $\pm$ 0.004	0.011 $\pm$ 0.001	0.106 $\pm$ 0.004	1.110 $\pm$ 0.068	1.054 $\pm$ 0.037	0.054	-0.047	-46.95	0.0947	0.1059	1.068	86.803 $\pm$ 1.459	0.396 $\pm$ 0.008	4.461 $\pm$ 0.890
UPI	beta31	0.316	0.320	0.004 $\pm$ 0.005	0.012 $\pm$ 0.016	0.013 $\pm$ 0.001	0.114 $\pm$ 0.004	0.013 $\pm$ 0.001	0.114 $\pm$ 0.005	0.130 $\pm$ 0.009	0.361 $\pm$ 0.015	0.318	0.002	0.54	0.1205	0.1205	1.655	97.398 $\pm$ 0.686	0.740 $\pm$ 0.036	64.312 $\pm$ 2.065
UPI	beta32	0.316	0.321	0.005 $\pm$ 0.005	0.017 $\pm$ 0.016	0.014 $\pm$ 0.001	0.118 $\pm$ 0.004	0.014 $\pm$ 0.001	0.118 $\pm$ 0.005	0.140 $\pm$ 0.010	0.374 $\pm$ 0.015	0.318	0.002	0.66	0.1265	0.1265	1.740	97.955 $\pm$ 0.610	0.806 $\pm$ 0.057	63.011 $\pm$ 2.081
UPI	beta33	0.139	0.184	0.045 $\pm$ 0.008	0.320 $\pm$ 0.061	0.039 $\pm$ 0.003	0.197 $\pm$ 0.009	0.041 $\pm$ 0.004	0.202 $\pm$ 0.010	2.102 $\pm$ 0.193	1.450 $\pm$ 0.073	0.147	0.008	5.90	0.1932	0.1934	1.784	98.885 $\pm$ 0.453	1.376 $\pm$ 0.076	5.204 $\pm$ 0.958
UPI	beta34	0.101	0.096	-0.005 $\pm$ 0.013	-0.052 $\pm$ 0.128	0.089 $\pm$ 0.008	0.299 $\pm$ 0.014	0.089 $\pm$ 0.008	0.299 $\pm$ 0.016	8.758 $\pm$ 0.829	2.959 $\pm$ 0.155	0.078	-0.023	-23.11	0.2340	0.2352	1.988	96.840 $\pm$ 0.754	2.331 $\pm$ 0.154	0.372 $\pm$ 0.262
UPI	beta35	0.101	0.084	-0.017 $\pm$ 0.012	-0.167 $\pm$ 0.121	0.080 $\pm$ 0.008	0.283 $\pm$ 0.014	0.080 $\pm$ 0.008	0.283 $\pm$ 0.015	7.868 $\pm$ 0.763	2.805 $\pm$ 0.150	0.080	-0.021	-20.35	0.2534	0.2543	1.980	96.283 $\pm$ 0.816	2.197 $\pm$ 0.145	0.000 $\pm$ 0.000

Note. MCSE = Monte Carlo Standard Error shown as $\backslash u00b1$ value. Coverage, Type I Error, and Power expressed as percentages. $\backslash u2014$ indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S50

Simulation 1 - Condition 50: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.4

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LMS	beta31	0.316	0.314	-0.002 $\pm$ 0.002	-0.005 $\pm$ 0.007	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.037 $\pm$ 0.002	0.192 $\pm$ 0.006	0.310	-0.006	-1.75	0.0614	0.0616	1.091	97.034 $\pm$ 0.584	0.260 $\pm$ 0.004	99.051 $\pm$ 0.334
LMS	beta32	0.316	0.311	-0.005 $\pm$ 0.002	-0.015 $\pm$ 0.007	0.004 $\pm$ 0.000	0.063 $\pm$ 0.002	0.004 $\pm$ 0.000	0.063 $\pm$ 0.002	0.040 $\pm$ 0.002	0.201 $\pm$ 0.006	0.310	-0.006	-1.82	0.0650	0.0653	1.032	94.187 $\pm$ 0.806	0.256 $\pm$ 0.003	99.288 $\pm$ 0.290
LMS	beta33	0.139	0.143	0.004 $\pm$ 0.003	0.027 $\pm$ 0.023	0.008 $\pm$ 0.001	0.091 $\pm$ 0.003	0.008 $\pm$ 0.001	0.091 $\pm$ 0.003	0.432 $\pm$ 0.028	0.658 $\pm$ 0.024	0.140	0.001	0.40	0.0929	0.0929	1.255	95.967 $\pm$ 0.678	0.450 $\pm$ 0.018	46.619 $\pm$ 1.718
LMS	beta34	0.101	0.066	-0.035 $\pm$ 0.003	-0.348 $\pm$ 0.025	0.005 $\pm$ 0.000	0.074 $\pm$ 0.003	0.007 $\pm$ 0.000	0.082 $\pm$ 0.002	0.656 $\pm$ 0.034	0.810 $\pm$ 0.024	0.063	-0.038	-37.14	0.0710	0.0803	1.262	84.816 $\pm$ 1.236	0.366 $\pm$ 0.017	1.068 $\pm$ 0.354
LMS	beta35	0.101	0.069	-0.032 $\pm$ 0.003	-0.318 $\pm$ 0.026	0.006 $\pm$ 0.000	0.076 $\pm$ 0.003	0.007 $\pm$ 0.000	0.083 $\pm$ 0.002	0.668 $\pm$ 0.034	0.817 $\pm$ 0.024	0.063	-0.038	-37.47	0.0705	0.0801	1.198	83.749 $\pm$ 1.271	0.357 $\pm$ 0.014	2.372 $\pm$ 0.524
LSAM	beta31	0.316	0.320	0.004 $\pm$ 0.002	0.011 $\pm$ 0.007	0.004 $\pm$ 0.000	0.064 $\pm$ 0.002	0.004 $\pm$ 0.000	0.064 $\pm$ 0.002	0.041 $\pm$ 0.002	0.202 $\pm$ 0.006	0.316	0.000	0.10	0.0633	0.0633	1.083	97.628 $\pm$ 0.524	0.271 $\pm$ 0.002	99.051 $\pm$ 0.334
LSAM	beta32	0.316	0.314	-0.002 $\pm$ 0.002	-0.006 $\pm$ 0.007	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.046 $\pm$ 0.002	0.214 $\pm$ 0.006	0.313	-0.003	-1.00	0.0683	0.0684	1.027	96.085 $\pm$ 0.668	0.272 $\pm$ 0.002	99.051 $\pm$ 0.334
LSAM	beta33	0.139	0.150	0.011 $\pm$ 0.003	0.082 $\pm$ 0.023	0.009 $\pm$ 0.001	0.094 $\pm$ 0.003	0.009 $\pm$ 0.001	0.094 $\pm$ 0.003	0.460 $\pm$ 0.028	0.678 $\pm$ 0.024	0.142	0.003	2.42	0.0897	0.0898	1.067	97.034 $\pm$ 0.584	0.392 $\pm$ 0.006	38.553 $\pm$ 1.676
LSAM	beta34	0.101	0.095	-0.006 $\pm$ 0.004	-0.060 $\pm$ 0.039	0.013 $\pm$ 0.001	0.116 $\pm$ 0.004	0.013 $\pm$ 0.001	0.116 $\pm$ 0.004	1.313 $\pm$ 0.090	1.146 $\pm$ 0.044	0.101	0.000	0.22	0.1180	0.1180	1.112	97.628 $\pm$ 0.524	0.504 $\pm$ 0.008	5.457 $\pm$ 0.782
LSAM	beta35	0.101	0.099	-0.002 $\pm$ 0.004	-0.017 $\pm$ 0.041	0.014 $\pm$ 0.001	0.120 $\pm$ 0.004	0.014 $\pm$ 0.001	0.120 $\pm$ 0.004	1.413 $\pm$ 0.090	1.189 $\pm$ 0.043	0.101	0.000	0.25	0.1141	0.1141	1.084	97.983 $\pm$ 0.484	0.511 $\pm$ 0.009	7.473 $\pm$ 0.906
QML	beta31	0.316	0.304	-0.012 $\pm$ 0.002	-0.038 $\pm$ 0.006	0.003 $\pm$ 0.000	0.056 $\pm$ 0.001	0.003 $\pm$ 0.000	0.057 $\pm$ 0.002	0.033 $\pm$ 0.002	0.182 $\pm$ 0.005	0.301	-0.015	-4.85	0.0560	0.0581	1.041	95.967 $\pm$ 0.678	0.229 $\pm$ 0.001	100.000 $\pm$ 0.000
QML	beta32	0.316	0.301	-0.015 $\pm$ 0.002	-0.048 $\pm$ 0.007	0.004 $\pm$ 0.000	0.060 $\pm$ 0.001	0.004 $\pm$ 0.000	0.062 $\pm$ 0.002	0.038 $\pm$ 0.002	0.195 $\pm$ 0.006	0.301	-0.015	-4.85	0.0588	0.0608	0.979	93.120 $\pm$ 0.872	0.230 $\pm$ 0.001	99.881 $\pm$ 0.119
QML	beta33	0.139	0.141	0.002 $\pm$ 0.003	0.013 $\pm$ 0.019	0.006 $\pm$ 0.000	0.077 $\pm$ 0.002	0.006 $\pm$ 0.000	0.077 $\pm$ 0.002	0.307 $\pm$ 0.015	0.554 $\pm$ 0.017	0.135	-0.004	-3.14	0.0778	0.0779	1.061	96.085 $\pm$ 0.668	0.320 $\pm$ 0.004	45.907 $\pm$ 1.716
QML	beta34	0.101	0.038	-0.063 $\pm$ 0.002	-0.620 $\pm$ 0.018	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.007 $\pm$ 0.000	0.082 $\pm$ 0.002	0.662 $\pm$ 0.025	0.814 $\pm$ 0.016	0.041	-0.060	-59.61	0.0560	0.0822	1.136	76.275 $\pm$ 1.465	0.237 $\pm$ 0.005	4.626 $\pm$ 0.723
QML	beta35	0.101	0.041	-0.060 $\pm$ 0.002	-0.596 $\pm$ 0.019	0.003 $\pm$ 0.000	0.056 $\pm$ 0.002	0.007 $\pm$ 0.000	0.082 $\pm$ 0.002	0.660 $\pm$ 0.026	0.812 $\pm$ 0.017	0.041	-0.060	-59.36	0.0567	0.0825	1.081	76.987 $\pm$ 1.450	0.236 $\pm$ 0.004	6.643 $\pm$ 0.858
UPI	beta31	0.316	0.320	0.004 $\pm$ 0.002	0.013 $\pm$ 0.007	0.004 $\pm$ 0.000	0.067 $\pm$ 0.002	0.004 $\pm$ 0.000	0.067 $\pm$ 0.002	0.045 $\pm$ 0.002	0.212 $\pm$ 0.007	0.317	0.001	0.23	0.0686	0.0686	1.169	97.983 $\pm$ 0.484	0.307 $\pm$ 0.007	97.153 $\pm$ 0.573
UPI	beta32	0.316	0.315	-0.001 $\pm$ 0.002	-0.004 $\pm$ 0.008	0.005 $\pm$ 0.000	0.071 $\pm$ 0.002	0.005 $\pm$ 0.000	0.071 $\pm$ 0.002	0.051 $\pm$ 0.003	0.226 $\pm$ 0.007	0.316	0.000	-0.13	0.0708	0.0708	1.110	96.204 $\pm$ 0.658	0.311 $\pm$ 0.008	97.153 $\pm$ 0.573
UPI	beta33	0.139	0.160	0.021 $\pm$ 0.004	0.151 $\pm$ 0.026	0.011 $\pm$ 0.001	0.105 $\pm$ 0.003	0.011 $\pm$ 0.001	0.107 $\pm$ 0.004	0.594 $\pm$ 0.041	0.771 $\pm$ 0.029	0.149	0.010	6.96	0.0967	0.0972	1.189	98.339 $\pm$ 0.440	0.490 $\pm$ 0.016	30.012 $\pm$ 1.579
UPI	beta34	0.101	0.095	-0.006 $\pm$ 0.005	-0.063 $\pm$ 0.050	0.021 $\pm$ 0.001	0.145 $\pm$ 0.005	0.021 $\pm$ 0.001	0.145 $\pm$ 0.006	2.072 $\pm$ 0.141	1.439 $\pm$ 0.055	0.096	-0.005	-5.20	0.1308	0.1309	1.345	97.509 $\pm$ 0.537	0.766 $\pm$ 0.027	0.712 $\pm$ 0.290
UPI	beta35	0.101	0.101	0.000 $\pm$ 0.005	0.002 $\pm$ 0.053	0.024 $\pm$ 0.002	0.154 $\pm$ 0.005	0.024 $\pm$ 0.002	0.154 $\pm$ 0.006	2.322 $\pm$ 0.153	1.524 $\pm$ 0.057	0.096	-0.005	-5.14	0.1306	0.1307	1.305	97.272 $\pm$ 0.561	0.788 $\pm$ 0.033	2.254 $\pm$ 0.511

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S51

Simulation 1 - Condition 51: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.6

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.315	-0.001 ± 0.002	-0.004 ± 0.007	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.051 ± 0.002	0.226 ± 0.006	0.313	-0.003	-0.99	0.0710	0.0710	1.043	95.135 ± 0.692	0.292 ± 0.007	98.861 ± 0.341
LMS	beta32	0.316	0.313	-0.003 ± 0.002	-0.009 ± 0.007	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.050 ± 0.003	0.225 ± 0.007	0.311	-0.005	-1.62	0.0684	0.0686	1.055	95.342 ± 0.678	0.294 ± 0.008	98.551 ± 0.385
LMS	beta33	0.139	0.139	0.000 ± 0.003	0.003 ± 0.022	0.009 ± 0.000	0.094 ± 0.003	0.009 ± 0.000	0.094 ± 0.003	0.459 ± 0.025	0.677 ± 0.022	0.147	0.008	6.05	0.0870	0.0874	1.074	95.549 ± 0.664	0.397 ± 0.012	43.064 ± 1.593
LMS	beta34	0.101	0.078	-0.023 ± 0.003	-0.231 ± 0.027	0.007 ± 0.000	0.086 ± 0.002	0.008 ± 0.000	0.089 ± 0.003	0.773 ± 0.038	0.879 ± 0.025	0.070	-0.031	-30.41	0.0794	0.0852	1.029	86.025 ± 1.116	0.346 ± 0.007	7.143 ± 0.829
LMS	beta35	0.101	0.079	-0.022 ± 0.003	-0.217 ± 0.027	0.007 ± 0.000	0.085 ± 0.003	0.008 ± 0.000	0.088 ± 0.003	0.757 ± 0.039	0.870 ± 0.026	0.070	-0.031	-30.66	0.0725	0.0788	1.134	88.716 ± 1.018	0.378 ± 0.028	5.072 ± 0.706
LSAM	beta31	0.316	0.319	0.003 ± 0.002	0.009 ± 0.008	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	0.056 ± 0.003	0.236 ± 0.007	0.320	0.004	1.16	0.0758	0.0759	1.003	95.859 ± 0.641	0.293 ± 0.001	98.551 ± 0.385
LSAM	beta32	0.316	0.317	0.001 ± 0.002	0.005 ± 0.007	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.052 ± 0.002	0.228 ± 0.007	0.318	0.002	0.61	0.0715	0.0715	1.035	95.756 ± 0.649	0.292 ± 0.001	98.551 ± 0.385
LSAM	beta33	0.139	0.147	0.008 ± 0.003	0.055 ± 0.021	0.008 ± 0.000	0.089 ± 0.002	0.008 ± 0.000	0.089 ± 0.003	0.413 ± 0.020	0.643 ± 0.019	0.146	0.007	5.24	0.0856	0.0859	1.000	95.549 ± 0.664	0.349 ± 0.003	42.857 ± 1.592
LSAM	beta34	0.101	0.099	-0.002 ± 0.004	-0.021 ± 0.035	0.012 ± 0.001	0.111 ± 0.003	0.012 ± 0.001	0.111 ± 0.003	1.213 ± 0.065	1.101 ± 0.035	0.100	-0.001	-0.75	0.1019	0.1019	0.993	95.031 ± 0.699	0.433 ± 0.004	12.629 ± 1.069
LSAM	beta35	0.101	0.100	-0.001 ± 0.003	-0.007 ± 0.035	0.012 ± 0.001	0.108 ± 0.003	0.012 ± 0.001	0.108 ± 0.003	1.150 ± 0.062	1.072 ± 0.034	0.101	0.000	-0.35	0.1060	0.1060	1.024	95.756 ± 0.649	0.435 ± 0.004	10.870 ± 1.001
QML	beta31	0.316	0.310	-0.006 ± 0.002	-0.019 ± 0.007	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.049 ± 0.002	0.221 ± 0.006	0.309	-0.007	-2.16	0.0685	0.0688	1.002	94.928 ± 0.706	0.274 ± 0.001	99.379 ± 0.253
QML	beta32	0.316	0.309	-0.007 ± 0.002	-0.024 ± 0.007	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.048 ± 0.002	0.218 ± 0.006	0.307	-0.009	-2.98	0.0672	0.0679	1.015	95.031 ± 0.699	0.273 ± 0.001	99.172 ± 0.292
QML	beta33	0.139	0.141	0.002 ± 0.003	0.017 ± 0.020	0.007 ± 0.000	0.086 ± 0.002	0.007 ± 0.000	0.086 ± 0.002	0.380 ± 0.018	0.617 ± 0.018	0.145	0.006	4.01	0.0834	0.0836	1.008	95.031 ± 0.699	0.339 ± 0.003	44.617 ± 1.599
QML	beta34	0.101	0.061	-0.040 ± 0.002	-0.395 ± 0.023	0.005 ± 0.000	0.073 ± 0.002	0.007 ± 0.000	0.083 ± 0.002	0.676 ± 0.028	0.822 ± 0.020	0.060	-0.041	-40.56	0.0737	0.0843	0.992	84.576 ± 1.162	0.283 ± 0.003	9.213 ± 0.931
QML	beta35	0.101	0.062	-0.039 ± 0.002	-0.390 ± 0.022	0.005 ± 0.000	0.070 ± 0.002	0.006 ± 0.000	0.080 ± 0.002	0.631 ± 0.027	0.794 ± 0.020	0.060	-0.041	-40.29	0.0652	0.0769	1.038	86.232 ± 1.109	0.285 ± 0.003	7.350 ± 0.840
UPI	beta31	0.316	0.319	0.003 ± 0.002	0.009 ± 0.008	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	0.057 ± 0.003	0.239 ± 0.007	0.319	0.003	1.06	0.0730	0.0731	0.996	95.445 ± 0.671	0.294 ± 0.002	98.551 ± 0.385
UPI	beta32	0.316	0.318	0.002 ± 0.002	0.006 ± 0.007	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.054 ± 0.003	0.231 ± 0.007	0.316	0.000	0.09	0.0715	0.0715	1.022	94.824 ± 0.713	0.293 ± 0.002	98.137 ± 0.435
UPI	beta33	0.139	0.151	0.012 ± 0.003	0.086 ± 0.022	0.009 ± 0.000	0.094 ± 0.003	0.009 ± 0.000	0.095 ± 0.003	0.469 ± 0.026	0.685 ± 0.022	0.148	0.009	6.59	0.0889	0.0893	1.018	95.963 ± 0.633	0.377 ± 0.003	39.027 ± 1.570
UPI	beta34	0.101	0.103	0.002 ± 0.004	0.018 ± 0.041	0.017 ± 0.001	0.130 ± 0.004	0.017 ± 0.001	0.130 ± 0.004	1.646 ± 0.095	1.283 ± 0.042	0.098	-0.003	-2.93	0.1161	0.1162	1.071	96.170 ± 0.618	0.544 ± 0.009	7.557 ± 0.850
UPI	beta35	0.101	0.099	-0.002 ± 0.004	-0.015 ± 0.039	0.015 ± 0.001	0.124 ± 0.003	0.015 ± 0.001	0.124 ± 0.004	1.500 ± 0.084	1.225 ± 0.040	0.098	-0.003	-3.15	0.1092	0.1093	1.099	96.377 ± 0.601	0.533 ± 0.008	5.901 ± 0.758

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S52

Simulation 1 - Condition 52: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.6

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.312	-0.004 ± 0.001	-0.012 ± 0.004	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.020 ± 0.001	0.140 ± 0.004	0.312	-0.004	-1.23	0.0426	0.0428	1.011	95.727 ± 0.645	0.175 ± 0.001	100.000 ± 0.000
LMS	beta32	0.316	0.313	-0.003 ± 0.001	-0.008 ± 0.004	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.019 ± 0.001	0.138 ± 0.004	0.313	-0.003	-1.04	0.0450	0.0451	1.026	95.931 ± 0.630	0.175 ± 0.001	100.000 ± 0.000
LMS	beta33	0.139	0.141	0.002 ± 0.002	0.013 ± 0.013	0.003 ± 0.000	0.057 ± 0.002	0.003 ± 0.000	0.057 ± 0.002	0.170 ± 0.009	0.412 ± 0.013	0.143	0.004	2.76	0.0547	0.0549	1.007	95.626 ± 0.652	0.226 ± 0.003	71.821 ± 1.435
LMS	beta34	0.101	0.071	-0.030 ± 0.002	-0.294 ± 0.016	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.058 ± 0.001	0.335 ± 0.013	0.579 ± 0.013	0.067	-0.034	-34.13	0.0466	0.0580	1.027	85.046 ± 1.137	0.203 ± 0.003	19.023 ± 1.252
LMS	beta35	0.101	0.074	-0.027 ± 0.002	-0.270 ± 0.016	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.058 ± 0.001	0.335 ± 0.012	0.579 ± 0.013	0.068	-0.033	-32.80	0.0476	0.0580	1.007	85.351 ± 1.128	0.204 ± 0.003	20.753 ± 1.293
LSAM	beta31	0.316	0.316	-0.000 ± 0.001	-0.000 ± 0.005	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.020 ± 0.001	0.142 ± 0.004	0.315	-0.001	-0.17	0.0439	0.0439	1.016	95.422 ± 0.667	0.179 ± 0.000	100.000 ± 0.000
LSAM	beta32	0.316	0.317	0.001 ± 0.001	0.002 ± 0.004	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.019 ± 0.001	0.138 ± 0.004	0.318	0.002	0.78	0.0455	0.0456	1.044	96.948 ± 0.549	0.179 ± 0.000	100.000 ± 0.000
LSAM	beta33	0.139	0.139	0.000 ± 0.002	0.002 ± 0.012	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.148 ± 0.007	0.385 ± 0.010	0.139	0.000	-0.05	0.0531	0.0531	1.002	95.219 ± 0.681	0.210 ± 0.001	72.431 ± 1.425
LSAM	beta34	0.101	0.099	-0.002 ± 0.002	-0.022 ± 0.021	0.004 ± 0.000	0.066 ± 0.001	0.004 ± 0.000	0.066 ± 0.002	0.429 ± 0.019	0.655 ± 0.018	0.097	-0.004	-3.89	0.0639	0.0640	1.003	95.219 ± 0.681	0.260 ± 0.001	29.908 ± 1.460
LSAM	beta35	0.101	0.102	0.001 ± 0.002	0.006 ± 0.021	0.004 ± 0.000	0.067 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	0.437 ± 0.021	0.661 ± 0.019	0.098	-0.003	-2.72	0.0664	0.0665	0.994	95.320 ± 0.674	0.260 ± 0.001	31.434 ± 1.481
QML	beta31	0.316	0.308	-0.008 ± 0.001	-0.026 ± 0.004	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.019 ± 0.001	0.139 ± 0.004	0.307	-0.009	-2.71	0.0416	0.0425	1.008	95.320 ± 0.674	0.171 ± 0.000	100.000 ± 0.000
QML	beta32	0.316	0.309	-0.007 ± 0.001	-0.023 ± 0.004	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.019 ± 0.001	0.136 ± 0.004	0.309	-0.007	-2.26	0.0434	0.0440	1.025	94.914 ± 0.701	0.171 ± 0.001	99.898 ± 0.102
QML	beta33	0.139	0.138	-0.001 ± 0.002	-0.008 ± 0.012	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.149 ± 0.007	0.386 ± 0.011	0.139	0.000	0.08	0.0520	0.0520	1.000	95.320 ± 0.674	0.210 ± 0.001	73.449 ± 1.409
QML	beta34	0.101	0.059	-0.042 ± 0.001	-0.413 ± 0.014	0.002 ± 0.000	0.044 ± 0.001	0.004 ± 0.000	0.060 ± 0.001	0.358 ± 0.013	0.598 ± 0.012	0.057	-0.044	-43.17	0.0423	0.0608	1.021	79.654 ± 1.284	0.175 ± 0.001	21.160 ± 1.303
QML	beta35	0.101	0.061	-0.040 ± 0.001	-0.394 ± 0.014	0.002 ± 0.000	0.045 ± 0.001	0.004 ± 0.000	0.060 ± 0.001	0.351 ± 0.012	0.592 ± 0.012	0.059	-0.042	-42.06	0.0439	0.0611	1.001	79.858 ± 1.279	0.175 ± 0.001	22.279 ± 1.327
UPI	beta31	0.316	0.316	-0.000 ± 0.001	-0.000 ± 0.005	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.020 ± 0.001	0.142 ± 0.004	0.316	0.000	0.04	0.0436	0.0436	0.998	94.710 ± 0.714	0.176 ± 0.000	100.000 ± 0.000
UPI	beta32	0.316	0.317	0.001 ± 0.001	0.002 ± 0.004	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.019 ± 0.001	0.139 ± 0.004	0.318	0.002	0.48	0.0460	0.0460	1.019	96.033 ± 0.623	0.176 ± 0.000	100.000 ± 0.000
UPI	beta33	0.139	0.141	0.002 ± 0.002	0.014 ± 0.013	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.158 ± 0.007	0.397 ± 0.011	0.139	0.000	0.27	0.0559	0.0559	1.002	95.829 ± 0.638	0.217 ± 0.001	72.431 ± 1.425
UPI	beta34	0.101	0.099	-0.002 ± 0.002	-0.017 ± 0.022	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.478 ± 0.024	0.691 ± 0.020	0.099	-0.002	-2.45	0.0673	0.0674	1.033	95.422 ± 0.667	0.283 ± 0.002	25.025 ± 1.382
UPI	beta35	0.101	0.102	0.001 ± 0.002	0.010 ± 0.022	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.475 ± 0.024	0.689 ± 0.020	0.099	-0.002	-2.28	0.0683	0.0684	1.031	96.134 ± 0.615	0.281 ± 0.002	26.144 ± 1.402

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S53

Simulation 1 - Condition 53: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.8

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.315	-0.001 ± 0.002	-0.004 ± 0.006	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.035 ± 0.002	0.188 ± 0.005	0.314	-0.002	-0.71	0.0593	0.0593	0.958	94.595 ± 0.715	0.224 ± 0.001	99.900 ± 0.100
LMS	beta32	0.316	0.314	-0.002 ± 0.002	-0.005 ± 0.006	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.035 ± 0.002	0.187 ± 0.006	0.313	-0.003	-0.90	0.0605	0.0606	0.971	95.495 ± 0.656	0.225 ± 0.001	99.700 ± 0.173
LMS	beta33	0.139	0.141	0.002 ± 0.002	0.015 ± 0.015	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.237 ± 0.011	0.487 ± 0.014	0.142	0.003	2.51	0.0596	0.0597	0.969	94.895 ± 0.696	0.257 ± 0.002	61.662 ± 1.538
LMS	beta34	0.101	0.084	-0.017 ± 0.002	-0.171 ± 0.020	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.432 ± 0.019	0.657 ± 0.017	0.080	-0.021	-20.75	0.0651	0.0684	0.977	90.991 ± 0.906	0.246 ± 0.002	21.722 ± 1.305
LMS	beta35	0.101	0.087	-0.014 ± 0.002	-0.137 ± 0.020	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.415 ± 0.019	0.644 ± 0.018	0.083	-0.018	-17.36	0.0610	0.0635	0.988	91.692 ± 0.873	0.246 ± 0.002	21.822 ± 1.307
LSAM	beta31	0.316	0.317	0.001 ± 0.002	0.005 ± 0.006	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.037 ± 0.002	0.191 ± 0.005	0.316	0.000	0.14	0.0589	0.0589	0.957	94.394 ± 0.728	0.227 ± 0.001	100.000 ± 0.000
LSAM	beta32	0.316	0.317	0.001 ± 0.002	0.003 ± 0.006	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	0.035 ± 0.002	0.188 ± 0.005	0.318	0.002	0.67	0.0614	0.0614	0.977	95.796 ± 0.635	0.228 ± 0.001	99.800 ± 0.141
LSAM	beta33	0.139	0.142	0.003 ± 0.002	0.019 ± 0.015	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	0.218 ± 0.010	0.467 ± 0.013	0.143	0.004	3.07	0.0604	0.0606	0.967	94.094 ± 0.746	0.246 ± 0.001	63.263 ± 1.525
LSAM	beta34	0.101	0.099	-0.002 ± 0.002	-0.024 ± 0.023	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.529 ± 0.024	0.727 ± 0.020	0.097	-0.004	-3.64	0.0751	0.0752	0.975	94.695 ± 0.709	0.281 ± 0.001	27.227 ± 1.408
LSAM	beta35	0.101	0.102	0.001 ± 0.002	0.015 ± 0.023	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.511 ± 0.024	0.715 ± 0.020	0.102	0.001	0.59	0.0714	0.0714	0.986	95.495 ± 0.656	0.279 ± 0.001	28.328 ± 1.426
QML	beta31	0.316	0.314	-0.002 ± 0.002	-0.008 ± 0.006	0.004 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	0.035 ± 0.002	0.188 ± 0.005	0.313	-0.003	-1.08	0.0588	0.0589	0.957	94.394 ± 0.728	0.223 ± 0.001	100.000 ± 0.000
QML	beta32	0.316	0.313	-0.003 ± 0.002	-0.009 ± 0.006	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.035 ± 0.002	0.186 ± 0.006	0.312	-0.004	-1.35	0.0596	0.0597	0.970	95.295 ± 0.670	0.224 ± 0.001	99.700 ± 0.173
QML	beta33	0.139	0.141	0.002 ± 0.002	0.011 ± 0.015	0.004 ± 0.000	0.067 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	0.230 ± 0.011	0.479 ± 0.014	0.142	0.003	1.84	0.0599	0.0600	0.967	94.094 ± 0.746	0.252 ± 0.002	62.062 ± 1.535
QML	beta34	0.101	0.080	-0.021 ± 0.002	-0.210 ± 0.019	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	0.417 ± 0.017	0.646 ± 0.016	0.078	-0.023	-23.26	0.0638	0.0680	0.977	90.490 ± 0.928	0.236 ± 0.002	22.422 ± 1.320
QML	beta35	0.101	0.083	-0.018 ± 0.002	-0.178 ± 0.019	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.395 ± 0.017	0.629 ± 0.016	0.080	-0.021	-20.53	0.0591	0.0627	0.989	91.191 ± 0.897	0.236 ± 0.002	23.223 ± 1.336
UPI	beta31	0.316	0.317	0.001 ± 0.002	0.004 ± 0.006	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.037 ± 0.002	0.192 ± 0.005	0.316	0.000	-0.01	0.0588	0.0588	0.918	93.293 ± 0.791	0.218 ± 0.001	100.000 ± 0.000
UPI	beta32	0.316	0.317	0.001 ± 0.002	0.004 ± 0.006	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.036 ± 0.002	0.189 ± 0.006	0.318	0.002	0.55	0.0607	0.0608	0.938	95.095 ± 0.683	0.219 ± 0.001	99.600 ± 0.200
UPI	beta33	0.139	0.142	0.003 ± 0.002	0.024 ± 0.015	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.226 ± 0.010	0.476 ± 0.013	0.143	0.004	2.94	0.0607	0.0609	0.962	93.393 ± 0.786	0.249 ± 0.001	64.665 ± 1.512
UPI	beta34	0.101	0.099	-0.002 ± 0.002	-0.016 ± 0.024	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	0.564 ± 0.026	0.751 ± 0.021	0.099	-0.002	-1.75	0.0785	0.0785	0.981	94.695 ± 0.709	0.292 ± 0.002	25.926 ± 1.386
UPI	beta35	0.101	0.103	0.002 ± 0.002	0.024 ± 0.024	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	0.552 ± 0.026	0.743 ± 0.021	0.102	0.001	0.74	0.0746	0.0746	0.985	94.895 ± 0.696	0.290 ± 0.002	26.627 ± 1.398

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S54

Simulation 1 - Condition 54: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.8

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.314	-0.002 ± 0.001	-0.006 ± 0.004	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.013 ± 0.001	0.115 ± 0.003	0.314	-0.002	-0.79	0.0361	0.0361	1.025	94.995 ± 0.690	0.146 ± 0.004	99.900 ± 0.100
LMS	beta32	0.316	0.312	-0.004 ± 0.001	-0.012 ± 0.004	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.014 ± 0.001	0.118 ± 0.003	0.311	-0.005	-1.61	0.0366	0.0369	1.005	93.794 ± 0.763	0.146 ± 0.005	99.900 ± 0.100
LMS	beta33	0.139	0.142	0.003 ± 0.001	0.022 ± 0.009	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.090 ± 0.004	0.301 ± 0.009	0.143	0.004	2.65	0.0416	0.0418	0.985	94.394 ± 0.728	0.161 ± 0.001	90.290 ± 0.937
LMS	beta34	0.101	0.084	-0.017 ± 0.001	-0.169 ± 0.012	0.001 ± 0.000	0.038 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.171 ± 0.007	0.414 ± 0.010	0.082	-0.019	-18.56	0.0372	0.0417	1.031	90.791 ± 0.915	0.154 ± 0.001	58.458 ± 1.559
LMS	beta35	0.101	0.083	-0.018 ± 0.001	-0.174 ± 0.012	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.181 ± 0.007	0.425 ± 0.010	0.083	-0.018	-17.58	0.0400	0.0438	1.002	90.891 ± 0.910	0.154 ± 0.001	55.856 ± 1.571
LSAM	beta31	0.316	0.317	0.001 ± 0.001	0.002 ± 0.004	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.013 ± 0.001	0.116 ± 0.003	0.317	0.001	0.31	0.0363	0.0364	1.000	94.795 ± 0.703	0.143 ± 0.000	100.000 ± 0.000
LSAM	beta32	0.316	0.315	-0.001 ± 0.001	-0.003 ± 0.004	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.014 ± 0.001	0.119 ± 0.003	0.314	-0.002	-0.69	0.0371	0.0372	0.971	93.594 ± 0.775	0.143 ± 0.000	100.000 ± 0.000
LSAM	beta33	0.139	0.140	0.001 ± 0.001	0.007 ± 0.009	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.083 ± 0.004	0.289 ± 0.008	0.140	0.001	0.41	0.0397	0.0397	0.989	94.494 ± 0.722	0.156 ± 0.001	92.593 ± 0.829
LSAM	beta34	0.101	0.102	0.001 ± 0.001	0.009 ± 0.014	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.192 ± 0.009	0.438 ± 0.012	0.103	0.002	1.95	0.0441	0.0442	1.019	95.796 ± 0.635	0.177 ± 0.001	62.963 ± 1.528
LSAM	beta35	0.101	0.101	0.000 ± 0.001	0.003 ± 0.014	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.202 ± 0.009	0.449 ± 0.012	0.102	0.001	0.96	0.0465	0.0465	0.996	95.095 ± 0.683	0.177 ± 0.001	60.961 ± 1.543
QML	beta31	0.316	0.313	-0.003 ± 0.001	-0.010 ± 0.004	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.013 ± 0.001	0.115 ± 0.003	0.312	-0.004	-1.19	0.0360	0.0362	0.999	95.395 ± 0.663	0.142 ± 0.001	100.000 ± 0.000
QML	beta32	0.316	0.311	-0.005 ± 0.001	-0.016 ± 0.004	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.014 ± 0.001	0.118 ± 0.003	0.310	-0.006	-2.03	0.0365	0.0371	0.974	93.694 ± 0.769	0.141 ± 0.000	100.000 ± 0.000
QML	beta33	0.139	0.141	0.002 ± 0.001	0.012 ± 0.009	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.088 ± 0.004	0.297 ± 0.008	0.141	0.002	1.58	0.0413	0.0414	0.985	94.394 ± 0.728	0.159 ± 0.001	90.390 ± 0.932
QML	beta34	0.101	0.081	-0.020 ± 0.001	-0.202 ± 0.012	0.001 ± 0.000	0.037 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.174 ± 0.007	0.418 ± 0.010	0.079	-0.022	-21.46	0.0364	0.0424	1.032	89.790 ± 0.958	0.149 ± 0.001	56.857 ± 1.567
QML	beta35	0.101	0.080	-0.021 ± 0.001	-0.207 ± 0.012	0.001 ± 0.000	0.038 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.184 ± 0.007	0.429 ± 0.010	0.080	-0.021	-20.86	0.0396	0.0448	0.999	88.989 ± 0.990	0.149 ± 0.001	54.755 ± 1.575
UPI	beta31	0.316	0.316	0.000 ± 0.001	0.001 ± 0.004	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.013 ± 0.001	0.116 ± 0.003	0.317	0.001	0.25	0.0365	0.0365	0.964	94.294 ± 0.734	0.138 ± 0.000	100.000 ± 0.000
UPI	beta32	0.316	0.315	-0.001 ± 0.001	-0.003 ± 0.004	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.014 ± 0.001	0.119 ± 0.003	0.314	-0.002	-0.54	0.0375	0.0376	0.936	92.492 ± 0.834	0.138 ± 0.000	100.000 ± 0.000
UPI	beta33	0.139	0.140	0.001 ± 0.001	0.009 ± 0.009	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.085 ± 0.004	0.291 ± 0.008	0.139	0.000	0.18	0.0395	0.0395	0.990	94.494 ± 0.722	0.157 ± 0.001	92.292 ± 0.844
UPI	beta34	0.101	0.102	0.001 ± 0.001	0.011 ± 0.014	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.198 ± 0.009	0.445 ± 0.013	0.103	0.002	1.53	0.0458	0.0458	1.023	95.896 ± 0.628	0.180 ± 0.001	62.062 ± 1.535
UPI	beta35	0.101	0.101	0.000 ± 0.001	0.005 ± 0.014	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.209 ± 0.009	0.457 ± 0.012	0.101	0.000	0.24	0.0470	0.0470	0.996	95.195 ± 0.677	0.180 ± 0.001	60.260 ± 1.548

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S55

Simulation 1 - Condition 55: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.4

Method	Param.	True	Mean-based										Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE	
LMS	beta31	0.316	0.319	0.003 ± 0.010	0.009 ± 0.030	0.018 ± 0.003	0.133 ± 0.012	0.018 ± 0.003	0.133 ± 0.013	0.177 ± 0.031	0.421 ± 0.041	0.310	-0.006	-1.89	0.1324	0.1325	1.305	96.954 ± 1.224	0.683 ± 0.108	75.635 ± 3.059	
LMS	beta32	0.316	0.324	0.008 ± 0.010	0.026 ± 0.030	0.018 ± 0.003	0.135 ± 0.010	0.018 ± 0.003	0.135 ± 0.011	0.182 ± 0.027	0.427 ± 0.036	0.318	0.002	0.48	0.1370	0.1370	1.097	95.939 ± 1.406	0.580 ± 0.043	74.112 ± 3.121	
LMS	beta33	0.139	0.118	-0.021 ± 0.026	-0.148 ± 0.188	0.135 ± 0.026	0.367 ± 0.036	0.134 ± 0.027	0.366 ± 0.039	6.951 ± 1.375	2.636 ± 0.282	0.187	0.048	34.29	0.2390	0.2437	1.081	91.878 ± 1.946	1.554 ± 0.191	35.025 ± 3.399	
LMS	beta34	0.101	0.142	0.041 ± 0.018	0.407 ± 0.180	0.065 ± 0.012	0.256 ± 0.024	0.067 ± 0.013	0.258 ± 0.028	6.541 ± 1.296	2.558 ± 0.273	0.082	-0.019	-18.52	0.1622	0.1633	1.194	83.756 ± 2.628	1.197 ± 0.204	5.584 ± 1.636	
LMS	beta35	0.101	0.105	0.004 ± 0.015	0.041 ± 0.147	0.043 ± 0.008	0.208 ± 0.020	0.043 ± 0.008	0.208 ± 0.022	4.238 ± 0.808	2.059 ± 0.213	0.067	-0.034	-33.69	0.1465	0.1504	1.223	85.787 ± 2.488	0.999 ± 0.149	3.046 ± 1.224	
LSAM	beta31	0.316	0.327	0.011 ± 0.010	0.036 ± 0.032	0.020 ± 0.002	0.141 ± 0.009	0.020 ± 0.003	0.141 ± 0.010	0.200 ± 0.026	0.447 ± 0.033	0.327	0.011	3.58	0.1437	0.1441	1.132	97.462 ± 1.121	0.627 ± 0.021	59.898 ± 3.492	
LSAM	beta32	0.316	0.328	0.012 ± 0.010	0.039 ± 0.030	0.018 ± 0.002	0.135 ± 0.007	0.018 ± 0.002	0.135 ± 0.008	0.184 ± 0.019	0.429 ± 0.027	0.316	0.000	-0.13	0.1426	0.1426	1.179	97.970 ± 1.005	0.625 ± 0.019	62.437 ± 3.450	
LSAM	beta33	0.139	0.279	0.140 ± 0.013	1.005 ± 0.093	0.033 ± 0.004	0.181 ± 0.011	0.052 ± 0.006	0.228 ± 0.014	2.696 ± 0.308	1.642 ± 0.101	0.284	0.145	104.14	0.1731	0.2256	1.103	96.954 ± 1.224	0.782 ± 0.036	35.533 ± 3.410	
LSAM	beta34	0.101	0.001	-0.100 ± 0.009	-0.992 ± 0.084	0.014 ± 0.002	0.120 ± 0.007	0.024 ± 0.003	0.156 ± 0.009	2.381 ± 0.263	1.543 ± 0.092	-0.008	-0.109	-108.31	0.1110	0.1558	1.078	84.264 ± 2.594	0.506 ± 0.021	1.523 ± 0.872	
LSAM	beta35	0.101	0.001	-0.100 ± 0.008	-0.988 ± 0.083	0.014 ± 0.002	0.118 ± 0.007	0.024 ± 0.002	0.154 ± 0.008	2.331 ± 0.209	1.527 ± 0.076	0.003	-0.098	-97.22	0.1104	0.1477	1.107	81.726 ± 2.753	0.511 ± 0.027	3.553 ± 1.319	
QML	beta31	0.316	0.292	-0.024 ± 0.006	-0.076 ± 0.020	0.008 ± 0.001	0.088 ± 0.005	0.008 ± 0.001	0.091 ± 0.006	0.083 ± 0.009	0.288 ± 0.019	0.294	-0.022	-7.04	0.0916	0.0943	1.103	94.924 ± 1.564	0.380 ± 0.005	90.355 ± 2.103	
QML	beta32	0.316	0.296	-0.020 ± 0.007	-0.064 ± 0.021	0.009 ± 0.001	0.094 ± 0.005	0.009 ± 0.001	0.096 ± 0.006	0.093 ± 0.010	0.304 ± 0.020	0.299	-0.017	-5.50	0.0957	0.0973	1.021	91.878 ± 1.946	0.377 ± 0.005	86.802 ± 2.411	
QML	beta33	0.139	0.178	0.039 ± 0.010	0.280 ± 0.071	0.019 ± 0.002	0.138 ± 0.008	0.020 ± 0.002	0.143 ± 0.009	1.060 ± 0.110	1.030 ± 0.064	0.201	0.062	44.76	0.1350	0.1487	0.993	94.416 ± 1.636	0.537 ± 0.014	38.579 ± 3.468	
QML	beta34	0.101	0.059	-0.042 ± 0.006	-0.413 ± 0.064	0.008 ± 0.001	0.091 ± 0.005	0.010 ± 0.001	0.100 ± 0.005	0.978 ± 0.089	0.989 ± 0.054	0.048	-0.053	-52.87	0.1038	0.1167	1.007	83.249 ± 2.661	0.359 ± 0.010	6.599 ± 1.769	
QML	beta35	0.101	0.056	-0.045 ± 0.007	-0.446 ± 0.065	0.009 ± 0.001	0.093 ± 0.007	0.011 ± 0.001	0.103 ± 0.006	1.037 ± 0.098	1.018 ± 0.057	0.052	-0.049	-48.97	0.0924	0.1048	0.935	81.218 ± 2.783	0.340 ± 0.011	7.614 ± 1.890	
UPI	beta31	0.316	0.319	0.003 ± 0.016	0.009 ± 0.051	0.052 ± 0.008	0.228 ± 0.017	0.052 ± 0.008	0.227 ± 0.019	0.517 ± 0.077	0.719 ± 0.060	0.329	0.013	4.11	0.2114	0.2118	1.802	92.893 ± 1.831	1.608 ± 0.212	37.056 ± 3.441	
UPI	beta32	0.316	0.327	0.011 ± 0.016	0.034 ± 0.050	0.050 ± 0.008	0.223 ± 0.019	0.050 ± 0.008	0.223 ± 0.021	0.498 ± 0.083	0.706 ± 0.065	0.324	0.008	2.45	0.2242	0.2243	1.543	92.386 ± 1.890	1.351 ± 0.149	36.548 ± 3.431	
UPI	beta33	0.139	0.252	0.113 ± 0.017	0.814 ± 0.119	0.054 ± 0.007	0.232 ± 0.016	0.066 ± 0.010	0.257 ± 0.020	3.431 ± 0.496	1.852 ± 0.146	0.237	0.098	70.85	0.2145	0.2360	1.588	98.985 ± 0.714	1.444 ± 0.137	10.152 ± 2.152	
UPI	beta34	0.101	-0.043	-0.144 ± 0.020	-1.425 ± 0.197	0.078 ± 0.012	0.280 ± 0.022	0.099 ± 0.014	0.314 ± 0.024	9.663 ± 1.374	3.109 ± 0.240	-0.013	-0.114	-112.80	0.2092	0.2382	1.676	88.325 ± 2.288	1.838 ± 0.237	0.508 ± 0.506	
UPI	beta35	0.101	-0.010	-0.111 ± 0.019	-1.102 ± 0.185	0.069 ± 0.011	0.263 ± 0.022	0.081 ± 0.012	0.285 ± 0.023	7.941 ± 1.155	2.818 ± 0.225	-0.007	-0.108	-106.76	0.2327	0.2565	1.466	79.695 ± 2.866	1.509 ± 0.192	2.030 ± 1.005	

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S56

Simulation 1 - Condition 56: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.4

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.310	-0.006 ± 0.005	-0.020 ± 0.016	0.005 ± 0.001	0.072 ± 0.006	0.005 ± 0.001	0.073 ± 0.006	0.053 ± 0.008	0.230 ± 0.019	0.311	-0.005	-1.67	0.0824	0.0825	1.140	95.283 ± 1.456	0.324 ± 0.053	98.113 ± 0.934
LMS	beta32	0.316	0.313	-0.003 ± 0.005	-0.010 ± 0.015	0.005 ± 0.001	0.071 ± 0.004	0.005 ± 0.001	0.071 ± 0.005	0.050 ± 0.006	0.223 ± 0.015	0.302	-0.014	-4.41	0.0767	0.0779	1.366	96.226 ± 1.309	0.378 ± 0.108	98.113 ± 0.934
LMS	beta33	0.139	0.223	0.084 ± 0.007	0.602 ± 0.047	0.009 ± 0.002	0.095 ± 0.008	0.016 ± 0.002	0.127 ± 0.009	0.829 ± 0.107	0.910 ± 0.062	0.211	0.072	51.46	0.1237	0.1429	1.606	79.717 ± 2.762	0.599 ± 0.126	66.509 ± 3.241
LMS	beta34	0.101	0.073	-0.028 ± 0.005	-0.281 ± 0.048	0.005 ± 0.001	0.071 ± 0.005	0.006 ± 0.001	0.076 ± 0.005	0.565 ± 0.066	0.752 ± 0.050	0.079	-0.022	-21.47	0.0980	0.1003	1.505	83.962 ± 2.520	0.417 ± 0.082	1.415 ± 0.811
LMS	beta35	0.101	0.064	-0.037 ± 0.004	-0.367 ± 0.044	0.004 ± 0.001	0.065 ± 0.007	0.006 ± 0.001	0.075 ± 0.007	0.548 ± 0.089	0.740 ± 0.065	0.059	-0.042	-41.39	0.0785	0.0890	1.615	83.962 ± 2.520	0.412 ± 0.113	3.302 ± 1.227
LSAM	beta31	0.316	0.316	0.000 ± 0.007	0.001 ± 0.021	0.009 ± 0.001	0.095 ± 0.005	0.009 ± 0.001	0.095 ± 0.006	0.091 ± 0.009	0.301 ± 0.018	0.320	0.004	1.30	0.0908	0.0909	0.944	92.925 ± 1.761	0.353 ± 0.005	93.868 ± 1.648
LSAM	beta32	0.316	0.332	0.016 ± 0.007	0.050 ± 0.022	0.010 ± 0.001	0.100 ± 0.006	0.010 ± 0.001	0.102 ± 0.008	0.103 ± 0.014	0.321 ± 0.024	0.329	0.013	3.97	0.0971	0.0979	0.921	93.396 ± 1.706	0.363 ± 0.005	96.226 ± 1.309
LSAM	beta33	0.139	0.268	0.129 ± 0.007	0.929 ± 0.048	0.009 ± 0.001	0.097 ± 0.005	0.026 ± 0.002	0.161 ± 0.007	1.347 ± 0.112	1.161 ± 0.051	0.265	0.126	90.33	0.1015	0.1615	1.030	80.660 ± 2.713	0.391 ± 0.008	81.604 ± 2.661
LSAM	beta34	0.101	0.012	-0.089 ± 0.004	-0.885 ± 0.044	0.004 ± 0.000	0.064 ± 0.003	0.012 ± 0.001	0.110 ± 0.004	1.184 ± 0.086	1.088 ± 0.042	0.008	-0.093	-92.26	0.0599	0.1108	1.011	67.925 ± 3.206	0.254 ± 0.005	5.189 ± 1.523
LSAM	beta35	0.101	-0.001	-0.102 ± 0.004	-1.013 ± 0.041	0.004 ± 0.000	0.061 ± 0.003	0.014 ± 0.001	0.119 ± 0.004	1.387 ± 0.101	1.178 ± 0.045	-0.001	-0.102	-101.07	0.0557	0.1163	1.062	62.736 ± 3.321	0.253 ± 0.005	3.774 ± 1.309
QML	beta31	0.316	0.291	-0.025 ± 0.004	-0.080 ± 0.014	0.004 ± 0.000	0.064 ± 0.003	0.005 ± 0.000	0.069 ± 0.004	0.047 ± 0.005	0.218 ± 0.013	0.298	-0.018	-5.72	0.0656	0.0680	0.917	91.981 ± 1.865	0.231 ± 0.002	100.000 ± 0.000
QML	beta32	0.316	0.295	-0.021 ± 0.005	-0.065 ± 0.014	0.004 ± 0.000	0.066 ± 0.004	0.005 ± 0.001	0.069 ± 0.004	0.047 ± 0.005	0.218 ± 0.014	0.292	-0.024	-7.66	0.0672	0.0715	0.899	93.868 ± 1.648	0.232 ± 0.002	100.000 ± 0.000
QML	beta33	0.139	0.195	0.056 ± 0.006	0.405 ± 0.040	0.007 ± 0.001	0.081 ± 0.004	0.010 ± 0.001	0.098 ± 0.005	0.499 ± 0.049	0.706 ± 0.038	0.198	0.059	42.21	0.0979	0.1141	1.063	86.792 ± 2.325	0.336 ± 0.008	64.151 ± 3.294
QML	beta34	0.101	0.052	-0.049 ± 0.004	-0.483 ± 0.036	0.003 ± 0.000	0.053 ± 0.003	0.005 ± 0.000	0.072 ± 0.003	0.511 ± 0.045	0.715 ± 0.034	0.052	-0.049	-48.67	0.0583	0.0763	1.054	79.717 ± 2.762	0.220 ± 0.006	8.019 ± 1.865
QML	beta35	0.101	0.046	-0.055 ± 0.003	-0.547 ± 0.030	0.002 ± 0.000	0.045 ± 0.002	0.005 ± 0.000	0.071 ± 0.003	0.493 ± 0.037	0.702 ± 0.028	0.038	-0.063	-62.04	0.0481	0.0790	1.218	76.887 ± 2.895	0.213 ± 0.005	8.962 ± 1.962
UPI	beta31	0.316	0.323	0.007 ± 0.015	0.024 ± 0.049	0.050 ± 0.006	0.223 ± 0.013	0.050 ± 0.006	0.223 ± 0.015	0.497 ± 0.058	0.705 ± 0.048	0.339	0.023	7.16	0.2300	0.2311	1.706	89.623 ± 2.095	1.492 ± 0.191	47.170 ± 3.429
UPI	beta32	0.316	0.298	-0.018 ± 0.019	-0.058 ± 0.059	0.074 ± 0.013	0.272 ± 0.025	0.074 ± 0.013	0.272 ± 0.026	0.738 ± 0.130	0.859 ± 0.083	0.324	0.008	2.62	0.2853	0.2854	1.473	82.547 ± 2.607	1.568 ± 0.227	42.925 ± 3.399
UPI	beta33	0.139	0.246	0.107 ± 0.014	0.773 ± 0.099	0.040 ± 0.006	0.201 ± 0.015	0.052 ± 0.008	0.228 ± 0.019	2.682 ± 0.422	1.638 ± 0.138	0.219	0.080	57.79	0.1664	0.1848	1.661	98.113 ± 0.934	1.310 ± 0.153	21.226 ± 2.808
UPI	beta34	0.101	-0.093	-0.194 ± 0.017	-1.919 ± 0.165	0.059 ± 0.009	0.242 ± 0.018	0.096 ± 0.013	0.310 ± 0.022	9.409 ± 1.272	3.067 ± 0.220	-0.070	-0.171	-168.92	0.2409	0.2952	1.538	85.849 ± 2.394	1.461 ± 0.148	2.358 ± 1.042
UPI	beta35	0.101	-0.087	-0.188 ± 0.020	-1.862 ± 0.202	0.089 ± 0.015	0.298 ± 0.026	0.124 ± 0.017	0.352 ± 0.027	12.115 ± 1.705	3.481 ± 0.263	-0.078	-0.179	-177.39	0.2617	0.3171	1.680	83.491 ± 2.550	1.960 ± 0.292	3.302 ± 1.227

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S57

Simulation 1 - Condition 57: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.6

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.309	-0.007 ± 0.002	-0.022 ± 0.007	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.051 ± 0.003	0.227 ± 0.007	0.308	-0.008	-2.59	0.0719	0.0724	1.031	95.954 ± 0.635	0.288 ± 0.002	97.925 ± 0.459
LMS	beta32	0.316	0.308	-0.008 ± 0.002	-0.026 ± 0.008	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	0.056 ± 0.003	0.237 ± 0.007	0.306	-0.010	-3.14	0.0713	0.0720	0.991	95.228 ± 0.687	0.290 ± 0.002	97.095 ± 0.541
LMS	beta33	0.139	0.157	0.018 ± 0.003	0.131 ± 0.023	0.010 ± 0.001	0.099 ± 0.003	0.010 ± 0.001	0.100 ± 0.003	0.521 ± 0.027	0.722 ± 0.022	0.163	0.024	16.92	0.0938	0.0967	1.038	93.257 ± 0.808	0.401 ± 0.007	47.822 ± 1.609
LMS	beta34	0.101	0.090	-0.011 ± 0.002	-0.111 ± 0.022	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.487 ± 0.025	0.698 ± 0.021	0.083	-0.018	-18.02	0.0656	0.0681	1.020	89.938 ± 0.969	0.278 ± 0.006	15.353 ± 1.161
LMS	beta35	0.101	0.092	-0.009 ± 0.002	-0.092 ± 0.022	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.484 ± 0.026	0.696 ± 0.022	0.084	-0.017	-17.19	0.0639	0.0662	1.023	91.286 ± 0.908	0.280 ± 0.005	16.390 ± 1.192
LSAM	beta31	0.316	0.314	-0.002 ± 0.002	-0.007 ± 0.008	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	0.057 ± 0.003	0.239 ± 0.007	0.311	-0.005	-1.68	0.0746	0.0748	1.026	96.577 ± 0.586	0.304 ± 0.001	97.718 ± 0.481
LSAM	beta32	0.316	0.313	-0.003 ± 0.003	-0.009 ± 0.008	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	0.063 ± 0.003	0.250 ± 0.007	0.309	-0.007	-2.28	0.0736	0.0739	0.987	94.606 ± 0.728	0.306 ± 0.001	97.822 ± 0.470
LSAM	beta33	0.139	0.185	0.046 ± 0.003	0.327 ± 0.021	0.008 ± 0.000	0.092 ± 0.002	0.011 ± 0.001	0.103 ± 0.003	0.545 ± 0.026	0.738 ± 0.020	0.180	0.041	29.82	0.0918	0.1007	1.001	92.531 ± 0.847	0.361 ± 0.003	55.913 ± 1.599
LSAM	beta34	0.101	0.066	-0.035 ± 0.002	-0.348 ± 0.020	0.004 ± 0.000	0.064 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.520 ± 0.022	0.721 ± 0.018	0.064	-0.037	-36.45	0.0666	0.0761	1.001	87.656 ± 1.059	0.250 ± 0.002	16.183 ± 1.186
LSAM	beta35	0.101	0.068	-0.033 ± 0.002	-0.324 ± 0.021	0.004 ± 0.000	0.066 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.529 ± 0.024	0.727 ± 0.019	0.066	-0.035	-34.65	0.0623	0.0715	0.974	88.485 ± 1.028	0.251 ± 0.002	18.361 ± 1.247
QML	beta31	0.316	0.304	-0.012 ± 0.002	-0.039 ± 0.007	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.050 ± 0.002	0.225 ± 0.006	0.303	-0.013	-4.20	0.0699	0.0712	1.023	95.539 ± 0.665	0.280 ± 0.001	98.340 ± 0.411
QML	beta32	0.316	0.302	-0.014 ± 0.002	-0.043 ± 0.007	0.005 ± 0.000	0.073 ± 0.002	0.006 ± 0.000	0.074 ± 0.002	0.055 ± 0.002	0.235 ± 0.006	0.302	-0.014	-4.58	0.0689	0.0704	0.990	94.606 ± 0.728	0.283 ± 0.002	97.822 ± 0.470
QML	beta33	0.139	0.154	0.015 ± 0.003	0.111 ± 0.021	0.008 ± 0.000	0.090 ± 0.002	0.008 ± 0.000	0.091 ± 0.003	0.433 ± 0.021	0.658 ± 0.019	0.157	0.018	13.00	0.0904	0.0922	1.027	93.983 ± 0.766	0.363 ± 0.011	47.718 ± 1.609
QML	beta34	0.101	0.080	-0.021 ± 0.002	-0.204 ± 0.020	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.412 ± 0.018	0.642 ± 0.017	0.077	-0.024	-24.19	0.0616	0.0663	1.024	88.278 ± 1.036	0.247 ± 0.007	19.710 ± 1.281
QML	beta35	0.101	0.082	-0.019 ± 0.002	-0.189 ± 0.020	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	0.407 ± 0.018	0.638 ± 0.017	0.076	-0.025	-24.27	0.0584	0.0634	1.032	89.627 ± 0.982	0.249 ± 0.009	21.266 ± 1.318
UPI	beta31	0.316	0.324	0.008 ± 0.003	0.025 ± 0.008	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	0.062 ± 0.003	0.248 ± 0.008	0.321	0.005	1.64	0.0760	0.0762	1.032	96.784 ± 0.568	0.316 ± 0.003	96.992 ± 0.550
UPI	beta32	0.316	0.321	0.005 ± 0.003	0.017 ± 0.008	0.006 ± 0.000	0.080 ± 0.002	0.006 ± 0.000	0.080 ± 0.002	0.065 ± 0.003	0.255 ± 0.007	0.319	0.003	0.84	0.0766	0.0766	0.999	95.228 ± 0.687	0.315 ± 0.002	97.718 ± 0.481
UPI	beta33	0.139	0.176	0.037 ± 0.003	0.267 ± 0.024	0.010 ± 0.001	0.102 ± 0.003	0.012 ± 0.001	0.108 ± 0.003	0.606 ± 0.032	0.778 ± 0.024	0.165	0.026	18.90	0.0934	0.0970	0.979	94.917 ± 0.707	0.390 ± 0.004	46.162 ± 1.606
UPI	beta34	0.101	0.081	-0.020 ± 0.003	-0.201 ± 0.025	0.006 ± 0.000	0.079 ± 0.002	0.007 ± 0.000	0.082 ± 0.003	0.652 ± 0.038	0.807 ± 0.027	0.080	-0.021	-20.50	0.0753	0.0781	1.002	92.946 ± 0.825	0.310 ± 0.006	19.502 ± 1.276
UPI	beta35	0.101	0.083	-0.018 ± 0.002	-0.179 ± 0.024	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	0.595 ± 0.031	0.771 ± 0.023	0.080	-0.021	-20.99	0.0673	0.0705	0.996	94.087 ± 0.760	0.296 ± 0.004	21.473 ± 1.323

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S58

Simulation 1 - Condition 58: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.6

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.310	-0.006 ± 0.001	-0.019 ± 0.005	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.021 ± 0.001	0.144 ± 0.004	0.311	-0.005	-1.66	0.0432	0.0435	1.054	94.864 ± 0.700	0.187 ± 0.005	99.799 ± 0.142
LMS	beta32	0.316	0.310	-0.006 ± 0.001	-0.020 ± 0.005	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.021 ± 0.001	0.144 ± 0.004	0.309	-0.007	-2.23	0.0430	0.0436	1.036	95.871 ± 0.631	0.183 ± 0.002	99.799 ± 0.142
LMS	beta33	0.139	0.165	0.026 ± 0.002	0.188 ± 0.013	0.003 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.213 ± 0.009	0.461 ± 0.012	0.166	0.027	19.74	0.0577	0.0639	1.034	90.534 ± 0.929	0.237 ± 0.002	76.636 ± 1.343
LMS	beta34	0.101	0.088	-0.013 ± 0.001	-0.126 ± 0.013	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.193 ± 0.009	0.439 ± 0.012	0.085	-0.016	-15.84	0.0403	0.0434	1.001	91.037 ± 0.906	0.167 ± 0.002	56.395 ± 1.574
LMS	beta35	0.101	0.088	-0.013 ± 0.001	-0.132 ± 0.013	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.189 ± 0.008	0.435 ± 0.011	0.084	-0.017	-16.79	0.0397	0.0432	1.019	89.930 ± 0.955	0.167 ± 0.002	55.186 ± 1.578
LSAM	beta31	0.316	0.313	-0.003 ± 0.001	-0.009 ± 0.005	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.022 ± 0.001	0.148 ± 0.004	0.312	-0.004	-1.17	0.0463	0.0464	1.037	95.871 ± 0.631	0.190 ± 0.001	100.000 ± 0.000
LSAM	beta32	0.316	0.313	-0.003 ± 0.001	-0.009 ± 0.005	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.022 ± 0.001	0.150 ± 0.004	0.313	-0.003	-1.09	0.0451	0.0452	1.031	95.972 ± 0.624	0.191 ± 0.001	100.000 ± 0.000
LSAM	beta33	0.139	0.179	0.040 ± 0.002	0.291 ± 0.013	0.003 ± 0.000	0.055 ± 0.001	0.005 ± 0.000	0.068 ± 0.002	0.241 ± 0.011	0.491 ± 0.012	0.177	0.038	27.65	0.0547	0.0669	1.013	88.218 ± 1.023	0.218 ± 0.001	88.721 ± 1.004
LSAM	beta34	0.101	0.069	-0.032 ± 0.001	-0.313 ± 0.012	0.002 ± 0.000	0.039 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.247 ± 0.010	0.497 ± 0.011	0.069	-0.032	-31.61	0.0390	0.0504	1.004	85.901 ± 1.104	0.154 ± 0.001	41.994 ± 1.566
LSAM	beta35	0.101	0.069	-0.032 ± 0.001	-0.319 ± 0.012	0.002 ± 0.000	0.039 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.253 ± 0.011	0.503 ± 0.012	0.069	-0.032	-32.09	0.0389	0.0506	1.005	84.693 ± 1.143	0.155 ± 0.001	40.383 ± 1.557
QML	beta31	0.316	0.304	-0.012 ± 0.001	-0.038 ± 0.004	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.021 ± 0.001	0.145 ± 0.004	0.305	-0.011	-3.57	0.0432	0.0447	1.023	94.159 ± 0.744	0.178 ± 0.001	99.899 ± 0.101
QML	beta32	0.316	0.303	-0.013 ± 0.001	-0.040 ± 0.004	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.021 ± 0.001	0.146 ± 0.004	0.303	-0.013	-4.14	0.0432	0.0452	1.018	95.166 ± 0.681	0.178 ± 0.001	99.899 ± 0.101
QML	beta33	0.139	0.157	0.018 ± 0.002	0.127 ± 0.013	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.175 ± 0.008	0.418 ± 0.011	0.157	0.018	13.08	0.0551	0.0580	1.021	93.353 ± 0.790	0.222 ± 0.001	76.234 ± 1.351
QML	beta34	0.101	0.082	-0.019 ± 0.001	-0.191 ± 0.012	0.001 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.182 ± 0.008	0.427 ± 0.010	0.080	-0.021	-21.19	0.0385	0.0441	1.006	89.023 ± 0.992	0.152 ± 0.001	57.200 ± 1.570
QML	beta35	0.101	0.081	-0.020 ± 0.001	-0.196 ± 0.012	0.001 ± 0.000	0.038 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.182 ± 0.007	0.426 ± 0.010	0.079	-0.022	-21.71	0.0376	0.0435	1.017	87.915 ± 1.034	0.153 ± 0.001	55.992 ± 1.575
UPI	beta31	0.316	0.317	0.001 ± 0.001	0.003 ± 0.005	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.022 ± 0.001	0.149 ± 0.004	0.316	0.000	0.12	0.0456	0.0456	1.024	95.569 ± 0.653	0.189 ± 0.001	100.000 ± 0.000
UPI	beta32	0.316	0.317	0.001 ± 0.002	0.003 ± 0.005	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.023 ± 0.001	0.151 ± 0.004	0.316	0.000	-0.07	0.0455	0.0455	1.015	95.871 ± 0.631	0.190 ± 0.001	100.000 ± 0.000
UPI	beta33	0.139	0.160	0.021 ± 0.002	0.152 ± 0.013	0.003 ± 0.000	0.057 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.191 ± 0.009	0.437 ± 0.012	0.160	0.021	15.02	0.0571	0.0608	1.012	94.562 ± 0.720	0.226 ± 0.001	79.557 ± 1.280
UPI	beta34	0.101	0.084	-0.017 ± 0.001	-0.167 ± 0.014	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.212 ± 0.010	0.461 ± 0.012	0.084	-0.017	-16.46	0.0415	0.0447	1.019	92.246 ± 0.849	0.173 ± 0.001	47.835 ± 1.585
UPI	beta35	0.101	0.084	-0.017 ± 0.001	-0.173 ± 0.014	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.217 ± 0.010	0.466 ± 0.013	0.083	-0.018	-17.77	0.0430	0.0466	1.006	93.656 ± 0.774	0.173 ± 0.001	48.238 ± 1.586

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S59

Simulation 1 - Condition 59: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.8

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.308	-0.008 ± 0.002	-0.025 ± 0.006	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.034 ± 0.002	0.183 ± 0.005	0.310	-0.006	-1.92	0.0599	0.0602	1.019	95.000 ± 0.689	0.229 ± 0.001	99.600 ± 0.200
LMS	beta32	0.316	0.310	-0.006 ± 0.002	-0.018 ± 0.006	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.036 ± 0.002	0.191 ± 0.005	0.311	-0.005	-1.59	0.0599	0.0602	0.975	95.100 ± 0.683	0.230 ± 0.001	99.500 ± 0.223
LMS	beta33	0.139	0.138	-0.001 ± 0.002	-0.004 ± 0.016	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.266 ± 0.013	0.516 ± 0.015	0.144	0.005	3.70	0.0691	0.0693	0.931	93.200 ± 0.796	0.262 ± 0.002	59.200 ± 1.554
LMS	beta34	0.101	0.098	-0.003 ± 0.002	-0.028 ± 0.016	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.242 ± 0.011	0.492 ± 0.014	0.095	-0.006	-5.71	0.0484	0.0487	0.931	92.200 ± 0.848	0.181 ± 0.002	56.700 ± 1.567
LMS	beta35	0.101	0.099	-0.002 ± 0.002	-0.023 ± 0.016	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.245 ± 0.012	0.495 ± 0.015	0.095	-0.006	-5.57	0.0464	0.0467	0.939	92.800 ± 0.817	0.184 ± 0.002	56.000 ± 1.570
LSAM	beta31	0.316	0.311	-0.005 ± 0.002	-0.014 ± 0.006	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.034 ± 0.002	0.185 ± 0.005	0.312	-0.004	-1.21	0.0593	0.0594	1.014	95.100 ± 0.683	0.232 ± 0.001	99.700 ± 0.173
LSAM	beta32	0.316	0.314	-0.002 ± 0.002	-0.008 ± 0.006	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.036 ± 0.002	0.191 ± 0.005	0.313	-0.003	-0.94	0.0600	0.0601	0.983	95.400 ± 0.662	0.232 ± 0.001	99.600 ± 0.200
LSAM	beta33	0.139	0.148	0.009 ± 0.002	0.063 ± 0.016	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.255 ± 0.012	0.505 ± 0.014	0.151	0.012	8.60	0.0688	0.0699	0.915	92.400 ± 0.838	0.250 ± 0.002	64.500 ± 1.513
LSAM	beta34	0.101	0.093	-0.008 ± 0.002	-0.079 ± 0.015	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.237 ± 0.010	0.487 ± 0.013	0.090	-0.011	-10.55	0.0497	0.0508	0.922	91.800 ± 0.868	0.176 ± 0.001	54.300 ± 1.575
LSAM	beta35	0.101	0.093	-0.008 ± 0.002	-0.075 ± 0.015	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.234 ± 0.011	0.484 ± 0.014	0.092	-0.009	-8.53	0.0463	0.0471	0.940	92.600 ± 0.828	0.178 ± 0.001	54.800 ± 1.574
QML	beta31	0.316	0.307	-0.009 ± 0.002	-0.029 ± 0.006	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.034 ± 0.002	0.183 ± 0.005	0.309	-0.007	-2.31	0.0595	0.0599	1.018	94.700 ± 0.708	0.229 ± 0.001	99.600 ± 0.200
QML	beta32	0.316	0.309	-0.007 ± 0.002	-0.022 ± 0.006	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.036 ± 0.002	0.191 ± 0.005	0.310	-0.006	-1.95	0.0601	0.0604	0.976	94.700 ± 0.708	0.229 ± 0.001	99.500 ± 0.223
QML	beta33	0.139	0.137	-0.002 ± 0.002	-0.012 ± 0.016	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.258 ± 0.012	0.508 ± 0.015	0.142	0.003	2.47	0.0689	0.0690	0.929	92.900 ± 0.812	0.257 ± 0.002	59.300 ± 1.554
QML	beta34	0.101	0.097	-0.004 ± 0.002	-0.044 ± 0.015	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.234 ± 0.011	0.483 ± 0.013	0.094	-0.007	-6.65	0.0478	0.0483	0.930	91.600 ± 0.877	0.177 ± 0.001	57.100 ± 1.565
QML	beta35	0.101	0.097	-0.004 ± 0.002	-0.039 ± 0.015	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.236 ± 0.012	0.486 ± 0.014	0.094	-0.007	-6.62	0.0459	0.0464	0.937	92.500 ± 0.833	0.180 ± 0.002	56.300 ± 1.569
UPI	beta31	0.316	0.316	0.000 ± 0.002	0.001 ± 0.006	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.034 ± 0.002	0.184 ± 0.005	0.319	0.003	0.99	0.0602	0.0603	0.979	94.800 ± 0.702	0.223 ± 0.001	99.800 ± 0.141
UPI	beta32	0.316	0.318	0.002 ± 0.002	0.007 ± 0.006	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.036 ± 0.002	0.191 ± 0.005	0.318	0.002	0.54	0.0616	0.0616	0.947	93.900 ± 0.757	0.224 ± 0.001	99.700 ± 0.173
UPI	beta33	0.139	0.146	0.007 ± 0.002	0.050 ± 0.016	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.253 ± 0.012	0.503 ± 0.014	0.149	0.010	7.08	0.0667	0.0674	0.923	92.600 ± 0.828	0.252 ± 0.002	62.800 ± 1.528
UPI	beta34	0.101	0.097	-0.004 ± 0.002	-0.041 ± 0.016	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.249 ± 0.011	0.499 ± 0.014	0.093	-0.008	-7.78	0.0491	0.0498	0.910	92.300 ± 0.843	0.179 ± 0.001	57.400 ± 1.564
UPI	beta35	0.101	0.097	-0.004 ± 0.002	-0.035 ± 0.015	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.241 ± 0.012	0.491 ± 0.015	0.096	-0.005	-5.04	0.0461	0.0464	0.941	93.100 ± 0.801	0.182 ± 0.001	55.400 ± 1.572

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S60

Simulation 1 - Condition 60: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.8

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LMS	beta31	0.316	0.312	-0.004 $\pm$ 0.001	-0.014 $\pm$ 0.004	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.014 $\pm$ 0.001	0.118 $\pm$ 0.003	0.312	-0.004	-1.32	0.0374	0.0377	1.002	95.200 $\pm$ 0.676	0.145 $\pm$ 0.000	100.000 $\pm$ 0.000
LMS	beta32	0.316	0.312	-0.004 $\pm$ 0.001	-0.013 $\pm$ 0.004	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.014 $\pm$ 0.001	0.120 $\pm$ 0.003	0.311	-0.005	-1.43	0.0373	0.0376	0.981	94.500 $\pm$ 0.721	0.145 $\pm$ 0.000	100.000 $\pm$ 0.000
LMS	beta33	0.139	0.139	-0.000 $\pm$ 0.001	-0.002 $\pm$ 0.009	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	0.087 $\pm$ 0.004	0.296 $\pm$ 0.008	0.139	0.000	-0.05	0.0402	0.0402	1.014	94.100 $\pm$ 0.745	0.163 $\pm$ 0.001	87.100 $\pm$ 1.060
LMS	beta34	0.101	0.099	-0.002 $\pm$ 0.001	-0.021 $\pm$ 0.009	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.084 $\pm$ 0.004	0.290 $\pm$ 0.008	0.097	-0.004	-4.40	0.0294	0.0297	1.004	94.700 $\pm$ 0.708	0.115 $\pm$ 0.001	95.800 $\pm$ 0.634
LMS	beta35	0.101	0.100	-0.001 $\pm$ 0.001	-0.009 $\pm$ 0.009	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.087 $\pm$ 0.004	0.295 $\pm$ 0.008	0.097	-0.004	-3.77	0.0296	0.0299	0.988	94.900 $\pm$ 0.696	0.115 $\pm$ 0.001	95.900 $\pm$ 0.627
LSAM	beta31	0.316	0.315	-0.001 $\pm$ 0.001	-0.004 $\pm$ 0.004	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.014 $\pm$ 0.001	0.118 $\pm$ 0.003	0.315	-0.001	-0.20	0.0382	0.0382	0.998	95.700 $\pm$ 0.641	0.146 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta32	0.316	0.315	-0.001 $\pm$ 0.001	-0.003 $\pm$ 0.004	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.014 $\pm$ 0.001	0.120 $\pm$ 0.003	0.315	-0.001	-0.34	0.0364	0.0364	0.987	94.600 $\pm$ 0.715	0.147 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta33	0.139	0.145	0.006 $\pm$ 0.001	0.044 $\pm$ 0.009	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.082 $\pm$ 0.004	0.287 $\pm$ 0.008	0.145	0.006	4.21	0.0379	0.0384	1.008	94.100 $\pm$ 0.745	0.156 $\pm$ 0.001	92.900 $\pm$ 0.812
LSAM	beta34	0.101	0.095	-0.006 $\pm$ 0.001	-0.059 $\pm$ 0.009	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.083 $\pm$ 0.004	0.288 $\pm$ 0.008	0.095	-0.006	-6.18	0.0295	0.0302	0.997	94.000 $\pm$ 0.751	0.111 $\pm$ 0.001	94.200 $\pm$ 0.739
LSAM	beta35	0.101	0.096	-0.005 $\pm$ 0.001	-0.050 $\pm$ 0.009	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.085 $\pm$ 0.004	0.291 $\pm$ 0.008	0.094	-0.007	-6.91	0.0285	0.0294	0.977	94.000 $\pm$ 0.751	0.111 $\pm$ 0.001	94.300 $\pm$ 0.733
QML	beta31	0.316	0.310	-0.006 $\pm$ 0.001	-0.018 $\pm$ 0.004	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.014 $\pm$ 0.001	0.118 $\pm$ 0.003	0.310	-0.006	-1.77	0.0375	0.0379	1.016	95.000 $\pm$ 0.689	0.147 $\pm$ 0.001	99.800 $\pm$ 0.141
QML	beta32	0.316	0.311	-0.005 $\pm$ 0.001	-0.017 $\pm$ 0.004	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.015 $\pm$ 0.001	0.121 $\pm$ 0.003	0.310	-0.006	-1.80	0.0374	0.0379	0.991	94.400 $\pm$ 0.727	0.147 $\pm$ 0.001	99.800 $\pm$ 0.141
QML	beta33	0.139	0.137	-0.002 $\pm$ 0.001	-0.015 $\pm$ 0.009	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	0.086 $\pm$ 0.004	0.293 $\pm$ 0.008	0.137	-0.002	-1.34	0.0399	0.0400	1.011	94.500 $\pm$ 0.721	0.161 $\pm$ 0.001	87.100 $\pm$ 1.060
QML	beta34	0.101	0.098	-0.003 $\pm$ 0.001	-0.033 $\pm$ 0.009	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.082 $\pm$ 0.004	0.287 $\pm$ 0.008	0.096	-0.005	-4.70	0.0294	0.0297	1.002	94.500 $\pm$ 0.721	0.113 $\pm$ 0.001	95.700 $\pm$ 0.641
QML	beta35	0.101	0.099	-0.002 $\pm$ 0.001	-0.022 $\pm$ 0.009	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.084 $\pm$ 0.004	0.291 $\pm$ 0.008	0.096	-0.005	-4.68	0.0289	0.0293	0.986	94.600 $\pm$ 0.715	0.113 $\pm$ 0.001	95.900 $\pm$ 0.627
UPI	beta31	0.316	0.317	0.001 $\pm$ 0.001	0.004 $\pm$ 0.004	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.014 $\pm$ 0.001	0.118 $\pm$ 0.003	0.317	0.001	0.47	0.0384	0.0384	0.965	94.700 $\pm$ 0.708	0.141 $\pm$ 0.000	100.000 $\pm$ 0.000
UPI	beta32	0.316	0.318	0.002 $\pm$ 0.001	0.006 $\pm$ 0.004	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.014 $\pm$ 0.001	0.119 $\pm$ 0.003	0.318	0.002	0.48	0.0361	0.0361	0.960	94.000 $\pm$ 0.751	0.141 $\pm$ 0.000	100.000 $\pm$ 0.000
UPI	beta33	0.139	0.141	0.002 $\pm$ 0.001	0.012 $\pm$ 0.009	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.083 $\pm$ 0.004	0.288 $\pm$ 0.008	0.141	0.002	1.09	0.0384	0.0385	0.994	94.800 $\pm$ 0.702	0.156 $\pm$ 0.001	91.500 $\pm$ 0.882
UPI	beta34	0.101	0.099	-0.002 $\pm$ 0.001	-0.016 $\pm$ 0.009	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.083 $\pm$ 0.004	0.288 $\pm$ 0.008	0.098	-0.003	-2.62	0.0292	0.0294	0.987	93.900 $\pm$ 0.757	0.113 $\pm$ 0.001	95.600 $\pm$ 0.649
UPI	beta35	0.101	0.100	-0.001 $\pm$ 0.001	-0.009 $\pm$ 0.009	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.086 $\pm$ 0.004	0.292 $\pm$ 0.008	0.097	-0.004	-4.04	0.0289	0.0292	0.970	94.300 $\pm$ 0.733	0.112 $\pm$ 0.001	95.500 $\pm$ 0.656

Note. MCSE = Monte Carlo Standard Error shown as $\backslash u00b1$ value. Coverage, Type I Error, and Power expressed as percentages. $\backslash u2014$ indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S61

Simulation 1 - Condition 61: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.4

Method	Param.	True	Mean-based										Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE	
LMS	beta31	0.316	0.326	0.010 ± 0.007	0.031 ± 0.024	0.033 ± 0.003	0.181 ± 0.007	0.033 ± 0.003	0.181 ± 0.008	0.330 ± 0.027	0.574 ± 0.026	0.340	0.024	7.56	0.1647	0.1664	1.109	95.424 ± 0.860	0.788 ± 0.032	57.627 ± 2.034	
LMS	beta32	0.316	0.315	-0.001 ± 0.008	-0.004 ± 0.024	0.034 ± 0.002	0.183 ± 0.006	0.034 ± 0.002	0.183 ± 0.008	0.336 ± 0.024	0.580 ± 0.024	0.342	0.026	8.20	0.1713	0.1733	1.092	94.237 ± 0.959	0.785 ± 0.032	55.424 ± 2.046	
LMS	beta33	0.139	-0.331	-0.470 ± 0.030	-3.384 ± 0.216	0.534 ± 0.027	0.731 ± 0.019	0.755 ± 0.051	0.869 ± 0.032	39.059 ± 2.623	6.250 ± 0.229	-0.040	-0.179	-128.78	0.6333	0.6581	0.779	82.034 ± 1.581	2.232 ± 0.136	24.576 ± 1.772	
LMS	beta34	0.101	0.460	0.359 ± 0.019	3.552 ± 0.192	0.221 ± 0.019	0.470 ± 0.020	0.349 ± 0.031	0.591 ± 0.027	34.247 ± 3.025	5.852 ± 0.271	0.268	0.167	164.98	0.2641	0.3123	0.893	95.424 ± 0.860	1.646 ± 0.113	23.051 ± 1.734	
LMS	beta35	0.101	0.466	0.365 ± 0.019	3.617 ± 0.185	0.206 ± 0.014	0.454 ± 0.016	0.339 ± 0.026	0.582 ± 0.024	33.252 ± 2.572	5.766 ± 0.235	0.283	0.182	180.50	0.2845	0.3379	0.925	93.559 ± 1.011	1.646 ± 0.107	26.102 ± 1.808	
LSAM	beta31	0.316	0.374	0.058 ± 0.050	0.183 ± 0.157	1.449 ± 0.098	1.204 ± 0.041	1.450 ± 0.099	1.204 ± 0.048	14.524 ± 0.991	3.811 ± 0.152	0.372	0.056	17.74	1.2516	1.2528	2.554	100.000 ± 0.000	12.051 ± 0.455	2.712 ± 0.669	
LSAM	beta32	0.316	0.406	0.090 ± 0.050	0.284 ± 0.159	1.488 ± 0.102	1.220 ± 0.042	1.494 ± 0.103	1.222 ± 0.049	14.959 ± 1.035	3.868 ± 0.156	0.395	0.079	24.93	1.2753	1.2777	2.533	99.661 ± 0.239	12.111 ± 0.462	2.881 ± 0.689	
LSAM	beta33	0.139	0.262	0.123 ± 0.014	0.883 ± 0.103	0.122 ± 0.011	0.349 ± 0.015	0.136 ± 0.012	0.369 ± 0.018	7.060 ± 0.645	2.657 ± 0.132	0.251	0.112	80.31	0.3092	0.3287	2.153	98.305 ± 0.531	2.942 ± 0.152	6.271 ± 0.998	
LSAM	beta34	0.101	0.041	-0.060 ± 0.020	-0.598 ± 0.194	0.226 ± 0.018	0.475 ± 0.020	0.229 ± 0.019	0.478 ± 0.022	22.432 ± 1.868	4.736 ± 0.220	0.051	-0.050	-49.37	0.3870	0.3902	2.385	99.492 ± 0.293	4.441 ± 0.195	2.881 ± 0.689	
LSAM	beta35	0.101	0.032	-0.069 ± 0.019	-0.680 ± 0.193	0.224 ± 0.018	0.473 ± 0.020	0.228 ± 0.019	0.477 ± 0.022	22.340 ± 1.846	4.726 ± 0.218	0.044	-0.057	-56.66	0.3875	0.3917	2.418	98.814 ± 0.446	4.482 ± 0.200	3.051 ± 0.708	
QML	beta31	0.316	0.340	0.024 ± 0.005	0.075 ± 0.014	0.012 ± 0.001	0.110 ± 0.003	0.013 ± 0.001	0.113 ± 0.004	0.127 ± 0.008	0.356 ± 0.013	0.343	0.027	8.64	0.1192	0.1223	1.100	96.441 ± 0.763	0.475 ± 0.005	81.525 ± 1.598	
QML	beta32	0.316	0.339	0.023 ± 0.005	0.071 ± 0.015	0.014 ± 0.001	0.119 ± 0.004	0.015 ± 0.001	0.121 ± 0.005	0.146 ± 0.009	0.382 ± 0.015	0.341	0.025	7.92	0.1175	0.1201	1.015	95.593 ± 0.845	0.473 ± 0.004	80.847 ± 1.620	
QML	beta33	0.139	0.088	-0.051 ± 0.008	-0.368 ± 0.055	0.035 ± 0.002	0.186 ± 0.006	0.037 ± 0.002	0.193 ± 0.007	1.923 ± 0.125	1.387 ± 0.052	0.104	-0.035	-25.03	0.1931	0.1962	0.988	94.237 ± 0.959	0.720 ± 0.018	14.915 ± 1.467	
QML	beta34	0.101	0.184	0.083 ± 0.004	0.823 ± 0.042	0.011 ± 0.001	0.102 ± 0.004	0.017 ± 0.001	0.132 ± 0.005	1.706 ± 0.120	1.306 ± 0.049	0.179	0.078	77.01	0.0960	0.1236	1.025	91.017 ± 1.177	0.412 ± 0.012	52.542 ± 2.056	
QML	beta35	0.101	0.190	0.089 ± 0.005	0.880 ± 0.045	0.012 ± 0.001	0.110 ± 0.004	0.020 ± 0.001	0.141 ± 0.005	1.952 ± 0.141	1.397 ± 0.053	0.175	0.074	73.12	0.1014	0.1255	0.963	90.847 ± 1.187	0.414 ± 0.010	50.847 ± 2.058	
UPI	beta31	0.316	0.368	0.052 ± 0.021	0.166 ± 0.067	0.264 ± 0.021	0.514 ± 0.020	0.267 ± 0.021	0.517 ± 0.023	2.672 ± 0.215	1.635 ± 0.074	0.311	-0.005	-1.51	0.4851	0.4851	2.095	99.661 ± 0.239	4.223 ± 0.205	6.780 ± 1.035	
UPI	beta32	0.316	0.311	-0.005 ± 0.022	-0.016 ± 0.068	0.273 ± 0.023	0.522 ± 0.022	0.272 ± 0.023	0.522 ± 0.024	2.728 ± 0.227	1.652 ± 0.077	0.292	-0.024	-7.65	0.4553	0.4559	2.102	99.831 ± 0.169	4.303 ± 0.210	6.102 ± 0.985	
UPI	beta33	0.139	0.156	0.017 ± 0.012	0.123 ± 0.083	0.078 ± 0.006	0.280 ± 0.012	0.078 ± 0.007	0.280 ± 0.013	4.054 ± 0.339	2.013 ± 0.094	0.147	0.008	5.95	0.2708	0.2709	1.890	98.644 ± 0.476	2.071 ± 0.111	3.220 ± 0.727	
UPI	beta34	0.101	0.055	-0.046 ± 0.008	-0.456 ± 0.077	0.036 ± 0.003	0.189 ± 0.008	0.038 ± 0.003	0.194 ± 0.009	3.698 ± 0.320	1.923 ± 0.092	0.066	-0.035	-34.24	0.1844	0.1876	2.168	99.322 ± 0.338	1.605 ± 0.083	2.373 ± 0.627	
UPI	beta35	0.101	0.078	-0.023 ± 0.008	-0.229 ± 0.084	0.042 ± 0.004	0.205 ± 0.009	0.042 ± 0.004	0.206 ± 0.010	4.165 ± 0.359	2.041 ± 0.098	0.084	-0.017	-16.74	0.1751	0.1759	2.099	99.153 ± 0.377	1.687 ± 0.088	3.220 ± 0.727	

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S62

Simulation 1 - Condition 62: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.4

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.324	0.008 ± 0.004	0.027 ± 0.014	0.012 ± 0.001	0.111 ± 0.004	0.012 ± 0.001	0.112 ± 0.004	0.125 ± 0.009	0.353 ± 0.014	0.340	0.024	7.68	0.0970	0.0999	1.068	93.730 ± 0.972	0.466 ± 0.026	79.421 ± 1.621
LMS	beta32	0.316	0.321	0.005 ± 0.005	0.015 ± 0.016	0.016 ± 0.001	0.125 ± 0.005	0.016 ± 0.001	0.125 ± 0.005	0.156 ± 0.012	0.395 ± 0.017	0.347	0.031	9.93	0.1045	0.1091	1.047	92.444 ± 1.060	0.513 ± 0.038	78.296 ± 1.653
LMS	beta33	0.139	-0.177	-0.316 ± 0.026	-2.272 ± 0.186	0.418 ± 0.025	0.646 ± 0.019	0.517 ± 0.039	0.719 ± 0.030	26.757 ± 2.031	5.173 ± 0.214	0.129	-0.010	-7.24	0.2981	0.2982	0.571	75.080 ± 1.734	1.446 ± 0.086	42.926 ± 1.985
LMS	beta34	0.101	0.367	0.266 ± 0.014	2.638 ± 0.139	0.123 ± 0.010	0.351 ± 0.015	0.194 ± 0.017	0.441 ± 0.020	19.030 ± 1.680	4.362 ± 0.201	0.217	0.116	115.06	0.1372	0.1798	0.622	85.370 ± 1.417	0.856 ± 0.058	61.576 ± 1.950
LMS	beta35	0.101	0.392	0.291 ± 0.016	2.885 ± 0.156	0.153 ± 0.011	0.392 ± 0.015	0.238 ± 0.020	0.488 ± 0.021	23.345 ± 1.946	4.832 ± 0.211	0.215	0.114	112.67	0.1389	0.1796	0.703	81.511 ± 1.557	1.079 ± 0.101	62.540 ± 1.941
LSAM	beta31	0.316	0.334	0.018 ± 0.022	0.057 ± 0.070	0.307 ± 0.027	0.554 ± 0.024	0.307 ± 0.027	0.554 ± 0.027	3.072 ± 0.266	1.753 ± 0.084	0.331	0.015	4.74	0.5630	0.5632	1.957	99.196 ± 0.358	4.250 ± 0.263	13.666 ± 1.377
LSAM	beta32	0.316	0.383	0.067 ± 0.023	0.212 ± 0.073	0.327 ± 0.028	0.572 ± 0.024	0.331 ± 0.028	0.575 ± 0.027	3.316 ± 0.283	1.821 ± 0.086	0.365	0.049	15.53	0.6037	0.6057	1.987	99.518 ± 0.278	4.456 ± 0.289	12.701 ± 1.335
LSAM	beta33	0.139	0.224	0.085 ± 0.006	0.612 ± 0.040	0.019 ± 0.002	0.138 ± 0.006	0.026 ± 0.002	0.162 ± 0.007	1.359 ± 0.105	1.166 ± 0.048	0.224	0.085	61.16	0.1509	0.1732	1.437	95.177 ± 0.859	0.778 ± 0.039	40.514 ± 1.968
LSAM	beta34	0.101	0.062	-0.039 ± 0.007	-0.386 ± 0.071	0.032 ± 0.003	0.180 ± 0.008	0.034 ± 0.003	0.184 ± 0.009	3.324 ± 0.300	1.823 ± 0.090	0.066	-0.035	-34.88	0.1790	0.1825	1.868	98.071 ± 0.552	1.319 ± 0.084	8.360 ± 1.110
LSAM	beta35	0.101	0.048	-0.053 ± 0.008	-0.524 ± 0.076	0.037 ± 0.003	0.192 ± 0.008	0.040 ± 0.003	0.199 ± 0.009	3.894 ± 0.335	1.973 ± 0.093	0.053	-0.048	-47.71	0.1956	0.2014	1.995	97.749 ± 0.595	1.504 ± 0.109	7.235 ± 1.039
QML	beta31	0.316	0.333	0.017 ± 0.003	0.055 ± 0.009	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.049 ± 0.003	0.221 ± 0.008	0.335	0.019	6.08	0.0702	0.0728	1.082	95.820 ± 0.802	0.287 ± 0.001	99.357 ± 0.321
QML	beta32	0.316	0.338	0.022 ± 0.003	0.070 ± 0.009	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.054 ± 0.003	0.232 ± 0.008	0.342	0.026	8.18	0.0716	0.0761	1.049	94.373 ± 0.924	0.287 ± 0.002	98.392 ± 0.504
QML	beta33	0.139	0.111	-0.028 ± 0.004	-0.202 ± 0.032	0.012 ± 0.001	0.112 ± 0.003	0.013 ± 0.001	0.115 ± 0.004	0.685 ± 0.042	0.827 ± 0.030	0.112	-0.027	-19.11	0.1055	0.1088	0.945	92.765 ± 1.039	0.414 ± 0.005	23.151 ± 1.691
QML	beta34	0.101	0.182	0.081 ± 0.002	0.807 ± 0.022	0.003 ± 0.000	0.055 ± 0.002	0.010 ± 0.000	0.098 ± 0.002	0.944 ± 0.043	0.972 ± 0.023	0.180	0.079	78.59	0.0600	0.0995	1.082	74.920 ± 1.738	0.232 ± 0.003	91.318 ± 1.129
QML	beta35	0.101	0.182	0.081 ± 0.003	0.802 ± 0.025	0.004 ± 0.000	0.064 ± 0.002	0.011 ± 0.001	0.103 ± 0.003	1.043 ± 0.052	1.021 ± 0.027	0.178	0.077	76.37	0.0620	0.0990	0.928	72.990 ± 1.780	0.232 ± 0.003	88.907 ± 1.259
UPI	beta31	0.316	0.338	0.022 ± 0.015	0.071 ± 0.049	0.148 ± 0.012	0.385 ± 0.015	0.148 ± 0.012	0.385 ± 0.017	1.483 ± 0.118	1.218 ± 0.054	0.318	0.002	0.60	0.3482	0.3482	1.852	99.518 ± 0.278	2.792 ± 0.159	13.344 ± 1.363
UPI	beta32	0.316	0.309	-0.007 ± 0.015	-0.021 ± 0.047	0.135 ± 0.011	0.367 ± 0.014	0.135 ± 0.011	0.367 ± 0.016	1.348 ± 0.106	1.161 ± 0.051	0.291	-0.025	-7.81	0.3257	0.3266	1.902	99.678 ± 0.227	2.738 ± 0.148	10.611 ± 1.235
UPI	beta33	0.139	0.175	0.036 ± 0.007	0.256 ± 0.047	0.027 ± 0.002	0.163 ± 0.007	0.028 ± 0.002	0.167 ± 0.008	1.439 ± 0.120	1.199 ± 0.055	0.158	0.019	13.82	0.1537	0.1549	1.694	98.071 ± 0.552	1.082 ± 0.059	13.826 ± 1.384
UPI	beta34	0.101	0.063	-0.038 ± 0.005	-0.371 ± 0.051	0.017 ± 0.001	0.129 ± 0.005	0.018 ± 0.001	0.134 ± 0.006	1.772 ± 0.132	1.331 ± 0.055	0.068	-0.033	-32.89	0.1184	0.1230	1.832	98.071 ± 0.552	0.928 ± 0.051	5.466 ± 0.911
UPI	beta35	0.101	0.074	-0.027 ± 0.005	-0.269 ± 0.051	0.017 ± 0.001	0.129 ± 0.005	0.017 ± 0.001	0.132 ± 0.006	1.713 ± 0.138	1.309 ± 0.059	0.076	-0.025	-24.66	0.1158	0.1185	1.851	99.035 ± 0.392	0.940 ± 0.054	6.431 ± 0.984

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S63

Simulation 1 - Condition 63: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.6

Method	Param.	True	Mean-based										Median-based					Inference			
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE	
LMS	beta31	0.316	0.318	0.002 $\pm$ 0.003	0.005 $\pm$ 0.011	0.010 $\pm$ 0.001	0.099 $\pm$ 0.003	0.010 $\pm$ 0.001	0.099 $\pm$ 0.003	0.098 $\pm$ 0.005	0.313 $\pm$ 0.010	0.320	0.004	1.30	0.1039	0.1040	0.997	94.622 $\pm$ 0.763	0.387 $\pm$ 0.005	85.698 $\pm$ 1.184	
LMS	beta32	0.316	0.326	0.010 $\pm$ 0.003	0.031 $\pm$ 0.009	0.008 $\pm$ 0.000	0.089 $\pm$ 0.002	0.008 $\pm$ 0.000	0.089 $\pm$ 0.003	0.080 $\pm$ 0.004	0.282 $\pm$ 0.008	0.328	0.012	3.81	0.0940	0.0948	1.088	96.682 $\pm$ 0.606	0.378 $\pm$ 0.004	89.245 $\pm$ 1.048	
LMS	beta33	0.139	0.117	-0.022 $\pm$ 0.004	-0.159 $\pm$ 0.028	0.014 $\pm$ 0.001	0.117 $\pm$ 0.003	0.014 $\pm$ 0.001	0.119 $\pm$ 0.004	0.731 $\pm$ 0.043	0.855 $\pm$ 0.029	0.119	-0.020	-14.75	0.1168	0.1186	1.027	94.851 $\pm$ 0.748	0.470 $\pm$ 0.013	27.460 $\pm$ 1.510	
LMS	beta34	0.101	0.153	0.052 $\pm$ 0.002	0.511 $\pm$ 0.022	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	0.007 $\pm$ 0.000	0.084 $\pm$ 0.003	0.693 $\pm$ 0.045	0.832 $\pm$ 0.029	0.148	0.047	46.50	0.0628	0.0784	1.021	93.021 $\pm$ 0.862	0.266 $\pm$ 0.008	73.455 $\pm$ 1.494	
LMS	beta35	0.101	0.148	0.047 $\pm$ 0.002	0.465 $\pm$ 0.020	0.003 $\pm$ 0.000	0.058 $\pm$ 0.002	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.549 $\pm$ 0.032	0.741 $\pm$ 0.023	0.144	0.043	42.61	0.0532	0.0685	1.081	93.478 $\pm$ 0.835	0.247 $\pm$ 0.006	74.943 $\pm$ 1.466	
LSAM	beta31	0.316	0.323	0.007 $\pm$ 0.006	0.023 $\pm$ 0.019	0.032 $\pm$ 0.002	0.178 $\pm$ 0.004	0.032 $\pm$ 0.002	0.178 $\pm$ 0.005	0.317 $\pm$ 0.016	0.563 $\pm$ 0.017	0.329	0.013	4.06	0.1797	0.1802	0.993	95.080 $\pm$ 0.732	0.693 $\pm$ 0.008	50.572 $\pm$ 1.691	
LSAM	beta32	0.316	0.329	0.013 $\pm$ 0.006	0.040 $\pm$ 0.018	0.027 $\pm$ 0.001	0.164 $\pm$ 0.004	0.027 $\pm$ 0.001	0.165 $\pm$ 0.005	0.272 $\pm$ 0.014	0.521 $\pm$ 0.016	0.329	0.013	4.17	0.1697	0.1702	1.049	96.796 $\pm$ 0.596	0.676 $\pm$ 0.007	52.975 $\pm$ 1.688	
LSAM	beta33	0.139	0.172	0.033 $\pm$ 0.003	0.238 $\pm$ 0.025	0.011 $\pm$ 0.001	0.103 $\pm$ 0.003	0.012 $\pm$ 0.001	0.109 $\pm$ 0.004	0.610 $\pm$ 0.036	0.781 $\pm$ 0.026	0.164	0.025	18.04	0.1028	0.1058	0.933	92.792 $\pm$ 0.875	0.379 $\pm$ 0.005	45.881 $\pm$ 1.686	
LSAM	beta34	0.101	0.088	-0.013 $\pm$ 0.002	-0.126 $\pm$ 0.024	0.005 $\pm$ 0.000	0.070 $\pm$ 0.002	0.005 $\pm$ 0.000	0.071 $\pm$ 0.002	0.500 $\pm$ 0.027	0.707 $\pm$ 0.022	0.091	-0.010	-9.97	0.0693	0.0700	0.943	90.847 $\pm$ 0.975	0.260 $\pm$ 0.005	38.101 $\pm$ 1.643	
LSAM	beta35	0.101	0.087	-0.014 $\pm$ 0.002	-0.134 $\pm$ 0.022	0.004 $\pm$ 0.000	0.064 $\pm$ 0.002	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	0.422 $\pm$ 0.024	0.649 $\pm$ 0.022	0.089	-0.012	-12.31	0.0619	0.0632	0.989	93.249 $\pm$ 0.849	0.249 $\pm$ 0.004	39.703 $\pm$ 1.655	
QML	beta31	0.316	0.320	0.004 $\pm$ 0.003	0.013 $\pm$ 0.010	0.009 $\pm$ 0.000	0.094 $\pm$ 0.002	0.009 $\pm$ 0.000	0.094 $\pm$ 0.003	0.088 $\pm$ 0.004	0.297 $\pm$ 0.009	0.320	0.004	1.17	0.1029	0.1029	0.975	94.394 $\pm$ 0.778	0.358 $\pm$ 0.002	90.732 $\pm$ 0.981	
QML	beta32	0.316	0.326	0.010 $\pm$ 0.003	0.032 $\pm$ 0.009	0.007 $\pm$ 0.000	0.085 $\pm$ 0.002	0.007 $\pm$ 0.000	0.086 $\pm$ 0.002	0.073 $\pm$ 0.003	0.271 $\pm$ 0.008	0.325	0.009	2.86	0.0920	0.0925	1.058	95.881 $\pm$ 0.672	0.353 $\pm$ 0.002	92.563 $\pm$ 0.887	
QML	beta33	0.139	0.110	-0.029 $\pm$ 0.003	-0.211 $\pm$ 0.025	0.010 $\pm$ 0.001	0.101 $\pm$ 0.003	0.011 $\pm$ 0.001	0.105 $\pm$ 0.003	0.575 $\pm$ 0.028	0.758 $\pm$ 0.022	0.110	-0.029	-21.18	0.1069	0.1109	0.965	93.249 $\pm$ 0.849	0.383 $\pm$ 0.005	26.545 $\pm$ 1.494	
QML	beta34	0.101	0.146	0.045 $\pm$ 0.002	0.442 $\pm$ 0.019	0.003 $\pm$ 0.000	0.056 $\pm$ 0.001	0.005 $\pm$ 0.000	0.072 $\pm$ 0.002	0.507 $\pm$ 0.026	0.712 $\pm$ 0.020	0.144	0.043	42.75	0.0575	0.0719	0.972	89.245 $\pm$ 1.048	0.215 $\pm$ 0.003	78.719 $\pm$ 1.384	
QML	beta35	0.101	0.143	0.042 $\pm$ 0.002	0.415 $\pm$ 0.018	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.005 $\pm$ 0.000	0.067 $\pm$ 0.002	0.441 $\pm$ 0.023	0.664 $\pm$ 0.018	0.141	0.040	39.73	0.0481	0.0626	1.020	89.588 $\pm$ 1.033	0.209 $\pm$ 0.003	80.435 $\pm$ 1.342	
UPI	beta31	0.316	0.327	0.011 $\pm$ 0.007	0.035 $\pm$ 0.022	0.044 $\pm$ 0.002	0.210 $\pm$ 0.006	0.044 $\pm$ 0.002	0.210 $\pm$ 0.007	0.442 $\pm$ 0.024	0.665 $\pm$ 0.022	0.329	0.013	4.20	0.2091	0.2095	1.090	95.881 $\pm$ 0.672	0.897 $\pm$ 0.022	44.279 $\pm$ 1.680	
UPI	beta32	0.316	0.335	0.019 $\pm$ 0.007	0.061 $\pm$ 0.021	0.038 $\pm$ 0.002	0.195 $\pm$ 0.006	0.038 $\pm$ 0.002	0.196 $\pm$ 0.006	0.384 $\pm$ 0.022	0.619 $\pm$ 0.020	0.335	0.019	5.93	0.1898	0.1907	1.156	97.140 $\pm$ 0.564	0.883 $\pm$ 0.023	47.025 $\pm$ 1.688	
UPI	beta33	0.139	0.163	0.024 $\pm$ 0.004	0.171 $\pm$ 0.028	0.013 $\pm$ 0.001	0.114 $\pm$ 0.004	0.014 $\pm$ 0.001	0.116 $\pm$ 0.004	0.699 $\pm$ 0.048	0.836 $\pm$ 0.032	0.153	0.014	10.40	0.1020	0.1030	1.040	95.195 $\pm$ 0.723	0.464 $\pm$ 0.012	34.325 $\pm$ 1.606	
UPI	beta34	0.101	0.087	-0.014 $\pm$ 0.003	-0.137 $\pm$ 0.028	0.007 $\pm$ 0.000	0.082 $\pm$ 0.003	0.007 $\pm$ 0.000	0.083 $\pm$ 0.003	0.682 $\pm$ 0.043	0.826 $\pm$ 0.029	0.087	-0.014	-13.99	0.0758	0.0771	1.038	92.449 $\pm$ 0.894	0.335 $\pm$ 0.010	35.240 $\pm$ 1.616	
UPI	beta35	0.101	0.087	-0.014 $\pm$ 0.003	-0.138 $\pm$ 0.026	0.006 $\pm$ 0.000	0.076 $\pm$ 0.002	0.006 $\pm$ 0.000	0.078 $\pm$ 0.003	0.589 $\pm$ 0.038	0.768 $\pm$ 0.028	0.088	-0.013	-12.95	0.0664	0.0677	1.100	94.851 $\pm$ 0.748	0.329 $\pm$ 0.010	35.240 $\pm$ 1.616	

Note. MCSE = Monte Carlo Standard Error shown as $\backslash$ u00b1 value. Coverage, Type I Error, and Power expressed as percentages. $\backslash$ u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S64

Simulation 1 - Condition 64: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.6

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.322	0.006 ± 0.002	0.019 ± 0.006	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.032 ± 0.002	0.180 ± 0.005	0.324	0.008	2.64	0.0574	0.0580	1.038	95.469 ± 0.675	0.230 ± 0.002	99.368 ± 0.257
LMS	beta32	0.316	0.320	0.004 ± 0.002	0.014 ± 0.006	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.034 ± 0.002	0.185 ± 0.005	0.322	0.006	1.99	0.0597	0.0600	1.008	95.047 ± 0.704	0.230 ± 0.002	98.841 ± 0.347
LMS	beta33	0.139	0.125	-0.014 ± 0.002	-0.103 ± 0.016	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.250 ± 0.014	0.500 ± 0.016	0.126	-0.013	-9.70	0.0649	0.0663	0.981	94.942 ± 0.711	0.262 ± 0.003	51.739 ± 1.622
LMS	beta34	0.101	0.146	0.045 ± 0.001	0.449 ± 0.012	0.001 ± 0.000	0.036 ± 0.001	0.003 ± 0.000	0.058 ± 0.001	0.331 ± 0.014	0.576 ± 0.013	0.143	0.042	41.83	0.0360	0.0555	0.984	78.820 ± 1.326	0.140 ± 0.002	98.736 ± 0.363
LMS	beta35	0.101	0.147	0.046 ± 0.001	0.454 ± 0.012	0.001 ± 0.000	0.038 ± 0.001	0.004 ± 0.000	0.060 ± 0.001	0.349 ± 0.016	0.591 ± 0.014	0.143	0.042	41.30	0.0366	0.0555	0.932	78.609 ± 1.331	0.139 ± 0.002	99.262 ± 0.278
LSAM	beta31	0.316	0.322	0.006 ± 0.003	0.018 ± 0.010	0.009 ± 0.000	0.097 ± 0.002	0.009 ± 0.000	0.097 ± 0.003	0.094 ± 0.005	0.307 ± 0.009	0.318	0.002	0.60	0.0971	0.0971	1.023	95.258 ± 0.690	0.388 ± 0.002	90.306 ± 0.960
LSAM	beta32	0.316	0.320	0.004 ± 0.003	0.012 ± 0.010	0.010 ± 0.000	0.098 ± 0.002	0.010 ± 0.000	0.098 ± 0.003	0.097 ± 0.005	0.312 ± 0.009	0.318	0.002	0.78	0.1003	0.1003	1.002	95.047 ± 0.704	0.387 ± 0.002	87.987 ± 1.055
LSAM	beta33	0.139	0.161	0.022 ± 0.002	0.161 ± 0.014	0.003 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.206 ± 0.011	0.454 ± 0.013	0.157	0.018	13.26	0.0558	0.0587	0.930	91.254 ± 0.917	0.215 ± 0.001	83.667 ± 1.200
LSAM	beta34	0.101	0.090	-0.011 ± 0.001	-0.106 ± 0.012	0.001 ± 0.000	0.038 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.153 ± 0.008	0.392 ± 0.012	0.092	-0.009	-9.36	0.0356	0.0368	0.914	89.463 ± 0.997	0.136 ± 0.002	71.970 ± 1.458
LSAM	beta35	0.101	0.091	-0.010 ± 0.001	-0.099 ± 0.012	0.001 ± 0.000	0.038 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.149 ± 0.007	0.385 ± 0.011	0.091	-0.010	-10.21	0.0360	0.0375	0.909	89.146 ± 1.010	0.134 ± 0.001	72.708 ± 1.446
QML	beta31	0.316	0.321	0.005 ± 0.002	0.017 ± 0.006	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.031 ± 0.001	0.176 ± 0.005	0.322	0.006	1.89	0.0549	0.0552	1.023	95.153 ± 0.697	0.222 ± 0.001	100.000 ± 0.000
QML	beta32	0.316	0.320	0.004 ± 0.002	0.011 ± 0.006	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.032 ± 0.002	0.180 ± 0.005	0.320	0.004	1.26	0.0590	0.0591	0.998	94.731 ± 0.725	0.222 ± 0.001	99.684 ± 0.182
QML	beta33	0.139	0.112	-0.027 ± 0.002	-0.192 ± 0.014	0.004 ± 0.000	0.062 ± 0.002	0.005 ± 0.000	0.067 ± 0.002	0.235 ± 0.012	0.485 ± 0.014	0.111	-0.028	-19.94	0.0602	0.0663	0.952	91.043 ± 0.927	0.231 ± 0.002	49.420 ± 1.623
QML	beta34	0.101	0.144	0.043 ± 0.001	0.427 ± 0.011	0.001 ± 0.000	0.034 ± 0.001	0.003 ± 0.000	0.055 ± 0.001	0.295 ± 0.011	0.543 ± 0.011	0.143	0.042	41.67	0.0334	0.0537	0.958	75.026 ± 1.405	0.128 ± 0.001	98.946 ± 0.331
QML	beta35	0.101	0.145	0.044 ± 0.001	0.434 ± 0.011	0.001 ± 0.000	0.036 ± 0.001	0.003 ± 0.000	0.056 ± 0.001	0.312 ± 0.013	0.559 ± 0.013	0.143	0.042	41.15	0.0350	0.0544	0.911	74.499 ± 1.415	0.127 ± 0.001	99.684 ± 0.182
UPI	beta31	0.316	0.319	0.003 ± 0.004	0.010 ± 0.011	0.012 ± 0.001	0.111 ± 0.003	0.012 ± 0.001	0.111 ± 0.003	0.122 ± 0.006	0.350 ± 0.011	0.315	-0.001	-0.40	0.1088	0.1088	1.034	95.785 ± 0.652	0.448 ± 0.004	80.717 ± 1.281
UPI	beta32	0.316	0.315	-0.001 ± 0.004	-0.003 ± 0.012	0.013 ± 0.001	0.112 ± 0.003	0.013 ± 0.001	0.112 ± 0.003	0.127 ± 0.006	0.356 ± 0.011	0.315	-0.001	-0.37	0.1134	0.1134	1.013	95.996 ± 0.636	0.447 ± 0.005	77.977 ± 1.345
UPI	beta33	0.139	0.151	0.012 ± 0.002	0.088 ± 0.014	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	0.197 ± 0.011	0.444 ± 0.014	0.147	0.008	5.59	0.0568	0.0573	0.978	94.310 ± 0.752	0.232 ± 0.003	77.766 ± 1.350
UPI	beta34	0.101	0.091	-0.010 ± 0.001	-0.096 ± 0.014	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.183 ± 0.011	0.428 ± 0.014	0.093	-0.008	-8.36	0.0372	0.0381	0.942	92.308 ± 0.865	0.156 ± 0.002	65.859 ± 1.539
UPI	beta35	0.101	0.093	-0.008 ± 0.001	-0.076 ± 0.013	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.173 ± 0.009	0.416 ± 0.012	0.092	-0.009	-8.84	0.0387	0.0398	0.943	91.359 ± 0.912	0.153 ± 0.002	68.388 ± 1.509

Note. MCSE = Monte Carlo Standard Error shown as $\backslash 00b1$ value. Coverage, Type I Error, and Power expressed as percentages. $\backslash 02014$ indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S65

Simulation 1 - Condition 65: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.8

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LMS	beta31	0.316	0.311	-0.005 $\pm$ 0.002	-0.015 $\pm$ 0.008	0.006 $\pm$ 0.000	0.077 $\pm$ 0.002	0.006 $\pm$ 0.000	0.077 $\pm$ 0.002	0.060 $\pm$ 0.003	0.245 $\pm$ 0.007	0.311	-0.005	-1.52	0.0766	0.0768	1.027	95.560 $\pm$ 0.654	0.311 $\pm$ 0.001	96.165 $\pm$ 0.610
LMS	beta32	0.316	0.319	0.003 $\pm$ 0.002	0.010 $\pm$ 0.008	0.006 $\pm$ 0.000	0.077 $\pm$ 0.002	0.006 $\pm$ 0.000	0.077 $\pm$ 0.002	0.060 $\pm$ 0.003	0.245 $\pm$ 0.007	0.320	0.004	1.12	0.0781	0.0782	1.027	95.560 $\pm$ 0.654	0.311 $\pm$ 0.002	97.175 $\pm$ 0.526
LMS	beta33	0.139	0.124	-0.015 $\pm$ 0.002	-0.107 $\pm$ 0.017	0.005 $\pm$ 0.000	0.072 $\pm$ 0.002	0.005 $\pm$ 0.000	0.074 $\pm$ 0.002	0.283 $\pm$ 0.015	0.532 $\pm$ 0.016	0.127	-0.012	-8.99	0.0652	0.0664	0.896	91.826 $\pm$ 0.870	0.254 $\pm$ 0.003	54.591 $\pm$ 1.582
LMS	beta34	0.101	0.121	0.020 $\pm$ 0.001	0.196 $\pm$ 0.013	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.197 $\pm$ 0.010	0.444 $\pm$ 0.013	0.120	0.019	19.15	0.0378	0.0425	0.889	91.019 $\pm$ 0.908	0.140 $\pm$ 0.002	89.203 $\pm$ 0.986
LMS	beta35	0.101	0.120	0.019 $\pm$ 0.001	0.189 $\pm$ 0.013	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.198 $\pm$ 0.011	0.445 $\pm$ 0.014	0.116	0.015	15.19	0.0357	0.0388	0.884	90.515 $\pm$ 0.931	0.141 $\pm$ 0.002	90.918 $\pm$ 0.913
LSAM	beta31	0.316	0.311	-0.005 $\pm$ 0.003	-0.014 $\pm$ 0.010	0.010 $\pm$ 0.000	0.099 $\pm$ 0.002	0.010 $\pm$ 0.000	0.099 $\pm$ 0.003	0.098 $\pm$ 0.005	0.313 $\pm$ 0.009	0.311	-0.005	-1.70	0.0971	0.0972	1.011	95.459 $\pm$ 0.661	0.392 $\pm$ 0.002	86.276 $\pm$ 1.093
LSAM	beta32	0.316	0.321	0.005 $\pm$ 0.003	0.014 $\pm$ 0.010	0.010 $\pm$ 0.000	0.100 $\pm$ 0.002	0.010 $\pm$ 0.000	0.100 $\pm$ 0.003	0.100 $\pm$ 0.004	0.316 $\pm$ 0.009	0.321	0.005	1.54	0.1001	0.1002	1.002	95.560 $\pm$ 0.654	0.392 $\pm$ 0.002	87.386 $\pm$ 1.055
LSAM	beta33	0.139	0.145	0.006 $\pm$ 0.002	0.042 $\pm$ 0.016	0.005 $\pm$ 0.000	0.070 $\pm$ 0.002	0.005 $\pm$ 0.000	0.071 $\pm$ 0.002	0.257 $\pm$ 0.013	0.507 $\pm$ 0.015	0.145	0.006	4.20	0.0644	0.0646	0.866	90.716 $\pm$ 0.922	0.239 $\pm$ 0.002	68.416 $\pm$ 1.477
LSAM	beta34	0.101	0.100	-0.001 $\pm$ 0.001	-0.012 $\pm$ 0.013	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.170 $\pm$ 0.009	0.413 $\pm$ 0.012	0.100	-0.001	-0.55	0.0393	0.0393	0.864	90.716 $\pm$ 0.922	0.141 $\pm$ 0.002	76.589 $\pm$ 1.345
LSAM	beta35	0.101	0.099	-0.002 $\pm$ 0.001	-0.024 $\pm$ 0.013	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.172 $\pm$ 0.009	0.415 $\pm$ 0.013	0.097	-0.004	-4.42	0.0379	0.0382	0.874	91.726 $\pm$ 0.875	0.144 $\pm$ 0.002	74.874 $\pm$ 1.378
QML	beta31	0.316	0.311	-0.005 $\pm$ 0.002	-0.015 $\pm$ 0.008	0.006 $\pm$ 0.000	0.077 $\pm$ 0.002	0.006 $\pm$ 0.000	0.077 $\pm$ 0.002	0.060 $\pm$ 0.003	0.245 $\pm$ 0.007	0.310	-0.006	-1.80	0.0762	0.0764	1.025	95.459 $\pm$ 0.661	0.311 $\pm$ 0.002	96.468 $\pm$ 0.586
QML	beta32	0.316	0.319	0.003 $\pm$ 0.002	0.009 $\pm$ 0.008	0.006 $\pm$ 0.000	0.077 $\pm$ 0.002	0.006 $\pm$ 0.000	0.078 $\pm$ 0.002	0.060 $\pm$ 0.003	0.245 $\pm$ 0.007	0.320	0.004	1.18	0.0774	0.0775	1.021	95.762 $\pm$ 0.640	0.310 $\pm$ 0.001	97.275 $\pm$ 0.517
QML	beta33	0.139	0.122	-0.017 $\pm$ 0.002	-0.123 $\pm$ 0.016	0.005 $\pm$ 0.000	0.072 $\pm$ 0.002	0.005 $\pm$ 0.000	0.074 $\pm$ 0.002	0.280 $\pm$ 0.015	0.529 $\pm$ 0.016	0.124	-0.015	-10.97	0.0647	0.0664	0.887	91.322 $\pm$ 0.894	0.249 $\pm$ 0.002	54.591 $\pm$ 1.582
QML	beta34	0.101	0.120	0.019 $\pm$ 0.001	0.189 $\pm$ 0.013	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.192 $\pm$ 0.010	0.438 $\pm$ 0.012	0.119	0.018	18.03	0.0369	0.0412	0.876	90.010 $\pm$ 0.953	0.137 $\pm$ 0.002	89.102 $\pm$ 0.990
QML	beta35	0.101	0.120	0.019 $\pm$ 0.001	0.185 $\pm$ 0.013	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.196 $\pm$ 0.011	0.442 $\pm$ 0.014	0.116	0.015	15.22	0.0354	0.0386	0.876	90.212 $\pm$ 0.944	0.139 $\pm$ 0.002	91.120 $\pm$ 0.904
UPI	beta31	0.316	0.316	0.000 $\pm$ 0.003	0.002 $\pm$ 0.010	0.010 $\pm$ 0.000	0.098 $\pm$ 0.002	0.010 $\pm$ 0.000	0.098 $\pm$ 0.003	0.095 $\pm$ 0.005	0.309 $\pm$ 0.009	0.312	-0.004	-1.30	0.0964	0.0965	1.004	95.257 $\pm$ 0.675	0.384 $\pm$ 0.002	89.707 $\pm$ 0.965
UPI	beta32	0.316	0.325	0.009 $\pm$ 0.003	0.029 $\pm$ 0.010	0.010 $\pm$ 0.000	0.098 $\pm$ 0.002	0.010 $\pm$ 0.000	0.098 $\pm$ 0.003	0.096 $\pm$ 0.004	0.310 $\pm$ 0.008	0.325	0.009	2.94	0.1007	0.1011	1.001	95.560 $\pm$ 0.654	0.383 $\pm$ 0.002	89.001 $\pm$ 0.994
UPI	beta33	0.139	0.143	0.004 $\pm$ 0.002	0.026 $\pm$ 0.016	0.005 $\pm$ 0.000	0.070 $\pm$ 0.002	0.005 $\pm$ 0.000	0.070 $\pm$ 0.002	0.256 $\pm$ 0.013	0.506 $\pm$ 0.015	0.142	0.003	2.50	0.0669	0.0670	0.878	92.331 $\pm$ 0.845	0.242 $\pm$ 0.003	66.498 $\pm$ 1.499
UPI	beta34	0.101	0.100	-0.001 $\pm$ 0.001	-0.014 $\pm$ 0.013	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.172 $\pm$ 0.009	0.415 $\pm$ 0.013	0.100	-0.001	-1.26	0.0389	0.0389	0.858	91.019 $\pm$ 0.908	0.141 $\pm$ 0.002	76.993 $\pm$ 1.337
UPI	beta35	0.101	0.098	-0.003 $\pm$ 0.001	-0.027 $\pm$ 0.013	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	0.167 $\pm$ 0.009	0.408 $\pm$ 0.013	0.097	-0.004	-3.51	0.0369	0.0370	0.878	91.120 $\pm$ 0.904	0.142 $\pm$ 0.002	74.672 $\pm$ 1.381

Note. MCSE = Monte Carlo Standard Error shown as $\backslash u00b1$ value. Coverage, Type I Error, and Power expressed as percentages. $\backslash u2014$ indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S66

Simulation 1 - Condition 66: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.8

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.315	-0.001 ± 0.002	-0.004 ± 0.005	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.026 ± 0.001	0.160 ± 0.004	0.316	0.000	-0.02	0.0501	0.0501	0.980	93.700 ± 0.768	0.195 ± 0.001	100.000 ± 0.000
LMS	beta32	0.316	0.316	0.000 ± 0.002	0.001 ± 0.005	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.024 ± 0.001	0.156 ± 0.004	0.318	0.002	0.74	0.0477	0.0478	1.008	95.300 ± 0.669	0.195 ± 0.001	100.000 ± 0.000
LMS	beta33	0.139	0.128	-0.011 ± 0.001	-0.082 ± 0.010	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.100 ± 0.004	0.316 ± 0.009	0.130	-0.009	-6.66	0.0425	0.0435	0.949	92.100 ± 0.853	0.158 ± 0.001	86.200 ± 1.091
LMS	beta34	0.101	0.120	0.019 ± 0.001	0.192 ± 0.007	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.090 ± 0.004	0.301 ± 0.008	0.119	0.018	18.13	0.0220	0.0286	0.926	86.800 ± 1.070	0.085 ± 0.001	99.700 ± 0.173
LMS	beta35	0.101	0.120	0.019 ± 0.001	0.187 ± 0.007	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.088 ± 0.004	0.296 ± 0.008	0.119	0.018	17.53	0.0227	0.0288	0.937	88.100 ± 1.024	0.085 ± 0.001	100.000 ± 0.000
LSAM	beta31	0.316	0.315	-0.001 ± 0.002	-0.004 ± 0.006	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.040 ± 0.002	0.200 ± 0.006	0.314	-0.002	-0.62	0.0619	0.0619	0.976	94.600 ± 0.715	0.241 ± 0.001	99.700 ± 0.173
LSAM	beta32	0.316	0.316	-0.000 ± 0.002	-0.001 ± 0.006	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.038 ± 0.002	0.194 ± 0.005	0.316	0.000	-0.02	0.0597	0.0597	1.005	95.500 ± 0.656	0.242 ± 0.001	99.900 ± 0.100
LSAM	beta33	0.139	0.144	0.005 ± 0.001	0.034 ± 0.009	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.084 ± 0.004	0.289 ± 0.008	0.143	0.004	3.01	0.0402	0.0404	0.927	91.400 ± 0.887	0.145 ± 0.001	95.400 ± 0.662
LSAM	beta34	0.101	0.100	-0.001 ± 0.001	-0.010 ± 0.007	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	0.053 ± 0.003	0.231 ± 0.007	0.099	-0.002	-1.62	0.0223	0.0223	0.910	93.500 ± 0.780	0.083 ± 0.001	98.400 ± 0.397
LSAM	beta35	0.101	0.100	-0.001 ± 0.001	-0.011 ± 0.007	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	0.053 ± 0.003	0.230 ± 0.007	0.099	-0.002	-1.54	0.0217	0.0218	0.925	92.500 ± 0.833	0.084 ± 0.001	98.500 ± 0.384
QML	beta31	0.316	0.315	-0.001 ± 0.002	-0.003 ± 0.005	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.026 ± 0.001	0.161 ± 0.005	0.316	0.000	-0.08	0.0495	0.0495	0.978	94.100 ± 0.745	0.195 ± 0.001	100.000 ± 0.000
QML	beta32	0.316	0.316	-0.000 ± 0.002	-0.001 ± 0.005	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.024 ± 0.001	0.156 ± 0.004	0.318	0.002	0.74	0.0482	0.0483	1.009	95.500 ± 0.656	0.195 ± 0.001	99.900 ± 0.100
QML	beta33	0.139	0.125	-0.014 ± 0.001	-0.100 ± 0.010	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.102 ± 0.005	0.319 ± 0.008	0.126	-0.013	-9.07	0.0419	0.0437	0.939	91.400 ± 0.887	0.155 ± 0.001	85.500 ± 1.113
QML	beta34	0.101	0.120	0.019 ± 0.001	0.187 ± 0.007	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.088 ± 0.004	0.296 ± 0.008	0.119	0.018	17.74	0.0217	0.0282	0.919	86.300 ± 1.087	0.084 ± 0.001	99.700 ± 0.173
QML	beta35	0.101	0.120	0.019 ± 0.001	0.186 ± 0.007	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.087 ± 0.004	0.295 ± 0.008	0.119	0.018	17.46	0.0224	0.0285	0.933	87.700 ± 1.039	0.084 ± 0.001	100.000 ± 0.000
UPI	beta31	0.316	0.316	0.000 ± 0.002	0.001 ± 0.006	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.040 ± 0.002	0.199 ± 0.006	0.316	0.000	0.03	0.0631	0.0631	0.959	93.900 ± 0.757	0.237 ± 0.001	99.800 ± 0.141
UPI	beta32	0.316	0.317	0.001 ± 0.002	0.003 ± 0.006	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.038 ± 0.002	0.194 ± 0.005	0.317	0.001	0.44	0.0591	0.0591	0.987	95.200 ± 0.676	0.238 ± 0.001	99.800 ± 0.141
UPI	beta33	0.139	0.141	0.002 ± 0.001	0.017 ± 0.009	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.083 ± 0.004	0.289 ± 0.008	0.141	0.002	1.48	0.0394	0.0394	0.921	92.200 ± 0.848	0.145 ± 0.001	95.500 ± 0.656
UPI	beta34	0.101	0.101	-0.000 ± 0.001	-0.003 ± 0.007	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	0.053 ± 0.003	0.230 ± 0.007	0.101	0.000	-0.07	0.0221	0.0221	0.907	93.300 ± 0.791	0.082 ± 0.001	98.600 ± 0.372
UPI	beta35	0.101	0.101	-0.000 ± 0.001	-0.002 ± 0.007	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	0.053 ± 0.003	0.231 ± 0.007	0.100	-0.001	-0.89	0.0217	0.0217	0.912	92.800 ± 0.817	0.083 ± 0.001	98.500 ± 0.384

Note. MCSE = Monte Carlo Standard Error shown as $\backslash 00b1$ value. Coverage, Type I Error, and Power expressed as percentages. $\backslash 02014$ indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S67

Simulation 1 - Condition 67: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.4

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.306	-0.010 ± 0.012	-0.031 ± 0.039	0.007 ± 0.002	0.082 ± 0.012	0.007 ± 0.002	0.082 ± 0.012	0.067 ± 0.017	0.259 ± 0.039	0.305	-0.011	-3.61	0.0879	0.0886	1.155	97.727 ± 2.247	0.372 ± 0.013	90.909 ± 4.334
LMS	beta32	0.316	0.309	-0.007 ± 0.013	-0.021 ± 0.040	0.007 ± 0.001	0.085 ± 0.008	0.007 ± 0.001	0.084 ± 0.010	0.071 ± 0.013	0.266 ± 0.033	0.302	-0.014	-4.34	0.1324	0.1331	1.132	93.182 ± 3.800	0.376 ± 0.017	90.909 ± 4.334
LMS	beta33	0.139	0.191	0.052 ± 0.023	0.377 ± 0.164	0.023 ± 0.008	0.151 ± 0.032	0.025 ± 0.007	0.158 ± 0.027	1.297 ± 0.374	1.139 ± 0.194	0.184	0.045	32.52	0.1940	0.1992	0.874	100.000 ± 0.000	0.517 ± 0.032	50.000 ± 7.538
LMS	beta34	0.101	-0.009	-0.110 ± 0.013	-1.087 ± 0.127	0.007 ± 0.003	0.085 ± 0.025	0.019 ± 0.003	0.138 ± 0.010	1.873 ± 0.254	1.368 ± 0.103	-0.002	-0.103	-101.54	0.0969	0.1411	1.056	70.455 ± 6.878	0.352 ± 0.021	9.091 ± 4.334
LMS	beta35	0.101	0.009	-0.092 ± 0.021	-0.913 ± 0.207	0.019 ± 0.013	0.139 ± 0.063	0.027 ± 0.010	0.165 ± 0.035	2.672 ± 0.960	1.635 ± 0.345	0.009	-0.092	-90.85	0.1342	0.1626	0.796	68.182 ± 7.022	0.432 ± 0.053	2.273 ± 2.247
LSAM	beta31	0.316	0.308	-0.008 ± 0.036	-0.026 ± 0.115	0.058 ± 0.017	0.240 ± 0.038	0.056 ± 0.017	0.238 ± 0.042	0.565 ± 0.167	0.752 ± 0.132	0.344	0.028	8.81	0.1501	0.1526	1.508	97.727 ± 2.247	1.420 ± 0.359	45.455 ± 7.507
LSAM	beta32	0.316	0.396	0.080 ± 0.035	0.254 ± 0.110	0.053 ± 0.013	0.231 ± 0.030	0.058 ± 0.015	0.242 ± 0.036	0.585 ± 0.147	0.765 ± 0.114	0.346	0.030	9.65	0.2319	0.2339	1.712	100.000 ± 0.000	1.548 ± 0.388	45.455 ± 7.507
LSAM	beta33	0.139	0.308	0.169 ± 0.030	1.218 ± 0.219	0.041 ± 0.010	0.202 ± 0.027	0.068 ± 0.018	0.262 ± 0.038	3.542 ± 0.927	1.882 ± 0.270	0.274	0.135	97.05	0.2086	0.2484	1.443	100.000 ± 0.000	1.141 ± 0.245	20.455 ± 6.081
LSAM	beta34	0.101	-0.063	-0.164 ± 0.027	-1.625 ± 0.268	0.032 ± 0.008	0.179 ± 0.024	0.058 ± 0.012	0.241 ± 0.027	5.717 ± 1.170	2.391 ± 0.271	-0.060	-0.161	-159.74	0.1716	0.2355	1.514	86.364 ± 5.174	1.064 ± 0.183	2.273 ± 2.247
LSAM	beta35	0.101	-0.127	-0.228 ± 0.045	-2.254 ± 0.442	0.088 ± 0.027	0.296 ± 0.050	0.138 ± 0.029	0.371 ± 0.044	13.480 ± 2.846	3.672 ± 0.437	-0.058	-0.159	-157.63	0.2628	0.3072	1.511	88.636 ± 4.785	1.754 ± 0.465	0.000 ± 0.000
QML	beta31	0.316	0.303	-0.013 ± 0.012	-0.040 ± 0.039	0.007 ± 0.002	0.083 ± 0.011	0.007 ± 0.002	0.083 ± 0.012	0.068 ± 0.016	0.261 ± 0.038	0.309	-0.007	-2.23	0.0788	0.0791	1.098	97.727 ± 2.247	0.355 ± 0.010	93.182 ± 3.800
QML	beta32	0.316	0.307	-0.009 ± 0.013	-0.030 ± 0.040	0.007 ± 0.001	0.083 ± 0.009	0.007 ± 0.001	0.083 ± 0.011	0.068 ± 0.013	0.261 ± 0.034	0.290	-0.026	-8.30	0.0811	0.0852	1.093	93.182 ± 3.800	0.355 ± 0.009	93.182 ± 3.800
QML	beta33	0.139	0.211	0.072 ± 0.017	0.520 ± 0.123	0.013 ± 0.002	0.113 ± 0.011	0.018 ± 0.003	0.133 ± 0.015	0.917 ± 0.178	0.958 ± 0.109	0.189	0.050	36.06	0.1479	0.1562	1.182	97.727 ± 2.247	0.524 ± 0.027	47.727 ± 7.530
QML	beta34	0.101	-0.038	-0.139 ± 0.010	-1.381 ± 0.099	0.004 ± 0.001	0.066 ± 0.008	0.024 ± 0.003	0.154 ± 0.009	2.326 ± 0.273	1.525 ± 0.092	-0.036	-0.137	-135.59	0.0820	0.1596	1.309	61.364 ± 7.341	0.339 ± 0.018	9.091 ± 4.334
QML	beta35	0.101	-0.029	-0.130 ± 0.012	-1.288 ± 0.119	0.006 ± 0.002	0.080 ± 0.013	0.023 ± 0.003	0.152 ± 0.010	2.265 ± 0.295	1.505 ± 0.104	-0.010	-0.111	-109.41	0.1101	0.1560	1.187	68.182 ± 7.022	0.370 ± 0.025	4.545 ± 3.140
UPI	beta31	0.316	0.262	-0.054 ± 0.039	-0.171 ± 0.125	0.068 ± 0.017	0.261 ± 0.037	0.070 ± 0.016	0.264 ± 0.037	0.698 ± 0.156	0.835 ± 0.117	0.332	0.016	5.03	0.4569	0.4572	1.588	84.091 ± 5.514	1.627 ± 0.297	20.455 ± 6.081
UPI	beta32	0.316	0.401	0.085 ± 0.045	0.269 ± 0.144	0.091 ± 0.021	0.302 ± 0.038	0.096 ± 0.026	0.310 ± 0.049	0.962 ± 0.256	0.981 ± 0.154	0.298	-0.018	-5.82	0.3957	0.3961	1.444	93.182 ± 3.800	1.707 ± 0.271	20.455 ± 6.081
UPI	beta33	0.139	0.186	0.047 ± 0.031	0.341 ± 0.226	0.044 ± 0.011	0.209 ± 0.029	0.045 ± 0.013	0.212 ± 0.037	2.320 ± 0.685	1.523 ± 0.263	0.142	0.003	1.86	0.2816	0.2816	2.280	100.000 ± 0.000	1.866 ± 0.444	4.545 ± 3.140
UPI	beta34	0.101	-0.129	-0.230 ± 0.072	-2.274 ± 0.709	0.226 ± 0.070	0.475 ± 0.082	0.274 ± 0.096	0.523 ± 0.103	26.813 ± 9.379	5.178 ± 1.020	-0.151	-0.252	-249.69	0.6406	0.6885	1.547	90.909 ± 4.334	2.882 ± 0.594	0.000 ± 0.000
UPI	beta35	0.101	-0.342	-0.443 ± 0.097	-4.390 ± 0.957	0.411 ± 0.140	0.641 ± 0.123	0.598 ± 0.208	0.773 ± 0.150	58.629 ± 20.437	7.657 ± 1.485	-0.143	-0.244	-241.33	0.6524	0.6964	1.842	88.636 ± 4.785	4.628 ± 0.973	0.000 ± 0.000

Note. MCSE = Monte Carlo Standard Error shown as $\backslash u00b1$ value. Coverage, Type I Error, and Power expressed as percentages. $\backslash u2014$ indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S68

Simulation 1 - Condition 68: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.4

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.321	0.005 ± 0.009	0.016 ± 0.029	0.004 ± 0.001	0.061 ± 0.007	0.004 ± 0.001	0.060 ± 0.008	0.037 ± 0.008	0.191 ± 0.027	0.316	0.000	0.08	0.0820	0.0820	0.964	97.727 ± 2.247	0.230 ± 0.007	100.000 ± 0.000
LMS	beta32	0.316	0.313	-0.003 ± 0.010	-0.010 ± 0.032	0.005 ± 0.001	0.068 ± 0.008	0.005 ± 0.001	0.067 ± 0.009	0.046 ± 0.009	0.213 ± 0.028	0.318	0.002	0.53	0.0890	0.0890	0.924	95.455 ± 3.140	0.247 ± 0.014	100.000 ± 0.000
LMS	beta33	0.139	0.239	0.100 ± 0.012	0.716 ± 0.089	0.007 ± 0.002	0.082 ± 0.017	0.016 ± 0.004	0.128 ± 0.017	0.851 ± 0.210	0.922 ± 0.124	0.203	0.064	46.37	0.1126	0.1297	1.046	79.545 ± 6.081	0.335 ± 0.024	86.364 ± 5.174
LMS	beta34	0.101	0.002	-0.099 ± 0.009	-0.982 ± 0.090	0.004 ± 0.001	0.060 ± 0.009	0.013 ± 0.001	0.116 ± 0.006	1.313 ± 0.125	1.146 ± 0.058	-0.008	-0.109	-108.02	0.0643	0.1266	1.107	45.455 ± 7.507	0.261 ± 0.039	4.545 ± 3.140
LMS	beta35	0.101	0.011	-0.090 ± 0.012	-0.891 ± 0.115	0.006 ± 0.003	0.077 ± 0.020	0.014 ± 0.004	0.118 ± 0.019	1.367 ± 0.390	1.169 ± 0.185	0.017	-0.084	-82.98	0.0842	0.1188	0.962	56.818 ± 7.467	0.291 ± 0.033	2.273 ± 2.247
LSAM	beta31	0.316	0.365	0.049 ± 0.017	0.156 ± 0.053	0.013 ± 0.002	0.112 ± 0.011	0.015 ± 0.003	0.121 ± 0.015	0.147 ± 0.029	0.383 ± 0.047	0.339	0.023	7.18	0.1324	0.1344	1.012	97.727 ± 2.247	0.444 ± 0.027	88.636 ± 4.785
LSAM	beta32	0.316	0.300	-0.016 ± 0.018	-0.051 ± 0.057	0.014 ± 0.003	0.119 ± 0.014	0.014 ± 0.003	0.119 ± 0.017	0.142 ± 0.032	0.377 ± 0.053	0.320	0.004	1.25	0.1198	0.1199	0.921	95.455 ± 3.140	0.431 ± 0.028	79.545 ± 6.081
LSAM	beta33	0.139	0.275	0.136 ± 0.013	0.979 ± 0.095	0.008 ± 0.002	0.088 ± 0.011	0.026 ± 0.004	0.161 ± 0.014	1.347 ± 0.220	1.161 ± 0.100	0.262	0.123	88.58	0.0839	0.1490	1.115	81.818 ± 5.815	0.383 ± 0.016	86.364 ± 5.174
LSAM	beta34	0.101	-0.084	-0.185 ± 0.015	-1.834 ± 0.153	0.011 ± 0.002	0.103 ± 0.012	0.045 ± 0.006	0.211 ± 0.016	4.375 ± 0.634	2.092 ± 0.158	-0.080	-0.181	-179.51	0.0917	0.2032	1.014	45.455 ± 7.507	0.409 ± 0.029	13.636 ± 5.174
LSAM	beta35	0.101	-0.027	-0.128 ± 0.016	-1.267 ± 0.160	0.011 ± 0.003	0.107 ± 0.015	0.028 ± 0.004	0.166 ± 0.013	2.705 ± 0.373	1.645 ± 0.126	-0.046	-0.147	-146.03	0.0898	0.1727	0.984	56.818 ± 7.467	0.413 ± 0.037	2.273 ± 2.247
QML	beta31	0.316	0.311	-0.005 ± 0.009	-0.016 ± 0.027	0.003 ± 0.001	0.057 ± 0.007	0.003 ± 0.001	0.057 ± 0.008	0.032 ± 0.007	0.180 ± 0.025	0.304	-0.012	-3.93	0.0711	0.0721	0.955	97.727 ± 2.247	0.215 ± 0.003	100.000 ± 0.000
QML	beta32	0.316	0.297	-0.019 ± 0.009	-0.060 ± 0.029	0.004 ± 0.001	0.061 ± 0.007	0.004 ± 0.001	0.063 ± 0.008	0.040 ± 0.007	0.200 ± 0.024	0.304	-0.012	-3.80	0.0598	0.0610	0.910	93.182 ± 3.800	0.218 ± 0.003	100.000 ± 0.000
QML	beta33	0.139	0.219	0.080 ± 0.009	0.572 ± 0.064	0.003 ± 0.001	0.059 ± 0.008	0.010 ± 0.001	0.099 ± 0.008	0.504 ± 0.075	0.710 ± 0.057	0.217	0.078	55.80	0.0593	0.0976	1.249	81.818 ± 5.815	0.289 ± 0.009	88.636 ± 4.785
QML	beta34	0.101	-0.026	-0.127 ± 0.008	-1.262 ± 0.074	0.002 ± 0.001	0.050 ± 0.007	0.019 ± 0.002	0.137 ± 0.006	1.830 ± 0.162	1.353 ± 0.061	-0.040	-0.141	-139.68	0.0419	0.1472	1.036	27.273 ± 6.714	0.202 ± 0.009	11.364 ± 4.785
QML	beta35	0.101	-0.011	-0.112 ± 0.007	-1.105 ± 0.071	0.002 ± 0.001	0.048 ± 0.006	0.015 ± 0.002	0.121 ± 0.007	1.438 ± 0.155	1.199 ± 0.066	-0.021	-0.122	-121.23	0.0530	0.1334	1.094	40.909 ± 7.412	0.204 ± 0.009	11.364 ± 4.785
UPI	beta31	0.316	0.314	-0.002 ± 0.071	-0.007 ± 0.225	0.223 ± 0.090	0.473 ± 0.112	0.218 ± 0.089	0.467 ± 0.114	2.186 ± 0.895	1.478 ± 0.362	0.356	0.040	12.52	0.5006	0.5022	1.504	81.818 ± 5.815	2.786 ± 0.634	27.273 ± 6.714
UPI	beta32	0.316	0.404	0.088 ± 0.128	0.280 ± 0.406	0.724 ± 0.301	0.851 ± 0.202	0.715 ± 0.307	0.846 ± 0.210	7.161 ± 3.078	2.676 ± 0.664	0.317	0.001	0.23	0.4987	0.4987	1.228	81.818 ± 5.815	4.094 ± 1.073	20.455 ± 6.081
UPI	beta33	0.139	0.116	-0.023 ± 0.048	-0.169 ± 0.345	0.101 ± 0.028	0.318 ± 0.048	0.099 ± 0.029	0.315 ± 0.054	5.139 ± 1.475	2.267 ± 0.386	0.129	-0.010	-6.96	0.3876	0.3877	1.932	100.000 ± 0.000	2.407 ± 0.452	18.182 ± 5.815
UPI	beta34	0.101	-0.826	-0.927 ± 0.360	-9.174 ± 3.560	5.690 ± 3.164	2.385 ± 0.813	6.419 ± 3.774	2.534 ± 0.889	629.265 ± 370.011	25.085 ± 8.802	-0.436	-0.537	-531.95	1.1663	1.2841	0.642	93.182 ± 3.800	6.004 ± 1.201	2.273 ± 2.247
UPI	beta35	0.101	-0.615	-0.716 ± 0.285	-7.089 ± 2.824	3.580 ± 1.586	1.892 ± 0.492	4.011 ± 1.941	2.003 ± 0.561	393.199 ± 190.287	19.829 ± 5.552	-0.248	-0.349	-345.60	1.0755	1.1307	0.889	86.364 ± 5.174	6.590 ± 1.635	0.000 ± 0.000

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S69

Simulation 1 - Condition 69: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.6

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.309	-0.007 ± 0.003	-0.023 ± 0.008	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.049 ± 0.003	0.221 ± 0.007	0.308	-0.008	-2.41	0.0722	0.0726	1.024	95.640 ± 0.779	0.280 ± 0.002	98.692 ± 0.433
LMS	beta32	0.316	0.315	-0.001 ± 0.003	-0.005 ± 0.009	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.052 ± 0.003	0.228 ± 0.008	0.313	-0.003	-1.02	0.0736	0.0736	0.992	96.076 ± 0.740	0.281 ± 0.003	98.110 ± 0.519
LMS	beta33	0.139	0.171	0.032 ± 0.003	0.229 ± 0.024	0.008 ± 0.000	0.087 ± 0.003	0.009 ± 0.000	0.092 ± 0.003	0.442 ± 0.025	0.664 ± 0.022	0.171	0.032	23.13	0.0827	0.0888	1.102	90.988 ± 1.092	0.375 ± 0.010	55.233 ± 1.896
LMS	beta34	0.101	0.049	-0.052 ± 0.003	-0.511 ± 0.031	0.007 ± 0.000	0.083 ± 0.003	0.010 ± 0.000	0.098 ± 0.003	0.933 ± 0.041	0.966 ± 0.025	0.037	-0.064	-63.52	0.0757	0.0992	1.036	77.907 ± 1.582	0.336 ± 0.011	5.087 ± 0.838
LMS	beta35	0.101	0.046	-0.055 ± 0.003	-0.545 ± 0.030	0.006 ± 0.000	0.078 ± 0.003	0.009 ± 0.000	0.096 ± 0.002	0.897 ± 0.039	0.947 ± 0.024	0.039	-0.062	-61.70	0.0724	0.0955	1.095	75.872 ± 1.631	0.336 ± 0.009	1.744 ± 0.499
LSAM	beta31	0.316	0.311	-0.005 ± 0.003	-0.017 ± 0.009	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.078 ± 0.003	0.061 ± 0.003	0.246 ± 0.008	0.311	-0.005	-1.53	0.0791	0.0792	1.014	94.913 ± 0.838	0.309 ± 0.002	96.802 ± 0.671
LSAM	beta32	0.316	0.317	0.001 ± 0.003	0.004 ± 0.010	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.079 ± 0.003	0.062 ± 0.003	0.249 ± 0.008	0.319	0.003	1.07	0.0797	0.0798	1.003	95.785 ± 0.766	0.310 ± 0.002	97.093 ± 0.641
LSAM	beta33	0.139	0.188	0.049 ± 0.003	0.356 ± 0.024	0.008 ± 0.000	0.089 ± 0.003	0.010 ± 0.001	0.102 ± 0.003	0.539 ± 0.032	0.734 ± 0.024	0.187	0.048	34.22	0.0822	0.0949	1.015	90.552 ± 1.115	0.355 ± 0.003	58.430 ± 1.879
LSAM	beta34	0.101	0.045	-0.056 ± 0.004	-0.557 ± 0.042	0.012 ± 0.001	0.111 ± 0.003	0.016 ± 0.001	0.125 ± 0.004	1.524 ± 0.078	1.234 ± 0.037	0.034	-0.067	-66.57	0.1069	0.1263	0.969	85.901 ± 1.327	0.423 ± 0.006	5.959 ± 0.903
LSAM	beta35	0.101	0.037	-0.064 ± 0.004	-0.633 ± 0.038	0.010 ± 0.001	0.101 ± 0.003	0.014 ± 0.001	0.119 ± 0.003	1.399 ± 0.071	1.183 ± 0.034	0.037	-0.064	-63.69	0.1019	0.1205	1.053	85.901 ± 1.327	0.417 ± 0.005	3.052 ± 0.656
QML	beta31	0.316	0.305	-0.011 ± 0.003	-0.035 ± 0.008	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.048 ± 0.002	0.218 ± 0.007	0.305	-0.011	-3.51	0.0717	0.0725	1.012	94.622 ± 0.860	0.271 ± 0.002	98.983 ± 0.383
QML	beta32	0.316	0.310	-0.006 ± 0.003	-0.018 ± 0.008	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.049 ± 0.003	0.222 ± 0.007	0.310	-0.006	-1.75	0.0724	0.0726	0.986	95.494 ± 0.791	0.271 ± 0.002	98.692 ± 0.433
QML	beta33	0.139	0.171	0.032 ± 0.003	0.232 ± 0.023	0.007 ± 0.000	0.082 ± 0.002	0.008 ± 0.000	0.088 ± 0.003	0.403 ± 0.022	0.635 ± 0.020	0.171	0.032	22.99	0.0796	0.0858	1.064	90.698 ± 1.107	0.343 ± 0.008	56.105 ± 1.892
QML	beta34	0.101	0.035	-0.066 ± 0.003	-0.658 ± 0.027	0.005 ± 0.000	0.072 ± 0.002	0.010 ± 0.000	0.098 ± 0.002	0.939 ± 0.038	0.969 ± 0.022	0.026	-0.075	-73.90	0.0691	0.1017	1.012	72.965 ± 1.693	0.285 ± 0.008	6.831 ± 0.962
QML	beta35	0.101	0.031	-0.070 ± 0.003	-0.693 ± 0.025	0.004 ± 0.000	0.066 ± 0.002	0.009 ± 0.000	0.096 ± 0.002	0.912 ± 0.037	0.955 ± 0.021	0.029	-0.072	-71.78	0.0683	0.0996	1.107	73.837 ± 1.676	0.288 ± 0.009	3.343 ± 0.685
UPI	beta31	0.316	0.320	0.004 ± 0.004	0.014 ± 0.012	0.009 ± 0.001	0.097 ± 0.003	0.009 ± 0.001	0.097 ± 0.004	0.094 ± 0.006	0.307 ± 0.011	0.324	0.008	2.48	0.0923	0.0926	1.331	97.384 ± 0.609	0.506 ± 0.020	80.233 ± 1.518
UPI	beta32	0.316	0.326	0.010 ± 0.004	0.033 ± 0.012	0.009 ± 0.001	0.097 ± 0.003	0.010 ± 0.001	0.098 ± 0.003	0.095 ± 0.006	0.309 ± 0.011	0.327	0.011	3.48	0.0958	0.0964	1.312	97.529 ± 0.592	0.500 ± 0.021	81.250 ± 1.488
UPI	beta33	0.139	0.197	0.058 ± 0.004	0.416 ± 0.026	0.009 ± 0.001	0.095 ± 0.003	0.012 ± 0.001	0.111 ± 0.004	0.637 ± 0.038	0.798 ± 0.026	0.194	0.055	39.72	0.0917	0.1070	1.276	95.058 ± 0.826	0.474 ± 0.010	45.494 ± 1.898
UPI	beta34	0.101	0.052	-0.049 ± 0.007	-0.484 ± 0.071	0.036 ± 0.003	0.189 ± 0.007	0.038 ± 0.003	0.195 ± 0.008	3.712 ± 0.274	1.927 ± 0.079	0.045	-0.056	-55.46	0.1727	0.1815	1.264	95.058 ± 0.826	0.934 ± 0.043	4.942 ± 0.826
UPI	beta35	0.101	0.050	-0.051 ± 0.007	-0.503 ± 0.066	0.030 ± 0.002	0.174 ± 0.006	0.033 ± 0.002	0.181 ± 0.007	3.203 ± 0.225	1.790 ± 0.070	0.055	-0.046	-45.73	0.1552	0.1619	1.259	93.750 ± 0.923	0.857 ± 0.036	3.924 ± 0.740

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S70

Simulation 1 - Condition 70: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.6

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LMS	beta31	0.316	0.311	-0.005 $\pm$ 0.002	-0.017 $\pm$ 0.005	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.020 $\pm$ 0.001	0.141 $\pm$ 0.005	0.311	-0.005	-1.71	0.0452	0.0455	0.983	94.902 $\pm$ 0.795	0.170 $\pm$ 0.001	100.000 $\pm$ 0.000
LMS	beta32	0.316	0.313	-0.003 $\pm$ 0.002	-0.010 $\pm$ 0.005	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.019 $\pm$ 0.001	0.138 $\pm$ 0.004	0.314	-0.002	-0.74	0.0435	0.0436	0.998	95.425 $\pm$ 0.755	0.171 $\pm$ 0.001	100.000 $\pm$ 0.000
LMS	beta33	0.139	0.175	0.036 $\pm$ 0.002	0.261 $\pm$ 0.014	0.003 $\pm$ 0.000	0.053 $\pm$ 0.002	0.004 $\pm$ 0.000	0.064 $\pm$ 0.002	0.212 $\pm$ 0.009	0.461 $\pm$ 0.012	0.178	0.039	28.31	0.0508	0.0643	1.017	87.451 $\pm$ 1.198	0.211 $\pm$ 0.002	87.582 $\pm$ 1.192
LMS	beta34	0.101	0.041	-0.060 $\pm$ 0.002	-0.590 $\pm$ 0.018	0.003 $\pm$ 0.000	0.050 $\pm$ 0.002	0.006 $\pm$ 0.000	0.078 $\pm$ 0.001	0.596 $\pm$ 0.020	0.772 $\pm$ 0.014	0.035	-0.066	-65.58	0.0456	0.0804	0.971	64.314 $\pm$ 1.732	0.191 $\pm$ 0.003	5.882 $\pm$ 0.851
LMS	beta35	0.101	0.043	-0.058 $\pm$ 0.002	-0.575 $\pm$ 0.018	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	0.006 $\pm$ 0.000	0.077 $\pm$ 0.001	0.576 $\pm$ 0.020	0.759 $\pm$ 0.014	0.036	-0.065	-64.85	0.0435	0.0786	0.977	66.144 $\pm$ 1.711	0.191 $\pm$ 0.002	6.928 $\pm$ 0.918
LSAM	beta31	0.316	0.312	-0.004 $\pm$ 0.002	-0.014 $\pm$ 0.006	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	0.002 $\pm$ 0.000	0.050 $\pm$ 0.002	0.025 $\pm$ 0.001	0.158 $\pm$ 0.005	0.312	-0.004	-1.31	0.0485	0.0487	0.969	94.118 $\pm$ 0.851	0.189 $\pm$ 0.001	100.000 $\pm$ 0.000
LSAM	beta32	0.316	0.313	-0.003 $\pm$ 0.002	-0.009 $\pm$ 0.005	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.023 $\pm$ 0.001	0.151 $\pm$ 0.005	0.311	-0.005	-1.71	0.0486	0.0489	1.008	95.163 $\pm$ 0.776	0.189 $\pm$ 0.001	100.000 $\pm$ 0.000
LSAM	beta33	0.139	0.180	0.041 $\pm$ 0.002	0.298 $\pm$ 0.014	0.003 $\pm$ 0.000	0.054 $\pm$ 0.001	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.239 $\pm$ 0.011	0.489 $\pm$ 0.013	0.181	0.042	29.99	0.0531	0.0675	0.991	88.235 $\pm$ 1.165	0.209 $\pm$ 0.001	91.111 $\pm$ 1.029
LSAM	beta34	0.101	0.041	-0.060 $\pm$ 0.002	-0.596 $\pm$ 0.024	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	0.008 $\pm$ 0.000	0.090 $\pm$ 0.002	0.786 $\pm$ 0.034	0.886 $\pm$ 0.021	0.036	-0.065	-64.34	0.0611	0.0892	0.966	79.085 $\pm$ 1.470	0.251 $\pm$ 0.002	9.804 $\pm$ 1.075
LSAM	beta35	0.101	0.044	-0.057 $\pm$ 0.002	-0.569 $\pm$ 0.024	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.008 $\pm$ 0.000	0.089 $\pm$ 0.002	0.772 $\pm$ 0.031	0.879 $\pm$ 0.020	0.038	-0.063	-62.87	0.0635	0.0898	0.954	80.261 $\pm$ 1.439	0.253 $\pm$ 0.002	9.804 $\pm$ 1.075
QML	beta31	0.316	0.307	-0.009 $\pm$ 0.002	-0.030 $\pm$ 0.005	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.020 $\pm$ 0.001	0.141 $\pm$ 0.005	0.307	-0.009	-2.92	0.0434	0.0443	0.979	93.856 $\pm$ 0.868	0.168 $\pm$ 0.001	100.000 $\pm$ 0.000
QML	beta32	0.316	0.309	-0.007 $\pm$ 0.002	-0.022 $\pm$ 0.005	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.019 $\pm$ 0.001	0.138 $\pm$ 0.004	0.309	-0.007	-2.07	0.0432	0.0437	0.995	95.033 $\pm$ 0.786	0.168 $\pm$ 0.001	100.000 $\pm$ 0.000
QML	beta33	0.139	0.172	0.033 $\pm$ 0.002	0.235 $\pm$ 0.013	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.193 $\pm$ 0.009	0.440 $\pm$ 0.011	0.174	0.035	25.08	0.0508	0.0616	1.012	88.627 $\pm$ 1.148	0.205 $\pm$ 0.001	87.582 $\pm$ 1.192
QML	beta34	0.101	0.030	-0.071 $\pm$ 0.002	-0.700 $\pm$ 0.016	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.007 $\pm$ 0.000	0.084 $\pm$ 0.001	0.690 $\pm$ 0.022	0.830 $\pm$ 0.014	0.026	-0.075	-73.78	0.0428	0.0859	0.985	56.471 $\pm$ 1.793	0.174 $\pm$ 0.002	8.366 $\pm$ 1.001
QML	beta35	0.101	0.031	-0.070 $\pm$ 0.002	-0.689 $\pm$ 0.016	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.007 $\pm$ 0.000	0.083 $\pm$ 0.001	0.675 $\pm$ 0.022	0.822 $\pm$ 0.014	0.026	-0.075	-73.85	0.0421	0.0857	0.974	57.908 $\pm$ 1.785	0.173 $\pm$ 0.001	7.712 $\pm$ 0.965
UPI	beta31	0.316	0.315	-0.001 $\pm$ 0.002	-0.004 $\pm$ 0.008	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	0.044 $\pm$ 0.003	0.210 $\pm$ 0.008	0.316	0.000	-0.04	0.0626	0.0626	1.263	97.124 $\pm$ 0.604	0.328 $\pm$ 0.016	93.203 $\pm$ 0.910
UPI	beta32	0.316	0.317	0.001 $\pm$ 0.002	0.004 $\pm$ 0.007	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.038 $\pm$ 0.002	0.194 $\pm$ 0.007	0.312	-0.004	-1.34	0.0601	0.0602	1.321	97.908 $\pm$ 0.517	0.318 $\pm$ 0.011	93.464 $\pm$ 0.894
UPI	beta33	0.139	0.183	0.044 $\pm$ 0.002	0.314 $\pm$ 0.016	0.004 $\pm$ 0.000	0.062 $\pm$ 0.002	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.294 $\pm$ 0.015	0.543 $\pm$ 0.016	0.186	0.047	33.63	0.0597	0.0758	1.160	94.118 $\pm$ 0.851	0.280 $\pm$ 0.006	77.778 $\pm$ 1.503
UPI	beta34	0.101	0.046	-0.055 $\pm$ 0.004	-0.545 $\pm$ 0.041	0.013 $\pm$ 0.001	0.115 $\pm$ 0.004	0.016 $\pm$ 0.001	0.128 $\pm$ 0.004	1.603 $\pm$ 0.096	1.266 $\pm$ 0.043	0.040	-0.061	-60.21	0.1062	0.1224	1.222	92.418 $\pm$ 0.957	0.553 $\pm$ 0.036	8.235 $\pm$ 0.994
UPI	beta35	0.101	0.051	-0.050 $\pm$ 0.004	-0.491 $\pm$ 0.041	0.013 $\pm$ 0.001	0.115 $\pm$ 0.004	0.016 $\pm$ 0.001	0.125 $\pm$ 0.004	1.531 $\pm$ 0.090	1.237 $\pm$ 0.041	0.047	-0.054	-53.96	0.0985	0.1126	1.173	94.248 $\pm$ 0.842	0.528 $\pm$ 0.015	7.190 $\pm$ 0.934

Note. MCSE = Monte Carlo Standard Error shown as $\backslash$ u00b1 value. Coverage, Type I Error, and Power expressed as percentages. $\backslash$ u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S71

Simulation 1 - Condition 71: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.8

Method	Param.	True	Mean-based									Median-based					Inference				
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE	
LMS	beta31	0.316	0.314	-0.002 $\pm$ 0.002	-0.006 $\pm$ 0.006	0.003 $\pm$ 0.000	0.058 $\pm$ 0.001	0.003 $\pm$ 0.000	0.058 $\pm$ 0.002	0.034 $\pm$ 0.002	0.185 $\pm$ 0.005	0.315	-0.001	-0.30	0.0543	0.0543	0.983	94.294 $\pm$ 0.734	0.225 $\pm$ 0.001	99.700 $\pm$ 0.173	
LMS	beta32	0.316	0.314	-0.002 $\pm$ 0.002	-0.005 $\pm$ 0.006	0.004 $\pm$ 0.000	0.060 $\pm$ 0.001	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	0.036 $\pm$ 0.002	0.189 $\pm$ 0.005	0.316	0.000	0.10	0.0599	0.0599	0.962	94.695 $\pm$ 0.709	0.226 $\pm$ 0.001	99.800 $\pm$ 0.141	
LMS	beta33	0.139	0.148	0.009 $\pm$ 0.002	0.065 $\pm$ 0.015	0.005 $\pm$ 0.000	0.067 $\pm$ 0.002	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.240 $\pm$ 0.011	0.489 $\pm$ 0.014	0.152	0.013	9.28	0.0683	0.0695	0.978	94.194 $\pm$ 0.740	0.259 $\pm$ 0.002	64.965 $\pm$ 1.509	
LMS	beta34	0.101	0.077	-0.024 $\pm$ 0.002	-0.237 $\pm$ 0.021	0.004 $\pm$ 0.000	0.065 $\pm$ 0.002	0.005 $\pm$ 0.000	0.070 $\pm$ 0.002	0.476 $\pm$ 0.020	0.690 $\pm$ 0.018	0.072	-0.029	-29.18	0.0631	0.0696	0.964	88.288 $\pm$ 1.017	0.247 $\pm$ 0.002	16.717 $\pm$ 1.181	
LMS	beta35	0.101	0.080	-0.021 $\pm$ 0.002	-0.209 $\pm$ 0.021	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.005 $\pm$ 0.000	0.071 $\pm$ 0.002	0.501 $\pm$ 0.024	0.708 $\pm$ 0.020	0.074	-0.027	-26.38	0.0678	0.0728	0.929	89.089 $\pm$ 0.986	0.249 $\pm$ 0.003	20.320 $\pm$ 1.273	
LSAM	beta31	0.316	0.316	-0.000 $\pm$ 0.002	-0.001 $\pm$ 0.006	0.004 $\pm$ 0.000	0.060 $\pm$ 0.001	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	0.035 $\pm$ 0.002	0.188 $\pm$ 0.005	0.318	0.002	0.52	0.0573	0.0573	0.984	94.895 $\pm$ 0.696	0.230 $\pm$ 0.001	99.700 $\pm$ 0.173	
LSAM	beta32	0.316	0.316	-0.000 $\pm$ 0.002	-0.001 $\pm$ 0.006	0.004 $\pm$ 0.000	0.061 $\pm$ 0.001	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.037 $\pm$ 0.002	0.192 $\pm$ 0.005	0.317	0.001	0.46	0.0599	0.0599	0.969	95.095 $\pm$ 0.683	0.231 $\pm$ 0.001	99.900 $\pm$ 0.100	
LSAM	beta33	0.139	0.150	0.011 $\pm$ 0.002	0.076 $\pm$ 0.015	0.004 $\pm$ 0.000	0.065 $\pm$ 0.001	0.004 $\pm$ 0.000	0.065 $\pm$ 0.002	0.221 $\pm$ 0.010	0.470 $\pm$ 0.013	0.152	0.013	9.52	0.0659	0.0673	0.978	95.095 $\pm$ 0.683	0.247 $\pm$ 0.001	66.867 $\pm$ 1.489	
LSAM	beta34	0.101	0.088	-0.013 $\pm$ 0.002	-0.126 $\pm$ 0.024	0.006 $\pm$ 0.000	0.076 $\pm$ 0.002	0.006 $\pm$ 0.000	0.077 $\pm$ 0.002	0.575 $\pm$ 0.026	0.758 $\pm$ 0.021	0.084	-0.017	-17.07	0.0742	0.0762	0.956	93.193 $\pm$ 0.797	0.283 $\pm$ 0.001	20.120 $\pm$ 1.268	
LSAM	beta35	0.101	0.091	-0.010 $\pm$ 0.002	-0.100 $\pm$ 0.024	0.006 $\pm$ 0.000	0.077 $\pm$ 0.002	0.006 $\pm$ 0.000	0.078 $\pm$ 0.002	0.591 $\pm$ 0.027	0.769 $\pm$ 0.021	0.088	-0.013	-12.44	0.0751	0.0761	0.935	93.093 $\pm$ 0.802	0.282 $\pm$ 0.001	23.524 $\pm$ 1.342	
QML	beta31	0.316	0.313	-0.003 $\pm$ 0.002	-0.010 $\pm$ 0.006	0.003 $\pm$ 0.000	0.058 $\pm$ 0.001	0.003 $\pm$ 0.000	0.058 $\pm$ 0.002	0.034 $\pm$ 0.002	0.184 $\pm$ 0.005	0.313	-0.003	-0.80	0.0545	0.0545	0.981	94.294 $\pm$ 0.734	0.224 $\pm$ 0.001	99.800 $\pm$ 0.141	
QML	beta32	0.316	0.313	-0.003 $\pm$ 0.002	-0.009 $\pm$ 0.006	0.004 $\pm$ 0.000	0.060 $\pm$ 0.001	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	0.036 $\pm$ 0.002	0.189 $\pm$ 0.005	0.315	-0.001	-0.45	0.0596	0.0596	0.962	94.394 $\pm$ 0.728	0.225 $\pm$ 0.001	99.800 $\pm$ 0.141	
QML	beta33	0.139	0.148	0.009 $\pm$ 0.002	0.064 $\pm$ 0.015	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	0.004 $\pm$ 0.000	0.067 $\pm$ 0.002	0.231 $\pm$ 0.010	0.480 $\pm$ 0.013	0.152	0.013	9.02	0.0676	0.0687	0.975	94.294 $\pm$ 0.734	0.253 $\pm$ 0.002	65.265 $\pm$ 1.506	
QML	beta34	0.101	0.073	-0.028 $\pm$ 0.002	-0.278 $\pm$ 0.020	0.004 $\pm$ 0.000	0.063 $\pm$ 0.002	0.005 $\pm$ 0.000	0.069 $\pm$ 0.002	0.467 $\pm$ 0.020	0.683 $\pm$ 0.017	0.069	-0.032	-31.94	0.0614	0.0693	0.963	87.487 $\pm$ 1.047	0.238 $\pm$ 0.002	17.618 $\pm$ 1.205	
QML	beta35	0.101	0.076	-0.025 $\pm$ 0.002	-0.252 $\pm$ 0.020	0.004 $\pm$ 0.000	0.065 $\pm$ 0.002	0.005 $\pm$ 0.000	0.070 $\pm$ 0.002	0.480 $\pm$ 0.020	0.693 $\pm$ 0.017	0.072	-0.029	-29.19	0.0661	0.0724	0.928	87.688 $\pm$ 1.040	0.237 $\pm$ 0.002	21.221 $\pm$ 1.294	
UPI	beta31	0.316	0.320	0.004 $\pm$ 0.002	0.013 $\pm$ 0.006	0.004 $\pm$ 0.000	0.060 $\pm$ 0.001	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	0.036 $\pm$ 0.002	0.190 $\pm$ 0.005	0.322	0.006	1.78	0.0584	0.0586	0.937	93.594 $\pm$ 0.775	0.221 $\pm$ 0.001	99.800 $\pm$ 0.141	
UPI	beta32	0.316	0.320	0.004 $\pm$ 0.002	0.012 $\pm$ 0.006	0.004 $\pm$ 0.000	0.061 $\pm$ 0.001	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.037 $\pm$ 0.002	0.194 $\pm$ 0.006	0.322	0.006	1.96	0.0601	0.0604	0.925	93.894 $\pm$ 0.758	0.222 $\pm$ 0.001	99.900 $\pm$ 0.100	
UPI	beta33	0.139	0.146	0.007 $\pm$ 0.002	0.054 $\pm$ 0.015	0.004 $\pm$ 0.000	0.065 $\pm$ 0.001	0.004 $\pm$ 0.000	0.065 $\pm$ 0.002	0.221 $\pm$ 0.010	0.470 $\pm$ 0.013	0.147	0.008	5.63	0.0654	0.0659	0.980	94.294 $\pm$ 0.734	0.249 $\pm$ 0.001	65.866 $\pm$ 1.500	
UPI	beta34	0.101	0.097	-0.004 $\pm$ 0.003	-0.036 $\pm$ 0.025	0.006 $\pm$ 0.000	0.080 $\pm$ 0.002	0.006 $\pm$ 0.000	0.080 $\pm$ 0.002	0.633 $\pm$ 0.029	0.796 $\pm$ 0.022	0.094	-0.007	-7.21	0.0787	0.0790	0.952	93.894 $\pm$ 0.758	0.300 $\pm$ 0.002	22.923 $\pm$ 1.330	
UPI	beta35	0.101	0.100	-0.001 $\pm$ 0.003	-0.006 $\pm$ 0.025	0.006 $\pm$ 0.000	0.080 $\pm$ 0.002	0.006 $\pm$ 0.000	0.080 $\pm$ 0.002	0.625 $\pm$ 0.029	0.790 $\pm$ 0.022	0.098	-0.003	-2.98	0.0821	0.0822	0.952	94.995 $\pm$ 0.690	0.298 $\pm$ 0.002	26.426 $\pm$ 1.395	

Note. MCSE = Monte Carlo Standard Error shown as $\backslash 00b1$ value. Coverage, Type I Error, and Power expressed as percentages. $\backslash 02014$ indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S72

Simulation 1 - Condition 72: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.8

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LMS	beta31	0.316	0.313	-0.003 ± 0.001	-0.010 ± 0.004	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.013 ± 0.001	0.116 ± 0.003	0.313	-0.003	-0.96	0.0373	0.0374	1.004	94.995 ± 0.690	0.143 ± 0.002	99.800 ± 0.141
LMS	beta32	0.316	0.311	-0.005 ± 0.001	-0.017 ± 0.004	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.014 ± 0.001	0.116 ± 0.003	0.310	-0.006	-1.77	0.0366	0.0371	1.002	94.595 ± 0.715	0.143 ± 0.001	99.800 ± 0.141
LMS	beta33	0.139	0.150	0.011 ± 0.001	0.083 ± 0.009	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.093 ± 0.004	0.304 ± 0.008	0.152	0.013	9.33	0.0377	0.0399	0.999	93.594 ± 0.775	0.159 ± 0.001	93.093 ± 0.802
LMS	beta34	0.101	0.075	-0.026 ± 0.001	-0.260 ± 0.012	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.218 ± 0.009	0.467 ± 0.011	0.073	-0.028	-27.48	0.0380	0.0471	1.002	86.486 ± 1.082	0.154 ± 0.001	46.346 ± 1.578
LMS	beta35	0.101	0.074	-0.027 ± 0.001	-0.263 ± 0.012	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.221 ± 0.009	0.470 ± 0.011	0.073	-0.028	-27.97	0.0380	0.0474	0.989	86.386 ± 1.085	0.153 ± 0.001	46.446 ± 1.578
LSAM	beta31	0.316	0.314	-0.002 ± 0.001	-0.005 ± 0.004	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.014 ± 0.001	0.118 ± 0.003	0.314	-0.002	-0.52	0.0372	0.0372	0.986	94.895 ± 0.696	0.144 ± 0.000	100.000 ± 0.000
LSAM	beta32	0.316	0.312	-0.004 ± 0.001	-0.012 ± 0.004	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.014 ± 0.001	0.119 ± 0.003	0.312	-0.004	-1.28	0.0385	0.0387	0.986	94.795 ± 0.703	0.144 ± 0.000	100.000 ± 0.000
LSAM	beta33	0.139	0.150	0.011 ± 0.001	0.076 ± 0.009	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.089 ± 0.004	0.298 ± 0.008	0.150	0.011	7.74	0.0385	0.0400	0.991	94.294 ± 0.734	0.156 ± 0.001	94.394 ± 0.728
LSAM	beta34	0.101	0.088	-0.013 ± 0.001	-0.126 ± 0.014	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.219 ± 0.010	0.468 ± 0.012	0.088	-0.013	-12.81	0.0445	0.0464	0.998	93.594 ± 0.775	0.178 ± 0.001	49.149 ± 1.582
LSAM	beta35	0.101	0.088	-0.013 ± 0.001	-0.130 ± 0.014	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.220 ± 0.009	0.469 ± 0.012	0.086	-0.015	-14.37	0.0448	0.0471	0.994	93.994 ± 0.752	0.178 ± 0.001	48.448 ± 1.581
QML	beta31	0.316	0.312	-0.004 ± 0.001	-0.014 ± 0.004	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.013 ± 0.001	0.116 ± 0.003	0.312	-0.004	-1.40	0.0372	0.0374	0.986	94.294 ± 0.734	0.141 ± 0.000	100.000 ± 0.000
QML	beta32	0.316	0.309	-0.007 ± 0.001	-0.021 ± 0.004	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.014 ± 0.001	0.117 ± 0.003	0.309	-0.007	-2.14	0.0364	0.0370	0.987	94.494 ± 0.722	0.141 ± 0.000	100.000 ± 0.000
QML	beta33	0.139	0.149	0.010 ± 0.001	0.075 ± 0.009	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.090 ± 0.004	0.300 ± 0.008	0.151	0.012	8.40	0.0378	0.0396	0.998	93.994 ± 0.752	0.158 ± 0.001	93.293 ± 0.791
QML	beta34	0.101	0.071	-0.030 ± 0.001	-0.293 ± 0.012	0.001 ± 0.000	0.038 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.228 ± 0.009	0.478 ± 0.010	0.070	-0.031	-30.21	0.0374	0.0483	0.999	84.685 ± 1.139	0.149 ± 0.001	45.145 ± 1.574
QML	beta35	0.101	0.071	-0.030 ± 0.001	-0.295 ± 0.012	0.001 ± 0.000	0.038 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.231 ± 0.009	0.481 ± 0.010	0.070	-0.031	-30.96	0.0368	0.0483	0.986	84.484 ± 1.145	0.148 ± 0.001	45.145 ± 1.574
UPI	beta31	0.316	0.317	0.001 ± 0.001	0.003 ± 0.004	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.014 ± 0.001	0.118 ± 0.003	0.317	0.001	0.36	0.0375	0.0375	0.951	94.494 ± 0.722	0.139 ± 0.000	100.000 ± 0.000
UPI	beta32	0.316	0.315	-0.001 ± 0.001	-0.005 ± 0.004	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.014 ± 0.001	0.118 ± 0.003	0.315	-0.001	-0.39	0.0384	0.0384	0.951	93.894 ± 0.758	0.139 ± 0.000	100.000 ± 0.000
UPI	beta33	0.139	0.144	0.005 ± 0.001	0.037 ± 0.009	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.086 ± 0.004	0.294 ± 0.008	0.144	0.005	3.64	0.0383	0.0386	0.982	94.695 ± 0.709	0.156 ± 0.001	93.293 ± 0.791
UPI	beta34	0.101	0.098	-0.003 ± 0.001	-0.034 ± 0.015	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.216 ± 0.010	0.465 ± 0.013	0.096	-0.005	-4.83	0.0465	0.0468	1.006	95.195 ± 0.677	0.185 ± 0.001	55.055 ± 1.574
UPI	beta35	0.101	0.097	-0.004 ± 0.001	-0.043 ± 0.015	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.222 ± 0.010	0.471 ± 0.013	0.095	-0.006	-6.08	0.0462	0.0466	0.990	94.595 ± 0.715	0.184 ± 0.001	54.054 ± 1.577

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S73

Simulation 1 - Condition 73: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.4

Method	Param.	True	Mean-based										Median-based					Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE		
LMS	beta31	0.316	0.312	-0.004 ± 0.003	-0.013 ± 0.010	0.009 ± 0.000	0.094 ± 0.002	0.009 ± 0.000	0.094 ± 0.003	0.088 ± 0.004	0.296 ± 0.009	0.306	-0.010	-3.03	0.0926	0.0930	1.009	95.000 ± 0.726	0.370 ± 0.002	—		
LMS	beta32	0.316	0.319	0.003 ± 0.003	0.009 ± 0.010	0.009 ± 0.000	0.097 ± 0.002	0.009 ± 0.000	0.097 ± 0.003	0.094 ± 0.004	0.307 ± 0.009	0.316	0.000	0.13	0.0992	0.0992	0.983	94.778 ± 0.742	0.373 ± 0.002	—		
LMS	beta33	0.000	-0.038	-0.038 ± 0.004	—	0.016 ± 0.001	0.126 ± 0.003	0.017 ± 0.001	0.132 ± 0.004	—	—	-0.035	-0.035	—	0.1309	0.1356	1.049	93.222 ± 0.838	0.519 ± 0.007	6.778 ± 0.838		
LMS	beta34	0.000	0.003	0.003 ± 0.003	—	0.007 ± 0.000	0.085 ± 0.002	0.007 ± 0.000	0.085 ± 0.003	—	—	0.004	0.004	—	0.0853	0.0854	1.013	94.889 ± 0.734	0.339 ± 0.005	5.111 ± 0.734		
LMS	beta35	0.000	0.003	0.003 ± 0.003	—	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.081 ± 0.002	—	—	0.002	0.002	—	0.0842	0.0843	1.068	95.667 ± 0.679	0.338 ± 0.006	4.333 ± 0.679		
LSAM	beta31	0.316	0.320	0.004 ± 0.004	0.012 ± 0.012	0.012 ± 0.001	0.110 ± 0.003	0.012 ± 0.001	0.110 ± 0.003	0.122 ± 0.006	0.349 ± 0.011	0.316	0.000	-0.03	0.1047	0.1047	1.038	96.333 ± 0.626	0.449 ± 0.005	—		
LSAM	beta32	0.316	0.326	0.010 ± 0.004	0.031 ± 0.012	0.013 ± 0.001	0.113 ± 0.003	0.013 ± 0.001	0.113 ± 0.004	0.129 ± 0.007	0.359 ± 0.011	0.327	0.011	3.52	0.1158	0.1164	1.012	95.333 ± 0.703	0.448 ± 0.005	—		
LSAM	beta33	0.000	0.002	0.002 ± 0.005	—	0.025 ± 0.002	0.159 ± 0.005	0.025 ± 0.002	0.159 ± 0.005	—	—	0.006	0.006	—	0.1509	0.1510	1.111	98.222 ± 0.440	0.691 ± 0.019	1.778 ± 0.440		
LSAM	beta34	0.000	-0.003	-0.003 ± 0.004	—	0.013 ± 0.001	0.116 ± 0.003	0.013 ± 0.001	0.116 ± 0.004	—	—	-0.004	-0.004	—	0.1064	0.1065	1.057	97.667 ± 0.503	0.479 ± 0.012	2.333 ± 0.503		
LSAM	beta35	0.000	-0.003	-0.003 ± 0.004	—	0.012 ± 0.001	0.109 ± 0.003	0.012 ± 0.001	0.109 ± 0.004	—	—	-0.003	-0.003	—	0.0980	0.0980	1.109	98.667 ± 0.382	0.475 ± 0.011	1.333 ± 0.382		
QML	beta31	0.316	0.312	-0.004 ± 0.003	-0.012 ± 0.010	0.009 ± 0.000	0.094 ± 0.002	0.009 ± 0.000	0.094 ± 0.003	0.088 ± 0.004	0.296 ± 0.009	0.306	-0.010	-3.29	0.0901	0.0907	1.007	94.778 ± 0.742	0.370 ± 0.002	—		
QML	beta32	0.316	0.319	0.003 ± 0.003	0.009 ± 0.010	0.009 ± 0.000	0.097 ± 0.002	0.009 ± 0.000	0.097 ± 0.003	0.094 ± 0.005	0.306 ± 0.009	0.316	0.000	-0.10	0.0990	0.0990	0.980	94.556 ± 0.756	0.372 ± 0.002	—		
QML	beta33	0.000	-0.038	-0.038 ± 0.004	—	0.016 ± 0.001	0.125 ± 0.003	0.017 ± 0.001	0.131 ± 0.004	—	—	-0.035	-0.035	—	0.1294	0.1340	1.049	93.778 ± 0.805	0.515 ± 0.007	6.222 ± 0.805		
QML	beta34	0.000	0.003	0.003 ± 0.003	—	0.007 ± 0.000	0.085 ± 0.002	0.007 ± 0.000	0.085 ± 0.003	—	—	0.003	0.003	—	0.0834	0.0835	1.014	94.889 ± 0.734	0.337 ± 0.004	5.111 ± 0.734		
QML	beta35	0.000	0.003	0.003 ± 0.003	—	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	—	—	0.002	0.002	—	0.0822	0.0822	1.065	95.556 ± 0.687	0.331 ± 0.004	4.444 ± 0.687		
UPI	beta31	0.316	0.320	0.004 ± 0.004	0.014 ± 0.012	0.013 ± 0.001	0.113 ± 0.003	0.013 ± 0.001	0.113 ± 0.004	0.128 ± 0.007	0.358 ± 0.011	0.317	0.001	0.17	0.1075	0.1075	1.095	97.000 ± 0.569	0.485 ± 0.008	—		
UPI	beta32	0.316	0.326	0.010 ± 0.004	0.032 ± 0.012	0.014 ± 0.001	0.116 ± 0.003	0.014 ± 0.001	0.117 ± 0.004	0.137 ± 0.008	0.370 ± 0.012	0.326	0.010	3.02	0.1185	0.1189	1.073	96.889 ± 0.579	0.489 ± 0.009	—		
UPI	beta33	0.000	0.006	0.006 ± 0.006	—	0.030 ± 0.002	0.174 ± 0.005	0.030 ± 0.002	0.174 ± 0.006	—	—	0.005	0.005	—	0.1608	0.1609	1.202	99.444 ± 0.248	0.820 ± 0.022	0.556 ± 0.248		
UPI	beta34	0.000	-0.004	-0.004 ± 0.004	—	0.016 ± 0.001	0.128 ± 0.004	0.016 ± 0.001	0.128 ± 0.005	—	—	-0.002	-0.002	—	0.1125	0.1126	1.243	98.778 ± 0.366	0.623 ± 0.020	1.222 ± 0.366		
UPI	beta35	0.000	0.000	-0.000 ± 0.004	—	0.017 ± 0.001	0.130 ± 0.004	0.017 ± 0.001	0.130 ± 0.005	—	—	0.001	0.001	—	0.1059	0.1059	1.252	99.222 ± 0.293	0.636 ± 0.020	0.778 ± 0.293		

Note. MCSE = Monte Carlo Standard Error shown as $\backslash u00b1$ value. Coverage, Type I Error, and Power expressed as percentages. $\backslash u2014$ indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S74

Simulation 1 - Condition 74: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.4

Method	Param.	True	Mean-based										Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE	
LMS	beta31	0.316	0.313	-0.003 ± 0.002	-0.008 ± 0.006	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.032 ± 0.002	0.179 ± 0.005	0.313	-0.003	-1.07	0.0550	0.0552	1.028	95.888 ± 0.629	0.228 ± 0.001	—	
LMS	beta32	0.316	0.317	0.001 ± 0.002	0.003 ± 0.006	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.033 ± 0.002	0.182 ± 0.005	0.315	-0.001	-0.32	0.0589	0.0589	1.014	94.885 ± 0.698	0.229 ± 0.001	—	
LMS	beta33	0.000	-0.043	-0.043 ± 0.003	—	0.007 ± 0.000	0.084 ± 0.002	0.009 ± 0.000	0.094 ± 0.002	—	—	-0.044	-0.044	—	0.0789	0.0901	0.971	90.471 ± 0.930	0.320 ± 0.002	9.529 ± 0.930	
LMS	beta34	0.000	0.006	0.006 ± 0.002	—	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	—	—	0.006	0.006	—	0.0532	0.0535	0.963	93.480 ± 0.782	0.205 ± 0.002	6.520 ± 0.782	
LMS	beta35	0.000	0.007	0.007 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	—	—	0.006	0.006	—	0.0483	0.0486	1.043	94.885 ± 0.698	0.207 ± 0.002	5.115 ± 0.698	
LSAM	beta31	0.316	0.318	0.002 ± 0.002	0.005 ± 0.006	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	0.039 ± 0.002	0.198 ± 0.006	0.319	0.003	0.81	0.0627	0.0627	1.023	95.687 ± 0.643	0.251 ± 0.001	—	
LSAM	beta32	0.316	0.321	0.005 ± 0.002	0.016 ± 0.006	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	0.040 ± 0.002	0.201 ± 0.006	0.319	0.003	0.88	0.0642	0.0642	1.013	95.587 ± 0.650	0.252 ± 0.001	—	
LSAM	beta33	0.000	-0.004	-0.004 ± 0.003	—	0.008 ± 0.000	0.089 ± 0.002	0.008 ± 0.000	0.089 ± 0.003	—	—	-0.003	-0.003	—	0.0810	0.0811	0.977	94.784 ± 0.704	0.340 ± 0.003	5.216 ± 0.704	
LSAM	beta34	0.000	0.001	0.001 ± 0.002	—	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	—	—	0.002	0.002	—	0.0592	0.0593	0.974	95.286 ± 0.671	0.237 ± 0.002	4.714 ± 0.671	
LSAM	beta35	0.000	0.001	0.001 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	—	—	0.000	0.000	—	0.0542	0.0542	1.033	96.690 ± 0.567	0.236 ± 0.002	3.310 ± 0.567	
QML	beta31	0.316	0.314	-0.002 ± 0.002	-0.008 ± 0.006	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.032 ± 0.002	0.179 ± 0.005	0.313	-0.003	-1.07	0.0549	0.0550	1.028	95.787 ± 0.636	0.228 ± 0.001	—	
QML	beta32	0.316	0.317	0.001 ± 0.002	0.002 ± 0.006	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.033 ± 0.002	0.182 ± 0.005	0.315	-0.001	-0.34	0.0587	0.0587	1.013	94.885 ± 0.698	0.229 ± 0.001	—	
QML	beta33	0.000	-0.043	-0.043 ± 0.003	—	0.007 ± 0.000	0.083 ± 0.002	0.009 ± 0.000	0.094 ± 0.002	—	—	-0.043	-0.043	—	0.0779	0.0890	0.977	90.772 ± 0.917	0.319 ± 0.002	9.228 ± 0.917	
QML	beta34	0.000	0.006	0.006 ± 0.002	—	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	—	—	0.006	0.006	—	0.0527	0.0531	0.975	94.082 ± 0.747	0.206 ± 0.002	5.918 ± 0.747	
QML	beta35	0.000	0.007	0.007 ± 0.002	—	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	—	—	0.006	0.006	—	0.0485	0.0488	1.048	95.286 ± 0.671	0.206 ± 0.002	4.714 ± 0.671	
UPI	beta31	0.316	0.318	0.002 ± 0.002	0.006 ± 0.006	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	0.039 ± 0.002	0.198 ± 0.006	0.318	0.002	0.68	0.0621	0.0621	1.038	96.389 ± 0.591	0.254 ± 0.001	—	
UPI	beta32	0.316	0.321	0.005 ± 0.002	0.016 ± 0.006	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	0.041 ± 0.002	0.202 ± 0.006	0.319	0.003	1.08	0.0629	0.0630	1.025	95.587 ± 0.650	0.255 ± 0.001	—	
UPI	beta33	0.000	-0.004	-0.004 ± 0.003	—	0.008 ± 0.000	0.092 ± 0.002	0.008 ± 0.000	0.092 ± 0.003	—	—	-0.004	-0.004	—	0.0810	0.0811	0.997	95.988 ± 0.622	0.358 ± 0.004	4.012 ± 0.622	
UPI	beta34	0.000	0.000	0.000 ± 0.002	—	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	—	—	0.002	0.002	—	0.0606	0.0606	1.015	96.891 ± 0.550	0.259 ± 0.003	3.109 ± 0.550	
UPI	beta35	0.000	0.002	0.002 ± 0.002	—	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	—	—	0.001	0.001	—	0.0573	0.0573	1.055	97.894 ± 0.455	0.256 ± 0.003	2.106 ± 0.455	

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S75

Simulation 1 - Condition 75: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.6

			Mean-based										Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE	
LMS	beta31	0.316	0.317	0.001 ± 0.002	0.004 ± 0.007	0.004 ± 0.000	0.066 ± 0.001	0.004 ± 0.000	0.066 ± 0.002	0.044 ± 0.002	0.209 ± 0.006	0.315	-0.001	-0.43	0.0702	0.0702	1.028	96.084 ± 0.615	0.266 ± 0.001		
LMS	beta32	0.316	0.314	-0.002 ± 0.002	-0.007 ± 0.007	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.067 ± 0.002	0.045 ± 0.002	0.213 ± 0.006	0.314	-0.002	-0.77	0.0663	0.0663	1.009	95.080 ± 0.685	0.266 ± 0.001	—	
LMS	beta33	0.000	-0.022	-0.022 ± 0.003	—	0.007 ± 0.000	0.085 ± 0.002	0.008 ± 0.000	0.088 ± 0.002	—	—	-0.022	-0.022	—	0.0844	0.0872	0.982	93.574 ± 0.777	0.327 ± 0.004	6.426 ± 0.777	
LMS	beta34	0.000	0.002	0.002 ± 0.002	—	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	—	—	0.000	0.000	—	0.0548	0.0548	0.989	93.474 ± 0.783	0.219 ± 0.002	6.526 ± 0.783	
LMS	beta35	0.000	0.006	0.006 ± 0.002	—	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	—	—	0.005	0.005	—	0.0540	0.0543	1.015	95.080 ± 0.685	0.221 ± 0.002	4.920 ± 0.685	
LSAM	beta31	0.316	0.321	0.005 ± 0.002	0.017 ± 0.007	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.050 ± 0.002	0.224 ± 0.006	0.318	0.002	0.54	0.0737	0.0737	1.014	95.281 ± 0.672	0.281 ± 0.001	—	
LSAM	beta32	0.316	0.316	0.000 ± 0.002	0.001 ± 0.007	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.051 ± 0.002	0.227 ± 0.006	0.314	-0.002	-0.53	0.0708	0.0708	0.999	94.880 ± 0.698	0.281 ± 0.001	—	
LSAM	beta33	0.000	0.000	-0.000 ± 0.003	—	0.008 ± 0.000	0.089 ± 0.002	0.008 ± 0.000	0.089 ± 0.003	—	—	-0.001	-0.001	—	0.0853	0.0853	0.957	94.378 ± 0.730	0.334 ± 0.003	5.622 ± 0.730	
LSAM	beta34	0.000	-0.001	-0.001 ± 0.002	—	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	—	—	-0.002	-0.002	—	0.0601	0.0601	0.957	93.775 ± 0.766	0.235 ± 0.002	6.225 ± 0.766	
LSAM	beta35	0.000	0.004	0.004 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	—	—	0.004	0.004	—	0.0587	0.0588	1.001	95.984 ± 0.622	0.239 ± 0.002	4.016 ± 0.622	
QML	beta31	0.316	0.317	0.001 ± 0.002	0.004 ± 0.007	0.004 ± 0.000	0.066 ± 0.001	0.004 ± 0.000	0.066 ± 0.002	0.044 ± 0.002	0.209 ± 0.006	0.315	-0.001	-0.20	0.0703	0.0703	1.026	95.984 ± 0.622	0.265 ± 0.001	—	
QML	beta32	0.316	0.314	-0.002 ± 0.002	-0.007 ± 0.007	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.067 ± 0.002	0.045 ± 0.002	0.213 ± 0.006	0.313	-0.003	-0.80	0.0662	0.0663	1.008	95.181 ± 0.679	0.266 ± 0.001	—	
QML	beta33	0.000	-0.022	-0.022 ± 0.003	—	0.007 ± 0.000	0.085 ± 0.002	0.008 ± 0.000	0.087 ± 0.002	—	—	-0.022	-0.022	—	0.0841	0.0868	0.974	93.574 ± 0.777	0.324 ± 0.003	6.426 ± 0.777	
QML	beta34	0.000	0.002	0.002 ± 0.002	—	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	—	—	0.000	0.000	—	0.0549	0.0549	0.983	93.474 ± 0.783	0.217 ± 0.002	6.526 ± 0.783	
QML	beta35	0.000	0.006	0.006 ± 0.002	—	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	—	—	0.005	0.005	—	0.0532	0.0534	1.015	95.181 ± 0.679	0.220 ± 0.002	4.819 ± 0.679	
UPI	beta31	0.316	0.321	0.005 ± 0.002	0.016 ± 0.007	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.051 ± 0.002	0.225 ± 0.006	0.318	0.002	0.75	0.0737	0.0737	1.016	95.582 ± 0.651	0.283 ± 0.001	—	
UPI	beta32	0.316	0.317	0.001 ± 0.002	0.003 ± 0.007	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.052 ± 0.002	0.228 ± 0.006	0.316	0.000	-0.12	0.0705	0.0705	1.002	95.181 ± 0.679	0.283 ± 0.001	—	
UPI	beta33	0.000	0.001	0.001 ± 0.003	—	0.008 ± 0.000	0.092 ± 0.002	0.008 ± 0.000	0.092 ± 0.003	—	—	0.000	0.000	—	0.0885	0.0885	0.959	95.482 ± 0.658	0.345 ± 0.003	4.518 ± 0.658	
UPI	beta34	0.000	-0.002	-0.002 ± 0.002	—	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	—	—	-0.004	-0.004	—	0.0620	0.0621	0.983	94.980 ± 0.692	0.249 ± 0.002	5.020 ± 0.692	
UPI	beta35	0.000	0.002	0.002 ± 0.002	—	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	—	—	0.003	0.003	—	0.0613	0.0614	1.025	96.486 ± 0.583	0.255 ± 0.003	3.514 ± 0.583	

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S76

Simulation 1 - Condition 76: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.6

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.314	-0.002 ± 0.001	-0.008 ± 0.004	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.019 ± 0.001	0.136 ± 0.004	0.312	-0.004	-1.15	0.0444	0.0445	0.985	94.400 ± 0.727	0.166 ± 0.000	—
LMS	beta32	0.316	0.314	-0.002 ± 0.001	-0.006 ± 0.004	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.018 ± 0.001	0.136 ± 0.004	0.313	-0.003	-1.04	0.0421	0.0423	0.992	94.100 ± 0.745	0.167 ± 0.000	—
LMS	beta33	0.000	-0.021	-0.021 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.056 ± 0.001	—	—	-0.022	-0.022	—	0.0513	0.0559	1.001	93.000 ± 0.807	0.205 ± 0.002	7.000 ± 0.807
LMS	beta34	0.000	0.003	0.003 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	0.005	0.005	—	0.0340	0.0344	0.991	93.900 ± 0.757	0.136 ± 0.001	6.100 ± 0.757
LMS	beta35	0.000	0.004	0.004 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	0.003	0.003	—	0.0367	0.0368	0.984	94.600 ± 0.715	0.138 ± 0.002	5.400 ± 0.715
LSAM	beta31	0.316	0.316	0.000 ± 0.001	0.001 ± 0.004	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.020 ± 0.001	0.142 ± 0.004	0.315	-0.001	-0.31	0.0463	0.0464	0.987	94.500 ± 0.721	0.174 ± 0.000	—
LSAM	beta32	0.316	0.317	0.001 ± 0.001	0.003 ± 0.005	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.020 ± 0.001	0.143 ± 0.004	0.316	0.000	-0.01	0.0431	0.0431	0.985	94.600 ± 0.715	0.174 ± 0.000	—
LSAM	beta33	0.000	-0.001	-0.001 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	—	—	-0.002	-0.002	—	0.0509	0.0510	0.984	94.400 ± 0.727	0.201 ± 0.001	5.600 ± 0.727
LSAM	beta34	0.000	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.002	0.002	—	0.0348	0.0349	0.992	94.800 ± 0.702	0.142 ± 0.001	5.200 ± 0.702
LSAM	beta35	0.000	0.002	0.002 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	0.002	0.002	—	0.0382	0.0383	0.960	94.800 ± 0.702	0.142 ± 0.001	5.200 ± 0.702
QML	beta31	0.316	0.314	-0.002 ± 0.001	-0.008 ± 0.004	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.019 ± 0.001	0.136 ± 0.004	0.312	-0.004	-1.14	0.0443	0.0444	0.984	94.200 ± 0.739	0.166 ± 0.000	—
QML	beta32	0.316	0.314	-0.002 ± 0.001	-0.006 ± 0.004	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.018 ± 0.001	0.136 ± 0.004	0.313	-0.003	-1.06	0.0419	0.0420	0.990	94.100 ± 0.745	0.167 ± 0.000	—
QML	beta33	0.000	-0.021	-0.021 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.056 ± 0.001	—	—	-0.022	-0.022	—	0.0513	0.0560	0.990	93.000 ± 0.807	0.202 ± 0.001	7.000 ± 0.807
QML	beta34	0.000	0.003	0.003 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	0.005	0.005	—	0.0339	0.0343	0.991	94.100 ± 0.745	0.135 ± 0.001	5.900 ± 0.745
QML	beta35	0.000	0.004	0.004 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	0.003	0.003	—	0.0366	0.0368	0.973	94.600 ± 0.715	0.136 ± 0.001	5.400 ± 0.715
UPI	beta31	0.316	0.316	0.000 ± 0.001	0.001 ± 0.005	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.020 ± 0.001	0.142 ± 0.004	0.315	-0.001	-0.30	0.0469	0.0469	0.986	94.600 ± 0.715	0.174 ± 0.000	—
UPI	beta32	0.316	0.317	0.001 ± 0.001	0.004 ± 0.005	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.020 ± 0.001	0.143 ± 0.004	0.315	-0.001	-0.26	0.0436	0.0436	0.985	94.500 ± 0.721	0.174 ± 0.000	—
UPI	beta33	0.000	0.000	-0.000 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	—	—	-0.002	-0.002	—	0.0512	0.0512	0.980	94.400 ± 0.727	0.204 ± 0.001	5.600 ± 0.727
UPI	beta34	0.000	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.001	0.001	—	0.0354	0.0354	1.012	95.400 ± 0.662	0.145 ± 0.001	4.600 ± 0.662
UPI	beta35	0.000	0.002	0.002 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	0.001	0.001	—	0.0387	0.0388	0.971	95.500 ± 0.656	0.145 ± 0.001	4.500 ± 0.656

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S77

Simulation 1 - Condition 77: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.8

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.317	0.001 ± 0.002	0.005 ± 0.006	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.032 ± 0.001	0.178 ± 0.005	0.317	0.001	0.38	0.0536	0.0537	0.990	94.995 ± 0.690	0.218 ± 0.001	—
LMS	beta32	0.316	0.314	-0.002 ± 0.002	-0.007 ± 0.006	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.031 ± 0.001	0.177 ± 0.005	0.312	-0.004	-1.24	0.0566	0.0568	0.989	94.494 ± 0.722	0.217 ± 0.001	—
LMS	beta33	0.000	-0.009	-0.009 ± 0.002	—	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	—	—	-0.010	-0.010	—	0.0642	0.0649	0.947	93.093 ± 0.802	0.244 ± 0.001	6.907 ± 0.802
LMS	beta34	0.000	0.000	-0.000 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	0.000	0.000	—	0.0444	0.0444	0.967	93.393 ± 0.786	0.169 ± 0.001	6.607 ± 0.786
LMS	beta35	0.000	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	—	—	0.004	0.004	—	0.0475	0.0476	0.975	95.095 ± 0.683	0.168 ± 0.001	4.905 ± 0.683
LSAM	beta31	0.316	0.319	0.003 ± 0.002	0.010 ± 0.006	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.033 ± 0.002	0.183 ± 0.005	0.318	0.002	0.64	0.0557	0.0558	0.986	94.895 ± 0.696	0.223 ± 0.001	—
LSAM	beta32	0.316	0.315	-0.001 ± 0.002	-0.002 ± 0.006	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.033 ± 0.002	0.183 ± 0.005	0.313	-0.003	-0.93	0.0574	0.0575	0.982	93.994 ± 0.752	0.222 ± 0.001	—
LSAM	beta33	0.000	-0.001	-0.001 ± 0.002	—	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	—	—	-0.001	-0.001	—	0.0633	0.0633	0.943	93.293 ± 0.791	0.245 ± 0.001	6.707 ± 0.791
LSAM	beta34	0.000	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	-0.001	-0.001	—	0.0460	0.0460	0.957	93.594 ± 0.775	0.173 ± 0.001	6.406 ± 0.775
LSAM	beta35	0.000	0.000	0.000 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	0.003	0.003	—	0.0486	0.0486	0.967	95.295 ± 0.670	0.173 ± 0.001	4.705 ± 0.670
QML	beta31	0.316	0.317	0.001 ± 0.002	0.005 ± 0.006	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.032 ± 0.001	0.178 ± 0.005	0.317	0.001	0.38	0.0536	0.0537	0.990	94.995 ± 0.690	0.218 ± 0.001	—
QML	beta32	0.316	0.314	-0.002 ± 0.002	-0.007 ± 0.006	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.031 ± 0.001	0.177 ± 0.005	0.312	-0.004	-1.24	0.0565	0.0567	0.988	94.494 ± 0.722	0.217 ± 0.001	—
QML	beta33	0.000	-0.009	-0.009 ± 0.002	—	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	—	—	-0.009	-0.009	—	0.0637	0.0644	0.951	93.193 ± 0.797	0.245 ± 0.001	6.807 ± 0.797
QML	beta34	0.000	0.000	-0.000 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	0.000	0.000	—	0.0442	0.0442	0.967	93.393 ± 0.786	0.169 ± 0.001	6.607 ± 0.786
QML	beta35	0.000	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	—	—	0.003	0.003	—	0.0475	0.0476	0.975	95.095 ± 0.683	0.168 ± 0.001	4.905 ± 0.683
UPI	beta31	0.316	0.319	0.003 ± 0.002	0.009 ± 0.006	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.033 ± 0.002	0.183 ± 0.005	0.318	0.002	0.78	0.0551	0.0551	0.982	94.394 ± 0.728	0.222 ± 0.001	—
UPI	beta32	0.316	0.315	-0.001 ± 0.002	-0.002 ± 0.006	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.033 ± 0.002	0.182 ± 0.005	0.313	-0.003	-1.04	0.0576	0.0577	0.980	93.794 ± 0.763	0.221 ± 0.001	—
UPI	beta33	0.000	-0.001	-0.001 ± 0.002	—	0.004 ± 0.000	0.067 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	—	—	-0.002	-0.002	—	0.0637	0.0637	0.944	93.093 ± 0.802	0.246 ± 0.001	6.907 ± 0.802
UPI	beta34	0.000	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	-0.001	-0.001	—	0.0459	0.0459	0.958	93.493 ± 0.780	0.175 ± 0.001	6.507 ± 0.780
UPI	beta35	0.000	0.000	0.000 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	0.001	0.001	—	0.0478	0.0479	0.975	95.596 ± 0.649	0.175 ± 0.001	4.404 ± 0.649

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S78

Simulation 1 - Condition 78: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.8

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.314	-0.002 ± 0.001	-0.006 ± 0.003	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.012 ± 0.001	0.110 ± 0.003	0.313	-0.003	-0.92	0.0354	0.0355	1.002	94.700 ± 0.708	0.137 ± 0.000	—
LMS	beta32	0.316	0.314	-0.002 ± 0.001	-0.006 ± 0.004	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.013 ± 0.001	0.112 ± 0.003	0.314	-0.002	-0.51	0.0337	0.0338	0.986	93.200 ± 0.796	0.137 ± 0.000	—
LMS	beta33	0.000	-0.008	-0.008 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	-0.007	-0.007	—	0.0392	0.0399	1.017	94.000 ± 0.751	0.157 ± 0.003	6.000 ± 0.751
LMS	beta34	0.000	0.002	0.002 ± 0.001	—	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	—	—	0.002	0.002	—	0.0267	0.0267	1.001	93.800 ± 0.763	0.107 ± 0.001	6.200 ± 0.763
LMS	beta35	0.000	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	0.002	0.002	—	0.0265	0.0266	1.005	94.000 ± 0.751	0.108 ± 0.001	6.000 ± 0.751
LSAM	beta31	0.316	0.315	-0.001 ± 0.001	-0.002 ± 0.004	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.013 ± 0.001	0.112 ± 0.003	0.314	-0.002	-0.61	0.0359	0.0360	0.999	94.500 ± 0.721	0.139 ± 0.000	—
LSAM	beta32	0.316	0.315	-0.001 ± 0.001	-0.002 ± 0.004	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.013 ± 0.001	0.114 ± 0.003	0.316	0.000	-0.07	0.0347	0.0347	0.985	93.800 ± 0.763	0.139 ± 0.000	—
LSAM	beta33	0.000	0.000	-0.000 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	0.000	0.000	—	0.0384	0.0384	0.992	94.800 ± 0.702	0.152 ± 0.001	5.200 ± 0.702
LSAM	beta34	0.000	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	0.001	0.001	—	0.0274	0.0274	0.988	93.800 ± 0.763	0.107 ± 0.000	6.200 ± 0.763
LSAM	beta35	0.000	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	0.001	0.001	—	0.0275	0.0275	0.980	93.600 ± 0.774	0.108 ± 0.000	6.400 ± 0.774
QML	beta31	0.316	0.314	-0.002 ± 0.001	-0.006 ± 0.003	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.012 ± 0.001	0.110 ± 0.003	0.313	-0.003	-0.94	0.0354	0.0355	1.005	94.700 ± 0.708	0.137 ± 0.001	—
QML	beta32	0.316	0.314	-0.002 ± 0.001	-0.007 ± 0.004	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.013 ± 0.001	0.112 ± 0.003	0.314	-0.002	-0.50	0.0337	0.0337	0.986	93.300 ± 0.791	0.137 ± 0.000	—
QML	beta33	0.000	-0.008	-0.008 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	-0.008	-0.008	—	0.0390	0.0398	1.017	94.000 ± 0.751	0.158 ± 0.003	6.000 ± 0.751
QML	beta34	0.000	0.002	0.002 ± 0.001	—	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	—	—	0.002	0.002	—	0.0268	0.0269	1.000	93.800 ± 0.763	0.107 ± 0.001	6.200 ± 0.763
QML	beta35	0.000	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	0.002	0.002	—	0.0267	0.0268	0.997	94.100 ± 0.745	0.108 ± 0.001	5.900 ± 0.745
UPI	beta31	0.316	0.315	-0.001 ± 0.001	-0.002 ± 0.004	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.013 ± 0.001	0.113 ± 0.003	0.314	-0.002	-0.51	0.0359	0.0360	0.997	94.200 ± 0.739	0.139 ± 0.000	—
UPI	beta32	0.316	0.315	-0.001 ± 0.001	-0.003 ± 0.004	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.013 ± 0.001	0.114 ± 0.003	0.316	0.000	-0.13	0.0343	0.0343	0.984	93.800 ± 0.763	0.139 ± 0.000	—
UPI	beta33	0.000	0.000	-0.000 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	0.000	0.000	—	0.0382	0.0382	0.988	94.700 ± 0.708	0.153 ± 0.001	5.300 ± 0.708
UPI	beta34	0.000	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	0.001	0.001	—	0.0272	0.0272	0.986	93.900 ± 0.757	0.108 ± 0.000	6.100 ± 0.757
UPI	beta35	0.000	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	0.000	0.000	—	0.0275	0.0275	0.977	93.700 ± 0.768	0.108 ± 0.000	6.300 ± 0.768

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S79

Simulation 1 - Condition 79: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.4

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.276	-0.040 ± 0.004	-0.127 ± 0.011	0.010 ± 0.000	0.102 ± 0.002	0.012 ± 0.001	0.110 ± 0.003	0.121 ± 0.005	0.347 ± 0.009	0.269	-0.047	-15.00	0.1040	0.1143	1.028	91.924 ± 0.939	0.412 ± 0.003	—
LMS	beta32	0.316	0.279	-0.037 ± 0.003	-0.116 ± 0.011	0.009 ± 0.000	0.097 ± 0.002	0.011 ± 0.000	0.104 ± 0.003	0.108 ± 0.005	0.329 ± 0.009	0.278	-0.038	-12.03	0.1044	0.1111	1.083	93.587 ± 0.844	0.412 ± 0.003	—
LMS	beta33	0.000	-0.041	-0.041 ± 0.004	—	0.014 ± 0.001	0.116 ± 0.003	0.015 ± 0.001	0.123 ± 0.004	—	—	-0.041	-0.041	—	0.1151	0.1222	1.038	94.893 ± 0.759	0.474 ± 0.006	5.107 ± 0.759
LMS	beta34	0.000	0.072	0.072 ± 0.002	—	0.005 ± 0.000	0.070 ± 0.002	0.010 ± 0.001	0.100 ± 0.003	—	—	0.064	0.064	—	0.0704	0.0950	1.029	86.580 ± 1.175	0.284 ± 0.005	13.420 ± 1.175
LMS	beta35	0.000	0.066	0.066 ± 0.002	—	0.005 ± 0.000	0.070 ± 0.002	0.009 ± 0.000	0.096 ± 0.003	—	—	0.063	0.063	—	0.0686	0.0929	1.009	85.273 ± 1.221	0.276 ± 0.005	14.727 ± 1.221
LSAM	beta31	0.316	0.318	0.002 ± 0.007	0.005 ± 0.023	0.043 ± 0.003	0.207 ± 0.006	0.043 ± 0.003	0.207 ± 0.007	0.428 ± 0.025	0.654 ± 0.022	0.307	-0.009	-2.94	0.2126	0.2128	1.190	97.625 ± 0.525	0.965 ± 0.032	—
LSAM	beta32	0.316	0.330	0.014 ± 0.007	0.045 ± 0.022	0.041 ± 0.002	0.203 ± 0.006	0.041 ± 0.002	0.203 ± 0.007	0.413 ± 0.022	0.642 ± 0.021	0.320	0.004	1.12	0.2103	0.2103	1.230	98.100 ± 0.471	0.977 ± 0.028	—
LSAM	beta33	0.000	0.011	0.011 ± 0.005	—	0.018 ± 0.001	0.134 ± 0.004	0.018 ± 0.001	0.134 ± 0.005	—	—	0.009	0.009	—	0.1272	0.1276	1.233	97.268 ± 0.562	0.646 ± 0.029	2.732 ± 0.562
LSAM	beta34	0.000	0.001	0.001 ± 0.003	—	0.007 ± 0.000	0.082 ± 0.003	0.007 ± 0.000	0.082 ± 0.003	—	—	0.002	0.002	—	0.0721	0.0721	1.207	98.219 ± 0.456	0.390 ± 0.021	1.781 ± 0.456
LSAM	beta35	0.000	-0.006	-0.006 ± 0.003	—	0.006 ± 0.000	0.080 ± 0.003	0.006 ± 0.000	0.081 ± 0.003	—	—	-0.002	-0.002	—	0.0710	0.0710	1.217	97.981 ± 0.485	0.384 ± 0.016	2.019 ± 0.485
QML	beta31	0.316	0.276	-0.040 ± 0.004	-0.126 ± 0.011	0.011 ± 0.001	0.103 ± 0.002	0.012 ± 0.001	0.110 ± 0.003	0.121 ± 0.005	0.348 ± 0.009	0.270	-0.046	-14.54	0.1056	0.1152	1.025	91.924 ± 0.939	0.412 ± 0.003	—
QML	beta32	0.316	0.279	-0.037 ± 0.003	-0.116 ± 0.011	0.009 ± 0.000	0.097 ± 0.002	0.011 ± 0.000	0.104 ± 0.003	0.108 ± 0.005	0.329 ± 0.009	0.277	-0.039	-12.38	0.1042	0.1113	1.080	93.824 ± 0.830	0.412 ± 0.003	—
QML	beta33	0.000	-0.040	-0.040 ± 0.004	—	0.014 ± 0.001	0.117 ± 0.004	0.015 ± 0.001	0.124 ± 0.004	—	—	-0.040	-0.040	—	0.1142	0.1209	1.061	94.893 ± 0.759	0.486 ± 0.014	5.107 ± 0.759
QML	beta34	0.000	0.071	0.071 ± 0.002	—	0.005 ± 0.000	0.070 ± 0.002	0.010 ± 0.001	0.100 ± 0.003	—	—	0.064	0.064	—	0.0693	0.0940	1.047	86.461 ± 1.179	0.289 ± 0.006	13.539 ± 1.179
QML	beta35	0.000	0.066	0.066 ± 0.002	—	0.005 ± 0.000	0.070 ± 0.002	0.009 ± 0.000	0.096 ± 0.003	—	—	0.062	0.062	—	0.0684	0.0921	1.017	85.273 ± 1.221	0.278 ± 0.006	14.727 ± 1.221
UPI	beta31	0.316	0.319	0.003 ± 0.007	0.010 ± 0.023	0.043 ± 0.003	0.207 ± 0.006	0.043 ± 0.003	0.207 ± 0.007	0.430 ± 0.026	0.656 ± 0.023	0.308	-0.008	-2.39	0.2122	0.2124	1.186	98.219 ± 0.456	0.964 ± 0.023	—
UPI	beta32	0.316	0.332	0.016 ± 0.007	0.051 ± 0.022	0.040 ± 0.002	0.200 ± 0.005	0.040 ± 0.002	0.201 ± 0.006	0.403 ± 0.021	0.635 ± 0.020	0.322	0.006	1.91	0.2100	0.2101	1.235	98.337 ± 0.441	0.969 ± 0.025	—
UPI	beta33	0.000	0.007	0.007 ± 0.005	—	0.023 ± 0.002	0.151 ± 0.005	0.023 ± 0.002	0.151 ± 0.006	—	—	0.009	0.009	—	0.1298	0.1301	1.443	98.931 ± 0.354	0.856 ± 0.036	1.069 ± 0.354
UPI	beta34	0.000	0.000	0.000 ± 0.003	—	0.007 ± 0.000	0.083 ± 0.003	0.007 ± 0.000	0.083 ± 0.003	—	—	-0.001	-0.001	—	0.0719	0.0719	1.223	98.100 ± 0.471	0.399 ± 0.012	1.900 ± 0.471
UPI	beta35	0.000	-0.007	-0.007 ± 0.003	—	0.007 ± 0.000	0.082 ± 0.003	0.007 ± 0.000	0.082 ± 0.003	—	—	-0.002	-0.002	—	0.0743	0.0744	1.204	98.219 ± 0.456	0.387 ± 0.013	1.781 ± 0.456

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S80

Simulation 1 - Condition 80: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.4

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.276	-0.040 ± 0.002	-0.126 ± 0.006	0.004 ± 0.000	0.062 ± 0.001	0.005 ± 0.000	0.074 ± 0.002	0.055 ± 0.002	0.234 ± 0.006	0.276	-0.040	-12.51	0.0649	0.0760	1.042	89.744 ± 0.972	0.255 ± 0.003	—
LMS	beta32	0.316	0.272	-0.044 ± 0.002	-0.139 ± 0.006	0.004 ± 0.000	0.064 ± 0.001	0.006 ± 0.000	0.078 ± 0.002	0.060 ± 0.002	0.246 ± 0.006	0.271	-0.045	-14.16	0.0660	0.0798	1.013	88.513 ± 1.021	0.254 ± 0.001	—
LMS	beta33	0.000	-0.046	-0.046 ± 0.002	—	0.005 ± 0.000	0.073 ± 0.002	0.007 ± 0.000	0.086 ± 0.002	—	—	-0.043	-0.043	—	0.0704	0.0826	1.034	91.897 ± 0.874	0.297 ± 0.003	8.103 ± 0.874
LMS	beta34	0.000	0.071	0.071 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.007 ± 0.000	0.084 ± 0.002	—	—	0.067	0.067	—	0.0423	0.0790	0.972	62.769 ± 1.548	0.170 ± 0.002	37.231 ± 1.548
LMS	beta35	0.000	0.069	0.069 ± 0.001	—	0.002 ± 0.000	0.044 ± 0.001	0.007 ± 0.000	0.082 ± 0.001	—	—	0.068	0.068	—	0.0410	0.0797	0.992	62.359 ± 1.552	0.170 ± 0.002	37.641 ± 1.552
LSAM	beta31	0.316	0.320	0.004 ± 0.004	0.012 ± 0.012	0.015 ± 0.001	0.122 ± 0.003	0.015 ± 0.001	0.122 ± 0.004	0.149 ± 0.007	0.386 ± 0.011	0.320	0.004	1.19	0.1206	0.1206	1.032	96.923 ± 0.553	0.493 ± 0.004	—
LSAM	beta32	0.316	0.313	-0.003 ± 0.004	-0.008 ± 0.013	0.015 ± 0.001	0.124 ± 0.003	0.015 ± 0.001	0.124 ± 0.003	0.155 ± 0.007	0.393 ± 0.011	0.314	-0.002	-0.61	0.1261	0.1262	1.018	96.718 ± 0.571	0.496 ± 0.004	—
LSAM	beta33	0.000	0.001	0.001 ± 0.002	—	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	—	—	0.000	0.000	—	0.0681	0.0681	1.033	96.103 ± 0.620	0.281 ± 0.003	3.897 ± 0.620
LSAM	beta34	0.000	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	—	—	0.001	0.001	—	0.0375	0.0375	0.977	96.205 ± 0.612	0.168 ± 0.002	3.795 ± 0.612
LSAM	beta35	0.000	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	—	—	0.000	0.000	—	0.0389	0.0389	0.998	96.923 ± 0.553	0.168 ± 0.002	3.077 ± 0.553
QML	beta31	0.316	0.276	-0.040 ± 0.002	-0.125 ± 0.006	0.004 ± 0.000	0.062 ± 0.001	0.005 ± 0.000	0.074 ± 0.002	0.055 ± 0.002	0.234 ± 0.006	0.277	-0.039	-12.24	0.0652	0.0758	1.059	89.846 ± 0.967	0.259 ± 0.006	—
QML	beta32	0.316	0.272	-0.044 ± 0.002	-0.138 ± 0.006	0.004 ± 0.000	0.064 ± 0.001	0.006 ± 0.000	0.078 ± 0.002	0.060 ± 0.002	0.245 ± 0.006	0.272	-0.044	-13.78	0.0659	0.0790	1.109	88.513 ± 1.021	0.279 ± 0.025	—
QML	beta33	0.000	-0.044	-0.044 ± 0.002	—	0.005 ± 0.000	0.073 ± 0.002	0.007 ± 0.000	0.085 ± 0.002	—	—	-0.042	-0.042	—	0.0701	0.0815	1.106	91.897 ± 0.874	0.315 ± 0.022	8.103 ± 0.874
QML	beta34	0.000	0.070	0.070 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.007 ± 0.000	0.083 ± 0.002	—	—	0.066	0.066	—	0.0416	0.0778	1.003	62.564 ± 1.550	0.175 ± 0.007	37.436 ± 1.550
QML	beta35	0.000	0.068	0.068 ± 0.001	—	0.002 ± 0.000	0.043 ± 0.001	0.007 ± 0.000	0.081 ± 0.001	—	—	0.067	0.067	—	0.0405	0.0784	1.033	62.872 ± 1.547	0.175 ± 0.007	37.128 ± 1.547
UPI	beta31	0.316	0.320	0.004 ± 0.004	0.013 ± 0.012	0.015 ± 0.001	0.121 ± 0.003	0.015 ± 0.001	0.121 ± 0.004	0.146 ± 0.007	0.382 ± 0.011	0.319	0.003	0.92	0.1202	0.1202	1.044	97.128 ± 0.535	0.495 ± 0.004	—
UPI	beta32	0.316	0.313	-0.003 ± 0.004	-0.008 ± 0.012	0.015 ± 0.001	0.123 ± 0.003	0.015 ± 0.001	0.123 ± 0.004	0.152 ± 0.007	0.390 ± 0.011	0.314	-0.002	-0.70	0.1231	0.1231	1.029	96.923 ± 0.553	0.497 ± 0.004	—
UPI	beta33	0.000	0.001	0.001 ± 0.002	—	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	—	—	-0.001	-0.001	—	0.0672	0.0672	1.140	98.051 ± 0.443	0.324 ± 0.005	1.949 ± 0.443
UPI	beta34	0.000	0.000	0.000 ± 0.001	—	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	—	—	-0.001	-0.001	—	0.0379	0.0379	1.000	96.718 ± 0.571	0.170 ± 0.002	3.282 ± 0.571
UPI	beta35	0.000	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	-0.001	-0.001	—	0.0377	0.0378	1.018	97.128 ± 0.535	0.169 ± 0.002	2.872 ± 0.535

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S81

Simulation 1 - Condition 81: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.6

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.283	-0.033 ± 0.003	-0.104 ± 0.008	0.007 ± 0.000	0.084 ± 0.002	0.008 ± 0.000	0.090 ± 0.002	0.082 ± 0.004	0.286 ± 0.007	0.280	-0.036	-11.25	0.0843	0.0915	1.012	94.010 ± 0.756	0.334 ± 0.001	—
LMS	beta32	0.316	0.293	-0.023 ± 0.003	-0.073 ± 0.009	0.007 ± 0.000	0.084 ± 0.002	0.008 ± 0.000	0.087 ± 0.002	0.077 ± 0.003	0.277 ± 0.007	0.292	-0.024	-7.56	0.0866	0.0898	1.009	93.807 ± 0.768	0.334 ± 0.002	—
LMS	beta33	0.000	-0.018	-0.018 ± 0.003	—	0.006 ± 0.000	0.080 ± 0.002	0.007 ± 0.000	0.082 ± 0.002	—	—	-0.017	-0.017	—	0.0780	0.0798	0.995	93.401 ± 0.791	0.310 ± 0.010	6.599 ± 0.791
LMS	beta34	0.000	0.036	0.036 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.003 ± 0.000	0.058 ± 0.001	—	—	0.035	0.035	—	0.0457	0.0574	0.958	84.873 ± 1.142	0.172 ± 0.003	15.127 ± 1.142
LMS	beta35	0.000	0.032	0.032 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	—	—	0.028	0.028	—	0.0478	0.0555	0.886	84.975 ± 1.139	0.169 ± 0.003	15.025 ± 1.139
LSAM	beta31	0.316	0.310	-0.006 ± 0.004	-0.018 ± 0.014	0.019 ± 0.001	0.137 ± 0.003	0.019 ± 0.001	0.137 ± 0.004	0.187 ± 0.009	0.433 ± 0.013	0.311	-0.005	-1.53	0.1306	0.1307	0.984	95.431 ± 0.665	0.527 ± 0.004	—
LSAM	beta32	0.316	0.328	0.012 ± 0.004	0.039 ± 0.014	0.019 ± 0.001	0.137 ± 0.003	0.019 ± 0.001	0.138 ± 0.004	0.190 ± 0.009	0.436 ± 0.012	0.329	0.013	4.20	0.1392	0.1398	0.975	95.330 ± 0.672	0.524 ± 0.004	—
LSAM	beta33	0.000	0.003	0.003 ± 0.003	—	0.007 ± 0.000	0.083 ± 0.002	0.007 ± 0.000	0.083 ± 0.002	—	—	0.003	0.003	—	0.0793	0.0793	0.952	95.025 ± 0.693	0.309 ± 0.003	4.975 ± 0.693
LSAM	beta34	0.000	0.002	0.002 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	—	—	0.002	0.002	—	0.0466	0.0467	0.945	94.213 ± 0.744	0.191 ± 0.003	5.787 ± 0.744
LSAM	beta35	0.000	-0.003	-0.003 ± 0.002	—	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	—	—	-0.002	-0.002	—	0.0502	0.0502	0.890	94.213 ± 0.744	0.188 ± 0.003	5.787 ± 0.744
QML	beta31	0.316	0.283	-0.033 ± 0.003	-0.104 ± 0.009	0.007 ± 0.000	0.085 ± 0.002	0.008 ± 0.000	0.091 ± 0.002	0.082 ± 0.004	0.287 ± 0.007	0.280	-0.036	-11.31	0.0846	0.0918	1.004	93.909 ± 0.762	0.333 ± 0.001	—
QML	beta32	0.316	0.293	-0.023 ± 0.003	-0.072 ± 0.009	0.007 ± 0.000	0.085 ± 0.002	0.008 ± 0.000	0.088 ± 0.002	0.077 ± 0.003	0.278 ± 0.007	0.292	-0.024	-7.56	0.0879	0.0911	0.997	93.706 ± 0.774	0.332 ± 0.001	—
QML	beta33	0.000	-0.018	-0.018 ± 0.003	—	0.006 ± 0.000	0.079 ± 0.002	0.007 ± 0.000	0.081 ± 0.002	—	—	-0.016	-0.016	—	0.0776	0.0794	0.965	93.401 ± 0.791	0.300 ± 0.003	6.599 ± 0.791
QML	beta34	0.000	0.036	0.036 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.003 ± 0.000	0.058 ± 0.001	—	—	0.035	0.035	—	0.0454	0.0573	0.948	84.670 ± 1.148	0.170 ± 0.002	15.330 ± 1.148
QML	beta35	0.000	0.032	0.032 ± 0.002	—	0.002 ± 0.000	0.048 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	—	—	0.028	0.028	—	0.0479	0.0556	0.875	84.975 ± 1.139	0.166 ± 0.002	15.025 ± 1.139
UPI	beta31	0.316	0.310	-0.006 ± 0.004	-0.018 ± 0.014	0.018 ± 0.001	0.135 ± 0.003	0.018 ± 0.001	0.135 ± 0.004	0.184 ± 0.009	0.428 ± 0.012	0.310	-0.006	-1.83	0.1287	0.1288	0.987	95.533 ± 0.658	0.524 ± 0.004	—
UPI	beta32	0.316	0.330	0.014 ± 0.004	0.044 ± 0.014	0.018 ± 0.001	0.136 ± 0.003	0.019 ± 0.001	0.137 ± 0.004	0.187 ± 0.009	0.432 ± 0.012	0.330	0.014	4.38	0.1350	0.1357	0.977	95.228 ± 0.679	0.521 ± 0.004	—
UPI	beta33	0.000	0.002	0.002 ± 0.003	—	0.007 ± 0.000	0.086 ± 0.002	0.007 ± 0.000	0.086 ± 0.003	—	—	0.003	0.003	—	0.0768	0.0768	1.027	95.431 ± 0.665	0.345 ± 0.006	4.569 ± 0.665
UPI	beta34	0.000	0.003	0.003 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.002	—	—	0.001	0.001	—	0.0455	0.0455	0.953	94.721 ± 0.713	0.192 ± 0.003	5.279 ± 0.713
UPI	beta35	0.000	-0.004	-0.004 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	—	—	-0.003	-0.003	—	0.0486	0.0486	0.906	94.010 ± 0.756	0.189 ± 0.003	5.990 ± 0.756

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S82

Simulation 1 - Condition 82: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.6

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.286	-0.030 ± 0.002	-0.095 ± 0.005	0.003 ± 0.000	0.052 ± 0.001	0.004 ± 0.000	0.060 ± 0.001	0.036 ± 0.002	0.189 ± 0.005	0.285	-0.031	-9.81	0.0534	0.0617	1.104	90.991 ± 0.906	0.224 ± 0.012	—
LMS	beta32	0.316	0.287	-0.029 ± 0.002	-0.092 ± 0.005	0.003 ± 0.000	0.052 ± 0.001	0.004 ± 0.000	0.060 ± 0.001	0.036 ± 0.002	0.189 ± 0.005	0.285	-0.031	-9.77	0.0512	0.0598	1.022	91.692 ± 0.873	0.209 ± 0.001	—
LMS	beta33	0.000	-0.015	-0.015 ± 0.001	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	—	—	-0.014	-0.014	—	0.0475	0.0494	1.090	94.094 ± 0.746	0.202 ± 0.014	5.906 ± 0.746
LMS	beta34	0.000	0.031	0.031 ± 0.001	—	0.001 ± 0.000	0.027 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	0.030	0.030	—	0.0257	0.0396	1.067	75.075 ± 1.369	0.112 ± 0.011	24.925 ± 1.369
LMS	beta35	0.000	0.032	0.032 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	0.030	0.030	—	0.0258	0.0395	0.972	76.877 ± 1.334	0.105 ± 0.003	23.123 ± 1.334
LSAM	beta31	0.316	0.316	0.000 ± 0.002	0.000 ± 0.008	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	0.062 ± 0.003	0.249 ± 0.007	0.317	0.001	0.17	0.0812	0.0813	1.000	94.995 ± 0.690	0.308 ± 0.001	—
LSAM	beta32	0.316	0.318	0.002 ± 0.003	0.005 ± 0.008	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	0.063 ± 0.003	0.251 ± 0.007	0.316	0.000	0.03	0.0748	0.0748	1.003	94.595 ± 0.715	0.311 ± 0.001	—
LSAM	beta33	0.000	0.000	-0.000 ± 0.001	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	—	—	0.001	0.001	—	0.0439	0.0440	1.004	94.995 ± 0.690	0.175 ± 0.001	5.005 ± 0.690
LSAM	beta34	0.000	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	—	—	0.000	0.000	—	0.0259	0.0259	0.953	95.796 ± 0.635	0.101 ± 0.001	4.204 ± 0.635
LSAM	beta35	0.000	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	—	—	0.000	0.000	—	0.0238	0.0238	0.960	94.094 ± 0.746	0.103 ± 0.001	5.906 ± 0.746
QML	beta31	0.316	0.286	-0.030 ± 0.002	-0.095 ± 0.005	0.003 ± 0.000	0.052 ± 0.001	0.004 ± 0.000	0.060 ± 0.001	0.036 ± 0.002	0.190 ± 0.005	0.285	-0.031	-9.70	0.0536	0.0618	1.009	90.991 ± 0.906	0.205 ± 0.001	—
QML	beta32	0.316	0.287	-0.029 ± 0.002	-0.091 ± 0.005	0.003 ± 0.000	0.052 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.036 ± 0.002	0.189 ± 0.005	0.285	-0.031	-9.69	0.0517	0.0601	1.006	91.391 ± 0.887	0.207 ± 0.001	—
QML	beta33	0.000	-0.015	-0.015 ± 0.001	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	—	—	-0.013	-0.013	—	0.0472	0.0491	1.013	94.094 ± 0.746	0.187 ± 0.002	5.906 ± 0.746
QML	beta34	0.000	0.031	0.031 ± 0.001	—	0.001 ± 0.000	0.027 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	0.030	0.030	—	0.0258	0.0397	0.953	75.175 ± 1.367	0.100 ± 0.001	24.825 ± 1.367
QML	beta35	0.000	0.031	0.031 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	0.030	0.030	—	0.0260	0.0397	0.944	76.777 ± 1.336	0.102 ± 0.001	23.223 ± 1.336
UPI	beta31	0.316	0.316	0.000 ± 0.002	0.001 ± 0.008	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	0.061 ± 0.003	0.248 ± 0.007	0.317	0.001	0.32	0.0841	0.0841	1.002	95.596 ± 0.649	0.308 ± 0.001	—
UPI	beta32	0.316	0.318	0.002 ± 0.003	0.006 ± 0.008	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	0.063 ± 0.003	0.250 ± 0.007	0.315	-0.001	-0.43	0.0738	0.0738	1.002	94.695 ± 0.709	0.311 ± 0.001	—
UPI	beta33	0.000	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	0.001	0.001	—	0.0443	0.0443	1.007	95.395 ± 0.663	0.181 ± 0.002	4.605 ± 0.663
UPI	beta34	0.000	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	—	—	0.001	0.001	—	0.0260	0.0260	0.951	96.096 ± 0.613	0.102 ± 0.001	3.904 ± 0.613
UPI	beta35	0.000	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	0.000	0.000	—	0.0241	0.0241	0.954	94.194 ± 0.740	0.104 ± 0.001	5.806 ± 0.740

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S83

Simulation 1 - Condition 83: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.8

Method	Param.	True	Mean-based										Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE	
LMS	beta31	0.316	0.301	-0.015 ± 0.002	-0.047 ± 0.008	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	0.059 ± 0.003	0.243 ± 0.007	0.301	-0.015	-4.78	0.0759	0.0774	1.012	94.895 ± 0.696	0.299 ± 0.001	—	
LMS	beta32	0.316	0.303	-0.013 ± 0.002	-0.040 ± 0.008	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	0.061 ± 0.003	0.247 ± 0.007	0.302	-0.014	-4.37	0.0759	0.0771	0.991	93.794 ± 0.763	0.299 ± 0.001	—	
LMS	beta33	0.000	-0.002	-0.002 ± 0.002	—	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	—	—	-0.002	-0.002	—	0.0608	0.0608	0.918	92.492 ± 0.834	0.223 ± 0.002	7.508 ± 0.834	
LMS	beta34	0.000	0.012	0.012 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.012	0.012	—	0.0320	0.0341	0.888	89.389 ± 0.974	0.122 ± 0.001	10.611 ± 0.974	
LMS	beta35	0.000	0.011	0.011 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.011	0.011	—	0.0334	0.0351	0.888	89.790 ± 0.958	0.121 ± 0.001	10.210 ± 0.958	
LSAM	beta31	0.316	0.317	0.001 ± 0.003	0.002 ± 0.009	0.009 ± 0.000	0.095 ± 0.002	0.009 ± 0.000	0.095 ± 0.003	0.089 ± 0.004	0.299 ± 0.008	0.318	0.002	0.70	0.0943	0.0943	0.999	95.395 ± 0.663	0.371 ± 0.002	—	
LSAM	beta32	0.316	0.319	0.003 ± 0.003	0.010 ± 0.010	0.009 ± 0.000	0.096 ± 0.002	0.009 ± 0.000	0.096 ± 0.003	0.092 ± 0.004	0.304 ± 0.009	0.317	0.001	0.34	0.0931	0.0931	0.986	93.594 ± 0.775	0.371 ± 0.002	—	
LSAM	beta33	0.000	0.003	0.003 ± 0.002	—	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	—	—	0.003	0.003	—	0.0595	0.0596	0.913	92.693 ± 0.823	0.223 ± 0.002	7.307 ± 0.823	
LSAM	beta34	0.000	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.000	0.000	—	0.0320	0.0320	0.893	91.792 ± 0.868	0.130 ± 0.002	8.208 ± 0.868	
LSAM	beta35	0.000	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.000	0.000	—	0.0345	0.0345	0.899	91.692 ± 0.873	0.131 ± 0.002	8.308 ± 0.873	
QML	beta31	0.316	0.301	-0.015 ± 0.002	-0.047 ± 0.008	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	0.060 ± 0.003	0.244 ± 0.007	0.301	-0.015	-4.78	0.0765	0.0779	1.004	94.795 ± 0.703	0.298 ± 0.001	—	
QML	beta32	0.316	0.303	-0.013 ± 0.002	-0.040 ± 0.008	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	0.061 ± 0.003	0.248 ± 0.007	0.302	-0.014	-4.29	0.0761	0.0773	0.986	93.794 ± 0.763	0.299 ± 0.001	—	
QML	beta33	0.000	-0.002	-0.002 ± 0.002	—	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	—	—	-0.002	-0.002	—	0.0610	0.0610	0.916	92.392 ± 0.839	0.222 ± 0.002	7.608 ± 0.839	
QML	beta34	0.000	0.012	0.012 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.012	0.012	—	0.0320	0.0340	0.880	89.289 ± 0.978	0.121 ± 0.001	10.711 ± 0.978	
QML	beta35	0.000	0.011	0.011 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.011	0.011	—	0.0338	0.0356	0.886	89.890 ± 0.954	0.121 ± 0.001	10.110 ± 0.954	
UPI	beta31	0.316	0.317	0.001 ± 0.003	0.003 ± 0.009	0.009 ± 0.000	0.094 ± 0.002	0.009 ± 0.000	0.094 ± 0.003	0.089 ± 0.004	0.298 ± 0.008	0.318	0.002	0.71	0.0919	0.0919	0.999	95.596 ± 0.649	0.369 ± 0.002	—	
UPI	beta32	0.316	0.319	0.003 ± 0.003	0.011 ± 0.010	0.009 ± 0.000	0.096 ± 0.002	0.009 ± 0.000	0.096 ± 0.003	0.092 ± 0.004	0.303 ± 0.009	0.318	0.002	0.52	0.0928	0.0928	0.986	93.594 ± 0.775	0.370 ± 0.002	—	
UPI	beta33	0.000	0.004	0.004 ± 0.002	—	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	—	—	0.005	0.005	—	0.0603	0.0605	0.920	93.393 ± 0.786	0.228 ± 0.002	6.607 ± 0.786	
UPI	beta34	0.000	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.000	0.000	—	0.0322	0.0322	0.898	91.892 ± 0.864	0.130 ± 0.002	8.108 ± 0.864	
UPI	beta35	0.000	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.001	-0.001	—	0.0348	0.0348	0.899	91.291 ± 0.892	0.131 ± 0.002	8.709 ± 0.892	

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S84

Simulation 1 - Condition 84: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, $N = 1000$, Reliability = 0.8

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.302	-0.014 ± 0.002	-0.043 ± 0.005	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.025 ± 0.001	0.157 ± 0.004	0.303	-0.013	-4.09	0.0464	0.0482	0.994	93.700 ± 0.768	0.186 ± 0.001	
LMS	beta32	0.316	0.301	-0.015 ± 0.002	-0.048 ± 0.005	0.002 ± 0.000	0.049 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.026 ± 0.001	0.162 ± 0.004	0.302	-0.014	-4.57	0.0476	0.0498	0.967	92.900 ± 0.812	0.186 ± 0.000	—
LMS	beta33	0.000	-0.003	-0.003 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.003	-0.003	—	0.0368	0.0369	0.916	92.200 ± 0.848	0.137 ± 0.001	7.800 ± 0.848
LMS	beta34	0.000	0.011	0.011 ± 0.001	—	0.000 ± 0.000	0.020 ± 0.000	0.001 ± 0.000	0.023 ± 0.001	—	—	0.010	0.010	—	0.0197	0.0223	0.919	89.500 ± 0.969	0.073 ± 0.001	10.500 ± 0.969
LMS	beta35	0.000	0.012	0.012 ± 0.001	—	0.000 ± 0.000	0.020 ± 0.000	0.001 ± 0.000	0.024 ± 0.001	—	—	0.012	0.012	—	0.0198	0.0229	0.907	87.800 ± 1.035	0.073 ± 0.001	12.200 ± 1.035
LSAM	beta31	0.316	0.317	0.001 ± 0.002	0.004 ± 0.006	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.035 ± 0.002	0.186 ± 0.005	0.318	0.002	0.53	0.0581	0.0581	0.981	94.100 ± 0.745	0.226 ± 0.001	—
LSAM	beta32	0.316	0.316	-0.000 ± 0.002	-0.001 ± 0.006	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.037 ± 0.002	0.191 ± 0.006	0.316	0.000	0.15	0.0573	0.0573	0.958	93.700 ± 0.768	0.227 ± 0.001	—
LSAM	beta33	0.000	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.001	0.001	—	0.0360	0.0360	0.912	92.300 ± 0.843	0.133 ± 0.001	7.700 ± 0.843
LSAM	beta34	0.000	0.000	-0.000 ± 0.001	—	0.000 ± 0.000	0.021 ± 0.000	0.000 ± 0.000	0.021 ± 0.001	—	—	0.000	0.000	—	0.0197	0.0197	0.922	93.200 ± 0.796	0.075 ± 0.001	6.800 ± 0.796
LSAM	beta35	0.000	0.000	0.000 ± 0.001	—	0.000 ± 0.000	0.021 ± 0.001	0.000 ± 0.000	0.021 ± 0.001	—	—	0.000	0.000	—	0.0207	0.0207	0.914	93.100 ± 0.801	0.076 ± 0.001	6.900 ± 0.801
QML	beta31	0.316	0.303	-0.013 ± 0.002	-0.043 ± 0.005	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.025 ± 0.001	0.158 ± 0.004	0.303	-0.013	-4.04	0.0468	0.0486	0.985	93.700 ± 0.768	0.185 ± 0.000	—
QML	beta32	0.316	0.301	-0.015 ± 0.002	-0.048 ± 0.005	0.002 ± 0.000	0.049 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.027 ± 0.001	0.163 ± 0.004	0.302	-0.014	-4.57	0.0479	0.0500	0.961	92.800 ± 0.817	0.186 ± 0.000	—
QML	beta33	0.000	-0.003	-0.003 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.003	-0.003	—	0.0367	0.0368	0.916	92.100 ± 0.853	0.137 ± 0.001	7.900 ± 0.853
QML	beta34	0.000	0.011	0.011 ± 0.001	—	0.000 ± 0.000	0.020 ± 0.000	0.001 ± 0.000	0.023 ± 0.001	—	—	0.011	0.011	—	0.0196	0.0223	0.909	89.600 ± 0.965	0.072 ± 0.001	10.400 ± 0.965
QML	beta35	0.000	0.012	0.012 ± 0.001	—	0.000 ± 0.000	0.021 ± 0.000	0.001 ± 0.000	0.024 ± 0.001	—	—	0.011	0.011	—	0.0200	0.0230	0.905	87.700 ± 1.039	0.073 ± 0.001	12.300 ± 1.039
UPI	beta31	0.316	0.318	0.002 ± 0.002	0.005 ± 0.006	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.034 ± 0.002	0.185 ± 0.005	0.318	0.002	0.56	0.0571	0.0572	0.981	94.300 ± 0.733	0.225 ± 0.001	—
UPI	beta32	0.316	0.316	-0.000 ± 0.002	-0.001 ± 0.006	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.036 ± 0.002	0.190 ± 0.006	0.316	0.000	0.12	0.0573	0.0573	0.961	94.200 ± 0.739	0.227 ± 0.001	—
UPI	beta33	0.000	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.000	0.000	—	0.0356	0.0356	0.923	93.100 ± 0.801	0.133 ± 0.001	6.900 ± 0.801
UPI	beta34	0.000	0.000	-0.000 ± 0.001	—	0.000 ± 0.000	0.021 ± 0.001	0.000 ± 0.000	0.021 ± 0.001	—	—	-0.001	-0.001	—	0.0197	0.0198	0.921	93.300 ± 0.791	0.074 ± 0.001	6.700 ± 0.791
UPI	beta35	0.000	0.000	0.000 ± 0.001	—	0.000 ± 0.000	0.021 ± 0.001	0.000 ± 0.000	0.021 ± 0.001	—	—	0.000	0.000	—	0.0202	0.0202	0.921	93.000 ± 0.807	0.076 ± 0.001	7.000 ± 0.807

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S85

Simulation 1 - Condition 85: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.4

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Type I (%) $\pm$ MCSE
LMS	beta31	0.316	0.315	-0.001 $\pm$ 0.004	-0.005 $\pm$ 0.013	0.008 $\pm$ 0.001	0.092 $\pm$ 0.003	0.008 $\pm$ 0.001	0.092 $\pm$ 0.004	0.084 $\pm$ 0.006	0.290 $\pm$ 0.012	0.315	-0.001	-0.24	0.0890	0.0890	1.008	95.098 $\pm$ 0.956	0.362 $\pm$ 0.003	—
LMS	beta32	0.316	0.314	-0.002 $\pm$ 0.004	-0.007 $\pm$ 0.013	0.009 $\pm$ 0.001	0.096 $\pm$ 0.003	0.009 $\pm$ 0.001	0.096 $\pm$ 0.004	0.092 $\pm$ 0.006	0.304 $\pm$ 0.012	0.312	-0.004	-1.26	0.0954	0.0954	0.973	95.098 $\pm$ 0.956	0.367 $\pm$ 0.003	—
LMS	beta33	0.000	-0.010	-0.010 $\pm$ 0.005	—	0.015 $\pm$ 0.001	0.122 $\pm$ 0.005	0.015 $\pm$ 0.001	0.122 $\pm$ 0.005	—	—	-0.014	-0.014	—	0.1204	0.1212	1.048	94.902 $\pm$ 0.974	0.500 $\pm$ 0.008	5.098 $\pm$ 0.974
LMS	beta34	0.000	0.001	0.001 $\pm$ 0.004	—	0.008 $\pm$ 0.001	0.088 $\pm$ 0.003	0.008 $\pm$ 0.001	0.088 $\pm$ 0.004	—	—	0.001	0.001	—	0.0930	0.0930	1.029	95.294 $\pm$ 0.938	0.357 $\pm$ 0.005	4.706 $\pm$ 0.938
LMS	beta35	0.000	-0.001	-0.001 $\pm$ 0.004	—	0.007 $\pm$ 0.000	0.085 $\pm$ 0.003	0.007 $\pm$ 0.000	0.085 $\pm$ 0.004	—	—	-0.001	-0.001	—	0.0886	0.0886	1.089	97.451 $\pm$ 0.698	0.364 $\pm$ 0.006	2.549 $\pm$ 0.698
LSAM	beta31	0.316	0.321	0.005 $\pm$ 0.005	0.014 $\pm$ 0.016	0.014 $\pm$ 0.001	0.117 $\pm$ 0.004	0.014 $\pm$ 0.001	0.117 $\pm$ 0.005	0.136 $\pm$ 0.010	0.369 $\pm$ 0.016	0.321	0.005	1.45	0.1170	0.1171	1.372	97.059 $\pm$ 0.748	0.627 $\pm$ 0.031	—
LSAM	beta32	0.316	0.323	0.007 $\pm$ 0.005	0.021 $\pm$ 0.016	0.014 $\pm$ 0.001	0.116 $\pm$ 0.004	0.014 $\pm$ 0.001	0.117 $\pm$ 0.005	0.136 $\pm$ 0.009	0.369 $\pm$ 0.015	0.319	0.003	0.86	0.1261	0.1261	1.304	97.451 $\pm$ 0.698	0.595 $\pm$ 0.024	—
LSAM	beta33	0.000	0.008	0.008 $\pm$ 0.008	—	0.031 $\pm$ 0.003	0.175 $\pm$ 0.008	0.031 $\pm$ 0.003	0.176 $\pm$ 0.009	—	—	0.005	0.005	—	0.1727	0.1728	1.540	98.824 $\pm$ 0.477	1.059 $\pm$ 0.076	1.176 $\pm$ 0.477
LSAM	beta34	0.000	0.015	0.015 $\pm$ 0.011	—	0.058 $\pm$ 0.005	0.240 $\pm$ 0.011	0.058 $\pm$ 0.005	0.240 $\pm$ 0.012	—	—	0.009	0.009	—	0.2188	0.2190	1.474	100.000 $\pm$ 0.000	1.386 $\pm$ 0.105	0.000 $\pm$ 0.000
LSAM	beta35	0.000	-0.007	-0.007 $\pm$ 0.010	—	0.051 $\pm$ 0.005	0.226 $\pm$ 0.010	0.051 $\pm$ 0.005	0.226 $\pm$ 0.011	—	—	-0.004	-0.004	—	0.1995	0.1995	1.586	99.804 $\pm$ 0.196	1.407 $\pm$ 0.104	0.196 $\pm$ 0.196
QML	beta31	0.316	0.315	-0.001 $\pm$ 0.004	-0.004 $\pm$ 0.013	0.008 $\pm$ 0.001	0.092 $\pm$ 0.003	0.008 $\pm$ 0.001	0.092 $\pm$ 0.004	0.084 $\pm$ 0.006	0.290 $\pm$ 0.012	0.316	0.000	-0.13	0.0879	0.0879	1.008	95.294 $\pm$ 0.938	0.363 $\pm$ 0.003	—
QML	beta32	0.316	0.314	-0.002 $\pm$ 0.004	-0.006 $\pm$ 0.013	0.009 $\pm$ 0.001	0.096 $\pm$ 0.003	0.009 $\pm$ 0.001	0.096 $\pm$ 0.004	0.092 $\pm$ 0.006	0.304 $\pm$ 0.012	0.313	-0.003	-1.05	0.0944	0.0945	0.974	94.706 $\pm$ 0.992	0.367 $\pm$ 0.003	—
QML	beta33	0.000	-0.010	-0.010 $\pm$ 0.005	—	0.015 $\pm$ 0.001	0.124 $\pm$ 0.005	0.015 $\pm$ 0.001	0.124 $\pm$ 0.006	—	—	-0.012	-0.012	—	0.1202	0.1208	1.041	95.882 $\pm$ 0.880	0.505 $\pm$ 0.008	4.118 $\pm$ 0.880
QML	beta34	0.000	0.001	0.001 $\pm$ 0.004	—	0.008 $\pm$ 0.001	0.089 $\pm$ 0.003	0.008 $\pm$ 0.001	0.089 $\pm$ 0.004	—	—	0.002	0.002	—	0.0941	0.0941	1.030	95.294 $\pm$ 0.938	0.359 $\pm$ 0.005	4.706 $\pm$ 0.938
QML	beta35	0.000	-0.001	-0.001 $\pm$ 0.004	—	0.007 $\pm$ 0.001	0.086 $\pm$ 0.003	0.007 $\pm$ 0.001	0.086 $\pm$ 0.004	—	—	-0.004	-0.004	—	0.0868	0.0869	1.084	97.255 $\pm$ 0.724	0.366 $\pm$ 0.006	2.745 $\pm$ 0.724
UPI	beta31	0.316	0.325	0.009 $\pm$ 0.005	0.028 $\pm$ 0.015	0.012 $\pm$ 0.001	0.110 $\pm$ 0.004	0.012 $\pm$ 0.001	0.110 $\pm$ 0.004	0.122 $\pm$ 0.008	0.349 $\pm$ 0.014	0.320	0.004	1.40	0.1205	0.1205	1.662	97.059 $\pm$ 0.748	0.718 $\pm$ 0.048	—
UPI	beta32	0.316	0.318	0.002 $\pm$ 0.005	0.006 $\pm$ 0.016	0.013 $\pm$ 0.001	0.115 $\pm$ 0.004	0.013 $\pm$ 0.001	0.115 $\pm$ 0.005	0.133 $\pm$ 0.009	0.365 $\pm$ 0.015	0.313	-0.003	-0.86	0.1227	0.1228	1.476	97.647 $\pm$ 0.671	0.667 $\pm$ 0.032	—
UPI	beta33	0.000	0.012	0.012 $\pm$ 0.007	—	0.029 $\pm$ 0.002	0.169 $\pm$ 0.007	0.029 $\pm$ 0.003	0.170 $\pm$ 0.008	—	—	0.005	0.005	—	0.1605	0.1606	1.745	99.804 $\pm$ 0.196	1.158 $\pm$ 0.062	0.196 $\pm$ 0.196
UPI	beta34	0.000	0.013	0.013 $\pm$ 0.011	—	0.061 $\pm$ 0.006	0.246 $\pm$ 0.012	0.061 $\pm$ 0.006	0.246 $\pm$ 0.013	—	—	0.005	0.005	—	0.2250	0.2251	2.130	99.804 $\pm$ 0.196	2.057 $\pm$ 0.143	0.196 $\pm$ 0.196
UPI	beta35	0.000	0.014	0.014 $\pm$ 0.011	—	0.059 $\pm$ 0.006	0.243 $\pm$ 0.012	0.059 $\pm$ 0.006	0.243 $\pm$ 0.013	—	—	0.004	0.004	—	0.2303	0.2304	2.178	99.804 $\pm$ 0.196	2.076 $\pm$ 0.159	0.196 $\pm$ 0.196

Note. MCSE = Monte Carlo Standard Error shown as $\backslash$ u00b1 value. Coverage, Type I Error, and Power expressed as percentages. $\backslash$ u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S86

Simulation 1 - Condition 86: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, $N = 1000$, Reliability = 0.4

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Type I (%) $\pm$ MCSE
LMS	beta31	0.316	0.316	-0.000 $\pm$ 0.002	-0.001 $\pm$ 0.006	0.003 $\pm$ 0.000	0.057 $\pm$ 0.001	0.003 $\pm$ 0.000	0.057 $\pm$ 0.002	0.032 $\pm$ 0.002	0.180 $\pm$ 0.005	0.314	-0.002	-0.77	0.0586	0.0586	1.020	96.102 $\pm$ 0.646	0.227 $\pm$ 0.001	—
LMS	beta32	0.316	0.317	0.001 $\pm$ 0.002	0.004 $\pm$ 0.006	0.004 $\pm$ 0.000	0.059 $\pm$ 0.001	0.003 $\pm$ 0.000	0.059 $\pm$ 0.002	0.035 $\pm$ 0.002	0.187 $\pm$ 0.005	0.315	-0.001	-0.35	0.0600	0.0600	0.982	94.321 $\pm$ 0.772	0.228 $\pm$ 0.001	—
LMS	beta33	0.000	-0.021	-0.021 $\pm$ 0.003	—	0.006 $\pm$ 0.000	0.077 $\pm$ 0.002	0.006 $\pm$ 0.000	0.080 $\pm$ 0.002	—	—	-0.018	-0.018	—	0.0761	0.0783	1.037	93.653 $\pm$ 0.814	0.315 $\pm$ 0.002	6.347 $\pm$ 0.814
LMS	beta34	0.000	-0.004	-0.004 $\pm$ 0.002	—	0.003 $\pm$ 0.000	0.056 $\pm$ 0.002	0.003 $\pm$ 0.000	0.056 $\pm$ 0.002	—	—	-0.006	-0.006	—	0.0520	0.0523	1.030	95.657 $\pm$ 0.680	0.225 $\pm$ 0.002	4.343 $\pm$ 0.680
LMS	beta35	0.000	0.002	0.002 $\pm$ 0.002	—	0.003 $\pm$ 0.000	0.056 $\pm$ 0.001	0.003 $\pm$ 0.000	0.056 $\pm$ 0.002	—	—	0.001	0.001	—	0.0545	0.0545	1.025	95.323 $\pm$ 0.705	0.226 $\pm$ 0.002	4.677 $\pm$ 0.705
LSAM	beta31	0.316	0.320	0.004 $\pm$ 0.002	0.014 $\pm$ 0.007	0.004 $\pm$ 0.000	0.065 $\pm$ 0.002	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	0.043 $\pm$ 0.002	0.207 $\pm$ 0.007	0.318	0.002	0.58	0.0663	0.0663	1.051	97.105 $\pm$ 0.560	0.270 $\pm$ 0.002	—
LSAM	beta32	0.316	0.319	0.003 $\pm$ 0.002	0.010 $\pm$ 0.007	0.004 $\pm$ 0.000	0.067 $\pm$ 0.002	0.004 $\pm$ 0.000	0.067 $\pm$ 0.002	0.045 $\pm$ 0.002	0.211 $\pm$ 0.006	0.317	0.001	0.46	0.0646	0.0646	1.035	95.880 $\pm$ 0.663	0.270 $\pm$ 0.003	—
LSAM	beta33	0.000	0.001	0.001 $\pm$ 0.003	—	0.008 $\pm$ 0.000	0.092 $\pm$ 0.003	0.008 $\pm$ 0.000	0.092 $\pm$ 0.003	—	—	0.003	0.003	—	0.0898	0.0898	1.078	96.882 $\pm$ 0.580	0.389 $\pm$ 0.007	3.118 $\pm$ 0.580
LSAM	beta34	0.000	-0.010	-0.010 $\pm$ 0.004	—	0.014 $\pm$ 0.001	0.120 $\pm$ 0.004	0.015 $\pm$ 0.001	0.121 $\pm$ 0.004	—	—	-0.009	-0.009	—	0.1063	0.1068	1.087	98.330 $\pm$ 0.428	0.512 $\pm$ 0.014	1.670 $\pm$ 0.428
LSAM	beta35	0.000	0.004	0.004 $\pm$ 0.004	—	0.015 $\pm$ 0.001	0.122 $\pm$ 0.004	0.015 $\pm$ 0.001	0.122 $\pm$ 0.004	—	—	0.007	0.007	—	0.1091	0.1094	1.046	98.107 $\pm$ 0.455	0.502 $\pm$ 0.009	1.893 $\pm$ 0.455
QML	beta31	0.316	0.316	0.000 $\pm$ 0.002	0.000 $\pm$ 0.006	0.003 $\pm$ 0.000	0.057 $\pm$ 0.001	0.003 $\pm$ 0.000	0.057 $\pm$ 0.002	0.032 $\pm$ 0.002	0.180 $\pm$ 0.005	0.314	-0.002	-0.70	0.0592	0.0593	1.018	96.214 $\pm$ 0.637	0.227 $\pm$ 0.001	—
QML	beta32	0.316	0.317	0.001 $\pm$ 0.002	0.005 $\pm$ 0.006	0.004 $\pm$ 0.000	0.059 $\pm$ 0.001	0.003 $\pm$ 0.000	0.059 $\pm$ 0.002	0.035 $\pm$ 0.002	0.187 $\pm$ 0.005	0.315	-0.001	-0.21	0.0596	0.0596	0.982	94.209 $\pm$ 0.779	0.228 $\pm$ 0.001	—
QML	beta33	0.000	-0.021	-0.021 $\pm$ 0.003	—	0.006 $\pm$ 0.000	0.077 $\pm$ 0.002	0.006 $\pm$ 0.000	0.080 $\pm$ 0.002	—	—	-0.018	-0.018	—	0.0766	0.0787	1.040	93.987 $\pm$ 0.793	0.316 $\pm$ 0.002	6.013 $\pm$ 0.793
QML	beta34	0.000	-0.004	-0.004 $\pm$ 0.002	—	0.003 $\pm$ 0.000	0.056 $\pm$ 0.002	0.003 $\pm$ 0.000	0.056 $\pm$ 0.002	—	—	-0.005	-0.005	—	0.0521	0.0523	1.035	95.880 $\pm$ 0.663	0.226 $\pm$ 0.002	4.120 $\pm$ 0.663
QML	beta35	0.000	0.002	0.002 $\pm$ 0.002	—	0.003 $\pm$ 0.000	0.056 $\pm$ 0.001	0.003 $\pm$ 0.000	0.056 $\pm$ 0.002	—	—	0.001	0.001	—	0.0540	0.0540	1.034	95.546 $\pm$ 0.688	0.227 $\pm$ 0.002	4.454 $\pm$ 0.688
UPI	beta31	0.316	0.321	0.005 $\pm$ 0.002	0.015 $\pm$ 0.007	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	0.044 $\pm$ 0.002	0.210 $\pm$ 0.007	0.320	0.004	1.37	0.0658	0.0659	1.174	98.107 $\pm$ 0.455	0.305 $\pm$ 0.006	—
UPI	beta32	0.316	0.320	0.004 $\pm$ 0.002	0.011 $\pm$ 0.007	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.005 $\pm$ 0.000	0.069 $\pm$ 0.002	0.047 $\pm$ 0.002	0.217 $\pm$ 0.007	0.318	0.002	0.75	0.0671	0.0671	1.140	96.437 $\pm$ 0.619	0.306 $\pm$ 0.006	—
UPI	beta33	0.000	0.002	0.002 $\pm$ 0.003	—	0.009 $\pm$ 0.001	0.097 $\pm$ 0.003	0.009 $\pm$ 0.001	0.097 $\pm$ 0.003	—	—	0.005	0.005	—	0.0886	0.0888	1.217	98.330 $\pm$ 0.428	0.461 $\pm$ 0.011	1.670 $\pm$ 0.428
UPI	beta34	0.000	-0.009	-0.009 $\pm$ 0.005	—	0.023 $\pm$ 0.002	0.153 $\pm$ 0.005	0.024 $\pm$ 0.002	0.153 $\pm$ 0.006	—	—	-0.010	-0.010	—	0.1259	0.1262	1.318	99.332 $\pm$ 0.272	0.791 $\pm$ 0.031	0.668 $\pm$ 0.272
UPI	beta35	0.000	0.008	0.008 $\pm$ 0.005	—	0.022 $\pm$ 0.001	0.149 $\pm$ 0.005	0.022 $\pm$ 0.001	0.149 $\pm$ 0.005	—	—	0.003	0.003	—	0.1262	0.1263	1.332	99.555 $\pm$ 0.222	0.778 $\pm$ 0.031	0.445 $\pm$ 0.222

Note. MCSE = Monte Carlo Standard Error shown as $\backslash u00b1$ value. Coverage, Type I Error, and Power expressed as percentages. $\backslash u2014$ indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S87*Simulation 1 - Condition 87: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.6*

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.314	-0.002 ± 0.002	-0.005 ± 0.007	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.047 ± 0.002	0.217 ± 0.006	0.314	-0.002	-0.62	0.0719	0.0720	0.980	94.556 ± 0.720	0.264 ± 0.001	—
LMS	beta32	0.316	0.316	0.000 ± 0.002	0.001 ± 0.007	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.047 ± 0.002	0.217 ± 0.006	0.315	-0.001	-0.27	0.0673	0.0673	0.980	95.060 ± 0.688	0.264 ± 0.001	—
LMS	beta33	0.000	-0.012	-0.012 ± 0.003	—	0.007 ± 0.000	0.087 ± 0.002	0.008 ± 0.000	0.087 ± 0.002	—	—	-0.015	-0.015	—	0.0832	0.0847	0.959	94.153 ± 0.745	0.325 ± 0.002	5.847 ± 0.745
LMS	beta34	0.000	0.002	0.002 ± 0.002	—	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	—	—	0.000	0.000	—	0.0695	0.0695	0.940	93.851 ± 0.763	0.264 ± 0.002	6.149 ± 0.763
LMS	beta35	0.000	-0.001	-0.001 ± 0.002	—	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	—	—	0.000	0.000	—	0.0653	0.0653	1.031	95.363 ± 0.668	0.265 ± 0.002	4.637 ± 0.668
LSAM	beta31	0.316	0.316	-0.000 ± 0.002	-0.001 ± 0.007	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.054 ± 0.002	0.232 ± 0.006	0.317	0.001	0.24	0.0759	0.0759	0.989	95.060 ± 0.688	0.284 ± 0.001	—
LSAM	beta32	0.316	0.318	0.002 ± 0.002	0.008 ± 0.007	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.053 ± 0.002	0.230 ± 0.006	0.318	0.002	0.65	0.0722	0.0723	0.996	95.766 ± 0.639	0.283 ± 0.001	—
LSAM	beta33	0.000	-0.002	-0.002 ± 0.003	—	0.009 ± 0.000	0.092 ± 0.002	0.009 ± 0.000	0.092 ± 0.003	—	—	-0.005	-0.005	—	0.0870	0.0871	0.963	94.254 ± 0.739	0.348 ± 0.003	5.746 ± 0.739
LSAM	beta34	0.000	0.005	0.005 ± 0.004	—	0.014 ± 0.001	0.117 ± 0.003	0.014 ± 0.001	0.117 ± 0.003	—	—	0.001	0.001	—	0.1076	0.1076	0.939	95.262 ± 0.675	0.431 ± 0.004	4.738 ± 0.675
LSAM	beta35	0.000	0.000	-0.000 ± 0.003	—	0.011 ± 0.001	0.104 ± 0.003	0.011 ± 0.001	0.104 ± 0.003	—	—	-0.001	-0.001	—	0.1013	0.1013	1.028	96.371 ± 0.594	0.420 ± 0.003	3.629 ± 0.594
QML	beta31	0.316	0.315	-0.001 ± 0.002	-0.005 ± 0.007	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.047 ± 0.002	0.217 ± 0.006	0.314	-0.002	-0.63	0.0719	0.0719	0.979	94.556 ± 0.720	0.263 ± 0.001	—
QML	beta32	0.316	0.316	0.000 ± 0.002	0.001 ± 0.007	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.047 ± 0.002	0.217 ± 0.006	0.316	0.000	-0.15	0.0669	0.0669	0.979	95.060 ± 0.688	0.263 ± 0.001	—
QML	beta33	0.000	-0.012	-0.012 ± 0.003	—	0.007 ± 0.000	0.086 ± 0.002	0.008 ± 0.000	0.087 ± 0.002	—	—	-0.016	-0.016	—	0.0836	0.0850	0.962	93.851 ± 0.763	0.326 ± 0.002	6.149 ± 0.763
QML	beta34	0.000	0.002	0.002 ± 0.002	—	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	—	—	0.000	0.000	—	0.0686	0.0686	0.946	93.851 ± 0.763	0.265 ± 0.002	6.149 ± 0.763
QML	beta35	0.000	0.000	-0.000 ± 0.002	—	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	—	—	0.000	0.000	—	0.0653	0.0653	1.031	95.968 ± 0.625	0.264 ± 0.002	4.032 ± 0.625
UPI	beta31	0.316	0.316	-0.000 ± 0.002	-0.000 ± 0.008	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	0.056 ± 0.003	0.236 ± 0.007	0.317	0.001	0.16	0.0752	0.0752	0.985	94.859 ± 0.701	0.289 ± 0.001	—
UPI	beta32	0.316	0.319	0.003 ± 0.002	0.009 ± 0.007	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.054 ± 0.002	0.232 ± 0.006	0.318	0.002	0.66	0.0724	0.0724	1.005	95.464 ± 0.661	0.288 ± 0.001	—
UPI	beta33	0.000	-0.002	-0.002 ± 0.003	—	0.009 ± 0.001	0.096 ± 0.003	0.009 ± 0.001	0.096 ± 0.003	—	—	-0.004	-0.004	—	0.0890	0.0891	0.979	95.867 ± 0.632	0.368 ± 0.003	4.133 ± 0.632
UPI	beta34	0.000	0.003	0.003 ± 0.004	—	0.018 ± 0.001	0.133 ± 0.004	0.018 ± 0.001	0.133 ± 0.004	—	—	0.004	0.004	—	0.1179	0.1180	1.001	97.278 ± 0.517	0.522 ± 0.008	2.722 ± 0.517
UPI	beta35	0.000	0.000	0.000 ± 0.004	—	0.014 ± 0.001	0.119 ± 0.003	0.014 ± 0.001	0.119 ± 0.004	—	—	0.000	0.000	—	0.1083	0.1083	1.103	97.077 ± 0.535	0.516 ± 0.009	2.923 ± 0.535

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S88

Simulation 1 - Condition 88: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.6

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.316	0.000 ± 0.001	0.001 ± 0.004	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.018 ± 0.001	0.135 ± 0.004	0.317	0.001	0.19	0.0442	0.0442	0.987	94.494 ± 0.722	0.165 ± 0.000	—
LMS	beta32	0.316	0.317	0.001 ± 0.001	0.005 ± 0.004	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.016 ± 0.001	0.127 ± 0.003	0.317	0.001	0.44	0.0401	0.0402	1.047	96.296 ± 0.598	0.165 ± 0.000	—
LMS	beta33	0.000	-0.008	-0.008 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	—	—	-0.005	-0.005	—	0.0475	0.0477	1.062	94.895 ± 0.696	0.204 ± 0.001	5.105 ± 0.696
LMS	beta34	0.000	-0.003	-0.003 ± 0.001	—	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	-0.004	-0.004	—	0.0422	0.0424	0.999	94.895 ± 0.696	0.164 ± 0.001	5.105 ± 0.696
LMS	beta35	0.000	0.000	0.000 ± 0.001	—	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	0.000	0.000	—	0.0426	0.0426	1.014	94.895 ± 0.696	0.167 ± 0.001	5.105 ± 0.696
LSAM	beta31	0.316	0.318	0.002 ± 0.001	0.006 ± 0.004	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.020 ± 0.001	0.141 ± 0.004	0.318	0.002	0.50	0.0451	0.0451	0.989	94.695 ± 0.709	0.172 ± 0.000	—
LSAM	beta32	0.316	0.318	0.002 ± 0.001	0.008 ± 0.004	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.018 ± 0.001	0.133 ± 0.004	0.318	0.002	0.52	0.0401	0.0402	1.053	96.697 ± 0.565	0.173 ± 0.000	—
LSAM	beta33	0.000	0.003	0.003 ± 0.002	—	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	—	—	0.004	0.004	—	0.0490	0.0492	1.038	95.295 ± 0.670	0.205 ± 0.001	4.705 ± 0.670
LSAM	beta34	0.000	-0.004	-0.004 ± 0.002	—	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	—	—	-0.003	-0.003	—	0.0643	0.0643	0.983	95.095 ± 0.683	0.251 ± 0.001	4.905 ± 0.683
LSAM	beta35	0.000	0.001	0.001 ± 0.002	—	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	—	—	0.001	0.001	—	0.0639	0.0640	0.994	95.696 ± 0.642	0.252 ± 0.001	4.304 ± 0.642
QML	beta31	0.316	0.316	0.000 ± 0.001	0.001 ± 0.004	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.018 ± 0.001	0.135 ± 0.004	0.317	0.001	0.24	0.0441	0.0441	0.986	94.494 ± 0.722	0.165 ± 0.000	—
QML	beta32	0.316	0.317	0.001 ± 0.001	0.005 ± 0.004	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.016 ± 0.001	0.127 ± 0.003	0.317	0.001	0.40	0.0400	0.0400	1.046	96.196 ± 0.605	0.165 ± 0.000	—
QML	beta33	0.000	-0.008	-0.008 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	—	—	-0.005	-0.005	—	0.0472	0.0475	1.065	94.795 ± 0.703	0.204 ± 0.001	5.205 ± 0.703
QML	beta34	0.000	-0.003	-0.003 ± 0.001	—	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	-0.004	-0.004	—	0.0421	0.0423	1.007	94.995 ± 0.690	0.164 ± 0.001	5.005 ± 0.690
QML	beta35	0.000	0.000	0.000 ± 0.001	—	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	0.000	0.000	—	0.0422	0.0422	1.009	94.995 ± 0.690	0.166 ± 0.001	5.005 ± 0.690
UPI	beta31	0.316	0.318	0.002 ± 0.001	0.006 ± 0.004	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.020 ± 0.001	0.141 ± 0.004	0.317	0.001	0.42	0.0444	0.0444	0.991	94.695 ± 0.709	0.173 ± 0.000	—
UPI	beta32	0.316	0.319	0.003 ± 0.001	0.008 ± 0.004	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.018 ± 0.001	0.134 ± 0.004	0.318	0.002	0.50	0.0412	0.0412	1.047	96.396 ± 0.590	0.174 ± 0.000	—
UPI	beta33	0.000	0.003	0.003 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	—	—	0.005	0.005	—	0.0491	0.0494	1.042	95.495 ± 0.656	0.208 ± 0.001	4.505 ± 0.656
UPI	beta34	0.000	-0.003	-0.003 ± 0.002	—	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	—	—	-0.004	-0.004	—	0.0674	0.0675	1.005	96.296 ± 0.598	0.269 ± 0.002	3.704 ± 0.598
UPI	beta35	0.000	0.000	0.000 ± 0.002	—	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	—	—	0.001	0.001	—	0.0668	0.0668	1.012	96.897 ± 0.549	0.272 ± 0.002	3.103 ± 0.549

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S89

Simulation 1 - Condition 89: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.8

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.315	-0.001 ± 0.002	-0.003 ± 0.005	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.028 ± 0.001	0.167 ± 0.005	0.315	-0.001	-0.18	0.0539	0.0539	1.038	95.900 ± 0.627	0.215 ± 0.001	—
LMS	beta32	0.316	0.318	0.002 ± 0.002	0.006 ± 0.005	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	0.029 ± 0.001	0.169 ± 0.005	0.317	0.001	0.23	0.0527	0.0527	1.028	95.100 ± 0.683	0.215 ± 0.000	—
LMS	beta33	0.000	-0.004	-0.004 ± 0.002	—	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	—	—	-0.001	-0.001	—	0.0614	0.0614	0.976	93.300 ± 0.791	0.245 ± 0.001	6.700 ± 0.791
LMS	beta34	0.000	0.002	0.002 ± 0.002	—	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	—	—	0.000	0.000	—	0.0579	0.0579	1.010	95.400 ± 0.662	0.224 ± 0.001	4.600 ± 0.662
LMS	beta35	0.000	-0.004	-0.004 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	—	—	-0.003	-0.003	—	0.0565	0.0565	0.989	94.100 ± 0.745	0.225 ± 0.001	5.900 ± 0.745
LSAM	beta31	0.316	0.315	-0.001 ± 0.002	-0.002 ± 0.005	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	0.029 ± 0.001	0.171 ± 0.005	0.315	-0.001	-0.21	0.0545	0.0545	1.041	96.100 ± 0.612	0.220 ± 0.001	—
LSAM	beta32	0.316	0.319	0.003 ± 0.002	0.009 ± 0.005	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.030 ± 0.001	0.173 ± 0.005	0.318	0.002	0.48	0.0535	0.0535	1.031	95.400 ± 0.662	0.220 ± 0.001	—
LSAM	beta33	0.000	-0.001	-0.001 ± 0.002	—	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	—	—	0.002	0.002	—	0.0603	0.0603	0.961	93.300 ± 0.791	0.245 ± 0.001	6.700 ± 0.791
LSAM	beta34	0.000	0.003	0.003 ± 0.002	—	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	—	—	0.001	0.001	—	0.0704	0.0704	1.007	95.600 ± 0.649	0.277 ± 0.001	4.400 ± 0.649
LSAM	beta35	0.000	-0.004	-0.004 ± 0.002	—	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	—	—	-0.004	-0.004	—	0.0719	0.0720	0.982	94.200 ± 0.739	0.278 ± 0.001	5.800 ± 0.739
QML	beta31	0.316	0.315	-0.001 ± 0.002	-0.003 ± 0.005	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.028 ± 0.001	0.167 ± 0.005	0.315	-0.001	-0.21	0.0539	0.0539	1.037	95.900 ± 0.627	0.215 ± 0.001	—
QML	beta32	0.316	0.318	0.002 ± 0.002	0.006 ± 0.005	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	0.029 ± 0.001	0.169 ± 0.005	0.317	0.001	0.16	0.0526	0.0526	1.028	95.100 ± 0.683	0.215 ± 0.000	—
QML	beta33	0.000	-0.004	-0.004 ± 0.002	—	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	—	—	-0.001	-0.001	—	0.0616	0.0616	0.977	93.300 ± 0.791	0.245 ± 0.001	6.700 ± 0.791
QML	beta34	0.000	0.002	0.002 ± 0.002	—	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	—	—	0.000	0.000	—	0.0580	0.0580	1.014	95.500 ± 0.656	0.224 ± 0.001	4.500 ± 0.656
QML	beta35	0.000	-0.004	-0.004 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	—	—	-0.003	-0.003	—	0.0565	0.0566	0.993	94.200 ± 0.739	0.225 ± 0.001	5.800 ± 0.739
UPI	beta31	0.316	0.315	-0.001 ± 0.002	-0.002 ± 0.005	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	0.029 ± 0.001	0.170 ± 0.005	0.316	0.000	-0.02	0.0551	0.0551	1.035	95.800 ± 0.634	0.219 ± 0.001	—
UPI	beta32	0.316	0.319	0.003 ± 0.002	0.009 ± 0.005	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.030 ± 0.001	0.173 ± 0.005	0.318	0.002	0.51	0.0543	0.0543	1.023	95.400 ± 0.662	0.219 ± 0.001	—
UPI	beta33	0.000	-0.001	-0.001 ± 0.002	—	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	—	—	0.001	0.001	—	0.0606	0.0606	0.955	93.800 ± 0.763	0.247 ± 0.001	6.200 ± 0.763
UPI	beta34	0.000	0.003	0.003 ± 0.002	—	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	—	—	0.003	0.003	—	0.0704	0.0705	1.032	95.800 ± 0.634	0.286 ± 0.001	4.200 ± 0.634
UPI	beta35	0.000	-0.004	-0.004 ± 0.002	—	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.074 ± 0.002	—	—	-0.002	-0.002	—	0.0719	0.0720	0.998	93.900 ± 0.757	0.288 ± 0.001	6.100 ± 0.757

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S90

Simulation 1 - Condition 90: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.8

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.316	-0.000 ± 0.001	-0.001 ± 0.003	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.011 ± 0.000	0.107 ± 0.003	0.316	0.000	0.08	0.0345	0.0345	1.022	95.900 ± 0.627	0.136 ± 0.000	—
LMS	beta32	0.316	0.315	-0.001 ± 0.001	-0.002 ± 0.003	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.012 ± 0.001	0.110 ± 0.003	0.316	0.000	-0.15	0.0349	0.0349	0.997	95.100 ± 0.683	0.136 ± 0.000	—
LMS	beta33	0.000	-0.003	-0.003 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	-0.004	-0.004	—	0.0419	0.0421	0.970	94.400 ± 0.727	0.156 ± 0.001	5.600 ± 0.727
LMS	beta34	0.000	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.001	-0.001	—	0.0375	0.0375	0.982	95.400 ± 0.662	0.143 ± 0.001	4.600 ± 0.662
LMS	beta35	0.000	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.003	-0.003	—	0.0352	0.0354	0.961	93.200 ± 0.796	0.142 ± 0.000	6.800 ± 0.796
LSAM	beta31	0.316	0.316	0.000 ± 0.001	0.000 ± 0.003	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.012 ± 0.001	0.109 ± 0.003	0.316	0.000	0.03	0.0341	0.0341	1.021	95.200 ± 0.676	0.138 ± 0.000	—
LSAM	beta32	0.316	0.316	-0.000 ± 0.001	-0.000 ± 0.004	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.013 ± 0.001	0.112 ± 0.003	0.316	0.000	-0.15	0.0361	0.0361	0.996	95.400 ± 0.662	0.138 ± 0.000	—
LSAM	beta33	0.000	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	0.000	0.000	—	0.0419	0.0419	0.966	94.100 ± 0.745	0.154 ± 0.000	5.900 ± 0.745
LSAM	beta34	0.000	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	-0.001	-0.001	—	0.0459	0.0459	0.974	94.800 ± 0.702	0.175 ± 0.000	5.200 ± 0.702
LSAM	beta35	0.000	-0.002	-0.002 ± 0.001	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	-0.004	-0.004	—	0.0438	0.0440	0.950	93.500 ± 0.780	0.175 ± 0.000	6.500 ± 0.780
QML	beta31	0.316	0.316	-0.000 ± 0.001	-0.001 ± 0.003	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.011 ± 0.000	0.107 ± 0.003	0.316	0.000	0.06	0.0347	0.0347	1.022	95.700 ± 0.641	0.136 ± 0.000	—
QML	beta32	0.316	0.315	-0.001 ± 0.001	-0.002 ± 0.003	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.012 ± 0.001	0.110 ± 0.003	0.315	-0.001	-0.16	0.0351	0.0351	0.996	95.100 ± 0.683	0.136 ± 0.000	—
QML	beta33	0.000	-0.003	-0.003 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	-0.004	-0.004	—	0.0419	0.0421	0.969	94.400 ± 0.727	0.155 ± 0.001	5.600 ± 0.727
QML	beta34	0.000	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.001	-0.001	—	0.0376	0.0376	0.983	95.400 ± 0.662	0.143 ± 0.001	4.600 ± 0.662
QML	beta35	0.000	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.004	-0.004	—	0.0355	0.0357	0.960	93.200 ± 0.796	0.142 ± 0.001	6.800 ± 0.796
UPI	beta31	0.316	0.316	0.000 ± 0.001	0.000 ± 0.003	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.012 ± 0.001	0.109 ± 0.003	0.316	0.000	0.03	0.0345	0.0345	1.018	95.600 ± 0.649	0.138 ± 0.000	—
UPI	beta32	0.316	0.316	-0.000 ± 0.001	-0.000 ± 0.004	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.013 ± 0.001	0.112 ± 0.003	0.316	0.000	-0.14	0.0358	0.0358	0.993	95.500 ± 0.656	0.138 ± 0.000	—
UPI	beta33	0.000	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	-0.001	-0.001	—	0.0412	0.0412	0.967	93.900 ± 0.757	0.155 ± 0.000	6.100 ± 0.757
UPI	beta34	0.000	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	0.000	0.000	—	0.0468	0.0468	0.972	94.600 ± 0.715	0.177 ± 0.001	5.400 ± 0.715
UPI	beta35	0.000	-0.002	-0.002 ± 0.001	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	-0.003	-0.003	—	0.0439	0.0440	0.955	93.800 ± 0.763	0.177 ± 0.001	6.200 ± 0.763

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S91*Simulation 1 - Condition 91: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.4*

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.326	0.010 ± 0.007	0.031 ± 0.022	0.011 ± 0.001	0.103 ± 0.007	0.011 ± 0.001	0.103 ± 0.008	0.107 ± 0.015	0.327 ± 0.026	0.321	0.005	1.63	0.1121	0.1122	0.953	95.575 ± 1.368	0.385 ± 0.010	
LMS	beta32	0.316	0.329	0.013 ± 0.007	0.040 ± 0.022	0.011 ± 0.002	0.104 ± 0.008	0.011 ± 0.002	0.104 ± 0.009	0.109 ± 0.018	0.331 ± 0.030	0.333	0.017	5.28	0.1180	0.1192	0.932	94.690 ± 1.492	0.380 ± 0.015	—
LMS	beta33	0.000	0.026	0.026 ± 0.008	—	0.016 ± 0.002	0.128 ± 0.006	0.017 ± 0.002	0.130 ± 0.008	—	—	0.040	0.040	—	0.1529	0.1580	1.121	94.690 ± 1.492	0.560 ± 0.046	5.310 ± 1.492
LMS	beta34	0.000	-0.051	-0.051 ± 0.006	—	0.009 ± 0.001	0.097 ± 0.008	0.012 ± 0.002	0.109 ± 0.009	—	—	-0.052	-0.052	—	0.0930	0.1067	0.925	90.265 ± 1.972	0.351 ± 0.020	9.735 ± 1.972
LMS	beta35	0.000	-0.040	-0.040 ± 0.005	—	0.006 ± 0.001	0.074 ± 0.004	0.007 ± 0.001	0.084 ± 0.005	—	—	-0.047	-0.047	—	0.0791	0.0921	1.344	88.496 ± 2.122	0.391 ± 0.079	11.504 ± 2.122
LSAM	beta31	0.316	0.341	0.025 ± 0.010	0.079 ± 0.032	0.023 ± 0.003	0.150 ± 0.009	0.023 ± 0.003	0.152 ± 0.011	0.231 ± 0.028	0.481 ± 0.033	0.332	0.016	5.14	0.1395	0.1405	1.068	97.345 ± 1.069	0.628 ± 0.017	—
LSAM	beta32	0.316	0.356	0.040 ± 0.010	0.125 ± 0.031	0.021 ± 0.003	0.146 ± 0.009	0.023 ± 0.003	0.151 ± 0.010	0.228 ± 0.028	0.478 ± 0.033	0.348	0.032	10.04	0.1564	0.1596	1.092	96.460 ± 1.229	0.625 ± 0.019	—
LSAM	beta33	0.000	0.112	0.112 ± 0.011	—	0.026 ± 0.003	0.160 ± 0.009	0.038 ± 0.004	0.196 ± 0.011	—	—	0.097	0.097	—	0.1465	0.1760	1.088	95.133 ± 1.431	0.683 ± 0.030	4.867 ± 1.431
LSAM	beta34	0.000	-0.076	-0.076 ± 0.008	—	0.013 ± 0.001	0.115 ± 0.006	0.019 ± 0.002	0.137 ± 0.008	—	—	-0.066	-0.066	—	0.1009	0.1208	1.076	94.690 ± 1.492	0.483 ± 0.021	5.310 ± 1.492
LSAM	beta35	0.000	-0.073	-0.073 ± 0.007	—	0.011 ± 0.001	0.105 ± 0.007	0.016 ± 0.002	0.128 ± 0.008	—	—	-0.068	-0.068	—	0.0847	0.1085	1.100	93.363 ± 1.656	0.455 ± 0.020	6.637 ± 1.656
QML	beta31	0.316	0.318	0.002 ± 0.006	0.005 ± 0.019	0.008 ± 0.001	0.092 ± 0.005	0.008 ± 0.001	0.092 ± 0.006	0.085 ± 0.008	0.291 ± 0.018	0.315	-0.001	-0.39	0.0944	0.0944	1.015	95.133 ± 1.431	0.366 ± 0.004	—
QML	beta32	0.316	0.321	0.005 ± 0.006	0.017 ± 0.018	0.008 ± 0.001	0.088 ± 0.004	0.008 ± 0.001	0.088 ± 0.005	0.077 ± 0.007	0.277 ± 0.016	0.327	0.011	3.52	0.1063	0.1069	1.049	95.575 ± 1.368	0.360 ± 0.004	—
QML	beta33	0.000	0.030	0.030 ± 0.008	—	0.015 ± 0.001	0.124 ± 0.006	0.016 ± 0.002	0.127 ± 0.007	—	—	0.034	0.034	—	0.1241	0.1286	1.014	96.018 ± 1.301	0.492 ± 0.011	3.982 ± 1.301
QML	beta34	0.000	-0.047	-0.047 ± 0.005	—	0.006 ± 0.001	0.080 ± 0.004	0.009 ± 0.001	0.093 ± 0.005	—	—	-0.050	-0.050	—	0.0805	0.0947	1.002	90.708 ± 1.931	0.315 ± 0.008	9.292 ± 1.931
QML	beta35	0.000	-0.043	-0.043 ± 0.005	—	0.006 ± 0.001	0.075 ± 0.004	0.007 ± 0.001	0.086 ± 0.005	—	—	-0.047	-0.047	—	0.0731	0.0871	1.035	90.708 ± 1.931	0.303 ± 0.008	9.292 ± 1.931
UPI	beta31	0.316	0.353	0.037 ± 0.017	0.116 ± 0.053	0.063 ± 0.010	0.251 ± 0.022	0.064 ± 0.010	0.253 ± 0.023	0.641 ± 0.105	0.801 ± 0.072	0.363	0.047	14.93	0.2504	0.2548	1.903	94.248 ± 1.549	1.872 ± 0.210	—
UPI	beta32	0.316	0.367	0.051 ± 0.018	0.163 ± 0.055	0.070 ± 0.009	0.264 ± 0.017	0.072 ± 0.010	0.268 ± 0.021	0.719 ± 0.101	0.848 ± 0.066	0.357	0.041	12.93	0.2759	0.2789	1.665	91.593 ± 1.846	1.721 ± 0.178	—
UPI	beta33	0.000	0.073	0.073 ± 0.015	—	0.054 ± 0.006	0.232 ± 0.014	0.059 ± 0.008	0.243 ± 0.017	—	—	0.028	0.028	—	0.2248	0.2266	1.742	99.558 ± 0.441	1.587 ± 0.149	0.442 ± 0.441
UPI	beta34	0.000	-0.117	-0.117 ± 0.017	—	0.064 ± 0.009	0.253 ± 0.018	0.077 ± 0.010	0.278 ± 0.021	—	—	-0.104	-0.104	—	0.2100	0.2343	1.845	99.558 ± 0.441	1.828 ± 0.170	0.442 ± 0.441
UPI	beta35	0.000	-0.128	-0.128 ± 0.019	—	0.078 ± 0.011	0.278 ± 0.020	0.094 ± 0.014	0.306 ± 0.024	—	—	-0.095	-0.095	—	0.2304	0.2493	1.551	99.558 ± 0.441	1.693 ± 0.158	0.442 ± 0.441

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S92

Simulation 1 - Condition 92: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.4

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.327	0.011 ± 0.004	0.036 ± 0.012	0.004 ± 0.000	0.059 ± 0.003	0.004 ± 0.000	0.060 ± 0.003	0.036 ± 0.003	0.191 ± 0.010	0.317	0.001	0.22	0.0687	0.0687	0.950	92.941 ± 1.604	0.221 ± 0.001	—
LMS	beta32	0.316	0.311	-0.005 ± 0.003	-0.016 ± 0.011	0.003 ± 0.000	0.056 ± 0.002	0.003 ± 0.000	0.056 ± 0.003	0.031 ± 0.003	0.177 ± 0.010	0.316	0.000	0.05	0.0658	0.0658	1.006	95.686 ± 1.272	0.220 ± 0.001	—
LMS	beta33	0.000	0.050	0.050 ± 0.005	—	0.006 ± 0.001	0.079 ± 0.004	0.009 ± 0.001	0.094 ± 0.004	—	—	0.057	0.057	—	0.0850	0.1021	0.930	87.843 ± 2.046	0.290 ± 0.004	12.157 ± 2.046
LMS	beta34	0.000	-0.055	-0.055 ± 0.003	—	0.002 ± 0.000	0.043 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	—	—	-0.057	-0.057	—	0.0549	0.0794	1.073	78.039 ± 2.592	0.180 ± 0.003	21.961 ± 2.592
LMS	beta35	0.000	-0.054	-0.054 ± 0.003	—	0.003 ± 0.000	0.052 ± 0.003	0.006 ± 0.000	0.075 ± 0.003	—	—	-0.055	-0.055	—	0.0540	0.0772	0.901	72.549 ± 2.795	0.183 ± 0.003	27.451 ± 2.795
LSAM	beta31	0.316	0.361	0.045 ± 0.006	0.141 ± 0.017	0.008 ± 0.001	0.088 ± 0.003	0.010 ± 0.001	0.098 ± 0.005	0.097 ± 0.008	0.312 ± 0.015	0.349	0.033	10.35	0.0904	0.0961	1.017	94.902 ± 1.377	0.350 ± 0.005	—
LSAM	beta32	0.316	0.336	0.020 ± 0.006	0.065 ± 0.018	0.008 ± 0.001	0.090 ± 0.004	0.008 ± 0.001	0.092 ± 0.005	0.084 ± 0.008	0.290 ± 0.016	0.335	0.019	5.91	0.0969	0.0987	0.992	96.471 ± 1.156	0.348 ± 0.005	—
LSAM	beta33	0.000	0.121	0.121 ± 0.006	—	0.009 ± 0.001	0.097 ± 0.005	0.024 ± 0.002	0.155 ± 0.007	—	—	0.121	0.121	—	0.0915	0.1514	0.930	77.255 ± 2.625	0.352 ± 0.007	22.745 ± 2.625
LSAM	beta34	0.000	-0.084	-0.084 ± 0.004	—	0.003 ± 0.000	0.058 ± 0.003	0.010 ± 0.001	0.102 ± 0.004	—	—	-0.084	-0.084	—	0.0634	0.1050	1.067	72.549 ± 2.795	0.244 ± 0.005	27.451 ± 2.795
LSAM	beta35	0.000	-0.079	-0.079 ± 0.004	—	0.005 ± 0.001	0.067 ± 0.004	0.011 ± 0.001	0.103 ± 0.005	—	—	-0.077	-0.077	—	0.0636	0.0998	0.913	76.078 ± 2.672	0.241 ± 0.005	23.922 ± 2.672
QML	beta31	0.316	0.329	0.013 ± 0.004	0.040 ± 0.012	0.004 ± 0.000	0.060 ± 0.003	0.004 ± 0.000	0.061 ± 0.003	0.038 ± 0.003	0.194 ± 0.010	0.318	0.002	0.52	0.0635	0.0635	0.951	93.333 ± 1.562	0.224 ± 0.002	—
QML	beta32	0.316	0.312	-0.004 ± 0.004	-0.014 ± 0.011	0.003 ± 0.000	0.057 ± 0.003	0.003 ± 0.000	0.057 ± 0.003	0.033 ± 0.003	0.180 ± 0.010	0.315	-0.001	-0.21	0.0596	0.0596	0.996	94.902 ± 1.377	0.222 ± 0.002	—
QML	beta33	0.000	0.052	0.052 ± 0.005	—	0.007 ± 0.001	0.083 ± 0.004	0.010 ± 0.001	0.098 ± 0.005	—	—	0.058	0.058	—	0.0848	0.1030	0.923	89.020 ± 1.958	0.300 ± 0.004	10.980 ± 1.958
QML	beta34	0.000	-0.059	-0.059 ± 0.003	—	0.002 ± 0.000	0.045 ± 0.002	0.006 ± 0.000	0.074 ± 0.003	—	—	-0.061	-0.061	—	0.0498	0.0785	1.059	78.039 ± 2.592	0.187 ± 0.003	21.961 ± 2.592
QML	beta35	0.000	-0.058	-0.058 ± 0.003	—	0.003 ± 0.000	0.054 ± 0.003	0.006 ± 0.000	0.079 ± 0.003	—	—	-0.059	-0.059	—	0.0543	0.0803	0.898	74.118 ± 2.743	0.189 ± 0.003	25.882 ± 2.743
UPI	beta31	0.316	0.379	0.063 ± 0.018	0.200 ± 0.057	0.081 ± 0.009	0.285 ± 0.016	0.085 ± 0.009	0.292 ± 0.018	0.853 ± 0.093	0.923 ± 0.058	0.400	0.084	26.56	0.2988	0.3103	1.333	84.706 ± 2.254	1.491 ± 0.160	—
UPI	beta32	0.316	0.315	-0.001 ± 0.017	-0.004 ± 0.054	0.075 ± 0.009	0.274 ± 0.017	0.075 ± 0.009	0.274 ± 0.019	0.750 ± 0.089	0.866 ± 0.059	0.356	0.040	12.74	0.3361	0.3385	1.284	83.137 ± 2.345	1.380 ± 0.166	—
UPI	beta33	0.000	0.052	0.052 ± 0.015	—	0.056 ± 0.007	0.236 ± 0.015	0.058 ± 0.008	0.241 ± 0.018	—	—	-0.013	-0.013	—	0.1914	0.1918	1.375	95.294 ± 1.326	1.271 ± 0.134	4.706 ± 1.326
UPI	beta34	0.000	-0.236	-0.236 ± 0.023	—	0.134 ± 0.018	0.366 ± 0.025	0.189 ± 0.026	0.435 ± 0.031	—	—	-0.200	-0.200	—	0.3241	0.3809	1.464	96.078 ± 1.216	2.101 ± 0.234	3.922 ± 1.216
UPI	beta35	0.000	-0.195	-0.195 ± 0.022	—	0.121 ± 0.016	0.348 ± 0.024	0.158 ± 0.022	0.398 ± 0.029	—	—	-0.182	-0.182	—	0.3346	0.3808	1.301	98.039 ± 0.868	1.773 ± 0.201	1.961 ± 0.868

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S93

Simulation 1 - Condition 93: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.6

Method	Param.	True	Mean-based										Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE	
LMS	beta31	0.316	0.313	-0.003 ± 0.002	-0.009 ± 0.007	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	0.044 ± 0.002	0.210 ± 0.006	0.309	-0.007	-2.17	0.0638	0.0642	1.021	94.939 ± 0.697	0.266 ± 0.001		
LMS	beta32	0.316	0.320	0.004 ± 0.002	0.012 ± 0.007	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.047 ± 0.002	0.217 ± 0.006	0.316	0.000	0.09	0.0674	0.0674	0.991	94.737 ± 0.710	0.266 ± 0.001	—	
LMS	beta33	0.000	0.007	0.007 ± 0.003	—	0.007 ± 0.000	0.086 ± 0.002	0.007 ± 0.000	0.086 ± 0.003	—	—	0.007	0.007	—	0.0836	0.0839	0.965	94.534 ± 0.723	0.326 ± 0.003	5.466 ± 0.723	
LMS	beta34	0.000	-0.017	-0.017 ± 0.002	—	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	—	—	-0.018	-0.018	—	0.0545	0.0575	0.990	93.117 ± 0.805	0.216 ± 0.002	6.883 ± 0.805	
LMS	beta35	0.000	-0.017	-0.017 ± 0.002	—	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	—	—	-0.020	-0.020	—	0.0553	0.0588	0.978	91.903 ± 0.868	0.216 ± 0.002	8.097 ± 0.868	
LSAM	beta31	0.316	0.318	0.002 ± 0.002	0.006 ± 0.007	0.005 ± 0.000	0.074 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.054 ± 0.003	0.233 ± 0.007	0.315	-0.001	-0.44	0.0706	0.0706	1.017	95.142 ± 0.684	0.293 ± 0.001	—	
LSAM	beta32	0.316	0.326	0.010 ± 0.002	0.030 ± 0.008	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	0.058 ± 0.003	0.240 ± 0.007	0.323	0.007	2.36	0.0715	0.0719	0.995	94.838 ± 0.704	0.294 ± 0.001	—	
LSAM	beta33	0.000	0.032	0.032 ± 0.003	—	0.008 ± 0.000	0.091 ± 0.002	0.009 ± 0.000	0.096 ± 0.003	—	—	0.034	0.034	—	0.0846	0.0911	0.952	93.826 ± 0.766	0.338 ± 0.003	6.174 ± 0.766	
LSAM	beta34	0.000	-0.024	-0.024 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	—	—	-0.025	-0.025	—	0.0576	0.0626	0.999	94.534 ± 0.723	0.235 ± 0.002	5.466 ± 0.723	
LSAM	beta35	0.000	-0.024	-0.024 ± 0.002	—	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	—	—	-0.025	-0.025	—	0.0601	0.0652	0.974	92.308 ± 0.848	0.236 ± 0.002	7.692 ± 0.848	
QML	beta31	0.316	0.313	-0.003 ± 0.002	-0.009 ± 0.007	0.004 ± 0.000	0.067 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	0.044 ± 0.002	0.211 ± 0.006	0.309	-0.007	-2.16	0.0641	0.0645	1.020	94.939 ± 0.697	0.266 ± 0.001	—	
QML	beta32	0.316	0.320	0.004 ± 0.002	0.012 ± 0.007	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.047 ± 0.002	0.217 ± 0.006	0.317	0.001	0.17	0.0677	0.0677	0.990	94.737 ± 0.710	0.266 ± 0.001	—	
QML	beta33	0.000	0.007	0.007 ± 0.003	—	0.007 ± 0.000	0.086 ± 0.002	0.007 ± 0.000	0.086 ± 0.003	—	—	0.007	0.007	—	0.0828	0.0831	0.971	94.636 ± 0.717	0.327 ± 0.002	5.364 ± 0.717	
QML	beta34	0.000	-0.017	-0.017 ± 0.002	—	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	—	—	-0.019	-0.019	—	0.0545	0.0576	0.993	93.219 ± 0.800	0.216 ± 0.002	6.781 ± 0.800	
QML	beta35	0.000	-0.017	-0.017 ± 0.002	—	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	—	—	-0.020	-0.020	—	0.0549	0.0584	0.985	91.802 ± 0.873	0.217 ± 0.002	8.198 ± 0.873	
UPI	beta31	0.316	0.318	0.002 ± 0.002	0.007 ± 0.008	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	0.056 ± 0.003	0.237 ± 0.007	0.316	0.000	0.09	0.0737	0.0737	1.085	95.850 ± 0.634	0.318 ± 0.008	—	
UPI	beta32	0.316	0.327	0.011 ± 0.002	0.035 ± 0.008	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	0.060 ± 0.003	0.245 ± 0.007	0.325	0.009	2.70	0.0722	0.0727	1.079	95.850 ± 0.634	0.324 ± 0.012	—	
UPI	beta33	0.000	0.024	0.024 ± 0.003	—	0.009 ± 0.001	0.096 ± 0.003	0.010 ± 0.001	0.099 ± 0.003	—	—	0.027	0.027	—	0.0864	0.0904	0.983	94.130 ± 0.748	0.368 ± 0.005	5.870 ± 0.748	
UPI	beta34	0.000	-0.016	-0.016 ± 0.002	—	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	—	—	-0.017	-0.017	—	0.0647	0.0669	1.065	95.850 ± 0.634	0.285 ± 0.006	4.150 ± 0.634	
UPI	beta35	0.000	-0.014	-0.014 ± 0.002	—	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	—	—	-0.015	-0.015	—	0.0697	0.0713	1.075	94.737 ± 0.710	0.301 ± 0.014	5.263 ± 0.710	

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S94

Simulation 1 - Condition 94: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.6

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.313	-0.003 ± 0.001	-0.009 ± 0.004	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.017 ± 0.001	0.131 ± 0.004	0.311	-0.005	-1.52	0.0399	0.0402	1.032	94.995 ± 0.690	0.168 ± 0.001	—
LMS	beta32	0.316	0.316	0.000 ± 0.001	0.001 ± 0.004	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.019 ± 0.001	0.137 ± 0.004	0.315	-0.001	-0.36	0.0456	0.0456	0.985	94.494 ± 0.722	0.168 ± 0.001	—
LMS	beta33	0.000	0.009	0.009 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	—	—	0.010	0.010	—	0.0524	0.0533	1.020	94.094 ± 0.746	0.208 ± 0.004	5.906 ± 0.746
LMS	beta34	0.000	-0.019	-0.019 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	-0.019	-0.019	—	0.0358	0.0406	1.019	90.591 ± 0.924	0.141 ± 0.005	9.409 ± 0.924
LMS	beta35	0.000	-0.021	-0.021 ± 0.001	—	0.001 ± 0.000	0.033 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	-0.021	-0.021	—	0.0319	0.0382	1.031	90.791 ± 0.915	0.135 ± 0.001	9.209 ± 0.915
LSAM	beta31	0.316	0.318	0.002 ± 0.001	0.005 ± 0.005	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.021 ± 0.001	0.144 ± 0.004	0.317	0.001	0.25	0.0449	0.0450	1.020	95.796 ± 0.635	0.182 ± 0.000	—
LSAM	beta32	0.316	0.321	0.005 ± 0.002	0.017 ± 0.005	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.023 ± 0.001	0.151 ± 0.004	0.320	0.004	1.39	0.0490	0.0492	0.980	94.394 ± 0.728	0.183 ± 0.000	—
LSAM	beta33	0.000	0.033	0.033 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	—	—	0.034	0.034	—	0.0516	0.0616	1.005	91.191 ± 0.897	0.203 ± 0.001	8.809 ± 0.897
LSAM	beta34	0.000	-0.025	-0.025 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	-0.025	-0.025	—	0.0374	0.0450	0.968	88.589 ± 1.006	0.142 ± 0.001	11.411 ± 1.006
LSAM	beta35	0.000	-0.027	-0.027 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	-0.027	-0.027	—	0.0340	0.0434	1.011	89.089 ± 0.986	0.142 ± 0.001	10.911 ± 0.986
QML	beta31	0.316	0.313	-0.003 ± 0.001	-0.009 ± 0.004	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.017 ± 0.001	0.132 ± 0.004	0.311	-0.005	-1.50	0.0399	0.0402	1.025	94.995 ± 0.690	0.167 ± 0.000	—
QML	beta32	0.316	0.316	0.000 ± 0.001	0.001 ± 0.004	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.019 ± 0.001	0.138 ± 0.004	0.315	-0.001	-0.34	0.0458	0.0458	0.981	94.394 ± 0.728	0.167 ± 0.000	—
QML	beta33	0.000	0.009	0.009 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	—	—	0.010	0.010	—	0.0526	0.0536	1.000	94.094 ± 0.746	0.204 ± 0.001	5.906 ± 0.746
QML	beta34	0.000	-0.019	-0.019 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	-0.019	-0.019	—	0.0359	0.0408	0.982	90.490 ± 0.928	0.136 ± 0.001	9.510 ± 0.928
QML	beta35	0.000	-0.021	-0.021 ± 0.001	—	0.001 ± 0.000	0.034 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	-0.021	-0.021	—	0.0321	0.0383	1.029	90.791 ± 0.915	0.135 ± 0.001	9.209 ± 0.915
UPI	beta31	0.316	0.317	0.001 ± 0.001	0.004 ± 0.005	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.022 ± 0.001	0.148 ± 0.004	0.316	0.000	-0.14	0.0453	0.0453	1.020	95.696 ± 0.642	0.188 ± 0.001	—
UPI	beta32	0.316	0.321	0.005 ± 0.002	0.014 ± 0.005	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.024 ± 0.001	0.154 ± 0.004	0.319	0.003	1.05	0.0480	0.0481	0.992	94.595 ± 0.715	0.188 ± 0.001	—
UPI	beta33	0.000	0.024	0.024 ± 0.002	—	0.003 ± 0.000	0.054 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	—	—	0.023	0.023	—	0.0520	0.0568	1.012	93.694 ± 0.769	0.216 ± 0.001	6.306 ± 0.769
UPI	beta34	0.000	-0.018	-0.018 ± 0.001	—	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	-0.016	-0.016	—	0.0401	0.0431	0.986	93.193 ± 0.797	0.163 ± 0.001	6.807 ± 0.797
UPI	beta35	0.000	-0.020	-0.020 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	-0.019	-0.019	—	0.0383	0.0428	1.019	94.394 ± 0.728	0.165 ± 0.002	5.606 ± 0.728

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S95

Simulation 1 - Condition 95: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.317	0.001 ± 0.002	0.002 ± 0.006	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.031 ± 0.001	0.177 ± 0.005	0.317	0.001	0.19	0.0550	0.0550	0.993	95.295 ± 0.670	0.218 ± 0.001	
LMS	beta32	0.316	0.313	-0.003 ± 0.002	-0.009 ± 0.006	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.032 ± 0.001	0.178 ± 0.005	0.312	-0.004	-1.37	0.0573	0.0574	0.991	95.095 ± 0.683	0.218 ± 0.001	—
LMS	beta33	0.000	0.002	0.002 ± 0.002	—	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	—	—	0.001	0.001	—	0.0646	0.0647	0.948	93.994 ± 0.752	0.243 ± 0.001	6.006 ± 0.752
LMS	beta34	0.000	-0.006	-0.006 ± 0.001	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	—	—	-0.007	-0.007	—	0.0444	0.0450	0.976	94.294 ± 0.734	0.169 ± 0.001	5.706 ± 0.734
LMS	beta35	0.000	-0.005	-0.005 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	-0.004	-0.004	—	0.0448	0.0450	0.962	92.693 ± 0.823	0.168 ± 0.001	7.307 ± 0.823
LSAM	beta31	0.316	0.318	0.002 ± 0.002	0.008 ± 0.006	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.033 ± 0.001	0.182 ± 0.005	0.318	0.002	0.51	0.0573	0.0574	0.991	94.995 ± 0.690	0.223 ± 0.001	—
LSAM	beta32	0.316	0.315	-0.001 ± 0.002	-0.004 ± 0.006	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.033 ± 0.002	0.183 ± 0.005	0.312	-0.004	-1.19	0.0602	0.0603	0.986	94.695 ± 0.709	0.223 ± 0.001	—
LSAM	beta33	0.000	0.011	0.011 ± 0.002	—	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	—	—	0.011	0.011	—	0.0643	0.0651	0.950	93.293 ± 0.791	0.244 ± 0.001	6.707 ± 0.791
LSAM	beta34	0.000	-0.008	-0.008 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	-0.009	-0.009	—	0.0460	0.0470	0.970	93.994 ± 0.752	0.173 ± 0.001	6.006 ± 0.752
LSAM	beta35	0.000	-0.006	-0.006 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	-0.006	-0.006	—	0.0461	0.0464	0.955	93.093 ± 0.802	0.172 ± 0.001	6.907 ± 0.802
QML	beta31	0.316	0.317	0.001 ± 0.002	0.002 ± 0.006	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.031 ± 0.001	0.177 ± 0.005	0.317	0.001	0.21	0.0550	0.0550	0.995	95.195 ± 0.677	0.218 ± 0.001	—
QML	beta32	0.316	0.313	-0.003 ± 0.002	-0.009 ± 0.006	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.032 ± 0.001	0.178 ± 0.005	0.312	-0.004	-1.35	0.0573	0.0575	0.990	95.095 ± 0.683	0.218 ± 0.001	—
QML	beta33	0.000	0.002	0.002 ± 0.002	—	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	—	—	0.001	0.001	—	0.0646	0.0646	0.975	93.894 ± 0.758	0.250 ± 0.006	6.106 ± 0.758
QML	beta34	0.000	-0.006	-0.006 ± 0.001	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	—	—	-0.007	-0.007	—	0.0442	0.0449	0.994	94.294 ± 0.734	0.171 ± 0.003	5.706 ± 0.734
QML	beta35	0.000	-0.005	-0.005 ± 0.001	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	-0.004	-0.004	—	0.0447	0.0449	0.969	92.793 ± 0.818	0.169 ± 0.002	7.207 ± 0.818
UPI	beta31	0.316	0.318	0.002 ± 0.002	0.007 ± 0.006	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.033 ± 0.001	0.181 ± 0.005	0.318	0.002	0.57	0.0560	0.0561	0.990	94.995 ± 0.690	0.221 ± 0.001	—
UPI	beta32	0.316	0.315	-0.001 ± 0.002	-0.004 ± 0.006	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.033 ± 0.001	0.182 ± 0.005	0.313	-0.003	-0.99	0.0595	0.0596	0.984	94.695 ± 0.709	0.222 ± 0.001	—
UPI	beta33	0.000	0.007	0.007 ± 0.002	—	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	—	—	0.007	0.007	—	0.0666	0.0670	0.948	93.994 ± 0.752	0.245 ± 0.001	6.006 ± 0.752
UPI	beta34	0.000	-0.004	-0.004 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	-0.006	-0.006	—	0.0465	0.0469	0.964	94.494 ± 0.722	0.176 ± 0.001	5.506 ± 0.722
UPI	beta35	0.000	-0.003	-0.003 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	-0.003	-0.003	—	0.0446	0.0447	0.960	93.193 ± 0.797	0.174 ± 0.001	6.807 ± 0.797

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S96

Simulation 1 - Condition 96: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.8

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.316	-0.000 ± 0.001	-0.001 ± 0.004	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.013 ± 0.001	0.113 ± 0.003	0.316	0.000	-0.15	0.0340	0.0340	0.984	93.694 ± 0.769	0.137 ± 0.000	—
LMS	beta32	0.316	0.315	-0.001 ± 0.001	-0.002 ± 0.004	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.012 ± 0.001	0.111 ± 0.003	0.316	0.000	-0.02	0.0364	0.0364	0.996	95.395 ± 0.663	0.137 ± 0.000	—
LMS	beta33	0.000	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.003	-0.003	—	0.0361	0.0362	1.032	95.395 ± 0.663	0.155 ± 0.001	4.605 ± 0.663
LMS	beta34	0.000	-0.005	-0.005 ± 0.001	—	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	—	—	-0.005	-0.005	—	0.0265	0.0271	1.012	94.194 ± 0.740	0.107 ± 0.001	5.806 ± 0.740
LMS	beta35	0.000	-0.004	-0.004 ± 0.001	—	0.001 ± 0.000	0.026 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	—	—	-0.004	-0.004	—	0.0275	0.0278	1.025	94.895 ± 0.696	0.106 ± 0.000	5.105 ± 0.696
LSAM	beta31	0.316	0.317	0.001 ± 0.001	0.004 ± 0.004	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.013 ± 0.001	0.115 ± 0.003	0.317	0.001	0.32	0.0351	0.0351	0.982	93.694 ± 0.769	0.140 ± 0.000	—
LSAM	beta32	0.316	0.317	0.001 ± 0.001	0.002 ± 0.004	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.013 ± 0.001	0.114 ± 0.003	0.317	0.001	0.26	0.0375	0.0375	0.997	95.896 ± 0.628	0.141 ± 0.000	—
LSAM	beta33	0.000	0.006	0.006 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	0.006	0.006	—	0.0358	0.0363	1.023	95.495 ± 0.656	0.152 ± 0.001	4.505 ± 0.656
LSAM	beta34	0.000	-0.006	-0.006 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	-0.007	-0.007	—	0.0273	0.0282	0.995	94.094 ± 0.746	0.108 ± 0.000	5.906 ± 0.746
LSAM	beta35	0.000	-0.005	-0.005 ± 0.001	—	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	—	—	-0.005	-0.005	—	0.0280	0.0284	1.022	94.695 ± 0.709	0.108 ± 0.000	5.305 ± 0.709
QML	beta31	0.316	0.316	-0.000 ± 0.001	-0.001 ± 0.004	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.013 ± 0.001	0.113 ± 0.003	0.316	0.000	-0.14	0.0339	0.0339	0.982	93.694 ± 0.769	0.137 ± 0.000	—
QML	beta32	0.316	0.315	-0.001 ± 0.001	-0.002 ± 0.004	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.012 ± 0.001	0.111 ± 0.003	0.316	0.000	-0.05	0.0364	0.0364	0.995	95.395 ± 0.663	0.137 ± 0.000	—
QML	beta33	0.000	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.003	-0.003	—	0.0362	0.0363	1.023	95.395 ± 0.663	0.153 ± 0.001	4.605 ± 0.663
QML	beta34	0.000	-0.005	-0.005 ± 0.001	—	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	—	—	-0.006	-0.006	—	0.0264	0.0270	0.999	94.194 ± 0.740	0.106 ± 0.000	5.806 ± 0.740
QML	beta35	0.000	-0.004	-0.004 ± 0.001	—	0.001 ± 0.000	0.026 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	—	—	-0.004	-0.004	—	0.0275	0.0278	1.026	94.895 ± 0.696	0.106 ± 0.000	5.105 ± 0.696
UPI	beta31	0.316	0.317	0.001 ± 0.001	0.003 ± 0.004	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.013 ± 0.001	0.115 ± 0.003	0.317	0.001	0.27	0.0348	0.0348	0.981	93.794 ± 0.763	0.140 ± 0.000	—
UPI	beta32	0.316	0.316	0.000 ± 0.001	0.001 ± 0.004	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.013 ± 0.001	0.113 ± 0.003	0.317	0.001	0.34	0.0377	0.0377	0.995	95.796 ± 0.635	0.140 ± 0.000	—
UPI	beta33	0.000	0.003	0.003 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	0.002	0.002	—	0.0368	0.0368	1.023	95.696 ± 0.642	0.152 ± 0.001	4.304 ± 0.642
UPI	beta34	0.000	-0.003	-0.003 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	-0.003	-0.003	—	0.0272	0.0273	0.997	94.494 ± 0.722	0.109 ± 0.000	5.506 ± 0.722
UPI	beta35	0.000	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	—	—	-0.002	-0.002	—	0.0287	0.0288	1.027	96.296 ± 0.598	0.109 ± 0.000	3.704 ± 0.598

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S97

Simulation 1 - Condition 97: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.4

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.280	-0.036 ± 0.005	-0.114 ± 0.015	0.011 ± 0.001	0.104 ± 0.003	0.012 ± 0.001	0.110 ± 0.004	0.120 ± 0.007	0.347 ± 0.012	0.278	-0.038	-12.17	0.1118	0.1182	1.084	95.294 ± 0.938	0.440 ± 0.009	—
LMS	beta32	0.316	0.286	-0.030 ± 0.005	-0.096 ± 0.015	0.012 ± 0.001	0.108 ± 0.004	0.013 ± 0.001	0.113 ± 0.004	0.127 ± 0.009	0.356 ± 0.014	0.293	-0.023	-7.23	0.1185	0.1207	1.011	92.549 ± 1.163	0.430 ± 0.005	—
LMS	beta33	0.000	-0.003	-0.003 ± 0.006	—	0.016 ± 0.001	0.127 ± 0.005	0.016 ± 0.001	0.127 ± 0.005	—	—	0.012	0.012	—	0.1428	0.1433	1.100	95.490 ± 0.919	0.546 ± 0.029	4.510 ± 0.919
LMS	beta34	0.000	0.039	0.039 ± 0.003	—	0.006 ± 0.000	0.075 ± 0.003	0.007 ± 0.001	0.084 ± 0.003	—	—	0.034	0.034	—	0.0767	0.0839	1.078	95.098 ± 0.956	0.316 ± 0.016	4.902 ± 0.956
LMS	beta35	0.000	0.040	0.040 ± 0.003	—	0.005 ± 0.000	0.073 ± 0.003	0.007 ± 0.001	0.083 ± 0.004	—	—	0.033	0.033	—	0.0738	0.0807	1.054	95.098 ± 0.956	0.302 ± 0.013	4.902 ± 0.956
LSAM	beta31	0.316	0.456	0.140 ± 0.051	0.443 ± 0.162	1.333 ± 0.102	1.154 ± 0.045	1.350 ± 0.104	1.162 ± 0.052	13.517 ± 1.039	3.677 ± 0.163	0.413	0.097	30.76	1.1809	1.1849	2.674	99.608 ± 0.277	12.103 ± 0.495	—
LSAM	beta32	0.316	0.466	0.150 ± 0.050	0.475 ± 0.159	1.295 ± 0.097	1.138 ± 0.043	1.315 ± 0.101	1.147 ± 0.051	13.171 ± 1.009	3.629 ± 0.161	0.450	0.134	42.40	1.1758	1.1834	2.641	99.412 ± 0.339	11.783 ± 0.481	—
LSAM	beta33	0.000	0.115	0.115 ± 0.013	—	0.088 ± 0.008	0.297 ± 0.014	0.102 ± 0.010	0.319 ± 0.017	—	—	0.088	0.088	—	0.2427	0.2580	2.452	99.412 ± 0.339	2.859 ± 0.157	0.588 ± 0.339
LSAM	beta34	0.000	-0.099	-0.099 ± 0.019	—	0.191 ± 0.018	0.438 ± 0.020	0.201 ± 0.019	0.448 ± 0.023	—	—	-0.049	-0.049	—	0.3486	0.3520	2.552	99.804 ± 0.196	4.378 ± 0.210	0.196 ± 0.196
LSAM	beta35	0.000	-0.101	-0.101 ± 0.019	—	0.187 ± 0.018	0.433 ± 0.020	0.197 ± 0.019	0.444 ± 0.024	—	—	-0.066	-0.066	—	0.3448	0.3512	2.481	99.608 ± 0.277	4.208 ± 0.202	0.392 ± 0.277
QML	beta31	0.316	0.282	-0.034 ± 0.005	-0.106 ± 0.015	0.011 ± 0.001	0.104 ± 0.003	0.012 ± 0.001	0.110 ± 0.004	0.120 ± 0.007	0.347 ± 0.012	0.278	-0.038	-11.87	0.1061	0.1125	1.057	94.902 ± 0.974	0.433 ± 0.004	—
QML	beta32	0.316	0.288	-0.028 ± 0.005	-0.089 ± 0.015	0.012 ± 0.001	0.109 ± 0.004	0.013 ± 0.001	0.112 ± 0.004	0.126 ± 0.008	0.355 ± 0.013	0.293	-0.023	-7.43	0.1167	0.1190	1.001	92.549 ± 1.163	0.427 ± 0.004	—
QML	beta33	0.000	0.001	0.001 ± 0.005	—	0.014 ± 0.001	0.119 ± 0.004	0.014 ± 0.001	0.119 ± 0.005	—	—	0.008	0.008	—	0.1314	0.1317	1.053	95.882 ± 0.880	0.490 ± 0.010	4.118 ± 0.880
QML	beta34	0.000	0.034	0.034 ± 0.003	—	0.005 ± 0.000	0.072 ± 0.003	0.006 ± 0.000	0.080 ± 0.003	—	—	0.031	0.031	—	0.0726	0.0790	1.023	94.706 ± 0.992	0.289 ± 0.007	5.294 ± 0.992
QML	beta35	0.000	0.036	0.036 ± 0.003	—	0.005 ± 0.000	0.070 ± 0.003	0.006 ± 0.000	0.078 ± 0.003	—	—	0.031	0.031	—	0.0673	0.0743	1.019	94.118 ± 1.042	0.278 ± 0.006	5.882 ± 1.042
UPI	beta31	0.316	0.440	0.124 ± 0.028	0.393 ± 0.088	0.392 ± 0.036	0.626 ± 0.029	0.406 ± 0.036	0.637 ± 0.032	4.068 ± 0.361	2.017 ± 0.100	0.477	0.161	50.84	0.6344	0.6545	1.756	97.843 ± 0.643	4.308 ± 0.232	—
UPI	beta32	0.316	0.431	0.115 ± 0.028	0.364 ± 0.089	0.405 ± 0.035	0.636 ± 0.028	0.417 ± 0.035	0.646 ± 0.031	4.176 ± 0.354	2.044 ± 0.098	0.439	0.123	39.05	0.6717	0.6829	1.837	98.039 ± 0.614	4.580 ± 0.237	—
UPI	beta33	0.000	0.041	0.041 ± 0.014	—	0.104 ± 0.010	0.323 ± 0.016	0.106 ± 0.010	0.325 ± 0.018	—	—	0.049	0.049	—	0.2837	0.2880	1.802	99.608 ± 0.277	2.282 ± 0.143	0.392 ± 0.277
UPI	beta34	0.000	-0.068	-0.068 ± 0.012	—	0.069 ± 0.007	0.263 ± 0.014	0.074 ± 0.007	0.272 ± 0.015	—	—	-0.076	-0.076	—	0.2405	0.2521	1.704	99.804 ± 0.196	1.757 ± 0.099	0.196 ± 0.196
UPI	beta35	0.000	-0.055	-0.055 ± 0.012	—	0.072 ± 0.007	0.269 ± 0.014	0.075 ± 0.007	0.274 ± 0.015	—	—	-0.068	-0.068	—	0.2469	0.2560	1.809	99.608 ± 0.277	1.908 ± 0.128	0.392 ± 0.277

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S98

Simulation 1 - Condition 98: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.4

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Type I (%) $\pm$ MCSE
LMS	beta31	0.316	0.286	-0.030 $\pm$ 0.003	-0.096 $\pm$ 0.008	0.004 $\pm$ 0.000	0.064 $\pm$ 0.002	0.005 $\pm$ 0.000	0.071 $\pm$ 0.002	0.051 $\pm$ 0.003	0.225 $\pm$ 0.007	0.285	-0.031	-9.89	0.0726	0.0791	1.037	92.919 $\pm$ 1.066	0.261 $\pm$ 0.002	—
LMS	beta32	0.316	0.287	-0.029 $\pm$ 0.003	-0.090 $\pm$ 0.008	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.005 $\pm$ 0.000	0.067 $\pm$ 0.002	0.045 $\pm$ 0.003	0.213 $\pm$ 0.007	0.289	-0.027	-8.56	0.0676	0.0728	1.087	95.509 $\pm$ 0.861	0.259 $\pm$ 0.001	—
LMS	beta33	0.000	0.014	0.014 $\pm$ 0.003	—	0.007 $\pm$ 0.000	0.081 $\pm$ 0.003	0.007 $\pm$ 0.000	0.082 $\pm$ 0.003	—	—	0.022	0.022	—	0.0810	0.0839	0.987	94.473 $\pm$ 0.950	0.312 $\pm$ 0.006	5.527 $\pm$ 0.950
LMS	beta34	0.000	0.031	0.031 $\pm$ 0.002	—	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.003 $\pm$ 0.000	0.053 $\pm$ 0.002	—	—	0.027	0.027	—	0.0443	0.0518	1.014	90.155 $\pm$ 1.238	0.169 $\pm$ 0.004	9.845 $\pm$ 1.238
LMS	beta35	0.000	0.033	0.033 $\pm$ 0.002	—	0.002 $\pm$ 0.000	0.042 $\pm$ 0.002	0.003 $\pm$ 0.000	0.054 $\pm$ 0.002	—	—	0.030	0.030	—	0.0438	0.0533	0.998	92.401 $\pm$ 1.101	0.166 $\pm$ 0.003	7.599 $\pm$ 1.101
LSAM	beta31	0.316	0.440	0.124 $\pm$ 0.023	0.391 $\pm$ 0.072	0.304 $\pm$ 0.027	0.551 $\pm$ 0.025	0.318 $\pm$ 0.029	0.564 $\pm$ 0.028	3.188 $\pm$ 0.287	1.786 $\pm$ 0.088	0.383	0.067	21.25	0.6078	0.6115	1.857	99.136 $\pm$ 0.385	4.012 $\pm$ 0.260	—
LSAM	beta32	0.316	0.413	0.097 $\pm$ 0.023	0.306 $\pm$ 0.073	0.308 $\pm$ 0.028	0.555 $\pm$ 0.025	0.317 $\pm$ 0.029	0.563 $\pm$ 0.028	3.173 $\pm$ 0.287	1.781 $\pm$ 0.089	0.404	0.088	27.76	0.5841	0.5907	1.829	99.827 $\pm$ 0.173	3.979 $\pm$ 0.258	—
LSAM	beta33	0.000	0.071	0.071 $\pm$ 0.005	—	0.013 $\pm$ 0.001	0.114 $\pm$ 0.005	0.018 $\pm$ 0.002	0.134 $\pm$ 0.006	—	—	0.067	0.067	—	0.1161	0.1342	1.509	95.164 $\pm$ 0.892	0.673 $\pm$ 0.037	4.836 $\pm$ 0.892
LSAM	beta34	0.000	-0.069	-0.069 $\pm$ 0.007	—	0.032 $\pm$ 0.003	0.178 $\pm$ 0.009	0.036 $\pm$ 0.003	0.191 $\pm$ 0.010	—	—	-0.042	-0.042	—	0.1740	0.1791	1.862	98.791 $\pm$ 0.454	1.298 $\pm$ 0.092	1.209 $\pm$ 0.454
LSAM	beta35	0.000	-0.061	-0.061 $\pm$ 0.008	—	0.033 $\pm$ 0.003	0.181 $\pm$ 0.009	0.036 $\pm$ 0.004	0.191 $\pm$ 0.010	—	—	-0.051	-0.051	—	0.1723	0.1797	1.785	99.655 $\pm$ 0.244	1.264 $\pm$ 0.088	0.345 $\pm$ 0.244
QML	beta31	0.316	0.287	-0.029 $\pm$ 0.003	-0.093 $\pm$ 0.009	0.004 $\pm$ 0.000	0.065 $\pm$ 0.002	0.005 $\pm$ 0.000	0.071 $\pm$ 0.002	0.051 $\pm$ 0.003	0.226 $\pm$ 0.008	0.287	-0.029	-9.29	0.0697	0.0756	1.032	93.264 $\pm$ 1.042	0.263 $\pm$ 0.001	—
QML	beta32	0.316	0.289	-0.027 $\pm$ 0.003	-0.087 $\pm$ 0.008	0.004 $\pm$ 0.000	0.062 $\pm$ 0.002	0.005 $\pm$ 0.000	0.067 $\pm$ 0.002	0.045 $\pm$ 0.003	0.213 $\pm$ 0.007	0.289	-0.027	-8.47	0.0650	0.0703	1.091	95.509 $\pm$ 0.861	0.263 $\pm$ 0.001	—
QML	beta33	0.000	0.012	0.012 $\pm$ 0.003	—	0.006 $\pm$ 0.000	0.078 $\pm$ 0.002	0.006 $\pm$ 0.000	0.079 $\pm$ 0.003	—	—	0.018	0.018	—	0.0744	0.0766	0.978	94.128 $\pm$ 0.977	0.299 $\pm$ 0.004	5.872 $\pm$ 0.977
QML	beta34	0.000	0.030	0.030 $\pm$ 0.002	—	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.003 $\pm$ 0.000	0.052 $\pm$ 0.002	—	—	0.027	0.027	—	0.0427	0.0504	0.997	88.601 $\pm$ 1.321	0.163 $\pm$ 0.002	11.399 $\pm$ 1.321
QML	beta35	0.000	0.032	0.032 $\pm$ 0.002	—	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	0.003 $\pm$ 0.000	0.052 $\pm$ 0.002	—	—	0.028	0.028	—	0.0417	0.0505	1.009	91.019 $\pm$ 1.188	0.164 $\pm$ 0.002	8.981 $\pm$ 1.188
UPI	beta31	0.316	0.526	0.210 $\pm$ 0.023	0.665 $\pm$ 0.072	0.302 $\pm$ 0.023	0.550 $\pm$ 0.021	0.346 $\pm$ 0.025	0.588 $\pm$ 0.024	3.463 $\pm$ 0.253	1.861 $\pm$ 0.076	0.586	0.270	85.40	0.5994	0.6574	1.537	93.092 $\pm$ 1.054	3.311 $\pm$ 0.162	—
UPI	beta32	0.316	0.524	0.208 $\pm$ 0.024	0.658 $\pm$ 0.077	0.345 $\pm$ 0.025	0.588 $\pm$ 0.021	0.388 $\pm$ 0.025	0.623 $\pm$ 0.024	3.883 $\pm$ 0.255	1.971 $\pm$ 0.074	0.574	0.258	81.70	0.6234	0.6748	1.498	94.991 $\pm$ 0.906	3.449 $\pm$ 0.180	—
UPI	beta33	0.000	0.104	0.104 $\pm$ 0.009	—	0.051 $\pm$ 0.004	0.225 $\pm$ 0.010	0.061 $\pm$ 0.005	0.248 $\pm$ 0.011	—	—	0.094	0.094	—	0.1929	0.2146	1.507	99.655 $\pm$ 0.244	1.328 $\pm$ 0.067	0.345 $\pm$ 0.244
UPI	beta34	0.000	-0.098	-0.098 $\pm$ 0.008	—	0.041 $\pm$ 0.004	0.202 $\pm$ 0.009	0.050 $\pm$ 0.004	0.224 $\pm$ 0.009	—	—	-0.109	-0.109	—	0.1842	0.2141	1.495	97.064 $\pm$ 0.702	1.181 $\pm$ 0.056	2.936 $\pm$ 0.702
UPI	beta35	0.000	-0.099	-0.099 $\pm$ 0.009	—	0.046 $\pm$ 0.004	0.216 $\pm$ 0.009	0.056 $\pm$ 0.004	0.237 $\pm$ 0.010	—	—	-0.109	-0.109	—	0.1991	0.2272	1.480	97.582 $\pm$ 0.638	1.251 $\pm$ 0.066	2.418 $\pm$ 0.638

Note. MCSE = Monte Carlo Standard Error shown as $\backslash$ u00b1 value. Coverage, Type I Error, and Power expressed as percentages. $\backslash$ u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S99

Simulation 1 - Condition 99: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, $N = 400$, Reliability = 0.6

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Type I (%) $\pm$ MCSE
LMS	beta31	0.316	0.293	-0.023 $\pm$ 0.003	-0.074 $\pm$ 0.009	0.008 $\pm$ 0.000	0.088 $\pm$ 0.002	0.008 $\pm$ 0.000	0.091 $\pm$ 0.003	0.082 $\pm$ 0.004	0.287 $\pm$ 0.008	0.295	-0.021	-6.80	0.0929	0.0954	0.986	93.296 $\pm$ 0.836	0.338 $\pm$ 0.001	—
LMS	beta32	0.316	0.296	-0.020 $\pm$ 0.003	-0.064 $\pm$ 0.009	0.007 $\pm$ 0.000	0.085 $\pm$ 0.002	0.008 $\pm$ 0.000	0.087 $\pm$ 0.002	0.076 $\pm$ 0.003	0.277 $\pm$ 0.008	0.291	-0.025	-7.85	0.0935	0.0967	1.015	94.860 $\pm$ 0.738	0.338 $\pm$ 0.002	—
LMS	beta33	0.000	0.002	0.002 $\pm$ 0.003	—	0.006 $\pm$ 0.000	0.079 $\pm$ 0.002	0.006 $\pm$ 0.000	0.079 $\pm$ 0.002	—	—	0.004	0.004	—	0.0841	0.0842	0.968	94.860 $\pm$ 0.738	0.299 $\pm$ 0.003	5.140 $\pm$ 0.738
LMS	beta34	0.000	0.022	0.022 $\pm$ 0.002	—	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.003 $\pm$ 0.000	0.050 $\pm$ 0.001	—	—	0.019	0.019	—	0.0453	0.0490	0.916	90.168 $\pm$ 0.995	0.163 $\pm$ 0.002	9.832 $\pm$ 0.995
LMS	beta35	0.000	0.020	0.020 $\pm$ 0.001	—	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	—	—	0.018	0.018	—	0.0425	0.0461	0.943	91.955 $\pm$ 0.909	0.166 $\pm$ 0.002	8.045 $\pm$ 0.909
LSAM	beta31	0.316	0.335	0.019 $\pm$ 0.006	0.060 $\pm$ 0.018	0.028 $\pm$ 0.001	0.168 $\pm$ 0.004	0.029 $\pm$ 0.001	0.169 $\pm$ 0.005	0.286 $\pm$ 0.014	0.535 $\pm$ 0.016	0.339	0.023	7.23	0.1683	0.1698	1.021	96.201 $\pm$ 0.639	0.673 $\pm$ 0.009	—
LSAM	beta32	0.316	0.343	0.027 $\pm$ 0.005	0.084 $\pm$ 0.017	0.027 $\pm$ 0.001	0.164 $\pm$ 0.004	0.027 $\pm$ 0.001	0.166 $\pm$ 0.005	0.275 $\pm$ 0.015	0.524 $\pm$ 0.017	0.335	0.019	6.12	0.1702	0.1713	1.047	97.095 $\pm$ 0.561	0.672 $\pm$ 0.009	—
LSAM	beta33	0.000	0.023	0.023 $\pm$ 0.003	—	0.007 $\pm$ 0.000	0.083 $\pm$ 0.002	0.007 $\pm$ 0.000	0.086 $\pm$ 0.003	—	—	0.021	0.021	—	0.0789	0.0815	0.990	94.972 $\pm$ 0.730	0.323 $\pm$ 0.004	5.028 $\pm$ 0.730
LSAM	beta34	0.000	-0.015	-0.015 $\pm$ 0.002	—	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.004 $\pm$ 0.000	0.063 $\pm$ 0.002	—	—	-0.015	-0.015	—	0.0584	0.0602	0.981	93.855 $\pm$ 0.803	0.235 $\pm$ 0.005	6.145 $\pm$ 0.803
LSAM	beta35	0.000	-0.018	-0.018 $\pm$ 0.002	—	0.004 $\pm$ 0.000	0.062 $\pm$ 0.002	0.004 $\pm$ 0.000	0.064 $\pm$ 0.002	—	—	-0.014	-0.014	—	0.0541	0.0560	0.981	95.419 $\pm$ 0.699	0.238 $\pm$ 0.004	4.581 $\pm$ 0.699
QML	beta31	0.316	0.293	-0.023 $\pm$ 0.003	-0.073 $\pm$ 0.009	0.008 $\pm$ 0.000	0.088 $\pm$ 0.002	0.008 $\pm$ 0.000	0.091 $\pm$ 0.003	0.083 $\pm$ 0.004	0.287 $\pm$ 0.008	0.294	-0.022	-6.82	0.0913	0.0939	0.988	93.520 $\pm$ 0.823	0.341 $\pm$ 0.002	—
QML	beta32	0.316	0.296	-0.020 $\pm$ 0.003	-0.063 $\pm$ 0.009	0.007 $\pm$ 0.000	0.086 $\pm$ 0.002	0.008 $\pm$ 0.000	0.088 $\pm$ 0.002	0.077 $\pm$ 0.003	0.278 $\pm$ 0.008	0.292	-0.024	-7.65	0.0916	0.0947	1.012	94.860 $\pm$ 0.738	0.339 $\pm$ 0.002	—
QML	beta33	0.000	0.002	0.002 $\pm$ 0.003	—	0.006 $\pm$ 0.000	0.079 $\pm$ 0.002	0.006 $\pm$ 0.000	0.079 $\pm$ 0.002	—	—	0.003	0.003	—	0.0811	0.0812	0.969	94.860 $\pm$ 0.738	0.299 $\pm$ 0.004	5.140 $\pm$ 0.738
QML	beta34	0.000	0.021	0.021 $\pm$ 0.002	—	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.003 $\pm$ 0.000	0.050 $\pm$ 0.001	—	—	0.019	0.019	—	0.0449	0.0486	0.926	89.832 $\pm$ 1.010	0.165 $\pm$ 0.003	10.168 $\pm$ 1.010
QML	beta35	0.000	0.019	0.019 $\pm$ 0.001	—	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	—	—	0.019	0.019	—	0.0409	0.0449	0.945	91.955 $\pm$ 0.909	0.166 $\pm$ 0.002	8.045 $\pm$ 0.909
UPI	beta31	0.316	0.339	0.023 $\pm$ 0.007	0.073 $\pm$ 0.022	0.042 $\pm$ 0.002	0.205 $\pm$ 0.006	0.043 $\pm$ 0.002	0.206 $\pm$ 0.007	0.426 $\pm$ 0.023	0.652 $\pm$ 0.021	0.336	0.020	6.19	0.2033	0.2043	1.177	96.983 $\pm$ 0.572	0.946 $\pm$ 0.047	—
UPI	beta32	0.316	0.356	0.040 $\pm$ 0.007	0.128 $\pm$ 0.022	0.043 $\pm$ 0.003	0.206 $\pm$ 0.006	0.044 $\pm$ 0.003	0.210 $\pm$ 0.007	0.442 $\pm$ 0.027	0.665 $\pm$ 0.023	0.342	0.026	8.07	0.1877	0.1894	1.185	97.654 $\pm$ 0.506	0.959 $\pm$ 0.047	—
UPI	beta33	0.000	0.027	0.027 $\pm$ 0.003	—	0.009 $\pm$ 0.001	0.095 $\pm$ 0.003	0.010 $\pm$ 0.001	0.099 $\pm$ 0.003	—	—	0.023	0.023	—	0.0847	0.0878	1.158	97.095 $\pm$ 0.561	0.431 $\pm$ 0.019	2.905 $\pm$ 0.561
UPI	beta34	0.000	-0.018	-0.018 $\pm$ 0.003	—	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.006 $\pm$ 0.000	0.077 $\pm$ 0.003	—	—	-0.014	-0.014	—	0.0662	0.0676	1.141	95.866 $\pm$ 0.665	0.335 $\pm$ 0.017	4.134 $\pm$ 0.665
UPI	beta35	0.000	-0.026	-0.026 $\pm$ 0.003	—	0.007 $\pm$ 0.000	0.081 $\pm$ 0.003	0.007 $\pm$ 0.001	0.085 $\pm$ 0.003	—	—	-0.017	-0.017	—	0.0625	0.0647	1.110	96.983 $\pm$ 0.572	0.353 $\pm$ 0.020	3.017 $\pm$ 0.572

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S100

Simulation 1 - Condition 100: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.6

Method	Param.	True	Mean-based									Median-based					Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE	
LMS	beta31	0.316	0.294	-0.022 ± 0.002	-0.069 ± 0.005	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.033 ± 0.002	0.181 ± 0.005	0.296	-0.020	-6.18	0.0561	0.0594	1.015	93.277 ± 0.812	0.210 ± 0.001		
LMS	beta32	0.316	0.293	-0.023 ± 0.002	-0.073 ± 0.006	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.034 ± 0.002	0.185 ± 0.005	0.292	-0.024	-7.60	0.0549	0.0600	0.998	92.647 ± 0.846	0.210 ± 0.001	—	
LMS	beta33	0.000	0.004	0.004 ± 0.002	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	—	—	0.003	0.003	—	0.0469	0.0471	1.007	95.063 ± 0.702	0.187 ± 0.001	4.937 ± 0.702	
LMS	beta34	0.000	0.019	0.019 ± 0.001	—	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	—	—	0.018	0.018	—	0.0259	0.0317	0.960	87.395 ± 1.076	0.100 ± 0.001	12.605 ± 1.076	
LMS	beta35	0.000	0.019	0.019 ± 0.001	—	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	—	—	0.017	0.017	—	0.0266	0.0315	0.935	87.605 ± 1.068	0.101 ± 0.001	12.395 ± 1.068	
LSAM	beta31	0.316	0.337	0.021 ± 0.003	0.066 ± 0.010	0.009 ± 0.000	0.094 ± 0.002	0.009 ± 0.000	0.096 ± 0.003	0.092 ± 0.005	0.303 ± 0.009	0.334	0.018	5.76	0.0951	0.0968	1.019	94.853 ± 0.716	0.374 ± 0.002	—	
LSAM	beta32	0.316	0.335	0.019 ± 0.003	0.059 ± 0.010	0.009 ± 0.000	0.095 ± 0.002	0.009 ± 0.000	0.096 ± 0.003	0.093 ± 0.004	0.305 ± 0.008	0.334	0.018	5.61	0.0962	0.0979	1.014	96.113 ± 0.626	0.376 ± 0.002	—	
LSAM	beta33	0.000	0.018	0.018 ± 0.002	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.050 ± 0.002	—	—	0.015	0.015	—	0.0437	0.0462	0.973	93.067 ± 0.823	0.177 ± 0.001	6.933 ± 0.823	
LSAM	beta34	0.000	-0.014	-0.014 ± 0.001	—	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	-0.012	-0.012	—	0.0301	0.0324	0.980	93.067 ± 0.823	0.122 ± 0.001	6.933 ± 0.823	
LSAM	beta35	0.000	-0.014	-0.014 ± 0.001	—	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	-0.014	-0.014	—	0.0306	0.0338	0.968	94.013 ± 0.769	0.123 ± 0.001	5.987 ± 0.769	
QML	beta31	0.316	0.294	-0.022 ± 0.002	-0.068 ± 0.005	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.033 ± 0.002	0.181 ± 0.005	0.297	-0.019	-6.05	0.0540	0.0573	1.010	93.172 ± 0.817	0.210 ± 0.001	—	
QML	beta32	0.316	0.293	-0.023 ± 0.002	-0.073 ± 0.006	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.035 ± 0.002	0.186 ± 0.005	0.293	-0.023	-7.33	0.0552	0.0599	0.994	92.332 ± 0.862	0.211 ± 0.001	—	
QML	beta33	0.000	0.004	0.004 ± 0.002	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	0.002	0.002	—	0.0463	0.0463	1.005	94.958 ± 0.709	0.187 ± 0.001	5.042 ± 0.709	
QML	beta34	0.000	0.019	0.019 ± 0.001	—	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	—	—	0.018	0.018	—	0.0254	0.0314	0.958	87.500 ± 1.072	0.100 ± 0.001	12.500 ± 1.072	
QML	beta35	0.000	0.019	0.019 ± 0.001	—	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	—	—	0.017	0.017	—	0.0260	0.0311	0.938	87.605 ± 1.068	0.101 ± 0.001	12.395 ± 1.068	
UPI	beta31	0.316	0.344	0.028 ± 0.004	0.088 ± 0.011	0.012 ± 0.001	0.111 ± 0.003	0.013 ± 0.001	0.115 ± 0.004	0.132 ± 0.007	0.363 ± 0.011	0.340	0.024	7.73	0.1090	0.1117	1.025	95.693 ± 0.658	0.447 ± 0.005	—	
UPI	beta32	0.316	0.341	0.025 ± 0.004	0.080 ± 0.011	0.013 ± 0.001	0.112 ± 0.003	0.013 ± 0.001	0.115 ± 0.003	0.132 ± 0.007	0.363 ± 0.011	0.338	0.022	7.05	0.1107	0.1129	1.019	96.744 ± 0.575	0.447 ± 0.005	—	
UPI	beta33	0.000	0.020	0.020 ± 0.002	—	0.002 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	—	—	0.017	0.017	—	0.0479	0.0508	0.994	94.538 ± 0.736	0.194 ± 0.002	5.462 ± 0.736	
UPI	beta34	0.000	-0.018	-0.018 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	-0.014	-0.014	—	0.0335	0.0364	0.991	94.538 ± 0.736	0.146 ± 0.002	5.462 ± 0.736	
UPI	beta35	0.000	-0.018	-0.018 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	—	—	-0.016	-0.016	—	0.0341	0.0378	0.959	95.063 ± 0.702	0.147 ± 0.002	4.937 ± 0.702	

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u002014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S101

Simulation 1 - Condition 101: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.8

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.298	-0.018 ± 0.002	-0.058 ± 0.008	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	0.063 ± 0.003	0.251 ± 0.007	0.296	-0.020	-6.47	0.0750	0.0778	1.017	94.076 ± 0.748	0.308 ± 0.005	
LMS	beta32	0.316	0.308	-0.008 ± 0.002	-0.026 ± 0.008	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	0.062 ± 0.003	0.250 ± 0.007	0.302	-0.014	-4.38	0.0798	0.0810	0.993	94.880 ± 0.698	0.306 ± 0.002	—
LMS	beta33	0.000	0.002	0.002 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	—	—	0.003	0.003	—	0.0570	0.0570	0.938	92.871 ± 0.815	0.224 ± 0.003	7.129 ± 0.815
LMS	beta34	0.000	0.011	0.011 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	0.009	0.009	—	0.0312	0.0325	0.928	89.357 ± 0.977	0.127 ± 0.004	10.643 ± 0.977
LMS	beta35	0.000	0.008	0.008 ± 0.001	—	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	—	—	0.009	0.009	—	0.0321	0.0333	0.961	92.169 ± 0.851	0.123 ± 0.001	7.831 ± 0.851
LSAM	beta31	0.316	0.313	-0.003 ± 0.003	-0.010 ± 0.010	0.010 ± 0.000	0.099 ± 0.003	0.010 ± 0.000	0.099 ± 0.003	0.099 ± 0.005	0.314 ± 0.009	0.311	-0.005	-1.64	0.0932	0.0933	0.994	95.382 ± 0.665	0.387 ± 0.002	—
LSAM	beta32	0.316	0.326	0.010 ± 0.003	0.032 ± 0.010	0.010 ± 0.000	0.101 ± 0.002	0.010 ± 0.000	0.101 ± 0.003	0.102 ± 0.005	0.320 ± 0.009	0.319	0.003	0.87	0.0999	0.0999	0.985	95.080 ± 0.685	0.388 ± 0.002	—
LSAM	beta33	0.000	0.008	0.008 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	—	—	0.007	0.007	—	0.0570	0.0574	0.930	92.972 ± 0.810	0.222 ± 0.002	7.028 ± 0.810
LSAM	beta34	0.000	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.002	-0.002	—	0.0364	0.0365	0.914	92.470 ± 0.836	0.136 ± 0.002	7.530 ± 0.836
LSAM	beta35	0.000	-0.005	-0.005 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.003	-0.003	—	0.0342	0.0344	0.958	94.177 ± 0.742	0.137 ± 0.002	5.823 ± 0.742
QML	beta31	0.316	0.298	-0.018 ± 0.002	-0.059 ± 0.008	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.080 ± 0.002	0.064 ± 0.003	0.253 ± 0.007	0.296	-0.020	-6.36	0.0754	0.0780	0.993	94.076 ± 0.748	0.302 ± 0.001	—
QML	beta32	0.316	0.308	-0.008 ± 0.003	-0.026 ± 0.008	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	0.063 ± 0.003	0.251 ± 0.007	0.302	-0.014	-4.39	0.0805	0.0817	0.981	94.779 ± 0.705	0.303 ± 0.001	—
QML	beta33	0.000	0.003	0.003 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	—	—	0.003	0.003	—	0.0569	0.0570	0.928	92.771 ± 0.821	0.221 ± 0.002	7.229 ± 0.821
QML	beta34	0.000	0.011	0.011 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	0.009	0.009	—	0.0314	0.0327	0.899	89.458 ± 0.973	0.123 ± 0.001	10.542 ± 0.973
QML	beta35	0.000	0.008	0.008 ± 0.001	—	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	—	—	0.009	0.009	—	0.0321	0.0333	0.959	92.068 ± 0.856	0.123 ± 0.001	7.932 ± 0.856
UPI	beta31	0.316	0.313	-0.003 ± 0.003	-0.011 ± 0.010	0.010 ± 0.000	0.099 ± 0.003	0.010 ± 0.000	0.099 ± 0.003	0.098 ± 0.005	0.312 ± 0.009	0.310	-0.006	-1.95	0.0928	0.0930	0.987	94.378 ± 0.730	0.382 ± 0.002	—
UPI	beta32	0.316	0.326	0.010 ± 0.003	0.031 ± 0.010	0.010 ± 0.000	0.100 ± 0.002	0.010 ± 0.000	0.101 ± 0.003	0.101 ± 0.005	0.319 ± 0.009	0.318	0.002	0.66	0.0981	0.0981	0.978	95.181 ± 0.679	0.384 ± 0.002	—
UPI	beta33	0.000	0.007	0.007 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	—	—	0.008	0.008	—	0.0565	0.0570	0.950	94.478 ± 0.724	0.224 ± 0.002	5.522 ± 0.724
UPI	beta34	0.000	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.002	-0.002	—	0.0352	0.0352	0.906	92.269 ± 0.846	0.135 ± 0.002	7.731 ± 0.846
UPI	beta35	0.000	-0.005	-0.005 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.003	-0.003	—	0.0345	0.0346	0.948	94.177 ± 0.742	0.136 ± 0.002	5.823 ± 0.742

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S102

Simulation 1 - Condition 102: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.8

Method	Param.	True	Mean-based										Median-based					Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE		
LMS	beta31	0.316	0.305	-0.011 ± 0.002	-0.034 ± 0.005	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.024 ± 0.001	0.156 ± 0.004	0.305	-0.011	-3.54	0.0463	0.0477	1.001	94.200 ± 0.739	0.188 ± 0.000			
LMS	beta32	0.316	0.305	-0.011 ± 0.002	-0.036 ± 0.005	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.024 ± 0.001	0.155 ± 0.004	0.305	-0.011	-3.48	0.0500	0.0512	1.005	93.800 ± 0.763	0.188 ± 0.000	—		
LMS	beta33	0.000	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	-0.001	-0.001	—	0.0362	0.0362	0.991	94.200 ± 0.739	0.138 ± 0.002	5.800 ± 0.739		
LMS	beta34	0.000	0.008	0.008 ± 0.001	—	0.000 ± 0.000	0.021 ± 0.001	0.000 ± 0.000	0.022 ± 0.001	—	—	0.008	0.008	—	0.0192	0.0207	0.903	91.000 ± 0.905	0.074 ± 0.001	9.000 ± 0.905		
LMS	beta35	0.000	0.009	0.009 ± 0.001	—	0.000 ± 0.000	0.020 ± 0.000	0.000 ± 0.000	0.022 ± 0.001	—	—	0.009	0.009	—	0.0185	0.0204	0.923	89.500 ± 0.969	0.073 ± 0.001	10.500 ± 0.969		
LSAM	beta31	0.316	0.322	0.006 ± 0.002	0.019 ± 0.006	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.037 ± 0.002	0.193 ± 0.005	0.321	0.005	1.69	0.0586	0.0589	0.983	94.300 ± 0.733	0.234 ± 0.001	—		
LSAM	beta32	0.316	0.321	0.005 ± 0.002	0.014 ± 0.006	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.036 ± 0.002	0.190 ± 0.005	0.321	0.005	1.49	0.0614	0.0616	0.992	94.800 ± 0.702	0.233 ± 0.001	—		
LSAM	beta33	0.000	0.003	0.003 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	0.004	0.004	—	0.0347	0.0349	0.971	94.200 ± 0.739	0.132 ± 0.001	5.800 ± 0.739		
LSAM	beta34	0.000	-0.004	-0.004 ± 0.001	—	0.000 ± 0.000	0.022 ± 0.001	0.000 ± 0.000	0.022 ± 0.001	—	—	-0.003	-0.003	—	0.0200	0.0203	0.898	91.700 ± 0.872	0.077 ± 0.001	8.300 ± 0.872		
LSAM	beta35	0.000	-0.003	-0.003 ± 0.001	—	0.000 ± 0.000	0.021 ± 0.001	0.000 ± 0.000	0.021 ± 0.001	—	—	-0.003	-0.003	—	0.0190	0.0192	0.922	92.900 ± 0.812	0.077 ± 0.001	7.100 ± 0.812		
QML	beta31	0.316	0.305	-0.011 ± 0.002	-0.035 ± 0.005	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.024 ± 0.001	0.156 ± 0.004	0.305	-0.011	-3.49	0.0464	0.0477	1.007	94.100 ± 0.745	0.190 ± 0.002	—		
QML	beta32	0.316	0.305	-0.011 ± 0.002	-0.036 ± 0.005	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.024 ± 0.001	0.156 ± 0.004	0.305	-0.011	-3.50	0.0501	0.0513	1.011	93.800 ± 0.763	0.190 ± 0.002	—		
QML	beta33	0.000	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	0.000	0.000	—	0.0361	0.0361	0.978	93.900 ± 0.757	0.136 ± 0.001	6.100 ± 0.757		
QML	beta34	0.000	0.008	0.008 ± 0.001	—	0.000 ± 0.000	0.021 ± 0.001	0.000 ± 0.000	0.022 ± 0.001	—	—	0.008	0.008	—	0.0192	0.0207	0.895	90.900 ± 0.909	0.073 ± 0.001	9.100 ± 0.909		
QML	beta35	0.000	0.009	0.009 ± 0.001	—	0.000 ± 0.000	0.020 ± 0.000	0.000 ± 0.000	0.022 ± 0.001	—	—	0.009	0.009	—	0.0183	0.0202	0.917	89.600 ± 0.965	0.073 ± 0.001	10.400 ± 0.965		
UPI	beta31	0.316	0.321	0.005 ± 0.002	0.016 ± 0.006	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.036 ± 0.002	0.191 ± 0.005	0.320	0.004	1.23	0.0582	0.0583	0.986	94.000 ± 0.751	0.232 ± 0.001	—		
UPI	beta32	0.316	0.320	0.004 ± 0.002	0.012 ± 0.006	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.036 ± 0.002	0.189 ± 0.005	0.319	0.003	1.10	0.0607	0.0608	0.993	94.500 ± 0.721	0.232 ± 0.001	—		
UPI	beta33	0.000	0.002	0.002 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	0.001	0.001	—	0.0343	0.0343	0.973	94.500 ± 0.721	0.132 ± 0.001	5.500 ± 0.721		
UPI	beta34	0.000	-0.003	-0.003 ± 0.001	—	0.000 ± 0.000	0.022 ± 0.001	0.000 ± 0.000	0.022 ± 0.001	—	—	-0.003	-0.003	—	0.0199	0.0202	0.904	92.000 ± 0.858	0.077 ± 0.001	8.000 ± 0.858		
UPI	beta35	0.000	-0.002	-0.002 ± 0.001	—	0.000 ± 0.000	0.021 ± 0.001	0.000 ± 0.000	0.021 ± 0.001	—	—	-0.002	-0.002	—	0.0190	0.0191	0.924	93.600 ± 0.774	0.076 ± 0.001	6.400 ± 0.774		

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u002014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S103

Simulation 1 - Condition 103: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, $N = 400$, Reliability = 0.4

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Type I (%) $\pm$ MCSE
LMS	beta31	0.316	0.305	-0.011 $\pm$ 0.010	-0.035 $\pm$ 0.033	0.007 $\pm$ 0.001	0.082 $\pm$ 0.008	0.007 $\pm$ 0.001	0.083 $\pm$ 0.010	0.068 $\pm$ 0.013	0.261 $\pm$ 0.030	0.305	-0.011	-3.60	0.1037	0.1043	1.061	92.188 $\pm$ 3.355	0.343 $\pm$ 0.007	—
LMS	beta32	0.316	0.322	0.006 $\pm$ 0.012	0.018 $\pm$ 0.038	0.009 $\pm$ 0.002	0.095 $\pm$ 0.009	0.009 $\pm$ 0.002	0.095 $\pm$ 0.011	0.090 $\pm$ 0.016	0.300 $\pm$ 0.033	0.343	0.027	8.50	0.1145	0.1176	0.985	95.312 $\pm$ 2.642	0.368 $\pm$ 0.020	—
LMS	beta33	0.000	0.054	0.054 $\pm$ 0.015	—	0.014 $\pm$ 0.003	0.120 $\pm$ 0.014	0.017 $\pm$ 0.003	0.131 $\pm$ 0.014	—	—	0.060	0.060	—	0.1395	0.1519	1.054	93.750 $\pm$ 3.026	0.496 $\pm$ 0.040	6.250 $\pm$ 3.026
LMS	beta34	0.000	-0.063	-0.063 $\pm$ 0.010	—	0.007 $\pm$ 0.001	0.082 $\pm$ 0.009	0.011 $\pm$ 0.002	0.103 $\pm$ 0.011	—	—	-0.061	-0.061	—	0.1113	0.1267	0.950	76.562 $\pm$ 5.295	0.307 $\pm$ 0.016	23.438 $\pm$ 5.295
LMS	beta35	0.000	-0.067	-0.067 $\pm$ 0.011	—	0.008 $\pm$ 0.001	0.088 $\pm$ 0.008	0.012 $\pm$ 0.002	0.111 $\pm$ 0.011	—	—	-0.064	-0.064	—	0.1123	0.1291	1.269	81.250 $\pm$ 4.879	0.439 $\pm$ 0.112	18.750 $\pm$ 4.879
LSAM	beta31	0.316	0.317	0.001 $\pm$ 0.022	0.004 $\pm$ 0.069	0.030 $\pm$ 0.006	0.174 $\pm$ 0.018	0.030 $\pm$ 0.006	0.172 $\pm$ 0.021	0.297 $\pm$ 0.059	0.545 $\pm$ 0.066	0.303	-0.013	-4.11	0.1482	0.1488	1.496	100.000 $\pm$ 0.000	1.018 $\pm$ 0.242	—
LSAM	beta32	0.316	0.380	0.064 $\pm$ 0.022	0.204 $\pm$ 0.069	0.031 $\pm$ 0.006	0.175 $\pm$ 0.018	0.034 $\pm$ 0.007	0.185 $\pm$ 0.022	0.345 $\pm$ 0.069	0.587 $\pm$ 0.068	0.387	0.071	22.45	0.1564	0.1718	1.410	100.000 $\pm$ 0.000	0.969 $\pm$ 0.196	—
LSAM	beta33	0.000	0.127	0.127 $\pm$ 0.018	—	0.022 $\pm$ 0.005	0.147 $\pm$ 0.019	0.038 $\pm$ 0.008	0.194 $\pm$ 0.023	—	—	0.100	0.100	—	0.1636	0.1915	1.311	100.000 $\pm$ 0.000	0.756 $\pm$ 0.069	0.000 $\pm$ 0.000
LSAM	beta34	0.000	-0.108	-0.108 $\pm$ 0.018	—	0.021 $\pm$ 0.005	0.146 $\pm$ 0.020	0.033 $\pm$ 0.007	0.181 $\pm$ 0.020	—	—	-0.079	-0.079	—	0.1297	0.1517	1.637	95.312 $\pm$ 2.642	0.957 $\pm$ 0.177	4.688 $\pm$ 2.642
LSAM	beta35	0.000	-0.161	-0.161 $\pm$ 0.026	—	0.044 $\pm$ 0.013	0.209 $\pm$ 0.034	0.069 $\pm$ 0.020	0.263 $\pm$ 0.041	—	—	-0.140	-0.140	—	0.1747	0.2240	1.330	96.875 $\pm$ 2.175	1.091 $\pm$ 0.268	3.125 $\pm$ 2.175
QML	beta31	0.316	0.306	-0.010 $\pm$ 0.011	-0.032 $\pm$ 0.034	0.007 $\pm$ 0.001	0.085 $\pm$ 0.009	0.007 $\pm$ 0.001	0.085 $\pm$ 0.010	0.072 $\pm$ 0.014	0.268 $\pm$ 0.032	0.304	-0.012	-3.70	0.0981	0.0988	1.046	92.188 $\pm$ 3.355	0.348 $\pm$ 0.007	—
QML	beta32	0.316	0.318	0.002 $\pm$ 0.011	0.005 $\pm$ 0.036	0.008 $\pm$ 0.001	0.092 $\pm$ 0.007	0.008 $\pm$ 0.001	0.091 $\pm$ 0.009	0.083 $\pm$ 0.013	0.288 $\pm$ 0.029	0.332	0.016	5.19	0.1052	0.1065	0.976	95.312 $\pm$ 2.642	0.351 $\pm$ 0.007	—
QML	beta33	0.000	0.068	0.068 $\pm$ 0.015	—	0.014 $\pm$ 0.003	0.119 $\pm$ 0.012	0.019 $\pm$ 0.004	0.136 $\pm$ 0.015	—	—	0.059	0.059	—	0.1336	0.1461	1.041	93.750 $\pm$ 3.026	0.485 $\pm$ 0.026	6.250 $\pm$ 3.026
QML	beta34	0.000	-0.071	-0.071 $\pm$ 0.011	—	0.008 $\pm$ 0.002	0.088 $\pm$ 0.010	0.013 $\pm$ 0.003	0.113 $\pm$ 0.013	—	—	-0.053	-0.053	—	0.1026	0.1154	0.940	79.688 $\pm$ 5.029	0.326 $\pm$ 0.016	20.312 $\pm$ 5.029
QML	beta35	0.000	-0.079	-0.079 $\pm$ 0.011	—	0.008 $\pm$ 0.001	0.092 $\pm$ 0.008	0.015 $\pm$ 0.003	0.121 $\pm$ 0.012	—	—	-0.068	-0.068	—	0.1138	0.1324	0.939	81.250 $\pm$ 4.879	0.337 $\pm$ 0.018	18.750 $\pm$ 4.879
UPI	beta31	0.316	0.322	0.006 $\pm$ 0.044	0.017 $\pm$ 0.139	0.123 $\pm$ 0.041	0.350 $\pm$ 0.067	0.121 $\pm$ 0.042	0.348 $\pm$ 0.069	1.210 $\pm$ 0.417	1.100 $\pm$ 0.220	0.327	0.011	3.42	0.3713	0.3715	1.606	84.375 $\pm$ 4.539	2.206 $\pm$ 0.422	—
UPI	beta32	0.316	0.300	-0.016 $\pm$ 0.052	-0.051 $\pm$ 0.165	0.174 $\pm$ 0.034	0.417 $\pm$ 0.043	0.171 $\pm$ 0.034	0.414 $\pm$ 0.050	1.715 $\pm$ 0.342	1.310 $\pm$ 0.159	0.340	0.024	7.59	0.3907	0.3915	1.738	84.375 $\pm$ 4.539	2.839 $\pm$ 0.585	—
UPI	beta33	0.000	0.011	0.011 $\pm$ 0.026	—	0.042 $\pm$ 0.011	0.206 $\pm$ 0.030	0.042 $\pm$ 0.012	0.204 $\pm$ 0.033	—	—	-0.025	-0.025	—	0.2027	0.2042	2.038	100.000 $\pm$ 0.000	1.642 $\pm$ 0.235	0.000 $\pm$ 0.000
UPI	beta34	0.000	-0.213	-0.213 $\pm$ 0.058	—	0.214 $\pm$ 0.051	0.463 $\pm$ 0.059	0.256 $\pm$ 0.068	0.506 $\pm$ 0.075	—	—	-0.102	-0.102	—	0.4095	0.4219	1.522	96.875 $\pm$ 2.175	2.759 $\pm$ 0.435	3.125 $\pm$ 2.175
UPI	beta35	0.000	-0.195	-0.195 $\pm$ 0.073	—	0.341 $\pm$ 0.097	0.584 $\pm$ 0.090	0.373 $\pm$ 0.107	0.611 $\pm$ 0.099	—	—	-0.176	-0.176	—	0.4514	0.4846	1.848	100.000 $\pm$ 0.000	4.228 $\pm$ 0.797	0.000 $\pm$ 0.000

Note. MCSE = Monte Carlo Standard Error shown as $\backslash u00b1$ value. Coverage, Type I Error, and Power expressed as percentages. $\backslash u0021$ indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S104

Simulation 1 - Condition 104: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.4

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.322	0.006 ± 0.007	0.019 ± 0.022	0.003 ± 0.000	0.052 ± 0.005	0.003 ± 0.000	0.052 ± 0.006	0.027 ± 0.005	0.165 ± 0.019	0.327	0.011	3.61	0.0653	0.0663	1.050	96.429 ± 2.480	0.215 ± 0.003	—
LMS	beta32	0.316	0.314	-0.002 ± 0.007	-0.008 ± 0.023	0.003 ± 0.001	0.054 ± 0.005	0.003 ± 0.001	0.053 ± 0.006	0.028 ± 0.005	0.168 ± 0.020	0.309	-0.007	-2.15	0.0558	0.0562	1.021	96.429 ± 2.480	0.214 ± 0.003	—
LMS	beta33	0.000	0.064	0.064 ± 0.008	—	0.004 ± 0.001	0.063 ± 0.006	0.008 ± 0.001	0.090 ± 0.008	—	—	0.052	0.052	—	0.0764	0.0924	1.134	89.286 ± 4.133	0.282 ± 0.009	10.714 ± 4.133
LMS	beta34	0.000	-0.073	-0.073 ± 0.007	—	0.003 ± 0.000	0.051 ± 0.004	0.008 ± 0.001	0.089 ± 0.006	—	—	-0.072	-0.072	—	0.0569	0.0915	0.970	60.714 ± 6.526	0.192 ± 0.006	39.286 ± 6.526
LMS	beta35	0.000	-0.060	-0.060 ± 0.005	—	0.002 ± 0.000	0.040 ± 0.004	0.005 ± 0.001	0.072 ± 0.005	—	—	-0.054	-0.054	—	0.0476	0.0721	1.265	80.357 ± 5.309	0.199 ± 0.007	19.643 ± 5.309
LSAM	beta31	0.316	0.357	0.041 ± 0.013	0.130 ± 0.041	0.010 ± 0.002	0.098 ± 0.009	0.011 ± 0.002	0.105 ± 0.012	0.111 ± 0.022	0.333 ± 0.039	0.353	0.037	11.86	0.0864	0.0942	1.038	96.429 ± 2.480	0.397 ± 0.015	—
LSAM	beta32	0.316	0.342	0.026 ± 0.014	0.081 ± 0.045	0.011 ± 0.002	0.107 ± 0.012	0.012 ± 0.003	0.109 ± 0.015	0.119 ± 0.027	0.344 ± 0.046	0.344	0.028	8.88	0.1091	0.1126	0.948	98.214 ± 1.770	0.397 ± 0.016	—
LSAM	beta33	0.000	0.105	0.105 ± 0.011	—	0.006 ± 0.001	0.080 ± 0.009	0.017 ± 0.003	0.131 ± 0.012	—	—	0.081	0.081	—	0.0941	0.1241	1.062	83.929 ± 4.908	0.333 ± 0.011	16.071 ± 4.908
LSAM	beta34	0.000	-0.125	-0.125 ± 0.013	—	0.010 ± 0.002	0.099 ± 0.008	0.025 ± 0.004	0.159 ± 0.012	—	—	-0.120	-0.120	—	0.0996	0.1561	0.950	73.214 ± 5.918	0.370 ± 0.018	26.786 ± 5.918
LSAM	beta35	0.000	-0.123	-0.123 ± 0.014	—	0.011 ± 0.002	0.103 ± 0.008	0.026 ± 0.004	0.160 ± 0.013	—	—	-0.113	-0.113	—	0.1196	0.1647	1.054	82.143 ± 5.118	0.426 ± 0.024	17.857 ± 5.118
QML	beta31	0.316	0.323	0.007 ± 0.007	0.022 ± 0.022	0.003 ± 0.000	0.052 ± 0.005	0.003 ± 0.001	0.052 ± 0.006	0.027 ± 0.005	0.165 ± 0.020	0.328	0.012	3.64	0.0568	0.0580	1.058	96.429 ± 2.480	0.217 ± 0.003	—
QML	beta32	0.316	0.314	-0.002 ± 0.007	-0.005 ± 0.023	0.003 ± 0.001	0.055 ± 0.005	0.003 ± 0.001	0.054 ± 0.006	0.029 ± 0.005	0.171 ± 0.020	0.310	-0.006	-2.02	0.0530	0.0534	1.013	94.643 ± 3.009	0.217 ± 0.003	—
QML	beta33	0.000	0.065	0.065 ± 0.009	—	0.004 ± 0.001	0.066 ± 0.007	0.008 ± 0.001	0.092 ± 0.009	—	—	0.059	0.059	—	0.0748	0.0954	1.116	92.857 ± 3.442	0.288 ± 0.008	7.143 ± 3.442
QML	beta34	0.000	-0.077	-0.077 ± 0.007	—	0.003 ± 0.000	0.051 ± 0.004	0.008 ± 0.001	0.092 ± 0.006	—	—	-0.074	-0.074	—	0.0597	0.0948	0.977	60.714 ± 6.526	0.195 ± 0.005	39.286 ± 6.526
QML	beta35	0.000	-0.065	-0.065 ± 0.006	—	0.002 ± 0.000	0.041 ± 0.004	0.006 ± 0.001	0.076 ± 0.005	—	—	-0.061	-0.061	—	0.0457	0.0760	1.260	82.143 ± 5.118	0.203 ± 0.006	17.857 ± 5.118
UPI	beta31	0.316	0.365	0.049 ± 0.079	0.156 ± 0.250	0.350 ± 0.096	0.591 ± 0.088	0.346 ± 0.101	0.588 ± 0.099	3.462 ± 1.011	1.861 ± 0.313	0.438	0.122	38.62	0.7469	0.7568	1.100	83.929 ± 4.908	2.551 ± 0.569	—
UPI	beta32	0.316	0.352	0.036 ± 0.083	0.113 ± 0.264	0.389 ± 0.118	0.623 ± 0.103	0.383 ± 0.118	0.619 ± 0.109	3.835 ± 1.181	1.958 ± 0.346	0.342	0.026	8.33	0.5574	0.5580	1.073	69.643 ± 6.144	2.622 ± 0.566	—
UPI	beta33	0.000	-0.077	-0.077 ± 0.049	—	0.134 ± 0.052	0.366 ± 0.079	0.137 ± 0.051	0.370 ± 0.077	—	—	-0.037	-0.037	—	0.2492	0.2519	1.117	98.214 ± 1.770	1.600 ± 0.340	1.786 ± 1.770
UPI	beta34	0.000	-0.780	-0.780 ± 0.240	—	3.224 ± 1.321	1.796 ± 0.421	3.775 ± 1.657	1.943 ± 0.482	—	—	-0.515	-0.515	—	1.2269	1.3306	0.704	94.643 ± 3.009	4.956 ± 0.974	5.357 ± 3.009
UPI	beta35	0.000	-0.750	-0.750 ± 0.213	—	2.548 ± 0.993	1.596 ± 0.360	3.066 ± 1.256	1.751 ± 0.410	—	—	-0.446	-0.446	—	1.3487	1.4207	1.081	98.214 ± 1.770	6.762 ± 1.258	1.786 ± 1.770

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S105

Simulation 1 - Condition 105: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, $N = 400$, Reliability = 0.6

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Type I (%) $\pm$ MCSE
LMS	beta31	0.316	0.318	0.002 $\pm$ 0.003	0.007 $\pm$ 0.008	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.046 $\pm$ 0.003	0.214 $\pm$ 0.007	0.315	-0.001	-0.23	0.0666	0.0666	1.002	94.783 $\pm$ 0.847	0.266 $\pm$ 0.001	—
LMS	beta32	0.316	0.317	0.001 $\pm$ 0.003	0.004 $\pm$ 0.008	0.005 $\pm$ 0.000	0.070 $\pm$ 0.002	0.005 $\pm$ 0.000	0.070 $\pm$ 0.002	0.048 $\pm$ 0.003	0.220 $\pm$ 0.008	0.315	-0.001	-0.19	0.0719	0.0719	0.965	94.783 $\pm$ 0.847	0.263 $\pm$ 0.001	—
LMS	beta33	0.000	0.016	0.016 $\pm$ 0.003	—	0.006 $\pm$ 0.000	0.079 $\pm$ 0.002	0.007 $\pm$ 0.000	0.081 $\pm$ 0.003	—	—	0.019	0.019	—	0.0755	0.0779	1.043	95.652 $\pm$ 0.776	0.324 $\pm$ 0.003	4.348 $\pm$ 0.776
LMS	beta34	0.000	-0.031	-0.031 $\pm$ 0.002	—	0.004 $\pm$ 0.000	0.065 $\pm$ 0.002	0.005 $\pm$ 0.000	0.072 $\pm$ 0.002	—	—	-0.034	-0.034	—	0.0607	0.0697	1.023	91.449 $\pm$ 1.065	0.259 $\pm$ 0.002	8.551 $\pm$ 1.065
LMS	beta35	0.000	-0.030	-0.030 $\pm$ 0.003	—	0.005 $\pm$ 0.000	0.067 $\pm$ 0.002	0.005 $\pm$ 0.000	0.074 $\pm$ 0.002	—	—	-0.036	-0.036	—	0.0688	0.0778	0.969	89.565 $\pm$ 1.164	0.255 $\pm$ 0.002	10.435 $\pm$ 1.164
LSAM	beta31	0.316	0.320	0.004 $\pm$ 0.003	0.013 $\pm$ 0.009	0.006 $\pm$ 0.000	0.077 $\pm$ 0.002	0.006 $\pm$ 0.000	0.077 $\pm$ 0.003	0.060 $\pm$ 0.003	0.245 $\pm$ 0.008	0.320	0.004	1.23	0.0771	0.0772	1.006	95.652 $\pm$ 0.776	0.305 $\pm$ 0.002	—
LSAM	beta32	0.316	0.322	0.006 $\pm$ 0.003	0.019 $\pm$ 0.010	0.006 $\pm$ 0.000	0.079 $\pm$ 0.002	0.006 $\pm$ 0.000	0.079 $\pm$ 0.003	0.063 $\pm$ 0.004	0.251 $\pm$ 0.009	0.317	0.001	0.38	0.0784	0.0784	0.981	95.942 $\pm$ 0.751	0.304 $\pm$ 0.002	—
LSAM	beta33	0.000	0.033	0.033 $\pm$ 0.003	—	0.007 $\pm$ 0.000	0.082 $\pm$ 0.002	0.008 $\pm$ 0.000	0.088 $\pm$ 0.003	—	—	0.034	0.034	—	0.0738	0.0811	1.063	95.507 $\pm$ 0.789	0.341 $\pm$ 0.003	4.493 $\pm$ 0.789
LSAM	beta34	0.000	-0.047	-0.047 $\pm$ 0.004	—	0.011 $\pm$ 0.001	0.103 $\pm$ 0.003	0.013 $\pm$ 0.001	0.113 $\pm$ 0.004	—	—	-0.051	-0.051	—	0.0897	0.1032	1.011	93.478 $\pm$ 0.940	0.409 $\pm$ 0.005	6.522 $\pm$ 0.940
LSAM	beta35	0.000	-0.050	-0.050 $\pm$ 0.004	—	0.011 $\pm$ 0.001	0.104 $\pm$ 0.003	0.013 $\pm$ 0.001	0.116 $\pm$ 0.004	—	—	-0.054	-0.054	—	0.0962	0.1102	0.974	91.304 $\pm$ 1.073	0.398 $\pm$ 0.005	8.696 $\pm$ 1.073
QML	beta31	0.316	0.318	0.002 $\pm$ 0.003	0.008 $\pm$ 0.008	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.046 $\pm$ 0.003	0.215 $\pm$ 0.007	0.316	0.000	-0.02	0.0663	0.0663	1.000	94.783 $\pm$ 0.847	0.266 $\pm$ 0.001	—
QML	beta32	0.316	0.317	0.001 $\pm$ 0.003	0.005 $\pm$ 0.008	0.005 $\pm$ 0.000	0.070 $\pm$ 0.002	0.005 $\pm$ 0.000	0.070 $\pm$ 0.002	0.049 $\pm$ 0.003	0.220 $\pm$ 0.008	0.315	-0.001	-0.21	0.0709	0.0709	0.965	94.928 $\pm$ 0.835	0.263 $\pm$ 0.001	—
QML	beta33	0.000	0.017	0.017 $\pm$ 0.003	—	0.006 $\pm$ 0.000	0.079 $\pm$ 0.002	0.007 $\pm$ 0.000	0.081 $\pm$ 0.003	—	—	0.016	0.016	—	0.0747	0.0765	1.039	95.652 $\pm$ 0.776	0.323 $\pm$ 0.003	4.348 $\pm$ 0.776
QML	beta34	0.000	-0.032	-0.032 $\pm$ 0.002	—	0.004 $\pm$ 0.000	0.065 $\pm$ 0.002	0.005 $\pm$ 0.000	0.072 $\pm$ 0.002	—	—	-0.035	-0.035	—	0.0590	0.0687	1.023	92.029 $\pm$ 1.031	0.260 $\pm$ 0.002	7.971 $\pm$ 1.031
QML	beta35	0.000	-0.031	-0.031 $\pm$ 0.003	—	0.005 $\pm$ 0.000	0.067 $\pm$ 0.002	0.005 $\pm$ 0.000	0.074 $\pm$ 0.002	—	—	-0.037	-0.037	—	0.0689	0.0783	0.974	90.000 $\pm$ 1.142	0.256 $\pm$ 0.002	10.000 $\pm$ 1.142
UPI	beta31	0.316	0.329	0.013 $\pm$ 0.004	0.042 $\pm$ 0.012	0.010 $\pm$ 0.001	0.102 $\pm$ 0.004	0.011 $\pm$ 0.001	0.103 $\pm$ 0.004	0.106 $\pm$ 0.007	0.325 $\pm$ 0.013	0.324	0.008	2.49	0.0973	0.0977	1.531	97.391 $\pm$ 0.607	0.612 $\pm$ 0.046	—
UPI	beta32	0.316	0.331	0.015 $\pm$ 0.004	0.048 $\pm$ 0.013	0.011 $\pm$ 0.001	0.105 $\pm$ 0.003	0.011 $\pm$ 0.001	0.106 $\pm$ 0.004	0.112 $\pm$ 0.007	0.335 $\pm$ 0.013	0.328	0.012	3.87	0.0937	0.0945	1.469	97.971 $\pm$ 0.537	0.604 $\pm$ 0.052	—
UPI	beta33	0.000	0.036	0.036 $\pm$ 0.004	—	0.009 $\pm$ 0.001	0.093 $\pm$ 0.003	0.010 $\pm$ 0.001	0.100 $\pm$ 0.004	—	—	0.034	0.034	—	0.0895	0.0957	1.315	97.681 $\pm$ 0.573	0.480 $\pm$ 0.016	2.319 $\pm$ 0.573
UPI	beta34	0.000	-0.056	-0.056 $\pm$ 0.007	—	0.033 $\pm$ 0.002	0.182 $\pm$ 0.007	0.036 $\pm$ 0.003	0.190 $\pm$ 0.008	—	—	-0.061	-0.061	—	0.1624	0.1734	1.357	97.536 $\pm$ 0.590	0.968 $\pm$ 0.053	2.464 $\pm$ 0.590
UPI	beta35	0.000	-0.043	-0.043 $\pm$ 0.007	—	0.030 $\pm$ 0.002	0.173 $\pm$ 0.006	0.032 $\pm$ 0.002	0.178 $\pm$ 0.007	—	—	-0.043	-0.043	—	0.1514	0.1574	1.372	97.826 $\pm$ 0.555	0.931 $\pm$ 0.055	2.174 $\pm$ 0.555

Note. MCSE = Monte Carlo Standard Error shown as $\pm$ value. Coverage, Type I Error, and Power expressed as percentages. $\pm$ 2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S106

Simulation 1 - Condition 106: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.6

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.316	-0.000 ± 0.002	-0.001 ± 0.005	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.019 ± 0.001	0.136 ± 0.004	0.316	0.000	-0.14	0.0448	0.0448	0.978	94.772 ± 0.815	0.165 ± 0.000	—
LMS	beta32	0.316	0.316	0.000 ± 0.002	0.001 ± 0.005	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.018 ± 0.001	0.133 ± 0.004	0.316	0.000	-0.09	0.0407	0.0407	0.999	94.504 ± 0.834	0.165 ± 0.000	—
LMS	beta33	0.000	0.020	0.020 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	—	—	0.024	0.024	—	0.0495	0.0550	1.015	93.164 ± 0.924	0.202 ± 0.002	6.836 ± 0.924
LMS	beta34	0.000	-0.032	-0.032 ± 0.002	—	0.002 ± 0.000	0.041 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	—	—	-0.034	-0.034	—	0.0400	0.0523	0.981	87.131 ± 1.226	0.160 ± 0.001	12.869 ± 1.226
LMS	beta35	0.000	-0.032	-0.032 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	—	—	-0.032	-0.032	—	0.0396	0.0512	1.018	86.863 ± 1.237	0.161 ± 0.001	13.137 ± 1.237
LSAM	beta31	0.316	0.318	0.002 ± 0.002	0.008 ± 0.006	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.023 ± 0.001	0.152 ± 0.005	0.318	0.002	0.67	0.0502	0.0503	0.986	94.906 ± 0.805	0.185 ± 0.001	—
LSAM	beta32	0.316	0.320	0.004 ± 0.002	0.012 ± 0.006	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.002	0.023 ± 0.001	0.152 ± 0.005	0.318	0.002	0.76	0.0488	0.0489	0.988	95.174 ± 0.785	0.185 ± 0.001	—
LSAM	beta33	0.000	0.035	0.035 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	—	—	0.038	0.038	—	0.0487	0.0616	1.015	89.678 ± 1.114	0.203 ± 0.001	10.322 ± 1.114
LSAM	beta34	0.000	-0.049	-0.049 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	—	—	-0.053	-0.053	—	0.0599	0.0799	0.989	85.925 ± 1.273	0.238 ± 0.002	14.075 ± 1.273
LSAM	beta35	0.000	-0.050	-0.050 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	—	—	-0.050	-0.050	—	0.0563	0.0755	0.989	86.461 ± 1.253	0.238 ± 0.002	13.539 ± 1.253
QML	beta31	0.316	0.316	-0.000 ± 0.002	-0.001 ± 0.005	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.019 ± 0.001	0.136 ± 0.004	0.316	0.000	-0.14	0.0448	0.0448	0.977	94.772 ± 0.815	0.165 ± 0.000	—
QML	beta32	0.316	0.317	0.001 ± 0.002	0.002 ± 0.005	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.018 ± 0.001	0.134 ± 0.004	0.316	0.000	0.01	0.0409	0.0409	0.997	94.504 ± 0.834	0.165 ± 0.000	—
QML	beta33	0.000	0.020	0.020 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	—	—	0.024	0.024	—	0.0498	0.0553	1.011	93.298 ± 0.916	0.202 ± 0.001	6.702 ± 0.916
QML	beta34	0.000	-0.032	-0.032 ± 0.002	—	0.002 ± 0.000	0.042 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	—	—	-0.034	-0.034	—	0.0404	0.0530	0.985	86.997 ± 1.231	0.160 ± 0.001	13.003 ± 1.231
QML	beta35	0.000	-0.032	-0.032 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	—	—	-0.033	-0.033	—	0.0395	0.0515	1.006	86.863 ± 1.237	0.160 ± 0.001	13.137 ± 1.237
UPI	beta31	0.316	0.326	0.010 ± 0.003	0.032 ± 0.009	0.006 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.075 ± 0.003	0.056 ± 0.004	0.237 ± 0.009	0.321	0.005	1.73	0.0721	0.0723	1.361	98.525 ± 0.441	0.396 ± 0.024	—
UPI	beta32	0.316	0.328	0.012 ± 0.003	0.038 ± 0.009	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.077 ± 0.003	0.060 ± 0.004	0.244 ± 0.009	0.325	0.009	2.95	0.0707	0.0713	1.293	97.855 ± 0.530	0.386 ± 0.018	—
UPI	beta33	0.000	0.049	0.049 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	—	—	0.050	0.050	—	0.0632	0.0807	1.200	94.236 ± 0.853	0.287 ± 0.007	5.764 ± 0.853
UPI	beta34	0.000	-0.066	-0.066 ± 0.004	—	0.014 ± 0.001	0.119 ± 0.004	0.019 ± 0.001	0.136 ± 0.005	—	—	-0.060	-0.060	—	0.1059	0.1217	1.252	96.649 ± 0.659	0.586 ± 0.033	3.351 ± 0.659
UPI	beta35	0.000	-0.067	-0.067 ± 0.004	—	0.014 ± 0.001	0.116 ± 0.004	0.018 ± 0.001	0.134 ± 0.005	—	—	-0.059	-0.059	—	0.1010	0.1169	1.279	95.576 ± 0.753	0.584 ± 0.028	4.424 ± 0.753

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S107

Simulation 1 - Condition 107: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.316	-0.000 ± 0.002	-0.001 ± 0.006	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.034 ± 0.001	0.184 ± 0.005	0.314	-0.002	-0.67	0.0577	0.0578	0.950	92.893 ± 0.813	0.216 ± 0.001	—
LMS	beta32	0.316	0.315	-0.001 ± 0.002	-0.003 ± 0.005	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.030 ± 0.001	0.173 ± 0.005	0.316	0.000	0.07	0.0558	0.0558	1.006	95.295 ± 0.670	0.215 ± 0.000	—
LMS	beta33	0.000	0.004	0.004 ± 0.002	—	0.004 ± 0.000	0.066 ± 0.001	0.004 ± 0.000	0.066 ± 0.002	—	—	0.004	0.004	—	0.0679	0.0680	0.953	93.193 ± 0.797	0.246 ± 0.002	6.807 ± 0.797
LMS	beta34	0.000	-0.009	-0.009 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	—	—	-0.008	-0.008	—	0.0578	0.0584	1.001	93.894 ± 0.758	0.226 ± 0.001	6.106 ± 0.758
LMS	beta35	0.000	-0.007	-0.007 ± 0.002	—	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	—	—	-0.006	-0.006	—	0.0545	0.0548	1.034	95.195 ± 0.677	0.224 ± 0.001	4.805 ± 0.677
LSAM	beta31	0.316	0.317	0.001 ± 0.002	0.002 ± 0.006	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.036 ± 0.002	0.189 ± 0.005	0.315	-0.001	-0.27	0.0598	0.0598	0.953	92.793 ± 0.818	0.223 ± 0.001	—
LSAM	beta32	0.316	0.315	-0.001 ± 0.002	-0.002 ± 0.006	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.031 ± 0.001	0.177 ± 0.005	0.318	0.002	0.52	0.0577	0.0578	1.009	95.295 ± 0.670	0.222 ± 0.001	—
LSAM	beta33	0.000	0.008	0.008 ± 0.002	—	0.004 ± 0.000	0.066 ± 0.001	0.004 ± 0.000	0.066 ± 0.002	—	—	0.008	0.008	—	0.0669	0.0674	0.945	93.193 ± 0.797	0.244 ± 0.001	6.807 ± 0.797
LSAM	beta34	0.000	-0.011	-0.011 ± 0.002	—	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	—	—	-0.008	-0.008	—	0.0720	0.0725	0.991	94.294 ± 0.734	0.278 ± 0.001	5.706 ± 0.734
LSAM	beta35	0.000	-0.008	-0.008 ± 0.002	—	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	—	—	-0.007	-0.007	—	0.0665	0.0669	1.021	95.596 ± 0.649	0.275 ± 0.001	4.404 ± 0.649
QML	beta31	0.316	0.316	-0.000 ± 0.002	-0.001 ± 0.006	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.034 ± 0.001	0.184 ± 0.005	0.314	-0.002	-0.75	0.0576	0.0577	0.948	92.893 ± 0.813	0.216 ± 0.001	—
QML	beta32	0.316	0.315	-0.001 ± 0.002	-0.003 ± 0.005	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.030 ± 0.001	0.173 ± 0.005	0.316	0.000	0.06	0.0559	0.0559	1.007	95.195 ± 0.677	0.216 ± 0.001	—
QML	beta33	0.000	0.004	0.004 ± 0.002	—	0.004 ± 0.000	0.066 ± 0.001	0.004 ± 0.000	0.066 ± 0.002	—	—	0.004	0.004	—	0.0674	0.0675	0.949	93.293 ± 0.791	0.245 ± 0.002	6.707 ± 0.791
QML	beta34	0.000	-0.009	-0.009 ± 0.002	—	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	—	—	-0.008	-0.008	—	0.0575	0.0581	1.000	93.994 ± 0.752	0.225 ± 0.001	6.006 ± 0.752
QML	beta35	0.000	-0.007	-0.007 ± 0.002	—	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	—	—	-0.006	-0.006	—	0.0542	0.0545	1.033	95.195 ± 0.677	0.224 ± 0.001	4.805 ± 0.677
UPI	beta31	0.316	0.317	0.001 ± 0.002	0.003 ± 0.006	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.036 ± 0.002	0.189 ± 0.005	0.316	0.000	-0.01	0.0605	0.0605	0.941	92.793 ± 0.818	0.221 ± 0.001	—
UPI	beta32	0.316	0.315	-0.001 ± 0.002	-0.002 ± 0.006	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.031 ± 0.001	0.177 ± 0.005	0.317	0.001	0.20	0.0576	0.0576	1.004	95.596 ± 0.649	0.220 ± 0.001	—
UPI	beta33	0.000	0.004	0.004 ± 0.002	—	0.004 ± 0.000	0.067 ± 0.001	0.005 ± 0.000	0.067 ± 0.002	—	—	0.004	0.004	—	0.0679	0.0680	0.937	93.193 ± 0.797	0.246 ± 0.001	6.807 ± 0.797
UPI	beta34	0.000	-0.006	-0.006 ± 0.002	—	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	—	—	-0.005	-0.005	—	0.0748	0.0750	1.003	95.195 ± 0.677	0.294 ± 0.002	4.805 ± 0.677
UPI	beta35	0.000	-0.001	-0.001 ± 0.002	—	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	—	—	-0.001	-0.001	—	0.0703	0.0703	1.017	95.395 ± 0.663	0.289 ± 0.002	4.605 ± 0.663

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S108

Simulation 1 - Condition 108: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.8

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.315	-0.001 ± 0.001	-0.004 ± 0.003	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.012 ± 0.001	0.110 ± 0.003	0.315	-0.001	-0.40	0.0348	0.0348	0.996	94.800 ± 0.702	0.136 ± 0.000	—
LMS	beta32	0.316	0.318	0.002 ± 0.001	0.006 ± 0.003	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.011 ± 0.001	0.106 ± 0.003	0.318	0.002	0.65	0.0324	0.0325	1.035	95.300 ± 0.669	0.136 ± 0.000	—
LMS	beta33	0.000	0.004	0.004 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	0.005	0.005	—	0.0408	0.0411	0.984	93.900 ± 0.757	0.154 ± 0.001	6.100 ± 0.757
LMS	beta34	0.000	-0.010	-0.010 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.010	-0.010	—	0.0344	0.0357	1.016	94.600 ± 0.715	0.142 ± 0.001	5.400 ± 0.715
LMS	beta35	0.000	-0.010	-0.010 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.010	-0.010	—	0.0391	0.0403	0.973	93.100 ± 0.801	0.142 ± 0.001	6.900 ± 0.801
LSAM	beta31	0.316	0.315	-0.001 ± 0.001	-0.003 ± 0.004	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.013 ± 0.001	0.113 ± 0.003	0.316	0.000	0.06	0.0360	0.0360	0.997	95.000 ± 0.689	0.139 ± 0.000	—
LSAM	beta32	0.316	0.318	0.002 ± 0.001	0.008 ± 0.003	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.012 ± 0.001	0.108 ± 0.003	0.318	0.002	0.75	0.0341	0.0342	1.035	95.400 ± 0.662	0.139 ± 0.000	—
LSAM	beta33	0.000	0.009	0.009 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	0.009	0.009	—	0.0405	0.0416	0.977	93.700 ± 0.768	0.153 ± 0.000	6.300 ± 0.768
LSAM	beta34	0.000	-0.012	-0.012 ± 0.001	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	-0.011	-0.011	—	0.0418	0.0432	1.009	94.700 ± 0.708	0.173 ± 0.000	5.300 ± 0.708
LSAM	beta35	0.000	-0.011	-0.011 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	-0.013	-0.013	—	0.0480	0.0496	0.961	93.200 ± 0.796	0.173 ± 0.000	6.800 ± 0.796
QML	beta31	0.316	0.315	-0.001 ± 0.001	-0.004 ± 0.003	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.012 ± 0.001	0.110 ± 0.003	0.315	-0.001	-0.42	0.0349	0.0349	0.997	94.900 ± 0.696	0.136 ± 0.000	—
QML	beta32	0.316	0.318	0.002 ± 0.001	0.006 ± 0.003	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.011 ± 0.001	0.106 ± 0.003	0.318	0.002	0.64	0.0324	0.0324	1.033	95.400 ± 0.662	0.135 ± 0.000	—
QML	beta33	0.000	0.004	0.004 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	0.005	0.005	—	0.0407	0.0410	0.981	93.900 ± 0.757	0.154 ± 0.000	6.100 ± 0.757
QML	beta34	0.000	-0.010	-0.010 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.010	-0.010	—	0.0342	0.0356	1.013	94.800 ± 0.702	0.141 ± 0.000	5.200 ± 0.702
QML	beta35	0.000	-0.010	-0.010 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.010	-0.010	—	0.0389	0.0402	0.965	93.100 ± 0.801	0.140 ± 0.000	6.900 ± 0.801
UPI	beta31	0.316	0.315	-0.001 ± 0.001	-0.002 ± 0.004	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.013 ± 0.001	0.112 ± 0.003	0.316	0.000	0.00	0.0362	0.0362	0.996	94.800 ± 0.702	0.138 ± 0.000	—
UPI	beta32	0.316	0.318	0.002 ± 0.001	0.008 ± 0.003	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.012 ± 0.001	0.109 ± 0.003	0.318	0.002	0.76	0.0339	0.0340	1.030	95.100 ± 0.683	0.138 ± 0.000	—
UPI	beta33	0.000	0.005	0.005 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	0.004	0.004	—	0.0405	0.0407	0.975	93.500 ± 0.780	0.154 ± 0.000	6.500 ± 0.780
UPI	beta34	0.000	-0.006	-0.006 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	-0.005	-0.005	—	0.0449	0.0452	1.008	94.900 ± 0.696	0.180 ± 0.001	5.100 ± 0.696
UPI	beta35	0.000	-0.005	-0.005 ± 0.001	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	-0.006	-0.006	—	0.0488	0.0492	0.968	94.600 ± 0.715	0.179 ± 0.001	5.400 ± 0.715

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S109

Simulation 1 - Condition 109: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.4

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.315	-0.001 ± 0.004	-0.003 ± 0.011	0.010 ± 0.001	0.099 ± 0.003	0.010 ± 0.001	0.099 ± 0.003	0.099 ± 0.006	0.315 ± 0.010	0.311	-0.005	-1.68	0.1062	0.1063	1.088	95.392 ± 0.740	0.424 ± 0.012	—
LMS	beta32	0.316	0.313	-0.003 ± 0.003	-0.008 ± 0.010	0.008 ± 0.000	0.091 ± 0.002	0.008 ± 0.000	0.091 ± 0.003	0.083 ± 0.004	0.288 ± 0.009	0.314	-0.002	-0.71	0.0941	0.0941	1.170	97.260 ± 0.576	0.418 ± 0.009	—
LMS	beta33	0.000	-0.095	-0.095 ± 0.008	—	0.053 ± 0.005	0.231 ± 0.010	0.062 ± 0.006	0.250 ± 0.013	—	—	-0.059	-0.059	—	0.1683	0.1784	1.053	95.890 ± 0.701	0.954 ± 0.065	4.110 ± 0.701
LMS	beta34	0.000	0.042	0.042 ± 0.005	—	0.020 ± 0.002	0.141 ± 0.006	0.022 ± 0.002	0.147 ± 0.008	—	—	0.023	0.023	—	0.1106	0.1129	1.016	95.641 ± 0.721	0.563 ± 0.035	4.359 ± 0.721
LMS	beta35	0.000	0.050	0.050 ± 0.005	—	0.020 ± 0.002	0.140 ± 0.006	0.022 ± 0.002	0.148 ± 0.007	—	—	0.031	0.031	—	0.1102	0.1145	1.079	95.392 ± 0.740	0.591 ± 0.042	4.608 ± 0.740
LSAM	beta31	0.316	0.322	0.006 ± 0.004	0.019 ± 0.012	0.011 ± 0.001	0.107 ± 0.003	0.011 ± 0.001	0.107 ± 0.003	0.115 ± 0.006	0.339 ± 0.011	0.316	0.000	0.13	0.1113	0.1113	1.041	97.011 ± 0.601	0.437 ± 0.004	—
LSAM	beta32	0.316	0.320	0.004 ± 0.004	0.013 ± 0.011	0.010 ± 0.001	0.102 ± 0.003	0.010 ± 0.001	0.102 ± 0.003	0.104 ± 0.006	0.323 ± 0.011	0.318	0.002	0.74	0.0986	0.0987	1.103	97.385 ± 0.563	0.441 ± 0.005	—
LSAM	beta33	0.000	0.013	0.013 ± 0.005	—	0.022 ± 0.001	0.147 ± 0.005	0.022 ± 0.001	0.147 ± 0.005	—	—	0.001	0.001	—	0.1414	0.1414	1.131	98.381 ± 0.445	0.650 ± 0.025	1.619 ± 0.445
LSAM	beta34	0.000	-0.011	-0.011 ± 0.004	—	0.011 ± 0.001	0.105 ± 0.003	0.011 ± 0.001	0.105 ± 0.004	—	—	-0.007	-0.007	—	0.0991	0.0993	1.104	98.257 ± 0.462	0.453 ± 0.015	1.743 ± 0.462
LSAM	beta35	0.000	0.000	-0.000 ± 0.004	—	0.012 ± 0.001	0.111 ± 0.004	0.012 ± 0.001	0.110 ± 0.004	—	—	0.005	0.005	—	0.1047	0.1048	1.055	98.007 ± 0.493	0.457 ± 0.013	1.993 ± 0.493
QML	beta31	0.316	0.315	-0.001 ± 0.003	-0.003 ± 0.010	0.009 ± 0.000	0.093 ± 0.002	0.009 ± 0.000	0.093 ± 0.003	0.087 ± 0.004	0.295 ± 0.009	0.313	-0.003	-1.01	0.0973	0.0974	1.023	95.517 ± 0.730	0.374 ± 0.003	—
QML	beta32	0.316	0.312	-0.004 ± 0.003	-0.013 ± 0.010	0.007 ± 0.000	0.085 ± 0.002	0.007 ± 0.000	0.085 ± 0.003	0.073 ± 0.004	0.270 ± 0.008	0.311	-0.005	-1.65	0.0890	0.0892	1.113	97.136 ± 0.589	0.372 ± 0.003	—
QML	beta33	0.000	-0.040	-0.040 ± 0.004	—	0.015 ± 0.001	0.122 ± 0.004	0.016 ± 0.001	0.128 ± 0.004	—	—	-0.049	-0.049	—	0.1274	0.1364	1.111	96.513 ± 0.647	0.531 ± 0.009	3.487 ± 0.647
QML	beta34	0.000	0.004	0.004 ± 0.003	—	0.007 ± 0.000	0.083 ± 0.003	0.007 ± 0.000	0.083 ± 0.003	—	—	0.005	0.005	—	0.0856	0.0858	1.078	96.513 ± 0.647	0.350 ± 0.008	3.487 ± 0.647
QML	beta35	0.000	0.012	0.012 ± 0.003	—	0.007 ± 0.000	0.084 ± 0.003	0.007 ± 0.000	0.084 ± 0.003	—	—	0.014	0.014	—	0.0839	0.0850	1.068	96.015 ± 0.690	0.350 ± 0.007	3.985 ± 0.690
UPI	beta31	0.316	0.322	0.006 ± 0.004	0.017 ± 0.012	0.012 ± 0.001	0.107 ± 0.003	0.012 ± 0.001	0.107 ± 0.003	0.115 ± 0.006	0.340 ± 0.010	0.318	0.002	0.58	0.1174	0.1174	1.107	97.509 ± 0.550	0.465 ± 0.006	—
UPI	beta32	0.316	0.322	0.006 ± 0.004	0.019 ± 0.011	0.010 ± 0.001	0.102 ± 0.003	0.010 ± 0.001	0.102 ± 0.003	0.104 ± 0.006	0.323 ± 0.011	0.321	0.005	1.48	0.0993	0.0995	1.166	97.758 ± 0.522	0.466 ± 0.005	—
UPI	beta33	0.000	0.016	0.016 ± 0.005	—	0.024 ± 0.002	0.155 ± 0.005	0.024 ± 0.002	0.155 ± 0.006	—	—	0.001	0.001	—	0.1468	0.1468	1.247	99.502 ± 0.248	0.756 ± 0.019	0.498 ± 0.248
UPI	beta34	0.000	-0.008	-0.008 ± 0.004	—	0.013 ± 0.001	0.114 ± 0.004	0.013 ± 0.001	0.114 ± 0.004	—	—	-0.002	-0.002	—	0.0995	0.0995	1.287	99.128 ± 0.328	0.574 ± 0.016	0.872 ± 0.328
UPI	beta35	0.000	-0.004	-0.004 ± 0.004	—	0.015 ± 0.001	0.122 ± 0.004	0.015 ± 0.001	0.122 ± 0.005	—	—	0.004	0.004	—	0.1065	0.1066	1.157	99.502 ± 0.248	0.552 ± 0.013	0.498 ± 0.248

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S110

Simulation 1 - Condition 110: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.4

Method	Param.	True	Mean-based										Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE	
LMS	beta31	0.316	0.318	0.002 ± 0.002	0.008 ± 0.006	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.036 ± 0.002	0.190 ± 0.006	0.315	-0.001	-0.41	0.0594	0.0594	1.051	96.272 ± 0.627	0.247 ± 0.003		
LMS	beta32	0.316	0.313	-0.003 ± 0.002	-0.009 ± 0.007	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	0.039 ± 0.002	0.197 ± 0.006	0.313	-0.003	-0.98	0.0659	0.0659	1.019	96.162 ± 0.636	0.248 ± 0.003	—	
LMS	beta33	0.000	-0.044	-0.044 ± 0.003	—	0.010 ± 0.001	0.101 ± 0.003	0.012 ± 0.001	0.110 ± 0.004	—	—	-0.043	-0.043	—	0.1028	0.1115	1.219	97.478 ± 0.519	0.482 ± 0.021	2.522 ± 0.519	
LMS	beta34	0.000	0.018	0.018 ± 0.002	—	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.066 ± 0.003	—	—	0.014	0.014	—	0.0621	0.0637	1.172	97.588 ± 0.508	0.294 ± 0.012	2.412 ± 0.508	
LMS	beta35	0.000	0.017	0.017 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	—	—	0.015	0.015	—	0.0615	0.0633	1.242	97.588 ± 0.508	0.295 ± 0.011	2.412 ± 0.508	
LSAM	beta31	0.316	0.322	0.006 ± 0.002	0.020 ± 0.007	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	0.039 ± 0.002	0.199 ± 0.006	0.319	0.003	0.98	0.0621	0.0622	1.018	95.504 ± 0.686	0.249 ± 0.001	—	
LSAM	beta32	0.316	0.316	0.000 ± 0.002	0.001 ± 0.007	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	0.041 ± 0.002	0.202 ± 0.006	0.317	0.001	0.19	0.0647	0.0647	0.991	95.285 ± 0.702	0.249 ± 0.001	—	
LSAM	beta33	0.000	0.009	0.009 ± 0.003	—	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.081 ± 0.003	—	—	-0.005	-0.005	—	0.0843	0.0844	1.041	96.272 ± 0.627	0.331 ± 0.003	3.728 ± 0.627	
LSAM	beta34	0.000	-0.006	-0.006 ± 0.002	—	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	—	—	-0.002	-0.002	—	0.0577	0.0577	1.032	96.601 ± 0.600	0.233 ± 0.002	3.399 ± 0.600	
LSAM	beta35	0.000	-0.003	-0.003 ± 0.002	—	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	—	—	0.000	0.000	—	0.0584	0.0584	1.073	97.588 ± 0.508	0.233 ± 0.002	2.412 ± 0.508	
QML	beta31	0.316	0.317	0.001 ± 0.002	0.005 ± 0.006	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.033 ± 0.002	0.182 ± 0.005	0.315	-0.001	-0.38	0.0575	0.0575	1.015	95.395 ± 0.694	0.229 ± 0.001	—	
QML	beta32	0.316	0.312	-0.004 ± 0.002	-0.012 ± 0.006	0.004 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	0.035 ± 0.002	0.188 ± 0.006	0.314	-0.002	-0.68	0.0600	0.0601	0.984	94.846 ± 0.732	0.229 ± 0.001	—	
QML	beta33	0.000	-0.034	-0.034 ± 0.003	—	0.006 ± 0.000	0.078 ± 0.002	0.007 ± 0.000	0.085 ± 0.002	—	—	-0.042	-0.042	—	0.0839	0.0937	1.032	94.408 ± 0.761	0.317 ± 0.003	5.592 ± 0.761	
QML	beta34	0.000	0.003	0.003 ± 0.002	—	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	—	—	0.006	0.006	—	0.0514	0.0517	1.052	96.162 ± 0.636	0.207 ± 0.002	3.838 ± 0.636	
QML	beta35	0.000	0.004	0.004 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	—	—	0.005	0.005	—	0.0528	0.0530	1.080	96.820 ± 0.581	0.206 ± 0.002	3.180 ± 0.581	
UPI	beta31	0.316	0.323	0.007 ± 0.002	0.021 ± 0.007	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	0.040 ± 0.002	0.200 ± 0.006	0.318	0.002	0.70	0.0613	0.0613	1.029	95.614 ± 0.678	0.254 ± 0.001	—	
UPI	beta32	0.316	0.316	0.000 ± 0.002	0.001 ± 0.007	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	0.041 ± 0.002	0.203 ± 0.006	0.318	0.002	0.60	0.0643	0.0643	1.003	95.504 ± 0.686	0.253 ± 0.001	—	
UPI	beta33	0.000	0.010	0.010 ± 0.003	—	0.007 ± 0.000	0.085 ± 0.002	0.007 ± 0.000	0.086 ± 0.003	—	—	-0.003	-0.003	—	0.0834	0.0835	1.051	97.149 ± 0.551	0.351 ± 0.004	2.851 ± 0.551	
UPI	beta34	0.000	-0.007	-0.007 ± 0.002	—	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	—	—	-0.003	-0.003	—	0.0582	0.0583	1.015	97.697 ± 0.497	0.254 ± 0.003	2.303 ± 0.497	
UPI	beta35	0.000	-0.004	-0.004 ± 0.002	—	0.004 ± 0.000	0.059 ± 0.002	0.004 ± 0.000	0.059 ± 0.002	—	—	0.000	0.000	—	0.0581	0.0581	1.103	98.575 ± 0.393	0.257 ± 0.004	1.425 ± 0.393	

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S111

Simulation 1 - Condition 111: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.6

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.318	0.002 ± 0.002	0.006 ± 0.007	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.047 ± 0.002	0.218 ± 0.006	0.318	0.002	0.70	0.0678	0.0679	1.008	95.306 ± 0.676	0.272 ± 0.001	—
LMS	beta32	0.316	0.316	0.000 ± 0.002	0.001 ± 0.007	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.051 ± 0.002	0.226 ± 0.007	0.311	-0.005	-1.55	0.0699	0.0701	0.970	94.694 ± 0.716	0.271 ± 0.001	—
LMS	beta33	0.000	-0.024	-0.024 ± 0.003	—	0.008 ± 0.000	0.090 ± 0.002	0.009 ± 0.000	0.093 ± 0.003	—	—	-0.018	-0.018	—	0.0871	0.0890	1.048	96.020 ± 0.624	0.368 ± 0.008	3.980 ± 0.624
LMS	beta34	0.000	0.008	0.008 ± 0.002	—	0.003 ± 0.000	0.059 ± 0.002	0.004 ± 0.000	0.059 ± 0.002	—	—	0.004	0.004	—	0.0562	0.0564	1.044	95.204 ± 0.683	0.241 ± 0.004	4.796 ± 0.683
LMS	beta35	0.000	0.011	0.011 ± 0.002	—	0.003 ± 0.000	0.059 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	—	—	0.006	0.006	—	0.0534	0.0538	1.055	95.000 ± 0.696	0.242 ± 0.005	5.000 ± 0.696
LSAM	beta31	0.316	0.321	0.005 ± 0.002	0.017 ± 0.007	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.052 ± 0.003	0.228 ± 0.007	0.324	0.008	2.40	0.0704	0.0708	0.990	94.796 ± 0.710	0.279 ± 0.001	—
LSAM	beta32	0.316	0.319	0.003 ± 0.002	0.009 ± 0.008	0.006 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.074 ± 0.002	0.055 ± 0.003	0.235 ± 0.007	0.316	0.000	-0.06	0.0724	0.0724	0.954	94.490 ± 0.729	0.277 ± 0.001	—
LSAM	beta33	0.000	0.004	0.004 ± 0.003	—	0.007 ± 0.000	0.086 ± 0.002	0.007 ± 0.000	0.086 ± 0.003	—	—	0.002	0.002	—	0.0835	0.0835	0.982	95.204 ± 0.683	0.331 ± 0.003	4.796 ± 0.683
LSAM	beta34	0.000	-0.004	-0.004 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	—	—	-0.002	-0.002	—	0.0583	0.0584	0.993	95.714 ± 0.647	0.234 ± 0.002	4.286 ± 0.647
LSAM	beta35	0.000	-0.001	-0.001 ± 0.002	—	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	—	—	-0.001	-0.001	—	0.0569	0.0569	1.007	95.306 ± 0.676	0.232 ± 0.002	4.694 ± 0.676
QML	beta31	0.316	0.318	0.002 ± 0.002	0.005 ± 0.007	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.046 ± 0.002	0.215 ± 0.006	0.318	0.002	0.78	0.0663	0.0663	1.003	95.000 ± 0.696	0.268 ± 0.001	—
QML	beta32	0.316	0.316	-0.000 ± 0.002	-0.001 ± 0.007	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.049 ± 0.002	0.222 ± 0.006	0.313	-0.003	-1.04	0.0689	0.0690	0.968	94.388 ± 0.735	0.267 ± 0.001	—
QML	beta33	0.000	-0.021	-0.021 ± 0.003	—	0.007 ± 0.000	0.083 ± 0.002	0.007 ± 0.000	0.085 ± 0.002	—	—	-0.020	-0.020	—	0.0840	0.0863	1.023	95.306 ± 0.676	0.331 ± 0.004	4.694 ± 0.676
QML	beta34	0.000	0.003	0.003 ± 0.002	—	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	—	—	0.001	0.001	—	0.0535	0.0536	1.029	94.694 ± 0.716	0.223 ± 0.002	5.306 ± 0.716
QML	beta35	0.000	0.006	0.006 ± 0.002	—	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	—	—	0.004	0.004	—	0.0534	0.0536	1.044	95.204 ± 0.683	0.222 ± 0.002	4.796 ± 0.683
UPI	beta31	0.316	0.321	0.005 ± 0.002	0.017 ± 0.007	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.053 ± 0.003	0.229 ± 0.007	0.321	0.005	1.67	0.0737	0.0739	0.990	95.000 ± 0.696	0.280 ± 0.001	—
UPI	beta32	0.316	0.319	0.003 ± 0.002	0.011 ± 0.008	0.006 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.074 ± 0.002	0.056 ± 0.003	0.236 ± 0.007	0.316	0.000	-0.09	0.0730	0.0730	0.957	94.694 ± 0.716	0.279 ± 0.001	—
UPI	beta33	0.000	0.005	0.005 ± 0.003	—	0.008 ± 0.000	0.089 ± 0.002	0.008 ± 0.000	0.089 ± 0.003	—	—	0.000	0.000	—	0.0837	0.0837	0.983	95.714 ± 0.647	0.345 ± 0.003	4.286 ± 0.647
UPI	beta34	0.000	-0.004	-0.004 ± 0.002	—	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	—	—	-0.003	-0.003	—	0.0573	0.0574	0.990	95.612 ± 0.654	0.248 ± 0.003	4.388 ± 0.654
UPI	beta35	0.000	-0.001	-0.001 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	—	—	-0.001	-0.001	—	0.0574	0.0574	1.030	96.020 ± 0.624	0.246 ± 0.002	3.980 ± 0.624

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S112

Simulation 1 - Condition 112: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.6

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Type I (%) $\pm$ MCSE
LMS	beta31	0.316	0.315	-0.001 $\pm$ 0.001	-0.003 $\pm$ 0.004	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.019 $\pm$ 0.001	0.136 $\pm$ 0.004	0.314	-0.002	-0.62	0.0444	0.0444	0.996	94.679 $\pm$ 0.711	0.168 $\pm$ 0.001	—
LMS	beta32	0.316	0.315	-0.001 $\pm$ 0.001	-0.004 $\pm$ 0.004	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.018 $\pm$ 0.001	0.134 $\pm$ 0.004	0.314	-0.002	-0.75	0.0410	0.0410	1.016	95.482 $\pm$ 0.658	0.169 $\pm$ 0.001	—
LMS	beta33	0.000	-0.023	-0.023 $\pm$ 0.002	—	0.003 $\pm$ 0.000	0.057 $\pm$ 0.001	0.004 $\pm$ 0.000	0.062 $\pm$ 0.002	—	—	-0.020	-0.020	—	0.0513	0.0549	0.997	94.980 $\pm$ 0.692	0.224 $\pm$ 0.003	5.020 $\pm$ 0.692
LMS	beta34	0.000	0.007	0.007 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	—	—	0.004	0.004	—	0.0361	0.0364	1.024	96.185 $\pm$ 0.607	0.148 $\pm$ 0.002	3.815 $\pm$ 0.607
LMS	beta35	0.000	0.008	0.008 $\pm$ 0.001	—	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	—	—	0.004	0.004	—	0.0356	0.0357	0.965	96.185 $\pm$ 0.607	0.147 $\pm$ 0.002	3.815 $\pm$ 0.607
LSAM	beta31	0.316	0.318	0.002 $\pm$ 0.001	0.006 $\pm$ 0.004	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.020 $\pm$ 0.001	0.142 $\pm$ 0.004	0.316	0.000	0.07	0.0454	0.0454	0.990	95.080 $\pm$ 0.685	0.174 $\pm$ 0.000	—
LSAM	beta32	0.316	0.316	0.000 $\pm$ 0.001	0.001 $\pm$ 0.004	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.019 $\pm$ 0.001	0.139 $\pm$ 0.004	0.317	0.001	0.17	0.0427	0.0427	1.010	95.683 $\pm$ 0.644	0.174 $\pm$ 0.000	—
LSAM	beta33	0.000	0.000	0.000 $\pm$ 0.002	—	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	—	—	0.000	0.000	—	0.0484	0.0484	0.995	94.076 $\pm$ 0.748	0.201 $\pm$ 0.001	5.924 $\pm$ 0.748
LSAM	beta34	0.000	-0.001	-0.001 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	—	—	-0.001	-0.001	—	0.0356	0.0356	0.987	95.080 $\pm$ 0.685	0.142 $\pm$ 0.001	4.920 $\pm$ 0.685
LSAM	beta35	0.000	0.001	0.001 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	—	—	0.000	0.000	—	0.0362	0.0362	0.964	94.277 $\pm$ 0.736	0.143 $\pm$ 0.001	5.723 $\pm$ 0.736
QML	beta31	0.316	0.315	-0.001 $\pm$ 0.001	-0.004 $\pm$ 0.004	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.018 $\pm$ 0.001	0.135 $\pm$ 0.004	0.314	-0.002	-0.64	0.0441	0.0441	1.000	94.578 $\pm$ 0.718	0.168 $\pm$ 0.001	—
QML	beta32	0.316	0.314	-0.002 $\pm$ 0.001	-0.006 $\pm$ 0.004	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.018 $\pm$ 0.001	0.134 $\pm$ 0.004	0.313	-0.003	-0.92	0.0409	0.0410	1.024	95.482 $\pm$ 0.658	0.169 $\pm$ 0.002	—
QML	beta33	0.000	-0.023	-0.023 $\pm$ 0.002	—	0.003 $\pm$ 0.000	0.054 $\pm$ 0.001	0.003 $\pm$ 0.000	0.058 $\pm$ 0.002	—	—	-0.021	-0.021	—	0.0493	0.0537	1.009	93.072 $\pm$ 0.805	0.212 $\pm$ 0.003	6.928 $\pm$ 0.805
QML	beta34	0.000	0.004	0.004 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	—	—	0.002	0.002	—	0.0348	0.0349	1.026	95.181 $\pm$ 0.679	0.142 $\pm$ 0.002	4.819 $\pm$ 0.679
QML	beta35	0.000	0.005	0.005 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	—	—	0.001	0.001	—	0.0351	0.0351	0.988	95.582 $\pm$ 0.651	0.142 $\pm$ 0.001	4.418 $\pm$ 0.651
UPI	beta31	0.316	0.318	0.002 $\pm$ 0.001	0.006 $\pm$ 0.004	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.020 $\pm$ 0.001	0.141 $\pm$ 0.004	0.317	0.001	0.24	0.0455	0.0455	0.993	94.980 $\pm$ 0.692	0.174 $\pm$ 0.000	—
UPI	beta32	0.316	0.317	0.001 $\pm$ 0.001	0.002 $\pm$ 0.004	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.019 $\pm$ 0.001	0.139 $\pm$ 0.004	0.316	0.000	0.15	0.0431	0.0431	1.013	95.181 $\pm$ 0.679	0.174 $\pm$ 0.000	—
UPI	beta33	0.000	0.000	0.000 $\pm$ 0.002	—	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	—	—	0.000	0.000	—	0.0494	0.0494	0.996	94.378 $\pm$ 0.730	0.203 $\pm$ 0.001	5.622 $\pm$ 0.730
UPI	beta34	0.000	-0.001	-0.001 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	—	—	-0.002	-0.002	—	0.0358	0.0358	0.995	95.582 $\pm$ 0.651	0.145 $\pm$ 0.001	4.418 $\pm$ 0.651
UPI	beta35	0.000	0.001	0.001 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	—	—	-0.001	-0.001	—	0.0361	0.0361	0.978	95.080 $\pm$ 0.685	0.145 $\pm$ 0.001	4.920 $\pm$ 0.685

Note. MCSE = Monte Carlo Standard Error shown as $\pm$ value. Coverage, Type I Error, and Power expressed as percentages. $\pm$ indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S113

Simulation 1 - Condition 113: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.315	-0.001 ± 0.002	-0.004 ± 0.006	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.031 ± 0.001	0.177 ± 0.005	0.314	-0.002	-0.54	0.0568	0.0568	1.003	95.796 ± 0.635	0.220 ± 0.003	
LMS	beta32	0.316	0.314	-0.002 ± 0.002	-0.006 ± 0.006	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.032 ± 0.001	0.178 ± 0.005	0.316	0.000	-0.13	0.0529	0.0529	1.009	94.895 ± 0.696	0.223 ± 0.004	—
LMS	beta33	0.000	-0.012	-0.012 ± 0.002	—	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	—	—	-0.009	-0.009	—	0.0676	0.0682	0.930	93.994 ± 0.752	0.252 ± 0.003	6.006 ± 0.752
LMS	beta34	0.000	0.004	0.004 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	0.002	0.002	—	0.0444	0.0444	0.979	93.894 ± 0.758	0.174 ± 0.003	6.106 ± 0.758
LMS	beta35	0.000	0.005	0.005 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	0.003	0.003	—	0.0455	0.0456	0.971	94.094 ± 0.746	0.174 ± 0.002	5.906 ± 0.746
LSAM	beta31	0.316	0.316	0.000 ± 0.002	0.000 ± 0.006	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.033 ± 0.001	0.180 ± 0.005	0.316	0.000	-0.05	0.0574	0.0574	0.991	95.395 ± 0.663	0.222 ± 0.001	—
LSAM	beta32	0.316	0.316	-0.000 ± 0.002	-0.001 ± 0.006	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.033 ± 0.001	0.182 ± 0.005	0.316	0.000	0.09	0.0536	0.0536	0.986	94.795 ± 0.703	0.222 ± 0.001	—
LSAM	beta33	0.000	-0.001	-0.001 ± 0.002	—	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.067 ± 0.002	—	—	0.001	0.001	—	0.0669	0.0669	0.919	93.293 ± 0.791	0.242 ± 0.002	6.707 ± 0.791
LSAM	beta34	0.000	0.000	0.000 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	0.000	0.000	—	0.0456	0.0456	0.961	93.794 ± 0.763	0.171 ± 0.001	6.206 ± 0.763
LSAM	beta35	0.000	0.002	0.002 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	0.000	0.000	—	0.0459	0.0459	0.958	93.694 ± 0.769	0.172 ± 0.001	6.306 ± 0.769
QML	beta31	0.316	0.315	-0.001 ± 0.002	-0.004 ± 0.006	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.031 ± 0.001	0.177 ± 0.005	0.314	-0.002	-0.55	0.0571	0.0571	0.992	95.696 ± 0.642	0.218 ± 0.001	—
QML	beta32	0.316	0.314	-0.002 ± 0.002	-0.006 ± 0.006	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.032 ± 0.001	0.178 ± 0.005	0.315	-0.001	-0.18	0.0524	0.0524	0.991	94.895 ± 0.696	0.219 ± 0.001	—
QML	beta33	0.000	-0.012	-0.012 ± 0.002	—	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	—	—	-0.009	-0.009	—	0.0674	0.0679	0.923	93.794 ± 0.763	0.247 ± 0.002	6.206 ± 0.763
QML	beta34	0.000	0.003	0.003 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	0.002	0.002	—	0.0446	0.0446	0.967	93.994 ± 0.752	0.170 ± 0.001	6.006 ± 0.752
QML	beta35	0.000	0.005	0.005 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	0.002	0.002	—	0.0456	0.0456	0.969	93.894 ± 0.758	0.172 ± 0.001	6.106 ± 0.758
UPI	beta31	0.316	0.316	0.000 ± 0.002	0.000 ± 0.006	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.032 ± 0.001	0.180 ± 0.005	0.316	0.000	0.12	0.0572	0.0572	0.990	95.095 ± 0.683	0.221 ± 0.001	—
UPI	beta32	0.316	0.316	-0.000 ± 0.002	-0.001 ± 0.006	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.033 ± 0.001	0.182 ± 0.005	0.316	0.000	0.05	0.0534	0.0534	0.985	94.795 ± 0.703	0.222 ± 0.001	—
UPI	beta33	0.000	-0.001	-0.001 ± 0.002	—	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	—	—	0.000	0.000	—	0.0681	0.0681	0.921	92.893 ± 0.813	0.245 ± 0.002	7.107 ± 0.813
UPI	beta34	0.000	0.000	0.000 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	-0.001	-0.001	—	0.0448	0.0448	0.960	93.694 ± 0.769	0.174 ± 0.001	6.306 ± 0.769
UPI	beta35	0.000	0.002	0.002 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	0.000	0.000	—	0.0454	0.0454	0.957	93.594 ± 0.775	0.174 ± 0.001	6.406 ± 0.775

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S114

Simulation 1 - Condition 114: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.8

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.316	0.000 ± 0.001	0.001 ± 0.004	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.013 ± 0.001	0.114 ± 0.003	0.317	0.001	0.22	0.0353	0.0353	0.968	93.493 ± 0.780	0.136 ± 0.000	—
LMS	beta32	0.316	0.316	0.000 ± 0.001	0.000 ± 0.004	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.013 ± 0.001	0.113 ± 0.003	0.318	0.002	0.54	0.0361	0.0361	0.978	94.995 ± 0.690	0.137 ± 0.000	—
LMS	beta33	0.000	-0.008	-0.008 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	-0.006	-0.006	—	0.0407	0.0411	0.963	94.194 ± 0.740	0.155 ± 0.001	5.806 ± 0.740
LMS	beta34	0.000	0.003	0.003 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	0.001	0.001	—	0.0259	0.0259	0.976	94.695 ± 0.709	0.107 ± 0.001	5.305 ± 0.709
LMS	beta35	0.000	0.003	0.003 ± 0.001	—	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	—	—	0.001	0.001	—	0.0271	0.0271	1.008	96.096 ± 0.613	0.107 ± 0.001	3.904 ± 0.613
LSAM	beta31	0.316	0.318	0.002 ± 0.001	0.005 ± 0.004	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.014 ± 0.001	0.116 ± 0.003	0.318	0.002	0.50	0.0360	0.0360	0.964	93.493 ± 0.780	0.139 ± 0.000	—
LSAM	beta32	0.316	0.317	0.001 ± 0.001	0.004 ± 0.004	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.013 ± 0.001	0.115 ± 0.003	0.320	0.004	1.11	0.0364	0.0366	0.975	94.895 ± 0.696	0.139 ± 0.000	—
LSAM	beta33	0.000	0.002	0.002 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	0.003	0.003	—	0.0396	0.0397	0.964	94.094 ± 0.746	0.152 ± 0.001	5.906 ± 0.746
LSAM	beta34	0.000	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	-0.001	-0.001	—	0.0266	0.0266	0.972	94.494 ± 0.722	0.107 ± 0.000	5.506 ± 0.722
LSAM	beta35	0.000	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	—	—	0.000	0.000	—	0.0275	0.0275	1.003	95.596 ± 0.649	0.107 ± 0.001	4.404 ± 0.649
QML	beta31	0.316	0.316	0.000 ± 0.001	0.001 ± 0.004	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.013 ± 0.001	0.114 ± 0.003	0.317	0.001	0.16	0.0353	0.0353	0.966	93.493 ± 0.780	0.136 ± 0.000	—
QML	beta32	0.316	0.316	-0.000 ± 0.001	-0.000 ± 0.004	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.013 ± 0.001	0.113 ± 0.003	0.317	0.001	0.45	0.0360	0.0360	0.976	95.095 ± 0.683	0.137 ± 0.000	—
QML	beta33	0.000	-0.008	-0.008 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	-0.006	-0.006	—	0.0406	0.0411	0.966	93.894 ± 0.758	0.155 ± 0.001	6.106 ± 0.758
QML	beta34	0.000	0.003	0.003 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	0.001	0.001	—	0.0259	0.0259	0.973	94.494 ± 0.722	0.106 ± 0.001	5.506 ± 0.722
QML	beta35	0.000	0.002	0.002 ± 0.001	—	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	—	—	0.001	0.001	—	0.0271	0.0271	1.006	95.996 ± 0.620	0.107 ± 0.001	4.004 ± 0.620
UPI	beta31	0.316	0.318	0.002 ± 0.001	0.005 ± 0.004	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.014 ± 0.001	0.117 ± 0.003	0.318	0.002	0.66	0.0360	0.0360	0.963	93.594 ± 0.775	0.139 ± 0.000	—
UPI	beta32	0.316	0.317	0.001 ± 0.001	0.004 ± 0.004	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.013 ± 0.001	0.115 ± 0.003	0.319	0.003	1.00	0.0361	0.0362	0.974	94.895 ± 0.696	0.139 ± 0.000	—
UPI	beta33	0.000	0.002	0.002 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	0.003	0.003	—	0.0390	0.0391	0.962	93.694 ± 0.769	0.152 ± 0.001	6.306 ± 0.769
UPI	beta34	0.000	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	-0.001	-0.001	—	0.0263	0.0263	0.971	94.094 ± 0.746	0.107 ± 0.001	5.906 ± 0.746
UPI	beta35	0.000	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	—	—	-0.001	-0.001	—	0.0275	0.0275	1.006	95.195 ± 0.677	0.108 ± 0.001	4.805 ± 0.677

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S115

Simulation 1 - Condition 115: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.4

Method	Param.	True	Mean-based										Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE	
LMS	beta31	0.316	0.274	-0.042 ± 0.004	-0.134 ± 0.013	0.013 ± 0.001	0.116 ± 0.004	0.015 ± 0.001	0.123 ± 0.005	0.152 ± 0.010	0.390 ± 0.014	0.274	-0.042	-13.44	0.1200	0.1273	1.091	94.702 ± 0.815	0.495 ± 0.018	—	
LMS	beta32	0.316	0.268	-0.048 ± 0.004	-0.151 ± 0.014	0.015 ± 0.001	0.123 ± 0.004	0.017 ± 0.001	0.132 ± 0.005	0.174 ± 0.012	0.417 ± 0.016	0.270	-0.046	-14.68	0.1206	0.1292	1.040	93.510 ± 0.897	0.501 ± 0.013	—	
LMS	beta33	0.000	-0.144	-0.144 ± 0.012	—	0.107 ± 0.009	0.327 ± 0.014	0.127 ± 0.012	0.357 ± 0.018	—	—	-0.075	-0.075	—	0.2013	0.2149	0.842	94.172 ± 0.853	1.077 ± 0.074	5.828 ± 0.853	
LMS	beta34	0.000	0.127	0.127 ± 0.006	—	0.026 ± 0.002	0.161 ± 0.007	0.042 ± 0.004	0.205 ± 0.009	—	—	0.091	0.091	—	0.1112	0.1439	0.927	93.642 ± 0.888	0.586 ± 0.038	6.358 ± 0.888	
LMS	beta35	0.000	0.134	0.134 ± 0.007	—	0.035 ± 0.003	0.187 ± 0.009	0.053 ± 0.005	0.230 ± 0.011	—	—	0.091	0.091	—	0.1191	0.1498	0.884	93.642 ± 0.888	0.647 ± 0.051	6.358 ± 0.888	
LSAM	beta31	0.316	0.331	0.015 ± 0.008	0.048 ± 0.026	0.050 ± 0.003	0.223 ± 0.007	0.050 ± 0.003	0.223 ± 0.008	0.498 ± 0.030	0.706 ± 0.025	0.319	0.003	1.09	0.2312	0.2312	1.097	97.881 ± 0.524	0.957 ± 0.019	—	
LSAM	beta32	0.316	0.325	0.009 ± 0.008	0.029 ± 0.026	0.053 ± 0.003	0.230 ± 0.007	0.053 ± 0.003	0.230 ± 0.008	0.529 ± 0.030	0.728 ± 0.025	0.316	0.000	-0.14	0.2327	0.2327	1.082	97.748 ± 0.540	0.975 ± 0.019	—	
LSAM	beta33	0.000	0.017	0.017 ± 0.005	—	0.022 ± 0.001	0.147 ± 0.005	0.022 ± 0.001	0.148 ± 0.006	—	—	0.004	0.004	—	0.1381	0.1381	1.115	98.013 ± 0.508	0.642 ± 0.019	1.987 ± 0.508	
LSAM	beta34	0.000	-0.006	-0.006 ± 0.003	—	0.006 ± 0.000	0.080 ± 0.003	0.006 ± 0.000	0.081 ± 0.003	—	—	-0.001	-0.001	—	0.0713	0.0713	1.156	98.278 ± 0.473	0.364 ± 0.010	1.722 ± 0.473	
LSAM	beta35	0.000	-0.006	-0.006 ± 0.003	—	0.008 ± 0.001	0.091 ± 0.003	0.008 ± 0.001	0.091 ± 0.003	—	—	-0.001	-0.001	—	0.0821	0.0821	1.077	98.411 ± 0.455	0.383 ± 0.011	1.589 ± 0.455	
QML	beta31	0.316	0.280	-0.036 ± 0.004	-0.114 ± 0.012	0.011 ± 0.001	0.103 ± 0.003	0.012 ± 0.001	0.109 ± 0.003	0.119 ± 0.006	0.345 ± 0.011	0.278	-0.038	-11.99	0.1110	0.1173	1.037	94.570 ± 0.825	0.419 ± 0.003	—	
QML	beta32	0.316	0.275	-0.041 ± 0.004	-0.130 ± 0.012	0.012 ± 0.001	0.108 ± 0.003	0.013 ± 0.001	0.115 ± 0.004	0.133 ± 0.007	0.364 ± 0.012	0.276	-0.040	-12.65	0.1047	0.1121	1.006	93.113 ± 0.922	0.424 ± 0.004	—	
QML	beta33	0.000	-0.046	-0.046 ± 0.004	—	0.015 ± 0.001	0.121 ± 0.004	0.017 ± 0.001	0.129 ± 0.004	—	—	-0.052	-0.052	—	0.1271	0.1374	1.085	96.689 ± 0.651	0.514 ± 0.010	3.311 ± 0.651	
QML	beta34	0.000	0.071	0.071 ± 0.002	—	0.004 ± 0.000	0.067 ± 0.002	0.009 ± 0.000	0.097 ± 0.003	—	—	0.070	0.070	—	0.0723	0.1008	1.117	92.053 ± 0.984	0.292 ± 0.006	7.947 ± 0.984	
QML	beta35	0.000	0.072	0.072 ± 0.003	—	0.006 ± 0.000	0.075 ± 0.002	0.011 ± 0.001	0.104 ± 0.003	—	—	0.070	0.070	—	0.0805	0.1066	1.035	90.331 ± 1.076	0.305 ± 0.007	9.669 ± 1.076	
UPI	beta31	0.316	0.329	0.013 ± 0.008	0.042 ± 0.026	0.050 ± 0.003	0.223 ± 0.007	0.050 ± 0.003	0.224 ± 0.008	0.501 ± 0.033	0.708 ± 0.026	0.324	0.008	2.52	0.2182	0.2183	1.145	97.881 ± 0.524	1.003 ± 0.026	—	
UPI	beta32	0.316	0.326	0.010 ± 0.008	0.033 ± 0.027	0.053 ± 0.003	0.231 ± 0.007	0.054 ± 0.003	0.231 ± 0.009	0.536 ± 0.035	0.732 ± 0.027	0.316	0.000	-0.15	0.2236	0.2236	1.138	97.881 ± 0.524	1.031 ± 0.027	—	
UPI	beta33	0.000	0.009	0.009 ± 0.006	—	0.025 ± 0.002	0.158 ± 0.006	0.025 ± 0.002	0.158 ± 0.007	—	—	-0.005	-0.005	—	0.1323	0.1324	1.481	98.940 ± 0.373	0.919 ± 0.050	1.060 ± 0.373	
UPI	beta34	0.000	-0.005	-0.005 ± 0.003	—	0.007 ± 0.000	0.082 ± 0.003	0.007 ± 0.000	0.082 ± 0.003	—	—	-0.001	-0.001	—	0.0734	0.0734	1.238	98.675 ± 0.416	0.399 ± 0.015	1.325 ± 0.416	
UPI	beta35	0.000	-0.005	-0.005 ± 0.003	—	0.008 ± 0.001	0.091 ± 0.003	0.008 ± 0.001	0.091 ± 0.003	—	—	0.003	0.003	—	0.0820	0.0820	1.153	98.543 ± 0.436	0.410 ± 0.013	1.457 ± 0.436	

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S116

Simulation 1 - Condition 116: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.4

Method	Param.	True	Mean-based									Median-based					Inference				
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Type I (%) $\pm$ MCSE	
LMS	beta31	0.316	0.272	-0.044 $\pm$ 0.002	-0.140 $\pm$ 0.007	0.004 $\pm$ 0.000	0.062 $\pm$ 0.002	0.006 $\pm$ 0.000	0.076 $\pm$ 0.002	0.058 $\pm$ 0.003	0.241 $\pm$ 0.006	0.269	-0.047	-14.79	0.0658	0.0807	1.165	91.373 $\pm$ 0.946	0.282 $\pm$ 0.011	—	
LMS	beta32	0.316	0.273	-0.043 $\pm$ 0.002	-0.136 $\pm$ 0.007	0.004 $\pm$ 0.000	0.064 $\pm$ 0.002	0.006 $\pm$ 0.000	0.077 $\pm$ 0.002	0.060 $\pm$ 0.003	0.245 $\pm$ 0.007	0.273	-0.043	-13.46	0.0659	0.0785	1.100	91.373 $\pm$ 0.946	0.278 $\pm$ 0.007	—	
LMS	beta33	0.000	-0.039	-0.039 $\pm$ 0.003	—	0.010 $\pm$ 0.001	0.098 $\pm$ 0.004	0.011 $\pm$ 0.001	0.105 $\pm$ 0.005	—	—	-0.041	-0.041	—	0.0954	0.1037	1.535	97.843 $\pm$ 0.489	0.589 $\pm$ 0.056	2.157 $\pm$ 0.489	
LMS	beta34	0.000	0.073	0.073 $\pm$ 0.002	—	0.003 $\pm$ 0.000	0.053 $\pm$ 0.002	0.008 $\pm$ 0.000	0.090 $\pm$ 0.002	—	—	0.071	0.071	—	0.0523	0.0880	1.422	86.833 $\pm$ 1.139	0.294 $\pm$ 0.023	13.167 $\pm$ 1.139	
LMS	beta35	0.000	0.076	0.076 $\pm$ 0.002	—	0.003 $\pm$ 0.000	0.057 $\pm$ 0.002	0.009 $\pm$ 0.000	0.095 $\pm$ 0.003	—	—	0.074	0.074	—	0.0568	0.0934	1.419	85.471 $\pm$ 1.187	0.317 $\pm$ 0.039	14.529 $\pm$ 1.187	
LSAM	beta31	0.316	0.319	0.003 $\pm$ 0.004	0.008 $\pm$ 0.012	0.013 $\pm$ 0.001	0.116 $\pm$ 0.003	0.013 $\pm$ 0.001	0.116 $\pm$ 0.004	0.134 $\pm$ 0.007	0.366 $\pm$ 0.011	0.318	0.002	0.68	0.1178	0.1178	1.071	97.276 $\pm$ 0.548	0.486 $\pm$ 0.004	—	
LSAM	beta32	0.316	0.323	0.007 $\pm$ 0.004	0.022 $\pm$ 0.013	0.015 $\pm$ 0.001	0.121 $\pm$ 0.003	0.015 $\pm$ 0.001	0.121 $\pm$ 0.004	0.148 $\pm$ 0.007	0.384 $\pm$ 0.012	0.315	-0.001	-0.39	0.1201	0.1201	1.025	96.708 $\pm$ 0.601	0.487 $\pm$ 0.004	—	
LSAM	beta33	0.000	0.014	0.014 $\pm$ 0.002	—	0.004 $\pm$ 0.000	0.065 $\pm$ 0.002	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	—	—	0.005	0.005	—	0.0645	0.0647	1.081	96.595 $\pm$ 0.611	0.274 $\pm$ 0.003	3.405 $\pm$ 0.611	
LSAM	beta34	0.000	-0.004	-0.004 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	—	—	-0.002	-0.002	—	0.0360	0.0360	1.094	96.595 $\pm$ 0.611	0.162 $\pm$ 0.002	3.405 $\pm$ 0.611	
LSAM	beta35	0.000	-0.005	-0.005 $\pm$ 0.001	—	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	—	—	-0.002	-0.002	—	0.0378	0.0379	0.994	96.254 $\pm$ 0.640	0.164 $\pm$ 0.002	3.746 $\pm$ 0.640	
QML	beta31	0.316	0.274	-0.042 $\pm$ 0.002	-0.134 $\pm$ 0.006	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.005 $\pm$ 0.000	0.074 $\pm$ 0.002	0.055 $\pm$ 0.002	0.234 $\pm$ 0.006	0.273	-0.043	-13.54	0.0616	0.0750	1.060	90.238 $\pm$ 1.000	0.252 $\pm$ 0.001	—	
QML	beta32	0.316	0.276	-0.040 $\pm$ 0.002	-0.128 $\pm$ 0.007	0.004 $\pm$ 0.000	0.063 $\pm$ 0.002	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.057 $\pm$ 0.003	0.238 $\pm$ 0.006	0.275	-0.041	-12.93	0.0633	0.0753	1.016	89.330 $\pm$ 1.040	0.252 $\pm$ 0.001	—	
QML	beta33	0.000	-0.031	-0.031 $\pm$ 0.002	—	0.004 $\pm$ 0.000	0.067 $\pm$ 0.002	0.005 $\pm$ 0.000	0.073 $\pm$ 0.002	—	—	-0.041	-0.041	—	0.0710	0.0819	1.142	96.254 $\pm$ 0.640	0.299 $\pm$ 0.003	3.746 $\pm$ 0.640	
QML	beta34	0.000	0.062	0.062 $\pm$ 0.001	—	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.005 $\pm$ 0.000	0.074 $\pm$ 0.001	—	—	0.062	0.062	—	0.0413	0.0745	1.100	75.596 $\pm$ 1.447	0.169 $\pm$ 0.002	24.404 $\pm$ 1.447	
QML	beta35	0.000	0.064	0.064 $\pm$ 0.001	—	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.006 $\pm$ 0.000	0.077 $\pm$ 0.002	—	—	0.064	0.064	—	0.0448	0.0784	1.025	74.347 $\pm$ 1.471	0.171 $\pm$ 0.002	25.653 $\pm$ 1.471	
UPI	beta31	0.316	0.320	0.004 $\pm$ 0.004	0.011 $\pm$ 0.012	0.013 $\pm$ 0.001	0.116 $\pm$ 0.003	0.013 $\pm$ 0.001	0.116 $\pm$ 0.004	0.134 $\pm$ 0.007	0.366 $\pm$ 0.011	0.317	0.001	0.27	0.1121	0.1121	1.075	97.049 $\pm$ 0.570	0.488 $\pm$ 0.004	—	
UPI	beta32	0.316	0.323	0.007 $\pm$ 0.004	0.023 $\pm$ 0.013	0.015 $\pm$ 0.001	0.120 $\pm$ 0.003	0.015 $\pm$ 0.001	0.121 $\pm$ 0.004	0.146 $\pm$ 0.007	0.382 $\pm$ 0.012	0.315	-0.001	-0.18	0.1217	0.1217	1.036	96.708 $\pm$ 0.601	0.489 $\pm$ 0.004	—	
UPI	beta33	0.000	0.013	0.013 $\pm$ 0.002	—	0.005 $\pm$ 0.000	0.070 $\pm$ 0.002	0.005 $\pm$ 0.000	0.071 $\pm$ 0.002	—	—	0.002	0.002	—	0.0641	0.0642	1.180	98.297 $\pm$ 0.436	0.323 $\pm$ 0.009	1.703 $\pm$ 0.436	
UPI	beta34	0.000	-0.004	-0.004 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	—	—	-0.001	-0.001	—	0.0365	0.0365	1.113	97.162 $\pm$ 0.559	0.165 $\pm$ 0.002	2.838 $\pm$ 0.559	
UPI	beta35	0.000	-0.005	-0.005 $\pm$ 0.001	—	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	—	—	-0.004	-0.004	—	0.0394	0.0396	1.010	96.595 $\pm$ 0.611	0.167 $\pm$ 0.002	3.405 $\pm$ 0.611	

Note. MCSE = Monte Carlo Standard Error shown as $\backslash$ u00b1 value. Coverage, Type I Error, and Power expressed as percentages. $\backslash$ u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S117

Simulation 1 - Condition 117: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.6

		Mean-based										Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.285	-0.031 ± 0.003	-0.098 ± 0.008	0.007 ± 0.000	0.081 ± 0.002	0.008 ± 0.000	0.087 ± 0.002	0.076 ± 0.004	0.275 ± 0.008	0.283	-0.033	-10.49	0.0784	0.0851	1.037	95.135 ± 0.692	0.331 ± 0.002	—
LMS	beta32	0.316	0.285	-0.031 ± 0.003	-0.097 ± 0.008	0.007 ± 0.000	0.082 ± 0.002	0.008 ± 0.000	0.088 ± 0.003	0.077 ± 0.004	0.278 ± 0.008	0.291	-0.025	-8.00	0.0775	0.0815	1.024	95.031 ± 0.699	0.331 ± 0.003	—
LMS	beta33	0.000	-0.023	-0.023 ± 0.003	—	0.008 ± 0.000	0.089 ± 0.003	0.008 ± 0.000	0.092 ± 0.003	—	—	-0.018	-0.018	—	0.0817	0.0837	1.015	96.170 ± 0.618	0.352 ± 0.008	3.830 ± 0.618
LMS	beta34	0.000	0.040	0.040 ± 0.002	—	0.003 ± 0.000	0.050 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	—	—	0.032	0.032	—	0.0449	0.0552	0.958	92.961 ± 0.823	0.188 ± 0.004	7.039 ± 0.823
LMS	beta35	0.000	0.038	0.038 ± 0.002	—	0.002 ± 0.000	0.048 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	—	—	0.032	0.032	—	0.0413	0.0524	0.996	91.925 ± 0.877	0.186 ± 0.004	8.075 ± 0.877
LSAM	beta31	0.316	0.321	0.005 ± 0.004	0.017 ± 0.013	0.018 ± 0.001	0.132 ± 0.003	0.018 ± 0.001	0.132 ± 0.004	0.175 ± 0.009	0.419 ± 0.013	0.314	-0.002	-0.56	0.1240	0.1241	1.016	96.066 ± 0.625	0.527 ± 0.004	—
LSAM	beta32	0.316	0.321	0.005 ± 0.004	0.015 ± 0.013	0.017 ± 0.001	0.132 ± 0.003	0.017 ± 0.001	0.132 ± 0.004	0.175 ± 0.009	0.418 ± 0.012	0.324	0.008	2.40	0.1282	0.1284	1.018	95.756 ± 0.649	0.527 ± 0.004	—
LSAM	beta33	0.000	0.003	0.003 ± 0.003	—	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.081 ± 0.003	—	—	0.001	0.001	—	0.0739	0.0739	0.978	94.410 ± 0.739	0.310 ± 0.004	5.590 ± 0.739
LSAM	beta34	0.000	-0.002	-0.002 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.002	—	—	-0.001	-0.001	—	0.0449	0.0449	0.953	94.513 ± 0.733	0.190 ± 0.003	5.487 ± 0.733
LSAM	beta35	0.000	-0.002	-0.002 ± 0.002	—	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.002	—	—	-0.002	-0.002	—	0.0422	0.0423	0.963	95.135 ± 0.692	0.190 ± 0.003	4.865 ± 0.692
QML	beta31	0.316	0.287	-0.029 ± 0.003	-0.092 ± 0.008	0.007 ± 0.000	0.082 ± 0.002	0.008 ± 0.000	0.087 ± 0.002	0.076 ± 0.003	0.275 ± 0.007	0.284	-0.032	-10.05	0.0786	0.0848	1.033	94.720 ± 0.719	0.332 ± 0.003	—
QML	beta32	0.316	0.287	-0.029 ± 0.003	-0.091 ± 0.008	0.007 ± 0.000	0.083 ± 0.002	0.008 ± 0.000	0.087 ± 0.002	0.076 ± 0.004	0.277 ± 0.008	0.291	-0.025	-7.77	0.0777	0.0815	1.016	94.513 ± 0.733	0.329 ± 0.002	—
QML	beta33	0.000	-0.023	-0.023 ± 0.003	—	0.006 ± 0.000	0.080 ± 0.002	0.007 ± 0.000	0.083 ± 0.002	—	—	-0.020	-0.020	—	0.0767	0.0792	1.003	94.617 ± 0.726	0.315 ± 0.004	5.383 ± 0.726
QML	beta34	0.000	0.036	0.036 ± 0.002	—	0.002 ± 0.000	0.047 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	—	—	0.031	0.031	—	0.0444	0.0541	0.951	90.994 ± 0.921	0.175 ± 0.003	9.006 ± 0.921
QML	beta35	0.000	0.035	0.035 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	—	—	0.031	0.031	—	0.0407	0.0509	0.974	90.994 ± 0.921	0.171 ± 0.003	9.006 ± 0.921
UPI	beta31	0.316	0.321	0.005 ± 0.004	0.017 ± 0.013	0.017 ± 0.001	0.132 ± 0.003	0.017 ± 0.001	0.132 ± 0.004	0.175 ± 0.009	0.419 ± 0.013	0.314	-0.002	-0.48	0.1237	0.1237	1.012	95.756 ± 0.649	0.525 ± 0.004	—
UPI	beta32	0.316	0.322	0.006 ± 0.004	0.018 ± 0.013	0.017 ± 0.001	0.131 ± 0.003	0.017 ± 0.001	0.132 ± 0.004	0.173 ± 0.008	0.416 ± 0.012	0.323	0.007	2.13	0.1250	0.1252	1.016	95.756 ± 0.649	0.524 ± 0.004	—
UPI	beta33	0.000	0.003	0.003 ± 0.003	—	0.007 ± 0.000	0.083 ± 0.002	0.007 ± 0.000	0.083 ± 0.003	—	—	0.000	0.000	—	0.0703	0.0703	1.068	95.859 ± 0.641	0.345 ± 0.006	4.141 ± 0.641
UPI	beta34	0.000	-0.002	-0.002 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.002	—	—	0.000	0.000	—	0.0442	0.0442	0.961	95.342 ± 0.678	0.192 ± 0.003	4.658 ± 0.678
UPI	beta35	0.000	-0.002	-0.002 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.002	—	—	-0.003	-0.003	—	0.0431	0.0432	0.954	95.859 ± 0.641	0.191 ± 0.003	4.141 ± 0.641

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S118

Simulation 1 - Condition 118: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.6

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.284	-0.032 ± 0.002	-0.103 ± 0.005	0.003 ± 0.000	0.050 ± 0.001	0.004 ± 0.000	0.060 ± 0.001	0.036 ± 0.002	0.189 ± 0.005	0.284	-0.032	-10.09	0.0503	0.0596	1.013	91.304 ± 0.896	0.199 ± 0.001	—
LMS	beta32	0.316	0.287	-0.029 ± 0.002	-0.091 ± 0.005	0.002 ± 0.000	0.049 ± 0.001	0.003 ± 0.000	0.057 ± 0.001	0.032 ± 0.002	0.179 ± 0.005	0.288	-0.028	-8.76	0.0486	0.0559	1.034	92.113 ± 0.857	0.198 ± 0.001	—
LMS	beta33	0.000	-0.012	-0.012 ± 0.002	—	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	—	—	-0.011	-0.011	—	0.0531	0.0543	0.959	95.956 ± 0.626	0.203 ± 0.002	4.044 ± 0.626
LMS	beta34	0.000	0.033	0.033 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	—	—	0.028	0.028	—	0.0256	0.0383	0.952	82.710 ± 1.202	0.106 ± 0.001	17.290 ± 1.202
LMS	beta35	0.000	0.031	0.031 ± 0.001	—	0.001 ± 0.000	0.026 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	0.029	0.029	—	0.0252	0.0386	1.006	85.541 ± 1.118	0.104 ± 0.001	14.459 ± 1.118
LSAM	beta31	0.316	0.315	-0.001 ± 0.002	-0.004 ± 0.008	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	0.059 ± 0.003	0.243 ± 0.006	0.315	-0.001	-0.21	0.0753	0.0753	1.020	96.259 ± 0.603	0.307 ± 0.001	—
LSAM	beta32	0.316	0.322	0.006 ± 0.002	0.019 ± 0.008	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	0.059 ± 0.003	0.242 ± 0.007	0.323	0.007	2.33	0.0755	0.0758	1.027	96.057 ± 0.619	0.307 ± 0.001	—
LSAM	beta33	0.000	0.002	0.002 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	0.001	0.001	—	0.0453	0.0453	0.944	94.135 ± 0.747	0.171 ± 0.001	5.865 ± 0.747
LSAM	beta34	0.000	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.026 ± 0.001	0.001 ± 0.000	0.026 ± 0.001	—	—	0.000	0.000	—	0.0238	0.0238	0.984	95.854 ± 0.634	0.101 ± 0.001	4.146 ± 0.634
LSAM	beta35	0.000	-0.003	-0.003 ± 0.001	—	0.001 ± 0.000	0.026 ± 0.001	0.001 ± 0.000	0.026 ± 0.001	—	—	-0.002	-0.002	—	0.0223	0.0223	0.993	95.349 ± 0.670	0.100 ± 0.001	4.651 ± 0.670
QML	beta31	0.316	0.285	-0.031 ± 0.002	-0.099 ± 0.005	0.003 ± 0.000	0.051 ± 0.001	0.004 ± 0.000	0.060 ± 0.001	0.036 ± 0.002	0.189 ± 0.005	0.285	-0.031	-9.90	0.0516	0.0604	1.010	91.203 ± 0.901	0.201 ± 0.001	—
QML	beta32	0.316	0.289	-0.027 ± 0.002	-0.087 ± 0.005	0.002 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.057 ± 0.001	0.032 ± 0.002	0.180 ± 0.005	0.289	-0.027	-8.43	0.0499	0.0566	1.028	92.012 ± 0.862	0.201 ± 0.001	—
QML	beta33	0.000	-0.014	-0.014 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	—	—	-0.014	-0.014	—	0.0508	0.0527	0.951	94.944 ± 0.697	0.190 ± 0.002	5.056 ± 0.697
QML	beta34	0.000	0.032	0.032 ± 0.001	—	0.001 ± 0.000	0.027 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	0.028	0.028	—	0.0252	0.0377	0.955	81.901 ± 1.224	0.102 ± 0.001	18.099 ± 1.224
QML	beta35	0.000	0.029	0.029 ± 0.001	—	0.001 ± 0.000	0.026 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	0.029	0.029	—	0.0250	0.0381	1.001	84.530 ± 1.150	0.101 ± 0.001	15.470 ± 1.150
UPI	beta31	0.316	0.315	-0.001 ± 0.002	-0.003 ± 0.008	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	0.059 ± 0.003	0.242 ± 0.006	0.314	-0.002	-0.52	0.0755	0.0756	1.020	95.956 ± 0.626	0.306 ± 0.001	—
UPI	beta32	0.316	0.322	0.006 ± 0.002	0.020 ± 0.008	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	0.058 ± 0.003	0.241 ± 0.006	0.324	0.008	2.64	0.0739	0.0743	1.030	95.956 ± 0.626	0.306 ± 0.001	—
UPI	beta33	0.000	0.001	0.001 ± 0.002	—	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	—	—	-0.001	-0.001	—	0.0479	0.0479	0.951	95.147 ± 0.683	0.179 ± 0.002	4.853 ± 0.683
UPI	beta34	0.000	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.026 ± 0.001	0.001 ± 0.000	0.026 ± 0.001	—	—	-0.001	-0.001	—	0.0238	0.0238	0.984	95.956 ± 0.626	0.101 ± 0.001	4.044 ± 0.626
UPI	beta35	0.000	-0.003	-0.003 ± 0.001	—	0.001 ± 0.000	0.026 ± 0.001	0.001 ± 0.000	0.026 ± 0.001	—	—	-0.002	-0.002	—	0.0225	0.0226	0.995	94.742 ± 0.710	0.101 ± 0.001	5.258 ± 0.710

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u00214 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S119

Simulation 1 - Condition 119: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.8

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.299	-0.017 ± 0.002	-0.053 ± 0.008	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	0.061 ± 0.003	0.247 ± 0.007	0.299	-0.017	-5.49	0.0783	0.0802	0.977	95.060 ± 0.688	0.292 ± 0.001	
LMS	beta32	0.316	0.301	-0.015 ± 0.003	-0.049 ± 0.008	0.006 ± 0.000	0.080 ± 0.002	0.007 ± 0.000	0.081 ± 0.002	0.066 ± 0.003	0.258 ± 0.007	0.299	-0.017	-5.43	0.0790	0.0808	0.931	93.246 ± 0.797	0.292 ± 0.001	—
LMS	beta33	0.000	-0.003	-0.003 ± 0.002	—	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	—	—	-0.001	-0.001	—	0.0558	0.0558	0.902	93.145 ± 0.802	0.224 ± 0.002	6.855 ± 0.802
LMS	beta34	0.000	0.013	0.013 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.009	0.009	—	0.0315	0.0327	0.898	92.843 ± 0.818	0.122 ± 0.002	7.157 ± 0.818
LMS	beta35	0.000	0.014	0.014 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.039 ± 0.001	—	—	0.009	0.009	—	0.0304	0.0318	0.853	90.927 ± 0.912	0.121 ± 0.002	9.073 ± 0.912
LSAM	beta31	0.316	0.316	0.000 ± 0.003	0.002 ± 0.010	0.010 ± 0.000	0.098 ± 0.002	0.010 ± 0.000	0.098 ± 0.003	0.097 ± 0.004	0.311 ± 0.009	0.316	0.000	-0.06	0.0986	0.0986	0.963	94.456 ± 0.727	0.371 ± 0.002	—
LSAM	beta32	0.316	0.317	0.001 ± 0.003	0.004 ± 0.010	0.011 ± 0.000	0.103 ± 0.002	0.011 ± 0.000	0.103 ± 0.003	0.105 ± 0.005	0.324 ± 0.009	0.316	0.000	0.12	0.1008	0.1008	0.921	93.448 ± 0.786	0.370 ± 0.002	—
LSAM	beta33	0.000	0.004	0.004 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	—	—	0.004	0.004	—	0.0571	0.0573	0.904	92.339 ± 0.844	0.217 ± 0.002	7.661 ± 0.844
LSAM	beta34	0.000	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.003	-0.003	—	0.0335	0.0336	0.906	93.548 ± 0.780	0.130 ± 0.002	6.452 ± 0.780
LSAM	beta35	0.000	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.002	-0.002	—	0.0322	0.0323	0.858	91.331 ± 0.893	0.129 ± 0.002	8.669 ± 0.893
QML	beta31	0.316	0.300	-0.016 ± 0.002	-0.051 ± 0.008	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	0.062 ± 0.003	0.249 ± 0.007	0.299	-0.017	-5.40	0.0794	0.0813	0.971	94.859 ± 0.701	0.294 ± 0.001	—
QML	beta32	0.316	0.301	-0.015 ± 0.003	-0.047 ± 0.008	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.082 ± 0.002	0.068 ± 0.003	0.260 ± 0.007	0.299	-0.017	-5.28	0.0801	0.0818	0.924	92.540 ± 0.834	0.293 ± 0.001	—
QML	beta33	0.000	-0.003	-0.003 ± 0.002	—	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	—	—	-0.001	-0.001	—	0.0563	0.0564	0.899	92.843 ± 0.818	0.221 ± 0.002	7.157 ± 0.818
QML	beta34	0.000	0.013	0.013 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.009	0.009	—	0.0316	0.0328	0.895	92.843 ± 0.818	0.121 ± 0.002	7.157 ± 0.818
QML	beta35	0.000	0.013	0.013 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.039 ± 0.001	—	—	0.009	0.009	—	0.0305	0.0319	0.851	90.524 ± 0.930	0.121 ± 0.002	9.476 ± 0.930
UPI	beta31	0.316	0.317	0.001 ± 0.003	0.002 ± 0.010	0.010 ± 0.000	0.098 ± 0.002	0.010 ± 0.000	0.098 ± 0.003	0.096 ± 0.004	0.310 ± 0.009	0.314	-0.002	-0.63	0.0965	0.0965	0.963	94.556 ± 0.720	0.370 ± 0.002	—
UPI	beta32	0.316	0.317	0.001 ± 0.003	0.004 ± 0.010	0.010 ± 0.000	0.102 ± 0.002	0.010 ± 0.000	0.102 ± 0.003	0.104 ± 0.005	0.323 ± 0.009	0.317	0.001	0.40	0.1007	0.1007	0.921	93.347 ± 0.791	0.368 ± 0.002	—
UPI	beta33	0.000	0.003	0.003 ± 0.002	—	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	—	—	0.002	0.002	—	0.0573	0.0574	0.929	93.145 ± 0.802	0.225 ± 0.003	6.855 ± 0.802
UPI	beta34	0.000	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.003	-0.003	—	0.0332	0.0334	0.909	93.952 ± 0.757	0.131 ± 0.002	6.048 ± 0.757
UPI	beta35	0.000	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.039 ± 0.001	0.001 ± 0.000	0.039 ± 0.001	—	—	-0.002	-0.002	—	0.0334	0.0335	0.856	91.532 ± 0.884	0.130 ± 0.002	8.468 ± 0.884

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S120

Simulation 1 - Condition 120: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.8

Method	Param.	True	Mean-based										Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE	
LMS	beta31	0.316	0.300	-0.016 ± 0.001	-0.052 ± 0.005	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.023 ± 0.001	0.153 ± 0.004	0.301	-0.015	-4.61	0.0455	0.0478	1.025	94.200 ± 0.739	0.182 ± 0.001		
LMS	beta32	0.316	0.299	-0.017 ± 0.001	-0.055 ± 0.005	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.024 ± 0.001	0.154 ± 0.004	0.298	-0.018	-5.84	0.0448	0.0485	1.019	93.100 ± 0.801	0.182 ± 0.001	—	
LMS	beta33	0.000	-0.003	-0.003 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.003	-0.003	—	0.0379	0.0381	0.956	94.900 ± 0.696	0.140 ± 0.001	5.100 ± 0.696	
LMS	beta34	0.000	0.012	0.012 ± 0.001	—	0.000 ± 0.000	0.020 ± 0.001	0.001 ± 0.000	0.024 ± 0.001	—	—	0.010	0.010	—	0.0181	0.0207	0.922	92.100 ± 0.853	0.073 ± 0.001	7.900 ± 0.853	
LMS	beta35	0.000	0.013	0.013 ± 0.001	—	0.000 ± 0.000	0.020 ± 0.000	0.001 ± 0.000	0.023 ± 0.001	—	—	0.012	0.012	—	0.0184	0.0218	0.950	92.400 ± 0.838	0.073 ± 0.001	7.600 ± 0.838	
LSAM	beta31	0.316	0.315	-0.001 ± 0.002	-0.003 ± 0.006	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.032 ± 0.001	0.180 ± 0.005	0.318	0.002	0.50	0.0570	0.0570	1.016	95.400 ± 0.662	0.226 ± 0.001	—	
LSAM	beta32	0.316	0.314	-0.002 ± 0.002	-0.007 ± 0.006	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.032 ± 0.001	0.180 ± 0.005	0.313	-0.003	-0.90	0.0547	0.0547	1.013	94.900 ± 0.696	0.225 ± 0.001	—	
LSAM	beta33	0.000	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	0.000	0.000	—	0.0350	0.0350	0.961	94.300 ± 0.733	0.132 ± 0.001	5.700 ± 0.733	
LSAM	beta34	0.000	0.000	0.000 ± 0.001	—	0.000 ± 0.000	0.020 ± 0.001	0.000 ± 0.000	0.020 ± 0.001	—	—	-0.001	-0.001	—	0.0179	0.0179	0.940	94.200 ± 0.739	0.075 ± 0.001	5.800 ± 0.739	
LSAM	beta35	0.000	0.001	0.001 ± 0.001	—	0.000 ± 0.000	0.020 ± 0.000	0.000 ± 0.000	0.020 ± 0.001	—	—	0.001	0.001	—	0.0182	0.0182	0.957	93.500 ± 0.780	0.074 ± 0.001	6.500 ± 0.780	
QML	beta31	0.316	0.300	-0.016 ± 0.001	-0.051 ± 0.005	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.024 ± 0.001	0.154 ± 0.004	0.302	-0.014	-4.46	0.0466	0.0487	1.015	94.200 ± 0.739	0.183 ± 0.001	—	
QML	beta32	0.316	0.299	-0.017 ± 0.001	-0.054 ± 0.005	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.024 ± 0.001	0.155 ± 0.004	0.298	-0.018	-5.75	0.0452	0.0487	1.010	93.100 ± 0.801	0.182 ± 0.001	—	
QML	beta33	0.000	-0.004	-0.004 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.004	-0.004	—	0.0378	0.0380	0.956	95.000 ± 0.689	0.139 ± 0.001	5.000 ± 0.689	
QML	beta34	0.000	0.012	0.012 ± 0.001	—	0.000 ± 0.000	0.020 ± 0.001	0.001 ± 0.000	0.024 ± 0.001	—	—	0.010	0.010	—	0.0182	0.0208	0.917	92.000 ± 0.858	0.073 ± 0.001	8.000 ± 0.858	
QML	beta35	0.000	0.013	0.013 ± 0.001	—	0.000 ± 0.000	0.020 ± 0.000	0.001 ± 0.000	0.023 ± 0.001	—	—	0.012	0.012	—	0.0188	0.0221	0.943	92.200 ± 0.848	0.072 ± 0.001	7.800 ± 0.848	
UPI	beta31	0.316	0.315	-0.001 ± 0.002	-0.003 ± 0.006	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.032 ± 0.001	0.180 ± 0.005	0.317	0.001	0.34	0.0556	0.0556	1.013	95.400 ± 0.662	0.226 ± 0.001	—	
UPI	beta32	0.316	0.314	-0.002 ± 0.002	-0.007 ± 0.006	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.032 ± 0.001	0.180 ± 0.005	0.312	-0.004	-1.12	0.0539	0.0540	1.011	95.100 ± 0.683	0.225 ± 0.001	—	
UPI	beta33	0.000	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	-0.001	-0.001	—	0.0358	0.0358	0.962	94.000 ± 0.751	0.134 ± 0.001	6.000 ± 0.751	
UPI	beta34	0.000	0.000	0.000 ± 0.001	—	0.000 ± 0.000	0.020 ± 0.001	0.000 ± 0.000	0.020 ± 0.001	—	—	0.000	0.000	—	0.0181	0.0181	0.937	94.500 ± 0.721	0.075 ± 0.001	5.500 ± 0.721	
UPI	beta35	0.000	0.001	0.001 ± 0.001	—	0.000 ± 0.000	0.020 ± 0.000	0.000 ± 0.000	0.020 ± 0.001	—	—	0.000	0.000	—	0.0178	0.0178	0.954	94.000 ± 0.751	0.074 ± 0.001	6.000 ± 0.751	

Note. MCSE = Monte Carlo Standard Error shown as $\backslash u00b1$ value. Coverage, Type I Error, and Power expressed as percentages. $\backslash u2014$ indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S121

Simulation 1 - Condition 121: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, $N = 400$, Reliability = 0.4

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Type I (%) $\pm$ MCSE
LMS	beta31	0.316	0.323	0.007 $\pm$ 0.005	0.023 $\pm$ 0.014	0.011 $\pm$ 0.001	0.103 $\pm$ 0.004	0.011 $\pm$ 0.001	0.103 $\pm$ 0.004	0.106 $\pm$ 0.007	0.326 $\pm$ 0.014	0.326	0.010	3.23	0.1029	0.1034	1.043	94.902 $\pm$ 0.974	0.420 $\pm$ 0.008	—
LMS	beta32	0.316	0.316	-0.000 $\pm$ 0.005	-0.001 $\pm$ 0.015	0.011 $\pm$ 0.001	0.106 $\pm$ 0.004	0.011 $\pm$ 0.001	0.106 $\pm$ 0.004	0.113 $\pm$ 0.008	0.336 $\pm$ 0.014	0.319	0.003	1.10	0.1038	0.1039	0.992	94.118 $\pm$ 1.042	0.413 $\pm$ 0.008	—
LMS	beta33	0.000	-0.081	-0.081 $\pm$ 0.011	—	0.059 $\pm$ 0.007	0.242 $\pm$ 0.014	0.065 $\pm$ 0.008	0.255 $\pm$ 0.017	—	—	-0.038	-0.038	—	0.1728	0.1769	0.871	94.706 $\pm$ 0.992	0.827 $\pm$ 0.045	5.294 $\pm$ 0.992
LMS	beta34	0.000	0.063	0.063 $\pm$ 0.008	—	0.032 $\pm$ 0.003	0.178 $\pm$ 0.010	0.035 $\pm$ 0.004	0.188 $\pm$ 0.012	—	—	0.032	0.032	—	0.1340	0.1377	0.861	94.314 $\pm$ 1.025	0.599 $\pm$ 0.034	5.686 $\pm$ 1.025
LMS	beta35	0.000	0.042	0.042 $\pm$ 0.006	—	0.019 $\pm$ 0.002	0.137 $\pm$ 0.007	0.020 $\pm$ 0.002	0.143 $\pm$ 0.009	—	—	0.023	0.023	—	0.1192	0.1214	1.077	95.098 $\pm$ 0.956	0.578 $\pm$ 0.050	4.902 $\pm$ 0.956
LSAM	beta31	0.316	0.321	0.005 $\pm$ 0.005	0.017 $\pm$ 0.016	0.012 $\pm$ 0.001	0.111 $\pm$ 0.004	0.012 $\pm$ 0.001	0.111 $\pm$ 0.004	0.123 $\pm$ 0.008	0.351 $\pm$ 0.014	0.326	0.010	3.02	0.1222	0.1226	1.215	97.451 $\pm$ 0.698	0.528 $\pm$ 0.018	—
LSAM	beta32	0.316	0.324	0.008 $\pm$ 0.005	0.026 $\pm$ 0.016	0.013 $\pm$ 0.001	0.114 $\pm$ 0.004	0.013 $\pm$ 0.001	0.114 $\pm$ 0.005	0.131 $\pm$ 0.009	0.362 $\pm$ 0.015	0.321	0.005	1.49	0.1190	0.1190	1.176	95.882 $\pm$ 0.880	0.527 $\pm$ 0.017	—
LSAM	beta33	0.000	0.018	0.018 $\pm$ 0.007	—	0.026 $\pm$ 0.002	0.163 $\pm$ 0.007	0.027 $\pm$ 0.002	0.163 $\pm$ 0.008	—	—	0.002	0.002	—	0.1628	0.1628	1.284	98.627 $\pm$ 0.515	0.818 $\pm$ 0.035	1.373 $\pm$ 0.515
LSAM	beta34	0.000	-0.001	-0.001 $\pm$ 0.009	—	0.044 $\pm$ 0.004	0.209 $\pm$ 0.010	0.044 $\pm$ 0.004	0.209 $\pm$ 0.011	—	—	0.001	0.001	—	0.2048	0.2048	1.251	99.216 $\pm$ 0.391	1.025 $\pm$ 0.053	0.784 $\pm$ 0.391
LSAM	beta35	0.000	-0.024	-0.024 $\pm$ 0.008	—	0.037 $\pm$ 0.003	0.192 $\pm$ 0.008	0.037 $\pm$ 0.003	0.193 $\pm$ 0.009	—	—	-0.006	-0.006	—	0.1871	0.1872	1.460	99.216 $\pm$ 0.391	1.098 $\pm$ 0.067	0.784 $\pm$ 0.391
QML	beta31	0.316	0.317	0.001 $\pm$ 0.004	0.002 $\pm$ 0.013	0.009 $\pm$ 0.001	0.093 $\pm$ 0.003	0.009 $\pm$ 0.001	0.093 $\pm$ 0.004	0.086 $\pm$ 0.006	0.294 $\pm$ 0.012	0.325	0.009	2.89	0.0953	0.0957	1.017	95.098 $\pm$ 0.956	0.371 $\pm$ 0.003	—
QML	beta32	0.316	0.317	0.001 $\pm$ 0.004	0.003 $\pm$ 0.014	0.010 $\pm$ 0.001	0.100 $\pm$ 0.003	0.010 $\pm$ 0.001	0.099 $\pm$ 0.004	0.099 $\pm$ 0.007	0.315 $\pm$ 0.013	0.315	-0.001	-0.23	0.0993	0.0993	0.945	93.725 $\pm$ 1.074	0.369 $\pm$ 0.003	—
QML	beta33	0.000	-0.021	-0.021 $\pm$ 0.006	—	0.016 $\pm$ 0.001	0.128 $\pm$ 0.005	0.017 $\pm$ 0.001	0.130 $\pm$ 0.006	—	—	-0.025	-0.025	—	0.1321	0.1344	1.035	95.490 $\pm$ 0.919	0.519 $\pm$ 0.011	4.510 $\pm$ 0.919
QML	beta34	0.000	0.008	0.008 $\pm$ 0.004	—	0.010 $\pm$ 0.001	0.098 $\pm$ 0.004	0.010 $\pm$ 0.001	0.098 $\pm$ 0.005	—	—	0.007	0.007	—	0.0964	0.0966	1.005	94.706 $\pm$ 0.992	0.386 $\pm$ 0.008	5.294 $\pm$ 0.992
QML	beta35	0.000	-0.001	-0.001 $\pm$ 0.004	—	0.007 $\pm$ 0.000	0.083 $\pm$ 0.003	0.007 $\pm$ 0.000	0.083 $\pm$ 0.003	—	—	0.001	0.001	—	0.0935	0.0936	1.144	96.471 $\pm$ 0.817	0.373 $\pm$ 0.007	3.529 $\pm$ 0.817
UPI	beta31	0.316	0.316	-0.000 $\pm$ 0.005	-0.000 $\pm$ 0.016	0.013 $\pm$ 0.001	0.112 $\pm$ 0.004	0.013 $\pm$ 0.001	0.112 $\pm$ 0.005	0.126 $\pm$ 0.008	0.355 $\pm$ 0.014	0.319	0.003	0.95	0.1236	0.1236	1.518	98.235 $\pm$ 0.583	0.668 $\pm$ 0.033	—
UPI	beta32	0.316	0.330	0.014 $\pm$ 0.005	0.043 $\pm$ 0.017	0.015 $\pm$ 0.001	0.124 $\pm$ 0.005	0.016 $\pm$ 0.001	0.125 $\pm$ 0.006	0.155 $\pm$ 0.013	0.394 $\pm$ 0.019	0.321	0.005	1.59	0.1200	0.1201	1.496	97.647 $\pm$ 0.671	0.727 $\pm$ 0.041	—
UPI	beta33	0.000	0.021	0.021 $\pm$ 0.007	—	0.029 $\pm$ 0.003	0.169 $\pm$ 0.008	0.029 $\pm$ 0.003	0.170 $\pm$ 0.009	—	—	0.011	0.011	—	0.1604	0.1608	1.904	99.608 $\pm$ 0.277	1.260 $\pm$ 0.093	0.392 $\pm$ 0.277
UPI	beta34	0.000	0.012	0.012 $\pm$ 0.011	—	0.067 $\pm$ 0.006	0.259 $\pm$ 0.012	0.067 $\pm$ 0.006	0.259 $\pm$ 0.014	—	—	0.007	0.007	—	0.2089	0.2090	1.997	99.804 $\pm$ 0.196	2.026 $\pm$ 0.140	0.196 $\pm$ 0.196
UPI	beta35	0.000	-0.027	-0.027 $\pm$ 0.011	—	0.056 $\pm$ 0.005	0.237 $\pm$ 0.011	0.057 $\pm$ 0.005	0.239 $\pm$ 0.012	—	—	-0.006	-0.006	—	0.2167	0.2168	2.057	99.804 $\pm$ 0.196	1.914 $\pm$ 0.139	0.196 $\pm$ 0.196

Note. MCSE = Monte Carlo Standard Error shown as $\backslash u00b1$ value. Coverage, Type I Error, and Power expressed as percentages. $\backslash u0014$ indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S122

Simulation 1 - Condition 122: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.4

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.318	0.002 ± 0.002	0.008 ± 0.007	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	0.040 ± 0.002	0.200 ± 0.007	0.318	0.002	0.67	0.0625	0.0625	1.004	95.329 ± 0.730	0.249 ± 0.003	—
LMS	beta32	0.316	0.316	0.000 ± 0.002	0.001 ± 0.007	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	0.038 ± 0.002	0.196 ± 0.006	0.314	-0.002	-0.69	0.0634	0.0635	1.019	96.048 ± 0.674	0.248 ± 0.003	—
LMS	beta33	0.000	-0.024	-0.024 ± 0.003	—	0.008 ± 0.001	0.090 ± 0.003	0.009 ± 0.001	0.093 ± 0.004	—	—	-0.026	-0.026	—	0.0888	0.0927	1.381	97.246 ± 0.566	0.487 ± 0.042	2.754 ± 0.566
LMS	beta34	0.000	0.018	0.018 ± 0.002	—	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.071 ± 0.003	—	—	0.014	0.014	—	0.0661	0.0676	1.308	97.485 ± 0.542	0.354 ± 0.034	2.515 ± 0.542
LMS	beta35	0.000	0.015	0.015 ± 0.002	—	0.005 ± 0.000	0.067 ± 0.003	0.005 ± 0.000	0.069 ± 0.003	—	—	0.015	0.015	—	0.0674	0.0690	1.233	95.808 ± 0.694	0.325 ± 0.017	4.192 ± 0.694
LSAM	beta31	0.316	0.319	0.003 ± 0.002	0.010 ± 0.007	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.044 ± 0.002	0.209 ± 0.007	0.319	0.003	1.02	0.0650	0.0651	1.001	95.210 ± 0.739	0.260 ± 0.002	—
LSAM	beta32	0.316	0.317	0.001 ± 0.002	0.003 ± 0.007	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	0.041 ± 0.002	0.203 ± 0.006	0.315	-0.001	-0.33	0.0645	0.0645	1.034	96.048 ± 0.674	0.260 ± 0.002	—
LSAM	beta33	0.000	0.010	0.010 ± 0.003	—	0.007 ± 0.000	0.082 ± 0.002	0.007 ± 0.000	0.082 ± 0.003	—	—	-0.002	-0.002	—	0.0849	0.0849	1.114	97.964 ± 0.489	0.357 ± 0.004	2.036 ± 0.489
LSAM	beta34	0.000	-0.013	-0.013 ± 0.004	—	0.012 ± 0.001	0.111 ± 0.003	0.013 ± 0.001	0.112 ± 0.004	—	—	-0.003	-0.003	—	0.1091	0.1091	1.096	98.683 ± 0.395	0.479 ± 0.008	1.317 ± 0.395
LSAM	beta35	0.000	-0.011	-0.011 ± 0.004	—	0.014 ± 0.001	0.118 ± 0.004	0.014 ± 0.001	0.119 ± 0.004	—	—	-0.003	-0.003	—	0.1145	0.1145	1.048	98.563 ± 0.412	0.487 ± 0.009	1.437 ± 0.412
QML	beta31	0.316	0.317	0.001 ± 0.002	0.002 ± 0.007	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	0.036 ± 0.002	0.189 ± 0.006	0.315	-0.001	-0.26	0.0596	0.0596	0.964	94.491 ± 0.790	0.226 ± 0.001	—
QML	beta32	0.316	0.315	-0.001 ± 0.002	-0.003 ± 0.006	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.034 ± 0.002	0.185 ± 0.006	0.313	-0.003	-0.88	0.0601	0.0602	0.984	95.449 ± 0.721	0.225 ± 0.001	—
QML	beta33	0.000	-0.014	-0.014 ± 0.002	—	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	—	—	-0.023	-0.023	—	0.0748	0.0782	1.120	96.407 ± 0.644	0.308 ± 0.003	3.593 ± 0.644
QML	beta34	0.000	-0.002	-0.002 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	—	—	0.000	0.000	—	0.0579	0.0579	1.085	96.048 ± 0.674	0.224 ± 0.002	3.952 ± 0.674
QML	beta35	0.000	-0.004	-0.004 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	—	—	0.000	0.000	—	0.0543	0.0543	1.093	94.970 ± 0.756	0.226 ± 0.003	5.030 ± 0.756
UPI	beta31	0.316	0.319	0.003 ± 0.002	0.010 ± 0.007	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.046 ± 0.003	0.214 ± 0.007	0.317	0.001	0.40	0.0669	0.0670	1.100	96.168 ± 0.664	0.292 ± 0.008	—
UPI	beta32	0.316	0.317	0.001 ± 0.002	0.004 ± 0.007	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.043 ± 0.002	0.207 ± 0.006	0.315	-0.001	-0.36	0.0675	0.0676	1.123	96.766 ± 0.612	0.287 ± 0.004	—
UPI	beta33	0.000	0.010	0.010 ± 0.003	—	0.007 ± 0.000	0.085 ± 0.002	0.007 ± 0.000	0.086 ± 0.003	—	—	0.001	0.001	—	0.0872	0.0872	1.264	98.323 ± 0.444	0.424 ± 0.008	1.677 ± 0.444
UPI	beta34	0.000	-0.014	-0.014 ± 0.005	—	0.021 ± 0.001	0.143 ± 0.005	0.021 ± 0.001	0.144 ± 0.006	—	—	-0.002	-0.002	—	0.1216	0.1216	1.360	99.521 ± 0.239	0.763 ± 0.033	0.479 ± 0.239
UPI	beta35	0.000	-0.007	-0.007 ± 0.005	—	0.021 ± 0.001	0.144 ± 0.005	0.021 ± 0.001	0.144 ± 0.005	—	—	0.002	0.002	—	0.1221	0.1221	1.382	99.641 ± 0.207	0.781 ± 0.049	0.359 ± 0.207

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S123

Simulation 1 - Condition 123: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.6

Method	Param.	True	Mean-based										Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE	
LMS	beta31	0.316	0.319	0.003 ± 0.002	0.009 ± 0.007	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.048 ± 0.002	0.220 ± 0.006	0.315	-0.001	-0.19	0.0665	0.0665	1.010	95.897 ± 0.635	0.275 ± 0.002		
LMS	beta32	0.316	0.320	0.004 ± 0.002	0.012 ± 0.007	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.050 ± 0.002	0.224 ± 0.007	0.323	0.007	2.08	0.0702	0.0705	0.994	94.769 ± 0.713	0.275 ± 0.002	—	
LMS	beta33	0.000	-0.017	-0.017 ± 0.003	—	0.009 ± 0.001	0.096 ± 0.003	0.009 ± 0.001	0.097 ± 0.003	—	—	-0.006	-0.006	—	0.0916	0.0918	1.066	96.000 ± 0.628	0.400 ± 0.013	4.000 ± 0.628	
LMS	beta34	0.000	0.018	0.018 ± 0.002	—	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.077 ± 0.003	—	—	0.008	0.008	—	0.0692	0.0697	1.130	95.487 ± 0.665	0.330 ± 0.014	4.513 ± 0.665	
LMS	beta35	0.000	0.015	0.015 ± 0.003	—	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.080 ± 0.003	—	—	0.006	0.006	—	0.0730	0.0733	1.059	95.282 ± 0.679	0.326 ± 0.009	4.718 ± 0.679	
LSAM	beta31	0.316	0.319	0.003 ± 0.002	0.010 ± 0.007	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.052 ± 0.002	0.227 ± 0.006	0.316	0.000	0.14	0.0678	0.0678	0.996	94.564 ± 0.726	0.280 ± 0.001	—	
LSAM	beta32	0.316	0.320	0.004 ± 0.002	0.014 ± 0.007	0.005 ± 0.000	0.074 ± 0.002	0.005 ± 0.000	0.074 ± 0.002	0.055 ± 0.003	0.234 ± 0.007	0.323	0.007	2.15	0.0745	0.0748	0.973	94.564 ± 0.726	0.282 ± 0.001	—	
LSAM	beta33	0.000	0.006	0.006 ± 0.003	—	0.008 ± 0.000	0.089 ± 0.002	0.008 ± 0.000	0.089 ± 0.003	—	—	0.010	0.010	—	0.0884	0.0890	0.975	94.872 ± 0.706	0.339 ± 0.003	5.128 ± 0.706	
LSAM	beta34	0.000	0.003	0.003 ± 0.003	—	0.010 ± 0.001	0.102 ± 0.002	0.010 ± 0.001	0.102 ± 0.003	—	—	0.002	0.002	—	0.0988	0.0988	1.035	96.513 ± 0.588	0.412 ± 0.003	3.487 ± 0.588	
LSAM	beta35	0.000	-0.005	-0.005 ± 0.003	—	0.012 ± 0.001	0.108 ± 0.003	0.012 ± 0.001	0.108 ± 0.003	—	—	-0.006	-0.006	—	0.1065	0.1067	0.995	96.410 ± 0.596	0.421 ± 0.004	3.590 ± 0.596	
QML	beta31	0.316	0.318	0.002 ± 0.002	0.006 ± 0.007	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.046 ± 0.002	0.215 ± 0.006	0.315	-0.001	-0.36	0.0642	0.0642	0.996	95.179 ± 0.686	0.265 ± 0.001	—	
QML	beta32	0.316	0.319	0.003 ± 0.002	0.009 ± 0.007	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.048 ± 0.002	0.220 ± 0.006	0.322	0.006	1.83	0.0697	0.0699	0.977	94.359 ± 0.739	0.266 ± 0.001	—	
QML	beta33	0.000	-0.011	-0.011 ± 0.003	—	0.007 ± 0.000	0.086 ± 0.002	0.008 ± 0.000	0.087 ± 0.003	—	—	-0.007	-0.007	—	0.0883	0.0885	0.981	95.282 ± 0.679	0.333 ± 0.003	4.718 ± 0.679	
QML	beta34	0.000	0.008	0.008 ± 0.002	—	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	—	—	0.003	0.003	—	0.0659	0.0659	1.043	94.974 ± 0.700	0.271 ± 0.003	5.026 ± 0.700	
QML	beta35	0.000	0.004	0.004 ± 0.002	—	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	—	—	0.000	0.000	—	0.0679	0.0679	1.009	94.667 ± 0.720	0.275 ± 0.003	5.333 ± 0.720	
UPI	beta31	0.316	0.319	0.003 ± 0.002	0.011 ± 0.007	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.053 ± 0.002	0.230 ± 0.006	0.318	0.002	0.53	0.0713	0.0713	1.002	94.974 ± 0.700	0.286 ± 0.002	—	
UPI	beta32	0.316	0.320	0.004 ± 0.002	0.014 ± 0.008	0.005 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.074 ± 0.002	0.055 ± 0.003	0.235 ± 0.007	0.322	0.006	2.03	0.0741	0.0744	0.986	94.872 ± 0.706	0.286 ± 0.001	—	
UPI	beta33	0.000	0.005	0.005 ± 0.003	—	0.008 ± 0.000	0.092 ± 0.002	0.009 ± 0.000	0.092 ± 0.003	—	—	0.008	0.008	—	0.0879	0.0882	0.997	95.897 ± 0.635	0.360 ± 0.004	4.103 ± 0.635	
UPI	beta34	0.000	0.002	0.002 ± 0.004	—	0.014 ± 0.001	0.119 ± 0.003	0.014 ± 0.001	0.119 ± 0.004	—	—	0.004	0.004	—	0.1058	0.1059	1.096	97.949 ± 0.454	0.510 ± 0.008	2.051 ± 0.454	
UPI	beta35	0.000	-0.003	-0.003 ± 0.004	—	0.014 ± 0.001	0.120 ± 0.003	0.014 ± 0.001	0.120 ± 0.004	—	—	-0.005	-0.005	—	0.1101	0.1102	1.102	97.641 ± 0.486	0.518 ± 0.008	2.359 ± 0.486	

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S124

Simulation 1 - Condition 124: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.318	0.002 ± 0.001	0.007 ± 0.004	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.018 ± 0.001	0.136 ± 0.004	0.316	0.000	0.13	0.0433	0.0433	1.057	95.477 ± 0.659	0.177 ± 0.009	—
LMS	beta32	0.316	0.316	0.000 ± 0.001	0.001 ± 0.004	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.016 ± 0.001	0.128 ± 0.004	0.316	0.000	-0.12	0.0399	0.0399	1.067	96.482 ± 0.584	0.169 ± 0.001	—
LMS	beta33	0.000	-0.015	-0.015 ± 0.002	—	0.003 ± 0.000	0.057 ± 0.002	0.004 ± 0.000	0.059 ± 0.002	—	—	-0.008	-0.008	—	0.0528	0.0535	1.014	96.583 ± 0.576	0.228 ± 0.004	3.417 ± 0.576
LMS	beta34	0.000	0.007	0.007 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	0.003	0.003	—	0.0439	0.0440	1.017	95.879 ± 0.630	0.183 ± 0.002	4.121 ± 0.630
LMS	beta35	0.000	0.009	0.009 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.047 ± 0.002	—	—	0.005	0.005	—	0.0433	0.0435	1.051	95.779 ± 0.637	0.190 ± 0.005	4.221 ± 0.637
LSAM	beta31	0.316	0.319	0.003 ± 0.001	0.009 ± 0.004	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.019 ± 0.001	0.139 ± 0.004	0.319	0.003	1.04	0.0435	0.0436	1.001	94.774 ± 0.706	0.173 ± 0.000	—
LSAM	beta32	0.316	0.317	0.001 ± 0.001	0.002 ± 0.004	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.018 ± 0.001	0.133 ± 0.004	0.316	0.000	-0.00	0.0430	0.0430	1.046	95.779 ± 0.637	0.173 ± 0.000	—
LSAM	beta33	0.000	-0.001	-0.001 ± 0.002	—	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	—	—	0.000	0.000	—	0.0508	0.0508	0.961	94.070 ± 0.749	0.204 ± 0.001	5.930 ± 0.749
LSAM	beta34	0.000	-0.002	-0.002 ± 0.002	—	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	—	—	-0.001	-0.001	—	0.0661	0.0661	0.988	94.774 ± 0.706	0.253 ± 0.001	5.226 ± 0.706
LSAM	beta35	0.000	0.002	0.002 ± 0.002	—	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	—	—	0.000	0.000	—	0.0628	0.0628	0.992	94.472 ± 0.724	0.253 ± 0.001	5.528 ± 0.724
QML	beta31	0.316	0.318	0.002 ± 0.001	0.005 ± 0.004	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.018 ± 0.001	0.134 ± 0.004	0.317	0.001	0.37	0.0426	0.0426	1.002	95.176 ± 0.679	0.167 ± 0.001	—
QML	beta32	0.316	0.316	-0.000 ± 0.001	-0.001 ± 0.004	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.016 ± 0.001	0.127 ± 0.003	0.315	-0.001	-0.37	0.0398	0.0398	1.056	96.683 ± 0.568	0.166 ± 0.000	—
QML	beta33	0.000	-0.014	-0.014 ± 0.002	—	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	—	—	-0.009	-0.009	—	0.0514	0.0521	0.979	94.472 ± 0.724	0.206 ± 0.001	5.528 ± 0.724
QML	beta34	0.000	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	—	—	-0.001	-0.001	—	0.0426	0.0426	1.003	95.176 ± 0.679	0.169 ± 0.001	4.824 ± 0.679
QML	beta35	0.000	0.003	0.003 ± 0.001	—	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	—	—	0.000	0.000	—	0.0419	0.0419	1.009	94.975 ± 0.693	0.170 ± 0.001	5.025 ± 0.693
UPI	beta31	0.316	0.319	0.003 ± 0.001	0.009 ± 0.004	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.020 ± 0.001	0.140 ± 0.004	0.319	0.003	0.94	0.0439	0.0440	1.001	94.171 ± 0.743	0.173 ± 0.000	—
UPI	beta32	0.316	0.317	0.001 ± 0.001	0.003 ± 0.004	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.018 ± 0.001	0.134 ± 0.004	0.316	0.000	0.07	0.0433	0.0433	1.044	95.879 ± 0.630	0.173 ± 0.000	—
UPI	beta33	0.000	-0.001	-0.001 ± 0.002	—	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	—	—	0.000	0.000	—	0.0516	0.0516	0.964	94.171 ± 0.743	0.207 ± 0.001	5.829 ± 0.743
UPI	beta34	0.000	-0.002	-0.002 ± 0.002	—	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	—	—	-0.001	-0.001	—	0.0645	0.0645	1.018	95.477 ± 0.659	0.272 ± 0.002	4.523 ± 0.659
UPI	beta35	0.000	0.000	0.000 ± 0.002	—	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	—	—	0.000	0.000	—	0.0659	0.0659	1.019	95.980 ± 0.623	0.271 ± 0.002	4.020 ± 0.623

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S125

Simulation 1 - Condition 125: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.8

Mean-based												Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.319	0.003 ± 0.002	0.008 ± 0.005	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.030 ± 0.001	0.174 ± 0.005	0.318	0.002	0.73	0.0557	0.0558	0.998	95.491 ± 0.657	0.214 ± 0.001	—
LMS	beta32	0.316	0.315	-0.001 ± 0.002	-0.004 ± 0.005	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	0.029 ± 0.001	0.170 ± 0.005	0.313	-0.003	-0.93	0.0547	0.0548	1.022	95.391 ± 0.664	0.215 ± 0.001	—
LMS	beta33	0.000	-0.003	-0.003 ± 0.002	—	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	—	—	-0.001	-0.001	—	0.0590	0.0590	1.000	93.788 ± 0.764	0.248 ± 0.002	6.212 ± 0.764
LMS	beta34	0.000	0.002	0.002 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	—	—	0.000	0.000	—	0.0611	0.0611	0.954	93.287 ± 0.792	0.230 ± 0.002	6.713 ± 0.792
LMS	beta35	0.000	0.004	0.004 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	—	—	0.000	0.000	—	0.0628	0.0628	0.978	95.391 ± 0.664	0.234 ± 0.003	4.609 ± 0.664
LSAM	beta31	0.316	0.319	0.003 ± 0.002	0.010 ± 0.006	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.031 ± 0.001	0.177 ± 0.005	0.319	0.003	0.96	0.0573	0.0574	0.998	95.090 ± 0.684	0.219 ± 0.001	—
LSAM	beta32	0.316	0.315	-0.001 ± 0.002	-0.003 ± 0.005	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.001	0.030 ± 0.001	0.173 ± 0.005	0.313	-0.003	-0.83	0.0555	0.0556	1.023	95.491 ± 0.657	0.219 ± 0.001	—
LSAM	beta33	0.000	0.004	0.004 ± 0.002	—	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	—	—	0.003	0.003	—	0.0602	0.0603	0.987	93.186 ± 0.798	0.240 ± 0.001	6.814 ± 0.798
LSAM	beta34	0.000	-0.004	-0.004 ± 0.002	—	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	—	—	-0.004	-0.004	—	0.0743	0.0744	0.942	92.886 ± 0.814	0.276 ± 0.001	7.114 ± 0.814
LSAM	beta35	0.000	-0.001	-0.001 ± 0.002	—	0.005 ± 0.000	0.074 ± 0.002	0.005 ± 0.000	0.074 ± 0.002	—	—	-0.003	-0.003	—	0.0756	0.0757	0.963	95.090 ± 0.684	0.278 ± 0.001	4.910 ± 0.684
QML	beta31	0.316	0.318	0.002 ± 0.002	0.008 ± 0.005	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.030 ± 0.001	0.173 ± 0.005	0.318	0.002	0.66	0.0558	0.0559	0.999	95.391 ± 0.664	0.214 ± 0.001	—
QML	beta32	0.316	0.315	-0.001 ± 0.002	-0.005 ± 0.005	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	0.029 ± 0.001	0.169 ± 0.005	0.313	-0.003	-1.05	0.0549	0.0550	1.022	95.391 ± 0.664	0.215 ± 0.001	—
QML	beta33	0.000	-0.003	-0.003 ± 0.002	—	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	—	—	-0.001	-0.001	—	0.0586	0.0586	1.019	93.788 ± 0.764	0.249 ± 0.006	6.212 ± 0.764
QML	beta34	0.000	0.000	-0.000 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	—	—	-0.002	-0.002	—	0.0602	0.0603	0.965	93.387 ± 0.787	0.229 ± 0.003	6.613 ± 0.787
QML	beta35	0.000	0.002	0.002 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	—	—	-0.001	-0.001	—	0.0624	0.0624	0.981	95.291 ± 0.671	0.230 ± 0.002	4.709 ± 0.671
UPI	beta31	0.316	0.319	0.003 ± 0.002	0.010 ± 0.006	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.031 ± 0.001	0.177 ± 0.005	0.318	0.002	0.78	0.0573	0.0573	0.991	94.890 ± 0.697	0.217 ± 0.001	—
UPI	beta32	0.316	0.315	-0.001 ± 0.002	-0.003 ± 0.005	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.001	0.030 ± 0.001	0.173 ± 0.005	0.313	-0.003	-1.08	0.0557	0.0558	1.016	94.990 ± 0.691	0.218 ± 0.001	—
UPI	beta33	0.000	0.003	0.003 ± 0.002	—	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	—	—	0.002	0.002	—	0.0599	0.0600	0.989	93.387 ± 0.787	0.242 ± 0.001	6.613 ± 0.787
UPI	beta34	0.000	-0.004	-0.004 ± 0.002	—	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	—	—	-0.004	-0.004	—	0.0743	0.0744	0.958	93.788 ± 0.764	0.286 ± 0.002	6.212 ± 0.764
UPI	beta35	0.000	-0.001	-0.001 ± 0.002	—	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	—	—	-0.002	-0.002	—	0.0742	0.0743	0.979	95.892 ± 0.628	0.287 ± 0.002	4.108 ± 0.628

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S126

Simulation 1 - Condition 126: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.8

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.316	0.000 ± 0.001	0.001 ± 0.003	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.012 ± 0.001	0.109 ± 0.003	0.316	0.000	0.07	0.0348	0.0348	1.000	95.000 ± 0.689	0.135 ± 0.001	—
LMS	beta32	0.316	0.316	-0.000 ± 0.001	-0.001 ± 0.003	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.011 ± 0.001	0.107 ± 0.003	0.316	0.000	0.01	0.0328	0.0328	1.029	95.800 ± 0.634	0.137 ± 0.001	—
LMS	beta33	0.000	-0.007	-0.007 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	-0.006	-0.006	—	0.0383	0.0388	1.033	95.100 ± 0.683	0.162 ± 0.005	4.900 ± 0.683
LMS	beta34	0.000	0.002	0.002 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.000	0.000	—	0.0364	0.0364	0.987	94.700 ± 0.708	0.143 ± 0.001	5.300 ± 0.708
LMS	beta35	0.000	0.004	0.004 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	0.002	0.002	—	0.0340	0.0341	1.134	96.400 ± 0.589	0.157 ± 0.012	3.600 ± 0.589
LSAM	beta31	0.316	0.317	0.001 ± 0.001	0.002 ± 0.004	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.012 ± 0.001	0.111 ± 0.003	0.316	0.000	0.10	0.0348	0.0348	0.998	94.800 ± 0.702	0.137 ± 0.000	—
LSAM	beta32	0.316	0.316	-0.000 ± 0.001	-0.000 ± 0.003	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.012 ± 0.001	0.109 ± 0.003	0.317	0.001	0.37	0.0335	0.0335	1.014	95.500 ± 0.656	0.138 ± 0.000	—
LSAM	beta33	0.000	-0.002	-0.002 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	-0.001	-0.001	—	0.0378	0.0378	0.998	94.800 ± 0.702	0.153 ± 0.001	5.200 ± 0.702
LSAM	beta34	0.000	0.000	0.000 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	0.000	0.000	—	0.0448	0.0448	0.987	94.400 ± 0.727	0.174 ± 0.001	5.600 ± 0.727
LSAM	beta35	0.000	0.003	0.003 ± 0.001	—	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	—	—	0.002	0.002	—	0.0434	0.0435	1.034	95.800 ± 0.634	0.174 ± 0.001	4.200 ± 0.634
QML	beta31	0.316	0.316	0.000 ± 0.001	0.000 ± 0.003	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.012 ± 0.001	0.109 ± 0.003	0.316	0.000	0.01	0.0346	0.0346	1.002	94.800 ± 0.702	0.136 ± 0.000	—
QML	beta32	0.316	0.316	-0.000 ± 0.001	-0.001 ± 0.003	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.011 ± 0.001	0.107 ± 0.003	0.316	0.000	-0.03	0.0326	0.0326	1.024	95.600 ± 0.649	0.136 ± 0.001	—
QML	beta33	0.000	-0.007	-0.007 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	-0.006	-0.006	—	0.0381	0.0387	1.062	95.100 ± 0.683	0.165 ± 0.007	4.900 ± 0.683
QML	beta34	0.000	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.001	-0.001	—	0.0366	0.0366	1.011	94.300 ± 0.733	0.146 ± 0.002	5.700 ± 0.733
QML	beta35	0.000	0.003	0.003 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	0.001	0.001	—	0.0340	0.0340	1.100	96.100 ± 0.612	0.151 ± 0.005	3.900 ± 0.612
UPI	beta31	0.316	0.317	0.001 ± 0.001	0.002 ± 0.004	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.012 ± 0.001	0.111 ± 0.003	0.316	0.000	0.12	0.0350	0.0350	0.995	94.700 ± 0.708	0.137 ± 0.000	—
UPI	beta32	0.316	0.316	-0.000 ± 0.001	-0.000 ± 0.003	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.012 ± 0.001	0.109 ± 0.003	0.317	0.001	0.34	0.0333	0.0333	1.013	95.400 ± 0.662	0.137 ± 0.000	—
UPI	beta33	0.000	-0.002	-0.002 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	-0.001	-0.001	—	0.0376	0.0376	0.998	94.700 ± 0.708	0.153 ± 0.001	5.300 ± 0.708
UPI	beta34	0.000	0.000	0.000 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	-0.001	-0.001	—	0.0445	0.0445	0.988	94.400 ± 0.727	0.176 ± 0.001	5.600 ± 0.727
UPI	beta35	0.000	0.003	0.003 ± 0.001	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	—	—	0.002	0.002	—	0.0436	0.0436	1.036	96.100 ± 0.612	0.177 ± 0.001	3.900 ± 0.612

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S127

Simulation 1 - Condition 127: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.4

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.333	0.017 ± 0.008	0.053 ± 0.026	0.011 ± 0.001	0.107 ± 0.007	0.012 ± 0.002	0.108 ± 0.008	0.117 ± 0.016	0.342 ± 0.027	0.332	0.016	4.91	0.1179	0.1189	1.140	96.000 ± 1.481	0.478 ± 0.045	—
LMS	beta32	0.316	0.336	0.020 ± 0.009	0.064 ± 0.028	0.014 ± 0.003	0.118 ± 0.013	0.014 ± 0.003	0.119 ± 0.014	0.142 ± 0.031	0.377 ± 0.045	0.331	0.015	4.69	0.1265	0.1273	0.950	96.000 ± 1.481	0.438 ± 0.029	—
LMS	beta33	0.000	0.027	0.027 ± 0.009	—	0.015 ± 0.002	0.122 ± 0.008	0.016 ± 0.002	0.125 ± 0.009	—	—	0.026	0.026	—	0.1701	0.1721	1.423	93.714 ± 1.835	0.681 ± 0.060	6.286 ± 1.835
LMS	beta34	0.000	-0.030	-0.030 ± 0.006	—	0.006 ± 0.001	0.080 ± 0.008	0.007 ± 0.001	0.085 ± 0.007	—	—	-0.026	-0.026	—	0.0936	0.0971	1.561	87.429 ± 2.506	0.487 ± 0.077	12.571 ± 2.506
LMS	beta35	0.000	-0.035	-0.035 ± 0.007	—	0.009 ± 0.002	0.097 ± 0.011	0.011 ± 0.002	0.103 ± 0.011	—	—	-0.041	-0.041	—	0.0994	0.1076	1.173	84.571 ± 2.731	0.446 ± 0.049	15.429 ± 2.731
LSAM	beta31	0.316	0.358	0.042 ± 0.011	0.134 ± 0.034	0.021 ± 0.002	0.144 ± 0.009	0.022 ± 0.003	0.149 ± 0.011	0.223 ± 0.028	0.472 ± 0.034	0.359	0.043	13.70	0.1417	0.1482	1.077	96.571 ± 1.376	0.606 ± 0.020	—
LSAM	beta32	0.316	0.350	0.034 ± 0.010	0.108 ± 0.033	0.019 ± 0.002	0.137 ± 0.008	0.020 ± 0.002	0.141 ± 0.009	0.199 ± 0.022	0.446 ± 0.030	0.358	0.042	13.14	0.1491	0.1548	1.110	98.286 ± 0.981	0.597 ± 0.017	—
LSAM	beta33	0.000	0.113	0.113 ± 0.010	—	0.019 ± 0.002	0.139 ± 0.009	0.032 ± 0.004	0.178 ± 0.012	—	—	0.121	0.121	—	0.1692	0.2079	1.163	96.571 ± 1.376	0.632 ± 0.023	3.429 ± 1.376
LSAM	beta34	0.000	-0.081	-0.081 ± 0.008	—	0.010 ± 0.002	0.102 ± 0.009	0.017 ± 0.003	0.130 ± 0.011	—	—	-0.075	-0.075	—	0.0979	0.1235	1.086	95.429 ± 1.579	0.433 ± 0.021	4.571 ± 1.579
LSAM	beta35	0.000	-0.074	-0.074 ± 0.007	—	0.009 ± 0.001	0.093 ± 0.005	0.014 ± 0.001	0.119 ± 0.006	—	—	-0.084	-0.084	—	0.1050	0.1345	1.165	92.571 ± 1.982	0.427 ± 0.017	7.429 ± 1.982
QML	beta31	0.316	0.324	0.008 ± 0.007	0.027 ± 0.023	0.009 ± 0.001	0.094 ± 0.006	0.009 ± 0.001	0.094 ± 0.007	0.089 ± 0.010	0.299 ± 0.021	0.325	0.009	2.92	0.0988	0.0993	0.994	94.286 ± 1.755	0.367 ± 0.006	—
QML	beta32	0.316	0.323	0.007 ± 0.007	0.022 ± 0.021	0.008 ± 0.001	0.089 ± 0.005	0.008 ± 0.001	0.089 ± 0.006	0.079 ± 0.009	0.281 ± 0.019	0.328	0.012	3.83	0.0984	0.0991	1.040	96.000 ± 1.481	0.362 ± 0.006	—
QML	beta33	0.000	0.033	0.033 ± 0.008	—	0.012 ± 0.001	0.109 ± 0.005	0.013 ± 0.001	0.113 ± 0.007	—	—	0.046	0.046	—	0.1395	0.1469	1.148	93.714 ± 1.835	0.489 ± 0.017	6.286 ± 1.835
QML	beta34	0.000	-0.048	-0.048 ± 0.005	—	0.004 ± 0.000	0.066 ± 0.004	0.007 ± 0.001	0.081 ± 0.004	—	—	-0.052	-0.052	—	0.0777	0.0933	1.167	89.714 ± 2.296	0.302 ± 0.010	10.286 ± 2.296
QML	beta35	0.000	-0.043	-0.043 ± 0.005	—	0.005 ± 0.001	0.072 ± 0.005	0.007 ± 0.001	0.084 ± 0.005	—	—	-0.056	-0.056	—	0.0830	0.1004	1.091	84.571 ± 2.731	0.309 ± 0.012	15.429 ± 2.731
UPI	beta31	0.316	0.357	0.041 ± 0.018	0.130 ± 0.056	0.054 ± 0.006	0.233 ± 0.014	0.056 ± 0.007	0.236 ± 0.017	0.559 ± 0.069	0.748 ± 0.055	0.348	0.032	10.22	0.2735	0.2754	2.319	90.857 ± 2.179	2.121 ± 0.335	—
UPI	beta32	0.316	0.355	0.039 ± 0.017	0.123 ± 0.053	0.049 ± 0.006	0.220 ± 0.013	0.050 ± 0.006	0.223 ± 0.016	0.499 ± 0.059	0.707 ± 0.050	0.375	0.059	18.57	0.2641	0.2706	2.276	93.714 ± 1.835	1.967 ± 0.267	—
UPI	beta33	0.000	0.068	0.068 ± 0.017	—	0.050 ± 0.008	0.223 ± 0.019	0.054 ± 0.010	0.232 ± 0.023	—	—	0.024	0.024	—	0.2220	0.2233	1.565	98.286 ± 0.981	1.368 ± 0.126	1.714 ± 0.981
UPI	beta34	0.000	-0.127	-0.127 ± 0.017	—	0.052 ± 0.010	0.228 ± 0.023	0.068 ± 0.013	0.261 ± 0.026	—	—	-0.090	-0.090	—	0.2415	0.2579	1.936	99.429 ± 0.570	1.733 ± 0.206	0.571 ± 0.570
UPI	beta35	0.000	-0.153	-0.153 ± 0.022	—	0.088 ± 0.017	0.297 ± 0.031	0.111 ± 0.022	0.333 ± 0.036	—	—	-0.115	-0.115	—	0.2587	0.2831	1.838	97.143 ± 1.259	2.137 ± 0.302	2.857 ± 1.259

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S128

Simulation 1 - Condition 128: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.4

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.325	0.009 ± 0.004	0.028 ± 0.012	0.003 ± 0.000	0.056 ± 0.005	0.003 ± 0.001	0.056 ± 0.005	0.032 ± 0.005	0.178 ± 0.016	0.319	0.003	0.96	0.0751	0.0751	0.994	96.059 ± 1.366	0.217 ± 0.002	
LMS	beta32	0.316	0.320	0.004 ± 0.004	0.012 ± 0.012	0.003 ± 0.000	0.056 ± 0.003	0.003 ± 0.000	0.056 ± 0.004	0.032 ± 0.003	0.178 ± 0.012	0.315	-0.001	-0.17	0.0821	0.0821	0.990	95.567 ± 1.445	0.218 ± 0.003	—
LMS	beta33	0.000	0.054	0.054 ± 0.004	—	0.004 ± 0.000	0.061 ± 0.004	0.007 ± 0.001	0.082 ± 0.004	—	—	0.040	0.040	—	0.0939	0.1019	1.201	84.729 ± 2.525	0.289 ± 0.015	15.271 ± 2.525
LMS	beta34	0.000	-0.047	-0.047 ± 0.003	—	0.002 ± 0.000	0.048 ± 0.004	0.005 ± 0.001	0.068 ± 0.004	—	—	-0.035	-0.035	—	0.0715	0.0798	0.945	70.936 ± 3.187	0.179 ± 0.008	29.064 ± 3.187
LMS	beta35	0.000	-0.048	-0.048 ± 0.003	—	0.002 ± 0.000	0.044 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	—	—	-0.043	-0.043	—	0.0659	0.0786	1.099	64.039 ± 3.368	0.189 ± 0.014	35.961 ± 3.368
LSAM	beta31	0.316	0.357	0.041 ± 0.006	0.131 ± 0.018	0.007 ± 0.001	0.082 ± 0.004	0.008 ± 0.001	0.092 ± 0.005	0.084 ± 0.009	0.290 ± 0.017	0.348	0.032	10.14	0.0921	0.0975	1.083	96.059 ± 1.366	0.348 ± 0.005	—
LSAM	beta32	0.316	0.349	0.033 ± 0.006	0.104 ± 0.019	0.007 ± 0.001	0.083 ± 0.005	0.008 ± 0.001	0.089 ± 0.006	0.080 ± 0.009	0.283 ± 0.018	0.348	0.032	10.25	0.0839	0.0899	1.041	95.074 ± 1.519	0.340 ± 0.005	—
LSAM	beta33	0.000	0.124	0.124 ± 0.006	—	0.006 ± 0.001	0.079 ± 0.006	0.022 ± 0.002	0.147 ± 0.007	—	—	0.116	0.116	—	0.0781	0.1398	1.096	77.340 ± 2.938	0.338 ± 0.007	22.660 ± 2.938
LSAM	beta34	0.000	-0.086	-0.086 ± 0.004	—	0.003 ± 0.000	0.058 ± 0.003	0.011 ± 0.001	0.103 ± 0.004	—	—	-0.080	-0.080	—	0.0606	0.1006	1.059	71.921 ± 3.154	0.240 ± 0.005	28.079 ± 3.154
LSAM	beta35	0.000	-0.080	-0.080 ± 0.004	—	0.003 ± 0.000	0.056 ± 0.003	0.009 ± 0.001	0.097 ± 0.004	—	—	-0.083	-0.083	—	0.0585	0.1019	1.049	68.966 ± 3.247	0.228 ± 0.005	31.034 ± 3.247
QML	beta31	0.316	0.325	0.009 ± 0.004	0.028 ± 0.012	0.003 ± 0.000	0.053 ± 0.003	0.003 ± 0.000	0.053 ± 0.004	0.029 ± 0.003	0.169 ± 0.011	0.325	0.009	2.69	0.0575	0.0581	1.050	96.059 ± 1.366	0.218 ± 0.002	—
QML	beta32	0.316	0.322	0.006 ± 0.004	0.018 ± 0.013	0.003 ± 0.000	0.057 ± 0.003	0.003 ± 0.000	0.057 ± 0.004	0.032 ± 0.003	0.179 ± 0.012	0.322	0.006	1.98	0.0583	0.0586	0.981	95.567 ± 1.445	0.217 ± 0.002	—
QML	beta33	0.000	0.055	0.055 ± 0.005	—	0.004 ± 0.000	0.066 ± 0.004	0.007 ± 0.001	0.086 ± 0.005	—	—	0.056	0.056	—	0.0673	0.0878	1.115	86.207 ± 2.420	0.287 ± 0.006	13.793 ± 2.420
QML	beta34	0.000	-0.057	-0.057 ± 0.003	—	0.002 ± 0.000	0.047 ± 0.002	0.005 ± 0.000	0.074 ± 0.003	—	—	-0.055	-0.055	—	0.0501	0.0742	0.988	70.936 ± 3.187	0.181 ± 0.004	29.064 ± 3.187
QML	beta35	0.000	-0.060	-0.060 ± 0.003	—	0.002 ± 0.000	0.046 ± 0.002	0.006 ± 0.000	0.075 ± 0.003	—	—	-0.061	-0.061	—	0.0465	0.0765	1.019	62.562 ± 3.397	0.182 ± 0.004	37.438 ± 3.397
UPI	beta31	0.316	0.356	0.040 ± 0.027	0.128 ± 0.084	0.143 ± 0.026	0.378 ± 0.035	0.144 ± 0.027	0.379 ± 0.038	1.440 ± 0.267	1.200 ± 0.122	0.370	0.054	17.11	0.3960	0.3996	1.406	85.714 ± 2.456	2.083 ± 0.265	—
UPI	beta32	0.316	0.383	0.067 ± 0.024	0.213 ± 0.077	0.121 ± 0.018	0.347 ± 0.026	0.125 ± 0.019	0.353 ± 0.030	1.248 ± 0.194	1.117 ± 0.096	0.371	0.055	17.50	0.3308	0.3354	1.525	90.148 ± 2.092	2.077 ± 0.286	—
UPI	beta33	0.000	0.104	0.104 ± 0.023	—	0.106 ± 0.021	0.326 ± 0.033	0.117 ± 0.025	0.341 ± 0.039	—	—	0.018	0.018	—	0.2467	0.2474	1.265	95.567 ± 1.445	1.616 ± 0.203	4.433 ± 1.445
UPI	beta34	0.000	-0.219	-0.219 ± 0.025	—	0.128 ± 0.020	0.357 ± 0.028	0.175 ± 0.025	0.418 ± 0.032	—	—	-0.213	-0.213	—	0.3382	0.3999	1.665	97.537 ± 1.088	2.331 ± 0.264	2.463 ± 1.088
UPI	beta35	0.000	-0.203	-0.203 ± 0.024	—	0.113 ± 0.021	0.336 ± 0.033	0.153 ± 0.024	0.392 ± 0.033	—	—	-0.208	-0.208	—	0.3422	0.4004	1.691	96.552 ± 1.281	2.223 ± 0.303	3.448 ± 1.281

Note. MCSE = Monte Carlo Standard Error shown as $\backslash u00b1$ value. Coverage, Type I Error, and Power expressed as percentages. $\backslash u2014$ indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S129

Simulation 1 - Condition 129: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.6

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Type I (%) $\pm$ MCSE
LMS	beta31	0.316	0.317	0.001 $\pm$ 0.002	0.002 $\pm$ 0.007	0.005 $\pm$ 0.000	0.069 $\pm$ 0.002	0.005 $\pm$ 0.000	0.069 $\pm$ 0.002	0.048 $\pm$ 0.002	0.220 $\pm$ 0.006	0.318	0.002	0.62	0.0704	0.0704	0.987	95.026 $\pm$ 0.700	0.269 $\pm$ 0.002	—
LMS	beta32	0.316	0.317	0.001 $\pm$ 0.002	0.003 $\pm$ 0.007	0.005 $\pm$ 0.000	0.069 $\pm$ 0.002	0.005 $\pm$ 0.000	0.069 $\pm$ 0.002	0.048 $\pm$ 0.002	0.219 $\pm$ 0.006	0.319	0.003	0.92	0.0692	0.0693	0.997	95.440 $\pm$ 0.672	0.270 $\pm$ 0.002	—
LMS	beta33	0.000	0.003	0.003 $\pm$ 0.003	—	0.008 $\pm$ 0.000	0.088 $\pm$ 0.002	0.008 $\pm$ 0.000	0.088 $\pm$ 0.003	—	—	0.010	0.010	—	0.0841	0.0847	1.068	94.301 $\pm$ 0.746	0.366 $\pm$ 0.008	5.699 $\pm$ 0.746
LMS	beta34	0.000	-0.012	-0.012 $\pm$ 0.002	—	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	0.004 $\pm$ 0.000	0.062 $\pm$ 0.002	—	—	-0.015	-0.015	—	0.0571	0.0590	1.017	90.466 $\pm$ 0.945	0.241 $\pm$ 0.006	9.534 $\pm$ 0.945
LMS	beta35	0.000	-0.015	-0.015 $\pm$ 0.002	—	0.003 $\pm$ 0.000	0.057 $\pm$ 0.002	0.003 $\pm$ 0.000	0.058 $\pm$ 0.002	—	—	-0.017	-0.017	—	0.0504	0.0532	1.052	91.295 $\pm$ 0.907	0.233 $\pm$ 0.004	8.705 $\pm$ 0.907
LSAM	beta31	0.316	0.322	0.006 $\pm$ 0.002	0.020 $\pm$ 0.008	0.006 $\pm$ 0.000	0.077 $\pm$ 0.002	0.006 $\pm$ 0.000	0.077 $\pm$ 0.002	0.059 $\pm$ 0.003	0.243 $\pm$ 0.007	0.324	0.008	2.58	0.0788	0.0792	0.978	94.819 $\pm$ 0.714	0.293 $\pm$ 0.001	—
LSAM	beta32	0.316	0.323	0.007 $\pm$ 0.002	0.023 $\pm$ 0.008	0.006 $\pm$ 0.000	0.077 $\pm$ 0.002	0.006 $\pm$ 0.000	0.078 $\pm$ 0.002	0.060 $\pm$ 0.003	0.246 $\pm$ 0.007	0.323	0.007	2.35	0.0765	0.0769	0.972	94.404 $\pm$ 0.740	0.295 $\pm$ 0.001	—
LSAM	beta33	0.000	0.038	0.038 $\pm$ 0.003	—	0.007 $\pm$ 0.000	0.086 $\pm$ 0.002	0.009 $\pm$ 0.000	0.094 $\pm$ 0.003	—	—	0.035	0.035	—	0.0864	0.0932	1.001	91.606 $\pm$ 0.893	0.338 $\pm$ 0.003	8.394 $\pm$ 0.893
LSAM	beta34	0.000	-0.029	-0.029 $\pm$ 0.002	—	0.004 $\pm$ 0.000	0.063 $\pm$ 0.002	0.005 $\pm$ 0.000	0.069 $\pm$ 0.002	—	—	-0.026	-0.026	—	0.0605	0.0659	0.956	90.052 $\pm$ 0.964	0.234 $\pm$ 0.002	9.948 $\pm$ 0.964
LSAM	beta35	0.000	-0.031	-0.031 $\pm$ 0.002	—	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	—	—	-0.031	-0.031	—	0.0553	0.0633	1.009	90.155 $\pm$ 0.959	0.237 $\pm$ 0.002	9.845 $\pm$ 0.959
QML	beta31	0.316	0.316	0.000 $\pm$ 0.002	0.001 $\pm$ 0.007	0.005 $\pm$ 0.000	0.069 $\pm$ 0.002	0.005 $\pm$ 0.000	0.069 $\pm$ 0.002	0.048 $\pm$ 0.002	0.219 $\pm$ 0.006	0.318	0.002	0.59	0.0694	0.0695	0.980	94.611 $\pm$ 0.727	0.266 $\pm$ 0.001	—
QML	beta32	0.316	0.317	0.001 $\pm$ 0.002	0.004 $\pm$ 0.007	0.005 $\pm$ 0.000	0.069 $\pm$ 0.002	0.005 $\pm$ 0.000	0.069 $\pm$ 0.002	0.048 $\pm$ 0.002	0.219 $\pm$ 0.006	0.317	0.001	0.36	0.0674	0.0674	0.988	95.130 $\pm$ 0.693	0.268 $\pm$ 0.001	—
QML	beta33	0.000	0.005	0.005 $\pm$ 0.003	—	0.007 $\pm$ 0.000	0.083 $\pm$ 0.002	0.007 $\pm$ 0.000	0.083 $\pm$ 0.002	—	—	0.008	0.008	—	0.0828	0.0831	1.028	93.886 $\pm$ 0.771	0.333 $\pm$ 0.003	6.114 $\pm$ 0.771
QML	beta34	0.000	-0.017	-0.017 $\pm$ 0.002	—	0.003 $\pm$ 0.000	0.057 $\pm$ 0.001	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	—	—	-0.018	-0.018	—	0.0560	0.0587	0.982	90.259 $\pm$ 0.955	0.220 $\pm$ 0.003	9.741 $\pm$ 0.955
QML	beta35	0.000	-0.019	-0.019 $\pm$ 0.002	—	0.003 $\pm$ 0.000	0.055 $\pm$ 0.001	0.003 $\pm$ 0.000	0.058 $\pm$ 0.002	—	—	-0.021	-0.021	—	0.0503	0.0545	1.031	90.674 $\pm$ 0.936	0.221 $\pm$ 0.002	9.326 $\pm$ 0.936
UPI	beta31	0.316	0.324	0.008 $\pm$ 0.003	0.024 $\pm$ 0.008	0.006 $\pm$ 0.000	0.080 $\pm$ 0.002	0.006 $\pm$ 0.000	0.080 $\pm$ 0.002	0.064 $\pm$ 0.003	0.253 $\pm$ 0.007	0.323	0.007	2.33	0.0782	0.0786	1.005	95.544 $\pm$ 0.664	0.314 $\pm$ 0.003	—
UPI	beta32	0.316	0.324	0.008 $\pm$ 0.003	0.025 $\pm$ 0.008	0.007 $\pm$ 0.000	0.081 $\pm$ 0.002	0.007 $\pm$ 0.000	0.081 $\pm$ 0.002	0.066 $\pm$ 0.003	0.256 $\pm$ 0.008	0.327	0.011	3.34	0.0781	0.0788	0.994	95.440 $\pm$ 0.672	0.314 $\pm$ 0.003	—
UPI	beta33	0.000	0.032	0.032 $\pm$ 0.003	—	0.009 $\pm$ 0.000	0.094 $\pm$ 0.002	0.010 $\pm$ 0.000	0.099 $\pm$ 0.003	—	—	0.030	0.030	—	0.0947	0.0992	1.006	94.819 $\pm$ 0.714	0.370 $\pm$ 0.004	5.181 $\pm$ 0.714
UPI	beta34	0.000	-0.022	-0.022 $\pm$ 0.002	—	0.005 $\pm$ 0.000	0.073 $\pm$ 0.002	0.006 $\pm$ 0.000	0.076 $\pm$ 0.002	—	—	-0.020	-0.020	—	0.0662	0.0691	1.009	94.301 $\pm$ 0.746	0.288 $\pm$ 0.004	5.699 $\pm$ 0.746
UPI	beta35	0.000	-0.024	-0.024 $\pm$ 0.002	—	0.005 $\pm$ 0.000	0.073 $\pm$ 0.002	0.006 $\pm$ 0.000	0.077 $\pm$ 0.002	—	—	-0.021	-0.021	—	0.0686	0.0719	1.020	94.715 $\pm$ 0.720	0.291 $\pm$ 0.004	5.285 $\pm$ 0.720

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S130

Simulation 1 - Condition 130: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability =

0.6

Method	Param.	True	Mean-based										Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE	
LMS	beta31	0.316	0.317	0.001 ± 0.001	0.004 ± 0.004	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.018 ± 0.001	0.133 ± 0.004	0.317	0.001	0.31	0.0422	0.0422	0.994	95.176 ± 0.679	0.164 ± 0.000	—	
LMS	beta32	0.316	0.314	-0.002 ± 0.001	-0.005 ± 0.004	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.018 ± 0.001	0.132 ± 0.004	0.315	-0.001	-0.35	0.0408	0.0408	1.002	95.276 ± 0.673	0.164 ± 0.001	—	
LMS	beta33	0.000	0.009	0.009 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	—	—	0.011	0.011	—	0.0507	0.0520	0.995	92.965 ± 0.811	0.206 ± 0.002	7.035 ± 0.811	
LMS	beta34	0.000	-0.018	-0.018 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	-0.020	-0.020	—	0.0320	0.0377	0.986	86.231 ± 1.092	0.135 ± 0.001	13.769 ± 1.092	
LMS	beta35	0.000	-0.015	-0.015 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.018	-0.018	—	0.0323	0.0368	0.996	89.045 ± 0.990	0.136 ± 0.001	10.955 ± 0.990	
LSAM	beta31	0.316	0.322	0.006 ± 0.001	0.020 ± 0.005	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.022 ± 0.001	0.149 ± 0.004	0.322	0.006	1.97	0.0467	0.0471	0.989	94.271 ± 0.737	0.180 ± 0.001	—	
LSAM	beta32	0.316	0.319	0.003 ± 0.001	0.009 ± 0.005	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.021 ± 0.001	0.145 ± 0.004	0.320	0.004	1.23	0.0447	0.0449	1.006	94.874 ± 0.699	0.181 ± 0.001	—	
LSAM	beta33	0.000	0.033	0.033 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	—	—	0.034	0.034	—	0.0503	0.0605	0.990	88.945 ± 0.994	0.201 ± 0.001	11.055 ± 0.994	
LSAM	beta34	0.000	-0.029	-0.029 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	-0.029	-0.029	—	0.0344	0.0448	0.973	83.819 ± 1.168	0.140 ± 0.001	16.181 ± 1.168	
LSAM	beta35	0.000	-0.025	-0.025 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	-0.026	-0.026	—	0.0357	0.0444	0.977	87.538 ± 1.047	0.141 ± 0.001	12.462 ± 1.047	
QML	beta31	0.316	0.317	0.001 ± 0.001	0.004 ± 0.004	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.018 ± 0.001	0.133 ± 0.004	0.316	0.000	0.10	0.0426	0.0426	0.992	95.276 ± 0.673	0.164 ± 0.000	—	
QML	beta32	0.316	0.314	-0.002 ± 0.001	-0.005 ± 0.004	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.018 ± 0.001	0.132 ± 0.004	0.315	-0.001	-0.28	0.0409	0.0409	1.004	94.975 ± 0.693	0.165 ± 0.000	—	
QML	beta33	0.000	0.008	0.008 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	—	—	0.010	0.010	—	0.0517	0.0526	0.994	93.769 ± 0.766	0.203 ± 0.001	6.231 ± 0.766	
QML	beta34	0.000	-0.020	-0.020 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	-0.022	-0.022	—	0.0325	0.0394	0.985	85.427 ± 1.119	0.134 ± 0.001	14.573 ± 1.119	
QML	beta35	0.000	-0.018	-0.018 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	-0.020	-0.020	—	0.0328	0.0383	0.996	88.844 ± 0.998	0.135 ± 0.001	11.156 ± 0.998	
UPI	beta31	0.316	0.322	0.006 ± 0.002	0.020 ± 0.005	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.023 ± 0.001	0.153 ± 0.004	0.322	0.006	1.74	0.0500	0.0503	0.992	95.176 ± 0.679	0.186 ± 0.001	—	
UPI	beta32	0.316	0.318	0.002 ± 0.001	0.007 ± 0.005	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.022 ± 0.001	0.149 ± 0.004	0.319	0.003	1.06	0.0464	0.0465	1.013	94.573 ± 0.718	0.186 ± 0.001	—	
UPI	beta33	0.000	0.025	0.025 ± 0.002	—	0.003 ± 0.000	0.057 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	—	—	0.024	0.024	—	0.0544	0.0594	0.963	91.658 ± 0.877	0.216 ± 0.001	8.342 ± 0.877	
UPI	beta34	0.000	-0.022	-0.022 ± 0.001	—	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	-0.022	-0.022	—	0.0395	0.0454	0.977	90.854 ± 0.914	0.161 ± 0.001	9.146 ± 0.914	
UPI	beta35	0.000	-0.018	-0.018 ± 0.001	—	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	-0.018	-0.018	—	0.0403	0.0441	0.986	92.362 ± 0.842	0.165 ± 0.001	7.638 ± 0.842	

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u002014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S131

Simulation 1 - Condition 131: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.8

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.317	0.001 ± 0.002	0.002 ± 0.006	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.032 ± 0.002	0.180 ± 0.005	0.318	0.002	0.48	0.0539	0.0539	0.968	94.300 ± 0.733	0.216 ± 0.001	—
LMS	beta32	0.316	0.317	0.001 ± 0.002	0.003 ± 0.005	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.030 ± 0.001	0.174 ± 0.005	0.315	-0.001	-0.35	0.0532	0.0532	1.012	94.900 ± 0.696	0.218 ± 0.001	—
LMS	beta33	0.000	-0.002	-0.002 ± 0.002	—	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	—	—	0.004	0.004	—	0.0658	0.0658	0.927	94.600 ± 0.715	0.248 ± 0.002	5.400 ± 0.715
LMS	beta34	0.000	-0.003	-0.003 ± 0.001	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	-0.007	-0.007	—	0.0436	0.0441	0.914	91.900 ± 0.863	0.169 ± 0.002	8.100 ± 0.863
LMS	beta35	0.000	-0.003	-0.003 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	-0.004	-0.004	—	0.0433	0.0435	0.953	92.700 ± 0.823	0.171 ± 0.002	7.300 ± 0.823
LSAM	beta31	0.316	0.318	0.002 ± 0.002	0.007 ± 0.006	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.034 ± 0.002	0.185 ± 0.005	0.319	0.003	1.06	0.0557	0.0558	0.967	93.300 ± 0.791	0.222 ± 0.001	—
LSAM	beta32	0.316	0.319	0.003 ± 0.002	0.009 ± 0.006	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.032 ± 0.002	0.180 ± 0.005	0.317	0.001	0.22	0.0543	0.0543	1.004	94.200 ± 0.739	0.223 ± 0.001	—
LSAM	beta33	0.000	0.010	0.010 ± 0.002	—	0.004 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.067 ± 0.002	—	—	0.012	0.012	—	0.0646	0.0657	0.923	93.400 ± 0.785	0.241 ± 0.002	6.600 ± 0.785
LSAM	beta34	0.000	-0.008	-0.008 ± 0.001	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	—	—	-0.010	-0.010	—	0.0437	0.0447	0.922	92.100 ± 0.853	0.170 ± 0.001	7.900 ± 0.853
LSAM	beta35	0.000	-0.007	-0.007 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	-0.006	-0.006	—	0.0431	0.0436	0.936	92.400 ± 0.838	0.170 ± 0.001	7.600 ± 0.838
QML	beta31	0.316	0.317	0.001 ± 0.002	0.002 ± 0.006	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.032 ± 0.002	0.180 ± 0.005	0.317	0.001	0.39	0.0543	0.0543	0.969	94.300 ± 0.733	0.216 ± 0.001	—
QML	beta32	0.316	0.317	0.001 ± 0.002	0.003 ± 0.006	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.030 ± 0.001	0.174 ± 0.005	0.315	-0.001	-0.37	0.0530	0.0531	1.011	94.900 ± 0.696	0.218 ± 0.001	—
QML	beta33	0.000	-0.002	-0.002 ± 0.002	—	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	—	—	0.003	0.003	—	0.0655	0.0656	0.926	94.700 ± 0.708	0.245 ± 0.002	5.300 ± 0.708
QML	beta34	0.000	-0.004	-0.004 ± 0.001	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	-0.007	-0.007	—	0.0436	0.0442	0.916	91.800 ± 0.868	0.168 ± 0.001	8.200 ± 0.868
QML	beta35	0.000	-0.003	-0.003 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	-0.004	-0.004	—	0.0434	0.0436	0.950	92.600 ± 0.828	0.169 ± 0.001	7.400 ± 0.828
UPI	beta31	0.316	0.318	0.002 ± 0.002	0.006 ± 0.006	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.034 ± 0.002	0.184 ± 0.005	0.318	0.002	0.75	0.0552	0.0553	0.964	93.300 ± 0.791	0.220 ± 0.001	—
UPI	beta32	0.316	0.319	0.003 ± 0.002	0.009 ± 0.006	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.032 ± 0.002	0.179 ± 0.005	0.317	0.001	0.18	0.0551	0.0551	0.999	94.300 ± 0.733	0.222 ± 0.001	—
UPI	beta33	0.000	0.006	0.006 ± 0.002	—	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	—	—	0.009	0.009	—	0.0685	0.0691	0.908	93.500 ± 0.780	0.243 ± 0.002	6.500 ± 0.780
UPI	beta34	0.000	-0.004	-0.004 ± 0.002	—	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	—	—	-0.005	-0.005	—	0.0432	0.0435	0.925	92.900 ± 0.812	0.173 ± 0.001	7.100 ± 0.812
UPI	beta35	0.000	-0.004	-0.004 ± 0.001	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	-0.005	-0.005	—	0.0447	0.0449	0.939	93.000 ± 0.807	0.173 ± 0.001	7.000 ± 0.807

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S132

Simulation 1 - Condition 132: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.8

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.314	-0.002 ± 0.001	-0.007 ± 0.004	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.013 ± 0.001	0.116 ± 0.003	0.314	-0.002	-0.62	0.0355	0.0355	0.955	93.900 ± 0.757	0.137 ± 0.000	
LMS	beta32	0.316	0.314	-0.002 ± 0.001	-0.007 ± 0.003	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.011 ± 0.000	0.107 ± 0.003	0.313	-0.003	-0.83	0.0332	0.0333	1.034	96.200 ± 0.605	0.137 ± 0.001	—
LMS	beta33	0.000	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	0.000	0.000	—	0.0389	0.0389	0.994	93.800 ± 0.763	0.159 ± 0.004	6.200 ± 0.763
LMS	beta34	0.000	-0.004	-0.004 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	-0.006	-0.006	—	0.0288	0.0293	0.968	93.500 ± 0.780	0.106 ± 0.001	6.500 ± 0.780
LMS	beta35	0.000	-0.004	-0.004 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	-0.006	-0.006	—	0.0278	0.0285	1.021	93.500 ± 0.780	0.112 ± 0.006	6.500 ± 0.780
LSAM	beta31	0.316	0.315	-0.001 ± 0.001	-0.002 ± 0.004	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.014 ± 0.001	0.119 ± 0.003	0.316	0.000	0.02	0.0367	0.0367	0.953	93.800 ± 0.763	0.140 ± 0.000	—
LSAM	beta32	0.316	0.315	-0.001 ± 0.001	-0.003 ± 0.003	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.012 ± 0.001	0.110 ± 0.003	0.315	-0.001	-0.33	0.0337	0.0337	1.028	96.300 ± 0.597	0.140 ± 0.000	—
LSAM	beta33	0.000	0.008	0.008 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	0.010	0.010	—	0.0376	0.0388	0.967	92.500 ± 0.833	0.152 ± 0.001	7.500 ± 0.833
LSAM	beta34	0.000	-0.007	-0.007 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	-0.007	-0.007	—	0.0287	0.0296	0.965	93.000 ± 0.807	0.108 ± 0.001	7.000 ± 0.807
LSAM	beta35	0.000	-0.006	-0.006 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	-0.008	-0.008	—	0.0286	0.0296	0.965	92.500 ± 0.833	0.107 ± 0.001	7.500 ± 0.833
QML	beta31	0.316	0.314	-0.002 ± 0.001	-0.007 ± 0.004	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.013 ± 0.001	0.116 ± 0.003	0.314	-0.002	-0.60	0.0353	0.0353	0.955	94.000 ± 0.751	0.137 ± 0.000	—
QML	beta32	0.316	0.314	-0.002 ± 0.001	-0.008 ± 0.003	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.012 ± 0.000	0.107 ± 0.003	0.313	-0.003	-0.84	0.0331	0.0332	1.030	96.300 ± 0.597	0.137 ± 0.000	—
QML	beta33	0.000	-0.002	-0.002 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	0.000	0.000	—	0.0387	0.0387	0.969	93.600 ± 0.774	0.154 ± 0.001	6.400 ± 0.774
QML	beta34	0.000	-0.005	-0.005 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	-0.006	-0.006	—	0.0287	0.0294	0.971	93.200 ± 0.796	0.106 ± 0.001	6.800 ± 0.796
QML	beta35	0.000	-0.004	-0.004 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	-0.006	-0.006	—	0.0278	0.0286	0.967	93.600 ± 0.774	0.106 ± 0.001	6.400 ± 0.774
UPI	beta31	0.316	0.315	-0.001 ± 0.001	-0.003 ± 0.004	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.014 ± 0.001	0.119 ± 0.003	0.316	0.000	-0.06	0.0359	0.0359	0.951	93.900 ± 0.757	0.140 ± 0.000	—
UPI	beta32	0.316	0.315	-0.001 ± 0.001	-0.004 ± 0.003	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.012 ± 0.001	0.110 ± 0.003	0.315	-0.001	-0.44	0.0340	0.0340	1.026	96.300 ± 0.597	0.140 ± 0.000	—
UPI	beta33	0.000	0.004	0.004 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	0.004	0.004	—	0.0392	0.0394	0.966	93.700 ± 0.768	0.152 ± 0.001	6.300 ± 0.768
UPI	beta34	0.000	-0.004	-0.004 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	-0.005	-0.005	—	0.0290	0.0293	0.958	93.200 ± 0.796	0.109 ± 0.001	6.800 ± 0.796
UPI	beta35	0.000	-0.003	-0.003 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	-0.003	-0.003	—	0.0294	0.0295	0.960	93.700 ± 0.768	0.108 ± 0.001	6.300 ± 0.768

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u00214 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S133

Simulation 1 - Condition 133: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, $N = 400$, Reliability = 0.4

Method	Param.	True	Mean-based										Median-based					Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE		
LMS	beta31	0.316	0.262	-0.054 ± 0.006	-0.172 ± 0.020	0.020 ± 0.002	0.140 ± 0.007	0.023 ± 0.002	0.150 ± 0.008	0.226 ± 0.023	0.475 ± 0.026	0.271	-0.045	-14.13	0.1249	0.1327	1.020	94.726 ± 1.007	0.560 ± 0.024			
LMS	beta32	0.316	0.253	-0.063 ± 0.006	-0.201 ± 0.020	0.019 ± 0.002	0.139 ± 0.007	0.023 ± 0.002	0.153 ± 0.008	0.234 ± 0.023	0.484 ± 0.025	0.270	-0.046	-14.64	0.1330	0.1408	1.071	94.523 ± 1.025	0.585 ± 0.030	—		
LMS	beta33	0.000	-0.168	-0.168 ± 0.022	—	0.248 ± 0.026	0.498 ± 0.027	0.276 ± 0.033	0.525 ± 0.033	—	—	-0.006	-0.006	—	0.2286	0.2287	0.762	92.089 ± 1.216	1.487 ± 0.149	7.911 ± 1.216		
LMS	beta34	0.000	0.160	0.160 ± 0.014	—	0.102 ± 0.012	0.319 ± 0.019	0.127 ± 0.017	0.356 ± 0.024	—	—	0.055	0.055	—	0.1165	0.1290	0.703	96.755 ± 0.798	0.878 ± 0.091	3.245 ± 0.798		
LMS	beta35	0.000	0.129	0.129 ± 0.012	—	0.066 ± 0.008	0.257 ± 0.016	0.082 ± 0.011	0.287 ± 0.020	—	—	0.056	0.056	—	0.1182	0.1307	0.944	95.538 ± 0.930	0.951 ± 0.124	4.462 ± 0.930		
LSAM	beta31	0.316	0.395	0.079 ± 0.053	0.249 ± 0.168	1.390 ± 0.108	1.179 ± 0.046	1.394 ± 0.110	1.181 ± 0.054	13.957 ± 1.104	3.736 ± 0.171	0.421	0.105	33.20	1.1468	1.1515	2.667	99.797 ± 0.203	12.325 ± 0.514	—		
LSAM	beta32	0.316	0.533	0.217 ± 0.053	0.687 ± 0.166	1.362 ± 0.104	1.167 ± 0.045	1.406 ± 0.111	1.186 ± 0.053	14.083 ± 1.107	3.753 ± 0.169	0.424	0.108	34.10	1.2151	1.2199	2.693	99.797 ± 0.203	12.321 ± 0.515	—		
LSAM	beta33	0.000	0.133	0.133 ± 0.013	—	0.089 ± 0.009	0.299 ± 0.014	0.107 ± 0.010	0.327 ± 0.017	—	—	0.113	0.113	—	0.2559	0.2795	2.449	99.189 ± 0.404	2.871 ± 0.150	0.811 ± 0.404		
LSAM	beta34	0.000	-0.082	-0.082 ± 0.020	—	0.205 ± 0.020	0.453 ± 0.022	0.211 ± 0.021	0.460 ± 0.025	—	—	-0.052	-0.052	—	0.3639	0.3676	2.498	99.797 ± 0.203	4.432 ± 0.217	0.203 ± 0.203		
LSAM	beta35	0.000	-0.125	-0.125 ± 0.020	—	0.195 ± 0.017	0.442 ± 0.019	0.210 ± 0.019	0.459 ± 0.023	—	—	-0.063	-0.063	—	0.3813	0.3865	2.582	100.000 ± 0.000	4.470 ± 0.215	0.000 ± 0.000		
QML	beta31	0.316	0.279	-0.037 ± 0.005	-0.118 ± 0.015	0.010 ± 0.001	0.102 ± 0.003	0.012 ± 0.001	0.108 ± 0.004	0.118 ± 0.008	0.343 ± 0.013	0.281	-0.035	-11.17	0.1069	0.1126	1.083	95.335 ± 0.950	0.432 ± 0.004	—		
QML	beta32	0.316	0.274	-0.042 ± 0.005	-0.133 ± 0.015	0.011 ± 0.001	0.103 ± 0.004	0.012 ± 0.001	0.111 ± 0.004	0.123 ± 0.008	0.351 ± 0.013	0.280	-0.036	-11.40	0.1118	0.1174	1.063	93.915 ± 1.077	0.429 ± 0.004	—		
QML	beta33	0.000	0.007	0.007 ± 0.006	—	0.018 ± 0.002	0.132 ± 0.006	0.018 ± 0.002	0.132 ± 0.007	—	—	0.010	0.010	—	0.1263	0.1266	0.998	96.146 ± 0.867	0.518 ± 0.012	3.854 ± 0.867		
QML	beta34	0.000	0.042	0.042 ± 0.003	—	0.006 ± 0.000	0.075 ± 0.003	0.007 ± 0.001	0.085 ± 0.004	—	—	0.033	0.033	—	0.0686	0.0760	1.014	96.146 ± 0.867	0.296 ± 0.008	3.854 ± 0.867		
QML	beta35	0.000	0.035	0.035 ± 0.003	—	0.006 ± 0.001	0.077 ± 0.003	0.007 ± 0.001	0.084 ± 0.004	—	—	0.030	0.030	—	0.0745	0.0802	0.966	95.740 ± 0.910	0.291 ± 0.008	4.260 ± 0.910		
UPI	beta31	0.316	0.456	0.140 ± 0.028	0.444 ± 0.089	0.393 ± 0.033	0.627 ± 0.026	0.412 ± 0.035	0.642 ± 0.031	4.122 ± 0.353	2.030 ± 0.098	0.442	0.126	39.78	0.6215	0.6341	1.840	97.566 ± 0.694	4.520 ± 0.258	—		
UPI	beta32	0.316	0.422	0.106 ± 0.031	0.335 ± 0.098	0.476 ± 0.044	0.690 ± 0.032	0.486 ± 0.046	0.697 ± 0.036	4.865 ± 0.457	2.206 ± 0.115	0.383	0.067	21.28	0.6659	0.6693	1.824	98.377 ± 0.569	4.931 ± 0.294	—		
UPI	beta33	0.000	0.042	0.042 ± 0.014	—	0.100 ± 0.010	0.316 ± 0.016	0.102 ± 0.010	0.319 ± 0.018	—	—	0.041	0.041	—	0.2813	0.2842	1.759	99.797 ± 0.203	2.181 ± 0.132	0.203 ± 0.203		
UPI	beta34	0.000	-0.071	-0.071 ± 0.011	—	0.065 ± 0.006	0.255 ± 0.012	0.070 ± 0.007	0.264 ± 0.014	—	—	-0.063	-0.063	—	0.2254	0.2339	1.785	98.986 ± 0.451	1.785 ± 0.109	1.014 ± 0.451		
UPI	beta35	0.000	-0.061	-0.061 ± 0.013	—	0.079 ± 0.007	0.280 ± 0.013	0.082 ± 0.008	0.287 ± 0.015	—	—	-0.056	-0.056	—	0.2488	0.2551	1.757	99.594 ± 0.286	1.931 ± 0.119	0.406 ± 0.286		

Note. MCSE = Monte Carlo Standard Error shown as $\backslash$ u00b1 value. Coverage, Type I Error, and Power expressed as percentages. $\backslash$ u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S134

Simulation 1 - Condition 134: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.4

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.278	-0.038 ± 0.003	-0.119 ± 0.009	0.004 ± 0.000	0.065 ± 0.002	0.006 ± 0.000	0.075 ± 0.003	0.057 ± 0.004	0.238 ± 0.009	0.284	-0.032	-10.25	0.0803	0.0866	1.060	93.478 ± 1.098	0.270 ± 0.006	
LMS	beta32	0.316	0.283	-0.033 ± 0.003	-0.104 ± 0.008	0.003 ± 0.000	0.059 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	0.045 ± 0.003	0.212 ± 0.008	0.283	-0.033	-10.59	0.0775	0.0844	1.217	96.640 ± 0.801	0.279 ± 0.016	—
LMS	beta33	0.000	0.020	0.020 ± 0.004	—	0.007 ± 0.001	0.087 ± 0.004	0.008 ± 0.001	0.089 ± 0.004	—	—	0.014	0.014	—	0.1098	0.1106	1.479	93.676 ± 1.082	0.502 ± 0.078	6.324 ± 1.082
LMS	beta34	0.000	0.038	0.038 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.002	0.004 ± 0.000	0.062 ± 0.003	—	—	0.037	0.037	—	0.0622	0.0725	1.318	95.652 ± 0.907	0.255 ± 0.035	4.348 ± 0.907
LMS	beta35	0.000	0.034	0.034 ± 0.002	—	0.002 ± 0.000	0.042 ± 0.003	0.003 ± 0.000	0.054 ± 0.003	—	—	0.039	0.039	—	0.0575	0.0696	1.472	96.245 ± 0.845	0.245 ± 0.040	3.755 ± 0.845
LSAM	beta31	0.316	0.404	0.088 ± 0.026	0.277 ± 0.083	0.352 ± 0.036	0.593 ± 0.031	0.359 ± 0.038	0.599 ± 0.034	3.592 ± 0.376	1.895 ± 0.108	0.404	0.088	27.74	0.6321	0.6381	1.744	99.012 ± 0.440	4.055 ± 0.268	—
LSAM	beta32	0.316	0.429	0.113 ± 0.026	0.357 ± 0.081	0.331 ± 0.035	0.575 ± 0.031	0.343 ± 0.035	0.585 ± 0.032	3.432 ± 0.347	1.853 ± 0.102	0.388	0.072	22.84	0.5769	0.5814	1.788	99.605 ± 0.279	4.029 ± 0.267	—
LSAM	beta33	0.000	0.068	0.068 ± 0.005	—	0.013 ± 0.001	0.113 ± 0.005	0.017 ± 0.001	0.132 ± 0.006	—	—	0.067	0.067	—	0.1218	0.1390	1.606	96.245 ± 0.845	0.712 ± 0.044	3.755 ± 0.845
LSAM	beta34	0.000	-0.056	-0.056 ± 0.009	—	0.038 ± 0.004	0.195 ± 0.010	0.041 ± 0.004	0.202 ± 0.011	—	—	-0.049	-0.049	—	0.1955	0.2014	1.708	99.012 ± 0.440	1.303 ± 0.090	0.988 ± 0.440
LSAM	beta35	0.000	-0.066	-0.066 ± 0.008	—	0.035 ± 0.004	0.186 ± 0.010	0.039 ± 0.004	0.197 ± 0.011	—	—	-0.044	-0.044	—	0.1718	0.1773	1.755	99.012 ± 0.440	1.279 ± 0.090	0.988 ± 0.440
QML	beta31	0.316	0.282	-0.034 ± 0.003	-0.108 ± 0.009	0.004 ± 0.000	0.066 ± 0.002	0.006 ± 0.000	0.074 ± 0.003	0.055 ± 0.003	0.235 ± 0.008	0.291	-0.025	-8.07	0.0684	0.0730	1.002	90.909 ± 1.278	0.259 ± 0.001	—
QML	beta32	0.316	0.285	-0.031 ± 0.003	-0.097 ± 0.008	0.004 ± 0.000	0.060 ± 0.002	0.005 ± 0.000	0.067 ± 0.002	0.046 ± 0.003	0.214 ± 0.008	0.287	-0.029	-9.23	0.0661	0.0723	1.094	95.652 ± 0.907	0.258 ± 0.002	—
QML	beta33	0.000	0.012	0.012 ± 0.003	—	0.005 ± 0.000	0.073 ± 0.003	0.005 ± 0.000	0.073 ± 0.003	—	—	0.007	0.007	—	0.0812	0.0815	1.052	96.245 ± 0.845	0.299 ± 0.004	3.755 ± 0.845
QML	beta34	0.000	0.033	0.033 ± 0.002	—	0.002 ± 0.000	0.043 ± 0.002	0.003 ± 0.000	0.054 ± 0.002	—	—	0.028	0.028	—	0.0457	0.0539	0.970	90.316 ± 1.315	0.165 ± 0.003	9.684 ± 1.315
QML	beta35	0.000	0.029	0.029 ± 0.002	—	0.001 ± 0.000	0.038 ± 0.001	0.002 ± 0.000	0.047 ± 0.002	—	—	0.031	0.031	—	0.0437	0.0536	1.099	92.688 ± 1.157	0.162 ± 0.003	7.312 ± 1.157
UPI	beta31	0.316	0.569	0.253 ± 0.027	0.800 ± 0.085	0.365 ± 0.032	0.604 ± 0.027	0.428 ± 0.036	0.654 ± 0.031	4.287 ± 0.364	2.070 ± 0.097	0.577	0.261	82.55	0.6069	0.6606	1.436	93.478 ± 1.098	3.400 ± 0.189	—
UPI	beta32	0.316	0.540	0.224 ± 0.025	0.709 ± 0.079	0.315 ± 0.026	0.561 ± 0.024	0.364 ± 0.028	0.603 ± 0.026	3.646 ± 0.278	1.909 ± 0.082	0.577	0.261	82.73	0.5903	0.6456	1.440	93.478 ± 1.098	3.166 ± 0.169	—
UPI	beta33	0.000	0.109	0.109 ± 0.011	—	0.056 ± 0.005	0.238 ± 0.012	0.068 ± 0.006	0.261 ± 0.013	—	—	0.085	0.085	—	0.1840	0.2027	1.453	99.802 ± 0.197	1.353 ± 0.080	0.198 ± 0.197
UPI	beta34	0.000	-0.117	-0.117 ± 0.010	—	0.050 ± 0.005	0.224 ± 0.011	0.064 ± 0.006	0.252 ± 0.012	—	—	-0.103	-0.103	—	0.1971	0.2225	1.458	97.431 ± 0.703	1.278 ± 0.073	2.569 ± 0.703
UPI	beta35	0.000	-0.110	-0.110 ± 0.009	—	0.043 ± 0.004	0.206 ± 0.010	0.054 ± 0.004	0.233 ± 0.010	—	—	-0.112	-0.112	—	0.1869	0.2178	1.449	97.036 ± 0.754	1.171 ± 0.063	2.964 ± 0.754

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u002014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S135

Simulation 1 - Condition 135: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.6

Method	Param.	True	Mean-based									Median-based					Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE	
LMS	beta31	0.316	0.290	-0.026 ± 0.003	-0.083 ± 0.009	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.085 ± 0.003	0.073 ± 0.004	0.271 ± 0.008	0.289	-0.027	-8.48	0.0815	0.0858	1.084	95.958 ± 0.669	0.346 ± 0.008		
LMS	beta32	0.316	0.291	-0.025 ± 0.003	-0.078 ± 0.009	0.006 ± 0.000	0.079 ± 0.002	0.007 ± 0.000	0.083 ± 0.003	0.069 ± 0.004	0.263 ± 0.008	0.295	-0.021	-6.72	0.0828	0.0855	1.080	96.074 ± 0.660	0.336 ± 0.005	—	
LMS	beta33	0.000	0.002	0.002 ± 0.003	—	0.007 ± 0.000	0.085 ± 0.003	0.007 ± 0.000	0.085 ± 0.003	—	—	0.000	0.000	—	0.0801	0.0801	1.058	93.764 ± 0.822	0.354 ± 0.013	6.236 ± 0.822	
LMS	beta34	0.000	0.024	0.024 ± 0.002	—	0.002 ± 0.000	0.046 ± 0.002	0.003 ± 0.000	0.052 ± 0.002	—	—	0.023	0.023	—	0.0442	0.0498	1.024	94.111 ± 0.800	0.186 ± 0.008	5.889 ± 0.800	
LMS	beta35	0.000	0.022	0.022 ± 0.002	—	0.002 ± 0.000	0.046 ± 0.002	0.003 ± 0.000	0.051 ± 0.002	—	—	0.015	0.015	—	0.0424	0.0452	1.000	95.381 ± 0.713	0.181 ± 0.005	4.619 ± 0.713	
LSAM	beta31	0.316	0.340	0.024 ± 0.005	0.074 ± 0.017	0.025 ± 0.001	0.158 ± 0.004	0.026 ± 0.001	0.160 ± 0.005	0.257 ± 0.014	0.507 ± 0.016	0.340	0.024	7.45	0.1510	0.1528	1.067	96.420 ± 0.631	0.663 ± 0.008	—	
LSAM	beta32	0.316	0.340	0.024 ± 0.005	0.077 ± 0.017	0.024 ± 0.001	0.154 ± 0.004	0.024 ± 0.001	0.156 ± 0.005	0.244 ± 0.013	0.494 ± 0.016	0.339	0.023	7.35	0.1577	0.1594	1.085	96.536 ± 0.621	0.657 ± 0.008	—	
LSAM	beta33	0.000	0.024	0.024 ± 0.003	—	0.007 ± 0.000	0.085 ± 0.003	0.008 ± 0.000	0.089 ± 0.003	—	—	0.022	0.022	—	0.0756	0.0788	0.954	92.610 ± 0.889	0.319 ± 0.004	7.390 ± 0.889	
LSAM	beta34	0.000	-0.018	-0.018 ± 0.002	—	0.004 ± 0.000	0.059 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	—	—	-0.017	-0.017	—	0.0546	0.0574	1.001	94.688 ± 0.762	0.232 ± 0.004	5.312 ± 0.762	
LSAM	beta35	0.000	-0.019	-0.019 ± 0.002	—	0.003 ± 0.000	0.057 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	—	—	-0.018	-0.018	—	0.0528	0.0558	1.027	94.919 ± 0.746	0.228 ± 0.005	5.081 ± 0.746	
QML	beta31	0.316	0.292	-0.024 ± 0.003	-0.076 ± 0.009	0.007 ± 0.000	0.083 ± 0.002	0.007 ± 0.000	0.086 ± 0.003	0.074 ± 0.004	0.272 ± 0.008	0.292	-0.024	-7.66	0.0817	0.0852	1.035	95.612 ± 0.696	0.335 ± 0.002	—	
QML	beta32	0.316	0.294	-0.022 ± 0.003	-0.071 ± 0.009	0.006 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.084 ± 0.003	0.070 ± 0.004	0.264 ± 0.008	0.297	-0.019	-6.08	0.0832	0.0854	1.054	95.266 ± 0.722	0.333 ± 0.002	—	
QML	beta33	0.000	0.001	0.001 ± 0.003	—	0.006 ± 0.000	0.080 ± 0.002	0.006 ± 0.000	0.080 ± 0.003	—	—	-0.003	-0.003	—	0.0763	0.0764	0.988	93.649 ± 0.829	0.309 ± 0.004	6.351 ± 0.829	
QML	beta34	0.000	0.021	0.021 ± 0.002	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.049 ± 0.002	—	—	0.020	0.020	—	0.0425	0.0468	0.967	93.187 ± 0.856	0.168 ± 0.003	6.813 ± 0.856	
QML	beta35	0.000	0.019	0.019 ± 0.002	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.048 ± 0.002	—	—	0.015	0.015	—	0.0410	0.0438	0.962	94.573 ± 0.770	0.166 ± 0.003	5.427 ± 0.770	
UPI	beta31	0.316	0.348	0.032 ± 0.007	0.102 ± 0.021	0.037 ± 0.002	0.191 ± 0.006	0.038 ± 0.002	0.194 ± 0.007	0.377 ± 0.023	0.614 ± 0.021	0.338	0.022	6.95	0.1757	0.1771	1.135	97.691 ± 0.510	0.852 ± 0.020	—	
UPI	beta32	0.316	0.351	0.035 ± 0.006	0.110 ± 0.020	0.036 ± 0.002	0.189 ± 0.006	0.037 ± 0.002	0.193 ± 0.007	0.371 ± 0.023	0.609 ± 0.021	0.348	0.032	10.15	0.1818	0.1846	1.153	97.691 ± 0.510	0.856 ± 0.023	—	
UPI	beta33	0.000	0.027	0.027 ± 0.003	—	0.009 ± 0.001	0.094 ± 0.003	0.010 ± 0.001	0.098 ± 0.003	—	—	0.023	0.023	—	0.0850	0.0880	1.063	95.035 ± 0.738	0.393 ± 0.007	4.965 ± 0.738	
UPI	beta34	0.000	-0.024	-0.024 ± 0.002	—	0.005 ± 0.000	0.073 ± 0.002	0.006 ± 0.000	0.076 ± 0.003	—	—	-0.016	-0.016	—	0.0642	0.0663	1.063	96.767 ± 0.601	0.302 ± 0.009	3.233 ± 0.601	
UPI	beta35	0.000	-0.025	-0.025 ± 0.002	—	0.005 ± 0.000	0.071 ± 0.002	0.006 ± 0.000	0.076 ± 0.003	—	—	-0.020	-0.020	—	0.0593	0.0626	1.099	96.536 ± 0.621	0.308 ± 0.011	3.464 ± 0.621	

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u00214 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S136

Simulation 1 - Condition 136: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.6

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.294	-0.022 ± 0.002	-0.071 ± 0.005	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.031 ± 0.001	0.176 ± 0.005	0.294	-0.022	-6.83	0.0546	0.0587	1.014	92.274 ± 0.881	0.202 ± 0.001	
LMS	beta32	0.316	0.291	-0.025 ± 0.002	-0.080 ± 0.005	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.034 ± 0.002	0.185 ± 0.005	0.290	-0.026	-8.13	0.0522	0.0582	0.979	91.730 ± 0.909	0.202 ± 0.001	—
LMS	beta33	0.000	0.005	0.005 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	—	—	0.007	0.007	—	0.0528	0.0533	0.969	94.015 ± 0.782	0.200 ± 0.002	5.985 ± 0.782
LMS	beta34	0.000	0.021	0.021 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	0.018	0.018	—	0.0282	0.0336	0.936	91.295 ± 0.930	0.103 ± 0.001	8.705 ± 0.930
LMS	beta35	0.000	0.021	0.021 ± 0.001	—	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	0.019	0.019	—	0.0277	0.0334	0.975	91.077 ± 0.940	0.104 ± 0.001	8.923 ± 0.940
LSAM	beta31	0.316	0.340	0.024 ± 0.003	0.077 ± 0.010	0.009 ± 0.000	0.097 ± 0.002	0.010 ± 0.000	0.099 ± 0.003	0.099 ± 0.005	0.315 ± 0.009	0.341	0.025	8.02	0.0972	0.1005	1.002	95.430 ± 0.689	0.379 ± 0.002	—
LSAM	beta32	0.316	0.334	0.018 ± 0.003	0.056 ± 0.010	0.010 ± 0.000	0.098 ± 0.002	0.010 ± 0.001	0.099 ± 0.003	0.099 ± 0.005	0.315 ± 0.009	0.333	0.017	5.30	0.0970	0.0984	0.984	93.798 ± 0.796	0.377 ± 0.002	—
LSAM	beta33	0.000	0.018	0.018 ± 0.002	—	0.002 ± 0.000	0.048 ± 0.001	0.003 ± 0.000	0.051 ± 0.002	—	—	0.017	0.017	—	0.0473	0.0502	0.937	92.709 ± 0.858	0.176 ± 0.001	7.291 ± 0.858
LSAM	beta34	0.000	-0.015	-0.015 ± 0.001	—	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.014	-0.014	—	0.0314	0.0343	0.945	91.839 ± 0.903	0.124 ± 0.001	8.161 ± 0.903
LSAM	beta35	0.000	-0.014	-0.014 ± 0.001	—	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	-0.012	-0.012	—	0.0301	0.0326	0.959	92.492 ± 0.869	0.123 ± 0.001	7.508 ± 0.869
QML	beta31	0.316	0.295	-0.021 ± 0.002	-0.066 ± 0.005	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.032 ± 0.002	0.179 ± 0.005	0.296	-0.020	-6.27	0.0553	0.0588	1.005	92.383 ± 0.875	0.207 ± 0.001	—
QML	beta32	0.316	0.292	-0.024 ± 0.002	-0.076 ± 0.006	0.003 ± 0.000	0.054 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	0.035 ± 0.002	0.188 ± 0.005	0.292	-0.024	-7.55	0.0541	0.0591	0.970	91.621 ± 0.914	0.207 ± 0.001	—
QML	beta33	0.000	0.002	0.002 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	—	—	0.003	0.003	—	0.0494	0.0495	0.969	94.015 ± 0.782	0.192 ± 0.002	5.985 ± 0.782
QML	beta34	0.000	0.020	0.020 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	—	—	0.018	0.018	—	0.0282	0.0334	0.936	91.186 ± 0.935	0.102 ± 0.001	8.814 ± 0.935
QML	beta35	0.000	0.020	0.020 ± 0.001	—	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	—	—	0.019	0.019	—	0.0269	0.0327	0.970	90.424 ± 0.971	0.103 ± 0.001	9.576 ± 0.971
UPI	beta31	0.316	0.348	0.032 ± 0.004	0.100 ± 0.012	0.014 ± 0.001	0.118 ± 0.003	0.015 ± 0.001	0.122 ± 0.004	0.148 ± 0.008	0.385 ± 0.012	0.348	0.032	9.99	0.1137	0.1180	1.000	96.083 ± 0.640	0.461 ± 0.006	—
UPI	beta32	0.316	0.340	0.024 ± 0.004	0.077 ± 0.012	0.014 ± 0.001	0.118 ± 0.003	0.015 ± 0.001	0.121 ± 0.004	0.145 ± 0.008	0.381 ± 0.012	0.336	0.020	6.34	0.1101	0.1119	0.995	94.668 ± 0.741	0.461 ± 0.006	—
UPI	beta33	0.000	0.019	0.019 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	—	—	0.017	0.017	—	0.0474	0.0504	0.964	94.668 ± 0.741	0.193 ± 0.002	5.332 ± 0.741
UPI	beta34	0.000	-0.019	-0.019 ± 0.001	—	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	-0.015	-0.015	—	0.0352	0.0385	0.928	93.689 ± 0.802	0.151 ± 0.003	6.311 ± 0.802
UPI	beta35	0.000	-0.017	-0.017 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.044 ± 0.002	—	—	-0.013	-0.013	—	0.0348	0.0371	0.953	94.559 ± 0.748	0.151 ± 0.003	5.441 ± 0.748

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u002014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S137

Simulation 1 - Condition 137: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.8

Method	Param.	True	Mean-based									Median-based					Inference				
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Type I (%) $\pm$ MCSE	
LMS	beta31	0.316	0.300	-0.016 $\pm$ 0.002	-0.050 $\pm$ 0.007	0.005 $\pm$ 0.000	0.074 $\pm$ 0.002	0.006 $\pm$ 0.000	0.076 $\pm$ 0.002	0.057 $\pm$ 0.002	0.240 $\pm$ 0.006	0.300	-0.016	-5.17	0.0765	0.0782	1.011	95.749 $\pm$ 0.642	0.294 $\pm$ 0.001	—	
LMS	beta32	0.316	0.299	-0.017 $\pm$ 0.002	-0.052 $\pm$ 0.007	0.005 $\pm$ 0.000	0.072 $\pm$ 0.002	0.006 $\pm$ 0.000	0.074 $\pm$ 0.002	0.055 $\pm$ 0.003	0.235 $\pm$ 0.007	0.302	-0.014	-4.51	0.0704	0.0718	1.038	95.445 $\pm$ 0.663	0.295 $\pm$ 0.001	—	
LMS	beta33	0.000	-0.002	-0.002 $\pm$ 0.002	—	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	—	—	-0.002	-0.002	—	0.0562	0.0562	0.950	94.231 $\pm$ 0.742	0.225 $\pm$ 0.003	5.769 $\pm$ 0.742	
LMS	beta34	0.000	0.010	0.010 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	—	—	0.006	0.006	—	0.0293	0.0300	0.916	92.814 $\pm$ 0.822	0.120 $\pm$ 0.002	7.186 $\pm$ 0.822	
LMS	beta35	0.000	0.013	0.013 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	—	—	0.009	0.009	—	0.0302	0.0316	0.938	92.308 $\pm$ 0.848	0.124 $\pm$ 0.002	7.692 $\pm$ 0.848	
LSAM	beta31	0.316	0.319	0.003 $\pm$ 0.003	0.010 $\pm$ 0.010	0.009 $\pm$ 0.000	0.097 $\pm$ 0.002	0.009 $\pm$ 0.000	0.097 $\pm$ 0.003	0.095 $\pm$ 0.004	0.308 $\pm$ 0.008	0.318	0.002	0.62	0.1009	0.1009	1.008	95.243 $\pm$ 0.677	0.384 $\pm$ 0.002	—	
LSAM	beta32	0.316	0.316	0.000 $\pm$ 0.003	0.002 $\pm$ 0.010	0.009 $\pm$ 0.000	0.095 $\pm$ 0.002	0.009 $\pm$ 0.000	0.095 $\pm$ 0.003	0.090 $\pm$ 0.004	0.300 $\pm$ 0.008	0.318	0.002	0.69	0.0898	0.0898	1.035	95.749 $\pm$ 0.642	0.385 $\pm$ 0.002	—	
LSAM	beta33	0.000	0.004	0.004 $\pm$ 0.002	—	0.003 $\pm$ 0.000	0.058 $\pm$ 0.002	0.003 $\pm$ 0.000	0.058 $\pm$ 0.002	—	—	0.003	0.003	—	0.0563	0.0564	0.958	94.231 $\pm$ 0.742	0.219 $\pm$ 0.002	5.769 $\pm$ 0.742	
LSAM	beta34	0.000	-0.004	-0.004 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	—	—	-0.006	-0.006	—	0.0316	0.0321	0.920	92.915 $\pm$ 0.816	0.133 $\pm$ 0.002	7.085 $\pm$ 0.816	
LSAM	beta35	0.000	-0.002	-0.002 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	—	—	-0.003	-0.003	—	0.0321	0.0322	0.966	92.814 $\pm$ 0.822	0.136 $\pm$ 0.002	7.186 $\pm$ 0.822	
QML	beta31	0.316	0.301	-0.015 $\pm$ 0.002	-0.048 $\pm$ 0.008	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.006 $\pm$ 0.000	0.077 $\pm$ 0.002	0.059 $\pm$ 0.003	0.242 $\pm$ 0.006	0.300	-0.016	-4.97	0.0785	0.0800	1.006	95.648 $\pm$ 0.649	0.296 $\pm$ 0.001	—	
QML	beta32	0.316	0.300	-0.016 $\pm$ 0.002	-0.051 $\pm$ 0.007	0.005 $\pm$ 0.000	0.073 $\pm$ 0.002	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.056 $\pm$ 0.003	0.238 $\pm$ 0.007	0.303	-0.013	-4.16	0.0722	0.0734	1.036	95.142 $\pm$ 0.684	0.298 $\pm$ 0.002	—	
QML	beta33	0.000	-0.003	-0.003 $\pm$ 0.002	—	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	—	—	-0.002	-0.002	—	0.0570	0.0570	0.954	94.534 $\pm$ 0.723	0.224 $\pm$ 0.003	5.466 $\pm$ 0.723	
QML	beta34	0.000	0.010	0.010 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	—	—	0.006	0.006	—	0.0294	0.0300	0.914	92.510 $\pm$ 0.837	0.120 $\pm$ 0.002	7.490 $\pm$ 0.837	
QML	beta35	0.000	0.012	0.012 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	—	—	0.009	0.009	—	0.0302	0.0315	0.946	92.206 $\pm$ 0.853	0.125 $\pm$ 0.002	7.794 $\pm$ 0.853	
UPI	beta31	0.316	0.319	0.003 $\pm$ 0.003	0.009 $\pm$ 0.010	0.009 $\pm$ 0.000	0.096 $\pm$ 0.002	0.009 $\pm$ 0.000	0.096 $\pm$ 0.003	0.092 $\pm$ 0.004	0.303 $\pm$ 0.008	0.319	0.003	0.86	0.0960	0.0960	1.011	95.040 $\pm$ 0.691	0.380 $\pm$ 0.002	—	
UPI	beta32	0.316	0.316	0.000 $\pm$ 0.003	0.001 $\pm$ 0.009	0.009 $\pm$ 0.000	0.094 $\pm$ 0.002	0.009 $\pm$ 0.000	0.094 $\pm$ 0.003	0.088 $\pm$ 0.004	0.297 $\pm$ 0.008	0.318	0.002	0.74	0.0896	0.0896	1.036	95.850 $\pm$ 0.634	0.381 $\pm$ 0.002	—	
UPI	beta33	0.000	0.003	0.003 $\pm$ 0.002	—	0.003 $\pm$ 0.000	0.059 $\pm$ 0.001	0.003 $\pm$ 0.000	0.059 $\pm$ 0.002	—	—	0.002	0.002	—	0.0557	0.0557	0.965	94.028 $\pm$ 0.754	0.223 $\pm$ 0.002	5.972 $\pm$ 0.754	
UPI	beta34	0.000	-0.004	-0.004 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	—	—	-0.004	-0.004	—	0.0308	0.0311	0.917	92.713 $\pm$ 0.827	0.131 $\pm$ 0.002	7.287 $\pm$ 0.827	
UPI	beta35	0.000	-0.001	-0.001 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	—	—	-0.002	-0.002	—	0.0328	0.0328	0.967	93.016 $\pm$ 0.811	0.135 $\pm$ 0.002	6.984 $\pm$ 0.811	

Note. MCSE = Monte Carlo Standard Error shown as $\backslash$ u00b1 value. Coverage, Type I Error, and Power expressed as percentages. $\backslash$ u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S138

Simulation 1 - Condition 138: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.8

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.301	-0.015 ± 0.001	-0.046 ± 0.005	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.023 ± 0.001	0.151 ± 0.004	0.302	-0.014	-4.42	0.0431	0.0454	1.030	94.389 ± 0.728	0.183 ± 0.001	
LMS	beta32	0.316	0.306	-0.010 ± 0.001	-0.032 ± 0.005	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.023 ± 0.001	0.153 ± 0.004	0.305	-0.011	-3.41	0.0470	0.0482	0.987	94.188 ± 0.741	0.183 ± 0.001	—
LMS	beta33	0.000	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.000	0.000	—	0.0362	0.0362	0.962	95.792 ± 0.636	0.139 ± 0.001	4.208 ± 0.636
LMS	beta34	0.000	0.009	0.009 ± 0.001	—	0.000 ± 0.000	0.020 ± 0.001	0.000 ± 0.000	0.022 ± 0.001	—	—	0.007	0.007	—	0.0189	0.0202	0.918	93.487 ± 0.781	0.073 ± 0.001	6.513 ± 0.781
LMS	beta35	0.000	0.009	0.009 ± 0.001	—	0.000 ± 0.000	0.021 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	—	—	0.007	0.007	—	0.0201	0.0214	0.889	92.184 ± 0.850	0.074 ± 0.001	7.816 ± 0.850
LSAM	beta31	0.316	0.318	0.002 ± 0.002	0.006 ± 0.006	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.034 ± 0.002	0.184 ± 0.005	0.317	0.001	0.36	0.0549	0.0549	1.019	95.491 ± 0.657	0.232 ± 0.001	—
LSAM	beta32	0.316	0.324	0.008 ± 0.002	0.024 ± 0.006	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.038 ± 0.002	0.194 ± 0.005	0.322	0.006	1.78	0.0606	0.0609	0.979	93.487 ± 0.781	0.234 ± 0.001	—
LSAM	beta33	0.000	0.003	0.003 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	0.003	0.003	—	0.0346	0.0348	0.958	94.589 ± 0.716	0.131 ± 0.001	5.411 ± 0.716
LSAM	beta34	0.000	-0.003	-0.003 ± 0.001	—	0.000 ± 0.000	0.021 ± 0.001	0.000 ± 0.000	0.021 ± 0.001	—	—	-0.004	-0.004	—	0.0189	0.0193	0.919	92.786 ± 0.819	0.076 ± 0.001	7.214 ± 0.819
LSAM	beta35	0.000	-0.003	-0.003 ± 0.001	—	0.000 ± 0.000	0.022 ± 0.001	0.000 ± 0.000	0.022 ± 0.001	—	—	-0.004	-0.004	—	0.0207	0.0211	0.905	92.285 ± 0.845	0.078 ± 0.001	7.715 ± 0.845
QML	beta31	0.316	0.302	-0.014 ± 0.001	-0.045 ± 0.005	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.023 ± 0.001	0.152 ± 0.004	0.302	-0.014	-4.36	0.0441	0.0462	1.038	94.188 ± 0.741	0.187 ± 0.003	—
QML	beta32	0.316	0.306	-0.010 ± 0.002	-0.031 ± 0.005	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.024 ± 0.001	0.155 ± 0.004	0.306	-0.010	-3.25	0.0476	0.0487	1.025	94.188 ± 0.741	0.193 ± 0.009	—
QML	beta33	0.000	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.001	-0.001	—	0.0359	0.0359	0.984	95.992 ± 0.621	0.141 ± 0.004	4.008 ± 0.621
QML	beta34	0.000	0.009	0.009 ± 0.001	—	0.000 ± 0.000	0.020 ± 0.001	0.000 ± 0.000	0.022 ± 0.001	—	—	0.007	0.007	—	0.0189	0.0201	0.912	93.487 ± 0.781	0.073 ± 0.001	6.513 ± 0.781
QML	beta35	0.000	0.009	0.009 ± 0.001	—	0.000 ± 0.000	0.021 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	—	—	0.007	0.007	—	0.0204	0.0216	0.966	92.084 ± 0.855	0.081 ± 0.007	7.916 ± 0.855
UPI	beta31	0.316	0.317	0.001 ± 0.002	0.003 ± 0.006	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.033 ± 0.001	0.183 ± 0.005	0.319	0.003	0.83	0.0554	0.0555	1.021	95.691 ± 0.643	0.231 ± 0.001	—
UPI	beta32	0.316	0.323	0.007 ± 0.002	0.021 ± 0.006	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.037 ± 0.002	0.192 ± 0.005	0.321	0.005	1.50	0.0621	0.0622	0.983	94.188 ± 0.741	0.232 ± 0.001	—
UPI	beta33	0.000	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	0.002	0.002	—	0.0348	0.0348	0.958	94.890 ± 0.697	0.132 ± 0.001	5.110 ± 0.697
UPI	beta34	0.000	-0.002	-0.002 ± 0.001	—	0.000 ± 0.000	0.021 ± 0.001	0.000 ± 0.000	0.021 ± 0.001	—	—	-0.003	-0.003	—	0.0190	0.0193	0.924	93.186 ± 0.798	0.076 ± 0.001	6.814 ± 0.798
UPI	beta35	0.000	-0.003	-0.003 ± 0.001	—	0.000 ± 0.000	0.022 ± 0.001	0.000 ± 0.000	0.022 ± 0.001	—	—	-0.003	-0.003	—	0.0208	0.0211	0.902	92.786 ± 0.819	0.078 ± 0.001	7.214 ± 0.819

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u002014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S139

Simulation 1 - Condition 139: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.4

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.323	0.007 ± 0.012	0.021 ± 0.040	0.007 ± 0.002	0.083 ± 0.013	0.007 ± 0.002	0.082 ± 0.014	0.068 ± 0.020	0.260 ± 0.046	0.335	0.019	6.15	0.1025	0.1043	1.190	95.455 ± 3.140	0.387 ± 0.032	—
LMS	beta32	0.316	0.308	-0.008 ± 0.015	-0.027 ± 0.046	0.009 ± 0.003	0.097 ± 0.014	0.009 ± 0.003	0.096 ± 0.016	0.093 ± 0.025	0.304 ± 0.050	0.293	-0.023	-7.40	0.1403	0.1423	0.939	93.182 ± 3.800	0.357 ± 0.023	—
LMS	beta33	0.000	0.035	0.035 ± 0.016	—	0.011 ± 0.002	0.105 ± 0.012	0.012 ± 0.003	0.110 ± 0.014	—	—	0.024	0.024	—	0.1980	0.1994	1.161	88.636 ± 4.785	0.480 ± 0.059	11.364 ± 4.785
LMS	beta34	0.000	-0.049	-0.049 ± 0.011	—	0.006 ± 0.002	0.075 ± 0.014	0.008 ± 0.002	0.089 ± 0.013	—	—	-0.042	-0.042	—	0.1472	0.1530	1.678	86.364 ± 5.174	0.491 ± 0.140	13.636 ± 5.174
LMS	beta35	0.000	-0.058	-0.058 ± 0.016	—	0.011 ± 0.007	0.104 ± 0.048	0.014 ± 0.009	0.118 ± 0.048	—	—	-0.028	-0.028	—	0.1312	0.1342	0.849	88.636 ± 4.785	0.346 ± 0.050	11.364 ± 4.785
LSAM	beta31	0.316	0.360	0.044 ± 0.023	0.141 ± 0.072	0.022 ± 0.007	0.150 ± 0.026	0.024 ± 0.008	0.155 ± 0.030	0.240 ± 0.080	0.490 ± 0.095	0.358	0.042	13.21	0.1810	0.1857	1.441	100.000 ± 0.000	0.847 ± 0.120	—
LSAM	beta32	0.316	0.355	0.039 ± 0.021	0.123 ± 0.068	0.020 ± 0.004	0.142 ± 0.016	0.021 ± 0.005	0.146 ± 0.020	0.213 ± 0.047	0.462 ± 0.062	0.363	0.047	14.94	0.1858	0.1917	1.489	100.000 ± 0.000	0.831 ± 0.095	—
LSAM	beta33	0.000	0.097	0.097 ± 0.022	—	0.021 ± 0.005	0.147 ± 0.021	0.030 ± 0.009	0.174 ± 0.028	—	—	0.084	0.084	—	0.1399	0.1633	1.220	95.455 ± 3.140	0.701 ± 0.084	4.545 ± 3.140
LSAM	beta34	0.000	-0.133	-0.133 ± 0.021	—	0.019 ± 0.007	0.139 ± 0.029	0.037 ± 0.012	0.191 ± 0.034	—	—	-0.104	-0.104	—	0.1370	0.1722	1.620	97.727 ± 2.247	0.883 ± 0.193	2.273 ± 2.247
LSAM	beta35	0.000	-0.107	-0.107 ± 0.020	—	0.017 ± 0.004	0.132 ± 0.018	0.029 ± 0.007	0.169 ± 0.024	—	—	-0.107	-0.107	—	0.1877	0.2158	1.506	97.727 ± 2.247	0.779 ± 0.127	2.273 ± 2.247
QML	beta31	0.316	0.318	0.002 ± 0.012	0.007 ± 0.039	0.007 ± 0.002	0.082 ± 0.014	0.007 ± 0.002	0.081 ± 0.015	0.065 ± 0.020	0.256 ± 0.046	0.322	0.006	1.97	0.0804	0.0807	1.062	95.455 ± 3.140	0.340 ± 0.009	—
QML	beta32	0.316	0.317	0.001 ± 0.013	0.004 ± 0.042	0.008 ± 0.002	0.088 ± 0.013	0.008 ± 0.002	0.087 ± 0.015	0.076 ± 0.021	0.275 ± 0.047	0.295	-0.021	-6.51	0.0956	0.0978	1.001	95.455 ± 3.140	0.345 ± 0.010	—
QML	beta33	0.000	0.043	0.043 ± 0.016	—	0.012 ± 0.002	0.109 ± 0.012	0.013 ± 0.003	0.116 ± 0.016	—	—	0.044	0.044	—	0.1122	0.1204	1.089	90.909 ± 4.334	0.465 ± 0.039	9.091 ± 4.334
QML	beta34	0.000	-0.062	-0.062 ± 0.009	—	0.004 ± 0.001	0.063 ± 0.008	0.008 ± 0.001	0.087 ± 0.008	—	—	-0.058	-0.058	—	0.0744	0.0941	1.247	84.091 ± 5.514	0.306 ± 0.027	15.909 ± 5.514
QML	beta35	0.000	-0.061	-0.061 ± 0.010	—	0.004 ± 0.001	0.064 ± 0.007	0.008 ± 0.002	0.088 ± 0.010	—	—	-0.056	-0.056	—	0.0943	0.1098	1.234	86.364 ± 5.174	0.310 ± 0.024	13.636 ± 5.174
UPI	beta31	0.316	0.356	0.040 ± 0.051	0.127 ± 0.161	0.114 ± 0.027	0.337 ± 0.044	0.113 ± 0.030	0.336 ± 0.053	1.131 ± 0.296	1.063 ± 0.168	0.397	0.081	25.61	0.5748	0.5805	1.635	86.364 ± 5.174	2.163 ± 0.572	—
UPI	beta32	0.316	0.353	0.037 ± 0.076	0.116 ± 0.242	0.257 ± 0.094	0.507 ± 0.106	0.252 ± 0.096	0.502 ± 0.112	2.528 ± 0.963	1.590 ± 0.353	0.313	-0.003	-0.83	0.4884	0.4884	1.656	84.091 ± 5.514	3.289 ± 1.000	—
UPI	beta33	0.000	-0.051	-0.051 ± 0.037	—	0.060 ± 0.024	0.246 ± 0.055	0.062 ± 0.026	0.248 ± 0.061	—	—	-0.030	-0.030	—	0.2688	0.2705	1.597	100.000 ± 0.000	1.539 ± 0.251	0.000 ± 0.000
UPI	beta34	0.000	-0.222	-0.222 ± 0.072	—	0.227 ± 0.069	0.476 ± 0.080	0.271 ± 0.093	0.520 ± 0.101	—	—	-0.189	-0.189	—	0.5684	0.5991	1.549	100.000 ± 0.000	2.890 ± 0.792	0.000 ± 0.000
UPI	beta35	0.000	-0.201	-0.201 ± 0.096	—	0.409 ± 0.168	0.639 ± 0.155	0.440 ± 0.192	0.663 ± 0.170	—	—	-0.207	-0.207	—	0.5030	0.5439	1.759	100.000 ± 0.000	4.406 ± 0.997	0.000 ± 0.000

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S140

Simulation 1 - Condition 140: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability =

0.4

Method	Param.	True	Mean-based									Median-based					Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE	
LMS	beta31	0.316	0.309	-0.007 ± 0.009	-0.023 ± 0.028	0.003 ± 0.001	0.056 ± 0.006	0.003 ± 0.001	0.055 ± 0.008	0.031 ± 0.007	0.176 ± 0.025	0.305	-0.011	-3.43	0.0760	0.0767	0.935	92.683 ± 4.067	0.204 ± 0.004		
LMS	beta32	0.316	0.313	-0.003 ± 0.009	-0.010 ± 0.027	0.003 ± 0.001	0.055 ± 0.008	0.003 ± 0.001	0.054 ± 0.009	0.029 ± 0.008	0.172 ± 0.029	0.317	0.001	0.38	0.0821	0.0821	0.986	95.122 ± 3.364	0.212 ± 0.005	—	
LMS	beta33	0.000	0.067	0.067 ± 0.011	—	0.005 ± 0.001	0.069 ± 0.009	0.009 ± 0.002	0.095 ± 0.012	—	—	0.043	0.043	—	0.1101	0.1182	0.945	78.049 ± 6.464	0.256 ± 0.016	21.951 ± 6.464	
LMS	beta34	0.000	-0.057	-0.057 ± 0.005	—	0.001 ± 0.000	0.034 ± 0.004	0.004 ± 0.001	0.066 ± 0.005	—	—	-0.041	-0.041	—	0.0747	0.0853	1.267	65.854 ± 7.406	0.168 ± 0.011	34.146 ± 7.406	
LMS	beta35	0.000	-0.053	-0.053 ± 0.007	—	0.002 ± 0.000	0.044 ± 0.005	0.005 ± 0.001	0.068 ± 0.007	—	—	-0.035	-0.035	—	0.0673	0.0757	1.118	65.854 ± 7.406	0.192 ± 0.019	34.146 ± 7.406	
LSAM	beta31	0.316	0.349	0.033 ± 0.017	0.105 ± 0.053	0.012 ± 0.002	0.108 ± 0.012	0.012 ± 0.003	0.111 ± 0.015	0.124 ± 0.027	0.352 ± 0.048	0.343	0.027	8.52	0.0926	0.0965	0.933	97.561 ± 2.409	0.394 ± 0.021	—	
LSAM	beta32	0.316	0.345	0.029 ± 0.015	0.092 ± 0.048	0.010 ± 0.003	0.098 ± 0.015	0.010 ± 0.003	0.101 ± 0.016	0.102 ± 0.028	0.320 ± 0.052	0.348	0.032	10.21	0.0739	0.0806	1.065	95.122 ± 3.364	0.409 ± 0.023	—	
LSAM	beta33	0.000	0.114	0.114 ± 0.014	—	0.008 ± 0.002	0.088 ± 0.011	0.020 ± 0.004	0.143 ± 0.016	—	—	0.100	0.100	—	0.0878	0.1329	0.989	87.805 ± 5.110	0.341 ± 0.023	12.195 ± 5.110	
LSAM	beta34	0.000	-0.126	-0.126 ± 0.014	—	0.008 ± 0.002	0.089 ± 0.010	0.024 ± 0.004	0.154 ± 0.015	—	—	-0.116	-0.116	—	0.0849	0.1437	1.048	78.049 ± 6.464	0.367 ± 0.033	21.951 ± 6.464	
LSAM	beta35	0.000	-0.119	-0.119 ± 0.012	—	0.006 ± 0.001	0.077 ± 0.005	0.020 ± 0.003	0.142 ± 0.011	—	—	-0.114	-0.114	—	0.0797	0.1395	1.207	78.049 ± 6.464	0.366 ± 0.027	21.951 ± 6.464	
QML	beta31	0.316	0.312	-0.004 ± 0.009	-0.014 ± 0.028	0.003 ± 0.001	0.057 ± 0.006	0.003 ± 0.001	0.056 ± 0.008	0.032 ± 0.007	0.178 ± 0.024	0.317	0.001	0.47	0.0584	0.0584	0.919	92.683 ± 4.067	0.204 ± 0.003	—	
QML	beta32	0.316	0.316	-0.000 ± 0.009	-0.000 ± 0.027	0.003 ± 0.001	0.056 ± 0.009	0.003 ± 0.001	0.055 ± 0.010	0.030 ± 0.009	0.174 ± 0.031	0.318	0.002	0.48	0.0601	0.0601	0.967	95.122 ± 3.364	0.211 ± 0.004	—	
QML	beta33	0.000	0.071	0.071 ± 0.012	—	0.006 ± 0.001	0.077 ± 0.010	0.011 ± 0.003	0.104 ± 0.014	—	—	0.075	0.075	—	0.0753	0.1064	0.897	80.488 ± 6.189	0.270 ± 0.013	19.512 ± 6.189	
QML	beta34	0.000	-0.072	-0.072 ± 0.006	—	0.001 ± 0.000	0.036 ± 0.004	0.006 ± 0.001	0.080 ± 0.006	—	—	-0.067	-0.067	—	0.0468	0.0818	1.217	63.415 ± 7.522	0.173 ± 0.007	36.585 ± 7.522	
QML	beta35	0.000	-0.070	-0.070 ± 0.008	—	0.003 ± 0.001	0.050 ± 0.006	0.007 ± 0.001	0.085 ± 0.009	—	—	-0.069	-0.069	—	0.0504	0.0856	0.986	63.415 ± 7.522	0.194 ± 0.016	36.585 ± 7.522	
UPI	beta31	0.316	0.360	0.044 ± 0.054	0.140 ± 0.172	0.121 ± 0.050	0.348 ± 0.087	0.120 ± 0.053	0.346 ± 0.094	1.201 ± 0.536	1.096 ± 0.296	0.404	0.088	27.70	0.6045	0.6108	1.363	87.805 ± 5.110	1.858 ± 0.452	—	
UPI	beta32	0.316	0.384	0.068 ± 0.097	0.215 ± 0.307	0.387 ± 0.184	0.622 ± 0.180	0.382 ± 0.194	0.618 ± 0.190	3.824 ± 1.946	1.956 ± 0.602	0.311	-0.005	-1.73	0.4865	0.4866	0.711	68.293 ± 7.267	1.733 ± 0.513	—	
UPI	beta33	0.000	-0.109	-0.109 ± 0.062	—	0.157 ± 0.065	0.396 ± 0.096	0.165 ± 0.074	0.406 ± 0.107	—	—	-0.040	-0.040	—	0.3589	0.3611	0.868	92.683 ± 4.067	1.348 ± 0.357	7.317 ± 4.067	
UPI	beta34	0.000	-0.285	-0.285 ± 0.202	—	1.673 ± 0.949	1.293 ± 0.445	1.714 ± 0.934	1.309 ± 0.419	—	—	-0.591	-0.591	—	1.3102	1.4374	0.887	97.561 ± 2.409	4.500 ± 1.229	2.439 ± 2.409	
UPI	beta35	0.000	-0.696	-0.696 ± 0.301	—	3.708 ± 1.868	1.926 ± 0.586	4.102 ± 2.255	2.025 ± 0.657	—	—	-0.454	-0.454	—	1.1927	1.2761	0.474	87.805 ± 5.110	3.579 ± 0.966	12.195 ± 5.110	

Note. MCSE = Monte Carlo Standard Error shown as $\backslash$ u00b1 value. Coverage, Type I Error, and Power expressed as percentages. $\backslash$ u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S141

Simulation 1 - Condition 141: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.6

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Type I (%) $\pm$ MCSE
LMS	beta31	0.316	0.315	-0.001 $\pm$ 0.003	-0.003 $\pm$ 0.009	0.005 $\pm$ 0.000	0.069 $\pm$ 0.002	0.005 $\pm$ 0.000	0.069 $\pm$ 0.003	0.047 $\pm$ 0.003	0.217 $\pm$ 0.008	0.309	-0.007	-2.31	0.0679	0.0683	1.001	95.320 $\pm$ 0.834	0.269 $\pm$ 0.002	—
LMS	beta32	0.316	0.318	0.002 $\pm$ 0.003	0.007 $\pm$ 0.008	0.004 $\pm$ 0.000	0.067 $\pm$ 0.002	0.004 $\pm$ 0.000	0.067 $\pm$ 0.002	0.045 $\pm$ 0.003	0.211 $\pm$ 0.008	0.321	0.005	1.66	0.0650	0.0652	1.018	95.476 $\pm$ 0.821	0.266 $\pm$ 0.002	—
LMS	beta33	0.000	0.010	0.010 $\pm$ 0.003	—	0.008 $\pm$ 0.001	0.088 $\pm$ 0.003	0.008 $\pm$ 0.001	0.089 $\pm$ 0.003	—	—	0.018	0.018	—	0.0812	0.0832	1.103	92.044 $\pm$ 1.069	0.381 $\pm$ 0.014	7.956 $\pm$ 1.069
LMS	beta34	0.000	-0.011	-0.011 $\pm$ 0.003	—	0.005 $\pm$ 0.000	0.071 $\pm$ 0.002	0.005 $\pm$ 0.000	0.072 $\pm$ 0.003	—	—	-0.017	-0.017	—	0.0698	0.0717	1.124	91.108 $\pm$ 1.124	0.313 $\pm$ 0.012	8.892 $\pm$ 1.124
LMS	beta35	0.000	-0.013	-0.013 $\pm$ 0.003	—	0.005 $\pm$ 0.000	0.072 $\pm$ 0.003	0.005 $\pm$ 0.000	0.073 $\pm$ 0.003	—	—	-0.020	-0.020	—	0.0676	0.0704	1.050	87.988 $\pm$ 1.284	0.297 $\pm$ 0.009	12.012 $\pm$ 1.284
LSAM	beta31	0.316	0.319	0.003 $\pm$ 0.003	0.009 $\pm$ 0.010	0.006 $\pm$ 0.000	0.078 $\pm$ 0.002	0.006 $\pm$ 0.000	0.078 $\pm$ 0.003	0.061 $\pm$ 0.004	0.247 $\pm$ 0.009	0.315	-0.001	-0.23	0.0785	0.0785	0.991	95.944 $\pm$ 0.779	0.303 $\pm$ 0.002	—
LSAM	beta32	0.316	0.322	0.006 $\pm$ 0.003	0.018 $\pm$ 0.009	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.006 $\pm$ 0.000	0.075 $\pm$ 0.003	0.056 $\pm$ 0.003	0.238 $\pm$ 0.008	0.324	0.008	2.58	0.0692	0.0696	1.022	95.476 $\pm$ 0.821	0.300 $\pm$ 0.002	—
LSAM	beta33	0.000	0.038	0.038 $\pm$ 0.003	—	0.008 $\pm$ 0.000	0.087 $\pm$ 0.003	0.009 $\pm$ 0.001	0.095 $\pm$ 0.003	—	—	0.038	0.038	—	0.0842	0.0922	0.992	91.420 $\pm$ 1.106	0.337 $\pm$ 0.004	8.580 $\pm$ 1.106
LSAM	beta34	0.000	-0.041	-0.041 $\pm$ 0.004	—	0.012 $\pm$ 0.001	0.108 $\pm$ 0.004	0.013 $\pm$ 0.001	0.115 $\pm$ 0.005	—	—	-0.042	-0.042	—	0.1018	0.1100	0.972	93.604 $\pm$ 0.966	0.410 $\pm$ 0.006	6.396 $\pm$ 0.966
LSAM	beta35	0.000	-0.043	-0.043 $\pm$ 0.004	—	0.010 $\pm$ 0.001	0.099 $\pm$ 0.003	0.012 $\pm$ 0.001	0.108 $\pm$ 0.003	—	—	-0.048	-0.048	—	0.0957	0.1071	1.000	90.952 $\pm$ 1.133	0.388 $\pm$ 0.005	9.048 $\pm$ 1.133
QML	beta31	0.316	0.315	-0.001 $\pm$ 0.003	-0.003 $\pm$ 0.009	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.047 $\pm$ 0.003	0.216 $\pm$ 0.008	0.311	-0.005	-1.63	0.0671	0.0673	0.983	95.008 $\pm$ 0.860	0.263 $\pm$ 0.001	—
QML	beta32	0.316	0.319	0.003 $\pm$ 0.003	0.008 $\pm$ 0.008	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	0.044 $\pm$ 0.003	0.210 $\pm$ 0.008	0.321	0.005	1.69	0.0643	0.0645	1.003	95.320 $\pm$ 0.834	0.261 $\pm$ 0.001	—
QML	beta33	0.000	0.014	0.014 $\pm$ 0.003	—	0.007 $\pm$ 0.000	0.083 $\pm$ 0.003	0.007 $\pm$ 0.000	0.084 $\pm$ 0.003	—	—	0.018	0.018	—	0.0789	0.0809	1.002	91.888 $\pm$ 1.078	0.325 $\pm$ 0.004	8.112 $\pm$ 1.078
QML	beta34	0.000	-0.021	-0.021 $\pm$ 0.003	—	0.004 $\pm$ 0.000	0.067 $\pm$ 0.002	0.005 $\pm$ 0.000	0.070 $\pm$ 0.002	—	—	-0.022	-0.022	—	0.0663	0.0698	1.022	90.796 $\pm$ 1.142	0.267 $\pm$ 0.003	9.204 $\pm$ 1.142
QML	beta35	0.000	-0.022	-0.022 $\pm$ 0.003	—	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	0.005 $\pm$ 0.000	0.070 $\pm$ 0.002	—	—	-0.027	-0.027	—	0.0646	0.0701	0.996	87.363 $\pm$ 1.312	0.260 $\pm$ 0.003	12.637 $\pm$ 1.312
UPI	beta31	0.316	0.325	0.009 $\pm$ 0.004	0.028 $\pm$ 0.013	0.011 $\pm$ 0.001	0.104 $\pm$ 0.004	0.011 $\pm$ 0.001	0.104 $\pm$ 0.004	0.109 $\pm$ 0.008	0.331 $\pm$ 0.014	0.320	0.004	1.17	0.1012	0.1013	1.379	97.660 $\pm$ 0.597	0.563 $\pm$ 0.027	—
UPI	beta32	0.316	0.333	0.017 $\pm$ 0.004	0.055 $\pm$ 0.013	0.011 $\pm$ 0.001	0.105 $\pm$ 0.004	0.011 $\pm$ 0.001	0.106 $\pm$ 0.004	0.113 $\pm$ 0.008	0.337 $\pm$ 0.014	0.332	0.016	5.00	0.1009	0.1021	1.425	97.348 $\pm$ 0.635	0.587 $\pm$ 0.040	—
UPI	beta33	0.000	0.043	0.043 $\pm$ 0.004	—	0.010 $\pm$ 0.001	0.100 $\pm$ 0.004	0.012 $\pm$ 0.001	0.109 $\pm$ 0.004	—	—	0.039	0.039	—	0.0915	0.0994	1.183	96.724 $\pm$ 0.703	0.463 $\pm$ 0.013	3.276 $\pm$ 0.703
UPI	beta34	0.000	-0.040	-0.040 $\pm$ 0.007	—	0.032 $\pm$ 0.002	0.179 $\pm$ 0.007	0.034 $\pm$ 0.002	0.183 $\pm$ 0.008	—	—	-0.041	-0.041	—	0.1601	0.1654	1.290	96.724 $\pm$ 0.703	0.904 $\pm$ 0.039	3.276 $\pm$ 0.703
UPI	beta35	0.000	-0.056	-0.056 $\pm$ 0.007	—	0.032 $\pm$ 0.003	0.180 $\pm$ 0.008	0.035 $\pm$ 0.003	0.188 $\pm$ 0.009	—	—	-0.051	-0.051	—	0.1655	0.1733	1.371	97.816 $\pm$ 0.577	0.965 $\pm$ 0.079	2.184 $\pm$ 0.577

Note. MCSE = Monte Carlo Standard Error shown as $\backslash u00b1$ value. Coverage, Type I Error, and Power expressed as percentages. $\backslash u2014$ indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S142

Simulation 1 - Condition 142: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.6

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.317	0.001 ± 0.001	0.004 ± 0.005	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.015 ± 0.001	0.124 ± 0.004	0.318	0.002	0.69	0.0383	0.0384	1.091	96.612 ± 0.666	0.167 ± 0.004	—
LMS	beta32	0.316	0.315	-0.001 ± 0.002	-0.002 ± 0.005	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.017 ± 0.001	0.132 ± 0.004	0.315	-0.001	-0.45	0.0427	0.0427	1.007	94.851 ± 0.813	0.165 ± 0.001	—
LMS	beta33	0.000	0.019	0.019 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	—	—	0.021	0.021	—	0.0503	0.0545	1.052	90.244 ± 1.092	0.209 ± 0.004	9.756 ± 1.092
LMS	beta34	0.000	-0.026	-0.026 ± 0.002	—	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	—	—	-0.029	-0.029	—	0.0404	0.0497	1.010	83.333 ± 1.372	0.165 ± 0.003	16.667 ± 1.372
LMS	beta35	0.000	-0.023	-0.023 ± 0.002	—	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	—	—	-0.027	-0.027	—	0.0389	0.0471	1.034	86.450 ± 1.260	0.172 ± 0.006	13.550 ± 1.260
LSAM	beta31	0.316	0.321	0.005 ± 0.002	0.015 ± 0.005	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.020 ± 0.001	0.142 ± 0.005	0.321	0.005	1.68	0.0437	0.0440	1.065	95.528 ± 0.761	0.186 ± 0.001	—
LSAM	beta32	0.316	0.318	0.002 ± 0.002	0.007 ± 0.006	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.023 ± 0.001	0.151 ± 0.005	0.319	0.003	0.88	0.0475	0.0475	0.997	94.444 ± 0.843	0.186 ± 0.001	—
LSAM	beta33	0.000	0.038	0.038 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	—	—	0.038	0.038	—	0.0492	0.0621	1.019	87.534 ± 1.216	0.203 ± 0.001	12.466 ± 1.216
LSAM	beta34	0.000	-0.052	-0.052 ± 0.002	—	0.004 ± 0.000	0.062 ± 0.002	0.007 ± 0.000	0.081 ± 0.002	—	—	-0.050	-0.050	—	0.0558	0.0750	0.988	85.366 ± 1.301	0.240 ± 0.002	14.634 ± 1.301
LSAM	beta35	0.000	-0.047	-0.047 ± 0.002	—	0.004 ± 0.000	0.063 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	—	—	-0.050	-0.050	—	0.0616	0.0793	0.968	85.908 ± 1.281	0.241 ± 0.002	14.092 ± 1.281
QML	beta31	0.316	0.317	0.001 ± 0.001	0.005 ± 0.005	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.015 ± 0.001	0.124 ± 0.004	0.319	0.003	0.83	0.0377	0.0378	1.093	96.341 ± 0.691	0.167 ± 0.005	—
QML	beta32	0.316	0.316	-0.000 ± 0.002	-0.001 ± 0.005	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.017 ± 0.001	0.132 ± 0.004	0.315	-0.001	-0.35	0.0422	0.0423	1.000	94.851 ± 0.813	0.164 ± 0.001	—
QML	beta33	0.000	0.019	0.019 ± 0.002	—	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	—	—	0.020	0.020	—	0.0503	0.0542	1.021	91.192 ± 1.043	0.201 ± 0.002	8.808 ± 1.043
QML	beta34	0.000	-0.032	-0.032 ± 0.002	—	0.002 ± 0.000	0.041 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	—	—	-0.033	-0.033	—	0.0411	0.0530	1.003	81.978 ± 1.415	0.162 ± 0.001	18.022 ± 1.415
QML	beta35	0.000	-0.029	-0.029 ± 0.002	—	0.002 ± 0.000	0.042 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	—	—	-0.031	-0.031	—	0.0389	0.0499	0.994	84.824 ± 1.321	0.163 ± 0.001	15.176 ± 1.321
UPI	beta31	0.316	0.327	0.011 ± 0.002	0.036 ± 0.008	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.003	0.047 ± 0.003	0.217 ± 0.008	0.326	0.010	3.17	0.0637	0.0645	1.409	98.374 ± 0.466	0.373 ± 0.016	—
UPI	beta32	0.316	0.328	0.012 ± 0.002	0.039 ± 0.008	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.047 ± 0.003	0.216 ± 0.007	0.327	0.011	3.60	0.0664	0.0674	1.433	98.238 ± 0.484	0.378 ± 0.015	—
UPI	beta33	0.000	0.049	0.049 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	—	—	0.046	0.046	—	0.0610	0.0767	1.201	92.412 ± 0.975	0.286 ± 0.008	7.588 ± 0.975
UPI	beta34	0.000	-0.066	-0.066 ± 0.004	—	0.013 ± 0.001	0.113 ± 0.004	0.017 ± 0.001	0.131 ± 0.004	—	—	-0.062	-0.062	—	0.0936	0.1121	1.316	95.257 ± 0.782	0.583 ± 0.030	4.743 ± 0.782
UPI	beta35	0.000	-0.065	-0.065 ± 0.004	—	0.013 ± 0.001	0.115 ± 0.004	0.017 ± 0.001	0.132 ± 0.004	—	—	-0.065	-0.065	—	0.1025	0.1212	1.241	94.715 ± 0.824	0.560 ± 0.021	5.285 ± 0.824

Note. MCSE = Monte Carlo Standard Error shown as $\backslash$ u00b1 value. Coverage, Type I Error, and Power expressed as percentages. $\backslash$ u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S143

Simulation 1 - Condition 143: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.8

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LMS	beta31	0.316	0.318	0.002 ± 0.002	0.006 ± 0.006	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.031 ± 0.001	0.175 ± 0.005	0.319	0.003	0.81	0.0554	0.0555	1.001	94.589 ± 0.716	0.217 ± 0.001	—
LMS	beta32	0.316	0.319	0.003 ± 0.002	0.008 ± 0.006	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.031 ± 0.001	0.177 ± 0.005	0.314	-0.002	-0.54	0.0555	0.0556	0.983	94.289 ± 0.735	0.216 ± 0.001	—
LMS	beta33	0.000	-0.002	-0.002 ± 0.002	—	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	—	—	0.000	0.000	—	0.0636	0.0636	0.978	94.790 ± 0.703	0.253 ± 0.002	5.210 ± 0.703
LMS	beta34	0.000	-0.002	-0.002 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	—	—	-0.008	-0.008	—	0.0595	0.0600	0.964	93.888 ± 0.758	0.232 ± 0.002	6.112 ± 0.758
LMS	beta35	0.000	-0.004	-0.004 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	—	—	-0.007	-0.007	—	0.0579	0.0584	0.974	93.086 ± 0.803	0.232 ± 0.002	6.914 ± 0.803
LSAM	beta31	0.316	0.318	0.002 ± 0.002	0.007 ± 0.006	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.032 ± 0.001	0.180 ± 0.005	0.319	0.003	0.80	0.0558	0.0559	1.000	94.589 ± 0.716	0.223 ± 0.001	—
LSAM	beta32	0.316	0.319	0.003 ± 0.002	0.009 ± 0.006	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.033 ± 0.001	0.183 ± 0.005	0.316	0.000	-0.05	0.0574	0.0574	0.983	93.888 ± 0.758	0.222 ± 0.001	—
LSAM	beta33	0.000	0.006	0.006 ± 0.002	—	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	—	—	0.007	0.007	—	0.0626	0.0630	0.979	93.988 ± 0.752	0.245 ± 0.002	6.012 ± 0.752
LSAM	beta34	0.000	-0.009	-0.009 ± 0.002	—	0.005 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.074 ± 0.002	—	—	-0.012	-0.012	—	0.0728	0.0739	0.955	93.186 ± 0.798	0.276 ± 0.001	6.814 ± 0.798
LSAM	beta35	0.000	-0.012	-0.012 ± 0.002	—	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.074 ± 0.002	—	—	-0.012	-0.012	—	0.0720	0.0729	0.966	92.986 ± 0.808	0.277 ± 0.001	7.014 ± 0.808
QML	beta31	0.316	0.318	0.002 ± 0.002	0.006 ± 0.006	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.031 ± 0.001	0.175 ± 0.005	0.319	0.003	0.84	0.0557	0.0557	1.000	94.790 ± 0.703	0.216 ± 0.001	—
QML	beta32	0.316	0.318	0.002 ± 0.002	0.008 ± 0.006	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.031 ± 0.001	0.177 ± 0.005	0.314	-0.002	-0.51	0.0550	0.0551	0.983	94.188 ± 0.741	0.216 ± 0.001	—
QML	beta33	0.000	-0.002	-0.002 ± 0.002	—	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	—	—	0.000	0.000	—	0.0635	0.0635	0.980	94.689 ± 0.710	0.250 ± 0.002	5.311 ± 0.710
QML	beta34	0.000	-0.004	-0.004 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	—	—	-0.009	-0.009	—	0.0590	0.0597	0.965	93.587 ± 0.775	0.229 ± 0.002	6.413 ± 0.775
QML	beta35	0.000	-0.006	-0.006 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	—	—	-0.008	-0.008	—	0.0572	0.0578	0.973	92.986 ± 0.808	0.228 ± 0.002	7.014 ± 0.808
UPI	beta31	0.316	0.318	0.002 ± 0.002	0.008 ± 0.006	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.033 ± 0.001	0.181 ± 0.005	0.319	0.003	1.00	0.0566	0.0567	0.988	94.389 ± 0.728	0.221 ± 0.001	—
UPI	beta32	0.316	0.319	0.003 ± 0.002	0.011 ± 0.006	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.033 ± 0.001	0.182 ± 0.005	0.317	0.001	0.26	0.0565	0.0565	0.979	94.088 ± 0.747	0.221 ± 0.001	—
UPI	beta33	0.000	0.003	0.003 ± 0.002	—	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	—	—	0.005	0.005	—	0.0646	0.0648	0.966	93.888 ± 0.758	0.247 ± 0.002	6.112 ± 0.758
UPI	beta34	0.000	-0.002	-0.002 ± 0.002	—	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	—	—	-0.003	-0.003	—	0.0764	0.0765	0.967	94.689 ± 0.710	0.293 ± 0.002	5.311 ± 0.710
UPI	beta35	0.000	-0.007	-0.007 ± 0.002	—	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	—	—	-0.007	-0.007	—	0.0757	0.0761	0.955	93.888 ± 0.758	0.292 ± 0.002	6.112 ± 0.758

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S144

Simulation 1 - Condition 144: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.8

Method	Param.	True	Mean-based									Median-based					Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE	
LMS	beta31	0.316	0.316	-0.000 ± 0.001	-0.001 ± 0.003	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.012 ± 0.001	0.109 ± 0.003	0.315	-0.001	-0.27	0.0322	0.0322	0.996	95.200 ± 0.676	0.135 ± 0.000		
LMS	beta32	0.316	0.316	-0.000 ± 0.001	-0.001 ± 0.003	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.012 ± 0.001	0.110 ± 0.003	0.316	0.000	0.14	0.0346	0.0346	0.990	95.500 ± 0.656	0.136 ± 0.001	—	
LMS	beta33	0.000	0.002	0.002 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	0.003	0.003	—	0.0394	0.0396	0.997	94.600 ± 0.715	0.156 ± 0.001	5.400 ± 0.715	
LMS	beta34	0.000	-0.007	-0.007 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.009	-0.009	—	0.0363	0.0375	1.020	94.200 ± 0.739	0.143 ± 0.001	5.800 ± 0.739	
LMS	beta35	0.000	-0.007	-0.007 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.009	-0.009	—	0.0349	0.0360	0.991	92.300 ± 0.843	0.143 ± 0.001	7.700 ± 0.843	
LSAM	beta31	0.316	0.316	-0.000 ± 0.001	-0.000 ± 0.004	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.013 ± 0.001	0.113 ± 0.003	0.315	-0.001	-0.19	0.0333	0.0333	0.996	94.700 ± 0.708	0.139 ± 0.000	—	
LSAM	beta32	0.316	0.316	-0.000 ± 0.001	-0.000 ± 0.004	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.013 ± 0.001	0.113 ± 0.003	0.316	0.000	0.02	0.0346	0.0346	0.993	95.500 ± 0.656	0.139 ± 0.000	—	
LSAM	beta33	0.000	0.008	0.008 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	0.008	0.008	—	0.0380	0.0387	0.997	93.900 ± 0.757	0.154 ± 0.001	6.100 ± 0.757	
LSAM	beta34	0.000	-0.012	-0.012 ± 0.001	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	-0.012	-0.012	—	0.0446	0.0462	1.009	94.900 ± 0.696	0.174 ± 0.001	5.100 ± 0.696	
LSAM	beta35	0.000	-0.011	-0.011 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	-0.011	-0.011	—	0.0436	0.0450	0.989	92.900 ± 0.812	0.174 ± 0.001	7.100 ± 0.812	
QML	beta31	0.316	0.316	-0.000 ± 0.001	-0.001 ± 0.003	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.012 ± 0.001	0.109 ± 0.003	0.315	-0.001	-0.27	0.0321	0.0321	0.995	95.200 ± 0.676	0.135 ± 0.000	—	
QML	beta32	0.316	0.315	-0.001 ± 0.001	-0.002 ± 0.003	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.012 ± 0.001	0.110 ± 0.003	0.316	0.000	0.10	0.0346	0.0346	0.989	95.500 ± 0.656	0.135 ± 0.000	—	
QML	beta33	0.000	0.002	0.002 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	0.003	0.003	—	0.0393	0.0394	0.998	94.600 ± 0.715	0.156 ± 0.001	5.400 ± 0.715	
QML	beta34	0.000	-0.008	-0.008 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.010	-0.010	—	0.0362	0.0377	1.019	94.200 ± 0.739	0.142 ± 0.001	5.800 ± 0.739	
QML	beta35	0.000	-0.008	-0.008 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.010	-0.010	—	0.0345	0.0359	0.991	92.100 ± 0.853	0.143 ± 0.001	7.900 ± 0.853	
UPI	beta31	0.316	0.316	0.000 ± 0.001	0.000 ± 0.004	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.013 ± 0.001	0.113 ± 0.003	0.315	-0.001	-0.18	0.0332	0.0332	0.991	94.600 ± 0.715	0.138 ± 0.000	—	
UPI	beta32	0.316	0.316	-0.000 ± 0.001	-0.000 ± 0.004	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.013 ± 0.001	0.113 ± 0.003	0.317	0.001	0.25	0.0352	0.0353	0.987	95.100 ± 0.683	0.138 ± 0.000	—	
UPI	beta33	0.000	0.003	0.003 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	0.003	0.003	—	0.0383	0.0385	0.993	94.500 ± 0.721	0.155 ± 0.001	5.500 ± 0.721	
UPI	beta34	0.000	-0.005	-0.005 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	-0.006	-0.006	—	0.0463	0.0466	1.001	95.000 ± 0.689	0.180 ± 0.001	5.000 ± 0.689	
UPI	beta35	0.000	-0.005	-0.005 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	-0.005	-0.005	—	0.0450	0.0452	0.988	94.200 ± 0.739	0.180 ± 0.001	5.800 ± 0.739	

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Complete Results Tables — Simulation 2

Table S145

Simulation 2 - Condition 1: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, $N = 400$, Reliability = 0.4

Method	Param.	True	Mean-based										Median-based					Inference		
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LSAM	beta41	0.20	0.203	0.003 $\pm$ 0.003	0.016 $\pm$ 0.016	0.008 $\pm$ 0.000	0.091 $\pm$ 0.003	0.008 $\pm$ 0.000	0.091 $\pm$ 0.003	0.205 $\pm$ 0.011	0.453 $\pm$ 0.015	0.201	0.001	0.68	0.0894	0.0894	1.033	96.179 $\pm$ 0.696	0.367 $\pm$ 0.003	60.343 $\pm$ 1.776
LSAM	beta42	0.20	0.202	0.002 $\pm$ 0.003	0.008 $\pm$ 0.017	0.009 $\pm$ 0.000	0.092 $\pm$ 0.003	0.009 $\pm$ 0.000	0.092 $\pm$ 0.003	0.213 $\pm$ 0.012	0.462 $\pm$ 0.015	0.198	-0.002	-0.87	0.0925	0.0925	1.042	96.838 $\pm$ 0.635	0.377 $\pm$ 0.004	60.870 $\pm$ 1.771
LSAM	beta43	0.20	0.208	0.008 $\pm$ 0.003	0.039 $\pm$ 0.015	0.007 $\pm$ 0.000	0.082 $\pm$ 0.002	0.007 $\pm$ 0.000	0.082 $\pm$ 0.003	0.169 $\pm$ 0.009	0.411 $\pm$ 0.013	0.210	0.010	5.08	0.0852	0.0858	1.045	95.916 $\pm$ 0.718	0.336 $\pm$ 0.003	70.751 $\pm$ 1.651
LSAM	beta44	0.11	0.111	0.001 $\pm$ 0.005	0.006 $\pm$ 0.043	0.017 $\pm$ 0.001	0.130 $\pm$ 0.004	0.017 $\pm$ 0.001	0.130 $\pm$ 0.005	1.394 $\pm$ 0.090	1.181 $\pm$ 0.044	0.105	-0.005	-4.41	0.1189	0.1190	1.100	97.892 $\pm$ 0.521	0.561 $\pm$ 0.010	11.858 $\pm$ 1.173
LSAM	beta45	0.11	0.113	0.003 $\pm$ 0.004	0.029 $\pm$ 0.035	0.011 $\pm$ 0.001	0.107 $\pm$ 0.003	0.011 $\pm$ 0.001	0.107 $\pm$ 0.004	0.940 $\pm$ 0.060	0.969 $\pm$ 0.036	0.110	0.000	-0.21	0.0955	0.0955	1.058	97.628 $\pm$ 0.552	0.442 $\pm$ 0.006	15.283 $\pm$ 1.306
LSAM	beta46	0.08	0.095	0.015 $\pm$ 0.004	0.187 $\pm$ 0.045	0.010 $\pm$ 0.001	0.099 $\pm$ 0.003	0.010 $\pm$ 0.001	0.100 $\pm$ 0.004	1.550 $\pm$ 0.103	1.245 $\pm$ 0.047	0.092	0.012	15.22	0.0958	0.0966	1.015	97.233 $\pm$ 0.595	0.392 $\pm$ 0.007	12.385 $\pm$ 1.196
LSAM	beta47	0.08	0.085	0.005 $\pm$ 0.003	0.062 $\pm$ 0.042	0.009 $\pm$ 0.001	0.092 $\pm$ 0.003	0.009 $\pm$ 0.001	0.093 $\pm$ 0.003	1.339 $\pm$ 0.087	1.157 $\pm$ 0.043	0.080	0.000	0.17	0.0889	0.0889	1.047	96.970 $\pm$ 0.622	0.380 $\pm$ 0.007	11.199 $\pm$ 1.145
LSAM	beta51	0.16	0.155	-0.005 $\pm$ 0.004	-0.031 $\pm$ 0.026	0.013 $\pm$ 0.001	0.115 $\pm$ 0.003	0.013 $\pm$ 0.001	0.115 $\pm$ 0.004	0.513 $\pm$ 0.029	0.716 $\pm$ 0.024	0.154	-0.006	-3.78	0.1054	0.1055	0.903	94.335 $\pm$ 0.839	0.406 $\pm$ 0.003	35.968 $\pm$ 1.742
LSAM	beta52	0.16	0.164	0.004 $\pm$ 0.003	0.028 $\pm$ 0.020	0.008 $\pm$ 0.000	0.087 $\pm$ 0.002	0.008 $\pm$ 0.000	0.087 $\pm$ 0.003	0.296 $\pm$ 0.016	0.544 $\pm$ 0.018	0.168	0.008	5.29	0.0847	0.0851	1.014	95.389 $\pm$ 0.761	0.346 $\pm$ 0.003	49.012 $\pm$ 1.815
LSAM	beta53	0.16	0.168	0.008 $\pm$ 0.003	0.047 $\pm$ 0.020	0.008 $\pm$ 0.000	0.090 $\pm$ 0.002	0.008 $\pm$ 0.000	0.090 $\pm$ 0.003	0.318 $\pm$ 0.016	0.564 $\pm$ 0.018	0.165	0.005	3.10	0.0906	0.0908	1.026	97.365 $\pm$ 0.581	0.362 $\pm$ 0.003	45.586 $\pm$ 1.808
LSAM	beta54	0.16	0.164	0.004 $\pm$ 0.003	0.027 $\pm$ 0.020	0.008 $\pm$ 0.000	0.088 $\pm$ 0.002	0.008 $\pm$ 0.000	0.088 $\pm$ 0.003	0.305 $\pm$ 0.017	0.552 $\pm$ 0.018	0.159	-0.001	-0.63	0.0856	0.0857	0.974	95.389 $\pm$ 0.761	0.337 $\pm$ 0.002	49.934 $\pm$ 1.815
LSAM	beta55	0.08	0.085	0.005 $\pm$ 0.005	0.056 $\pm$ 0.059	0.017 $\pm$ 0.001	0.130 $\pm$ 0.004	0.017 $\pm$ 0.001	0.130 $\pm$ 0.005	2.651 $\pm$ 0.165	1.628 $\pm$ 0.059	0.080	0.000	0.15	0.1227	0.1227	1.029	96.970 $\pm$ 0.622	0.525 $\pm$ 0.008	8.564 $\pm$ 1.016
LSAM	beta56	0.08	0.085	0.005 $\pm$ 0.004	0.056 $\pm$ 0.055	0.015 $\pm$ 0.001	0.122 $\pm$ 0.004	0.015 $\pm$ 0.001	0.122 $\pm$ 0.004	2.331 $\pm$ 0.148	1.527 $\pm$ 0.056	0.092	0.012	14.43	0.1142	0.1148	0.973	95.652 $\pm$ 0.740	0.466 $\pm$ 0.007	11.726 $\pm$ 1.168
LSAM	beta57	0.06	0.072	0.012 $\pm$ 0.003	0.192 $\pm$ 0.056	0.009 $\pm$ 0.001	0.093 $\pm$ 0.003	0.009 $\pm$ 0.001	0.094 $\pm$ 0.003	2.448 $\pm$ 0.145	1.564 $\pm$ 0.054	0.070	0.010	16.59	0.0879	0.0885	0.990	96.706 $\pm$ 0.648	0.362 $\pm$ 0.006	10.145 $\pm$ 1.096
LSAM	beta58	0.06	0.060	0.000 $\pm$ 0.003	0.008 $\pm$ 0.043	0.005 $\pm$ 0.000	0.071 $\pm$ 0.002	0.005 $\pm$ 0.000	0.071 $\pm$ 0.003	1.403 $\pm$ 0.092	1.185 $\pm$ 0.045	0.060	0.000	0.27	0.0630	0.0630	1.024	97.365 $\pm$ 0.581	0.285 $\pm$ 0.004	9.223 $\pm$ 1.050
QML	beta41	0.20	0.185	-0.015 $\pm$ 0.003	-0.077 $\pm$ 0.014	0.006 $\pm$ 0.000	0.079 $\pm$ 0.002	0.006 $\pm$ 0.000	0.080 $\pm$ 0.002	0.160 $\pm$ 0.008	0.400 $\pm$ 0.012	0.183	-0.017	-8.47	0.0762	0.0781	0.968	92.885 $\pm$ 0.933	0.298 $\pm$ 0.002	70.619 $\pm$ 1.653
QML	beta42	0.20	0.188	-0.012 $\pm$ 0.003	-0.059 $\pm$ 0.014	0.006 $\pm$ 0.000	0.078 $\pm$ 0.002	0.006 $\pm$ 0.000	0.079 $\pm$ 0.003	0.157 $\pm$ 0.008	0.396 $\pm$ 0.013	0.188	-0.012	-6.07	0.0772	0.0781	1.009	94.993 $\pm$ 0.792	0.310 $\pm$ 0.002	69.697 $\pm$ 1.668
QML	beta43	0.20	0.186	-0.014 $\pm$ 0.003	-0.069 $\pm$ 0.013	0.005 $\pm$ 0.000	0.074 $\pm$ 0.002	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.140 $\pm$ 0.007	0.374 $\pm$ 0.012	0.189	-0.011	-5.44	0.0721	0.0729	1.016	93.808 $\pm$ 0.875	0.293 $\pm$ 0.002	71.673 $\pm$ 1.636
QML	beta44	0.11	0.097	-0.013 $\pm$ 0.004	-0.115 $\pm$ 0.034	0.010 $\pm$ 0.001	0.102 $\pm$ 0.003	0.011 $\pm$ 0.001	0.103 $\pm$ 0.003	0.871 $\pm$ 0.046	0.933 $\pm$ 0.030	0.094	-0.016	-14.52	0.1069	0.1081	1.065	95.125 $\pm$ 0.782	0.426 $\pm$ 0.005	15.020 $\pm$ 1.297
QML	beta45	0.11	0.103	-0.007 $\pm$ 0.003	-0.065 $\pm$ 0.028	0.007 $\pm$ 0.000	0.083 $\pm$ 0.002	0.007 $\pm$ 0.000	0.084 $\pm$ 0.003	0.578 $\pm$ 0.030	0.760 $\pm$ 0.024	0.102	-0.008	-7.21	0.0812	0.0816	1.060	95.784 $\pm$ 0.729	0.346 $\pm$ 0.003	21.080 $\pm$ 1.481
QML	beta46	0.08	0.085	0.005 $\pm$ 0.003	0.061 $\pm$ 0.033	0.005 $\pm$ 0.000	0.072 $\pm$ 0.002	0.005 $\pm$ 0.000	0.072 $\pm$ 0.003	0.807 $\pm$ 0.051	0.898 $\pm$ 0.033	0.083	0.003	4.04	0.0683	0.0683	1.020	95.652 $\pm$ 0.740	0.287 $\pm$ 0.004	19.631 $\pm$ 1.442
QML	beta47	0.08	0.075	-0.005 $\pm$ 0.003	-0.059 $\pm$ 0.031	0.005 $\pm$ 0.000	0.069 $\pm$ 0.002	0.005 $\pm$ 0.000	0.069 $\pm$ 0.003	0.751 $\pm$ 0.047	0.867 $\pm$ 0.031	0.076	-0.004	-5.09	0.0644	0.0645	1.021	94.335 $\pm$ 0.839	0.277 $\pm$ 0.004	18.182 $\pm$ 1.400
QML	beta51	0.16	0.180	0.020 $\pm$ 0.004	0.128 $\pm$ 0.025	0.013 $\pm$ 0.001	0.112 $\pm$ 0.003	0.013 $\pm$ 0.001	0.114 $\pm$ 0.004	0.507 $\pm$ 0.028	0.712 $\pm$ 0.024	0.177	0.017	10.93	0.1047	0.1062	0.958	94.598 $\pm$ 0.821	0.421 $\pm$ 0.003	40.316 $\pm$ 1.781
QML	beta52	0.16	0.150	-0.010 $\pm$ 0.003	-0.066 $\pm$ 0.021	0.008 $\pm$ 0.000	0.091 $\pm$ 0.002	0.008 $\pm$ 0.000	0.092 $\pm$ 0.003	0.328 $\pm$ 0.017	0.572 $\pm$ 0.018	0.150	-0.010	-6.54	0.0857	0.0864	1.037	94.862 $\pm$ 0.801	0.370 $\pm$ 0.004	37.945 $\pm$ 1.761
QML	beta53	0.16	0.136	-0.024 $\pm$ 0.003	-0.152 $\pm$ 0.019	0.007 $\pm$ 0.000	0.085 $\pm$ 0.002	0.008 $\pm$ 0.000	0.088 $\pm$ 0.003	0.305 $\pm$ 0.015	0.552 $\pm$ 0.017	0.134	-0.026	-16.31	0.0879	0.0917	1.056	95.784 $\pm$ 0.729	0.352 $\pm$ 0.003	33.992 $\pm$ 1.719
QML	beta54	0.16	0.161	0.001 $\pm$ 0.003	0.006 $\pm$ 0.018	0.006 $\pm$ 0.000	0.079 $\pm$ 0.002	0.006 $\pm$ 0.000	0.079 $\pm$ 0.003	0.242 $\pm$ 0.013	0.492 $\pm$ 0.016	0.159	-0.001	-0.35	0.0783	0.0783	0.997	95.784 $\pm$ 0.729	0.308 $\pm$ 0.002	53.491 $\pm$ 1.810
QML	beta55	0.08	0.093	0.013 $\pm$ 0.004	0.161 $\pm$ 0.054	0.014 $\pm$ 0.001	0.118 $\pm$ 0.003	0.014 $\pm$ 0.001	0.119 $\pm$ 0.004	2.196 $\pm$ 0.116	1.482 $\pm$ 0.047	0.093	0.013	16.18	0.1194	0.1201	1.062	96.706 $\pm$ 0.648	0.491 $\pm$ 0.006	11.858 $\pm$ 1.173
QML	beta56	0.08	0.083	0.003 $\pm$ 0.004	0.037 $\pm$ 0.048	0.011 $\pm$ 0.001	0.106 $\pm$ 0.003	0.011 $\pm$ 0.001	0.106 $\pm$ 0.003	1.760 $\pm$ 0.095	1.327 $\pm$ 0.043	0.089	0.009	10.70	0.1052	0.1056	1.027	95.520 $\pm$ 0.751	0.427 $\pm$ 0.004	10.935 $\pm$ 1.133
QML	beta57	0.06	0.051	-0.009 $\pm$ 0.003	-0.158 $\pm$ 0.045	0.005 $\pm$ 0.000	0.074 $\pm$ 0.002	0.006 $\pm$ 0.000	0.074 $\pm$ 0.002	1.540 $\pm$ 0.084	1.241 $\pm$ 0.041	0.049	-0.011	-17.97	0.0731	0.0739	0.999	92.622 $\pm$ 0.949	0.290 $\pm$ 0.005	11.726 $\pm$ 1.168
QML	beta58	0.06	0.048	-0.012 $\pm$ 0.002	-0.201 $\pm$ 0.035	0.003 $\pm$ 0.000	0.057 $\pm$ 0.002	0.003 $\pm$ 0.000	0.058 $\pm$ 0.002	0.944 $\pm$ 0.056	0.971 $\pm$ 0.033	0.049	-0.011	-17.57	0.0518	0.0528	1.018	94.335 $\pm$ 0.839	0.228 $\pm$ 0.003	13.043 $\pm$ 1.222
UPI	beta41	0.20	0.207	0.007 $\pm$ 0.004	0.036 $\pm$ 0.018	0.009 $\pm$ 0.001	0.097 $\pm$ 0.003	0.010 $\pm$ 0.001	0.098 $\pm$ 0.003	0.238 $\pm$ 0.014	0.488 $\pm$ 0.017	0.206	0.006	2.98	0.1010	0.1012	1.262	97.760 $\pm$ 0.537	0.481 $\pm$ 0.011	47.694 $\pm$ 1.813
UPI	beta42	0.20	0.202	0.002 $\pm$ 0.004	0.009 $\pm$ 0.018	0.010 $\pm$ 0.001	0.100 $\pm$ 0.003	0.010 $\pm$ 0.001	0.100 $\pm$ 0.003	0.250 $\pm$ 0.014	0.500 $\pm$ 0.017	0.199	-0.001	-0.30	0.0959	0.0959	1.274	98.024 $\pm$ 0.505	0.499 $\pm$ 0.012	46.640 $\pm$ 1.811
UPI	beta43	0.20	0.207	0.007 $\pm$ 0.003	0.034 $\pm$ 0.016	0.008 $\pm$ 0.000	0.089 $\pm$ 0.003	0.008 $\pm$ 0.000	0.089 $\pm$ 0.003	0.197 $\pm$ 0.012	0.444 $\pm$ 0.016	0.208	0.008	4.12	0.0914	0.0918	1.213	97.760 $\pm$ 0.537	0.422 $\pm$ 0.008	56.653 $\pm$ 1.799

Table S145

Simulation 2 - Condition 1: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.4 (continued)

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.086	0.006 ± 0.004	0.076 ± 0.056	0.015 ± 0.001	0.123 ± 0.005	0.015 ± 0.001	0.123 ± 0.005	2.374 ± 0.183	1.541 ± 0.066	0.075	-0.005	-5.70	0.1021	0.1022	1.444	99.209 ± 0.321	0.697 ± 0.030	4.084 ± 0.718
UPI	beta51	0.16	0.165	0.005 ± 0.004	0.032 ± 0.025	0.012 ± 0.001	0.111 ± 0.003	0.012 ± 0.001	0.111 ± 0.004	0.482 ± 0.027	0.694 ± 0.023	0.161	0.001	0.44	0.1033	0.1033	1.015	95.916 ± 0.718	0.442 ± 0.006	34.124 ± 1.721
UPI	beta52	0.16	0.166	0.006 ± 0.003	0.037 ± 0.020	0.008 ± 0.000	0.087 ± 0.002	0.008 ± 0.000	0.087 ± 0.003	0.296 ± 0.016	0.544 ± 0.018	0.171	0.011	6.81	0.0846	0.0853	1.145	97.233 ± 0.595	0.390 ± 0.006	43.347 ± 1.799
UPI	beta53	0.16	0.163	0.003 ± 0.003	0.020 ± 0.020	0.008 ± 0.000	0.089 ± 0.002	0.008 ± 0.000	0.089 ± 0.003	0.313 ± 0.015	0.559 ± 0.017	0.163	0.003	2.11	0.0912	0.0912	1.174	98.155 ± 0.488	0.412 ± 0.006	35.441 ± 1.736
UPI	beta54	0.16	0.162	0.002 ± 0.003	0.015 ± 0.020	0.008 ± 0.000	0.088 ± 0.002	0.008 ± 0.000	0.088 ± 0.003	0.300 ± 0.016	0.548 ± 0.018	0.160	0.000	0.07	0.0825	0.0825	1.095	96.838 ± 0.635	0.376 ± 0.005	40.580 ± 1.782
UPI	beta55	0.08	0.088	0.008 ± 0.005	0.104 ± 0.066	0.021 ± 0.001	0.145 ± 0.005	0.021 ± 0.001	0.146 ± 0.005	3.309 ± 0.215	1.819 ± 0.068	0.081	0.001	1.82	0.1324	0.1324	1.272	98.682 ± 0.414	0.725 ± 0.022	2.899 ± 0.609
UPI	beta56	0.08	0.080	0.000 ± 0.005	0.006 ± 0.061	0.018 ± 0.001	0.134 ± 0.005	0.018 ± 0.001	0.134 ± 0.005	2.797 ± 0.196	1.672 ± 0.066	0.081	0.001	0.78	0.1188	0.1188	1.243	98.419 ± 0.453	0.652 ± 0.017	3.426 ± 0.660
UPI	beta57	0.06	0.070	0.010 ± 0.004	0.165 ± 0.066	0.012 ± 0.001	0.108 ± 0.004	0.012 ± 0.001	0.109 ± 0.004	3.285 ± 0.238	1.813 ± 0.073	0.069	0.009	15.18	0.0977	0.0982	1.259	96.970 ± 0.622	0.535 ± 0.022	3.030 ± 0.622
UPI	beta58	0.06	0.064	0.004 ± 0.003	0.075 ± 0.050	0.007 ± 0.000	0.083 ± 0.003	0.007 ± 0.001	0.083 ± 0.003	1.902 ± 0.139	1.379 ± 0.056	0.061	0.001	2.37	0.0751	0.0752	1.367	98.024 ± 0.505	0.443 ± 0.014	2.635 ± 0.581

Note. MCSE = Monte Carlo Standard Error shown as \u000b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S146

Simulation 2 - Condition 2: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, $N = 1000$, Reliability = 0.4

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LSAM	beta41	0.20	0.201	0.001 $\pm$ 0.002	0.007 $\pm$ 0.008	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.065 $\pm$ 0.003	0.255 $\pm$ 0.007	0.201	0.001	0.35	0.0527	0.0527	1.028	96.450 $\pm$ 0.589	0.206 $\pm$ 0.001	97.566 $\pm$ 0.491
LSAM	beta42	0.20	0.203	0.003 $\pm$ 0.002	0.017 $\pm$ 0.008	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.003 $\pm$ 0.000	0.052 $\pm$ 0.002	0.067 $\pm$ 0.003	0.258 $\pm$ 0.008	0.201	0.001	0.55	0.0522	0.0522	1.042	94.929 $\pm$ 0.699	0.211 $\pm$ 0.001	97.667 $\pm$ 0.481
LSAM	beta43	0.20	0.201	0.001 $\pm$ 0.002	0.006 $\pm$ 0.008	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.059 $\pm$ 0.003	0.243 $\pm$ 0.007	0.198	-0.002	-0.77	0.0469	0.0469	1.012	95.639 $\pm$ 0.650	0.193 $\pm$ 0.001	99.087 $\pm$ 0.303
LSAM	beta44	0.11	0.116	0.006 $\pm$ 0.002	0.054 $\pm$ 0.022	0.006 $\pm$ 0.000	0.074 $\pm$ 0.002	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.460 $\pm$ 0.024	0.678 $\pm$ 0.020	0.116	0.006	5.24	0.0719	0.0722	0.990	96.653 $\pm$ 0.573	0.289 $\pm$ 0.002	39.757 $\pm$ 1.551
LSAM	beta45	0.11	0.113	0.003 $\pm$ 0.002	0.024 $\pm$ 0.017	0.004 $\pm$ 0.000	0.060 $\pm$ 0.001	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	0.299 $\pm$ 0.014	0.547 $\pm$ 0.016	0.111	0.001	0.46	0.0560	0.0560	1.013	96.146 $\pm$ 0.613	0.239 $\pm$ 0.002	46.653 $\pm$ 1.589
LSAM	beta46	0.08	0.083	0.003 $\pm$ 0.002	0.040 $\pm$ 0.020	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.003 $\pm$ 0.000	0.051 $\pm$ 0.002	0.406 $\pm$ 0.021	0.637 $\pm$ 0.020	0.082	0.002	2.70	0.0494	0.0494	1.019	95.335 $\pm$ 0.672	0.203 $\pm$ 0.002	36.511 $\pm$ 1.533
LSAM	beta47	0.08	0.084	0.004 $\pm$ 0.002	0.048 $\pm$ 0.020	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.381 $\pm$ 0.019	0.617 $\pm$ 0.018	0.082	0.002	2.32	0.0464	0.0465	1.018	96.045 $\pm$ 0.621	0.197 $\pm$ 0.002	40.264 $\pm$ 1.562
LSAM	beta51	0.16	0.156	-0.004 $\pm$ 0.002	-0.024 $\pm$ 0.012	0.004 $\pm$ 0.000	0.062 $\pm$ 0.001	0.004 $\pm$ 0.000	0.062 $\pm$ 0.002	0.151 $\pm$ 0.007	0.388 $\pm$ 0.011	0.153	-0.007	-4.36	0.0589	0.0593	0.989	95.030 $\pm$ 0.692	0.240 $\pm$ 0.001	73.428 $\pm$ 1.407
LSAM	beta52	0.16	0.161	0.001 $\pm$ 0.002	0.009 $\pm$ 0.010	0.003 $\pm$ 0.000	0.050 $\pm$ 0.001	0.003 $\pm$ 0.000	0.050 $\pm$ 0.001	0.099 $\pm$ 0.005	0.314 $\pm$ 0.009	0.163	0.003	1.59	0.0482	0.0482	1.006	94.828 $\pm$ 0.705	0.198 $\pm$ 0.001	89.959 $\pm$ 0.957
LSAM	beta53	0.16	0.163	0.003 $\pm$ 0.002	0.019 $\pm$ 0.010	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.108 $\pm$ 0.005	0.329 $\pm$ 0.009	0.162	0.002	1.04	0.0521	0.0521	1.019	94.929 $\pm$ 0.699	0.210 $\pm$ 0.001	88.032 $\pm$ 1.034
LSAM	beta54	0.16	0.162	0.002 $\pm$ 0.002	0.011 $\pm$ 0.010	0.003 $\pm$ 0.000	0.050 $\pm$ 0.001	0.003 $\pm$ 0.000	0.050 $\pm$ 0.001	0.098 $\pm$ 0.005	0.313 $\pm$ 0.009	0.162	0.002	1.05	0.0496	0.0497	1.002	95.233 $\pm$ 0.679	0.197 $\pm$ 0.001	91.176 $\pm$ 0.903
LSAM	beta55	0.08	0.082	0.002 $\pm$ 0.002	0.022 $\pm$ 0.029	0.005 $\pm$ 0.000	0.073 $\pm$ 0.002	0.005 $\pm$ 0.000	0.073 $\pm$ 0.002	0.835 $\pm$ 0.039	0.914 $\pm$ 0.026	0.079	-0.001	-1.05	0.0725	0.0725	0.960	94.422 $\pm$ 0.731	0.275 $\pm$ 0.002	24.442 $\pm$ 1.369
LSAM	beta56	0.08	0.081	0.001 $\pm$ 0.002	0.010 $\pm$ 0.027	0.004 $\pm$ 0.000	0.067 $\pm$ 0.002	0.004 $\pm$ 0.000	0.067 $\pm$ 0.002	0.699 $\pm$ 0.032	0.836 $\pm$ 0.024	0.080	0.000	-0.57	0.0649	0.0649	0.976	94.828 $\pm$ 0.705	0.256 $\pm$ 0.002	24.037 $\pm$ 1.361
LSAM	beta57	0.06	0.066	0.006 $\pm$ 0.002	0.102 $\pm$ 0.026	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.002	0.659 $\pm$ 0.037	0.812 $\pm$ 0.026	0.064	0.004	6.38	0.0454	0.0455	1.002	96.755 $\pm$ 0.564	0.190 $\pm$ 0.002	26.978 $\pm$ 1.413
LSAM	beta58	0.06	0.062	0.002 $\pm$ 0.001	0.042 $\pm$ 0.022	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	0.465 $\pm$ 0.025	0.682 $\pm$ 0.021	0.061	0.001	1.43	0.0388	0.0388	1.000	95.335 $\pm$ 0.672	0.160 $\pm$ 0.001	34.178 $\pm$ 1.511
QML	beta41	0.20	0.184	-0.016 $\pm$ 0.001	-0.082 $\pm$ 0.007	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.058 $\pm$ 0.003	0.242 $\pm$ 0.006	0.184	-0.016	-8.11	0.0480	0.0506	1.034	94.219 $\pm$ 0.743	0.184 $\pm$ 0.001	98.479 $\pm$ 0.390
QML	beta42	0.20	0.190	-0.010 $\pm$ 0.001	-0.048 $\pm$ 0.007	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.056 $\pm$ 0.002	0.236 $\pm$ 0.006	0.189	-0.011	-5.67	0.0457	0.0471	1.046	95.132 $\pm$ 0.685	0.189 $\pm$ 0.001	98.783 $\pm$ 0.349
QML	beta43	0.20	0.182	-0.018 $\pm$ 0.001	-0.089 $\pm$ 0.007	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.058 $\pm$ 0.002	0.240 $\pm$ 0.006	0.181	-0.019	-9.62	0.0439	0.0480	1.021	93.306 $\pm$ 0.796	0.178 $\pm$ 0.001	98.986 $\pm$ 0.319
QML	beta44	0.11	0.103	-0.007 $\pm$ 0.002	-0.063 $\pm$ 0.018	0.004 $\pm$ 0.000	0.063 $\pm$ 0.001	0.004 $\pm$ 0.000	0.063 $\pm$ 0.002	0.333 $\pm$ 0.016	0.577 $\pm$ 0.016	0.103	-0.007	-6.59	0.0637	0.0641	1.062	95.030 $\pm$ 0.692	0.263 $\pm$ 0.011	38.235 $\pm$ 1.548
QML	beta45	0.11	0.107	-0.003 $\pm$ 0.002	-0.030 $\pm$ 0.015	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.231 $\pm$ 0.011	0.481 $\pm$ 0.013	0.107	-0.003	-2.33	0.0531	0.0531	1.049	94.726 $\pm$ 0.712	0.217 $\pm$ 0.007	53.347 $\pm$ 1.589
QML	beta46	0.08	0.078	-0.002 $\pm$ 0.001	-0.023 $\pm$ 0.016	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.255 $\pm$ 0.012	0.505 $\pm$ 0.014	0.079	-0.001	-1.13	0.0393	0.0393	1.193	95.436 $\pm$ 0.665	0.189 $\pm$ 0.021	45.233 $\pm$ 1.585
QML	beta47	0.08	0.077	-0.003 $\pm$ 0.001	-0.044 $\pm$ 0.016	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	0.258 $\pm$ 0.012	0.508 $\pm$ 0.014	0.075	-0.005	-6.28	0.0389	0.0392	1.042	95.335 $\pm$ 0.672	0.165 $\pm$ 0.003	45.335 $\pm$ 1.585
QML	beta51	0.16	0.178	0.018 $\pm$ 0.002	0.112 $\pm$ 0.012	0.004 $\pm$ 0.000	0.063 $\pm$ 0.002	0.004 $\pm$ 0.000	0.065 $\pm$ 0.002	0.166 $\pm$ 0.008	0.407 $\pm$ 0.011	0.177	0.017	10.44	0.0618	0.0640	1.028	94.422 $\pm$ 0.731	0.252 $\pm$ 0.003	80.933 $\pm$ 1.251
QML	beta52	0.16	0.147	-0.013 $\pm$ 0.002	-0.082 $\pm$ 0.011	0.003 $\pm$ 0.000	0.055 $\pm$ 0.001	0.003 $\pm$ 0.000	0.057 $\pm$ 0.002	0.125 $\pm$ 0.006	0.353 $\pm$ 0.010	0.150	-0.010	-6.44	0.0525	0.0535	1.015	94.016 $\pm$ 0.755	0.219 $\pm$ 0.003	75.963 $\pm$ 1.361
QML	beta53	0.16	0.136	-0.024 $\pm$ 0.002	-0.152 $\pm$ 0.010	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.003 $\pm$ 0.000	0.056 $\pm$ 0.001	0.125 $\pm$ 0.005	0.353 $\pm$ 0.009	0.135	-0.025	-15.84	0.0505	0.0565	1.042	93.205 $\pm$ 0.801	0.208 $\pm$ 0.001	74.746 $\pm$ 1.384
QML	beta54	0.16	0.160	-0.000 $\pm$ 0.002	-0.001 $\pm$ 0.009	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.088 $\pm$ 0.004	0.297 $\pm$ 0.008	0.159	-0.001	-0.83	0.0472	0.0473	1.011	94.219 $\pm$ 0.743	0.189 $\pm$ 0.001	92.394 $\pm$ 0.844
QML	beta55	0.08	0.094	0.014 $\pm$ 0.002	0.179 $\pm$ 0.029	0.005 $\pm$ 0.000	0.073 $\pm$ 0.002	0.006 $\pm$ 0.000	0.074 $\pm$ 0.002	0.861 $\pm$ 0.038	0.928 $\pm$ 0.025	0.094	0.014	17.26	0.0745	0.0758	1.016	94.219 $\pm$ 0.743	0.290 $\pm$ 0.005	27.485 $\pm$ 1.422
QML	beta56	0.08	0.077	-0.003 $\pm$ 0.002	-0.039 $\pm$ 0.026	0.004 $\pm$ 0.000	0.065 $\pm$ 0.001	0.004 $\pm$ 0.000	0.065 $\pm$ 0.002	0.667 $\pm$ 0.030	0.817 $\pm$ 0.023	0.075	-0.005	-5.82	0.0646	0.0648	0.987	94.219 $\pm$ 0.743	0.253 $\pm$ 0.003	24.239 $\pm$ 1.365
QML	beta57	0.06	0.050	-0.010 $\pm$ 0.001	-0.174 $\pm$ 0.022	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.526 $\pm$ 0.024	0.725 $\pm$ 0.020	0.049	-0.011	-19.05	0.0412	0.0427	1.042	93.915 $\pm$ 0.761	0.173 $\pm$ 0.003	20.081 $\pm$ 1.276
QML	beta58	0.06	0.051	-0.009 $\pm$ 0.001	-0.145 $\pm$ 0.019	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.001 $\pm$ 0.												

Table S146

Simulation 2 - Condition 2: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.4 (continued)

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.084	0.004 ± 0.002	0.054 ± 0.021	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.456 ± 0.026	0.675 ± 0.022	0.081	0.001	1.22	0.0513	0.0513	1.121	97.667 ± 0.481	0.237 ± 0.003	26.166 ± 1.400
UPI	beta51	0.16	0.158	-0.002 ± 0.002	-0.012 ± 0.012	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.144 ± 0.007	0.380 ± 0.011	0.156	-0.004	-2.46	0.0581	0.0582	1.015	95.335 ± 0.672	0.242 ± 0.001	75.355 ± 1.372
UPI	beta52	0.16	0.162	0.002 ± 0.002	0.013 ± 0.010	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.099 ± 0.005	0.315 ± 0.009	0.164	0.004	2.51	0.0495	0.0497	1.020	95.639 ± 0.650	0.201 ± 0.001	89.148 ± 0.991
UPI	beta53	0.16	0.162	0.002 ± 0.002	0.012 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.106 ± 0.005	0.325 ± 0.009	0.162	0.002	1.13	0.0513	0.0514	1.053	95.943 ± 0.628	0.215 ± 0.001	87.525 ± 1.052
UPI	beta54	0.16	0.161	0.001 ± 0.002	0.009 ± 0.010	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.097 ± 0.005	0.311 ± 0.009	0.160	0.000	0.21	0.0487	0.0487	1.027	96.146 ± 0.613	0.200 ± 0.001	90.264 ± 0.944
UPI	beta55	0.08	0.084	0.004 ± 0.002	0.046 ± 0.030	0.006 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.074 ± 0.002	0.865 ± 0.043	0.930 ± 0.028	0.080	0.000	0.54	0.0731	0.0731	1.026	96.653 ± 0.573	0.299 ± 0.003	19.270 ± 1.256
UPI	beta56	0.08	0.080	0.000 ± 0.002	0.006 ± 0.027	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.735 ± 0.037	0.858 ± 0.025	0.077	-0.003	-3.96	0.0647	0.0647	1.049	96.552 ± 0.581	0.282 ± 0.003	19.473 ± 1.261
UPI	beta57	0.06	0.065	0.005 ± 0.002	0.091 ± 0.027	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	0.747 ± 0.042	0.864 ± 0.028	0.062	0.002	2.87	0.0500	0.0501	1.049	96.957 ± 0.547	0.212 ± 0.002	19.371 ± 1.259
UPI	beta58	0.06	0.063	0.003 ± 0.001	0.050 ± 0.024	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.046 ± 0.002	0.576 ± 0.035	0.759 ± 0.026	0.060	0.000	-0.09	0.0428	0.0428	1.067	96.957 ± 0.547	0.190 ± 0.002	23.022 ± 1.341

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S147

Simulation 2 - Condition 3: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, $N = 400$, Reliability = 0.6

Method	Param.	True	Mean-based										Median-based					Inference		
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LSAM	beta41	0.20	0.201	0.001 $\pm$ 0.002	0.005 $\pm$ 0.010	0.004 $\pm$ 0.000	0.060 $\pm$ 0.001	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	0.091 $\pm$ 0.004	0.302 $\pm$ 0.008	0.198	-0.002	-0.80	0.0600	0.0600	0.985	94.361 $\pm$ 0.732	0.233 $\pm$ 0.001	92.649 $\pm$ 0.828
LSAM	beta42	0.20	0.201	0.001 $\pm$ 0.002	0.006 $\pm$ 0.010	0.004 $\pm$ 0.000	0.061 $\pm$ 0.001	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.092 $\pm$ 0.004	0.303 $\pm$ 0.009	0.201	0.001	0.68	0.0632	0.0632	1.004	95.569 $\pm$ 0.653	0.239 $\pm$ 0.001	91.742 $\pm$ 0.873
LSAM	beta43	0.20	0.201	0.001 $\pm$ 0.002	0.005 $\pm$ 0.009	0.003 $\pm$ 0.000	0.056 $\pm$ 0.001	0.003 $\pm$ 0.000	0.056 $\pm$ 0.002	0.077 $\pm$ 0.004	0.278 $\pm$ 0.008	0.201	0.001	0.61	0.0570	0.0570	1.009	95.468 $\pm$ 0.660	0.220 $\pm$ 0.001	95.670 $\pm$ 0.646
LSAM	beta44	0.11	0.111	0.001 $\pm$ 0.002	0.008 $\pm$ 0.021	0.005 $\pm$ 0.000	0.073 $\pm$ 0.002	0.005 $\pm$ 0.000	0.073 $\pm$ 0.002	0.439 $\pm$ 0.021	0.663 $\pm$ 0.019	0.107	-0.003	-2.85	0.0678	0.0679	0.978	93.656 $\pm$ 0.774	0.280 $\pm$ 0.002	36.757 $\pm$ 1.530
LSAM	beta45	0.11	0.113	0.003 $\pm$ 0.002	0.024 $\pm$ 0.018	0.004 $\pm$ 0.000	0.062 $\pm$ 0.001	0.004 $\pm$ 0.000	0.062 $\pm$ 0.002	0.320 $\pm$ 0.014	0.566 $\pm$ 0.016	0.115	0.005	4.25	0.0608	0.0610	0.988	94.965 $\pm$ 0.694	0.241 $\pm$ 0.001	48.036 $\pm$ 1.585
LSAM	beta46	0.08	0.082	0.002 $\pm$ 0.002	0.026 $\pm$ 0.021	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.003 $\pm$ 0.000	0.053 $\pm$ 0.002	0.432 $\pm$ 0.021	0.657 $\pm$ 0.019	0.081	0.001	0.92	0.0530	0.0530	0.956	94.058 $\pm$ 0.750	0.197 $\pm$ 0.002	38.771 $\pm$ 1.546
LSAM	beta47	0.08	0.083	0.003 $\pm$ 0.002	0.034 $\pm$ 0.021	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.003 $\pm$ 0.000	0.052 $\pm$ 0.002	0.421 $\pm$ 0.022	0.649 $\pm$ 0.020	0.079	-0.001	-1.86	0.0499	0.0499	0.957	93.454 $\pm$ 0.785	0.195 $\pm$ 0.002	39.074 $\pm$ 1.548
LSAM	beta51	0.16	0.162	0.002 $\pm$ 0.002	0.015 $\pm$ 0.014	0.005 $\pm$ 0.000	0.071 $\pm$ 0.002	0.005 $\pm$ 0.000	0.071 $\pm$ 0.002	0.197 $\pm$ 0.009	0.443 $\pm$ 0.012	0.160	0.000	0.13	0.0706	0.0706	0.994	94.562 $\pm$ 0.720	0.276 $\pm$ 0.001	63.041 $\pm$ 1.532
LSAM	beta52	0.16	0.162	0.002 $\pm$ 0.002	0.012 $\pm$ 0.011	0.003 $\pm$ 0.000	0.058 $\pm$ 0.001	0.003 $\pm$ 0.000	0.058 $\pm$ 0.002	0.131 $\pm$ 0.006	0.362 $\pm$ 0.010	0.158	-0.002	-1.17	0.0544	0.0544	1.015	95.267 $\pm$ 0.674	0.231 $\pm$ 0.001	79.255 $\pm$ 1.287
LSAM	beta53	0.16	0.160	0.000 $\pm$ 0.002	0.003 $\pm$ 0.012	0.004 $\pm$ 0.000	0.063 $\pm$ 0.001	0.004 $\pm$ 0.000	0.063 $\pm$ 0.002	0.154 $\pm$ 0.007	0.393 $\pm$ 0.011	0.162	0.002	1.25	0.0613	0.0614	0.986	94.663 $\pm$ 0.713	0.243 $\pm$ 0.001	74.723 $\pm$ 1.379
LSAM	beta54	0.16	0.158	-0.002 $\pm$ 0.002	-0.014 $\pm$ 0.012	0.003 $\pm$ 0.000	0.059 $\pm$ 0.001	0.003 $\pm$ 0.000	0.059 $\pm$ 0.002	0.134 $\pm$ 0.006	0.366 $\pm$ 0.010	0.158	-0.002	-1.37	0.0606	0.0606	0.988	94.965 $\pm$ 0.694	0.227 $\pm$ 0.001	77.140 $\pm$ 1.333
LSAM	beta55	0.08	0.085	0.005 $\pm$ 0.002	0.068 $\pm$ 0.029	0.005 $\pm$ 0.000	0.074 $\pm$ 0.002	0.006 $\pm$ 0.000	0.074 $\pm$ 0.002	0.861 $\pm$ 0.043	0.928 $\pm$ 0.027	0.084	0.004	5.25	0.0696	0.0697	0.993	94.864 $\pm$ 0.700	0.288 $\pm$ 0.002	21.752 $\pm$ 1.309
LSAM	beta56	0.08	0.082	0.002 $\pm$ 0.002	0.019 $\pm$ 0.029	0.005 $\pm$ 0.000	0.073 $\pm$ 0.002	0.005 $\pm$ 0.000	0.073 $\pm$ 0.002	0.831 $\pm$ 0.041	0.912 $\pm$ 0.027	0.077	-0.003	-3.98	0.0702	0.0703	0.933	94.562 $\pm$ 0.720	0.267 $\pm$ 0.002	23.867 $\pm$ 1.353
LSAM	beta57	0.06	0.058	-0.002 $\pm$ 0.002	-0.031 $\pm$ 0.027	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.729 $\pm$ 0.033	0.854 $\pm$ 0.024	0.057	-0.003	-5.38	0.0508	0.0509	0.966	93.857 $\pm$ 0.762	0.194 $\pm$ 0.001	22.558 $\pm$ 1.326
LSAM	beta58	0.06	0.062	0.002 $\pm$ 0.001	0.036 $\pm$ 0.024	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.553 $\pm$ 0.028	0.744 $\pm$ 0.022	0.060	0.000	0.67	0.0417	0.0417	0.957	95.267 $\pm$ 0.674	0.167 $\pm$ 0.001	32.226 $\pm$ 1.483
QML	beta41	0.20	0.191	-0.009 $\pm$ 0.002	-0.044 $\pm$ 0.009	0.003 $\pm$ 0.000	0.056 $\pm$ 0.001	0.003 $\pm$ 0.000	0.057 $\pm$ 0.002	0.081 $\pm$ 0.003	0.284 $\pm$ 0.008	0.188	-0.012	-6.04	0.0567	0.0580	0.998	94.663 $\pm$ 0.713	0.220 $\pm$ 0.001	92.951 $\pm$ 0.812
QML	beta42	0.20	0.195	-0.005 $\pm$ 0.002	-0.023 $\pm$ 0.009	0.003 $\pm$ 0.000	0.058 $\pm$ 0.001	0.003 $\pm$ 0.000	0.058 $\pm$ 0.002	0.083 $\pm$ 0.004	0.289 $\pm$ 0.008	0.194	-0.006	-2.97	0.0591	0.0594	1.001	95.368 $\pm$ 0.667	0.226 $\pm$ 0.001	92.548 $\pm$ 0.833
QML	beta43	0.20	0.190	-0.010 $\pm$ 0.002	-0.051 $\pm$ 0.008	0.003 $\pm$ 0.000	0.054 $\pm$ 0.001	0.003 $\pm$ 0.000	0.054 $\pm$ 0.002	0.074 $\pm$ 0.003	0.272 $\pm$ 0.008	0.190	-0.010	-5.22	0.0550	0.0560	1.014	94.562 $\pm$ 0.720	0.213 $\pm$ 0.001	94.864 $\pm$ 0.700
QML	beta44	0.11	0.102	-0.008 $\pm$ 0.002	-0.072 $\pm$ 0.020	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.386 $\pm$ 0.018	0.621 $\pm$ 0.018	0.099	-0.011	-9.95	0.0634	0.0643	0.996	93.454 $\pm$ 0.785	0.265 $\pm$ 0.002	33.736 $\pm$ 1.500
QML	beta45	0.11	0.110	0.000 $\pm$ 0.002	0.003 $\pm$ 0.017	0.003 $\pm$ 0.000	0.058 $\pm$ 0.001	0.003 $\pm$ 0.000	0.058 $\pm$ 0.002	0.279 $\pm$ 0.013	0.529 $\pm$ 0.015	0.113	0.003	2.63	0.0587	0.0587	1.001	95.166 $\pm$ 0.681	0.228 $\pm$ 0.001	49.245 $\pm$ 1.587
QML	beta46	0.08	0.078	-0.002 $\pm$ 0.001	-0.025 $\pm$ 0.018	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.337 $\pm$ 0.015	0.580 $\pm$ 0.016	0.077	-0.003	-3.62	0.0466	0.0467	0.991	94.260 $\pm$ 0.738	0.180 $\pm$ 0.001	40.584 $\pm$ 1.558
QML	beta47	0.08	0.079	-0.001 $\pm$ 0.001	-0.015 $\pm$ 0.018	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.333 $\pm$ 0.016	0.577 $\pm$ 0.017	0.077	-0.003	-4.02	0.0442	0.0443	0.980	92.951 $\pm$ 0.812	0.177 $\pm$ 0.001	42.095 $\pm$ 1.567
QML	beta51	0.16	0.176	0.016 $\pm$ 0.002	0.099 $\pm$ 0.014	0.005 $\pm$ 0.000	0.072 $\pm$ 0.002	0.005 $\pm$ 0.000	0.074 $\pm$ 0.002	0.215 $\pm$ 0.010	0.463 $\pm$ 0.012	0.174	0.014	8.97	0.0701	0.0715	1.016	94.361 $\pm$ 0.732	0.289 $\pm$ 0.001	66.667 $\pm$ 1.496
QML	beta52	0.16	0.143	-0.017 $\pm$ 0.002	-0.104 $\pm$ 0.012	0.004 $\pm$ 0.000	0.061 $\pm$ 0.001	0.004 $\pm$ 0.000	0.063 $\pm$ 0.002	0.157 $\pm$ 0.007	0.396 $\pm$ 0.010	0.142	-0.018	-11.40	0.0587	0.0615	1.042	95.166 $\pm$ 0.681	0.250 $\pm$ 0.001	63.041 $\pm$ 1.532
QML	beta53	0.16	0.136	-0.024 $\pm$ 0.002	-0.149 $\pm$ 0.013	0.004 $\pm$ 0.000	0.064 $\pm$ 0.002	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.183 $\pm$ 0.009	0.427 $\pm$ 0.012	0.138	-0.022	-13.88	0.0619	0.0658	0.992	92.850 $\pm$ 0.818	0.249 $\pm$ 0.001	58.409 $\pm$ 1.564
QML	beta54	0.16	0.156	-0.004 $\pm$ 0.002	-0.023 $\pm$ 0.011	0.003 $\pm$ 0.000	0.057 $\pm$ 0.001	0.003 $\pm$ 0.000	0.057 $\pm$ 0.002	0.128 $\pm$ 0.006	0.358 $\pm$ 0.010	0.155	-0.005	-3.12	0.0584	0.0586	0.991	94.965 $\pm$ 0.694	0.222 $\pm$ 0.001	78.651 $\pm$ 1.300
QML	beta55	0.08	0.092	0.012 $\pm$ 0.002	0.154 $\pm$ 0.029	0.006 $\pm$ 0.000	0.074 $\pm$ 0.002	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.883 $\pm$ 0.040	0.940 $\pm$ 0.026	0.092	0.012	14.87	0.0708	0.0718	1.022	94.965 $\pm$ 0.694	0.297 $\pm$ 0.002	23.364 $\pm$ 1.343
QML	beta56	0.08	0.081	0.001 $\pm$ 0.002	0.017 $\pm$ 0.028	0.005 $\pm$ 0.000	0.072 $\pm$ 0.002	0.005 $\pm$ 0.000	0.072 $\pm$ 0.002	0.804 $\pm$ 0.037	0.897 $\pm$ 0.025	0.083	0.003	3.78	0.0708	0.0709	0.963	94.260 $\pm$ 0.738	0.271 $\pm$ 0.002	22.659 $\pm$ 1.328
QML	beta57	0.06	0.049	-0.011 $\pm$ 0.001	-0.182 $\pm$ 0.025	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.648 $\pm$ 0.030	0.805 $\pm$ 0.022	0.049	-0.011	-18.06	0.0452	0.0464	0.989	93.253 $\pm$ 0.796	0.183 $\pm$ 0.001	19.637 $\pm$ 1.261
QML	beta58	0.06	0.056	-0.004 $\pm$ 0.001	-0.073 $\pm$ 0.021	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	0.462 $\pm$ 0.023	0.680 $\pm$ 0.020	0.055	-0.005	-8.99	0.0402	0.0406	0.978	95.368 $\pm$ 0.667	0.156 $\pm$ 0.001	30.111 $\pm$ 1.456
UPI	beta41	0.20	0.201	0.001 $\pm$ 0.002	0.006 $\pm$ 0.010	0.004 $\pm$ 0.000	0.060 $\pm$ 0.001	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	0.091 $\pm$ 0.004	0.302 $\pm$ 0.008	0.198	-0.002	-0.82	0.0606	0.0606	0.974	93.656 $\pm$ 0.774	0.231 $\pm$ 0.001	93.958 $\pm$ 0.756
UPI	beta42	0.20	0.201	0.001 $\pm$ 0.002	0.005 $\pm$ 0.010	0.004 $\pm$ 0.000	0.061 $\pm$ 0.001	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.094 $\pm$ 0.004	0.306 $\pm$ 0.009	0.201	0.001	0.34	0.0622	0.0622	0.995	95.770 $\pm$ 0.639	0.239 $\pm$ 0.001	91.944 $\pm$ 0.864
UPI	beta43	0.20	0.201	0.001 $\pm$ 0.002	0.004 $\pm$ 0.009	0.003 $\pm$ 0.000	0.056 $\pm$ 0.001	0.003 $\pm$ 0.000	0.056 $\pm$ 0.002	0.079 $\pm$ 0.004	0.280 $\pm$ 0.008	0.201	0.001	0.31	0.0586	0.0586	1.016	95.770 $\pm$ 0.639	0.224 $\pm$ 0.001	95.267 $\pm$ 0.674

Table S147

Simulation 2 - Condition 3: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.6 (continued)

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.084	0.004 ± 0.002	0.051 ± 0.022	0.003 ± 0.000	0.056 ± 0.002	0.003 ± 0.000	0.056 ± 0.002	0.487 ± 0.029	0.698 ± 0.024	0.079	-0.001	-1.29	0.0496	0.0496	0.997	94.763 ± 0.707	0.218 ± 0.002	31.722 ± 1.477
UPI	beta51	0.16	0.164	0.004 ± 0.002	0.024 ± 0.014	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.193 ± 0.008	0.439 ± 0.012	0.162	0.002	1.14	0.0691	0.0691	1.011	94.763 ± 0.707	0.278 ± 0.001	64.350 ± 1.520
UPI	beta52	0.16	0.162	0.002 ± 0.002	0.016 ± 0.012	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.133 ± 0.006	0.364 ± 0.010	0.158	-0.002	-0.98	0.0535	0.0535	1.013	95.267 ± 0.674	0.231 ± 0.001	79.355 ± 1.284
UPI	beta53	0.16	0.160	0.000 ± 0.002	0.000 ± 0.012	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.154 ± 0.007	0.393 ± 0.011	0.162	0.002	1.28	0.0622	0.0622	0.998	94.663 ± 0.713	0.246 ± 0.001	73.313 ± 1.404
UPI	beta54	0.16	0.157	-0.003 ± 0.002	-0.016 ± 0.012	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.135 ± 0.006	0.367 ± 0.010	0.157	-0.003	-1.79	0.0596	0.0596	0.990	94.663 ± 0.713	0.228 ± 0.001	76.636 ± 1.343
UPI	beta55	0.08	0.086	0.006 ± 0.002	0.078 ± 0.030	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	0.895 ± 0.046	0.946 ± 0.029	0.085	0.005	6.75	0.0688	0.0691	1.030	95.166 ± 0.681	0.305 ± 0.002	18.530 ± 1.233
UPI	beta56	0.08	0.081	0.001 ± 0.002	0.012 ± 0.030	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	0.874 ± 0.046	0.935 ± 0.029	0.080	0.000	-0.14	0.0714	0.0714	0.965	95.468 ± 0.660	0.283 ± 0.002	21.249 ± 1.298
UPI	beta57	0.06	0.058	-0.002 ± 0.002	-0.038 ± 0.029	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.808 ± 0.044	0.899 ± 0.028	0.057	-0.003	-4.83	0.0501	0.0502	0.980	94.663 ± 0.713	0.207 ± 0.002	19.738 ± 1.263
UPI	beta58	0.06	0.063	0.003 ± 0.002	0.053 ± 0.025	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.627 ± 0.033	0.792 ± 0.024	0.061	0.001	1.76	0.0459	0.0459	0.988	96.173 ± 0.609	0.184 ± 0.002	27.190 ± 1.412

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S148

Simulation 2 - Condition 4: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, $N = 1000$, Reliability = 0.6

Method	Param.	True	Mean-based										Median-based					Inference		
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LSAM	beta41	0.20	0.203	0.003 $\pm$ 0.001	0.013 $\pm$ 0.006	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.034 $\pm$ 0.002	0.184 $\pm$ 0.005	0.202	0.002	0.76	0.0355	0.0356	1.001	95.000 $\pm$ 0.689	0.144 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta42	0.20	0.198	-0.002 $\pm$ 0.001	-0.010 $\pm$ 0.006	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.034 $\pm$ 0.002	0.185 $\pm$ 0.005	0.198	-0.002	-1.04	0.0373	0.0374	1.013	95.000 $\pm$ 0.689	0.147 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta43	0.20	0.202	0.002 $\pm$ 0.001	0.009 $\pm$ 0.006	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.033 $\pm$ 0.002	0.183 $\pm$ 0.005	0.201	0.001	0.74	0.0356	0.0357	0.963	93.800 $\pm$ 0.763	0.138 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta44	0.11	0.111	0.001 $\pm$ 0.001	0.012 $\pm$ 0.013	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.157 $\pm$ 0.007	0.396 $\pm$ 0.011	0.112	0.002	2.00	0.0444	0.0445	0.988	95.800 $\pm$ 0.634	0.169 $\pm$ 0.001	74.200 $\pm$ 1.384
LSAM	beta45	0.11	0.112	0.002 $\pm$ 0.001	0.020 $\pm$ 0.011	0.001 $\pm$ 0.000	0.039 $\pm$ 0.001	0.001 $\pm$ 0.000	0.039 $\pm$ 0.001	0.123 $\pm$ 0.006	0.351 $\pm$ 0.010	0.112	0.002	1.51	0.0363	0.0363	0.981	94.400 $\pm$ 0.727	0.148 $\pm$ 0.001	83.400 $\pm$ 1.177
LSAM	beta46	0.08	0.081	0.001 $\pm$ 0.001	0.014 $\pm$ 0.012	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.154 $\pm$ 0.007	0.392 $\pm$ 0.011	0.080	0.000	0.60	0.0314	0.0314	0.968	94.900 $\pm$ 0.696	0.119 $\pm$ 0.001	75.300 $\pm$ 1.364
LSAM	beta47	0.08	0.081	0.001 $\pm$ 0.001	0.011 $\pm$ 0.012	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.143 $\pm$ 0.007	0.379 $\pm$ 0.011	0.081	0.001	1.21	0.0304	0.0304	0.988	94.000 $\pm$ 0.751	0.117 $\pm$ 0.001	77.900 $\pm$ 1.312
LSAM	beta51	0.16	0.160	0.000 $\pm$ 0.001	0.001 $\pm$ 0.009	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.078 $\pm$ 0.004	0.279 $\pm$ 0.008	0.160	0.000	-0.01	0.0437	0.0437	0.980	94.600 $\pm$ 0.715	0.172 $\pm$ 0.000	95.100 $\pm$ 0.683
LSAM	beta52	0.16	0.160	-0.000 $\pm$ 0.001	-0.002 $\pm$ 0.007	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.053 $\pm$ 0.002	0.231 $\pm$ 0.006	0.159	-0.001	-0.52	0.0355	0.0355	0.985	94.600 $\pm$ 0.715	0.143 $\pm$ 0.000	99.500 $\pm$ 0.223
LSAM	beta53	0.16	0.161	0.001 $\pm$ 0.001	0.008 $\pm$ 0.007	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.056 $\pm$ 0.003	0.237 $\pm$ 0.007	0.161	0.001	0.72	0.0380	0.0381	1.011	95.300 $\pm$ 0.669	0.150 $\pm$ 0.000	98.800 $\pm$ 0.344
LSAM	beta54	0.16	0.161	0.001 $\pm$ 0.001	0.007 $\pm$ 0.007	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.046 $\pm$ 0.002	0.216 $\pm$ 0.006	0.161	0.001	0.76	0.0352	0.0352	1.047	96.000 $\pm$ 0.620	0.142 $\pm$ 0.000	99.700 $\pm$ 0.173
LSAM	beta55	0.08	0.082	0.002 $\pm$ 0.001	0.029 $\pm$ 0.018	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.324 $\pm$ 0.015	0.569 $\pm$ 0.016	0.080	0.000	0.43	0.0454	0.0454	0.973	95.500 $\pm$ 0.656	0.174 $\pm$ 0.001	46.400 $\pm$ 1.577
LSAM	beta56	0.08	0.078	-0.002 $\pm$ 0.001	-0.022 $\pm$ 0.017	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.291 $\pm$ 0.012	0.540 $\pm$ 0.014	0.076	-0.004	-5.14	0.0440	0.0442	0.972	94.200 $\pm$ 0.739	0.164 $\pm$ 0.001	45.700 $\pm$ 1.575
LSAM	beta57	0.06	0.059	-0.001 $\pm$ 0.001	-0.012 $\pm$ 0.016	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.263 $\pm$ 0.013	0.513 $\pm$ 0.015	0.059	-0.001	-1.15	0.0298	0.0298	0.968	94.800 $\pm$ 0.702	0.117 $\pm$ 0.001	51.700 $\pm$ 1.580
LSAM	beta58	0.06	0.062	0.002 $\pm$ 0.001	0.032 $\pm$ 0.014	0.001 $\pm$ 0.000	0.027 $\pm$ 0.001	0.001 $\pm$ 0.000	0.027 $\pm$ 0.001	0.197 $\pm$ 0.010	0.444 $\pm$ 0.013	0.062	0.002	3.73	0.0256	0.0257	0.986	94.600 $\pm$ 0.715	0.103 $\pm$ 0.000	68.300 $\pm$ 1.471
QML	beta41	0.20	0.193	-0.007 $\pm$ 0.001	-0.033 $\pm$ 0.006	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.033 $\pm$ 0.001	0.180 $\pm$ 0.005	0.192	-0.008	-3.79	0.0347	0.0355	1.007	94.400 $\pm$ 0.727	0.140 $\pm$ 0.002	99.800 $\pm$ 0.141
QML	beta42	0.20	0.192	-0.008 $\pm$ 0.001	-0.038 $\pm$ 0.006	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.033 $\pm$ 0.002	0.183 $\pm$ 0.005	0.192	-0.008	-4.16	0.0355	0.0364	1.012	94.100 $\pm$ 0.745	0.142 $\pm$ 0.001	99.800 $\pm$ 0.141
QML	beta43	0.20	0.191	-0.009 $\pm$ 0.001	-0.047 $\pm$ 0.006	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.034 $\pm$ 0.002	0.183 $\pm$ 0.005	0.190	-0.010	-5.23	0.0340	0.0355	0.969	93.100 $\pm$ 0.801	0.135 $\pm$ 0.001	99.800 $\pm$ 0.141
QML	beta44	0.11	0.105	-0.005 $\pm$ 0.001	-0.045 $\pm$ 0.012	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.152 $\pm$ 0.007	0.390 $\pm$ 0.010	0.105	-0.005	-4.21	0.0430	0.0433	0.981	94.600 $\pm$ 0.715	0.164 $\pm$ 0.001	70.700 $\pm$ 1.439
QML	beta45	0.11	0.110	0.000 $\pm$ 0.001	0.003 $\pm$ 0.011	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.112 $\pm$ 0.005	0.335 $\pm$ 0.009	0.110	0.000	0.24	0.0355	0.0355	0.990	94.600 $\pm$ 0.715	0.143 $\pm$ 0.001	85.600 $\pm$ 1.110
QML	beta46	0.08	0.079	-0.001 $\pm$ 0.001	-0.011 $\pm$ 0.011	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.129 $\pm$ 0.006	0.360 $\pm$ 0.010	0.079	-0.001	-1.80	0.0301	0.0301	0.988	95.300 $\pm$ 0.669	0.111 $\pm$ 0.001	78.800 $\pm$ 1.293
QML	beta47	0.08	0.078	-0.002 $\pm$ 0.001	-0.025 $\pm$ 0.011	0.001 $\pm$ 0.000	0.028 $\pm$ 0.001	0.001 $\pm$ 0.000	0.028 $\pm$ 0.001	0.125 $\pm$ 0.006	0.354 $\pm$ 0.010	0.079	-0.001	-1.20	0.0277	0.0278	0.991	93.800 $\pm$ 0.763	0.110 $\pm$ 0.001	80.100 $\pm$ 1.263
QML	beta51	0.16	0.172	0.012 $\pm$ 0.001	0.078 $\pm$ 0.009	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.088 $\pm$ 0.004	0.296 $\pm$ 0.008	0.173	0.013	7.98	0.0459	0.0476	0.995	95.000 $\pm$ 0.689	0.178 $\pm$ 0.001	96.400 $\pm$ 0.589
QML	beta52	0.16	0.142	-0.018 $\pm$ 0.001	-0.114 $\pm$ 0.008	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.073 $\pm$ 0.003	0.271 $\pm$ 0.007	0.142	-0.018	-11.42	0.0390	0.0430	1.000	93.000 $\pm$ 0.807	0.154 $\pm$ 0.001	94.900 $\pm$ 0.696
QML	beta53	0.16	0.138	-0.022 $\pm$ 0.001	-0.139 $\pm$ 0.008	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.076 $\pm$ 0.003	0.276 $\pm$ 0.007	0.139	-0.021	-13.42	0.0389	0.0445	1.028	91.500 $\pm$ 0.882	0.154 $\pm$ 0.000	93.400 $\pm$ 0.785
QML	beta54	0.16	0.160	-0.000 $\pm$ 0.001	-0.000 $\pm$ 0.007	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.045 $\pm$ 0.002	0.212 $\pm$ 0.006	0.160	0.000	0.26	0.0340	0.0340	1.048	95.600 $\pm$ 0.649	0.139 $\pm$ 0.000	99.700 $\pm$ 0.173
QML	beta55	0.08	0.091	0.011 $\pm$ 0.001	0.138 $\pm$ 0.019	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.368 $\pm$ 0.017	0.606 $\pm$ 0.017	0.091	0.011	13.13	0.0467	0.0479	0.984	94.400 $\pm$ 0.727	0.182 $\pm$ 0.001	50.900 $\pm$ 1.581
QML	beta56	0.08	0.078	-0.002 $\pm$ 0.001	-0.024 $\pm$ 0.017	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.300 $\pm$ 0.012	0.548 $\pm$ 0.014	0.077	-0.003	-3.69	0.0471	0.0472	0.980	95.200 $\pm$ 0.676	0.168 $\pm$ 0.001	43.800 $\pm$ 1.569
QML	beta57	0.06	0.051	-0.009 $\pm$ 0.001	-0.156 $\pm$ 0.016	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.268 $\pm$ 0.012	0.518 $\pm$ 0.014	0.050	-0.010	-17.40	0.0292	0.0310	0.969	92.800 $\pm$ 0.817	0.113 $\pm$ 0.001	42.000 $\pm$ 1.561
QML	beta58	0.06	0.056	-0.004 $\pm$ 0.001	-0.060 $\pm$ 0.013	0.001 $\pm$ 0.000	0.025 $\pm$ 0.001	0.001 $\pm$ 0.000	0.025 $\pm$ 0.001	0.177 $\pm$ 0.009	0.421 $\pm$ 0.012	0.057	-0.003	-5.47	0.0242	0.0244	0.993	94.200 $\pm$ 0.739	0.097 $\pm$ 0.000	64.700 $\pm$ 1.511
UPI	beta41	0.20	0.203	0.003 $\pm$ 0.001	0.013 $\pm$ 0.006	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.034 $\pm$ 0.002	0.186 $\pm$ 0.005	0.202	0.002	0.78	0.0353	0.0353	0.968	93.700 $\pm$ 0.768	0.140 $\pm$ 0.000	100.000 $\pm$ 0.000
UPI	beta42	0.20	0.198	-0.002 $\pm$ 0.001	-0.011 $\pm$ 0.006	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.034 $\pm$ 0.002	0.185 $\pm$ 0.005	0.197	-0.003	-1.31	0.0378	0.0379	0.999	94.900 $\pm$ 0.696	0.145 $\pm$ 0.000	100.000 $\pm$ 0.000
UPI	beta43	0.20	0.202	0.002 $\pm$ 0.001	0.008 $\pm$ 0.006	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.033 $\pm$ 0.002	0.182 $\pm$ 0.005	0.201	0.001	0.47	0.0360	0.0360	0.963	94.200 $\pm$ 0.739	0.138 $\pm$ 0.000	100.000 $\pm$ 0.00

Table S148

Simulation 2 - Condition 4: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.6 (continued)

Method	Param.	True	Mean-based								Median-based					Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.081	0.001 ± 0.001	0.017 ± 0.013	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.158 ± 0.007	0.398 ± 0.011	0.081	0.001	1.85	0.0321	0.0322	0.993	94.400 ± 0.727	0.124 ± 0.001	75.600 ± 1.358
UPI	beta51	0.16	0.161	0.001 ± 0.001	0.005 ± 0.009	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.078 ± 0.004	0.279 ± 0.008	0.160	0.000	0.30	0.0450	0.0450	0.976	94.800 ± 0.702	0.171 ± 0.000	95.400 ± 0.662
UPI	beta52	0.16	0.160	-0.000 ± 0.001	-0.001 ± 0.007	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.053 ± 0.002	0.230 ± 0.006	0.159	-0.001	-0.47	0.0359	0.0359	0.980	94.300 ± 0.733	0.141 ± 0.000	99.500 ± 0.223
UPI	beta53	0.16	0.161	0.001 ± 0.001	0.007 ± 0.007	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.055 ± 0.002	0.235 ± 0.006	0.161	0.001	0.83	0.0382	0.0382	1.021	95.600 ± 0.649	0.151 ± 0.000	98.900 ± 0.330
UPI	beta54	0.16	0.161	0.001 ± 0.001	0.007 ± 0.007	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.046 ± 0.002	0.215 ± 0.006	0.161	0.001	0.86	0.0351	0.0352	1.045	96.000 ± 0.620	0.141 ± 0.000	99.700 ± 0.173
UPI	beta55	0.08	0.082	0.002 ± 0.001	0.027 ± 0.018	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.333 ± 0.016	0.577 ± 0.017	0.080	0.000	-0.39	0.0449	0.0449	0.981	95.900 ± 0.627	0.177 ± 0.001	45.000 ± 1.573
UPI	beta56	0.08	0.078	-0.002 ± 0.001	-0.028 ± 0.017	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.298 ± 0.013	0.546 ± 0.015	0.077	-0.003	-3.85	0.0444	0.0445	0.980	95.000 ± 0.689	0.168 ± 0.001	43.700 ± 1.569
UPI	beta57	0.06	0.059	-0.001 ± 0.001	-0.014 ± 0.017	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.275 ± 0.013	0.525 ± 0.015	0.058	-0.002	-3.75	0.0308	0.0309	0.976	94.600 ± 0.715	0.120 ± 0.001	49.900 ± 1.581
UPI	beta58	0.06	0.062	0.002 ± 0.001	0.035 ± 0.014	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	0.205 ± 0.010	0.453 ± 0.013	0.062	0.002	2.83	0.0268	0.0268	1.009	95.200 ± 0.676	0.107 ± 0.001	65.800 ± 1.500

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S149

Simulation 2 - Condition 5: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, $N = 400$, Reliability = 0.8

Method	Param.	True	Mean-based										Median-based					Inference			
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE	
LSAM	beta41	0.20	0.199	-0.001 $\pm$ 0.002	-0.005 $\pm$ 0.008	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.059 $\pm$ 0.003	0.243 $\pm$ 0.007	0.198	-0.002	-1.03	0.0487	0.0488	0.984	94.600 $\pm$ 0.715	0.187 $\pm$ 0.000	98.700 $\pm$ 0.358	
LSAM	beta42	0.20	0.201	0.001 $\pm$ 0.002	0.007 $\pm$ 0.008	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	0.062 $\pm$ 0.003	0.249 $\pm$ 0.007	0.202	0.002	0.95	0.0502	0.0502	0.974	94.200 $\pm$ 0.739	0.190 $\pm$ 0.000	98.300 $\pm$ 0.409	
LSAM	beta43	0.20	0.201	0.001 $\pm$ 0.001	0.004 $\pm$ 0.007	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.056 $\pm$ 0.002	0.236 $\pm$ 0.006	0.202	0.002	1.14	0.0488	0.0489	0.960	93.400 $\pm$ 0.785	0.178 $\pm$ 0.000	98.800 $\pm$ 0.344	
LSAM	beta44	0.11	0.114	0.004 $\pm$ 0.002	0.033 $\pm$ 0.015	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.236 $\pm$ 0.011	0.486 $\pm$ 0.013	0.114	0.004	3.37	0.0531	0.0532	0.962	93.700 $\pm$ 0.768	0.201 $\pm$ 0.001	59.800 $\pm$ 1.550	
LSAM	beta45	0.11	0.110	-0.000 $\pm$ 0.002	-0.003 $\pm$ 0.014	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.199 $\pm$ 0.009	0.446 $\pm$ 0.013	0.110	0.000	0.12	0.0485	0.0485	0.928	93.300 $\pm$ 0.791	0.179 $\pm$ 0.001	68.200 $\pm$ 1.473	
LSAM	beta46	0.08	0.080	0.000 $\pm$ 0.001	0.005 $\pm$ 0.015	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.215 $\pm$ 0.010	0.463 $\pm$ 0.013	0.080	0.000	0.41	0.0377	0.0377	0.971	94.600 $\pm$ 0.715	0.141 $\pm$ 0.001	62.100 $\pm$ 1.534	
LSAM	beta47	0.08	0.080	0.000 $\pm$ 0.001	0.001 $\pm$ 0.015	0.001 $\pm$ 0.000	0.039 $\pm$ 0.001	0.001 $\pm$ 0.000	0.039 $\pm$ 0.001	0.234 $\pm$ 0.011	0.483 $\pm$ 0.014	0.079	-0.001	-0.66	0.0392	0.0392	0.923	94.100 $\pm$ 0.745	0.140 $\pm$ 0.001	60.200 $\pm$ 1.548	
LSAM	beta51	0.16	0.161	0.001 $\pm$ 0.002	0.005 $\pm$ 0.011	0.003 $\pm$ 0.000	0.058 $\pm$ 0.001	0.003 $\pm$ 0.000	0.058 $\pm$ 0.002	0.130 $\pm$ 0.006	0.360 $\pm$ 0.010	0.160	0.000	-0.07	0.0580	0.0580	0.976	93.800 $\pm$ 0.763	0.221 $\pm$ 0.001	80.400 $\pm$ 1.255	
LSAM	beta52	0.16	0.160	-0.000 $\pm$ 0.001	-0.001 $\pm$ 0.009	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.084 $\pm$ 0.004	0.290 $\pm$ 0.008	0.159	-0.001	-0.55	0.0466	0.0467	1.024	94.500 $\pm$ 0.721	0.186 $\pm$ 0.000	92.000 $\pm$ 0.858	
LSAM	beta53	0.16	0.158	-0.002 $\pm$ 0.002	-0.010 $\pm$ 0.010	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	0.097 $\pm$ 0.004	0.312 $\pm$ 0.009	0.156	-0.004	-2.44	0.0485	0.0487	0.995	94.000 $\pm$ 0.751	0.194 $\pm$ 0.001	90.000 $\pm$ 0.949	
LSAM	beta54	0.16	0.161	0.001 $\pm$ 0.002	0.004 $\pm$ 0.009	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.090 $\pm$ 0.004	0.300 $\pm$ 0.008	0.160	0.000	-0.01	0.0484	0.0484	0.984	93.800 $\pm$ 0.763	0.185 $\pm$ 0.000	92.900 $\pm$ 0.812	
LSAM	beta55	0.08	0.081	0.001 $\pm$ 0.002	0.007 $\pm$ 0.023	0.003 $\pm$ 0.000	0.059 $\pm$ 0.001	0.003 $\pm$ 0.000	0.059 $\pm$ 0.002	0.536 $\pm$ 0.025	0.732 $\pm$ 0.021	0.080	0.000	-0.57	0.0575	0.0575	0.921	93.000 $\pm$ 0.807	0.211 $\pm$ 0.001	34.300 $\pm$ 1.501	
LSAM	beta56	0.08	0.082	0.002 $\pm$ 0.002	0.019 $\pm$ 0.021	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.003 $\pm$ 0.000	0.053 $\pm$ 0.002	0.442 $\pm$ 0.021	0.665 $\pm$ 0.019	0.084	0.004	4.96	0.0511	0.0512	0.963	94.000 $\pm$ 0.751	0.201 $\pm$ 0.001	37.600 $\pm$ 1.532	
LSAM	beta57	0.06	0.061	0.001 $\pm$ 0.001	0.022 $\pm$ 0.020	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.403 $\pm$ 0.019	0.635 $\pm$ 0.018	0.061	0.001	1.05	0.0368	0.0368	0.938	92.800 $\pm$ 0.817	0.140 $\pm$ 0.001	41.200 $\pm$ 1.556	
LSAM	beta58	0.06	0.060	0.000 $\pm$ 0.001	0.007 $\pm$ 0.018	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.323 $\pm$ 0.014	0.569 $\pm$ 0.015	0.060	0.000	0.55	0.0341	0.0341	0.939	93.100 $\pm$ 0.801	0.126 $\pm$ 0.001	48.300 $\pm$ 1.580	
QML	beta41	0.20	0.195	-0.005 $\pm$ 0.002	-0.025 $\pm$ 0.008	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.057 $\pm$ 0.002	0.239 $\pm$ 0.006	0.195	-0.005	-2.55	0.0467	0.0470	0.986	94.700 $\pm$ 0.708	0.184 $\pm$ 0.000	99.000 $\pm$ 0.315	
QML	beta42	0.20	0.199	-0.001 $\pm$ 0.002	-0.005 $\pm$ 0.008	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.060 $\pm$ 0.003	0.245 $\pm$ 0.007	0.199	-0.001	-0.68	0.0491	0.0491	0.975	95.000 $\pm$ 0.689	0.187 $\pm$ 0.001	98.300 $\pm$ 0.409	
QML	beta43	0.20	0.196	-0.004 $\pm$ 0.001	-0.019 $\pm$ 0.007	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.055 $\pm$ 0.002	0.235 $\pm$ 0.006	0.197	-0.003	-1.35	0.0482	0.0483	0.959	94.200 $\pm$ 0.739	0.176 $\pm$ 0.000	98.700 $\pm$ 0.358	
QML	beta44	0.11	0.111	0.001 $\pm$ 0.002	0.006 $\pm$ 0.015	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.226 $\pm$ 0.010	0.476 $\pm$ 0.013	0.110	0.000	-0.20	0.0516	0.0516	0.971	93.900 $\pm$ 0.757	0.199 $\pm$ 0.001	58.300 $\pm$ 1.559	
QML	beta45	0.11	0.109	-0.001 $\pm$ 0.002	-0.010 $\pm$ 0.014	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.189 $\pm$ 0.009	0.435 $\pm$ 0.012	0.109	-0.001	-0.61	0.0468	0.0468	0.941	93.200 $\pm$ 0.796	0.177 $\pm$ 0.001	68.700 $\pm$ 1.466	
QML	beta46	0.08	0.079	-0.001 $\pm$ 0.001	-0.010 $\pm$ 0.014	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.199 $\pm$ 0.009	0.446 $\pm$ 0.012	0.080	0.000	-0.30	0.0368	0.0368	0.981	95.200 $\pm$ 0.676	0.137 $\pm$ 0.001	63.000 $\pm$ 1.527	
QML	beta47	0.08	0.079	-0.001 $\pm$ 0.001	-0.017 $\pm$ 0.015	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.215 $\pm$ 0.010	0.464 $\pm$ 0.013	0.077	-0.003	-3.15	0.0379	0.0380	0.934	94.000 $\pm$ 0.751	0.136 $\pm$ 0.001	61.500 $\pm$ 1.539	
QML	beta51	0.16	0.167	0.007 $\pm$ 0.002	0.041 $\pm$ 0.012	0.003 $\pm$ 0.000	0.058 $\pm$ 0.001	0.003 $\pm$ 0.000	0.059 $\pm$ 0.002	0.134 $\pm$ 0.006	0.366 $\pm$ 0.010	0.167	0.007	4.15	0.0575	0.0579	0.989	94.000 $\pm$ 0.751	0.226 $\pm$ 0.001	81.600 $\pm$ 1.225	
QML	beta52	0.16	0.142	-0.018 $\pm$ 0.002	-0.111 $\pm$ 0.010	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.109 $\pm$ 0.005	0.331 $\pm$ 0.009	0.142	-0.018	-11.24	0.0478	0.0510	1.017	93.100 $\pm$ 0.801	0.199 $\pm$ 0.001	80.400 $\pm$ 1.255	
QML	beta53	0.16	0.137	-0.023 $\pm$ 0.002	-0.142 $\pm$ 0.010	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.003 $\pm$ 0.000	0.057 $\pm$ 0.001	0.125 $\pm$ 0.005	0.354 $\pm$ 0.009	0.135	-0.025	-15.82	0.0487	0.0549	0.998	91.800 $\pm$ 0.868	0.203 $\pm$ 0.001	76.800 $\pm$ 1.335	
QML	beta54	0.16	0.160	-0.000 $\pm$ 0.002	-0.001 $\pm$ 0.009	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.088 $\pm$ 0.004	0.297 $\pm$ 0.008	0.159	-0.001	-0.79	0.0478	0.0478	0.986	94.100 $\pm$ 0.745	0.184 $\pm$ 0.000	93.300 $\pm$ 0.791	
QML	beta55	0.08	0.084	0.004 $\pm$ 0.002	0.052 $\pm$ 0.024	0.004 $\pm$ 0.000	0.060 $\pm$ 0.001	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	0.556 $\pm$ 0.026	0.745 $\pm$ 0.021	0.083	0.003	3.53	0.0583	0.0584	0.925	93.200 $\pm$ 0.796	0.216 $\pm$ 0.001	36.300 $\pm$ 1.521	
QML	beta56	0.08	0.082	0.002 $\pm$ 0.002	0.024 $\pm$ 0.021	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.003 $\pm$ 0.000	0.053 $\pm$ 0.002	0.447 $\pm$ 0.021	0.669 $\pm$ 0.019	0.083	0.003	4.14	0.0520	0.0521	0.973	94.300 $\pm$ 0.733	0.204 $\pm$ 0.001	36.700 $\pm$ 1.524	
QML	beta57	0.06	0.057	-0.003 $\pm$ 0.001	-0.048 $\pm$ 0.019	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.380 $\pm$ 0.017	0.616 $\pm$ 0.017	0.056	-0.004	-7.12	0.0360	0.0362	0.950	92.300 $\pm$ 0.843	0.137 $\pm$ 0.001	37.800 $\pm$ 1.533	
QML	beta58	0.06	0.058	-0.002 $\pm$ 0.001	-0.032 $\pm$ 0.018	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.001 $\pm$ 0.000	td												

Table S149

Simulation 2 - Condition 5: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.8 (continued)

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.081	0.001 ± 0.001	0.008 ± 0.016	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.246 ± 0.012	0.496 ± 0.014	0.080	0.000	0.14	0.0406	0.0406	0.928	93.800 ± 0.763	0.144 ± 0.001	59.500 ± 1.552
UPI	beta51	0.16	0.161	0.001 ± 0.002	0.008 ± 0.011	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.129 ± 0.006	0.359 ± 0.010	0.161	0.001	0.46	0.0582	0.0582	0.978	93.700 ± 0.768	0.220 ± 0.001	80.200 ± 1.260
UPI	beta52	0.16	0.160	0.000 ± 0.001	0.000 ± 0.009	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.084 ± 0.004	0.289 ± 0.008	0.159	-0.001	-0.54	0.0461	0.0461	1.011	94.900 ± 0.696	0.183 ± 0.000	92.700 ± 0.823
UPI	beta53	0.16	0.158	-0.002 ± 0.002	-0.010 ± 0.010	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.097 ± 0.004	0.312 ± 0.009	0.157	-0.003	-1.60	0.0497	0.0498	0.995	93.900 ± 0.757	0.194 ± 0.001	89.600 ± 0.965
UPI	beta54	0.16	0.161	0.001 ± 0.002	0.004 ± 0.010	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.090 ± 0.004	0.301 ± 0.008	0.161	0.001	0.41	0.0476	0.0476	0.974	93.900 ± 0.757	0.184 ± 0.000	93.200 ± 0.796
UPI	beta55	0.08	0.081	0.001 ± 0.002	0.006 ± 0.023	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.539 ± 0.026	0.734 ± 0.021	0.079	-0.001	-0.89	0.0554	0.0554	0.930	93.500 ± 0.780	0.214 ± 0.001	33.900 ± 1.497
UPI	beta56	0.08	0.082	0.002 ± 0.002	0.021 ± 0.021	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.456 ± 0.022	0.675 ± 0.020	0.083	0.003	3.69	0.0505	0.0506	0.967	94.000 ± 0.751	0.205 ± 0.001	36.600 ± 1.523
UPI	beta57	0.06	0.061	0.001 ± 0.001	0.021 ± 0.020	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.410 ± 0.019	0.640 ± 0.018	0.061	0.001	1.06	0.0382	0.0382	0.946	93.700 ± 0.768	0.142 ± 0.001	38.600 ± 1.539
UPI	beta58	0.06	0.060	0.000 ± 0.001	0.007 ± 0.018	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.331 ± 0.015	0.575 ± 0.016	0.059	-0.001	-1.49	0.0345	0.0345	0.948	93.400 ± 0.785	0.128 ± 0.001	47.000 ± 1.578

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S150

Simulation 2 - Condition 6: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, $N = 1000$, Reliability = 0.8

Method	Param.	True	Mean-based										Median-based					Inference		
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LSAM	beta41	0.20	0.200	0.000 $\pm$ 0.001	0.002 $\pm$ 0.005	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.022 $\pm$ 0.001	0.150 $\pm$ 0.004	0.202	0.002	0.75	0.0300	0.0300	1.008	95.600 $\pm$ 0.649	0.118 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta42	0.20	0.202	0.002 $\pm$ 0.001	0.009 $\pm$ 0.005	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.024 $\pm$ 0.001	0.155 $\pm$ 0.004	0.202	0.002	0.78	0.0303	0.0303	0.986	94.300 $\pm$ 0.733	0.119 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta43	0.20	0.200	-0.000 $\pm$ 0.001	-0.001 $\pm$ 0.004	0.001 $\pm$ 0.000	0.027 $\pm$ 0.001	0.001 $\pm$ 0.000	0.027 $\pm$ 0.001	0.019 $\pm$ 0.001	0.136 $\pm$ 0.004	0.199	-0.001	-0.28	0.0278	0.0278	1.047	96.200 $\pm$ 0.605	0.112 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta44	0.11	0.112	0.002 $\pm$ 0.001	0.020 $\pm$ 0.009	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	0.087 $\pm$ 0.004	0.295 $\pm$ 0.008	0.113	0.003	2.94	0.0316	0.0318	0.988	93.700 $\pm$ 0.768	0.126 $\pm$ 0.000	92.800 $\pm$ 0.817
LSAM	beta45	0.11	0.110	-0.000 $\pm$ 0.001	-0.002 $\pm$ 0.008	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.070 $\pm$ 0.003	0.264 $\pm$ 0.007	0.109	-0.001	-1.26	0.0285	0.0285	0.983	94.900 $\pm$ 0.696	0.112 $\pm$ 0.000	97.200 $\pm$ 0.522
LSAM	beta46	0.08	0.081	0.001 $\pm$ 0.001	0.007 $\pm$ 0.009	0.001 $\pm$ 0.000	0.024 $\pm$ 0.001	0.001 $\pm$ 0.000	0.024 $\pm$ 0.001	0.087 $\pm$ 0.004	0.295 $\pm$ 0.008	0.080	0.000	-0.32	0.0234	0.0234	0.959	94.000 $\pm$ 0.751	0.089 $\pm$ 0.000	93.700 $\pm$ 0.768
LSAM	beta47	0.08	0.080	0.000 $\pm$ 0.001	0.003 $\pm$ 0.009	0.001 $\pm$ 0.000	0.023 $\pm$ 0.001	0.001 $\pm$ 0.000	0.023 $\pm$ 0.001	0.086 $\pm$ 0.004	0.293 $\pm$ 0.008	0.080	0.000	-0.22	0.0238	0.0238	0.952	94.100 $\pm$ 0.745	0.087 $\pm$ 0.000	94.100 $\pm$ 0.745
LSAM	beta51	0.16	0.159	-0.001 $\pm$ 0.001	-0.008 $\pm$ 0.007	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.050 $\pm$ 0.002	0.223 $\pm$ 0.006	0.159	-0.001	-0.34	0.0344	0.0344	1.001	94.200 $\pm$ 0.739	0.140 $\pm$ 0.000	99.000 $\pm$ 0.315
LSAM	beta52	0.16	0.160	-0.000 $\pm$ 0.001	-0.002 $\pm$ 0.006	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.032 $\pm$ 0.001	0.179 $\pm$ 0.005	0.159	-0.001	-0.66	0.0280	0.0280	1.049	96.100 $\pm$ 0.612	0.118 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta53	0.16	0.161	0.001 $\pm$ 0.001	0.003 $\pm$ 0.006	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.037 $\pm$ 0.002	0.191 $\pm$ 0.005	0.160	0.000	0.23	0.0295	0.0295	1.020	95.500 $\pm$ 0.656	0.122 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta54	0.16	0.162	0.002 $\pm$ 0.001	0.015 $\pm$ 0.006	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.034 $\pm$ 0.001	0.184 $\pm$ 0.005	0.161	0.001	0.81	0.0308	0.0308	1.009	96.100 $\pm$ 0.612	0.116 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta55	0.08	0.081	0.001 $\pm$ 0.001	0.009 $\pm$ 0.013	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.171 $\pm$ 0.008	0.414 $\pm$ 0.011	0.080	0.000	0.38	0.0320	0.0320	1.021	95.300 $\pm$ 0.669	0.132 $\pm$ 0.000	67.300 $\pm$ 1.483
LSAM	beta56	0.08	0.078	-0.002 $\pm$ 0.001	-0.019 $\pm$ 0.013	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.177 $\pm$ 0.008	0.421 $\pm$ 0.012	0.080	0.000	-0.16	0.0331	0.0331	0.959	93.200 $\pm$ 0.796	0.126 $\pm$ 0.000	69.100 $\pm$ 1.461
LSAM	beta57	0.06	0.060	0.000 $\pm$ 0.001	0.007 $\pm$ 0.012	0.001 $\pm$ 0.000	0.023 $\pm$ 0.001	0.001 $\pm$ 0.000	0.023 $\pm$ 0.001	0.147 $\pm$ 0.007	0.383 $\pm$ 0.011	0.061	0.001	2.33	0.0231	0.0232	0.980	94.200 $\pm$ 0.739	0.088 $\pm$ 0.000	76.000 $\pm$ 1.351
LSAM	beta58	0.06	0.060	0.000 $\pm$ 0.001	0.006 $\pm$ 0.011	0.000 $\pm$ 0.000	0.020 $\pm$ 0.000	0.000 $\pm$ 0.000	0.020 $\pm$ 0.001	0.113 $\pm$ 0.005	0.336 $\pm$ 0.009	0.060	0.000	0.53	0.0205	0.0205	0.998	94.800 $\pm$ 0.702	0.079 $\pm$ 0.000	85.400 $\pm$ 1.117
QML	beta41	0.20	0.196	-0.004 $\pm$ 0.001	-0.018 $\pm$ 0.005	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.022 $\pm$ 0.001	0.149 $\pm$ 0.004	0.197	-0.003	-1.37	0.0299	0.0300	1.005	94.900 $\pm$ 0.696	0.116 $\pm$ 0.000	100.000 $\pm$ 0.000
QML	beta42	0.20	0.200	-0.000 $\pm$ 0.001	-0.002 $\pm$ 0.005	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.023 $\pm$ 0.001	0.153 $\pm$ 0.004	0.200	0.000	-0.11	0.0293	0.0293	0.983	94.000 $\pm$ 0.751	0.118 $\pm$ 0.000	100.000 $\pm$ 0.000
QML	beta43	0.20	0.195	-0.005 $\pm$ 0.001	-0.024 $\pm$ 0.004	0.001 $\pm$ 0.000	0.027 $\pm$ 0.001	0.001 $\pm$ 0.000	0.027 $\pm$ 0.001	0.019 $\pm$ 0.001	0.137 $\pm$ 0.004	0.194	-0.006	-2.94	0.0280	0.0286	1.049	95.700 $\pm$ 0.641	0.111 $\pm$ 0.000	100.000 $\pm$ 0.000
QML	beta44	0.11	0.110	-0.000 $\pm$ 0.001	-0.001 $\pm$ 0.009	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	0.086 $\pm$ 0.004	0.293 $\pm$ 0.008	0.111	0.001	0.96	0.0315	0.0315	0.993	93.700 $\pm$ 0.768	0.125 $\pm$ 0.000	92.100 $\pm$ 0.853
QML	beta45	0.11	0.109	-0.001 $\pm$ 0.001	-0.007 $\pm$ 0.008	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.069 $\pm$ 0.003	0.262 $\pm$ 0.007	0.108	-0.002	-1.75	0.0289	0.0290	0.987	94.400 $\pm$ 0.727	0.111 $\pm$ 0.000	96.900 $\pm$ 0.548
QML	beta46	0.08	0.080	0.000 $\pm$ 0.001	0.001 $\pm$ 0.009	0.001 $\pm$ 0.000	0.023 $\pm$ 0.001	0.001 $\pm$ 0.000	0.023 $\pm$ 0.001	0.084 $\pm$ 0.004	0.290 $\pm$ 0.008	0.080	0.000	-0.48	0.0222	0.0222	0.958	94.000 $\pm$ 0.751	0.087 $\pm$ 0.000	94.500 $\pm$ 0.721
QML	beta47	0.08	0.079	-0.001 $\pm$ 0.001	-0.011 $\pm$ 0.009	0.001 $\pm$ 0.000	0.023 $\pm$ 0.000	0.001 $\pm$ 0.000	0.023 $\pm$ 0.001	0.080 $\pm$ 0.003	0.283 $\pm$ 0.008	0.079	-0.001	-1.34	0.0232	0.0233	0.963	94.200 $\pm$ 0.739	0.086 $\pm$ 0.000	94.200 $\pm$ 0.739
QML	beta51	0.16	0.164	0.004 $\pm$ 0.001	0.025 $\pm$ 0.007	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.052 $\pm$ 0.002	0.227 $\pm$ 0.006	0.164	0.004	2.68	0.0344	0.0347	1.006	94.400 $\pm$ 0.727	0.143 $\pm$ 0.000	99.300 $\pm$ 0.264
QML	beta52	0.16	0.142	-0.018 $\pm$ 0.001	-0.112 $\pm$ 0.006	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.049 $\pm$ 0.002	0.222 $\pm$ 0.005	0.143	-0.017	-10.87	0.0301	0.0348	1.045	91.900 $\pm$ 0.863	0.126 $\pm$ 0.000	99.500 $\pm$ 0.223
QML	beta53	0.16	0.140	-0.020 $\pm$ 0.001	-0.124 $\pm$ 0.006	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.056 $\pm$ 0.002	0.236 $\pm$ 0.006	0.141	-0.019	-12.18	0.0317	0.0372	1.014	90.300 $\pm$ 0.936	0.127 $\pm$ 0.000	99.700 $\pm$ 0.173
QML	beta54	0.16	0.162	0.002 $\pm$ 0.001	0.012 $\pm$ 0.006	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.034 $\pm$ 0.001	0.183 $\pm$ 0.005	0.161	0.001	0.61	0.0304	0.0304	1.010	95.700 $\pm$ 0.641	0.116 $\pm$ 0.000	100.000 $\pm$ 0.000
QML	beta55	0.08	0.085	0.005 $\pm$ 0.001	0.056 $\pm$ 0.013	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.181 $\pm$ 0.008	0.426 $\pm$ 0.012	0.084	0.004	5.33	0.0328	0.0331	1.027	95.000 $\pm$ 0.689	0.136 $\pm$ 0.000	68.400 $\pm$ 1.470
QML	beta56	0.08	0.079	-0.001 $\pm$ 0.001	-0.011 $\pm$ 0.014	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.182 $\pm$ 0.008	0.427 $\pm$ 0.012	0.081	0.001	0.95	0.0333	0.0333	0.961	93.800 $\pm$ 0.763	0.129 $\pm$ 0.000	67.400 $\pm$ 1.482
QML	beta57	0.06	0.057	-0.003 $\pm$ 0.001	-0.055 $\pm$ 0.012	0.001 $\pm$ 0.000	0.023 $\pm$ 0.001	0.001 $\pm$ 0.000	0.023 $\pm$ 0.001	0.146 $\pm$ 0.007	0.382 $\pm$ 0.010	0.058	-0.002	-4.09	0.0232	0.0234	0.981	93.900 $\pm$ 0.757	0.087 $\pm$ 0.000	72.800 $\pm$ 1.407
QML	beta58	0.06	0.058	-0.002 $\pm$ 0.001	-0.028 $\pm$ 0.010	0.000 $\pm$ 0.000	0.020 $\pm$ 0.000	0.000 $\pm$ 0.000	0.020 $\pm$ 0.001	0.108 $\pm$ 0.005	0.329 $\pm$ 0.009	0.058	-0.002	-2.67	0.0196	0.0197	1.007	94.800 $\pm$ 0.702	0.078 $\pm$ 0.000	83.700 $\pm$ 1.168
UPI	beta41	0.20	0.200	0.000 $\pm$ 0.001	0.002 $\pm$ 0.005	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.022 $\pm$ 0.001	0.150 $\pm$ 0.004	0.202	0.002	1.00	0.0297	0.0298	0.962	94.600 $\pm$ 0.715	0.113 $\pm$ 0.000	100.000 $\pm$ 0.000
UPI	beta42	0.20	0.202	0.002 $\pm$ 0.001	0.008 $\pm$ 0.005	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.024 $\pm$ 0.001	0.155 $\pm$ 0.004	0.202	0.002	0.80	0.0303	0.0304	0.958	93.100 $\pm$ 0.801	0.116 $\pm$ 0.000	100.000 $\pm$ 0.000
UPI	beta43	0.20	0.200	-0.000 $\pm$ 0.001	-0.001 $\pm$ 0.004	0.001 $\pm$ 0.000	0.027 $\pm$ 0.001	0.001 $\pm$ 0.000	0.027 $\pm$ 0.001	0.019 $\pm$ 0.001	0.136 $\pm$ 0.004	0.199	-0.001	-0.30	0.0278	0.0278	1.040	95.800 $\pm$ 0.634	0.111 $\pm$ 0.000	100.000 $\pm$

Table S150

Simulation 2 - Condition 6: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.8 (continued)

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.080	0.000 ± 0.001	0.004 ± 0.009	0.001 ± 0.000	0.024 ± 0.001	0.001 ± 0.000	0.024 ± 0.001	0.090 ± 0.004	0.300 ± 0.008	0.079	-0.001	-0.96	0.0241	0.0241	0.941	93.900 ± 0.757	0.089 ± 0.000	94.100 ± 0.745
UPI	beta51	0.16	0.159	-0.001 ± 0.001	-0.007 ± 0.007	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.050 ± 0.002	0.223 ± 0.006	0.159	-0.001	-0.42	0.0336	0.0336	0.998	93.700 ± 0.768	0.139 ± 0.000	99.100 ± 0.299
UPI	beta52	0.16	0.160	-0.000 ± 0.001	-0.001 ± 0.006	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.032 ± 0.001	0.179 ± 0.005	0.159	-0.001	-0.42	0.0280	0.0280	1.033	95.600 ± 0.649	0.116 ± 0.000	100.000 ± 0.000
UPI	beta53	0.16	0.161	0.001 ± 0.001	0.003 ± 0.006	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.037 ± 0.002	0.192 ± 0.005	0.161	0.001	0.36	0.0295	0.0295	1.016	95.600 ± 0.649	0.122 ± 0.000	100.000 ± 0.000
UPI	beta54	0.16	0.163	0.003 ± 0.001	0.016 ± 0.006	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.034 ± 0.001	0.184 ± 0.005	0.161	0.001	0.72	0.0310	0.0310	1.001	96.100 ± 0.612	0.115 ± 0.000	100.000 ± 0.000
UPI	beta55	0.08	0.081	0.001 ± 0.001	0.008 ± 0.013	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.171 ± 0.008	0.414 ± 0.011	0.080	0.000	-0.14	0.0323	0.0323	1.026	95.300 ± 0.669	0.133 ± 0.000	66.600 ± 1.491
UPI	beta56	0.08	0.078	-0.002 ± 0.001	-0.021 ± 0.013	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.177 ± 0.008	0.420 ± 0.012	0.080	0.000	-0.27	0.0325	0.0325	0.966	93.600 ± 0.774	0.127 ± 0.000	67.800 ± 1.478
UPI	beta57	0.06	0.060	0.000 ± 0.001	0.006 ± 0.012	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	0.148 ± 0.007	0.384 ± 0.011	0.061	0.001	1.74	0.0233	0.0233	0.985	94.500 ± 0.721	0.089 ± 0.000	75.100 ± 1.367
UPI	beta58	0.06	0.060	0.000 ± 0.001	0.004 ± 0.011	0.000 ± 0.000	0.020 ± 0.000	0.000 ± 0.000	0.020 ± 0.001	0.114 ± 0.005	0.338 ± 0.009	0.060	0.000	0.81	0.0204	0.0204	1.003	95.400 ± 0.662	0.080 ± 0.000	84.900 ± 1.132

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S151

Simulation 2 - Condition 7: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, $N = 400$, Reliability = 0.4

Method	Param.	True	Mean-based										Median-based					Inference		
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LSAM	beta41	0.20	0.201	0.001 $\pm$ 0.007	0.003 $\pm$ 0.033	0.030 $\pm$ 0.002	0.172 $\pm$ 0.005	0.030 $\pm$ 0.002	0.172 $\pm$ 0.006	0.739 $\pm$ 0.046	0.860 $\pm$ 0.031	0.203	0.003	1.50	0.1639	0.1639	1.008	97.376 $\pm$ 0.610	0.680 $\pm$ 0.012	26.968 $\pm$ 1.694
LSAM	beta42	0.20	0.202	0.002 $\pm$ 0.006	0.008 $\pm$ 0.032	0.027 $\pm$ 0.002	0.166 $\pm$ 0.005	0.027 $\pm$ 0.002	0.166 $\pm$ 0.006	0.685 $\pm$ 0.042	0.828 $\pm$ 0.030	0.206	0.006	3.03	0.1640	0.1641	1.073	97.668 $\pm$ 0.576	0.697 $\pm$ 0.011	25.073 $\pm$ 1.655
LSAM	beta43	0.20	0.197	-0.003 $\pm$ 0.003	-0.013 $\pm$ 0.017	0.008 $\pm$ 0.000	0.088 $\pm$ 0.003	0.008 $\pm$ 0.000	0.087 $\pm$ 0.003	0.191 $\pm$ 0.012	0.437 $\pm$ 0.016	0.196	-0.004	-1.85	0.0848	0.0848	1.040	96.793 $\pm$ 0.673	0.357 $\pm$ 0.004	66.035 $\pm$ 1.808
LSAM	beta44	0.11	0.125	0.015 $\pm$ 0.006	0.137 $\pm$ 0.052	0.022 $\pm$ 0.002	0.148 $\pm$ 0.005	0.022 $\pm$ 0.002	0.149 $\pm$ 0.006	1.837 $\pm$ 0.130	1.355 $\pm$ 0.054	0.114	0.004	3.97	0.1289	0.1290	1.076	96.793 $\pm$ 0.673	0.626 $\pm$ 0.021	13.411 $\pm$ 1.301
LSAM	beta45	0.11	0.123	0.013 $\pm$ 0.005	0.122 $\pm$ 0.046	0.018 $\pm$ 0.001	0.134 $\pm$ 0.005	0.018 $\pm$ 0.001	0.134 $\pm$ 0.005	1.494 $\pm$ 0.103	1.222 $\pm$ 0.048	0.116	0.006	5.63	0.1268	0.1270	1.037	96.793 $\pm$ 0.673	0.544 $\pm$ 0.014	15.743 $\pm$ 1.391
LSAM	beta46	0.08	0.085	0.005 $\pm$ 0.003	0.064 $\pm$ 0.038	0.006 $\pm$ 0.000	0.080 $\pm$ 0.003	0.006 $\pm$ 0.000	0.080 $\pm$ 0.003	0.996 $\pm$ 0.071	0.998 $\pm$ 0.040	0.084	0.004	5.22	0.0716	0.0717	0.997	95.190 $\pm$ 0.817	0.312 $\pm$ 0.009	26.968 $\pm$ 1.694
LSAM	beta47	0.08	0.087	0.007 $\pm$ 0.003	0.082 $\pm$ 0.034	0.005 $\pm$ 0.000	0.072 $\pm$ 0.003	0.005 $\pm$ 0.000	0.072 $\pm$ 0.003	0.806 $\pm$ 0.058	0.898 $\pm$ 0.037	0.080	0.000	0.40	0.0653	0.0653	1.068	96.647 $\pm$ 0.687	0.300 $\pm$ 0.008	27.988 $\pm$ 1.714
LSAM	beta51	0.16	0.159	-0.001 $\pm$ 0.004	-0.004 $\pm$ 0.024	0.010 $\pm$ 0.001	0.102 $\pm$ 0.003	0.010 $\pm$ 0.001	0.102 $\pm$ 0.004	0.408 $\pm$ 0.024	0.639 $\pm$ 0.022	0.159	-0.001	-0.64	0.1012	0.1012	1.010	95.627 $\pm$ 0.781	0.405 $\pm$ 0.003	35.131 $\pm$ 1.823
LSAM	beta52	0.16	0.161	0.001 $\pm$ 0.006	0.003 $\pm$ 0.039	0.027 $\pm$ 0.002	0.163 $\pm$ 0.005	0.027 $\pm$ 0.002	0.163 $\pm$ 0.006	1.043 $\pm$ 0.069	1.021 $\pm$ 0.039	0.155	-0.005	-2.98	0.1549	0.1550	1.056	97.376 $\pm$ 0.610	0.677 $\pm$ 0.012	16.764 $\pm$ 1.426
LSAM	beta53	0.16	0.161	0.001 $\pm$ 0.005	0.004 $\pm$ 0.030	0.016 $\pm$ 0.001	0.125 $\pm$ 0.004	0.016 $\pm$ 0.001	0.125 $\pm$ 0.004	0.613 $\pm$ 0.035	0.783 $\pm$ 0.027	0.157	-0.003	-1.65	0.1219	0.1220	0.995	96.647 $\pm$ 0.687	0.489 $\pm$ 0.006	27.259 $\pm$ 1.700
LSAM	beta54	0.16	0.159	-0.001 $\pm$ 0.006	-0.005 $\pm$ 0.036	0.023 $\pm$ 0.002	0.153 $\pm$ 0.005	0.023 $\pm$ 0.002	0.153 $\pm$ 0.006	0.911 $\pm$ 0.063	0.955 $\pm$ 0.038	0.160	0.000	0.21	0.1474	0.1474	1.040	97.085 $\pm$ 0.642	0.623 $\pm$ 0.010	16.035 $\pm$ 1.401
LSAM	beta55	0.08	0.077	-0.003 $\pm$ 0.004	-0.031 $\pm$ 0.055	0.013 $\pm$ 0.001	0.115 $\pm$ 0.004	0.013 $\pm$ 0.001	0.115 $\pm$ 0.004	2.062 $\pm$ 0.128	1.436 $\pm$ 0.052	0.076	-0.004	-4.59	0.1047	0.1048	1.065	96.501 $\pm$ 0.702	0.480 $\pm$ 0.011	11.808 $\pm$ 1.232
LSAM	beta56	0.08	0.086	0.006 $\pm$ 0.003	0.075 $\pm$ 0.033	0.005 $\pm$ 0.000	0.069 $\pm$ 0.002	0.005 $\pm$ 0.000	0.069 $\pm$ 0.003	0.751 $\pm$ 0.052	0.867 $\pm$ 0.034	0.087	0.007	9.25	0.0667	0.0671	1.008	95.481 $\pm$ 0.793	0.273 $\pm$ 0.006	31.924 $\pm$ 1.780
LSAM	beta57	0.06	0.065	0.005 $\pm$ 0.005	0.084 $\pm$ 0.076	0.014 $\pm$ 0.001	0.119 $\pm$ 0.004	0.014 $\pm$ 0.001	0.119 $\pm$ 0.005	3.914 $\pm$ 0.276	1.978 $\pm$ 0.080	0.064	0.004	6.62	0.1057	0.1058	1.060	97.668 $\pm$ 0.576	0.493 $\pm$ 0.012	7.434 $\pm$ 1.002
LSAM	beta58	0.06	0.062	0.002 $\pm$ 0.002	0.041 $\pm$ 0.038	0.004 $\pm$ 0.000	0.059 $\pm$ 0.002	0.004 $\pm$ 0.000	0.059 $\pm$ 0.002	0.980 $\pm$ 0.071	0.990 $\pm$ 0.040	0.061	0.001	1.83	0.0518	0.0518	1.011	96.647 $\pm$ 0.687	0.235 $\pm$ 0.005	23.032 $\pm$ 1.608
QML	beta41	0.20	0.238	0.038 $\pm$ 0.004	0.189 $\pm$ 0.018	0.009 $\pm$ 0.000	0.092 $\pm$ 0.003	0.010 $\pm$ 0.001	0.100 $\pm$ 0.003	0.249 $\pm$ 0.014	0.499 $\pm$ 0.017	0.237	0.037	18.55	0.0918	0.0990	0.950	94.169 $\pm$ 0.895	0.344 $\pm$ 0.003	79.155 $\pm$ 1.551
QML	beta42	0.20	0.237	0.037 $\pm$ 0.003	0.183 $\pm$ 0.017	0.008 $\pm$ 0.000	0.091 $\pm$ 0.003	0.010 $\pm$ 0.001	0.098 $\pm$ 0.003	0.240 $\pm$ 0.014	0.490 $\pm$ 0.017	0.237	0.037	18.68	0.0917	0.0990	1.093	94.752 $\pm$ 0.851	0.390 $\pm$ 0.037	76.385 $\pm$ 1.622
QML	beta43	0.20	0.184	-0.016 $\pm$ 0.003	-0.082 $\pm$ 0.015	0.006 $\pm$ 0.000	0.076 $\pm$ 0.002	0.006 $\pm$ 0.000	0.078 $\pm$ 0.003	0.151 $\pm$ 0.009	0.389 $\pm$ 0.013	0.177	-0.023	-11.26	0.0753	0.0786	0.998	94.169 $\pm$ 0.895	0.298 $\pm$ 0.002	69.242 $\pm$ 1.762
QML	beta44	0.11	0.031	-0.079 $\pm$ 0.004	-0.716 $\pm$ 0.039	0.012 $\pm$ 0.001	0.111 $\pm$ 0.003	0.018 $\pm$ 0.001	0.136 $\pm$ 0.004	1.529 $\pm$ 0.087	1.236 $\pm$ 0.039	0.030	-0.080	-72.40	0.1180	0.1424	0.974	88.630 $\pm$ 1.212	0.424 $\pm$ 0.007	8.017 $\pm$ 1.037
QML	beta45	0.11	0.098	-0.012 $\pm$ 0.004	-0.105 $\pm$ 0.032	0.009 $\pm$ 0.000	0.094 $\pm$ 0.002	0.009 $\pm$ 0.000	0.094 $\pm$ 0.003	0.734 $\pm$ 0.039	0.857 $\pm$ 0.028	0.099	-0.011	-9.95	0.0958	0.0964	0.959	91.691 $\pm$ 1.054	0.352 $\pm$ 0.005	22.157 $\pm$ 1.586
QML	beta46	0.08	0.158	0.078 $\pm$ 0.003	0.979 $\pm$ 0.033	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.011 $\pm$ 0.001	0.104 $\pm$ 0.003	1.685 $\pm$ 0.084	1.298 $\pm$ 0.034	0.155	0.075	93.95	0.0670	0.1007	1.005	83.382 $\pm$ 1.421	0.269 $\pm$ 0.004	70.700 $\pm$ 1.738
QML	beta47	0.08	0.157	0.077 $\pm$ 0.003	0.965 $\pm$ 0.032	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.011 $\pm$ 0.001	0.103 $\pm$ 0.003	1.650 $\pm$ 0.094	1.285 $\pm$ 0.038	0.152	0.072	90.20	0.0640	0.0965	0.986	83.965 $\pm$ 1.401	0.262 $\pm$ 0.006	75.510 $\pm$ 1.642
QML	beta51	0.16	0.184	0.024 $\pm$ 0.004	0.152 $\pm$ 0.027	0.013 $\pm$ 0.001	0.115 $\pm$ 0.003	0.014 $\pm$ 0.001	0.117 $\pm$ 0.004	0.536 $\pm$ 0.029	0.732 $\pm$ 0.024	0.178	0.018	11.56	0.1175	0.1190	1.013	94.315 $\pm$ 0.884	0.455 $\pm$ 0.005	37.026 $\pm$ 1.844
QML	beta52	0.16	0.160	-0.000 $\pm$ 0.004	-0.000 $\pm$ 0.027	0.013 $\pm$ 0.001	0.113 $\pm$ 0.004	0.013 $\pm$ 0.001	0.113 $\pm$ 0.004	0.498 $\pm$ 0.031	0.706 $\pm$ 0.026	0.157	-0.003	-1.64	0.1162	0.1163	1.004	94.461 $\pm$ 0.873	0.445 $\pm$ 0.009	31.924 $\pm$ 1.780
QML	beta53	0.16	0.108	-0.052 $\pm$ 0.004	-0.323 $\pm$ 0.025	0.011 $\pm$ 0.001	0.107 $\pm$ 0.003	0.014 $\pm$ 0.001	0.119 $\pm$ 0.003	0.549 $\pm$ 0.027	0.741 $\pm$ 0.021	0.101	-0.059	-36.75	0.1080	0.1230	0.979	91.545 $\pm$ 1.062	0.409 $\pm$ 0.003	19.971 $\pm$ 1.526
QML	beta54	0.16	0.178	0.018 $\pm$ 0.003	0.110 $\pm$ 0.021	0.008 $\pm$ 0.000	0.087 $\pm$ 0.002	0.008 $\pm$ 0.000	0.088 $\pm$ 0.003	0.305 $\pm$ 0.016	0.552 $\pm$ 0.018	0.179	0.019	11.94	0.0897	0.0917	1.036	94.606 $\pm$ 0.862	0.352 $\pm$ 0.005	54.082 $\pm$ 1.903
QML	beta55	0.08	0.075	-0.005 $\pm$ 0.005	-0.056 $\pm$ 0.063	0.017 $\pm$ 0.001	0.132 $\pm$ 0.004	0.017 $\pm$ 0.001	0.132 $\pm$ 0.005	2.714 $\pm$ 0.175	1.647 $\pm$ 0.062	0.072	-0.008	-9.57	0.1366	0.1368	0.898	89.359 $\pm$ 1.177	0.464 $\pm$ 0.009	18.805 $\pm$ 1.492
QML	beta56	0.08	0.121	0.041 $\pm$ 0.003	0.508 $\pm$ 0.036	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.007 $\pm$ 0.000	0.085 $\pm$ 0.003	1.128 $\pm$ 0.060	1.062 $\pm$ 0.032	0.118	0.038	48.07	0.0789	0.0878	0.936	88.338 $\pm$ 1.225	0.274 $\pm$ 0.004	45.044 $\pm$ 1.900
QML	beta57	0.06	0.134	0.074 $\pm$ 0.004	1.230 $\pm$ 0.073	0.013 $\pm$ 0.001	0.115 $\pm$ 0.003	0.019 $\pm$ 0.001	0.137 $\pm$ 0.004	5.185 $\pm$ 0.293	2.277 $\pm$ 0.072	0.133	0.073	121.67	0.1174	0.1383	0.964	86.589 $\pm$ 1.301	0.435 $\pm$ 0.009	32.507 $\pm$ 1.788
QML	beta58	0.06	0.108	0.048 $\pm$ 0.002	0.798 $\pm$ 0.034	0.003 $\pm$ 0.000	0.054 $\pm$ 0.002	0.005 $\pm$ 0.000	0.072 $\pm$ 0.002	1.440 $\pm$ 0.083	1.200 $\pm$ 0.037	0.105	0.045	74.79	0.0491	0.0665	0.985	88.776 $\pm$ 1.205	0.208 $\pm$ 0.004	60.933 $\pm$ 1.863
UPI	beta41	0.20	0.215	0.015 $\pm$ 0.007	0.075 $\pm$ 0.036	0.036 $\pm$ 0.003	0.190 $\pm$ 0.007	0.036 $\pm$ 0.003	0.191 $\pm$ 0.008	0.910 $\pm$ 0.067	0.954 $\pm$ 0.040	0.216	0.016	8.17	0.1820	0.1827	1.308	98.542 $\pm$ 0.458	0.976 $\pm$ 0.027	20.700 $\pm$ 1.547
UPI	beta42	0.20	0.223	0.023 $\pm$ 0.007	0.116 $\pm$ 0.037	0.037 $\pm$ 0.002	0.192 $\pm$ 0.006	0.037 $\pm$ 0.002	0.193 $\pm$ 0.007	0.931 $\pm$ 0.062	0.965 $\pm$ 0.037	0.226	0.026	13.12	0.1796	0.1815	1.347	98.542 $\pm$ 0.458	1.013 $\pm$ 0.039	16.181 $\pm$ 1.406
UPI	beta43	0.20	0.196	-0.004 $\pm$ 0.004	-0.018 $\pm$ 0.018	0.009 $\pm$ 0.001	0.094 $\pm$ 0.003	0.009 $\pm$ 0.001	0.094 $\pm$ 0.003	0.220 $\pm$ 0.013	0.469 $\pm$ 0.016	0.193	-0.007	-3.34	0.0974	0.0976	1.209	96.939 $\pm$ 0.658	0.444 $\pm$ 0.012	54.665 $\pm$ 1.

Table S151

Simulation 2 - Condition 7: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.4
(continued)

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.079	-0.001 ± 0.003	-0.007 ± 0.040	0.007 ± 0.001	0.085 ± 0.003	0.007 ± 0.001	0.085 ± 0.004	1.119 ± 0.084	1.058 ± 0.045	0.072	-0.008	-9.41	0.0712	0.0716	1.342	95.918 ± 0.755	0.445 ± 0.022	18.805 ± 1.492
UPI	beta51	0.16	0.169	0.009 ± 0.004	0.057 ± 0.025	0.011 ± 0.001	0.105 ± 0.003	0.011 ± 0.001	0.105 ± 0.004	0.430 ± 0.024	0.656 ± 0.022	0.161	0.001	0.37	0.1102	0.1102	1.060	96.210 ± 0.729	0.435 ± 0.004	34.694 ± 1.817
UPI	beta52	0.16	0.183	0.023 ± 0.007	0.141 ± 0.044	0.035 ± 0.003	0.186 ± 0.007	0.035 ± 0.003	0.187 ± 0.008	1.372 ± 0.108	1.171 ± 0.052	0.175	0.015	9.45	0.1855	0.1861	1.205	97.522 ± 0.594	0.879 ± 0.030	16.327 ± 1.411
UPI	beta53	0.16	0.155	-0.005 ± 0.005	-0.031 ± 0.032	0.018 ± 0.001	0.135 ± 0.004	0.018 ± 0.001	0.135 ± 0.005	0.717 ± 0.044	0.847 ± 0.031	0.156	-0.004	-2.79	0.1327	0.1327	1.150	97.668 ± 0.576	0.610 ± 0.013	19.534 ± 1.514
UPI	beta54	0.16	0.167	0.007 ± 0.007	0.046 ± 0.041	0.030 ± 0.002	0.172 ± 0.006	0.030 ± 0.002	0.172 ± 0.006	1.160 ± 0.074	1.077 ± 0.040	0.165	0.005	3.24	0.1710	0.1711	1.297	97.959 ± 0.540	0.876 ± 0.034	12.974 ± 1.283
UPI	beta55	0.08	0.090	0.010 ± 0.005	0.129 ± 0.061	0.017 ± 0.001	0.129 ± 0.005	0.017 ± 0.001	0.129 ± 0.005	2.598 ± 0.184	1.612 ± 0.065	0.084	0.004	4.64	0.1219	0.1219	1.269	98.251 ± 0.501	0.640 ± 0.025	11.516 ± 1.219
UPI	beta56	0.08	0.082	0.002 ± 0.003	0.029 ± 0.035	0.005 ± 0.000	0.074 ± 0.003	0.005 ± 0.000	0.074 ± 0.003	0.856 ± 0.061	0.925 ± 0.037	0.083	0.003	3.42	0.0701	0.0701	1.186	97.376 ± 0.610	0.344 ± 0.011	23.469 ± 1.618
UPI	beta57	0.06	0.048	-0.012 ± 0.005	-0.205 ± 0.088	0.019 ± 0.002	0.138 ± 0.006	0.019 ± 0.002	0.139 ± 0.006	5.350 ± 0.433	2.313 ± 0.104	0.049	-0.011	-18.89	0.1157	0.1163	1.201	97.813 ± 0.558	0.651 ± 0.029	4.519 ± 0.793
UPI	beta58	0.06	0.059	-0.001 ± 0.003	-0.018 ± 0.045	0.005 ± 0.000	0.071 ± 0.003	0.005 ± 0.000	0.071 ± 0.003	1.400 ± 0.111	1.183 ± 0.052	0.058	-0.002	-2.67	0.0616	0.0616	1.234	96.793 ± 0.673	0.344 ± 0.016	16.035 ± 1.401

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S152

Simulation 2 - Condition 8: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, $N = 1000$, Reliability = 0.4

Method	Param.	True	Mean-based										Median-based					Inference		
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LSAM	beta41	0.20	0.202	0.002 $\pm$ 0.003	0.011 $\pm$ 0.016	0.010 $\pm$ 0.000	0.098 $\pm$ 0.002	0.010 $\pm$ 0.000	0.098 $\pm$ 0.003	0.238 $\pm$ 0.012	0.488 $\pm$ 0.015	0.198	-0.002	-0.87	0.0973	0.0973	1.020	95.781 $\pm$ 0.653	0.391 $\pm$ 0.003	54.641 $\pm$ 1.617
LSAM	beta42	0.20	0.199	-0.001 $\pm$ 0.003	-0.007 $\pm$ 0.017	0.011 $\pm$ 0.001	0.107 $\pm$ 0.003	0.011 $\pm$ 0.001	0.107 $\pm$ 0.003	0.286 $\pm$ 0.014	0.535 $\pm$ 0.016	0.205	0.005	2.52	0.1048	0.1049	0.979	95.464 $\pm$ 0.676	0.411 $\pm$ 0.003	54.008 $\pm$ 1.619
LSAM	beta43	0.20	0.200	0.000 $\pm$ 0.002	0.001 $\pm$ 0.009	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.003 $\pm$ 0.000	0.052 $\pm$ 0.002	0.069 $\pm$ 0.003	0.262 $\pm$ 0.008	0.199	-0.001	-0.71	0.0523	0.0523	1.007	95.992 $\pm$ 0.637	0.207 $\pm$ 0.001	96.624 $\pm$ 0.587
LSAM	beta44	0.11	0.119	0.009 $\pm$ 0.003	0.078 $\pm$ 0.023	0.006 $\pm$ 0.000	0.078 $\pm$ 0.002	0.006 $\pm$ 0.000	0.079 $\pm$ 0.002	0.515 $\pm$ 0.027	0.717 $\pm$ 0.022	0.114	0.004	3.34	0.0777	0.0778	0.984	95.570 $\pm$ 0.668	0.303 $\pm$ 0.003	36.709 $\pm$ 1.565
LSAM	beta45	0.11	0.116	0.006 $\pm$ 0.002	0.057 $\pm$ 0.021	0.005 $\pm$ 0.000	0.070 $\pm$ 0.002	0.005 $\pm$ 0.000	0.071 $\pm$ 0.002	0.411 $\pm$ 0.022	0.641 $\pm$ 0.020	0.114	0.004	3.92	0.0662	0.0663	1.005	95.042 $\pm$ 0.705	0.277 $\pm$ 0.003	40.401 $\pm$ 1.594
LSAM	beta46	0.08	0.081	0.001 $\pm$ 0.001	0.012 $\pm$ 0.017	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.277 $\pm$ 0.014	0.526 $\pm$ 0.016	0.081	0.001	1.05	0.0396	0.0396	0.933	93.460 $\pm$ 0.803	0.154 $\pm$ 0.002	59.810 $\pm$ 1.592
LSAM	beta47	0.08	0.082	0.002 $\pm$ 0.001	0.029 $\pm$ 0.017	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	0.266 $\pm$ 0.014	0.515 $\pm$ 0.016	0.080	0.000	-0.31	0.0394	0.0394	0.950	95.042 $\pm$ 0.705	0.153 $\pm$ 0.002	60.338 $\pm$ 1.589
LSAM	beta51	0.16	0.155	-0.005 $\pm$ 0.002	-0.030 $\pm$ 0.013	0.004 $\pm$ 0.000	0.062 $\pm$ 0.001	0.004 $\pm$ 0.000	0.062 $\pm$ 0.002	0.152 $\pm$ 0.007	0.390 $\pm$ 0.011	0.155	-0.005	-2.88	0.0615	0.0617	1.013	95.675 $\pm$ 0.661	0.247 $\pm$ 0.001	70.464 $\pm$ 1.482
LSAM	beta52	0.16	0.165	0.005 $\pm$ 0.003	0.028 $\pm$ 0.021	0.011 $\pm$ 0.001	0.105 $\pm$ 0.003	0.011 $\pm$ 0.001	0.105 $\pm$ 0.003	0.429 $\pm$ 0.023	0.655 $\pm$ 0.021	0.159	-0.001	-0.71	0.1026	0.1026	0.998	96.730 $\pm$ 0.578	0.410 $\pm$ 0.003	38.924 $\pm$ 1.584
LSAM	beta53	0.16	0.163	0.003 $\pm$ 0.002	0.019 $\pm$ 0.015	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.221 $\pm$ 0.011	0.470 $\pm$ 0.014	0.164	0.004	2.52	0.0753	0.0754	0.994	95.886 $\pm$ 0.645	0.293 $\pm$ 0.001	59.599 $\pm$ 1.594
LSAM	beta54	0.16	0.160	0.000 $\pm$ 0.003	0.003 $\pm$ 0.020	0.009 $\pm$ 0.000	0.097 $\pm$ 0.002	0.009 $\pm$ 0.000	0.097 $\pm$ 0.003	0.365 $\pm$ 0.017	0.604 $\pm$ 0.017	0.165	0.005	2.86	0.0978	0.0979	0.990	95.675 $\pm$ 0.661	0.375 $\pm$ 0.003	41.245 $\pm$ 1.599
LSAM	beta55	0.08	0.081	0.001 $\pm$ 0.002	0.014 $\pm$ 0.028	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.729 $\pm$ 0.041	0.854 $\pm$ 0.028	0.082	0.002	3.00	0.0616	0.0616	0.949	95.570 $\pm$ 0.668	0.254 $\pm$ 0.003	31.013 $\pm$ 1.502
LSAM	beta56	0.08	0.083	0.003 $\pm$ 0.001	0.038 $\pm$ 0.016	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.245 $\pm$ 0.014	0.495 $\pm$ 0.016	0.081	0.001	0.90	0.0350	0.0350	0.916	94.304 $\pm$ 0.753	0.142 $\pm$ 0.002	66.456 $\pm$ 1.533
LSAM	beta57	0.06	0.062	0.002 $\pm$ 0.002	0.026 $\pm$ 0.038	0.005 $\pm$ 0.000	0.071 $\pm$ 0.002	0.005 $\pm$ 0.000	0.071 $\pm$ 0.002	1.387 $\pm$ 0.078	1.178 $\pm$ 0.038	0.059	-0.001	-1.54	0.0629	0.0629	0.974	95.886 $\pm$ 0.645	0.270 $\pm$ 0.003	14.135 $\pm$ 1.131
LSAM	beta58	0.06	0.062	0.002 $\pm$ 0.001	0.041 $\pm$ 0.019	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.326 $\pm$ 0.017	0.571 $\pm$ 0.017	0.061	0.001	0.90	0.0329	0.0329	0.965	94.831 $\pm$ 0.719	0.129 $\pm$ 0.001	53.059 $\pm$ 1.621
QML	beta41	0.20	0.236	0.036 $\pm$ 0.002	0.179 $\pm$ 0.008	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.004 $\pm$ 0.000	0.063 $\pm$ 0.002	0.099 $\pm$ 0.005	0.315 $\pm$ 0.008	0.234	0.034	17.10	0.0512	0.0615	1.042	90.717 $\pm$ 0.942	0.212 $\pm$ 0.002	99.262 $\pm$ 0.278
QML	beta42	0.20	0.231	0.031 $\pm$ 0.002	0.154 $\pm$ 0.009	0.003 $\pm$ 0.000	0.054 $\pm$ 0.001	0.004 $\pm$ 0.000	0.063 $\pm$ 0.002	0.098 $\pm$ 0.005	0.313 $\pm$ 0.008	0.230	0.030	15.25	0.0542	0.0622	1.023	92.616 $\pm$ 0.849	0.218 $\pm$ 0.002	98.418 $\pm$ 0.405
QML	beta43	0.20	0.184	-0.016 $\pm$ 0.001	-0.079 $\pm$ 0.007	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.059 $\pm$ 0.003	0.242 $\pm$ 0.007	0.184	-0.016	-8.03	0.0460	0.0487	1.020	94.515 $\pm$ 0.740	0.183 $\pm$ 0.001	97.890 $\pm$ 0.467
QML	beta44	0.11	0.030	-0.080 $\pm$ 0.002	-0.724 $\pm$ 0.020	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.011 $\pm$ 0.000	0.105 $\pm$ 0.002	0.908 $\pm$ 0.035	0.953 $\pm$ 0.020	0.031	-0.079	-71.63	0.0642	0.1016	0.983	75.422 $\pm$ 1.398	0.263 $\pm$ 0.006	10.021 $\pm$ 0.975
QML	beta45	0.11	0.099	-0.011 $\pm$ 0.002	-0.100 $\pm$ 0.017	0.003 $\pm$ 0.000	0.056 $\pm$ 0.001	0.003 $\pm$ 0.000	0.057 $\pm$ 0.002	0.269 $\pm$ 0.014	0.519 $\pm$ 0.016	0.102	-0.008	-7.62	0.0513	0.0520	1.025	94.093 $\pm$ 0.766	0.225 $\pm$ 0.007	45.781 $\pm$ 1.618
QML	beta46	0.08	0.158	0.078 $\pm$ 0.001	0.969 $\pm$ 0.016	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	0.008 $\pm$ 0.000	0.087 $\pm$ 0.001	1.196 $\pm$ 0.037	1.094 $\pm$ 0.017	0.155	0.075	93.65	0.0396	0.0847	1.051	51.899 $\pm$ 1.623	0.167 $\pm$ 0.006	98.207 $\pm$ 0.431
QML	beta47	0.08	0.154	0.074 $\pm$ 0.001	0.925 $\pm$ 0.017	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	0.007 $\pm$ 0.000	0.084 $\pm$ 0.001	1.113 $\pm$ 0.035	1.055 $\pm$ 0.017	0.150	0.070	87.76	0.0397	0.0807	0.961	52.004 $\pm$ 1.623	0.153 $\pm$ 0.002	98.101 $\pm$ 0.443
QML	beta51	0.16	0.174	0.014 $\pm$ 0.002	0.085 $\pm$ 0.014	0.005 $\pm$ 0.000	0.071 $\pm$ 0.002	0.005 $\pm$ 0.000	0.072 $\pm$ 0.002	0.201 $\pm$ 0.009	0.449 $\pm$ 0.012	0.174	0.014	8.45	0.0707	0.0720	1.080	94.093 $\pm$ 0.766	0.299 $\pm$ 0.014	68.987 $\pm$ 1.502
QML	beta52	0.16	0.165	0.005 $\pm$ 0.002	0.034 $\pm$ 0.014	0.005 $\pm$ 0.000	0.071 $\pm$ 0.002	0.005 $\pm$ 0.000	0.071 $\pm$ 0.002	0.198 $\pm$ 0.009	0.445 $\pm$ 0.013	0.165	0.005	3.26	0.0703	0.0705	1.042	93.565 $\pm$ 0.797	0.290 $\pm$ 0.014	68.460 $\pm$ 1.509
QML	beta53	0.16	0.107	-0.053 $\pm$ 0.002	-0.328 $\pm$ 0.013	0.004 $\pm$ 0.000	0.063 $\pm$ 0.001	0.007 $\pm$ 0.000	0.082 $\pm$ 0.002	0.264 $\pm$ 0.011	0.514 $\pm$ 0.012	0.104	-0.056	-34.80	0.0624	0.0836	1.003	86.392 $\pm$ 1.114	0.249 $\pm$ 0.002	40.084 $\pm$ 1.592
QML	beta54	0.16	0.178	0.018 $\pm$ 0.002	0.112 $\pm$ 0.011	0.003 $\pm$ 0.000	0.054 $\pm$ 0.001	0.003 $\pm$ 0.000	0.057 $\pm$ 0.002	0.125 $\pm$ 0.006	0.354 $\pm$ 0.009	0.178	0.018	11.33	0.0571	0.0599	1.034	95.042 $\pm$ 0.705	0.218 $\pm$ 0.004	91.245 $\pm$ 0.918
QML	beta55	0.08	0.071	-0.009 $\pm$ 0.003	-0.118 $\pm$ 0.035	0.008 $\pm$ 0.000	0.087 $\pm$ 0.002	0.008 $\pm$ 0.000	0.087 $\pm$ 0.002	1.185 $\pm$ 0.055	1.089 $\pm$ 0.031	0.062	-0.018	-22.60	0.0832	0.0851	0.993	87.025 $\pm$ 1.091	0.337 $\pm$ 0.024	19.198 $\pm$ 1.279
QML	beta56	0.08	0.124	0.044 $\pm$ 0.001	0.544 $\pm$ 0.018	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.004 $\pm$ 0.000	0.062 $\pm$ 0.001	0.610 $\pm$ 0.024	0.781 $\pm$ 0.017	0.123	0.043	54.25	0.0428	0.0609	1.000	78.481 $\pm$ 1.335	0.176 $\pm$ 0.007	80.591 $\pm$ 1.285
QML	beta57	0.06	0.142	0.082 $\pm$ 0.003	1.365 $\pm$ 0.042	0.006 $\pm$ 0.000	0.078 $\pm$ 0.002	0.013 $\pm$ 0.000	0.113 $\pm$ 0.002	3.567 $\pm$ 0.132	1.889 $\pm$ 0.038	0.145	0.085	142.35	0.0750	0.1137	1.054	70.042 $\pm$ 1.488	0.324 $\pm$ 0.027	57.806 $\pm$ 1.604
QML	beta58	0.06	0.113	0.053 $\pm$ 0.001	0.875 $\pm$ 0.017	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	0.004 $\pm$ 0.000	0.061 $\pm$ 0.001	1.043 $\pm$ 0.035	1.021 $\pm$ 0.018	0.111	0.051	85.68	0.0315	0.0603	1.008	63.608 $\pm$ 1.563	0.125 $\pm$ 0.001	96.519 $\pm$ 0.595
UPI	beta41	0.20	0.211	0.011 $\pm$ 0.003	0.057 $\pm$ 0.016	0.010 $\pm$ 0.000	0.098 $\pm$ 0.002	0.010 $\pm$ 0.000	0.098 $\pm$ 0.003	0.241 $\pm$ 0.012	0.491 $\pm$ 0.015	0.208	0.008	3.85	0.0955	0.0958	1.103	96.519 $\pm$ 0.595	0.422 $\pm$ 0.005	55.063 $\pm$ 1.616
UPI	beta42	0.20	0.206	0.006 $\pm$ 0.003	0.028 $\pm$ 0.017	0.011 $\pm$ 0.001	0.105 $\pm$ 0.003	0.011 $\pm$ 0.001	0.105 $\pm$ 0.003	0.276 $\pm$ 0.014	0.525 $\pm$ 0.016	0.207	0.007	3.39	0.0952	0.0955	1.063	96.308 $\pm$ 0.612	0.438 $\pm$ 0.005	52.426 $\pm$ 1.622
UPI	beta43	0.20	0.199	-0.001 $\pm$ 0.002	-0.003 $\pm$ 0.008	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.065 $\pm$ 0.003	0.255 $\pm$ 0.007	0.200	0.000	-0.02	0.0508	0.0508	1.039	96.097 $\pm$ 0.629	0.208 $\pm$ 0.002	95.464 $\pm$ 0

Table S152

Simulation 2 - Condition 8: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.4
(continued)

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.080	-0.000 ± 0.001	-0.000 ± 0.017	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.267 ± 0.015	0.516 ± 0.017	0.077	-0.003	-3.65	0.0371	0.0372	1.017	94.831 ± 0.719	0.165 ± 0.003	55.485 ± 1.614
UPI	beta51	0.16	0.159	-0.001 ± 0.002	-0.004 ± 0.012	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.140 ± 0.006	0.374 ± 0.010	0.159	-0.001	-0.53	0.0600	0.0600	1.036	95.992 ± 0.637	0.243 ± 0.001	73.523 ± 1.433
UPI	beta52	0.16	0.176	0.016 ± 0.003	0.102 ± 0.020	0.010 ± 0.001	0.097 ± 0.003	0.010 ± 0.001	0.099 ± 0.003	0.381 ± 0.020	0.617 ± 0.019	0.175	0.015	9.45	0.0928	0.0940	1.080	96.414 ± 0.604	0.413 ± 0.005	42.300 ± 1.605
UPI	beta53	0.16	0.162	0.002 ± 0.002	0.011 ± 0.015	0.005 ± 0.000	0.074 ± 0.002	0.005 ± 0.000	0.074 ± 0.002	0.213 ± 0.011	0.462 ± 0.014	0.162	0.002	1.38	0.0756	0.0756	1.015	95.570 ± 0.668	0.294 ± 0.002	59.494 ± 1.594
UPI	beta54	0.16	0.159	-0.001 ± 0.003	-0.009 ± 0.019	0.009 ± 0.000	0.095 ± 0.002	0.009 ± 0.000	0.095 ± 0.003	0.353 ± 0.017	0.594 ± 0.017	0.163	0.003	1.92	0.0972	0.0973	1.061	97.468 ± 0.510	0.395 ± 0.005	39.135 ± 1.585
UPI	beta55	0.08	0.086	0.006 ± 0.002	0.074 ± 0.025	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	0.605 ± 0.034	0.778 ± 0.025	0.087	0.007	9.28	0.0555	0.0559	1.048	96.624 ± 0.587	0.254 ± 0.003	30.591 ± 1.497
UPI	beta56	0.08	0.081	0.001 ± 0.001	0.018 ± 0.016	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.251 ± 0.014	0.501 ± 0.016	0.079	-0.001	-0.92	0.0368	0.0368	0.941	94.198 ± 0.759	0.148 ± 0.002	61.287 ± 1.582
UPI	beta57	0.06	0.053	-0.007 ± 0.002	-0.109 ± 0.033	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	1.050 ± 0.053	1.025 ± 0.031	0.051	-0.009	-14.35	0.0576	0.0582	1.114	96.730 ± 0.578	0.267 ± 0.004	10.865 ± 1.011
UPI	beta58	0.06	0.062	0.002 ± 0.001	0.041 ± 0.019	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.338 ± 0.018	0.582 ± 0.018	0.061	0.001	0.90	0.0317	0.0317	1.015	95.253 ± 0.691	0.139 ± 0.002	49.156 ± 1.624

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S153

Simulation 2 - Condition 9: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, $N = 400$, Reliability = 0.6

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LSAM	beta41	0.20	0.203	0.003 $\pm$ 0.003	0.015 $\pm$ 0.017	0.011 $\pm$ 0.000	0.103 $\pm$ 0.002	0.011 $\pm$ 0.001	0.103 $\pm$ 0.003	0.267 $\pm$ 0.013	0.517 $\pm$ 0.015	0.202	0.002	0.95	0.1002	0.1002	1.023	95.889 $\pm$ 0.637	0.415 $\pm$ 0.002	50.462 $\pm$ 1.603
LSAM	beta42	0.20	0.199	-0.001 $\pm$ 0.003	-0.005 $\pm$ 0.017	0.011 $\pm$ 0.001	0.106 $\pm$ 0.003	0.011 $\pm$ 0.001	0.106 $\pm$ 0.003	0.283 $\pm$ 0.014	0.532 $\pm$ 0.015	0.202	0.002	1.21	0.1040	0.1040	1.025	96.300 $\pm$ 0.605	0.428 $\pm$ 0.003	48.510 $\pm$ 1.602
LSAM	beta43	0.20	0.200	-0.000 $\pm$ 0.002	-0.000 $\pm$ 0.010	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	0.091 $\pm$ 0.005	0.301 $\pm$ 0.009	0.201	0.001	0.48	0.0564	0.0564	0.989	94.142 $\pm$ 0.753	0.234 $\pm$ 0.001	91.161 $\pm$ 0.910
LSAM	beta44	0.11	0.115	0.005 $\pm$ 0.003	0.049 $\pm$ 0.024	0.007 $\pm$ 0.000	0.083 $\pm$ 0.002	0.007 $\pm$ 0.000	0.083 $\pm$ 0.003	0.574 $\pm$ 0.031	0.758 $\pm$ 0.024	0.113	0.003	2.93	0.0774	0.0775	0.939	94.245 $\pm$ 0.747	0.306 $\pm$ 0.003	34.635 $\pm$ 1.525
LSAM	beta45	0.11	0.122	0.012 $\pm$ 0.002	0.113 $\pm$ 0.023	0.006 $\pm$ 0.000	0.078 $\pm$ 0.002	0.006 $\pm$ 0.000	0.079 $\pm$ 0.002	0.514 $\pm$ 0.027	0.717 $\pm$ 0.022	0.122	0.012	10.51	0.0770	0.0779	0.922	94.450 $\pm$ 0.734	0.282 $\pm$ 0.003	44.399 $\pm$ 1.593
LSAM	beta46	0.08	0.081	0.001 $\pm$ 0.002	0.018 $\pm$ 0.019	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.348 $\pm$ 0.019	0.590 $\pm$ 0.019	0.082	0.002	1.97	0.0426	0.0426	0.904	93.525 $\pm$ 0.789	0.167 $\pm$ 0.002	54.779 $\pm$ 1.596
LSAM	beta47	0.08	0.081	0.001 $\pm$ 0.001	0.018 $\pm$ 0.018	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.314 $\pm$ 0.017	0.560 $\pm$ 0.018	0.078	-0.002	-2.49	0.0409	0.0409	0.912	92.909 $\pm$ 0.823	0.160 $\pm$ 0.002	56.629 $\pm$ 1.589
LSAM	beta51	0.16	0.160	-0.000 $\pm$ 0.002	-0.001 $\pm$ 0.014	0.005 $\pm$ 0.000	0.072 $\pm$ 0.002	0.005 $\pm$ 0.000	0.072 $\pm$ 0.002	0.204 $\pm$ 0.009	0.452 $\pm$ 0.012	0.159	-0.001	-0.47	0.0706	0.0706	1.001	95.375 $\pm$ 0.673	0.284 $\pm$ 0.001	61.048 $\pm$ 1.563
LSAM	beta52	0.16	0.168	0.008 $\pm$ 0.004	0.051 $\pm$ 0.022	0.012 $\pm$ 0.001	0.109 $\pm$ 0.003	0.012 $\pm$ 0.001	0.110 $\pm$ 0.003	0.470 $\pm$ 0.022	0.685 $\pm$ 0.019	0.169	0.009	5.58	0.1104	0.1107	0.994	96.095 $\pm$ 0.621	0.426 $\pm$ 0.003	37.718 $\pm$ 1.554
LSAM	beta53	0.16	0.158	-0.002 $\pm$ 0.003	-0.012 $\pm$ 0.017	0.007 $\pm$ 0.000	0.084 $\pm$ 0.002	0.007 $\pm$ 0.000	0.084 $\pm$ 0.002	0.275 $\pm$ 0.013	0.524 $\pm$ 0.015	0.156	-0.004	-2.37	0.0801	0.0802	0.988	94.553 $\pm$ 0.728	0.325 $\pm$ 0.002	49.024 $\pm$ 1.603
LSAM	beta54	0.16	0.160	0.000 $\pm$ 0.003	0.002 $\pm$ 0.021	0.011 $\pm$ 0.001	0.107 $\pm$ 0.003	0.011 $\pm$ 0.001	0.107 $\pm$ 0.003	0.447 $\pm$ 0.022	0.669 $\pm$ 0.020	0.163	0.003	1.72	0.1056	0.1056	0.957	94.347 $\pm$ 0.740	0.402 $\pm$ 0.002	36.485 $\pm$ 1.543
LSAM	beta55	0.08	0.081	0.001 $\pm$ 0.002	0.018 $\pm$ 0.030	0.006 $\pm$ 0.000	0.074 $\pm$ 0.002	0.005 $\pm$ 0.000	0.074 $\pm$ 0.002	0.859 $\pm$ 0.046	0.927 $\pm$ 0.029	0.082	0.002	1.97	0.0688	0.0689	0.924	93.731 $\pm$ 0.777	0.269 $\pm$ 0.003	27.749 $\pm$ 1.435
LSAM	beta56	0.08	0.080	0.000 $\pm$ 0.002	0.003 $\pm$ 0.019	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.002	0.354 $\pm$ 0.019	0.595 $\pm$ 0.019	0.081	0.001	1.43	0.0424	0.0424	0.884	90.647 $\pm$ 0.933	0.165 $\pm$ 0.002	54.471 $\pm$ 1.597
LSAM	beta57	0.06	0.060	-0.000 $\pm$ 0.002	-0.002 $\pm$ 0.039	0.005 $\pm$ 0.000	0.073 $\pm$ 0.002	0.005 $\pm$ 0.000	0.073 $\pm$ 0.002	1.491 $\pm$ 0.081	1.221 $\pm$ 0.038	0.059	-0.001	-1.82	0.0647	0.0647	0.940	94.245 $\pm$ 0.747	0.270 $\pm$ 0.003	16.444 $\pm$ 1.188
LSAM	beta58	0.06	0.061	0.001 $\pm$ 0.001	0.023 $\pm$ 0.022	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.483 $\pm$ 0.027	0.695 $\pm$ 0.023	0.062	0.002	2.94	0.0363	0.0364	0.882	93.320 $\pm$ 0.800	0.144 $\pm$ 0.002	47.174 $\pm$ 1.600
QML	beta41	0.20	0.218	0.018 $\pm$ 0.002	0.088 $\pm$ 0.011	0.005 $\pm$ 0.000	0.067 $\pm$ 0.002	0.005 $\pm$ 0.000	0.070 $\pm$ 0.002	0.121 $\pm$ 0.006	0.349 $\pm$ 0.010	0.218	0.018	9.09	0.0691	0.0714	1.036	95.272 $\pm$ 0.680	0.274 $\pm$ 0.001	87.667 $\pm$ 1.054
QML	beta42	0.20	0.211	0.011 $\pm$ 0.002	0.056 $\pm$ 0.011	0.005 $\pm$ 0.000	0.069 $\pm$ 0.002	0.005 $\pm$ 0.000	0.070 $\pm$ 0.002	0.122 $\pm$ 0.006	0.350 $\pm$ 0.010	0.212	0.012	5.99	0.0697	0.0707	1.035	94.964 $\pm$ 0.701	0.280 $\pm$ 0.001	83.350 $\pm$ 1.194
QML	beta43	0.20	0.190	-0.010 $\pm$ 0.002	-0.050 $\pm$ 0.009	0.003 $\pm$ 0.000	0.056 $\pm$ 0.001	0.003 $\pm$ 0.000	0.057 $\pm$ 0.002	0.080 $\pm$ 0.004	0.283 $\pm$ 0.008	0.192	-0.008	-3.77	0.0547	0.0552	0.994	93.525 $\pm$ 0.789	0.217 $\pm$ 0.001	92.806 $\pm$ 0.828
QML	beta44	0.11	0.070	-0.040 $\pm$ 0.002	-0.367 $\pm$ 0.021	0.005 $\pm$ 0.000	0.072 $\pm$ 0.002	0.007 $\pm$ 0.000	0.082 $\pm$ 0.002	0.557 $\pm$ 0.026	0.746 $\pm$ 0.020	0.070	-0.040	-36.48	0.0693	0.0801	0.957	89.620 $\pm$ 0.978	0.268 $\pm$ 0.002	20.966 $\pm$ 1.305
QML	beta45	0.11	0.106	-0.004 $\pm$ 0.002	-0.040 $\pm$ 0.019	0.004 $\pm$ 0.000	0.064 $\pm$ 0.001	0.004 $\pm$ 0.000	0.064 $\pm$ 0.002	0.339 $\pm$ 0.016	0.582 $\pm$ 0.016	0.107	-0.003	-3.11	0.0646	0.0647	0.953	94.245 $\pm$ 0.747	0.239 $\pm$ 0.002	45.221 $\pm$ 1.596
QML	beta46	0.08	0.114	0.034 $\pm$ 0.001	0.429 $\pm$ 0.017	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.003 $\pm$ 0.000	0.055 $\pm$ 0.001	0.468 $\pm$ 0.022	0.684 $\pm$ 0.017	0.113	0.033	41.61	0.0395	0.0517	0.938	85.303 $\pm$ 1.135	0.157 $\pm$ 0.002	82.528 $\pm$ 1.217
QML	beta47	0.08	0.115	0.035 $\pm$ 0.001	0.434 $\pm$ 0.016	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.434 $\pm$ 0.021	0.658 $\pm$ 0.017	0.113	0.033	41.49	0.0381	0.0506	0.948	86.023 $\pm$ 1.112	0.147 $\pm$ 0.002	85.817 $\pm$ 1.118
QML	beta51	0.16	0.178	0.018 $\pm$ 0.002	0.111 $\pm$ 0.015	0.005 $\pm$ 0.000	0.073 $\pm$ 0.002	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.218 $\pm$ 0.010	0.467 $\pm$ 0.013	0.178	0.018	11.39	0.0708	0.0731	1.037	95.478 $\pm$ 0.666	0.295 $\pm$ 0.003	66.290 $\pm$ 1.515
QML	beta52	0.16	0.153	-0.007 $\pm$ 0.002	-0.047 $\pm$ 0.015	0.005 $\pm$ 0.000	0.073 $\pm$ 0.002	0.005 $\pm$ 0.000	0.074 $\pm$ 0.002	0.212 $\pm$ 0.009	0.460 $\pm$ 0.012	0.152	-0.008	-4.82	0.0769	0.0773	1.013	95.272 $\pm$ 0.680	0.291 $\pm$ 0.002	54.676 $\pm$ 1.596
QML	beta53	0.16	0.122	-0.038 $\pm$ 0.002	-0.236 $\pm$ 0.015	0.006 $\pm$ 0.000	0.076 $\pm$ 0.002	0.007 $\pm$ 0.000	0.085 $\pm$ 0.002	0.281 $\pm$ 0.012	0.530 $\pm$ 0.013	0.121	-0.039	-24.59	0.0780	0.0874	0.984	92.189 $\pm$ 0.860	0.293 $\pm$ 0.001	38.129 $\pm$ 1.557
QML	beta54	0.16	0.165	0.005 $\pm$ 0.002	0.033 $\pm$ 0.015	0.005 $\pm$ 0.000	0.073 $\pm$ 0.002	0.005 $\pm$ 0.000	0.073 $\pm$ 0.002	0.208 $\pm$ 0.010	0.456 $\pm$ 0.013	0.167	0.007	4.45	0.0757	0.0760	0.970	94.347 $\pm$ 0.740	0.277 $\pm$ 0.001	65.262 $\pm$ 1.526
QML	beta55	0.08	0.087	0.007 $\pm$ 0.002	0.081 $\pm$ 0.030	0.006 $\pm$ 0.000	0.074 $\pm$ 0.002	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.869 $\pm$ 0.043	0.932 $\pm$ 0.028	0.087	0.007	8.42	0.0688	0.0692	0.973	91.470 $\pm$ 0.895	0.284 $\pm$ 0.007	31.038 $\pm$ 1.483
QML	beta56	0.08	0.094	0.014 $\pm$ 0.002	0.181 $\pm$ 0.019	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	0.388 $\pm$ 0.019	0.623 $\pm$ 0.018	0.096	0.016	19.70	0.0444	0.0471	0.896	88.695 $\pm$ 1.015	0.168 $\pm$ 0.003	63.104 $\pm$ 1.547
QML	beta57	0.06	0.085	0.025 $\pm$ 0.002	0.411 $\pm$ 0.035	0.004 $\pm$ 0.000	0.065 $\pm$ 0.002	0.005 $\pm$ 0.000	0.070 $\pm$ 0.002	1.352 $\pm$ 0.068	1.163 $\pm$ 0.034	0.084	0.024	40.58	0.0600	0.0648	1.015	90.956 $\pm$ 0.919	0.260 $\pm$ 0.006	30.216 $\pm$ 1.472
QML	beta58	0.06	0.084	0.024 $\pm$ 0.001	0.392 $\pm$ 0.019	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.505 $\pm$ 0.023	0.711 $\pm$ 0.018	0.083	0.023	38.62	0.0326	0.0400	0.920	86.331 $\pm$ 1.101	0.128 $\pm$ 0.001	72.662 $\pm$ 1.429
UPI	beta41	0.20	0.213	0.013 $\pm$ 0.003	0.065 $\pm$ 0.016	0.010 $\pm$ 0.001	0.102 $\pm$ 0.003	0.010 $\pm$ 0.001	0.102 $\pm$ 0.003	0.262 $\pm$ 0.013	0.512 $\pm$ 0.015	0.216	0.016	8.16	0.0989	0.1003	1.048	96.608 $\pm$ 0.580	0.417 $\pm$ 0.003	54.368 $\pm$ 1.597
UPI	beta42	0.20	0.206	0.006 $\pm$ 0.003	0.032 $\pm$ 0.017	0.011 $\pm$ 0.001	0.105 $\pm$ 0.003	0.011 $\pm$ 0.001	0.105 $\pm$ 0.003	0.277 $\pm$ 0.013	0.527 $\pm$ 0.015	0.206	0.006	3.00	0.1071	0.1073	1.052	96.506 $\pm$ 0.589	0.434 $\pm$ 0.003	50.874 $\pm$ 1.603
UPI	beta43	0.20	0.196	-0.004 $\pm$ 0.002	-0.018 $\pm$ 0.010	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.092 $\pm$ 0.005	0.303 $\pm$ 0.009	0.199	-0.001	-0.60	0.0563	0.0563	0.972	93.628 $\pm$ 0.783	0.230 $\pm$ 0.001	89.620 $\pm$ 0.978</

Table S153

Simulation 2 - Condition 9: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.6
(continued)

Method	Param.	True	Mean-based										Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE	
UPI	beta47	0.08	0.079	-0.001 ± 0.001	-0.013 ± 0.017	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.294 ± 0.016	0.542 ± 0.017	0.077	-0.003	-3.96	0.0401	0.0402	0.970	93.114 ± 0.812	0.165 ± 0.002	54.162 ± 1.597	
UPI	beta51	0.16	0.164	0.004 ± 0.002	0.023 ± 0.014	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.198 ± 0.009	0.445 ± 0.012	0.164	0.004	2.20	0.0699	0.0700	0.997	94.758 ± 0.714	0.278 ± 0.001	64.234 ± 1.537	
UPI	beta52	0.16	0.177	0.017 ± 0.003	0.105 ± 0.022	0.012 ± 0.001	0.109 ± 0.003	0.012 ± 0.001	0.110 ± 0.003	0.472 ± 0.023	0.687 ± 0.020	0.177	0.017	10.81	0.1080	0.1094	0.975	94.553 ± 0.728	0.415 ± 0.003	41.829 ± 1.581	
UPI	beta53	0.16	0.157	-0.003 ± 0.003	-0.021 ± 0.016	0.007 ± 0.000	0.082 ± 0.002	0.007 ± 0.000	0.082 ± 0.002	0.262 ± 0.012	0.511 ± 0.014	0.157	-0.003	-1.58	0.0786	0.0786	1.006	95.067 ± 0.694	0.323 ± 0.002	50.257 ± 1.603	
UPI	beta54	0.16	0.161	0.001 ± 0.003	0.009 ± 0.021	0.011 ± 0.001	0.106 ± 0.003	0.011 ± 0.001	0.106 ± 0.003	0.438 ± 0.021	0.662 ± 0.019	0.164	0.004	2.50	0.1061	0.1062	0.977	95.170 ± 0.687	0.406 ± 0.003	37.307 ± 1.550	
UPI	beta55	0.08	0.085	0.005 ± 0.002	0.059 ± 0.030	0.005 ± 0.000	0.074 ± 0.002	0.005 ± 0.000	0.074 ± 0.002	0.857 ± 0.048	0.926 ± 0.030	0.084	0.004	4.70	0.0663	0.0664	0.924	93.628 ± 0.783	0.268 ± 0.003	28.160 ± 1.442	
UPI	beta56	0.08	0.079	-0.001 ± 0.002	-0.009 ± 0.019	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.002	0.361 ± 0.020	0.601 ± 0.019	0.079	-0.001	-1.20	0.0421	0.0421	0.889	91.367 ± 0.900	0.168 ± 0.002	52.107 ± 1.602	
UPI	beta57	0.06	0.054	-0.006 ± 0.002	-0.093 ± 0.038	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	1.425 ± 0.078	1.194 ± 0.038	0.054	-0.006	-9.82	0.0657	0.0660	0.936	93.628 ± 0.783	0.262 ± 0.003	14.697 ± 1.135	
UPI	beta58	0.06	0.061	0.001 ± 0.001	0.009 ± 0.022	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.483 ± 0.027	0.695 ± 0.022	0.060	0.000	-0.81	0.0357	0.0357	0.903	92.703 ± 0.834	0.148 ± 0.002	45.118 ± 1.595	

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S154

Simulation 2 - Condition 10: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.6

Method	Param.	True	Mean-based										Median-based					Inference		
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LSAM	beta41	0.20	0.202	0.002 $\pm$ 0.002	0.008 $\pm$ 0.010	0.004 $\pm$ 0.000	0.065 $\pm$ 0.002	0.004 $\pm$ 0.000	0.065 $\pm$ 0.002	0.105 $\pm$ 0.005	0.325 $\pm$ 0.009	0.200	0.000	0.07	0.0644	0.0644	0.964	94.944 $\pm$ 0.697	0.245 $\pm$ 0.001	88.777 $\pm$ 1.004
LSAM	beta42	0.20	0.197	-0.003 $\pm$ 0.002	-0.013 $\pm$ 0.010	0.004 $\pm$ 0.000	0.064 $\pm$ 0.002	0.004 $\pm$ 0.000	0.064 $\pm$ 0.002	0.103 $\pm$ 0.005	0.321 $\pm$ 0.009	0.196	-0.004	-2.06	0.0617	0.0618	1.010	95.854 $\pm$ 0.634	0.254 $\pm$ 0.001	86.350 $\pm$ 1.092
LSAM	beta43	0.20	0.201	0.001 $\pm$ 0.001	0.003 $\pm$ 0.006	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.033 $\pm$ 0.002	0.182 $\pm$ 0.005	0.201	0.001	0.57	0.0356	0.0356	1.001	95.046 $\pm$ 0.690	0.143 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta44	0.11	0.112	0.002 $\pm$ 0.001	0.014 $\pm$ 0.013	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.171 $\pm$ 0.009	0.413 $\pm$ 0.012	0.111	0.001	0.62	0.0432	0.0432	0.967	93.933 $\pm$ 0.759	0.172 $\pm$ 0.001	71.689 $\pm$ 1.433
LSAM	beta45	0.11	0.113	0.003 $\pm$ 0.001	0.025 $\pm$ 0.012	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.152 $\pm$ 0.007	0.390 $\pm$ 0.011	0.114	0.004	3.88	0.0436	0.0438	0.969	94.338 $\pm$ 0.735	0.163 $\pm$ 0.001	76.744 $\pm$ 1.343
LSAM	beta46	0.08	0.081	0.001 $\pm$ 0.001	0.015 $\pm$ 0.010	0.001 $\pm$ 0.000	0.025 $\pm$ 0.001	0.001 $\pm$ 0.000	0.025 $\pm$ 0.001	0.101 $\pm$ 0.005	0.317 $\pm$ 0.009	0.080	0.000	0.12	0.0241	0.0241	0.933	94.034 $\pm$ 0.753	0.093 $\pm$ 0.001	90.192 $\pm$ 0.946
LSAM	beta47	0.08	0.081	0.001 $\pm$ 0.001	0.010 $\pm$ 0.010	0.001 $\pm$ 0.000	0.025 $\pm$ 0.001	0.001 $\pm$ 0.000	0.025 $\pm$ 0.001	0.095 $\pm$ 0.005	0.308 $\pm$ 0.009	0.080	0.000	0.33	0.0242	0.0242	0.935	93.933 $\pm$ 0.759	0.090 $\pm$ 0.001	91.405 $\pm$ 0.891
LSAM	beta51	0.16	0.159	-0.001 $\pm$ 0.001	-0.005 $\pm$ 0.008	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.071 $\pm$ 0.003	0.267 $\pm$ 0.007	0.159	-0.001	-0.76	0.0423	0.0423	1.051	95.753 $\pm$ 0.641	0.176 $\pm$ 0.000	94.439 $\pm$ 0.729
LSAM	beta52	0.16	0.161	0.001 $\pm$ 0.002	0.005 $\pm$ 0.013	0.004 $\pm$ 0.000	0.065 $\pm$ 0.001	0.004 $\pm$ 0.000	0.065 $\pm$ 0.002	0.163 $\pm$ 0.007	0.403 $\pm$ 0.011	0.161	0.001	0.33	0.0650	0.0650	1.001	94.742 $\pm$ 0.710	0.253 $\pm$ 0.001	70.981 $\pm$ 1.443
LSAM	beta53	0.16	0.159	-0.001 $\pm$ 0.002	-0.004 $\pm$ 0.010	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.093 $\pm$ 0.004	0.305 $\pm$ 0.009	0.160	0.000	0.04	0.0473	0.0473	1.029	95.450 $\pm$ 0.663	0.197 $\pm$ 0.001	89.687 $\pm$ 0.967
LSAM	beta54	0.16	0.161	0.001 $\pm$ 0.002	0.009 $\pm$ 0.013	0.004 $\pm$ 0.000	0.064 $\pm$ 0.002	0.004 $\pm$ 0.000	0.064 $\pm$ 0.002	0.161 $\pm$ 0.008	0.402 $\pm$ 0.012	0.162	0.002	1.07	0.0623	0.0623	0.954	94.338 $\pm$ 0.735	0.240 $\pm$ 0.001	73.711 $\pm$ 1.400
LSAM	beta55	0.08	0.080	0.000 $\pm$ 0.001	0.001 $\pm$ 0.015	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.226 $\pm$ 0.011	0.475 $\pm$ 0.014	0.079	-0.001	-1.33	0.0366	0.0366	0.983	94.237 $\pm$ 0.741	0.147 $\pm$ 0.001	58.847 $\pm$ 1.565
LSAM	beta56	0.08	0.080	-0.000 $\pm$ 0.001	-0.003 $\pm$ 0.010	0.001 $\pm$ 0.000	0.024 $\pm$ 0.001	0.001 $\pm$ 0.000	0.024 $\pm$ 0.001	0.093 $\pm$ 0.005	0.304 $\pm$ 0.010	0.080	0.000	0.04	0.0220	0.0220	0.943	92.720 $\pm$ 0.826	0.090 $\pm$ 0.001	90.091 $\pm$ 0.950
LSAM	beta57	0.06	0.061	0.001 $\pm$ 0.001	0.021 $\pm$ 0.021	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.423 $\pm$ 0.022	0.650 $\pm$ 0.020	0.061	0.001	1.33	0.0374	0.0374	0.987	94.338 $\pm$ 0.735	0.151 $\pm$ 0.001	37.513 $\pm$ 1.540
LSAM	beta58	0.06	0.060	-0.000 $\pm$ 0.001	-0.004 $\pm$ 0.011	0.000 $\pm$ 0.000	0.022 $\pm$ 0.001	0.000 $\pm$ 0.000	0.022 $\pm$ 0.001	0.130 $\pm$ 0.007	0.361 $\pm$ 0.011	0.060	0.000	-0.79	0.0210	0.0210	0.964	95.248 $\pm$ 0.677	0.082 $\pm$ 0.001	81.699 $\pm$ 1.230
QML	beta41	0.20	0.217	0.017 $\pm$ 0.001	0.085 $\pm$ 0.007	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.056 $\pm$ 0.003	0.237 $\pm$ 0.007	0.216	0.016	8.09	0.0445	0.0474	0.994	92.821 $\pm$ 0.821	0.172 $\pm$ 0.002	99.697 $\pm$ 0.175
QML	beta42	0.20	0.210	0.010 $\pm$ 0.001	0.051 $\pm$ 0.007	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.049 $\pm$ 0.002	0.222 $\pm$ 0.006	0.209	0.009	4.51	0.0427	0.0437	1.032	94.944 $\pm$ 0.697	0.175 $\pm$ 0.002	99.697 $\pm$ 0.175
QML	beta43	0.20	0.192	-0.008 $\pm$ 0.001	-0.041 $\pm$ 0.005	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.031 $\pm$ 0.001	0.177 $\pm$ 0.005	0.192	-0.008	-4.04	0.0347	0.0356	1.011	93.225 $\pm$ 0.799	0.136 $\pm$ 0.000	100.000 $\pm$ 0.000
QML	beta44	0.11	0.072	-0.038 $\pm$ 0.001	-0.344 $\pm$ 0.013	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.003 $\pm$ 0.000	0.058 $\pm$ 0.001	0.281 $\pm$ 0.012	0.530 $\pm$ 0.012	0.074	-0.036	-32.83	0.0432	0.0563	0.950	83.215 $\pm$ 1.188	0.165 $\pm$ 0.001	43.175 $\pm$ 1.575
QML	beta45	0.11	0.103	-0.007 $\pm$ 0.001	-0.065 $\pm$ 0.011	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.129 $\pm$ 0.006	0.359 $\pm$ 0.010	0.104	-0.006	-5.87	0.0395	0.0400	0.983	93.832 $\pm$ 0.765	0.150 $\pm$ 0.001	76.744 $\pm$ 1.343
QML	beta46	0.08	0.115	0.035 $\pm$ 0.001	0.436 $\pm$ 0.010	0.001 $\pm$ 0.000	0.025 $\pm$ 0.001	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.285 $\pm$ 0.010	0.533 $\pm$ 0.010	0.114	0.034	42.20	0.0224	0.0405	0.982	71.183 $\pm$ 1.440	0.094 $\pm$ 0.001	99.191 $\pm$ 0.285
QML	beta47	0.08	0.114	0.034 $\pm$ 0.001	0.429 $\pm$ 0.010	0.001 $\pm$ 0.000	0.024 $\pm$ 0.001	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.276 $\pm$ 0.010	0.525 $\pm$ 0.010	0.114	0.034	41.93	0.0242	0.0414	0.945	67.442 $\pm$ 1.490	0.090 $\pm$ 0.001	99.596 $\pm$ 0.202
QML	beta51	0.16	0.175	0.015 $\pm$ 0.001	0.092 $\pm$ 0.009	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.086 $\pm$ 0.004	0.293 $\pm$ 0.008	0.175	0.015	9.33	0.0453	0.0477	1.054	95.248 $\pm$ 0.677	0.184 $\pm$ 0.001	96.360 $\pm$ 0.596
QML	beta52	0.16	0.147	-0.013 $\pm$ 0.001	-0.082 $\pm$ 0.009	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.087 $\pm$ 0.004	0.295 $\pm$ 0.008	0.147	-0.013	-8.21	0.0451	0.0469	1.007	93.933 $\pm$ 0.759	0.179 $\pm$ 0.001	89.181 $\pm$ 0.988
QML	beta53	0.16	0.124	-0.036 $\pm$ 0.001	-0.228 $\pm$ 0.009	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.003 $\pm$ 0.000	0.057 $\pm$ 0.001	0.128 $\pm$ 0.005	0.358 $\pm$ 0.008	0.124	-0.036	-22.48	0.0440	0.0569	1.049	87.664 $\pm$ 1.046	0.182 $\pm$ 0.001	78.362 $\pm$ 1.309
QML	beta54	0.16	0.165	0.005 $\pm$ 0.001	0.032 $\pm$ 0.009	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.086 $\pm$ 0.004	0.293 $\pm$ 0.008	0.165	0.005	2.91	0.0449	0.0451	0.957	93.630 $\pm$ 0.777	0.175 $\pm$ 0.003	95.349 $\pm$ 0.670
QML	beta55	0.08	0.081	0.001 $\pm$ 0.001	0.018 $\pm$ 0.018	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.338 $\pm$ 0.015	0.582 $\pm$ 0.016	0.080	0.000	-0.01	0.0473	0.0473	0.963	91.709 $\pm$ 0.877	0.176 $\pm$ 0.002	48.028 $\pm$ 1.589
QML	beta56	0.08	0.096	0.016 $\pm$ 0.001	0.204 $\pm$ 0.011	0.001 $\pm$ 0.000	0.027 $\pm$ 0.001	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.154 $\pm$ 0.007	0.393 $\pm$ 0.010	0.096	0.016	19.71	0.0260	0.0304	0.945	86.350 $\pm$ 1.092	0.099 $\pm$ 0.001	94.034 $\pm$ 0.753
QML	beta57	0.06	0.092	0.032 $\pm$ 0.001	0.539 $\pm$ 0.023	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.791 $\pm$ 0.035	0.890 $\pm$ 0.021	0.093	0.033	55.61	0.0401	0.0522	0.992	86.754 $\pm$ 1.078	0.165 $\pm$ 0.002	62.184 $\pm$ 1.542
QML	beta58	0.06	0.084	0.024 $\pm$ 0.001	0.396 $\pm$ 0.011	0.000 $\pm$ 0.000	0.021 $\pm$ 0.000	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.274 $\pm$ 0.010	0.523 $\pm$ 0.011	0.084	0.024	39.57	0.0197	0.0309	0.971	76.036 $\pm$ 1.357	0.078 $\pm$ 0.001	97.270 $\pm$ 0.518
UPI	beta41	0.20	0.207	0.007 $\pm$ 0.002	0.036 $\pm$ 0.010	0.004 $\pm$ 0.000	0.065 $\pm$ 0.002	0.004 $\pm$ 0.000	0.065 $\pm$ 0.002	0.106 $\pm$ 0.005	0.325 $\pm$ 0.009	0.206	0.006	3.12	0.0639	0.0642	0.968	94.338 $\pm$ 0.735	0.245 $\pm$ 0.001	90.091 $\pm$ 0.950
UPI	beta42	0.20	0.200	-0.000 $\pm$ 0.002	-0.001 $\pm$ 0.010	0.004 $\pm$ 0.000	0.063 $\pm$ 0.001	0.004 $\pm$ 0.000	0.063 $\pm$ 0.002	0.099 $\pm$ 0.004	0.315 $\pm$ 0.009	0.199	-0.001	-0.61	0.0623	0.0623	1.024	96.158 $\pm$ 0.611	0.253 $\pm$ 0.001	86.350 $\pm$ 1.092
UPI	beta43	0.20	0.200	-0.000 $\pm$ 0.001	-0.002 $\pm$ 0.006	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.033 $\pm$ 0.002	0.182 $\pm$ 0.005	0.199	-0.001	-0.36	0.0360	0.0360	0.985	94.135 $\pm$ 0.747	0.141 $\pm$ 0.001	100.000 $\pm$ 0.000</

Table S154

Simulation 2 - Condition 10: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.6
(continued)

Method	Param.	True	Mean-based										Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE	
UPI	beta47	0.08	0.080	0.000 ± 0.001	0.001 ± 0.010	0.001 ± 0.000	0.025 ± 0.001	0.001 ± 0.000	0.025 ± 0.001	0.095 ± 0.005	0.308 ± 0.009	0.080	0.000	-0.56	0.0241	0.0241	0.939	92.315 ± 0.847	0.091 ± 0.001	91.709 ± 0.877	
UPI	beta51	0.16	0.161	0.001 ± 0.001	0.007 ± 0.008	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.071 ± 0.003	0.267 ± 0.007	0.160	0.000	-0.08	0.0417	0.0417	1.041	95.248 ± 0.677	0.174 ± 0.000	94.843 ± 0.703	
UPI	beta52	0.16	0.165	0.005 ± 0.002	0.033 ± 0.013	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.159 ± 0.007	0.398 ± 0.011	0.166	0.006	4.04	0.0627	0.0630	0.991	94.338 ± 0.735	0.247 ± 0.001	73.812 ± 1.398	
UPI	beta53	0.16	0.159	-0.001 ± 0.002	-0.008 ± 0.010	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.091 ± 0.004	0.302 ± 0.008	0.158	-0.002	-1.06	0.0455	0.0455	1.024	95.349 ± 0.670	0.194 ± 0.001	90.394 ± 0.937	
UPI	beta54	0.16	0.161	0.001 ± 0.002	0.009 ± 0.013	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	0.158 ± 0.007	0.398 ± 0.011	0.162	0.002	1.16	0.0625	0.0625	0.960	94.439 ± 0.729	0.240 ± 0.001	74.419 ± 1.387	
UPI	beta55	0.08	0.082	0.002 ± 0.001	0.019 ± 0.015	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.225 ± 0.011	0.474 ± 0.014	0.080	0.000	0.14	0.0361	0.0361	0.976	93.529 ± 0.782	0.145 ± 0.001	59.757 ± 1.559	
UPI	beta56	0.08	0.079	-0.001 ± 0.001	-0.010 ± 0.010	0.001 ± 0.000	0.025 ± 0.001	0.001 ± 0.000	0.025 ± 0.001	0.096 ± 0.005	0.310 ± 0.010	0.079	-0.001	-1.43	0.0227	0.0227	0.936	92.214 ± 0.852	0.091 ± 0.001	89.080 ± 0.992	
UPI	beta57	0.06	0.059	-0.001 ± 0.001	-0.020 ± 0.021	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.423 ± 0.021	0.650 ± 0.019	0.057	-0.003	-4.71	0.0373	0.0374	0.956	94.034 ± 0.753	0.146 ± 0.001	36.502 ± 1.531	
UPI	beta58	0.06	0.060	-0.000 ± 0.001	-0.006 ± 0.012	0.000 ± 0.000	0.022 ± 0.001	0.000 ± 0.000	0.022 ± 0.001	0.133 ± 0.007	0.365 ± 0.011	0.060	0.000	-0.54	0.0208	0.0208	0.967	94.641 ± 0.716	0.083 ± 0.001	81.193 ± 1.243	

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S155

Simulation 2 - Condition 11: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, $N = 400$, Reliability = 0.8

Method	Param.	True	Mean-based										Median-based					Inference		
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LSAM	beta41	0.20	0.203	0.003 $\pm$ 0.002	0.016 $\pm$ 0.012	0.005 $\pm$ 0.000	0.073 $\pm$ 0.002	0.005 $\pm$ 0.000	0.074 $\pm$ 0.002	0.135 $\pm$ 0.006	0.368 $\pm$ 0.010	0.204	0.004	1.80	0.0720	0.0721	1.042	95.657 $\pm$ 0.648	0.300 $\pm$ 0.001	76.566 $\pm$ 1.346
LSAM	beta42	0.20	0.201	0.001 $\pm$ 0.003	0.004 $\pm$ 0.013	0.006 $\pm$ 0.000	0.080 $\pm$ 0.002	0.006 $\pm$ 0.000	0.080 $\pm$ 0.002	0.160 $\pm$ 0.007	0.400 $\pm$ 0.011	0.205	0.005	2.46	0.0774	0.0775	0.990	94.545 $\pm$ 0.722	0.310 $\pm$ 0.001	71.818 $\pm$ 1.430
LSAM	beta43	0.20	0.199	-0.001 $\pm$ 0.002	-0.007 $\pm$ 0.008	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.056 $\pm$ 0.002	0.237 $\pm$ 0.006	0.199	-0.001	-0.41	0.0474	0.0474	0.981	94.545 $\pm$ 0.722	0.182 $\pm$ 0.001	98.485 $\pm$ 0.388
LSAM	beta44	0.11	0.114	0.004 $\pm$ 0.002	0.038 $\pm$ 0.017	0.003 $\pm$ 0.000	0.057 $\pm$ 0.001	0.003 $\pm$ 0.000	0.057 $\pm$ 0.002	0.272 $\pm$ 0.013	0.522 $\pm$ 0.015	0.114	0.004	3.71	0.0541	0.0543	0.901	92.121 $\pm$ 0.856	0.202 $\pm$ 0.002	60.000 $\pm$ 1.557
LSAM	beta45	0.11	0.112	0.002 $\pm$ 0.002	0.017 $\pm$ 0.016	0.003 $\pm$ 0.000	0.056 $\pm$ 0.001	0.003 $\pm$ 0.000	0.056 $\pm$ 0.002	0.255 $\pm$ 0.012	0.505 $\pm$ 0.015	0.110	0.000	-0.18	0.0544	0.0544	0.880	91.414 $\pm$ 0.890	0.192 $\pm$ 0.002	60.909 $\pm$ 1.551
LSAM	beta46	0.08	0.080	-0.000 $\pm$ 0.001	-0.004 $\pm$ 0.012	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.150 $\pm$ 0.007	0.388 $\pm$ 0.011	0.079	-0.001	-0.76	0.0279	0.0279	0.938	93.333 $\pm$ 0.793	0.114 $\pm$ 0.001	77.071 $\pm$ 1.336
LSAM	beta47	0.08	0.080	-0.000 $\pm$ 0.001	-0.002 $\pm$ 0.013	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	0.162 $\pm$ 0.008	0.402 $\pm$ 0.012	0.077	-0.003	-3.16	0.0294	0.0295	0.894	92.424 $\pm$ 0.841	0.113 $\pm$ 0.001	74.949 $\pm$ 1.377
LSAM	beta51	0.16	0.157	-0.003 $\pm$ 0.002	-0.020 $\pm$ 0.012	0.003 $\pm$ 0.000	0.058 $\pm$ 0.001	0.003 $\pm$ 0.000	0.058 $\pm$ 0.002	0.133 $\pm$ 0.006	0.365 $\pm$ 0.010	0.157	-0.003	-1.83	0.0588	0.0588	1.002	95.051 $\pm$ 0.689	0.229 $\pm$ 0.001	76.768 $\pm$ 1.342
LSAM	beta52	0.16	0.163	0.003 $\pm$ 0.002	0.019 $\pm$ 0.015	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.222 $\pm$ 0.010	0.471 $\pm$ 0.013	0.162	0.002	1.41	0.0764	0.0764	1.039	96.263 $\pm$ 0.603	0.307 $\pm$ 0.001	57.879 $\pm$ 1.569
LSAM	beta53	0.16	0.158	-0.002 $\pm$ 0.002	-0.009 $\pm$ 0.013	0.004 $\pm$ 0.000	0.064 $\pm$ 0.001	0.004 $\pm$ 0.000	0.064 $\pm$ 0.002	0.161 $\pm$ 0.007	0.402 $\pm$ 0.011	0.159	-0.001	-0.86	0.0651	0.0652	0.985	95.152 $\pm$ 0.683	0.248 $\pm$ 0.001	70.404 $\pm$ 1.451
LSAM	beta54	0.16	0.159	-0.001 $\pm$ 0.002	-0.007 $\pm$ 0.015	0.005 $\pm$ 0.000	0.073 $\pm$ 0.002	0.005 $\pm$ 0.000	0.073 $\pm$ 0.002	0.208 $\pm$ 0.009	0.457 $\pm$ 0.013	0.156	-0.004	-2.30	0.0708	0.0709	1.030	95.253 $\pm$ 0.676	0.295 $\pm$ 0.001	56.869 $\pm$ 1.574
LSAM	beta55	0.08	0.079	-0.001 $\pm$ 0.002	-0.018 $\pm$ 0.020	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.408 $\pm$ 0.019	0.639 $\pm$ 0.018	0.077	-0.003	-3.25	0.0514	0.0515	0.925	92.727 $\pm$ 0.825	0.185 $\pm$ 0.001	39.697 $\pm$ 1.555
LSAM	beta56	0.08	0.083	0.003 $\pm$ 0.001	0.035 $\pm$ 0.013	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.180 $\pm$ 0.010	0.424 $\pm$ 0.013	0.081	0.001	1.68	0.0299	0.0300	0.930	92.727 $\pm$ 0.825	0.123 $\pm$ 0.001	73.030 $\pm$ 1.410
LSAM	beta57	0.06	0.061	0.001 $\pm$ 0.002	0.018 $\pm$ 0.025	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.642 $\pm$ 0.031	0.801 $\pm$ 0.023	0.061	0.001	2.44	0.0440	0.0441	0.947	92.929 $\pm$ 0.815	0.179 $\pm$ 0.001	30.404 $\pm$ 1.462
LSAM	beta58	0.06	0.062	0.002 $\pm$ 0.001	0.032 $\pm$ 0.015	0.001 $\pm$ 0.000	0.028 $\pm$ 0.001	0.001 $\pm$ 0.000	0.028 $\pm$ 0.001	0.216 $\pm$ 0.011	0.465 $\pm$ 0.014	0.061	0.001	2.07	0.0255	0.0255	0.938	93.737 $\pm$ 0.770	0.102 $\pm$ 0.001	67.677 $\pm$ 1.486
QML	beta41	0.20	0.209	0.009 $\pm$ 0.002	0.044 $\pm$ 0.010	0.004 $\pm$ 0.000	0.061 $\pm$ 0.001	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.094 $\pm$ 0.004	0.307 $\pm$ 0.008	0.209	0.009	4.41	0.0604	0.0610	1.045	95.556 $\pm$ 0.655	0.249 $\pm$ 0.001	90.606 $\pm$ 0.927
QML	beta42	0.20	0.205	0.005 $\pm$ 0.002	0.025 $\pm$ 0.010	0.004 $\pm$ 0.000	0.065 $\pm$ 0.001	0.004 $\pm$ 0.000	0.065 $\pm$ 0.002	0.106 $\pm$ 0.005	0.326 $\pm$ 0.009	0.209	0.009	4.61	0.0646	0.0652	0.998	94.848 $\pm$ 0.703	0.254 $\pm$ 0.001	87.576 $\pm$ 1.048
QML	beta43	0.20	0.195	-0.005 $\pm$ 0.001	-0.027 $\pm$ 0.007	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.053 $\pm$ 0.002	0.231 $\pm$ 0.006	0.194	-0.006	-2.93	0.0462	0.0465	0.988	95.253 $\pm$ 0.676	0.178 $\pm$ 0.001	98.182 $\pm$ 0.425
QML	beta44	0.11	0.098	-0.012 $\pm$ 0.002	-0.110 $\pm$ 0.016	0.003 $\pm$ 0.000	0.055 $\pm$ 0.001	0.003 $\pm$ 0.000	0.056 $\pm$ 0.002	0.262 $\pm$ 0.013	0.512 $\pm$ 0.015	0.098	-0.012	-10.53	0.0545	0.0557	0.917	92.020 $\pm$ 0.861	0.198 $\pm$ 0.001	51.515 $\pm$ 1.588
QML	beta45	0.11	0.106	-0.004 $\pm$ 0.002	-0.035 $\pm$ 0.015	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.227 $\pm$ 0.010	0.476 $\pm$ 0.013	0.105	-0.005	-4.38	0.0519	0.0522	0.896	91.717 $\pm$ 0.876	0.184 $\pm$ 0.001	59.596 $\pm$ 1.560
QML	beta46	0.08	0.091	0.011 $\pm$ 0.001	0.143 $\pm$ 0.012	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	0.156 $\pm$ 0.007	0.395 $\pm$ 0.011	0.091	0.011	14.25	0.0269	0.0292	0.946	92.323 $\pm$ 0.846	0.109 $\pm$ 0.001	87.172 $\pm$ 1.063
QML	beta47	0.08	0.092	0.012 $\pm$ 0.001	0.150 $\pm$ 0.012	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	0.161 $\pm$ 0.009	0.401 $\pm$ 0.012	0.090	0.010	11.97	0.0269	0.0285	0.911	90.808 $\pm$ 0.918	0.106 $\pm$ 0.001	87.879 $\pm$ 1.037
QML	beta51	0.16	0.165	0.005 $\pm$ 0.002	0.031 $\pm$ 0.012	0.003 $\pm$ 0.000	0.059 $\pm$ 0.001	0.003 $\pm$ 0.000	0.059 $\pm$ 0.002	0.135 $\pm$ 0.006	0.368 $\pm$ 0.010	0.165	0.005	2.82	0.0580	0.0581	1.005	95.354 $\pm$ 0.669	0.231 $\pm$ 0.001	80.404 $\pm$ 1.262
QML	beta52	0.16	0.148	-0.012 $\pm$ 0.002	-0.077 $\pm$ 0.012	0.004 $\pm$ 0.000	0.062 $\pm$ 0.001	0.004 $\pm$ 0.000	0.063 $\pm$ 0.002	0.157 $\pm$ 0.007	0.396 $\pm$ 0.011	0.146	-0.014	-8.49	0.0641	0.0655	1.030	95.152 $\pm$ 0.683	0.251 $\pm$ 0.001	63.838 $\pm$ 1.527
QML	beta53	0.16	0.133	-0.027 $\pm$ 0.002	-0.166 $\pm$ 0.012	0.004 $\pm$ 0.000	0.062 $\pm$ 0.001	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.179 $\pm$ 0.008	0.423 $\pm$ 0.011	0.133	-0.027	-16.60	0.0628	0.0682	1.016	93.333 $\pm$ 0.793	0.248 $\pm$ 0.004	58.384 $\pm$ 1.567
QML	beta54	0.16	0.160	0.000 $\pm$ 0.002	0.001 $\pm$ 0.012	0.004 $\pm$ 0.000	0.061 $\pm$ 0.001	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.147 $\pm$ 0.007	0.383 $\pm$ 0.011	0.157	-0.003	-2.12	0.0599	0.0600	1.038	95.253 $\pm$ 0.676	0.250 $\pm$ 0.001	72.222 $\pm$ 1.424
QML	beta55	0.08	0.081	0.001 $\pm$ 0.002	0.012 $\pm$ 0.020	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.415 $\pm$ 0.020	0.644 $\pm$ 0.019	0.079	-0.001	-0.73	0.0507	0.0507	0.935	92.020 $\pm$ 0.861	0.189 $\pm$ 0.002	41.212 $\pm$ 1.564
QML	beta56	0.08	0.088	0.008 $\pm$ 0.001	0.104 $\pm$ 0.013	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.190 $\pm$ 0.010	0.436 $\pm$ 0.013	0.086	0.006	7.98	0.0302	0.0309	0.945	91.313 $\pm$ 0.895	0.125 $\pm$ 0.002	77.475 $\pm$ 1.328
QML	beta57	0.06	0.070	0.010 $\pm$ 0.001	0.164 $\pm$ 0.024	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.597 $\pm$ 0.028	0.773 $\pm$ 0.021	0.072	0.012	19.47	0.0436	0.0451	0.972	92.828 $\pm$ 0.820	0.173 $\pm$ 0.002	40.404 $\pm$ 1.560
QML	beta58	0.06	0.071	0.011 $\pm$ 0.001	0.180 $\pm$ 0.014	0.001 $\pm$ 0.000	0.026 $\pm$ 0.001	0.001 $\pm$ 0.000	0.028 $\pm$ 0.001	0.224 $\pm$ 0.010	0.473 $\pm$ 0.013	0.070	0.010	17.10	0.0245	0.0266	0.943	90.606 $\pm$ 0.927	0.097 $\pm$ 0.001	80.707 $\pm$ 1.254
UPI	beta41	0.20	0.209	0.009 $\pm$ 0.002	0.044 $\pm$ 0.012	0.005 $\pm$ 0.000	0.073 $\pm$ 0.002	0.005 $\pm$ 0.000	0.074 $\pm$ 0.002	0.136 $\pm$ 0.006	0.369 $\pm$ 0.010	0.209	0.009	4.47	0.0705	0.0711	1.025	95.657 $\pm$ 0.648	0.294 $\pm$ 0.001	78.889 $\pm$ 1.297
UPI	beta42	0.20	0.203	0.003 $\pm$ 0.003	0.013 $\pm$ 0.013	0.006 $\pm$ 0.000	0.080 $\pm$ 0.002	0.006 $\pm$ 0.000	0.080 $\pm$ 0.002	0.158 $\pm$ 0.007	0.398 $\pm$ 0.011	0.206	0.006	2.78	0.0814	0.0816	0.979	94.646 $\pm$ 0.715	0.306 $\pm$ 0.001	72.424 $\pm$ 1.420
UPI	beta43	0.20	0.197	-0.003 $\pm$ 0.002	-0.013 $\pm$ 0.008	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.057 $\pm$ 0.003	0.238 $\pm$ 0.007	0.198	-0.002	-1.14	0.0478	0.0478	0.958	93.131 $\pm$ 0.804	0.179 $\pm$ 0.001	98.384 $\pm$ 0.401

Table S155

Simulation 2 - Condition 11: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.8
(continued)

Method	Param.	True	Mean-based										Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE	
UPI	beta47	0.08	0.079	-0.001 ± 0.001	-0.009 ± 0.013	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.167 ± 0.009	0.408 ± 0.012	0.077	-0.003	-3.49	0.0295	0.0297	0.879	91.313 ± 0.895	0.113 ± 0.001	74.848 ± 1.379	
UPI	beta51	0.16	0.158	-0.002 ± 0.002	-0.012 ± 0.012	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.132 ± 0.006	0.364 ± 0.010	0.157	-0.003	-1.57	0.0579	0.0580	0.991	94.848 ± 0.703	0.226 ± 0.001	78.081 ± 1.315	
UPI	beta52	0.16	0.167	0.007 ± 0.002	0.041 ± 0.015	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	0.219 ± 0.010	0.468 ± 0.013	0.167	0.007	4.19	0.0748	0.0751	1.029	96.061 ± 0.618	0.301 ± 0.001	59.798 ± 1.558	
UPI	beta53	0.16	0.158	-0.002 ± 0.002	-0.013 ± 0.013	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.161 ± 0.007	0.401 ± 0.011	0.160	0.000	-0.09	0.0642	0.0642	0.979	95.152 ± 0.683	0.246 ± 0.001	71.414 ± 1.436	
UPI	beta54	0.16	0.160	-0.000 ± 0.002	-0.002 ± 0.014	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.206 ± 0.009	0.453 ± 0.012	0.158	-0.002	-1.08	0.0710	0.0710	1.032	95.152 ± 0.683	0.294 ± 0.001	56.869 ± 1.574	
UPI	beta55	0.08	0.079	-0.001 ± 0.002	-0.009 ± 0.021	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.420 ± 0.020	0.648 ± 0.018	0.077	-0.003	-3.48	0.0525	0.0526	0.911	92.323 ± 0.846	0.185 ± 0.002	41.414 ± 1.566	
UPI	beta56	0.08	0.082	0.002 ± 0.001	0.030 ± 0.014	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.183 ± 0.010	0.428 ± 0.013	0.080	0.000	0.34	0.0301	0.0301	0.924	93.030 ± 0.809	0.124 ± 0.001	72.828 ± 1.414	
UPI	beta57	0.06	0.059	-0.001 ± 0.002	-0.011 ± 0.025	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.639 ± 0.031	0.799 ± 0.023	0.059	-0.001	-1.11	0.0462	0.0462	0.942	92.626 ± 0.831	0.177 ± 0.001	28.384 ± 1.433	
UPI	beta58	0.06	0.062	0.002 ± 0.001	0.029 ± 0.015	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.221 ± 0.011	0.470 ± 0.014	0.061	0.001	0.88	0.0251	0.0251	0.937	93.434 ± 0.787	0.103 ± 0.001	66.667 ± 1.498	

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S156

Simulation 2 - Condition 12: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, $N = 1000$, Reliability = 0.8

Method	Param.	True	Mean-based										Median-based					Inference		
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LSAM	beta41	0.20	0.200	0.000 $\pm$ 0.001	0.000 $\pm$ 0.007	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.053 $\pm$ 0.003	0.231 $\pm$ 0.007	0.200	0.000	0.06	0.0441	0.0441	1.008	95.195 $\pm$ 0.677	0.183 $\pm$ 0.000	98.699 $\pm$ 0.359
LSAM	beta42	0.20	0.201	0.001 $\pm$ 0.002	0.007 $\pm$ 0.008	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.061 $\pm$ 0.003	0.247 $\pm$ 0.007	0.202	0.002	0.78	0.0476	0.0476	0.978	94.394 $\pm$ 0.728	0.189 $\pm$ 0.001	97.497 $\pm$ 0.494
LSAM	beta43	0.20	0.198	-0.002 $\pm$ 0.001	-0.008 $\pm$ 0.005	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.023 $\pm$ 0.001	0.152 $\pm$ 0.004	0.198	-0.002	-0.98	0.0303	0.0304	0.973	94.394 $\pm$ 0.728	0.116 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta44	0.11	0.112	0.002 $\pm$ 0.001	0.015 $\pm$ 0.010	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.091 $\pm$ 0.004	0.302 $\pm$ 0.009	0.111	0.001	0.88	0.0332	0.0332	0.932	93.393 $\pm$ 0.786	0.121 $\pm$ 0.001	93.694 $\pm$ 0.769
LSAM	beta45	0.11	0.112	0.002 $\pm$ 0.001	0.014 $\pm$ 0.009	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.082 $\pm$ 0.004	0.286 $\pm$ 0.008	0.110	0.000	0.03	0.0306	0.0306	0.948	93.493 $\pm$ 0.780	0.117 $\pm$ 0.001	95.095 $\pm$ 0.683
LSAM	beta46	0.08	0.080	0.000 $\pm$ 0.001	0.001 $\pm$ 0.007	0.000 $\pm$ 0.000	0.018 $\pm$ 0.000	0.000 $\pm$ 0.000	0.018 $\pm$ 0.001	0.050 $\pm$ 0.002	0.223 $\pm$ 0.007	0.080	0.000	-0.28	0.0170	0.0170	0.926	93.093 $\pm$ 0.802	0.065 $\pm$ 0.001	98.599 $\pm$ 0.372
LSAM	beta47	0.08	0.080	-0.000 $\pm$ 0.001	-0.001 $\pm$ 0.007	0.000 $\pm$ 0.000	0.018 $\pm$ 0.000	0.000 $\pm$ 0.000	0.018 $\pm$ 0.001	0.049 $\pm$ 0.002	0.221 $\pm$ 0.007	0.079	-0.001	-0.67	0.0164	0.0164	0.926	91.892 $\pm$ 0.864	0.064 $\pm$ 0.001	98.498 $\pm$ 0.385
LSAM	beta51	0.16	0.161	0.001 $\pm$ 0.001	0.004 $\pm$ 0.007	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.054 $\pm$ 0.002	0.232 $\pm$ 0.006	0.160	0.000	-0.30	0.0388	0.0388	0.989	95.395 $\pm$ 0.663	0.144 $\pm$ 0.000	99.299 $\pm$ 0.264
LSAM	beta52	0.16	0.157	-0.003 $\pm$ 0.002	-0.017 $\pm$ 0.009	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.090 $\pm$ 0.004	0.300 $\pm$ 0.008	0.157	-0.003	-1.60	0.0481	0.0481	1.004	95.295 $\pm$ 0.670	0.189 $\pm$ 0.000	89.189 $\pm$ 0.982
LSAM	beta53	0.16	0.162	0.002 $\pm$ 0.001	0.010 $\pm$ 0.008	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	0.066 $\pm$ 0.003	0.256 $\pm$ 0.007	0.161	0.001	0.73	0.0411	0.0411	0.961	92.793 $\pm$ 0.818	0.154 $\pm$ 0.000	97.598 $\pm$ 0.484
LSAM	beta54	0.16	0.160	0.000 $\pm$ 0.001	0.002 $\pm$ 0.009	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.085 $\pm$ 0.004	0.291 $\pm$ 0.008	0.160	0.000	-0.03	0.0455	0.0455	0.996	94.595 $\pm$ 0.715	0.182 $\pm$ 0.001	92.192 $\pm$ 0.849
LSAM	beta55	0.08	0.079	-0.001 $\pm$ 0.001	-0.012 $\pm$ 0.011	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.131 $\pm$ 0.006	0.361 $\pm$ 0.010	0.079	-0.001	-1.57	0.0266	0.0267	0.925	93.093 $\pm$ 0.802	0.105 $\pm$ 0.001	81.782 $\pm$ 1.221
LSAM	beta56	0.08	0.080	0.000 $\pm$ 0.001	0.001 $\pm$ 0.008	0.000 $\pm$ 0.000	0.019 $\pm$ 0.000	0.000 $\pm$ 0.000	0.019 $\pm$ 0.001	0.057 $\pm$ 0.003	0.239 $\pm$ 0.007	0.080	0.000	-0.55	0.0186	0.0186	0.913	92.492 $\pm$ 0.834	0.068 $\pm$ 0.001	97.397 $\pm$ 0.504
LSAM	beta57	0.06	0.062	0.002 $\pm$ 0.001	0.026 $\pm$ 0.015	0.001 $\pm$ 0.000	0.028 $\pm$ 0.001	0.001 $\pm$ 0.000	0.028 $\pm$ 0.001	0.224 $\pm$ 0.010	0.474 $\pm$ 0.013	0.061	0.001	2.32	0.0290	0.0291	0.951	93.994 $\pm$ 0.752	0.106 $\pm$ 0.001	62.362 $\pm$ 1.533
LSAM	beta58	0.06	0.061	0.001 $\pm$ 0.001	0.011 $\pm$ 0.009	0.000 $\pm$ 0.000	0.016 $\pm$ 0.000	0.000 $\pm$ 0.000	0.016 $\pm$ 0.000	0.073 $\pm$ 0.004	0.270 $\pm$ 0.008	0.060	0.000	0.78	0.0149	0.0149	0.952	92.693 $\pm$ 0.823	0.061 $\pm$ 0.001	94.094 $\pm$ 0.746
QML	beta41	0.20	0.206	0.006 $\pm$ 0.001	0.028 $\pm$ 0.006	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.039 $\pm$ 0.002	0.197 $\pm$ 0.006	0.206	0.006	2.80	0.0373	0.0377	1.014	94.695 $\pm$ 0.709	0.155 $\pm$ 0.000	100.000 $\pm$ 0.000
QML	beta42	0.20	0.205	0.005 $\pm$ 0.001	0.026 $\pm$ 0.006	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	0.043 $\pm$ 0.002	0.207 $\pm$ 0.006	0.205	0.005	2.62	0.0405	0.0408	0.983	94.094 $\pm$ 0.746	0.158 $\pm$ 0.000	99.800 $\pm$ 0.141
QML	beta43	0.20	0.195	-0.005 $\pm$ 0.001	-0.026 $\pm$ 0.005	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.023 $\pm$ 0.001	0.151 $\pm$ 0.004	0.195	-0.005	-2.54	0.0297	0.0301	0.974	93.894 $\pm$ 0.758	0.113 $\pm$ 0.000	100.000 $\pm$ 0.000
QML	beta44	0.11	0.098	-0.012 $\pm$ 0.001	-0.111 $\pm$ 0.009	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.102 $\pm$ 0.005	0.319 $\pm$ 0.008	0.098	-0.012	-10.61	0.0330	0.0350	0.940	90.390 $\pm$ 0.932	0.121 $\pm$ 0.001	86.186 $\pm$ 1.092
QML	beta45	0.11	0.108	-0.002 $\pm$ 0.001	-0.016 $\pm$ 0.009	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.077 $\pm$ 0.004	0.277 $\pm$ 0.008	0.109	-0.001	-1.24	0.0300	0.0300	0.961	93.994 $\pm$ 0.752	0.115 $\pm$ 0.001	94.494 $\pm$ 0.722
QML	beta46	0.08	0.092	0.012 $\pm$ 0.001	0.153 $\pm$ 0.007	0.000 $\pm$ 0.000	0.017 $\pm$ 0.000	0.000 $\pm$ 0.000	0.021 $\pm$ 0.001	0.071 $\pm$ 0.003	0.266 $\pm$ 0.007	0.092	0.012	15.01	0.0170	0.0208	0.947	88.689 $\pm$ 1.002	0.065 $\pm$ 0.000	99.900 $\pm$ 0.100
QML	beta47	0.08	0.093	0.013 $\pm$ 0.001	0.157 $\pm$ 0.007	0.000 $\pm$ 0.000	0.017 $\pm$ 0.000	0.000 $\pm$ 0.000	0.021 $\pm$ 0.001	0.070 $\pm$ 0.003	0.265 $\pm$ 0.007	0.092	0.012	15.49	0.0157	0.0200	0.946	85.586 $\pm$ 1.111	0.063 $\pm$ 0.000	99.800 $\pm$ 0.141
QML	beta51	0.16	0.169	0.009 $\pm$ 0.001	0.054 $\pm$ 0.007	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.057 $\pm$ 0.003	0.240 $\pm$ 0.006	0.167	0.007	4.43	0.0386	0.0393	0.989	94.294 $\pm$ 0.734	0.145 $\pm$ 0.000	99.399 $\pm$ 0.244
QML	beta52	0.16	0.141	-0.019 $\pm$ 0.001	-0.118 $\pm$ 0.008	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.075 $\pm$ 0.003	0.273 $\pm$ 0.007	0.141	-0.019	-11.94	0.0395	0.0438	1.009	92.192 $\pm$ 0.849	0.156 $\pm$ 0.000	93.994 $\pm$ 0.752
QML	beta53	0.16	0.136	-0.024 $\pm$ 0.001	-0.148 $\pm$ 0.008	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.084 $\pm$ 0.004	0.291 $\pm$ 0.007	0.135	-0.025	-15.34	0.0389	0.0460	0.968	89.990 $\pm$ 0.950	0.152 $\pm$ 0.000	93.493 $\pm$ 0.780
QML	beta54	0.16	0.161	0.001 $\pm$ 0.001	0.004 $\pm$ 0.008	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.062 $\pm$ 0.003	0.249 $\pm$ 0.007	0.161	0.001	0.37	0.0380	0.0380	0.996	93.694 $\pm$ 0.769	0.156 $\pm$ 0.000	97.297 $\pm$ 0.513
QML	beta55	0.08	0.082	0.002 $\pm$ 0.001	0.028 $\pm$ 0.012	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.144 $\pm$ 0.007	0.380 $\pm$ 0.011	0.083	0.003	3.23	0.0294	0.0295	0.921	92.593 $\pm$ 0.829	0.110 $\pm$ 0.001	80.080 $\pm$ 1.264
QML	beta56	0.08	0.086	0.006 $\pm$ 0.001	0.074 $\pm$ 0.008	0.000 $\pm$ 0.000	0.020 $\pm$ 0.000	0.000 $\pm$ 0.000	0.020 $\pm$ 0.001	0.065 $\pm$ 0.003	0.255 $\pm$ 0.007	0.086	0.006	7.31	0.0185	0.0194	0.922	90.791 $\pm$ 0.915	0.071 $\pm$ 0.001	97.998 $\pm$ 0.443
QML	beta57	0.06	0.071	0.011 $\pm$ 0.001	0.177 $\pm$ 0.015	0.001 $\pm$ 0.000	0.028 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.247 $\pm$ 0.011	0.497 $\pm$ 0.013	0.070	0.010	17.03	0.0296	0.0313	0.964	92.893 $\pm$ 0.813	0.105 $\pm$ 0.001	73.974 $\pm$ 1.388
QML	beta58	0.06	0.070	0.010 $\pm$ 0.000	0.166 $\pm$ 0.008	0.000 $\pm$ 0.000	0.016 $\pm$ 0.000	0.000 $\pm$ 0.000	0.019 $\pm$ 0.000	0.096 $\pm$ 0.004	0.310 $\pm$ 0.008	0.070	0.010	15.95	0.0143	0.0172	0.955	88.188 $\pm$ 1.021	0.059 $\pm$ 0.000	98.899 $\pm$ 0.330
UPI	beta41	0.20	0.202	0.002 $\pm$ 0.001	0.011 $\pm$ 0.007	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.052 $\pm$ 0.002	0.229 $\pm$ 0.006	0.203	0.003	1.36	0.0423	0.0424	0.999	94.595 $\pm$ 0.715	0.179 $\pm$ 0.001	99.099 $\pm$ 0.299
UPI	beta42	0.20	0.203	0.003 $\pm$ 0.002	0.016 $\pm$ 0.008	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.060 $\pm$ 0.003	0.246 $\pm$ 0.007	0.203	0.003	1.33	0.0485	0.0486	0.970	94.194 $\pm$ 0.740	0.186 $\pm$ 0.001	97.497 $\pm$ 0.494
UPI	beta43	0.20	0.198	-0.002 $\pm$ 0.001	-0.012 $\pm$ 0.005	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.023 $\pm$ 0.001	0.152 $\pm$ 0.004	0.197	-0.003	-1.55	0.0304	0.0306	0.962	93.894 $\pm$ 0.758	0.114 $\pm$ 0.000	100.000 $\pm$

Table S156

Simulation 2 - Condition 12: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.8
(continued)

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.079	-0.001 ± 0.001	-0.009 ± 0.007	0.000 ± 0.000	0.018 ± 0.000	0.000 ± 0.000	0.018 ± 0.001	0.050 ± 0.003	0.223 ± 0.007	0.079	-0.001	-0.71	0.0167	0.0168	0.923	91.491 ± 0.883	0.064 ± 0.001	98.198 ± 0.421
UPI	beta51	0.16	0.161	0.001 ± 0.001	0.009 ± 0.007	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.054 ± 0.002	0.231 ± 0.006	0.160	0.000	-0.24	0.0385	0.0385	0.981	94.895 ± 0.696	0.142 ± 0.000	99.299 ± 0.264
UPI	beta52	0.16	0.159	-0.001 ± 0.002	-0.006 ± 0.009	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.090 ± 0.004	0.300 ± 0.008	0.159	-0.001	-0.71	0.0488	0.0488	0.989	94.795 ± 0.703	0.186 ± 0.000	90.691 ± 0.919
UPI	beta53	0.16	0.161	0.001 ± 0.001	0.008 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.065 ± 0.003	0.256 ± 0.007	0.161	0.001	0.54	0.0423	0.0423	0.954	92.893 ± 0.813	0.153 ± 0.000	97.998 ± 0.443
UPI	beta54	0.16	0.161	0.001 ± 0.001	0.004 ± 0.009	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.084 ± 0.004	0.290 ± 0.008	0.160	0.000	0.30	0.0441	0.0441	0.996	94.695 ± 0.709	0.181 ± 0.001	92.492 ± 0.834
UPI	beta55	0.08	0.079	-0.001 ± 0.001	-0.009 ± 0.011	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.130 ± 0.006	0.361 ± 0.010	0.080	0.000	-0.57	0.0274	0.0274	0.924	92.993 ± 0.808	0.105 ± 0.001	81.281 ± 1.234
UPI	beta56	0.08	0.080	-0.000 ± 0.001	-0.003 ± 0.008	0.000 ± 0.000	0.019 ± 0.000	0.000 ± 0.000	0.019 ± 0.001	0.058 ± 0.003	0.240 ± 0.007	0.080	0.000	-0.54	0.0185	0.0185	0.911	92.292 ± 0.844	0.069 ± 0.001	97.598 ± 0.484
UPI	beta57	0.06	0.061	0.001 ± 0.001	0.015 ± 0.015	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.227 ± 0.010	0.476 ± 0.013	0.061	0.001	0.90	0.0289	0.0289	0.940	93.894 ± 0.758	0.105 ± 0.001	61.962 ± 1.536
UPI	beta58	0.06	0.061	0.001 ± 0.001	0.011 ± 0.009	0.000 ± 0.000	0.016 ± 0.000	0.000 ± 0.000	0.016 ± 0.000	0.074 ± 0.004	0.273 ± 0.008	0.060	0.000	0.12	0.0155	0.0155	0.952	92.292 ± 0.844	0.061 ± 0.001	94.194 ± 0.740

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S157

Simulation 2 - Condition 13: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.4

Method	Param.	True	Mean-based										Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE	
LSAM	beta41	0.20	0.202	0.002 ± 0.005	0.008 ± 0.024	0.007 ± 0.001	0.084 ± 0.004	0.007 ± 0.001	0.083 ± 0.005	0.174 ± 0.016	0.417 ± 0.023	0.198	-0.002	-1.22	0.0847	0.0848	1.265	98.644 ± 0.673	0.414 ± 0.009	54.237 ± 2.901	
LSAM	beta42	0.20	0.202	0.002 ± 0.005	0.009 ± 0.026	0.008 ± 0.001	0.090 ± 0.004	0.008 ± 0.001	0.090 ± 0.005	0.203 ± 0.016	0.450 ± 0.023	0.200	0.000	0.17	0.0847	0.0847	1.218	97.966 ± 0.822	0.430 ± 0.011	50.847 ± 2.911	
LSAM	beta43	0.20	0.202	0.002 ± 0.005	0.011 ± 0.025	0.007 ± 0.001	0.086 ± 0.004	0.007 ± 0.001	0.086 ± 0.005	0.185 ± 0.018	0.431 ± 0.025	0.195	-0.005	-2.62	0.0842	0.0844	1.085	96.949 ± 1.001	0.367 ± 0.007	64.407 ± 2.788	
LSAM	beta44	0.11	0.126	0.016 ± 0.009	0.142 ± 0.079	0.022 ± 0.002	0.149 ± 0.007	0.022 ± 0.002	0.149 ± 0.008	1.846 ± 0.170	1.359 ± 0.074	0.119	0.009	8.21	0.1354	0.1357	1.133	97.966 ± 0.822	0.661 ± 0.025	13.559 ± 1.993	
LSAM	beta45	0.11	0.125	0.015 ± 0.007	0.132 ± 0.063	0.014 ± 0.001	0.120 ± 0.005	0.014 ± 0.001	0.120 ± 0.006	1.194 ± 0.101	1.093 ± 0.056	0.106	-0.004	-3.76	0.1162	0.1163	1.062	96.610 ± 1.054	0.497 ± 0.010	15.593 ± 2.112	
LSAM	beta46	0.08	0.084	0.004 ± 0.011	0.049 ± 0.132	0.033 ± 0.004	0.182 ± 0.010	0.033 ± 0.004	0.181 ± 0.011	5.138 ± 0.548	2.267 ± 0.139	0.084	0.004	4.51	0.1656	0.1656	1.266	98.983 ± 0.584	0.901 ± 0.040	1.017 ± 0.584	
LSAM	beta47	0.08	0.063	-0.017 ± 0.011	-0.215 ± 0.134	0.034 ± 0.004	0.184 ± 0.010	0.034 ± 0.004	0.184 ± 0.011	5.296 ± 0.556	2.301 ± 0.139	0.070	-0.010	-12.14	0.1628	0.1631	1.111	97.966 ± 0.822	0.800 ± 0.036	1.017 ± 0.584	
LSAM	beta51	0.16	0.157	-0.003 ± 0.007	-0.021 ± 0.043	0.014 ± 0.001	0.120 ± 0.006	0.014 ± 0.001	0.119 ± 0.007	0.557 ± 0.057	0.746 ± 0.045	0.155	-0.005	-3.32	0.1081	0.1083	0.944	95.593 ± 1.195	0.442 ± 0.007	34.237 ± 2.763	
LSAM	beta52	0.16	0.160	-0.000 ± 0.006	-0.002 ± 0.036	0.010 ± 0.001	0.098 ± 0.005	0.010 ± 0.001	0.098 ± 0.006	0.373 ± 0.041	0.611 ± 0.038	0.157	-0.003	-2.03	0.0906	0.0907	1.098	97.288 ± 0.946	0.421 ± 0.013	33.898 ± 2.756	
LSAM	beta53	0.16	0.168	0.008 ± 0.006	0.052 ± 0.036	0.010 ± 0.001	0.100 ± 0.005	0.010 ± 0.001	0.100 ± 0.006	0.391 ± 0.038	0.625 ± 0.036	0.160	0.000	0.06	0.0945	0.0945	1.045	96.271 ± 1.103	0.409 ± 0.008	37.627 ± 2.821	
LSAM	beta54	0.16	0.165	0.005 ± 0.005	0.031 ± 0.032	0.008 ± 0.001	0.089 ± 0.004	0.008 ± 0.001	0.089 ± 0.004	0.306 ± 0.025	0.553 ± 0.028	0.168	0.008	4.94	0.0962	0.0965	1.141	97.627 ± 0.886	0.396 ± 0.009	40.000 ± 2.852	
LSAM	beta55	0.08	0.078	-0.002 ± 0.010	-0.021 ± 0.120	0.027 ± 0.003	0.165 ± 0.009	0.027 ± 0.003	0.164 ± 0.010	4.219 ± 0.457	2.054 ± 0.127	0.075	-0.005	-6.22	0.1333	0.1334	1.090	97.288 ± 0.946	0.703 ± 0.024	5.763 ± 1.357	
LSAM	beta56	0.08	0.076	-0.004 ± 0.008	-0.048 ± 0.105	0.021 ± 0.002	0.144 ± 0.008	0.021 ± 0.002	0.144 ± 0.009	3.228 ± 0.340	1.797 ± 0.109	0.070	-0.010	-12.15	0.1203	0.1207	1.083	97.288 ± 0.946	0.611 ± 0.019	5.085 ± 1.279	
LSAM	beta57	0.06	0.074	0.014 ± 0.013	0.234 ± 0.211	0.047 ± 0.005	0.217 ± 0.013	0.047 ± 0.006	0.217 ± 0.015	13.082 ± 1.560	3.617 ± 0.242	0.050	-0.010	-16.53	0.1685	0.1688	1.023	97.966 ± 0.822	0.870 ± 0.045	1.356 ± 0.673	
LSAM	beta58	0.06	0.069	0.009 ± 0.009	0.153 ± 0.157	0.026 ± 0.003	0.162 ± 0.009	0.026 ± 0.003	0.162 ± 0.010	7.250 ± 0.828	2.693 ± 0.174	0.053	-0.007	-11.54	0.1334	0.1336	1.089	98.305 ± 0.752	0.689 ± 0.028	1.356 ± 0.673	
QML	beta41	0.20	0.191	-0.009 ± 0.004	-0.043 ± 0.020	0.005 ± 0.000	0.067 ± 0.003	0.005 ± 0.000	0.068 ± 0.003	0.114 ± 0.010	0.338 ± 0.017	0.192	-0.008	-4.04	0.0706	0.0710	1.101	95.593 ± 1.195	0.290 ± 0.003	79.661 ± 2.344	
QML	beta42	0.20	0.190	-0.010 ± 0.004	-0.050 ± 0.021	0.005 ± 0.000	0.073 ± 0.003	0.005 ± 0.000	0.073 ± 0.003	0.134 ± 0.010	0.366 ± 0.017	0.188	-0.012	-5.78	0.0765	0.0773	1.057	96.949 ± 1.001	0.301 ± 0.003	72.881 ± 2.588	
QML	beta43	0.20	0.187	-0.013 ± 0.004	-0.063 ± 0.021	0.005 ± 0.000	0.072 ± 0.003	0.005 ± 0.000	0.073 ± 0.004	0.132 ± 0.011	0.363 ± 0.018	0.185	-0.015	-7.65	0.0728	0.0744	1.035	94.915 ± 1.279	0.290 ± 0.003	73.220 ± 2.578	
QML	beta44	0.11	0.123	0.013 ± 0.006	0.116 ± 0.055	0.011 ± 0.001	0.104 ± 0.005	0.011 ± 0.001	0.105 ± 0.006	0.906 ± 0.081	0.952 ± 0.051	0.115	0.005	4.78	0.1083	0.1084	1.026	93.898 ± 1.394	0.419 ± 0.010	24.068 ± 2.489	
QML	beta45	0.11	0.111	0.001 ± 0.005	0.011 ± 0.049	0.009 ± 0.001	0.092 ± 0.004	0.009 ± 0.001	0.092 ± 0.005	0.704 ± 0.062	0.839 ± 0.045	0.103	-0.007	-6.47	0.0948	0.0950	0.991	93.220 ± 1.464	0.359 ± 0.010	26.102 ± 2.557	
QML	beta46	0.08	0.022	-0.058 ± 0.004	-0.726 ± 0.049	0.005 ± 0.000	0.067 ± 0.003	0.008 ± 0.001	0.089 ± 0.004	1.236 ± 0.103	1.112 ± 0.050	0.025	-0.055	-68.13	0.0675	0.0868	1.178	90.847 ± 1.679	0.312 ± 0.010	2.712 ± 0.946	
QML	beta47	0.08	0.023	-0.057 ± 0.004	-0.714 ± 0.051	0.005 ± 0.000	0.070 ± 0.004	0.008 ± 0.001	0.090 ± 0.004	1.275 ± 0.093	1.129 ± 0.046	0.024	-0.056	-69.50	0.0699	0.0894	1.092	86.780 ± 1.972	0.300 ± 0.012	5.085 ± 1.279	
QML	beta51	0.16	0.182	0.022 ± 0.006	0.140 ± 0.039	0.011 ± 0.001	0.107 ± 0.006	0.012 ± 0.001	0.109 ± 0.007	0.464 ± 0.050	0.681 ± 0.042	0.177	0.017	10.50	0.1009	0.1023	0.983	96.610 ± 1.054	0.412 ± 0.004	41.356 ± 2.867	
QML	beta52	0.16	0.148	-0.012 ± 0.005	-0.072 ± 0.030	0.007 ± 0.001	0.082 ± 0.004	0.007 ± 0.001	0.082 ± 0.005	0.265 ± 0.027	0.514 ± 0.031	0.153	-0.007	-4.63	0.0816	0.0819	1.049	95.932 ± 1.150	0.336 ± 0.006	44.068 ± 2.891	
QML	beta53	0.16	0.151	-0.009 ± 0.005	-0.057 ± 0.031	0.007 ± 0.001	0.085 ± 0.004	0.007 ± 0.001	0.085 ± 0.005	0.283 ± 0.026	0.532 ± 0.029	0.142	-0.018	-11.30	0.0799	0.0819	0.995	94.576 ± 1.319	0.331 ± 0.005	40.678 ± 2.860	
QML	beta54	0.16	0.160	0.000 ± 0.004	0.001 ± 0.027	0.006 ± 0.000	0.075 ± 0.003	0.006 ± 0.000	0.075 ± 0.004	0.218 ± 0.017	0.467 ± 0.023	0.163	0.003	1.81	0.0818	0.0818	1.039	95.593 ± 1.195	0.305 ± 0.003	58.305 ± 2.871	
QML	beta55	0.08	0.085	0.005 ± 0.008	0.061 ± 0.094	0.017 ± 0.002	0.129 ± 0.007	0.017 ± 0.002	0.129 ± 0.008	2.600 ± 0.277	1.612 ± 0.099	0.090	0.010	12.63	0.1300	0.1304	1.009	95.932 ± 1.150	0.511 ± 0.010	11.186 ± 1.835	
QML	beta56	0.08	0.068	-0.012 ± 0.006	-0.145 ± 0.078	0.011 ± 0.001	0.107 ± 0.005	0.011 ± 0.001	0.107 ± 0.006	1.796 ± 0.183	1.340 ± 0.079	0.070	-0.010	-13.12	0.1087	0.1092	1.047	94.915 ± 1.279	0.438 ± 0.007	6.780 ± 1.464	
QML	beta57	0.06	0.019	-0.041 ± 0.005	-0.681 ± 0.081	0.007 ± 0.001	0.083 ± 0.004	0.009 ± 0.001	0.093 ± 0.004	2.389 ± 0.198	1.545 ± 0.074	0.016	-0.044	-72.72	0.0810	0.0920	0.959	89.831 ± 1.760	0.313 ± 0.007	5.085 ± 1.279	
QML	beta58	0.06	0.028	-0.032 ± 0.004	-0.534 ± 0.065	0.004 ± 0.000	0.067 ± 0.003	0.005 ± 0.000	0.074 ± 0.004	1.524 ± 0.127	1.234 ± 0.060	0.023	-0.037	-61.62	0.0678	0.0772	1.001	93.898 ± 1.394	0.262 ± 0.006	7.458 ± 1.530	
UPI	beta41	0.20	0.199	-0.001 ± 0.007	-0.003 ± 0.037	0.016 ± 0.002	0.127 ± 0.008	0.016 ± 0.002	0.126 ± 0.009	0.400 ± 0.051	0.632 ± 0.044	0.190	-0.010	-4.93	0.1187	0.1192	1.551	100.000 ± 0.000	0.770 ± 0.070	36.271 ± 2.799	
UPI	beta42	0.20	0.209	0.009 ± 0.007	0.045 ± 0.037	0.016 ± 0.002	0.126 ± 0.008	0.016 ± 0.002	0.126 ± 0.009	0.396 ± 0.051	0.629 ± 0.045	0.202	0.002	0.76	0.1230	0.1230	1.623	99.322 ± 0.478	0.800 ± 0.067	27.797 ± 2.608	
UPI	beta43	0.20	0.200	0.000 ± 0.007	0.000 ± 0.033	0.013 ± 0.002	0.115 ± 0.008	0.013 ± 0.002	0.115 ± 0.008	0.329 ± 0.042	0.574 ± 0.041	0.194	-0.006	-2.79	0.1062	0.1064	1.520	98.305 ± 0.752	0.685 ± 0.068	37.966 ± 2.826	
UPI	beta44	0.11	0.161	0.051 ± 0.014	0.465 ± 0.129	0.059 ± 0.008	0.243 ± 0.016	0.062 ± 0.008	0.248 ± 0.018	5.088 ± 0.656	2.256 ± 0.160	0.140	0.000	27.46	0.2225	0.2226	1.565	100.000 ± 0.000	1.492 ± 0.097	4.746 ± 1.826	
UPI	beta45	0.11	0.128	0.018 ± 0.011	0.163 ± 0.101	0.036 ± 0.004	0.191 ± 0.012	0.037 ± 0.004	0.191 ± 0.013	3.026 ± 0.358	1.740 ± 0.115	0.121	0.011	10.17	0.1724	0.1727	1.683	98.983 ± 0.584	1.259 ± 0.106	3.390 ± 1.054	
UPI	beta46	0.08	0.076	-0.004 ± 0.023	-0.053 ± 0.288	0.156 ± 0.020	0.395 ± 0.026	0.156 ± 0.020	0.395 ± 0.028	24.323 ± 3.165	4.932 ± 0.355	0.052	-0.028	-35.21	0.3005	0.3019	1.877	97.966 ± 0.822	2.908 ± 0.288	0.000 ± 0.000	

Table S157

Simulation 2 - Condition 13: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.4 (continued)

Method	Param.	True	Mean-based										Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE	
UPI	beta47	0.08	0.027	-0.053 ± 0.020	-0.657 ± 0.256	0.124 ± 0.016	0.352 ± 0.023	0.126 ± 0.016	0.355 ± 0.025	19.687 ± 2.525	4.437 ± 0.314	0.045	-0.035	-43.65	0.2848	0.2869	1.965	98.644 ± 0.673	2.709 ± 0.249	0.339 ± 0.338	
UPI	beta51	0.16	0.168	0.008 ± 0.007	0.052 ± 0.043	0.014 ± 0.002	0.118 ± 0.007	0.014 ± 0.002	0.118 ± 0.008	0.545 ± 0.061	0.738 ± 0.047	0.164	0.004	2.19	0.1246	0.1247	1.243	98.305 ± 0.752	0.575 ± 0.027	23.729 ± 2.477	
UPI	beta52	0.16	0.155	-0.005 ± 0.005	-0.033 ± 0.034	0.009 ± 0.001	0.094 ± 0.004	0.009 ± 0.001	0.094 ± 0.005	0.342 ± 0.028	0.585 ± 0.030	0.155	-0.005	-3.10	0.1044	0.1045	1.593	98.305 ± 0.752	0.584 ± 0.038	23.390 ± 2.465	
UPI	beta53	0.16	0.162	0.002 ± 0.006	0.010 ± 0.038	0.011 ± 0.001	0.105 ± 0.005	0.011 ± 0.001	0.105 ± 0.006	0.427 ± 0.039	0.654 ± 0.036	0.144	-0.016	-9.85	0.1120	0.1131	1.410	97.627 ± 0.886	0.579 ± 0.029	24.068 ± 2.489	
UPI	beta54	0.16	0.161	0.001 ± 0.006	0.004 ± 0.036	0.010 ± 0.001	0.098 ± 0.005	0.010 ± 0.001	0.098 ± 0.006	0.375 ± 0.036	0.612 ± 0.035	0.162	0.002	1.43	0.1115	0.1115	1.449	97.966 ± 0.822	0.557 ± 0.031	28.814 ± 2.637	
UPI	beta55	0.08	0.103	0.023 ± 0.011	0.284 ± 0.131	0.033 ± 0.004	0.180 ± 0.010	0.033 ± 0.004	0.182 ± 0.012	5.154 ± 0.599	2.270 ± 0.148	0.090	0.010	12.84	0.1632	0.1635	1.714	99.661 ± 0.338	1.213 ± 0.070	1.017 ± 0.584	
UPI	beta56	0.08	0.073	-0.007 ± 0.009	-0.082 ± 0.116	0.026 ± 0.003	0.160 ± 0.009	0.026 ± 0.003	0.160 ± 0.010	3.988 ± 0.437	1.997 ± 0.125	0.068	-0.012	-15.46	0.1415	0.1421	1.808	98.644 ± 0.673	1.133 ± 0.084	1.695 ± 0.752	
UPI	beta57	0.06	0.033	-0.027 ± 0.016	-0.445 ± 0.260	0.072 ± 0.010	0.268 ± 0.018	0.072 ± 0.010	0.269 ± 0.020	20.126 ± 2.678	4.486 ± 0.329	0.019	-0.041	-68.38	0.2282	0.2319	1.808	97.288 ± 0.946	1.902 ± 0.165	0.000 ± 0.000	
UPI	beta58	0.06	0.037	-0.023 ± 0.013	-0.384 ± 0.215	0.049 ± 0.006	0.221 ± 0.013	0.049 ± 0.006	0.222 ± 0.014	13.712 ± 1.542	3.703 ± 0.236	0.030	-0.030	-49.90	0.1741	0.1767	1.918	98.305 ± 0.752	1.665 ± 0.137	0.339 ± 0.338	

Note. MCSE = Monte Carlo Standard Error shown as \u000b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S158

Simulation 2 - Condition 14: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.4

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LSAM	beta41	0.20	0.203	0.003 $\pm$ 0.002	0.017 $\pm$ 0.010	0.003 $\pm$ 0.000	0.054 $\pm$ 0.001	0.003 $\pm$ 0.000	0.054 $\pm$ 0.002	0.073 $\pm$ 0.004	0.269 $\pm$ 0.009	0.203	0.003	1.61	0.0528	0.0529	1.025	95.206 $\pm$ 0.780	0.216 $\pm$ 0.002	96.671 $\pm$ 0.655
LSAM	beta42	0.20	0.202	0.002 $\pm$ 0.002	0.012 $\pm$ 0.010	0.003 $\pm$ 0.000	0.054 $\pm$ 0.001	0.003 $\pm$ 0.000	0.054 $\pm$ 0.002	0.072 $\pm$ 0.004	0.269 $\pm$ 0.009	0.200	0.000	-0.04	0.0519	0.0519	1.054	96.937 $\pm$ 0.629	0.222 $\pm$ 0.002	95.606 $\pm$ 0.748
LSAM	beta43	0.20	0.203	0.003 $\pm$ 0.002	0.016 $\pm$ 0.009	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	0.002 $\pm$ 0.000	0.050 $\pm$ 0.002	0.062 $\pm$ 0.003	0.249 $\pm$ 0.008	0.200	0.000	-0.04	0.0482	0.0482	1.034	96.272 $\pm$ 0.691	0.202 $\pm$ 0.001	98.269 $\pm$ 0.476
LSAM	beta44	0.11	0.111	0.001 $\pm$ 0.003	0.010 $\pm$ 0.026	0.006 $\pm$ 0.000	0.078 $\pm$ 0.002	0.006 $\pm$ 0.000	0.078 $\pm$ 0.003	0.500 $\pm$ 0.029	0.707 $\pm$ 0.025	0.112	0.002	1.55	0.0739	0.0740	1.038	97.337 $\pm$ 0.588	0.317 $\pm$ 0.006	32.357 $\pm$ 1.707
LSAM	beta45	0.11	0.114	0.004 $\pm$ 0.002	0.035 $\pm$ 0.020	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.313 $\pm$ 0.018	0.559 $\pm$ 0.019	0.114	0.004	3.69	0.0568	0.0570	1.061	96.005 $\pm$ 0.715	0.255 $\pm$ 0.002	43.808 $\pm$ 1.810
LSAM	beta46	0.08	0.084	0.004 $\pm$ 0.004	0.055 $\pm$ 0.048	0.011 $\pm$ 0.001	0.106 $\pm$ 0.003	0.011 $\pm$ 0.001	0.106 $\pm$ 0.004	1.749 $\pm$ 0.108	1.322 $\pm$ 0.047	0.082	0.002	2.78	0.0994	0.0994	1.035	97.071 $\pm$ 0.615	0.429 $\pm$ 0.007	6.525 $\pm$ 0.901
LSAM	beta47	0.08	0.091	0.011 $\pm$ 0.004	0.138 $\pm$ 0.047	0.011 $\pm$ 0.001	0.103 $\pm$ 0.003	0.011 $\pm$ 0.001	0.104 $\pm$ 0.004	1.674 $\pm$ 0.106	1.294 $\pm$ 0.047	0.076	-0.004	-4.81	0.0957	0.0958	1.091	98.136 $\pm$ 0.494	0.440 $\pm$ 0.011	7.057 $\pm$ 0.935
LSAM	beta51	0.16	0.156	-0.004 $\pm$ 0.002	-0.026 $\pm$ 0.015	0.004 $\pm$ 0.000	0.065 $\pm$ 0.002	0.004 $\pm$ 0.000	0.065 $\pm$ 0.002	0.164 $\pm$ 0.008	0.404 $\pm$ 0.013	0.152	-0.008	-5.21	0.0633	0.0639	1.001	95.739 $\pm$ 0.737	0.254 $\pm$ 0.001	68.575 $\pm$ 1.694
LSAM	beta52	0.16	0.160	-0.000 $\pm$ 0.002	-0.000 $\pm$ 0.011	0.003 $\pm$ 0.000	0.050 $\pm$ 0.001	0.003 $\pm$ 0.000	0.050 $\pm$ 0.002	0.099 $\pm$ 0.005	0.314 $\pm$ 0.010	0.160	0.000	-0.19	0.0505	0.0505	1.066	95.872 $\pm$ 0.726	0.210 $\pm$ 0.001	87.217 $\pm$ 1.218
LSAM	beta53	0.16	0.163	0.003 $\pm$ 0.002	0.019 $\pm$ 0.013	0.003 $\pm$ 0.000	0.058 $\pm$ 0.002	0.003 $\pm$ 0.000	0.058 $\pm$ 0.002	0.132 $\pm$ 0.007	0.363 $\pm$ 0.012	0.162	0.002	1.55	0.0563	0.0563	0.981	95.872 $\pm$ 0.726	0.224 $\pm$ 0.002	84.421 $\pm$ 1.323
LSAM	beta54	0.16	0.161	0.001 $\pm$ 0.002	0.006 $\pm$ 0.012	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.003 $\pm$ 0.000	0.053 $\pm$ 0.002	0.111 $\pm$ 0.006	0.333 $\pm$ 0.011	0.161	0.001	0.44	0.0526	0.0527	1.005	94.940 $\pm$ 0.800	0.210 $\pm$ 0.001	87.617 $\pm$ 1.202
LSAM	beta55	0.08	0.082	0.002 $\pm$ 0.003	0.021 $\pm$ 0.039	0.007 $\pm$ 0.000	0.085 $\pm$ 0.003	0.007 $\pm$ 0.000	0.085 $\pm$ 0.003	1.133 $\pm$ 0.067	1.064 $\pm$ 0.037	0.076	-0.004	-4.89	0.0803	0.0803	1.004	96.538 $\pm$ 0.667	0.335 $\pm$ 0.003	15.979 $\pm$ 1.337
LSAM	beta56	0.08	0.085	0.005 $\pm$ 0.003	0.063 $\pm$ 0.035	0.006 $\pm$ 0.000	0.076 $\pm$ 0.002	0.006 $\pm$ 0.000	0.076 $\pm$ 0.002	0.914 $\pm$ 0.049	0.956 $\pm$ 0.031	0.084	0.004	5.17	0.0746	0.0747	1.030	96.138 $\pm$ 0.703	0.308 $\pm$ 0.003	19.574 $\pm$ 1.448
LSAM	beta57	0.06	0.069	0.009 $\pm$ 0.004	0.157 $\pm$ 0.061	0.010 $\pm$ 0.001	0.100 $\pm$ 0.003	0.010 $\pm$ 0.001	0.100 $\pm$ 0.004	2.783 $\pm$ 0.179	1.668 $\pm$ 0.062	0.067	0.007	11.41	0.0963	0.0966	1.041	97.603 $\pm$ 0.558	0.407 $\pm$ 0.006	5.593 $\pm$ 0.838
LSAM	beta58	0.06	0.066	0.006 $\pm$ 0.003	0.094 $\pm$ 0.056	0.009 $\pm$ 0.001	0.093 $\pm$ 0.003	0.009 $\pm$ 0.001	0.093 $\pm$ 0.003	2.401 $\pm$ 0.156	1.549 $\pm$ 0.058	0.064	0.004	7.15	0.0833	0.0834	1.012	98.003 $\pm$ 0.511	0.368 $\pm$ 0.007	8.655 $\pm$ 1.026
QML	beta41	0.20	0.191	-0.009 $\pm$ 0.002	-0.047 $\pm$ 0.009	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.057 $\pm$ 0.003	0.238 $\pm$ 0.007	0.190	-0.010	-5.23	0.0469	0.0480	0.974	91.478 $\pm$ 1.019	0.178 $\pm$ 0.001	99.068 $\pm$ 0.351
QML	beta42	0.20	0.195	-0.005 $\pm$ 0.002	-0.023 $\pm$ 0.009	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.002	0.056 $\pm$ 0.003	0.237 $\pm$ 0.008	0.192	-0.008	-3.99	0.0449	0.0456	1.001	95.872 $\pm$ 0.726	0.185 $\pm$ 0.001	98.935 $\pm$ 0.375
QML	beta43	0.20	0.188	-0.012 $\pm$ 0.002	-0.062 $\pm$ 0.008	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.055 $\pm$ 0.003	0.234 $\pm$ 0.007	0.188	-0.012	-6.22	0.0444	0.0461	0.998	93.875 $\pm$ 0.875	0.177 $\pm$ 0.001	98.802 $\pm$ 0.397
QML	beta44	0.11	0.114	0.004 $\pm$ 0.002	0.036 $\pm$ 0.021	0.004 $\pm$ 0.000	0.064 $\pm$ 0.002	0.004 $\pm$ 0.000	0.064 $\pm$ 0.002	0.343 $\pm$ 0.018	0.586 $\pm$ 0.019	0.114	0.004	3.43	0.0594	0.0595	0.980	94.274 $\pm$ 0.848	0.247 $\pm$ 0.002	45.539 $\pm$ 1.817
QML	beta45	0.11	0.107	-0.003 $\pm$ 0.002	-0.024 $\pm$ 0.017	0.003 $\pm$ 0.000	0.050 $\pm$ 0.001	0.003 $\pm$ 0.000	0.050 $\pm$ 0.002	0.211 $\pm$ 0.011	0.459 $\pm$ 0.015	0.102	-0.008	-7.06	0.0510	0.0516	1.044	95.206 $\pm$ 0.780	0.207 $\pm$ 0.001	53.262 $\pm$ 1.821
QML	beta46	0.08	0.026	-0.054 $\pm$ 0.002	-0.675 $\pm$ 0.020	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.005 $\pm$ 0.000	0.070 $\pm$ 0.002	0.765 $\pm$ 0.031	0.875 $\pm$ 0.019	0.026	-0.054	-67.82	0.0459	0.0710	1.041	76.698 $\pm$ 1.543	0.182 $\pm$ 0.002	7.856 $\pm$ 0.982
QML	beta47	0.08	0.030	-0.050 $\pm$ 0.002	-0.628 $\pm$ 0.019	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.004 $\pm$ 0.000	0.065 $\pm$ 0.001	0.667 $\pm$ 0.028	0.816 $\pm$ 0.018	0.028	-0.052	-64.89	0.0416	0.0665	1.081	77.630 $\pm$ 1.521	0.177 $\pm$ 0.002	8.123 $\pm$ 0.997
QML	beta51	0.16	0.180	0.020 $\pm$ 0.002	0.123 $\pm$ 0.014	0.004 $\pm$ 0.000	0.062 $\pm$ 0.002	0.004 $\pm$ 0.000	0.065 $\pm$ 0.002	0.167 $\pm$ 0.009	0.408 $\pm$ 0.013	0.176	0.016	10.03	0.0631	0.0652	0.996	94.807 $\pm$ 0.810	0.243 $\pm$ 0.001	84.154 $\pm$ 1.333
QML	beta52	0.16	0.150	-0.010 $\pm$ 0.002	-0.060 $\pm$ 0.011	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.002	0.093 $\pm$ 0.005	0.304 $\pm$ 0.010	0.151	-0.009	-5.81	0.0453	0.0463	1.039	94.008 $\pm$ 0.866	0.195 $\pm$ 0.001	87.617 $\pm$ 1.202
QML	beta53	0.16	0.143	-0.017 $\pm$ 0.002	-0.105 $\pm$ 0.012	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.003 $\pm$ 0.000	0.054 $\pm$ 0.002	0.115 $\pm$ 0.006	0.340 $\pm$ 0.010	0.143	-0.017	-10.82	0.0499	0.0529	0.971	94.274 $\pm$ 0.848	0.197 $\pm$ 0.001	82.956 $\pm$ 1.372
QML	beta54	0.16	0.157	-0.003 $\pm$ 0.002	-0.017 $\pm$ 0.011	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.002	0.091 $\pm$ 0.005	0.302 $\pm$ 0.010	0.159	-0.001	-0.88	0.0507	0.0507	0.978	93.209 $\pm$ 0.918	0.185 $\pm$ 0.001	92.277 $\pm$ 0.974
QML	beta55	0.08	0.086	0.006 $\pm$ 0.003	0.073 $\pm$ 0.033	0.005 $\pm$ 0.000	0.073 $\pm$ 0.002	0.005 $\pm$ 0.000	0.073 $\pm$ 0.002	0.829 $\pm$ 0.047	0.911 $\pm$ 0.031	0.085	0.005	5.76	0.0705	0.0706	1.042	96.405 $\pm$ 0.679	0.297 $\pm$ 0.002	18.109 $\pm$ 1.405
QML	beta56	0.08	0.075	-0.005 $\pm$ 0.002	-0.056 $\pm$ 0.030	0.004 $\pm$ 0.000	0.065 $\pm$ 0.002	0.004 $\pm$ 0.000	0.065 $\pm$ 0.002	0.670 $\pm$ 0.036	0.819 $\pm$ 0.027	0.079	-0.001	-1.47	0.0658	0.0658	1.007	93.742 $\pm$ 0.884	0.258 $\pm$ 0.001	19.707 $\pm$ 1.452
QML	beta57	0.06	0.021	-0.039 $\pm$ 0.002	-0.643 $\pm$ 0.028	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	1.022 $\pm$ 0.051	1.011 $\pm$ 0.027	0.022	-0.038	-62.70	0.0467	0.0600	1.032	86.285 $\pm$ 1.255	0.189 $\pm$ 0.002	6.924 $\pm$ 0.926
QML	beta58	0.06	0.030	-0.030 $\pm$ 0.001	-0.499 $\pm$ 0.025	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	0.003 $\pm$ 0.0.												

Table S158

Simulation 2 - Condition 14: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.4 (continued)

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.093	0.013 ± 0.005	0.166 ± 0.061	0.018 ± 0.001	0.133 ± 0.005	0.018 ± 0.001	0.133 ± 0.006	2.783 ± 0.209	1.668 ± 0.070	0.081	0.001	0.82	0.1120	0.1120	1.611	98.802 ± 0.397	0.839 ± 0.035	1.065 ± 0.375
UPI	beta51	0.16	0.159	-0.001 ± 0.002	-0.009 ± 0.015	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	0.162 ± 0.009	0.402 ± 0.013	0.156	-0.004	-2.50	0.0629	0.0630	1.087	96.138 ± 0.703	0.274 ± 0.003	66.312 ± 1.725
UPI	beta52	0.16	0.160	-0.000 ± 0.002	-0.001 ± 0.011	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.002	0.098 ± 0.005	0.314 ± 0.010	0.161	0.001	0.66	0.0495	0.0496	1.189	96.671 ± 0.655	0.234 ± 0.003	81.625 ± 1.413
UPI	beta53	0.16	0.160	0.000 ± 0.002	0.001 ± 0.014	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	0.140 ± 0.009	0.375 ± 0.013	0.160	0.000	-0.30	0.0581	0.0581	1.089	96.804 ± 0.642	0.256 ± 0.004	74.967 ± 1.581
UPI	beta54	0.16	0.160	0.000 ± 0.002	0.001 ± 0.012	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.114 ± 0.006	0.338 ± 0.011	0.160	0.000	-0.24	0.0553	0.0553	1.131	96.272 ± 0.691	0.240 ± 0.005	80.160 ± 1.451
UPI	beta55	0.08	0.084	0.004 ± 0.003	0.046 ± 0.041	0.008 ± 0.000	0.089 ± 0.003	0.008 ± 0.000	0.089 ± 0.003	1.233 ± 0.077	1.110 ± 0.040	0.078	-0.002	-2.16	0.0829	0.0829	1.220	98.668 ± 0.418	0.425 ± 0.010	9.188 ± 1.054
UPI	beta56	0.08	0.084	0.004 ± 0.003	0.055 ± 0.036	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.078 ± 0.003	0.950 ± 0.053	0.974 ± 0.033	0.084	0.004	5.30	0.0803	0.0804	1.236	98.269 ± 0.476	0.377 ± 0.007	9.987 ± 1.094
UPI	beta57	0.06	0.068	0.008 ± 0.004	0.130 ± 0.073	0.014 ± 0.001	0.119 ± 0.004	0.014 ± 0.001	0.120 ± 0.005	3.975 ± 0.278	1.994 ± 0.079	0.061	0.001	1.79	0.1085	0.1085	1.420	97.736 ± 0.543	0.665 ± 0.024	1.598 ± 0.458
UPI	beta58	0.06	0.063	0.003 ± 0.004	0.052 ± 0.071	0.014 ± 0.001	0.118 ± 0.005	0.014 ± 0.001	0.117 ± 0.005	3.833 ± 0.294	1.958 ± 0.084	0.053	-0.007	-12.10	0.0971	0.0974	1.470	97.470 ± 0.573	0.677 ± 0.028	0.932 ± 0.351

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S159

Simulation 2 - Condition 15: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.6

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LSAM	beta41	0.20	0.201	0.001 $\pm$ 0.002	0.005 $\pm$ 0.010	0.003 $\pm$ 0.000	0.059 $\pm$ 0.001	0.003 $\pm$ 0.000	0.059 $\pm$ 0.002	0.087 $\pm$ 0.004	0.294 $\pm$ 0.008	0.201	0.001	0.63	0.0593	0.0594	1.020	95.886 $\pm$ 0.645	0.235 $\pm$ 0.001	92.827 $\pm$ 0.838
LSAM	beta42	0.20	0.200	-0.000 $\pm$ 0.002	-0.001 $\pm$ 0.010	0.003 $\pm$ 0.000	0.059 $\pm$ 0.001	0.003 $\pm$ 0.000	0.059 $\pm$ 0.002	0.087 $\pm$ 0.004	0.295 $\pm$ 0.009	0.198	-0.002	-1.03	0.0600	0.0600	1.037	95.886 $\pm$ 0.645	0.240 $\pm$ 0.001	92.827 $\pm$ 0.838
LSAM	beta43	0.20	0.203	0.003 $\pm$ 0.002	0.017 $\pm$ 0.009	0.003 $\pm$ 0.000	0.055 $\pm$ 0.001	0.003 $\pm$ 0.000	0.055 $\pm$ 0.002	0.076 $\pm$ 0.004	0.276 $\pm$ 0.008	0.201	0.001	0.74	0.0566	0.0567	1.032	95.675 $\pm$ 0.661	0.223 $\pm$ 0.001	96.624 $\pm$ 0.587
LSAM	beta44	0.11	0.111	0.001 $\pm$ 0.002	0.011 $\pm$ 0.021	0.005 $\pm$ 0.000	0.072 $\pm$ 0.002	0.005 $\pm$ 0.000	0.072 $\pm$ 0.002	0.430 $\pm$ 0.020	0.656 $\pm$ 0.019	0.110	0.000	-0.01	0.0734	0.0734	0.993	96.203 $\pm$ 0.621	0.281 $\pm$ 0.002	36.709 $\pm$ 1.561
LSAM	beta45	0.11	0.110	0.000 $\pm$ 0.002	0.002 $\pm$ 0.018	0.004 $\pm$ 0.000	0.062 $\pm$ 0.001	0.004 $\pm$ 0.000	0.062 $\pm$ 0.002	0.321 $\pm$ 0.015	0.567 $\pm$ 0.016	0.107	-0.003	-2.89	0.0614	0.0615	1.019	96.414 $\pm$ 0.604	0.249 $\pm$ 0.001	43.565 $\pm$ 1.610
LSAM	beta46	0.08	0.083	0.003 $\pm$ 0.003	0.032 $\pm$ 0.036	0.008 $\pm$ 0.000	0.089 $\pm$ 0.002	0.008 $\pm$ 0.000	0.089 $\pm$ 0.003	1.224 $\pm$ 0.062	1.106 $\pm$ 0.033	0.084	0.004	4.59	0.0865	0.0866	1.041	96.730 $\pm$ 0.578	0.361 $\pm$ 0.003	11.814 $\pm$ 1.048
LSAM	beta47	0.08	0.079	-0.001 $\pm$ 0.003	-0.015 $\pm$ 0.037	0.008 $\pm$ 0.000	0.091 $\pm$ 0.002	0.008 $\pm$ 0.000	0.091 $\pm$ 0.003	1.303 $\pm$ 0.065	1.141 $\pm$ 0.034	0.078	-0.002	-2.81	0.0884	0.0885	0.990	94.620 $\pm$ 0.733	0.354 $\pm$ 0.003	12.447 $\pm$ 1.072
LSAM	beta51	0.16	0.160	0.000 $\pm$ 0.002	0.001 $\pm$ 0.015	0.005 $\pm$ 0.000	0.073 $\pm$ 0.002	0.005 $\pm$ 0.000	0.073 $\pm$ 0.002	0.207 $\pm$ 0.009	0.455 $\pm$ 0.013	0.158	-0.002	-0.96	0.0714	0.0714	0.988	94.093 $\pm$ 0.766	0.282 $\pm$ 0.001	61.181 $\pm$ 1.583
LSAM	beta52	0.16	0.163	0.003 $\pm$ 0.002	0.020 $\pm$ 0.012	0.004 $\pm$ 0.000	0.060 $\pm$ 0.001	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	0.142 $\pm$ 0.007	0.377 $\pm$ 0.011	0.162	0.002	1.04	0.0567	0.0567	1.000	94.620 $\pm$ 0.733	0.237 $\pm$ 0.001	77.848 $\pm$ 1.349
LSAM	beta53	0.16	0.158	-0.002 $\pm$ 0.002	-0.013 $\pm$ 0.013	0.004 $\pm$ 0.000	0.064 $\pm$ 0.001	0.004 $\pm$ 0.000	0.064 $\pm$ 0.002	0.159 $\pm$ 0.007	0.399 $\pm$ 0.011	0.158	-0.002	-1.40	0.0661	0.0661	0.971	94.620 $\pm$ 0.733	0.243 $\pm$ 0.001	71.308 $\pm$ 1.469
LSAM	beta54	0.16	0.159	-0.001 $\pm$ 0.002	-0.008 $\pm$ 0.012	0.003 $\pm$ 0.000	0.057 $\pm$ 0.001	0.003 $\pm$ 0.000	0.057 $\pm$ 0.002	0.128 $\pm$ 0.006	0.358 $\pm$ 0.010	0.160	0.000	-0.17	0.0559	0.0559	1.044	95.781 $\pm$ 0.653	0.235 $\pm$ 0.001	76.688 $\pm$ 1.373
LSAM	beta55	0.08	0.079	-0.001 $\pm$ 0.003	-0.018 $\pm$ 0.035	0.007 $\pm$ 0.000	0.086 $\pm$ 0.002	0.007 $\pm$ 0.000	0.086 $\pm$ 0.003	1.168 $\pm$ 0.065	1.081 $\pm$ 0.035	0.081	0.001	1.25	0.0866	0.0866	0.958	94.831 $\pm$ 0.719	0.325 $\pm$ 0.002	18.143 $\pm$ 1.252
LSAM	beta56	0.08	0.080	0.000 $\pm$ 0.003	0.003 $\pm$ 0.032	0.006 $\pm$ 0.000	0.079 $\pm$ 0.002	0.006 $\pm$ 0.000	0.079 $\pm$ 0.002	0.970 $\pm$ 0.044	0.985 $\pm$ 0.028	0.081	0.001	1.54	0.0800	0.0801	0.978	94.198 $\pm$ 0.759	0.302 $\pm$ 0.002	19.726 $\pm$ 1.292
LSAM	beta57	0.06	0.064	0.004 $\pm$ 0.003	0.072 $\pm$ 0.049	0.008 $\pm$ 0.000	0.091 $\pm$ 0.002	0.008 $\pm$ 0.000	0.091 $\pm$ 0.003	2.293 $\pm$ 0.113	1.514 $\pm$ 0.045	0.061	0.001	0.93	0.0844	0.0844	1.007	96.941 $\pm$ 0.559	0.359 $\pm$ 0.003	8.861 $\pm$ 0.923
LSAM	beta58	0.06	0.055	-0.005 $\pm$ 0.003	-0.075 $\pm$ 0.043	0.006 $\pm$ 0.000	0.080 $\pm$ 0.002	0.006 $\pm$ 0.000	0.080 $\pm$ 0.002	1.795 $\pm$ 0.083	1.340 $\pm$ 0.038	0.054	-0.006	-9.60	0.0794	0.0796	1.028	96.624 $\pm$ 0.587	0.324 $\pm$ 0.002	8.755 $\pm$ 0.918
QML	beta41	0.20	0.195	-0.005 $\pm$ 0.002	-0.026 $\pm$ 0.009	0.003 $\pm$ 0.000	0.055 $\pm$ 0.001	0.003 $\pm$ 0.000	0.055 $\pm$ 0.002	0.077 $\pm$ 0.003	0.277 $\pm$ 0.008	0.195	-0.005	-2.38	0.0563	0.0565	0.998	95.042 $\pm$ 0.705	0.216 $\pm$ 0.001	95.148 $\pm$ 0.698
QML	beta42	0.20	0.195	-0.005 $\pm$ 0.002	-0.027 $\pm$ 0.009	0.003 $\pm$ 0.000	0.055 $\pm$ 0.001	0.003 $\pm$ 0.000	0.056 $\pm$ 0.002	0.078 $\pm$ 0.004	0.278 $\pm$ 0.008	0.192	-0.008	-4.13	0.0570	0.0576	1.018	95.886 $\pm$ 0.645	0.221 $\pm$ 0.001	94.726 $\pm$ 0.726
QML	beta43	0.20	0.194	-0.006 $\pm$ 0.002	-0.031 $\pm$ 0.009	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.003 $\pm$ 0.000	0.053 $\pm$ 0.002	0.070 $\pm$ 0.003	0.265 $\pm$ 0.008	0.192	-0.008	-3.90	0.0521	0.0526	1.031	95.148 $\pm$ 0.698	0.213 $\pm$ 0.001	96.097 $\pm$ 0.629
QML	beta44	0.11	0.115	0.005 $\pm$ 0.002	0.045 $\pm$ 0.020	0.005 $\pm$ 0.000	0.067 $\pm$ 0.002	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.377 $\pm$ 0.017	0.614 $\pm$ 0.017	0.113	0.003	3.16	0.0703	0.0704	0.995	95.781 $\pm$ 0.653	0.263 $\pm$ 0.002	41.772 $\pm$ 1.602
QML	beta45	0.11	0.109	-0.001 $\pm$ 0.002	-0.008 $\pm$ 0.017	0.003 $\pm$ 0.000	0.057 $\pm$ 0.001	0.003 $\pm$ 0.000	0.057 $\pm$ 0.002	0.271 $\pm$ 0.013	0.521 $\pm$ 0.015	0.109	-0.001	-1.33	0.0594	0.0594	1.032	96.519 $\pm$ 0.595	0.232 $\pm$ 0.001	46.624 $\pm$ 1.620
QML	beta46	0.08	0.041	-0.039 $\pm$ 0.002	-0.485 $\pm$ 0.022	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	0.677 $\pm$ 0.031	0.823 $\pm$ 0.021	0.044	-0.036	-44.97	0.0524	0.0636	1.051	88.502 $\pm$ 1.036	0.219 $\pm$ 0.003	10.021 $\pm$ 0.975
QML	beta47	0.08	0.042	-0.038 $\pm$ 0.002	-0.476 $\pm$ 0.022	0.003 $\pm$ 0.000	0.054 $\pm$ 0.001	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	0.690 $\pm$ 0.030	0.831 $\pm$ 0.020	0.043	-0.037	-46.00	0.0506	0.0626	1.009	88.080 $\pm$ 1.052	0.215 $\pm$ 0.002	12.447 $\pm$ 1.072
QML	beta51	0.16	0.175	0.015 $\pm$ 0.002	0.092 $\pm$ 0.014	0.005 $\pm$ 0.000	0.071 $\pm$ 0.002	0.005 $\pm$ 0.000	0.073 $\pm$ 0.002	0.206 $\pm$ 0.009	0.453 $\pm$ 0.013	0.174	0.014	8.78	0.0697	0.0711	1.028	95.042 $\pm$ 0.705	0.286 $\pm$ 0.001	67.722 $\pm$ 1.519
QML	beta52	0.16	0.152	-0.008 $\pm$ 0.002	-0.052 $\pm$ 0.012	0.004 $\pm$ 0.000	0.060 $\pm$ 0.001	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.144 $\pm$ 0.007	0.380 $\pm$ 0.011	0.150	-0.010	-6.22	0.0570	0.0579	0.994	93.882 $\pm$ 0.778	0.235 $\pm$ 0.001	72.363 $\pm$ 1.452
QML	beta53	0.16	0.139	-0.021 $\pm$ 0.002	-0.132 $\pm$ 0.012	0.004 $\pm$ 0.000	0.062 $\pm$ 0.001	0.004 $\pm$ 0.000	0.065 $\pm$ 0.002	0.165 $\pm$ 0.007	0.406 $\pm$ 0.011	0.139	-0.021	-13.01	0.0642	0.0675	0.983	93.143 $\pm$ 0.821	0.237 $\pm$ 0.001	63.291 $\pm$ 1.565
QML	beta54	0.16	0.156	-0.004 $\pm$ 0.002	-0.022 $\pm$ 0.011	0.003 $\pm$ 0.000	0.054 $\pm$ 0.001	0.003 $\pm$ 0.000	0.054 $\pm$ 0.002	0.116 $\pm$ 0.005	0.340 $\pm$ 0.010	0.157	-0.003	-1.69	0.0532	0.0533	1.046	96.308 $\pm$ 0.612	0.223 $\pm$ 0.001	79.536 $\pm$ 1.310
QML	beta55	0.08	0.082	0.002 $\pm$ 0.003	0.020 $\pm$ 0.033	0.007 $\pm$ 0.000	0.081 $\pm$ 0.002	0.007 $\pm$ 0.000	0.081 $\pm$ 0.002	1.030 $\pm$ 0.053	1.015 $\pm$ 0.031	0.086	0.006	7.76	0.0796	0.0798	0.990	95.464 $\pm$ 0.676	0.315 $\pm$ 0.002	17.405 $\pm$ 1.231
QML	beta56	0.08	0.077	-0.003 $\pm$ 0.002	-0.032 $\pm$ 0.030	0.005 $\pm$ 0.000	0.074 $\pm$ 0.002	0.005 $\pm$ 0.000	0.074 $\pm$ 0.002	0.851 $\pm$ 0.039	0.923 $\pm$ 0.026	0.079	-0.001	-1.79	0.0737	0.0737	0.993	94.726 $\pm$ 0.726	0.287 $\pm$ 0.001	18.354 $\pm$ 1.257
QML	beta57	0.06	0.033	-0.027 $\pm$ 0.002	-0.454 $\pm$ 0.030	0.003 $\pm$ 0.000	0.056 $\pm$ 0.001	0.004 $\pm$ 0.000	0.062 $\pm$ 0.002	1.078 $\pm$ 0.048	1.038 $\pm$ 0.027	0.034	-0.026	-43.89	0.0552	0.0612	1.028	92.405 $\pm$ 0.860	0.226 $\pm$ 0.002	8.439 $\pm$ 0.903
QML	beta58	0.06	0.034	-0.026 $\pm$ 0.002	-0.440 $\pm$ 0.026	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.003 $\pm$ 0.000</												

Table S159

Simulation 2 - Condition 15: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.6 (continued)

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.083	0.003 ± 0.004	0.032 ± 0.046	0.013 ± 0.001	0.112 ± 0.003	0.013 ± 0.001	0.112 ± 0.004	1.975 ± 0.114	1.405 ± 0.047	0.077	-0.003	-3.89	0.1001	0.1002	1.156	97.574 ± 0.500	0.509 ± 0.011	5.696 ± 0.753
UPI	beta51	0.16	0.163	0.003 ± 0.002	0.017 ± 0.014	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.199 ± 0.009	0.446 ± 0.013	0.161	0.001	0.78	0.0700	0.0700	1.029	95.042 ± 0.705	0.288 ± 0.002	61.392 ± 1.581
UPI	beta52	0.16	0.163	0.003 ± 0.002	0.018 ± 0.012	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.143 ± 0.007	0.378 ± 0.011	0.162	0.002	1.06	0.0574	0.0574	1.014	94.726 ± 0.726	0.240 ± 0.001	76.899 ± 1.369
UPI	beta53	0.16	0.156	-0.004 ± 0.002	-0.026 ± 0.013	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.160 ± 0.007	0.400 ± 0.011	0.156	-0.004	-2.71	0.0654	0.0655	1.008	95.042 ± 0.705	0.252 ± 0.001	68.143 ± 1.513
UPI	beta54	0.16	0.158	-0.002 ± 0.002	-0.012 ± 0.012	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.128 ± 0.006	0.358 ± 0.010	0.158	-0.002	-1.02	0.0567	0.0567	1.064	95.675 ± 0.661	0.239 ± 0.001	75.000 ± 1.406
UPI	beta55	0.08	0.081	0.001 ± 0.003	0.011 ± 0.035	0.007 ± 0.000	0.086 ± 0.002	0.007 ± 0.000	0.086 ± 0.003	1.162 ± 0.063	1.078 ± 0.034	0.084	0.004	4.57	0.0828	0.0828	1.033	95.886 ± 0.645	0.349 ± 0.003	12.447 ± 1.072
UPI	beta56	0.08	0.083	0.003 ± 0.003	0.032 ± 0.034	0.007 ± 0.000	0.084 ± 0.002	0.007 ± 0.000	0.084 ± 0.003	1.097 ± 0.056	1.047 ± 0.032	0.082	0.002	2.68	0.0779	0.0780	1.008	95.675 ± 0.661	0.331 ± 0.003	15.190 ± 1.166
UPI	beta57	0.06	0.060	-0.000 ± 0.003	-0.003 ± 0.054	0.010 ± 0.001	0.099 ± 0.003	0.010 ± 0.001	0.099 ± 0.003	2.745 ± 0.149	1.657 ± 0.052	0.057	-0.003	-4.57	0.0930	0.0931	1.141	97.679 ± 0.489	0.445 ± 0.008	4.219 ± 0.653
UPI	beta58	0.06	0.054	-0.006 ± 0.003	-0.102 ± 0.050	0.009 ± 0.000	0.093 ± 0.003	0.009 ± 0.000	0.093 ± 0.003	2.425 ± 0.129	1.557 ± 0.049	0.049	-0.011	-18.07	0.0851	0.0858	1.182	96.835 ± 0.569	0.432 ± 0.008	3.903 ± 0.629

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S160

Simulation 2 - Condition 16: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.6

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LSAM	beta41	0.20	0.199	-0.001 $\pm$ 0.001	-0.006 $\pm$ 0.006	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.032 $\pm$ 0.001	0.178 $\pm$ 0.005	0.199	-0.001	-0.29	0.0343	0.0343	1.020	95.596 $\pm$ 0.649	0.143 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta42	0.20	0.202	0.002 $\pm$ 0.001	0.011 $\pm$ 0.006	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.033 $\pm$ 0.001	0.181 $\pm$ 0.005	0.202	0.002	1.02	0.0346	0.0347	1.029	95.696 $\pm$ 0.642	0.146 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta43	0.20	0.200	-0.000 $\pm$ 0.001	-0.001 $\pm$ 0.006	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.031 $\pm$ 0.001	0.177 $\pm$ 0.005	0.200	0.000	-0.15	0.0355	0.0355	0.988	94.795 $\pm$ 0.703	0.137 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta44	0.11	0.112	0.002 $\pm$ 0.001	0.016 $\pm$ 0.012	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.151 $\pm$ 0.007	0.388 $\pm$ 0.011	0.113	0.003	2.68	0.0394	0.0395	1.000	94.194 $\pm$ 0.740	0.167 $\pm$ 0.001	76.376 $\pm$ 1.344
LSAM	beta45	0.11	0.110	0.000 $\pm$ 0.001	0.004 $\pm$ 0.011	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.122 $\pm$ 0.005	0.350 $\pm$ 0.009	0.110	0.000	0.19	0.0386	0.0386	0.991	94.595 $\pm$ 0.715	0.149 $\pm$ 0.000	84.184 $\pm$ 1.154
LSAM	beta46	0.08	0.081	0.001 $\pm$ 0.002	0.015 $\pm$ 0.021	0.003 $\pm$ 0.000	0.054 $\pm$ 0.001	0.003 $\pm$ 0.000	0.054 $\pm$ 0.002	0.459 $\pm$ 0.022	0.678 $\pm$ 0.019	0.081	0.001	1.45	0.0535	0.0535	0.990	95.495 $\pm$ 0.656	0.210 $\pm$ 0.001	32.232 $\pm$ 1.479
LSAM	beta47	0.08	0.080	0.000 $\pm$ 0.002	0.000 $\pm$ 0.021	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.433 $\pm$ 0.020	0.658 $\pm$ 0.018	0.080	0.000	0.47	0.0510	0.0510	1.016	95.295 $\pm$ 0.670	0.210 $\pm$ 0.001	31.331 $\pm$ 1.468
LSAM	beta51	0.16	0.159	-0.001 $\pm$ 0.001	-0.009 $\pm$ 0.009	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.085 $\pm$ 0.004	0.291 $\pm$ 0.008	0.159	-0.001	-0.53	0.0465	0.0465	0.963	93.193 $\pm$ 0.797	0.176 $\pm$ 0.000	93.594 $\pm$ 0.775
LSAM	beta52	0.16	0.161	0.001 $\pm$ 0.001	0.003 $\pm$ 0.007	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.052 $\pm$ 0.002	0.229 $\pm$ 0.006	0.161	0.001	0.36	0.0369	0.0369	1.000	94.595 $\pm$ 0.715	0.144 $\pm$ 0.000	99.199 $\pm$ 0.282
LSAM	beta53	0.16	0.161	0.001 $\pm$ 0.001	0.009 $\pm$ 0.008	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.057 $\pm$ 0.003	0.240 $\pm$ 0.007	0.160	0.000	0.31	0.0372	0.0372	0.999	95.195 $\pm$ 0.677	0.150 $\pm$ 0.000	98.999 $\pm$ 0.315
LSAM	beta54	0.16	0.161	0.001 $\pm$ 0.001	0.005 $\pm$ 0.007	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.053 $\pm$ 0.002	0.229 $\pm$ 0.006	0.160	0.000	0.28	0.0363	0.0363	0.996	95.596 $\pm$ 0.649	0.143 $\pm$ 0.000	99.299 $\pm$ 0.264
LSAM	beta55	0.08	0.083	0.003 $\pm$ 0.002	0.042 $\pm$ 0.019	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.354 $\pm$ 0.016	0.595 $\pm$ 0.017	0.082	0.002	2.30	0.0475	0.0475	1.047	96.096 $\pm$ 0.613	0.195 $\pm$ 0.001	39.039 $\pm$ 1.543
LSAM	beta56	0.08	0.076	-0.004 $\pm$ 0.002	-0.056 $\pm$ 0.019	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.370 $\pm$ 0.017	0.609 $\pm$ 0.017	0.075	-0.005	-6.60	0.0448	0.0451	0.971	94.595 $\pm$ 0.715	0.185 $\pm$ 0.001	35.135 $\pm$ 1.510
LSAM	beta57	0.06	0.061	0.001 $\pm$ 0.002	0.023 $\pm$ 0.029	0.003 $\pm$ 0.000	0.054 $\pm$ 0.001	0.003 $\pm$ 0.000	0.054 $\pm$ 0.002	0.819 $\pm$ 0.039	0.905 $\pm$ 0.026	0.062	0.002	3.89	0.0515	0.0516	0.990	94.895 $\pm$ 0.696	0.211 $\pm$ 0.001	19.820 $\pm$ 1.261
LSAM	beta58	0.06	0.061	0.001 $\pm$ 0.002	0.014 $\pm$ 0.026	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.664 $\pm$ 0.029	0.815 $\pm$ 0.022	0.059	-0.001	-2.26	0.0487	0.0487	1.021	96.597 $\pm$ 0.574	0.196 $\pm$ 0.001	22.523 $\pm$ 1.322
QML	beta41	0.20	0.193	-0.007 $\pm$ 0.001	-0.036 $\pm$ 0.005	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.031 $\pm$ 0.001	0.175 $\pm$ 0.005	0.195	-0.005	-2.71	0.0325	0.0329	1.002	94.394 $\pm$ 0.728	0.135 $\pm$ 0.000	100.000 $\pm$ 0.000
QML	beta42	0.20	0.198	-0.002 $\pm$ 0.001	-0.010 $\pm$ 0.006	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.031 $\pm$ 0.001	0.175 $\pm$ 0.005	0.199	-0.001	-0.62	0.0353	0.0354	1.010	95.195 $\pm$ 0.677	0.139 $\pm$ 0.000	100.000 $\pm$ 0.000
QML	beta43	0.20	0.190	-0.010 $\pm$ 0.001	-0.048 $\pm$ 0.005	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.031 $\pm$ 0.001	0.177 $\pm$ 0.005	0.190	-0.010	-4.82	0.0337	0.0351	0.994	93.493 $\pm$ 0.780	0.133 $\pm$ 0.000	99.900 $\pm$ 0.100
QML	beta44	0.11	0.115	0.005 $\pm$ 0.001	0.045 $\pm$ 0.012	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.143 $\pm$ 0.007	0.378 $\pm$ 0.011	0.116	0.006	5.07	0.0379	0.0383	1.005	94.595 $\pm$ 0.715	0.163 $\pm$ 0.001	79.479 $\pm$ 1.278
QML	beta45	0.11	0.110	0.000 $\pm$ 0.001	0.004 $\pm$ 0.011	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.111 $\pm$ 0.005	0.333 $\pm$ 0.009	0.110	0.000	-0.04	0.0368	0.0368	0.997	95.195 $\pm$ 0.677	0.143 $\pm$ 0.000	87.387 $\pm$ 1.050
QML	beta46	0.08	0.042	-0.038 $\pm$ 0.001	-0.476 $\pm$ 0.014	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.413 $\pm$ 0.015	0.643 $\pm$ 0.013	0.041	-0.039	-48.13	0.0337	0.0512	0.986	78.078 $\pm$ 1.309	0.134 $\pm$ 0.001	23.824 $\pm$ 1.348
QML	beta47	0.08	0.044	-0.036 $\pm$ 0.001	-0.447 $\pm$ 0.013	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.373 $\pm$ 0.014	0.611 $\pm$ 0.013	0.044	-0.036	-44.55	0.0315	0.0476	1.014	81.381 $\pm$ 1.232	0.133 $\pm$ 0.001	24.925 $\pm$ 1.369
QML	beta51	0.16	0.172	0.012 $\pm$ 0.001	0.077 $\pm$ 0.009	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.089 $\pm$ 0.004	0.299 $\pm$ 0.008	0.172	0.012	7.47	0.0455	0.0470	0.979	93.393 $\pm$ 0.786	0.177 $\pm$ 0.000	96.597 $\pm$ 0.574
QML	beta52	0.16	0.148	-0.012 $\pm$ 0.001	-0.075 $\pm$ 0.007	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.057 $\pm$ 0.003	0.239 $\pm$ 0.006	0.147	-0.013	-7.84	0.0354	0.0376	1.013	94.294 $\pm$ 0.734	0.144 $\pm$ 0.000	98.398 $\pm$ 0.397
QML	beta53	0.16	0.144	-0.016 $\pm$ 0.001	-0.103 $\pm$ 0.007	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	0.065 $\pm$ 0.003	0.256 $\pm$ 0.006	0.142	-0.018	-11.37	0.0371	0.0413	1.002	92.593 $\pm$ 0.829	0.147 $\pm$ 0.000	97.397 $\pm$ 0.504
QML	beta54	0.16	0.159	-0.001 $\pm$ 0.001	-0.005 $\pm$ 0.007	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.049 $\pm$ 0.002	0.222 $\pm$ 0.006	0.159	-0.001	-0.63	0.0350	0.0351	1.000	95.395 $\pm$ 0.663	0.139 $\pm$ 0.000	99.600 $\pm$ 0.200
QML	beta55	0.08	0.086	0.006 $\pm$ 0.001	0.070 $\pm$ 0.018	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.338 $\pm$ 0.015	0.582 $\pm$ 0.016	0.087	0.007	8.71	0.0464	0.0469	1.072	95.896 $\pm$ 0.628	0.194 $\pm$ 0.001	40.240 $\pm$ 1.552
QML	beta56	0.08	0.073	-0.007 $\pm$ 0.001	-0.090 $\pm$ 0.018	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.339 $\pm$ 0.016	0.582 $\pm$ 0.016	0.071	-0.009	-10.75	0.0443	0.0451	0.991	94.494 $\pm$ 0.722	0.179 $\pm$ 0.001	33.734 $\pm$ 1.496
QML	beta57	0.06	0.033	-0.027 $\pm$ 0.001	-0.455 $\pm$ 0.019	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.551 $\pm$ 0.024	0.742 $\pm$ 0.018	0.034	-0.026	-43.29	0.0339	0.0427	1.007	86.987 $\pm$ 1.064	0.139 $\pm$ 0.001	14.915 $\pm$ 1.127
QML	beta58	0.06	0.037	-0.023 $\pm$ 0.001	-0.376 $\pm$ 0.016	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.407 $\pm$ 0.017	0.638 $\pm$ 0.015	0.037	-0.023	-38.56	0.0313	0.0389	1.025	88.889 $\pm$ 0.994	0.124 $\pm$ 0.001	22.222 $\pm$ 1.315
UPI	beta41	0.20	0.199	-0.001 $\pm$ 0.001	-0.006 $\pm$ 0.006	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.032 $\pm$ 0.002	0.179 $\pm$ 0.005	0.200	0.000	0.03	0.0355	0.0355	0.990	94.895 $\pm$ 0.696	0.139 $\pm$ 0.000	100.000 $\pm$ 0.000
UPI	beta42	0.20	0.202	0.002 $\pm$ 0.001	0.011 $\pm$ 0.006	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.033 $\pm$ 0.001	0.182 $\pm$ 0.005	0.202	0.002	1.12	0.0350	0.0351	1.009	94.995 $\pm$ 0.690	0.144 $\pm$ 0.000	100.000 $\pm$ 0.000
UPI	beta43	0.20	0.200	-0.000 $\pm$ 0.001	-0.000 $\pm$ 0.006	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.032 $\pm$ 0.001	0.177 $\pm$ 0.005	0.200	0.000	0.16	0.0355	0.0355	0.988	95.195 $\pm$ 0.677	0.138 $\pm$ 0.000	100.000 $\pm$ 0.00

Table S160
Simulation 2 - Condition 16: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.6 (continued)

Method	Param.	True	Mean-based										Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE	
UPI	beta47	0.08	0.082	0.002 ± 0.002	0.020 ± 0.022	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.495 ± 0.023	0.703 ± 0.020	0.080	0.000	-0.54	0.0527	0.0527	1.052	95.896 ± 0.628	0.232 ± 0.002	25.425 ± 1.378	
UPI	beta51	0.16	0.159	-0.001 ± 0.001	-0.006 ± 0.009	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.084 ± 0.004	0.289 ± 0.008	0.158	-0.002	-1.13	0.0469	0.0470	0.966	94.094 ± 0.746	0.175 ± 0.000	93.493 ± 0.780	
UPI	beta52	0.16	0.161	0.001 ± 0.001	0.003 ± 0.007	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.052 ± 0.002	0.228 ± 0.006	0.161	0.001	0.77	0.0372	0.0373	0.995	94.294 ± 0.734	0.142 ± 0.000	99.299 ± 0.264	
UPI	beta53	0.16	0.161	0.001 ± 0.001	0.006 ± 0.008	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.057 ± 0.003	0.239 ± 0.007	0.160	0.000	-0.19	0.0373	0.0373	1.008	95.796 ± 0.635	0.151 ± 0.000	98.899 ± 0.330	
UPI	beta54	0.16	0.161	0.001 ± 0.001	0.005 ± 0.007	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.052 ± 0.002	0.229 ± 0.006	0.160	0.000	-0.12	0.0369	0.0369	0.996	95.796 ± 0.635	0.143 ± 0.000	99.399 ± 0.244	
UPI	beta55	0.08	0.084	0.004 ± 0.002	0.047 ± 0.019	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.362 ± 0.017	0.602 ± 0.017	0.084	0.004	4.52	0.0483	0.0485	1.064	96.096 ± 0.613	0.200 ± 0.001	37.437 ± 1.531	
UPI	beta56	0.08	0.076	-0.004 ± 0.002	-0.047 ± 0.019	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.377 ± 0.018	0.614 ± 0.017	0.075	-0.005	-6.14	0.0464	0.0466	0.993	94.795 ± 0.703	0.191 ± 0.001	33.734 ± 1.496	
UPI	beta57	0.06	0.061	0.001 ± 0.002	0.018 ± 0.030	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.881 ± 0.043	0.939 ± 0.028	0.062	0.002	3.90	0.0547	0.0548	1.029	95.896 ± 0.628	0.227 ± 0.002	15.516 ± 1.145	
UPI	beta58	0.06	0.060	0.000 ± 0.002	0.004 ± 0.027	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.752 ± 0.035	0.867 ± 0.024	0.057	-0.003	-4.21	0.0512	0.0513	1.067	96.897 ± 0.549	0.218 ± 0.002	16.416 ± 1.172	

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S161

Simulation 2 - Condition 17: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, $N = 400$, Reliability = 0.8

Method	Param.	True	Mean-based										Median-based					Inference		
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LSAM	beta41	0.20	0.197	-0.003 $\pm$ 0.001	-0.013 $\pm$ 0.007	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.056 $\pm$ 0.003	0.237 $\pm$ 0.007	0.197	-0.003	-1.50	0.0480	0.0481	1.002	95.395 $\pm$ 0.663	0.186 $\pm$ 0.000	98.599 $\pm$ 0.372
LSAM	beta42	0.20	0.202	0.002 $\pm$ 0.002	0.011 $\pm$ 0.008	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	0.062 $\pm$ 0.003	0.249 $\pm$ 0.007	0.203	0.003	1.71	0.0494	0.0495	0.967	94.394 $\pm$ 0.728	0.189 $\pm$ 0.000	97.998 $\pm$ 0.443
LSAM	beta43	0.20	0.201	0.001 $\pm$ 0.001	0.007 $\pm$ 0.007	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.053 $\pm$ 0.002	0.230 $\pm$ 0.006	0.200	0.000	-0.14	0.0473	0.0473	0.983	94.795 $\pm$ 0.703	0.177 $\pm$ 0.000	99.199 $\pm$ 0.282
LSAM	beta44	0.11	0.111	0.001 $\pm$ 0.002	0.008 $\pm$ 0.015	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.229 $\pm$ 0.010	0.479 $\pm$ 0.013	0.110	0.000	-0.06	0.0524	0.0524	0.963	94.094 $\pm$ 0.746	0.199 $\pm$ 0.001	59.159 $\pm$ 1.555
LSAM	beta45	0.11	0.111	0.001 $\pm$ 0.002	0.007 $\pm$ 0.014	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.190 $\pm$ 0.009	0.435 $\pm$ 0.012	0.111	0.001	0.57	0.0462	0.0462	0.970	93.994 $\pm$ 0.752	0.182 $\pm$ 0.001	67.668 $\pm$ 1.480
LSAM	beta46	0.08	0.078	-0.002 $\pm$ 0.002	-0.024 $\pm$ 0.024	0.004 $\pm$ 0.000	0.062 $\pm$ 0.001	0.004 $\pm$ 0.000	0.062 $\pm$ 0.002	0.591 $\pm$ 0.025	0.769 $\pm$ 0.020	0.078	-0.002	-2.60	0.0622	0.0622	0.952	94.595 $\pm$ 0.715	0.230 $\pm$ 0.001	27.728 $\pm$ 1.416
LSAM	beta47	0.08	0.082	0.002 $\pm$ 0.002	0.020 $\pm$ 0.023	0.003 $\pm$ 0.000	0.058 $\pm$ 0.001	0.003 $\pm$ 0.000	0.058 $\pm$ 0.002	0.530 $\pm$ 0.026	0.728 $\pm$ 0.021	0.081	0.001	1.36	0.0582	0.0583	1.001	94.895 $\pm$ 0.696	0.229 $\pm$ 0.001	28.228 $\pm$ 1.424
LSAM	beta51	0.16	0.161	0.001 $\pm$ 0.002	0.004 $\pm$ 0.012	0.004 $\pm$ 0.000	0.060 $\pm$ 0.001	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	0.140 $\pm$ 0.006	0.374 $\pm$ 0.010	0.160	0.000	0.04	0.0597	0.0597	0.957	93.994 $\pm$ 0.752	0.225 $\pm$ 0.001	78.879 $\pm$ 1.291
LSAM	beta52	0.16	0.160	-0.000 $\pm$ 0.002	-0.001 $\pm$ 0.010	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.093 $\pm$ 0.004	0.305 $\pm$ 0.008	0.160	0.000	-0.24	0.0480	0.0480	0.975	93.994 $\pm$ 0.752	0.187 $\pm$ 0.000	91.191 $\pm$ 0.897
LSAM	beta53	0.16	0.160	-0.000 $\pm$ 0.002	-0.002 $\pm$ 0.010	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	0.096 $\pm$ 0.004	0.310 $\pm$ 0.008	0.159	-0.001	-0.60	0.0509	0.0509	0.992	94.394 $\pm$ 0.728	0.193 $\pm$ 0.000	90.290 $\pm$ 0.937
LSAM	beta54	0.16	0.161	0.001 $\pm$ 0.002	0.004 $\pm$ 0.010	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.093 $\pm$ 0.004	0.306 $\pm$ 0.008	0.161	0.001	0.89	0.0478	0.0478	0.974	94.194 $\pm$ 0.740	0.187 $\pm$ 0.000	92.192 $\pm$ 0.849
LSAM	beta55	0.08	0.084	0.004 $\pm$ 0.002	0.049 $\pm$ 0.024	0.004 $\pm$ 0.000	0.061 $\pm$ 0.001	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.589 $\pm$ 0.026	0.767 $\pm$ 0.021	0.086	0.006	7.86	0.0612	0.0615	0.969	93.794 $\pm$ 0.763	0.233 $\pm$ 0.001	29.730 $\pm$ 1.446
LSAM	beta56	0.08	0.079	-0.001 $\pm$ 0.002	-0.016 $\pm$ 0.023	0.004 $\pm$ 0.000	0.059 $\pm$ 0.001	0.004 $\pm$ 0.000	0.059 $\pm$ 0.002	0.550 $\pm$ 0.023	0.741 $\pm$ 0.019	0.081	0.001	0.90	0.0602	0.0602	0.962	94.294 $\pm$ 0.734	0.224 $\pm$ 0.001	29.930 $\pm$ 1.449
LSAM	beta57	0.06	0.057	-0.003 $\pm$ 0.002	-0.043 $\pm$ 0.031	0.004 $\pm$ 0.000	0.059 $\pm$ 0.001	0.004 $\pm$ 0.000	0.059 $\pm$ 0.002	0.977 $\pm$ 0.044	0.988 $\pm$ 0.027	0.057	-0.003	-4.64	0.0586	0.0587	0.999	95.596 $\pm$ 0.649	0.232 $\pm$ 0.001	15.916 $\pm$ 1.157
LSAM	beta58	0.06	0.056	-0.004 $\pm$ 0.002	-0.067 $\pm$ 0.030	0.003 $\pm$ 0.000	0.056 $\pm$ 0.001	0.003 $\pm$ 0.000	0.057 $\pm$ 0.002	0.889 $\pm$ 0.041	0.943 $\pm$ 0.026	0.055	-0.005	-8.86	0.0566	0.0568	0.983	94.094 $\pm$ 0.746	0.218 $\pm$ 0.001	18.519 $\pm$ 1.229
QML	beta41	0.20	0.195	-0.005 $\pm$ 0.001	-0.026 $\pm$ 0.007	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.054 $\pm$ 0.002	0.231 $\pm$ 0.006	0.194	-0.006	-2.82	0.0465	0.0468	1.001	95.095 $\pm$ 0.683	0.181 $\pm$ 0.000	98.999 $\pm$ 0.315
QML	beta42	0.20	0.201	0.001 $\pm$ 0.002	0.003 $\pm$ 0.008	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.059 $\pm$ 0.003	0.242 $\pm$ 0.007	0.201	0.001	0.69	0.0481	0.0482	0.966	94.394 $\pm$ 0.728	0.184 $\pm$ 0.000	98.398 $\pm$ 0.397
QML	beta43	0.20	0.197	-0.003 $\pm$ 0.001	-0.017 $\pm$ 0.007	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.051 $\pm$ 0.002	0.226 $\pm$ 0.006	0.195	-0.005	-2.61	0.0461	0.0464	0.988	95.195 $\pm$ 0.677	0.175 $\pm$ 0.000	99.299 $\pm$ 0.264
QML	beta44	0.11	0.113	0.003 $\pm$ 0.002	0.023 $\pm$ 0.015	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.226 $\pm$ 0.010	0.475 $\pm$ 0.013	0.111	0.001	0.97	0.0521	0.0521	0.964	93.594 $\pm$ 0.775	0.197 $\pm$ 0.001	60.060 $\pm$ 1.550
QML	beta45	0.11	0.112	0.002 $\pm$ 0.001	0.014 $\pm$ 0.014	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.185 $\pm$ 0.009	0.430 $\pm$ 0.012	0.111	0.001	1.30	0.0463	0.0463	0.972	94.194 $\pm$ 0.740	0.180 $\pm$ 0.001	69.770 $\pm$ 1.453
QML	beta46	0.08	0.058	-0.022 $\pm$ 0.002	-0.275 $\pm$ 0.020	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	0.003 $\pm$ 0.000	0.054 $\pm$ 0.001	0.460 $\pm$ 0.019	0.678 $\pm$ 0.017	0.058	-0.022	-27.73	0.0508	0.0555	0.949	90.390 $\pm$ 0.932	0.185 $\pm$ 0.001	25.025 $\pm$ 1.370
QML	beta47	0.08	0.062	-0.018 $\pm$ 0.001	-0.222 $\pm$ 0.018	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.382 $\pm$ 0.017	0.618 $\pm$ 0.016	0.061	-0.019	-23.30	0.0453	0.0490	1.008	92.993 $\pm$ 0.808	0.183 $\pm$ 0.001	26.126 $\pm$ 1.390
QML	beta51	0.16	0.167	0.007 $\pm$ 0.002	0.047 $\pm$ 0.012	0.004 $\pm$ 0.000	0.060 $\pm$ 0.001	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.145 $\pm$ 0.006	0.380 $\pm$ 0.010	0.166	0.006	4.01	0.0600	0.0603	0.969	94.094 $\pm$ 0.746	0.229 $\pm$ 0.001	82.282 $\pm$ 1.208
QML	beta52	0.16	0.144	-0.016 $\pm$ 0.002	-0.098 $\pm$ 0.010	0.003 $\pm$ 0.000	0.050 $\pm$ 0.001	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.109 $\pm$ 0.005	0.330 $\pm$ 0.009	0.144	-0.016	-9.83	0.0497	0.0522	0.972	92.693 $\pm$ 0.823	0.192 $\pm$ 0.001	82.783 $\pm$ 1.194
QML	beta53	0.16	0.141	-0.019 $\pm$ 0.002	-0.116 $\pm$ 0.010	0.003 $\pm$ 0.000	0.050 $\pm$ 0.001	0.003 $\pm$ 0.000	0.054 $\pm$ 0.001	0.112 $\pm$ 0.005	0.335 $\pm$ 0.009	0.141	-0.019	-12.18	0.0514	0.0549	0.995	92.993 $\pm$ 0.808	0.197 $\pm$ 0.001	80.480 $\pm$ 1.254
QML	beta54	0.16	0.160	-0.000 $\pm$ 0.002	-0.003 $\pm$ 0.010	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.091 $\pm$ 0.004	0.302 $\pm$ 0.008	0.161	0.001	0.63	0.0473	0.0474	0.974	94.194 $\pm$ 0.740	0.184 $\pm$ 0.001	92.693 $\pm$ 0.823
QML	beta55	0.08	0.086	0.006 $\pm$ 0.002	0.073 $\pm$ 0.024	0.004 $\pm$ 0.000	0.061 $\pm$ 0.001	0.004 $\pm$ 0.000	0.062 $\pm$ 0.002	0.596 $\pm$ 0.026	0.772 $\pm$ 0.021	0.089	0.009	11.40	0.0626	0.0633	0.983	93.493 $\pm$ 0.780	0.237 $\pm$ 0.002	30.731 $\pm$ 1.460
QML	beta56	0.08	0.078	-0.002 $\pm$ 0.002	-0.020 $\pm$ 0.023	0.003 $\pm$ 0.000	0.059 $\pm$ 0.001	0.003 $\pm$ 0.000	0.059 $\pm$ 0.002	0.540 $\pm$ 0.023	0.735 $\pm$ 0.019	0.079	-0.001	-1.16	0.0600	0.0600	0.966	94.394 $\pm$ 0.728	0.223 $\pm$ 0.001	30.731 $\pm$ 1.460
QML	beta57	0.06	0.042	-0.018 $\pm$ 0.002	-0.304 $\pm$ 0.025	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.726 $\pm$ 0.033	0.852 $\pm$ 0.023	0.041	-0.019	-31.10	0.0482	0.0517	1.034	92.793 $\pm$ 0.818	0.194 $\pm$ 0.004	14.314 $\pm$ 1.108
QML	beta58	0.06	0.044	-0.016 $\pm$ 0.001	-0.263 $\pm$ 0.024	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.633 $\pm$ 0.028	0.796 $\pm$ 0.021	0.042	-0.018	-30.01	0.0452	0.0486	0.994	92.893 $\pm$ 0.813	0.176 $\pm$ 0.001	17.518 $\pm$ 1.203
UPI	beta41	0.20	0.198	-0.002 $\pm$ 0.001	-0.011 $\pm$ 0.007	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.056 $\pm$ 0.003	0.237 $\pm$ 0.007	0.198	-0.002	-0.98	0.0483	0.0484	0.953	93.694 $\pm$ 0.769	0.177 $\pm$ 0.000	98.799 $\pm$ 0.345
UPI	beta42	0.20	0.202	0.002 $\pm$ 0.002	0.010 $\pm$ 0.008	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	0.062 $\pm$ 0.003	0.249 $\pm$ 0.007	0.203	0.003	1.70	0.0504	0.0505	0.938	93.794 $\pm$ 0.763	0.183 $\pm$ 0.000	98.298 $\pm$ 0.409
UPI	beta43	0.20	0.201	0.001 $\pm$ 0.001	0.007 $\pm$ 0.007	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.053 $\pm$ 0.002	0.230 $\pm$ 0.006	0.200	0.000	-0.15	0.0475	0.0475	0.978	94.695 $\pm$ 0.709	0.177 $\pm$ 0.000	99.399 $\pm$ 0.244

Table S161
Simulation 2 - Condition 17: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.8 (continued)

Method	Param.	True	Mean-based										Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE	
UPI	beta47	0.08	0.082	0.002 ± 0.002	0.021 ± 0.024	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	0.577 ± 0.029	0.760 ± 0.023	0.079	-0.001	-1.17	0.0584	0.0584	1.010	95.596 ± 0.649	0.241 ± 0.001	25.425 ± 1.378	
UPI	beta51	0.16	0.161	0.001 ± 0.002	0.006 ± 0.012	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.138 ± 0.006	0.372 ± 0.010	0.160	0.000	-0.24	0.0596	0.0596	0.962	94.294 ± 0.734	0.225 ± 0.001	79.580 ± 1.275	
UPI	beta52	0.16	0.160	0.000 ± 0.002	0.000 ± 0.010	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.094 ± 0.004	0.306 ± 0.008	0.160	0.000	-0.19	0.0481	0.0481	0.957	93.694 ± 0.769	0.184 ± 0.000	91.992 ± 0.859	
UPI	beta53	0.16	0.159	-0.001 ± 0.002	-0.004 ± 0.010	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.096 ± 0.004	0.311 ± 0.008	0.160	0.000	-0.15	0.0514	0.0515	0.992	94.394 ± 0.728	0.193 ± 0.000	90.090 ± 0.945	
UPI	beta54	0.16	0.161	0.001 ± 0.002	0.005 ± 0.010	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.094 ± 0.004	0.306 ± 0.008	0.162	0.002	0.95	0.0488	0.0488	0.962	93.794 ± 0.763	0.185 ± 0.000	92.092 ± 0.854	
UPI	beta55	0.08	0.083	0.003 ± 0.002	0.043 ± 0.024	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.585 ± 0.026	0.765 ± 0.021	0.085	0.005	6.60	0.0593	0.0595	0.983	94.094 ± 0.746	0.236 ± 0.001	28.028 ± 1.421	
UPI	beta56	0.08	0.078	-0.002 ± 0.002	-0.021 ± 0.024	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.560 ± 0.024	0.749 ± 0.020	0.079	-0.001	-0.91	0.0609	0.0609	0.969	94.795 ± 0.703	0.227 ± 0.001	30.230 ± 1.453	
UPI	beta57	0.06	0.056	-0.004 ± 0.002	-0.064 ± 0.032	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	1.004 ± 0.046	1.002 ± 0.028	0.055	-0.005	-8.09	0.0580	0.0582	1.021	96.096 ± 0.613	0.240 ± 0.001	14.815 ± 1.124	
UPI	beta58	0.06	0.056	-0.004 ± 0.002	-0.059 ± 0.031	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.948 ± 0.045	0.973 ± 0.028	0.055	-0.005	-7.82	0.0590	0.0592	0.998	95.796 ± 0.635	0.228 ± 0.001	16.817 ± 1.183	

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S162

Simulation 2 - Condition 18: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.8

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LSAM	beta41	0.20	0.198	-0.002 $\pm$ 0.001	-0.009 $\pm$ 0.005	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.022 $\pm$ 0.001	0.147 $\pm$ 0.004	0.198	-0.002	-1.07	0.0296	0.0297	1.012	95.400 $\pm$ 0.662	0.117 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta42	0.20	0.202	0.002 $\pm$ 0.001	0.010 $\pm$ 0.005	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.023 $\pm$ 0.001	0.151 $\pm$ 0.004	0.203	0.003	1.40	0.0287	0.0289	0.998	94.700 $\pm$ 0.708	0.118 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta43	0.20	0.201	0.001 $\pm$ 0.001	0.004 $\pm$ 0.005	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.021 $\pm$ 0.001	0.144 $\pm$ 0.004	0.201	0.001	0.25	0.0279	0.0279	0.990	95.200 $\pm$ 0.676	0.112 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta44	0.11	0.112	0.002 $\pm$ 0.001	0.017 $\pm$ 0.009	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	0.082 $\pm$ 0.004	0.287 $\pm$ 0.008	0.114	0.004	3.54	0.0303	0.0305	1.012	94.400 $\pm$ 0.727	0.125 $\pm$ 0.000	94.000 $\pm$ 0.751
LSAM	beta45	0.11	0.111	0.001 $\pm$ 0.001	0.005 $\pm$ 0.009	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.078 $\pm$ 0.003	0.279 $\pm$ 0.008	0.110	0.000	0.16	0.0302	0.0302	0.950	92.700 $\pm$ 0.823	0.114 $\pm$ 0.000	95.500 $\pm$ 0.656
LSAM	beta46	0.08	0.082	0.002 $\pm$ 0.001	0.025 $\pm$ 0.015	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.220 $\pm$ 0.010	0.469 $\pm$ 0.013	0.082	0.002	2.39	0.0382	0.0383	0.975	94.800 $\pm$ 0.702	0.143 $\pm$ 0.000	59.800 $\pm$ 1.550
LSAM	beta47	0.08	0.079	-0.001 $\pm$ 0.001	-0.014 $\pm$ 0.015	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.216 $\pm$ 0.009	0.465 $\pm$ 0.012	0.078	-0.002	-2.94	0.0370	0.0371	0.980	94.300 $\pm$ 0.733	0.143 $\pm$ 0.000	57.600 $\pm$ 1.563
LSAM	beta51	0.16	0.159	-0.001 $\pm$ 0.001	-0.008 $\pm$ 0.007	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.050 $\pm$ 0.002	0.223 $\pm$ 0.006	0.157	-0.003	-1.66	0.0341	0.0342	1.021	95.700 $\pm$ 0.641	0.143 $\pm$ 0.000	99.300 $\pm$ 0.264
LSAM	beta52	0.16	0.159	-0.001 $\pm$ 0.001	-0.004 $\pm$ 0.006	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.036 $\pm$ 0.002	0.188 $\pm$ 0.005	0.159	-0.001	-0.78	0.0307	0.0307	0.995	95.000 $\pm$ 0.689	0.118 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta53	0.16	0.161	0.001 $\pm$ 0.001	0.009 $\pm$ 0.006	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.036 $\pm$ 0.002	0.190 $\pm$ 0.005	0.162	0.002	1.00	0.0301	0.0302	1.019	95.700 $\pm$ 0.641	0.122 $\pm$ 0.000	99.900 $\pm$ 0.100
LSAM	beta54	0.16	0.160	0.000 $\pm$ 0.001	0.001 $\pm$ 0.006	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.028 $\pm$ 0.001	0.032 $\pm$ 0.001	0.178 $\pm$ 0.005	0.160	0.000	-0.25	0.0287	0.0287	1.049	96.300 $\pm$ 0.597	0.117 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta55	0.08	0.080	-0.000 $\pm$ 0.001	-0.000 $\pm$ 0.015	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.221 $\pm$ 0.010	0.471 $\pm$ 0.013	0.080	0.000	-0.25	0.0377	0.0377	0.996	95.100 $\pm$ 0.683	0.147 $\pm$ 0.000	55.900 $\pm$ 1.570
LSAM	beta56	0.08	0.081	0.001 $\pm$ 0.001	0.011 $\pm$ 0.015	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.223 $\pm$ 0.009	0.472 $\pm$ 0.013	0.081	0.001	1.38	0.0370	0.0370	0.953	92.300 $\pm$ 0.843	0.141 $\pm$ 0.000	60.800 $\pm$ 1.544
LSAM	beta57	0.06	0.061	0.001 $\pm$ 0.001	0.023 $\pm$ 0.019	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.358 $\pm$ 0.017	0.598 $\pm$ 0.017	0.062	0.002	2.82	0.0349	0.0349	1.038	95.500 $\pm$ 0.656	0.146 $\pm$ 0.000	35.200 $\pm$ 1.510
LSAM	beta58	0.06	0.060	0.000 $\pm$ 0.001	0.003 $\pm$ 0.018	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.330 $\pm$ 0.015	0.575 $\pm$ 0.016	0.059	-0.001	-1.00	0.0341	0.0341	1.013	95.300 $\pm$ 0.669	0.137 $\pm$ 0.000	38.400 $\pm$ 1.538
QML	beta41	0.20	0.196	-0.004 $\pm$ 0.001	-0.022 $\pm$ 0.005	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.021 $\pm$ 0.001	0.145 $\pm$ 0.004	0.195	-0.005	-2.54	0.0285	0.0289	1.036	95.200 $\pm$ 0.676	0.117 $\pm$ 0.002	99.700 $\pm$ 0.173
QML	beta42	0.20	0.200	0.000 $\pm$ 0.001	0.001 $\pm$ 0.005	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.022 $\pm$ 0.001	0.148 $\pm$ 0.004	0.201	0.001	0.62	0.0285	0.0285	1.029	94.500 $\pm$ 0.721	0.120 $\pm$ 0.003	99.700 $\pm$ 0.173
QML	beta43	0.20	0.196	-0.004 $\pm$ 0.001	-0.020 $\pm$ 0.005	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.021 $\pm$ 0.001	0.144 $\pm$ 0.004	0.197	-0.003	-1.67	0.0279	0.0281	0.989	94.700 $\pm$ 0.708	0.111 $\pm$ 0.001	99.900 $\pm$ 0.100
QML	beta44	0.11	0.114	0.004 $\pm$ 0.001	0.033 $\pm$ 0.009	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	0.082 $\pm$ 0.004	0.287 $\pm$ 0.008	0.115	0.005	4.43	0.0306	0.0309	1.035	94.500 $\pm$ 0.721	0.127 $\pm$ 0.003	95.000 $\pm$ 0.689
QML	beta45	0.11	0.111	0.001 $\pm$ 0.001	0.010 $\pm$ 0.009	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.075 $\pm$ 0.003	0.274 $\pm$ 0.007	0.111	0.001	0.98	0.0290	0.0291	0.961	93.200 $\pm$ 0.796	0.114 $\pm$ 0.000	96.100 $\pm$ 0.612
QML	beta46	0.08	0.062	-0.018 $\pm$ 0.001	-0.231 $\pm$ 0.012	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.196 $\pm$ 0.008	0.443 $\pm$ 0.011	0.061	-0.019	-23.24	0.0307	0.0359	0.978	88.300 $\pm$ 1.016	0.116 $\pm$ 0.000	54.400 $\pm$ 1.575
QML	beta47	0.08	0.061	-0.019 $\pm$ 0.001	-0.243 $\pm$ 0.012	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.198 $\pm$ 0.008	0.445 $\pm$ 0.011	0.060	-0.020	-24.95	0.0297	0.0358	1.048	88.800 $\pm$ 0.997	0.123 $\pm$ 0.008	53.800 $\pm$ 1.577
QML	beta51	0.16	0.165	0.005 $\pm$ 0.001	0.034 $\pm$ 0.007	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.051 $\pm$ 0.002	0.226 $\pm$ 0.006	0.164	0.004	2.21	0.0338	0.0339	1.034	95.500 $\pm$ 0.656	0.145 $\pm$ 0.001	99.500 $\pm$ 0.223
QML	beta52	0.16	0.145	-0.015 $\pm$ 0.001	-0.097 $\pm$ 0.006	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.046 $\pm$ 0.002	0.214 $\pm$ 0.005	0.143	-0.017	-10.55	0.0318	0.0360	1.011	92.700 $\pm$ 0.823	0.121 $\pm$ 0.001	99.400 $\pm$ 0.244
QML	beta53	0.16	0.143	-0.017 $\pm$ 0.001	-0.109 $\pm$ 0.006	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.050 $\pm$ 0.002	0.223 $\pm$ 0.006	0.144	-0.016	-10.19	0.0317	0.0357	1.014	90.900 $\pm$ 0.909	0.124 $\pm$ 0.001	99.400 $\pm$ 0.244
QML	beta54	0.16	0.159	-0.001 $\pm$ 0.001	-0.004 $\pm$ 0.006	0.001 $\pm$ 0.000	0.028 $\pm$ 0.001	0.001 $\pm$ 0.000	0.028 $\pm$ 0.001	0.031 $\pm$ 0.001	0.176 $\pm$ 0.005	0.159	-0.001	-0.81	0.0281	0.0282	1.059	96.500 $\pm$ 0.581	0.117 $\pm$ 0.001	99.800 $\pm$ 0.141
QML	beta55	0.08	0.081	0.001 $\pm$ 0.001	0.017 $\pm$ 0.015	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.225 $\pm$ 0.010	0.475 $\pm$ 0.013	0.081	0.001	1.50	0.0380	0.0380	0.998	95.100 $\pm$ 0.683	0.149 $\pm$ 0.000	57.300 $\pm$ 1.564
QML	beta56	0.08	0.081	0.001 $\pm$ 0.001	0.007 $\pm$ 0.015	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.220 $\pm$ 0.009	0.469 $\pm$ 0.012	0.081	0.001	1.18	0.0371	0.0371	0.956	92.600 $\pm$ 0.828	0.141 $\pm$ 0.000	60.600 $\pm$ 1.545
QML	beta57	0.06	0.046	-0.014 $\pm$ 0.001	-0.235 $\pm$ 0.015	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	0.291 $\pm$ 0.013	0.539 $\pm$ 0.014	0.046	-0.014	-23.24	0.0284	0.0316	1.046	94.000 $\pm$ 0.751	0.119 $\pm$ 0.000	31.200 $\pm$ 1.465
QML	beta58	0.06	0.048	-0.012 $\pm$ 0.001	-0.202 $\pm$ 0.015	0.001 $\pm$ 0.000	0.028 $\pm$ 0.001	0.0.												

Table S162
Simulation 2 - Condition 18: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.8 (continued)

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.079	-0.001 ± 0.001	-0.011 ± 0.015	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.223 ± 0.010	0.472 ± 0.013	0.077	-0.003	-3.17	0.0372	0.0373	0.984	95.000 ± 0.689	0.146 ± 0.000	56.400 ± 1.568
UPI	beta51	0.16	0.159	-0.001 ± 0.001	-0.007 ± 0.007	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.049 ± 0.002	0.222 ± 0.006	0.157	-0.003	-1.61	0.0336	0.0337	1.021	95.700 ± 0.641	0.142 ± 0.000	99.500 ± 0.223
UPI	beta52	0.16	0.159	-0.001 ± 0.001	-0.004 ± 0.006	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.035 ± 0.002	0.188 ± 0.005	0.159	-0.001	-0.61	0.0304	0.0304	0.980	94.500 ± 0.721	0.116 ± 0.000	100.000 ± 0.000
UPI	beta53	0.16	0.161	0.001 ± 0.001	0.009 ± 0.006	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.036 ± 0.002	0.191 ± 0.005	0.162	0.002	0.98	0.0304	0.0304	1.017	95.500 ± 0.656	0.122 ± 0.000	99.900 ± 0.100
UPI	beta54	0.16	0.160	0.000 ± 0.001	0.001 ± 0.006	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.032 ± 0.001	0.178 ± 0.005	0.160	0.000	-0.22	0.0284	0.0284	1.041	96.100 ± 0.612	0.116 ± 0.000	100.000 ± 0.000
UPI	beta55	0.08	0.080	-0.000 ± 0.001	-0.000 ± 0.015	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.223 ± 0.010	0.472 ± 0.013	0.080	0.000	-0.13	0.0376	0.0376	1.000	94.600 ± 0.715	0.148 ± 0.000	55.800 ± 1.570
UPI	beta56	0.08	0.081	0.001 ± 0.001	0.012 ± 0.015	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.223 ± 0.009	0.472 ± 0.012	0.081	0.001	1.29	0.0377	0.0377	0.960	93.200 ± 0.796	0.142 ± 0.000	60.100 ± 1.549
UPI	beta57	0.06	0.061	0.001 ± 0.001	0.022 ± 0.019	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.361 ± 0.017	0.601 ± 0.017	0.061	0.001	1.92	0.0340	0.0340	1.048	95.900 ± 0.627	0.148 ± 0.000	34.800 ± 1.506
UPI	beta58	0.06	0.060	0.000 ± 0.001	0.003 ± 0.018	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.342 ± 0.015	0.585 ± 0.016	0.059	-0.001	-1.33	0.0350	0.0350	1.018	95.300 ± 0.669	0.140 ± 0.000	36.800 ± 1.525

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S163

Simulation 2 - Condition 19: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, $N = 400$, Reliability = 0.4

Method	Param.	True	Mean-based										Median-based				Inference			
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LSAM	beta41	0.20	0.202	0.002 $\pm$ 0.010	0.012 $\pm$ 0.051	0.011 $\pm$ 0.002	0.107 $\pm$ 0.008	0.011 $\pm$ 0.002	0.106 $\pm$ 0.009	0.283 $\pm$ 0.040	0.532 $\pm$ 0.047	0.198	-0.002	-1.06	0.1112	0.1112	1.068	97.273 $\pm$ 1.553	0.448 $\pm$ 0.012	41.818 $\pm$ 4.703
LSAM	beta42	0.20	0.220	0.020 $\pm$ 0.011	0.100 $\pm$ 0.054	0.013 $\pm$ 0.002	0.114 $\pm$ 0.007	0.013 $\pm$ 0.002	0.115 $\pm$ 0.009	0.332 $\pm$ 0.042	0.576 $\pm$ 0.046	0.220	0.020	10.21	0.1067	0.1086	1.117	97.273 $\pm$ 1.553	0.499 $\pm$ 0.023	45.455 $\pm$ 4.748
LSAM	beta43	0.20	0.188	-0.012 $\pm$ 0.008	-0.059 $\pm$ 0.038	0.006 $\pm$ 0.001	0.079 $\pm$ 0.005	0.006 $\pm$ 0.001	0.080 $\pm$ 0.006	0.160 $\pm$ 0.018	0.400 $\pm$ 0.029	0.198	-0.002	-1.18	0.0881	0.0882	1.102	98.182 $\pm$ 1.274	0.343 $\pm$ 0.010	61.818 $\pm$ 4.632
LSAM	beta44	0.11	0.177	0.067 $\pm$ 0.013	0.613 $\pm$ 0.121	0.020 $\pm$ 0.003	0.140 $\pm$ 0.012	0.024 $\pm$ 0.004	0.155 $\pm$ 0.015	1.977 $\pm$ 0.342	1.406 $\pm$ 0.136	0.169	0.059	53.19	0.1319	0.1443	1.150	94.545 $\pm$ 2.165	0.630 $\pm$ 0.038	26.364 $\pm$ 4.201
LSAM	beta45	0.11	0.127	0.017 $\pm$ 0.010	0.157 $\pm$ 0.094	0.012 $\pm$ 0.002	0.109 $\pm$ 0.009	0.012 $\pm$ 0.002	0.110 $\pm$ 0.011	0.996 $\pm$ 0.176	0.998 $\pm$ 0.102	0.119	0.009	8.03	0.1107	0.1110	1.057	98.182 $\pm$ 1.274	0.451 $\pm$ 0.028	20.909 $\pm$ 3.877
LSAM	beta46	0.08	0.030	-0.050 $\pm$ 0.009	-0.619 $\pm$ 0.110	0.009 $\pm$ 0.001	0.092 $\pm$ 0.007	0.011 $\pm$ 0.001	0.104 $\pm$ 0.007	1.704 $\pm$ 0.192	1.305 $\pm$ 0.089	0.024	-0.056	-70.61	0.0823	0.0998	1.000	86.364 $\pm$ 3.272	0.362 $\pm$ 0.016	3.636 $\pm$ 1.785
LSAM	beta47	0.08	0.016	-0.064 $\pm$ 0.008	-0.804 $\pm$ 0.097	0.007 $\pm$ 0.001	0.081 $\pm$ 0.007	0.011 $\pm$ 0.001	0.104 $\pm$ 0.008	1.674 $\pm$ 0.231	1.294 $\pm$ 0.098	0.013	-0.067	-83.85	0.0748	0.1005	1.170	89.091 $\pm$ 2.972	0.374 $\pm$ 0.020	3.636 $\pm$ 1.785
LSAM	beta51	0.16	0.195	0.035 $\pm$ 0.010	0.220 $\pm$ 0.064	0.011 $\pm$ 0.002	0.107 $\pm$ 0.010	0.013 $\pm$ 0.002	0.112 $\pm$ 0.012	0.491 $\pm$ 0.092	0.700 $\pm$ 0.074	0.181	0.021	13.05	0.1053	0.1074	0.949	93.636 $\pm$ 2.327	0.398 $\pm$ 0.007	50.000 $\pm$ 4.767
LSAM	beta52	0.16	0.142	-0.018 $\pm$ 0.009	-0.111 $\pm$ 0.059	0.010 $\pm$ 0.001	0.100 $\pm$ 0.007	0.010 $\pm$ 0.001	0.101 $\pm$ 0.008	0.396 $\pm$ 0.050	0.629 $\pm$ 0.050	0.144	-0.016	-10.27	0.0963	0.0977	0.966	92.727 $\pm$ 2.476	0.377 $\pm$ 0.015	34.545 $\pm$ 4.534
LSAM	beta53	0.16	0.155	-0.005 $\pm$ 0.009	-0.030 $\pm$ 0.054	0.008 $\pm$ 0.001	0.090 $\pm$ 0.008	0.008 $\pm$ 0.001	0.090 $\pm$ 0.009	0.313 $\pm$ 0.052	0.560 $\pm$ 0.055	0.157	-0.003	-1.67	0.0944	0.0944	1.012	96.364 $\pm$ 1.785	0.356 $\pm$ 0.007	40.000 $\pm$ 4.671
LSAM	beta54	0.16	0.140	-0.020 $\pm$ 0.008	-0.123 $\pm$ 0.051	0.007 $\pm$ 0.001	0.086 $\pm$ 0.006	0.008 $\pm$ 0.001	0.088 $\pm$ 0.008	0.301 $\pm$ 0.046	0.549 $\pm$ 0.049	0.142	-0.018	-11.14	0.0872	0.0890	1.109	95.455 $\pm$ 1.986	0.374 $\pm$ 0.015	35.455 $\pm$ 4.561
LSAM	beta55	0.08	0.100	0.020 $\pm$ 0.014	0.247 $\pm$ 0.176	0.022 $\pm$ 0.003	0.147 $\pm$ 0.011	0.022 $\pm$ 0.003	0.148 $\pm$ 0.014	3.424 $\pm$ 0.529	1.851 $\pm$ 0.170	0.106	0.026	31.94	0.1314	0.1338	0.921	96.364 $\pm$ 1.785	0.532 $\pm$ 0.024	10.000 $\pm$ 2.860
LSAM	beta56	0.08	0.076	-0.004 $\pm$ 0.011	-0.051 $\pm$ 0.135	0.013 $\pm$ 0.002	0.114 $\pm$ 0.008	0.013 $\pm$ 0.002	0.113 $\pm$ 0.009	2.002 $\pm$ 0.260	1.415 $\pm$ 0.116	0.072	-0.008	-10.03	0.1118	0.1121	1.034	94.545 $\pm$ 2.165	0.461 $\pm$ 0.019	9.091 $\pm$ 2.741
LSAM	beta57	0.06	0.026	-0.034 $\pm$ 0.007	-0.572 $\pm$ 0.119	0.006 $\pm$ 0.001	0.075 $\pm$ 0.006	0.007 $\pm$ 0.001	0.082 $\pm$ 0.006	1.875 $\pm$ 0.244	1.369 $\pm$ 0.106	0.026	-0.034	-56.90	0.0827	0.0894	1.011	95.455 $\pm$ 1.986	0.297 $\pm$ 0.012	5.455 $\pm$ 2.165
LSAM	beta58	0.06	0.029	-0.031 $\pm$ 0.007	-0.511 $\pm$ 0.113	0.005 $\pm$ 0.001	0.071 $\pm$ 0.007	0.006 $\pm$ 0.001	0.077 $\pm$ 0.008	1.654 $\pm$ 0.287	1.286 $\pm$ 0.126	0.022	-0.038	-63.70	0.0596	0.0708	0.961	89.091 $\pm$ 2.972	0.268 $\pm$ 0.014	6.364 $\pm$ 2.327
QML	beta41	0.20	0.185	-0.015 $\pm$ 0.008	-0.075 $\pm$ 0.038	0.006 $\pm$ 0.001	0.079 $\pm$ 0.007	0.006 $\pm$ 0.001	0.080 $\pm$ 0.008	0.161 $\pm$ 0.026	0.401 $\pm$ 0.038	0.181	-0.019	-9.75	0.0793	0.0817	0.940	92.727 $\pm$ 2.476	0.292 $\pm$ 0.005	67.273 $\pm$ 4.474
QML	beta42	0.20	0.190	-0.010 $\pm$ 0.008	-0.050 $\pm$ 0.038	0.006 $\pm$ 0.001	0.080 $\pm$ 0.005	0.006 $\pm$ 0.001	0.080 $\pm$ 0.006	0.160 $\pm$ 0.020	0.400 $\pm$ 0.032	0.188	-0.012	-5.85	0.0803	0.0812	0.951	93.636 $\pm$ 2.327	0.297 $\pm$ 0.005	69.091 $\pm$ 4.406
QML	beta43	0.20	0.175	-0.025 $\pm$ 0.006	-0.123 $\pm$ 0.031	0.004 $\pm$ 0.001	0.065 $\pm$ 0.004	0.005 $\pm$ 0.001	0.069 $\pm$ 0.005	0.120 $\pm$ 0.013	0.346 $\pm$ 0.024	0.177	-0.023	-11.38	0.0813	0.0844	1.100	96.364 $\pm$ 1.785	0.281 $\pm$ 0.004	67.273 $\pm$ 4.474
QML	beta44	0.11	0.137	0.027 $\pm$ 0.010	0.242 $\pm$ 0.093	0.011 $\pm$ 0.002	0.107 $\pm$ 0.008	0.012 $\pm$ 0.002	0.110 $\pm$ 0.010	0.996 $\pm$ 0.145	0.998 $\pm$ 0.087	0.140	0.030	26.94	0.0922	0.0968	1.006	93.636 $\pm$ 2.327	0.422 $\pm$ 0.017	28.182 $\pm$ 4.289
QML	beta45	0.11	0.114	0.004 $\pm$ 0.008	0.033 $\pm$ 0.070	0.007 $\pm$ 0.001	0.081 $\pm$ 0.005	0.007 $\pm$ 0.001	0.081 $\pm$ 0.006	0.542 $\pm$ 0.064	0.736 $\pm$ 0.057	0.122	0.012	11.19	0.0939	0.0947	1.021	95.455 $\pm$ 1.986	0.325 $\pm$ 0.007	28.182 $\pm$ 4.289
QML	beta46	0.08	0.052	-0.028 $\pm$ 0.007	-0.352 $\pm$ 0.085	0.005 $\pm$ 0.001	0.071 $\pm$ 0.005	0.006 $\pm$ 0.001	0.076 $\pm$ 0.005	0.905 $\pm$ 0.093	0.951 $\pm$ 0.063	0.041	-0.039	-48.74	0.0770	0.0863	0.969	90.000 $\pm$ 2.860	0.270 $\pm$ 0.012	13.636 $\pm$ 3.272
QML	beta47	0.08	0.037	-0.043 $\pm$ 0.006	-0.542 $\pm$ 0.069	0.003 $\pm$ 0.001	0.058 $\pm$ 0.005	0.005 $\pm$ 0.001	0.072 $\pm$ 0.005	0.816 $\pm$ 0.095	0.903 $\pm$ 0.060	0.039	-0.041	-50.70	0.0512	0.0653	1.106	88.182 $\pm$ 3.078	0.252 $\pm$ 0.009	5.455 $\pm$ 2.165
QML	beta51	0.16	0.201	0.041 $\pm$ 0.011	0.255 $\pm$ 0.066	0.012 $\pm$ 0.002	0.110 $\pm$ 0.009	0.014 $\pm$ 0.002	0.117 $\pm$ 0.011	0.535 $\pm$ 0.085	0.732 $\pm$ 0.067	0.182	0.022	13.78	0.1167	0.1188	1.011	93.636 $\pm$ 2.327	0.437 $\pm$ 0.011	48.182 $\pm$ 4.764
QML	beta52	0.16	0.144	-0.016 $\pm$ 0.010	-0.098 $\pm$ 0.060	0.010 $\pm$ 0.002	0.100 $\pm$ 0.008	0.010 $\pm$ 0.001	0.101 $\pm$ 0.009	0.397 $\pm$ 0.058	0.630 $\pm$ 0.056	0.140	-0.020	-12.24	0.1094	0.1112	0.886	90.909 $\pm$ 2.741	0.347 $\pm$ 0.010	39.091 $\pm$ 4.652
QML	beta53	0.16	0.137	-0.023 $\pm$ 0.007	-0.142 $\pm$ 0.045	0.006 $\pm$ 0.001	0.076 $\pm$ 0.006	0.006 $\pm$ 0.001	0.079 $\pm$ 0.006	0.245 $\pm$ 0.029	0.495 $\pm$ 0.037	0.132	-0.028	-17.57	0.0874	0.0919	1.096	97.273 $\pm$ 1.553	0.327 $\pm$ 0.005	39.091 $\pm$ 4.652
QML	beta54	0.16	0.153	-0.007 $\pm$ 0.007	-0.046 $\pm$ 0.042	0.005 $\pm$ 0.001	0.071 $\pm$ 0.004	0.005 $\pm$ 0.001	0.071 $\pm$ 0.006	0.195 $\pm$ 0.024	0.441 $\pm$ 0.035	0.154	-0.006	-3.61	0.0757	0.0760	1.104	97.273 $\pm$ 1.553	0.305 $\pm$ 0.005	54.545 $\pm$ 4.748
QML	beta55	0.08	0.082	0.002 $\pm$ 0.013	0.024 $\pm$ 0.163	0.019 $\pm$ 0.002	0.137 $\pm$ 0.009	0.018 $\pm$ 0.002	0.136 $\pm$ 0.011	2.888 $\pm$ 0.373	1.699 $\pm$ 0.139	0.111	0.031	38.97	0.1408	0.1442	0.949	93.636 $\pm$ 2.327	0.508 $\pm$ 0.018	10.909 $\pm$ 2.972
QML	beta56	0.08	0.081	0.001 $\pm$ 0.010	0.008 $\pm$ 0.121	0.010 $\pm$ 0.002	0.102 $\pm$ 0.008	0.010 $\pm$ 0.002	0.101 $\pm$ 0.009	1.604 $\pm$ 0.237	1.266 $\pm$ 0.114	0.084	0.004	4.49	0.1112	0.1113	1.040	91.818 $\pm$ 2.613	0.415 $\pm$ 0.011	13.636 $\pm$ 3.272
QML	beta57	0.06	0.032	-0.028 $\pm$ 0.008	-0.462 $\pm$ 0.127	0.006 $\pm$ 0.001	0.080 $\pm$ 0.008	0.007 $\pm$ 0.001	0.084 $\pm$ 0.008	1.976 $\pm$ 0.301	1.406 $\pm$ 0.125	0.027	-0.033	-55.83	0.0745	0.0817	0.942	90.000 $\pm$ 2.860	0.296 $\pm$ 0.014	6.364 $\pm$ 2.327
QML	beta58	0.06	0.032	-0.028 $\pm$ 0.005	-0.468 $\pm$ 0.091	0.003 $\pm$ 0.000	0.057 $\pm$ 0.004	0.004 $\pm$ 0.001	0.064 $\pm$ 0.005	1.128 $\pm$ 0.164	1.062 $\pm$ 0.089	0.026	-0.034	-56.91	0.0514	0.0617	0.972	85.455 $\pm$ 3.362	0.219 $\pm$ 0.006	6.364 $\pm$ 2.327
UPI	beta41	0.20	0.202	0.002 $\pm$ 0.019	0.011 $\pm$ 0.094	0.039 $\pm$ 0.007	0.198 $\pm$ 0.019	0.039 $\pm$ 0.007	0.197 $\pm$ 0.021	0.967 $\pm$ 0.179	0.983 $\pm$ 0.105	0.234	0.034	16.82	0.2053	0.2081	1.482	98.182 $\pm$ 1.274	1.148 $\pm$ 0.140	32.727 $\pm$ 4.474
UPI	beta42	0.20	0.267	0.067 $\pm$ 0.023	0.335 $\pm$ 0.117	0.060 $\pm$ 0.011	0.245 $\pm$ 0.024	0.064 $\pm$ 0.013	0.253 $\pm$ 0.028	1.597 $\pm$ 0.317	1.264 $\pm$ 0.140	0.237	0.037	18.68	0.1915	0.1951	1.508	91.818 $\pm$ 2.613	1.447 $\pm$ 0.198	24.545 $\pm$ 4.103
UPI	beta43	0.20	0.160	-0.040 $\pm$ 0.011	-0.201 $\pm$ 0.055	0.013 $\pm$ 0.002	0.116 $\pm$ 0.011	0.015 $\pm$ 0.003	0.122 $\pm$ 0.012	0.371 $\pm$ 0.066	0.609 $\pm$ 0.062	0.158	-0.042	-20.91	0.1267	0.1334	1.267	87.273 $\pm$ 3.178	0.574 $\pm$ 0.049	27.273 $\pm$

Table S163

Simulation 2 - Condition 19: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.4
(continued)

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	-0.052	-0.132 ± 0.023	-1.653 ± 0.289	0.059 ± 0.011	0.242 ± 0.024	0.076 ± 0.015	0.275 ± 0.030	11.823 ± 2.332	3.438 ± 0.370	-0.008	-0.088	-109.98	0.2104	0.2281	1.593	93.636 ± 2.327	1.513 ± 0.231	0.000 ± 0.000
UPI	beta51	0.16	0.240	0.080 ± 0.010	0.499 ± 0.062	0.011 ± 0.002	0.105 ± 0.008	0.017 ± 0.003	0.131 ± 0.011	0.673 ± 0.100	0.820 ± 0.067	0.224	0.064	39.84	0.1192	0.1352	1.194	93.636 ± 2.327	0.490 ± 0.020	59.091 ± 4.688
UPI	beta52	0.16	0.151	-0.009 ± 0.012	-0.058 ± 0.077	0.017 ± 0.003	0.130 ± 0.010	0.017 ± 0.003	0.130 ± 0.012	0.657 ± 0.102	0.811 ± 0.075	0.155	-0.005	-3.06	0.1156	0.1157	1.302	94.545 ± 2.165	0.664 ± 0.064	23.636 ± 4.051
UPI	beta53	0.16	0.142	-0.018 ± 0.010	-0.111 ± 0.064	0.012 ± 0.002	0.107 ± 0.009	0.012 ± 0.002	0.108 ± 0.009	0.459 ± 0.066	0.677 ± 0.059	0.125	-0.035	-21.91	0.1099	0.1154	1.211	90.909 ± 2.741	0.510 ± 0.038	20.909 ± 3.877
UPI	beta54	0.16	0.129	-0.031 ± 0.011	-0.196 ± 0.068	0.013 ± 0.002	0.114 ± 0.009	0.014 ± 0.002	0.118 ± 0.010	0.541 ± 0.078	0.736 ± 0.064	0.113	-0.047	-29.26	0.1184	0.1273	1.629	91.818 ± 2.613	0.728 ± 0.097	12.727 ± 3.178
UPI	beta55	0.08	0.140	0.060 ± 0.016	0.744 ± 0.205	0.030 ± 0.005	0.172 ± 0.016	0.033 ± 0.006	0.181 ± 0.019	5.124 ± 0.944	2.264 ± 0.234	0.143	0.063	78.75	0.1387	0.1523	1.353	98.182 ± 1.274	0.911 ± 0.075	5.455 ± 2.165
UPI	beta56	0.08	0.061	-0.019 ± 0.013	-0.234 ± 0.166	0.019 ± 0.003	0.139 ± 0.011	0.019 ± 0.003	0.140 ± 0.013	3.046 ± 0.467	1.745 ± 0.160	0.055	-0.025	-31.19	0.1360	0.1382	1.292	99.091 ± 0.905	0.704 ± 0.058	8.182 ± 2.613
UPI	beta57	0.06	0.026	-0.034 ± 0.014	-0.570 ± 0.229	0.021 ± 0.005	0.144 ± 0.017	0.022 ± 0.005	0.147 ± 0.017	6.025 ± 1.272	2.455 ± 0.291	0.020	-0.040	-67.34	0.1329	0.1389	1.294	91.818 ± 2.613	0.730 ± 0.088	0.909 ± 0.905
UPI	beta58	0.06	0.013	-0.047 ± 0.013	-0.784 ± 0.220	0.019 ± 0.003	0.139 ± 0.013	0.021 ± 0.004	0.146 ± 0.014	5.904 ± 1.004	2.430 ± 0.236	0.011	-0.049	-81.38	0.1014	0.1125	1.650	88.182 ± 3.078	0.897 ± 0.152	1.818 ± 1.274

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S164

Simulation 2 - Condition 20: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.4

Method	Param.	True	Mean-based										Median-based				Inference			
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LSAM	beta41	0.20	0.211	0.011 $\pm$ 0.006	0.057 $\pm$ 0.029	0.004 $\pm$ 0.000	0.066 $\pm$ 0.004	0.004 $\pm$ 0.000	0.067 $\pm$ 0.005	0.112 $\pm$ 0.012	0.335 $\pm$ 0.023	0.203	0.003	1.36	0.0628	0.0629	1.008	96.269 $\pm$ 1.637	0.262 $\pm$ 0.004	91.045 $\pm$ 2.467
LSAM	beta42	0.20	0.190	-0.010 $\pm$ 0.006	-0.050 $\pm$ 0.028	0.004 $\pm$ 0.001	0.066 $\pm$ 0.004	0.004 $\pm$ 0.001	0.066 $\pm$ 0.005	0.110 $\pm$ 0.014	0.332 $\pm$ 0.026	0.194	-0.006	-3.07	0.0744	0.0746	1.048	95.522 $\pm$ 1.787	0.270 $\pm$ 0.004	82.090 $\pm$ 3.312
LSAM	beta43	0.20	0.186	-0.014 $\pm$ 0.004	-0.072 $\pm$ 0.021	0.002 $\pm$ 0.000	0.048 $\pm$ 0.002	0.002 $\pm$ 0.000	0.050 $\pm$ 0.003	0.061 $\pm$ 0.006	0.248 $\pm$ 0.015	0.188	-0.012	-6.14	0.0554	0.0568	1.078	94.776 $\pm$ 1.922	0.201 $\pm$ 0.002	97.761 $\pm$ 1.278
LSAM	beta44	0.11	0.179	0.069 $\pm$ 0.007	0.623 $\pm$ 0.068	0.007 $\pm$ 0.001	0.086 $\pm$ 0.005	0.012 $\pm$ 0.001	0.110 $\pm$ 0.007	0.996 $\pm$ 0.122	0.998 $\pm$ 0.067	0.171	0.061	55.85	0.0815	0.1021	0.936	86.567 $\pm$ 2.946	0.316 $\pm$ 0.007	60.448 $\pm$ 4.224
LSAM	beta45	0.11	0.125	0.015 $\pm$ 0.005	0.137 $\pm$ 0.049	0.004 $\pm$ 0.000	0.063 $\pm$ 0.003	0.004 $\pm$ 0.000	0.064 $\pm$ 0.005	0.340 $\pm$ 0.038	0.583 $\pm$ 0.041	0.123	0.013	11.37	0.0639	0.0652	1.020	97.761 $\pm$ 1.278	0.250 $\pm$ 0.005	53.731 $\pm$ 4.307
LSAM	beta46	0.08	0.016	-0.064 $\pm$ 0.005	-0.802 $\pm$ 0.059	0.003 $\pm$ 0.000	0.055 $\pm$ 0.003	0.007 $\pm$ 0.001	0.084 $\pm$ 0.005	1.110 $\pm$ 0.111	1.054 $\pm$ 0.056	0.023	-0.057	-71.46	0.0520	0.0773	0.928	69.403 $\pm$ 3.981	0.200 $\pm$ 0.005	3.731 $\pm$ 1.637
LSAM	beta47	0.08	0.028	-0.052 $\pm$ 0.004	-0.652 $\pm$ 0.049	0.002 $\pm$ 0.000	0.046 $\pm$ 0.003	0.005 $\pm$ 0.001	0.069 $\pm$ 0.004	0.748 $\pm$ 0.082	0.865 $\pm$ 0.050	0.025	-0.055	-68.34	0.0449	0.0707	1.094	84.328 $\pm$ 3.140	0.195 $\pm$ 0.005	7.463 $\pm$ 2.270
LSAM	beta51	0.16	0.188	0.028 $\pm$ 0.005	0.178 $\pm$ 0.030	0.003 $\pm$ 0.000	0.055 $\pm$ 0.003	0.004 $\pm$ 0.000	0.062 $\pm$ 0.004	0.150 $\pm$ 0.018	0.388 $\pm$ 0.026	0.191	0.031	19.14	0.0577	0.0653	1.064	93.284 $\pm$ 2.162	0.231 $\pm$ 0.002	90.299 $\pm$ 2.557
LSAM	beta52	0.16	0.146	-0.014 $\pm$ 0.005	-0.088 $\pm$ 0.033	0.004 $\pm$ 0.000	0.061 $\pm$ 0.004	0.004 $\pm$ 0.000	0.063 $\pm$ 0.004	0.153 $\pm$ 0.017	0.391 $\pm$ 0.027	0.141	-0.019	-12.11	0.0598	0.0629	0.888	90.299 $\pm$ 2.557	0.213 $\pm$ 0.002	75.733 $\pm$ 3.722
LSAM	beta53	0.16	0.144	-0.016 $\pm$ 0.005	-0.099 $\pm$ 0.029	0.003 $\pm$ 0.000	0.054 $\pm$ 0.003	0.003 $\pm$ 0.000	0.056 $\pm$ 0.004	0.121 $\pm$ 0.014	0.348 $\pm$ 0.025	0.138	-0.022	-14.03	0.0491	0.0540	1.013	94.776 $\pm$ 1.922	0.213 $\pm$ 0.002	77.612 $\pm$ 3.601
LSAM	beta54	0.16	0.132	-0.028 $\pm$ 0.005	-0.174 $\pm$ 0.030	0.003 $\pm$ 0.000	0.056 $\pm$ 0.004	0.004 $\pm$ 0.000	0.063 $\pm$ 0.005	0.154 $\pm$ 0.019	0.392 $\pm$ 0.028	0.137	-0.023	-14.11	0.0476	0.0527	0.976	92.537 $\pm$ 2.270	0.216 $\pm$ 0.003	70.896 $\pm$ 3.924
LSAM	beta55	0.08	0.107	0.027 $\pm$ 0.006	0.342 $\pm$ 0.076	0.005 $\pm$ 0.001	0.070 $\pm$ 0.004	0.006 $\pm$ 0.001	0.075 $\pm$ 0.005	0.875 $\pm$ 0.099	0.936 $\pm$ 0.064	0.107	0.027	33.51	0.0786	0.0830	0.989	94.030 $\pm$ 2.047	0.271 $\pm$ 0.005	33.582 $\pm$ 4.080
LSAM	beta56	0.08	0.073	-0.007 $\pm$ 0.006	-0.092 $\pm$ 0.077	0.005 $\pm$ 0.001	0.071 $\pm$ 0.005	0.005 $\pm$ 0.001	0.071 $\pm$ 0.006	0.798 $\pm$ 0.113	0.893 $\pm$ 0.075	0.075	-0.005	-6.61	0.0684	0.0686	0.909	91.045 $\pm$ 2.467	0.254 $\pm$ 0.005	18.657 $\pm$ 3.365
LSAM	beta57	0.06	0.018	-0.042 $\pm$ 0.004	-0.703 $\pm$ 0.060	0.002 $\pm$ 0.000	0.042 $\pm$ 0.003	0.004 $\pm$ 0.000	0.059 $\pm$ 0.004	0.977 $\pm$ 0.108	0.989 $\pm$ 0.059	0.017	-0.043	-71.55	0.0396	0.0584	0.957	79.104 $\pm$ 3.512	0.157 $\pm$ 0.003	5.224 $\pm$ 1.922
LSAM	beta58	0.06	0.030	-0.030 $\pm$ 0.003	-0.501 $\pm$ 0.052	0.001 $\pm$ 0.000	0.036 $\pm$ 0.002	0.002 $\pm$ 0.000	0.047 $\pm$ 0.003	0.617 $\pm$ 0.069	0.785 $\pm$ 0.049	0.028	-0.032	-53.61	0.0318	0.0452	0.964	83.582 $\pm$ 3.200	0.138 $\pm$ 0.003	14.925 $\pm$ 3.078
QML	beta41	0.20	0.185	-0.015 $\pm$ 0.004	-0.077 $\pm$ 0.021	0.002 $\pm$ 0.000	0.048 $\pm$ 0.003	0.003 $\pm$ 0.000	0.050 $\pm$ 0.004	0.063 $\pm$ 0.007	0.251 $\pm$ 0.018	0.180	-0.020	-9.83	0.0461	0.0502	0.993	93.284 $\pm$ 2.162	0.187 $\pm$ 0.002	97.015 $\pm$ 1.470
QML	beta42	0.20	0.176	-0.024 $\pm$ 0.004	-0.120 $\pm$ 0.022	0.002 $\pm$ 0.000	0.050 $\pm$ 0.003	0.003 $\pm$ 0.000	0.055 $\pm$ 0.004	0.076 $\pm$ 0.009	0.276 $\pm$ 0.020	0.178	-0.022	-11.13	0.0550	0.0593	0.979	91.791 $\pm$ 2.371	0.192 $\pm$ 0.002	97.015 $\pm$ 1.470
QML	beta43	0.20	0.174	-0.026 $\pm$ 0.004	-0.132 $\pm$ 0.019	0.002 $\pm$ 0.000	0.045 $\pm$ 0.002	0.003 $\pm$ 0.000	0.052 $\pm$ 0.003	0.067 $\pm$ 0.007	0.259 $\pm$ 0.015	0.171	-0.029	-14.42	0.0467	0.0549	1.004	89.552 $\pm$ 2.642	0.176 $\pm$ 0.001	98.507 $\pm$ 1.047
QML	beta44	0.11	0.145	0.035 $\pm$ 0.006	0.315 $\pm$ 0.056	0.005 $\pm$ 0.001	0.071 $\pm$ 0.004	0.006 $\pm$ 0.001	0.079 $\pm$ 0.005	0.516 $\pm$ 0.058	0.718 $\pm$ 0.048	0.147	0.037	33.60	0.0680	0.0774	0.929	86.567 $\pm$ 2.946	0.260 $\pm$ 0.005	63.433 $\pm$ 4.161
QML	beta45	0.11	0.118	0.008 $\pm$ 0.005	0.069 $\pm$ 0.043	0.003 $\pm$ 0.000	0.054 $\pm$ 0.003	0.003 $\pm$ 0.000	0.055 $\pm$ 0.004	0.247 $\pm$ 0.023	0.497 $\pm$ 0.032	0.115	0.005	4.12	0.0659	0.0660	1.010	98.507 $\pm$ 1.047	0.215 $\pm$ 0.003	61.940 $\pm$ 4.194
QML	beta46	0.08	0.043	-0.037 $\pm$ 0.004	-0.468 $\pm$ 0.050	0.002 $\pm$ 0.000	0.046 $\pm$ 0.003	0.004 $\pm$ 0.000	0.059 $\pm$ 0.003	0.549 $\pm$ 0.051	0.741 $\pm$ 0.040	0.043	-0.037	-45.99	0.0428	0.0564	0.915	79.104 $\pm$ 3.512	0.165 $\pm$ 0.004	13.433 $\pm$ 2.946
QML	beta47	0.08	0.044	-0.036 $\pm$ 0.003	-0.446 $\pm$ 0.039	0.001 $\pm$ 0.000	0.036 $\pm$ 0.002	0.003 $\pm$ 0.000	0.050 $\pm$ 0.003	0.398 $\pm$ 0.042	0.631 $\pm$ 0.036	0.041	-0.039	-49.24	0.0410	0.0568	1.146	84.328 $\pm$ 3.140	0.161 $\pm$ 0.003	14.925 $\pm$ 3.078
QML	beta51	0.16	0.196	0.036 $\pm$ 0.005	0.222 $\pm$ 0.031	0.003 $\pm$ 0.000	0.057 $\pm$ 0.004	0.005 $\pm$ 0.001	0.067 $\pm$ 0.004	0.177 $\pm$ 0.020	0.420 $\pm$ 0.027	0.202	0.042	26.16	0.0519	0.0667	1.096	94.030 $\pm$ 2.047	0.246 $\pm$ 0.003	88.060 $\pm$ 2.801
QML	beta52	0.16	0.141	-0.019 $\pm$ 0.005	-0.121 $\pm$ 0.032	0.003 $\pm$ 0.000	0.059 $\pm$ 0.004	0.004 $\pm$ 0.000	0.062 $\pm$ 0.004	0.148 $\pm$ 0.017	0.385 $\pm$ 0.027	0.135	-0.025	-15.72	0.0679	0.0724	0.875	88.060 $\pm$ 2.801	0.201 $\pm$ 0.003	77.612 $\pm$ 3.601
QML	beta53	0.16	0.132	-0.028 $\pm$ 0.004	-0.172 $\pm$ 0.028	0.003 $\pm$ 0.000	0.052 $\pm$ 0.003	0.003 $\pm$ 0.000	0.059 $\pm$ 0.004	0.134 $\pm$ 0.014	0.367 $\pm$ 0.023	0.126	-0.034	-21.43	0.0543	0.0642	0.984	93.284 $\pm$ 2.162	0.200 $\pm$ 0.002	73.881 $\pm$ 3.795
QML	beta54	0.16	0.143	-0.017 $\pm$ 0.004	-0.105 $\pm$ 0.025	0.002 $\pm$ 0.000	0.047 $\pm$ 0.003	0.002 $\pm$ 0.000	0.050 $\pm$ 0.004	0.096 $\pm$ 0.012	0.310 $\pm$ 0.022	0.149	-0.011	-7.08	0.0400	0.0416	1.005	94.030 $\pm$ 2.047	0.185 $\pm$ 0.002	85.821 $\pm$ 3.013
QML	beta55	0.08	0.110	0.030 $\pm$ 0.006	0.370 $\pm$ 0.078	0.005 $\pm$ 0.001	0.073 $\pm$ 0.004	0.006 $\pm$ 0.001	0.078 $\pm$ 0.006	0.956 $\pm$ 0.114	0.978 $\pm$ 0.069	0.115	0.035	44.36	0.0747	0.0827	1.018	91.791 $\pm$ 2.371	0.290 $\pm$ 0.005	32.836 $\pm$ 4.057
QML	beta56	0.08	0.071	-0.009 $\pm$ 0.006	-0.115 $\pm$ 0.074	0.005 $\pm$ 0.001	0.068 $\pm$ 0.005	0.005 $\pm$ 0.001	0.069 $\pm$ 0.005	0.740 $\pm$ 0.092	0.860 $\pm$ 0.065	0.079	-0.001	-0.69	0.0710	0.0710	0.944	93.284 $\pm$ 2.162	0.253 $\pm$ 0.004	21.642 $\pm$ 3.557
QML	beta57	0.06	0.018	-0.042 $\pm$ 0.003	-0.704 $\pm$ 0.058	0.002 $\pm$ 0.000	0.040 $\pm$ 0.003	0.003 $\pm$ 0.000	0.058 $\pm$ 0.003	0.946 $\pm$ 0.090	0.972 $\pm$ 0.051	0.015	-0.045	-74.70	0.0396	0.0598	1.035	76.119 $\pm$ 3.683	0.164 $\pm$ 0.005	4.478 $\pm$ 1.787
QML	beta58	0.06	0.035	-0.025 $\pm$ 0.003	-0.408 $\pm$ 0.049	0.001 $\pm$ 0.000	0.034 $\pm$ 0.002	0.002 $\pm$ 0.000	0.042 $\pm$ 0.003	0.486 $\pm$ 0.056	0.697 $\pm$ 0.045	0.035	-0.025	-42.25	0.0326	0.0413	0.983	84.328 $\pm$ 3.140	0.131 $\pm$ 0.003	17.910 $\pm$ 3.312
UPI	beta41	0.20	0.230	0.030 $\pm$ 0.015	0.148 $\pm$ 0.076	0.031 $\pm$ 0.006	0.175 $\pm$ 0.017	0.031 $\pm$ 0.006	0.177 $\pm$ 0.019	0.785 $\pm$ 0.149	0.886 $\pm$ 0.095	0.223	0.023	11.36	0.1709	0.1724	1.389	88.060 $\pm$ 2.801	0.955 $\pm$ 0.178	41.045 $\pm$ 4.249
UPI	beta42	0.20	0.223	0.023 $\pm$ 0.014	0.114 $\pm$ 0.070	0.026 $\pm$ 0.004	0.161 $\pm$ 0.012	0.026 $\pm$ 0.004	0.163 $\pm$ 0.014	0.660 $\pm$ 0.093	0.813 $\pm$ 0.068	0.229	0.029	14.33	0.1892	0.1913	1.300	94.030 $\pm$ 2.047	0.823 $\pm$ 0.093	42.537 $\pm$ 4.271
UPI	beta43	0.20	0.152	-0.048 $\pm$ 0.008	-0.240 $\pm$ 0.038	0.008 $\pm$ 0.001	0.088 $\pm$ 0.006	0.010 $\pm$ 0.001	0.100 $\pm$ 0.008	0.251 $\pm$ 0.033	0.501 $\pm$ 0.038	0.128	-0.072	-35.86	0.1151	0.1356	1.065	88.060 $\pm$ 2.801	0.368 $\pm$ 0.030	54.478

Table S164

Simulation 2 - Condition 20: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.4
(continued)

Method	Param.	True	Mean-based										Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE	
UPI	beta47	0.08	-0.046	-0.126 ± 0.017	-1.571 ± 0.210	0.038 ± 0.008	0.194 ± 0.021	0.053 ± 0.009	0.231 ± 0.022	8.312 ± 1.475	2.883 ± 0.277	-0.045	-0.125	-156.28	0.1867	0.2247	1.323	91.791 ± 2.371	1.007 ± 0.148	4.478 ± 1.787	
UPI	beta51	0.16	0.242	0.082 ± 0.005	0.514 ± 0.032	0.003 ± 0.000	0.059 ± 0.003	0.010 ± 0.001	0.101 ± 0.005	0.399 ± 0.037	0.631 ± 0.031	0.253	0.093	58.01	0.0612	0.1112	1.199	76.866 ± 3.643	0.276 ± 0.012	95.522 ± 1.787	
UPI	beta52	0.16	0.157	-0.003 ± 0.008	-0.021 ± 0.052	0.009 ± 0.001	0.096 ± 0.007	0.009 ± 0.001	0.095 ± 0.008	0.356 ± 0.052	0.597 ± 0.051	0.163	0.003	2.17	0.0926	0.0926	1.116	88.806 ± 2.724	0.419 ± 0.035	42.537 ± 4.271	
UPI	beta53	0.16	0.115	-0.045 ± 0.006	-0.282 ± 0.035	0.004 ± 0.000	0.066 ± 0.003	0.006 ± 0.001	0.079 ± 0.005	0.247 ± 0.026	0.496 ± 0.030	0.109	-0.051	-31.95	0.0815	0.0962	1.155	85.821 ± 3.013	0.297 ± 0.028	47.015 ± 4.312	
UPI	beta54	0.16	0.119	-0.041 ± 0.008	-0.258 ± 0.051	0.009 ± 0.002	0.094 ± 0.010	0.011 ± 0.002	0.103 ± 0.010	0.412 ± 0.069	0.642 ± 0.060	0.114	-0.046	-28.82	0.0934	0.1041	1.239	83.582 ± 3.200	0.459 ± 0.057	32.090 ± 4.033	
UPI	beta55	0.08	0.133	0.053 ± 0.007	0.662 ± 0.082	0.006 ± 0.001	0.076 ± 0.005	0.009 ± 0.001	0.092 ± 0.007	1.335 ± 0.188	1.155 ± 0.089	0.131	0.051	63.64	0.0900	0.1034	1.175	95.522 ± 1.787	0.350 ± 0.015	39.552 ± 4.224	
UPI	beta56	0.08	0.075	-0.005 ± 0.007	-0.059 ± 0.086	0.006 ± 0.001	0.079 ± 0.006	0.006 ± 0.001	0.079 ± 0.007	0.976 ± 0.135	0.988 ± 0.082	0.073	-0.007	-8.30	0.0829	0.0831	1.092	97.761 ± 1.278	0.339 ± 0.014	15.672 ± 3.140	
UPI	beta57	0.06	-0.031	-0.091 ± 0.011	-1.522 ± 0.187	0.017 ± 0.003	0.130 ± 0.010	0.025 ± 0.004	0.158 ± 0.014	6.942 ± 1.130	2.635 ± 0.231	-0.009	-0.069	-115.80	0.0971	0.1194	1.032	82.836 ± 3.257	0.524 ± 0.054	5.224 ± 1.922	
UPI	beta58	0.06	-0.020	-0.080 ± 0.009	-1.330 ± 0.148	0.011 ± 0.002	0.103 ± 0.009	0.017 ± 0.003	0.130 ± 0.011	4.681 ± 0.748	2.164 ± 0.185	-0.004	-0.064	-106.48	0.0986	0.1175	1.358	84.328 ± 3.140	0.547 ± 0.064	2.239 ± 1.278	

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S165

Simulation 2 - Condition 21: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, $N = 400$, Reliability = 0.6

Method	Param.	True	Mean-based										Median-based					Inference		
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LSAM	beta41	0.20	0.196	-0.004 $\pm$ 0.002	-0.020 $\pm$ 0.010	0.004 $\pm$ 0.000	0.065 $\pm$ 0.001	0.004 $\pm$ 0.000	0.065 $\pm$ 0.002	0.105 $\pm$ 0.005	0.325 $\pm$ 0.009	0.197	-0.003	-1.27	0.0656	0.0657	0.950	94.523 $\pm$ 0.725	0.241 $\pm$ 0.001	88.337 $\pm$ 1.022
LSAM	beta42	0.20	0.199	-0.001 $\pm$ 0.002	-0.006 $\pm$ 0.010	0.004 $\pm$ 0.000	0.063 $\pm$ 0.001	0.004 $\pm$ 0.000	0.063 $\pm$ 0.002	0.098 $\pm$ 0.004	0.314 $\pm$ 0.009	0.200	0.000	-0.15	0.0626	0.0626	1.001	95.943 $\pm$ 0.628	0.246 $\pm$ 0.001	87.627 $\pm$ 1.049
LSAM	beta43	0.20	0.202	0.002 $\pm$ 0.002	0.008 $\pm$ 0.009	0.003 $\pm$ 0.000	0.058 $\pm$ 0.001	0.003 $\pm$ 0.000	0.058 $\pm$ 0.002	0.083 $\pm$ 0.004	0.289 $\pm$ 0.008	0.201	0.001	0.61	0.0554	0.0554	0.984	94.828 $\pm$ 0.705	0.223 $\pm$ 0.001	94.929 $\pm$ 0.699
LSAM	beta44	0.11	0.128	0.018 $\pm$ 0.002	0.166 $\pm$ 0.022	0.006 $\pm$ 0.000	0.078 $\pm$ 0.002	0.006 $\pm$ 0.000	0.080 $\pm$ 0.002	0.526 $\pm$ 0.026	0.725 $\pm$ 0.021	0.128	0.018	16.26	0.0719	0.0741	0.953	93.712 $\pm$ 0.773	0.290 $\pm$ 0.002	44.828 $\pm$ 1.584
LSAM	beta45	0.11	0.116	0.006 $\pm$ 0.002	0.054 $\pm$ 0.019	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	0.364 $\pm$ 0.018	0.603 $\pm$ 0.018	0.114	0.004	3.65	0.0645	0.0646	0.952	93.915 $\pm$ 0.761	0.247 $\pm$ 0.001	46.349 $\pm$ 1.588
LSAM	beta46	0.08	0.061	-0.019 $\pm$ 0.002	-0.231 $\pm$ 0.021	0.003 $\pm$ 0.000	0.054 $\pm$ 0.001	0.003 $\pm$ 0.000	0.057 $\pm$ 0.002	0.508 $\pm$ 0.026	0.713 $\pm$ 0.021	0.063	-0.017	-21.54	0.0516	0.0544	0.956	91.886 $\pm$ 0.870	0.202 $\pm$ 0.002	25.659 $\pm$ 1.391
LSAM	beta47	0.08	0.065	-0.015 $\pm$ 0.002	-0.191 $\pm$ 0.022	0.003 $\pm$ 0.000	0.054 $\pm$ 0.001	0.003 $\pm$ 0.000	0.056 $\pm$ 0.002	0.492 $\pm$ 0.023	0.701 $\pm$ 0.019	0.063	-0.017	-20.75	0.0547	0.0572	0.930	91.886 $\pm$ 0.870	0.197 $\pm$ 0.002	27.282 $\pm$ 1.418
LSAM	beta51	0.16	0.168	0.008 $\pm$ 0.002	0.050 $\pm$ 0.014	0.005 $\pm$ 0.000	0.073 $\pm$ 0.002	0.005 $\pm$ 0.000	0.073 $\pm$ 0.002	0.208 $\pm$ 0.010	0.456 $\pm$ 0.013	0.167	0.007	4.43	0.0698	0.0702	0.958	94.422 $\pm$ 0.731	0.272 $\pm$ 0.001	67.343 $\pm$ 1.493
LSAM	beta52	0.16	0.151	-0.009 $\pm$ 0.002	-0.054 $\pm$ 0.012	0.003 $\pm$ 0.000	0.059 $\pm$ 0.001	0.004 $\pm$ 0.000	0.059 $\pm$ 0.002	0.137 $\pm$ 0.006	0.370 $\pm$ 0.010	0.152	-0.008	-4.95	0.0599	0.0605	1.003	94.929 $\pm$ 0.699	0.230 $\pm$ 0.001	74.645 $\pm$ 1.385
LSAM	beta53	0.16	0.162	0.002 $\pm$ 0.002	0.013 $\pm$ 0.013	0.004 $\pm$ 0.000	0.065 $\pm$ 0.001	0.004 $\pm$ 0.000	0.065 $\pm$ 0.002	0.165 $\pm$ 0.007	0.406 $\pm$ 0.011	0.161	0.001	0.49	0.0667	0.0667	0.958	94.523 $\pm$ 0.725	0.244 $\pm$ 0.001	74.949 $\pm$ 1.380
LSAM	beta54	0.16	0.154	-0.006 $\pm$ 0.002	-0.035 $\pm$ 0.012	0.004 $\pm$ 0.000	0.060 $\pm$ 0.001	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	0.142 $\pm$ 0.007	0.377 $\pm$ 0.011	0.154	-0.006	-3.49	0.0591	0.0593	0.973	93.915 $\pm$ 0.761	0.229 $\pm$ 0.001	75.558 $\pm$ 1.369
LSAM	beta55	0.08	0.086	0.006 $\pm$ 0.002	0.075 $\pm$ 0.030	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.878 $\pm$ 0.043	0.937 $\pm$ 0.027	0.086	0.006	7.24	0.0695	0.0697	0.976	94.219 $\pm$ 0.743	0.286 $\pm$ 0.002	24.544 $\pm$ 1.370
LSAM	beta56	0.08	0.082	0.002 $\pm$ 0.002	0.022 $\pm$ 0.028	0.005 $\pm$ 0.000	0.070 $\pm$ 0.002	0.005 $\pm$ 0.000	0.070 $\pm$ 0.002	0.757 $\pm$ 0.036	0.870 $\pm$ 0.025	0.080	0.000	0.23	0.0659	0.0659	0.996	94.625 $\pm$ 0.718	0.272 $\pm$ 0.002	22.414 $\pm$ 1.328
LSAM	beta57	0.06	0.048	-0.012 $\pm$ 0.002	-0.206 $\pm$ 0.027	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.769 $\pm$ 0.034	0.877 $\pm$ 0.024	0.045	-0.015	-24.95	0.0501	0.0523	0.941	91.684 $\pm$ 0.879	0.189 $\pm$ 0.001	19.270 $\pm$ 1.256
LSAM	beta58	0.06	0.047	-0.013 $\pm$ 0.001	-0.208 $\pm$ 0.023	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.587 $\pm$ 0.027	0.766 $\pm$ 0.021	0.048	-0.012	-19.78	0.0427	0.0443	0.948	91.684 $\pm$ 0.879	0.164 $\pm$ 0.001	21.501 $\pm$ 1.308
QML	beta41	0.20	0.188	-0.012 $\pm$ 0.002	-0.061 $\pm$ 0.010	0.004 $\pm$ 0.000	0.060 $\pm$ 0.001	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.094 $\pm$ 0.004	0.306 $\pm$ 0.008	0.190	-0.010	-4.95	0.0606	0.0614	0.949	92.799 $\pm$ 0.823	0.224 $\pm$ 0.001	89.452 $\pm$ 0.978
QML	beta42	0.20	0.195	-0.005 $\pm$ 0.002	-0.026 $\pm$ 0.009	0.003 $\pm$ 0.000	0.058 $\pm$ 0.001	0.003 $\pm$ 0.000	0.058 $\pm$ 0.002	0.085 $\pm$ 0.004	0.291 $\pm$ 0.008	0.194	-0.006	-2.92	0.0553	0.0556	1.002	94.828 $\pm$ 0.705	0.228 $\pm$ 0.001	91.481 $\pm$ 0.889
QML	beta43	0.20	0.191	-0.009 $\pm$ 0.002	-0.044 $\pm$ 0.009	0.003 $\pm$ 0.000	0.055 $\pm$ 0.001	0.003 $\pm$ 0.000	0.056 $\pm$ 0.002	0.077 $\pm$ 0.003	0.278 $\pm$ 0.008	0.192	-0.008	-4.09	0.0519	0.0525	0.989	93.915 $\pm$ 0.761	0.213 $\pm$ 0.001	94.828 $\pm$ 0.705
QML	beta44	0.11	0.118	0.008 $\pm$ 0.002	0.076 $\pm$ 0.020	0.005 $\pm$ 0.000	0.070 $\pm$ 0.002	0.005 $\pm$ 0.000	0.071 $\pm$ 0.002	0.411 $\pm$ 0.019	0.641 $\pm$ 0.018	0.118	0.008	7.55	0.0666	0.0671	0.986	93.813 $\pm$ 0.767	0.271 $\pm$ 0.002	42.901 $\pm$ 1.576
QML	beta45	0.11	0.111	0.001 $\pm$ 0.002	0.013 $\pm$ 0.018	0.004 $\pm$ 0.000	0.061 $\pm$ 0.001	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.303 $\pm$ 0.015	0.550 $\pm$ 0.016	0.112	0.002	1.53	0.0581	0.0581	0.979	94.118 $\pm$ 0.749	0.232 $\pm$ 0.001	48.073 $\pm$ 1.591
QML	beta46	0.08	0.065	-0.015 $\pm$ 0.002	-0.191 $\pm$ 0.019	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	0.386 $\pm$ 0.018	0.622 $\pm$ 0.017	0.065	-0.015	-19.07	0.0461	0.0486	0.994	91.785 $\pm$ 0.874	0.184 $\pm$ 0.001	29.817 $\pm$ 1.457
QML	beta47	0.08	0.066	-0.014 $\pm$ 0.002	-0.179 $\pm$ 0.019	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.382 $\pm$ 0.016	0.618 $\pm$ 0.016	0.064	-0.016	-19.89	0.0482	0.0507	0.960	92.596 $\pm$ 0.834	0.178 $\pm$ 0.001	32.353 $\pm$ 1.490
QML	beta51	0.16	0.177	0.017 $\pm$ 0.002	0.108 $\pm$ 0.015	0.005 $\pm$ 0.000	0.073 $\pm$ 0.002	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.222 $\pm$ 0.011	0.472 $\pm$ 0.013	0.176	0.016	10.21	0.0703	0.0722	0.987	94.726 $\pm$ 0.712	0.284 $\pm$ 0.001	69.777 $\pm$ 1.462
QML	beta52	0.16	0.139	-0.021 $\pm$ 0.002	-0.129 $\pm$ 0.012	0.004 $\pm$ 0.000	0.060 $\pm$ 0.001	0.004 $\pm$ 0.000	0.064 $\pm$ 0.002	0.159 $\pm$ 0.007	0.399 $\pm$ 0.011	0.139	-0.021	-13.29	0.0613	0.0649	1.031	94.016 $\pm$ 0.755	0.244 $\pm$ 0.002	62.677 $\pm$ 1.540
QML	beta53	0.16	0.142	-0.018 $\pm$ 0.002	-0.115 $\pm$ 0.013	0.004 $\pm$ 0.000	0.064 $\pm$ 0.001	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	0.171 $\pm$ 0.008	0.413 $\pm$ 0.011	0.142	-0.018	-11.38	0.0627	0.0652	0.986	93.408 $\pm$ 0.790	0.246 $\pm$ 0.001	61.765 $\pm$ 1.548
QML	beta54	0.16	0.156	-0.004 $\pm$ 0.002	-0.025 $\pm$ 0.012	0.003 $\pm$ 0.000	0.058 $\pm$ 0.001	0.003 $\pm$ 0.000	0.058 $\pm$ 0.002	0.132 $\pm$ 0.007	0.363 $\pm$ 0.011	0.157	-0.003	-2.16	0.0546	0.0547	0.976	94.320 $\pm$ 0.737	0.222 $\pm$ 0.001	78.905 $\pm$ 1.299
QML	beta55	0.08	0.090	0.010 $\pm$ 0.002	0.124 $\pm$ 0.030	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.006 $\pm$ 0.000	0.076 $\pm$ 0.002	0.897 $\pm$ 0.041	0.947 $\pm$ 0.026	0.091	0.011	13.20	0.0722	0.0729	1.012	94.929 $\pm$ 0.699	0.298 $\pm$ 0.002	24.341 $\pm$ 1.367
QML	beta56	0.08	0.081	0.001 $\pm$ 0.002	0.009 $\pm$ 0.027	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.727 $\pm$ 0.034	0.853 $\pm$ 0.024	0.079	-0.001	-1.84	0.0644	0.0645	1.018	95.132 $\pm$ 0.685	0.272 $\pm$ 0.002	20.994 $\pm$ 1.297
QML	beta57	0.06	0.042	-0.018 $\pm$ 0.002	-0.299 $\pm$ 0.025	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.728 $\pm$ 0.031	0.854 $\pm$ 0.022	0.041	-0.019	-31.69	0.0463	0.0501	0.976	90.872 $\pm$ 0.917	0.184 $\pm$ 0.002	16.126 $\pm$ 1.171
QML	beta58	0.06	0.045	-0.015 $\pm$ 0.001	-0.245 $\pm$ 0.022	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.523 $\pm$ 0.023	0.723 $\pm$ 0.019	0.045	-0.015	-24.53	0.0400	0.0426	0.966	91.481 $\pm$ 0.889	0.155 $\pm$ 0.001	22.414 $\pm$ 1.328
UPI	beta41	0.20	0.208	0.008 $\pm$ 0.002	0.042 $\pm$ 0.010	0.004 $\pm$ 0.000	0.065 $\pm$ 0.002	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	0.109 $\pm$ 0.005	0.329 $\pm$ 0.009	0.211	0.011	5.45	0.0632	0.0642	0.953	93.915 $\pm$ 0.761	0.244 $\pm$ 0.003	91.075 $\pm$ 0.908
UPI	beta42	0.20	0.209	0.009 $\pm$ 0.002	0.045 $\pm$ 0.010	0.004 $\pm$ 0.000	0.064 $\pm$ 0.001	0.004 $\pm$ 0.000	0.065 $\pm$ 0.002	0.104 $\pm$ 0.005	0.323 $\pm$ 0.009	0.208	0.008	4.03	0.0615	0.0620	1.007	95.132 $\pm$ 0.685	0.253 $\pm$ 0.003	89.858 $\pm$ 0.961
UPI	beta43	0.20	0.203	0.003 $\pm$ 0.002	0.014 $\pm$ 0.009	0.003 $\pm$ 0.000	0.059 $\pm$ 0.001	0.003 $\pm$ 0.000	0.059 $\pm$ 0.002	0.086 $\pm$ 0.004	0.294 $\pm$ 0.008	0.200	0.000	0.19	0.0575	0.0575	0.981	94.726 $\pm$ 0.712	0.226 $\pm$ 0.001	95.538 $\pm$

Table S165

Simulation 2 - Condition 21: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.6
(continued)

Method	Param.	True	Mean-based										Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE	
UPI	beta47	0.08	0.073	-0.007 ± 0.002	-0.088 ± 0.023	0.003 ± 0.000	0.057 ± 0.002	0.003 ± 0.000	0.058 ± 0.002	0.521 ± 0.027	0.722 ± 0.022	0.071	-0.009	-11.62	0.0541	0.0549	1.022	94.118 ± 0.749	0.230 ± 0.004	26.572 ± 1.407	
UPI	beta51	0.16	0.169	0.009 ± 0.002	0.054 ± 0.014	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.204 ± 0.010	0.452 ± 0.013	0.167	0.007	4.26	0.0701	0.0704	0.977	95.233 ± 0.679	0.275 ± 0.001	67.444 ± 1.492	
UPI	beta52	0.16	0.161	0.001 ± 0.002	0.009 ± 0.012	0.004 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	0.138 ± 0.007	0.372 ± 0.011	0.162	0.002	1.03	0.0586	0.0586	1.009	95.335 ± 0.672	0.235 ± 0.001	78.195 ± 1.315	
UPI	beta53	0.16	0.163	0.003 ± 0.002	0.021 ± 0.013	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	0.164 ± 0.007	0.405 ± 0.011	0.162	0.002	1.49	0.0653	0.0654	0.974	94.422 ± 0.731	0.247 ± 0.001	73.834 ± 1.400	
UPI	beta54	0.16	0.160	0.000 ± 0.002	0.003 ± 0.012	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.143 ± 0.007	0.378 ± 0.011	0.161	0.001	0.56	0.0576	0.0576	1.004	94.422 ± 0.731	0.238 ± 0.002	76.572 ± 1.349	
UPI	beta55	0.08	0.080	0.000 ± 0.002	0.006 ± 0.030	0.006 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.074 ± 0.002	0.863 ± 0.042	0.929 ± 0.027	0.081	0.001	0.89	0.0704	0.0704	1.010	95.030 ± 0.692	0.294 ± 0.002	19.574 ± 1.264	
UPI	beta56	0.08	0.083	0.003 ± 0.002	0.041 ± 0.028	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.786 ± 0.039	0.887 ± 0.026	0.079	-0.001	-1.11	0.0681	0.0681	1.010	95.740 ± 0.643	0.281 ± 0.002	21.602 ± 1.311	
UPI	beta57	0.06	0.058	-0.002 ± 0.002	-0.041 ± 0.029	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.844 ± 0.041	0.919 ± 0.027	0.055	-0.005	-9.03	0.0515	0.0518	0.985	94.422 ± 0.731	0.213 ± 0.002	18.763 ± 1.243	
UPI	beta58	0.06	0.053	-0.007 ± 0.002	-0.114 ± 0.027	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.002	0.732 ± 0.038	0.856 ± 0.026	0.052	-0.008	-12.95	0.0503	0.0509	1.034	94.118 ± 0.749	0.206 ± 0.004	19.878 ± 1.271	

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S166

Simulation 2 - Condition 22: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.6

Method	Param.	True	Mean-based										Median-based					Inference		
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LSAM	beta41	0.20	0.195	-0.005 $\pm$ 0.001	-0.024 $\pm$ 0.006	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.040 $\pm$ 0.002	0.199 $\pm$ 0.006	0.195	-0.005	-2.42	0.0386	0.0389	0.956	93.994 $\pm$ 0.752	0.148 $\pm$ 0.000	99.900 $\pm$ 0.100
LSAM	beta42	0.20	0.197	-0.003 $\pm$ 0.001	-0.015 $\pm$ 0.006	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.038 $\pm$ 0.002	0.195 $\pm$ 0.005	0.196	-0.004	-2.21	0.0395	0.0397	1.001	94.194 $\pm$ 0.740	0.152 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta43	0.20	0.201	0.001 $\pm$ 0.001	0.003 $\pm$ 0.005	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.030 $\pm$ 0.001	0.173 $\pm$ 0.005	0.200	0.000	0.17	0.0354	0.0354	1.024	96.096 $\pm$ 0.613	0.139 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta44	0.11	0.129	0.019 $\pm$ 0.001	0.169 $\pm$ 0.013	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.190 $\pm$ 0.008	0.436 $\pm$ 0.011	0.128	0.018	16.04	0.0457	0.0490	0.999	93.493 $\pm$ 0.780	0.173 $\pm$ 0.001	83.083 $\pm$ 1.186
LSAM	beta45	0.11	0.114	0.004 $\pm$ 0.001	0.035 $\pm$ 0.011	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.130 $\pm$ 0.006	0.360 $\pm$ 0.010	0.114	0.004	3.77	0.0406	0.0408	0.970	94.795 $\pm$ 0.703	0.150 $\pm$ 0.001	83.784 $\pm$ 1.166
LSAM	beta46	0.08	0.062	-0.018 $\pm$ 0.001	-0.224 $\pm$ 0.012	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.192 $\pm$ 0.009	0.439 $\pm$ 0.012	0.064	-0.016	-19.99	0.0292	0.0333	1.007	89.489 $\pm$ 0.970	0.119 $\pm$ 0.001	57.257 $\pm$ 1.565
LSAM	beta47	0.08	0.062	-0.018 $\pm$ 0.001	-0.221 $\pm$ 0.012	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.191 $\pm$ 0.008	0.437 $\pm$ 0.010	0.062	-0.018	-22.91	0.0298	0.0350	1.005	89.890 $\pm$ 0.954	0.119 $\pm$ 0.001	55.856 $\pm$ 1.571
LSAM	beta51	0.16	0.172	0.012 $\pm$ 0.001	0.075 $\pm$ 0.008	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.072 $\pm$ 0.003	0.268 $\pm$ 0.008	0.171	0.011	7.01	0.0405	0.0420	1.051	95.095 $\pm$ 0.683	0.170 $\pm$ 0.000	98.398 $\pm$ 0.397
LSAM	beta52	0.16	0.153	-0.007 $\pm$ 0.001	-0.046 $\pm$ 0.007	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.053 $\pm$ 0.002	0.231 $\pm$ 0.006	0.152	-0.008	-4.78	0.0360	0.0368	1.002	94.494 $\pm$ 0.722	0.142 $\pm$ 0.000	98.599 $\pm$ 0.372
LSAM	beta53	0.16	0.156	-0.004 $\pm$ 0.001	-0.026 $\pm$ 0.007	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.056 $\pm$ 0.002	0.237 $\pm$ 0.006	0.154	-0.006	-3.45	0.0377	0.0381	1.021	95.095 $\pm$ 0.683	0.151 $\pm$ 0.000	98.498 $\pm$ 0.385
LSAM	beta54	0.16	0.152	-0.008 $\pm$ 0.001	-0.047 $\pm$ 0.007	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.054 $\pm$ 0.002	0.231 $\pm$ 0.006	0.153	-0.007	-4.11	0.0378	0.0383	1.005	94.194 $\pm$ 0.740	0.143 $\pm$ 0.000	98.599 $\pm$ 0.372
LSAM	beta55	0.08	0.088	0.008 $\pm$ 0.001	0.101 $\pm$ 0.017	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.311 $\pm$ 0.015	0.558 $\pm$ 0.016	0.087	0.007	8.66	0.0409	0.0415	0.995	94.494 $\pm$ 0.722	0.171 $\pm$ 0.001	53.353 $\pm$ 1.578
LSAM	beta56	0.08	0.079	-0.001 $\pm$ 0.001	-0.012 $\pm$ 0.016	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	0.264 $\pm$ 0.012	0.513 $\pm$ 0.015	0.078	-0.002	-3.02	0.0395	0.0396	1.013	94.795 $\pm$ 0.703	0.163 $\pm$ 0.001	48.048 $\pm$ 1.581
LSAM	beta57	0.06	0.045	-0.015 $\pm$ 0.001	-0.249 $\pm$ 0.016	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.304 $\pm$ 0.013	0.551 $\pm$ 0.014	0.044	-0.016	-27.05	0.0291	0.0334	0.973	89.690 $\pm$ 0.962	0.113 $\pm$ 0.001	34.535 $\pm$ 1.504
LSAM	beta58	0.06	0.050	-0.010 $\pm$ 0.001	-0.159 $\pm$ 0.014	0.001 $\pm$ 0.000	0.027 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.227 $\pm$ 0.011	0.477 $\pm$ 0.013	0.051	-0.009	-15.20	0.0255	0.0271	0.963	90.691 $\pm$ 0.919	0.102 $\pm$ 0.000	51.552 $\pm$ 1.581
QML	beta41	0.20	0.188	-0.012 $\pm$ 0.001	-0.059 $\pm$ 0.006	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.038 $\pm$ 0.002	0.196 $\pm$ 0.005	0.188	-0.012	-6.15	0.0363	0.0383	0.946	91.792 $\pm$ 0.868	0.138 $\pm$ 0.000	99.900 $\pm$ 0.100
QML	beta42	0.20	0.193	-0.007 $\pm$ 0.001	-0.034 $\pm$ 0.006	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.035 $\pm$ 0.001	0.187 $\pm$ 0.005	0.193	-0.007	-3.66	0.0376	0.0383	0.988	93.994 $\pm$ 0.752	0.142 $\pm$ 0.000	100.000 $\pm$ 0.000
QML	beta43	0.20	0.191	-0.009 $\pm$ 0.001	-0.044 $\pm$ 0.005	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.030 $\pm$ 0.001	0.174 $\pm$ 0.005	0.191	-0.009	-4.48	0.0328	0.0340	1.018	94.094 $\pm$ 0.746	0.134 $\pm$ 0.000	100.000 $\pm$ 0.000
QML	beta44	0.11	0.120	0.010 $\pm$ 0.001	0.095 $\pm$ 0.012	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.160 $\pm$ 0.007	0.400 $\pm$ 0.010	0.120	0.010	9.35	0.0448	0.0459	0.996	94.094 $\pm$ 0.746	0.167 $\pm$ 0.001	80.280 $\pm$ 1.259
QML	beta45	0.11	0.112	0.002 $\pm$ 0.001	0.016 $\pm$ 0.011	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.122 $\pm$ 0.005	0.349 $\pm$ 0.009	0.113	0.003	2.47	0.0386	0.0387	0.961	94.094 $\pm$ 0.746	0.145 $\pm$ 0.001	83.684 $\pm$ 1.169
QML	beta46	0.08	0.065	-0.015 $\pm$ 0.001	-0.182 $\pm$ 0.011	0.001 $\pm$ 0.000	0.027 $\pm$ 0.001	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.150 $\pm$ 0.007	0.388 $\pm$ 0.011	0.066	-0.014	-17.02	0.0251	0.0286	1.038	92.492 $\pm$ 0.834	0.112 $\pm$ 0.001	67.067 $\pm$ 1.487
QML	beta47	0.08	0.064	-0.016 $\pm$ 0.001	-0.199 $\pm$ 0.011	0.001 $\pm$ 0.000	0.028 $\pm$ 0.001	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	0.160 $\pm$ 0.007	0.400 $\pm$ 0.010	0.064	-0.016	-19.86	0.0270	0.0313	1.014	90.090 $\pm$ 0.945	0.110 $\pm$ 0.001	64.765 $\pm$ 1.511
QML	beta51	0.16	0.181	0.021 $\pm$ 0.001	0.128 $\pm$ 0.008	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.086 $\pm$ 0.004	0.294 $\pm$ 0.008	0.180	0.020	12.46	0.0400	0.0447	1.057	93.994 $\pm$ 0.752	0.175 $\pm$ 0.000	98.699 $\pm$ 0.359
QML	beta52	0.16	0.140	-0.020 $\pm$ 0.001	-0.122 $\pm$ 0.008	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.072 $\pm$ 0.003	0.268 $\pm$ 0.007	0.140	-0.020	-12.47	0.0374	0.0423	0.998	90.991 $\pm$ 0.906	0.149 $\pm$ 0.000	95.696 $\pm$ 0.642
QML	beta53	0.16	0.136	-0.024 $\pm$ 0.001	-0.151 $\pm$ 0.008	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.079 $\pm$ 0.003	0.281 $\pm$ 0.006	0.134	-0.026	-16.15	0.0375	0.0455	1.018	90.591 $\pm$ 0.924	0.152 $\pm$ 0.000	94.494 $\pm$ 0.722
QML	beta54	0.16	0.155	-0.005 $\pm$ 0.001	-0.031 $\pm$ 0.007	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.049 $\pm$ 0.002	0.222 $\pm$ 0.006	0.155	-0.005	-2.87	0.0359	0.0362	1.010	95.095 $\pm$ 0.683	0.140 $\pm$ 0.000	99.099 $\pm$ 0.299
QML	beta55	0.08	0.092	0.012 $\pm$ 0.001	0.153 $\pm$ 0.018	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.356 $\pm$ 0.016	0.597 $\pm$ 0.016	0.093	0.013	15.67	0.0440	0.0457	1.004	93.694 $\pm$ 0.769	0.182 $\pm$ 0.001	50.851 $\pm$ 1.582
QML	beta56	0.08	0.079	-0.001 $\pm$ 0.001	-0.017 $\pm$ 0.016	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.269 $\pm$ 0.012	0.519 $\pm$ 0.014	0.078	-0.002	-2.90	0.0402	0.0403	1.023	94.795 $\pm$ 0.703	0.166 $\pm$ 0.001	45.145 $\pm$ 1.574
QML	beta57	0.06	0.041	-0.019 $\pm$ 0.001	-0.312 $\pm$ 0.015	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.326 $\pm$ 0.013	0.571 $\pm$ 0.013	0.040	-0.020	-32.88	0.0285	0.0347	0.997	88.488 $\pm$ 1.010	0.112 $\pm$ 0.001	29.429 $\pm$ 1.442
QML	beta58	0.06	0.048	-0.012 $\pm$ 0.001	-0.193 $\pm$ 0.014	0.001 $\pm$ 0.000	0.026 $\pm$ 0.001	0.001 $\pm$ 0.000	0.028 $\pm$ 0.001	0.225 $\pm$ 0.010	0.474 $\pm$ 0.013	0.048	-0.012	-19.21	0.0247	0.0272	0.958	89.590 $\pm$ 0.966	0.098 $\pm$ 0.000	50.951 $\pm$ 1.582
UPI	beta41	0.20	0.202	0.002 $\pm$ 0.001	0.012 $\pm$ 0.006	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.040 $\pm$ 0.002	0.199 $\pm$ 0.006	0.203	0.003	1.32	0.0378	0.0379	0.938	93.994 $\pm$ 0.752	0.146 $\pm$ 0.000	99.800 $\pm$ 0.141
UPI	beta42	0.20	0.202	0.002 $\pm$ 0.001	0.012 $\pm$ 0.006	0.001 $\pm$ 0.000	0.039 $\pm$ 0.001	0.001 $\pm$ 0.000	0.039 $\pm$ 0.001	0.037 $\pm$ 0.002	0.193 $\pm$ 0.005	0.202	0.002	0.86	0.0388	0.0388	0.998	94.294 $\pm$ 0.734	0.151 $\pm$ 0.000	100.000 $\pm$ 0.000
UPI	beta43	0.20	0.202	0.002 $\pm$ 0.001	0.011 $\pm$ 0.005	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.030 $\pm$ 0.001	0.173 $\pm$ 0.005	0.202	0.002	0.96	0.0352	0.0353	1.026	96.396 $\pm$ 0.590	0.139 $\pm$ 0.000	100.000 $\pm</$

Table S166

Simulation 2 - Condition 22: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.6
(continued)

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.072	-0.008 ± 0.001	-0.100 ± 0.013	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.177 ± 0.008	0.421 ± 0.012	0.071	-0.009	-11.85	0.0309	0.0324	1.010	94.595 ± 0.715	0.130 ± 0.001	60.761 ± 1.545
UPI	beta51	0.16	0.169	0.009 ± 0.001	0.054 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.069 ± 0.003	0.262 ± 0.007	0.168	0.008	5.17	0.0390	0.0398	1.056	95.696 ± 0.642	0.170 ± 0.000	97.898 ± 0.454
UPI	beta52	0.16	0.160	-0.000 ± 0.001	-0.001 ± 0.007	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.052 ± 0.002	0.228 ± 0.006	0.158	-0.002	-0.98	0.0357	0.0357	0.997	94.695 ± 0.709	0.143 ± 0.000	99.099 ± 0.299
UPI	beta53	0.16	0.158	-0.002 ± 0.001	-0.013 ± 0.007	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.056 ± 0.002	0.236 ± 0.006	0.157	-0.003	-1.86	0.0371	0.0372	1.021	95.896 ± 0.628	0.151 ± 0.000	98.999 ± 0.315
UPI	beta54	0.16	0.158	-0.002 ± 0.001	-0.014 ± 0.007	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.052 ± 0.002	0.229 ± 0.006	0.157	-0.003	-1.86	0.0375	0.0376	1.006	95.395 ± 0.663	0.144 ± 0.000	98.799 ± 0.345
UPI	beta55	0.08	0.084	0.004 ± 0.001	0.047 ± 0.018	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.311 ± 0.015	0.558 ± 0.016	0.083	0.003	3.80	0.0430	0.0431	0.991	94.595 ± 0.715	0.173 ± 0.001	48.148 ± 1.581
UPI	beta56	0.08	0.079	-0.001 ± 0.001	-0.009 ± 0.016	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.263 ± 0.013	0.513 ± 0.015	0.078	-0.002	-2.29	0.0389	0.0389	1.022	94.895 ± 0.696	0.164 ± 0.001	47.347 ± 1.580
UPI	beta57	0.06	0.054	-0.006 ± 0.001	-0.098 ± 0.018	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.319 ± 0.016	0.565 ± 0.017	0.051	-0.009	-14.20	0.0311	0.0322	0.942	93.293 ± 0.791	0.123 ± 0.001	39.339 ± 1.546
UPI	beta58	0.06	0.056	-0.004 ± 0.001	-0.065 ± 0.016	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.254 ± 0.012	0.504 ± 0.015	0.055	-0.005	-8.42	0.0290	0.0294	0.968	93.093 ± 0.802	0.114 ± 0.001	50.050 ± 1.582

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S167

Simulation 2 - Condition 23: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.8

Method	Param.	True	Mean-based										Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE	
LSAM	beta41	0.20	0.200	-0.000 ± 0.002	-0.002 ± 0.008	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.061 ± 0.003	0.247 ± 0.007	0.201	0.001	0.39	0.0489	0.0489	0.971	94.800 ± 0.702	0.188 ± 0.000	98.600 ± 0.372	
LSAM	beta42	0.20	0.198	-0.002 ± 0.002	-0.008 ± 0.008	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.060 ± 0.003	0.244 ± 0.007	0.197	-0.003	-1.54	0.0482	0.0483	0.993	94.700 ± 0.708	0.190 ± 0.000	98.400 ± 0.397	
LSAM	beta43	0.20	0.200	-0.000 ± 0.001	-0.001 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.055 ± 0.002	0.234 ± 0.006	0.200	0.000	0.11	0.0474	0.0474	0.970	95.000 ± 0.689	0.178 ± 0.000	99.600 ± 0.200	
LSAM	beta44	0.11	0.114	0.004 ± 0.002	0.038 ± 0.015	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	0.241 ± 0.011	0.491 ± 0.013	0.116	0.006	5.26	0.0540	0.0543	0.946	93.200 ± 0.796	0.200 ± 0.001	61.400 ± 1.539	
LSAM	beta45	0.11	0.111	0.001 ± 0.002	0.009 ± 0.014	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.199 ± 0.010	0.446 ± 0.013	0.112	0.002	2.01	0.0461	0.0462	0.933	93.200 ± 0.796	0.179 ± 0.001	67.100 ± 1.486	
LSAM	beta46	0.08	0.078	-0.002 ± 0.001	-0.023 ± 0.016	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.243 ± 0.011	0.493 ± 0.013	0.077	-0.003	-3.30	0.0393	0.0394	0.918	92.800 ± 0.817	0.142 ± 0.001	58.700 ± 1.557	
LSAM	beta47	0.08	0.077	-0.003 ± 0.001	-0.039 ± 0.015	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.213 ± 0.010	0.462 ± 0.013	0.077	-0.003	-4.12	0.0360	0.0362	0.968	93.000 ± 0.807	0.140 ± 0.001	58.900 ± 1.556	
LSAM	beta51	0.16	0.163	0.003 ± 0.002	0.016 ± 0.011	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.123 ± 0.006	0.351 ± 0.010	0.165	0.005	2.94	0.0562	0.0564	1.011	95.100 ± 0.683	0.223 ± 0.001	82.400 ± 1.204	
LSAM	beta52	0.16	0.159	-0.001 ± 0.001	-0.006 ± 0.009	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.087 ± 0.004	0.295 ± 0.008	0.160	0.000	-0.21	0.0464	0.0464	1.014	94.600 ± 0.715	0.187 ± 0.000	92.100 ± 0.853	
LSAM	beta53	0.16	0.161	0.001 ± 0.002	0.005 ± 0.010	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.094 ± 0.004	0.307 ± 0.008	0.161	0.001	0.77	0.0499	0.0500	1.014	95.600 ± 0.649	0.195 ± 0.001	90.500 ± 0.927	
LSAM	beta54	0.16	0.157	-0.003 ± 0.002	-0.016 ± 0.009	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.090 ± 0.004	0.300 ± 0.008	0.157	-0.003	-2.14	0.0481	0.0482	0.985	94.600 ± 0.715	0.185 ± 0.000	91.000 ± 0.905	
LSAM	beta55	0.08	0.083	0.003 ± 0.002	0.034 ± 0.023	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.509 ± 0.023	0.714 ± 0.019	0.083	0.003	3.97	0.0574	0.0575	0.954	93.700 ± 0.768	0.213 ± 0.001	35.100 ± 1.509	
LSAM	beta56	0.08	0.083	0.003 ± 0.002	0.034 ± 0.021	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	0.431 ± 0.022	0.657 ± 0.019	0.082	0.002	2.57	0.0509	0.0510	0.991	94.900 ± 0.696	0.204 ± 0.001	38.000 ± 1.535	
LSAM	beta57	0.06	0.059	-0.001 ± 0.001	-0.016 ± 0.020	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.384 ± 0.017	0.619 ± 0.017	0.059	-0.001	-2.40	0.0376	0.0376	0.963	94.200 ± 0.739	0.140 ± 0.001	38.700 ± 1.540	
LSAM	beta58	0.06	0.057	-0.003 ± 0.001	-0.049 ± 0.018	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.324 ± 0.015	0.570 ± 0.016	0.056	-0.004	-6.37	0.0341	0.0343	0.952	93.500 ± 0.780	0.127 ± 0.001	43.400 ± 1.567	
QML	beta41	0.20	0.196	-0.004 ± 0.002	-0.019 ± 0.008	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.059 ± 0.003	0.243 ± 0.007	0.197	-0.003	-1.65	0.0480	0.0481	0.969	94.400 ± 0.727	0.184 ± 0.001	98.800 ± 0.344	
QML	beta42	0.20	0.197	-0.003 ± 0.002	-0.017 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.057 ± 0.003	0.239 ± 0.007	0.195	-0.005	-2.44	0.0471	0.0474	0.996	94.800 ± 0.702	0.186 ± 0.001	98.500 ± 0.384	
QML	beta43	0.20	0.195	-0.005 ± 0.001	-0.024 ± 0.007	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.054 ± 0.002	0.232 ± 0.006	0.195	-0.005	-2.27	0.0475	0.0477	0.973	94.500 ± 0.721	0.176 ± 0.000	99.500 ± 0.223	
QML	beta44	0.11	0.111	0.001 ± 0.002	0.013 ± 0.015	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.231 ± 0.010	0.480 ± 0.013	0.113	0.003	2.96	0.0528	0.0529	0.955	93.300 ± 0.791	0.198 ± 0.001	60.200 ± 1.548	
QML	beta45	0.11	0.110	-0.000 ± 0.002	-0.001 ± 0.014	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.193 ± 0.009	0.440 ± 0.013	0.111	0.001	0.62	0.0458	0.0458	0.935	92.600 ± 0.828	0.177 ± 0.001	67.800 ± 1.478	
QML	beta46	0.08	0.078	-0.002 ± 0.001	-0.030 ± 0.015	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.226 ± 0.010	0.476 ± 0.013	0.077	-0.003	-3.30	0.0385	0.0386	0.926	93.100 ± 0.801	0.138 ± 0.001	61.400 ± 1.539	
QML	beta47	0.08	0.076	-0.004 ± 0.001	-0.052 ± 0.014	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.198 ± 0.010	0.445 ± 0.013	0.075	-0.005	-5.70	0.0344	0.0347	0.975	93.100 ± 0.801	0.135 ± 0.001	60.600 ± 1.545	
QML	beta51	0.16	0.168	0.008 ± 0.002	0.049 ± 0.011	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.128 ± 0.006	0.358 ± 0.010	0.171	0.011	6.79	0.0554	0.0564	1.024	95.300 ± 0.669	0.228 ± 0.001	83.400 ± 1.177	
QML	beta52	0.16	0.142	-0.018 ± 0.002	-0.114 ± 0.010	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.112 ± 0.005	0.334 ± 0.009	0.141	-0.019	-12.16	0.0507	0.0543	1.013	94.100 ± 0.745	0.200 ± 0.001	78.900 ± 1.290	
QML	beta53	0.16	0.140	-0.020 ± 0.002	-0.124 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.055 ± 0.001	0.116 ± 0.005	0.341 ± 0.009	0.142	-0.018	-11.39	0.0528	0.0558	1.019	93.400 ± 0.785	0.203 ± 0.001	77.300 ± 1.325	
QML	beta54	0.16	0.157	-0.003 ± 0.002	-0.017 ± 0.009	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.089 ± 0.004	0.298 ± 0.008	0.156	-0.004	-2.46	0.0481	0.0483	0.985	95.000 ± 0.689	0.184 ± 0.000	91.600 ± 0.877	
QML	beta55	0.08	0.086	0.006 ± 0.002	0.074 ± 0.023	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.530 ± 0.023	0.728 ± 0.020	0.085	0.005	6.42	0.0582	0.0584	0.959	94.000 ± 0.751	0.218 ± 0.001	36.500 ± 1.522	
QML	beta56	0.08	0.083	0.003 ± 0.002	0.038 ± 0.021	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	0.431 ± 0.021	0.657 ± 0.019	0.084	0.004	5.28	0.0506	0.0508	1.002	95.200 ± 0.676	0.206 ± 0.001	37.100 ± 1.528	
QML	beta57	0.06	0.055	-0.005 ± 0.001	-0.078 ± 0.019	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.368 ± 0.016	0.607 ± 0.016	0.055	-0.005	-8.85	0.0374	0.0377	0.979	94.300 ± 0.733	0.139 ± 0.001	35.900 ± 1.517	
QML	beta58	0.06	0.055	-0.005 ± 0.001	-0.086 ± 0.017	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.313 ± 0.015	0.559 ± 0.016	0.054	-0.006	-9.63	0.0329	0.0334	0.955	93.700 ± 0.768	0.124 ± 0.001	42.100 ± 1.561	
UPI	beta41	0.20	0.206	0.006 ± 0.002	0.029 ± 0.008	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.062 ± 0.003	0.249 ± 0.007	0.207	0.007	3.60	0.0489	0.0494	0.924	93.000 ± 0.807	0.179 ± 0.000	99.100 ± 0.299	
UPI	beta42	0.20	0.203	0.003 ± 0.002	0.016 ± 0.008	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.060 ± 0.003	0.244 ± 0.007	0.201	0.001	0.73	0.0480	0.0480	0.967	93.700 ± 0.768	0.185 ± 0.000	98.900 ± 0.330	
UPI	beta43	0.20	0.202	0.002 ± 0.001	0.008 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.055 ± 0.002	0.235 ± 0.006	0.201	0.001	0.74	0.0479	0.0480	0.961	94.500 ± 0.721	0.177 ± 0.000	99.700 ± 0.173	
UPI	beta44	0.11	0.114	0.004 ± 0.002	0.034 ± 0.016	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.055 ± 0.001	0.246 ± 0.011	0.496 ± 0.013	0.115	0.005	4.99	0.0555	0.0558	0.948	93.800 ± 0.763	0.202 ± 0.001	61.100 ± 1.542	
UPI	beta45	0.11	0.111	0.001 ± 0.002	0.008 ± 0.014	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.203 ± 0.010	0.450 ± 0.013	0.113	0.003	2.54	0.0462	0.0463	0.937	93.300 ± 0.791	0.182 ± 0.001	66.400 ± 1.494	
UPI	beta46	0.08	0.080	-0.000 ± 0.001	-0.003 ± 0.016	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.252 ± 0.012	0.502 ± 0.014	0.079	-0.001	-0.97	0.0401	0.0401	0.915	92.400 ± 0.838	0.144 ± 0.001	58.000 ± 1.564	

Table S167

Simulation 2 - Condition 23: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.8
(continued)

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.079	-0.001 ± 0.001	-0.017 ± 0.015	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.216 ± 0.011	0.465 ± 0.014	0.079	-0.001	-1.51	0.0347	0.0347	0.977	94.300 ± 0.733	0.142 ± 0.001	58.400 ± 1.559
UPI	beta51	0.16	0.164	0.004 ± 0.002	0.026 ± 0.011	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.123 ± 0.006	0.351 ± 0.010	0.166	0.006	3.74	0.0550	0.0553	1.011	94.700 ± 0.708	0.222 ± 0.001	84.100 ± 1.156
UPI	beta52	0.16	0.163	0.003 ± 0.001	0.020 ± 0.009	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.087 ± 0.004	0.295 ± 0.008	0.164	0.004	2.63	0.0466	0.0468	0.998	94.800 ± 0.702	0.184 ± 0.000	93.300 ± 0.791
UPI	beta53	0.16	0.162	0.002 ± 0.002	0.011 ± 0.010	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.093 ± 0.004	0.305 ± 0.008	0.163	0.003	1.80	0.0496	0.0497	1.017	95.800 ± 0.634	0.195 ± 0.001	91.000 ± 0.905
UPI	beta54	0.16	0.160	-0.000 ± 0.002	-0.001 ± 0.009	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.090 ± 0.004	0.300 ± 0.008	0.158	-0.002	-0.97	0.0478	0.0478	0.976	94.700 ± 0.708	0.184 ± 0.000	92.300 ± 0.843
UPI	beta55	0.08	0.081	0.001 ± 0.002	0.016 ± 0.022	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.499 ± 0.022	0.706 ± 0.019	0.081	0.001	1.20	0.0542	0.0542	0.963	93.200 ± 0.796	0.213 ± 0.001	34.000 ± 1.498
UPI	beta56	0.08	0.083	0.003 ± 0.002	0.039 ± 0.021	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	0.429 ± 0.021	0.655 ± 0.019	0.082	0.002	3.04	0.0499	0.0500	0.999	94.800 ± 0.702	0.205 ± 0.001	36.600 ± 1.523
UPI	beta57	0.06	0.060	0.000 ± 0.001	0.005 ± 0.020	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.389 ± 0.018	0.624 ± 0.017	0.061	0.001	0.85	0.0380	0.0380	0.970	94.200 ± 0.739	0.142 ± 0.001	38.000 ± 1.535
UPI	beta58	0.06	0.059	-0.001 ± 0.001	-0.018 ± 0.018	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.339 ± 0.016	0.583 ± 0.016	0.058	-0.002	-3.70	0.0356	0.0356	0.952	94.200 ± 0.739	0.130 ± 0.001	42.200 ± 1.562

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S168

Simulation 2 - Condition 24: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.8

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LSAM	beta41	0.20	0.198	-0.002 $\pm$ 0.001	-0.012 $\pm$ 0.005	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.022 $\pm$ 0.001	0.149 $\pm$ 0.004	0.198	-0.002	-1.21	0.0314	0.0315	1.015	95.395 $\pm$ 0.663	0.119 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta42	0.20	0.198	-0.002 $\pm$ 0.001	-0.009 $\pm$ 0.005	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.024 $\pm$ 0.001	0.154 $\pm$ 0.004	0.199	-0.001	-0.36	0.0327	0.0327	0.993	95.796 $\pm$ 0.635	0.120 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta43	0.20	0.200	0.000 $\pm$ 0.001	0.001 $\pm$ 0.005	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.021 $\pm$ 0.001	0.143 $\pm$ 0.004	0.200	0.000	-0.10	0.0271	0.0271	1.000	94.595 $\pm$ 0.715	0.112 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta44	0.11	0.112	0.002 $\pm$ 0.001	0.021 $\pm$ 0.010	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.093 $\pm$ 0.004	0.306 $\pm$ 0.008	0.113	0.003	2.58	0.0337	0.0338	0.963	93.994 $\pm$ 0.752	0.127 $\pm$ 0.000	91.692 $\pm$ 0.873
LSAM	beta45	0.11	0.112	0.002 $\pm$ 0.001	0.017 $\pm$ 0.008	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.069 $\pm$ 0.003	0.263 $\pm$ 0.008	0.113	0.003	2.93	0.0278	0.0280	0.995	95.295 $\pm$ 0.670	0.112 $\pm$ 0.000	97.397 $\pm$ 0.504
LSAM	beta46	0.08	0.076	-0.004 $\pm$ 0.001	-0.053 $\pm$ 0.009	0.001 $\pm$ 0.000	0.023 $\pm$ 0.001	0.001 $\pm$ 0.000	0.023 $\pm$ 0.001	0.085 $\pm$ 0.004	0.291 $\pm$ 0.008	0.076	-0.004	-5.50	0.0228	0.0233	0.989	94.094 $\pm$ 0.746	0.089 $\pm$ 0.000	90.891 $\pm$ 0.910
LSAM	beta47	0.08	0.076	-0.004 $\pm$ 0.001	-0.050 $\pm$ 0.009	0.001 $\pm$ 0.000	0.023 $\pm$ 0.001	0.001 $\pm$ 0.000	0.023 $\pm$ 0.001	0.085 $\pm$ 0.004	0.291 $\pm$ 0.008	0.076	-0.004	-5.62	0.0235	0.0239	0.979	94.494 $\pm$ 0.722	0.088 $\pm$ 0.000	91.491 $\pm$ 0.883
LSAM	beta51	0.16	0.165	0.005 $\pm$ 0.001	0.031 $\pm$ 0.007	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.056 $\pm$ 0.003	0.237 $\pm$ 0.007	0.164	0.004	2.20	0.0375	0.0376	0.951	93.393 $\pm$ 0.786	0.140 $\pm$ 0.000	99.099 $\pm$ 0.299
LSAM	beta52	0.16	0.158	-0.002 $\pm$ 0.001	-0.015 $\pm$ 0.006	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.038 $\pm$ 0.002	0.194 $\pm$ 0.005	0.158	-0.002	-1.54	0.0305	0.0306	0.968	94.494 $\pm$ 0.722	0.118 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta53	0.16	0.158	-0.002 $\pm$ 0.001	-0.015 $\pm$ 0.006	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	0.041 $\pm$ 0.002	0.203 $\pm$ 0.006	0.158	-0.002	-1.55	0.0321	0.0322	0.969	94.695 $\pm$ 0.709	0.123 $\pm$ 0.000	99.800 $\pm$ 0.141
LSAM	beta54	0.16	0.158	-0.002 $\pm$ 0.001	-0.015 $\pm$ 0.006	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.036 $\pm$ 0.002	0.189 $\pm$ 0.005	0.157	-0.003	-2.00	0.0301	0.0303	0.988	95.095 $\pm$ 0.683	0.117 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta55	0.08	0.084	0.004 $\pm$ 0.001	0.047 $\pm$ 0.014	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.206 $\pm$ 0.009	0.454 $\pm$ 0.013	0.083	0.003	4.23	0.0353	0.0355	0.936	92.392 $\pm$ 0.839	0.132 $\pm$ 0.000	68.969 $\pm$ 1.464
LSAM	beta56	0.08	0.080	0.000 $\pm$ 0.001	0.003 $\pm$ 0.013	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.174 $\pm$ 0.008	0.417 $\pm$ 0.012	0.079	-0.001	-0.71	0.0330	0.0330	0.976	93.894 $\pm$ 0.758	0.128 $\pm$ 0.000	69.469 $\pm$ 1.457
LSAM	beta57	0.06	0.056	-0.004 $\pm$ 0.001	-0.075 $\pm$ 0.012	0.001 $\pm$ 0.000	0.023 $\pm$ 0.001	0.001 $\pm$ 0.000	0.023 $\pm$ 0.001	0.152 $\pm$ 0.007	0.390 $\pm$ 0.011	0.056	-0.004	-6.07	0.0239	0.0241	0.972	93.393 $\pm$ 0.786	0.088 $\pm$ 0.000	68.268 $\pm$ 1.473
LSAM	beta58	0.06	0.056	-0.004 $\pm$ 0.001	-0.069 $\pm$ 0.011	0.000 $\pm$ 0.000	0.020 $\pm$ 0.000	0.000 $\pm$ 0.000	0.020 $\pm$ 0.001	0.116 $\pm$ 0.005	0.341 $\pm$ 0.010	0.056	-0.004	-7.38	0.0195	0.0200	1.011	95.395 $\pm$ 0.663	0.080 $\pm$ 0.000	80.080 $\pm$ 1.264
QML	beta41	0.20	0.194	-0.006 $\pm$ 0.001	-0.029 $\pm$ 0.005	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.022 $\pm$ 0.001	0.149 $\pm$ 0.004	0.194	-0.006	-2.97	0.0308	0.0314	1.016	95.596 $\pm$ 0.649	0.117 $\pm$ 0.000	99.900 $\pm$ 0.100
QML	beta42	0.20	0.197	-0.003 $\pm$ 0.001	-0.017 $\pm$ 0.005	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.023 $\pm$ 0.001	0.152 $\pm$ 0.004	0.197	-0.003	-1.51	0.0319	0.0321	0.992	95.395 $\pm$ 0.663	0.118 $\pm$ 0.000	100.000 $\pm$ 0.000
QML	beta43	0.20	0.196	-0.004 $\pm$ 0.001	-0.022 $\pm$ 0.004	0.001 $\pm$ 0.000	0.028 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.021 $\pm$ 0.001	0.143 $\pm$ 0.004	0.195	-0.005	-2.36	0.0270	0.0275	1.001	94.494 $\pm$ 0.722	0.111 $\pm$ 0.000	100.000 $\pm$ 0.000
QML	beta44	0.11	0.110	-0.000 $\pm$ 0.001	-0.002 $\pm$ 0.010	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.092 $\pm$ 0.004	0.303 $\pm$ 0.008	0.111	0.001	0.68	0.0333	0.0333	0.963	93.894 $\pm$ 0.758	0.126 $\pm$ 0.000	90.290 $\pm$ 0.937
QML	beta45	0.11	0.111	0.001 $\pm$ 0.001	0.012 $\pm$ 0.008	0.001 $\pm$ 0.000	0.028 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.067 $\pm$ 0.003	0.259 $\pm$ 0.007	0.113	0.003	2.37	0.0274	0.0275	1.001	95.996 $\pm$ 0.620	0.112 $\pm$ 0.000	97.497 $\pm$ 0.494
QML	beta46	0.08	0.076	-0.004 $\pm$ 0.001	-0.054 $\pm$ 0.009	0.000 $\pm$ 0.000	0.022 $\pm$ 0.001	0.001 $\pm$ 0.000	0.023 $\pm$ 0.001	0.081 $\pm$ 0.004	0.284 $\pm$ 0.008	0.076	-0.004	-4.90	0.0228	0.0232	0.992	94.194 $\pm$ 0.740	0.087 $\pm$ 0.000	91.992 $\pm$ 0.859
QML	beta47	0.08	0.075	-0.005 $\pm$ 0.001	-0.057 $\pm$ 0.009	0.000 $\pm$ 0.000	0.022 $\pm$ 0.000	0.001 $\pm$ 0.000	0.023 $\pm$ 0.001	0.081 $\pm$ 0.003	0.285 $\pm$ 0.007	0.075	-0.005	-6.33	0.0234	0.0239	0.981	94.494 $\pm$ 0.722	0.086 $\pm$ 0.000	92.593 $\pm$ 0.829
QML	beta51	0.16	0.170	0.010 $\pm$ 0.001	0.061 $\pm$ 0.008	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.061 $\pm$ 0.003	0.246 $\pm$ 0.007	0.168	0.008	5.07	0.0386	0.0394	0.955	92.893 $\pm$ 0.813	0.143 $\pm$ 0.000	99.199 $\pm$ 0.282
QML	beta52	0.16	0.141	-0.019 $\pm$ 0.001	-0.121 $\pm$ 0.007	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.057 $\pm$ 0.002	0.238 $\pm$ 0.006	0.140	-0.020	-12.47	0.0317	0.0375	0.968	90.190 $\pm$ 0.941	0.125 $\pm$ 0.000	98.699 $\pm$ 0.359
QML	beta53	0.16	0.138	-0.022 $\pm$ 0.001	-0.140 $\pm$ 0.007	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.064 $\pm$ 0.003	0.252 $\pm$ 0.006	0.137	-0.023	-14.21	0.0333	0.0403	0.971	88.388 $\pm$ 1.014	0.128 $\pm$ 0.000	98.098 $\pm$ 0.432
QML	beta54	0.16	0.158	-0.002 $\pm$ 0.001	-0.015 $\pm$ 0.006	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.035 $\pm$ 0.001	0.188 $\pm$ 0.005	0.156	-0.004	-2.20	0.0297	0.0299	0.988	95.295 $\pm$ 0.670	0.116 $\pm$ 0.000	100.000 $\pm$ 0.000
QML	beta55	0.08	0.087	0.007 $\pm$ 0.001	0.087 $\pm$ 0.015	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.221 $\pm$ 0.010	0.470 $\pm$ 0.013	0.086	0.006	7.72	0.0371	0.0376	0.942	92.492 $\pm$ 0.834	0.137 $\pm$ 0.000	69.469 $\pm$ 1.457
QML	beta56	0.08	0.081	0.001 $\pm$ 0.001	0.010 $\pm$ 0.013	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.180 $\pm$ 0.008	0.424 $\pm$ 0.012	0.081	0.001	0.79	0.0333	0.0333	0.977	93.894 $\pm$ 0.758	0.130 $\pm$ 0.000	68.569 $\pm$ 1.469
QML	beta57	0.06	0.053	-0.007 $\pm$ 0.001	-0.124 $\pm$ 0.012	0.001 $\pm$ 0.000	0.023 $\pm$ 0.001	0.001 $\pm$ 0.000	0.024 $\pm$ 0.001	0.159 $\pm$ 0.007	0.398 $\pm$ 0.011	0.053	-0.007	-11.33	0.0233	0.0243	0.978	93.493 $\pm$ 0.780	0.087 $\pm$ 0.001	65.065 $\pm$ 1.508
QML	beta58	0.06	0.054	-0.006 $\pm$ 0.001	-0.096 $\pm$ 0.010	0.000 $\pm$ 0.000	0.020 $\pm$ 0.000	0.000 $\pm$												

Table S168

Simulation 2 - Condition 24: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.8
(continued)

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.078	-0.002 ± 0.001	-0.021 ± 0.009	0.001 ± 0.000	0.024 ± 0.001	0.001 ± 0.000	0.024 ± 0.001	0.087 ± 0.004	0.294 ± 0.008	0.078	-0.002	-2.90	0.0237	0.0238	0.966	94.094 ± 0.746	0.089 ± 0.000	92.192 ± 0.849
UPI	beta51	0.16	0.165	0.005 ± 0.001	0.029 ± 0.007	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.056 ± 0.003	0.237 ± 0.007	0.163	0.003	2.12	0.0373	0.0374	0.945	92.993 ± 0.808	0.140 ± 0.000	99.099 ± 0.299
UPI	beta52	0.16	0.160	0.000 ± 0.001	0.001 ± 0.006	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.037 ± 0.002	0.193 ± 0.005	0.160	0.000	-0.09	0.0299	0.0299	0.955	93.193 ± 0.797	0.116 ± 0.000	100.000 ± 0.000
UPI	beta53	0.16	0.158	-0.002 ± 0.001	-0.011 ± 0.006	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.041 ± 0.002	0.203 ± 0.006	0.158	-0.002	-1.21	0.0319	0.0319	0.966	94.695 ± 0.709	0.123 ± 0.000	99.700 ± 0.173
UPI	beta54	0.16	0.159	-0.001 ± 0.001	-0.003 ± 0.006	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.035 ± 0.001	0.188 ± 0.005	0.159	-0.001	-0.86	0.0297	0.0298	0.984	94.895 ± 0.696	0.116 ± 0.000	100.000 ± 0.000
UPI	beta55	0.08	0.082	0.002 ± 0.001	0.030 ± 0.014	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.203 ± 0.009	0.450 ± 0.013	0.082	0.002	2.17	0.0344	0.0344	0.938	92.593 ± 0.829	0.132 ± 0.000	66.867 ± 1.489
UPI	beta56	0.08	0.080	0.000 ± 0.001	0.002 ± 0.013	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.172 ± 0.008	0.415 ± 0.011	0.079	-0.001	-1.26	0.0329	0.0329	0.979	94.394 ± 0.728	0.127 ± 0.000	69.570 ± 1.456
UPI	beta57	0.06	0.058	-0.002 ± 0.001	-0.038 ± 0.012	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	0.150 ± 0.007	0.387 ± 0.011	0.058	-0.002	-3.71	0.0229	0.0230	0.977	94.394 ± 0.728	0.088 ± 0.000	72.272 ± 1.416
UPI	beta58	0.06	0.057	-0.003 ± 0.001	-0.043 ± 0.011	0.000 ± 0.000	0.020 ± 0.000	0.000 ± 0.000	0.021 ± 0.001	0.118 ± 0.006	0.343 ± 0.010	0.057	-0.003	-4.43	0.0204	0.0206	1.008	95.696 ± 0.642	0.081 ± 0.000	80.380 ± 1.256

Note. MCSE = Monte Carlo Standard Error shown as \u000b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S169

Simulation 2 - Condition 25: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, $N = 400$, Reliability = 0.4

Method	Param.	True	Mean-based										Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE	
LSAM	beta41	0.20	0.199	-0.001 ± 0.010	-0.005 ± 0.049	0.059 ± 0.004	0.243 ± 0.009	0.059 ± 0.004	0.243 ± 0.010	1.473 ± 0.103	1.214 ± 0.049	0.198	-0.002	-1.04	0.2239	0.2239	1.073	98.058 ± 0.555	1.022 ± 0.024	13.592 ± 1.379	
LSAM	beta42	0.20	0.197	-0.003 ± 0.010	-0.013 ± 0.050	0.061 ± 0.004	0.246 ± 0.008	0.061 ± 0.004	0.246 ± 0.010	1.514 ± 0.103	1.230 ± 0.049	0.193	-0.007	-3.47	0.2233	0.2234	1.051	97.896 ± 0.577	1.014 ± 0.021	12.945 ± 1.350	
LSAM	beta43	0.20	0.195	-0.005 ± 0.003	-0.024 ± 0.017	0.007 ± 0.000	0.085 ± 0.003	0.007 ± 0.000	0.085 ± 0.003	0.182 ± 0.011	0.427 ± 0.016	0.195	-0.005	-2.49	0.0822	0.0824	1.004	95.793 ± 0.808	0.336 ± 0.003	67.638 ± 1.882	
LSAM	beta44	0.11	0.130	0.020 ± 0.005	0.183 ± 0.047	0.016 ± 0.001	0.128 ± 0.005	0.017 ± 0.001	0.129 ± 0.006	1.377 ± 0.105	1.173 ± 0.050	0.129	0.019	17.13	0.1129	0.1145	1.011	96.602 ± 0.729	0.506 ± 0.011	19.417 ± 1.591	
LSAM	beta45	0.11	0.106	-0.004 ± 0.004	-0.038 ± 0.041	0.012 ± 0.001	0.112 ± 0.004	0.012 ± 0.001	0.112 ± 0.004	1.029 ± 0.067	1.014 ± 0.039	0.097	-0.013	-12.06	0.0958	0.0967	0.983	94.498 ± 0.917	0.430 ± 0.009	15.858 ± 1.469	
LSAM	beta46	0.08	0.063	-0.017 ± 0.004	-0.208 ± 0.050	0.010 ± 0.001	0.100 ± 0.004	0.010 ± 0.001	0.102 ± 0.004	1.616 ± 0.114	1.271 ± 0.051	0.063	-0.017	-21.78	0.0865	0.0882	1.084	95.307 ± 0.851	0.426 ± 0.012	15.049 ± 1.438	
LSAM	beta47	0.08	0.063	-0.017 ± 0.004	-0.210 ± 0.050	0.010 ± 0.001	0.100 ± 0.004	0.010 ± 0.001	0.101 ± 0.004	1.602 ± 0.126	1.266 ± 0.056	0.062	-0.018	-23.02	0.0818	0.0838	1.008	96.764 ± 0.712	0.395 ± 0.010	17.638 ± 1.533	
LSAM	beta51	0.16	0.174	0.014 ± 0.004	0.085 ± 0.024	0.009 ± 0.001	0.095 ± 0.003	0.009 ± 0.001	0.095 ± 0.003	0.356 ± 0.020	0.597 ± 0.021	0.170	0.010	6.34	0.0863	0.0868	0.980	94.337 ± 0.930	0.363 ± 0.003	47.573 ± 2.009	
LSAM	beta52	0.16	0.215	0.055 ± 0.011	0.345 ± 0.067	0.071 ± 0.005	0.267 ± 0.009	0.074 ± 0.005	0.272 ± 0.011	2.899 ± 0.201	1.703 ± 0.068	0.207	0.047	29.18	0.2351	0.2397	1.079	97.411 ± 0.639	1.130 ± 0.029	10.680 ± 1.242	
LSAM	beta53	0.16	0.146	-0.014 ± 0.004	-0.089 ± 0.028	0.012 ± 0.001	0.110 ± 0.004	0.012 ± 0.001	0.111 ± 0.004	0.483 ± 0.032	0.695 ± 0.027	0.142	-0.018	-11.44	0.1114	0.1129	1.100	97.249 ± 0.658	0.476 ± 0.006	23.786 ± 1.713	
LSAM	beta54	0.16	0.145	-0.015 ± 0.008	-0.096 ± 0.053	0.044 ± 0.003	0.210 ± 0.007	0.044 ± 0.003	0.210 ± 0.008	1.727 ± 0.120	1.314 ± 0.053	0.143	-0.017	-10.47	0.1923	0.1930	1.091	96.278 ± 0.761	0.897 ± 0.022	12.298 ± 1.321	
LSAM	beta55	0.08	0.131	0.051 ± 0.005	0.641 ± 0.059	0.014 ± 0.001	0.117 ± 0.004	0.016 ± 0.001	0.128 ± 0.005	2.549 ± 0.176	1.596 ± 0.062	0.123	0.043	53.79	0.1027	0.1113	1.014	94.175 ± 0.942	0.465 ± 0.012	27.184 ± 1.790	
LSAM	beta56	0.08	0.073	-0.007 ± 0.003	-0.085 ± 0.032	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.003	0.641 ± 0.047	0.801 ± 0.033	0.076	-0.004	-5.30	0.0566	0.0567	0.964	93.689 ± 0.978	0.241 ± 0.005	33.819 ± 1.903	
LSAM	beta57	0.06	-0.004	-0.064 ± 0.007	-1.059 ± 0.109	0.027 ± 0.002	0.163 ± 0.006	0.031 ± 0.002	0.175 ± 0.007	8.509 ± 0.632	2.917 ± 0.120	0.006	-0.054	-89.91	0.1329	0.1434	1.021	94.013 ± 0.954	0.653 ± 0.020	2.104 ± 0.577	
LSAM	beta58	0.06	0.048	-0.012 ± 0.003	-0.192 ± 0.053	0.006 ± 0.000	0.079 ± 0.003	0.006 ± 0.000	0.080 ± 0.003	1.757 ± 0.133	1.326 ± 0.057	0.048	-0.012	-20.79	0.0695	0.0706	1.077	94.984 ± 0.878	0.333 ± 0.010	15.049 ± 1.438	
QML	beta41	0.20	0.229	0.029 ± 0.004	0.143 ± 0.019	0.009 ± 0.000	0.093 ± 0.003	0.009 ± 0.001	0.097 ± 0.003	0.235 ± 0.014	0.485 ± 0.017	0.229	0.029	14.40	0.0994	0.1035	1.030	95.469 ± 0.837	0.375 ± 0.010	69.741 ± 1.848	
QML	beta42	0.20	0.223	0.023 ± 0.004	0.114 ± 0.019	0.009 ± 0.001	0.094 ± 0.003	0.009 ± 0.001	0.097 ± 0.003	0.235 ± 0.014	0.485 ± 0.017	0.221	0.021	10.51	0.0985	0.1007	1.024	96.117 ± 0.777	0.379 ± 0.004	66.505 ± 1.899	
QML	beta43	0.20	0.190	-0.010 ± 0.003	-0.049 ± 0.016	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.078 ± 0.003	0.153 ± 0.009	0.391 ± 0.014	0.195	-0.005	-2.66	0.0754	0.0756	0.991	94.175 ± 0.942	0.302 ± 0.006	73.139 ± 1.783	
QML	beta44	0.11	0.075	-0.035 ± 0.005	-0.316 ± 0.044	0.015 ± 0.001	0.121 ± 0.004	0.016 ± 0.001	0.126 ± 0.005	1.310 ± 0.093	1.145 ± 0.046	0.077	-0.033	-29.89	0.1156	0.1202	1.001	93.366 ± 1.001	0.475 ± 0.007	12.783 ± 1.343	
QML	beta45	0.11	0.097	-0.013 ± 0.004	-0.115 ± 0.035	0.009 ± 0.001	0.097 ± 0.003	0.010 ± 0.001	0.098 ± 0.004	0.791 ± 0.050	0.889 ± 0.033	0.098	-0.012	-10.51	0.1001	0.1008	0.985	92.557 ± 1.056	0.375 ± 0.005	21.197 ± 1.644	
QML	beta46	0.08	0.137	0.057 ± 0.003	0.714 ± 0.037	0.005 ± 0.000	0.073 ± 0.003	0.009 ± 0.001	0.093 ± 0.003	1.352 ± 0.089	1.163 ± 0.041	0.132	0.052	65.12	0.0740	0.0905	0.990	90.777 ± 1.164	0.285 ± 0.005	51.618 ± 2.010	
QML	beta47	0.08	0.132	0.052 ± 0.003	0.656 ± 0.036	0.005 ± 0.000	0.071 ± 0.003	0.008 ± 0.001	0.089 ± 0.003	1.224 ± 0.087	1.107 ± 0.042	0.120	0.040	49.46	0.0629	0.0743	0.954	92.233 ± 1.077	0.267 ± 0.005	55.663 ± 1.998	
QML	beta51	0.16	0.193	0.033 ± 0.005	0.208 ± 0.030	0.014 ± 0.001	0.119 ± 0.004	0.015 ± 0.001	0.124 ± 0.004	0.599 ± 0.034	0.774 ± 0.026	0.192	0.032	20.00	0.1238	0.1279	0.957	92.557 ± 1.056	0.448 ± 0.005	42.557 ± 1.989	
QML	beta52	0.16	0.148	-0.012 ± 0.004	-0.073 ± 0.028	0.012 ± 0.001	0.110 ± 0.004	0.012 ± 0.001	0.110 ± 0.004	0.476 ± 0.031	0.690 ± 0.026	0.149	-0.011	-6.93	0.1079	0.1085	1.000	94.660 ± 0.904	0.431 ± 0.007	28.479 ± 1.815	
QML	beta53	0.16	0.117	-0.043 ± 0.004	-0.267 ± 0.026	0.011 ± 0.001	0.105 ± 0.004	0.013 ± 0.001	0.113 ± 0.004	0.499 ± 0.031	0.706 ± 0.025	0.113	-0.047	-29.06	0.1052	0.1150	1.000	94.013 ± 0.954	0.410 ± 0.004	18.123 ± 1.550	
QML	beta54	0.16	0.174	0.014 ± 0.004	0.085 ± 0.024	0.009 ± 0.001	0.095 ± 0.003	0.009 ± 0.001	0.096 ± 0.004	0.362 ± 0.023	0.601 ± 0.023	0.172	0.012	7.69	0.0993	0.1000	1.008	94.822 ± 0.891	0.377 ± 0.008	45.469 ± 2.003	
QML	beta55	0.08	0.104	0.024 ± 0.006	0.296 ± 0.078	0.024 ± 0.001	0.156 ± 0.004	0.025 ± 0.001	0.157 ± 0.005	3.861 ± 0.216	1.965 ± 0.067	0.094	0.014	17.72	0.1693	0.1699	0.813	83.333 ± 1.499	0.496 ± 0.010	25.728 ± 1.758	
QML	beta56	0.08	0.112	0.032 ± 0.003	0.399 ± 0.041	0.007 ± 0.000	0.081 ± 0.003	0.008 ± 0.000	0.087 ± 0.003	1.176 ± 0.068	1.084 ± 0.037	0.113	0.033	41.85	0.0773	0.0843	0.885	88.188 ± 1.298	0.280 ± 0.004	44.337 ± 1.998	
QML	beta57	0.06	0.104	0.044 ± 0.006	0.737 ± 0.098	0.021 ± 0.001	0.146 ± 0.004	0.023 ± 0.001	0.152 ± 0.005	6.426 ± 0.388	2.535 ± 0.090	0.106	0.046	76.48	0.1508	0.1576	0.806	82.201 ± 1.539	0.460 ± 0.009	25.243 ± 1.747	
QML	beta58	0.06	0.098	0.038 ± 0.002	0.636 ± 0.038	0.003 ± 0.000	0.057 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	1.299 ± 0.085	1.140 ± 0.040	0.095	0.035	58.76	0.0573	0.0673	0.974	90.777 ± 1.164	0.217 ± 0.004	48.058 ± 2.010	
UPI	beta41	0.20	0.285	0.085 ± 0.012	0.426 ± 0.062	0.096 ± 0.007	0.309 ± 0.011	0.103 ± 0.007	0.321 ± 0.013	2.570 ± 0.180	1.603 ± 0.064	0.276	0.076	37.95	0.2871	0.2970	1.341	98.058 ± 0.555	1.626 ± 0.065	11.812 ± 1.298	
UPI	beta42	0.20	0.290	0.090 ± 0.011	0.451 ± 0.055	0.076 ± 0.006	0.276 ± 0.011	0.084 ± 0.006	0.290 ± 0.012	2.101 ± 0.162	1.449 ± 0.062	0.266	0.066	32.79	0.2652	0.2732	1.260	98.544 ± 0.482	1.362 ± 0.061	14.887 ± 1.432	
UPI	beta43	0.20	0.171	-0.029 ± 0.004	-0.143 ± 0.021	0.011 ± 0.001	0.105 ± 0.004	0.012 ± 0.001	0.109 ± 0.004	0.295 ± 0.020	0.543 ± 0.022	0.176	-0.024	-12.20	0.1041	0.1070	1.221	96.278 ± 0.761	0.502 ± 0.016	40.129 ± 1.972	
UPI	beta44	0.11	0.173	0.063 ± 0.008	0.569 ± 0.076	0.043 ± 0.003	0.208 ± 0.008	0.047 ± 0.004	0.217 ± 0.010	3.877 ± 0.310	1.969 ± 0.087	0.162	0.052	46.92	0.1824	0.1896	1.540	98.220 ± 0.532	1.253 ± 0.065	5.016 ± 0.978	
UPI	beta45	0.11	0.109	-0.001 ± 0.007	-0.008 ± 0.066	0.032 ± 0.003	0.180 ± 0.007	0.032 ± 0.003	0.180 ± 0.008	2.673 ± 0.219	1.635 ± 0.075	0.098	-0.012	-11.33	0.1479	0.1484	1.467	97.573 ± 0.619	1.035 ± 0.050	3.883 ± 0.777	
UPI	beta46	0.08	0.029	-0.051 ± 0.006	-0.633 ± 0.069	0.019 ± 0.002	0.137 ± 0.006	0.021 ± 0.002	0.146 ± 0.006	3.334 ± 0.268	1.826 ± 0.081	0.040	-0.040	-50.27	0.1226	0.1290	1.354	97.573 ± 0.619	0.728 ± 0.031	8.091 ± 1.092	

Table S169

Simulation 2 - Condition 25: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.4
(continued)

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.029	-0.051 ± 0.005	-0.636 ± 0.057	0.013 ± 0.001	0.113 ± 0.005	0.015 ± 0.001	0.124 ± 0.006	2.414 ± 0.203	1.554 ± 0.071	0.040	-0.040	-49.56	0.0984	0.1061	1.244	96.440 ± 0.745	0.554 ± 0.026	9.709 ± 1.191
UPI	beta51	0.16	0.204	0.044 ± 0.004	0.276 ± 0.026	0.011 ± 0.001	0.105 ± 0.003	0.013 ± 0.001	0.114 ± 0.004	0.508 ± 0.028	0.713 ± 0.023	0.209	0.049	30.57	0.1077	0.1183	1.211	95.793 ± 0.808	0.500 ± 0.013	42.395 ± 1.988
UPI	beta52	0.16	0.252	0.092 ± 0.014	0.573 ± 0.085	0.114 ± 0.009	0.337 ± 0.014	0.122 ± 0.010	0.349 ± 0.016	4.762 ± 0.401	2.182 ± 0.101	0.214	0.054	33.57	0.3176	0.3221	1.532	99.029 ± 0.394	2.025 ± 0.101	7.605 ± 1.066
UPI	beta53	0.16	0.107	-0.053 ± 0.006	-0.330 ± 0.040	0.025 ± 0.002	0.159 ± 0.006	0.028 ± 0.002	0.167 ± 0.007	1.089 ± 0.081	1.044 ± 0.043	0.109	-0.051	-32.17	0.1537	0.1621	1.407	98.058 ± 0.555	0.875 ± 0.035	7.929 ± 1.087
UPI	beta54	0.16	0.248	0.088 ± 0.018	0.551 ± 0.115	0.208 ± 0.017	0.456 ± 0.018	0.216 ± 0.017	0.464 ± 0.021	8.424 ± 0.672	2.902 ± 0.129	0.224	0.064	39.71	0.4060	0.4110	1.551	99.191 ± 0.360	2.774 ± 0.145	3.074 ± 0.694
UPI	beta55	0.08	0.149	0.069 ± 0.007	0.864 ± 0.085	0.029 ± 0.002	0.170 ± 0.007	0.034 ± 0.003	0.183 ± 0.008	5.245 ± 0.402	2.290 ± 0.097	0.132	0.052	65.42	0.1553	0.1639	1.531	98.544 ± 0.482	1.019 ± 0.051	9.547 ± 1.182
UPI	beta56	0.08	0.076	-0.004 ± 0.004	-0.054 ± 0.044	0.008 ± 0.001	0.088 ± 0.003	0.008 ± 0.001	0.088 ± 0.004	1.218 ± 0.093	1.104 ± 0.048	0.075	-0.005	-6.12	0.0850	0.0851	1.362	97.411 ± 0.639	0.471 ± 0.017	15.372 ± 1.451
UPI	beta57	0.06	-0.025	-0.085 ± 0.008	-1.414 ± 0.135	0.040 ± 0.003	0.201 ± 0.008	0.048 ± 0.004	0.218 ± 0.010	13.213 ± 1.078	3.635 ± 0.161	-0.008	-0.068	-113.32	0.1721	0.1850	1.511	98.544 ± 0.482	1.191 ± 0.060	0.324 ± 0.228
UPI	beta58	0.06	0.006	-0.054 ± 0.007	-0.892 ± 0.111	0.027 ± 0.002	0.165 ± 0.007	0.030 ± 0.002	0.174 ± 0.008	8.368 ± 0.692	2.893 ± 0.131	0.015	-0.045	-75.18	0.1382	0.1454	1.514	98.706 ± 0.455	0.980 ± 0.053	3.398 ± 0.729

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S170

Simulation 2 - Condition 26: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, $N = 1000$, Reliability = 0.4

Method	Param.	True	Mean-based										Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE	
LSAM	beta41	0.20	0.218	0.018 ± 0.006	0.088 ± 0.031	0.033 ± 0.002	0.181 ± 0.005	0.033 ± 0.002	0.182 ± 0.006	0.826 ± 0.049	0.909 ± 0.031	0.211	0.011	5.71	0.1660	0.1663	1.037	97.275 ± 0.560	0.736 ± 0.013	26.303 ± 1.516	
LSAM	beta42	0.20	0.203	0.003 ± 0.007	0.016 ± 0.033	0.036 ± 0.002	0.191 ± 0.005	0.036 ± 0.002	0.191 ± 0.006	0.908 ± 0.051	0.953 ± 0.031	0.202	0.002	1.23	0.1804	0.1804	1.026	95.972 ± 0.677	0.767 ± 0.012	21.564 ± 1.416	
LSAM	beta43	0.20	0.199	-0.001 ± 0.002	-0.005 ± 0.010	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.077 ± 0.004	0.277 ± 0.009	0.199	-0.001	-0.47	0.0535	0.0535	1.015	95.498 ± 0.714	0.221 ± 0.002	93.720 ± 0.835	
LSAM	beta44	0.11	0.139	0.029 ± 0.003	0.265 ± 0.026	0.007 ± 0.000	0.084 ± 0.002	0.008 ± 0.000	0.089 ± 0.003	0.650 ± 0.038	0.806 ± 0.027	0.137	0.027	24.83	0.0797	0.0843	0.960	94.313 ± 0.797	0.315 ± 0.004	45.261 ± 1.713	
LSAM	beta45	0.11	0.116	0.006 ± 0.003	0.058 ± 0.023	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.074 ± 0.003	0.447 ± 0.028	0.668 ± 0.024	0.113	0.003	2.94	0.0698	0.0699	0.958	93.483 ± 0.850	0.275 ± 0.004	39.692 ± 1.684	
LSAM	beta46	0.08	0.058	-0.022 ± 0.002	-0.279 ± 0.030	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.074 ± 0.003	0.846 ± 0.053	0.920 ± 0.032	0.060	-0.020	-24.75	0.0601	0.0633	0.997	94.668 ± 0.773	0.274 ± 0.006	22.630 ± 1.440	
LSAM	beta47	0.08	0.060	-0.020 ± 0.002	-0.246 ± 0.031	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.074 ± 0.003	0.856 ± 0.053	0.925 ± 0.032	0.059	-0.021	-25.86	0.0633	0.0666	0.973	93.720 ± 0.835	0.272 ± 0.005	24.408 ± 1.479	
LSAM	beta51	0.16	0.175	0.015 ± 0.002	0.096 ± 0.013	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	0.148 ± 0.007	0.384 ± 0.011	0.176	0.016	9.93	0.0584	0.0605	0.997	94.787 ± 0.765	0.233 ± 0.001	84.716 ± 1.239	
LSAM	beta52	0.16	0.231	0.071 ± 0.007	0.445 ± 0.044	0.043 ± 0.003	0.207 ± 0.006	0.048 ± 0.003	0.219 ± 0.008	1.865 ± 0.117	1.366 ± 0.048	0.215	0.055	34.30	0.1852	0.1932	1.115	97.275 ± 0.560	0.903 ± 0.020	20.261 ± 1.384	
LSAM	beta53	0.16	0.141	-0.019 ± 0.003	-0.116 ± 0.017	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.081 ± 0.002	0.253 ± 0.013	0.503 ± 0.015	0.143	-0.017	-10.60	0.0815	0.0833	1.052	96.090 ± 0.667	0.323 ± 0.003	44.194 ± 1.709	
LSAM	beta54	0.16	0.138	-0.022 ± 0.005	-0.137 ± 0.034	0.025 ± 0.002	0.158 ± 0.005	0.025 ± 0.002	0.159 ± 0.006	0.991 ± 0.063	0.995 ± 0.036	0.139	-0.021	-13.25	0.1488	0.1503	1.091	97.038 ± 0.584	0.675 ± 0.015	17.773 ± 1.316	
LSAM	beta55	0.08	0.138	0.058 ± 0.003	0.729 ± 0.034	0.006 ± 0.000	0.080 ± 0.002	0.010 ± 0.001	0.099 ± 0.003	1.519 ± 0.081	1.232 ± 0.036	0.136	0.056	70.10	0.0705	0.0901	1.076	90.758 ± 0.997	0.335 ± 0.008	48.223 ± 1.720	
LSAM	beta56	0.08	0.070	-0.010 ± 0.001	-0.124 ± 0.016	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.229 ± 0.013	0.479 ± 0.016	0.070	-0.010	-12.81	0.0367	0.0381	1.030	94.194 ± 0.805	0.149 ± 0.003	51.777 ± 1.720	
LSAM	beta57	0.06	-0.009	-0.069 ± 0.004	-1.143 ± 0.065	0.013 ± 0.001	0.114 ± 0.004	0.018 ± 0.001	0.133 ± 0.004	4.920 ± 0.293	2.218 ± 0.072	-0.004	-0.064	-106.79	0.0995	0.1184	1.101	91.943 ± 0.937	0.493 ± 0.012	1.896 ± 0.469	
LSAM	beta58	0.06	0.051	-0.009 ± 0.002	-0.158 ± 0.033	0.003 ± 0.000	0.057 ± 0.002	0.003 ± 0.000	0.057 ± 0.002	0.916 ± 0.056	0.957 ± 0.033	0.050	-0.010	-16.50	0.0487	0.0497	1.048	94.431 ± 0.789	0.233 ± 0.006	21.327 ± 1.410	
QML	beta41	0.20	0.239	0.039 ± 0.002	0.197 ± 0.010	0.003 ± 0.000	0.058 ± 0.001	0.005 ± 0.000	0.070 ± 0.002	0.123 ± 0.005	0.350 ± 0.009	0.240	0.040	19.87	0.0577	0.0700	0.999	89.218 ± 1.068	0.227 ± 0.001	97.393 ± 0.548	
QML	beta42	0.20	0.221	0.021 ± 0.002	0.107 ± 0.010	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.100 ± 0.005	0.317 ± 0.009	0.219	0.019	9.61	0.0608	0.0638	0.985	93.128 ± 0.871	0.231 ± 0.001	95.024 ± 0.749	
QML	beta43	0.20	0.192	-0.008 ± 0.002	-0.038 ± 0.008	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.056 ± 0.003	0.236 ± 0.007	0.193	-0.007	-3.58	0.0463	0.0468	1.005	94.194 ± 0.805	0.184 ± 0.001	97.749 ± 0.511	
QML	beta44	0.11	0.076	-0.034 ± 0.003	-0.305 ± 0.026	0.007 ± 0.000	0.082 ± 0.002	0.008 ± 0.000	0.088 ± 0.003	0.645 ± 0.033	0.803 ± 0.024	0.080	-0.030	-26.95	0.0824	0.0875	0.920	89.218 ± 1.068	0.295 ± 0.004	23.697 ± 1.464	
QML	beta45	0.11	0.106	-0.004 ± 0.002	-0.040 ± 0.019	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	0.321 ± 0.018	0.566 ± 0.018	0.107	-0.003	-3.09	0.0633	0.0634	0.962	93.246 ± 0.864	0.235 ± 0.002	44.076 ± 1.709	
QML	beta46	0.08	0.136	0.056 ± 0.002	0.696 ± 0.019	0.002 ± 0.000	0.045 ± 0.001	0.005 ± 0.000	0.072 ± 0.002	0.804 ± 0.034	0.897 ± 0.020	0.134	0.054	67.41	0.0441	0.0697	0.990	74.052 ± 1.509	0.176 ± 0.003	87.322 ± 1.145	
QML	beta47	0.08	0.129	0.049 ± 0.002	0.617 ± 0.019	0.002 ± 0.000	0.044 ± 0.001	0.004 ± 0.000	0.066 ± 0.001	0.678 ± 0.029	0.823 ± 0.019	0.125	0.045	55.78	0.0445	0.0630	0.935	75.829 ± 1.474	0.160 ± 0.002	88.507 ± 1.098	
QML	beta51	0.16	0.184	0.024 ± 0.003	0.150 ± 0.016	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.080 ± 0.002	0.251 ± 0.013	0.501 ± 0.015	0.182	0.022	13.75	0.0746	0.0778	0.982	91.351 ± 0.968	0.295 ± 0.006	70.616 ± 1.568	
QML	beta52	0.16	0.153	-0.007 ± 0.002	-0.046 ± 0.015	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.199 ± 0.010	0.446 ± 0.013	0.153	-0.007	-4.61	0.0708	0.0712	0.988	93.602 ± 0.842	0.275 ± 0.004	60.782 ± 1.681	
QML	beta53	0.16	0.117	-0.043 ± 0.002	-0.269 ± 0.014	0.004 ± 0.000	0.064 ± 0.001	0.006 ± 0.000	0.077 ± 0.002	0.232 ± 0.010	0.482 ± 0.012	0.118	-0.042	-26.53	0.0661	0.0785	0.993	88.744 ± 1.088	0.249 ± 0.001	46.090 ± 1.716	
QML	beta54	0.16	0.173	0.013 ± 0.002	0.081 ± 0.012	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.131 ± 0.006	0.362 ± 0.010	0.172	0.012	7.50	0.0595	0.0607	1.046	94.905 ± 0.757	0.231 ± 0.001	84.834 ± 1.235	
QML	beta55	0.08	0.084	0.004 ± 0.004	0.046 ± 0.051	0.014 ± 0.001	0.119 ± 0.003	0.014 ± 0.001	0.119 ± 0.003	2.225 ± 0.097	1.491 ± 0.041	0.070	-0.010	-11.97	0.1266	0.1269	0.832	77.014 ± 1.448	0.389 ± 0.012	27.488 ± 1.537	
QML	beta56	0.08	0.117	0.037 ± 0.002	0.464 ± 0.021	0.002 ± 0.000	0.048 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.577 ± 0.027	0.759 ± 0.020	0.116	0.036	45.29	0.0474	0.0596	0.956	81.517 ± 1.336	0.180 ± 0.003	71.090 ± 1.560	
QML	beta57	0.06	0.130	0.070 ± 0.004	1.161 ± 0.064	0.012 ± 0.001	0.112 ± 0.002	0.017 ± 0.001	0.132 ± 0.003	4.804 ± 0.197	2.192 ± 0.053	0.139	0.079	131.67	0.1145	0.1391	0.865	71.564 ± 1.553	0.378 ± 0.012	44.668 ± 1.711	
QML	beta58	0.06	0.100	0.040 ± 0.001	0.668 ± 0.020	0.001 ± 0.000	0.035 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.785 ± 0.036	0.886 ± 0.021	0.100	0.040	66.53	0.0345	0.0528	0.977	77.725 ± 1.432	0.134 ± 0.001	82.109 ± 1.319	
UPI	beta41	0.20	0.241	0.041 ± 0.006	0.204 ± 0.028	0.026 ± 0.001	0.162 ± 0.004	0.028 ± 0.002	0.167 ± 0.005	0.698 ± 0.039	0.835 ± 0.027	0.240	0.040	19.91	0.1564	0.1614	1.152	96.919 ± 0.595	0.732 ± 0.021	30.332 ± 1.582	
UPI	beta42	0.20	0.235	0.035 ± 0.005	0.176 ± 0.026	0.022 ± 0.001	0.149 ± 0.004	0.023 ± 0.001	0.153 ± 0.005	0.584 ± 0.033	0.764 ± 0.025	0.236	0.036	17.80	0.1440	0.1484	1.114	96.919 ± 0.595	0.650 ± 0.011	34.005 ± 1.631	
UPI	beta43	0.20	0.176	-0.024 ± 0.002	-0.119 ± 0.010	0.003 ± 0.000	0.056 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.093 ± 0.005	0.305 ± 0.009	0.176	-0.024	-12.07	0.0550	0.0601	1.072	93.009 ± 0.878	0.236 ± 0.004	84.716 ± 1.239	
UPI	beta44	0.11	0.146	0.036 ± 0.004	0.330 ± 0.033	0.011 ± 0.001	0.105 ± 0.003	0.012 ± 0.001	0.111 ± 0.004	1.026 ± 0.064	1.013 ± 0.035	0.136	0.026	23.89	0.0992	0.1026	1.232	97.156 ± 0.572	0.509 ± 0.022	24.171 ± 1.471	
UPI	beta45	0.11	0.129	0.019 ± 0.003	0.171 ± 0.030	0.009 ± 0.001	0.095 ± 0.003	0.009 ± 0.001	0.097 ± 0.004	0.771 ± 0.053	0.878 ± 0.034	0.120	0.010	9.21	0.0878	0.0884	1.206	95.616 ± 0.705	0.448 ± 0.021	24.052 ± 1.474	
UPI	beta46	0.08	0.048	-0.032 ± 0.002	-0.394 ± 0.028	0.004 ± 0.000	0.066 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.834 ± 0.050	0.913 ± 0.030	0.050	-0.030	-37.91	0.0619	0.0690	1.131	95.616 ± 0.705	0.292 ± 0.010	18.720 ± 1.343	

Table S170

Simulation 2 - Condition 26: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.4
(continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.049	-0.031 ± 0.002	-0.387 ± 0.024	0.003 ± 0.000	0.055 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	0.626 ± 0.038	0.791 ± 0.026	0.052	-0.028	-34.52	0.0506	0.0576	1.141	93.365 ± 0.857	0.247 ± 0.006	20.024 ± 1.377
UPI	beta51	0.16	0.196	0.036 ± 0.002	0.225 ± 0.014	0.004 ± 0.000	0.063 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.206 ± 0.010	0.453 ± 0.013	0.198	0.038	23.85	0.0613	0.0722	1.073	92.891 ± 0.885	0.265 ± 0.003	85.545 ± 1.210
UPI	beta52	0.16	0.212	0.052 ± 0.007	0.327 ± 0.044	0.042 ± 0.003	0.206 ± 0.006	0.045 ± 0.003	0.212 ± 0.008	1.763 ± 0.111	1.328 ± 0.047	0.207	0.047	29.26	0.1797	0.1857	1.400	98.223 ± 0.455	1.131 ± 0.050	16.825 ± 1.288
UPI	beta53	0.16	0.102	-0.058 ± 0.003	-0.363 ± 0.021	0.009 ± 0.000	0.096 ± 0.003	0.013 ± 0.001	0.112 ± 0.003	0.494 ± 0.026	0.703 ± 0.021	0.107	-0.053	-33.19	0.0932	0.1072	1.194	92.773 ± 0.891	0.451 ± 0.014	24.645 ± 1.483
UPI	beta54	0.16	0.274	0.114 ± 0.011	0.710 ± 0.069	0.103 ± 0.007	0.321 ± 0.010	0.116 ± 0.008	0.340 ± 0.012	4.522 ± 0.299	2.126 ± 0.078	0.245	0.085	53.23	0.2533	0.2672	1.350	98.815 ± 0.372	1.698 ± 0.072	9.005 ± 0.985
UPI	beta55	0.08	0.132	0.052 ± 0.003	0.656 ± 0.040	0.008 ± 0.001	0.092 ± 0.003	0.011 ± 0.001	0.106 ± 0.003	1.750 ± 0.095	1.323 ± 0.040	0.126	0.046	57.65	0.0830	0.0950	1.424	96.919 ± 0.595	0.513 ± 0.021	28.318 ± 1.551
UPI	beta56	0.08	0.079	-0.001 ± 0.002	-0.018 ± 0.021	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.002	0.372 ± 0.023	0.610 ± 0.021	0.079	-0.001	-1.11	0.0442	0.0442	1.245	95.616 ± 0.705	0.238 ± 0.009	42.062 ± 1.699
UPI	beta57	0.06	-0.003	-0.063 ± 0.004	-1.051 ± 0.062	0.012 ± 0.001	0.108 ± 0.004	0.016 ± 0.001	0.125 ± 0.004	4.314 ± 0.252	2.077 ± 0.066	0.002	-0.058	-97.05	0.0904	0.1075	1.429	96.090 ± 0.667	0.603 ± 0.027	0.474 ± 0.236
UPI	beta58	0.06	0.000	-0.060 ± 0.004	-1.000 ± 0.059	0.011 ± 0.001	0.103 ± 0.003	0.014 ± 0.001	0.119 ± 0.004	3.929 ± 0.248	1.982 ± 0.068	0.010	-0.050	-82.62	0.0821	0.0959	1.350	95.616 ± 0.705	0.544 ± 0.024	5.213 ± 0.765

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S171

Simulation 2 - Condition 27: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, $N = 400$, Reliability = 0.6

Method	Param.	True	Mean-based										Median-based					Inference		
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LSAM	beta41	0.20	0.209	0.009 $\pm$ 0.005	0.045 $\pm$ 0.023	0.019 $\pm$ 0.001	0.139 $\pm$ 0.004	0.019 $\pm$ 0.001	0.139 $\pm$ 0.004	0.481 $\pm$ 0.026	0.694 $\pm$ 0.022	0.210	0.010	5.22	0.1376	0.1380	0.961	96.179 $\pm$ 0.633	0.522 $\pm$ 0.005	39.956 $\pm$ 1.618
LSAM	beta42	0.20	0.199	-0.001 $\pm$ 0.005	-0.007 $\pm$ 0.023	0.020 $\pm$ 0.001	0.140 $\pm$ 0.004	0.020 $\pm$ 0.001	0.140 $\pm$ 0.005	0.493 $\pm$ 0.027	0.702 $\pm$ 0.023	0.202	0.002	0.76	0.1299	0.1299	0.980	96.070 $\pm$ 0.642	0.540 $\pm$ 0.006	35.808 $\pm$ 1.584
LSAM	beta43	0.20	0.204	0.004 $\pm$ 0.002	0.022 $\pm$ 0.010	0.004 $\pm$ 0.000	0.062 $\pm$ 0.001	0.004 $\pm$ 0.000	0.062 $\pm$ 0.002	0.097 $\pm$ 0.004	0.311 $\pm$ 0.009	0.204	0.004	2.20	0.0640	0.0642	0.991	95.306 $\pm$ 0.699	0.241 $\pm$ 0.001	90.830 $\pm$ 0.954
LSAM	beta44	0.11	0.131	0.021 $\pm$ 0.003	0.190 $\pm$ 0.027	0.008 $\pm$ 0.000	0.090 $\pm$ 0.003	0.009 $\pm$ 0.001	0.092 $\pm$ 0.003	0.705 $\pm$ 0.042	0.839 $\pm$ 0.028	0.127	0.017	15.72	0.0844	0.0861	0.921	92.576 $\pm$ 0.866	0.325 $\pm$ 0.004	38.755 $\pm$ 1.610
LSAM	beta45	0.11	0.115	0.005 $\pm$ 0.003	0.044 $\pm$ 0.025	0.007 $\pm$ 0.000	0.085 $\pm$ 0.002	0.007 $\pm$ 0.000	0.085 $\pm$ 0.003	0.597 $\pm$ 0.035	0.773 $\pm$ 0.026	0.109	-0.001	-0.63	0.0747	0.0747	0.869	93.341 $\pm$ 0.824	0.289 $\pm$ 0.003	39.192 $\pm$ 1.613
LSAM	beta46	0.08	0.075	-0.005 $\pm$ 0.002	-0.065 $\pm$ 0.025	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.588 $\pm$ 0.035	0.767 $\pm$ 0.026	0.075	-0.005	-6.65	0.0551	0.0554	0.885	91.812 $\pm$ 0.906	0.212 $\pm$ 0.004	39.520 $\pm$ 1.615
LSAM	beta47	0.08	0.077	-0.003 $\pm$ 0.002	-0.036 $\pm$ 0.025	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	0.572 $\pm$ 0.034	0.756 $\pm$ 0.026	0.076	-0.004	-5.24	0.0525	0.0526	0.871	92.795 $\pm$ 0.854	0.206 $\pm$ 0.003	43.886 $\pm$ 1.640
LSAM	beta51	0.16	0.165	0.005 $\pm$ 0.002	0.028 $\pm$ 0.015	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.220 $\pm$ 0.011	0.469 $\pm$ 0.014	0.167	0.007	4.51	0.0740	0.0743	0.974	94.760 $\pm$ 0.736	0.286 $\pm$ 0.001	62.336 $\pm$ 1.601
LSAM	beta52	0.16	0.165	0.005 $\pm$ 0.005	0.031 $\pm$ 0.030	0.021 $\pm$ 0.001	0.144 $\pm$ 0.004	0.021 $\pm$ 0.001	0.144 $\pm$ 0.005	0.812 $\pm$ 0.044	0.901 $\pm$ 0.029	0.162	0.002	1.16	0.1322	0.1322	0.977	96.070 $\pm$ 0.642	0.553 $\pm$ 0.005	24.017 $\pm$ 1.411
LSAM	beta53	0.16	0.161	0.001 $\pm$ 0.003	0.009 $\pm$ 0.019	0.008 $\pm$ 0.000	0.091 $\pm$ 0.002	0.008 $\pm$ 0.000	0.091 $\pm$ 0.003	0.322 $\pm$ 0.014	0.567 $\pm$ 0.016	0.165	0.005	2.98	0.0943	0.0944	0.961	94.651 $\pm$ 0.743	0.342 $\pm$ 0.002	48.472 $\pm$ 1.651
LSAM	beta54	0.16	0.156	-0.004 $\pm$ 0.004	-0.024 $\pm$ 0.024	0.013 $\pm$ 0.001	0.115 $\pm$ 0.003	0.013 $\pm$ 0.001	0.115 $\pm$ 0.003	0.516 $\pm$ 0.025	0.718 $\pm$ 0.021	0.159	-0.001	-0.85	0.1142	0.1142	1.058	96.943 $\pm$ 0.569	0.477 $\pm$ 0.004	26.528 $\pm$ 1.459
LSAM	beta55	0.08	0.095	0.015 $\pm$ 0.003	0.188 $\pm$ 0.033	0.007 $\pm$ 0.000	0.081 $\pm$ 0.002	0.007 $\pm$ 0.000	0.082 $\pm$ 0.003	1.062 $\pm$ 0.058	1.031 $\pm$ 0.033	0.094	0.014	17.25	0.0756	0.0768	0.941	92.576 $\pm$ 0.866	0.299 $\pm$ 0.004	30.786 $\pm$ 1.525
LSAM	beta56	0.08	0.078	-0.002 $\pm$ 0.002	-0.022 $\pm$ 0.020	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.002	0.378 $\pm$ 0.021	0.615 $\pm$ 0.020	0.077	-0.003	-3.26	0.0445	0.0445	0.887	92.686 $\pm$ 0.860	0.171 $\pm$ 0.002	49.454 $\pm$ 1.652
LSAM	beta57	0.06	0.049	-0.011 $\pm$ 0.003	-0.182 $\pm$ 0.053	0.009 $\pm$ 0.001	0.096 $\pm$ 0.003	0.009 $\pm$ 0.001	0.096 $\pm$ 0.003	2.567 $\pm$ 0.145	1.602 $\pm$ 0.052	0.046	-0.014	-23.10	0.0850	0.0861	0.915	94.978 $\pm$ 0.722	0.343 $\pm$ 0.005	9.061 $\pm$ 0.948
LSAM	beta58	0.06	0.059	-0.001 $\pm$ 0.002	-0.020 $\pm$ 0.025	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.574 $\pm$ 0.031	0.758 $\pm$ 0.024	0.056	-0.004	-6.06	0.0410	0.0412	0.952	93.559 $\pm$ 0.811	0.170 $\pm$ 0.003	36.245 $\pm$ 1.588
QML	beta41	0.20	0.220	0.020 $\pm$ 0.002	0.098 $\pm$ 0.012	0.006 $\pm$ 0.000	0.076 $\pm$ 0.002	0.006 $\pm$ 0.000	0.078 $\pm$ 0.002	0.152 $\pm$ 0.007	0.390 $\pm$ 0.011	0.222	0.022	10.88	0.0762	0.0792	0.960	93.559 $\pm$ 0.811	0.284 $\pm$ 0.001	83.952 $\pm$ 1.213
QML	beta42	0.20	0.211	0.011 $\pm$ 0.002	0.055 $\pm$ 0.012	0.006 $\pm$ 0.000	0.074 $\pm$ 0.002	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.141 $\pm$ 0.007	0.375 $\pm$ 0.011	0.214	0.014	7.12	0.0729	0.0743	0.993	94.869 $\pm$ 0.729	0.289 $\pm$ 0.001	81.004 $\pm$ 1.296
QML	beta43	0.20	0.195	-0.005 $\pm$ 0.002	-0.023 $\pm$ 0.009	0.003 $\pm$ 0.000	0.056 $\pm$ 0.001	0.003 $\pm$ 0.000	0.056 $\pm$ 0.002	0.079 $\pm$ 0.004	0.281 $\pm$ 0.008	0.195	-0.005	-2.46	0.0573	0.0575	0.992	94.760 $\pm$ 0.736	0.218 $\pm$ 0.001	92.576 $\pm$ 0.866
QML	beta44	0.11	0.085	-0.025 $\pm$ 0.003	-0.224 $\pm$ 0.023	0.006 $\pm$ 0.000	0.076 $\pm$ 0.002	0.006 $\pm$ 0.000	0.080 $\pm$ 0.002	0.528 $\pm$ 0.025	0.727 $\pm$ 0.020	0.087	-0.023	-21.06	0.0747	0.0782	0.951	91.594 $\pm$ 0.917	0.284 $\pm$ 0.003	24.672 $\pm$ 1.424
QML	beta45	0.11	0.103	-0.007 $\pm$ 0.002	-0.060 $\pm$ 0.020	0.005 $\pm$ 0.000	0.067 $\pm$ 0.002	0.005 $\pm$ 0.000	0.067 $\pm$ 0.002	0.376 $\pm$ 0.018	0.613 $\pm$ 0.018	0.102	-0.008	-7.28	0.0669	0.0674	0.934	93.013 $\pm$ 0.842	0.246 $\pm$ 0.002	41.376 $\pm$ 1.627
QML	beta46	0.08	0.108	0.028 $\pm$ 0.002	0.345 $\pm$ 0.019	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.003 $\pm$ 0.000	0.054 $\pm$ 0.002	0.452 $\pm$ 0.024	0.672 $\pm$ 0.020	0.107	0.027	33.61	0.0436	0.0512	0.910	88.210 $\pm$ 1.066	0.165 $\pm$ 0.002	73.144 $\pm$ 1.464
QML	beta47	0.08	0.108	0.028 $\pm$ 0.001	0.351 $\pm$ 0.018	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.003 $\pm$ 0.000	0.053 $\pm$ 0.002	0.432 $\pm$ 0.024	0.657 $\pm$ 0.020	0.106	0.026	33.11	0.0417	0.0494	0.886	86.245 $\pm$ 1.138	0.154 $\pm$ 0.002	77.511 $\pm$ 1.379
QML	beta51	0.16	0.181	0.021 $\pm$ 0.003	0.133 $\pm$ 0.016	0.006 $\pm$ 0.000	0.076 $\pm$ 0.002	0.006 $\pm$ 0.000	0.079 $\pm$ 0.002	0.242 $\pm$ 0.012	0.492 $\pm$ 0.014	0.181	0.021	13.35	0.0740	0.0770	0.998	93.996 $\pm$ 0.785	0.297 $\pm$ 0.002	67.467 $\pm$ 1.548
QML	beta52	0.16	0.144	-0.016 $\pm$ 0.003	-0.101 $\pm$ 0.016	0.006 $\pm$ 0.000	0.076 $\pm$ 0.002	0.006 $\pm$ 0.000	0.078 $\pm$ 0.002	0.238 $\pm$ 0.011	0.488 $\pm$ 0.014	0.142	-0.018	-11.28	0.0756	0.0777	0.985	93.996 $\pm$ 0.785	0.295 $\pm$ 0.001	48.472 $\pm$ 1.651
QML	beta53	0.16	0.129	-0.031 $\pm$ 0.003	-0.195 $\pm$ 0.016	0.006 $\pm$ 0.000	0.078 $\pm$ 0.002	0.007 $\pm$ 0.000	0.084 $\pm$ 0.002	0.274 $\pm$ 0.012	0.524 $\pm$ 0.014	0.131	-0.029	-18.27	0.0787	0.0840	0.969	91.703 $\pm$ 0.911	0.295 $\pm$ 0.001	41.266 $\pm$ 1.627
QML	beta54	0.16	0.162	0.002 $\pm$ 0.002	0.012 $\pm$ 0.014	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.180 $\pm$ 0.009	0.424 $\pm$ 0.012	0.162	0.002	1.20	0.0693	0.0693	1.083	96.943 $\pm$ 0.569	0.288 $\pm$ 0.001	60.371 $\pm$ 1.616
QML	beta55	0.08	0.093	0.013 $\pm$ 0.003	0.158 $\pm$ 0.033	0.006 $\pm$ 0.000	0.080 $\pm$ 0.002	0.007 $\pm$ 0.000	0.081 $\pm$ 0.002	1.031 $\pm$ 0.052	1.015 $\pm$ 0.030	0.092	0.012	14.82	0.0741	0.0751	0.960	89.410 $\pm$ 1.017	0.302 $\pm$ 0.005	30.459 $\pm$ 1.521
QML	beta56	0.08	0.094	0.014 $\pm$ 0.002	0.175 $\pm$ 0.021	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	0.003 $\pm$ 0.000	0.052 $\pm$ 0.002	0.418 $\pm$ 0.022	0.647 $\pm$ 0.020	0.093	0.013	16.48	0.0452	0.0471	0.883	89.520 $\pm$ 1.012	0.173 $\pm$ 0.002	61.245 $\pm$ 1.610
QML	beta57	0.06	0.080	0.020 $\pm$ 0.002	0.331 $\pm$ 0.040	0.005 $\pm$ 0.000	0.072 $\pm$ 0.002	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	1.562 $\pm$ 0.080	1.250 $\pm$ 0.038	0.078	0.018	30.80	0.0710	0.0734	0.995	90.721 $\pm$ 0.959	0.282 $\pm$ 0.005	25.983 $\pm$ 1.449
QML	beta58	0.06	0.081	0.021 $\pm$ 0.001	0.357 $\pm$ 0.020	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.490 $\pm$ 0.024	0.700 $\pm$ 0.019	0.079	0.019	32.29	0.0353	0.0403	0.940	88.646 $\pm$ 1.048	0.133 $\pm$ 0.002	69.214 $\pm$ 1.525
UPI	beta41	0.20	0.229	0.029 $\pm$ 0.004	0.147 $\pm$ 0.022	0.018 $\pm$ 0.001	0.135 $\pm$ 0.004	0.019 $\pm$ 0.001	0.138 $\pm$ 0.004	0.478 $\pm$ 0.027	0.691 $\pm$ 0.022	0.231	0.031	15.40	0.1273	0.1310	1.042	95.197 $\pm$ 0.707	0.552 $\pm$ 0.012	46.943 $\pm$ 1.649
UPI	beta42	0.20	0.221	0.021 $\pm$ 0.004	0.105 $\pm$ 0.022	0.018 $\pm$ 0.001	0.136 $\pm$ 0.003	0.019 $\pm$ 0.001	0.137 $\pm$ 0.004	0.472 $\pm$ 0.023	0.687 $\pm$ 0.020	0.222	0.022	10.80	0.1347	0.1364	1.057	96.397 $\pm$ 0.616	0.563 $\pm$ 0.009	41.485 $\pm$ 1.628
UPI	beta43	0.20	0.199	-0.001 $\pm$ 0.002	-0.005 $\pm$ 0.010	0.004 $\pm$ 0.000	0.063 $\pm$ 0.002	0.004 $\pm$ 0.000	0.063 $\pm$ 0.002	0.101 $\pm$ 0.005	0.317 $\pm$ 0.009	0.200	0.000	-0.19	0.0646	0.0646	0.957	93.996 $\pm$ 0.785	0.238 $\pm$ 0.002	88

Table S171

Simulation 2 - Condition 27: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.6
(continued)

Method	Param.	True	Mean-based										Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE	
UPI	beta47	0.08	0.070	-0.010 ± 0.002	-0.124 ± 0.024	0.003 ± 0.000	0.057 ± 0.002	0.003 ± 0.000	0.058 ± 0.002	0.526 ± 0.031	0.726 ± 0.024	0.073	-0.007	-8.46	0.0493	0.0497	0.974	93.231 ± 0.830	0.219 ± 0.005	40.611 ± 1.623	
UPI	beta51	0.16	0.175	0.015 ± 0.002	0.093 ± 0.015	0.005 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	0.222 ± 0.011	0.471 ± 0.014	0.175	0.015	9.67	0.0721	0.0738	0.969	94.651 ± 0.743	0.281 ± 0.001	68.996 ± 1.528	
UPI	beta52	0.16	0.184	0.024 ± 0.005	0.150 ± 0.028	0.019 ± 0.001	0.137 ± 0.004	0.019 ± 0.001	0.139 ± 0.004	0.756 ± 0.042	0.870 ± 0.028	0.184	0.024	14.77	0.1339	0.1360	1.057	97.162 ± 0.549	0.568 ± 0.011	29.913 ± 1.513	
UPI	beta53	0.16	0.155	-0.005 ± 0.003	-0.029 ± 0.018	0.008 ± 0.000	0.089 ± 0.002	0.008 ± 0.000	0.089 ± 0.003	0.309 ± 0.014	0.555 ± 0.016	0.154	-0.006	-3.97	0.0919	0.0921	0.972	95.087 ± 0.714	0.338 ± 0.002	46.397 ± 1.648	
UPI	beta54	0.16	0.155	-0.005 ± 0.004	-0.028 ± 0.026	0.015 ± 0.001	0.124 ± 0.003	0.015 ± 0.001	0.124 ± 0.004	0.601 ± 0.033	0.775 ± 0.025	0.156	-0.004	-2.42	0.1189	0.1189	1.150	97.380 ± 0.528	0.559 ± 0.010	21.943 ± 1.367	
UPI	beta55	0.08	0.099	0.019 ± 0.003	0.242 ± 0.034	0.007 ± 0.000	0.083 ± 0.003	0.007 ± 0.000	0.085 ± 0.003	1.137 ± 0.071	1.066 ± 0.037	0.096	0.016	19.89	0.0716	0.0733	1.002	95.415 ± 0.691	0.327 ± 0.007	26.965 ± 1.466	
UPI	beta56	0.08	0.077	-0.003 ± 0.002	-0.044 ± 0.020	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.002	0.377 ± 0.022	0.614 ± 0.020	0.075	-0.005	-6.32	0.0431	0.0434	0.916	92.576 ± 0.866	0.176 ± 0.003	48.144 ± 1.651	
UPI	beta57	0.06	0.039	-0.021 ± 0.003	-0.355 ± 0.050	0.008 ± 0.001	0.092 ± 0.003	0.009 ± 0.001	0.094 ± 0.003	2.455 ± 0.146	1.567 ± 0.053	0.037	-0.023	-38.87	0.0815	0.0848	1.016	95.306 ± 0.699	0.365 ± 0.009	7.751 ± 0.884	
UPI	beta58	0.06	0.058	-0.002 ± 0.002	-0.036 ± 0.027	0.002 ± 0.000	0.050 ± 0.002	0.002 ± 0.000	0.050 ± 0.002	0.682 ± 0.044	0.826 ± 0.030	0.058	-0.002	-3.33	0.0423	0.0423	1.039	94.869 ± 0.729	0.202 ± 0.006	33.297 ± 1.557	

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S172

Simulation 2 - Condition 28: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.6

Method	Param.	True	Mean-based										Median-based					Inference		
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LSAM	beta41	0.20	0.201	0.001 $\pm$ 0.002	0.006 $\pm$ 0.012	0.006 $\pm$ 0.000	0.078 $\pm$ 0.002	0.006 $\pm$ 0.000	0.078 $\pm$ 0.002	0.151 $\pm$ 0.007	0.389 $\pm$ 0.011	0.205	0.005	2.30	0.0733	0.0734	0.968	94.567 $\pm$ 0.719	0.295 $\pm$ 0.002	75.855 $\pm$ 1.357
LSAM	beta42	0.20	0.200	-0.000 $\pm$ 0.003	-0.001 $\pm$ 0.013	0.006 $\pm$ 0.000	0.079 $\pm$ 0.002	0.006 $\pm$ 0.000	0.079 $\pm$ 0.002	0.155 $\pm$ 0.007	0.394 $\pm$ 0.011	0.198	-0.002	-0.99	0.0790	0.0791	0.991	95.573 $\pm$ 0.652	0.307 $\pm$ 0.002	73.944 $\pm$ 1.392
LSAM	beta43	0.20	0.201	0.001 $\pm$ 0.001	0.005 $\pm$ 0.006	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.034 $\pm$ 0.002	0.185 $\pm$ 0.005	0.202	0.002	1.10	0.0357	0.0358	1.014	94.970 $\pm$ 0.693	0.147 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta44	0.11	0.121	0.011 $\pm$ 0.002	0.103 $\pm$ 0.014	0.003 $\pm$ 0.000	0.050 $\pm$ 0.001	0.003 $\pm$ 0.000	0.051 $\pm$ 0.002	0.217 $\pm$ 0.012	0.466 $\pm$ 0.014	0.119	0.009	8.01	0.0473	0.0481	0.930	92.052 $\pm$ 0.858	0.182 $\pm$ 0.001	74.748 $\pm$ 1.378
LSAM	beta45	0.11	0.114	0.004 $\pm$ 0.001	0.037 $\pm$ 0.014	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.185 $\pm$ 0.009	0.430 $\pm$ 0.013	0.115	0.005	4.73	0.0463	0.0466	0.913	93.058 $\pm$ 0.806	0.169 $\pm$ 0.001	76.459 $\pm$ 1.346
LSAM	beta46	0.08	0.076	-0.004 $\pm$ 0.001	-0.046 $\pm$ 0.012	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.144 $\pm$ 0.007	0.379 $\pm$ 0.011	0.076	-0.004	-5.30	0.0283	0.0286	0.932	93.260 $\pm$ 0.795	0.110 $\pm$ 0.001	75.553 $\pm$ 1.363
LSAM	beta47	0.08	0.075	-0.005 $\pm$ 0.001	-0.059 $\pm$ 0.012	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.147 $\pm$ 0.007	0.383 $\pm$ 0.011	0.074	-0.006	-7.52	0.0298	0.0304	0.906	92.354 $\pm$ 0.843	0.108 $\pm$ 0.001	75.553 $\pm$ 1.363
LSAM	beta51	0.16	0.164	0.004 $\pm$ 0.001	0.022 $\pm$ 0.009	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.083 $\pm$ 0.004	0.288 $\pm$ 0.008	0.164	0.004	2.72	0.0461	0.0463	0.970	94.064 $\pm$ 0.749	0.175 $\pm$ 0.000	95.272 $\pm$ 0.673
LSAM	beta52	0.16	0.167	0.007 $\pm$ 0.003	0.045 $\pm$ 0.016	0.006 $\pm$ 0.000	0.081 $\pm$ 0.002	0.007 $\pm$ 0.000	0.081 $\pm$ 0.002	0.255 $\pm$ 0.012	0.505 $\pm$ 0.014	0.167	0.007	4.47	0.0806	0.0809	0.997	95.070 $\pm$ 0.687	0.315 $\pm$ 0.002	55.936 $\pm$ 1.575
LSAM	beta53	0.16	0.161	0.001 $\pm$ 0.002	0.008 $\pm$ 0.011	0.003 $\pm$ 0.000	0.054 $\pm$ 0.001	0.003 $\pm$ 0.000	0.054 $\pm$ 0.002	0.114 $\pm$ 0.005	0.337 $\pm$ 0.010	0.160	0.000	0.28	0.0511	0.0511	0.972	94.668 $\pm$ 0.713	0.206 $\pm$ 0.001	86.117 $\pm$ 1.097
LSAM	beta54	0.16	0.155	-0.005 $\pm$ 0.002	-0.031 $\pm$ 0.015	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.220 $\pm$ 0.010	0.470 $\pm$ 0.013	0.153	-0.007	-4.16	0.0739	0.0742	0.959	94.465 $\pm$ 0.725	0.282 $\pm$ 0.001	58.350 $\pm$ 1.564
LSAM	beta55	0.08	0.096	0.016 $\pm$ 0.001	0.199 $\pm$ 0.017	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.331 $\pm$ 0.016	0.576 $\pm$ 0.016	0.097	0.017	20.67	0.0426	0.0457	0.940	91.851 $\pm$ 0.868	0.159 $\pm$ 0.001	66.801 $\pm$ 1.494
LSAM	beta56	0.08	0.076	-0.004 $\pm$ 0.001	-0.045 $\pm$ 0.010	0.001 $\pm$ 0.000	0.026 $\pm$ 0.001	0.001 $\pm$ 0.000	0.026 $\pm$ 0.001	0.107 $\pm$ 0.005	0.327 $\pm$ 0.010	0.076	-0.004	-4.54	0.0252	0.0255	0.899	91.247 $\pm$ 0.896	0.091 $\pm$ 0.001	86.620 $\pm$ 1.080
LSAM	beta57	0.06	0.045	-0.015 $\pm$ 0.002	-0.249 $\pm$ 0.026	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.738 $\pm$ 0.034	0.859 $\pm$ 0.024	0.043	-0.017	-28.48	0.0481	0.0510	0.951	92.052 $\pm$ 0.858	0.184 $\pm$ 0.002	18.813 $\pm$ 1.240
LSAM	beta58	0.06	0.058	-0.002 $\pm$ 0.001	-0.036 $\pm$ 0.014	0.001 $\pm$ 0.000	0.027 $\pm$ 0.001	0.001 $\pm$ 0.000	0.027 $\pm$ 0.001	0.198 $\pm$ 0.010	0.445 $\pm$ 0.013	0.057	-0.003	-5.14	0.0238	0.0240	0.927	91.751 $\pm$ 0.873	0.097 $\pm$ 0.001	67.103 $\pm$ 1.490
QML	beta41	0.20	0.217	0.017 $\pm$ 0.001	0.083 $\pm$ 0.007	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.060 $\pm$ 0.003	0.245 $\pm$ 0.006	0.217	0.017	8.30	0.0460	0.0489	0.992	93.561 $\pm$ 0.778	0.179 $\pm$ 0.001	99.396 $\pm$ 0.246
QML	beta42	0.20	0.210	0.010 $\pm$ 0.001	0.048 $\pm$ 0.007	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.054 $\pm$ 0.002	0.233 $\pm$ 0.006	0.210	0.010	5.12	0.0469	0.0480	1.020	94.869 $\pm$ 0.700	0.182 $\pm$ 0.001	99.095 $\pm$ 0.300
QML	beta43	0.20	0.193	-0.007 $\pm$ 0.001	-0.037 $\pm$ 0.005	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.031 $\pm$ 0.001	0.177 $\pm$ 0.005	0.194	-0.006	-3.10	0.0344	0.0350	1.010	93.763 $\pm$ 0.767	0.137 $\pm$ 0.000	100.000 $\pm$ 0.000
QML	beta44	0.11	0.085	-0.025 $\pm$ 0.001	-0.230 $\pm$ 0.013	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.231 $\pm$ 0.010	0.480 $\pm$ 0.012	0.085	-0.025	-23.15	0.0464	0.0529	0.957	89.940 $\pm$ 0.954	0.174 $\pm$ 0.001	49.497 $\pm$ 1.586
QML	beta45	0.11	0.106	-0.004 $\pm$ 0.001	-0.036 $\pm$ 0.012	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.147 $\pm$ 0.007	0.383 $\pm$ 0.011	0.108	-0.002	-1.63	0.0398	0.0399	0.939	92.555 $\pm$ 0.833	0.154 $\pm$ 0.001	76.962 $\pm$ 1.336
QML	beta46	0.08	0.109	0.029 $\pm$ 0.001	0.358 $\pm$ 0.010	0.001 $\pm$ 0.000	0.026 $\pm$ 0.001	0.001 $\pm$ 0.000	0.039 $\pm$ 0.001	0.232 $\pm$ 0.009	0.482 $\pm$ 0.010	0.108	0.028	34.65	0.0248	0.0372	0.963	78.773 $\pm$ 1.297	0.097 $\pm$ 0.001	99.296 $\pm$ 0.265
QML	beta47	0.08	0.109	0.029 $\pm$ 0.001	0.357 $\pm$ 0.010	0.001 $\pm$ 0.000	0.026 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.230 $\pm$ 0.009	0.479 $\pm$ 0.010	0.108	0.028	34.57	0.0250	0.0373	0.937	75.755 $\pm$ 1.359	0.094 $\pm$ 0.001	98.893 $\pm$ 0.332
QML	beta51	0.16	0.178	0.018 $\pm$ 0.002	0.114 $\pm$ 0.009	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.100 $\pm$ 0.004	0.317 $\pm$ 0.008	0.179	0.019	12.12	0.0476	0.0514	0.993	93.360 $\pm$ 0.790	0.184 $\pm$ 0.001	96.177 $\pm$ 0.608
QML	beta52	0.16	0.141	-0.019 $\pm$ 0.001	-0.116 $\pm$ 0.009	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	0.096 $\pm$ 0.004	0.309 $\pm$ 0.008	0.141	-0.019	-11.73	0.0469	0.0505	1.002	93.461 $\pm$ 0.784	0.180 $\pm$ 0.001	86.821 $\pm$ 1.073
QML	beta53	0.16	0.128	-0.032 $\pm$ 0.002	-0.198 $\pm$ 0.010	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.003 $\pm$ 0.000	0.058 $\pm$ 0.001	0.131 $\pm$ 0.006	0.362 $\pm$ 0.009	0.129	-0.031	-19.60	0.0455	0.0552	0.965	87.425 $\pm$ 1.052	0.183 $\pm$ 0.001	78.471 $\pm$ 1.304
QML	beta54	0.16	0.162	0.002 $\pm$ 0.002	0.009 $\pm$ 0.009	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.088 $\pm$ 0.004	0.297 $\pm$ 0.008	0.160	0.000	-0.19	0.0453	0.0453	0.975	94.266 $\pm$ 0.737	0.181 $\pm$ 0.001	94.366 $\pm$ 0.731
QML	beta55	0.08	0.092	0.012 $\pm$ 0.002	0.154 $\pm$ 0.020	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.413 $\pm$ 0.018	0.643 $\pm$ 0.017	0.095	0.015	18.54	0.0503	0.0525	0.978	89.738 $\pm$ 0.963	0.191 $\pm$ 0.004	53.924 $\pm$ 1.581
QML	beta56	0.08	0.094	0.014 $\pm$ 0.001	0.172 $\pm$ 0.011	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	0.158 $\pm$ 0.008	0.397 $\pm$ 0.011	0.093	0.013	16.39	0.0270	0.0300	0.908	89.336 $\pm$ 0.979	0.102 $\pm$ 0.001	90.946 $\pm$ 0.910
QML	beta57	0.06	0.082	0.022 $\pm$ 0.001	0.362 $\pm$ 0.025	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.748 $\pm$ 0.034	0.865 $\pm$ 0.023	0.079	0.019	31.68	0.0468	0.0505	0.994	89.537 $\pm$ 0.971	0.184 $\pm$ 0.003	45.573 $\pm$ 1.580
QML	beta58	0.06	0.081	0.021 $\pm$ 0.001	0.357 $\pm$ 0.012	0.000 $\pm$ 0.000	0.022 $\pm$ 0.001	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.263 $\pm$ 0.011	0.513 $\pm$ 0.011	0.081	0.021	34.61	0.0210	0.0295	0.947	82.193 $\pm$ 1.213	0.082 $\pm$ 0.001	95.674 $\pm$ 0.645
UPI	beta41	0.20	0.209	0.009 $\pm$ 0.002	0.044 $\pm$ 0.012	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.142 $\pm$ 0.006	0.377 $\pm$ 0.010	0.211	0.011	5.42	0.0749	0.0757	1.013	96.076 $\pm$ 0.616	0.297 $\pm$ 0.002	77.968 $\pm$ 1.315
UPI	beta42	0.20	0.204	0.004 $\pm$ 0.002	0.019 $\pm$ 0.012	0.006 $\pm$ 0.000	0.077 $\pm$ 0.002	0.006 $\pm$ 0.000	0.077 $\pm$ 0.002	0.150 $\pm$ 0.008	0.387 $\pm$ 0.012	0.203	0.003	1.38	0.0720	0.0720	1.020	96.278 $\pm$ 0.600	0.309 $\pm$ 0.002	75.050 $\pm$ 1.373
UPI	beta43	0.20	0.198	-0.002 $\pm$ 0.001	-0.010 $\pm$ 0.006	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.034 $\pm$ 0.002	0.185 $\pm$ 0.005	0.198	-0.002	-0.90	0.0372	0.0372	0.986	93.863 $\pm$ 0.761	0.143 $\pm$ 0.001	100.000 $\pm$

Table S172

Simulation 2 - Condition 28: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.6
(continued)

Method	Param.	True	Mean-based										Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE	
UPI	beta47	0.08	0.075	-0.005 ± 0.001	-0.059 ± 0.012	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.140 ± 0.008	0.374 ± 0.012	0.075	-0.005	-5.84	0.0271	0.0275	0.939	91.147 ± 0.901	0.109 ± 0.001	76.157 ± 1.352	
UPI	beta51	0.16	0.168	0.008 ± 0.001	0.052 ± 0.009	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.083 ± 0.004	0.288 ± 0.008	0.169	0.009	5.34	0.0459	0.0467	0.971	94.064 ± 0.749	0.173 ± 0.000	96.479 ± 0.585	
UPI	beta52	0.16	0.172	0.012 ± 0.002	0.078 ± 0.016	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.080 ± 0.002	0.247 ± 0.012	0.497 ± 0.014	0.172	0.012	7.67	0.0781	0.0791	1.036	96.076 ± 0.616	0.319 ± 0.002	57.545 ± 1.568	
UPI	beta53	0.16	0.158	-0.002 ± 0.002	-0.013 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.106 ± 0.005	0.326 ± 0.009	0.157	-0.003	-2.04	0.0515	0.0516	0.976	94.668 ± 0.713	0.200 ± 0.001	85.211 ± 1.126	
UPI	beta54	0.16	0.152	-0.008 ± 0.003	-0.048 ± 0.016	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.081 ± 0.002	0.257 ± 0.012	0.507 ± 0.014	0.152	-0.008	-4.92	0.0787	0.0791	0.989	95.272 ± 0.673	0.313 ± 0.003	51.509 ± 1.585	
UPI	beta55	0.08	0.093	0.013 ± 0.001	0.158 ± 0.018	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.341 ± 0.016	0.584 ± 0.016	0.092	0.012	15.08	0.0417	0.0434	0.974	93.561 ± 0.778	0.172 ± 0.002	60.563 ± 1.550	
UPI	beta56	0.08	0.077	-0.003 ± 0.001	-0.037 ± 0.011	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	0.112 ± 0.006	0.335 ± 0.010	0.077	-0.003	-4.26	0.0250	0.0252	0.901	90.543 ± 0.928	0.094 ± 0.001	85.312 ± 1.123	
UPI	beta57	0.06	0.045	-0.015 ± 0.002	-0.246 ± 0.026	0.002 ± 0.000	0.049 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.736 ± 0.036	0.858 ± 0.025	0.044	-0.016	-27.33	0.0469	0.0497	0.992	94.064 ± 0.749	0.192 ± 0.002	18.712 ± 1.237	
UPI	beta58	0.06	0.058	-0.002 ± 0.001	-0.034 ± 0.015	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.231 ± 0.012	0.481 ± 0.015	0.058	-0.002	-4.01	0.0248	0.0249	0.955	93.159 ± 0.801	0.108 ± 0.001	61.771 ± 1.541	

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S173

Simulation 2 - Condition 29: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, $N = 400$, Reliability = 0.8

Method	Param.	True	Mean-based										Median-based					Inference		
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LSAM	beta41	0.20	0.195	-0.005 $\pm$ 0.003	-0.027 $\pm$ 0.013	0.006 $\pm$ 0.000	0.080 $\pm$ 0.002	0.006 $\pm$ 0.000	0.080 $\pm$ 0.002	0.161 $\pm$ 0.007	0.401 $\pm$ 0.011	0.196	-0.004	-1.82	0.0789	0.0790	0.978	94.584 $\pm$ 0.717	0.307 $\pm$ 0.001	70.913 $\pm$ 1.438
LSAM	beta42	0.20	0.199	-0.001 $\pm$ 0.003	-0.005 $\pm$ 0.013	0.007 $\pm$ 0.000	0.084 $\pm$ 0.002	0.007 $\pm$ 0.000	0.084 $\pm$ 0.002	0.175 $\pm$ 0.008	0.419 $\pm$ 0.012	0.201	0.001	0.52	0.0814	0.0814	0.970	94.483 $\pm$ 0.723	0.319 $\pm$ 0.001	69.107 $\pm$ 1.463
LSAM	beta43	0.20	0.202	0.002 $\pm$ 0.001	0.008 $\pm$ 0.007	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.054 $\pm$ 0.002	0.233 $\pm$ 0.006	0.202	0.002	1.05	0.0459	0.0459	1.012	95.186 $\pm$ 0.678	0.185 $\pm$ 0.001	98.796 $\pm$ 0.345
LSAM	beta44	0.11	0.112	0.002 $\pm$ 0.002	0.015 $\pm$ 0.016	0.003 $\pm$ 0.000	0.057 $\pm$ 0.001	0.003 $\pm$ 0.000	0.057 $\pm$ 0.002	0.269 $\pm$ 0.013	0.518 $\pm$ 0.015	0.112	0.002	2.04	0.0539	0.0540	0.909	92.477 $\pm$ 0.835	0.203 $\pm$ 0.002	59.478 $\pm$ 1.555
LSAM	beta45	0.11	0.111	0.001 $\pm$ 0.002	0.009 $\pm$ 0.016	0.003 $\pm$ 0.000	0.054 $\pm$ 0.001	0.003 $\pm$ 0.000	0.054 $\pm$ 0.002	0.245 $\pm$ 0.012	0.495 $\pm$ 0.014	0.108	-0.002	-1.80	0.0522	0.0523	0.901	92.277 $\pm$ 0.845	0.192 $\pm$ 0.002	63.892 $\pm$ 1.521
LSAM	beta46	0.08	0.081	0.001 $\pm$ 0.001	0.013 $\pm$ 0.014	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.188 $\pm$ 0.010	0.434 $\pm$ 0.013	0.081	0.001	1.64	0.0326	0.0326	0.846	90.572 $\pm$ 0.925	0.115 $\pm$ 0.001	76.630 $\pm$ 1.340
LSAM	beta47	0.08	0.081	0.001 $\pm$ 0.001	0.014 $\pm$ 0.013	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.167 $\pm$ 0.009	0.409 $\pm$ 0.013	0.080	0.000	0.19	0.0292	0.0292	0.888	92.177 $\pm$ 0.850	0.114 $\pm$ 0.001	76.028 $\pm$ 1.352
LSAM	beta51	0.16	0.162	0.002 $\pm$ 0.002	0.013 $\pm$ 0.012	0.004 $\pm$ 0.000	0.060 $\pm$ 0.001	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	0.142 $\pm$ 0.006	0.377 $\pm$ 0.010	0.162	0.002	1.10	0.0578	0.0579	0.968	93.681 $\pm$ 0.771	0.229 $\pm$ 0.001	78.536 $\pm$ 1.300
LSAM	beta52	0.16	0.160	-0.000 $\pm$ 0.003	-0.002 $\pm$ 0.016	0.007 $\pm$ 0.000	0.081 $\pm$ 0.002	0.007 $\pm$ 0.000	0.081 $\pm$ 0.002	0.257 $\pm$ 0.012	0.507 $\pm$ 0.014	0.159	-0.001	-0.54	0.0781	0.0781	0.994	95.787 $\pm$ 0.636	0.316 $\pm$ 0.001	52.457 $\pm$ 1.582
LSAM	beta53	0.16	0.161	0.001 $\pm$ 0.002	0.007 $\pm$ 0.013	0.004 $\pm$ 0.000	0.065 $\pm$ 0.001	0.004 $\pm$ 0.000	0.065 $\pm$ 0.002	0.164 $\pm$ 0.007	0.405 $\pm$ 0.011	0.165	0.005	2.99	0.0632	0.0634	0.984	95.286 $\pm$ 0.671	0.250 $\pm$ 0.001	71.815 $\pm$ 1.425
LSAM	beta54	0.16	0.153	-0.007 $\pm$ 0.002	-0.042 $\pm$ 0.016	0.007 $\pm$ 0.000	0.083 $\pm$ 0.002	0.007 $\pm$ 0.000	0.083 $\pm$ 0.002	0.268 $\pm$ 0.012	0.518 $\pm$ 0.015	0.154	-0.006	-3.91	0.0776	0.0778	0.941	92.879 $\pm$ 0.815	0.305 $\pm$ 0.001	51.254 $\pm$ 1.583
LSAM	beta55	0.08	0.083	0.003 $\pm$ 0.002	0.043 $\pm$ 0.021	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.003 $\pm$ 0.000	0.053 $\pm$ 0.002	0.435 $\pm$ 0.021	0.659 $\pm$ 0.019	0.083	0.003	3.35	0.0494	0.0494	0.897	92.979 $\pm$ 0.809	0.185 $\pm$ 0.001	47.242 $\pm$ 1.581
LSAM	beta56	0.08	0.081	0.001 $\pm$ 0.001	0.008 $\pm$ 0.014	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.201 $\pm$ 0.010	0.448 $\pm$ 0.014	0.080	0.000	-0.22	0.0322	0.0322	0.850	89.669 $\pm$ 0.964	0.119 $\pm$ 0.001	72.317 $\pm$ 1.417
LSAM	beta57	0.06	0.058	-0.002 $\pm$ 0.002	-0.032 $\pm$ 0.027	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.727 $\pm$ 0.034	0.853 $\pm$ 0.024	0.058	-0.002	-3.34	0.0508	0.0508	0.913	93.480 $\pm$ 0.782	0.183 $\pm$ 0.001	27.783 $\pm$ 1.419
LSAM	beta58	0.06	0.061	0.001 $\pm$ 0.001	0.014 $\pm$ 0.016	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.271 $\pm$ 0.015	0.521 $\pm$ 0.016	0.061	0.001	1.13	0.0283	0.0283	0.871	89.569 $\pm$ 0.968	0.107 $\pm$ 0.001	61.785 $\pm$ 1.539
QML	beta41	0.20	0.201	0.001 $\pm$ 0.002	0.006 $\pm$ 0.010	0.004 $\pm$ 0.000	0.063 $\pm$ 0.001	0.004 $\pm$ 0.000	0.063 $\pm$ 0.002	0.100 $\pm$ 0.004	0.317 $\pm$ 0.009	0.203	0.003	1.26	0.0651	0.0651	1.008	95.386 $\pm$ 0.664	0.250 $\pm$ 0.001	86.760 $\pm$ 1.073
QML	beta42	0.20	0.203	0.003 $\pm$ 0.002	0.015 $\pm$ 0.010	0.004 $\pm$ 0.000	0.066 $\pm$ 0.001	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	0.109 $\pm$ 0.005	0.331 $\pm$ 0.009	0.204	0.004	1.95	0.0650	0.0652	0.989	94.885 $\pm$ 0.698	0.256 $\pm$ 0.001	87.262 $\pm$ 1.056
QML	beta43	0.20	0.198	-0.002 $\pm$ 0.001	-0.012 $\pm$ 0.007	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.052 $\pm$ 0.002	0.227 $\pm$ 0.006	0.198	-0.002	-0.90	0.0436	0.0436	1.008	94.584 $\pm$ 0.717	0.180 $\pm$ 0.001	98.997 $\pm$ 0.316
QML	beta44	0.11	0.096	-0.014 $\pm$ 0.002	-0.129 $\pm$ 0.016	0.003 $\pm$ 0.000	0.054 $\pm$ 0.001	0.003 $\pm$ 0.000	0.056 $\pm$ 0.002	0.258 $\pm$ 0.012	0.508 $\pm$ 0.014	0.095	-0.015	-13.35	0.0560	0.0578	0.928	92.076 $\pm$ 0.855	0.197 $\pm$ 0.001	51.454 $\pm$ 1.583
QML	beta45	0.11	0.106	-0.004 $\pm$ 0.002	-0.038 $\pm$ 0.015	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.219 $\pm$ 0.010	0.468 $\pm$ 0.013	0.105	-0.005	-4.44	0.0507	0.0509	0.916	92.377 $\pm$ 0.840	0.184 $\pm$ 0.001	63.490 $\pm$ 1.525
QML	beta46	0.08	0.092	0.012 $\pm$ 0.001	0.153 $\pm$ 0.012	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.176 $\pm$ 0.009	0.420 $\pm$ 0.012	0.091	0.011	13.96	0.0297	0.0317	0.886	89.669 $\pm$ 0.964	0.109 $\pm$ 0.001	86.359 $\pm$ 1.087
QML	beta47	0.08	0.093	0.013 $\pm$ 0.001	0.166 $\pm$ 0.012	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	0.161 $\pm$ 0.008	0.402 $\pm$ 0.011	0.092	0.012	15.37	0.0278	0.0304	0.920	89.168 $\pm$ 0.984	0.106 $\pm$ 0.001	89.468 $\pm$ 0.972
QML	beta51	0.16	0.170	0.010 $\pm$ 0.002	0.063 $\pm$ 0.012	0.004 $\pm$ 0.000	0.061 $\pm$ 0.001	0.004 $\pm$ 0.000	0.062 $\pm$ 0.002	0.148 $\pm$ 0.007	0.385 $\pm$ 0.011	0.169	0.009	5.78	0.0572	0.0580	0.966	93.380 $\pm$ 0.787	0.230 $\pm$ 0.001	82.046 $\pm$ 1.216
QML	beta52	0.16	0.143	-0.017 $\pm$ 0.002	-0.106 $\pm$ 0.013	0.004 $\pm$ 0.000	0.064 $\pm$ 0.001	0.004 $\pm$ 0.000	0.067 $\pm$ 0.002	0.173 $\pm$ 0.008	0.416 $\pm$ 0.011	0.141	-0.019	-11.97	0.0630	0.0658	1.001	95.085 $\pm$ 0.685	0.253 $\pm$ 0.001	61.083 $\pm$ 1.544
QML	beta53	0.16	0.136	-0.024 $\pm$ 0.002	-0.150 $\pm$ 0.013	0.004 $\pm$ 0.000	0.064 $\pm$ 0.001	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.180 $\pm$ 0.008	0.424 $\pm$ 0.011	0.137	-0.023	-14.34	0.0616	0.0658	0.983	92.277 $\pm$ 0.845	0.245 $\pm$ 0.001	57.974 $\pm$ 1.563
QML	beta54	0.16	0.155	-0.005 $\pm$ 0.002	-0.031 $\pm$ 0.013	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.181 $\pm$ 0.008	0.426 $\pm$ 0.012	0.154	-0.006	-3.56	0.0678	0.0680	0.952	92.678 $\pm$ 0.825	0.254 $\pm$ 0.001	65.296 $\pm$ 1.508
QML	beta55	0.08	0.086	0.006 $\pm$ 0.002	0.074 $\pm$ 0.021	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.003 $\pm$ 0.000	0.053 $\pm$ 0.002	0.433 $\pm$ 0.022	0.658 $\pm$ 0.019	0.085	0.005	6.09	0.0502	0.0504	0.908	92.377 $\pm$ 0.840	0.186 $\pm$ 0.002	46.540 $\pm$ 1.580
QML	beta56	0.08	0.086	0.006 $\pm$ 0.001	0.078 $\pm$ 0.014	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.208 $\pm$ 0.011	0.456 $\pm$ 0.014	0.086	0.006	7.24	0.0331	0.0336	0.848	88.465 $\pm$ 1.012	0.120 $\pm$ 0.001	76.730 $\pm$ 1.338
QML	beta57	0.06	0.066	0.006 $\pm$ 0.001	0.104 $\pm$ 0.025	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.632 $\pm$ 0.030	0.795 $\pm$ 0.022	0.068	0.008	12.60	0.0457	0.0463	0.927	93.480 $\pm$ 0.782	0.172 $\pm$ 0.001	36.810 $\pm$ 1.527
QML	beta58	0.06	0.069	0.009 $\pm$ 0.001	0.152 $\pm$ 0.015	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.252 $\pm$ 0.014	0.502 $\pm$ 0.016	0.068	0.008	13.37	0.0261	0.0273	0.876	89.067 $\pm$ 0.988	0.099 $\pm$ 0.001	76.830 $\pm$ 1.336
UPI	beta41	0.20	0.205	0.005 $\pm$ 0.002	0.026 $\pm$ 0.012	0.006 $\pm$ 0.000	0.077 $\pm$ 0.002	0.006 $\pm$ 0.000	0.078 $\pm$ 0.002	0.151 $\pm$ 0.007	0.388 $\pm$ 0.011	0.207	0.007	3.31	0.0757	0.0760	0.973	94.684 $\pm$ 0.711	0.295 $\pm$ 0.001	78.034 $\pm$ 1.311
UPI	beta42	0.20	0.206	0.006 $\pm$ 0.003	0.031 $\pm$ 0.013	0.007 $\pm$ 0.000	0.081 $\pm$ 0.002	0.007 $\pm$ 0.000	0.081 $\pm$ 0.002	0.165 $\pm$ 0.007	0.407 $\pm$ 0.011	0.205	0.005	2.33	0.0819	0.0821	0.973	95.286 $\pm$ 0.671	0.310 $\pm$ 0.001	74.323 $\pm$ 1.384
UPI	beta43	0.20	0.202	0.002 $\pm$ 0.001	0.008 $\pm$ 0.007	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.053 $\pm$ 0.002	0.231 $\pm$ 0.006	0.202	0.002	0.96	0.0469	0.0469	0.999	94.483 $\pm$ 0.723	0.181 $\pm$ 0.001	98.997 $\pm$ 0.31

Table S173

Simulation 2 - Condition 29: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.8
(continued)

Method	Param.	True	Mean-based										Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE	
UPI	beta47	0.08	0.080	-0.000 ± 0.001	-0.002 ± 0.013	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.159 ± 0.009	0.399 ± 0.012	0.078	-0.002	-2.25	0.0294	0.0295	0.890	90.672 ± 0.921	0.111 ± 0.001	77.633 ± 1.320	
UPI	beta51	0.16	0.166	0.006 ± 0.002	0.039 ± 0.012	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.143 ± 0.006	0.378 ± 0.010	0.164	0.004	2.80	0.0572	0.0573	0.956	93.581 ± 0.776	0.226 ± 0.001	82.046 ± 1.216	
UPI	beta52	0.16	0.166	0.006 ± 0.002	0.039 ± 0.016	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	0.243 ± 0.011	0.493 ± 0.014	0.162	0.002	1.44	0.0765	0.0766	0.985	94.784 ± 0.704	0.303 ± 0.001	58.275 ± 1.562	
UPI	beta53	0.16	0.160	-0.000 ± 0.002	-0.002 ± 0.013	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.160 ± 0.007	0.400 ± 0.011	0.161	0.001	0.89	0.0625	0.0625	0.983	94.483 ± 0.723	0.247 ± 0.001	70.812 ± 1.440	
UPI	beta54	0.16	0.155	-0.005 ± 0.003	-0.028 ± 0.016	0.007 ± 0.000	0.082 ± 0.002	0.007 ± 0.000	0.082 ± 0.002	0.261 ± 0.012	0.511 ± 0.014	0.155	-0.005	-2.91	0.0798	0.0799	0.941	92.778 ± 0.820	0.301 ± 0.001	53.962 ± 1.579	
UPI	beta55	0.08	0.082	0.002 ± 0.002	0.030 ± 0.021	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.420 ± 0.020	0.648 ± 0.018	0.082	0.002	2.40	0.0518	0.0518	0.897	92.377 ± 0.840	0.182 ± 0.001	46.439 ± 1.579	
UPI	beta56	0.08	0.081	0.001 ± 0.001	0.012 ± 0.014	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.196 ± 0.010	0.442 ± 0.013	0.079	-0.001	-0.84	0.0323	0.0323	0.852	89.067 ± 0.988	0.118 ± 0.001	74.122 ± 1.387	
UPI	beta57	0.06	0.057	-0.003 ± 0.002	-0.054 ± 0.026	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.691 ± 0.033	0.831 ± 0.024	0.057	-0.003	-5.15	0.0476	0.0477	0.907	93.380 ± 0.787	0.177 ± 0.001	29.589 ± 1.446	
UPI	beta58	0.06	0.060	0.000 ± 0.001	0.006 ± 0.016	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.264 ± 0.014	0.514 ± 0.016	0.060	0.000	-0.58	0.0281	0.0281	0.873	90.371 ± 0.934	0.106 ± 0.001	62.487 ± 1.533	

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S174

Simulation 2 - Condition 30: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, $N = 1000$, Reliability = 0.8

Method	Param.	True	Mean-based										Median-based					Inference		
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LSAM	beta41	0.20	0.199	-0.001 $\pm$ 0.001	-0.003 $\pm$ 0.007	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.056 $\pm$ 0.002	0.237 $\pm$ 0.006	0.198	-0.002	-1.07	0.0483	0.0484	1.013	95.700 $\pm$ 0.641	0.188 $\pm$ 0.001	98.600 $\pm$ 0.372
LSAM	beta42	0.20	0.200	-0.000 $\pm$ 0.002	-0.001 $\pm$ 0.008	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.065 $\pm$ 0.003	0.255 $\pm$ 0.007	0.202	0.002	0.90	0.0514	0.0514	0.977	93.900 $\pm$ 0.757	0.195 $\pm$ 0.001	96.800 $\pm$ 0.557
LSAM	beta43	0.20	0.201	0.001 $\pm$ 0.001	0.005 $\pm$ 0.005	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.022 $\pm$ 0.001	0.150 $\pm$ 0.004	0.201	0.001	0.68	0.0307	0.0307	0.986	94.700 $\pm$ 0.708	0.116 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta44	0.11	0.116	0.006 $\pm$ 0.001	0.054 $\pm$ 0.009	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.093 $\pm$ 0.004	0.304 $\pm$ 0.009	0.114	0.004	3.39	0.0323	0.0325	0.957	93.400 $\pm$ 0.785	0.124 $\pm$ 0.001	94.000 $\pm$ 0.751
LSAM	beta45	0.11	0.110	-0.000 $\pm$ 0.001	-0.002 $\pm$ 0.009	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.078 $\pm$ 0.004	0.280 $\pm$ 0.008	0.110	0.000	-0.00	0.0302	0.0302	0.984	94.000 $\pm$ 0.751	0.119 $\pm$ 0.001	93.600 $\pm$ 0.774
LSAM	beta46	0.08	0.079	-0.001 $\pm$ 0.001	-0.012 $\pm$ 0.007	0.000 $\pm$ 0.000	0.018 $\pm$ 0.000	0.000 $\pm$ 0.000	0.018 $\pm$ 0.001	0.051 $\pm$ 0.002	0.226 $\pm$ 0.006	0.079	-0.001	-1.00	0.0178	0.0178	0.937	93.800 $\pm$ 0.763	0.066 $\pm$ 0.001	98.200 $\pm$ 0.420
LSAM	beta47	0.08	0.079	-0.001 $\pm$ 0.001	-0.015 $\pm$ 0.007	0.000 $\pm$ 0.000	0.018 $\pm$ 0.000	0.000 $\pm$ 0.000	0.019 $\pm$ 0.001	0.054 $\pm$ 0.003	0.232 $\pm$ 0.007	0.079	-0.001	-0.97	0.0171	0.0171	0.924	91.400 $\pm$ 0.887	0.067 $\pm$ 0.001	97.700 $\pm$ 0.474
LSAM	beta51	0.16	0.161	0.001 $\pm$ 0.001	0.008 $\pm$ 0.007	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.054 $\pm$ 0.003	0.233 $\pm$ 0.007	0.160	0.000	0.01	0.0371	0.0371	0.983	95.100 $\pm$ 0.683	0.144 $\pm$ 0.000	99.500 $\pm$ 0.223
LSAM	beta52	0.16	0.161	0.001 $\pm$ 0.002	0.004 $\pm$ 0.010	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.104 $\pm$ 0.005	0.322 $\pm$ 0.009	0.158	-0.002	-1.18	0.0515	0.0516	0.964	94.500 $\pm$ 0.721	0.195 $\pm$ 0.000	88.000 $\pm$ 1.028
LSAM	beta53	0.16	0.159	-0.001 $\pm$ 0.001	-0.007 $\pm$ 0.008	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	0.064 $\pm$ 0.003	0.254 $\pm$ 0.007	0.161	0.001	0.37	0.0380	0.0380	0.977	93.300 $\pm$ 0.791	0.156 $\pm$ 0.000	96.800 $\pm$ 0.557
LSAM	beta54	0.16	0.156	-0.004 $\pm$ 0.002	-0.024 $\pm$ 0.010	0.003 $\pm$ 0.000	0.050 $\pm$ 0.001	0.003 $\pm$ 0.000	0.050 $\pm$ 0.001	0.099 $\pm$ 0.004	0.315 $\pm$ 0.009	0.157	-0.003	-2.12	0.0509	0.0510	0.951	94.500 $\pm$ 0.721	0.187 $\pm$ 0.001	88.600 $\pm$ 1.005
LSAM	beta55	0.08	0.083	0.003 $\pm$ 0.001	0.036 $\pm$ 0.012	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.135 $\pm$ 0.006	0.367 $\pm$ 0.010	0.082	0.002	2.27	0.0296	0.0297	0.940	92.500 $\pm$ 0.833	0.108 $\pm$ 0.001	81.900 $\pm$ 1.218
LSAM	beta56	0.08	0.080	-0.000 $\pm$ 0.001	-0.005 $\pm$ 0.008	0.000 $\pm$ 0.000	0.020 $\pm$ 0.000	0.000 $\pm$ 0.000	0.020 $\pm$ 0.001	0.060 $\pm$ 0.003	0.245 $\pm$ 0.007	0.080	0.000	-0.20	0.0188	0.0188	0.905	92.700 $\pm$ 0.823	0.070 $\pm$ 0.001	96.900 $\pm$ 0.548
LSAM	beta57	0.06	0.058	-0.002 $\pm$ 0.001	-0.041 $\pm$ 0.015	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.242 $\pm$ 0.012	0.492 $\pm$ 0.014	0.058	-0.002	-3.36	0.0282	0.0283	0.951	92.800 $\pm$ 0.817	0.110 $\pm$ 0.001	55.700 $\pm$ 1.571
LSAM	beta58	0.06	0.061	0.001 $\pm$ 0.001	0.013 $\pm$ 0.009	0.000 $\pm$ 0.000	0.017 $\pm$ 0.000	0.000 $\pm$ 0.000	0.017 $\pm$ 0.001	0.085 $\pm$ 0.004	0.291 $\pm$ 0.009	0.060	0.000	0.23	0.0162	0.0162	0.914	92.300 $\pm$ 0.843	0.062 $\pm$ 0.001	93.500 $\pm$ 0.780
QML	beta41	0.20	0.205	0.005 $\pm$ 0.001	0.026 $\pm$ 0.006	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.039 $\pm$ 0.002	0.197 $\pm$ 0.005	0.204	0.004	2.21	0.0404	0.0406	1.103	94.600 $\pm$ 0.715	0.169 $\pm$ 0.011	99.700 $\pm$ 0.173
QML	beta42	0.20	0.203	0.003 $\pm$ 0.001	0.017 $\pm$ 0.007	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.044 $\pm$ 0.002	0.209 $\pm$ 0.006	0.206	0.006	2.77	0.0417	0.0421	1.037	94.600 $\pm$ 0.715	0.169 $\pm$ 0.008	99.600 $\pm$ 0.200
QML	beta43	0.20	0.197	-0.003 $\pm$ 0.001	-0.013 $\pm$ 0.005	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.022 $\pm$ 0.001	0.147 $\pm$ 0.004	0.197	-0.003	-1.44	0.0305	0.0306	1.040	94.900 $\pm$ 0.696	0.119 $\pm$ 0.006	99.800 $\pm$ 0.141
QML	beta44	0.11	0.103	-0.007 $\pm$ 0.001	-0.067 $\pm$ 0.009	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.093 $\pm$ 0.004	0.305 $\pm$ 0.008	0.101	-0.009	-7.84	0.0313	0.0325	0.965	93.300 $\pm$ 0.791	0.124 $\pm$ 0.001	89.000 $\pm$ 0.989
QML	beta45	0.11	0.107	-0.003 $\pm$ 0.001	-0.030 $\pm$ 0.009	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.076 $\pm$ 0.003	0.275 $\pm$ 0.008	0.107	-0.003	-2.83	0.0305	0.0306	0.989	93.600 $\pm$ 0.774	0.117 $\pm$ 0.001	93.500 $\pm$ 0.780
QML	beta46	0.08	0.091	0.011 $\pm$ 0.001	0.134 $\pm$ 0.007	0.000 $\pm$ 0.000	0.017 $\pm$ 0.000	0.000 $\pm$ 0.000	0.020 $\pm$ 0.001	0.065 $\pm$ 0.003	0.255 $\pm$ 0.006	0.091	0.011	13.78	0.0177	0.0209	0.960	89.400 $\pm$ 0.973	0.065 $\pm$ 0.000	99.900 $\pm$ 0.100
QML	beta47	0.08	0.091	0.011 $\pm$ 0.001	0.140 $\pm$ 0.007	0.000 $\pm$ 0.000	0.017 $\pm$ 0.000	0.000 $\pm$ 0.000	0.021 $\pm$ 0.001	0.066 $\pm$ 0.003	0.257 $\pm$ 0.006	0.091	0.011	14.14	0.0164	0.0200	0.961	88.200 $\pm$ 1.020	0.065 $\pm$ 0.000	99.600 $\pm$ 0.200
QML	beta51	0.16	0.169	0.009 $\pm$ 0.001	0.057 $\pm$ 0.007	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.001 $\pm$ 0.000	0.039 $\pm$ 0.001	0.059 $\pm$ 0.003	0.242 $\pm$ 0.007	0.167	0.007	4.61	0.0366	0.0373	1.019	94.500 $\pm$ 0.721	0.150 $\pm$ 0.005	99.500 $\pm$ 0.223
QML	beta52	0.16	0.143	-0.017 $\pm$ 0.001	-0.108 $\pm$ 0.008	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.080 $\pm$ 0.003	0.284 $\pm$ 0.007	0.142	-0.018	-11.52	0.0427	0.0465	0.980	91.800 $\pm$ 0.868	0.161 $\pm$ 0.003	93.700 $\pm$ 0.768
QML	beta53	0.16	0.134	-0.026 $\pm$ 0.001	-0.161 $\pm$ 0.008	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.087 $\pm$ 0.004	0.296 $\pm$ 0.007	0.135	-0.025	-15.67	0.0372	0.0449	0.987	87.800 $\pm$ 1.035	0.154 $\pm$ 0.001	91.700 $\pm$ 0.872
QML	beta54	0.16	0.157	-0.003 $\pm$ 0.001	-0.017 $\pm$ 0.008	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.070 $\pm$ 0.003	0.265 $\pm$ 0.007	0.157	-0.003	-1.72	0.0428	0.0429	0.964	94.700 $\pm$ 0.708	0.160 $\pm$ 0.001	96.300 $\pm$ 0.597
QML	beta55	0.08	0.086	0.006 $\pm$ 0.001	0.069 $\pm$ 0.012	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.153 $\pm$ 0.007	0.391 $\pm$ 0.010	0.085	0.005	6.21	0.0310	0.0314	0.944	91.600 $\pm$ 0.877	0.114 $\pm$ 0.001	79.800 $\pm$ 1.270
QML	beta56	0.08	0.085	0.005 $\pm$ 0.001	0.067 $\pm$ 0.008	0.000 $\pm$ 0.000	0.020 $\pm$ 0.000	0.000 $\pm$ 0.000	0.021 $\pm$ 0.001	0.069 $\pm$ 0.003	0.262 $\pm$ 0.007	0.086	0.006	7.48	0.0202	0.0210	0.906	90.400 $\pm$ 0.932	0.072 $\pm$ 0.001	98.200 $\pm$ 0.420
QML	beta57	0.06	0.067	0.007 $\pm$ 0.001	0.121 $\pm$ 0.015	0.001 $\pm$ 0.000	0.028 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.238 $\pm$ 0.011	0.488 $\pm$ 0.013	0.067	0.007	11.81	0.0284	0.0293	0.978	92.700 $\pm$ 0.823	0.109 $\pm$ 0.001	68.100 $\pm$ 1.474
QML	beta58	0.06	0.070	0.010 $\pm$ 0.001	0.165 $\pm$ 0.009	0.000 $\pm$ 0.000	0.016 $\pm$ 0.000	0.000 $\pm$ 0.000	0.019 $\pm$ 0.000	0.102 $\pm$ 0.005	0.319 $\pm$ 0.008	0.070	0.010	16.30	0.0157	0.0185	0.927	86.900 $\pm$ 1.067	0.060 $\pm$ 0.000	97.700 $\pm$ 0.474
UPI	beta41	0.20	0.204	0.004 $\pm$ 0.001	0.019 $\pm$ 0.007	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.056 $\pm$ 0.002	0.238 $\pm$ 0.006	0.202	0.002	1.16	0.0486	0.0486	0.982	94.300 $\pm$ 0.733	0.183 $\pm$ 0.001	98.800 $\pm$ 0.344
UPI	beta42	0.20	0.202	0.002 $\pm$ 0.002	0.011 $\pm$ 0.008	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.065 $\pm$ 0.003	0.255 $\pm$ 0.007	0.205	0.005	2.40	0.0516	0.0518	0.957	93.700 $\pm$ 0.768	0.191 $\pm$ 0.001	97.500 $\pm$ 0.494
UPI	beta43	0.20	0.201	0.001 $\pm$ 0.001	0.007 $\pm$ 0.005	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.023 $\pm$ 0.001	0.151 $\pm$ 0.004	0.202	0.002	0.91	0.0308	0.0309	0.962	93.900 $\pm$ 0.757	0.114 $\pm$ 0.000	100.000 $\pm$ 0.

Table S174

Simulation 2 - Condition 30: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.8
(continued)

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.079	-0.001 ± 0.001	-0.015 ± 0.007	0.000 ± 0.000	0.018 ± 0.000	0.000 ± 0.000	0.018 ± 0.001	0.053 ± 0.003	0.230 ± 0.007	0.079	-0.001	-0.79	0.0170	0.0170	0.917	91.300 ± 0.891	0.066 ± 0.001	97.800 ± 0.464
UPI	beta51	0.16	0.163	0.003 ± 0.001	0.018 ± 0.007	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.054 ± 0.003	0.233 ± 0.007	0.162	0.002	1.07	0.0357	0.0358	0.977	94.700 ± 0.708	0.142 ± 0.000	99.600 ± 0.200
UPI	beta52	0.16	0.163	0.003 ± 0.002	0.018 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.101 ± 0.005	0.317 ± 0.009	0.161	0.001	0.43	0.0524	0.0524	0.956	94.000 ± 0.751	0.190 ± 0.000	91.000 ± 0.905
UPI	beta53	0.16	0.158	-0.002 ± 0.001	-0.010 ± 0.008	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.063 ± 0.003	0.251 ± 0.007	0.160	0.000	0.28	0.0371	0.0371	0.976	93.800 ± 0.763	0.154 ± 0.000	96.900 ± 0.548
UPI	beta54	0.16	0.157	-0.003 ± 0.002	-0.019 ± 0.010	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.097 ± 0.004	0.311 ± 0.009	0.157	-0.003	-1.86	0.0493	0.0494	0.954	94.500 ± 0.721	0.186 ± 0.001	89.100 ± 0.985
UPI	beta55	0.08	0.082	0.002 ± 0.001	0.023 ± 0.012	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.133 ± 0.006	0.364 ± 0.010	0.081	0.001	0.69	0.0290	0.0290	0.937	92.900 ± 0.812	0.107 ± 0.001	81.000 ± 1.241
UPI	beta56	0.08	0.080	-0.000 ± 0.001	-0.006 ± 0.008	0.000 ± 0.000	0.020 ± 0.000	0.000 ± 0.000	0.020 ± 0.001	0.060 ± 0.003	0.246 ± 0.007	0.080	0.000	-0.22	0.0191	0.0191	0.899	92.100 ± 0.853	0.069 ± 0.001	97.000 ± 0.539
UPI	beta57	0.06	0.058	-0.002 ± 0.001	-0.036 ± 0.015	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.234 ± 0.011	0.483 ± 0.014	0.059	-0.001	-1.48	0.0290	0.0291	0.946	92.800 ± 0.817	0.107 ± 0.001	57.900 ± 1.561
UPI	beta58	0.06	0.061	0.001 ± 0.001	0.016 ± 0.009	0.000 ± 0.000	0.017 ± 0.000	0.000 ± 0.000	0.017 ± 0.001	0.083 ± 0.004	0.287 ± 0.009	0.061	0.001	1.07	0.0156	0.0156	0.923	92.800 ± 0.817	0.062 ± 0.001	93.700 ± 0.768

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S175

Simulation 2 - Condition 31: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.4

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.209	0.009 ± 0.032	0.045 ± 0.159	0.012 ± 0.007	0.110 ± 0.050	0.011 ± 0.007	0.106 ± 0.044	0.281 ± 0.163	0.530 ± 0.220	0.204	0.004	1.88	0.0920	0.0921	1.227	91.667 ± 7.979	0.531 ± 0.075	33.333 ± 13.608
LSAM	beta42	0.20	0.215	0.015 ± 0.034	0.075 ± 0.170	0.014 ± 0.003	0.118 ± 0.020	0.013 ± 0.004	0.114 ± 0.026	0.323 ± 0.090	0.568 ± 0.132	0.206	0.006	3.06	0.1524	0.1525	1.291	100.000 ± 0.000	0.595 ± 0.094	50.000 ± 14.434
LSAM	beta43	0.20	0.181	-0.019 ± 0.022	-0.097 ± 0.109	0.006 ± 0.002	0.076 ± 0.015	0.006 ± 0.002	0.075 ± 0.017	0.140 ± 0.041	0.375 ± 0.085	0.228	0.028	14.21	0.0857	0.0903	1.190	100.000 ± 0.000	0.353 ± 0.028	50.000 ± 14.434
LSAM	beta44	0.11	0.184	0.074 ± 0.034	0.671 ± 0.310	0.014 ± 0.004	0.118 ± 0.022	0.018 ± 0.005	0.135 ± 0.025	1.505 ± 0.405	1.227 ± 0.225	0.196	0.086	77.83	0.1397	0.1639	1.042	91.667 ± 7.979	0.482 ± 0.051	50.000 ± 14.434
LSAM	beta45	0.11	0.096	-0.014 ± 0.034	-0.131 ± 0.306	0.014 ± 0.004	0.117 ± 0.022	0.013 ± 0.004	0.113 ± 0.027	1.048 ± 0.328	1.024 ± 0.249	0.128	0.018	16.50	0.0584	0.0612	0.951	100.000 ± 0.000	0.435 ± 0.031	16.667 ± 10.758
LSAM	beta46	0.08	-0.057	-0.137 ± 0.039	-1.707 ± 0.484	0.018 ± 0.008	0.134 ± 0.043	0.035 ± 0.012	0.187 ± 0.040	5.488 ± 1.934	2.343 ± 0.498	-0.055	-0.135	-169.08	0.1181	0.1795	0.902	58.333 ± 14.232	0.474 ± 0.087	0.000 ± 0.000
LSAM	beta47	0.08	-0.057	-0.137 ± 0.033	-1.712 ± 0.416	0.013 ± 0.003	0.115 ± 0.018	0.031 ± 0.010	0.176 ± 0.030	4.836 ± 1.499	2.199 ± 0.379	-0.038	-0.118	-147.08	0.1847	0.2190	1.316	83.333 ± 10.758	0.595 ± 0.104	0.000 ± 0.000
LSAM	beta51	0.16	0.223	0.063 ± 0.024	0.391 ± 0.151	0.007 ± 0.002	0.084 ± 0.019	0.010 ± 0.003	0.102 ± 0.020	0.404 ± 0.130	0.635 ± 0.128	0.197	0.037	23.38	0.1002	0.1069	1.313	100.000 ± 0.000	0.430 ± 0.026	50.000 ± 14.434
LSAM	beta52	0.16	0.175	0.015 ± 0.023	0.095 ± 0.145	0.006 ± 0.002	0.080 ± 0.021	0.006 ± 0.002	0.078 ± 0.021	0.239 ± 0.089	0.489 ± 0.133	0.177	0.017	10.82	0.0688	0.0710	1.343	100.000 ± 0.000	0.422 ± 0.025	33.333 ± 13.608
LSAM	beta53	0.16	0.098	-0.062 ± 0.023	-0.386 ± 0.143	0.006 ± 0.001	0.079 ± 0.013	0.010 ± 0.002	0.098 ± 0.016	0.373 ± 0.095	0.611 ± 0.099	0.096	-0.064	-39.91	0.1060	0.1237	1.152	100.000 ± 0.000	0.357 ± 0.022	16.667 ± 10.758
LSAM	beta54	0.16	0.163	0.003 ± 0.039	0.019 ± 0.247	0.019 ± 0.006	0.137 ± 0.029	0.017 ± 0.006	0.131 ± 0.033	0.669 ± 0.219	0.818 ± 0.207	0.177	0.017	10.39	0.1286	0.1297	0.828	91.667 ± 7.979	0.443 ± 0.038	41.667 ± 14.232
LSAM	beta55	0.08	0.157	0.077 ± 0.042	0.966 ± 0.523	0.021 ± 0.009	0.145 ± 0.044	0.025 ± 0.013	0.159 ± 0.055	3.943 ± 2.057	1.986 ± 0.686	0.125	0.045	56.09	0.1794	0.1850	1.078	91.667 ± 7.979	0.613 ± 0.051	8.333 ± 7.979
LSAM	beta56	0.08	0.067	-0.013 ± 0.040	-0.163 ± 0.505	0.020 ± 0.003	0.140 ± 0.017	0.018 ± 0.003	0.135 ± 0.026	2.832 ± 0.475	1.683 ± 0.328	0.069	-0.011	-14.19	0.1425	0.1430	0.964	91.667 ± 7.979	0.529 ± 0.041	8.333 ± 7.979
LSAM	beta57	0.06	-0.015	-0.075 ± 0.026	-1.248 ± 0.436	0.008 ± 0.003	0.091 ± 0.022	0.013 ± 0.005	0.115 ± 0.025	3.648 ± 1.322	1.910 ± 0.418	0.000	-0.060	-99.22	0.0809	0.1005	1.066	100.000 ± 0.000	0.379 ± 0.041	0.000 ± 0.000
LSAM	beta58	0.06	-0.006	-0.066 ± 0.031	-1.107 ± 0.511	0.011 ± 0.005	0.106 ± 0.037	0.015 ± 0.008	0.121 ± 0.045	4.097 ± 2.310	2.024 ± 0.754	-0.014	-0.074	-123.33	0.0667	0.0996	1.092	83.333 ± 10.758	0.454 ± 0.068	0.000 ± 0.000
QML	beta41	0.20	0.180	-0.020 ± 0.021	-0.099 ± 0.107	0.005 ± 0.002	0.074 ± 0.022	0.005 ± 0.003	0.074 ± 0.026	0.136 ± 0.069	0.369 ± 0.130	0.167	-0.033	-16.52	0.0711	0.0784	0.953	91.667 ± 7.979	0.277 ± 0.012	75.000 ± 12.500
QML	beta42	0.20	0.187	-0.013 ± 0.019	-0.065 ± 0.094	0.004 ± 0.001	0.065 ± 0.009	0.004 ± 0.001	0.064 ± 0.014	0.102 ± 0.025	0.319 ± 0.068	0.184	-0.016	-7.96	0.0680	0.0698	1.122	100.000 ± 0.000	0.287 ± 0.012	66.667 ± 13.608
QML	beta43	0.20	0.169	-0.031 ± 0.021	-0.154 ± 0.104	0.005 ± 0.002	0.072 ± 0.015	0.006 ± 0.002	0.076 ± 0.018	0.144 ± 0.048	0.379 ± 0.089	0.189	-0.011	-5.56	0.0984	0.0990	1.033	100.000 ± 0.000	0.293 ± 0.022	66.667 ± 13.608
QML	beta44	0.11	0.174	0.064 ± 0.026	0.584 ± 0.237	0.008 ± 0.003	0.090 ± 0.021	0.012 ± 0.003	0.108 ± 0.020	0.961 ± 0.273	0.980 ± 0.178	0.166	0.056	51.36	0.0973	0.1125	1.050	91.667 ± 7.979	0.372 ± 0.028	58.333 ± 14.232
QML	beta45	0.11	0.085	-0.025 ± 0.028	-0.223 ± 0.253	0.009 ± 0.002	0.096 ± 0.017	0.009 ± 0.003	0.096 ± 0.023	0.754 ± 0.234	0.869 ± 0.207	0.101	-0.009	-7.99	0.0661	0.0667	0.941	91.667 ± 7.979	0.356 ± 0.028	33.333 ± 13.608
QML	beta46	0.08	-0.057	-0.137 ± 0.023	-1.714 ± 0.288	0.006 ± 0.002	0.080 ± 0.016	0.025 ± 0.006	0.157 ± 0.022	3.851 ± 0.989	1.962 ± 0.269	-0.038	-0.118	-147.08	0.0846	0.1449	0.824	33.333 ± 13.608	0.258 ± 0.031	16.667 ± 10.758
QML	beta47	0.08	-0.040	-0.120 ± 0.019	-1.504 ± 0.231	0.004 ± 0.001	0.064 ± 0.012	0.018 ± 0.004	0.135 ± 0.015	2.852 ± 0.617	1.689 ± 0.187	-0.036	-0.116	-144.52	0.0829	0.1423	1.046	50.000 ± 14.434	0.263 ± 0.027	8.333 ± 7.979
QML	beta51	0.16	0.234	0.074 ± 0.027	0.460 ± 0.171	0.009 ± 0.004	0.095 ± 0.028	0.014 ± 0.005	0.117 ± 0.027	0.535 ± 0.195	0.731 ± 0.166	0.217	0.057	35.41	0.1077	0.1217	1.133	91.667 ± 7.979	0.422 ± 0.022	58.333 ± 14.232
QML	beta52	0.16	0.197	0.037 ± 0.018	0.229 ± 0.115	0.004 ± 0.001	0.064 ± 0.011	0.005 ± 0.002	0.071 ± 0.016	0.198 ± 0.069	0.445 ± 0.102	0.161	0.001	0.75	0.0814	0.0814	1.356	100.000 ± 0.000	0.339 ± 0.016	75.000 ± 12.500
QML	beta53	0.16	0.122	-0.038 ± 0.018	-0.240 ± 0.110	0.004 ± 0.001	0.061 ± 0.011	0.005 ± 0.002	0.070 ± 0.016	0.190 ± 0.068	0.436 ± 0.098	0.118	-0.042	-26.26	0.0660	0.0782	1.346	100.000 ± 0.000	0.321 ± 0.016	16.667 ± 10.758
QML	beta54	0.16	0.137	-0.023 ± 0.025	-0.147 ± 0.155	0.007 ± 0.001	0.086 ± 0.011	0.007 ± 0.002	0.086 ± 0.019	0.286 ± 0.079	0.535 ± 0.117	0.167	0.007	4.69	0.0803	0.0806	0.972	83.333 ± 10.758	0.328 ± 0.034	41.667 ± 14.232
QML	beta55	0.08	0.147	0.067 ± 0.039	0.835 ± 0.488	0.018 ± 0.006	0.135 ± 0.034	0.021 ± 0.010	0.146 ± 0.045	3.321 ± 1.576	1.822 ± 0.566	0.109	0.029	35.68	0.1287	0.1319	0.975	91.667 ± 7.979	0.517 ± 0.034	16.667 ± 10.758
QML	beta56	0.08	0.069	-0.011 ± 0.034	-0.133 ± 0.424	0.014 ± 0.004	0.118 ± 0.022	0.013 ± 0.004	0.113 ± 0.027	1.996 ± 0.579	1.413 ± 0.335	0.059	-0.021	-26.48	0.1038	0.1060	0.991	91.667 ± 7.979	0.457 ± 0.035	0.000 ± 0.000
QML	beta57	0.06	-0.014	-0.074 ± 0.015	-1.237 ± 0.248	0.003 ± 0.001	0.052 ± 0.015	0.008 ± 0.003	0.089 ± 0.019	2.210 ± 0.803	1.487 ± 0.309	-0.010	-0.070	-115.88	0.0392	0.0798	1.182	75.000 ± 12.500	0.239 ± 0.025	8.333 ± 7.979
QML	beta58	0.06	-0.017	-0.077 ± 0.016	-1.287 ± 0.275	0.003 ± 0.001	0.057 ± 0.012	0.009 ± 0.003	0.095 ± 0.019	2.486 ± 0.882	1.577 ± 0.313	-0.009	-0.069	-115.28	0.0586	0.0907	1.090	75.000 ± 12.500	0.244 ± 0.041	16.667 ± 10.758
UPI	beta41	0.20	0.195	-0.005 ± 0.085	-0.027 ± 0.424	0.086 ± 0.037	0.294 ± 0.090	0.079 ± 0.036	0.281 ± 0.091	1.980 ± 0.904	1.407 ± 0.455	0.111	-0.089	-44.72	0.2234	0.2406	1.210	83.333 ± 10.758	1.394 ± 0.453	16.667 ± 10.758
UPI	beta42	0.20	0.275	0.075 ± 0.104	0.376 ± 0.522	0.131 ± 0.061	0.362 ± 0.125	0.126 ± 0.070	0.354 ± 0.138	3.141 ± 1.758	1.772 ± 0.688	0.190	-0.010	-5.19	0.2663	0.2665	1.547	100.000 ± 0.000	2.193 ± 1.235	8.333 ± 7.979
UPI	beta43	0.20	0.106	-0.094 ± 0.026	-0.471 ± 0.129	0.008 ± 0.002	0.090 ± 0.020	0.016 ± 0.003	0.127 ± 0.014	0.405 ± 0.076	0.637 ± 0.072	0.077	-0.123	-61.62	0.0873	0.1511	1.618	75.000 ± 12.500	0.568 ± 0.132	16.667 ± 10.758
UPI	beta44	0.11	0.213	0.103 ± 0.041	0.936 ± 0.371	0.020 ± 0.008	0.141 ± 0.042	0.029 ± 0.014	0.170 ± 0.052	2.386 ± 1.153	1.545 ± 0.471	0.173	0.063	57.58	0.1525	0.1652	2.024	100.000 ± 0.000	1.120 ± 0.200	8.333 ± 7.979
UPI	beta45	0.11	-0.012	-0.122 ± 0.040	-1.110 ± 0.365	0.019 ± 0.005	0.139 ± 0.027	0.033 ± 0.013	0.181 ± 0.043	2.702 ± 1.078	1.644 ± 0.388	0.041	-0.069	-62.96	0.1550	0.1697	1.848	100.000 ± 0.000	1.009 ± 0.267	0.000 ± 0.000
UPI	beta46	0.08	-0.039	-0.119 ± 0.096	-1.490 ± 1.202	0.111 ± 0.062	0.333 ± 0.148	0.116 ± 0.080	0.341 ± 0.167	18.121 ± 12.468	4.257 ± 2.090	0.000	-0.080	-100.17	0.1962	0.2119	1.439	75.000 ± 12.500	1.880 ± 0.599	0.000 ± 0.000

Table S175

Simulation 2 - Condition 31: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.4
(continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	-0.203	-0.283 ± 0.118	-3.541 ± 1.472	0.166 ± 0.052	0.408 ± 0.091	0.233 ± 0.105	0.482 ± 0.133	36.364 ± 16.384	6.030 ± 1.667	-0.172	-0.252	-314.73	0.4470	0.5130	1.643	100.000 ± 0.000	2.626 ± 1.101	0.000 ± 0.000
UPI	beta51	0.16	0.256	0.096 ± 0.025	0.603 ± 0.156	0.007 ± 0.003	0.086 ± 0.027	0.016 ± 0.007	0.127 ± 0.032	0.630 ± 0.265	0.794 ± 0.201	0.266	0.106	65.94	0.0711	0.1272	1.858	91.667 ± 7.979	0.629 ± 0.135	66.667 ± 13.608
UPI	beta52	0.16	0.136	-0.024 ± 0.045	-0.148 ± 0.282	0.024 ± 0.006	0.156 ± 0.026	0.023 ± 0.006	0.152 ± 0.034	0.897 ± 0.232	0.947 ± 0.210	0.088	-0.072	-45.27	0.2050	0.2174	1.554	100.000 ± 0.000	0.953 ± 0.290	16.667 ± 10.758
UPI	beta53	0.16	0.123	-0.037 ± 0.033	-0.229 ± 0.205	0.013 ± 0.002	0.114 ± 0.015	0.013 ± 0.003	0.115 ± 0.023	0.516 ± 0.118	0.718 ± 0.141	0.057	-0.103	-64.32	0.1470	0.1794	0.970	75.000 ± 12.500	0.433 ± 0.069	33.333 ± 13.608
UPI	beta54	0.16	0.201	0.041 ± 0.109	0.258 ± 0.680	0.142 ± 0.078	0.377 ± 0.164	0.132 ± 0.085	0.363 ± 0.168	5.148 ± 3.302	2.269 ± 1.050	0.191	0.031	19.41	0.2750	0.2768	1.096	83.333 ± 10.758	1.619 ± 0.561	0.000 ± 0.000
UPI	beta55	0.08	0.140	0.060 ± 0.051	0.754 ± 0.633	0.031 ± 0.008	0.175 ± 0.034	0.032 ± 0.010	0.178 ± 0.040	4.977 ± 1.528	2.231 ± 0.504	0.119	0.039	48.92	0.1349	0.1405	1.282	91.667 ± 7.979	0.881 ± 0.126	8.333 ± 7.979
UPI	beta56	0.08	0.124	0.044 ± 0.054	0.553 ± 0.672	0.035 ± 0.013	0.186 ± 0.052	0.034 ± 0.016	0.184 ± 0.061	5.271 ± 2.552	2.296 ± 0.765	0.036	-0.044	-55.30	0.2144	0.2190	1.175	91.667 ± 7.979	0.858 ± 0.161	0.000 ± 0.000
UPI	beta57	0.06	-0.033	-0.093 ± 0.084	-1.547 ± 1.399	0.085 ± 0.040	0.291 ± 0.104	0.086 ± 0.051	0.293 ± 0.120	23.924 ± 14.045	4.891 ± 2.001	-0.031	-0.091	-151.63	0.1992	0.2190	1.581	91.667 ± 7.979	1.803 ± 0.599	0.000 ± 0.000
UPI	beta58	0.06	-0.186	-0.246 ± 0.103	-4.108 ± 1.724	0.128 ± 0.079	0.358 ± 0.187	0.178 ± 0.125	0.422 ± 0.207	49.563 ± 34.713	7.040 ± 3.455	-0.120	-0.180	-299.33	0.2897	0.3408	1.036	83.333 ± 10.758	1.455 ± 0.466	0.000 ± 0.000

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S176

Simulation 2 - Condition 32: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.4

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LSAM	beta41	0.20	0.196	-0.004 $\pm$ 0.026	-0.021 $\pm$ 0.132	0.007 $\pm$ 0.003	0.083 $\pm$ 0.030	0.006 $\pm$ 0.003	0.079 $\pm$ 0.031	0.156 $\pm$ 0.081	0.395 $\pm$ 0.153	0.214	0.014	7.10	0.0584	0.0601	0.810	90.000 $\pm$ 9.487	0.264 $\pm$ 0.016	80.000 $\pm$ 12.649
LSAM	beta42	0.20	0.229	0.029 $\pm$ 0.028	0.145 $\pm$ 0.142	0.008 $\pm$ 0.004	0.090 $\pm$ 0.037	0.008 $\pm$ 0.005	0.090 $\pm$ 0.042	0.203 $\pm$ 0.130	0.451 $\pm$ 0.209	0.215	0.015	7.66	0.0549	0.0570	0.833	80.000 $\pm$ 12.649	0.294 $\pm$ 0.017	80.000 $\pm$ 12.649
LSAM	beta43	0.20	0.174	-0.026 $\pm$ 0.020	-0.129 $\pm$ 0.100	0.004 $\pm$ 0.001	0.063 $\pm$ 0.016	0.004 $\pm$ 0.001	0.065 $\pm$ 0.014	0.107 $\pm$ 0.027	0.327 $\pm$ 0.070	0.160	-0.040	-19.91	0.0429	0.0586	0.764	80.000 $\pm$ 12.649	0.190 $\pm$ 0.007	100.000 $\pm$ 0.000
LSAM	beta44	0.11	0.202	0.092 $\pm$ 0.027	0.833 $\pm$ 0.244	0.007 $\pm$ 0.002	0.085 $\pm$ 0.019	0.015 $\pm$ 0.004	0.122 $\pm$ 0.018	1.231 $\pm$ 0.308	1.109 $\pm$ 0.160	0.180	0.070	63.43	0.0805	0.1065	0.815	70.000 $\pm$ 14.491	0.272 $\pm$ 0.016	80.000 $\pm$ 12.649
LSAM	beta45	0.11	0.119	0.009 $\pm$ 0.022	0.080 $\pm$ 0.199	0.005 $\pm$ 0.002	0.069 $\pm$ 0.022	0.004 $\pm$ 0.002	0.066 $\pm$ 0.021	0.362 $\pm$ 0.144	0.602 $\pm$ 0.187	0.118	0.008	7.03	0.0702	0.0706	0.926	90.000 $\pm$ 9.487	0.251 $\pm$ 0.008	60.000 $\pm$ 15.492
LSAM	beta46	0.08	-0.011	-0.091 $\pm$ 0.023	-1.141 $\pm$ 0.292	0.005 $\pm$ 0.003	0.074 $\pm$ 0.028	0.013 $\pm$ 0.003	0.115 $\pm$ 0.017	2.070 $\pm$ 0.532	1.439 $\pm$ 0.210	0.001	-0.079	-99.23	0.0788	0.1118	0.818	50.000 $\pm$ 15.811	0.237 $\pm$ 0.025	0.000 $\pm$ 0.000
LSAM	beta47	0.08	-0.039	-0.119 $\pm$ 0.029	-1.494 $\pm$ 0.364	0.008 $\pm$ 0.003	0.092 $\pm$ 0.025	0.022 $\pm$ 0.007	0.148 $\pm$ 0.025	3.420 $\pm$ 1.022	1.849 $\pm$ 0.310	-0.032	-0.112	-140.16	0.0787	0.1370	0.757	50.000 $\pm$ 15.811	0.273 $\pm$ 0.043	20.000 $\pm$ 12.649
LSAM	beta51	0.16	0.148	-0.012 $\pm$ 0.023	-0.075 $\pm$ 0.143	0.005 $\pm$ 0.001	0.072 $\pm$ 0.009	0.005 $\pm$ 0.001	0.069 $\pm$ 0.014	0.189 $\pm$ 0.026	0.434 $\pm$ 0.090	0.170	0.010	6.15	0.0730	0.0737	0.798	100.000 $\pm$ 0.000	0.226 $\pm$ 0.006	50.000 $\pm$ 15.811
LSAM	beta52	0.16	0.164	0.004 $\pm$ 0.018	0.026 $\pm$ 0.110	0.003 $\pm$ 0.001	0.056 $\pm$ 0.012	0.003 $\pm$ 0.001	0.053 $\pm$ 0.014	0.110 $\pm$ 0.030	0.332 $\pm$ 0.085	0.168	0.008	4.75	0.0544	0.0549	1.080	100.000 $\pm$ 0.000	0.236 $\pm$ 0.010	80.000 $\pm$ 12.649
LSAM	beta53	0.16	0.140	-0.020 $\pm$ 0.015	-0.124 $\pm$ 0.095	0.002 $\pm$ 0.001	0.048 $\pm$ 0.017	0.002 $\pm$ 0.001	0.050 $\pm$ 0.015	0.097 $\pm$ 0.041	0.312 $\pm$ 0.093	0.133	-0.027	-16.63	0.0447	0.0520	1.155	90.000 $\pm$ 9.487	0.219 $\pm$ 0.011	90.000 $\pm$ 9.487
LSAM	beta54	0.16	0.126	-0.034 $\pm$ 0.021	-0.213 $\pm$ 0.133	0.005 $\pm$ 0.002	0.067 $\pm$ 0.026	0.005 $\pm$ 0.003	0.072 $\pm$ 0.032	0.205 $\pm$ 0.130	0.452 $\pm$ 0.202	0.142	-0.018	-11.12	0.0468	0.0501	0.821	90.000 $\pm$ 9.487	0.217 $\pm$ 0.007	70.000 $\pm$ 14.491
LSAM	beta55	0.08	0.122	0.042 $\pm$ 0.020	0.528 $\pm$ 0.244	0.004 $\pm$ 0.001	0.062 $\pm$ 0.013	0.005 $\pm$ 0.002	0.072 $\pm$ 0.014	0.815 $\pm$ 0.244	0.903 $\pm$ 0.179	0.114	0.034	41.95	0.0629	0.0713	1.158	100.000 $\pm$ 0.000	0.280 $\pm$ 0.012	50.000 $\pm$ 15.811
LSAM	beta56	0.08	0.063	-0.017 $\pm$ 0.022	-0.217 $\pm$ 0.275	0.005 $\pm$ 0.001	0.070 $\pm$ 0.012	0.005 $\pm$ 0.001	0.068 $\pm$ 0.015	0.727 $\pm$ 0.172	0.852 $\pm$ 0.192	0.080	0.000	-0.14	0.0575	0.0575	1.102	100.000 $\pm$ 0.000	0.300 $\pm$ 0.017	10.000 $\pm$ 9.487
LSAM	beta57	0.06	-0.025	-0.085 $\pm$ 0.018	-1.425 $\pm$ 0.293	0.003 $\pm$ 0.001	0.056 $\pm$ 0.013	0.010 $\pm$ 0.003	0.100 $\pm$ 0.016	2.802 $\pm$ 0.803	1.674 $\pm$ 0.259	-0.011	-0.071	-118.55	0.0555	0.0902	0.948	60.000 $\pm$ 15.492	0.206 $\pm$ 0.016	0.000 $\pm$ 0.000
LSAM	beta58	0.06	0.013	-0.047 $\pm$ 0.018	-0.783 $\pm$ 0.308	0.003 $\pm$ 0.002	0.058 $\pm$ 0.022	0.005 $\pm$ 0.001	0.073 $\pm$ 0.013	1.469 $\pm$ 0.372	1.212 $\pm$ 0.212	-0.008	-0.068	-112.60	0.0423	0.0797	0.858	70.000 $\pm$ 14.491	0.197 $\pm$ 0.018	0.000 $\pm$ 0.000
QML	beta41	0.20	0.182	-0.018 $\pm$ 0.015	-0.089 $\pm$ 0.076	0.002 $\pm$ 0.001	0.048 $\pm$ 0.009	0.002 $\pm$ 0.001	0.049 $\pm$ 0.012	0.060 $\pm$ 0.018	0.244 $\pm$ 0.058	0.183	-0.017	-8.53	0.0537	0.0564	0.904	100.000 $\pm$ 0.000	0.170 $\pm$ 0.004	100.000 $\pm$ 0.000
QML	beta42	0.20	0.195	-0.005 $\pm$ 0.013	-0.023 $\pm$ 0.066	0.002 $\pm$ 0.001	0.042 $\pm$ 0.012	0.002 $\pm$ 0.001	0.040 $\pm$ 0.012	0.040 $\pm$ 0.016	0.199 $\pm$ 0.061	0.196	-0.004	-1.94	0.0466	0.0468	1.121	100.000 $\pm$ 0.000	0.184 $\pm$ 0.005	100.000 $\pm$ 0.000
QML	beta43	0.20	0.174	-0.026 $\pm$ 0.016	-0.129 $\pm$ 0.079	0.002 $\pm$ 0.001	0.050 $\pm$ 0.011	0.003 $\pm$ 0.001	0.054 $\pm$ 0.014	0.073 $\pm$ 0.028	0.270 $\pm$ 0.070	0.178	-0.022	-10.91	0.0563	0.0604	0.847	80.000 $\pm$ 12.649	0.166 $\pm$ 0.004	100.000 $\pm$ 0.000
QML	beta44	0.11	0.180	0.070 $\pm$ 0.025	0.640 $\pm$ 0.231	0.006 $\pm$ 0.002	0.080 $\pm$ 0.021	0.011 $\pm$ 0.003	0.104 $\pm$ 0.017	0.890 $\pm$ 0.231	0.943 $\pm$ 0.156	0.187	0.077	70.26	0.0713	0.1052	0.704	60.000 $\pm$ 15.492	0.222 $\pm$ 0.009	80.000 $\pm$ 12.649
QML	beta45	0.11	0.105	-0.005 $\pm$ 0.021	-0.043 $\pm$ 0.187	0.004 $\pm$ 0.001	0.065 $\pm$ 0.011	0.004 $\pm$ 0.001	0.062 $\pm$ 0.015	0.316 $\pm$ 0.075	0.562 $\pm$ 0.133	0.107	-0.003	-2.48	0.0705	0.0706	0.766	90.000 $\pm$ 9.487	0.195 $\pm$ 0.007	60.000 $\pm$ 15.492
QML	beta46	0.08	-0.023	-0.103 $\pm$ 0.018	-1.284 $\pm$ 0.222	0.003 $\pm$ 0.001	0.056 $\pm$ 0.017	0.013 $\pm$ 0.004	0.116 $\pm$ 0.017	2.093 $\pm$ 0.578	1.447 $\pm$ 0.215	-0.022	-0.102	-127.96	0.0435	0.1112	0.682	10.000 $\pm$ 9.487	0.150 $\pm$ 0.010	20.000 $\pm$ 12.649
QML	beta47	0.08	-0.034	-0.114 $\pm$ 0.015	-1.429 $\pm$ 0.194	0.002 $\pm$ 0.001	0.049 $\pm$ 0.013	0.015 $\pm$ 0.003	0.123 $\pm$ 0.012	2.380 $\pm$ 0.470	1.543 $\pm$ 0.150	-0.022	-0.102	-128.05	0.0560	0.1167	0.803	20.000 $\pm$ 12.649	0.154 $\pm$ 0.013	20.000 $\pm$ 12.649
QML	beta51	0.16	0.166	0.006 $\pm$ 0.025	0.038 $\pm$ 0.153	0.006 $\pm$ 0.001	0.078 $\pm$ 0.009	0.005 $\pm$ 0.001	0.074 $\pm$ 0.016	0.213 $\pm$ 0.035	0.462 $\pm$ 0.100	0.189	0.029	18.14	0.0846	0.0894	0.742	90.000 $\pm$ 9.487	0.226 $\pm$ 0.007	80.000 $\pm$ 12.649
QML	beta52	0.16	0.154	-0.006 $\pm$ 0.012	-0.039 $\pm$ 0.076	0.001 $\pm$ 0.001	0.038 $\pm$ 0.013	0.001 $\pm$ 0.001	0.037 $\pm$ 0.014	0.053 $\pm$ 0.028	0.231 $\pm$ 0.088	0.163	0.003	1.89	0.0474	0.0475	1.201	90.000 $\pm$ 9.487	0.181 $\pm$ 0.005	90.000 $\pm$ 9.487
QML	beta53	0.16	0.148	-0.012 $\pm$ 0.016	-0.077 $\pm$ 0.099	0.003 $\pm$ 0.001	0.050 $\pm$ 0.021	0.002 $\pm$ 0.001	0.049 $\pm$ 0.017	0.094 $\pm$ 0.043	0.307 $\pm$ 0.104	0.135	-0.025	-15.66	0.0413	0.0483	0.999	90.000 $\pm$ 9.487	0.196 $\pm$ 0.008	90.000 $\pm$ 9.487
QML	beta54	0.16	0.125	-0.035 $\pm$ 0.016	-0.221 $\pm$ 0.097	0.002 $\pm$ 0.001	0.049 $\pm$ 0.017	0.003 $\pm$ 0.002	0.059 $\pm$ 0.022	0.134 $\pm$ 0.076	0.366 $\pm$ 0.137	0.150	-0.010	-6.15	0.0391	0.0403	0.875	90.000 $\pm$ 9.487	0.169 $\pm$ 0.003	90.000 $\pm$ 9.487
QML	beta55	0.08	0.120	0.040 $\pm$ 0.014	0.506 $\pm$ 0.171	0.002 $\pm$ 0.001	0.043 $\pm$ 0.012	0.003 $\pm$ 0.001	0.058 $\pm$ 0.011	0.519 $\pm$ 0.155	0.720 $\pm$ 0.131	0.115	0.035	44.04	0.0604	0.0699	1.559	100.000 $\pm$ 0.000	0.264 $\pm$ 0.013	40.000 $\pm$ 15.492
QML	beta56	0.08	0.063	-0.017 $\pm$ 0.020	-0.218 $\pm$ 0.246	0.004 $\pm$ 0.001	0.062 $\pm$ 0.013	0.004 $\pm$ 0.001	0.062 $\pm$ 0.013	0.592 $\pm$ 0.122	0.770 $\pm$ 0.165	0.073	-0.007	-9.14	0.0550	0.0554	1.091	100.000 $\pm$ 0.000	0.266 $\pm$ 0.014	10.000 $\pm$ 9.487
QML	beta57	0.06	-0.005	-0.065 $\pm$ 0.012	-1.077 $\pm$ 0.202	0.001 $\pm$ 0.000	0.038 $\pm$ 0.009	0.005 $\pm$ 0.001	0.074 $\pm$ 0.010	1.525 $\pm$ 0.402	1.235 $\pm$ 0.171	-0.008	-0.068	-114.00	0.0384	0.0784	0.991	60.000 $\pm$ 15.492	0.149 $\pm$ 0.011	0.000 $\pm$ 0.000
QML	beta58	0.06	0.017	-0.043 $\pm$ 0.011	-0.717 $\pm$ 0.185	0.001 $\pmath$														

Table S176

Simulation 2 - Condition 32: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.4
(continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	-0.466	-0.546 ± 0.193	-6.830 ± 2.410	0.372 ± 0.146	0.610 ± 0.185	0.633 ± 0.351	0.796 ± 0.269	98.902 ± 54.891	9.945 ± 3.358	-0.392	-0.472	-590.03	0.5770	0.7455	1.598	90.000 ± 9.487	3.819 ± 1.662	0.000 ± 0.000
UPI	beta51	0.16	0.228	0.068 ± 0.031	0.428 ± 0.195	0.010 ± 0.003	0.099 ± 0.027	0.013 ± 0.004	0.116 ± 0.024	0.527 ± 0.163	0.726 ± 0.148	0.223	0.063	39.32	0.1011	0.1191	0.707	60.000 ± 15.492	0.274 ± 0.033	90.000 ± 9.487
UPI	beta52	0.16	0.116	-0.044 ± 0.028	-0.273 ± 0.172	0.008 ± 0.002	0.087 ± 0.017	0.009 ± 0.004	0.094 ± 0.027	0.342 ± 0.150	0.585 ± 0.169	0.102	-0.058	-36.01	0.1252	0.1378	1.205	100.000 ± 0.000	0.412 ± 0.048	20.000 ± 12.649
UPI	beta53	0.16	0.154	-0.006 ± 0.034	-0.038 ± 0.215	0.012 ± 0.006	0.109 ± 0.042	0.011 ± 0.005	0.103 ± 0.039	0.418 ± 0.211	0.646 ± 0.245	0.114	-0.046	-28.76	0.1067	0.1162	0.633	80.000 ± 12.649	0.270 ± 0.047	60.000 ± 15.492
UPI	beta54	0.16	0.036	-0.124 ± 0.020	-0.774 ± 0.123	0.004 ± 0.001	0.062 ± 0.014	0.019 ± 0.005	0.137 ± 0.018	0.737 ± 0.188	0.858 ± 0.115	0.123	-0.037	-22.89	0.1302	0.1353	0.876	30.000 ± 14.491	0.214 ± 0.044	10.000 ± 9.487
UPI	beta55	0.08	0.133	0.053 ± 0.017	0.666 ± 0.211	0.003 ± 0.001	0.053 ± 0.010	0.005 ± 0.002	0.074 ± 0.016	0.845 ± 0.310	0.919 ± 0.196	0.143	0.063	78.75	0.0845	0.1054	1.364	100.000 ± 0.000	0.286 ± 0.025	50.000 ± 15.811
UPI	beta56	0.08	0.075	-0.005 ± 0.020	-0.061 ± 0.248	0.004 ± 0.001	0.063 ± 0.014	0.004 ± 0.001	0.060 ± 0.017	0.558 ± 0.198	0.747 ± 0.215	0.108	0.028	35.50	0.0726	0.0779	1.323	90.000 ± 9.487	0.326 ± 0.025	0.000 ± 0.000
UPI	beta57	0.06	0.001	-0.059 ± 0.063	-0.978 ± 1.053	0.040 ± 0.016	0.200 ± 0.062	0.039 ± 0.016	0.198 ± 0.060	10.929 ± 4.566	3.306 ± 1.007	-0.007	-0.067	-110.94	0.1268	0.1432	0.749	100.000 ± 0.000	0.587 ± 0.141	0.000 ± 0.000
UPI	beta58	0.06	0.032	-0.028 ± 0.013	-0.468 ± 0.220	0.002 ± 0.001	0.042 ± 0.015	0.002 ± 0.001	0.049 ± 0.009	0.656 ± 0.164	0.810 ± 0.149	-0.005	-0.065	-108.71	0.1031	0.1220	1.548	80.000 ± 12.649	0.254 ± 0.043	0.000 ± 0.000

Note. MCSE = Monte Carlo Standard Error shown as \u000b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S177

Simulation 2 - Condition 33: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.6

Method	Param.	True	Mean-based										Median-based					Inference		
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LSAM	beta41	0.20	0.198	-0.002 $\pm$ 0.002	-0.012 $\pm$ 0.012	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.092 $\pm$ 0.005	0.304 $\pm$ 0.010	0.199	-0.001	-0.31	0.0608	0.0608	1.019	96.147 $\pm$ 0.788	0.243 $\pm$ 0.001	90.117 $\pm$ 1.221
LSAM	beta42	0.20	0.199	-0.001 $\pm$ 0.003	-0.004 $\pm$ 0.013	0.004 $\pm$ 0.000	0.064 $\pm$ 0.002	0.004 $\pm$ 0.000	0.064 $\pm$ 0.002	0.102 $\pm$ 0.006	0.320 $\pm$ 0.012	0.200	0.000	0.05	0.0622	0.0622	0.999	95.645 $\pm$ 0.835	0.251 $\pm$ 0.001	88.275 $\pm$ 1.317
LSAM	beta43	0.20	0.195	-0.005 $\pm$ 0.002	-0.023 $\pm$ 0.012	0.004 $\pm$ 0.000	0.059 $\pm$ 0.002	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	0.089 $\pm$ 0.005	0.298 $\pm$ 0.010	0.194	-0.006	-3.13	0.0566	0.0570	0.964	92.797 $\pm$ 1.058	0.224 $\pm$ 0.001	93.132 $\pm$ 1.035
LSAM	beta44	0.11	0.131	0.021 $\pm$ 0.003	0.195 $\pm$ 0.025	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.005 $\pm$ 0.000	0.071 $\pm$ 0.003	0.415 $\pm$ 0.027	0.645 $\pm$ 0.024	0.129	0.019	17.53	0.0650	0.0678	1.075	95.980 $\pm$ 0.804	0.285 $\pm$ 0.002	45.394 $\pm$ 2.038
LSAM	beta45	0.11	0.116	0.006 $\pm$ 0.003	0.051 $\pm$ 0.025	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.005 $\pm$ 0.000	0.068 $\pm$ 0.003	0.382 $\pm$ 0.024	0.618 $\pm$ 0.024	0.117	0.007	6.43	0.0653	0.0657	0.947	93.970 $\pm$ 0.974	0.252 $\pm$ 0.002	44.221 $\pm$ 2.033
LSAM	beta46	0.08	0.048	-0.032 $\pm$ 0.003	-0.402 $\pm$ 0.043	0.007 $\pm$ 0.000	0.084 $\pm$ 0.003	0.008 $\pm$ 0.000	0.090 $\pm$ 0.003	1.260 $\pm$ 0.070	1.122 $\pm$ 0.037	0.041	-0.039	-48.74	0.0766	0.0859	1.025	92.127 $\pm$ 1.102	0.337 $\pm$ 0.003	5.025 $\pm$ 0.894
LSAM	beta47	0.08	0.049	-0.031 $\pm$ 0.004	-0.391 $\pm$ 0.046	0.008 $\pm$ 0.001	0.090 $\pm$ 0.003	0.009 $\pm$ 0.001	0.096 $\pm$ 0.004	1.425 $\pm$ 0.090	1.194 $\pm$ 0.044	0.044	-0.036	-45.28	0.0794	0.0873	0.973	92.295 $\pm$ 1.091	0.344 $\pm$ 0.004	7.203 $\pm$ 1.058
LSAM	beta51	0.16	0.172	0.012 $\pm$ 0.003	0.077 $\pm$ 0.017	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.005 $\pm$ 0.000	0.069 $\pm$ 0.003	0.185 $\pm$ 0.012	0.431 $\pm$ 0.016	0.168	0.008	5.06	0.0667	0.0672	1.037	94.137 $\pm$ 0.961	0.276 $\pm$ 0.001	70.854 $\pm$ 1.860
LSAM	beta52	0.16	0.157	-0.003 $\pm$ 0.003	-0.017 $\pm$ 0.016	0.004 $\pm$ 0.000	0.062 $\pm$ 0.002	0.004 $\pm$ 0.000	0.062 $\pm$ 0.002	0.152 $\pm$ 0.009	0.390 $\pm$ 0.014	0.153	-0.007	-4.49	0.0576	0.0581	0.973	94.975 $\pm$ 0.894	0.238 $\pm$ 0.001	76.047 $\pm$ 1.747
LSAM	beta53	0.16	0.152	-0.008 $\pm$ 0.003	-0.053 $\pm$ 0.016	0.004 $\pm$ 0.000	0.064 $\pm$ 0.002	0.004 $\pm$ 0.000	0.064 $\pm$ 0.002	0.162 $\pm$ 0.010	0.402 $\pm$ 0.015	0.153	-0.007	-4.64	0.0612	0.0617	0.979	94.137 $\pm$ 0.961	0.245 $\pm$ 0.001	69.682 $\pm$ 1.881
LSAM	beta54	0.16	0.157	-0.003 $\pm$ 0.002	-0.018 $\pm$ 0.016	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.144 $\pm$ 0.008	0.379 $\pm$ 0.013	0.156	-0.004	-2.72	0.0627	0.0628	0.997	94.472 $\pm$ 0.935	0.237 $\pm$ 0.001	73.869 $\pm$ 1.798
LSAM	beta55	0.08	0.094	0.014 $\pm$ 0.004	0.178 $\pm$ 0.044	0.007 $\pm$ 0.000	0.086 $\pm$ 0.003	0.008 $\pm$ 0.000	0.087 $\pm$ 0.003	1.183 $\pm$ 0.075	1.088 $\pm$ 0.041	0.091	0.011	13.95	0.0833	0.0841	0.953	92.797 $\pm$ 1.058	0.321 $\pm$ 0.003	21.776 $\pm$ 1.689
LSAM	beta56	0.08	0.079	-0.001 $\pm$ 0.003	-0.017 $\pm$ 0.041	0.006 $\pm$ 0.000	0.080 $\pm$ 0.002	0.006 $\pm$ 0.000	0.080 $\pm$ 0.003	0.999 $\pm$ 0.055	1.000 $\pm$ 0.034	0.077	-0.003	-3.31	0.0792	0.0793	0.962	94.640 $\pm$ 0.922	0.302 $\pm$ 0.002	19.598 $\pm$ 1.625
LSAM	beta57	0.06	0.031	-0.029 $\pm$ 0.003	-0.475 $\pm$ 0.057	0.007 $\pm$ 0.000	0.083 $\pm$ 0.003	0.008 $\pm$ 0.000	0.088 $\pm$ 0.003	2.134 $\pm$ 0.123	1.461 $\pm$ 0.050	0.029	-0.031	-51.08	0.0779	0.0837	1.004	93.300 $\pm$ 1.023	0.327 $\pm$ 0.003	5.360 $\pm$ 0.922
LSAM	beta58	0.06	0.037	-0.023 $\pm$ 0.003	-0.389 $\pm$ 0.051	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.006 $\pm$ 0.000	0.079 $\pm$ 0.003	1.726 $\pm$ 0.107	1.314 $\pm$ 0.048	0.034	-0.026	-42.54	0.0698	0.0743	1.051	93.802 $\pm$ 0.987	0.310 $\pm$ 0.004	5.193 $\pm$ 0.908
QML	beta41	0.20	0.193	-0.007 $\pm$ 0.002	-0.034 $\pm$ 0.011	0.003 $\pm$ 0.000	0.055 $\pm$ 0.001	0.003 $\pm$ 0.000	0.055 $\pm$ 0.002	0.076 $\pm$ 0.004	0.276 $\pm$ 0.009	0.193	-0.007	-3.72	0.0571	0.0575	1.035	95.310 $\pm$ 0.865	0.222 $\pm$ 0.006	93.132 $\pm$ 1.035
QML	beta42	0.20	0.197	-0.003 $\pm$ 0.002	-0.013 $\pm$ 0.012	0.003 $\pm$ 0.000	0.058 $\pm$ 0.002	0.003 $\pm$ 0.000	0.058 $\pm$ 0.002	0.085 $\pm$ 0.005	0.292 $\pm$ 0.011	0.197	-0.003	-1.41	0.0580	0.0581	0.979	94.472 $\pm$ 0.935	0.224 $\pm$ 0.001	94.305 $\pm$ 0.948
QML	beta43	0.20	0.187	-0.013 $\pm$ 0.002	-0.063 $\pm$ 0.012	0.003 $\pm$ 0.000	0.057 $\pm$ 0.002	0.003 $\pm$ 0.000	0.058 $\pm$ 0.002	0.085 $\pm$ 0.005	0.292 $\pm$ 0.010	0.183	-0.017	-8.68	0.0545	0.0572	0.947	92.797 $\pm$ 1.058	0.212 $\pm$ 0.001	94.137 $\pm$ 0.961
QML	beta44	0.11	0.129	0.019 $\pm$ 0.003	0.170 $\pm$ 0.023	0.004 $\pm$ 0.000	0.063 $\pm$ 0.002	0.004 $\pm$ 0.000	0.065 $\pm$ 0.003	0.353 $\pm$ 0.024	0.594 $\pm$ 0.023	0.125	0.015	13.41	0.0614	0.0632	1.082	95.980 $\pm$ 0.804	0.266 $\pm$ 0.002	48.409 $\pm$ 2.045
QML	beta45	0.11	0.114	0.004 $\pm$ 0.003	0.035 $\pm$ 0.023	0.004 $\pm$ 0.000	0.062 $\pm$ 0.002	0.004 $\pm$ 0.000	0.062 $\pm$ 0.002	0.321 $\pm$ 0.020	0.566 $\pm$ 0.021	0.117	0.007	6.15	0.0622	0.0625	0.964	94.472 $\pm$ 0.935	0.235 $\pm$ 0.002	47.906 $\pm$ 2.045
QML	beta46	0.08	0.026	-0.054 $\pm$ 0.002	-0.673 $\pm$ 0.026	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.005 $\pm$ 0.000	0.074 $\pm$ 0.002	0.858 $\pm$ 0.038	0.926 $\pm$ 0.022	0.023	-0.057	-71.16	0.0503	0.0760	1.083	82.747 $\pm$ 1.546	0.216 $\pm$ 0.002	4.858 $\pm$ 0.880
QML	beta47	0.08	0.029	-0.051 $\pm$ 0.002	-0.639 $\pm$ 0.029	0.003 $\pm$ 0.000	0.056 $\pm$ 0.002	0.006 $\pm$ 0.000	0.076 $\pm$ 0.002	0.895 $\pm$ 0.047	0.946 $\pm$ 0.027	0.026	-0.054	-67.21	0.0523	0.0750	0.986	82.747 $\pm$ 1.546	0.216 $\pm$ 0.002	8.543 $\pm$ 1.144
QML	beta51	0.16	0.184	0.024 $\pm$ 0.003	0.149 $\pm$ 0.017	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.005 $\pm$ 0.000	0.072 $\pm$ 0.003	0.202 $\pm$ 0.013	0.449 $\pm$ 0.016	0.182	0.022	13.45	0.0649	0.0684	1.059	94.305 $\pm$ 0.948	0.282 $\pm$ 0.001	76.047 $\pm$ 1.747
QML	beta52	0.16	0.149	-0.011 $\pm$ 0.002	-0.070 $\pm$ 0.015	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.146 $\pm$ 0.008	0.382 $\pm$ 0.013	0.145	-0.015	-9.64	0.0581	0.0601	0.978	94.305 $\pm$ 0.948	0.230 $\pm$ 0.001	72.027 $\pm$ 1.837
QML	beta53	0.16	0.138	-0.022 $\pm$ 0.002	-0.139 $\pm$ 0.016	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.004 $\pm$ 0.000	0.065 $\pm$ 0.002	0.165 $\pm$ 0.010	0.406 $\pm$ 0.014	0.137	-0.023	-14.17	0.0617	0.0657	0.992	93.635 $\pm$ 0.999	0.237 $\pm$ 0.002	64.657 $\pm$ 1.956
QML	beta54	0.16	0.158	-0.002 $\pm$ 0.002	-0.014 $\pm$ 0.015	0.003 $\pm$ 0.000	0.058 $\pm$ 0.002	0.003 $\pm$ 0.000	0.058 $\pm$ 0.002	0.132 $\pm$ 0.007	0.364 $\pm$ 0.013	0.154	-0.006	-3.66	0.0585	0.0588	1.008	95.477 $\pm$ 0.850	0.230 $\pm$ 0.008	79.397 $\pm$ 1.655
QML	beta55	0.08	0.094	0.014 $\pm$ 0.003	0.175 $\pm$ 0.042	0.007 $\pm$ 0.000	0.082 $\pm$ 0.002	0.007 $\pm$ 0.000	0.084 $\pm$ 0.003	1.092 $\pm$ 0.066	1.045 $\pm$ 0.038	0.090	0.010	11.93	0.0834	0.0839	0.974	93.132 $\pm$ 1.035	0.315 $\pm$ 0.002	22.948 $\pm$ 1.721
QML	beta56	0.08	0.075	-0.005 $\pm$ 0.003	-0.060 $\pm$ 0.038	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.872 $\pm$ 0.045	0.934 $\pm$ 0.031	0.073	-0.007	-8.72	0.0813	0.0816	0.979	94.472 $\pm$ 0.935	0.286 $\pm$ 0.002	19.765 $\pm$ 1.630
QML	beta57	0.06	0.017	-0.043 $\pm$ 0.002	-0.716 $\pm$ 0.038	0.003 $\pm$ 0.000	0.056 $\pm$ 0.002	0.005 $\pm$ 0.000	0.071 $\pm$ 0.002	1.392 $\pm$ 0.075	1.180 $\pm$ 0.035	0.014	-0.046	-76.82	0.0535	0.0706	1.007	85.092 $\pm$ 1.458	0.222 $\pm$ 0.002	6.198 $\pm$ 0.987
QML	beta58	0.06	0.026	-0.034 $\pm$ 0.002	-0.572 $\pm$ 0.032	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.003 $\pm$ 0.000	0.059 $\pm$ 0.002	0.952 $\pm$ 0.052	0.976 $\pm$ 0.030	0.021	-0.039	-64.73	0.0460	0.0602	1.066	88.610 $\pm$ 1.300	0.198 $\pm$ 0.002	5.863 $\pm$ 0.961
UPI	beta41	0.20	0.211	0.011 $\pm$ 0.003	0.054 $\pm$ 0.015	0.005 $\pm$ 0.000	0.072 $\pm$ 0.003	0.005 $\pm$ 0.000	0.073 $\pm$ 0.003	0.132 $\pm$ 0.010	0.363 $\pm$ 0.016	0.207	0.007	3.46	0.0708	0.0712	1.177	96.147 $\pm$ 0.788	0.331 $\pm$ 0.024	81.240 $\pm$ 1.598
UPI	beta42	0.20	0.214	0.014 $\pm$ 0.003	0.069 $\pm$ 0.016	0.006 $\pm$ 0.000	0.077 $\pm$ 0.003	0.006 $\pm$ 0.000	0.078 $\pm$ 0.003	0.151 $\pm$ 0.011	0.389 $\pm$ 0.016	0.213	0.013	6.52	0.0730	0.0741	1.204	96.817 $\pm$ 0.718	0.362 $\pm$ 0.013	75.377 $\pm$ 1.763
UPI	beta43	0.20	0.190	-0.010 $\pm$ 0.003	-0.050 $\pm$ 0.013	0.004 $\pm$ 0.000	0.064 $\pm$ 0.002	0.004 $\pm$ 0.000	0.065 $\pm$ 0.002	0.105 $\pm$ 0.006	0.324 $\pm$ 0.011	0.188	-0.012	-5.92	0.0657	0.0667	1.059	93.970 $\pm$ 0.974	0.266 $\pm$ 0.006	83.752 $\pm$ 1.510

Table S177

Simulation 2 - Condition 33: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.6
(continued)

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.050	-0.030 ± 0.006	-0.370 ± 0.076	0.022 ± 0.002	0.148 ± 0.005	0.023 ± 0.002	0.151 ± 0.006	3.547 ± 0.251	1.883 ± 0.077	0.043	-0.037	-45.71	0.1358	0.1407	1.193	94.137 ± 0.961	0.691 ± 0.028	4.858 ± 0.880
UPI	beta51	0.16	0.179	0.019 ± 0.003	0.116 ± 0.017	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.070 ± 0.003	0.190 ± 0.012	0.436 ± 0.016	0.175	0.015	9.20	0.0696	0.0711	1.099	95.812 ± 0.820	0.290 ± 0.002	70.687 ± 1.863
UPI	beta52	0.16	0.164	0.004 ± 0.003	0.027 ± 0.017	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.171 ± 0.011	0.413 ± 0.015	0.161	0.001	0.48	0.0614	0.0614	1.128	96.315 ± 0.771	0.292 ± 0.008	67.672 ± 1.914
UPI	beta53	0.16	0.148	-0.012 ± 0.003	-0.078 ± 0.016	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.166 ± 0.010	0.407 ± 0.015	0.146	-0.014	-8.90	0.0644	0.0660	1.112	95.310 ± 0.865	0.279 ± 0.004	57.956 ± 2.020
UPI	beta54	0.16	0.161	0.001 ± 0.003	0.008 ± 0.018	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.003	0.183 ± 0.013	0.428 ± 0.017	0.159	-0.001	-0.36	0.0638	0.0638	1.192	96.985 ± 0.700	0.320 ± 0.012	63.484 ± 1.971
UPI	beta55	0.08	0.097	0.017 ± 0.003	0.210 ± 0.043	0.007 ± 0.000	0.084 ± 0.003	0.007 ± 0.000	0.085 ± 0.003	1.134 ± 0.075	1.065 ± 0.041	0.092	0.012	15.31	0.0817	0.0827	1.094	95.812 ± 0.820	0.358 ± 0.005	18.258 ± 1.581
UPI	beta56	0.08	0.078	-0.002 ± 0.003	-0.027 ± 0.042	0.007 ± 0.000	0.082 ± 0.002	0.007 ± 0.000	0.082 ± 0.003	1.044 ± 0.059	1.022 ± 0.036	0.073	-0.007	-8.60	0.0825	0.0827	1.049	96.650 ± 0.736	0.336 ± 0.004	14.740 ± 1.451
UPI	beta57	0.06	0.049	-0.011 ± 0.005	-0.180 ± 0.077	0.013 ± 0.001	0.113 ± 0.004	0.013 ± 0.001	0.114 ± 0.005	3.591 ± 0.251	1.895 ± 0.077	0.043	-0.017	-28.43	0.1091	0.1104	1.176	94.305 ± 0.948	0.522 ± 0.020	5.863 ± 0.961
UPI	beta58	0.06	0.041	-0.019 ± 0.005	-0.316 ± 0.082	0.015 ± 0.001	0.120 ± 0.005	0.015 ± 0.001	0.122 ± 0.005	4.126 ± 0.325	2.031 ± 0.090	0.034	-0.026	-44.15	0.0999	0.1033	1.301	94.137 ± 0.961	0.615 ± 0.029	3.350 ± 0.736

Note. MCSE = Monte Carlo Standard Error shown as \u000b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S178

Simulation 2 - Condition 34: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.6

Method	Param.	True	Mean-based										Median-based					Inference		
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LSAM	beta41	0.20	0.198	-0.002 $\pm$ 0.001	-0.009 $\pm$ 0.007	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.036 $\pm$ 0.002	0.189 $\pm$ 0.006	0.199	-0.001	-0.37	0.0384	0.0384	1.002	95.448 $\pm$ 0.799	0.148 $\pm$ 0.000	99.853 $\pm$ 0.147
LSAM	beta42	0.20	0.194	-0.006 $\pm$ 0.002	-0.032 $\pm$ 0.008	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.040 $\pm$ 0.002	0.199 $\pm$ 0.007	0.194	-0.006	-3.08	0.0373	0.0378	0.989	94.126 $\pm$ 0.901	0.153 $\pm$ 0.000	99.853 $\pm$ 0.147
LSAM	beta43	0.20	0.198	-0.002 $\pm$ 0.001	-0.011 $\pm$ 0.007	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.033 $\pm$ 0.002	0.181 $\pm$ 0.006	0.197	-0.003	-1.26	0.0361	0.0362	0.979	94.567 $\pm$ 0.869	0.139 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta44	0.11	0.127	0.017 $\pm$ 0.002	0.158 $\pm$ 0.015	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.002	0.187 $\pm$ 0.011	0.432 $\pm$ 0.015	0.126	0.016	14.88	0.0453	0.0482	0.983	92.217 $\pm$ 1.027	0.171 $\pm$ 0.001	82.379 $\pm$ 1.460
LSAM	beta45	0.11	0.117	0.007 $\pm$ 0.001	0.065 $\pm$ 0.014	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.130 $\pm$ 0.007	0.360 $\pm$ 0.012	0.116	0.006	5.74	0.0391	0.0396	1.001	95.154 $\pm$ 0.823	0.153 $\pm$ 0.001	85.609 $\pm$ 1.345
LSAM	beta46	0.08	0.043	-0.037 $\pm$ 0.002	-0.463 $\pm$ 0.025	0.003 $\pm$ 0.000	0.053 $\pm$ 0.002	0.004 $\pm$ 0.000	0.064 $\pm$ 0.002	0.645 $\pm$ 0.031	0.803 $\pm$ 0.022	0.041	-0.039	-49.20	0.0494	0.0631	1.009	86.344 $\pm$ 1.316	0.208 $\pm$ 0.001	10.866 $\pm$ 1.193
LSAM	beta47	0.08	0.044	-0.036 $\pm$ 0.002	-0.446 $\pm$ 0.025	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.004 $\pm$ 0.000	0.063 $\pm$ 0.002	0.621 $\pm$ 0.030	0.788 $\pm$ 0.022	0.043	-0.037	-46.35	0.0535	0.0651	0.987	85.316 $\pm$ 1.356	0.201 $\pm$ 0.001	12.041 $\pm$ 1.247
LSAM	beta51	0.16	0.170	0.010 $\pm$ 0.002	0.063 $\pm$ 0.010	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.073 $\pm$ 0.004	0.269 $\pm$ 0.008	0.170	0.010	6.33	0.0415	0.0427	1.049	95.888 $\pm$ 0.761	0.173 $\pm$ 0.000	97.504 $\pm$ 0.598
LSAM	beta52	0.16	0.154	-0.006 $\pm$ 0.001	-0.040 $\pm$ 0.009	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.058 $\pm$ 0.003	0.240 $\pm$ 0.008	0.153	-0.007	-4.56	0.0367	0.0374	0.986	94.420 $\pm$ 0.880	0.146 $\pm$ 0.000	97.944 $\pm$ 0.544
LSAM	beta53	0.16	0.155	-0.005 $\pm$ 0.001	-0.030 $\pm$ 0.009	0.001 $\pm$ 0.000	0.039 $\pm$ 0.001	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.059 $\pm$ 0.003	0.243 $\pm$ 0.008	0.154	-0.006	-3.52	0.0403	0.0407	1.000	95.742 $\pm$ 0.774	0.151 $\pm$ 0.000	98.238 $\pm$ 0.504
LSAM	beta54	0.16	0.155	-0.005 $\pm$ 0.001	-0.033 $\pm$ 0.009	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.056 $\pm$ 0.003	0.237 $\pm$ 0.008	0.155	-0.005	-3.41	0.0382	0.0386	0.992	95.154 $\pm$ 0.823	0.146 $\pm$ 0.000	97.944 $\pm$ 0.544
LSAM	beta55	0.08	0.093	0.013 $\pm$ 0.002	0.164 $\pm$ 0.024	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.003 $\pm$ 0.000	0.053 $\pm$ 0.002	0.433 $\pm$ 0.024	0.658 $\pm$ 0.022	0.092	0.012	14.57	0.0525	0.0537	0.977	94.420 $\pm$ 0.880	0.196 $\pm$ 0.001	46.549 $\pm$ 1.911
LSAM	beta56	0.08	0.083	0.003 $\pm$ 0.002	0.032 $\pm$ 0.023	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.002	0.360 $\pm$ 0.019	0.600 $\pm$ 0.020	0.083	0.003	3.62	0.0486	0.0487	0.984	95.154 $\pm$ 0.823	0.185 $\pm$ 0.001	41.410 $\pm$ 1.888
LSAM	beta57	0.06	0.032	-0.028 $\pm$ 0.002	-0.461 $\pm$ 0.034	0.003 $\pm$ 0.000	0.054 $\pm$ 0.002	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	1.010 $\pm$ 0.053	1.005 $\pm$ 0.030	0.031	-0.029	-48.62	0.0539	0.0613	0.953	88.253 $\pm$ 1.234	0.200 $\pm$ 0.001	8.517 $\pm$ 1.070
LSAM	beta58	0.06	0.035	-0.025 $\pm$ 0.002	-0.423 $\pm$ 0.032	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	0.003 $\pm$ 0.000	0.056 $\pm$ 0.002	0.867 $\pm$ 0.043	0.931 $\pm$ 0.027	0.030	-0.030	-50.39	0.0479	0.0566	0.943	87.372 $\pm$ 1.273	0.184 $\pm$ 0.001	10.866 $\pm$ 1.193
QML	beta41	0.20	0.195	-0.005 $\pm$ 0.001	-0.026 $\pm$ 0.007	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.031 $\pm$ 0.002	0.177 $\pm$ 0.006	0.194	-0.006	-2.79	0.0356	0.0361	0.989	95.595 $\pm$ 0.786	0.136 $\pm$ 0.000	100.000 $\pm$ 0.000
QML	beta42	0.20	0.192	-0.008 $\pm$ 0.001	-0.041 $\pm$ 0.007	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.035 $\pm$ 0.002	0.187 $\pm$ 0.006	0.190	-0.010	-5.00	0.0338	0.0353	0.970	92.805 $\pm$ 0.990	0.139 $\pm$ 0.000	100.000 $\pm$ 0.000
QML	beta43	0.20	0.191	-0.009 $\pm$ 0.001	-0.046 $\pm$ 0.007	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.032 $\pm$ 0.002	0.178 $\pm$ 0.006	0.191	-0.009	-4.73	0.0353	0.0365	0.985	93.686 $\pm$ 0.932	0.133 $\pm$ 0.000	100.000 $\pm$ 0.000
QML	beta44	0.11	0.125	0.015 $\pm$ 0.002	0.140 $\pm$ 0.015	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.165 $\pm$ 0.009	0.406 $\pm$ 0.013	0.124	0.014	13.18	0.0429	0.0453	1.002	92.952 $\pm$ 0.981	0.165 $\pm$ 0.001	85.169 $\pm$ 1.362
QML	beta45	0.11	0.115	0.005 $\pm$ 0.001	0.048 $\pm$ 0.013	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.115 $\pm$ 0.006	0.339 $\pm$ 0.011	0.114	0.004	3.90	0.0373	0.0376	1.015	95.595 $\pm$ 0.786	0.147 $\pm$ 0.001	87.665 $\pm$ 1.260
QML	beta46	0.08	0.024	-0.056 $\pm$ 0.001	-0.706 $\pm$ 0.016	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.004 $\pm$ 0.000	0.066 $\pm$ 0.001	0.672 $\pm$ 0.024	0.820 $\pm$ 0.015	0.023	-0.057	-71.55	0.0331	0.0661	1.039	61.380 $\pm$ 1.866	0.136 $\pm$ 0.001	8.811 $\pm$ 1.086
QML	beta47	0.08	0.027	-0.053 $\pm$ 0.001	-0.661 $\pm$ 0.016	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.004 $\pm$ 0.000	0.063 $\pm$ 0.001	0.619 $\pm$ 0.023	0.787 $\pm$ 0.016	0.026	-0.054	-67.65	0.0348	0.0643	0.980	62.849 $\pm$ 1.852	0.131 $\pm$ 0.001	11.601 $\pm$ 1.227
QML	beta51	0.16	0.180	0.020 $\pm$ 0.002	0.126 $\pm$ 0.010	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.086 $\pm$ 0.004	0.293 $\pm$ 0.009	0.180	0.020	12.78	0.0419	0.0466	1.054	94.420 $\pm$ 0.880	0.175 $\pm$ 0.000	98.532 $\pm$ 0.461
QML	beta52	0.16	0.146	-0.014 $\pm$ 0.001	-0.088 $\pm$ 0.009	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.061 $\pm$ 0.003	0.247 $\pm$ 0.008	0.147	-0.013	-8.19	0.0357	0.0380	0.979	91.777 $\pm$ 1.053	0.142 $\pm$ 0.000	98.385 $\pm$ 0.483
QML	beta53	0.16	0.141	-0.019 $\pm$ 0.001	-0.116 $\pm$ 0.009	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.067 $\pm$ 0.003	0.260 $\pm$ 0.007	0.141	-0.019	-11.60	0.0394	0.0436	0.998	92.658 $\pm$ 0.999	0.146 $\pm$ 0.000	96.916 $\pm$ 0.662
QML	beta54	0.16	0.156	-0.004 $\pm$ 0.001	-0.023 $\pm$ 0.008	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.049 $\pm$ 0.003	0.221 $\pm$ 0.007	0.155	-0.005	-3.00	0.0353	0.0356	1.006	95.448 $\pm$ 0.799	0.139 $\pm$ 0.000	99.266 $\pm$ 0.327
QML	beta55	0.08	0.093	0.013 $\pm$ 0.002	0.167 $\pm$ 0.024	0.003 $\pm$ 0.000	0.050 $\pm$ 0.001	0.003 $\pm$ 0.000	0.052 $\pm$ 0.002	0.419 $\pm$ 0.022	0.647 $\pm$ 0.021	0.091	0.011	13.82	0.0521	0.0532	0.997	94.860 $\pm$ 0.846	0.196 $\pm$ 0.001	45.962 $\pm$ 1.910
QML	beta56	0.08	0.079	-0.001 $\pm$ 0.002	-0.015 $\pm$ 0.022	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.002	0.332 $\pm$ 0.018	0.576 $\pm$ 0.019	0.079	-0.001	-1.47	0.0452	0.0452	0.985	95.448 $\pm$ 0.799	0.178 $\pm$ 0.001	41.116 $\pm$ 1.886
QML	beta57	0.06	0.017	-0.043 $\pm$ 0.001	-0.722 $\pm$ 0.023	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.003 $\pm$ 0.000	0.056 $\pm$ 0.001	0.881 $\pm$ 0.037	0.938 $\pm$ 0.021	0.017	-0.043	-71.97	0.0371	0.0569	0.983	72.981 $\pm$ 1.702	0.139 $\pm$ 0.001	6.461 $\pm$ 0.942
QML	beta58	0.06	0.024	-0.036 $\pm$ 0.001	-0.593 $\pm$ 0.021	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.649 $\pm$ 0.028	0.806 $\pm$ 0.019	0.022	-0.038	-63.83	0.0321	0.0500	0.958	75.918 $\pm$ 1.639	0.123 $\pm$ 0.001	12.628 $\pm$ 1.273
UPI	beta41	0.20	0.205	0.005 $\pm$ 0.002	0.026 $\pm$ 0.008	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.002 $\pm$ 0.000	0.044 $\pm$ 0.002	0.049 $\pm$ 0.003	0.222 $\pm$ 0.008	0.206	0.006	2.96	0.0429	0.0433	1.232	95.448 $\pm$ 0.799	0.213 $\pm$ 0.013	95.595 $\pm$ 0.786
UPI	beta42	0.20	0.198	-0.002 $\pm$ 0.002	-0.008 $\pm$ 0.009	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.002 $\pm$ 0.000	0.045 $\pm$ 0.002	0.051 $\pm$ 0.003	0.226 $\pm$ 0.008	0.198	-0.002	-0.96	0.0432	0.0432	1.233	96.476 $\pm$ 0.707	0.219 $\pm$ 0.008	93.245 $\pm$ 0.962
UPI	beta43	0.20	0.195	-0.005 $\pm$ 0.001	-0.026 $\pm$ 0.007	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.001 $\pm$ 0.000	0.039 $\pm$ 0.001	0.037 $\pm$ 0.002	0.193 $\pm$ 0.006	0.193	-0.007	-3.49	0.0394	0.0400	1.067	95.154 $\pm$ 0.823	0.160 $\pm$ 0.004	98.972 $\pm$

Table S178

Simulation 2 - Condition 34: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.6
(continued)

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.048	-0.032 ± 0.003	-0.402 ± 0.039	0.007 ± 0.000	0.082 ± 0.003	0.008 ± 0.000	0.088 ± 0.003	1.216 ± 0.075	1.103 ± 0.039	0.043	-0.037	-46.27	0.0748	0.0834	1.142	93.686 ± 0.932	0.368 ± 0.012	8.517 ± 1.070
UPI	beta51	0.16	0.173	0.013 ± 0.002	0.079 ± 0.010	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.075 ± 0.004	0.273 ± 0.008	0.174	0.014	8.58	0.0423	0.0444	1.103	96.623 ± 0.692	0.181 ± 0.002	96.916 ± 0.662
UPI	beta52	0.16	0.159	-0.001 ± 0.002	-0.003 ± 0.010	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.065 ± 0.004	0.254 ± 0.009	0.160	0.000	0.15	0.0416	0.0416	1.200	96.476 ± 0.707	0.192 ± 0.010	93.392 ± 0.952
UPI	beta53	0.16	0.151	-0.009 ± 0.002	-0.058 ± 0.009	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.064 ± 0.003	0.252 ± 0.008	0.148	-0.012	-7.32	0.0422	0.0438	1.118	96.329 ± 0.721	0.172 ± 0.005	94.714 ± 0.857
UPI	beta54	0.16	0.157	-0.003 ± 0.002	-0.020 ± 0.011	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.002	0.076 ± 0.005	0.275 ± 0.011	0.156	-0.004	-2.58	0.0439	0.0441	1.115	96.329 ± 0.721	0.192 ± 0.004	89.427 ± 1.178
UPI	beta55	0.08	0.095	0.015 ± 0.002	0.183 ± 0.025	0.003 ± 0.000	0.053 ± 0.002	0.003 ± 0.000	0.055 ± 0.002	0.474 ± 0.029	0.688 ± 0.024	0.092	0.012	15.40	0.0519	0.0533	1.049	95.007 ± 0.835	0.219 ± 0.004	42.144 ± 1.892
UPI	beta56	0.08	0.082	0.002 ± 0.002	0.026 ± 0.023	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.002	0.372 ± 0.019	0.610 ± 0.020	0.083	0.003	3.49	0.0494	0.0495	1.025	95.888 ± 0.761	0.196 ± 0.001	37.445 ± 1.855
UPI	beta57	0.06	0.043	-0.017 ± 0.003	-0.283 ± 0.049	0.006 ± 0.000	0.076 ± 0.003	0.006 ± 0.000	0.078 ± 0.003	1.699 ± 0.113	1.303 ± 0.050	0.042	-0.018	-29.51	0.0699	0.0721	1.124	93.392 ± 0.952	0.337 ± 0.019	8.076 ± 1.044
UPI	beta58	0.06	0.034	-0.026 ± 0.003	-0.432 ± 0.051	0.006 ± 0.000	0.079 ± 0.003	0.007 ± 0.000	0.083 ± 0.003	1.935 ± 0.138	1.391 ± 0.056	0.028	-0.032	-53.66	0.0665	0.0739	1.075	93.392 ± 0.952	0.335 ± 0.009	7.489 ± 1.009

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S179

Simulation 2 - Condition 35: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, $N = 400$, Reliability = 0.8

Method	Param.	True	Mean-based										Median-based					Inference		
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LSAM	beta41	0.20	0.197	-0.003 $\pm$ 0.001	-0.013 $\pm$ 0.007	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.056 $\pm$ 0.002	0.236 $\pm$ 0.006	0.198	-0.002	-0.79	0.0469	0.0470	1.011	95.090 $\pm$ 0.684	0.187 $\pm$ 0.000	98.697 $\pm$ 0.359
LSAM	beta42	0.20	0.200	0.000 $\pm$ 0.002	0.000 $\pm$ 0.008	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.057 $\pm$ 0.003	0.238 $\pm$ 0.006	0.198	-0.002	-1.06	0.0487	0.0487	1.015	95.291 $\pm$ 0.671	0.190 $\pm$ 0.000	98.998 $\pm$ 0.315
LSAM	beta43	0.20	0.201	0.001 $\pm$ 0.001	0.007 $\pm$ 0.007	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.050 $\pm$ 0.002	0.225 $\pm$ 0.006	0.201	0.001	0.57	0.0441	0.0441	1.007	95.291 $\pm$ 0.671	0.177 $\pm$ 0.000	99.299 $\pm$ 0.264
LSAM	beta44	0.11	0.114	0.004 $\pm$ 0.002	0.036 $\pm$ 0.015	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.003 $\pm$ 0.000	0.053 $\pm$ 0.002	0.236 $\pm$ 0.011	0.485 $\pm$ 0.014	0.114	0.004	3.69	0.0514	0.0516	0.960	94.188 $\pm$ 0.741	0.201 $\pm$ 0.001	61.022 $\pm$ 1.544
LSAM	beta45	0.11	0.112	0.002 $\pm$ 0.002	0.022 $\pm$ 0.014	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.193 $\pm$ 0.009	0.439 $\pm$ 0.013	0.109	-0.001	-0.71	0.0472	0.0472	0.965	94.389 $\pm$ 0.728	0.182 $\pm$ 0.001	66.633 $\pm$ 1.493
LSAM	beta46	0.08	0.071	-0.009 $\pm$ 0.002	-0.118 $\pm$ 0.023	0.003 $\pm$ 0.000	0.058 $\pm$ 0.001	0.003 $\pm$ 0.000	0.058 $\pm$ 0.002	0.532 $\pm$ 0.025	0.729 $\pm$ 0.020	0.070	-0.010	-12.62	0.0570	0.0579	1.023	94.589 $\pm$ 0.716	0.231 $\pm$ 0.001	20.842 $\pm$ 1.286
LSAM	beta47	0.08	0.071	-0.009 $\pm$ 0.002	-0.113 $\pm$ 0.023	0.003 $\pm$ 0.000	0.058 $\pm$ 0.001	0.003 $\pm$ 0.000	0.059 $\pm$ 0.002	0.544 $\pm$ 0.025	0.738 $\pm$ 0.020	0.068	-0.012	-14.71	0.0575	0.0587	1.000	94.289 $\pm$ 0.735	0.229 $\pm$ 0.001	22.645 $\pm$ 1.325
LSAM	beta51	0.16	0.166	0.006 $\pm$ 0.002	0.035 $\pm$ 0.012	0.003 $\pm$ 0.000	0.059 $\pm$ 0.001	0.004 $\pm$ 0.000	0.059 $\pm$ 0.002	0.137 $\pm$ 0.006	0.371 $\pm$ 0.010	0.167	0.007	4.42	0.0607	0.0611	0.971	94.990 $\pm$ 0.691	0.225 $\pm$ 0.001	81.463 $\pm$ 1.230
LSAM	beta52	0.16	0.160	0.000 $\pm$ 0.002	0.001 $\pm$ 0.010	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.092 $\pm$ 0.004	0.303 $\pm$ 0.008	0.162	0.002	1.47	0.0486	0.0487	0.989	95.190 $\pm$ 0.677	0.188 $\pm$ 0.000	89.980 $\pm$ 0.950
LSAM	beta53	0.16	0.158	-0.002 $\pm$ 0.002	-0.012 $\pm$ 0.010	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.104 $\pm$ 0.005	0.322 $\pm$ 0.009	0.157	-0.003	-1.90	0.0517	0.0518	0.960	93.687 $\pm$ 0.770	0.194 $\pm$ 0.000	87.275 $\pm$ 1.055
LSAM	beta54	0.16	0.159	-0.001 $\pm$ 0.002	-0.009 $\pm$ 0.010	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.090 $\pm$ 0.004	0.300 $\pm$ 0.008	0.159	-0.001	-0.68	0.0486	0.0487	0.995	94.489 $\pm$ 0.722	0.187 $\pm$ 0.000	91.884 $\pm$ 0.864
LSAM	beta55	0.08	0.083	0.003 $\pm$ 0.002	0.036 $\pm$ 0.024	0.004 $\pm$ 0.000	0.061 $\pm$ 0.001	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.586 $\pm$ 0.025	0.765 $\pm$ 0.020	0.081	0.001	1.39	0.0638	0.0638	0.978	93.788 $\pm$ 0.764	0.235 $\pm$ 0.001	29.359 $\pm$ 1.442
LSAM	beta56	0.08	0.081	0.001 $\pm$ 0.002	0.016 $\pm$ 0.023	0.003 $\pm$ 0.000	0.059 $\pm$ 0.001	0.003 $\pm$ 0.000	0.059 $\pm$ 0.002	0.540 $\pm$ 0.024	0.735 $\pm$ 0.020	0.081	0.001	1.11	0.0599	0.0599	0.969	94.389 $\pm$ 0.728	0.223 $\pm$ 0.001	29.659 $\pm$ 1.446
LSAM	beta57	0.06	0.053	-0.007 $\pm$ 0.002	-0.125 $\pm$ 0.032	0.004 $\pm$ 0.000	0.061 $\pm$ 0.001	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	1.040 $\pm$ 0.049	1.020 $\pm$ 0.029	0.052	-0.008	-14.13	0.0566	0.0572	0.978	94.088 $\pm$ 0.747	0.233 $\pm$ 0.001	14.729 $\pm$ 1.122
LSAM	beta58	0.06	0.055	-0.005 $\pm$ 0.002	-0.081 $\pm$ 0.029	0.003 $\pm$ 0.000	0.055 $\pm$ 0.001	0.003 $\pm$ 0.000	0.055 $\pm$ 0.002	0.847 $\pm$ 0.041	0.921 $\pm$ 0.026	0.057	-0.003	-4.98	0.0540	0.0541	1.008	94.088 $\pm$ 0.747	0.218 $\pm$ 0.001	15.431 $\pm$ 1.143
QML	beta41	0.20	0.196	-0.004 $\pm$ 0.001	-0.022 $\pm$ 0.007	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.053 $\pm$ 0.002	0.231 $\pm$ 0.006	0.196	-0.004	-2.07	0.0448	0.0450	1.006	95.090 $\pm$ 0.684	0.182 $\pm$ 0.000	99.098 $\pm$ 0.299
QML	beta42	0.20	0.199	-0.001 $\pm$ 0.001	-0.006 $\pm$ 0.007	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.054 $\pm$ 0.002	0.233 $\pm$ 0.006	0.197	-0.003	-1.63	0.0484	0.0485	1.008	95.291 $\pm$ 0.671	0.184 $\pm$ 0.000	99.299 $\pm$ 0.264
QML	beta43	0.20	0.197	-0.003 $\pm$ 0.001	-0.016 $\pm$ 0.007	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.049 $\pm$ 0.002	0.221 $\pm$ 0.006	0.197	-0.003	-1.46	0.0432	0.0433	1.010	95.291 $\pm$ 0.671	0.175 $\pm$ 0.000	99.198 $\pm$ 0.282
QML	beta44	0.11	0.115	0.005 $\pm$ 0.002	0.049 $\pm$ 0.015	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.003 $\pm$ 0.000	0.053 $\pm$ 0.002	0.232 $\pm$ 0.011	0.482 $\pm$ 0.014	0.115	0.005	4.23	0.0510	0.0512	0.970	94.188 $\pm$ 0.741	0.201 $\pm$ 0.002	63.026 $\pm$ 1.528
QML	beta45	0.11	0.113	0.003 $\pm$ 0.002	0.024 $\pm$ 0.014	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.188 $\pm$ 0.009	0.434 $\pm$ 0.012	0.109	-0.001	-0.85	0.0465	0.0465	0.966	93.888 $\pm$ 0.758	0.181 $\pm$ 0.001	67.735 $\pm$ 1.480
QML	beta46	0.08	0.053	-0.027 $\pm$ 0.001	-0.333 $\pm$ 0.018	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.443 $\pm$ 0.019	0.665 $\pm$ 0.016	0.053	-0.027	-33.85	0.0447	0.0523	1.028	90.481 $\pm$ 0.929	0.186 $\pm$ 0.001	19.138 $\pm$ 1.245
QML	beta47	0.08	0.055	-0.025 $\pm$ 0.001	-0.318 $\pm$ 0.018	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.441 $\pm$ 0.019	0.664 $\pm$ 0.016	0.054	-0.026	-32.73	0.0464	0.0533	1.019	90.281 $\pm$ 0.938	0.186 $\pm$ 0.003	22.846 $\pm$ 1.329
QML	beta51	0.16	0.171	0.011 $\pm$ 0.002	0.071 $\pm$ 0.012	0.004 $\pm$ 0.000	0.059 $\pm$ 0.001	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	0.143 $\pm$ 0.006	0.378 $\pm$ 0.010	0.174	0.014	8.52	0.0612	0.0627	0.983	94.790 $\pm$ 0.703	0.229 $\pm$ 0.001	81.964 $\pm$ 1.217
QML	beta52	0.16	0.147	-0.013 $\pm$ 0.002	-0.083 $\pm$ 0.010	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.003 $\pm$ 0.000	0.050 $\pm$ 0.001	0.098 $\pm$ 0.004	0.313 $\pm$ 0.008	0.146	-0.014	-8.52	0.0484	0.0503	1.010	94.489 $\pm$ 0.722	0.191 $\pm$ 0.001	84.870 $\pm$ 1.134
QML	beta53	0.16	0.141	-0.019 $\pm$ 0.002	-0.119 $\pm$ 0.010	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.003 $\pm$ 0.000	0.055 $\pm$ 0.001	0.118 $\pm$ 0.005	0.344 $\pm$ 0.009	0.140	-0.020	-12.28	0.0521	0.0557	0.968	92.184 $\pm$ 0.850	0.196 $\pm$ 0.001	77.956 $\pm$ 1.312
QML	beta54	0.16	0.158	-0.002 $\pm$ 0.001	-0.012 $\pm$ 0.009	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.087 $\pm$ 0.004	0.296 $\pm$ 0.008	0.159	-0.001	-0.42	0.0480	0.0480	0.994	94.589 $\pm$ 0.716	0.184 $\pm$ 0.000	92.585 $\pm$ 0.829
QML	beta55	0.08	0.084	0.004 $\pm$ 0.002	0.048 $\pm$ 0.024	0.004 $\pm$ 0.000	0.061 $\pm$ 0.001	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.579 $\pm$ 0.024	0.761 $\pm$ 0.020	0.082	0.002	1.89	0.0637	0.0637	0.992	94.589 $\pm$ 0.716	0.236 $\pm$ 0.001	29.259 $\pm$ 1.440
QML	beta56	0.08	0.081	0.001 $\pm$ 0.002	0.007 $\pm$ 0.023	0.003 $\pm$ 0.000	0.058 $\pm$ 0.001	0.003 $\pm$ 0.000	0.058 $\pm$ 0.002	0.527 $\pm$ 0.023	0.726 $\pm$ 0.020	0.079	-0.001	-1.15	0.0582	0.0582	0.976	94.289 $\pm$ 0.735	0.222 $\pm$ 0.001	29.559 $\pm$ 1.444
QML	beta57	0.06	0.039	-0.021 $\pm$ 0.002	-0.350 $\pm$ 0.026	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.003 $\pm$ 0.000	0.054 $\pm$ 0.001	0.802 $\pm$ 0.037	0.896 $\pm$ 0.024	0.038	-0.022	-36.08	0.0469	0.0516	0.983	92.685 $\pm$ 0.824	0.191 $\pm$ 0.001	13.327 $\pm$ 1.076
QML	beta58	0.06	0.044	-0.016 $\pm$ 0.001	-0.265 $\pm$ 0.023	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.612 $\pm$ 0.029	0.782 $\pm$ 0.021	0.046	-0.014	-23.99	0.0441	0.0464	1.013	92.485 $\pm$ 0.835	0.175 $\pm$ 0.001	14.930 $\pm$ 1.128
UPI	beta41	0.20	0.204	0.004 $\pm$ 0.002	0.018 $\pm$ 0.008	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.057 $\pm$ 0.002	0.238 $\pm$ 0.006	0.204	0.004	2.05	0.0487	0.0489	0.958	93.186 $\pm$ 0.798	0.178 $\pm$ 0.000	99.198 $\pm$ 0.282
UPI	beta42	0.20	0.205	0.005 $\pm$ 0.002	0.023 $\pm$ 0.008	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.058 $\pm$ 0.002	0.240 $\pm$ 0.006	0.204	0.004	1.95	0.0480	0.0481	0.980	94.689 $\pm$ 0.710	0.184 $\pm$ 0.000	99.499 $\pm$ 0.223
UPI	beta43	0.20	0.203	0.003 $\pm$ 0.001	0.016 $\pm$ 0.007	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.050 $\pm$ 0.002	0.224 $\pm$ 0.006	0.203	0.003	1.69	0.0443	0.0444	1.007	95.291 $\pm$ 0.671	0.176 $\pm$ 0.000	99.599 $\pm$ 0.200

Table S179

Simulation 2 - Condition 35: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.8
(continued)

Method	Param.	True	Mean-based										Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE	
UPI	beta47	0.08	0.076	-0.004 ± 0.002	-0.052 ± 0.024	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.566 ± 0.027	0.752 ± 0.021	0.075	-0.005	-5.96	0.0585	0.0587	1.027	94.890 ± 0.697	0.242 ± 0.001	22.946 ± 1.331	
UPI	beta51	0.16	0.167	0.007 ± 0.002	0.043 ± 0.012	0.003 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	0.137 ± 0.006	0.370 ± 0.010	0.167	0.007	4.42	0.0607	0.0611	0.974	94.990 ± 0.691	0.225 ± 0.001	81.764 ± 1.222	
UPI	beta52	0.16	0.164	0.004 ± 0.002	0.024 ± 0.010	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.092 ± 0.004	0.304 ± 0.008	0.164	0.004	2.79	0.0495	0.0497	0.974	94.489 ± 0.722	0.185 ± 0.000	91.784 ± 0.869	
UPI	beta53	0.16	0.159	-0.001 ± 0.002	-0.005 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.104 ± 0.005	0.322 ± 0.009	0.158	-0.002	-0.96	0.0518	0.0518	0.958	93.788 ± 0.764	0.193 ± 0.001	88.477 ± 1.011	
UPI	beta54	0.16	0.161	0.001 ± 0.002	0.005 ± 0.010	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.090 ± 0.004	0.301 ± 0.008	0.162	0.002	1.19	0.0489	0.0490	0.982	93.788 ± 0.764	0.185 ± 0.000	92.485 ± 0.835	
UPI	beta55	0.08	0.082	0.002 ± 0.002	0.020 ± 0.024	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.587 ± 0.025	0.766 ± 0.020	0.080	0.000	-0.40	0.0639	0.0639	0.980	94.389 ± 0.728	0.236 ± 0.001	28.056 ± 1.422	
UPI	beta56	0.08	0.081	0.001 ± 0.002	0.014 ± 0.023	0.004 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	0.549 ± 0.025	0.741 ± 0.020	0.080	0.000	0.57	0.0596	0.0596	0.969	94.088 ± 0.747	0.225 ± 0.001	28.557 ± 1.430	
UPI	beta57	0.06	0.057	-0.003 ± 0.002	-0.049 ± 0.033	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	1.102 ± 0.053	1.050 ± 0.030	0.056	-0.004	-7.29	0.0561	0.0562	0.985	94.790 ± 0.703	0.243 ± 0.001	13.928 ± 1.096	
UPI	beta58	0.06	0.060	-0.000 ± 0.002	-0.008 ± 0.031	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.932 ± 0.044	0.965 ± 0.027	0.061	0.001	1.42	0.0569	0.0570	1.028	95.792 ± 0.636	0.234 ± 0.001	13.828 ± 1.093	

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S180

Simulation 2 - Condition 36: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, $N = 1000$, Reliability = 0.8

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LSAM	beta41	0.20	0.198	-0.002 $\pm$ 0.001	-0.008 $\pm$ 0.005	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.021 $\pm$ 0.001	0.146 $\pm$ 0.004	0.198	-0.002	-0.92	0.0289	0.0290	1.030	95.600 $\pm$ 0.649	0.117 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta42	0.20	0.199	-0.001 $\pm$ 0.001	-0.006 $\pm$ 0.005	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.023 $\pm$ 0.001	0.153 $\pm$ 0.004	0.199	-0.001	-0.58	0.0313	0.0313	0.993	95.600 $\pm$ 0.649	0.119 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta43	0.20	0.200	-0.000 $\pm$ 0.001	-0.000 $\pm$ 0.004	0.001 $\pm$ 0.000	0.028 $\pm$ 0.001	0.001 $\pm$ 0.000	0.028 $\pm$ 0.001	0.020 $\pm$ 0.001	0.141 $\pm$ 0.004	0.201	0.001	0.55	0.0275	0.0275	1.006	95.200 $\pm$ 0.676	0.112 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta44	0.11	0.115	0.005 $\pm$ 0.001	0.048 $\pm$ 0.009	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.089 $\pm$ 0.004	0.298 $\pm$ 0.008	0.114	0.004	3.79	0.0339	0.0341	0.988	94.300 $\pm$ 0.733	0.126 $\pm$ 0.000	94.900 $\pm$ 0.696
LSAM	beta45	0.11	0.113	0.003 $\pm$ 0.001	0.024 $\pm$ 0.008	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.071 $\pm$ 0.003	0.266 $\pm$ 0.007	0.113	0.003	2.58	0.0297	0.0299	1.005	95.200 $\pm$ 0.676	0.115 $\pm$ 0.000	97.000 $\pm$ 0.539
LSAM	beta46	0.08	0.072	-0.008 $\pm$ 0.001	-0.101 $\pm$ 0.014	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.219 $\pm$ 0.010	0.468 $\pm$ 0.012	0.069	-0.011	-13.16	0.0355	0.0370	1.010	94.200 $\pm$ 0.739	0.145 $\pm$ 0.000	48.100 $\pm$ 1.580
LSAM	beta47	0.08	0.072	-0.008 $\pm$ 0.001	-0.103 $\pm$ 0.014	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.208 $\pm$ 0.010	0.456 $\pm$ 0.013	0.071	-0.009	-10.86	0.0359	0.0369	1.035	95.500 $\pm$ 0.656	0.144 $\pm$ 0.000	48.900 $\pm$ 1.581
LSAM	beta51	0.16	0.162	0.002 $\pm$ 0.001	0.015 $\pm$ 0.007	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.054 $\pm$ 0.002	0.233 $\pm$ 0.006	0.161	0.001	0.75	0.0367	0.0367	0.972	95.000 $\pm$ 0.689	0.142 $\pm$ 0.000	99.100 $\pm$ 0.299
LSAM	beta52	0.16	0.160	-0.000 $\pm$ 0.001	-0.002 $\pm$ 0.006	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.036 $\pm$ 0.002	0.188 $\pm$ 0.005	0.160	0.000	0.23	0.0288	0.0288	0.995	94.700 $\pm$ 0.708	0.118 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta53	0.16	0.158	-0.002 $\pm$ 0.001	-0.013 $\pm$ 0.006	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	0.039 $\pm$ 0.002	0.198 $\pm$ 0.005	0.158	-0.002	-1.38	0.0315	0.0316	0.983	94.900 $\pm$ 0.696	0.122 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta54	0.16	0.157	-0.003 $\pm$ 0.001	-0.017 $\pm$ 0.006	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.036 $\pm$ 0.002	0.191 $\pm$ 0.005	0.158	-0.002	-1.36	0.0301	0.0302	0.982	94.700 $\pm$ 0.708	0.117 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta55	0.08	0.082	0.002 $\pm$ 0.001	0.030 $\pm$ 0.014	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.210 $\pm$ 0.010	0.458 $\pm$ 0.013	0.082	0.002	2.70	0.0360	0.0361	1.022	95.200 $\pm$ 0.676	0.147 $\pm$ 0.000	60.400 $\pm$ 1.547
LSAM	beta56	0.08	0.080	-0.000 $\pm$ 0.001	-0.002 $\pm$ 0.014	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.207 $\pm$ 0.010	0.455 $\pm$ 0.013	0.079	-0.001	-1.63	0.0351	0.0351	0.986	94.500 $\pm$ 0.721	0.141 $\pm$ 0.000	60.200 $\pm$ 1.548
LSAM	beta57	0.06	0.053	-0.007 $\pm$ 0.001	-0.125 $\pm$ 0.020	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.411 $\pm$ 0.020	0.641 $\pm$ 0.018	0.053	-0.007	-11.10	0.0378	0.0384	0.983	94.000 $\pm$ 0.751	0.145 $\pm$ 0.000	29.900 $\pm$ 1.448
LSAM	beta58	0.06	0.056	-0.004 $\pm$ 0.001	-0.071 $\pm$ 0.019	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.350 $\pm$ 0.015	0.591 $\pm$ 0.016	0.055	-0.005	-8.00	0.0352	0.0355	0.987	94.400 $\pm$ 0.727	0.136 $\pm$ 0.000	35.400 $\pm$ 1.512
QML	beta41	0.20	0.197	-0.003 $\pm$ 0.001	-0.017 $\pm$ 0.005	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.021 $\pm$ 0.001	0.144 $\pm$ 0.004	0.196	-0.004	-1.83	0.0285	0.0287	1.026	95.800 $\pm$ 0.634	0.115 $\pm$ 0.000	100.000 $\pm$ 0.000
QML	beta42	0.20	0.198	-0.002 $\pm$ 0.001	-0.011 $\pm$ 0.005	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.022 $\pm$ 0.001	0.150 $\pm$ 0.004	0.197	-0.003	-1.45	0.0307	0.0308	0.989	95.300 $\pm$ 0.669	0.116 $\pm$ 0.000	100.000 $\pm$ 0.000
QML	beta43	0.20	0.195	-0.005 $\pm$ 0.001	-0.023 $\pm$ 0.004	0.001 $\pm$ 0.000	0.028 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.020 $\pm$ 0.001	0.143 $\pm$ 0.004	0.197	-0.003	-1.73	0.0271	0.0274	0.999	94.600 $\pm$ 0.715	0.110 $\pm$ 0.000	100.000 $\pm$ 0.000
QML	beta44	0.11	0.117	0.007 $\pm$ 0.001	0.060 $\pm$ 0.009	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.090 $\pm$ 0.004	0.299 $\pm$ 0.008	0.116	0.006	5.25	0.0331	0.0336	0.989	94.600 $\pm$ 0.715	0.125 $\pm$ 0.000	95.500 $\pm$ 0.656
QML	beta45	0.11	0.113	0.003 $\pm$ 0.001	0.027 $\pm$ 0.008	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.070 $\pm$ 0.003	0.264 $\pm$ 0.007	0.113	0.003	2.71	0.0297	0.0299	1.005	95.400 $\pm$ 0.662	0.114 $\pm$ 0.000	97.300 $\pm$ 0.513
QML	beta46	0.08	0.055	-0.025 $\pm$ 0.001	-0.317 $\pm$ 0.012	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.236 $\pm$ 0.009	0.485 $\pm$ 0.010	0.053	-0.027	-33.48	0.0285	0.0391	1.015	86.100 $\pm$ 1.094	0.117 $\pm$ 0.000	44.100 $\pm$ 1.570
QML	beta47	0.08	0.056	-0.024 $\pm$ 0.001	-0.303 $\pm$ 0.011	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.221 $\pm$ 0.009	0.470 $\pm$ 0.011	0.056	-0.024	-30.17	0.0299	0.0384	1.030	87.600 $\pm$ 1.042	0.116 $\pm$ 0.000	46.400 $\pm$ 1.577
QML	beta51	0.16	0.168	0.008 $\pm$ 0.001	0.052 $\pm$ 0.007	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.001 $\pm$ 0.000	0.039 $\pm$ 0.001	0.058 $\pm$ 0.003	0.241 $\pm$ 0.006	0.167	0.007	4.56	0.0377	0.0384	0.974	93.900 $\pm$ 0.757	0.144 $\pm$ 0.000	99.500 $\pm$ 0.223
QML	beta52	0.16	0.146	-0.014 $\pm$ 0.001	-0.088 $\pm$ 0.006	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.044 $\pm$ 0.002	0.211 $\pm$ 0.006	0.147	-0.013	-8.20	0.0302	0.0329	0.995	92.700 $\pm$ 0.823	0.120 $\pm$ 0.000	99.400 $\pm$ 0.244
QML	beta53	0.16	0.141	-0.019 $\pm$ 0.001	-0.121 $\pm$ 0.006	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.053 $\pm$ 0.002	0.231 $\pm$ 0.005	0.139	-0.021	-13.09	0.0317	0.0380	0.993	90.500 $\pm$ 0.927	0.123 $\pm$ 0.000	99.800 $\pm$ 0.141
QML	beta54	0.16	0.157	-0.003 $\pm$ 0.001	-0.018 $\pm$ 0.006	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.036 $\pm$ 0.002	0.189 $\pm$ 0.005	0.157	-0.003	-1.70	0.0300	0.0301	0.977	94.600 $\pm$ 0.715	0.116 $\pm$ 0.000	100.000 $\pm$ 0.000
QML	beta55	0.08	0.084	0.004 $\pm$ 0.001	0.044 $\pm$ 0.015	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.212 $\pm$ 0.010	0.460 $\pm$ 0.013	0.084	0.004	4.85	0.0359	0.0361	1.030	95.200 $\pm$ 0.676	0.148 $\pm$ 0.000	60.700 $\pm$ 1.545
QML	beta56	0.08	0.079	-0.001 $\pm$ 0.001	-0.011 $\pm$ 0.014	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.204 $\pm$ 0.010	0.452 $\pm$ 0.013	0.078	-0.002	-2.22	0.0350	0.0350	0.988	95.100 $\pm$ 0.683	0.140 $\pm$ 0.000	60.300 $\pm$ 1.547
QML	beta57	0.06	0.039	-0.021 $\pm$ 0.001	-0.345 $\pm$ 0.016	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.383 $\pm$ 0.018	0.619 $\pm$ 0.016	0.040	-0.020	-34.16	0.0303	0.0366	0.987	90.000 $\pm$ 0.949	0.119 $\pm$ 0.000	25.100 $\pm$ 1.371
QML	beta58	0.06	0.045	-0.015 $\pm$ 0.001	-0.252 $\pm$ 0.015	0.001 $\pm$ 0.000	0.028 $\pm$ 0.001	0.0												

Table S180

Simulation 2 - Condition 36: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.8
(continued)

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.077	-0.003 ± 0.001	-0.039 ± 0.014	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.209 ± 0.009	0.458 ± 0.012	0.076	-0.004	-4.86	0.0362	0.0364	1.033	95.700 ± 0.641	0.148 ± 0.000	52.700 ± 1.579
UPI	beta51	0.16	0.162	0.002 ± 0.001	0.014 ± 0.007	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.054 ± 0.002	0.233 ± 0.006	0.161	0.001	0.35	0.0381	0.0381	0.968	94.600 ± 0.715	0.141 ± 0.000	99.100 ± 0.299
UPI	beta52	0.16	0.162	0.002 ± 0.001	0.013 ± 0.006	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.036 ± 0.002	0.189 ± 0.005	0.162	0.002	1.40	0.0293	0.0294	0.977	94.100 ± 0.745	0.116 ± 0.000	100.000 ± 0.000
UPI	beta53	0.16	0.159	-0.001 ± 0.001	-0.008 ± 0.006	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.039 ± 0.002	0.197 ± 0.005	0.158	-0.002	-1.19	0.0313	0.0314	0.982	95.200 ± 0.676	0.121 ± 0.000	100.000 ± 0.000
UPI	beta54	0.16	0.159	-0.001 ± 0.001	-0.006 ± 0.006	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.036 ± 0.002	0.191 ± 0.005	0.159	-0.001	-0.39	0.0300	0.0300	0.968	94.600 ± 0.715	0.116 ± 0.000	100.000 ± 0.000
UPI	beta55	0.08	0.081	0.001 ± 0.001	0.013 ± 0.014	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.210 ± 0.010	0.458 ± 0.013	0.082	0.002	2.12	0.0368	0.0368	1.021	95.700 ± 0.641	0.147 ± 0.000	60.200 ± 1.548
UPI	beta56	0.08	0.080	-0.000 ± 0.001	-0.006 ± 0.014	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.203 ± 0.009	0.450 ± 0.013	0.078	-0.002	-2.73	0.0350	0.0351	0.996	94.900 ± 0.696	0.141 ± 0.000	59.700 ± 1.551
UPI	beta57	0.06	0.057	-0.003 ± 0.001	-0.058 ± 0.020	0.001 ± 0.000	0.039 ± 0.001	0.001 ± 0.000	0.039 ± 0.001	0.415 ± 0.020	0.644 ± 0.018	0.057	-0.003	-5.15	0.0374	0.0375	0.985	94.500 ± 0.721	0.149 ± 0.000	32.000 ± 1.475
UPI	beta58	0.06	0.060	-0.000 ± 0.001	-0.001 ± 0.019	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.366 ± 0.016	0.605 ± 0.016	0.059	-0.001	-1.94	0.0367	0.0367	0.992	94.800 ± 0.702	0.141 ± 0.000	38.700 ± 1.540

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S181

Simulation 2 - Condition 37: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.4

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LSAM	beta41	0.20	0.200	-0.000 $\pm$ 0.003	-0.002 $\pm$ 0.016	0.007 $\pm$ 0.000	0.082 $\pm$ 0.002	0.007 $\pm$ 0.000	0.082 $\pm$ 0.003	0.168 $\pm$ 0.010	0.410 $\pm$ 0.014	0.202	0.002	0.92	0.0804	0.0804	1.111	96.407 $\pm$ 0.720	0.357 $\pm$ 0.003	66.467 $\pm$ 1.827
LSAM	beta42	0.20	0.204	0.004 $\pm$ 0.004	0.019 $\pm$ 0.018	0.008 $\pm$ 0.000	0.091 $\pm$ 0.002	0.008 $\pm$ 0.000	0.091 $\pm$ 0.003	0.206 $\pm$ 0.011	0.454 $\pm$ 0.015	0.202	0.002	1.02	0.0939	0.0939	1.048	97.455 $\pm$ 0.609	0.373 $\pm$ 0.004	61.377 $\pm$ 1.884
LSAM	beta43	0.20	0.208	0.008 $\pm$ 0.003	0.039 $\pm$ 0.016	0.007 $\pm$ 0.000	0.085 $\pm$ 0.003	0.007 $\pm$ 0.000	0.085 $\pm$ 0.003	0.183 $\pm$ 0.011	0.427 $\pm$ 0.016	0.204	0.004	2.16	0.0804	0.0805	1.012	95.210 $\pm$ 0.826	0.338 $\pm$ 0.003	72.305 $\pm$ 1.731
LSAM	beta44	0.11	0.117	0.007 $\pm$ 0.005	0.063 $\pm$ 0.047	0.017 $\pm$ 0.001	0.132 $\pm$ 0.004	0.018 $\pm$ 0.001	0.132 $\pm$ 0.005	1.447 $\pm$ 0.096	1.203 $\pm$ 0.046	0.109	-0.001	-0.89	0.1259	0.1259	1.065	98.054 $\pm$ 0.534	0.552 $\pm$ 0.011	13.323 $\pm$ 1.315
LSAM	beta45	0.11	0.112	0.002 $\pm$ 0.004	0.017 $\pm$ 0.038	0.012 $\pm$ 0.001	0.109 $\pm$ 0.004	0.012 $\pm$ 0.001	0.109 $\pm$ 0.004	0.986 $\pm$ 0.065	0.993 $\pm$ 0.038	0.110	0.000	0.31	0.1003	0.1003	1.042	97.904 $\pm$ 0.554	0.447 $\pm$ 0.007	17.814 $\pm$ 1.480
LSAM	beta46	0.08	0.087	0.007 $\pm$ 0.004	0.091 $\pm$ 0.046	0.009 $\pm$ 0.001	0.094 $\pm$ 0.003	0.009 $\pm$ 0.001	0.094 $\pm$ 0.003	1.392 $\pm$ 0.087	1.180 $\pm$ 0.043	0.086	0.006	7.69	0.0937	0.0939	1.046	97.156 $\pm$ 0.643	0.386 $\pm$ 0.008	11.078 $\pm$ 1.214
LSAM	beta47	0.08	0.085	0.005 $\pm$ 0.004	0.063 $\pm$ 0.045	0.009 $\pm$ 0.001	0.094 $\pm$ 0.003	0.009 $\pm$ 0.001	0.094 $\pm$ 0.004	1.373 $\pm$ 0.096	1.172 $\pm$ 0.047	0.079	-0.001	-0.70	0.0837	0.0837	1.020	97.455 $\pm$ 0.609	0.375 $\pm$ 0.008	11.826 $\pm$ 1.249
LSAM	beta51	0.16	0.160	-0.000 $\pm$ 0.004	-0.001 $\pm$ 0.026	0.012 $\pm$ 0.001	0.109 $\pm$ 0.003	0.012 $\pm$ 0.001	0.109 $\pm$ 0.004	0.466 $\pm$ 0.027	0.682 $\pm$ 0.024	0.160	0.000	-0.26	0.1019	0.1019	0.972	94.162 $\pm$ 0.907	0.416 $\pm$ 0.004	34.431 $\pm$ 1.838
LSAM	beta52	0.16	0.159	-0.001 $\pm$ 0.003	-0.005 $\pm$ 0.020	0.007 $\pm$ 0.000	0.084 $\pm$ 0.003	0.007 $\pm$ 0.000	0.084 $\pm$ 0.003	0.276 $\pm$ 0.016	0.526 $\pm$ 0.019	0.159	-0.001	-0.87	0.0844	0.0844	1.036	96.707 $\pm$ 0.690	0.342 $\pm$ 0.003	47.754 $\pm$ 1.933
LSAM	beta53	0.16	0.167	0.007 $\pm$ 0.004	0.044 $\pm$ 0.023	0.009 $\pm$ 0.001	0.093 $\pm$ 0.003	0.009 $\pm$ 0.001	0.094 $\pm$ 0.003	0.342 $\pm$ 0.021	0.585 $\pm$ 0.021	0.163	0.003	1.58	0.0933	0.0933	0.996	94.910 $\pm$ 0.850	0.364 $\pm$ 0.003	46.257 $\pm$ 1.929
LSAM	beta54	0.16	0.163	0.003 $\pm$ 0.004	0.016 $\pm$ 0.022	0.008 $\pm$ 0.001	0.091 $\pm$ 0.003	0.008 $\pm$ 0.001	0.091 $\pm$ 0.003	0.325 $\pm$ 0.020	0.570 $\pm$ 0.021	0.158	-0.002	-1.07	0.0863	0.0863	0.964	95.509 $\pm$ 0.801	0.345 $\pm$ 0.003	48.503 $\pm$ 1.934
LSAM	beta55	0.08	0.086	0.006 $\pm$ 0.006	0.080 $\pm$ 0.072	0.022 $\pm$ 0.001	0.148 $\pm$ 0.005	0.022 $\pm$ 0.001	0.148 $\pm$ 0.006	3.418 $\pm$ 0.232	1.849 $\pm$ 0.072	0.091	0.011	13.27	0.1394	0.1398	0.978	96.557 $\pm$ 0.705	0.567 $\pm$ 0.012	10.329 $\pm$ 1.178
LSAM	beta56	0.08	0.083	0.003 $\pm$ 0.005	0.038 $\pm$ 0.062	0.017 $\pm$ 0.001	0.129 $\pm$ 0.004	0.017 $\pm$ 0.001	0.129 $\pm$ 0.005	2.586 $\pm$ 0.178	1.608 $\pm$ 0.064	0.085	0.005	5.94	0.1141	0.1142	1.020	96.407 $\pm$ 0.720	0.514 $\pm$ 0.010	8.084 $\pm$ 1.055
LSAM	beta57	0.06	0.063	0.003 $\pm$ 0.003	0.058 $\pm$ 0.055	0.007 $\pm$ 0.000	0.085 $\pm$ 0.003	0.007 $\pm$ 0.001	0.085 $\pm$ 0.003	2.006 $\pm$ 0.140	1.416 $\pm$ 0.057	0.059	-0.001	-1.68	0.0816	0.0816	1.081	97.455 $\pm$ 0.609	0.360 $\pm$ 0.007	4.641 $\pm$ 0.814
LSAM	beta58	0.06	0.064	0.004 $\pm$ 0.003	0.070 $\pm$ 0.047	0.005 $\pm$ 0.000	0.072 $\pm$ 0.003	0.005 $\pm$ 0.000	0.073 $\pm$ 0.003	1.462 $\pm$ 0.108	1.209 $\pm$ 0.051	0.058	-0.002	-3.18	0.0668	0.0669	1.045	97.006 $\pm$ 0.659	0.297 $\pm$ 0.006	9.132 $\pm$ 1.115
QML	beta41	0.20	0.183	-0.017 $\pm$ 0.003	-0.087 $\pm$ 0.014	0.005 $\pm$ 0.000	0.073 $\pm$ 0.002	0.006 $\pm$ 0.000	0.075 $\pm$ 0.003	0.140 $\pm$ 0.008	0.374 $\pm$ 0.013	0.184	-0.016	-7.96	0.0706	0.0724	1.070	94.162 $\pm$ 0.907	0.305 $\pm$ 0.002	69.760 $\pm$ 1.777
QML	beta42	0.20	0.189	-0.011 $\pm$ 0.003	-0.053 $\pm$ 0.015	0.006 $\pm$ 0.000	0.078 $\pm$ 0.002	0.006 $\pm$ 0.000	0.079 $\pm$ 0.003	0.156 $\pm$ 0.008	0.395 $\pm$ 0.013	0.191	-0.009	-4.38	0.0799	0.0803	1.036	94.910 $\pm$ 0.850	0.318 $\pm$ 0.002	67.515 $\pm$ 1.812
QML	beta43	0.20	0.187	-0.013 $\pm$ 0.003	-0.063 $\pm$ 0.015	0.006 $\pm$ 0.000	0.076 $\pm$ 0.002	0.006 $\pm$ 0.000	0.077 $\pm$ 0.003	0.149 $\pm$ 0.009	0.385 $\pm$ 0.014	0.187	-0.013	-6.68	0.0708	0.0720	0.986	92.814 $\pm$ 0.999	0.294 $\pm$ 0.002	72.605 $\pm$ 1.726
QML	beta44	0.11	0.089	-0.021 $\pm$ 0.004	-0.192 $\pm$ 0.039	0.012 $\pm$ 0.001	0.110 $\pm$ 0.004	0.013 $\pm$ 0.001	0.112 $\pm$ 0.004	1.043 $\pm$ 0.068	1.021 $\pm$ 0.038	0.092	-0.018	-16.11	0.1143	0.1157	1.030	96.257 $\pm$ 0.734	0.446 $\pm$ 0.008	17.066 $\pm$ 1.456
QML	beta45	0.11	0.104	-0.006 $\pm$ 0.003	-0.058 $\pm$ 0.031	0.008 $\pm$ 0.000	0.087 $\pm$ 0.002	0.008 $\pm$ 0.000	0.087 $\pm$ 0.003	0.628 $\pm$ 0.036	0.793 $\pm$ 0.027	0.104	-0.006	-5.61	0.0883	0.0885	1.066	96.856 $\pm$ 0.675	0.364 $\pm$ 0.005	20.659 $\pm$ 1.566
QML	beta46	0.08	0.087	0.007 $\pm$ 0.003	0.087 $\pm$ 0.037	0.006 $\pm$ 0.000	0.076 $\pm$ 0.003	0.006 $\pm$ 0.000	0.076 $\pm$ 0.003	0.901 $\pm$ 0.064	0.949 $\pm$ 0.038	0.084	0.004	4.88	0.0765	0.0766	1.020	95.359 $\pm$ 0.814	0.303 $\pm$ 0.006	15.419 $\pm$ 1.397
QML	beta47	0.08	0.082	0.002 $\pm$ 0.003	0.029 $\pm$ 0.035	0.005 $\pm$ 0.000	0.072 $\pm$ 0.002	0.005 $\pm$ 0.000	0.072 $\pm$ 0.003	0.818 $\pm$ 0.051	0.905 $\pm$ 0.033	0.077	-0.003	-3.62	0.0685	0.0685	1.046	93.563 $\pm$ 0.950	0.297 $\pm$ 0.007	15.569 $\pm$ 1.403
QML	beta51	0.16	0.182	0.022 $\pm$ 0.004	0.138 $\pm$ 0.027	0.012 $\pm$ 0.001	0.110 $\pm$ 0.003	0.013 $\pm$ 0.001	0.112 $\pm$ 0.004	0.489 $\pm$ 0.030	0.699 $\pm$ 0.025	0.179	0.019	11.98	0.1036	0.1053	1.096	97.006 $\pm$ 0.659	0.471 $\pm$ 0.009	36.228 $\pm$ 1.860
QML	beta52	0.16	0.141	-0.019 $\pm$ 0.004	-0.118 $\pm$ 0.023	0.009 $\pm$ 0.001	0.096 $\pm$ 0.003	0.010 $\pm$ 0.001	0.098 $\pm$ 0.004	0.376 $\pm$ 0.024	0.613 $\pm$ 0.023	0.140	-0.020	-12.41	0.0916	0.0937	1.048	94.760 $\pm$ 0.862	0.396 $\pm$ 0.007	31.737 $\pm$ 1.801
QML	beta53	0.16	0.141	-0.019 $\pm$ 0.003	-0.120 $\pm$ 0.021	0.008 $\pm$ 0.000	0.087 $\pm$ 0.003	0.008 $\pm$ 0.000	0.089 $\pm$ 0.003	0.309 $\pm$ 0.017	0.556 $\pm$ 0.019	0.132	-0.028	-17.40	0.0884	0.0927	1.066	95.808 $\pm$ 0.775	0.363 $\pm$ 0.007	33.982 $\pm$ 1.833
QML	beta54	0.16	0.159	-0.001 $\pm$ 0.003	-0.006 $\pm$ 0.021	0.007 $\pm$ 0.000	0.085 $\pm$ 0.002	0.007 $\pm$ 0.000	0.085 $\pm$ 0.003	0.281 $\pm$ 0.016	0.530 $\pm$ 0.019	0.156	-0.004	-2.35	0.0811	0.0812	0.974	95.060 $\pm$ 0.838	0.324 $\pm$ 0.003	49.251 $\pm$ 1.934
QML	beta55	0.08	0.096	0.016 $\pm$ 0.005	0.197 $\pm$ 0.063	0.017 $\pm$ 0.001	0.130 $\pm$ 0.004	0.017 $\pm$ 0.001	0.131 $\pm$ 0.005	2.675 $\pm$ 0.167	1.636 $\pm$ 0.060	0.102	0.022	27.74	0.1272	0.1292	1.007	95.808 $\pm$ 0.775	0.513 $\pm$ 0.008	14.371 $\pm$ 1.357
QML	beta56	0.08	0.076	-0.004 $\pm$ 0.004	-0.055 $\pm$ 0.055	0.013 $\pm$ 0.001	0.113 $\pm$ 0.003	0.013 $\pm$ 0.001	0.113 $\pm$ 0.004	2.012 $\pm$ 0.120	1.418 $\pm$ 0.051	0.077	-0.003	-4.03	0.1063	0.1063	1.050	94.910 $\pm$ 0.850	0.467 $\pm$ 0.018	8.234 $\pm$ 1.064
QML	beta57	0.06	0.047	-0.013 $\pm$ 0.003	-0.224 $\pm$ 0.048	0.005 $\pm$ 0.000	0.074 $\pm$ 0.003	0.006 $\pm$ 0.000	0.075 $\pm$ 0.003	1.558 $\pm$ 0.099	1.248 $\pm$ 0.046	0.040	-0.020	-33.47	0.0693	0.0721	1.068	92.964 $\pm$ 0.990	0.309 $\pm$ 0.007	3.743 $\pm$ 0.734
QML	beta58	0.06	0.055	-0.005 $\pm$ 0.002	-0.088 $\pm$ 0.040	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.004 $\pm$ 0.000												

Table S181

Simulation 2 - Condition 37: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.4
(continued)

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.092	0.012 ± 0.005	0.144 ± 0.065	0.018 ± 0.002	0.134 ± 0.006	0.018 ± 0.002	0.135 ± 0.006	2.838 ± 0.242	1.685 ± 0.079	0.081	0.001	1.85	0.1180	0.1180	1.422	98.204 ± 0.514	0.749 ± 0.035	3.144 ± 0.675
UPI	beta51	0.16	0.170	0.010 ± 0.004	0.061 ± 0.025	0.011 ± 0.001	0.105 ± 0.003	0.011 ± 0.001	0.105 ± 0.004	0.432 ± 0.025	0.658 ± 0.023	0.167	0.007	4.68	0.0998	0.1001	1.125	97.156 ± 0.643	0.462 ± 0.006	33.982 ± 1.833
UPI	beta52	0.16	0.156	-0.004 ± 0.003	-0.023 ± 0.020	0.007 ± 0.000	0.083 ± 0.003	0.007 ± 0.000	0.083 ± 0.003	0.270 ± 0.016	0.520 ± 0.019	0.161	0.001	0.59	0.0902	0.0902	1.209	98.204 ± 0.514	0.394 ± 0.007	39.371 ± 1.890
UPI	beta53	0.16	0.164	0.004 ± 0.004	0.025 ± 0.023	0.009 ± 0.000	0.094 ± 0.003	0.009 ± 0.000	0.094 ± 0.003	0.342 ± 0.019	0.585 ± 0.020	0.158	-0.002	-0.95	0.0967	0.0967	1.138	96.557 ± 0.705	0.417 ± 0.006	35.479 ± 1.851
UPI	beta54	0.16	0.161	0.001 ± 0.004	0.005 ± 0.022	0.009 ± 0.001	0.093 ± 0.003	0.009 ± 0.001	0.093 ± 0.003	0.335 ± 0.021	0.579 ± 0.022	0.156	-0.004	-2.25	0.0913	0.0914	1.140	97.156 ± 0.643	0.414 ± 0.011	36.527 ± 1.863
UPI	beta55	0.08	0.092	0.012 ± 0.007	0.151 ± 0.083	0.029 ± 0.002	0.171 ± 0.007	0.029 ± 0.002	0.171 ± 0.007	4.573 ± 0.353	2.138 ± 0.093	0.087	0.007	8.80	0.1457	0.1458	1.309	98.204 ± 0.514	0.876 ± 0.038	3.593 ± 0.720
UPI	beta56	0.08	0.083	0.003 ± 0.006	0.041 ± 0.071	0.021 ± 0.001	0.146 ± 0.005	0.021 ± 0.001	0.146 ± 0.006	3.319 ± 0.233	1.822 ± 0.073	0.075	-0.005	-6.24	0.1226	0.1227	1.354	98.353 ± 0.492	0.774 ± 0.033	3.743 ± 0.734
UPI	beta57	0.06	0.058	-0.002 ± 0.004	-0.040 ± 0.065	0.010 ± 0.001	0.101 ± 0.004	0.010 ± 0.001	0.101 ± 0.004	2.811 ± 0.217	1.676 ± 0.073	0.048	-0.012	-19.17	0.0807	0.0816	1.378	97.006 ± 0.659	0.544 ± 0.020	1.198 ± 0.421
UPI	beta58	0.06	0.063	0.003 ± 0.003	0.042 ± 0.056	0.007 ± 0.001	0.086 ± 0.003	0.007 ± 0.001	0.086 ± 0.004	2.068 ± 0.156	1.438 ± 0.061	0.055	-0.005	-8.61	0.0751	0.0753	1.428	97.605 ± 0.592	0.483 ± 0.019	2.096 ± 0.554

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S182

Simulation 2 - Condition 38: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.4

Method	Param.	True	Mean-based										Median-based					Inference		
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LSAM	beta41	0.20	0.202	0.002 $\pm$ 0.002	0.010 $\pm$ 0.008	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.060 $\pm$ 0.003	0.245 $\pm$ 0.007	0.201	0.001	0.67	0.0472	0.0472	1.063	96.917 $\pm$ 0.554	0.204 $\pm$ 0.001	97.636 $\pm$ 0.487
LSAM	beta42	0.20	0.204	0.004 $\pm$ 0.002	0.021 $\pm$ 0.009	0.003 $\pm$ 0.000	0.054 $\pm$ 0.001	0.003 $\pm$ 0.000	0.054 $\pm$ 0.002	0.073 $\pm$ 0.004	0.270 $\pm$ 0.008	0.202	0.002	1.05	0.0515	0.0515	0.998	95.786 $\pm$ 0.644	0.211 $\pm$ 0.001	98.253 $\pm$ 0.420
LSAM	beta43	0.20	0.203	0.003 $\pm$ 0.002	0.014 $\pm$ 0.008	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	0.062 $\pm$ 0.003	0.248 $\pm$ 0.007	0.200	0.000	0.19	0.0499	0.0499	0.994	95.375 $\pm$ 0.673	0.193 $\pm$ 0.001	99.075 $\pm$ 0.307
LSAM	beta44	0.11	0.113	0.003 $\pm$ 0.002	0.025 $\pm$ 0.021	0.005 $\pm$ 0.000	0.072 $\pm$ 0.002	0.005 $\pm$ 0.000	0.072 $\pm$ 0.002	0.423 $\pm$ 0.020	0.650 $\pm$ 0.019	0.112	0.002	1.38	0.0723	0.0723	1.010	96.711 $\pm$ 0.572	0.283 $\pm$ 0.002	40.596 $\pm$ 1.574
LSAM	beta45	0.11	0.113	0.003 $\pm$ 0.002	0.030 $\pm$ 0.018	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.310 $\pm$ 0.016	0.557 $\pm$ 0.017	0.112	0.002	1.97	0.0548	0.0548	0.993	95.992 $\pm$ 0.629	0.238 $\pm$ 0.002	48.304 $\pm$ 1.602
LSAM	beta46	0.08	0.084	0.004 $\pm$ 0.002	0.053 $\pm$ 0.021	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.003 $\pm$ 0.000	0.052 $\pm$ 0.002	0.429 $\pm$ 0.023	0.655 $\pm$ 0.021	0.083	0.003	3.22	0.0494	0.0495	0.961	94.861 $\pm$ 0.708	0.197 $\pm$ 0.002	39.054 $\pm$ 1.564
LSAM	beta47	0.08	0.082	0.002 $\pm$ 0.002	0.028 $\pm$ 0.020	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.373 $\pm$ 0.019	0.611 $\pm$ 0.018	0.080	0.000	-0.33	0.0473	0.0473	1.023	96.608 $\pm$ 0.580	0.196 $\pm$ 0.002	38.849 $\pm$ 1.563
LSAM	beta51	0.16	0.160	-0.000 $\pm$ 0.002	-0.000 $\pm$ 0.013	0.004 $\pm$ 0.000	0.063 $\pm$ 0.001	0.004 $\pm$ 0.000	0.063 $\pm$ 0.002	0.153 $\pm$ 0.007	0.391 $\pm$ 0.011	0.162	0.002	1.29	0.0613	0.0613	0.988	94.450 $\pm$ 0.734	0.243 $\pm$ 0.001	74.923 $\pm$ 1.390
LSAM	beta52	0.16	0.160	0.000 $\pm$ 0.002	0.002 $\pm$ 0.010	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.101 $\pm$ 0.005	0.317 $\pm$ 0.009	0.160	0.000	0.01	0.0505	0.0505	0.993	94.450 $\pm$ 0.734	0.198 $\pm$ 0.001	90.339 $\pm$ 0.947
LSAM	beta53	0.16	0.164	0.004 $\pm$ 0.002	0.025 $\pm$ 0.011	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.003 $\pm$ 0.000	0.053 $\pm$ 0.002	0.109 $\pm$ 0.005	0.330 $\pm$ 0.010	0.163	0.003	1.97	0.0502	0.0503	1.025	95.478 $\pm$ 0.666	0.211 $\pm$ 0.001	87.667 $\pm$ 1.054
LSAM	beta54	0.16	0.159	-0.001 $\pm$ 0.002	-0.006 $\pm$ 0.010	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	0.097 $\pm$ 0.005	0.312 $\pm$ 0.009	0.159	-0.001	-0.80	0.0494	0.0494	1.003	94.964 $\pm$ 0.701	0.196 $\pm$ 0.001	90.442 $\pm$ 0.943
LSAM	beta55	0.08	0.080	0.000 $\pm$ 0.002	0.002 $\pm$ 0.029	0.005 $\pm$ 0.000	0.073 $\pm$ 0.002	0.005 $\pm$ 0.000	0.073 $\pm$ 0.002	0.824 $\pm$ 0.042	0.908 $\pm$ 0.027	0.080	0.000	-0.44	0.0684	0.0684	1.006	95.581 $\pm$ 0.659	0.287 $\pm$ 0.002	21.377 $\pm$ 1.314
LSAM	beta56	0.08	0.083	0.003 $\pm$ 0.002	0.040 $\pm$ 0.027	0.004 $\pm$ 0.000	0.067 $\pm$ 0.002	0.004 $\pm$ 0.000	0.067 $\pm$ 0.002	0.700 $\pm$ 0.033	0.837 $\pm$ 0.024	0.084	0.004	5.24	0.0659	0.0660	1.010	94.861 $\pm$ 0.708	0.265 $\pm$ 0.002	24.769 $\pm$ 1.384
LSAM	beta57	0.06	0.062	0.002 $\pm$ 0.002	0.030 $\pm$ 0.026	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.002	0.651 $\pm$ 0.035	0.807 $\pm$ 0.025	0.059	-0.001	-1.05	0.0456	0.0456	0.998	95.272 $\pm$ 0.680	0.189 $\pm$ 0.002	23.124 $\pm$ 1.352
LSAM	beta58	0.06	0.062	0.002 $\pm$ 0.001	0.035 $\pm$ 0.022	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	0.463 $\pm$ 0.024	0.681 $\pm$ 0.021	0.061	0.001	0.96	0.0396	0.0396	0.998	95.581 $\pm$ 0.659	0.160 $\pm$ 0.001	32.580 $\pm$ 1.502
QML	beta41	0.20	0.186	-0.014 $\pm$ 0.002	-0.070 $\pm$ 0.008	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.060 $\pm$ 0.003	0.245 $\pm$ 0.006	0.186	-0.014	-7.07	0.0473	0.0494	1.019	93.628 $\pm$ 0.783	0.188 $\pm$ 0.001	97.842 $\pm$ 0.466
QML	beta42	0.20	0.192	-0.008 $\pm$ 0.002	-0.041 $\pm$ 0.008	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.066 $\pm$ 0.003	0.256 $\pm$ 0.007	0.190	-0.010	-5.11	0.0501	0.0512	0.983	94.347 $\pm$ 0.740	0.195 $\pm$ 0.001	98.253 $\pm$ 0.420
QML	beta43	0.20	0.183	-0.017 $\pm$ 0.001	-0.083 $\pm$ 0.007	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	0.061 $\pm$ 0.003	0.248 $\pm$ 0.006	0.181	-0.019	-9.25	0.0464	0.0499	0.978	92.806 $\pm$ 0.828	0.179 $\pm$ 0.001	98.458 $\pm$ 0.395
QML	beta44	0.11	0.097	-0.013 $\pm$ 0.002	-0.118 $\pm$ 0.019	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.377 $\pm$ 0.019	0.614 $\pm$ 0.018	0.099	-0.011	-10.18	0.0625	0.0634	1.016	95.272 $\pm$ 0.680	0.264 $\pm$ 0.003	36.280 $\pm$ 1.541
QML	beta45	0.11	0.106	-0.004 $\pm$ 0.002	-0.038 $\pm$ 0.016	0.003 $\pm$ 0.000	0.056 $\pm$ 0.001	0.003 $\pm$ 0.000	0.056 $\pm$ 0.002	0.262 $\pm$ 0.012	0.512 $\pm$ 0.014	0.105	-0.005	-4.46	0.0553	0.0555	1.007	95.170 $\pm$ 0.687	0.222 $\pm$ 0.001	48.510 $\pm$ 1.602
QML	beta46	0.08	0.084	0.004 $\pm$ 0.001	0.046 $\pm$ 0.018	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.320 $\pm$ 0.016	0.566 $\pm$ 0.017	0.081	0.001	0.72	0.0436	0.0436	0.998	94.245 $\pm$ 0.747	0.177 $\pm$ 0.002	45.118 $\pm$ 1.595
QML	beta47	0.08	0.079	-0.001 $\pm$ 0.001	-0.018 $\pm$ 0.017	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.285 $\pm$ 0.013	0.534 $\pm$ 0.015	0.076	-0.004	-4.46	0.0418	0.0420	1.025	94.964 $\pm$ 0.701	0.172 $\pm$ 0.002	41.829 $\pm$ 1.581
QML	beta51	0.16	0.180	0.020 $\pm$ 0.002	0.126 $\pm$ 0.013	0.004 $\pm$ 0.000	0.064 $\pm$ 0.001	0.005 $\pm$ 0.000	0.067 $\pm$ 0.002	0.177 $\pm$ 0.008	0.421 $\pm$ 0.011	0.180	0.020	12.72	0.0647	0.0678	1.035	95.889 $\pm$ 0.637	0.261 $\pm$ 0.002	78.931 $\pm$ 1.307
QML	beta52	0.16	0.146	-0.014 $\pm$ 0.002	-0.086 $\pm$ 0.011	0.003 $\pm$ 0.000	0.057 $\pm$ 0.001	0.003 $\pm$ 0.000	0.059 $\pm$ 0.002	0.135 $\pm$ 0.006	0.368 $\pm$ 0.010	0.146	-0.014	-8.65	0.0560	0.0577	1.010	93.628 $\pm$ 0.783	0.227 $\pm$ 0.002	74.101 $\pm$ 1.404
QML	beta53	0.16	0.137	-0.023 $\pm$ 0.002	-0.147 $\pm$ 0.010	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.003 $\pm$ 0.000	0.057 $\pm$ 0.001	0.128 $\pm$ 0.006	0.357 $\pm$ 0.009	0.138	-0.022	-13.96	0.0522	0.0568	1.016	91.881 $\pm$ 0.876	0.208 $\pm$ 0.001	73.484 $\pm$ 1.415
QML	beta54	0.16	0.157	-0.003 $\pm$ 0.002	-0.016 $\pm$ 0.010	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.095 $\pm$ 0.004	0.308 $\pm$ 0.008	0.157	-0.003	-1.99	0.0518	0.0519	0.999	94.861 $\pm$ 0.708	0.193 $\pm$ 0.001	91.161 $\pm$ 0.910
QML	beta55	0.08	0.090	0.010 $\pm$ 0.002	0.131 $\pm$ 0.030	0.005 $\pm$ 0.000	0.074 $\pm$ 0.002	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.873 $\pm$ 0.041	0.934 $\pm$ 0.026	0.091	0.011	13.70	0.0729	0.0738	1.034	94.964 $\pm$ 0.701	0.300 $\pm$ 0.003	24.358 $\pm$ 1.376
QML	beta56	0.08	0.079	-0.001 $\pm$ 0.002	-0.009 $\pm$ 0.026	0.004 $\pm$ 0.000	0.065 $\pm$ 0.001	0.004 $\pm$ 0.000	0.065 $\pm$ 0.002	0.660 $\pm$ 0.029	0.813 $\pm$ 0.022	0.079	-0.001	-0.82	0.0649	0.0649	1.014	94.553 $\pm$ 0.728	0.258 $\pm$ 0.002	22.508 $\pm$ 1.339
QML	beta57	0.06	0.049	-0.011 $\pm$ 0.002	-0.175 $\pm$ 0.025	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.646 $\pm$ 0.032	0.804 $\pm$ 0.024	0.047	-0.013	-22.26	0.0452	0.0471	0.999	91.264 $\pm$ 0.905	0.184 $\pm$ 0.002	13.566 $\pm$ 1.098
QML	beta58	0.06	0.053	-0.007 $\pm$ 0.001	-0.119 $\pm$ 0.020	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.409 $\pm$ 0.018	0.639 $\pm$ 0.017	0.051	-0.009	-15.56	0.0362	0.0374	1.005	92.806 $\pm$ 0.828	0.148 $\pm$ 0.001	24.666 $\pm$ 1.382
UPI	beta41	0.20	0.202	0.002 $\pm$ 0.002	0.010 $\pm$ 0.008	0.003 $\pm$ 0.000	0.050 $\pm$ 0.001	0.003 $\pm$ 0.000	0.050 $\pm$ 0.001	0.063 $\pm$ 0.003	0.251 $\pm$ 0.007	0.201	0.001	0.68	0.0507	0.0507	1.104	97.431 $\pm$ 0.507	0.217 $\pm$ 0.002	96.095 $\pm$ 0.621
UPI	beta42	0.20	0.204	0.004 $\pm$ 0.002	0.022 $\pm$ 0.009	0.003 $\pm$ 0.000	0.054 $\pm$ 0.001	0.003 $\pm$ 0.000	0.054 $\pm$ 0.002	0.074 $\pm$ 0.004	0.272 $\pm$ 0.008	0.203	0.003	1.46	0.0543	0.0543	1.055	96.095 $\pm$ 0.621	0.224 $\pm$ 0.002	96.197 $\pm$ 0.613
UPI	beta43	0.20	0.203	0.003 $\pm$ 0.002	0.015 $\pm$ 0.008	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.065 $\pm$ 0.003	0.256 $\pm$ 0.007	0.200	0.000	0.12	0.0521	0.0521	1.019	96.300 $\pm$ 0.605	0.204 $\pm$ 0.001	98.150 $\pm$ 0.432

Table S182

Simulation 2 - Condition 38: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.4
(continued)

Method	Param.	True	Mean-based										Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE	
UPI	beta47	0.08	0.083	0.003 ± 0.002	0.042 ± 0.023	0.003 ± 0.000	0.058 ± 0.002	0.003 ± 0.000	0.058 ± 0.002	0.527 ± 0.031	0.726 ± 0.024	0.079	-0.001	-0.76	0.0502	0.0502	1.132	97.122 ± 0.536	0.257 ± 0.005	22.405 ± 1.337	
UPI	beta51	0.16	0.163	0.003 ± 0.002	0.016 ± 0.012	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.146 ± 0.007	0.382 ± 0.011	0.164	0.004	2.42	0.0585	0.0586	1.028	95.992 ± 0.629	0.246 ± 0.001	76.362 ± 1.362	
UPI	beta52	0.16	0.161	0.001 ± 0.002	0.005 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.100 ± 0.005	0.316 ± 0.009	0.160	0.000	0.00	0.0511	0.0511	1.014	94.758 ± 0.714	0.201 ± 0.001	90.134 ± 0.956	
UPI	beta53	0.16	0.163	0.003 ± 0.002	0.018 ± 0.011	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	0.108 ± 0.005	0.328 ± 0.009	0.163	0.003	1.65	0.0518	0.0519	1.051	95.786 ± 0.644	0.216 ± 0.001	86.639 ± 1.091	
UPI	beta54	0.16	0.159	-0.001 ± 0.002	-0.005 ± 0.010	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.098 ± 0.005	0.313 ± 0.009	0.158	-0.002	-1.17	0.0491	0.0492	1.021	94.656 ± 0.721	0.200 ± 0.001	90.853 ± 0.924	
UPI	beta55	0.08	0.081	0.001 ± 0.002	0.014 ± 0.029	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.843 ± 0.046	0.918 ± 0.029	0.079	-0.001	-1.76	0.0689	0.0690	1.096	97.020 ± 0.545	0.316 ± 0.003	15.827 ± 1.170	
UPI	beta56	0.08	0.080	-0.000 ± 0.002	-0.006 ± 0.027	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.718 ± 0.034	0.847 ± 0.024	0.081	0.001	0.94	0.0696	0.0696	1.104	96.197 ± 0.613	0.293 ± 0.003	18.294 ± 1.239	
UPI	beta57	0.06	0.062	0.002 ± 0.002	0.033 ± 0.027	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.002	0.721 ± 0.041	0.849 ± 0.028	0.060	0.000	0.72	0.0481	0.0481	1.086	96.300 ± 0.605	0.217 ± 0.003	14.902 ± 1.142	
UPI	beta58	0.06	0.063	0.003 ± 0.001	0.052 ± 0.024	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.566 ± 0.033	0.752 ± 0.025	0.059	-0.001	-1.68	0.0428	0.0428	1.068	96.403 ± 0.597	0.189 ± 0.003	22.097 ± 1.330	

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S183

Simulation 2 - Condition 39: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.6

Method	Param.	True	Mean-based										Median-based					Inference		
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LSAM	beta41	0.20	0.203	0.003 $\pm$ 0.002	0.015 $\pm$ 0.009	0.003 $\pm$ 0.000	0.058 $\pm$ 0.001	0.003 $\pm$ 0.000	0.058 $\pm$ 0.002	0.084 $\pm$ 0.004	0.289 $\pm$ 0.009	0.201	0.001	0.66	0.0549	0.0549	1.027	95.749 $\pm$ 0.642	0.233 $\pm$ 0.001	93.826 $\pm$ 0.766
LSAM	beta42	0.20	0.204	0.004 $\pm$ 0.002	0.019 $\pm$ 0.010	0.004 $\pm$ 0.000	0.062 $\pm$ 0.001	0.004 $\pm$ 0.000	0.062 $\pm$ 0.002	0.096 $\pm$ 0.005	0.311 $\pm$ 0.009	0.204	0.004	1.93	0.0617	0.0618	0.977	96.154 $\pm$ 0.612	0.238 $\pm$ 0.001	91.599 $\pm$ 0.883
LSAM	beta43	0.20	0.199	-0.001 $\pm$ 0.002	-0.005 $\pm$ 0.009	0.003 $\pm$ 0.000	0.058 $\pm$ 0.001	0.003 $\pm$ 0.000	0.058 $\pm$ 0.002	0.084 $\pm$ 0.004	0.291 $\pm$ 0.008	0.198	-0.002	-0.96	0.0613	0.0614	0.963	94.231 $\pm$ 0.742	0.220 $\pm$ 0.001	93.927 $\pm$ 0.760
LSAM	beta44	0.11	0.109	-0.001 $\pm$ 0.002	-0.008 $\pm$ 0.021	0.005 $\pm$ 0.000	0.073 $\pm$ 0.002	0.005 $\pm$ 0.000	0.073 $\pm$ 0.002	0.443 $\pm$ 0.023	0.666 $\pm$ 0.020	0.106	-0.004	-3.86	0.0670	0.0672	0.977	95.040 $\pm$ 0.691	0.281 $\pm$ 0.002	37.753 $\pm$ 1.542
LSAM	beta45	0.11	0.115	0.005 $\pm$ 0.002	0.041 $\pm$ 0.019	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	0.360 $\pm$ 0.017	0.600 $\pm$ 0.017	0.114	0.004	3.69	0.0668	0.0669	0.930	92.814 $\pm$ 0.822	0.240 $\pm$ 0.002	47.470 $\pm$ 1.589
LSAM	beta46	0.08	0.084	0.004 $\pm$ 0.002	0.050 $\pm$ 0.020	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.003 $\pm$ 0.000	0.051 $\pm$ 0.002	0.408 $\pm$ 0.021	0.639 $\pm$ 0.019	0.082	0.002	2.47	0.0472	0.0473	0.989	95.344 $\pm$ 0.670	0.198 $\pm$ 0.002	38.158 $\pm$ 1.545
LSAM	beta47	0.08	0.084	0.004 $\pm$ 0.002	0.055 $\pm$ 0.020	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.414 $\pm$ 0.020	0.643 $\pm$ 0.019	0.080	0.000	-0.39	0.0492	0.0492	0.958	94.332 $\pm$ 0.736	0.193 $\pm$ 0.002	40.486 $\pm$ 1.562
LSAM	beta51	0.16	0.159	-0.001 $\pm$ 0.002	-0.009 $\pm$ 0.015	0.006 $\pm$ 0.000	0.076 $\pm$ 0.002	0.006 $\pm$ 0.000	0.076 $\pm$ 0.002	0.224 $\pm$ 0.010	0.473 $\pm$ 0.013	0.157	-0.003	-1.80	0.0732	0.0733	0.937	92.814 $\pm$ 0.822	0.278 $\pm$ 0.001	60.425 $\pm$ 1.556
LSAM	beta52	0.16	0.161	0.001 $\pm$ 0.002	0.009 $\pm$ 0.012	0.004 $\pm$ 0.000	0.061 $\pm$ 0.001	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.145 $\pm$ 0.006	0.380 $\pm$ 0.010	0.160	0.000	-0.21	0.0638	0.0638	0.958	95.344 $\pm$ 0.670	0.229 $\pm$ 0.001	79.049 $\pm$ 1.295
LSAM	beta53	0.16	0.159	-0.001 $\pm$ 0.002	-0.007 $\pm$ 0.013	0.004 $\pm$ 0.000	0.064 $\pm$ 0.001	0.004 $\pm$ 0.000	0.064 $\pm$ 0.002	0.161 $\pm$ 0.007	0.401 $\pm$ 0.011	0.157	-0.003	-1.69	0.0615	0.0616	0.952	94.332 $\pm$ 0.736	0.240 $\pm$ 0.001	73.381 $\pm$ 1.406
LSAM	beta54	0.16	0.163	0.003 $\pm$ 0.002	0.018 $\pm$ 0.012	0.004 $\pm$ 0.000	0.060 $\pm$ 0.001	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	0.140 $\pm$ 0.006	0.374 $\pm$ 0.010	0.163	0.003	1.90	0.0620	0.0621	0.959	94.534 $\pm$ 0.723	0.225 $\pm$ 0.001	78.947 $\pm$ 1.297
LSAM	beta55	0.08	0.083	0.003 $\pm$ 0.003	0.043 $\pm$ 0.032	0.006 $\pm$ 0.000	0.079 $\pm$ 0.002	0.006 $\pm$ 0.000	0.080 $\pm$ 0.002	0.988 $\pm$ 0.049	0.994 $\pm$ 0.029	0.083	0.003	4.00	0.0750	0.0751	0.949	94.433 $\pm$ 0.729	0.296 $\pm$ 0.003	23.178 $\pm$ 1.342
LSAM	beta56	0.08	0.081	0.001 $\pm$ 0.002	0.011 $\pm$ 0.028	0.005 $\pm$ 0.000	0.070 $\pm$ 0.002	0.005 $\pm$ 0.000	0.070 $\pm$ 0.002	0.769 $\pm$ 0.037	0.877 $\pm$ 0.025	0.079	-0.001	-1.08	0.0682	0.0682	0.990	94.332 $\pm$ 0.736	0.272 $\pm$ 0.002	23.279 $\pm$ 1.345
LSAM	beta57	0.06	0.063	0.003 $\pm$ 0.002	0.052 $\pm$ 0.027	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.711 $\pm$ 0.034	0.843 $\pm$ 0.024	0.063	0.003	4.73	0.0482	0.0482	0.978	94.534 $\pm$ 0.723	0.194 $\pm$ 0.002	23.988 $\pm$ 1.358
LSAM	beta58	0.06	0.060	-0.000 $\pm$ 0.001	-0.003 $\pm$ 0.022	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.499 $\pm$ 0.025	0.706 $\pm$ 0.021	0.058	-0.002	-2.93	0.0401	0.0401	0.986	94.534 $\pm$ 0.723	0.164 $\pm$ 0.001	27.733 $\pm$ 1.424
QML	beta41	0.20	0.194	-0.006 $\pm$ 0.002	-0.029 $\pm$ 0.009	0.003 $\pm$ 0.000	0.057 $\pm$ 0.001	0.003 $\pm$ 0.000	0.057 $\pm$ 0.002	0.082 $\pm$ 0.004	0.287 $\pm$ 0.008	0.192	-0.008	-4.02	0.0547	0.0553	1.068	95.547 $\pm$ 0.656	0.239 $\pm$ 0.010	92.004 $\pm$ 0.863
QML	beta42	0.20	0.199	-0.001 $\pm$ 0.002	-0.007 $\pm$ 0.010	0.004 $\pm$ 0.000	0.061 $\pm$ 0.001	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.092 $\pm$ 0.004	0.304 $\pm$ 0.009	0.198	-0.002	-0.85	0.0605	0.0605	0.995	95.951 $\pm$ 0.627	0.237 $\pm$ 0.003	91.599 $\pm$ 0.883
QML	beta43	0.20	0.188	-0.012 $\pm$ 0.002	-0.060 $\pm$ 0.009	0.003 $\pm$ 0.000	0.055 $\pm$ 0.001	0.003 $\pm$ 0.000	0.056 $\pm$ 0.002	0.080 $\pm$ 0.003	0.282 $\pm$ 0.008	0.186	-0.014	-6.77	0.0565	0.0581	0.987	93.623 $\pm$ 0.777	0.214 $\pm$ 0.002	93.016 $\pm$ 0.811
QML	beta44	0.11	0.098	-0.012 $\pm$ 0.002	-0.113 $\pm$ 0.021	0.005 $\pm$ 0.000	0.073 $\pm$ 0.002	0.005 $\pm$ 0.000	0.074 $\pm$ 0.002	0.447 $\pm$ 0.024	0.669 $\pm$ 0.021	0.100	-0.010	-8.84	0.0667	0.0674	0.997	95.445 $\pm$ 0.663	0.284 $\pm$ 0.003	35.223 $\pm$ 1.520
QML	beta45	0.11	0.111	0.001 $\pm$ 0.002	0.005 $\pm$ 0.018	0.004 $\pm$ 0.000	0.064 $\pm$ 0.001	0.004 $\pm$ 0.000	0.064 $\pm$ 0.002	0.334 $\pm$ 0.016	0.578 $\pm$ 0.016	0.110	0.000	-0.32	0.0642	0.0642	0.957	93.826 $\pm$ 0.766	0.239 $\pm$ 0.002	46.356 $\pm$ 1.586
QML	beta46	0.08	0.085	0.005 $\pm$ 0.002	0.059 $\pm$ 0.019	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.373 $\pm$ 0.019	0.611 $\pm$ 0.018	0.081	0.001	1.35	0.0465	0.0465	1.011	95.648 $\pm$ 0.649	0.193 $\pm$ 0.002	38.866 $\pm$ 1.551
QML	beta47	0.08	0.084	0.004 $\pm$ 0.002	0.055 $\pm$ 0.020	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	0.002 $\pm$ 0.000	0.050 $\pm$ 0.002	0.388 $\pm$ 0.020	0.623 $\pm$ 0.019	0.078	-0.002	-1.88	0.0487	0.0488	0.978	95.040 $\pm$ 0.691	0.190 $\pm$ 0.002	38.057 $\pm$ 1.545
QML	beta51	0.16	0.170	0.010 $\pm$ 0.002	0.065 $\pm$ 0.015	0.006 $\pm$ 0.000	0.078 $\pm$ 0.002	0.006 $\pm$ 0.000	0.078 $\pm$ 0.002	0.239 $\pm$ 0.011	0.489 $\pm$ 0.014	0.171	0.011	6.94	0.0742	0.0750	0.991	94.332 $\pm$ 0.736	0.302 $\pm$ 0.003	61.134 $\pm$ 1.551
QML	beta52	0.16	0.143	-0.017 $\pm$ 0.002	-0.105 $\pm$ 0.013	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.182 $\pm$ 0.008	0.427 $\pm$ 0.012	0.139	-0.021	-13.18	0.0684	0.0716	1.011	92.915 $\pm$ 0.816	0.262 $\pm$ 0.003	59.514 $\pm$ 1.562
QML	beta53	0.16	0.134	-0.026 $\pm$ 0.002	-0.162 $\pm$ 0.013	0.004 $\pm$ 0.000	0.064 $\pm$ 0.001	0.005 $\pm$ 0.000	0.069 $\pm$ 0.002	0.186 $\pm$ 0.008	0.431 $\pm$ 0.011	0.134	-0.026	-16.21	0.0652	0.0702	0.984	92.510 $\pm$ 0.837	0.247 $\pm$ 0.002	58.097 $\pm$ 1.570
QML	beta54	0.16	0.162	0.002 $\pm$ 0.002	0.015 $\pm$ 0.012	0.004 $\pm$ 0.000	0.060 $\pm$ 0.001	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	0.140 $\pm$ 0.006	0.374 $\pm$ 0.010	0.162	0.002	1.06	0.0603	0.0603	0.975	94.636 $\pm$ 0.717	0.229 $\pm$ 0.002	78.947 $\pm$ 1.297
QML	beta55	0.08	0.089	0.009 $\pm$ 0.003	0.116 $\pm$ 0.031	0.006 $\pm$ 0.000	0.079 $\pm$ 0.002	0.006 $\pm$ 0.000	0.080 $\pm$ 0.002	0.991 $\pm$ 0.046	0.996 $\pm$ 0.028	0.093	0.013	15.86	0.0765	0.0775	0.996	95.243 $\pm$ 0.677	0.309 $\pm$ 0.003	23.178 $\pm$ 1.342
QML	beta56	0.08	0.081	0.001 $\pm$ 0.002	0.013 $\pm$ 0.027	0.005 $\pm$ 0.000	0.069 $\pm$ 0.002	0.005 $\pm$ 0.000	0.069 $\pm$ 0.002	0.744 $\pm$ 0.036	0.863 $\pm$ 0.025	0.079	-0.001	-1.71	0.0658	0.0658	1.027	95.142 $\pm$ 0.684	0.278 $\pm$ 0.003	21.053 $\pm$ 1.297
QML	beta57	0.06	0.056	-0.004 $\pm$ 0.002	-0.073 $\pm$ 0.026	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.678 $\pm$ 0.031	0.824 $\pm$ 0.023	0.053	-0.007	-12.01	0.0484	0.0489	1.016	93.016 $\pm$ 0.811	0.196 $\pm$ 0.002	16.194 $\pm$ 1.172
QML	beta58	0.06	0.056	-0.004 $\pm$ 0.001	-0.065 $\pm$ 0.022	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.490 $\pm$ 0.026	0.700 $\pm$ 0.021	0.053	-0.007	-11.65	0.0387	0.0394	0.989	94.028 $\pm$ 0.754	0.162 $\pm$ 0.002	22.571 $\pm$ 1.330
UPI	beta41	0.20	0.204	0.004 $\pm$ 0.002	0.019 $\pm$ 0.009	0.003 $\pm$ 0.000	0.059 $\pm$ 0.001	0.003 $\pm$ 0.000	0.059 $\pm$ 0.002	0.087 $\pm$ 0.004	0.295 $\pm$ 0.009	0.202	0.002	0.86	0.0564	0.0564	1.015	95.850 $\pm$ 0.634	0.234 $\pm$ 0.001	92.713 $\pm$ 0.827
UPI	beta42	0.20	0.204	0.004 $\pm$ 0.002	0.022 $\pm$ 0.010	0.004 $\pm$ 0.000	0.062 $\pm$ 0.001	0.004 $\pm$ 0.000	0.062 $\pm$ 0.002	0.098 $\pm$ 0.005	0.312 $\pm$ 0.009	0.202	0.002	1.16	0.0601	0.0601	0.987	96.255 $\pm$ 0.604	0.241 $\pm$ 0.001	91.700 $\pm$ 0.878
UPI	beta43	0.20	0.199	-0.001 $\pm$ 0.002	-0.005 $\pm$ 0.009	0.003 $\pm$ 0.000	0.059 $\pm$ 0.001	0.003 $\pm$ 0.000	0.059 $\pm$ 0.002	0.087 $\pm$ 0.004	0.295 $\pm$ 0.008	0.198	-0.002	-0.97	0.0619	0.0619	0.975	94.433 $\pm$ 0.729	0.225 $\pm$ 0.001	93.522 $\pm$ 0.783

Table S183

Simulation 2 - Condition 39: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.6
(continued)

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.087	0.007 ± 0.002	0.083 ± 0.023	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.509 ± 0.027	0.714 ± 0.022	0.081	0.001	1.79	0.0541	0.0541	1.020	94.636 ± 0.717	0.227 ± 0.003	28.644 ± 1.438
UPI	beta51	0.16	0.160	0.000 ± 0.002	0.001 ± 0.015	0.005 ± 0.000	0.074 ± 0.002	0.005 ± 0.000	0.074 ± 0.002	0.213 ± 0.010	0.461 ± 0.013	0.159	-0.001	-0.82	0.0717	0.0718	0.971	92.713 ± 0.827	0.281 ± 0.001	61.134 ± 1.551
UPI	beta52	0.16	0.162	0.002 ± 0.002	0.014 ± 0.012	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.144 ± 0.006	0.379 ± 0.010	0.161	0.001	0.59	0.0635	0.0635	0.959	95.547 ± 0.656	0.228 ± 0.001	80.061 ± 1.271
UPI	beta53	0.16	0.159	-0.001 ± 0.002	-0.006 ± 0.013	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.161 ± 0.007	0.401 ± 0.011	0.157	-0.003	-1.82	0.0623	0.0623	0.964	93.826 ± 0.766	0.243 ± 0.001	72.470 ± 1.421
UPI	beta54	0.16	0.163	0.003 ± 0.002	0.019 ± 0.012	0.004 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	0.137 ± 0.006	0.371 ± 0.010	0.163	0.003	1.85	0.0605	0.0606	0.975	94.636 ± 0.717	0.226 ± 0.001	79.352 ± 1.288
UPI	beta55	0.08	0.081	0.001 ± 0.003	0.009 ± 0.032	0.006 ± 0.000	0.080 ± 0.002	0.006 ± 0.000	0.080 ± 0.002	1.006 ± 0.051	1.003 ± 0.030	0.082	0.002	2.74	0.0757	0.0757	0.989	96.154 ± 0.612	0.311 ± 0.003	19.737 ± 1.266
UPI	beta56	0.08	0.080	0.000 ± 0.002	0.006 ± 0.029	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.815 ± 0.040	0.903 ± 0.026	0.078	-0.002	-2.79	0.0660	0.0661	1.029	95.648 ± 0.649	0.291 ± 0.003	18.219 ± 1.228
UPI	beta57	0.06	0.063	0.003 ± 0.002	0.057 ± 0.027	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	0.749 ± 0.039	0.866 ± 0.027	0.063	0.003	5.73	0.0493	0.0494	1.021	95.040 ± 0.691	0.207 ± 0.002	19.636 ± 1.264
UPI	beta58	0.06	0.060	0.000 ± 0.001	0.006 ± 0.024	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.571 ± 0.029	0.756 ± 0.023	0.057	-0.003	-5.75	0.0426	0.0427	1.012	94.433 ± 0.729	0.180 ± 0.002	22.571 ± 1.330

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S184

Simulation 2 - Condition 40: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.6

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LSAM	beta41	0.20	0.199	-0.001 $\pm$ 0.001	-0.004 $\pm$ 0.006	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.034 $\pm$ 0.002	0.183 $\pm$ 0.005	0.199	-0.001	-0.66	0.0357	0.0357	1.002	95.100 $\pm$ 0.683	0.144 $\pm$ 0.000	99.800 $\pm$ 0.141
LSAM	beta42	0.20	0.201	0.001 $\pm$ 0.001	0.003 $\pm$ 0.006	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.034 $\pm$ 0.002	0.184 $\pm$ 0.005	0.201	0.001	0.68	0.0356	0.0356	1.018	95.700 $\pm$ 0.641	0.147 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta43	0.20	0.201	0.001 $\pm$ 0.001	0.004 $\pm$ 0.006	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.032 $\pm$ 0.001	0.179 $\pm$ 0.005	0.200	0.000	0.13	0.0331	0.0331	0.980	93.800 $\pm$ 0.763	0.137 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta44	0.11	0.110	-0.000 $\pm$ 0.001	-0.004 $\pm$ 0.012	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.149 $\pm$ 0.007	0.386 $\pm$ 0.010	0.109	-0.001	-0.90	0.0423	0.0423	1.011	95.400 $\pm$ 0.662	0.169 $\pm$ 0.001	72.600 $\pm$ 1.410
LSAM	beta45	0.11	0.110	0.000 $\pm$ 0.001	0.003 $\pm$ 0.011	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.115 $\pm$ 0.005	0.339 $\pm$ 0.009	0.111	0.001	0.53	0.0380	0.0380	1.004	95.700 $\pm$ 0.641	0.147 $\pm$ 0.001	83.800 $\pm$ 1.165
LSAM	beta46	0.08	0.081	0.001 $\pm$ 0.001	0.009 $\pm$ 0.012	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.149 $\pm$ 0.007	0.387 $\pm$ 0.011	0.078	-0.002	-2.37	0.0299	0.0300	0.974	94.500 $\pm$ 0.721	0.118 $\pm$ 0.001	77.600 $\pm$ 1.318
LSAM	beta47	0.08	0.082	0.002 $\pm$ 0.001	0.020 $\pm$ 0.012	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.146 $\pm$ 0.007	0.382 $\pm$ 0.011	0.081	0.001	1.47	0.0294	0.0294	0.977	94.100 $\pm$ 0.745	0.117 $\pm$ 0.001	80.200 $\pm$ 1.260
LSAM	beta51	0.16	0.160	0.000 $\pm$ 0.001	0.000 $\pm$ 0.009	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.079 $\pm$ 0.003	0.281 $\pm$ 0.008	0.160	0.000	0.05	0.0470	0.0470	0.981	95.000 $\pm$ 0.689	0.173 $\pm$ 0.001	95.200 $\pm$ 0.676
LSAM	beta52	0.16	0.162	0.002 $\pm$ 0.001	0.013 $\pm$ 0.007	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.054 $\pm$ 0.002	0.231 $\pm$ 0.007	0.163	0.003	1.58	0.0346	0.0347	0.984	94.000 $\pm$ 0.751	0.143 $\pm$ 0.000	99.300 $\pm$ 0.264
LSAM	beta53	0.16	0.161	0.001 $\pm$ 0.001	0.006 $\pm$ 0.008	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.064 $\pm$ 0.003	0.252 $\pm$ 0.007	0.161	0.001	0.87	0.0410	0.0410	0.945	94.000 $\pm$ 0.751	0.150 $\pm$ 0.000	98.800 $\pm$ 0.344
LSAM	beta54	0.16	0.160	0.000 $\pm$ 0.001	0.003 $\pm$ 0.007	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.049 $\pm$ 0.002	0.221 $\pm$ 0.006	0.160	0.000	0.15	0.0356	0.0356	1.021	95.800 $\pm$ 0.634	0.141 $\pm$ 0.000	99.600 $\pm$ 0.200
LSAM	beta55	0.08	0.080	-0.000 $\pm$ 0.001	-0.004 $\pm$ 0.018	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.323 $\pm$ 0.014	0.568 $\pm$ 0.015	0.079	-0.001	-0.92	0.0450	0.0450	0.979	94.800 $\pm$ 0.702	0.175 $\pm$ 0.001	43.900 $\pm$ 1.569
LSAM	beta56	0.08	0.081	0.001 $\pm$ 0.001	0.018 $\pm$ 0.017	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.301 $\pm$ 0.014	0.549 $\pm$ 0.015	0.080	0.000	0.36	0.0419	0.0419	0.960	94.400 $\pm$ 0.727	0.165 $\pm$ 0.001	50.500 $\pm$ 1.581
LSAM	beta57	0.06	0.060	0.000 $\pm$ 0.001	0.005 $\pm$ 0.016	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.261 $\pm$ 0.012	0.511 $\pm$ 0.014	0.059	-0.001	-1.22	0.0300	0.0300	0.970	94.400 $\pm$ 0.727	0.117 $\pm$ 0.001	54.000 $\pm$ 1.576
LSAM	beta58	0.06	0.061	0.001 $\pm$ 0.001	0.017 $\pm$ 0.014	0.001 $\pm$ 0.000	0.026 $\pm$ 0.001	0.001 $\pm$ 0.000	0.026 $\pm$ 0.001	0.193 $\pm$ 0.009	0.439 $\pm$ 0.012	0.060	0.000	-0.22	0.0268	0.0268	0.980	95.500 $\pm$ 0.656	0.101 $\pm$ 0.000	66.500 $\pm$ 1.493
QML	beta41	0.20	0.190	-0.010 $\pm$ 0.001	-0.048 $\pm$ 0.006	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.036 $\pm$ 0.002	0.189 $\pm$ 0.005	0.190	-0.010	-5.22	0.0354	0.0369	0.999	93.600 $\pm$ 0.774	0.143 $\pm$ 0.001	99.700 $\pm$ 0.173
QML	beta42	0.20	0.195	-0.005 $\pm$ 0.001	-0.024 $\pm$ 0.006	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.034 $\pm$ 0.002	0.184 $\pm$ 0.005	0.197	-0.003	-1.74	0.0361	0.0363	1.020	94.900 $\pm$ 0.696	0.146 $\pm$ 0.001	99.700 $\pm$ 0.173
QML	beta43	0.20	0.190	-0.010 $\pm$ 0.001	-0.048 $\pm$ 0.006	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.033 $\pm$ 0.001	0.181 $\pm$ 0.005	0.190	-0.010	-5.19	0.0325	0.0341	0.981	93.100 $\pm$ 0.801	0.134 $\pm$ 0.000	100.000 $\pm$ 0.000
QML	beta44	0.11	0.102	-0.008 $\pm$ 0.001	-0.073 $\pm$ 0.012	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.159 $\pm$ 0.007	0.399 $\pm$ 0.011	0.103	-0.007	-6.64	0.0424	0.0431	1.026	95.800 $\pm$ 0.634	0.174 $\pm$ 0.001	63.400 $\pm$ 1.523
QML	beta45	0.11	0.108	-0.002 $\pm$ 0.001	-0.014 $\pm$ 0.011	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.119 $\pm$ 0.005	0.345 $\pm$ 0.009	0.109	-0.001	-1.01	0.0387	0.0387	1.008	95.400 $\pm$ 0.662	0.150 $\pm$ 0.001	80.400 $\pm$ 1.255
QML	beta46	0.08	0.081	0.001 $\pm$ 0.001	0.007 $\pm$ 0.012	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.152 $\pm$ 0.007	0.390 $\pm$ 0.011	0.077	-0.003	-3.72	0.0305	0.0307	0.967	93.900 $\pm$ 0.757	0.118 $\pm$ 0.001	80.100 $\pm$ 1.263
QML	beta47	0.08	0.080	0.000 $\pm$ 0.001	0.001 $\pm$ 0.012	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.141 $\pm$ 0.007	0.375 $\pm$ 0.011	0.079	-0.001	-1.08	0.0286	0.0286	0.986	93.500 $\pm$ 0.780	0.116 $\pm$ 0.001	82.700 $\pm$ 1.196
QML	beta51	0.16	0.171	0.011 $\pm$ 0.001	0.070 $\pm$ 0.009	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.090 $\pm$ 0.004	0.300 $\pm$ 0.008	0.171	0.011	6.92	0.0481	0.0494	0.992	94.100 $\pm$ 0.745	0.181 $\pm$ 0.001	95.900 $\pm$ 0.627
QML	beta52	0.16	0.145	-0.015 $\pm$ 0.001	-0.091 $\pm$ 0.008	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.075 $\pm$ 0.004	0.274 $\pm$ 0.008	0.144	-0.016	-9.90	0.0382	0.0413	0.979	91.500 $\pm$ 0.882	0.159 $\pm$ 0.001	94.200 $\pm$ 0.739
QML	beta53	0.16	0.137	-0.023 $\pm$ 0.001	-0.146 $\pm$ 0.008	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.086 $\pm$ 0.004	0.293 $\pm$ 0.007	0.138	-0.022	-13.51	0.0414	0.0467	0.950	88.000 $\pm$ 1.028	0.151 $\pm$ 0.000	93.700 $\pm$ 0.768
QML	beta54	0.16	0.160	-0.000 $\pm$ 0.001	-0.000 $\pm$ 0.007	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.050 $\pm$ 0.002	0.224 $\pm$ 0.006	0.160	0.000	0.01	0.0345	0.0345	1.020	95.900 $\pm$ 0.627	0.143 $\pm$ 0.000	99.500 $\pm$ 0.223
QML	beta55	0.08	0.087	0.007 $\pm$ 0.002	0.084 $\pm$ 0.019	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.368 $\pm$ 0.017	0.606 $\pm$ 0.017	0.085	0.005	6.42	0.0477	0.0479	0.980	94.100 $\pm$ 0.745	0.185 $\pm$ 0.001	45.700 $\pm$ 1.575
QML	beta56	0.08	0.082	0.002 $\pm$ 0.001	0.026 $\pm$ 0.018	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.318 $\pm$ 0.015	0.564 $\pm$ 0.016	0.083	0.003	3.57	0.0444	0.0445	0.960	93.900 $\pm$ 0.757	0.170 $\pm$ 0.001	48.800 $\pm$ 1.581
QML	beta57	0.06	0.053	-0.007 $\pm$ 0.001	-0.114 $\pm$ 0.016	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	0.279 $\pm$ 0.012	0.529 $\pm$ 0.014	0.051	-0.009	-15.11	0.0299	0.0312	0.977	91.900 $\pm$ 0.863	0.119 $\pm$ 0.001	41.000 $\pm$ 1.555
QML	beta58	0.06	0.057	-0.003 $\pm$ 0.001	-0.051 $\pm$ 0.014	0.001 $\pm$ 0.000	0.027 $\pm$ 0.001	0.001 $\pm</$												

Table S184

Simulation 2 - Condition 40: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.6
(continued)

Method	Param.	True	Mean-based										Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE	
UPI	beta47	0.08	0.082	0.002 ± 0.001	0.024 ± 0.013	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.162 ± 0.008	0.403 ± 0.012	0.081	0.001	1.39	0.0310	0.0310	0.994	94.400 ± 0.727	0.125 ± 0.001	77.900 ± 1.312	
UPI	beta51	0.16	0.161	0.001 ± 0.001	0.005 ± 0.009	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.078 ± 0.003	0.279 ± 0.008	0.161	0.001	0.48	0.0471	0.0471	0.985	94.700 ± 0.708	0.173 ± 0.001	95.500 ± 0.656	
UPI	beta52	0.16	0.163	0.003 ± 0.001	0.016 ± 0.007	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.053 ± 0.003	0.231 ± 0.007	0.163	0.003	1.64	0.0342	0.0343	0.978	94.200 ± 0.739	0.142 ± 0.000	99.200 ± 0.282	
UPI	beta53	0.16	0.161	0.001 ± 0.001	0.006 ± 0.008	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.064 ± 0.003	0.253 ± 0.007	0.162	0.002	1.32	0.0411	0.0412	0.947	94.000 ± 0.751	0.150 ± 0.000	98.600 ± 0.372	
UPI	beta54	0.16	0.161	0.001 ± 0.001	0.004 ± 0.007	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.049 ± 0.002	0.221 ± 0.006	0.160	0.000	0.22	0.0355	0.0355	1.015	95.500 ± 0.656	0.141 ± 0.000	99.600 ± 0.200	
UPI	beta55	0.08	0.080	-0.000 ± 0.001	-0.002 ± 0.018	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.332 ± 0.015	0.576 ± 0.016	0.079	-0.001	-1.86	0.0443	0.0443	0.997	96.600 ± 0.573	0.180 ± 0.001	41.800 ± 1.560	
UPI	beta56	0.08	0.081	0.001 ± 0.001	0.012 ± 0.018	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.308 ± 0.014	0.555 ± 0.015	0.081	0.001	0.83	0.0434	0.0434	0.982	94.800 ± 0.702	0.171 ± 0.001	47.700 ± 1.579	
UPI	beta57	0.06	0.060	-0.000 ± 0.001	-0.005 ± 0.016	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.272 ± 0.014	0.521 ± 0.016	0.059	-0.001	-2.15	0.0314	0.0314	0.978	94.600 ± 0.715	0.120 ± 0.001	50.000 ± 1.581	
UPI	beta58	0.06	0.061	0.001 ± 0.001	0.016 ± 0.014	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	0.198 ± 0.010	0.445 ± 0.013	0.060	0.000	0.66	0.0264	0.0264	1.007	95.800 ± 0.634	0.105 ± 0.001	63.900 ± 1.519	

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S185

Simulation 2 - Condition 41: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.8

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.197	-0.003 ± 0.002	-0.017 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.058 ± 0.003	0.241 ± 0.007	0.197	-0.003	-1.59	0.0472	0.0473	0.994	95.792 ± 0.636	0.188 ± 0.001	97.495 ± 0.495
LSAM	beta42	0.20	0.202	0.002 ± 0.002	0.010 ± 0.008	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.059 ± 0.003	0.244 ± 0.007	0.198	-0.002	-0.77	0.0497	0.0498	0.987	93.988 ± 0.752	0.188 ± 0.001	98.497 ± 0.385
LSAM	beta43	0.20	0.201	0.001 ± 0.001	0.005 ± 0.007	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.054 ± 0.002	0.232 ± 0.006	0.200	0.000	0.19	0.0472	0.0472	0.970	93.888 ± 0.758	0.176 ± 0.001	99.198 ± 0.282
LSAM	beta44	0.11	0.108	-0.002 ± 0.002	-0.015 ± 0.016	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.250 ± 0.012	0.500 ± 0.014	0.107	-0.003	-2.35	0.0533	0.0534	0.925	93.387 ± 0.787	0.199 ± 0.001	58.717 ± 1.558
LSAM	beta45	0.11	0.112	0.002 ± 0.002	0.020 ± 0.014	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.192 ± 0.009	0.439 ± 0.012	0.113	0.003	2.57	0.0488	0.0489	0.937	93.487 ± 0.781	0.177 ± 0.001	69.439 ± 1.458
LSAM	beta46	0.08	0.081	0.001 ± 0.001	0.014 ± 0.015	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.221 ± 0.010	0.470 ± 0.013	0.079	-0.001	-0.65	0.0370	0.0370	0.946	93.387 ± 0.787	0.139 ± 0.001	64.529 ± 1.514
LSAM	beta47	0.08	0.079	-0.001 ± 0.001	-0.009 ± 0.015	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.210 ± 0.010	0.459 ± 0.013	0.079	-0.001	-1.07	0.0361	0.0361	0.965	93.487 ± 0.781	0.139 ± 0.001	61.924 ± 1.537
LSAM	beta51	0.16	0.160	-0.000 ± 0.002	-0.000 ± 0.012	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.143 ± 0.006	0.378 ± 0.010	0.160	0.000	-0.27	0.0628	0.0628	0.930	93.687 ± 0.770	0.220 ± 0.001	80.862 ± 1.245
LSAM	beta52	0.16	0.161	0.001 ± 0.002	0.004 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.100 ± 0.004	0.316 ± 0.009	0.160	0.000	0.27	0.0488	0.0488	0.944	93.587 ± 0.775	0.187 ± 0.001	89.980 ± 0.950
LSAM	beta53	0.16	0.161	0.001 ± 0.002	0.004 ± 0.011	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.111 ± 0.005	0.333 ± 0.009	0.160	0.000	-0.17	0.0532	0.0532	0.934	93.587 ± 0.775	0.195 ± 0.001	88.076 ± 1.026
LSAM	beta54	0.16	0.159	-0.001 ± 0.002	-0.007 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.100 ± 0.004	0.316 ± 0.008	0.158	-0.002	-1.41	0.0505	0.0505	0.931	93.186 ± 0.798	0.184 ± 0.001	90.281 ± 0.938
LSAM	beta55	0.08	0.081	0.001 ± 0.002	0.012 ± 0.024	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.584 ± 0.027	0.764 ± 0.022	0.082	0.002	2.24	0.0595	0.0595	0.904	93.287 ± 0.792	0.217 ± 0.002	35.271 ± 1.512
LSAM	beta56	0.08	0.078	-0.002 ± 0.002	-0.023 ± 0.022	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.497 ± 0.023	0.705 ± 0.020	0.077	-0.003	-3.35	0.0526	0.0527	0.926	91.784 ± 0.869	0.205 ± 0.002	35.070 ± 1.511
LSAM	beta57	0.06	0.060	0.000 ± 0.001	0.002 ± 0.020	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.398 ± 0.019	0.631 ± 0.018	0.059	-0.001	-1.68	0.0394	0.0395	0.955	94.188 ± 0.741	0.142 ± 0.001	39.880 ± 1.550
LSAM	beta58	0.06	0.060	0.000 ± 0.001	0.002 ± 0.018	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.319 ± 0.014	0.565 ± 0.015	0.059	-0.001	-2.05	0.0342	0.0342	0.942	92.385 ± 0.840	0.125 ± 0.001	47.295 ± 1.580
QML	beta41	0.20	0.192	-0.008 ± 0.002	-0.038 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.060 ± 0.003	0.245 ± 0.007	0.194	-0.006	-3.06	0.0478	0.0482	1.012	95.591 ± 0.650	0.192 ± 0.005	96.293 ± 0.598
QML	beta42	0.20	0.200	-0.000 ± 0.002	-0.001 ± 0.008	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.059 ± 0.003	0.244 ± 0.007	0.196	-0.004	-2.10	0.0503	0.0504	1.011	94.489 ± 0.722	0.193 ± 0.004	98.297 ± 0.410
QML	beta43	0.20	0.197	-0.003 ± 0.001	-0.017 ± 0.007	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.053 ± 0.002	0.231 ± 0.006	0.196	-0.004	-2.18	0.0466	0.0468	0.975	93.888 ± 0.758	0.176 ± 0.002	98.898 ± 0.330
QML	beta44	0.11	0.104	-0.006 ± 0.002	-0.056 ± 0.016	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.266 ± 0.013	0.516 ± 0.015	0.103	-0.007	-6.07	0.0538	0.0542	0.927	93.287 ± 0.792	0.205 ± 0.002	54.008 ± 1.578
QML	beta45	0.11	0.111	0.001 ± 0.002	0.014 ± 0.014	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.199 ± 0.009	0.446 ± 0.012	0.111	0.001	0.93	0.0500	0.0500	0.939	93.788 ± 0.764	0.181 ± 0.001	68.637 ± 1.469
QML	beta46	0.08	0.082	0.002 ± 0.001	0.020 ± 0.015	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.224 ± 0.011	0.473 ± 0.013	0.079	-0.001	-1.24	0.0359	0.0359	0.957	93.287 ± 0.792	0.142 ± 0.001	63.427 ± 1.525
QML	beta47	0.08	0.079	-0.001 ± 0.001	-0.007 ± 0.015	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.214 ± 0.010	0.462 ± 0.013	0.078	-0.002	-1.90	0.0375	0.0375	0.966	92.886 ± 0.814	0.140 ± 0.001	61.523 ± 1.540
QML	beta51	0.16	0.164	0.004 ± 0.002	0.026 ± 0.012	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.145 ± 0.006	0.381 ± 0.010	0.164	0.004	2.63	0.0651	0.0653	0.966	95.090 ± 0.684	0.230 ± 0.001	80.361 ± 1.258
QML	beta52	0.16	0.143	-0.017 ± 0.002	-0.104 ± 0.011	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.130 ± 0.006	0.360 ± 0.010	0.143	-0.017	-10.43	0.0546	0.0571	0.950	92.084 ± 0.855	0.206 ± 0.002	77.154 ± 1.329
QML	beta53	0.16	0.140	-0.020 ± 0.002	-0.123 ± 0.011	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.058 ± 0.001	0.131 ± 0.006	0.362 ± 0.009	0.139	-0.021	-12.91	0.0551	0.0588	0.946	91.283 ± 0.893	0.202 ± 0.001	75.952 ± 1.353
QML	beta54	0.16	0.159	-0.001 ± 0.002	-0.009 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.101 ± 0.004	0.318 ± 0.008	0.157	-0.003	-1.76	0.0504	0.0504	0.935	93.086 ± 0.803	0.187 ± 0.001	89.479 ± 0.971
QML	beta55	0.08	0.084	0.004 ± 0.002	0.049 ± 0.024	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.598 ± 0.028	0.774 ± 0.022	0.084	0.004	4.52	0.0619	0.0620	0.924	93.988 ± 0.752	0.224 ± 0.002	36.072 ± 1.520
QML	beta56	0.08	0.079	-0.001 ± 0.002	-0.014 ± 0.022	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.497 ± 0.023	0.705 ± 0.020	0.078	-0.002	-2.69	0.0526	0.0526	0.945	92.986 ± 0.808	0.209 ± 0.002	32.966 ± 1.488
QML	beta57	0.06	0.057	-0.003 ± 0.001	-0.044 ± 0.020	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.408 ± 0.020	0.639 ± 0.019	0.056	-0.004	-6.67	0.0383	0.0385	0.962	94.088 ± 0.747	0.144 ± 0.001	33.267 ± 1.491
QML	beta58	0.06	0.059	-0.001 ± 0.001	-0.023 ± 0.018	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.328 ± 0.014	0.573 ± 0.016	0.056	-0.004	-5.96	0.0335	0.0337	0.943	91.984 ± 0.860	0.127 ± 0.001	43.888 ± 1.571
UPI	beta41	0.20	0.196	-0.004 ± 0.002	-0.018 ± 0.008	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.059 ± 0.003	0.243 ± 0.007	0.196	-0.004	-1.95	0.0470	0.0472	0.948	95.190 ± 0.677	0.181 ± 0.001	97.395 ± 0.504
UPI	beta42	0.20	0.202	0.002 ± 0.002	0.011 ± 0.008	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.060 ± 0.003	0.245 ± 0.007	0.198	-0.002	-1.02	0.0499	0.0499	0.961	93.487 ± 0.781	0.184 ± 0.001	98.297 ± 0.410
UPI	beta43	0.20	0.201	0.001 ± 0.001	0.005 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.054 ± 0.002	0.233 ± 0.006	0.200	0.000	-0.14	0.0474	0.0474	0.963	93.687 ± 0.770	0.176 ± 0.001	99.098 ± 0.299
UPI	beta44	0.11	0.109	-0.001 ± 0.002	-0.010 ± 0.017	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.282 ± 0.014	0.531 ± 0.016	0.108	-0.002	-1.75	0.0568	0.0568	0.908	93.186 ± 0.798	0.208 ± 0.001	56.012 ± 1.571
UPI	beta45	0.11	0.113	0.003 ± 0.002	0.024 ± 0.014	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.205 ± 0.009	0.453 ± 0.012	0.112	0.002	1.75	0.0508	0.0509	0.940	93.988 ± 0.752	0.183 ± 0.001	67.134 ± 1.487
UPI	beta46	0.08	0.081	0.001 ± 0.001	0.013 ± 0.015	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.237 ± 0.011	0.487 ± 0.014	0.079	-0.001	-1.07	0.0362	0.0362	0.954	93.888 ± 0.758	0.146 ± 0.001	60.521 ± 1.544

Table S185

Simulation 2 - Condition 41: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.8
(continued)

Method	Param.	True	Mean-based										Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE	
UPI	beta47	0.08	0.080	-0.000 ± 0.001	-0.002 ± 0.015	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.231 ± 0.011	0.480 ± 0.014	0.078	-0.002	-2.36	0.0373	0.0373	0.966	94.188 ± 0.741	0.146 ± 0.001	58.617 ± 1.559	
UPI	beta51	0.16	0.160	0.000 ± 0.002	0.001 ± 0.012	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.140 ± 0.006	0.374 ± 0.010	0.161	0.001	0.46	0.0643	0.0643	0.947	94.489 ± 0.722	0.222 ± 0.001	79.659 ± 1.274	
UPI	beta52	0.16	0.161	0.001 ± 0.002	0.005 ± 0.010	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.100 ± 0.005	0.316 ± 0.009	0.161	0.001	0.88	0.0486	0.0486	0.933	93.287 ± 0.792	0.185 ± 0.001	90.481 ± 0.929	
UPI	beta53	0.16	0.161	0.001 ± 0.002	0.005 ± 0.011	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.110 ± 0.005	0.332 ± 0.009	0.159	-0.001	-0.33	0.0523	0.0523	0.938	93.587 ± 0.775	0.196 ± 0.001	88.076 ± 1.026	
UPI	beta54	0.16	0.159	-0.001 ± 0.002	-0.006 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.100 ± 0.004	0.316 ± 0.008	0.158	-0.002	-1.33	0.0503	0.0504	0.925	93.086 ± 0.803	0.183 ± 0.001	90.581 ± 0.925	
UPI	beta55	0.08	0.081	0.001 ± 0.002	0.008 ± 0.024	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	0.598 ± 0.029	0.773 ± 0.022	0.081	0.001	0.89	0.0623	0.0623	0.914	93.487 ± 0.781	0.222 ± 0.002	33.768 ± 1.497	
UPI	beta56	0.08	0.078	-0.002 ± 0.002	-0.019 ± 0.022	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.501 ± 0.023	0.707 ± 0.020	0.077	-0.003	-4.21	0.0538	0.0539	0.946	92.786 ± 0.819	0.210 ± 0.002	33.267 ± 1.491	
UPI	beta57	0.06	0.060	0.000 ± 0.001	0.002 ± 0.020	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.401 ± 0.019	0.633 ± 0.018	0.059	-0.001	-1.26	0.0385	0.0385	0.967	93.788 ± 0.764	0.144 ± 0.001	37.575 ± 1.533	
UPI	beta58	0.06	0.061	0.001 ± 0.001	0.012 ± 0.018	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.332 ± 0.015	0.576 ± 0.016	0.059	-0.001	-1.79	0.0325	0.0325	0.954	93.186 ± 0.798	0.129 ± 0.001	45.090 ± 1.575	

Note. MCSE = Monte Carlo Standard Error shown as \u000b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S186

Simulation 2 - Condition 42: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.8

Method	Param.	True	Mean-based										Median-based					Inference		
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LSAM	beta41	0.20	0.200	0.000 $\pm$ 0.001	0.000 $\pm$ 0.005	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.022 $\pm$ 0.001	0.150 $\pm$ 0.004	0.200	0.000	0.10	0.0290	0.0290	0.998	95.000 $\pm$ 0.689	0.117 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta42	0.20	0.200	-0.000 $\pm$ 0.001	-0.001 $\pm$ 0.005	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.023 $\pm$ 0.001	0.150 $\pm$ 0.004	0.200	0.000	0.17	0.0299	0.0299	1.011	95.400 $\pm$ 0.662	0.119 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta43	0.20	0.200	-0.000 $\pm$ 0.001	-0.000 $\pm$ 0.005	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.021 $\pm$ 0.001	0.143 $\pm$ 0.004	0.200	0.000	-0.04	0.0287	0.0287	0.995	94.700 $\pm$ 0.708	0.112 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta44	0.11	0.110	0.000 $\pm$ 0.001	0.001 $\pm$ 0.009	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	0.086 $\pm$ 0.004	0.292 $\pm$ 0.008	0.111	0.001	0.48	0.0315	0.0315	0.982	94.000 $\pm$ 0.751	0.124 $\pm$ 0.000	91.500 $\pm$ 0.882
LSAM	beta45	0.11	0.111	0.001 $\pm$ 0.001	0.010 $\pm$ 0.008	0.001 $\pm$ 0.000	0.028 $\pm$ 0.001	0.001 $\pm$ 0.000	0.028 $\pm$ 0.001	0.067 $\pm$ 0.003	0.259 $\pm$ 0.007	0.112	0.002	1.48	0.0290	0.0291	0.998	94.800 $\pm$ 0.702	0.111 $\pm$ 0.000	96.600 $\pm$ 0.573
LSAM	beta46	0.08	0.080	-0.000 $\pm$ 0.001	-0.003 $\pm$ 0.009	0.000 $\pm$ 0.000	0.022 $\pm$ 0.001	0.000 $\pm$ 0.000	0.022 $\pm$ 0.001	0.074 $\pm$ 0.003	0.273 $\pm$ 0.008	0.080	0.000	-0.51	0.0213	0.0213	1.021	95.600 $\pm$ 0.649	0.087 $\pm$ 0.000	96.900 $\pm$ 0.548
LSAM	beta47	0.08	0.080	0.000 $\pm$ 0.001	0.005 $\pm$ 0.009	0.001 $\pm$ 0.000	0.023 $\pm$ 0.001	0.001 $\pm$ 0.000	0.023 $\pm$ 0.001	0.080 $\pm$ 0.004	0.282 $\pm$ 0.008	0.080	0.000	-0.09	0.0212	0.0212	0.980	93.200 $\pm$ 0.796	0.087 $\pm$ 0.000	95.400 $\pm$ 0.662
LSAM	beta51	0.16	0.161	0.001 $\pm$ 0.001	0.009 $\pm$ 0.008	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.058 $\pm$ 0.002	0.240 $\pm$ 0.006	0.160	0.000	0.10	0.0383	0.0383	0.932	92.100 $\pm$ 0.853	0.140 $\pm$ 0.000	99.400 $\pm$ 0.244
LSAM	beta52	0.16	0.160	-0.000 $\pm$ 0.001	-0.001 $\pm$ 0.006	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.033 $\pm$ 0.002	0.183 $\pm$ 0.005	0.159	-0.001	-0.69	0.0278	0.0278	1.023	95.800 $\pm$ 0.634	0.117 $\pm$ 0.000	99.900 $\pm$ 0.100
LSAM	beta53	0.16	0.159	-0.001 $\pm$ 0.001	-0.005 $\pm$ 0.006	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.038 $\pm$ 0.002	0.194 $\pm$ 0.005	0.160	0.000	0.08	0.0314	0.0314	1.005	95.100 $\pm$ 0.683	0.122 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta54	0.16	0.160	-0.000 $\pm$ 0.001	-0.001 $\pm$ 0.006	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.038 $\pm$ 0.002	0.196 $\pm$ 0.005	0.160	0.000	-0.16	0.0320	0.0320	0.949	93.800 $\pm$ 0.763	0.116 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta55	0.08	0.082	0.002 $\pm$ 0.001	0.022 $\pm$ 0.014	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.194 $\pm$ 0.009	0.441 $\pm$ 0.012	0.082	0.002	2.18	0.0352	0.0352	0.958	93.300 $\pm$ 0.791	0.132 $\pm$ 0.001	68.000 $\pm$ 1.475
LSAM	beta56	0.08	0.079	-0.001 $\pm$ 0.001	-0.010 $\pm$ 0.014	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.183 $\pm$ 0.008	0.428 $\pm$ 0.012	0.079	-0.001	-1.05	0.0343	0.0344	0.940	92.700 $\pm$ 0.823	0.126 $\pm$ 0.001	67.600 $\pm$ 1.480
LSAM	beta57	0.06	0.060	-0.000 $\pm$ 0.001	-0.005 $\pm$ 0.012	0.001 $\pm$ 0.000	0.023 $\pm$ 0.001	0.001 $\pm$ 0.000	0.023 $\pm$ 0.001	0.144 $\pm$ 0.007	0.379 $\pm$ 0.011	0.060	0.000	-0.37	0.0215	0.0215	0.976	94.000 $\pm$ 0.751	0.087 $\pm$ 0.000	79.200 $\pm$ 1.283
LSAM	beta58	0.06	0.060	0.000 $\pm$ 0.001	0.004 $\pm$ 0.011	0.000 $\pm$ 0.000	0.020 $\pm$ 0.000	0.000 $\pm$ 0.000	0.020 $\pm$ 0.001	0.114 $\pm$ 0.006	0.338 $\pm$ 0.010	0.060	0.000	-0.82	0.0207	0.0207	0.992	94.200 $\pm$ 0.739	0.079 $\pm$ 0.000	87.700 $\pm$ 1.039
QML	beta41	0.20	0.196	-0.004 $\pm$ 0.001	-0.020 $\pm$ 0.005	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.023 $\pm$ 0.001	0.152 $\pm$ 0.004	0.196	-0.004	-1.93	0.0293	0.0295	1.004	94.600 $\pm$ 0.715	0.118 $\pm$ 0.001	99.900 $\pm$ 0.100
QML	beta42	0.20	0.198	-0.002 $\pm$ 0.001	-0.011 $\pm$ 0.005	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.023 $\pm$ 0.001	0.152 $\pm$ 0.004	0.198	-0.002	-0.92	0.0304	0.0305	1.008	95.100 $\pm$ 0.683	0.120 $\pm$ 0.000	100.000 $\pm$ 0.000
QML	beta43	0.20	0.195	-0.005 $\pm$ 0.001	-0.023 $\pm$ 0.005	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.021 $\pm$ 0.001	0.144 $\pm$ 0.004	0.196	-0.004	-2.02	0.0292	0.0294	0.992	94.700 $\pm$ 0.708	0.111 $\pm$ 0.000	100.000 $\pm$ 0.000
QML	beta44	0.11	0.107	-0.003 $\pm$ 0.001	-0.026 $\pm$ 0.010	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.092 $\pm$ 0.004	0.303 $\pm$ 0.008	0.107	-0.003	-2.41	0.0322	0.0324	0.985	94.200 $\pm$ 0.739	0.128 $\pm$ 0.001	87.600 $\pm$ 1.042
QML	beta45	0.11	0.111	0.001 $\pm$ 0.001	0.005 $\pm$ 0.008	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.070 $\pm$ 0.003	0.265 $\pm$ 0.007	0.110	0.000	0.25	0.0298	0.0298	1.000	94.900 $\pm$ 0.696	0.114 $\pm$ 0.000	96.200 $\pm$ 0.605
QML	beta46	0.08	0.080	-0.000 $\pm$ 0.001	-0.000 $\pm$ 0.009	0.001 $\pm$ 0.000	0.022 $\pm$ 0.001	0.001 $\pm$ 0.000	0.022 $\pm$ 0.001	0.079 $\pm$ 0.004	0.280 $\pm$ 0.008	0.079	-0.001	-0.88	0.0218	0.0218	1.012	94.700 $\pm$ 0.708	0.089 $\pm$ 0.000	96.500 $\pm$ 0.581
QML	beta47	0.08	0.080	-0.000 $\pm$ 0.001	-0.001 $\pm$ 0.009	0.001 $\pm$ 0.000	0.023 $\pm$ 0.001	0.001 $\pm$ 0.000	0.023 $\pm$ 0.001	0.081 $\pm$ 0.004	0.285 $\pm$ 0.008	0.079	-0.001	-1.05	0.0212	0.0212	0.986	93.000 $\pm$ 0.807	0.088 $\pm$ 0.000	96.100 $\pm$ 0.612
QML	beta51	0.16	0.166	0.006 $\pm$ 0.001	0.039 $\pm$ 0.008	0.001 $\pm$ 0.000	0.039 $\pm$ 0.001	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.060 $\pm$ 0.003	0.245 $\pm$ 0.007	0.166	0.006	3.82	0.0388	0.0392	0.947	93.200 $\pm$ 0.796	0.144 $\pm$ 0.001	99.600 $\pm$ 0.200
QML	beta52	0.16	0.142	-0.018 $\pm$ 0.001	-0.111 $\pm$ 0.006	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.051 $\pm$ 0.002	0.226 $\pm$ 0.005	0.141	-0.019	-11.64	0.0293	0.0347	1.028	91.900 $\pm$ 0.863	0.127 $\pm$ 0.000	99.500 $\pm$ 0.223
QML	beta53	0.16	0.139	-0.021 $\pm$ 0.001	-0.132 $\pm$ 0.006	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	0.001 $\pm$ 0.000	0.039 $\pm$ 0.001	0.058 $\pm$ 0.003	0.242 $\pm$ 0.006	0.139	-0.021	-13.10	0.0323	0.0385	0.996	89.500 $\pm$ 0.969	0.127 $\pm$ 0.000	98.700 $\pm$ 0.358
QML	beta54	0.16	0.160	-0.000 $\pm$ 0.001	-0.002 $\pm$ 0.006	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	0.039 $\pm$ 0.002	0.198 $\pm$ 0.005	0.160	0.000	-0.17	0.0326	0.0326	0.952	93.300 $\pm$ 0.791	0.118 $\pm$ 0.001	99.900 $\pm$ 0.100
QML	beta55	0.08	0.085	0.005 $\pm$ 0.001	0.064 $\pm$ 0.014	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.206 $\pm$ 0.010	0.454 $\pm$ 0.013	0.085	0.005	6.41	0.0361	0.0364	0.965	93.400 $\pm$ 0.785	0.136 $\pm$ 0.001	69.500 $\pm$ 1.456
QML	beta56	0.08	0.080	-0.000 $\pm$ 0.001	-0.001 $\pm$ 0.014	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.189 $\pm$ 0.008	0.435 $\pm$ 0.012	0.080	0.000	-0.25	0.0343	0.0343	0.946	93.100 $\pm$ 0.801	0.129 $\pm$ 0.001	67.900 $\pm$ 1.476
QML	beta57	0.06	0.057	-0.003 $\pm$ 0.001	-0.056 $\pm$ 0.012	0.001 $\pm$ 0.000	0.023 $\pm$ 0.001	0.001 $\pm$ 0.000	0.023 $\pm$ 0.001	0.150 $\pm$ 0.007	0.388 $\pm$ 0.011	0.057	-0.003	-5.36	0.0217	0.0220	0.978	93.400 $\pm$ 0.785	0.088 $\pm$ 0.000	73.400 $\pm$ 1.397
QML	beta58	0.06	0.059	-0.001 $\pm$ 0.001	-0.023 $\pm$ 0.011	0.000 $\pm$ 0.000	0.021 $\pm$ 0.001	0.000 $\pm$ 0.000	0.021 $\pm$ 0.001	0.119 $\pm$ 0.006	0.345 $\pm$ 0.010	0.058	-0.002	-3.69	0.0201	0.0202	0.993	93.700 $\pm$ 0.768	0.080 $\pm$ 0.000	85.200 $\pm$ 1.123
UPI	beta41	0.20	0.200	0.000 $\pm$ 0.001	0.000 $\pm$ 0.005	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.023 $\pm$ 0.001	0.150 $\pm$ 0.004	0.200	0.000	-0.20	0.0291	0.0291	0.956	93.800 $\pm$ 0.763	0.113 $\pm$ 0.000	100.000 $\pm$ 0.000
UPI	beta42	0.20	0.200	-0.000 $\pm$ 0.001	-0.001 $\pm$ 0.005	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.023 $\pm$ 0.001	0.151 $\pm$ 0.004	0.200	0.000	0.25	0.0299	0.0299	0.982	94.600 $\pm$ 0.715	0.116 $\pm$ 0.000	100.000 $\pm$ 0.000
UPI	beta43	0.20	0.200	-0.000 $\pm$ 0.001	-0.000 $\pm$ 0.005	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.021 $\pm$ 0.001	0.143 $\pm$ 0.004	0.200	0.000	-0.12	0.0289	0.0289	0.988	94.600 $\pm$ 0.715	0.111 $\pm$ 0.000	100.000 $\pm$ 0.0

Table S186

Simulation 2 - Condition 42: Full Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.8
(continued)

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.081	0.001 ± 0.001	0.007 ± 0.009	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	0.081 ± 0.004	0.285 ± 0.008	0.080	0.000	-0.30	0.0216	0.0216	0.992	93.600 ± 0.774	0.089 ± 0.000	95.000 ± 0.689
UPI	beta51	0.16	0.162	0.002 ± 0.001	0.009 ± 0.008	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.057 ± 0.002	0.239 ± 0.006	0.161	0.001	0.32	0.0383	0.0383	0.934	92.600 ± 0.828	0.140 ± 0.000	99.400 ± 0.244
UPI	beta52	0.16	0.160	0.000 ± 0.001	0.000 ± 0.006	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.033 ± 0.002	0.183 ± 0.005	0.159	-0.001	-0.62	0.0279	0.0279	1.009	95.500 ± 0.656	0.116 ± 0.000	99.900 ± 0.100
UPI	beta53	0.16	0.159	-0.001 ± 0.001	-0.004 ± 0.006	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.038 ± 0.002	0.195 ± 0.005	0.160	0.000	-0.12	0.0308	0.0308	1.003	94.900 ± 0.696	0.122 ± 0.000	100.000 ± 0.000
UPI	beta54	0.16	0.160	-0.000 ± 0.001	-0.001 ± 0.006	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.038 ± 0.002	0.196 ± 0.005	0.160	0.000	-0.26	0.0323	0.0323	0.940	93.200 ± 0.796	0.115 ± 0.000	100.000 ± 0.000
UPI	beta55	0.08	0.082	0.002 ± 0.001	0.022 ± 0.014	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.201 ± 0.009	0.448 ± 0.013	0.081	0.001	1.65	0.0350	0.0350	0.953	93.900 ± 0.757	0.134 ± 0.001	67.500 ± 1.481
UPI	beta56	0.08	0.079	-0.001 ± 0.001	-0.015 ± 0.013	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.182 ± 0.008	0.426 ± 0.012	0.080	0.000	-0.37	0.0335	0.0335	0.957	93.200 ± 0.796	0.128 ± 0.001	66.900 ± 1.488
UPI	beta57	0.06	0.060	-0.000 ± 0.001	-0.003 ± 0.012	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	0.147 ± 0.007	0.384 ± 0.011	0.059	-0.001	-1.42	0.0223	0.0223	0.976	93.900 ± 0.757	0.088 ± 0.000	78.700 ± 1.295
UPI	beta58	0.06	0.060	0.000 ± 0.001	0.007 ± 0.011	0.000 ± 0.000	0.021 ± 0.000	0.000 ± 0.000	0.021 ± 0.001	0.118 ± 0.006	0.343 ± 0.010	0.060	0.000	0.09	0.0202	0.0202	0.991	94.400 ± 0.727	0.080 ± 0.000	87.000 ± 1.063

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S187

Simulation 2 - Condition 43: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, $N = 400$, Reliability = 0.4

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LSAM	beta41	0.20	0.203	0.003 $\pm$ 0.006	0.017 $\pm$ 0.030	0.023 $\pm$ 0.001	0.150 $\pm$ 0.005	0.023 $\pm$ 0.001	0.150 $\pm$ 0.006	0.564 $\pm$ 0.037	0.751 $\pm$ 0.029	0.210	0.010	4.78	0.1539	0.1542	1.115	98.295 $\pm$ 0.510	0.657 $\pm$ 0.011	29.302 $\pm$ 1.792
LSAM	beta42	0.20	0.199	-0.001 $\pm$ 0.007	-0.003 $\pm$ 0.033	0.028 $\pm$ 0.002	0.167 $\pm$ 0.005	0.028 $\pm$ 0.002	0.167 $\pm$ 0.006	0.694 $\pm$ 0.045	0.833 $\pm$ 0.032	0.202	0.002	1.11	0.1581	0.1582	1.049	97.054 $\pm$ 0.666	0.686 $\pm$ 0.012	26.357 $\pm$ 1.735
LSAM	beta43	0.20	0.202	0.002 $\pm$ 0.003	0.011 $\pm$ 0.017	0.007 $\pm$ 0.000	0.086 $\pm$ 0.003	0.007 $\pm$ 0.000	0.086 $\pm$ 0.003	0.186 $\pm$ 0.011	0.431 $\pm$ 0.015	0.203	0.003	1.63	0.0825	0.0825	1.008	96.434 $\pm$ 0.730	0.341 $\pm$ 0.004	67.132 $\pm$ 1.850
LSAM	beta44	0.11	0.127	0.017 $\pm$ 0.005	0.157 $\pm$ 0.049	0.018 $\pm$ 0.001	0.136 $\pm$ 0.005	0.019 $\pm$ 0.001	0.137 $\pm$ 0.006	1.549 $\pm$ 0.111	1.245 $\pm$ 0.051	0.116	0.006	5.87	0.1204	0.1206	1.088	97.674 $\pm$ 0.593	0.580 $\pm$ 0.015	15.969 $\pm$ 1.442
LSAM	beta45	0.11	0.115	0.005 $\pm$ 0.005	0.045 $\pm$ 0.046	0.016 $\pm$ 0.001	0.127 $\pm$ 0.005	0.016 $\pm$ 0.001	0.127 $\pm$ 0.005	1.341 $\pm$ 0.099	1.158 $\pm$ 0.049	0.108	-0.002	-2.04	0.1139	0.1140	1.022	94.729 $\pm$ 0.880	0.510 $\pm$ 0.013	14.419 $\pm$ 1.383
LSAM	beta46	0.08	0.080	0.000 $\pm$ 0.003	0.002 $\pm$ 0.035	0.005 $\pm$ 0.000	0.072 $\pm$ 0.003	0.005 $\pm$ 0.000	0.072 $\pm$ 0.003	0.807 $\pm$ 0.058	0.899 $\pm$ 0.037	0.078	-0.002	-2.07	0.0684	0.0684	1.031	95.349 $\pm$ 0.829	0.291 $\pm$ 0.007	24.031 $\pm$ 1.682
LSAM	beta47	0.08	0.081	0.001 $\pm$ 0.003	0.018 $\pm$ 0.037	0.006 $\pm$ 0.000	0.075 $\pm$ 0.003	0.006 $\pm$ 0.000	0.075 $\pm$ 0.003	0.884 $\pm$ 0.068	0.940 $\pm$ 0.041	0.078	-0.002	-2.28	0.0683	0.0683	0.971	96.899 $\pm$ 0.683	0.286 $\pm$ 0.007	29.767 $\pm$ 1.800
LSAM	beta51	0.16	0.149	-0.011 $\pm$ 0.004	-0.068 $\pm$ 0.027	0.012 $\pm$ 0.001	0.111 $\pm$ 0.003	0.012 $\pm$ 0.001	0.111 $\pm$ 0.004	0.481 $\pm$ 0.030	0.694 $\pm$ 0.025	0.154	-0.006	-3.47	0.1012	0.1013	0.954	95.349 $\pm$ 0.829	0.413 $\pm$ 0.003	31.318 $\pm$ 1.826
LSAM	beta52	0.16	0.161	0.001 $\pm$ 0.006	0.004 $\pm$ 0.038	0.024 $\pm$ 0.001	0.156 $\pm$ 0.005	0.024 $\pm$ 0.001	0.156 $\pm$ 0.006	0.954 $\pm$ 0.058	0.977 $\pm$ 0.036	0.163	0.003	2.13	0.1523	0.1523	1.084	98.760 $\pm$ 0.436	0.665 $\pm$ 0.011	18.140 $\pm$ 1.517
LSAM	beta53	0.16	0.161	0.001 $\pm$ 0.005	0.004 $\pm$ 0.029	0.014 $\pm$ 0.001	0.119 $\pm$ 0.004	0.014 $\pm$ 0.001	0.119 $\pm$ 0.004	0.552 $\pm$ 0.034	0.743 $\pm$ 0.027	0.161	0.001	0.55	0.1211	0.1211	1.027	96.279 $\pm$ 0.745	0.479 $\pm$ 0.005	26.977 $\pm$ 1.748
LSAM	beta54	0.16	0.173	0.013 $\pm$ 0.006	0.083 $\pm$ 0.037	0.023 $\pm$ 0.001	0.151 $\pm$ 0.005	0.023 $\pm$ 0.001	0.151 $\pm$ 0.006	0.891 $\pm$ 0.056	0.944 $\pm$ 0.035	0.172	0.012	7.52	0.1455	0.1460	1.036	96.899 $\pm$ 0.683	0.611 $\pm$ 0.009	22.171 $\pm$ 1.636
LSAM	beta55	0.08	0.076	-0.004 $\pm$ 0.005	-0.044 $\pm$ 0.060	0.015 $\pm$ 0.001	0.122 $\pm$ 0.005	0.015 $\pm$ 0.001	0.122 $\pm$ 0.005	2.313 $\pm$ 0.181	1.521 $\pm$ 0.067	0.076	-0.004	-4.83	0.1133	0.1134	1.015	97.054 $\pm$ 0.666	0.484 $\pm$ 0.011	13.488 $\pm$ 1.345
LSAM	beta56	0.08	0.086	0.006 $\pm$ 0.003	0.078 $\pm$ 0.035	0.005 $\pm$ 0.000	0.070 $\pm$ 0.003	0.005 $\pm$ 0.000	0.070 $\pm$ 0.003	0.773 $\pm$ 0.055	0.879 $\pm$ 0.036	0.084	0.004	4.56	0.0630	0.0631	0.991	94.884 $\pm$ 0.868	0.272 $\pm$ 0.006	30.698 $\pm$ 1.816
LSAM	beta57	0.06	0.068	0.008 $\pm$ 0.005	0.129 $\pm$ 0.076	0.013 $\pm$ 0.001	0.116 $\pm$ 0.004	0.013 $\pm$ 0.001	0.116 $\pm$ 0.005	3.745 $\pm$ 0.275	1.935 $\pm$ 0.081	0.066	0.006	10.13	0.1115	0.1117	1.055	98.295 $\pm$ 0.510	0.480 $\pm$ 0.011	5.891 $\pm$ 0.927
LSAM	beta58	0.06	0.060	-0.000 $\pm$ 0.002	-0.008 $\pm$ 0.037	0.003 $\pm$ 0.000	0.056 $\pm$ 0.002	0.003 $\pm$ 0.000	0.056 $\pm$ 0.002	0.867 $\pm$ 0.059	0.931 $\pm$ 0.037	0.058	-0.002	-3.34	0.0530	0.0530	1.041	94.729 $\pm$ 0.880	0.228 $\pm$ 0.005	23.101 $\pm$ 1.660
QML	beta41	0.20	0.235	0.035 $\pm$ 0.003	0.174 $\pm$ 0.017	0.008 $\pm$ 0.000	0.088 $\pm$ 0.003	0.009 $\pm$ 0.000	0.094 $\pm$ 0.003	0.222 $\pm$ 0.012	0.471 $\pm$ 0.015	0.241	0.041	20.26	0.0918	0.1003	1.053	94.729 $\pm$ 0.880	0.362 $\pm$ 0.003	74.419 $\pm$ 1.718
QML	beta42	0.20	0.231	0.031 $\pm$ 0.004	0.157 $\pm$ 0.018	0.008 $\pm$ 0.001	0.091 $\pm$ 0.003	0.009 $\pm$ 0.001	0.096 $\pm$ 0.003	0.230 $\pm$ 0.014	0.480 $\pm$ 0.017	0.230	0.030	14.94	0.0915	0.0962	1.033	94.419 $\pm$ 0.904	0.367 $\pm$ 0.003	71.473 $\pm$ 1.778
QML	beta43	0.20	0.190	-0.010 $\pm$ 0.003	-0.052 $\pm$ 0.015	0.006 $\pm$ 0.000	0.077 $\pm$ 0.002	0.006 $\pm$ 0.000	0.078 $\pm$ 0.003	0.152 $\pm$ 0.009	0.390 $\pm$ 0.014	0.190	-0.010	-5.21	0.0752	0.0759	0.969	93.953 $\pm$ 0.938	0.294 $\pm$ 0.002	72.713 $\pm$ 1.754
QML	beta44	0.11	0.037	-0.073 $\pm$ 0.005	-0.662 $\pm$ 0.043	0.015 $\pm$ 0.001	0.121 $\pm$ 0.004	0.020 $\pm$ 0.001	0.141 $\pm$ 0.005	1.647 $\pm$ 0.107	1.283 $\pm$ 0.046	0.041	-0.069	-63.04	0.1189	0.1376	0.972	91.163 $\pm$ 1.118	0.461 $\pm$ 0.007	9.457 $\pm$ 1.152
QML	beta45	0.11	0.090	-0.020 $\pm$ 0.004	-0.182 $\pm$ 0.035	0.010 $\pm$ 0.001	0.099 $\pm$ 0.003	0.010 $\pm$ 0.001	0.100 $\pm$ 0.004	0.835 $\pm$ 0.052	0.914 $\pm$ 0.033	0.089	-0.021	-19.22	0.0972	0.0995	0.960	92.403 $\pm$ 1.043	0.371 $\pm$ 0.005	16.899 $\pm$ 1.476
QML	beta46	0.08	0.162	0.082 $\pm$ 0.003	1.025 $\pm$ 0.037	0.006 $\pm$ 0.000	0.075 $\pm$ 0.003	0.012 $\pm$ 0.001	0.111 $\pm$ 0.003	1.935 $\pm$ 0.116	1.391 $\pm$ 0.044	0.156	0.076	95.54	0.0724	0.1053	1.011	89.612 $\pm$ 1.201	0.298 $\pm$ 0.006	64.961 $\pm$ 1.879
QML	beta47	0.08	0.154	0.074 $\pm$ 0.003	0.929 $\pm$ 0.033	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.010 $\pm$ 0.001	0.101 $\pm$ 0.003	1.582 $\pm$ 0.090	1.258 $\pm$ 0.038	0.148	0.068	85.00	0.0654	0.0943	1.029	89.922 $\pm$ 1.185	0.274 $\pm$ 0.005	68.372 $\pm$ 1.831
QML	beta51	0.16	0.169	0.009 $\pm$ 0.005	0.057 $\pm$ 0.031	0.016 $\pm$ 0.001	0.125 $\pm$ 0.004	0.016 $\pm$ 0.001	0.125 $\pm$ 0.005	0.610 $\pm$ 0.039	0.781 $\pm$ 0.029	0.165	0.005	3.20	0.1217	0.1218	0.978	94.884 $\pm$ 0.868	0.478 $\pm$ 0.005	29.922 $\pm$ 1.803
QML	beta52	0.16	0.165	0.005 $\pm$ 0.005	0.034 $\pm$ 0.031	0.016 $\pm$ 0.001	0.128 $\pm$ 0.004	0.016 $\pm$ 0.001	0.128 $\pm$ 0.005	0.640 $\pm$ 0.038	0.800 $\pm$ 0.029	0.159	-0.001	-0.33	0.1323	0.1323	0.927	94.574 $\pm$ 0.892	0.465 $\pm$ 0.006	32.403 $\pm$ 1.843
QML	beta53	0.16	0.113	-0.047 $\pm$ 0.004	-0.292 $\pm$ 0.026	0.011 $\pm$ 0.001	0.104 $\pm$ 0.003	0.013 $\pm$ 0.001	0.114 $\pm$ 0.004	0.510 $\pm$ 0.029	0.714 $\pm$ 0.024	0.114	-0.046	-28.87	0.1038	0.1136	0.997	91.163 $\pm$ 1.118	0.408 $\pm$ 0.004	19.225 $\pm$ 1.552
QML	beta54	0.16	0.184	0.024 $\pm$ 0.004	0.151 $\pm$ 0.023	0.009 $\pm$ 0.001	0.095 $\pm$ 0.003	0.010 $\pm$ 0.001	0.098 $\pm$ 0.003	0.377 $\pm$ 0.021	0.614 $\pm$ 0.021	0.183	0.023	14.33	0.0972	0.0998	0.977	94.884 $\pm$ 0.868	0.365 $\pm$ 0.003	53.488 $\pm$ 1.964
QML	beta55	0.08	0.070	-0.010 $\pm$ 0.006	-0.119 $\pm$ 0.070	0.020 $\pm$ 0.001	0.142 $\pm$ 0.005	0.020 $\pm$ 0.001	0.142 $\pm$ 0.005	3.157 $\pm$ 0.205	1.777 $\pm$ 0.068	0.063	-0.017	-21.47	0.1307	0.1318	0.858	86.822 $\pm$ 1.332	0.477 $\pm$ 0.011	18.915 $\pm$ 1.542
QML	beta56	0.08	0.121	0.041 $\pm$ 0.003	0.511 $\pm$ 0.040	0.006 $\pm$ 0.000	0.080 $\pm$ 0.003	0.008 $\pm$ 0.000	0.090 $\pm$ 0.003	1.266 $\pm$ 0.076	1.125 $\pm$ 0.038	0.123	0.043	53.84	0.0758	0.0872	0.907	88.837 $\pm$ 1.240	0.285 $\pm$ 0.005	48.682 $\pm$ 1.968
QML	beta57	0.06	0.139	0.079 $\pm$ 0.005	1.321 $\pm$ 0.084	0.017 $\pm$ 0.001	0.129 $\pm$ 0.004	0.023 $\pm$ 0.001	0.151 $\pm$ 0.005	6.342 $\pm$ 0.395	2.518 $\pm$ 0.087	0.138	0.078	130.27	0.1233	0.1460	0.905	88.372 $\pm$ 1.262	0.457 $\pm$ 0.010	25.891 $\pm$ 1.725
QML	beta58	0.06	0.112	0.052 $\pm$ 0.002	0.863 $\pm$ 0.038	0.003 $\pm$ 0.000	0.058 $\pm$ 0.002	0.006 $\pm$ 0.000	0.078 $\pm$ 0.003	1.693 $\pm$ 0.111	1.301 $\pm$ 0.045	0.104	0.044	72.90	0.0524	0.0683	0.997	91.163 $\pm$ 1.118	0.229 $\pm$ 0.005	53.953 $\pm$ 1.963
UPI	beta41	0.20	0.223	0.023 $\pm$ 0.007	0.114 $\pm$ 0.037	0.036 $\pm$ 0.003	0.190 $\pm$ 0.008	0.036 $\pm$ 0.003	0.191 $\pm$ 0.009	0.912 $\pm$ 0.074	0.955 $\pm$ 0.043	0.226	0.026	13.04	0.1950	0.1967	1.412	98.915 $\pm$ 0.408	1.051 $\pm$ 0.045	22.016 $\pm$ 1.632
UPI	beta42	0.20	0.209	0.009 $\pm$ 0.007	0.044 $\pm$ 0.035	0.031 $\pm$ 0.002	0.177 $\pm$ 0.007	0.031 $\pm$ 0.002	0.177 $\pm$ 0.007	0.782 $\pm$ 0.058	0.884 $\pm$ 0.037	0.200	0.000	0.12	0.1888	0.1888	1.453	99.070 $\pm$ 0.378	1.007 $\pm$ 0.039	18.140 $\pm$ 1.517
UPI	beta43	0.20	0.203	0.003 $\pm$ 0.004	0.015 $\pm$ 0.018	0.008 $\pm$ 0.001	0.090 $\pm$ 0.003	0.008 $\pm$ 0.001	0.090 $\pm$ 0.003	0.204 $\pm$ 0.013	0.451 $\pm$ 0.017	0.208	0.008	3.79	0.0989	0.0992	1.242	97.054 $\pm$ 0.666	0.440 $\pm$ 0.012	54.419 $\pm$

Table S187

Simulation 2 - Condition 43: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.4

(continued)

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
UPI	beta47	0.08	0.078	-0.002 $\pm$ 0.003	-0.022 $\pm$ 0.040	0.007 $\pm$ 0.001	0.081 $\pm$ 0.003	0.007 $\pm$ 0.001	0.081 $\pm$ 0.004	1.034 $\pm$ 0.082	1.017 $\pm$ 0.045	0.078	-0.002	-2.01	0.0763	0.0763	1.342	97.674 $\pm$ 0.593	0.428 $\pm$ 0.017	20.620 $\pm$ 1.593
UPI	beta51	0.16	0.157	-0.003 $\pm$ 0.004	-0.021 $\pm$ 0.028	0.013 $\pm$ 0.001	0.112 $\pm$ 0.004	0.013 $\pm$ 0.001	0.112 $\pm$ 0.004	0.489 $\pm$ 0.031	0.700 $\pm$ 0.026	0.158	-0.002	-1.28	0.1071	0.1071	1.033	95.969 $\pm$ 0.774	0.453 $\pm$ 0.007	29.147 $\pm$ 1.789
UPI	beta52	0.16	0.181	0.021 $\pm$ 0.007	0.131 $\pm$ 0.046	0.035 $\pm$ 0.003	0.187 $\pm$ 0.007	0.035 $\pm$ 0.003	0.188 $\pm$ 0.008	1.380 $\pm$ 0.100	1.175 $\pm$ 0.048	0.182	0.022	13.48	0.1777	0.1790	1.216	99.070 $\pm$ 0.378	0.891 $\pm$ 0.026	16.124 $\pm$ 1.448
UPI	beta53	0.16	0.158	-0.002 $\pm$ 0.005	-0.011 $\pm$ 0.031	0.016 $\pm$ 0.001	0.125 $\pm$ 0.004	0.016 $\pm$ 0.001	0.125 $\pm$ 0.005	0.612 $\pm$ 0.039	0.782 $\pm$ 0.029	0.159	-0.001	-0.90	0.1338	0.1339	1.203	98.450 $\pm$ 0.486	0.590 $\pm$ 0.012	20.465 $\pm$ 1.589
UPI	beta54	0.16	0.180	0.020 $\pm$ 0.007	0.126 $\pm$ 0.043	0.031 $\pm$ 0.002	0.176 $\pm$ 0.006	0.031 $\pm$ 0.002	0.177 $\pm$ 0.007	1.224 $\pm$ 0.088	1.107 $\pm$ 0.045	0.173	0.013	8.06	0.1722	0.1727	1.210	98.295 $\pm$ 0.510	0.835 $\pm$ 0.020	15.504 $\pm$ 1.425
UPI	beta55	0.08	0.081	0.001 $\pm$ 0.005	0.009 $\pm$ 0.067	0.019 $\pm$ 0.001	0.136 $\pm$ 0.005	0.019 $\pm$ 0.001	0.136 $\pm$ 0.006	2.904 $\pm$ 0.228	1.704 $\pm$ 0.075	0.079	-0.001	-1.69	0.1239	0.1239	1.214	97.364 $\pm$ 0.631	0.649 $\pm$ 0.023	9.147 $\pm$ 1.135
UPI	beta56	0.08	0.084	0.004 $\pm$ 0.003	0.044 $\pm$ 0.037	0.006 $\pm$ 0.000	0.075 $\pm$ 0.003	0.006 $\pm$ 0.000	0.075 $\pm$ 0.003	0.887 $\pm$ 0.068	0.942 $\pm$ 0.041	0.082	0.002	2.97	0.0730	0.0730	1.178	96.434 $\pm$ 0.730	0.348 $\pm$ 0.012	22.016 $\pm$ 1.632
UPI	beta57	0.06	0.058	-0.002 $\pm$ 0.005	-0.038 $\pm$ 0.086	0.017 $\pm$ 0.001	0.131 $\pm$ 0.005	0.017 $\pm$ 0.001	0.131 $\pm$ 0.006	4.742 $\pm$ 0.376	2.178 $\pm$ 0.097	0.056	-0.004	-7.19	0.1167	0.1168	1.283	98.915 $\pm$ 0.408	0.658 $\pm$ 0.027	3.411 $\pm$ 0.715
UPI	beta58	0.06	0.057	-0.003 $\pm$ 0.003	-0.050 $\pm$ 0.043	0.004 $\pm$ 0.000	0.065 $\pm$ 0.002	0.004 $\pm$ 0.000	0.065 $\pm$ 0.003	1.170 $\pm$ 0.084	1.082 $\pm$ 0.044	0.055	-0.005	-9.16	0.0584	0.0587	1.246	96.899 $\pm$ 0.683	0.317 $\pm$ 0.010	15.349 $\pm$ 1.419

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S188

Simulation 2 - Condition 44: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, $N = 1000$, Reliability = 0.4

Method	Param.	True	Mean-based										Median-based					Inference		
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LSAM	beta41	0.20	0.199	-0.001 $\pm$ 0.003	-0.004 $\pm$ 0.017	0.011 $\pm$ 0.001	0.105 $\pm$ 0.003	0.011 $\pm$ 0.001	0.105 $\pm$ 0.003	0.275 $\pm$ 0.015	0.524 $\pm$ 0.016	0.197	-0.003	-1.46	0.1013	0.1013	0.939	94.105 $\pm$ 0.771	0.386 $\pm$ 0.003	55.734 $\pm$ 1.626
LSAM	beta42	0.20	0.200	0.000 $\pm$ 0.003	0.000 $\pm$ 0.017	0.011 $\pm$ 0.001	0.106 $\pm$ 0.003	0.011 $\pm$ 0.001	0.106 $\pm$ 0.003	0.278 $\pm$ 0.015	0.528 $\pm$ 0.016	0.198	-0.002	-0.86	0.0990	0.0990	0.994	95.820 $\pm$ 0.655	0.411 $\pm$ 0.004	54.984 $\pm$ 1.629
LSAM	beta43	0.20	0.203	0.003 $\pm$ 0.002	0.014 $\pm$ 0.008	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.003 $\pm$ 0.000	0.052 $\pm$ 0.002	0.067 $\pm$ 0.003	0.259 $\pm$ 0.008	0.202	0.002	1.20	0.0512	0.0513	1.027	95.284 $\pm$ 0.694	0.208 $\pm$ 0.001	96.677 $\pm$ 0.587
LSAM	beta44	0.11	0.118	0.008 $\pm$ 0.003	0.076 $\pm$ 0.023	0.006 $\pm$ 0.000	0.078 $\pm$ 0.002	0.006 $\pm$ 0.000	0.078 $\pm$ 0.002	0.507 $\pm$ 0.027	0.712 $\pm$ 0.022	0.117	0.007	6.60	0.0723	0.0726	1.001	94.427 $\pm$ 0.751	0.306 $\pm$ 0.004	34.405 $\pm$ 1.555
LSAM	beta45	0.11	0.118	0.008 $\pm$ 0.002	0.074 $\pm$ 0.022	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.006 $\pm$ 0.000	0.076 $\pm$ 0.003	0.474 $\pm$ 0.028	0.688 $\pm$ 0.023	0.114	0.004	3.41	0.0689	0.0690	0.961	95.713 $\pm$ 0.663	0.284 $\pm$ 0.003	41.372 $\pm$ 1.612
LSAM	beta46	0.08	0.082	0.002 $\pm$ 0.001	0.025 $\pm$ 0.017	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.273 $\pm$ 0.016	0.523 $\pm$ 0.018	0.080	0.000	0.53	0.0376	0.0376	0.925	92.819 $\pm$ 0.845	0.151 $\pm$ 0.002	60.236 $\pm$ 1.602
LSAM	beta47	0.08	0.083	0.003 $\pm$ 0.001	0.035 $\pm$ 0.017	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.273 $\pm$ 0.016	0.523 $\pm$ 0.017	0.082	0.002	2.33	0.0354	0.0354	0.937	94.427 $\pm$ 0.751	0.153 $\pm$ 0.002	64.094 $\pm$ 1.571
LSAM	beta51	0.16	0.157	-0.003 $\pm$ 0.002	-0.021 $\pm$ 0.014	0.004 $\pm$ 0.000	0.067 $\pm$ 0.002	0.004 $\pm$ 0.000	0.067 $\pm$ 0.002	0.174 $\pm$ 0.009	0.418 $\pm$ 0.012	0.158	-0.002	-1.47	0.0638	0.0639	0.968	94.427 $\pm$ 0.751	0.253 $\pm$ 0.001	69.453 $\pm$ 1.508
LSAM	beta52	0.16	0.163	0.003 $\pm$ 0.003	0.021 $\pm$ 0.021	0.011 $\pm$ 0.001	0.103 $\pm$ 0.002	0.011 $\pm$ 0.001	0.103 $\pm$ 0.003	0.413 $\pm$ 0.020	0.642 $\pm$ 0.019	0.159	-0.001	-0.47	0.1005	0.1005	1.010	96.463 $\pm$ 0.605	0.407 $\pm$ 0.003	39.228 $\pm$ 1.598
LSAM	beta53	0.16	0.162	0.002 $\pm$ 0.002	0.014 $\pm$ 0.015	0.006 $\pm$ 0.000	0.076 $\pm$ 0.002	0.006 $\pm$ 0.000	0.076 $\pm$ 0.002	0.223 $\pm$ 0.010	0.472 $\pm$ 0.013	0.161	0.001	0.33	0.0739	0.0739	0.991	95.498 $\pm$ 0.679	0.294 $\pm$ 0.002	59.378 $\pm$ 1.608
LSAM	beta54	0.16	0.164	0.004 $\pm$ 0.003	0.026 $\pm$ 0.019	0.009 $\pm$ 0.000	0.092 $\pm$ 0.002	0.009 $\pm$ 0.000	0.092 $\pm$ 0.003	0.334 $\pm$ 0.018	0.578 $\pm$ 0.018	0.163	0.003	1.60	0.0926	0.0927	1.047	97.213 $\pm$ 0.539	0.379 $\pm$ 0.003	40.943 $\pm$ 1.610
LSAM	beta55	0.08	0.082	0.002 $\pm$ 0.002	0.022 $\pm$ 0.029	0.005 $\pm$ 0.000	0.071 $\pm$ 0.002	0.005 $\pm$ 0.000	0.071 $\pm$ 0.002	0.783 $\pm$ 0.045	0.885 $\pm$ 0.029	0.080	0.000	-0.29	0.0658	0.0658	0.930	95.498 $\pm$ 0.679	0.258 $\pm$ 0.003	33.226 $\pm$ 1.542
LSAM	beta56	0.08	0.083	0.003 $\pm$ 0.001	0.036 $\pm$ 0.016	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.001 $\pm$ 0.000	0.039 $\pm$ 0.001	0.232 $\pm$ 0.013	0.482 $\pm$ 0.016	0.082	0.002	2.98	0.0347	0.0347	0.937	93.355 $\pm$ 0.815	0.141 $\pm$ 0.002	66.881 $\pm$ 1.541
LSAM	beta57	0.06	0.062	0.002 $\pm$ 0.002	0.036 $\pm$ 0.039	0.005 $\pm$ 0.000	0.072 $\pm$ 0.002	0.005 $\pm$ 0.000	0.072 $\pm$ 0.002	1.444 $\pm$ 0.079	1.202 $\pm$ 0.038	0.060	0.000	-0.32	0.0633	0.0633	0.955	95.391 $\pm$ 0.686	0.270 $\pm$ 0.003	13.826 $\pm$ 1.130
LSAM	beta58	0.06	0.062	0.002 $\pm$ 0.001	0.027 $\pm$ 0.018	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.313 $\pm$ 0.017	0.560 $\pm$ 0.018	0.061	0.001	1.01	0.0302	0.0302	1.010	94.855 $\pm$ 0.723	0.133 $\pm$ 0.002	51.125 $\pm$ 1.637
QML	beta41	0.20	0.235	0.035 $\pm$ 0.002	0.173 $\pm$ 0.010	0.003 $\pm$ 0.000	0.058 $\pm$ 0.001	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.114 $\pm$ 0.005	0.338 $\pm$ 0.009	0.235	0.035	17.61	0.0567	0.0667	0.975	89.711 $\pm$ 0.995	0.222 $\pm$ 0.002	97.749 $\pm$ 0.486
QML	beta42	0.20	0.230	0.030 $\pm$ 0.002	0.151 $\pm$ 0.009	0.003 $\pm$ 0.000	0.057 $\pm$ 0.001	0.004 $\pm$ 0.000	0.064 $\pm$ 0.002	0.103 $\pm$ 0.005	0.320 $\pm$ 0.008	0.230	0.030	15.23	0.0539	0.0619	1.024	92.605 $\pm$ 0.857	0.227 $\pm$ 0.001	96.249 $\pm$ 0.622
QML	beta43	0.20	0.186	-0.014 $\pm$ 0.001	-0.068 $\pm$ 0.007	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.056 $\pm$ 0.003	0.237 $\pm$ 0.007	0.185	-0.015	-7.36	0.0416	0.0442	1.026	93.355 $\pm$ 0.815	0.183 $\pm$ 0.001	97.749 $\pm$ 0.486
QML	beta44	0.11	0.028	-0.082 $\pm$ 0.002	-0.741 $\pm$ 0.021	0.005 $\pm$ 0.000	0.071 $\pm$ 0.002	0.012 $\pm$ 0.000	0.108 $\pm$ 0.002	0.966 $\pm$ 0.041	0.983 $\pm$ 0.022	0.032	-0.078	-70.87	0.0680	0.1034	0.994	78.135 $\pm$ 1.353	0.277 $\pm$ 0.003	9.539 $\pm$ 0.962
QML	beta45	0.11	0.097	-0.013 $\pm$ 0.002	-0.116 $\pm$ 0.018	0.004 $\pm$ 0.000	0.061 $\pm$ 0.001	0.004 $\pm$ 0.000	0.063 $\pm$ 0.002	0.325 $\pm$ 0.016	0.570 $\pm$ 0.017	0.096	-0.014	-12.66	0.0583	0.0599	0.952	92.176 $\pm$ 0.879	0.229 $\pm$ 0.002	40.836 $\pm$ 1.609
QML	beta46	0.08	0.158	0.078 $\pm$ 0.001	0.979 $\pm$ 0.018	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.008 $\pm$ 0.000	0.089 $\pm$ 0.002	1.246 $\pm$ 0.042	1.116 $\pm$ 0.019	0.155	0.075	93.61	0.0424	0.0861	1.018	59.057 $\pm$ 1.610	0.171 $\pm$ 0.002	98.071 $\pm$ 0.450
QML	beta47	0.08	0.158	0.078 $\pm$ 0.001	0.973 $\pm$ 0.017	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.008 $\pm$ 0.000	0.089 $\pm$ 0.001	1.229 $\pm$ 0.040	1.108 $\pm$ 0.019	0.155	0.075	93.52	0.0412	0.0854	0.976	53.162 $\pm$ 1.634	0.162 $\pm$ 0.002	98.499 $\pm$ 0.398
QML	beta51	0.16	0.173	0.013 $\pm$ 0.002	0.084 $\pm$ 0.015	0.006 $\pm$ 0.000	0.076 $\pm$ 0.002	0.006 $\pm$ 0.000	0.077 $\pm$ 0.002	0.231 $\pm$ 0.011	0.481 $\pm$ 0.014	0.173	0.013	8.25	0.0753	0.0765	0.980	93.891 $\pm$ 0.784	0.291 $\pm$ 0.003	64.094 $\pm$ 1.571
QML	beta52	0.16	0.167	0.007 $\pm$ 0.002	0.046 $\pm$ 0.015	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.006 $\pm$ 0.000	0.076 $\pm$ 0.002	0.223 $\pm$ 0.012	0.472 $\pm$ 0.015	0.165	0.005	2.84	0.0731	0.0733	0.996	95.606 $\pm$ 0.671	0.294 $\pm$ 0.004	63.880 $\pm$ 1.573
QML	beta53	0.16	0.107	-0.053 $\pm$ 0.002	-0.329 $\pm$ 0.013	0.004 $\pm$ 0.000	0.063 $\pm$ 0.001	0.007 $\pm$ 0.000	0.082 $\pm$ 0.002	0.263 $\pm$ 0.010	0.513 $\pm$ 0.011	0.107	-0.053	-33.02	0.0630	0.0822	0.982	83.816 $\pm$ 1.206	0.243 $\pm$ 0.001	41.908 $\pm$ 1.615
QML	beta54	0.16	0.179	0.019 $\pm$ 0.002	0.121 $\pm$ 0.011	0.003 $\pm$ 0.000	0.054 $\pm$ 0.001	0.003 $\pm$ 0.000	0.057 $\pm$ 0.002	0.129 $\pm$ 0.007	0.359 $\pm$ 0.011	0.178	0.018	11.27	0.0558	0.0587	1.051	94.641 $\pm$ 0.737	0.223 $\pm$ 0.001	89.496 $\pm$ 1.004
QML	beta55	0.08	0.066	-0.014 $\pm$ 0.003	-0.180 $\pm$ 0.038	0.009 $\pm$ 0.000	0.093 $\pm$ 0.002	0.009 $\pm$ 0.000	0.094 $\pm$ 0.003	1.388 $\pm$ 0.069	1.178 $\pm$ 0.035	0.065	-0.015	-18.93	0.0851	0.0865	0.888	86.066 $\pm$ 1.134	0.324 $\pm$ 0.007	22.079 $\pm$ 1.358
QML	beta56	0.08	0.125	0.045 $\pm$ 0.002	0.557 $\pm$ 0.019	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.004 $\pm$ 0.000	0.064 $\pm$ 0.002	0.650 $\pm$ 0.030	0.806 $\pm$ 0.020	0.125	0.045	56.82	0.0445	0.0636	0.927	80.600 $\pm$ 1.295	0.170 $\pm$ 0.002	81.886 $\pm$ 1.261
QML	beta57	0.06	0.147	0.087 $\pm$ 0.003	1.451 $\pm$ 0.047	0.007 $\pm$ 0.000	0.086 $\pm$ 0.002	0.015 $\pm$ 0.001	0.123 $\pm$ 0.003	4.179 $\pm$ 0.174	2.044 $\pm$ 0.046	0.146	0.086	142.75	0.0864	0.1216	0.923	74.384 $\pm$ 1.429	0.313 $\pm$ 0.006	52.626 $\pm$ 1.635
QML	beta58	0.06	0.114	0.054 $\pm$ 0.001	0.892 $\pm$ 0.018	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.004 $\pm$ 0.000	0.063 $\pm$ 0.001	1.105 $\pm$ 0.040	1.051 $\pm$ 0.020	0.112	0.052	87.35	0.0319	0.0614	1.025	69.668 $\pm$ 1.505	0.134 $\pm$ 0.001	95.284 $\pm$ 0.694
UPI	beta41	0.20	0.209	0.009 $\pm$ 0.003	0.045 $\pm$ 0.016	0.009 $\pm$ 0.001	0.096 $\pm$ 0.003	0.009 $\pm$ 0.001	0.096 $\pm$ 0.003	0.231 $\pm$ 0.014	0.481 $\pm$ 0.016	0.206	0.006	3.00	0.0941	0.0943	1.090	97.320 $\pm$ 0.529	0.409 $\pm$ 0.005	58.842 $\pm$ 1.611
UPI	beta42	0.20	0.205	0.005 $\pm$ 0.003	0.023 $\pm$ 0.016	0.010 $\pm$ 0.001	0.098 $\pm$ 0.003	0.010 $\pm$ 0.001	0.098 $\pm$ 0.003	0.239 $\pm$ 0.013	0.489 $\pm$ 0.016	0.205	0.005	2.31	0.0888	0.0889	1.131	97.320 $\pm$ 0.529	0.433 $\pm$ 0.007	55.841 $\pm$ 1.626
UPI	beta43	0.20	0.201	0.001 $\pm$ 0.002	0.007 $\pm$ 0.008	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.067 $\pm$ 0.003	0.260 $\pm$ 0.007	0.202	0.002	0.95	0.0526	0.0526	1.038	94.427 $\pm$ 0.751	0.211 $\pm$ 0.002	95.498 $\pm</$

Table S188

Simulation 2 - Condition 44: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.4
(continued)

Method	Param.	True	Mean-based										Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE	
UPI	beta47	0.08	0.081	0.001 ± 0.001	0.013 ± 0.017	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.266 ± 0.016	0.515 ± 0.018	0.079	-0.001	-1.49	0.0365	0.0365	1.052	94.962 ± 0.716	0.170 ± 0.003	57.235 ± 1.620	
UPI	beta51	0.16	0.163	0.003 ± 0.002	0.016 ± 0.013	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.165 ± 0.008	0.406 ± 0.012	0.163	0.003	2.04	0.0638	0.0639	0.977	94.212 ± 0.764	0.249 ± 0.001	72.347 ± 1.464	
UPI	beta52	0.16	0.178	0.018 ± 0.003	0.110 ± 0.020	0.009 ± 0.000	0.096 ± 0.002	0.010 ± 0.000	0.098 ± 0.003	0.372 ± 0.018	0.610 ± 0.018	0.176	0.016	10.21	0.0970	0.0983	1.072	96.999 ± 0.559	0.404 ± 0.004	45.016 ± 1.629	
UPI	beta53	0.16	0.158	-0.002 ± 0.002	-0.013 ± 0.015	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.210 ± 0.009	0.458 ± 0.013	0.158	-0.002	-1.32	0.0750	0.0750	1.018	96.034 ± 0.639	0.293 ± 0.002	58.842 ± 1.611	
UPI	beta54	0.16	0.163	0.003 ± 0.003	0.021 ± 0.019	0.008 ± 0.000	0.092 ± 0.002	0.008 ± 0.000	0.092 ± 0.003	0.329 ± 0.017	0.574 ± 0.018	0.162	0.002	0.97	0.0912	0.0912	1.097	97.320 ± 0.529	0.395 ± 0.004	39.550 ± 1.601	
UPI	beta55	0.08	0.088	0.008 ± 0.002	0.097 ± 0.027	0.004 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.067 ± 0.002	0.708 ± 0.040	0.841 ± 0.027	0.087	0.007	8.48	0.0612	0.0616	0.975	95.713 ± 0.663	0.256 ± 0.004	35.584 ± 1.567	
UPI	beta56	0.08	0.081	0.001 ± 0.001	0.009 ± 0.016	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.238 ± 0.014	0.488 ± 0.016	0.079	-0.001	-1.49	0.0362	0.0362	0.952	93.998 ± 0.778	0.146 ± 0.002	62.915 ± 1.581	
UPI	beta57	0.06	0.052	-0.008 ± 0.002	-0.129 ± 0.036	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	1.191 ± 0.063	1.091 ± 0.034	0.052	-0.008	-13.43	0.0577	0.0583	1.034	95.927 ± 0.647	0.264 ± 0.004	11.147 ± 1.030	
UPI	beta58	0.06	0.061	0.001 ± 0.001	0.012 ± 0.019	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.322 ± 0.018	0.567 ± 0.018	0.060	0.000	0.57	0.0304	0.0304	1.044	95.177 ± 0.701	0.139 ± 0.002	47.267 ± 1.634	

Note. MCSE = Monte Carlo Standard Error shown as \u000b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S189

Simulation 2 - Condition 45: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, $N = 400$, Reliability = 0.6

Method	Param.	True	Mean-based										Median-based					Inference		
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LSAM	beta41	0.20	0.209	0.009 $\pm$ 0.003	0.047 $\pm$ 0.017	0.012 $\pm$ 0.001	0.108 $\pm$ 0.003	0.012 $\pm$ 0.001	0.108 $\pm$ 0.003	0.294 $\pm$ 0.014	0.542 $\pm$ 0.016	0.211	0.011	5.37	0.1067	0.1072	0.965	95.073 $\pm$ 0.701	0.409 $\pm$ 0.003	55.136 $\pm$ 1.610
LSAM	beta42	0.20	0.199	-0.001 $\pm$ 0.004	-0.007 $\pm$ 0.018	0.013 $\pm$ 0.001	0.113 $\pm$ 0.003	0.013 $\pm$ 0.001	0.113 $\pm$ 0.003	0.319 $\pm$ 0.015	0.565 $\pm$ 0.016	0.198	-0.002	-1.03	0.1136	0.1136	0.970	95.388 $\pm$ 0.679	0.430 $\pm$ 0.003	50.000 $\pm$ 1.619
LSAM	beta43	0.20	0.202	0.002 $\pm$ 0.002	0.010 $\pm$ 0.010	0.004 $\pm$ 0.000	0.060 $\pm$ 0.001	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	0.090 $\pm$ 0.004	0.300 $\pm$ 0.009	0.202	0.002	1.03	0.0586	0.0587	0.982	94.969 $\pm$ 0.708	0.231 $\pm$ 0.001	92.034 $\pm$ 0.877
LSAM	beta44	0.11	0.120	0.010 $\pm$ 0.003	0.088 $\pm$ 0.025	0.007 $\pm$ 0.000	0.084 $\pm$ 0.002	0.007 $\pm$ 0.000	0.084 $\pm$ 0.003	0.585 $\pm$ 0.032	0.765 $\pm$ 0.024	0.114	0.004	3.72	0.0801	0.0802	0.911	94.130 $\pm$ 0.761	0.299 $\pm$ 0.003	38.365 $\pm$ 1.574
LSAM	beta45	0.11	0.113	0.003 $\pm$ 0.002	0.029 $\pm$ 0.022	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.460 $\pm$ 0.024	0.678 $\pm$ 0.021	0.109	-0.001	-0.94	0.0724	0.0724	0.933	92.348 $\pm$ 0.861	0.273 $\pm$ 0.003	39.518 $\pm$ 1.583
LSAM	beta46	0.08	0.077	-0.003 $\pm$ 0.001	-0.033 $\pm$ 0.019	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.334 $\pm$ 0.018	0.578 $\pm$ 0.018	0.076	-0.004	-5.61	0.0440	0.0443	0.893	91.509 $\pm$ 0.902	0.162 $\pm$ 0.002	53.669 $\pm$ 1.614
LSAM	beta47	0.08	0.082	0.002 $\pm$ 0.002	0.022 $\pm$ 0.019	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.002	0.356 $\pm$ 0.021	0.597 $\pm$ 0.020	0.080	0.000	0.15	0.0424	0.0424	0.872	92.662 $\pm$ 0.844	0.163 $\pm$ 0.002	58.071 $\pm$ 1.598
LSAM	beta51	0.16	0.155	-0.005 $\pm$ 0.002	-0.030 $\pm$ 0.015	0.006 $\pm$ 0.000	0.076 $\pm$ 0.002	0.006 $\pm$ 0.000	0.076 $\pm$ 0.002	0.225 $\pm$ 0.012	0.475 $\pm$ 0.015	0.154	-0.006	-4.02	0.0692	0.0695	0.961	93.187 $\pm$ 0.816	0.286 $\pm$ 0.002	57.862 $\pm$ 1.599
LSAM	beta52	0.16	0.163	0.003 $\pm$ 0.003	0.019 $\pm$ 0.021	0.011 $\pm$ 0.001	0.105 $\pm$ 0.003	0.011 $\pm$ 0.001	0.105 $\pm$ 0.003	0.432 $\pm$ 0.021	0.657 $\pm$ 0.019	0.171	0.011	6.89	0.0932	0.0939	1.007	95.597 $\pm$ 0.664	0.415 $\pm$ 0.003	38.994 $\pm$ 1.579
LSAM	beta53	0.16	0.159	-0.001 $\pm$ 0.003	-0.006 $\pm$ 0.018	0.008 $\pm$ 0.000	0.088 $\pm$ 0.002	0.008 $\pm$ 0.000	0.088 $\pm$ 0.002	0.304 $\pm$ 0.014	0.551 $\pm$ 0.015	0.152	-0.008	-5.12	0.0913	0.0917	0.937	94.130 $\pm$ 0.761	0.324 $\pm$ 0.002	48.428 $\pm$ 1.618
LSAM	beta54	0.16	0.163	0.003 $\pm$ 0.004	0.016 $\pm$ 0.022	0.012 $\pm$ 0.001	0.108 $\pm$ 0.002	0.012 $\pm$ 0.001	0.108 $\pm$ 0.003	0.456 $\pm$ 0.021	0.676 $\pm$ 0.019	0.162	0.002	1.08	0.1023	0.1023	0.939	94.549 $\pm$ 0.735	0.398 $\pm$ 0.003	40.042 $\pm$ 1.586
LSAM	beta55	0.08	0.082	0.002 $\pm$ 0.002	0.020 $\pm$ 0.030	0.006 $\pm$ 0.000	0.074 $\pm$ 0.002	0.006 $\pm$ 0.000	0.074 $\pm$ 0.002	0.860 $\pm$ 0.046	0.927 $\pm$ 0.029	0.080	0.000	0.28	0.0729	0.0729	0.926	94.549 $\pm$ 0.735	0.269 $\pm$ 0.004	29.874 $\pm$ 1.482
LSAM	beta56	0.08	0.081	0.001 $\pm$ 0.002	0.013 $\pm$ 0.019	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.348 $\pm$ 0.018	0.590 $\pm$ 0.018	0.081	0.001	0.70	0.0440	0.0440	0.888	92.243 $\pm$ 0.866	0.164 $\pm$ 0.002	52.935 $\pm$ 1.616
LSAM	beta57	0.06	0.061	0.001 $\pm$ 0.002	0.008 $\pm$ 0.039	0.005 $\pm$ 0.000	0.072 $\pm$ 0.002	0.005 $\pm$ 0.000	0.072 $\pm$ 0.002	1.423 $\pm$ 0.080	1.193 $\pm$ 0.039	0.058	-0.002	-4.12	0.0639	0.0640	0.932	95.178 $\pm$ 0.694	0.262 $\pm$ 0.003	14.570 $\pm$ 1.142
LSAM	beta58	0.06	0.061	0.001 $\pm$ 0.001	0.010 $\pm$ 0.022	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.479 $\pm$ 0.027	0.692 $\pm$ 0.023	0.058	-0.002	-2.50	0.0370	0.0371	0.857	91.614 $\pm$ 0.897	0.140 $\pm$ 0.002	47.589 $\pm$ 1.617
QML	beta41	0.20	0.220	0.020 $\pm$ 0.002	0.102 $\pm$ 0.012	0.005 $\pm$ 0.000	0.073 $\pm$ 0.002	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.142 $\pm$ 0.006	0.377 $\pm$ 0.010	0.218	0.018	9.13	0.0741	0.0763	0.983	94.025 $\pm$ 0.767	0.280 $\pm$ 0.002	84.382 $\pm$ 1.175
QML	beta42	0.20	0.211	0.011 $\pm$ 0.002	0.054 $\pm$ 0.012	0.005 $\pm$ 0.000	0.074 $\pm$ 0.002	0.006 $\pm$ 0.000	0.074 $\pm$ 0.002	0.139 $\pm$ 0.006	0.372 $\pm$ 0.010	0.215	0.015	7.29	0.0742	0.0757	0.999	95.283 $\pm$ 0.686	0.289 $\pm$ 0.002	79.140 $\pm$ 1.315
QML	beta43	0.20	0.192	-0.008 $\pm$ 0.002	-0.039 $\pm$ 0.009	0.003 $\pm$ 0.000	0.055 $\pm$ 0.001	0.003 $\pm$ 0.000	0.056 $\pm$ 0.002	0.078 $\pm$ 0.004	0.279 $\pm$ 0.008	0.190	-0.010	-4.89	0.0518	0.0527	0.985	94.025 $\pm$ 0.767	0.214 $\pm$ 0.001	94.025 $\pm$ 0.767
QML	beta44	0.11	0.072	-0.038 $\pm$ 0.002	-0.343 $\pm$ 0.022	0.006 $\pm$ 0.000	0.076 $\pm$ 0.002	0.007 $\pm$ 0.000	0.085 $\pm$ 0.002	0.593 $\pm$ 0.029	0.770 $\pm$ 0.021	0.074	-0.036	-32.57	0.0716	0.0801	0.946	90.042 $\pm$ 0.969	0.282 $\pm$ 0.003	22.432 $\pm$ 1.351
QML	beta45	0.11	0.098	-0.012 $\pm$ 0.002	-0.105 $\pm$ 0.019	0.004 $\pm$ 0.000	0.065 $\pm$ 0.002	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	0.363 $\pm$ 0.018	0.603 $\pm$ 0.018	0.098	-0.012	-10.81	0.0665	0.0676	0.950	93.187 $\pm$ 0.816	0.243 $\pm$ 0.002	39.099 $\pm$ 1.580
QML	beta46	0.08	0.112	0.032 $\pm$ 0.001	0.395 $\pm$ 0.017	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.432 $\pm$ 0.022	0.657 $\pm$ 0.018	0.109	0.029	36.36	0.0395	0.0490	0.977	90.776 $\pm$ 0.937	0.161 $\pm$ 0.002	80.398 $\pm$ 1.285
QML	beta47	0.08	0.117	0.037 $\pm$ 0.001	0.462 $\pm$ 0.018	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.003 $\pm$ 0.000	0.057 $\pm$ 0.002	0.514 $\pm$ 0.028	0.717 $\pm$ 0.021	0.112	0.032	40.53	0.0407	0.0520	0.919	89.727 $\pm$ 0.983	0.158 $\pm$ 0.002	86.059 $\pm$ 1.121
QML	beta51	0.16	0.170	0.010 $\pm$ 0.003	0.065 $\pm$ 0.016	0.006 $\pm$ 0.000	0.078 $\pm$ 0.002	0.006 $\pm$ 0.000	0.078 $\pm$ 0.002	0.240 $\pm$ 0.012	0.490 $\pm$ 0.015	0.169	0.009	5.53	0.0722	0.0728	0.996	93.291 $\pm$ 0.810	0.304 $\pm$ 0.002	61.321 $\pm$ 1.577
QML	beta52	0.16	0.151	-0.009 $\pm$ 0.002	-0.057 $\pm$ 0.015	0.006 $\pm$ 0.000	0.074 $\pm$ 0.002	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.220 $\pm$ 0.010	0.469 $\pm$ 0.013	0.154	-0.006	-3.66	0.0734	0.0736	1.018	94.969 $\pm$ 0.708	0.297 $\pm$ 0.002	55.241 $\pm$ 1.610
QML	beta53	0.16	0.124	-0.036 $\pm$ 0.003	-0.225 $\pm$ 0.016	0.006 $\pm$ 0.000	0.078 $\pm$ 0.002	0.007 $\pm$ 0.000	0.086 $\pm$ 0.002	0.286 $\pm$ 0.013	0.535 $\pm$ 0.014	0.119	-0.041	-25.70	0.0776	0.0878	0.947	90.671 $\pm$ 0.942	0.288 $\pm$ 0.001	38.994 $\pm$ 1.579
QML	beta54	0.16	0.166	0.006 $\pm$ 0.002	0.036 $\pm$ 0.015	0.006 $\pm$ 0.000	0.074 $\pm$ 0.002	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.218 $\pm$ 0.010	0.466 $\pm$ 0.013	0.167	0.007	4.57	0.0739	0.0743	0.960	94.235 $\pm$ 0.755	0.280 $\pm$ 0.002	64.151 $\pm$ 1.553
QML	beta55	0.08	0.078	-0.002 $\pm$ 0.003	-0.025 $\pm$ 0.033	0.006 $\pm$ 0.000	0.081 $\pm$ 0.002	0.006 $\pm$ 0.000	0.081 $\pm$ 0.003	1.015 $\pm$ 0.056	1.007 $\pm$ 0.032	0.080	0.000	0.32	0.0749	0.0749	0.933	91.824 $\pm$ 0.887	0.295 $\pm$ 0.005	29.874 $\pm$ 1.482
QML	beta56	0.08	0.098	0.018 $\pm$ 0.002	0.221 $\pm$ 0.019	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.404 $\pm$ 0.021	0.636 $\pm$ 0.019	0.099	0.019	23.34	0.0439	0.0477	0.908	90.776 $\pm$ 0.937	0.170 $\pm$ 0.002	62.893 $\pm$ 1.564
QML	beta57	0.06	0.092	0.032 $\pm$ 0.002	0.541 $\pm$ 0.040	0.005 $\pm$ 0.000	0.074 $\pm$ 0.002	0.007 $\pm$ 0.000	0.081 $\pm$ 0.003	1.812 $\pm$ 0.117	1.346 $\pm$ 0.047	0.087	0.027	45.17	0.0634	0.0689	0.943	93.920 $\pm$ 0.774	0.274 $\pm$ 0.005	27.778 $\pm$ 1.450
QML	beta58	0.06	0.084	0.024 $\pm$ 0.001	0.402 $\pm$ 0.020	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.547 $\pm$ 0.030	0.740 $\pm$ 0.022	0.082	0.022	36.89	0.0344	0.0409	0.893	90.252 $\pm$ 0.960	0.130 $\pm$ 0.002	73.899 $\pm$ 1.422
UPI	beta41	0.20	0.217	0.017 $\pm$ 0.003	0.086 $\pm$ 0.016	0.010 $\pm$ 0.000	0.098 $\pm$ 0.002	0.010 $\pm$ 0.000	0.099 $\pm$ 0.003	0.246 $\pm$ 0.011	0.496 $\pm$ 0.014	0.222	0.022	11.24	0.1007	0.1032	1.040	96.017 $\pm$ 0.633	0.399 $\pm$ 0.004	62.788 $\pm$ 1.565
UPI	beta42	0.20	0.205	0.005 $\pm$ 0.003	0.026 $\pm$ 0.017	0.011 $\pm$ 0.001	0.105 $\pm$ 0.003	0.011 $\pm$ 0.001	0.105 $\pm$ 0.003	0.278 $\pm$ 0.014	0.527 $\pm$ 0.016	0.203	0.003	1.57	0.1014	0.1014	1.018	96.122 $\pm$ 0.625	0.421 $\pm$ 0.005	53.774 $\pm$ 1.614
UPI	beta43	0.20	0.199	-0.001 $\pm$ 0.002	-0.005 $\pm$ 0.010	0.004 $\pm$ 0.000	0.061 $\pm$ 0.001	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.093 $\pm$ 0.004	0.304 $\pm$ 0.009	0.198	-0.002	-0.79	0.0606	0.0606	0.977	94.235 $\pm$ 0.755	0.233 $\pm$ 0.002	91.090

Table S189

Simulation 2 - Condition 45: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.6
(continued)

Method	Param.	True	Mean-based										Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE	
UPI	beta47	0.08	0.080	-0.000 ± 0.002	-0.000 ± 0.020	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.002	0.370 ± 0.021	0.608 ± 0.020	0.076	-0.004	-4.68	0.0412	0.0414	0.911	92.558 ± 0.850	0.174 ± 0.004	54.717 ± 1.612	
UPI	beta51	0.16	0.160	-0.000 ± 0.002	-0.002 ± 0.015	0.006 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.074 ± 0.002	0.216 ± 0.011	0.465 ± 0.014	0.158	-0.002	-1.54	0.0690	0.0690	0.965	93.920 ± 0.774	0.281 ± 0.001	62.579 ± 1.567	
UPI	beta52	0.16	0.173	0.013 ± 0.003	0.082 ± 0.021	0.010 ± 0.001	0.102 ± 0.003	0.011 ± 0.001	0.102 ± 0.003	0.410 ± 0.020	0.641 ± 0.019	0.177	0.017	10.81	0.0969	0.0985	1.016	95.178 ± 0.694	0.405 ± 0.003	43.816 ± 1.606	
UPI	beta53	0.16	0.158	-0.002 ± 0.003	-0.015 ± 0.018	0.008 ± 0.000	0.088 ± 0.002	0.008 ± 0.000	0.088 ± 0.002	0.299 ± 0.014	0.547 ± 0.015	0.152	-0.008	-5.12	0.0877	0.0880	0.934	94.235 ± 0.755	0.321 ± 0.002	49.266 ± 1.619	
UPI	beta54	0.16	0.163	0.003 ± 0.003	0.019 ± 0.021	0.011 ± 0.001	0.106 ± 0.002	0.011 ± 0.001	0.106 ± 0.003	0.438 ± 0.020	0.662 ± 0.019	0.161	0.001	0.89	0.1029	0.1029	0.967	95.388 ± 0.679	0.402 ± 0.003	40.042 ± 1.586	
UPI	beta55	0.08	0.085	0.005 ± 0.002	0.059 ± 0.028	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	0.760 ± 0.039	0.872 ± 0.026	0.081	0.001	1.33	0.0659	0.0659	0.979	94.759 ± 0.722	0.267 ± 0.004	31.237 ± 1.501	
UPI	beta56	0.08	0.079	-0.001 ± 0.002	-0.010 ± 0.019	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.002	0.348 ± 0.019	0.590 ± 0.019	0.077	-0.003	-3.29	0.0426	0.0426	0.900	91.614 ± 0.897	0.167 ± 0.002	51.258 ± 1.618	
UPI	beta57	0.06	0.054	-0.006 ± 0.002	-0.101 ± 0.035	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	1.169 ± 0.064	1.081 ± 0.035	0.052	-0.008	-12.82	0.0576	0.0581	1.003	94.759 ± 0.722	0.254 ± 0.004	12.893 ± 1.085	
UPI	beta58	0.06	0.060	-0.000 ± 0.001	-0.005 ± 0.022	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.473 ± 0.027	0.688 ± 0.023	0.057	-0.003	-4.62	0.0375	0.0376	0.880	91.824 ± 0.887	0.142 ± 0.002	44.340 ± 1.608	

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S190

Simulation 2 - Condition 46: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.6

Method	Param.	True	Mean-based										Median-based					Inference		
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LSAM	beta41	0.20	0.200	-0.000 $\pm$ 0.002	-0.000 $\pm$ 0.010	0.004 $\pm$ 0.000	0.061 $\pm$ 0.001	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.092 $\pm$ 0.004	0.303 $\pm$ 0.009	0.202	0.002	0.89	0.0596	0.0597	1.037	94.885 $\pm$ 0.698	0.246 $\pm$ 0.001	88.867 $\pm$ 0.996
LSAM	beta42	0.20	0.202	0.002 $\pm$ 0.002	0.008 $\pm$ 0.010	0.004 $\pm$ 0.000	0.065 $\pm$ 0.002	0.004 $\pm$ 0.000	0.065 $\pm$ 0.002	0.105 $\pm$ 0.005	0.324 $\pm$ 0.009	0.202	0.002	1.04	0.0622	0.0622	1.006	94.684 $\pm$ 0.711	0.255 $\pm$ 0.001	87.563 $\pm$ 1.045
LSAM	beta43	0.20	0.198	-0.002 $\pm$ 0.001	-0.009 $\pm$ 0.006	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.032 $\pm$ 0.001	0.179 $\pm$ 0.005	0.198	-0.002	-1.10	0.0355	0.0356	1.020	94.885 $\pm$ 0.698	0.143 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta44	0.11	0.113	0.003 $\pm$ 0.001	0.026 $\pm$ 0.013	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.172 $\pm$ 0.008	0.414 $\pm$ 0.011	0.113	0.003	2.71	0.0461	0.0462	0.953	94.383 $\pm$ 0.729	0.170 $\pm$ 0.001	74.323 $\pm$ 1.384
LSAM	beta45	0.11	0.113	0.003 $\pm$ 0.001	0.029 $\pm$ 0.013	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.161 $\pm$ 0.007	0.401 $\pm$ 0.011	0.111	0.001	0.62	0.0438	0.0438	0.938	93.581 $\pm$ 0.776	0.162 $\pm$ 0.001	77.533 $\pm$ 1.322
LSAM	beta46	0.08	0.081	0.001 $\pm$ 0.001	0.007 $\pm$ 0.010	0.001 $\pm$ 0.000	0.025 $\pm$ 0.001	0.001 $\pm$ 0.000	0.024 $\pm$ 0.001	0.094 $\pm$ 0.005	0.306 $\pm$ 0.009	0.080	0.000	0.45	0.0244	0.0244	0.958	93.480 $\pm$ 0.782	0.092 $\pm$ 0.001	91.976 $\pm$ 0.860
LSAM	beta47	0.08	0.080	-0.000 $\pm$ 0.001	-0.001 $\pm$ 0.010	0.001 $\pm$ 0.000	0.024 $\pm$ 0.001	0.001 $\pm$ 0.000	0.024 $\pm$ 0.001	0.091 $\pm$ 0.005	0.301 $\pm$ 0.009	0.080	0.000	-0.38	0.0217	0.0218	0.958	93.781 $\pm$ 0.765	0.091 $\pm$ 0.001	90.873 $\pm$ 0.912
LSAM	beta51	0.16	0.161	0.001 $\pm$ 0.001	0.004 $\pm$ 0.009	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.078 $\pm$ 0.003	0.279 $\pm$ 0.008	0.162	0.002	1.14	0.0468	0.0469	1.017	95.687 $\pm$ 0.643	0.178 $\pm$ 0.001	94.283 $\pm$ 0.735
LSAM	beta52	0.16	0.164	0.004 $\pm$ 0.002	0.023 $\pm$ 0.013	0.004 $\pm$ 0.000	0.067 $\pm$ 0.001	0.004 $\pm$ 0.000	0.067 $\pm$ 0.002	0.175 $\pm$ 0.008	0.418 $\pm$ 0.011	0.166	0.006	3.68	0.0693	0.0696	0.967	94.784 $\pm$ 0.704	0.253 $\pm$ 0.001	70.110 $\pm$ 1.450
LSAM	beta53	0.16	0.160	-0.000 $\pm$ 0.002	-0.003 $\pm$ 0.010	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	0.096 $\pm$ 0.005	0.310 $\pm$ 0.009	0.159	-0.001	-0.63	0.0482	0.0483	1.015	95.787 $\pm$ 0.636	0.197 $\pm$ 0.001	89.268 $\pm$ 0.980
LSAM	beta54	0.16	0.161	0.001 $\pm$ 0.002	0.007 $\pm$ 0.012	0.004 $\pm$ 0.000	0.061 $\pm$ 0.001	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.147 $\pm$ 0.006	0.383 $\pm$ 0.010	0.161	0.001	0.70	0.0595	0.0595	0.995	94.584 $\pm$ 0.717	0.239 $\pm$ 0.001	74.624 $\pm$ 1.378
LSAM	beta55	0.08	0.080	0.000 $\pm$ 0.001	0.003 $\pm$ 0.016	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	0.264 $\pm$ 0.012	0.514 $\pm$ 0.015	0.080	0.000	0.22	0.0414	0.0414	0.924	93.380 $\pm$ 0.787	0.149 $\pm$ 0.001	57.874 $\pm$ 1.564
LSAM	beta56	0.08	0.081	0.001 $\pm$ 0.001	0.012 $\pm$ 0.010	0.001 $\pm$ 0.000	0.025 $\pm$ 0.001	0.001 $\pm$ 0.000	0.025 $\pm$ 0.001	0.100 $\pm$ 0.005	0.316 $\pm$ 0.009	0.081	0.001	0.74	0.0240	0.0240	0.907	91.374 $\pm$ 0.889	0.090 $\pm$ 0.001	89.669 $\pm$ 0.964
LSAM	beta57	0.06	0.060	0.000 $\pm$ 0.001	0.007 $\pm$ 0.021	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.449 $\pm$ 0.021	0.670 $\pm$ 0.019	0.059	-0.001	-1.40	0.0401	0.0401	0.964	94.383 $\pm$ 0.729	0.152 $\pm$ 0.001	36.209 $\pm$ 1.522
LSAM	beta58	0.06	0.060	0.000 $\pm$ 0.001	0.004 $\pm$ 0.012	0.001 $\pm$ 0.000	0.022 $\pm$ 0.001	0.001 $\pm$ 0.000	0.022 $\pm$ 0.001	0.139 $\pm$ 0.007	0.373 $\pm$ 0.011	0.061	0.001	1.71	0.0206	0.0206	0.919	92.879 $\pm$ 0.815	0.081 $\pm$ 0.001	82.548 $\pm$ 1.202
QML	beta41	0.20	0.215	0.015 $\pm$ 0.001	0.074 $\pm$ 0.007	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.052 $\pm$ 0.002	0.229 $\pm$ 0.006	0.216	0.016	8.09	0.0419	0.0449	1.045	94.082 $\pm$ 0.747	0.178 $\pm$ 0.001	99.398 $\pm$ 0.245
QML	beta42	0.20	0.213	0.013 $\pm$ 0.001	0.064 $\pm$ 0.007	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.055 $\pm$ 0.002	0.235 $\pm$ 0.006	0.212	0.012	6.23	0.0443	0.0460	1.021	93.581 $\pm$ 0.776	0.181 $\pm$ 0.002	99.298 $\pm$ 0.264
QML	beta43	0.20	0.190	-0.010 $\pm$ 0.001	-0.052 $\pm$ 0.005	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.031 $\pm$ 0.001	0.175 $\pm$ 0.005	0.189	-0.011	-5.25	0.0332	0.0349	1.027	93.681 $\pm$ 0.771	0.135 $\pm$ 0.001	99.900 $\pm$ 0.100
QML	beta44	0.11	0.074	-0.036 $\pm$ 0.002	-0.331 $\pm$ 0.014	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.004 $\pm$ 0.000	0.060 $\pm$ 0.001	0.295 $\pm$ 0.013	0.543 $\pm$ 0.013	0.076	-0.034	-31.28	0.0462	0.0576	0.945	84.855 $\pm$ 1.135	0.175 $\pm$ 0.001	41.926 $\pm$ 1.563
QML	beta45	0.11	0.103	-0.007 $\pm$ 0.001	-0.066 $\pm$ 0.012	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.151 $\pm$ 0.007	0.388 $\pm$ 0.011	0.103	-0.007	-6.74	0.0412	0.0418	0.945	93.280 $\pm$ 0.793	0.156 $\pm$ 0.001	73.019 $\pm$ 1.406
QML	beta46	0.08	0.116	0.036 $\pm$ 0.001	0.444 $\pm$ 0.011	0.001 $\pm$ 0.000	0.027 $\pm$ 0.001	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.309 $\pm$ 0.012	0.556 $\pm$ 0.011	0.114	0.034	42.24	0.0240	0.0415	0.941	74.022 $\pm$ 1.389	0.099 $\pm$ 0.001	99.599 $\pm$ 0.200
QML	beta47	0.08	0.115	0.035 $\pm$ 0.001	0.432 $\pm$ 0.010	0.001 $\pm$ 0.000	0.024 $\pm$ 0.001	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.279 $\pm$ 0.010	0.528 $\pm$ 0.010	0.112	0.032	40.50	0.0217	0.0390	0.982	72.417 $\pm$ 1.415	0.094 $\pm$ 0.001	100.000 $\pm$ 0.000
QML	beta51	0.16	0.174	0.014 $\pm$ 0.001	0.088 $\pm$ 0.009	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.094 $\pm$ 0.004	0.307 $\pm$ 0.008	0.176	0.016	9.99	0.0478	0.0504	1.021	95.587 $\pm$ 0.650	0.189 $\pm$ 0.001	95.286 $\pm$ 0.671
QML	beta52	0.16	0.149	-0.011 $\pm$ 0.002	-0.067 $\pm$ 0.010	0.003 $\pm$ 0.000	0.050 $\pm$ 0.001	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.102 $\pm$ 0.005	0.320 $\pm$ 0.009	0.151	-0.009	-5.93	0.0521	0.0529	0.939	92.578 $\pm$ 0.830	0.184 $\pm$ 0.001	87.362 $\pm$ 1.052
QML	beta53	0.16	0.124	-0.036 $\pm$ 0.001	-0.224 $\pm$ 0.009	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.003 $\pm$ 0.000	0.058 $\pm$ 0.001	0.130 $\pm$ 0.005	0.360 $\pm$ 0.008	0.123	-0.037	-23.14	0.0437	0.0573	1.007	87.462 $\pm$ 1.049	0.178 $\pm$ 0.001	80.040 $\pm$ 1.266
QML	beta54	0.16	0.166	0.006 $\pm$ 0.001	0.035 $\pm$ 0.009	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.079 $\pm$ 0.003	0.281 $\pm$ 0.008	0.166	0.006	3.64	0.0426	0.0430	1.002	94.383 $\pm$ 0.729	0.175 $\pm$ 0.001	94.383 $\pm$ 0.729
QML	beta55	0.08	0.078	-0.002 $\pm$ 0.002	-0.027 $\pm$ 0.020	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.417 $\pm$ 0.020	0.646 $\pm$ 0.018	0.079	-0.001	-1.07	0.0471	0.0471	0.906	90.471 $\pm$ 0.930	0.183 $\pm$ 0.002	46.138 $\pm$ 1.579
QML	beta56	0.08	0.098	0.018 $\pm$ 0.001	0.229 $\pm$ 0.011	0.001 $\pm$ 0.000	0.028 $\pm$ 0.001	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.176 $\pm$ 0.008	0.420 $\pm$ 0.011	0.098	0.018	21.91	0.0272	0.0324	0.918	86.660 $\pm$ 1.077	0.101 $\pm$ 0.001	94.684 $\pm$ 0.711
QML	beta57	0.06	0.096	0.036 $\pm$ 0.002	0.597 $\pm$ 0.025	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.004 $\pm$ 0.000	0.059 $\pm$ 0.002	0.980 $\pm$ 0.049	0.990 $\pm$ 0.027	0.091	0.031	51.97	0.0451	0.0548	0.950	90.271 $\pm$ 0.939	0.177 $\pm$ 0.002	58.776 $\pm$ 1.559
QML	beta58	0.06	0.085	0.025 $\pm$ 0.001	0.413 $\pm$ 0.012	0.001 $\pm$ 0.000	0.023 $\pm$ 0.001	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.313 $\pm$ 0.013	0.560 $\pm$ 0.013	0.084	0.024	40.04	0.0209	0.0318	0.919	79.739 $\pm$ 1.273	0.082 $\pm$ 0.001	98.295 $\pm$ 0.410
UPI	beta41	0.20	0.207	0.007 $\pm$ 0.002	0.033 $\pm$ 0.009	0.003 $\pm$ 0.000	0.056 $\pm$ 0.001	0.003 $\pm$ 0.000	0.056 $\pm$ 0.002	0.080 $\pm$ 0.003	0.282 $\pm$ 0.008	0.206	0.006	2.77	0.0581	0.0584	1.015	94.885 $\pm$ 0.698	0.223 $\pm$ 0.001	93.681 $\pm$ 0.771
UPI	beta42	0.20	0.203	0.003 $\pm$ 0.002	0.015 $\pm$ 0.009	0.003 $\pm$ 0.000	0.058 $\pm$ 0.001	0.003 $\pm$ 0.000	0.058 $\pm$ 0.002	0.084 $\pm$ 0.004	0.289 $\pm$ 0.008	0.204	0.004	1.84	0.0579	0.0580	1.020	95.186 $\pm$ 0.678	0.231 $\pm$ 0.001	91.575 $\pm$ 0.880
UPI	beta43	0.20	0.197	-0.003 $\pm$ 0.001	-0.014 $\pm$ 0.006	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.032 $\pm$ 0.001	0.179 $\pm$ 0.005	0.197	-0.003	-1.64	0.0364	0.0365	1.005	94.784 $\pm$ 0.704	0.141 $\pm$ 0.001	100.000 $\pm$ 0.00

Table S190

Simulation 2 - Condition 46: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.6
(continued)

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.079	-0.001 ± 0.001	-0.010 ± 0.010	0.001 ± 0.000	0.024 ± 0.001	0.001 ± 0.000	0.024 ± 0.001	0.093 ± 0.005	0.306 ± 0.009	0.078	-0.002	-2.69	0.0218	0.0220	0.935	92.578 ± 0.830	0.090 ± 0.001	91.675 ± 0.875
UPI	beta51	0.16	0.162	0.002 ± 0.001	0.015 ± 0.009	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.076 ± 0.003	0.276 ± 0.007	0.164	0.004	2.76	0.0455	0.0457	1.021	95.386 ± 0.664	0.176 ± 0.001	95.386 ± 0.664
UPI	beta52	0.16	0.169	0.009 ± 0.002	0.054 ± 0.013	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.066 ± 0.002	0.170 ± 0.007	0.413 ± 0.011	0.168	0.008	4.69	0.0655	0.0660	0.962	94.483 ± 0.723	0.247 ± 0.001	75.226 ± 1.367
UPI	beta53	0.16	0.159	-0.001 ± 0.002	-0.007 ± 0.010	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.090 ± 0.004	0.301 ± 0.008	0.158	-0.002	-1.05	0.0465	0.0465	1.024	95.386 ± 0.664	0.193 ± 0.001	91.274 ± 0.894
UPI	beta54	0.16	0.161	0.001 ± 0.002	0.009 ± 0.012	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.145 ± 0.006	0.381 ± 0.010	0.162	0.002	1.44	0.0587	0.0588	0.999	94.784 ± 0.704	0.239 ± 0.001	75.226 ± 1.367
UPI	beta55	0.08	0.081	0.001 ± 0.001	0.012 ± 0.016	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.246 ± 0.012	0.496 ± 0.014	0.081	0.001	1.51	0.0395	0.0396	0.944	93.681 ± 0.771	0.147 ± 0.001	60.181 ± 1.550
UPI	beta56	0.08	0.080	0.000 ± 0.001	0.000 ± 0.010	0.001 ± 0.000	0.025 ± 0.001	0.001 ± 0.000	0.025 ± 0.001	0.101 ± 0.005	0.318 ± 0.009	0.080	0.000	-0.51	0.0239	0.0239	0.911	91.775 ± 0.870	0.091 ± 0.001	89.468 ± 0.972
UPI	beta57	0.06	0.058	-0.002 ± 0.001	-0.036 ± 0.021	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.422 ± 0.020	0.649 ± 0.019	0.057	-0.003	-4.55	0.0379	0.0380	0.964	94.082 ± 0.747	0.147 ± 0.001	35.105 ± 1.512
UPI	beta58	0.06	0.060	-0.000 ± 0.001	-0.002 ± 0.012	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	0.141 ± 0.007	0.375 ± 0.011	0.060	0.000	-0.04	0.0215	0.0215	0.921	92.377 ± 0.840	0.081 ± 0.001	82.447 ± 1.205

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S191

Simulation 2 - Condition 47: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, $N = 400$, Reliability = 0.8

Method	Param.	True	Mean-based										Median-based					Inference		
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LSAM	beta41	0.20	0.201	0.001 $\pm$ 0.003	0.004 $\pm$ 0.013	0.006 $\pm$ 0.000	0.079 $\pm$ 0.002	0.006 $\pm$ 0.000	0.079 $\pm$ 0.002	0.156 $\pm$ 0.007	0.395 $\pm$ 0.011	0.200	0.000	-0.18	0.0802	0.0802	0.963	93.401 $\pm$ 0.791	0.298 $\pm$ 0.001	73.807 $\pm$ 1.401
LSAM	beta42	0.20	0.202	0.002 $\pm$ 0.003	0.011 $\pm$ 0.013	0.006 $\pm$ 0.000	0.080 $\pm$ 0.002	0.006 $\pm$ 0.000	0.080 $\pm$ 0.002	0.162 $\pm$ 0.007	0.402 $\pm$ 0.011	0.203	0.003	1.62	0.0758	0.0759	0.976	94.416 $\pm$ 0.732	0.308 $\pm$ 0.001	73.401 $\pm$ 1.408
LSAM	beta43	0.20	0.201	0.001 $\pm$ 0.002	0.004 $\pm$ 0.008	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.058 $\pm$ 0.003	0.241 $\pm$ 0.007	0.201	0.001	0.57	0.0499	0.0499	0.963	94.315 $\pm$ 0.738	0.182 $\pm$ 0.001	98.782 $\pm$ 0.350
LSAM	beta44	0.11	0.112	0.002 $\pm$ 0.002	0.021 $\pm$ 0.016	0.003 $\pm$ 0.000	0.055 $\pm$ 0.001	0.003 $\pm$ 0.000	0.055 $\pm$ 0.002	0.254 $\pm$ 0.013	0.504 $\pm$ 0.015	0.110	0.000	-0.13	0.0541	0.0541	0.917	91.980 $\pm$ 0.865	0.199 $\pm$ 0.002	60.305 $\pm$ 1.559
LSAM	beta45	0.11	0.112	0.002 $\pm$ 0.002	0.016 $\pm$ 0.015	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.003 $\pm$ 0.000	0.051 $\pm$ 0.002	0.219 $\pm$ 0.011	0.468 $\pm$ 0.014	0.112	0.002	1.58	0.0482	0.0483	0.932	93.503 $\pm$ 0.785	0.188 $\pm$ 0.002	66.497 $\pm$ 1.504
LSAM	beta46	0.08	0.079	-0.001 $\pm$ 0.001	-0.008 $\pm$ 0.013	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	0.156 $\pm$ 0.008	0.395 $\pm$ 0.012	0.078	-0.002	-2.51	0.0296	0.0297	0.899	90.761 $\pm$ 0.923	0.111 $\pm$ 0.001	77.665 $\pm$ 1.327
LSAM	beta47	0.08	0.080	-0.000 $\pm$ 0.001	-0.003 $\pm$ 0.012	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.152 $\pm$ 0.008	0.390 $\pm$ 0.012	0.078	-0.002	-2.07	0.0276	0.0277	0.901	92.386 $\pm$ 0.845	0.110 $\pm$ 0.001	79.594 $\pm$ 1.284
LSAM	beta51	0.16	0.160	0.000 $\pm$ 0.002	0.002 $\pm$ 0.012	0.004 $\pm$ 0.000	0.060 $\pm$ 0.001	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	0.141 $\pm$ 0.007	0.375 $\pm$ 0.011	0.159	-0.001	-0.70	0.0618	0.0618	0.978	94.416 $\pm$ 0.732	0.230 $\pm$ 0.001	78.071 $\pm$ 1.318
LSAM	beta52	0.16	0.164	0.004 $\pm$ 0.003	0.022 $\pm$ 0.016	0.006 $\pm$ 0.000	0.079 $\pm$ 0.002	0.006 $\pm$ 0.000	0.079 $\pm$ 0.002	0.245 $\pm$ 0.012	0.495 $\pm$ 0.014	0.164	0.004	2.39	0.0783	0.0784	0.979	95.127 $\pm$ 0.686	0.304 $\pm$ 0.001	57.462 $\pm$ 1.575
LSAM	beta53	0.16	0.157	-0.003 $\pm$ 0.002	-0.021 $\pm$ 0.013	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	0.172 $\pm$ 0.008	0.415 $\pm$ 0.012	0.158	-0.002	-1.33	0.0656	0.0656	0.959	94.213 $\pm$ 0.744	0.249 $\pm$ 0.001	70.761 $\pm$ 1.449
LSAM	beta54	0.16	0.159	-0.001 $\pm$ 0.002	-0.007 $\pm$ 0.015	0.006 $\pm$ 0.000	0.077 $\pm$ 0.002	0.006 $\pm$ 0.000	0.077 $\pm$ 0.002	0.233 $\pm$ 0.010	0.482 $\pm$ 0.013	0.161	0.001	0.38	0.0770	0.0770	0.982	94.619 $\pm$ 0.719	0.297 $\pm$ 0.001	57.462 $\pm$ 1.575
LSAM	beta55	0.08	0.080	-0.000 $\pm$ 0.002	-0.004 $\pm$ 0.022	0.003 $\pm$ 0.000	0.054 $\pm$ 0.001	0.003 $\pm$ 0.000	0.054 $\pm$ 0.002	0.463 $\pm$ 0.023	0.680 $\pm$ 0.020	0.080	0.000	0.13	0.0506	0.0506	0.860	91.168 $\pm$ 0.904	0.183 $\pm$ 0.002	45.685 $\pm$ 1.587
LSAM	beta56	0.08	0.083	0.003 $\pm$ 0.001	0.035 $\pm$ 0.014	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.204 $\pm$ 0.011	0.452 $\pm$ 0.015	0.081	0.001	1.51	0.0313	0.0313	0.878	92.487 $\pm$ 0.840	0.124 $\pm$ 0.002	71.980 $\pm$ 1.431
LSAM	beta57	0.06	0.060	-0.000 $\pm$ 0.002	-0.003 $\pm$ 0.026	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.681 $\pm$ 0.033	0.825 $\pm$ 0.024	0.058	-0.002	-3.74	0.0473	0.0474	0.892	91.675 $\pm$ 0.880	0.173 $\pm$ 0.002	32.284 $\pm$ 1.490
LSAM	beta58	0.06	0.061	0.001 $\pm$ 0.001	0.016 $\pm$ 0.016	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.247 $\pm$ 0.012	0.497 $\pm$ 0.015	0.060	0.000	0.07	0.0274	0.0274	0.893	92.081 $\pm$ 0.860	0.104 $\pm$ 0.001	64.772 $\pm$ 1.522
QML	beta41	0.20	0.206	0.006 $\pm$ 0.002	0.029 $\pm$ 0.010	0.004 $\pm$ 0.000	0.065 $\pm$ 0.001	0.004 $\pm$ 0.000	0.065 $\pm$ 0.002	0.107 $\pm$ 0.005	0.327 $\pm$ 0.009	0.206	0.006	3.09	0.0648	0.0651	0.973	94.213 $\pm$ 0.744	0.249 $\pm$ 0.001	87.005 $\pm$ 1.071
QML	beta42	0.20	0.206	0.006 $\pm$ 0.002	0.028 $\pm$ 0.010	0.004 $\pm$ 0.000	0.066 $\pm$ 0.001	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	0.108 $\pm$ 0.005	0.329 $\pm$ 0.009	0.207	0.007	3.55	0.0648	0.0652	0.985	94.518 $\pm$ 0.725	0.253 $\pm$ 0.001	85.990 $\pm$ 1.106
QML	beta43	0.20	0.197	-0.003 $\pm$ 0.001	-0.015 $\pm$ 0.007	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.055 $\pm$ 0.003	0.234 $\pm$ 0.007	0.198	-0.002	-1.02	0.0494	0.0495	0.964	93.706 $\pm$ 0.774	0.177 $\pm$ 0.001	98.883 $\pm$ 0.335
QML	beta44	0.11	0.096	-0.014 $\pm$ 0.002	-0.132 $\pm$ 0.016	0.003 $\pm$ 0.000	0.054 $\pm$ 0.001	0.003 $\pm$ 0.000	0.056 $\pm$ 0.002	0.261 $\pm$ 0.013	0.511 $\pm$ 0.015	0.094	-0.016	-14.59	0.0537	0.0560	0.939	92.284 $\pm$ 0.850	0.200 $\pm$ 0.002	50.964 $\pm$ 1.593
QML	beta45	0.11	0.106	-0.004 $\pm$ 0.002	-0.034 $\pm$ 0.014	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	0.205 $\pm$ 0.010	0.452 $\pm$ 0.014	0.106	-0.004	-3.23	0.0450	0.0452	0.946	93.401 $\pm$ 0.791	0.184 $\pm$ 0.002	64.772 $\pm$ 1.522
QML	beta46	0.08	0.091	0.011 $\pm$ 0.001	0.143 $\pm$ 0.012	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	0.160 $\pm$ 0.009	0.400 $\pm$ 0.013	0.089	0.009	11.71	0.0273	0.0289	0.928	91.371 $\pm$ 0.895	0.109 $\pm$ 0.001	91.066 $\pm$ 0.909
QML	beta47	0.08	0.093	0.013 $\pm$ 0.001	0.160 $\pm$ 0.012	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.166 $\pm$ 0.009	0.407 $\pm$ 0.012	0.091	0.011	13.49	0.0280	0.0300	0.914	92.183 $\pm$ 0.855	0.107 $\pm$ 0.001	91.574 $\pm$ 0.885
QML	beta51	0.16	0.168	0.008 $\pm$ 0.002	0.047 $\pm$ 0.012	0.004 $\pm$ 0.000	0.060 $\pm$ 0.001	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.143 $\pm$ 0.007	0.378 $\pm$ 0.011	0.168	0.008	5.09	0.0621	0.0626	0.991	95.431 $\pm$ 0.665	0.233 $\pm$ 0.001	80.508 $\pm$ 1.262
QML	beta52	0.16	0.147	-0.013 $\pm$ 0.002	-0.079 $\pm$ 0.013	0.004 $\pm$ 0.000	0.064 $\pm$ 0.002	0.004 $\pm$ 0.000	0.065 $\pm$ 0.002	0.166 $\pm$ 0.008	0.407 $\pm$ 0.012	0.146	-0.014	-9.02	0.0622	0.0639	1.000	94.924 $\pm$ 0.699	0.251 $\pm$ 0.001	63.959 $\pm$ 1.530
QML	beta53	0.16	0.132	-0.028 $\pm$ 0.002	-0.173 $\pm$ 0.013	0.004 $\pm$ 0.000	0.064 $\pm$ 0.001	0.005 $\pm$ 0.000	0.069 $\pm$ 0.002	0.188 $\pm$ 0.009	0.433 $\pm$ 0.011	0.131	-0.029	-18.23	0.0616	0.0681	0.969	91.371 $\pm$ 0.895	0.241 $\pm$ 0.001	57.970 $\pm$ 1.573
QML	beta54	0.16	0.160	-0.000 $\pm$ 0.002	-0.002 $\pm$ 0.013	0.004 $\pm$ 0.000	0.065 $\pm$ 0.001	0.004 $\pm$ 0.000	0.065 $\pm$ 0.002	0.163 $\pm$ 0.007	0.404 $\pm$ 0.011	0.162	0.002	1.00	0.0660	0.0660	0.994	95.127 $\pm$ 0.686	0.252 $\pm$ 0.001	70.863 $\pm$ 1.448
QML	beta55	0.08	0.080	-0.000 $\pm$ 0.002	-0.001 $\pm$ 0.022	0.003 $\pm$ 0.000	0.055 $\pm$ 0.001	0.003 $\pm$ 0.000	0.055 $\pm$ 0.002	0.472 $\pm$ 0.024	0.687 $\pm$ 0.021	0.080	0.000	-0.38	0.0519	0.0519	0.885	91.777 $\pm$ 0.875	0.191 $\pm$ 0.002	45.381 $\pm$ 1.586
QML	beta56	0.08	0.089	0.009 $\pm$ 0.001	0.107 $\pm$ 0.014	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.211 $\pm$ 0.012	0.459 $\pm$ 0.014	0.087	0.007	8.17	0.0324	0.0330	0.894	91.980 $\pm$ 0.865	0.125 $\pm$ 0.002	76.853 $\pm$ 1.344
QML	beta57	0.06	0.070	0.010 $\pm$ 0.002	0.173 $\pm$ 0.025	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.650 $\pm$ 0.033	0.806 $\pm$ 0.024	0.068	0.008	13.68	0.0445	0.0453	0.929	92.995 $\pm$ 0.813	0.172 $\pm$ 0.002	39.188 $\pm$ 1.555
QML	beta58	0.06	0.070	0.010 $\pm$ 0.001	0.171 $\pm$ 0.015	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.255 $\pm$ 0.014	0.505 $\pm$ 0.015	0.069	0.009	14.85	0.0248	0.0263	0.894	90.660 $\pm$ 0.927	0.100 $\pm$ 0.001	77.665 $\pm$ 1.327
UPI	beta41	0.20	0.206	0.006 $\pm$ 0.002	0.029 $\pm$ 0.011	0.005 $\pm$ 0.000	0.071 $\pm$ 0.002	0.005 $\pm$ 0.000	0.072 $\pm$ 0.002	0.128 $\pm$ 0.006	0.358 $\pm$ 0.010	0.204	0.004	2.20	0.0729	0.0730	0.964	93.706 $\pm$ 0.774	0.270 $\pm$ 0.001	83.858 $\pm$ 1.172
UPI	beta42	0.20	0.204	0.004 $\pm$ 0.002	0.022 $\pm$ 0.012	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.139 $\pm$ 0.006	0.373 $\pm$ 0.010	0.206	0.006	2.92	0.0716	0.0719	0.958	93.807 $\pm$ 0.768	0.280 $\pm$ 0.001	81.624 $\pm$ 1.234
UPI	beta43	0.20	0.199	-0.001 $\pm$ 0.002	-0.004 $\pm$ 0.008	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.059 $\pm$ 0.003	0.242 $\pm$ 0.007	0.199	-0.001	-0.33	0.0498	0.0498	0.943	93.807 $\pm$ 0.768	0.179 $\pm$ 0.001	98.680 $\pm$ 0.364

Table S191

Simulation 2 - Condition 47: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.8
(continued)

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.079	-0.001 ± 0.001	-0.012 ± 0.013	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.158 ± 0.009	0.398 ± 0.013	0.077	-0.003	-3.68	0.0284	0.0286	0.873	90.051 ± 0.954	0.109 ± 0.001	78.376 ± 1.312
UPI	beta51	0.16	0.161	0.001 ± 0.002	0.009 ± 0.012	0.004 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	0.137 ± 0.006	0.370 ± 0.011	0.159	-0.001	-0.48	0.0599	0.0599	0.982	94.416 ± 0.732	0.228 ± 0.001	79.391 ± 1.289
UPI	beta52	0.16	0.168	0.008 ± 0.002	0.050 ± 0.015	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	0.237 ± 0.011	0.486 ± 0.014	0.167	0.007	4.64	0.0764	0.0767	0.981	94.924 ± 0.699	0.298 ± 0.001	60.102 ± 1.560
UPI	beta53	0.16	0.156	-0.004 ± 0.002	-0.027 ± 0.013	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	0.167 ± 0.008	0.409 ± 0.011	0.157	-0.003	-1.64	0.0663	0.0663	0.964	94.518 ± 0.725	0.247 ± 0.001	70.964 ± 1.446
UPI	beta54	0.16	0.159	-0.001 ± 0.002	-0.005 ± 0.015	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	0.234 ± 0.010	0.483 ± 0.013	0.160	0.000	0.24	0.0783	0.0783	0.977	94.924 ± 0.699	0.296 ± 0.001	58.173 ± 1.572
UPI	beta55	0.08	0.081	0.001 ± 0.002	0.010 ± 0.021	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.454 ± 0.023	0.674 ± 0.020	0.079	-0.001	-0.63	0.0488	0.0488	0.872	91.980 ± 0.865	0.184 ± 0.002	45.279 ± 1.586
UPI	beta56	0.08	0.082	0.002 ± 0.001	0.029 ± 0.014	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.203 ± 0.011	0.451 ± 0.014	0.081	0.001	1.08	0.0319	0.0319	0.884	90.964 ± 0.913	0.125 ± 0.002	72.386 ± 1.425
UPI	beta57	0.06	0.058	-0.002 ± 0.002	-0.040 ± 0.026	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.646 ± 0.030	0.804 ± 0.023	0.058	-0.002	-3.52	0.0454	0.0455	0.913	92.589 ± 0.835	0.173 ± 0.002	30.761 ± 1.470
UPI	beta58	0.06	0.061	0.001 ± 0.001	0.011 ± 0.016	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.250 ± 0.013	0.500 ± 0.015	0.059	-0.001	-1.69	0.0274	0.0274	0.894	91.269 ± 0.899	0.105 ± 0.001	65.076 ± 1.519

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S192

Simulation 2 - Condition 48: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, $N = 1000$, Reliability = 0.8

Method	Param.	True	Mean-based										Median-based					Inference		
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LSAM	beta41	0.20	0.201	0.001 $\pm$ 0.002	0.004 $\pm$ 0.008	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.056 $\pm$ 0.002	0.237 $\pm$ 0.006	0.201	0.001	0.27	0.0472	0.0472	0.984	94.478 $\pm$ 0.724	0.183 $\pm$ 0.001	98.494 $\pm$ 0.386
LSAM	beta42	0.20	0.199	-0.001 $\pm$ 0.002	-0.006 $\pm$ 0.008	0.003 $\pm$ 0.000	0.050 $\pm$ 0.001	0.003 $\pm$ 0.000	0.050 $\pm$ 0.001	0.063 $\pm$ 0.003	0.251 $\pm$ 0.007	0.200	0.000	-0.17	0.0493	0.0493	0.960	93.675 $\pm$ 0.771	0.189 $\pm$ 0.001	96.586 $\pm$ 0.575
LSAM	beta43	0.20	0.200	-0.000 $\pm$ 0.001	-0.002 $\pm$ 0.005	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.023 $\pm$ 0.001	0.151 $\pm$ 0.004	0.199	-0.001	-0.34	0.0304	0.0304	0.969	93.775 $\pm$ 0.766	0.115 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta44	0.11	0.109	-0.001 $\pm$ 0.001	-0.011 $\pm$ 0.009	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.088 $\pm$ 0.004	0.297 $\pm$ 0.008	0.109	-0.001	-1.34	0.0325	0.0326	0.945	93.675 $\pm$ 0.771	0.121 $\pm$ 0.001	90.663 $\pm$ 0.922
LSAM	beta45	0.11	0.111	0.001 $\pm$ 0.001	0.006 $\pm$ 0.009	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.080 $\pm$ 0.004	0.282 $\pm$ 0.008	0.111	0.001	0.59	0.0287	0.0287	0.952	93.173 $\pm$ 0.799	0.116 $\pm$ 0.001	94.578 $\pm$ 0.718
LSAM	beta46	0.08	0.081	0.001 $\pm$ 0.001	0.008 $\pm$ 0.007	0.000 $\pm$ 0.000	0.018 $\pm$ 0.000	0.000 $\pm$ 0.000	0.018 $\pm$ 0.001	0.051 $\pm$ 0.002	0.226 $\pm$ 0.007	0.080	0.000	0.31	0.0171	0.0171	0.910	91.165 $\pm$ 0.899	0.064 $\pm$ 0.001	98.896 $\pm$ 0.331
LSAM	beta47	0.08	0.080	0.000 $\pm$ 0.001	0.001 $\pm$ 0.007	0.000 $\pm$ 0.000	0.018 $\pm$ 0.000	0.000 $\pm$ 0.000	0.018 $\pm$ 0.001	0.050 $\pm$ 0.003	0.224 $\pm$ 0.007	0.079	-0.001	-0.82	0.0159	0.0159	0.915	92.269 $\pm$ 0.846	0.064 $\pm$ 0.001	99.598 $\pm$ 0.200
LSAM	beta51	0.16	0.159	-0.001 $\pm$ 0.001	-0.007 $\pm$ 0.007	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.054 $\pm$ 0.002	0.232 $\pm$ 0.007	0.158	-0.002	-1.31	0.0349	0.0349	0.994	94.277 $\pm$ 0.736	0.144 $\pm$ 0.001	99.699 $\pm$ 0.174
LSAM	beta52	0.16	0.160	-0.000 $\pm$ 0.002	-0.002 $\pm$ 0.010	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.094 $\pm$ 0.004	0.307 $\pm$ 0.009	0.160	0.000	0.10	0.0471	0.0471	0.976	94.779 $\pm$ 0.705	0.188 $\pm$ 0.001	89.759 $\pm$ 0.961
LSAM	beta53	0.16	0.161	0.001 $\pm$ 0.001	0.005 $\pm$ 0.008	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	0.064 $\pm$ 0.003	0.254 $\pm$ 0.007	0.160	0.000	-0.29	0.0391	0.0391	0.973	94.177 $\pm$ 0.742	0.155 $\pm$ 0.000	98.394 $\pm$ 0.398
LSAM	beta54	0.16	0.162	0.002 $\pm$ 0.002	0.010 $\pm$ 0.009	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.089 $\pm$ 0.004	0.298 $\pm$ 0.008	0.164	0.004	2.24	0.0488	0.0489	0.974	94.177 $\pm$ 0.742	0.182 $\pm$ 0.001	92.369 $\pm$ 0.841
LSAM	beta55	0.08	0.080	-0.000 $\pm$ 0.001	-0.004 $\pm$ 0.012	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.133 $\pm$ 0.007	0.365 $\pm$ 0.011	0.080	0.000	-0.47	0.0283	0.0283	0.927	92.871 $\pm$ 0.815	0.106 $\pm$ 0.001	80.723 $\pm$ 1.250
LSAM	beta56	0.08	0.081	0.001 $\pm$ 0.001	0.015 $\pm$ 0.008	0.000 $\pm$ 0.000	0.020 $\pm$ 0.001	0.000 $\pm$ 0.000	0.020 $\pm$ 0.001	0.061 $\pm$ 0.003	0.246 $\pm$ 0.008	0.081	0.001	1.18	0.0178	0.0178	0.892	90.863 $\pm$ 0.913	0.069 $\pm$ 0.001	97.992 $\pm$ 0.444
LSAM	beta57	0.06	0.061	0.001 $\pm$ 0.001	0.009 $\pm$ 0.015	0.001 $\pm$ 0.000	0.028 $\pm$ 0.001	0.001 $\pm$ 0.000	0.028 $\pm$ 0.001	0.224 $\pm$ 0.011	0.474 $\pm$ 0.014	0.060	0.000	-0.41	0.0285	0.0285	0.950	94.177 $\pm$ 0.742	0.106 $\pm$ 0.001	62.349 $\pm$ 1.535
LSAM	beta58	0.06	0.060	-0.000 $\pm$ 0.001	-0.002 $\pm$ 0.009	0.000 $\pm$ 0.000	0.017 $\pm$ 0.000	0.000 $\pm$ 0.000	0.017 $\pm$ 0.001	0.081 $\pm$ 0.004	0.284 $\pm$ 0.009	0.059	-0.001	-2.19	0.0161	0.0161	0.898	92.570 $\pm$ 0.831	0.060 $\pm$ 0.001	94.779 $\pm$ 0.705
QML	beta41	0.20	0.206	0.006 $\pm$ 0.001	0.030 $\pm$ 0.006	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.041 $\pm$ 0.002	0.202 $\pm$ 0.005	0.206	0.006	3.23	0.0390	0.0395	0.997	95.181 $\pm$ 0.679	0.156 $\pm$ 0.001	99.598 $\pm$ 0.200
QML	beta42	0.20	0.202	0.002 $\pm$ 0.001	0.012 $\pm$ 0.007	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.044 $\pm$ 0.002	0.209 $\pm$ 0.006	0.204	0.004	1.78	0.0406	0.0407	0.969	93.574 $\pm$ 0.777	0.159 $\pm$ 0.000	99.398 $\pm$ 0.245
QML	beta43	0.20	0.196	-0.004 $\pm$ 0.001	-0.020 $\pm$ 0.005	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.022 $\pm$ 0.001	0.147 $\pm$ 0.004	0.195	-0.005	-2.39	0.0293	0.0297	0.977	93.273 $\pm$ 0.794	0.112 $\pm$ 0.000	100.000 $\pm$ 0.000
QML	beta44	0.11	0.094	-0.016 $\pm$ 0.001	-0.141 $\pm$ 0.010	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.114 $\pm$ 0.005	0.338 $\pm$ 0.009	0.094	-0.016	-14.17	0.0334	0.0368	0.953	90.763 $\pm$ 0.917	0.126 $\pm$ 0.001	82.631 $\pm$ 1.200
QML	beta45	0.11	0.107	-0.003 $\pm$ 0.001	-0.025 $\pm$ 0.009	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.082 $\pm$ 0.004	0.286 $\pm$ 0.008	0.107	-0.003	-2.44	0.0306	0.0307	0.951	93.173 $\pm$ 0.799	0.117 $\pm$ 0.001	92.972 $\pm$ 0.810
QML	beta46	0.08	0.093	0.013 $\pm$ 0.001	0.165 $\pm$ 0.007	0.000 $\pm$ 0.000	0.018 $\pm$ 0.000	0.000 $\pm$ 0.000	0.022 $\pm$ 0.001	0.078 $\pm$ 0.004	0.279 $\pm$ 0.007	0.093	0.013	16.51	0.0180	0.0223	0.935	88.253 $\pm$ 1.020	0.066 $\pm$ 0.001	100.000 $\pm$ 0.000
QML	beta47	0.08	0.093	0.013 $\pm$ 0.001	0.162 $\pm$ 0.007	0.000 $\pm$ 0.000	0.018 $\pm$ 0.000	0.000 $\pm$ 0.000	0.022 $\pm$ 0.001	0.075 $\pm$ 0.004	0.274 $\pm$ 0.007	0.092	0.012	15.37	0.0161	0.0202	0.934	89.759 $\pm$ 0.961	0.065 $\pm$ 0.001	100.000 $\pm$ 0.000
QML	beta51	0.16	0.167	0.007 $\pm$ 0.001	0.042 $\pm$ 0.007	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.056 $\pm$ 0.003	0.238 $\pm$ 0.007	0.166	0.006	3.87	0.0360	0.0365	0.997	94.578 $\pm$ 0.718	0.146 $\pm$ 0.001	100.000 $\pm$ 0.000
QML	beta52	0.16	0.143	-0.017 $\pm$ 0.001	-0.107 $\pm$ 0.008	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.077 $\pm$ 0.004	0.277 $\pm$ 0.007	0.143	-0.017	-10.53	0.0407	0.0441	0.978	92.671 $\pm$ 0.826	0.157 $\pm$ 0.000	93.173 $\pm$ 0.799
QML	beta53	0.16	0.136	-0.024 $\pm$ 0.001	-0.152 $\pm$ 0.008	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.085 $\pm$ 0.004	0.291 $\pm$ 0.007	0.135	-0.025	-15.82	0.0387	0.0462	0.966	88.454 $\pm$ 1.013	0.151 $\pm$ 0.000	94.177 $\pm$ 0.742
QML	beta54	0.16	0.162	0.002 $\pm$ 0.001	0.009 $\pm$ 0.008	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	0.065 $\pm$ 0.003	0.256 $\pm$ 0.007	0.164	0.004	2.35	0.0407	0.0409	0.978	94.478 $\pm$ 0.724	0.157 $\pm$ 0.000	97.390 $\pm$ 0.505
QML	beta55	0.08	0.082	0.002 $\pm$ 0.001	0.026 $\pm$ 0.012	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.152 $\pm$ 0.007	0.390 $\pm$ 0.011	0.083	0.003	3.23	0.0299	0.0300	0.934	93.072 $\pm$ 0.805	0.114 $\pm$ 0.001	78.715 $\pm$ 1.297
QML	beta56	0.08	0.087	0.007 $\pm$ 0.001	0.090 $\pm$ 0.008	0.000 $\pm$ 0.000	0.020 $\pm$ 0.001	0.000 $\pm$ 0.000	0.022 $\pm$ 0.001	0.073 $\pm$ 0.004	0.270 $\pm$ 0.008	0.087	0.007	8.60	0.0190	0.0202	0.897	91.265 $\pm$ 0.895	0.072 $\pm$ 0.001	98.795 $\pm$ 0.346
QML	beta57	0.06	0.070	0.010 $\pm$ 0.001	0.172 $\pm$ 0.015	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.258 $\pm$ 0.013	0.508 $\pm$ 0.015	0.069	0.009	15.59	0.0285	0.0300	0.966	94.578 $\pm$ 0.718	0.109 $\pm$ 0.001	75.402 $\pm$ 1.365
QML	beta58	0.06	0.069	0.009 $\pm$ 0.001	0.157 $\pm$ 0.009	0.000 $\pm$ 0.000	0.017 $\pm$ 0.000	0.000 $\pm$ 0.000	0.019 $\pm$ 0.001	0.104 $\pm$ 0.005	0.323 $\pm$ 0.009	0.068	0.008	13.44	0.0163	0.0182	0.899	88.956 $\pm$ 0.993	0.060 $\pm$ 0.001	99.096 $\pm$ 0.300
UPI	beta41	0.20	0.203	0.003 $\pm$ 0.001	0.016 $\pm$ 0.007	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.046 $\pm$ 0.002	0.215 $\pm$ 0.006	0.203	0.003	1.58	0.0414	0.0416	0.968	93.574 $\pm$ 0.777	0.163 $\pm$ 0.001	99.699 $\pm$ 0.174
UPI	beta42	0.20	0.200	0.000 $\pm$ 0.001	0.002 $\pm$ 0.007	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.051 $\pm$ 0.002	0.227 $\pm$ 0.007	0.202	0.002	0.78	0.0442	0.0443	0.951	93.373 $\pm$ 0.788	0.169 $\pm$ 0.001	98.594 $\pm$ 0.373
UPI	beta43	0.20	0.199	-0.001 $\pm$ 0.001	-0.006 $\pm$ 0.005	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.023 $\pm$ 0.001	0.152 $\pm$ 0.004	0.198	-0.002	-1.10	0.0299	0.0300	0.953	93.675 $\pm$ 0.771	0.113 $\pm$ 0.001	100.000 $\pm</$

Table S192

Simulation 2 - Condition 48: Full Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.8
(continued)

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.080	-0.000 ± 0.001	-0.005 ± 0.007	0.000 ± 0.000	0.018 ± 0.000	0.000 ± 0.000	0.018 ± 0.001	0.049 ± 0.002	0.222 ± 0.007	0.079	-0.001	-1.64	0.0164	0.0164	0.908	91.165 ± 0.899	0.063 ± 0.001	99.598 ± 0.200
UPI	beta51	0.16	0.160	-0.000 ± 0.001	-0.002 ± 0.007	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.053 ± 0.002	0.231 ± 0.006	0.158	-0.002	-0.97	0.0355	0.0355	0.987	94.177 ± 0.742	0.143 ± 0.001	99.699 ± 0.174
UPI	beta52	0.16	0.162	0.002 ± 0.002	0.011 ± 0.010	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.091 ± 0.004	0.302 ± 0.009	0.162	0.002	1.54	0.0470	0.0471	0.975	94.277 ± 0.736	0.185 ± 0.001	91.064 ± 0.904
UPI	beta53	0.16	0.160	0.000 ± 0.001	0.002 ± 0.008	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.062 ± 0.003	0.250 ± 0.007	0.159	-0.001	-0.60	0.0388	0.0389	0.980	94.478 ± 0.724	0.154 ± 0.001	98.193 ± 0.422
UPI	beta54	0.16	0.162	0.002 ± 0.002	0.012 ± 0.009	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.088 ± 0.004	0.297 ± 0.008	0.163	0.003	2.03	0.0476	0.0477	0.972	94.578 ± 0.718	0.181 ± 0.001	92.369 ± 0.841
UPI	beta55	0.08	0.080	-0.000 ± 0.001	-0.003 ± 0.011	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.132 ± 0.006	0.363 ± 0.010	0.079	-0.001	-0.70	0.0274	0.0274	0.930	93.474 ± 0.783	0.106 ± 0.001	80.522 ± 1.255
UPI	beta56	0.08	0.081	0.001 ± 0.001	0.013 ± 0.008	0.000 ± 0.000	0.020 ± 0.001	0.000 ± 0.000	0.020 ± 0.001	0.062 ± 0.003	0.249 ± 0.008	0.081	0.001	1.00	0.0177	0.0177	0.887	90.964 ± 0.908	0.069 ± 0.001	97.691 ± 0.476
UPI	beta57	0.06	0.060	-0.000 ± 0.001	-0.005 ± 0.015	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.221 ± 0.011	0.470 ± 0.014	0.059	-0.001	-1.05	0.0276	0.0277	0.949	94.177 ± 0.742	0.105 ± 0.001	63.052 ± 1.529
UPI	beta58	0.06	0.060	-0.000 ± 0.001	-0.007 ± 0.009	0.000 ± 0.000	0.017 ± 0.000	0.000 ± 0.000	0.017 ± 0.001	0.081 ± 0.004	0.285 ± 0.008	0.058	-0.002	-2.67	0.0165	0.0166	0.899	91.566 ± 0.881	0.060 ± 0.001	94.880 ± 0.698

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S193

Simulation 2 - Condition 49: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.4

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LSAM	beta41	0.20	0.192	-0.008 $\pm$ 0.005	-0.038 $\pm$ 0.026	0.007 $\pm$ 0.001	0.085 $\pm$ 0.004	0.007 $\pm$ 0.001	0.086 $\pm$ 0.005	0.183 $\pm$ 0.017	0.428 $\pm$ 0.024	0.196	-0.004	-1.78	0.0808	0.0809	1.143	97.026 $\pm$ 1.036	0.383 $\pm$ 0.008	58.736 $\pm$ 3.002
LSAM	beta42	0.20	0.203	0.003 $\pm$ 0.005	0.014 $\pm$ 0.025	0.007 $\pm$ 0.001	0.082 $\pm$ 0.004	0.007 $\pm$ 0.001	0.082 $\pm$ 0.005	0.170 $\pm$ 0.016	0.412 $\pm$ 0.023	0.204	0.004	1.82	0.0809	0.0810	1.236	97.770 $\pm$ 0.900	0.399 $\pm$ 0.008	57.993 $\pm$ 3.009
LSAM	beta43	0.20	0.201	0.001 $\pm$ 0.005	0.006 $\pm$ 0.026	0.007 $\pm$ 0.001	0.087 $\pm$ 0.004	0.007 $\pm$ 0.001	0.086 $\pm$ 0.005	0.187 $\pm$ 0.016	0.432 $\pm$ 0.023	0.205	0.005	2.74	0.0899	0.0901	1.048	96.283 $\pm$ 1.154	0.356 $\pm$ 0.007	60.595 $\pm$ 2.979
LSAM	beta44	0.11	0.112	0.002 $\pm$ 0.008	0.019 $\pm$ 0.075	0.018 $\pm$ 0.002	0.135 $\pm$ 0.008	0.018 $\pm$ 0.002	0.134 $\pm$ 0.009	1.491 $\pm$ 0.166	1.221 $\pm$ 0.078	0.103	-0.007	-6.41	0.1083	0.1085	1.168	98.513 $\pm$ 0.738	0.616 $\pm$ 0.028	11.896 $\pm$ 1.974
LSAM	beta45	0.11	0.113	0.003 $\pm$ 0.007	0.027 $\pm$ 0.060	0.012 $\pm$ 0.001	0.109 $\pm$ 0.005	0.012 $\pm$ 0.001	0.109 $\pm$ 0.006	0.976 $\pm$ 0.096	0.988 $\pm$ 0.058	0.114	0.004	3.71	0.1067	0.1068	1.150	99.257 $\pm$ 0.524	0.491 $\pm$ 0.017	13.011 $\pm$ 2.051
LSAM	beta46	0.08	0.061	-0.019 $\pm$ 0.010	-0.237 $\pm$ 0.131	0.029 $\pm$ 0.003	0.171 $\pm$ 0.009	0.030 $\pm$ 0.003	0.172 $\pm$ 0.011	4.628 $\pm$ 0.500	2.151 $\pm$ 0.134	0.082	0.002	2.44	0.1475	0.1475	1.248	99.257 $\pm$ 0.524	0.838 $\pm$ 0.044	1.115 $\pm$ 0.640
LSAM	beta47	0.08	0.096	0.016 $\pm$ 0.011	0.198 $\pm$ 0.132	0.030 $\pm$ 0.003	0.173 $\pm$ 0.010	0.030 $\pm$ 0.003	0.174 $\pm$ 0.011	4.706 $\pm$ 0.536	2.169 $\pm$ 0.141	0.082	0.002	2.32	0.1366	0.1366	1.136	98.141 $\pm$ 0.823	0.771 $\pm$ 0.032	1.487 $\pm$ 0.738
LSAM	beta51	0.16	0.162	0.002 $\pm$ 0.007	0.013 $\pm$ 0.045	0.014 $\pm$ 0.001	0.118 $\pm$ 0.005	0.014 $\pm$ 0.001	0.117 $\pm$ 0.006	0.538 $\pm$ 0.048	0.734 $\pm$ 0.040	0.154	-0.006	-3.85	0.1218	0.1219	0.977	95.539 $\pm$ 1.259	0.450 $\pm$ 0.008	34.944 $\pm$ 2.907
LSAM	beta52	0.16	0.166	0.006 $\pm$ 0.006	0.040 $\pm$ 0.036	0.009 $\pm$ 0.001	0.096 $\pm$ 0.005	0.009 $\pm$ 0.001	0.096 $\pm$ 0.006	0.357 $\pm$ 0.037	0.597 $\pm$ 0.036	0.164	0.004	2.59	0.0874	0.0875	1.060	98.141 $\pm$ 0.823	0.397 $\pm$ 0.010	44.610 $\pm$ 3.031
LSAM	beta53	0.16	0.162	0.002 $\pm$ 0.005	0.012 $\pm$ 0.034	0.008 $\pm$ 0.001	0.090 $\pm$ 0.005	0.008 $\pm$ 0.001	0.090 $\pm$ 0.005	0.315 $\pm$ 0.032	0.561 $\pm$ 0.033	0.168	0.008	4.70	0.0995	0.0998	1.131	96.654 $\pm$ 1.096	0.399 $\pm$ 0.009	38.290 $\pm$ 2.964
LSAM	beta54	0.16	0.163	0.003 $\pm$ 0.005	0.017 $\pm$ 0.034	0.008 $\pm$ 0.001	0.089 $\pm$ 0.004	0.008 $\pm$ 0.001	0.089 $\pm$ 0.005	0.310 $\pm$ 0.030	0.557 $\pm$ 0.032	0.158	-0.002	-1.01	0.0893	0.0893	1.116	97.398 $\pm$ 0.971	0.390 $\pm$ 0.008	41.264 $\pm$ 3.002
LSAM	beta55	0.08	0.099	0.019 $\pm$ 0.010	0.232 $\pm$ 0.126	0.027 $\pm$ 0.003	0.166 $\pm$ 0.008	0.028 $\pm$ 0.003	0.166 $\pm$ 0.009	4.322 $\pm$ 0.414	2.079 $\pm$ 0.119	0.082	0.002	2.74	0.1514	0.1514	1.130	97.770 $\pm$ 0.900	0.734 $\pm$ 0.028	8.178 $\pm$ 1.671
LSAM	beta56	0.08	0.070	-0.010 $\pm$ 0.009	-0.121 $\pm$ 0.115	0.023 $\pm$ 0.002	0.151 $\pm$ 0.008	0.023 $\pm$ 0.002	0.151 $\pm$ 0.009	3.575 $\pm$ 0.371	1.891 $\pm$ 0.115	0.080	0.000	-0.21	0.1411	0.1411	1.059	98.141 $\pm$ 0.823	0.628 $\pm$ 0.020	4.461 $\pm$ 1.259
LSAM	beta57	0.06	0.072	0.012 $\pm$ 0.011	0.204 $\pm$ 0.176	0.030 $\pm$ 0.004	0.173 $\pm$ 0.011	0.030 $\pm$ 0.004	0.173 $\pm$ 0.012	8.361 $\pm$ 1.030	2.892 $\pm$ 0.201	0.067	0.007	11.25	0.1600	0.1601	1.225	98.513 $\pm$ 0.738	0.832 $\pm$ 0.051	1.115 $\pm$ 0.640
LSAM	beta58	0.06	0.075	0.015 $\pm$ 0.009	0.252 $\pm$ 0.157	0.024 $\pm$ 0.002	0.155 $\pm$ 0.008	0.024 $\pm$ 0.003	0.155 $\pm$ 0.009	6.685 $\pm$ 0.698	2.586 $\pm$ 0.157	0.062	0.002	2.63	0.1332	0.1332	1.163	98.885 $\pm$ 0.640	0.705 $\pm$ 0.029	2.602 $\pm$ 0.971
QML	beta41	0.20	0.184	-0.016 $\pm$ 0.005	-0.080 $\pm$ 0.023	0.006 $\pm$ 0.001	0.076 $\pm$ 0.004	0.006 $\pm$ 0.001	0.077 $\pm$ 0.004	0.150 $\pm$ 0.014	0.387 $\pm$ 0.022	0.183	-0.017	-8.29	0.0793	0.0810	0.996	92.937 $\pm$ 1.562	0.296 $\pm$ 0.004	73.234 $\pm$ 2.699
QML	beta42	0.20	0.193	-0.007 $\pm$ 0.004	-0.034 $\pm$ 0.021	0.005 $\pm$ 0.000	0.069 $\pm$ 0.004	0.005 $\pm$ 0.000	0.069 $\pm$ 0.004	0.120 $\pm$ 0.012	0.346 $\pm$ 0.020	0.194	-0.006	-3.19	0.0702	0.0705	1.132	95.911 $\pm$ 1.207	0.306 $\pm$ 0.003	77.695 $\pm$ 2.538
QML	beta43	0.20	0.185	-0.015 $\pm$ 0.005	-0.073 $\pm$ 0.023	0.006 $\pm$ 0.000	0.076 $\pm$ 0.003	0.006 $\pm$ 0.000	0.078 $\pm$ 0.004	0.151 $\pm$ 0.012	0.388 $\pm$ 0.019	0.188	-0.012	-5.79	0.0793	0.0801	0.959	94.424 $\pm$ 1.399	0.287 $\pm$ 0.003	72.491 $\pm$ 2.723
QML	beta44	0.11	0.108	-0.002 $\pm$ 0.006	-0.021 $\pm$ 0.055	0.010 $\pm$ 0.001	0.100 $\pm$ 0.005	0.010 $\pm$ 0.001	0.100 $\pm$ 0.006	0.818 $\pm$ 0.077	0.905 $\pm$ 0.051	0.102	-0.008	-7.06	0.1066	0.1069	1.141	95.167 $\pm$ 1.308	0.446 $\pm$ 0.024	20.074 $\pm$ 2.442
QML	beta45	0.11	0.105	-0.005 $\pm$ 0.005	-0.041 $\pm$ 0.047	0.007 $\pm$ 0.001	0.085 $\pm$ 0.004	0.007 $\pm$ 0.001	0.085 $\pm$ 0.005	0.599 $\pm$ 0.052	0.774 $\pm$ 0.041	0.112	0.002	1.56	0.0831	0.0831	1.053	96.283 $\pm$ 1.154	0.352 $\pm$ 0.006	22.305 $\pm$ 2.538
QML	beta46	0.08	0.025	-0.055 $\pm$ 0.004	-0.684 $\pm$ 0.054	0.005 $\pm$ 0.001	0.071 $\pm$ 0.004	0.008 $\pm$ 0.001	0.089 $\pm$ 0.004	1.244 $\pm$ 0.094	1.115 $\pm$ 0.048	0.029	-0.051	-64.14	0.0695	0.0864	1.246	86.989 $\pm$ 2.051	0.345 $\pm$ 0.020	1.859 $\pm$ 0.823
QML	beta47	0.08	0.043	-0.037 $\pm$ 0.004	-0.467 $\pm$ 0.055	0.005 $\pm$ 0.000	0.073 $\pm$ 0.003	0.007 $\pm$ 0.000	0.082 $\pm$ 0.004	1.040 $\pm$ 0.076	1.020 $\pm$ 0.045	0.033	-0.047	-59.32	0.0730	0.0871	1.093	86.617 $\pm$ 2.076	0.311 $\pm$ 0.014	3.346 $\pm$ 1.096
QML	beta51	0.16	0.182	0.022 $\pm$ 0.007	0.137 $\pm$ 0.042	0.012 $\pm$ 0.001	0.110 $\pm$ 0.006	0.013 $\pm$ 0.001	0.112 $\pm$ 0.007	0.489 $\pm$ 0.055	0.699 $\pm$ 0.045	0.179	0.019	11.61	0.1139	0.1154	1.068	98.513 $\pm$ 0.738	0.460 $\pm$ 0.011	39.777 $\pm$ 2.984
QML	beta52	0.16	0.160	0.000 $\pm$ 0.005	0.001 $\pm$ 0.034	0.008 $\pm$ 0.001	0.090 $\pm$ 0.005	0.008 $\pm$ 0.001	0.090 $\pm$ 0.006	0.315 $\pm$ 0.034	0.562 $\pm$ 0.035	0.158	-0.002	-1.08	0.0930	0.0930	1.076	96.654 $\pm$ 1.096	0.380 $\pm$ 0.013	42.751 $\pm$ 3.016
QML	beta53	0.16	0.145	-0.015 $\pm$ 0.005	-0.091 $\pm$ 0.030	0.006 $\pm$ 0.001	0.079 $\pm$ 0.004	0.006 $\pm$ 0.001	0.080 $\pm$ 0.004	0.253 $\pm$ 0.021	0.503 $\pm$ 0.026	0.147	-0.013	-8.39	0.0853	0.0864	1.077	95.911 $\pm$ 1.207	0.335 $\pm$ 0.005	42.379 $\pm$ 3.013
QML	beta54	0.16	0.163	0.003 $\pm$ 0.005	0.021 $\pm$ 0.030	0.006 $\pm$ 0.001	0.079 $\pm$ 0.004	0.006 $\pm$ 0.001	0.079 $\pm$ 0.004	0.246 $\pm$ 0.022	0.496 $\pm$ 0.027	0.162	0.002	1.35	0.0765	0.0765	1.030	95.911 $\pm$ 1.207	0.321 $\pm$ 0.004	53.160 $\pm$ 3.042
QML	beta55	0.08	0.083	0.003 $\pm$ 0.008	0.034 $\pm$ 0.104	0.019 $\pm$ 0.002	0.137 $\pm$ 0.007	0.019 $\pm$ 0.002	0.136 $\pm$ 0.008	2.903 $\pm$ 0.277	1.704 $\pm$ 0.097	0.070	-0.010	-12.50	0.1244	0.1248	1.046	95.539 $\pm$ 1.259	0.560 $\pm$ 0.015	11.896 $\pm$ 1.974
QML	beta56	0.08	0.061	-0.019 $\pm$ 0.007	-0.237 $\pm$ 0.085	0.012 $\pm$ 0.001	0.111 $\pm$ 0.005	0.013 $\pm$ 0.001	0.113 $\pm$ 0.006	1.981 $\pm$ 0.188	1.407 $\pm$ 0.079	0.067	-0.013	-16.68	0.1278	0.1285	1.076	95.539 $\pm$ 1.259	0.469 $\pm$ 0.010	5.576 $\pm$ 1.399
QML	beta57	0.06	0.031	-0.029 $\pm$ 0.005	-0.477 $\pm$ 0.089	0.008 $\pm$ 0.001	0.088 $\pm$ 0.005	0.008 $\pm$ 0.001	0.092 $\pm$ 0.005	2.359 $\pm$ 0.220	1.536 $\pm$ 0.084	0.029	-0.031	-51.71	0.0853	0.0907	1.025	86.245 $\pm$ 2.100	0.353 $\pm$ 0.012	3.346 $\pm$ 1.096
QML	beta58	0.06	0.043	-0.017 $\pm$ 0.004	-0.291 $\pm$ 0.070	0.005 $\pm$ 0.000	0.069 $\pm$ 0.003	0.005 $\pm$ 0.000	0.071 <											

Table S193

Simulation 2 - Condition 49: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.4
(continued)

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.165	0.085 ± 0.026	1.066 ± 0.325	0.181 ± 0.024	0.426 ± 0.029	0.188 ± 0.026	0.434 ± 0.033	29.373 ± 4.112	5.420 ± 0.415	0.085	0.005	6.21	0.2862	0.2863	1.978	98.885 ± 0.640	3.302 ± 0.330	0.372 ± 0.371
UPI	beta51	0.16	0.165	0.005 ± 0.007	0.029 ± 0.046	0.014 ± 0.002	0.120 ± 0.007	0.014 ± 0.002	0.120 ± 0.008	0.563 ± 0.067	0.750 ± 0.051	0.160	0.000	0.27	0.1218	0.1218	1.508	99.628 ± 0.371	0.710 ± 0.068	22.305 ± 2.538
UPI	beta52	0.16	0.161	0.001 ± 0.006	0.007 ± 0.039	0.010 ± 0.001	0.102 ± 0.005	0.010 ± 0.001	0.102 ± 0.006	0.407 ± 0.041	0.638 ± 0.038	0.167	0.007	4.40	0.1060	0.1063	1.647	99.257 ± 0.524	0.660 ± 0.063	27.138 ± 2.711
UPI	beta53	0.16	0.159	-0.001 ± 0.006	-0.004 ± 0.039	0.010 ± 0.001	0.102 ± 0.006	0.010 ± 0.001	0.102 ± 0.007	0.407 ± 0.050	0.638 ± 0.044	0.162	0.002	1.41	0.1177	0.1177	1.598	98.513 ± 0.738	0.640 ± 0.039	17.844 ± 2.334
UPI	beta54	0.16	0.162	0.002 ± 0.006	0.011 ± 0.039	0.010 ± 0.001	0.101 ± 0.006	0.010 ± 0.001	0.101 ± 0.007	0.400 ± 0.050	0.633 ± 0.045	0.149	-0.011	-7.11	0.0925	0.0932	1.554	99.257 ± 0.524	0.618 ± 0.037	22.305 ± 2.538
UPI	beta55	0.08	0.100	0.020 ± 0.014	0.245 ± 0.173	0.051 ± 0.007	0.227 ± 0.015	0.052 ± 0.007	0.227 ± 0.016	8.057 ± 1.048	2.838 ± 0.206	0.071	-0.009	-11.40	0.2035	0.2037	1.778	99.628 ± 0.371	1.579 ± 0.138	0.000 ± 0.000
UPI	beta56	0.08	0.086	0.006 ± 0.012	0.071 ± 0.151	0.039 ± 0.005	0.198 ± 0.012	0.039 ± 0.005	0.197 ± 0.013	6.080 ± 0.715	2.466 ± 0.165	0.081	0.001	0.83	0.1795	0.1795	1.739	99.628 ± 0.371	1.347 ± 0.103	0.743 ± 0.524
UPI	beta57	0.06	0.074	0.014 ± 0.018	0.237 ± 0.296	0.085 ± 0.011	0.291 ± 0.020	0.085 ± 0.011	0.291 ± 0.022	23.506 ± 3.184	4.848 ± 0.364	0.059	-0.001	-0.87	0.2127	0.2127	1.757	99.257 ± 0.524	2.005 ± 0.193	0.000 ± 0.000
UPI	beta58	0.06	0.069	0.009 ± 0.016	0.152 ± 0.263	0.067 ± 0.008	0.259 ± 0.016	0.067 ± 0.008	0.258 ± 0.018	18.562 ± 2.301	4.308 ± 0.300	0.051	-0.009	-14.76	0.2128	0.2130	2.066	99.257 ± 0.524	2.096 ± 0.179	0.743 ± 0.524

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S194

Simulation 2 - Condition 50: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.4

Method	Param.	True	Mean-based										Median-based					Inference		
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.201	0.001 ± 0.002	0.004 ± 0.010	0.003 ± 0.000	0.052 ± 0.002	0.003 ± 0.000	0.052 ± 0.002	0.068 ± 0.004	0.260 ± 0.009	0.201	0.001	0.32	0.0526	0.0526	1.035	96.291 ± 0.728	0.211 ± 0.001	96.439 ± 0.714
LSAM	beta42	0.20	0.204	0.004 ± 0.002	0.022 ± 0.010	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.002	0.063 ± 0.003	0.251 ± 0.008	0.206	0.006	2.86	0.0552	0.0555	1.098	97.033 ± 0.654	0.216 ± 0.001	96.291 ± 0.728
LSAM	beta43	0.20	0.199	-0.001 ± 0.002	-0.007 ± 0.009	0.002 ± 0.000	0.049 ± 0.002	0.002 ± 0.000	0.049 ± 0.002	0.061 ± 0.004	0.246 ± 0.009	0.199	-0.001	-0.31	0.0457	0.0457	1.021	95.104 ± 0.831	0.197 ± 0.001	98.071 ± 0.530
LSAM	beta44	0.11	0.114	0.004 ± 0.003	0.033 ± 0.026	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.003	0.465 ± 0.028	0.682 ± 0.024	0.110	0.000	-0.08	0.0747	0.0747	1.018	96.736 ± 0.684	0.299 ± 0.004	37.092 ± 1.861
LSAM	beta45	0.11	0.111	0.001 ± 0.002	0.010 ± 0.022	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	0.322 ± 0.019	0.567 ± 0.020	0.114	0.004	3.83	0.0625	0.0626	1.019	95.846 ± 0.769	0.249 ± 0.002	44.065 ± 1.912
LSAM	beta46	0.08	0.088	0.008 ± 0.004	0.102 ± 0.048	0.010 ± 0.001	0.101 ± 0.003	0.010 ± 0.001	0.101 ± 0.004	1.592 ± 0.105	1.262 ± 0.048	0.088	0.008	9.78	0.0956	0.0959	1.038	97.181 ± 0.638	0.410 ± 0.006	8.754 ± 1.089
LSAM	beta47	0.08	0.076	-0.004 ± 0.004	-0.055 ± 0.047	0.010 ± 0.001	0.098 ± 0.003	0.010 ± 0.001	0.098 ± 0.004	1.493 ± 0.099	1.222 ± 0.047	0.075	-0.005	-6.22	0.0919	0.0920	1.042	97.774 ± 0.568	0.399 ± 0.007	5.341 ± 0.866
LSAM	beta51	0.16	0.159	-0.001 ± 0.003	-0.006 ± 0.016	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.183 ± 0.011	0.427 ± 0.016	0.155	-0.005	-3.15	0.0662	0.0664	0.965	93.769 ± 0.931	0.259 ± 0.002	69.139 ± 1.779
LSAM	beta52	0.16	0.161	0.001 ± 0.002	0.009 ± 0.013	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	0.110 ± 0.006	0.331 ± 0.011	0.160	0.000	0.30	0.0588	0.0589	1.016	96.291 ± 0.728	0.211 ± 0.001	87.834 ± 1.259
LSAM	beta53	0.16	0.164	0.004 ± 0.002	0.027 ± 0.014	0.003 ± 0.000	0.056 ± 0.002	0.003 ± 0.000	0.056 ± 0.002	0.124 ± 0.007	0.352 ± 0.012	0.166	0.006	3.48	0.0569	0.0572	1.002	95.549 ± 0.794	0.221 ± 0.001	85.015 ± 1.375
LSAM	beta54	0.16	0.159	-0.001 ± 0.002	-0.007 ± 0.012	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.002	0.103 ± 0.005	0.321 ± 0.011	0.160	0.000	-0.08	0.0516	0.0516	1.032	96.291 ± 0.728	0.208 ± 0.001	87.537 ± 1.272
LSAM	beta55	0.08	0.084	0.004 ± 0.003	0.046 ± 0.044	0.008 ± 0.001	0.091 ± 0.003	0.008 ± 0.001	0.091 ± 0.003	1.281 ± 0.084	1.132 ± 0.043	0.082	0.002	3.07	0.0822	0.0822	0.989	96.884 ± 0.669	0.351 ± 0.004	16.172 ± 1.418
LSAM	beta56	0.08	0.082	0.002 ± 0.003	0.030 ± 0.039	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.081 ± 0.003	1.028 ± 0.059	1.014 ± 0.035	0.078	-0.002	-2.60	0.0866	0.0867	1.001	95.846 ± 0.769	0.318 ± 0.003	16.914 ± 1.444
LSAM	beta57	0.06	0.064	0.004 ± 0.004	0.068 ± 0.068	0.011 ± 0.001	0.107 ± 0.004	0.011 ± 0.001	0.107 ± 0.004	3.152 ± 0.216	1.775 ± 0.070	0.063	0.003	5.19	0.0966	0.0967	0.971	96.291 ± 0.728	0.406 ± 0.007	3.412 ± 0.699
LSAM	beta58	0.06	0.063	0.003 ± 0.003	0.050 ± 0.055	0.007 ± 0.000	0.085 ± 0.002	0.007 ± 0.000	0.085 ± 0.003	2.006 ± 0.116	1.416 ± 0.049	0.062	0.002	4.15	0.0842	0.0842	1.052	97.033 ± 0.654	0.350 ± 0.005	6.825 ± 0.971
QML	beta41	0.20	0.192	-0.008 ± 0.002	-0.040 ± 0.009	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.047 ± 0.002	0.055 ± 0.003	0.234 ± 0.008	0.189	-0.011	-5.35	0.0475	0.0487	1.011	95.104 ± 0.831	0.183 ± 0.002	98.665 ± 0.442
QML	beta42	0.20	0.196	-0.004 ± 0.002	-0.022 ± 0.009	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.002	0.052 ± 0.003	0.229 ± 0.008	0.196	-0.004	-1.77	0.0504	0.0505	1.047	95.401 ± 0.807	0.187 ± 0.001	98.368 ± 0.488
QML	beta43	0.20	0.184	-0.016 ± 0.002	-0.080 ± 0.009	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.048 ± 0.002	0.057 ± 0.003	0.240 ± 0.008	0.184	-0.016	-7.79	0.0436	0.0463	0.991	92.582 ± 1.009	0.175 ± 0.001	98.516 ± 0.466
QML	beta44	0.11	0.110	0.000 ± 0.002	0.002 ± 0.022	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	0.312 ± 0.016	0.558 ± 0.018	0.110	0.000	0.19	0.0662	0.0662	1.041	97.774 ± 0.568	0.251 ± 0.002	44.807 ± 1.916
QML	beta45	0.11	0.106	-0.004 ± 0.002	-0.033 ± 0.019	0.003 ± 0.000	0.055 ± 0.002	0.003 ± 0.000	0.055 ± 0.002	0.247 ± 0.014	0.497 ± 0.017	0.106	-0.004	-3.28	0.0549	0.0551	1.003	94.510 ± 0.877	0.215 ± 0.001	50.000 ± 1.926
QML	beta46	0.08	0.032	-0.048 ± 0.002	-0.595 ± 0.023	0.002 ± 0.000	0.047 ± 0.001	0.004 ± 0.000	0.067 ± 0.002	0.699 ± 0.032	0.836 ± 0.021	0.030	-0.050	-62.59	0.0435	0.0663	1.037	78.932 ± 1.571	0.191 ± 0.002	6.380 ± 0.941
QML	beta47	0.08	0.029	-0.051 ± 0.002	-0.636 ± 0.021	0.002 ± 0.000	0.044 ± 0.001	0.005 ± 0.000	0.068 ± 0.001	0.712 ± 0.029	0.844 ± 0.019	0.028	-0.052	-64.66	0.0444	0.0682	1.035	72.849 ± 1.713	0.180 ± 0.002	4.896 ± 0.831
QML	beta51	0.16	0.180	0.020 ± 0.003	0.125 ± 0.016	0.004 ± 0.000	0.066 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.188 ± 0.012	0.433 ± 0.016	0.180	0.020	12.39	0.0614	0.0645	0.977	93.472 ± 0.951	0.254 ± 0.002	82.196 ± 1.474
QML	beta52	0.16	0.154	-0.006 ± 0.002	-0.039 ± 0.012	0.003 ± 0.000	0.051 ± 0.002	0.003 ± 0.000	0.052 ± 0.002	0.105 ± 0.006	0.323 ± 0.011	0.149	-0.011	-6.64	0.0529	0.0540	1.009	95.104 ± 0.831	0.203 ± 0.001	86.202 ± 1.328
QML	beta53	0.16	0.147	-0.013 ± 0.002	-0.082 ± 0.012	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	0.104 ± 0.005	0.323 ± 0.010	0.145	-0.015	-9.38	0.0522	0.0543	1.002	93.620 ± 0.941	0.197 ± 0.001	83.976 ± 1.413
QML	beta54	0.16	0.156	-0.004 ± 0.002	-0.023 ± 0.012	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.002	0.091 ± 0.005	0.301 ± 0.010	0.156	-0.004	-2.66	0.0496	0.0498	1.012	95.549 ± 0.794	0.191 ± 0.001	92.136 ± 1.037
QML	beta55	0.08	0.081	0.001 ± 0.003	0.019 ± 0.037	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.076 ± 0.003	0.914 ± 0.052	0.956 ± 0.033	0.083	0.003	3.75	0.0722	0.0723	1.034	95.697 ± 0.782	0.310 ± 0.003	18.991 ± 1.511
QML	beta56	0.08	0.073	-0.007 ± 0.003	-0.093 ± 0.033	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.725 ± 0.037	0.851 ± 0.027	0.072	-0.008	-10.27	0.0721	0.0726	1.003	94.955 ± 0.843	0.266 ± 0.002	18.694 ± 1.502
QML	beta57	0.06	0.025	-0.035 ± 0.002	-0.579 ± 0.032	0.003 ± 0.000	0.050 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	1.041 ± 0.050	1.020 ± 0.028	0.025	-0.035	-59.07	0.0492	0.0606	1.018	82.938 ± 1.449	0.201 ± 0.003	4.154 ± 0.769
QML	beta58	0.06	0.032	-0.028 ± 0.002	-0.471 ± 0.027	0.002 ± 0.000	0.041 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.698 ± 0.031	0.835 ± 0.022	0.029	-0.031	-51.98	0.0443	0.0542	1.028	85.015 ± 1.375	0.167 ± 0.002	7.270 ± 1.000
UPI	beta41	0.20	0.202	0.002 ± 0.002	0.010 ± 0.011	0.003 ± 0.000	0.058 ± 0.002	0.003 ± 0.000	0.058 ± 0.002	0.084 ± 0.006	0.290 ± 0.012	0.205	0.005	2.34	0.0670	0.0671	1.246	97.478 ± 0.604	0.283 ± 0.008	85.460 ± 1.358
UPI	beta42	0.20	0.204	0.004 ± 0.002	0.019 ± 0.011	0.003 ± 0.000	0.057 ± 0.002	0.003 ± 0.000	0.057 ± 0.002	0.082 ± 0.006	0.287 ± 0.011	0.203	0.003	1.41	0.0620	0.0620	1.341	98.516 ± 0.466	0.301 ± 0.013	84.718 ± 1.386
UPI	beta43	0.20	0.200	-0.000 ± 0.002	-0.001 ± 0.010	0.003 ± 0.000	0.054 ± 0.002	0.003 ± 0.000	0.054 ± 0.002	0.072 ± 0.005	0.269 ± 0.010	0.200	0.000	-0.17	0.0575	0.0575	1.193	96.736 ± 0.684	0.252 ± 0.006	90.950 ± 1.105
UPI	beta44	0.11	0.117	0.007 ± 0.004	0.068 ± 0.032	0.008 ± 0.001	0.092 ± 0.004	0.009 ± 0.001	0.092 ± 0.004	0.705 ± 0.056	0.840 ± 0.037	0.111	0.001	0.95	0.0889	0.0889	1.379	99.555 ± 0.256	0.498 ± 0.017	18.694 ± 1.502
UPI	beta45	0.11	0.111	0.001 ± 0.003	0.013 ± 0.025	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.003	0.430 ± 0.027	0.656 ± 0.024	0.110	0.000	0.23	0.0785	0.0785	1.292	98.368 ± 0.488	0.365 ± 0.008	24.629 ± 1.660
UPI	beta46	0.08	0.100	0.020 ± 0.006	0.247 ± 0.072	0.022 ± 0.002	0.150 ± 0.006	0.023 ± 0.002	0.151 ± 0.007	3.555 ± 0.290	1.886 ± 0.085	0.085	0.005	6.76	0.1385	0.1386	1.664	98.368 ± 0.488	0.976 ± 0.057	1.632 ± 0.488

Table S194

Simulation 2 - Condition 50: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.4
(continued)

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.087	0.007 ± 0.006	0.089 ± 0.076	0.025 ± 0.002	0.157 ± 0.007	0.025 ± 0.002	0.157 ± 0.007	3.863 ± 0.329	1.965 ± 0.092	0.075	-0.005	-6.66	0.1367	0.1368	1.558	98.368 ± 0.488	0.960 ± 0.046	0.445 ± 0.256
UPI	beta51	0.16	0.162	0.002 ± 0.003	0.012 ± 0.016	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.003	0.178 ± 0.012	0.422 ± 0.016	0.159	-0.001	-0.60	0.0644	0.0644	1.098	95.252 ± 0.819	0.291 ± 0.008	65.282 ± 1.834
UPI	beta52	0.16	0.161	0.001 ± 0.002	0.009 ± 0.013	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	0.112 ± 0.006	0.334 ± 0.011	0.162	0.002	1.03	0.0569	0.0569	1.158	97.626 ± 0.586	0.243 ± 0.005	82.047 ± 1.478
UPI	beta53	0.16	0.162	0.002 ± 0.002	0.012 ± 0.014	0.003 ± 0.000	0.057 ± 0.002	0.003 ± 0.000	0.057 ± 0.002	0.128 ± 0.007	0.358 ± 0.012	0.162	0.002	1.50	0.0629	0.0630	1.186	97.329 ± 0.621	0.266 ± 0.008	73.887 ± 1.692
UPI	beta54	0.16	0.159	-0.001 ± 0.002	-0.009 ± 0.012	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	0.105 ± 0.006	0.324 ± 0.011	0.160	0.000	-0.18	0.0508	0.0508	1.173	97.626 ± 0.586	0.239 ± 0.006	81.602 ± 1.492
UPI	beta55	0.08	0.086	0.006 ± 0.004	0.070 ± 0.045	0.009 ± 0.001	0.094 ± 0.003	0.009 ± 0.001	0.094 ± 0.004	1.375 ± 0.093	1.173 ± 0.046	0.084	0.004	5.60	0.0881	0.0882	1.249	98.665 ± 0.442	0.459 ± 0.017	7.418 ± 1.009
UPI	beta56	0.08	0.084	0.004 ± 0.003	0.046 ± 0.043	0.008 ± 0.001	0.090 ± 0.003	0.008 ± 0.001	0.090 ± 0.004	1.272 ± 0.087	1.128 ± 0.044	0.078	-0.002	-2.83	0.0853	0.0853	1.188	98.220 ± 0.509	0.420 ± 0.010	10.089 ± 1.160
UPI	beta57	0.06	0.065	0.005 ± 0.005	0.080 ± 0.083	0.017 ± 0.001	0.130 ± 0.005	0.017 ± 0.001	0.130 ± 0.006	4.668 ± 0.357	2.160 ± 0.093	0.059	-0.001	-0.94	0.1112	0.1112	1.436	98.665 ± 0.442	0.730 ± 0.036	0.593 ± 0.296
UPI	beta58	0.06	0.058	-0.002 ± 0.004	-0.032 ± 0.069	0.011 ± 0.001	0.107 ± 0.004	0.011 ± 0.001	0.107 ± 0.004	3.179 ± 0.232	1.783 ± 0.074	0.048	-0.012	-20.79	0.0863	0.0872	1.481	97.033 ± 0.654	0.621 ± 0.030	1.335 ± 0.442

Note. MCSE = Monte Carlo Standard Error shown as \u000b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S195

Simulation 2 - Condition 51: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.6

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LSAM	beta41	0.20	0.201	0.001 $\pm$ 0.002	0.005 $\pm$ 0.010	0.003 $\pm$ 0.000	0.058 $\pm$ 0.001	0.003 $\pm$ 0.000	0.058 $\pm$ 0.002	0.085 $\pm$ 0.004	0.292 $\pm$ 0.008	0.197	-0.003	-1.57	0.0631	0.0632	1.021	96.858 $\pm$ 0.574	0.233 $\pm$ 0.001	92.849 $\pm$ 0.848
LSAM	beta42	0.20	0.199	-0.001 $\pm$ 0.002	-0.007 $\pm$ 0.010	0.004 $\pm$ 0.000	0.060 $\pm$ 0.001	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	0.091 $\pm$ 0.004	0.301 $\pm$ 0.009	0.198	-0.002	-1.24	0.0595	0.0596	1.009	95.450 $\pm$ 0.686	0.238 $\pm$ 0.001	91.224 $\pm$ 0.931
LSAM	beta43	0.20	0.202	0.002 $\pm$ 0.002	0.009 $\pm$ 0.009	0.003 $\pm$ 0.000	0.057 $\pm$ 0.001	0.003 $\pm$ 0.000	0.057 $\pm$ 0.002	0.080 $\pm$ 0.004	0.283 $\pm$ 0.008	0.201	0.001	0.44	0.0544	0.0544	0.995	94.800 $\pm$ 0.731	0.221 $\pm$ 0.001	94.800 $\pm$ 0.731
LSAM	beta44	0.11	0.108	-0.002 $\pm$ 0.002	-0.021 $\pm$ 0.021	0.005 $\pm$ 0.000	0.071 $\pm$ 0.002	0.005 $\pm$ 0.000	0.071 $\pm$ 0.002	0.421 $\pm$ 0.021	0.649 $\pm$ 0.019	0.111	0.001	1.02	0.0703	0.0703	0.989	95.233 $\pm$ 0.701	0.277 $\pm$ 0.002	38.137 $\pm$ 1.599
LSAM	beta45	0.11	0.114	0.004 $\pm$ 0.002	0.038 $\pm$ 0.019	0.004 $\pm$ 0.000	0.063 $\pm$ 0.001	0.004 $\pm$ 0.000	0.063 $\pm$ 0.002	0.332 $\pm$ 0.016	0.576 $\pm$ 0.017	0.114	0.004	3.64	0.0631	0.0633	0.995	96.316 $\pm$ 0.620	0.247 $\pm$ 0.001	46.154 $\pm$ 1.641
LSAM	beta46	0.08	0.079	-0.001 $\pm$ 0.003	-0.012 $\pm$ 0.038	0.008 $\pm$ 0.000	0.092 $\pm$ 0.002	0.008 $\pm$ 0.000	0.092 $\pm$ 0.003	1.310 $\pm$ 0.065	1.144 $\pm$ 0.034	0.080	0.000	-0.01	0.0902	0.0902	0.977	94.800 $\pm$ 0.731	0.351 $\pm$ 0.003	12.893 $\pm$ 1.103
LSAM	beta47	0.08	0.086	0.006 $\pm$ 0.003	0.071 $\pm$ 0.037	0.008 $\pm$ 0.000	0.091 $\pm$ 0.002	0.008 $\pm$ 0.000	0.091 $\pm$ 0.003	1.288 $\pm$ 0.069	1.135 $\pm$ 0.036	0.084	0.004	4.46	0.0839	0.0840	0.982	94.908 $\pm$ 0.724	0.349 $\pm$ 0.003	13.001 $\pm$ 1.107
LSAM	beta51	0.16	0.159	-0.001 $\pm$ 0.003	-0.005 $\pm$ 0.017	0.007 $\pm$ 0.000	0.081 $\pm$ 0.002	0.007 $\pm$ 0.000	0.081 $\pm$ 0.002	0.257 $\pm$ 0.013	0.507 $\pm$ 0.015	0.155	-0.005	-3.26	0.0777	0.0779	0.904	91.874 $\pm$ 0.899	0.287 $\pm$ 0.002	57.421 $\pm$ 1.628
LSAM	beta52	0.16	0.164	0.004 $\pm$ 0.002	0.026 $\pm$ 0.013	0.004 $\pm$ 0.000	0.063 $\pm$ 0.001	0.004 $\pm$ 0.000	0.063 $\pm$ 0.002	0.155 $\pm$ 0.007	0.394 $\pm$ 0.011	0.161	0.001	0.79	0.0589	0.0590	0.962	94.475 $\pm$ 0.752	0.237 $\pm$ 0.001	78.332 $\pm$ 1.356
LSAM	beta53	0.16	0.163	0.003 $\pm$ 0.002	0.017 $\pm$ 0.013	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	0.168 $\pm$ 0.008	0.410 $\pm$ 0.012	0.161	0.001	0.91	0.0650	0.0650	0.953	93.391 $\pm$ 0.818	0.245 $\pm$ 0.001	73.348 $\pm$ 1.455
LSAM	beta54	0.16	0.163	0.003 $\pm$ 0.002	0.018 $\pm$ 0.012	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	0.142 $\pm$ 0.007	0.377 $\pm$ 0.011	0.163	0.003	1.80	0.0564	0.0564	0.993	94.908 $\pm$ 0.724	0.235 $\pm$ 0.001	78.873 $\pm$ 1.344
LSAM	beta55	0.08	0.082	0.002 $\pm$ 0.003	0.025 $\pm$ 0.037	0.008 $\pm$ 0.000	0.091 $\pm$ 0.002	0.008 $\pm$ 0.000	0.091 $\pm$ 0.003	1.296 $\pm$ 0.062	1.138 $\pm$ 0.033	0.084	0.004	4.67	0.0904	0.0905	0.942	94.691 $\pm$ 0.738	0.336 $\pm$ 0.003	20.152 $\pm$ 1.320
LSAM	beta56	0.08	0.078	-0.002 $\pm$ 0.003	-0.019 $\pm$ 0.036	0.008 $\pm$ 0.000	0.087 $\pm$ 0.002	0.008 $\pm$ 0.000	0.087 $\pm$ 0.003	1.186 $\pm$ 0.058	1.089 $\pm$ 0.032	0.078	-0.002	-2.65	0.0866	0.0866	0.924	93.716 $\pm$ 0.799	0.316 $\pm$ 0.003	19.285 $\pm$ 1.299
LSAM	beta57	0.06	0.061	0.001 $\pm$ 0.003	0.014 $\pm$ 0.051	0.009 $\pm$ 0.000	0.093 $\pm$ 0.002	0.009 $\pm$ 0.000	0.093 $\pm$ 0.003	2.390 $\pm$ 0.119	1.546 $\pm$ 0.046	0.060	0.000	0.37	0.0887	0.0887	0.976	95.125 $\pm$ 0.709	0.355 $\pm$ 0.003	8.451 $\pm$ 0.916
LSAM	beta58	0.06	0.066	0.006 $\pm$ 0.003	0.104 $\pm$ 0.045	0.007 $\pm$ 0.000	0.082 $\pm$ 0.002	0.007 $\pm$ 0.000	0.082 $\pm$ 0.003	1.878 $\pm$ 0.100	1.370 $\pm$ 0.043	0.063	0.003	4.39	0.0804	0.0805	1.010	96.100 $\pm$ 0.637	0.325 $\pm$ 0.003	9.426 $\pm$ 0.962
QML	beta41	0.20	0.194	-0.006 $\pm$ 0.002	-0.031 $\pm$ 0.009	0.003 $\pm$ 0.000	0.056 $\pm$ 0.001	0.003 $\pm$ 0.000	0.057 $\pm$ 0.002	0.081 $\pm$ 0.004	0.284 $\pm$ 0.008	0.191	-0.009	-4.38	0.0613	0.0619	0.996	96.533 $\pm$ 0.602	0.221 $\pm$ 0.001	93.283 $\pm$ 0.824
QML	beta42	0.20	0.195	-0.005 $\pm$ 0.002	-0.024 $\pm$ 0.009	0.003 $\pm$ 0.000	0.057 $\pm$ 0.001	0.003 $\pm$ 0.000	0.057 $\pm$ 0.002	0.082 $\pm$ 0.004	0.287 $\pm$ 0.008	0.194	-0.006	-3.24	0.0557	0.0561	1.005	95.558 $\pm$ 0.678	0.225 $\pm$ 0.001	93.283 $\pm$ 0.824
QML	beta43	0.20	0.193	-0.007 $\pm$ 0.002	-0.036 $\pm$ 0.009	0.003 $\pm$ 0.000	0.054 $\pm$ 0.001	0.003 $\pm$ 0.000	0.055 $\pm$ 0.002	0.075 $\pm$ 0.004	0.274 $\pm$ 0.008	0.192	-0.008	-4.16	0.0547	0.0553	0.989	93.933 $\pm$ 0.786	0.211 $\pm$ 0.001	94.150 $\pm$ 0.773
QML	beta44	0.11	0.106	-0.004 $\pm$ 0.002	-0.038 $\pm$ 0.021	0.005 $\pm$ 0.000	0.070 $\pm$ 0.002	0.005 $\pm$ 0.000	0.070 $\pm$ 0.002	0.410 $\pm$ 0.022	0.640 $\pm$ 0.020	0.110	0.000	-0.03	0.0721	0.0721	0.990	96.100 $\pm$ 0.637	0.273 $\pm$ 0.003	41.928 $\pm$ 1.624
QML	beta45	0.11	0.112	0.002 $\pm$ 0.002	0.022 $\pm$ 0.018	0.004 $\pm$ 0.000	0.062 $\pm$ 0.001	0.004 $\pm$ 0.000	0.062 $\pm$ 0.002	0.314 $\pm$ 0.015	0.560 $\pm$ 0.016	0.110	0.000	-0.26	0.0603	0.0603	0.996	95.666 $\pm$ 0.670	0.241 $\pm$ 0.002	48.646 $\pm$ 1.645
QML	beta46	0.08	0.047	-0.033 $\pm$ 0.002	-0.408 $\pm$ 0.025	0.004 $\pm$ 0.000	0.060 $\pm$ 0.001	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.724 $\pm$ 0.031	0.851 $\pm$ 0.021	0.045	-0.035	-43.14	0.0606	0.0698	0.992	86.241 $\pm$ 1.134	0.232 $\pm$ 0.003	9.317 $\pm$ 0.957
QML	beta47	0.08	0.051	-0.029 $\pm$ 0.002	-0.359 $\pm$ 0.024	0.003 $\pm$ 0.000	0.058 $\pm$ 0.002	0.004 $\pm$ 0.000	0.065 $\pm$ 0.002	0.660 $\pm$ 0.029	0.812 $\pm$ 0.021	0.048	-0.032	-39.71	0.0540	0.0626	0.978	86.891 $\pm$ 1.111	0.224 $\pm$ 0.002	10.726 $\pm$ 1.019
QML	beta51	0.16	0.172	0.012 $\pm$ 0.003	0.077 $\pm$ 0.017	0.006 $\pm$ 0.000	0.080 $\pm$ 0.002	0.007 $\pm$ 0.000	0.081 $\pm$ 0.002	0.257 $\pm$ 0.013	0.507 $\pm$ 0.015	0.167	0.007	4.42	0.0780	0.0783	1.019	93.174 $\pm$ 0.830	0.321 $\pm$ 0.017	61.538 $\pm$ 1.601
QML	beta52	0.16	0.152	-0.008 $\pm$ 0.002	-0.053 $\pm$ 0.014	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	0.172 $\pm$ 0.008	0.415 $\pm$ 0.012	0.149	-0.011	-6.66	0.0627	0.0636	0.973	93.933 $\pm$ 0.786	0.252 $\pm$ 0.004	69.881 $\pm$ 1.510
QML	beta53	0.16	0.144	-0.016 $\pm$ 0.002	-0.102 $\pm$ 0.013	0.004 $\pm$ 0.000	0.062 $\pm$ 0.001	0.004 $\pm$ 0.000	0.064 $\pm$ 0.002	0.162 $\pm$ 0.007	0.402 $\pm$ 0.011	0.143	-0.017	-10.70	0.0603	0.0627	0.988	93.283 $\pm$ 0.824	0.241 $\pm$ 0.003	66.956 $\pm$ 1.548
QML	beta54	0.16	0.162	0.002 $\pm$ 0.002	0.011 $\pm$ 0.012	0.003 $\pm$ 0.000	0.059 $\pm$ 0.002	0.003 $\pm$ 0.000	0.059 $\pm$ 0.002	0.137 $\pm$ 0.007	0.370 $\pm$ 0.011	0.162	0.002	1.08	0.0558	0.0559	1.015	95.016 $\pm$ 0.716	0.235 $\pm$ 0.004	79.523 $\pm$ 1.328
QML	beta55	0.08	0.081	0.001 $\pm$ 0.003	0.013 $\pm$ 0.036	0.008 $\pm$ 0.000	0.088 $\pm$ 0.002	0.008 $\pm$ 0.000	0.088 $\pm$ 0.003	1.211 $\pm$ 0.057	1.101 $\pm$ 0.032	0.084	0.004	5.14	0.0887	0.0888	0.976	94.691 $\pm$ 0.738	0.337 $\pm$ 0.004	20.477 $\pm$ 1.328
QML	beta56	0.08	0.075	-0.005 $\pm$ 0.003	-0.067 $\pm$ 0.033	0.006 $\pm$ 0.000	0.080 $\pm$ 0.002	0.006 $\pm$ 0.000	0.081 $\pm$ 0.002	1.015 $\pm$ 0.049	1.008 $\pm$ 0.029	0.073	-0.007	-9.33	0.0815	0.0819	0.948	93.283 $\pm$ 0.824	0.299 $\pm$ 0.002	17.551 $\pm$ 1.252
QML	beta57	0.06	0.037	-0.023 $\pm$ 0.002	-0.390 $\pm$ 0.034	0.004 $\pm$ 0.000	0.062 $\pm$ 0.002	0.004 $\pm$ 0.000	0.067 $\pm$ 0.002	1.235 $\pm$ 0.057	1.111 $\pm$ 0.030	0.032	-0.028	-46.61	0.0603	0.0664	1.025	87.974 $\pm$ 1.071	0.251 $\pm$ 0.006	5.092 $\pm$ 0.724
QML	beta58	0.06	0.044	-0.016 $\pm$ 0.002	-0.258 $\pm$ 0.030	0.003 $\pm$ 0.000	0.054 $\pm$ 0.001	0.003 $\pm$ 0.000												

Table S195

Simulation 2 - Condition 51: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.6
(continued)

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.094	0.014 ± 0.004	0.178 ± 0.050	0.015 ± 0.001	0.122 ± 0.004	0.015 ± 0.001	0.122 ± 0.004	2.342 ± 0.147	1.530 ± 0.054	0.084	0.004	5.20	0.1088	0.1089	1.287	98.050 ± 0.455	0.614 ± 0.022	4.009 ± 0.646
UPI	beta51	0.16	0.160	0.000 ± 0.003	0.002 ± 0.016	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	0.243 ± 0.012	0.493 ± 0.015	0.157	-0.003	-2.18	0.0765	0.0766	0.968	93.716 ± 0.799	0.300 ± 0.003	57.421 ± 1.628
UPI	beta52	0.16	0.164	0.004 ± 0.002	0.025 ± 0.013	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.154 ± 0.007	0.393 ± 0.011	0.162	0.002	1.23	0.0639	0.0639	0.980	94.800 ± 0.731	0.241 ± 0.001	76.815 ± 1.389
UPI	beta53	0.16	0.162	0.002 ± 0.002	0.013 ± 0.014	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.168 ± 0.008	0.410 ± 0.012	0.162	0.002	1.19	0.0652	0.0652	1.003	94.475 ± 0.752	0.258 ± 0.002	71.073 ± 1.492
UPI	beta54	0.16	0.163	0.003 ± 0.002	0.021 ± 0.012	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.143 ± 0.007	0.378 ± 0.011	0.162	0.002	1.52	0.0572	0.0573	1.016	95.016 ± 0.716	0.241 ± 0.001	77.790 ± 1.368
UPI	beta55	0.08	0.082	0.002 ± 0.003	0.028 ± 0.039	0.009 ± 0.000	0.094 ± 0.002	0.009 ± 0.000	0.094 ± 0.003	1.370 ± 0.069	1.171 ± 0.035	0.080	0.000	0.21	0.0933	0.0933	1.017	96.641 ± 0.593	0.373 ± 0.004	14.626 ± 1.163
UPI	beta56	0.08	0.078	-0.002 ± 0.003	-0.022 ± 0.037	0.008 ± 0.000	0.089 ± 0.002	0.008 ± 0.000	0.089 ± 0.003	1.238 ± 0.063	1.113 ± 0.034	0.074	-0.006	-6.88	0.0825	0.0827	1.014	95.883 ± 0.654	0.354 ± 0.005	14.518 ± 1.160
UPI	beta57	0.06	0.067	0.007 ± 0.004	0.115 ± 0.059	0.012 ± 0.001	0.108 ± 0.003	0.012 ± 0.001	0.108 ± 0.004	3.239 ± 0.187	1.800 ± 0.060	0.059	-0.001	-1.03	0.0952	0.0952	1.128	97.725 ± 0.491	0.477 ± 0.012	4.550 ± 0.686
UPI	beta58	0.06	0.065	0.005 ± 0.003	0.092 ± 0.050	0.008 ± 0.000	0.092 ± 0.002	0.008 ± 0.000	0.092 ± 0.003	2.337 ± 0.128	1.529 ± 0.049	0.060	0.000	-0.10	0.0861	0.0861	1.248	97.291 ± 0.534	0.448 ± 0.012	4.117 ± 0.654

Note. MCSE = Monte Carlo Standard Error shown as \u000b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S196

Simulation 2 - Condition 52: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.6

Method	Param.	True	Mean-based										Median-based					Inference		
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LSAM	beta41	0.20	0.201	0.001 $\pm$ 0.001	0.006 $\pm$ 0.006	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.032 $\pm$ 0.002	0.179 $\pm$ 0.005	0.200	0.000	-0.23	0.0343	0.0343	1.018	95.181 $\pm$ 0.679	0.143 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta42	0.20	0.199	-0.001 $\pm$ 0.001	-0.003 $\pm$ 0.006	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.032 $\pm$ 0.001	0.179 $\pm$ 0.005	0.199	-0.001	-0.70	0.0345	0.0345	1.039	96.084 $\pm$ 0.615	0.146 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta43	0.20	0.202	0.002 $\pm$ 0.001	0.011 $\pm$ 0.005	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.029 $\pm$ 0.001	0.172 $\pm$ 0.005	0.202	0.002	1.17	0.0341	0.0342	1.018	94.578 $\pm$ 0.718	0.137 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta44	0.11	0.109	-0.001 $\pm$ 0.001	-0.005 $\pm$ 0.013	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.167 $\pm$ 0.008	0.409 $\pm$ 0.012	0.110	0.000	0.31	0.0434	0.0434	0.942	94.378 $\pm$ 0.730	0.166 $\pm$ 0.001	72.590 $\pm$ 1.413
LSAM	beta45	0.11	0.109	-0.001 $\pm$ 0.001	-0.009 $\pm$ 0.011	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.125 $\pm$ 0.006	0.354 $\pm$ 0.010	0.108	-0.002	-1.44	0.0384	0.0384	0.978	95.181 $\pm$ 0.679	0.149 $\pm$ 0.001	81.024 $\pm$ 1.242
LSAM	beta46	0.08	0.084	0.004 $\pm$ 0.002	0.045 $\pm$ 0.022	0.003 $\pm$ 0.000	0.056 $\pm$ 0.001	0.003 $\pm$ 0.000	0.056 $\pm$ 0.002	0.493 $\pm$ 0.024	0.702 $\pm$ 0.020	0.082	0.002	2.69	0.0556	0.0556	0.965	94.378 $\pm$ 0.730	0.212 $\pm$ 0.001	35.542 $\pm$ 1.517
LSAM	beta47	0.08	0.084	0.004 $\pm$ 0.002	0.045 $\pm$ 0.021	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.433 $\pm$ 0.020	0.658 $\pm$ 0.018	0.081	0.001	1.35	0.0521	0.0522	1.016	96.185 $\pm$ 0.607	0.209 $\pm$ 0.001	32.530 $\pm$ 1.484
LSAM	beta51	0.16	0.160	0.000 $\pm$ 0.002	0.001 $\pm$ 0.010	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.091 $\pm$ 0.004	0.302 $\pm$ 0.008	0.160	0.000	0.13	0.0487	0.0487	0.930	93.072 $\pm$ 0.805	0.176 $\pm$ 0.001	94.076 $\pm$ 0.748
LSAM	beta52	0.16	0.159	-0.001 $\pm$ 0.001	-0.005 $\pm$ 0.007	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.052 $\pm$ 0.002	0.228 $\pm$ 0.006	0.159	-0.001	-0.50	0.0353	0.0353	1.003	94.980 $\pm$ 0.692	0.143 $\pm$ 0.000	99.297 $\pm$ 0.265
LSAM	beta53	0.16	0.162	0.002 $\pm$ 0.001	0.012 $\pm$ 0.008	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.057 $\pm$ 0.003	0.238 $\pm$ 0.007	0.161	0.001	0.86	0.0380	0.0380	1.006	94.880 $\pm$ 0.698	0.150 $\pm$ 0.000	98.996 $\pm$ 0.316
LSAM	beta54	0.16	0.161	0.001 $\pm$ 0.001	0.008 $\pm$ 0.007	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.052 $\pm$ 0.002	0.227 $\pm$ 0.006	0.161	0.001	0.42	0.0372	0.0372	1.001	94.880 $\pm$ 0.698	0.143 $\pm$ 0.000	99.297 $\pm$ 0.265
LSAM	beta55	0.08	0.078	-0.002 $\pm$ 0.002	-0.023 $\pm$ 0.021	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.426 $\pm$ 0.019	0.653 $\pm$ 0.018	0.078	-0.002	-2.80	0.0520	0.0520	0.962	93.675 $\pm$ 0.771	0.197 $\pm$ 0.001	36.546 $\pm$ 1.526
LSAM	beta56	0.08	0.082	0.002 $\pm$ 0.002	0.025 $\pm$ 0.020	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.381 $\pm$ 0.018	0.617 $\pm$ 0.017	0.081	0.001	1.03	0.0461	0.0461	0.962	93.574 $\pm$ 0.777	0.186 $\pm$ 0.001	41.767 $\pm$ 1.563
LSAM	beta57	0.06	0.064	0.004 $\pm$ 0.002	0.064 $\pm$ 0.030	0.003 $\pm$ 0.000	0.056 $\pm$ 0.001	0.003 $\pm$ 0.000	0.056 $\pm$ 0.002	0.883 $\pm$ 0.040	0.940 $\pm$ 0.026	0.062	0.002	3.32	0.0596	0.0597	0.966	94.679 $\pm$ 0.711	0.213 $\pm$ 0.001	20.984 $\pm$ 1.290
LSAM	beta58	0.06	0.061	0.001 $\pm$ 0.002	0.020 $\pm$ 0.026	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	0.689 $\pm$ 0.033	0.830 $\pm$ 0.024	0.061	0.001	1.67	0.0489	0.0489	0.993	95.281 $\pm$ 0.672	0.194 $\pm$ 0.001	21.486 $\pm$ 1.301
QML	beta41	0.20	0.195	-0.005 $\pm$ 0.001	-0.023 $\pm$ 0.005	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.030 $\pm$ 0.001	0.174 $\pm$ 0.005	0.195	-0.005	-2.73	0.0329	0.0334	1.029	94.679 $\pm$ 0.711	0.139 $\pm$ 0.002	99.900 $\pm$ 0.100
QML	beta42	0.20	0.195	-0.005 $\pm$ 0.001	-0.024 $\pm$ 0.006	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.032 $\pm$ 0.001	0.179 $\pm$ 0.005	0.193	-0.007	-3.36	0.0352	0.0358	1.019	95.884 $\pm$ 0.630	0.141 $\pm$ 0.001	99.900 $\pm$ 0.100
QML	beta43	0.20	0.193	-0.007 $\pm$ 0.001	-0.034 $\pm$ 0.005	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.029 $\pm$ 0.001	0.171 $\pm$ 0.005	0.193	-0.007	-3.45	0.0320	0.0327	1.010	94.076 $\pm$ 0.748	0.133 $\pm$ 0.000	99.900 $\pm$ 0.100
QML	beta44	0.11	0.111	0.001 $\pm$ 0.001	0.012 $\pm$ 0.013	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.161 $\pm$ 0.008	0.401 $\pm$ 0.012	0.113	0.003	3.02	0.0431	0.0433	0.961	94.478 $\pm$ 0.724	0.166 $\pm$ 0.001	74.398 $\pm$ 1.383
QML	beta45	0.11	0.109	-0.001 $\pm$ 0.001	-0.011 $\pm$ 0.011	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.117 $\pm$ 0.006	0.342 $\pm$ 0.010	0.109	-0.001	-1.00	0.0372	0.0372	1.003	95.080 $\pm$ 0.685	0.148 $\pm$ 0.001	82.831 $\pm$ 1.195
QML	beta46	0.08	0.046	-0.034 $\pm$ 0.001	-0.425 $\pm$ 0.014	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	0.388 $\pm$ 0.015	0.623 $\pm$ 0.013	0.045	-0.035	-43.83	0.0363	0.0504	0.981	78.916 $\pm$ 1.293	0.140 $\pm$ 0.001	21.787 $\pm$ 1.308
QML	beta47	0.08	0.049	-0.031 $\pm$ 0.001	-0.386 $\pm$ 0.014	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.334 $\pm$ 0.012	0.578 $\pm$ 0.012	0.047	-0.033	-41.43	0.0339	0.0474	1.031	81.526 $\pm$ 1.230	0.139 $\pm$ 0.001	23.695 $\pm$ 1.347
QML	beta51	0.16	0.173	0.013 $\pm$ 0.002	0.084 $\pm$ 0.010	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	0.098 $\pm$ 0.004	0.312 $\pm$ 0.008	0.173	0.013	7.94	0.0487	0.0504	0.957	92.671 $\pm$ 0.826	0.181 $\pm$ 0.001	96.787 $\pm$ 0.559
QML	beta52	0.16	0.147	-0.013 $\pm$ 0.001	-0.079 $\pm$ 0.007	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.062 $\pm$ 0.003	0.249 $\pm$ 0.007	0.147	-0.013	-8.42	0.0367	0.0391	0.998	92.269 $\pm$ 0.846	0.148 $\pm$ 0.001	97.590 $\pm$ 0.486
QML	beta53	0.16	0.142	-0.018 $\pm$ 0.001	-0.110 $\pm$ 0.007	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	0.065 $\pm$ 0.003	0.255 $\pm$ 0.006	0.141	-0.019	-11.77	0.0378	0.0422	1.009	92.369 $\pm$ 0.841	0.146 $\pm$ 0.000	97.189 $\pm$ 0.524
QML	beta54	0.16	0.160	-0.000 $\pm$ 0.001	-0.002 $\pm$ 0.007	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.052 $\pm$ 0.002	0.228 $\pm$ 0.006	0.158	-0.002	-1.06	0.0366	0.0367	0.992	94.980 $\pm$ 0.692	0.142 $\pm$ 0.000	99.096 $\pm$ 0.300
QML	beta55	0.08	0.079	-0.001 $\pm$ 0.002	-0.009 $\pm$ 0.020	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.404 $\pm$ 0.018	0.636 $\pm$ 0.017	0.080	0.000	-0.28	0.0509	0.0509	0.983	94.177 $\pm$ 0.742	0.196 $\pm$ 0.001	37.751 $\pm$ 1.536
QML	beta56	0.08	0.079	-0.001 $\pm$ 0.002	-0.013 $\pm$ 0.019	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.354 $\pm$ 0.016	0.595 $\pm$ 0.017	0.078	-0.002	-2.74	0.0457	0.0458	0.967	93.976 $\pm$ 0.754	0.180 $\pm$ 0.001	40.863 $\pm$ 1.558
QML	beta57	0.06	0.035	-0.025 $\pm$ 0.001	-0.409 $\pm$ 0.020	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.559 $\pm$ 0.021	0.747 $\pm$ 0.017	0.033	-0.027	-45.34	0.0382	0.0469	0.981	84.839 $\pm$ 1.136	0.144 $\pm$ 0.001	11.446 $\pm$ 1.009
QML	beta58	0.06	0.039	-0.021 $\pm$ 0.001	-0.342 $\pm$ 0.018	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.426 $\pm$ 0.017	0.652 $\pm$ 0.015	0.038	-0.022	-36.37	0.0325	0.0391	0.989	85.542 $\pm$ 1.114	0.129 $\pm$ 0.001	17.570 $\pm$ 1.206
UPI	beta41	0.20	0.201	0.001 $\pm$ 0.001	0.007 $\pm$ 0.006	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.031 $\pm$ 0.001	0.177 $\pm$ 0.005	0.201	0.001	0.42	0.0354	0.0355	0.998	95.181 $\pm$ 0.679	0.139 $\pm$ 0.000	100.000 $\pm$ 0.000
UPI	beta42	0.20	0.199	-0.001 $\pm$ 0.001	-0.003 $\pm$ 0.006	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.031 $\pm$ 0.001	0.177 $\pm$ 0.005	0.199	-0.001	-0.42	0.0345	0.0345	1.030	95.984 $\pm$ 0.622	0.143 $\pm$ 0.000	100.000 $\pm$ 0.000
UPI	beta43	0.20	0.203	0.003 $\pm$ 0.001	0.014 $\pm$ 0.005	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.030 $\pm$ 0.001	0.173 $\pm$ 0.005	0.204	0.004	1.90	0.0341	0.0343	1.023	94.880 $\pm$ 0.698	0.139 $\pm$ 0.000	100.000 $\pm$ 0.00

Table S196

Simulation 2 - Condition 52: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.6
(continued)

Method	Param.	True	Mean-based										Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE	
UPI	beta47	0.08	0.088	0.008 ± 0.002	0.103 ± 0.024	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	0.574 ± 0.032	0.758 ± 0.025	0.084	0.004	5.60	0.0551	0.0553	1.077	97.289 ± 0.515	0.254 ± 0.003	23.293 ± 1.339	
UPI	beta51	0.16	0.160	0.000 ± 0.002	0.002 ± 0.010	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.090 ± 0.004	0.300 ± 0.008	0.160	0.000	0.01	0.0487	0.0487	0.939	93.273 ± 0.794	0.177 ± 0.001	94.177 ± 0.742	
UPI	beta52	0.16	0.159	-0.001 ± 0.001	-0.005 ± 0.007	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.052 ± 0.002	0.227 ± 0.006	0.160	0.000	-0.23	0.0357	0.0357	0.997	94.478 ± 0.724	0.142 ± 0.000	99.398 ± 0.245	
UPI	beta53	0.16	0.162	0.002 ± 0.001	0.012 ± 0.008	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.057 ± 0.003	0.238 ± 0.007	0.160	0.000	0.09	0.0384	0.0384	1.013	95.181 ± 0.679	0.151 ± 0.000	98.896 ± 0.331	
UPI	beta54	0.16	0.161	0.001 ± 0.001	0.009 ± 0.007	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.052 ± 0.002	0.228 ± 0.007	0.161	0.001	0.59	0.0369	0.0370	0.999	94.578 ± 0.718	0.143 ± 0.000	99.197 ± 0.283	
UPI	beta55	0.08	0.078	-0.002 ± 0.002	-0.029 ± 0.021	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.437 ± 0.020	0.661 ± 0.019	0.077	-0.003	-4.01	0.0519	0.0520	0.980	94.980 ± 0.692	0.203 ± 0.001	34.438 ± 1.506	
UPI	beta56	0.08	0.082	0.002 ± 0.002	0.029 ± 0.020	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.400 ± 0.019	0.632 ± 0.018	0.080	0.000	0.09	0.0471	0.0471	0.977	94.980 ± 0.692	0.194 ± 0.001	38.153 ± 1.539	
UPI	beta57	0.06	0.065	0.005 ± 0.002	0.077 ± 0.031	0.004 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.986 ± 0.049	0.993 ± 0.029	0.063	0.003	5.65	0.0600	0.0601	0.997	95.884 ± 0.630	0.232 ± 0.002	16.165 ± 1.166	
UPI	beta58	0.06	0.062	0.002 ± 0.002	0.032 ± 0.029	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.826 ± 0.044	0.909 ± 0.028	0.058	-0.002	-3.80	0.0513	0.0513	1.007	95.984 ± 0.622	0.215 ± 0.002	16.867 ± 1.187	

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S197

Simulation 2 - Condition 53: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.8

Method	Param.	True	Mean-based										Median-based					Inference		
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LSAM	beta41	0.20	0.200	-0.000 $\pm$ 0.001	-0.001 $\pm$ 0.007	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.054 $\pm$ 0.002	0.232 $\pm$ 0.006	0.200	0.000	-0.21	0.0482	0.0482	1.024	96.200 $\pm$ 0.605	0.187 $\pm$ 0.001	98.600 $\pm$ 0.372
LSAM	beta42	0.20	0.202	0.002 $\pm$ 0.002	0.008 $\pm$ 0.008	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.065 $\pm$ 0.003	0.255 $\pm$ 0.007	0.202	0.002	0.87	0.0480	0.0480	0.942	93.200 $\pm$ 0.796	0.189 $\pm$ 0.001	97.100 $\pm$ 0.531
LSAM	beta43	0.20	0.201	0.001 $\pm$ 0.001	0.005 $\pm$ 0.007	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.050 $\pm$ 0.002	0.223 $\pm$ 0.006	0.201	0.001	0.64	0.0445	0.0445	1.013	95.100 $\pm$ 0.683	0.177 $\pm$ 0.001	98.900 $\pm$ 0.330
LSAM	beta44	0.11	0.108	-0.002 $\pm$ 0.002	-0.017 $\pm$ 0.015	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.226 $\pm$ 0.010	0.475 $\pm$ 0.013	0.111	0.001	1.10	0.0533	0.0533	0.972	95.200 $\pm$ 0.676	0.199 $\pm$ 0.001	58.600 $\pm$ 1.558
LSAM	beta45	0.11	0.113	0.003 $\pm$ 0.001	0.027 $\pm$ 0.013	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.176 $\pm$ 0.008	0.420 $\pm$ 0.012	0.113	0.003	2.80	0.0427	0.0428	1.008	95.300 $\pm$ 0.669	0.182 $\pm$ 0.001	69.000 $\pm$ 1.463
LSAM	beta46	0.08	0.084	0.004 $\pm$ 0.002	0.050 $\pm$ 0.023	0.003 $\pm$ 0.000	0.058 $\pm$ 0.001	0.003 $\pm$ 0.000	0.058 $\pm$ 0.002	0.526 $\pm$ 0.025	0.726 $\pm$ 0.021	0.083	0.003	3.29	0.0561	0.0561	1.017	95.600 $\pm$ 0.649	0.231 $\pm$ 0.001	27.200 $\pm$ 1.407
LSAM	beta47	0.08	0.080	-0.000 $\pm$ 0.002	-0.001 $\pm$ 0.024	0.004 $\pm$ 0.000	0.060 $\pm$ 0.001	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	0.569 $\pm$ 0.025	0.755 $\pm$ 0.021	0.078	-0.002	-2.03	0.0605	0.0605	0.969	95.000 $\pm$ 0.689	0.229 $\pm$ 0.001	25.600 $\pm$ 1.380
LSAM	beta51	0.16	0.158	-0.002 $\pm$ 0.002	-0.013 $\pm$ 0.012	0.004 $\pm$ 0.000	0.060 $\pm$ 0.001	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	0.139 $\pm$ 0.006	0.373 $\pm$ 0.010	0.157	-0.003	-1.99	0.0629	0.0630	0.958	95.000 $\pm$ 0.689	0.224 $\pm$ 0.001	77.800 $\pm$ 1.314
LSAM	beta52	0.16	0.159	-0.001 $\pm$ 0.002	-0.005 $\pm$ 0.010	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.092 $\pm$ 0.004	0.304 $\pm$ 0.008	0.159	-0.001	-0.83	0.0493	0.0493	0.991	94.900 $\pm$ 0.696	0.189 $\pm$ 0.001	90.400 $\pm$ 0.932
LSAM	beta53	0.16	0.162	0.002 $\pm$ 0.002	0.012 $\pm$ 0.010	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.102 $\pm$ 0.005	0.319 $\pm$ 0.009	0.161	0.001	0.56	0.0498	0.0498	0.968	93.200 $\pm$ 0.796	0.194 $\pm$ 0.001	89.700 $\pm$ 0.961
LSAM	beta54	0.16	0.162	0.002 $\pm$ 0.002	0.014 $\pm$ 0.009	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.089 $\pm$ 0.004	0.299 $\pm$ 0.008	0.162	0.002	1.16	0.0465	0.0465	1.005	94.700 $\pm$ 0.708	0.188 $\pm$ 0.001	91.500 $\pm$ 0.882
LSAM	beta55	0.08	0.079	-0.001 $\pm$ 0.002	-0.009 $\pm$ 0.026	0.004 $\pm$ 0.000	0.067 $\pm$ 0.002	0.004 $\pm$ 0.000	0.067 $\pm$ 0.002	0.695 $\pm$ 0.035	0.834 $\pm$ 0.025	0.081	0.001	0.91	0.0641	0.0641	0.899	93.500 $\pm$ 0.780	0.235 $\pm$ 0.002	31.800 $\pm$ 1.473
LSAM	beta56	0.08	0.082	0.002 $\pm$ 0.002	0.030 $\pm$ 0.025	0.004 $\pm$ 0.000	0.064 $\pm$ 0.002	0.004 $\pm$ 0.000	0.064 $\pm$ 0.002	0.631 $\pm$ 0.030	0.795 $\pm$ 0.023	0.082	0.002	2.65	0.0622	0.0623	0.904	93.300 $\pm$ 0.791	0.225 $\pm$ 0.001	33.500 $\pm$ 1.493
LSAM	beta57	0.06	0.064	0.004 $\pm$ 0.002	0.060 $\pm$ 0.033	0.004 $\pm$ 0.000	0.062 $\pm$ 0.001	0.004 $\pm$ 0.000	0.062 $\pm$ 0.002	1.079 $\pm$ 0.052	1.039 $\pm$ 0.030	0.061	0.001	1.54	0.0580	0.0580	0.972	93.000 $\pm$ 0.807	0.237 $\pm$ 0.001	16.600 $\pm$ 1.177
LSAM	beta58	0.06	0.057	-0.003 $\pm$ 0.002	-0.045 $\pm$ 0.030	0.003 $\pm$ 0.000	0.057 $\pm$ 0.001	0.003 $\pm$ 0.000	0.057 $\pm$ 0.002	0.891 $\pm$ 0.040	0.944 $\pm$ 0.026	0.057	-0.003	-4.81	0.0553	0.0554	0.979	94.500 $\pm$ 0.721	0.217 $\pm$ 0.001	17.000 $\pm$ 1.188
QML	beta41	0.20	0.198	-0.002 $\pm$ 0.001	-0.012 $\pm$ 0.007	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.053 $\pm$ 0.002	0.231 $\pm$ 0.006	0.197	-0.003	-1.36	0.0458	0.0459	1.026	96.000 $\pm$ 0.620	0.186 $\pm$ 0.001	98.700 $\pm$ 0.358
QML	beta42	0.20	0.200	-0.000 $\pm$ 0.002	-0.000 $\pm$ 0.008	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.064 $\pm$ 0.003	0.254 $\pm$ 0.007	0.200	0.000	-0.20	0.0472	0.0472	0.947	93.300 $\pm$ 0.791	0.188 $\pm$ 0.002	97.100 $\pm$ 0.531
QML	beta43	0.20	0.196	-0.004 $\pm$ 0.001	-0.019 $\pm$ 0.007	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.049 $\pm$ 0.002	0.221 $\pm$ 0.006	0.198	-0.002	-1.22	0.0438	0.0439	1.012	94.800 $\pm$ 0.702	0.175 $\pm$ 0.001	98.900 $\pm$ 0.330
QML	beta44	0.11	0.109	-0.001 $\pm$ 0.002	-0.011 $\pm$ 0.015	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.003 $\pm$ 0.000	0.053 $\pm$ 0.002	0.235 $\pm$ 0.011	0.485 $\pm$ 0.014	0.112	0.002	1.95	0.0539	0.0540	0.980	95.600 $\pm$ 0.649	0.205 $\pm$ 0.001	58.000 $\pm$ 1.561
QML	beta45	0.11	0.113	0.003 $\pm$ 0.001	0.024 $\pm$ 0.013	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.179 $\pm$ 0.008	0.424 $\pm$ 0.012	0.112	0.002	2.03	0.0425	0.0426	1.014	95.900 $\pm$ 0.627	0.185 $\pm$ 0.001	67.800 $\pm$ 1.478
QML	beta46	0.08	0.065	-0.015 $\pm$ 0.002	-0.184 $\pm$ 0.019	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.401 $\pm$ 0.018	0.634 $\pm$ 0.017	0.064	-0.016	-20.58	0.0461	0.0489	1.021	92.000 $\pm$ 0.858	0.194 $\pm$ 0.001	20.800 $\pm$ 1.283
QML	beta47	0.08	0.064	-0.016 $\pm$ 0.002	-0.203 $\pm$ 0.020	0.003 $\pm$ 0.000	0.050 $\pm$ 0.001	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.434 $\pm$ 0.018	0.659 $\pm$ 0.016	0.061	-0.019	-23.91	0.0516	0.0550	0.971	91.000 $\pm$ 0.905	0.191 $\pm$ 0.001	22.000 $\pm$ 1.310
QML	beta51	0.16	0.164	0.004 $\pm$ 0.002	0.025 $\pm$ 0.012	0.004 $\pm$ 0.000	0.060 $\pm$ 0.001	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	0.141 $\pm$ 0.006	0.375 $\pm$ 0.010	0.163	0.003	1.67	0.0644	0.0644	0.990	95.300 $\pm$ 0.669	0.232 $\pm$ 0.001	78.400 $\pm$ 1.301
QML	beta52	0.16	0.145	-0.015 $\pm$ 0.002	-0.096 $\pm$ 0.010	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.105 $\pm$ 0.005	0.325 $\pm$ 0.008	0.143	-0.017	-10.69	0.0514	0.0542	1.013	94.200 $\pm$ 0.739	0.197 $\pm$ 0.001	82.600 $\pm$ 1.199
QML	beta53	0.16	0.143	-0.017 $\pm$ 0.002	-0.107 $\pm$ 0.010	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.112 $\pm$ 0.005	0.334 $\pm$ 0.009	0.142	-0.018	-11.46	0.0493	0.0526	0.988	93.600 $\pm$ 0.774	0.196 $\pm$ 0.001	81.400 $\pm$ 1.230
QML	beta54	0.16	0.162	0.002 $\pm$ 0.002	0.010 $\pm$ 0.009	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.090 $\pm$ 0.004	0.300 $\pm$ 0.008	0.162	0.002	1.05	0.0465	0.0465	1.002	95.000 $\pm$ 0.689	0.189 $\pm$ 0.001	91.400 $\pm$ 0.887
QML	beta55	0.08	0.080	0.000 $\pm$ 0.002	0.002 $\pm$ 0.026	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	0.683 $\pm$ 0.034	0.827 $\pm$ 0.025	0.081	0.001	0.98	0.0636	0.0636	0.924	94.100 $\pm$ 0.745	0.240 $\pm$ 0.002	30.100 $\pm$ 1.451
QML	beta56	0.08	0.082	0.002 $\pm$ 0.002	0.019 $\pm$ 0.025	0.004 $\pm$ 0.000	0.062 $\pm$ 0.001	0.004 $\pm$ 0.000	0.062 $\pm$ 0.002	0.607 $\pm$ 0.029	0.779 $\pm$ 0.022	0.082	0.002	2.77	0.0606	0.0606	0.926	93.900 $\pm$ 0.757	0.226 $\pm$ 0.001	31.800 $\pm$ 1.473
QML	beta57	0.06	0.049	-0.011 $\pm$ 0.002	-0.184 $\pm$ 0.027	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.777 $\pm$ 0.034	0.881 $\pm$ 0.024	0.045	-0.015	-24.95	0.0486	0.0508	0.983	92.200 $\pm$ 0.848	0.199 $\pm$ 0.001	12.700 $\pm$ 1.053
QML	beta58	0.06	0.047	-0.013 $\pm$ 0.001	-0.212 $\pm$ 0.025	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.658 $\pm$ 0.029	0.811 $\pm$ 0.021	0.046	-0.014	-23.79	0.0475	0.0496	0.983	92.100 $\pm$ 0.853	0.181 $\pm$ 0.001	14.200 $\pm$ 1.104
UPI	beta41	0.20	0.200	-0.000 $\pm$ 0.001	-0.001 $\pm$ 0.007	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.053 $\pm$ 0.002	0.230 $\pm$ 0.006	0.199	-0.001	-0.52	0.0465	0.0465	0.978	95.500 $\pm$ 0.656	0.177 $\pm$ 0.001	99.100 $\pm$ 0.299
UPI	beta42	0.20	0.202	0.002 $\pm$ 0.002	0.008 $\pm$ 0.008	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.064 $\pm$ 0.003	0.254 $\pm$ 0.007	0.202	0.002	0.97	0.0480	0.0481	0.913	92.300 $\pm$ 0.843	0.182 $\pm$ 0.001	97.900 $\pm$ 0.453
UPI	beta43	0.20	0.201	0.001 $\pm$ 0.001	0.006 $\pm$ 0.007	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.051 $\pm$ 0.002	0.225 $\pm$ 0.006	0.202	0.002	0.80	0.0446	0.0446	1.004	95.100 $\pm$ 0.683	0.177 $\pm$ 0.001	99.100 $\pm$ 0.299

Table S197

Simulation 2 - Condition 53: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.8
(continued)

Method	Param.	True	Mean-based										Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE	
UPI	beta47	0.08	0.082	0.002 ± 0.002	0.019 ± 0.026	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.674 ± 0.032	0.821 ± 0.023	0.078	-0.002	-2.76	0.0668	0.0669	0.983	95.100 ± 0.683	0.253 ± 0.002	21.400 ± 1.297	
UPI	beta51	0.16	0.158	-0.002 ± 0.002	-0.011 ± 0.012	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.136 ± 0.006	0.369 ± 0.010	0.158	-0.002	-1.56	0.0630	0.0630	0.976	95.100 ± 0.683	0.226 ± 0.001	77.200 ± 1.327	
UPI	beta52	0.16	0.159	-0.001 ± 0.002	-0.004 ± 0.010	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.092 ± 0.004	0.303 ± 0.008	0.159	-0.001	-0.86	0.0482	0.0482	0.975	94.400 ± 0.727	0.185 ± 0.001	91.400 ± 0.887	
UPI	beta53	0.16	0.162	0.002 ± 0.002	0.012 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.102 ± 0.005	0.319 ± 0.009	0.160	0.000	0.11	0.0507	0.0507	0.969	93.600 ± 0.774	0.194 ± 0.001	89.800 ± 0.957	
UPI	beta54	0.16	0.162	0.002 ± 0.002	0.014 ± 0.009	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.089 ± 0.004	0.298 ± 0.008	0.162	0.002	1.17	0.0460	0.0460	0.998	94.600 ± 0.715	0.187 ± 0.001	91.600 ± 0.877	
UPI	beta55	0.08	0.080	-0.000 ± 0.002	-0.004 ± 0.027	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.715 ± 0.037	0.846 ± 0.026	0.082	0.002	2.13	0.0632	0.0632	0.907	94.200 ± 0.739	0.241 ± 0.002	30.500 ± 1.456	
UPI	beta56	0.08	0.082	0.002 ± 0.002	0.028 ± 0.025	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	0.646 ± 0.031	0.804 ± 0.023	0.082	0.002	2.78	0.0613	0.0613	0.918	94.400 ± 0.727	0.231 ± 0.002	31.200 ± 1.465	
UPI	beta57	0.06	0.063	0.003 ± 0.002	0.057 ± 0.033	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	1.122 ± 0.054	1.059 ± 0.031	0.060	0.000	0.44	0.0592	0.0592	0.989	93.800 ± 0.763	0.246 ± 0.001	15.000 ± 1.129	
UPI	beta58	0.06	0.058	-0.002 ± 0.002	-0.038 ± 0.031	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.960 ± 0.044	0.980 ± 0.028	0.058	-0.002	-4.01	0.0581	0.0581	0.991	94.700 ± 0.708	0.228 ± 0.001	14.900 ± 1.126	

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S198

Simulation 2 - Condition 54: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.8

Method	Param.	True	Mean-based										Median-based					Inference		
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LSAM	beta41	0.20	0.200	-0.000 $\pm$ 0.001	-0.001 $\pm$ 0.005	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.022 $\pm$ 0.001	0.148 $\pm$ 0.004	0.200	0.000	-0.10	0.0288	0.0288	1.004	95.195 $\pm$ 0.677	0.117 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta42	0.20	0.199	-0.001 $\pm$ 0.001	-0.006 $\pm$ 0.005	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.021 $\pm$ 0.001	0.145 $\pm$ 0.004	0.200	0.000	-0.14	0.0291	0.0291	1.033	95.295 $\pm$ 0.670	0.118 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta43	0.20	0.200	0.000 $\pm$ 0.001	0.001 $\pm$ 0.005	0.001 $\pm$ 0.000	0.028 $\pm$ 0.001	0.001 $\pm$ 0.000	0.028 $\pm$ 0.001	0.020 $\pm$ 0.001	0.142 $\pm$ 0.004	0.199	-0.001	-0.45	0.0289	0.0289	0.995	95.295 $\pm$ 0.670	0.111 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta44	0.11	0.110	0.000 $\pm$ 0.001	0.004 $\pm$ 0.009	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	0.084 $\pm$ 0.004	0.290 $\pm$ 0.008	0.111	0.001	0.70	0.0314	0.0314	0.993	94.194 $\pm$ 0.740	0.124 $\pm$ 0.000	91.592 $\pm$ 0.878
LSAM	beta45	0.11	0.110	0.000 $\pm$ 0.001	0.002 $\pm$ 0.009	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.076 $\pm$ 0.003	0.275 $\pm$ 0.008	0.111	0.001	1.18	0.0299	0.0299	0.961	93.493 $\pm$ 0.780	0.114 $\pm$ 0.000	95.095 $\pm$ 0.683
LSAM	beta46	0.08	0.081	0.001 $\pm$ 0.001	0.012 $\pm$ 0.014	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.208 $\pm$ 0.009	0.456 $\pm$ 0.013	0.082	0.002	2.34	0.0351	0.0351	1.001	94.695 $\pm$ 0.709	0.143 $\pm$ 0.000	62.262 $\pm$ 1.534
LSAM	beta47	0.08	0.077	-0.003 $\pm$ 0.001	-0.036 $\pm$ 0.015	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.214 $\pm$ 0.009	0.462 $\pm$ 0.012	0.075	-0.005	-6.01	0.0377	0.0380	0.986	94.494 $\pm$ 0.722	0.143 $\pm$ 0.000	55.355 $\pm$ 1.573
LSAM	beta51	0.16	0.162	0.002 $\pm$ 0.001	0.013 $\pm$ 0.007	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.056 $\pm$ 0.003	0.237 $\pm$ 0.007	0.162	0.002	1.27	0.0374	0.0375	0.958	93.994 $\pm$ 0.752	0.142 $\pm$ 0.001	99.499 $\pm$ 0.223
LSAM	beta52	0.16	0.158	-0.002 $\pm$ 0.001	-0.013 $\pm$ 0.006	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.033 $\pm$ 0.001	0.182 $\pm$ 0.005	0.160	0.000	0.23	0.0292	0.0292	1.033	95.195 $\pm$ 0.677	0.117 $\pm$ 0.000	99.900 $\pm$ 0.100
LSAM	beta53	0.16	0.160	-0.000 $\pm$ 0.001	-0.000 $\pm$ 0.006	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.036 $\pm$ 0.002	0.191 $\pm$ 0.005	0.161	0.001	0.81	0.0305	0.0306	1.010	96.096 $\pm$ 0.613	0.121 $\pm$ 0.000	99.800 $\pm$ 0.141
LSAM	beta54	0.16	0.160	-0.000 $\pm$ 0.001	-0.002 $\pm$ 0.006	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.034 $\pm$ 0.002	0.185 $\pm$ 0.005	0.161	0.001	0.57	0.0300	0.0300	1.006	94.795 $\pm$ 0.703	0.117 $\pm$ 0.000	99.900 $\pm$ 0.100
LSAM	beta55	0.08	0.079	-0.001 $\pm$ 0.001	-0.009 $\pm$ 0.016	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.249 $\pm$ 0.011	0.499 $\pm$ 0.014	0.080	0.000	0.05	0.0405	0.0405	0.936	94.294 $\pm$ 0.734	0.147 $\pm$ 0.001	57.558 $\pm$ 1.564
LSAM	beta56	0.08	0.079	-0.001 $\pm$ 0.001	-0.009 $\pm$ 0.015	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.213 $\pm$ 0.009	0.462 $\pm$ 0.012	0.079	-0.001	-1.10	0.0384	0.0384	0.971	94.695 $\pm$ 0.709	0.141 $\pm$ 0.001	59.259 $\pm$ 1.555
LSAM	beta57	0.06	0.060	0.000 $\pm$ 0.001	0.004 $\pm$ 0.020	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.388 $\pm$ 0.018	0.623 $\pm$ 0.017	0.060	0.000	-0.49	0.0368	0.0368	0.991	94.995 $\pm$ 0.690	0.145 $\pm$ 0.000	36.837 $\pm$ 1.526
LSAM	beta58	0.06	0.060	-0.000 $\pm$ 0.001	-0.005 $\pm$ 0.018	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.323 $\pm$ 0.014	0.569 $\pm$ 0.015	0.061	0.001	1.39	0.0349	0.0349	1.017	95.195 $\pm$ 0.677	0.136 $\pm$ 0.000	40.741 $\pm$ 1.555
QML	beta41	0.20	0.197	-0.003 $\pm$ 0.001	-0.013 $\pm$ 0.005	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.022 $\pm$ 0.001	0.148 $\pm$ 0.004	0.198	-0.002	-1.23	0.0289	0.0290	1.051	94.795 $\pm$ 0.703	0.121 $\pm$ 0.005	99.900 $\pm$ 0.100
QML	beta42	0.20	0.197	-0.003 $\pm$ 0.001	-0.015 $\pm$ 0.005	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.021 $\pm$ 0.001	0.145 $\pm$ 0.004	0.198	-0.002	-1.14	0.0290	0.0291	1.069	95.596 $\pm$ 0.649	0.121 $\pm$ 0.004	99.900 $\pm$ 0.100
QML	beta43	0.20	0.196	-0.004 $\pm$ 0.001	-0.022 $\pm$ 0.004	0.001 $\pm$ 0.000	0.028 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.020 $\pm$ 0.001	0.143 $\pm$ 0.004	0.195	-0.005	-2.75	0.0279	0.0285	1.017	94.895 $\pm$ 0.696	0.113 $\pm$ 0.003	99.900 $\pm$ 0.100
QML	beta44	0.11	0.111	0.001 $\pm$ 0.001	0.012 $\pm$ 0.009	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.087 $\pm$ 0.004	0.296 $\pm$ 0.008	0.112	0.002	1.66	0.0324	0.0324	0.999	94.294 $\pm$ 0.734	0.127 $\pm$ 0.001	89.690 $\pm$ 0.962
QML	beta45	0.11	0.111	0.001 $\pm$ 0.001	0.006 $\pm$ 0.009	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.077 $\pm$ 0.003	0.278 $\pm$ 0.008	0.111	0.001	1.14	0.0300	0.0300	0.969	94.394 $\pm$ 0.728	0.116 $\pm$ 0.000	94.795 $\pm$ 0.703
QML	beta46	0.08	0.062	-0.018 $\pm$ 0.001	-0.219 $\pm$ 0.012	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.196 $\pm$ 0.008	0.443 $\pm$ 0.011	0.062	-0.018	-22.73	0.0299	0.0350	1.001	88.989 $\pm$ 0.990	0.121 $\pm$ 0.000	52.352 $\pm$ 1.580
QML	beta47	0.08	0.060	-0.020 $\pm$ 0.001	-0.249 $\pm$ 0.012	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.211 $\pm$ 0.008	0.460 $\pm$ 0.010	0.058	-0.022	-27.65	0.0310	0.0381	0.978	86.086 $\pm$ 1.095	0.119 $\pm$ 0.000	48.248 $\pm$ 1.581
QML	beta51	0.16	0.168	0.008 $\pm$ 0.001	0.051 $\pm$ 0.007	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.001 $\pm$ 0.000	0.039 $\pm$ 0.001	0.058 $\pm$ 0.003	0.242 $\pm$ 0.007	0.167	0.007	4.47	0.0370	0.0377	0.995	94.795 $\pm$ 0.703	0.147 $\pm$ 0.003	99.499 $\pm$ 0.223
QML	beta52	0.16	0.144	-0.016 $\pm$ 0.001	-0.103 $\pm$ 0.006	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.045 $\pm$ 0.002	0.212 $\pm$ 0.005	0.145	-0.015	-9.59	0.0292	0.0329	1.077	91.592 $\pm$ 0.878	0.126 $\pm$ 0.004	99.600 $\pm$ 0.200
QML	beta53	0.16	0.142	-0.018 $\pm$ 0.001	-0.114 $\pm$ 0.006	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.050 $\pm$ 0.002	0.223 $\pm$ 0.006	0.143	-0.017	-10.61	0.0294	0.0339	1.013	89.489 $\pm$ 0.970	0.122 $\pm$ 0.000	99.199 $\pm$ 0.282
QML	beta54	0.16	0.159	-0.001 $\pm$ 0.001	-0.007 $\pm$ 0.006	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.035 $\pm$ 0.002	0.186 $\pm$ 0.005	0.160	0.000	-0.01	0.0299	0.0299	1.010	94.895 $\pm$ 0.696	0.118 $\pm$ 0.000	99.800 $\pm$ 0.141
QML	beta55	0.08	0.080	0.000 $\pm$ 0.001	0.005 $\pm$ 0.016	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.248 $\pm$ 0.011	0.498 $\pm$ 0.014	0.080	0.000	0.54	0.0406	0.0406	0.952	94.595 $\pm$ 0.715	0.149 $\pm$ 0.001	57.758 $\pm$ 1.563
QML	beta56	0.08	0.079	-0.001 $\pm$ 0.001	-0.016 $\pm$ 0.015	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.210 $\pm$ 0.009	0.459 $\pm$ 0.012	0.078	-0.002	-2.80	0.0371	0.0372	0.977	94.695 $\pm$ 0.709	0.140 $\pm$ 0.001	58.759 $\pm$ 1.557
QML	beta57	0.06	0.046	-0.014 $\pm$ 0.001	-0.239 $\pm$ 0.016	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.326 $\pm$ 0.014	0.571 $\pm$ 0.014	0.046	-0.014	-24.06	0.0302	0.0335	0.997	89.790 $\pm$ 0.958	0.122 $\pm$ 0.001	27.828 $\pm$ 1.418
QML	beta58	0.06	0.048	-0.012 $\pm$ 0.001	-0.195 $\pm$ 0.015	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.264 $\pm$ 0.011	0.514 $\pm$ 0.013	0.048	-0.012	-20.00	0.0290	0.0314	1.016	91.391 $\pm$ 0.887	0.114 $\pm$ 0.000	37.137 $\pm$ 1.529
UPI	beta41	0.20	0.200	-0.000 $\pm$ 0.001	-0.001 $\pm$ 0.005	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.022 $\pm$ 0.001	0.147 $\pm$ 0.004	0.200	0.000	0.02	0.0284	0.0284	0.958	93.894 $\pm$ 0.758	0.110 $\pm$ 0.000	100.000 $\pm$ 0.000
UPI	beta42	0.20	0.199	-0.001 $\pm$ 0.001	-0.006 $\pm$ 0.005	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.021 $\pm$ 0.001	0.144 $\pm$ 0.004	0.200	0.000	0.11	0.0288	0.0288	1.004	94.895 $\pm$ 0.696	0.113 $\pm$ 0.000	100.000 $\pm$ 0.000
UPI	beta43	0.20	0.200	0.000 $\pm$ 0.001	0.002 $\pm$ 0.005	0.001 $\pm$ 0.000	0.028 $\pm$ 0.001	0.001 $\pm$ 0.000	0.028 $\pm$ 0.001	0.020 $\pm$ 0.001	0.142 $\pm$ 0.004	0.199	-0.001	-0.35	0.0282	0.0282	0.992	95.295 $\pm$ 0.670	0.111 $\pm$ 0.000	100.000 $\pm$ 0.00

Table S198

Simulation 2 - Condition 54: Full Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.8
(continued)

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.078	-0.002 ± 0.001	-0.029 ± 0.015	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.223 ± 0.009	0.472 ± 0.012	0.076	-0.004	-5.13	0.0380	0.0382	1.007	95.596 ± 0.649	0.149 ± 0.001	52.953 ± 1.579
UPI	beta51	0.16	0.162	0.002 ± 0.001	0.014 ± 0.007	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.055 ± 0.003	0.235 ± 0.007	0.162	0.002	1.19	0.0369	0.0369	0.961	94.494 ± 0.722	0.142 ± 0.001	99.499 ± 0.223
UPI	beta52	0.16	0.158	-0.002 ± 0.001	-0.012 ± 0.006	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.033 ± 0.001	0.181 ± 0.005	0.160	0.000	-0.15	0.0295	0.0295	1.018	95.095 ± 0.683	0.116 ± 0.000	99.900 ± 0.100
UPI	beta53	0.16	0.160	-0.000 ± 0.001	-0.000 ± 0.006	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.037 ± 0.002	0.191 ± 0.005	0.162	0.002	1.09	0.0297	0.0297	1.006	96.096 ± 0.613	0.121 ± 0.000	99.800 ± 0.141
UPI	beta54	0.16	0.160	-0.000 ± 0.001	-0.002 ± 0.006	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.034 ± 0.002	0.185 ± 0.005	0.161	0.001	0.63	0.0300	0.0300	0.996	94.394 ± 0.728	0.116 ± 0.000	100.000 ± 0.000
UPI	beta55	0.08	0.079	-0.001 ± 0.001	-0.011 ± 0.016	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.251 ± 0.011	0.501 ± 0.014	0.080	0.000	-0.33	0.0400	0.0400	0.943	93.994 ± 0.752	0.148 ± 0.001	56.256 ± 1.569
UPI	beta56	0.08	0.079	-0.001 ± 0.001	-0.009 ± 0.015	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.215 ± 0.009	0.463 ± 0.012	0.078	-0.002	-2.53	0.0379	0.0380	0.979	95.295 ± 0.670	0.142 ± 0.001	58.859 ± 1.557
UPI	beta57	0.06	0.060	0.000 ± 0.001	0.005 ± 0.020	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.408 ± 0.019	0.639 ± 0.018	0.060	0.000	0.05	0.0374	0.0374	0.984	95.696 ± 0.642	0.148 ± 0.001	35.636 ± 1.515
UPI	beta58	0.06	0.060	0.000 ± 0.001	0.006 ± 0.018	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.331 ± 0.014	0.576 ± 0.015	0.062	0.002	2.75	0.0349	0.0349	1.029	95.495 ± 0.656	0.139 ± 0.000	39.940 ± 1.550

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S199

Simulation 2 - Condition 55: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, $N = 400$, Reliability = 0.4

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.199	-0.001 ± 0.011	-0.003 ± 0.055	0.013 ± 0.003	0.113 ± 0.012	0.013 ± 0.003	0.113 ± 0.013	0.318 ± 0.063	0.564 ± 0.064	0.211	0.011	5.34	0.1022	0.1028	1.014	94.340 ± 2.244	0.451 ± 0.017	42.453 ± 4.801
LSAM	beta42	0.20	0.204	0.004 ± 0.012	0.021 ± 0.059	0.015 ± 0.003	0.122 ± 0.012	0.015 ± 0.003	0.122 ± 0.013	0.370 ± 0.067	0.608 ± 0.064	0.195	-0.005	-2.71	0.1220	0.1221	0.999	97.170 ± 1.611	0.478 ± 0.017	40.566 ± 4.769
LSAM	beta43	0.20	0.194	-0.006 ± 0.008	-0.031 ± 0.041	0.007 ± 0.001	0.084 ± 0.006	0.007 ± 0.001	0.084 ± 0.007	0.176 ± 0.022	0.419 ± 0.034	0.188	-0.012	-5.91	0.0868	0.0876	1.016	95.283 ± 2.059	0.334 ± 0.006	66.038 ± 4.600
LSAM	beta44	0.11	0.175	0.065 ± 0.012	0.591 ± 0.113	0.016 ± 0.003	0.128 ± 0.013	0.020 ± 0.004	0.143 ± 0.015	1.683 ± 0.315	1.297 ± 0.134	0.163	0.053	47.85	0.1150	0.1264	1.113	96.226 ± 1.851	0.557 ± 0.023	17.925 ± 3.725
LSAM	beta45	0.11	0.118	0.008 ± 0.012	0.072 ± 0.105	0.014 ± 0.003	0.119 ± 0.011	0.014 ± 0.003	0.119 ± 0.013	1.171 ± 0.213	1.082 ± 0.114	0.116	0.006	5.55	0.1058	0.1060	0.937	94.340 ± 2.244	0.438 ± 0.021	21.698 ± 4.004
LSAM	beta46	0.08	0.034	-0.046 ± 0.009	-0.579 ± 0.114	0.009 ± 0.002	0.094 ± 0.009	0.011 ± 0.002	0.104 ± 0.008	1.706 ± 0.237	1.306 ± 0.106	0.029	-0.051	-63.31	0.0824	0.0967	1.003	93.396 ± 2.412	0.370 ± 0.023	3.774 ± 1.851
LSAM	beta47	0.08	0.027	-0.053 ± 0.008	-0.659 ± 0.103	0.007 ± 0.001	0.085 ± 0.006	0.010 ± 0.001	0.100 ± 0.007	1.552 ± 0.192	1.246 ± 0.090	0.021	-0.059	-74.09	0.0849	0.1036	1.089	88.679 ± 3.077	0.363 ± 0.017	2.830 ± 1.611
LSAM	beta51	0.16	0.200	0.040 ± 0.011	0.251 ± 0.066	0.012 ± 0.002	0.109 ± 0.008	0.013 ± 0.002	0.116 ± 0.011	0.524 ± 0.082	0.724 ± 0.066	0.194	0.034	21.56	0.1021	0.1077	0.968	91.509 ± 2.707	0.414 ± 0.009	47.170 ± 4.849
LSAM	beta52	0.16	0.144	-0.016 ± 0.009	-0.098 ± 0.055	0.008 ± 0.001	0.090 ± 0.007	0.008 ± 0.001	0.091 ± 0.008	0.326 ± 0.046	0.571 ± 0.049	0.136	-0.024	-15.28	0.0934	0.0965	1.025	93.396 ± 2.412	0.363 ± 0.010	37.736 ± 4.708
LSAM	beta53	0.16	0.146	-0.014 ± 0.009	-0.084 ± 0.054	0.008 ± 0.001	0.089 ± 0.006	0.008 ± 0.001	0.090 ± 0.008	0.316 ± 0.043	0.562 ± 0.047	0.160	0.000	-0.23	0.0933	0.0933	1.060	94.340 ± 2.244	0.371 ± 0.008	33.962 ± 4.600
LSAM	beta54	0.16	0.144	-0.016 ± 0.009	-0.103 ± 0.059	0.009 ± 0.001	0.097 ± 0.007	0.010 ± 0.001	0.098 ± 0.009	0.371 ± 0.054	0.609 ± 0.054	0.148	-0.012	-7.41	0.0870	0.0878	1.004	95.283 ± 2.059	0.380 ± 0.009	32.075 ± 4.534
LSAM	beta55	0.08	0.094	0.014 ± 0.014	0.169 ± 0.173	0.020 ± 0.004	0.143 ± 0.013	0.020 ± 0.004	0.143 ± 0.015	3.179 ± 0.563	1.783 ± 0.185	0.089	0.009	10.96	0.1287	0.1290	0.933	94.340 ± 2.244	0.522 ± 0.026	14.151 ± 3.385
LSAM	beta56	0.08	0.084	0.004 ± 0.011	0.055 ± 0.140	0.013 ± 0.002	0.115 ± 0.008	0.013 ± 0.002	0.115 ± 0.010	2.049 ± 0.271	1.431 ± 0.119	0.079	-0.001	-1.26	0.1059	0.1059	1.111	98.113 ± 1.322	0.501 ± 0.028	8.491 ± 2.707
LSAM	beta57	0.06	0.032	-0.028 ± 0.007	-0.466 ± 0.114	0.005 ± 0.001	0.070 ± 0.006	0.006 ± 0.001	0.076 ± 0.007	1.585 ± 0.233	1.259 ± 0.108	0.029	-0.031	-52.30	0.0665	0.0735	1.036	92.453 ± 2.566	0.286 ± 0.010	0.943 ± 0.939
LSAM	beta58	0.06	0.026	-0.034 ± 0.006	-0.559 ± 0.108	0.004 ± 0.001	0.067 ± 0.007	0.006 ± 0.001	0.074 ± 0.008	1.534 ± 0.282	1.238 ± 0.126	0.030	-0.030	-49.44	0.0516	0.0595	0.937	85.849 ± 3.385	0.245 ± 0.010	1.887 ± 1.322
QML	beta41	0.20	0.175	-0.025 ± 0.007	-0.124 ± 0.037	0.006 ± 0.001	0.075 ± 0.005	0.006 ± 0.001	0.079 ± 0.007	0.156 ± 0.021	0.395 ± 0.033	0.187	-0.013	-6.37	0.0836	0.0845	1.053	91.509 ± 2.707	0.311 ± 0.006	66.038 ± 4.600
QML	beta42	0.20	0.180	-0.020 ± 0.008	-0.099 ± 0.042	0.008 ± 0.001	0.087 ± 0.006	0.008 ± 0.001	0.089 ± 0.007	0.196 ± 0.025	0.443 ± 0.035	0.180	-0.020	-9.86	0.0785	0.0810	0.934	96.226 ± 1.851	0.318 ± 0.006	58.491 ± 4.786
QML	beta43	0.20	0.183	-0.017 ± 0.007	-0.086 ± 0.035	0.005 ± 0.001	0.073 ± 0.005	0.006 ± 0.001	0.074 ± 0.006	0.138 ± 0.018	0.372 ± 0.031	0.181	-0.019	-9.56	0.0713	0.0738	1.020	95.283 ± 2.059	0.291 ± 0.005	73.585 ± 4.282
QML	beta44	0.11	0.122	0.012 ± 0.010	0.111 ± 0.089	0.010 ± 0.002	0.101 ± 0.008	0.010 ± 0.002	0.101 ± 0.009	0.844 ± 0.133	0.919 ± 0.086	0.124	0.014	12.75	0.0981	0.0991	1.148	93.396 ± 2.412	0.454 ± 0.016	23.585 ± 4.123
QML	beta45	0.11	0.117	0.007 ± 0.009	0.061 ± 0.084	0.009 ± 0.001	0.095 ± 0.007	0.009 ± 0.001	0.095 ± 0.008	0.742 ± 0.106	0.861 ± 0.076	0.122	0.012	10.58	0.0937	0.0945	0.993	97.170 ± 1.611	0.370 ± 0.011	22.642 ± 4.065
QML	beta46	0.08	0.063	-0.017 ± 0.006	-0.210 ± 0.074	0.004 ± 0.001	0.061 ± 0.005	0.004 ± 0.001	0.063 ± 0.005	0.623 ± 0.086	0.789 ± 0.066	0.056	-0.024	-30.45	0.0675	0.0718	1.208	91.509 ± 2.707	0.290 ± 0.013	10.377 ± 2.962
QML	beta47	0.08	0.054	-0.026 ± 0.007	-0.325 ± 0.088	0.005 ± 0.001	0.072 ± 0.007	0.006 ± 0.001	0.076 ± 0.007	0.913 ± 0.137	0.956 ± 0.084	0.048	-0.032	-40.28	0.0614	0.0694	1.020	85.849 ± 3.385	0.289 ± 0.013	3.774 ± 1.851
QML	beta51	0.16	0.201	0.041 ± 0.011	0.259 ± 0.071	0.014 ± 0.002	0.116 ± 0.010	0.015 ± 0.003	0.123 ± 0.012	0.591 ± 0.101	0.769 ± 0.076	0.195	0.035	21.69	0.1142	0.1193	1.017	92.453 ± 2.566	0.464 ± 0.014	44.340 ± 4.825
QML	beta52	0.16	0.149	-0.011 ± 0.009	-0.067 ± 0.054	0.008 ± 0.001	0.088 ± 0.006	0.008 ± 0.001	0.089 ± 0.007	0.307 ± 0.039	0.554 ± 0.045	0.138	-0.022	-13.70	0.0898	0.0924	1.038	92.453 ± 2.566	0.360 ± 0.010	41.509 ± 4.786
QML	beta53	0.16	0.129	-0.031 ± 0.008	-0.192 ± 0.051	0.007 ± 0.001	0.084 ± 0.006	0.008 ± 0.001	0.089 ± 0.007	0.312 ± 0.043	0.558 ± 0.046	0.152	-0.008	-5.18	0.0946	0.0950	1.040	94.340 ± 2.244	0.344 ± 0.007	32.075 ± 4.534
QML	beta54	0.16	0.152	-0.008 ± 0.007	-0.052 ± 0.045	0.005 ± 0.001	0.073 ± 0.006	0.005 ± 0.001	0.074 ± 0.007	0.212 ± 0.031	0.460 ± 0.041	0.157	-0.003	-2.09	0.0672	0.0673	1.097	96.226 ± 1.851	0.316 ± 0.006	49.057 ± 4.856
QML	beta55	0.08	0.090	0.010 ± 0.013	0.119 ± 0.160	0.017 ± 0.003	0.132 ± 0.010	0.017 ± 0.003	0.132 ± 0.012	2.711 ± 0.391	1.647 ± 0.146	0.085	0.005	6.22	0.1457	0.1458	1.015	91.509 ± 2.707	0.525 ± 0.022	12.264 ± 3.186
QML	beta56	0.08	0.073	-0.007 ± 0.009	-0.084 ± 0.118	0.010 ± 0.001	0.097 ± 0.007	0.009 ± 0.001	0.097 ± 0.008	1.478 ± 0.200	1.216 ± 0.103	0.072	-0.008	-9.45	0.1031	0.1034	1.140	93.396 ± 2.412	0.436 ± 0.013	10.377 ± 2.962
QML	beta57	0.06	0.037	-0.023 ± 0.007	-0.383 ± 0.120	0.005 ± 0.001	0.074 ± 0.007	0.006 ± 0.001	0.077 ± 0.007	1.660 ± 0.252	1.288 ± 0.116	0.023	-0.037	-62.29	0.0573	0.0684	1.017	88.679 ± 3.077	0.296 ± 0.016	3.774 ± 1.851
QML	beta58	0.06	0.036	-0.024 ± 0.006	-0.408 ± 0.096	0.004 ± 0.001	0.060 ± 0.005	0.004 ± 0.001	0.064 ± 0.005	1.144 ± 0.160	1.069 ± 0.088	0.038	-0.022	-37.15	0.0595	0.0635	1.013	84.906 ± 3.477	0.237 ± 0.012	1.887 ± 1.322
UPI	beta41	0.20	0.203	0.003 ± 0.018	0.016 ± 0.089	0.034 ± 0.008	0.183 ± 0.022	0.033 ± 0.008	0.183 ± 0.024	0.833 ± 0.193	0.913 ± 0.118	0.220	0.020	10.13	0.1719	0.1731	1.488	95.283 ± 2.059	1.070 ± 0.157	39.623 ± 4.751
UPI	beta42	0.20	0.249	0.049 ± 0.020	0.243 ± 0.099	0.041 ± 0.009	0.204 ± 0.023	0.043 ± 0.009	0.208 ± 0.024	1.085 ± 0.222	1.042 ± 0.120	0.246	0.046	23.08	0.1808	0.1866	1.389	94.340 ± 2.244	1.108 ± 0.130	32.075 ± 4.534
UPI	beta43	0.20	0.177	-0.023 ± 0.011	-0.113 ± 0.055	0.013 ± 0.002	0.114 ± 0.008	0.013 ± 0.002	0.115 ± 0.010	0.333 ± 0.047	0.577 ± 0.050	0.180	-0.020	-9.94	0.1509	0.1522	1.186	85.849 ± 3.385	0.528 ± 0.032	26.415 ± 4.282
UPI	beta44	0.11	0.189	0.079 ± 0.022	0.718 ± 0.199	0.051 ± 0.010	0.226 ± 0.024	0.057 ± 0.012	0.238 ± 0.028	4.670 ± 0.982	2.161 ± 0.251	0.176	0.066	59.65	0.2134	0.2232	1.313	100.000 ± 0.000	1.159 ± 0.105	9.434 ± 2.839
UPI	beta45	0.11	0.122	0.012 ± 0.012	0.106 ± 0.111	0.016 ± 0.003	0.126 ± 0.011	0.016 ± 0.003	0.126 ± 0.013	1.302 ± 0.221	1.141 ± 0.114	0.125	0.015	13.37	0.1283	0.1291	1.470	98.113 ± 1.322	0.724 ± 0.048	6.604 ± 2.412
UPI	beta46	0.08	0.031	-0.049 ± 0.021	-0.607 ± 0.258	0.045 ± 0.011	0.212 ± 0.027	0.047 ± 0.011	0.217 ± 0.027	7.341 ± 1.672	2.709 ± 0.343	0.025	-0.055	-69.28	0.2086	0.2159	1.263	92.453 ± 2.566	1.051 ± 0.117	2.830 ± 1.611

Table S199

Simulation 2 - Condition 55: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.4
(continued)

Method	Param.	True	Mean-based										Median-based				Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	-0.006	-0.086 ± 0.023	-1.079 ± 0.284	0.055 ± 0.010	0.234 ± 0.022	0.062 ± 0.011	0.249 ± 0.026	9.655 ± 1.784	3.107 ± 0.322	-0.002	-0.082	-102.88	0.1955	0.2121	1.259	87.736 ± 3.186	1.156 ± 0.136	3.774 ± 1.851
UPI	beta51	0.16	0.244	0.084 ± 0.011	0.526 ± 0.068	0.013 ± 0.002	0.112 ± 0.011	0.020 ± 0.004	0.140 ± 0.014	0.765 ± 0.145	0.874 ± 0.089	0.230	0.070	43.93	0.1095	0.1301	1.045	92.453 ± 2.566	0.460 ± 0.017	59.434 ± 4.769
UPI	beta52	0.16	0.159	-0.001 ± 0.012	-0.006 ± 0.077	0.016 ± 0.004	0.126 ± 0.015	0.016 ± 0.004	0.126 ± 0.016	0.617 ± 0.139	0.785 ± 0.099	0.164	0.004	2.20	0.1045	0.1045	1.190	95.283 ± 2.059	0.589 ± 0.062	33.019 ± 4.568
UPI	beta53	0.16	0.136	-0.024 ± 0.010	-0.148 ± 0.060	0.010 ± 0.002	0.098 ± 0.009	0.010 ± 0.002	0.100 ± 0.009	0.394 ± 0.059	0.628 ± 0.056	0.133	-0.027	-16.89	0.1063	0.1097	1.211	92.453 ± 2.566	0.466 ± 0.030	19.811 ± 3.871
UPI	beta54	0.16	0.143	-0.017 ± 0.010	-0.104 ± 0.061	0.010 ± 0.001	0.101 ± 0.007	0.010 ± 0.001	0.101 ± 0.008	0.402 ± 0.047	0.634 ± 0.049	0.142	-0.018	-11.18	0.0944	0.0961	1.471	93.396 ± 2.412	0.580 ± 0.042	17.925 ± 3.725
UPI	beta55	0.08	0.117	0.037 ± 0.014	0.464 ± 0.172	0.020 ± 0.004	0.142 ± 0.014	0.021 ± 0.004	0.146 ± 0.017	3.335 ± 0.669	1.826 ± 0.207	0.095	0.015	18.39	0.1319	0.1327	1.177	98.113 ± 1.322	0.655 ± 0.042	9.434 ± 2.839
UPI	beta56	0.08	0.070	-0.010 ± 0.013	-0.128 ± 0.159	0.017 ± 0.002	0.131 ± 0.010	0.017 ± 0.003	0.131 ± 0.012	2.663 ± 0.392	1.632 ± 0.146	0.066	-0.014	-17.63	0.1242	0.1249	1.360	98.113 ± 1.322	0.697 ± 0.073	3.774 ± 1.851
UPI	beta57	0.06	0.006	-0.054 ± 0.012	-0.906 ± 0.203	0.016 ± 0.003	0.125 ± 0.014	0.019 ± 0.004	0.136 ± 0.017	5.151 ± 1.145	2.270 ± 0.276	0.013	-0.047	-78.58	0.0978	0.1086	1.244	93.396 ± 2.412	0.612 ± 0.074	0.000 ± 0.000
UPI	beta58	0.06	0.011	-0.049 ± 0.011	-0.811 ± 0.181	0.012 ± 0.002	0.112 ± 0.010	0.015 ± 0.002	0.121 ± 0.011	4.093 ± 0.628	2.023 ± 0.179	0.006	-0.054	-90.13	0.1078	0.1206	1.508	84.906 ± 3.477	0.660 ± 0.104	0.943 ± 0.939

Note. MCSE = Monte Carlo Standard Error shown as \u000b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S200

Simulation 2 - Condition 56: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, $N = 1000$, Reliability = 0.4

Method	Param.	True	Mean-based										Median-based				Inference			
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LSAM	beta41	0.20	0.205	0.005 $\pm$ 0.006	0.023 $\pm$ 0.028	0.005 $\pm$ 0.001	0.068 $\pm$ 0.005	0.005 $\pm$ 0.001	0.068 $\pm$ 0.005	0.115 $\pm$ 0.016	0.339 $\pm$ 0.027	0.202	0.002	1.07	0.0613	0.0613	0.961	91.034 $\pm$ 2.373	0.255 $\pm$ 0.003	92.414 $\pm$ 2.199
LSAM	beta42	0.20	0.200	-0.000 $\pm$ 0.006	-0.002 $\pm$ 0.028	0.005 $\pm$ 0.001	0.068 $\pm$ 0.004	0.005 $\pm$ 0.001	0.068 $\pm$ 0.005	0.114 $\pm$ 0.013	0.338 $\pm$ 0.025	0.203	0.003	1.39	0.0654	0.0655	1.004	95.172 $\pm$ 1.780	0.267 $\pm$ 0.003	82.759 $\pm$ 3.137
LSAM	beta43	0.20	0.192	-0.008 $\pm$ 0.005	-0.039 $\pm$ 0.023	0.003 $\pm$ 0.000	0.056 $\pm$ 0.003	0.003 $\pm$ 0.000	0.056 $\pm$ 0.004	0.078 $\pm$ 0.009	0.280 $\pm$ 0.021	0.195	-0.005	-2.66	0.0524	0.0527	0.908	93.103 $\pm$ 2.104	0.198 $\pm$ 0.002	97.241 $\pm$ 1.360
LSAM	beta44	0.11	0.182	0.072 $\pm$ 0.006	0.658 $\pm$ 0.056	0.006 $\pm$ 0.001	0.075 $\pm$ 0.005	0.011 $\pm$ 0.001	0.104 $\pm$ 0.006	0.890 $\pm$ 0.094	0.943 $\pm$ 0.054	0.183	0.073	66.13	0.0699	0.1008	1.039	90.345 $\pm$ 2.453	0.304 $\pm$ 0.006	67.586 $\pm$ 3.887
LSAM	beta45	0.11	0.122	0.012 $\pm$ 0.005	0.113 $\pm$ 0.046	0.004 $\pm$ 0.000	0.060 $\pm$ 0.004	0.004 $\pm$ 0.000	0.061 $\pm$ 0.005	0.311 $\pm$ 0.040	0.558 $\pm$ 0.043	0.125	0.015	13.47	0.0600	0.0618	1.026	97.241 $\pm$ 1.360	0.242 $\pm$ 0.004	53.103 $\pm$ 4.144
LSAM	beta46	0.08	0.022	-0.058 $\pm$ 0.005	-0.726 $\pm$ 0.057	0.003 $\pm$ 0.000	0.055 $\pm$ 0.003	0.006 $\pm$ 0.001	0.080 $\pm$ 0.005	0.995 $\pm$ 0.106	0.997 $\pm$ 0.057	0.017	-0.063	-79.14	0.0557	0.0843	0.923	71.034 $\pm$ 3.767	0.199 $\pm$ 0.004	5.517 $\pm$ 1.896
LSAM	beta47	0.08	0.021	-0.059 $\pm$ 0.004	-0.733 $\pm$ 0.056	0.003 $\pm$ 0.000	0.054 $\pm$ 0.003	0.006 $\pm$ 0.001	0.080 $\pm$ 0.004	0.992 $\pm$ 0.090	0.996 $\pm$ 0.049	0.021	-0.059	-73.29	0.0511	0.0778	0.912	66.207 $\pm$ 3.928	0.193 $\pm$ 0.005	6.897 $\pm$ 2.104
LSAM	beta51	0.16	0.187	0.027 $\pm$ 0.006	0.169 $\pm$ 0.036	0.005 $\pm$ 0.001	0.070 $\pm$ 0.004	0.006 $\pm$ 0.001	0.075 $\pm$ 0.005	0.220 $\pm$ 0.026	0.469 $\pm$ 0.033	0.191	0.031	19.17	0.0702	0.0766	0.871	90.345 $\pm$ 2.453	0.240 $\pm$ 0.002	81.379 $\pm$ 3.233
LSAM	beta52	0.16	0.153	-0.007 $\pm$ 0.005	-0.042 $\pm$ 0.033	0.004 $\pm$ 0.000	0.063 $\pm$ 0.003	0.004 $\pm$ 0.000	0.063 $\pm$ 0.004	0.155 $\pm$ 0.016	0.393 $\pm$ 0.027	0.153	-0.007	-4.09	0.0601	0.0604	0.889	92.414 $\pm$ 2.199	0.219 $\pm$ 0.002	77.241 $\pm$ 3.482
LSAM	beta53	0.16	0.145	-0.015 $\pm$ 0.005	-0.095 $\pm$ 0.031	0.004 $\pm$ 0.000	0.059 $\pm$ 0.004	0.004 $\pm$ 0.000	0.061 $\pm$ 0.004	0.146 $\pm$ 0.018	0.383 $\pm$ 0.028	0.141	-0.019	-11.76	0.0581	0.0611	0.934	91.724 $\pm$ 2.288	0.218 $\pm$ 0.002	73.793 $\pm$ 3.652
LSAM	beta54	0.16	0.134	-0.026 $\pm$ 0.005	-0.160 $\pm$ 0.029	0.003 $\pm$ 0.000	0.056 $\pm$ 0.003	0.004 $\pm$ 0.000	0.061 $\pm$ 0.004	0.145 $\pm$ 0.016	0.381 $\pm$ 0.025	0.140	-0.020	-12.71	0.0589	0.0623	1.001	93.103 $\pm$ 2.104	0.218 $\pm$ 0.003	66.207 $\pm$ 3.928
LSAM	beta55	0.08	0.111	0.031 $\pm$ 0.006	0.390 $\pm$ 0.080	0.006 $\pm$ 0.001	0.077 $\pm$ 0.005	0.007 $\pm$ 0.001	0.082 $\pm$ 0.006	1.063 $\pm$ 0.142	1.031 $\pm$ 0.079	0.112	0.032	40.05	0.0753	0.0818	0.955	94.483 $\pm$ 1.896	0.287 $\pm$ 0.006	35.172 $\pm$ 3.965
LSAM	beta56	0.08	0.068	-0.012 $\pm$ 0.006	-0.148 $\pm$ 0.072	0.005 $\pm$ 0.001	0.069 $\pm$ 0.004	0.005 $\pm$ 0.001	0.070 $\pm$ 0.005	0.762 $\pm$ 0.097	0.873 $\pm$ 0.067	0.072	-0.008	-9.56	0.0672	0.0676	1.008	95.862 $\pm$ 1.654	0.273 $\pm$ 0.006	17.241 $\pm$ 3.137
LSAM	beta57	0.06	0.014	-0.046 $\pm$ 0.004	-0.767 $\pm$ 0.065	0.002 $\pm$ 0.000	0.047 $\pm$ 0.003	0.004 $\pm$ 0.000	0.065 $\pm$ 0.004	1.190 $\pm$ 0.120	1.091 $\pm$ 0.060	0.011	-0.049	-81.71	0.0490	0.0693	0.907	76.552 $\pm$ 3.518	0.166 $\pm$ 0.003	6.897 $\pm$ 2.104
LSAM	beta58	0.06	0.034	-0.026 $\pm$ 0.003	-0.438 $\pm$ 0.052	0.001 $\pm$ 0.000	0.038 $\pm$ 0.002	0.002 $\pm$ 0.000	0.046 $\pm$ 0.003	0.584 $\pm$ 0.056	0.764 $\pm$ 0.043	0.030	-0.030	-49.70	0.0399	0.0498	0.977	82.069 $\pm$ 3.186	0.144 $\pm$ 0.003	11.724 $\pm$ 2.672
QML	beta41	0.20	0.182	-0.018 $\pm$ 0.004	-0.092 $\pm$ 0.021	0.003 $\pm$ 0.000	0.051 $\pm$ 0.004	0.003 $\pm$ 0.000	0.054 $\pm$ 0.004	0.074 $\pm$ 0.010	0.271 $\pm$ 0.022	0.175	-0.025	-12.49	0.0444	0.0509	0.927	92.414 $\pm$ 2.199	0.186 $\pm$ 0.002	95.862 $\pm$ 1.654
QML	beta42	0.20	0.182	-0.018 $\pm$ 0.004	-0.091 $\pm$ 0.021	0.002 $\pm$ 0.000	0.050 $\pm$ 0.003	0.003 $\pm$ 0.000	0.053 $\pm$ 0.004	0.070 $\pm$ 0.009	0.265 $\pm$ 0.019	0.179	-0.021	-10.69	0.0524	0.0566	0.978	93.103 $\pm$ 2.104	0.191 $\pm$ 0.002	95.172 $\pm$ 1.780
QML	beta43	0.20	0.183	-0.017 $\pm$ 0.004	-0.087 $\pm$ 0.020	0.002 $\pm$ 0.000	0.049 $\pm$ 0.003	0.003 $\pm$ 0.000	0.051 $\pm$ 0.003	0.066 $\pm$ 0.007	0.257 $\pm$ 0.017	0.183	-0.017	-8.32	0.0504	0.0531	0.927	88.966 $\pm$ 2.602	0.177 $\pm$ 0.002	97.241 $\pm$ 1.360
QML	beta44	0.11	0.147	0.037 $\pm$ 0.005	0.337 $\pm$ 0.048	0.004 $\pm$ 0.001	0.063 $\pm$ 0.004	0.005 $\pm$ 0.001	0.073 $\pm$ 0.004	0.444 $\pm$ 0.044	0.666 $\pm$ 0.039	0.148	0.038	34.75	0.0608	0.0718	1.022	94.483 $\pm$ 1.896	0.254 $\pm$ 0.004	68.276 $\pm$ 3.865
QML	beta45	0.11	0.114	0.004 $\pm$ 0.005	0.034 $\pm$ 0.043	0.003 $\pm$ 0.000	0.058 $\pm$ 0.003	0.003 $\pm$ 0.000	0.058 $\pm$ 0.004	0.273 $\pm$ 0.031	0.523 $\pm$ 0.037	0.118	0.008	7.18	0.0601	0.0606	0.974	95.862 $\pm$ 1.654	0.220 $\pm$ 0.003	59.310 $\pm$ 4.080
QML	beta46	0.08	0.047	-0.033 $\pm$ 0.004	-0.408 $\pm$ 0.050	0.002 $\pm$ 0.000	0.048 $\pm$ 0.003	0.003 $\pm$ 0.000	0.058 $\pm$ 0.003	0.520 $\pm$ 0.050	0.721 $\pm$ 0.040	0.043	-0.037	-46.21	0.0494	0.0617	0.898	78.621 $\pm$ 3.405	0.168 $\pm$ 0.004	16.552 $\pm$ 3.086
QML	beta47	0.08	0.041	-0.039 $\pm$ 0.004	-0.487 $\pm$ 0.046	0.002 $\pm$ 0.000	0.044 $\pm$ 0.003	0.003 $\pm$ 0.000	0.059 $\pm$ 0.003	0.541 $\pm$ 0.049	0.736 $\pm$ 0.038	0.039	-0.041	-51.55	0.0456	0.0615	0.926	74.483 $\pm$ 3.620	0.161 $\pm$ 0.004	13.103 $\pm$ 2.802
QML	beta51	0.16	0.194	0.034 $\pm$ 0.006	0.213 $\pm$ 0.038	0.005 $\pm$ 0.001	0.073 $\pm$ 0.005	0.006 $\pm$ 0.001	0.080 $\pm$ 0.005	0.251 $\pm$ 0.029	0.501 $\pm$ 0.034	0.197	0.037	22.84	0.0701	0.0790	0.922	88.966 $\pm$ 2.602	0.263 $\pm$ 0.004	80.690 $\pm$ 3.278
QML	beta52	0.16	0.146	-0.014 $\pm$ 0.005	-0.090 $\pm$ 0.030	0.003 $\pm$ 0.000	0.058 $\pm$ 0.004	0.004 $\pm$ 0.000	0.060 $\pm$ 0.004	0.140 $\pm$ 0.015	0.374 $\pm$ 0.025	0.145	-0.015	-9.20	0.0572	0.0591	0.923	91.034 $\pm$ 2.373	0.211 $\pm$ 0.004	79.310 $\pm$ 3.364
QML	beta53	0.16	0.132	-0.028 $\pm$ 0.004	-0.173 $\pm$ 0.027	0.003 $\pm$ 0.000	0.053 $\pm$ 0.003	0.004 $\pm$ 0.000	0.059 $\pm$ 0.004	0.138 $\pm$ 0.017	0.371 $\pm$ 0.026	0.129	-0.031	-19.39	0.0549	0.0630	0.981	89.655 $\pm$ 2.529	0.203 $\pm$ 0.002	74.483 $\pm$ 3.620
QML	beta54	0.16	0.147	-0.013 $\pm$ 0.004	-0.081 $\pm$ 0.026	0.003 $\pm$ 0.000	0.050 $\pm$ 0.003	0.003 $\pm$ 0.000	0.052 $\pm$ 0.004	0.104 $\pm$ 0.013	0.323 $\pm$ 0.024	0.151	-0.009	-5.40	0.0478	0.0486	0.972	93.103 $\pm$ 2.104	0.191 $\pm$ 0.002	86.207 $\pm$ 2.864
QML	beta55	0.08	0.109	0.029 $\pm$ 0.006	0.360 $\pm$ 0.081	0.006 $\pm$ 0.001	0.078 $\pm$ 0.005	0.007 $\pm$ 0.001	0.083 $\pm$ 0.006	1.078 $\pm$ 0.123	1.038 $\pm$ 0.071	0.111	0.031	39.04	0.0798	0.0857	1.025	93.793 $\pm$ 2.004	0.314 $\pm$ 0.008	28.966 $\pm$ 3.767
QML	beta56	0.08	0.070	-0.010 $\pm$ 0.005	-0.125 $\pm$ 0.065	0.004 $\pm$ 0.000	0.062 $\pm$ 0.003	0.004 $\pm$ 0.000	0.063 $\pm$ 0.004	0.615 $\pm$ 0.061	0.784 $\pm$ 0.051	0.078	-0.002	-2.40	0.0702	0.0702	1.064	95.172 $\pm$ 1.780	0.259 $\pm$ 0.004	15.862 $\pm$ 3.034
QML	beta57	0.06	0.022	-0.038 $\pm$ 0.004	-0.638 $\pm$ 0.065	0.002 $\pm$ 0.000	0.047 $\pm$ 0.003	0.004 $\pm$ 0.000	0.060 $\pm$ 0.003	1.009 $\pm$ 0.092	1.005 $\pm$ 0.053	0.013	-0.047	-77.70	0.0489	0.0675	0.973	75.862 $\pm$ 3.554	0.178 $\pm$ 0.006	2.759 $\pm$ 1.360
QML	beta58	0.06	0.037	-0.023 $\pm$ 0.003	-0.379 $\pm$ 0.052	0.001 $\pm$ 0.000	0.037 $\pm$ 0.002	0.002 $\pm$ 0.000	0.044 $\pm$ 0.002	0.530 $\pm$ 0.049	0.728 $\pm$ 0.041	0.032	-0.028	-47.34	0.0375	0.0470	0.959	82.759 $\pm$ 3.137	0.141 $\pm$ 0.003	11.724 $\pm$ 2.672
UPI	beta41	0.20	0.228	0.028 $\pm$ 0.015	0.138 $\pm$ 0.076	0.034 $\pm$ 0.005	0.184 $\pm$ 0.015	0.034 $\pm$ 0.006	0.185 $\pm$ 0.017	0.857 $\pm$ 0.142	0.926 $\pm$ 0.087	0.214	0.014	7.11	0.1703	0.1709	1.493	95.862 $\pm$ 1.654	1.075 $\pm$ 0.169	31.724 $\pm$ 3.865
UPI	beta42	0.20	0.252	0.052 $\pm$ 0.017	0.260 $\pm$ 0.087	0.044 $\pm$ 0.008	0.210 $\pm$ 0.018	0.046 $\pm$ 0.008	0.216 $\pm$ 0.021	1.161 $\pm$ 0.200	1.078 $\pm$ 0.104	0.239	0.039	19.39	0.1909	0.1948	1.184	93.793 $\pm$ 2.004	0.974 $\pm$ 0.120	40.690 $\pm$ 4.080
UPI	beta43	0.20	0.147	-0.053 $\pm$ 0.008	-0.264 $\pm$ 0.040	0.009 $\pm$ 0.001	0.096 $\pm$ 0.006	0.012 $\pm$ 0.002	0.109 $\pm$ 0.008	0.297 $\pm$ 0.040	0.545 $\pm$ 0.041	0.147	-0.053	-26.28	0.1267	0.1371	0.979	80.690 $\pm$ 3.278	0.367 $\pm$ 0.029	49.655

Table S200

Simulation 2 - Condition 56: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.4
(continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	-0.051	-0.131 ± 0.016	-1.641 ± 0.196	0.036 ± 0.006	0.189 ± 0.017	0.053 ± 0.006	0.229 ± 0.016	8.211 ± 1.003	2.866 ± 0.195	-0.033	-0.113	-140.76	0.1758	0.2088	1.220	88.276 ± 2.672	0.902 ± 0.104	4.138 ± 1.654
UPI	beta51	0.16	0.242	0.082 ± 0.006	0.514 ± 0.037	0.005 ± 0.001	0.071 ± 0.004	0.012 ± 0.001	0.109 ± 0.005	0.462 ± 0.042	0.680 ± 0.034	0.242	0.082	51.14	0.0810	0.1152	1.035	76.552 ± 3.518	0.290 ± 0.012	87.586 ± 2.738
UPI	beta52	0.16	0.159	-0.001 ± 0.008	-0.009 ± 0.050	0.009 ± 0.002	0.097 ± 0.009	0.009 ± 0.002	0.097 ± 0.010	0.367 ± 0.064	0.606 ± 0.060	0.163	0.003	1.67	0.1003	0.1003	1.639	92.414 ± 2.199	0.625 ± 0.189	38.621 ± 4.043
UPI	beta53	0.16	0.126	-0.034 ± 0.006	-0.211 ± 0.037	0.005 ± 0.001	0.072 ± 0.004	0.006 ± 0.001	0.079 ± 0.005	0.243 ± 0.026	0.493 ± 0.031	0.125	-0.035	-22.07	0.0826	0.0898	1.068	90.345 ± 2.453	0.299 ± 0.019	48.276 ± 4.150
UPI	beta54	0.16	0.115	-0.045 ± 0.007	-0.280 ± 0.044	0.007 ± 0.001	0.086 ± 0.006	0.009 ± 0.001	0.096 ± 0.007	0.363 ± 0.043	0.603 ± 0.041	0.105	-0.055	-34.20	0.0932	0.1081	1.126	81.379 ± 3.233	0.378 ± 0.022	26.897 ± 3.682
UPI	beta55	0.08	0.131	0.051 ± 0.008	0.639 ± 0.098	0.009 ± 0.001	0.095 ± 0.007	0.011 ± 0.002	0.107 ± 0.008	1.794 ± 0.249	1.339 ± 0.104	0.133	0.053	66.42	0.0888	0.1034	1.069	93.793 ± 2.004	0.396 ± 0.016	28.966 ± 3.767
UPI	beta56	0.08	0.072	-0.008 ± 0.007	-0.103 ± 0.082	0.006 ± 0.001	0.079 ± 0.005	0.006 ± 0.001	0.079 ± 0.006	0.971 ± 0.129	0.986 ± 0.078	0.066	-0.014	-17.23	0.0772	0.0784	1.196	95.862 ± 1.654	0.369 ± 0.015	8.966 ± 2.373
UPI	beta57	0.06	-0.039	-0.099 ± 0.011	-1.644 ± 0.191	0.019 ± 0.004	0.138 ± 0.013	0.029 ± 0.005	0.169 ± 0.017	7.969 ± 1.475	2.823 ± 0.278	-0.009	-0.069	-114.66	0.1187	0.1372	1.315	85.517 ± 2.923	0.712 ± 0.197	0.690 ± 0.687
UPI	beta58	0.06	-0.009	-0.069 ± 0.008	-1.157 ± 0.139	0.010 ± 0.002	0.101 ± 0.008	0.015 ± 0.002	0.122 ± 0.010	4.131 ± 0.631	2.033 ± 0.168	0.002	-0.058	-96.17	0.0797	0.0984	1.092	82.759 ± 3.137	0.431 ± 0.044	1.379 ± 0.969

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S201

Simulation 2 - Condition 57: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, $N = 400$, Reliability = 0.6

Method	Param.	True	Mean-based										Median-based					Inference		
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LSAM	beta41	0.20	0.197	-0.003 $\pm$ 0.002	-0.015 $\pm$ 0.010	0.004 $\pm$ 0.000	0.063 $\pm$ 0.001	0.004 $\pm$ 0.000	0.063 $\pm$ 0.002	0.099 $\pm$ 0.005	0.314 $\pm$ 0.009	0.198	-0.002	-0.84	0.0620	0.0620	0.976	93.821 $\pm$ 0.773	0.240 $\pm$ 0.001	89.289 $\pm$ 0.992
LSAM	beta42	0.20	0.197	-0.003 $\pm$ 0.002	-0.015 $\pm$ 0.010	0.004 $\pm$ 0.000	0.062 $\pm$ 0.001	0.004 $\pm$ 0.000	0.062 $\pm$ 0.002	0.095 $\pm$ 0.005	0.309 $\pm$ 0.009	0.197	-0.003	-1.48	0.0595	0.0596	1.016	95.469 $\pm$ 0.667	0.246 $\pm$ 0.001	88.465 $\pm$ 1.025
LSAM	beta43	0.20	0.200	-0.000 $\pm$ 0.002	-0.002 $\pm$ 0.010	0.004 $\pm$ 0.000	0.059 $\pm$ 0.001	0.004 $\pm$ 0.000	0.059 $\pm$ 0.002	0.088 $\pm$ 0.004	0.297 $\pm$ 0.008	0.198	-0.002	-1.21	0.0613	0.0613	0.958	94.645 $\pm$ 0.722	0.223 $\pm$ 0.001	94.130 $\pm$ 0.754
LSAM	beta44	0.11	0.132	0.022 $\pm$ 0.002	0.201 $\pm$ 0.022	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.006 $\pm$ 0.000	0.078 $\pm$ 0.002	0.506 $\pm$ 0.024	0.711 $\pm$ 0.020	0.132	0.022	20.41	0.0748	0.0781	0.973	93.821 $\pm$ 0.773	0.286 $\pm$ 0.002	47.065 $\pm$ 1.602
LSAM	beta45	0.11	0.112	0.002 $\pm$ 0.002	0.015 $\pm$ 0.019	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	0.358 $\pm$ 0.019	0.599 $\pm$ 0.019	0.107	-0.003	-2.49	0.0654	0.0655	0.952	94.439 $\pm$ 0.735	0.246 $\pm$ 0.002	43.357 $\pm$ 1.590
LSAM	beta46	0.08	0.062	-0.018 $\pm$ 0.002	-0.221 $\pm$ 0.022	0.003 $\pm$ 0.000	0.055 $\pm$ 0.001	0.003 $\pm$ 0.000	0.058 $\pm$ 0.002	0.522 $\pm$ 0.024	0.723 $\pm$ 0.020	0.061	-0.019	-23.46	0.0518	0.0551	0.928	90.525 $\pm$ 0.940	0.200 $\pm$ 0.002	24.511 $\pm$ 1.380
LSAM	beta47	0.08	0.060	-0.020 $\pm$ 0.002	-0.255 $\pm$ 0.021	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.003 $\pm$ 0.000	0.056 $\pm$ 0.002	0.495 $\pm$ 0.023	0.704 $\pm$ 0.019	0.056	-0.024	-29.63	0.0499	0.0552	0.947	89.495 $\pm$ 0.984	0.195 $\pm$ 0.002	21.833 $\pm$ 1.326
LSAM	beta51	0.16	0.174	0.014 $\pm$ 0.002	0.085 $\pm$ 0.015	0.006 $\pm$ 0.000	0.074 $\pm$ 0.002	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.222 $\pm$ 0.011	0.472 $\pm$ 0.013	0.173	0.013	8.00	0.0714	0.0726	0.940	93.100 $\pm$ 0.813	0.273 $\pm$ 0.001	71.679 $\pm$ 1.446
LSAM	beta52	0.16	0.150	-0.010 $\pm$ 0.002	-0.066 $\pm$ 0.012	0.003 $\pm$ 0.000	0.059 $\pm$ 0.001	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	0.139 $\pm$ 0.007	0.373 $\pm$ 0.011	0.151	-0.009	-5.82	0.0572	0.0579	0.994	94.542 $\pm$ 0.729	0.229 $\pm$ 0.001	73.018 $\pm$ 1.424
LSAM	beta53	0.16	0.161	0.001 $\pm$ 0.002	0.003 $\pm$ 0.013	0.004 $\pm$ 0.000	0.063 $\pm$ 0.001	0.004 $\pm$ 0.000	0.063 $\pm$ 0.002	0.153 $\pm$ 0.007	0.391 $\pm$ 0.011	0.160	0.000	0.03	0.0624	0.0624	0.990	93.924 $\pm$ 0.767	0.243 $\pm$ 0.001	74.047 $\pm$ 1.407
LSAM	beta54	0.16	0.153	-0.007 $\pm$ 0.002	-0.046 $\pm$ 0.012	0.004 $\pm$ 0.000	0.061 $\pm$ 0.001	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.146 $\pm$ 0.007	0.382 $\pm$ 0.011	0.154	-0.006	-3.86	0.0582	0.0585	0.968	93.615 $\pm$ 0.785	0.230 $\pm$ 0.001	74.871 $\pm$ 1.392
LSAM	beta55	0.08	0.092	0.012 $\pm$ 0.003	0.148 $\pm$ 0.032	0.006 $\pm$ 0.000	0.079 $\pm$ 0.002	0.006 $\pm$ 0.000	0.080 $\pm$ 0.002	0.992 $\pm$ 0.051	0.996 $\pm$ 0.030	0.093	0.013	16.06	0.0719	0.0730	0.954	94.439 $\pm$ 0.735	0.295 $\pm$ 0.002	26.571 $\pm$ 1.418
LSAM	beta56	0.08	0.080	-0.000 $\pm$ 0.002	-0.000 $\pm$ 0.030	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.883 $\pm$ 0.045	0.940 $\pm$ 0.028	0.077	-0.003	-4.33	0.0708	0.0709	0.943	94.027 $\pm$ 0.761	0.278 $\pm$ 0.002	21.833 $\pm$ 1.326
LSAM	beta57	0.06	0.046	-0.014 $\pm$ 0.002	-0.237 $\pm$ 0.027	0.003 $\pm$ 0.000	0.050 $\pm$ 0.001	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.763 $\pm$ 0.035	0.873 $\pm$ 0.024	0.042	-0.018	-30.06	0.0500	0.0531	0.953	91.555 $\pm$ 0.892	0.189 $\pm$ 0.002	15.963 $\pm$ 1.175
LSAM	beta58	0.06	0.051	-0.009 $\pm$ 0.001	-0.143 $\pm$ 0.025	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.604 $\pm$ 0.029	0.777 $\pm$ 0.022	0.049	-0.011	-17.89	0.0429	0.0443	0.920	90.422 $\pm$ 0.944	0.165 $\pm$ 0.001	22.142 $\pm$ 1.332
QML	beta41	0.20	0.190	-0.010 $\pm$ 0.002	-0.052 $\pm$ 0.010	0.004 $\pm$ 0.000	0.060 $\pm$ 0.001	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.092 $\pm$ 0.005	0.304 $\pm$ 0.009	0.191	-0.009	-4.62	0.0593	0.0600	0.978	93.615 $\pm$ 0.785	0.230 $\pm$ 0.001	89.598 $\pm$ 0.980
QML	beta42	0.20	0.192	-0.008 $\pm$ 0.002	-0.038 $\pm$ 0.010	0.004 $\pm$ 0.000	0.059 $\pm$ 0.001	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	0.090 $\pm$ 0.004	0.300 $\pm$ 0.009	0.192	-0.008	-4.12	0.0572	0.0578	1.001	94.954 $\pm$ 0.702	0.233 $\pm$ 0.001	90.319 $\pm$ 0.949
QML	beta43	0.20	0.190	-0.010 $\pm$ 0.002	-0.048 $\pm$ 0.009	0.003 $\pm$ 0.000	0.057 $\pm$ 0.001	0.003 $\pm$ 0.000	0.058 $\pm$ 0.002	0.083 $\pm$ 0.004	0.288 $\pm$ 0.008	0.189	-0.011	-5.74	0.0583	0.0594	0.964	94.027 $\pm$ 0.761	0.215 $\pm$ 0.001	93.821 $\pm$ 0.773
QML	beta44	0.11	0.118	0.008 $\pm$ 0.002	0.077 $\pm$ 0.022	0.006 $\pm$ 0.000	0.074 $\pm$ 0.002	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.463 $\pm$ 0.023	0.681 $\pm$ 0.020	0.121	0.011	9.62	0.0722	0.0730	0.970	94.748 $\pm$ 0.716	0.283 $\pm$ 0.003	44.799 $\pm$ 1.596
QML	beta45	0.11	0.107	-0.003 $\pm$ 0.002	-0.032 $\pm$ 0.019	0.004 $\pm$ 0.000	0.064 $\pm$ 0.002	0.004 $\pm$ 0.000	0.064 $\pm$ 0.002	0.344 $\pm$ 0.018	0.586 $\pm$ 0.018	0.106	-0.004	-3.70	0.0619	0.0620	0.968	94.542 $\pm$ 0.729	0.244 $\pm$ 0.002	42.327 $\pm$ 1.586
QML	beta46	0.08	0.069	-0.011 $\pm$ 0.002	-0.136 $\pm$ 0.020	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.425 $\pm$ 0.020	0.652 $\pm$ 0.018	0.066	-0.014	-17.21	0.0499	0.0517	0.966	90.628 $\pm$ 0.935	0.193 $\pm$ 0.002	25.541 $\pm$ 1.399
QML	beta47	0.08	0.065	-0.015 $\pm$ 0.002	-0.182 $\pm$ 0.020	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.435 $\pm$ 0.019	0.659 $\pm$ 0.018	0.060	-0.020	-25.08	0.0487	0.0526	0.953	88.260 $\pm$ 1.033	0.189 $\pm$ 0.002	23.378 $\pm$ 1.358
QML	beta51	0.16	0.181	0.021 $\pm$ 0.002	0.129 $\pm$ 0.015	0.006 $\pm$ 0.000	0.076 $\pm$ 0.002	0.006 $\pm$ 0.000	0.079 $\pm$ 0.002	0.242 $\pm$ 0.012	0.492 $\pm$ 0.014	0.178	0.018	11.21	0.0738	0.0759	0.984	93.615 $\pm$ 0.785	0.293 $\pm$ 0.002	69.104 $\pm$ 1.483
QML	beta52	0.16	0.137	-0.023 $\pm$ 0.002	-0.141 $\pm$ 0.012	0.004 $\pm$ 0.000	0.062 $\pm$ 0.002	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	0.170 $\pm$ 0.008	0.413 $\pm$ 0.011	0.137	-0.023	-14.46	0.0576	0.0621	1.025	93.615 $\pm$ 0.785	0.249 $\pm$ 0.002	59.011 $\pm$ 1.578
QML	beta53	0.16	0.141	-0.019 $\pm$ 0.002	-0.121 $\pm$ 0.012	0.004 $\pm$ 0.000	0.062 $\pm$ 0.001	0.004 $\pm$ 0.000	0.064 $\pm$ 0.002	0.162 $\pm$ 0.007	0.403 $\pm$ 0.011	0.138	-0.022	-13.54	0.0601	0.0639	1.008	93.203 $\pm$ 0.808	0.243 $\pm$ 0.001	62.307 $\pm$ 1.555
QML	beta54	0.16	0.156	-0.004 $\pm$ 0.002	-0.026 $\pm$ 0.012	0.004 $\pm$ 0.000	0.060 $\pm$ 0.001	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	0.143 $\pm$ 0.007	0.378 $\pm$ 0.011	0.156	-0.004	-2.30	0.0586	0.0588	0.970	94.130 $\pm$ 0.754	0.230 $\pm$ 0.001	76.004 $\pm$ 1.370
QML	beta55	0.08	0.092	0.012 $\pm$ 0.003	0.151 $\pm$ 0.032	0.006 $\pm$ 0.000	0.079 $\pm$ 0.002	0.006 $\pm$ 0.000	0.080 $\pm$ 0.002	0.990 $\pm$ 0.047	0.995 $\pm$ 0.028	0.097	0.017	21.86	0.0746	0.0767	0.992	94.439 $\pm$ 0.735	0.306 $\pm$ 0.003	24.099 $\pm$ 1.373
QML	beta56	0.08	0.079	-0.001 $\pm$ 0.002	-0.007 $\pm$ 0.029	0.005 $\pm$ 0.000	0.073 $\pm$ 0.002	0.005 $\pm$ 0.000	0.073 $\pm$ 0.002	0.835 $\pm$ 0.041	0.914 $\pm$ 0.027	0.076	-0.004	-5.38	0.0720	0.0721	0.973	94.439 $\pm$ 0.735	0.279 $\pm$ 0.002	19.258 $\pm$ 1.265
QML	beta57	0.06	0.043	-0.017 $\pm$ 0.002	-0.280 $\pm$ 0.026	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.751 $\pm$ 0.032	0.867 $\pm$ 0.022	0.038	-0.022	-35.93	0.0468	0.0515	0.986	89.907 $\pm$ 0.967	0.190 $\pm$ 0.002	11.226 $\pm$ 1.013
QML	beta58	0.06	0.051	-0.009 $\pm$ 0.001	-0.148 $\pm$ 0.024	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.571 $\pm$ 0.026	0.756 $\pm$ 0.021	0.048	-0.012	-20.13	0.0430	0.0446	0.948	89.701 $\pm$ 0.975	0.165 $\pm$ 0.002	18.744 $\pm$ 1.252
UPI	beta41	0.20	0.209	0.009 $\pm$ 0.002	0.047 $\pm$ 0.010	0.004 $\pm$ 0.000	0.065 $\pm$ 0.002	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	0.108 $\pm$ 0.005	0.329 $\pm$ 0.010	0.210	0.010	4.89	0.0661	0.0668	0.963	94.027 $\pm$ 0.761	0.246 $\pm$ 0.002	90.422 $\pm$ 0.944
UPI	beta42	0.20	0.206	0.006 $\pm$ 0.002	0.031 $\pm$ 0.010	0.004 $\pm$ 0.000	0.062 $\pm$ 0.001	0.004 $\pm$ 0.000	0.062 $\pm$ 0.002	0.096 $\pm$ 0.005	0.309 $\pm$ 0.009	0.204	0.004	1.88	0.0582	0.0584	1.054	96.087 $\pm$ 0.622	0.254 $\pm$ 0.002	89.392 $\pm$ 0.988
UPI	beta43	0.20	0.202	0.002 $\pm$ 0.002	0.012 $\pm$ 0.010	0.004 $\pm$ 0.000	0.061 $\pm$ 0.001	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.093 $\pm$ 0.004	0.305 $\pm$ 0.009	0.200	0.000	0.01	0.0622	0.0622	0.967	95.160 $\pm$ 0.689	0.231 $\pm$ 0.001	93.821 $\pm$

Table S201

Simulation 2 - Condition 57: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.6
(continued)

Method	Param.	True	Mean-based										Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE	
UPI	beta47	0.08	0.070	-0.010 ± 0.002	-0.125 ± 0.025	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	0.619 ± 0.031	0.787 ± 0.023	0.065	-0.015	-18.61	0.0580	0.0599	1.023	92.997 ± 0.819	0.249 ± 0.007	20.906 ± 1.305	
UPI	beta51	0.16	0.174	0.014 ± 0.002	0.088 ± 0.015	0.005 ± 0.000	0.073 ± 0.002	0.006 ± 0.000	0.074 ± 0.002	0.216 ± 0.010	0.465 ± 0.013	0.171	0.011	7.11	0.0729	0.0738	0.969	93.615 ± 0.785	0.278 ± 0.002	70.752 ± 1.460	
UPI	beta52	0.16	0.159	-0.001 ± 0.002	-0.005 ± 0.012	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.138 ± 0.007	0.372 ± 0.011	0.159	-0.001	-0.86	0.0599	0.0599	0.999	94.748 ± 0.716	0.233 ± 0.001	76.828 ± 1.354	
UPI	beta53	0.16	0.162	0.002 ± 0.002	0.014 ± 0.013	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.152 ± 0.007	0.390 ± 0.011	0.160	0.000	0.13	0.0628	0.0628	1.008	94.645 ± 0.722	0.247 ± 0.001	74.253 ± 1.403	
UPI	beta54	0.16	0.159	-0.001 ± 0.002	-0.003 ± 0.012	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.147 ± 0.007	0.383 ± 0.011	0.160	0.000	-0.09	0.0604	0.0604	0.985	93.924 ± 0.767	0.237 ± 0.001	75.077 ± 1.388	
UPI	beta55	0.08	0.086	0.006 ± 0.002	0.076 ± 0.031	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	0.951 ± 0.047	0.975 ± 0.029	0.090	0.010	12.89	0.0748	0.0755	0.997	95.572 ± 0.660	0.304 ± 0.003	22.142 ± 1.332	
UPI	beta56	0.08	0.080	0.000 ± 0.002	0.004 ± 0.031	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	0.910 ± 0.050	0.954 ± 0.030	0.075	-0.005	-6.41	0.0711	0.0712	0.971	94.748 ± 0.716	0.290 ± 0.003	18.023 ± 1.234	
UPI	beta57	0.06	0.056	-0.004 ± 0.002	-0.064 ± 0.029	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.797 ± 0.041	0.893 ± 0.027	0.053	-0.007	-11.44	0.0527	0.0531	1.018	93.718 ± 0.779	0.213 ± 0.002	16.581 ± 1.194	
UPI	beta58	0.06	0.058	-0.002 ± 0.002	-0.029 ± 0.028	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	0.763 ± 0.042	0.873 ± 0.028	0.055	-0.005	-7.65	0.0490	0.0492	0.978	93.512 ± 0.790	0.201 ± 0.003	21.112 ± 1.310	

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S202

Simulation 2 - Condition 58: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.6

Method	Param.	True	Mean-based										Median-based					Inference		
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LSAM	beta41	0.20	0.192	-0.008 $\pm$ 0.001	-0.038 $\pm$ 0.006	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.001 $\pm$ 0.000	0.039 $\pm$ 0.001	0.037 $\pm$ 0.002	0.193 $\pm$ 0.005	0.191	-0.009	-4.54	0.0365	0.0376	1.002	94.684 $\pm$ 0.711	0.149 $\pm$ 0.000	99.900 $\pm$ 0.100
LSAM	beta42	0.20	0.195	-0.005 $\pm$ 0.001	-0.025 $\pm$ 0.006	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.040 $\pm$ 0.002	0.200 $\pm$ 0.005	0.194	-0.006	-3.03	0.0415	0.0419	0.978	94.985 $\pm$ 0.691	0.152 $\pm$ 0.000	99.699 $\pm$ 0.173
LSAM	beta43	0.20	0.201	0.001 $\pm$ 0.001	0.006 $\pm$ 0.006	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.032 $\pm$ 0.001	0.179 $\pm$ 0.005	0.200	0.000	0.02	0.0334	0.0334	0.988	93.681 $\pm$ 0.771	0.139 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta44	0.11	0.127	0.017 $\pm$ 0.001	0.150 $\pm$ 0.013	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.191 $\pm$ 0.008	0.437 $\pm$ 0.011	0.126	0.016	14.82	0.0456	0.0485	0.976	92.778 $\pm$ 0.820	0.173 $\pm$ 0.001	79.940 $\pm$ 1.268
LSAM	beta45	0.11	0.113	0.003 $\pm$ 0.001	0.032 $\pm$ 0.011	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.121 $\pm$ 0.005	0.347 $\pm$ 0.009	0.112	0.002	2.11	0.0395	0.0396	1.014	95.085 $\pm$ 0.685	0.151 $\pm$ 0.001	84.554 $\pm$ 1.145
LSAM	beta46	0.08	0.064	-0.016 $\pm$ 0.001	-0.205 $\pm$ 0.012	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.194 $\pm$ 0.008	0.440 $\pm$ 0.011	0.064	-0.016	-20.31	0.0296	0.0337	0.984	89.368 $\pm$ 0.976	0.120 $\pm$ 0.001	55.466 $\pm$ 1.574
LSAM	beta47	0.08	0.063	-0.017 $\pm$ 0.001	-0.219 $\pm$ 0.012	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.195 $\pm$ 0.009	0.442 $\pm$ 0.011	0.063	-0.017	-21.71	0.0296	0.0343	0.983	88.867 $\pm$ 0.996	0.119 $\pm$ 0.001	55.868 $\pm$ 1.573
LSAM	beta51	0.16	0.170	0.010 $\pm$ 0.001	0.063 $\pm$ 0.009	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.086 $\pm$ 0.004	0.294 $\pm$ 0.008	0.169	0.009	5.36	0.0465	0.0472	0.954	93.681 $\pm$ 0.771	0.172 $\pm$ 0.001	96.991 $\pm$ 0.541
LSAM	beta52	0.16	0.153	-0.007 $\pm$ 0.001	-0.045 $\pm$ 0.007	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.054 $\pm$ 0.002	0.232 $\pm$ 0.006	0.153	-0.007	-4.23	0.0361	0.0367	1.000	94.684 $\pm$ 0.711	0.143 $\pm$ 0.000	98.395 $\pm$ 0.398
LSAM	beta53	0.16	0.157	-0.003 $\pm$ 0.001	-0.018 $\pm$ 0.008	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.057 $\pm$ 0.003	0.238 $\pm$ 0.007	0.157	-0.003	-1.93	0.0377	0.0379	1.011	94.483 $\pm$ 0.723	0.151 $\pm$ 0.000	98.596 $\pm$ 0.373
LSAM	beta54	0.16	0.151	-0.009 $\pm$ 0.001	-0.055 $\pm$ 0.007	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.057 $\pm$ 0.003	0.240 $\pm$ 0.006	0.150	-0.010	-5.99	0.0362	0.0374	0.973	94.483 $\pm$ 0.723	0.142 $\pm$ 0.000	98.094 $\pm$ 0.433
LSAM	beta55	0.08	0.093	0.013 $\pm$ 0.001	0.162 $\pm$ 0.018	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.350 $\pm$ 0.016	0.592 $\pm$ 0.016	0.093	0.013	16.67	0.0452	0.0471	0.976	94.684 $\pm$ 0.711	0.174 $\pm$ 0.001	55.266 $\pm$ 1.575
LSAM	beta56	0.08	0.076	-0.004 $\pm$ 0.001	-0.045 $\pm$ 0.018	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.321 $\pm$ 0.015	0.567 $\pm$ 0.016	0.076	-0.004	-5.50	0.0440	0.0442	0.930	92.879 $\pm$ 0.815	0.165 $\pm$ 0.001	45.135 $\pm$ 1.576
LSAM	beta57	0.06	0.046	-0.014 $\pm$ 0.001	-0.225 $\pm$ 0.016	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	0.291 $\pm$ 0.013	0.540 $\pm$ 0.014	0.046	-0.014	-22.85	0.0310	0.0339	0.988	90.171 $\pm$ 0.943	0.114 $\pm$ 0.001	36.710 $\pm$ 1.527
LSAM	beta58	0.06	0.049	-0.011 $\pm$ 0.001	-0.186 $\pm$ 0.013	0.001 $\pm$ 0.000	0.025 $\pm$ 0.001	0.001 $\pm$ 0.000	0.028 $\pm$ 0.001	0.213 $\pm$ 0.009	0.461 $\pm$ 0.012	0.049	-0.011	-18.95	0.0246	0.0271	1.016	90.672 $\pm$ 0.921	0.101 $\pm$ 0.001	47.041 $\pm$ 1.581
QML	beta41	0.20	0.186	-0.014 $\pm$ 0.001	-0.071 $\pm$ 0.006	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.039 $\pm$ 0.002	0.198 $\pm$ 0.005	0.185	-0.015	-7.40	0.0357	0.0386	0.989	92.979 $\pm$ 0.809	0.144 $\pm$ 0.001	99.799 $\pm$ 0.142
QML	beta42	0.20	0.191	-0.009 $\pm$ 0.001	-0.043 $\pm$ 0.006	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.038 $\pm$ 0.002	0.195 $\pm$ 0.005	0.191	-0.009	-4.58	0.0391	0.0402	0.976	94.283 $\pm$ 0.735	0.146 $\pm$ 0.000	99.699 $\pm$ 0.173
QML	beta43	0.20	0.192	-0.008 $\pm$ 0.001	-0.039 $\pm$ 0.006	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.032 $\pm$ 0.001	0.178 $\pm$ 0.005	0.190	-0.010	-5.01	0.0338	0.0353	0.984	93.480 $\pm$ 0.782	0.134 $\pm$ 0.000	100.000 $\pm$ 0.000
QML	beta44	0.11	0.117	0.007 $\pm$ 0.001	0.066 $\pm$ 0.013	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.176 $\pm$ 0.008	0.420 $\pm$ 0.011	0.118	0.008	7.33	0.0456	0.0463	0.982	93.781 $\pm$ 0.765	0.176 $\pm$ 0.001	74.223 $\pm$ 1.385
QML	beta45	0.11	0.110	0.000 $\pm$ 0.001	0.002 $\pm$ 0.011	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.121 $\pm$ 0.006	0.348 $\pm$ 0.010	0.110	0.000	0.10	0.0388	0.0388	1.015	94.784 $\pm$ 0.704	0.153 $\pm$ 0.001	81.745 $\pm$ 1.223
QML	beta46	0.08	0.068	-0.012 $\pm$ 0.001	-0.145 $\pm$ 0.012	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.170 $\pm$ 0.007	0.413 $\pm$ 0.010	0.067	-0.013	-16.19	0.0294	0.0321	0.989	90.171 $\pm$ 0.943	0.120 $\pm$ 0.001	62.387 $\pm$ 1.534
QML	beta47	0.08	0.066	-0.014 $\pm$ 0.001	-0.171 $\pm$ 0.012	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.172 $\pm$ 0.007	0.414 $\pm$ 0.010	0.065	-0.015	-19.28	0.0293	0.0331	0.983	88.766 $\pm$ 1.000	0.116 $\pm$ 0.001	62.889 $\pm$ 1.530
QML	beta51	0.16	0.178	0.018 $\pm$ 0.001	0.113 $\pm$ 0.009	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.003 $\pm$ 0.000	0.050 $\pm$ 0.001	0.098 $\pm$ 0.004	0.314 $\pm$ 0.008	0.177	0.017	10.48	0.0478	0.0506	0.977	92.778 $\pm$ 0.820	0.180 $\pm$ 0.001	96.891 $\pm$ 0.550
QML	beta52	0.16	0.140	-0.020 $\pm$ 0.001	-0.126 $\pm$ 0.008	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.075 $\pm$ 0.003	0.274 $\pm$ 0.007	0.139	-0.021	-13.35	0.0377	0.0433	1.008	91.174 $\pm$ 0.898	0.154 $\pm$ 0.001	94.885 $\pm$ 0.698
QML	beta53	0.16	0.137	-0.023 $\pm$ 0.001	-0.141 $\pm$ 0.008	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.076 $\pm$ 0.003	0.277 $\pm$ 0.007	0.136	-0.024	-15.05	0.0370	0.0442	1.006	90.471 $\pm$ 0.930	0.150 $\pm$ 0.000	94.283 $\pm$ 0.735
QML	beta54	0.16	0.154	-0.006 $\pm$ 0.001	-0.038 $\pm$ 0.007	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.056 $\pm$ 0.002	0.237 $\pm$ 0.006	0.154	-0.006	-4.03	0.0370	0.0375	0.971	94.684 $\pm$ 0.711	0.142 $\pm$ 0.000	98.596 $\pm$ 0.373
QML	beta55	0.08	0.096	0.016 $\pm$ 0.002	0.204 $\pm$ 0.019	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.401 $\pm$ 0.017	0.633 $\pm$ 0.016	0.098	0.018	22.39	0.0502	0.0533	0.991	94.483 $\pm$ 0.723	0.186 $\pm$ 0.001	54.865 $\pm$ 1.576
QML	beta56	0.08	0.077	-0.003 $\pm$ 0.001	-0.037 $\pm$ 0.018	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.337 $\pm$ 0.015	0.580 $\pm$ 0.016	0.077	-0.003	-3.86	0.0465	0.0466	0.931	92.377 $\pm$ 0.840	0.169 $\pm$ 0.001	43.129 $\pm$ 1.568
QML	beta57	0.06	0.044	-0.016 $\pm$ 0.001	-0.267 $\pm$ 0.016	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.328 $\pm$ 0.014	0.573 $\pm$ 0.014	0.042	-0.018	-29.19	0.0303	0.0350	0.990	87.362 $\pm$ 1.052	0.118 $\pm$ 0.001	28.686 $\pm$ 1.432
QML	beta58	0.06	0.048	-0.012 $\pm$ 0.001	-0.204 $\pm$ 0.013	0.001 $\pm$ 0.000	0.025 $\pm$ 0.001	0.001 $\pm$ 0.000	0.028 $\pm$ 0.001	0.219 $\pm$ 0.009	0.467 $\pm$ 0.011	0.047	-0.013	-22.47	0.0245	0.0280	1.034	90.171 $\pm$ 0.943	0.102 $\pm$ 0.001	43.430 $\pm$ 1.570
UPI	beta41	0.20	0.200	-0.000 $\pm$ 0.001	-0.002 $\pm$ 0.006	0.001 $\pm$ 0.000	0.039 $\pm$ 0.001	0.001 $\pm$ 0.000	0.039 $\pm$ 0.001	0.037 $\pm$ 0.002	0.193 $\pm$ 0.005	0.199	-0.001	-0.37	0.0361	0.0361	0.975	94.283 $\pm$ 0.735	0.148 $\pm$ 0.000	99.799 $\pm$ 0.142
UPI	beta42	0.20	0.200	-0.000 $\pm$ 0.001	-0.000 $\pm$ 0.006	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.038 $\pm$ 0.002	0.195 $\pm$ 0.005	0.200	0.000	-0.14	0.0400	0.0400	0.983	95.386 $\pm$ 0.664	0.150 $\pm$ 0.000	99.799 $\pm$ 0.142
UPI	beta43	0.20	0.203	0.003 $\pm$ 0.001	0.014 $\pm$ 0.006	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.033 $\pm$ 0.002	0.181 $\pm$ 0.005	0.201	0.001	0.70	0.0338	0.0339	0.983	93.882 $\pm$ 0.759	0.139 $\pm$ 0.000	100.000 $\pm</$

Table S202

Simulation 2 - Condition 58: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.6
(continued)

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.073	-0.007 ± 0.001	-0.084 ± 0.013	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.184 ± 0.009	0.429 ± 0.012	0.072	-0.008	-9.59	0.0323	0.0332	1.014	94.183 ± 0.741	0.134 ± 0.001	58.275 ± 1.562
UPI	beta51	0.16	0.167	0.007 ± 0.001	0.044 ± 0.009	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.082 ± 0.004	0.286 ± 0.008	0.166	0.006	3.97	0.0459	0.0463	0.970	93.882 ± 0.759	0.172 ± 0.001	96.389 ± 0.591
UPI	beta52	0.16	0.160	-0.000 ± 0.001	-0.002 ± 0.007	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.053 ± 0.002	0.230 ± 0.006	0.159	-0.001	-0.46	0.0354	0.0354	0.997	94.183 ± 0.741	0.144 ± 0.000	99.198 ± 0.283
UPI	beta53	0.16	0.159	-0.001 ± 0.001	-0.004 ± 0.008	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.057 ± 0.003	0.238 ± 0.006	0.159	-0.001	-0.75	0.0372	0.0372	1.010	94.584 ± 0.717	0.151 ± 0.000	98.897 ± 0.331
UPI	beta54	0.16	0.156	-0.004 ± 0.001	-0.022 ± 0.007	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.055 ± 0.002	0.235 ± 0.006	0.156	-0.004	-2.64	0.0364	0.0366	0.981	95.186 ± 0.678	0.144 ± 0.000	98.395 ± 0.398
UPI	beta55	0.08	0.087	0.007 ± 0.001	0.093 ± 0.018	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.327 ± 0.014	0.572 ± 0.015	0.087	0.007	8.73	0.0488	0.0493	0.997	95.386 ± 0.664	0.176 ± 0.001	51.153 ± 1.583
UPI	beta56	0.08	0.076	-0.004 ± 0.001	-0.045 ± 0.018	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.333 ± 0.015	0.577 ± 0.016	0.075	-0.005	-6.63	0.0445	0.0448	0.926	92.578 ± 0.830	0.167 ± 0.001	43.631 ± 1.571
UPI	beta57	0.06	0.055	-0.005 ± 0.001	-0.076 ± 0.016	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.276 ± 0.012	0.526 ± 0.014	0.055	-0.005	-7.61	0.0315	0.0318	1.015	94.082 ± 0.747	0.124 ± 0.001	43.230 ± 1.569
UPI	beta58	0.06	0.055	-0.005 ± 0.001	-0.091 ± 0.015	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.235 ± 0.011	0.485 ± 0.014	0.053	-0.007	-11.54	0.0269	0.0278	1.010	93.380 ± 0.787	0.113 ± 0.001	49.147 ± 1.583

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S203

Simulation 2 - Condition 59: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.8

Method	Param.	True	Mean-based										Median-based					Inference		
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LSAM	beta41	0.20	0.199	-0.001 $\pm$ 0.002	-0.006 $\pm$ 0.008	0.003 $\pm$ 0.000	0.050 $\pm$ 0.001	0.003 $\pm$ 0.000	0.050 $\pm$ 0.001	0.063 $\pm$ 0.003	0.250 $\pm$ 0.007	0.200	0.000	-0.17	0.0490	0.0490	0.958	94.088 $\pm$ 0.747	0.188 $\pm$ 0.001	96.794 $\pm$ 0.558
LSAM	beta42	0.20	0.197	-0.003 $\pm$ 0.002	-0.014 $\pm$ 0.008	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.061 $\pm$ 0.003	0.247 $\pm$ 0.007	0.196	-0.004	-1.94	0.0486	0.0487	0.987	93.988 $\pm$ 0.752	0.191 $\pm$ 0.001	96.994 $\pm$ 0.541
LSAM	beta43	0.20	0.201	0.001 $\pm$ 0.001	0.006 $\pm$ 0.007	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.054 $\pm$ 0.002	0.232 $\pm$ 0.006	0.202	0.002	1.00	0.0466	0.0466	0.972	94.589 $\pm$ 0.716	0.177 $\pm$ 0.001	98.798 $\pm$ 0.345
LSAM	beta44	0.11	0.117	0.007 $\pm$ 0.002	0.061 $\pm$ 0.015	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.235 $\pm$ 0.011	0.485 $\pm$ 0.013	0.116	0.006	5.09	0.0519	0.0522	0.964	93.788 $\pm$ 0.764	0.200 $\pm$ 0.001	63.126 $\pm$ 1.527
LSAM	beta45	0.11	0.111	0.001 $\pm$ 0.002	0.009 $\pm$ 0.014	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.186 $\pm$ 0.009	0.432 $\pm$ 0.012	0.112	0.002	1.48	0.0460	0.0460	0.963	93.687 $\pm$ 0.770	0.179 $\pm$ 0.001	67.335 $\pm$ 1.485
LSAM	beta46	0.08	0.078	-0.002 $\pm$ 0.001	-0.031 $\pm$ 0.015	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.237 $\pm$ 0.011	0.487 $\pm$ 0.014	0.076	-0.004	-4.43	0.0370	0.0372	0.924	91.583 $\pm$ 0.879	0.141 $\pm$ 0.001	58.317 $\pm$ 1.561
LSAM	beta47	0.08	0.076	-0.004 $\pm$ 0.001	-0.044 $\pm$ 0.015	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.226 $\pm$ 0.010	0.476 $\pm$ 0.013	0.075	-0.005	-6.02	0.0379	0.0382	0.937	93.487 $\pm$ 0.781	0.139 $\pm$ 0.001	58.216 $\pm$ 1.561
LSAM	beta51	0.16	0.160	0.000 $\pm$ 0.002	0.002 $\pm$ 0.012	0.004 $\pm$ 0.000	0.061 $\pm$ 0.001	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.145 $\pm$ 0.007	0.381 $\pm$ 0.011	0.160	0.000	-0.27	0.0605	0.0605	0.922	91.884 $\pm$ 0.864	0.220 $\pm$ 0.001	82.565 $\pm$ 1.201
LSAM	beta52	0.16	0.161	0.001 $\pm$ 0.002	0.004 $\pm$ 0.010	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.090 $\pm$ 0.004	0.301 $\pm$ 0.009	0.161	0.001	0.60	0.0465	0.0465	0.994	95.391 $\pm$ 0.664	0.188 $\pm$ 0.001	91.082 $\pm$ 0.902
LSAM	beta53	0.16	0.156	-0.004 $\pm$ 0.002	-0.022 $\pm$ 0.010	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	0.096 $\pm$ 0.004	0.310 $\pm$ 0.008	0.157	-0.003	-1.81	0.0486	0.0487	1.006	95.291 $\pm$ 0.671	0.195 $\pm$ 0.001	88.176 $\pm$ 1.022
LSAM	beta54	0.16	0.156	-0.004 $\pm$ 0.002	-0.022 $\pm$ 0.010	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.093 $\pm$ 0.004	0.305 $\pm$ 0.008	0.156	-0.004	-2.29	0.0522	0.0523	0.967	94.990 $\pm$ 0.691	0.185 $\pm$ 0.001	88.778 $\pm$ 0.999
LSAM	beta55	0.08	0.082	0.002 $\pm$ 0.002	0.021 $\pm$ 0.023	0.003 $\pm$ 0.000	0.057 $\pm$ 0.001	0.003 $\pm$ 0.000	0.057 $\pm$ 0.002	0.505 $\pm$ 0.024	0.711 $\pm$ 0.020	0.082	0.002	2.14	0.0562	0.0562	0.959	94.289 $\pm$ 0.735	0.214 $\pm$ 0.002	37.074 $\pm$ 1.529
LSAM	beta56	0.08	0.081	0.001 $\pm$ 0.002	0.016 $\pm$ 0.023	0.003 $\pm$ 0.000	0.057 $\pm$ 0.001	0.003 $\pm$ 0.000	0.057 $\pm$ 0.002	0.513 $\pm$ 0.025	0.716 $\pm$ 0.021	0.077	-0.003	-3.42	0.0544	0.0545	0.908	91.583 $\pm$ 0.879	0.204 $\pm$ 0.001	35.371 $\pm$ 1.513
LSAM	beta57	0.06	0.058	-0.002 $\pm$ 0.001	-0.032 $\pm$ 0.019	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.377 $\pm$ 0.017	0.614 $\pm$ 0.017	0.058	-0.002	-2.69	0.0365	0.0365	0.982	93.587 $\pm$ 0.775	0.142 $\pm$ 0.001	35.872 $\pm$ 1.518
LSAM	beta58	0.06	0.057	-0.003 $\pm$ 0.001	-0.044 $\pm$ 0.017	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.301 $\pm$ 0.014	0.549 $\pm$ 0.015	0.056	-0.004	-7.30	0.0326	0.0328	0.966	93.287 $\pm$ 0.792	0.124 $\pm$ 0.001	44.890 $\pm$ 1.574
QML	beta41	0.20	0.195	-0.005 $\pm$ 0.002	-0.024 $\pm$ 0.008	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	0.003 $\pm$ 0.000	0.050 $\pm$ 0.001	0.063 $\pm$ 0.003	0.250 $\pm$ 0.007	0.197	-0.003	-1.58	0.0493	0.0494	0.964	93.587 $\pm$ 0.775	0.188 $\pm$ 0.001	96.493 $\pm$ 0.582
QML	beta42	0.20	0.195	-0.005 $\pm$ 0.002	-0.024 $\pm$ 0.008	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.061 $\pm$ 0.003	0.247 $\pm$ 0.007	0.195	-0.005	-2.57	0.0490	0.0493	0.985	94.589 $\pm$ 0.716	0.190 $\pm$ 0.001	96.693 $\pm$ 0.566
QML	beta43	0.20	0.197	-0.003 $\pm$ 0.001	-0.015 $\pm$ 0.007	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.053 $\pm$ 0.002	0.229 $\pm$ 0.006	0.198	-0.002	-1.03	0.0469	0.0470	0.975	94.289 $\pm$ 0.735	0.175 $\pm$ 0.001	98.697 $\pm$ 0.359
QML	beta44	0.11	0.112	0.002 $\pm$ 0.002	0.021 $\pm$ 0.015	0.003 $\pm$ 0.000	0.054 $\pm$ 0.001	0.003 $\pm$ 0.000	0.054 $\pm$ 0.001	0.237 $\pm$ 0.011	0.487 $\pm$ 0.013	0.112	0.002	1.60	0.0545	0.0545	0.975	94.389 $\pm$ 0.728	0.205 $\pm$ 0.001	59.419 $\pm$ 1.554
QML	beta45	0.11	0.110	-0.000 $\pm$ 0.002	-0.001 $\pm$ 0.014	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.191 $\pm$ 0.010	0.437 $\pm$ 0.013	0.111	0.001	0.69	0.0450	0.0450	0.969	93.888 $\pm$ 0.758	0.183 $\pm$ 0.001	65.932 $\pm$ 1.500
QML	beta46	0.08	0.079	-0.001 $\pm$ 0.001	-0.017 $\pm$ 0.015	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.238 $\pm$ 0.012	0.488 $\pm$ 0.014	0.077	-0.003	-3.60	0.0357	0.0358	0.931	91.984 $\pm$ 0.860	0.142 $\pm$ 0.001	59.018 $\pm$ 1.557
QML	beta47	0.08	0.077	-0.003 $\pm$ 0.001	-0.034 $\pm$ 0.015	0.001 $\pm$ 0.000	0.039 $\pm$ 0.001	0.001 $\pm$ 0.000	0.039 $\pm$ 0.001	0.233 $\pm$ 0.011	0.482 $\pm$ 0.014	0.076	-0.004	-5.62	0.0368	0.0371	0.929	92.485 $\pm$ 0.835	0.140 $\pm$ 0.001	59.920 $\pm$ 1.551
QML	beta51	0.16	0.164	0.004 $\pm$ 0.002	0.026 $\pm$ 0.012	0.004 $\pm$ 0.000	0.061 $\pm$ 0.001	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.147 $\pm$ 0.007	0.384 $\pm$ 0.011	0.164	0.004	2.49	0.0596	0.0598	0.951	93.687 $\pm$ 0.770	0.229 $\pm$ 0.001	82.365 $\pm$ 1.206
QML	beta52	0.16	0.144	-0.016 $\pm$ 0.002	-0.098 $\pm$ 0.010	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.003 $\pm$ 0.000	0.054 $\pm$ 0.001	0.114 $\pm$ 0.005	0.337 $\pm$ 0.009	0.144	-0.016	-10.30	0.0515	0.0541	1.002	93.988 $\pm$ 0.752	0.203 $\pm$ 0.001	80.361 $\pm$ 1.258
QML	beta53	0.16	0.136	-0.024 $\pm$ 0.002	-0.150 $\pm$ 0.010	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.003 $\pm$ 0.000	0.057 $\pm$ 0.001	0.125 $\pm$ 0.005	0.354 $\pm$ 0.009	0.136	-0.024	-15.20	0.0509	0.0564	1.001	92.084 $\pm$ 0.855	0.201 $\pm$ 0.001	74.950 $\pm$ 1.372
QML	beta54	0.16	0.157	-0.003 $\pm$ 0.002	-0.021 $\pm$ 0.010	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.094 $\pm$ 0.004	0.307 $\pm$ 0.008	0.157	-0.003	-2.06	0.0530	0.0532	0.974	95.691 $\pm$ 0.643	0.187 $\pm$ 0.001	88.176 $\pm$ 1.022
QML	beta55	0.08	0.084	0.004 $\pm$ 0.002	0.048 $\pm$ 0.023	0.003 $\pm$ 0.000	0.058 $\pm$ 0.001	0.003 $\pm$ 0.000	0.058 $\pm$ 0.002	0.523 $\pm$ 0.025	0.723 $\pm$ 0.021	0.085	0.005	6.49	0.0589	0.0591	0.970	94.790 $\pm$ 0.703	0.220 $\pm$ 0.002	36.874 $\pm$ 1.527
QML	beta56	0.08	0.082	0.002 $\pm$ 0.002	0.023 $\pm$ 0.023	0.003 $\pm$ 0.000	0.058 $\pm$ 0.001	0.003 $\pm$ 0.000	0.058 $\pm$ 0.002	0.518 $\pm$ 0.026	0.720 $\pm$ 0.021	0.080	0.000	-0.26	0.0529	0.0529	0.922	91.884 $\pm$ 0.864	0.208 $\pm$ 0.002	33.868 $\pm$ 1.498
QML	beta57	0.06	0.056	-0.004 $\pm$ 0.001	-0.068 $\pm$ 0.020	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.395 $\pm$ 0.019	0.629 $\pm$ 0.018	0.055	-0.005	-8.14	0.0343	0.0346	0.977	92.585 $\pm$ 0.829	0.144 $\pm$ 0.001	30.060 $\pm$ 1.451
QML	beta58	0.06	0.056	-0.004 $\pm$ 0.001	-0.063 $\pm$ 0.018	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.316 $\pm$ 0.015	0.562 $\pm$ 0.016	0.054	-0.006	-10.81	0.0327	0.0334	0.958	92.385 $\pm$ 0.840	0.126 $\pm$ 0.001	41.583 $\pm$ 1.560
UPI	beta41	0.20	0.205	0.005 $\pm$ 0.002	0.023 $\pm$ 0.008	0.003 $\pm$ 0.000	0.050 $\pm$ 0.001	0.003 $\pm$ 0.000	0.050 $\pm$ 0.001	0.063 $\pm$ 0.003	0.252 $\pm$ 0.007	0.204	0.004	2.09	0.0492	0.0494	0.915	92.184 $\pm$ 0.850	0.180 $\pm$ 0.001	97.896 $\pm$ 0.454
UPI	beta42	0.20	0.202	0.002 $\pm$ 0.002	0.010 $\pm$ 0.008	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.060 $\pm$ 0.003	0.245 $\pm$ 0.007	0.202	0.002	0.78	0.0483	0.0483	0.963	93.487 $\pm$ 0.781	0.185 $\pm$ 0.001	97.695 $\pm$ 0.475
UPI	beta43	0.20	0.203	0.003 $\pm$ 0.001	0.015 $\pm$ 0.007	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.055 $\pm$ 0.002	0.234 $\pm$ 0.006	0.204	0.004	1.76	0.0475	0.0476	0.964	94.589 $\pm$ 0.716	0.176 $\pm$ 0.001	98.998 $\pm$ 0.315

Table S203

Simulation 2 - Condition 59: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.8
(continued)

Method	Param.	True	Mean-based										Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE	
UPI	beta47	0.08	0.079	-0.001 ± 0.001	-0.010 ± 0.016	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.245 ± 0.012	0.495 ± 0.014	0.078	-0.002	-2.36	0.0383	0.0384	0.933	93.788 ± 0.764	0.145 ± 0.001	58.517 ± 1.560	
UPI	beta51	0.16	0.162	0.002 ± 0.002	0.011 ± 0.012	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.142 ± 0.007	0.377 ± 0.011	0.161	0.001	0.34	0.0580	0.0581	0.936	92.986 ± 0.808	0.221 ± 0.001	83.968 ± 1.161	
UPI	beta52	0.16	0.165	0.005 ± 0.002	0.031 ± 0.009	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.091 ± 0.004	0.301 ± 0.009	0.165	0.005	3.31	0.0456	0.0460	0.983	95.190 ± 0.677	0.185 ± 0.001	93.287 ± 0.792	
UPI	beta53	0.16	0.157	-0.003 ± 0.002	-0.017 ± 0.010	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.095 ± 0.004	0.309 ± 0.008	0.157	-0.003	-1.76	0.0493	0.0494	1.005	95.591 ± 0.650	0.194 ± 0.001	88.577 ± 1.007	
UPI	beta54	0.16	0.159	-0.001 ± 0.002	-0.007 ± 0.010	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.092 ± 0.004	0.303 ± 0.008	0.159	-0.001	-0.76	0.0516	0.0516	0.961	95.190 ± 0.677	0.183 ± 0.001	90.581 ± 0.925	
UPI	beta55	0.08	0.080	0.000 ± 0.002	0.003 ± 0.023	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.507 ± 0.025	0.712 ± 0.021	0.081	0.001	1.39	0.0566	0.0567	0.962	95.090 ± 0.684	0.215 ± 0.002	36.072 ± 1.520	
UPI	beta56	0.08	0.081	0.001 ± 0.002	0.006 ± 0.023	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.512 ± 0.025	0.716 ± 0.021	0.078	-0.002	-2.97	0.0539	0.0540	0.915	92.084 ± 0.855	0.205 ± 0.002	34.168 ± 1.501	
UPI	beta57	0.06	0.060	0.000 ± 0.001	0.000 ± 0.020	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.388 ± 0.018	0.623 ± 0.018	0.059	-0.001	-1.97	0.0363	0.0363	0.980	94.188 ± 0.741	0.144 ± 0.001	36.273 ± 1.522	
UPI	beta58	0.06	0.059	-0.001 ± 0.001	-0.012 ± 0.018	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.320 ± 0.015	0.566 ± 0.016	0.058	-0.002	-4.11	0.0322	0.0323	0.962	93.287 ± 0.792	0.128 ± 0.001	45.090 ± 1.575	

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S204

Simulation 2 - Condition 60: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.8

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LSAM	beta41	0.20	0.197	-0.003 $\pm$ 0.001	-0.017 $\pm$ 0.005	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.023 $\pm$ 0.001	0.153 $\pm$ 0.004	0.197	-0.003	-1.59	0.0310	0.0312	0.995	95.700 $\pm$ 0.641	0.118 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta42	0.20	0.198	-0.002 $\pm$ 0.001	-0.010 $\pm$ 0.005	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.022 $\pm$ 0.001	0.150 $\pm$ 0.004	0.198	-0.002	-1.11	0.0297	0.0298	1.024	95.500 $\pm$ 0.656	0.120 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta43	0.20	0.200	0.000 $\pm$ 0.001	0.002 $\pm$ 0.005	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.021 $\pm$ 0.001	0.143 $\pm$ 0.004	0.201	0.001	0.26	0.0284	0.0284	0.999	95.100 $\pm$ 0.683	0.112 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta44	0.11	0.113	0.003 $\pm$ 0.001	0.029 $\pm$ 0.010	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.093 $\pm$ 0.004	0.305 $\pm$ 0.009	0.113	0.003	2.76	0.0333	0.0334	0.959	94.200 $\pm$ 0.739	0.125 $\pm$ 0.001	92.000 $\pm$ 0.858
LSAM	beta45	0.11	0.111	0.001 $\pm$ 0.001	0.008 $\pm$ 0.009	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.074 $\pm$ 0.003	0.273 $\pm$ 0.008	0.112	0.002	1.45	0.0293	0.0293	0.955	93.200 $\pm$ 0.796	0.112 $\pm$ 0.000	96.000 $\pm$ 0.620
LSAM	beta46	0.08	0.076	-0.004 $\pm$ 0.001	-0.053 $\pm$ 0.009	0.001 $\pm$ 0.000	0.023 $\pm$ 0.001	0.001 $\pm$ 0.000	0.024 $\pm$ 0.001	0.088 $\pm$ 0.004	0.297 $\pm$ 0.008	0.075	-0.005	-5.83	0.0231	0.0236	0.959	93.100 $\pm$ 0.801	0.088 $\pm$ 0.000	93.700 $\pm$ 0.768
LSAM	beta47	0.08	0.077	-0.003 $\pm$ 0.001	-0.036 $\pm$ 0.009	0.001 $\pm$ 0.000	0.023 $\pm$ 0.001	0.001 $\pm$ 0.000	0.023 $\pm$ 0.001	0.084 $\pm$ 0.004	0.290 $\pm$ 0.008	0.077	-0.003	-3.45	0.0226	0.0227	0.972	92.600 $\pm$ 0.828	0.088 $\pm$ 0.000	94.300 $\pm$ 0.733
LSAM	beta51	0.16	0.162	0.002 $\pm$ 0.001	0.012 $\pm$ 0.007	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.049 $\pm$ 0.002	0.222 $\pm$ 0.006	0.162	0.002	1.51	0.0369	0.0369	1.008	95.400 $\pm$ 0.662	0.140 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta52	0.16	0.158	-0.002 $\pm$ 0.001	-0.012 $\pm$ 0.006	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.038 $\pm$ 0.002	0.195 $\pm$ 0.005	0.158	-0.002	-1.38	0.0319	0.0319	0.965	94.400 $\pm$ 0.727	0.118 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta53	0.16	0.159	-0.001 $\pm$ 0.001	-0.009 $\pm$ 0.006	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.037 $\pm$ 0.002	0.192 $\pm$ 0.005	0.159	-0.001	-0.90	0.0325	0.0325	1.020	95.800 $\pm$ 0.634	0.123 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta54	0.16	0.159	-0.001 $\pm$ 0.001	-0.008 $\pm$ 0.006	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.036 $\pm$ 0.001	0.189 $\pm$ 0.005	0.157	-0.003	-1.63	0.0334	0.0335	0.984	94.900 $\pm$ 0.696	0.117 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta55	0.08	0.083	0.003 $\pm$ 0.001	0.040 $\pm$ 0.013	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.182 $\pm$ 0.008	0.426 $\pm$ 0.012	0.082	0.002	2.94	0.0341	0.0342	1.003	94.400 $\pm$ 0.727	0.134 $\pm$ 0.001	68.600 $\pm$ 1.468
LSAM	beta56	0.08	0.081	0.001 $\pm$ 0.001	0.007 $\pm$ 0.013	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.172 $\pm$ 0.008	0.414 $\pm$ 0.012	0.080	0.000	-0.56	0.0322	0.0322	0.982	94.200 $\pm$ 0.739	0.128 $\pm$ 0.001	71.400 $\pm$ 1.429
LSAM	beta57	0.06	0.057	-0.003 $\pm$ 0.001	-0.056 $\pm$ 0.012	0.001 $\pm$ 0.000	0.023 $\pm$ 0.001	0.001 $\pm$ 0.000	0.023 $\pm$ 0.001	0.149 $\pm$ 0.007	0.386 $\pm$ 0.011	0.057	-0.003	-5.64	0.0226	0.0228	0.977	92.600 $\pm$ 0.828	0.088 $\pm$ 0.000	72.100 $\pm$ 1.418
LSAM	beta58	0.06	0.058	-0.002 $\pm$ 0.001	-0.037 $\pm$ 0.011	0.000 $\pm$ 0.000	0.021 $\pm$ 0.000	0.000 $\pm$ 0.000	0.021 $\pm$ 0.001	0.118 $\pm$ 0.005	0.344 $\pm$ 0.010	0.057	-0.003	-4.34	0.0211	0.0213	0.978	93.700 $\pm$ 0.768	0.079 $\pm$ 0.000	84.200 $\pm$ 1.153
QML	beta41	0.20	0.193	-0.007 $\pm$ 0.001	-0.033 $\pm$ 0.005	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.024 $\pm$ 0.001	0.155 $\pm$ 0.004	0.193	-0.007	-3.27	0.0308	0.0315	1.003	95.100 $\pm$ 0.683	0.119 $\pm$ 0.001	99.800 $\pm$ 0.141
QML	beta42	0.20	0.196	-0.004 $\pm$ 0.001	-0.018 $\pm$ 0.005	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.023 $\pm$ 0.001	0.151 $\pm$ 0.004	0.196	-0.004	-1.89	0.0291	0.0294	1.026	95.200 $\pm$ 0.676	0.121 $\pm$ 0.001	99.900 $\pm$ 0.100
QML	beta43	0.20	0.196	-0.004 $\pm$ 0.001	-0.020 $\pm$ 0.005	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.021 $\pm$ 0.001	0.144 $\pm$ 0.004	0.197	-0.003	-1.72	0.0283	0.0285	0.998	94.800 $\pm$ 0.702	0.112 $\pm$ 0.000	100.000 $\pm$ 0.000
QML	beta44	0.11	0.110	-0.000 $\pm$ 0.001	-0.000 $\pm$ 0.010	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.097 $\pm$ 0.004	0.312 $\pm$ 0.009	0.110	0.000	0.41	0.0343	0.0343	0.960	94.400 $\pm$ 0.727	0.129 $\pm$ 0.001	88.100 $\pm$ 1.024
QML	beta45	0.11	0.110	0.000 $\pm$ 0.001	0.001 $\pm$ 0.009	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.078 $\pm$ 0.004	0.279 $\pm$ 0.008	0.110	0.000	0.39	0.0304	0.0304	0.958	94.000 $\pm$ 0.751	0.115 $\pm$ 0.000	95.000 $\pm$ 0.689
QML	beta46	0.08	0.076	-0.004 $\pm$ 0.001	-0.044 $\pm$ 0.009	0.001 $\pm$ 0.000	0.024 $\pm$ 0.001	0.001 $\pm$ 0.000	0.024 $\pm$ 0.001	0.088 $\pm$ 0.004	0.297 $\pm$ 0.008	0.075	-0.005	-5.84	0.0233	0.0238	0.969	92.400 $\pm$ 0.838	0.089 $\pm$ 0.000	94.800 $\pm$ 0.702
QML	beta47	0.08	0.077	-0.003 $\pm$ 0.001	-0.033 $\pm$ 0.009	0.001 $\pm$ 0.000	0.023 $\pm$ 0.001	0.001 $\pm$ 0.000	0.023 $\pm$ 0.001	0.086 $\pm$ 0.004	0.293 $\pm$ 0.008	0.076	-0.004	-4.71	0.0229	0.0232	0.971	93.000 $\pm$ 0.807	0.089 $\pm$ 0.001	95.700 $\pm$ 0.641
QML	beta51	0.16	0.166	0.006 $\pm$ 0.001	0.040 $\pm$ 0.007	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.052 $\pm$ 0.002	0.228 $\pm$ 0.006	0.167	0.007	4.41	0.0374	0.0381	1.022	96.000 $\pm$ 0.620	0.144 $\pm$ 0.001	100.000 $\pm$ 0.000
QML	beta52	0.16	0.141	-0.019 $\pm$ 0.001	-0.118 $\pm$ 0.007	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.057 $\pm$ 0.002	0.239 $\pm$ 0.006	0.140	-0.020	-12.47	0.0345	0.0398	0.976	89.900 $\pm$ 0.953	0.127 $\pm$ 0.000	99.000 $\pm$ 0.315
QML	beta53	0.16	0.138	-0.022 $\pm$ 0.001	-0.136 $\pm$ 0.006	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	0.001 $\pm$ 0.000	0.039 $\pm$ 0.001	0.058 $\pm$ 0.002	0.241 $\pm$ 0.006	0.139	-0.021	-13.33	0.0346	0.0407	1.008	90.300 $\pm$ 0.936	0.126 $\pm$ 0.000	99.000 $\pm$ 0.315
QML	beta54	0.16	0.159	-0.001 $\pm$ 0.001	-0.007 $\pm$ 0.006	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.036 $\pm$ 0.001	0.191 $\pm$ 0.005	0.157	-0.003	-1.64	0.0327	0.0328	0.988	95.200 $\pm$ 0.676	0.118 $\pm$ 0.000	100.000 $\pm$ 0.000
QML	beta55	0.08	0.086	0.006 $\pm$ 0.001	0.076 $\pm$ 0.014	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.195 $\pm$ 0.009	0.442 $\pm$ 0.012	0.085	0.005	6.66	0.0345	0.0349	1.007	94.900 $\pm$ 0.696	0.137 $\pm$ 0.001	68.900 $\pm$ 1.464
QML	beta56	0.08	0.081	0.001 $\pm$ 0.001	0.019 $\pm$ 0.013	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.176 $\pm$ 0.008	0.420 $\pm$ 0.012	0.080	0.000	0.56	0.0320	0.0320	0.995	94.400 $\pm$ 0.727	0.131 $\pm$ 0.001	70.700 $\pm$ 1.439
QML	beta57	0.06	0.054	-0.006 $\pm$ 0.001	-0.099 $\pm$ 0.012	0.001 $\pm$ 0.000	0.023 $\pm$ 0.001	0.001 $\pm$ 0.000	0.024 $\pm$ 0.001	0.159 $\pm$ 0.007	0.399 $\pm$ 0.011	0.053	-0.007	-11.10	0.0229	0.0239	0.979	91.600 $\pm$ 0.877	0.089 $\pm$ 0.000	68.200 $\pm$ 1.473
QML	beta58	0.06	0.057	-0.003 $\pm$ 0.001	-0.058 $\pm$ 0.011	0.000 $\pm$ 0.000	0.021 $\pm$ 0.001	0.000 $\pm$ 0.000	0.021 $\pm$ 0.001	0.126 $\pm$ 0.006	0.355 $\pm$ 0.010	0.055	-0.005	-7.53	0.0212	0.0217	0.974	93.800 $\pm$ 0.763	0.080 $\pm$ 0.000	81.700 $\pm$ 1.223
UPI	beta41	0.20	0.200	0.000 $\pm$ 0.001	0.000 $\pm$ 0.005	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.023 $\pm$ 0.001	0.151 $\pm$ 0.004	0.200	0.000	-0.06	0.0305	0.0305	0.954	95.100 $\pm$ 0.683	0.113 $\pm$ 0.000	100.000 $\pm$ 0.000
UPI	beta42	0.20	0.201	0.001 $\pm$ 0.001	0.004 $\pm$ 0.005	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.022 $\pm$ 0.001	0.149 $\pm$ 0.004	0.201	0.001	0.73	0.0290	0.0291	1.001	94.300 $\pm$ 0.733	0.117 $\pm$ 0.000	100.000 $\pm$ 0.000
UPI	beta43	0.20	0.201	0.001 $\pm$ 0.001	0.006 $\pm$ 0.005	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.021 $\pm$ 0.001	0.144 $\pm$ 0.004	0.201	0.001	0.47	0.0293	0.0293	0.988	95.100 $\pm$ 0.683	0.111 $\pm$ 0.000	100.000 $\pm$

Table S204

Simulation 2 - Condition 60: Full Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.8
(continued)

Method	Param.	True	Mean-based										Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE	
UPI	beta47	0.08	0.080	-0.000 ± 0.001	-0.002 ± 0.009	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	0.086 ± 0.004	0.293 ± 0.008	0.079	-0.001	-1.18	0.0237	0.0237	0.977	93.500 ± 0.780	0.090 ± 0.000	95.400 ± 0.662	
UPI	beta51	0.16	0.162	0.002 ± 0.001	0.010 ± 0.007	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.049 ± 0.002	0.221 ± 0.006	0.162	0.002	1.11	0.0375	0.0376	1.008	96.000 ± 0.620	0.140 ± 0.000	100.000 ± 0.000	
UPI	beta52	0.16	0.161	0.001 ± 0.001	0.005 ± 0.006	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.038 ± 0.002	0.194 ± 0.005	0.160	0.000	0.07	0.0315	0.0315	0.953	94.000 ± 0.751	0.116 ± 0.000	100.000 ± 0.000	
UPI	beta53	0.16	0.159	-0.001 ± 0.001	-0.004 ± 0.006	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.037 ± 0.002	0.192 ± 0.005	0.159	-0.001	-0.49	0.0319	0.0319	1.017	95.600 ± 0.649	0.122 ± 0.000	100.000 ± 0.000	
UPI	beta54	0.16	0.161	0.001 ± 0.001	0.004 ± 0.006	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.036 ± 0.001	0.189 ± 0.005	0.160	0.000	-0.17	0.0329	0.0329	0.976	95.000 ± 0.689	0.116 ± 0.000	100.000 ± 0.000	
UPI	beta55	0.08	0.081	0.001 ± 0.001	0.019 ± 0.014	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.183 ± 0.008	0.427 ± 0.012	0.080	0.000	0.15	0.0328	0.0328	0.998	94.400 ± 0.727	0.134 ± 0.001	66.400 ± 1.494	
UPI	beta56	0.08	0.081	0.001 ± 0.001	0.009 ± 0.013	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.170 ± 0.008	0.412 ± 0.012	0.079	-0.001	-1.00	0.0315	0.0315	0.989	94.100 ± 0.745	0.128 ± 0.001	71.500 ± 1.427	
UPI	beta57	0.06	0.059	-0.001 ± 0.001	-0.017 ± 0.012	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	0.150 ± 0.007	0.387 ± 0.011	0.059	-0.001	-2.14	0.0231	0.0232	0.971	92.900 ± 0.812	0.088 ± 0.000	75.300 ± 1.364	
UPI	beta58	0.06	0.059	-0.001 ± 0.001	-0.010 ± 0.011	0.000 ± 0.000	0.021 ± 0.000	0.000 ± 0.000	0.021 ± 0.001	0.120 ± 0.006	0.346 ± 0.010	0.059	-0.001	-2.33	0.0210	0.0210	0.978	93.500 ± 0.780	0.080 ± 0.000	85.900 ± 1.101	

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S205

Simulation 2 - Condition 61: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, $N = 400$, Reliability = 0.4

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
LSAM	beta41	0.20	0.215	0.015 ± 0.009	0.077 ± 0.046	0.051 ± 0.003	0.227 ± 0.008	0.052 ± 0.003	0.227 ± 0.009	1.291 ± 0.085	1.136 ± 0.044	0.218	0.018	9.15	0.2155	0.2163	1.065	97.411 ± 0.639	0.947 ± 0.022	16.505 ± 1.493
LSAM	beta42	0.20	0.204	0.004 ± 0.009	0.020 ± 0.047	0.053 ± 0.004	0.231 ± 0.008	0.053 ± 0.004	0.231 ± 0.009	1.335 ± 0.090	1.155 ± 0.046	0.202	0.002	1.11	0.2211	0.2211	1.073	98.058 ± 0.555	0.973 ± 0.021	16.343 ± 1.487
LSAM	beta43	0.20	0.192	-0.008 ± 0.003	-0.040 ± 0.016	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.079 ± 0.003	0.158 ± 0.009	0.397 ± 0.014	0.195	-0.005	-2.57	0.0811	0.0812	1.078	95.631 ± 0.822	0.335 ± 0.004	66.828 ± 1.894
LSAM	beta44	0.11	0.127	0.017 ± 0.005	0.159 ± 0.046	0.016 ± 0.001	0.126 ± 0.005	0.016 ± 0.001	0.127 ± 0.005	1.341 ± 0.099	1.158 ± 0.049	0.122	0.012	10.97	0.1218	0.1224	0.994	96.117 ± 0.777	0.492 ± 0.010	19.094 ± 1.581
LSAM	beta45	0.11	0.116	0.006 ± 0.004	0.055 ± 0.040	0.012 ± 0.001	0.109 ± 0.003	0.012 ± 0.001	0.109 ± 0.004	0.984 ± 0.063	0.992 ± 0.038	0.118	0.008	7.46	0.1084	0.1087	1.007	93.366 ± 1.001	0.430 ± 0.009	18.123 ± 1.550
LSAM	beta46	0.08	0.056	-0.024 ± 0.004	-0.297 ± 0.050	0.010 ± 0.001	0.099 ± 0.004	0.010 ± 0.001	0.101 ± 0.004	1.604 ± 0.112	1.267 ± 0.051	0.058	-0.022	-27.29	0.0848	0.0876	0.994	94.660 ± 0.904	0.384 ± 0.011	15.858 ± 1.469
LSAM	beta47	0.08	0.062	-0.018 ± 0.004	-0.226 ± 0.047	0.009 ± 0.001	0.093 ± 0.003	0.009 ± 0.001	0.095 ± 0.004	1.405 ± 0.104	1.185 ± 0.050	0.065	-0.015	-18.22	0.0791	0.0804	1.043	94.822 ± 0.891	0.381 ± 0.011	16.181 ± 1.481
LSAM	beta51	0.16	0.173	0.013 ± 0.004	0.084 ± 0.024	0.009 ± 0.001	0.095 ± 0.003	0.009 ± 0.001	0.095 ± 0.004	0.356 ± 0.023	0.596 ± 0.022	0.172	0.012	7.45	0.0969	0.0976	0.986	95.631 ± 0.822	0.366 ± 0.003	46.602 ± 2.007
LSAM	beta52	0.16	0.218	0.058 ± 0.010	0.364 ± 0.061	0.059 ± 0.004	0.243 ± 0.009	0.063 ± 0.005	0.250 ± 0.011	2.444 ± 0.187	1.563 ± 0.067	0.194	0.034	21.05	0.2164	0.2190	1.131	98.382 ± 0.508	1.079 ± 0.028	11.489 ± 1.283
LSAM	beta53	0.16	0.143	-0.017 ± 0.004	-0.105 ± 0.027	0.012 ± 0.001	0.108 ± 0.004	0.012 ± 0.001	0.109 ± 0.004	0.467 ± 0.030	0.683 ± 0.026	0.144	-0.016	-9.87	0.1031	0.1043	1.105	96.440 ± 0.745	0.468 ± 0.006	22.330 ± 1.675
LSAM	beta54	0.16	0.139	-0.021 ± 0.008	-0.132 ± 0.052	0.043 ± 0.003	0.208 ± 0.007	0.043 ± 0.003	0.209 ± 0.008	1.699 ± 0.113	1.303 ± 0.051	0.145	-0.015	-9.36	0.1807	0.1814	1.116	96.926 ± 0.694	0.909 ± 0.024	10.194 ± 1.217
LSAM	beta55	0.08	0.139	0.059 ± 0.004	0.732 ± 0.055	0.012 ± 0.001	0.109 ± 0.004	0.015 ± 0.001	0.124 ± 0.005	2.397 ± 0.164	1.548 ± 0.059	0.137	0.057	70.76	0.0980	0.1132	1.097	93.851 ± 0.966	0.470 ± 0.014	30.421 ± 1.851
LSAM	beta56	0.08	0.068	-0.012 ± 0.002	-0.146 ± 0.031	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.063 ± 0.003	0.623 ± 0.045	0.789 ± 0.032	0.071	-0.009	-11.65	0.0537	0.0545	1.004	94.175 ± 0.942	0.244 ± 0.006	28.964 ± 1.825
LSAM	beta57	0.06	-0.006	-0.066 ± 0.006	-1.102 ± 0.093	0.019 ± 0.002	0.139 ± 0.006	0.024 ± 0.002	0.154 ± 0.007	6.570 ± 0.511	2.563 ± 0.109	0.008	-0.052	-86.92	0.1149	0.1262	1.132	95.307 ± 0.851	0.616 ± 0.019	1.942 ± 0.555
LSAM	beta58	0.06	0.049	-0.011 ± 0.003	-0.178 ± 0.056	0.007 ± 0.001	0.084 ± 0.003	0.007 ± 0.001	0.084 ± 0.004	1.968 ± 0.149	1.403 ± 0.060	0.045	-0.015	-25.29	0.0644	0.0662	1.060	94.175 ± 0.942	0.347 ± 0.012	13.269 ± 1.365
QML	beta41	0.20	0.233	0.033 ± 0.004	0.166 ± 0.019	0.009 ± 0.001	0.094 ± 0.003	0.010 ± 0.001	0.100 ± 0.003	0.248 ± 0.014	0.498 ± 0.017	0.233	0.033	16.42	0.0980	0.1034	1.025	94.337 ± 0.930	0.377 ± 0.003	69.903 ± 1.845
QML	beta42	0.20	0.226	0.026 ± 0.004	0.131 ± 0.020	0.010 ± 0.001	0.098 ± 0.003	0.010 ± 0.001	0.101 ± 0.004	0.255 ± 0.015	0.505 ± 0.018	0.221	0.021	10.66	0.0967	0.0990	1.014	94.498 ± 0.917	0.388 ± 0.003	65.696 ± 1.910
QML	beta43	0.20	0.192	-0.008 ± 0.003	-0.038 ± 0.014	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.130 ± 0.007	0.360 ± 0.012	0.194	-0.006	-2.93	0.0773	0.0775	1.064	95.631 ± 0.822	0.299 ± 0.002	73.139 ± 1.783
QML	beta44	0.11	0.065	-0.045 ± 0.005	-0.407 ± 0.048	0.017 ± 0.001	0.130 ± 0.004	0.019 ± 0.001	0.138 ± 0.005	1.563 ± 0.099	1.250 ± 0.046	0.069	-0.041	-36.89	0.1274	0.1337	0.989	93.851 ± 0.966	0.505 ± 0.009	12.783 ± 1.343
QML	beta45	0.11	0.109	-0.001 ± 0.004	-0.011 ± 0.039	0.011 ± 0.001	0.105 ± 0.003	0.011 ± 0.001	0.105 ± 0.004	0.918 ± 0.053	0.958 ± 0.034	0.111	0.001	0.90	0.1061	0.1061	0.961	91.909 ± 1.097	0.397 ± 0.006	21.197 ± 1.644
QML	beta46	0.08	0.138	0.058 ± 0.003	0.721 ± 0.040	0.006 ± 0.000	0.080 ± 0.003	0.010 ± 0.001	0.098 ± 0.004	1.515 ± 0.106	1.231 ± 0.046	0.129	0.049	61.65	0.0783	0.0926	0.959	92.233 ± 1.077	0.300 ± 0.006	49.676 ± 2.011
QML	beta47	0.08	0.130	0.050 ± 0.003	0.631 ± 0.036	0.005 ± 0.000	0.072 ± 0.003	0.008 ± 0.001	0.088 ± 0.003	1.204 ± 0.086	1.097 ± 0.042	0.125	0.045	56.26	0.0667	0.0804	0.999	93.528 ± 0.990	0.281 ± 0.006	50.485 ± 2.011
QML	beta51	0.16	0.181	0.021 ± 0.005	0.128 ± 0.032	0.016 ± 0.001	0.128 ± 0.004	0.017 ± 0.001	0.130 ± 0.005	0.658 ± 0.040	0.811 ± 0.030	0.180	0.020	12.31	0.1261	0.1276	0.944	94.498 ± 0.917	0.474 ± 0.006	39.159 ± 1.963
QML	beta52	0.16	0.156	-0.004 ± 0.005	-0.025 ± 0.030	0.014 ± 0.001	0.119 ± 0.004	0.014 ± 0.001	0.119 ± 0.005	0.557 ± 0.036	0.746 ± 0.029	0.149	-0.011	-6.73	0.1145	0.1150	0.975	94.498 ± 0.917	0.457 ± 0.007	26.699 ± 1.780
QML	beta53	0.16	0.126	-0.034 ± 0.004	-0.211 ± 0.026	0.011 ± 0.001	0.103 ± 0.003	0.012 ± 0.001	0.108 ± 0.004	0.455 ± 0.027	0.674 ± 0.024	0.126	-0.034	-21.16	0.0963	0.1021	1.009	92.718 ± 1.045	0.406 ± 0.004	22.977 ± 1.692
QML	beta54	0.16	0.169	0.009 ± 0.004	0.054 ± 0.024	0.009 ± 0.001	0.094 ± 0.003	0.009 ± 0.001	0.094 ± 0.003	0.348 ± 0.020	0.590 ± 0.021	0.175	0.015	9.60	0.0994	0.1006	1.042	95.631 ± 0.822	0.384 ± 0.004	44.013 ± 1.997
QML	beta55	0.08	0.091	0.011 ± 0.006	0.137 ± 0.081	0.026 ± 0.002	0.161 ± 0.005	0.026 ± 0.002	0.161 ± 0.006	4.053 ± 0.239	2.013 ± 0.072	0.095	0.015	18.70	0.1735	0.1742	0.849	83.981 ± 1.475	0.535 ± 0.013	25.405 ± 1.751
QML	beta56	0.08	0.107	0.027 ± 0.003	0.336 ± 0.038	0.006 ± 0.000	0.076 ± 0.002	0.007 ± 0.000	0.081 ± 0.003	1.026 ± 0.067	1.013 ± 0.038	0.108	0.028	35.32	0.0748	0.0799	0.957	91.586 ± 1.117	0.287 ± 0.005	36.731 ± 1.939
QML	beta57	0.06	0.120	0.060 ± 0.006	1.003 ± 0.102	0.023 ± 0.001	0.152 ± 0.005	0.027 ± 0.002	0.163 ± 0.006	7.396 ± 0.485	2.720 ± 0.101	0.116	0.056	94.04	0.1603	0.1700	0.857	84.466 ± 1.457	0.510 ± 0.013	21.036 ± 1.639
QML	beta58	0.06	0.104	0.044 ± 0.003	0.727 ± 0.043	0.004 ± 0.000	0.065 ± 0.002	0.006 ± 0.000	0.078 ± 0.003	1.694 ± 0.127	1.302 ± 0.052	0.101	0.041	68.26	0.0607	0.0733	0.984	94.175 ± 0.942	0.250 ± 0.007	41.100 ± 1.979
UPI	beta41	0.20	0.292	0.092 ± 0.013	0.461 ± 0.066	0.106 ± 0.009	0.326 ± 0.013	0.115 ± 0.009	0.339 ± 0.015	2.868 ± 0.222	1.693 ± 0.073	0.277	0.077	38.53	0.3169	0.3262	1.299	98.058 ± 0.555	1.660 ± 0.082	12.945 ± 1.350
UPI	beta42	0.20	0.314	0.114 ± 0.012	0.570 ± 0.062	0.094 ± 0.007	0.306 ± 0.011	0.107 ± 0.008	0.327 ± 0.014	2.668 ± 0.204	1.633 ± 0.069	0.297	0.097	48.32	0.2944	0.3099	1.191	98.058 ± 0.555	1.431 ± 0.053	14.725 ± 1.425
UPI	beta43	0.20	0.166	-0.034 ± 0.004	-0.169 ± 0.021	0.011 ± 0.001	0.104 ± 0.004	0.012 ± 0.001	0.109 ± 0.004	0.299 ± 0.020	0.547 ± 0.021	0.165	-0.035	-17.34	0.1085	0.1139	1.320	96.602 ± 0.729	0.538 ± 0.030	35.761 ± 1.928
UPI	beta44	0.11	0.180	0.070 ± 0.010	0.639 ± 0.095	0.068 ± 0.006	0.261 ± 0.011	0.073 ± 0.006	0.270 ± 0.012	6.020 ± 0.501	2.454 ± 0.112	0.165	0.055	49.98	0.2197	0.2264	1.296	98.382 ± 0.508	1.325 ± 0.058	6.149 ± 0.966

Table S205

Simulation 2 - Condition 61: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.4 (continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta45	0.11	0.134	0.024 ± 0.008	0.220 ± 0.074	0.040 ± 0.003	0.201 ± 0.009	0.041 ± 0.004	0.202 ± 0.010	3.386 ± 0.299	1.840 ± 0.089	0.114	0.004	3.78	0.1908	0.1908	1.337	95.469 ± 0.837	1.054 ± 0.051	5.825 ± 0.942
UPI	beta46	0.08	0.022	-0.058 ± 0.006	-0.731 ± 0.076	0.023 ± 0.002	0.152 ± 0.007	0.026 ± 0.002	0.163 ± 0.007	4.133 ± 0.350	2.033 ± 0.094	0.035	-0.045	-56.43	0.1251	0.1330	1.253	95.631 ± 0.822	0.746 ± 0.038	6.958 ± 1.023
UPI	beta47	0.08	0.023	-0.057 ± 0.005	-0.718 ± 0.062	0.015 ± 0.001	0.124 ± 0.005	0.019 ± 0.001	0.137 ± 0.006	2.917 ± 0.220	1.708 ± 0.071	0.032	-0.048	-59.44	0.1144	0.1239	1.175	96.764 ± 0.712	0.571 ± 0.019	7.443 ± 1.056
UPI	beta51	0.16	0.203	0.043 ± 0.004	0.267 ± 0.028	0.012 ± 0.001	0.110 ± 0.003	0.014 ± 0.001	0.117 ± 0.004	0.539 ± 0.034	0.734 ± 0.027	0.202	0.042	26.07	0.1134	0.1208	1.187	96.117 ± 0.777	0.509 ± 0.010	39.968 ± 1.970
UPI	beta52	0.16	0.253	0.093 ± 0.013	0.583 ± 0.082	0.106 ± 0.008	0.325 ± 0.013	0.114 ± 0.009	0.338 ± 0.014	4.460 ± 0.341	2.112 ± 0.090	0.238	0.078	48.64	0.3036	0.3134	1.437	98.220 ± 0.532	1.831 ± 0.093	7.282 ± 1.045
UPI	beta53	0.16	0.111	-0.049 ± 0.006	-0.308 ± 0.039	0.025 ± 0.002	0.157 ± 0.006	0.027 ± 0.002	0.164 ± 0.007	1.051 ± 0.074	1.025 ± 0.041	0.104	-0.056	-34.98	0.1501	0.1602	1.319	97.087 ± 0.676	0.810 ± 0.032	9.061 ± 1.155
UPI	beta54	0.16	0.242	0.082 ± 0.019	0.514 ± 0.118	0.219 ± 0.019	0.468 ± 0.020	0.226 ± 0.019	0.475 ± 0.022	8.816 ± 0.757	2.969 ± 0.141	0.222	0.062	38.80	0.3863	0.3913	1.382	98.706 ± 0.455	2.537 ± 0.114	3.398 ± 0.729
UPI	beta55	0.08	0.156	0.076 ± 0.007	0.952 ± 0.090	0.032 ± 0.003	0.178 ± 0.007	0.038 ± 0.003	0.194 ± 0.008	5.868 ± 0.458	2.422 ± 0.103	0.150	0.070	87.38	0.1597	0.1744	1.373	98.544 ± 0.482	0.960 ± 0.043	10.356 ± 1.226
UPI	beta56	0.08	0.073	-0.007 ± 0.004	-0.094 ± 0.046	0.009 ± 0.001	0.092 ± 0.003	0.009 ± 0.001	0.092 ± 0.004	1.336 ± 0.097	1.156 ± 0.048	0.070	-0.010	-12.24	0.0832	0.0837	1.254	97.249 ± 0.658	0.453 ± 0.016	14.887 ± 1.432
UPI	beta57	0.06	-0.026	-0.086 ± 0.008	-1.426 ± 0.127	0.036 ± 0.003	0.189 ± 0.008	0.043 ± 0.003	0.208 ± 0.009	11.965 ± 0.910	3.459 ± 0.144	-0.011	-0.071	-118.87	0.1722	0.1864	1.417	98.058 ± 0.555	1.051 ± 0.052	0.971 ± 0.394
UPI	beta58	0.06	0.005	-0.055 ± 0.007	-0.914 ± 0.111	0.028 ± 0.002	0.166 ± 0.007	0.031 ± 0.002	0.175 ± 0.008	8.493 ± 0.688	2.914 ± 0.130	0.013	-0.047	-77.63	0.1312	0.1393	1.434	96.440 ± 0.745	0.934 ± 0.058	3.722 ± 0.761

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S206

Simulation 2 - Condition 62: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, $N = 1000$, Reliability = 0.4

Method	Param.	True	Mean-based										Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE	
LSAM	beta41	0.20	0.220	0.020 ± 0.006	0.100 ± 0.030	0.032 ± 0.002	0.178 ± 0.005	0.032 ± 0.002	0.179 ± 0.006	0.805 ± 0.050	0.897 ± 0.032	0.208	0.008	4.02	0.1622	0.1624	1.013	96.391 ± 0.636	0.709 ± 0.012	27.590 ± 1.525	
LSAM	beta42	0.20	0.203	0.003 ± 0.006	0.015 ± 0.031	0.033 ± 0.002	0.181 ± 0.005	0.033 ± 0.002	0.181 ± 0.006	0.823 ± 0.048	0.907 ± 0.031	0.198	-0.002	-1.20	0.1638	0.1639	1.044	97.206 ± 0.562	0.742 ± 0.012	23.865 ± 1.454	
LSAM	beta43	0.20	0.200	-0.000 ± 0.002	-0.000 ± 0.009	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	0.067 ± 0.003	0.259 ± 0.008	0.197	-0.003	-1.30	0.0506	0.0507	1.060	96.391 ± 0.636	0.215 ± 0.001	95.925 ± 0.675	
LSAM	beta44	0.11	0.137	0.027 ± 0.003	0.248 ± 0.024	0.006 ± 0.000	0.078 ± 0.002	0.007 ± 0.000	0.083 ± 0.003	0.565 ± 0.030	0.752 ± 0.023	0.133	0.023	20.57	0.0774	0.0806	0.991	95.111 ± 0.736	0.303 ± 0.004	45.169 ± 1.698	
LSAM	beta45	0.11	0.110	0.000 ± 0.002	0.001 ± 0.022	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.420 ± 0.023	0.648 ± 0.021	0.107	-0.003	-2.83	0.0664	0.0665	0.948	94.645 ± 0.768	0.265 ± 0.003	40.978 ± 1.678	
LSAM	beta46	0.08	0.056	-0.024 ± 0.002	-0.296 ± 0.029	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.825 ± 0.050	0.908 ± 0.031	0.061	-0.019	-23.71	0.0631	0.0659	0.968	92.782 ± 0.883	0.261 ± 0.005	25.495 ± 1.487	
LSAM	beta47	0.08	0.063	-0.017 ± 0.002	-0.218 ± 0.028	0.004 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.744 ± 0.047	0.863 ± 0.031	0.063	-0.017	-21.58	0.0578	0.0604	1.004	94.529 ± 0.776	0.263 ± 0.005	24.680 ± 1.471	
LSAM	beta51	0.16	0.174	0.014 ± 0.002	0.085 ± 0.013	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	0.157 ± 0.008	0.397 ± 0.012	0.173	0.013	8.33	0.0628	0.0642	0.973	94.296 ± 0.791	0.237 ± 0.001	82.654 ± 1.292	
LSAM	beta52	0.16	0.223	0.063 ± 0.007	0.396 ± 0.045	0.044 ± 0.003	0.209 ± 0.006	0.048 ± 0.003	0.218 ± 0.008	1.863 ± 0.115	1.365 ± 0.047	0.220	0.060	37.67	0.1801	0.1899	1.055	98.137 ± 0.461	0.865 ± 0.017	20.722 ± 1.383	
LSAM	beta53	0.16	0.146	-0.014 ± 0.003	-0.087 ± 0.016	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	0.229 ± 0.012	0.478 ± 0.015	0.144	-0.016	-9.96	0.0724	0.0741	1.096	96.391 ± 0.636	0.324 ± 0.003	45.634 ± 1.699	
LSAM	beta54	0.16	0.141	-0.019 ± 0.005	-0.116 ± 0.034	0.025 ± 0.002	0.160 ± 0.005	0.026 ± 0.002	0.161 ± 0.006	1.007 ± 0.060	1.004 ± 0.035	0.139	-0.021	-12.87	0.1381	0.1396	1.071	97.439 ± 0.539	0.670 ± 0.016	16.531 ± 1.267	
LSAM	beta55	0.08	0.134	0.054 ± 0.003	0.678 ± 0.034	0.007 ± 0.000	0.081 ± 0.002	0.009 ± 0.001	0.097 ± 0.003	1.475 ± 0.082	1.214 ± 0.037	0.131	0.051	63.20	0.0736	0.0893	1.017	90.803 ± 0.986	0.322 ± 0.006	45.867 ± 1.700	
LSAM	beta56	0.08	0.068	-0.012 ± 0.001	-0.148 ± 0.016	0.001 ± 0.000	0.037 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.240 ± 0.015	0.490 ± 0.017	0.070	-0.010	-12.56	0.0338	0.0352	0.972	92.899 ± 0.876	0.142 ± 0.002	54.366 ± 1.699	
LSAM	beta57	0.06	-0.004	-0.064 ± 0.004	-1.060 ± 0.065	0.013 ± 0.001	0.114 ± 0.004	0.017 ± 0.001	0.131 ± 0.004	4.759 ± 0.279	2.181 ± 0.070	-0.001	-0.061	-101.51	0.0981	0.1155	1.045	92.317 ± 0.909	0.469 ± 0.010	1.513 ± 0.417	
LSAM	beta58	0.06	0.051	-0.009 ± 0.002	-0.157 ± 0.032	0.003 ± 0.000	0.056 ± 0.002	0.003 ± 0.000	0.057 ± 0.002	0.889 ± 0.057	0.943 ± 0.034	0.053	-0.007	-11.98	0.0456	0.0462	1.065	96.042 ± 0.665	0.233 ± 0.007	22.352 ± 1.421	
QML	beta41	0.20	0.239	0.039 ± 0.002	0.197 ± 0.010	0.004 ± 0.000	0.061 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.130 ± 0.006	0.361 ± 0.010	0.237	0.037	18.55	0.0573	0.0682	1.017	89.057 ± 1.065	0.241 ± 0.006	96.158 ± 0.656	
QML	beta42	0.20	0.223	0.023 ± 0.002	0.116 ± 0.010	0.004 ± 0.000	0.059 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	0.101 ± 0.005	0.318 ± 0.009	0.223	0.023	11.52	0.0583	0.0627	1.026	94.179 ± 0.799	0.238 ± 0.001	94.179 ± 0.799	
QML	beta43	0.20	0.194	-0.006 ± 0.002	-0.029 ± 0.008	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.054 ± 0.003	0.233 ± 0.007	0.193	-0.007	-3.63	0.0443	0.0449	1.014	94.412 ± 0.784	0.184 ± 0.001	98.836 ± 0.366	
QML	beta44	0.11	0.073	-0.037 ± 0.003	-0.335 ± 0.026	0.007 ± 0.000	0.085 ± 0.002	0.009 ± 0.000	0.093 ± 0.003	0.713 ± 0.035	0.845 ± 0.024	0.077	-0.033	-29.89	0.0874	0.0934	0.939	91.967 ± 0.927	0.314 ± 0.003	21.769 ± 1.408	
QML	beta45	0.11	0.104	-0.006 ± 0.002	-0.052 ± 0.020	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.331 ± 0.015	0.575 ± 0.016	0.102	-0.008	-6.95	0.0640	0.0644	0.983	95.111 ± 0.736	0.243 ± 0.002	43.073 ± 1.690	
QML	beta46	0.08	0.137	0.057 ± 0.002	0.711 ± 0.020	0.002 ± 0.000	0.046 ± 0.001	0.005 ± 0.000	0.073 ± 0.002	0.842 ± 0.035	0.918 ± 0.020	0.135	0.055	68.63	0.0434	0.0700	1.007	80.093 ± 1.362	0.183 ± 0.002	85.797 ± 1.191	
QML	beta47	0.08	0.134	0.054 ± 0.002	0.670 ± 0.020	0.002 ± 0.000	0.046 ± 0.001	0.005 ± 0.000	0.071 ± 0.002	0.777 ± 0.034	0.881 ± 0.020	0.130	0.050	61.98	0.0430	0.0656	0.942	79.045 ± 1.389	0.169 ± 0.002	89.290 ± 1.055	
QML	beta51	0.16	0.183	0.023 ± 0.003	0.143 ± 0.017	0.007 ± 0.000	0.082 ± 0.002	0.007 ± 0.000	0.085 ± 0.003	0.280 ± 0.014	0.529 ± 0.016	0.183	0.023	14.40	0.0792	0.0825	0.940	91.735 ± 0.940	0.300 ± 0.005	67.753 ± 1.595	
QML	beta52	0.16	0.151	-0.009 ± 0.003	-0.053 ± 0.016	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	0.226 ± 0.011	0.475 ± 0.014	0.149	-0.011	-6.92	0.0732	0.0740	0.965	93.015 ± 0.870	0.286 ± 0.004	57.159 ± 1.688	
QML	beta53	0.16	0.121	-0.039 ± 0.002	-0.246 ± 0.014	0.004 ± 0.000	0.063 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	0.218 ± 0.010	0.467 ± 0.013	0.120	-0.040	-24.82	0.0617	0.0734	0.991	89.290 ± 1.055	0.247 ± 0.002	50.291 ± 1.706	
QML	beta54	0.16	0.173	0.013 ± 0.002	0.083 ± 0.013	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.146 ± 0.007	0.382 ± 0.011	0.170	0.010	6.14	0.0594	0.0602	1.001	93.597 ± 0.835	0.234 ± 0.001	82.771 ± 1.288	
QML	beta55	0.08	0.079	-0.001 ± 0.004	-0.016 ± 0.053	0.015 ± 0.001	0.123 ± 0.003	0.015 ± 0.001	0.123 ± 0.003	2.379 ± 0.101	1.543 ± 0.042	0.065	-0.015	-18.36	0.1358	0.1366	0.781	76.717 ± 1.442	0.378 ± 0.009	27.008 ± 1.515	
QML	beta56	0.08	0.115	0.035 ± 0.002	0.435 ± 0.021	0.002 ± 0.000	0.049 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.567 ± 0.026	0.753 ± 0.019	0.116	0.036	44.77	0.0510	0.0623	0.912	82.305 ± 1.302	0.176 ± 0.002	71.595 ± 1.539	
QML	beta57	0.06	0.134	0.074 ± 0.004	1.238 ± 0.067	0.014 ± 0.001	0.118 ± 0.003	0.019 ± 0.001	0.139 ± 0.003	5.368 ± 0.230	2.317 ± 0.057	0.141	0.081	134.53	0.1322	0.1549	0.806	72.643 ± 1.521	0.371 ± 0.009	41.444 ± 1.681	
QML	beta58	0.06	0.103	0.043 ± 0.001	0.711 ± 0.020	0.001 ± 0.000	0.035 ± 0.001	0.003 ± 0.000	0.055 ± 0.001	0.843 ± 0.036	0.918 ± 0.021	0.101	0.041	68.74	0.0353	0.0543	1.032	82.421 ± 1.299	0.141 ± 0.002	85.914 ± 1.187	
UPI	beta41	0.20	0.241	0.041 ± 0.006	0.207 ± 0.029	0.029 ± 0.002	0.171 ± 0.005	0.031 ± 0.002	0.176 ± 0.006	0.776 ± 0.046	0.881 ± 0.030	0.235	0.035	17.28	0.1510	0.1549	1.133	98.137 ± 0.461	0.761 ± 0.014	28.405 ± 1.539	
UPI	beta42	0.20	0.241	0.041 ± 0.006	0.203 ± 0.028	0.026 ± 0.001	0.161 ± 0.005	0.028 ± 0.002	0.166 ± 0.006	0.691 ± 0.040	0.831 ± 0.028	0.243	0.043	21.42	0.1547	0.1605	1.067	96.158 ± 0.656	0.675 ± 0.011	35.856 ± 1.636	
UPI	beta43	0.20	0.178	-0.022 ± 0.002	-0.110 ± 0.010	0.003 ± 0.000	0.059 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	0.099 ± 0.005	0.315 ± 0.009	0.176	-0.024	-12.20	0.0576	0.0626	1.033	93.132 ± 0.863	0.239 ± 0.003	84.866 ± 1.223	
UPI	beta44	0.11	0.151	0.041 ± 0.004	0.374 ± 0.036	0.014 ± 0.001	0.117 ± 0.004	0.015 ± 0.001	0.124 ± 0.005	1.265 ± 0.084	1.125 ± 0.041	0.138	0.028	25.28	0.1016	0.1053	1.098	97.555 ± 0.527	0.503 ± 0.013	22.235 ± 1.435	

Table S206

Simulation 2 - Condition 62: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.4 (continued)

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta45	0.11	0.122	0.012 ± 0.003	0.107 ± 0.028	0.008 ± 0.001	0.091 ± 0.003	0.008 ± 0.001	0.092 ± 0.004	0.699 ± 0.048	0.836 ± 0.032	0.118	0.008	7.19	0.0853	0.0857	1.246	96.857 ± 0.595	0.446 ± 0.020	23.283 ± 1.442
UPI	beta46	0.08	0.048	-0.032 ± 0.002	-0.395 ± 0.030	0.005 ± 0.000	0.071 ± 0.002	0.006 ± 0.000	0.078 ± 0.003	0.946 ± 0.057	0.973 ± 0.032	0.056	-0.024	-29.66	0.0638	0.0681	1.105	95.227 ± 0.727	0.308 ± 0.007	17.113 ± 1.285
UPI	beta47	0.08	0.050	-0.030 ± 0.002	-0.381 ± 0.027	0.004 ± 0.000	0.064 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.779 ± 0.048	0.883 ± 0.030	0.052	-0.028	-35.33	0.0568	0.0635	1.048	93.714 ± 0.828	0.262 ± 0.005	21.769 ± 1.408
UPI	beta51	0.16	0.197	0.037 ± 0.002	0.232 ± 0.014	0.004 ± 0.000	0.063 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.211 ± 0.010	0.459 ± 0.012	0.199	0.039	24.24	0.0633	0.0742	1.082	93.830 ± 0.821	0.269 ± 0.003	83.236 ± 1.275
UPI	beta52	0.16	0.219	0.059 ± 0.007	0.370 ± 0.041	0.036 ± 0.002	0.191 ± 0.006	0.040 ± 0.002	0.199 ± 0.007	1.555 ± 0.096	1.247 ± 0.043	0.211	0.051	31.68	0.1742	0.1814	1.395	98.137 ± 0.461	1.042 ± 0.039	18.161 ± 1.315
UPI	beta53	0.16	0.111	-0.049 ± 0.003	-0.308 ± 0.019	0.008 ± 0.000	0.090 ± 0.003	0.011 ± 0.001	0.103 ± 0.003	0.411 ± 0.024	0.641 ± 0.020	0.112	-0.048	-29.78	0.0818	0.0947	1.221	94.179 ± 0.799	0.431 ± 0.011	23.632 ± 1.449
UPI	beta54	0.16	0.248	0.088 ± 0.010	0.548 ± 0.062	0.085 ± 0.005	0.292 ± 0.009	0.093 ± 0.006	0.305 ± 0.011	3.629 ± 0.232	1.905 ± 0.068	0.216	0.056	34.80	0.2561	0.2621	1.385	98.487 ± 0.417	1.586 ± 0.069	10.361 ± 1.040
UPI	beta55	0.08	0.131	0.051 ± 0.003	0.642 ± 0.037	0.008 ± 0.000	0.088 ± 0.003	0.010 ± 0.001	0.102 ± 0.003	1.614 ± 0.097	1.271 ± 0.042	0.128	0.048	60.14	0.0825	0.0955	1.393	96.740 ± 0.606	0.479 ± 0.018	29.453 ± 1.555
UPI	beta56	0.08	0.074	-0.006 ± 0.002	-0.070 ± 0.019	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.326 ± 0.018	0.571 ± 0.019	0.075	-0.005	-6.37	0.0421	0.0425	1.234	95.576 ± 0.702	0.219 ± 0.008	41.909 ± 1.683
UPI	beta57	0.06	-0.004	-0.064 ± 0.003	-1.070 ± 0.057	0.010 ± 0.001	0.101 ± 0.003	0.014 ± 0.001	0.120 ± 0.004	3.980 ± 0.239	1.995 ± 0.065	0.001	-0.059	-98.75	0.0888	0.1068	1.407	95.693 ± 0.693	0.557 ± 0.022	0.931 ± 0.328
UPI	beta58	0.06	0.010	-0.050 ± 0.003	-0.835 ± 0.054	0.009 ± 0.001	0.094 ± 0.003	0.011 ± 0.001	0.107 ± 0.004	3.156 ± 0.192	1.777 ± 0.059	0.018	-0.042	-69.32	0.0850	0.0946	1.350	95.576 ± 0.702	0.498 ± 0.020	5.239 ± 0.760

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S207

Simulation 2 - Condition 63: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability =

0.6

Method	Param.	True	Mean-based										Median-based					Inference			
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE	
LSAM	beta41	0.20	0.196	-0.004 $\pm$ 0.004	-0.018 $\pm$ 0.021	0.017 $\pm$ 0.001	0.129 $\pm$ 0.003	0.017 $\pm$ 0.001	0.129 $\pm$ 0.004	0.418 $\pm$ 0.020	0.647 $\pm$ 0.019	0.195	-0.005	-2.45	0.1337	0.1338	1.001	96.378 $\pm$ 0.619	0.507 $\pm$ 0.004	37.322 $\pm$ 1.602	
LSAM	beta42	0.20	0.204	0.004 $\pm$ 0.004	0.019 $\pm$ 0.022	0.018 $\pm$ 0.001	0.135 $\pm$ 0.003	0.018 $\pm$ 0.001	0.135 $\pm$ 0.004	0.456 $\pm$ 0.022	0.675 $\pm$ 0.020	0.209	0.009	4.55	0.1313	0.1316	1.005	96.268 $\pm$ 0.628	0.532 $\pm$ 0.005	37.212 $\pm$ 1.601	
LSAM	beta43	0.20	0.202	0.002 $\pm$ 0.002	0.008 $\pm$ 0.011	0.004 $\pm$ 0.000	0.063 $\pm$ 0.002	0.004 $\pm$ 0.000	0.063 $\pm$ 0.002	0.101 $\pm$ 0.005	0.317 $\pm$ 0.009	0.201	0.001	0.47	0.0640	0.0640	0.970	94.841 $\pm$ 0.733	0.241 $\pm$ 0.001	90.889 $\pm$ 0.953	
LSAM	beta44	0.11	0.125	0.015 $\pm$ 0.003	0.133 $\pm$ 0.027	0.008 $\pm$ 0.000	0.088 $\pm$ 0.002	0.008 $\pm$ 0.000	0.089 $\pm$ 0.003	0.657 $\pm$ 0.033	0.810 $\pm$ 0.024	0.123	0.013	11.41	0.0866	0.0875	0.932	93.304 $\pm$ 0.828	0.321 $\pm$ 0.004	38.749 $\pm$ 1.614	
LSAM	beta45	0.11	0.116	0.006 $\pm$ 0.003	0.056 $\pm$ 0.026	0.007 $\pm$ 0.000	0.085 $\pm$ 0.002	0.007 $\pm$ 0.000	0.085 $\pm$ 0.003	0.597 $\pm$ 0.034	0.773 $\pm$ 0.025	0.111	0.001	1.24	0.0752	0.0753	0.889	92.426 $\pm$ 0.877	0.296 $\pm$ 0.004	38.858 $\pm$ 1.615	
LSAM	beta46	0.08	0.078	-0.002 $\pm$ 0.002	-0.031 $\pm$ 0.024	0.003 $\pm$ 0.000	0.058 $\pm$ 0.002	0.003 $\pm$ 0.000	0.058 $\pm$ 0.002	0.519 $\pm$ 0.030	0.720 $\pm$ 0.024	0.075	-0.005	-6.05	0.0543	0.0546	0.895	94.182 $\pm$ 0.776	0.202 $\pm$ 0.003	42.371 $\pm$ 1.637	
LSAM	beta47	0.08	0.077	-0.003 $\pm$ 0.002	-0.041 $\pm$ 0.023	0.003 $\pm$ 0.000	0.056 $\pm$ 0.002	0.003 $\pm$ 0.000	0.056 $\pm$ 0.002	0.485 $\pm$ 0.027	0.696 $\pm$ 0.023	0.075	-0.005	-5.64	0.0534	0.0536	0.919	93.853 $\pm$ 0.796	0.200 $\pm$ 0.003	40.724 $\pm$ 1.628	
LSAM	beta51	0.16	0.164	0.004 $\pm$ 0.003	0.026 $\pm$ 0.016	0.006 $\pm$ 0.000	0.078 $\pm$ 0.002	0.006 $\pm$ 0.000	0.078 $\pm$ 0.002	0.240 $\pm$ 0.011	0.490 $\pm$ 0.014	0.164	0.004	2.31	0.0787	0.0788	0.938	94.292 $\pm$ 0.769	0.288 $\pm$ 0.002	61.032 $\pm$ 1.616	
LSAM	beta52	0.16	0.170	0.010 $\pm$ 0.005	0.060 $\pm$ 0.029	0.020 $\pm$ 0.001	0.142 $\pm$ 0.004	0.020 $\pm$ 0.001	0.142 $\pm$ 0.004	0.790 $\pm$ 0.040	0.889 $\pm$ 0.027	0.170	0.010	5.97	0.1363	0.1366	0.975	95.170 $\pm$ 0.710	0.542 $\pm$ 0.005	27.333 $\pm$ 1.477	
LSAM	beta53	0.16	0.155	-0.005 $\pm$ 0.003	-0.029 $\pm$ 0.019	0.008 $\pm$ 0.000	0.090 $\pm$ 0.002	0.008 $\pm$ 0.000	0.090 $\pm$ 0.003	0.315 $\pm$ 0.015	0.561 $\pm$ 0.017	0.154	-0.006	-3.77	0.0885	0.0887	0.973	94.731 $\pm$ 0.740	0.342 $\pm$ 0.002	43.578 $\pm$ 1.643	
LSAM	beta54	0.16	0.156	-0.004 $\pm$ 0.004	-0.023 $\pm$ 0.025	0.014 $\pm$ 0.001	0.120 $\pm$ 0.003	0.014 $\pm$ 0.001	0.120 $\pm$ 0.004	0.565 $\pm$ 0.030	0.752 $\pm$ 0.023	0.157	-0.003	-1.87	0.1175	0.1175	1.010	95.719 $\pm$ 0.671	0.477 $\pm$ 0.004	27.662 $\pm$ 1.482	
LSAM	beta55	0.08	0.095	0.015 $\pm$ 0.003	0.189 $\pm$ 0.035	0.007 $\pm$ 0.000	0.083 $\pm$ 0.002	0.007 $\pm$ 0.000	0.085 $\pm$ 0.003	1.119 $\pm$ 0.062	1.058 $\pm$ 0.034	0.097	0.017	21.14	0.0741	0.0760	0.907	93.633 $\pm$ 0.809	0.296 $\pm$ 0.004	33.150 $\pm$ 1.560	
LSAM	beta56	0.08	0.079	-0.001 $\pm$ 0.002	-0.014 $\pm$ 0.020	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.002	0.363 $\pm$ 0.021	0.602 $\pm$ 0.020	0.076	-0.004	-5.15	0.0440	0.0442	0.897	91.109 $\pm$ 0.943	0.169 $\pm$ 0.002	49.286 $\pm$ 1.656	
LSAM	beta57	0.06	0.045	-0.015 $\pm$ 0.003	-0.253 $\pm$ 0.051	0.008 $\pm$ 0.000	0.092 $\pm$ 0.003	0.009 $\pm$ 0.000	0.093 $\pm$ 0.003	2.393 $\pm$ 0.131	1.547 $\pm$ 0.049	0.042	-0.018	-30.54	0.0829	0.0849	0.922	92.755 $\pm$ 0.859	0.331 $\pm$ 0.005	9.221 $\pm$ 0.959	
LSAM	beta58	0.06	0.057	-0.003 $\pm$ 0.002	-0.049 $\pm$ 0.026	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.002	0.606 $\pm$ 0.034	0.779 $\pm$ 0.026	0.057	-0.003	-4.50	0.0419	0.0420	0.942	93.743 $\pm$ 0.802	0.172 $\pm$ 0.003	35.785 $\pm$ 1.588	
QML	beta41	0.20	0.211	0.011 $\pm$ 0.002	0.055 $\pm$ 0.012	0.005 $\pm$ 0.000	0.073 $\pm$ 0.002	0.005 $\pm$ 0.000	0.074 $\pm$ 0.002	0.137 $\pm$ 0.007	0.371 $\pm$ 0.011	0.209	0.009	4.53	0.0775	0.0780	1.012	95.609 $\pm$ 0.679	0.291 $\pm$ 0.002	80.132 $\pm$ 1.322	
QML	beta42	0.20	0.211	0.011 $\pm$ 0.002	0.055 $\pm$ 0.012	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.142 $\pm$ 0.007	0.377 $\pm$ 0.011	0.215	0.015	7.52	0.0722	0.0737	1.014	95.939 $\pm$ 0.654	0.296 $\pm$ 0.002	78.595 $\pm$ 1.359	
QML	beta43	0.20	0.192	-0.008 $\pm$ 0.002	-0.038 $\pm$ 0.009	0.003 $\pm$ 0.000	0.057 $\pm$ 0.001	0.003 $\pm$ 0.000	0.058 $\pm$ 0.002	0.083 $\pm$ 0.004	0.288 $\pm$ 0.008	0.193	-0.007	-3.35	0.0562	0.0566	0.964	93.414 $\pm$ 0.822	0.216 $\pm$ 0.001	93.085 $\pm$ 0.841	
QML	beta44	0.11	0.081	-0.029 $\pm$ 0.003	-0.264 $\pm$ 0.024	0.006 $\pm$ 0.000	0.080 $\pm$ 0.002	0.007 $\pm$ 0.000	0.085 $\pm$ 0.002	0.604 $\pm$ 0.028	0.777 $\pm$ 0.022	0.084	-0.026	-23.76	0.0843	0.0883	0.955	92.536 $\pm$ 0.871	0.301 $\pm$ 0.004	25.576 $\pm$ 1.445	
QML	beta45	0.11	0.102	-0.008 $\pm$ 0.002	-0.068 $\pm$ 0.022	0.005 $\pm$ 0.000	0.072 $\pm$ 0.002	0.005 $\pm$ 0.000	0.072 $\pm$ 0.002	0.429 $\pm$ 0.021	0.655 $\pm$ 0.019	0.104	-0.006	-5.08	0.0657	0.0660	0.919	91.328 $\pm$ 0.932	0.258 $\pm$ 0.003	37.870 $\pm$ 1.607	
QML	beta46	0.08	0.110	0.030 $\pm$ 0.002	0.376 $\pm$ 0.019	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.003 $\pm$ 0.000	0.055 $\pm$ 0.002	0.477 $\pm$ 0.025	0.691 $\pm$ 0.020	0.107	0.027	33.16	0.0473	0.0542	0.936	90.560 $\pm$ 0.969	0.170 $\pm$ 0.003	74.204 $\pm$ 1.450	
QML	beta47	0.08	0.112	0.032 $\pm$ 0.002	0.397 $\pm$ 0.019	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.003 $\pm$ 0.000	0.056 $\pm$ 0.002	0.484 $\pm$ 0.029	0.696 $\pm$ 0.022	0.107	0.027	34.35	0.0430	0.0510	0.904	90.121 $\pm$ 0.989	0.162 $\pm$ 0.002	79.802 $\pm$ 1.330	
QML	beta51	0.16	0.177	0.017 $\pm$ 0.003	0.103 $\pm$ 0.017	0.006 $\pm$ 0.000	0.080 $\pm$ 0.002	0.007 $\pm$ 0.000	0.082 $\pm$ 0.002	0.260 $\pm$ 0.013	0.510 $\pm$ 0.015	0.175	0.015	9.46	0.0793	0.0807	0.973	95.170 $\pm$ 0.710	0.305 $\pm$ 0.002	62.678 $\pm$ 1.602	
QML	beta52	0.16	0.145	-0.015 $\pm$ 0.003	-0.095 $\pm$ 0.017	0.007 $\pm$ 0.000	0.081 $\pm$ 0.002	0.007 $\pm$ 0.000	0.082 $\pm$ 0.002	0.263 $\pm$ 0.012	0.513 $\pm$ 0.014	0.144	-0.016	-10.06	0.0830	0.0845	0.961	93.743 $\pm$ 0.802	0.304 $\pm$ 0.002	48.189 $\pm$ 1.655	
QML	beta53	0.16	0.125	-0.035 $\pm$ 0.003	-0.221 $\pm$ 0.016	0.006 $\pm$ 0.000	0.077 $\pm$ 0.002	0.007 $\pm$ 0.000	0.085 $\pm$ 0.002	0.282 $\pm$ 0.013	0.531 $\pm$ 0.014	0.125	-0.035	-21.92	0.0800	0.0874	0.961	91.109 $\pm$ 0.943	0.291 $\pm$ 0.002	40.285 $\pm$ 1.625	
QML	beta54	0.16	0.162	0.002 $\pm$ 0.002	0.013 $\pm$ 0.015	0.005 $\pm$ 0.000	0.073 $\pm$ 0.002	0.005 $\pm$ 0.000	0.073 $\pm$ 0.002	0.210 $\pm$ 0.011	0.458 $\pm$ 0.014	0.164	0.004	2.44	0.0719	0.0720	1.016	95.499 $\pm$ 0.687	0.292 $\pm$ 0.001	59.276 $\pm$ 1.628	
QML	beta55	0.08	0.084	0.004 $\pm$ 0.003	0.048 $\pm$ 0.038	0.008 $\pm$ 0.000	0.091 $\pm$ 0.002	0.008 $\pm$ 0.000	0.091 $\pm$ 0.003	1.285 $\pm$ 0.070	1.134 $\pm$ 0.036	0.090	0.010	12.10	0.0810	0.0816	0.875	89.352 $\pm$ 1.022	0.311 $\pm$ 0.006	33.150 $\pm$ 1.560	
QML	beta56	0.08	0.096	0.016 $\pm$ 0.002	0.204 $\pm$ 0.021	0.003 $\pm$ 0.000	0.050 $\pm$ 0.002	0.003 $\pm$ 0.000	0.053 $\pm$ 0.002	0.438 $\pm$ 0.027	0.662 $\pm$ 0.023	0.093	0.013	15.83	0.0437	0.0455	0.879	89.682 $\pm$ 1.008	0.174 $\pm$ 0.002	61.251 $\pm$ 1.614	
QML	beta57	0.06	0.086	0.026 $\pm$ 0.003	0.427 $\pm$ 0.044	0.006 $\pm$ 0.000	0.080 $\pm$ 0.002	0.007 $\pm$ 0.000	0.084 $\pm$ 0.003	1.948 $\pm$ 0.112	1.396 $\pm$ 0.045	0.078	0.018	29.43	0.0738	0.0759	0.933	91.218 $\pm$ 0.938	0.292 $\pm$ 0.006	21.515 $\pm$ 1.361	
QML	beta58	0.06	0.081	0.021 $\pm$ 0.001	0.346 $\pm$ 0.021	0.001 $\pm$ 0.000	0.039 $\pm$ 0.001	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.534 $\pm$ 0.029	0.731 $\pm$ 0.022	0.078	0.018	30.54	0.0360	0.0404</					

Table S207

Simulation 2 - Condition 63: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.6 (continued)

Method	Param.	True	Mean-based										Median-based					Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE		
UPI	beta45	0.11	0.111	0.001 ± 0.003	0.013 ± 0.027	0.008 ± 0.000	0.088 ± 0.003	0.008 ± 0.000	0.088 ± 0.003	0.640 ± 0.036	0.800 ± 0.026	0.109	-0.001	-0.68	0.0780	0.0780	0.984	93.743 ± 0.802	0.339 ± 0.007	31.065 ± 1.533		
UPI	beta46	0.08	0.068	-0.012 ± 0.002	-0.144 ± 0.026	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	0.616 ± 0.039	0.785 ± 0.028	0.069	-0.011	-13.53	0.0525	0.0536	0.948	93.853 ± 0.796	0.230 ± 0.005	34.248 ± 1.572		
UPI	beta47	0.08	0.074	-0.006 ± 0.002	-0.078 ± 0.024	0.003 ± 0.000	0.057 ± 0.002	0.003 ± 0.000	0.058 ± 0.002	0.522 ± 0.034	0.722 ± 0.026	0.072	-0.008	-9.89	0.0497	0.0504	0.973	92.536 ± 0.871	0.219 ± 0.005	38.090 ± 1.609		
UPI	beta51	0.16	0.174	0.014 ± 0.003	0.087 ± 0.016	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	0.230 ± 0.011	0.479 ± 0.014	0.175	0.015	9.13	0.0745	0.0759	0.955	94.951 ± 0.725	0.282 ± 0.002	68.825 ± 1.535		
UPI	beta52	0.16	0.182	0.022 ± 0.004	0.137 ± 0.026	0.016 ± 0.001	0.127 ± 0.003	0.017 ± 0.001	0.129 ± 0.004	0.649 ± 0.033	0.805 ± 0.024	0.185	0.025	15.36	0.1241	0.1265	1.109	95.499 ± 0.687	0.552 ± 0.010	30.626 ± 1.527		
UPI	beta53	0.16	0.150	-0.010 ± 0.003	-0.061 ± 0.018	0.007 ± 0.000	0.085 ± 0.002	0.007 ± 0.000	0.085 ± 0.002	0.283 ± 0.013	0.532 ± 0.015	0.149	-0.011	-7.16	0.0845	0.0852	1.020	94.512 ± 0.755	0.338 ± 0.003	41.932 ± 1.635		
UPI	beta54	0.16	0.160	0.000 ± 0.004	0.001 ± 0.027	0.017 ± 0.001	0.130 ± 0.003	0.017 ± 0.001	0.130 ± 0.004	0.656 ± 0.033	0.810 ± 0.025	0.159	-0.001	-0.49	0.1260	0.1260	1.120	96.926 ± 0.572	0.569 ± 0.013	24.259 ± 1.420		
UPI	beta55	0.08	0.093	0.013 ± 0.003	0.167 ± 0.033	0.006 ± 0.000	0.080 ± 0.002	0.007 ± 0.000	0.081 ± 0.003	1.028 ± 0.055	1.014 ± 0.032	0.094	0.014	17.14	0.0743	0.0755	1.004	94.182 ± 0.776	0.315 ± 0.005	28.979 ± 1.503		
UPI	beta56	0.08	0.078	-0.002 ± 0.002	-0.022 ± 0.020	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.002	0.354 ± 0.020	0.595 ± 0.020	0.077	-0.003	-3.67	0.0428	0.0429	0.949	91.548 ± 0.922	0.177 ± 0.003	48.299 ± 1.656		
UPI	beta57	0.06	0.041	-0.019 ± 0.003	-0.310 ± 0.045	0.007 ± 0.000	0.082 ± 0.002	0.007 ± 0.000	0.084 ± 0.003	1.962 ± 0.104	1.401 ± 0.043	0.036	-0.024	-39.89	0.0740	0.0778	1.063	94.512 ± 0.755	0.342 ± 0.008	8.342 ± 0.916		
UPI	beta58	0.06	0.055	-0.005 ± 0.002	-0.088 ± 0.028	0.003 ± 0.000	0.051 ± 0.002	0.003 ± 0.000	0.051 ± 0.002	0.727 ± 0.043	0.852 ± 0.029	0.056	-0.004	-6.82	0.0434	0.0436	1.051	95.390 ± 0.695	0.210 ± 0.006	31.614 ± 1.541		

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S208

Simulation 2 - Condition 64: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, $N = 1000$, Reliability = 0.6

Method	Param.	True	Mean-based										Median-based					Inference		
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LSAM	beta41	0.20	0.197	-0.003 $\pm$ 0.002	-0.017 $\pm$ 0.012	0.006 $\pm$ 0.000	0.076 $\pm$ 0.002	0.006 $\pm$ 0.000	0.076 $\pm$ 0.002	0.145 $\pm$ 0.006	0.381 $\pm$ 0.010	0.194	-0.006	-3.10	0.0776	0.0778	0.967	94.898 $\pm$ 0.703	0.289 $\pm$ 0.002	74.490 $\pm$ 1.392
LSAM	beta42	0.20	0.204	0.004 $\pm$ 0.003	0.020 $\pm$ 0.013	0.006 $\pm$ 0.000	0.080 $\pm$ 0.002	0.006 $\pm$ 0.000	0.080 $\pm$ 0.002	0.161 $\pm$ 0.008	0.401 $\pm$ 0.012	0.202	0.002	0.99	0.0771	0.0771	0.972	94.490 $\pm$ 0.729	0.305 $\pm$ 0.002	73.469 $\pm$ 1.410
LSAM	beta43	0.20	0.201	0.001 $\pm$ 0.001	0.005 $\pm$ 0.006	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.038 $\pm$ 0.002	0.194 $\pm$ 0.006	0.201	0.001	0.48	0.0380	0.0380	0.967	95.306 $\pm$ 0.676	0.147 $\pm$ 0.001	99.898 $\pm$ 0.102
LSAM	beta44	0.11	0.124	0.014 $\pm$ 0.002	0.124 $\pm$ 0.015	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.003 $\pm$ 0.000	0.053 $\pm$ 0.002	0.233 $\pm$ 0.012	0.483 $\pm$ 0.014	0.123	0.013	11.65	0.0480	0.0496	0.891	92.551 $\pm$ 0.839	0.179 $\pm$ 0.001	77.551 $\pm$ 1.333
LSAM	beta45	0.11	0.112	0.002 $\pm$ 0.001	0.014 $\pm$ 0.014	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.182 $\pm$ 0.009	0.426 $\pm$ 0.013	0.112	0.002	1.50	0.0437	0.0437	0.917	92.551 $\pm$ 0.839	0.169 $\pm$ 0.001	73.061 $\pm$ 1.417
LSAM	beta46	0.08	0.076	-0.004 $\pm$ 0.001	-0.046 $\pm$ 0.012	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.142 $\pm$ 0.007	0.377 $\pm$ 0.011	0.077	-0.003	-4.27	0.0292	0.0294	0.895	91.020 $\pm$ 0.913	0.105 $\pm$ 0.001	77.857 $\pm$ 1.326
LSAM	beta47	0.08	0.074	-0.006 $\pm$ 0.001	-0.072 $\pm$ 0.012	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.144 $\pm$ 0.007	0.380 $\pm$ 0.011	0.075	-0.005	-6.63	0.0261	0.0267	0.918	92.245 $\pm$ 0.854	0.107 $\pm$ 0.001	76.837 $\pm$ 1.348
LSAM	beta51	0.16	0.167	0.007 $\pm$ 0.001	0.043 $\pm$ 0.009	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.083 $\pm$ 0.004	0.287 $\pm$ 0.008	0.167	0.007	4.15	0.0478	0.0482	0.989	95.306 $\pm$ 0.676	0.176 $\pm$ 0.001	96.837 $\pm$ 0.559
LSAM	beta52	0.16	0.167	0.007 $\pm$ 0.003	0.045 $\pm$ 0.016	0.007 $\pm$ 0.000	0.082 $\pm$ 0.002	0.007 $\pm$ 0.000	0.082 $\pm$ 0.002	0.262 $\pm$ 0.013	0.512 $\pm$ 0.015	0.169	0.009	5.38	0.0763	0.0768	0.968	93.571 $\pm$ 0.783	0.309 $\pm$ 0.002	57.449 $\pm$ 1.579
LSAM	beta53	0.16	0.158	-0.002 $\pm$ 0.002	-0.014 $\pm$ 0.010	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.108 $\pm$ 0.005	0.328 $\pm$ 0.009	0.156	-0.004	-2.56	0.0542	0.0543	0.996	94.898 $\pm$ 0.703	0.205 $\pm$ 0.001	85.204 $\pm$ 1.134
LSAM	beta54	0.16	0.156	-0.004 $\pm$ 0.002	-0.027 $\pm$ 0.015	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.219 $\pm$ 0.011	0.468 $\pm$ 0.014	0.155	-0.005	-3.19	0.0783	0.0785	0.960	94.490 $\pm$ 0.729	0.281 $\pm$ 0.001	59.388 $\pm$ 1.569
LSAM	beta55	0.08	0.096	0.016 $\pm$ 0.001	0.195 $\pm$ 0.017	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.314 $\pm$ 0.014	0.560 $\pm$ 0.015	0.097	0.017	21.59	0.0409	0.0444	0.969	92.143 $\pm$ 0.860	0.160 $\pm$ 0.001	66.633 $\pm$ 1.506
LSAM	beta56	0.08	0.076	-0.004 $\pm$ 0.001	-0.048 $\pm$ 0.011	0.001 $\pm$ 0.000	0.027 $\pm$ 0.001	0.001 $\pm$ 0.000	0.027 $\pm$ 0.001	0.115 $\pm$ 0.006	0.339 $\pm$ 0.010	0.075	-0.005	-5.86	0.0244	0.0249	0.863	90.612 $\pm$ 0.932	0.091 $\pm$ 0.001	85.510 $\pm$ 1.124
LSAM	beta57	0.06	0.046	-0.014 $\pm$ 0.002	-0.237 $\pm$ 0.026	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	0.694 $\pm$ 0.033	0.833 $\pm$ 0.023	0.042	-0.018	-29.74	0.0446	0.0480	0.958	91.837 $\pm$ 0.875	0.180 $\pm$ 0.002	17.347 $\pm$ 1.210
LSAM	beta58	0.06	0.057	-0.003 $\pm$ 0.001	-0.053 $\pm$ 0.014	0.001 $\pm$ 0.000	0.026 $\pm$ 0.001	0.001 $\pm$ 0.000	0.026 $\pm$ 0.001	0.186 $\pm$ 0.010	0.431 $\pm$ 0.013	0.057	-0.003	-5.01	0.0249	0.0250	0.952	94.388 $\pm$ 0.735	0.096 $\pm$ 0.001	65.612 $\pm$ 1.517
QML	beta41	0.20	0.213	0.013 $\pm$ 0.001	0.063 $\pm$ 0.007	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.058 $\pm$ 0.003	0.241 $\pm$ 0.007	0.210	0.010	5.12	0.0463	0.0474	0.989	92.959 $\pm$ 0.817	0.181 $\pm$ 0.001	98.980 $\pm$ 0.321
QML	beta42	0.20	0.212	0.012 $\pm$ 0.001	0.059 $\pm$ 0.007	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.058 $\pm$ 0.003	0.240 $\pm$ 0.007	0.211	0.011	5.73	0.0455	0.0469	1.011	94.082 $\pm$ 0.754	0.184 $\pm$ 0.001	99.184 $\pm$ 0.287
QML	beta43	0.20	0.193	-0.007 $\pm$ 0.001	-0.033 $\pm$ 0.006	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.033 $\pm$ 0.002	0.181 $\pm$ 0.005	0.193	-0.007	-3.54	0.0337	0.0345	0.971	93.673 $\pm$ 0.778	0.136 $\pm$ 0.000	100.000 $\pm$ 0.000
QML	beta44	0.11	0.087	-0.023 $\pm$ 0.002	-0.213 $\pm$ 0.014	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.003 $\pm$ 0.000	0.055 $\pm$ 0.001	0.246 $\pm$ 0.011	0.496 $\pm$ 0.013	0.088	-0.022	-19.78	0.0474	0.0522	0.947	90.000 $\pm$ 0.958	0.183 $\pm$ 0.001	48.571 $\pm$ 1.597
QML	beta45	0.11	0.103	-0.007 $\pm$ 0.001	-0.062 $\pm$ 0.012	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.156 $\pm$ 0.007	0.395 $\pm$ 0.011	0.102	-0.008	-7.24	0.0422	0.0430	0.960	92.551 $\pm$ 0.839	0.162 $\pm$ 0.001	71.531 $\pm$ 1.442
QML	beta46	0.08	0.108	0.028 $\pm$ 0.001	0.354 $\pm$ 0.011	0.001 $\pm$ 0.000	0.027 $\pm$ 0.001	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.237 $\pm$ 0.010	0.487 $\pm$ 0.011	0.108	0.028	34.56	0.0261	0.0380	0.952	82.041 $\pm$ 1.226	0.100 $\pm$ 0.001	98.469 $\pm$ 0.392
QML	beta47	0.08	0.108	0.028 $\pm$ 0.001	0.354 $\pm$ 0.010	0.001 $\pm$ 0.000	0.025 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.222 $\pm$ 0.008	0.471 $\pm$ 0.010	0.107	0.027	34.19	0.0253	0.0373	0.991	79.694 $\pm$ 1.285	0.097 $\pm$ 0.001	99.184 $\pm$ 0.287
QML	beta51	0.16	0.180	0.020 $\pm$ 0.001	0.128 $\pm$ 0.009	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.102 $\pm$ 0.005	0.319 $\pm$ 0.008	0.180	0.020	12.74	0.0510	0.0549	1.028	94.082 $\pm$ 0.754	0.189 $\pm$ 0.001	95.918 $\pm$ 0.632
QML	beta52	0.16	0.142	-0.018 $\pm$ 0.002	-0.114 $\pm$ 0.010	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.106 $\pm$ 0.005	0.325 $\pm$ 0.008	0.141	-0.019	-12.10	0.0474	0.0512	0.960	91.633 $\pm$ 0.885	0.183 $\pm$ 0.001	84.286 $\pm$ 1.163
QML	beta53	0.16	0.126	-0.034 $\pm$ 0.001	-0.214 $\pm$ 0.009	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.003 $\pm$ 0.000	0.057 $\pm$ 0.001	0.128 $\pm$ 0.005	0.358 $\pm$ 0.008	0.125	-0.035	-21.77	0.0457	0.0575	0.993	87.143 $\pm$ 1.069	0.179 $\pm$ 0.001	79.388 $\pm$ 1.292
QML	beta54	0.16	0.161	0.001 $\pm$ 0.002	0.004 $\pm$ 0.010	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.090 $\pm$ 0.004	0.300 $\pm$ 0.008	0.161	0.001	0.91	0.0489	0.0489	0.971	95.000 $\pm$ 0.696	0.183 $\pm$ 0.001	91.122 $\pm$ 0.909
QML	beta55	0.08	0.091	0.011 $\pm$ 0.002	0.135 $\pm$ 0.021	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.003 $\pm$ 0.000	0.054 $\pm$ 0.002	0.461 $\pm$ 0.024	0.679 $\pm$ 0.020	0.093	0.013	16.22	0.0522	0.0538	0.948	89.490 $\pm$ 0.980	0.198 $\pm$ 0.003	53.367 $\pm$ 1.594
QML	beta56	0.08	0.094	0.014 $\pm$ 0.001	0.174 $\pm$ 0.012	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.168 $\pm$ 0.008	0.409 $\pm$ 0.011	0.093	0.013	16.50	0.0279	0.0309	0.883	88.061 $\pm$ 1.036	0.103 $\pm$ 0.001	91.429 $\pm$ 0.894
QML	beta57	0.06	0.083	0.023 $\pm$ 0.002	0.391 $\pm$ 0.027	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.003 $\pm$ 0.000	0.056 $\pm$ 0.002	0.869 $\pm$ 0.048	0.932 $\pm$ 0.029	0.078	0.018	30.23	0.0473	0.0506	0.960	92.041 $\pm$ 0.865	0.191 $\pm$ 0.003	43.163 $\pm$ 1.582
QML	beta58	0.06	0.081	0.021 $\pm$ 0.001	0.353 $\pm$ 0.012	0.000 $\pm$ 0.000	0.022 $\pm$ 0.001	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.261 $\pm$ 0.012	0.511 $\pm$ 0.012	0.081	0.021	34.67	0.0218	0.0301	0.987	85.102 $\pm$ 1.137	0.086 $\pm$ 0.001	96.429 $\pm$ 0.593
UPI	beta41	0.20	0.206	0.006 $\pm$ 0.002	0.029 $\pm$ 0.011	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.115 $\pm$ 0.006	0.340 $\pm$ 0.010	0.204	0.004	1.79	0.0670	0.0671	1.022	96.122 $\pm$ 0.617	0.271 $\pm$ 0.003	84.286 $\pm$ 1.163
UPI	beta42	0.20	0.208	0.008 $\pm$ 0.002	0.040 $\pm$ 0.011	0.005 $\pm$ 0.000	0.072 $\pm$ 0.002	0.005 $\pm$ 0.000	0.072 $\pm$ 0.002	0.131 $\pm$ 0.007	0.362 $\pm$ 0.011	0.209	0.009	4.57	0.0689	0.0695	1.024	95.918 $\pm$ 0.632	0.289 $\pm$ 0.003	80.714 $\pm$ 1.260
UPI	beta43	0.20	0.198	-0.002 $\pm$ 0.001	-0.011 $\pm$ 0.006	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.038 $\pm$ 0.002	0.196 $\pm$ 0.006	0.198	-0.002	-0.84	0.0380	0.0381	0.942	93.980 $\pm$ 0.760	0.144 $\pm$ 0.001	9

Table S208

Simulation 2 - Condition 64: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.6 (continued)

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta45	0.11	0.111	0.001 ± 0.002	0.006 ± 0.014	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.197 ± 0.010	0.443 ± 0.013	0.109	-0.001	-0.48	0.0465	0.0465	0.934	93.061 ± 0.812	0.179 ± 0.002	69.898 ± 1.465
UPI	beta46	0.08	0.074	-0.006 ± 0.001	-0.071 ± 0.012	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.146 ± 0.007	0.381 ± 0.011	0.075	-0.005	-6.47	0.0291	0.0295	0.934	90.918 ± 0.918	0.110 ± 0.002	74.592 ± 1.391
UPI	beta47	0.08	0.074	-0.006 ± 0.001	-0.081 ± 0.012	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.151 ± 0.009	0.388 ± 0.013	0.073	-0.007	-9.37	0.0262	0.0273	0.945	92.245 ± 0.854	0.113 ± 0.002	73.776 ± 1.405
UPI	beta51	0.16	0.172	0.012 ± 0.001	0.075 ± 0.009	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.083 ± 0.004	0.288 ± 0.008	0.171	0.011	6.73	0.0464	0.0477	0.998	95.612 ± 0.654	0.174 ± 0.001	98.061 ± 0.440
UPI	beta52	0.16	0.173	0.013 ± 0.003	0.080 ± 0.016	0.007 ± 0.000	0.082 ± 0.002	0.007 ± 0.000	0.083 ± 0.003	0.271 ± 0.014	0.521 ± 0.016	0.173	0.013	7.99	0.0742	0.0753	0.976	95.306 ± 0.676	0.315 ± 0.003	61.735 ± 1.553
UPI	beta53	0.16	0.154	-0.006 ± 0.002	-0.035 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.106 ± 0.005	0.326 ± 0.009	0.154	-0.006	-4.03	0.0518	0.0522	0.978	94.184 ± 0.748	0.199 ± 0.001	86.633 ± 1.087
UPI	beta54	0.16	0.153	-0.007 ± 0.003	-0.045 ± 0.016	0.006 ± 0.000	0.080 ± 0.002	0.007 ± 0.000	0.081 ± 0.002	0.255 ± 0.013	0.505 ± 0.015	0.153	-0.007	-4.37	0.0800	0.0804	0.990	95.510 ± 0.661	0.312 ± 0.003	52.347 ± 1.595
UPI	beta55	0.08	0.092	0.012 ± 0.001	0.145 ± 0.018	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.321 ± 0.017	0.567 ± 0.017	0.093	0.013	15.65	0.0415	0.0433	0.992	94.184 ± 0.748	0.170 ± 0.002	58.878 ± 1.572
UPI	beta56	0.08	0.077	-0.003 ± 0.001	-0.041 ± 0.011	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	0.116 ± 0.006	0.341 ± 0.010	0.076	-0.004	-4.49	0.0254	0.0256	0.875	90.816 ± 0.923	0.093 ± 0.001	86.837 ± 1.080
UPI	beta57	0.06	0.046	-0.014 ± 0.002	-0.232 ± 0.026	0.002 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	0.740 ± 0.040	0.860 ± 0.027	0.044	-0.016	-26.03	0.0453	0.0479	0.965	92.755 ± 0.828	0.188 ± 0.002	17.857 ± 1.223
UPI	beta58	0.06	0.057	-0.003 ± 0.001	-0.049 ± 0.015	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.215 ± 0.012	0.464 ± 0.014	0.057	-0.003	-5.83	0.0253	0.0256	0.981	94.490 ± 0.729	0.106 ± 0.001	60.612 ± 1.561

Note. MCSE = Monte Carlo Standard Error shown as % value. Coverage, Type I Error, and Power expressed as percentages. % indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S209

Simulation 2 - Condition 65: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.8

Method	Param.	True	Mean-based										Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE	
LSAM	beta41	0.20	0.198	-0.002 ± 0.003	-0.011 ± 0.013	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	0.155 ± 0.007	0.394 ± 0.011	0.197	-0.003	-1.74	0.0787	0.0787	0.999	95.648 ± 0.649	0.309 ± 0.001	72.470 ± 1.421	
LSAM	beta42	0.20	0.199	-0.001 ± 0.003	-0.007 ± 0.014	0.007 ± 0.000	0.085 ± 0.002	0.007 ± 0.000	0.085 ± 0.002	0.182 ± 0.009	0.427 ± 0.012	0.197	-0.003	-1.41	0.0804	0.0804	0.961	94.130 ± 0.748	0.322 ± 0.002	67.611 ± 1.489	
LSAM	beta43	0.20	0.201	0.001 ± 0.002	0.005 ± 0.008	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.059 ± 0.003	0.243 ± 0.007	0.200	0.000	0.17	0.0488	0.0488	0.976	93.522 ± 0.783	0.186 ± 0.001	98.988 ± 0.318	
LSAM	beta44	0.11	0.115	0.005 ± 0.002	0.044 ± 0.017	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.277 ± 0.014	0.526 ± 0.016	0.114	0.004	3.60	0.0540	0.0542	0.900	92.004 ± 0.863	0.204 ± 0.002	61.538 ± 1.548	
LSAM	beta45	0.11	0.113	0.003 ± 0.002	0.025 ± 0.016	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.252 ± 0.014	0.502 ± 0.016	0.109	-0.001	-0.81	0.0517	0.0517	0.891	93.117 ± 0.805	0.193 ± 0.002	65.081 ± 1.517	
LSAM	beta46	0.08	0.082	0.002 ± 0.001	0.021 ± 0.013	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.179 ± 0.010	0.423 ± 0.013	0.082	0.002	1.93	0.0306	0.0306	0.859	90.587 ± 0.929	0.114 ± 0.001	78.846 ± 1.299	
LSAM	beta47	0.08	0.079	-0.001 ± 0.001	-0.009 ± 0.014	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.182 ± 0.010	0.427 ± 0.014	0.078	-0.002	-2.97	0.0318	0.0319	0.863	90.486 ± 0.933	0.116 ± 0.001	76.113 ± 1.357	
LSAM	beta51	0.16	0.162	0.002 ± 0.002	0.013 ± 0.012	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.147 ± 0.007	0.383 ± 0.011	0.162	0.002	1.41	0.0597	0.0598	0.948	93.725 ± 0.772	0.228 ± 0.001	79.757 ± 1.278	
LSAM	beta52	0.16	0.159	-0.001 ± 0.003	-0.006 ± 0.016	0.007 ± 0.000	0.082 ± 0.002	0.007 ± 0.000	0.082 ± 0.002	0.265 ± 0.011	0.515 ± 0.014	0.156	-0.004	-2.59	0.0804	0.0805	0.979	94.939 ± 0.697	0.316 ± 0.001	50.202 ± 1.591	
LSAM	beta53	0.16	0.159	-0.001 ± 0.002	-0.005 ± 0.013	0.004 ± 0.000	0.067 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	0.173 ± 0.008	0.416 ± 0.012	0.161	0.001	0.47	0.0672	0.0672	0.956	94.534 ± 0.723	0.250 ± 0.001	70.951 ± 1.444	
LSAM	beta54	0.16	0.157	-0.003 ± 0.003	-0.019 ± 0.016	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.081 ± 0.002	0.256 ± 0.011	0.506 ± 0.014	0.159	-0.001	-0.62	0.0829	0.0829	0.966	94.939 ± 0.697	0.306 ± 0.002	53.239 ± 1.587	
LSAM	beta55	0.08	0.083	0.003 ± 0.002	0.033 ± 0.022	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.484 ± 0.024	0.696 ± 0.021	0.083	0.003	4.36	0.0534	0.0535	0.852	90.789 ± 0.920	0.186 ± 0.002	47.267 ± 1.588	
LSAM	beta56	0.08	0.081	0.001 ± 0.001	0.010 ± 0.014	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.197 ± 0.010	0.444 ± 0.014	0.079	-0.001	-0.65	0.0321	0.0321	0.878	92.004 ± 0.863	0.122 ± 0.002	72.672 ± 1.418	
LSAM	beta57	0.06	0.059	-0.001 ± 0.002	-0.020 ± 0.028	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.754 ± 0.035	0.869 ± 0.024	0.059	-0.001	-2.48	0.0512	0.0512	0.890	90.486 ± 0.933	0.182 ± 0.002	28.846 ± 1.441	
LSAM	beta58	0.06	0.061	0.001 ± 0.001	0.009 ± 0.017	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.274 ± 0.015	0.524 ± 0.017	0.058	-0.002	-3.35	0.0269	0.0269	0.890	91.903 ± 0.868	0.110 ± 0.001	61.032 ± 1.552	
QML	beta41	0.20	0.203	0.003 ± 0.002	0.015 ± 0.010	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	0.104 ± 0.005	0.323 ± 0.009	0.203	0.003	1.57	0.0646	0.0647	1.013	96.053 ± 0.619	0.256 ± 0.002	85.526 ± 1.119	
QML	beta42	0.20	0.202	0.002 ± 0.002	0.009 ± 0.011	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.118 ± 0.005	0.344 ± 0.010	0.201	0.001	0.67	0.0653	0.0653	0.966	93.522 ± 0.783	0.260 ± 0.001	83.806 ± 1.172	
QML	beta43	0.20	0.197	-0.003 ± 0.001	-0.013 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.055 ± 0.002	0.234 ± 0.006	0.196	-0.004	-1.91	0.0463	0.0465	0.982	93.320 ± 0.794	0.180 ± 0.001	99.190 ± 0.285	
QML	beta44	0.11	0.099	-0.011 ± 0.002	-0.100 ± 0.017	0.003 ± 0.000	0.058 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	0.290 ± 0.015	0.539 ± 0.016	0.098	-0.012	-10.60	0.0553	0.0565	0.905	91.903 ± 0.868	0.207 ± 0.002	49.798 ± 1.591	
QML	beta45	0.11	0.107	-0.003 ± 0.002	-0.024 ± 0.015	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	0.234 ± 0.011	0.484 ± 0.014	0.106	-0.004	-3.19	0.0475	0.0476	0.914	92.814 ± 0.822	0.190 ± 0.002	61.437 ± 1.549	
QML	beta46	0.08	0.094	0.014 ± 0.001	0.174 ± 0.013	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.193 ± 0.011	0.440 ± 0.014	0.093	0.013	16.74	0.0303	0.0331	0.873	90.283 ± 0.942	0.111 ± 0.001	88.158 ± 1.028	
QML	beta47	0.08	0.092	0.012 ± 0.001	0.154 ± 0.013	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.182 ± 0.010	0.426 ± 0.013	0.089	0.009	11.77	0.0292	0.0306	0.881	90.587 ± 0.929	0.110 ± 0.001	88.360 ± 1.020	
QML	beta51	0.16	0.169	0.009 ± 0.002	0.059 ± 0.012	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.154 ± 0.007	0.392 ± 0.011	0.169	0.009	5.72	0.0627	0.0633	0.958	93.826 ± 0.766	0.233 ± 0.001	82.085 ± 1.220	
QML	beta52	0.16	0.142	-0.018 ± 0.002	-0.116 ± 0.013	0.004 ± 0.000	0.066 ± 0.001	0.005 ± 0.000	0.068 ± 0.002	0.182 ± 0.008	0.427 ± 0.011	0.144	-0.016	-10.13	0.0670	0.0689	0.995	93.320 ± 0.794	0.257 ± 0.001	57.895 ± 1.571	
QML	beta53	0.16	0.135	-0.025 ± 0.002	-0.158 ± 0.013	0.004 ± 0.000	0.063 ± 0.001	0.005 ± 0.000	0.068 ± 0.002	0.180 ± 0.008	0.425 ± 0.011	0.135	-0.025	-15.80	0.0631	0.0680	0.977	91.194 ± 0.902	0.242 ± 0.001	60.121 ± 1.558	
QML	beta54	0.16	0.158	-0.002 ± 0.002	-0.013 ± 0.013	0.004 ± 0.000	0.067 ± 0.001	0.004 ± 0.000	0.067 ± 0.002	0.175 ± 0.007	0.418 ± 0.011	0.159	-0.001	-0.61	0.0679	0.0679	0.990	94.534 ± 0.723	0.260 ± 0.004	66.498 ± 1.502	
QML	beta55	0.08	0.083	0.003 ± 0.002	0.044 ± 0.023	0.003 ± 0.000	0.057 ± 0.002	0.003 ± 0.000	0.057 ± 0.002	0.516 ± 0.027	0.718 ± 0.022	0.083	0.003	4.01	0.0513	0.0514	0.886	91.397 ± 0.892	0.199 ± 0.004	46.660 ± 1.587	
QML	beta56	0.08	0.087	0.007 ± 0.001	0.083 ± 0.014	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.211 ± 0.011	0.459 ± 0.014	0.085	0.005	6.41	0.0326	0.0330	0.886	91.599 ± 0.883	0.126 ± 0.002	76.923 ± 1.340	
QML	beta57	0.06	0.069	0.009 ± 0.002	0.154 ± 0.027	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	0.745 ± 0.041	0.863 ± 0.027	0.067	0.007	10.85	0.0504	0.0509	0.921	93.117 ± 0.805	0.184 ± 0.004	35.526 ± 1.523	
QML	beta58	0.06	0.070	0.010 ± 0.001	0.160 ± 0.016	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.272 ± 0.016	0.521 ± 0.017	0.066	0.006	10.46	0.0255	0.0263	0.886	91.903 ± 0.868	0.103 ± 0.001	75.607 ± 1.366	
UPI	beta41	0.20	0.208	0.008 ± 0.002	0.042 ± 0.011	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.126 ± 0.006	0.355 ± 0.010	0.208	0.008	3.90	0.0691	0.0696	0.990	94.534 ± 0.723	0.273 ± 0.001	84.109 ± 1.163	
UPI	beta42	0.20	0.205	0.005 ± 0.002	0.023 ± 0.012	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	0.146 ± 0.007	0.382 ± 0.011	0.204	0.004	1.83	0.0750	0.0751	0.964	94.332 ± 0.736	0.288 ± 0.002	78.036 ± 1.317	
UPI	beta43	0.20	0.201	0.001 ± 0.002	0.007 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.058 ± 0.003	0.240 ± 0.007	0.201	0.001	0.45	0.0475	0.0476	0.970	93.016 ± 0.811	0.182 ± 0.001	99.089 ± 0.302	
UPI	beta44	0.11	0.111	0.001 ± 0.002	0.012 ± 0.017	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.274 ± 0.014	0.523 ± 0.016	0.111	0.001	0.58	0.0534	0.0534	0.915	91.802 ± 0.873	0.206 ± 0.002	58.300 ± 1.569	

Table S209

Simulation 2 - Condition 65: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.8 (continued)

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta45	0.11	0.112	0.002 ± 0.002	0.017 ± 0.016	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.259 ± 0.014	0.509 ± 0.016	0.110	0.000	-0.15	0.0513	0.0513	0.901	92.409 ± 0.843	0.198 ± 0.002	62.449 ± 1.541
UPI	beta46	0.08	0.081	0.001 ± 0.001	0.008 ± 0.013	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.167 ± 0.009	0.409 ± 0.013	0.079	-0.001	-0.71	0.0310	0.0310	0.854	90.081 ± 0.951	0.110 ± 0.001	78.340 ± 1.311
UPI	beta47	0.08	0.079	-0.001 ± 0.001	-0.017 ± 0.013	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.176 ± 0.010	0.419 ± 0.013	0.077	-0.003	-4.04	0.0299	0.0300	0.851	89.372 ± 0.980	0.112 ± 0.002	76.619 ± 1.347
UPI	beta51	0.16	0.166	0.006 ± 0.002	0.040 ± 0.012	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.145 ± 0.006	0.381 ± 0.010	0.166	0.006	3.60	0.0607	0.0610	0.946	94.231 ± 0.742	0.225 ± 0.001	82.186 ± 1.217
UPI	beta52	0.16	0.165	0.005 ± 0.003	0.033 ± 0.016	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	0.243 ± 0.011	0.493 ± 0.013	0.166	0.006	3.74	0.0787	0.0789	0.984	94.939 ± 0.697	0.304 ± 0.001	56.073 ± 1.579
UPI	beta53	0.16	0.158	-0.002 ± 0.002	-0.011 ± 0.013	0.004 ± 0.000	0.066 ± 0.001	0.004 ± 0.000	0.066 ± 0.002	0.171 ± 0.008	0.413 ± 0.011	0.159	-0.001	-0.54	0.0677	0.0677	0.950	94.028 ± 0.754	0.246 ± 0.001	71.356 ± 1.438
UPI	beta54	0.16	0.158	-0.002 ± 0.003	-0.010 ± 0.016	0.006 ± 0.000	0.081 ± 0.002	0.006 ± 0.000	0.081 ± 0.002	0.253 ± 0.011	0.503 ± 0.013	0.161	0.001	0.61	0.0813	0.0813	0.960	94.433 ± 0.729	0.303 ± 0.002	54.858 ± 1.583
UPI	beta55	0.08	0.082	0.002 ± 0.002	0.021 ± 0.022	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.464 ± 0.023	0.681 ± 0.020	0.082	0.002	2.14	0.0524	0.0524	0.857	90.992 ± 0.911	0.183 ± 0.002	47.368 ± 1.589
UPI	beta56	0.08	0.081	0.001 ± 0.001	0.008 ± 0.014	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.198 ± 0.010	0.445 ± 0.013	0.079	-0.001	-1.84	0.0318	0.0318	0.864	90.688 ± 0.925	0.121 ± 0.002	73.988 ± 1.396
UPI	beta57	0.06	0.058	-0.002 ± 0.002	-0.039 ± 0.026	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.685 ± 0.033	0.828 ± 0.024	0.057	-0.003	-4.50	0.0511	0.0512	0.905	92.308 ± 0.848	0.176 ± 0.002	28.239 ± 1.432
UPI	beta58	0.06	0.060	0.000 ± 0.001	0.007 ± 0.017	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.269 ± 0.014	0.519 ± 0.016	0.058	-0.002	-3.20	0.0270	0.0271	0.891	91.599 ± 0.883	0.109 ± 0.001	62.348 ± 1.541

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S210

Simulation 2 - Condition 66: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, $N = 1000$, Reliability = 0.8

Method	Param.	True	Mean-based										Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE	
LSAM	beta41	0.20	0.198	-0.002 ± 0.002	-0.008 ± 0.008	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.059 ± 0.003	0.244 ± 0.007	0.198	-0.002	-1.21	0.0464	0.0464	0.983	94.378 ± 0.730	0.188 ± 0.001	97.490 ± 0.496	
LSAM	beta42	0.20	0.200	-0.000 ± 0.002	-0.002 ± 0.009	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	0.073 ± 0.003	0.270 ± 0.007	0.200	0.000	-0.08	0.0547	0.0547	0.921	92.570 ± 0.831	0.195 ± 0.001	96.285 ± 0.599	
LSAM	beta43	0.20	0.202	0.002 ± 0.001	0.012 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.021 ± 0.001	0.145 ± 0.004	0.203	0.003	1.33	0.0288	0.0289	1.018	95.582 ± 0.651	0.115 ± 0.000	100.000 ± 0.000	
LSAM	beta44	0.11	0.112	0.002 ± 0.001	0.020 ± 0.009	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.090 ± 0.004	0.300 ± 0.008	0.111	0.001	1.17	0.0325	0.0325	0.949	93.775 ± 0.766	0.123 ± 0.001	93.675 ± 0.771	
LSAM	beta45	0.11	0.112	0.002 ± 0.001	0.019 ± 0.009	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.079 ± 0.004	0.282 ± 0.008	0.112	0.002	1.52	0.0321	0.0321	0.965	93.474 ± 0.783	0.117 ± 0.001	94.679 ± 0.711	
LSAM	beta46	0.08	0.079	-0.001 ± 0.001	-0.010 ± 0.007	0.000 ± 0.000	0.019 ± 0.001	0.000 ± 0.000	0.019 ± 0.001	0.056 ± 0.003	0.237 ± 0.007	0.078	-0.002	-2.15	0.0178	0.0179	0.908	91.767 ± 0.871	0.067 ± 0.001	98.092 ± 0.433	
LSAM	beta47	0.08	0.079	-0.001 ± 0.001	-0.015 ± 0.008	0.000 ± 0.000	0.019 ± 0.000	0.000 ± 0.000	0.019 ± 0.001	0.059 ± 0.003	0.243 ± 0.007	0.079	-0.001	-1.48	0.0177	0.0177	0.881	90.562 ± 0.926	0.067 ± 0.001	97.992 ± 0.444	
LSAM	beta51	0.16	0.163	0.003 ± 0.001	0.016 ± 0.007	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.053 ± 0.002	0.231 ± 0.006	0.162	0.002	0.95	0.0375	0.0375	1.000	94.478 ± 0.724	0.145 ± 0.000	99.598 ± 0.200	
LSAM	beta52	0.16	0.163	0.003 ± 0.002	0.019 ± 0.010	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.096 ± 0.004	0.310 ± 0.008	0.163	0.003	2.07	0.0516	0.0517	0.998	95.683 ± 0.644	0.194 ± 0.001	89.157 ± 0.985	
LSAM	beta53	0.16	0.161	0.001 ± 0.001	0.005 ± 0.008	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.060 ± 0.002	0.245 ± 0.006	0.161	0.001	0.93	0.0407	0.0407	1.011	95.683 ± 0.644	0.155 ± 0.000	98.695 ± 0.360	
LSAM	beta54	0.16	0.157	-0.003 ± 0.002	-0.016 ± 0.009	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.089 ± 0.004	0.298 ± 0.009	0.156	-0.004	-2.76	0.0423	0.0425	0.998	94.478 ± 0.724	0.186 ± 0.001	90.663 ± 0.922	
LSAM	beta55	0.08	0.083	0.003 ± 0.001	0.039 ± 0.011	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.131 ± 0.006	0.362 ± 0.010	0.083	0.003	3.47	0.0287	0.0288	0.966	95.281 ± 0.672	0.109 ± 0.001	81.928 ± 1.219	
LSAM	beta56	0.08	0.079	-0.001 ± 0.001	-0.008 ± 0.008	0.000 ± 0.000	0.020 ± 0.001	0.000 ± 0.000	0.020 ± 0.001	0.060 ± 0.003	0.245 ± 0.007	0.078	-0.002	-2.46	0.0175	0.0176	0.928	92.169 ± 0.851	0.071 ± 0.001	97.992 ± 0.444	
LSAM	beta57	0.06	0.056	-0.004 ± 0.001	-0.068 ± 0.015	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.227 ± 0.010	0.476 ± 0.013	0.055	-0.005	-8.00	0.0279	0.0283	0.987	94.076 ± 0.748	0.109 ± 0.001	51.305 ± 1.584	
LSAM	beta58	0.06	0.060	0.000 ± 0.001	0.003 ± 0.009	0.000 ± 0.000	0.017 ± 0.000	0.000 ± 0.000	0.017 ± 0.001	0.083 ± 0.004	0.289 ± 0.009	0.060	0.000	-0.25	0.0161	0.0161	0.918	93.273 ± 0.794	0.062 ± 0.001	93.876 ± 0.760	
QML	beta41	0.20	0.204	0.004 ± 0.001	0.020 ± 0.006	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.040 ± 0.002	0.201 ± 0.006	0.203	0.003	1.69	0.0397	0.0399	1.004	94.478 ± 0.724	0.157 ± 0.000	99.699 ± 0.174	
QML	beta42	0.20	0.203	0.003 ± 0.001	0.015 ± 0.007	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.049 ± 0.002	0.222 ± 0.006	0.204	0.004	1.76	0.0437	0.0438	0.927	92.269 ± 0.846	0.161 ± 0.000	99.498 ± 0.224	
QML	beta43	0.20	0.199	-0.001 ± 0.001	-0.006 ± 0.004	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.020 ± 0.001	0.140 ± 0.004	0.198	-0.002	-0.82	0.0271	0.0272	1.018	95.281 ± 0.672	0.112 ± 0.000	100.000 ± 0.000	
QML	beta44	0.11	0.099	-0.011 ± 0.001	-0.102 ± 0.010	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.106 ± 0.005	0.326 ± 0.009	0.099	-0.011	-9.58	0.0359	0.0374	0.952	92.169 ± 0.851	0.127 ± 0.001	84.237 ± 1.155	
QML	beta45	0.11	0.109	-0.001 ± 0.001	-0.012 ± 0.009	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.075 ± 0.003	0.275 ± 0.008	0.110	0.000	-0.24	0.0295	0.0295	0.996	94.177 ± 0.742	0.118 ± 0.001	94.378 ± 0.730	
QML	beta46	0.08	0.091	0.011 ± 0.001	0.140 ± 0.007	0.000 ± 0.000	0.019 ± 0.000	0.000 ± 0.000	0.022 ± 0.001	0.073 ± 0.004	0.270 ± 0.008	0.090	0.010	12.71	0.0175	0.0202	0.928	91.064 ± 0.904	0.067 ± 0.001	99.900 ± 0.100	
QML	beta47	0.08	0.091	0.011 ± 0.001	0.143 ± 0.007	0.000 ± 0.000	0.018 ± 0.000	0.000 ± 0.000	0.022 ± 0.001	0.073 ± 0.003	0.271 ± 0.007	0.091	0.011	13.38	0.0174	0.0204	0.914	88.353 ± 1.016	0.066 ± 0.001	99.900 ± 0.100	
QML	beta51	0.16	0.170	0.010 ± 0.001	0.065 ± 0.007	0.001 ± 0.000	0.038 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.059 ± 0.003	0.243 ± 0.007	0.170	0.010	6.30	0.0379	0.0392	0.993	94.177 ± 0.742	0.146 ± 0.001	99.699 ± 0.174	
QML	beta52	0.16	0.144	-0.016 ± 0.001	-0.098 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.074 ± 0.003	0.273 ± 0.007	0.146	-0.014	-9.06	0.0410	0.0435	0.989	92.972 ± 0.810	0.158 ± 0.000	92.871 ± 0.815	
QML	beta53	0.16	0.136	-0.024 ± 0.001	-0.147 ± 0.008	0.001 ± 0.000	0.038 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.079 ± 0.003	0.280 ± 0.007	0.136	-0.024	-14.91	0.0409	0.0474	1.009	90.361 ± 0.935	0.151 ± 0.000	94.679 ± 0.711	
QML	beta54	0.16	0.158	-0.002 ± 0.001	-0.011 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.064 ± 0.003	0.253 ± 0.008	0.158	-0.002	-1.41	0.0369	0.0370	0.998	94.679 ± 0.711	0.159 ± 0.000	97.390 ± 0.505	
QML	beta55	0.08	0.085	0.005 ± 0.001	0.066 ± 0.012	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.156 ± 0.008	0.395 ± 0.011	0.085	0.005	6.76	0.0302	0.0307	0.967	93.876 ± 0.760	0.118 ± 0.001	78.012 ± 1.312	
QML	beta56	0.08	0.085	0.005 ± 0.001	0.068 ± 0.008	0.000 ± 0.000	0.020 ± 0.001	0.000 ± 0.000	0.021 ± 0.001	0.070 ± 0.004	0.265 ± 0.008	0.084	0.004	5.44	0.0195	0.0200	0.921	93.775 ± 0.766	0.074 ± 0.001	98.394 ± 0.398	
QML	beta57	0.06	0.066	0.006 ± 0.001	0.100 ± 0.015	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.236 ± 0.012	0.486 ± 0.014	0.064	0.004	7.32	0.0280	0.0284	1.000	95.281 ± 0.672	0.112 ± 0.001	65.060 ± 1.511	
QML	beta58	0.06	0.069	0.009 ± 0.001	0.157 ± 0.009	0.000 ± 0.000	0.017 ± 0.000	0.000 ± 0.000	0.020 ± 0.001	0.106 ± 0.005	0.326 ± 0.009	0.068	0.008	14.15	0.0164	0.0185	0.909	90.863 ± 0.913	0.061 ± 0.001	98.394 ± 0.398	
UPI	beta41	0.20	0.203	0.003 ± 0.001	0.014 ± 0.007	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.048 ± 0.002	0.219 ± 0.006	0.204	0.004	1.84	0.0425	0.0427	0.964	93.273 ± 0.794	0.165 ± 0.001	98.695 ± 0.360	
UPI	beta42	0.20	0.203	0.003 ± 0.002	0.014 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.057 ± 0.003	0.238 ± 0.007	0.202	0.002	0.92	0.0476	0.0476	0.920	92.570 ± 0.831	0.172 ± 0.001	98.494 ± 0.386	
UPI	beta43	0.20	0.202	0.002 ± 0.001	0.012 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.021 ± 0.001	0.144 ± 0.004	0.203	0.003	1.29	0.0284	0.0286	1.004	95.382 ± 0.665	0.113 ± 0.000	100.000 ± 0.000	
UPI	beta44	0.11	0.110	0.000 ± 0.001	0.001 ± 0.009	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.089 ± 0.004	0.299 ± 0.008	0.111	0.001	0.48	0.0331	0.0331	0.947	94.277 ± 0.736	0.122 ± 0.001	93.072 ± 0.801	

Table S210

Simulation 2 - Condition 66: Full Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.8 (continued)

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta45	0.11	0.111	0.001 ± 0.001	0.011 ± 0.009	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.080 ± 0.004	0.282 ± 0.008	0.112	0.002	2.08	0.0324	0.0325	0.968	93.775 ± 0.766	0.118 ± 0.001	94.779 ± 0.705
UPI	beta46	0.08	0.079	-0.001 ± 0.001	-0.017 ± 0.007	0.000 ± 0.000	0.019 ± 0.001	0.000 ± 0.000	0.019 ± 0.001	0.054 ± 0.003	0.233 ± 0.007	0.078	-0.002	-2.57	0.0173	0.0174	0.895	91.265 ± 0.895	0.065 ± 0.001	98.795 ± 0.346
UPI	beta47	0.08	0.078	-0.002 ± 0.001	-0.020 ± 0.008	0.000 ± 0.000	0.019 ± 0.000	0.000 ± 0.000	0.019 ± 0.001	0.057 ± 0.003	0.238 ± 0.007	0.078	-0.002	-2.14	0.0174	0.0175	0.865	88.956 ± 0.993	0.064 ± 0.001	98.494 ± 0.386
UPI	beta51	0.16	0.164	0.004 ± 0.001	0.028 ± 0.007	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.054 ± 0.002	0.233 ± 0.006	0.164	0.004	2.22	0.0371	0.0373	0.986	94.177 ± 0.742	0.143 ± 0.001	99.699 ± 0.174
UPI	beta52	0.16	0.166	0.006 ± 0.002	0.035 ± 0.010	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.094 ± 0.004	0.307 ± 0.008	0.168	0.008	4.79	0.0478	0.0484	0.988	95.181 ± 0.679	0.189 ± 0.001	90.964 ± 0.908
UPI	beta53	0.16	0.160	0.000 ± 0.001	0.001 ± 0.008	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.059 ± 0.002	0.244 ± 0.006	0.160	0.000	0.11	0.0402	0.0402	1.004	95.683 ± 0.644	0.153 ± 0.001	98.795 ± 0.346
UPI	beta54	0.16	0.158	-0.002 ± 0.001	-0.013 ± 0.009	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.087 ± 0.004	0.295 ± 0.009	0.157	-0.003	-2.11	0.0425	0.0426	1.001	94.980 ± 0.692	0.185 ± 0.001	91.064 ± 0.904
UPI	beta55	0.08	0.082	0.002 ± 0.001	0.025 ± 0.011	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.126 ± 0.006	0.355 ± 0.010	0.082	0.002	2.36	0.0282	0.0282	0.971	95.783 ± 0.637	0.108 ± 0.001	81.827 ± 1.222
UPI	beta56	0.08	0.079	-0.001 ± 0.001	-0.009 ± 0.008	0.000 ± 0.000	0.019 ± 0.000	0.000 ± 0.000	0.019 ± 0.001	0.059 ± 0.003	0.243 ± 0.007	0.078	-0.002	-2.20	0.0177	0.0178	0.926	92.169 ± 0.851	0.071 ± 0.001	98.494 ± 0.386
UPI	beta57	0.06	0.056	-0.004 ± 0.001	-0.068 ± 0.015	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.219 ± 0.010	0.468 ± 0.013	0.054	-0.006	-10.19	0.0268	0.0275	0.983	94.578 ± 0.718	0.107 ± 0.001	54.819 ± 1.577
UPI	beta58	0.06	0.060	0.000 ± 0.001	0.003 ± 0.009	0.000 ± 0.000	0.017 ± 0.000	0.000 ± 0.000	0.017 ± 0.001	0.083 ± 0.004	0.287 ± 0.009	0.060	0.000	-0.17	0.0164	0.0164	0.923	93.273 ± 0.794	0.062 ± 0.001	94.076 ± 0.748

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S211

Simulation 2 - Condition 67: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, $N = 400$, Reliability = 0.4

Method	Param.	True	Mean-based										Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE	
LSAM	beta41	0.20	0.287	0.087 ± 0.043	0.435 ± 0.217	0.021 ± 0.009	0.144 ± 0.051	0.026 ± 0.007	0.162 ± 0.031	0.659 ± 0.166	0.812 ± 0.155	0.248	0.048	23.81	0.0960	0.1072	1.399	81.818 ± 11.629	0.789 ± 0.326	72.727 ± 13.428	
LSAM	beta42	0.20	0.192	-0.008 ± 0.038	-0.042 ± 0.192	0.016 ± 0.006	0.127 ± 0.039	0.015 ± 0.006	0.122 ± 0.038	0.371 ± 0.151	0.609 ± 0.190	0.222	0.022	10.76	0.1154	0.1174	1.583	90.909 ± 8.668	0.791 ± 0.271	18.182 ± 11.629	
LSAM	beta43	0.20	0.200	0.000 ± 0.030	0.000 ± 0.151	0.010 ± 0.002	0.100 ± 0.015	0.009 ± 0.002	0.095 ± 0.021	0.228 ± 0.047	0.477 ± 0.104	0.173	-0.027	-13.28	0.1009	0.1044	0.864	90.909 ± 8.668	0.339 ± 0.018	54.545 ± 15.013	
LSAM	beta44	0.11	0.195	0.085 ± 0.037	0.776 ± 0.338	0.015 ± 0.006	0.123 ± 0.039	0.021 ± 0.011	0.145 ± 0.051	1.747 ± 0.950	1.322 ± 0.461	0.176	0.066	59.88	0.1347	0.1500	0.997	90.909 ± 8.668	0.482 ± 0.054	27.273 ± 13.428	
LSAM	beta45	0.11	0.180	0.070 ± 0.023	0.641 ± 0.205	0.006 ± 0.001	0.075 ± 0.015	0.010 ± 0.004	0.100 ± 0.023	0.829 ± 0.324	0.911 ± 0.211	0.124	0.014	12.60	0.0975	0.0985	1.574	100.000 ± 0.000	0.461 ± 0.039	27.273 ± 13.428	
LSAM	beta46	0.08	-0.087	-0.167 ± 0.025	-2.090 ± 0.313	0.007 ± 0.002	0.083 ± 0.022	0.034 ± 0.010	0.185 ± 0.029	5.349 ± 1.545	2.313 ± 0.365	-0.030	-0.110	-136.88	0.0856	0.1390	1.623	90.909 ± 8.668	0.529 ± 0.150	0.000 ± 0.000	
LSAM	beta47	0.08	-0.049	-0.129 ± 0.026	-1.612 ± 0.319	0.007 ± 0.002	0.085 ± 0.016	0.023 ± 0.006	0.152 ± 0.021	3.617 ± 0.944	1.902 ± 0.262	-0.024	-0.104	-130.56	0.1475	0.1808	2.637	81.818 ± 11.629	0.875 ± 0.381	0.000 ± 0.000	
LSAM	beta51	0.16	0.157	-0.003 ± 0.031	-0.019 ± 0.194	0.011 ± 0.005	0.103 ± 0.032	0.010 ± 0.005	0.098 ± 0.033	0.378 ± 0.179	0.615 ± 0.208	0.206	0.046	28.91	0.0977	0.1081	1.113	100.000 ± 0.000	0.450 ± 0.035	27.273 ± 13.428	
LSAM	beta52	0.16	0.164	0.004 ± 0.032	0.023 ± 0.203	0.012 ± 0.002	0.108 ± 0.015	0.011 ± 0.002	0.103 ± 0.022	0.413 ± 0.080	0.642 ± 0.138	0.179	0.019	11.94	0.1245	0.1260	1.208	100.000 ± 0.000	0.510 ± 0.041	18.182 ± 11.629	
LSAM	beta53	0.16	0.113	-0.047 ± 0.025	-0.293 ± 0.159	0.007 ± 0.002	0.084 ± 0.017	0.009 ± 0.003	0.093 ± 0.021	0.338 ± 0.113	0.582 ± 0.132	0.121	-0.039	-24.15	0.1102	0.1168	1.295	100.000 ± 0.000	0.428 ± 0.034	27.273 ± 13.428	
LSAM	beta54	0.16	0.188	0.028 ± 0.039	0.177 ± 0.246	0.017 ± 0.005	0.130 ± 0.025	0.016 ± 0.005	0.128 ± 0.032	0.635 ± 0.208	0.797 ± 0.200	0.173	0.013	8.40	0.0896	0.0906	0.930	100.000 ± 0.000	0.475 ± 0.040	36.364 ± 14.504	
LSAM	beta55	0.08	0.078	-0.002 ± 0.053	-0.021 ± 0.660	0.031 ± 0.014	0.175 ± 0.062	0.028 ± 0.014	0.167 ± 0.060	4.352 ± 2.146	2.086 ± 0.755	0.087	0.007	8.72	0.0990	0.0993	1.018	81.818 ± 11.629	0.698 ± 0.073	9.091 ± 8.668	
LSAM	beta56	0.08	0.169	0.089 ± 0.050	1.112 ± 0.624	0.027 ± 0.012	0.166 ± 0.054	0.033 ± 0.018	0.181 ± 0.067	5.134 ± 2.865	2.266 ± 0.839	0.117	0.037	46.57	0.1076	0.1138	1.161	100.000 ± 0.000	0.754 ± 0.125	0.000 ± 0.000	
LSAM	beta57	0.06	-0.065	-0.125 ± 0.016	-2.088 ± 0.269	0.003 ± 0.001	0.054 ± 0.014	0.018 ± 0.005	0.135 ± 0.020	5.084 ± 1.357	2.255 ± 0.326	-0.046	-0.106	-175.87	0.0565	0.1197	2.029	81.818 ± 11.629	0.426 ± 0.049	0.000 ± 0.000	
LSAM	beta58	0.06	-0.053	-0.113 ± 0.039	-1.880 ± 0.653	0.017 ± 0.005	0.130 ± 0.029	0.028 ± 0.013	0.168 ± 0.046	7.801 ± 3.633	2.793 ± 0.772	-0.021	-0.081	-134.83	0.0928	0.1231	1.070	90.909 ± 8.668	0.545 ± 0.114	0.000 ± 0.000	
QML	beta41	0.20	0.220	0.020 ± 0.032	0.100 ± 0.158	0.011 ± 0.006	0.105 ± 0.040	0.010 ± 0.005	0.102 ± 0.037	0.258 ± 0.134	0.508 ± 0.183	0.221	0.021	10.26	0.0631	0.0663	0.703	81.818 ± 11.629	0.288 ± 0.018	90.909 ± 8.668	
QML	beta42	0.20	0.166	-0.034 ± 0.034	-0.170 ± 0.170	0.013 ± 0.004	0.113 ± 0.025	0.013 ± 0.004	0.113 ± 0.026	0.319 ± 0.093	0.564 ± 0.131	0.203	0.003	1.67	0.0847	0.0847	0.709	63.636 ± 14.504	0.314 ± 0.020	54.545 ± 15.013	
QML	beta43	0.20	0.217	0.017 ± 0.032	0.085 ± 0.161	0.011 ± 0.001	0.107 ± 0.012	0.011 ± 0.001	0.104 ± 0.020	0.268 ± 0.035	0.518 ± 0.100	0.168	-0.032	-16.11	0.1015	0.1065	0.705	81.818 ± 11.629	0.296 ± 0.015	63.636 ± 14.504	
QML	beta44	0.11	0.177	0.067 ± 0.032	0.605 ± 0.295	0.012 ± 0.004	0.108 ± 0.030	0.015 ± 0.008	0.122 ± 0.042	1.236 ± 0.676	1.112 ± 0.382	0.172	0.062	56.64	0.1068	0.1237	0.938	90.909 ± 8.668	0.396 ± 0.043	36.364 ± 14.504	
QML	beta45	0.11	0.139	0.029 ± 0.025	0.263 ± 0.225	0.007 ± 0.003	0.082 ± 0.024	0.007 ± 0.004	0.083 ± 0.030	0.576 ± 0.302	0.759 ± 0.274	0.100	-0.010	-8.64	0.0878	0.0883	1.112	100.000 ± 0.000	0.358 ± 0.034	27.273 ± 13.428	
QML	beta46	0.08	-0.044	-0.124 ± 0.014	-1.551 ± 0.174	0.002 ± 0.001	0.046 ± 0.014	0.017 ± 0.004	0.132 ± 0.016	2.708 ± 0.630	1.646 ± 0.206	-0.034	-0.114	-142.89	0.0738	0.1361	1.352	54.545 ± 15.013	0.244 ± 0.028	0.000 ± 0.000	
QML	beta47	0.08	-0.032	-0.112 ± 0.016	-1.400 ± 0.201	0.003 ± 0.001	0.053 ± 0.014	0.015 ± 0.003	0.123 ± 0.014	2.365 ± 0.523	1.538 ± 0.175	-0.029	-0.109	-136.82	0.0518	0.1211	1.258	63.636 ± 14.504	0.263 ± 0.026	9.091 ± 8.668	
QML	beta51	0.16	0.205	0.045 ± 0.036	0.279 ± 0.226	0.014 ± 0.008	0.120 ± 0.057	0.015 ± 0.011	0.123 ± 0.065	0.586 ± 0.422	0.766 ± 0.404	0.181	0.021	13.34	0.0775	0.0804	1.038	100.000 ± 0.000	0.487 ± 0.035	36.364 ± 14.504	
QML	beta52	0.16	0.156	-0.004 ± 0.030	-0.026 ± 0.188	0.010 ± 0.003	0.100 ± 0.025	0.009 ± 0.003	0.095 ± 0.027	0.353 ± 0.129	0.595 ± 0.167	0.153	-0.007	-4.60	0.1091	0.1094	0.950	100.000 ± 0.000	0.371 ± 0.047	36.364 ± 14.504	
QML	beta53	0.16	0.126	-0.034 ± 0.020	-0.211 ± 0.128	0.005 ± 0.001	0.068 ± 0.013	0.005 ± 0.001	0.073 ± 0.014	0.209 ± 0.052	0.457 ± 0.088	0.119	-0.041	-25.61	0.0887	0.0977	1.296	100.000 ± 0.000	0.345 ± 0.023	27.273 ± 13.428	
QML	beta54	0.16	0.170	0.010 ± 0.027	0.064 ± 0.167	0.008 ± 0.003	0.089 ± 0.023	0.007 ± 0.002	0.085 ± 0.023	0.284 ± 0.097	0.533 ± 0.143	0.172	0.012	7.35	0.0681	0.0692	0.924	100.000 ± 0.000	0.322 ± 0.009	72.727 ± 13.428	
QML	beta55	0.08	0.080	0.000 ± 0.052	0.002 ± 0.646	0.029 ± 0.017	0.171 ± 0.085	0.027 ± 0.017	0.164 ± 0.078	4.177 ± 2.671	2.044 ± 0.974	0.102	0.022	27.64	0.1286	0.1304	0.919	100.000 ± 0.000	0.617 ± 0.083	9.091 ± 8.668	
QML	beta56	0.08	0.085	0.005 ± 0.036	0.067 ± 0.453	0.014 ± 0.007	0.120 ± 0.040	0.013 ± 0.007	0.115 ± 0.041	2.057 ± 1.033	1.434 ± 0.507	0.103	0.023	28.67	0.0713	0.0749	1.059	90.909 ± 8.668	0.499 ± 0.040	9.091 ± 8.668	
QML	beta57	0.06	-0.043	-0.103 ± 0.012	-1.710 ± 0.193	0.001 ± 0.000	0.038 ± 0.009	0.012 ± 0.003	0.109 ± 0.013	3.296 ± 0.749	1.816 ± 0.219	-0.028	-0.088	-146.25	0.0446	0.0984	1.956	54.545 ± 15.013	0.294 ± 0.041	0.000 ± 0.000	
QML	beta58	0.06	0.004	-0.056 ± 0.019	-0.935 ± 0.316	0.004 ± 0.001	0.063 ± 0.011	0.007 ± 0.002	0.082 ± 0.016	1.872 ± 0.634	1.368 ± 0.274	-0.005	-0.065	-108.66	0.0813	0.1042	1.104	72.727 ± 13.428	0.272 ± 0.027	0.000 ± 0.000	
UPI	beta41	0.20	0.237	0.037 ± 0.119	0.185 ± 0.593	0.155 ± 0.073	0.394 ± 0.145	0.142 ± 0.079	0.377 ± 0.151	3.555 ± 1.973	1.885 ± 0.753	0.257	0.057	28.70	0.6889	0.6913	1.382	90.909 ± 8.668	2.132 ± 0.343	9.091 ± 8.668	
UPI	beta42	0.20	0.254	0.054 ± 0.092	0.268 ± 0.461	0.094 ± 0.022	0.306 ± 0.054	0.088 ± 0.027	0.297 ± 0.073	2.201 ± 0.668	1.484 ± 0.366	0.207	0.007	3.29	0.4840	0.4840	1.357	90.909 ± 8.668	1.628 ± 0.378	0.000 ± 0.000	
UPI	beta43	0.20	0.146	-0.054 ± 0.028	-0.270 ± 0.142	0.009 ± 0.002	0.094 ± 0.015	0.011 ± 0.003	0.105 ± 0.020	0.276 ± 0.073	0.525 ± 0.100	0.166	-0.034	-17.18	0.1913	0.1943	1.926	90.909 ± 8.668	0.713 ± 0.117	9.091 ± 8.668	
UPI	beta44	0.11	0.049	-0.061 ± 0.063	-0.556 ± 0.574	0.044 ± 0.025	0.210 ± 0.099	0.044 ± 0.031	0.209 ± 0.111	3.609 ± 2.587	1.900 ± 1.007	0.136	0.026	-47.08	0.2171	0.2187	1.916	100.000 ± 0.000	1.328 ± 0.379	0.000 ± 0.000	
UPI	beta45	0.11	0.188	0.078 ± 0.062	0.712 ± 0.567	0.043 ± 0.024	0.207 ± 0.100	0.045 ± 0.033	0.212 ± 0.114	3.716 ± 2.716	1.928 ± 1.035	0.114	0.004	3.99	0.1528	0.1528	1.646	100.000 ± 0.000	1.333 ± 0.348	0.000 ± 0.000	
UPI	beta46	0.08	0.065	-0.015 ± 0.181	-0.183 ± 2.263	0.361 ± 0.117	0.600 ± 0.141	0.328 ± 0.115	0.573 ± 0.157	51.245 ± 18.025	7.159 ± 1.966	-0.096	-0.176	-219.48	1.1847	1.1976	1.697	100.000 ± 0.000	3.995 ± 1.192	0.000 ± 0.000	

Table S211

Simulation 2 - Condition 67: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.4
(continued)

Method	Param.	True	Mean-based										Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE	
UPI	beta47	0.08	-0.204	-0.284 ± 0.111	-3.550 ± 1.391	0.136 ± 0.045	0.369 ± 0.090	0.205 ± 0.093	0.452 ± 0.127	31.954 ± 14.562	5.653 ± 1.587	-0.032	-0.112	-140.44	0.4317	0.4460	1.462	90.909 ± 8.668	2.116 ± 0.683	0.000 ± 0.000	
UPI	beta51	0.16	0.224	0.064 ± 0.046	0.400 ± 0.288	0.023 ± 0.007	0.153 ± 0.035	0.025 ± 0.011	0.159 ± 0.045	0.992 ± 0.410	0.996 ± 0.278	0.213	0.053	33.01	0.1766	0.1844	1.315	100.000 ± 0.000	0.789 ± 0.083	18.182 ± 11.629	
UPI	beta52	0.16	0.312	0.152 ± 0.074	0.952 ± 0.463	0.060 ± 0.015	0.245 ± 0.046	0.078 ± 0.033	0.279 ± 0.074	3.046 ± 1.297	1.745 ± 0.464	0.250	0.090	56.26	0.3783	0.3889	1.516	81.818 ± 11.629	1.459 ± 0.268	9.091 ± 8.668	
UPI	beta53	0.16	0.058	-0.102 ± 0.032	-0.635 ± 0.201	0.011 ± 0.005	0.107 ± 0.034	0.021 ± 0.006	0.144 ± 0.028	0.807 ± 0.253	0.898 ± 0.172	0.049	-0.111	-69.23	0.1063	0.1535	1.124	63.636 ± 14.504	0.470 ± 0.122	0.000 ± 0.000	
UPI	beta54	0.16	0.176	0.016 ± 0.053	0.101 ± 0.333	0.031 ± 0.011	0.177 ± 0.048	0.029 ± 0.013	0.169 ± 0.055	1.117 ± 0.490	1.057 ± 0.341	0.110	-0.050	-31.38	0.1511	0.1592	1.849	90.909 ± 8.668	1.279 ± 0.383	0.000 ± 0.000	
UPI	beta55	0.08	0.175	0.095 ± 0.123	1.186 ± 1.538	0.167 ± 0.064	0.408 ± 0.114	0.160 ± 0.074	0.401 ± 0.131	25.069 ± 11.637	5.007 ± 1.641	0.061	-0.019	-24.03	0.3056	0.3062	1.384	100.000 ± 0.000	2.214 ± 0.487	0.000 ± 0.000	
UPI	beta56	0.08	0.073	-0.007 ± 0.088	-0.087 ± 1.100	0.085 ± 0.035	0.292 ± 0.087	0.078 ± 0.035	0.278 ± 0.090	12.111 ± 5.469	3.480 ± 1.129	0.016	-0.064	-79.69	0.1728	0.1842	0.993	100.000 ± 0.000	1.137 ± 0.309	0.000 ± 0.000	
UPI	beta57	0.06	-0.291	-0.351 ± 0.072	-5.847 ± 1.206	0.058 ± 0.029	0.240 ± 0.095	0.175 ± 0.073	0.419 ± 0.103	48.726 ± 20.156	6.980 ± 1.719	-0.176	-0.236	-393.09	0.3200	0.3975	2.507	90.909 ± 8.668	2.359 ± 0.661	0.000 ± 0.000	
UPI	beta58	0.06	-0.003	-0.063 ± 0.041	-1.049 ± 0.681	0.018 ± 0.004	0.136 ± 0.024	0.021 ± 0.006	0.144 ± 0.032	5.745 ± 1.772	2.397 ± 0.530	-0.033	-0.093	-154.29	0.1765	0.1993	1.960	100.000 ± 0.000	1.042 ± 0.232	0.000 ± 0.000	

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S212

Simulation 2 - Condition 68: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.4

Method	Param.	True	Mean-based										Median-based					Inference			
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE	
LSAM	beta41	0.20	0.184	-0.016 $\pm$ 0.031	-0.078 $\pm$ 0.154	0.010 $\pm$ 0.003	0.102 $\pm$ 0.023	0.010 $\pm$ 0.003	0.099 $\pm$ 0.024	0.244 $\pm$ 0.069	0.494 $\pm$ 0.121	0.207	0.007	3.43	0.0784	0.0787	0.790	81.818 $\pm$ 11.629	0.317 $\pm$ 0.025	54.545 $\pm$ 15.013	
LSAM	beta42	0.20	0.229	0.029 $\pm$ 0.028	0.143 $\pm$ 0.142	0.009 $\pm$ 0.002	0.094 $\pm$ 0.016	0.009 $\pm$ 0.002	0.094 $\pm$ 0.020	0.222 $\pm$ 0.051	0.471 $\pm$ 0.098	0.213	0.013	6.28	0.0926	0.0934	0.858	100.000 $\pm$ 0.000	0.317 $\pm$ 0.019	81.818 $\pm$ 11.629	
LSAM	beta43	0.20	0.162	-0.038 $\pm$ 0.014	-0.190 $\pm$ 0.070	0.002 $\pm$ 0.001	0.046 $\pm$ 0.010	0.003 $\pm$ 0.001	0.058 $\pm$ 0.010	0.085 $\pm$ 0.024	0.291 $\pm$ 0.050	0.173	-0.027	-13.58	0.0606	0.0664	1.134	100.000 $\pm$ 0.000	0.205 $\pm$ 0.007	90.909 $\pm$ 8.668	
LSAM	beta44	0.11	0.133	0.023 $\pm$ 0.021	0.205 $\pm$ 0.192	0.005 $\pm$ 0.001	0.070 $\pm$ 0.012	0.005 $\pm$ 0.002	0.071 $\pm$ 0.017	0.411 $\pm$ 0.137	0.641 $\pm$ 0.158	0.149	0.039	35.41	0.0698	0.0799	1.072	100.000 $\pm$ 0.000	0.294 $\pm$ 0.023	45.455 $\pm$ 15.013	
LSAM	beta45	0.11	0.123	0.013 $\pm$ 0.019	0.115 $\pm$ 0.169	0.004 $\pm$ 0.001	0.062 $\pm$ 0.015	0.004 $\pm$ 0.001	0.060 $\pm$ 0.015	0.300 $\pm$ 0.092	0.547 $\pm$ 0.136	0.135	0.025	23.00	0.0580	0.0633	0.997	90.909 $\pm$ 8.668	0.241 $\pm$ 0.009	45.455 $\pm$ 15.013	
LSAM	beta46	0.08	0.013	-0.067 $\pm$ 0.031	-0.831 $\pm$ 0.382	0.010 $\pm$ 0.002	0.101 $\pm$ 0.016	0.014 $\pm$ 0.005	0.117 $\pm$ 0.028	2.147 $\pm$ 0.806	1.465 $\pm$ 0.346	-0.027	-0.107	-133.20	0.0744	0.1300	0.937	72.727 $\pm$ 13.428	0.372 $\pm$ 0.053	0.000 $\pm$ 0.000	
LSAM	beta47	0.08	-0.021	-0.101 $\pm$ 0.021	-1.267 $\pm$ 0.258	0.005 $\pm$ 0.001	0.068 $\pm$ 0.014	0.015 $\pm$ 0.004	0.121 $\pm$ 0.015	2.269 $\pm$ 0.551	1.506 $\pm$ 0.192	-0.037	-0.117	-146.79	0.0771	0.1405	1.103	72.727 $\pm$ 13.428	0.295 $\pm$ 0.047	9.091 $\pm$ 8.668	
LSAM	beta51	0.16	0.167	0.007 $\pm$ 0.013	0.041 $\pm$ 0.082	0.002 $\pm$ 0.001	0.044 $\pm$ 0.012	0.002 $\pm$ 0.001	0.042 $\pm$ 0.014	0.069 $\pm$ 0.031	0.263 $\pm$ 0.086	0.165	0.005	3.15	0.0395	0.0398	1.456	100.000 $\pm$ 0.000	0.249 $\pm$ 0.006	81.818 $\pm$ 11.629	
LSAM	beta52	0.16	0.184	0.024 $\pm$ 0.025	0.148 $\pm$ 0.155	0.007 $\pm$ 0.002	0.082 $\pm$ 0.015	0.007 $\pm$ 0.002	0.082 $\pm$ 0.019	0.261 $\pm$ 0.080	0.511 $\pm$ 0.121	0.165	0.005	2.82	0.0648	0.0649	0.846	100.000 $\pm$ 0.000	0.272 $\pm$ 0.014	63.636 $\pm$ 14.504	
LSAM	beta53	0.16	0.144	-0.016 $\pm$ 0.022	-0.102 $\pm$ 0.136	0.005 $\pm$ 0.002	0.072 $\pm$ 0.021	0.005 $\pm$ 0.002	0.071 $\pm$ 0.019	0.195 $\pm$ 0.068	0.442 $\pm$ 0.119	0.147	-0.013	-8.33	0.0431	0.0451	0.804	81.818 $\pm$ 11.629	0.227 $\pm$ 0.007	63.636 $\pm$ 14.504	
LSAM	beta54	0.16	0.171	0.011 $\pm$ 0.012	0.067 $\pm$ 0.078	0.002 $\pm$ 0.000	0.041 $\pm$ 0.008	0.002 $\pm$ 0.000	0.041 $\pm$ 0.008	0.065 $\pm$ 0.014	0.255 $\pm$ 0.053	0.157	-0.003	-2.03	0.0465	0.0466	1.584	100.000 $\pm$ 0.000	0.256 $\pm$ 0.011	72.727 $\pm$ 13.428	
LSAM	beta55	0.08	0.099	0.019 $\pm$ 0.030	0.238 $\pm$ 0.377	0.010 $\pm$ 0.003	0.100 $\pm$ 0.021	0.009 $\pm$ 0.003	0.097 $\pm$ 0.023	1.477 $\pm$ 0.400	1.215 $\pm$ 0.284	0.119	0.039	49.03	0.0799	0.0890	0.779	90.909 $\pm$ 8.668	0.305 $\pm$ 0.022	45.455 $\pm$ 15.013	
LSAM	beta56	0.08	0.076	-0.004 $\pm$ 0.023	-0.049 $\pm$ 0.292	0.006 $\pm$ 0.001	0.078 $\pm$ 0.012	0.005 $\pm$ 0.001	0.074 $\pm$ 0.016	0.857 $\pm$ 0.182	0.926 $\pm$ 0.203	0.077	-0.003	-4.12	0.0793	0.0794	0.947	90.909 $\pm$ 8.668	0.288 $\pm$ 0.024	9.091 $\pm$ 8.668	
LSAM	beta57	0.06	-0.047	-0.107 $\pm$ 0.025	-1.791 $\pm$ 0.422	0.007 $\pm$ 0.002	0.084 $\pm$ 0.019	0.018 $\pm$ 0.007	0.134 $\pm$ 0.029	4.992 $\pm$ 1.889	2.234 $\pm$ 0.480	-0.030	-0.090	-149.34	0.0440	0.0998	0.877	72.727 $\pm$ 13.428	0.289 $\pm$ 0.024	18.182 $\pm$ 11.629	
LSAM	beta58	0.06	0.005	-0.055 $\pm$ 0.009	-0.922 $\pm$ 0.146	0.001 $\pm$ 0.000	0.029 $\pm$ 0.008	0.004 $\pm$ 0.001	0.062 $\pm$ 0.010	1.066 $\pm$ 0.306	1.032 $\pm$ 0.161	0.005	-0.055	-91.95	0.0452	0.0714	2.057	81.818 $\pm$ 11.629	0.235 $\pm$ 0.014	0.000 $\pm$ 0.000	
QML	beta41	0.20	0.182	-0.018 $\pm$ 0.016	-0.088 $\pm$ 0.079	0.003 $\pm$ 0.001	0.052 $\pm$ 0.008	0.003 $\pm$ 0.001	0.053 $\pm$ 0.013	0.070 $\pm$ 0.024	0.264 $\pm$ 0.065	0.201	0.001	0.67	0.0409	0.0409	0.907	81.818 $\pm$ 11.629	0.185 $\pm$ 0.006	100.000 $\pm$ 0.000	
QML	beta42	0.20	0.212	0.012 $\pm$ 0.018	0.058 $\pm$ 0.092	0.004 $\pm$ 0.001	0.061 $\pm$ 0.011	0.004 $\pm$ 0.001	0.060 $\pm$ 0.015	0.089 $\pm$ 0.027	0.298 $\pm$ 0.074	0.193	-0.007	-3.72	0.0394	0.0401	0.816	90.909 $\pm$ 8.668	0.196 $\pm$ 0.005	100.000 $\pm$ 0.000	
QML	beta43	0.20	0.168	-0.032 $\pm$ 0.011	-0.160 $\pm$ 0.057	0.001 $\pm$ 0.000	0.038 $\pm$ 0.008	0.002 $\pm$ 0.001	0.048 $\pm$ 0.008	0.059 $\pm$ 0.015	0.242 $\pm$ 0.038	0.177	-0.023	-11.70	0.0491	0.0544	1.185	100.000 $\pm$ 0.000	0.177 $\pm$ 0.004	100.000 $\pm$ 0.000	
QML	beta44	0.11	0.122	0.012 $\pm$ 0.021	0.113 $\pm$ 0.195	0.005 $\pm$ 0.002	0.071 $\pm$ 0.019	0.005 $\pm$ 0.002	0.069 $\pm$ 0.022	0.391 $\pm$ 0.176	0.625 $\pm$ 0.203	0.158	0.048	43.26	0.0707	0.0852	0.999	90.909 $\pm$ 8.668	0.278 $\pm$ 0.017	45.455 $\pm$ 15.013	
QML	beta45	0.11	0.118	0.008 $\pm$ 0.022	0.075 $\pm$ 0.199	0.005 $\pm$ 0.002	0.073 $\pm$ 0.016	0.005 $\pm$ 0.002	0.070 $\pm$ 0.018	0.402 $\pm$ 0.136	0.634 $\pm$ 0.167	0.113	0.003	2.75	0.0484	0.0485	0.811	81.818 $\pm$ 11.629	0.231 $\pm$ 0.013	54.545 $\pm$ 15.013	
QML	beta46	0.08	0.018	-0.062 $\pm$ 0.016	-0.780 $\pm$ 0.205	0.003 $\pm$ 0.001	0.054 $\pm$ 0.009	0.007 $\pm$ 0.002	0.081 $\pm$ 0.013	1.030 $\pm$ 0.297	1.015 $\pm$ 0.163	-0.022	-0.102	-127.45	0.0480	0.1127	0.955	54.545 $\pm$ 15.013	0.204 $\pm$ 0.025	9.091 $\pm$ 8.668	
QML	beta47	0.08	-0.001	-0.081 $\pm$ 0.011	-1.013 $\pm$ 0.137	0.001 $\pm$ 0.000	0.036 $\pm$ 0.008	0.008 $\pm$ 0.002	0.088 $\pm$ 0.010	1.212 $\pm$ 0.269	1.101 $\pm$ 0.126	-0.014	-0.094	-118.03	0.0407	0.1028	1.156	45.455 $\pm$ 15.013	0.164 $\pm$ 0.011	0.000 $\pm$ 0.000	
QML	beta51	0.16	0.175	0.015 $\pm$ 0.013	0.096 $\pm$ 0.083	0.002 $\pm$ 0.001	0.044 $\pm$ 0.010	0.002 $\pm$ 0.001	0.045 $\pm$ 0.011	0.078 $\pm$ 0.026	0.279 $\pm$ 0.069	0.189	0.029	18.12	0.0485	0.0565	1.462	100.000 $\pm$ 0.000	0.252 $\pm$ 0.008	90.909 $\pm$ 8.668	
QML	beta52	0.16	0.166	0.006 $\pm$ 0.017	0.036 $\pm$ 0.104	0.003 $\pm$ 0.001	0.055 $\pm$ 0.012	0.003 $\pm$ 0.001	0.053 $\pm$ 0.013	0.109 $\pm$ 0.031	0.330 $\pm$ 0.082	0.150	-0.010	-6.50	0.0663	0.0671	0.916	90.909 $\pm$ 8.668	0.198 $\pm$ 0.008	81.818 $\pm$ 11.629	
QML	beta53	0.16	0.151	-0.009 $\pm$ 0.020	-0.054 $\pm$ 0.122	0.004 $\pm$ 0.001	0.065 $\pm$ 0.014	0.004 $\pm$ 0.001	0.063 $\pm$ 0.015	0.153 $\pm$ 0.041	0.391 $\pm$ 0.094	0.147	-0.013	-7.96	0.0539	0.0554	0.810	90.909 $\pm$ 8.668	0.206 $\pm$ 0.008	72.727 $\pm$ 13.428	
QML	beta54	0.16	0.175	0.015 $\pm$ 0.012	0.096 $\pm$ 0.075	0.002 $\pm$ 0.000	0.040 $\pm$ 0.007	0.002 $\pm$ 0.001	0.041 $\pm$ 0.009	0.065 $\pm$ 0.020	0.254 $\pm$ 0.058	0.160	0.000	0.02	0.0415	0.0415	1.244	100.000 $\pm$ 0.000	0.193 $\pm$ 0.005	100.000 $\pm$ 0.000	
QML	beta55	0.08	0.084	0.004 $\pm$ 0.031	0.056 $\pm$ 0.385	0.010 $\pm$ 0.004	0.102 $\pm$ 0.026	0.010 $\pm$ 0.003	0.097 $\pm$ 0.027	1.485 $\pm$ 0.524	1.218 $\pm$ 0.343	0.119	0.039	48.98	0.0790	0.0882	0.752	90.909 $\pm$ 8.668	0.301 $\pm$ 0.014	27.273 $\pm$ 13.428	
QML	beta56	0.08	0.056	-0.024 $\pm$ 0.021	-0.295 $\pm$ 0.265	0.005 $\pm$ 0.001	0.070 $\pm$ 0.009	0.005 $\pm$ 0.001	0.071 $\pm$ 0.016	0.791 $\pm$ 0.220	0.890 $\pm$ 0.195	0.069	-0.011	-13.20	0.0700	0.0708	0.937	81.818 $\pm$ 11.629	0.259 $\pm$ 0.015	0.000 $\pm$ 0.000	
QML	beta57	0.06	-0.024	-0.084 $\pm$ 0.017	-1.407 $\pm$ 0.281	0.003 $\pm$ 0.001	0.056 $\pm$ 0.010	0.010 $\pm$ 0.003	0.100 $\pm$ 0.014	2.772 $\pm$ 0.720	1.665 $\pm$ 0.228	-0.028	-0.088	-146.02	0.0387	0.0958	0.779	36.364 $\pm$ 14.504	0.171 $\pm$ 0.016	27.273 $\pm$ 13.428	
QML	beta58	0.06	0.018	-0.042 $\pm$ 0.006	-0.703 $\pm$ 0.106	0.000 $\pm$ 0															

Table S212

Simulation 2 - Condition 68: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.4
(continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	-0.338	-0.418 ± 0.144	-5.221 ± 1.800	0.228 ± 0.059	0.478 ± 0.093	0.382 ± 0.173	0.618 ± 0.165	59.667 ± 27.032	7.724 ± 2.059	-0.071	-0.151	-188.85	0.3483	0.3797	1.770	100.000 ± 0.000	3.314 ± 1.557	9.091 ± 8.668
UPI	beta51	0.16	0.186	0.026 ± 0.033	0.162 ± 0.207	0.012 ± 0.005	0.110 ± 0.034	0.012 ± 0.004	0.108 ± 0.030	0.454 ± 0.167	0.674 ± 0.187	0.205	0.045	27.90	0.2059	0.2107	0.953	81.818 ± 11.629	0.410 ± 0.065	63.636 ± 14.504
UPI	beta52	0.16	0.402	0.242 ± 0.109	1.510 ± 0.679	0.130 ± 0.055	0.360 ± 0.117	0.176 ± 0.099	0.420 ± 0.152	6.889 ± 3.849	2.625 ± 0.953	0.377	0.217	135.63	0.5280	0.5709	0.857	100.000 ± 0.000	1.210 ± 0.283	0.000 ± 0.000
UPI	beta53	0.16	0.083	-0.077 ± 0.023	-0.480 ± 0.146	0.006 ± 0.001	0.078 ± 0.008	0.011 ± 0.003	0.107 ± 0.017	0.445 ± 0.124	0.667 ± 0.106	0.023	-0.137	-85.84	0.1357	0.1931	1.153	81.818 ± 11.629	0.351 ± 0.068	27.273 ± 13.428
UPI	beta54	0.16	0.044	-0.116 ± 0.033	-0.725 ± 0.209	0.012 ± 0.005	0.111 ± 0.033	0.025 ± 0.009	0.157 ± 0.033	0.961 ± 0.339	0.980 ± 0.204	0.024	-0.136	-84.74	0.1510	0.2029	0.997	90.909 ± 8.668	0.432 ± 0.082	9.091 ± 8.668
UPI	beta55	0.08	0.130	0.050 ± 0.029	0.623 ± 0.358	0.009 ± 0.005	0.095 ± 0.044	0.011 ± 0.007	0.103 ± 0.051	1.667 ± 1.154	1.291 ± 0.638	0.121	0.041	51.23	0.1284	0.1347	1.184	100.000 ± 0.000	0.440 ± 0.068	9.091 ± 8.668
UPI	beta56	0.08	0.050	-0.030 ± 0.032	-0.373 ± 0.405	0.012 ± 0.004	0.107 ± 0.029	0.011 ± 0.005	0.107 ± 0.034	1.781 ± 0.794	1.334 ± 0.421	0.064	-0.016	-20.25	0.1662	0.1670	1.207	90.909 ± 8.668	0.509 ± 0.072	9.091 ± 8.668
UPI	beta57	0.06	-0.334	-0.394 ± 0.154	-6.564 ± 2.568	0.261 ± 0.131	0.511 ± 0.207	0.393 ± 0.235	0.627 ± 0.248	109.035 ± 65.324	10.442 ± 4.139	-0.407	-0.467	-778.18	0.7094	0.8493	1.119	100.000 ± 0.000	2.241 ± 0.817	0.000 ± 0.000
UPI	beta58	0.06	0.069	0.009 ± 0.029	0.155 ± 0.484	0.009 ± 0.005	0.096 ± 0.036	0.009 ± 0.004	0.092 ± 0.034	2.369 ± 1.243	1.539 ± 0.565	0.059	-0.001	-1.40	0.1152	0.1152	1.439	100.000 ± 0.000	0.544 ± 0.149	0.000 ± 0.000

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S213

Simulation 2 - Condition 69: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.6

Method	Param.	True	Mean-based										Median-based					Inference		
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LSAM	beta41	0.20	0.202	0.002 $\pm$ 0.002	0.008 $\pm$ 0.012	0.003 $\pm$ 0.000	0.057 $\pm$ 0.002	0.003 $\pm$ 0.000	0.057 $\pm$ 0.002	0.081 $\pm$ 0.005	0.284 $\pm$ 0.010	0.196	-0.004	-1.84	0.0579	0.0580	1.077	97.043 $\pm$ 0.728	0.240 $\pm$ 0.001	89.834 $\pm$ 1.299
LSAM	beta42	0.20	0.196	-0.004 $\pm$ 0.003	-0.019 $\pm$ 0.013	0.004 $\pm$ 0.000	0.062 $\pm$ 0.002	0.004 $\pm$ 0.000	0.062 $\pm$ 0.002	0.096 $\pm$ 0.006	0.309 $\pm$ 0.012	0.200	0.000	-0.13	0.0613	0.0613	1.027	96.118 $\pm$ 0.830	0.249 $\pm$ 0.002	86.876 $\pm$ 1.452
LSAM	beta43	0.20	0.200	-0.000 $\pm$ 0.003	-0.002 $\pm$ 0.013	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	0.090 $\pm$ 0.005	0.299 $\pm$ 0.011	0.200	0.000	0.09	0.0616	0.0616	0.944	94.270 $\pm$ 0.999	0.222 $\pm$ 0.001	94.455 $\pm$ 0.984
LSAM	beta44	0.11	0.134	0.024 $\pm$ 0.003	0.216 $\pm$ 0.028	0.005 $\pm$ 0.000	0.072 $\pm$ 0.002	0.006 $\pm$ 0.000	0.076 $\pm$ 0.003	0.474 $\pm$ 0.031	0.689 $\pm$ 0.026	0.135	0.025	22.45	0.0722	0.0763	1.005	94.085 $\pm$ 1.014	0.284 $\pm$ 0.003	50.092 $\pm$ 2.150
LSAM	beta45	0.11	0.118	0.008 $\pm$ 0.003	0.076 $\pm$ 0.026	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	0.004 $\pm$ 0.000	0.067 $\pm$ 0.003	0.367 $\pm$ 0.023	0.606 $\pm$ 0.023	0.118	0.008	7.57	0.0644	0.0650	0.950	95.379 $\pm$ 0.903	0.246 $\pm$ 0.002	51.017 $\pm$ 2.149
LSAM	beta46	0.08	0.040	-0.040 $\pm$ 0.004	-0.494 $\pm$ 0.046	0.007 $\pm$ 0.000	0.086 $\pm$ 0.003	0.009 $\pm$ 0.001	0.095 $\pm$ 0.003	1.401 $\pm$ 0.087	1.184 $\pm$ 0.042	0.041	-0.039	-48.67	0.0861	0.0945	1.014	88.540 $\pm$ 1.370	0.342 $\pm$ 0.004	4.806 $\pm$ 0.920
LSAM	beta47	0.08	0.050	-0.030 $\pm$ 0.004	-0.374 $\pm$ 0.050	0.009 $\pm$ 0.001	0.093 $\pm$ 0.004	0.009 $\pm$ 0.001	0.097 $\pm$ 0.004	1.476 $\pm$ 0.098	1.215 $\pm$ 0.047	0.043	-0.037	-46.16	0.0810	0.0890	0.933	89.834 $\pm$ 1.299	0.339 $\pm$ 0.004	5.730 $\pm$ 0.999
LSAM	beta51	0.16	0.168	0.008 $\pm$ 0.004	0.051 $\pm$ 0.022	0.007 $\pm$ 0.000	0.083 $\pm$ 0.003	0.007 $\pm$ 0.000	0.083 $\pm$ 0.003	0.271 $\pm$ 0.019	0.520 $\pm$ 0.021	0.166	0.006	3.94	0.0782	0.0785	0.866	91.128 $\pm$ 1.222	0.282 $\pm$ 0.002	64.695 $\pm$ 2.055
LSAM	beta52	0.16	0.154	-0.006 $\pm$ 0.002	-0.036 $\pm$ 0.015	0.003 $\pm$ 0.000	0.057 $\pm$ 0.002	0.003 $\pm$ 0.000	0.057 $\pm$ 0.002	0.127 $\pm$ 0.007	0.356 $\pm$ 0.012	0.155	-0.005	-3.32	0.0609	0.0611	1.082	97.412 $\pm$ 0.683	0.240 $\pm$ 0.001	71.534 $\pm$ 1.940
LSAM	beta53	0.16	0.154	-0.006 $\pm$ 0.003	-0.038 $\pm$ 0.017	0.004 $\pm$ 0.000	0.063 $\pm$ 0.002	0.004 $\pm$ 0.000	0.064 $\pm$ 0.003	0.158 $\pm$ 0.010	0.397 $\pm$ 0.016	0.153	-0.007	-4.15	0.0606	0.0610	0.992	93.900 $\pm$ 1.029	0.246 $\pm$ 0.001	69.501 $\pm$ 1.979
LSAM	beta54	0.16	0.155	-0.005 $\pm$ 0.003	-0.033 $\pm$ 0.016	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.146 $\pm$ 0.009	0.383 $\pm$ 0.014	0.153	-0.007	-4.24	0.0568	0.0572	0.997	95.564 $\pm$ 0.885	0.239 $\pm$ 0.001	72.828 $\pm$ 1.913
LSAM	beta55	0.08	0.093	0.013 $\pm$ 0.004	0.157 $\pm$ 0.051	0.009 $\pm$ 0.001	0.095 $\pm$ 0.003	0.009 $\pm$ 0.001	0.096 $\pm$ 0.004	1.443 $\pm$ 0.102	1.201 $\pm$ 0.050	0.093	0.013	16.45	0.0877	0.0886	0.901	93.900 $\pm$ 1.029	0.337 $\pm$ 0.004	23.105 $\pm$ 1.812
LSAM	beta56	0.08	0.086	0.006 $\pm$ 0.004	0.075 $\pm$ 0.045	0.007 $\pm$ 0.000	0.083 $\pm$ 0.003	0.007 $\pm$ 0.000	0.083 $\pm$ 0.003	1.081 $\pm$ 0.073	1.040 $\pm$ 0.042	0.084	0.004	5.31	0.0830	0.0831	0.960	94.640 $\pm$ 0.968	0.312 $\pm$ 0.003	19.039 $\pm$ 1.688
LSAM	beta57	0.06	0.036	-0.024 $\pm$ 0.004	-0.396 $\pm$ 0.065	0.008 $\pm$ 0.001	0.090 $\pm$ 0.003	0.009 $\pm$ 0.001	0.093 $\pm$ 0.003	2.405 $\pm$ 0.150	1.551 $\pm$ 0.058	0.028	-0.032	-53.42	0.0832	0.0892	0.953	90.203 $\pm$ 1.278	0.336 $\pm$ 0.005	5.360 $\pm$ 0.968
LSAM	beta58	0.06	0.033	-0.027 $\pm$ 0.003	-0.451 $\pm$ 0.056	0.006 $\pm$ 0.000	0.078 $\pm$ 0.003	0.007 $\pm$ 0.000	0.082 $\pm$ 0.003	1.871 $\pm$ 0.126	1.368 $\pm$ 0.053	0.032	-0.028	-46.21	0.0744	0.0794	0.993	92.237 $\pm$ 1.150	0.302 $\pm$ 0.004	4.991 $\pm$ 0.936
QML	beta41	0.20	0.196	-0.004 $\pm$ 0.002	-0.018 $\pm$ 0.011	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.003 $\pm$ 0.000	0.051 $\pm$ 0.002	0.065 $\pm$ 0.004	0.254 $\pm$ 0.009	0.193	-0.007	-3.56	0.0553	0.0557	1.100	96.488 $\pm$ 0.791	0.219 $\pm$ 0.001	94.455 $\pm$ 0.984
QML	beta42	0.20	0.195	-0.005 $\pm$ 0.002	-0.026 $\pm$ 0.012	0.003 $\pm$ 0.000	0.056 $\pm$ 0.002	0.003 $\pm$ 0.000	0.056 $\pm$ 0.002	0.079 $\pm$ 0.005	0.281 $\pm$ 0.011	0.197	-0.003	-1.38	0.0561	0.0562	1.025	95.194 $\pm$ 0.920	0.225 $\pm$ 0.001	92.052 $\pm$ 1.163
QML	beta43	0.20	0.193	-0.007 $\pm$ 0.002	-0.037 $\pm$ 0.012	0.003 $\pm$ 0.000	0.057 $\pm$ 0.002	0.003 $\pm$ 0.000	0.057 $\pm$ 0.002	0.082 $\pm$ 0.005	0.286 $\pm$ 0.010	0.193	-0.007	-3.29	0.0582	0.0586	0.945	94.640 $\pm$ 0.968	0.210 $\pm$ 0.001	94.640 $\pm$ 0.968
QML	beta44	0.11	0.127	0.017 $\pm$ 0.003	0.155 $\pm$ 0.027	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.005 $\pm$ 0.000	0.070 $\pm$ 0.003	0.409 $\pm$ 0.028	0.639 $\pm$ 0.026	0.130	0.020	17.96	0.0664	0.0692	1.015	94.824 $\pm$ 0.952	0.272 $\pm$ 0.003	49.723 $\pm$ 2.150
QML	beta45	0.11	0.116	0.006 $\pm$ 0.003	0.058 $\pm$ 0.025	0.004 $\pm$ 0.000	0.064 $\pm$ 0.002	0.004 $\pm$ 0.000	0.064 $\pm$ 0.003	0.339 $\pm$ 0.023	0.582 $\pm$ 0.023	0.113	0.003	3.04	0.0635	0.0636	0.967	95.009 $\pm$ 0.936	0.242 $\pm$ 0.002	48.799 $\pm$ 2.149
QML	beta46	0.08	0.030	-0.050 $\pm$ 0.002	-0.627 $\pm$ 0.031	0.003 $\pm$ 0.000	0.057 $\pm$ 0.002	0.006 $\pm$ 0.000	0.076 $\pm$ 0.002	0.907 $\pm$ 0.046	0.952 $\pm$ 0.027	0.026	-0.054	-66.92	0.0570	0.0782	1.018	78.004 $\pm$ 1.781	0.229 $\pm$ 0.003	4.436 $\pm$ 0.885
QML	beta47	0.08	0.035	-0.045 $\pm$ 0.003	-0.560 $\pm$ 0.032	0.004 $\pm$ 0.000	0.059 $\pm$ 0.002	0.006 $\pm$ 0.000	0.074 $\pm$ 0.002	0.861 $\pm$ 0.042	0.928 $\pm$ 0.026	0.031	-0.049	-60.72	0.0564	0.0745	0.965	77.634 $\pm$ 1.792	0.224 $\pm$ 0.003	5.730 $\pm$ 0.999
QML	beta51	0.16	0.178	0.018 $\pm$ 0.003	0.116 $\pm$ 0.022	0.007 $\pm$ 0.000	0.081 $\pm$ 0.003	0.007 $\pm$ 0.000	0.083 $\pm$ 0.003	0.271 $\pm$ 0.019	0.521 $\pm$ 0.021	0.176	0.016	9.86	0.0772	0.0788	0.932	92.976 $\pm$ 1.099	0.297 $\pm$ 0.003	65.619 $\pm$ 2.042
QML	beta52	0.16	0.147	-0.013 $\pm$ 0.003	-0.083 $\pm$ 0.016	0.003 $\pm$ 0.000	0.059 $\pm$ 0.002	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	0.141 $\pm$ 0.007	0.375 $\pm$ 0.013	0.145	-0.015	-9.26	0.0624	0.0641	1.043	96.303 $\pm$ 0.811	0.239 $\pm$ 0.002	68.022 $\pm$ 2.005
QML	beta53	0.16	0.140	-0.020 $\pm$ 0.003	-0.128 $\pm$ 0.016	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	0.004 $\pm$ 0.000	0.063 $\pm$ 0.002	0.155 $\pm$ 0.010	0.394 $\pm$ 0.015	0.140	-0.020	-12.57	0.0606	0.0639	1.008	93.530 $\pm$ 1.058	0.236 $\pm$ 0.001	65.619 $\pm$ 2.042
QML	beta54	0.16	0.155	-0.005 $\pm$ 0.002	-0.029 $\pm$ 0.015	0.003 $\pm$ 0.000	0.057 $\pm$ 0.002	0.003 $\pm$ 0.000	0.057 $\pm$ 0.002	0.127 $\pm$ 0.008	0.356 $\pm$ 0.013	0.153	-0.007	-4.37	0.0552	0.0557	1.012	95.009 $\pm$ 0.936	0.226 $\pm$ 0.001	79.113 $\pm$ 1.748
QML	beta55	0.08	0.090	0.010 $\pm$ 0.004	0.130 $\pm$ 0.048	0.008 $\pm$ 0.001	0.088 $\pm$ 0.003	0.008 $\pm$ 0.001	0.089 $\pm$ 0.004	1.236 $\pm$ 0.082	1.112 $\pm$ 0.044	0.092	0.012	15.15	0.0856	0.0864	0.963	93.530 $\pm$ 1.058	0.334 $\pm$ 0.004	21.996 $\pm$ 1.781
QML	beta56	0.08	0.079	-0.001 $\pm$ 0.003	-0.006 $\pm$ 0.040	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.006 $\pm$ 0.000	0.075 $\pm$ 0.003	0.868 $\pm$ 0.054	0.932 $\pm$ 0.036	0.078	-0.002	-1.93	0.0745	0.0745	1.004	94.824 $\pm$ 0.952	0.294 $\pm$ 0.003	17.745 $\pm$ 1.643
QML	beta57	0.06	0.022	-0.038 $\pm$ 0.003	-0.626 $\pm$ 0.044	0.004 $\pm$ 0.000	0.062 $\pm$ 0.002	0.005 $\pm$ 0.000	0.072 $\pm$ 0.002	1.452 $\pm$ 0.077	1.205 $\pm$ 0.037	0.017	-0.043	-71.14	0.0579	0.0719	0.984	82.810 $\pm$ 1.622	0.239 $\pm$ 0.004	4.251 $\pm$ 0.867
QML	beta58	0.06	0.026	-0.034 $\pm$ 0.002	-0.562 $\pm$ 0.036	0.003 $\pm$ 0.000	0.050 $\pm$ 0.002	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	1.011 $\pm$ 0.052	1.005 $\pm$ 0.030	0.023	-0.037	-62.22	0.0480	0.0608	1.042	84.288 $\pm$ 1.565	0.205 $\pm$ 0.003	4.991 $\pm$ 0.936
UPI	beta41	0.20	0.218	0.018 $\pm$ 0.003	0.088 $\pm$ 0.014	0.004 $\pm$ 0.000	0.065 $\pm$ 0.003	0.005 $\pm$ 0.000	0.068 $\pm$ 0.003	0.115 $\pm$ 0.009	0.339 $\pm$ 0.015	0.213	0.013	6.53	0.0685	0.0697	1.266	97.782 $\pm$ 0.633	0.325 $\pm$ 0.017	83.179 $\pm$ 1.608
UPI	beta42	0.20	0.208	0.008 $\pm$ 0.003	0.038 $\pm$ 0.015	0.005 $\pm$ 0.000	0.071 $\pm$ 0.003	0.005 $\pm$ 0.000	0.072 $\pm$ 0.003	0.128 $\pm$ 0.010	0.358 $\pm$ 0.016	0.211	0.011	5.66	0.0741	0.0750	1.306	97.412 $\pm$ 0.683	0.365 $\pm$ 0.019	74.677 $\pm$ 1.870
UPI	beta43	0.20	0.197	-0.003 $\pm$ 0.003	-0.014 $\pm$ 0.014	0.004 $\pm$ 0.000	0.064 $\pm$ 0.002	0.004 $\pm$ 0.000	0.064 $\pm$ 0.002	0.101 $\pm$ 0.006	0.319 $\pm$ 0.011	0.193	-0.007	-3.53	0.0717	0.0721	1.080	95.194 $\pm$ 0.920	0.270 $\pm$ 0.005	84.473 $\pm$ 1.557

Table S213

Simulation 2 - Condition 69: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.6
(continued)

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.065	-0.015 ± 0.007	-0.182 ± 0.082	0.023 ± 0.002	0.152 ± 0.007	0.023 ± 0.002	0.153 ± 0.007	3.647 ± 0.314	1.910 ± 0.092	0.059	-0.021	-25.99	0.1340	0.1356	1.273	95.194 ± 0.920	0.760 ± 0.051	3.512 ± 0.791
UPI	beta51	0.16	0.174	0.014 ± 0.003	0.090 ± 0.021	0.006 ± 0.000	0.079 ± 0.003	0.006 ± 0.000	0.080 ± 0.003	0.251 ± 0.018	0.501 ± 0.021	0.172	0.012	7.32	0.0754	0.0763	0.965	92.791 ± 1.112	0.299 ± 0.003	63.956 ± 2.064
UPI	beta52	0.16	0.164	0.004 ± 0.003	0.024 ± 0.017	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.062 ± 0.003	0.152 ± 0.011	0.390 ± 0.016	0.164	0.004	2.31	0.0671	0.0672	1.204	98.336 ± 0.550	0.294 ± 0.008	67.468 ± 2.014
UPI	beta53	0.16	0.148	-0.012 ± 0.003	-0.075 ± 0.017	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.164 ± 0.011	0.405 ± 0.016	0.151	-0.009	-5.66	0.0657	0.0663	1.113	95.009 ± 0.936	0.278 ± 0.007	60.259 ± 2.104
UPI	beta54	0.16	0.160	0.000 ± 0.003	0.000 ± 0.019	0.005 ± 0.000	0.070 ± 0.003	0.005 ± 0.000	0.070 ± 0.003	0.192 ± 0.014	0.438 ± 0.019	0.160	0.000	-0.24	0.0717	0.0717	1.208	97.597 ± 0.658	0.332 ± 0.015	60.628 ± 2.101
UPI	beta55	0.08	0.092	0.012 ± 0.004	0.146 ± 0.051	0.009 ± 0.001	0.095 ± 0.003	0.009 ± 0.001	0.096 ± 0.004	1.434 ± 0.091	1.198 ± 0.046	0.094	0.014	17.37	0.0976	0.0986	1.035	94.640 ± 0.968	0.386 ± 0.008	18.854 ± 1.682
UPI	beta56	0.08	0.086	0.006 ± 0.004	0.081 ± 0.047	0.008 ± 0.001	0.088 ± 0.003	0.008 ± 0.001	0.088 ± 0.004	1.207 ± 0.090	1.099 ± 0.047	0.083	0.003	3.99	0.0824	0.0825	1.042	97.043 ± 0.728	0.358 ± 0.007	14.233 ± 1.502
UPI	beta57	0.06	0.051	-0.009 ± 0.005	-0.155 ± 0.085	0.014 ± 0.001	0.119 ± 0.004	0.014 ± 0.001	0.120 ± 0.005	3.969 ± 0.288	1.992 ± 0.084	0.052	-0.008	-13.93	0.1128	0.1131	1.168	94.455 ± 0.984	0.546 ± 0.020	4.436 ± 0.885
UPI	beta58	0.06	0.034	-0.026 ± 0.005	-0.431 ± 0.090	0.016 ± 0.001	0.125 ± 0.005	0.016 ± 0.001	0.128 ± 0.006	4.532 ± 0.385	2.129 ± 0.101	0.032	-0.028	-45.95	0.1034	0.1070	1.352	95.194 ± 0.920	0.664 ± 0.053	3.697 ± 0.811

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S214

Simulation 2 - Condition 70: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, $N = 1000$, Reliability = 0.6

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LSAM	beta41	0.20	0.198	-0.002 $\pm$ 0.001	-0.011 $\pm$ 0.007	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.031 $\pm$ 0.002	0.176 $\pm$ 0.006	0.197	-0.003	-1.71	0.0340	0.0342	1.072	96.724 $\pm$ 0.703	0.148 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta42	0.20	0.195	-0.005 $\pm$ 0.001	-0.023 $\pm$ 0.007	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.036 $\pm$ 0.002	0.190 $\pm$ 0.006	0.197	-0.003	-1.48	0.0364	0.0366	1.030	95.320 $\pm$ 0.834	0.152 $\pm$ 0.001	100.000 $\pm$ 0.000
LSAM	beta43	0.20	0.200	0.000 $\pm$ 0.001	0.001 $\pm$ 0.007	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.031 $\pm$ 0.002	0.176 $\pm$ 0.006	0.200	0.000	-0.11	0.0369	0.0369	1.007	95.788 $\pm$ 0.793	0.139 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta44	0.11	0.130	0.020 $\pm$ 0.002	0.183 $\pm$ 0.015	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.002	0.184 $\pm$ 0.010	0.429 $\pm$ 0.014	0.130	0.020	17.78	0.0400	0.0445	1.013	91.888 $\pm$ 1.078	0.170 $\pm$ 0.001	83.619 $\pm$ 1.462
LSAM	beta45	0.11	0.114	0.004 $\pm$ 0.002	0.039 $\pm$ 0.014	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.126 $\pm$ 0.007	0.354 $\pm$ 0.012	0.113	0.003	2.89	0.0400	0.0401	0.994	95.944 $\pm$ 0.779	0.151 $\pm$ 0.001	82.839 $\pm$ 1.489
LSAM	beta46	0.08	0.041	-0.039 $\pm$ 0.002	-0.485 $\pm$ 0.026	0.003 $\pm$ 0.000	0.053 $\pm$ 0.002	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	0.677 $\pm$ 0.034	0.823 $\pm$ 0.023	0.040	-0.040	-50.50	0.0511	0.0651	0.977	84.867 $\pm$ 1.415	0.204 $\pm$ 0.001	9.984 $\pm$ 1.184
LSAM	beta47	0.08	0.046	-0.034 $\pm$ 0.002	-0.420 $\pm$ 0.026	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.004 $\pm$ 0.000	0.063 $\pm$ 0.002	0.611 $\pm$ 0.029	0.782 $\pm$ 0.022	0.042	-0.038	-48.02	0.0531	0.0655	0.992	86.271 $\pm$ 1.359	0.205 $\pm$ 0.002	10.452 $\pm$ 1.208
LSAM	beta51	0.16	0.170	0.010 $\pm$ 0.002	0.061 $\pm$ 0.011	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.002	0.083 $\pm$ 0.005	0.289 $\pm$ 0.010	0.169	0.009	5.64	0.0455	0.0464	0.980	94.384 $\pm$ 0.909	0.174 $\pm$ 0.001	97.192 $\pm$ 0.653
LSAM	beta52	0.16	0.152	-0.008 $\pm$ 0.001	-0.052 $\pm$ 0.009	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.053 $\pm$ 0.003	0.231 $\pm$ 0.008	0.151	-0.009	-5.40	0.0376	0.0386	1.037	95.320 $\pm$ 0.834	0.146 $\pm$ 0.000	97.816 $\pm$ 0.577
LSAM	beta53	0.16	0.155	-0.005 $\pm$ 0.002	-0.029 $\pm$ 0.010	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.062 $\pm$ 0.003	0.248 $\pm$ 0.008	0.154	-0.006	-3.76	0.0395	0.0400	0.979	93.916 $\pm$ 0.944	0.152 $\pm$ 0.000	97.816 $\pm$ 0.577
LSAM	beta54	0.16	0.154	-0.006 $\pm$ 0.001	-0.037 $\pm$ 0.009	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.055 $\pm$ 0.003	0.234 $\pm$ 0.008	0.152	-0.008	-5.23	0.0375	0.0384	1.008	95.632 $\pm$ 0.807	0.146 $\pm$ 0.000	98.596 $\pm$ 0.465
LSAM	beta55	0.08	0.091	0.011 $\pm$ 0.002	0.133 $\pm$ 0.025	0.003 $\pm$ 0.000	0.050 $\pm$ 0.001	0.003 $\pm$ 0.000	0.051 $\pm$ 0.002	0.409 $\pm$ 0.024	0.639 $\pm$ 0.022	0.090	0.010	12.41	0.0473	0.0483	1.007	94.852 $\pm$ 0.873	0.198 $\pm$ 0.001	44.618 $\pm$ 1.963
LSAM	beta56	0.08	0.080	-0.000 $\pm$ 0.002	-0.005 $\pm$ 0.026	0.003 $\pm$ 0.000	0.054 $\pm$ 0.002	0.003 $\pm$ 0.000	0.054 $\pm$ 0.002	0.448 $\pm$ 0.026	0.669 $\pm$ 0.024	0.082	0.002	2.10	0.0500	0.0501	0.892	93.448 $\pm$ 0.977	0.187 $\pm$ 0.001	40.562 $\pm$ 1.939
LSAM	beta57	0.06	0.032	-0.028 $\pm$ 0.002	-0.461 $\pm$ 0.034	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.003 $\pm$ 0.000	0.058 $\pm$ 0.002	0.933 $\pm$ 0.047	0.966 $\pm$ 0.029	0.027	-0.033	-55.01	0.0514	0.0611	0.994	88.768 $\pm$ 1.247	0.199 $\pm$ 0.001	8.736 $\pm$ 1.115
LSAM	beta58	0.06	0.035	-0.025 $\pm$ 0.002	-0.421 $\pm$ 0.030	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.738 $\pm$ 0.036	0.859 $\pm$ 0.025	0.032	-0.028	-47.12	0.0463	0.0542	1.037	90.952 $\pm$ 1.133	0.183 $\pm$ 0.001	7.956 $\pm$ 1.069
QML	beta41	0.20	0.194	-0.006 $\pm$ 0.001	-0.028 $\pm$ 0.006	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.027 $\pm$ 0.002	0.166 $\pm$ 0.006	0.194	-0.006	-3.01	0.0326	0.0332	1.069	96.100 $\pm$ 0.765	0.137 $\pm$ 0.001	99.844 $\pm$ 0.156
QML	beta42	0.20	0.194	-0.006 $\pm$ 0.001	-0.031 $\pm$ 0.007	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.031 $\pm$ 0.002	0.177 $\pm$ 0.006	0.193	-0.007	-3.35	0.0325	0.0332	1.026	95.476 $\pm$ 0.821	0.140 $\pm$ 0.000	100.000 $\pm$ 0.000
QML	beta43	0.20	0.193	-0.007 $\pm$ 0.001	-0.034 $\pm$ 0.007	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.029 $\pm$ 0.002	0.172 $\pm$ 0.006	0.194	-0.006	-3.09	0.0335	0.0341	1.011	95.164 $\pm$ 0.847	0.133 $\pm$ 0.000	100.000 $\pm$ 0.000
QML	beta44	0.11	0.127	0.017 $\pm$ 0.002	0.159 $\pm$ 0.015	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.163 $\pm$ 0.009	0.404 $\pm$ 0.013	0.128	0.018	16.61	0.0386	0.0427	1.038	92.356 $\pm$ 1.049	0.166 $\pm$ 0.001	83.463 $\pm$ 1.467
QML	beta45	0.11	0.112	0.002 $\pm$ 0.001	0.023 $\pm$ 0.014	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.119 $\pm$ 0.006	0.345 $\pm$ 0.011	0.110	0.000	0.42	0.0409	0.0409	1.006	96.724 $\pm$ 0.703	0.149 $\pm$ 0.001	84.399 $\pm$ 1.433
QML	beta46	0.08	0.024	-0.056 $\pm$ 0.001	-0.696 $\pm$ 0.017	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.004 $\pm$ 0.000	0.066 $\pm$ 0.001	0.675 $\pm$ 0.024	0.821 $\pm$ 0.016	0.024	-0.056	-69.80	0.0350	0.0659	1.012	61.154 $\pm$ 1.925	0.138 $\pm$ 0.001	5.772 $\pm$ 0.921
QML	beta47	0.08	0.029	-0.051 $\pm$ 0.001	-0.633 $\pm$ 0.017	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.004 $\pm$ 0.000	0.062 $\pm$ 0.001	0.592 $\pm$ 0.022	0.769 $\pm$ 0.015	0.025	-0.055	-68.54	0.0352	0.0652	1.003	65.055 $\pm$ 1.883	0.137 $\pm$ 0.001	9.204 $\pm$ 1.142
QML	beta51	0.16	0.180	0.020 $\pm$ 0.002	0.122 $\pm$ 0.011	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.002	0.095 $\pm$ 0.005	0.309 $\pm$ 0.010	0.178	0.018	11.53	0.0448	0.0484	0.997	92.824 $\pm$ 1.019	0.178 $\pm$ 0.001	97.660 $\pm$ 0.597
QML	beta52	0.16	0.144	-0.016 $\pm$ 0.001	-0.098 $\pm$ 0.009	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.001 $\pm$ 0.000	0.039 $\pm$ 0.001	0.058 $\pm$ 0.003	0.241 $\pm$ 0.007	0.145	-0.015	-9.61	0.0370	0.0401	1.039	92.980 $\pm$ 1.009	0.144 $\pm$ 0.001	98.440 $\pm$ 0.489
QML	beta53	0.16	0.142	-0.018 $\pm$ 0.001	-0.111 $\pm$ 0.009	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	0.067 $\pm$ 0.003	0.259 $\pm$ 0.008	0.142	-0.018	-11.49	0.0369	0.0412	0.986	91.888 $\pm$ 1.078	0.145 $\pm$ 0.000	97.348 $\pm$ 0.635
QML	beta54	0.16	0.156	-0.004 $\pm$ 0.001	-0.025 $\pm$ 0.009	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.050 $\pm$ 0.003	0.223 $\pm$ 0.007	0.156	-0.004	-2.71	0.0358	0.0361	1.009	95.788 $\pm$ 0.793	0.141 $\pm$ 0.000	99.064 $\pm$ 0.380
QML	beta55	0.08	0.089	0.009 $\pm$ 0.002	0.115 $\pm$ 0.024	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.002	0.382 $\pm$ 0.022	0.618 $\pm$ 0.021	0.089	0.009	10.87	0.0461	0.0469	1.049	95.632 $\pm$ 0.807	0.200 $\pm$ 0.002	44.618 $\pm$ 1.963
QML	beta56	0.08	0.076	-0.004 $\pm$ 0.002	-0.050 $\pm$ 0.025	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.003 $\pm$ 0.000	0.051 $\pm$ 0.002	0.402 $\pm$ 0.022	0.634 $\pm$ 0.022	0.077	-0.003	-3.37	0.0478	0.0479	0.921	92.356 $\pm$ 1.049	0.183 $\pm$ 0.002	39.938 $\pm$ 1.934
QML	beta57	0.06	0.020	-0.040 $\pm$ 0.001	-0.672 $\pm$ 0.024	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.003 $\pm$ 0.000	0.055 $\pm$ 0.001	0.826 $\pm$ 0.034	0.909 $\pm$ 0.020	0.016	-0.044	-72.59	0.0369	0.0571	0.992	71.763 $\pm$ 1.778	0.143 $\pm$ 0.001	4.836 $\pm$ 0.847
QML	beta58	0.06	0.026	-0.034 $\pm$ 0.001	-0.566 $\pm$ 0.021	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.593 $\pm$ 0.026	0.770 $\pm$ 0.018	0.024	-0.036	-60.31	0.0318	0.0482	1.035	76.443 $\pm$ 1.676	0.127 $\pm$ 0.001	6.552 $\pm$ 0.977
UPI	beta41	0.20	0.204	0.004 $\pm$ 0.002	0.019 $\pm$ 0.008	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.038 $\pm$ 0.002	0.196 $\pm$ 0.007	0.205	0.005	2.43	0.0418	0.0421	1.475	97.192 $\pm$ 0.653	0.225 $\pm$ 0.015	94.384 $\pm$ 0.909
UPI	beta42	0.20	0.201	0.001 $\pm$ 0.002	0.004 $\pm$ 0.009	0.002 $\pm$ 0.000	0.044 $\pm$ 0.002	0.002 $\pm$ 0.000	0.044 $\pm$ 0.002	0.047 $\pm$ 0.003	0.218 $\pm$ 0.009	0.199	-0.001	-0.58	0.0430	0.0431	1.383	97.348 $\pm$ 0.635	0.236 $\pm$ 0.015	93.136 $\pm$ 0.999
UPI	beta43	0.20	0.198	-0.002 $\pm$ 0.001	-0.012 $\pm$ 0.007	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.035 $\pm$ 0.002	0.186 $\pm$ 0.006	0.197	-0.003	-1.65	0.0421	0.0423	1.099	96.256 $\pm$ 0.750	0.160 $\pm$ 0.003	98.752 $\pm$

Table S214

Simulation 2 - Condition 70: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.6
(continued)

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.052	-0.028 ± 0.003	-0.350 ± 0.044	0.008 ± 0.001	0.089 ± 0.003	0.009 ± 0.001	0.093 ± 0.004	1.346 ± 0.091	1.160 ± 0.045	0.043	-0.037	-45.89	0.0847	0.0923	1.233	93.136 ± 0.999	0.428 ± 0.030	7.644 ± 1.049
UPI	beta51	0.16	0.172	0.012 ± 0.002	0.077 ± 0.011	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.047 ± 0.002	0.086 ± 0.005	0.293 ± 0.010	0.172	0.012	7.34	0.0475	0.0490	1.022	93.760 ± 0.955	0.181 ± 0.002	95.944 ± 0.779
UPI	beta52	0.16	0.155	-0.005 ± 0.002	-0.029 ± 0.010	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.061 ± 0.003	0.247 ± 0.008	0.156	-0.004	-2.38	0.0401	0.0403	1.216	96.568 ± 0.719	0.187 ± 0.005	92.044 ± 1.069
UPI	beta53	0.16	0.152	-0.008 ± 0.002	-0.050 ± 0.010	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.066 ± 0.004	0.257 ± 0.009	0.148	-0.012	-7.55	0.0418	0.0435	1.075	95.164 ± 0.847	0.170 ± 0.002	94.228 ± 0.921
UPI	beta54	0.16	0.156	-0.004 ± 0.002	-0.022 ± 0.011	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.002	0.073 ± 0.005	0.271 ± 0.010	0.155	-0.005	-2.82	0.0425	0.0428	1.220	97.036 ± 0.670	0.207 ± 0.010	89.860 ± 1.192
UPI	beta55	0.08	0.090	0.010 ± 0.002	0.120 ± 0.025	0.003 ± 0.000	0.051 ± 0.002	0.003 ± 0.000	0.052 ± 0.002	0.421 ± 0.025	0.649 ± 0.023	0.089	0.009	11.77	0.0509	0.0518	1.110	96.412 ± 0.735	0.222 ± 0.004	36.817 ± 1.905
UPI	beta56	0.08	0.081	0.001 ± 0.002	0.013 ± 0.027	0.003 ± 0.000	0.054 ± 0.002	0.003 ± 0.000	0.054 ± 0.002	0.460 ± 0.025	0.679 ± 0.023	0.084	0.004	5.35	0.0554	0.0556	0.944	93.604 ± 0.966	0.201 ± 0.002	37.754 ± 1.915
UPI	beta57	0.06	0.043	-0.017 ± 0.003	-0.281 ± 0.045	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.071 ± 0.003	1.403 ± 0.083	1.185 ± 0.042	0.040	-0.020	-32.57	0.0751	0.0776	1.184	94.852 ± 0.873	0.321 ± 0.010	6.084 ± 0.944
UPI	beta58	0.06	0.036	-0.024 ± 0.003	-0.403 ± 0.050	0.006 ± 0.000	0.076 ± 0.003	0.006 ± 0.000	0.080 ± 0.003	1.759 ± 0.118	1.326 ± 0.051	0.030	-0.030	-49.50	0.0653	0.0717	1.309	94.540 ± 0.897	0.389 ± 0.029	7.020 ± 1.009

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S215

Simulation 2 - Condition 71: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, $N = 400$, Reliability = 0.8

Method	Param.	True	Mean-based										Median-based					Inference		
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LSAM	beta41	0.20	0.199	-0.001 $\pm$ 0.001	-0.007 $\pm$ 0.007	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.056 $\pm$ 0.002	0.236 $\pm$ 0.006	0.199	-0.001	-0.59	0.0496	0.0496	1.005	95.796 $\pm$ 0.635	0.186 $\pm$ 0.001	98.398 $\pm$ 0.397
LSAM	beta42	0.20	0.202	0.002 $\pm$ 0.002	0.008 $\pm$ 0.008	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.060 $\pm$ 0.003	0.246 $\pm$ 0.007	0.203	0.003	1.36	0.0482	0.0482	0.977	94.494 $\pm$ 0.722	0.188 $\pm$ 0.001	97.898 $\pm$ 0.454
LSAM	beta43	0.20	0.198	-0.002 $\pm$ 0.001	-0.009 $\pm$ 0.007	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.053 $\pm$ 0.002	0.230 $\pm$ 0.006	0.197	-0.003	-1.64	0.0460	0.0462	0.975	94.995 $\pm$ 0.690	0.176 $\pm$ 0.001	98.999 $\pm$ 0.315
LSAM	beta44	0.11	0.112	0.002 $\pm$ 0.002	0.015 $\pm$ 0.015	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.222 $\pm$ 0.010	0.471 $\pm$ 0.013	0.115	0.005	4.60	0.0501	0.0504	0.975	94.394 $\pm$ 0.728	0.198 $\pm$ 0.001	62.062 $\pm$ 1.535
LSAM	beta45	0.11	0.113	0.003 $\pm$ 0.001	0.031 $\pm$ 0.014	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.184 $\pm$ 0.008	0.429 $\pm$ 0.011	0.116	0.006	5.08	0.0494	0.0497	0.974	95.195 $\pm$ 0.677	0.180 $\pm$ 0.001	69.770 $\pm$ 1.453
LSAM	beta46	0.08	0.073	-0.007 $\pm$ 0.002	-0.083 $\pm$ 0.023	0.003 $\pm$ 0.000	0.059 $\pm$ 0.001	0.003 $\pm$ 0.000	0.059 $\pm$ 0.002	0.544 $\pm$ 0.024	0.737 $\pm$ 0.020	0.070	-0.010	-12.86	0.0578	0.0587	0.997	93.894 $\pm$ 0.758	0.229 $\pm$ 0.001	22.923 $\pm$ 1.330
LSAM	beta47	0.08	0.073	-0.007 $\pm$ 0.002	-0.082 $\pm$ 0.024	0.004 $\pm$ 0.000	0.060 $\pm$ 0.001	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	0.562 $\pm$ 0.025	0.749 $\pm$ 0.021	0.071	-0.009	-11.22	0.0580	0.0587	0.973	93.694 $\pm$ 0.769	0.227 $\pm$ 0.001	23.223 $\pm$ 1.336
LSAM	beta51	0.16	0.161	0.001 $\pm$ 0.002	0.008 $\pm$ 0.012	0.004 $\pm$ 0.000	0.061 $\pm$ 0.001	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.146 $\pm$ 0.007	0.382 $\pm$ 0.011	0.160	0.000	0.28	0.0607	0.0607	0.924	93.393 $\pm$ 0.786	0.222 $\pm$ 0.001	81.982 $\pm$ 1.216
LSAM	beta52	0.16	0.156	-0.004 $\pm$ 0.002	-0.023 $\pm$ 0.010	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	0.096 $\pm$ 0.004	0.310 $\pm$ 0.009	0.156	-0.004	-2.56	0.0501	0.0503	0.961	94.895 $\pm$ 0.696	0.186 $\pm$ 0.001	89.489 $\pm$ 0.970
LSAM	beta53	0.16	0.162	0.002 $\pm$ 0.002	0.012 $\pm$ 0.010	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.093 $\pm$ 0.004	0.304 $\pm$ 0.008	0.160	0.000	-0.20	0.0491	0.0491	1.004	95.896 $\pm$ 0.628	0.192 $\pm$ 0.001	90.691 $\pm$ 0.919
LSAM	beta54	0.16	0.160	0.000 $\pm$ 0.001	0.001 $\pm$ 0.009	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.083 $\pm$ 0.004	0.289 $\pm$ 0.008	0.160	0.000	-0.30	0.0464	0.0464	1.025	95.395 $\pm$ 0.663	0.186 $\pm$ 0.001	91.391 $\pm$ 0.887
LSAM	beta55	0.08	0.081	0.001 $\pm$ 0.002	0.016 $\pm$ 0.025	0.004 $\pm$ 0.000	0.062 $\pm$ 0.001	0.004 $\pm$ 0.000	0.062 $\pm$ 0.002	0.604 $\pm$ 0.028	0.777 $\pm$ 0.022	0.083	0.003	4.28	0.0587	0.0588	0.952	93.293 $\pm$ 0.791	0.232 $\pm$ 0.002	31.932 $\pm$ 1.475
LSAM	beta56	0.08	0.083	0.003 $\pm$ 0.002	0.032 $\pm$ 0.024	0.004 $\pm$ 0.000	0.061 $\pm$ 0.001	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.587 $\pm$ 0.028	0.766 $\pm$ 0.022	0.083	0.003	3.21	0.0584	0.0585	0.930	93.493 $\pm$ 0.780	0.223 $\pm$ 0.001	31.932 $\pm$ 1.475
LSAM	beta57	0.06	0.054	-0.006 $\pm$ 0.002	-0.098 $\pm$ 0.033	0.004 $\pm$ 0.000	0.062 $\pm$ 0.002	0.004 $\pm$ 0.000	0.062 $\pm$ 0.002	1.065 $\pm$ 0.052	1.032 $\pm$ 0.030	0.052	-0.008	-14.04	0.0630	0.0635	0.949	93.393 $\pm$ 0.786	0.229 $\pm$ 0.001	14.615 $\pm$ 1.118
LSAM	beta58	0.06	0.057	-0.003 $\pm$ 0.002	-0.047 $\pm$ 0.030	0.003 $\pm$ 0.000	0.057 $\pm$ 0.001	0.003 $\pm$ 0.000	0.057 $\pm$ 0.002	0.893 $\pm$ 0.041	0.945 $\pm$ 0.026	0.055	-0.005	-8.44	0.0561	0.0564	0.968	93.994 $\pm$ 0.752	0.215 $\pm$ 0.001	16.717 $\pm$ 1.181
QML	beta41	0.20	0.197	-0.003 $\pm$ 0.001	-0.017 $\pm$ 0.007	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.055 $\pm$ 0.002	0.234 $\pm$ 0.006	0.197	-0.003	-1.57	0.0486	0.0487	0.999	95.395 $\pm$ 0.663	0.183 $\pm$ 0.001	98.198 $\pm$ 0.421
QML	beta42	0.20	0.200	0.000 $\pm$ 0.002	0.002 $\pm$ 0.008	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.059 $\pm$ 0.003	0.242 $\pm$ 0.007	0.201	0.001	0.28	0.0476	0.0476	0.974	94.294 $\pm$ 0.734	0.185 $\pm$ 0.001	98.098 $\pm$ 0.432
QML	beta43	0.20	0.194	-0.006 $\pm$ 0.001	-0.030 $\pm$ 0.007	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.052 $\pm$ 0.002	0.229 $\pm$ 0.006	0.193	-0.007	-3.33	0.0446	0.0451	0.973	94.795 $\pm$ 0.703	0.173 $\pm$ 0.001	98.699 $\pm$ 0.359
QML	beta44	0.11	0.111	0.001 $\pm$ 0.002	0.011 $\pm$ 0.015	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.231 $\pm$ 0.011	0.481 $\pm$ 0.013	0.116	0.006	5.43	0.0542	0.0545	0.981	94.394 $\pm$ 0.728	0.203 $\pm$ 0.001	60.861 $\pm$ 1.544
QML	beta45	0.11	0.113	0.003 $\pm$ 0.002	0.030 $\pm$ 0.014	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.187 $\pm$ 0.008	0.433 $\pm$ 0.012	0.115	0.005	4.57	0.0490	0.0492	0.982	94.895 $\pm$ 0.696	0.183 $\pm$ 0.001	67.768 $\pm$ 1.479
QML	beta46	0.08	0.059	-0.021 $\pm$ 0.002	-0.264 $\pm$ 0.019	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.441 $\pm$ 0.018	0.664 $\pm$ 0.016	0.055	-0.025	-31.73	0.0485	0.0547	1.011	90.390 $\pm$ 0.932	0.193 $\pm$ 0.001	18.218 $\pm$ 1.221
QML	beta47	0.08	0.060	-0.020 $\pm$ 0.002	-0.253 $\pm$ 0.019	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.440 $\pm$ 0.019	0.663 $\pm$ 0.017	0.057	-0.023	-28.18	0.0461	0.0513	0.986	90.791 $\pm$ 0.915	0.190 $\pm$ 0.001	19.720 $\pm$ 1.259
QML	beta51	0.16	0.167	0.007 $\pm$ 0.002	0.043 $\pm$ 0.012	0.004 $\pm$ 0.000	0.062 $\pm$ 0.001	0.004 $\pm$ 0.000	0.062 $\pm$ 0.002	0.150 $\pm$ 0.007	0.387 $\pm$ 0.011	0.165	0.005	3.30	0.0614	0.0616	0.950	93.994 $\pm$ 0.752	0.229 $\pm$ 0.001	83.083 $\pm$ 1.186
QML	beta52	0.16	0.143	-0.017 $\pm$ 0.002	-0.109 $\pm$ 0.010	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.003 $\pm$ 0.000	0.054 $\pm$ 0.001	0.112 $\pm$ 0.005	0.335 $\pm$ 0.009	0.142	-0.018	-11.23	0.0508	0.0539	0.969	92.092 $\pm$ 0.854	0.193 $\pm$ 0.001	81.782 $\pm$ 1.221
QML	beta53	0.16	0.144	-0.016 $\pm$ 0.002	-0.101 $\pm$ 0.010	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.102 $\pm$ 0.005	0.319 $\pm$ 0.009	0.142	-0.018	-11.07	0.0459	0.0492	1.017	94.494 $\pm$ 0.722	0.193 $\pm$ 0.001	83.884 $\pm$ 1.163
QML	beta54	0.16	0.160	0.000 $\pm$ 0.001	0.001 $\pm$ 0.009	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.083 $\pm$ 0.004	0.288 $\pm$ 0.008	0.160	0.000	-0.24	0.0471	0.0471	1.025	95.696 $\pm$ 0.642	0.185 $\pm$ 0.001	92.092 $\pm$ 0.854
QML	beta55	0.08	0.081	0.001 $\pm$ 0.002	0.012 $\pm$ 0.025	0.004 $\pm$ 0.000	0.063 $\pm$ 0.001	0.004 $\pm$ 0.000	0.063 $\pm$ 0.002	0.612 $\pm$ 0.029	0.782 $\pm$ 0.022	0.083	0.003	3.74	0.0602	0.0602	0.965	93.694 $\pm$ 0.769	0.237 $\pm$ 0.002	30.330 $\pm$ 1.454
QML	beta56	0.08	0.082	0.002 $\pm$ 0.002	0.025 $\pm$ 0.024	0.004 $\pm$ 0.000	0.061 $\pm$ 0.001	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.574 $\pm$ 0.028	0.758 $\pm$ 0.022	0.081	0.001	1.01	0.0582	0.0582	0.940	93.093 $\pm$ 0.802	0.223 $\pm$ 0.001	30.230 $\pm$ 1.453
QML	beta57	0.06	0.042	-0.018 $\pm$ 0.002	-0.296 $\pm$ 0.027	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.003 $\pm$ 0.000	0.055 $\pm$ 0.001	0.827 $\pm$ 0.038	0.910 $\pm$ 0.024	0.040	-0.020	-33.17	0.0508	0.0546	0.955	90.390 $\pm$ 0.932	0.193 $\pm$ 0.001	12.713 $\pm$ 1.054
QML	beta58	0.06	0.048	-0.012 $\pm$ 0.001	-0.204 $\pm$ 0.025	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.656 $\pm$ 0.029	0.810 $\pm$ 0.022	0.045	-0.015	-25.21	0.0464	0.0488	0.974	91.992 $\pm$ 0.859	0.180 $\pm$ 0.001	14.615 $\pm$ 1.118
UPI	beta41	0.20	0.204	0.004 $\pm$ 0.001	0.022 $\pm$ 0.007	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.055 $\pm$ 0.002	0.234 $\pm$ 0.006	0.205	0.005	2.26	0.0479	0.0482	0.962	94.494 $\pm$ 0.722	0.176 $\pm$ 0.001	99.399 $\pm$ 0.244
UPI	beta42	0.20	0.206	0.006 $\pm$ 0.002	0.032 $\pm$ 0.008	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.060 $\pm$ 0.003	0.245 $\pm$ 0.007	0.207	0.007	3.25	0.0472	0.0476	0.950	93.894 $\pm$ 0.758	0.181 $\pm$ 0.001	98.298 $\pm$ 0.409
UPI	beta43	0.20	0.200	0.000 $\pm$ 0.001	0.001 $\pm$ 0.007	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.053 $\pm$ 0.002	0.230 $\pm$ 0.006	0.199	-0.001	-0.44	0.0463	0.0463	0.973	94.595 $\pm$ 0.715	0.176 $\pm$ 0.001	99.099 $\pm$ 0

Table S215

Simulation 2 - Condition 71: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.8
(continued)

Method	Param.	True	Mean-based										Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE	
UPI	beta47	0.08	0.080	-0.000 ± 0.002	-0.001 ± 0.025	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.633 ± 0.029	0.795 ± 0.022	0.078	-0.002	-3.10	0.0613	0.0614	1.018	95.896 ± 0.628	0.254 ± 0.002	22.523 ± 1.322	
UPI	beta51	0.16	0.163	0.003 ± 0.002	0.018 ± 0.012	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.143 ± 0.007	0.378 ± 0.011	0.160	0.000	-0.31	0.0603	0.0603	0.942	93.594 ± 0.775	0.223 ± 0.001	82.082 ± 1.213	
UPI	beta52	0.16	0.160	0.000 ± 0.002	0.000 ± 0.010	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.095 ± 0.004	0.309 ± 0.009	0.159	-0.001	-0.32	0.0511	0.0511	0.945	94.595 ± 0.715	0.183 ± 0.001	91.592 ± 0.878	
UPI	beta53	0.16	0.163	0.003 ± 0.002	0.016 ± 0.010	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.092 ± 0.004	0.304 ± 0.008	0.161	0.001	0.62	0.0484	0.0484	1.004	95.796 ± 0.635	0.191 ± 0.001	91.391 ± 0.887	
UPI	beta54	0.16	0.163	0.003 ± 0.001	0.017 ± 0.009	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.083 ± 0.004	0.288 ± 0.008	0.162	0.002	1.22	0.0477	0.0478	1.018	95.395 ± 0.663	0.183 ± 0.001	93.794 ± 0.763	
UPI	beta55	0.08	0.080	-0.000 ± 0.002	-0.003 ± 0.025	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.604 ± 0.028	0.777 ± 0.022	0.079	-0.001	-0.93	0.0575	0.0575	0.961	93.694 ± 0.769	0.234 ± 0.002	29.229 ± 1.439	
UPI	beta56	0.08	0.082	0.002 ± 0.002	0.030 ± 0.024	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	0.592 ± 0.030	0.770 ± 0.023	0.081	0.001	1.48	0.0572	0.0572	0.935	92.793 ± 0.818	0.226 ± 0.001	31.031 ± 1.464	
UPI	beta57	0.06	0.059	-0.001 ± 0.002	-0.021 ± 0.033	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	1.118 ± 0.055	1.058 ± 0.031	0.058	-0.002	-4.07	0.0605	0.0605	0.966	93.794 ± 0.763	0.240 ± 0.001	14.815 ± 1.124	
UPI	beta58	0.06	0.061	0.001 ± 0.002	0.024 ± 0.031	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.948 ± 0.046	0.974 ± 0.028	0.059	-0.001	-2.17	0.0575	0.0575	1.004	94.494 ± 0.722	0.230 ± 0.001	17.117 ± 1.192	

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S216

Simulation 2 - Condition 72: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.8

Method	Param.	True	Mean-based										Median-based					Inference		
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Power (%) $\pm$ MCSE
LSAM	beta41	0.20	0.200	-0.000 $\pm$ 0.001	-0.002 $\pm$ 0.005	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.022 $\pm$ 0.001	0.149 $\pm$ 0.004	0.201	0.001	0.68	0.0303	0.0303	1.000	95.500 $\pm$ 0.656	0.117 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta42	0.20	0.198	-0.002 $\pm$ 0.001	-0.008 $\pm$ 0.005	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.023 $\pm$ 0.001	0.151 $\pm$ 0.004	0.199	-0.001	-0.49	0.0289	0.0289	1.001	95.400 $\pm$ 0.662	0.118 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta43	0.20	0.200	0.000 $\pm$ 0.001	0.001 $\pm$ 0.004	0.001 $\pm$ 0.000	0.028 $\pm$ 0.001	0.001 $\pm$ 0.000	0.028 $\pm$ 0.001	0.019 $\pm$ 0.001	0.138 $\pm$ 0.004	0.200	0.000	0.07	0.0275	0.0275	1.035	95.500 $\pm$ 0.656	0.112 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta44	0.11	0.116	0.006 $\pm$ 0.001	0.053 $\pm$ 0.009	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.090 $\pm$ 0.004	0.299 $\pm$ 0.008	0.115	0.005	4.52	0.0316	0.0319	0.983	94.000 $\pm$ 0.751	0.125 $\pm$ 0.000	93.700 $\pm$ 0.768
LSAM	beta45	0.11	0.112	0.002 $\pm$ 0.001	0.021 $\pm$ 0.009	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.075 $\pm$ 0.003	0.274 $\pm$ 0.007	0.113	0.003	2.32	0.0297	0.0298	0.972	93.100 $\pm$ 0.801	0.115 $\pm$ 0.000	96.100 $\pm$ 0.612
LSAM	beta46	0.08	0.069	-0.011 $\pm$ 0.001	-0.135 $\pm$ 0.015	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.237 $\pm$ 0.010	0.487 $\pm$ 0.013	0.068	-0.012	-14.88	0.0377	0.0395	0.984	92.800 $\pm$ 0.817	0.144 $\pm$ 0.000	45.700 $\pm$ 1.575
LSAM	beta47	0.08	0.071	-0.009 $\pm$ 0.001	-0.115 $\pm$ 0.015	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.001 $\pm$ 0.000	0.039 $\pm$ 0.001	0.234 $\pm$ 0.010	0.483 $\pm$ 0.013	0.070	-0.010	-12.86	0.0377	0.0391	0.973	93.100 $\pm$ 0.801	0.143 $\pm$ 0.000	48.700 $\pm$ 1.581
LSAM	beta51	0.16	0.160	0.000 $\pm$ 0.001	0.003 $\pm$ 0.008	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.056 $\pm$ 0.003	0.238 $\pm$ 0.007	0.158	-0.002	-1.08	0.0348	0.0349	0.950	93.100 $\pm$ 0.801	0.142 $\pm$ 0.000	99.200 $\pm$ 0.282
LSAM	beta52	0.16	0.160	0.000 $\pm$ 0.001	0.001 $\pm$ 0.006	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.034 $\pm$ 0.001	0.185 $\pm$ 0.005	0.161	0.001	0.68	0.0296	0.0296	1.014	95.900 $\pm$ 0.627	0.118 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta53	0.16	0.159	-0.001 $\pm$ 0.001	-0.005 $\pm$ 0.006	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.038 $\pm$ 0.002	0.195 $\pm$ 0.005	0.158	-0.002	-1.05	0.0303	0.0304	0.992	94.600 $\pm$ 0.715	0.122 $\pm$ 0.000	100.000 $\pm$ 0.000
LSAM	beta54	0.16	0.158	-0.002 $\pm$ 0.001	-0.015 $\pm$ 0.006	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.036 $\pm$ 0.002	0.189 $\pm$ 0.005	0.158	-0.002	-1.49	0.0299	0.0300	0.992	94.900 $\pm$ 0.696	0.117 $\pm$ 0.000	99.900 $\pm$ 0.100
LSAM	beta55	0.08	0.082	0.002 $\pm$ 0.001	0.024 $\pm$ 0.015	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.236 $\pm$ 0.011	0.486 $\pm$ 0.014	0.082	0.002	2.69	0.0373	0.0373	0.968	94.700 $\pm$ 0.708	0.147 $\pm$ 0.001	60.200 $\pm$ 1.548
LSAM	beta56	0.08	0.083	0.003 $\pm$ 0.001	0.032 $\pm$ 0.015	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.214 $\pm$ 0.009	0.462 $\pm$ 0.012	0.083	0.003	4.25	0.0381	0.0383	0.973	94.800 $\pm$ 0.702	0.141 $\pm$ 0.001	64.200 $\pm$ 1.516
LSAM	beta57	0.06	0.052	-0.008 $\pm$ 0.001	-0.127 $\pm$ 0.020	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.417 $\pm$ 0.018	0.646 $\pm$ 0.017	0.051	-0.009	-15.29	0.0375	0.0386	0.975	93.600 $\pm$ 0.774	0.145 $\pm$ 0.000	27.800 $\pm$ 1.417
LSAM	beta58	0.06	0.054	-0.006 $\pm$ 0.001	-0.100 $\pm$ 0.020	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.391 $\pm$ 0.017	0.625 $\pm$ 0.017	0.052	-0.008	-13.27	0.0374	0.0382	0.938	92.800 $\pm$ 0.817	0.136 $\pm$ 0.000	34.300 $\pm$ 1.501
QML	beta41	0.20	0.198	-0.002 $\pm$ 0.001	-0.012 $\pm$ 0.005	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.022 $\pm$ 0.001	0.147 $\pm$ 0.004	0.199	-0.001	-0.39	0.0289	0.0289	0.998	94.900 $\pm$ 0.696	0.115 $\pm$ 0.000	100.000 $\pm$ 0.000
QML	beta42	0.20	0.197	-0.003 $\pm$ 0.001	-0.013 $\pm$ 0.005	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.023 $\pm$ 0.001	0.150 $\pm$ 0.004	0.198	-0.002	-1.15	0.0288	0.0289	0.994	95.700 $\pm$ 0.641	0.117 $\pm$ 0.000	100.000 $\pm$ 0.000
QML	beta43	0.20	0.196	-0.004 $\pm$ 0.001	-0.020 $\pm$ 0.004	0.001 $\pm$ 0.000	0.027 $\pm$ 0.001	0.001 $\pm$ 0.000	0.028 $\pm$ 0.001	0.019 $\pm$ 0.001	0.138 $\pm$ 0.004	0.195	-0.005	-2.46	0.0278	0.0282	1.030	95.600 $\pm$ 0.649	0.110 $\pm$ 0.000	100.000 $\pm$ 0.000
QML	beta44	0.11	0.116	0.006 $\pm$ 0.001	0.059 $\pm$ 0.009	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.093 $\pm$ 0.004	0.306 $\pm$ 0.008	0.116	0.006	5.70	0.0322	0.0328	0.981	93.600 $\pm$ 0.774	0.127 $\pm$ 0.001	92.200 $\pm$ 0.848
QML	beta45	0.11	0.113	0.003 $\pm$ 0.001	0.023 $\pm$ 0.009	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.076 $\pm$ 0.003	0.275 $\pm$ 0.007	0.113	0.003	2.85	0.0295	0.0297	0.980	93.900 $\pm$ 0.757	0.116 $\pm$ 0.000	95.800 $\pm$ 0.634
QML	beta46	0.08	0.053	-0.027 $\pm$ 0.001	-0.332 $\pm$ 0.012	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	0.261 $\pm$ 0.010	0.511 $\pm$ 0.011	0.052	-0.028	-34.48	0.0307	0.0413	0.988	82.100 $\pm$ 1.212	0.120 $\pm$ 0.000	38.900 $\pm$ 1.542
QML	beta47	0.08	0.056	-0.024 $\pm$ 0.001	-0.301 $\pm$ 0.012	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.238 $\pm$ 0.009	0.488 $\pm$ 0.011	0.055	-0.025	-31.74	0.0316	0.0406	0.986	84.400 $\pm$ 1.147	0.119 $\pm$ 0.000	44.800 $\pm$ 1.573
QML	beta51	0.16	0.166	0.006 $\pm$ 0.001	0.039 $\pm$ 0.008	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.001 $\pm$ 0.000	0.039 $\pm$ 0.001	0.058 $\pm$ 0.003	0.241 $\pm$ 0.007	0.164	0.004	2.64	0.0355	0.0357	0.964	93.400 $\pm$ 0.785	0.144 $\pm$ 0.000	99.600 $\pm$ 0.200
QML	beta52	0.16	0.147	-0.013 $\pm$ 0.001	-0.084 $\pm$ 0.006	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.043 $\pm$ 0.002	0.206 $\pm$ 0.005	0.146	-0.014	-8.53	0.0301	0.0331	1.020	92.600 $\pm$ 0.828	0.121 $\pm$ 0.000	99.900 $\pm$ 0.100
QML	beta53	0.16	0.142	-0.018 $\pm$ 0.001	-0.115 $\pm$ 0.006	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.051 $\pm$ 0.002	0.226 $\pm$ 0.005	0.142	-0.018	-11.56	0.0308	0.0359	0.995	91.100 $\pm$ 0.900	0.122 $\pm$ 0.000	99.700 $\pm$ 0.173
QML	beta54	0.16	0.157	-0.003 $\pm$ 0.001	-0.016 $\pm$ 0.006	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.036 $\pm$ 0.002	0.189 $\pm$ 0.005	0.158	-0.002	-1.48	0.0303	0.0304	0.992	94.700 $\pm$ 0.708	0.117 $\pm$ 0.000	99.900 $\pm$ 0.100
QML	beta55	0.08	0.083	0.003 $\pm$ 0.001	0.033 $\pm$ 0.015	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.236 $\pm$ 0.011	0.486 $\pm$ 0.014	0.083	0.003	3.93	0.0385	0.0386	0.980	95.000 $\pm$ 0.689	0.149 $\pm$ 0.001	60.100 $\pm$ 1.549
QML	beta56	0.08	0.082	0.002 $\pm$ 0.001	0.020 $\pm$ 0.014	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.208 $\pm$ 0.009	0.456 $\pm$ 0.012	0.082	0.002	2.92	0.0368	0.0368	0.983	94.700 $\pm$ 0.708	0.140 $\pm$ 0.001	63.800 $\pm$ 1.520
QML	beta57	0.06	0.040	-0.020 $\pm$ 0.001	-0.336 $\pm$ 0.017	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.395 $\pm$ 0.015	0.629 $\pm$ 0.014	0.038	-0.022	-35.86	0.0311	0.0378	0.973	86.500 $\pm$ 1.081	0.121 $\pm$ 0.001	21.500 $\pm$ 1.299
QML	beta58	0.06	0.044	-0.016 $\pm$ 0.001	-0.265 $\pm$ 0.016	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.333 $\pm$ 0.013	0.577 $\pm$ 0.014	0.042	-0.018	-30.00	0.0304	0.0353	0.940	88.400 $\pm$ 1.013	0.113 $\pm$ 0.000	30.900 $\pm$ 1.461
UPI	beta41	0.20	0.203	0.003 $\pm$ 0.001	0.013 $\pm$ 0.005	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.022 $\pm$ 0.001	0.148 $\pm$ 0.004	0.204	0.004	1.93	0.0296	0.0299	0.951	93.900 $\pm$ 0.757	0.110 $\pm$ 0.000	100.000 $\pm$ 0.000
UPI	beta42	0.20	0.201	0.001 $\pm$ 0.001	0.005 $\pm$ 0.005	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.022 $\pm$ 0.001	0.149 $\pm$ 0.004	0.201	0.001	0.35	0.0291	0.0291	0.974	94.500 $\pm$ 0.721	0.113 $\pm$ 0.000	100.000 $\pm$ 0.000
UPI	beta43	0.20	0.201	0.001 $\pm$ 0.001	0.007 $\pm$ 0.004	0.001 $\pm$ 0.000	0.027 $\pm$ 0.001	0.001 $\pm$ 0.000	0.027 $\pm$ 0.001	0.019 $\pm$ 0.001	0.137 $\pm$ 0.004	0.201	0.001	0.26	0.0270	0.0270	1.031	95.300 $\pm$ 0.669	0.111 $\pm$ 0.000	100.000

Table S216

Simulation 2 - Condition 72: Full Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.8
(continued)

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Power (%) ± MCSE
UPI	beta47	0.08	0.077	-0.003 ± 0.001	-0.044 ± 0.016	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.244 ± 0.011	0.494 ± 0.014	0.077	-0.003	-3.86	0.0400	0.0401	0.978	94.000 ± 0.751	0.151 ± 0.001	51.800 ± 1.580
UPI	beta51	0.16	0.160	0.000 ± 0.001	0.002 ± 0.007	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.056 ± 0.003	0.237 ± 0.007	0.159	-0.001	-0.72	0.0347	0.0347	0.953	92.800 ± 0.817	0.141 ± 0.000	99.100 ± 0.299
UPI	beta52	0.16	0.162	0.002 ± 0.001	0.014 ± 0.006	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.034 ± 0.002	0.186 ± 0.005	0.163	0.003	2.13	0.0299	0.0301	0.996	95.500 ± 0.656	0.116 ± 0.000	100.000 ± 0.000
UPI	beta53	0.16	0.160	0.000 ± 0.001	0.001 ± 0.006	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.038 ± 0.002	0.195 ± 0.005	0.159	-0.001	-0.59	0.0303	0.0303	0.991	94.800 ± 0.702	0.121 ± 0.000	100.000 ± 0.000
UPI	beta54	0.16	0.159	-0.001 ± 0.001	-0.003 ± 0.006	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.035 ± 0.002	0.188 ± 0.005	0.160	0.000	-0.22	0.0296	0.0296	0.984	94.700 ± 0.708	0.116 ± 0.000	99.900 ± 0.100
UPI	beta55	0.08	0.080	-0.000 ± 0.001	-0.002 ± 0.015	0.001 ± 0.000	0.039 ± 0.001	0.001 ± 0.000	0.039 ± 0.001	0.233 ± 0.011	0.483 ± 0.014	0.080	0.000	0.18	0.0382	0.0382	0.971	94.600 ± 0.715	0.147 ± 0.001	58.400 ± 1.559
UPI	beta56	0.08	0.082	0.002 ± 0.001	0.030 ± 0.015	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.213 ± 0.009	0.462 ± 0.012	0.083	0.003	3.43	0.0390	0.0391	0.976	94.800 ± 0.702	0.141 ± 0.001	64.300 ± 1.515
UPI	beta57	0.06	0.058	-0.002 ± 0.001	-0.040 ± 0.020	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.418 ± 0.019	0.647 ± 0.018	0.056	-0.004	-6.06	0.0384	0.0385	0.984	94.400 ± 0.727	0.149 ± 0.001	30.200 ± 1.452
UPI	beta58	0.06	0.058	-0.002 ± 0.001	-0.036 ± 0.020	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.400 ± 0.017	0.632 ± 0.017	0.056	-0.004	-6.31	0.0387	0.0389	0.951	94.000 ± 0.751	0.141 ± 0.001	35.300 ± 1.511

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S217

Simulation 2 - Condition 73: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.4

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Type I (%) $\pm$ MCSE
LSAM	beta41	0.20	0.201	0.001 $\pm$ 0.003	0.006 $\pm$ 0.015	0.007 $\pm$ 0.000	0.082 $\pm$ 0.002	0.007 $\pm$ 0.000	0.082 $\pm$ 0.003	0.170 $\pm$ 0.010	0.412 $\pm$ 0.014	0.201	0.001	0.47	0.0843	0.0843	1.033	95.448 $\pm$ 0.774	0.334 $\pm$ 0.003	—
LSAM	beta42	0.20	0.203	0.003 $\pm$ 0.003	0.017 $\pm$ 0.015	0.007 $\pm$ 0.000	0.081 $\pm$ 0.002	0.007 $\pm$ 0.000	0.081 $\pm$ 0.003	0.164 $\pm$ 0.009	0.405 $\pm$ 0.013	0.201	0.001	0.70	0.0805	0.0805	1.082	96.966 $\pm$ 0.637	0.344 $\pm$ 0.003	—
LSAM	beta43	0.20	0.204	0.004 $\pm$ 0.003	0.022 $\pm$ 0.015	0.006 $\pm$ 0.000	0.080 $\pm$ 0.002	0.006 $\pm$ 0.000	0.080 $\pm$ 0.003	0.162 $\pm$ 0.009	0.402 $\pm$ 0.014	0.201	0.001	0.29	0.0799	0.0799	0.989	94.207 $\pm$ 0.868	0.312 $\pm$ 0.002	—
LSAM	beta44	0.00	0.001	0.001 $\pm$ 0.005	—	0.016 $\pm$ 0.001	0.125 $\pm$ 0.004	0.016 $\pm$ 0.001	0.125 $\pm$ 0.005	—	—	0.002	0.002	—	0.1123	0.1124	1.038	98.069 $\pm$ 0.511	0.508 $\pm$ 0.010	1.931 $\pm$ 0.511
LSAM	beta45	0.00	0.002	0.002 $\pm$ 0.004	—	0.011 $\pm$ 0.001	0.103 $\pm$ 0.004	0.011 $\pm$ 0.001	0.103 $\pm$ 0.004	—	—	0.001	0.001	—	0.0883	0.0883	1.016	96.552 $\pm$ 0.678	0.409 $\pm$ 0.007	3.448 $\pm$ 0.678
LSAM	beta46	0.00	0.001	0.001 $\pm$ 0.003	—	0.008 $\pm$ 0.001	0.088 $\pm$ 0.003	0.008 $\pm$ 0.001	0.088 $\pm$ 0.003	—	—	0.003	0.003	—	0.0761	0.0762	1.040	97.379 $\pm$ 0.593	0.361 $\pm$ 0.008	2.621 $\pm$ 0.593
LSAM	beta47	0.00	0.002	0.002 $\pm$ 0.003	—	0.007 $\pm$ 0.000	0.083 $\pm$ 0.003	0.007 $\pm$ 0.000	0.083 $\pm$ 0.003	—	—	0.002	0.002	—	0.0747	0.0747	1.043	98.345 $\pm$ 0.474	0.339 $\pm$ 0.006	1.655 $\pm$ 0.474
LSAM	beta51	0.16	0.168	0.008 $\pm$ 0.004	0.050 $\pm$ 0.026	0.013 $\pm$ 0.001	0.113 $\pm$ 0.003	0.013 $\pm$ 0.001	0.114 $\pm$ 0.004	0.504 $\pm$ 0.028	0.710 $\pm$ 0.024	0.165	0.005	3.17	0.1099	0.1100	0.920	94.345 $\pm$ 0.858	0.409 $\pm$ 0.003	—
LSAM	beta52	0.16	0.163	0.003 $\pm$ 0.003	0.019 $\pm$ 0.020	0.007 $\pm$ 0.000	0.086 $\pm$ 0.003	0.007 $\pm$ 0.000	0.086 $\pm$ 0.003	0.291 $\pm$ 0.017	0.539 $\pm$ 0.019	0.163	0.003	1.77	0.0799	0.0800	0.997	95.034 $\pm$ 0.807	0.337 $\pm$ 0.003	—
LSAM	beta53	0.16	0.154	-0.006 $\pm$ 0.003	-0.038 $\pm$ 0.020	0.007 $\pm$ 0.000	0.085 $\pm$ 0.002	0.007 $\pm$ 0.000	0.085 $\pm$ 0.003	0.283 $\pm$ 0.016	0.532 $\pm$ 0.018	0.149	-0.011	-7.08	0.0812	0.0820	1.034	96.000 $\pm$ 0.728	0.344 $\pm$ 0.003	—
LSAM	beta54	0.16	0.161	0.001 $\pm$ 0.003	0.008 $\pm$ 0.020	0.007 $\pm$ 0.000	0.086 $\pm$ 0.002	0.007 $\pm$ 0.000	0.086 $\pm$ 0.003	0.290 $\pm$ 0.017	0.538 $\pm$ 0.018	0.160	0.000	0.24	0.0856	0.0856	0.980	94.897 $\pm$ 0.817	0.331 $\pm$ 0.003	—
LSAM	beta55	0.00	-0.014	-0.014 $\pm$ 0.006	—	0.024 $\pm$ 0.002	0.156 $\pm$ 0.005	0.025 $\pm$ 0.002	0.157 $\pm$ 0.006	—	—	-0.011	-0.011	—	0.1522	0.1526	0.996	96.552 $\pm$ 0.678	0.610 $\pm$ 0.012	3.448 $\pm$ 0.678
LSAM	beta56	0.00	-0.001	-0.001 $\pm$ 0.005	—	0.016 $\pm$ 0.001	0.126 $\pm$ 0.004	0.016 $\pm$ 0.001	0.126 $\pm$ 0.004	—	—	-0.001	-0.001	—	0.1208	0.1208	1.007	96.966 $\pm$ 0.637	0.497 $\pm$ 0.008	3.034 $\pm$ 0.637
LSAM	beta57	0.00	0.011	0.011 $\pm$ 0.004	—	0.009 $\pm$ 0.001	0.095 $\pm$ 0.003	0.009 $\pm$ 0.001	0.096 $\pm$ 0.004	—	—	0.012	0.012	—	0.0880	0.0888	0.967	97.241 $\pm$ 0.608	0.362 $\pm$ 0.007	2.759 $\pm$ 0.608
LSAM	beta58	0.00	0.001	0.001 $\pm$ 0.003	—	0.005 $\pm$ 0.000	0.071 $\pm$ 0.002	0.005 $\pm$ 0.000	0.071 $\pm$ 0.003	—	—	0.003	0.003	—	0.0619	0.0619	1.011	97.517 $\pm$ 0.578	0.282 $\pm$ 0.004	2.483 $\pm$ 0.578
QML	beta41	0.20	0.197	-0.003 $\pm$ 0.003	-0.017 $\pm$ 0.013	0.005 $\pm$ 0.000	0.071 $\pm$ 0.002	0.005 $\pm$ 0.000	0.071 $\pm$ 0.002	0.124 $\pm$ 0.007	0.353 $\pm$ 0.011	0.195	-0.005	-2.41	0.0711	0.0712	1.048	94.345 $\pm$ 0.858	0.290 $\pm$ 0.002	—
QML	beta42	0.20	0.199	-0.001 $\pm$ 0.003	-0.005 $\pm$ 0.013	0.005 $\pm$ 0.000	0.071 $\pm$ 0.002	0.005 $\pm$ 0.000	0.071 $\pm$ 0.002	0.127 $\pm$ 0.007	0.357 $\pm$ 0.011	0.196	-0.004	-1.75	0.0724	0.0725	1.068	95.724 $\pm$ 0.751	0.299 $\pm$ 0.002	—
QML	beta43	0.20	0.199	-0.001 $\pm$ 0.003	-0.007 $\pm$ 0.014	0.005 $\pm$ 0.000	0.074 $\pm$ 0.002	0.005 $\pm$ 0.000	0.074 $\pm$ 0.002	0.136 $\pm$ 0.008	0.368 $\pm$ 0.012	0.195	-0.005	-2.32	0.0736	0.0737	0.982	94.483 $\pm$ 0.848	0.284 $\pm$ 0.002	—
QML	beta44	0.00	-0.014	-0.014 $\pm$ 0.004	—	0.010 $\pm$ 0.001	0.101 $\pm$ 0.003	0.010 $\pm$ 0.001	0.101 $\pm$ 0.004	—	—	-0.016	-0.016	—	0.0965	0.0979	1.060	95.586 $\pm$ 0.763	0.418 $\pm$ 0.008	4.414 $\pm$ 0.763
QML	beta45	0.00	-0.013	-0.013 $\pm$ 0.003	—	0.007 $\pm$ 0.000	0.083 $\pm$ 0.003	0.007 $\pm$ 0.000	0.084 $\pm$ 0.003	—	—	-0.016	-0.016	—	0.0780	0.0795	0.997	95.034 $\pm$ 0.807	0.323 $\pm$ 0.003	4.966 $\pm$ 0.807
QML	beta46	0.00	-0.001	-0.001 $\pm$ 0.002	—	0.004 $\pm$ 0.000	0.065 $\pm$ 0.002	0.004 $\pm$ 0.000	0.065 $\pm$ 0.002	—	—	-0.001	-0.001	—	0.0603	0.0603	1.094	96.414 $\pm$ 0.691	0.279 $\pm$ 0.005	3.586 $\pm$ 0.691
QML	beta47	0.00	-0.003	-0.003 $\pm$ 0.002	—	0.004 $\pm$ 0.000	0.063 $\pm$ 0.002	0.004 $\pm$ 0.000	0.063 $\pm$ 0.002	—	—	-0.003	-0.003	—	0.0583	0.0584	1.074	96.000 $\pm$ 0.728	0.264 $\pm$ 0.006	4.000 $\pm$ 0.728
QML	beta51	0.16	0.172	0.012 $\pm$ 0.004	0.074 $\pm$ 0.025	0.012 $\pm$ 0.001	0.109 $\pm$ 0.003	0.012 $\pm$ 0.001	0.110 $\pm$ 0.004	0.470 $\pm$ 0.027	0.685 $\pm$ 0.024	0.169	0.009	5.41	0.1077	0.1081	1.001	96.276 $\pm$ 0.703	0.428 $\pm$ 0.004	—
QML	beta52	0.16	0.159	-0.001 $\pm$ 0.003	-0.005 $\pm$ 0.019	0.006 $\pm$ 0.000	0.081 $\pm$ 0.002	0.006 $\pm$ 0.000	0.080 $\pm$ 0.003	0.253 $\pm$ 0.014	0.503 $\pm$ 0.017	0.159	-0.001	-0.53	0.0784	0.0784	1.044	94.483 $\pm$ 0.848	0.330 $\pm$ 0.004	—
QML	beta53	0.16	0.151	-0.009 $\pm$ 0.003	-0.057 $\pm$ 0.018	0.006 $\pm$ 0.000	0.078 $\pm$ 0.002	0.006 $\pm$ 0.000	0.079 $\pm$ 0.003	0.243 $\pm$ 0.013	0.493 $\pm$ 0.016	0.147	-0.013	-7.98	0.0786	0.0796	1.057	96.000 $\pm$ 0.728	0.325 $\pm$ 0.002	—
QML	beta54	0.16	0.157	-0.003 $\pm$ 0.003	-0.021 $\pm$ 0.018	0.006 $\pm$ 0.000	0.077 $\pm$ 0.002	0.006 $\pm$ 0.000	0.077 $\pm$ 0.003	0.232 $\pm$ 0.013	0.482 $\pm$ 0.016	0.156	-0.004	-2.53	0.0754	0.0755	1.007	94.345 $\pm$ 0.858	0.304 $\pm$ 0.002	—
QML	beta55	0.00	-0.014	-0.014 $\pm$ 0.005	—	0.015 $\pm$ 0.001	0.124 $\pm$ 0.003	0.016 $\pm$ 0.001	0.125 $\pm$ 0.004	—	—	-0.011	-0.011	—	0.1304	0.1309	1.074	95.172 $\pm$ 0.796	0.521 $\pm$ 0.008	4.828 $\pm$ 0.796
QML	beta56	0.00	-0.014	-0.014 $\pm$ 0.004	—	0.011 $\pm$ 0.001	0.105 $\pm$ 0.003	0.011 $\pm$ 0.001	0.105 $\pm$ 0.003	—	—	-0.014	-0.014	—	0.1034	0.1043	1.011	95.724 $\pm$ 0.751	0.414 $\pm$ 0.004	4.276 $\pm$ 0.751
QML	beta57	0.00	0.003	0.003 $\pm$ 0.003	—	0.005 $\pm$ 0.000	0.073 $\pm$ 0.002	0.005 $\pm$ 0.000	0.073 $\pm$ 0.003	—	—	0.003	0.003	—	0.0693	0.0694	1.018	94.897 $\pm$ 0.817	0.291 $\pm$ 0.005	5.103 $\pm$ 0.817
QML	beta58	0.00	-0.004	-0.004 $\pm$ 0.002	—	0.003 $\pm$ 0.000	0.056 $\pm$ 0.002	0.003 $\pm$ 0.000	0.056 $\pm$ 0.002	—	—	-0.003	-0.003	—	0.0512	0.0513	1.016	96.276 $\pm$ 0.703	0.222 $\pm$ 0.003	3.724 $\pm$ 0.703
UPI	beta41	0.20	0.204	0.004 $\pm$ 0.003	0.021 $\pm$ 0.016	0.007 $\pm$ 0.000	0.086 $\pm$ 0.003	0.007 $\pm$ 0.000	0.086 $\pm$ 0.003	0.184 $\pm$ 0.011	0.429 $\pm$ 0.015	0.207	0.007	3.56	0.0935	0.0938	1.245	97.655 $\pm$ 0.562	0.418 $\pm$ 0.010	—
UPI	beta42	0.20	0.204	0.004 $\pm$ 0.003	0.020 $\pm$ 0.016	0.007 $\pm$ 0.000	0.085 $\pm$ 0.002	0.007 $\pm$ 0.000	0.085 $\pm$ 0.003	0.179 $\pm$ 0.010	0.423 $\pm$ 0.014	0.200	0.000	0.23	0.0896	0.0896	1.301	98.483 $\pm$ 0.454	0.432 $\pm$ 0.010	—
UPI	beta43	0.20	0.204	0.004 $\pm$ 0.003	0.019 $\pm$ 0.015	0.007 $\pm$ 0.000	0.082 $\pm$ 0.002	0.007 $\pm$ 0.000	0.082 $\pm$ 0.003	0.169 $\pm$ 0.010	0.411 $\pm$ 0.014	0.200	0.000	-0.03	0.0815	0.0815	1.153	96.828 $\pm$ 0.651	0.371 $\pm$ 0.007	—
UPI	beta44	0.00	-0.001	-0.001 $\pm$ 0.006	—	0.023 $\pm$ 0.002	0.151 $\pm$ 0.006	0.023 $\pm$ 0.002	0.151 $\pm$ 0.006	—	—	-0.009	-0.009	—	0.1292	0.1295	1.479	99.586 $\pm$ 0.238	0.874 $\pm$ 0.031	0.414 $\pm$ 0.238
UPI	beta45	0.00	0.002	0.002 $\pm$ 0.005	—	0.015 $\pm$ 0.001	0.124 $\pm$ 0.005	0.015 $\pm$ 0.001	0.124 $\pm$ 0.005	—	—	0.003	0.003	—	0.1102	0.1103	1.449	99.172 $\pm$ 0.336	0.705 $\pm$ 0.032	0.828 $\pm$ 0.336
UPI	beta46	0.00	-0.002	-0.002 $\pm$ 0.004	—	0.012 $\pm$ 0.001	0.109 $\pm$ 0.004	0.012 $\pm$ 0.001	0.109 $\pm$ 0.005	—	—	0.001	0.001	—	0.0946	0.0946	1.481	99.586 $\pm$ 0.238	0.631 $\pm$ 0.026	0.414 $\pm$ 0.238

Table S217

Simulation 2 - Condition 73: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.4 (continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	0.005	0.005 ± 0.004	—	0.010 ± 0.001	0.097 ± 0.004	0.010 ± 0.001	0.098 ± 0.004	—	—	0.007	0.007	—	0.0889	0.0892	1.551	99.586 ± 0.238	0.593 ± 0.023	0.414 ± 0.238
UPI	beta51	0.16	0.167	0.007 ± 0.004	0.046 ± 0.025	0.012 ± 0.001	0.108 ± 0.003	0.012 ± 0.001	0.108 ± 0.004	0.455 ± 0.026	0.674 ± 0.023	0.165	0.005	2.91	0.1034	0.1036	1.027	96.552 ± 0.678	0.434 ± 0.004	—
UPI	beta52	0.16	0.163	0.003 ± 0.003	0.018 ± 0.020	0.008 ± 0.000	0.087 ± 0.003	0.008 ± 0.000	0.087 ± 0.003	0.294 ± 0.017	0.542 ± 0.019	0.161	0.001	0.42	0.0827	0.0827	1.122	96.414 ± 0.691	0.381 ± 0.006	—
UPI	beta53	0.16	0.155	-0.005 ± 0.003	-0.031 ± 0.020	0.007 ± 0.000	0.085 ± 0.002	0.007 ± 0.000	0.085 ± 0.003	0.285 ± 0.015	0.534 ± 0.017	0.148	-0.012	-7.46	0.0848	0.0857	1.162	96.966 ± 0.637	0.389 ± 0.006	—
UPI	beta54	0.16	0.162	0.002 ± 0.003	0.014 ± 0.020	0.008 ± 0.000	0.087 ± 0.003	0.008 ± 0.000	0.087 ± 0.003	0.294 ± 0.019	0.542 ± 0.020	0.161	0.001	0.36	0.0843	0.0843	1.097	95.724 ± 0.751	0.373 ± 0.005	—
UPI	beta55	0.00	-0.012	-0.012 ± 0.006	—	0.029 ± 0.002	0.169 ± 0.006	0.029 ± 0.002	0.169 ± 0.007	—	—	-0.008	-0.008	—	0.1517	0.1519	1.334	99.172 ± 0.336	0.884 ± 0.036	0.828 ± 0.336
UPI	beta56	0.00	-0.004	-0.004 ± 0.005	—	0.020 ± 0.001	0.141 ± 0.005	0.020 ± 0.001	0.141 ± 0.006	—	—	-0.005	-0.005	—	0.1277	0.1278	1.283	99.586 ± 0.238	0.710 ± 0.022	0.414 ± 0.238
UPI	beta57	0.00	0.007	0.007 ± 0.004	—	0.011 ± 0.001	0.107 ± 0.004	0.011 ± 0.001	0.107 ± 0.004	—	—	0.007	0.007	—	0.0914	0.0916	1.340	99.172 ± 0.336	0.562 ± 0.028	0.828 ± 0.336
UPI	beta58	0.00	0.000	-0.000 ± 0.003	—	0.006 ± 0.000	0.080 ± 0.003	0.006 ± 0.000	0.080 ± 0.003	—	—	0.000	0.000	—	0.0689	0.0689	1.388	98.621 ± 0.433	0.434 ± 0.014	1.379 ± 0.433

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S218

Simulation 2 - Condition 74: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, $N = 1000$, Reliability = 0.4

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.201	0.001 ± 0.002	0.004 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.058 ± 0.003	0.242 ± 0.007	0.201	0.001	0.54	0.0479	0.0479	1.014	95.441 ± 0.664	0.192 ± 0.001	—
LSAM	beta42	0.20	0.204	0.004 ± 0.002	0.019 ± 0.008	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.064 ± 0.003	0.253 ± 0.007	0.202	0.002	1.16	0.0477	0.0478	1.009	95.745 ± 0.642	0.199 ± 0.001	—
LSAM	beta43	0.20	0.199	-0.001 ± 0.002	-0.004 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.058 ± 0.003	0.241 ± 0.007	0.196	-0.004	-1.82	0.0494	0.0495	0.977	93.718 ± 0.772	0.184 ± 0.001	—
LSAM	beta44	0.00	0.000	0.000 ± 0.002	—	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	—	—	0.002	0.002	—	0.0630	0.0631	0.991	96.353 ± 0.597	0.265 ± 0.002	3.647 ± 0.597
LSAM	beta45	0.00	-0.001	-0.001 ± 0.002	—	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	—	—	-0.001	-0.001	—	0.0508	0.0508	1.001	96.251 ± 0.605	0.219 ± 0.001	3.749 ± 0.605
LSAM	beta46	0.00	0.000	-0.000 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	—	—	0.000	0.000	—	0.0463	0.0463	0.969	95.846 ± 0.635	0.185 ± 0.002	4.154 ± 0.635
LSAM	beta47	0.00	0.002	0.002 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	0.001	0.001	—	0.0458	0.0458	1.026	96.353 ± 0.597	0.180 ± 0.002	3.647 ± 0.597
LSAM	beta51	0.16	0.160	0.000 ± 0.002	0.000 ± 0.013	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.155 ± 0.007	0.393 ± 0.011	0.160	0.000	-0.31	0.0611	0.0611	0.977	94.529 ± 0.724	0.241 ± 0.001	—
LSAM	beta52	0.16	0.161	0.001 ± 0.002	0.005 ± 0.010	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.097 ± 0.004	0.311 ± 0.009	0.160	0.000	-0.07	0.0505	0.0505	0.999	96.049 ± 0.620	0.195 ± 0.001	—
LSAM	beta53	0.16	0.161	0.001 ± 0.002	0.006 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	0.104 ± 0.005	0.322 ± 0.009	0.159	-0.001	-0.41	0.0524	0.0524	0.997	95.339 ± 0.671	0.202 ± 0.001	—
LSAM	beta54	0.16	0.163	0.003 ± 0.002	0.020 ± 0.010	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.098 ± 0.004	0.313 ± 0.009	0.163	0.003	1.82	0.0517	0.0517	0.976	94.833 ± 0.705	0.192 ± 0.001	—
LSAM	beta55	0.00	0.000	-0.000 ± 0.003	—	0.007 ± 0.000	0.085 ± 0.002	0.007 ± 0.000	0.085 ± 0.003	—	—	0.002	0.002	—	0.0803	0.0803	0.957	95.137 ± 0.685	0.318 ± 0.002	4.863 ± 0.685
LSAM	beta56	0.00	0.004	0.004 ± 0.002	—	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	—	—	0.004	0.004	—	0.0632	0.0633	0.992	95.745 ± 0.642	0.264 ± 0.002	4.255 ± 0.642
LSAM	beta57	0.00	-0.001	-0.001 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.002	—	—	-0.002	-0.002	—	0.0479	0.0479	0.967	95.441 ± 0.664	0.193 ± 0.002	4.559 ± 0.664
LSAM	beta58	0.00	0.000	-0.000 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	0.000	0.000	—	0.0383	0.0383	0.968	95.035 ± 0.691	0.155 ± 0.001	4.965 ± 0.691
QML	beta41	0.20	0.197	-0.003 ± 0.001	-0.015 ± 0.007	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.051 ± 0.002	0.225 ± 0.006	0.197	-0.003	-1.47	0.0446	0.0447	1.013	95.339 ± 0.671	0.178 ± 0.001	—
QML	beta42	0.20	0.201	0.001 ± 0.002	0.005 ± 0.008	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.056 ± 0.003	0.236 ± 0.007	0.200	0.000	-0.21	0.0459	0.0459	1.001	95.643 ± 0.650	0.185 ± 0.001	—
QML	beta43	0.20	0.196	-0.004 ± 0.001	-0.018 ± 0.007	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.052 ± 0.002	0.229 ± 0.006	0.195	-0.005	-2.72	0.0472	0.0475	0.988	94.225 ± 0.743	0.177 ± 0.001	—
QML	beta44	0.00	-0.014	-0.014 ± 0.002	—	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	—	—	-0.016	-0.016	—	0.0649	0.0669	1.015	94.428 ± 0.730	0.246 ± 0.002	5.572 ± 0.730
QML	beta45	0.00	-0.015	-0.015 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	—	—	-0.016	-0.016	—	0.0493	0.0518	1.021	94.225 ± 0.743	0.202 ± 0.001	5.775 ± 0.743
QML	beta46	0.00	0.000	-0.000 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	-0.002	-0.002	—	0.0408	0.0408	1.014	95.643 ± 0.650	0.162 ± 0.001	4.357 ± 0.650
QML	beta47	0.00	-0.003	-0.003 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.003	-0.003	—	0.0382	0.0383	1.045	95.339 ± 0.671	0.156 ± 0.001	4.661 ± 0.671
QML	beta51	0.16	0.162	0.002 ± 0.002	0.012 ± 0.012	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.152 ± 0.007	0.390 ± 0.011	0.163	0.003	1.71	0.0600	0.0600	1.004	95.542 ± 0.657	0.245 ± 0.001	—
QML	beta52	0.16	0.158	-0.002 ± 0.002	-0.010 ± 0.010	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.090 ± 0.004	0.300 ± 0.008	0.158	-0.002	-1.12	0.0490	0.0491	1.020	95.643 ± 0.650	0.192 ± 0.001	—
QML	beta53	0.16	0.159	-0.001 ± 0.002	-0.007 ± 0.010	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.096 ± 0.004	0.310 ± 0.009	0.157	-0.003	-1.92	0.0494	0.0495	1.005	95.339 ± 0.671	0.195 ± 0.001	—
QML	beta54	0.16	0.161	0.001 ± 0.002	0.005 ± 0.010	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.090 ± 0.004	0.300 ± 0.008	0.160	0.000	-0.17	0.0490	0.0490	0.985	95.137 ± 0.685	0.186 ± 0.001	—
QML	beta55	0.00	-0.003	-0.003 ± 0.002	—	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	—	—	-0.003	-0.003	—	0.0752	0.0753	1.005	95.238 ± 0.678	0.304 ± 0.002	4.762 ± 0.678
QML	beta56	0.00	-0.009	-0.009 ± 0.002	—	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	—	—	-0.008	-0.008	—	0.0597	0.0602	1.000	94.529 ± 0.724	0.243 ± 0.001	5.471 ± 0.724
QML	beta57	0.00	-0.006	-0.006 ± 0.001	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	-0.005	-0.005	—	0.0430	0.0433	1.009	94.529 ± 0.724	0.175 ± 0.001	5.471 ± 0.724
QML	beta58	0.00	-0.005	-0.005 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	-0.004	-0.004	—	0.0342	0.0345	0.994	94.124 ± 0.749	0.137 ± 0.001	5.876 ± 0.749
UPI	beta41	0.20	0.200	0.000 ± 0.002	0.002 ± 0.008	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.059 ± 0.003	0.243 ± 0.007	0.200	0.000	0.06	0.0469	0.0469	1.053	96.657 ± 0.572	0.200 ± 0.001	—
UPI	beta42	0.20	0.204	0.004 ± 0.002	0.021 ± 0.008	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.066 ± 0.003	0.256 ± 0.007	0.203	0.003	1.74	0.0496	0.0498	1.039	95.846 ± 0.635	0.208 ± 0.001	—
UPI	beta43	0.20	0.200	-0.000 ± 0.002	-0.002 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.058 ± 0.003	0.241 ± 0.007	0.197	-0.003	-1.56	0.0490	0.0491	1.003	94.529 ± 0.724	0.190 ± 0.001	—
UPI	beta44	0.00	0.000	-0.000 ± 0.002	—	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	—	—	0.001	0.001	—	0.0695	0.0695	1.061	98.075 ± 0.437	0.313 ± 0.004	1.925 ± 0.437
UPI	beta45	0.00	0.000	-0.000 ± 0.002	—	0.003 ± 0.000	0.059 ± 0.002	0.003 ± 0.000	0.059 ± 0.002	—	—	-0.001	-0.001	—	0.0535	0.0535	1.086	98.176 ± 0.426	0.251 ± 0.003	1.824 ± 0.426
UPI	beta46	0.00	0.000	0.000 ± 0.002	—	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	—	—	-0.001	-0.001	—	0.0474	0.0474	1.045	97.771 ± 0.470	0.223 ± 0.003	2.229 ± 0.470

Table S218

Simulation 2 - Condition 74: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.4 (continued)

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	0.001	0.001 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.002	—	—	0.001	0.001	—	0.0452	0.0452	1.097	97.974 ± 0.448	0.213 ± 0.003	2.026 ± 0.448
UPI	beta51	0.16	0.160	-0.000 ± 0.002	-0.001 ± 0.012	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.150 ± 0.007	0.387 ± 0.011	0.160	0.000	-0.15	0.0598	0.0598	0.999	95.441 ± 0.664	0.243 ± 0.001	—
UPI	beta52	0.16	0.161	0.001 ± 0.002	0.009 ± 0.010	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.096 ± 0.004	0.311 ± 0.008	0.161	0.001	0.87	0.0500	0.0500	1.023	96.555 ± 0.581	0.199 ± 0.001	—
UPI	beta53	0.16	0.161	0.001 ± 0.002	0.006 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.103 ± 0.005	0.322 ± 0.009	0.161	0.001	0.43	0.0541	0.0541	1.020	96.150 ± 0.612	0.206 ± 0.001	—
UPI	beta54	0.16	0.163	0.003 ± 0.002	0.020 ± 0.010	0.002 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.098 ± 0.004	0.313 ± 0.009	0.163	0.003	2.16	0.0506	0.0507	0.998	95.238 ± 0.678	0.195 ± 0.001	—
UPI	beta55	0.00	0.000	0.000 ± 0.003	—	0.007 ± 0.000	0.086 ± 0.002	0.007 ± 0.000	0.086 ± 0.003	—	—	-0.001	-0.001	—	0.0778	0.0778	1.033	97.163 ± 0.528	0.350 ± 0.004	2.837 ± 0.528
UPI	beta56	0.00	0.004	0.004 ± 0.002	—	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	—	—	0.005	0.005	—	0.0647	0.0649	1.059	97.568 ± 0.490	0.288 ± 0.003	2.432 ± 0.490
UPI	beta57	0.00	-0.001	-0.001 ± 0.002	—	0.003 ± 0.000	0.055 ± 0.002	0.003 ± 0.000	0.055 ± 0.002	—	—	-0.002	-0.002	—	0.0489	0.0490	1.008	97.264 ± 0.519	0.219 ± 0.003	2.736 ± 0.519
UPI	beta58	0.00	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	—	—	0.000	0.000	—	0.0400	0.0400	1.070	97.467 ± 0.500	0.178 ± 0.002	2.533 ± 0.500

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S219

Simulation 2 - Condition 75: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.6

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.200	-0.000 ± 0.002	-0.001 ± 0.009	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.081 ± 0.004	0.285 ± 0.008	0.201	0.001	0.73	0.0569	0.0569	0.983	94.679 ± 0.711	0.219 ± 0.001	—
LSAM	beta42	0.20	0.204	0.004 ± 0.002	0.018 ± 0.009	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.079 ± 0.004	0.280 ± 0.008	0.200	0.000	0.06	0.0555	0.0555	1.033	96.486 ± 0.583	0.227 ± 0.001	—
LSAM	beta43	0.20	0.202	0.002 ± 0.002	0.011 ± 0.009	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.001	0.075 ± 0.003	0.274 ± 0.007	0.201	0.001	0.28	0.0573	0.0573	1.000	94.980 ± 0.692	0.214 ± 0.001	—
LSAM	beta44	0.00	-0.002	-0.002 ± 0.002	—	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	—	—	-0.005	-0.005	—	0.0686	0.0687	0.974	94.679 ± 0.711	0.267 ± 0.002	5.321 ± 0.711
LSAM	beta45	0.00	0.003	0.003 ± 0.002	—	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	—	—	0.003	0.003	—	0.0612	0.0613	0.915	93.173 ± 0.799	0.228 ± 0.001	6.827 ± 0.799
LSAM	beta46	0.00	0.001	0.001 ± 0.002	—	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	—	—	0.001	0.001	—	0.0464	0.0464	0.992	95.382 ± 0.665	0.188 ± 0.001	4.618 ± 0.665
LSAM	beta47	0.00	-0.001	-0.001 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	—	—	0.000	0.000	—	0.0469	0.0469	0.947	94.076 ± 0.748	0.184 ± 0.001	5.924 ± 0.748
LSAM	beta51	0.16	0.161	0.001 ± 0.002	0.005 ± 0.015	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.210 ± 0.010	0.458 ± 0.013	0.161	0.001	0.89	0.0734	0.0734	0.977	94.880 ± 0.698	0.281 ± 0.001	—
LSAM	beta52	0.16	0.161	0.001 ± 0.002	0.004 ± 0.012	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.143 ± 0.007	0.378 ± 0.011	0.161	0.001	0.38	0.0581	0.0581	0.950	93.072 ± 0.805	0.225 ± 0.001	—
LSAM	beta53	0.16	0.160	-0.000 ± 0.002	-0.003 ± 0.012	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.142 ± 0.006	0.377 ± 0.010	0.156	-0.004	-2.42	0.0607	0.0608	0.988	94.980 ± 0.692	0.234 ± 0.001	—
LSAM	beta54	0.16	0.161	0.001 ± 0.002	0.009 ± 0.012	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.135 ± 0.006	0.367 ± 0.010	0.160	0.000	0.27	0.0538	0.0538	0.973	93.976 ± 0.754	0.224 ± 0.001	—
LSAM	beta55	0.00	-0.002	-0.002 ± 0.003	—	0.007 ± 0.000	0.086 ± 0.002	0.007 ± 0.000	0.086 ± 0.002	—	—	-0.005	-0.005	—	0.0841	0.0842	0.946	93.574 ± 0.777	0.320 ± 0.002	6.426 ± 0.777
LSAM	beta56	0.00	0.001	0.001 ± 0.002	—	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	—	—	0.000	0.000	—	0.0663	0.0663	0.977	94.378 ± 0.730	0.276 ± 0.002	5.622 ± 0.730
LSAM	beta57	0.00	0.001	0.001 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	—	—	-0.001	-0.001	—	0.0518	0.0518	0.952	93.675 ± 0.771	0.193 ± 0.001	6.325 ± 0.771
LSAM	beta58	0.00	-0.003	-0.003 ± 0.001	—	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	—	—	-0.003	-0.003	—	0.0408	0.0409	0.961	94.880 ± 0.698	0.164 ± 0.001	5.120 ± 0.698
QML	beta41	0.20	0.197	-0.003 ± 0.002	-0.014 ± 0.009	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	0.074 ± 0.003	0.271 ± 0.007	0.197	-0.003	-1.28	0.0545	0.0546	0.994	94.076 ± 0.748	0.211 ± 0.001	—
QML	beta42	0.20	0.202	0.002 ± 0.002	0.009 ± 0.008	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	0.072 ± 0.003	0.268 ± 0.007	0.199	-0.001	-0.52	0.0537	0.0537	1.043	96.386 ± 0.591	0.219 ± 0.001	—
QML	beta43	0.20	0.200	0.000 ± 0.002	0.001 ± 0.008	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.070 ± 0.003	0.264 ± 0.007	0.199	-0.001	-0.62	0.0543	0.0543	1.011	94.980 ± 0.692	0.210 ± 0.001	—
QML	beta44	0.00	-0.010	-0.010 ± 0.002	—	0.004 ± 0.000	0.066 ± 0.001	0.004 ± 0.000	0.067 ± 0.002	—	—	-0.012	-0.012	—	0.0662	0.0672	0.998	94.578 ± 0.718	0.257 ± 0.002	5.422 ± 0.718
QML	beta45	0.00	-0.003	-0.003 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	—	—	-0.002	-0.002	—	0.0597	0.0598	0.931	92.972 ± 0.810	0.221 ± 0.002	7.028 ± 0.810
QML	beta46	0.00	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	—	—	0.001	0.001	—	0.0417	0.0417	1.014	95.181 ± 0.679	0.174 ± 0.001	4.819 ± 0.679
QML	beta47	0.00	-0.003	-0.003 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	-0.004	-0.004	—	0.0426	0.0428	0.961	93.373 ± 0.788	0.170 ± 0.001	6.627 ± 0.788
QML	beta51	0.16	0.162	0.002 ± 0.002	0.012 ± 0.014	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.204 ± 0.009	0.452 ± 0.012	0.163	0.003	1.78	0.0733	0.0734	1.014	96.185 ± 0.607	0.287 ± 0.001	—
QML	beta52	0.16	0.159	-0.001 ± 0.002	-0.004 ± 0.012	0.004 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	0.138 ± 0.007	0.371 ± 0.011	0.158	-0.002	-0.96	0.0573	0.0573	0.965	93.474 ± 0.783	0.225 ± 0.001	—
QML	beta53	0.16	0.158	-0.002 ± 0.002	-0.010 ± 0.012	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.136 ± 0.006	0.368 ± 0.010	0.156	-0.004	-2.24	0.0603	0.0604	1.002	96.185 ± 0.607	0.232 ± 0.001	—
QML	beta54	0.16	0.160	-0.000 ± 0.002	-0.000 ± 0.011	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.129 ± 0.006	0.359 ± 0.010	0.160	0.000	-0.30	0.0526	0.0526	0.982	93.775 ± 0.766	0.221 ± 0.001	—
QML	beta55	0.00	-0.004	-0.004 ± 0.003	—	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.081 ± 0.002	—	—	-0.005	-0.005	—	0.0814	0.0815	0.986	94.177 ± 0.742	0.313 ± 0.002	5.823 ± 0.742
QML	beta56	0.00	-0.004	-0.004 ± 0.002	—	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.067 ± 0.002	—	—	-0.005	-0.005	—	0.0631	0.0632	1.015	94.378 ± 0.730	0.267 ± 0.002	5.622 ± 0.730
QML	beta57	0.00	-0.002	-0.002 ± 0.002	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	-0.002	-0.002	—	0.0473	0.0473	0.976	93.574 ± 0.777	0.182 ± 0.001	6.426 ± 0.777
QML	beta58	0.00	-0.005	-0.005 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	-0.004	-0.004	—	0.0387	0.0389	0.990	94.880 ± 0.698	0.153 ± 0.001	5.120 ± 0.698
UPI	beta41	0.20	0.200	-0.000 ± 0.002	-0.001 ± 0.009	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.082 ± 0.004	0.286 ± 0.008	0.201	0.001	0.49	0.0582	0.0582	0.994	94.880 ± 0.698	0.223 ± 0.001	—
UPI	beta42	0.20	0.203	0.003 ± 0.002	0.017 ± 0.009	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.079 ± 0.004	0.281 ± 0.008	0.202	0.002	0.78	0.0547	0.0548	1.049	97.490 ± 0.496	0.231 ± 0.001	—
UPI	beta43	0.20	0.202	0.002 ± 0.002	0.012 ± 0.009	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.001	0.075 ± 0.003	0.274 ± 0.007	0.200	0.000	0.10	0.0570	0.0570	1.015	95.181 ± 0.679	0.218 ± 0.001	—
UPI	beta44	0.00	0.000	-0.000 ± 0.002	—	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	—	—	-0.003	-0.003	—	0.0695	0.0695	1.013	95.783 ± 0.637	0.290 ± 0.002	4.217 ± 0.637
UPI	beta45	0.00	0.002	0.002 ± 0.002	—	0.004 ± 0.000	0.067 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	—	—	0.004	0.004	—	0.0623	0.0624	0.946	94.076 ± 0.748	0.247 ± 0.002	5.924 ± 0.748
UPI	beta46	0.00	0.001	0.001 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.002	—	—	0.001	0.001	—	0.0454	0.0454	1.041	95.783 ± 0.637	0.206 ± 0.002	4.217 ± 0.637

Table S219

Simulation 2 - Condition 75: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.6 (continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.002	-0.002 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	—	—	-0.001	-0.001	—	0.0491	0.0491	0.989	95.382 ± 0.665	0.201 ± 0.002	4.618 ± 0.665
UPI	beta51	0.16	0.161	0.001 ± 0.002	0.006 ± 0.014	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.201 ± 0.009	0.449 ± 0.012	0.162	0.002	1.22	0.0732	0.0732	1.010	95.281 ± 0.672	0.284 ± 0.001	—
UPI	beta52	0.16	0.161	0.001 ± 0.002	0.004 ± 0.012	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.144 ± 0.007	0.379 ± 0.011	0.159	-0.001	-0.38	0.0588	0.0588	0.958	93.173 ± 0.799	0.228 ± 0.001	—
UPI	beta53	0.16	0.160	-0.000 ± 0.002	-0.003 ± 0.012	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.141 ± 0.006	0.376 ± 0.010	0.156	-0.004	-2.38	0.0600	0.0601	1.007	95.783 ± 0.637	0.237 ± 0.001	—
UPI	beta54	0.16	0.161	0.001 ± 0.002	0.008 ± 0.012	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.134 ± 0.006	0.366 ± 0.011	0.159	-0.001	-0.51	0.0528	0.0528	0.987	94.076 ± 0.748	0.227 ± 0.001	—
UPI	beta55	0.00	-0.001	-0.001 ± 0.003	—	0.008 ± 0.000	0.088 ± 0.002	0.008 ± 0.000	0.088 ± 0.003	—	—	-0.001	-0.001	—	0.0829	0.0829	0.979	95.683 ± 0.644	0.337 ± 0.003	4.317 ± 0.644
UPI	beta56	0.00	0.001	0.001 ± 0.002	—	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	—	—	0.000	0.000	—	0.0690	0.0690	1.013	95.382 ± 0.665	0.290 ± 0.002	4.618 ± 0.665
UPI	beta57	0.00	0.000	-0.000 ± 0.002	—	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	—	—	-0.002	-0.002	—	0.0516	0.0516	0.974	94.980 ± 0.692	0.205 ± 0.002	5.020 ± 0.692
UPI	beta58	0.00	-0.002	-0.002 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	-0.002	-0.002	—	0.0416	0.0416	1.025	96.386 ± 0.591	0.179 ± 0.003	3.614 ± 0.591

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S220

Simulation 2 - Condition 76: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, $N = 1000$, Reliability = 0.6

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Type I (%) $\pm$ MCSE
LSAM	beta41	0.20	0.200	-0.000 $\pm$ 0.001	-0.000 $\pm$ 0.005	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.028 $\pm$ 0.001	0.166 $\pm$ 0.005	0.199	-0.001	-0.49	0.0319	0.0319	1.040	96.000 $\pm$ 0.620	0.135 $\pm$ 0.000	—
LSAM	beta42	0.20	0.199	-0.001 $\pm$ 0.001	-0.005 $\pm$ 0.006	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.031 $\pm$ 0.001	0.175 $\pm$ 0.005	0.199	-0.001	-0.32	0.0340	0.0341	1.022	95.100 $\pm$ 0.683	0.140 $\pm$ 0.000	—
LSAM	beta43	0.20	0.200	0.000 $\pm$ 0.001	0.001 $\pm$ 0.005	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.029 $\pm$ 0.001	0.170 $\pm$ 0.005	0.201	0.001	0.42	0.0328	0.0329	0.999	94.000 $\pm$ 0.751	0.133 $\pm$ 0.000	—
LSAM	beta44	0.00	-0.001	-0.001 $\pm$ 0.001	—	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	—	—	0.000	0.000	—	0.0422	0.0422	0.974	93.900 $\pm$ 0.757	0.159 $\pm$ 0.001	6.100 $\pm$ 0.757
LSAM	beta45	0.00	-0.001	-0.001 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	—	—	-0.002	-0.002	—	0.0334	0.0334	1.023	94.600 $\pm$ 0.715	0.139 $\pm$ 0.000	5.400 $\pm$ 0.715
LSAM	beta46	0.00	-0.001	-0.001 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	—	—	-0.001	-0.001	—	0.0285	0.0285	0.978	94.600 $\pm$ 0.715	0.111 $\pm$ 0.001	5.400 $\pm$ 0.715
LSAM	beta47	0.00	0.001	0.001 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.028 $\pm$ 0.001	0.001 $\pm$ 0.000	0.028 $\pm$ 0.001	—	—	0.001	0.001	—	0.0278	0.0278	1.009	95.800 $\pm$ 0.634	0.111 $\pm$ 0.001	4.200 $\pm$ 0.634
LSAM	beta51	0.16	0.162	0.002 $\pm$ 0.001	0.011 $\pm$ 0.009	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.078 $\pm$ 0.004	0.280 $\pm$ 0.008	0.162	0.002	0.95	0.0444	0.0444	1.005	95.300 $\pm$ 0.669	0.176 $\pm$ 0.000	—
LSAM	beta52	0.16	0.160	0.000 $\pm$ 0.001	0.003 $\pm$ 0.007	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.049 $\pm$ 0.002	0.221 $\pm$ 0.006	0.160	0.000	0.23	0.0340	0.0340	1.008	95.000 $\pm$ 0.689	0.140 $\pm$ 0.000	—
LSAM	beta53	0.16	0.160	-0.000 $\pm$ 0.001	-0.002 $\pm$ 0.007	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.054 $\pm$ 0.002	0.233 $\pm$ 0.006	0.159	-0.001	-0.68	0.0364	0.0364	0.993	95.000 $\pm$ 0.689	0.145 $\pm$ 0.000	—
LSAM	beta54	0.16	0.161	0.001 $\pm$ 0.001	0.006 $\pm$ 0.007	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.049 $\pm$ 0.002	0.221 $\pm$ 0.006	0.161	0.001	0.86	0.0354	0.0354	1.006	96.100 $\pm$ 0.612	0.139 $\pm$ 0.000	—
LSAM	beta55	0.00	-0.001	-0.001 $\pm$ 0.002	—	0.003 $\pm$ 0.000	0.050 $\pm$ 0.001	0.003 $\pm$ 0.000	0.050 $\pm$ 0.001	—	—	0.002	0.002	—	0.0494	0.0494	0.991	95.200 $\pm$ 0.676	0.195 $\pm$ 0.001	4.800 $\pm$ 0.676
LSAM	beta56	0.00	0.003	0.003 $\pm$ 0.001	—	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	—	—	0.002	0.002	—	0.0475	0.0476	0.947	93.600 $\pm$ 0.774	0.169 $\pm$ 0.001	6.400 $\pm$ 0.774
LSAM	beta57	0.00	0.001	0.001 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	—	—	0.000	0.000	—	0.0302	0.0302	0.991	95.100 $\pm$ 0.683	0.118 $\pm$ 0.001	4.900 $\pm$ 0.683
LSAM	beta58	0.00	-0.001	-0.001 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.027 $\pm$ 0.001	0.001 $\pm$ 0.000	0.027 $\pm$ 0.001	—	—	-0.001	-0.001	—	0.0264	0.0264	0.967	95.100 $\pm$ 0.683	0.101 $\pm$ 0.000	4.900 $\pm$ 0.683
QML	beta41	0.20	0.198	-0.002 $\pm$ 0.001	-0.011 $\pm$ 0.005	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	0.026 $\pm$ 0.001	0.161 $\pm$ 0.005	0.197	-0.003	-1.40	0.0312	0.0313	1.046	96.300 $\pm$ 0.597	0.132 $\pm$ 0.000	—
QML	beta42	0.20	0.198	-0.002 $\pm$ 0.001	-0.011 $\pm$ 0.005	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.029 $\pm$ 0.001	0.170 $\pm$ 0.005	0.199	-0.001	-0.73	0.0338	0.0338	1.021	95.400 $\pm$ 0.662	0.136 $\pm$ 0.000	—
QML	beta43	0.20	0.199	-0.001 $\pm$ 0.001	-0.006 $\pm$ 0.005	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.028 $\pm$ 0.001	0.167 $\pm$ 0.005	0.199	-0.001	-0.56	0.0321	0.0321	1.002	93.700 $\pm$ 0.768	0.131 $\pm$ 0.000	—
QML	beta44	0.00	-0.009	-0.009 $\pm$ 0.001	—	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	—	—	-0.010	-0.010	—	0.0421	0.0431	0.991	93.400 $\pm$ 0.785	0.160 $\pm$ 0.001	6.600 $\pm$ 0.785
QML	beta45	0.00	-0.008	-0.008 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	—	—	-0.008	-0.008	—	0.0328	0.0337	1.013	94.500 $\pm$ 0.721	0.137 $\pm$ 0.001	5.500 $\pm$ 0.721
QML	beta46	0.00	-0.001	-0.001 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.027 $\pm$ 0.001	0.001 $\pm$ 0.000	0.027 $\pm$ 0.001	—	—	0.000	0.000	—	0.0273	0.0273	1.004	95.300 $\pm$ 0.669	0.108 $\pm$ 0.001	4.700 $\pm$ 0.669
QML	beta47	0.00	-0.001	-0.001 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.027 $\pm$ 0.001	0.001 $\pm$ 0.000	0.027 $\pm$ 0.001	—	—	-0.001	-0.001	—	0.0267	0.0267	1.019	95.600 $\pm$ 0.649	0.106 $\pm$ 0.001	4.400 $\pm$ 0.649
QML	beta51	0.16	0.162	0.002 $\pm$ 0.001	0.014 $\pm$ 0.009	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.078 $\pm$ 0.003	0.279 $\pm$ 0.008	0.162	0.002	1.00	0.0454	0.0454	1.018	95.600 $\pm$ 0.649	0.178 $\pm$ 0.000	—
QML	beta52	0.16	0.160	-0.000 $\pm$ 0.001	-0.002 $\pm$ 0.007	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.047 $\pm$ 0.002	0.218 $\pm$ 0.006	0.159	-0.001	-0.53	0.0344	0.0345	1.018	95.500 $\pm$ 0.656	0.139 $\pm$ 0.000	—
QML	beta53	0.16	0.158	-0.002 $\pm$ 0.001	-0.010 $\pm$ 0.007	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.053 $\pm$ 0.002	0.230 $\pm$ 0.006	0.157	-0.003	-1.80	0.0357	0.0358	0.996	94.200 $\pm$ 0.739	0.143 $\pm$ 0.000	—
QML	beta54	0.16	0.160	0.000 $\pm$ 0.001	0.000 $\pm$ 0.007	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.047 $\pm$ 0.002	0.218 $\pm$ 0.006	0.160	0.000	0.03	0.0357	0.0357	1.009	96.100 $\pm$ 0.612	0.138 $\pm$ 0.000	—
QML	beta55	0.00	-0.003	-0.003 $\pm$ 0.002	—	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	—	—	-0.001	-0.001	—	0.0487	0.0487	1.000	95.000 $\pm$ 0.689	0.194 $\pm$ 0.001	5.000 $\pm$ 0.689
QML	beta56	0.00	-0.003	-0.003 $\pm$ 0.001	—	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	—	—	-0.003	-0.003	—	0.0438	0.0439	0.951	93.400 $\pm$ 0.785	0.166 $\pm$ 0.001	6.600 $\pm$ 0.785
QML	beta57	0.00	-0.001	-0.001 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	—	—	-0.001	-0.001	—	0.0292	0.0292	1.002	95.100 $\pm$ 0.683	0.114 $\pm$ 0.001	4.900 $\pm$ 0.683
QML	beta58	0.00	-0.003	-0.003 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.025 $\pm$ 0.001	0.001 $\pm$ 0.000	0.025 $\pm$ 0.001	—	—	-0.003	-0.003	—	0.0251	0.0252	0.981	94.600 $\pm$ 0.715	0.096 $\pm$ 0.000	5.400 $\pm$ 0.715
UPI	beta41	0.20	0.200	-0.000 $\pm$ 0.001	-0.000 $\pm$ 0.005	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.028 $\pm$ 0.001	0.166 $\pm$ 0.005	0.200	0.000	-0.21	0.0321	0.0321	1.043	96.100 $\pm$ 0.612	0.136 $\pm$ 0.000	—
UPI	beta42	0.20	0.199	-0.001 $\pm$ 0.001	-0.005 $\pm$ 0.006	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.031 $\pm$ 0.001	0.175 $\pm$ 0.005	0.200	0.000	-0.18	0.0349	0.0349	1.026	95.300 $\pm$ 0.669	0.141 $\pm$ 0.000	—
UPI	beta43	0.20	0.200	0.000 $\pm$ 0.001	0.002 $\pm$ 0.005	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.029 $\pm$ 0.001	0.169 $\pm$ 0.005	0.201	0.001	0.38	0.0326	0.0326	1.007	94.300 $\pm$ 0.733	0.134 $\pm$ 0.000	—
UPI	beta44	0.00	-0.001	-0.001 $\pm$ 0.001	—	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	—	—	0.001	0.001	—	0.0404	0.0404	0.992	95.200 $\pm$ 0.676	0.163 $\pm$ 0.001	4.800 $\pm$ 0.676
UPI	beta45	0.00	-0.002	-0.002 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	—	—	-0.001	-0.001	—	0.0338	0.0338	1.032	95.500 $\pm$ 0.656	0.142 $\pm$ 0.001	4.500 $\pm$ 0.656
UPI	beta46	0.00	-0.001	-0.001 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	—	—	-0.001	-0.001	—	0.0296	0.0296	0.981	95.700 $\pm$ 0.641	0.115 $\pm$ 0.001	4.300 $\pm$ 0.641

Table S220

Simulation 2 - Condition 76: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.6 (continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	0.001	0.001	—	0.0283	0.0284	1.021	96.200 ± 0.605	0.115 ± 0.001	3.800 ± 0.605
UPI	beta51	0.16	0.162	0.002 ± 0.001	0.010 ± 0.009	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.078 ± 0.003	0.279 ± 0.008	0.162	0.002	1.15	0.0449	0.0450	1.014	95.500 ± 0.656	0.177 ± 0.000	—
UPI	beta52	0.16	0.161	0.001 ± 0.001	0.004 ± 0.007	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.049 ± 0.002	0.221 ± 0.006	0.161	0.001	0.32	0.0340	0.0340	1.012	95.200 ± 0.676	0.140 ± 0.000	—
UPI	beta53	0.16	0.160	-0.000 ± 0.001	-0.002 ± 0.007	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.054 ± 0.002	0.233 ± 0.006	0.159	-0.001	-0.74	0.0371	0.0371	0.998	94.900 ± 0.696	0.146 ± 0.000	—
UPI	beta54	0.16	0.161	0.001 ± 0.001	0.006 ± 0.007	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.049 ± 0.002	0.221 ± 0.006	0.161	0.001	0.66	0.0361	0.0361	1.009	95.900 ± 0.627	0.140 ± 0.000	—
UPI	beta55	0.00	-0.001	-0.001 ± 0.002	—	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	—	—	0.001	0.001	—	0.0477	0.0477	1.010	96.100 ± 0.612	0.197 ± 0.001	3.900 ± 0.612
UPI	beta56	0.00	0.003	0.003 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	0.003	0.003	—	0.0472	0.0472	0.958	94.000 ± 0.751	0.172 ± 0.001	6.000 ± 0.751
UPI	beta57	0.00	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	0.000	0.000	—	0.0306	0.0306	0.999	95.700 ± 0.641	0.120 ± 0.001	4.300 ± 0.641
UPI	beta58	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	—	—	-0.001	-0.001	—	0.0256	0.0257	1.001	95.400 ± 0.662	0.104 ± 0.001	4.600 ± 0.662

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u0214 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S221

Simulation 2 - Condition 77: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.8

Method	Param.	True	Mean-based										Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE	
LSAM	beta41	0.20	0.201	0.001 ± 0.001	0.005 ± 0.007	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.053 ± 0.002	0.230 ± 0.006	0.201	0.001	0.60	0.0466	0.0467	0.983	95.000 ± 0.689	0.177 ± 0.000	—	
LSAM	beta42	0.20	0.201	0.001 ± 0.002	0.004 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.057 ± 0.002	0.238 ± 0.006	0.204	0.004	1.82	0.0459	0.0460	0.977	93.900 ± 0.757	0.182 ± 0.000	—	
LSAM	beta43	0.20	0.203	0.003 ± 0.001	0.014 ± 0.007	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.050 ± 0.002	0.224 ± 0.006	0.205	0.005	2.31	0.0453	0.0455	0.992	94.500 ± 0.721	0.174 ± 0.000	—	
LSAM	beta44	0.00	0.000	-0.000 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	—	—	-0.002	-0.002	—	0.0509	0.0510	0.946	93.700 ± 0.768	0.196 ± 0.001	6.300 ± 0.768	
LSAM	beta45	0.00	0.003	0.003 ± 0.002	—	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	—	—	0.003	0.003	—	0.0472	0.0473	0.929	92.400 ± 0.838	0.175 ± 0.001	7.600 ± 0.838	
LSAM	beta46	0.00	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.000	0.000	—	0.0342	0.0342	0.956	93.600 ± 0.774	0.137 ± 0.001	6.400 ± 0.774	
LSAM	beta47	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.000	0.000	—	0.0368	0.0368	0.951	92.900 ± 0.812	0.136 ± 0.001	7.100 ± 0.812	
LSAM	beta51	0.16	0.157	-0.003 ± 0.002	-0.016 ± 0.012	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.142 ± 0.006	0.377 ± 0.010	0.158	-0.002	-1.54	0.0623	0.0623	0.968	94.000 ± 0.751	0.229 ± 0.001	—	
LSAM	beta52	0.16	0.160	0.000 ± 0.001	0.002 ± 0.009	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.085 ± 0.004	0.291 ± 0.008	0.160	0.000	0.16	0.0450	0.0450	1.004	94.700 ± 0.708	0.184 ± 0.000	—	
LSAM	beta53	0.16	0.160	-0.000 ± 0.002	-0.002 ± 0.010	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.099 ± 0.004	0.314 ± 0.008	0.158	-0.002	-1.23	0.0495	0.0496	0.963	93.700 ± 0.768	0.190 ± 0.001	—	
LSAM	beta54	0.16	0.161	0.001 ± 0.001	0.008 ± 0.009	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.085 ± 0.004	0.292 ± 0.008	0.161	0.001	0.67	0.0448	0.0448	1.003	94.000 ± 0.751	0.184 ± 0.000	—	
LSAM	beta55	0.00	0.002	0.002 ± 0.002	—	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	—	—	0.004	0.004	—	0.0629	0.0630	0.948	93.500 ± 0.780	0.237 ± 0.001	6.500 ± 0.780	
LSAM	beta56	0.00	0.001	0.001 ± 0.002	—	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	—	—	-0.002	-0.002	—	0.0537	0.0538	0.956	94.000 ± 0.751	0.209 ± 0.001	6.000 ± 0.751	
LSAM	beta57	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.003	-0.003	—	0.0371	0.0372	0.948	93.300 ± 0.791	0.143 ± 0.001	6.700 ± 0.791	
LSAM	beta58	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	—	—	-0.002	-0.002	—	0.0345	0.0345	0.950	94.200 ± 0.739	0.126 ± 0.001	5.800 ± 0.739	
QML	beta41	0.20	0.200	-0.000 ± 0.001	-0.001 ± 0.007	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.051 ± 0.002	0.227 ± 0.006	0.200	0.000	-0.03	0.0449	0.0449	0.983	94.700 ± 0.708	0.175 ± 0.000	—	
QML	beta42	0.20	0.200	-0.000 ± 0.001	-0.000 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.055 ± 0.002	0.233 ± 0.006	0.203	0.003	1.31	0.0452	0.0453	0.988	94.000 ± 0.751	0.181 ± 0.001	—	
QML	beta43	0.20	0.202	0.002 ± 0.001	0.011 ± 0.007	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.049 ± 0.002	0.221 ± 0.006	0.204	0.004	1.85	0.0451	0.0452	0.998	94.500 ± 0.721	0.173 ± 0.000	—	
QML	beta44	0.00	-0.004	-0.004 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	—	—	-0.005	-0.005	—	0.0506	0.0509	0.965	94.200 ± 0.739	0.197 ± 0.001	5.800 ± 0.739	
QML	beta45	0.00	0.000	0.000 ± 0.001	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	0.000	0.000	—	0.0458	0.0458	0.943	93.000 ± 0.807	0.174 ± 0.001	7.000 ± 0.807	
QML	beta46	0.00	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	0.000	0.000	—	0.0339	0.0339	0.974	94.100 ± 0.745	0.135 ± 0.001	5.900 ± 0.745	
QML	beta47	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	-0.001	-0.001	—	0.0355	0.0355	0.962	92.900 ± 0.812	0.133 ± 0.001	7.100 ± 0.812	
QML	beta51	0.16	0.158	-0.002 ± 0.002	-0.014 ± 0.012	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.141 ± 0.006	0.376 ± 0.010	0.157	-0.003	-1.61	0.0607	0.0607	0.984	94.400 ± 0.727	0.232 ± 0.001	—	
QML	beta52	0.16	0.160	-0.000 ± 0.001	-0.001 ± 0.009	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.085 ± 0.004	0.291 ± 0.008	0.160	0.000	-0.26	0.0444	0.0444	1.011	94.900 ± 0.696	0.184 ± 0.000	—	
QML	beta53	0.16	0.159	-0.001 ± 0.002	-0.005 ± 0.010	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.097 ± 0.004	0.312 ± 0.008	0.157	-0.003	-1.85	0.0495	0.0496	0.970	94.000 ± 0.751	0.190 ± 0.001	—	
QML	beta54	0.16	0.161	0.001 ± 0.001	0.004 ± 0.009	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.084 ± 0.004	0.290 ± 0.008	0.161	0.001	0.61	0.0451	0.0451	1.006	94.400 ± 0.727	0.183 ± 0.001	—	
QML	beta55	0.00	0.001	0.001 ± 0.002	—	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	—	—	0.004	0.004	—	0.0629	0.0630	0.956	93.600 ± 0.774	0.237 ± 0.001	6.400 ± 0.774	
QML	beta56	0.00	-0.002	-0.002 ± 0.002	—	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	—	—	-0.004	-0.004	—	0.0546	0.0548	0.966	94.400 ± 0.727	0.209 ± 0.001	5.600 ± 0.727	
QML	beta57	0.00	-0.003	-0.003 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.003	-0.003	—	0.0363	0.0364	0.959	93.300 ± 0.791	0.141 ± 0.001	6.700 ± 0.791	
QML	beta58	0.00	-0.003	-0.003 ± 0.001	—	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	—	—	-0.003	-0.003	—	0.0334	0.0335	0.955	94.500 ± 0.721	0.124 ± 0.001	5.500 ± 0.721	
UPI	beta41	0.20	0.201	0.001 ± 0.001	0.005 ± 0.007	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.053 ± 0.002	0.231 ± 0.006	0.202	0.002	0.89	0.0467	0.0467	0.979	95.100 ± 0.683	0.177 ± 0.000	—	
UPI	beta42	0.20	0.201	0.001 ± 0.002	0.004 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.057 ± 0.003	0.239 ± 0.006	0.203	0.003	1.60	0.0458	0.0459	0.975	94.200 ± 0.739	0.183 ± 0.000	—	
UPI	beta43	0.20	0.203	0.003 ± 0.001	0.015 ± 0.007	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.050 ± 0.002	0.224 ± 0.006	0.205	0.005	2.34	0.0459	0.0461	0.996	94.500 ± 0.721	0.175 ± 0.000	—	
UPI	beta44	0.00	0.000	0.000 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	—	—	-0.003	-0.003	—	0.0513	0.0513	0.955	94.200 ± 0.739	0.200 ± 0.001	5.800 ± 0.739	
UPI	beta45	0.00	0.003	0.003 ± 0.002	—	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	—	—	0.003	0.003	—	0.0499	0.0499	0.938	93.100 ± 0.801	0.178 ± 0.001	6.900 ± 0.801	
UPI	beta46	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.000	0.000	—	0.0346	0.0346	0.971	93.900 ± 0.757	0.140 ± 0.001	6.100 ± 0.757	

Table S221

Simulation 2 - Condition 77: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.8 (continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.001	0.001	—	0.0370	0.0370	0.957	92.700 ± 0.823	0.140 ± 0.001	7.300 ± 0.823
UPI	beta51	0.16	0.157	-0.003 ± 0.002	-0.016 ± 0.012	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.141 ± 0.006	0.375 ± 0.010	0.157	-0.003	-1.98	0.0611	0.0612	0.978	94.300 ± 0.733	0.230 ± 0.001	—
UPI	beta52	0.16	0.160	0.000 ± 0.001	0.003 ± 0.009	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.085 ± 0.004	0.291 ± 0.008	0.160	0.000	0.11	0.0443	0.0443	1.007	94.500 ± 0.721	0.184 ± 0.000	—
UPI	beta53	0.16	0.160	-0.000 ± 0.002	-0.001 ± 0.010	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.099 ± 0.004	0.314 ± 0.008	0.158	-0.002	-0.99	0.0503	0.0504	0.966	93.700 ± 0.768	0.191 ± 0.001	—
UPI	beta54	0.16	0.161	0.001 ± 0.001	0.008 ± 0.009	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.085 ± 0.004	0.292 ± 0.008	0.161	0.001	0.74	0.0440	0.0440	1.006	94.100 ± 0.745	0.184 ± 0.000	—
UPI	beta55	0.00	0.002	0.002 ± 0.002	—	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	—	—	0.003	0.003	—	0.0635	0.0635	0.957	93.800 ± 0.763	0.239 ± 0.001	6.200 ± 0.763
UPI	beta56	0.00	0.000	0.000 ± 0.002	—	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	—	—	-0.003	-0.003	—	0.0534	0.0535	0.968	94.600 ± 0.715	0.213 ± 0.001	5.400 ± 0.715
UPI	beta57	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.039 ± 0.001	0.001 ± 0.000	0.039 ± 0.001	—	—	-0.001	-0.001	—	0.0365	0.0365	0.955	93.600 ± 0.774	0.145 ± 0.001	6.400 ± 0.774
UPI	beta58	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	—	—	-0.002	-0.002	—	0.0342	0.0343	0.958	94.800 ± 0.702	0.128 ± 0.001	5.200 ± 0.702

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S222

Simulation 2 - Condition 78: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.8

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.200	-0.000 ± 0.001	-0.002 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.021 ± 0.001	0.146 ± 0.004	0.200	0.000	0.12	0.0290	0.0290	0.975	93.700 ± 0.768	0.111 ± 0.000	—
LSAM	beta42	0.20	0.200	-0.000 ± 0.001	-0.002 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.022 ± 0.001	0.147 ± 0.004	0.200	0.000	-0.18	0.0281	0.0281	0.991	94.300 ± 0.733	0.115 ± 0.000	—
LSAM	beta43	0.20	0.200	-0.000 ± 0.001	-0.002 ± 0.004	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.020 ± 0.001	0.142 ± 0.004	0.199	-0.001	-0.52	0.0289	0.0289	0.985	95.000 ± 0.689	0.110 ± 0.000	—
LSAM	beta44	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	—	—	-0.002	-0.002	—	0.0326	0.0326	0.968	94.500 ± 0.721	0.122 ± 0.000	5.500 ± 0.721
LSAM	beta45	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	-0.001	-0.001	—	0.0310	0.0310	0.949	94.700 ± 0.708	0.109 ± 0.000	5.300 ± 0.708
LSAM	beta46	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.022 ± 0.001	0.001 ± 0.000	0.022 ± 0.001	—	—	0.001	0.001	—	0.0221	0.0221	0.977	94.200 ± 0.739	0.086 ± 0.000	5.800 ± 0.739
LSAM	beta47	0.00	0.001	0.001 ± 0.001	—	0.000 ± 0.000	0.022 ± 0.001	0.001 ± 0.000	0.022 ± 0.001	—	—	0.001	0.001	—	0.0221	0.0221	0.973	93.400 ± 0.785	0.085 ± 0.000	6.600 ± 0.785
LSAM	beta51	0.16	0.159	-0.001 ± 0.001	-0.004 ± 0.007	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.054 ± 0.002	0.233 ± 0.006	0.159	-0.001	-0.62	0.0387	0.0387	0.990	94.300 ± 0.733	0.145 ± 0.000	—
LSAM	beta52	0.16	0.161	0.001 ± 0.001	0.004 ± 0.006	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.034 ± 0.002	0.184 ± 0.005	0.161	0.001	0.51	0.0277	0.0278	1.000	95.100 ± 0.683	0.116 ± 0.000	—
LSAM	beta53	0.16	0.162	0.002 ± 0.001	0.013 ± 0.006	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.036 ± 0.002	0.190 ± 0.005	0.161	0.001	0.67	0.0307	0.0307	1.007	94.800 ± 0.702	0.120 ± 0.000	—
LSAM	beta54	0.16	0.159	-0.001 ± 0.001	-0.004 ± 0.006	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.033 ± 0.001	0.183 ± 0.005	0.158	-0.002	-1.11	0.0294	0.0294	1.006	94.700 ± 0.708	0.115 ± 0.000	—
LSAM	beta55	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	0.000	0.000	—	0.0394	0.0394	1.008	95.100 ± 0.683	0.149 ± 0.000	4.900 ± 0.683
LSAM	beta56	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	—	—	0.000	0.000	—	0.0328	0.0328	0.995	93.700 ± 0.768	0.132 ± 0.000	6.300 ± 0.768
LSAM	beta57	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	—	—	0.000	0.000	—	0.0242	0.0242	0.979	94.000 ± 0.751	0.090 ± 0.000	6.000 ± 0.751
LSAM	beta58	0.00	0.001	0.001 ± 0.001	—	0.000 ± 0.000	0.021 ± 0.000	0.000 ± 0.000	0.021 ± 0.001	—	—	0.001	0.001	—	0.0198	0.0199	0.965	93.100 ± 0.801	0.079 ± 0.000	6.900 ± 0.801
QML	beta41	0.20	0.199	-0.001 ± 0.001	-0.006 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.021 ± 0.001	0.143 ± 0.004	0.199	-0.001	-0.53	0.0289	0.0290	0.979	93.700 ± 0.768	0.110 ± 0.000	—
QML	beta42	0.20	0.199	-0.001 ± 0.001	-0.004 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.021 ± 0.001	0.146 ± 0.004	0.199	-0.001	-0.31	0.0277	0.0277	0.991	94.100 ± 0.745	0.113 ± 0.000	—
QML	beta43	0.20	0.199	-0.001 ± 0.001	-0.006 ± 0.004	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.020 ± 0.001	0.141 ± 0.004	0.198	-0.002	-0.93	0.0293	0.0294	0.987	95.100 ± 0.683	0.109 ± 0.000	—
QML	beta44	0.00	-0.005	-0.005 ± 0.001	—	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	—	—	-0.005	-0.005	—	0.0321	0.0325	0.975	93.900 ± 0.757	0.123 ± 0.001	6.100 ± 0.757
QML	beta45	0.00	-0.003	-0.003 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	-0.003	-0.003	—	0.0309	0.0310	0.951	94.400 ± 0.727	0.109 ± 0.000	5.600 ± 0.727
QML	beta46	0.00	0.001	0.001 ± 0.001	—	0.000 ± 0.000	0.022 ± 0.000	0.000 ± 0.000	0.022 ± 0.001	—	—	0.001	0.001	—	0.0214	0.0214	0.991	94.100 ± 0.745	0.085 ± 0.001	5.900 ± 0.745
QML	beta47	0.00	0.000	0.000 ± 0.001	—	0.000 ± 0.000	0.022 ± 0.000	0.000 ± 0.000	0.022 ± 0.001	—	—	0.000	0.000	—	0.0216	0.0216	0.980	93.600 ± 0.774	0.084 ± 0.000	6.400 ± 0.774
QML	beta51	0.16	0.160	-0.000 ± 0.001	-0.003 ± 0.007	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.055 ± 0.002	0.234 ± 0.006	0.160	0.000	-0.17	0.0382	0.0382	0.994	94.500 ± 0.721	0.146 ± 0.000	—
QML	beta52	0.16	0.160	0.000 ± 0.001	0.002 ± 0.006	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.034 ± 0.002	0.184 ± 0.005	0.161	0.001	0.55	0.0282	0.0282	1.000	95.000 ± 0.689	0.115 ± 0.000	—
QML	beta53	0.16	0.162	0.002 ± 0.001	0.010 ± 0.006	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.036 ± 0.002	0.188 ± 0.005	0.161	0.001	0.49	0.0302	0.0302	1.009	94.600 ± 0.715	0.119 ± 0.000	—
QML	beta54	0.16	0.159	-0.001 ± 0.001	-0.006 ± 0.006	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.033 ± 0.001	0.182 ± 0.005	0.158	-0.002	-1.30	0.0290	0.0290	1.009	94.900 ± 0.696	0.115 ± 0.000	—
QML	beta55	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.002	-0.002	—	0.0395	0.0395	1.016	95.200 ± 0.676	0.150 ± 0.001	4.800 ± 0.676
QML	beta56	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	—	—	-0.002	-0.002	—	0.0328	0.0329	1.002	93.800 ± 0.763	0.132 ± 0.000	6.200 ± 0.763
QML	beta57	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	—	—	-0.001	-0.001	—	0.0237	0.0237	0.986	93.700 ± 0.768	0.089 ± 0.000	6.300 ± 0.768
QML	beta58	0.00	0.000	0.000 ± 0.001	—	0.000 ± 0.000	0.020 ± 0.000	0.000 ± 0.000	0.020 ± 0.001	—	—	0.000	0.000	—	0.0197	0.0197	0.973	93.400 ± 0.785	0.078 ± 0.000	6.600 ± 0.785
UPI	beta41	0.20	0.200	-0.000 ± 0.001	-0.002 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.021 ± 0.001	0.145 ± 0.004	0.200	0.000	0.11	0.0295	0.0295	0.975	94.000 ± 0.751	0.111 ± 0.000	—
UPI	beta42	0.20	0.200	-0.000 ± 0.001	-0.002 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.022 ± 0.001	0.147 ± 0.004	0.200	0.000	-0.15	0.0279	0.0279	0.991	94.300 ± 0.733	0.115 ± 0.000	—
UPI	beta43	0.20	0.200	-0.000 ± 0.001	-0.002 ± 0.004	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.020 ± 0.001	0.142 ± 0.004	0.199	-0.001	-0.50	0.0288	0.0288	0.987	95.200 ± 0.676	0.110 ± 0.000	—
UPI	beta44	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	—	—	-0.002	-0.002	—	0.0328	0.0329	0.969	94.700 ± 0.708	0.123 ± 0.000	5.300 ± 0.708
UPI	beta45	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	0.000	0.000	—	0.0305	0.0305	0.952	94.300 ± 0.733	0.110 ± 0.000	5.700 ± 0.733
UPI	beta46	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	—	—	0.001	0.001	—	0.0218	0.0218	0.979	94.400 ± 0.727	0.086 ± 0.000	5.600 ± 0.727

Table S222

Simulation 2 - Condition 78: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.8 (continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	0.001	0.001 ± 0.001	—	0.000 ± 0.000	0.022 ± 0.001	0.000 ± 0.000	0.022 ± 0.001	—	—	0.001	0.001	—	0.0217	0.0217	0.982	93.800 ± 0.763	0.086 ± 0.000	6.200 ± 0.763
UPI	beta51	0.16	0.159	-0.001 ± 0.001	-0.004 ± 0.007	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.054 ± 0.002	0.233 ± 0.006	0.159	-0.001	-0.54	0.0387	0.0387	0.993	94.600 ± 0.715	0.145 ± 0.000	—
UPI	beta52	0.16	0.161	0.001 ± 0.001	0.004 ± 0.006	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.034 ± 0.002	0.184 ± 0.005	0.161	0.001	0.35	0.0280	0.0280	1.001	95.200 ± 0.676	0.116 ± 0.000	—
UPI	beta53	0.16	0.162	0.002 ± 0.001	0.013 ± 0.006	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.036 ± 0.002	0.190 ± 0.005	0.161	0.001	0.84	0.0312	0.0312	1.009	95.000 ± 0.689	0.120 ± 0.000	—
UPI	beta54	0.16	0.159	-0.001 ± 0.001	-0.004 ± 0.006	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.033 ± 0.001	0.183 ± 0.005	0.158	-0.002	-0.99	0.0293	0.0294	1.007	94.600 ± 0.715	0.116 ± 0.000	—
UPI	beta55	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	0.000	0.000	—	0.0395	0.0395	1.016	95.700 ± 0.641	0.150 ± 0.000	4.300 ± 0.641
UPI	beta56	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	—	—	0.000	0.000	—	0.0328	0.0328	0.997	94.300 ± 0.733	0.133 ± 0.000	5.700 ± 0.733
UPI	beta57	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.024 ± 0.001	0.001 ± 0.000	0.024 ± 0.001	—	—	0.001	0.001	—	0.0238	0.0238	0.980	93.700 ± 0.768	0.091 ± 0.000	6.300 ± 0.768
UPI	beta58	0.00	0.001	0.001 ± 0.001	—	0.000 ± 0.000	0.021 ± 0.000	0.000 ± 0.000	0.021 ± 0.001	—	—	0.002	0.002	—	0.0199	0.0199	0.974	93.400 ± 0.785	0.080 ± 0.000	6.600 ± 0.785

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u02014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S223

Simulation 2 - Condition 79: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.4

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.205	0.005 ± 0.007	0.025 ± 0.034	0.027 ± 0.002	0.164 ± 0.006	0.027 ± 0.002	0.164 ± 0.007	0.671 ± 0.047	0.819 ± 0.033	0.200	0.000	0.07	0.1655	0.1655	1.070	97.470 ± 0.645	0.688 ± 0.015	—
LSAM	beta42	0.20	0.195	-0.005 ± 0.007	-0.026 ± 0.036	0.031 ± 0.002	0.176 ± 0.006	0.031 ± 0.002	0.176 ± 0.007	0.770 ± 0.054	0.878 ± 0.036	0.205	0.005	2.26	0.1735	0.1735	1.059	96.459 ± 0.759	0.729 ± 0.014	—
LSAM	beta43	0.20	0.203	0.003 ± 0.003	0.014 ± 0.017	0.007 ± 0.000	0.085 ± 0.003	0.007 ± 0.000	0.085 ± 0.003	0.179 ± 0.012	0.424 ± 0.017	0.199	-0.001	-0.72	0.0837	0.0837	1.051	96.965 ± 0.705	0.349 ± 0.005	—
LSAM	beta44	0.00	0.009	0.009 ± 0.006	—	0.019 ± 0.002	0.136 ± 0.006	0.019 ± 0.002	0.136 ± 0.006	—	—	0.011	0.011	—	0.1189	0.1194	1.070	98.145 ± 0.554	0.571 ± 0.016	1.855 ± 0.554
LSAM	beta45	0.00	0.001	0.001 ± 0.005	—	0.015 ± 0.001	0.121 ± 0.005	0.015 ± 0.001	0.121 ± 0.005	—	—	0.003	0.003	—	0.1096	0.1097	1.054	98.145 ± 0.554	0.499 ± 0.013	1.855 ± 0.554
LSAM	beta46	0.00	0.001	0.001 ± 0.003	—	0.005 ± 0.000	0.071 ± 0.003	0.005 ± 0.000	0.071 ± 0.003	—	—	-0.004	-0.004	—	0.0634	0.0635	1.036	97.808 ± 0.601	0.288 ± 0.008	2.192 ± 0.601
LSAM	beta47	0.00	0.002	0.002 ± 0.003	—	0.006 ± 0.000	0.075 ± 0.003	0.006 ± 0.000	0.075 ± 0.003	—	—	-0.001	-0.001	—	0.0636	0.0636	0.992	97.976 ± 0.578	0.293 ± 0.008	2.024 ± 0.578
LSAM	beta51	0.16	0.167	0.007 ± 0.005	0.041 ± 0.028	0.012 ± 0.001	0.110 ± 0.003	0.012 ± 0.001	0.110 ± 0.004	0.474 ± 0.029	0.689 ± 0.025	0.166	0.006	3.77	0.1053	0.1055	0.941	94.772 ± 0.914	0.406 ± 0.003	—
LSAM	beta52	0.16	0.156	-0.004 ± 0.006	-0.028 ± 0.040	0.024 ± 0.002	0.155 ± 0.005	0.024 ± 0.002	0.154 ± 0.006	0.932 ± 0.060	0.966 ± 0.037	0.154	-0.006	-3.74	0.1553	0.1554	1.054	96.965 ± 0.705	0.639 ± 0.009	—
LSAM	beta53	0.16	0.160	0.000 ± 0.005	0.002 ± 0.030	0.014 ± 0.001	0.117 ± 0.004	0.014 ± 0.001	0.117 ± 0.005	0.533 ± 0.038	0.730 ± 0.030	0.154	-0.006	-3.74	0.1137	0.1138	1.028	96.796 ± 0.723	0.471 ± 0.006	—
LSAM	beta54	0.16	0.161	0.001 ± 0.007	0.009 ± 0.041	0.025 ± 0.002	0.159 ± 0.006	0.025 ± 0.002	0.159 ± 0.006	0.982 ± 0.069	0.991 ± 0.041	0.157	-0.003	-1.66	0.1503	0.1504	1.055	95.953 ± 0.809	0.656 ± 0.012	—
LSAM	beta55	0.00	-0.008	-0.008 ± 0.006	—	0.023 ± 0.002	0.151 ± 0.006	0.023 ± 0.002	0.151 ± 0.007	—	—	-0.009	-0.009	—	0.1386	0.1389	1.118	98.482 ± 0.502	0.662 ± 0.015	1.518 ± 0.502
LSAM	beta56	0.00	0.006	0.006 ± 0.005	—	0.015 ± 0.001	0.124 ± 0.005	0.015 ± 0.001	0.124 ± 0.005	—	—	0.008	0.008	—	0.1140	0.1143	1.039	97.470 ± 0.645	0.505 ± 0.010	2.530 ± 0.645
LSAM	beta57	0.00	0.006	0.006 ± 0.003	—	0.005 ± 0.000	0.071 ± 0.003	0.005 ± 0.000	0.071 ± 0.003	—	—	0.005	0.005	—	0.0652	0.0653	1.081	97.976 ± 0.578	0.302 ± 0.007	2.024 ± 0.578
LSAM	beta58	0.00	-0.002	-0.002 ± 0.003	—	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.062 ± 0.003	—	—	0.001	0.001	—	0.0525	0.0525	1.013	97.639 ± 0.623	0.247 ± 0.007	2.361 ± 0.623
QML	beta41	0.20	0.175	-0.025 ± 0.003	-0.124 ± 0.016	0.006 ± 0.000	0.080 ± 0.002	0.007 ± 0.000	0.084 ± 0.003	0.176 ± 0.009	0.419 ± 0.014	0.175	-0.025	-12.38	0.0836	0.0872	1.037	93.929 ± 0.981	0.326 ± 0.003	—
QML	beta42	0.20	0.171	-0.029 ± 0.004	-0.147 ± 0.018	0.008 ± 0.000	0.087 ± 0.003	0.008 ± 0.001	0.092 ± 0.003	0.212 ± 0.013	0.460 ± 0.017	0.176	-0.024	-11.78	0.0869	0.0900	1.002	92.243 ± 1.098	0.343 ± 0.003	—
QML	beta43	0.20	0.204	0.004 ± 0.003	0.022 ± 0.015	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.003	0.133 ± 0.008	0.364 ± 0.014	0.204	0.004	1.79	0.0746	0.0747	1.017	95.278 ± 0.871	0.290 ± 0.002	—
QML	beta44	0.00	-0.019	-0.019 ± 0.004	—	0.010 ± 0.001	0.102 ± 0.004	0.011 ± 0.001	0.104 ± 0.004	—	—	-0.017	-0.017	—	0.1009	0.1023	1.011	95.278 ± 0.871	0.405 ± 0.008	4.722 ± 0.871
QML	beta45	0.00	0.006	0.006 ± 0.003	—	0.007 ± 0.000	0.084 ± 0.003	0.007 ± 0.000	0.084 ± 0.003	—	—	0.009	0.009	—	0.0832	0.0837	0.986	95.784 ± 0.825	0.324 ± 0.004	4.216 ± 0.825
QML	beta46	0.00	0.041	0.041 ± 0.002	—	0.003 ± 0.000	0.059 ± 0.002	0.005 ± 0.000	0.072 ± 0.003	—	—	0.037	0.037	—	0.0555	0.0666	0.964	90.556 ± 1.201	0.223 ± 0.004	9.444 ± 1.201
QML	beta47	0.00	0.045	0.045 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.002	0.006 ± 0.000	0.075 ± 0.003	—	—	0.037	0.037	—	0.0533	0.0651	0.961	87.184 ± 1.373	0.227 ± 0.006	12.816 ± 1.373
QML	beta51	0.16	0.177	0.017 ± 0.004	0.104 ± 0.028	0.012 ± 0.001	0.108 ± 0.003	0.012 ± 0.001	0.109 ± 0.004	0.465 ± 0.030	0.682 ± 0.026	0.175	0.015	9.17	0.1056	0.1066	1.002	95.447 ± 0.856	0.424 ± 0.004	—
QML	beta52	0.16	0.137	-0.023 ± 0.004	-0.145 ± 0.023	0.008 ± 0.000	0.088 ± 0.003	0.008 ± 0.001	0.091 ± 0.003	0.324 ± 0.020	0.569 ± 0.021	0.141	-0.019	-12.12	0.0927	0.0947	1.044	94.772 ± 0.914	0.361 ± 0.003	—
QML	beta53	0.16	0.158	-0.002 ± 0.004	-0.015 ± 0.023	0.008 ± 0.000	0.088 ± 0.003	0.008 ± 0.000	0.088 ± 0.003	0.304 ± 0.019	0.551 ± 0.020	0.155	-0.005	-3.41	0.0880	0.0882	1.048	96.121 ± 0.793	0.363 ± 0.003	—
QML	beta54	0.16	0.131	-0.029 ± 0.004	-0.179 ± 0.022	0.007 ± 0.000	0.086 ± 0.003	0.008 ± 0.000	0.090 ± 0.003	0.318 ± 0.019	0.564 ± 0.020	0.133	-0.027	-16.71	0.0859	0.0900	1.041	94.098 ± 0.968	0.349 ± 0.003	—
QML	beta55	0.00	-0.023	-0.023 ± 0.005	—	0.014 ± 0.001	0.119 ± 0.004	0.015 ± 0.001	0.121 ± 0.005	—	—	-0.025	-0.025	—	0.1234	0.1259	1.109	95.110 ± 0.886	0.515 ± 0.011	4.890 ± 0.886
QML	beta56	0.00	0.026	0.026 ± 0.004	—	0.009 ± 0.001	0.097 ± 0.003	0.010 ± 0.001	0.100 ± 0.004	—	—	0.024	0.024	—	0.0917	0.0949	0.989	94.941 ± 0.900	0.376 ± 0.005	5.059 ± 0.900
QML	beta57	0.00	0.035	0.035 ± 0.003	—	0.004 ± 0.000	0.062 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	—	—	0.034	0.034	—	0.0606	0.0694	1.079	93.929 ± 0.981	0.264 ± 0.006	6.071 ± 0.981
QML	beta58	0.00	0.029	0.029 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.002	0.003 ± 0.000	0.056 ± 0.002	—	—	0.027	0.027	—	0.0436	0.0510	0.978	91.906 ± 1.120	0.186 ± 0.003	8.094 ± 1.120
UPI	beta41	0.20	0.207	0.007 ± 0.008	0.035 ± 0.038	0.034 ± 0.003	0.183 ± 0.007	0.034 ± 0.003	0.183 ± 0.008	0.840 ± 0.064	0.917 ± 0.040	0.206	0.006	3.08	0.1946	0.1947	1.390	98.482 ± 0.502	0.999 ± 0.040	—
UPI	beta42	0.20	0.198	-0.002 ± 0.008	-0.012 ± 0.041	0.040 ± 0.003	0.199 ± 0.008	0.040 ± 0.003	0.199 ± 0.009	0.991 ± 0.076	0.995 ± 0.043	0.207	0.007	3.45	0.1979	0.1980	1.384	97.976 ± 0.578	1.081 ± 0.046	—
UPI	beta43	0.20	0.204	0.004 ± 0.003	0.020 ± 0.017	0.007 ± 0.000	0.083 ± 0.003	0.007 ± 0.000	0.083 ± 0.003	0.171 ± 0.012	0.414 ± 0.017	0.202	0.002	0.77	0.0874	0.0874	1.346	97.302 ± 0.665	0.436 ± 0.016	—
UPI	beta44	0.00	0.012	0.012 ± 0.006	—	0.024 ± 0.002	0.154 ± 0.007	0.024 ± 0.002	0.154 ± 0.008	—	—	0.005	0.005	—	0.1401	0.1402	1.788	99.157 ± 0.375	1.079 ± 0.070	0.843 ± 0.375
UPI	beta45	0.00	-0.008	-0.008 ± 0.006	—	0.020 ± 0.002	0.141 ± 0.006	0.020 ± 0.002	0.141 ± 0.007	—	—	-0.002	-0.002	—	0.1263	0.1263	1.603	99.157 ± 0.375	0.884 ± 0.061	0.843 ± 0.375
UPI	beta46	0.00	-0.001	-0.001 ± 0.003	—	0.006 ± 0.001	0.080 ± 0.003	0.006 ± 0.001	0.080 ± 0.004	—	—	-0.003	-0.003	—	0.0768	0.0769	1.469	99.325 ± 0.336	0.463 ± 0.028	0.675 ± 0.336

Table S223

Simulation 2 - Condition 79: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.4
(continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	0.003	0.003 ± 0.004	—	0.008 ± 0.001	0.087 ± 0.004	0.008 ± 0.001	0.087 ± 0.004	—	—	-0.002	-0.002	—	0.0728	0.0729	1.323	99.157 ± 0.375	0.450 ± 0.024	0.843 ± 0.375
UPI	beta51	0.16	0.165	0.005 ± 0.004	0.032 ± 0.027	0.011 ± 0.001	0.107 ± 0.003	0.011 ± 0.001	0.107 ± 0.004	0.444 ± 0.026	0.667 ± 0.024	0.166	0.006	3.68	0.1031	0.1032	1.037	95.953 ± 0.809	0.433 ± 0.004	—
UPI	beta52	0.16	0.167	0.007 ± 0.006	0.045 ± 0.039	0.024 ± 0.002	0.154 ± 0.005	0.024 ± 0.002	0.154 ± 0.006	0.921 ± 0.060	0.960 ± 0.037	0.170	0.010	6.51	0.1707	0.1710	1.276	98.651 ± 0.474	0.768 ± 0.019	—
UPI	beta53	0.16	0.164	0.004 ± 0.005	0.024 ± 0.031	0.014 ± 0.001	0.120 ± 0.004	0.014 ± 0.001	0.120 ± 0.005	0.558 ± 0.037	0.747 ± 0.029	0.158	-0.002	-1.45	0.1226	0.1226	1.178	97.470 ± 0.645	0.552 ± 0.015	—
UPI	beta54	0.16	0.156	-0.004 ± 0.006	-0.026 ± 0.040	0.024 ± 0.002	0.156 ± 0.005	0.024 ± 0.002	0.156 ± 0.006	0.946 ± 0.059	0.973 ± 0.036	0.160	0.000	0.09	0.1501	0.1501	1.268	97.639 ± 0.623	0.774 ± 0.017	—
UPI	beta55	0.00	-0.010	-0.010 ± 0.007	—	0.026 ± 0.002	0.162 ± 0.006	0.026 ± 0.002	0.162 ± 0.007	—	—	-0.007	-0.007	—	0.1523	0.1525	1.507	99.494 ± 0.291	0.956 ± 0.037	0.506 ± 0.291
UPI	beta56	0.00	0.002	0.002 ± 0.006	—	0.019 ± 0.001	0.137 ± 0.005	0.019 ± 0.001	0.137 ± 0.006	—	—	0.000	0.000	—	0.1251	0.1251	1.303	98.651 ± 0.474	0.698 ± 0.027	1.349 ± 0.474
UPI	beta57	0.00	0.000	-0.000 ± 0.003	—	0.005 ± 0.000	0.073 ± 0.003	0.005 ± 0.000	0.073 ± 0.003	—	—	0.001	0.001	—	0.0646	0.0646	1.363	98.145 ± 0.554	0.390 ± 0.015	1.855 ± 0.554
UPI	beta58	0.00	0.000	-0.000 ± 0.003	—	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.003	—	—	-0.001	-0.001	—	0.0537	0.0538	1.208	98.820 ± 0.444	0.299 ± 0.010	1.180 ± 0.444

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S224

Simulation 2 - Condition 80: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.4

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Type I (%) $\pm$ MCSE
LSAM	beta41	0.20	0.197	-0.003 $\pm$ 0.003	-0.017 $\pm$ 0.015	0.008 $\pm$ 0.000	0.092 $\pm$ 0.002	0.008 $\pm$ 0.000	0.092 $\pm$ 0.003	0.212 $\pm$ 0.010	0.460 $\pm$ 0.013	0.194	-0.006	-3.12	0.0951	0.0953	1.020	96.394 $\pm$ 0.607	0.368 $\pm$ 0.003	—
LSAM	beta42	0.20	0.210	0.010 $\pm$ 0.003	0.052 $\pm$ 0.015	0.009 $\pm$ 0.000	0.095 $\pm$ 0.002	0.009 $\pm$ 0.000	0.096 $\pm$ 0.003	0.228 $\pm$ 0.011	0.478 $\pm$ 0.014	0.211	0.011	5.43	0.0944	0.0950	1.053	96.925 $\pm$ 0.562	0.392 $\pm$ 0.003	—
LSAM	beta43	0.20	0.198	-0.002 $\pm$ 0.002	-0.010 $\pm$ 0.008	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	0.062 $\pm$ 0.003	0.249 $\pm$ 0.007	0.200	0.000	-0.19	0.0492	0.0493	1.010	94.698 $\pm$ 0.730	0.197 $\pm$ 0.001	—
LSAM	beta44	0.00	0.002	0.002 $\pm$ 0.002	—	0.004 $\pm$ 0.000	0.062 $\pm$ 0.002	0.004 $\pm$ 0.000	0.062 $\pm$ 0.002	—	—	0.000	0.000	—	0.0600	0.0600	1.026	96.819 $\pm$ 0.572	0.250 $\pm$ 0.003	3.181 $\pm$ 0.572
LSAM	beta45	0.00	0.005	0.005 $\pm$ 0.002	—	0.004 $\pm$ 0.000	0.064 $\pm$ 0.002	0.004 $\pm$ 0.000	0.064 $\pm$ 0.002	—	—	0.004	0.004	—	0.0540	0.0542	0.925	95.546 $\pm$ 0.672	0.232 $\pm$ 0.002	4.454 $\pm$ 0.672
LSAM	beta46	0.00	0.001	0.001 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	—	—	0.002	0.002	—	0.0334	0.0334	1.002	97.031 $\pm$ 0.553	0.134 $\pm$ 0.002	2.969 $\pm$ 0.553
LSAM	beta47	0.00	-0.002	-0.002 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	—	—	-0.002	-0.002	—	0.0309	0.0310	1.012	96.501 $\pm$ 0.598	0.134 $\pm$ 0.002	3.499 $\pm$ 0.598
LSAM	beta51	0.16	0.161	0.001 $\pm$ 0.002	0.007 $\pm$ 0.013	0.004 $\pm$ 0.000	0.063 $\pm$ 0.001	0.004 $\pm$ 0.000	0.063 $\pm$ 0.002	0.155 $\pm$ 0.007	0.394 $\pm$ 0.011	0.163	0.003	1.58	0.0648	0.0648	0.980	95.864 $\pm$ 0.648	0.242 $\pm$ 0.001	—
LSAM	beta52	0.16	0.156	-0.004 $\pm$ 0.003	-0.023 $\pm$ 0.019	0.009 $\pm$ 0.000	0.093 $\pm$ 0.002	0.009 $\pm$ 0.000	0.093 $\pm$ 0.003	0.336 $\pm$ 0.017	0.580 $\pm$ 0.018	0.160	0.000	0.05	0.0846	0.0846	0.998	95.334 $\pm$ 0.687	0.363 $\pm$ 0.003	—
LSAM	beta53	0.16	0.164	0.004 $\pm$ 0.002	0.026 $\pm$ 0.014	0.005 $\pm$ 0.000	0.069 $\pm$ 0.002	0.005 $\pm$ 0.000	0.069 $\pm$ 0.002	0.188 $\pm$ 0.009	0.434 $\pm$ 0.012	0.165	0.005	2.82	0.0695	0.0696	1.023	96.607 $\pm$ 0.590	0.278 $\pm$ 0.002	—
LSAM	beta54	0.16	0.161	0.001 $\pm$ 0.003	0.007 $\pm$ 0.019	0.009 $\pm$ 0.000	0.093 $\pm$ 0.002	0.009 $\pm$ 0.000	0.093 $\pm$ 0.003	0.336 $\pm$ 0.017	0.579 $\pm$ 0.018	0.159	-0.001	-0.42	0.0856	0.0856	0.990	95.016 $\pm$ 0.709	0.360 $\pm$ 0.002	—
LSAM	beta55	0.00	0.000	-0.000 $\pm$ 0.003	—	0.007 $\pm$ 0.000	0.084 $\pm$ 0.002	0.007 $\pm$ 0.000	0.084 $\pm$ 0.003	—	—	-0.003	-0.003	—	0.0816	0.0817	0.998	97.137 $\pm$ 0.543	0.328 $\pm$ 0.003	2.863 $\pm$ 0.543
LSAM	beta56	0.00	-0.002	-0.002 $\pm$ 0.002	—	0.004 $\pm$ 0.000	0.067 $\pm$ 0.002	0.004 $\pm$ 0.000	0.067 $\pm$ 0.002	—	—	-0.004	-0.004	—	0.0669	0.0670	1.030	96.288 $\pm$ 0.616	0.271 $\pm$ 0.003	3.712 $\pm$ 0.616
LSAM	beta57	0.00	0.001	0.001 $\pm$ 0.001	—	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	—	—	0.000	0.000	—	0.0350	0.0350	0.945	95.546 $\pm$ 0.672	0.153 $\pm$ 0.002	4.454 $\pm$ 0.672
LSAM	beta58	0.00	0.000	0.000 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	—	—	0.000	0.000	—	0.0289	0.0289	0.963	96.076 $\pm$ 0.632	0.121 $\pm$ 0.001	3.924 $\pm$ 0.632
QML	beta41	0.20	0.171	-0.029 $\pm$ 0.002	-0.146 $\pm$ 0.008	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.003 $\pm$ 0.000	0.059 $\pm$ 0.001	0.087 $\pm$ 0.004	0.296 $\pm$ 0.007	0.169	-0.031	-15.26	0.0524	0.0606	0.997	88.017 $\pm$ 1.058	0.201 $\pm$ 0.001	—
QML	beta42	0.20	0.178	-0.022 $\pm$ 0.002	-0.110 $\pm$ 0.008	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.003 $\pm$ 0.000	0.056 $\pm$ 0.001	0.078 $\pm$ 0.003	0.280 $\pm$ 0.007	0.177	-0.023	-11.47	0.0508	0.0557	1.073	94.168 $\pm$ 0.763	0.216 $\pm$ 0.007	—
QML	beta43	0.20	0.203	0.003 $\pm$ 0.001	0.014 $\pm$ 0.007	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.051 $\pm$ 0.002	0.226 $\pm$ 0.007	0.203	0.003	1.32	0.0444	0.0444	1.019	94.804 $\pm$ 0.723	0.180 $\pm$ 0.001	—
QML	beta44	0.00	-0.024	-0.024 $\pm$ 0.002	—	0.003 $\pm$ 0.000	0.057 $\pm$ 0.001	0.004 $\pm$ 0.000	0.062 $\pm$ 0.002	—	—	-0.025	-0.025	—	0.0562	0.0615	1.132	93.849 $\pm$ 0.782	0.252 $\pm$ 0.018	6.151 $\pm$ 0.782
QML	beta45	0.00	0.010	0.010 $\pm$ 0.002	—	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.003 $\pm$ 0.000	0.053 $\pm$ 0.002	—	—	0.010	0.010	—	0.0494	0.0504	0.986	94.168 $\pm$ 0.763	0.203 $\pm$ 0.004	5.832 $\pm$ 0.763
QML	beta46	0.00	0.039	0.039 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	—	—	0.037	0.037	—	0.0325	0.0492	1.031	77.413 $\pm$ 1.362	0.133 $\pm$ 0.004	22.587 $\pm$ 1.362
QML	beta47	0.00	0.040	0.040 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	—	—	0.038	0.038	—	0.0325	0.0497	1.088	77.519 $\pm$ 1.359	0.143 $\pm$ 0.015	22.481 $\pm$ 1.359
QML	beta51	0.16	0.167	0.007 $\pm$ 0.002	0.043 $\pm$ 0.013	0.004 $\pm$ 0.000	0.062 $\pm$ 0.001	0.004 $\pm$ 0.000	0.062 $\pm$ 0.002	0.151 $\pm$ 0.007	0.389 $\pm$ 0.011	0.168	0.008	5.27	0.0624	0.0630	1.027	96.501 $\pm$ 0.598	0.249 $\pm$ 0.003	—
QML	beta52	0.16	0.138	-0.022 $\pm$ 0.002	-0.137 $\pm$ 0.010	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.003 $\pm$ 0.000	0.056 $\pm$ 0.001	0.121 $\pm$ 0.005	0.347 $\pm$ 0.009	0.139	-0.021	-13.34	0.0542	0.0583	1.061	92.895 $\pm$ 0.837	0.212 $\pm$ 0.002	—
QML	beta53	0.16	0.157	-0.003 $\pm$ 0.002	-0.017 $\pm$ 0.011	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.107 $\pm$ 0.005	0.327 $\pm$ 0.009	0.156	-0.004	-2.52	0.0520	0.0522	1.070	96.076 $\pm$ 0.632	0.220 $\pm$ 0.004	—
QML	beta54	0.16	0.137	-0.023 $\pm$ 0.002	-0.146 $\pm$ 0.011	0.003 $\pm$ 0.000	0.054 $\pm$ 0.001	0.003 $\pm$ 0.000	0.058 $\pm$ 0.002	0.134 $\pm$ 0.006	0.366 $\pm$ 0.010	0.137	-0.023	-14.63	0.0512	0.0563	0.999	91.516 $\pm$ 0.907	0.210 $\pm$ 0.001	—
QML	beta55	0.00	-0.024	-0.024 $\pm$ 0.002	—	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.006 $\pm$ 0.000	0.078 $\pm$ 0.002	—	—	-0.025	-0.025	—	0.0742	0.0782	1.028	93.425 $\pm$ 0.807	0.301 $\pm$ 0.005	6.575 $\pm$ 0.807
QML	beta56	0.00	0.025	0.025 $\pm$ 0.002	—	0.003 $\pm$ 0.000	0.054 $\pm$ 0.001	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	—	—	0.027	0.027	—	0.0521	0.0589	1.032	93.531 $\pm$ 0.801	0.220 $\pm$ 0.002	6.469 $\pm$ 0.801
QML	beta57	0.00	0.035	0.035 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.039 $\pm$ 0.001	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	—	—	0.034	0.034	—	0.0371	0.0507	1.025	86.426 $\pm$ 1.115	0.155 $\pm$ 0.003	13.574 $\pm$ 1.115
QML	beta58	0.00	0.029	0.029 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.028 $\pm$ 0.001	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	—	—	0.027	0.027	—	0.0274	0.0386	0.981	81.654 $\pm$ 1.260	0.108 $\pm$ 0.001	18.346 $\pm$ 1.260
UPI	beta41	0.20	0.196	-0.004 $\pm$ 0.003	-0.020 $\pm$ 0.016	0.009 $\pm$ 0.000	0.095 $\pm$ 0.002	0.009 $\pm$ 0.000	0.095 $\pm$ 0.003	0.228 $\pm$ 0.011	0.477 $\pm$ 0.014	0.195	-0.005	-2.46	0.0929	0.0930	1.124	96.925 $\pm$ 0.562	0.420 $\pm$ 0.008	—
UPI	beta42	0.20	0.213	0.013 $\pm$ 0.003	0.067 $\pm$ 0.016	0.010 $\pm$ 0.001	0.098 $\pm$ 0.003	0.010 $\pm$ 0.001	0.099 $\pm$ 0.003	0.244 $\pm$ 0.013	0.494 $\pm$ 0.015	0.211	0.011	5.43	0.0949	0.0956	1.191	97.243 $\pm$ 0.533	0.457 $\pm$ 0.018	—
UPI	beta43	0.20	0.198	-0.002 $\pm$ 0.002	-0.010 $\pm$ 0.008	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.058 $\pm$ 0.003	0.240 $\pm$ 0.007	0.198	-0.002	-1.23	0.0471	0.0472	1.055	95.334 $\pm$ 0.687	0.198 $\pm$ 0.004	—
UPI	beta44	0.00	0.003	0.003 $\pm$ 0.002	—	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	—	—	0.001	0.001	—	0.0580	0.0580	1.312	98.409 $\pm$ 0.407	0.340 $\pm$ 0.031	1.591 $\pm$ 0.407
UPI	beta45	0.00	0.003	0.003 $\pm$ 0.002	—	0.004 $\pm$ 0.000	0.065 $\pm$ 0.002	0.004 $\pm$ 0.000	0.065 $\pm$ 0.002	—	—	0.002	0.002	—	0.0566	0.0567	1.210	97.985 $\pm$ 0.458	0.310 $\pm$ 0.014	2.015 $\pm$ 0.458
UPI	beta46	0.00	0.000	0.000 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	—	—	0.000	0.000	—	0.0316	0.0316	1.143	97.773 $\pm$ 0.481	0.159 $\pm$ 0.005	2.227 $\pm$ 0.481

Table S224

Simulation 2 - Condition 80: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.4
(continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.004	-0.004 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	-0.003	-0.003	—	0.0303	0.0304	1.200	97.031 ± 0.553	0.165 ± 0.012	2.969 ± 0.553
UPI	beta51	0.16	0.161	0.001 ± 0.002	0.007 ± 0.012	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.147 ± 0.007	0.383 ± 0.011	0.162	0.002	1.01	0.0625	0.0625	1.015	96.076 ± 0.632	0.244 ± 0.001	—
UPI	beta52	0.16	0.159	-0.001 ± 0.003	-0.007 ± 0.018	0.008 ± 0.000	0.090 ± 0.002	0.008 ± 0.000	0.090 ± 0.003	0.319 ± 0.016	0.565 ± 0.017	0.162	0.002	1.15	0.0832	0.0832	1.054	96.288 ± 0.616	0.374 ± 0.004	—
UPI	beta53	0.16	0.163	0.003 ± 0.002	0.016 ± 0.014	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.186 ± 0.009	0.431 ± 0.012	0.163	0.003	1.89	0.0702	0.0703	1.042	96.288 ± 0.616	0.282 ± 0.002	—
UPI	beta54	0.16	0.161	0.001 ± 0.003	0.008 ± 0.019	0.008 ± 0.000	0.091 ± 0.002	0.008 ± 0.000	0.091 ± 0.003	0.325 ± 0.016	0.570 ± 0.017	0.160	0.000	0.12	0.0869	0.0869	1.032	95.652 ± 0.664	0.369 ± 0.003	—
UPI	beta55	0.00	-0.002	-0.002 ± 0.003	—	0.007 ± 0.000	0.085 ± 0.002	0.007 ± 0.000	0.085 ± 0.003	—	—	-0.001	-0.001	—	0.0771	0.0771	1.102	97.773 ± 0.481	0.366 ± 0.007	2.227 ± 0.481
UPI	beta56	0.00	-0.003	-0.003 ± 0.002	—	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	—	—	-0.004	-0.004	—	0.0636	0.0637	1.120	97.455 ± 0.513	0.301 ± 0.005	2.545 ± 0.513
UPI	beta57	0.00	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	-0.001	-0.001	—	0.0348	0.0348	1.004	96.819 ± 0.572	0.160 ± 0.003	3.181 ± 0.572
UPI	beta58	0.00	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	—	—	-0.001	-0.001	—	0.0274	0.0274	0.999	96.394 ± 0.607	0.125 ± 0.002	3.606 ± 0.607

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S225

Simulation 2 - Condition 81: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.6

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.209	0.009 ± 0.003	0.044 ± 0.016	0.010 ± 0.000	0.102 ± 0.002	0.010 ± 0.000	0.102 ± 0.003	0.260 ± 0.012	0.510 ± 0.015	0.209	0.009	4.39	0.0998	0.1002	1.010	96.350 ± 0.606	0.402 ± 0.003	—
LSAM	beta42	0.20	0.196	-0.004 ± 0.003	-0.019 ± 0.017	0.012 ± 0.001	0.108 ± 0.002	0.012 ± 0.001	0.108 ± 0.003	0.292 ± 0.013	0.540 ± 0.015	0.194	-0.006	-3.08	0.1094	0.1096	0.997	95.516 ± 0.668	0.422 ± 0.003	—
LSAM	beta43	0.20	0.202	0.002 ± 0.002	0.010 ± 0.010	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.087 ± 0.004	0.295 ± 0.008	0.202	0.002	1.11	0.0608	0.0608	0.989	94.578 ± 0.731	0.229 ± 0.001	—
LSAM	beta44	0.00	0.006	0.006 ± 0.002	—	0.005 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.074 ± 0.002	—	—	0.004	0.004	—	0.0704	0.0705	0.945	94.891 ± 0.711	0.275 ± 0.003	5.109 ± 0.711
LSAM	beta45	0.00	0.000	-0.000 ± 0.002	—	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	—	—	0.000	0.000	—	0.0696	0.0696	0.919	93.639 ± 0.788	0.262 ± 0.003	6.361 ± 0.788
LSAM	beta46	0.00	-0.003	-0.003 ± 0.001	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	—	—	-0.002	-0.002	—	0.0395	0.0395	0.904	93.118 ± 0.817	0.155 ± 0.002	6.882 ± 0.817
LSAM	beta47	0.00	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	0.001	0.001	—	0.0368	0.0368	0.933	94.682 ± 0.725	0.153 ± 0.002	5.318 ± 0.725
LSAM	beta51	0.16	0.158	-0.002 ± 0.002	-0.011 ± 0.016	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	0.233 ± 0.011	0.482 ± 0.014	0.159	-0.001	-0.52	0.0753	0.0753	0.934	92.701 ± 0.840	0.282 ± 0.001	—
LSAM	beta52	0.16	0.163	0.003 ± 0.003	0.020 ± 0.021	0.011 ± 0.001	0.105 ± 0.003	0.011 ± 0.001	0.105 ± 0.003	0.431 ± 0.021	0.657 ± 0.019	0.162	0.002	1.51	0.1049	0.1049	0.975	95.412 ± 0.676	0.402 ± 0.002	—
LSAM	beta53	0.16	0.159	-0.001 ± 0.003	-0.009 ± 0.016	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	0.238 ± 0.012	0.488 ± 0.014	0.161	0.001	0.62	0.0768	0.0768	0.982	94.786 ± 0.718	0.301 ± 0.002	—
LSAM	beta54	0.16	0.168	0.008 ± 0.003	0.053 ± 0.021	0.011 ± 0.001	0.104 ± 0.002	0.011 ± 0.001	0.104 ± 0.003	0.426 ± 0.020	0.652 ± 0.019	0.163	0.003	1.74	0.1007	0.1007	0.983	95.099 ± 0.697	0.401 ± 0.003	—
LSAM	beta55	0.00	-0.002	-0.002 ± 0.003	—	0.009 ± 0.000	0.094 ± 0.002	0.009 ± 0.000	0.094 ± 0.003	—	—	0.001	0.001	—	0.0860	0.0861	0.945	94.578 ± 0.731	0.347 ± 0.003	5.422 ± 0.731
LSAM	beta56	0.00	0.003	0.003 ± 0.003	—	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	—	—	0.002	0.002	—	0.0753	0.0753	0.929	94.473 ± 0.738	0.289 ± 0.003	5.527 ± 0.738
LSAM	beta57	0.00	0.000	-0.000 ± 0.002	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.002	—	—	-0.002	-0.002	—	0.0418	0.0419	0.925	94.473 ± 0.738	0.171 ± 0.002	5.527 ± 0.738
LSAM	beta58	0.00	-0.003	-0.003 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	-0.003	-0.003	—	0.0355	0.0356	0.930	94.995 ± 0.704	0.144 ± 0.002	5.005 ± 0.704
QML	beta41	0.20	0.185	-0.015 ± 0.002	-0.076 ± 0.011	0.004 ± 0.000	0.066 ± 0.001	0.005 ± 0.000	0.068 ± 0.002	0.114 ± 0.005	0.338 ± 0.009	0.185	-0.015	-7.51	0.0676	0.0692	1.075	94.369 ± 0.744	0.277 ± 0.011	—
QML	beta42	0.20	0.178	-0.022 ± 0.002	-0.112 ± 0.011	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.134 ± 0.006	0.366 ± 0.010	0.176	-0.024	-12.12	0.0702	0.0742	1.026	93.222 ± 0.812	0.280 ± 0.005	—
QML	beta43	0.20	0.206	0.006 ± 0.002	0.028 ± 0.009	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.076 ± 0.003	0.276 ± 0.008	0.206	0.006	3.01	0.0561	0.0564	1.019	94.369 ± 0.744	0.219 ± 0.004	—
QML	beta44	0.00	-0.005	-0.005 ± 0.002	—	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	—	—	-0.005	-0.005	—	0.0629	0.0631	0.956	94.265 ± 0.751	0.246 ± 0.002	5.735 ± 0.751
QML	beta45	0.00	0.000	-0.000 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	—	—	0.002	0.002	—	0.0605	0.0605	0.941	93.743 ± 0.782	0.227 ± 0.002	6.257 ± 0.782
QML	beta46	0.00	0.018	0.018 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	0.016	0.016	—	0.0342	0.0379	0.900	89.260 ± 1.000	0.134 ± 0.002	10.740 ± 1.000
QML	beta47	0.00	0.021	0.021 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	0.019	0.019	—	0.0342	0.0391	0.918	88.217 ± 1.041	0.132 ± 0.002	11.783 ± 1.041
QML	beta51	0.16	0.161	0.001 ± 0.002	0.009 ± 0.015	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	0.224 ± 0.010	0.473 ± 0.013	0.164	0.004	2.30	0.0754	0.0755	0.992	93.535 ± 0.794	0.295 ± 0.009	—
QML	beta52	0.16	0.147	-0.013 ± 0.002	-0.082 ± 0.015	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.074 ± 0.002	0.212 ± 0.010	0.460 ± 0.013	0.150	-0.010	-6.28	0.0731	0.0738	0.997	93.743 ± 0.782	0.283 ± 0.006	—
QML	beta53	0.16	0.158	-0.002 ± 0.002	-0.014 ± 0.013	0.004 ± 0.000	0.066 ± 0.001	0.004 ± 0.000	0.066 ± 0.002	0.168 ± 0.008	0.410 ± 0.011	0.158	-0.002	-1.27	0.0639	0.0639	1.016	94.369 ± 0.744	0.261 ± 0.003	—
QML	beta54	0.16	0.148	-0.012 ± 0.002	-0.073 ± 0.014	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.206 ± 0.009	0.453 ± 0.012	0.146	-0.014	-8.75	0.0731	0.0744	0.997	94.369 ± 0.744	0.280 ± 0.003	—
QML	beta55	0.00	-0.010	-0.010 ± 0.003	—	0.007 ± 0.000	0.084 ± 0.002	0.007 ± 0.000	0.085 ± 0.003	—	—	-0.008	-0.008	—	0.0805	0.0809	0.959	93.535 ± 0.794	0.317 ± 0.003	6.465 ± 0.794
QML	beta56	0.00	0.015	0.015 ± 0.002	—	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	—	—	0.013	0.013	—	0.0680	0.0693	0.935	92.492 ± 0.851	0.251 ± 0.002	7.508 ± 0.851
QML	beta57	0.00	0.016	0.016 ± 0.001	—	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	0.016	0.016	—	0.0402	0.0434	0.951	92.284 ± 0.862	0.157 ± 0.002	7.716 ± 0.862
QML	beta58	0.00	0.012	0.012 ± 0.001	—	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	0.011	0.011	—	0.0319	0.0337	0.950	91.867 ± 0.883	0.123 ± 0.002	8.133 ± 0.883
UPI	beta41	0.20	0.210	0.010 ± 0.003	0.048 ± 0.016	0.010 ± 0.000	0.102 ± 0.002	0.011 ± 0.000	0.102 ± 0.003	0.263 ± 0.012	0.512 ± 0.015	0.209	0.009	4.38	0.1000	0.1004	1.062	96.767 ± 0.571	0.425 ± 0.004	—
UPI	beta42	0.20	0.196	-0.004 ± 0.003	-0.022 ± 0.017	0.012 ± 0.001	0.108 ± 0.002	0.012 ± 0.001	0.108 ± 0.003	0.291 ± 0.013	0.539 ± 0.015	0.192	-0.008	-4.22	0.1054	0.1057	1.055	96.142 ± 0.622	0.446 ± 0.005	—
UPI	beta43	0.20	0.202	0.002 ± 0.002	0.010 ± 0.009	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.084 ± 0.004	0.289 ± 0.008	0.201	0.001	0.30	0.0602	0.0602	0.997	94.682 ± 0.725	0.226 ± 0.002	—
UPI	beta44	0.00	0.005	0.005 ± 0.003	—	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.078 ± 0.003	—	—	0.005	0.005	—	0.0704	0.0706	1.067	96.663 ± 0.580	0.325 ± 0.006	3.337 ± 0.580
UPI	beta45	0.00	-0.001	-0.001 ± 0.002	—	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.003	—	—	0.001	0.001	—	0.0675	0.0675	1.069	96.455 ± 0.597	0.318 ± 0.008	3.545 ± 0.597
UPI	beta46	0.00	-0.003	-0.003 ± 0.001	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	-0.002	-0.002	—	0.0373	0.0374	0.965	95.203 ± 0.690	0.168 ± 0.003	4.797 ± 0.690

Table S225

Simulation 2 - Condition 81: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.6
(continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	0.002	0.002 ± 0.001	—	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	0.002	0.002	—	0.0384	0.0385	0.996	95.829 ± 0.646	0.166 ± 0.003	4.171 ± 0.646
UPI	beta51	0.16	0.158	-0.002 ± 0.002	-0.011 ± 0.015	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	0.224 ± 0.010	0.474 ± 0.013	0.159	-0.001	-0.37	0.0756	0.0756	0.958	93.535 ± 0.794	0.285 ± 0.001	—
UPI	beta52	0.16	0.161	0.001 ± 0.003	0.008 ± 0.021	0.011 ± 0.001	0.104 ± 0.003	0.011 ± 0.001	0.104 ± 0.003	0.423 ± 0.021	0.651 ± 0.019	0.162	0.002	1.43	0.1045	0.1046	0.983	95.308 ± 0.683	0.401 ± 0.003	—
UPI	beta53	0.16	0.160	-0.000 ± 0.002	-0.001 ± 0.015	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	0.224 ± 0.011	0.473 ± 0.014	0.161	0.001	0.40	0.0730	0.0730	1.008	94.995 ± 0.704	0.299 ± 0.002	—
UPI	beta54	0.16	0.168	0.008 ± 0.003	0.048 ± 0.021	0.011 ± 0.000	0.103 ± 0.002	0.011 ± 0.000	0.103 ± 0.003	0.415 ± 0.019	0.644 ± 0.018	0.162	0.002	1.20	0.1032	0.1032	1.000	95.203 ± 0.690	0.403 ± 0.003	—
UPI	beta55	0.00	-0.003	-0.003 ± 0.003	—	0.009 ± 0.000	0.094 ± 0.002	0.009 ± 0.000	0.094 ± 0.003	—	—	-0.002	-0.002	—	0.0837	0.0837	1.019	95.308 ± 0.683	0.375 ± 0.005	4.692 ± 0.683
UPI	beta56	0.00	0.002	0.002 ± 0.003	—	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.081 ± 0.003	—	—	0.000	0.000	—	0.0741	0.0741	0.978	95.412 ± 0.676	0.311 ± 0.004	4.588 ± 0.676
UPI	beta57	0.00	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	-0.001	-0.001	—	0.0407	0.0407	0.960	94.265 ± 0.751	0.174 ± 0.002	5.735 ± 0.751
UPI	beta58	0.00	-0.003	-0.003 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	-0.002	-0.002	—	0.0347	0.0348	0.955	95.412 ± 0.676	0.146 ± 0.002	4.588 ± 0.676

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S226

Simulation 2 - Condition 82: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.6

Method	Param.	True	Mean-based										Median-based					Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE		
LSAM	beta41	0.20	0.200	0.000 ± 0.002	0.001 ± 0.009	0.004 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	0.088 ± 0.004	0.297 ± 0.009	0.198	-0.002	-0.96	0.0586	0.0586	1.009	95.281 ± 0.672	0.235 ± 0.001	—		
LSAM	beta42	0.20	0.202	0.002 ± 0.002	0.011 ± 0.010	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.094 ± 0.004	0.307 ± 0.008	0.202	0.002	1.12	0.0579	0.0579	1.033	95.984 ± 0.622	0.248 ± 0.001	—		
LSAM	beta43	0.20	0.200	0.000 ± 0.001	0.001 ± 0.006	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.034 ± 0.001	0.183 ± 0.005	0.201	0.001	0.29	0.0385	0.0385	0.966	94.378 ± 0.730	0.139 ± 0.000	—		
LSAM	beta44	0.00	0.003	0.003 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	0.002	0.002	—	0.0367	0.0367	0.961	94.378 ± 0.730	0.148 ± 0.001	5.622 ± 0.730		
LSAM	beta45	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.000	0.000	—	0.0367	0.0367	0.980	95.181 ± 0.679	0.142 ± 0.001	4.819 ± 0.679		
LSAM	beta46	0.00	0.000	-0.000 ± 0.001	—	0.000 ± 0.000	0.021 ± 0.001	0.000 ± 0.000	0.021 ± 0.001	—	—	0.000	0.000	—	0.0191	0.0191	0.958	93.876 ± 0.760	0.080 ± 0.001	6.124 ± 0.760		
LSAM	beta47	0.00	-0.001	-0.001 ± 0.001	—	0.000 ± 0.000	0.022 ± 0.001	0.000 ± 0.000	0.022 ± 0.001	—	—	0.000	0.000	—	0.0200	0.0200	0.975	93.976 ± 0.754	0.082 ± 0.001	6.024 ± 0.754		
LSAM	beta51	0.16	0.162	0.002 ± 0.001	0.012 ± 0.009	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.081 ± 0.004	0.285 ± 0.008	0.163	0.003	1.77	0.0449	0.0450	0.989	94.478 ± 0.724	0.176 ± 0.000	—		
LSAM	beta52	0.16	0.161	0.001 ± 0.002	0.005 ± 0.012	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.150 ± 0.007	0.387 ± 0.011	0.161	0.001	0.64	0.0599	0.0599	0.970	94.679 ± 0.711	0.235 ± 0.001	—		
LSAM	beta53	0.16	0.161	0.001 ± 0.001	0.004 ± 0.009	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.084 ± 0.004	0.290 ± 0.008	0.161	0.001	0.92	0.0459	0.0459	1.008	94.880 ± 0.698	0.183 ± 0.001	—		
LSAM	beta54	0.16	0.162	0.002 ± 0.002	0.012 ± 0.012	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.140 ± 0.006	0.374 ± 0.010	0.160	0.000	0.08	0.0619	0.0619	1.002	95.080 ± 0.685	0.235 ± 0.001	—		
LSAM	beta55	0.00	-0.001	-0.001 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.002	—	—	-0.002	-0.002	—	0.0497	0.0497	0.988	94.980 ± 0.692	0.197 ± 0.001	5.020 ± 0.692		
LSAM	beta56	0.00	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	—	—	-0.002	-0.002	—	0.0441	0.0442	0.987	94.378 ± 0.730	0.170 ± 0.001	5.622 ± 0.730		
LSAM	beta57	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.024 ± 0.001	0.001 ± 0.000	0.024 ± 0.001	—	—	0.000	0.000	—	0.0227	0.0227	0.965	94.378 ± 0.730	0.091 ± 0.001	5.622 ± 0.730		
LSAM	beta58	0.00	0.000	-0.000 ± 0.001	—	0.000 ± 0.000	0.021 ± 0.001	0.000 ± 0.000	0.021 ± 0.001	—	—	0.000	0.000	—	0.0185	0.0185	0.923	92.871 ± 0.815	0.077 ± 0.001	7.129 ± 0.815		
QML	beta41	0.20	0.181	-0.019 ± 0.001	-0.097 ± 0.007	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.052 ± 0.002	0.229 ± 0.006	0.180	-0.020	-10.09	0.0412	0.0459	1.015	92.369 ± 0.841	0.165 ± 0.001	—		
QML	beta42	0.20	0.182	-0.018 ± 0.001	-0.092 ± 0.007	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.052 ± 0.002	0.227 ± 0.006	0.183	-0.017	-8.71	0.0402	0.0438	1.042	92.972 ± 0.810	0.170 ± 0.000	—		
QML	beta43	0.20	0.205	0.005 ± 0.001	0.023 ± 0.006	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.031 ± 0.001	0.177 ± 0.005	0.204	0.004	2.05	0.0363	0.0365	0.971	94.779 ± 0.705	0.134 ± 0.001	—		
QML	beta44	0.00	-0.006	-0.006 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	-0.007	-0.007	—	0.0374	0.0381	0.977	93.675 ± 0.771	0.151 ± 0.002	6.325 ± 0.771		
QML	beta45	0.00	0.002	0.002 ± 0.001	—	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	—	—	0.002	0.002	—	0.0341	0.0342	1.059	94.880 ± 0.698	0.143 ± 0.007	5.120 ± 0.698		
QML	beta46	0.00	0.018	0.018 ± 0.001	—	0.000 ± 0.000	0.021 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	0.017	0.017	—	0.0191	0.0255	0.956	85.241 ± 1.124	0.079 ± 0.001	14.759 ± 1.124		
QML	beta47	0.00	0.018	0.018 ± 0.001	—	0.000 ± 0.000	0.021 ± 0.000	0.001 ± 0.000	0.027 ± 0.001	—	—	0.017	0.017	—	0.0210	0.0272	0.950	84.839 ± 1.136	0.078 ± 0.001	15.161 ± 1.136		
QML	beta51	0.16	0.164	0.004 ± 0.001	0.025 ± 0.009	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.081 ± 0.004	0.284 ± 0.008	0.165	0.005	2.89	0.0441	0.0444	0.998	94.578 ± 0.718	0.177 ± 0.000	—		
QML	beta52	0.16	0.144	-0.016 ± 0.001	-0.098 ± 0.009	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.086 ± 0.004	0.293 ± 0.008	0.145	-0.015	-9.27	0.0445	0.0469	0.979	91.466 ± 0.885	0.170 ± 0.000	—		
QML	beta53	0.16	0.160	0.000 ± 0.001	0.001 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.064 ± 0.003	0.253 ± 0.007	0.161	0.001	0.35	0.0405	0.0405	1.015	95.482 ± 0.658	0.161 ± 0.002	—		
QML	beta54	0.16	0.145	-0.015 ± 0.001	-0.094 ± 0.008	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.081 ± 0.003	0.284 ± 0.007	0.143	-0.017	-10.70	0.0447	0.0479	1.016	94.578 ± 0.718	0.171 ± 0.001	—		
QML	beta55	0.00	-0.009	-0.009 ± 0.002	—	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	—	—	-0.010	-0.010	—	0.0481	0.0490	1.022	94.980 ± 0.692	0.194 ± 0.002	5.020 ± 0.692		
QML	beta56	0.00	0.011	0.011 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	0.011	0.011	—	0.0391	0.0407	1.020	93.574 ± 0.777	0.158 ± 0.004	6.426 ± 0.777		
QML	beta57	0.00	0.015	0.015 ± 0.001	—	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	—	—	0.014	0.014	—	0.0231	0.0271	1.008	90.161 ± 0.944	0.092 ± 0.001	9.839 ± 0.944		
QML	beta58	0.00	0.013	0.013 ± 0.001	—	0.000 ± 0.000	0.020 ± 0.000	0.001 ± 0.000	0.024 ± 0.001	—	—	0.013	0.013	—	0.0179	0.0220	0.927	88.454 ± 1.013	0.072 ± 0.001	11.546 ± 1.013		
UPI	beta41	0.20	0.200	0.000 ± 0.002	0.002 ± 0.009	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.087 ± 0.004	0.295 ± 0.008	0.198	-0.002	-1.01	0.0582	0.0583	1.025	96.486 ± 0.583	0.237 ± 0.001	—		
UPI	beta42	0.20	0.202	0.002 ± 0.002	0.011 ± 0.010	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.093 ± 0.004	0.306 ± 0.009	0.201	0.001	0.44	0.0581	0.0581	1.044	95.984 ± 0.622	0.250 ± 0.001	—		
UPI	beta43	0.20	0.200	0.000 ± 0.001	0.002 ± 0.006	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.032 ± 0.001	0.178 ± 0.005	0.200	0.000	0.08	0.0366	0.0366	0.962	93.876 ± 0.760	0.134 ± 0.001	—		
UPI	beta44	0.00	0.002	0.002 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	0.001	0.001	—	0.0369	0.0369	1.000	95.783 ± 0.637	0.154 ± 0.001	4.217 ± 0.637		
UPI	beta45	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	0.000	0.000	—	0.0377	0.0377	1.020	95.683 ± 0.644	0.151 ± 0.001	4.317 ± 0.644		
UPI	beta46	0.00	0.000	-0.000 ± 0.001	—	0.000 ± 0.000	0.021 ± 0.001	0.000 ± 0.000	0.021 ± 0.001	—	—	0.000	0.000	—	0.0190	0.0190	0.977	94.478 ± 0.724	0.081 ± 0.001	5.522 ± 0.724		

Table S226

Simulation 2 - Condition 82: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.6
(continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.001	-0.001 ± 0.001	—	0.000 ± 0.000	0.022 ± 0.001	0.000 ± 0.000	0.022 ± 0.001	—	—	0.000	0.000	—	0.0200	0.0200	0.986	94.578 ± 0.718	0.083 ± 0.001	5.422 ± 0.718
UPI	beta51	0.16	0.162	0.002 ± 0.001	0.011 ± 0.009	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.080 ± 0.004	0.282 ± 0.008	0.163	0.003	1.58	0.0447	0.0448	1.000	94.980 ± 0.692	0.177 ± 0.000	—
UPI	beta52	0.16	0.161	0.001 ± 0.002	0.008 ± 0.012	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.147 ± 0.007	0.383 ± 0.011	0.160	0.000	0.29	0.0594	0.0594	0.973	95.080 ± 0.685	0.234 ± 0.001	—
UPI	beta53	0.16	0.161	0.001 ± 0.001	0.004 ± 0.009	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.082 ± 0.004	0.286 ± 0.008	0.161	0.001	0.93	0.0447	0.0447	1.006	95.181 ± 0.679	0.181 ± 0.001	—
UPI	beta54	0.16	0.161	0.001 ± 0.002	0.009 ± 0.012	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.140 ± 0.006	0.374 ± 0.010	0.161	0.001	0.33	0.0606	0.0606	1.001	94.980 ± 0.692	0.235 ± 0.001	—
UPI	beta55	0.00	-0.001	-0.001 ± 0.002	—	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.002	—	—	-0.002	-0.002	—	0.0476	0.0477	1.017	95.984 ± 0.622	0.200 ± 0.001	4.016 ± 0.622
UPI	beta56	0.00	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	—	—	-0.001	-0.001	—	0.0438	0.0438	1.000	94.578 ± 0.718	0.173 ± 0.001	5.422 ± 0.718
UPI	beta57	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.024 ± 0.001	0.001 ± 0.000	0.024 ± 0.001	—	—	0.000	0.000	—	0.0219	0.0219	0.978	95.582 ± 0.651	0.091 ± 0.001	4.418 ± 0.651
UPI	beta58	0.00	0.000	-0.000 ± 0.001	—	0.000 ± 0.000	0.021 ± 0.001	0.000 ± 0.000	0.021 ± 0.001	—	—	0.000	0.000	—	0.0183	0.0183	0.924	93.273 ± 0.794	0.077 ± 0.001	6.727 ± 0.794

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S227

Simulation 2 - Condition 83: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.8

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.198	-0.002 ± 0.002	-0.009 ± 0.012	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	0.147 ± 0.007	0.384 ± 0.011	0.197	-0.003	-1.35	0.0718	0.0718	0.966	94.467 ± 0.725	0.291 ± 0.001	—
LSAM	beta42	0.20	0.200	-0.000 ± 0.002	-0.001 ± 0.012	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	0.148 ± 0.007	0.384 ± 0.011	0.201	0.001	0.39	0.0756	0.0756	1.003	95.573 ± 0.652	0.302 ± 0.001	—
LSAM	beta43	0.20	0.202	0.002 ± 0.002	0.011 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.057 ± 0.003	0.238 ± 0.007	0.202	0.002	1.18	0.0467	0.0468	0.963	93.058 ± 0.806	0.180 ± 0.001	—
LSAM	beta44	0.00	0.000	-0.000 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	—	—	0.000	0.000	—	0.0474	0.0474	0.940	92.354 ± 0.843	0.191 ± 0.002	7.646 ± 0.843
LSAM	beta45	0.00	0.001	0.001 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.002	—	—	0.000	0.000	—	0.0512	0.0512	0.903	92.857 ± 0.817	0.181 ± 0.002	7.143 ± 0.817
LSAM	beta46	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	—	—	0.001	0.001	—	0.0311	0.0311	0.849	92.052 ± 0.858	0.108 ± 0.001	7.948 ± 0.858
LSAM	beta47	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	0.000	0.000	—	0.0254	0.0254	0.894	92.857 ± 0.817	0.106 ± 0.001	7.143 ± 0.817
LSAM	beta51	0.16	0.159	-0.001 ± 0.002	-0.004 ± 0.012	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.147 ± 0.007	0.384 ± 0.011	0.159	-0.001	-0.69	0.0625	0.0625	0.953	94.165 ± 0.743	0.230 ± 0.001	—
LSAM	beta52	0.16	0.160	-0.000 ± 0.003	-0.001 ± 0.016	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.079 ± 0.003	0.244 ± 0.014	0.494 ± 0.016	0.159	-0.001	-0.60	0.0754	0.0754	0.953	94.467 ± 0.725	0.295 ± 0.001	—
LSAM	beta53	0.16	0.159	-0.001 ± 0.002	-0.008 ± 0.012	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.153 ± 0.007	0.391 ± 0.011	0.159	-0.001	-0.75	0.0647	0.0648	0.931	92.958 ± 0.812	0.228 ± 0.001	—
LSAM	beta54	0.16	0.159	-0.001 ± 0.002	-0.006 ± 0.015	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	0.219 ± 0.010	0.468 ± 0.013	0.157	-0.003	-1.64	0.0757	0.0757	0.999	94.970 ± 0.693	0.293 ± 0.001	—
LSAM	beta55	0.00	0.001	0.001 ± 0.002	—	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	—	—	-0.001	-0.001	—	0.0659	0.0659	0.928	93.561 ± 0.778	0.246 ± 0.002	6.439 ± 0.778
LSAM	beta56	0.00	0.002	0.002 ± 0.002	—	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	—	—	0.001	0.001	—	0.0557	0.0557	0.948	93.461 ± 0.784	0.213 ± 0.002	6.539 ± 0.784
LSAM	beta57	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	—	—	0.000	0.000	—	0.0337	0.0337	0.903	92.555 ± 0.833	0.121 ± 0.001	7.445 ± 0.833
LSAM	beta58	0.00	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	0.001	0.001	—	0.0253	0.0253	0.925	92.254 ± 0.848	0.101 ± 0.001	7.746 ± 0.848
QML	beta41	0.20	0.188	-0.012 ± 0.002	-0.059 ± 0.010	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	0.100 ± 0.005	0.316 ± 0.009	0.188	-0.012	-5.87	0.0596	0.0608	1.118	94.467 ± 0.725	0.273 ± 0.031	—
QML	beta42	0.20	0.190	-0.010 ± 0.002	-0.050 ± 0.010	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.099 ± 0.004	0.314 ± 0.009	0.190	-0.010	-5.22	0.0623	0.0632	1.032	95.171 ± 0.680	0.251 ± 0.003	—
QML	beta43	0.20	0.204	0.004 ± 0.001	0.022 ± 0.007	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.054 ± 0.003	0.233 ± 0.007	0.205	0.005	2.45	0.0458	0.0460	1.000	93.260 ± 0.795	0.182 ± 0.006	—
QML	beta44	0.00	-0.004	-0.004 ± 0.002	—	0.002 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	—	—	-0.002	-0.002	—	0.0471	0.0471	0.952	92.656 ± 0.827	0.186 ± 0.002	7.344 ± 0.827
QML	beta45	0.00	0.001	0.001 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	—	—	-0.001	-0.001	—	0.0504	0.0504	0.910	93.159 ± 0.801	0.173 ± 0.001	6.841 ± 0.801
QML	beta46	0.00	0.008	0.008 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	0.008	0.008	—	0.0292	0.0303	0.850	89.638 ± 0.967	0.100 ± 0.001	10.362 ± 0.967
QML	beta47	0.00	0.008	0.008 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	0.007	0.007	—	0.0248	0.0257	0.894	91.147 ± 0.901	0.098 ± 0.001	8.853 ± 0.901
QML	beta51	0.16	0.160	-0.000 ± 0.002	-0.001 ± 0.012	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.146 ± 0.007	0.382 ± 0.011	0.160	0.000	-0.30	0.0629	0.0629	0.985	94.366 ± 0.731	0.236 ± 0.003	—
QML	beta52	0.16	0.151	-0.009 ± 0.002	-0.054 ± 0.013	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	0.174 ± 0.009	0.417 ± 0.013	0.150	-0.010	-6.39	0.0635	0.0643	1.013	94.266 ± 0.737	0.263 ± 0.014	—
QML	beta53	0.16	0.159	-0.001 ± 0.002	-0.004 ± 0.011	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.131 ± 0.006	0.362 ± 0.010	0.158	-0.002	-1.24	0.0603	0.0604	0.969	93.461 ± 0.784	0.220 ± 0.006	—
QML	beta54	0.16	0.151	-0.009 ± 0.002	-0.058 ± 0.013	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.160 ± 0.007	0.400 ± 0.011	0.148	-0.012	-7.34	0.0651	0.0661	1.041	95.372 ± 0.666	0.258 ± 0.009	—
QML	beta55	0.00	-0.002	-0.002 ± 0.002	—	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	—	—	-0.005	-0.005	—	0.0656	0.0658	0.940	93.461 ± 0.784	0.239 ± 0.002	6.539 ± 0.784
QML	beta56	0.00	0.006	0.006 ± 0.002	—	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	—	—	0.005	0.005	—	0.0534	0.0537	0.953	93.360 ± 0.790	0.202 ± 0.001	6.640 ± 0.790
QML	beta57	0.00	0.006	0.006 ± 0.001	—	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	—	—	0.005	0.005	—	0.0325	0.0330	0.914	92.052 ± 0.858	0.116 ± 0.001	7.948 ± 0.858
QML	beta58	0.00	0.006	0.006 ± 0.001	—	0.001 ± 0.000	0.026 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	—	—	0.006	0.006	—	0.0241	0.0248	0.936	91.650 ± 0.877	0.095 ± 0.001	8.350 ± 0.877
UPI	beta41	0.20	0.199	-0.001 ± 0.002	-0.004 ± 0.012	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	0.145 ± 0.007	0.381 ± 0.011	0.197	-0.003	-1.33	0.0716	0.0717	0.976	94.467 ± 0.725	0.292 ± 0.001	—
UPI	beta42	0.20	0.200	-0.000 ± 0.002	-0.002 ± 0.012	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	0.147 ± 0.007	0.383 ± 0.011	0.201	0.001	0.75	0.0726	0.0726	1.010	96.378 ± 0.593	0.304 ± 0.001	—
UPI	beta43	0.20	0.202	0.002 ± 0.002	0.011 ± 0.008	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.056 ± 0.003	0.237 ± 0.007	0.203	0.003	1.63	0.0479	0.0480	0.957	93.058 ± 0.806	0.178 ± 0.001	—
UPI	beta44	0.00	0.000	0.000 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	—	—	0.001	0.001	—	0.0479	0.0479	0.961	93.561 ± 0.778	0.196 ± 0.002	6.439 ± 0.778
UPI	beta45	0.00	0.001	0.001 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	—	—	0.000	0.000	—	0.0513	0.0513	0.927	93.964 ± 0.755	0.188 ± 0.002	6.036 ± 0.755
UPI	beta46	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	—	—	0.000	0.000	—	0.0307	0.0307	0.863	92.555 ± 0.833	0.109 ± 0.001	7.445 ± 0.833

Table S227

Simulation 2 - Condition 83: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.8
(continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	-0.001	-0.001	—	0.0258	0.0258	0.901	93.461 ± 0.784	0.108 ± 0.001	6.539 ± 0.784
UPI	beta51	0.16	0.159	-0.001 ± 0.002	-0.005 ± 0.012	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.146 ± 0.007	0.382 ± 0.011	0.159	-0.001	-0.73	0.0627	0.0628	0.962	94.366 ± 0.731	0.231 ± 0.001	—
UPI	beta52	0.16	0.160	0.000 ± 0.002	0.002 ± 0.015	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.078 ± 0.003	0.237 ± 0.013	0.487 ± 0.016	0.158	-0.002	-1.44	0.0744	0.0745	0.960	94.769 ± 0.706	0.293 ± 0.001	—
UPI	beta53	0.16	0.159	-0.001 ± 0.002	-0.007 ± 0.012	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.149 ± 0.007	0.386 ± 0.010	0.160	0.000	0.14	0.0632	0.0632	0.934	93.159 ± 0.801	0.227 ± 0.001	—
UPI	beta54	0.16	0.159	-0.001 ± 0.002	-0.005 ± 0.015	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	0.217 ± 0.010	0.466 ± 0.013	0.157	-0.003	-2.02	0.0720	0.0721	1.001	94.970 ± 0.693	0.293 ± 0.001	—
UPI	beta55	0.00	0.001	0.001 ± 0.002	—	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	—	—	-0.001	-0.001	—	0.0668	0.0668	0.925	93.763 ± 0.767	0.250 ± 0.002	6.237 ± 0.767
UPI	beta56	0.00	0.001	0.001 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	—	—	0.001	0.001	—	0.0568	0.0568	0.951	93.561 ± 0.778	0.217 ± 0.002	6.439 ± 0.778
UPI	beta57	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	—	—	0.000	0.000	—	0.0329	0.0329	0.913	92.656 ± 0.827	0.121 ± 0.001	7.344 ± 0.827
UPI	beta58	0.00	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	0.000	0.000	—	0.0248	0.0248	0.921	92.153 ± 0.853	0.101 ± 0.001	7.847 ± 0.853

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S228

Simulation 2 - Condition 84: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.8

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Type I (%) $\pm$ MCSE
LSAM	beta41	0.20	0.198	-0.002 $\pm$ 0.001	-0.010 $\pm$ 0.007	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.053 $\pm$ 0.002	0.230 $\pm$ 0.006	0.198	-0.002	-0.83	0.0455	0.0455	0.981	94.289 $\pm$ 0.735	0.177 $\pm$ 0.000	—
LSAM	beta42	0.20	0.200	0.000 $\pm$ 0.002	0.002 $\pm$ 0.008	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.059 $\pm$ 0.003	0.243 $\pm$ 0.007	0.201	0.001	0.61	0.0469	0.0470	0.968	94.188 $\pm$ 0.741	0.184 $\pm$ 0.001	—
LSAM	beta43	0.20	0.200	-0.000 $\pm$ 0.001	-0.002 $\pm$ 0.005	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.021 $\pm$ 0.001	0.146 $\pm$ 0.004	0.199	-0.001	-0.30	0.0282	0.0282	0.987	94.790 $\pm$ 0.703	0.113 $\pm$ 0.000	—
LSAM	beta44	0.00	0.002	0.002 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	—	—	0.001	0.001	—	0.0288	0.0289	0.979	94.289 $\pm$ 0.735	0.112 $\pm$ 0.001	5.711 $\pm$ 0.735
LSAM	beta45	0.00	0.000	-0.000 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	—	—	-0.001	-0.001	—	0.0302	0.0302	0.951	93.888 $\pm$ 0.758	0.109 $\pm$ 0.001	6.112 $\pm$ 0.758
LSAM	beta46	0.00	0.001	0.001 $\pm$ 0.001	—	0.000 $\pm$ 0.000	0.017 $\pm$ 0.000	0.000 $\pm$ 0.000	0.017 $\pm$ 0.000	—	—	0.002	0.002	—	0.0165	0.0166	0.928	92.685 $\pm$ 0.824	0.061 $\pm$ 0.001	7.315 $\pm$ 0.824
LSAM	beta47	0.00	0.000	-0.000 $\pm$ 0.001	—	0.000 $\pm$ 0.000	0.017 $\pm$ 0.000	0.000 $\pm$ 0.000	0.017 $\pm$ 0.001	—	—	0.000	0.000	—	0.0153	0.0153	0.889	92.285 $\pm$ 0.845	0.060 $\pm$ 0.001	7.715 $\pm$ 0.845
LSAM	beta51	0.16	0.160	0.000 $\pm$ 0.001	0.002 $\pm$ 0.007	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.053 $\pm$ 0.002	0.230 $\pm$ 0.006	0.160	0.000	0.23	0.0362	0.0362	1.004	94.990 $\pm$ 0.691	0.145 $\pm$ 0.000	—
LSAM	beta52	0.16	0.162	0.002 $\pm$ 0.001	0.011 $\pm$ 0.009	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.085 $\pm$ 0.004	0.292 $\pm$ 0.008	0.161	0.001	0.66	0.0479	0.0480	0.984	95.190 $\pm$ 0.677	0.180 $\pm$ 0.000	—
LSAM	beta53	0.16	0.159	-0.001 $\pm$ 0.001	-0.005 $\pm$ 0.008	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.057 $\pm$ 0.003	0.239 $\pm$ 0.007	0.158	-0.002	-1.47	0.0381	0.0381	0.959	94.389 $\pm$ 0.728	0.144 $\pm$ 0.000	—
LSAM	beta54	0.16	0.159	-0.001 $\pm$ 0.001	-0.009 $\pm$ 0.009	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.079 $\pm$ 0.003	0.281 $\pm$ 0.007	0.158	-0.002	-1.36	0.0450	0.0451	1.019	94.890 $\pm$ 0.697	0.179 $\pm$ 0.001	—
LSAM	beta55	0.00	-0.001	-0.001 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	—	—	-0.002	-0.002	—	0.0392	0.0392	0.989	95.591 $\pm$ 0.650	0.149 $\pm$ 0.001	4.409 $\pm$ 0.650
LSAM	beta56	0.00	0.002	0.002 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	—	—	0.003	0.003	—	0.0343	0.0344	0.954	93.988 $\pm$ 0.752	0.128 $\pm$ 0.001	6.012 $\pm$ 0.752
LSAM	beta57	0.00	0.000	0.000 $\pm$ 0.001	—	0.000 $\pm$ 0.000	0.019 $\pm$ 0.000	0.000 $\pm$ 0.000	0.019 $\pm$ 0.001	—	—	0.000	0.000	—	0.0184	0.0184	0.940	93.287 $\pm$ 0.792	0.069 $\pm$ 0.001	6.713 $\pm$ 0.792
LSAM	beta58	0.00	0.000	0.000 $\pm$ 0.000	—	0.000 $\pm$ 0.000	0.016 $\pm$ 0.000	0.000 $\pm$ 0.000	0.016 $\pm$ 0.000	—	—	0.000	0.000	—	0.0144	0.0144	0.960	94.088 $\pm$ 0.747	0.059 $\pm$ 0.001	5.912 $\pm$ 0.747
QML	beta41	0.20	0.188	-0.012 $\pm$ 0.001	-0.058 $\pm$ 0.006	0.001 $\pm$ 0.000	0.039 $\pm$ 0.001	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.041 $\pm$ 0.002	0.202 $\pm$ 0.005	0.189	-0.011	-5.30	0.0378	0.0393	0.986	93.788 $\pm$ 0.764	0.150 $\pm$ 0.000	—
QML	beta42	0.20	0.191	-0.009 $\pm$ 0.001	-0.046 $\pm$ 0.006	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	0.043 $\pm$ 0.002	0.207 $\pm$ 0.006	0.192	-0.008	-4.07	0.0391	0.0399	0.975	93.487 $\pm$ 0.781	0.154 $\pm$ 0.000	—
QML	beta43	0.20	0.202	0.002 $\pm$ 0.001	0.010 $\pm$ 0.005	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.021 $\pm$ 0.001	0.144 $\pm$ 0.004	0.202	0.002	0.95	0.0278	0.0279	0.988	93.888 $\pm$ 0.758	0.111 $\pm$ 0.000	—
QML	beta44	0.00	-0.001	-0.001 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	—	—	-0.001	-0.001	—	0.0287	0.0287	0.984	93.287 $\pm$ 0.792	0.113 $\pm$ 0.001	6.713 $\pm$ 0.792
QML	beta45	0.00	0.000	0.000 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	—	—	0.000	0.000	—	0.0287	0.0287	0.966	94.289 $\pm$ 0.735	0.108 $\pm$ 0.001	5.711 $\pm$ 0.735
QML	beta46	0.00	0.008	0.008 $\pm$ 0.001	—	0.000 $\pm$ 0.000	0.016 $\pm$ 0.000	0.000 $\pm$ 0.000	0.018 $\pm$ 0.000	—	—	0.008	0.008	—	0.0166	0.0184	0.932	90.080 $\pm$ 0.946	0.059 $\pm$ 0.001	9.920 $\pm$ 0.946
QML	beta47	0.00	0.007	0.007 $\pm$ 0.001	—	0.000 $\pm$ 0.000	0.017 $\pm$ 0.000	0.000 $\pm$ 0.000	0.018 $\pm$ 0.001	—	—	0.006	0.006	—	0.0148	0.0160	0.881	89.780 $\pm$ 0.959	0.058 $\pm$ 0.000	10.220 $\pm$ 0.959
QML	beta51	0.16	0.161	0.001 $\pm$ 0.001	0.005 $\pm$ 0.007	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.053 $\pm$ 0.002	0.230 $\pm$ 0.006	0.161	0.001	0.53	0.0361	0.0362	1.011	94.990 $\pm$ 0.691	0.146 $\pm$ 0.000	—
QML	beta52	0.16	0.153	-0.007 $\pm$ 0.001	-0.042 $\pm$ 0.008	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	0.064 $\pm$ 0.003	0.254 $\pm$ 0.007	0.153	-0.007	-4.32	0.0411	0.0417	0.984	93.988 $\pm$ 0.752	0.155 $\pm$ 0.000	—
QML	beta53	0.16	0.160	-0.000 $\pm$ 0.001	-0.002 $\pm$ 0.007	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.050 $\pm$ 0.002	0.224 $\pm$ 0.006	0.158	-0.002	-1.27	0.0355	0.0356	0.966	95.090 $\pm$ 0.684	0.135 $\pm$ 0.000	—
QML	beta54	0.16	0.151	-0.009 $\pm$ 0.001	-0.059 $\pm$ 0.008	0.001 $\pm$ 0.000	0.039 $\pm$ 0.001	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.061 $\pm$ 0.003	0.248 $\pm$ 0.006	0.151	-0.009	-5.49	0.0392	0.0402	1.019	94.589 $\pm$ 0.716	0.154 $\pm$ 0.000	—
QML	beta55	0.00	-0.003	-0.003 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	—	—	-0.003	-0.003	—	0.0380	0.0382	1.001	95.291 $\pm$ 0.671	0.149 $\pm$ 0.001	4.709 $\pm$ 0.671
QML	beta56	0.00	0.007	0.007 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	—	—	0.007	0.007	—	0.0335	0.0342	0.960	93.186 $\pm$ 0.798	0.124 $\pm$ 0.001	6.814 $\pm$ 0.798
QML	beta57	0.00	0.006	0.006 $\pm$ 0.001	—	0.000 $\pm$ 0.000	0.019 $\pm$ 0.000	0.000 $\pm$ 0.000	0.019 $\pm$ 0.001	—	—	0.006	0.006	—	0.0185	0.0195	0.941	91.984 $\pm$ 0.860	0.068 $\pm$ 0.001	8.016 $\pm$ 0.860
QML	beta58	0.00	0.006	0.006 $\pm$ 0.000	—	0.000 $\pm$ 0.000	0.015 $\pm$ 0.000	0.000 $\pm$ 0.000	0.016 $\pm$ 0.000	—	—	0.005	0.005	—	0.0149	0.0158	0.957	91.784 $\pm$ 0.869	0.057 $\pm$ 0.001	8.216 $\pm$ 0.869
UPI	beta41	0.20	0.199	-0.001 $\pm$ 0.001	-0.007 $\pm$ 0.007	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.053 $\pm$ 0.002	0.230 $\pm$ 0.006	0.199	-0.001	-0.56	0.0467	0.0467	0.985	94.489 $\pm$ 0.722	0.177 $\pm$ 0.001	—
UPI	beta42	0.20	0.200	0.000 $\pm$ 0.002	0.001 $\pm$ 0.008	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.058 $\pm$ 0.003	0.241 $\pm$ 0.007	0.201	0.001	0.57	0.0472	0.0473	0.976	94.589 $\pm$ 0.716	0.184 $\pm$ 0.001	—
UPI	beta43	0.20	0.200	-0.000 $\pm$ 0.001	-0.001 $\pm$ 0.005	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.021 $\pm$ 0.001	0.146 $\pm$ 0.004	0.200	0.000	-0.07	0.0278	0.0278	0.979	94.790 $\pm$ 0.703	0.112 $\pm$ 0.000	—
UPI	beta44	0.00	0.002	0.002 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	—	—	0.001	0.001	—	0.0292	0.0293	0.989	94.489 $\pm$ 0.722	0.114 $\pm$ 0.001	5.511 $\pm$ 0.722
UPI	beta45	0.00	0.000	-0.000 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	—	—	0.000	0.000	—	0.0295	0.0295	0.965	94.389 $\pm$ 0.728	0.110 $\pm$ 0.001	5.611 $\pm$ 0.728
UPI	beta46	0.00	0.000	0.000 $\pm$ 0.001	—	0.000 $\pm$ 0.000	0.017 $\pm$ 0.000	0.000 $\pm$ 0.000	0.017 $\pm$ 0.000	—	—	0.001	0.001	—	0.0162	0.0162	0.929	93.186 $\pm$ 0.798	0.061 $\pm$ 0.001	6.814 $\pm$ 0.798

Table S228

Simulation 2 - Condition 84: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.8
(continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	0.000	-0.000 ± 0.001	—	0.000 ± 0.000	0.017 ± 0.000	0.000 ± 0.000	0.017 ± 0.001	—	—	-0.001	-0.001	—	0.0153	0.0153	0.895	93.086 ± 0.803	0.060 ± 0.001	6.914 ± 0.803
UPI	beta51	0.16	0.160	0.000 ± 0.001	0.002 ± 0.007	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.053 ± 0.002	0.230 ± 0.006	0.160	0.000	0.31	0.0365	0.0365	1.009	95.090 ± 0.684	0.145 ± 0.000	—
UPI	beta52	0.16	0.162	0.002 ± 0.001	0.011 ± 0.009	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.084 ± 0.004	0.290 ± 0.008	0.161	0.001	0.39	0.0467	0.0467	0.988	95.691 ± 0.643	0.180 ± 0.000	—
UPI	beta53	0.16	0.159	-0.001 ± 0.001	-0.004 ± 0.008	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.057 ± 0.003	0.239 ± 0.007	0.159	-0.001	-0.66	0.0378	0.0378	0.956	93.687 ± 0.770	0.143 ± 0.000	—
UPI	beta54	0.16	0.158	-0.002 ± 0.001	-0.010 ± 0.009	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.078 ± 0.003	0.280 ± 0.007	0.159	-0.001	-0.82	0.0434	0.0435	1.019	94.790 ± 0.703	0.179 ± 0.001	—
UPI	beta55	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.039 ± 0.001	0.001 ± 0.000	0.039 ± 0.001	—	—	-0.001	-0.001	—	0.0390	0.0390	0.994	94.790 ± 0.703	0.150 ± 0.001	5.210 ± 0.703
UPI	beta56	0.00	0.002	0.002 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	0.002	0.002	—	0.0334	0.0334	0.955	93.687 ± 0.770	0.130 ± 0.001	6.313 ± 0.770
UPI	beta57	0.00	0.000	0.000 ± 0.001	—	0.000 ± 0.000	0.019 ± 0.000	0.000 ± 0.000	0.019 ± 0.001	—	—	0.000	0.000	—	0.0183	0.0183	0.947	94.088 ± 0.747	0.069 ± 0.001	5.912 ± 0.747
UPI	beta58	0.00	0.000	0.000 ± 0.000	—	0.000 ± 0.000	0.016 ± 0.000	0.000 ± 0.000	0.016 ± 0.000	—	—	0.000	0.000	—	0.0148	0.0148	0.959	93.888 ± 0.758	0.059 ± 0.001	6.112 ± 0.758

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S229

Simulation 2 - Condition 85: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.4

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.199	-0.001 ± 0.005	-0.004 ± 0.026	0.008 ± 0.001	0.089 ± 0.004	0.008 ± 0.001	0.089 ± 0.005	0.199 ± 0.019	0.446 ± 0.025	0.202	0.002	1.16	0.0901	0.0901	1.110	96.066 ± 1.113	0.389 ± 0.010	—
LSAM	beta42	0.20	0.195	-0.005 ± 0.005	-0.024 ± 0.025	0.008 ± 0.001	0.089 ± 0.004	0.008 ± 0.001	0.089 ± 0.004	0.198 ± 0.016	0.445 ± 0.022	0.196	-0.004	-2.04	0.0882	0.0883	1.180	98.033 ± 0.795	0.411 ± 0.013	—
LSAM	beta43	0.20	0.196	-0.004 ± 0.005	-0.018 ± 0.024	0.007 ± 0.001	0.084 ± 0.004	0.007 ± 0.001	0.084 ± 0.004	0.175 ± 0.015	0.419 ± 0.021	0.199	-0.001	-0.62	0.0883	0.0883	1.100	97.049 ± 0.969	0.361 ± 0.007	—
LSAM	beta44	0.00	-0.005	-0.005 ± 0.008	—	0.018 ± 0.002	0.133 ± 0.007	0.018 ± 0.002	0.133 ± 0.008	—	—	-0.003	-0.003	—	0.1327	0.1328	1.236	98.361 ± 0.727	0.643 ± 0.028	1.639 ± 0.727
LSAM	beta45	0.00	0.002	0.002 ± 0.006	—	0.012 ± 0.001	0.111 ± 0.006	0.012 ± 0.001	0.111 ± 0.007	—	—	0.003	0.003	—	0.1104	0.1104	1.125	99.016 ± 0.565	0.488 ± 0.018	0.984 ± 0.565
LSAM	beta46	0.00	0.006	0.006 ± 0.010	—	0.029 ± 0.003	0.171 ± 0.009	0.029 ± 0.003	0.171 ± 0.010	—	—	0.001	0.001	—	0.1580	0.1580	1.191	99.016 ± 0.565	0.801 ± 0.033	0.984 ± 0.565
LSAM	beta47	0.00	0.005	0.005 ± 0.010	—	0.033 ± 0.004	0.181 ± 0.010	0.033 ± 0.004	0.181 ± 0.012	—	—	0.002	0.002	—	0.1594	0.1594	1.198	99.672 ± 0.327	0.849 ± 0.055	0.328 ± 0.327
LSAM	beta51	0.16	0.165	0.005 ± 0.006	0.032 ± 0.040	0.012 ± 0.001	0.111 ± 0.005	0.012 ± 0.001	0.111 ± 0.006	0.478 ± 0.046	0.691 ± 0.039	0.163	0.003	1.69	0.1097	0.1097	0.984	97.377 ± 0.915	0.427 ± 0.005	—
LSAM	beta52	0.16	0.164	0.004 ± 0.005	0.022 ± 0.034	0.009 ± 0.001	0.095 ± 0.004	0.009 ± 0.001	0.095 ± 0.005	0.354 ± 0.030	0.595 ± 0.030	0.158	-0.002	-1.43	0.0918	0.0918	1.030	95.410 ± 1.198	0.384 ± 0.009	—
LSAM	beta53	0.16	0.157	-0.003 ± 0.005	-0.018 ± 0.033	0.009 ± 0.001	0.093 ± 0.005	0.009 ± 0.001	0.093 ± 0.006	0.338 ± 0.035	0.582 ± 0.034	0.157	-0.003	-1.72	0.0918	0.0919	1.033	97.377 ± 0.915	0.377 ± 0.007	—
LSAM	beta54	0.16	0.157	-0.003 ± 0.005	-0.018 ± 0.033	0.009 ± 0.001	0.093 ± 0.005	0.009 ± 0.001	0.093 ± 0.006	0.339 ± 0.037	0.582 ± 0.036	0.156	-0.004	-2.53	0.0894	0.0895	1.083	97.049 ± 0.969	0.396 ± 0.011	—
LSAM	beta55	0.00	0.002	0.002 ± 0.009	—	0.024 ± 0.002	0.156 ± 0.008	0.024 ± 0.002	0.156 ± 0.009	—	—	0.010	0.010	—	0.1560	0.1563	1.115	98.361 ± 0.727	0.682 ± 0.022	1.639 ± 0.727
LSAM	beta56	0.00	0.002	0.002 ± 0.008	—	0.020 ± 0.002	0.141 ± 0.008	0.020 ± 0.002	0.140 ± 0.009	—	—	0.000	0.000	—	0.1303	0.1303	1.067	98.033 ± 0.795	0.588 ± 0.016	1.967 ± 0.795
LSAM	beta57	0.00	-0.010	-0.010 ± 0.011	—	0.035 ± 0.004	0.186 ± 0.011	0.035 ± 0.004	0.186 ± 0.013	—	—	-0.008	-0.008	—	0.1504	0.1506	1.060	100.000 ± 0.000	0.773 ± 0.036	0.000 ± 0.000
LSAM	beta58	0.00	0.005	0.005 ± 0.010	—	0.030 ± 0.003	0.172 ± 0.010	0.030 ± 0.003	0.172 ± 0.011	—	—	-0.002	-0.002	—	0.1424	0.1424	1.108	99.016 ± 0.565	0.748 ± 0.036	0.984 ± 0.565
QML	beta41	0.20	0.196	-0.004 ± 0.004	-0.020 ± 0.022	0.006 ± 0.000	0.077 ± 0.003	0.006 ± 0.000	0.077 ± 0.004	0.147 ± 0.012	0.383 ± 0.020	0.191	-0.009	-4.52	0.0850	0.0855	0.951	93.770 ± 1.384	0.286 ± 0.003	—
QML	beta42	0.20	0.195	-0.005 ± 0.004	-0.027 ± 0.021	0.006 ± 0.000	0.074 ± 0.003	0.006 ± 0.000	0.074 ± 0.004	0.138 ± 0.011	0.372 ± 0.019	0.199	-0.001	-0.37	0.0778	0.0778	1.033	96.393 ± 1.068	0.301 ± 0.003	—
QML	beta43	0.20	0.196	-0.004 ± 0.004	-0.022 ± 0.020	0.005 ± 0.000	0.071 ± 0.003	0.005 ± 0.000	0.071 ± 0.004	0.126 ± 0.010	0.354 ± 0.018	0.200	0.000	0.11	0.0756	0.0756	1.037	96.393 ± 1.068	0.288 ± 0.003	—
QML	beta44	0.00	-0.012	-0.012 ± 0.005	—	0.009 ± 0.001	0.094 ± 0.004	0.009 ± 0.001	0.095 ± 0.005	—	—	-0.005	-0.005	—	0.1037	0.1038	1.105	96.721 ± 1.020	0.408 ± 0.007	3.279 ± 1.020
QML	beta45	0.00	-0.006	-0.006 ± 0.005	—	0.007 ± 0.001	0.086 ± 0.003	0.007 ± 0.001	0.086 ± 0.004	—	—	0.001	0.001	—	0.0992	0.0992	1.009	96.393 ± 1.068	0.339 ± 0.005	3.607 ± 1.068
QML	beta46	0.00	0.005	0.005 ± 0.004	—	0.006 ± 0.001	0.077 ± 0.004	0.006 ± 0.001	0.077 ± 0.004	—	—	-0.003	-0.003	—	0.0785	0.0786	1.016	95.410 ± 1.198	0.306 ± 0.007	4.590 ± 1.198
QML	beta47	0.00	-0.001	-0.001 ± 0.004	—	0.005 ± 0.000	0.073 ± 0.003	0.005 ± 0.000	0.073 ± 0.004	—	—	-0.004	-0.004	—	0.0746	0.0748	1.023	97.377 ± 0.915	0.294 ± 0.005	2.623 ± 0.915
QML	beta51	0.16	0.170	0.010 ± 0.006	0.065 ± 0.037	0.011 ± 0.001	0.104 ± 0.005	0.011 ± 0.001	0.105 ± 0.006	0.428 ± 0.039	0.654 ± 0.035	0.164	0.004	2.69	0.1045	0.1046	1.030	97.377 ± 0.915	0.421 ± 0.006	—
QML	beta52	0.16	0.159	-0.001 ± 0.005	-0.004 ± 0.032	0.008 ± 0.001	0.088 ± 0.004	0.008 ± 0.001	0.088 ± 0.005	0.303 ± 0.027	0.550 ± 0.029	0.157	-0.003	-1.85	0.0912	0.0912	0.946	93.443 ± 1.417	0.327 ± 0.005	—
QML	beta53	0.16	0.158	-0.002 ± 0.005	-0.013 ± 0.030	0.007 ± 0.001	0.083 ± 0.004	0.007 ± 0.001	0.083 ± 0.004	0.271 ± 0.023	0.520 ± 0.027	0.164	0.004	2.77	0.0872	0.0873	1.000	95.082 ± 1.238	0.327 ± 0.004	—
QML	beta54	0.16	0.154	-0.006 ± 0.004	-0.040 ± 0.027	0.006 ± 0.000	0.076 ± 0.003	0.006 ± 0.000	0.076 ± 0.004	0.227 ± 0.018	0.477 ± 0.023	0.152	-0.008	-5.03	0.0783	0.0788	1.021	95.082 ± 1.238	0.305 ± 0.003	—
QML	beta55	0.00	-0.002	-0.002 ± 0.007	—	0.016 ± 0.002	0.127 ± 0.006	0.016 ± 0.002	0.127 ± 0.007	—	—	-0.003	-0.003	—	0.1279	0.1280	1.041	96.393 ± 1.068	0.519 ± 0.014	3.607 ± 1.068
QML	beta56	0.00	-0.008	-0.008 ± 0.006	—	0.012 ± 0.001	0.112 ± 0.005	0.012 ± 0.001	0.112 ± 0.006	—	—	-0.009	-0.009	—	0.1133	0.1137	0.972	95.738 ± 1.157	0.425 ± 0.006	4.262 ± 1.157
QML	beta57	0.00	-0.007	-0.007 ± 0.005	—	0.006 ± 0.001	0.079 ± 0.005	0.006 ± 0.001	0.080 ± 0.005	—	—	-0.004	-0.004	—	0.0803	0.0803	1.023	97.049 ± 0.969	0.319 ± 0.010	2.951 ± 0.969
QML	beta58	0.00	-0.001	-0.001 ± 0.004	—	0.004 ± 0.000	0.063 ± 0.003	0.004 ± 0.000	0.063 ± 0.003	—	—	-0.005	-0.005	—	0.0655	0.0657	1.065	97.377 ± 0.915	0.265 ± 0.005	2.623 ± 0.915
UPI	beta41	0.20	0.204	0.004 ± 0.006	0.022 ± 0.029	0.010 ± 0.001	0.102 ± 0.005	0.010 ± 0.001	0.102 ± 0.006	0.261 ± 0.026	0.511 ± 0.029	0.202	0.002	0.95	0.1143	0.1143	1.596	98.033 ± 0.795	0.640 ± 0.031	—
UPI	beta42	0.20	0.196	-0.004 ± 0.006	-0.019 ± 0.031	0.011 ± 0.001	0.107 ± 0.005	0.011 ± 0.001	0.107 ± 0.006	0.284 ± 0.027	0.533 ± 0.030	0.202	0.002	1.16	0.1129	0.1130	1.577	98.689 ± 0.651	0.659 ± 0.036	—
UPI	beta43	0.20	0.198	-0.002 ± 0.005	-0.008 ± 0.027	0.009 ± 0.001	0.096 ± 0.005	0.009 ± 0.001	0.096 ± 0.006	0.230 ± 0.024	0.479 ± 0.029	0.199	-0.001	-0.60	0.1032	0.1032	1.565	99.344 ± 0.462	0.589 ± 0.038	—
UPI	beta44	0.00	0.001	0.001 ± 0.010	—	0.029 ± 0.004	0.170 ± 0.011	0.029 ± 0.004	0.169 ± 0.012	—	—	-0.003	-0.003	—	0.1702	0.1703	1.877	100.000 ± 0.000	1.248 ± 0.081	0.000 ± 0.000
UPI	beta45	0.00	-0.011	-0.011 ± 0.009	—	0.025 ± 0.003	0.159 ± 0.009	0.025 ± 0.003	0.159 ± 0.010	—	—	-0.013	-0.013	—	0.1509	0.1515	1.800	99.672 ± 0.327	1.123 ± 0.074	0.328 ± 0.327
UPI	beta46	0.00	0.016	0.016 ± 0.017	—	0.088 ± 0.011	0.296 ± 0.019	0.088 ± 0.011	0.296 ± 0.020	—	—	0.007	0.007	—	0.2514	0.2515	1.859	100.000 ± 0.000	2.158 ± 0.143	0.000 ± 0.000

Table S229
Simulation 2 - Condition 85: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.4 (continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	0.015	0.015 ± 0.017	—	0.093 ± 0.011	0.306 ± 0.018	0.093 ± 0.011	0.305 ± 0.020	—	—	0.001	0.001	—	0.2430	0.2430	1.924	100.000 ± 0.000	2.305 ± 0.195	0.000 ± 0.000
UPI	beta51	0.16	0.166	0.006 ± 0.006	0.040 ± 0.039	0.012 ± 0.001	0.108 ± 0.005	0.012 ± 0.001	0.108 ± 0.006	0.454 ± 0.044	0.674 ± 0.038	0.159	-0.001	-0.83	0.1069	0.1069	1.268	98.033 ± 0.795	0.536 ± 0.019	—
UPI	beta52	0.16	0.163	0.003 ± 0.006	0.017 ± 0.037	0.011 ± 0.001	0.103 ± 0.004	0.011 ± 0.001	0.103 ± 0.005	0.412 ± 0.034	0.642 ± 0.032	0.163	0.003	1.64	0.1139	0.1139	1.409	96.721 ± 1.020	0.568 ± 0.034	—
UPI	beta53	0.16	0.154	-0.006 ± 0.006	-0.036 ± 0.036	0.010 ± 0.001	0.101 ± 0.005	0.010 ± 0.001	0.101 ± 0.006	0.396 ± 0.040	0.629 ± 0.037	0.148	-0.012	-7.68	0.1071	0.1078	1.483	98.033 ± 0.795	0.585 ± 0.036	—
UPI	beta54	0.16	0.163	0.003 ± 0.006	0.017 ± 0.035	0.010 ± 0.001	0.098 ± 0.005	0.010 ± 0.001	0.098 ± 0.005	0.372 ± 0.035	0.610 ± 0.034	0.160	0.000	-0.23	0.0989	0.0989	1.480	98.033 ± 0.795	0.567 ± 0.029	—
UPI	beta55	0.00	-0.017	-0.017 ± 0.011	—	0.035 ± 0.004	0.187 ± 0.011	0.035 ± 0.004	0.187 ± 0.013	—	—	-0.012	-0.012	—	0.1813	0.1817	1.749	99.672 ± 0.327	1.282 ± 0.105	0.328 ± 0.327
UPI	beta56	0.00	0.013	0.013 ± 0.010	—	0.032 ± 0.003	0.179 ± 0.010	0.032 ± 0.004	0.179 ± 0.011	—	—	0.002	0.002	—	0.1691	0.1691	1.575	99.672 ± 0.327	1.102 ± 0.067	0.328 ± 0.327
UPI	beta57	0.00	-0.019	-0.019 ± 0.014	—	0.062 ± 0.007	0.249 ± 0.014	0.062 ± 0.007	0.249 ± 0.016	—	—	-0.008	-0.008	—	0.2000	0.2001	1.694	100.000 ± 0.000	1.653 ± 0.134	0.000 ± 0.000
UPI	beta58	0.00	0.012	0.012 ± 0.015	—	0.068 ± 0.008	0.260 ± 0.016	0.067 ± 0.008	0.260 ± 0.018	—	—	-0.004	-0.004	—	0.1918	0.1918	1.922	100.000 ± 0.000	1.959 ± 0.176	0.000 ± 0.000

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S230

Simulation 2 - Condition 86: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, $N = 1000$, Reliability = 0.4

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Type I (%) $\pm$ MCSE
LSAM	beta41	0.20	0.201	0.001 $\pm$ 0.002	0.003 $\pm$ 0.009	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.002	0.058 $\pm$ 0.003	0.241 $\pm$ 0.008	0.199	-0.001	-0.34	0.0476	0.0476	1.059	96.038 $\pm$ 0.721	0.200 $\pm$ 0.001	—
LSAM	beta42	0.20	0.205	0.005 $\pm$ 0.002	0.023 $\pm$ 0.009	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.003 $\pm$ 0.000	0.051 $\pm$ 0.002	0.065 $\pm$ 0.004	0.254 $\pm$ 0.009	0.205	0.005	2.37	0.0482	0.0484	1.059	95.219 $\pm$ 0.789	0.210 $\pm$ 0.001	—
LSAM	beta43	0.20	0.202	0.002 $\pm$ 0.002	0.009 $\pm$ 0.009	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.058 $\pm$ 0.003	0.241 $\pm$ 0.007	0.200	0.000	0.17	0.0515	0.0515	1.008	95.492 $\pm$ 0.767	0.191 $\pm$ 0.001	—
LSAM	beta44	0.00	0.002	0.002 $\pm$ 0.003	—	0.005 $\pm$ 0.000	0.069 $\pm$ 0.002	0.005 $\pm$ 0.000	0.069 $\pm$ 0.002	—	—	0.002	0.002	—	0.0693	0.0693	1.052	96.995 $\pm$ 0.631	0.285 $\pm$ 0.003	3.005 $\pm$ 0.631
LSAM	beta45	0.00	0.001	0.001 $\pm$ 0.002	—	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	—	—	0.001	0.001	—	0.0597	0.0597	0.997	96.175 $\pm$ 0.709	0.233 $\pm$ 0.002	3.825 $\pm$ 0.709
LSAM	beta46	0.00	0.001	0.001 $\pm$ 0.004	—	0.009 $\pm$ 0.001	0.096 $\pm$ 0.003	0.009 $\pm$ 0.001	0.096 $\pm$ 0.004	—	—	-0.002	-0.002	—	0.0874	0.0874	1.034	98.361 $\pm$ 0.469	0.387 $\pm$ 0.006	1.639 $\pm$ 0.469
LSAM	beta47	0.00	-0.005	-0.005 $\pm$ 0.003	—	0.009 $\pm$ 0.001	0.094 $\pm$ 0.003	0.009 $\pm$ 0.001	0.094 $\pm$ 0.003	—	—	-0.004	-0.004	—	0.0846	0.0847	1.070	97.541 $\pm$ 0.572	0.395 $\pm$ 0.006	2.459 $\pm$ 0.572
LSAM	beta51	0.16	0.163	0.003 $\pm$ 0.003	0.021 $\pm$ 0.016	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.183 $\pm$ 0.010	0.428 $\pm$ 0.014	0.162	0.002	1.20	0.0671	0.0672	0.930	93.579 $\pm$ 0.906	0.249 $\pm$ 0.001	—
LSAM	beta52	0.16	0.162	0.002 $\pm$ 0.002	0.012 $\pm$ 0.011	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	0.002 $\pm$ 0.000	0.050 $\pm$ 0.002	0.097 $\pm$ 0.006	0.311 $\pm$ 0.011	0.160	0.000	0.06	0.0487	0.0487	1.039	96.448 $\pm$ 0.684	0.203 $\pm$ 0.001	—
LSAM	beta53	0.16	0.159	-0.001 $\pm$ 0.002	-0.003 $\pm$ 0.013	0.003 $\pm$ 0.000	0.055 $\pm$ 0.001	0.003 $\pm$ 0.000	0.055 $\pm$ 0.002	0.117 $\pm$ 0.006	0.343 $\pm$ 0.011	0.158	-0.002	-1.54	0.0562	0.0562	0.977	95.492 $\pm$ 0.767	0.210 $\pm$ 0.001	—
LSAM	beta54	0.16	0.161	0.001 $\pm$ 0.002	0.005 $\pm$ 0.011	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.002	0.093 $\pm$ 0.005	0.305 $\pm$ 0.010	0.159	-0.001	-0.80	0.0500	0.0500	1.054	97.404 $\pm$ 0.588	0.201 $\pm$ 0.001	—
LSAM	beta55	0.00	0.002	0.002 $\pm$ 0.003	—	0.007 $\pm$ 0.000	0.086 $\pm$ 0.002	0.007 $\pm$ 0.000	0.086 $\pm$ 0.003	—	—	-0.001	-0.001	—	0.0795	0.0795	1.003	95.765 $\pm$ 0.744	0.338 $\pm$ 0.004	4.235 $\pm$ 0.744
LSAM	beta56	0.00	0.002	0.002 $\pm$ 0.003	—	0.006 $\pm$ 0.000	0.078 $\pm$ 0.002	0.006 $\pm$ 0.000	0.078 $\pm$ 0.003	—	—	0.001	0.001	—	0.0750	0.0751	0.984	96.311 $\pm$ 0.697	0.302 $\pm$ 0.003	3.689 $\pm$ 0.697
LSAM	beta57	0.00	-0.002	-0.002 $\pm$ 0.004	—	0.009 $\pm$ 0.001	0.096 $\pm$ 0.003	0.009 $\pm$ 0.001	0.096 $\pm$ 0.004	—	—	-0.002	-0.002	—	0.0857	0.0857	1.043	97.541 $\pm$ 0.572	0.391 $\pm$ 0.007	2.459 $\pm$ 0.572
LSAM	beta58	0.00	0.000	0.000 $\pm$ 0.003	—	0.008 $\pm$ 0.000	0.087 $\pm$ 0.003	0.008 $\pm$ 0.000	0.087 $\pm$ 0.003	—	—	0.002	0.002	—	0.0794	0.0794	1.028	97.814 $\pm$ 0.540	0.350 $\pm$ 0.005	2.186 $\pm$ 0.540
QML	beta41	0.20	0.198	-0.002 $\pm$ 0.002	-0.008 $\pm$ 0.008	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.049 $\pm$ 0.003	0.221 $\pm$ 0.007	0.196	-0.004	-2.00	0.0440	0.0442	1.013	95.355 $\pm$ 0.778	0.175 $\pm$ 0.001	—
QML	beta42	0.20	0.201	0.001 $\pm$ 0.002	0.006 $\pm$ 0.009	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.002	0.054 $\pm$ 0.003	0.233 $\pm$ 0.008	0.202	0.002	0.99	0.0422	0.0422	1.009	94.262 $\pm$ 0.860	0.184 $\pm$ 0.001	—
QML	beta43	0.20	0.200	0.000 $\pm$ 0.002	0.002 $\pm$ 0.008	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.049 $\pm$ 0.002	0.222 $\pm$ 0.007	0.198	-0.002	-1.11	0.0490	0.0490	1.006	95.628 $\pm$ 0.756	0.175 $\pm$ 0.001	—
QML	beta44	0.00	-0.008	-0.008 $\pm$ 0.002	—	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	—	—	-0.006	-0.006	—	0.0631	0.0634	1.023	95.082 $\pm$ 0.799	0.243 $\pm$ 0.002	4.918 $\pm$ 0.799
QML	beta45	0.00	-0.005	-0.005 $\pm$ 0.002	—	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.003 $\pm$ 0.000	0.051 $\pm$ 0.002	—	—	-0.004	-0.004	—	0.0511	0.0513	1.002	94.536 $\pm$ 0.840	0.200 $\pm$ 0.001	5.464 $\pm$ 0.840
QML	beta46	0.00	-0.001	-0.001 $\pm$ 0.002	—	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.002 $\pm$ 0.000	0.045 $\pm$ 0.002	—	—	-0.001	-0.001	—	0.0439	0.0439	1.033	95.355 $\pm$ 0.778	0.180 $\pm$ 0.002	4.645 $\pm$ 0.778
QML	beta47	0.00	-0.007	-0.007 $\pm$ 0.002	—	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	—	—	-0.008	-0.008	—	0.0417	0.0425	1.045	96.038 $\pm$ 0.721	0.177 $\pm$ 0.001	3.962 $\pm$ 0.721
QML	beta51	0.16	0.165	0.005 $\pm$ 0.002	0.033 $\pm$ 0.015	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	0.171 $\pm$ 0.009	0.413 $\pm$ 0.014	0.164	0.004	2.42	0.0668	0.0669	0.958	94.536 $\pm$ 0.840	0.248 $\pm$ 0.001	—
QML	beta52	0.16	0.160	0.000 $\pm$ 0.002	0.000 $\pm$ 0.011	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.002	0.087 $\pm$ 0.005	0.296 $\pm$ 0.010	0.159	-0.001	-0.51	0.0468	0.0468	1.025	95.355 $\pm$ 0.778	0.190 $\pm$ 0.001	—
QML	beta53	0.16	0.158	-0.002 $\pm$ 0.002	-0.011 $\pm$ 0.012	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.003 $\pm$ 0.000	0.051 $\pm$ 0.002	0.101 $\pm$ 0.005	0.318 $\pm$ 0.010	0.157	-0.003	-2.00	0.0546	0.0547	0.977	94.672 $\pm$ 0.830	0.195 $\pm$ 0.001	—
QML	beta54	0.16	0.159	-0.001 $\pm$ 0.002	-0.004 $\pm$ 0.011	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.083 $\pm$ 0.004	0.288 $\pm$ 0.009	0.158	-0.002	-1.10	0.0470	0.0470	1.027	96.721 $\pm$ 0.658	0.186 $\pm$ 0.001	—
QML	beta55	0.00	0.000	0.000 $\pm$ 0.003	—	0.006 $\pm$ 0.000	0.079 $\pm$ 0.002	0.006 $\pm$ 0.000	0.079 $\pm$ 0.003	—	—	-0.002	-0.002	—	0.0767	0.0767	0.984	95.082 $\pm$ 0.799	0.303 $\pm$ 0.002	4.918 $\pm$ 0.799
QML	beta56	0.00	-0.006	-0.006 $\pm$ 0.002	—	0.004 $\pm$ 0.000	0.067 $\pm$ 0.002	0.005 $\pm$ 0.000	0.067 $\pm$ 0.002	—	—	-0.008	-0.008	—	0.0682	0.0687	0.977	95.219 $\pm$ 0.789	0.257 $\pm$ 0.001	4.781 $\pm$ 0.789
QML	beta57	0.00	-0.004	-0.004 $\pm$ 0.002	—	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.002	—	—	-0.005	-0.005	—	0.0457	0.0460	1.041	94.945 $\pm$ 0.810	0.190 $\pm$ 0.002	5.055 $\pm$ 0.810
QML	beta58	0.00	-0.005	-0.005 $\pm$ 0.001	—	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	—	—	-0.004	-0.004	—	0.0385	0.0386	1.028	95.219 $\pm$ 0.789	0.158 $\pm$ 0.001	4.781 $\pm$ 0.789
UPI	beta41	0.20	0.202	0.002 $\pm$ 0.002	0.008 $\pm$ 0.009	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	0.002 $\pm$ 0.000	0.050 $\pm$ 0.002	0.062 $\pm$ 0.004	0.249 $\pm$ 0.009	0.202	0.002	0.86	0.0523	0.0523	1.241	97.404 $\pm$ 0.588	0.243 $\pm$ 0.005	—
UPI	beta42	0.20	0.205	0.005 $\pm$ 0.002	0.025 $\pm$ 0.010	0.003 $\pm$ 0.000	0.053 $\pm$ 0.002	0.003 $\pm$ 0.000	0.053 $\pm$ 0.002	0.070 $\pm$ 0.004	0.265 $\pm$ 0.009	0.205	0.005	2.38	0.0528	0.0531	1.238	96.038 $\pm$ 0.721	0.256 $\pm$ 0.006	—
UPI	beta43	0.20	0.202	0.002 $\pm$ 0.002	0.009 $\pm$ 0.009	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.002	0.060 $\pm$ 0.003	0.245 $\pm$ 0.008	0.201	0.001	0.70	0.0547	0.0547	1.178	97.678 $\pm$ 0.557	0.226 $\pm$ 0.006	—
UPI	beta44	0.00	0.000	0.000 $\pm$ 0.003	—	0.006 $\pm$ 0.000	0.078 $\pm$ 0.002	0.006 $\pm$ 0.000	0.077 $\pm$ 0.003	—	—	0.001	0.001	—	0.0802	0.0802	1.389	99.044 $\pm$ 0.360	0.422 $\pm$ 0.017	0.956 $\pm$ 0.360
UPI	beta45	0.00	0.001	0.001 $\pm$ 0.002	—	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	—	—	0.002	0.002	—	0.0696	0.0697	1.193	98.361 $\pm$ 0.469	0.310 $\pm$ 0.006	1.639 $\pm$ 0.469
UPI	beta46	0.00	0.003	0.003 $\pm$ 0.004	—	0.014 $\pm$ 0.001	0.119 $\pm$ 0.004	0.014 $\pm$ 0.001	0.119 $\pm$ 0.005	—	—	0.001	0.001	—	0.1018	0.1018	1.491	99.727 $\pm$ 0.193	0.695 $\pm$ 0.027	0.273 $\pm$ 0.193

Table S230

Simulation 2 - Condition 86: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.4
(continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.009	-0.009 ± 0.005	—	0.017 ± 0.001	0.129 ± 0.005	0.017 ± 0.001	0.129 ± 0.005	—	—	-0.005	-0.005	—	0.1101	0.1102	1.640	99.727 ± 0.193	0.828 ± 0.045	0.273 ± 0.193
UPI	beta51	0.16	0.164	0.004 ± 0.002	0.022 ± 0.015	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.171 ± 0.009	0.413 ± 0.013	0.162	0.002	1.54	0.0685	0.0685	1.004	95.082 ± 0.799	0.260 ± 0.002	—
UPI	beta52	0.16	0.162	0.002 ± 0.002	0.012 ± 0.012	0.003 ± 0.000	0.051 ± 0.002	0.003 ± 0.000	0.051 ± 0.002	0.100 ± 0.006	0.316 ± 0.011	0.161	0.001	0.37	0.0486	0.0486	1.138	97.404 ± 0.588	0.226 ± 0.004	—
UPI	beta53	0.16	0.159	-0.001 ± 0.002	-0.007 ± 0.013	0.003 ± 0.000	0.054 ± 0.002	0.003 ± 0.000	0.054 ± 0.002	0.116 ± 0.006	0.340 ± 0.011	0.158	-0.002	-1.12	0.0551	0.0551	1.084	96.311 ± 0.697	0.231 ± 0.003	—
UPI	beta54	0.16	0.161	0.001 ± 0.002	0.005 ± 0.011	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.093 ± 0.005	0.305 ± 0.009	0.159	-0.001	-0.69	0.0488	0.0489	1.168	97.541 ± 0.572	0.223 ± 0.004	—
UPI	beta55	0.00	0.003	0.003 ± 0.003	—	0.008 ± 0.000	0.089 ± 0.003	0.008 ± 0.000	0.089 ± 0.003	—	—	-0.001	-0.001	—	0.0837	0.0837	1.187	98.224 ± 0.488	0.414 ± 0.009	1.776 ± 0.488
UPI	beta56	0.00	0.001	0.001 ± 0.003	—	0.007 ± 0.000	0.083 ± 0.002	0.007 ± 0.000	0.083 ± 0.003	—	—	0.002	0.002	—	0.0810	0.0810	1.144	97.541 ± 0.572	0.374 ± 0.008	2.459 ± 0.572
UPI	beta57	0.00	-0.005	-0.005 ± 0.004	—	0.011 ± 0.001	0.107 ± 0.004	0.011 ± 0.001	0.107 ± 0.004	—	—	-0.001	-0.001	—	0.0956	0.0956	1.435	99.863 ± 0.137	0.600 ± 0.023	0.137 ± 0.137
UPI	beta58	0.00	-0.005	-0.005 ± 0.004	—	0.011 ± 0.001	0.103 ± 0.004	0.011 ± 0.001	0.103 ± 0.004	—	—	0.000	0.000	—	0.0857	0.0857	1.542	99.317 ± 0.304	0.625 ± 0.036	0.683 ± 0.304

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S231

Simulation 2 - Condition 87: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.203	0.003 ± 0.002	0.013 ± 0.009	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.078 ± 0.004	0.279 ± 0.008	0.203	0.003	1.26	0.0537	0.0538	1.018	95.657 ± 0.655	0.222 ± 0.001	—
LSAM	beta42	0.20	0.200	-0.000 ± 0.002	-0.000 ± 0.009	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.084 ± 0.004	0.290 ± 0.008	0.200	0.000	-0.12	0.0588	0.0588	1.014	94.933 ± 0.705	0.231 ± 0.001	—
LSAM	beta43	0.20	0.199	-0.001 ± 0.002	-0.004 ± 0.009	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.079 ± 0.003	0.280 ± 0.008	0.200	0.000	-0.22	0.0577	0.0577	0.984	94.623 ± 0.725	0.216 ± 0.001	—
LSAM	beta44	0.00	0.002	0.002 ± 0.002	—	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	—	—	0.003	0.003	—	0.0705	0.0705	0.966	95.657 ± 0.655	0.269 ± 0.002	4.343 ± 0.655
LSAM	beta45	0.00	-0.001	-0.001 ± 0.002	—	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	—	—	-0.003	-0.003	—	0.0624	0.0624	0.969	95.243 ± 0.684	0.237 ± 0.001	4.757 ± 0.684
LSAM	beta46	0.00	-0.002	-0.002 ± 0.003	—	0.008 ± 0.000	0.089 ± 0.002	0.008 ± 0.000	0.089 ± 0.003	—	—	0.002	0.002	—	0.0844	0.0845	0.980	95.863 ± 0.640	0.342 ± 0.003	4.137 ± 0.640
LSAM	beta47	0.00	-0.002	-0.002 ± 0.003	—	0.008 ± 0.000	0.090 ± 0.002	0.008 ± 0.000	0.090 ± 0.003	—	—	0.000	0.000	—	0.0845	0.0845	0.980	95.140 ± 0.692	0.344 ± 0.003	4.860 ± 0.692
LSAM	beta51	0.16	0.159	-0.001 ± 0.002	-0.008 ± 0.015	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	0.220 ± 0.010	0.469 ± 0.013	0.159	-0.001	-0.53	0.0765	0.0765	0.970	94.209 ± 0.751	0.285 ± 0.001	—
LSAM	beta52	0.16	0.162	0.002 ± 0.002	0.012 ± 0.012	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.135 ± 0.006	0.368 ± 0.010	0.162	0.002	1.45	0.0604	0.0605	1.001	95.863 ± 0.640	0.231 ± 0.001	—
LSAM	beta53	0.16	0.163	0.003 ± 0.002	0.020 ± 0.012	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	0.151 ± 0.007	0.389 ± 0.011	0.162	0.002	1.14	0.0610	0.0610	0.974	94.726 ± 0.719	0.237 ± 0.001	—
LSAM	beta54	0.16	0.160	-0.000 ± 0.002	-0.002 ± 0.012	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.140 ± 0.006	0.374 ± 0.011	0.160	0.000	-0.20	0.0585	0.0585	0.984	95.036 ± 0.698	0.231 ± 0.001	—
LSAM	beta55	0.00	0.002	0.002 ± 0.003	—	0.007 ± 0.000	0.083 ± 0.002	0.007 ± 0.000	0.083 ± 0.002	—	—	-0.002	-0.002	—	0.0793	0.0793	1.007	94.933 ± 0.705	0.327 ± 0.002	5.067 ± 0.705
LSAM	beta56	0.00	0.000	-0.000 ± 0.003	—	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	—	—	0.000	0.000	—	0.0782	0.0782	0.991	95.243 ± 0.684	0.305 ± 0.002	4.757 ± 0.684
LSAM	beta57	0.00	0.000	0.000 ± 0.003	—	0.009 ± 0.000	0.093 ± 0.002	0.009 ± 0.000	0.093 ± 0.003	—	—	0.002	0.002	—	0.0874	0.0874	0.958	94.623 ± 0.725	0.349 ± 0.003	5.377 ± 0.725
LSAM	beta58	0.00	-0.001	-0.001 ± 0.003	—	0.008 ± 0.000	0.088 ± 0.002	0.008 ± 0.000	0.088 ± 0.003	—	—	-0.001	-0.001	—	0.0816	0.0816	0.933	94.312 ± 0.745	0.321 ± 0.002	5.688 ± 0.745
QML	beta41	0.20	0.201	0.001 ± 0.002	0.006 ± 0.009	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	0.070 ± 0.003	0.265 ± 0.008	0.201	0.001	0.27	0.0504	0.0504	1.006	95.243 ± 0.684	0.209 ± 0.001	—
QML	beta42	0.20	0.199	-0.001 ± 0.002	-0.007 ± 0.009	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.074 ± 0.004	0.271 ± 0.008	0.198	-0.002	-1.09	0.0551	0.0551	1.024	94.933 ± 0.705	0.218 ± 0.001	—
QML	beta43	0.20	0.198	-0.002 ± 0.002	-0.010 ± 0.009	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	0.072 ± 0.003	0.268 ± 0.007	0.199	-0.001	-0.73	0.0559	0.0560	0.993	94.726 ± 0.719	0.209 ± 0.001	—
QML	beta44	0.00	-0.003	-0.003 ± 0.002	—	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	—	—	-0.001	-0.001	—	0.0678	0.0678	0.990	95.450 ± 0.670	0.257 ± 0.002	4.550 ± 0.670
QML	beta45	0.00	-0.005	-0.005 ± 0.002	—	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	—	—	-0.005	-0.005	—	0.0597	0.0599	0.978	95.036 ± 0.698	0.225 ± 0.001	4.964 ± 0.698
QML	beta46	0.00	-0.002	-0.002 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	—	—	-0.002	-0.002	—	0.0530	0.0530	1.024	96.174 ± 0.617	0.213 ± 0.001	3.826 ± 0.617
QML	beta47	0.00	-0.004	-0.004 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	—	—	-0.002	-0.002	—	0.0512	0.0512	1.011	94.933 ± 0.705	0.211 ± 0.001	5.067 ± 0.705
QML	beta51	0.16	0.160	-0.000 ± 0.002	-0.001 ± 0.015	0.005 ± 0.000	0.074 ± 0.002	0.005 ± 0.000	0.074 ± 0.002	0.213 ± 0.009	0.461 ± 0.013	0.164	0.004	2.50	0.0747	0.0748	1.002	95.036 ± 0.698	0.290 ± 0.001	—
QML	beta52	0.16	0.162	0.002 ± 0.002	0.011 ± 0.012	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.130 ± 0.006	0.361 ± 0.010	0.162	0.002	1.53	0.0595	0.0595	1.002	95.450 ± 0.670	0.227 ± 0.001	—
QML	beta53	0.16	0.162	0.002 ± 0.002	0.014 ± 0.012	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	0.145 ± 0.007	0.381 ± 0.011	0.160	0.000	-0.04	0.0596	0.0596	0.974	94.416 ± 0.738	0.233 ± 0.001	—
QML	beta54	0.16	0.158	-0.002 ± 0.002	-0.012 ± 0.012	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.130 ± 0.006	0.361 ± 0.010	0.157	-0.003	-1.91	0.0578	0.0579	0.982	95.243 ± 0.684	0.223 ± 0.001	—
QML	beta55	0.00	0.001	0.001 ± 0.003	—	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	—	—	-0.001	-0.001	—	0.0752	0.0752	1.071	95.243 ± 0.684	0.328 ± 0.009	4.757 ± 0.684
QML	beta56	0.00	-0.005	-0.005 ± 0.002	—	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	—	—	-0.004	-0.004	—	0.0736	0.0737	1.036	95.863 ± 0.640	0.290 ± 0.003	4.137 ± 0.640
QML	beta57	0.00	-0.003	-0.003 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	—	—	-0.001	-0.001	—	0.0561	0.0561	0.994	93.899 ± 0.770	0.227 ± 0.003	6.101 ± 0.770
QML	beta58	0.00	-0.004	-0.004 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	—	—	-0.005	-0.005	—	0.0529	0.0531	0.965	93.278 ± 0.805	0.201 ± 0.001	6.722 ± 0.805
UPI	beta41	0.20	0.203	0.003 ± 0.002	0.017 ± 0.009	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.082 ± 0.004	0.286 ± 0.008	0.202	0.002	1.21	0.0552	0.0553	1.046	95.760 ± 0.648	0.234 ± 0.003	—
UPI	beta42	0.20	0.200	-0.000 ± 0.002	-0.002 ± 0.010	0.004 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	0.088 ± 0.004	0.296 ± 0.009	0.200	0.000	-0.20	0.0572	0.0572	1.058	95.036 ± 0.698	0.246 ± 0.003	—
UPI	beta43	0.20	0.200	-0.000 ± 0.002	-0.002 ± 0.009	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.081 ± 0.004	0.285 ± 0.008	0.201	0.001	0.45	0.0596	0.0596	1.022	95.450 ± 0.670	0.228 ± 0.002	—
UPI	beta44	0.00	0.002	0.002 ± 0.002	—	0.005 ± 0.000	0.074 ± 0.002	0.005 ± 0.000	0.074 ± 0.002	—	—	0.004	0.004	—	0.0725	0.0725	1.059	96.484 ± 0.592	0.307 ± 0.004	3.516 ± 0.592
UPI	beta45	0.00	-0.001	-0.001 ± 0.002	—	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	—	—	-0.001	-0.001	—	0.0639	0.0639	1.053	96.484 ± 0.592	0.270 ± 0.005	3.516 ± 0.592
UPI	beta46	0.00	-0.003	-0.003 ± 0.003	—	0.010 ± 0.001	0.101 ± 0.003	0.010 ± 0.001	0.101 ± 0.003	—	—	0.000	0.000	—	0.0928	0.0928	1.222	98.552 ± 0.384	0.484 ± 0.019	1.448 ± 0.384

Table S231

Simulation 2 - Condition 87: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.6 (continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.001	-0.001 ± 0.004	—	0.012 ± 0.001	0.109 ± 0.003	0.012 ± 0.001	0.109 ± 0.004	—	—	0.002	0.002	—	0.0915	0.0915	1.152	98.139 ± 0.435	0.494 ± 0.012	1.861 ± 0.435
UPI	beta51	0.16	0.159	-0.001 ± 0.002	-0.009 ± 0.015	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.210 ± 0.009	0.459 ± 0.013	0.160	0.000	0.05	0.0729	0.0729	1.009	95.140 ± 0.692	0.290 ± 0.001	—
UPI	beta52	0.16	0.162	0.002 ± 0.002	0.013 ± 0.012	0.004 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	0.138 ± 0.006	0.371 ± 0.010	0.162	0.002	0.94	0.0618	0.0619	1.012	95.657 ± 0.655	0.235 ± 0.001	—
UPI	beta53	0.16	0.163	0.003 ± 0.002	0.021 ± 0.013	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	0.154 ± 0.008	0.393 ± 0.012	0.161	0.001	0.91	0.0602	0.0603	0.997	94.829 ± 0.712	0.245 ± 0.001	—
UPI	beta54	0.16	0.160	-0.000 ± 0.002	-0.001 ± 0.012	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.141 ± 0.007	0.375 ± 0.011	0.162	0.002	1.19	0.0583	0.0583	1.002	94.933 ± 0.705	0.236 ± 0.001	—
UPI	beta55	0.00	0.001	0.001 ± 0.003	—	0.008 ± 0.000	0.087 ± 0.002	0.008 ± 0.000	0.087 ± 0.003	—	—	-0.001	-0.001	—	0.0826	0.0826	1.040	96.794 ± 0.566	0.354 ± 0.003	3.206 ± 0.566
UPI	beta56	0.00	0.001	0.001 ± 0.003	—	0.007 ± 0.000	0.082 ± 0.002	0.007 ± 0.000	0.082 ± 0.002	—	—	0.002	0.002	—	0.0811	0.0811	1.031	97.001 ± 0.548	0.332 ± 0.003	2.999 ± 0.548
UPI	beta57	0.00	0.001	0.001 ± 0.003	—	0.011 ± 0.001	0.103 ± 0.003	0.011 ± 0.001	0.103 ± 0.003	—	—	0.002	0.002	—	0.0906	0.0907	1.097	96.381 ± 0.601	0.444 ± 0.011	3.619 ± 0.601
UPI	beta58	0.00	0.001	0.001 ± 0.003	—	0.010 ± 0.001	0.102 ± 0.003	0.010 ± 0.001	0.102 ± 0.003	—	—	0.001	0.001	—	0.0872	0.0872	1.114	97.828 ± 0.469	0.444 ± 0.010	2.172 ± 0.469

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S232

Simulation 2 - Condition 88: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.6

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Type I (%) $\pm$ MCSE
LSAM	beta41	0.20	0.200	-0.000 $\pm$ 0.001	-0.000 $\pm$ 0.006	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.032 $\pm$ 0.001	0.179 $\pm$ 0.005	0.199	-0.001	-0.52	0.0338	0.0338	0.963	94.194 $\pm$ 0.740	0.135 $\pm$ 0.000	—
LSAM	beta42	0.20	0.200	-0.000 $\pm$ 0.001	-0.002 $\pm$ 0.006	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.033 $\pm$ 0.001	0.181 $\pm$ 0.005	0.198	-0.002	-0.79	0.0355	0.0355	0.984	94.595 $\pm$ 0.715	0.140 $\pm$ 0.000	—
LSAM	beta43	0.20	0.203	0.003 $\pm$ 0.001	0.016 $\pm$ 0.005	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.029 $\pm$ 0.001	0.171 $\pm$ 0.005	0.202	0.002	1.24	0.0349	0.0350	1.001	96.196 $\pm$ 0.605	0.133 $\pm$ 0.000	—
LSAM	beta44	0.00	-0.001	-0.001 $\pm$ 0.001	—	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	—	—	-0.002	-0.002	—	0.0400	0.0401	1.020	95.195 $\pm$ 0.677	0.162 $\pm$ 0.001	4.805 $\pm$ 0.677
LSAM	beta45	0.00	-0.002	-0.002 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	—	—	-0.001	-0.001	—	0.0359	0.0359	0.989	94.494 $\pm$ 0.722	0.144 $\pm$ 0.000	5.506 $\pm$ 0.722
LSAM	beta46	0.00	0.001	0.001 $\pm$ 0.002	—	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	—	—	0.001	0.001	—	0.0509	0.0509	0.993	95.395 $\pm$ 0.663	0.204 $\pm$ 0.001	4.605 $\pm$ 0.663
LSAM	beta47	0.00	0.000	-0.000 $\pm$ 0.002	—	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	—	—	-0.001	-0.001	—	0.0545	0.0545	0.978	95.295 $\pm$ 0.670	0.203 $\pm$ 0.001	4.705 $\pm$ 0.670
LSAM	beta51	0.16	0.160	0.000 $\pm$ 0.001	0.002 $\pm$ 0.009	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.076 $\pm$ 0.003	0.275 $\pm$ 0.007	0.160	0.000	-0.03	0.0448	0.0448	1.024	95.996 $\pm$ 0.620	0.177 $\pm$ 0.000	—
LSAM	beta52	0.16	0.162	0.002 $\pm$ 0.001	0.015 $\pm$ 0.007	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.051 $\pm$ 0.002	0.227 $\pm$ 0.006	0.163	0.003	1.98	0.0362	0.0364	0.992	95.495 $\pm$ 0.656	0.141 $\pm$ 0.000	—
LSAM	beta53	0.16	0.158	-0.002 $\pm$ 0.001	-0.014 $\pm$ 0.007	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.056 $\pm$ 0.003	0.236 $\pm$ 0.007	0.157	-0.003	-2.09	0.0376	0.0377	0.986	94.294 $\pm$ 0.734	0.146 $\pm$ 0.000	—
LSAM	beta54	0.16	0.162	0.002 $\pm$ 0.001	0.012 $\pm$ 0.007	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.049 $\pm$ 0.002	0.222 $\pm$ 0.006	0.160	0.000	0.14	0.0346	0.0346	1.014	95.696 $\pm$ 0.642	0.141 $\pm$ 0.000	—
LSAM	beta55	0.00	-0.002	-0.002 $\pm$ 0.002	—	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	—	—	-0.002	-0.002	—	0.0493	0.0493	1.023	95.696 $\pm$ 0.642	0.197 $\pm$ 0.001	4.304 $\pm$ 0.642
LSAM	beta56	0.00	0.001	0.001 $\pm$ 0.002	—	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	—	—	0.000	0.000	—	0.0493	0.0493	0.960	94.194 $\pm$ 0.740	0.185 $\pm$ 0.001	5.806 $\pm$ 0.740
LSAM	beta57	0.00	0.000	-0.000 $\pm$ 0.002	—	0.003 $\pm$ 0.000	0.054 $\pm$ 0.001	0.003 $\pm$ 0.000	0.054 $\pm$ 0.002	—	—	0.001	0.001	—	0.0551	0.0551	0.978	94.494 $\pm$ 0.722	0.207 $\pm$ 0.001	5.506 $\pm$ 0.722
LSAM	beta58	0.00	0.000	0.000 $\pm$ 0.002	—	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	—	—	0.002	0.002	—	0.0512	0.0513	0.958	94.795 $\pm$ 0.703	0.194 $\pm$ 0.001	5.205 $\pm$ 0.703
QML	beta41	0.20	0.199	-0.001 $\pm$ 0.001	-0.005 $\pm$ 0.006	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.030 $\pm$ 0.001	0.174 $\pm$ 0.005	0.198	-0.002	-0.95	0.0341	0.0342	0.964	94.294 $\pm$ 0.734	0.131 $\pm$ 0.000	—
QML	beta42	0.20	0.199	-0.001 $\pm$ 0.001	-0.006 $\pm$ 0.006	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.030 $\pm$ 0.001	0.174 $\pm$ 0.005	0.197	-0.003	-1.37	0.0349	0.0350	0.991	94.995 $\pm$ 0.690	0.136 $\pm$ 0.000	—
QML	beta43	0.20	0.202	0.002 $\pm$ 0.001	0.012 $\pm$ 0.005	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.028 $\pm$ 0.001	0.167 $\pm$ 0.004	0.202	0.002	1.21	0.0346	0.0347	1.005	96.496 $\pm$ 0.582	0.131 $\pm$ 0.000	—
QML	beta44	0.00	-0.005	-0.005 $\pm$ 0.001	—	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	—	—	-0.005	-0.005	—	0.0389	0.0393	1.023	95.395 $\pm$ 0.663	0.161 $\pm$ 0.001	4.605 $\pm$ 0.663
QML	beta45	0.00	-0.005	-0.005 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	—	—	-0.005	-0.005	—	0.0356	0.0359	0.993	94.595 $\pm$ 0.715	0.140 $\pm$ 0.001	5.405 $\pm$ 0.715
QML	beta46	0.00	-0.001	-0.001 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	—	—	-0.001	-0.001	—	0.0336	0.0337	1.023	95.195 $\pm$ 0.677	0.135 $\pm$ 0.002	4.805 $\pm$ 0.677
QML	beta47	0.00	-0.003	-0.003 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	—	—	-0.003	-0.003	—	0.0355	0.0356	0.994	94.394 $\pm$ 0.728	0.132 $\pm$ 0.001	5.606 $\pm$ 0.728
QML	beta51	0.16	0.161	0.001 $\pm$ 0.001	0.006 $\pm$ 0.009	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.074 $\pm$ 0.003	0.272 $\pm$ 0.007	0.161	0.001	0.41	0.0433	0.0433	1.050	96.296 $\pm$ 0.598	0.179 $\pm$ 0.001	—
QML	beta52	0.16	0.162	0.002 $\pm$ 0.001	0.012 $\pm$ 0.007	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.049 $\pm$ 0.002	0.222 $\pm$ 0.006	0.163	0.003	1.67	0.0357	0.0358	1.003	95.195 $\pm$ 0.677	0.139 $\pm$ 0.000	—
QML	beta53	0.16	0.157	-0.003 $\pm$ 0.001	-0.017 $\pm$ 0.007	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.053 $\pm$ 0.002	0.230 $\pm$ 0.006	0.157	-0.003	-1.99	0.0369	0.0371	0.998	94.895 $\pm$ 0.696	0.143 $\pm$ 0.000	—
QML	beta54	0.16	0.161	0.001 $\pm$ 0.001	0.008 $\pm$ 0.007	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.047 $\pm$ 0.002	0.217 $\pm$ 0.006	0.160	0.000	-0.23	0.0341	0.0341	1.017	95.596 $\pm$ 0.649	0.139 $\pm$ 0.000	—
QML	beta55	0.00	-0.003	-0.003 $\pm$ 0.002	—	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	—	—	-0.002	-0.002	—	0.0483	0.0483	1.035	95.395 $\pm$ 0.663	0.195 $\pm$ 0.001	4.605 $\pm$ 0.663
QML	beta56	0.00	-0.003	-0.003 $\pm$ 0.001	—	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	—	—	-0.002	-0.002	—	0.0474	0.0474	0.966	94.394 $\pm$ 0.728	0.177 $\pm$ 0.001	5.606 $\pm$ 0.728
QML	beta57	0.00	-0.002	-0.002 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	—	—	-0.002	-0.002	—	0.0366	0.0367	1.007	95.195 $\pm$ 0.677	0.138 $\pm$ 0.001	4.805 $\pm$ 0.677
QML	beta58	0.00	-0.003	-0.003 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	—	—	-0.002	-0.002	—	0.0326	0.0327	0.978	94.194 $\pm$ 0.740	0.125 $\pm$ 0.001	5.806 $\pm$ 0.740
UPI	beta41	0.20	0.200	0.000 $\pm$ 0.001	0.000 $\pm$ 0.006	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.032 $\pm$ 0.001	0.180 $\pm$ 0.005	0.199	-0.001	-0.45	0.0340	0.0340	0.960	94.394 $\pm$ 0.728	0.136 $\pm$ 0.000	—
UPI	beta42	0.20	0.199	-0.001 $\pm$ 0.001	-0.003 $\pm$ 0.006	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.033 $\pm$ 0.001	0.181 $\pm$ 0.005	0.198	-0.002	-0.83	0.0354	0.0354	0.989	94.895 $\pm$ 0.696	0.141 $\pm$ 0.000	—
UPI	beta43	0.20	0.203	0.003 $\pm$ 0.001	0.017 $\pm$ 0.005	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.029 $\pm$ 0.001	0.172 $\pm$ 0.005	0.203	0.003	1.37	0.0346	0.0347	1.005	96.396 $\pm$ 0.590	0.135 $\pm$ 0.000	—
UPI	beta44	0.00	-0.001	-0.001 $\pm$ 0.001	—	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	—	—	-0.002	-0.002	—	0.0409	0.0409	1.029	95.796 $\pm$ 0.635	0.167 $\pm$ 0.001	4.204 $\pm$ 0.635
UPI	beta45	0.00	-0.002	-0.002 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	—	—	-0.001	-0.001	—	0.0358	0.0358	1.001	95.195 $\pm$ 0.677	0.148 $\pm$ 0.001	4.805 $\pm$ 0.677
UPI	beta46	0.00	0.001	0.001 $\pm$ 0.002	—	0.003 $\pm$ 0.000	0.054 $\pm$ 0.001	0.003 $\pm$ 0.000	0.054 $\pm$ 0.002	—	—	0.000	0.000	—	0.0518	0.0518	1.051	96.396 $\pm$ 0.590	0.223 $\pm$ 0.002	3.604 $\pm$ 0.590

Table S232

Simulation 2 - Condition 88: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.6
(continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	0.000	-0.000 ± 0.002	—	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	—	—	-0.002	-0.002	—	0.0566	0.0566	1.022	96.496 ± 0.582	0.223 ± 0.002	3.504 ± 0.582
UPI	beta51	0.16	0.160	0.000 ± 0.001	0.002 ± 0.009	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.074 ± 0.003	0.272 ± 0.007	0.160	0.000	-0.05	0.0435	0.0435	1.039	96.196 ± 0.605	0.177 ± 0.000	—
UPI	beta52	0.16	0.162	0.002 ± 0.001	0.014 ± 0.007	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.051 ± 0.002	0.227 ± 0.006	0.163	0.003	2.01	0.0361	0.0362	0.993	95.095 ± 0.683	0.141 ± 0.000	—
UPI	beta53	0.16	0.158	-0.002 ± 0.001	-0.014 ± 0.007	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.055 ± 0.003	0.236 ± 0.007	0.157	-0.003	-2.10	0.0365	0.0366	0.996	94.795 ± 0.703	0.147 ± 0.000	—
UPI	beta54	0.16	0.162	0.002 ± 0.001	0.013 ± 0.007	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.049 ± 0.002	0.222 ± 0.006	0.161	0.001	0.36	0.0351	0.0351	1.015	95.996 ± 0.620	0.141 ± 0.000	—
UPI	beta55	0.00	-0.002	-0.002 ± 0.002	—	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	—	—	-0.001	-0.001	—	0.0499	0.0499	1.025	95.796 ± 0.635	0.201 ± 0.001	4.204 ± 0.635
UPI	beta56	0.00	0.001	0.001 ± 0.002	—	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	—	—	0.002	0.002	—	0.0498	0.0498	0.961	94.595 ± 0.715	0.189 ± 0.001	5.405 ± 0.715
UPI	beta57	0.00	-0.001	-0.001 ± 0.002	—	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	—	—	0.002	0.002	—	0.0552	0.0552	1.014	96.196 ± 0.605	0.221 ± 0.001	3.804 ± 0.605
UPI	beta58	0.00	0.000	-0.000 ± 0.002	—	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	—	—	0.001	0.001	—	0.0496	0.0496	1.006	96.496 ± 0.582	0.213 ± 0.002	3.504 ± 0.582

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S233

Simulation 2 - Condition 89: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, $N = 400$, Reliability = 0.8

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.201	0.001 ± 0.001	0.003 ± 0.007	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.050 ± 0.002	0.224 ± 0.006	0.201	0.001	0.45	0.0433	0.0433	1.003	94.700 ± 0.708	0.176 ± 0.000	—
LSAM	beta42	0.20	0.201	0.001 ± 0.001	0.007 ± 0.007	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.054 ± 0.002	0.231 ± 0.006	0.202	0.002	1.01	0.0481	0.0481	1.000	94.400 ± 0.727	0.181 ± 0.000	—
LSAM	beta43	0.20	0.200	-0.000 ± 0.001	-0.001 ± 0.007	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.050 ± 0.002	0.223 ± 0.006	0.200	0.000	0.21	0.0421	0.0421	0.996	94.900 ± 0.696	0.174 ± 0.000	—
LSAM	beta44	0.00	0.001	0.001 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	—	—	0.000	0.000	—	0.0529	0.0529	0.945	93.500 ± 0.780	0.196 ± 0.001	6.500 ± 0.780
LSAM	beta45	0.00	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	-0.001	-0.001	—	0.0465	0.0465	0.984	93.800 ± 0.763	0.179 ± 0.001	6.200 ± 0.763
LSAM	beta46	0.00	-0.001	-0.001 ± 0.002	—	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	—	—	-0.001	-0.001	—	0.0557	0.0557	1.012	95.500 ± 0.656	0.226 ± 0.001	4.500 ± 0.656
LSAM	beta47	0.00	0.001	0.001 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	—	—	0.003	0.003	—	0.0560	0.0561	0.985	94.300 ± 0.733	0.225 ± 0.001	5.700 ± 0.733
LSAM	beta51	0.16	0.159	-0.001 ± 0.002	-0.005 ± 0.012	0.004 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	0.138 ± 0.007	0.371 ± 0.011	0.157	-0.003	-1.90	0.0584	0.0585	0.983	94.800 ± 0.702	0.229 ± 0.001	—
LSAM	beta52	0.16	0.162	0.002 ± 0.002	0.010 ± 0.010	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.091 ± 0.004	0.301 ± 0.008	0.161	0.001	0.45	0.0478	0.0478	0.974	94.600 ± 0.715	0.184 ± 0.000	—
LSAM	beta53	0.16	0.159	-0.001 ± 0.002	-0.005 ± 0.009	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.088 ± 0.004	0.297 ± 0.008	0.160	0.000	-0.21	0.0478	0.0478	1.015	95.400 ± 0.662	0.189 ± 0.000	—
LSAM	beta54	0.16	0.159	-0.001 ± 0.002	-0.006 ± 0.010	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.091 ± 0.004	0.302 ± 0.009	0.160	0.000	-0.06	0.0477	0.0477	0.972	93.600 ± 0.774	0.184 ± 0.000	—
LSAM	beta55	0.00	0.000	0.000 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	—	—	-0.003	-0.003	—	0.0591	0.0592	0.990	94.400 ± 0.727	0.237 ± 0.001	5.600 ± 0.727
LSAM	beta56	0.00	0.000	-0.000 ± 0.002	—	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	—	—	-0.003	-0.003	—	0.0571	0.0572	0.967	93.400 ± 0.785	0.224 ± 0.001	6.600 ± 0.785
LSAM	beta57	0.00	-0.001	-0.001 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	—	—	-0.001	-0.001	—	0.0616	0.0616	0.959	93.700 ± 0.768	0.230 ± 0.001	6.300 ± 0.768
LSAM	beta58	0.00	0.001	0.001 ± 0.002	—	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	—	—	0.001	0.001	—	0.0581	0.0581	0.986	94.600 ± 0.715	0.216 ± 0.001	5.400 ± 0.715
QML	beta41	0.20	0.200	-0.000 ± 0.001	-0.000 ± 0.007	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.048 ± 0.002	0.220 ± 0.006	0.201	0.001	0.37	0.0433	0.0433	1.007	94.600 ± 0.715	0.174 ± 0.000	—
QML	beta42	0.20	0.201	0.001 ± 0.001	0.005 ± 0.007	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.052 ± 0.002	0.228 ± 0.006	0.202	0.002	0.97	0.0463	0.0464	1.007	94.300 ± 0.733	0.180 ± 0.001	—
QML	beta43	0.20	0.199	-0.001 ± 0.001	-0.003 ± 0.007	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.049 ± 0.002	0.220 ± 0.006	0.199	-0.001	-0.27	0.0418	0.0418	0.999	95.000 ± 0.689	0.173 ± 0.000	—
QML	beta44	0.00	-0.001	-0.001 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	—	—	-0.002	-0.002	—	0.0522	0.0522	0.952	94.100 ± 0.745	0.196 ± 0.001	5.900 ± 0.745
QML	beta45	0.00	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	-0.002	-0.002	—	0.0460	0.0460	1.003	93.900 ± 0.757	0.180 ± 0.002	6.100 ± 0.757
QML	beta46	0.00	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	-0.001	-0.001	—	0.0448	0.0449	1.024	95.400 ± 0.662	0.182 ± 0.001	4.600 ± 0.662
QML	beta47	0.00	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	0.001	0.001	—	0.0442	0.0442	0.993	94.400 ± 0.727	0.180 ± 0.001	5.600 ± 0.727
QML	beta51	0.16	0.160	-0.000 ± 0.002	-0.003 ± 0.012	0.004 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.137 ± 0.007	0.370 ± 0.011	0.157	-0.003	-1.79	0.0574	0.0575	1.000	95.200 ± 0.676	0.232 ± 0.001	—
QML	beta52	0.16	0.161	0.001 ± 0.002	0.009 ± 0.009	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.090 ± 0.004	0.300 ± 0.008	0.161	0.001	0.41	0.0485	0.0485	0.979	94.800 ± 0.702	0.184 ± 0.000	—
QML	beta53	0.16	0.159	-0.001 ± 0.001	-0.007 ± 0.009	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.087 ± 0.004	0.295 ± 0.008	0.160	0.000	-0.11	0.0469	0.0469	1.023	95.400 ± 0.662	0.189 ± 0.001	—
QML	beta54	0.16	0.159	-0.001 ± 0.002	-0.007 ± 0.009	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.089 ± 0.004	0.299 ± 0.008	0.159	-0.001	-0.53	0.0466	0.0466	0.974	94.000 ± 0.751	0.183 ± 0.000	—
QML	beta55	0.00	0.000	-0.000 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	—	—	-0.003	-0.003	—	0.0582	0.0583	1.009	94.500 ± 0.721	0.239 ± 0.002	5.500 ± 0.721
QML	beta56	0.00	-0.002	-0.002 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	—	—	-0.004	-0.004	—	0.0542	0.0543	0.977	93.600 ± 0.774	0.222 ± 0.001	6.400 ± 0.774
QML	beta57	0.00	-0.002	-0.002 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	—	—	-0.002	-0.002	—	0.0510	0.0510	0.985	93.900 ± 0.757	0.191 ± 0.003	6.100 ± 0.757
QML	beta58	0.00	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	-0.001	-0.001	—	0.0453	0.0453	0.991	95.000 ± 0.689	0.175 ± 0.001	5.000 ± 0.689
UPI	beta41	0.20	0.201	0.001 ± 0.001	0.003 ± 0.007	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.050 ± 0.002	0.224 ± 0.006	0.201	0.001	0.64	0.0444	0.0444	1.000	94.700 ± 0.708	0.175 ± 0.000	—
UPI	beta42	0.20	0.201	0.001 ± 0.001	0.007 ± 0.007	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.053 ± 0.002	0.231 ± 0.006	0.202	0.002	1.15	0.0475	0.0476	0.999	95.100 ± 0.683	0.181 ± 0.000	—
UPI	beta43	0.20	0.200	-0.000 ± 0.001	-0.002 ± 0.007	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.050 ± 0.002	0.223 ± 0.006	0.200	0.000	0.15	0.0427	0.0427	0.999	94.700 ± 0.708	0.175 ± 0.000	—
UPI	beta44	0.00	0.002	0.002 ± 0.002	—	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	—	—	0.001	0.001	—	0.0535	0.0535	0.948	93.200 ± 0.796	0.200 ± 0.001	6.800 ± 0.796
UPI	beta45	0.00	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	0.001	0.001	—	0.0453	0.0453	0.988	94.200 ± 0.739	0.182 ± 0.001	5.800 ± 0.739
UPI	beta46	0.00	-0.001	-0.001 ± 0.002	—	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	—	—	-0.001	-0.001	—	0.0572	0.0572	1.022	96.000 ± 0.620	0.236 ± 0.001	4.000 ± 0.620

Table S233
Simulation 2 - Condition 89: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 400, Reliability = 0.8 (continued)

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	0.000	-0.000 ± 0.002	—	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	—	—	0.001	0.001	—	0.0552	0.0552	1.016	94.900 ± 0.696	0.235 ± 0.001	5.100 ± 0.696
UPI	beta51	0.16	0.159	-0.001 ± 0.002	-0.005 ± 0.012	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.136 ± 0.007	0.368 ± 0.011	0.157	-0.003	-2.01	0.0577	0.0578	0.997	95.200 ± 0.676	0.230 ± 0.001	—
UPI	beta52	0.16	0.162	0.002 ± 0.002	0.010 ± 0.010	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.091 ± 0.004	0.301 ± 0.008	0.161	0.001	0.53	0.0485	0.0485	0.970	94.600 ± 0.715	0.183 ± 0.000	—
UPI	beta53	0.16	0.159	-0.001 ± 0.002	-0.005 ± 0.009	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.089 ± 0.004	0.298 ± 0.008	0.159	-0.001	-0.37	0.0480	0.0480	1.017	95.400 ± 0.662	0.190 ± 0.000	—
UPI	beta54	0.16	0.159	-0.001 ± 0.002	-0.007 ± 0.010	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.091 ± 0.004	0.302 ± 0.008	0.159	-0.001	-0.38	0.0483	0.0483	0.971	93.400 ± 0.785	0.184 ± 0.000	—
UPI	beta55	0.00	0.000	0.000 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	—	—	-0.002	-0.002	—	0.0612	0.0612	0.998	94.500 ± 0.721	0.240 ± 0.001	5.500 ± 0.721
UPI	beta56	0.00	0.000	0.000 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	—	—	-0.002	-0.002	—	0.0567	0.0567	0.974	93.800 ± 0.763	0.228 ± 0.001	6.200 ± 0.763
UPI	beta57	0.00	-0.001	-0.001 ± 0.002	—	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	—	—	-0.001	-0.001	—	0.0595	0.0595	0.973	94.200 ± 0.739	0.238 ± 0.001	5.800 ± 0.739
UPI	beta58	0.00	0.000	0.000 ± 0.002	—	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	—	—	0.001	0.001	—	0.0557	0.0557	1.013	96.200 ± 0.605	0.227 ± 0.001	3.800 ± 0.605

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S234

Simulation 2 - Condition 90: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.8

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.201	0.001 ± 0.001	0.005 ± 0.004	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.020 ± 0.001	0.140 ± 0.004	0.201	0.001	0.56	0.0278	0.0278	1.007	94.800 ± 0.702	0.111 ± 0.000	—
LSAM	beta42	0.20	0.201	0.001 ± 0.001	0.007 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.021 ± 0.001	0.145 ± 0.004	0.201	0.001	0.60	0.0309	0.0309	0.999	94.500 ± 0.721	0.114 ± 0.000	—
LSAM	beta43	0.20	0.201	0.001 ± 0.001	0.003 ± 0.004	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	0.019 ± 0.001	0.137 ± 0.004	0.200	0.000	0.14	0.0272	0.0272	1.022	96.200 ± 0.605	0.110 ± 0.000	—
LSAM	beta44	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	—	—	0.002	0.002	—	0.0319	0.0319	0.980	94.400 ± 0.727	0.123 ± 0.000	5.600 ± 0.727
LSAM	beta45	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	0.000	0.000	—	0.0293	0.0293	1.003	95.000 ± 0.689	0.112 ± 0.000	5.000 ± 0.689
LSAM	beta46	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	-0.002	-0.002	—	0.0363	0.0364	0.995	95.200 ± 0.676	0.142 ± 0.000	4.800 ± 0.676
LSAM	beta47	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	0.000	0.000	—	0.0360	0.0360	0.993	95.300 ± 0.669	0.141 ± 0.000	4.700 ± 0.669
LSAM	beta51	0.16	0.161	0.001 ± 0.001	0.004 ± 0.007	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.056 ± 0.002	0.236 ± 0.006	0.160	0.000	0.21	0.0372	0.0372	0.981	93.800 ± 0.763	0.145 ± 0.000	—
LSAM	beta52	0.16	0.159	-0.001 ± 0.001	-0.003 ± 0.006	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.036 ± 0.002	0.190 ± 0.005	0.159	-0.001	-0.42	0.0302	0.0302	0.971	94.100 ± 0.745	0.116 ± 0.000	—
LSAM	beta53	0.16	0.162	0.002 ± 0.001	0.013 ± 0.006	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.037 ± 0.002	0.193 ± 0.005	0.163	0.003	1.57	0.0322	0.0323	0.987	95.500 ± 0.656	0.120 ± 0.000	—
LSAM	beta54	0.16	0.160	-0.000 ± 0.001	-0.003 ± 0.006	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.033 ± 0.001	0.182 ± 0.005	0.160	0.000	-0.25	0.0303	0.0303	1.013	94.800 ± 0.702	0.116 ± 0.000	—
LSAM	beta55	0.00	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.039 ± 0.001	0.001 ± 0.000	0.039 ± 0.001	—	—	0.000	0.000	—	0.0394	0.0394	0.985	94.800 ± 0.702	0.149 ± 0.000	5.200 ± 0.702
LSAM	beta56	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	0.000	0.000	—	0.0360	0.0360	1.012	95.800 ± 0.634	0.142 ± 0.000	4.200 ± 0.634
LSAM	beta57	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.001	-0.001	—	0.0373	0.0373	0.972	94.600 ± 0.715	0.145 ± 0.000	5.400 ± 0.715
LSAM	beta58	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	-0.001	-0.001	—	0.0352	0.0353	0.988	95.900 ± 0.627	0.137 ± 0.000	4.100 ± 0.627
QML	beta41	0.20	0.201	0.001 ± 0.001	0.003 ± 0.004	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.019 ± 0.001	0.139 ± 0.004	0.201	0.001	0.33	0.0273	0.0273	1.007	94.900 ± 0.696	0.109 ± 0.000	—
QML	beta42	0.20	0.201	0.001 ± 0.001	0.005 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.020 ± 0.001	0.143 ± 0.004	0.200	0.000	0.22	0.0306	0.0306	1.003	95.100 ± 0.683	0.112 ± 0.000	—
QML	beta43	0.20	0.200	0.000 ± 0.001	0.001 ± 0.004	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	0.018 ± 0.001	0.136 ± 0.004	0.200	0.000	0.02	0.0273	0.0273	1.024	96.500 ± 0.581	0.109 ± 0.000	—
QML	beta44	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	—	—	-0.001	-0.001	—	0.0319	0.0319	1.002	94.800 ± 0.702	0.126 ± 0.002	5.200 ± 0.702
QML	beta45	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	-0.001	-0.001	—	0.0295	0.0296	1.011	94.900 ± 0.696	0.113 ± 0.001	5.100 ± 0.696
QML	beta46	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	-0.002	-0.002	—	0.0288	0.0289	1.012	94.800 ± 0.702	0.116 ± 0.001	5.200 ± 0.702
QML	beta47	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	-0.002	-0.002	—	0.0293	0.0293	1.006	95.400 ± 0.662	0.114 ± 0.001	4.600 ± 0.662
QML	beta51	0.16	0.161	0.001 ± 0.001	0.005 ± 0.007	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.055 ± 0.002	0.235 ± 0.006	0.160	0.000	-0.05	0.0371	0.0371	0.988	94.100 ± 0.745	0.146 ± 0.000	—
QML	beta52	0.16	0.159	-0.001 ± 0.001	-0.004 ± 0.006	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.036 ± 0.002	0.189 ± 0.005	0.160	0.000	-0.28	0.0300	0.0300	0.977	94.700 ± 0.708	0.116 ± 0.000	—
QML	beta53	0.16	0.162	0.002 ± 0.001	0.011 ± 0.006	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.037 ± 0.002	0.192 ± 0.005	0.162	0.002	1.22	0.0315	0.0315	0.991	95.800 ± 0.634	0.119 ± 0.000	—
QML	beta54	0.16	0.159	-0.001 ± 0.001	-0.004 ± 0.006	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.033 ± 0.001	0.181 ± 0.005	0.160	0.000	-0.31	0.0296	0.0296	1.016	95.100 ± 0.683	0.115 ± 0.000	—
QML	beta55	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.039 ± 0.001	0.001 ± 0.000	0.039 ± 0.001	—	—	0.000	0.000	—	0.0393	0.0393	0.994	94.900 ± 0.696	0.150 ± 0.000	5.100 ± 0.696
QML	beta56	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	-0.002	-0.002	—	0.0349	0.0349	1.021	96.200 ± 0.605	0.141 ± 0.001	3.800 ± 0.605
QML	beta57	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	-0.002	-0.002	—	0.0299	0.0300	0.982	94.400 ± 0.727	0.119 ± 0.000	5.600 ± 0.727
QML	beta58	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	-0.002	-0.002	—	0.0286	0.0287	0.998	95.700 ± 0.641	0.111 ± 0.001	4.300 ± 0.641
UPI	beta41	0.20	0.201	0.001 ± 0.001	0.005 ± 0.004	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.020 ± 0.001	0.140 ± 0.004	0.201	0.001	0.44	0.0283	0.0284	1.004	94.700 ± 0.708	0.110 ± 0.000	—
UPI	beta42	0.20	0.201	0.001 ± 0.001	0.006 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.021 ± 0.001	0.145 ± 0.004	0.201	0.001	0.51	0.0308	0.0308	0.997	94.500 ± 0.721	0.114 ± 0.000	—
UPI	beta43	0.20	0.201	0.001 ± 0.001	0.003 ± 0.004	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	0.019 ± 0.001	0.137 ± 0.004	0.200	0.000	0.08	0.0274	0.0274	1.022	96.200 ± 0.605	0.110 ± 0.000	—
UPI	beta44	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	—	—	0.001	0.001	—	0.0323	0.0323	0.977	94.300 ± 0.733	0.124 ± 0.000	5.700 ± 0.733
UPI	beta45	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	0.000	0.000	—	0.0306	0.0306	1.007	95.100 ± 0.683	0.113 ± 0.000	4.900 ± 0.683
UPI	beta46	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.002	-0.002	—	0.0353	0.0354	0.992	95.200 ± 0.676	0.144 ± 0.000	4.800 ± 0.676

Table S234

Simulation 2 - Condition 90: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Normal, N = 1000, Reliability = 0.8
(continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.001	0.001	—	0.0360	0.0360	0.999	95.700 ± 0.641	0.143 ± 0.000	4.300 ± 0.641
UPI	beta51	0.16	0.161	0.001 ± 0.001	0.003 ± 0.007	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.055 ± 0.002	0.235 ± 0.006	0.160	0.000	-0.18	0.0369	0.0369	0.987	94.000 ± 0.751	0.145 ± 0.000	—
UPI	beta52	0.16	0.159	-0.001 ± 0.001	-0.003 ± 0.006	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.036 ± 0.002	0.190 ± 0.005	0.160	0.000	-0.30	0.0304	0.0304	0.969	93.700 ± 0.768	0.115 ± 0.000	—
UPI	beta53	0.16	0.162	0.002 ± 0.001	0.013 ± 0.006	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.037 ± 0.002	0.193 ± 0.005	0.163	0.003	1.63	0.0323	0.0325	0.989	95.700 ± 0.641	0.120 ± 0.000	—
UPI	beta54	0.16	0.160	-0.000 ± 0.001	-0.003 ± 0.006	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.033 ± 0.001	0.182 ± 0.005	0.159	-0.001	-0.32	0.0304	0.0304	1.012	94.900 ± 0.696	0.116 ± 0.000	—
UPI	beta55	0.00	0.000	0.000 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.001 ± 0.000	0.039 ± 0.001	—	—	0.000	0.000	—	0.0394	0.0394	0.989	94.500 ± 0.721	0.150 ± 0.000	5.500 ± 0.721
UPI	beta56	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	-0.001	-0.001	—	0.0365	0.0366	1.013	95.700 ± 0.641	0.142 ± 0.000	4.300 ± 0.641
UPI	beta57	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.001	-0.001	—	0.0374	0.0374	0.979	94.600 ± 0.715	0.147 ± 0.000	5.400 ± 0.715
UPI	beta58	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	-0.001	-0.001	—	0.0354	0.0354	0.989	95.900 ± 0.627	0.139 ± 0.000	4.100 ± 0.627

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S235

Simulation 2 - Condition 91: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.4

Method	Param.	True	Mean-based										Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE	
LSAM	beta41	0.20	0.220	0.020 ± 0.010	0.102 ± 0.051	0.013 ± 0.002	0.113 ± 0.007	0.013 ± 0.002	0.114 ± 0.009	0.325 ± 0.044	0.570 ± 0.047	0.214	0.014	7.06	0.1087	0.1096	0.957	95.968 ± 1.767	0.423 ± 0.012	—	
LSAM	beta42	0.20	0.233	0.033 ± 0.010	0.164 ± 0.049	0.012 ± 0.002	0.109 ± 0.008	0.013 ± 0.002	0.113 ± 0.009	0.320 ± 0.045	0.566 ± 0.047	0.230	0.030	15.14	0.1104	0.1145	1.121	99.194 ± 0.803	0.478 ± 0.015	—	
LSAM	beta43	0.20	0.180	-0.020 ± 0.007	-0.100 ± 0.033	0.005 ± 0.001	0.074 ± 0.005	0.006 ± 0.001	0.076 ± 0.006	0.146 ± 0.018	0.382 ± 0.029	0.174	-0.026	-12.80	0.0839	0.0877	1.122	96.774 ± 1.587	0.326 ± 0.007	—	
LSAM	beta44	0.00	0.059	0.059 ± 0.014	—	0.023 ± 0.004	0.152 ± 0.013	0.026 ± 0.004	0.162 ± 0.015	—	—	0.064	0.064	—	0.1129	0.1298	0.960	95.161 ± 1.927	0.571 ± 0.033	4.839 ± 1.927	
LSAM	beta45	0.00	-0.008	-0.008 ± 0.009	—	0.010 ± 0.002	0.100 ± 0.009	0.010 ± 0.002	0.100 ± 0.010	—	—	0.007	0.007	—	0.1021	0.1024	1.013	96.774 ± 1.587	0.398 ± 0.014	3.226 ± 1.587	
LSAM	beta46	0.00	-0.048	-0.048 ± 0.009	—	0.010 ± 0.002	0.102 ± 0.011	0.013 ± 0.003	0.112 ± 0.012	—	—	-0.052	-0.052	—	0.0897	0.1037	0.853	91.129 ± 2.553	0.342 ± 0.018	8.871 ± 2.553	
LSAM	beta47	0.00	-0.049	-0.049 ± 0.008	—	0.008 ± 0.001	0.091 ± 0.008	0.011 ± 0.002	0.103 ± 0.009	—	—	-0.053	-0.053	—	0.0788	0.0951	1.110	94.355 ± 2.073	0.394 ± 0.024	5.645 ± 2.073	
LSAM	beta51	0.16	0.150	-0.010 ± 0.010	-0.063 ± 0.065	0.013 ± 0.002	0.116 ± 0.008	0.013 ± 0.002	0.116 ± 0.009	0.526 ± 0.070	0.726 ± 0.059	0.146	-0.014	-9.02	0.1075	0.1084	0.908	93.548 ± 2.206	0.414 ± 0.006	—	
LSAM	beta52	0.16	0.177	0.017 ± 0.009	0.106 ± 0.056	0.010 ± 0.001	0.100 ± 0.007	0.010 ± 0.001	0.101 ± 0.008	0.400 ± 0.051	0.633 ± 0.050	0.184	0.024	14.99	0.0989	0.1018	1.020	96.774 ± 1.587	0.401 ± 0.011	—	
LSAM	beta53	0.16	0.144	-0.016 ± 0.009	-0.099 ± 0.057	0.010 ± 0.002	0.102 ± 0.008	0.011 ± 0.002	0.103 ± 0.009	0.412 ± 0.060	0.642 ± 0.056	0.137	-0.023	-14.29	0.0902	0.0930	0.877	91.129 ± 2.553	0.350 ± 0.006	—	
LSAM	beta54	0.16	0.175	0.015 ± 0.007	0.095 ± 0.045	0.006 ± 0.001	0.080 ± 0.005	0.007 ± 0.001	0.081 ± 0.006	0.257 ± 0.031	0.507 ± 0.038	0.178	0.018	11.43	0.0916	0.0934	1.154	98.387 ± 1.131	0.362 ± 0.007	—	
LSAM	beta55	0.00	0.042	0.042 ± 0.014	—	0.023 ± 0.003	0.151 ± 0.010	0.024 ± 0.003	0.156 ± 0.013	—	—	0.033	0.033	—	0.1474	0.1511	1.004	95.161 ± 1.927	0.594 ± 0.021	4.839 ± 1.927	
LSAM	beta56	0.00	0.009	0.009 ± 0.009	—	0.011 ± 0.001	0.105 ± 0.007	0.011 ± 0.001	0.105 ± 0.008	—	—	0.012	0.012	—	0.1007	0.1014	1.186	98.387 ± 1.131	0.489 ± 0.016	1.613 ± 1.131	
LSAM	beta57	0.00	-0.031	-0.031 ± 0.007	—	0.006 ± 0.001	0.079 ± 0.006	0.007 ± 0.001	0.084 ± 0.006	—	—	-0.038	-0.038	—	0.0768	0.0858	1.068	96.774 ± 1.587	0.329 ± 0.016	3.226 ± 1.587	
LSAM	beta58	0.00	-0.029	-0.029 ± 0.005	—	0.003 ± 0.000	0.056 ± 0.004	0.004 ± 0.000	0.063 ± 0.004	—	—	-0.034	-0.034	—	0.0585	0.0676	1.072	96.774 ± 1.587	0.237 ± 0.008	3.226 ± 1.587	
QML	beta41	0.20	0.202	0.002 ± 0.007	0.011 ± 0.036	0.007 ± 0.001	0.081 ± 0.005	0.006 ± 0.001	0.080 ± 0.006	0.161 ± 0.020	0.402 ± 0.031	0.193	-0.007	-3.43	0.0717	0.0720	0.905	93.548 ± 2.206	0.286 ± 0.005	—	
QML	beta42	0.20	0.213	0.013 ± 0.007	0.067 ± 0.034	0.006 ± 0.001	0.076 ± 0.005	0.006 ± 0.001	0.077 ± 0.006	0.149 ± 0.018	0.386 ± 0.029	0.220	0.020	9.81	0.0798	0.0822	1.022	95.968 ± 1.767	0.306 ± 0.005	—	
QML	beta43	0.20	0.188	-0.012 ± 0.006	-0.059 ± 0.029	0.004 ± 0.001	0.065 ± 0.005	0.004 ± 0.001	0.066 ± 0.005	0.107 ± 0.014	0.328 ± 0.027	0.189	-0.011	-5.56	0.0726	0.0735	1.091	96.774 ± 1.587	0.277 ± 0.004	—	
QML	beta44	0.00	0.015	0.015 ± 0.010	—	0.012 ± 0.002	0.108 ± 0.007	0.012 ± 0.002	0.109 ± 0.009	—	—	0.016	0.016	—	0.1121	0.1132	0.976	94.355 ± 2.073	0.413 ± 0.016	5.645 ± 2.073	
QML	beta45	0.00	-0.026	-0.026 ± 0.006	—	0.005 ± 0.001	0.072 ± 0.004	0.006 ± 0.001	0.076 ± 0.005	—	—	-0.009	-0.009	—	0.0827	0.0832	1.097	94.355 ± 2.073	0.309 ± 0.007	5.645 ± 2.073	
QML	beta46	0.00	-0.027	-0.027 ± 0.006	—	0.004 ± 0.001	0.064 ± 0.005	0.005 ± 0.001	0.069 ± 0.005	—	—	-0.030	-0.030	—	0.0678	0.0741	0.996	90.323 ± 2.655	0.248 ± 0.010	9.677 ± 2.655	
QML	beta47	0.00	-0.031	-0.031 ± 0.005	—	0.003 ± 0.000	0.058 ± 0.003	0.004 ± 0.000	0.065 ± 0.004	—	—	-0.040	-0.040	—	0.0538	0.0669	1.142	93.548 ± 2.206	0.258 ± 0.010	6.452 ± 2.206	
QML	beta51	0.16	0.163	0.003 ± 0.010	0.021 ± 0.065	0.013 ± 0.002	0.115 ± 0.008	0.013 ± 0.002	0.115 ± 0.010	0.514 ± 0.073	0.717 ± 0.061	0.158	-0.002	-0.94	0.1079	0.1079	0.973	94.355 ± 2.073	0.439 ± 0.013	—	
QML	beta52	0.16	0.158	-0.002 ± 0.008	-0.014 ± 0.048	0.007 ± 0.001	0.085 ± 0.006	0.007 ± 0.001	0.084 ± 0.007	0.278 ± 0.037	0.527 ± 0.043	0.163	0.003	1.57	0.0834	0.0835	1.039	96.774 ± 1.587	0.345 ± 0.013	—	
QML	beta53	0.16	0.148	-0.012 ± 0.009	-0.072 ± 0.055	0.009 ± 0.001	0.097 ± 0.008	0.010 ± 0.001	0.098 ± 0.009	0.372 ± 0.056	0.610 ± 0.054	0.140	-0.020	-12.40	0.0959	0.0979	0.855	93.548 ± 2.206	0.326 ± 0.006	—	
QML	beta54	0.16	0.159	-0.001 ± 0.006	-0.006 ± 0.037	0.004 ± 0.001	0.067 ± 0.004	0.004 ± 0.001	0.066 ± 0.005	0.173 ± 0.023	0.415 ± 0.034	0.159	-0.001	-0.68	0.0719	0.0720	1.138	96.774 ± 1.587	0.298 ± 0.005	—	
QML	beta55	0.00	0.024	0.024 ± 0.012	—	0.019 ± 0.003	0.139 ± 0.010	0.020 ± 0.003	0.140 ± 0.012	—	—	0.025	0.025	—	0.1382	0.1404	0.970	94.355 ± 2.073	0.528 ± 0.022	5.645 ± 2.073	
QML	beta56	0.00	-0.012	-0.012 ± 0.008	—	0.008 ± 0.001	0.088 ± 0.006	0.008 ± 0.001	0.088 ± 0.008	—	—	-0.012	-0.012	—	0.0895	0.0903	1.200	99.194 ± 0.803	0.413 ± 0.010	0.806 ± 0.803	
QML	beta57	0.00	-0.022	-0.022 ± 0.006	—	0.005 ± 0.001	0.072 ± 0.005	0.006 ± 0.001	0.075 ± 0.006	—	—	-0.029	-0.029	—	0.0731	0.0787	0.990	91.935 ± 2.445	0.281 ± 0.016	8.065 ± 2.445	
QML	beta58	0.00	-0.020	-0.020 ± 0.005	—	0.003 ± 0.000	0.053 ± 0.003	0.003 ± 0.000	0.056 ± 0.004	—	—	-0.031	-0.031	—	0.0548	0.0631	0.985	93.548 ± 2.206	0.204 ± 0.007	6.452 ± 2.206	
UPI	beta41	0.20	0.219	0.019 ± 0.013	0.095 ± 0.066	0.022 ± 0.004	0.148 ± 0.015	0.022 ± 0.005	0.149 ± 0.017	0.552 ± 0.113	0.743 ± 0.085	0.202	0.002	1.16	0.1612	0.1612	1.422	95.161 ± 1.927	0.825 ± 0.069	—	
UPI	beta42	0.20	0.262	0.062 ± 0.019	0.312 ± 0.096	0.046 ± 0.010	0.214 ± 0.024	0.049 ± 0.010	0.222 ± 0.025	1.229 ± 0.256	1.109 ± 0.127	0.270	0.070	34.85	0.2220	0.2327	1.396	91.935 ± 2.445	1.170 ± 0.173	—	
UPI	beta43	0.20	0.155	-0.045 ± 0.009	-0.224 ± 0.045	0.010 ± 0.001	0.101 ± 0.007	0.012 ± 0.001	0.110 ± 0.007	0.304 ± 0.031	0.551 ± 0.035	0.152	-0.048	-23.78	0.1276	0.1362	1.277	89.516 ± 2.751	0.506 ± 0.038	—	
UPI	beta44	0.00	0.043	0.043 ± 0.018	—	0.039 ± 0.007	0.198 ± 0.017	0.041 ± 0.007	0.202 ± 0.020	—	—	0.058	0.058	—	0.1903	0.1989	1.473	100.000 ± 0.000	1.144 ± 0.094	0.000 ± 0.000	
UPI	beta45	0.00	-0.033	-0.033 ± 0.011	—	0.016 ± 0.003	0.127 ± 0.012	0.017 ± 0.003	0.131 ± 0.013	—	—	-0.028	-0.028	—	0.1203	0.1236	1.445	99.194 ± 0.803	0.719 ± 0.055	0.806 ± 0.803	
UPI	beta46	0.00	-0.042	-0.042 ± 0.013	—	0.021 ± 0.004	0.147 ± 0.016	0.023 ± 0.005	0.152 ± 0.018	—	—	-0.046	-0.046	—	0.1584	0.1650	1.409	99.194 ± 0.803	0.810 ± 0.082	0.806 ± 0.803	

Table S235

Simulation 2 - Condition 91: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.4
(continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.073	-0.073 ± 0.019	—	0.043 ± 0.008	0.207 ± 0.020	0.048 ± 0.008	0.219 ± 0.022	—	—	-0.060	-0.060	—	0.1724	0.1826	1.562	99.194 ± 0.803	1.268 ± 0.184	0.806 ± 0.803
UPI	beta51	0.16	0.194	0.034 ± 0.011	0.213 ± 0.066	0.014 ± 0.002	0.117 ± 0.008	0.015 ± 0.002	0.122 ± 0.009	0.579 ± 0.072	0.761 ± 0.058	0.204	0.044	27.74	0.1255	0.1332	1.091	96.774 ± 1.587	0.502 ± 0.023	—
UPI	beta52	0.16	0.177	0.017 ± 0.011	0.106 ± 0.070	0.015 ± 0.003	0.124 ± 0.013	0.016 ± 0.003	0.125 ± 0.014	0.608 ± 0.125	0.780 ± 0.090	0.180	0.020	12.30	0.1205	0.1221	1.396	94.355 ± 2.073	0.679 ± 0.076	—
UPI	beta53	0.16	0.140	-0.020 ± 0.010	-0.127 ± 0.061	0.012 ± 0.001	0.108 ± 0.007	0.012 ± 0.001	0.109 ± 0.008	0.467 ± 0.056	0.683 ± 0.051	0.128	-0.032	-19.91	0.1235	0.1275	1.093	91.129 ± 2.553	0.462 ± 0.030	—
UPI	beta54	0.16	0.154	-0.006 ± 0.011	-0.035 ± 0.068	0.015 ± 0.003	0.121 ± 0.011	0.014 ± 0.003	0.120 ± 0.012	0.565 ± 0.101	0.752 ± 0.077	0.157	-0.003	-1.92	0.1185	0.1186	1.331	90.323 ± 2.655	0.629 ± 0.056	—
UPI	beta55	0.00	0.051	0.051 ± 0.015	—	0.027 ± 0.003	0.166 ± 0.011	0.030 ± 0.004	0.173 ± 0.014	—	—	0.028	0.028	—	0.1548	0.1574	1.508	99.194 ± 0.803	0.978 ± 0.084	0.806 ± 0.803
UPI	beta56	0.00	-0.015	-0.015 ± 0.013	—	0.021 ± 0.003	0.145 ± 0.012	0.021 ± 0.003	0.145 ± 0.014	—	—	-0.004	-0.004	—	0.1503	0.1504	1.432	100.000 ± 0.000	0.814 ± 0.069	0.000 ± 0.000
UPI	beta57	0.00	-0.048	-0.048 ± 0.010	—	0.014 ± 0.002	0.116 ± 0.009	0.016 ± 0.003	0.125 ± 0.012	—	—	-0.047	-0.047	—	0.1027	0.1131	1.435	100.000 ± 0.000	0.654 ± 0.058	0.000 ± 0.000
UPI	beta58	0.00	-0.054	-0.054 ± 0.011	—	0.015 ± 0.002	0.121 ± 0.010	0.017 ± 0.003	0.132 ± 0.013	—	—	-0.036	-0.036	—	0.0976	0.1040	1.367	99.194 ± 0.803	0.649 ± 0.063	0.806 ± 0.803

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S236

Simulation 2 - Condition 92: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.4

Method	Param.	True	Mean-based										Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE	
LSAM	beta41	0.20	0.216	0.016 ± 0.005	0.080 ± 0.024	0.004 ± 0.000	0.060 ± 0.004	0.004 ± 0.000	0.061 ± 0.004	0.094 ± 0.011	0.307 ± 0.021	0.222	0.022	10.76	0.0596	0.0633	1.006	95.597 ± 1.627	0.235 ± 0.003	—	
LSAM	beta42	0.20	0.233	0.033 ± 0.006	0.164 ± 0.029	0.005 ± 0.001	0.072 ± 0.004	0.006 ± 0.001	0.079 ± 0.006	0.157 ± 0.020	0.396 ± 0.029	0.225	0.025	12.41	0.0745	0.0785	0.898	94.340 ± 1.833	0.254 ± 0.004	—	
LSAM	beta43	0.20	0.172	-0.028 ± 0.004	-0.140 ± 0.018	0.002 ± 0.000	0.044 ± 0.002	0.003 ± 0.000	0.052 ± 0.003	0.068 ± 0.007	0.262 ± 0.015	0.175	-0.025	-12.50	0.0495	0.0554	1.095	92.453 ± 2.095	0.190 ± 0.002	—	
LSAM	beta44	0.00	0.049	0.049 ± 0.005	—	0.004 ± 0.000	0.065 ± 0.004	0.007 ± 0.001	0.081 ± 0.006	—	—	0.045	0.045	—	0.0607	0.0754	1.096	95.597 ± 1.627	0.278 ± 0.007	4.403 ± 1.627	
LSAM	beta45	0.00	0.013	0.013 ± 0.004	—	0.003 ± 0.000	0.054 ± 0.004	0.003 ± 0.000	0.056 ± 0.004	—	—	0.013	0.013	—	0.0545	0.0559	1.012	96.226 ± 1.511	0.215 ± 0.004	3.774 ± 1.511	
LSAM	beta46	0.00	-0.042	-0.042 ± 0.004	—	0.002 ± 0.000	0.044 ± 0.003	0.004 ± 0.000	0.061 ± 0.003	—	—	-0.047	-0.047	—	0.0428	0.0635	0.974	80.503 ± 3.142	0.169 ± 0.004	19.497 ± 3.142	
LSAM	beta47	0.00	-0.052	-0.052 ± 0.004	—	0.003 ± 0.000	0.054 ± 0.004	0.006 ± 0.001	0.075 ± 0.005	—	—	-0.046	-0.046	—	0.0471	0.0660	0.882	78.616 ± 3.252	0.187 ± 0.005	21.384 ± 3.252	
LSAM	beta51	0.16	0.150	-0.010 ± 0.006	-0.064 ± 0.034	0.005 ± 0.000	0.069 ± 0.004	0.005 ± 0.000	0.070 ± 0.005	0.191 ± 0.019	0.437 ± 0.028	0.146	-0.014	-8.65	0.0747	0.0759	0.898	93.711 ± 1.925	0.244 ± 0.002	—	
LSAM	beta52	0.16	0.178	0.018 ± 0.005	0.113 ± 0.028	0.003 ± 0.000	0.057 ± 0.003	0.004 ± 0.000	0.060 ± 0.004	0.139 ± 0.016	0.373 ± 0.026	0.175	0.015	9.22	0.0601	0.0619	1.014	96.226 ± 1.511	0.227 ± 0.003	—	
LSAM	beta53	0.16	0.145	-0.015 ± 0.004	-0.092 ± 0.027	0.003 ± 0.000	0.054 ± 0.004	0.003 ± 0.000	0.056 ± 0.004	0.124 ± 0.015	0.352 ± 0.026	0.139	-0.021	-13.36	0.0503	0.0547	0.972	94.969 ± 1.734	0.208 ± 0.002	—	
LSAM	beta54	0.16	0.172	0.012 ± 0.004	0.076 ± 0.027	0.003 ± 0.000	0.054 ± 0.003	0.003 ± 0.000	0.056 ± 0.004	0.121 ± 0.012	0.348 ± 0.022	0.179	0.019	11.72	0.0548	0.0579	1.011	96.226 ± 1.511	0.216 ± 0.002	—	
LSAM	beta55	0.00	0.039	0.039 ± 0.007	—	0.007 ± 0.001	0.084 ± 0.006	0.009 ± 0.001	0.093 ± 0.006	—	—	0.034	0.034	—	0.0795	0.0866	0.951	95.597 ± 1.627	0.315 ± 0.006	4.403 ± 1.627	
LSAM	beta56	0.00	0.006	0.006 ± 0.005	—	0.004 ± 0.001	0.064 ± 0.005	0.004 ± 0.001	0.064 ± 0.006	—	—	0.008	0.008	—	0.0578	0.0583	1.047	95.597 ± 1.627	0.263 ± 0.005	4.403 ± 1.627	
LSAM	beta57	0.00	-0.038	-0.038 ± 0.004	—	0.002 ± 0.000	0.047 ± 0.003	0.004 ± 0.000	0.061 ± 0.003	—	—	-0.038	-0.038	—	0.0422	0.0570	0.924	86.164 ± 2.738	0.171 ± 0.004	13.836 ± 2.738	
LSAM	beta58	0.00	-0.027	-0.027 ± 0.003	—	0.001 ± 0.000	0.033 ± 0.002	0.002 ± 0.000	0.042 ± 0.002	—	—	-0.028	-0.028	—	0.0378	0.0471	1.068	87.421 ± 2.630	0.137 ± 0.003	12.579 ± 2.630	
QML	beta41	0.20	0.201	0.001 ± 0.003	0.007 ± 0.017	0.002 ± 0.000	0.043 ± 0.003	0.002 ± 0.000	0.043 ± 0.003	0.046 ± 0.006	0.215 ± 0.016	0.206	0.006	2.81	0.0476	0.0479	1.036	95.597 ± 1.627	0.175 ± 0.001	—	
QML	beta42	0.20	0.210	0.010 ± 0.004	0.048 ± 0.020	0.003 ± 0.000	0.052 ± 0.003	0.003 ± 0.000	0.052 ± 0.004	0.068 ± 0.008	0.261 ± 0.018	0.212	0.012	5.85	0.0516	0.0529	0.910	93.711 ± 1.925	0.184 ± 0.001	—	
QML	beta43	0.20	0.184	-0.016 ± 0.003	-0.082 ± 0.017	0.002 ± 0.000	0.042 ± 0.002	0.002 ± 0.000	0.045 ± 0.003	0.050 ± 0.005	0.223 ± 0.014	0.186	-0.014	-6.87	0.0457	0.0478	1.057	92.453 ± 2.095	0.173 ± 0.001	—	
QML	beta44	0.00	0.019	0.019 ± 0.004	—	0.003 ± 0.000	0.055 ± 0.003	0.003 ± 0.000	0.058 ± 0.004	—	—	0.021	0.021	—	0.0605	0.0641	1.115	95.597 ± 1.627	0.242 ± 0.004	4.403 ± 1.627	
QML	beta45	0.00	-0.005	-0.005 ± 0.004	—	0.002 ± 0.000	0.048 ± 0.003	0.002 ± 0.000	0.048 ± 0.004	—	—	-0.007	-0.007	—	0.0502	0.0507	1.016	96.226 ± 1.511	0.192 ± 0.003	3.774 ± 1.511	
QML	beta46	0.00	-0.031	-0.031 ± 0.003	—	0.001 ± 0.000	0.038 ± 0.002	0.002 ± 0.000	0.050 ± 0.003	—	—	-0.036	-0.036	—	0.0401	0.0541	0.997	81.132 ± 3.103	0.150 ± 0.003	18.868 ± 3.103	
QML	beta47	0.00	-0.039	-0.039 ± 0.003	—	0.002 ± 0.000	0.041 ± 0.002	0.003 ± 0.000	0.057 ± 0.003	—	—	-0.037	-0.037	—	0.0387	0.0537	0.932	77.987 ± 3.286	0.151 ± 0.003	22.013 ± 3.286	
QML	beta51	0.16	0.158	-0.002 ± 0.005	-0.010 ± 0.033	0.004 ± 0.000	0.066 ± 0.003	0.004 ± 0.000	0.066 ± 0.004	0.170 ± 0.017	0.412 ± 0.027	0.163	0.003	1.94	0.0753	0.0753	0.946	94.969 ± 1.734	0.245 ± 0.002	—	
QML	beta52	0.16	0.164	0.004 ± 0.004	0.026 ± 0.023	0.002 ± 0.000	0.046 ± 0.002	0.002 ± 0.000	0.046 ± 0.003	0.082 ± 0.009	0.286 ± 0.019	0.163	0.003	1.87	0.0568	0.0569	1.066	96.855 ± 1.384	0.191 ± 0.002	—	
QML	beta53	0.16	0.152	-0.008 ± 0.004	-0.050 ± 0.025	0.003 ± 0.000	0.050 ± 0.004	0.003 ± 0.000	0.051 ± 0.004	0.100 ± 0.013	0.317 ± 0.025	0.148	-0.012	-7.53	0.0503	0.0517	0.988	95.597 ± 1.627	0.194 ± 0.002	—	
QML	beta54	0.16	0.161	0.001 ± 0.004	0.006 ± 0.022	0.002 ± 0.000	0.044 ± 0.003	0.002 ± 0.000	0.044 ± 0.003	0.077 ± 0.009	0.277 ± 0.019	0.164	0.004	2.41	0.0472	0.0474	1.037	96.226 ± 1.511	0.181 ± 0.002	—	
QML	beta55	0.00	0.032	0.032 ± 0.006	—	0.006 ± 0.001	0.075 ± 0.004	0.007 ± 0.001	0.081 ± 0.005	—	—	0.033	0.033	—	0.0802	0.0866	1.016	96.226 ± 1.511	0.300 ± 0.005	3.774 ± 1.511	
QML	beta56	0.00	-0.013	-0.013 ± 0.004	—	0.003 ± 0.000	0.056 ± 0.004	0.003 ± 0.000	0.058 ± 0.004	—	—	-0.008	-0.008	—	0.0623	0.0628	1.098	96.226 ± 1.511	0.243 ± 0.003	3.774 ± 1.511	
QML	beta57	0.00	-0.033	-0.033 ± 0.003	—	0.002 ± 0.000	0.042 ± 0.003	0.003 ± 0.000	0.053 ± 0.003	—	—	-0.037	-0.037	—	0.0435	0.0573	0.979	83.648 ± 2.933	0.160 ± 0.005	16.352 ± 2.933	
QML	beta58	0.00	-0.023	-0.023 ± 0.002	—	0.001 ± 0.000	0.029 ± 0.002	0.001 ± 0.000	0.037 ± 0.002	—	—	-0.022	-0.022	—	0.0328	0.0397	1.095	85.535 ± 2.790	0.123 ± 0.002	14.465 ± 2.790	
UPI	beta41	0.20	0.282	0.082 ± 0.018	0.409 ± 0.091	0.053 ± 0.009	0.230 ± 0.020	0.059 ± 0.011	0.244 ± 0.024	1.484 ± 0.272	1.218 ± 0.122	0.273	0.073	36.30	0.2163	0.2282	1.018	89.937 ± 2.386	0.919 ± 0.081	—	
UPI	beta42	0.20	0.292	0.092 ± 0.021	0.462 ± 0.105	0.070 ± 0.013	0.265 ± 0.025	0.079 ± 0.016	0.280 ± 0.030	1.964 ± 0.392	1.401 ± 0.151	0.273	0.073	36.35	0.2471	0.2576	1.128	88.679 ± 2.513	1.174 ± 0.156	—	
UPI	beta43	0.20	0.135	-0.065 ± 0.008	-0.327 ± 0.041	0.011 ± 0.001	0.103 ± 0.006	0.015 ± 0.002	0.122 ± 0.008	0.372 ± 0.046	0.610 ± 0.042	0.120	-0.080	-40.20	0.1216	0.1457	1.039	81.761 ± 3.062	0.421 ± 0.041	—	
UPI	beta44	0.00	0.084	0.084 ± 0.015	—	0.035 ± 0.007	0.186 ± 0.020	0.041 ± 0.008	0.203 ± 0.022	—	—	0.063	0.063	—	0.1522	0.1647	1.126	97.484 ± 1.242	0.821 ± 0.085	2.516 ± 1.242	
UPI	beta45	0.00	0.003	0.003 ± 0.007	—	0.008 ± 0.001	0.092 ± 0.008	0.008 ± 0.001	0.092 ± 0.009	—	—	0.003	0.003	—	0.0935	0.0936	1.172	99.371 ± 0.627	0.423 ± 0.038	0.629 ± 0.627	
UPI	beta46	0.00	-0.170	-0.170 ± 0.020	—	0.065 ± 0.012	0.254 ± 0.023	0.093 ± 0.017	0.305 ± 0.029	—	—	-0.112	-0.112	—	0.1713	0.2048	1.285	98.742 ± 0.884	1.280 ± 0.183	1.258 ± 0.884	

Table S236

Simulation 2 - Condition 92: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.4
(continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.152	-0.152 ± 0.019	—	0.060 ± 0.010	0.245 ± 0.022	0.083 ± 0.014	0.287 ± 0.026	—	—	-0.105	-0.105	—	0.1982	0.2243	1.351	96.226 ± 1.511	1.297 ± 0.168	3.774 ± 1.511
UPI	beta51	0.16	0.205	0.045 ± 0.007	0.283 ± 0.044	0.008 ± 0.001	0.089 ± 0.005	0.010 ± 0.001	0.099 ± 0.006	0.386 ± 0.039	0.621 ± 0.037	0.205	0.045	28.34	0.0885	0.0995	1.014	84.906 ± 2.839	0.353 ± 0.024	—
UPI	beta52	0.16	0.215	0.055 ± 0.013	0.345 ± 0.078	0.025 ± 0.004	0.158 ± 0.015	0.028 ± 0.005	0.167 ± 0.018	1.090 ± 0.214	1.044 ± 0.112	0.184	0.024	14.93	0.1469	0.1488	1.026	89.937 ± 2.386	0.636 ± 0.065	—
UPI	beta53	0.16	0.133	-0.027 ± 0.006	-0.166 ± 0.039	0.006 ± 0.001	0.078 ± 0.005	0.007 ± 0.001	0.083 ± 0.006	0.266 ± 0.032	0.516 ± 0.036	0.121	-0.039	-24.47	0.0896	0.0977	0.990	91.195 ± 2.247	0.304 ± 0.015	—
UPI	beta54	0.16	0.139	-0.021 ± 0.008	-0.132 ± 0.049	0.010 ± 0.001	0.098 ± 0.006	0.010 ± 0.001	0.100 ± 0.007	0.391 ± 0.043	0.626 ± 0.042	0.136	-0.024	-15.17	0.1167	0.1192	1.135	84.277 ± 2.887	0.437 ± 0.040	—
UPI	beta55	0.00	0.052	0.052 ± 0.009	—	0.013 ± 0.002	0.116 ± 0.009	0.016 ± 0.003	0.127 ± 0.012	—	—	0.034	0.034	—	0.1081	0.1134	1.127	97.484 ± 1.242	0.512 ± 0.035	2.516 ± 1.242
UPI	beta56	0.00	-0.018	-0.018 ± 0.006	—	0.006 ± 0.001	0.076 ± 0.004	0.006 ± 0.001	0.078 ± 0.005	—	—	-0.014	-0.014	—	0.0868	0.0880	1.332	100.000 ± 0.000	0.396 ± 0.017	0.000 ± 0.000
UPI	beta57	0.00	-0.132	-0.132 ± 0.015	—	0.036 ± 0.006	0.190 ± 0.018	0.053 ± 0.009	0.231 ± 0.022	—	—	-0.096	-0.096	—	0.1541	0.1813	1.191	100.000 ± 0.000	0.886 ± 0.111	0.000 ± 0.000
UPI	beta58	0.00	-0.063	-0.063 ± 0.009	—	0.014 ± 0.002	0.117 ± 0.010	0.017 ± 0.003	0.132 ± 0.012	—	—	-0.040	-0.040	—	0.1013	0.1089	1.162	99.371 ± 0.627	0.531 ± 0.049	0.629 ± 0.627

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S237

Simulation 2 - Condition 93: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.6

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Type I (%) $\pm$ MCSE
LSAM	beta41	0.20	0.206	0.006 $\pm$ 0.002	0.028 $\pm$ 0.009	0.003 $\pm$ 0.000	0.059 $\pm$ 0.001	0.003 $\pm$ 0.000	0.059 $\pm$ 0.002	0.087 $\pm$ 0.004	0.296 $\pm$ 0.009	0.203	0.003	1.38	0.0542	0.0542	0.985	94.646 $\pm$ 0.715	0.227 $\pm$ 0.001	—
LSAM	beta42	0.20	0.205	0.005 $\pm$ 0.002	0.023 $\pm$ 0.009	0.004 $\pm$ 0.000	0.059 $\pm$ 0.001	0.004 $\pm$ 0.000	0.059 $\pm$ 0.002	0.088 $\pm$ 0.004	0.297 $\pm$ 0.008	0.204	0.004	2.06	0.0586	0.0588	1.019	95.556 $\pm$ 0.655	0.236 $\pm$ 0.001	—
LSAM	beta43	0.20	0.196	-0.004 $\pm$ 0.002	-0.020 $\pm$ 0.009	0.003 $\pm$ 0.000	0.054 $\pm$ 0.001	0.003 $\pm$ 0.000	0.054 $\pm$ 0.001	0.072 $\pm$ 0.003	0.269 $\pm$ 0.007	0.196	-0.004	-2.11	0.0545	0.0547	1.020	95.253 $\pm$ 0.676	0.215 $\pm$ 0.001	—
LSAM	beta44	0.00	0.015	0.015 $\pm$ 0.002	—	0.005 $\pm$ 0.000	0.073 $\pm$ 0.002	0.005 $\pm$ 0.000	0.074 $\pm$ 0.002	—	—	0.014	0.014	—	0.0695	0.0710	0.953	93.535 $\pm$ 0.782	0.271 $\pm$ 0.002	6.465 $\pm$ 0.782
LSAM	beta45	0.00	0.002	0.002 $\pm$ 0.002	—	0.004 $\pm$ 0.000	0.063 $\pm$ 0.001	0.004 $\pm$ 0.000	0.063 $\pm$ 0.002	—	—	0.003	0.003	—	0.0630	0.0631	0.940	94.444 $\pm$ 0.728	0.231 $\pm$ 0.001	5.556 $\pm$ 0.728
LSAM	beta46	0.00	-0.014	-0.014 $\pm$ 0.002	—	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.003 $\pm$ 0.000	0.050 $\pm$ 0.001	—	—	-0.015	-0.015	—	0.0449	0.0474	0.988	93.535 $\pm$ 0.782	0.187 $\pm$ 0.001	6.465 $\pm$ 0.782
LSAM	beta47	0.00	-0.016	-0.016 $\pm$ 0.002	—	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.003 $\pm$ 0.000	0.050 $\pm$ 0.001	—	—	-0.016	-0.016	—	0.0459	0.0485	0.986	93.535 $\pm$ 0.782	0.185 $\pm$ 0.001	6.465 $\pm$ 0.782
LSAM	beta51	0.16	0.162	0.002 $\pm$ 0.002	0.013 $\pm$ 0.015	0.005 $\pm$ 0.000	0.074 $\pm$ 0.002	0.005 $\pm$ 0.000	0.074 $\pm$ 0.002	0.213 $\pm$ 0.010	0.461 $\pm$ 0.013	0.158	-0.002	-1.19	0.0728	0.0728	0.963	92.828 $\pm$ 0.820	0.278 $\pm$ 0.001	—
LSAM	beta52	0.16	0.165	0.005 $\pm$ 0.002	0.033 $\pm$ 0.012	0.003 $\pm$ 0.000	0.059 $\pm$ 0.001	0.003 $\pm$ 0.000	0.059 $\pm$ 0.002	0.136 $\pm$ 0.006	0.369 $\pm$ 0.010	0.165	0.005	3.41	0.0600	0.0603	1.000	95.354 $\pm$ 0.669	0.231 $\pm$ 0.001	—
LSAM	beta53	0.16	0.153	-0.007 $\pm$ 0.002	-0.041 $\pm$ 0.012	0.004 $\pm$ 0.000	0.062 $\pm$ 0.001	0.004 $\pm$ 0.000	0.062 $\pm$ 0.002	0.151 $\pm$ 0.007	0.389 $\pm$ 0.011	0.155	-0.005	-3.37	0.0625	0.0628	0.968	94.242 $\pm$ 0.740	0.235 $\pm$ 0.001	—
LSAM	beta54	0.16	0.161	0.001 $\pm$ 0.002	0.006 $\pm$ 0.012	0.003 $\pm$ 0.000	0.059 $\pm$ 0.001	0.003 $\pm$ 0.000	0.059 $\pm$ 0.002	0.136 $\pm$ 0.006	0.369 $\pm$ 0.010	0.160	0.000	-0.02	0.0579	0.0579	0.982	95.051 $\pm$ 0.689	0.227 $\pm$ 0.001	—
LSAM	beta55	0.00	0.012	0.012 $\pm$ 0.003	—	0.007 $\pm$ 0.000	0.085 $\pm$ 0.002	0.007 $\pm$ 0.000	0.086 $\pm$ 0.002	—	—	0.016	0.016	—	0.0831	0.0846	0.948	93.636 $\pm$ 0.776	0.318 $\pm$ 0.002	6.364 $\pm$ 0.776
LSAM	beta56	0.00	0.002	0.002 $\pm$ 0.002	—	0.005 $\pm$ 0.000	0.074 $\pm$ 0.002	0.005 $\pm$ 0.000	0.074 $\pm$ 0.002	—	—	-0.002	-0.002	—	0.0740	0.0741	0.959	93.939 $\pm$ 0.758	0.277 $\pm$ 0.002	6.061 $\pm$ 0.758
LSAM	beta57	0.00	-0.014	-0.014 $\pm$ 0.002	—	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	—	—	-0.014	-0.014	—	0.0476	0.0495	0.983	93.838 $\pm$ 0.764	0.191 $\pm$ 0.001	6.162 $\pm$ 0.764
LSAM	beta58	0.00	-0.010	-0.010 $\pm$ 0.001	—	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	—	—	-0.010	-0.010	—	0.0428	0.0439	0.962	94.242 $\pm$ 0.740	0.161 $\pm$ 0.001	5.758 $\pm$ 0.740
QML	beta41	0.20	0.202	0.002 $\pm$ 0.002	0.009 $\pm$ 0.009	0.003 $\pm$ 0.000	0.055 $\pm$ 0.001	0.003 $\pm$ 0.000	0.055 $\pm$ 0.002	0.075 $\pm$ 0.004	0.274 $\pm$ 0.008	0.199	-0.001	-0.49	0.0523	0.0523	0.991	94.747 $\pm$ 0.709	0.213 $\pm$ 0.001	—
QML	beta42	0.20	0.201	0.001 $\pm$ 0.002	0.005 $\pm$ 0.009	0.003 $\pm$ 0.000	0.055 $\pm$ 0.001	0.003 $\pm$ 0.000	0.055 $\pm$ 0.002	0.077 $\pm$ 0.004	0.277 $\pm$ 0.008	0.200	0.000	-0.01	0.0548	0.0548	1.012	95.152 $\pm$ 0.683	0.220 $\pm$ 0.001	—
QML	beta43	0.20	0.196	-0.004 $\pm$ 0.002	-0.020 $\pm$ 0.008	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.069 $\pm$ 0.003	0.262 $\pm$ 0.007	0.197	-0.003	-1.32	0.0536	0.0536	1.020	95.455 $\pm$ 0.662	0.209 $\pm$ 0.001	—
QML	beta44	0.00	0.005	0.005 $\pm$ 0.002	—	0.005 $\pm$ 0.000	0.067 $\pm$ 0.002	0.005 $\pm$ 0.000	0.067 $\pm$ 0.002	—	—	0.003	0.003	—	0.0685	0.0685	0.988	94.141 $\pm$ 0.746	0.260 $\pm$ 0.002	5.859 $\pm$ 0.746
QML	beta45	0.00	-0.005	-0.005 $\pm$ 0.002	—	0.003 $\pm$ 0.000	0.059 $\pm$ 0.001	0.004 $\pm$ 0.000	0.059 $\pm$ 0.002	—	—	-0.004	-0.004	—	0.0611	0.0612	0.959	94.242 $\pm$ 0.740	0.222 $\pm$ 0.001	5.758 $\pm$ 0.740
QML	beta46	0.00	-0.011	-0.011 $\pm$ 0.001	—	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	—	—	-0.012	-0.012	—	0.0441	0.0458	1.005	94.040 $\pm$ 0.752	0.176 $\pm$ 0.002	5.960 $\pm$ 0.752
QML	beta47	0.00	-0.015	-0.015 $\pm$ 0.001	—	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	—	—	-0.014	-0.014	—	0.0430	0.0454	1.006	92.727 $\pm$ 0.825	0.170 $\pm$ 0.001	7.273 $\pm$ 0.825
QML	beta51	0.16	0.164	0.004 $\pm$ 0.002	0.022 $\pm$ 0.015	0.005 $\pm$ 0.000	0.073 $\pm$ 0.002	0.005 $\pm$ 0.000	0.073 $\pm$ 0.002	0.210 $\pm$ 0.010	0.458 $\pm$ 0.013	0.160	0.000	0.05	0.0710	0.0710	0.990	94.242 $\pm$ 0.740	0.284 $\pm$ 0.001	—
QML	beta52	0.16	0.163	0.003 $\pm$ 0.002	0.019 $\pm$ 0.011	0.003 $\pm$ 0.000	0.057 $\pm$ 0.001	0.003 $\pm$ 0.000	0.057 $\pm$ 0.002	0.128 $\pm$ 0.005	0.357 $\pm$ 0.010	0.164	0.004	2.40	0.0587	0.0589	1.014	95.758 $\pm$ 0.641	0.227 $\pm$ 0.001	—
QML	beta53	0.16	0.154	-0.006 $\pm$ 0.002	-0.040 $\pm$ 0.012	0.004 $\pm$ 0.000	0.060 $\pm$ 0.001	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.143 $\pm$ 0.006	0.378 $\pm$ 0.010	0.153	-0.007	-4.28	0.0612	0.0616	0.976	94.444 $\pm$ 0.728	0.231 $\pm$ 0.001	—
QML	beta54	0.16	0.159	-0.001 $\pm$ 0.002	-0.007 $\pm$ 0.011	0.003 $\pm$ 0.000	0.057 $\pm$ 0.001	0.003 $\pm$ 0.000	0.057 $\pm$ 0.002	0.126 $\pm$ 0.006	0.355 $\pm$ 0.010	0.157	-0.003	-2.01	0.0556	0.0557	0.986	94.747 $\pm$ 0.709	0.220 $\pm$ 0.001	—
QML	beta55	0.00	0.009	0.009 $\pm$ 0.003	—	0.007 $\pm$ 0.000	0.082 $\pm$ 0.002	0.007 $\pm$ 0.000	0.083 $\pm$ 0.002	—	—	0.013	0.013	—	0.0818	0.0828	0.966	93.737 $\pm$ 0.770	0.312 $\pm$ 0.002	6.263 $\pm$ 0.770
QML	beta56	0.00	-0.005	-0.005 $\pm$ 0.002	—	0.005 $\pm$ 0.000	0.070 $\pm$ 0.002	0.005 $\pm$ 0.000	0.070 $\pm$ 0.002	—	—	-0.007	-0.007	—	0.0691	0.0694	0.968	94.646 $\pm$ 0.715	0.266 $\pm$ 0.001	5.354 $\pm$ 0.715
QML	beta57	0.00	-0.013	-0.013 $\pm$ 0.001	—	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	—	—	-0.014	-0.014	—	0.0447	0.0469	1.008	93.030 $\pm$ 0.809	0.183 $\pm$ 0.001	6.970 $\pm$ 0.809
QML	beta58	0.00	-0.010	-0.010 $\pm$ 0.001	—	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	—	—	-0.010	-0.010	—	0.0403	0.0415	0.973	93.434 $\pm$ 0.787	0.151 $\pm$ 0.001	6.566 $\pm$ 0.787
UPI	beta41	0.20	0.205	0.005 $\pm$ 0.002	0.027 $\pm$ 0.009	0.004 $\pm$ 0.000	0.059 $\pm$ 0.001	0.004 $\pm$ 0.000	0.059 $\pm$ 0.002	0.088 $\pm$ 0.004	0.297 $\pm$ 0.008	0.202	0.002	0.75	0.0559	0.0559	1.010	94.949 $\pm$ 0.696	0.234 $\pm$ 0.001	—
UPI	beta42	0.20	0.205	0.005 $\pm$ 0.002	0.025 $\pm$ 0.009	0.004 $\pm$ 0.000	0.060 $\pm$ 0.001	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	0.090 $\pm$ 0.004	0.299 $\pm$ 0.009	0.204	0.004	1.92	0.0603	0.0605	1.046	95.758 $\pm$ 0.641	0.245 $\pm$ 0.001	—
UPI	beta43	0.20	0.196	-0.004 $\pm$ 0.002	-0.020 $\pm$ 0.009	0.003 $\pm$ 0.000	0.054 $\pm$ 0.001	0.003 $\pm$ 0.000	0.054 $\pm$ 0.002	0.074 $\pm$ 0.003	0.272 $\pm$ 0.008	0.196	-0.004	-1.80	0.0549	0.0550	1.036	94.848 $\pm$ 0.703	0.220 $\pm$ 0.001	—
UPI	beta44	0.00	0.011	0.011 $\pm$ 0.002	—	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	—	—	0.008	0.008	—	0.0721	0.0725	0.999	94.949 $\pm$ 0.696	0.292 $\pm$ 0.003	5.051 $\pm$ 0.696
UPI	beta45	0.00	0.000	-0.000 $\pm$ 0.002	—	0.004 $\pm$ 0.000	0.063 $\pm$ 0.002	0.004 $\pm$ 0.000	0.063 $\pm$ 0.002	—	—	0.002	0.002	—	0.0602	0.0603	1.011	95.758 $\pm$ 0.641	0.248 $\pm$ 0.002	4.242 $\pm$ 0.641
UPI	beta46	0.00	-0.008	-0.008 $\pm$ 0.002	—	0.003 $\pm$ 0.000	0.055 $\pm$ 0.001	0.003 $\pm$ 0.000	0.056 $\pm$ 0.002	—	—	-0.009	-0.009	—	0.0534	0.0541	1.014	94.646 $\pm$ 0.715	0.220 $\pm$ 0.003	5.354 $\pm$ 0.715

Table S237

Simulation 2 - Condition 93: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.6
(continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.011	-0.011 ± 0.002	—	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	—	—	-0.008	-0.008	—	0.0498	0.0505	1.034	96.667 ± 0.571	0.220 ± 0.004	3.333 ± 0.571
UPI	beta51	0.16	0.162	0.002 ± 0.002	0.013 ± 0.014	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.206 ± 0.010	0.454 ± 0.013	0.158	-0.002	-1.02	0.0692	0.0692	0.991	94.444 ± 0.728	0.282 ± 0.001	—
UPI	beta52	0.16	0.165	0.005 ± 0.002	0.031 ± 0.012	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.134 ± 0.006	0.366 ± 0.010	0.164	0.004	2.78	0.0577	0.0579	1.026	95.556 ± 0.655	0.235 ± 0.001	—
UPI	beta53	0.16	0.155	-0.005 ± 0.002	-0.033 ± 0.012	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.150 ± 0.007	0.387 ± 0.011	0.155	-0.005	-3.17	0.0617	0.0619	0.986	94.444 ± 0.728	0.239 ± 0.001	—
UPI	beta54	0.16	0.162	0.002 ± 0.002	0.011 ± 0.012	0.004 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	0.137 ± 0.006	0.371 ± 0.010	0.162	0.002	1.12	0.0578	0.0578	1.007	94.949 ± 0.696	0.234 ± 0.001	—
UPI	beta55	0.00	0.010	0.010 ± 0.003	—	0.007 ± 0.000	0.086 ± 0.002	0.007 ± 0.000	0.086 ± 0.003	—	—	0.013	0.013	—	0.0796	0.0807	0.986	94.646 ± 0.715	0.331 ± 0.003	5.354 ± 0.715
UPI	beta56	0.00	0.001	0.001 ± 0.002	—	0.005 ± 0.000	0.074 ± 0.002	0.005 ± 0.000	0.074 ± 0.002	—	—	-0.001	-0.001	—	0.0685	0.0685	0.997	95.455 ± 0.662	0.287 ± 0.002	4.545 ± 0.662
UPI	beta57	0.00	-0.007	-0.007 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	—	—	-0.007	-0.007	—	0.0515	0.0520	1.020	96.465 ± 0.587	0.212 ± 0.002	3.535 ± 0.587
UPI	beta58	0.00	-0.006	-0.006 ± 0.002	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	—	—	-0.005	-0.005	—	0.0455	0.0457	1.042	95.960 ± 0.626	0.194 ± 0.003	4.040 ± 0.626

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S238

Simulation 2 - Condition 94: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.6

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Type I (%) $\pm$ MCSE
LSAM	beta41	0.20	0.201	0.001 $\pm$ 0.001	0.007 $\pm$ 0.006	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.032 $\pm$ 0.001	0.180 $\pm$ 0.005	0.203	0.003	1.46	0.0352	0.0353	0.992	94.589 $\pm$ 0.716	0.140 $\pm$ 0.000	—
LSAM	beta42	0.20	0.205	0.005 $\pm$ 0.001	0.026 $\pm$ 0.006	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.038 $\pm$ 0.002	0.194 $\pm$ 0.006	0.203	0.003	1.63	0.0372	0.0374	0.964	93.888 $\pm$ 0.758	0.145 $\pm$ 0.000	—
LSAM	beta43	0.20	0.195	-0.005 $\pm$ 0.001	-0.023 $\pm$ 0.006	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.031 $\pm$ 0.001	0.177 $\pm$ 0.005	0.195	-0.005	-2.71	0.0363	0.0367	0.969	93.487 $\pm$ 0.781	0.133 $\pm$ 0.000	—
LSAM	beta44	0.00	0.015	0.015 $\pm$ 0.001	—	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	—	—	0.013	0.013	—	0.0420	0.0438	0.951	93.287 $\pm$ 0.792	0.160 $\pm$ 0.001	6.713 $\pm$ 0.792
LSAM	beta45	0.00	0.006	0.006 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	—	—	0.005	0.005	—	0.0369	0.0373	0.978	94.890 $\pm$ 0.697	0.139 $\pm$ 0.001	5.110 $\pm$ 0.697
LSAM	beta46	0.00	-0.013	-0.013 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	—	—	-0.013	-0.013	—	0.0279	0.0309	0.961	91.383 $\pm$ 0.888	0.111 $\pm$ 0.001	8.617 $\pm$ 0.888
LSAM	beta47	0.00	-0.016	-0.016 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	—	—	-0.016	-0.016	—	0.0294	0.0337	0.960	89.379 $\pm$ 0.975	0.110 $\pm$ 0.001	10.621 $\pm$ 0.975
LSAM	beta51	0.16	0.161	0.001 $\pm$ 0.001	0.009 $\pm$ 0.009	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.076 $\pm$ 0.003	0.276 $\pm$ 0.008	0.161	0.001	0.38	0.0436	0.0436	1.022	96.192 $\pm$ 0.606	0.177 $\pm$ 0.000	—
LSAM	beta52	0.16	0.164	0.004 $\pm$ 0.001	0.024 $\pm$ 0.007	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.054 $\pm$ 0.002	0.232 $\pm$ 0.006	0.164	0.004	2.28	0.0349	0.0351	0.987	94.188 $\pm$ 0.741	0.143 $\pm$ 0.000	—
LSAM	beta53	0.16	0.156	-0.004 $\pm$ 0.001	-0.022 $\pm$ 0.007	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.053 $\pm$ 0.002	0.231 $\pm$ 0.006	0.158	-0.002	-1.51	0.0368	0.0369	1.015	94.890 $\pm$ 0.697	0.146 $\pm$ 0.000	—
LSAM	beta54	0.16	0.161	0.001 $\pm$ 0.001	0.006 $\pm$ 0.007	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.052 $\pm$ 0.002	0.228 $\pm$ 0.006	0.160	0.000	-0.09	0.0365	0.0365	0.989	94.990 $\pm$ 0.691	0.141 $\pm$ 0.000	—
LSAM	beta55	0.00	0.013	0.013 $\pm$ 0.002	—	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	—	—	0.013	0.013	—	0.0503	0.0520	0.982	94.088 $\pm$ 0.747	0.195 $\pm$ 0.001	5.912 $\pm$ 0.747
LSAM	beta56	0.00	0.001	0.001 $\pm$ 0.001	—	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	—	—	0.001	0.001	—	0.0409	0.0410	0.990	94.389 $\pm$ 0.728	0.170 $\pm$ 0.001	5.611 $\pm$ 0.728
LSAM	beta57	0.00	-0.014	-0.014 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	—	—	-0.014	-0.014	—	0.0281	0.0314	0.973	91.283 $\pm$ 0.893	0.114 $\pm$ 0.001	8.717 $\pm$ 0.893
LSAM	beta58	0.00	-0.010	-0.010 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.027 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	—	—	-0.010	-0.010	—	0.0260	0.0278	0.917	92.585 $\pm$ 0.829	0.099 $\pm$ 0.000	7.415 $\pm$ 0.829
QML	beta41	0.20	0.198	-0.002 $\pm$ 0.001	-0.008 $\pm$ 0.005	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.029 $\pm$ 0.001	0.170 $\pm$ 0.005	0.200	0.000	-0.13	0.0336	0.0336	0.992	94.790 $\pm$ 0.703	0.133 $\pm$ 0.000	—
QML	beta42	0.20	0.202	0.002 $\pm$ 0.001	0.010 $\pm$ 0.006	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.033 $\pm$ 0.002	0.181 $\pm$ 0.005	0.201	0.001	0.40	0.0350	0.0351	0.964	93.186 $\pm$ 0.798	0.137 $\pm$ 0.000	—
QML	beta43	0.20	0.196	-0.004 $\pm$ 0.001	-0.022 $\pm$ 0.005	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.030 $\pm$ 0.001	0.172 $\pm$ 0.005	0.195	-0.005	-2.41	0.0355	0.0358	0.973	94.188 $\pm$ 0.741	0.130 $\pm$ 0.000	—
QML	beta44	0.00	0.005	0.005 $\pm$ 0.001	—	0.002 $\pm$ 0.000	0.042 $\pm$ 0.001	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	—	—	0.004	0.004	—	0.0409	0.0411	0.966	94.890 $\pm$ 0.697	0.161 $\pm$ 0.001	5.110 $\pm$ 0.697
QML	beta45	0.00	-0.002	-0.002 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	—	—	-0.002	-0.002	—	0.0358	0.0358	0.982	94.389 $\pm$ 0.728	0.137 $\pm$ 0.001	5.611 $\pm$ 0.728
QML	beta46	0.00	-0.011	-0.011 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.028 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	—	—	-0.012	-0.012	—	0.0275	0.0299	0.978	91.984 $\pm$ 0.860	0.107 $\pm$ 0.001	8.016 $\pm$ 0.860
QML	beta47	0.00	-0.015	-0.015 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.027 $\pm$ 0.001	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	—	—	-0.016	-0.016	—	0.0274	0.0318	0.980	88.978 $\pm$ 0.991	0.105 $\pm$ 0.001	11.022 $\pm$ 0.991
QML	beta51	0.16	0.162	0.002 $\pm$ 0.001	0.015 $\pm$ 0.009	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.075 $\pm$ 0.003	0.275 $\pm$ 0.008	0.162	0.002	0.99	0.0437	0.0437	1.034	96.693 $\pm$ 0.566	0.178 $\pm$ 0.000	—
QML	beta52	0.16	0.162	0.002 $\pm$ 0.001	0.014 $\pm$ 0.007	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.051 $\pm$ 0.002	0.226 $\pm$ 0.006	0.162	0.002	1.10	0.0357	0.0357	0.987	94.389 $\pm$ 0.728	0.140 $\pm$ 0.000	—
QML	beta53	0.16	0.157	-0.003 $\pm$ 0.001	-0.020 $\pm$ 0.007	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.051 $\pm$ 0.002	0.225 $\pm$ 0.006	0.158	-0.002	-1.22	0.0355	0.0355	1.019	95.491 $\pm$ 0.657	0.144 $\pm$ 0.000	—
QML	beta54	0.16	0.160	-0.000 $\pm$ 0.001	-0.003 $\pm$ 0.007	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.048 $\pm$ 0.002	0.220 $\pm$ 0.006	0.159	-0.001	-0.44	0.0351	0.0351	0.999	95.291 $\pm$ 0.671	0.138 $\pm$ 0.000	—
QML	beta55	0.00	0.010	0.010 $\pm$ 0.002	—	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	—	—	0.010	0.010	—	0.0495	0.0505	0.992	94.389 $\pm$ 0.728	0.194 $\pm$ 0.001	5.611 $\pm$ 0.728
QML	beta56	0.00	-0.005	-0.005 $\pm$ 0.001	—	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	—	—	-0.004	-0.004	—	0.0410	0.0412	0.997	94.589 $\pm$ 0.716	0.167 $\pm$ 0.001	5.411 $\pm$ 0.716
QML	beta57	0.00	-0.014	-0.014 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	—	—	-0.014	-0.014	—	0.0274	0.0308	0.987	91.182 $\pm$ 0.898	0.112 $\pm$ 0.001	8.818 $\pm$ 0.898
QML	beta58	0.00	-0.011	-0.011 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.026 $\pm$ 0.001	0.001 $\pm$ 0.000	0.028 $\pm$ 0.001	—	—	-0.011	-0.011	—	0.0252	0.0276	0.934	91.383 $\pm$ 0.888	0.095 $\pm$ 0.001	8.617 $\pm$ 0.888
UPI	beta41	0.20	0.201	0.001 $\pm$ 0.001	0.003 $\pm$ 0.006	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.032 $\pm$ 0.001	0.180 $\pm$ 0.005	0.202	0.002	1.01	0.0359	0.0359	1.000	95.391 $\pm$ 0.664	0.141 $\pm$ 0.000	—
UPI	beta42	0.20	0.205	0.005 $\pm$ 0.001	0.027 $\pm$ 0.006	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.038 $\pm$ 0.002	0.194 $\pm$ 0.006	0.205	0.005	2.27	0.0366	0.0369	0.975	94.289 $\pm$ 0.735	0.147 $\pm$ 0.000	—
UPI	beta43	0.20	0.196	-0.004 $\pm$ 0.001	-0.022 $\pm$ 0.006	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.032 $\pm$ 0.001	0.178 $\pm$ 0.005	0.195	-0.005	-2.40	0.0357	0.0361	0.969	93.587 $\pm$ 0.775	0.134 $\pm$ 0.000	—
UPI	beta44	0.00	0.011	0.011 $\pm$ 0.001	—	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	—	—	0.010	0.010	—	0.0430	0.0442	0.959	94.188 $\pm$ 0.741	0.167 $\pm$ 0.001	5.812 $\pm$ 0.741
UPI	beta45	0.00	0.004	0.004 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	—	—	0.005	0.005	—	0.0355	0.0359	0.993	95.090 $\pm$ 0.684	0.142 $\pm$ 0.001	4.910 $\pm$ 0.684
UPI	beta46	0.00	-0.008	-0.008 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	—	—	-0.009	-0.009	—	0.0297	0.0310	0.994	95.090 $\pm$ 0.684	0.123 $\pm$ 0.001	4.910 $\pm$ 0.684

Table S238

Simulation 2 - Condition 94: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.6
(continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.012	-0.012 ± 0.001	—	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	—	—	-0.012	-0.012	—	0.0314	0.0337	0.974	93.086 ± 0.803	0.121 ± 0.001	6.914 ± 0.803
UPI	beta51	0.16	0.162	0.002 ± 0.001	0.013 ± 0.009	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.075 ± 0.003	0.273 ± 0.008	0.161	0.001	0.92	0.0428	0.0428	1.033	96.293 ± 0.598	0.177 ± 0.000	—
UPI	beta52	0.16	0.163	0.003 ± 0.001	0.017 ± 0.007	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.052 ± 0.002	0.229 ± 0.006	0.162	0.002	1.33	0.0355	0.0356	0.999	94.389 ± 0.728	0.143 ± 0.000	—
UPI	beta53	0.16	0.157	-0.003 ± 0.001	-0.018 ± 0.007	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.053 ± 0.002	0.231 ± 0.006	0.158	-0.002	-1.51	0.0369	0.0370	1.017	94.289 ± 0.735	0.147 ± 0.000	—
UPI	beta54	0.16	0.161	0.001 ± 0.001	0.009 ± 0.007	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.051 ± 0.002	0.227 ± 0.006	0.160	0.000	-0.10	0.0363	0.0363	1.007	95.291 ± 0.671	0.143 ± 0.000	—
UPI	beta55	0.00	0.009	0.009 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	—	—	0.010	0.010	—	0.0485	0.0495	0.974	94.188 ± 0.741	0.198 ± 0.001	5.812 ± 0.741
UPI	beta56	0.00	0.000	0.000 ± 0.001	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	—	—	0.001	0.001	—	0.0400	0.0400	0.997	95.190 ± 0.677	0.171 ± 0.001	4.810 ± 0.677
UPI	beta57	0.00	-0.009	-0.009 ± 0.001	—	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	—	—	-0.008	-0.008	—	0.0304	0.0315	0.993	93.888 ± 0.758	0.124 ± 0.001	6.112 ± 0.758
UPI	beta58	0.00	-0.007	-0.007 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	-0.008	-0.008	—	0.0284	0.0295	0.934	93.788 ± 0.764	0.110 ± 0.001	6.212 ± 0.764

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S239

Simulation 2 - Condition 95: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.8

			Mean-based									Median-based					Inference				
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE	
LSAM	beta41	0.20	0.200	0.000 ± 0.002	0.002 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.058 ± 0.002	0.240 ± 0.006	0.200	0.000	0.23	0.0501	0.0501	0.946	94.100 ± 0.745	0.178 ± 0.000	—	
LSAM	beta42	0.20	0.204	0.004 ± 0.002	0.020 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.058 ± 0.002	0.240 ± 0.006	0.204	0.004	1.79	0.0502	0.0503	0.976	94.200 ± 0.739	0.183 ± 0.000	—	
LSAM	beta43	0.20	0.200	0.000 ± 0.001	0.000 ± 0.007	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.051 ± 0.002	0.226 ± 0.006	0.201	0.001	0.44	0.0434	0.0434	0.987	94.300 ± 0.733	0.175 ± 0.000	—	
LSAM	beta44	0.00	0.004	0.004 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	—	—	0.002	0.002	—	0.0486	0.0486	0.980	94.200 ± 0.739	0.196 ± 0.001	5.800 ± 0.739	
LSAM	beta45	0.00	0.000	-0.000 ± 0.001	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	0.000	0.000	—	0.0465	0.0465	0.951	94.900 ± 0.696	0.175 ± 0.001	5.100 ± 0.696	
LSAM	beta46	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.002	-0.002	—	0.0352	0.0352	0.935	92.900 ± 0.812	0.137 ± 0.001	7.100 ± 0.812	
LSAM	beta47	0.00	-0.004	-0.004 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	—	-0.006	-0.006	—	0.0372	0.0376	0.954	93.500 ± 0.780	0.136 ± 0.001	6.500 ± 0.780
LSAM	beta51	0.16	0.161	0.001 ± 0.002	0.005 ± 0.012	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.136 ± 0.006	0.369 ± 0.010	0.161	0.001	0.61	0.0574	0.0574	0.987	94.300 ± 0.733	0.229 ± 0.001	—	
LSAM	beta52	0.16	0.160	0.000 ± 0.001	0.003 ± 0.009	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.084 ± 0.004	0.290 ± 0.008	0.160	0.000	-0.02	0.0444	0.0444	1.013	94.800 ± 0.702	0.184 ± 0.000	—	
LSAM	beta53	0.16	0.159	-0.001 ± 0.001	-0.008 ± 0.009	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.087 ± 0.004	0.295 ± 0.008	0.157	-0.003	-1.70	0.0477	0.0478	1.028	96.100 ± 0.612	0.191 ± 0.001	—	
LSAM	beta54	0.16	0.163	0.003 ± 0.001	0.017 ± 0.009	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.084 ± 0.004	0.290 ± 0.008	0.162	0.002	1.35	0.0483	0.0483	1.014	96.000 ± 0.620	0.184 ± 0.000	—	
LSAM	beta55	0.00	-0.002	-0.002 ± 0.002	—	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	—	—	-0.006	-0.006	—	0.0621	0.0623	0.959	93.600 ± 0.774	0.236 ± 0.001	6.400 ± 0.774	
LSAM	beta56	0.00	0.002	0.002 ± 0.002	—	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	—	—	0.002	0.002	—	0.0517	0.0517	0.960	93.600 ± 0.774	0.210 ± 0.001	6.400 ± 0.774	
LSAM	beta57	0.00	-0.002	-0.002 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	-0.001	-0.001	—	0.0376	0.0376	0.931	92.000 ± 0.858	0.144 ± 0.001	8.000 ± 0.858	
LSAM	beta58	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	—	—	—	-0.002	-0.002	—	0.0320	0.0320	0.997	94.700 ± 0.708	0.126 ± 0.001	5.300 ± 0.708
QML	beta41	0.20	0.199	-0.001 ± 0.001	-0.004 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.056 ± 0.002	0.236 ± 0.006	0.200	0.000	0.01	0.0489	0.0489	0.948	94.300 ± 0.733	0.175 ± 0.000	—	
QML	beta42	0.20	0.203	0.003 ± 0.001	0.016 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.056 ± 0.002	0.236 ± 0.006	0.203	0.003	1.66	0.0501	0.0503	0.977	94.500 ± 0.721	0.180 ± 0.000	—	
QML	beta43	0.20	0.199	-0.001 ± 0.001	-0.003 ± 0.007	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.050 ± 0.002	0.224 ± 0.006	0.200	0.000	0.15	0.0426	0.0426	0.990	94.300 ± 0.733	0.174 ± 0.000	—	
QML	beta44	0.00	0.000	0.000 ± 0.002	—	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	—	—	0.000	0.000	—	0.0481	0.0481	0.985	94.500 ± 0.721	0.195 ± 0.001	5.500 ± 0.721	
QML	beta45	0.00	-0.003	-0.003 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	-0.003	-0.003	—	0.0445	0.0446	0.958	94.600 ± 0.715	0.173 ± 0.001	5.400 ± 0.715	
QML	beta46	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	-0.002	-0.002	—	0.0339	0.0339	0.943	93.200 ± 0.796	0.134 ± 0.001	6.800 ± 0.796	
QML	beta47	0.00	-0.005	-0.005 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	-0.006	-0.006	—	0.0365	0.0370	0.959	93.700 ± 0.768	0.132 ± 0.001	6.300 ± 0.768	
QML	beta51	0.16	0.161	0.001 ± 0.002	0.007 ± 0.012	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.135 ± 0.006	0.368 ± 0.010	0.162	0.002	0.95	0.0565	0.0565	1.001	94.900 ± 0.696	0.231 ± 0.001	—	
QML	beta52	0.16	0.160	-0.000 ± 0.001	-0.000 ± 0.009	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.084 ± 0.004	0.289 ± 0.008	0.159	-0.001	-0.57	0.0450	0.0450	1.016	94.800 ± 0.702	0.184 ± 0.000	—	
QML	beta53	0.16	0.158	-0.002 ± 0.001	-0.012 ± 0.009	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.086 ± 0.004	0.293 ± 0.008	0.157	-0.003	-2.01	0.0478	0.0479	1.034	95.900 ± 0.627	0.190 ± 0.001	—	
QML	beta54	0.16	0.162	0.002 ± 0.001	0.015 ± 0.009	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.082 ± 0.003	0.286 ± 0.008	0.161	0.001	0.89	0.0474	0.0474	1.021	96.000 ± 0.620	0.183 ± 0.000	—	
QML	beta55	0.00	-0.003	-0.003 ± 0.002	—	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	—	—	-0.007	-0.007	—	0.0623	0.0628	0.972	94.000 ± 0.751	0.236 ± 0.001	6.000 ± 0.751	
QML	beta56	0.00	0.000	-0.000 ± 0.002	—	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	—	—	0.000	0.000	—	0.0508	0.0508	0.967	93.500 ± 0.780	0.208 ± 0.001	6.500 ± 0.780	
QML	beta57	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.002	-0.002	—	0.0362	0.0362	0.948	92.100 ± 0.853	0.141 ± 0.001	7.900 ± 0.853	
QML	beta58	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	-0.003	-0.003	—	0.0303	0.0304	1.010	95.000 ± 0.689	0.123 ± 0.001	5.000 ± 0.689	
UPI	beta41	0.20	0.200	0.000 ± 0.002	0.002 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.058 ± 0.002	0.240 ± 0.006	0.201	0.001	0.39	0.0497	0.0497	0.943	93.900 ± 0.757	0.178 ± 0.000	—	
UPI	beta42	0.20	0.204	0.004 ± 0.002	0.020 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.057 ± 0.002	0.239 ± 0.006	0.203	0.003	1.75	0.0506	0.0507	0.980	94.300 ± 0.733	0.183 ± 0.001	—	
UPI	beta43	0.20	0.200	0.000 ± 0.001	0.002 ± 0.007	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.051 ± 0.002	0.226 ± 0.006	0.201	0.001	0.52	0.0424	0.0424	0.989	94.100 ± 0.745	0.176 ± 0.000	—	
UPI	beta44	0.00	0.003	0.003 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	—	—	0.002	0.002	—	0.0486	0.0486	0.976	94.500 ± 0.721	0.198 ± 0.001	5.500 ± 0.721	
UPI	beta45	0.00	-0.001	-0.001 ± 0.002	—	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	—	—	-0.001	-0.001	—	0.0453	0.0454	0.943	93.800 ± 0.763	0.177 ± 0.001	6.200 ± 0.763	
UPI	beta46	0.00	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.001	-0.001	—	0.0350	0.0350	0.938	92.400 ± 0.838	0.140 ± 0.001	7.600 ± 0.838	

Table S239

Simulation 2 - Condition 95: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.8
(continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.003	-0.003 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.004	-0.004	—	0.0384	0.0387	0.957	94.000 ± 0.751	0.140 ± 0.001	6.000 ± 0.751
UPI	beta51	0.16	0.161	0.001 ± 0.002	0.004 ± 0.012	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.134 ± 0.006	0.366 ± 0.010	0.160	0.000	0.26	0.0559	0.0559	1.000	94.600 ± 0.715	0.230 ± 0.001	—
UPI	beta52	0.16	0.160	0.000 ± 0.001	0.002 ± 0.009	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.083 ± 0.004	0.289 ± 0.008	0.159	-0.001	-0.33	0.0436	0.0436	1.015	95.000 ± 0.689	0.184 ± 0.000	—
UPI	beta53	0.16	0.159	-0.001 ± 0.001	-0.004 ± 0.009	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.087 ± 0.004	0.295 ± 0.008	0.157	-0.003	-1.65	0.0479	0.0480	1.031	96.100 ± 0.612	0.191 ± 0.001	—
UPI	beta54	0.16	0.163	0.003 ± 0.001	0.018 ± 0.009	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.084 ± 0.004	0.290 ± 0.008	0.162	0.002	0.99	0.0480	0.0480	1.016	95.700 ± 0.641	0.184 ± 0.000	—
UPI	beta55	0.00	-0.004	-0.004 ± 0.002	—	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	—	—	-0.005	-0.005	—	0.0636	0.0639	0.956	93.900 ± 0.757	0.237 ± 0.001	6.100 ± 0.757
UPI	beta56	0.00	0.002	0.002 ± 0.002	—	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	—	—	0.002	0.002	—	0.0525	0.0526	0.961	93.900 ± 0.757	0.211 ± 0.001	6.100 ± 0.757
UPI	beta57	0.00	0.000	0.000 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	0.001	0.001	—	0.0387	0.0387	0.936	92.700 ± 0.823	0.146 ± 0.001	7.300 ± 0.823
UPI	beta58	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	—	—	0.000	0.000	—	0.0319	0.0319	0.997	95.100 ± 0.683	0.128 ± 0.001	4.900 ± 0.683

Note. MCSE = Monte Carlo Standard Error shown as \u000b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S240

Simulation 2 - Condition 96: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.8

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.201	0.001 ± 0.001	0.004 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.021 ± 0.001	0.146 ± 0.004	0.201	0.001	0.51	0.0293	0.0293	0.974	94.600 ± 0.715	0.112 ± 0.000	—
LSAM	beta42	0.20	0.200	0.000 ± 0.001	0.002 ± 0.004	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.020 ± 0.001	0.142 ± 0.004	0.200	0.000	0.22	0.0281	0.0281	1.035	95.800 ± 0.634	0.115 ± 0.000	—
LSAM	beta43	0.20	0.201	0.001 ± 0.001	0.006 ± 0.005	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.022 ± 0.001	0.149 ± 0.004	0.200	0.000	0.19	0.0294	0.0294	0.940	93.300 ± 0.791	0.110 ± 0.000	—
LSAM	beta44	0.00	0.003	0.003 ± 0.001	—	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	—	—	0.003	0.003	—	0.0308	0.0309	0.994	94.700 ± 0.708	0.122 ± 0.000	5.300 ± 0.708
LSAM	beta45	0.00	0.002	0.002 ± 0.001	—	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	—	—	0.000	0.000	—	0.0275	0.0275	1.024	95.800 ± 0.634	0.109 ± 0.000	4.200 ± 0.634
LSAM	beta46	0.00	-0.003	-0.003 ± 0.001	—	0.001 ± 0.000	0.022 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	—	—	-0.003	-0.003	—	0.0221	0.0224	0.976	94.900 ± 0.696	0.086 ± 0.000	5.100 ± 0.696
LSAM	beta47	0.00	-0.002	-0.002 ± 0.001	—	0.000 ± 0.000	0.022 ± 0.001	0.000 ± 0.000	0.022 ± 0.001	—	—	-0.003	-0.003	—	0.0219	0.0221	0.986	94.200 ± 0.739	0.085 ± 0.000	5.800 ± 0.739
LSAM	beta51	0.16	0.162	0.002 ± 0.001	0.011 ± 0.007	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.054 ± 0.002	0.231 ± 0.006	0.161	0.001	0.89	0.0366	0.0367	0.998	94.000 ± 0.751	0.145 ± 0.000	—
LSAM	beta52	0.16	0.158	-0.002 ± 0.001	-0.015 ± 0.006	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.036 ± 0.002	0.190 ± 0.005	0.157	-0.003	-1.89	0.0298	0.0299	0.975	94.200 ± 0.739	0.116 ± 0.000	—
LSAM	beta53	0.16	0.158	-0.002 ± 0.001	-0.010 ± 0.006	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.036 ± 0.002	0.191 ± 0.005	0.159	-0.001	-0.75	0.0297	0.0297	1.001	94.600 ± 0.715	0.120 ± 0.000	—
LSAM	beta54	0.16	0.160	-0.000 ± 0.001	-0.003 ± 0.006	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.036 ± 0.002	0.189 ± 0.005	0.159	-0.001	-0.92	0.0287	0.0287	0.978	94.200 ± 0.739	0.116 ± 0.000	—
LSAM	beta55	0.00	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	0.001	0.001	—	0.0369	0.0369	0.978	93.500 ± 0.780	0.149 ± 0.000	6.500 ± 0.780
LSAM	beta56	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	—	—	0.001	0.001	—	0.0327	0.0327	1.001	95.300 ± 0.669	0.132 ± 0.000	4.700 ± 0.669
LSAM	beta57	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.024 ± 0.001	0.001 ± 0.000	0.024 ± 0.001	—	—	-0.002	-0.002	—	0.0234	0.0234	0.955	94.300 ± 0.733	0.090 ± 0.000	5.700 ± 0.733
LSAM	beta58	0.00	-0.003	-0.003 ± 0.001	—	0.000 ± 0.000	0.021 ± 0.001	0.000 ± 0.000	0.021 ± 0.001	—	—	-0.002	-0.002	—	0.0217	0.0218	0.955	93.200 ± 0.796	0.079 ± 0.000	6.800 ± 0.796
QML	beta41	0.20	0.200	-0.000 ± 0.001	-0.001 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.021 ± 0.001	0.144 ± 0.004	0.200	0.000	-0.01	0.0285	0.0285	0.976	95.000 ± 0.689	0.110 ± 0.000	—
QML	beta42	0.20	0.200	-0.000 ± 0.001	-0.002 ± 0.004	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.019 ± 0.001	0.140 ± 0.004	0.200	0.000	-0.08	0.0276	0.0276	1.037	95.900 ± 0.627	0.113 ± 0.000	—
QML	beta43	0.20	0.201	0.001 ± 0.001	0.003 ± 0.005	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.022 ± 0.001	0.148 ± 0.004	0.200	0.000	-0.08	0.0299	0.0299	0.941	93.400 ± 0.785	0.109 ± 0.000	—
QML	beta44	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	-0.001	-0.001	—	0.0300	0.0300	0.997	95.400 ± 0.662	0.123 ± 0.000	4.600 ± 0.662
QML	beta45	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	—	—	-0.002	-0.002	—	0.0275	0.0276	1.033	95.500 ± 0.656	0.110 ± 0.001	4.500 ± 0.656
QML	beta46	0.00	-0.003	-0.003 ± 0.001	—	0.000 ± 0.000	0.022 ± 0.001	0.000 ± 0.000	0.022 ± 0.001	—	—	-0.003	-0.003	—	0.0218	0.0221	0.980	95.100 ± 0.683	0.085 ± 0.000	4.900 ± 0.683
QML	beta47	0.00	-0.003	-0.003 ± 0.001	—	0.000 ± 0.000	0.021 ± 0.000	0.000 ± 0.000	0.022 ± 0.001	—	—	-0.004	-0.004	—	0.0212	0.0216	0.991	94.000 ± 0.751	0.083 ± 0.000	6.000 ± 0.751
QML	beta51	0.16	0.162	0.002 ± 0.001	0.011 ± 0.007	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.053 ± 0.002	0.231 ± 0.006	0.161	0.001	0.93	0.0367	0.0367	1.004	94.200 ± 0.739	0.145 ± 0.000	—
QML	beta52	0.16	0.157	-0.003 ± 0.001	-0.017 ± 0.006	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.036 ± 0.002	0.190 ± 0.005	0.157	-0.003	-1.94	0.0298	0.0300	0.975	94.300 ± 0.733	0.116 ± 0.000	—
QML	beta53	0.16	0.158	-0.002 ± 0.001	-0.012 ± 0.006	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.036 ± 0.002	0.190 ± 0.005	0.159	-0.001	-0.75	0.0301	0.0301	1.001	94.400 ± 0.727	0.119 ± 0.000	—
QML	beta54	0.16	0.159	-0.001 ± 0.001	-0.005 ± 0.006	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.035 ± 0.002	0.187 ± 0.005	0.158	-0.002	-1.09	0.0286	0.0286	0.981	94.100 ± 0.745	0.115 ± 0.000	—
QML	beta55	0.00	0.000	0.000 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	0.000	0.000	—	0.0371	0.0371	0.984	93.500 ± 0.780	0.150 ± 0.001	6.500 ± 0.780
QML	beta56	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	—	—	-0.001	-0.001	—	0.0324	0.0324	1.006	95.100 ± 0.683	0.131 ± 0.000	4.900 ± 0.683
QML	beta57	0.00	-0.003	-0.003 ± 0.001	—	0.001 ± 0.000	0.024 ± 0.001	0.001 ± 0.000	0.024 ± 0.001	—	—	-0.003	-0.003	—	0.0229	0.0231	0.965	94.300 ± 0.733	0.089 ± 0.000	5.700 ± 0.733
QML	beta58	0.00	-0.003	-0.003 ± 0.001	—	0.000 ± 0.000	0.021 ± 0.000	0.000 ± 0.000	0.021 ± 0.001	—	—	-0.003	-0.003	—	0.0211	0.0213	0.962	93.600 ± 0.774	0.078 ± 0.000	6.400 ± 0.774
UPI	beta41	0.20	0.201	0.001 ± 0.001	0.003 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.021 ± 0.001	0.146 ± 0.004	0.201	0.001	0.50	0.0290	0.0291	0.973	94.700 ± 0.708	0.111 ± 0.000	—
UPI	beta42	0.20	0.200	0.000 ± 0.001	0.001 ± 0.004	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.020 ± 0.001	0.141 ± 0.004	0.200	0.000	0.17	0.0282	0.0282	1.037	96.100 ± 0.612	0.115 ± 0.000	—
UPI	beta43	0.20	0.202	0.002 ± 0.001	0.008 ± 0.005	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.022 ± 0.001	0.150 ± 0.004	0.201	0.001	0.51	0.0300	0.0300	0.940	93.400 ± 0.785	0.110 ± 0.000	—
UPI	beta44	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	—	—	0.001	0.001	—	0.0310	0.0310	0.996	95.400 ± 0.662	0.123 ± 0.000	4.600 ± 0.662
UPI	beta45	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	—	—	0.001	0.001	—	0.0273	0.0273	1.031	95.900 ± 0.627	0.110 ± 0.000	4.100 ± 0.627
UPI	beta46	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	—	—	-0.001	-0.001	—	0.0230	0.0231	0.978	95.000 ± 0.689	0.086 ± 0.000	5.000 ± 0.689

Table S240

Simulation 2 - Condition 96: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.8
(continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.022 ± 0.001	0.001 ± 0.000	0.022 ± 0.001	—	—	-0.001	-0.001	—	0.0225	0.0225	0.980	94.000 ± 0.751	0.086 ± 0.000	6.000 ± 0.751
UPI	beta51	0.16	0.162	0.002 ± 0.001	0.011 ± 0.007	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.053 ± 0.002	0.231 ± 0.006	0.161	0.001	0.82	0.0363	0.0364	1.001	94.200 ± 0.739	0.145 ± 0.000	—
UPI	beta52	0.16	0.158	-0.002 ± 0.001	-0.015 ± 0.006	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.036 ± 0.002	0.190 ± 0.005	0.157	-0.003	-1.74	0.0301	0.0302	0.975	93.900 ± 0.757	0.116 ± 0.000	—
UPI	beta53	0.16	0.159	-0.001 ± 0.001	-0.007 ± 0.006	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.037 ± 0.002	0.191 ± 0.005	0.159	-0.001	-0.45	0.0300	0.0300	0.999	94.800 ± 0.702	0.120 ± 0.000	—
UPI	beta54	0.16	0.159	-0.001 ± 0.001	-0.003 ± 0.006	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.035 ± 0.002	0.188 ± 0.005	0.159	-0.001	-0.86	0.0286	0.0287	0.979	93.900 ± 0.757	0.116 ± 0.000	—
UPI	beta55	0.00	0.000	0.000 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	0.001	0.001	—	0.0367	0.0367	0.975	93.300 ± 0.791	0.149 ± 0.000	6.700 ± 0.791
UPI	beta56	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	—	—	0.001	0.001	—	0.0321	0.0321	1.005	94.900 ± 0.696	0.132 ± 0.000	5.100 ± 0.696
UPI	beta57	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.024 ± 0.001	0.001 ± 0.000	0.024 ± 0.001	—	—	0.000	0.000	—	0.0227	0.0227	0.953	94.400 ± 0.727	0.090 ± 0.000	5.600 ± 0.727
UPI	beta58	0.00	-0.001	-0.001 ± 0.001	—	0.000 ± 0.000	0.021 ± 0.001	0.000 ± 0.000	0.021 ± 0.001	—	—	-0.001	-0.001	—	0.0214	0.0214	0.967	94.400 ± 0.727	0.080 ± 0.000	5.600 ± 0.727

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S241

Simulation 2 - Condition 97: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, $N = 400$, Reliability = 0.4

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Type I (%) $\pm$ MCSE
LSAM	beta41	0.20	0.256	0.056 $\pm$ 0.014	0.282 $\pm$ 0.070	0.101 $\pm$ 0.008	0.317 $\pm$ 0.013	0.104 $\pm$ 0.009	0.322 $\pm$ 0.015	2.589 $\pm$ 0.214	1.609 $\pm$ 0.075	0.237	0.037	18.72	0.2760	0.2786	1.233	98.459 $\pm$ 0.541	1.533 $\pm$ 0.051	—
LSAM	beta42	0.20	0.259	0.059 $\pm$ 0.015	0.296 $\pm$ 0.074	0.114 $\pm$ 0.009	0.337 $\pm$ 0.013	0.117 $\pm$ 0.010	0.342 $\pm$ 0.016	2.922 $\pm$ 0.239	1.709 $\pm$ 0.079	0.237	0.037	18.60	0.2859	0.2883	1.235	99.615 $\pm$ 0.272	1.631 $\pm$ 0.052	—
LSAM	beta43	0.20	0.176	-0.024 $\pm$ 0.004	-0.121 $\pm$ 0.020	0.008 $\pm$ 0.001	0.092 $\pm$ 0.003	0.009 $\pm$ 0.001	0.095 $\pm$ 0.004	0.225 $\pm$ 0.015	0.475 $\pm$ 0.019	0.172	-0.028	-13.77	0.0923	0.0963	1.109	95.954 $\pm$ 0.865	0.399 $\pm$ 0.007	—
LSAM	beta44	0.00	0.034	0.034 $\pm$ 0.006	—	0.017 $\pm$ 0.001	0.131 $\pm$ 0.006	0.018 $\pm$ 0.002	0.135 $\pm$ 0.006	—	—	0.033	0.033	—	0.1154	0.1200	1.173	97.881 $\pm$ 0.632	0.600 $\pm$ 0.018	2.119 $\pm$ 0.632
LSAM	beta45	0.00	0.015	0.015 $\pm$ 0.005	—	0.012 $\pm$ 0.001	0.112 $\pm$ 0.005	0.013 $\pm$ 0.001	0.113 $\pm$ 0.005	—	—	0.010	0.010	—	0.1035	0.1040	1.119	98.266 $\pm$ 0.573	0.490 $\pm$ 0.014	1.734 $\pm$ 0.573
LSAM	beta46	0.00	-0.035	-0.035 $\pm$ 0.006	—	0.017 $\pm$ 0.001	0.130 $\pm$ 0.006	0.018 $\pm$ 0.002	0.134 $\pm$ 0.006	—	—	-0.028	-0.028	—	0.1026	0.1065	1.207	98.844 $\pm$ 0.469	0.613 $\pm$ 0.024	1.156 $\pm$ 0.469
LSAM	beta47	0.00	-0.035	-0.035 $\pm$ 0.006	—	0.016 $\pm$ 0.001	0.128 $\pm$ 0.005	0.018 $\pm$ 0.001	0.133 $\pm$ 0.006	—	—	-0.027	-0.027	—	0.1030	0.1065	1.232	99.422 $\pm$ 0.333	0.619 $\pm$ 0.022	0.578 $\pm$ 0.333
LSAM	beta51	0.16	0.149	-0.011 $\pm$ 0.005	-0.066 $\pm$ 0.030	0.012 $\pm$ 0.001	0.108 $\pm$ 0.003	0.012 $\pm$ 0.001	0.109 $\pm$ 0.004	0.460 $\pm$ 0.028	0.678 $\pm$ 0.025	0.153	-0.007	-4.55	0.1057	0.1060	0.952	95.183 $\pm$ 0.940	0.404 $\pm$ 0.004	—
LSAM	beta52	0.16	0.240	0.080 $\pm$ 0.012	0.501 $\pm$ 0.074	0.072 $\pm$ 0.005	0.269 $\pm$ 0.010	0.078 $\pm$ 0.006	0.280 $\pm$ 0.012	3.062 $\pm$ 0.240	1.750 $\pm$ 0.078	0.233	0.073	45.92	0.2588	0.2691	1.130	98.844 $\pm$ 0.469	1.190 $\pm$ 0.034	—
LSAM	beta53	0.16	0.139	-0.021 $\pm$ 0.006	-0.131 $\pm$ 0.036	0.017 $\pm$ 0.001	0.131 $\pm$ 0.005	0.018 $\pm$ 0.001	0.132 $\pm$ 0.005	0.684 $\pm$ 0.046	0.827 $\pm$ 0.033	0.134	-0.026	-16.14	0.1282	0.1308	1.026	95.376 $\pm$ 0.922	0.526 $\pm$ 0.010	—
LSAM	beta54	0.16	0.190	0.030 $\pm$ 0.012	0.190 $\pm$ 0.073	0.072 $\pm$ 0.006	0.268 $\pm$ 0.011	0.072 $\pm$ 0.006	0.269 $\pm$ 0.012	2.826 $\pm$ 0.230	1.681 $\pm$ 0.078	0.188	0.028	17.49	0.2259	0.2276	1.085	97.110 $\pm$ 0.735	1.138 $\pm$ 0.035	—
LSAM	beta55	0.00	0.039	0.039 $\pm$ 0.006	—	0.020 $\pm$ 0.002	0.141 $\pm$ 0.006	0.021 $\pm$ 0.002	0.146 $\pm$ 0.007	—	—	0.032	0.032	—	0.1286	0.1325	1.094	99.615 $\pm$ 0.272	0.605 $\pm$ 0.015	0.385 $\pm$ 0.272
LSAM	beta56	0.00	0.000	0.000 $\pm$ 0.005	—	0.013 $\pm$ 0.001	0.116 $\pm$ 0.004	0.013 $\pm$ 0.001	0.115 $\pm$ 0.005	—	—	0.002	0.002	—	0.1050	0.1050	1.061	97.688 $\pm$ 0.660	0.481 $\pm$ 0.011	2.312 $\pm$ 0.660
LSAM	beta57	0.00	-0.043	-0.043 $\pm$ 0.005	—	0.014 $\pm$ 0.001	0.117 $\pm$ 0.005	0.016 $\pm$ 0.001	0.125 $\pm$ 0.006	—	—	-0.034	-0.034	—	0.0941	0.1000	1.124	98.844 $\pm$ 0.469	0.517 $\pm$ 0.018	1.156 $\pm$ 0.469
LSAM	beta58	0.00	-0.018	-0.018 $\pm$ 0.004	—	0.010 $\pm$ 0.001	0.101 $\pm$ 0.005	0.010 $\pm$ 0.001	0.102 $\pm$ 0.005	—	—	-0.015	-0.015	—	0.0793	0.0806	1.073	97.495 $\pm$ 0.686	0.423 $\pm$ 0.016	2.505 $\pm$ 0.686
QML	beta41	0.20	0.189	-0.011 $\pm$ 0.004	-0.056 $\pm$ 0.019	0.008 $\pm$ 0.001	0.089 $\pm$ 0.003	0.008 $\pm$ 0.001	0.089 $\pm$ 0.004	0.199 $\pm$ 0.014	0.446 $\pm$ 0.018	0.189	-0.011	-5.25	0.0889	0.0895	1.019	94.990 $\pm$ 0.958	0.354 $\pm$ 0.003	—
QML	beta42	0.20	0.179	-0.021 $\pm$ 0.004	-0.106 $\pm$ 0.020	0.008 $\pm$ 0.001	0.090 $\pm$ 0.003	0.009 $\pm$ 0.001	0.092 $\pm$ 0.003	0.214 $\pm$ 0.013	0.462 $\pm$ 0.017	0.178	-0.022	-10.96	0.0925	0.0951	1.042	95.568 $\pm$ 0.903	0.368 $\pm$ 0.004	—
QML	beta43	0.20	0.201	0.001 $\pm$ 0.003	0.005 $\pm$ 0.017	0.006 $\pm$ 0.000	0.079 $\pm$ 0.003	0.006 $\pm$ 0.000	0.079 $\pm$ 0.003	0.155 $\pm$ 0.010	0.394 $\pm$ 0.016	0.199	-0.001	-0.55	0.0777	0.0777	0.943	94.027 $\pm$ 1.040	0.291 $\pm$ 0.003	—
QML	beta44	0.00	-0.005	-0.005 $\pm$ 0.005	—	0.011 $\pm$ 0.001	0.105 $\pm$ 0.004	0.011 $\pm$ 0.001	0.105 $\pm$ 0.005	—	—	0.002	0.002	—	0.1034	0.1034	1.045	95.568 $\pm$ 0.903	0.432 $\pm$ 0.011	4.432 $\pm$ 0.903
QML	beta45	0.00	0.012	0.012 $\pm$ 0.004	—	0.008 $\pm$ 0.001	0.088 $\pm$ 0.004	0.008 $\pm$ 0.001	0.088 $\pm$ 0.004	—	—	0.009	0.009	—	0.0881	0.0886	1.000	95.183 $\pm$ 0.940	0.344 $\pm$ 0.006	4.817 $\pm$ 0.940
QML	beta46	0.00	0.022	0.022 $\pm$ 0.003	—	0.004 $\pm$ 0.000	0.059 $\pm$ 0.003	0.004 $\pm$ 0.000	0.063 $\pm$ 0.003	—	—	0.016	0.016	—	0.0580	0.0603	1.054	94.412 $\pm$ 1.008	0.245 $\pm$ 0.007	5.588 $\pm$ 1.008
QML	beta47	0.00	0.027	0.027 $\pm$ 0.003	—	0.004 $\pm$ 0.000	0.059 $\pm$ 0.002	0.004 $\pm$ 0.000	0.065 $\pm$ 0.003	—	—	0.021	0.021	—	0.0562	0.0598	1.032	94.798 $\pm$ 0.975	0.240 $\pm$ 0.007	5.202 $\pm$ 0.975
QML	beta51	0.16	0.169	0.009 $\pm$ 0.005	0.056 $\pm$ 0.028	0.011 $\pm$ 0.001	0.104 $\pm$ 0.003	0.011 $\pm$ 0.001	0.104 $\pm$ 0.004	0.422 $\pm$ 0.025	0.650 $\pm$ 0.024	0.169	0.009	5.38	0.0976	0.0980	1.037	96.532 $\pm$ 0.803	0.422 $\pm$ 0.005	—
QML	beta52	0.16	0.153	-0.007 $\pm$ 0.004	-0.042 $\pm$ 0.025	0.009 $\pm$ 0.001	0.093 $\pm$ 0.003	0.009 $\pm$ 0.001	0.093 $\pm$ 0.004	0.338 $\pm$ 0.021	0.581 $\pm$ 0.022	0.153	-0.007	-4.39	0.1002	0.1004	1.020	96.146 $\pm$ 0.845	0.371 $\pm$ 0.003	—
QML	beta53	0.16	0.159	-0.001 $\pm$ 0.004	-0.007 $\pm$ 0.026	0.009 $\pm$ 0.001	0.096 $\pm$ 0.004	0.009 $\pm$ 0.001	0.096 $\pm$ 0.004	0.359 $\pm$ 0.027	0.599 $\pm$ 0.026	0.153	-0.007	-4.52	0.0918	0.0921	0.966	94.220 $\pm$ 1.024	0.363 $\pm$ 0.004	—
QML	beta54	0.16	0.140	-0.020 $\pm$ 0.004	-0.124 $\pm$ 0.026	0.009 $\pm$ 0.001	0.096 $\pm$ 0.003	0.010 $\pm$ 0.001	0.098 $\pm$ 0.004	0.375 $\pm$ 0.024	0.612 $\pm$ 0.023	0.140	-0.020	-12.59	0.0922	0.0943	0.974	93.256 $\pm$ 1.101	0.367 $\pm$ 0.003	—
QML	beta55	0.00	0.006	0.006 $\pm$ 0.005	—	0.015 $\pm$ 0.001	0.122 $\pm$ 0.004	0.015 $\pm$ 0.001	0.122 $\pm$ 0.005	—	—	0.008	0.008	—	0.1227	0.1229	1.121	96.532 $\pm$ 0.803	0.535 $\pm$ 0.016	3.468 $\pm$ 0.803
QML	beta56	0.00	0.021	0.021 $\pm$ 0.004	—	0.010 $\pm$ 0.001	0.098 $\pm$ 0.004	0.010 $\pm$ 0.001	0.100 $\pm$ 0.004	—	—	0.017	0.017	—	0.0986	0.1001	1.032	96.339 $\pm$ 0.824	0.397 $\pm$ 0.007	3.661 $\pm$ 0.824
QML	beta57	0.00	0.013	0.013 $\pm$ 0.003	—	0.004 $\pm$ 0.000	0.066 $\pm$ 0.003	0.005 $\pm$ 0.000	0.068 $\pm$ 0.003	—	—	0.005	0.005	—	0.0636	0.0638	1.080	96.532 $\pm$ 0.803	0.281 $\pm$ 0.010	3.468 $\pm$ 0.803
QML	beta58	0.00	0.019	0.019 $\pm$ 0.002	—	0.002 $\pm$ 0.000	0.050 $\pm$ 0.002	0.003 $\pm$ 0.000	0.053 $\pm$ 0.002	—	—	0.015	0.015	—	0.0495	0.0517	0.970	93.642 $\pm$ 1.071	0.190 $\pm$ 0.004	6.358 $\pm$ 1.071
UPI	beta41	0.20	0.378	0.178 $\pm$ 0.026	0.888 $\pm$ 0.132	0.361 $\pm$ 0.031	0.601 $\pm$ 0.026	0.392 $\pm$ 0.034	0.626 $\pm$ 0.030	9.794 $\pm$ 0.847	3.130 $\pm$ 0.150	0.367	0.167	83.26	0.5515	0.5761	1.395	99.422 $\pm$ 0.333	3.285 $\pm$ 0.177	—
UPI	beta42	0.20	0.311	0.111 $\pm$ 0.021	0.556 $\pm$ 0.104	0.224 $\pm$ 0.021	0.473 $\pm$ 0.022	0.236 $\pm$ 0.023	0.486 $\pm$ 0.026	5.900 $\pm$ 0.568	2.429 $\pm$ 0.128	0.264	0.064	31.80	0.4490	0.4535	1.442	99.037 $\pm$ 0.429	2.676 $\pm$ 0.130	—
UPI	beta43	0.20	0.155	-0.045 $\pm$ 0.005	-0.224 $\pm$ 0.026	0.014 $\pm$ 0.001	0.119 $\pm$ 0.005	0.016 $\pm$ 0.001	0.127 $\pm$ 0.005	0.401 $\pm$ 0.029	0.633 $\pm$ 0.026	0.146	-0.054	-27.00	0.1192	0.1309	1.432	96.532 $\pm$ 0.803	0.665 $\pm$ 0.028	—
UPI	beta44	0.00	0.049	0.049 $\pm$ 0.011	—	0.060 $\pm$ 0.005	0.246 $\pm$ 0.011	0.063 $\pm$ 0.006	0.250 $\pm$ 0.013	—	—	0.041	0.041	—	0.2233	0.2270	1.743	99.037 $\pm$ 0.429	1.678 $\pm$ 0.102	0.963 $\pm$ 0.429
UPI	beta45	0.00	0.028	0.028 $\pm$ 0.009	—	0.040 $\pm$ 0.004	0.199 $\pm$ 0.009	0.040 $\pm$ 0.004	0.201 $\pm$ 0.010	—	—	0.023	0.023	—	0.1704	0.1720	1.649	99.422 $\pm$ 0.333	1.288 $\pm$ 0.076	0.578 $\pm$ 0.333
UPI	beta46	0.00	-0.080	-0.080 $\pm$ 0.011	—	0.058 $\pm$ 0.005	0.242 $\pm$ 0.011	0.065 $\pm$ 0.006	0.254 $\pm$ 0.013	—	—	-0.069	-0.069	—	0.2053	0.2166	1.407	99.807 $\pm$ 0.192	1.332 $\pm$ 0.074	0.193 $\pm$ 0.192

Table S241

Simulation 2 - Condition 97: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.4
(continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.053	-0.053 ± 0.008	—	0.029 ± 0.003	0.171 ± 0.008	0.032 ± 0.003	0.179 ± 0.009	—	—	-0.043	-0.043	—	0.1516	0.1575	1.518	99.229 ± 0.384	1.017 ± 0.051	0.771 ± 0.384
UPI	beta51	0.16	0.159	-0.001 ± 0.005	-0.006 ± 0.033	0.014 ± 0.001	0.119 ± 0.004	0.014 ± 0.001	0.119 ± 0.005	0.554 ± 0.039	0.744 ± 0.031	0.155	-0.005	-3.25	0.1241	0.1242	1.303	98.459 ± 0.541	0.609 ± 0.018	—
UPI	beta52	0.16	0.249	0.089 ± 0.018	0.559 ± 0.110	0.161 ± 0.014	0.401 ± 0.017	0.168 ± 0.015	0.410 ± 0.020	6.577 ± 0.582	2.565 ± 0.126	0.212	0.052	32.42	0.3797	0.3832	1.649	98.266 ± 0.573	2.591 ± 0.158	—
UPI	beta53	0.16	0.126	-0.034 ± 0.007	-0.211 ± 0.041	0.023 ± 0.002	0.150 ± 0.005	0.024 ± 0.002	0.154 ± 0.006	0.923 ± 0.064	0.961 ± 0.039	0.116	-0.044	-27.63	0.1425	0.1492	1.538	96.339 ± 0.824	0.905 ± 0.043	—
UPI	beta54	0.16	0.250	0.090 ± 0.020	0.563 ± 0.123	0.200 ± 0.018	0.447 ± 0.021	0.207 ± 0.020	0.455 ± 0.024	8.096 ± 0.782	2.845 ± 0.151	0.181	0.021	13.36	0.4148	0.4154	1.549	99.229 ± 0.384	2.713 ± 0.152	—
UPI	beta55	0.00	0.046	0.046 ± 0.010	—	0.056 ± 0.005	0.237 ± 0.011	0.058 ± 0.005	0.241 ± 0.012	—	—	0.035	0.035	—	0.1999	0.2030	1.683	99.615 ± 0.272	1.562 ± 0.095	0.385 ± 0.272
UPI	beta56	0.00	0.004	0.004 ± 0.008	—	0.029 ± 0.003	0.171 ± 0.008	0.029 ± 0.003	0.171 ± 0.009	—	—	0.006	0.006	—	0.1570	0.1571	1.523	100.000 ± 0.000	1.022 ± 0.046	0.000 ± 0.000
UPI	beta57	0.00	-0.041	-0.041 ± 0.008	—	0.030 ± 0.003	0.173 ± 0.008	0.032 ± 0.003	0.178 ± 0.009	—	—	-0.034	-0.034	—	0.1481	0.1520	1.655	100.000 ± 0.000	1.123 ± 0.076	0.000 ± 0.000
UPI	beta58	0.00	-0.037	-0.037 ± 0.007	—	0.026 ± 0.002	0.160 ± 0.008	0.027 ± 0.003	0.164 ± 0.009	—	—	-0.015	-0.015	—	0.1328	0.1336	1.526	99.807 ± 0.192	0.956 ± 0.055	0.193 ± 0.192

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S242

Simulation 2 - Condition 98: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.4

Method	Param.	True	Mean-based										Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE	
LSAM	beta41	0.20	0.251	0.051 ± 0.010	0.253 ± 0.052	0.080 ± 0.005	0.282 ± 0.009	0.082 ± 0.005	0.286 ± 0.011	2.051 ± 0.135	1.432 ± 0.054	0.257	0.057	28.73	0.2461	0.2528	1.301	99.174 ± 0.336	1.439 ± 0.051	—	
LSAM	beta42	0.20	0.310	0.110 ± 0.012	0.550 ± 0.059	0.100 ± 0.006	0.316 ± 0.010	0.112 ± 0.007	0.334 ± 0.012	2.792 ± 0.184	1.671 ± 0.062	0.281	0.081	40.63	0.2730	0.2848	1.266	99.862 ± 0.138	1.567 ± 0.050	—	
LSAM	beta43	0.20	0.175	-0.025 ± 0.002	-0.126 ± 0.011	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	0.112 ± 0.006	0.334 ± 0.011	0.175	-0.025	-12.72	0.0640	0.0689	1.180	95.868 ± 0.739	0.286 ± 0.005	—	
LSAM	beta44	0.00	0.036	0.036 ± 0.003	—	0.006 ± 0.000	0.079 ± 0.002	0.007 ± 0.000	0.086 ± 0.003	—	—	0.035	0.035	—	0.0748	0.0824	1.209	96.970 ± 0.636	0.372 ± 0.010	3.030 ± 0.636	
LSAM	beta45	0.00	0.016	0.016 ± 0.002	—	0.004 ± 0.000	0.066 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	—	—	0.015	0.015	—	0.0639	0.0655	1.218	98.072 ± 0.510	0.315 ± 0.006	1.928 ± 0.510	
LSAM	beta46	0.00	-0.033	-0.033 ± 0.004	—	0.010 ± 0.001	0.098 ± 0.003	0.011 ± 0.001	0.103 ± 0.004	—	—	-0.031	-0.031	—	0.0808	0.0865	1.309	98.898 ± 0.387	0.502 ± 0.020	1.102 ± 0.387	
LSAM	beta47	0.00	-0.053	-0.053 ± 0.004	—	0.013 ± 0.001	0.112 ± 0.004	0.015 ± 0.001	0.124 ± 0.005	—	—	-0.041	-0.041	—	0.0879	0.0970	1.227	99.174 ± 0.336	0.540 ± 0.019	0.826 ± 0.336	
LSAM	beta51	0.16	0.150	-0.010 ± 0.002	-0.065 ± 0.015	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.167 ± 0.009	0.409 ± 0.013	0.147	-0.013	-7.91	0.0685	0.0696	1.032	96.006 ± 0.727	0.262 ± 0.001	—	
LSAM	beta52	0.16	0.240	0.080 ± 0.008	0.499 ± 0.048	0.044 ± 0.003	0.209 ± 0.007	0.050 ± 0.003	0.223 ± 0.008	1.946 ± 0.127	1.395 ± 0.051	0.228	0.068	42.79	0.2053	0.2164	1.151	97.521 ± 0.577	0.941 ± 0.028	—	
LSAM	beta53	0.16	0.130	-0.030 ± 0.003	-0.188 ± 0.019	0.007 ± 0.000	0.082 ± 0.002	0.008 ± 0.000	0.087 ± 0.003	0.298 ± 0.016	0.546 ± 0.018	0.125	-0.035	-21.71	0.0800	0.0872	1.098	96.281 ± 0.702	0.353 ± 0.005	—	
LSAM	beta54	0.16	0.230	0.070 ± 0.007	0.440 ± 0.045	0.038 ± 0.002	0.194 ± 0.006	0.043 ± 0.003	0.206 ± 0.007	1.664 ± 0.107	1.290 ± 0.047	0.222	0.062	38.56	0.1712	0.1820	1.158	98.072 ± 0.510	0.881 ± 0.025	—	
LSAM	beta55	0.00	0.037	0.037 ± 0.004	—	0.010 ± 0.001	0.098 ± 0.003	0.011 ± 0.001	0.105 ± 0.004	—	—	0.032	0.032	—	0.0935	0.0989	1.073	95.455 ± 0.773	0.414 ± 0.009	4.545 ± 0.773	
LSAM	beta56	0.00	0.008	0.008 ± 0.003	—	0.006 ± 0.000	0.080 ± 0.002	0.006 ± 0.000	0.080 ± 0.003	—	—	0.008	0.008	—	0.0759	0.0763	0.985	96.006 ± 0.727	0.308 ± 0.004	3.994 ± 0.727	
LSAM	beta57	0.00	-0.043	-0.043 ± 0.003	—	0.008 ± 0.001	0.089 ± 0.003	0.010 ± 0.001	0.099 ± 0.004	—	—	-0.035	-0.035	—	0.0726	0.0805	1.113	97.796 ± 0.545	0.390 ± 0.014	2.204 ± 0.545	
LSAM	beta58	0.00	-0.033	-0.033 ± 0.003	—	0.005 ± 0.000	0.071 ± 0.003	0.006 ± 0.000	0.078 ± 0.003	—	—	-0.027	-0.027	—	0.0540	0.0602	1.120	97.796 ± 0.545	0.312 ± 0.011	2.204 ± 0.545	
QML	beta41	0.20	0.183	-0.017 ± 0.002	-0.086 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.075 ± 0.004	0.274 ± 0.008	0.182	-0.018	-8.83	0.0542	0.0570	1.057	94.628 ± 0.837	0.216 ± 0.003	—	
QML	beta42	0.20	0.182	-0.018 ± 0.002	-0.090 ± 0.010	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.082 ± 0.004	0.286 ± 0.009	0.182	-0.018	-9.08	0.0585	0.0612	1.040	94.353 ± 0.857	0.221 ± 0.002	—	
QML	beta43	0.20	0.206	0.006 ± 0.002	0.029 ± 0.008	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.051 ± 0.003	0.225 ± 0.007	0.203	0.003	1.38	0.0440	0.0441	1.030	95.179 ± 0.795	0.180 ± 0.001	—	
QML	beta44	0.00	0.002	0.002 ± 0.002	—	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	—	—	0.001	0.001	—	0.0595	0.0595	0.994	93.939 ± 0.886	0.240 ± 0.002	6.061 ± 0.886	
QML	beta45	0.00	0.014	0.014 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	—	—	0.012	0.012	—	0.0526	0.0540	1.030	94.628 ± 0.837	0.204 ± 0.002	5.372 ± 0.837	
QML	beta46	0.00	0.019	0.019 ± 0.001	—	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.016	0.016	—	0.0313	0.0351	1.047	94.628 ± 0.837	0.130 ± 0.002	5.372 ± 0.837	
QML	beta47	0.00	0.020	0.020 ± 0.001	—	0.001 ± 0.000	0.033 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	0.016	0.016	—	0.0306	0.0345	0.998	92.700 ± 0.965	0.131 ± 0.002	7.300 ± 0.965	
QML	beta51	0.16	0.166	0.006 ± 0.002	0.036 ± 0.014	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	0.143 ± 0.008	0.378 ± 0.012	0.162	0.002	1.17	0.0626	0.0626	1.059	96.143 ± 0.715	0.250 ± 0.004	—	
QML	beta52	0.16	0.143	-0.017 ± 0.002	-0.105 ± 0.013	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.131 ± 0.006	0.362 ± 0.011	0.143	-0.017	-10.74	0.0594	0.0618	1.020	94.353 ± 0.857	0.222 ± 0.001	—	
QML	beta53	0.16	0.156	-0.004 ± 0.002	-0.023 ± 0.012	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	0.105 ± 0.006	0.324 ± 0.010	0.155	-0.005	-3.29	0.0514	0.0517	1.045	95.179 ± 0.795	0.212 ± 0.001	—	
QML	beta54	0.16	0.144	-0.016 ± 0.002	-0.100 ± 0.012	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.115 ± 0.006	0.339 ± 0.010	0.149	-0.011	-6.96	0.0552	0.0563	1.091	95.730 ± 0.750	0.222 ± 0.001	—	
QML	beta55	0.00	0.003	0.003 ± 0.003	—	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.077 ± 0.003	—	—	0.001	0.001	—	0.0752	0.0752	1.007	94.215 ± 0.866	0.306 ± 0.004	5.785 ± 0.866	
QML	beta56	0.00	0.024	0.024 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	—	—	0.019	0.019	—	0.0587	0.0619	0.953	91.322 ± 1.045	0.226 ± 0.002	8.678 ± 1.045	
QML	beta57	0.00	0.016	0.016 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	0.014	0.014	—	0.0389	0.0413	1.045	93.664 ± 0.904	0.160 ± 0.003	6.336 ± 0.904	
QML	beta58	0.00	0.013	0.013 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	—	—	0.011	0.011	—	0.0281	0.0303	0.966	93.388 ± 0.922	0.115 ± 0.001	6.612 ± 0.922	
UPI	beta41	0.20	0.416	0.216 ± 0.013	1.082 ± 0.066	0.128 ± 0.008	0.357 ± 0.012	0.174 ± 0.011	0.418 ± 0.014	4.359 ± 0.267	2.088 ± 0.070	0.414	0.214	106.92	0.3693	0.4268	1.350	96.281 ± 0.702	1.891 ± 0.078	—	
UPI	beta42	0.20	0.291	0.091 ± 0.011	0.455 ± 0.057	0.093 ± 0.007	0.305 ± 0.012	0.101 ± 0.008	0.318 ± 0.014	2.526 ± 0.200	1.589 ± 0.069	0.262	0.062	31.19	0.2844	0.2911	1.289	97.658 ± 0.561	1.540 ± 0.063	—	
UPI	beta43	0.20	0.149	-0.051 ± 0.002	-0.254 ± 0.012	0.004 ± 0.000	0.064 ± 0.002	0.007 ± 0.000	0.082 ± 0.003	0.167 ± 0.010	0.409 ± 0.013	0.146	-0.054	-26.85	0.0641	0.0836	1.232	91.322 ± 1.045	0.309 ± 0.008	—	
UPI	beta44	0.00	0.054	0.054 ± 0.004	—	0.013 ± 0.001	0.114 ± 0.004	0.016 ± 0.001	0.126 ± 0.005	—	—	0.049	0.049	—	0.1054	0.1160	1.407	99.174 ± 0.336	0.628 ± 0.019	0.826 ± 0.336	
UPI	beta45	0.00	0.030	0.030 ± 0.003	—	0.007 ± 0.000	0.086 ± 0.003	0.008 ± 0.001	0.091 ± 0.003	—	—	0.023	0.023	—	0.0741	0.0776	1.569	99.449 ± 0.275	0.529 ± 0.024	0.551 ± 0.275	
UPI	beta46	0.00	-0.088	-0.088 ± 0.005	—	0.015 ± 0.001	0.124 ± 0.004	0.023 ± 0.001	0.152 ± 0.005	—	—	-0.084	-0.084	—	0.1282	0.1531	1.374	97.245 ± 0.607	0.666 ± 0.027	2.755 ± 0.607	

Table S242

Simulation 2 - Condition 98: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.4
(continued)

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.048	-0.048 ± 0.004	—	0.010 ± 0.001	0.098 ± 0.004	0.012 ± 0.001	0.109 ± 0.005	—	—	-0.038	-0.038	—	0.0884	0.0961	1.311	98.347 ± 0.473	0.503 ± 0.019	1.653 ± 0.473
UPI	beta51	0.16	0.154	-0.006 ± 0.003	-0.039 ± 0.016	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.186 ± 0.010	0.431 ± 0.014	0.152	-0.008	-4.86	0.0741	0.0745	1.232	97.658 ± 0.561	0.332 ± 0.005	—
UPI	beta52	0.16	0.285	0.125 ± 0.013	0.783 ± 0.081	0.122 ± 0.009	0.350 ± 0.012	0.138 ± 0.010	0.371 ± 0.014	5.388 ± 0.372	2.321 ± 0.089	0.241	0.081	50.83	0.3029	0.3137	1.443	98.623 ± 0.433	1.979 ± 0.101	—
UPI	beta53	0.16	0.112	-0.048 ± 0.003	-0.300 ± 0.020	0.007 ± 0.000	0.086 ± 0.003	0.010 ± 0.001	0.098 ± 0.003	0.376 ± 0.023	0.613 ± 0.020	0.107	-0.053	-33.25	0.0822	0.0979	1.266	94.628 ± 0.837	0.425 ± 0.012	—
UPI	beta54	0.16	0.264	0.104 ± 0.014	0.649 ± 0.089	0.147 ± 0.010	0.383 ± 0.013	0.157 ± 0.012	0.397 ± 0.016	6.148 ± 0.453	2.480 ± 0.101	0.220	0.060	37.30	0.3331	0.3384	1.390	97.383 ± 0.592	2.088 ± 0.112	—
UPI	beta55	0.00	0.055	0.055 ± 0.005	—	0.020 ± 0.002	0.142 ± 0.005	0.023 ± 0.002	0.152 ± 0.006	—	—	0.046	0.046	—	0.1235	0.1317	1.398	99.311 ± 0.307	0.776 ± 0.032	0.689 ± 0.307
UPI	beta56	0.00	0.007	0.007 ± 0.003	—	0.008 ± 0.000	0.088 ± 0.003	0.008 ± 0.000	0.088 ± 0.003	—	—	0.011	0.011	—	0.0885	0.0891	1.287	99.587 ± 0.238	0.444 ± 0.012	0.413 ± 0.238
UPI	beta57	0.00	-0.057	-0.057 ± 0.005	—	0.017 ± 0.001	0.131 ± 0.005	0.020 ± 0.001	0.143 ± 0.005	—	—	-0.040	-0.040	—	0.1082	0.1153	1.445	99.174 ± 0.336	0.740 ± 0.036	0.826 ± 0.336
UPI	beta58	0.00	-0.040	-0.040 ± 0.004	—	0.014 ± 0.001	0.119 ± 0.004	0.016 ± 0.001	0.125 ± 0.005	—	—	-0.026	-0.026	—	0.1002	0.1035	1.424	99.174 ± 0.336	0.662 ± 0.038	0.826 ± 0.336

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S243

Simulation 2 - Condition 99: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, $N = 400$, Reliability = 0.6

Method	Param.	True	Mean-based										Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE	
LSAM	beta41	0.20	0.211	0.011 ± 0.004	0.057 ± 0.022	0.017 ± 0.001	0.130 ± 0.003	0.017 ± 0.001	0.130 ± 0.004	0.425 ± 0.022	0.652 ± 0.020	0.210	0.010	5.16	0.1258	0.1262	0.968	95.526 ± 0.691	0.493 ± 0.005	—	
LSAM	beta42	0.20	0.224	0.024 ± 0.005	0.119 ± 0.023	0.018 ± 0.001	0.136 ± 0.004	0.019 ± 0.001	0.138 ± 0.005	0.473 ± 0.028	0.688 ± 0.023	0.219	0.019	9.38	0.1291	0.1305	0.990	96.197 ± 0.640	0.526 ± 0.005	—	
LSAM	beta43	0.20	0.195	-0.005 ± 0.002	-0.026 ± 0.010	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.085 ± 0.004	0.292 ± 0.009	0.193	-0.007	-3.72	0.0569	0.0574	1.020	95.302 ± 0.708	0.233 ± 0.001	—	
LSAM	beta44	0.00	0.008	0.008 ± 0.002	—	0.005 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	—	—	0.010	0.010	—	0.0654	0.0661	0.960	93.848 ± 0.804	0.279 ± 0.004	6.152 ± 0.804	
LSAM	beta45	0.00	0.003	0.003 ± 0.002	—	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	—	—	0.002	0.002	—	0.0651	0.0651	0.969	94.631 ± 0.754	0.261 ± 0.003	5.369 ± 0.754	
LSAM	beta46	0.00	-0.008	-0.008 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.002	0.003 ± 0.000	0.053 ± 0.002	—	—	-0.008	-0.008	—	0.0449	0.0456	0.896	94.743 ± 0.746	0.186 ± 0.003	5.257 ± 0.746	
LSAM	beta47	0.00	-0.011	-0.011 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.002	0.003 ± 0.000	0.052 ± 0.002	—	—	-0.009	-0.009	—	0.0451	0.0460	0.974	95.190 ± 0.716	0.193 ± 0.003	4.810 ± 0.716	
LSAM	beta51	0.16	0.161	0.001 ± 0.002	0.004 ± 0.015	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.209 ± 0.011	0.457 ± 0.014	0.158	-0.002	-1.35	0.0728	0.0728	0.996	94.631 ± 0.754	0.286 ± 0.001	—	
LSAM	beta52	0.16	0.178	0.018 ± 0.004	0.114 ± 0.026	0.016 ± 0.001	0.125 ± 0.003	0.016 ± 0.001	0.127 ± 0.004	0.627 ± 0.032	0.792 ± 0.024	0.176	0.016	10.05	0.1212	0.1222	0.947	93.177 ± 0.843	0.466 ± 0.004	—	
LSAM	beta53	0.16	0.155	-0.005 ± 0.003	-0.034 ± 0.017	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.081 ± 0.003	0.257 ± 0.015	0.507 ± 0.017	0.153	-0.007	-4.08	0.0807	0.0809	0.980	94.743 ± 0.746	0.311 ± 0.002	—	
LSAM	beta54	0.16	0.181	0.021 ± 0.004	0.134 ± 0.025	0.014 ± 0.001	0.119 ± 0.003	0.015 ± 0.001	0.121 ± 0.004	0.572 ± 0.028	0.756 ± 0.022	0.182	0.022	13.84	0.1196	0.1217	0.993	95.078 ± 0.723	0.464 ± 0.004	—	
LSAM	beta55	0.00	0.014	0.014 ± 0.003	—	0.009 ± 0.000	0.093 ± 0.003	0.009 ± 0.001	0.095 ± 0.003	—	—	0.011	0.011	—	0.0869	0.0876	0.948	94.407 ± 0.769	0.347 ± 0.003	5.593 ± 0.769	
LSAM	beta56	0.00	0.001	0.001 ± 0.003	—	0.007 ± 0.000	0.081 ± 0.002	0.007 ± 0.000	0.081 ± 0.003	—	—	0.001	0.001	—	0.0718	0.0718	0.927	94.183 ± 0.783	0.294 ± 0.003	5.817 ± 0.783	
LSAM	beta57	0.00	-0.013	-0.013 ± 0.002	—	0.003 ± 0.000	0.056 ± 0.002	0.003 ± 0.000	0.057 ± 0.002	—	—	-0.010	-0.010	—	0.0515	0.0525	0.895	94.072 ± 0.790	0.195 ± 0.003	5.928 ± 0.790	
LSAM	beta58	0.00	-0.009	-0.009 ± 0.002	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	-0.009	-0.009	—	0.0421	0.0430	0.940	93.289 ± 0.837	0.166 ± 0.003	6.711 ± 0.837	
QML	beta41	0.20	0.184	-0.016 ± 0.002	-0.081 ± 0.012	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.133 ± 0.006	0.365 ± 0.010	0.187	-0.013	-6.58	0.0737	0.0749	0.979	94.407 ± 0.769	0.273 ± 0.001	—	
QML	beta42	0.20	0.190	-0.010 ± 0.002	-0.052 ± 0.012	0.005 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.074 ± 0.002	0.138 ± 0.007	0.372 ± 0.011	0.187	-0.013	-6.46	0.0732	0.0743	0.979	93.736 ± 0.810	0.283 ± 0.001	—	
QML	beta43	0.20	0.203	0.003 ± 0.002	0.015 ± 0.009	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.072 ± 0.003	0.268 ± 0.008	0.201	0.001	0.27	0.0528	0.0528	1.011	95.414 ± 0.700	0.212 ± 0.001	—	
QML	beta44	0.00	-0.003	-0.003 ± 0.002	—	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	—	—	-0.003	-0.003	—	0.0611	0.0612	0.962	93.512 ± 0.824	0.244 ± 0.002	6.488 ± 0.824	
QML	beta45	0.00	0.002	0.002 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	—	—	0.001	0.001	—	0.0596	0.0596	0.977	93.289 ± 0.837	0.221 ± 0.002	6.711 ± 0.837	
QML	beta46	0.00	0.014	0.014 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	0.013	0.013	—	0.0366	0.0387	0.872	90.940 ± 0.960	0.133 ± 0.002	9.060 ± 0.960	
QML	beta47	0.00	0.013	0.013 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	0.011	0.011	—	0.0347	0.0365	0.954	93.400 ± 0.830	0.135 ± 0.002	6.600 ± 0.830	
QML	beta51	0.16	0.165	0.005 ± 0.002	0.030 ± 0.015	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.201 ± 0.010	0.448 ± 0.014	0.161	0.001	0.49	0.0685	0.0685	1.025	95.638 ± 0.683	0.288 ± 0.001	—	
QML	beta52	0.16	0.149	-0.011 ± 0.003	-0.067 ± 0.016	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	0.226 ± 0.011	0.476 ± 0.014	0.151	-0.009	-5.46	0.0724	0.0730	0.956	92.506 ± 0.881	0.283 ± 0.001	—	
QML	beta53	0.16	0.159	-0.001 ± 0.002	-0.006 ± 0.014	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.168 ± 0.009	0.410 ± 0.013	0.160	0.000	-0.16	0.0663	0.0663	0.993	95.414 ± 0.700	0.256 ± 0.001	—	
QML	beta54	0.16	0.152	-0.008 ± 0.002	-0.052 ± 0.015	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.207 ± 0.010	0.455 ± 0.013	0.154	-0.006	-3.88	0.0730	0.0732	0.999	94.743 ± 0.746	0.284 ± 0.001	—	
QML	beta55	0.00	0.004	0.004 ± 0.003	—	0.007 ± 0.000	0.082 ± 0.002	0.007 ± 0.000	0.082 ± 0.003	—	—	0.002	0.002	—	0.0822	0.0822	0.977	95.414 ± 0.700	0.314 ± 0.003	4.586 ± 0.700	
QML	beta56	0.00	0.011	0.011 ± 0.002	—	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	—	—	0.011	0.011	—	0.0648	0.0657	0.945	93.289 ± 0.837	0.253 ± 0.002	6.711 ± 0.837	
QML	beta57	0.00	0.007	0.007 ± 0.001	—	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	—	—	0.006	0.006	—	0.0395	0.0400	0.935	93.289 ± 0.837	0.156 ± 0.002	6.711 ± 0.837	
QML	beta58	0.00	0.009	0.009 ± 0.001	—	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	0.006	0.006	—	0.0334	0.0339	0.920	90.716 ± 0.971	0.124 ± 0.001	9.284 ± 0.971	
UPI	beta41	0.20	0.217	0.017 ± 0.004	0.084 ± 0.022	0.018 ± 0.001	0.132 ± 0.003	0.018 ± 0.001	0.133 ± 0.004	0.445 ± 0.023	0.667 ± 0.021	0.219	0.019	9.40	0.1364	0.1377	1.174	97.763 ± 0.495	0.609 ± 0.014	—	
UPI	beta42	0.20	0.241	0.041 ± 0.005	0.204 ± 0.024	0.021 ± 0.001	0.146 ± 0.004	0.023 ± 0.001	0.151 ± 0.005	0.572 ± 0.034	0.756 ± 0.025	0.236	0.036	18.13	0.1420	0.1465	1.178	97.092 ± 0.562	0.672 ± 0.018	—	
UPI	beta43	0.20	0.188	-0.012 ± 0.002	-0.059 ± 0.009	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.083 ± 0.004	0.289 ± 0.008	0.184	-0.016	-7.77	0.0598	0.0617	1.063	95.302 ± 0.708	0.236 ± 0.003	—	
UPI	beta44	0.00	0.012	0.012 ± 0.003	—	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.076 ± 0.003	—	—	0.013	0.013	—	0.0723	0.0734	1.183	95.973 ± 0.657	0.348 ± 0.013	4.027 ± 0.657	
UPI	beta45	0.00	0.002	0.002 ± 0.002	—	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	—	—	0.002	0.002	—	0.0652	0.0652	1.121	96.980 ± 0.572	0.304 ± 0.006	3.020 ± 0.572	
UPI	beta46	0.00	-0.012	-0.012 ± 0.002	—	0.003 ± 0.000	0.054 ± 0.002	0.003 ± 0.000	0.056 ± 0.002	—	—	-0.009	-0.009	—	0.0483	0.0491	1.111	96.756 ± 0.593	0.237 ± 0.008	3.244 ± 0.593	

Table S243

Simulation 2 - Condition 99: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.6
(continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.020	-0.020 ± 0.002	—	0.003 ± 0.000	0.056 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	—	—	-0.013	-0.013	—	0.0499	0.0516	1.160	97.315 ± 0.541	0.256 ± 0.009	2.685 ± 0.541
UPI	beta51	0.16	0.162	0.002 ± 0.002	0.013 ± 0.015	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.198 ± 0.010	0.445 ± 0.013	0.158	-0.002	-1.25	0.0675	0.0675	1.029	95.749 ± 0.675	0.287 ± 0.001	—
UPI	beta52	0.16	0.178	0.018 ± 0.004	0.112 ± 0.026	0.015 ± 0.001	0.124 ± 0.003	0.016 ± 0.001	0.125 ± 0.004	0.612 ± 0.032	0.783 ± 0.024	0.174	0.014	9.00	0.1259	0.1267	1.014	95.190 ± 0.716	0.493 ± 0.006	—
UPI	beta53	0.16	0.150	-0.010 ± 0.003	-0.063 ± 0.016	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.078 ± 0.003	0.237 ± 0.014	0.487 ± 0.016	0.150	-0.010	-6.42	0.0751	0.0758	1.014	94.966 ± 0.731	0.307 ± 0.002	—
UPI	beta54	0.16	0.191	0.031 ± 0.004	0.194 ± 0.027	0.016 ± 0.001	0.127 ± 0.003	0.017 ± 0.001	0.131 ± 0.004	0.666 ± 0.036	0.816 ± 0.026	0.195	0.035	21.68	0.1250	0.1297	1.079	96.421 ± 0.621	0.537 ± 0.008	—
UPI	beta55	0.00	0.017	0.017 ± 0.003	—	0.009 ± 0.001	0.096 ± 0.003	0.009 ± 0.001	0.097 ± 0.003	—	—	0.015	0.015	—	0.0859	0.0871	1.037	96.421 ± 0.621	0.389 ± 0.006	3.579 ± 0.621
UPI	beta56	0.00	0.001	0.001 ± 0.003	—	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.079 ± 0.003	—	—	0.001	0.001	—	0.0695	0.0696	1.006	95.973 ± 0.657	0.310 ± 0.004	4.027 ± 0.657
UPI	beta57	0.00	-0.013	-0.013 ± 0.002	—	0.003 ± 0.000	0.056 ± 0.002	0.003 ± 0.000	0.058 ± 0.002	—	—	-0.011	-0.011	—	0.0499	0.0512	0.975	96.085 ± 0.649	0.216 ± 0.004	3.915 ± 0.649
UPI	beta58	0.00	-0.014	-0.014 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.001	0.003 ± 0.000	0.051 ± 0.002	—	—	-0.013	-0.013	—	0.0427	0.0446	1.026	94.519 ± 0.761	0.197 ± 0.005	5.481 ± 0.761

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S244

Simulation 2 - Condition 100: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.6

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.218	0.018 ± 0.002	0.090 ± 0.011	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.127 ± 0.006	0.357 ± 0.010	0.222	0.022	10.98	0.0680	0.0715	1.012	94.479 ± 0.730	0.274 ± 0.001	—
LSAM	beta42	0.20	0.208	0.008 ± 0.002	0.041 ± 0.012	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.074 ± 0.002	0.136 ± 0.006	0.369 ± 0.010	0.205	0.005	2.56	0.0715	0.0717	1.016	95.501 ± 0.663	0.293 ± 0.002	—
LSAM	beta43	0.20	0.195	-0.005 ± 0.001	-0.023 ± 0.006	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.032 ± 0.001	0.180 ± 0.005	0.195	-0.005	-2.73	0.0339	0.0343	1.007	94.376 ± 0.737	0.141 ± 0.000	—
LSAM	beta44	0.00	0.008	0.008 ± 0.001	—	0.001 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	0.008	0.008	—	0.0367	0.0376	0.975	94.581 ± 0.724	0.148 ± 0.001	5.419 ± 0.724
LSAM	beta45	0.00	0.002	0.002 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	0.001	0.001	—	0.0369	0.0369	0.967	93.354 ± 0.796	0.144 ± 0.001	6.646 ± 0.796
LSAM	beta46	0.00	-0.009	-0.009 ± 0.001	—	0.001 ± 0.000	0.024 ± 0.001	0.001 ± 0.000	0.026 ± 0.001	—	—	-0.008	-0.008	—	0.0232	0.0246	0.976	93.149 ± 0.808	0.093 ± 0.001	6.851 ± 0.808
LSAM	beta47	0.00	-0.006	-0.006 ± 0.001	—	0.001 ± 0.000	0.026 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	—	—	-0.006	-0.006	—	0.0241	0.0249	0.955	92.638 ± 0.835	0.097 ± 0.001	7.362 ± 0.835
LSAM	beta51	0.16	0.159	-0.001 ± 0.001	-0.003 ± 0.009	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.085 ± 0.004	0.291 ± 0.008	0.160	0.000	-0.16	0.0457	0.0457	0.967	93.763 ± 0.773	0.177 ± 0.000	—
LSAM	beta52	0.16	0.174	0.014 ± 0.002	0.089 ± 0.014	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	0.188 ± 0.008	0.434 ± 0.012	0.174	0.014	8.79	0.0634	0.0650	1.004	95.399 ± 0.670	0.268 ± 0.001	—
LSAM	beta53	0.16	0.152	-0.008 ± 0.002	-0.053 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.103 ± 0.005	0.321 ± 0.009	0.151	-0.009	-5.43	0.0516	0.0523	0.941	92.638 ± 0.835	0.187 ± 0.001	—
LSAM	beta54	0.16	0.176	0.016 ± 0.002	0.099 ± 0.013	0.004 ± 0.000	0.065 ± 0.002	0.005 ± 0.000	0.067 ± 0.002	0.177 ± 0.009	0.421 ± 0.012	0.176	0.016	10.13	0.0646	0.0666	1.037	95.808 ± 0.641	0.266 ± 0.001	—
LSAM	beta55	0.00	0.010	0.010 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	—	—	0.011	0.011	—	0.0509	0.0521	0.989	94.479 ± 0.730	0.197 ± 0.001	5.521 ± 0.730
LSAM	beta56	0.00	0.004	0.004 ± 0.001	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	—	—	0.003	0.003	—	0.0415	0.0417	0.970	93.558 ± 0.785	0.169 ± 0.001	6.442 ± 0.785
LSAM	beta57	0.00	-0.009	-0.009 ± 0.001	—	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	-0.008	-0.008	—	0.0255	0.0268	0.961	93.763 ± 0.773	0.102 ± 0.001	6.237 ± 0.773
LSAM	beta58	0.00	-0.007	-0.007 ± 0.001	—	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.024 ± 0.001	—	—	-0.006	-0.006	—	0.0203	0.0211	0.988	94.581 ± 0.724	0.088 ± 0.001	5.419 ± 0.724
QML	beta41	0.20	0.188	-0.012 ± 0.001	-0.060 ± 0.007	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.049 ± 0.002	0.220 ± 0.006	0.191	-0.009	-4.36	0.0410	0.0419	1.015	93.763 ± 0.773	0.169 ± 0.000	—
QML	beta42	0.20	0.182	-0.018 ± 0.001	-0.088 ± 0.007	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.056 ± 0.002	0.237 ± 0.006	0.183	-0.017	-8.74	0.0450	0.0483	1.014	93.456 ± 0.791	0.175 ± 0.000	—
QML	beta43	0.20	0.202	0.002 ± 0.001	0.012 ± 0.005	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.028 ± 0.001	0.168 ± 0.005	0.202	0.002	0.89	0.0321	0.0322	1.009	95.399 ± 0.670	0.133 ± 0.000	—
QML	beta44	0.00	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	-0.001	-0.001	—	0.0379	0.0379	0.967	94.581 ± 0.724	0.149 ± 0.001	5.419 ± 0.724
QML	beta45	0.00	0.003	0.003 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	0.002	0.002	—	0.0351	0.0351	0.962	93.661 ± 0.779	0.136 ± 0.001	6.339 ± 0.779
QML	beta46	0.00	0.011	0.011 ± 0.001	—	0.000 ± 0.000	0.021 ± 0.000	0.001 ± 0.000	0.024 ± 0.001	—	—	0.011	0.011	—	0.0211	0.0239	0.957	91.207 ± 0.906	0.078 ± 0.001	8.793 ± 0.906
QML	beta47	0.00	0.013	0.013 ± 0.001	—	0.000 ± 0.000	0.022 ± 0.000	0.001 ± 0.000	0.025 ± 0.001	—	—	0.013	0.013	—	0.0221	0.0254	0.938	88.241 ± 1.030	0.079 ± 0.001	11.759 ± 1.030
QML	beta51	0.16	0.162	0.002 ± 0.001	0.014 ± 0.009	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.083 ± 0.004	0.289 ± 0.008	0.162	0.002	1.22	0.0455	0.0456	0.977	94.479 ± 0.730	0.177 ± 0.000	—
QML	beta52	0.16	0.149	-0.011 ± 0.001	-0.069 ± 0.009	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.080 ± 0.004	0.284 ± 0.008	0.150	-0.010	-6.51	0.0426	0.0439	1.013	93.967 ± 0.761	0.175 ± 0.000	—
QML	beta53	0.16	0.155	-0.005 ± 0.001	-0.030 ± 0.009	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.074 ± 0.003	0.272 ± 0.007	0.155	-0.005	-3.22	0.0432	0.0435	0.935	92.229 ± 0.856	0.159 ± 0.001	—
QML	beta54	0.16	0.150	-0.010 ± 0.001	-0.061 ± 0.009	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.074 ± 0.003	0.273 ± 0.007	0.151	-0.009	-5.67	0.0432	0.0441	1.052	95.706 ± 0.648	0.175 ± 0.000	—
QML	beta55	0.00	0.002	0.002 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	—	—	0.003	0.003	—	0.0490	0.0491	0.995	93.967 ± 0.761	0.191 ± 0.001	6.033 ± 0.761
QML	beta56	0.00	0.014	0.014 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	0.014	0.014	—	0.0388	0.0411	0.983	92.843 ± 0.824	0.154 ± 0.001	7.157 ± 0.824
QML	beta57	0.00	0.009	0.009 ± 0.001	—	0.001 ± 0.000	0.024 ± 0.001	0.001 ± 0.000	0.025 ± 0.001	—	—	0.008	0.008	—	0.0229	0.0242	0.982	92.127 ± 0.861	0.092 ± 0.001	7.873 ± 0.861
QML	beta58	0.00	0.008	0.008 ± 0.001	—	0.000 ± 0.000	0.019 ± 0.000	0.000 ± 0.000	0.021 ± 0.001	—	—	0.008	0.008	—	0.0177	0.0195	0.994	93.661 ± 0.779	0.074 ± 0.001	6.339 ± 0.779
UPI	beta41	0.20	0.222	0.022 ± 0.002	0.110 ± 0.012	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	0.155 ± 0.007	0.394 ± 0.011	0.222	0.022	11.13	0.0740	0.0773	1.042	96.524 ± 0.586	0.309 ± 0.003	—
UPI	beta42	0.20	0.218	0.018 ± 0.003	0.088 ± 0.013	0.006 ± 0.000	0.079 ± 0.002	0.007 ± 0.000	0.081 ± 0.002	0.163 ± 0.008	0.404 ± 0.011	0.218	0.018	9.12	0.0766	0.0787	1.072	96.115 ± 0.618	0.331 ± 0.003	—
UPI	beta43	0.20	0.191	-0.009 ± 0.001	-0.047 ± 0.006	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.032 ± 0.001	0.180 ± 0.005	0.190	-0.010	-5.24	0.0340	0.0356	1.001	93.252 ± 0.802	0.137 ± 0.001	—
UPI	beta44	0.00	0.009	0.009 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	0.007	0.007	—	0.0385	0.0392	1.007	95.603 ± 0.656	0.159 ± 0.002	4.397 ± 0.656

Table S244
Simulation 2 - Condition 100: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.6 (continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta45	0.00	0.003	0.003 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	0.002	0.002	—	0.0378	0.0379	0.994	94.785 ± 0.711	0.152 ± 0.002	5.215 ± 0.711
UPI	beta46	0.00	-0.012	-0.012 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	-0.009	-0.009	—	0.0236	0.0252	0.981	95.092 ± 0.691	0.107 ± 0.002	4.908 ± 0.691
UPI	beta47	0.00	-0.010	-0.010 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	-0.009	-0.009	—	0.0259	0.0274	1.002	94.990 ± 0.698	0.113 ± 0.002	5.010 ± 0.698
UPI	beta51	0.16	0.160	-0.000 ± 0.001	-0.000 ± 0.009	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.083 ± 0.004	0.289 ± 0.008	0.159	-0.001	-0.42	0.0451	0.0451	0.977	94.274 ± 0.743	0.177 ± 0.000	—
UPI	beta52	0.16	0.177	0.017 ± 0.002	0.105 ± 0.014	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.211 ± 0.010	0.459 ± 0.013	0.175	0.015	9.17	0.0707	0.0722	1.014	96.115 ± 0.618	0.284 ± 0.002	—
UPI	beta53	0.16	0.148	-0.012 ± 0.002	-0.075 ± 0.010	0.002 ± 0.000	0.049 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.101 ± 0.005	0.317 ± 0.009	0.148	-0.012	-7.26	0.0493	0.0507	0.951	93.456 ± 0.791	0.184 ± 0.001	—
UPI	beta54	0.16	0.182	0.022 ± 0.002	0.138 ± 0.015	0.005 ± 0.000	0.073 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	0.227 ± 0.012	0.477 ± 0.015	0.178	0.018	11.25	0.0667	0.0691	1.032	95.808 ± 0.641	0.295 ± 0.002	—
UPI	beta55	0.00	0.012	0.012 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	—	—	0.012	0.012	—	0.0528	0.0542	1.002	94.990 ± 0.698	0.208 ± 0.002	5.010 ± 0.698
UPI	beta56	0.00	0.004	0.004 ± 0.001	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	—	—	0.005	0.005	—	0.0417	0.0420	0.990	94.888 ± 0.704	0.172 ± 0.001	5.112 ± 0.704
UPI	beta57	0.00	-0.010	-0.010 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	-0.009	-0.009	—	0.0261	0.0277	0.965	93.354 ± 0.796	0.111 ± 0.001	6.646 ± 0.796
UPI	beta58	0.00	-0.010	-0.010 ± 0.001	—	0.001 ± 0.000	0.026 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	-0.008	-0.008	—	0.0210	0.0223	0.974	95.808 ± 0.641	0.099 ± 0.001	4.192 ± 0.641

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u002014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S245

Simulation 2 - Condition 101: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.8

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.202	0.002 ± 0.002	0.008 ± 0.012	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	0.153 ± 0.007	0.391 ± 0.011	0.202	0.002	1.20	0.0785	0.0785	0.979	94.737 ± 0.710	0.300 ± 0.001	—
LSAM	beta42	0.20	0.204	0.004 ± 0.003	0.022 ± 0.013	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	0.157 ± 0.007	0.397 ± 0.011	0.203	0.003	1.73	0.0798	0.0799	1.018	95.850 ± 0.634	0.316 ± 0.001	—
LSAM	beta43	0.20	0.200	-0.000 ± 0.002	-0.001 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.058 ± 0.003	0.242 ± 0.007	0.199	-0.001	-0.32	0.0477	0.0477	0.953	94.534 ± 0.723	0.181 ± 0.001	—
LSAM	beta44	0.00	-0.002	-0.002 ± 0.002	—	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	—	—	0.000	0.000	—	0.0520	0.0520	0.904	92.308 ± 0.848	0.193 ± 0.002	7.692 ± 0.848
LSAM	beta45	0.00	0.001	0.001 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.002	—	—	0.001	0.001	—	0.0459	0.0459	0.953	92.915 ± 0.816	0.184 ± 0.002	7.085 ± 0.816
LSAM	beta46	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	-0.002	-0.002	—	0.0272	0.0272	0.911	92.713 ± 0.827	0.110 ± 0.001	7.287 ± 0.827
LSAM	beta47	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	—	—	-0.001	-0.001	—	0.0292	0.0293	0.904	91.194 ± 0.902	0.112 ± 0.001	8.806 ± 0.902
LSAM	beta51	0.16	0.161	0.001 ± 0.002	0.009 ± 0.012	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.142 ± 0.006	0.377 ± 0.010	0.161	0.001	0.69	0.0605	0.0605	0.967	94.433 ± 0.729	0.229 ± 0.001	—
LSAM	beta52	0.16	0.163	0.003 ± 0.003	0.019 ± 0.016	0.006 ± 0.000	0.080 ± 0.002	0.006 ± 0.000	0.080 ± 0.002	0.252 ± 0.012	0.502 ± 0.014	0.162	0.002	1.08	0.0778	0.0778	0.961	94.028 ± 0.754	0.303 ± 0.001	—
LSAM	beta53	0.16	0.159	-0.001 ± 0.002	-0.006 ± 0.012	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.144 ± 0.006	0.379 ± 0.010	0.159	-0.001	-0.36	0.0608	0.0608	0.972	94.231 ± 0.742	0.231 ± 0.001	—
LSAM	beta54	0.16	0.162	0.002 ± 0.003	0.010 ± 0.016	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	0.245 ± 0.011	0.495 ± 0.014	0.158	-0.002	-1.37	0.0819	0.0819	0.965	95.142 ± 0.684	0.300 ± 0.001	—
LSAM	beta55	0.00	0.004	0.004 ± 0.002	—	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	—	—	0.002	0.002	—	0.0664	0.0665	0.922	92.308 ± 0.848	0.246 ± 0.002	7.692 ± 0.848
LSAM	beta56	0.00	0.001	0.001 ± 0.002	—	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	—	—	0.001	0.001	—	0.0549	0.0549	0.931	91.903 ± 0.868	0.214 ± 0.002	8.097 ± 0.868
LSAM	beta57	0.00	-0.003	-0.003 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	-0.002	-0.002	—	0.0310	0.0311	0.899	92.611 ± 0.832	0.122 ± 0.001	7.389 ± 0.832
LSAM	beta58	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	-0.001	-0.001	—	0.0267	0.0267	0.880	92.814 ± 0.822	0.103 ± 0.001	7.186 ± 0.822
QML	beta41	0.20	0.190	-0.010 ± 0.002	-0.048 ± 0.010	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.101 ± 0.004	0.319 ± 0.008	0.191	-0.009	-4.47	0.0640	0.0646	0.993	94.028 ± 0.754	0.245 ± 0.002	—
QML	beta42	0.20	0.193	-0.007 ± 0.002	-0.033 ± 0.010	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.099 ± 0.004	0.314 ± 0.009	0.193	-0.007	-3.60	0.0638	0.0642	1.044	95.344 ± 0.670	0.256 ± 0.003	—
QML	beta43	0.20	0.202	0.002 ± 0.002	0.012 ± 0.008	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.056 ± 0.003	0.236 ± 0.007	0.201	0.001	0.71	0.0473	0.0473	0.989	94.636 ± 0.717	0.183 ± 0.007	—
QML	beta44	0.00	-0.005	-0.005 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	—	—	-0.002	-0.002	—	0.0503	0.0504	0.909	91.903 ± 0.868	0.186 ± 0.001	8.097 ± 0.868
QML	beta45	0.00	0.000	0.000 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	0.000	0.000	—	0.0432	0.0432	0.968	93.421 ± 0.789	0.175 ± 0.001	6.579 ± 0.789
QML	beta46	0.00	0.006	0.006 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	0.006	0.006	—	0.0243	0.0249	0.909	92.105 ± 0.858	0.100 ± 0.001	7.895 ± 0.858
QML	beta47	0.00	0.007	0.007 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	0.007	0.007	—	0.0282	0.0289	0.893	90.688 ± 0.925	0.101 ± 0.001	9.312 ± 0.925
QML	beta51	0.16	0.162	0.002 ± 0.002	0.013 ± 0.012	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.142 ± 0.006	0.376 ± 0.010	0.162	0.002	1.03	0.0600	0.0600	0.993	94.737 ± 0.710	0.234 ± 0.003	—
QML	beta52	0.16	0.153	-0.007 ± 0.002	-0.042 ± 0.013	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	0.174 ± 0.008	0.417 ± 0.011	0.153	-0.007	-4.37	0.0653	0.0657	0.971	94.130 ± 0.748	0.253 ± 0.001	—
QML	beta53	0.16	0.160	0.000 ± 0.002	0.000 ± 0.011	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.122 ± 0.005	0.349 ± 0.009	0.160	0.000	0.17	0.0563	0.0563	0.991	94.332 ± 0.736	0.217 ± 0.002	—
QML	beta54	0.16	0.152	-0.008 ± 0.002	-0.049 ± 0.013	0.004 ± 0.000	0.066 ± 0.001	0.004 ± 0.000	0.066 ± 0.002	0.170 ± 0.007	0.412 ± 0.011	0.150	-0.010	-6.00	0.0677	0.0684	0.981	95.344 ± 0.670	0.252 ± 0.001	—
QML	beta55	0.00	0.001	0.001 ± 0.002	—	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	—	—	0.000	0.000	—	0.0654	0.0654	0.947	92.611 ± 0.832	0.242 ± 0.004	7.389 ± 0.832
QML	beta56	0.00	0.005	0.005 ± 0.002	—	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	—	—	0.005	0.005	—	0.0523	0.0526	0.941	91.296 ± 0.897	0.203 ± 0.002	8.704 ± 0.897
QML	beta57	0.00	0.003	0.003 ± 0.001	—	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	—	—	0.004	0.004	—	0.0290	0.0293	0.922	93.016 ± 0.811	0.115 ± 0.001	6.984 ± 0.811
QML	beta58	0.00	0.004	0.004 ± 0.001	—	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	—	—	0.004	0.004	—	0.0252	0.0255	0.894	91.700 ± 0.878	0.094 ± 0.001	8.300 ± 0.878
UPI	beta41	0.20	0.202	0.002 ± 0.002	0.008 ± 0.012	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	0.148 ± 0.007	0.385 ± 0.011	0.201	0.001	0.65	0.0773	0.0773	0.982	94.737 ± 0.710	0.296 ± 0.001	—
UPI	beta42	0.20	0.206	0.006 ± 0.002	0.030 ± 0.012	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	0.154 ± 0.007	0.393 ± 0.011	0.204	0.004	2.00	0.0792	0.0793	1.021	96.255 ± 0.604	0.314 ± 0.001	—
UPI	beta43	0.20	0.199	-0.001 ± 0.002	-0.004 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.058 ± 0.003	0.241 ± 0.007	0.198	-0.002	-1.09	0.0490	0.0491	0.940	94.130 ± 0.748	0.178 ± 0.001	—
UPI	beta44	0.00	-0.002	-0.002 ± 0.002	—	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	—	—	-0.001	-0.001	—	0.0523	0.0523	0.927	93.117 ± 0.805	0.195 ± 0.002	6.883 ± 0.805
UPI	beta45	0.00	0.001	0.001 ± 0.002	—	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	—	—	0.000	0.000	—	0.0448	0.0448	0.979	94.636 ± 0.717	0.186 ± 0.002	5.364 ± 0.717
UPI	beta46	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	-0.002	-0.002	—	0.0277	0.0278	0.925	93.219 ± 0.800	0.109 ± 0.001	6.781 ± 0.800

Table S245

Simulation 2 - Condition 101: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.8
(continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	-0.002	-0.002	—	0.0284	0.0285	0.913	92.004 ± 0.863	0.112 ± 0.001	7.996 ± 0.863
UPI	beta51	0.16	0.162	0.002 ± 0.002	0.010 ± 0.012	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.142 ± 0.006	0.377 ± 0.010	0.162	0.002	0.99	0.0599	0.0599	0.973	94.636 ± 0.717	0.230 ± 0.001	—
UPI	beta52	0.16	0.162	0.002 ± 0.002	0.013 ± 0.016	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	0.238 ± 0.011	0.487 ± 0.014	0.161	0.001	0.91	0.0749	0.0750	0.969	94.737 ± 0.710	0.296 ± 0.001	—
UPI	beta53	0.16	0.159	-0.001 ± 0.002	-0.005 ± 0.012	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.140 ± 0.006	0.374 ± 0.010	0.159	-0.001	-0.88	0.0599	0.0599	0.973	94.028 ± 0.754	0.228 ± 0.001	—
UPI	beta54	0.16	0.161	0.001 ± 0.002	0.009 ± 0.016	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	0.238 ± 0.011	0.487 ± 0.014	0.158	-0.002	-1.39	0.0802	0.0802	0.974	95.040 ± 0.691	0.298 ± 0.001	—
UPI	beta55	0.00	0.004	0.004 ± 0.002	—	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.067 ± 0.002	—	—	0.004	0.004	—	0.0654	0.0656	0.932	93.016 ± 0.811	0.245 ± 0.002	6.984 ± 0.811
UPI	beta56	0.00	0.000	0.000 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	—	—	-0.001	-0.001	—	0.0552	0.0552	0.938	91.700 ± 0.878	0.214 ± 0.002	8.300 ± 0.878
UPI	beta57	0.00	-0.003	-0.003 ± 0.001	—	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	—	—	-0.002	-0.002	—	0.0308	0.0308	0.914	92.713 ± 0.827	0.120 ± 0.001	7.287 ± 0.827
UPI	beta58	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	-0.001	-0.001	—	0.0268	0.0268	0.882	92.409 ± 0.843	0.102 ± 0.001	7.591 ± 0.843

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S246

Simulation 2 - Condition 102: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.8

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.204	0.004 ± 0.002	0.018 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.057 ± 0.003	0.239 ± 0.007	0.202	0.002	1.17	0.0445	0.0446	0.971	94.995 ± 0.690	0.181 ± 0.001	—
LSAM	beta42	0.20	0.203	0.003 ± 0.002	0.017 ± 0.008	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.062 ± 0.003	0.250 ± 0.007	0.203	0.003	1.30	0.0509	0.0510	0.973	94.494 ± 0.722	0.190 ± 0.001	—
LSAM	beta43	0.20	0.198	-0.002 ± 0.001	-0.009 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.021 ± 0.001	0.144 ± 0.004	0.198	-0.002	-1.24	0.0286	0.0288	1.000	94.895 ± 0.696	0.113 ± 0.000	—
LSAM	beta44	0.00	0.002	0.002 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	0.001	0.001	—	0.0291	0.0291	0.944	92.693 ± 0.823	0.112 ± 0.001	7.307 ± 0.823
LSAM	beta45	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	0.001	0.001	—	0.0277	0.0277	0.987	94.494 ± 0.722	0.111 ± 0.001	5.506 ± 0.722
LSAM	beta46	0.00	-0.002	-0.002 ± 0.001	—	0.000 ± 0.000	0.017 ± 0.000	0.000 ± 0.000	0.017 ± 0.001	—	—	-0.001	-0.001	—	0.0152	0.0153	0.958	94.494 ± 0.722	0.062 ± 0.001	5.506 ± 0.722
LSAM	beta47	0.00	-0.002	-0.002 ± 0.001	—	0.000 ± 0.000	0.018 ± 0.000	0.000 ± 0.000	0.018 ± 0.001	—	—	-0.002	-0.002	—	0.0166	0.0167	0.905	92.993 ± 0.808	0.063 ± 0.001	7.007 ± 0.808
LSAM	beta51	0.16	0.159	-0.001 ± 0.001	-0.004 ± 0.007	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.054 ± 0.002	0.232 ± 0.006	0.159	-0.001	-0.57	0.0366	0.0367	0.999	94.895 ± 0.696	0.145 ± 0.000	—
LSAM	beta52	0.16	0.164	0.004 ± 0.002	0.026 ± 0.009	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.089 ± 0.004	0.298 ± 0.008	0.166	0.006	3.53	0.0484	0.0487	0.987	94.895 ± 0.696	0.184 ± 0.001	—
LSAM	beta53	0.16	0.159	-0.001 ± 0.001	-0.005 ± 0.007	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.050 ± 0.002	0.224 ± 0.006	0.158	-0.002	-1.23	0.0357	0.0357	1.031	95.996 ± 0.620	0.145 ± 0.000	—
LSAM	beta54	0.16	0.161	0.001 ± 0.001	0.003 ± 0.009	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.083 ± 0.004	0.287 ± 0.008	0.163	0.003	1.68	0.0435	0.0436	1.017	95.095 ± 0.683	0.183 ± 0.001	—
LSAM	beta55	0.00	0.003	0.003 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	0.005	0.005	—	0.0378	0.0381	0.969	93.594 ± 0.775	0.148 ± 0.001	6.406 ± 0.775
LSAM	beta56	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	—	—	0.002	0.002	—	0.0331	0.0331	0.999	94.695 ± 0.709	0.130 ± 0.001	5.305 ± 0.709
LSAM	beta57	0.00	-0.003	-0.003 ± 0.001	—	0.000 ± 0.000	0.019 ± 0.000	0.000 ± 0.000	0.019 ± 0.001	—	—	-0.003	-0.003	—	0.0172	0.0175	0.942	93.694 ± 0.769	0.070 ± 0.001	6.306 ± 0.769
LSAM	beta58	0.00	-0.001	-0.001 ± 0.001	—	0.000 ± 0.000	0.017 ± 0.000	0.000 ± 0.000	0.017 ± 0.000	—	—	-0.001	-0.001	—	0.0163	0.0163	0.919	92.392 ± 0.839	0.061 ± 0.001	7.608 ± 0.839
QML	beta41	0.20	0.193	-0.007 ± 0.001	-0.037 ± 0.006	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.040 ± 0.002	0.201 ± 0.005	0.192	-0.008	-3.94	0.0384	0.0392	0.976	93.794 ± 0.763	0.151 ± 0.000	—
QML	beta42	0.20	0.193	-0.007 ± 0.001	-0.036 ± 0.006	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.043 ± 0.002	0.207 ± 0.006	0.193	-0.007	-3.74	0.0408	0.0415	0.980	94.695 ± 0.709	0.157 ± 0.000	—
QML	beta43	0.20	0.201	0.001 ± 0.001	0.005 ± 0.004	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.020 ± 0.001	0.141 ± 0.004	0.200	0.000	0.13	0.0284	0.0284	1.001	95.495 ± 0.656	0.111 ± 0.000	—
QML	beta44	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	-0.001	-0.001	—	0.0293	0.0293	0.950	93.393 ± 0.786	0.114 ± 0.001	6.607 ± 0.786
QML	beta45	0.00	0.002	0.002 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	0.001	0.001	—	0.0267	0.0267	0.992	94.795 ± 0.703	0.109 ± 0.001	5.205 ± 0.703
QML	beta46	0.00	0.005	0.005 ± 0.000	—	0.000 ± 0.000	0.016 ± 0.000	0.000 ± 0.000	0.016 ± 0.000	—	—	0.005	0.005	—	0.0153	0.0162	0.969	92.893 ± 0.813	0.060 ± 0.001	7.107 ± 0.813
QML	beta47	0.00	0.005	0.005 ± 0.001	—	0.000 ± 0.000	0.017 ± 0.000	0.000 ± 0.000	0.017 ± 0.000	—	—	0.004	0.004	—	0.0158	0.0164	0.915	91.091 ± 0.901	0.059 ± 0.001	8.909 ± 0.901
QML	beta51	0.16	0.160	-0.000 ± 0.001	-0.001 ± 0.007	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.054 ± 0.002	0.232 ± 0.006	0.159	-0.001	-0.39	0.0367	0.0367	1.005	94.795 ± 0.703	0.146 ± 0.000	—
QML	beta52	0.16	0.155	-0.005 ± 0.001	-0.032 ± 0.008	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.065 ± 0.003	0.254 ± 0.007	0.156	-0.004	-2.76	0.0408	0.0410	0.988	93.894 ± 0.758	0.156 ± 0.000	—
QML	beta53	0.16	0.160	0.000 ± 0.001	0.000 ± 0.007	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.044 ± 0.002	0.209 ± 0.006	0.158	-0.002	-0.97	0.0330	0.0330	1.041	95.796 ± 0.635	0.136 ± 0.001	—
QML	beta54	0.16	0.152	-0.008 ± 0.001	-0.052 ± 0.008	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.061 ± 0.003	0.248 ± 0.007	0.153	-0.007	-4.42	0.0368	0.0374	1.034	94.094 ± 0.746	0.157 ± 0.001	—
QML	beta55	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	0.002	0.002	—	0.0377	0.0378	0.982	94.294 ± 0.734	0.148 ± 0.001	5.706 ± 0.734
QML	beta56	0.00	0.005	0.005 ± 0.001	—	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	—	—	0.006	0.006	—	0.0305	0.0311	1.011	94.294 ± 0.734	0.127 ± 0.001	5.706 ± 0.734
QML	beta57	0.00	0.004	0.004 ± 0.001	—	0.000 ± 0.000	0.018 ± 0.000	0.000 ± 0.000	0.019 ± 0.001	—	—	0.003	0.003	—	0.0166	0.0168	0.947	93.093 ± 0.802	0.068 ± 0.001	6.907 ± 0.802
QML	beta58	0.00	0.005	0.005 ± 0.001	—	0.000 ± 0.000	0.016 ± 0.000	0.000 ± 0.000	0.017 ± 0.000	—	—	0.004	0.004	—	0.0158	0.0164	0.926	91.391 ± 0.887	0.058 ± 0.001	8.609 ± 0.887
UPI	beta41	0.20	0.203	0.003 ± 0.002	0.016 ± 0.008	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.056 ± 0.003	0.238 ± 0.007	0.202	0.002	1.14	0.0448	0.0449	0.969	95.195 ± 0.677	0.180 ± 0.001	—
UPI	beta42	0.20	0.203	0.003 ± 0.002	0.015 ± 0.008	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.061 ± 0.003	0.247 ± 0.007	0.202	0.002	1.23	0.0502	0.0502	0.977	94.795 ± 0.703	0.189 ± 0.001	—
UPI	beta43	0.20	0.198	-0.002 ± 0.001	-0.009 ± 0.005	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.020 ± 0.001	0.143 ± 0.004	0.197	-0.003	-1.51	0.0286	0.0287	1.000	94.094 ± 0.746	0.112 ± 0.000	—
UPI	beta44	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	0.001	0.001	—	0.0290	0.0290	0.942	93.493 ± 0.780	0.113 ± 0.001	6.507 ± 0.780

Table S246

Simulation 2 - Condition 102: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.8 (continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta45	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	0.001	0.001	—	0.0268	0.0269	0.993	95.095 ± 0.683	0.110 ± 0.001	4.905 ± 0.683
UPI	beta46	0.00	-0.002	-0.002 ± 0.001	—	0.000 ± 0.000	0.017 ± 0.000	0.000 ± 0.000	0.017 ± 0.001	—	—	-0.001	-0.001	—	0.0153	0.0153	0.955	93.894 ± 0.758	0.062 ± 0.001	6.106 ± 0.758
UPI	beta47	0.00	-0.002	-0.002 ± 0.001	—	0.000 ± 0.000	0.017 ± 0.000	0.000 ± 0.000	0.018 ± 0.001	—	—	-0.001	-0.001	—	0.0166	0.0166	0.909	92.993 ± 0.808	0.062 ± 0.001	7.007 ± 0.808
UPI	beta51	0.16	0.159	-0.001 ± 0.001	-0.003 ± 0.007	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.054 ± 0.002	0.231 ± 0.006	0.159	-0.001	-0.75	0.0367	0.0368	1.001	94.895 ± 0.696	0.145 ± 0.000	—
UPI	beta52	0.16	0.164	0.004 ± 0.001	0.023 ± 0.009	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.087 ± 0.004	0.295 ± 0.008	0.165	0.005	2.89	0.0473	0.0475	0.988	95.295 ± 0.670	0.182 ± 0.001	—
UPI	beta53	0.16	0.159	-0.001 ± 0.001	-0.004 ± 0.007	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.050 ± 0.002	0.224 ± 0.006	0.158	-0.002	-1.23	0.0352	0.0353	1.022	95.996 ± 0.620	0.144 ± 0.000	—
UPI	beta54	0.16	0.160	-0.000 ± 0.001	-0.001 ± 0.009	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.082 ± 0.004	0.287 ± 0.008	0.162	0.002	1.04	0.0439	0.0440	1.015	94.494 ± 0.722	0.183 ± 0.001	—
UPI	beta55	0.00	0.003	0.003 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	0.004	0.004	—	0.0384	0.0386	0.967	93.894 ± 0.758	0.148 ± 0.001	6.106 ± 0.758
UPI	beta56	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	—	—	0.001	0.001	—	0.0331	0.0331	0.995	94.595 ± 0.715	0.130 ± 0.001	5.405 ± 0.715
UPI	beta57	0.00	-0.002	-0.002 ± 0.001	—	0.000 ± 0.000	0.019 ± 0.000	0.000 ± 0.000	0.019 ± 0.001	—	—	-0.002	-0.002	—	0.0178	0.0180	0.936	94.394 ± 0.728	0.070 ± 0.001	5.606 ± 0.728
UPI	beta58	0.00	0.000	-0.000 ± 0.001	—	0.000 ± 0.000	0.017 ± 0.000	0.000 ± 0.000	0.017 ± 0.000	—	—	-0.001	-0.001	—	0.0165	0.0165	0.918	92.593 ± 0.829	0.060 ± 0.001	7.407 ± 0.829

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S247

Simulation 2 - Condition 103: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, $N = 400$, Reliability = 0.4

Method	Param.	True	Mean-based										Median-based					Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE		
LSAM	beta41	0.20	0.296	0.096 ± 0.040	0.479 ± 0.198	0.024 ± 0.008	0.153 ± 0.034	0.031 ± 0.014	0.177 ± 0.049	0.779 ± 0.360	0.883 ± 0.246	0.251	0.051	25.63	0.1145	0.1254	0.980	93.333 ± 6.441	0.590 ± 0.085	—		
LSAM	beta42	0.20	0.156	-0.044 ± 0.032	-0.221 ± 0.162	0.016 ± 0.005	0.126 ± 0.025	0.017 ± 0.006	0.129 ± 0.030	0.418 ± 0.142	0.647 ± 0.149	0.210	0.010	5.00	0.1517	0.1521	1.115	100.000 ± 0.000	0.550 ± 0.060	—		
LSAM	beta43	0.20	0.162	-0.038 ± 0.018	-0.192 ± 0.090	0.005 ± 0.002	0.070 ± 0.018	0.006 ± 0.002	0.078 ± 0.016	0.151 ± 0.045	0.388 ± 0.078	0.162	-0.038	-19.09	0.0656	0.0759	1.119	93.333 ± 6.441	0.307 ± 0.014	—		
LSAM	beta44	0.00	0.013	0.013 ± 0.035	—	0.019 ± 0.005	0.137 ± 0.025	0.018 ± 0.006	0.133 ± 0.030	—	—	0.023	0.023	—	0.1279	0.1300	0.938	93.333 ± 6.441	0.505 ± 0.065	6.667 ± 6.441		
LSAM	beta45	0.00	0.041	0.041 ± 0.018	—	0.005 ± 0.001	0.069 ± 0.012	0.006 ± 0.002	0.078 ± 0.019	—	—	0.008	0.008	—	0.0908	0.0911	1.440	100.000 ± 0.000	0.392 ± 0.039	0.000 ± 0.000		
LSAM	beta46	0.00	-0.144	-0.144 ± 0.041	—	0.025 ± 0.008	0.157 ± 0.035	0.044 ± 0.019	0.209 ± 0.051	—	—	-0.106	-0.106	—	0.1199	0.1602	1.001	100.000 ± 0.000	0.616 ± 0.106	0.000 ± 0.000		
LSAM	beta47	0.00	0.015	0.015 ± 0.048	—	0.035 ± 0.011	0.187 ± 0.037	0.033 ± 0.011	0.182 ± 0.043	—	—	-0.017	-0.017	—	0.1168	0.1180	0.921	93.333 ± 6.441	0.677 ± 0.098	6.667 ± 6.441		
LSAM	beta51	0.16	0.137	-0.023 ± 0.023	-0.144 ± 0.146	0.008 ± 0.002	0.091 ± 0.016	0.008 ± 0.002	0.091 ± 0.017	0.321 ± 0.077	0.566 ± 0.108	0.100	-0.060	-37.19	0.0866	0.1050	1.165	100.000 ± 0.000	0.414 ± 0.016	—		
LSAM	beta52	0.16	0.154	-0.006 ± 0.021	-0.038 ± 0.132	0.007 ± 0.002	0.082 ± 0.020	0.006 ± 0.002	0.079 ± 0.020	0.246 ± 0.090	0.496 ± 0.128	0.191	0.031	19.37	0.1178	0.1218	1.600	100.000 ± 0.000	0.513 ± 0.104	—		
LSAM	beta53	0.16	0.185	0.025 ± 0.023	0.157 ± 0.146	0.008 ± 0.002	0.091 ± 0.015	0.008 ± 0.002	0.091 ± 0.018	0.324 ± 0.088	0.569 ± 0.114	0.177	0.017	10.75	0.0895	0.0911	1.002	100.000 ± 0.000	0.356 ± 0.015	—		
LSAM	beta54	0.16	0.162	0.002 ± 0.024	0.014 ± 0.152	0.009 ± 0.002	0.094 ± 0.016	0.008 ± 0.002	0.091 ± 0.019	0.322 ± 0.089	0.568 ± 0.121	0.146	-0.014	-8.66	0.1362	0.1369	1.140	100.000 ± 0.000	0.420 ± 0.050	—		
LSAM	beta55	0.00	0.072	0.072 ± 0.044	—	0.029 ± 0.010	0.169 ± 0.040	0.032 ± 0.013	0.178 ± 0.045	—	—	0.053	0.053	—	0.1310	0.1414	1.125	93.333 ± 6.441	0.746 ± 0.170	6.667 ± 6.441		
LSAM	beta56	0.00	-0.043	-0.043 ± 0.030	—	0.013 ± 0.004	0.115 ± 0.022	0.014 ± 0.004	0.119 ± 0.022	—	—	-0.018	-0.018	—	0.1135	0.1149	1.332	100.000 ± 0.000	0.603 ± 0.057	0.000 ± 0.000		
LSAM	beta57	0.00	-0.034	-0.034 ± 0.030	—	0.013 ± 0.005	0.115 ± 0.030	0.014 ± 0.006	0.116 ± 0.035	—	—	-0.069	-0.069	—	0.1129	0.1324	1.289	100.000 ± 0.000	0.581 ± 0.149	0.000 ± 0.000		
LSAM	beta58	0.00	-0.015	-0.015 ± 0.016	—	0.004 ± 0.001	0.063 ± 0.009	0.004 ± 0.001	0.062 ± 0.012	—	—	-0.043	-0.043	—	0.0801	0.0911	1.565	100.000 ± 0.000	0.385 ± 0.044	0.000 ± 0.000		
QML	beta41	0.20	0.222	0.022 ± 0.019	0.112 ± 0.097	0.006 ± 0.003	0.075 ± 0.026	0.006 ± 0.003	0.076 ± 0.030	0.145 ± 0.087	0.381 ± 0.152	0.211	0.011	5.47	0.0469	0.0481	0.902	93.333 ± 6.441	0.267 ± 0.009	—		
QML	beta42	0.20	0.171	-0.029 ± 0.010	-0.143 ± 0.051	0.002 ± 0.000	0.040 ± 0.008	0.002 ± 0.001	0.048 ± 0.011	0.057 ± 0.022	0.239 ± 0.056	0.202	0.002	1.16	0.0668	0.0668	1.875	100.000 ± 0.000	0.292 ± 0.026	—		
QML	beta43	0.20	0.176	-0.024 ± 0.015	-0.118 ± 0.075	0.003 ± 0.001	0.058 ± 0.015	0.004 ± 0.001	0.061 ± 0.013	0.092 ± 0.028	0.303 ± 0.064	0.163	-0.037	-18.33	0.0757	0.0841	1.162	93.333 ± 6.441	0.263 ± 0.013	—		
QML	beta44	0.00	-0.011	-0.011 ± 0.023	—	0.008 ± 0.003	0.091 ± 0.019	0.008 ± 0.003	0.089 ± 0.021	—	—	0.053	0.053	—	0.1106	0.1227	0.932	93.333 ± 6.441	0.333 ± 0.021	6.667 ± 6.441		
QML	beta45	0.00	0.017	0.017 ± 0.012	—	0.002 ± 0.001	0.045 ± 0.010	0.002 ± 0.001	0.047 ± 0.011	—	—	0.001	0.001	—	0.0801	0.0801	1.610	100.000 ± 0.000	0.284 ± 0.018	0.000 ± 0.000		
QML	beta46	0.00	-0.057	-0.057 ± 0.013	—	0.003 ± 0.001	0.050 ± 0.010	0.006 ± 0.002	0.075 ± 0.012	—	—	-0.055	-0.055	—	0.0842	0.1004	1.187	80.000 ± 10.328	0.233 ± 0.015	20.000 ± 10.328		
QML	beta47	0.00	-0.026	-0.026 ± 0.017	—	0.004 ± 0.001	0.066 ± 0.012	0.005 ± 0.002	0.069 ± 0.017	—	—	-0.027	-0.027	—	0.0841	0.0884	0.992	86.667 ± 8.777	0.256 ± 0.032	13.333 ± 8.777		
QML	beta51	0.16	0.163	0.003 ± 0.022	0.019 ± 0.139	0.007 ± 0.002	0.086 ± 0.016	0.007 ± 0.002	0.083 ± 0.019	0.272 ± 0.082	0.521 ± 0.118	0.116	-0.044	-27.70	0.1083	0.1170	1.202	100.000 ± 0.000	0.406 ± 0.020	—		
QML	beta52	0.16	0.129	-0.031 ± 0.013	-0.192 ± 0.084	0.003 ± 0.001	0.052 ± 0.008	0.003 ± 0.001	0.059 ± 0.009	0.137 ± 0.031	0.370 ± 0.058	0.155	-0.005	-2.90	0.0959	0.0960	1.514	100.000 ± 0.000	0.310 ± 0.016	—		
QML	beta53	0.16	0.181	0.021 ± 0.025	0.129 ± 0.154	0.009 ± 0.003	0.096 ± 0.019	0.009 ± 0.003	0.095 ± 0.022	0.350 ± 0.116	0.592 ± 0.137	0.184	0.024	15.18	0.0929	0.0960	0.831	86.667 ± 8.777	0.312 ± 0.016	—		
QML	beta54	0.16	0.163	0.003 ± 0.020	0.017 ± 0.124	0.006 ± 0.003	0.077 ± 0.024	0.006 ± 0.003	0.075 ± 0.025	0.217 ± 0.106	0.466 ± 0.154	0.154	-0.006	-3.85	0.0723	0.0725	0.951	93.333 ± 6.441	0.287 ± 0.010	—		
QML	beta55	0.00	0.051	0.051 ± 0.028	—	0.012 ± 0.003	0.110 ± 0.019	0.014 ± 0.005	0.118 ± 0.026	—	—	0.032	0.032	—	0.1569	0.1601	1.091	93.333 ± 6.441	0.471 ± 0.045	6.667 ± 6.441		
QML	beta56	0.00	-0.063	-0.063 ± 0.024	—	0.008 ± 0.002	0.091 ± 0.017	0.012 ± 0.003	0.108 ± 0.018	—	—	-0.047	-0.047	—	0.1038	0.1139	1.269	100.000 ± 0.000	0.453 ± 0.052	0.000 ± 0.000		
QML	beta57	0.00	-0.012	-0.012 ± 0.017	—	0.004 ± 0.002	0.065 ± 0.019	0.004 ± 0.002	0.064 ± 0.017	—	—	-0.052	-0.052	—	0.0628	0.0814	0.942	100.000 ± 0.000	0.239 ± 0.025	0.000 ± 0.000		
QML	beta58	0.00	-0.005	-0.005 ± 0.012	—	0.002 ± 0.001	0.045 ± 0.009	0.002 ± 0.001	0.044 ± 0.010	—	—	-0.023	-0.023	—	0.0504	0.0553	1.212	100.000 ± 0.000	0.214 ± 0.019	0.000 ± 0.000		
UPI	beta41	0.20	0.331	0.131 ± 0.078	0.656 ± 0.390	0.091 ± 0.032	0.302 ± 0.071	0.102 ± 0.048	0.320 ± 0.094	2.557 ± 1.190	1.599 ± 0.470	0.250	0.050	24.78	0.3084	0.3124	1.457	86.667 ± 8.777	1.724 ± 0.607	—		
UPI	beta42	0.20	0.221	0.021 ± 0.070	0.106 ± 0.348	0.073 ± 0.015	0.269 ± 0.037	0.068 ± 0.015	0.261 ± 0.049	1.703 ± 0.370	1.305 ± 0.247	0.246	0.046	22.88	0.3931	0.3958	2.146	93.333 ± 6.441	2.266 ± 0.700	—		
UPI	beta43	0.20	0.101	-0.099 ± 0.023	-0.497 ± 0.113	0.008 ± 0.001	0.088 ± 0.010	0.017 ± 0.005	0.131 ± 0.020	0.427 ± 0.121	0.654 ± 0.102	0.143	-0.057	-28.72	0.1609	0.1708	1.376	80.000 ± 10.328	0.474 ± 0.051	—		
UPI	beta44	0.00	-0.001	-0.001 ± 0.048	—	0.035 ± 0.019	0.187 ± 0.075	0.033 ± 0.019	0.181 ± 0.072	—	—	-0.018	-0.018	—	0.1502	0.1512	1.626	100.000 ± 0.000	1.194 ± 0.311	0.000 ± 0.000		
UPI	beta45	0.00	-0.042	-0.042 ± 0.048	—	0.034 ± 0.019	0.185 ± 0.078	0.034 ± 0.022	0.184 ± 0.084	—	—	-0.033	-0.033	—	0.1621	0.1653	1.508	100.000 ± 0.000	1.095 ± 0.332	0.000 ± 0.000		
UPI	beta46	0.00	-0.365	-0.365 ± 0.198	—	0.589 ± 0.343	0.768 ± 0.342	0.683 ± 0.467	0.826 ± 0.388	—	—	-0.116	-0.116	—	0.4133	0.4293	1.626	100.000 ± 0.000	4.893 ± 1.885	0.000 ± 0.000		

Table S247

Simulation 2 - Condition 103: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.4
(continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.032	-0.032 ± 0.126	—	0.240 ± 0.110	0.489 ± 0.156	0.225 ± 0.114	0.474 ± 0.161	—	—	-0.049	-0.049	—	0.3447	0.3482	1.543	100.000 ± 0.000	2.961 ± 0.663	0.000 ± 0.000
UPI	beta51	0.16	0.200	0.040 ± 0.032	0.250 ± 0.202	0.016 ± 0.003	0.125 ± 0.016	0.016 ± 0.004	0.127 ± 0.025	0.634 ± 0.166	0.796 ± 0.154	0.152	-0.008	-5.02	0.1726	0.1728	1.738	100.000 ± 0.000	0.853 ± 0.205	—
UPI	beta52	0.16	0.254	0.094 ± 0.080	0.591 ± 0.500	0.096 ± 0.041	0.310 ± 0.090	0.099 ± 0.039	0.314 ± 0.081	3.855 ± 1.507	1.963 ± 0.509	0.325	0.165	103.15	0.3973	0.4302	1.409	100.000 ± 0.000	1.713 ± 0.432	—
UPI	beta53	0.16	0.137	-0.023 ± 0.035	-0.145 ± 0.220	0.019 ± 0.004	0.136 ± 0.021	0.018 ± 0.003	0.134 ± 0.024	0.697 ± 0.130	0.835 ± 0.148	0.087	-0.073	-45.60	0.1455	0.1627	1.768	86.667 ± 8.777	0.944 ± 0.268	—
UPI	beta54	0.16	0.133	-0.027 ± 0.054	-0.170 ± 0.335	0.043 ± 0.011	0.207 ± 0.034	0.041 ± 0.010	0.202 ± 0.040	1.596 ± 0.389	1.263 ± 0.250	0.108	-0.052	-32.33	0.1979	0.2046	1.712	93.333 ± 6.441	1.392 ± 0.526	—
UPI	beta55	0.00	0.096	0.096 ± 0.058	—	0.051 ± 0.027	0.226 ± 0.087	0.057 ± 0.035	0.239 ± 0.099	—	—	0.048	0.048	—	0.1827	0.1888	1.476	100.000 ± 0.000	1.309 ± 0.306	0.000 ± 0.000
UPI	beta56	0.00	-0.093	-0.093 ± 0.040	—	0.024 ± 0.006	0.154 ± 0.027	0.031 ± 0.011	0.175 ± 0.038	—	—	-0.073	-0.073	—	0.1961	0.2094	2.660	100.000 ± 0.000	1.603 ± 0.425	0.000 ± 0.000
UPI	beta57	0.00	-0.333	-0.333 ± 0.242	—	0.881 ± 0.584	0.939 ± 0.502	0.933 ± 0.722	0.966 ± 0.548	—	—	-0.247	-0.247	—	0.6822	0.7254	1.421	100.000 ± 0.000	5.228 ± 2.285	0.000 ± 0.000
UPI	beta58	0.00	-0.154	-0.154 ± 0.080	—	0.096 ± 0.022	0.311 ± 0.048	0.114 ± 0.041	0.337 ± 0.077	—	—	-0.068	-0.068	—	0.3672	0.3734	1.875	100.000 ± 0.000	2.282 ± 0.586	0.000 ± 0.000

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S248

Simulation 2 - Condition 104: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, $N = 1000$, Reliability = 0.4

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.230	0.030 ± 0.021	0.148 ± 0.105	0.005 ± 0.001	0.069 ± 0.011	0.005 ± 0.002	0.072 ± 0.017	0.131 ± 0.042	0.362 ± 0.084	0.244	0.044	22.21	0.0653	0.0790	1.096	100.000 ± 0.000	0.298 ± 0.021	—
LSAM	beta42	0.20	0.221	0.021 ± 0.018	0.103 ± 0.089	0.003 ± 0.001	0.059 ± 0.008	0.004 ± 0.001	0.060 ± 0.013	0.090 ± 0.024	0.299 ± 0.064	0.216	0.016	7.90	0.0623	0.0643	1.109	100.000 ± 0.000	0.257 ± 0.014	—
LSAM	beta43	0.20	0.187	-0.013 ± 0.018	-0.063 ± 0.089	0.003 ± 0.001	0.059 ± 0.011	0.003 ± 0.001	0.057 ± 0.015	0.083 ± 0.028	0.287 ± 0.073	0.187	-0.013	-6.45	0.0629	0.0642	0.824	81.818 ± 11.629	0.190 ± 0.010	—
LSAM	beta44	0.00	0.049	0.049 ± 0.024	—	0.006 ± 0.001	0.078 ± 0.011	0.008 ± 0.003	0.089 ± 0.021	—	—	0.056	0.056	—	0.0807	0.0981	1.019	100.000 ± 0.000	0.312 ± 0.023	0.000 ± 0.000
LSAM	beta45	0.00	0.017	0.017 ± 0.014	—	0.002 ± 0.000	0.048 ± 0.007	0.002 ± 0.000	0.049 ± 0.009	—	—	0.008	0.008	—	0.0692	0.0696	1.281	100.000 ± 0.000	0.240 ± 0.011	0.000 ± 0.000
LSAM	beta46	0.00	-0.108	-0.108 ± 0.024	—	0.006 ± 0.003	0.080 ± 0.023	0.018 ± 0.005	0.132 ± 0.021	—	—	-0.094	-0.094	—	0.0625	0.1130	1.089	100.000 ± 0.000	0.341 ± 0.053	0.000 ± 0.000
LSAM	beta47	0.00	-0.027	-0.027 ± 0.022	—	0.006 ± 0.002	0.074 ± 0.019	0.006 ± 0.003	0.076 ± 0.023	—	—	-0.068	-0.068	—	0.0723	0.0993	0.934	90.909 ± 8.668	0.272 ± 0.022	9.091 ± 8.668
LSAM	beta51	0.16	0.159	-0.001 ± 0.012	-0.005 ± 0.072	0.001 ± 0.001	0.038 ± 0.011	0.001 ± 0.001	0.037 ± 0.012	0.052 ± 0.023	0.229 ± 0.075	0.151	-0.009	-5.64	0.0671	0.0677	1.699	100.000 ± 0.000	0.256 ± 0.007	—
LSAM	beta52	0.16	0.204	0.044 ± 0.024	0.274 ± 0.151	0.006 ± 0.002	0.080 ± 0.020	0.008 ± 0.003	0.088 ± 0.025	0.303 ± 0.132	0.550 ± 0.154	0.199	0.039	24.12	0.0627	0.0736	0.908	90.909 ± 8.668	0.285 ± 0.023	—
LSAM	beta53	0.16	0.108	-0.052 ± 0.012	-0.324 ± 0.077	0.002 ± 0.001	0.041 ± 0.009	0.004 ± 0.001	0.065 ± 0.011	0.164 ± 0.052	0.405 ± 0.072	0.144	-0.016	-10.03	0.0478	0.0505	1.258	90.909 ± 8.668	0.200 ± 0.006	—
LSAM	beta54	0.16	0.167	0.007 ± 0.014	0.043 ± 0.086	0.002 ± 0.001	0.046 ± 0.013	0.002 ± 0.001	0.044 ± 0.012	0.076 ± 0.026	0.275 ± 0.075	0.179	0.019	12.13	0.0557	0.0590	1.411	100.000 ± 0.000	0.253 ± 0.014	—
LSAM	beta55	0.00	0.032	0.032 ± 0.018	—	0.004 ± 0.001	0.061 ± 0.014	0.004 ± 0.002	0.067 ± 0.020	—	—	0.036	0.036	—	0.0535	0.0647	1.324	100.000 ± 0.000	0.316 ± 0.016	0.000 ± 0.000
LSAM	beta56	0.00	-0.026	-0.026 ± 0.017	—	0.003 ± 0.001	0.055 ± 0.009	0.003 ± 0.001	0.059 ± 0.013	—	—	-0.021	-0.021	—	0.0576	0.0613	1.369	100.000 ± 0.000	0.296 ± 0.016	0.000 ± 0.000
LSAM	beta57	0.00	-0.065	-0.065 ± 0.010	—	0.001 ± 0.000	0.034 ± 0.007	0.005 ± 0.001	0.072 ± 0.011	—	—	-0.069	-0.069	—	0.0435	0.0815	2.249	90.909 ± 8.668	0.301 ± 0.040	9.091 ± 8.668
LSAM	beta58	0.00	-0.027	-0.027 ± 0.023	—	0.006 ± 0.001	0.077 ± 0.014	0.006 ± 0.002	0.078 ± 0.017	—	—	-0.040	-0.040	—	0.0726	0.0826	0.898	90.909 ± 8.668	0.271 ± 0.033	9.091 ± 8.668
QML	beta41	0.20	0.187	-0.013 ± 0.015	-0.065 ± 0.075	0.003 ± 0.001	0.050 ± 0.009	0.002 ± 0.001	0.049 ± 0.013	0.061 ± 0.020	0.247 ± 0.063	0.195	-0.005	-2.38	0.0554	0.0556	0.932	90.909 ± 8.668	0.183 ± 0.006	—
QML	beta42	0.20	0.219	0.019 ± 0.012	0.097 ± 0.060	0.002 ± 0.001	0.040 ± 0.010	0.002 ± 0.001	0.043 ± 0.011	0.045 ± 0.017	0.213 ± 0.053	0.211	0.011	5.30	0.0444	0.0457	1.125	100.000 ± 0.000	0.176 ± 0.005	—
QML	beta43	0.20	0.200	0.000 ± 0.020	0.000 ± 0.098	0.004 ± 0.001	0.065 ± 0.012	0.004 ± 0.001	0.062 ± 0.015	0.095 ± 0.027	0.308 ± 0.076	0.186	-0.014	-6.90	0.0654	0.0668	0.680	90.909 ± 8.668	0.172 ± 0.004	—
QML	beta44	0.00	0.017	0.017 ± 0.013	—	0.002 ± 0.001	0.044 ± 0.010	0.002 ± 0.001	0.046 ± 0.014	—	—	0.003	0.003	—	0.0472	0.0473	1.434	100.000 ± 0.000	0.248 ± 0.014	0.000 ± 0.000
QML	beta45	0.00	0.001	0.001 ± 0.010	—	0.001 ± 0.000	0.033 ± 0.005	0.001 ± 0.000	0.031 ± 0.007	—	—	0.006	0.006	—	0.0414	0.0419	1.560	100.000 ± 0.000	0.201 ± 0.007	0.000 ± 0.000
QML	beta46	0.00	-0.055	-0.055 ± 0.014	—	0.002 ± 0.001	0.047 ± 0.018	0.005 ± 0.001	0.071 ± 0.010	—	—	-0.051	-0.051	—	0.0456	0.0687	1.031	81.818 ± 11.629	0.188 ± 0.019	18.182 ± 11.629
QML	beta47	0.00	-0.008	-0.008 ± 0.013	—	0.002 ± 0.001	0.042 ± 0.012	0.002 ± 0.001	0.040 ± 0.014	—	—	-0.036	-0.036	—	0.0396	0.0536	0.973	90.909 ± 8.668	0.159 ± 0.006	9.091 ± 8.668
QML	beta51	0.16	0.170	0.010 ± 0.013	0.063 ± 0.080	0.002 ± 0.001	0.043 ± 0.011	0.002 ± 0.001	0.042 ± 0.012	0.069 ± 0.028	0.262 ± 0.077	0.168	0.008	4.97	0.0477	0.0483	1.512	100.000 ± 0.000	0.253 ± 0.007	—
QML	beta52	0.16	0.176	0.016 ± 0.019	0.098 ± 0.118	0.004 ± 0.001	0.062 ± 0.015	0.004 ± 0.002	0.061 ± 0.017	0.148 ± 0.059	0.384 ± 0.109	0.170	0.010	5.98	0.0388	0.0399	0.794	81.818 ± 11.629	0.194 ± 0.009	—
QML	beta53	0.16	0.118	-0.042 ± 0.010	-0.262 ± 0.060	0.001 ± 0.000	0.032 ± 0.006	0.003 ± 0.001	0.052 ± 0.009	0.105 ± 0.031	0.324 ± 0.054	0.145	-0.015	-9.13	0.0517	0.0537	1.489	100.000 ± 0.000	0.186 ± 0.004	—
QML	beta54	0.16	0.159	-0.001 ± 0.010	-0.009 ± 0.061	0.001 ± 0.000	0.032 ± 0.010	0.001 ± 0.000	0.031 ± 0.011	0.038 ± 0.018	0.194 ± 0.066	0.163	0.003	1.98	0.0392	0.0394	1.425	100.000 ± 0.000	0.182 ± 0.006	—
QML	beta55	0.00	0.037	0.037 ± 0.017	—	0.003 ± 0.001	0.057 ± 0.015	0.004 ± 0.002	0.066 ± 0.022	—	—	0.034	0.034	—	0.0725	0.0802	1.410	90.909 ± 8.668	0.317 ± 0.018	9.091 ± 8.668
QML	beta56	0.00	-0.045	-0.045 ± 0.012	—	0.002 ± 0.000	0.040 ± 0.007	0.003 ± 0.001	0.059 ± 0.012	—	—	-0.026	-0.026	—	0.0391	0.0468	1.632	100.000 ± 0.000	0.254 ± 0.008	0.000 ± 0.000
QML	beta57	0.00	-0.031	-0.031 ± 0.007	—	0.001 ± 0.000	0.023 ± 0.004	0.001 ± 0.001	0.038 ± 0.008	—	—	-0.041	-0.041	—	0.0353	0.0538	1.902	81.818 ± 11.629	0.170 ± 0.011	18.182 ± 11.629
QML	beta58	0.00	-0.018	-0.018 ± 0.014	—	0.002 ± 0.000	0.047 ± 0.007	0.002 ± 0.001	0.048 ± 0.011	—	—	-0.033	-0.033	—	0.0448	0.0554	0.796	81.818 ± 11.629	0.146 ± 0.011	18.182 ± 11.629
UPI	beta41	0.20	0.358	0.158 ± 0.109	0.790 ± 0.543	0.130 ± 0.035	0.360 ± 0.071	0.143 ± 0.061	0.378 ± 0.106	3.575 ± 1.519	1.891 ± 0.528	0.361	0.161	80.31	0.5801	0.6019	1.031	100.000 ± 0.000	1.456 ± 0.216	—
UPI	beta42	0.20	0.341	0.141 ± 0.113	0.704 ± 0.567	0.142 ± 0.055	0.376 ± 0.107	0.149 ± 0.084	0.385 ± 0.142	3.714 ± 2.100	1.927 ± 0.712	0.150	-0.050	-25.06	0.3866	0.3898	0.905	100.000 ± 0.000	1.335 ± 0.309	—
UPI	beta43	0.20	0.129	-0.071 ± 0.052	-0.357 ± 0.258	0.029 ± 0.010	0.171 ± 0.043	0.032 ± 0.007	0.178 ± 0.036	0.793 ± 0.183	0.890 ± 0.178	0.081	-0.119	-59.66	0.1913	0.2255	0.711	63.636 ± 14.504	0.476 ± 0.089	—
UPI	beta44	0.00	0.063	0.063 ± 0.031	—	0.010 ± 0.003	0.101 ± 0.022	0.013 ± 0.006	0.116 ± 0.031	—	—	0.029	0.029	—	0.1845	0.1868	1.870	100.000 ± 0.000	0.743 ± 0.110	0.000 ± 0.000
UPI	beta45	0.00	0.040	0.040 ± 0.015	—	0.002 ± 0.000	0.050 ± 0.006	0.004 ± 0.001	0.062 ± 0.013	—	—	0.015	0.015	—	0.1022	0.1033	3.029	100.000 ± 0.000	0.590 ± 0.116	0.000 ± 0.000
UPI	beta46	0.00	-0.562	-0.562 ± 0.185	—	0.375 ± 0.200	0.612 ± 0.270	0.657 ± 0.387	0.810 ± 0.311	—	—	-0.457	-0.457	—	0.7018	0.8372	1.615	100.000 ± 0.000	3.874 ± 1.758	0.000 ± 0.000

Table S248

Simulation 2 - Condition 104: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.4
(continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.143	-0.143 ± 0.092	—	0.092 ± 0.030	0.303 ± 0.071	0.104 ± 0.046	0.323 ± 0.094	—	—	-0.010	-0.010	—	0.3440	0.3441	1.146	100.000 ± 0.000	1.363 ± 0.311	0.000 ± 0.000
UPI	beta51	0.16	0.225	0.065 ± 0.033	0.407 ± 0.206	0.012 ± 0.003	0.109 ± 0.020	0.015 ± 0.005	0.123 ± 0.027	0.589 ± 0.195	0.767 ± 0.171	0.230	0.070	43.52	0.1511	0.1664	1.386	90.909 ± 8.668	0.593 ± 0.106	—
UPI	beta52	0.16	0.313	0.153 ± 0.075	0.956 ± 0.468	0.062 ± 0.014	0.248 ± 0.042	0.079 ± 0.030	0.282 ± 0.068	3.103 ± 1.188	1.761 ± 0.427	0.331	0.171	106.60	0.3646	0.4025	1.443	100.000 ± 0.000	1.404 ± 0.353	—
UPI	beta53	0.16	0.094	-0.066 ± 0.013	-0.411 ± 0.084	0.002 ± 0.000	0.045 ± 0.007	0.006 ± 0.001	0.078 ± 0.010	0.240 ± 0.058	0.489 ± 0.062	0.071	-0.089	-55.54	0.0888	0.1256	2.711	72.727 ± 13.428	0.473 ± 0.128	—
UPI	beta54	0.16	0.122	-0.038 ± 0.065	-0.238 ± 0.404	0.046 ± 0.020	0.215 ± 0.070	0.043 ± 0.017	0.208 ± 0.061	1.692 ± 0.654	1.301 ± 0.384	0.059	-0.101	-63.19	0.3199	0.3355	0.935	81.818 ± 11.629	0.786 ± 0.280	—
UPI	beta55	0.00	0.058	0.058 ± 0.049	—	0.027 ± 0.015	0.163 ± 0.078	0.028 ± 0.020	0.166 ± 0.088	—	—	0.040	0.040	—	0.1476	0.1528	1.362	100.000 ± 0.000	0.872 ± 0.253	0.000 ± 0.000
UPI	beta56	0.00	-0.044	-0.044 ± 0.029	—	0.009 ± 0.003	0.095 ± 0.027	0.010 ± 0.005	0.100 ± 0.033	—	—	-0.006	-0.006	—	0.1035	0.1037	2.124	100.000 ± 0.000	0.788 ± 0.167	0.000 ± 0.000
UPI	beta57	0.00	-0.494	-0.494 ± 0.183	—	0.368 ± 0.167	0.607 ± 0.218	0.579 ± 0.324	0.761 ± 0.273	—	—	-0.747	-0.747	—	0.9547	1.2124	1.562	100.000 ± 0.000	3.716 ± 1.621	0.000 ± 0.000
UPI	beta58	0.00	-0.035	-0.035 ± 0.042	—	0.019 ± 0.008	0.138 ± 0.044	0.018 ± 0.010	0.136 ± 0.053	—	—	0.016	0.016	—	0.1370	0.1379	2.132	100.000 ± 0.000	1.152 ± 0.525	0.000 ± 0.000

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S249

Simulation 2 - Condition 105: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.6

Method	Param.	True	Mean-based										Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE	
LSAM	beta41	0.20	0.202	0.002 ± 0.002	0.012 ± 0.012	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	0.089 ± 0.005	0.298 ± 0.010	0.202	0.002	0.85	0.0625	0.0625	0.995	95.372 ± 0.854	0.233 ± 0.001	—	
LSAM	beta42	0.20	0.203	0.003 ± 0.003	0.016 ± 0.013	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	0.100 ± 0.006	0.316 ± 0.012	0.204	0.004	1.80	0.0632	0.0633	0.990	95.207 ± 0.869	0.245 ± 0.001	—	
LSAM	beta43	0.20	0.194	-0.006 ± 0.002	-0.031 ± 0.011	0.003 ± 0.000	0.054 ± 0.002	0.003 ± 0.000	0.054 ± 0.002	0.073 ± 0.004	0.271 ± 0.009	0.193	-0.007	-3.25	0.0582	0.0586	1.030	95.537 ± 0.839	0.218 ± 0.001	—	
LSAM	beta44	0.00	0.015	0.015 ± 0.003	—	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.073 ± 0.003	—	—	0.012	0.012	—	0.0692	0.0703	0.967	94.215 ± 0.949	0.271 ± 0.002	5.785 ± 0.949	
LSAM	beta45	0.00	0.006	0.006 ± 0.003	—	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	—	—	0.006	0.006	—	0.0636	0.0639	0.968	94.711 ± 0.910	0.236 ± 0.002	5.289 ± 0.910	
LSAM	beta46	0.00	-0.028	-0.028 ± 0.003	—	0.007 ± 0.000	0.084 ± 0.003	0.008 ± 0.000	0.089 ± 0.003	—	—	-0.030	-0.030	—	0.0816	0.0869	0.981	93.058 ± 1.033	0.325 ± 0.004	6.942 ± 1.033	
LSAM	beta47	0.00	-0.029	-0.029 ± 0.003	—	0.007 ± 0.000	0.084 ± 0.003	0.008 ± 0.001	0.089 ± 0.003	—	—	-0.031	-0.031	—	0.0756	0.0815	0.988	93.719 ± 0.986	0.324 ± 0.004	6.281 ± 0.986	
LSAM	beta51	0.16	0.159	-0.001 ± 0.003	-0.007 ± 0.020	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.078 ± 0.003	0.235 ± 0.014	0.485 ± 0.018	0.156	-0.004	-2.50	0.0737	0.0738	0.925	93.719 ± 0.986	0.281 ± 0.001	—	
LSAM	beta52	0.16	0.163	0.003 ± 0.002	0.019 ± 0.015	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	0.142 ± 0.008	0.377 ± 0.014	0.163	0.003	1.63	0.0584	0.0585	1.005	95.702 ± 0.825	0.238 ± 0.001	—	
LSAM	beta53	0.16	0.157	-0.003 ± 0.003	-0.020 ± 0.017	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.166 ± 0.010	0.407 ± 0.015	0.157	-0.003	-1.92	0.0642	0.0643	0.937	93.388 ± 1.010	0.239 ± 0.001	—	
LSAM	beta54	0.16	0.163	0.003 ± 0.002	0.017 ± 0.015	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	0.140 ± 0.009	0.374 ± 0.014	0.161	0.001	0.77	0.0575	0.0575	1.007	95.207 ± 0.869	0.236 ± 0.001	—	
LSAM	beta55	0.00	0.010	0.010 ± 0.003	—	0.007 ± 0.000	0.086 ± 0.002	0.007 ± 0.000	0.086 ± 0.003	—	—	0.009	0.009	—	0.0841	0.0846	0.966	94.215 ± 0.949	0.324 ± 0.003	5.785 ± 0.949	
LSAM	beta56	0.00	-0.001	-0.001 ± 0.003	—	0.005 ± 0.000	0.074 ± 0.002	0.005 ± 0.000	0.074 ± 0.003	—	—	0.001	0.001	—	0.0696	0.0696	1.048	95.537 ± 0.839	0.304 ± 0.002	4.463 ± 0.839	
LSAM	beta57	0.00	-0.026	-0.026 ± 0.004	—	0.008 ± 0.000	0.090 ± 0.003	0.009 ± 0.000	0.093 ± 0.003	—	—	-0.026	-0.026	—	0.0872	0.0909	0.919	93.058 ± 1.033	0.324 ± 0.004	6.942 ± 1.033	
LSAM	beta58	0.00	-0.022	-0.022 ± 0.003	—	0.006 ± 0.000	0.080 ± 0.003	0.007 ± 0.000	0.083 ± 0.003	—	—	-0.025	-0.025	—	0.0771	0.0812	0.952	94.380 ± 0.936	0.299 ± 0.003	5.620 ± 0.936	
QML	beta41	0.20	0.200	-0.000 ± 0.002	-0.002 ± 0.011	0.003 ± 0.000	0.054 ± 0.002	0.003 ± 0.000	0.054 ± 0.002	0.074 ± 0.004	0.272 ± 0.010	0.199	-0.001	-0.48	0.0532	0.0532	0.984	93.884 ± 0.974	0.210 ± 0.001	—	
QML	beta42	0.20	0.200	0.000 ± 0.002	0.002 ± 0.011	0.003 ± 0.000	0.057 ± 0.002	0.003 ± 0.000	0.056 ± 0.002	0.080 ± 0.005	0.282 ± 0.010	0.200	0.000	0.17	0.0549	0.0549	0.988	94.876 ± 0.896	0.219 ± 0.001	—	
QML	beta43	0.20	0.195	-0.005 ± 0.002	-0.023 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	0.067 ± 0.004	0.258 ± 0.009	0.195	-0.005	-2.32	0.0546	0.0548	1.030	96.033 ± 0.794	0.208 ± 0.001	—	
QML	beta44	0.00	0.008	0.008 ± 0.003	—	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	—	—	0.006	0.006	—	0.0664	0.0667	1.001	95.207 ± 0.869	0.258 ± 0.002	4.793 ± 0.869	
QML	beta45	0.00	0.000	0.000 ± 0.002	—	0.003 ± 0.000	0.059 ± 0.002	0.003 ± 0.000	0.059 ± 0.002	—	—	-0.003	-0.003	—	0.0593	0.0593	0.971	93.884 ± 0.974	0.224 ± 0.001	6.116 ± 0.974	
QML	beta46	0.00	-0.018	-0.018 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.002	0.003 ± 0.000	0.056 ± 0.002	—	—	-0.020	-0.020	—	0.0564	0.0596	1.000	91.736 ± 1.119	0.209 ± 0.002	8.264 ± 1.119	
QML	beta47	0.00	-0.021	-0.021 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.002	0.003 ± 0.000	0.056 ± 0.002	—	—	-0.022	-0.022	—	0.0521	0.0565	1.006	93.719 ± 0.986	0.207 ± 0.002	6.281 ± 0.986	
QML	beta51	0.16	0.161	0.001 ± 0.003	0.005 ± 0.019	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.003	0.227 ± 0.015	0.476 ± 0.018	0.159	-0.001	-0.71	0.0718	0.0718	0.956	94.380 ± 0.936	0.286 ± 0.001	—	
QML	beta52	0.16	0.160	0.000 ± 0.002	0.000 ± 0.014	0.003 ± 0.000	0.057 ± 0.002	0.003 ± 0.000	0.057 ± 0.002	0.126 ± 0.007	0.355 ± 0.012	0.159	-0.001	-0.86	0.0558	0.0559	1.021	95.868 ± 0.809	0.227 ± 0.001	—	
QML	beta53	0.16	0.160	-0.000 ± 0.003	-0.002 ± 0.016	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	0.151 ± 0.010	0.388 ± 0.015	0.159	-0.001	-0.45	0.0625	0.0625	0.950	93.884 ± 0.974	0.231 ± 0.001	—	
QML	beta54	0.16	0.160	-0.000 ± 0.002	-0.000 ± 0.014	0.003 ± 0.000	0.055 ± 0.002	0.003 ± 0.000	0.055 ± 0.002	0.119 ± 0.007	0.345 ± 0.013	0.159	-0.001	-0.66	0.0527	0.0527	1.018	96.198 ± 0.777	0.221 ± 0.001	—	
QML	beta55	0.00	0.008	0.008 ± 0.003	—	0.007 ± 0.000	0.083 ± 0.002	0.007 ± 0.000	0.083 ± 0.003	—	—	0.010	0.010	—	0.0818	0.0825	0.976	93.223 ± 1.022	0.317 ± 0.003	6.777 ± 1.022	
QML	beta56	0.00	-0.005	-0.005 ± 0.003	—	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.003	—	—	-0.008	-0.008	—	0.0678	0.0682	1.060	95.372 ± 0.854	0.286 ± 0.002	4.628 ± 0.854	
QML	beta57	0.00	-0.018	-0.018 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	—	—	-0.019	-0.019	—	0.0595	0.0626	0.916	90.579 ± 1.188	0.217 ± 0.002	9.421 ± 1.188	
QML	beta58	0.00	-0.015	-0.015 ± 0.002	—	0.002 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	—	—	-0.016	-0.016	—	0.0497	0.0523	0.986	93.554 ± 0.998	0.192 ± 0.002	6.446 ± 0.998	
UPI	beta41	0.20	0.204	0.004 ± 0.003	0.020 ± 0.014	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.067 ± 0.003	0.113 ± 0.008	0.337 ± 0.013	0.202	0.002	0.90	0.0661	0.0661	1.283	97.521 ± 0.632	0.338 ± 0.012	—	
UPI	beta42	0.20	0.218	0.018 ± 0.003	0.091 ± 0.017	0.007 ± 0.001	0.083 ± 0.003	0.007 ± 0.001	0.085 ± 0.004	0.182 ± 0.015	0.426 ± 0.019	0.217	0.017	8.66	0.0733	0.0753	1.209	96.694 ± 0.727	0.395 ± 0.015	—	
UPI	beta43	0.20	0.186	-0.014 ± 0.002	-0.072 ± 0.012	0.004 ± 0.000	0.059 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	0.093 ± 0.006	0.305 ± 0.011	0.180	-0.020	-9.77	0.0605	0.0636	1.139	95.868 ± 0.809	0.265 ± 0.004	—	
UPI	beta44	0.00	0.019	0.019 ± 0.003	—	0.006 ± 0.000	0.080 ± 0.003	0.007 ± 0.000	0.083 ± 0.003	—	—	0.018	0.018	—	0.0778	0.0799	1.161	97.025 ± 0.691	0.366 ± 0.009	2.975 ± 0.691	
UPI	beta45	0.00	0.005	0.005 ± 0.003	—	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.003	—	—	0.004	0.004	—	0.0671	0.0672	1.100	95.537 ± 0.839	0.291 ± 0.006	4.463 ± 0.839	
UPI	beta46	0.00	-0.023	-0.023 ± 0.005	—	0.016 ± 0.001	0.125 ± 0.005	0.016 ± 0.001	0.127 ± 0.005	—	—	-0.024	-0.024	—	0.1140	0.1164	1.337	98.182 ± 0.543	0.653 ± 0.028	1.818 ± 0.543	

Table S249
Simulation 2 - Condition 105: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.6
(continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.042	-0.042 ± 0.006	—	0.022 ± 0.002	0.148 ± 0.006	0.024 ± 0.002	0.153 ± 0.007	—	—	-0.039	-0.039	—	0.1192	0.1254	1.339	97.025 ± 0.691	0.776 ± 0.046	2.975 ± 0.691
UPI	beta51	0.16	0.158	-0.002 ± 0.003	-0.015 ± 0.020	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.077 ± 0.003	0.233 ± 0.015	0.483 ± 0.018	0.153	-0.007	-4.56	0.0782	0.0785	0.991	94.545 ± 0.923	0.300 ± 0.002	—
UPI	beta52	0.16	0.167	0.007 ± 0.003	0.045 ± 0.016	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.166 ± 0.010	0.407 ± 0.015	0.165	0.005	3.26	0.0669	0.0671	1.209	97.190 ± 0.672	0.307 ± 0.008	—
UPI	beta53	0.16	0.154	-0.006 ± 0.003	-0.036 ± 0.017	0.004 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.067 ± 0.003	0.177 ± 0.012	0.420 ± 0.016	0.154	-0.006	-3.77	0.0680	0.0682	1.062	94.711 ± 0.910	0.279 ± 0.004	—
UPI	beta54	0.16	0.168	0.008 ± 0.003	0.050 ± 0.018	0.005 ± 0.000	0.070 ± 0.003	0.005 ± 0.000	0.071 ± 0.003	0.196 ± 0.014	0.443 ± 0.018	0.166	0.006	3.81	0.0639	0.0642	1.243	97.521 ± 0.632	0.343 ± 0.011	—
UPI	beta55	0.00	0.011	0.011 ± 0.004	—	0.008 ± 0.000	0.087 ± 0.003	0.008 ± 0.000	0.088 ± 0.003	—	—	0.013	0.013	—	0.0842	0.0852	1.111	96.033 ± 0.794	0.381 ± 0.009	3.967 ± 0.794
UPI	beta56	0.00	0.002	0.002 ± 0.003	—	0.006 ± 0.000	0.079 ± 0.003	0.006 ± 0.000	0.079 ± 0.003	—	—	0.003	0.003	—	0.0793	0.0794	1.139	97.686 ± 0.611	0.352 ± 0.006	2.314 ± 0.611
UPI	beta57	0.00	-0.022	-0.022 ± 0.005	—	0.016 ± 0.001	0.127 ± 0.005	0.016 ± 0.001	0.128 ± 0.006	—	—	-0.016	-0.016	—	0.1109	0.1121	1.182	96.033 ± 0.794	0.586 ± 0.032	3.967 ± 0.794
UPI	beta58	0.00	-0.023	-0.023 ± 0.005	—	0.016 ± 0.001	0.127 ± 0.005	0.017 ± 0.001	0.129 ± 0.006	—	—	-0.021	-0.021	—	0.1102	0.1122	1.282	97.851 ± 0.590	0.636 ± 0.028	2.149 ± 0.590

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S250

Simulation 2 - Condition 106: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.6

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.203	0.003 ± 0.001	0.015 ± 0.007	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.032 ± 0.002	0.178 ± 0.006	0.202	0.002	1.09	0.0351	0.0351	1.018	95.646 ± 0.777	0.142 ± 0.000	—
LSAM	beta42	0.20	0.205	0.005 ± 0.001	0.023 ± 0.007	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.036 ± 0.002	0.190 ± 0.006	0.206	0.006	2.85	0.0376	0.0381	0.998	94.485 ± 0.870	0.148 ± 0.000	—
LSAM	beta43	0.20	0.196	-0.004 ± 0.001	-0.018 ± 0.007	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.031 ± 0.002	0.176 ± 0.006	0.194	-0.006	-2.79	0.0348	0.0352	0.977	94.630 ± 0.859	0.134 ± 0.000	—
LSAM	beta44	0.00	0.016	0.016 ± 0.002	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	—	—	0.015	0.015	—	0.0400	0.0428	1.002	92.888 ± 0.979	0.160 ± 0.001	7.112 ± 0.979
LSAM	beta45	0.00	0.006	0.006 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.007	0.007	—	0.0382	0.0388	1.009	96.226 ± 0.726	0.144 ± 0.001	3.774 ± 0.726
LSAM	beta46	0.00	-0.028	-0.028 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.002	0.003 ± 0.000	0.059 ± 0.002	—	—	-0.030	-0.030	—	0.0501	0.0583	0.947	89.985 ± 1.144	0.192 ± 0.001	10.015 ± 1.144
LSAM	beta47	0.00	-0.034	-0.034 ± 0.002	—	0.002 ± 0.000	0.047 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	—	—	-0.034	-0.034	—	0.0455	0.0567	1.027	89.260 ± 1.180	0.191 ± 0.001	10.740 ± 1.180
LSAM	beta51	0.16	0.159	-0.001 ± 0.002	-0.007 ± 0.011	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.078 ± 0.004	0.280 ± 0.009	0.159	-0.001	-0.48	0.0433	0.0433	1.012	93.904 ± 0.911	0.178 ± 0.000	—
LSAM	beta52	0.16	0.163	0.003 ± 0.001	0.016 ± 0.009	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.051 ± 0.003	0.225 ± 0.008	0.162	0.002	1.08	0.0365	0.0365	1.035	94.775 ± 0.848	0.146 ± 0.000	—
LSAM	beta53	0.16	0.155	-0.005 ± 0.001	-0.030 ± 0.009	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.054 ± 0.003	0.232 ± 0.008	0.156	-0.004	-2.61	0.0345	0.0348	1.015	94.340 ± 0.880	0.147 ± 0.000	—
LSAM	beta54	0.16	0.165	0.005 ± 0.001	0.033 ± 0.009	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.057 ± 0.003	0.238 ± 0.008	0.165	0.005	3.20	0.0367	0.0371	0.979	94.340 ± 0.880	0.145 ± 0.000	—
LSAM	beta55	0.00	0.011	0.011 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	—	—	0.010	0.010	—	0.0511	0.0521	0.965	93.324 ± 0.951	0.196 ± 0.001	6.676 ± 0.951
LSAM	beta56	0.00	0.003	0.003 ± 0.002	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.002	—	—	0.002	0.002	—	0.0455	0.0455	0.997	94.630 ± 0.859	0.183 ± 0.001	5.370 ± 0.859
LSAM	beta57	0.00	-0.024	-0.024 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	—	—	-0.025	-0.025	—	0.0496	0.0555	1.000	93.904 ± 0.911	0.194 ± 0.001	6.096 ± 0.911
LSAM	beta58	0.00	-0.023	-0.023 ± 0.002	—	0.002 ± 0.000	0.048 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	—	—	-0.024	-0.024	—	0.0485	0.0540	0.957	90.421 ± 1.121	0.179 ± 0.001	9.579 ± 1.121
QML	beta41	0.20	0.201	0.001 ± 0.001	0.005 ± 0.006	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.027 ± 0.001	0.166 ± 0.005	0.201	0.001	0.70	0.0324	0.0324	1.011	94.340 ± 0.880	0.131 ± 0.000	—
QML	beta42	0.20	0.201	0.001 ± 0.001	0.007 ± 0.007	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.030 ± 0.002	0.174 ± 0.006	0.201	0.001	0.60	0.0356	0.0356	0.994	94.340 ± 0.880	0.136 ± 0.000	—
QML	beta43	0.20	0.199	-0.001 ± 0.001	-0.006 ± 0.006	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.029 ± 0.002	0.170 ± 0.006	0.197	-0.003	-1.34	0.0333	0.0334	0.975	94.775 ± 0.848	0.130 ± 0.000	—
QML	beta44	0.00	0.010	0.010 ± 0.002	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	0.010	0.010	—	0.0404	0.0417	1.005	93.904 ± 0.911	0.158 ± 0.001	6.096 ± 0.911
QML	beta45	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	0.001	0.001	—	0.0371	0.0371	1.010	95.791 ± 0.765	0.140 ± 0.001	4.209 ± 0.765
QML	beta46	0.00	-0.019	-0.019 ± 0.001	—	0.001 ± 0.000	0.034 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	-0.021	-0.021	—	0.0348	0.0405	0.967	89.115 ± 1.187	0.129 ± 0.001	10.885 ± 1.187
QML	beta47	0.00	-0.025	-0.025 ± 0.001	—	0.001 ± 0.000	0.031 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	-0.025	-0.025	—	0.0306	0.0395	1.032	88.534 ± 1.214	0.127 ± 0.001	11.466 ± 1.214
QML	beta51	0.16	0.160	0.000 ± 0.002	0.003 ± 0.011	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.076 ± 0.004	0.276 ± 0.009	0.161	0.001	0.71	0.0422	0.0423	1.030	94.920 ± 0.837	0.179 ± 0.000	—
QML	beta52	0.16	0.161	0.001 ± 0.001	0.007 ± 0.008	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.047 ± 0.003	0.216 ± 0.007	0.161	0.001	0.53	0.0339	0.0339	1.037	94.775 ± 0.848	0.140 ± 0.000	—
QML	beta53	0.16	0.158	-0.002 ± 0.001	-0.013 ± 0.009	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.051 ± 0.003	0.225 ± 0.008	0.158	-0.002	-1.06	0.0336	0.0337	1.016	94.485 ± 0.870	0.143 ± 0.000	—
QML	beta54	0.16	0.163	0.003 ± 0.001	0.017 ± 0.008	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.049 ± 0.003	0.222 ± 0.008	0.162	0.002	1.53	0.0342	0.0343	0.993	94.485 ± 0.870	0.138 ± 0.000	—
QML	beta55	0.00	0.011	0.011 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	—	—	0.010	0.010	—	0.0506	0.0517	0.982	93.033 ± 0.970	0.195 ± 0.001	6.967 ± 0.970
QML	beta56	0.00	-0.001	-0.001 ± 0.002	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.002	—	—	-0.002	-0.002	—	0.0433	0.0433	0.999	94.630 ± 0.859	0.176 ± 0.001	5.370 ± 0.859
QML	beta57	0.00	-0.017	-0.017 ± 0.001	—	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.019	-0.019	—	0.0346	0.0393	1.012	92.017 ± 1.033	0.134 ± 0.001	7.983 ± 1.033
QML	beta58	0.00	-0.016	-0.016 ± 0.001	—	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	-0.017	-0.017	—	0.0311	0.0354	0.974	89.840 ± 1.151	0.120 ± 0.001	10.160 ± 1.151
UPI	beta41	0.20	0.206	0.006 ± 0.002	0.031 ± 0.008	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.044 ± 0.002	0.048 ± 0.003	0.219 ± 0.008	0.205	0.005	2.34	0.0448	0.0451	1.284	97.097 ± 0.640	0.219 ± 0.010	—
UPI	beta42	0.20	0.215	0.015 ± 0.002	0.075 ± 0.009	0.002 ± 0.000	0.049 ± 0.002	0.003 ± 0.000	0.052 ± 0.002	0.067 ± 0.005	0.258 ± 0.010	0.212	0.012	6.08	0.0493	0.0508	1.211	97.678 ± 0.574	0.234 ± 0.008	—
UPI	beta43	0.20	0.189	-0.011 ± 0.001	-0.054 ± 0.007	0.001 ± 0.000	0.037 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.038 ± 0.002	0.194 ± 0.006	0.186	-0.014	-6.88	0.0385	0.0409	1.080	95.065 ± 0.825	0.158 ± 0.003	—
UPI	beta44	0.00	0.021	0.021 ± 0.002	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.049 ± 0.002	—	—	0.021	0.021	—	0.0466	0.0513	1.122	93.904 ± 0.911	0.194 ± 0.003	6.096 ± 0.911
UPI	beta45	0.00	0.008	0.008 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.039 ± 0.001	—	—	0.008	0.008	—	0.0401	0.0409	1.097	96.662 ± 0.684	0.162 ± 0.003	3.338 ± 0.684
UPI	beta46	0.00	-0.030	-0.030 ± 0.003	—	0.007 ± 0.000	0.082 ± 0.003	0.008 ± 0.001	0.087 ± 0.003	—	—	-0.030	-0.030	—	0.0733	0.0792	1.181	95.356 ± 0.802	0.381 ± 0.019	4.644 ± 0.802

Table S250

Simulation 2 - Condition 106: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.6
(continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.049	-0.049 ± 0.003	—	0.007 ± 0.001	0.082 ± 0.003	0.009 ± 0.001	0.095 ± 0.004	—	—	-0.045	-0.045	—	0.0724	0.0854	1.179	94.485 ± 0.870	0.379 ± 0.014	5.515 ± 0.870
UPI	beta51	0.16	0.157	-0.003 ± 0.002	-0.016 ± 0.011	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.079 ± 0.004	0.282 ± 0.009	0.156	-0.004	-2.38	0.0460	0.0461	1.042	94.920 ± 0.837	0.184 ± 0.001	—
UPI	beta52	0.16	0.166	0.006 ± 0.002	0.040 ± 0.010	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.002	0.072 ± 0.005	0.269 ± 0.010	0.167	0.007	4.11	0.0412	0.0417	1.122	96.372 ± 0.712	0.187 ± 0.003	—
UPI	beta53	0.16	0.149	-0.011 ± 0.001	-0.067 ± 0.009	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.064 ± 0.004	0.254 ± 0.009	0.148	-0.012	-7.29	0.0395	0.0412	1.080	95.065 ± 0.825	0.166 ± 0.002	—
UPI	beta54	0.16	0.174	0.014 ± 0.002	0.087 ± 0.011	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.048 ± 0.002	0.090 ± 0.006	0.300 ± 0.011	0.172	0.012	7.24	0.0437	0.0452	1.178	95.501 ± 0.790	0.212 ± 0.007	—
UPI	beta55	0.00	0.014	0.014 ± 0.002	—	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	—	—	0.013	0.013	—	0.0554	0.0568	1.015	94.340 ± 0.880	0.215 ± 0.002	5.660 ± 0.880
UPI	beta56	0.00	0.004	0.004 ± 0.002	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.048 ± 0.002	—	—	0.007	0.007	—	0.0471	0.0476	1.054	96.372 ± 0.712	0.196 ± 0.001	3.628 ± 0.712
UPI	beta57	0.00	-0.021	-0.021 ± 0.003	—	0.005 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.077 ± 0.003	—	—	-0.024	-0.024	—	0.0692	0.0734	1.082	95.356 ± 0.802	0.313 ± 0.008	4.644 ± 0.802
UPI	beta58	0.00	-0.034	-0.034 ± 0.003	—	0.006 ± 0.000	0.077 ± 0.003	0.007 ± 0.001	0.085 ± 0.003	—	—	-0.033	-0.033	—	0.0656	0.0732	1.206	95.646 ± 0.777	0.366 ± 0.015	4.354 ± 0.777

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S251

Simulation 2 - Condition 107: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.200	-0.000 ± 0.001	-0.001 ± 0.007	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.054 ± 0.002	0.232 ± 0.006	0.199	-0.001	-0.72	0.0465	0.0465	0.967	94.000 ± 0.751	0.176 ± 0.000	—
LSAM	beta42	0.20	0.201	0.001 ± 0.001	0.006 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.055 ± 0.002	0.235 ± 0.006	0.198	-0.002	-0.85	0.0477	0.0477	0.987	94.900 ± 0.696	0.182 ± 0.000	—
LSAM	beta43	0.20	0.200	-0.000 ± 0.001	-0.000 ± 0.007	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.052 ± 0.002	0.227 ± 0.006	0.199	-0.001	-0.34	0.0473	0.0474	0.975	95.000 ± 0.689	0.174 ± 0.000	—
LSAM	beta44	0.00	0.002	0.002 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	—	—	0.002	0.002	—	0.0492	0.0492	0.979	93.800 ± 0.763	0.195 ± 0.001	6.200 ± 0.763
LSAM	beta45	0.00	0.003	0.003 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	0.002	0.002	—	0.0463	0.0463	0.985	94.700 ± 0.708	0.177 ± 0.001	5.300 ± 0.708
LSAM	beta46	0.00	-0.006	-0.006 ± 0.002	—	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	—	—	-0.008	-0.008	—	0.0567	0.0572	0.994	95.400 ± 0.662	0.224 ± 0.001	4.600 ± 0.662
LSAM	beta47	0.00	-0.005	-0.005 ± 0.002	—	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	—	—	-0.005	-0.005	—	0.0555	0.0558	0.986	94.100 ± 0.745	0.221 ± 0.001	5.900 ± 0.745
LSAM	beta51	0.16	0.158	-0.002 ± 0.002	-0.011 ± 0.012	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.147 ± 0.006	0.383 ± 0.010	0.157	-0.003	-1.71	0.0603	0.0604	0.956	93.900 ± 0.757	0.230 ± 0.001	—
LSAM	beta52	0.16	0.163	0.003 ± 0.001	0.022 ± 0.009	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.086 ± 0.004	0.294 ± 0.008	0.164	0.004	2.49	0.0461	0.0463	1.006	94.700 ± 0.708	0.185 ± 0.000	—
LSAM	beta53	0.16	0.160	-0.000 ± 0.002	-0.001 ± 0.010	0.003 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.098 ± 0.004	0.312 ± 0.009	0.158	-0.002	-1.10	0.0517	0.0518	0.966	93.800 ± 0.763	0.189 ± 0.000	—
LSAM	beta54	0.16	0.159	-0.001 ± 0.001	-0.003 ± 0.009	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.083 ± 0.004	0.289 ± 0.008	0.160	0.000	0.17	0.0447	0.0447	1.019	94.400 ± 0.727	0.185 ± 0.000	—
LSAM	beta55	0.00	0.001	0.001 ± 0.002	—	0.004 ± 0.000	0.066 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	—	—	0.003	0.003	—	0.0659	0.0659	0.929	92.900 ± 0.812	0.239 ± 0.001	7.100 ± 0.812
LSAM	beta56	0.00	0.000	0.000 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	—	—	0.001	0.001	—	0.0584	0.0584	0.945	92.900 ± 0.812	0.227 ± 0.001	7.100 ± 0.812
LSAM	beta57	0.00	-0.005	-0.005 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	—	—	-0.004	-0.004	—	0.0591	0.0593	0.980	94.000 ± 0.751	0.229 ± 0.001	6.000 ± 0.751
LSAM	beta58	0.00	-0.004	-0.004 ± 0.002	—	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	—	—	-0.005	-0.005	—	0.0579	0.0581	0.968	94.000 ± 0.751	0.216 ± 0.001	6.000 ± 0.751
QML	beta41	0.20	0.199	-0.001 ± 0.001	-0.003 ± 0.007	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.052 ± 0.002	0.227 ± 0.006	0.198	-0.002	-1.13	0.0450	0.0450	0.972	94.100 ± 0.745	0.173 ± 0.001	—
QML	beta42	0.20	0.201	0.001 ± 0.001	0.004 ± 0.007	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.053 ± 0.002	0.230 ± 0.006	0.199	-0.001	-0.65	0.0458	0.0458	0.985	94.800 ± 0.702	0.178 ± 0.000	—
QML	beta43	0.20	0.200	-0.000 ± 0.001	-0.001 ± 0.007	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.050 ± 0.002	0.224 ± 0.006	0.200	0.000	-0.19	0.0471	0.0471	0.978	94.900 ± 0.696	0.172 ± 0.001	—
QML	beta44	0.00	0.000	0.000 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	—	—	0.000	0.000	—	0.0500	0.0500	0.981	94.000 ± 0.751	0.194 ± 0.001	6.000 ± 0.751
QML	beta45	0.00	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	0.000	0.000	—	0.0457	0.0457	0.995	94.900 ± 0.696	0.177 ± 0.001	5.100 ± 0.696
QML	beta46	0.00	-0.005	-0.005 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	-0.006	-0.006	—	0.0458	0.0462	1.010	95.200 ± 0.676	0.181 ± 0.001	4.800 ± 0.676
QML	beta47	0.00	-0.006	-0.006 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	-0.005	-0.005	—	0.0444	0.0447	0.993	94.500 ± 0.721	0.178 ± 0.001	5.500 ± 0.721
QML	beta51	0.16	0.159	-0.001 ± 0.002	-0.008 ± 0.012	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.145 ± 0.006	0.381 ± 0.010	0.159	-0.001	-0.74	0.0592	0.0592	0.969	94.400 ± 0.727	0.232 ± 0.001	—
QML	beta52	0.16	0.163	0.003 ± 0.001	0.021 ± 0.009	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.086 ± 0.004	0.293 ± 0.008	0.164	0.004	2.32	0.0463	0.0464	1.010	94.800 ± 0.702	0.185 ± 0.001	—
QML	beta53	0.16	0.160	-0.000 ± 0.002	-0.000 ± 0.010	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.096 ± 0.004	0.311 ± 0.009	0.159	-0.001	-0.93	0.0508	0.0509	0.968	94.000 ± 0.751	0.189 ± 0.000	—
QML	beta54	0.16	0.159	-0.001 ± 0.001	-0.006 ± 0.009	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.082 ± 0.004	0.286 ± 0.008	0.159	-0.001	-0.58	0.0449	0.0449	1.020	94.400 ± 0.727	0.183 ± 0.000	—
QML	beta55	0.00	0.001	0.001 ± 0.002	—	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	—	—	0.001	0.001	—	0.0653	0.0653	0.937	93.400 ± 0.785	0.238 ± 0.001	6.600 ± 0.785
QML	beta56	0.00	-0.001	-0.001 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	—	—	0.000	0.000	—	0.0567	0.0567	0.950	92.600 ± 0.828	0.223 ± 0.001	7.400 ± 0.828
QML	beta57	0.00	-0.005	-0.005 ± 0.002	—	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	—	—	-0.005	-0.005	—	0.0480	0.0483	0.985	94.200 ± 0.739	0.187 ± 0.001	5.800 ± 0.739
QML	beta58	0.00	-0.004	-0.004 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	-0.005	-0.005	—	0.0462	0.0464	0.977	94.100 ± 0.745	0.174 ± 0.001	5.900 ± 0.745
UPI	beta41	0.20	0.200	-0.000 ± 0.001	-0.001 ± 0.007	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.053 ± 0.002	0.230 ± 0.006	0.199	-0.001	-0.65	0.0465	0.0465	0.967	94.000 ± 0.751	0.175 ± 0.000	—
UPI	beta42	0.20	0.201	0.001 ± 0.001	0.007 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.055 ± 0.002	0.234 ± 0.006	0.199	-0.001	-0.54	0.0475	0.0476	0.982	94.900 ± 0.696	0.181 ± 0.000	—
UPI	beta43	0.20	0.200	0.000 ± 0.001	0.000 ± 0.007	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.052 ± 0.002	0.227 ± 0.006	0.200	0.000	-0.10	0.0471	0.0471	0.977	95.100 ± 0.683	0.174 ± 0.000	—
UPI	beta44	0.00	0.001	0.001 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.002	—	—	-0.001	-0.001	—	0.0502	0.0502	0.977	94.000 ± 0.751	0.197 ± 0.001	6.000 ± 0.751
UPI	beta45	0.00	0.002	0.002 ± 0.001	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	0.001	0.001	—	0.0468	0.0468	0.976	93.800 ± 0.763	0.179 ± 0.001	6.200 ± 0.763
UPI	beta46	0.00	-0.003	-0.003 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	—	—	-0.003	-0.003	—	0.0589	0.0590	1.016	94.900 ± 0.696	0.237 ± 0.001	5.100 ± 0.696

Table S251

Simulation 2 - Condition 107: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, N = 400, Reliability = 0.8
(continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.003	-0.003 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	—	—	-0.003	-0.003	—	0.0597	0.0598	1.011	94.900 ± 0.696	0.236 ± 0.001	5.100 ± 0.696
UPI	beta51	0.16	0.158	-0.002 ± 0.002	-0.011 ± 0.012	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.145 ± 0.006	0.380 ± 0.010	0.158	-0.002	-1.40	0.0585	0.0586	0.967	94.000 ± 0.751	0.231 ± 0.001	—
UPI	beta52	0.16	0.163	0.003 ± 0.001	0.021 ± 0.009	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.086 ± 0.004	0.293 ± 0.008	0.164	0.004	2.43	0.0449	0.0450	0.999	94.300 ± 0.733	0.183 ± 0.000	—
UPI	beta53	0.16	0.160	0.000 ± 0.002	0.003 ± 0.010	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.098 ± 0.004	0.313 ± 0.009	0.158	-0.002	-0.95	0.0519	0.0519	0.967	94.100 ± 0.745	0.190 ± 0.000	—
UPI	beta54	0.16	0.160	-0.000 ± 0.001	-0.002 ± 0.009	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.084 ± 0.004	0.289 ± 0.008	0.160	0.000	0.01	0.0458	0.0458	1.014	94.500 ± 0.721	0.184 ± 0.000	—
UPI	beta55	0.00	0.000	-0.000 ± 0.002	—	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	—	—	0.002	0.002	—	0.0677	0.0677	0.940	93.000 ± 0.807	0.240 ± 0.001	7.000 ± 0.807
UPI	beta56	0.00	0.000	0.000 ± 0.002	—	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	—	—	0.001	0.001	—	0.0579	0.0579	0.949	92.800 ± 0.817	0.229 ± 0.001	7.200 ± 0.817
UPI	beta57	0.00	-0.001	-0.001 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	—	—	-0.001	-0.001	—	0.0606	0.0606	0.997	94.100 ± 0.745	0.238 ± 0.001	5.900 ± 0.745
UPI	beta58	0.00	-0.002	-0.002 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	—	—	-0.002	-0.002	—	0.0584	0.0584	1.017	95.000 ± 0.689	0.230 ± 0.001	5.000 ± 0.689

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S252

Simulation 2 - Condition 108: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.8

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.200	0.000 ± 0.001	0.002 ± 0.004	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.020 ± 0.001	0.141 ± 0.004	0.199	-0.001	-0.47	0.0275	0.0275	1.002	95.500 ± 0.656	0.111 ± 0.000	—
LSAM	beta42	0.20	0.200	-0.000 ± 0.001	-0.001 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.022 ± 0.001	0.147 ± 0.004	0.200	0.000	-0.13	0.0296	0.0296	0.990	94.800 ± 0.702	0.114 ± 0.000	—
LSAM	beta43	0.20	0.199	-0.001 ± 0.001	-0.003 ± 0.004	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	0.019 ± 0.001	0.137 ± 0.004	0.199	-0.001	-0.40	0.0260	0.0260	1.015	95.000 ± 0.689	0.109 ± 0.000	—
LSAM	beta44	0.00	0.004	0.004 ± 0.001	—	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	—	—	0.005	0.005	—	0.0308	0.0312	0.997	94.300 ± 0.733	0.123 ± 0.000	5.700 ± 0.733
LSAM	beta45	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	-0.001	-0.001	—	0.0284	0.0284	0.998	95.000 ± 0.689	0.112 ± 0.000	5.000 ± 0.689
LSAM	beta46	0.00	-0.009	-0.009 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.011	-0.011	—	0.0368	0.0383	0.971	93.500 ± 0.780	0.140 ± 0.000	6.500 ± 0.780
LSAM	beta47	0.00	-0.006	-0.006 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.005	-0.005	—	0.0363	0.0366	0.970	93.600 ± 0.774	0.140 ± 0.000	6.400 ± 0.774
LSAM	beta51	0.16	0.159	-0.001 ± 0.001	-0.006 ± 0.007	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.055 ± 0.003	0.235 ± 0.007	0.158	-0.002	-1.17	0.0369	0.0370	0.982	94.600 ± 0.715	0.145 ± 0.000	—
LSAM	beta52	0.16	0.161	0.001 ± 0.001	0.007 ± 0.006	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.034 ± 0.001	0.184 ± 0.005	0.161	0.001	0.84	0.0298	0.0298	1.002	95.600 ± 0.649	0.116 ± 0.000	—
LSAM	beta53	0.16	0.160	0.000 ± 0.001	0.003 ± 0.006	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.037 ± 0.002	0.193 ± 0.005	0.160	0.000	-0.25	0.0303	0.0303	0.986	94.300 ± 0.733	0.119 ± 0.000	—
LSAM	beta54	0.16	0.160	-0.000 ± 0.001	-0.003 ± 0.006	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.036 ± 0.002	0.190 ± 0.005	0.160	0.000	0.02	0.0302	0.0302	0.969	93.100 ± 0.801	0.116 ± 0.000	—
LSAM	beta55	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	0.002	0.002	—	0.0348	0.0349	1.015	95.200 ± 0.676	0.150 ± 0.000	4.800 ± 0.676
LSAM	beta56	0.00	0.002	0.002 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.001	0.001	—	0.0369	0.0369	0.988	94.700 ± 0.708	0.141 ± 0.000	5.300 ± 0.708
LSAM	beta57	0.00	-0.008	-0.008 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.008	-0.008	—	0.0389	0.0396	0.973	93.600 ± 0.774	0.143 ± 0.000	6.400 ± 0.774
LSAM	beta58	0.00	-0.005	-0.005 ± 0.001	—	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	—	—	-0.005	-0.005	—	0.0353	0.0357	1.007	96.000 ± 0.620	0.135 ± 0.000	4.000 ± 0.620
QML	beta41	0.20	0.200	-0.000 ± 0.001	-0.000 ± 0.004	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.019 ± 0.001	0.139 ± 0.004	0.199	-0.001	-0.60	0.0268	0.0268	1.000	95.200 ± 0.676	0.109 ± 0.000	—
QML	beta42	0.20	0.200	-0.000 ± 0.001	-0.002 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.021 ± 0.001	0.145 ± 0.004	0.199	-0.001	-0.34	0.0289	0.0289	0.991	94.500 ± 0.721	0.112 ± 0.000	—
QML	beta43	0.20	0.199	-0.001 ± 0.001	-0.003 ± 0.004	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	0.019 ± 0.001	0.137 ± 0.004	0.199	-0.001	-0.33	0.0260	0.0260	1.013	94.700 ± 0.708	0.109 ± 0.000	—
QML	beta44	0.00	0.002	0.002 ± 0.001	—	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	0.003	0.003	—	0.0304	0.0305	1.000	94.800 ± 0.702	0.123 ± 0.000	5.200 ± 0.702
QML	beta45	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	-0.003	-0.003	—	0.0281	0.0282	1.011	95.000 ± 0.689	0.113 ± 0.001	5.000 ± 0.689
QML	beta46	0.00	-0.008	-0.008 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	-0.009	-0.009	—	0.0302	0.0316	0.974	93.400 ± 0.785	0.114 ± 0.000	6.600 ± 0.785
QML	beta47	0.00	-0.006	-0.006 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	-0.006	-0.006	—	0.0293	0.0299	0.980	93.500 ± 0.780	0.113 ± 0.000	6.500 ± 0.780
QML	beta51	0.16	0.159	-0.001 ± 0.001	-0.005 ± 0.007	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.055 ± 0.003	0.235 ± 0.007	0.158	-0.002	-1.16	0.0369	0.0369	0.988	94.800 ± 0.702	0.146 ± 0.000	—
QML	beta52	0.16	0.161	0.001 ± 0.001	0.005 ± 0.006	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.034 ± 0.001	0.183 ± 0.005	0.161	0.001	0.68	0.0297	0.0297	1.007	95.500 ± 0.656	0.116 ± 0.000	—
QML	beta53	0.16	0.161	0.001 ± 0.001	0.004 ± 0.006	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.037 ± 0.002	0.192 ± 0.005	0.160	0.000	-0.02	0.0303	0.0303	0.987	94.100 ± 0.745	0.119 ± 0.000	—
QML	beta54	0.16	0.159	-0.001 ± 0.001	-0.004 ± 0.006	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.036 ± 0.002	0.189 ± 0.005	0.160	0.000	-0.10	0.0297	0.0297	0.969	93.000 ± 0.807	0.115 ± 0.000	—
QML	beta55	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.001	0.001	—	0.0348	0.0348	1.044	95.300 ± 0.669	0.153 ± 0.002	4.700 ± 0.669
QML	beta56	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	-0.001	-0.001	—	0.0370	0.0370	1.000	95.300 ± 0.669	0.141 ± 0.001	4.700 ± 0.669
QML	beta57	0.00	-0.007	-0.007 ± 0.001	—	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	—	—	-0.008	-0.008	—	0.0316	0.0326	0.987	93.200 ± 0.796	0.119 ± 0.001	6.800 ± 0.796
QML	beta58	0.00	-0.005	-0.005 ± 0.001	—	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	-0.005	-0.005	—	0.0284	0.0289	1.022	95.700 ± 0.641	0.109 ± 0.000	4.300 ± 0.641
UPI	beta41	0.20	0.200	0.000 ± 0.001	0.002 ± 0.004	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.020 ± 0.001	0.141 ± 0.004	0.199	-0.001	-0.60	0.0271	0.0271	0.999	95.100 ± 0.683	0.110 ± 0.000	—
UPI	beta42	0.20	0.200	-0.000 ± 0.001	-0.001 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.021 ± 0.001	0.146 ± 0.004	0.200	0.000	-0.25	0.0289	0.0289	0.992	95.000 ± 0.689	0.114 ± 0.000	—
UPI	beta43	0.20	0.200	-0.000 ± 0.001	-0.001 ± 0.004	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	0.019 ± 0.001	0.137 ± 0.004	0.200	0.000	-0.16	0.0262	0.0262	1.016	94.800 ± 0.702	0.109 ± 0.000	—
UPI	beta44	0.00	0.002	0.002 ± 0.001	—	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	—	—	0.003	0.003	—	0.0311	0.0313	0.994	94.500 ± 0.721	0.123 ± 0.000	5.500 ± 0.721
UPI	beta45	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	-0.002	-0.002	—	0.0293	0.0293	0.987	94.400 ± 0.727	0.112 ± 0.000	5.600 ± 0.727
UPI	beta46	0.00	-0.005	-0.005 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.006	-0.006	—	0.0376	0.0382	0.985	94.700 ± 0.708	0.144 ± 0.000	5.300 ± 0.708

Table S252

Simulation 2 - Condition 108: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Normal, N = 1000, Reliability = 0.8
(continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.002	-0.002	—	0.0369	0.0370	0.976	94.800 ± 0.702	0.144 ± 0.000	5.200 ± 0.702
UPI	beta51	0.16	0.159	-0.001 ± 0.001	-0.006 ± 0.007	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.055 ± 0.003	0.235 ± 0.007	0.158	-0.002	-1.28	0.0371	0.0372	0.987	94.800 ± 0.702	0.145 ± 0.000	—
UPI	beta52	0.16	0.161	0.001 ± 0.001	0.006 ± 0.006	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.034 ± 0.001	0.184 ± 0.005	0.161	0.001	0.74	0.0293	0.0294	1.000	95.100 ± 0.683	0.115 ± 0.000	—
UPI	beta53	0.16	0.161	0.001 ± 0.001	0.006 ± 0.006	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.037 ± 0.002	0.193 ± 0.005	0.160	0.000	0.11	0.0301	0.0301	0.985	94.200 ± 0.739	0.119 ± 0.000	—
UPI	beta54	0.16	0.160	-0.000 ± 0.001	-0.003 ± 0.006	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.036 ± 0.002	0.190 ± 0.005	0.160	0.000	0.01	0.0294	0.0294	0.967	93.300 ± 0.791	0.115 ± 0.000	—
UPI	beta55	0.00	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	0.001	0.001	—	0.0341	0.0341	1.011	94.600 ± 0.715	0.150 ± 0.000	5.400 ± 0.715
UPI	beta56	0.00	0.002	0.002 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.001	0.001	—	0.0369	0.0369	0.989	95.100 ± 0.683	0.142 ± 0.000	4.900 ± 0.683
UPI	beta57	0.00	-0.005	-0.005 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	-0.005	-0.005	—	0.0405	0.0408	0.960	93.900 ± 0.757	0.146 ± 0.000	6.100 ± 0.757
UPI	beta58	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	-0.003	-0.003	—	0.0355	0.0356	1.015	96.300 ± 0.597	0.139 ± 0.000	3.700 ± 0.597

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S253

Simulation 2 - Condition 109: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.209	0.009 ± 0.003	0.044 ± 0.017	0.007 ± 0.000	0.082 ± 0.003	0.007 ± 0.000	0.082 ± 0.003	0.168 ± 0.011	0.410 ± 0.016	0.210	0.010	5.13	0.0817	0.0823	1.044	96.672 ± 0.732	0.334 ± 0.003	—
LSAM	beta42	0.20	0.197	-0.003 ± 0.004	-0.013 ± 0.018	0.008 ± 0.001	0.088 ± 0.003	0.008 ± 0.001	0.088 ± 0.003	0.192 ± 0.013	0.438 ± 0.017	0.193	-0.007	-3.37	0.0863	0.0866	1.005	95.341 ± 0.860	0.345 ± 0.003	—
LSAM	beta43	0.20	0.205	0.005 ± 0.003	0.024 ± 0.016	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.003	0.146 ± 0.009	0.382 ± 0.014	0.208	0.008	3.97	0.0806	0.0810	1.050	97.005 ± 0.695	0.314 ± 0.003	—
LSAM	beta44	0.00	0.000	0.000 ± 0.005	—	0.014 ± 0.001	0.120 ± 0.004	0.014 ± 0.001	0.119 ± 0.005	—	—	-0.004	-0.004	—	0.1155	0.1156	1.070	97.837 ± 0.593	0.502 ± 0.009	2.163 ± 0.593
LSAM	beta45	0.00	-0.002	-0.002 ± 0.004	—	0.011 ± 0.001	0.104 ± 0.004	0.011 ± 0.001	0.104 ± 0.004	—	—	0.002	0.002	—	0.1013	0.1013	0.996	97.671 ± 0.615	0.407 ± 0.007	2.329 ± 0.615
LSAM	beta46	0.00	0.002	0.002 ± 0.004	—	0.007 ± 0.001	0.086 ± 0.003	0.007 ± 0.001	0.086 ± 0.004	—	—	0.001	0.001	—	0.0810	0.0810	1.036	98.669 ± 0.467	0.350 ± 0.007	1.331 ± 0.467
LSAM	beta47	0.00	0.000	-0.000 ± 0.003	—	0.007 ± 0.001	0.084 ± 0.003	0.007 ± 0.001	0.083 ± 0.003	—	—	0.002	0.002	—	0.0796	0.0797	1.038	98.003 ± 0.571	0.340 ± 0.007	1.997 ± 0.571
LSAM	beta51	0.16	0.162	0.002 ± 0.005	0.010 ± 0.029	0.013 ± 0.001	0.114 ± 0.003	0.013 ± 0.001	0.114 ± 0.004	0.510 ± 0.031	0.714 ± 0.026	0.165	0.005	2.92	0.1204	0.1205	0.923	93.178 ± 1.028	0.414 ± 0.004	—
LSAM	beta52	0.16	0.163	0.003 ± 0.004	0.019 ± 0.022	0.008 ± 0.000	0.087 ± 0.003	0.008 ± 0.000	0.087 ± 0.003	0.295 ± 0.019	0.543 ± 0.021	0.161	0.001	0.56	0.0859	0.0859	0.988	96.339 ± 0.766	0.337 ± 0.003	—
LSAM	beta53	0.16	0.163	0.003 ± 0.004	0.019 ± 0.022	0.007 ± 0.000	0.086 ± 0.003	0.007 ± 0.000	0.086 ± 0.003	0.289 ± 0.019	0.537 ± 0.021	0.161	0.001	0.67	0.0838	0.0838	1.014	96.007 ± 0.799	0.342 ± 0.003	—
LSAM	beta54	0.16	0.164	0.004 ± 0.004	0.027 ± 0.022	0.007 ± 0.000	0.086 ± 0.003	0.007 ± 0.000	0.086 ± 0.003	0.291 ± 0.018	0.539 ± 0.020	0.163	0.003	2.05	0.0899	0.0900	0.980	94.842 ± 0.902	0.331 ± 0.003	—
LSAM	beta55	0.00	0.002	0.002 ± 0.007	—	0.026 ± 0.002	0.162 ± 0.006	0.026 ± 0.002	0.162 ± 0.007	—	—	0.001	0.001	—	0.1505	0.1505	1.007	97.671 ± 0.615	0.639 ± 0.013	2.329 ± 0.615
LSAM	beta56	0.00	0.000	-0.000 ± 0.005	—	0.018 ± 0.001	0.133 ± 0.005	0.018 ± 0.001	0.133 ± 0.005	—	—	-0.003	-0.003	—	0.1242	0.1242	0.994	96.839 ± 0.714	0.518 ± 0.009	3.161 ± 0.714
LSAM	beta57	0.00	-0.004	-0.004 ± 0.004	—	0.010 ± 0.001	0.097 ± 0.004	0.010 ± 0.001	0.097 ± 0.004	—	—	0.000	0.000	—	0.0867	0.0867	0.975	97.504 ± 0.636	0.372 ± 0.008	2.496 ± 0.636
LSAM	beta58	0.00	-0.001	-0.001 ± 0.003	—	0.005 ± 0.000	0.073 ± 0.003	0.005 ± 0.000	0.073 ± 0.003	—	—	-0.001	-0.001	—	0.0696	0.0696	0.988	97.005 ± 0.695	0.283 ± 0.005	2.995 ± 0.695
QML	beta41	0.20	0.203	0.003 ± 0.003	0.017 ± 0.015	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.003	0.128 ± 0.008	0.358 ± 0.014	0.202	0.002	1.21	0.0739	0.0739	1.058	96.506 ± 0.749	0.297 ± 0.002	—
QML	beta42	0.20	0.194	-0.006 ± 0.003	-0.030 ± 0.015	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.003	0.142 ± 0.009	0.377 ± 0.014	0.191	-0.009	-4.38	0.0738	0.0743	1.035	95.341 ± 0.860	0.306 ± 0.002	—
QML	beta43	0.20	0.202	0.002 ± 0.003	0.008 ± 0.014	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.069 ± 0.003	0.121 ± 0.007	0.347 ± 0.013	0.204	0.004	1.81	0.0741	0.0742	1.062	97.171 ± 0.676	0.289 ± 0.002	—
QML	beta44	0.00	-0.023	-0.023 ± 0.004	—	0.010 ± 0.001	0.101 ± 0.003	0.011 ± 0.001	0.103 ± 0.004	—	—	-0.022	-0.022	—	0.0986	0.1011	1.078	97.504 ± 0.636	0.426 ± 0.007	2.496 ± 0.636
QML	beta45	0.00	-0.014	-0.014 ± 0.004	—	0.008 ± 0.000	0.088 ± 0.003	0.008 ± 0.000	0.089 ± 0.003	—	—	-0.011	-0.011	—	0.0963	0.0968	0.993	96.506 ± 0.749	0.344 ± 0.005	3.494 ± 0.749
QML	beta46	0.00	0.010	0.010 ± 0.003	—	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.071 ± 0.003	—	—	0.006	0.006	—	0.0700	0.0703	1.056	97.171 ± 0.676	0.290 ± 0.006	2.829 ± 0.676
QML	beta47	0.00	0.001	0.001 ± 0.003	—	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.003	—	—	0.000	0.000	—	0.0630	0.0630	1.046	96.506 ± 0.749	0.272 ± 0.005	3.494 ± 0.749
QML	beta51	0.16	0.167	0.007 ± 0.005	0.043 ± 0.028	0.012 ± 0.001	0.112 ± 0.003	0.012 ± 0.001	0.112 ± 0.004	0.487 ± 0.030	0.698 ± 0.026	0.165	0.005	3.10	0.1094	0.1095	1.008	95.507 ± 0.845	0.441 ± 0.004	—
QML	beta52	0.16	0.159	-0.001 ± 0.003	-0.006 ± 0.021	0.007 ± 0.000	0.082 ± 0.002	0.007 ± 0.000	0.081 ± 0.003	0.259 ± 0.016	0.509 ± 0.019	0.157	-0.003	-1.58	0.0830	0.0830	1.054	96.672 ± 0.732	0.337 ± 0.004	—
QML	beta53	0.16	0.161	0.001 ± 0.003	0.006 ± 0.020	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.079 ± 0.003	0.246 ± 0.015	0.496 ± 0.018	0.158	-0.002	-1.10	0.0767	0.0767	1.055	96.007 ± 0.799	0.328 ± 0.002	—
QML	beta54	0.16	0.159	-0.001 ± 0.003	-0.006 ± 0.021	0.007 ± 0.000	0.082 ± 0.003	0.007 ± 0.000	0.081 ± 0.003	0.259 ± 0.016	0.509 ± 0.019	0.160	0.000	-0.27	0.0830	0.0830	0.970	95.341 ± 0.860	0.310 ± 0.002	—
QML	beta55	0.00	-0.010	-0.010 ± 0.005	—	0.017 ± 0.001	0.131 ± 0.004	0.017 ± 0.001	0.131 ± 0.005	—	—	-0.017	-0.017	—	0.1330	0.1340	1.070	97.171 ± 0.676	0.548 ± 0.010	2.829 ± 0.676
QML	beta56	0.00	-0.013	-0.013 ± 0.004	—	0.011 ± 0.001	0.106 ± 0.003	0.011 ± 0.001	0.107 ± 0.004	—	—	-0.010	-0.010	—	0.1063	0.1068	1.023	96.506 ± 0.749	0.425 ± 0.005	3.494 ± 0.749
QML	beta57	0.00	0.000	-0.000 ± 0.003	—	0.006 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.074 ± 0.003	—	—	-0.001	-0.001	—	0.0738	0.0738	1.056	95.175 ± 0.874	0.307 ± 0.007	4.825 ± 0.874
QML	beta58	0.00	-0.002	-0.002 ± 0.002	—	0.003 ± 0.000	0.059 ± 0.002	0.003 ± 0.000	0.059 ± 0.002	—	—	-0.003	-0.003	—	0.0570	0.0571	1.012	94.509 ± 0.929	0.235 ± 0.004	5.491 ± 0.929
UPI	beta41	0.20	0.211	0.011 ± 0.004	0.053 ± 0.019	0.008 ± 0.001	0.091 ± 0.003	0.008 ± 0.001	0.091 ± 0.004	0.209 ± 0.014	0.457 ± 0.018	0.216	0.016	8.13	0.1087	0.1099	1.285	98.336 ± 0.522	0.457 ± 0.014	—
UPI	beta42	0.20	0.197	-0.003 ± 0.004	-0.016 ± 0.020	0.009 ± 0.001	0.096 ± 0.003	0.009 ± 0.001	0.096 ± 0.003	0.230 ± 0.014	0.480 ± 0.017	0.187	-0.013	-6.47	0.1139	0.1147	1.298	98.170 ± 0.547	0.488 ± 0.016	—
UPI	beta43	0.20	0.208	0.008 ± 0.003	0.041 ± 0.017	0.007 ± 0.001	0.082 ± 0.003	0.007 ± 0.001	0.083 ± 0.004	0.171 ± 0.013	0.414 ± 0.018	0.212	0.012	6.09	0.0984	0.0992	1.288	98.336 ± 0.522	0.416 ± 0.010	—
UPI	beta44	0.00	0.010	0.010 ± 0.007	—	0.031 ± 0.003	0.175 ± 0.008	0.031 ± 0.003	0.175 ± 0.009	—	—	-0.014	-0.014	—	0.1724	0.1730	1.492	99.834 ± 0.166	1.022 ± 0.046	0.166 ± 0.166
UPI	beta45	0.00	-0.003	-0.003 ± 0.006	—	0.022 ± 0.002	0.149 ± 0.006	0.022 ± 0.002	0.149 ± 0.007	—	—	0.003	0.003	—	0.1518	0.1518	1.362	99.667 ± 0.235	0.794 ± 0.031	0.333 ± 0.235
UPI	beta46	0.00	0.002	0.002 ± 0.005	—	0.016 ± 0.002	0.127 ± 0.006	0.016 ± 0.002	0.127 ± 0.007	—	—	0.004	0.004	—	0.1183	0.1184	1.493	99.667 ± 0.235	0.741 ± 0.034	0.333 ± 0.235

Table S253

Simulation 2 - Condition 109: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.4
(continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	0.001	0.001 ± 0.005	—	0.016 ± 0.001	0.126 ± 0.005	0.016 ± 0.001	0.126 ± 0.006	—	—	0.010	0.010	—	0.1101	0.1106	1.503	99.834 ± 0.166	0.743 ± 0.044	0.166 ± 0.166
UPI	beta51	0.16	0.163	0.003 ± 0.004	0.018 ± 0.028	0.012 ± 0.001	0.110 ± 0.003	0.012 ± 0.001	0.110 ± 0.004	0.469 ± 0.029	0.685 ± 0.025	0.161	0.001	0.32	0.1163	0.1163	1.037	96.339 ± 0.766	0.446 ± 0.005	—
UPI	beta52	0.16	0.164	0.004 ± 0.003	0.023 ± 0.021	0.007 ± 0.000	0.083 ± 0.003	0.007 ± 0.000	0.083 ± 0.003	0.270 ± 0.017	0.520 ± 0.020	0.160	0.000	0.22	0.0909	0.0909	1.181	98.336 ± 0.522	0.385 ± 0.007	—
UPI	beta53	0.16	0.163	0.003 ± 0.003	0.020 ± 0.022	0.007 ± 0.000	0.085 ± 0.003	0.007 ± 0.000	0.085 ± 0.003	0.281 ± 0.019	0.530 ± 0.021	0.163	0.003	2.16	0.0891	0.0891	1.182	97.504 ± 0.636	0.393 ± 0.007	—
UPI	beta54	0.16	0.164	0.004 ± 0.004	0.027 ± 0.022	0.008 ± 0.000	0.088 ± 0.003	0.008 ± 0.001	0.088 ± 0.003	0.301 ± 0.020	0.548 ± 0.021	0.165	0.005	3.02	0.0918	0.0919	1.114	96.339 ± 0.766	0.383 ± 0.006	—
UPI	beta55	0.00	0.003	0.003 ± 0.007	—	0.028 ± 0.002	0.166 ± 0.007	0.027 ± 0.002	0.166 ± 0.007	—	—	-0.004	-0.004	—	0.1506	0.1507	1.461	99.501 ± 0.287	0.950 ± 0.038	0.499 ± 0.287
UPI	beta56	0.00	0.001	0.001 ± 0.006	—	0.019 ± 0.002	0.139 ± 0.006	0.019 ± 0.002	0.139 ± 0.006	—	—	-0.010	-0.010	—	0.1322	0.1326	1.424	99.667 ± 0.235	0.775 ± 0.031	0.333 ± 0.235
UPI	beta57	0.00	-0.005	-0.005 ± 0.005	—	0.012 ± 0.001	0.111 ± 0.005	0.012 ± 0.001	0.111 ± 0.005	—	—	-0.002	-0.002	—	0.0989	0.0989	1.363	98.835 ± 0.438	0.591 ± 0.026	1.165 ± 0.438
UPI	beta58	0.00	-0.001	-0.001 ± 0.003	—	0.007 ± 0.001	0.081 ± 0.003	0.007 ± 0.001	0.081 ± 0.004	—	—	-0.004	-0.004	—	0.0713	0.0715	1.332	99.501 ± 0.287	0.423 ± 0.014	0.499 ± 0.287

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S254

Simulation 2 - Condition 110: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.4

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.201	0.001 ± 0.002	0.007 ± 0.008	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.056 ± 0.003	0.237 ± 0.007	0.203	0.003	1.55	0.0496	0.0497	1.021	95.124 ± 0.694	0.190 ± 0.001	—
LSAM	beta42	0.20	0.201	0.001 ± 0.002	0.004 ± 0.008	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	0.068 ± 0.003	0.261 ± 0.008	0.201	0.001	0.47	0.0530	0.0530	0.963	94.191 ± 0.753	0.197 ± 0.001	—
LSAM	beta43	0.20	0.201	0.001 ± 0.001	0.006 ± 0.007	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.053 ± 0.002	0.229 ± 0.007	0.200	0.000	-0.16	0.0452	0.0452	1.024	95.954 ± 0.635	0.184 ± 0.001	—
LSAM	beta44	0.00	0.003	0.003 ± 0.002	—	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	—	—	0.003	0.003	—	0.0649	0.0649	1.034	97.407 ± 0.512	0.261 ± 0.002	2.593 ± 0.512
LSAM	beta45	0.00	-0.002	-0.002 ± 0.002	—	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	—	—	-0.004	-0.004	—	0.0542	0.0543	0.976	95.851 ± 0.642	0.216 ± 0.001	4.149 ± 0.642
LSAM	beta46	0.00	-0.002	-0.002 ± 0.002	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	-0.005	-0.005	—	0.0454	0.0456	0.983	95.747 ± 0.650	0.183 ± 0.002	4.253 ± 0.650
LSAM	beta47	0.00	0.000	-0.000 ± 0.001	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	—	—	0.000	0.000	—	0.0448	0.0448	1.013	96.266 ± 0.611	0.176 ± 0.001	3.734 ± 0.611
LSAM	beta51	0.16	0.160	0.000 ± 0.002	0.001 ± 0.012	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	0.149 ± 0.007	0.386 ± 0.011	0.159	-0.001	-0.60	0.0623	0.0623	1.007	95.228 ± 0.687	0.244 ± 0.001	—
LSAM	beta52	0.16	0.162	0.002 ± 0.002	0.014 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.002	0.101 ± 0.005	0.317 ± 0.009	0.161	0.001	0.47	0.0489	0.0489	0.978	94.606 ± 0.728	0.194 ± 0.001	—
LSAM	beta53	0.16	0.162	0.002 ± 0.002	0.010 ± 0.010	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.093 ± 0.004	0.304 ± 0.009	0.162	0.002	0.97	0.0469	0.0470	1.051	97.095 ± 0.541	0.201 ± 0.001	—
LSAM	beta54	0.16	0.163	0.003 ± 0.002	0.019 ± 0.010	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.089 ± 0.004	0.298 ± 0.009	0.164	0.004	2.51	0.0481	0.0483	1.032	96.266 ± 0.611	0.192 ± 0.001	—
LSAM	beta55	0.00	0.005	0.005 ± 0.003	—	0.008 ± 0.000	0.088 ± 0.002	0.008 ± 0.000	0.088 ± 0.003	—	—	0.006	0.006	—	0.0827	0.0829	0.957	96.680 ± 0.577	0.331 ± 0.003	3.320 ± 0.577
LSAM	beta56	0.00	-0.002	-0.002 ± 0.002	—	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	—	—	-0.005	-0.005	—	0.0632	0.0635	0.977	96.058 ± 0.627	0.270 ± 0.002	3.942 ± 0.627
LSAM	beta57	0.00	-0.003	-0.003 ± 0.002	—	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.002	—	—	-0.004	-0.004	—	0.0467	0.0469	0.981	96.577 ± 0.586	0.193 ± 0.002	3.423 ± 0.586
LSAM	beta58	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.039 ± 0.001	0.001 ± 0.000	0.039 ± 0.001	—	—	-0.001	-0.001	—	0.0375	0.0375	1.020	95.851 ± 0.642	0.155 ± 0.001	4.149 ± 0.642
QML	beta41	0.20	0.199	-0.001 ± 0.001	-0.007 ± 0.007	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.048 ± 0.002	0.219 ± 0.006	0.200	0.000	-0.12	0.0445	0.0445	1.051	95.228 ± 0.687	0.181 ± 0.001	—
QML	beta42	0.20	0.198	-0.002 ± 0.002	-0.009 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.058 ± 0.003	0.242 ± 0.007	0.197	-0.003	-1.50	0.0488	0.0489	0.983	94.813 ± 0.714	0.186 ± 0.001	—
QML	beta43	0.20	0.198	-0.002 ± 0.001	-0.008 ± 0.007	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.048 ± 0.002	0.219 ± 0.006	0.197	-0.003	-1.73	0.0432	0.0434	1.041	96.058 ± 0.627	0.178 ± 0.001	—
QML	beta44	0.00	-0.017	-0.017 ± 0.002	—	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	—	—	-0.013	-0.013	—	0.0622	0.0636	1.053	97.303 ± 0.522	0.256 ± 0.004	2.697 ± 0.522
QML	beta45	0.00	-0.015	-0.015 ± 0.002	—	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	—	—	-0.016	-0.016	—	0.0542	0.0565	0.983	94.502 ± 0.734	0.207 ± 0.002	5.498 ± 0.734
QML	beta46	0.00	0.002	0.002 ± 0.001	—	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	—	—	0.000	0.000	—	0.0410	0.0410	1.004	95.124 ± 0.694	0.169 ± 0.002	4.876 ± 0.694
QML	beta47	0.00	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	-0.002	-0.002	—	0.0384	0.0385	1.035	95.021 ± 0.701	0.161 ± 0.002	4.979 ± 0.701
QML	beta51	0.16	0.163	0.003 ± 0.002	0.016 ± 0.012	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.144 ± 0.007	0.380 ± 0.011	0.161	0.001	0.60	0.0614	0.0614	1.056	96.266 ± 0.611	0.251 ± 0.001	—
QML	beta52	0.16	0.160	-0.000 ± 0.002	-0.001 ± 0.010	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.095 ± 0.005	0.309 ± 0.009	0.160	0.000	0.25	0.0475	0.0475	0.999	95.124 ± 0.694	0.193 ± 0.001	—
QML	beta53	0.16	0.159	-0.001 ± 0.002	-0.004 ± 0.009	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.086 ± 0.004	0.293 ± 0.008	0.158	-0.002	-1.24	0.0450	0.0450	1.062	96.266 ± 0.611	0.195 ± 0.001	—
QML	beta54	0.16	0.161	0.001 ± 0.001	0.007 ± 0.009	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.082 ± 0.004	0.286 ± 0.008	0.160	0.000	-0.01	0.0458	0.0458	1.046	96.369 ± 0.602	0.188 ± 0.001	—
QML	beta55	0.00	-0.003	-0.003 ± 0.003	—	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	—	—	-0.003	-0.003	—	0.0764	0.0764	1.034	95.954 ± 0.635	0.316 ± 0.004	4.046 ± 0.635
QML	beta56	0.00	-0.014	-0.014 ± 0.002	—	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	—	—	-0.016	-0.016	—	0.0602	0.0623	0.996	93.154 ± 0.813	0.247 ± 0.002	6.846 ± 0.813
QML	beta57	0.00	-0.004	-0.004 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	-0.006	-0.006	—	0.0431	0.0435	1.025	95.228 ± 0.687	0.183 ± 0.003	4.772 ± 0.687
QML	beta58	0.00	-0.005	-0.005 ± 0.001	—	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	-0.006	-0.006	—	0.0331	0.0337	1.042	94.191 ± 0.753	0.140 ± 0.001	5.809 ± 0.753
UPI	beta41	0.20	0.203	0.003 ± 0.002	0.014 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.058 ± 0.003	0.241 ± 0.007	0.205	0.005	2.33	0.0498	0.0500	1.066	96.369 ± 0.602	0.201 ± 0.001	—
UPI	beta42	0.20	0.200	0.000 ± 0.002	0.002 ± 0.008	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	0.069 ± 0.003	0.263 ± 0.008	0.200	0.000	-0.24	0.0536	0.0536	1.010	95.436 ± 0.672	0.209 ± 0.001	—
UPI	beta43	0.20	0.202	0.002 ± 0.002	0.008 ± 0.008	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.055 ± 0.003	0.234 ± 0.007	0.199	-0.001	-0.60	0.0468	0.0468	1.055	96.369 ± 0.602	0.193 ± 0.001	—
UPI	beta44	0.00	0.003	0.003 ± 0.002	—	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	—	—	0.002	0.002	—	0.0667	0.0668	1.154	98.963 ± 0.326	0.339 ± 0.005	1.037 ± 0.326
UPI	beta45	0.00	-0.001	-0.001 ± 0.002	—	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	—	—	-0.002	-0.002	—	0.0568	0.0569	1.073	98.340 ± 0.411	0.262 ± 0.003	1.660 ± 0.411
UPI	beta46	0.00	-0.001	-0.001 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	—	—	-0.004	-0.004	—	0.0495	0.0496	1.140	98.548 ± 0.385	0.238 ± 0.004	1.452 ± 0.385

Table S254

Simulation 2 - Condition 110: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.4
(continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	0.001	0.001 ± 0.002	—	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.002	—	—	0.001	0.001	—	0.0464	0.0464	1.146	98.651 ± 0.371	0.225 ± 0.004	1.349 ± 0.371
UPI	beta51	0.16	0.160	-0.000 ± 0.002	-0.001 ± 0.012	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.140 ± 0.007	0.375 ± 0.011	0.157	-0.003	-1.67	0.0607	0.0608	1.043	96.162 ± 0.619	0.245 ± 0.001	—
UPI	beta52	0.16	0.162	0.002 ± 0.002	0.014 ± 0.010	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.098 ± 0.005	0.313 ± 0.009	0.161	0.001	0.72	0.0484	0.0484	1.014	95.124 ± 0.694	0.199 ± 0.001	—
UPI	beta53	0.16	0.162	0.002 ± 0.002	0.011 ± 0.010	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.093 ± 0.004	0.305 ± 0.009	0.160	0.000	-0.17	0.0456	0.0456	1.069	96.577 ± 0.586	0.205 ± 0.001	—
UPI	beta54	0.16	0.163	0.003 ± 0.002	0.020 ± 0.010	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.089 ± 0.004	0.298 ± 0.009	0.164	0.004	2.62	0.0476	0.0478	1.052	96.473 ± 0.594	0.196 ± 0.001	—
UPI	beta55	0.00	0.004	0.004 ± 0.003	—	0.008 ± 0.000	0.089 ± 0.003	0.008 ± 0.000	0.089 ± 0.003	—	—	0.004	0.004	—	0.0815	0.0816	1.057	98.548 ± 0.385	0.370 ± 0.006	1.452 ± 0.385
UPI	beta56	0.00	-0.002	-0.002 ± 0.002	—	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	—	—	-0.002	-0.002	—	0.0675	0.0676	1.086	98.237 ± 0.424	0.299 ± 0.004	1.763 ± 0.424
UPI	beta57	0.00	-0.003	-0.003 ± 0.002	—	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.002	—	—	-0.001	-0.001	—	0.0476	0.0476	1.093	98.237 ± 0.424	0.216 ± 0.003	1.763 ± 0.424
UPI	beta58	0.00	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	0.000	0.000	—	0.0379	0.0379	1.085	98.029 ± 0.448	0.177 ± 0.002	1.971 ± 0.448

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S255

Simulation 2 - Condition 111: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.6

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Type I (%) $\pm$ MCSE
LSAM	beta41	0.20	0.202	0.002 $\pm$ 0.002	0.010 $\pm$ 0.009	0.003 $\pm$ 0.000	0.055 $\pm$ 0.001	0.003 $\pm$ 0.000	0.055 $\pm$ 0.002	0.076 $\pm$ 0.003	0.275 $\pm$ 0.008	0.200	0.000	0.22	0.0535	0.0535	1.013	94.383 $\pm$ 0.729	0.218 $\pm$ 0.001	—
LSAM	beta42	0.20	0.202	0.002 $\pm$ 0.002	0.010 $\pm$ 0.010	0.004 $\pm$ 0.000	0.060 $\pm$ 0.001	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	0.090 $\pm$ 0.004	0.300 $\pm$ 0.008	0.200	0.000	-0.09	0.0613	0.0613	0.955	94.584 $\pm$ 0.717	0.225 $\pm$ 0.001	—
LSAM	beta43	0.20	0.200	-0.000 $\pm$ 0.002	-0.001 $\pm$ 0.009	0.003 $\pm$ 0.000	0.056 $\pm$ 0.001	0.003 $\pm$ 0.000	0.056 $\pm$ 0.002	0.079 $\pm$ 0.004	0.281 $\pm$ 0.008	0.197	-0.003	-1.53	0.0557	0.0557	0.968	93.882 $\pm$ 0.759	0.213 $\pm$ 0.001	—
LSAM	beta44	0.00	0.004	0.004 $\pm$ 0.002	—	0.005 $\pm$ 0.000	0.071 $\pm$ 0.002	0.005 $\pm$ 0.000	0.071 $\pm$ 0.002	—	—	0.006	0.006	—	0.0693	0.0696	0.955	94.784 $\pm$ 0.704	0.267 $\pm$ 0.002	5.216 $\pm$ 0.704
LSAM	beta45	0.00	0.002	0.002 $\pm$ 0.002	—	0.004 $\pm$ 0.000	0.060 $\pm$ 0.001	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	—	—	0.002	0.002	—	0.0592	0.0592	0.975	95.085 $\pm$ 0.685	0.229 $\pm$ 0.002	4.915 $\pm$ 0.685
LSAM	beta46	0.00	-0.001	-0.001 $\pm$ 0.001	—	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	—	—	-0.003	-0.003	—	0.0453	0.0454	1.018	95.687 $\pm$ 0.643	0.188 $\pm$ 0.002	4.313 $\pm$ 0.643
LSAM	beta47	0.00	-0.002	-0.002 $\pm$ 0.001	—	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	—	—	-0.001	-0.001	—	0.0432	0.0432	0.997	95.386 $\pm$ 0.664	0.182 $\pm$ 0.002	4.614 $\pm$ 0.664
LSAM	beta51	0.16	0.162	0.002 $\pm$ 0.002	0.011 $\pm$ 0.015	0.006 $\pm$ 0.000	0.077 $\pm$ 0.002	0.006 $\pm$ 0.000	0.077 $\pm$ 0.002	0.231 $\pm$ 0.011	0.481 $\pm$ 0.014	0.162	0.002	1.11	0.0758	0.0759	0.937	92.778 $\pm$ 0.820	0.282 $\pm$ 0.001	—
LSAM	beta52	0.16	0.159	-0.001 $\pm$ 0.002	-0.008 $\pm$ 0.011	0.003 $\pm$ 0.000	0.057 $\pm$ 0.001	0.003 $\pm$ 0.000	0.057 $\pm$ 0.002	0.126 $\pm$ 0.005	0.355 $\pm$ 0.010	0.157	-0.003	-1.66	0.0571	0.0572	1.017	96.389 $\pm$ 0.591	0.226 $\pm$ 0.001	—
LSAM	beta53	0.16	0.161	0.001 $\pm$ 0.002	0.009 $\pm$ 0.012	0.004 $\pm$ 0.000	0.061 $\pm$ 0.001	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.147 $\pm$ 0.007	0.384 $\pm$ 0.011	0.161	0.001	0.73	0.0613	0.0613	0.968	94.082 $\pm$ 0.747	0.233 $\pm$ 0.001	—
LSAM	beta54	0.16	0.163	0.003 $\pm$ 0.002	0.018 $\pm$ 0.012	0.003 $\pm$ 0.000	0.059 $\pm$ 0.001	0.003 $\pm$ 0.000	0.059 $\pm$ 0.002	0.134 $\pm$ 0.007	0.366 $\pm$ 0.011	0.161	0.001	0.47	0.0553	0.0553	0.978	94.383 $\pm$ 0.729	0.225 $\pm$ 0.001	—
LSAM	beta55	0.00	-0.003	-0.003 $\pm$ 0.003	—	0.008 $\pm$ 0.000	0.090 $\pm$ 0.002	0.008 $\pm$ 0.000	0.090 $\pm$ 0.003	—	—	-0.001	-0.001	—	0.0826	0.0826	0.945	93.280 $\pm$ 0.793	0.335 $\pm$ 0.003	6.720 $\pm$ 0.793
LSAM	beta56	0.00	0.007	0.007 $\pm$ 0.003	—	0.007 $\pm$ 0.000	0.081 $\pm$ 0.002	0.007 $\pm$ 0.000	0.082 $\pm$ 0.002	—	—	0.004	0.004	—	0.0745	0.0746	0.899	92.678 $\pm$ 0.825	0.286 $\pm$ 0.002	7.322 $\pm$ 0.825
LSAM	beta57	0.00	0.001	0.001 $\pm$ 0.002	—	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.003 $\pm$ 0.000	0.053 $\pm$ 0.002	—	—	-0.001	-0.001	—	0.0517	0.0517	0.954	94.584 $\pm$ 0.717	0.198 $\pm$ 0.002	5.416 $\pm$ 0.717
LSAM	beta58	0.00	0.001	0.001 $\pm$ 0.001	—	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	—	—	0.000	0.000	—	0.0425	0.0425	0.970	94.584 $\pm$ 0.717	0.167 $\pm$ 0.001	5.416 $\pm$ 0.717
QML	beta41	0.20	0.199	-0.001 $\pm$ 0.002	-0.003 $\pm$ 0.008	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.071 $\pm$ 0.003	0.266 $\pm$ 0.007	0.198	-0.002	-1.12	0.0525	0.0526	1.044	95.286 $\pm$ 0.671	0.218 $\pm$ 0.004	—
QML	beta42	0.20	0.200	-0.000 $\pm$ 0.002	-0.001 $\pm$ 0.009	0.003 $\pm$ 0.000	0.058 $\pm$ 0.001	0.003 $\pm$ 0.000	0.058 $\pm$ 0.002	0.084 $\pm$ 0.004	0.289 $\pm$ 0.008	0.199	-0.001	-0.67	0.0590	0.0591	0.975	94.584 $\pm$ 0.717	0.221 $\pm$ 0.002	—
QML	beta43	0.20	0.198	-0.002 $\pm$ 0.002	-0.008 $\pm$ 0.009	0.003 $\pm$ 0.000	0.055 $\pm$ 0.001	0.003 $\pm$ 0.000	0.055 $\pm$ 0.002	0.076 $\pm$ 0.004	0.275 $\pm$ 0.008	0.197	-0.003	-1.52	0.0537	0.0538	0.983	94.383 $\pm$ 0.729	0.212 $\pm$ 0.001	—
QML	beta44	0.00	-0.009	-0.009 $\pm$ 0.002	—	0.005 $\pm$ 0.000	0.069 $\pm$ 0.002	0.005 $\pm$ 0.000	0.070 $\pm$ 0.002	—	—	-0.006	-0.006	—	0.0634	0.0637	1.094	94.985 $\pm$ 0.691	0.297 $\pm$ 0.021	5.015 $\pm$ 0.691
QML	beta45	0.00	-0.005	-0.005 $\pm$ 0.002	—	0.003 $\pm$ 0.000	0.058 $\pm$ 0.002	0.003 $\pm$ 0.000	0.059 $\pm$ 0.002	—	—	-0.007	-0.007	—	0.0579	0.0583	1.046	95.186 $\pm$ 0.678	0.240 $\pm$ 0.008	4.814 $\pm$ 0.678
QML	beta46	0.00	0.003	0.003 $\pm$ 0.001	—	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	—	—	0.001	0.001	—	0.0425	0.0425	1.220	95.888 $\pm$ 0.629	0.210 $\pm$ 0.025	4.112 $\pm$ 0.629
QML	beta47	0.00	-0.001	-0.001 $\pm$ 0.001	—	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	—	—	-0.002	-0.002	—	0.0416	0.0416	1.088	94.985 $\pm$ 0.691	0.183 $\pm$ 0.007	5.015 $\pm$ 0.691
QML	beta51	0.16	0.163	0.003 $\pm$ 0.002	0.017 $\pm$ 0.015	0.006 $\pm$ 0.000	0.076 $\pm$ 0.002	0.006 $\pm$ 0.000	0.076 $\pm$ 0.002	0.224 $\pm$ 0.010	0.473 $\pm$ 0.013	0.165	0.005	3.22	0.0748	0.0749	0.988	94.584 $\pm$ 0.717	0.293 $\pm$ 0.002	—
QML	beta52	0.16	0.158	-0.002 $\pm$ 0.002	-0.015 $\pm$ 0.011	0.003 $\pm$ 0.000	0.056 $\pm$ 0.001	0.003 $\pm$ 0.000	0.056 $\pm$ 0.002	0.123 $\pm$ 0.005	0.351 $\pm$ 0.009	0.156	-0.004	-2.59	0.0582	0.0584	1.042	96.389 $\pm$ 0.591	0.229 $\pm$ 0.001	—
QML	beta53	0.16	0.160	-0.000 $\pm$ 0.002	-0.001 $\pm$ 0.012	0.004 $\pm$ 0.000	0.060 $\pm$ 0.001	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	0.142 $\pm$ 0.007	0.377 $\pm$ 0.011	0.160	0.000	-0.20	0.0586	0.0586	0.981	94.082 $\pm$ 0.747	0.232 $\pm$ 0.001	—
QML	beta54	0.16	0.162	0.002 $\pm$ 0.002	0.009 $\pm$ 0.011	0.003 $\pm$ 0.000	0.057 $\pm$ 0.001	0.003 $\pm$ 0.000	0.057 $\pm$ 0.002	0.129 $\pm$ 0.006	0.359 $\pm$ 0.011	0.161	0.001	0.35	0.0527	0.0527	0.987	94.684 $\pm$ 0.711	0.223 $\pm$ 0.001	—
QML	beta55	0.00	-0.009	-0.009 $\pm$ 0.003	—	0.007 $\pm$ 0.000	0.083 $\pm$ 0.002	0.007 $\pm$ 0.000	0.083 $\pm$ 0.002	—	—	-0.007	-0.007	—	0.0768	0.0771	1.012	94.283 $\pm$ 0.735	0.328 $\pm$ 0.003	5.717 $\pm$ 0.735
QML	beta56	0.00	0.001	0.001 $\pm$ 0.002	—	0.006 $\pm$ 0.000	0.076 $\pm$ 0.002	0.006 $\pm$ 0.000	0.076 $\pm$ 0.002	—	—	-0.002	-0.002	—	0.0720	0.0720	0.923	93.882 $\pm$ 0.759	0.274 $\pm$ 0.002	6.118 $\pm$ 0.759
QML	beta57	0.00	0.002	0.002 $\pm$ 0.002	—	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.002	—	—	-0.001	-0.001	—	0.0473	0.0473	0.999	94.483 $\pm$ 0.723	0.193 $\pm$ 0.002	5.517 $\pm$ 0.723
QML	beta58	0.00	0.001	0.001 $\pm$ 0.001	—	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	—	—	-0.003	-0.003	—	0.0382	0.0383	0.981	94.283 $\pm$ 0.735	0.159 $\pm$ 0.002	5.717 $\pm$ 0.735
UPI	beta41	0.20	0.202	0.002 $\pm$ 0.002	0.010 $\pm$ 0.009	0.003 $\pm$ 0.000	0.055 $\pm$ 0.001	0.003 $\pm$ 0.000	0.055 $\pm$ 0.002	0.076 $\pm$ 0.004	0.276 $\pm$ 0.008	0.202	0.002	1.19	0.0506	0.0507	1.043	94.684 $\pm$ 0.711	0.226 $\pm$ 0.001	—
UPI	beta42	0.20	0.202	0.002 $\pm$ 0.002	0.008 $\pm$ 0.009	0.004 $\pm$ 0.000	0.060 $\pm$ 0.001	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	0.089 $\pm$ 0.004	0.298 $\pm$ 0.008	0.201	0.001	0.35	0.0620	0.0620	0.990	94.985 $\pm$ 0.691	0.231 $\pm$ 0.001	—
UPI	beta43	0.20	0.200	0.000 $\pm$ 0.002	0.002 $\pm$ 0.009	0.003 $\pm$ 0.000	0.057 $\pm$ 0.001	0.003 $\pm$ 0.000	0.057 $\pm$ 0.002	0.081 $\pm$ 0.004	0.284 $\pm$ 0.008	0.197	-0.003	-1.25	0.0557	0.0558	0.990	94.784 $\pm$ 0.704	0.221 $\pm$ 0.001	—
UPI	beta44	0.00	0.001	0.001 $\pm$ 0.003	—	0.006 $\pm$ 0.000	0.080 $\pm$ 0.002	0.006 $\pm$ 0.000	0.080 $\pm$ 0.003	—	—	-0.001	-0.001	—	0.0692	0.0692	1.000	95.787 $\pm$ 0.636	0.313 $\pm$ 0.004	4.213 $\pm$ 0.636
UPI	beta45	0.00	0.003	0.003 $\pm$ 0.002	—	0.004 $\pm$ 0.000	0.065 $\pm$ 0.002	0.004 $\pm$ 0.000	0.065 $\pm$ 0.002	—	—	0.004	0.004	—	0.0608	0.0609	1.016	96.189 $\pm$ 0.606	0.260 $\pm$ 0.002	3.811 $\pm$ 0.606
UPI	beta46	0.00	0.001	0.001 $\pm$ 0.002	—	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.003 $\pm$ 0.000	0.053 $\pm$ 0.002	—	—	0.000	0.000	—	0.0499	0.0499	1.070	96.489 $\pm$ 0.583	0.224 $\pm$ 0.003	3.511 $\pm$ 0.583

Table S255

Simulation 2 - Condition 111: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.6
(continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	0.000	-0.000 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.002	—	—	-0.001	-0.001	—	0.0481	0.0481	1.088	97.392 ± 0.505	0.217 ± 0.003	2.608 ± 0.505
UPI	beta51	0.16	0.162	0.002 ± 0.002	0.011 ± 0.015	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	0.219 ± 0.010	0.468 ± 0.013	0.163	0.003	1.94	0.0714	0.0715	0.977	94.283 ± 0.735	0.286 ± 0.001	—
UPI	beta52	0.16	0.159	-0.001 ± 0.002	-0.006 ± 0.011	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.127 ± 0.006	0.356 ± 0.010	0.157	-0.003	-2.10	0.0582	0.0583	1.026	96.389 ± 0.591	0.229 ± 0.001	—
UPI	beta53	0.16	0.161	0.001 ± 0.002	0.008 ± 0.012	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.148 ± 0.007	0.384 ± 0.011	0.160	0.000	-0.04	0.0600	0.0600	0.981	94.283 ± 0.735	0.236 ± 0.001	—
UPI	beta54	0.16	0.163	0.003 ± 0.002	0.018 ± 0.012	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.134 ± 0.007	0.367 ± 0.011	0.162	0.002	1.30	0.0564	0.0564	0.990	94.684 ± 0.711	0.228 ± 0.001	—
UPI	beta55	0.00	-0.004	-0.004 ± 0.003	—	0.008 ± 0.000	0.091 ± 0.002	0.008 ± 0.000	0.091 ± 0.003	—	—	0.000	0.000	—	0.0829	0.0829	0.987	94.885 ± 0.698	0.352 ± 0.004	5.115 ± 0.698
UPI	beta56	0.00	0.006	0.006 ± 0.003	—	0.007 ± 0.000	0.084 ± 0.002	0.007 ± 0.000	0.084 ± 0.003	—	—	0.005	0.005	—	0.0767	0.0769	0.937	95.286 ± 0.671	0.307 ± 0.003	4.714 ± 0.671
UPI	beta57	0.00	0.001	0.001 ± 0.002	—	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	—	—	-0.002	-0.002	—	0.0510	0.0510	1.001	95.787 ± 0.636	0.212 ± 0.002	4.213 ± 0.636
UPI	beta58	0.00	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	0.000	0.000	—	0.0426	0.0426	1.009	95.787 ± 0.636	0.181 ± 0.002	4.213 ± 0.636

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S256

Simulation 2 - Condition 112: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.6

Method	Param.	True	Mean-based									Median-based					Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE	
LSAM	beta41	0.20	0.202	0.002 ± 0.001	0.010 ± 0.005	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.029 ± 0.001	0.171 ± 0.005	0.201	0.001	0.59	0.0330	0.0331	1.015	93.982 ± 0.753	0.135 ± 0.000	—	
LSAM	beta42	0.20	0.201	0.001 ± 0.001	0.007 ± 0.006	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.032 ± 0.001	0.180 ± 0.005	0.201	0.001	0.39	0.0369	0.0369	0.996	94.985 ± 0.691	0.140 ± 0.000	—	
LSAM	beta43	0.20	0.200	-0.000 ± 0.001	-0.001 ± 0.006	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.032 ± 0.001	0.179 ± 0.005	0.199	-0.001	-0.31	0.0361	0.0361	0.947	93.280 ± 0.793	0.133 ± 0.000	—	
LSAM	beta44	0.00	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	0.001	0.001	—	0.0392	0.0393	1.002	94.885 ± 0.698	0.159 ± 0.001	5.115 ± 0.698	
LSAM	beta45	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	0.001	0.001	—	0.0354	0.0354	0.980	94.885 ± 0.698	0.138 ± 0.001	5.115 ± 0.698	
LSAM	beta46	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	-0.002	-0.002	—	0.0261	0.0262	1.007	95.186 ± 0.678	0.111 ± 0.001	4.814 ± 0.678	
LSAM	beta47	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	0.002	0.002	—	0.0282	0.0283	1.022	95.286 ± 0.671	0.111 ± 0.001	4.714 ± 0.671	
LSAM	beta51	0.16	0.158	-0.002 ± 0.001	-0.013 ± 0.009	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.079 ± 0.004	0.280 ± 0.008	0.156	-0.004	-2.67	0.0430	0.0432	0.998	94.784 ± 0.704	0.175 ± 0.001	—	
LSAM	beta52	0.16	0.162	0.002 ± 0.001	0.014 ± 0.007	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.051 ± 0.002	0.226 ± 0.006	0.161	0.001	0.46	0.0367	0.0367	0.985	94.584 ± 0.717	0.140 ± 0.000	—	
LSAM	beta53	0.16	0.161	0.001 ± 0.001	0.005 ± 0.008	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.059 ± 0.003	0.242 ± 0.007	0.158	-0.002	-1.28	0.0379	0.0380	0.952	93.982 ± 0.753	0.145 ± 0.000	—	
LSAM	beta54	0.16	0.160	0.000 ± 0.001	0.001 ± 0.007	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.053 ± 0.002	0.229 ± 0.006	0.160	0.000	-0.22	0.0371	0.0371	0.970	94.383 ± 0.729	0.140 ± 0.000	—	
LSAM	beta55	0.00	-0.001	-0.001 ± 0.002	—	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	—	—	-0.001	-0.001	—	0.0489	0.0489	1.000	94.483 ± 0.723	0.195 ± 0.001	5.517 ± 0.723	
LSAM	beta56	0.00	0.000	0.000 ± 0.001	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	—	—	0.000	0.000	—	0.0424	0.0424	0.980	94.985 ± 0.691	0.169 ± 0.001	5.015 ± 0.691	
LSAM	beta57	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	0.000	0.000	—	0.0280	0.0280	1.011	94.784 ± 0.704	0.117 ± 0.001	5.216 ± 0.704	
LSAM	beta58	0.00	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.025 ± 0.001	0.001 ± 0.000	0.025 ± 0.001	—	—	0.000	0.000	—	0.0234	0.0234	1.040	95.085 ± 0.685	0.101 ± 0.001	4.915 ± 0.685	
QML	beta41	0.20	0.200	-0.000 ± 0.001	-0.001 ± 0.005	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.028 ± 0.001	0.166 ± 0.005	0.200	0.000	-0.15	0.0311	0.0311	1.019	94.684 ± 0.711	0.133 ± 0.000	—	
QML	beta42	0.20	0.200	0.000 ± 0.001	0.000 ± 0.006	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.031 ± 0.001	0.175 ± 0.005	0.200	0.000	0.10	0.0364	0.0364	1.001	94.985 ± 0.691	0.137 ± 0.001	—	
QML	beta43	0.20	0.199	-0.001 ± 0.001	-0.007 ± 0.006	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.031 ± 0.001	0.176 ± 0.005	0.198	-0.002	-1.04	0.0351	0.0352	0.955	93.180 ± 0.798	0.132 ± 0.000	—	
QML	beta44	0.00	-0.008	-0.008 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	-0.008	-0.008	—	0.0401	0.0409	1.013	95.286 ± 0.671	0.164 ± 0.002	4.714 ± 0.671	
QML	beta45	0.00	-0.006	-0.006 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	-0.006	-0.006	—	0.0350	0.0355	0.998	94.283 ± 0.735	0.139 ± 0.001	5.717 ± 0.735	
QML	beta46	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	0.000	0.000	—	0.0254	0.0254	1.017	94.985 ± 0.691	0.110 ± 0.001	5.015 ± 0.691	
QML	beta47	0.00	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	—	—	0.000	0.000	—	0.0266	0.0266	1.058	95.587 ± 0.650	0.111 ± 0.002	4.413 ± 0.650	
QML	beta51	0.16	0.159	-0.001 ± 0.001	-0.009 ± 0.009	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.077 ± 0.004	0.278 ± 0.008	0.157	-0.003	-1.70	0.0425	0.0426	1.024	94.985 ± 0.691	0.178 ± 0.001	—	
QML	beta52	0.16	0.161	0.001 ± 0.001	0.008 ± 0.007	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.050 ± 0.002	0.224 ± 0.006	0.160	0.000	-0.14	0.0374	0.0374	0.993	95.085 ± 0.685	0.139 ± 0.000	—	
QML	beta53	0.16	0.160	-0.000 ± 0.001	-0.003 ± 0.008	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.057 ± 0.003	0.238 ± 0.007	0.158	-0.002	-1.42	0.0376	0.0376	0.960	94.383 ± 0.729	0.143 ± 0.000	—	
QML	beta54	0.16	0.159	-0.001 ± 0.001	-0.004 ± 0.007	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.051 ± 0.002	0.225 ± 0.006	0.158	-0.002	-1.26	0.0363	0.0364	0.980	94.784 ± 0.704	0.138 ± 0.000	—	
QML	beta55	0.00	-0.004	-0.004 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	—	—	-0.005	-0.005	—	0.0485	0.0487	1.021	94.985 ± 0.691	0.196 ± 0.001	5.015 ± 0.691	
QML	beta56	0.00	-0.006	-0.006 ± 0.001	—	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	—	—	-0.006	-0.006	—	0.0411	0.0416	0.989	94.483 ± 0.723	0.166 ± 0.001	5.517 ± 0.723	
QML	beta57	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	-0.002	-0.002	—	0.0275	0.0275	1.032	94.684 ± 0.711	0.116 ± 0.001	5.316 ± 0.711	
QML	beta58	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.024 ± 0.001	0.001 ± 0.000	0.024 ± 0.001	—	—	-0.003	-0.003	—	0.0220	0.0221	1.065	95.085 ± 0.685	0.099 ± 0.001	4.915 ± 0.685	
UPI	beta41	0.20	0.202	0.002 ± 0.001	0.010 ± 0.005	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.029 ± 0.001	0.171 ± 0.005	0.202	0.002	0.75	0.0320	0.0320	1.022	94.383 ± 0.729	0.136 ± 0.000	—	
UPI	beta42	0.20	0.201	0.001 ± 0.001	0.007 ± 0.006	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.032 ± 0.001	0.178 ± 0.005	0.201	0.001	0.44	0.0362	0.0363	1.008	95.787 ± 0.636	0.141 ± 0.000	—	
UPI	beta43	0.20	0.200	0.000 ± 0.001	0.000 ± 0.006	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.032 ± 0.001	0.180 ± 0.005	0.199	-0.001	-0.39	0.0360	0.0361	0.953	93.280 ± 0.793	0.134 ± 0.000	—	
UPI	beta44	0.00	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	0.001	0.001	—	0.0417	0.0417	1.019	95.186 ± 0.678	0.168 ± 0.001	4.814 ± 0.678	
UPI	beta45	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.003	0.003	—	0.0355	0.0356	1.007	95.687 ± 0.643	0.145 ± 0.001	4.313 ± 0.643	
UPI	beta46	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	-0.001	-0.001	—	0.0270	0.0270	1.028	95.988 ± 0.622	0.118 ± 0.001	4.012 ± 0.622	

Table S256

Simulation 2 - Condition 112: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.6
(continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	0.002	0.002	—	0.0283	0.0284	1.055	96.389 ± 0.591	0.118 ± 0.001	3.611 ± 0.591
UPI	beta51	0.16	0.158	-0.002 ± 0.001	-0.014 ± 0.009	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.077 ± 0.004	0.277 ± 0.008	0.157	-0.003	-1.87	0.0426	0.0427	1.017	94.985 ± 0.691	0.176 ± 0.001	—
UPI	beta52	0.16	0.162	0.002 ± 0.001	0.014 ± 0.007	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.051 ± 0.002	0.226 ± 0.006	0.161	0.001	0.60	0.0373	0.0373	0.992	94.885 ± 0.698	0.140 ± 0.000	—
UPI	beta53	0.16	0.161	0.001 ± 0.001	0.005 ± 0.008	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.059 ± 0.003	0.242 ± 0.007	0.158	-0.002	-1.49	0.0380	0.0381	0.958	94.183 ± 0.741	0.146 ± 0.000	—
UPI	beta54	0.16	0.160	0.000 ± 0.001	0.001 ± 0.007	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.053 ± 0.002	0.230 ± 0.006	0.159	-0.001	-0.44	0.0369	0.0369	0.971	94.283 ± 0.735	0.140 ± 0.000	—
UPI	beta55	0.00	-0.001	-0.001 ± 0.002	—	0.003 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	—	—	0.000	0.000	—	0.0483	0.0483	1.015	95.286 ± 0.671	0.199 ± 0.001	4.714 ± 0.671
UPI	beta56	0.00	0.000	-0.000 ± 0.001	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	—	—	0.001	0.001	—	0.0428	0.0428	1.000	95.486 ± 0.657	0.172 ± 0.001	4.514 ± 0.657
UPI	beta57	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	0.000	0.000	—	0.0295	0.0295	1.017	95.787 ± 0.636	0.120 ± 0.001	4.213 ± 0.636
UPI	beta58	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.025 ± 0.001	0.001 ± 0.000	0.025 ± 0.001	—	—	0.000	0.000	—	0.0238	0.0238	1.053	95.988 ± 0.622	0.104 ± 0.001	4.012 ± 0.622

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u02014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S257

Simulation 2 - Condition 113: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, $N = 400$, Reliability = 0.8

Method	Param.	True	Mean-based										Median-based				Inference			
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Type I (%) $\pm$ MCSE
LSAM	beta41	0.20	0.198	-0.002 $\pm$ 0.002	-0.008 $\pm$ 0.008	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.057 $\pm$ 0.003	0.238 $\pm$ 0.007	0.198	-0.002	-0.92	0.0478	0.0478	0.946	93.079 $\pm$ 0.804	0.176 $\pm$ 0.001	—
LSAM	beta42	0.20	0.199	-0.001 $\pm$ 0.001	-0.004 $\pm$ 0.007	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.054 $\pm$ 0.003	0.231 $\pm$ 0.007	0.200	0.000	0.04	0.0458	0.0458	0.997	94.684 $\pm$ 0.711	0.181 $\pm$ 0.001	—
LSAM	beta43	0.20	0.201	0.001 $\pm$ 0.001	0.007 $\pm$ 0.007	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.049 $\pm$ 0.002	0.222 $\pm$ 0.006	0.201	0.001	0.51	0.0416	0.0416	1.001	94.784 $\pm$ 0.704	0.174 $\pm$ 0.001	—
LSAM	beta44	0.00	0.002	0.002 $\pm$ 0.002	—	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	—	—	0.002	0.002	—	0.0495	0.0496	0.966	93.380 $\pm$ 0.787	0.195 $\pm$ 0.001	6.620 $\pm$ 0.787
LSAM	beta45	0.00	-0.002	-0.002 $\pm$ 0.002	—	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	—	—	-0.004	-0.004	—	0.0474	0.0475	0.911	91.474 $\pm$ 0.884	0.173 $\pm$ 0.001	8.526 $\pm$ 0.884
LSAM	beta46	0.00	0.001	0.001 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	—	—	0.000	0.000	—	0.0382	0.0382	0.933	93.380 $\pm$ 0.787	0.136 $\pm$ 0.001	6.620 $\pm$ 0.787
LSAM	beta47	0.00	-0.001	-0.001 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	—	—	-0.003	-0.003	—	0.0319	0.0321	0.960	93.882 $\pm$ 0.759	0.134 $\pm$ 0.001	6.118 $\pm$ 0.759
LSAM	beta51	0.16	0.158	-0.002 $\pm$ 0.002	-0.011 $\pm$ 0.012	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	0.138 $\pm$ 0.007	0.372 $\pm$ 0.011	0.155	-0.005	-3.00	0.0571	0.0573	0.968	94.283 $\pm$ 0.735	0.226 $\pm$ 0.001	—
LSAM	beta52	0.16	0.162	0.002 $\pm$ 0.002	0.013 $\pm$ 0.010	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.094 $\pm$ 0.004	0.307 $\pm$ 0.009	0.161	0.001	0.35	0.0463	0.0463	0.954	93.480 $\pm$ 0.782	0.184 $\pm$ 0.001	—
LSAM	beta53	0.16	0.157	-0.003 $\pm$ 0.002	-0.017 $\pm$ 0.010	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.092 $\pm$ 0.004	0.303 $\pm$ 0.008	0.157	-0.003	-1.73	0.0461	0.0462	0.996	95.186 $\pm$ 0.678	0.189 $\pm$ 0.001	—
LSAM	beta54	0.16	0.162	0.002 $\pm$ 0.001	0.015 $\pm$ 0.009	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.087 $\pm$ 0.004	0.294 $\pm$ 0.008	0.162	0.002	1.14	0.0461	0.0461	0.992	95.888 $\pm$ 0.629	0.183 $\pm$ 0.001	—
LSAM	beta55	0.00	-0.002	-0.002 $\pm$ 0.002	—	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	—	—	0.000	0.000	—	0.0653	0.0653	0.918	93.480 $\pm$ 0.782	0.239 $\pm$ 0.002	6.520 $\pm$ 0.782
LSAM	beta56	0.00	0.002	0.002 $\pm$ 0.002	—	0.003 $\pm$ 0.000	0.057 $\pm$ 0.001	0.003 $\pm$ 0.000	0.057 $\pm$ 0.002	—	—	0.002	0.002	—	0.0546	0.0546	0.928	93.480 $\pm$ 0.782	0.209 $\pm$ 0.001	6.520 $\pm$ 0.782
LSAM	beta57	0.00	0.000	0.000 $\pm$ 0.001	—	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	—	—	-0.002	-0.002	—	0.0366	0.0367	0.952	93.882 $\pm$ 0.759	0.145 $\pm$ 0.001	6.118 $\pm$ 0.759
LSAM	beta58	0.00	-0.001	-0.001 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	—	—	-0.003	-0.003	—	0.0328	0.0329	0.933	93.581 $\pm$ 0.776	0.126 $\pm$ 0.001	6.419 $\pm$ 0.776
QML	beta41	0.20	0.197	-0.003 $\pm$ 0.001	-0.013 $\pm$ 0.007	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.055 $\pm$ 0.003	0.235 $\pm$ 0.007	0.198	-0.002	-1.12	0.0468	0.0469	0.953	93.480 $\pm$ 0.782	0.176 $\pm$ 0.001	—
QML	beta42	0.20	0.198	-0.002 $\pm$ 0.001	-0.008 $\pm$ 0.007	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.053 $\pm$ 0.003	0.229 $\pm$ 0.007	0.199	-0.001	-0.30	0.0452	0.0452	0.997	94.684 $\pm$ 0.711	0.179 $\pm$ 0.001	—
QML	beta43	0.20	0.201	0.001 $\pm$ 0.001	0.005 $\pm$ 0.007	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.048 $\pm$ 0.002	0.220 $\pm$ 0.006	0.201	0.001	0.59	0.0423	0.0423	1.003	95.085 $\pm$ 0.685	0.173 $\pm$ 0.001	—
QML	beta44	0.00	-0.003	-0.003 $\pm$ 0.002	—	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	—	—	-0.001	-0.001	—	0.0482	0.0482	0.973	93.982 $\pm$ 0.753	0.197 $\pm$ 0.001	6.018 $\pm$ 0.753
QML	beta45	0.00	-0.005	-0.005 $\pm$ 0.002	—	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	—	—	-0.006	-0.006	—	0.0477	0.0481	0.921	91.374 $\pm$ 0.889	0.174 $\pm$ 0.001	8.626 $\pm$ 0.889
QML	beta46	0.00	0.003	0.003 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	—	—	0.001	0.001	—	0.0372	0.0373	0.943	93.982 $\pm$ 0.753	0.136 $\pm$ 0.001	6.018 $\pm$ 0.753
QML	beta47	0.00	0.000	-0.000 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	—	—	-0.004	-0.004	—	0.0322	0.0324	0.961	94.082 $\pm$ 0.747	0.132 $\pm$ 0.001	5.918 $\pm$ 0.747
QML	beta51	0.16	0.159	-0.001 $\pm$ 0.002	-0.008 $\pm$ 0.012	0.004 $\pm$ 0.000	0.059 $\pm$ 0.002	0.004 $\pm$ 0.000	0.059 $\pm$ 0.002	0.137 $\pm$ 0.007	0.370 $\pm$ 0.011	0.155	-0.005	-3.11	0.0565	0.0567	0.997	94.885 $\pm$ 0.698	0.231 $\pm$ 0.001	—
QML	beta52	0.16	0.162	0.002 $\pm$ 0.002	0.010 $\pm$ 0.010	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.094 $\pm$ 0.004	0.306 $\pm$ 0.009	0.161	0.001	0.49	0.0465	0.0465	0.962	93.581 $\pm$ 0.776	0.185 $\pm$ 0.001	—
QML	beta53	0.16	0.157	-0.003 $\pm$ 0.002	-0.020 $\pm$ 0.009	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.090 $\pm$ 0.004	0.300 $\pm$ 0.008	0.157	-0.003	-1.72	0.0456	0.0457	1.005	95.386 $\pm$ 0.664	0.189 $\pm$ 0.001	—
QML	beta54	0.16	0.162	0.002 $\pm$ 0.001	0.012 $\pm$ 0.009	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.085 $\pm$ 0.004	0.291 $\pm$ 0.008	0.161	0.001	0.65	0.0459	0.0459	0.998	95.687 $\pm$ 0.643	0.182 $\pm$ 0.001	—
QML	beta55	0.00	-0.004	-0.004 $\pm$ 0.002	—	0.004 $\pm$ 0.000	0.065 $\pm$ 0.002	0.004 $\pm$ 0.000	0.065 $\pm$ 0.002	—	—	-0.002	-0.002	—	0.0643	0.0643	0.938	93.079 $\pm$ 0.804	0.239 $\pm$ 0.002	6.921 $\pm$ 0.804
QML	beta56	0.00	0.000	-0.000 $\pm$ 0.002	—	0.003 $\pm$ 0.000	0.056 $\pm$ 0.001	0.003 $\pm$ 0.000	0.056 $\pm$ 0.002	—	—	-0.002	-0.002	—	0.0531	0.0531	0.938	93.581 $\pm$ 0.776	0.207 $\pm$ 0.001	6.419 $\pm$ 0.776
QML	beta57	0.00	0.001	0.001 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	—	—	-0.002	-0.002	—	0.0354	0.0354	0.961	93.480 $\pm$ 0.782	0.144 $\pm$ 0.001	6.520 $\pm$ 0.782
QML	beta58	0.00	-0.001	-0.001 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	—	—	-0.004	-0.004	—	0.0319	0.0321	0.942	92.377 $\pm$ 0.840	0.124 $\pm$ 0.001	7.623 $\pm$ 0.840
UPI	beta41	0.20	0.198	-0.002 $\pm$ 0.002	-0.008 $\pm$ 0.008	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.057 $\pm$ 0.003	0.238 $\pm$ 0.007	0.199	-0.001	-0.70	0.0479	0.0479	0.949	92.879 $\pm$ 0.815	0.177 $\pm$ 0.001	—
UPI	beta42	0.20	0.199	-0.001 $\pm$ 0.001	-0.004 $\pm$ 0.007	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.053 $\pm$ 0.003	0.230 $\pm$ 0.007	0.200	0.000	0.05	0.0461	0.0461	1.004	95.085 $\pm$ 0.685	0.181 $\pm$ 0.001	—
UPI	beta43	0.20	0.201	0.001 $\pm$ 0.001	0.007 $\pm$ 0.007	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	0.049 $\pm$ 0.002	0.222 $\pm$ 0.006	0.201	0.001	0.56	0.0416	0.0416	1.008	94.885 $\pm$ 0.698	0.175 $\pm$ 0.001	—
UPI	beta44	0.00	0.002	0.002 $\pm$ 0.002	—	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	0.003 $\pm$ 0.000	0.053 $\pm$ 0.002	—	—	0.003	0.003	—	0.0495	0.0496	0.971	93.180 $\pm$ 0.798	0.203 $\pm$ 0.001	6.820 $\pm$ 0.798
UPI	beta45	0.00	-0.003	-0.003 $\pm$ 0.002	—	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	—	—	-0.003	-0.003	—	0.0480	0.0481	0.914	91.575 $\pm$ 0.880	0.179 $\pm$ 0.001	8.425 $\pm$ 0.880
UPI	beta46	0.00	0.001	0.001 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.039 $\pm$ 0.001	0.001 $\pm$ 0.000	0.039 $\pm$ 0.001	—	—	0.000	0.000	—	0.0394	0.0394	0.933	93.781 $\pm$ 0.765	0.141 $\pm$ 0.001	6.219 $\pm$ 0.765

Table S257

Simulation 2 - Condition 113: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.8
(continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.003	-0.003	—	0.0336	0.0338	0.970	94.383 ± 0.729	0.140 ± 0.001	5.617 ± 0.729
UPI	beta51	0.16	0.158	-0.002 ± 0.002	-0.011 ± 0.012	0.003 ± 0.000	0.059 ± 0.002	0.003 ± 0.000	0.059 ± 0.002	0.136 ± 0.007	0.369 ± 0.011	0.155	-0.005	-2.92	0.0549	0.0551	0.992	94.684 ± 0.711	0.229 ± 0.001	—
UPI	beta52	0.16	0.162	0.002 ± 0.002	0.013 ± 0.010	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.094 ± 0.004	0.307 ± 0.009	0.161	0.001	0.72	0.0462	0.0462	0.955	93.280 ± 0.793	0.184 ± 0.001	—
UPI	beta53	0.16	0.157	-0.003 ± 0.002	-0.017 ± 0.010	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.092 ± 0.004	0.303 ± 0.008	0.158	-0.002	-1.49	0.0461	0.0461	0.999	95.286 ± 0.671	0.190 ± 0.001	—
UPI	beta54	0.16	0.162	0.002 ± 0.001	0.014 ± 0.009	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.087 ± 0.004	0.294 ± 0.008	0.162	0.002	1.37	0.0467	0.0468	0.994	95.587 ± 0.650	0.183 ± 0.001	—
UPI	beta55	0.00	-0.002	-0.002 ± 0.002	—	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	—	—	0.000	0.000	—	0.0642	0.0642	0.939	94.383 ± 0.729	0.244 ± 0.002	5.617 ± 0.729
UPI	beta56	0.00	0.002	0.002 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	—	—	0.000	0.000	—	0.0527	0.0527	0.941	94.183 ± 0.741	0.213 ± 0.001	5.817 ± 0.741
UPI	beta57	0.00	0.000	-0.000 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	-0.001	-0.001	—	0.0374	0.0375	0.962	94.483 ± 0.723	0.146 ± 0.001	5.517 ± 0.723
UPI	beta58	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	—	—	-0.002	-0.002	—	0.0332	0.0333	0.950	94.283 ± 0.735	0.128 ± 0.001	5.717 ± 0.735

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S258

Simulation 2 - Condition 114: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, $N = 1000$, Reliability = 0.8

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Type I (%) $\pm$ MCSE
LSAM	beta41	0.20	0.200	0.000 $\pm$ 0.001	0.000 $\pm$ 0.005	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.022 $\pm$ 0.001	0.149 $\pm$ 0.004	0.200	0.000	0.09	0.0293	0.0294	0.951	94.100 $\pm$ 0.745	0.111 $\pm$ 0.000	—
LSAM	beta42	0.20	0.200	0.000 $\pm$ 0.001	0.001 $\pm$ 0.005	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.023 $\pm$ 0.001	0.153 $\pm$ 0.004	0.200	0.000	-0.20	0.0289	0.0289	0.952	93.200 $\pm$ 0.796	0.114 $\pm$ 0.000	—
LSAM	beta43	0.20	0.198	-0.002 $\pm$ 0.001	-0.010 $\pm$ 0.004	0.001 $\pm$ 0.000	0.028 $\pm$ 0.001	0.001 $\pm$ 0.000	0.028 $\pm$ 0.001	0.020 $\pm$ 0.001	0.140 $\pm$ 0.004	0.198	-0.002	-0.91	0.0283	0.0283	1.005	94.500 $\pm$ 0.721	0.110 $\pm$ 0.000	—
LSAM	beta44	0.00	0.002	0.002 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	—	—	0.002	0.002	—	0.0331	0.0331	0.941	93.800 $\pm$ 0.763	0.121 $\pm$ 0.001	6.200 $\pm$ 0.763
LSAM	beta45	0.00	-0.001	-0.001 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	—	—	0.000	0.000	—	0.0284	0.0284	0.976	94.000 $\pm$ 0.751	0.110 $\pm$ 0.000	6.000 $\pm$ 0.751
LSAM	beta46	0.00	-0.001	-0.001 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.023 $\pm$ 0.000	0.001 $\pm$ 0.000	0.023 $\pm$ 0.001	—	—	-0.001	-0.001	—	0.0230	0.0230	0.965	94.100 $\pm$ 0.745	0.085 $\pm$ 0.000	5.900 $\pm$ 0.745
LSAM	beta47	0.00	-0.001	-0.001 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.023 $\pm$ 0.001	0.001 $\pm$ 0.000	0.023 $\pm$ 0.001	—	—	-0.002	-0.002	—	0.0227	0.0228	0.944	93.600 $\pm$ 0.774	0.085 $\pm$ 0.000	6.400 $\pm$ 0.774
LSAM	beta51	0.16	0.161	0.001 $\pm$ 0.001	0.005 $\pm$ 0.008	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.056 $\pm$ 0.003	0.237 $\pm$ 0.007	0.160	0.000	-0.01	0.0371	0.0371	0.971	93.800 $\pm$ 0.763	0.145 $\pm$ 0.001	—
LSAM	beta52	0.16	0.160	-0.000 $\pm$ 0.001	-0.000 $\pm$ 0.006	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.035 $\pm$ 0.002	0.187 $\pm$ 0.005	0.159	-0.001	-0.33	0.0281	0.0281	0.985	94.500 $\pm$ 0.721	0.115 $\pm$ 0.000	—
LSAM	beta53	0.16	0.160	0.000 $\pm$ 0.001	0.001 $\pm$ 0.006	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.037 $\pm$ 0.002	0.191 $\pm$ 0.005	0.159	-0.001	-0.61	0.0318	0.0318	0.996	94.300 $\pm$ 0.733	0.120 $\pm$ 0.000	—
LSAM	beta54	0.16	0.160	0.000 $\pm$ 0.001	0.003 $\pm$ 0.006	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.034 $\pm$ 0.001	0.185 $\pm$ 0.005	0.161	0.001	0.64	0.0306	0.0306	0.991	95.500 $\pm$ 0.656	0.115 $\pm$ 0.000	—
LSAM	beta55	0.00	0.001	0.001 $\pm$ 0.001	—	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	—	—	0.000	0.000	—	0.0397	0.0397	0.964	94.700 $\pm$ 0.708	0.150 $\pm$ 0.001	5.300 $\pm$ 0.708
LSAM	beta56	0.00	0.002	0.002 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	—	—	0.001	0.001	—	0.0346	0.0346	0.954	94.500 $\pm$ 0.721	0.133 $\pm$ 0.001	5.500 $\pm$ 0.721
LSAM	beta57	0.00	0.000	0.000 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.023 $\pm$ 0.001	0.001 $\pm$ 0.000	0.023 $\pm$ 0.001	—	—	-0.001	-0.001	—	0.0224	0.0224	1.004	95.300 $\pm$ 0.669	0.090 $\pm$ 0.000	4.700 $\pm$ 0.669
LSAM	beta58	0.00	0.000	0.000 $\pm$ 0.001	—	0.000 $\pm$ 0.000	0.021 $\pm$ 0.000	0.000 $\pm$ 0.000	0.021 $\pm$ 0.001	—	—	0.000	0.000	—	0.0211	0.0211	0.961	94.100 $\pm$ 0.745	0.079 $\pm$ 0.000	5.900 $\pm$ 0.745
QML	beta41	0.20	0.199	-0.001 $\pm$ 0.001	-0.005 $\pm$ 0.005	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.021 $\pm$ 0.001	0.147 $\pm$ 0.004	0.199	-0.001	-0.50	0.0289	0.0289	0.958	93.800 $\pm$ 0.763	0.110 $\pm$ 0.000	—
QML	beta42	0.20	0.199	-0.001 $\pm$ 0.001	-0.003 $\pm$ 0.005	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.023 $\pm$ 0.001	0.150 $\pm$ 0.004	0.199	-0.001	-0.51	0.0287	0.0287	0.958	93.600 $\pm$ 0.774	0.113 $\pm$ 0.000	—
QML	beta43	0.20	0.198	-0.002 $\pm$ 0.001	-0.012 $\pm$ 0.004	0.001 $\pm$ 0.000	0.028 $\pm$ 0.001	0.001 $\pm$ 0.000	0.028 $\pm$ 0.001	0.019 $\pm$ 0.001	0.139 $\pm$ 0.004	0.198	-0.002	-0.97	0.0280	0.0281	1.008	94.400 $\pm$ 0.727	0.110 $\pm$ 0.000	—
QML	beta44	0.00	-0.002	-0.002 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	—	—	-0.002	-0.002	—	0.0329	0.0329	0.955	94.000 $\pm$ 0.751	0.124 $\pm$ 0.001	6.000 $\pm$ 0.751
QML	beta45	0.00	-0.003	-0.003 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	—	—	-0.002	-0.002	—	0.0282	0.0283	0.989	93.900 $\pm$ 0.757	0.111 $\pm$ 0.001	6.100 $\pm$ 0.757
QML	beta46	0.00	-0.001	-0.001 $\pm$ 0.001	—	0.000 $\pm$ 0.000	0.022 $\pm$ 0.000	0.000 $\pm$ 0.000	0.022 $\pm$ 0.001	—	—	-0.001	-0.001	—	0.0226	0.0227	0.975	93.800 $\pm$ 0.763	0.085 $\pm$ 0.000	6.200 $\pm$ 0.763
QML	beta47	0.00	-0.002	-0.002 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.022 $\pm$ 0.001	0.001 $\pm$ 0.000	0.023 $\pm$ 0.001	—	—	-0.003	-0.003	—	0.0218	0.0220	0.956	92.800 $\pm$ 0.817	0.084 $\pm$ 0.001	7.200 $\pm$ 0.817
QML	beta51	0.16	0.161	0.001 $\pm$ 0.001	0.005 $\pm$ 0.007	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.056 $\pm$ 0.003	0.237 $\pm$ 0.007	0.160	0.000	0.29	0.0374	0.0374	0.984	94.200 $\pm$ 0.739	0.146 $\pm$ 0.001	—
QML	beta52	0.16	0.160	-0.000 $\pm$ 0.001	-0.002 $\pm$ 0.006	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.034 $\pm$ 0.002	0.186 $\pm$ 0.005	0.159	-0.001	-0.64	0.0277	0.0277	0.990	94.700 $\pm$ 0.708	0.115 $\pm$ 0.000	—
QML	beta53	0.16	0.160	-0.000 $\pm$ 0.001	-0.002 $\pm$ 0.006	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.036 $\pm$ 0.002	0.190 $\pm$ 0.005	0.159	-0.001	-0.84	0.0306	0.0307	1.001	94.700 $\pm$ 0.708	0.119 $\pm$ 0.000	—
QML	beta54	0.16	0.160	0.000 $\pm$ 0.001	0.000 $\pm$ 0.006	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.034 $\pm$ 0.001	0.185 $\pm$ 0.005	0.161	0.001	0.48	0.0303	0.0303	0.992	95.500 $\pm$ 0.656	0.115 $\pm$ 0.000	—
QML	beta55	0.00	-0.001	-0.001 $\pm$ 0.001	—	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	0.002 $\pm$ 0.000	0.039 $\pm$ 0.001	—	—	-0.001	-0.001	—	0.0386	0.0386	0.976	94.800 $\pm$ 0.702	0.151 $\pm$ 0.001	5.200 $\pm$ 0.702
QML	beta56	0.00	-0.001	-0.001 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	—	—	-0.001	-0.001	—	0.0341	0.0341	0.972	94.900 $\pm$ 0.696	0.133 $\pm$ 0.001	5.100 $\pm$ 0.696
QML	beta57	0.00	0.000	-0.000 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.022 $\pm$ 0.001	0.001 $\pm$ 0.000	0.022 $\pm$ 0.001	—	—	-0.001	-0.001	—	0.0222	0.0223	1.017	94.700 $\pm$ 0.708	0.089 $\pm$ 0.001	5.300 $\pm$ 0.708
QML	beta58	0.00	-0.001	-0.001 $\pm$ 0.001	—	0.000 $\pm$ 0.000	0.021 $\pm$ 0.000	0.000 $\pm$ 0.000	0.021 $\pm$ 0.001	—	—	-0.001	-0.001	—	0.0204	0.0204	0.973	93.600 $\pm$ 0.774	0.078 $\pm$ 0.001	6.400 $\pm$ 0.774
UPI	beta41	0.20	0.200	0.000 $\pm$ 0.001	0.000 $\pm$ 0.005	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.022 $\pm$ 0.001	0.148 $\pm$ 0.004	0.200	0.000	0.17	0.0288	0.0288	0.953	93.900 $\pm$ 0.757	0.111 $\pm$ 0.000	—
UPI	beta42	0.20	0.200	0.000 $\pm$ 0.001	0.001 $\pm$ 0.005	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.023 $\pm$ 0.001	0.152 $\pm$ 0.004	0.200	0.000	-0.20	0.0287	0.0287	0.952	93.800 $\pm$ 0.763	0.114 $\pm$ 0.000	—
UPI	beta43	0.20	0.198	-0.002 $\pm$ 0.001	-0.010 $\pm$ 0.004	0.001 $\pm$ 0.000	0.028 $\pm$ 0.001	0.001 $\pm$ 0.000	0.028 $\pm$ 0.001	0.020 $\pm$ 0.001	0.140 $\pm$ 0.004	0.198	-0.002	-0.77	0.0280	0.0281	1.008	94.500 $\pm$ 0.721	0.110 $\pm$ 0.000	—
UPI	beta44	0.00	0.002	0.002 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	—	—	0.001	0.001	—	0.0332	0.0332	0.943	93.900 $\pm$ 0.757	0.124 $\pm$ 0.001	6.100 $\pm$ 0.757
UPI	beta45	0.00	-0.001	-0.001 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	—	—	0.000	0.000	—	0.0285	0.0285	0.982	94.200 $\pm$ 0.739	0.111 $\pm$ 0.001	5.800 $\pm$ 0.739
UPI	beta46	0.00	-0.001	-0.001 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.023 $\pm$ 0.000	0.001 $\pm$ 0.000	0.023 $\pm$ 0.001	—	—	-0.001	-0.001	—	0.0232	0.0233	0.977	94.000 $\pm$ 0.751	0.087 $\pm$ 0.000	6.000 $\pm$ 0.751

Table S258

Simulation 2 - Condition 114: Linear Model, Distr = Normal, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.8
(continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	—	—	-0.002	-0.002	—	0.0234	0.0235	0.951	93.100 ± 0.801	0.086 ± 0.000	6.900 ± 0.801
UPI	beta51	0.16	0.161	0.001 ± 0.001	0.004 ± 0.007	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.056 ± 0.003	0.236 ± 0.007	0.160	0.000	0.12	0.0372	0.0372	0.981	94.200 ± 0.739	0.146 ± 0.001	—
UPI	beta52	0.16	0.160	0.000 ± 0.001	0.000 ± 0.006	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.035 ± 0.002	0.186 ± 0.005	0.159	-0.001	-0.40	0.0285	0.0285	0.987	94.500 ± 0.721	0.115 ± 0.000	—
UPI	beta53	0.16	0.160	0.000 ± 0.001	0.001 ± 0.006	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.037 ± 0.002	0.191 ± 0.005	0.159	-0.001	-0.55	0.0317	0.0317	0.998	94.500 ± 0.721	0.120 ± 0.000	—
UPI	beta54	0.16	0.160	0.000 ± 0.001	0.003 ± 0.006	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.034 ± 0.001	0.185 ± 0.005	0.161	0.001	0.64	0.0306	0.0306	0.994	95.600 ± 0.649	0.115 ± 0.000	—
UPI	beta55	0.00	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	0.000	0.000	—	0.0385	0.0385	0.972	94.200 ± 0.739	0.151 ± 0.001	5.800 ± 0.739
UPI	beta56	0.00	0.002	0.002 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	0.002	0.002	—	0.0350	0.0351	0.964	94.600 ± 0.715	0.133 ± 0.001	5.400 ± 0.715
UPI	beta57	0.00	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	—	—	0.000	0.000	—	0.0230	0.0230	1.006	95.100 ± 0.683	0.090 ± 0.000	4.900 ± 0.683
UPI	beta58	0.00	0.000	0.000 ± 0.001	—	0.000 ± 0.000	0.021 ± 0.000	0.000 ± 0.000	0.021 ± 0.001	—	—	0.000	0.000	—	0.0209	0.0209	0.961	93.900 ± 0.757	0.079 ± 0.000	6.100 ± 0.757

Note. MCSE = Monte Carlo Standard Error shown as \u000b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S259

Simulation 2 - Condition 115: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.4

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.201	0.001 ± 0.007	0.006 ± 0.033	0.022 ± 0.002	0.150 ± 0.005	0.022 ± 0.002	0.150 ± 0.006	0.560 ± 0.040	0.748 ± 0.031	0.202	0.002	0.77	0.1533	0.1533	1.099	98.211 ± 0.591	0.645 ± 0.011	—
LSAM	beta42	0.20	0.217	0.017 ± 0.007	0.083 ± 0.036	0.026 ± 0.002	0.160 ± 0.006	0.026 ± 0.002	0.160 ± 0.007	0.643 ± 0.048	0.802 ± 0.035	0.212	0.012	5.86	0.1530	0.1535	1.096	98.608 ± 0.522	0.686 ± 0.014	—
LSAM	beta43	0.20	0.198	-0.002 ± 0.004	-0.011 ± 0.018	0.007 ± 0.000	0.082 ± 0.003	0.007 ± 0.000	0.082 ± 0.003	0.166 ± 0.011	0.408 ± 0.017	0.198	-0.002	-0.90	0.0848	0.0848	1.044	95.229 ± 0.950	0.334 ± 0.004	—
LSAM	beta44	0.00	0.002	0.002 ± 0.006	—	0.016 ± 0.001	0.128 ± 0.006	0.016 ± 0.001	0.128 ± 0.006	—	—	0.004	0.004	—	0.1152	0.1153	1.077	97.416 ± 0.707	0.542 ± 0.018	2.584 ± 0.707
LSAM	beta45	0.00	0.007	0.007 ± 0.005	—	0.012 ± 0.001	0.109 ± 0.005	0.012 ± 0.001	0.109 ± 0.005	—	—	0.005	0.005	—	0.0985	0.0987	1.069	98.807 ± 0.484	0.458 ± 0.013	1.193 ± 0.484
LSAM	beta46	0.00	-0.001	-0.001 ± 0.003	—	0.005 ± 0.000	0.070 ± 0.003	0.005 ± 0.000	0.070 ± 0.003	—	—	-0.006	-0.006	—	0.0653	0.0655	1.014	98.410 ± 0.558	0.279 ± 0.007	1.590 ± 0.558
LSAM	beta47	0.00	-0.005	-0.005 ± 0.003	—	0.004 ± 0.000	0.062 ± 0.003	0.004 ± 0.000	0.063 ± 0.003	—	—	-0.005	-0.005	—	0.0562	0.0564	1.118	98.608 ± 0.522	0.274 ± 0.008	1.392 ± 0.522
LSAM	beta51	0.16	0.160	0.000 ± 0.005	0.001 ± 0.032	0.013 ± 0.001	0.115 ± 0.004	0.013 ± 0.001	0.115 ± 0.005	0.514 ± 0.037	0.717 ± 0.031	0.161	0.001	0.49	0.1127	0.1127	0.917	93.241 ± 1.119	0.413 ± 0.004	—
LSAM	beta52	0.16	0.149	-0.011 ± 0.007	-0.071 ± 0.042	0.023 ± 0.002	0.151 ± 0.005	0.023 ± 0.002	0.151 ± 0.006	0.895 ± 0.060	0.946 ± 0.038	0.154	-0.006	-3.80	0.1540	0.1542	1.080	97.217 ± 0.733	0.639 ± 0.012	—
LSAM	beta53	0.16	0.160	-0.000 ± 0.005	-0.001 ± 0.032	0.014 ± 0.001	0.117 ± 0.004	0.014 ± 0.001	0.116 ± 0.005	0.530 ± 0.039	0.728 ± 0.031	0.160	0.000	0.16	0.1128	0.1128	1.015	97.217 ± 0.733	0.464 ± 0.006	—
LSAM	beta54	0.16	0.166	0.006 ± 0.007	0.035 ± 0.042	0.023 ± 0.002	0.151 ± 0.006	0.023 ± 0.002	0.151 ± 0.006	0.892 ± 0.065	0.945 ± 0.041	0.159	-0.001	-0.43	0.1580	0.1580	1.060	97.018 ± 0.758	0.628 ± 0.010	—
LSAM	beta55	0.00	0.017	0.017 ± 0.008	—	0.030 ± 0.002	0.172 ± 0.007	0.030 ± 0.002	0.173 ± 0.008	—	—	0.010	0.010	—	0.1566	0.1569	1.049	97.813 ± 0.652	0.709 ± 0.021	2.187 ± 0.652
LSAM	beta56	0.00	0.000	0.000 ± 0.006	—	0.017 ± 0.001	0.130 ± 0.005	0.017 ± 0.001	0.130 ± 0.006	—	—	0.002	0.002	—	0.1169	0.1169	1.064	98.410 ± 0.558	0.543 ± 0.013	1.590 ± 0.558
LSAM	beta57	0.00	0.001	0.001 ± 0.003	—	0.006 ± 0.001	0.078 ± 0.003	0.006 ± 0.001	0.078 ± 0.004	—	—	0.001	0.001	—	0.0684	0.0684	1.054	97.813 ± 0.652	0.322 ± 0.009	2.187 ± 0.652
LSAM	beta58	0.00	0.001	0.001 ± 0.003	—	0.003 ± 0.000	0.058 ± 0.002	0.003 ± 0.000	0.058 ± 0.003	—	—	0.001	0.001	—	0.0499	0.0499	1.038	96.819 ± 0.782	0.237 ± 0.006	3.181 ± 0.782
QML	beta41	0.20	0.172	-0.028 ± 0.003	-0.138 ± 0.017	0.006 ± 0.000	0.076 ± 0.002	0.007 ± 0.000	0.081 ± 0.003	0.165 ± 0.011	0.406 ± 0.015	0.174	-0.026	-12.91	0.0811	0.0851	1.121	95.229 ± 0.950	0.336 ± 0.004	—
QML	beta42	0.20	0.181	-0.019 ± 0.004	-0.096 ± 0.018	0.007 ± 0.000	0.082 ± 0.003	0.007 ± 0.000	0.084 ± 0.003	0.176 ± 0.012	0.419 ± 0.017	0.174	-0.026	-13.08	0.0824	0.0865	1.071	94.831 ± 0.987	0.343 ± 0.003	—
QML	beta43	0.20	0.203	0.003 ± 0.003	0.017 ± 0.016	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.003	0.127 ± 0.008	0.357 ± 0.014	0.206	0.006	2.78	0.0730	0.0732	1.037	94.831 ± 0.987	0.290 ± 0.003	—
QML	beta44	0.00	-0.031	-0.031 ± 0.004	—	0.010 ± 0.001	0.101 ± 0.003	0.011 ± 0.001	0.105 ± 0.004	—	—	-0.025	-0.025	—	0.1022	0.1052	1.099	96.223 ± 0.850	0.433 ± 0.010	3.777 ± 0.850
QML	beta45	0.00	0.007	0.007 ± 0.004	—	0.007 ± 0.000	0.082 ± 0.003	0.007 ± 0.000	0.082 ± 0.003	—	—	0.001	0.001	—	0.0827	0.0827	1.062	98.211 ± 0.591	0.340 ± 0.006	1.789 ± 0.591
QML	beta46	0.00	0.046	0.046 ± 0.003	—	0.004 ± 0.000	0.064 ± 0.003	0.006 ± 0.001	0.079 ± 0.004	—	—	0.040	0.040	—	0.0605	0.0724	1.020	92.644 ± 1.164	0.257 ± 0.009	7.356 ± 1.164
QML	beta47	0.00	0.043	0.043 ± 0.003	—	0.003 ± 0.000	0.056 ± 0.002	0.005 ± 0.000	0.071 ± 0.003	—	—	0.039	0.039	—	0.0543	0.0667	1.074	95.427 ± 0.931	0.237 ± 0.006	4.573 ± 0.931
QML	beta51	0.16	0.168	0.008 ± 0.005	0.048 ± 0.031	0.013 ± 0.001	0.113 ± 0.004	0.013 ± 0.001	0.113 ± 0.004	0.497 ± 0.032	0.705 ± 0.028	0.170	0.010	6.20	0.1164	0.1168	1.009	95.427 ± 0.931	0.445 ± 0.006	—
QML	beta52	0.16	0.132	-0.028 ± 0.004	-0.172 ± 0.024	0.007 ± 0.000	0.086 ± 0.003	0.008 ± 0.001	0.091 ± 0.003	0.320 ± 0.020	0.566 ± 0.021	0.136	-0.024	-14.70	0.0880	0.0911	1.101	96.223 ± 0.850	0.373 ± 0.006	—
QML	beta53	0.16	0.154	-0.006 ± 0.004	-0.037 ± 0.025	0.008 ± 0.001	0.089 ± 0.003	0.008 ± 0.001	0.089 ± 0.004	0.308 ± 0.021	0.555 ± 0.022	0.151	-0.009	-5.32	0.0929	0.0933	1.031	96.421 ± 0.828	0.359 ± 0.004	—
QML	beta54	0.16	0.137	-0.023 ± 0.004	-0.142 ± 0.024	0.008 ± 0.000	0.087 ± 0.003	0.008 ± 0.001	0.090 ± 0.003	0.316 ± 0.020	0.562 ± 0.022	0.131	-0.029	-18.21	0.0881	0.0927	1.024	96.421 ± 0.828	0.349 ± 0.005	—
QML	beta55	0.00	-0.017	-0.017 ± 0.006	—	0.017 ± 0.001	0.130 ± 0.004	0.017 ± 0.001	0.131 ± 0.005	—	—	-0.020	-0.020	—	0.1328	0.1342	1.090	95.626 ± 0.912	0.555 ± 0.016	4.374 ± 0.912
QML	beta56	0.00	0.027	0.027 ± 0.004	—	0.009 ± 0.001	0.092 ± 0.003	0.009 ± 0.001	0.096 ± 0.004	—	—	0.023	0.023	—	0.0921	0.0948	1.074	95.825 ± 0.892	0.388 ± 0.007	4.175 ± 0.892
QML	beta57	0.00	0.037	0.037 ± 0.003	—	0.005 ± 0.000	0.067 ± 0.003	0.006 ± 0.000	0.077 ± 0.003	—	—	0.028	0.028	—	0.0622	0.0683	1.133	95.626 ± 0.912	0.299 ± 0.012	4.374 ± 0.912
QML	beta58	0.00	0.034	0.034 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.002	0.004 ± 0.000	0.063 ± 0.003	—	—	0.029	0.029	—	0.0458	0.0541	0.950	93.241 ± 1.119	0.197 ± 0.006	6.759 ± 1.119
UPI	beta41	0.20	0.209	0.009 ± 0.009	0.045 ± 0.044	0.039 ± 0.004	0.197 ± 0.009	0.039 ± 0.004	0.197 ± 0.010	0.974 ± 0.091	0.987 ± 0.051	0.207	0.007	3.33	0.2051	0.2052	1.588	99.602 ± 0.281	1.229 ± 0.080	—
UPI	beta42	0.20	0.217	0.017 ± 0.009	0.083 ± 0.045	0.041 ± 0.004	0.203 ± 0.009	0.041 ± 0.004	0.203 ± 0.010	1.031 ± 0.093	1.015 ± 0.051	0.207	0.007	3.50	0.2199	0.2200	1.551	99.404 ± 0.343	1.232 ± 0.064	—
UPI	beta43	0.20	0.200	-0.000 ± 0.004	-0.000 ± 0.019	0.007 ± 0.000	0.085 ± 0.003	0.007 ± 0.000	0.085 ± 0.003	0.181 ± 0.012	0.425 ± 0.017	0.197	-0.003	-1.69	0.1047	0.1048	1.316	96.620 ± 0.806	0.439 ± 0.016	—
UPI	beta44	0.00	0.005	0.005 ± 0.008	—	0.034 ± 0.003	0.183 ± 0.009	0.034 ± 0.003	0.183 ± 0.010	—	—	0.001	0.001	—	0.1516	0.1516	1.735	99.801 ± 0.199	1.246 ± 0.088	0.199 ± 0.199
UPI	beta45	0.00	0.006	0.006 ± 0.007	—	0.021 ± 0.002	0.146 ± 0.008	0.021 ± 0.002	0.146 ± 0.008	—	—	0.007	0.007	—	0.1366	0.1368	1.691	99.801 ± 0.199	0.971 ± 0.092	0.199 ± 0.199
UPI	beta46	0.00	-0.003	-0.003 ± 0.004	—	0.008 ± 0.001	0.091 ± 0.004	0.008 ± 0.001	0.091 ± 0.005	—	—	-0.006	-0.006	—	0.0952	0.0954	1.628	99.602 ± 0.281	0.581 ± 0.046	0.398 ± 0.281

Table S259

Simulation 2 - Condition 115: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.4
(continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.005	-0.005 ± 0.004	—	0.007 ± 0.001	0.084 ± 0.005	0.007 ± 0.001	0.084 ± 0.005	—	—	-0.001	-0.001	—	0.0839	0.0839	1.562	99.404 ± 0.343	0.516 ± 0.030	0.596 ± 0.343
UPI	beta51	0.16	0.159	-0.001 ± 0.005	-0.005 ± 0.031	0.012 ± 0.001	0.111 ± 0.004	0.012 ± 0.001	0.111 ± 0.004	0.484 ± 0.031	0.696 ± 0.027	0.157	-0.003	-2.15	0.1185	0.1186	1.031	95.825 ± 0.892	0.451 ± 0.007	—
UPI	beta52	0.16	0.149	-0.011 ± 0.007	-0.070 ± 0.047	0.028 ± 0.002	0.167 ± 0.007	0.028 ± 0.002	0.167 ± 0.008	1.093 ± 0.093	1.046 ± 0.050	0.154	-0.006	-3.76	0.1700	0.1701	1.289	97.813 ± 0.652	0.844 ± 0.035	—
UPI	beta53	0.16	0.160	0.000 ± 0.005	0.003 ± 0.032	0.013 ± 0.001	0.113 ± 0.005	0.013 ± 0.001	0.113 ± 0.005	0.502 ± 0.040	0.708 ± 0.033	0.160	0.000	-0.02	0.1212	0.1212	1.259	97.813 ± 0.652	0.560 ± 0.015	—
UPI	beta54	0.16	0.171	0.011 ± 0.007	0.068 ± 0.046	0.027 ± 0.002	0.164 ± 0.007	0.027 ± 0.002	0.165 ± 0.008	1.059 ± 0.087	1.029 ± 0.048	0.157	-0.003	-1.72	0.1656	0.1656	1.245	98.410 ± 0.558	0.803 ± 0.027	—
UPI	beta55	0.00	0.010	0.010 ± 0.008	—	0.036 ± 0.003	0.190 ± 0.009	0.036 ± 0.003	0.190 ± 0.010	—	—	0.004	0.004	—	0.1504	0.1505	1.359	98.807 ± 0.484	1.011 ± 0.053	1.193 ± 0.484
UPI	beta56	0.00	0.003	0.003 ± 0.006	—	0.017 ± 0.001	0.129 ± 0.005	0.017 ± 0.001	0.129 ± 0.006	—	—	-0.001	-0.001	—	0.1273	0.1274	1.619	98.608 ± 0.522	0.817 ± 0.051	1.392 ± 0.522
UPI	beta57	0.00	0.003	0.003 ± 0.004	—	0.007 ± 0.001	0.084 ± 0.004	0.007 ± 0.001	0.084 ± 0.004	—	—	-0.001	-0.001	—	0.0756	0.0756	1.358	98.012 ± 0.622	0.445 ± 0.022	1.988 ± 0.622
UPI	beta58	0.00	-0.002	-0.002 ± 0.003	—	0.004 ± 0.000	0.064 ± 0.003	0.004 ± 0.000	0.064 ± 0.003	—	—	0.002	0.002	—	0.0541	0.0541	1.245	97.813 ± 0.652	0.311 ± 0.012	2.187 ± 0.652

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S260

Simulation 2 - Condition 116: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, $N = 1000$, Reliability = 0.4

Method	Param.	True	Mean-based										Median-based					Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE		
LSAM	beta41	0.20	0.197	-0.003 ± 0.003	-0.015 ± 0.015	0.009 ± 0.000	0.093 ± 0.002	0.009 ± 0.000	0.093 ± 0.003	0.217 ± 0.010	0.465 ± 0.013	0.199	-0.001	-0.40	0.0943	0.0943	1.005	94.731 ± 0.740	0.367 ± 0.003	—		
LSAM	beta42	0.20	0.206	0.006 ± 0.003	0.030 ± 0.017	0.010 ± 0.001	0.100 ± 0.003	0.010 ± 0.001	0.100 ± 0.003	0.249 ± 0.013	0.499 ± 0.015	0.205	0.005	2.56	0.0972	0.0973	0.998	96.378 ± 0.619	0.390 ± 0.003	—		
LSAM	beta43	0.20	0.199	-0.001 ± 0.002	-0.003 ± 0.008	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.064 ± 0.003	0.253 ± 0.007	0.200	0.000	0.10	0.0494	0.0494	0.987	94.731 ± 0.740	0.196 ± 0.001	—		
LSAM	beta44	0.00	-0.001	-0.001 ± 0.002	—	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	—	—	-0.003	-0.003	—	0.0624	0.0625	1.000	96.707 ± 0.591	0.250 ± 0.003	3.293 ± 0.591		
LSAM	beta45	0.00	0.003	0.003 ± 0.002	—	0.003 ± 0.000	0.059 ± 0.002	0.004 ± 0.000	0.059 ± 0.002	—	—	0.001	0.001	—	0.0567	0.0567	0.996	96.268 ± 0.628	0.231 ± 0.003	3.732 ± 0.628		
LSAM	beta46	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	—	—	0.000	0.000	—	0.0316	0.0316	1.003	96.378 ± 0.619	0.133 ± 0.002	3.622 ± 0.619		
LSAM	beta47	0.00	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	0.000	0.000	—	0.0298	0.0298	0.958	96.487 ± 0.610	0.134 ± 0.002	3.513 ± 0.610		
LSAM	beta51	0.16	0.164	0.004 ± 0.002	0.024 ± 0.014	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.169 ± 0.008	0.411 ± 0.012	0.163	0.003	2.17	0.0650	0.0651	0.958	93.963 ± 0.789	0.247 ± 0.001	—		
LSAM	beta52	0.16	0.160	-0.000 ± 0.003	-0.003 ± 0.019	0.009 ± 0.000	0.094 ± 0.002	0.009 ± 0.000	0.094 ± 0.003	0.345 ± 0.017	0.588 ± 0.017	0.159	-0.001	-0.81	0.0934	0.0934	0.999	96.926 ± 0.572	0.368 ± 0.003	—		
LSAM	beta53	0.16	0.158	-0.002 ± 0.002	-0.011 ± 0.014	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.068 ± 0.002	0.178 ± 0.009	0.422 ± 0.012	0.158	-0.002	-1.30	0.0673	0.0673	1.049	96.378 ± 0.619	0.278 ± 0.002	—		
LSAM	beta54	0.16	0.167	0.007 ± 0.003	0.045 ± 0.020	0.009 ± 0.000	0.095 ± 0.002	0.009 ± 0.000	0.095 ± 0.003	0.352 ± 0.018	0.593 ± 0.018	0.159	-0.001	-0.85	0.0889	0.0889	0.988	96.158 ± 0.637	0.367 ± 0.003	—		
LSAM	beta55	0.00	-0.001	-0.001 ± 0.003	—	0.008 ± 0.000	0.091 ± 0.003	0.008 ± 0.000	0.090 ± 0.003	—	—	-0.005	-0.005	—	0.0834	0.0836	0.991	96.378 ± 0.619	0.352 ± 0.005	3.622 ± 0.619		
LSAM	beta56	0.00	0.002	0.002 ± 0.002	—	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	—	—	0.001	0.001	—	0.0659	0.0659	1.017	97.036 ± 0.562	0.283 ± 0.003	2.964 ± 0.562		
LSAM	beta57	0.00	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	-0.001	-0.001	—	0.0364	0.0365	1.004	96.268 ± 0.628	0.157 ± 0.002	3.732 ± 0.628		
LSAM	beta58	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	—	—	-0.002	-0.002	—	0.0296	0.0296	0.958	95.939 ± 0.654	0.124 ± 0.002	4.061 ± 0.654		
QML	beta41	0.20	0.170	-0.030 ± 0.002	-0.151 ± 0.009	0.003 ± 0.000	0.051 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.089 ± 0.004	0.298 ± 0.008	0.170	-0.030	-15.08	0.0519	0.0600	1.027	91.109 ± 0.943	0.207 ± 0.006	—		
QML	beta42	0.20	0.176	-0.024 ± 0.002	-0.122 ± 0.009	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.086 ± 0.004	0.292 ± 0.008	0.177	-0.023	-11.74	0.0530	0.0579	1.019	92.755 ± 0.859	0.213 ± 0.003	—		
QML	beta43	0.20	0.204	0.004 ± 0.002	0.022 ± 0.008	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.054 ± 0.003	0.232 ± 0.007	0.206	0.006	2.82	0.0444	0.0447	1.001	94.731 ± 0.740	0.182 ± 0.003	—		
QML	beta44	0.00	-0.031	-0.031 ± 0.002	—	0.004 ± 0.000	0.063 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	—	—	-0.029	-0.029	—	0.0586	0.0654	1.005	94.182 ± 0.776	0.249 ± 0.003	5.818 ± 0.776		
QML	beta45	0.00	0.007	0.007 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	—	—	0.006	0.006	—	0.0526	0.0530	1.004	95.390 ± 0.695	0.205 ± 0.002	4.610 ± 0.695		
QML	beta46	0.00	0.043	0.043 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.003 ± 0.000	0.056 ± 0.001	—	—	0.042	0.042	—	0.0338	0.0540	0.991	81.120 ± 1.297	0.137 ± 0.002	18.880 ± 1.297		
QML	beta47	0.00	0.045	0.045 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.003 ± 0.000	0.057 ± 0.001	—	—	0.041	0.041	—	0.0305	0.0508	0.955	80.571 ± 1.311	0.135 ± 0.002	19.429 ± 1.311		
QML	beta51	0.16	0.171	0.011 ± 0.002	0.066 ± 0.013	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	0.167 ± 0.008	0.409 ± 0.012	0.173	0.013	8.04	0.0630	0.0643	1.006	95.499 ± 0.687	0.255 ± 0.002	—		
QML	beta52	0.16	0.140	-0.020 ± 0.002	-0.126 ± 0.011	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.132 ± 0.006	0.364 ± 0.010	0.139	-0.021	-12.84	0.0550	0.0587	1.024	94.402 ± 0.762	0.219 ± 0.004	—		
QML	beta53	0.16	0.153	-0.007 ± 0.002	-0.043 ± 0.011	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	0.109 ± 0.005	0.330 ± 0.010	0.152	-0.008	-5.01	0.0536	0.0542	1.043	95.829 ± 0.662	0.214 ± 0.001	—		
QML	beta54	0.16	0.138	-0.022 ± 0.002	-0.140 ± 0.011	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.132 ± 0.006	0.364 ± 0.010	0.134	-0.026	-16.36	0.0542	0.0602	1.000	94.731 ± 0.740	0.211 ± 0.001	—		
QML	beta55	0.00	-0.027	-0.027 ± 0.003	—	0.006 ± 0.000	0.078 ± 0.002	0.007 ± 0.000	0.083 ± 0.002	—	—	-0.029	-0.029	—	0.0766	0.0820	1.059	95.170 ± 0.710	0.325 ± 0.011	4.830 ± 0.710		
QML	beta56	0.00	0.028	0.028 ± 0.002	—	0.003 ± 0.000	0.056 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	—	—	0.029	0.029	—	0.0543	0.0613	1.036	94.182 ± 0.776	0.227 ± 0.003	5.818 ± 0.776		
QML	beta57	0.00	0.037	0.037 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.003 ± 0.000	0.055 ± 0.001	—	—	0.033	0.033	—	0.0391	0.0511	1.064	90.889 ± 0.953	0.170 ± 0.004	9.111 ± 0.953		
QML	beta58	0.00	0.028	0.028 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	0.026	0.026	—	0.0283	0.0382	0.963	87.925 ± 1.080	0.115 ± 0.001	12.075 ± 1.080		
UPI	beta41	0.20	0.200	0.000 ± 0.003	0.001 ± 0.017	0.010 ± 0.001	0.100 ± 0.003	0.010 ± 0.001	0.100 ± 0.003	0.251 ± 0.013	0.501 ± 0.016	0.204	0.004	1.90	0.0968	0.0969	1.133	97.585 ± 0.509	0.445 ± 0.010	—		
UPI	beta42	0.20	0.205	0.005 ± 0.003	0.024 ± 0.017	0.011 ± 0.001	0.105 ± 0.003	0.011 ± 0.001	0.105 ± 0.003	0.275 ± 0.014	0.525 ± 0.016	0.201	0.001	0.41	0.1017	0.1017	1.137	97.805 ± 0.485	0.467 ± 0.010	—		
UPI	beta43	0.20	0.200	0.000 ± 0.002	0.000 ± 0.008	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.061 ± 0.003	0.246 ± 0.007	0.199	-0.001	-0.47	0.0479	0.0479	1.028	95.280 ± 0.703	0.199 ± 0.002	—		
UPI	beta44	0.00	-0.003	-0.003 ± 0.002	—	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	—	—	-0.005	-0.005	—	0.0644	0.0646	1.243	98.244 ± 0.435	0.348 ± 0.010	1.756 ± 0.435		

Table S260

Simulation 2 - Condition 116: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.4 (continued)

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta45	0.00	0.003	0.003 ± 0.002	—	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	—	—	0.002	0.002	—	0.0539	0.0540	1.271	97.914 ± 0.473	0.316 ± 0.008	2.086 ± 0.473
UPI	beta46	0.00	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.000	0.000	—	0.0320	0.0320	1.153	98.024 ± 0.461	0.168 ± 0.004	1.976 ± 0.461
UPI	beta47	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	0.000	0.000	—	0.0312	0.0312	1.111	98.353 ± 0.422	0.167 ± 0.004	1.647 ± 0.422
UPI	beta51	0.16	0.164	0.004 ± 0.002	0.023 ± 0.013	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	0.159 ± 0.008	0.399 ± 0.012	0.165	0.005	3.12	0.0635	0.0637	0.991	94.951 ± 0.725	0.248 ± 0.001	—
UPI	beta52	0.16	0.162	0.002 ± 0.003	0.013 ± 0.019	0.009 ± 0.000	0.094 ± 0.002	0.009 ± 0.000	0.093 ± 0.003	0.341 ± 0.017	0.584 ± 0.017	0.163	0.003	2.09	0.0911	0.0912	1.040	97.256 ± 0.541	0.381 ± 0.003	—
UPI	beta53	0.16	0.159	-0.001 ± 0.002	-0.009 ± 0.014	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.169 ± 0.008	0.411 ± 0.012	0.157	-0.003	-2.02	0.0636	0.0637	1.092	96.597 ± 0.601	0.281 ± 0.002	—
UPI	beta54	0.16	0.167	0.007 ± 0.003	0.043 ± 0.019	0.009 ± 0.000	0.093 ± 0.002	0.009 ± 0.000	0.093 ± 0.003	0.338 ± 0.017	0.581 ± 0.018	0.160	0.000	-0.12	0.0866	0.0866	1.039	96.707 ± 0.591	0.378 ± 0.003	—
UPI	beta55	0.00	-0.001	-0.001 ± 0.003	—	0.009 ± 0.001	0.094 ± 0.003	0.009 ± 0.001	0.094 ± 0.003	—	—	-0.003	-0.003	—	0.0819	0.0820	1.068	98.134 ± 0.448	0.392 ± 0.008	1.866 ± 0.448
UPI	beta56	0.00	-0.001	-0.001 ± 0.002	—	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	—	—	0.000	0.000	—	0.0633	0.0633	1.125	98.134 ± 0.448	0.317 ± 0.007	1.866 ± 0.448
UPI	beta57	0.00	0.000	-0.000 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	-0.001	-0.001	—	0.0363	0.0364	1.063	97.256 ± 0.541	0.165 ± 0.002	2.744 ± 0.541
UPI	beta58	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	—	—	-0.002	-0.002	—	0.0295	0.0295	1.010	96.597 ± 0.601	0.129 ± 0.002	3.403 ± 0.601

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S261

Simulation 2 - Condition 117: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, $N = 400$, Reliability = 0.6

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.203	0.003 ± 0.003	0.013 ± 0.016	0.009 ± 0.000	0.097 ± 0.002	0.009 ± 0.000	0.097 ± 0.003	0.237 ± 0.012	0.486 ± 0.015	0.202	0.002	1.05	0.0950	0.0950	1.044	95.690 ± 0.667	0.398 ± 0.003	—
LSAM	beta42	0.20	0.200	0.000 ± 0.003	0.001 ± 0.017	0.011 ± 0.001	0.104 ± 0.002	0.011 ± 0.001	0.104 ± 0.003	0.269 ± 0.013	0.519 ± 0.015	0.201	0.001	0.36	0.1051	0.1051	1.021	94.504 ± 0.748	0.415 ± 0.003	—
LSAM	beta43	0.20	0.199	-0.001 ± 0.002	-0.003 ± 0.010	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.086 ± 0.004	0.293 ± 0.009	0.198	-0.002	-0.95	0.0562	0.0563	0.974	93.966 ± 0.782	0.224 ± 0.001	—
LSAM	beta44	0.00	0.002	0.002 ± 0.002	—	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	—	—	0.001	0.001	—	0.0664	0.0664	0.951	95.259 ± 0.698	0.269 ± 0.003	4.741 ± 0.698
LSAM	beta45	0.00	0.006	0.006 ± 0.002	—	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	—	—	0.007	0.007	—	0.0642	0.0645	0.936	95.366 ± 0.690	0.253 ± 0.003	4.634 ± 0.690
LSAM	beta46	0.00	-0.002	-0.002 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	0.000	0.000	—	0.0347	0.0347	0.978	95.474 ± 0.682	0.153 ± 0.002	4.526 ± 0.682
LSAM	beta47	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.002	-0.002	—	0.0354	0.0355	0.979	95.151 ± 0.705	0.148 ± 0.002	4.849 ± 0.705
LSAM	beta51	0.16	0.161	0.001 ± 0.003	0.005 ± 0.016	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	0.228 ± 0.011	0.478 ± 0.014	0.161	0.001	0.45	0.0751	0.0751	0.945	93.642 ± 0.801	0.283 ± 0.002	—
LSAM	beta52	0.16	0.157	-0.003 ± 0.003	-0.021 ± 0.022	0.011 ± 0.001	0.107 ± 0.003	0.011 ± 0.001	0.107 ± 0.003	0.443 ± 0.023	0.666 ± 0.020	0.154	-0.006	-3.55	0.1042	0.1043	0.954	94.504 ± 0.748	0.398 ± 0.003	—
LSAM	beta53	0.16	0.162	0.002 ± 0.003	0.011 ± 0.016	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	0.229 ± 0.011	0.479 ± 0.014	0.158	-0.002	-1.29	0.0762	0.0762	1.008	95.259 ± 0.698	0.303 ± 0.002	—
LSAM	beta54	0.16	0.163	0.003 ± 0.003	0.019 ± 0.021	0.011 ± 0.001	0.104 ± 0.002	0.011 ± 0.001	0.104 ± 0.003	0.423 ± 0.020	0.651 ± 0.019	0.167	0.007	4.32	0.1023	0.1026	0.964	93.966 ± 0.782	0.393 ± 0.003	—
LSAM	beta55	0.00	0.000	0.000 ± 0.003	—	0.009 ± 0.001	0.097 ± 0.003	0.009 ± 0.001	0.097 ± 0.003	—	—	-0.003	-0.003	—	0.0896	0.0897	0.940	95.582 ± 0.675	0.359 ± 0.004	4.418 ± 0.675
LSAM	beta56	0.00	-0.003	-0.003 ± 0.003	—	0.007 ± 0.000	0.084 ± 0.002	0.007 ± 0.000	0.084 ± 0.003	—	—	-0.002	-0.002	—	0.0803	0.0803	0.921	93.427 ± 0.813	0.304 ± 0.004	6.573 ± 0.813
LSAM	beta57	0.00	0.000	0.000 ± 0.002	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	0.000	0.000	—	0.0426	0.0426	0.929	95.043 ± 0.713	0.169 ± 0.002	4.957 ± 0.713
LSAM	beta58	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.001	-0.001	—	0.0349	0.0349	0.938	94.935 ± 0.720	0.139 ± 0.002	5.065 ± 0.720
QML	beta41	0.20	0.179	-0.021 ± 0.002	-0.106 ± 0.010	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	0.112 ± 0.006	0.335 ± 0.010	0.180	-0.020	-9.85	0.0656	0.0685	1.058	95.043 ± 0.713	0.264 ± 0.001	—
QML	beta42	0.20	0.179	-0.021 ± 0.002	-0.106 ± 0.011	0.005 ± 0.000	0.068 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.128 ± 0.006	0.358 ± 0.010	0.179	-0.021	-10.40	0.0668	0.0700	1.017	94.828 ± 0.727	0.272 ± 0.002	—
QML	beta43	0.20	0.204	0.004 ± 0.002	0.020 ± 0.009	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.076 ± 0.004	0.276 ± 0.008	0.204	0.004	1.99	0.0522	0.0523	0.979	94.397 ± 0.755	0.211 ± 0.001	—
QML	beta44	0.00	-0.012	-0.012 ± 0.002	—	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	—	—	-0.011	-0.011	—	0.0635	0.0645	0.997	94.612 ± 0.741	0.255 ± 0.003	5.388 ± 0.741
QML	beta45	0.00	0.005	0.005 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	—	—	0.004	0.004	—	0.0567	0.0568	0.960	94.935 ± 0.720	0.227 ± 0.003	5.065 ± 0.720
QML	beta46	0.00	0.021	0.021 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	0.018	0.018	—	0.0326	0.0371	1.002	93.534 ± 0.807	0.140 ± 0.002	6.466 ± 0.807
QML	beta47	0.00	0.021	0.021 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	0.017	0.017	—	0.0320	0.0361	0.950	93.858 ± 0.788	0.133 ± 0.002	6.142 ± 0.788
QML	beta51	0.16	0.163	0.003 ± 0.002	0.020 ± 0.015	0.006 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.074 ± 0.002	0.216 ± 0.011	0.464 ± 0.014	0.163	0.003	1.92	0.0741	0.0742	1.009	94.935 ± 0.720	0.294 ± 0.003	—
QML	beta52	0.16	0.140	-0.020 ± 0.002	-0.122 ± 0.015	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.074 ± 0.002	0.214 ± 0.011	0.462 ± 0.013	0.138	-0.022	-13.53	0.0703	0.0736	0.978	94.504 ± 0.748	0.274 ± 0.001	—
QML	beta53	0.16	0.160	0.000 ± 0.002	0.000 ± 0.013	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.157 ± 0.007	0.396 ± 0.011	0.158	-0.002	-1.18	0.0631	0.0631	1.026	95.474 ± 0.682	0.255 ± 0.002	—
QML	beta54	0.16	0.144	-0.016 ± 0.002	-0.102 ± 0.015	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.211 ± 0.010	0.459 ± 0.013	0.146	-0.014	-8.58	0.0703	0.0716	0.964	93.319 ± 0.820	0.271 ± 0.001	—
QML	beta55	0.00	-0.013	-0.013 ± 0.003	—	0.007 ± 0.000	0.085 ± 0.002	0.007 ± 0.000	0.086 ± 0.003	—	—	-0.014	-0.014	—	0.0813	0.0825	1.001	95.582 ± 0.675	0.332 ± 0.004	4.418 ± 0.675
QML	beta56	0.00	0.010	0.010 ± 0.002	—	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	—	—	0.009	0.009	—	0.0652	0.0659	0.942	93.319 ± 0.820	0.254 ± 0.003	6.681 ± 0.820
QML	beta57	0.00	0.019	0.019 ± 0.001	—	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	0.015	0.015	—	0.0401	0.0429	0.996	94.720 ± 0.734	0.163 ± 0.003	5.280 ± 0.734
QML	beta58	0.00	0.015	0.015 ± 0.001	—	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	0.014	0.014	—	0.0313	0.0341	0.952	93.858 ± 0.788	0.120 ± 0.002	6.142 ± 0.788
UPI	beta41	0.20	0.205	0.005 ± 0.003	0.027 ± 0.016	0.010 ± 0.001	0.100 ± 0.003	0.010 ± 0.001	0.101 ± 0.003	0.253 ± 0.014	0.503 ± 0.016	0.201	0.001	0.75	0.0943	0.0943	1.131	97.522 ± 0.510	0.446 ± 0.006	—
UPI	beta42	0.20	0.200	0.000 ± 0.003	0.002 ± 0.017	0.011 ± 0.001	0.107 ± 0.003	0.011 ± 0.001	0.107 ± 0.003	0.284 ± 0.015	0.533 ± 0.016	0.201	0.001	0.50	0.1029	0.1029	1.110	97.091 ± 0.552	0.464 ± 0.006	—
UPI	beta43	0.20	0.200	-0.000 ± 0.002	-0.001 ± 0.010	0.003 ± 0.000	0.058 ± 0.002	0.003 ± 0.000	0.058 ± 0.002	0.084 ± 0.004	0.289 ± 0.009	0.196	-0.004	-2.05	0.0556	0.0558	0.998	95.151 ± 0.705	0.227 ± 0.002	—
UPI	beta44	0.00	0.000	0.000 ± 0.003	—	0.006 ± 0.000	0.080 ± 0.002	0.006 ± 0.000	0.080 ± 0.003	—	—	-0.001	-0.001	—	0.0695	0.0695	1.108	96.444 ± 0.608	0.346 ± 0.008	3.556 ± 0.608
UPI	beta45	0.00	0.006	0.006 ± 0.003	—	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.077 ± 0.003	—	—	0.006	0.006	—	0.0657	0.0660	1.130	97.629 ± 0.499	0.339 ± 0.025	2.371 ± 0.499
UPI	beta46	0.00	-0.003	-0.003 ± 0.001	—	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	-0.002	-0.002	—	0.0378	0.0379	1.096	97.091 ± 0.552	0.180 ± 0.004	2.909 ± 0.552

Table S261

Simulation 2 - Condition 117: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.6
(continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	0.000	-0.000 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	-0.002	-0.002	—	0.0358	0.0358	1.085	96.659 ± 0.590	0.171 ± 0.003	3.341 ± 0.590
UPI	beta51	0.16	0.160	-0.000 ± 0.002	-0.001 ± 0.015	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.208 ± 0.010	0.456 ± 0.014	0.159	-0.001	-0.53	0.0712	0.0712	0.999	95.043 ± 0.713	0.286 ± 0.002	—
UPI	beta52	0.16	0.159	-0.001 ± 0.003	-0.006 ± 0.021	0.011 ± 0.001	0.104 ± 0.003	0.011 ± 0.001	0.104 ± 0.003	0.425 ± 0.022	0.652 ± 0.020	0.159	-0.001	-0.90	0.1022	0.1022	0.974	95.582 ± 0.675	0.398 ± 0.003	—
UPI	beta53	0.16	0.162	0.002 ± 0.002	0.013 ± 0.015	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	0.218 ± 0.010	0.467 ± 0.013	0.156	-0.004	-2.50	0.0774	0.0775	1.027	96.228 ± 0.625	0.301 ± 0.002	—
UPI	beta54	0.16	0.162	0.002 ± 0.003	0.011 ± 0.021	0.010 ± 0.000	0.102 ± 0.002	0.010 ± 0.000	0.102 ± 0.003	0.406 ± 0.019	0.637 ± 0.018	0.165	0.005	2.82	0.1045	0.1046	0.992	94.504 ± 0.748	0.396 ± 0.003	—
UPI	beta55	0.00	0.003	0.003 ± 0.003	—	0.010 ± 0.001	0.100 ± 0.003	0.010 ± 0.001	0.100 ± 0.003	—	—	-0.003	-0.003	—	0.0860	0.0860	1.016	96.552 ± 0.599	0.397 ± 0.007	3.448 ± 0.599
UPI	beta56	0.00	-0.005	-0.005 ± 0.003	—	0.007 ± 0.000	0.084 ± 0.002	0.007 ± 0.000	0.084 ± 0.003	—	—	-0.003	-0.003	—	0.0772	0.0773	0.996	94.289 ± 0.762	0.327 ± 0.005	5.711 ± 0.762
UPI	beta57	0.00	-0.001	-0.001 ± 0.002	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.002	—	—	-0.001	-0.001	—	0.0421	0.0421	0.945	95.797 ± 0.659	0.172 ± 0.002	4.203 ± 0.659
UPI	beta58	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.001	-0.001	—	0.0333	0.0333	0.974	95.151 ± 0.705	0.141 ± 0.002	4.849 ± 0.705

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S262

Simulation 2 - Condition 118: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, $N = 1000$, Reliability = 0.6

Method	Param.	True	Mean-based										Median-based					Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE		
LSAM	beta41	0.20	0.200	-0.000 ± 0.002	-0.001 ± 0.009	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.089 ± 0.004	0.298 ± 0.009	0.200	0.000	0.15	0.0589	0.0589	1.006	96.360 ± 0.596	0.235 ± 0.001	—		
LSAM	beta42	0.20	0.197	-0.003 ± 0.002	-0.017 ± 0.010	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.096 ± 0.004	0.310 ± 0.009	0.197	-0.003	-1.55	0.0640	0.0641	1.028	96.360 ± 0.596	0.249 ± 0.001	—		
LSAM	beta43	0.20	0.201	0.001 ± 0.001	0.007 ± 0.006	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.032 ± 0.002	0.180 ± 0.005	0.200	0.000	0.02	0.0369	0.0369	0.983	94.135 ± 0.747	0.139 ± 0.001	—		
LSAM	beta44	0.00	-0.002	-0.002 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	-0.002	-0.002	—	0.0373	0.0373	0.965	95.147 ± 0.683	0.147 ± 0.001	4.853 ± 0.683		
LSAM	beta45	0.00	0.002	0.002 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.001	-0.001	—	0.0362	0.0363	0.945	95.551 ± 0.656	0.142 ± 0.001	4.449 ± 0.656		
LSAM	beta46	0.00	0.000	0.000 ± 0.001	—	0.000 ± 0.000	0.022 ± 0.001	0.000 ± 0.000	0.022 ± 0.001	—	—	-0.002	-0.002	—	0.0183	0.0184	0.969	95.046 ± 0.690	0.082 ± 0.001	4.954 ± 0.690		
LSAM	beta47	0.00	0.001	0.001 ± 0.001	—	0.000 ± 0.000	0.022 ± 0.001	0.000 ± 0.000	0.022 ± 0.001	—	—	0.002	0.002	—	0.0207	0.0207	0.975	95.147 ± 0.683	0.083 ± 0.001	4.853 ± 0.683		
LSAM	beta51	0.16	0.162	0.002 ± 0.002	0.010 ± 0.009	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.088 ± 0.004	0.297 ± 0.008	0.160	0.000	-0.12	0.0481	0.0481	0.948	94.135 ± 0.747	0.176 ± 0.001	—		
LSAM	beta52	0.16	0.159	-0.001 ± 0.002	-0.007 ± 0.012	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.146 ± 0.007	0.381 ± 0.011	0.158	-0.002	-1.17	0.0608	0.0608	0.986	95.046 ± 0.690	0.236 ± 0.001	—		
LSAM	beta53	0.16	0.162	0.002 ± 0.002	0.010 ± 0.010	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.092 ± 0.004	0.303 ± 0.008	0.161	0.001	0.46	0.0483	0.0483	0.970	94.135 ± 0.747	0.185 ± 0.001	—		
LSAM	beta54	0.16	0.159	-0.001 ± 0.002	-0.006 ± 0.012	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.135 ± 0.006	0.367 ± 0.010	0.159	-0.001	-0.51	0.0575	0.0575	1.024	95.652 ± 0.648	0.236 ± 0.001	—		
LSAM	beta55	0.00	0.002	0.002 ± 0.002	—	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	—	—	0.002	0.002	—	0.0498	0.0498	0.944	94.135 ± 0.747	0.201 ± 0.002	5.865 ± 0.747		
LSAM	beta56	0.00	0.000	-0.000 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	0.000	0.000	—	0.0435	0.0435	0.957	94.540 ± 0.722	0.173 ± 0.001	5.460 ± 0.722		
LSAM	beta57	0.00	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.025 ± 0.001	0.001 ± 0.000	0.025 ± 0.001	—	—	0.001	0.001	—	0.0233	0.0233	0.930	94.135 ± 0.747	0.092 ± 0.001	5.865 ± 0.747		
LSAM	beta58	0.00	0.000	0.000 ± 0.001	—	0.000 ± 0.000	0.021 ± 0.001	0.000 ± 0.000	0.021 ± 0.001	—	—	0.000	0.000	—	0.0191	0.0191	0.951	94.237 ± 0.741	0.078 ± 0.001	5.763 ± 0.741		
QML	beta41	0.20	0.179	-0.021 ± 0.001	-0.105 ± 0.007	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.053 ± 0.003	0.230 ± 0.006	0.179	-0.021	-10.43	0.0417	0.0466	1.013	94.135 ± 0.747	0.163 ± 0.001	—		
QML	beta42	0.20	0.177	-0.023 ± 0.001	-0.113 ± 0.007	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.057 ± 0.003	0.239 ± 0.006	0.178	-0.022	-11.05	0.0426	0.0480	1.025	93.529 ± 0.782	0.169 ± 0.001	—		
QML	beta43	0.20	0.206	0.006 ± 0.001	0.029 ± 0.005	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.030 ± 0.001	0.174 ± 0.005	0.204	0.004	2.11	0.0338	0.0341	0.986	94.742 ± 0.710	0.133 ± 0.000	—		
QML	beta44	0.00	-0.012	-0.012 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	—	—	-0.011	-0.011	—	0.0396	0.0411	0.960	94.237 ± 0.741	0.156 ± 0.001	5.763 ± 0.741		
QML	beta45	0.00	0.003	0.003 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.002	0.002	—	0.0359	0.0359	0.967	94.843 ± 0.703	0.138 ± 0.001	5.157 ± 0.703		
QML	beta46	0.00	0.020	0.020 ± 0.001	—	0.000 ± 0.000	0.022 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	0.017	0.017	—	0.0198	0.0261	0.961	88.170 ± 1.027	0.082 ± 0.001	11.830 ± 1.027		
QML	beta47	0.00	0.020	0.020 ± 0.001	—	0.000 ± 0.000	0.022 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	0.019	0.019	—	0.0213	0.0284	0.941	86.249 ± 1.095	0.082 ± 0.001	13.751 ± 1.095		
QML	beta51	0.16	0.164	0.004 ± 0.001	0.024 ± 0.009	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.087 ± 0.004	0.294 ± 0.008	0.162	0.002	1.03	0.0475	0.0476	0.973	94.843 ± 0.703	0.179 ± 0.001	—		
QML	beta52	0.16	0.143	-0.017 ± 0.001	-0.108 ± 0.009	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.084 ± 0.004	0.290 ± 0.008	0.143	-0.017	-10.82	0.0418	0.0453	0.988	92.417 ± 0.842	0.167 ± 0.000	—		
QML	beta53	0.16	0.161	0.001 ± 0.001	0.003 ± 0.008	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.067 ± 0.003	0.259 ± 0.007	0.159	-0.001	-0.62	0.0420	0.0421	0.974	93.933 ± 0.759	0.159 ± 0.001	—		
QML	beta54	0.16	0.142	-0.018 ± 0.001	-0.111 ± 0.008	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.081 ± 0.004	0.285 ± 0.008	0.143	-0.017	-10.86	0.0408	0.0443	1.024	93.529 ± 0.782	0.169 ± 0.001	—		
QML	beta55	0.00	-0.009	-0.009 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	—	—	-0.009	-0.009	—	0.0500	0.0507	0.993	94.135 ± 0.747	0.203 ± 0.002	5.865 ± 0.747		
QML	beta56	0.00	0.013	0.013 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	—	—	0.012	0.012	—	0.0403	0.0420	0.978	94.439 ± 0.729	0.158 ± 0.001	5.561 ± 0.729		
QML	beta57	0.00	0.017	0.017 ± 0.001	—	0.001 ± 0.000	0.026 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	0.016	0.016	—	0.0245	0.0291	0.947	91.709 ± 0.877	0.097 ± 0.001	8.291 ± 0.877		
QML	beta58	0.00	0.014	0.014 ± 0.001	—	0.000 ± 0.000	0.020 ± 0.001	0.001 ± 0.000	0.025 ± 0.001	—	—	0.013	0.013	—	0.0192	0.0231	0.924	91.001 ± 0.910	0.074 ± 0.001	8.999 ± 0.910		
UPI	beta41	0.20	0.201	0.001 ± 0.002	0.004 ± 0.009	0.004 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	0.088 ± 0.004	0.297 ± 0.009	0.200	0.000	-0.01	0.0573	0.0573	1.028	95.956 ± 0.626	0.239 ± 0.001	—		
UPI	beta42	0.20	0.197	-0.003 ± 0.002	-0.016 ± 0.010	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.091 ± 0.004	0.302 ± 0.008	0.197	-0.003	-1.64	0.0600	0.0601	1.067	96.764 ± 0.563	0.253 ± 0.001	—		
UPI	beta43	0.20	0.201	0.001 ± 0.001	0.007 ± 0.006	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.031 ± 0.001	0.176 ± 0.005	0.201	0.001	0.27	0.0359	0.0359	0.984	94.135 ± 0.747	0.136 ± 0.001	—		
UPI	beta44	0.00	-0.002	-0.002 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	-0.002	-0.002	—	0.0380	0.0381	0.992	95.551 ± 0.656	0.158 ± 0.002	4.449 ± 0.656		

Table S262

Simulation 2 - Condition 118: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.6 (continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta45	0.00	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	-0.001	-0.001	—	0.0365	0.0365	0.970	96.158 ± 0.611	0.153 ± 0.002	3.842 ± 0.611
UPI	beta46	0.00	0.000	-0.000 ± 0.001	—	0.000 ± 0.000	0.022 ± 0.001	0.000 ± 0.000	0.022 ± 0.001	—	—	-0.002	-0.002	—	0.0199	0.0200	0.976	95.248 ± 0.677	0.085 ± 0.001	4.752 ± 0.677
UPI	beta47	0.00	0.001	0.001 ± 0.001	—	0.000 ± 0.000	0.022 ± 0.001	0.000 ± 0.000	0.022 ± 0.001	—	—	0.001	0.001	—	0.0204	0.0205	0.999	95.753 ± 0.641	0.086 ± 0.001	4.247 ± 0.641
UPI	beta51	0.16	0.161	0.001 ± 0.001	0.009 ± 0.009	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.085 ± 0.004	0.292 ± 0.008	0.160	0.000	-0.05	0.0470	0.0470	0.970	94.338 ± 0.735	0.177 ± 0.001	—
UPI	beta52	0.16	0.159	-0.001 ± 0.002	-0.007 ± 0.012	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.141 ± 0.007	0.376 ± 0.011	0.158	-0.002	-1.53	0.0612	0.0613	0.997	94.843 ± 0.703	0.235 ± 0.001	—
UPI	beta53	0.16	0.162	0.002 ± 0.001	0.012 ± 0.009	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.086 ± 0.004	0.294 ± 0.008	0.162	0.002	1.05	0.0474	0.0474	0.988	94.742 ± 0.710	0.182 ± 0.001	—
UPI	beta54	0.16	0.159	-0.001 ± 0.002	-0.006 ± 0.012	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.132 ± 0.006	0.363 ± 0.010	0.159	-0.001	-0.68	0.0568	0.0568	1.034	96.259 ± 0.603	0.236 ± 0.001	—
UPI	beta55	0.00	0.001	0.001 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	—	—	0.000	0.000	—	0.0504	0.0504	0.989	95.753 ± 0.641	0.205 ± 0.002	4.247 ± 0.641
UPI	beta56	0.00	0.000	-0.000 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	0.000	0.000	—	0.0425	0.0425	1.009	95.854 ± 0.634	0.177 ± 0.002	4.146 ± 0.634
UPI	beta57	0.00	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.025 ± 0.001	0.001 ± 0.000	0.025 ± 0.001	—	—	0.001	0.001	—	0.0233	0.0234	0.950	94.540 ± 0.722	0.093 ± 0.001	5.460 ± 0.722
UPI	beta58	0.00	0.000	0.000 ± 0.001	—	0.000 ± 0.000	0.021 ± 0.001	0.000 ± 0.000	0.021 ± 0.001	—	—	0.000	0.000	—	0.0192	0.0192	0.966	94.742 ± 0.710	0.078 ± 0.001	5.258 ± 0.710

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u002014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S263

Simulation 2 - Condition 119: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, $N = 400$, Reliability = 0.8

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Type I (%) $\pm$ MCSE
LSAM	beta41	0.20	0.198	-0.002 $\pm$ 0.002	-0.012 $\pm$ 0.012	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.139 $\pm$ 0.006	0.373 $\pm$ 0.010	0.199	-0.001	-0.72	0.0773	0.0773	0.989	94.785 $\pm$ 0.711	0.289 $\pm$ 0.001	—
LSAM	beta42	0.20	0.201	0.001 $\pm$ 0.002	0.007 $\pm$ 0.012	0.006 $\pm$ 0.000	0.077 $\pm$ 0.002	0.006 $\pm$ 0.000	0.077 $\pm$ 0.002	0.149 $\pm$ 0.007	0.386 $\pm$ 0.011	0.200	0.000	0.20	0.0749	0.0749	0.992	96.012 $\pm$ 0.626	0.300 $\pm$ 0.001	—
LSAM	beta43	0.20	0.202	0.002 $\pm$ 0.002	0.009 $\pm$ 0.008	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.056 $\pm$ 0.003	0.237 $\pm$ 0.007	0.200	0.000	0.24	0.0456	0.0456	0.964	93.661 $\pm$ 0.779	0.179 $\pm$ 0.001	—
LSAM	beta44	0.00	-0.001	-0.001 $\pm$ 0.002	—	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.003 $\pm$ 0.000	0.052 $\pm$ 0.002	—	—	-0.001	-0.001	—	0.0466	0.0466	0.921	93.354 $\pm$ 0.796	0.187 $\pm$ 0.002	6.646 $\pm$ 0.796
LSAM	beta45	0.00	0.000	-0.000 $\pm$ 0.002	—	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	—	—	0.002	0.002	—	0.0464	0.0465	0.946	94.070 $\pm$ 0.755	0.182 $\pm$ 0.002	5.930 $\pm$ 0.755
LSAM	beta46	0.00	0.000	0.000 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	—	—	0.000	0.000	—	0.0268	0.0268	0.900	93.149 $\pm$ 0.808	0.105 $\pm$ 0.001	6.851 $\pm$ 0.808
LSAM	beta47	0.00	0.000	0.000 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	—	—	0.001	0.001	—	0.0267	0.0267	0.924	93.763 $\pm$ 0.773	0.105 $\pm$ 0.001	6.237 $\pm$ 0.773
LSAM	beta51	0.16	0.161	0.001 $\pm$ 0.002	0.007 $\pm$ 0.012	0.004 $\pm$ 0.000	0.062 $\pm$ 0.001	0.004 $\pm$ 0.000	0.062 $\pm$ 0.002	0.148 $\pm$ 0.007	0.385 $\pm$ 0.011	0.162	0.002	1.09	0.0618	0.0618	0.939	93.354 $\pm$ 0.796	0.227 $\pm$ 0.001	—
LSAM	beta52	0.16	0.163	0.003 $\pm$ 0.003	0.021 $\pm$ 0.016	0.006 $\pm$ 0.000	0.079 $\pm$ 0.002	0.006 $\pm$ 0.000	0.079 $\pm$ 0.002	0.245 $\pm$ 0.011	0.494 $\pm$ 0.014	0.162	0.002	1.54	0.0796	0.0796	0.947	92.638 $\pm$ 0.835	0.294 $\pm$ 0.001	—
LSAM	beta53	0.16	0.160	0.000 $\pm$ 0.002	0.001 $\pm$ 0.012	0.004 $\pm$ 0.000	0.061 $\pm$ 0.001	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.146 $\pm$ 0.007	0.382 $\pm$ 0.011	0.160	0.000	-0.22	0.0622	0.0622	0.952	93.456 $\pm$ 0.791	0.228 $\pm$ 0.001	—
LSAM	beta54	0.16	0.162	0.002 $\pm$ 0.002	0.013 $\pm$ 0.015	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.221 $\pm$ 0.011	0.470 $\pm$ 0.014	0.159	-0.001	-0.90	0.0723	0.0723	0.980	94.274 $\pm$ 0.743	0.289 $\pm$ 0.001	—
LSAM	beta55	0.00	-0.001	-0.001 $\pm$ 0.002	—	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	0.005 $\pm$ 0.000	0.068 $\pm$ 0.002	—	—	-0.001	-0.001	—	0.0674	0.0674	0.928	93.763 $\pm$ 0.773	0.247 $\pm$ 0.002	6.237 $\pm$ 0.773
LSAM	beta56	0.00	-0.001	-0.001 $\pm$ 0.002	—	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	0.004 $\pm$ 0.000	0.060 $\pm$ 0.002	—	—	-0.004	-0.004	—	0.0552	0.0553	0.882	90.798 $\pm$ 0.924	0.209 $\pm$ 0.002	9.202 $\pm$ 0.924
LSAM	beta57	0.00	-0.001	-0.001 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	—	—	-0.001	-0.001	—	0.0319	0.0319	0.871	93.047 $\pm$ 0.813	0.117 $\pm$ 0.001	6.953 $\pm$ 0.813
LSAM	beta58	0.00	-0.002	-0.002 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.027 $\pm$ 0.001	0.001 $\pm$ 0.000	0.027 $\pm$ 0.001	—	—	-0.002	-0.002	—	0.0241	0.0242	0.934	93.763 $\pm$ 0.773	0.099 $\pm$ 0.001	6.237 $\pm$ 0.773
QML	beta41	0.20	0.187	-0.013 $\pm$ 0.002	-0.066 $\pm$ 0.010	0.004 $\pm$ 0.000	0.061 $\pm$ 0.001	0.004 $\pm$ 0.000	0.062 $\pm$ 0.002	0.096 $\pm$ 0.004	0.310 $\pm$ 0.008	0.187	-0.013	-6.63	0.0625	0.0639	0.999	94.683 $\pm$ 0.717	0.238 $\pm$ 0.001	—
QML	beta42	0.20	0.190	-0.010 $\pm$ 0.002	-0.048 $\pm$ 0.010	0.004 $\pm$ 0.000	0.062 $\pm$ 0.001	0.004 $\pm$ 0.000	0.063 $\pm$ 0.002	0.098 $\pm$ 0.005	0.313 $\pm$ 0.009	0.190	-0.010	-5.10	0.0620	0.0628	1.005	95.501 $\pm$ 0.663	0.244 $\pm$ 0.001	—
QML	beta43	0.20	0.204	0.004 $\pm$ 0.001	0.021 $\pm$ 0.007	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.054 $\pm$ 0.002	0.232 $\pm$ 0.006	0.203	0.003	1.68	0.0444	0.0446	0.965	93.354 $\pm$ 0.796	0.175 $\pm$ 0.001	—
QML	beta44	0.00	-0.006	-0.006 $\pm$ 0.002	—	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.003 $\pm$ 0.000	0.051 $\pm$ 0.002	—	—	-0.006	-0.006	—	0.0461	0.0464	0.923	93.456 $\pm$ 0.791	0.185 $\pm$ 0.002	6.544 $\pm$ 0.791
QML	beta45	0.00	0.000	-0.000 $\pm$ 0.002	—	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	—	—	0.001	0.001	—	0.0438	0.0438	0.949	93.865 $\pm$ 0.767	0.175 $\pm$ 0.002	6.135 $\pm$ 0.767
QML	beta46	0.00	0.009	0.009 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.028 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	—	—	0.007	0.007	—	0.0243	0.0254	0.898	92.740 $\pm$ 0.830	0.099 $\pm$ 0.001	7.260 $\pm$ 0.830
QML	beta47	0.00	0.009	0.009 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.027 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	—	—	0.007	0.007	—	0.0253	0.0263	0.918	93.661 $\pm$ 0.779	0.098 $\pm$ 0.001	6.339 $\pm$ 0.779
QML	beta51	0.16	0.162	0.002 $\pm$ 0.002	0.010 $\pm$ 0.012	0.004 $\pm$ 0.000	0.061 $\pm$ 0.001	0.004 $\pm$ 0.000	0.061 $\pm$ 0.002	0.145 $\pm$ 0.007	0.381 $\pm$ 0.011	0.163	0.003	1.57	0.0615	0.0616	0.970	94.479 $\pm$ 0.730	0.232 $\pm$ 0.001	—
QML	beta52	0.16	0.154	-0.006 $\pm$ 0.002	-0.039 $\pm$ 0.013	0.004 $\pm$ 0.000	0.066 $\pm$ 0.001	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	0.170 $\pm$ 0.008	0.412 $\pm$ 0.011	0.153	-0.007	-4.11	0.0654	0.0657	0.953	93.763 $\pm$ 0.773	0.246 $\pm$ 0.001	—
QML	beta53	0.16	0.161	0.001 $\pm$ 0.002	0.006 $\pm$ 0.011	0.003 $\pm$ 0.000	0.056 $\pm$ 0.001	0.003 $\pm$ 0.000	0.056 $\pm$ 0.002	0.124 $\pm$ 0.006	0.352 $\pm$ 0.010	0.160	0.000	0.17	0.0572	0.0572	0.957	93.558 $\pm$ 0.785	0.211 $\pm$ 0.001	—
QML	beta54	0.16	0.152	-0.008 $\pm$ 0.002	-0.049 $\pm$ 0.012	0.004 $\pm$ 0.000	0.063 $\pm$ 0.001	0.004 $\pm$ 0.000	0.063 $\pm$ 0.002	0.155 $\pm$ 0.007	0.394 $\pm$ 0.011	0.151	-0.009	-5.78	0.0613	0.0620	0.989	94.683 $\pm$ 0.717	0.242 $\pm$ 0.001	—
QML	beta55	0.00	-0.005	-0.005 $\pm$ 0.002	—	0.004 $\pm$ 0.000	0.065 $\pm$ 0.002	0.004 $\pm$ 0.000	0.065 $\pm$ 0.002	—	—	-0.004	-0.004	—	0.0634	0.0635	0.944	94.172 $\pm$ 0.749	0.239 $\pm$ 0.002	5.828 $\pm$ 0.749
QML	beta56	0.00	0.003	0.003 $\pm$ 0.002	—	0.003 $\pm$ 0.000	0.056 $\pm$ 0.001	0.003 $\pm$ 0.000	0.056 $\pm$ 0.002	—	—	0.000	0.000	—	0.0528	0.0528	0.890	90.900 $\pm$ 0.920	0.196 $\pm$ 0.002	9.100 $\pm$ 0.920
QML	beta57	0.00	0.006	0.006 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	—	—	0.005	0.005	—	0.0314	0.0319	0.886	92.536 $\pm$ 0.840	0.113 $\pm$ 0.001	7.464 $\pm$ 0.840
QML	beta58	0.00	0.005	0.005 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.025 $\pm$ 0.001	0.001 $\pm$ 0.000	0.025 $\pm$ 0.001	—	—	0.003	0.003	—	0.0219	0.0221	0.943	94.376 $\pm$ 0.737	0.092 $\pm$ 0.001	5.624 $\pm$ 0.737
UPI	beta41	0.20	0.199	-0.001 $\pm$ 0.002	-0.005 $\pm$ 0.012	0.005 $\pm$ 0.000	0.073 $\pm$ 0.002	0.005 $\pm$ 0.000	0.073 $\pm$ 0.002	0.134 $\pm$ 0.006	0.367 $\pm$ 0.010	0.195	-0.005	-2.26	0.0753	0.0754	1.009	95.603 $\pm$ 0.656	0.290 $\pm$ 0.002	—
UPI	beta42	0.20	0.201	0.001 $\pm$ 0.002	0.007 $\pm$ 0.012	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.006 $\pm$ 0.000	0.075 $\pm$ 0.002	0.142 $\pm$ 0.007	0.377 $\pm$ 0.011	0.199	-0.001	-0.54	0.0719	0.0719	1.025	96.626 $\pm$ 0.577	0.303 $\pm$ 0.002	—
UPI	beta43	0.20	0.202	0.002 $\pm$ 0.002	0.008 $\pm$ 0.008	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.055 $\pm$ 0.002	0.236 $\pm$ 0.007	0.200	0.000	-0.21	0.0454	0.0454	0.964	93.865 $\pm$ 0.767	0.178 $\pm$ 0.001	—
UPI	beta44	0.00	-0.001	-0.001 $\pm$ 0.002	—	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.003 $\pm$ 0.000	0.052 $\pm$ 0.002	—	—	-0.002	-0.002	—	0.0489	0.0489	0.963	94.479 $\pm$ 0.730	0.196 $\pm$ 0.002	5.521 $\pm$ 0.730
UPI	beta45	0.00	-0.001	-0.001 $\pm$ 0.002	—	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	0.002 $\pm$ 0.000	0.050 $\pm$ 0.002	—	—	0.000	0.000	—	0.0448	0.0448	0.971	94.479 $\pm$ 0.730	0.189 $\pm$ 0.002	5.521 $\pm$ 0.730
UPI	beta46	0.00	0.000	-0.000 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	—	—	-0.001	-0.001	—	0.0260	0.0260	0.918	94.172 $\pm$ 0.749	0.108 $\pm$ 0.001	5.828 $\pm$ 0.749

Table S263

Simulation 2 - Condition 119: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.8
(continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	0.001	0.001	—	0.0268	0.0268	0.953	94.581 ± 0.724	0.108 ± 0.001	5.419 ± 0.724
UPI	beta51	0.16	0.161	0.001 ± 0.002	0.005 ± 0.012	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.144 ± 0.007	0.380 ± 0.011	0.161	0.001	0.86	0.0615	0.0615	0.965	94.479 ± 0.730	0.230 ± 0.001	—
UPI	beta52	0.16	0.164	0.004 ± 0.003	0.022 ± 0.016	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	0.244 ± 0.011	0.494 ± 0.014	0.163	0.003	1.59	0.0769	0.0770	0.944	92.945 ± 0.819	0.292 ± 0.001	—
UPI	beta53	0.16	0.160	-0.000 ± 0.002	-0.000 ± 0.012	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.142 ± 0.006	0.377 ± 0.010	0.160	0.000	-0.12	0.0610	0.0610	0.956	93.354 ± 0.796	0.226 ± 0.001	—
UPI	beta54	0.16	0.162	0.002 ± 0.002	0.015 ± 0.015	0.006 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.074 ± 0.002	0.216 ± 0.010	0.465 ± 0.013	0.159	-0.001	-0.90	0.0736	0.0736	0.989	94.376 ± 0.737	0.289 ± 0.001	—
UPI	beta55	0.00	-0.003	-0.003 ± 0.002	—	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.067 ± 0.002	—	—	-0.003	-0.003	—	0.0651	0.0651	0.956	94.070 ± 0.755	0.252 ± 0.002	5.930 ± 0.755
UPI	beta56	0.00	-0.002	-0.002 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	—	—	-0.004	-0.004	—	0.0546	0.0548	0.902	91.207 ± 0.906	0.212 ± 0.002	8.793 ± 0.906
UPI	beta57	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	—	—	-0.001	-0.001	—	0.0315	0.0315	0.877	93.354 ± 0.796	0.117 ± 0.001	6.646 ± 0.796
UPI	beta58	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	—	—	-0.002	-0.002	—	0.0235	0.0236	0.945	94.888 ± 0.704	0.099 ± 0.001	5.112 ± 0.704

Note. MCSE = Monte Carlo Standard Error shown as \u000b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S264

Simulation 2 - Condition 120: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.8

			Mean-based										Median-based					Inference				
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE		
LSAM	beta41	0.20	0.201	0.001 ± 0.001	0.006 ± 0.007	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.054 ± 0.002	0.231 ± 0.006	0.200	0.000	0.08	0.0466	0.0466	0.972	95.090 ± 0.684	0.176 ± 0.001	—		
LSAM	beta42	0.20	0.199	-0.001 ± 0.001	-0.006 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.054 ± 0.002	0.233 ± 0.006	0.200	0.000	-0.10	0.0474	0.0474	1.013	95.391 ± 0.664	0.185 ± 0.001	—		
LSAM	beta43	0.20	0.201	0.001 ± 0.001	0.004 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.022 ± 0.001	0.147 ± 0.004	0.201	0.001	0.37	0.0281	0.0281	0.978	94.289 ± 0.735	0.112 ± 0.000	—		
LSAM	beta44	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	0.000	0.000	—	0.0294	0.0294	0.952	94.389 ± 0.728	0.112 ± 0.001	5.611 ± 0.728		
LSAM	beta45	0.00	0.002	0.002 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	0.001	0.001	—	0.0277	0.0277	0.937	93.487 ± 0.781	0.109 ± 0.001	6.513 ± 0.781		
LSAM	beta46	0.00	0.000	-0.000 ± 0.001	—	0.000 ± 0.000	0.016 ± 0.000	0.000 ± 0.000	0.016 ± 0.000	—	—	-0.001	-0.001	—	0.0153	0.0153	0.927	93.888 ± 0.758	0.060 ± 0.001	6.112 ± 0.758		
LSAM	beta47	0.00	0.000	0.000 ± 0.001	—	0.000 ± 0.000	0.017 ± 0.000	0.000 ± 0.000	0.017 ± 0.000	—	—	0.000	0.000	—	0.0160	0.0160	0.945	93.587 ± 0.775	0.061 ± 0.001	6.413 ± 0.775		
LSAM	beta51	0.16	0.160	0.000 ± 0.001	0.001 ± 0.008	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.056 ± 0.002	0.237 ± 0.006	0.159	-0.001	-0.56	0.0369	0.0369	0.969	93.788 ± 0.764	0.144 ± 0.000	—		
LSAM	beta52	0.16	0.158	-0.002 ± 0.002	-0.010 ± 0.010	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.090 ± 0.004	0.300 ± 0.008	0.157	-0.003	-1.99	0.0469	0.0470	0.952	94.188 ± 0.741	0.179 ± 0.001	—		
LSAM	beta53	0.16	0.162	0.002 ± 0.001	0.011 ± 0.007	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.055 ± 0.002	0.234 ± 0.006	0.160	0.000	0.30	0.0366	0.0366	0.986	94.589 ± 0.716	0.145 ± 0.001	—		
LSAM	beta54	0.16	0.160	0.000 ± 0.001	0.001 ± 0.009	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.085 ± 0.004	0.292 ± 0.008	0.161	0.001	0.55	0.0476	0.0476	0.978	94.389 ± 0.728	0.179 ± 0.001	—		
LSAM	beta55	0.00	0.002	0.002 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	0.003	0.003	—	0.0353	0.0354	0.949	94.689 ± 0.710	0.148 ± 0.001	5.311 ± 0.710		
LSAM	beta56	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	-0.002	-0.002	—	0.0355	0.0355	0.913	94.188 ± 0.741	0.129 ± 0.001	5.812 ± 0.741		
LSAM	beta57	0.00	0.000	-0.000 ± 0.001	—	0.000 ± 0.000	0.020 ± 0.001	0.000 ± 0.000	0.020 ± 0.001	—	—	-0.001	-0.001	—	0.0178	0.0179	0.887	91.884 ± 0.864	0.068 ± 0.001	8.116 ± 0.864		
LSAM	beta58	0.00	0.000	0.000 ± 0.001	—	0.000 ± 0.000	0.016 ± 0.000	0.000 ± 0.000	0.016 ± 0.000	—	—	-0.001	-0.001	—	0.0149	0.0149	0.900	92.585 ± 0.829	0.058 ± 0.001	7.415 ± 0.829		
QML	beta41	0.20	0.191	-0.009 ± 0.001	-0.046 ± 0.006	0.001 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.039 ± 0.002	0.198 ± 0.005	0.190	-0.010	-4.76	0.0388	0.0399	0.976	94.489 ± 0.722	0.147 ± 0.000	—		
QML	beta42	0.20	0.189	-0.011 ± 0.001	-0.055 ± 0.006	0.001 ± 0.000	0.038 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.039 ± 0.002	0.198 ± 0.005	0.191	-0.009	-4.66	0.0395	0.0406	1.019	94.890 ± 0.697	0.152 ± 0.000	—		
QML	beta43	0.20	0.203	0.003 ± 0.001	0.016 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.021 ± 0.001	0.144 ± 0.004	0.203	0.003	1.37	0.0279	0.0280	0.979	94.389 ± 0.728	0.110 ± 0.000	—		
QML	beta44	0.00	-0.003	-0.003 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	-0.003	-0.003	—	0.0303	0.0304	0.959	94.088 ± 0.747	0.115 ± 0.001	5.912 ± 0.747		
QML	beta45	0.00	0.002	0.002 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	0.001	0.001	—	0.0281	0.0282	0.938	93.587 ± 0.775	0.108 ± 0.001	6.413 ± 0.775		
QML	beta46	0.00	0.007	0.007 ± 0.001	—	0.000 ± 0.000	0.016 ± 0.000	0.000 ± 0.000	0.018 ± 0.001	—	—	0.006	0.006	—	0.0153	0.0163	0.929	92.385 ± 0.840	0.058 ± 0.001	7.615 ± 0.840		
QML	beta47	0.00	0.007	0.007 ± 0.001	—	0.000 ± 0.000	0.016 ± 0.000	0.000 ± 0.000	0.018 ± 0.001	—	—	0.006	0.006	—	0.0156	0.0168	0.931	93.086 ± 0.803	0.059 ± 0.001	6.914 ± 0.803		
QML	beta51	0.16	0.161	0.001 ± 0.001	0.004 ± 0.007	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.056 ± 0.002	0.236 ± 0.006	0.160	0.000	-0.31	0.0369	0.0369	0.983	94.589 ± 0.716	0.146 ± 0.000	—		
QML	beta52	0.16	0.150	-0.010 ± 0.001	-0.061 ± 0.008	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	0.068 ± 0.003	0.261 ± 0.007	0.149	-0.011	-6.95	0.0398	0.0413	0.958	93.788 ± 0.764	0.152 ± 0.000	—		
QML	beta53	0.16	0.162	0.002 ± 0.001	0.013 ± 0.007	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.047 ± 0.002	0.217 ± 0.006	0.161	0.001	0.71	0.0333	0.0333	0.985	94.589 ± 0.716	0.134 ± 0.000	—		
QML	beta54	0.16	0.152	-0.008 ± 0.001	-0.051 ± 0.008	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.064 ± 0.003	0.254 ± 0.007	0.153	-0.007	-4.46	0.0407	0.0413	0.981	94.289 ± 0.735	0.153 ± 0.000	—		
QML	beta55	0.00	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	-0.001	-0.001	—	0.0353	0.0354	0.968	95.190 ± 0.677	0.148 ± 0.001	4.810 ± 0.677		
QML	beta56	0.00	0.004	0.004 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	0.004	0.004	—	0.0346	0.0348	0.906	93.186 ± 0.798	0.123 ± 0.001	6.814 ± 0.798		
QML	beta57	0.00	0.006	0.006 ± 0.001	—	0.000 ± 0.000	0.020 ± 0.001	0.000 ± 0.000	0.020 ± 0.001	—	—	0.005	0.005	—	0.0180	0.0186	0.890	92.685 ± 0.824	0.068 ± 0.001	7.315 ± 0.824		
QML	beta58	0.00	0.005	0.005 ± 0.001	—	0.000 ± 0.000	0.016 ± 0.000	0.000 ± 0.000	0.017 ± 0.001	—	—	0.004	0.004	—	0.0150	0.0156	0.892	92.685 ± 0.824	0.056 ± 0.001	7.315 ± 0.824		
UPI	beta41	0.20	0.202	0.002 ± 0.001	0.008 ± 0.007	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.052 ± 0.002	0.227 ± 0.006	0.200	0.000	-0.17	0.0452	0.0452	0.983	95.090 ± 0.684	0.175 ± 0.001	—		
UPI	beta42	0.20	0.199	-0.001 ± 0.001	-0.005 ± 0.007	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.053 ± 0.002	0.230 ± 0.006	0.200	0.000	-0.16	0.0464	0.0464	1.022	96.092 ± 0.613	0.184 ± 0.001	—		
UPI	beta43	0.20	0.201	0.001 ± 0.001	0.004 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.021 ± 0.001	0.146 ± 0.004	0.200	0.000	-0.17	0.0285	0.0285	0.978	94.289 ± 0.735	0.112 ± 0.000	—		
UPI	beta44	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	-0.001	-0.001	—	0.0292	0.0292	0.945	94.289 ± 0.735	0.114 ± 0.001	5.711 ± 0.735		

Table S264

Simulation 2 - Condition 120: Linear Model, Distr = Right-skewed, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.8 (continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta45	0.00	0.002	0.002 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	0.001	0.001	—	0.0277	0.0277	0.948	93.888 ± 0.758	0.111 ± 0.001	6.112 ± 0.758
UPI	beta46	0.00	-0.001	-0.001 ± 0.001	—	0.000 ± 0.000	0.017 ± 0.000	0.000 ± 0.000	0.017 ± 0.001	—	—	-0.001	-0.001	—	0.0151	0.0151	0.931	94.589 ± 0.716	0.060 ± 0.001	5.411 ± 0.716
UPI	beta47	0.00	0.000	0.000 ± 0.001	—	0.000 ± 0.000	0.017 ± 0.000	0.000 ± 0.000	0.017 ± 0.001	—	—	0.000	0.000	—	0.0157	0.0157	0.953	93.988 ± 0.752	0.062 ± 0.001	6.012 ± 0.752
UPI	beta51	0.16	0.160	0.000 ± 0.001	0.000 ± 0.007	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.055 ± 0.002	0.236 ± 0.006	0.159	-0.001	-0.64	0.0370	0.0370	0.981	94.489 ± 0.722	0.145 ± 0.000	—
UPI	beta52	0.16	0.158	-0.002 ± 0.002	-0.010 ± 0.009	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.089 ± 0.004	0.299 ± 0.008	0.156	-0.004	-2.29	0.0466	0.0467	0.956	93.888 ± 0.758	0.179 ± 0.001	—
UPI	beta53	0.16	0.162	0.002 ± 0.001	0.012 ± 0.007	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.054 ± 0.002	0.232 ± 0.006	0.160	0.000	-0.04	0.0360	0.0360	0.988	95.090 ± 0.684	0.144 ± 0.001	—
UPI	beta54	0.16	0.160	0.000 ± 0.001	0.001 ± 0.009	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.086 ± 0.004	0.293 ± 0.008	0.160	0.000	-0.10	0.0476	0.0476	0.975	94.088 ± 0.747	0.179 ± 0.001	—
UPI	beta55	0.00	0.002	0.002 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	0.002	0.002	—	0.0363	0.0363	0.963	95.391 ± 0.664	0.150 ± 0.001	4.609 ± 0.664
UPI	beta56	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	-0.001	-0.001	—	0.0358	0.0359	0.923	94.088 ± 0.747	0.130 ± 0.001	5.912 ± 0.747
UPI	beta57	0.00	0.000	-0.000 ± 0.001	—	0.000 ± 0.000	0.020 ± 0.001	0.000 ± 0.000	0.020 ± 0.001	—	—	-0.001	-0.001	—	0.0181	0.0181	0.892	92.184 ± 0.850	0.068 ± 0.001	7.816 ± 0.850
UPI	beta58	0.00	0.000	0.000 ± 0.001	—	0.000 ± 0.000	0.017 ± 0.000	0.000 ± 0.000	0.017 ± 0.001	—	—	-0.001	-0.001	—	0.0150	0.0150	0.897	92.886 ± 0.814	0.058 ± 0.001	7.114 ± 0.814

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S265

Simulation 2 - Condition 121: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, $N = 400$, Reliability = 0.4

Method	Param.	True	Mean-based										Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE	
LSAM	beta41	0.20	0.206	0.006 ± 0.006	0.031 ± 0.028	0.008 ± 0.001	0.088 ± 0.004	0.008 ± 0.001	0.088 ± 0.005	0.195 ± 0.019	0.442 ± 0.026	0.197	-0.003	-1.39	0.0840	0.0840	1.084	95.918 ± 1.264	0.375 ± 0.011	—	
LSAM	beta42	0.20	0.202	0.002 ± 0.006	0.011 ± 0.029	0.008 ± 0.001	0.091 ± 0.005	0.008 ± 0.001	0.091 ± 0.005	0.205 ± 0.021	0.453 ± 0.027	0.197	-0.003	-1.72	0.0871	0.0872	1.099	94.694 ± 1.432	0.391 ± 0.010	—	
LSAM	beta43	0.20	0.204	0.004 ± 0.005	0.021 ± 0.027	0.007 ± 0.001	0.084 ± 0.004	0.007 ± 0.001	0.084 ± 0.005	0.175 ± 0.016	0.418 ± 0.023	0.199	-0.001	-0.29	0.0892	0.0892	1.065	99.184 ± 0.575	0.349 ± 0.008	—	
LSAM	beta44	0.00	-0.007	-0.007 ± 0.007	—	0.014 ± 0.002	0.117 ± 0.007	0.014 ± 0.002	0.117 ± 0.008	—	—	0.001	0.001	—	0.1124	0.1124	1.322	98.367 ± 0.810	0.604 ± 0.024	1.633 ± 0.810	
LSAM	beta45	0.00	0.002	0.002 ± 0.006	—	0.010 ± 0.001	0.100 ± 0.005	0.010 ± 0.001	0.100 ± 0.006	—	—	0.003	0.003	—	0.1037	0.1037	1.159	98.367 ± 0.810	0.453 ± 0.013	1.633 ± 0.810	
LSAM	beta46	0.00	0.014	0.014 ± 0.010	—	0.023 ± 0.003	0.152 ± 0.009	0.023 ± 0.003	0.153 ± 0.010	—	—	0.015	0.015	—	0.1495	0.1503	1.250	100.000 ± 0.000	0.746 ± 0.035	0.000 ± 0.000	
LSAM	beta47	0.00	-0.006	-0.006 ± 0.010	—	0.024 ± 0.002	0.156 ± 0.008	0.024 ± 0.002	0.155 ± 0.009	—	—	-0.006	-0.006	—	0.1454	0.1455	1.265	99.592 ± 0.407	0.771 ± 0.032	0.408 ± 0.407	
LSAM	beta51	0.16	0.161	0.001 ± 0.008	0.004 ± 0.050	0.016 ± 0.002	0.125 ± 0.007	0.016 ± 0.002	0.125 ± 0.008	0.608 ± 0.071	0.780 ± 0.052	0.166	0.006	3.89	0.1300	0.1301	0.884	93.061 ± 1.623	0.433 ± 0.006	—	
LSAM	beta52	0.16	0.153	-0.007 ± 0.005	-0.046 ± 0.032	0.006 ± 0.001	0.079 ± 0.004	0.006 ± 0.001	0.079 ± 0.005	0.246 ± 0.023	0.496 ± 0.028	0.153	-0.007	-4.59	0.0799	0.0802	1.163	98.367 ± 0.810	0.361 ± 0.006	—	
LSAM	beta53	0.16	0.167	0.007 ± 0.005	0.044 ± 0.033	0.007 ± 0.001	0.083 ± 0.004	0.007 ± 0.001	0.083 ± 0.005	0.271 ± 0.024	0.521 ± 0.028	0.167	0.007	4.54	0.0906	0.0909	1.146	98.367 ± 0.810	0.374 ± 0.008	—	
LSAM	beta54	0.16	0.165	0.005 ± 0.006	0.032 ± 0.039	0.010 ± 0.001	0.098 ± 0.005	0.010 ± 0.001	0.098 ± 0.006	0.376 ± 0.038	0.614 ± 0.037	0.159	-0.001	-0.71	0.0894	0.0894	0.970	94.286 ± 1.483	0.373 ± 0.007	—	
LSAM	beta55	0.00	-0.029	-0.029 ± 0.011	—	0.029 ± 0.003	0.169 ± 0.010	0.029 ± 0.003	0.171 ± 0.012	—	—	-0.021	-0.021	—	0.1499	0.1513	1.013	99.184 ± 0.575	0.671 ± 0.022	0.816 ± 0.575	
LSAM	beta56	0.00	0.035	0.035 ± 0.010	—	0.025 ± 0.003	0.159 ± 0.009	0.027 ± 0.003	0.163 ± 0.010	—	—	0.023	0.023	—	0.1428	0.1446	0.984	95.918 ± 1.264	0.615 ± 0.020	4.082 ± 1.264	
LSAM	beta57	0.00	0.015	0.015 ± 0.010	—	0.024 ± 0.003	0.154 ± 0.011	0.024 ± 0.003	0.155 ± 0.012	—	—	0.008	0.008	—	0.1371	0.1374	1.091	98.367 ± 0.810	0.660 ± 0.024	1.633 ± 0.810	
LSAM	beta58	0.00	-0.006	-0.006 ± 0.010	—	0.024 ± 0.003	0.155 ± 0.008	0.024 ± 0.003	0.154 ± 0.010	—	—	0.005	0.005	—	0.1573	0.1574	1.127	99.592 ± 0.407	0.683 ± 0.028	0.408 ± 0.407	
QML	beta41	0.20	0.206	0.006 ± 0.005	0.029 ± 0.025	0.006 ± 0.001	0.078 ± 0.005	0.006 ± 0.001	0.078 ± 0.005	0.154 ± 0.018	0.392 ± 0.026	0.202	0.002	0.94	0.0750	0.0751	1.003	95.102 ± 1.379	0.308 ± 0.009	—	
QML	beta42	0.20	0.201	0.001 ± 0.005	0.003 ± 0.025	0.006 ± 0.001	0.079 ± 0.004	0.006 ± 0.001	0.079 ± 0.005	0.156 ± 0.017	0.395 ± 0.025	0.195	-0.005	-2.58	0.0769	0.0771	0.991	94.694 ± 1.432	0.307 ± 0.004	—	
QML	beta43	0.20	0.202	0.002 ± 0.005	0.011 ± 0.023	0.005 ± 0.000	0.072 ± 0.003	0.005 ± 0.000	0.072 ± 0.004	0.130 ± 0.011	0.361 ± 0.019	0.200	0.000	-0.16	0.0829	0.0829	1.030	96.735 ± 1.135	0.292 ± 0.003	—	
QML	beta44	0.00	-0.022	-0.022 ± 0.006	—	0.008 ± 0.001	0.090 ± 0.004	0.008 ± 0.001	0.092 ± 0.006	—	—	-0.015	-0.015	—	0.1013	0.1024	1.245	95.918 ± 1.264	0.437 ± 0.013	4.082 ± 1.264	
QML	beta45	0.00	-0.011	-0.011 ± 0.005	—	0.007 ± 0.001	0.085 ± 0.005	0.007 ± 0.001	0.086 ± 0.006	—	—	-0.008	-0.008	—	0.0861	0.0865	1.075	97.551 ± 0.987	0.360 ± 0.009	2.449 ± 0.987	
QML	beta46	0.00	0.018	0.018 ± 0.005	—	0.006 ± 0.001	0.074 ± 0.004	0.006 ± 0.001	0.076 ± 0.005	—	—	0.008	0.008	—	0.0718	0.0722	1.191	99.592 ± 0.407	0.347 ± 0.016	0.408 ± 0.407	
QML	beta47	0.00	0.004	0.004 ± 0.005	—	0.005 ± 0.001	0.072 ± 0.004	0.005 ± 0.001	0.072 ± 0.004	—	—	0.002	0.002	—	0.0737	0.0737	1.161	97.959 ± 0.903	0.328 ± 0.013	2.041 ± 0.903	
QML	beta51	0.16	0.164	0.004 ± 0.007	0.022 ± 0.045	0.013 ± 0.001	0.114 ± 0.006	0.013 ± 0.001	0.113 ± 0.007	0.503 ± 0.050	0.709 ± 0.042	0.164	0.004	2.22	0.1060	0.1061	1.088	95.510 ± 1.323	0.485 ± 0.020	—	
QML	beta52	0.16	0.153	-0.007 ± 0.005	-0.041 ± 0.033	0.007 ± 0.001	0.082 ± 0.005	0.007 ± 0.001	0.082 ± 0.005	0.262 ± 0.030	0.512 ± 0.034	0.159	-0.001	-0.92	0.0809	0.0809	1.230	97.959 ± 0.903	0.395 ± 0.026	—	
QML	beta53	0.16	0.164	0.004 ± 0.005	0.025 ± 0.030	0.006 ± 0.000	0.076 ± 0.003	0.006 ± 0.000	0.076 ± 0.004	0.225 ± 0.019	0.474 ± 0.025	0.165	0.005	3.44	0.0877	0.0879	1.149	99.592 ± 0.407	0.342 ± 0.008	—	
QML	beta54	0.16	0.163	0.003 ± 0.006	0.018 ± 0.036	0.008 ± 0.001	0.091 ± 0.004	0.008 ± 0.001	0.090 ± 0.005	0.319 ± 0.030	0.565 ± 0.032	0.160	0.000	0.15	0.0843	0.0843	0.900	94.286 ± 1.483	0.319 ± 0.004	—	
QML	beta55	0.00	-0.036	-0.036 ± 0.009	—	0.020 ± 0.002	0.141 ± 0.008	0.021 ± 0.002	0.145 ± 0.009	—	—	-0.028	-0.028	—	0.1404	0.1433	1.147	97.959 ± 0.903	0.635 ± 0.044	2.041 ± 0.903	
QML	beta56	0.00	0.016	0.016 ± 0.007	—	0.013 ± 0.001	0.114 ± 0.005	0.013 ± 0.001	0.115 ± 0.006	—	—	0.006	0.006	—	0.1106	0.1108	1.012	95.918 ± 1.264	0.453 ± 0.008	4.082 ± 1.264	
QML	beta57	0.00	0.014	0.014 ± 0.006	—	0.008 ± 0.001	0.092 ± 0.006	0.009 ± 0.001	0.093 ± 0.007	—	—	0.010	0.010	—	0.0828	0.0834	1.109	96.327 ± 1.202	0.401 ± 0.032	3.673 ± 1.202	
QML	beta58	0.00	-0.002	-0.002 ± 0.004	—	0.005 ± 0.000	0.068 ± 0.003	0.005 ± 0.000	0.068 ± 0.004	—	—	-0.004	-0.004	—	0.0646	0.0647	1.048	95.918 ± 1.264	0.279 ± 0.007	4.082 ± 1.264	
UPI	beta41	0.20	0.213	0.013 ± 0.007	0.067 ± 0.037	0.014 ± 0.001	0.117 ± 0.006	0.014 ± 0.002	0.117 ± 0.008	0.345 ± 0.038	0.587 ± 0.038	0.203	0.003	1.62	0.1327	0.1327	1.577	97.551 ± 0.987	0.723 ± 0.050	—	
UPI	beta42	0.20	0.202	0.002 ± 0.008	0.009 ± 0.039	0.015 ± 0.002	0.122 ± 0.009	0.015 ± 0.002	0.121 ± 0.009	0.368 ± 0.051	0.607 ± 0.047	0.198	-0.002	-0.95	0.1289	0.1289	1.579	97.551 ± 0.987	0.753 ± 0.053	—	
UPI	beta43	0.20	0.209	0.009 ± 0.007	0.044 ± 0.033	0.010 ± 0.001	0.102 ± 0.005	0.010 ± 0.001	0.102 ± 0.006	0.261 ± 0.027	0.511 ± 0.032	0.205	0.005	2.69	0.1251	0.1252	1.719	99.184 ± 0.575	0.688 ± 0.046	—	
UPI	beta44	0.00	-0.014	-0.014 ± 0.013	—	0.042 ± 0.006	0.204 ± 0.014	0.042 ± 0.006	0.204 ± 0.016	—	—	-0.012	-0.012	—	0.1870	0.1874	2.013	99.592 ± 0.407	1.608 ± 0.118	0.408 ± 0.407	
UPI	beta45	0.00	-0.009	-0.009 ± 0.013	—	0.041 ± 0.006	0.202 ± 0.015	0.041 ± 0.006	0.202 ± 0.016	—	—	-0.008	-0.008	—	0.1766	0.1768	1.744	100.000 ± 0.000	1.382 ± 0.125	0.000 ± 0.000	
UPI	beta46	0.00	0.061	0.061 ± 0.020	—	0.101 ± 0.013	0.318 ± 0.021	0.105 ± 0.014	0.323 ± 0.024	—	—	0.028	0.028	—	0.2881	0.2895	1.994	100.000 ± 0.000	2.487 ± 0.210	0.000 ± 0.000	

Table S265

Simulation 2 - Condition 121: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.4
(continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	0.003	0.003 ± 0.023	—	0.130 ± 0.018	0.360 ± 0.025	0.129 ± 0.018	0.360 ± 0.027	—	—	0.010	0.010	—	0.2807	0.2809	1.837	100.000 ± 0.000	2.596 ± 0.277	0.000 ± 0.000
UPI	beta51	0.16	0.161	0.001 ± 0.008	0.007 ± 0.051	0.016 ± 0.002	0.127 ± 0.008	0.016 ± 0.002	0.127 ± 0.009	0.632 ± 0.080	0.795 ± 0.057	0.166	0.006	3.61	0.1304	0.1305	1.269	96.735 ± 1.135	0.634 ± 0.047	—
UPI	beta52	0.16	0.152	-0.008 ± 0.006	-0.049 ± 0.039	0.010 ± 0.001	0.098 ± 0.007	0.010 ± 0.001	0.098 ± 0.007	0.373 ± 0.050	0.611 ± 0.046	0.149	-0.011	-6.92	0.0963	0.0969	1.647	98.776 ± 0.703	0.630 ± 0.078	—
UPI	beta53	0.16	0.165	0.005 ± 0.006	0.029 ± 0.038	0.009 ± 0.001	0.094 ± 0.006	0.009 ± 0.001	0.094 ± 0.007	0.345 ± 0.043	0.588 ± 0.041	0.166	0.006	3.90	0.1026	0.1028	1.536	100.000 ± 0.000	0.567 ± 0.030	—
UPI	beta54	0.16	0.165	0.005 ± 0.007	0.033 ± 0.042	0.011 ± 0.001	0.106 ± 0.005	0.011 ± 0.001	0.105 ± 0.006	0.434 ± 0.041	0.659 ± 0.038	0.160	0.000	-0.25	0.1054	0.1054	1.511	98.776 ± 0.703	0.625 ± 0.066	—
UPI	beta55	0.00	-0.018	-0.018 ± 0.013	—	0.039 ± 0.005	0.197 ± 0.013	0.039 ± 0.005	0.197 ± 0.014	—	—	-0.012	-0.012	—	0.1805	0.1808	1.609	99.592 ± 0.407	1.240 ± 0.076	0.408 ± 0.407
UPI	beta56	0.00	0.039	0.039 ± 0.011	—	0.032 ± 0.004	0.178 ± 0.011	0.033 ± 0.004	0.182 ± 0.013	—	—	0.011	0.011	—	0.1567	0.1571	1.746	99.184 ± 0.575	1.220 ± 0.098	0.816 ± 0.575
UPI	beta57	0.00	-0.013	-0.013 ± 0.015	—	0.056 ± 0.007	0.237 ± 0.015	0.056 ± 0.007	0.237 ± 0.017	—	—	0.001	0.001	—	0.1894	0.1894	1.801	99.592 ± 0.407	1.671 ± 0.158	0.408 ± 0.407
UPI	beta58	0.00	-0.021	-0.021 ± 0.016	—	0.066 ± 0.008	0.258 ± 0.016	0.066 ± 0.008	0.258 ± 0.019	—	—	0.009	0.009	—	0.1980	0.1983	1.939	100.000 ± 0.000	1.957 ± 0.230	0.000 ± 0.000

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S266

Simulation 2 - Condition 122: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.4

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.204	0.004 ± 0.002	0.018 ± 0.010	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.002	0.062 ± 0.004	0.249 ± 0.009	0.201	0.001	0.34	0.0512	0.0512	1.025	96.353 ± 0.731	0.200 ± 0.001	—
LSAM	beta42	0.20	0.206	0.006 ± 0.002	0.028 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	0.067 ± 0.004	0.259 ± 0.009	0.208	0.008	4.13	0.0530	0.0537	1.024	95.897 ± 0.773	0.207 ± 0.001	—
LSAM	beta43	0.20	0.198	-0.002 ± 0.002	-0.012 ± 0.009	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.002	0.055 ± 0.003	0.234 ± 0.008	0.199	-0.001	-0.51	0.0471	0.0471	1.031	96.201 ± 0.745	0.189 ± 0.001	—
LSAM	beta44	0.00	0.007	0.007 ± 0.003	—	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.003	—	—	0.006	0.006	—	0.0682	0.0685	1.034	96.960 ± 0.669	0.286 ± 0.004	3.040 ± 0.669
LSAM	beta45	0.00	0.001	0.001 ± 0.002	—	0.003 ± 0.000	0.056 ± 0.002	0.003 ± 0.000	0.056 ± 0.002	—	—	0.001	0.001	—	0.0541	0.0541	1.071	98.176 ± 0.522	0.235 ± 0.002	1.824 ± 0.522
LSAM	beta46	0.00	-0.005	-0.005 ± 0.004	—	0.009 ± 0.001	0.096 ± 0.003	0.009 ± 0.001	0.096 ± 0.004	—	—	-0.004	-0.004	—	0.0881	0.0882	1.012	97.416 ± 0.618	0.383 ± 0.007	2.584 ± 0.618
LSAM	beta47	0.00	-0.005	-0.005 ± 0.004	—	0.009 ± 0.001	0.095 ± 0.003	0.009 ± 0.001	0.096 ± 0.004	—	—	-0.002	-0.002	—	0.0873	0.0873	0.998	97.568 ± 0.600	0.374 ± 0.005	2.432 ± 0.600
LSAM	beta51	0.16	0.157	-0.003 ± 0.003	-0.020 ± 0.016	0.004 ± 0.000	0.067 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	0.174 ± 0.011	0.418 ± 0.015	0.156	-0.004	-2.37	0.0678	0.0679	0.951	94.073 ± 0.921	0.249 ± 0.001	—
LSAM	beta52	0.16	0.163	0.003 ± 0.002	0.019 ± 0.013	0.003 ± 0.000	0.055 ± 0.002	0.003 ± 0.000	0.055 ± 0.002	0.120 ± 0.007	0.346 ± 0.012	0.160	0.000	0.02	0.0558	0.0558	0.945	93.921 ± 0.932	0.205 ± 0.001	—
LSAM	beta53	0.16	0.162	0.002 ± 0.002	0.012 ± 0.013	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	0.109 ± 0.006	0.330 ± 0.011	0.161	0.001	0.71	0.0547	0.0547	1.008	94.985 ± 0.851	0.209 ± 0.001	—
LSAM	beta54	0.16	0.161	0.001 ± 0.002	0.009 ± 0.013	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.002	0.103 ± 0.006	0.321 ± 0.011	0.157	-0.003	-1.88	0.0509	0.0510	0.998	94.225 ± 0.909	0.201 ± 0.001	—
LSAM	beta55	0.00	-0.005	-0.005 ± 0.004	—	0.008 ± 0.001	0.091 ± 0.003	0.008 ± 0.001	0.091 ± 0.004	—	—	-0.003	-0.003	—	0.0914	0.0914	0.991	96.960 ± 0.669	0.354 ± 0.005	3.040 ± 0.669
LSAM	beta56	0.00	0.003	0.003 ± 0.003	—	0.006 ± 0.000	0.077 ± 0.003	0.006 ± 0.000	0.077 ± 0.003	—	—	-0.001	-0.001	—	0.0711	0.0711	1.039	96.201 ± 0.745	0.313 ± 0.003	3.799 ± 0.745
LSAM	beta57	0.00	0.003	0.003 ± 0.004	—	0.010 ± 0.001	0.099 ± 0.003	0.010 ± 0.001	0.099 ± 0.004	—	—	0.002	0.002	—	0.0901	0.0902	0.991	97.568 ± 0.600	0.384 ± 0.006	2.432 ± 0.600
LSAM	beta58	0.00	-0.003	-0.003 ± 0.003	—	0.007 ± 0.001	0.086 ± 0.003	0.007 ± 0.001	0.086 ± 0.004	—	—	-0.004	-0.004	—	0.0821	0.0822	1.026	98.480 ± 0.477	0.347 ± 0.006	1.520 ± 0.477
QML	beta41	0.20	0.201	0.001 ± 0.002	0.007 ± 0.009	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.002	0.050 ± 0.003	0.223 ± 0.008	0.202	0.002	1.05	0.0464	0.0464	1.022	95.289 ± 0.826	0.179 ± 0.001	—
QML	beta42	0.20	0.204	0.004 ± 0.002	0.020 ± 0.009	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.002	0.056 ± 0.003	0.236 ± 0.008	0.206	0.006	2.80	0.0480	0.0483	1.002	95.897 ± 0.773	0.185 ± 0.001	—
QML	beta43	0.20	0.196	-0.004 ± 0.002	-0.018 ± 0.008	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.048 ± 0.003	0.218 ± 0.007	0.197	-0.003	-1.66	0.0439	0.0440	1.032	95.441 ± 0.813	0.176 ± 0.001	—
QML	beta44	0.00	-0.005	-0.005 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	—	—	-0.004	-0.004	—	0.0577	0.0578	1.031	96.353 ± 0.731	0.248 ± 0.003	3.647 ± 0.731
QML	beta45	0.00	-0.009	-0.009 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.002	—	—	-0.009	-0.009	—	0.0517	0.0525	1.044	97.416 ± 0.618	0.207 ± 0.002	2.584 ± 0.618
QML	beta46	0.00	0.000	0.000 ± 0.002	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.002	—	—	-0.001	-0.001	—	0.0469	0.0469	1.033	96.049 ± 0.759	0.187 ± 0.002	3.951 ± 0.759
QML	beta47	0.00	-0.003	-0.003 ± 0.002	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.002	—	—	-0.003	-0.003	—	0.0463	0.0464	1.015	94.225 ± 0.909	0.182 ± 0.002	5.775 ± 0.909
QML	beta51	0.16	0.158	-0.002 ± 0.002	-0.015 ± 0.016	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	0.160 ± 0.010	0.400 ± 0.014	0.156	-0.004	-2.28	0.0620	0.0621	1.007	94.681 ± 0.875	0.253 ± 0.002	—
QML	beta52	0.16	0.163	0.003 ± 0.002	0.016 ± 0.013	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	0.109 ± 0.006	0.330 ± 0.011	0.159	-0.001	-0.61	0.0552	0.0553	0.950	93.769 ± 0.942	0.196 ± 0.002	—
QML	beta53	0.16	0.161	0.001 ± 0.002	0.004 ± 0.012	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.002	0.096 ± 0.005	0.309 ± 0.010	0.161	0.001	0.33	0.0522	0.0522	1.004	94.225 ± 0.909	0.195 ± 0.001	—
QML	beta54	0.16	0.160	-0.000 ± 0.002	-0.003 ± 0.012	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.002	0.092 ± 0.005	0.303 ± 0.010	0.157	-0.003	-1.96	0.0491	0.0492	0.975	94.225 ± 0.909	0.185 ± 0.001	—
QML	beta55	0.00	-0.010	-0.010 ± 0.003	—	0.006 ± 0.000	0.080 ± 0.002	0.007 ± 0.000	0.081 ± 0.003	—	—	-0.010	-0.010	—	0.0810	0.0816	1.008	95.897 ± 0.773	0.318 ± 0.005	4.103 ± 0.773
QML	beta56	0.00	-0.005	-0.005 ± 0.002	—	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	—	—	-0.008	-0.008	—	0.0584	0.0590	1.060	96.049 ± 0.759	0.262 ± 0.003	3.951 ± 0.759
QML	beta57	0.00	0.001	0.001 ± 0.002	—	0.003 ± 0.000	0.050 ± 0.002	0.003 ± 0.000	0.050 ± 0.002	—	—	-0.002	-0.002	—	0.0491	0.0491	1.013	95.593 ± 0.800	0.200 ± 0.003	4.407 ± 0.800
QML	beta58	0.00	-0.003	-0.003 ± 0.002	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	-0.006	-0.006	—	0.0376	0.0381	1.057	95.745 ± 0.787	0.162 ± 0.002	4.255 ± 0.787
UPI	beta41	0.20	0.205	0.005 ± 0.002	0.023 ± 0.011	0.003 ± 0.000	0.055 ± 0.002	0.003 ± 0.000	0.055 ± 0.002	0.076 ± 0.006	0.275 ± 0.012	0.202	0.002	1.16	0.0574	0.0574	1.318	98.480 ± 0.477	0.284 ± 0.013	—
UPI	beta42	0.20	0.206	0.006 ± 0.002	0.031 ± 0.011	0.003 ± 0.000	0.057 ± 0.002	0.003 ± 0.000	0.057 ± 0.002	0.081 ± 0.006	0.284 ± 0.011	0.210	0.010	5.02	0.0581	0.0589	1.319	98.024 ± 0.543	0.292 ± 0.012	—
UPI	beta43	0.20	0.199	-0.001 ± 0.002	-0.007 ± 0.010	0.002 ± 0.000	0.050 ± 0.002	0.002 ± 0.000	0.050 ± 0.002	0.062 ± 0.004	0.248 ± 0.009	0.198	-0.002	-1.18	0.0552	0.0552	1.294	98.024 ± 0.543	0.252 ± 0.008	—
UPI	beta44	0.00	0.006	0.006 ± 0.004	—	0.009 ± 0.001	0.092 ± 0.003	0.009 ± 0.001	0.093 ± 0.004	—	—	0.006	0.006	—	0.0883	0.0885	1.604	98.936 ± 0.400	0.581 ± 0.055	1.064 ± 0.400
UPI	beta45	0.00	0.001	0.001 ± 0.003	—	0.004 ± 0.000	0.067 ± 0.002	0.004 ± 0.000	0.066 ± 0.003	—	—	-0.002	-0.002	—	0.0707	0.0707	1.402	99.240 ± 0.339	0.366 ± 0.011	0.760 ± 0.339
UPI	beta46	0.00	-0.008	-0.008 ± 0.006	—	0.021 ± 0.002	0.145 ± 0.006	0.021 ± 0.002	0.146 ± 0.007	—	—	-0.002	-0.002	—	0.1321	0.1321	1.658	100.000 ± 0.000	0.945 ± 0.054	0.000 ± 0.000

Table S266

Simulation 2 - Condition 122: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.4
(continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	0.002	0.002 ± 0.006	—	0.022 ± 0.002	0.149 ± 0.006	0.022 ± 0.002	0.149 ± 0.007	—	—	0.003	0.003	—	0.1308	0.1309	1.783	99.848 ± 0.152	1.045 ± 0.086	0.152 ± 0.152
UPI	beta51	0.16	0.157	-0.003 ± 0.003	-0.020 ± 0.016	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	0.161 ± 0.010	0.401 ± 0.014	0.156	-0.004	-2.63	0.0653	0.0655	1.041	95.137 ± 0.839	0.262 ± 0.002	—
UPI	beta52	0.16	0.163	0.003 ± 0.002	0.019 ± 0.013	0.003 ± 0.000	0.055 ± 0.002	0.003 ± 0.000	0.055 ± 0.002	0.117 ± 0.007	0.342 ± 0.012	0.160	0.000	0.06	0.0563	0.0563	1.052	94.529 ± 0.887	0.226 ± 0.004	—
UPI	beta53	0.16	0.162	0.002 ± 0.002	0.010 ± 0.013	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	0.107 ± 0.006	0.327 ± 0.011	0.161	0.001	0.61	0.0550	0.0550	1.126	96.809 ± 0.685	0.231 ± 0.003	—
UPI	beta54	0.16	0.162	0.002 ± 0.002	0.011 ± 0.013	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.002	0.103 ± 0.006	0.322 ± 0.011	0.158	-0.002	-1.40	0.0502	0.0503	1.102	94.985 ± 0.851	0.222 ± 0.003	—
UPI	beta55	0.00	-0.007	-0.007 ± 0.004	—	0.009 ± 0.001	0.096 ± 0.004	0.009 ± 0.001	0.096 ± 0.004	—	—	-0.007	-0.007	—	0.0896	0.0899	1.190	98.480 ± 0.477	0.448 ± 0.012	1.520 ± 0.477
UPI	beta56	0.00	0.003	0.003 ± 0.003	—	0.007 ± 0.000	0.081 ± 0.003	0.007 ± 0.000	0.081 ± 0.003	—	—	-0.002	-0.002	—	0.0802	0.0802	1.267	99.240 ± 0.339	0.404 ± 0.008	0.760 ± 0.339
UPI	beta57	0.00	0.005	0.005 ± 0.005	—	0.014 ± 0.001	0.118 ± 0.005	0.014 ± 0.001	0.118 ± 0.005	—	—	0.009	0.009	—	0.1004	0.1008	1.370	99.544 ± 0.263	0.632 ± 0.024	0.456 ± 0.263
UPI	beta58	0.00	0.002	0.002 ± 0.004	—	0.010 ± 0.001	0.101 ± 0.004	0.010 ± 0.001	0.101 ± 0.004	—	—	-0.001	-0.001	—	0.0882	0.0882	1.574	100.000 ± 0.000	0.625 ± 0.032	0.000 ± 0.000

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S267

Simulation 2 - Condition 123: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.6

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.202	0.002 ± 0.002	0.009 ± 0.009	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.077 ± 0.003	0.277 ± 0.008	0.202	0.002	1.06	0.0570	0.0570	1.017	95.983 ± 0.647	0.221 ± 0.001	—
LSAM	beta42	0.20	0.202	0.002 ± 0.002	0.012 ± 0.009	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.079 ± 0.004	0.281 ± 0.008	0.202	0.002	0.91	0.0575	0.0575	1.036	96.091 ± 0.639	0.229 ± 0.001	—
LSAM	beta43	0.20	0.204	0.004 ± 0.002	0.020 ± 0.009	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.078 ± 0.004	0.280 ± 0.008	0.204	0.004	1.84	0.0575	0.0576	0.984	95.223 ± 0.703	0.216 ± 0.001	—
LSAM	beta44	0.00	0.000	0.000 ± 0.002	—	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	—	—	-0.001	-0.001	—	0.0682	0.0682	0.936	94.245 ± 0.767	0.267 ± 0.002	5.755 ± 0.767
LSAM	beta45	0.00	0.002	0.002 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	—	—	0.003	0.003	—	0.0571	0.0572	0.978	93.811 ± 0.794	0.235 ± 0.001	6.189 ± 0.794
LSAM	beta46	0.00	0.002	0.002 ± 0.003	—	0.008 ± 0.000	0.089 ± 0.002	0.008 ± 0.000	0.089 ± 0.003	—	—	-0.002	-0.002	—	0.0890	0.0890	0.977	95.223 ± 0.703	0.339 ± 0.003	4.777 ± 0.703
LSAM	beta47	0.00	0.000	0.000 ± 0.003	—	0.008 ± 0.000	0.088 ± 0.002	0.008 ± 0.000	0.088 ± 0.003	—	—	-0.002	-0.002	—	0.0794	0.0794	0.967	95.005 ± 0.718	0.334 ± 0.003	4.995 ± 0.718
LSAM	beta51	0.16	0.159	-0.001 ± 0.003	-0.004 ± 0.016	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	0.241 ± 0.012	0.490 ± 0.015	0.160	0.000	0.29	0.0783	0.0783	0.933	93.811 ± 0.794	0.287 ± 0.002	—
LSAM	beta52	0.16	0.160	-0.000 ± 0.002	-0.003 ± 0.012	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.136 ± 0.007	0.368 ± 0.011	0.160	0.000	0.29	0.0588	0.0588	1.004	94.354 ± 0.761	0.232 ± 0.001	—
LSAM	beta53	0.16	0.158	-0.002 ± 0.002	-0.010 ± 0.013	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.145 ± 0.007	0.381 ± 0.011	0.159	-0.001	-0.65	0.0611	0.0611	0.989	95.114 ± 0.710	0.236 ± 0.001	—
LSAM	beta54	0.16	0.165	0.005 ± 0.002	0.029 ± 0.012	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.128 ± 0.006	0.358 ± 0.010	0.163	0.003	2.08	0.0564	0.0565	1.027	96.200 ± 0.630	0.230 ± 0.001	—
LSAM	beta55	0.00	-0.002	-0.002 ± 0.003	—	0.009 ± 0.000	0.093 ± 0.002	0.009 ± 0.000	0.093 ± 0.003	—	—	-0.001	-0.001	—	0.0857	0.0857	0.939	94.354 ± 0.761	0.341 ± 0.003	5.646 ± 0.761
LSAM	beta56	0.00	0.001	0.001 ± 0.003	—	0.007 ± 0.000	0.085 ± 0.002	0.007 ± 0.000	0.085 ± 0.003	—	—	0.004	0.004	—	0.0830	0.0831	0.947	94.680 ± 0.740	0.316 ± 0.003	5.320 ± 0.740
LSAM	beta57	0.00	0.001	0.001 ± 0.003	—	0.008 ± 0.000	0.092 ± 0.002	0.008 ± 0.000	0.092 ± 0.003	—	—	0.000	0.000	—	0.0827	0.0827	0.976	95.657 ± 0.672	0.351 ± 0.003	4.343 ± 0.672
LSAM	beta58	0.00	-0.004	-0.004 ± 0.003	—	0.007 ± 0.000	0.083 ± 0.002	0.007 ± 0.000	0.084 ± 0.003	—	—	-0.002	-0.002	—	0.0807	0.0807	0.981	95.114 ± 0.710	0.321 ± 0.003	4.886 ± 0.710
QML	beta41	0.20	0.201	0.001 ± 0.002	0.006 ± 0.009	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	0.071 ± 0.003	0.266 ± 0.007	0.202	0.002	1.16	0.0541	0.0542	1.019	95.874 ± 0.655	0.212 ± 0.001	—
QML	beta42	0.20	0.201	0.001 ± 0.002	0.003 ± 0.009	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.072 ± 0.003	0.268 ± 0.008	0.200	0.000	0.19	0.0543	0.0543	1.039	95.765 ± 0.664	0.218 ± 0.001	—
QML	beta43	0.20	0.203	0.003 ± 0.002	0.017 ± 0.009	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.073 ± 0.003	0.270 ± 0.008	0.203	0.003	1.67	0.0555	0.0556	0.992	95.223 ± 0.703	0.210 ± 0.001	—
QML	beta44	0.00	-0.008	-0.008 ± 0.002	—	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	—	—	-0.005	-0.005	—	0.0666	0.0668	0.966	95.005 ± 0.718	0.265 ± 0.003	4.995 ± 0.718
QML	beta45	0.00	-0.003	-0.003 ± 0.002	—	0.004 ± 0.000	0.059 ± 0.002	0.004 ± 0.000	0.059 ± 0.002	—	—	-0.002	-0.002	—	0.0585	0.0585	1.001	94.897 ± 0.725	0.232 ± 0.002	5.103 ± 0.725
QML	beta46	0.00	0.006	0.006 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.002	0.003 ± 0.000	0.058 ± 0.002	—	—	0.001	0.001	—	0.0541	0.0541	1.000	95.223 ± 0.703	0.226 ± 0.002	4.777 ± 0.703
QML	beta47	0.00	0.002	0.002 ± 0.002	—	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	—	—	-0.003	-0.003	—	0.0526	0.0526	1.002	94.354 ± 0.761	0.219 ± 0.002	5.646 ± 0.761
QML	beta51	0.16	0.160	0.000 ± 0.003	0.001 ± 0.016	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	0.233 ± 0.011	0.483 ± 0.014	0.162	0.002	1.41	0.0785	0.0786	0.987	95.331 ± 0.695	0.299 ± 0.002	—
QML	beta52	0.16	0.160	-0.000 ± 0.002	-0.003 ± 0.012	0.004 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	0.138 ± 0.007	0.371 ± 0.011	0.160	0.000	0.02	0.0598	0.0598	1.005	94.897 ± 0.725	0.234 ± 0.002	—
QML	beta53	0.16	0.159	-0.001 ± 0.002	-0.009 ± 0.012	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.135 ± 0.006	0.367 ± 0.011	0.159	-0.001	-0.49	0.0571	0.0571	1.007	95.331 ± 0.695	0.232 ± 0.001	—
QML	beta54	0.16	0.164	0.004 ± 0.002	0.022 ± 0.011	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.120 ± 0.005	0.347 ± 0.010	0.164	0.004	2.65	0.0558	0.0559	1.026	96.851 ± 0.575	0.223 ± 0.001	—
QML	beta55	0.00	-0.008	-0.008 ± 0.003	—	0.007 ± 0.000	0.086 ± 0.002	0.007 ± 0.000	0.087 ± 0.003	—	—	-0.007	-0.007	—	0.0818	0.0821	0.991	94.788 ± 0.732	0.335 ± 0.004	5.212 ± 0.732
QML	beta56	0.00	-0.003	-0.003 ± 0.002	—	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	—	—	-0.002	-0.002	—	0.0737	0.0738	0.984	95.874 ± 0.655	0.292 ± 0.002	4.126 ± 0.655
QML	beta57	0.00	0.003	0.003 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	—	—	-0.001	-0.001	—	0.0574	0.0574	1.012	94.788 ± 0.732	0.237 ± 0.003	5.212 ± 0.732
QML	beta58	0.00	-0.001	-0.001 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	—	—	-0.004	-0.004	—	0.0510	0.0511	1.012	94.897 ± 0.725	0.201 ± 0.002	5.103 ± 0.725
UPI	beta41	0.20	0.203	0.003 ± 0.002	0.013 ± 0.009	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.082 ± 0.004	0.286 ± 0.008	0.205	0.005	2.57	0.0584	0.0586	1.104	97.503 ± 0.514	0.247 ± 0.004	—
UPI	beta42	0.20	0.202	0.002 ± 0.002	0.010 ± 0.009	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.081 ± 0.004	0.285 ± 0.008	0.202	0.002	1.01	0.0603	0.0603	1.151	96.960 ± 0.566	0.257 ± 0.005	—
UPI	beta43	0.20	0.205	0.005 ± 0.002	0.025 ± 0.009	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.082 ± 0.004	0.287 ± 0.008	0.203	0.003	1.73	0.0577	0.0578	1.077	95.548 ± 0.680	0.241 ± 0.006	—
UPI	beta44	0.00	-0.002	-0.002 ± 0.003	—	0.006 ± 0.000	0.080 ± 0.002	0.006 ± 0.000	0.080 ± 0.003	—	—	0.001	0.001	—	0.0790	0.0790	1.098	96.960 ± 0.566	0.343 ± 0.007	3.040 ± 0.566
UPI	beta45	0.00	0.003	0.003 ± 0.002	—	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.067 ± 0.002	—	—	0.002	0.002	—	0.0638	0.0638	1.094	96.417 ± 0.612	0.288 ± 0.008	3.583 ± 0.612
UPI	beta46	0.00	0.007	0.007 ± 0.004	—	0.014 ± 0.001	0.118 ± 0.004	0.014 ± 0.001	0.118 ± 0.004	—	—	0.004	0.004	—	0.0998	0.0999	1.259	99.240 ± 0.286	0.582 ± 0.017	0.760 ± 0.286

Table S267

Simulation 2 - Condition 123: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.6
(continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	0.005	0.005 ± 0.004	—	0.012 ± 0.001	0.110 ± 0.003	0.012 ± 0.001	0.110 ± 0.004	—	—	0.005	0.005	—	0.0987	0.0988	1.400	98.914 ± 0.341	0.604 ± 0.037	1.086 ± 0.341
UPI	beta51	0.16	0.160	-0.000 ± 0.003	-0.003 ± 0.016	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	0.225 ± 0.011	0.475 ± 0.014	0.160	0.000	-0.20	0.0760	0.0760	0.992	94.680 ± 0.740	0.295 ± 0.002	—
UPI	beta52	0.16	0.160	-0.000 ± 0.002	-0.001 ± 0.012	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.136 ± 0.007	0.369 ± 0.011	0.161	0.001	0.78	0.0581	0.0581	1.030	95.005 ± 0.718	0.238 ± 0.001	—
UPI	beta53	0.16	0.158	-0.002 ± 0.002	-0.012 ± 0.013	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.145 ± 0.007	0.381 ± 0.011	0.161	0.001	0.40	0.0600	0.0600	1.034	96.091 ± 0.639	0.247 ± 0.002	—
UPI	beta54	0.16	0.165	0.005 ± 0.002	0.031 ± 0.012	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.128 ± 0.006	0.357 ± 0.010	0.164	0.004	2.55	0.0575	0.0577	1.065	96.417 ± 0.612	0.238 ± 0.002	—
UPI	beta55	0.00	0.000	0.000 ± 0.003	—	0.008 ± 0.000	0.092 ± 0.002	0.008 ± 0.000	0.092 ± 0.003	—	—	0.003	0.003	—	0.0852	0.0853	1.048	95.765 ± 0.664	0.378 ± 0.005	4.235 ± 0.664
UPI	beta56	0.00	0.000	-0.000 ± 0.003	—	0.008 ± 0.000	0.088 ± 0.002	0.008 ± 0.000	0.087 ± 0.003	—	—	0.001	0.001	—	0.0841	0.0841	1.027	96.743 ± 0.585	0.352 ± 0.004	3.257 ± 0.585
UPI	beta57	0.00	0.000	-0.000 ± 0.003	—	0.010 ± 0.001	0.102 ± 0.003	0.010 ± 0.001	0.102 ± 0.003	—	—	-0.001	-0.001	—	0.0921	0.0921	1.182	98.588 ± 0.389	0.471 ± 0.012	1.412 ± 0.389
UPI	beta58	0.00	-0.003	-0.003 ± 0.003	—	0.010 ± 0.001	0.098 ± 0.003	0.010 ± 0.001	0.098 ± 0.003	—	—	-0.003	-0.003	—	0.0873	0.0873	1.171	97.828 ± 0.480	0.450 ± 0.009	2.172 ± 0.480

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S268

Simulation 2 - Condition 124: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.6

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.205	0.005 ± 0.001	0.023 ± 0.006	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.033 ± 0.002	0.181 ± 0.005	0.203	0.003	1.56	0.0345	0.0346	0.966	94.188 ± 0.741	0.136 ± 0.000	—
LSAM	beta42	0.20	0.202	0.002 ± 0.001	0.009 ± 0.005	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.030 ± 0.001	0.173 ± 0.005	0.201	0.001	0.62	0.0346	0.0346	1.039	95.591 ± 0.650	0.141 ± 0.000	—
LSAM	beta43	0.20	0.199	-0.001 ± 0.001	-0.005 ± 0.005	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.028 ± 0.001	0.169 ± 0.005	0.200	0.000	-0.22	0.0329	0.0329	1.009	95.190 ± 0.677	0.133 ± 0.000	—
LSAM	beta44	0.00	0.000	0.000 ± 0.001	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	—	—	0.000	0.000	—	0.0448	0.0448	0.944	94.088 ± 0.747	0.162 ± 0.001	5.912 ± 0.747
LSAM	beta45	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	0.001	0.001	—	0.0373	0.0373	0.970	93.888 ± 0.758	0.143 ± 0.000	6.112 ± 0.758
LSAM	beta46	0.00	-0.001	-0.001 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	—	—	-0.001	-0.001	—	0.0510	0.0510	0.996	94.389 ± 0.728	0.204 ± 0.001	5.611 ± 0.728
LSAM	beta47	0.00	-0.002	-0.002 ± 0.002	—	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	—	—	-0.004	-0.004	—	0.0523	0.0524	0.968	93.788 ± 0.764	0.204 ± 0.001	6.212 ± 0.764
LSAM	beta51	0.16	0.163	0.003 ± 0.001	0.018 ± 0.009	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.086 ± 0.004	0.293 ± 0.008	0.163	0.003	1.56	0.0463	0.0464	0.965	95.391 ± 0.664	0.177 ± 0.001	—
LSAM	beta52	0.16	0.161	0.001 ± 0.001	0.004 ± 0.007	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.053 ± 0.002	0.230 ± 0.006	0.161	0.001	0.48	0.0368	0.0368	0.976	95.591 ± 0.650	0.141 ± 0.000	—
LSAM	beta53	0.16	0.161	0.001 ± 0.001	0.008 ± 0.007	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.054 ± 0.002	0.233 ± 0.006	0.160	0.000	0.20	0.0381	0.0381	0.998	95.792 ± 0.636	0.146 ± 0.000	—
LSAM	beta54	0.16	0.162	0.002 ± 0.001	0.011 ± 0.007	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.048 ± 0.002	0.219 ± 0.006	0.161	0.001	0.33	0.0349	0.0349	1.023	94.990 ± 0.691	0.140 ± 0.000	—
LSAM	beta55	0.00	0.000	0.000 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	—	—	0.000	0.000	—	0.0509	0.0509	0.954	93.587 ± 0.775	0.197 ± 0.001	6.413 ± 0.775
LSAM	beta56	0.00	0.000	0.000 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	—	—	0.000	0.000	—	0.0495	0.0495	0.968	94.489 ± 0.722	0.185 ± 0.001	5.511 ± 0.722
LSAM	beta57	0.00	-0.002	-0.002 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	—	—	-0.004	-0.004	—	0.0523	0.0524	1.003	95.391 ± 0.664	0.208 ± 0.001	4.609 ± 0.664
LSAM	beta58	0.00	0.000	0.000 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	—	—	-0.001	-0.001	—	0.0492	0.0492	0.989	94.990 ± 0.691	0.192 ± 0.001	5.010 ± 0.691
QML	beta41	0.20	0.204	0.004 ± 0.001	0.018 ± 0.006	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.031 ± 0.001	0.175 ± 0.005	0.203	0.003	1.37	0.0340	0.0341	0.971	94.890 ± 0.697	0.133 ± 0.000	—
QML	beta42	0.20	0.201	0.001 ± 0.001	0.007 ± 0.005	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.028 ± 0.001	0.168 ± 0.005	0.201	0.001	0.37	0.0328	0.0328	1.035	95.190 ± 0.677	0.137 ± 0.000	—
QML	beta43	0.20	0.199	-0.001 ± 0.001	-0.007 ± 0.005	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	0.027 ± 0.001	0.166 ± 0.005	0.199	-0.001	-0.65	0.0325	0.0325	1.009	95.090 ± 0.684	0.131 ± 0.000	—
QML	beta44	0.00	-0.006	-0.006 ± 0.001	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	—	—	-0.006	-0.006	—	0.0456	0.0460	0.958	94.990 ± 0.691	0.164 ± 0.001	5.010 ± 0.691
QML	beta45	0.00	-0.004	-0.004 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.004	-0.004	—	0.0358	0.0359	0.981	94.389 ± 0.728	0.143 ± 0.001	5.611 ± 0.728
QML	beta46	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	—	—	-0.001	-0.001	—	0.0314	0.0314	1.027	94.990 ± 0.691	0.137 ± 0.001	5.010 ± 0.691
QML	beta47	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	-0.004	-0.004	—	0.0343	0.0345	0.962	94.389 ± 0.728	0.134 ± 0.001	5.611 ± 0.728
QML	beta51	0.16	0.163	0.003 ± 0.001	0.019 ± 0.009	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.084 ± 0.004	0.290 ± 0.008	0.162	0.002	1.43	0.0469	0.0470	0.990	95.190 ± 0.677	0.180 ± 0.001	—
QML	beta52	0.16	0.160	0.000 ± 0.001	0.003 ± 0.007	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.052 ± 0.002	0.228 ± 0.006	0.161	0.001	0.46	0.0369	0.0369	0.981	95.591 ± 0.650	0.140 ± 0.000	—
QML	beta53	0.16	0.161	0.001 ± 0.001	0.007 ± 0.007	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.052 ± 0.002	0.227 ± 0.006	0.161	0.001	0.36	0.0378	0.0378	1.008	95.591 ± 0.650	0.144 ± 0.000	—
QML	beta54	0.16	0.161	0.001 ± 0.001	0.008 ± 0.007	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.046 ± 0.002	0.216 ± 0.006	0.160	0.000	0.11	0.0346	0.0346	1.026	94.990 ± 0.691	0.139 ± 0.000	—
QML	beta55	0.00	-0.003	-0.003 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	—	—	-0.001	-0.001	—	0.0510	0.0510	0.970	94.389 ± 0.728	0.197 ± 0.001	5.611 ± 0.728
QML	beta56	0.00	-0.003	-0.003 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	-0.003	-0.003	—	0.0479	0.0480	0.986	95.491 ± 0.657	0.177 ± 0.001	4.509 ± 0.657
QML	beta57	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	-0.004	-0.004	—	0.0346	0.0348	1.009	94.489 ± 0.722	0.141 ± 0.001	5.511 ± 0.722
QML	beta58	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	—	—	-0.002	-0.002	—	0.0320	0.0321	1.005	94.188 ± 0.741	0.126 ± 0.001	5.812 ± 0.741
UPI	beta41	0.20	0.205	0.005 ± 0.001	0.025 ± 0.006	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.033 ± 0.002	0.181 ± 0.005	0.204	0.004	2.05	0.0344	0.0346	0.975	94.689 ± 0.710	0.137 ± 0.000	—
UPI	beta42	0.20	0.202	0.002 ± 0.001	0.009 ± 0.005	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.029 ± 0.001	0.171 ± 0.005	0.201	0.001	0.64	0.0337	0.0337	1.056	95.892 ± 0.628	0.141 ± 0.000	—
UPI	beta43	0.20	0.199	-0.001 ± 0.001	-0.003 ± 0.005	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.029 ± 0.001	0.170 ± 0.005	0.200	0.000	-0.18	0.0328	0.0328	1.020	95.491 ± 0.657	0.136 ± 0.000	—
UPI	beta44	0.00	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	-0.001	-0.001	—	0.0470	0.0470	0.968	95.391 ± 0.664	0.174 ± 0.001	4.609 ± 0.664
UPI	beta45	0.00	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	0.002	0.002	—	0.0379	0.0380	0.991	95.090 ± 0.684	0.151 ± 0.001	4.910 ± 0.684
UPI	beta46	0.00	0.001	0.001 ± 0.002	—	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	—	—	-0.001	-0.001	—	0.0542	0.0543	1.089	96.794 ± 0.558	0.241 ± 0.002	3.206 ± 0.558

Table S268

Simulation 2 - Condition 124: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.6
(continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	0.000	-0.000 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	—	—	0.000	0.000	—	0.0564	0.0564	1.041	96.593 ± 0.574	0.245 ± 0.002	3.407 ± 0.574
UPI	beta51	0.16	0.163	0.003 ± 0.001	0.017 ± 0.009	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.083 ± 0.004	0.288 ± 0.008	0.162	0.002	1.05	0.0460	0.0461	0.987	95.190 ± 0.677	0.178 ± 0.001	—
UPI	beta52	0.16	0.161	0.001 ± 0.001	0.005 ± 0.007	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.052 ± 0.002	0.229 ± 0.006	0.161	0.001	0.71	0.0375	0.0375	0.984	95.691 ± 0.643	0.141 ± 0.000	—
UPI	beta53	0.16	0.161	0.001 ± 0.001	0.008 ± 0.007	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.054 ± 0.002	0.233 ± 0.006	0.160	0.000	0.24	0.0376	0.0376	1.008	95.792 ± 0.636	0.147 ± 0.000	—
UPI	beta54	0.16	0.162	0.002 ± 0.001	0.011 ± 0.007	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.048 ± 0.002	0.219 ± 0.006	0.160	0.000	0.16	0.0350	0.0350	1.026	95.691 ± 0.643	0.141 ± 0.000	—
UPI	beta55	0.00	0.000	0.000 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	—	—	0.000	0.000	—	0.0506	0.0506	0.968	94.088 ± 0.747	0.201 ± 0.001	5.912 ± 0.747
UPI	beta56	0.00	0.000	-0.000 ± 0.002	—	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	—	—	-0.002	-0.002	—	0.0497	0.0498	0.977	95.792 ± 0.636	0.191 ± 0.001	4.208 ± 0.636
UPI	beta57	0.00	-0.003	-0.003 ± 0.002	—	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	—	—	-0.003	-0.003	—	0.0539	0.0540	1.042	96.092 ± 0.613	0.223 ± 0.002	3.908 ± 0.613
UPI	beta58	0.00	0.000	0.000 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	—	—	-0.001	-0.001	—	0.0494	0.0495	1.031	96.693 ± 0.566	0.212 ± 0.002	3.307 ± 0.566

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S269

Simulation 2 - Condition 125: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.8

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.199	-0.001 ± 0.001	-0.007 ± 0.007	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.048 ± 0.002	0.220 ± 0.006	0.199	-0.001	-0.57	0.0448	0.0448	1.023	96.389 ± 0.591	0.176 ± 0.001	—
LSAM	beta42	0.20	0.202	0.002 ± 0.001	0.008 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.054 ± 0.003	0.233 ± 0.007	0.202	0.002	0.96	0.0461	0.0462	0.994	94.183 ± 0.741	0.181 ± 0.001	—
LSAM	beta43	0.20	0.199	-0.001 ± 0.001	-0.005 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.054 ± 0.002	0.233 ± 0.006	0.200	0.000	0.01	0.0447	0.0447	0.952	93.079 ± 0.804	0.174 ± 0.001	—
LSAM	beta44	0.00	-0.001	-0.001 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	—	—	0.001	0.001	—	0.0495	0.0495	0.956	93.882 ± 0.759	0.195 ± 0.001	6.118 ± 0.759
LSAM	beta45	0.00	0.000	-0.000 ± 0.001	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	0.001	0.001	—	0.0459	0.0460	0.977	94.985 ± 0.691	0.178 ± 0.001	5.015 ± 0.691
LSAM	beta46	0.00	0.003	0.003 ± 0.002	—	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	—	—	0.004	0.004	—	0.0604	0.0605	0.936	94.383 ± 0.729	0.226 ± 0.001	5.617 ± 0.729
LSAM	beta47	0.00	-0.002	-0.002 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	—	—	-0.005	-0.005	—	0.0583	0.0585	0.977	95.085 ± 0.685	0.223 ± 0.001	4.915 ± 0.685
LSAM	beta51	0.16	0.162	0.002 ± 0.002	0.010 ± 0.012	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	0.151 ± 0.008	0.389 ± 0.012	0.161	0.001	0.50	0.0598	0.0598	0.922	92.879 ± 0.815	0.225 ± 0.001	—
LSAM	beta52	0.16	0.160	-0.000 ± 0.002	-0.003 ± 0.009	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.088 ± 0.004	0.296 ± 0.008	0.158	-0.002	-1.23	0.0474	0.0475	0.987	94.885 ± 0.698	0.183 ± 0.001	—
LSAM	beta53	0.16	0.162	0.002 ± 0.002	0.011 ± 0.010	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.092 ± 0.004	0.303 ± 0.008	0.163	0.003	1.96	0.0468	0.0469	0.991	93.781 ± 0.765	0.188 ± 0.001	—
LSAM	beta54	0.16	0.160	0.000 ± 0.001	0.002 ± 0.009	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.087 ± 0.004	0.294 ± 0.008	0.160	0.000	0.17	0.0462	0.0462	0.988	94.383 ± 0.729	0.183 ± 0.001	—
LSAM	beta55	0.00	0.001	0.001 ± 0.002	—	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	—	—	0.001	0.001	—	0.0624	0.0624	0.954	94.082 ± 0.747	0.236 ± 0.002	5.918 ± 0.747
LSAM	beta56	0.00	0.000	0.000 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	—	—	-0.003	-0.003	—	0.0588	0.0589	0.953	94.684 ± 0.711	0.224 ± 0.001	5.316 ± 0.711
LSAM	beta57	0.00	0.001	0.001 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	—	—	0.000	0.000	—	0.0594	0.0594	0.952	94.283 ± 0.735	0.229 ± 0.001	5.717 ± 0.735
LSAM	beta58	0.00	-0.004	-0.004 ± 0.002	—	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	—	—	-0.003	-0.003	—	0.0536	0.0537	0.972	93.380 ± 0.787	0.214 ± 0.001	6.620 ± 0.787
QML	beta41	0.20	0.198	-0.002 ± 0.001	-0.008 ± 0.007	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	0.047 ± 0.002	0.217 ± 0.006	0.199	-0.001	-0.60	0.0445	0.0445	1.028	95.888 ± 0.629	0.175 ± 0.001	—
QML	beta42	0.20	0.201	0.001 ± 0.001	0.007 ± 0.007	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.052 ± 0.002	0.228 ± 0.006	0.202	0.002	0.99	0.0456	0.0456	1.000	94.483 ± 0.723	0.179 ± 0.001	—
QML	beta43	0.20	0.199	-0.001 ± 0.001	-0.006 ± 0.007	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.053 ± 0.002	0.230 ± 0.006	0.200	0.000	-0.10	0.0451	0.0451	0.958	93.280 ± 0.793	0.173 ± 0.001	—
QML	beta44	0.00	-0.005	-0.005 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	—	—	-0.001	-0.001	—	0.0500	0.0500	0.960	94.383 ± 0.729	0.199 ± 0.001	5.617 ± 0.729
QML	beta45	0.00	-0.002	-0.002 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	-0.001	-0.001	—	0.0470	0.0470	0.995	95.386 ± 0.664	0.180 ± 0.001	4.614 ± 0.664
QML	beta46	0.00	0.005	0.005 ± 0.002	—	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	—	—	0.003	0.003	—	0.0504	0.0505	0.951	94.283 ± 0.735	0.187 ± 0.001	5.717 ± 0.735
QML	beta47	0.00	-0.001	-0.001 ± 0.002	—	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	—	—	-0.006	-0.006	—	0.0471	0.0475	0.979	95.085 ± 0.685	0.183 ± 0.001	4.915 ± 0.685
QML	beta51	0.16	0.162	0.002 ± 0.002	0.012 ± 0.012	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	0.150 ± 0.008	0.387 ± 0.012	0.161	0.001	0.65	0.0597	0.0597	0.951	93.781 ± 0.765	0.231 ± 0.001	—
QML	beta52	0.16	0.159	-0.001 ± 0.002	-0.003 ± 0.009	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.088 ± 0.004	0.297 ± 0.008	0.158	-0.002	-1.06	0.0474	0.0474	0.993	94.784 ± 0.704	0.185 ± 0.001	—
QML	beta53	0.16	0.162	0.002 ± 0.002	0.011 ± 0.010	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.091 ± 0.004	0.301 ± 0.008	0.163	0.003	1.64	0.0463	0.0464	0.996	94.082 ± 0.747	0.188 ± 0.001	—
QML	beta54	0.16	0.160	-0.000 ± 0.001	-0.000 ± 0.009	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.085 ± 0.004	0.291 ± 0.008	0.160	0.000	-0.13	0.0464	0.0464	0.990	94.784 ± 0.704	0.181 ± 0.001	—
QML	beta55	0.00	-0.001	-0.001 ± 0.002	—	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	—	—	0.000	0.000	—	0.0607	0.0607	0.968	94.383 ± 0.729	0.238 ± 0.002	5.617 ± 0.729
QML	beta56	0.00	-0.001	-0.001 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	—	—	-0.002	-0.002	—	0.0581	0.0581	0.968	95.286 ± 0.671	0.222 ± 0.001	4.714 ± 0.671
QML	beta57	0.00	0.002	0.002 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	—	—	-0.001	-0.001	—	0.0481	0.0481	0.959	95.085 ± 0.685	0.190 ± 0.001	4.915 ± 0.685
QML	beta58	0.00	-0.003	-0.003 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	-0.003	-0.003	—	0.0435	0.0436	0.987	93.681 ± 0.771	0.174 ± 0.001	6.319 ± 0.771
UPI	beta41	0.20	0.199	-0.001 ± 0.001	-0.007 ± 0.007	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.048 ± 0.002	0.219 ± 0.006	0.199	-0.001	-0.34	0.0444	0.0445	1.021	95.888 ± 0.629	0.176 ± 0.001	—
UPI	beta42	0.20	0.202	0.002 ± 0.001	0.009 ± 0.007	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.054 ± 0.003	0.232 ± 0.007	0.201	0.001	0.60	0.0461	0.0461	0.996	94.283 ± 0.735	0.181 ± 0.001	—
UPI	beta43	0.20	0.199	-0.001 ± 0.001	-0.004 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.055 ± 0.002	0.234 ± 0.006	0.201	0.001	0.58	0.0452	0.0452	0.959	93.180 ± 0.798	0.176 ± 0.001	—
UPI	beta44	0.00	-0.001	-0.001 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.053 ± 0.001	—	—	0.001	0.001	—	0.0505	0.0505	0.977	94.183 ± 0.741	0.203 ± 0.001	5.817 ± 0.741
UPI	beta45	0.00	0.000	-0.000 ± 0.002	—	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	—	—	0.000	0.000	—	0.0490	0.0490	0.983	94.584 ± 0.717	0.184 ± 0.001	5.416 ± 0.717
UPI	beta46	0.00	0.004	0.004 ± 0.002	—	0.004 ± 0.000	0.065 ± 0.001	0.004 ± 0.000	0.065 ± 0.002	—	—	0.005	0.005	—	0.0642	0.0644	0.962	95.085 ± 0.685	0.246 ± 0.002	4.915 ± 0.685

Table S269

Simulation 2 - Condition 125: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 400, Reliability = 0.8
(continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.002	-0.002 ± 0.002	—	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	—	—	-0.007	-0.007	—	0.0613	0.0618	1.021	96.289 ± 0.599	0.246 ± 0.002	3.711 ± 0.599
UPI	beta51	0.16	0.162	0.002 ± 0.002	0.009 ± 0.012	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	0.148 ± 0.007	0.385 ± 0.012	0.160	0.000	0.26	0.0595	0.0595	0.947	93.681 ± 0.771	0.229 ± 0.001	—
UPI	beta52	0.16	0.160	-0.000 ± 0.002	-0.003 ± 0.009	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.088 ± 0.004	0.296 ± 0.008	0.158	-0.002	-1.12	0.0470	0.0471	0.984	94.784 ± 0.704	0.183 ± 0.001	—
UPI	beta53	0.16	0.162	0.002 ± 0.002	0.012 ± 0.010	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.092 ± 0.004	0.303 ± 0.008	0.163	0.003	2.05	0.0460	0.0461	0.995	93.581 ± 0.776	0.189 ± 0.001	—
UPI	beta54	0.16	0.160	0.000 ± 0.001	0.003 ± 0.009	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.086 ± 0.004	0.294 ± 0.008	0.160	0.000	0.24	0.0460	0.0460	0.987	94.684 ± 0.711	0.182 ± 0.001	—
UPI	beta55	0.00	0.001	0.001 ± 0.002	—	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	—	—	0.002	0.002	—	0.0588	0.0588	0.969	94.885 ± 0.698	0.240 ± 0.002	5.115 ± 0.698
UPI	beta56	0.00	0.000	0.000 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	—	—	-0.002	-0.002	—	0.0599	0.0600	0.979	94.584 ± 0.717	0.229 ± 0.002	5.416 ± 0.717
UPI	beta57	0.00	0.002	0.002 ± 0.002	—	0.004 ± 0.000	0.063 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	—	—	-0.001	-0.001	—	0.0607	0.0607	0.958	94.784 ± 0.704	0.238 ± 0.001	5.216 ± 0.704
UPI	beta58	0.00	-0.004	-0.004 ± 0.002	—	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	—	—	-0.002	-0.002	—	0.0561	0.0561	0.998	94.684 ± 0.711	0.224 ± 0.001	5.316 ± 0.711

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S270

Simulation 2 - Condition 126: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, $N = 1000$, Reliability = 0.8

Method	Param.	True	Mean-based										Median-based					Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE		
LSAM	beta41	0.20	0.200	0.000 ± 0.001	0.002 ± 0.004	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.019 ± 0.001	0.138 ± 0.004	0.200	0.000	-0.11	0.0282	0.0282	1.017	95.495 ± 0.656	0.110 ± 0.000	—		
LSAM	beta42	0.20	0.200	-0.000 ± 0.001	-0.002 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.021 ± 0.001	0.146 ± 0.004	0.199	-0.001	-0.28	0.0289	0.0289	0.993	95.696 ± 0.642	0.114 ± 0.000	—		
LSAM	beta43	0.20	0.200	-0.000 ± 0.001	-0.001 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.021 ± 0.001	0.144 ± 0.004	0.200	0.000	-0.15	0.0299	0.0299	0.969	93.694 ± 0.769	0.110 ± 0.000	—		
LSAM	beta44	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	-0.001	-0.001	—	0.0310	0.0311	1.021	95.495 ± 0.656	0.123 ± 0.000	4.505 ± 0.656		
LSAM	beta45	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	-0.001	-0.001	—	0.0311	0.0311	0.968	94.194 ± 0.740	0.112 ± 0.000	5.806 ± 0.740		
LSAM	beta46	0.00	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.001	0.001	—	0.0382	0.0382	0.976	94.494 ± 0.722	0.142 ± 0.000	5.506 ± 0.722		
LSAM	beta47	0.00	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	0.000	0.000	—	0.0368	0.0368	0.991	94.695 ± 0.709	0.141 ± 0.000	5.305 ± 0.709		
LSAM	beta51	0.16	0.159	-0.001 ± 0.001	-0.007 ± 0.007	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.052 ± 0.002	0.228 ± 0.006	0.159	-0.001	-0.47	0.0369	0.0369	1.007	94.995 ± 0.690	0.144 ± 0.001	—		
LSAM	beta52	0.16	0.160	-0.000 ± 0.001	-0.000 ± 0.006	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.038 ± 0.002	0.194 ± 0.005	0.159	-0.001	-0.36	0.0309	0.0309	0.950	93.594 ± 0.775	0.115 ± 0.000	—		
LSAM	beta53	0.16	0.161	0.001 ± 0.001	0.008 ± 0.006	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.038 ± 0.002	0.196 ± 0.005	0.161	0.001	0.66	0.0321	0.0321	0.975	94.995 ± 0.690	0.120 ± 0.000	—		
LSAM	beta54	0.16	0.161	0.001 ± 0.001	0.007 ± 0.006	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.033 ± 0.002	0.182 ± 0.005	0.161	0.001	0.44	0.0285	0.0285	1.016	94.995 ± 0.690	0.116 ± 0.000	—		
LSAM	beta55	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	0.001	0.001	—	0.0359	0.0359	1.002	94.795 ± 0.703	0.148 ± 0.001	5.205 ± 0.703		
LSAM	beta56	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.000	0.000	—	0.0346	0.0346	0.982	93.794 ± 0.763	0.141 ± 0.001	6.206 ± 0.763		
LSAM	beta57	0.00	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	0.000	0.000	—	0.0360	0.0360	0.977	94.595 ± 0.715	0.145 ± 0.000	5.405 ± 0.715		
LSAM	beta58	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	-0.001	-0.001	—	0.0364	0.0364	0.955	93.493 ± 0.780	0.136 ± 0.000	6.507 ± 0.780		
QML	beta41	0.20	0.200	0.000 ± 0.001	0.001 ± 0.004	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	0.019 ± 0.001	0.137 ± 0.004	0.200	0.000	-0.07	0.0279	0.0279	1.020	95.495 ± 0.656	0.110 ± 0.000	—		
QML	beta42	0.20	0.199	-0.001 ± 0.001	-0.003 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.021 ± 0.001	0.144 ± 0.004	0.199	-0.001	-0.54	0.0288	0.0288	0.994	95.395 ± 0.663	0.113 ± 0.000	—		
QML	beta43	0.20	0.200	-0.000 ± 0.001	-0.002 ± 0.005	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.020 ± 0.001	0.142 ± 0.004	0.199	-0.001	-0.42	0.0292	0.0292	0.976	93.994 ± 0.752	0.109 ± 0.000	—		
QML	beta44	0.00	-0.003	-0.003 ± 0.001	—	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	-0.003	-0.003	—	0.0318	0.0319	1.039	95.896 ± 0.628	0.126 ± 0.001	4.104 ± 0.628		
QML	beta45	0.00	-0.003	-0.003 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	-0.003	-0.003	—	0.0306	0.0308	0.987	94.394 ± 0.728	0.114 ± 0.001	5.606 ± 0.728		
QML	beta46	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	0.001	0.001	—	0.0314	0.0314	1.023	94.795 ± 0.703	0.122 ± 0.005	5.205 ± 0.703		
QML	beta47	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	-0.001	-0.001	—	0.0305	0.0305	1.011	95.395 ± 0.663	0.116 ± 0.001	4.605 ± 0.663		
QML	beta51	0.16	0.159	-0.001 ± 0.001	-0.006 ± 0.007	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.051 ± 0.002	0.226 ± 0.006	0.160	0.000	-0.21	0.0367	0.0367	1.023	95.495 ± 0.656	0.145 ± 0.001	—		
QML	beta52	0.16	0.160	-0.000 ± 0.001	-0.001 ± 0.006	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.037 ± 0.002	0.193 ± 0.005	0.160	0.000	-0.27	0.0299	0.0299	0.955	93.994 ± 0.752	0.115 ± 0.000	—		
QML	beta53	0.16	0.161	0.001 ± 0.001	0.007 ± 0.006	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.038 ± 0.002	0.194 ± 0.005	0.161	0.001	0.65	0.0316	0.0316	0.979	94.695 ± 0.709	0.119 ± 0.000	—		
QML	beta54	0.16	0.161	0.001 ± 0.001	0.006 ± 0.006	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.033 ± 0.002	0.181 ± 0.005	0.161	0.001	0.32	0.0282	0.0282	1.017	94.895 ± 0.696	0.115 ± 0.000	—		
QML	beta55	0.00	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	0.001	0.001	—	0.0367	0.0367	1.019	95.395 ± 0.663	0.150 ± 0.002	4.605 ± 0.663		
QML	beta56	0.00	-0.003	-0.003 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	-0.002	-0.002	—	0.0335	0.0336	0.995	93.994 ± 0.752	0.139 ± 0.001	6.006 ± 0.752		
QML	beta57	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	-0.001	-0.001	—	0.0304	0.0304	0.986	94.895 ± 0.696	0.119 ± 0.001	5.105 ± 0.696		
QML	beta58	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	-0.002	-0.002	—	0.0303	0.0303	0.965	92.893 ± 0.813	0.111 ± 0.000	7.107 ± 0.813		
UPI	beta41	0.20	0.200	0.000 ± 0.001	0.002 ± 0.004	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.019 ± 0.001	0.138 ± 0.004	0.200	0.000	0.01	0.0281	0.0281	1.016	95.495 ± 0.656	0.110 ± 0.000	—		
UPI	beta42	0.20	0.200	-0.000 ± 0.001	-0.001 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.021 ± 0.001	0.145 ± 0.004	0.200	0.000	-0.21	0.0287	0.0287	0.995	95.896 ± 0.628	0.113 ± 0.000	—		
UPI	beta43	0.20	0.200	-0.000 ± 0.001	-0.000 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.021 ± 0.001	0.144 ± 0.004	0.200	0.000	-0.25	0.0297	0.0297	0.972	93.794 ± 0.763	0.110 ± 0.000	—		
UPI	beta44	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	0.000	0.000	—	0.0322	0.0322	1.025	95.495 ± 0.656	0.125 ± 0.000	4.505 ± 0.656		
UPI	beta45	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	-0.002	-0.002	—	0.0309	0.0310	0.969	94.494 ± 0.722	0.114 ± 0.000	5.506 ± 0.722		
UPI	beta46	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	0.001	0.001	—	0.0388	0.0388	0.977	94.995 ± 0.690	0.147 ± 0.001	5.005 ± 0.690		

Table S270

Simulation 2 - Condition 126: Linear Model, Distr = Uniform, Epsilon = Normal, Zeta = Right-skewed, N = 1000, Reliability = 0.8
(continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.001	0.001	—	0.0386	0.0386	1.008	95.696 ± 0.642	0.147 ± 0.001	4.304 ± 0.642
UPI	beta51	0.16	0.159	-0.001 ± 0.001	-0.007 ± 0.007	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.051 ± 0.002	0.226 ± 0.006	0.159	-0.001	-0.44	0.0367	0.0367	1.020	95.395 ± 0.663	0.145 ± 0.001	—
UPI	beta52	0.16	0.160	-0.000 ± 0.001	-0.001 ± 0.006	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.037 ± 0.002	0.193 ± 0.005	0.159	-0.001	-0.36	0.0310	0.0310	0.951	93.894 ± 0.758	0.115 ± 0.000	—
UPI	beta53	0.16	0.161	0.001 ± 0.001	0.008 ± 0.006	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.038 ± 0.002	0.196 ± 0.005	0.161	0.001	0.61	0.0318	0.0318	0.976	94.695 ± 0.709	0.120 ± 0.000	—
UPI	beta54	0.16	0.161	0.001 ± 0.001	0.007 ± 0.006	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.033 ± 0.002	0.182 ± 0.005	0.161	0.001	0.48	0.0282	0.0282	1.016	95.095 ± 0.683	0.116 ± 0.000	—
UPI	beta55	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	0.001	0.001	—	0.0361	0.0361	1.009	95.395 ± 0.663	0.149 ± 0.001	4.605 ± 0.663
UPI	beta56	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.001	0.001	—	0.0353	0.0353	0.984	94.094 ± 0.746	0.142 ± 0.001	5.906 ± 0.746
UPI	beta57	0.00	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	0.001	0.001	—	0.0388	0.0388	0.980	95.095 ± 0.683	0.146 ± 0.001	4.905 ± 0.683
UPI	beta58	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.000	0.000	—	0.0369	0.0369	0.967	94.194 ± 0.740	0.139 ± 0.000	5.806 ± 0.740

Note. MCSE = Monte Carlo Standard Error shown as \u000b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S271

Simulation 2 - Condition 127: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, $N = 400$, Reliability = 0.4

Method	Param.	True	Mean-based										Median-based					Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE		
LSAM	beta41	0.20	0.213	0.013 ± 0.009	0.065 ± 0.045	0.009 ± 0.001	0.094 ± 0.007	0.009 ± 0.001	0.094 ± 0.008	0.222 ± 0.033	0.471 ± 0.042	0.209	0.009	4.41	0.0881	0.0886	1.137	95.327 ± 2.040	0.418 ± 0.013	—		
LSAM	beta42	0.20	0.227	0.027 ± 0.008	0.133 ± 0.042	0.008 ± 0.001	0.087 ± 0.007	0.008 ± 0.001	0.091 ± 0.007	0.207 ± 0.027	0.454 ± 0.037	0.216	0.016	8.13	0.0893	0.0908	1.332	100.000 ± 0.000	0.456 ± 0.020	—		
LSAM	beta43	0.20	0.191	-0.009 ± 0.007	-0.043 ± 0.037	0.006 ± 0.001	0.076 ± 0.005	0.006 ± 0.001	0.076 ± 0.006	0.145 ± 0.020	0.380 ± 0.032	0.186	-0.014	-6.95	0.0698	0.0711	1.101	98.131 ± 1.309	0.328 ± 0.008	—		
LSAM	beta44	0.00	0.042	0.042 ± 0.013	—	0.018 ± 0.003	0.134 ± 0.011	0.019 ± 0.003	0.139 ± 0.013	—	—	0.038	0.038	—	0.1193	0.1252	1.041	91.589 ± 2.683	0.545 ± 0.039	8.411 ± 2.683		
LSAM	beta45	0.00	0.026	0.026 ± 0.010	—	0.011 ± 0.002	0.105 ± 0.009	0.012 ± 0.002	0.107 ± 0.011	—	—	0.026	0.026	—	0.0943	0.0978	0.960	95.327 ± 2.040	0.394 ± 0.014	4.673 ± 2.040		
LSAM	beta46	0.00	-0.034	-0.034 ± 0.008	—	0.007 ± 0.001	0.081 ± 0.006	0.008 ± 0.001	0.087 ± 0.007	—	—	-0.049	-0.049	—	0.0795	0.0935	1.113	97.196 ± 1.596	0.353 ± 0.020	2.804 ± 1.596		
LSAM	beta47	0.00	-0.051	-0.051 ± 0.008	—	0.007 ± 0.001	0.081 ± 0.007	0.009 ± 0.001	0.096 ± 0.009	—	—	-0.037	-0.037	—	0.0734	0.0821	1.077	91.589 ± 2.683	0.344 ± 0.023	8.411 ± 2.683		
LSAM	beta51	0.16	0.142	-0.018 ± 0.010	-0.114 ± 0.060	0.010 ± 0.001	0.099 ± 0.007	0.010 ± 0.001	0.100 ± 0.009	0.390 ± 0.054	0.625 ± 0.053	0.135	-0.025	-15.80	0.1036	0.1066	1.044	95.327 ± 2.040	0.404 ± 0.008	—		
LSAM	beta52	0.16	0.183	0.023 ± 0.010	0.144 ± 0.063	0.011 ± 0.001	0.104 ± 0.007	0.011 ± 0.001	0.106 ± 0.009	0.443 ± 0.058	0.665 ± 0.054	0.180	0.020	12.49	0.0932	0.0953	0.956	95.327 ± 2.040	0.392 ± 0.010	—		
LSAM	beta53	0.16	0.145	-0.015 ± 0.009	-0.091 ± 0.055	0.008 ± 0.001	0.090 ± 0.008	0.008 ± 0.001	0.091 ± 0.009	0.325 ± 0.052	0.570 ± 0.054	0.136	-0.024	-14.98	0.0898	0.0929	0.965	92.523 ± 2.543	0.342 ± 0.006	—		
LSAM	beta54	0.16	0.186	0.026 ± 0.009	0.162 ± 0.058	0.009 ± 0.002	0.096 ± 0.009	0.010 ± 0.002	0.099 ± 0.010	0.386 ± 0.068	0.621 ± 0.063	0.193	0.033	20.68	0.0964	0.1019	1.001	96.262 ± 1.834	0.378 ± 0.010	—		
LSAM	beta55	0.00	0.040	0.040 ± 0.013	—	0.019 ± 0.004	0.137 ± 0.014	0.020 ± 0.004	0.142 ± 0.017	—	—	0.046	0.046	—	0.1339	0.1415	1.055	99.065 ± 0.930	0.567 ± 0.029	0.935 ± 0.930		
LSAM	beta56	0.00	-0.005	-0.005 ± 0.011	—	0.013 ± 0.002	0.115 ± 0.010	0.013 ± 0.002	0.114 ± 0.011	—	—	-0.003	-0.003	—	0.1166	0.1166	1.080	96.262 ± 1.834	0.486 ± 0.023	3.738 ± 1.834		
LSAM	beta57	0.00	-0.033	-0.033 ± 0.009	—	0.008 ± 0.001	0.089 ± 0.008	0.009 ± 0.002	0.095 ± 0.010	—	—	-0.031	-0.031	—	0.0808	0.0864	0.895	92.523 ± 2.543	0.313 ± 0.014	7.477 ± 2.543		
LSAM	beta58	0.00	-0.032	-0.032 ± 0.006	—	0.004 ± 0.000	0.061 ± 0.004	0.005 ± 0.001	0.068 ± 0.006	—	—	-0.031	-0.031	—	0.0599	0.0675	1.081	91.589 ± 2.683	0.257 ± 0.010	8.411 ± 2.683		
QML	beta41	0.20	0.197	-0.003 ± 0.007	-0.017 ± 0.033	0.005 ± 0.001	0.069 ± 0.005	0.005 ± 0.001	0.069 ± 0.006	0.119 ± 0.017	0.345 ± 0.030	0.200	0.000	0.15	0.0650	0.0650	1.089	95.327 ± 2.040	0.295 ± 0.006	—		
QML	beta42	0.20	0.207	0.007 ± 0.006	0.036 ± 0.029	0.004 ± 0.000	0.060 ± 0.004	0.004 ± 0.000	0.060 ± 0.005	0.090 ± 0.012	0.299 ± 0.025	0.202	0.002	1.18	0.0581	0.0581	1.273	99.065 ± 0.930	0.298 ± 0.005	—		
QML	beta43	0.20	0.199	-0.001 ± 0.007	-0.006 ± 0.035	0.005 ± 0.001	0.073 ± 0.005	0.005 ± 0.001	0.073 ± 0.006	0.133 ± 0.017	0.364 ± 0.030	0.190	-0.010	-5.06	0.0660	0.0668	0.993	98.131 ± 1.309	0.285 ± 0.005	—		
QML	beta44	0.00	0.001	0.001 ± 0.011	—	0.013 ± 0.002	0.112 ± 0.007	0.013 ± 0.002	0.112 ± 0.009	—	—	0.004	0.004	—	0.1206	0.1206	0.948	90.654 ± 2.814	0.418 ± 0.016	9.346 ± 2.814		
QML	beta45	0.00	-0.001	-0.001 ± 0.009	—	0.008 ± 0.001	0.090 ± 0.008	0.008 ± 0.001	0.090 ± 0.009	—	—	-0.004	-0.004	—	0.0837	0.0838	0.960	95.327 ± 2.040	0.339 ± 0.012	4.673 ± 2.040		
QML	beta46	0.00	-0.015	-0.015 ± 0.007	—	0.005 ± 0.001	0.074 ± 0.007	0.006 ± 0.001	0.075 ± 0.008	—	—	-0.029	-0.029	—	0.0587	0.0656	0.988	95.327 ± 2.040	0.286 ± 0.016	4.673 ± 2.040		
QML	beta47	0.00	-0.031	-0.031 ± 0.006	—	0.004 ± 0.001	0.067 ± 0.005	0.005 ± 0.001	0.073 ± 0.006	—	—	-0.033	-0.033	—	0.0625	0.0707	0.948	84.112 ± 3.534	0.247 ± 0.010	15.888 ± 3.534		
QML	beta51	0.16	0.162	0.002 ± 0.009	0.010 ± 0.059	0.010 ± 0.001	0.098 ± 0.007	0.009 ± 0.001	0.097 ± 0.009	0.369 ± 0.055	0.607 ± 0.055	0.149	-0.011	-6.86	0.0995	0.1001	1.122	98.131 ± 1.309	0.429 ± 0.012	—		
QML	beta52	0.16	0.168	0.008 ± 0.008	0.049 ± 0.048	0.006 ± 0.001	0.079 ± 0.006	0.006 ± 0.001	0.079 ± 0.007	0.244 ± 0.034	0.494 ± 0.043	0.165	0.005	3.04	0.0740	0.0741	1.078	100.000 ± 0.000	0.334 ± 0.014	—		
QML	beta53	0.16	0.151	-0.009 ± 0.008	-0.054 ± 0.051	0.007 ± 0.001	0.085 ± 0.006	0.007 ± 0.001	0.085 ± 0.007	0.280 ± 0.039	0.529 ± 0.046	0.147	-0.013	-8.40	0.0805	0.0816	0.982	95.327 ± 2.040	0.326 ± 0.006	—		
QML	beta54	0.16	0.167	0.007 ± 0.007	0.043 ± 0.044	0.005 ± 0.001	0.072 ± 0.005	0.005 ± 0.001	0.072 ± 0.006	0.204 ± 0.026	0.451 ± 0.037	0.167	0.007	4.49	0.0717	0.0721	1.056	98.131 ± 1.309	0.299 ± 0.005	—		
QML	beta55	0.00	0.028	0.028 ± 0.012	—	0.015 ± 0.003	0.123 ± 0.012	0.016 ± 0.003	0.126 ± 0.014	—	—	0.030	0.030	—	0.1193	0.1229	1.049	99.065 ± 0.930	0.506 ± 0.020	0.935 ± 0.930		
QML	beta56	0.00	-0.022	-0.022 ± 0.009	—	0.009 ± 0.001	0.095 ± 0.008	0.010 ± 0.001	0.098 ± 0.009	—	—	-0.022	-0.022	—	0.0948	0.0973	1.055	94.393 ± 2.224	0.394 ± 0.012	5.607 ± 2.224		
QML	beta57	0.00	-0.025	-0.025 ± 0.007	—	0.005 ± 0.001	0.069 ± 0.006	0.005 ± 0.001	0.073 ± 0.007	—	—	-0.027	-0.027	—	0.0679	0.0730	1.035	88.785 ± 3.051	0.280 ± 0.015	11.215 ± 3.051		
QML	beta58	0.00	-0.022	-0.022 ± 0.004	—	0.002 ± 0.000	0.046 ± 0.003	0.003 ± 0.000	0.050 ± 0.004	—	—	-0.024	-0.024	—	0.0525	0.0578	1.178	90.654 ± 2.814	0.210 ± 0.008	9.346 ± 2.814		
UPI	beta41	0.20	0.216	0.016 ± 0.022	0.081 ± 0.112	0.054 ± 0.012	0.232 ± 0.028	0.053 ± 0.012	0.231 ± 0.030	1.334 ± 0.309	1.155 ± 0.151	0.231	0.031	15.48	0.2201	0.2222	1.619	95.327 ± 2.040	1.469 ± 0.239	—		
UPI	beta42	0.20	0.317	0.117 ± 0.026	0.586 ± 0.129	0.072 ± 0.016	0.268 ± 0.032	0.085 ± 0.019	0.291 ± 0.036	2.117 ± 0.481	1.455 ± 0.180	0.278	0.078	38.89	0.2510	0.2628	1.500	96.262 ± 1.834	1.573 ± 0.219	—		
UPI	beta43	0.20	0.143	-0.057 ± 0.013	-0.286 ± 0.064	0.018 ± 0.003	0.133 ± 0.012	0.021 ± 0.003	0.144 ± 0.014	0.517 ± 0.086	0.719 ± 0.068	0.136	-0.064	-32.12	0.1526	0.1655	1.387	85.047 ± 3.448	0.721 ± 0.081	—		
UPI	beta44	0.00	0.082	0.082 ± 0.021	—	0.046 ± 0.007	0.215 ± 0.017	0.052 ± 0.009	0.229 ± 0.023	—	—	0.056	0.056	—	0.2079	0.2153	1.861	100.000 ± 0.000	1.568 ± 0.243	0.000 ± 0.000		
UPI	beta45	0.00	0.022	0.022 ± 0.016	—	0.029 ± 0.006	0.170 ± 0.019	0.029 ± 0.006	0.170 ± 0.020	—	—	0.000	0.000	—	0.1703	0.1703	1.495	100.000 ± 0.000	0.995 ± 0.091	0.000 ± 0.000		
UPI	beta46	0.00	-0.065	-0.065 ± 0.024	—	0.061 ± 0.014	0.246 ± 0.029	0.064 ± 0.014	0.253 ± 0.031	—	—	-0.065	-0.065	—	0.2398	0.2484	1.569	100.000 ± 0.000	1.513 ± 0.234	0.000 ± 0.000		

Table S271

Simulation 2 - Condition 127: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.4
(continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.130	-0.130 ± 0.028	—	0.083 ± 0.020	0.288 ± 0.037	0.099 ± 0.025	0.315 ± 0.044	—	—	-0.119	-0.119	—	0.2722	0.2972	1.481	97.196 ± 1.596	1.674 ± 0.348	2.804 ± 1.596
UPI	beta51	0.16	0.184	0.024 ± 0.010	0.149 ± 0.065	0.012 ± 0.002	0.108 ± 0.008	0.012 ± 0.002	0.110 ± 0.010	0.471 ± 0.068	0.686 ± 0.060	0.183	0.023	14.20	0.1087	0.1111	1.358	95.327 ± 2.040	0.573 ± 0.046	—
UPI	beta52	0.16	0.184	0.024 ± 0.014	0.148 ± 0.087	0.021 ± 0.005	0.144 ± 0.019	0.021 ± 0.005	0.145 ± 0.021	0.824 ± 0.210	0.908 ± 0.129	0.185	0.025	15.48	0.1398	0.1419	1.785	96.262 ± 1.834	1.007 ± 0.207	—
UPI	beta53	0.16	0.142	-0.018 ± 0.010	-0.112 ± 0.061	0.010 ± 0.001	0.101 ± 0.006	0.010 ± 0.001	0.102 ± 0.008	0.410 ± 0.050	0.640 ± 0.050	0.119	-0.041	-25.60	0.1167	0.1236	1.436	91.589 ± 2.683	0.570 ± 0.059	—
UPI	beta54	0.16	0.165	0.005 ± 0.016	0.032 ± 0.098	0.026 ± 0.006	0.162 ± 0.019	0.026 ± 0.006	0.161 ± 0.020	1.012 ± 0.228	1.006 ± 0.127	0.158	-0.002	-0.99	0.1437	0.1437	1.245	90.654 ± 2.814	0.789 ± 0.140	—
UPI	beta55	0.00	0.015	0.015 ± 0.015	—	0.024 ± 0.007	0.156 ± 0.023	0.024 ± 0.006	0.156 ± 0.023	—	—	0.036	0.036	—	0.1512	0.1555	1.411	100.000 ± 0.000	0.864 ± 0.068	0.000 ± 0.000
UPI	beta56	0.00	-0.014	-0.014 ± 0.012	—	0.015 ± 0.003	0.124 ± 0.013	0.015 ± 0.003	0.124 ± 0.015	—	—	-0.017	-0.017	—	0.1390	0.1401	1.611	100.000 ± 0.000	0.781 ± 0.071	0.000 ± 0.000
UPI	beta57	0.00	-0.051	-0.051 ± 0.016	—	0.027 ± 0.007	0.164 ± 0.022	0.029 ± 0.008	0.171 ± 0.024	—	—	-0.041	-0.041	—	0.1136	0.1208	1.406	100.000 ± 0.000	0.903 ± 0.192	0.000 ± 0.000
UPI	beta58	0.00	-0.051	-0.051 ± 0.012	—	0.016 ± 0.003	0.125 ± 0.012	0.018 ± 0.004	0.135 ± 0.016	—	—	-0.030	-0.030	—	0.1175	0.1212	1.872	97.196 ± 1.596	0.921 ± 0.162	2.804 ± 1.596

Note. MCSE = Monte Carlo Standard Error shown as \u000b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S272

Simulation 2 - Condition 128: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.4

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.207	0.007 ± 0.004	0.036 ± 0.022	0.002 ± 0.000	0.049 ± 0.003	0.002 ± 0.000	0.050 ± 0.004	0.062 ± 0.008	0.248 ± 0.020	0.211	0.011	5.31	0.0516	0.0527	1.222	97.541 ± 1.402	0.236 ± 0.004	—
LSAM	beta42	0.20	0.222	0.022 ± 0.006	0.108 ± 0.028	0.004 ± 0.000	0.062 ± 0.004	0.004 ± 0.001	0.065 ± 0.005	0.107 ± 0.014	0.327 ± 0.025	0.221	0.021	10.71	0.0598	0.0635	1.036	94.262 ± 2.106	0.252 ± 0.004	—
LSAM	beta43	0.20	0.187	-0.013 ± 0.004	-0.067 ± 0.020	0.002 ± 0.000	0.044 ± 0.003	0.002 ± 0.000	0.045 ± 0.003	0.051 ± 0.006	0.227 ± 0.017	0.184	-0.016	-7.93	0.0463	0.0490	1.098	97.541 ± 1.402	0.187 ± 0.002	—
LSAM	beta44	0.00	0.037	0.037 ± 0.006	—	0.005 ± 0.001	0.072 ± 0.005	0.006 ± 0.001	0.080 ± 0.006	—	—	0.054	0.054	—	0.0672	0.0862	0.940	95.082 ± 1.958	0.264 ± 0.006	4.918 ± 1.958
LSAM	beta45	0.00	0.016	0.016 ± 0.005	—	0.003 ± 0.000	0.050 ± 0.003	0.003 ± 0.000	0.053 ± 0.004	—	—	0.020	0.020	—	0.0460	0.0501	1.101	97.541 ± 1.402	0.217 ± 0.004	2.459 ± 1.402
LSAM	beta46	0.00	-0.039	-0.039 ± 0.004	—	0.002 ± 0.000	0.040 ± 0.003	0.003 ± 0.000	0.056 ± 0.003	—	—	-0.047	-0.047	—	0.0344	0.0586	1.109	87.705 ± 2.973	0.174 ± 0.005	12.295 ± 2.973
LSAM	beta47	0.00	-0.043	-0.043 ± 0.004	—	0.002 ± 0.000	0.041 ± 0.002	0.003 ± 0.000	0.059 ± 0.004	—	—	-0.049	-0.049	—	0.0482	0.0685	1.085	85.246 ± 3.211	0.174 ± 0.005	14.754 ± 3.211
LSAM	beta51	0.16	0.164	0.004 ± 0.005	0.024 ± 0.031	0.003 ± 0.000	0.054 ± 0.004	0.003 ± 0.000	0.054 ± 0.004	0.113 ± 0.015	0.337 ± 0.028	0.167	0.007	4.07	0.0544	0.0548	1.164	96.721 ± 1.612	0.246 ± 0.003	—
LSAM	beta52	0.16	0.177	0.017 ± 0.005	0.109 ± 0.031	0.003 ± 0.000	0.055 ± 0.003	0.003 ± 0.000	0.058 ± 0.004	0.131 ± 0.016	0.362 ± 0.027	0.165	0.005	3.16	0.0524	0.0526	1.047	97.541 ± 1.402	0.227 ± 0.003	—
LSAM	beta53	0.16	0.139	-0.021 ± 0.005	-0.131 ± 0.030	0.003 ± 0.000	0.052 ± 0.003	0.003 ± 0.000	0.056 ± 0.004	0.123 ± 0.016	0.351 ± 0.026	0.140	-0.020	-12.22	0.0504	0.0541	0.989	92.623 ± 2.367	0.203 ± 0.002	—
LSAM	beta54	0.16	0.168	0.008 ± 0.004	0.053 ± 0.026	0.002 ± 0.000	0.045 ± 0.003	0.002 ± 0.000	0.046 ± 0.003	0.083 ± 0.009	0.288 ± 0.021	0.175	0.015	9.52	0.0499	0.0522	1.195	98.361 ± 1.150	0.213 ± 0.002	—
LSAM	beta55	0.00	0.042	0.042 ± 0.008	—	0.008 ± 0.001	0.088 ± 0.008	0.009 ± 0.002	0.097 ± 0.009	—	—	0.030	0.030	—	0.0776	0.0831	0.936	90.984 ± 2.593	0.323 ± 0.008	9.016 ± 2.593
LSAM	beta56	0.00	0.004	0.004 ± 0.006	—	0.004 ± 0.001	0.063 ± 0.005	0.004 ± 0.001	0.063 ± 0.006	—	—	0.003	0.003	—	0.0592	0.0592	1.105	97.541 ± 1.402	0.275 ± 0.006	2.459 ± 1.402
LSAM	beta57	0.00	-0.038	-0.038 ± 0.004	—	0.002 ± 0.000	0.040 ± 0.003	0.003 ± 0.000	0.056 ± 0.004	—	—	-0.040	-0.040	—	0.0402	0.0570	1.064	87.705 ± 2.973	0.168 ± 0.004	12.295 ± 2.973
LSAM	beta58	0.00	-0.023	-0.023 ± 0.003	—	0.001 ± 0.000	0.034 ± 0.003	0.002 ± 0.000	0.041 ± 0.003	—	—	-0.025	-0.025	—	0.0371	0.0446	1.000	91.803 ± 2.484	0.135 ± 0.003	8.197 ± 2.484
QML	beta41	0.20	0.194	-0.006 ± 0.003	-0.031 ± 0.017	0.001 ± 0.000	0.037 ± 0.002	0.001 ± 0.000	0.037 ± 0.003	0.034 ± 0.004	0.185 ± 0.015	0.198	-0.002	-1.09	0.0403	0.0404	1.201	97.541 ± 1.402	0.173 ± 0.002	—
QML	beta42	0.20	0.203	0.003 ± 0.004	0.016 ± 0.020	0.002 ± 0.000	0.044 ± 0.003	0.002 ± 0.000	0.044 ± 0.004	0.049 ± 0.006	0.220 ± 0.018	0.201	0.001	0.65	0.0513	0.0514	1.036	93.443 ± 2.241	0.179 ± 0.002	—
QML	beta43	0.20	0.194	-0.006 ± 0.004	-0.029 ± 0.019	0.002 ± 0.000	0.041 ± 0.003	0.002 ± 0.000	0.041 ± 0.003	0.042 ± 0.005	0.206 ± 0.016	0.194	-0.006	-3.08	0.0446	0.0450	1.081	97.541 ± 1.402	0.173 ± 0.002	—
QML	beta44	0.00	0.006	0.006 ± 0.006	—	0.004 ± 0.000	0.062 ± 0.004	0.004 ± 0.000	0.062 ± 0.005	—	—	0.022	0.022	—	0.0617	0.0656	0.960	96.721 ± 1.612	0.235 ± 0.006	3.279 ± 1.612
QML	beta45	0.00	-0.003	-0.003 ± 0.004	—	0.002 ± 0.000	0.048 ± 0.003	0.002 ± 0.000	0.048 ± 0.004	—	—	-0.001	-0.001	—	0.0527	0.0527	1.060	95.902 ± 1.795	0.198 ± 0.004	4.098 ± 1.795
QML	beta46	0.00	-0.028	-0.028 ± 0.003	—	0.001 ± 0.000	0.038 ± 0.003	0.002 ± 0.000	0.047 ± 0.003	—	—	-0.034	-0.034	—	0.0340	0.0478	1.030	85.246 ± 3.211	0.152 ± 0.005	14.754 ± 3.211
QML	beta47	0.00	-0.033	-0.033 ± 0.003	—	0.001 ± 0.000	0.032 ± 0.002	0.002 ± 0.000	0.046 ± 0.002	—	—	-0.039	-0.039	—	0.0352	0.0527	1.135	85.246 ± 3.211	0.142 ± 0.003	14.754 ± 3.211
QML	beta51	0.16	0.169	0.009 ± 0.005	0.058 ± 0.031	0.003 ± 0.000	0.055 ± 0.004	0.003 ± 0.000	0.055 ± 0.004	0.119 ± 0.015	0.345 ± 0.027	0.172	0.012	7.44	0.0595	0.0607	1.161	98.361 ± 1.150	0.249 ± 0.003	—
QML	beta52	0.16	0.165	0.005 ± 0.004	0.030 ± 0.026	0.002 ± 0.000	0.046 ± 0.003	0.002 ± 0.000	0.046 ± 0.004	0.084 ± 0.012	0.289 ± 0.025	0.156	-0.004	-2.29	0.0516	0.0517	1.052	96.721 ± 1.612	0.191 ± 0.003	—
QML	beta53	0.16	0.146	-0.014 ± 0.004	-0.087 ± 0.028	0.002 ± 0.000	0.049 ± 0.003	0.003 ± 0.000	0.051 ± 0.004	0.102 ± 0.012	0.319 ± 0.023	0.149	-0.011	-7.03	0.0518	0.0530	1.000	95.902 ± 1.795	0.193 ± 0.002	—
QML	beta54	0.16	0.159	-0.001 ± 0.004	-0.005 ± 0.022	0.002 ± 0.000	0.040 ± 0.002	0.002 ± 0.000	0.039 ± 0.003	0.060 ± 0.007	0.246 ± 0.018	0.166	0.006	3.72	0.0410	0.0414	1.168	98.361 ± 1.150	0.181 ± 0.002	—
QML	beta55	0.00	0.032	0.032 ± 0.007	—	0.006 ± 0.001	0.080 ± 0.007	0.007 ± 0.001	0.086 ± 0.008	—	—	0.023	0.023	—	0.0765	0.0798	0.945	91.803 ± 2.484	0.298 ± 0.007	8.197 ± 2.484
QML	beta56	0.00	-0.013	-0.013 ± 0.005	—	0.003 ± 0.000	0.056 ± 0.004	0.003 ± 0.000	0.058 ± 0.005	—	—	-0.013	-0.013	—	0.0583	0.0598	1.121	95.082 ± 1.958	0.248 ± 0.005	4.918 ± 1.958
QML	beta57	0.00	-0.034	-0.034 ± 0.003	—	0.001 ± 0.000	0.035 ± 0.002	0.002 ± 0.000	0.049 ± 0.003	—	—	-0.034	-0.034	—	0.0369	0.0505	1.092	81.967 ± 3.481	0.152 ± 0.005	18.033 ± 3.481
QML	beta58	0.00	-0.020	-0.020 ± 0.003	—	0.001 ± 0.000	0.033 ± 0.003	0.001 ± 0.000	0.038 ± 0.003	—	—	-0.022	-0.022	—	0.0344	0.0408	1.027	86.066 ± 3.135	0.131 ± 0.005	13.934 ± 3.135
UPI	beta41	0.20	0.263	0.063 ± 0.027	0.314 ± 0.137	0.091 ± 0.018	0.302 ± 0.030	0.095 ± 0.019	0.308 ± 0.035	2.366 ± 0.480	1.538 ± 0.174	0.235	0.035	17.64	0.2875	0.2897	1.200	90.164 ± 2.696	1.423 ± 0.191	—
UPI	beta42	0.20	0.277	0.077 ± 0.025	0.385 ± 0.125	0.076 ± 0.013	0.276 ± 0.025	0.082 ± 0.015	0.286 ± 0.029	2.043 ± 0.363	1.429 ± 0.144	0.281	0.081	40.67	0.2853	0.2967	1.303	92.623 ± 2.367	1.412 ± 0.155	—
UPI	beta43	0.20	0.131	-0.069 ± 0.009	-0.346 ± 0.046	0.010 ± 0.001	0.101 ± 0.006	0.015 ± 0.002	0.122 ± 0.008	0.372 ± 0.043	0.610 ± 0.040	0.112	-0.088	-43.90	0.1411	0.1662	1.280	81.967 ± 3.481	0.506 ± 0.062	—
UPI	beta44	0.00	0.098	0.098 ± 0.018	—	0.041 ± 0.008	0.202 ± 0.020	0.050 ± 0.010	0.224 ± 0.024	—	—	0.081	0.081	—	0.1656	0.1844	1.287	97.541 ± 1.402	1.021 ± 0.110	2.459 ± 1.402

Table S272

Simulation 2 - Condition 128: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.4 (continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta45	0.00	0.027	0.027 ± 0.010	—	0.011 ± 0.002	0.107 ± 0.011	0.012 ± 0.003	0.109 ± 0.013	—	—	0.019	0.019	—	0.0867	0.0887	1.427	100.000 ± 0.000	0.596 ± 0.062	0.000 ± 0.000
UPI	beta46	0.00	-0.124	-0.124 ± 0.023	—	0.065 ± 0.012	0.255 ± 0.024	0.080 ± 0.014	0.283 ± 0.027	—	—	-0.101	-0.101	—	0.2202	0.2424	1.366	99.180 ± 0.816	1.367 ± 0.160	0.820 ± 0.816
UPI	beta47	0.00	-0.122	-0.122 ± 0.020	—	0.047 ± 0.008	0.217 ± 0.019	0.062 ± 0.010	0.249 ± 0.022	—	—	-0.116	-0.116	—	0.1953	0.2269	1.703	97.541 ± 1.402	1.450 ± 0.264	2.459 ± 1.402
UPI	beta51	0.16	0.210	0.050 ± 0.008	0.315 ± 0.047	0.007 ± 0.001	0.083 ± 0.008	0.009 ± 0.001	0.097 ± 0.008	0.368 ± 0.051	0.607 ± 0.047	0.206	0.046	28.97	0.0830	0.0950	1.220	85.246 ± 3.211	0.398 ± 0.039	—
UPI	beta52	0.16	0.205	0.045 ± 0.013	0.284 ± 0.080	0.020 ± 0.003	0.141 ± 0.013	0.022 ± 0.004	0.148 ± 0.016	0.855 ± 0.164	0.924 ± 0.099	0.206	0.046	28.97	0.1681	0.1743	1.294	93.443 ± 2.241	0.717 ± 0.100	—
UPI	beta53	0.16	0.124	-0.036 ± 0.007	-0.223 ± 0.045	0.006 ± 0.001	0.079 ± 0.005	0.007 ± 0.001	0.086 ± 0.006	0.291 ± 0.036	0.539 ± 0.039	0.116	-0.044	-27.42	0.0975	0.1069	1.012	85.246 ± 3.211	0.313 ± 0.023	—
UPI	beta54	0.16	0.141	-0.019 ± 0.011	-0.116 ± 0.071	0.016 ± 0.003	0.125 ± 0.014	0.016 ± 0.003	0.126 ± 0.014	0.616 ± 0.123	0.785 ± 0.088	0.127	-0.033	-20.74	0.1235	0.1279	1.098	86.885 ± 3.056	0.537 ± 0.069	—
UPI	beta55	0.00	0.052	0.052 ± 0.011	—	0.015 ± 0.003	0.122 ± 0.011	0.017 ± 0.003	0.132 ± 0.013	—	—	0.044	0.044	—	0.1118	0.1202	1.224	100.000 ± 0.000	0.586 ± 0.044	0.000 ± 0.000
UPI	beta56	0.00	-0.012	-0.012 ± 0.008	—	0.008 ± 0.001	0.088 ± 0.007	0.008 ± 0.001	0.089 ± 0.008	—	—	-0.003	-0.003	—	0.0985	0.0985	1.336	99.180 ± 0.816	0.463 ± 0.033	0.820 ± 0.816
UPI	beta57	0.00	-0.087	-0.087 ± 0.012	—	0.017 ± 0.003	0.130 ± 0.011	0.024 ± 0.004	0.156 ± 0.015	—	—	-0.086	-0.086	—	0.1337	0.1588	1.470	99.180 ± 0.816	0.750 ± 0.077	0.820 ± 0.816
UPI	beta58	0.00	-0.062	-0.062 ± 0.013	—	0.019 ± 0.003	0.139 ± 0.011	0.023 ± 0.004	0.152 ± 0.014	—	—	-0.028	-0.028	—	0.1170	0.1204	1.223	97.541 ± 1.402	0.668 ± 0.083	2.459 ± 1.402

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S273

Simulation 2 - Condition 129: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.6

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.204	0.004 ± 0.002	0.018 ± 0.009	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.085 ± 0.004	0.292 ± 0.008	0.202	0.002	0.80	0.0575	0.0575	0.987	94.753 ± 0.715	0.225 ± 0.001	—
LSAM	beta42	0.20	0.204	0.004 ± 0.002	0.021 ± 0.010	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.093 ± 0.004	0.305 ± 0.009	0.203	0.003	1.62	0.0598	0.0599	0.980	94.959 ± 0.702	0.234 ± 0.001	—
LSAM	beta43	0.20	0.197	-0.003 ± 0.002	-0.014 ± 0.009	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.081 ± 0.004	0.285 ± 0.008	0.197	-0.003	-1.62	0.0556	0.0556	0.960	93.930 ± 0.766	0.215 ± 0.001	—
LSAM	beta44	0.00	0.014	0.014 ± 0.002	—	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	—	—	0.013	0.013	—	0.0737	0.0748	0.953	93.210 ± 0.807	0.267 ± 0.002	6.790 ± 0.807
LSAM	beta45	0.00	0.007	0.007 ± 0.002	—	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	—	—	0.008	0.008	—	0.0590	0.0596	0.953	94.547 ± 0.728	0.230 ± 0.002	5.453 ± 0.728
LSAM	beta46	0.00	-0.015	-0.015 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.001	0.003 ± 0.000	0.052 ± 0.001	—	—	-0.017	-0.017	—	0.0457	0.0486	0.948	91.872 ± 0.876	0.184 ± 0.002	8.128 ± 0.876
LSAM	beta47	0.00	-0.016	-0.016 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	—	—	-0.017	-0.017	—	0.0467	0.0496	0.946	91.255 ± 0.906	0.181 ± 0.002	8.745 ± 0.906
LSAM	beta51	0.16	0.168	0.008 ± 0.002	0.049 ± 0.015	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	0.225 ± 0.011	0.475 ± 0.014	0.165	0.005	2.85	0.0717	0.0718	0.951	93.621 ± 0.784	0.282 ± 0.002	—
LSAM	beta52	0.16	0.163	0.003 ± 0.002	0.020 ± 0.012	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.135 ± 0.006	0.367 ± 0.010	0.162	0.002	1.23	0.0568	0.0569	1.008	95.062 ± 0.695	0.232 ± 0.001	—
LSAM	beta53	0.16	0.154	-0.006 ± 0.002	-0.039 ± 0.012	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.141 ± 0.007	0.376 ± 0.011	0.150	-0.010	-6.13	0.0594	0.0602	0.994	94.342 ± 0.741	0.233 ± 0.001	—
LSAM	beta54	0.16	0.163	0.003 ± 0.002	0.017 ± 0.012	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.141 ± 0.007	0.375 ± 0.011	0.163	0.003	2.05	0.0605	0.0606	0.969	94.753 ± 0.715	0.228 ± 0.001	—
LSAM	beta55	0.00	0.010	0.010 ± 0.003	—	0.009 ± 0.000	0.094 ± 0.003	0.009 ± 0.000	0.094 ± 0.003	—	—	0.009	0.009	—	0.0906	0.0910	0.891	93.004 ± 0.818	0.328 ± 0.003	6.996 ± 0.818
LSAM	beta56	0.00	0.005	0.005 ± 0.003	—	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	—	—	0.004	0.004	—	0.0757	0.0758	0.921	93.313 ± 0.801	0.281 ± 0.002	6.687 ± 0.801
LSAM	beta57	0.00	-0.013	-0.013 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.001	0.003 ± 0.000	0.051 ± 0.002	—	—	-0.013	-0.013	—	0.0471	0.0489	0.983	93.930 ± 0.766	0.191 ± 0.002	6.070 ± 0.766
LSAM	beta58	0.00	-0.012	-0.012 ± 0.001	—	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	-0.013	-0.013	—	0.0397	0.0419	0.963	92.798 ± 0.829	0.163 ± 0.001	7.202 ± 0.829
QML	beta41	0.20	0.200	-0.000 ± 0.002	-0.001 ± 0.009	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.074 ± 0.003	0.271 ± 0.008	0.198	-0.002	-1.19	0.0534	0.0534	1.008	95.165 ± 0.688	0.214 ± 0.001	—
QML	beta42	0.20	0.201	0.001 ± 0.002	0.003 ± 0.009	0.003 ± 0.000	0.056 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.078 ± 0.004	0.280 ± 0.008	0.200	0.000	-0.12	0.0550	0.0550	0.998	95.473 ± 0.667	0.219 ± 0.001	—
QML	beta43	0.20	0.197	-0.003 ± 0.002	-0.013 ± 0.009	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	0.076 ± 0.004	0.275 ± 0.008	0.197	-0.003	-1.53	0.0543	0.0544	0.975	94.136 ± 0.754	0.210 ± 0.001	—
QML	beta44	0.00	-0.001	-0.001 ± 0.002	—	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	—	—	0.001	0.001	—	0.0689	0.0689	0.987	94.547 ± 0.728	0.268 ± 0.003	5.453 ± 0.728
QML	beta45	0.00	-0.001	-0.001 ± 0.002	—	0.004 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	—	—	-0.002	-0.002	—	0.0557	0.0558	0.976	94.444 ± 0.735	0.227 ± 0.002	5.556 ± 0.735
QML	beta46	0.00	-0.008	-0.008 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	-0.012	-0.012	—	0.0434	0.0451	0.997	92.387 ± 0.851	0.180 ± 0.002	7.613 ± 0.851
QML	beta47	0.00	-0.012	-0.012 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	-0.015	-0.015	—	0.0437	0.0464	0.981	89.918 ± 0.966	0.172 ± 0.002	10.082 ± 0.966
QML	beta51	0.16	0.170	0.010 ± 0.002	0.063 ± 0.015	0.006 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	0.220 ± 0.011	0.469 ± 0.014	0.166	0.006	3.52	0.0720	0.0722	1.010	94.959 ± 0.702	0.295 ± 0.002	—
QML	beta52	0.16	0.160	0.000 ± 0.002	0.002 ± 0.011	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.125 ± 0.006	0.354 ± 0.010	0.160	0.000	0.16	0.0548	0.0548	1.034	95.679 ± 0.652	0.229 ± 0.001	—
QML	beta53	0.16	0.154	-0.006 ± 0.002	-0.036 ± 0.012	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	0.136 ± 0.007	0.369 ± 0.011	0.153	-0.007	-4.10	0.0580	0.0583	1.001	94.444 ± 0.735	0.231 ± 0.001	—
QML	beta54	0.16	0.160	0.000 ± 0.002	0.002 ± 0.012	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.130 ± 0.006	0.360 ± 0.010	0.161	0.001	0.85	0.0579	0.0579	0.979	94.959 ± 0.702	0.221 ± 0.001	—
QML	beta55	0.00	0.002	0.002 ± 0.003	—	0.008 ± 0.000	0.088 ± 0.002	0.008 ± 0.000	0.088 ± 0.003	—	—	0.004	0.004	—	0.0852	0.0853	1.003	94.959 ± 0.702	0.348 ± 0.023	5.041 ± 0.702
QML	beta56	0.00	-0.001	-0.001 ± 0.002	—	0.005 ± 0.000	0.074 ± 0.002	0.005 ± 0.000	0.074 ± 0.002	—	—	-0.003	-0.003	—	0.0720	0.0721	0.959	93.827 ± 0.772	0.277 ± 0.005	6.173 ± 0.772
QML	beta57	0.00	-0.009	-0.009 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	-0.012	-0.012	—	0.0426	0.0443	1.089	92.695 ± 0.835	0.194 ± 0.009	7.305 ± 0.835
QML	beta58	0.00	-0.010	-0.010 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	-0.013	-0.013	—	0.0373	0.0394	0.978	92.078 ± 0.866	0.156 ± 0.001	7.922 ± 0.866
UPI	beta41	0.20	0.204	0.004 ± 0.002	0.020 ± 0.009	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.083 ± 0.004	0.288 ± 0.008	0.203	0.003	1.54	0.0578	0.0579	1.045	95.576 ± 0.660	0.235 ± 0.001	—
UPI	beta42	0.20	0.206	0.006 ± 0.002	0.028 ± 0.010	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	0.093 ± 0.005	0.306 ± 0.009	0.205	0.005	2.29	0.0598	0.0600	1.031	95.885 ± 0.637	0.246 ± 0.002	—
UPI	beta43	0.20	0.197	-0.003 ± 0.002	-0.015 ± 0.009	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.083 ± 0.004	0.288 ± 0.008	0.196	-0.004	-2.19	0.0564	0.0566	0.999	94.547 ± 0.728	0.226 ± 0.001	—
UPI	beta44	0.00	0.010	0.010 ± 0.003	—	0.006 ± 0.000	0.080 ± 0.002	0.006 ± 0.000	0.080 ± 0.002	—	—	0.008	0.008	—	0.0757	0.0761	1.014	95.679 ± 0.652	0.317 ± 0.005	4.321 ± 0.652
UPI	beta45	0.00	0.007	0.007 ± 0.002	—	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	—	—	0.008	0.008	—	0.0639	0.0644	1.002	95.370 ± 0.674	0.256 ± 0.002	4.630 ± 0.674
UPI	beta46	0.00	-0.011	-0.011 ± 0.002	—	0.004 ± 0.000	0.059 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	—	—	-0.011	-0.011	—	0.0565	0.0576	1.013	95.370 ± 0.674	0.235 ± 0.003	4.630 ± 0.674

Table S273

Simulation 2 - Condition 129: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.6
(continued)

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.013	-0.013 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	—	—	-0.013	-0.013	—	0.0536	0.0552	1.044	95.576 ± 0.660	0.238 ± 0.004	4.424 ± 0.660
UPI	beta51	0.16	0.168	0.008 ± 0.002	0.050 ± 0.015	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.211 ± 0.011	0.459 ± 0.014	0.163	0.003	1.99	0.0696	0.0697	1.000	95.062 ± 0.695	0.287 ± 0.002	—
UPI	beta52	0.16	0.162	0.002 ± 0.002	0.015 ± 0.012	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.131 ± 0.006	0.362 ± 0.010	0.165	0.005	3.00	0.0553	0.0555	1.033	95.988 ± 0.629	0.235 ± 0.001	—
UPI	beta53	0.16	0.155	-0.005 ± 0.002	-0.034 ± 0.012	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.142 ± 0.007	0.377 ± 0.011	0.153	-0.007	-4.20	0.0587	0.0591	1.006	95.267 ± 0.681	0.237 ± 0.001	—
UPI	beta54	0.16	0.164	0.004 ± 0.002	0.025 ± 0.012	0.004 ± 0.000	0.060 ± 0.001	0.004 ± 0.000	0.060 ± 0.002	0.141 ± 0.007	0.375 ± 0.011	0.164	0.004	2.74	0.0598	0.0600	1.006	95.782 ± 0.645	0.236 ± 0.001	—
UPI	beta55	0.00	0.004	0.004 ± 0.003	—	0.009 ± 0.000	0.094 ± 0.002	0.009 ± 0.000	0.094 ± 0.003	—	—	0.005	0.005	—	0.0923	0.0924	0.934	94.959 ± 0.702	0.344 ± 0.004	5.041 ± 0.702
UPI	beta56	0.00	0.004	0.004 ± 0.003	—	0.006 ± 0.000	0.079 ± 0.002	0.006 ± 0.000	0.079 ± 0.002	—	—	0.001	0.001	—	0.0740	0.0740	0.964	95.165 ± 0.688	0.297 ± 0.003	4.835 ± 0.688
UPI	beta57	0.00	-0.005	-0.005 ± 0.002	—	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	—	—	-0.005	-0.005	—	0.0507	0.0510	1.012	95.988 ± 0.629	0.214 ± 0.002	4.012 ± 0.629
UPI	beta58	0.00	-0.008	-0.008 ± 0.002	—	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.049 ± 0.002	—	—	-0.007	-0.007	—	0.0423	0.0429	1.045	96.708 ± 0.572	0.198 ± 0.003	3.292 ± 0.572

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S274

Simulation 2 - Condition 130: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, $N = 1000$, Reliability = 0.6

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Type I (%) $\pm$ MCSE
LSAM	beta41	0.20	0.204	0.004 $\pm$ 0.001	0.018 $\pm$ 0.006	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.034 $\pm$ 0.002	0.186 $\pm$ 0.005	0.203	0.003	1.35	0.0369	0.0370	0.966	94.689 $\pm$ 0.710	0.140 $\pm$ 0.000	—
LSAM	beta42	0.20	0.203	0.003 $\pm$ 0.001	0.016 $\pm$ 0.006	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.035 $\pm$ 0.002	0.187 $\pm$ 0.005	0.203	0.003	1.64	0.0363	0.0365	0.995	95.491 $\pm$ 0.657	0.145 $\pm$ 0.000	—
LSAM	beta43	0.20	0.197	-0.003 $\pm$ 0.001	-0.017 $\pm$ 0.005	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.028 $\pm$ 0.001	0.166 $\pm$ 0.005	0.195	-0.005	-2.71	0.0315	0.0320	1.036	95.892 $\pm$ 0.628	0.134 $\pm$ 0.000	—
LSAM	beta44	0.00	0.014	0.014 $\pm$ 0.001	—	0.002 $\pm$ 0.000	0.041 $\pm$ 0.001	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	—	—	0.014	0.014	—	0.0408	0.0433	1.004	92.986 $\pm$ 0.808	0.160 $\pm$ 0.001	7.014 $\pm$ 0.808
LSAM	beta45	0.00	0.004	0.004 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	—	—	0.004	0.004	—	0.0362	0.0365	0.968	94.489 $\pm$ 0.722	0.140 $\pm$ 0.001	5.511 $\pm$ 0.722
LSAM	beta46	0.00	-0.014	-0.014 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	—	—	-0.013	-0.013	—	0.0290	0.0317	0.986	91.182 $\pm$ 0.898	0.111 $\pm$ 0.001	8.818 $\pm$ 0.898
LSAM	beta47	0.00	-0.014	-0.014 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.028 $\pm$ 0.001	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	—	—	-0.014	-0.014	—	0.0280	0.0314	0.985	91.082 $\pm$ 0.902	0.109 $\pm$ 0.001	8.918 $\pm$ 0.902
LSAM	beta51	0.16	0.159	-0.001 $\pm$ 0.001	-0.007 $\pm$ 0.009	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.083 $\pm$ 0.004	0.287 $\pm$ 0.008	0.159	-0.001	-0.90	0.0454	0.0454	0.974	94.188 $\pm$ 0.741	0.176 $\pm$ 0.001	—
LSAM	beta52	0.16	0.162	0.002 $\pm$ 0.001	0.012 $\pm$ 0.007	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.049 $\pm$ 0.002	0.222 $\pm$ 0.006	0.161	0.001	0.63	0.0345	0.0345	1.025	94.890 $\pm$ 0.697	0.142 $\pm$ 0.000	—
LSAM	beta53	0.16	0.153	-0.007 $\pm$ 0.001	-0.042 $\pm$ 0.007	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.038 $\pm$ 0.001	0.057 $\pm$ 0.003	0.238 $\pm$ 0.007	0.153	-0.007	-4.22	0.0363	0.0369	0.991	94.088 $\pm$ 0.747	0.145 $\pm$ 0.000	—
LSAM	beta54	0.16	0.162	0.002 $\pm$ 0.001	0.013 $\pm$ 0.007	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.051 $\pm$ 0.002	0.227 $\pm$ 0.006	0.162	0.002	1.21	0.0337	0.0337	0.997	94.489 $\pm$ 0.722	0.141 $\pm$ 0.000	—
LSAM	beta55	0.00	0.013	0.013 $\pm$ 0.002	—	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.003 $\pm$ 0.000	0.053 $\pm$ 0.001	—	—	0.012	0.012	—	0.0483	0.0497	0.967	94.188 $\pm$ 0.741	0.194 $\pm$ 0.001	5.812 $\pm$ 0.741
LSAM	beta56	0.00	0.000	-0.000 $\pm$ 0.001	—	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	—	—	0.000	0.000	—	0.0430	0.0430	0.931	93.587 $\pm$ 0.775	0.169 $\pm$ 0.001	6.413 $\pm$ 0.775
LSAM	beta57	0.00	-0.014	-0.014 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.031 $\pm$ 0.001	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	—	—	-0.014	-0.014	—	0.0295	0.0327	0.954	89.379 $\pm$ 0.975	0.114 $\pm$ 0.001	10.621 $\pm$ 0.975
LSAM	beta58	0.00	-0.011	-0.011 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.026 $\pm$ 0.001	0.001 $\pm$ 0.000	0.028 $\pm$ 0.001	—	—	-0.012	-0.012	—	0.0244	0.0271	0.984	91.383 $\pm$ 0.888	0.099 $\pm$ 0.001	8.617 $\pm$ 0.888
QML	beta41	0.20	0.200	0.000 $\pm$ 0.001	0.001 $\pm$ 0.005	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.030 $\pm$ 0.001	0.173 $\pm$ 0.005	0.199	-0.001	-0.53	0.0339	0.0339	0.978	95.291 $\pm$ 0.671	0.133 $\pm$ 0.000	—
QML	beta42	0.20	0.200	0.000 $\pm$ 0.001	0.002 $\pm$ 0.006	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.030 $\pm$ 0.001	0.174 $\pm$ 0.005	0.201	0.001	0.51	0.0341	0.0341	0.999	95.792 $\pm$ 0.636	0.136 $\pm$ 0.000	—
QML	beta43	0.20	0.197	-0.003 $\pm$ 0.001	-0.016 $\pm$ 0.005	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	0.026 $\pm$ 0.001	0.162 $\pm$ 0.004	0.195	-0.005	-2.40	0.0307	0.0311	1.040	95.792 $\pm$ 0.636	0.132 $\pm$ 0.000	—
QML	beta44	0.00	0.003	0.003 $\pm$ 0.001	—	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	0.002 $\pm$ 0.000	0.040 $\pm$ 0.001	—	—	0.004	0.004	—	0.0406	0.0408	1.019	95.190 $\pm$ 0.677	0.161 $\pm$ 0.001	4.810 $\pm$ 0.677
QML	beta45	0.00	-0.004	-0.004 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	—	—	-0.003	-0.003	—	0.0353	0.0355	0.965	93.788 $\pm$ 0.764	0.139 $\pm$ 0.001	6.212 $\pm$ 0.764
QML	beta46	0.00	-0.010	-0.010 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.027 $\pm$ 0.001	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	—	—	-0.010	-0.010	—	0.0276	0.0293	1.008	91.182 $\pm$ 0.898	0.108 $\pm$ 0.001	8.818 $\pm$ 0.898
QML	beta47	0.00	-0.012	-0.012 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.027 $\pm$ 0.001	0.001 $\pm$ 0.000	0.030 $\pm$ 0.001	—	—	-0.013	-0.013	—	0.0260	0.0292	0.994	90.080 $\pm$ 0.946	0.105 $\pm$ 0.001	9.920 $\pm$ 0.946
QML	beta51	0.16	0.160	-0.000 $\pm$ 0.001	-0.001 $\pm$ 0.009	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.081 $\pm$ 0.004	0.285 $\pm$ 0.008	0.159	-0.001	-0.35	0.0447	0.0447	0.993	94.890 $\pm$ 0.697	0.178 $\pm$ 0.001	—
QML	beta52	0.16	0.160	0.000 $\pm$ 0.001	0.001 $\pm$ 0.007	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.001 $\pm$ 0.000	0.034 $\pm$ 0.001	0.045 $\pm$ 0.002	0.212 $\pm$ 0.006	0.159	-0.001	-0.73	0.0325	0.0325	1.041	95.491 $\pm$ 0.657	0.138 $\pm$ 0.000	—
QML	beta53	0.16	0.154	-0.006 $\pm$ 0.001	-0.038 $\pm$ 0.007	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.054 $\pm$ 0.003	0.233 $\pm$ 0.006	0.154	-0.006	-3.89	0.0354	0.0359	0.993	94.088 $\pm$ 0.747	0.143 $\pm$ 0.000	—
QML	beta54	0.16	0.161	0.001 $\pm$ 0.001	0.003 $\pm$ 0.007	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.047 $\pm$ 0.002	0.216 $\pm$ 0.006	0.161	0.001	0.44	0.0331	0.0331	1.008	94.689 $\pm$ 0.710	0.137 $\pm$ 0.000	—
QML	beta55	0.00	0.008	0.008 $\pm$ 0.002	—	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	0.003 $\pm$ 0.000	0.050 $\pm$ 0.001	—	—	0.008	0.008	—	0.0495	0.0502	0.997	94.990 $\pm$ 0.691	0.193 $\pm$ 0.001	5.010 $\pm$ 0.691
QML	beta56	0.00	-0.007	-0.007 $\pm$ 0.001	—	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	—	—	-0.006	-0.006	—	0.0414	0.0418	0.934	92.886 $\pm$ 0.814	0.165 $\pm$ 0.001	7.114 $\pm$ 0.814
QML	beta57	0.00	-0.013	-0.013 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.029 $\pm$ 0.001	0.001 $\pm$ 0.000	0.032 $\pm$ 0.001	—	—	-0.014	-0.014	—	0.0272	0.0307	0.976	88.277 $\pm$ 1.018	0.111 $\pm$ 0.001	11.723 $\pm$ 1.018
QML	beta58	0.00	-0.011	-0.011 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.025 $\pm$ 0.001	0.001 $\pm$ 0.000	0.027 $\pm$ 0.001	—	—	-0.012	-0.012	—	0.0238	0.0266	0.999	90.281 $\pm$ 0.938	0.096 $\pm$ 0.001	9.719 $\pm$ 0.938
UPI	beta41	0.20	0.203	0.003 $\pm$ 0.001	0.013 $\pm$ 0.006	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.033 $\pm$ 0.001	0.181 $\pm$ 0.005	0.202	0.002	0.99	0.0363	0.0364	0.986	95.591 $\pm$ 0.650	0.140 $\pm$ 0.000	—
UPI	beta42	0.20	0.204	0.004 $\pm$ 0.001	0.018 $\pm$ 0.006	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.033 $\pm$ 0.002	0.182 $\pm$ 0.005	0.204	0.004	1.78	0.0365	0.0366	1.016	95.892 $\pm$ 0.628	0.144 $\pm$ 0.000	—
UPI	beta43	0.20	0.197	-0.003 $\pm$ 0.001	-0.015 $\pm$ 0.005	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.028 $\pm$ 0.001	0.167 $\pm$ 0.005	0.195	-0.005	-2.32	0.0319	0.0323	1.040	95.691 $\pm$ 0.643	0.136 $\pm$ 0.000	—
UPI	beta44	0.00	0.010	0.010 $\pm$ 0.001	—	0.002 $\pm$ 0.000	0.043 $\pm$ 0.001	0.002 $\pm$ 0.000	0.044 $\pm$ 0.001	—	—	0.011	0.011	—	0.0428	0.0442	1.013	93.988 $\pm$ 0.752	0.172 $\pm$ 0.001	6.012 $\pm$ 0.752

Table S274

Simulation 2 - Condition 130: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.6 (continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta45	0.00	0.002	0.002 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	0.003	0.003	—	0.0359	0.0360	1.000	94.990 ± 0.691	0.147 ± 0.001	5.010 ± 0.691
UPI	beta46	0.00	-0.008	-0.008 ± 0.001	—	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	—	—	-0.007	-0.007	—	0.0313	0.0321	0.997	94.990 ± 0.691	0.128 ± 0.001	5.010 ± 0.691
UPI	beta47	0.00	-0.010	-0.010 ± 0.001	—	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	—	—	-0.010	-0.010	—	0.0305	0.0322	1.003	93.687 ± 0.770	0.126 ± 0.001	6.313 ± 0.770
UPI	beta51	0.16	0.159	-0.001 ± 0.001	-0.004 ± 0.009	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.081 ± 0.004	0.285 ± 0.008	0.159	-0.001	-0.72	0.0442	0.0442	0.985	94.689 ± 0.710	0.176 ± 0.001	—
UPI	beta52	0.16	0.161	0.001 ± 0.001	0.007 ± 0.007	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.048 ± 0.002	0.219 ± 0.006	0.160	0.000	-0.20	0.0345	0.0345	1.038	95.892 ± 0.628	0.143 ± 0.000	—
UPI	beta53	0.16	0.154	-0.006 ± 0.001	-0.038 ± 0.007	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.056 ± 0.003	0.237 ± 0.007	0.154	-0.006	-3.86	0.0363	0.0369	0.996	93.988 ± 0.752	0.146 ± 0.000	—
UPI	beta54	0.16	0.162	0.002 ± 0.001	0.014 ± 0.007	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.051 ± 0.002	0.226 ± 0.006	0.162	0.002	1.32	0.0343	0.0343	1.008	95.090 ± 0.684	0.143 ± 0.000	—
UPI	beta55	0.00	0.010	0.010 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	—	—	0.010	0.010	—	0.0487	0.0497	0.988	95.491 ± 0.657	0.199 ± 0.001	4.509 ± 0.657
UPI	beta56	0.00	-0.002	-0.002 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	-0.002	-0.002	—	0.0429	0.0430	0.943	93.687 ± 0.770	0.171 ± 0.001	6.313 ± 0.770
UPI	beta57	0.00	-0.009	-0.009 ± 0.001	—	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	—	—	-0.009	-0.009	—	0.0310	0.0324	0.958	92.986 ± 0.808	0.124 ± 0.001	7.014 ± 0.808
UPI	beta58	0.00	-0.008	-0.008 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	-0.009	-0.009	—	0.0258	0.0272	1.011	94.188 ± 0.741	0.110 ± 0.001	5.812 ± 0.741

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u002014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S275

Simulation 2 - Condition 131: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.8

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias $\pm$ MCSE	Rel. Bias $\pm$ MCSE	Var. $\pm$ MCSE	SD $\pm$ MCSE	MSE $\pm$ MCSE	RMSE $\pm$ MCSE	Rel. MSE $\pm$ MCSE	Rel. RMSE $\pm$ MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) $\pm$ MCSE	CI Width $\pm$ MCSE	Type I (%) $\pm$ MCSE
LSAM	beta41	0.20	0.199	-0.001 $\pm$ 0.001	-0.006 $\pm$ 0.007	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.052 $\pm$ 0.002	0.228 $\pm$ 0.006	0.200	0.000	0.04	0.0441	0.0441	0.989	95.100 $\pm$ 0.683	0.177 $\pm$ 0.001	—
LSAM	beta42	0.20	0.201	0.001 $\pm$ 0.001	0.004 $\pm$ 0.007	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.054 $\pm$ 0.002	0.232 $\pm$ 0.006	0.201	0.001	0.38	0.0455	0.0455	1.000	94.700 $\pm$ 0.708	0.182 $\pm$ 0.001	—
LSAM	beta43	0.20	0.198	-0.002 $\pm$ 0.001	-0.009 $\pm$ 0.007	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.053 $\pm$ 0.002	0.231 $\pm$ 0.006	0.198	-0.002	-0.99	0.0453	0.0454	0.952	92.900 $\pm$ 0.812	0.172 $\pm$ 0.001	—
LSAM	beta44	0.00	0.003	0.003 $\pm$ 0.002	—	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	—	—	0.000	0.000	—	0.0509	0.0509	0.979	95.200 $\pm$ 0.676	0.195 $\pm$ 0.001	4.800 $\pm$ 0.676
LSAM	beta45	0.00	0.000	0.000 $\pm$ 0.001	—	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	—	—	0.000	0.000	—	0.0440	0.0440	0.960	94.000 $\pm$ 0.751	0.171 $\pm$ 0.001	6.000 $\pm$ 0.751
LSAM	beta46	0.00	-0.004	-0.004 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	—	—	-0.005	-0.005	—	0.0350	0.0354	0.984	93.300 $\pm$ 0.791	0.137 $\pm$ 0.001	6.700 $\pm$ 0.791
LSAM	beta47	0.00	-0.004	-0.004 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	—	—	-0.004	-0.004	—	0.0359	0.0362	0.969	93.900 $\pm$ 0.757	0.135 $\pm$ 0.001	6.100 $\pm$ 0.757
LSAM	beta51	0.16	0.161	0.001 $\pm$ 0.002	0.007 $\pm$ 0.012	0.004 $\pm$ 0.000	0.062 $\pm$ 0.001	0.004 $\pm$ 0.000	0.062 $\pm$ 0.002	0.151 $\pm$ 0.007	0.388 $\pm$ 0.011	0.161	0.001	0.58	0.0622	0.0622	0.935	93.900 $\pm$ 0.757	0.228 $\pm$ 0.001	—
LSAM	beta52	0.16	0.162	0.002 $\pm$ 0.001	0.015 $\pm$ 0.009	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.002 $\pm$ 0.000	0.047 $\pm$ 0.001	0.085 $\pm$ 0.004	0.291 $\pm$ 0.008	0.164	0.004	2.28	0.0434	0.0435	1.000	95.200 $\pm$ 0.676	0.183 $\pm$ 0.001	—
LSAM	beta53	0.16	0.159	-0.001 $\pm$ 0.002	-0.005 $\pm$ 0.010	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	0.002 $\pm$ 0.000	0.050 $\pm$ 0.001	0.097 $\pm$ 0.005	0.311 $\pm$ 0.009	0.158	-0.002	-1.25	0.0486	0.0486	0.972	94.300 $\pm$ 0.733	0.190 $\pm$ 0.001	—
LSAM	beta54	0.16	0.158	-0.002 $\pm$ 0.002	-0.010 $\pm$ 0.010	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.094 $\pm$ 0.004	0.306 $\pm$ 0.009	0.160	0.000	0.05	0.0467	0.0467	0.953	93.600 $\pm$ 0.774	0.183 $\pm$ 0.001	—
LSAM	beta55	0.00	0.005	0.005 $\pm$ 0.002	—	0.004 $\pm$ 0.000	0.067 $\pm$ 0.002	0.005 $\pm$ 0.000	0.067 $\pm$ 0.002	—	—	0.006	0.006	—	0.0637	0.0640	0.907	92.900 $\pm$ 0.812	0.238 $\pm$ 0.002	7.100 $\pm$ 0.812
LSAM	beta56	0.00	0.003	0.003 $\pm$ 0.002	—	0.003 $\pm$ 0.000	0.055 $\pm$ 0.001	0.003 $\pm$ 0.000	0.055 $\pm$ 0.002	—	—	0.003	0.003	—	0.0497	0.0498	0.972	94.700 $\pm$ 0.708	0.210 $\pm$ 0.001	5.300 $\pm$ 0.708
LSAM	beta57	0.00	-0.004	-0.004 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	—	—	-0.006	-0.006	—	0.0370	0.0374	0.992	93.000 $\pm$ 0.807	0.141 $\pm$ 0.001	7.000 $\pm$ 0.807
LSAM	beta58	0.00	-0.003	-0.003 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	—	—	-0.003	-0.003	—	0.0337	0.0338	0.957	93.000 $\pm$ 0.807	0.125 $\pm$ 0.001	7.000 $\pm$ 0.807
QML	beta41	0.20	0.198	-0.002 $\pm$ 0.001	-0.012 $\pm$ 0.007	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.050 $\pm$ 0.002	0.224 $\pm$ 0.006	0.199	-0.001	-0.39	0.0433	0.0434	0.997	95.400 $\pm$ 0.662	0.175 $\pm$ 0.001	—
QML	beta42	0.20	0.200	0.000 $\pm$ 0.001	0.000 $\pm$ 0.007	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.052 $\pm$ 0.002	0.228 $\pm$ 0.006	0.200	0.000	0.16	0.0444	0.0444	1.003	94.700 $\pm$ 0.708	0.179 $\pm$ 0.001	—
QML	beta43	0.20	0.198	-0.002 $\pm$ 0.001	-0.011 $\pm$ 0.007	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.052 $\pm$ 0.002	0.229 $\pm$ 0.006	0.197	-0.003	-1.29	0.0457	0.0457	0.956	93.100 $\pm$ 0.801	0.172 $\pm$ 0.001	—
QML	beta44	0.00	-0.002	-0.002 $\pm$ 0.002	—	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	0.003 $\pm$ 0.000	0.051 $\pm$ 0.001	—	—	-0.003	-0.003	—	0.0500	0.0501	0.984	95.400 $\pm$ 0.662	0.197 $\pm$ 0.001	4.600 $\pm$ 0.662
QML	beta45	0.00	-0.002	-0.002 $\pm$ 0.001	—	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	—	—	-0.002	-0.002	—	0.0447	0.0448	0.966	94.400 $\pm$ 0.727	0.172 $\pm$ 0.001	5.600 $\pm$ 0.727
QML	beta46	0.00	-0.002	-0.002 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	—	—	-0.004	-0.004	—	0.0335	0.0337	0.991	93.500 $\pm$ 0.780	0.135 $\pm$ 0.001	6.500 $\pm$ 0.780
QML	beta47	0.00	-0.003	-0.003 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	—	—	-0.004	-0.004	—	0.0346	0.0349	0.974	93.900 $\pm$ 0.757	0.133 $\pm$ 0.001	6.100 $\pm$ 0.757
QML	beta51	0.16	0.161	0.001 $\pm$ 0.002	0.009 $\pm$ 0.012	0.004 $\pm$ 0.000	0.062 $\pm$ 0.001	0.004 $\pm$ 0.000	0.062 $\pm$ 0.002	0.149 $\pm$ 0.007	0.386 $\pm$ 0.011	0.161	0.001	0.50	0.0609	0.0609	0.967	94.600 $\pm$ 0.715	0.234 $\pm$ 0.001	—
QML	beta52	0.16	0.162	0.002 $\pm$ 0.001	0.012 $\pm$ 0.009	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.084 $\pm$ 0.004	0.289 $\pm$ 0.008	0.163	0.003	1.75	0.0446	0.0447	1.010	95.500 $\pm$ 0.656	0.183 $\pm$ 0.001	—
QML	beta53	0.16	0.159	-0.001 $\pm$ 0.002	-0.006 $\pm$ 0.010	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.002 $\pm$ 0.000	0.049 $\pm$ 0.001	0.095 $\pm$ 0.004	0.309 $\pm$ 0.009	0.158	-0.002	-1.47	0.0480	0.0481	0.977	94.400 $\pm$ 0.727	0.189 $\pm$ 0.001	—
QML	beta54	0.16	0.158	-0.002 $\pm$ 0.002	-0.013 $\pm$ 0.010	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.002 $\pm$ 0.000	0.048 $\pm$ 0.001	0.092 $\pm$ 0.004	0.303 $\pm$ 0.009	0.160	0.000	-0.14	0.0458	0.0458	0.960	93.800 $\pm$ 0.763	0.182 $\pm$ 0.001	—
QML	beta55	0.00	0.003	0.003 $\pm$ 0.002	—	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	0.004 $\pm$ 0.000	0.066 $\pm$ 0.002	—	—	0.004	0.004	—	0.0632	0.0633	0.942	94.000 $\pm$ 0.751	0.243 $\pm$ 0.003	6.000 $\pm$ 0.751
QML	beta56	0.00	0.001	0.001 $\pm$ 0.002	—	0.003 $\pm$ 0.000	0.054 $\pm$ 0.001	0.003 $\pm$ 0.000	0.054 $\pm$ 0.002	—	—	0.000	0.000	—	0.0501	0.0501	0.992	95.000 $\pm$ 0.689	0.211 $\pm$ 0.002	5.000 $\pm$ 0.689
QML	beta57	0.00	-0.004	-0.004 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.035 $\pm$ 0.001	0.001 $\pm$ 0.000	0.036 $\pm$ 0.001	—	—	-0.005	-0.005	—	0.0363	0.0367	1.018	92.800 $\pm$ 0.817	0.142 $\pm$ 0.001	7.200 $\pm$ 0.817
QML	beta58	0.00	-0.003	-0.003 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	0.001 $\pm$ 0.000	0.033 $\pm$ 0.001	—	—	-0.003	-0.003	—	0.0328	0.0329	0.961	93.000 $\pm$ 0.807	0.123 $\pm$ 0.001	7.000 $\pm$ 0.807
UPI	beta41	0.20	0.199	-0.001 $\pm$ 0.001	-0.006 $\pm$ 0.007	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.002 $\pm$ 0.000	0.045 $\pm$ 0.001	0.052 $\pm$ 0.002	0.227 $\pm$ 0.006	0.200	0.000	-0.20	0.0443	0.0443	0.993	95.000 $\pm$ 0.689	0.177 $\pm$ 0.001	—
UPI	beta42	0.20	0.201	0.001 $\pm$ 0.001	0.004 $\pm$ 0.007	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.053 $\pm$ 0.002	0.231 $\pm$ 0.006	0.200	0.000	-0.08	0.0455	0.0455	1.002	94.700 $\pm$ 0.708	0.181 $\pm$ 0.001	—
UPI	beta43	0.20	0.198	-0.002 $\pm$ 0.001	-0.008 $\pm$ 0.007	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.054 $\pm$ 0.002	0.232 $\pm$ 0.006	0.198	-0.002	-1.15	0.0459	0.0460	0.955	92.900 $\pm$ 0.812	0.173 $\pm$ 0.001	—
UPI	beta44	0.00	0.002	0.002 $\pm$ 0.002	—	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	0.003 $\pm$ 0.000	0.052 $\pm$ 0.001	—	—	0.000	0.000	—	0.0512	0.0512	0.984	94.900 $\pm$ 0.696	0.202 $\pm$ 0.001	5.100 $\pm$ 0.696
UPI	beta45	0.00	0.000	0.000 $\pm$ 0.001	—	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	0.002 $\pm$ 0.000	0.046 $\pm$ 0.001	—	—	-0.001	-0.001	—	0.0430	0.0430	0.974	94.000 $\pm$ 0.751	0.176 $\pm$ 0.001	6.000 $\pm$ 0.751
UPI	beta46	0.00	-0.003	-0.003 $\pm$ 0.001	—	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	0.001 $\pm$ 0.000	0.037 $\pm$ 0.001	—	—	-0.004	-0.004	—	0.0366	0.0369	0.988	94.000 $\pm$ 0.751	0.143 $\pm$ 0.001	6.000 $\pm$ 0.751

Table S275

Simulation 2 - Condition 131: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.8
(continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.003	-0.003 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.004	-0.004	—	0.0363	0.0366	0.973	94.400 ± 0.727	0.141 ± 0.001	5.600 ± 0.727
UPI	beta51	0.16	0.161	0.001 ± 0.002	0.007 ± 0.012	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.147 ± 0.007	0.383 ± 0.011	0.161	0.001	0.50	0.0616	0.0616	0.963	94.200 ± 0.739	0.232 ± 0.001	—
UPI	beta52	0.16	0.162	0.002 ± 0.001	0.014 ± 0.009	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.084 ± 0.004	0.290 ± 0.008	0.163	0.003	1.98	0.0434	0.0435	1.005	95.600 ± 0.649	0.182 ± 0.001	—
UPI	beta53	0.16	0.160	-0.000 ± 0.002	-0.002 ± 0.010	0.002 ± 0.000	0.050 ± 0.001	0.002 ± 0.000	0.050 ± 0.001	0.096 ± 0.004	0.310 ± 0.009	0.158	-0.002	-1.10	0.0476	0.0476	0.977	94.300 ± 0.733	0.190 ± 0.001	—
UPI	beta54	0.16	0.158	-0.002 ± 0.002	-0.010 ± 0.010	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.093 ± 0.004	0.306 ± 0.009	0.160	0.000	0.24	0.0463	0.0463	0.956	93.700 ± 0.768	0.183 ± 0.001	—
UPI	beta55	0.00	0.004	0.004 ± 0.002	—	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	—	—	0.003	0.003	—	0.0627	0.0628	0.942	94.100 ± 0.745	0.242 ± 0.002	5.900 ± 0.745
UPI	beta56	0.00	0.003	0.003 ± 0.002	—	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	—	—	0.002	0.002	—	0.0490	0.0490	0.988	95.200 ± 0.676	0.213 ± 0.002	4.800 ± 0.676
UPI	beta57	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	-0.004	-0.004	—	0.0368	0.0370	1.010	94.800 ± 0.702	0.144 ± 0.001	5.200 ± 0.702
UPI	beta58	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	—	—	-0.002	-0.002	—	0.0339	0.0340	0.966	93.300 ± 0.791	0.128 ± 0.001	6.700 ± 0.791

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u00202014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S276

Simulation 2 - Condition 132: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.8

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.201	0.001 ± 0.001	0.004 ± 0.004	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.019 ± 0.001	0.139 ± 0.004	0.202	0.002	0.86	0.0287	0.0288	1.027	96.200 ± 0.605	0.112 ± 0.000	—
LSAM	beta42	0.20	0.201	0.001 ± 0.001	0.004 ± 0.005	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.023 ± 0.001	0.151 ± 0.004	0.200	0.000	0.17	0.0291	0.0291	0.969	94.700 ± 0.708	0.115 ± 0.000	—
LSAM	beta43	0.20	0.199	-0.001 ± 0.001	-0.006 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.021 ± 0.001	0.144 ± 0.004	0.199	-0.001	-0.33	0.0301	0.0301	0.972	94.700 ± 0.708	0.110 ± 0.000	—
LSAM	beta44	0.00	0.003	0.003 ± 0.001	—	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	—	—	0.003	0.003	—	0.0342	0.0344	0.965	94.000 ± 0.751	0.123 ± 0.001	6.000 ± 0.751
LSAM	beta45	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	0.001	0.001	—	0.0283	0.0283	0.962	94.400 ± 0.727	0.109 ± 0.000	5.600 ± 0.727
LSAM	beta46	0.00	-0.003	-0.003 ± 0.001	—	0.000 ± 0.000	0.022 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	—	—	-0.003	-0.003	—	0.0218	0.0221	0.977	93.500 ± 0.780	0.085 ± 0.000	6.500 ± 0.780
LSAM	beta47	0.00	-0.003	-0.003 ± 0.001	—	0.000 ± 0.000	0.022 ± 0.001	0.000 ± 0.000	0.022 ± 0.001	—	—	-0.003	-0.003	—	0.0214	0.0216	0.997	93.300 ± 0.791	0.085 ± 0.000	6.700 ± 0.791
LSAM	beta51	0.16	0.161	0.001 ± 0.001	0.008 ± 0.008	0.001 ± 0.000	0.039 ± 0.001	0.001 ± 0.000	0.039 ± 0.001	0.058 ± 0.003	0.241 ± 0.007	0.162	0.002	1.11	0.0375	0.0376	0.958	94.300 ± 0.733	0.145 ± 0.001	—
LSAM	beta52	0.16	0.162	0.002 ± 0.001	0.010 ± 0.006	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.035 ± 0.002	0.186 ± 0.005	0.162	0.002	1.35	0.0287	0.0288	0.994	94.900 ± 0.696	0.116 ± 0.000	—
LSAM	beta53	0.16	0.157	-0.003 ± 0.001	-0.016 ± 0.006	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.040 ± 0.002	0.199 ± 0.006	0.157	-0.003	-1.83	0.0305	0.0306	0.960	92.800 ± 0.817	0.119 ± 0.000	—
LSAM	beta54	0.16	0.160	0.000 ± 0.001	0.003 ± 0.006	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.037 ± 0.002	0.192 ± 0.006	0.161	0.001	0.51	0.0289	0.0289	0.961	94.200 ± 0.739	0.116 ± 0.000	—
LSAM	beta55	0.00	0.003	0.003 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.039 ± 0.001	—	—	0.002	0.002	—	0.0378	0.0379	0.992	94.700 ± 0.708	0.149 ± 0.001	5.300 ± 0.708
LSAM	beta56	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	—	—	0.001	0.001	—	0.0333	0.0333	0.973	94.600 ± 0.715	0.131 ± 0.001	5.400 ± 0.715
LSAM	beta57	0.00	-0.003	-0.003 ± 0.001	—	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.024 ± 0.001	—	—	-0.004	-0.004	—	0.0241	0.0245	0.977	93.700 ± 0.768	0.090 ± 0.000	6.300 ± 0.768
LSAM	beta58	0.00	-0.002	-0.002 ± 0.001	—	0.000 ± 0.000	0.021 ± 0.000	0.000 ± 0.000	0.021 ± 0.001	—	—	-0.003	-0.003	—	0.0202	0.0205	0.979	93.600 ± 0.774	0.079 ± 0.000	6.400 ± 0.774
QML	beta41	0.20	0.200	-0.000 ± 0.001	-0.001 ± 0.004	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	0.019 ± 0.001	0.137 ± 0.004	0.200	0.000	0.16	0.0282	0.0282	1.029	96.500 ± 0.581	0.110 ± 0.000	—
QML	beta42	0.20	0.200	0.000 ± 0.001	0.001 ± 0.005	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.022 ± 0.001	0.148 ± 0.004	0.200	0.000	-0.23	0.0284	0.0284	0.973	94.800 ± 0.702	0.113 ± 0.000	—
QML	beta43	0.20	0.198	-0.002 ± 0.001	-0.008 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.021 ± 0.001	0.144 ± 0.004	0.199	-0.001	-0.66	0.0299	0.0299	0.971	94.800 ± 0.702	0.109 ± 0.000	—
QML	beta44	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	—	—	0.000	0.000	—	0.0348	0.0348	0.971	95.000 ± 0.689	0.125 ± 0.001	5.000 ± 0.689
QML	beta45	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	-0.001	-0.001	—	0.0283	0.0283	0.963	93.600 ± 0.774	0.109 ± 0.001	6.400 ± 0.774
QML	beta46	0.00	-0.002	-0.002 ± 0.001	—	0.000 ± 0.000	0.022 ± 0.000	0.000 ± 0.000	0.022 ± 0.001	—	—	-0.003	-0.003	—	0.0213	0.0214	0.986	93.600 ± 0.774	0.085 ± 0.001	6.400 ± 0.774
QML	beta47	0.00	-0.003	-0.003 ± 0.001	—	0.000 ± 0.000	0.021 ± 0.001	0.000 ± 0.000	0.022 ± 0.001	—	—	-0.003	-0.003	—	0.0205	0.0208	1.007	93.000 ± 0.807	0.084 ± 0.001	7.000 ± 0.807
QML	beta51	0.16	0.161	0.001 ± 0.001	0.009 ± 0.008	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.039 ± 0.001	0.058 ± 0.003	0.241 ± 0.007	0.161	0.001	0.89	0.0382	0.0383	0.969	94.600 ± 0.715	0.146 ± 0.001	—
QML	beta52	0.16	0.161	0.001 ± 0.001	0.007 ± 0.006	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.034 ± 0.002	0.184 ± 0.005	0.161	0.001	0.89	0.0286	0.0287	0.999	94.800 ± 0.702	0.116 ± 0.000	—
QML	beta53	0.16	0.157	-0.003 ± 0.001	-0.018 ± 0.006	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.039 ± 0.002	0.198 ± 0.006	0.157	-0.003	-2.11	0.0295	0.0297	0.961	92.700 ± 0.823	0.119 ± 0.000	—
QML	beta54	0.16	0.160	0.000 ± 0.001	0.001 ± 0.006	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.036 ± 0.002	0.190 ± 0.006	0.161	0.001	0.33	0.0287	0.0287	0.964	94.700 ± 0.708	0.115 ± 0.000	—
QML	beta55	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	0.001	0.001	—	0.0379	0.0379	0.999	94.800 ± 0.702	0.150 ± 0.001	5.200 ± 0.702
QML	beta56	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	—	—	-0.002	-0.002	—	0.0338	0.0338	0.983	94.600 ± 0.715	0.131 ± 0.001	5.400 ± 0.715
QML	beta57	0.00	-0.003	-0.003 ± 0.001	—	0.001 ± 0.000	0.023 ± 0.001	0.001 ± 0.000	0.023 ± 0.001	—	—	-0.005	-0.005	—	0.0230	0.0234	0.984	93.200 ± 0.796	0.089 ± 0.001	6.800 ± 0.796
QML	beta58	0.00	-0.003	-0.003 ± 0.001	—	0.000 ± 0.000	0.020 ± 0.000	0.000 ± 0.000	0.020 ± 0.001	—	—	-0.004	-0.004	—	0.0199	0.0202	0.984	93.000 ± 0.807	0.078 ± 0.000	7.000 ± 0.807
UPI	beta41	0.20	0.201	0.001 ± 0.001	0.004 ± 0.004	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.019 ± 0.001	0.138 ± 0.004	0.202	0.002	0.75	0.0283	0.0284	1.027	96.200 ± 0.605	0.111 ± 0.000	—
UPI	beta42	0.20	0.201	0.001 ± 0.001	0.003 ± 0.005	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.023 ± 0.001	0.150 ± 0.004	0.200	0.000	0.14	0.0289	0.0289	0.971	94.300 ± 0.733	0.114 ± 0.000	—
UPI	beta43	0.20	0.199	-0.001 ± 0.001	-0.003 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.021 ± 0.001	0.145 ± 0.004	0.199	-0.001	-0.36	0.0302	0.0302	0.970	94.900 ± 0.696	0.110 ± 0.000	—
UPI	beta44	0.00	0.002	0.002 ± 0.001	—	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	—	—	0.003	0.003	—	0.0346	0.0347	0.969	94.600 ± 0.715	0.124 ± 0.001	5.400 ± 0.715

Table S276

Simulation 2 - Condition 132: Linear Model, Distr = Normal, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.8 (continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta45	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	0.001	0.001	—	0.0292	0.0293	0.964	94.900 ± 0.696	0.110 ± 0.000	5.100 ± 0.696
UPI	beta46	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.023 ± 0.000	0.001 ± 0.000	0.023 ± 0.001	—	—	-0.003	-0.003	—	0.0217	0.0218	0.989	94.800 ± 0.702	0.087 ± 0.000	5.200 ± 0.702
UPI	beta47	0.00	-0.001	-0.001 ± 0.001	—	0.000 ± 0.000	0.022 ± 0.001	0.000 ± 0.000	0.022 ± 0.001	—	—	-0.002	-0.002	—	0.0217	0.0217	1.008	94.800 ± 0.702	0.087 ± 0.000	5.200 ± 0.702
UPI	beta51	0.16	0.161	0.001 ± 0.001	0.008 ± 0.008	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.058 ± 0.003	0.240 ± 0.007	0.162	0.002	1.11	0.0382	0.0383	0.966	94.300 ± 0.733	0.146 ± 0.001	—
UPI	beta52	0.16	0.161	0.001 ± 0.001	0.009 ± 0.006	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.034 ± 0.002	0.185 ± 0.005	0.162	0.002	1.10	0.0284	0.0285	0.996	95.000 ± 0.689	0.116 ± 0.000	—
UPI	beta53	0.16	0.158	-0.002 ± 0.001	-0.013 ± 0.006	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	0.039 ± 0.002	0.198 ± 0.006	0.158	-0.002	-1.42	0.0301	0.0302	0.963	92.500 ± 0.833	0.119 ± 0.000	—
UPI	beta54	0.16	0.160	0.000 ± 0.001	0.002 ± 0.006	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	0.036 ± 0.002	0.191 ± 0.006	0.161	0.001	0.36	0.0288	0.0288	0.964	94.500 ± 0.721	0.116 ± 0.000	—
UPI	beta55	0.00	0.002	0.002 ± 0.001	—	0.001 ± 0.000	0.039 ± 0.001	0.001 ± 0.000	0.039 ± 0.001	—	—	0.001	0.001	—	0.0390	0.0390	0.991	94.800 ± 0.702	0.150 ± 0.001	5.200 ± 0.702
UPI	beta56	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	—	—	0.001	0.001	—	0.0331	0.0331	0.983	94.500 ± 0.721	0.132 ± 0.001	5.500 ± 0.721
UPI	beta57	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.024 ± 0.001	0.001 ± 0.000	0.024 ± 0.001	—	—	-0.002	-0.002	—	0.0248	0.0249	0.968	93.700 ± 0.768	0.091 ± 0.000	6.300 ± 0.768
UPI	beta58	0.00	-0.001	-0.001 ± 0.001	—	0.000 ± 0.000	0.021 ± 0.000	0.000 ± 0.000	0.021 ± 0.001	—	—	-0.002	-0.002	—	0.0202	0.0203	0.985	94.200 ± 0.739	0.080 ± 0.000	5.800 ± 0.739

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u002014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S277
Simulation 2 - Condition 133: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.4

Method	Param.	True	Mean-based										Median-based					Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE		
LSAM	beta41	0.20	0.246	0.046 ± 0.013	0.230 ± 0.067	0.083 ± 0.007	0.289 ± 0.012	0.085 ± 0.007	0.292 ± 0.014	2.135 ± 0.184	1.461 ± 0.071	0.231	0.031	15.53	0.2608	0.2627	1.251	99.351 ± 0.374	1.416 ± 0.052	—		
LSAM	beta42	0.20	0.262	0.062 ± 0.014	0.310 ± 0.070	0.089 ± 0.007	0.299 ± 0.011	0.093 ± 0.007	0.305 ± 0.013	2.324 ± 0.173	1.524 ± 0.067	0.257	0.057	28.27	0.2854	0.2910	1.225	99.351 ± 0.374	1.435 ± 0.045	—		
LSAM	beta43	0.20	0.173	-0.027 ± 0.004	-0.134 ± 0.021	0.008 ± 0.001	0.088 ± 0.003	0.009 ± 0.001	0.092 ± 0.004	0.213 ± 0.015	0.461 ± 0.019	0.176	-0.024	-12.20	0.0865	0.0899	1.109	95.022 ± 1.012	0.384 ± 0.007	—		
LSAM	beta44	0.00	0.036	0.036 ± 0.006	—	0.014 ± 0.001	0.120 ± 0.006	0.016 ± 0.001	0.125 ± 0.006	—	—	0.029	0.029	—	0.1114	0.1151	1.269	99.351 ± 0.374	0.599 ± 0.024	0.649 ± 0.374		
LSAM	beta45	0.00	0.010	0.010 ± 0.005	—	0.013 ± 0.001	0.116 ± 0.005	0.014 ± 0.001	0.116 ± 0.006	—	—	0.014	0.014	—	0.1007	0.1017	1.107	98.701 ± 0.527	0.503 ± 0.018	1.299 ± 0.527		
LSAM	beta46	0.00	-0.033	-0.033 ± 0.005	—	0.013 ± 0.001	0.115 ± 0.005	0.014 ± 0.001	0.120 ± 0.006	—	—	-0.025	-0.025	—	0.0955	0.0987	1.225	99.567 ± 0.305	0.554 ± 0.023	0.433 ± 0.305		
LSAM	beta47	0.00	-0.033	-0.033 ± 0.005	—	0.012 ± 0.001	0.112 ± 0.005	0.013 ± 0.001	0.116 ± 0.005	—	—	-0.028	-0.028	—	0.0942	0.0984	1.221	98.485 ± 0.568	0.534 ± 0.019	1.515 ± 0.568		
LSAM	beta51	0.16	0.165	0.005 ± 0.005	0.030 ± 0.033	0.013 ± 0.001	0.113 ± 0.005	0.013 ± 0.001	0.113 ± 0.006	0.503 ± 0.043	0.709 ± 0.035	0.153	-0.007	-4.67	0.1149	0.1152	0.917	96.537 ± 0.851	0.408 ± 0.005	—		
LSAM	beta52	0.16	0.225	0.065 ± 0.013	0.405 ± 0.079	0.074 ± 0.007	0.273 ± 0.013	0.078 ± 0.007	0.280 ± 0.015	3.066 ± 0.293	1.751 ± 0.093	0.221	0.061	37.89	0.2541	0.2613	1.097	98.918 ± 0.481	1.173 ± 0.040	—		
LSAM	beta53	0.16	0.132	-0.028 ± 0.005	-0.175 ± 0.034	0.013 ± 0.001	0.116 ± 0.004	0.014 ± 0.001	0.119 ± 0.005	0.557 ± 0.042	0.746 ± 0.032	0.126	-0.034	-21.48	0.1133	0.1184	1.109	96.537 ± 0.851	0.505 ± 0.010	—		
LSAM	beta54	0.16	0.192	0.032 ± 0.012	0.201 ± 0.074	0.064 ± 0.005	0.253 ± 0.011	0.065 ± 0.006	0.255 ± 0.013	2.544 ± 0.220	1.595 ± 0.078	0.197	0.037	23.01	0.2280	0.2310	1.132	98.918 ± 0.481	1.124 ± 0.036	—		
LSAM	beta55	0.00	0.041	0.041 ± 0.007	—	0.025 ± 0.002	0.158 ± 0.007	0.027 ± 0.002	0.163 ± 0.008	—	—	0.031	0.031	—	0.1320	0.1357	1.064	97.186 ± 0.769	0.660 ± 0.019	2.814 ± 0.769		
LSAM	beta56	0.00	0.011	0.011 ± 0.006	—	0.014 ± 0.001	0.119 ± 0.005	0.014 ± 0.001	0.119 ± 0.006	—	—	0.008	0.008	—	0.1117	0.1120	1.067	97.186 ± 0.769	0.497 ± 0.014	2.814 ± 0.769		
LSAM	beta57	0.00	-0.042	-0.042 ± 0.006	—	0.014 ± 0.001	0.120 ± 0.006	0.016 ± 0.002	0.127 ± 0.007	—	—	-0.032	-0.032	—	0.0962	0.1015	1.090	99.351 ± 0.374	0.511 ± 0.021	0.649 ± 0.374		
LSAM	beta58	0.00	-0.022	-0.022 ± 0.005	—	0.010 ± 0.001	0.098 ± 0.005	0.010 ± 0.001	0.100 ± 0.005	—	—	-0.018	-0.018	—	0.0759	0.0781	1.113	99.134 ± 0.431	0.428 ± 0.017	0.866 ± 0.431		
QML	beta41	0.20	0.174	-0.026 ± 0.004	-0.129 ± 0.020	0.007 ± 0.001	0.085 ± 0.003	0.008 ± 0.001	0.089 ± 0.004	0.198 ± 0.015	0.445 ± 0.020	0.179	-0.021	-10.29	0.0867	0.0891	1.067	96.970 ± 0.798	0.357 ± 0.005	—		
QML	beta42	0.20	0.186	-0.014 ± 0.004	-0.070 ± 0.021	0.008 ± 0.001	0.090 ± 0.003	0.008 ± 0.001	0.091 ± 0.004	0.208 ± 0.014	0.456 ± 0.019	0.186	-0.014	-7.10	0.0905	0.0916	1.057	96.753 ± 0.825	0.374 ± 0.008	—		
QML	beta43	0.20	0.201	0.001 ± 0.003	0.005 ± 0.017	0.005 ± 0.000	0.074 ± 0.002	0.005 ± 0.000	0.074 ± 0.003	0.137 ± 0.009	0.370 ± 0.015	0.202	0.002	0.98	0.0762	0.0762	1.019	95.238 ± 0.991	0.296 ± 0.005	—		
QML	beta44	0.00	-0.004	-0.004 ± 0.005	—	0.010 ± 0.001	0.102 ± 0.004	0.010 ± 0.001	0.102 ± 0.004	—	—	0.000	0.000	—	0.1069	0.1069	1.163	97.619 ± 0.709	0.465 ± 0.035	2.381 ± 0.709		
QML	beta45	0.00	0.007	0.007 ± 0.004	—	0.008 ± 0.001	0.090 ± 0.003	0.008 ± 0.001	0.090 ± 0.004	—	—	0.010	0.010	—	0.0856	0.0862	0.986	96.104 ± 0.900	0.349 ± 0.007	3.896 ± 0.900		
QML	beta46	0.00	0.025	0.025 ± 0.003	—	0.003 ± 0.000	0.058 ± 0.002	0.004 ± 0.000	0.063 ± 0.003	—	—	0.014	0.014	—	0.0543	0.0560	1.148	96.753 ± 0.825	0.262 ± 0.023	3.247 ± 0.825		
QML	beta47	0.00	0.030	0.030 ± 0.003	—	0.003 ± 0.000	0.058 ± 0.002	0.004 ± 0.000	0.065 ± 0.003	—	—	0.022	0.022	—	0.0568	0.0609	1.118	94.372 ± 1.072	0.253 ± 0.014	5.628 ± 1.072		
QML	beta51	0.16	0.177	0.017 ± 0.005	0.105 ± 0.032	0.012 ± 0.001	0.111 ± 0.004	0.013 ± 0.001	0.113 ± 0.005	0.495 ± 0.039	0.703 ± 0.032	0.167	0.007	4.20	0.1165	0.1167	1.088	97.403 ± 0.740	0.475 ± 0.040	—		
QML	beta52	0.16	0.143	-0.017 ± 0.004	-0.107 ± 0.027	0.009 ± 0.001	0.092 ± 0.003	0.009 ± 0.001	0.094 ± 0.004	0.345 ± 0.025	0.587 ± 0.025	0.150	-0.010	-6.28	0.0964	0.0969	1.125	96.104 ± 0.900	0.408 ± 0.032	—		
QML	beta53	0.16	0.154	-0.006 ± 0.004	-0.039 ± 0.026	0.008 ± 0.001	0.090 ± 0.003	0.008 ± 0.001	0.090 ± 0.004	0.318 ± 0.021	0.564 ± 0.023	0.154	-0.006	-3.97	0.0956	0.0958	1.051	96.104 ± 0.900	0.371 ± 0.017	—		
QML	beta54	0.16	0.137	-0.023 ± 0.004	-0.144 ± 0.024	0.007 ± 0.000	0.084 ± 0.003	0.008 ± 0.001	0.087 ± 0.004	0.294 ± 0.021	0.542 ± 0.023	0.145	-0.015	-9.20	0.0919	0.0930	1.143	97.186 ± 0.769	0.375 ± 0.013	—		
QML	beta55	0.00	-0.002	-0.002 ± 0.007	—	0.020 ± 0.002	0.140 ± 0.006	0.020 ± 0.002	0.140 ± 0.007	—	—	0.004	0.004	—	0.1272	0.1273	1.077	93.939 ± 1.110	0.590 ± 0.057	6.061 ± 1.110		
QML	beta56	0.00	0.027	0.027 ± 0.005	—	0.010 ± 0.001	0.101 ± 0.004	0.011 ± 0.001	0.105 ± 0.005	—	—	0.018	0.018	—	0.0935	0.0953	1.052	92.424 ± 1.231	0.418 ± 0.030	7.576 ± 1.231		
QML	beta57	0.00	0.019	0.019 ± 0.003	—	0.005 ± 0.000	0.072 ± 0.003	0.006 ± 0.000	0.074 ± 0.004	—	—	0.007	0.007	—	0.0630	0.0634	1.120	96.970 ± 0.798	0.316 ± 0.028	3.030 ± 0.798		
QML	beta58	0.00	0.019	0.019 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.002	0.003 ± 0.000	0.053 ± 0.003	—	—	0.014	0.014	—	0.0481	0.0502	1.043	97.403 ± 0.740	0.202 ± 0.006	2.597 ± 0.740		
UPI	beta41	0.20	0.427	0.227 ± 0.028	1.134 ± 0.138	0.350 ± 0.030	0.591 ± 0.026	0.400 ± 0.035	0.633 ± 0.031	10.008 ± 0.883	3.164 ± 0.154	0.405	0.205	102.68	0.6351	0.6675	1.541	98.918 ± 0.481	3.572 ± 0.205	—		
UPI	beta42	0.20	0.316	0.116 ± 0.025	0.581 ± 0.124	0.284 ± 0.030	0.533 ± 0.029	0.297 ± 0.031	0.545 ± 0.031	7.430 ± 0.779	2.726 ± 0.156	0.266	0.066	32.92	0.5128	0.5170	1.449	99.351 ± 0.374	3.028 ± 0.167	—		
UPI	beta43	0.20	0.144	-0.056 ± 0.006	-0.280 ± 0.030	0.016 ± 0.002	0.127 ± 0.006	0.019 ± 0.002	0.139 ± 0.002	0.480 ± 0.043	0.693 ± 0.034	0.143	-0.057	-28.64	0.1282	0.1405	1.456	95.022 ± 1.012	0.724 ± 0.033	—		
UPI	beta44	0.00	0.088	0.088 ± 0.015	—	0.102 ± 0.010	0.320 ± 0.017	0.110 ± 0.011	0.331 ± 0.019	—	—	0.061	0.061	—	0.2713	0.2781	1.644	100.000 ± 0.000	2.060 ± 0.152	0.000 ± 0.000		

Table S277

Simulation 2 - Condition 133: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.4 (continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta45	0.00	0.034	0.034 ± 0.011	—	0.051 ± 0.005	0.226 ± 0.011	0.052 ± 0.005	0.228 ± 0.012	—	—	0.011	0.011	—	0.1903	0.1907	1.633	100.000 ± 0.000	1.448 ± 0.085	0.000 ± 0.000
UPI	beta46	0.00	-0.110	-0.110 ± 0.012	—	0.068 ± 0.007	0.261 ± 0.014	0.080 ± 0.009	0.283 ± 0.016	—	—	-0.081	-0.081	—	0.2413	0.2544	1.467	99.567 ± 0.305	1.501 ± 0.097	0.433 ± 0.305
UPI	beta47	0.00	-0.059	-0.059 ± 0.010	—	0.043 ± 0.004	0.207 ± 0.011	0.046 ± 0.005	0.215 ± 0.012	—	—	-0.033	-0.033	—	0.1849	0.1878	1.403	99.784 ± 0.216	1.141 ± 0.062	0.216 ± 0.216
UPI	beta51	0.16	0.169	0.009 ± 0.006	0.053 ± 0.038	0.017 ± 0.001	0.130 ± 0.005	0.017 ± 0.001	0.130 ± 0.006	0.664 ± 0.055	0.815 ± 0.039	0.164	0.004	2.43	0.1387	0.1387	1.293	98.268 ± 0.607	0.660 ± 0.023	—
UPI	beta52	0.16	0.247	0.087 ± 0.018	0.547 ± 0.111	0.147 ± 0.014	0.383 ± 0.018	0.154 ± 0.014	0.392 ± 0.020	6.016 ± 0.559	2.453 ± 0.127	0.201	0.041	25.93	0.3653	0.3677	1.645	99.567 ± 0.305	2.469 ± 0.145	—
UPI	beta53	0.16	0.120	-0.040 ± 0.007	-0.252 ± 0.041	0.020 ± 0.002	0.140 ± 0.006	0.021 ± 0.002	0.146 ± 0.007	0.830 ± 0.073	0.911 ± 0.045	0.112	-0.048	-29.70	0.1373	0.1453	1.529	98.268 ± 0.607	0.841 ± 0.040	—
UPI	beta54	0.16	0.235	0.075 ± 0.021	0.470 ± 0.132	0.207 ± 0.020	0.455 ± 0.023	0.212 ± 0.022	0.460 ± 0.026	8.275 ± 0.854	2.877 ± 0.163	0.179	0.019	11.82	0.3905	0.3910	1.508	98.485 ± 0.568	2.688 ± 0.155	—
UPI	beta55	0.00	0.038	0.038 ± 0.013	—	0.073 ± 0.007	0.270 ± 0.013	0.074 ± 0.007	0.272 ± 0.014	—	—	0.026	0.026	—	0.2209	0.2223	1.559	99.351 ± 0.374	1.649 ± 0.097	0.649 ± 0.374
UPI	beta56	0.00	0.015	0.015 ± 0.008	—	0.032 ± 0.003	0.179 ± 0.008	0.032 ± 0.003	0.180 ± 0.009	—	—	0.008	0.008	—	0.1497	0.1499	1.539	99.784 ± 0.216	1.082 ± 0.057	0.216 ± 0.216
UPI	beta57	0.00	-0.045	-0.045 ± 0.008	—	0.028 ± 0.003	0.167 ± 0.008	0.030 ± 0.003	0.173 ± 0.009	—	—	-0.025	-0.025	—	0.1485	0.1505	1.610	99.784 ± 0.216	1.057 ± 0.063	0.216 ± 0.216
UPI	beta58	0.00	-0.035	-0.035 ± 0.008	—	0.028 ± 0.003	0.168 ± 0.009	0.029 ± 0.003	0.171 ± 0.010	—	—	-0.013	-0.013	—	0.1258	0.1265	1.452	99.784 ± 0.216	0.956 ± 0.057	0.216 ± 0.216

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u002014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S278

Simulation 2 - Condition 134: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.4

Method	Param.	True	Mean-based									Median-based					Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE	
LSAM	beta41	0.20	0.278	0.078 ± 0.010	0.391 ± 0.052	0.077 ± 0.005	0.277 ± 0.009	0.083 ± 0.006	0.288 ± 0.011	2.072 ± 0.143	1.440 ± 0.056	0.256	0.056	28.13	0.2484	0.2547	1.218	99.163 ± 0.340	1.324 ± 0.039	—	
LSAM	beta42	0.20	0.276	0.076 ± 0.011	0.381 ± 0.053	0.081 ± 0.005	0.285 ± 0.009	0.087 ± 0.006	0.295 ± 0.011	2.173 ± 0.145	1.474 ± 0.056	0.262	0.062	31.09	0.2570	0.2644	1.251	99.303 ± 0.311	1.397 ± 0.041	—	
LSAM	beta43	0.20	0.173	-0.027 ± 0.002	-0.137 ± 0.011	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.109 ± 0.007	0.330 ± 0.011	0.168	-0.032	-16.07	0.0645	0.0721	1.175	96.932 ± 0.644	0.277 ± 0.004	—	
LSAM	beta44	0.00	0.032	0.032 ± 0.003	—	0.006 ± 0.000	0.076 ± 0.002	0.007 ± 0.000	0.082 ± 0.003	—	—	0.031	0.031	—	0.0749	0.0811	1.190	97.490 ± 0.584	0.353 ± 0.009	2.510 ± 0.584	
LSAM	beta45	0.00	0.009	0.009 ± 0.003	—	0.005 ± 0.000	0.067 ± 0.002	0.005 ± 0.000	0.068 ± 0.003	—	—	0.011	0.011	—	0.0629	0.0638	1.130	97.350 ± 0.600	0.298 ± 0.006	2.650 ± 0.600	
LSAM	beta46	0.00	-0.043	-0.043 ± 0.004	—	0.010 ± 0.001	0.100 ± 0.004	0.012 ± 0.001	0.109 ± 0.004	—	—	-0.033	-0.033	—	0.0790	0.0858	1.184	98.326 ± 0.479	0.464 ± 0.016	1.674 ± 0.479	
LSAM	beta47	0.00	-0.042	-0.042 ± 0.004	—	0.010 ± 0.001	0.099 ± 0.004	0.012 ± 0.001	0.107 ± 0.004	—	—	-0.033	-0.033	—	0.0824	0.0888	1.221	99.303 ± 0.311	0.474 ± 0.016	0.697 ± 0.311	
LSAM	beta51	0.16	0.151	-0.009 ± 0.003	-0.055 ± 0.016	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.196 ± 0.011	0.443 ± 0.015	0.150	-0.010	-5.95	0.0688	0.0694	0.958	94.003 ± 0.887	0.264 ± 0.002	—	
LSAM	beta52	0.16	0.247	0.087 ± 0.008	0.545 ± 0.049	0.045 ± 0.003	0.211 ± 0.007	0.052 ± 0.004	0.228 ± 0.009	2.039 ± 0.139	1.428 ± 0.054	0.234	0.074	46.26	0.1854	0.1996	1.145	98.326 ± 0.479	0.949 ± 0.028	—	
LSAM	beta53	0.16	0.122	-0.038 ± 0.003	-0.238 ± 0.019	0.007 ± 0.000	0.081 ± 0.003	0.008 ± 0.000	0.090 ± 0.003	0.314 ± 0.019	0.560 ± 0.019	0.121	-0.039	-24.37	0.0812	0.0900	1.113	96.932 ± 0.644	0.354 ± 0.005	—	
LSAM	beta54	0.16	0.238	0.078 ± 0.008	0.485 ± 0.047	0.041 ± 0.003	0.202 ± 0.006	0.047 ± 0.003	0.216 ± 0.008	1.822 ± 0.116	1.350 ± 0.049	0.230	0.070	44.03	0.1935	0.2060	1.124	98.326 ± 0.479	0.889 ± 0.028	—	
LSAM	beta55	0.00	0.042	0.042 ± 0.004	—	0.011 ± 0.001	0.103 ± 0.003	0.012 ± 0.001	0.111 ± 0.004	—	—	0.040	0.040	—	0.0958	0.1038	1.024	96.932 ± 0.644	0.412 ± 0.008	3.068 ± 0.644	
LSAM	beta56	0.00	0.011	0.011 ± 0.003	—	0.006 ± 0.000	0.076 ± 0.003	0.006 ± 0.000	0.076 ± 0.003	—	—	0.010	0.010	—	0.0682	0.0690	1.057	98.047 ± 0.517	0.314 ± 0.004	1.953 ± 0.517	
LSAM	beta57	0.00	-0.047	-0.047 ± 0.003	—	0.008 ± 0.001	0.090 ± 0.003	0.010 ± 0.001	0.101 ± 0.004	—	—	-0.036	-0.036	—	0.0699	0.0786	1.105	97.071 ± 0.630	0.388 ± 0.013	2.929 ± 0.630	
LSAM	beta58	0.00	-0.035	-0.035 ± 0.003	—	0.005 ± 0.000	0.072 ± 0.003	0.006 ± 0.000	0.080 ± 0.003	—	—	-0.027	-0.027	—	0.0601	0.0660	1.128	98.466 ± 0.459	0.317 ± 0.013	1.534 ± 0.459	
QML	beta41	0.20	0.181	-0.019 ± 0.002	-0.095 ± 0.010	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	0.085 ± 0.005	0.292 ± 0.009	0.181	-0.019	-9.38	0.0579	0.0609	0.988	93.863 ± 0.896	0.214 ± 0.002	—	
QML	beta42	0.20	0.180	-0.020 ± 0.002	-0.101 ± 0.010	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.056 ± 0.002	0.079 ± 0.004	0.281 ± 0.009	0.185	-0.015	-7.71	0.0584	0.0604	1.068	96.095 ± 0.723	0.219 ± 0.002	—	
QML	beta43	0.20	0.203	0.003 ± 0.002	0.014 ± 0.008	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.051 ± 0.003	0.226 ± 0.007	0.200	0.000	-0.22	0.0477	0.0477	1.007	95.816 ± 0.748	0.179 ± 0.001	—	
QML	beta44	0.00	-0.001	-0.001 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	—	—	-0.003	-0.003	—	0.0630	0.0630	1.196	96.095 ± 0.723	0.283 ± 0.036	3.905 ± 0.723	
QML	beta45	0.00	0.009	0.009 ± 0.002	—	0.003 ± 0.000	0.054 ± 0.002	0.003 ± 0.000	0.055 ± 0.002	—	—	0.011	0.011	—	0.0503	0.0515	1.045	95.676 ± 0.760	0.221 ± 0.013	4.324 ± 0.760	
QML	beta46	0.00	0.020	0.020 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.002 ± 0.000	0.041 ± 0.002	—	—	0.016	0.016	—	0.0333	0.0371	1.009	94.282 ± 0.867	0.143 ± 0.006	5.718 ± 0.867	
QML	beta47	0.00	0.021	0.021 ± 0.001	—	0.001 ± 0.000	0.033 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	0.017	0.017	—	0.0312	0.0355	1.088	94.561 ± 0.847	0.142 ± 0.011	5.439 ± 0.847	
QML	beta51	0.16	0.169	0.009 ± 0.002	0.057 ± 0.015	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	0.155 ± 0.009	0.394 ± 0.013	0.166	0.006	3.92	0.0577	0.0580	1.023	94.840 ± 0.826	0.250 ± 0.002	—	
QML	beta52	0.16	0.143	-0.017 ± 0.002	-0.103 ± 0.012	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	0.114 ± 0.006	0.338 ± 0.011	0.148	-0.012	-7.70	0.0531	0.0545	1.077	95.119 ± 0.805	0.217 ± 0.001	—	
QML	beta53	0.16	0.153	-0.007 ± 0.002	-0.046 ± 0.012	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.002	0.102 ± 0.005	0.319 ± 0.010	0.152	-0.008	-4.82	0.0510	0.0516	1.059	95.537 ± 0.771	0.210 ± 0.003	—	
QML	beta54	0.16	0.142	-0.018 ± 0.002	-0.111 ± 0.013	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.128 ± 0.007	0.358 ± 0.012	0.148	-0.012	-7.78	0.0548	0.0562	1.013	95.258 ± 0.794	0.216 ± 0.001	—	
QML	beta55	0.00	0.004	0.004 ± 0.003	—	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	—	—	0.005	0.005	—	0.0786	0.0787	1.026	95.258 ± 0.794	0.312 ± 0.005	4.742 ± 0.794	
QML	beta56	0.00	0.025	0.025 ± 0.002	—	0.003 ± 0.000	0.055 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	—	—	0.026	0.026	—	0.0535	0.0593	1.088	95.258 ± 0.794	0.233 ± 0.006	4.742 ± 0.794	
QML	beta57	0.00	0.013	0.013 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	0.011	0.011	—	0.0372	0.0389	1.054	96.095 ± 0.723	0.162 ± 0.003	3.905 ± 0.723	
QML	beta58	0.00	0.016	0.016 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	—	—	0.012	0.012	—	0.0281	0.0307	1.012	94.142 ± 0.877	0.115 ± 0.002	5.858 ± 0.877	
UPI	beta41	0.20	0.414	0.214 ± 0.019	1.069 ± 0.094	0.253 ± 0.020	0.503 ± 0.020	0.298 ± 0.023	0.546 ± 0.023	7.449 ± 0.573	2.729 ± 0.114	0.377	0.177	88.71	0.4578	0.4909	1.202	97.071 ± 0.630	2.367 ± 0.096	—	
UPI	beta42	0.20	0.311	0.111 ± 0.014	0.554 ± 0.070	0.141 ± 0.011	0.375 ± 0.014	0.153 ± 0.011	0.391 ± 0.016	3.816 ± 0.276	1.954 ± 0.079	0.291	0.091	45.44	0.3477	0.3594	1.244	95.955 ± 0.736	1.828 ± 0.073	—	
UPI	beta43	0.20	0.144	-0.056 ± 0.003	-0.280 ± 0.013	0.005 ± 0.000	0.069 ± 0.002	0.008 ± 0.000	0.089 ± 0.003	0.199 ± 0.012	0.446 ± 0.014	0.142	-0.058	-29.24	0.0698	0.0911	1.260	90.377 ± 1.101	0.343 ± 0.011	—	
UPI	beta44	0.00	0.058	0.058 ± 0.005	—	0.016 ± 0.001	0.125 ± 0.005	0.019 ± 0.001	0.138 ± 0.005	—	—	0.047	0.047	—	0.1099	0.1194	1.443	99.582 ± 0.241	0.708 ± 0.026	0.418 ± 0.241	

Table S278

Simulation 2 - Condition 134: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.4 (continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta45	0.00	0.020	0.020 ± 0.004	—	0.010 ± 0.001	0.100 ± 0.004	0.010 ± 0.001	0.102 ± 0.004	—	—	0.013	0.013	—	0.0840	0.0851	1.426	99.163 ± 0.340	0.561 ± 0.020	0.837 ± 0.340
UPI	beta46	0.00	-0.090	-0.090 ± 0.007	—	0.031 ± 0.002	0.176 ± 0.006	0.039 ± 0.003	0.198 ± 0.008	—	—	-0.070	-0.070	—	0.1513	0.1669	1.209	98.745 ± 0.416	0.833 ± 0.033	1.255 ± 0.416
UPI	beta47	0.00	-0.054	-0.054 ± 0.004	—	0.014 ± 0.001	0.117 ± 0.004	0.017 ± 0.001	0.129 ± 0.005	—	—	-0.048	-0.048	—	0.1070	0.1172	1.272	97.350 ± 0.600	0.583 ± 0.021	2.650 ± 0.600
UPI	beta51	0.16	0.157	-0.003 ± 0.003	-0.017 ± 0.018	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.003	0.226 ± 0.014	0.475 ± 0.018	0.155	-0.005	-3.41	0.0724	0.0726	1.172	96.513 ± 0.685	0.349 ± 0.008	—
UPI	beta52	0.16	0.256	0.096 ± 0.014	0.603 ± 0.087	0.139 ± 0.010	0.373 ± 0.014	0.148 ± 0.011	0.385 ± 0.016	5.790 ± 0.432	2.406 ± 0.100	0.222	0.062	39.00	0.3214	0.3274	1.355	97.629 ± 0.568	1.980 ± 0.092	—
UPI	beta53	0.16	0.109	-0.051 ± 0.003	-0.317 ± 0.020	0.008 ± 0.000	0.088 ± 0.003	0.010 ± 0.001	0.101 ± 0.003	0.399 ± 0.024	0.632 ± 0.021	0.107	-0.053	-33.11	0.0840	0.0993	1.285	94.142 ± 0.877	0.441 ± 0.012	—
UPI	beta54	0.16	0.286	0.126 ± 0.015	0.788 ± 0.092	0.154 ± 0.010	0.393 ± 0.012	0.170 ± 0.011	0.412 ± 0.015	6.637 ± 0.442	2.576 ± 0.096	0.215	0.055	34.61	0.3441	0.3485	1.312	97.490 ± 0.584	2.020 ± 0.081	—
UPI	beta55	0.00	0.047	0.047 ± 0.005	—	0.018 ± 0.001	0.133 ± 0.004	0.020 ± 0.001	0.141 ± 0.005	—	—	0.033	0.033	—	0.1202	0.1247	1.491	98.884 ± 0.392	0.779 ± 0.030	1.116 ± 0.392
UPI	beta56	0.00	0.012	0.012 ± 0.003	—	0.008 ± 0.001	0.089 ± 0.003	0.008 ± 0.001	0.090 ± 0.003	—	—	0.011	0.011	—	0.0800	0.0807	1.326	99.442 ± 0.278	0.462 ± 0.014	0.558 ± 0.278
UPI	beta57	0.00	-0.047	-0.047 ± 0.005	—	0.019 ± 0.001	0.136 ± 0.005	0.021 ± 0.002	0.144 ± 0.006	—	—	-0.029	-0.029	—	0.1089	0.1127	1.382	99.024 ± 0.367	0.738 ± 0.034	0.976 ± 0.367
UPI	beta58	0.00	-0.047	-0.047 ± 0.005	—	0.015 ± 0.001	0.123 ± 0.004	0.017 ± 0.001	0.131 ± 0.005	—	—	-0.025	-0.025	—	0.1045	0.1075	1.308	99.024 ± 0.367	0.630 ± 0.025	0.976 ± 0.367

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S279

Simulation 2 - Condition 135: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.6

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.214	0.014 ± 0.004	0.071 ± 0.021	0.014 ± 0.001	0.118 ± 0.003	0.014 ± 0.001	0.119 ± 0.004	0.355 ± 0.019	0.596 ± 0.019	0.210	0.010	4.89	0.1257	0.1261	1.026	97.121 ± 0.592	0.476 ± 0.005	—
LSAM	beta42	0.20	0.220	0.020 ± 0.005	0.098 ± 0.023	0.017 ± 0.001	0.129 ± 0.003	0.017 ± 0.001	0.130 ± 0.004	0.424 ± 0.021	0.651 ± 0.020	0.220	0.020	9.76	0.1322	0.1337	0.999	95.620 ± 0.724	0.504 ± 0.005	—
LSAM	beta43	0.20	0.196	-0.004 ± 0.002	-0.022 ± 0.010	0.003 ± 0.000	0.058 ± 0.002	0.003 ± 0.000	0.058 ± 0.002	0.083 ± 0.005	0.289 ± 0.009	0.194	-0.006	-3.12	0.0547	0.0551	1.017	95.995 ± 0.694	0.230 ± 0.001	—
LSAM	beta44	0.00	0.011	0.011 ± 0.003	—	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.073 ± 0.003	—	—	0.012	0.012	—	0.0658	0.0669	0.958	94.368 ± 0.816	0.272 ± 0.003	5.632 ± 0.816
LSAM	beta45	0.00	0.007	0.007 ± 0.003	—	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.072 ± 0.003	—	—	0.002	0.002	—	0.0671	0.0671	0.920	95.119 ± 0.762	0.257 ± 0.003	4.881 ± 0.762
LSAM	beta46	0.00	-0.009	-0.009 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.001	0.003 ± 0.000	0.050 ± 0.002	—	—	-0.007	-0.007	—	0.0460	0.0466	0.922	94.743 ± 0.790	0.178 ± 0.003	5.257 ± 0.790
LSAM	beta47	0.00	-0.012	-0.012 ± 0.002	—	0.002 ± 0.000	0.050 ± 0.002	0.003 ± 0.000	0.051 ± 0.002	—	—	-0.010	-0.010	—	0.0431	0.0442	0.919	94.118 ± 0.832	0.179 ± 0.003	5.882 ± 0.832
LSAM	beta51	0.16	0.160	-0.000 ± 0.003	-0.002 ± 0.017	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	0.237 ± 0.012	0.487 ± 0.015	0.159	-0.001	-0.82	0.0778	0.0778	0.931	93.742 ± 0.857	0.285 ± 0.002	—
LSAM	beta52	0.16	0.177	0.017 ± 0.004	0.106 ± 0.026	0.014 ± 0.001	0.119 ± 0.003	0.014 ± 0.001	0.120 ± 0.004	0.563 ± 0.029	0.750 ± 0.023	0.174	0.014	9.02	0.1231	0.1240	0.990	95.620 ± 0.724	0.461 ± 0.004	—
LSAM	beta53	0.16	0.151	-0.009 ± 0.003	-0.054 ± 0.018	0.006 ± 0.000	0.080 ± 0.002	0.006 ± 0.000	0.080 ± 0.003	0.253 ± 0.013	0.503 ± 0.016	0.148	-0.012	-7.70	0.0821	0.0830	0.994	94.493 ± 0.807	0.312 ± 0.002	—
LSAM	beta54	0.16	0.173	0.013 ± 0.004	0.079 ± 0.026	0.014 ± 0.001	0.118 ± 0.003	0.014 ± 0.001	0.118 ± 0.004	0.546 ± 0.031	0.739 ± 0.025	0.175	0.015	9.54	0.1153	0.1163	0.995	95.244 ± 0.753	0.459 ± 0.004	—
LSAM	beta55	0.00	0.013	0.013 ± 0.003	—	0.009 ± 0.001	0.094 ± 0.003	0.009 ± 0.001	0.095 ± 0.003	—	—	0.014	0.014	—	0.0881	0.0892	0.964	94.368 ± 0.816	0.355 ± 0.005	5.632 ± 0.816
LSAM	beta56	0.00	0.000	-0.000 ± 0.003	—	0.007 ± 0.000	0.083 ± 0.002	0.007 ± 0.000	0.083 ± 0.003	—	—	-0.002	-0.002	—	0.0759	0.0759	0.942	94.743 ± 0.790	0.305 ± 0.004	5.257 ± 0.790
LSAM	beta57	0.00	-0.013	-0.013 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.002	0.003 ± 0.000	0.054 ± 0.002	—	—	-0.013	-0.013	—	0.0495	0.0511	0.932	94.869 ± 0.781	0.192 ± 0.003	5.131 ± 0.781
LSAM	beta58	0.00	-0.007	-0.007 ± 0.002	—	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.045 ± 0.002	—	—	-0.005	-0.005	—	0.0379	0.0383	0.949	94.368 ± 0.816	0.164 ± 0.003	5.632 ± 0.816
QML	beta41	0.20	0.183	-0.017 ± 0.002	-0.083 ± 0.011	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.110 ± 0.006	0.332 ± 0.010	0.183	-0.017	-8.25	0.0671	0.0691	1.076	96.746 ± 0.628	0.271 ± 0.002	—
QML	beta42	0.20	0.185	-0.015 ± 0.002	-0.075 ± 0.012	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.129 ± 0.006	0.359 ± 0.011	0.185	-0.015	-7.47	0.0740	0.0754	1.010	95.119 ± 0.762	0.278 ± 0.002	—
QML	beta43	0.20	0.203	0.003 ± 0.002	0.015 ± 0.009	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	0.068 ± 0.004	0.261 ± 0.009	0.202	0.002	1.12	0.0517	0.0518	1.031	95.870 ± 0.704	0.211 ± 0.001	—
QML	beta44	0.00	-0.004	-0.004 ± 0.002	—	0.004 ± 0.000	0.065 ± 0.002	0.004 ± 0.000	0.065 ± 0.002	—	—	-0.001	-0.001	—	0.0642	0.0643	0.989	95.745 ± 0.714	0.252 ± 0.004	4.255 ± 0.714
QML	beta45	0.00	0.005	0.005 ± 0.002	—	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	—	—	0.001	0.001	—	0.0596	0.0596	0.935	94.493 ± 0.807	0.227 ± 0.003	5.507 ± 0.807
QML	beta46	0.00	0.016	0.016 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.002 ± 0.000	0.042 ± 0.001	—	—	0.013	0.013	—	0.0380	0.0402	0.927	93.116 ± 0.896	0.139 ± 0.003	6.884 ± 0.896
QML	beta47	0.00	0.014	0.014 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	0.010	0.010	—	0.0332	0.0347	0.925	92.491 ± 0.932	0.134 ± 0.003	7.509 ± 0.932
QML	beta51	0.16	0.162	0.002 ± 0.003	0.015 ± 0.017	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	0.222 ± 0.011	0.471 ± 0.015	0.161	0.001	0.82	0.0733	0.0734	0.988	95.494 ± 0.734	0.292 ± 0.002	—
QML	beta52	0.16	0.147	-0.013 ± 0.003	-0.080 ± 0.016	0.005 ± 0.000	0.072 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.207 ± 0.012	0.454 ± 0.015	0.147	-0.013	-7.84	0.0717	0.0728	0.994	95.494 ± 0.734	0.279 ± 0.002	—
QML	beta53	0.16	0.155	-0.005 ± 0.002	-0.029 ± 0.014	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.002	0.169 ± 0.008	0.411 ± 0.013	0.154	-0.006	-3.76	0.0651	0.0654	0.990	93.867 ± 0.849	0.254 ± 0.002	—
QML	beta54	0.16	0.145	-0.015 ± 0.002	-0.094 ± 0.016	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.201 ± 0.010	0.449 ± 0.014	0.150	-0.010	-6.17	0.0699	0.0706	1.010	94.743 ± 0.790	0.278 ± 0.001	—
QML	beta55	0.00	-0.004	-0.004 ± 0.003	—	0.007 ± 0.000	0.082 ± 0.003	0.007 ± 0.000	0.082 ± 0.003	—	—	-0.003	-0.003	—	0.0793	0.0794	1.005	95.244 ± 0.753	0.324 ± 0.005	4.756 ± 0.753
QML	beta56	0.00	0.013	0.013 ± 0.002	—	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.070 ± 0.002	—	—	0.008	0.008	—	0.0642	0.0646	0.936	93.617 ± 0.865	0.253 ± 0.003	6.383 ± 0.865
QML	beta57	0.00	0.010	0.010 ± 0.001	—	0.002 ± 0.000	0.042 ± 0.001	0.002 ± 0.000	0.043 ± 0.002	—	—	0.005	0.005	—	0.0383	0.0387	0.975	93.867 ± 0.849	0.160 ± 0.004	6.133 ± 0.849
QML	beta58	0.00	0.011	0.011 ± 0.001	—	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	0.009	0.009	—	0.0304	0.0315	0.936	93.992 ± 0.841	0.124 ± 0.002	6.008 ± 0.841
UPI	beta41	0.20	0.224	0.024 ± 0.005	0.120 ± 0.024	0.018 ± 0.001	0.135 ± 0.004	0.019 ± 0.001	0.137 ± 0.005	0.470 ± 0.030	0.685 ± 0.025	0.219	0.019	9.72	0.1333	0.1347	1.244	98.999 ± 0.352	0.658 ± 0.021	—
UPI	beta42	0.20	0.233	0.033 ± 0.005	0.165 ± 0.026	0.021 ± 0.001	0.146 ± 0.005	0.022 ± 0.002	0.150 ± 0.006	0.562 ± 0.039	0.750 ± 0.029	0.229	0.029	14.38	0.1464	0.1492	1.234	98.999 ± 0.352	0.708 ± 0.023	—
UPI	beta43	0.20	0.190	-0.010 ± 0.002	-0.049 ± 0.010	0.003 ± 0.000	0.058 ± 0.002	0.003 ± 0.000	0.059 ± 0.002	0.087 ± 0.005	0.295 ± 0.009	0.186	-0.014	-6.90	0.0586	0.0602	1.066	95.620 ± 0.724	0.243 ± 0.003	—
UPI	beta44	0.00	0.013	0.013 ± 0.003	—	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.076 ± 0.003	—	—	0.012	0.012	—	0.0749	0.0758	1.222	97.372 ± 0.566	0.358 ± 0.009	2.628 ± 0.566

Table S279

Simulation 2 - Condition 135: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.6 (continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta45	0.00	0.007	0.007 ± 0.003	—	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.003	—	—	0.010	0.010	—	0.0702	0.0709	1.140	97.747 ± 0.525	0.337 ± 0.010	2.253 ± 0.525
UPI	beta46	0.00	-0.014	-0.014 ± 0.002	—	0.003 ± 0.000	0.057 ± 0.002	0.003 ± 0.000	0.059 ± 0.002	—	—	-0.010	-0.010	—	0.0488	0.0498	1.155	97.622 ± 0.539	0.259 ± 0.011	2.378 ± 0.539
UPI	beta47	0.00	-0.019	-0.019 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.002	0.004 ± 0.000	0.061 ± 0.003	—	—	-0.014	-0.014	—	0.0492	0.0513	1.162	97.872 ± 0.511	0.265 ± 0.010	2.128 ± 0.511
UPI	beta51	0.16	0.159	-0.001 ± 0.003	-0.008 ± 0.017	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	0.224 ± 0.012	0.473 ± 0.015	0.156	-0.004	-2.21	0.0731	0.0732	0.974	94.869 ± 0.781	0.289 ± 0.002	—
UPI	beta52	0.16	0.177	0.017 ± 0.004	0.107 ± 0.026	0.014 ± 0.001	0.119 ± 0.003	0.014 ± 0.001	0.120 ± 0.004	0.564 ± 0.032	0.751 ± 0.025	0.178	0.018	11.55	0.1233	0.1247	1.056	95.995 ± 0.694	0.493 ± 0.007	—
UPI	beta53	0.16	0.149	-0.011 ± 0.003	-0.072 ± 0.017	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	0.233 ± 0.012	0.483 ± 0.015	0.145	-0.015	-9.56	0.0758	0.0773	1.034	94.368 ± 0.816	0.310 ± 0.003	—
UPI	beta54	0.16	0.181	0.021 ± 0.004	0.130 ± 0.028	0.016 ± 0.001	0.126 ± 0.003	0.016 ± 0.001	0.127 ± 0.004	0.633 ± 0.035	0.795 ± 0.026	0.176	0.016	10.11	0.1218	0.1229	1.078	96.746 ± 0.628	0.531 ± 0.009	—
UPI	beta55	0.00	0.009	0.009 ± 0.003	—	0.009 ± 0.001	0.092 ± 0.003	0.009 ± 0.001	0.093 ± 0.003	—	—	0.008	0.008	—	0.0863	0.0867	1.096	96.496 ± 0.651	0.396 ± 0.008	3.504 ± 0.651
UPI	beta56	0.00	0.001	0.001 ± 0.003	—	0.006 ± 0.000	0.080 ± 0.002	0.006 ± 0.000	0.080 ± 0.003	—	—	0.002	0.002	—	0.0729	0.0730	1.017	96.370 ± 0.662	0.319 ± 0.005	3.630 ± 0.662
UPI	beta57	0.00	-0.012	-0.012 ± 0.002	—	0.003 ± 0.000	0.054 ± 0.002	0.003 ± 0.000	0.055 ± 0.002	—	—	-0.011	-0.011	—	0.0487	0.0499	1.016	96.621 ± 0.639	0.214 ± 0.005	3.379 ± 0.639
UPI	beta58	0.00	-0.010	-0.010 ± 0.002	—	0.002 ± 0.000	0.048 ± 0.002	0.002 ± 0.000	0.049 ± 0.002	—	—	-0.009	-0.009	—	0.0385	0.0395	1.041	96.496 ± 0.651	0.195 ± 0.005	3.504 ± 0.651

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S280

Simulation 2 - Condition 136: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.6

			Mean-based										Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE	
LSAM	beta41	0.20	0.208	0.008 ± 0.002	0.042 ± 0.011	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	0.127 ± 0.006	0.357 ± 0.010	0.208	0.008	4.22	0.0716	0.0721	0.993	95.272 ± 0.680	0.276 ± 0.001	—	
LSAM	beta42	0.20	0.219	0.019 ± 0.002	0.095 ± 0.012	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	0.152 ± 0.007	0.390 ± 0.011	0.215	0.015	7.60	0.0752	0.0767	0.992	94.656 ± 0.721	0.295 ± 0.002	—	
LSAM	beta43	0.20	0.195	-0.005 ± 0.001	-0.025 ± 0.006	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.035 ± 0.002	0.188 ± 0.005	0.193	-0.007	-3.33	0.0381	0.0387	0.969	93.525 ± 0.789	0.142 ± 0.001	—	
LSAM	beta44	0.00	0.007	0.007 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	0.006	0.006	—	0.0372	0.0377	0.959	93.422 ± 0.795	0.150 ± 0.001	6.578 ± 0.795	
LSAM	beta45	0.00	0.003	0.003 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	0.000	0.000	—	0.0362	0.0362	0.966	95.478 ± 0.666	0.144 ± 0.001	4.522 ± 0.666	
LSAM	beta46	0.00	-0.007	-0.007 ± 0.001	—	0.001 ± 0.000	0.025 ± 0.001	0.001 ± 0.000	0.026 ± 0.001	—	—	-0.007	-0.007	—	0.0236	0.0245	0.959	93.422 ± 0.795	0.094 ± 0.001	6.578 ± 0.795	
LSAM	beta47	0.00	-0.010	-0.010 ± 0.001	—	0.001 ± 0.000	0.026 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	-0.008	-0.008	—	0.0242	0.0256	0.959	92.909 ± 0.823	0.097 ± 0.001	7.091 ± 0.823	
LSAM	beta51	0.16	0.157	-0.003 ± 0.001	-0.019 ± 0.009	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.079 ± 0.004	0.280 ± 0.008	0.156	-0.004	-2.48	0.0445	0.0447	1.005	94.758 ± 0.714	0.176 ± 0.001	—	
LSAM	beta52	0.16	0.177	0.017 ± 0.002	0.104 ± 0.014	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.201 ± 0.009	0.448 ± 0.012	0.178	0.018	11.07	0.0713	0.0734	0.987	95.067 ± 0.694	0.270 ± 0.001	—	
LSAM	beta53	0.16	0.152	-0.008 ± 0.002	-0.052 ± 0.009	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.090 ± 0.004	0.299 ± 0.008	0.154	-0.006	-4.01	0.0469	0.0473	1.029	94.964 ± 0.701	0.190 ± 0.001	—	
LSAM	beta54	0.16	0.176	0.016 ± 0.002	0.103 ± 0.014	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.072 ± 0.002	0.202 ± 0.009	0.450 ± 0.012	0.180	0.020	12.27	0.0703	0.0730	0.984	94.347 ± 0.740	0.270 ± 0.001	—	
LSAM	beta55	0.00	0.011	0.011 ± 0.002	—	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	—	—	0.010	0.010	—	0.0503	0.0513	0.995	94.758 ± 0.714	0.201 ± 0.001	5.242 ± 0.714	
LSAM	beta56	0.00	0.003	0.003 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	0.002	0.002	—	0.0414	0.0414	0.970	94.142 ± 0.753	0.174 ± 0.001	5.858 ± 0.753	
LSAM	beta57	0.00	-0.010	-0.010 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	-0.009	-0.009	—	0.0253	0.0270	0.959	93.628 ± 0.783	0.104 ± 0.001	6.372 ± 0.783	
LSAM	beta58	0.00	-0.007	-0.007 ± 0.001	—	0.001 ± 0.000	0.024 ± 0.001	0.001 ± 0.000	0.025 ± 0.001	—	—	-0.008	-0.008	—	0.0224	0.0238	0.962	93.011 ± 0.817	0.090 ± 0.001	6.989 ± 0.817	
QML	beta41	0.20	0.182	-0.018 ± 0.001	-0.090 ± 0.007	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.054 ± 0.002	0.231 ± 0.006	0.183	-0.017	-8.44	0.0451	0.0481	1.014	93.834 ± 0.771	0.169 ± 0.002	—	
QML	beta42	0.20	0.188	-0.012 ± 0.001	-0.061 ± 0.007	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.052 ± 0.002	0.228 ± 0.006	0.187	-0.013	-6.70	0.0461	0.0480	0.998	94.553 ± 0.728	0.172 ± 0.001	—	
QML	beta43	0.20	0.203	0.003 ± 0.001	0.013 ± 0.006	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.030 ± 0.001	0.173 ± 0.005	0.201	0.001	0.71	0.0355	0.0355	0.986	94.245 ± 0.747	0.134 ± 0.001	—	
QML	beta44	0.00	-0.002	-0.002 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	-0.002	-0.002	—	0.0377	0.0377	0.981	94.553 ± 0.728	0.154 ± 0.002	5.447 ± 0.728	
QML	beta45	0.00	0.004	0.004 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	0.002	0.002	—	0.0350	0.0350	0.993	95.170 ± 0.687	0.139 ± 0.001	4.830 ± 0.687	
QML	beta46	0.00	0.013	0.013 ± 0.001	—	0.000 ± 0.000	0.021 ± 0.001	0.001 ± 0.000	0.025 ± 0.001	—	—	0.011	0.011	—	0.0210	0.0236	0.976	93.525 ± 0.789	0.080 ± 0.001	6.475 ± 0.789	
QML	beta47	0.00	0.012	0.012 ± 0.001	—	0.000 ± 0.000	0.021 ± 0.001	0.001 ± 0.000	0.024 ± 0.001	—	—	0.011	0.011	—	0.0205	0.0234	0.974	93.320 ± 0.800	0.079 ± 0.001	6.680 ± 0.800	
QML	beta51	0.16	0.159	-0.001 ± 0.001	-0.003 ± 0.009	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.076 ± 0.003	0.275 ± 0.008	0.159	-0.001	-0.65	0.0445	0.0445	1.054	95.375 ± 0.673	0.182 ± 0.004	—	
QML	beta52	0.16	0.149	-0.011 ± 0.001	-0.066 ± 0.009	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.081 ± 0.004	0.284 ± 0.008	0.151	-0.009	-5.89	0.0446	0.0455	1.003	94.861 ± 0.708	0.174 ± 0.002	—	
QML	beta53	0.16	0.156	-0.004 ± 0.001	-0.023 ± 0.008	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.062 ± 0.003	0.248 ± 0.007	0.157	-0.003	-1.67	0.0403	0.0404	1.028	95.375 ± 0.673	0.159 ± 0.001	—	
QML	beta54	0.16	0.150	-0.010 ± 0.001	-0.062 ± 0.009	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.083 ± 0.004	0.288 ± 0.008	0.152	-0.008	-5.02	0.0462	0.0469	0.996	94.039 ± 0.759	0.176 ± 0.002	—	
QML	beta55	0.00	-0.001	-0.001 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.002	—	—	0.001	0.001	—	0.0477	0.0477	1.012	95.992 ± 0.629	0.196 ± 0.002	4.008 ± 0.629	
QML	beta56	0.00	0.014	0.014 ± 0.001	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	—	—	0.012	0.012	—	0.0385	0.0403	0.964	93.011 ± 0.817	0.155 ± 0.001	6.989 ± 0.817	
QML	beta57	0.00	0.009	0.009 ± 0.001	—	0.001 ± 0.000	0.025 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	—	—	0.007	0.007	—	0.0228	0.0239	0.961	94.142 ± 0.753	0.095 ± 0.001	5.858 ± 0.753	
QML	beta58	0.00	0.009	0.009 ± 0.001	—	0.000 ± 0.000	0.020 ± 0.001	0.000 ± 0.000	0.022 ± 0.001	—	—	0.007	0.007	—	0.0184	0.0198	0.970	93.525 ± 0.789	0.077 ± 0.002	6.475 ± 0.789	
UPI	beta41	0.20	0.214	0.014 ± 0.002	0.068 ± 0.012	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	0.148 ± 0.007	0.385 ± 0.011	0.212	0.012	5.92	0.0740	0.0749	1.046	97.020 ± 0.545	0.311 ± 0.003	—	
UPI	beta42	0.20	0.231	0.031 ± 0.003	0.157 ± 0.013	0.006 ± 0.000	0.080 ± 0.002	0.007 ± 0.000	0.086 ± 0.002	0.183 ± 0.009	0.428 ± 0.012	0.226	0.026	12.78	0.0838	0.0876	1.085	96.197 ± 0.613	0.339 ± 0.004	—	
UPI	beta43	0.20	0.190	-0.010 ± 0.001	-0.051 ± 0.006	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.036 ± 0.002	0.189 ± 0.005	0.189	-0.011	-5.51	0.0366	0.0383	0.977	93.422 ± 0.795	0.139 ± 0.001	—	
UPI	beta44	0.00	0.008	0.008 ± 0.001	—	0.002 ± 0.000	0.043 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	—	—	0.006	0.006	—	0.0384	0.0389	1.002	95.683 ± 0.652	0.167 ± 0.002	4.317 ± 0.652	

Table S280

Simulation 2 - Condition 136: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.6 (continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta45	0.00	0.004	0.004 ± 0.001	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	—	—	0.000	0.000	—	0.0366	0.0366	1.015	96.300 ± 0.605	0.156 ± 0.002	3.700 ± 0.605
UPI	beta46	0.00	-0.010	-0.010 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	-0.008	-0.008	—	0.0240	0.0254	0.995	95.375 ± 0.673	0.111 ± 0.002	4.625 ± 0.673
UPI	beta47	0.00	-0.015	-0.015 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	—	—	-0.012	-0.012	—	0.0269	0.0296	1.053	93.834 ± 0.771	0.120 ± 0.002	6.166 ± 0.771
UPI	beta51	0.16	0.157	-0.003 ± 0.001	-0.016 ± 0.009	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.001	0.076 ± 0.003	0.276 ± 0.008	0.157	-0.003	-2.05	0.0450	0.0452	1.024	94.964 ± 0.701	0.177 ± 0.001	—
UPI	beta52	0.16	0.179	0.019 ± 0.002	0.119 ± 0.015	0.005 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	0.228 ± 0.011	0.478 ± 0.013	0.179	0.019	12.10	0.0726	0.0751	0.992	95.478 ± 0.666	0.288 ± 0.002	—
UPI	beta53	0.16	0.148	-0.012 ± 0.001	-0.074 ± 0.009	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.088 ± 0.004	0.297 ± 0.008	0.149	-0.011	-6.81	0.0462	0.0475	1.034	93.011 ± 0.817	0.187 ± 0.001	—
UPI	beta54	0.16	0.184	0.024 ± 0.002	0.147 ± 0.015	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.080 ± 0.002	0.248 ± 0.012	0.498 ± 0.014	0.184	0.024	14.77	0.0752	0.0788	1.009	94.656 ± 0.721	0.302 ± 0.003	—
UPI	beta55	0.00	0.011	0.011 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	—	—	0.010	0.010	—	0.0488	0.0499	1.020	95.581 ± 0.659	0.210 ± 0.002	4.419 ± 0.659
UPI	beta56	0.00	0.003	0.003 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	0.003	0.003	—	0.0427	0.0429	0.996	94.861 ± 0.708	0.178 ± 0.002	5.139 ± 0.708
UPI	beta57	0.00	-0.011	-0.011 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	—	—	-0.011	-0.011	—	0.0269	0.0290	0.977	94.553 ± 0.728	0.113 ± 0.001	5.447 ± 0.728
UPI	beta58	0.00	-0.010	-0.010 ± 0.001	—	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	-0.010	-0.010	—	0.0241	0.0259	0.976	93.936 ± 0.765	0.102 ± 0.002	6.064 ± 0.765

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S281

Simulation 2 - Condition 137: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.8

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.199	-0.001 ± 0.002	-0.007 ± 0.012	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.002	0.142 ± 0.006	0.377 ± 0.010	0.197	-0.003	-1.55	0.0746	0.0747	1.006	95.030 ± 0.692	0.298 ± 0.002	—
LSAM	beta42	0.20	0.206	0.006 ± 0.003	0.028 ± 0.013	0.006 ± 0.000	0.080 ± 0.002	0.006 ± 0.000	0.080 ± 0.002	0.159 ± 0.007	0.399 ± 0.011	0.205	0.005	2.53	0.0800	0.0802	1.002	96.045 ± 0.621	0.313 ± 0.002	—
LSAM	beta43	0.20	0.198	-0.002 ± 0.002	-0.008 ± 0.008	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.056 ± 0.003	0.237 ± 0.007	0.194	-0.006	-3.18	0.0459	0.0463	0.969	94.625 ± 0.718	0.180 ± 0.001	—
LSAM	beta44	0.00	0.001	0.001 ± 0.002	—	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.002	—	—	0.000	0.000	—	0.0452	0.0452	0.957	94.828 ± 0.705	0.189 ± 0.002	5.172 ± 0.705
LSAM	beta45	0.00	0.000	0.000 ± 0.002	—	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	-0.002	-0.002	—	0.0450	0.0451	0.981	93.813 ± 0.767	0.183 ± 0.002	6.187 ± 0.767
LSAM	beta46	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	-0.002	-0.002	—	0.0285	0.0286	0.917	94.219 ± 0.743	0.109 ± 0.001	5.781 ± 0.743
LSAM	beta47	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	-0.003	-0.003	—	0.0276	0.0278	0.901	92.596 ± 0.834	0.110 ± 0.001	7.404 ± 0.834
LSAM	beta51	0.16	0.164	0.004 ± 0.002	0.026 ± 0.012	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.150 ± 0.007	0.388 ± 0.011	0.161	0.001	0.87	0.0618	0.0618	0.938	94.118 ± 0.749	0.228 ± 0.001	—
LSAM	beta52	0.16	0.158	-0.002 ± 0.002	-0.011 ± 0.015	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	0.226 ± 0.011	0.475 ± 0.014	0.157	-0.003	-1.72	0.0758	0.0758	1.004	95.740 ± 0.643	0.299 ± 0.001	—
LSAM	beta53	0.16	0.160	-0.000 ± 0.002	-0.001 ± 0.012	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.149 ± 0.007	0.386 ± 0.011	0.157	-0.003	-2.14	0.0597	0.0598	0.946	93.205 ± 0.801	0.229 ± 0.001	—
LSAM	beta54	0.16	0.166	0.006 ± 0.002	0.036 ± 0.015	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.077 ± 0.002	0.229 ± 0.011	0.479 ± 0.014	0.161	0.001	0.39	0.0733	0.0733	0.991	94.422 ± 0.731	0.297 ± 0.001	—
LSAM	beta55	0.00	0.005	0.005 ± 0.002	—	0.005 ± 0.000	0.070 ± 0.002	0.005 ± 0.000	0.071 ± 0.002	—	—	0.005	0.005	—	0.0690	0.0692	0.905	93.408 ± 0.790	0.250 ± 0.002	6.592 ± 0.790
LSAM	beta56	0.00	-0.002	-0.002 ± 0.002	—	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	—	—	-0.003	-0.003	—	0.0530	0.0530	0.900	93.002 ± 0.812	0.212 ± 0.002	6.998 ± 0.812
LSAM	beta57	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.032 ± 0.001	—	—	-0.003	-0.003	—	0.0289	0.0290	0.970	94.219 ± 0.743	0.120 ± 0.001	5.781 ± 0.743
LSAM	beta58	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	-0.003	-0.003	—	0.0257	0.0259	0.870	91.886 ± 0.870	0.101 ± 0.001	8.114 ± 0.870
QML	beta41	0.20	0.187	-0.013 ± 0.002	-0.064 ± 0.010	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	0.096 ± 0.004	0.310 ± 0.008	0.189	-0.011	-5.75	0.0614	0.0624	1.013	95.233 ± 0.679	0.241 ± 0.001	—
QML	beta42	0.20	0.193	-0.007 ± 0.002	-0.036 ± 0.010	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.099 ± 0.004	0.315 ± 0.008	0.193	-0.007	-3.60	0.0623	0.0627	1.012	95.740 ± 0.643	0.248 ± 0.001	—
QML	beta43	0.20	0.201	0.001 ± 0.001	0.006 ± 0.007	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.053 ± 0.002	0.230 ± 0.006	0.197	-0.003	-1.62	0.0444	0.0446	0.973	94.422 ± 0.731	0.175 ± 0.001	—
QML	beta44	0.00	-0.003	-0.003 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	—	—	-0.004	-0.004	—	0.0434	0.0436	0.972	94.828 ± 0.705	0.187 ± 0.002	5.172 ± 0.705
QML	beta45	0.00	0.000	-0.000 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	-0.003	-0.003	—	0.0434	0.0435	0.984	93.915 ± 0.761	0.176 ± 0.002	6.085 ± 0.761
QML	beta46	0.00	0.008	0.008 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	0.005	0.005	—	0.0263	0.0267	0.903	93.408 ± 0.790	0.101 ± 0.001	6.592 ± 0.790
QML	beta47	0.00	0.007	0.007 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	0.004	0.004	—	0.0251	0.0254	0.895	93.103 ± 0.807	0.100 ± 0.001	6.897 ± 0.807
QML	beta51	0.16	0.165	0.005 ± 0.002	0.029 ± 0.012	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.147 ± 0.007	0.383 ± 0.011	0.162	0.002	1.28	0.0605	0.0605	0.970	94.523 ± 0.725	0.233 ± 0.001	—
QML	beta52	0.16	0.149	-0.011 ± 0.002	-0.069 ± 0.012	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.158 ± 0.008	0.398 ± 0.011	0.149	-0.011	-6.90	0.0632	0.0642	1.005	95.538 ± 0.658	0.247 ± 0.001	—
QML	beta53	0.16	0.161	0.001 ± 0.002	0.006 ± 0.011	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	0.126 ± 0.006	0.355 ± 0.010	0.157	-0.003	-1.69	0.0544	0.0545	0.950	93.611 ± 0.779	0.212 ± 0.001	—
QML	beta54	0.16	0.155	-0.005 ± 0.002	-0.032 ± 0.012	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.153 ± 0.007	0.391 ± 0.011	0.152	-0.008	-4.86	0.0607	0.0612	1.006	95.132 ± 0.685	0.246 ± 0.001	—
QML	beta55	0.00	0.001	0.001 ± 0.002	—	0.004 ± 0.000	0.067 ± 0.002	0.004 ± 0.000	0.067 ± 0.002	—	—	0.001	0.001	—	0.0666	0.0666	0.933	94.118 ± 0.749	0.244 ± 0.003	5.882 ± 0.749
QML	beta56	0.00	0.002	0.002 ± 0.002	—	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	—	—	0.001	0.001	—	0.0497	0.0497	0.917	93.103 ± 0.807	0.199 ± 0.002	6.897 ± 0.807
QML	beta57	0.00	0.005	0.005 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	0.004	0.004	—	0.0280	0.0282	0.981	94.219 ± 0.743	0.114 ± 0.001	5.781 ± 0.743
QML	beta58	0.00	0.004	0.004 ± 0.001	—	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	0.002	0.002	—	0.0238	0.0238	0.866	92.089 ± 0.860	0.092 ± 0.001	7.911 ± 0.860
UPI	beta41	0.20	0.199	-0.001 ± 0.002	-0.003 ± 0.012	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.002	0.132 ± 0.006	0.364 ± 0.010	0.197	-0.003	-1.55	0.0705	0.0706	1.034	96.247 ± 0.605	0.295 ± 0.002	—
UPI	beta42	0.20	0.206	0.006 ± 0.002	0.029 ± 0.012	0.006 ± 0.000	0.078 ± 0.002	0.006 ± 0.000	0.078 ± 0.002	0.152 ± 0.007	0.390 ± 0.011	0.206	0.006	3.23	0.0752	0.0755	1.024	95.943 ± 0.628	0.312 ± 0.002	—
UPI	beta43	0.20	0.198	-0.002 ± 0.002	-0.008 ± 0.008	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.056 ± 0.003	0.236 ± 0.007	0.194	-0.006	-2.78	0.0449	0.0452	0.964	93.915 ± 0.761	0.178 ± 0.001	—
UPI	beta44	0.00	0.000	-0.000 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.052 ± 0.002	—	—	-0.002	-0.002	—	0.0474	0.0475	0.965	95.436 ± 0.665	0.196 ± 0.002	4.564 ± 0.665

Table S281

Simulation 2 - Condition 137: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.8 (continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta45	0.00	-0.001	-0.001 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.002	—	—	-0.003	-0.003	—	0.0464	0.0465	0.976	94.523 ± 0.725	0.188 ± 0.002	5.477 ± 0.725
UPI	beta46	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	-0.003	-0.003	—	0.0279	0.0280	0.921	94.320 ± 0.737	0.110 ± 0.001	5.680 ± 0.737
UPI	beta47	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	-0.003	-0.003	—	0.0271	0.0272	0.914	94.422 ± 0.731	0.111 ± 0.001	5.578 ± 0.731
UPI	beta51	0.16	0.164	0.004 ± 0.002	0.026 ± 0.012	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.147 ± 0.007	0.383 ± 0.011	0.162	0.002	1.37	0.0614	0.0614	0.961	94.219 ± 0.743	0.230 ± 0.001	—
UPI	beta52	0.16	0.158	-0.002 ± 0.002	-0.014 ± 0.015	0.006 ± 0.000	0.074 ± 0.002	0.006 ± 0.000	0.074 ± 0.002	0.216 ± 0.010	0.465 ± 0.013	0.158	-0.002	-1.43	0.0745	0.0745	1.007	95.943 ± 0.628	0.294 ± 0.001	—
UPI	beta53	0.16	0.160	0.000 ± 0.002	0.001 ± 0.012	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	0.145 ± 0.007	0.381 ± 0.011	0.156	-0.004	-2.26	0.0587	0.0588	0.944	93.306 ± 0.796	0.226 ± 0.001	—
UPI	beta54	0.16	0.165	0.005 ± 0.002	0.033 ± 0.015	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.076 ± 0.002	0.223 ± 0.011	0.472 ± 0.013	0.162	0.002	1.11	0.0723	0.0723	0.998	95.030 ± 0.692	0.295 ± 0.001	—
UPI	beta55	0.00	0.005	0.005 ± 0.002	—	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.002	—	—	0.003	0.003	—	0.0675	0.0676	0.934	94.422 ± 0.731	0.252 ± 0.002	5.578 ± 0.731
UPI	beta56	0.00	-0.003	-0.003 ± 0.002	—	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	—	—	-0.003	-0.003	—	0.0544	0.0545	0.912	93.408 ± 0.790	0.211 ± 0.002	6.592 ± 0.790
UPI	beta57	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	-0.004	-0.004	—	0.0274	0.0277	0.971	94.625 ± 0.718	0.118 ± 0.001	5.375 ± 0.718
UPI	beta58	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	-0.003	-0.003	—	0.0249	0.0251	0.881	92.394 ± 0.844	0.101 ± 0.001	7.606 ± 0.844

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u002014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S282

Simulation 2 - Condition 138: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, $N = 1000$, Reliability = 0.8

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.204	0.004 ± 0.001	0.018 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.056 ± 0.003	0.237 ± 0.007	0.205	0.005	2.68	0.0474	0.0477	0.984	95.491 ± 0.657	0.182 ± 0.001	—
LSAM	beta42	0.20	0.202	0.002 ± 0.002	0.012 ± 0.008	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.060 ± 0.003	0.245 ± 0.007	0.203	0.003	1.43	0.0510	0.0511	0.985	95.591 ± 0.650	0.189 ± 0.001	—
LSAM	beta43	0.20	0.200	-0.000 ± 0.001	-0.001 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.021 ± 0.001	0.145 ± 0.004	0.199	-0.001	-0.69	0.0292	0.0293	0.992	95.992 ± 0.621	0.113 ± 0.000	—
LSAM	beta44	0.00	0.002	0.002 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	0.000	0.000	—	0.0283	0.0283	0.980	93.988 ± 0.752	0.111 ± 0.001	6.012 ± 0.752
LSAM	beta45	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	0.001	0.001	—	0.0264	0.0264	0.982	93.687 ± 0.770	0.109 ± 0.001	6.313 ± 0.770
LSAM	beta46	0.00	-0.002	-0.002 ± 0.001	—	0.000 ± 0.000	0.017 ± 0.000	0.000 ± 0.000	0.017 ± 0.000	—	—	-0.002	-0.002	—	0.0160	0.0162	0.927	92.786 ± 0.819	0.062 ± 0.001	7.214 ± 0.819
LSAM	beta47	0.00	-0.002	-0.002 ± 0.001	—	0.000 ± 0.000	0.017 ± 0.000	0.000 ± 0.000	0.018 ± 0.001	—	—	-0.002	-0.002	—	0.0161	0.0163	0.904	92.685 ± 0.824	0.062 ± 0.001	7.315 ± 0.824
LSAM	beta51	0.16	0.159	-0.001 ± 0.001	-0.008 ± 0.008	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.060 ± 0.003	0.244 ± 0.006	0.158	-0.002	-1.31	0.0400	0.0401	0.942	93.487 ± 0.781	0.144 ± 0.001	—
LSAM	beta52	0.16	0.164	0.004 ± 0.001	0.023 ± 0.009	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.081 ± 0.004	0.285 ± 0.008	0.163	0.003	2.09	0.0453	0.0454	1.033	95.591 ± 0.650	0.184 ± 0.001	—
LSAM	beta53	0.16	0.157	-0.003 ± 0.001	-0.016 ± 0.008	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.058 ± 0.003	0.240 ± 0.007	0.157	-0.003	-1.79	0.0400	0.0401	0.959	93.587 ± 0.775	0.144 ± 0.001	—
LSAM	beta54	0.16	0.165	0.005 ± 0.001	0.034 ± 0.009	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.088 ± 0.004	0.296 ± 0.008	0.165	0.005	3.39	0.0478	0.0481	0.995	94.489 ± 0.722	0.184 ± 0.001	—
LSAM	beta55	0.00	0.003	0.003 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	0.004	0.004	—	0.0410	0.0412	0.936	93.888 ± 0.758	0.149 ± 0.001	6.112 ± 0.758
LSAM	beta56	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	0.001	0.001	—	0.0336	0.0336	0.951	94.088 ± 0.747	0.130 ± 0.001	5.912 ± 0.747
LSAM	beta57	0.00	-0.002	-0.002 ± 0.001	—	0.000 ± 0.000	0.019 ± 0.000	0.000 ± 0.000	0.019 ± 0.001	—	—	-0.003	-0.003	—	0.0175	0.0178	0.941	93.587 ± 0.775	0.070 ± 0.001	6.413 ± 0.775
LSAM	beta58	0.00	-0.002	-0.002 ± 0.001	—	0.000 ± 0.000	0.017 ± 0.000	0.000 ± 0.000	0.017 ± 0.001	—	—	-0.002	-0.002	—	0.0157	0.0158	0.911	92.685 ± 0.824	0.061 ± 0.001	7.315 ± 0.824
QML	beta41	0.20	0.192	-0.008 ± 0.001	-0.038 ± 0.006	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.039 ± 0.002	0.198 ± 0.005	0.193	-0.007	-3.30	0.0395	0.0401	0.983	95.090 ± 0.684	0.150 ± 0.000	—
QML	beta42	0.20	0.192	-0.008 ± 0.001	-0.041 ± 0.006	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	0.041 ± 0.002	0.202 ± 0.006	0.192	-0.008	-4.08	0.0404	0.0413	0.995	95.591 ± 0.650	0.155 ± 0.001	—
QML	beta43	0.20	0.203	0.003 ± 0.001	0.013 ± 0.004	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.020 ± 0.001	0.142 ± 0.004	0.202	0.002	0.98	0.0284	0.0285	0.996	95.291 ± 0.671	0.111 ± 0.000	—
QML	beta44	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	-0.002	-0.002	—	0.0289	0.0289	0.990	94.289 ± 0.735	0.114 ± 0.001	5.711 ± 0.735
QML	beta45	0.00	0.002	0.002 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	0.001	0.001	—	0.0262	0.0262	0.993	93.687 ± 0.770	0.109 ± 0.001	6.313 ± 0.770
QML	beta46	0.00	0.006	0.006 ± 0.001	—	0.000 ± 0.000	0.016 ± 0.000	0.000 ± 0.000	0.017 ± 0.001	—	—	0.005	0.005	—	0.0156	0.0163	0.924	92.786 ± 0.819	0.060 ± 0.001	7.214 ± 0.819
QML	beta47	0.00	0.006	0.006 ± 0.001	—	0.000 ± 0.000	0.017 ± 0.000	0.000 ± 0.000	0.018 ± 0.001	—	—	0.004	0.004	—	0.0158	0.0163	0.896	91.784 ± 0.869	0.059 ± 0.001	8.216 ± 0.869
QML	beta51	0.16	0.159	-0.001 ± 0.001	-0.005 ± 0.008	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.059 ± 0.002	0.243 ± 0.006	0.158	-0.002	-1.15	0.0399	0.0399	0.955	93.888 ± 0.758	0.146 ± 0.001	—
QML	beta52	0.16	0.154	-0.006 ± 0.001	-0.037 ± 0.007	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.057 ± 0.002	0.239 ± 0.006	0.154	-0.006	-3.45	0.0377	0.0381	1.039	95.491 ± 0.657	0.154 ± 0.000	—
QML	beta53	0.16	0.159	-0.001 ± 0.001	-0.008 ± 0.007	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.049 ± 0.002	0.221 ± 0.006	0.158	-0.002	-1.42	0.0363	0.0364	0.965	94.088 ± 0.747	0.133 ± 0.000	—
QML	beta54	0.16	0.156	-0.004 ± 0.001	-0.027 ± 0.008	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.061 ± 0.003	0.246 ± 0.007	0.156	-0.004	-2.33	0.0393	0.0395	1.005	94.689 ± 0.710	0.154 ± 0.000	—
QML	beta55	0.00	0.000	-0.000 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	0.002	0.002	—	0.0391	0.0392	0.943	94.389 ± 0.728	0.147 ± 0.001	5.611 ± 0.728
QML	beta56	0.00	0.006	0.006 ± 0.001	—	0.001 ± 0.000	0.033 ± 0.001	0.001 ± 0.000	0.033 ± 0.001	—	—	0.004	0.004	—	0.0318	0.0321	0.963	94.489 ± 0.722	0.124 ± 0.001	5.511 ± 0.722
QML	beta57	0.00	0.004	0.004 ± 0.001	—	0.000 ± 0.000	0.018 ± 0.000	0.000 ± 0.000	0.019 ± 0.001	—	—	0.003	0.003	—	0.0169	0.0171	0.939	93.788 ± 0.764	0.068 ± 0.001	6.212 ± 0.764
QML	beta58	0.00	0.004	0.004 ± 0.001	—	0.000 ± 0.000	0.016 ± 0.000	0.000 ± 0.000	0.017 ± 0.001	—	—	0.003	0.003	—	0.0145	0.0148	0.904	92.585 ± 0.829	0.058 ± 0.001	7.415 ± 0.829
UPI	beta41	0.20	0.203	0.003 ± 0.001	0.017 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.054 ± 0.003	0.233 ± 0.007	0.205	0.005	2.33	0.0455	0.0457	0.983	94.990 ± 0.691	0.179 ± 0.001	—
UPI	beta42	0.20	0.203	0.003 ± 0.002	0.013 ± 0.008	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.057 ± 0.003	0.240 ± 0.007	0.204	0.004	1.77	0.0482	0.0484	0.994	95.792 ± 0.636	0.187 ± 0.001	—
UPI	beta43	0.20	0.200	-0.000 ± 0.001	-0.002 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.021 ± 0.001	0.144 ± 0.004	0.198	-0.002	-0.84	0.0290	0.0291	0.995	95.892 ± 0.628	0.112 ± 0.000	—
UPI	beta44	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	0.000	0.000	—	0.0284	0.0284	0.975	94.088 ± 0.747	0.113 ± 0.001	5.912 ± 0.747

Table S282

Simulation 2 - Condition 138: Linear Model, Distr = Right-skewed, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.8 (continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta45	0.00	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	-0.001	-0.001	—	0.0271	0.0271	0.983	94.188 ± 0.741	0.111 ± 0.001	5.812 ± 0.741
UPI	beta46	0.00	-0.001	-0.001 ± 0.001	—	0.000 ± 0.000	0.017 ± 0.000	0.000 ± 0.000	0.017 ± 0.001	—	—	-0.002	-0.002	—	0.0162	0.0163	0.921	93.687 ± 0.770	0.062 ± 0.001	6.313 ± 0.770
UPI	beta47	0.00	-0.001	-0.001 ± 0.001	—	0.000 ± 0.000	0.017 ± 0.000	0.000 ± 0.000	0.018 ± 0.001	—	—	-0.002	-0.002	—	0.0156	0.0157	0.902	92.585 ± 0.829	0.062 ± 0.001	7.415 ± 0.829
UPI	beta51	0.16	0.159	-0.001 ± 0.001	-0.008 ± 0.008	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.039 ± 0.001	0.059 ± 0.002	0.243 ± 0.006	0.158	-0.002	-1.22	0.0401	0.0401	0.953	94.088 ± 0.747	0.145 ± 0.000	—
UPI	beta52	0.16	0.163	0.003 ± 0.001	0.018 ± 0.009	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.079 ± 0.003	0.281 ± 0.008	0.163	0.003	1.65	0.0446	0.0447	1.038	95.792 ± 0.636	0.183 ± 0.001	—
UPI	beta53	0.16	0.158	-0.002 ± 0.001	-0.015 ± 0.008	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.057 ± 0.003	0.240 ± 0.007	0.158	-0.002	-1.38	0.0393	0.0393	0.954	93.387 ± 0.787	0.143 ± 0.001	—
UPI	beta54	0.16	0.165	0.005 ± 0.001	0.032 ± 0.009	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.087 ± 0.004	0.295 ± 0.008	0.165	0.005	3.20	0.0471	0.0474	0.996	95.190 ± 0.677	0.183 ± 0.001	—
UPI	beta55	0.00	0.002	0.002 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	0.004	0.004	—	0.0386	0.0387	0.943	93.888 ± 0.758	0.149 ± 0.001	6.112 ± 0.758
UPI	beta56	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	0.001	0.001	—	0.0324	0.0324	0.960	94.289 ± 0.735	0.130 ± 0.001	5.711 ± 0.735
UPI	beta57	0.00	-0.002	-0.002 ± 0.001	—	0.000 ± 0.000	0.019 ± 0.000	0.000 ± 0.000	0.019 ± 0.001	—	—	-0.003	-0.003	—	0.0176	0.0178	0.943	93.587 ± 0.775	0.069 ± 0.001	6.413 ± 0.775
UPI	beta58	0.00	-0.001	-0.001 ± 0.001	—	0.000 ± 0.000	0.017 ± 0.000	0.000 ± 0.000	0.017 ± 0.001	—	—	-0.002	-0.002	—	0.0153	0.0154	0.910	92.685 ± 0.824	0.061 ± 0.001	7.315 ± 0.824

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S283

Simulation 2 - Condition 139: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.4

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.297	0.097 ± 0.036	0.483 ± 0.178	0.016 ± 0.004	0.128 ± 0.024	0.025 ± 0.007	0.157 ± 0.029	0.613 ± 0.178	0.783 ± 0.143	0.250	0.050	24.90	0.1592	0.1668	1.424	100.000 ± 0.000	0.716 ± 0.193	—
LSAM	beta42	0.20	0.226	0.026 ± 0.032	0.131 ± 0.160	0.013 ± 0.005	0.115 ± 0.030	0.013 ± 0.006	0.114 ± 0.035	0.324 ± 0.144	0.569 ± 0.174	0.230	0.030	14.82	0.0991	0.1034	1.319	100.000 ± 0.000	0.596 ± 0.127	—
LSAM	beta43	0.20	0.175	-0.025 ± 0.024	-0.123 ± 0.122	0.008 ± 0.003	0.088 ± 0.020	0.008 ± 0.002	0.088 ± 0.020	0.194 ± 0.057	0.440 ± 0.098	0.174	-0.026	-12.92	0.1128	0.1158	0.901	92.308 ± 7.391	0.311 ± 0.016	—
LSAM	beta44	0.00	0.088	0.088 ± 0.037	—	0.018 ± 0.006	0.133 ± 0.032	0.024 ± 0.009	0.155 ± 0.037	—	—	0.034	0.034	—	0.1103	0.1155	1.215	92.308 ± 7.391	0.634 ± 0.135	7.692 ± 7.391
LSAM	beta45	0.00	0.044	0.044 ± 0.032	—	0.013 ± 0.004	0.115 ± 0.022	0.014 ± 0.004	0.119 ± 0.025	—	—	0.021	0.021	—	0.1102	0.1122	1.061	92.308 ± 7.391	0.477 ± 0.043	7.692 ± 7.391
LSAM	beta46	0.00	-0.167	-0.167 ± 0.056	—	0.041 ± 0.013	0.203 ± 0.045	0.066 ± 0.025	0.257 ± 0.058	—	—	-0.068	-0.068	—	0.1173	0.1356	1.193	100.000 ± 0.000	0.951 ± 0.260	0.000 ± 0.000
LSAM	beta47	0.00	-0.057	-0.057 ± 0.032	—	0.013 ± 0.005	0.116 ± 0.032	0.016 ± 0.006	0.125 ± 0.033	—	—	-0.062	-0.062	—	0.1015	0.1187	1.186	100.000 ± 0.000	0.538 ± 0.094	0.000 ± 0.000
LSAM	beta51	0.16	0.090	-0.070 ± 0.038	-0.440 ± 0.238	0.019 ± 0.005	0.137 ± 0.024	0.022 ± 0.009	0.150 ± 0.036	0.874 ± 0.337	0.935 ± 0.227	0.108	-0.052	-32.65	0.1016	0.1143	0.845	92.308 ± 7.391	0.455 ± 0.032	—
LSAM	beta52	0.16	0.244	0.084 ± 0.039	0.528 ± 0.243	0.020 ± 0.008	0.140 ± 0.042	0.025 ± 0.009	0.159 ± 0.037	0.985 ± 0.363	0.992 ± 0.234	0.215	0.055	34.13	0.1064	0.1195	0.996	92.308 ± 7.391	0.546 ± 0.059	—
LSAM	beta53	0.16	0.123	-0.037 ± 0.018	-0.234 ± 0.115	0.004 ± 0.001	0.066 ± 0.014	0.005 ± 0.002	0.074 ± 0.020	0.213 ± 0.093	0.462 ± 0.128	0.144	-0.016	-10.30	0.0702	0.0721	1.509	100.000 ± 0.000	0.392 ± 0.063	—
LSAM	beta54	0.16	0.212	0.052 ± 0.038	0.324 ± 0.235	0.018 ± 0.002	0.136 ± 0.014	0.020 ± 0.005	0.140 ± 0.026	0.770 ± 0.177	0.878 ± 0.161	0.244	0.084	52.19	0.1912	0.2086	0.821	92.308 ± 7.391	0.437 ± 0.036	—
LSAM	beta55	0.00	0.058	0.058 ± 0.041	—	0.022 ± 0.006	0.149 ± 0.027	0.024 ± 0.009	0.155 ± 0.039	—	—	0.054	0.054	—	0.1660	0.1747	1.291	100.000 ± 0.000	0.755 ± 0.078	0.000 ± 0.000
LSAM	beta56	0.00	0.009	0.009 ± 0.039	—	0.020 ± 0.007	0.141 ± 0.032	0.018 ± 0.007	0.136 ± 0.036	—	—	0.034	0.034	—	0.1393	0.1433	1.053	84.615 ± 10.007	0.582 ± 0.068	15.385 ± 10.007
LSAM	beta57	0.00	-0.133	-0.133 ± 0.052	—	0.035 ± 0.011	0.186 ± 0.041	0.050 ± 0.017	0.223 ± 0.047	—	—	-0.079	-0.079	—	0.1246	0.1473	1.088	100.000 ± 0.000	0.793 ± 0.195	0.000 ± 0.000
LSAM	beta58	0.00	-0.052	-0.052 ± 0.029	—	0.011 ± 0.003	0.103 ± 0.022	0.013 ± 0.003	0.112 ± 0.020	—	—	-0.074	-0.074	—	0.0723	0.1032	1.287	100.000 ± 0.000	0.521 ± 0.068	0.000 ± 0.000
QML	beta41	0.20	0.229	0.029 ± 0.021	0.144 ± 0.103	0.006 ± 0.002	0.074 ± 0.017	0.006 ± 0.002	0.077 ± 0.019	0.148 ± 0.051	0.384 ± 0.093	0.216	0.016	7.95	0.0790	0.0806	1.043	100.000 ± 0.000	0.303 ± 0.013	—
QML	beta42	0.20	0.217	0.017 ± 0.015	0.084 ± 0.074	0.003 ± 0.001	0.054 ± 0.019	0.003 ± 0.001	0.054 ± 0.014	0.073 ± 0.027	0.271 ± 0.071	0.225	0.025	12.40	0.0574	0.0625	1.326	100.000 ± 0.000	0.278 ± 0.013	—
QML	beta43	0.20	0.193	-0.007 ± 0.024	-0.033 ± 0.120	0.008 ± 0.003	0.087 ± 0.021	0.007 ± 0.002	0.083 ± 0.022	0.174 ± 0.060	0.417 ± 0.109	0.184	-0.016	-8.03	0.0874	0.0888	0.809	84.615 ± 10.007	0.275 ± 0.012	—
QML	beta44	0.00	0.034	0.034 ± 0.030	—	0.012 ± 0.005	0.110 ± 0.035	0.012 ± 0.006	0.111 ± 0.036	—	—	-0.006	-0.006	—	0.0890	0.0892	1.045	92.308 ± 7.391	0.449 ± 0.068	7.692 ± 7.391
QML	beta45	0.00	0.022	0.022 ± 0.033	—	0.014 ± 0.004	0.117 ± 0.025	0.013 ± 0.005	0.115 ± 0.030	—	—	0.007	0.007	—	0.1138	0.1140	0.757	84.615 ± 10.007	0.349 ± 0.028	15.385 ± 10.007
QML	beta46	0.00	-0.068	-0.068 ± 0.021	—	0.006 ± 0.001	0.075 ± 0.009	0.010 ± 0.003	0.099 ± 0.017	—	—	-0.041	-0.041	—	0.0774	0.0875	1.149	69.231 ± 12.801	0.336 ± 0.047	30.769 ± 12.801
QML	beta47	0.00	-0.038	-0.038 ± 0.015	—	0.003 ± 0.001	0.055 ± 0.012	0.004 ± 0.001	0.065 ± 0.012	—	—	-0.028	-0.028	—	0.0469	0.0545	1.233	100.000 ± 0.000	0.265 ± 0.032	0.000 ± 0.000
QML	beta51	0.16	0.121	-0.039 ± 0.028	-0.241 ± 0.175	0.010 ± 0.003	0.101 ± 0.022	0.011 ± 0.005	0.105 ± 0.031	0.427 ± 0.196	0.654 ± 0.193	0.163	0.003	1.57	0.0929	0.0929	1.158	84.615 ± 10.007	0.459 ± 0.045	—
QML	beta52	0.16	0.200	0.040 ± 0.023	0.249 ± 0.145	0.007 ± 0.002	0.083 ± 0.018	0.008 ± 0.003	0.090 ± 0.023	0.314 ± 0.124	0.560 ± 0.144	0.168	0.008	4.92	0.0807	0.0811	1.087	92.308 ± 7.391	0.356 ± 0.036	—
QML	beta53	0.16	0.157	-0.003 ± 0.022	-0.020 ± 0.138	0.006 ± 0.002	0.079 ± 0.016	0.006 ± 0.002	0.076 ± 0.018	0.228 ± 0.072	0.478 ± 0.115	0.150	-0.010	-6.44	0.0589	0.0598	1.056	100.000 ± 0.000	0.329 ± 0.026	—
QML	beta54	0.16	0.170	0.010 ± 0.026	0.063 ± 0.162	0.009 ± 0.002	0.093 ± 0.013	0.008 ± 0.002	0.090 ± 0.019	0.317 ± 0.076	0.563 ± 0.117	0.156	-0.004	-2.35	0.1146	0.1146	0.824	100.000 ± 0.000	0.301 ± 0.021	—
QML	beta55	0.00	0.028	0.028 ± 0.034	—	0.015 ± 0.005	0.123 ± 0.026	0.015 ± 0.005	0.122 ± 0.031	—	—	-0.016	-0.016	—	0.1173	0.1183	1.293	92.308 ± 7.391	0.623 ± 0.082	7.692 ± 7.391
QML	beta56	0.00	-0.008	-0.008 ± 0.034	—	0.015 ± 0.006	0.121 ± 0.035	0.014 ± 0.006	0.117 ± 0.035	—	—	-0.013	-0.013	—	0.0906	0.0916	1.005	100.000 ± 0.000	0.477 ± 0.056	0.000 ± 0.000
QML	beta57	0.00	-0.034	-0.034 ± 0.025	—	0.008 ± 0.004	0.091 ± 0.035	0.009 ± 0.003	0.093 ± 0.024	—	—	-0.028	-0.028	—	0.0781	0.0828	0.951	92.308 ± 7.391	0.338 ± 0.053	7.692 ± 7.391
QML	beta58	0.00	-0.009	-0.009 ± 0.016	—	0.003 ± 0.001	0.056 ± 0.020	0.003 ± 0.001	0.055 ± 0.017	—	—	-0.031	-0.031	—	0.0541	0.0623	1.186	100.000 ± 0.000	0.262 ± 0.038	0.000 ± 0.000
UPI	beta41	0.20	0.327	0.127 ± 0.167	0.634 ± 0.836	0.364 ± 0.187	0.603 ± 0.234	0.352 ± 0.218	0.593 ± 0.256	8.797 ± 5.445	2.966 ± 1.282	0.310	0.110	55.14	0.8308	0.8381	0.716	100.000 ± 0.000	1.694 ± 0.365	—
UPI	beta42	0.20	0.151	-0.049 ± 0.056	-0.244 ± 0.278	0.040 ± 0.013	0.201 ± 0.045	0.039 ± 0.016	0.199 ± 0.056	0.987 ± 0.392	0.994 ± 0.279	0.192	-0.008	-3.85	0.2840	0.2841	2.491	76.923 ± 11.685	1.958 ± 0.836	—
UPI	beta43	0.20	0.185	-0.015 ± 0.039	-0.073 ± 0.194	0.019 ± 0.006	0.140 ± 0.027	0.018 ± 0.006	0.135 ± 0.032	0.455 ± 0.138	0.674 ± 0.158	0.154	-0.046	-23.18	0.1300	0.1381	1.975	100.000 ± 0.000	1.081 ± 0.407	—
UPI	beta44	0.00	0.144	0.144 ± 0.070	—	0.063 ± 0.015	0.251 ± 0.041	0.079 ± 0.030	0.280 ± 0.066	—	—	0.051	0.051	—	0.2608	0.2657	1.481	100.000 ± 0.000	1.455 ± 0.332	0.000 ± 0.000
UPI	beta45	0.00	-0.142	-0.142 ± 0.068	—	0.060 ± 0.022	0.245 ± 0.061	0.076 ± 0.038	0.275 ± 0.085	—	—	-0.043	-0.043	—	0.1785	0.1837	1.373	100.000 ± 0.000	1.321 ± 0.329	0.000 ± 0.000
UPI	beta46	0.00	-0.310	-0.310 ± 0.240	—	0.747 ± 0.414	0.864 ± 0.378	0.785 ± 0.541	0.886 ± 0.431	—	—	-0.266	-0.266	—	0.6773	0.7278	1.141	100.000 ± 0.000	3.864 ± 1.432	0.000 ± 0.000

Table S283

Simulation 2 - Condition 139: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.4 (continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	0.107	0.107 ± 0.118	—	0.180 ± 0.104	0.424 ± 0.196	0.177 ± 0.124	0.421 ± 0.213	—	—	-0.005	-0.005	—	0.4962	0.4963	2.260	100.000 ± 0.000	3.753 ± 1.910	0.000 ± 0.000
UPI	beta51	0.16	0.252	0.092 ± 0.044	0.575 ± 0.273	0.025 ± 0.007	0.157 ± 0.029	0.031 ± 0.009	0.177 ± 0.033	1.224 ± 0.337	1.106 ± 0.206	0.248	0.088	55.21	0.2027	0.2211	1.268	92.308 ± 7.391	0.782 ± 0.172	—
UPI	beta52	0.16	0.145	-0.015 ± 0.040	-0.096 ± 0.251	0.021 ± 0.007	0.145 ± 0.035	0.020 ± 0.008	0.140 ± 0.039	0.763 ± 0.301	0.874 ± 0.246	0.169	0.009	5.86	0.1618	0.1620	2.009	100.000 ± 0.000	1.138 ± 0.218	—
UPI	beta53	0.16	0.044	-0.116 ± 0.056	-0.728 ± 0.347	0.040 ± 0.025	0.200 ± 0.101	0.051 ± 0.036	0.225 ± 0.112	1.974 ± 1.392	1.405 ± 0.699	0.089	-0.071	-44.21	0.1869	0.1999	1.155	69.231 ± 12.801	0.907 ± 0.391	—
UPI	beta54	0.16	0.195	0.035 ± 0.135	0.220 ± 0.841	0.235 ± 0.129	0.485 ± 0.205	0.218 ± 0.123	0.467 ± 0.184	8.530 ± 4.789	2.921 ± 1.150	0.250	0.090	55.95	0.3557	0.3668	1.527	92.308 ± 7.391	2.902 ± 1.212	—
UPI	beta55	0.00	0.095	0.095 ± 0.071	—	0.065 ± 0.026	0.256 ± 0.071	0.070 ± 0.033	0.264 ± 0.084	—	—	0.066	0.066	—	0.2310	0.2402	2.735	100.000 ± 0.000	2.742 ± 1.063	0.000 ± 0.000
UPI	beta56	0.00	-0.060	-0.060 ± 0.040	—	0.021 ± 0.005	0.146 ± 0.025	0.023 ± 0.007	0.152 ± 0.033	—	—	-0.031	-0.031	—	0.1330	0.1365	2.064	100.000 ± 0.000	1.178 ± 0.287	0.000 ± 0.000
UPI	beta57	0.00	-0.164	-0.164 ± 0.145	—	0.273 ± 0.113	0.523 ± 0.153	0.279 ± 0.106	0.528 ± 0.137	—	—	-0.030	-0.030	—	0.4262	0.4273	2.065	100.000 ± 0.000	4.230 ± 1.482	0.000 ± 0.000
UPI	beta58	0.00	-0.178	-0.178 ± 0.106	—	0.147 ± 0.046	0.383 ± 0.085	0.167 ± 0.050	0.408 ± 0.085	—	—	-0.154	-0.154	—	0.3838	0.4137	2.414	100.000 ± 0.000	3.622 ± 1.520	0.000 ± 0.000

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S284

Simulation 2 - Condition 140: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.4

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.248	0.048 ± 0.019	0.239 ± 0.097	0.003 ± 0.001	0.058 ± 0.010	0.005 ± 0.002	0.073 ± 0.016	0.132 ± 0.047	0.363 ± 0.079	0.228	0.028	14.10	0.0850	0.0896	1.273	100.000 ± 0.000	0.289 ± 0.019	—
LSAM	beta42	0.20	0.168	-0.032 ± 0.035	-0.159 ± 0.176	0.011 ± 0.003	0.106 ± 0.020	0.011 ± 0.003	0.105 ± 0.027	0.274 ± 0.082	0.524 ± 0.133	0.202	0.002	0.84	0.0697	0.0697	0.784	88.889 ± 10.476	0.325 ± 0.026	—
LSAM	beta43	0.20	0.196	-0.004 ± 0.027	-0.022 ± 0.136	0.007 ± 0.001	0.082 ± 0.014	0.006 ± 0.001	0.077 ± 0.019	0.149 ± 0.035	0.385 ± 0.097	0.185	-0.015	-7.72	0.0735	0.0751	0.639	77.778 ± 13.858	0.204 ± 0.010	—
LSAM	beta44	0.00	0.037	0.037 ± 0.024	—	0.005 ± 0.001	0.073 ± 0.016	0.006 ± 0.003	0.079 ± 0.023	—	—	0.019	0.019	—	0.0605	0.0633	0.836	100.000 ± 0.000	0.240 ± 0.023	0.000 ± 0.000
LSAM	beta45	0.00	0.020	0.020 ± 0.018	—	0.003 ± 0.001	0.053 ± 0.020	0.003 ± 0.001	0.054 ± 0.015	—	—	0.049	0.049	—	0.0611	0.0780	1.000	88.889 ± 10.476	0.207 ± 0.011	11.111 ± 10.476
LSAM	beta46	0.00	-0.121	-0.121 ± 0.016	—	0.002 ± 0.001	0.048 ± 0.011	0.017 ± 0.004	0.130 ± 0.015	—	—	-0.082	-0.082	—	0.0631	0.1033	1.689	77.778 ± 13.858	0.320 ± 0.034	22.222 ± 13.858
LSAM	beta47	0.00	-0.026	-0.026 ± 0.031	—	0.009 ± 0.002	0.092 ± 0.022	0.008 ± 0.004	0.091 ± 0.029	—	—	-0.058	-0.058	—	0.0842	0.1025	0.909	88.889 ± 10.476	0.329 ± 0.028	11.111 ± 10.476
LSAM	beta51	0.16	0.118	-0.042 ± 0.025	-0.265 ± 0.159	0.006 ± 0.001	0.076 ± 0.015	0.007 ± 0.003	0.084 ± 0.025	0.273 ± 0.126	0.522 ± 0.157	0.137	-0.023	-14.45	0.0714	0.0751	0.827	77.778 ± 13.858	0.248 ± 0.006	—
LSAM	beta52	0.16	0.221	0.061 ± 0.024	0.382 ± 0.148	0.005 ± 0.001	0.071 ± 0.015	0.008 ± 0.003	0.091 ± 0.020	0.322 ± 0.116	0.568 ± 0.123	0.212	0.052	32.30	0.0501	0.0720	0.977	88.889 ± 10.476	0.272 ± 0.015	—
LSAM	beta53	0.16	0.165	0.005 ± 0.020	0.034 ± 0.124	0.004 ± 0.001	0.059 ± 0.017	0.003 ± 0.001	0.056 ± 0.017	0.124 ± 0.046	0.351 ± 0.108	0.138	-0.022	-13.54	0.0617	0.0654	0.907	100.000 ± 0.000	0.211 ± 0.007	—
LSAM	beta54	0.16	0.151	-0.009 ± 0.017	-0.055 ± 0.108	0.003 ± 0.001	0.052 ± 0.017	0.002 ± 0.001	0.050 ± 0.016	0.097 ± 0.038	0.311 ± 0.098	0.163	0.003	1.71	0.0353	0.0354	1.104	100.000 ± 0.000	0.225 ± 0.008	—
LSAM	beta55	0.00	0.046	0.046 ± 0.020	—	0.004 ± 0.001	0.060 ± 0.015	0.005 ± 0.002	0.073 ± 0.020	—	—	0.028	0.028	—	0.0709	0.0763	1.360	100.000 ± 0.000	0.322 ± 0.028	0.000 ± 0.000
LSAM	beta56	0.00	-0.022	-0.022 ± 0.014	—	0.002 ± 0.001	0.043 ± 0.009	0.002 ± 0.001	0.046 ± 0.013	—	—	-0.017	-0.017	—	0.0632	0.0654	1.659	100.000 ± 0.000	0.281 ± 0.017	0.000 ± 0.000
LSAM	beta57	0.00	-0.073	-0.073 ± 0.025	—	0.006 ± 0.002	0.076 ± 0.017	0.011 ± 0.004	0.102 ± 0.025	—	—	-0.064	-0.064	—	0.0444	0.0777	0.958	88.889 ± 10.476	0.285 ± 0.040	11.111 ± 10.476
LSAM	beta58	0.00	-0.057	-0.057 ± 0.016	—	0.002 ± 0.001	0.048 ± 0.011	0.005 ± 0.002	0.073 ± 0.014	—	—	-0.044	-0.044	—	0.0796	0.0909	1.085	77.778 ± 13.858	0.203 ± 0.015	22.222 ± 13.858
QML	beta41	0.20	0.192	-0.008 ± 0.010	-0.041 ± 0.048	0.001 ± 0.000	0.029 ± 0.007	0.001 ± 0.000	0.029 ± 0.007	0.020 ± 0.006	0.143 ± 0.037	0.193	-0.007	-3.45	0.0515	0.0520	1.449	100.000 ± 0.000	0.165 ± 0.006	—
QML	beta42	0.20	0.187	-0.013 ± 0.020	-0.064 ± 0.100	0.004 ± 0.001	0.060 ± 0.011	0.003 ± 0.001	0.058 ± 0.015	0.084 ± 0.025	0.290 ± 0.077	0.206	0.006	2.80	0.0795	0.0797	0.788	88.889 ± 10.476	0.185 ± 0.008	—
QML	beta43	0.20	0.202	0.002 ± 0.023	0.011 ± 0.113	0.005 ± 0.001	0.068 ± 0.009	0.004 ± 0.001	0.064 ± 0.015	0.103 ± 0.015	0.321 ± 0.074	0.191	-0.009	-4.27	0.0941	0.0944	0.636	88.889 ± 10.476	0.170 ± 0.006	—
QML	beta44	0.00	0.026	0.026 ± 0.018	—	0.003 ± 0.001	0.054 ± 0.014	0.003 ± 0.001	0.057 ± 0.017	—	—	0.013	0.013	—	0.0482	0.0501	1.013	100.000 ± 0.000	0.213 ± 0.017	0.000 ± 0.000
QML	beta45	0.00	0.016	0.016 ± 0.017	—	0.003 ± 0.001	0.052 ± 0.017	0.003 ± 0.001	0.051 ± 0.014	—	—	0.033	0.033	—	0.0598	0.0682	0.888	88.889 ± 10.476	0.181 ± 0.008	11.111 ± 10.476
QML	beta46	0.00	-0.061	-0.061 ± 0.008	—	0.001 ± 0.000	0.023 ± 0.006	0.004 ± 0.001	0.065 ± 0.007	—	—	-0.043	-0.043	—	0.0393	0.0585	1.491	55.556 ± 16.563	0.135 ± 0.009	44.444 ± 16.563
QML	beta47	0.00	-0.034	-0.034 ± 0.018	—	0.003 ± 0.001	0.055 ± 0.013	0.004 ± 0.002	0.062 ± 0.018	—	—	-0.042	-0.042	—	0.0633	0.0761	0.841	77.778 ± 13.858	0.182 ± 0.018	22.222 ± 13.858
QML	beta51	0.16	0.134	-0.026 ± 0.021	-0.164 ± 0.131	0.004 ± 0.001	0.063 ± 0.013	0.004 ± 0.002	0.065 ± 0.020	0.164 ± 0.072	0.405 ± 0.122	0.147	-0.013	-8.06	0.0867	0.0877	1.001	88.889 ± 10.476	0.247 ± 0.008	—
QML	beta52	0.16	0.189	0.029 ± 0.016	0.183 ± 0.099	0.002 ± 0.001	0.047 ± 0.012	0.003 ± 0.001	0.053 ± 0.012	0.111 ± 0.036	0.333 ± 0.074	0.188	0.028	17.25	0.0534	0.0601	1.005	100.000 ± 0.000	0.186 ± 0.005	—
QML	beta53	0.16	0.184	0.024 ± 0.017	0.149 ± 0.107	0.003 ± 0.001	0.051 ± 0.014	0.003 ± 0.001	0.054 ± 0.013	0.113 ± 0.034	0.337 ± 0.079	0.171	0.011	6.71	0.0775	0.0782	0.988	100.000 ± 0.000	0.198 ± 0.006	—
QML	beta54	0.16	0.136	-0.024 ± 0.014	-0.152 ± 0.085	0.002 ± 0.001	0.041 ± 0.015	0.002 ± 0.001	0.046 ± 0.011	0.081 ± 0.029	0.285 ± 0.071	0.140	-0.020	-12.58	0.0526	0.0563	1.077	100.000 ± 0.000	0.173 ± 0.005	—
QML	beta55	0.00	0.033	0.033 ± 0.017	—	0.003 ± 0.001	0.052 ± 0.015	0.003 ± 0.002	0.059 ± 0.018	—	—	0.054	0.054	—	0.0440	0.0695	1.391	100.000 ± 0.000	0.282 ± 0.016	0.000 ± 0.000
QML	beta56	0.00	-0.023	-0.023 ± 0.015	—	0.002 ± 0.000	0.046 ± 0.008	0.002 ± 0.001	0.049 ± 0.011	—	—	-0.051	-0.051	—	0.0715	0.0880	1.486	100.000 ± 0.000	0.266 ± 0.016	0.000 ± 0.000
QML	beta57	0.00	-0.033	-0.033 ± 0.012	—	0.001 ± 0.000	0.035 ± 0.010	0.002 ± 0.001	0.047 ± 0.008	—	—	-0.034	-0.034	—	0.0415	0.0538	1.166	88.889 ± 10.476	0.160 ± 0.020	11.111 ± 10.476
QML	beta58	0.00	-0.035	-0.035 ± 0.009	—	0.001 ± 0.000	0.026 ± 0.006	0.002 ± 0.001	0.043 ± 0.007	—	—	-0.034	-0.034	—	0.0469	0.0578	1.215	88.889 ± 10.476	0.123 ± 0.005	11.111 ± 10.476
UPI	beta41	0.20	0.610	0.410 ± 0.298	2.050 ± 1.492	0.802 ± 0.221	0.895 ± 0.202	0.881 ± 0.352	0.939 ± 0.265	22.021 ± 8.791	4.693 ± 1.327	0.366	0.166	83.15	1.0042	1.0178	0.897	100.000 ± 0.000	3.148 ± 0.770	—
UPI	beta42	0.20	0.568	0.368 ± 0.329	1.841 ± 1.645	0.975 ± 0.362	0.987 ± 0.309	1.002 ± 0.580	1.001 ± 0.405	25.047 ± 14.501	5.005 ± 2.026	0.047	-0.153	-76.59	0.5069	0.5295	0.787	88.889 ± 10.476	3.046 ± 1.051	—
UPI	beta43	0.20	-0.004	-0.204 ± 0.061	-1.020 ± 0.304	0.033 ± 0.012	0.182 ± 0.054	0.071 ± 0.033	0.267 ± 0.074	1.780 ± 0.816	1.334 ± 0.370	0.019	-0.181	-90.44	0.2440	0.3037	1.366	88.889 ± 10.476	0.977 ± 0.277	—
UPI	beta44	0.00	0.100	0.100 ± 0.031	—	0.009 ± 0.002	0.092 ± 0.021	0.018 ± 0.005	0.133 ± 0.022	—	—	0.094	0.094	—	0.1623	0.1875	2.588	100.000 ± 0.000	0.937 ± 0.215	0.000 ± 0.000

Table S284

Simulation 2 - Condition 140: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.4 (continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta45	0.00	0.089	0.089 ± 0.023	—	0.005 ± 0.002	0.068 ± 0.019	0.012 ± 0.005	0.110 ± 0.028	—	—	0.030	0.030	—	0.1013	0.1057	3.102	100.000 ± 0.000	0.827 ± 0.298	0.000 ± 0.000
UPI	beta46	0.00	-0.697	-0.697 ± 0.293	—	0.771 ± 0.191	0.878 ± 0.187	1.172 ± 0.375	1.082 ± 0.216	—	—	-0.450	-0.450	—	1.4810	1.5479	1.353	100.000 ± 0.000	4.659 ± 1.114	0.000 ± 0.000
UPI	beta47	0.00	-0.472	-0.472 ± 0.458	—	1.892 ± 1.045	1.375 ± 0.688	1.904 ± 1.394	1.380 ± 0.773	—	—	0.003	0.003	—	0.7508	0.7508	1.051	100.000 ± 0.000	5.667 ± 3.031	0.000 ± 0.000
UPI	beta51	0.16	0.000	-0.160 ± 0.062	-0.999 ± 0.387	0.034 ± 0.006	0.186 ± 0.030	0.056 ± 0.019	0.237 ± 0.049	2.194 ± 0.757	1.481 ± 0.307	0.136	-0.024	-15.29	0.2456	0.2468	1.284	100.000 ± 0.000	0.934 ± 0.229	—
UPI	beta52	0.16	0.888	0.728 ± 0.222	4.547 ± 1.389	0.444 ± NaN	0.667 ± 0.055	0.924 ± 0.299	0.961 ± 0.171	36.108 ± 11.661	6.009 ± 1.069	0.398	0.238	148.51	0.5932	0.6390	1.125	100.000 ± 0.000	2.941 ± 1.006	—
UPI	beta53	0.16	0.059	-0.101 ± 0.050	-0.628 ± 0.315	0.023 ± 0.005	0.151 ± 0.029	0.030 ± 0.012	0.174 ± 0.045	1.189 ± 0.466	1.090 ± 0.280	0.018	-0.142	-88.99	0.1155	0.1834	0.815	77.778 ± 13.858	0.483 ± 0.125	—
UPI	beta54	0.16	0.078	-0.082 ± 0.041	-0.515 ± 0.256	0.015 ± 0.005	0.123 ± 0.035	0.020 ± 0.010	0.142 ± 0.048	0.789 ± 0.406	0.888 ± 0.302	0.141	-0.019	-11.63	0.2338	0.2345	1.280	77.778 ± 13.858	0.616 ± 0.170	—
UPI	beta55	0.00	0.124	0.124 ± 0.062	—	0.034 ± 0.013	0.185 ± 0.058	0.046 ± 0.026	0.214 ± 0.080	—	—	0.027	0.027	—	0.1301	0.1328	0.946	88.889 ± 10.476	0.685 ± 0.114	11.111 ± 10.476
UPI	beta56	0.00	-0.004	-0.004 ± 0.040	—	0.015 ± 0.005	0.121 ± 0.036	0.013 ± 0.005	0.114 ± 0.038	—	—	-0.020	-0.020	—	0.0911	0.0934	1.110	100.000 ± 0.000	0.528 ± 0.076	0.000 ± 0.000
UPI	beta57	0.00	-0.877	-0.877 ± 0.235	—	0.499 ± 0.100	0.706 ± 0.123	1.213 ± 0.441	1.101 ± 0.223	—	—	-0.459	-0.459	—	0.7402	0.8709	1.585	100.000 ± 0.000	4.387 ± 1.578	0.000 ± 0.000
UPI	beta58	0.00	-0.045	-0.045 ± 0.050	—	0.022 ± 0.010	0.149 ± 0.056	0.022 ± 0.013	0.148 ± 0.065	—	—	-0.063	-0.063	—	0.2010	0.2108	1.194	100.000 ± 0.000	0.697 ± 0.218	0.000 ± 0.000

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u002f0 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S285

Simulation 2 - Condition 141: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, $N = 400$, Reliability = 0.6

Method	Param.	True	Mean-based										Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE	
LSAM	beta41	0.20	0.207	0.007 ± 0.002	0.036 ± 0.012	0.003 ± 0.000	0.057 ± 0.002	0.003 ± 0.000	0.057 ± 0.002	0.081 ± 0.005	0.285 ± 0.011	0.206	0.006	3.00	0.0581	0.0584	1.045	96.863 ± 0.749	0.232 ± 0.001	—	
LSAM	beta42	0.20	0.203	0.003 ± 0.003	0.015 ± 0.013	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	0.096 ± 0.006	0.310 ± 0.012	0.200	0.000	0.10	0.0621	0.0621	0.998	95.387 ± 0.901	0.242 ± 0.002	—	
LSAM	beta43	0.20	0.195	-0.005 ± 0.002	-0.023 ± 0.012	0.003 ± 0.000	0.057 ± 0.002	0.003 ± 0.000	0.057 ± 0.002	0.080 ± 0.005	0.283 ± 0.011	0.196	-0.004	-2.09	0.0560	0.0561	0.982	94.280 ± 0.997	0.218 ± 0.001	—	
LSAM	beta44	0.00	0.015	0.015 ± 0.003	—	0.005 ± 0.000	0.071 ± 0.002	0.005 ± 0.000	0.073 ± 0.003	—	—	0.015	0.015	—	0.0703	0.0719	0.972	94.834 ± 0.951	0.272 ± 0.003	5.166 ± 0.951	
LSAM	beta45	0.00	0.004	0.004 ± 0.003	—	0.004 ± 0.000	0.066 ± 0.002	0.004 ± 0.000	0.066 ± 0.003	—	—	0.005	0.005	—	0.0640	0.0642	0.910	92.620 ± 1.123	0.236 ± 0.002	7.380 ± 1.123	
LSAM	beta46	0.00	-0.026	-0.026 ± 0.004	—	0.007 ± 0.000	0.086 ± 0.003	0.008 ± 0.001	0.089 ± 0.003	—	—	-0.028	-0.028	—	0.0788	0.0835	0.969	93.727 ± 1.042	0.325 ± 0.004	6.273 ± 1.042	
LSAM	beta47	0.00	-0.023	-0.023 ± 0.004	—	0.007 ± 0.001	0.086 ± 0.003	0.008 ± 0.001	0.089 ± 0.004	—	—	-0.027	-0.027	—	0.0794	0.0839	0.964	93.727 ± 1.042	0.325 ± 0.004	6.273 ± 1.042	
LSAM	beta51	0.16	0.157	-0.003 ± 0.003	-0.017 ± 0.021	0.006 ± 0.000	0.077 ± 0.002	0.006 ± 0.000	0.077 ± 0.003	0.232 ± 0.015	0.481 ± 0.018	0.156	-0.004	-2.59	0.0789	0.0790	0.941	93.173 ± 1.083	0.284 ± 0.002	—	
LSAM	beta52	0.16	0.163	0.003 ± 0.003	0.018 ± 0.016	0.003 ± 0.000	0.059 ± 0.002	0.003 ± 0.000	0.059 ± 0.002	0.134 ± 0.008	0.366 ± 0.013	0.163	0.003	2.03	0.0604	0.0605	1.039	97.048 ± 0.727	0.239 ± 0.001	—	
LSAM	beta53	0.16	0.157	-0.003 ± 0.003	-0.018 ± 0.017	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	0.152 ± 0.009	0.389 ± 0.014	0.156	-0.004	-2.24	0.0619	0.0620	0.975	95.018 ± 0.935	0.238 ± 0.001	—	
LSAM	beta54	0.16	0.163	0.003 ± 0.003	0.020 ± 0.016	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.061 ± 0.002	0.147 ± 0.010	0.383 ± 0.015	0.166	0.006	3.89	0.0579	0.0583	0.981	94.096 ± 1.012	0.235 ± 0.001	—	
LSAM	beta55	0.00	0.008	0.008 ± 0.004	—	0.008 ± 0.001	0.090 ± 0.003	0.008 ± 0.001	0.090 ± 0.004	—	—	0.009	0.009	—	0.0886	0.0891	0.943	95.941 ± 0.848	0.333 ± 0.004	4.059 ± 0.848	
LSAM	beta56	0.00	0.002	0.002 ± 0.004	—	0.007 ± 0.000	0.082 ± 0.003	0.007 ± 0.000	0.082 ± 0.003	—	—	0.004	0.004	—	0.0807	0.0808	0.949	94.465 ± 0.982	0.304 ± 0.003	5.535 ± 0.982	
LSAM	beta57	0.00	-0.019	-0.019 ± 0.004	—	0.007 ± 0.001	0.086 ± 0.003	0.008 ± 0.001	0.088 ± 0.003	—	—	-0.022	-0.022	—	0.0829	0.0858	0.966	93.911 ± 1.027	0.327 ± 0.004	6.089 ± 1.027	
LSAM	beta58	0.00	-0.018	-0.018 ± 0.003	—	0.005 ± 0.000	0.074 ± 0.003	0.006 ± 0.000	0.076 ± 0.003	—	—	-0.020	-0.020	—	0.0761	0.0786	1.036	95.018 ± 0.935	0.300 ± 0.004	4.982 ± 0.935	
QML	beta41	0.20	0.204	0.004 ± 0.002	0.021 ± 0.011	0.003 ± 0.000	0.052 ± 0.002	0.003 ± 0.000	0.052 ± 0.002	0.067 ± 0.004	0.259 ± 0.010	0.203	0.003	1.45	0.0547	0.0548	1.044	96.863 ± 0.749	0.212 ± 0.001	—	
QML	beta42	0.20	0.201	0.001 ± 0.002	0.004 ± 0.012	0.003 ± 0.000	0.057 ± 0.002	0.003 ± 0.000	0.057 ± 0.002	0.081 ± 0.005	0.285 ± 0.011	0.201	0.001	0.36	0.0596	0.0596	0.974	94.649 ± 0.967	0.218 ± 0.001	—	
QML	beta43	0.20	0.197	-0.003 ± 0.002	-0.015 ± 0.012	0.003 ± 0.000	0.055 ± 0.002	0.003 ± 0.000	0.055 ± 0.002	0.077 ± 0.005	0.277 ± 0.011	0.200	0.000	-0.23	0.0555	0.0555	0.967	94.096 ± 1.012	0.210 ± 0.001	—	
QML	beta44	0.00	0.003	0.003 ± 0.003	—	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.069 ± 0.003	—	—	0.004	0.004	—	0.0642	0.0643	0.989	94.096 ± 1.012	0.269 ± 0.004	5.904 ± 1.012	
QML	beta45	0.00	-0.002	-0.002 ± 0.003	—	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.003	—	—	0.001	0.001	—	0.0628	0.0628	0.922	93.173 ± 1.083	0.231 ± 0.002	6.827 ± 1.083	
QML	beta46	0.00	-0.010	-0.010 ± 0.002	—	0.003 ± 0.000	0.056 ± 0.002	0.003 ± 0.000	0.057 ± 0.002	—	—	-0.014	-0.014	—	0.0538	0.0556	1.012	92.620 ± 1.123	0.221 ± 0.003	7.380 ± 1.123	
QML	beta47	0.00	-0.012	-0.012 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.002	0.003 ± 0.000	0.059 ± 0.002	—	—	-0.018	-0.018	—	0.0538	0.0567	0.957	91.328 ± 1.209	0.217 ± 0.003	8.672 ± 1.209	
QML	beta51	0.16	0.160	0.000 ± 0.003	0.001 ± 0.021	0.006 ± 0.000	0.076 ± 0.002	0.006 ± 0.000	0.076 ± 0.003	0.228 ± 0.014	0.477 ± 0.018	0.159	-0.001	-0.33	0.0787	0.0787	0.981	95.018 ± 0.935	0.294 ± 0.002	—	
QML	beta52	0.16	0.161	0.001 ± 0.002	0.004 ± 0.015	0.003 ± 0.000	0.055 ± 0.002	0.003 ± 0.000	0.055 ± 0.002	0.120 ± 0.007	0.346 ± 0.012	0.163	0.003	1.72	0.0594	0.0594	1.062	97.048 ± 0.727	0.231 ± 0.002	—	
QML	beta53	0.16	0.159	-0.001 ± 0.003	-0.004 ± 0.016	0.004 ± 0.000	0.060 ± 0.002	0.004 ± 0.000	0.060 ± 0.002	0.142 ± 0.009	0.377 ± 0.014	0.161	0.001	0.34	0.0590	0.0590	0.982	95.203 ± 0.918	0.233 ± 0.001	—	
QML	beta54	0.16	0.161	0.001 ± 0.002	0.008 ± 0.015	0.003 ± 0.000	0.057 ± 0.002	0.003 ± 0.000	0.057 ± 0.002	0.129 ± 0.009	0.359 ± 0.015	0.163	0.003	2.19	0.0553	0.0554	0.974	94.834 ± 0.951	0.219 ± 0.001	—	
QML	beta55	0.00	0.003	0.003 ± 0.004	—	0.007 ± 0.000	0.086 ± 0.003	0.007 ± 0.000	0.086 ± 0.003	—	—	0.002	0.002	—	0.0859	0.0859	0.984	96.494 ± 0.790	0.330 ± 0.004	3.506 ± 0.790	
QML	beta56	0.00	-0.003	-0.003 ± 0.003	—	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.003	—	—	-0.002	-0.002	—	0.0763	0.0764	0.977	92.989 ± 1.097	0.287 ± 0.003	7.011 ± 1.097	
QML	beta57	0.00	-0.009	-0.009 ± 0.003	—	0.004 ± 0.000	0.059 ± 0.002	0.004 ± 0.000	0.060 ± 0.003	—	—	-0.016	-0.016	—	0.0582	0.0603	0.991	91.697 ± 1.185	0.230 ± 0.004	8.303 ± 1.185	
QML	beta58	0.00	-0.009	-0.009 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.002	0.002 ± 0.000	0.050 ± 0.002	—	—	-0.013	-0.013	—	0.0521	0.0537	1.033	91.697 ± 1.185	0.197 ± 0.003	8.303 ± 1.185	
UPI	beta41	0.20	0.210	0.010 ± 0.003	0.048 ± 0.014	0.005 ± 0.000	0.067 ± 0.003	0.005 ± 0.000	0.068 ± 0.003	0.115 ± 0.009	0.339 ± 0.015	0.209	0.009	4.72	0.0704	0.0710	1.355	98.339 ± 0.549	0.357 ± 0.021	—	
UPI	beta42	0.20	0.216	0.016 ± 0.003	0.081 ± 0.015	0.005 ± 0.000	0.071 ± 0.003	0.005 ± 0.000	0.073 ± 0.004	0.132 ± 0.011	0.364 ± 0.018	0.214	0.014	6.96	0.0801	0.0813	1.406	96.494 ± 0.790	0.391 ± 0.018	—	
UPI	beta43	0.20	0.188	-0.012 ± 0.003	-0.060 ± 0.013	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	0.096 ± 0.006	0.310 ± 0.012	0.186	-0.014	-6.79	0.0707	0.0720	1.153	96.679 ± 0.770	0.275 ± 0.006	—	
UPI	beta44	0.00	0.020	0.020 ± 0.004	—	0.007 ± 0.001	0.086 ± 0.003	0.008 ± 0.001	0.089 ± 0.004	—	—	0.019	0.019	—	0.0868	0.0889	1.173	97.417 ± 0.681	0.397 ± 0.012	2.583 ± 0.681	
UPI	beta45	0.00	0.002	0.002 ± 0.003	—	0.005 ± 0.000	0.073 ± 0.002	0.005 ± 0.000	0.073 ± 0.003	—	—	0.004	0.004	—	0.0789	0.0790	1.085	95.941 ± 0.848	0.310 ± 0.014	4.059 ± 0.848	
UPI	beta46	0.00	-0.012	-0.012 ± 0.006	—	0.019 ± 0.002	0.139 ± 0.006	0.019 ± 0.002	0.139 ± 0.007	—	—	-0.011	-0.011	—	0.1243	0.1249	1.330	97.970 ± 0.606	0.725 ± 0.049	2.030 ± 0.606	

Table S285

Simulation 2 - Condition 141: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.6 (continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.032	-0.032 ± 0.006	—	0.021 ± 0.002	0.145 ± 0.006	0.022 ± 0.002	0.149 ± 0.007	—	—	-0.037	-0.037	—	0.1344	0.1395	1.408	97.417 ± 0.681	0.802 ± 0.055	2.583 ± 0.681
UPI	beta51	0.16	0.157	-0.003 ± 0.003	-0.016 ± 0.020	0.006 ± 0.000	0.075 ± 0.002	0.006 ± 0.000	0.075 ± 0.003	0.219 ± 0.014	0.468 ± 0.018	0.155	-0.005	-3.37	0.0800	0.0802	1.045	94.834 ± 0.951	0.307 ± 0.006	—
UPI	beta52	0.16	0.165	0.005 ± 0.003	0.031 ± 0.017	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	0.149 ± 0.010	0.387 ± 0.015	0.168	0.008	4.82	0.0657	0.0662	1.195	98.155 ± 0.578	0.289 ± 0.005	—
UPI	beta53	0.16	0.155	-0.005 ± 0.003	-0.034 ± 0.017	0.004 ± 0.000	0.064 ± 0.002	0.004 ± 0.000	0.064 ± 0.002	0.159 ± 0.010	0.399 ± 0.015	0.151	-0.009	-5.55	0.0646	0.0652	1.084	95.203 ± 0.918	0.270 ± 0.004	—
UPI	beta54	0.16	0.167	0.007 ± 0.003	0.043 ± 0.019	0.005 ± 0.000	0.069 ± 0.002	0.005 ± 0.000	0.070 ± 0.003	0.190 ± 0.013	0.436 ± 0.018	0.173	0.013	8.21	0.0662	0.0675	1.210	95.572 ± 0.884	0.329 ± 0.012	—
UPI	beta55	0.00	0.010	0.010 ± 0.004	—	0.008 ± 0.001	0.089 ± 0.003	0.008 ± 0.001	0.090 ± 0.003	—	—	0.014	0.014	—	0.0902	0.0913	1.146	97.601 ± 0.657	0.401 ± 0.015	2.399 ± 0.657
UPI	beta56	0.00	0.000	0.000 ± 0.004	—	0.007 ± 0.000	0.082 ± 0.003	0.007 ± 0.000	0.082 ± 0.003	—	—	0.004	0.004	—	0.0878	0.0879	1.096	97.232 ± 0.705	0.354 ± 0.008	2.768 ± 0.705
UPI	beta57	0.00	-0.007	-0.007 ± 0.005	—	0.013 ± 0.001	0.114 ± 0.005	0.013 ± 0.001	0.114 ± 0.005	—	—	-0.016	-0.016	—	0.1109	0.1120	1.237	97.601 ± 0.657	0.553 ± 0.038	2.399 ± 0.657
UPI	beta58	0.00	-0.020	-0.020 ± 0.005	—	0.016 ± 0.001	0.125 ± 0.005	0.016 ± 0.001	0.126 ± 0.006	—	—	-0.024	-0.024	—	0.0992	0.1021	1.414	98.893 ± 0.449	0.692 ± 0.046	1.107 ± 0.449

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S286

Simulation 2 - Condition 142: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability =

0.6

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.203	0.003 ± 0.001	0.013 ± 0.007	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.034 ± 0.002	0.185 ± 0.007	0.201	0.001	0.73	0.0353	0.0354	0.979	94.557 ± 0.895	0.142 ± 0.000	—
LSAM	beta42	0.20	0.204	0.004 ± 0.001	0.020 ± 0.007	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.035 ± 0.002	0.187 ± 0.007	0.203	0.003	1.34	0.0384	0.0385	1.010	95.334 ± 0.832	0.148 ± 0.000	—
LSAM	beta43	0.20	0.193	-0.007 ± 0.001	-0.033 ± 0.007	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.032 ± 0.002	0.179 ± 0.006	0.194	-0.006	-2.93	0.0348	0.0353	0.974	94.401 ± 0.907	0.134 ± 0.000	—
LSAM	beta44	0.00	0.014	0.014 ± 0.002	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.043 ± 0.001	—	—	0.015	0.015	—	0.0407	0.0434	0.991	94.246 ± 0.918	0.160 ± 0.001	5.754 ± 0.918
LSAM	beta45	0.00	0.006	0.006 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	0.006	0.006	—	0.0358	0.0363	0.987	94.246 ± 0.918	0.144 ± 0.001	5.754 ± 0.918
LSAM	beta46	0.00	-0.029	-0.029 ± 0.002	—	0.003 ± 0.000	0.050 ± 0.002	0.003 ± 0.000	0.058 ± 0.002	—	—	-0.028	-0.028	—	0.0473	0.0550	0.980	89.269 ± 1.221	0.192 ± 0.001	10.731 ± 1.221
LSAM	beta47	0.00	-0.028	-0.028 ± 0.002	—	0.002 ± 0.000	0.047 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	—	—	-0.027	-0.027	—	0.0467	0.0539	1.037	89.891 ± 1.189	0.191 ± 0.001	10.109 ± 1.189
LSAM	beta51	0.16	0.159	-0.001 ± 0.002	-0.004 ± 0.011	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.002	0.079 ± 0.004	0.280 ± 0.010	0.158	-0.002	-1.56	0.0428	0.0428	1.003	94.246 ± 0.918	0.177 ± 0.001	—
LSAM	beta52	0.16	0.166	0.006 ± 0.001	0.038 ± 0.009	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.054 ± 0.003	0.233 ± 0.008	0.165	0.005	3.33	0.0362	0.0366	1.013	95.956 ± 0.777	0.146 ± 0.000	—
LSAM	beta53	0.16	0.155	-0.005 ± 0.002	-0.031 ± 0.009	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	0.058 ± 0.003	0.240 ± 0.008	0.154	-0.006	-3.99	0.0388	0.0393	0.985	94.868 ± 0.870	0.147 ± 0.000	—
LSAM	beta54	0.16	0.161	0.001 ± 0.001	0.007 ± 0.009	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.050 ± 0.003	0.223 ± 0.008	0.161	0.001	0.80	0.0337	0.0337	1.034	95.801 ± 0.791	0.144 ± 0.000	—
LSAM	beta55	0.00	0.013	0.013 ± 0.002	—	0.003 ± 0.000	0.053 ± 0.001	0.003 ± 0.000	0.054 ± 0.002	—	—	0.014	0.014	—	0.0512	0.0531	0.953	91.913 ± 1.075	0.197 ± 0.001	8.087 ± 1.075
LSAM	beta56	0.00	0.001	0.001 ± 0.002	—	0.002 ± 0.000	0.048 ± 0.002	0.002 ± 0.000	0.048 ± 0.002	—	—	0.000	0.000	—	0.0468	0.0468	0.979	95.023 ± 0.858	0.185 ± 0.001	4.977 ± 0.858
LSAM	beta57	0.00	-0.020	-0.020 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.055 ± 0.002	—	—	-0.023	-0.023	—	0.0489	0.0539	0.963	91.757 ± 1.085	0.196 ± 0.001	8.243 ± 1.085
LSAM	beta58	0.00	-0.019	-0.019 ± 0.002	—	0.002 ± 0.000	0.047 ± 0.001	0.003 ± 0.000	0.050 ± 0.002	—	—	-0.023	-0.023	—	0.0441	0.0497	0.988	92.379 ± 1.046	0.181 ± 0.001	7.621 ± 1.046
QML	beta41	0.20	0.200	0.000 ± 0.001	0.002 ± 0.007	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.028 ± 0.002	0.168 ± 0.006	0.199	-0.001	-0.65	0.0328	0.0328	0.987	94.868 ± 0.870	0.130 ± 0.000	—
QML	beta42	0.20	0.202	0.002 ± 0.001	0.010 ± 0.007	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.029 ± 0.002	0.171 ± 0.006	0.201	0.001	0.32	0.0350	0.0350	1.000	95.645 ± 0.805	0.134 ± 0.000	—
QML	beta43	0.20	0.196	-0.004 ± 0.001	-0.021 ± 0.007	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	0.030 ± 0.002	0.173 ± 0.006	0.196	-0.004	-2.12	0.0333	0.0336	0.972	93.935 ± 0.941	0.131 ± 0.000	—
QML	beta44	0.00	0.006	0.006 ± 0.002	—	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	—	—	0.009	0.009	—	0.0389	0.0399	1.003	94.557 ± 0.895	0.159 ± 0.001	5.443 ± 0.895
QML	beta45	0.00	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	0.000	0.000	—	0.0352	0.0352	0.991	95.801 ± 0.791	0.141 ± 0.001	4.199 ± 0.791
QML	beta46	0.00	-0.017	-0.017 ± 0.001	—	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.018	-0.018	—	0.0319	0.0365	0.990	88.802 ± 1.244	0.132 ± 0.001	11.198 ± 1.244
QML	beta47	0.00	-0.019	-0.019 ± 0.001	—	0.001 ± 0.000	0.032 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.020	-0.020	—	0.0329	0.0383	1.023	87.558 ± 1.302	0.129 ± 0.001	12.442 ± 1.302
QML	beta51	0.16	0.161	0.001 ± 0.002	0.004 ± 0.011	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.044 ± 0.002	0.077 ± 0.004	0.277 ± 0.010	0.159	-0.001	-0.81	0.0416	0.0416	1.030	94.557 ± 0.895	0.179 ± 0.001	—
QML	beta52	0.16	0.164	0.004 ± 0.001	0.028 ± 0.009	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	0.050 ± 0.003	0.224 ± 0.008	0.164	0.004	2.27	0.0356	0.0358	1.003	95.179 ± 0.845	0.140 ± 0.001	—
QML	beta53	0.16	0.157	-0.003 ± 0.001	-0.017 ± 0.009	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.055 ± 0.003	0.234 ± 0.008	0.155	-0.005	-2.83	0.0368	0.0371	0.980	94.868 ± 0.870	0.144 ± 0.000	—
QML	beta54	0.16	0.159	-0.001 ± 0.001	-0.006 ± 0.008	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	0.045 ± 0.003	0.212 ± 0.008	0.159	-0.001	-0.66	0.0323	0.0323	1.028	96.112 ± 0.762	0.137 ± 0.000	—
QML	beta55	0.00	0.011	0.011 ± 0.002	—	0.003 ± 0.000	0.052 ± 0.001	0.003 ± 0.000	0.053 ± 0.002	—	—	0.012	0.012	—	0.0513	0.0526	0.987	92.379 ± 1.046	0.200 ± 0.003	7.621 ± 1.046
QML	beta56	0.00	-0.003	-0.003 ± 0.002	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.002	—	—	-0.005	-0.005	—	0.0442	0.0444	1.000	94.868 ± 0.870	0.178 ± 0.001	5.132 ± 0.870
QML	beta57	0.00	-0.013	-0.013 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.015	-0.015	—	0.0355	0.0387	0.982	89.736 ± 1.197	0.140 ± 0.002	10.264 ± 1.197
QML	beta58	0.00	-0.012	-0.012 ± 0.001	—	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	—	—	-0.014	-0.014	—	0.0304	0.0337	0.996	91.135 ± 1.121	0.123 ± 0.001	8.865 ± 1.121
UPI	beta41	0.20	0.206	0.006 ± 0.002	0.031 ± 0.009	0.002 ± 0.000	0.044 ± 0.001	0.002 ± 0.000	0.045 ± 0.002	0.050 ± 0.003	0.223 ± 0.008	0.206	0.006	3.03	0.0486	0.0490	1.287	97.356 ± 0.633	0.223 ± 0.009	—
UPI	beta42	0.20	0.214	0.014 ± 0.002	0.072 ± 0.009	0.002 ± 0.000	0.048 ± 0.002	0.003 ± 0.000	0.050 ± 0.002	0.063 ± 0.005	0.251 ± 0.011	0.211	0.011	5.60	0.0494	0.0506	1.299	97.356 ± 0.633	0.245 ± 0.009	—
UPI	beta43	0.20	0.186	-0.014 ± 0.002	-0.071 ± 0.008	0.001 ± 0.000	0.038 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.042 ± 0.002	0.204 ± 0.006	0.183	-0.017	-8.34	0.0428	0.0460	1.093	94.868 ± 0.870	0.164 ± 0.003	—
UPI	beta44	0.00	0.020	0.020 ± 0.002	—	0.002 ± 0.000	0.047 ± 0.001	0.003 ± 0.000	0.051 ± 0.002	—	—	0.022	0.022	—	0.0468	0.0519	1.125	95.645 ± 0.805	0.206 ± 0.003	4.355 ± 0.805

Table S286

Simulation 2 - Condition 142: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.6 (continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta45	0.00	0.007	0.007 ± 0.002	—	0.002 ± 0.000	0.039 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	0.006	0.006	—	0.0392	0.0397	1.067	95.023 ± 0.858	0.164 ± 0.002	4.977 ± 0.858
UPI	beta46	0.00	-0.031	-0.031 ± 0.003	—	0.008 ± 0.001	0.087 ± 0.003	0.009 ± 0.001	0.092 ± 0.004	—	—	-0.032	-0.032	—	0.0752	0.0816	1.212	94.868 ± 0.870	0.415 ± 0.018	5.132 ± 0.870
UPI	beta47	0.00	-0.040	-0.040 ± 0.003	—	0.008 ± 0.001	0.089 ± 0.003	0.009 ± 0.001	0.097 ± 0.004	—	—	-0.039	-0.039	—	0.0818	0.0907	1.229	96.423 ± 0.732	0.427 ± 0.016	3.577 ± 0.732
UPI	beta51	0.16	0.158	-0.002 ± 0.002	-0.014 ± 0.011	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.002	0.078 ± 0.004	0.279 ± 0.009	0.154	-0.006	-3.92	0.0462	0.0466	1.067	95.179 ± 0.845	0.186 ± 0.002	—
UPI	beta52	0.16	0.169	0.009 ± 0.002	0.058 ± 0.010	0.002 ± 0.000	0.041 ± 0.001	0.002 ± 0.000	0.042 ± 0.002	0.070 ± 0.005	0.264 ± 0.010	0.169	0.009	5.34	0.0417	0.0426	1.199	97.512 ± 0.614	0.194 ± 0.007	—
UPI	beta53	0.16	0.150	-0.010 ± 0.002	-0.064 ± 0.010	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.041 ± 0.001	0.066 ± 0.004	0.257 ± 0.009	0.145	-0.015	-9.56	0.0432	0.0459	1.064	95.179 ± 0.845	0.166 ± 0.002	—
UPI	beta54	0.16	0.169	0.009 ± 0.002	0.054 ± 0.011	0.002 ± 0.000	0.044 ± 0.002	0.002 ± 0.000	0.045 ± 0.002	0.080 ± 0.006	0.282 ± 0.011	0.167	0.007	4.33	0.0445	0.0450	1.223	96.579 ± 0.717	0.212 ± 0.006	—
UPI	beta55	0.00	0.015	0.015 ± 0.002	—	0.003 ± 0.000	0.055 ± 0.002	0.003 ± 0.000	0.057 ± 0.002	—	—	0.016	0.016	—	0.0564	0.0586	1.007	94.401 ± 0.907	0.217 ± 0.002	5.599 ± 0.907
UPI	beta56	0.00	0.003	0.003 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.002	0.002 ± 0.000	0.049 ± 0.002	—	—	0.003	0.003	—	0.0487	0.0488	1.032	96.267 ± 0.748	0.198 ± 0.001	3.733 ± 0.748
UPI	beta57	0.00	-0.020	-0.020 ± 0.003	—	0.005 ± 0.000	0.074 ± 0.003	0.006 ± 0.000	0.076 ± 0.003	—	—	-0.021	-0.021	—	0.0702	0.0733	1.139	95.023 ± 0.858	0.329 ± 0.017	4.977 ± 0.858
UPI	beta58	0.00	-0.031	-0.031 ± 0.003	—	0.006 ± 0.000	0.077 ± 0.003	0.007 ± 0.000	0.083 ± 0.003	—	—	-0.032	-0.032	—	0.0705	0.0773	1.207	97.045 ± 0.668	0.367 ± 0.012	2.955 ± 0.668

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u002014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S287
Simulation 2 - Condition 143: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.8

			Mean-based									Median-based					Inference			
Method	Param.	True	Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.201	0.001 ± 0.001	0.004 ± 0.007	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.052 ± 0.002	0.228 ± 0.006	0.200	0.000	-0.18	0.0464	0.0464	0.993	95.195 ± 0.677	0.177 ± 0.001	—
LSAM	beta42	0.20	0.201	0.001 ± 0.001	0.005 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.056 ± 0.002	0.237 ± 0.006	0.201	0.001	0.27	0.0477	0.0477	0.979	94.795 ± 0.703	0.182 ± 0.001	—
LSAM	beta43	0.20	0.202	0.002 ± 0.001	0.008 ± 0.007	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.053 ± 0.002	0.229 ± 0.006	0.201	0.001	0.50	0.0456	0.0456	0.964	94.294 ± 0.734	0.173 ± 0.001	—
LSAM	beta44	0.00	0.006	0.006 ± 0.002	—	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	—	—	0.006	0.006	—	0.0538	0.0541	0.919	92.492 ± 0.834	0.195 ± 0.001	7.508 ± 0.834
LSAM	beta45	0.00	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	-0.001	-0.001	—	0.0411	0.0412	1.006	94.995 ± 0.690	0.177 ± 0.001	5.005 ± 0.690
LSAM	beta46	0.00	-0.010	-0.010 ± 0.002	—	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	—	—	-0.011	-0.011	—	0.0547	0.0558	0.997	93.594 ± 0.775	0.225 ± 0.001	6.406 ± 0.775
LSAM	beta47	0.00	-0.009	-0.009 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	—	—	-0.011	-0.011	—	0.0552	0.0563	0.983	93.694 ± 0.769	0.223 ± 0.001	6.306 ± 0.769
LSAM	beta51	0.16	0.159	-0.001 ± 0.002	-0.007 ± 0.013	0.004 ± 0.000	0.064 ± 0.001	0.004 ± 0.000	0.064 ± 0.002	0.159 ± 0.007	0.399 ± 0.011	0.156	-0.004	-2.31	0.0657	0.0658	0.904	92.893 ± 0.813	0.226 ± 0.001	—
LSAM	beta52	0.16	0.162	0.002 ± 0.002	0.013 ± 0.010	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.092 ± 0.004	0.303 ± 0.008	0.162	0.002	1.26	0.0487	0.0487	0.975	93.994 ± 0.752	0.185 ± 0.001	—
LSAM	beta53	0.16	0.158	-0.002 ± 0.002	-0.010 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.100 ± 0.004	0.316 ± 0.009	0.159	-0.001	-0.47	0.0500	0.0500	0.956	94.294 ± 0.734	0.190 ± 0.001	—
LSAM	beta54	0.16	0.163	0.003 ± 0.002	0.018 ± 0.010	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.093 ± 0.004	0.304 ± 0.008	0.165	0.005	3.15	0.0478	0.0481	0.972	94.194 ± 0.740	0.185 ± 0.001	—
LSAM	beta55	0.00	0.000	-0.000 ± 0.002	—	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.062 ± 0.002	—	—	0.002	0.002	—	0.0610	0.0610	0.988	95.395 ± 0.663	0.239 ± 0.002	4.605 ± 0.663
LSAM	beta56	0.00	-0.001	-0.001 ± 0.002	—	0.003 ± 0.000	0.059 ± 0.001	0.003 ± 0.000	0.059 ± 0.002	—	—	-0.003	-0.003	—	0.0561	0.0562	0.987	93.894 ± 0.758	0.228 ± 0.002	6.106 ± 0.758
LSAM	beta57	0.00	-0.006	-0.006 ± 0.002	—	0.004 ± 0.000	0.059 ± 0.001	0.004 ± 0.000	0.059 ± 0.002	—	—	-0.005	-0.005	—	0.0588	0.0590	0.992	94.194 ± 0.740	0.230 ± 0.001	5.806 ± 0.740
LSAM	beta58	0.00	-0.003	-0.003 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	—	—	-0.004	-0.004	—	0.0591	0.0592	0.950	94.094 ± 0.746	0.215 ± 0.001	5.906 ± 0.746
QML	beta41	0.20	0.200	0.000 ± 0.001	0.001 ± 0.007	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.050 ± 0.002	0.223 ± 0.006	0.200	0.000	-0.24	0.0453	0.0453	0.997	95.195 ± 0.677	0.174 ± 0.001	—
QML	beta42	0.20	0.201	0.001 ± 0.001	0.004 ± 0.007	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.053 ± 0.002	0.231 ± 0.006	0.200	0.000	0.14	0.0476	0.0476	0.985	95.195 ± 0.677	0.178 ± 0.001	—
QML	beta43	0.20	0.202	0.002 ± 0.001	0.009 ± 0.007	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	0.051 ± 0.002	0.227 ± 0.006	0.201	0.001	0.69	0.0458	0.0459	0.969	94.294 ± 0.734	0.172 ± 0.001	—
QML	beta44	0.00	0.002	0.002 ± 0.002	—	0.003 ± 0.000	0.054 ± 0.001	0.003 ± 0.000	0.054 ± 0.001	—	—	0.003	0.003	—	0.0543	0.0544	0.939	93.193 ± 0.797	0.198 ± 0.001	6.807 ± 0.797
QML	beta45	0.00	-0.001	-0.001 ± 0.001	—	0.002 ± 0.000	0.045 ± 0.001	0.002 ± 0.000	0.045 ± 0.001	—	—	-0.002	-0.002	—	0.0418	0.0419	1.013	95.395 ± 0.663	0.178 ± 0.001	4.605 ± 0.663
QML	beta46	0.00	-0.006	-0.006 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	-0.008	-0.008	—	0.0444	0.0452	1.019	93.894 ± 0.758	0.185 ± 0.001	6.106 ± 0.758
QML	beta47	0.00	-0.006	-0.006 ± 0.001	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	—	—	-0.009	-0.009	—	0.0455	0.0463	1.004	93.694 ± 0.769	0.185 ± 0.003	6.306 ± 0.769
QML	beta51	0.16	0.159	-0.001 ± 0.002	-0.003 ± 0.013	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.157 ± 0.007	0.396 ± 0.011	0.157	-0.003	-2.10	0.0655	0.0656	0.938	93.894 ± 0.758	0.233 ± 0.001	—
QML	beta52	0.16	0.162	0.002 ± 0.002	0.010 ± 0.009	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.090 ± 0.004	0.300 ± 0.008	0.162	0.002	1.29	0.0480	0.0480	0.989	94.394 ± 0.728	0.186 ± 0.001	—
QML	beta53	0.16	0.159	-0.001 ± 0.002	-0.006 ± 0.010	0.003 ± 0.000	0.050 ± 0.001	0.003 ± 0.000	0.050 ± 0.001	0.098 ± 0.004	0.313 ± 0.009	0.159	-0.001	-0.85	0.0493	0.0494	0.970	94.394 ± 0.728	0.190 ± 0.001	—
QML	beta54	0.16	0.162	0.002 ± 0.002	0.015 ± 0.010	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.091 ± 0.004	0.301 ± 0.008	0.165	0.005	2.90	0.0476	0.0479	0.973	94.595 ± 0.715	0.184 ± 0.001	—
QML	beta55	0.00	-0.002	-0.002 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	—	—	-0.001	-0.001	—	0.0597	0.0597	1.036	96.096 ± 0.613	0.247 ± 0.005	3.904 ± 0.613
QML	beta56	0.00	-0.003	-0.003 ± 0.002	—	0.003 ± 0.000	0.057 ± 0.001	0.003 ± 0.000	0.057 ± 0.002	—	—	-0.003	-0.003	—	0.0557	0.0558	1.020	95.195 ± 0.677	0.229 ± 0.003	4.805 ± 0.677
QML	beta57	0.00	-0.004	-0.004 ± 0.002	—	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	—	—	-0.005	-0.005	—	0.0484	0.0486	1.053	94.094 ± 0.746	0.201 ± 0.009	5.906 ± 0.746
QML	beta58	0.00	-0.002	-0.002 ± 0.001	—	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	—	—	-0.004	-0.004	—	0.0478	0.0480	0.958	93.794 ± 0.763	0.177 ± 0.001	6.206 ± 0.763
UPI	beta41	0.20	0.201	0.001 ± 0.001	0.003 ± 0.007	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.052 ± 0.002	0.228 ± 0.007	0.200	0.000	-0.25	0.0468	0.0468	0.986	94.995 ± 0.690	0.176 ± 0.001	—
UPI	beta42	0.20	0.201	0.001 ± 0.001	0.006 ± 0.007	0.002 ± 0.000	0.047 ± 0.001	0.002 ± 0.000	0.047 ± 0.001	0.055 ± 0.002	0.235 ± 0.006	0.202	0.002	0.80	0.0464	0.0465	0.980	94.895 ± 0.696	0.181 ± 0.001	—
UPI	beta43	0.20	0.202	0.002 ± 0.001	0.010 ± 0.007	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	0.053 ± 0.002	0.230 ± 0.006	0.201	0.001	0.55	0.0457	0.0457	0.969	94.394 ± 0.728	0.175 ± 0.001	—
UPI	beta44	0.00	0.005	0.005 ± 0.002	—	0.003 ± 0.000	0.055 ± 0.001	0.003 ± 0.000	0.055 ± 0.001	—	—	0.003	0.003	—	0.0561	0.0562	0.932	93.093 ± 0.802	0.202 ± 0.001	6.907 ± 0.802
UPI	beta45	0.00	0.000	0.000 ± 0.001	—	0.002 ± 0.000	0.046 ± 0.001	0.002 ± 0.000	0.046 ± 0.001	—	—	-0.002	-0.002	—	0.0452	0.0452	1.001	94.795 ± 0.703	0.182 ± 0.001	5.205 ± 0.703
UPI	beta46	0.00	-0.007	-0.007 ± 0.002	—	0.004 ± 0.000	0.062 ± 0.002	0.004 ± 0.000	0.063 ± 0.002	—	—	-0.008	-0.008	—	0.0606	0.0611	1.026	94.995 ± 0.690	0.250 ± 0.002	5.005 ± 0.690

Table S287

Simulation 2 - Condition 143: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 400, Reliability = 0.8 (continued)

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta47	0.00	-0.006	-0.006 ± 0.002	—	0.004 ± 0.000	0.062 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	—	—	-0.006	-0.006	—	0.0616	0.0619	1.017	95.195 ± 0.677	0.249 ± 0.002	4.805 ± 0.677
UPI	beta51	0.16	0.159	-0.001 ± 0.002	-0.008 ± 0.012	0.004 ± 0.000	0.063 ± 0.001	0.004 ± 0.000	0.063 ± 0.002	0.156 ± 0.007	0.394 ± 0.011	0.157	-0.003	-2.14	0.0641	0.0642	0.930	93.594 ± 0.775	0.230 ± 0.001	—
UPI	beta52	0.16	0.162	0.002 ± 0.002	0.012 ± 0.010	0.002 ± 0.000	0.048 ± 0.001	0.002 ± 0.000	0.048 ± 0.001	0.092 ± 0.004	0.303 ± 0.008	0.161	0.001	0.75	0.0491	0.0491	0.971	93.794 ± 0.763	0.184 ± 0.001	—
UPI	beta53	0.16	0.159	-0.001 ± 0.002	-0.006 ± 0.010	0.003 ± 0.000	0.051 ± 0.001	0.003 ± 0.000	0.051 ± 0.001	0.100 ± 0.004	0.316 ± 0.009	0.160	0.000	-0.11	0.0508	0.0508	0.959	94.995 ± 0.690	0.190 ± 0.001	—
UPI	beta54	0.16	0.163	0.003 ± 0.002	0.019 ± 0.010	0.002 ± 0.000	0.049 ± 0.001	0.002 ± 0.000	0.049 ± 0.001	0.093 ± 0.004	0.305 ± 0.009	0.165	0.005	3.15	0.0489	0.0491	0.968	94.494 ± 0.722	0.185 ± 0.001	—
UPI	beta55	0.00	-0.002	-0.002 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	—	—	0.001	0.001	—	0.0599	0.0599	1.006	95.796 ± 0.635	0.242 ± 0.002	4.204 ± 0.635
UPI	beta56	0.00	-0.001	-0.001 ± 0.002	—	0.003 ± 0.000	0.058 ± 0.001	0.003 ± 0.000	0.058 ± 0.002	—	—	-0.003	-0.003	—	0.0557	0.0557	1.008	94.695 ± 0.709	0.230 ± 0.002	5.305 ± 0.709
UPI	beta57	0.00	-0.003	-0.003 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.002	0.004 ± 0.000	0.062 ± 0.002	—	—	-0.003	-0.003	—	0.0610	0.0610	1.001	95.796 ± 0.635	0.241 ± 0.001	4.204 ± 0.635
UPI	beta58	0.00	0.000	0.000 ± 0.002	—	0.004 ± 0.000	0.061 ± 0.001	0.004 ± 0.000	0.061 ± 0.002	—	—	0.000	0.000	—	0.0600	0.0600	0.973	95.395 ± 0.663	0.232 ± 0.002	4.605 ± 0.663

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u2014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Table S288

Simulation 2 - Condition 144: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.8

Method	Param.	True	Mean-based									Median-based					Inference			
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
LSAM	beta41	0.20	0.200	-0.000 ± 0.001	-0.001 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.020 ± 0.001	0.143 ± 0.004	0.201	0.001	0.41	0.0295	0.0295	0.991	95.200 ± 0.676	0.111 ± 0.000	—
LSAM	beta42	0.20	0.200	-0.000 ± 0.001	-0.001 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.021 ± 0.001	0.146 ± 0.004	0.200	0.000	-0.04	0.0302	0.0302	1.000	95.100 ± 0.683	0.114 ± 0.000	—
LSAM	beta43	0.20	0.200	-0.000 ± 0.001	-0.002 ± 0.004	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	0.019 ± 0.001	0.137 ± 0.004	0.199	-0.001	-0.31	0.0285	0.0285	1.019	95.000 ± 0.689	0.110 ± 0.000	—
LSAM	beta44	0.00	0.003	0.003 ± 0.001	—	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	0.002	0.002	—	0.0309	0.0310	1.023	95.000 ± 0.689	0.123 ± 0.000	5.000 ± 0.689
LSAM	beta45	0.00	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	0.000	0.000	—	0.0276	0.0276	1.016	95.000 ± 0.689	0.112 ± 0.000	5.000 ± 0.689
LSAM	beta46	0.00	-0.008	-0.008 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.008	-0.008	—	0.0366	0.0375	0.979	92.600 ± 0.828	0.140 ± 0.000	7.400 ± 0.828
LSAM	beta47	0.00	-0.006	-0.006 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	—	—	-0.006	-0.006	—	0.0362	0.0366	0.972	94.300 ± 0.733	0.140 ± 0.000	5.700 ± 0.733
LSAM	beta51	0.16	0.160	0.000 ± 0.001	0.001 ± 0.007	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.054 ± 0.002	0.233 ± 0.006	0.160	0.000	0.00	0.0377	0.0377	0.985	94.700 ± 0.708	0.144 ± 0.000	—
LSAM	beta52	0.16	0.161	0.001 ± 0.001	0.005 ± 0.006	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.034 ± 0.002	0.185 ± 0.005	0.160	0.000	-0.01	0.0290	0.0290	1.000	94.900 ± 0.696	0.116 ± 0.000	—
LSAM	beta53	0.16	0.159	-0.001 ± 0.001	-0.003 ± 0.006	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.034 ± 0.001	0.186 ± 0.005	0.159	-0.001	-0.58	0.0304	0.0305	1.029	95.700 ± 0.641	0.120 ± 0.000	—
LSAM	beta54	0.16	0.159	-0.001 ± 0.001	-0.008 ± 0.006	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.035 ± 0.001	0.187 ± 0.005	0.159	-0.001	-0.45	0.0314	0.0314	0.989	95.400 ± 0.662	0.116 ± 0.000	—
LSAM	beta55	0.00	0.003	0.003 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	0.003	0.003	—	0.0410	0.0412	0.956	93.800 ± 0.763	0.148 ± 0.001	6.200 ± 0.763
LSAM	beta56	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	0.002	0.002	—	0.0345	0.0345	0.988	95.700 ± 0.641	0.141 ± 0.001	4.300 ± 0.641
LSAM	beta57	0.00	-0.008	-0.008 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.008	-0.008	—	0.0384	0.0392	0.992	94.500 ± 0.721	0.143 ± 0.000	5.500 ± 0.721
LSAM	beta58	0.00	-0.005	-0.005 ± 0.001	—	0.001 ± 0.000	0.034 ± 0.001	0.001 ± 0.000	0.034 ± 0.001	—	—	-0.004	-0.004	—	0.0336	0.0338	1.007	94.100 ± 0.745	0.135 ± 0.000	5.900 ± 0.745
QML	beta41	0.20	0.199	-0.001 ± 0.001	-0.004 ± 0.004	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.020 ± 0.001	0.141 ± 0.004	0.200	0.000	-0.03	0.0291	0.0291	0.999	95.100 ± 0.683	0.110 ± 0.001	—
QML	beta42	0.20	0.200	-0.000 ± 0.001	-0.002 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.021 ± 0.001	0.143 ± 0.004	0.200	0.000	0.03	0.0297	0.0297	1.009	95.100 ± 0.683	0.113 ± 0.001	—
QML	beta43	0.20	0.200	-0.000 ± 0.001	-0.001 ± 0.004	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	0.019 ± 0.001	0.136 ± 0.004	0.199	-0.001	-0.27	0.0292	0.0292	1.030	95.600 ± 0.649	0.110 ± 0.001	—
QML	beta44	0.00	0.000	0.000 ± 0.001	—	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	0.000	0.000	—	0.0309	0.0309	1.027	95.500 ± 0.656	0.124 ± 0.001	4.500 ± 0.656
QML	beta45	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	-0.001	-0.001	—	0.0278	0.0278	1.020	95.500 ± 0.656	0.112 ± 0.000	4.500 ± 0.656
QML	beta46	0.00	-0.006	-0.006 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	-0.006	-0.006	—	0.0302	0.0309	0.999	92.100 ± 0.853	0.116 ± 0.001	7.900 ± 0.853
QML	beta47	0.00	-0.005	-0.005 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	—	—	-0.006	-0.006	—	0.0280	0.0286	0.980	93.300 ± 0.791	0.115 ± 0.001	6.700 ± 0.791
QML	beta51	0.16	0.160	0.000 ± 0.001	0.002 ± 0.007	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.054 ± 0.002	0.232 ± 0.006	0.160	0.000	-0.19	0.0372	0.0372	1.032	94.800 ± 0.702	0.150 ± 0.005	—
QML	beta52	0.16	0.160	0.000 ± 0.001	0.003 ± 0.006	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.033 ± 0.002	0.183 ± 0.005	0.160	0.000	-0.26	0.0279	0.0279	1.020	94.900 ± 0.696	0.117 ± 0.002	—
QML	beta53	0.16	0.160	-0.000 ± 0.001	-0.002 ± 0.006	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.034 ± 0.001	0.185 ± 0.005	0.159	-0.001	-0.43	0.0305	0.0305	1.027	95.400 ± 0.662	0.119 ± 0.000	—
QML	beta54	0.16	0.159	-0.001 ± 0.001	-0.009 ± 0.006	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.034 ± 0.001	0.185 ± 0.005	0.159	-0.001	-0.76	0.0304	0.0304	1.031	95.600 ± 0.649	0.119 ± 0.005	—
QML	beta55	0.00	0.002	0.002 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	0.003	0.003	—	0.0405	0.0406	0.965	93.600 ± 0.774	0.150 ± 0.001	6.400 ± 0.774
QML	beta56	0.00	-0.001	-0.001 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	0.001	0.001	—	0.0346	0.0346	1.001	95.400 ± 0.662	0.140 ± 0.001	4.600 ± 0.662
QML	beta57	0.00	-0.007	-0.007 ± 0.001	—	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	-0.007	-0.007	—	0.0311	0.0319	1.004	93.700 ± 0.768	0.118 ± 0.001	6.300 ± 0.768
QML	beta58	0.00	-0.004	-0.004 ± 0.001	—	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	—	—	-0.004	-0.004	—	0.0281	0.0285	1.018	93.900 ± 0.757	0.110 ± 0.000	6.100 ± 0.757
UPI	beta41	0.20	0.200	-0.000 ± 0.001	-0.001 ± 0.004	0.001 ± 0.000	0.028 ± 0.001	0.001 ± 0.000	0.028 ± 0.001	0.020 ± 0.001	0.142 ± 0.004	0.200	0.000	0.09	0.0302	0.0302	0.988	95.200 ± 0.676	0.110 ± 0.000	—
UPI	beta42	0.20	0.200	-0.000 ± 0.001	-0.001 ± 0.005	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.021 ± 0.001	0.144 ± 0.004	0.200	0.000	-0.17	0.0295	0.0295	0.998	94.800 ± 0.702	0.113 ± 0.000	—
UPI	beta43	0.20	0.200	0.000 ± 0.001	0.001 ± 0.004	0.001 ± 0.000	0.027 ± 0.001	0.001 ± 0.000	0.027 ± 0.001	0.019 ± 0.001	0.137 ± 0.004	0.199	-0.001	-0.28	0.0285	0.0285	1.020	95.200 ± 0.676	0.110 ± 0.000	—
UPI	beta44	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.031 ± 0.001	0.001 ± 0.000	0.031 ± 0.001	—	—	0.001	0.001	—	0.0311	0.0311	1.019	95.000 ± 0.689	0.125 ± 0.000	5.000 ± 0.689

Table S288

Simulation 2 - Condition 144: Linear Model, Distr = Uniform, Epsilon = Right-skewed, Zeta = Right-skewed, N = 1000, Reliability = 0.8 (continued)

Method	Param.	True	Mean-based									Median-based				Inference				
			Mean	Bias ± MCSE	Rel. Bias ± MCSE	Var. ± MCSE	SD ± MCSE	MSE ± MCSE	RMSE ± MCSE	Rel. MSE ± MCSE	Rel. RMSE ± MCSE	Median	Bias	% Rel. Bias	MAD	RMSE	SE/SD	Cover. (%) ± MCSE	CI Width ± MCSE	Type I (%) ± MCSE
UPI	beta45	0.00	0.000	-0.000 ± 0.001	—	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	—	—	0.000	0.000	—	0.0288	0.0288	1.009	95.700 ± 0.641	0.113 ± 0.000	4.300 ± 0.641
UPI	beta46	0.00	-0.005	-0.005 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.005	-0.005	—	0.0380	0.0383	0.994	94.400 ± 0.727	0.148 ± 0.001	5.600 ± 0.727
UPI	beta47	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.038 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.002	-0.002	—	0.0368	0.0369	0.992	94.600 ± 0.715	0.149 ± 0.001	5.400 ± 0.715
UPI	beta51	0.16	0.160	0.000 ± 0.001	0.001 ± 0.007	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.037 ± 0.001	0.054 ± 0.002	0.232 ± 0.006	0.159	-0.001	-0.44	0.0375	0.0375	0.996	94.700 ± 0.708	0.145 ± 0.001	—
UPI	beta52	0.16	0.161	0.001 ± 0.001	0.004 ± 0.006	0.001 ± 0.000	0.029 ± 0.001	0.001 ± 0.000	0.029 ± 0.001	0.034 ± 0.002	0.184 ± 0.005	0.160	0.000	-0.02	0.0283	0.0283	1.000	94.700 ± 0.708	0.115 ± 0.000	—
UPI	beta53	0.16	0.160	0.000 ± 0.001	0.000 ± 0.006	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.034 ± 0.001	0.186 ± 0.005	0.160	0.000	-0.17	0.0306	0.0306	1.029	95.400 ± 0.662	0.120 ± 0.000	—
UPI	beta54	0.16	0.159	-0.001 ± 0.001	-0.008 ± 0.006	0.001 ± 0.000	0.030 ± 0.001	0.001 ± 0.000	0.030 ± 0.001	0.035 ± 0.001	0.187 ± 0.005	0.159	-0.001	-0.33	0.0311	0.0311	0.987	95.400 ± 0.662	0.116 ± 0.000	—
UPI	beta55	0.00	0.001	0.001 ± 0.001	—	0.002 ± 0.000	0.040 ± 0.001	0.002 ± 0.000	0.040 ± 0.001	—	—	0.002	0.002	—	0.0407	0.0408	0.952	93.600 ± 0.774	0.149 ± 0.001	6.400 ± 0.774
UPI	beta56	0.00	0.001	0.001 ± 0.001	—	0.001 ± 0.000	0.036 ± 0.001	0.001 ± 0.000	0.036 ± 0.001	—	—	0.002	0.002	—	0.0345	0.0345	0.994	95.500 ± 0.656	0.142 ± 0.001	4.500 ± 0.656
UPI	beta57	0.00	-0.004	-0.004 ± 0.001	—	0.001 ± 0.000	0.037 ± 0.001	0.001 ± 0.000	0.038 ± 0.001	—	—	-0.004	-0.004	—	0.0382	0.0385	0.993	95.300 ± 0.669	0.146 ± 0.001	4.700 ± 0.669
UPI	beta58	0.00	-0.002	-0.002 ± 0.001	—	0.001 ± 0.000	0.035 ± 0.001	0.001 ± 0.000	0.035 ± 0.001	—	—	-0.002	-0.002	—	0.0343	0.0344	1.008	95.000 ± 0.689	0.140 ± 0.001	5.000 ± 0.689

Note. MCSE = Monte Carlo Standard Error shown as \u00b1 value. Coverage, Type I Error, and Power expressed as percentages. \u002014 indicates relative measures are undefined when true value equals 0 and non-existent results for Type I error rate which was evaluated only for nonlinear effects.

Additional Distributional Conditions: Differences from Baseline

Figure S1

Additional Distributional Conditions Difference: Relative Bias for Simulation 1 (Epsilon)

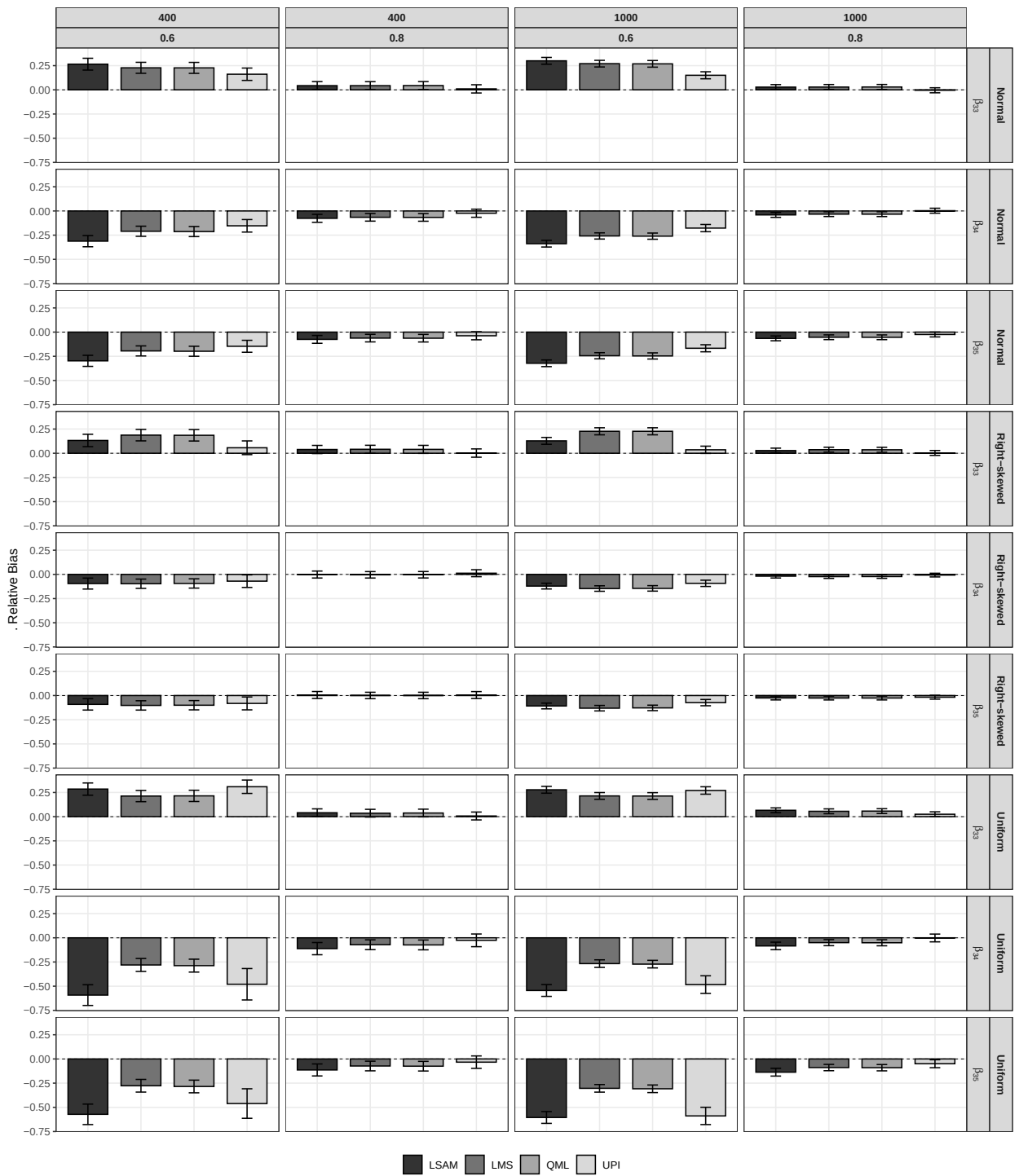
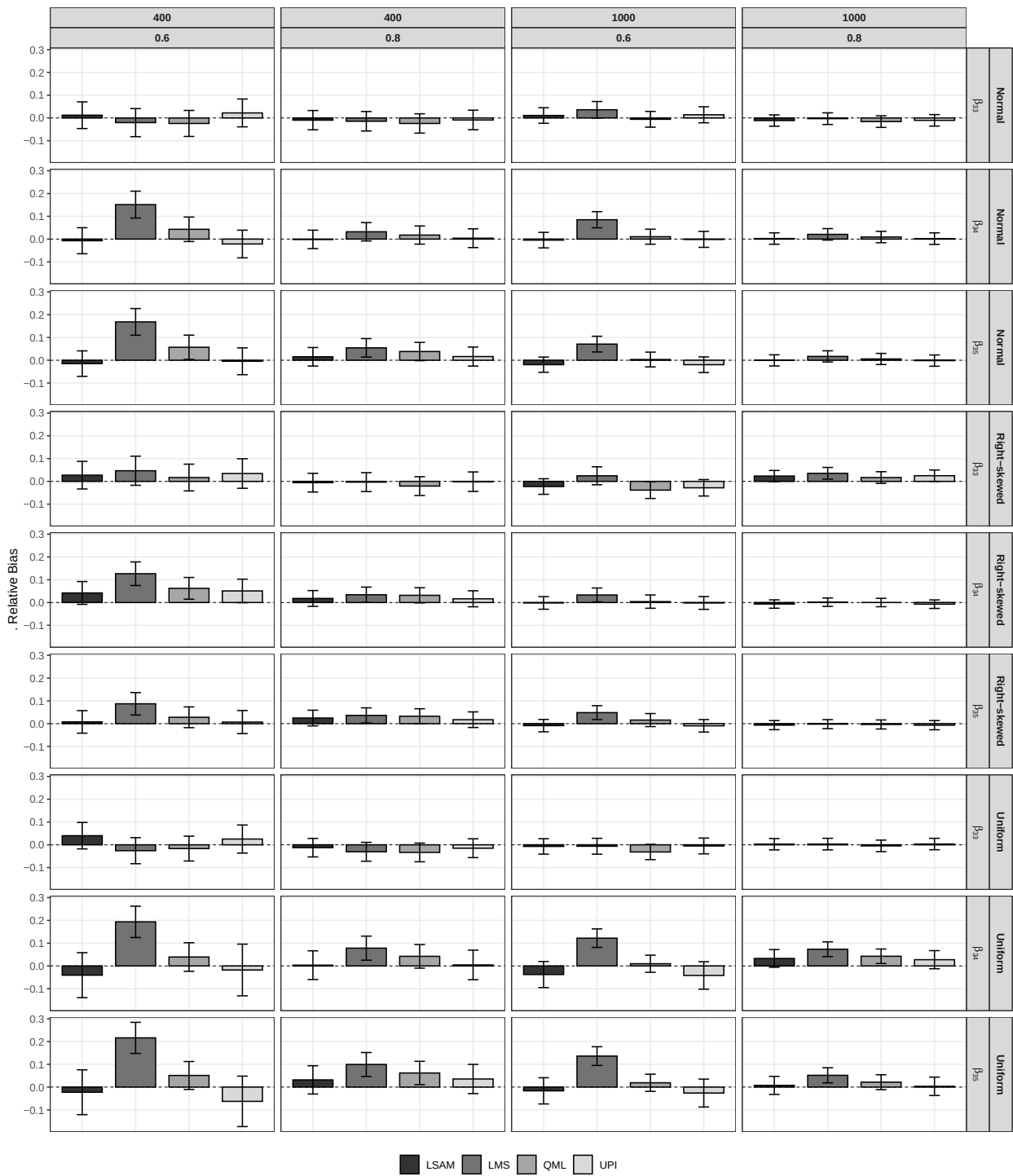


Figure S2

Additional Distributional Conditions Difference: Relative Bias for Simulation 1 (Zeta)



Additional Distributional Conditions Difference: Relative Bias for Simulation 1 (Epsilon & Zeta)

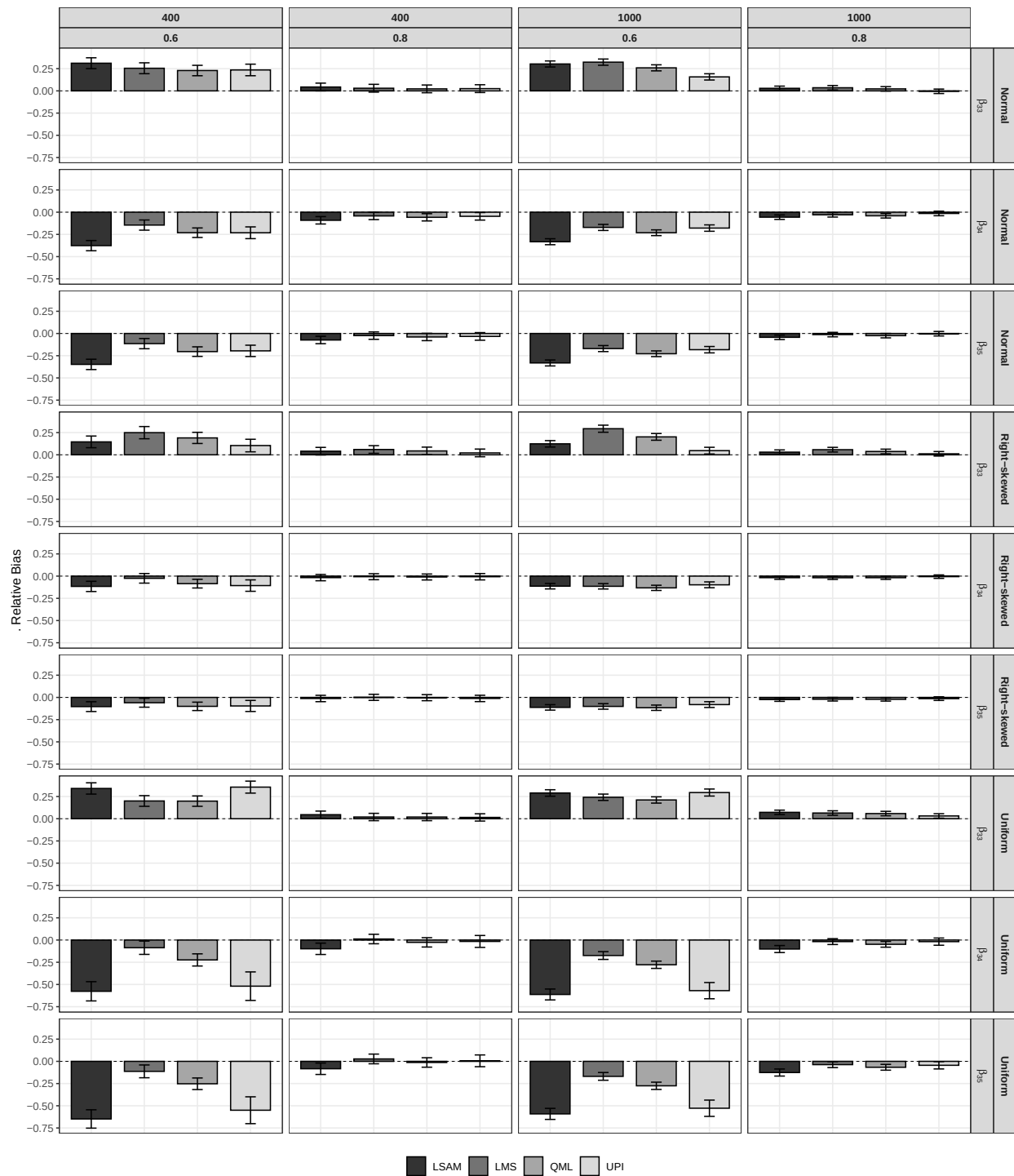


Figure S4

Additional Distributional Conditions Difference: Relative RMSE for Simulation 1 (Epsilon)

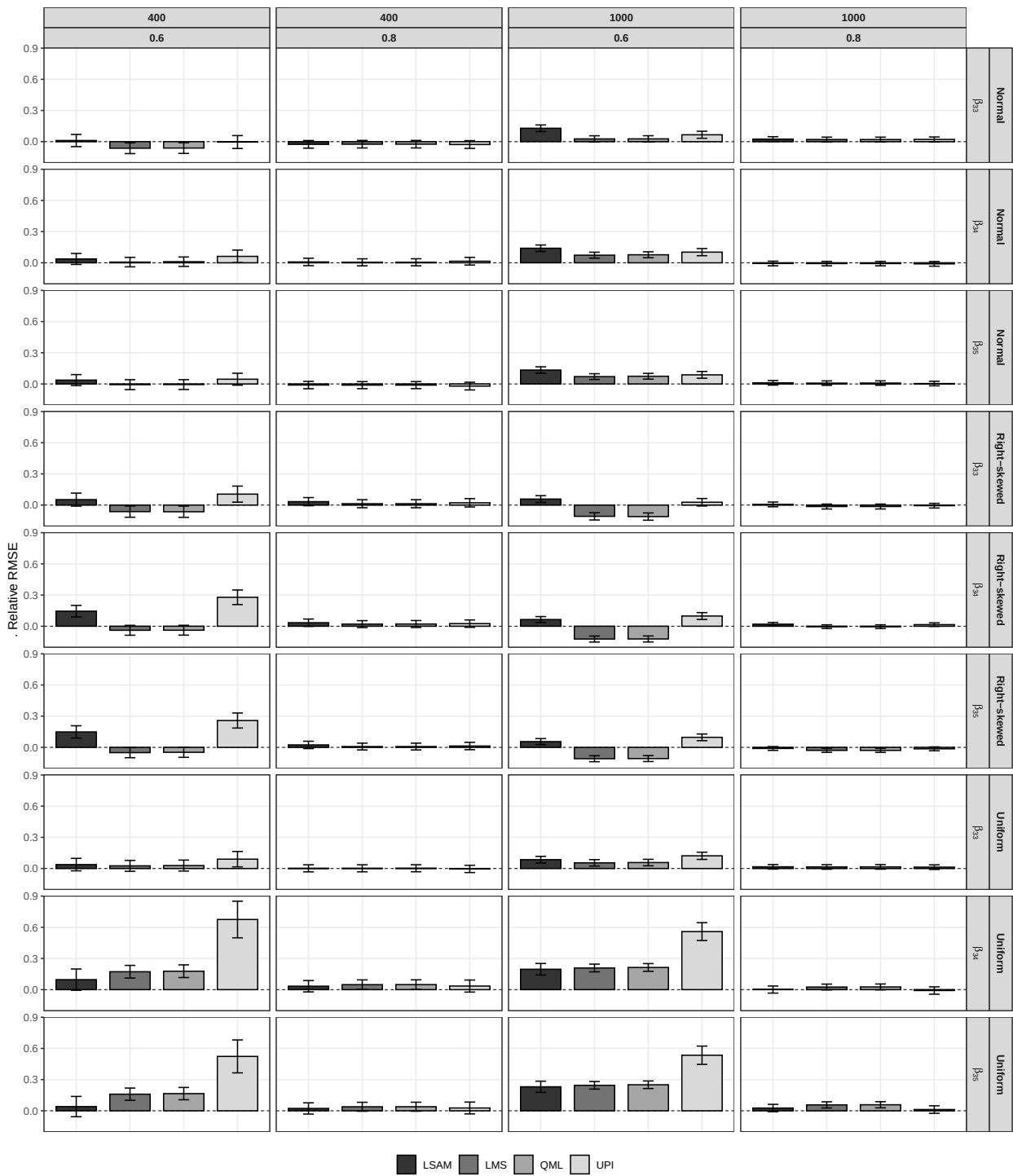


Figure S5

Additional Distributional Conditions Difference: Relative RMSE for Simulation 1 (Zeta)

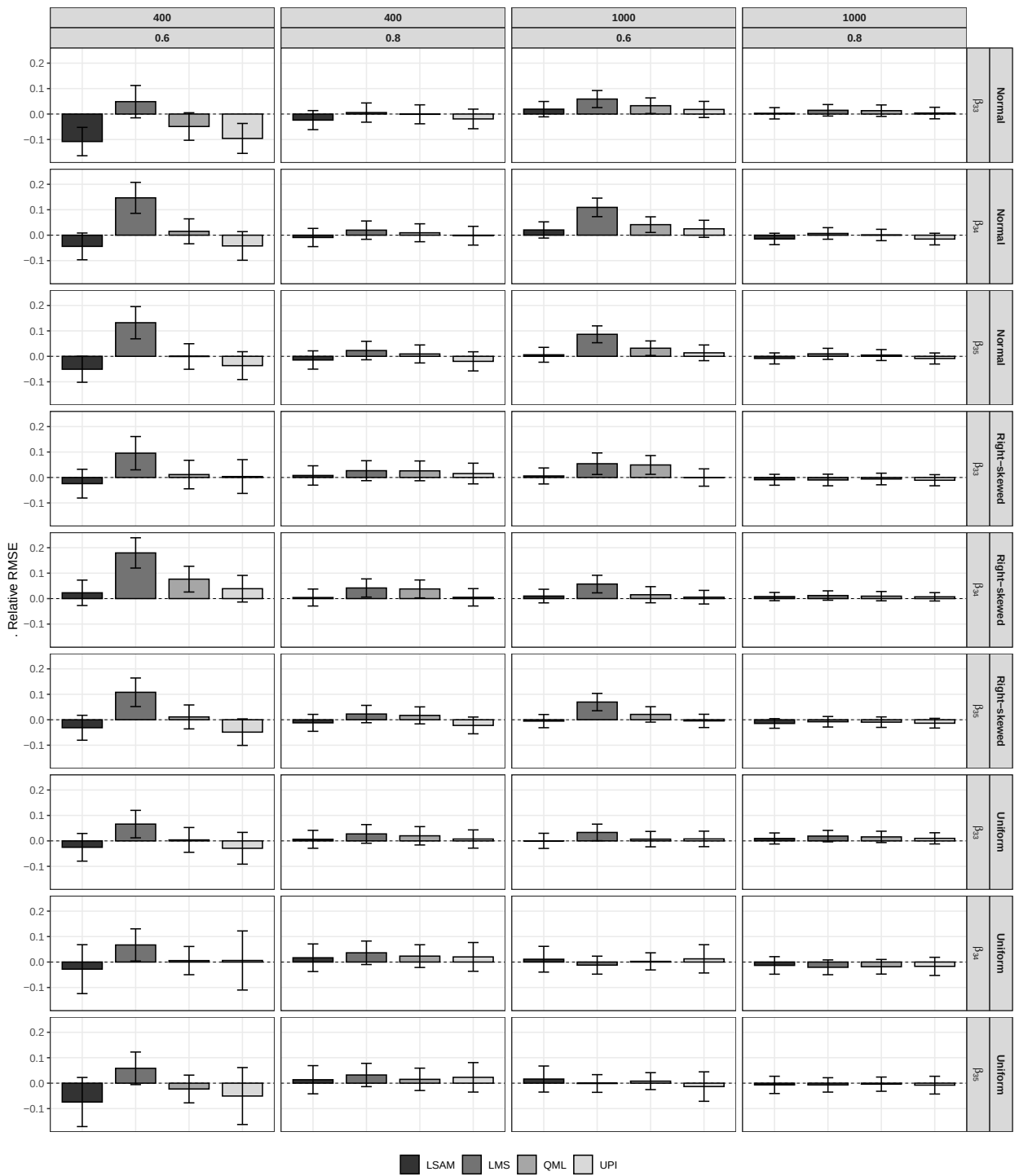


Figure S6

Additional Distributional Conditions Difference: Relative RMSE for Simulation 1 (Epsilon & Zeta)

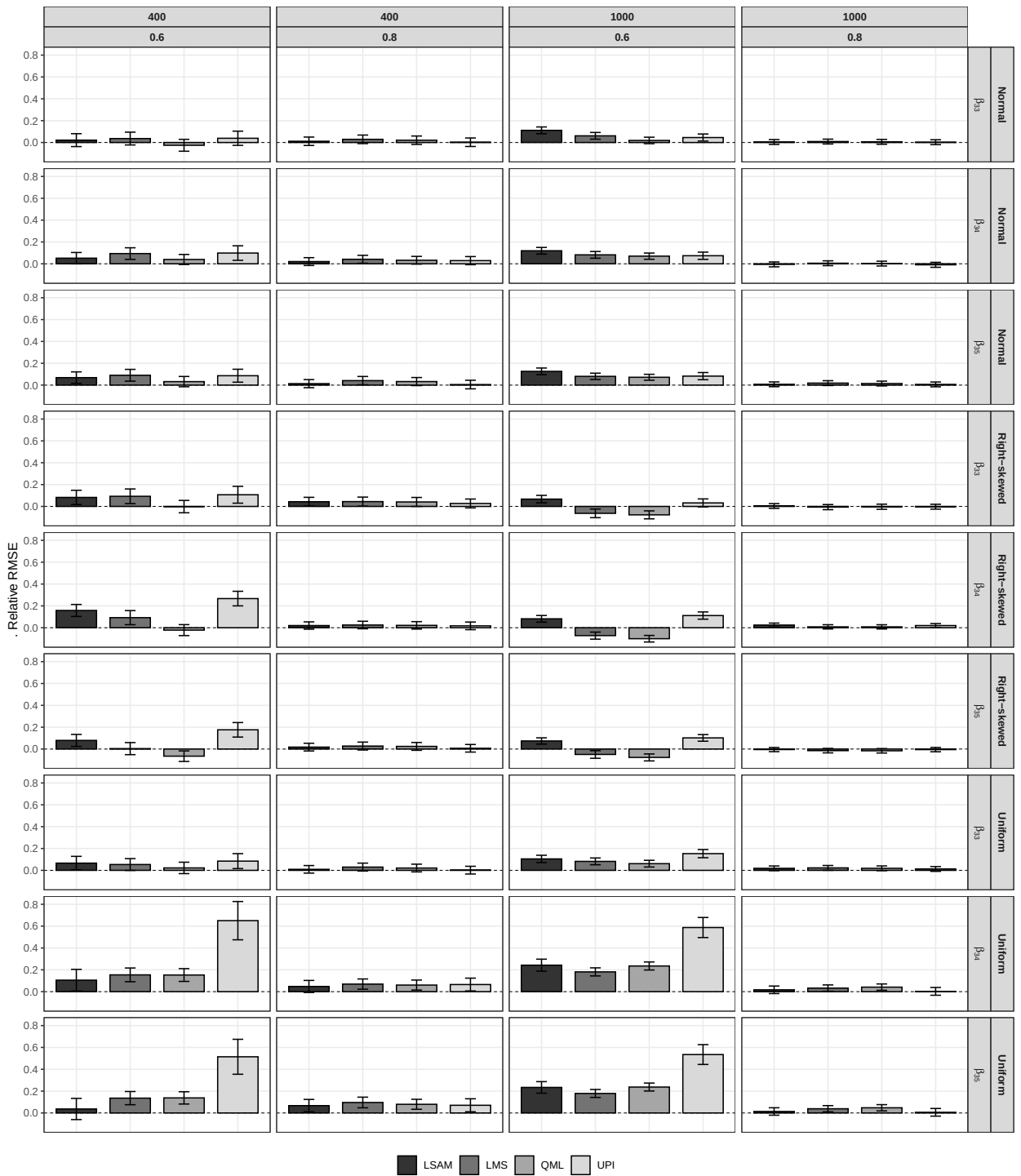


Figure S7

Additional Distributional Conditions Difference: Coverage Rate for Simulation 1 (Epsilon)

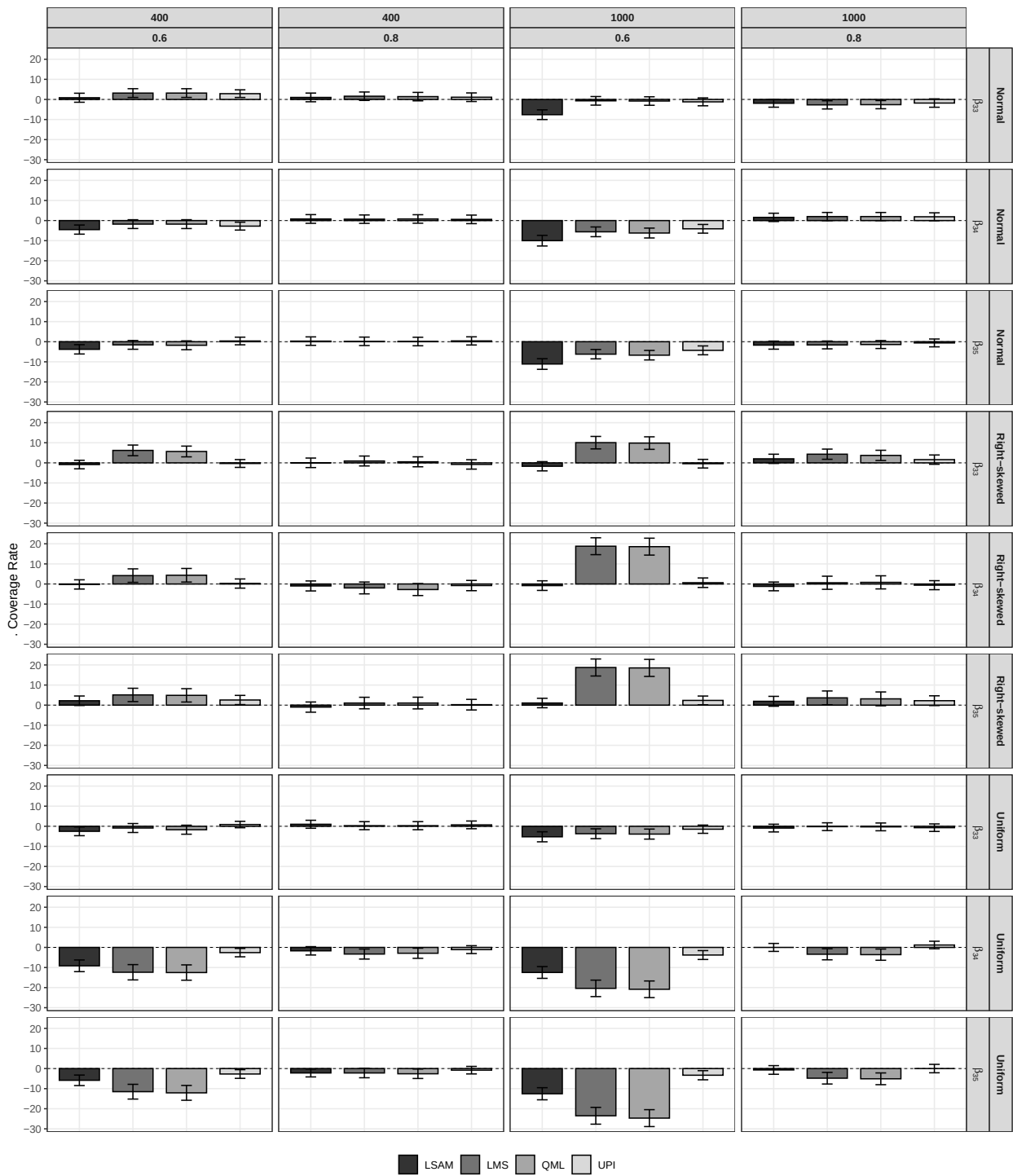


Figure S8

Additional Distributional Conditions Difference: Coverage Rate for Simulation 1 (Zeta)

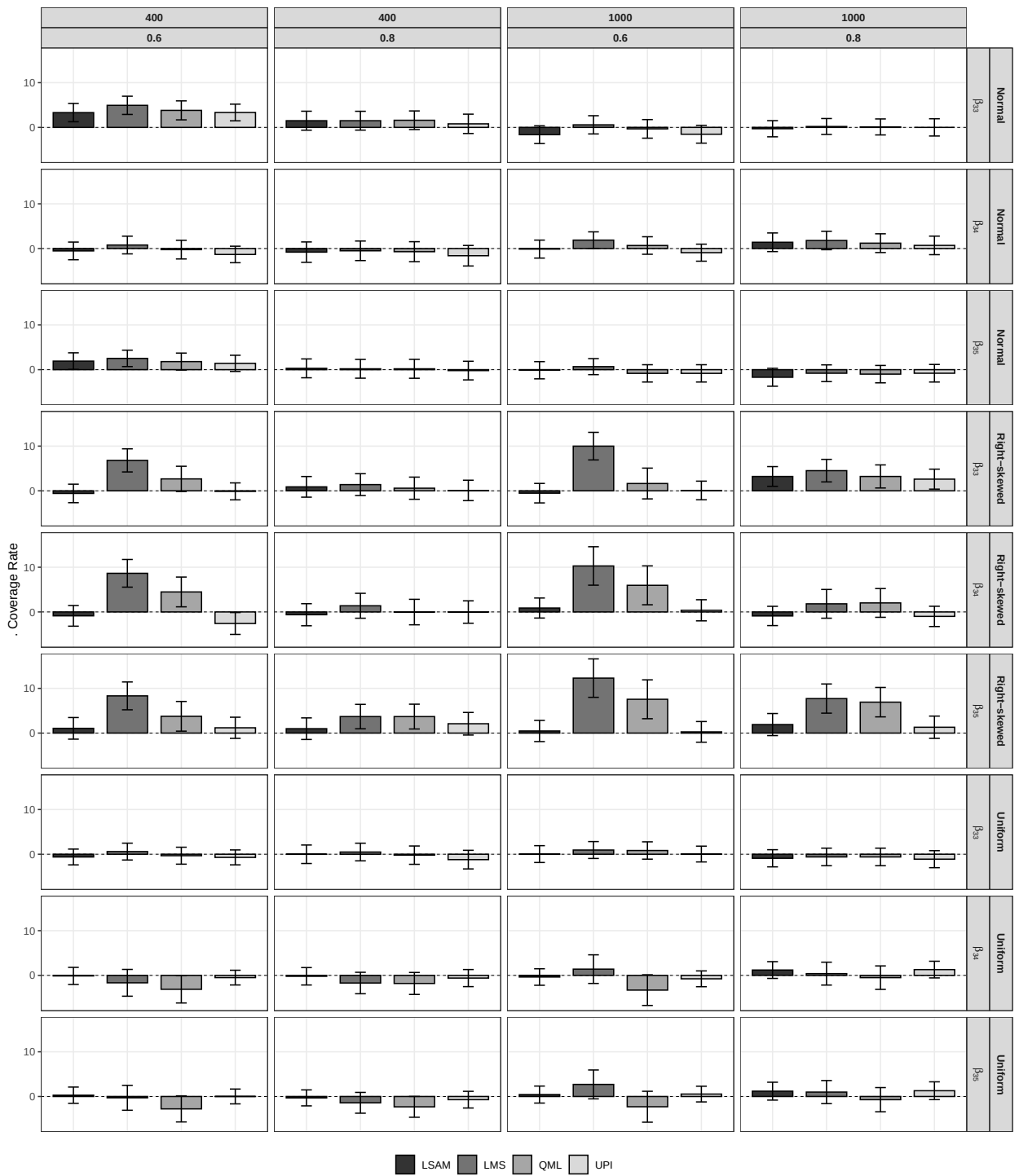


Figure S9

Additional Distributional Conditions Difference: Coverage Rate for Simulation 1 (Epsilon & Zeta)

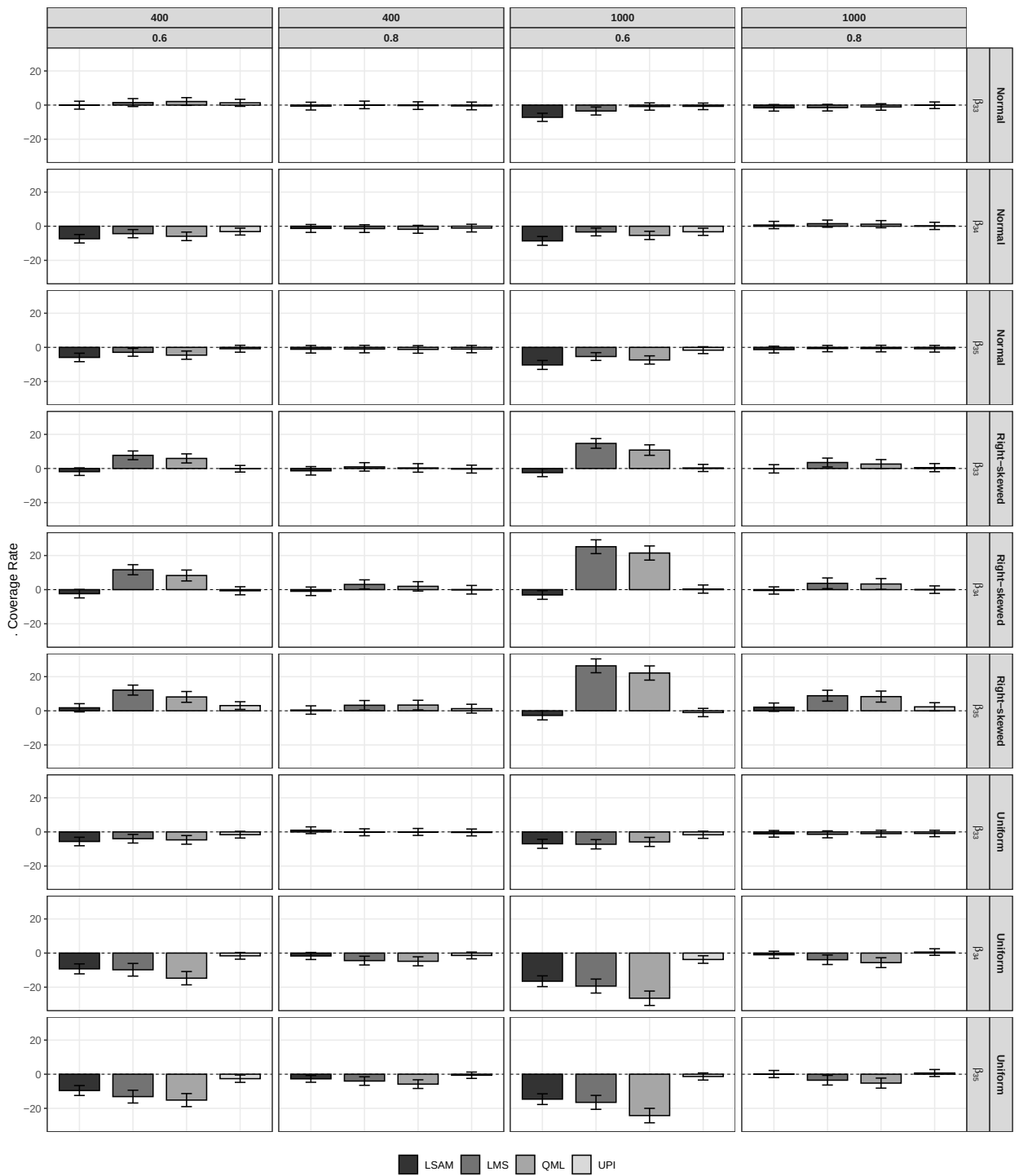


Figure S10

Additional Distributional Conditions Difference: SE/SD Ratio for Simulation 1 (Epsilon)

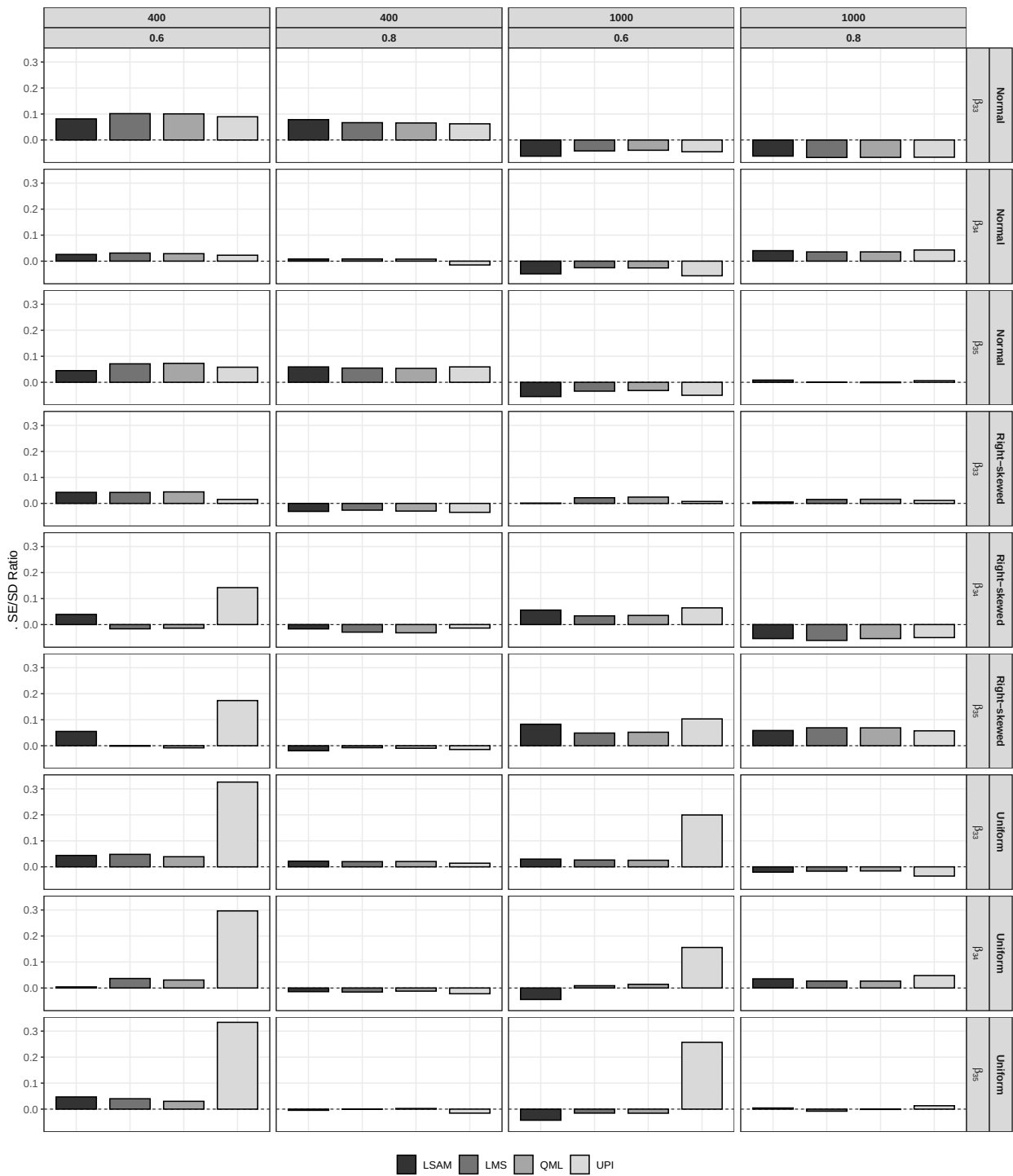


Figure S11

Additional Distributional Conditions Difference: SE/SD Ratio for Simulation 1 (Zeta)

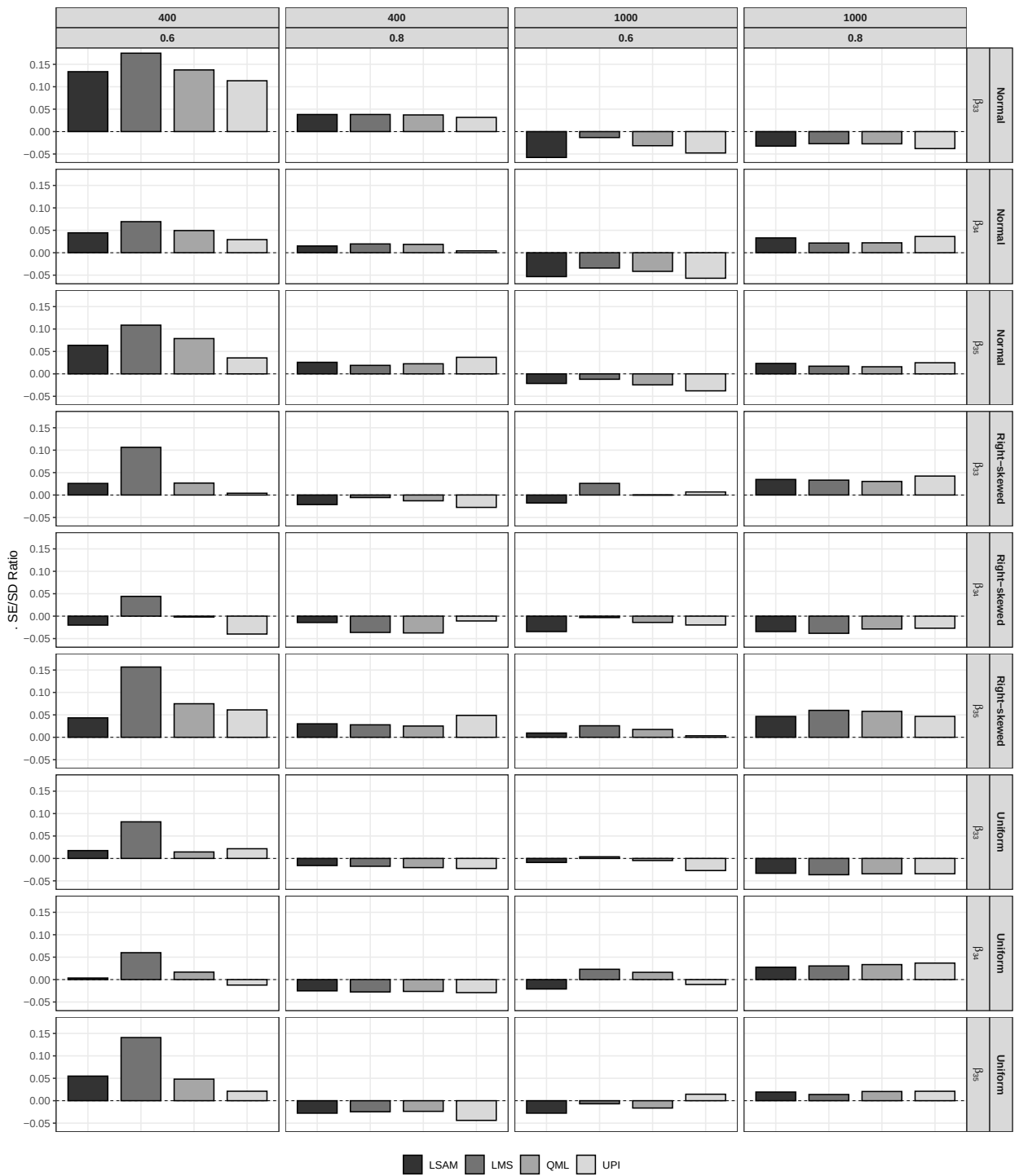


Figure S12

Additional Distributional Conditions Difference: SE/SD Ratio for Simulation 1 (Epsilon & Zeta)

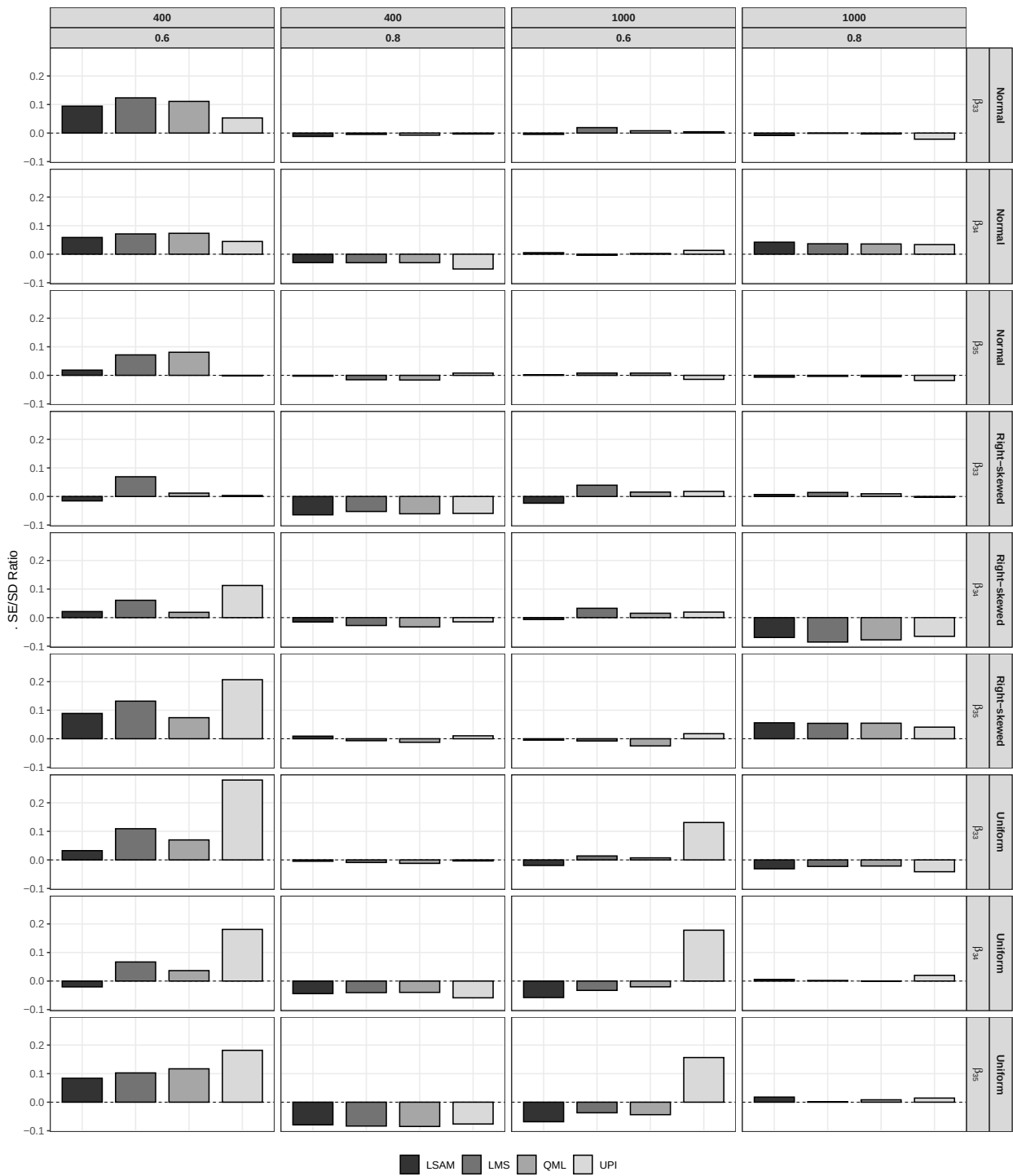


Figure S13

Additional Distributional Conditions Difference: Power for Simulation 1 (Epsilon)

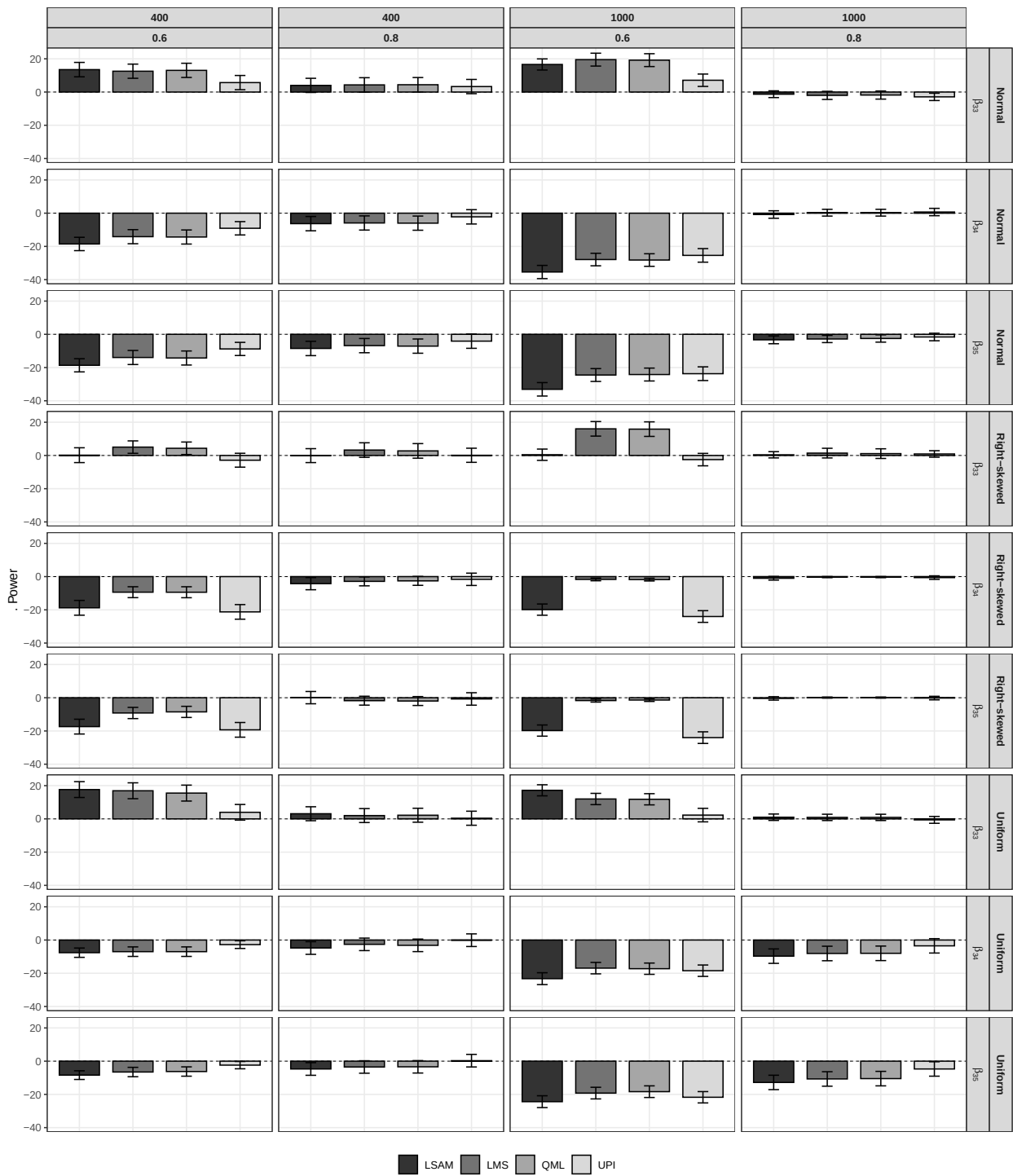
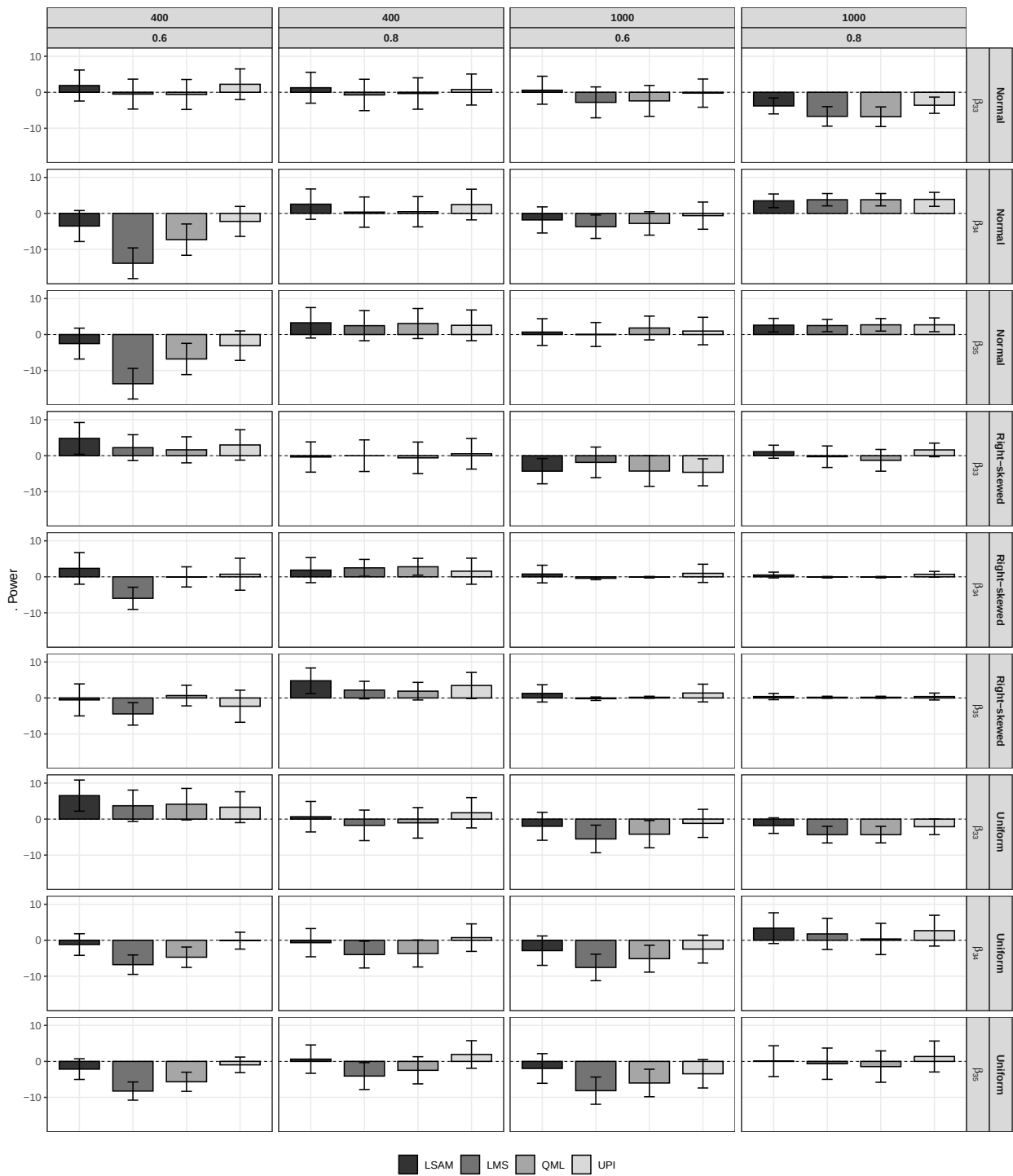


Figure S14

Additional Distributional Conditions Difference: Power for Simulation 1 (Zeta)



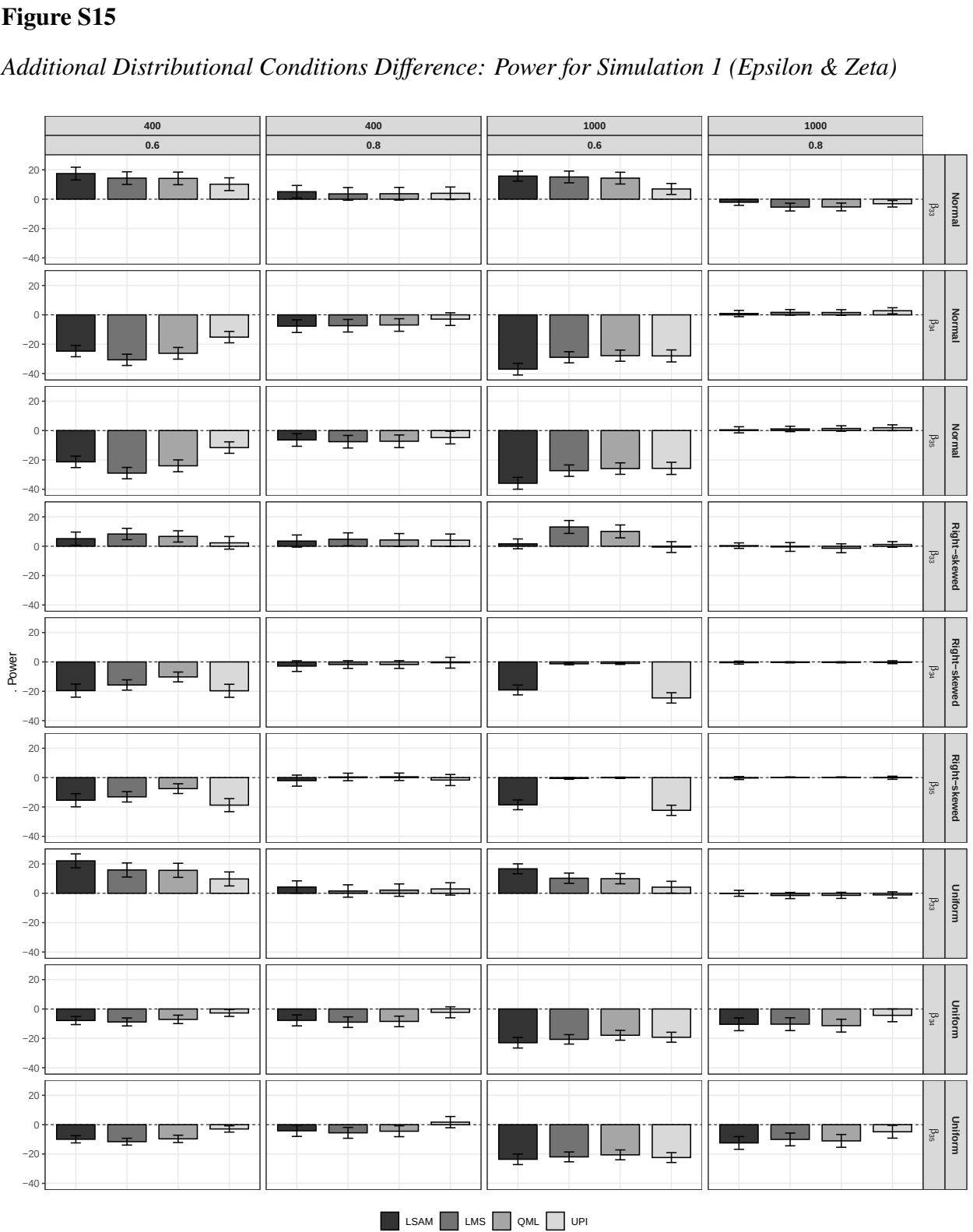


Figure S16

Additional Distributional Conditions Difference: Type I Error for Simulation 1 (Epsilon)

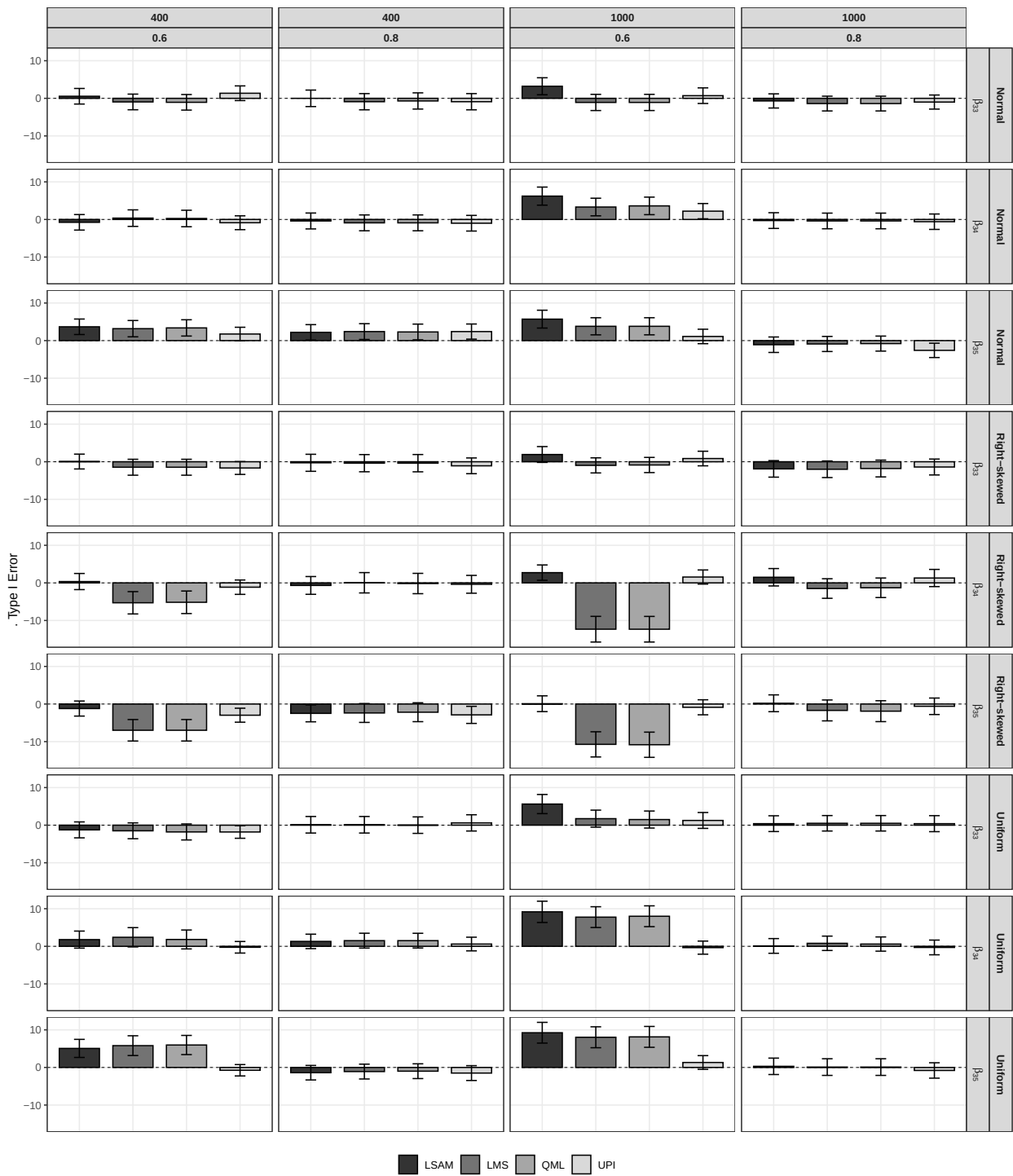


Figure S17

Additional Distributional Conditions Difference: Type I Error for Simulation 1 (Zeta)

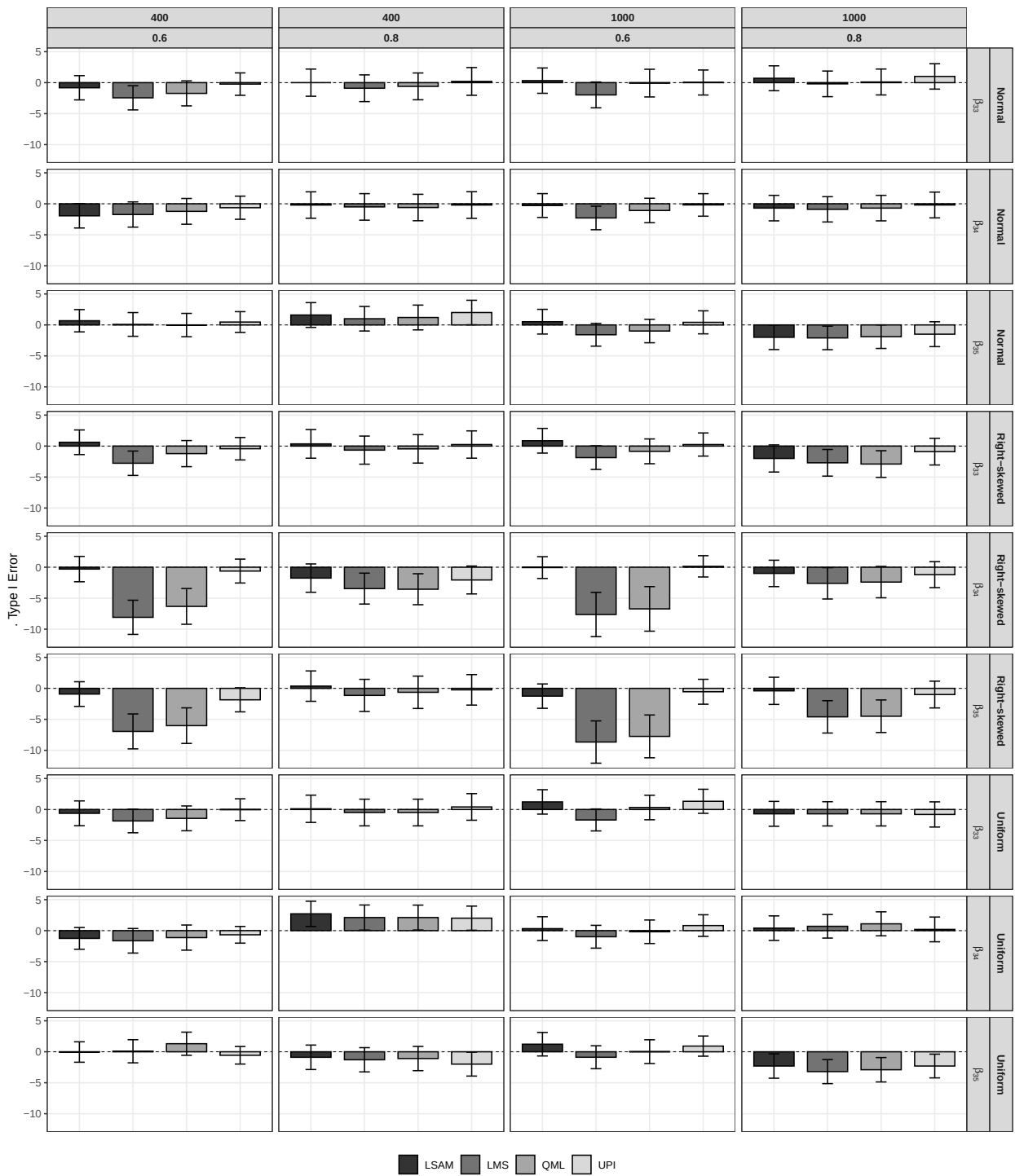


Figure S18

Additional Distributional Conditions Difference: Type I Error for Simulation 1 (Epsilon & Zeta)

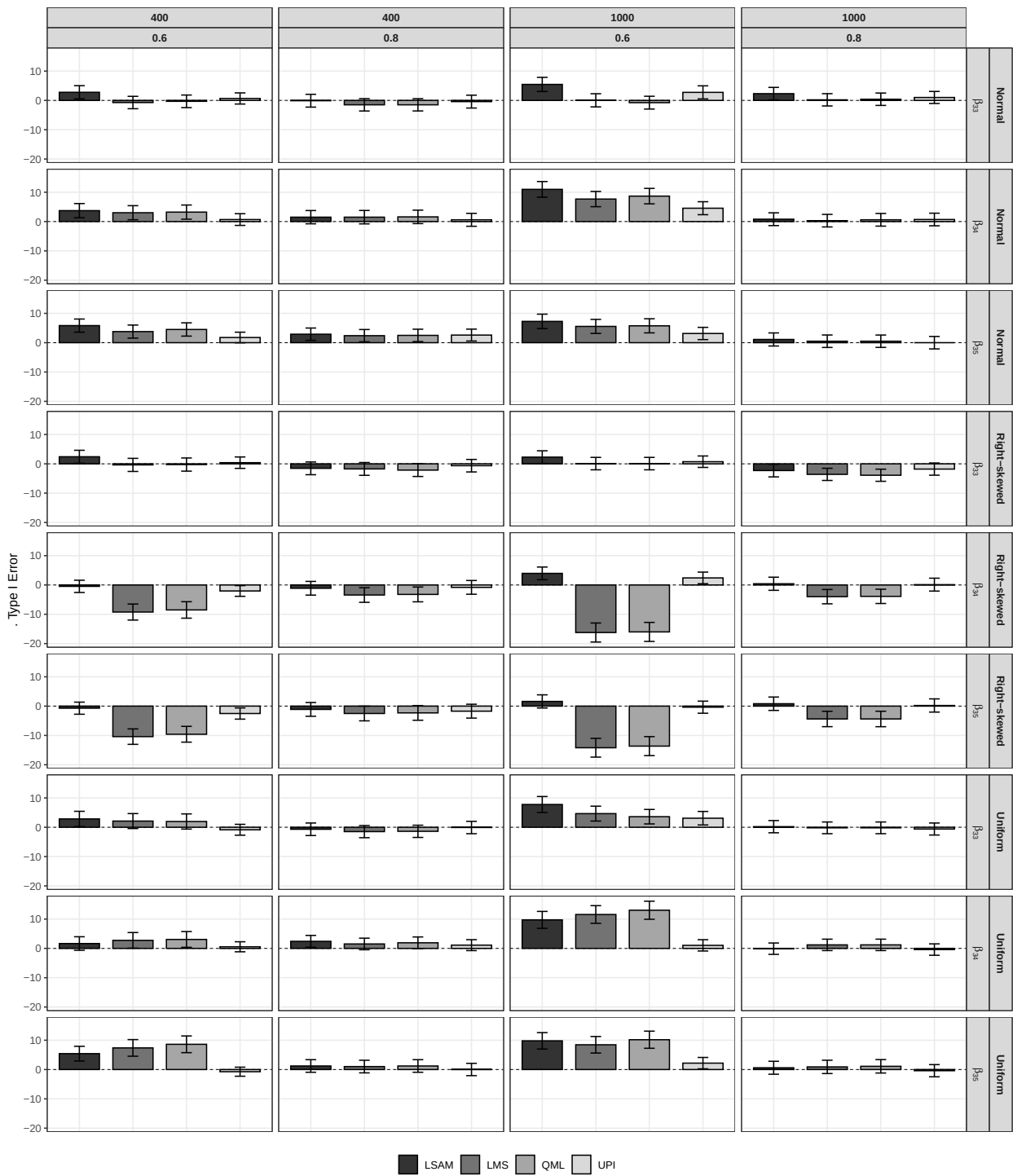


Figure S19

Additional Distributional Conditions Difference: Relative Bias for Simulation 2 – Normal (Epsilon)

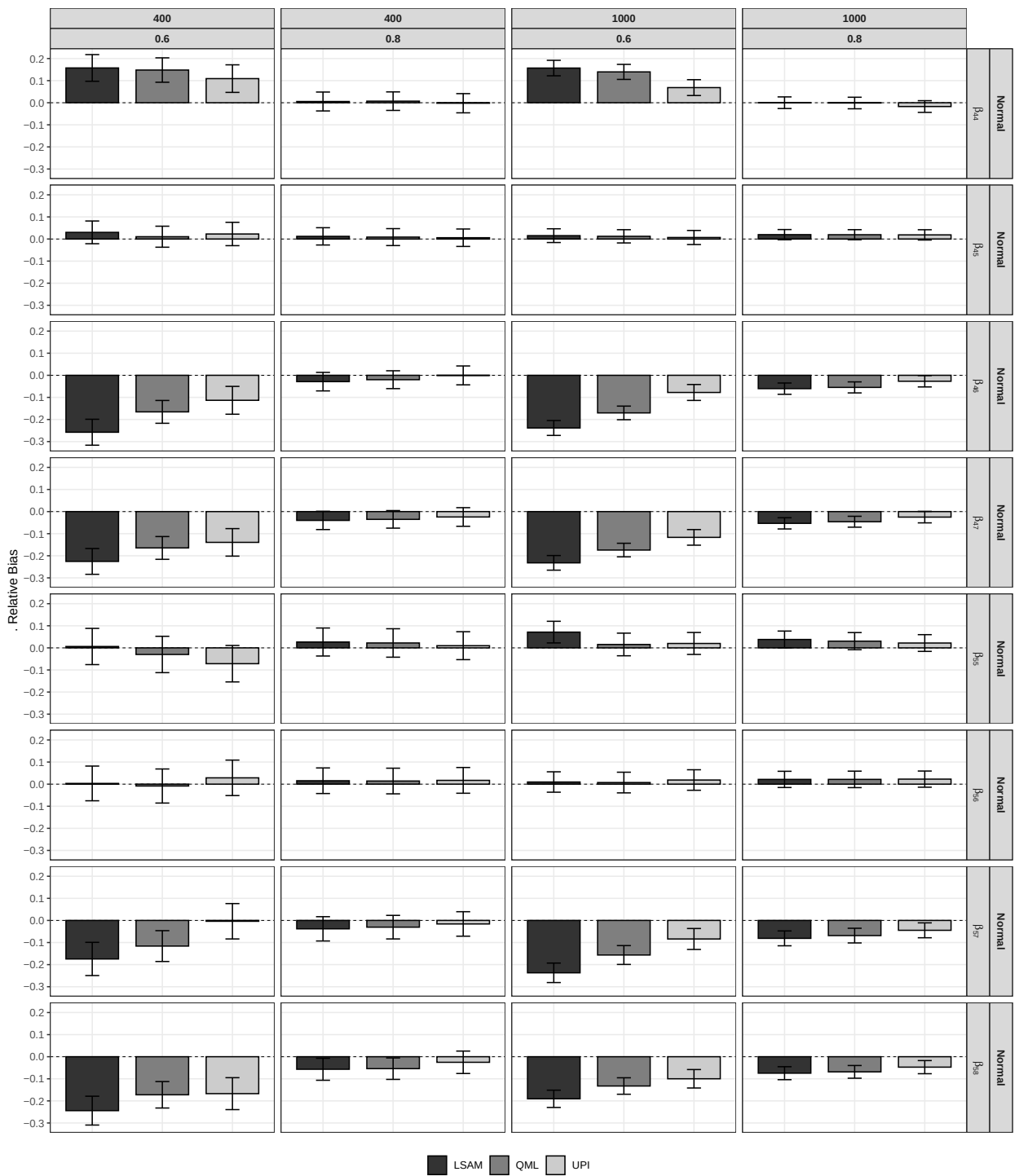


Figure S20

Additional Distributional Conditions Difference: Relative Bias for Simulation 2 – Normal (Zeta)

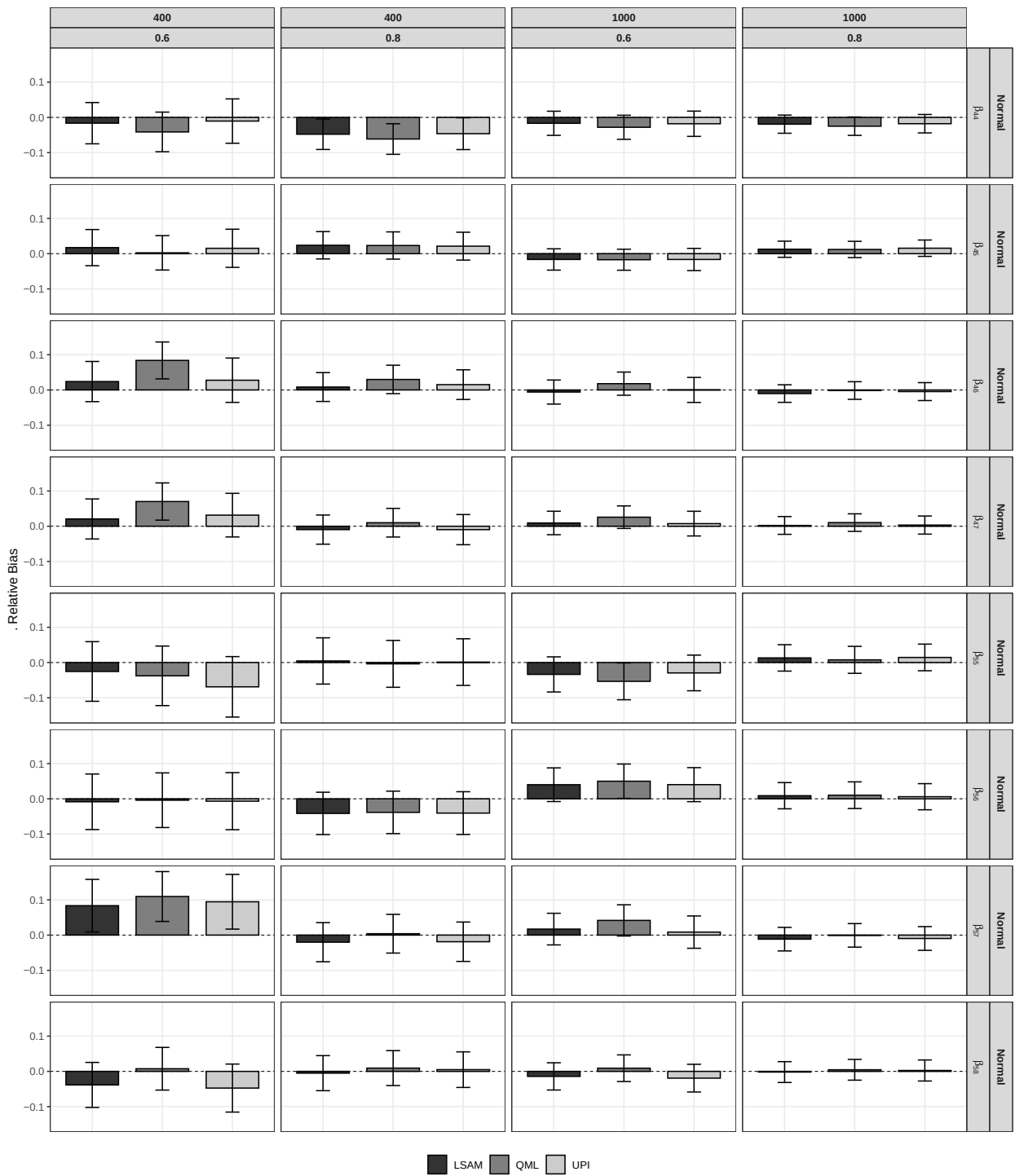


Figure S21

Additional Distributional Conditions Difference: Relative Bias for Simulation 2 – Normal
(Epsilon & Zeta)

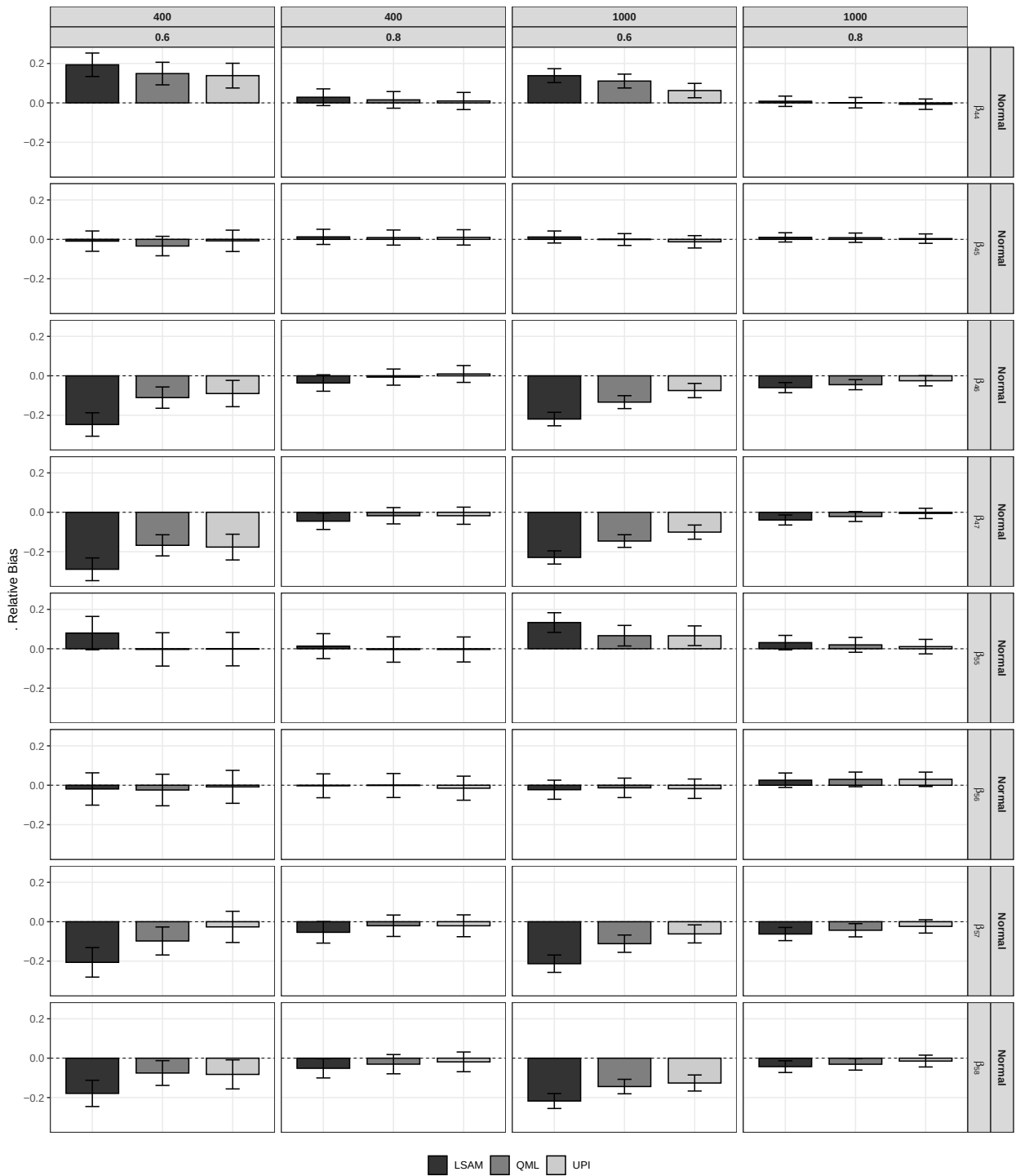


Figure S22

Additional Distributional Conditions Difference: Relative RMSE for Simulation 2 – Normal (Epsilon)

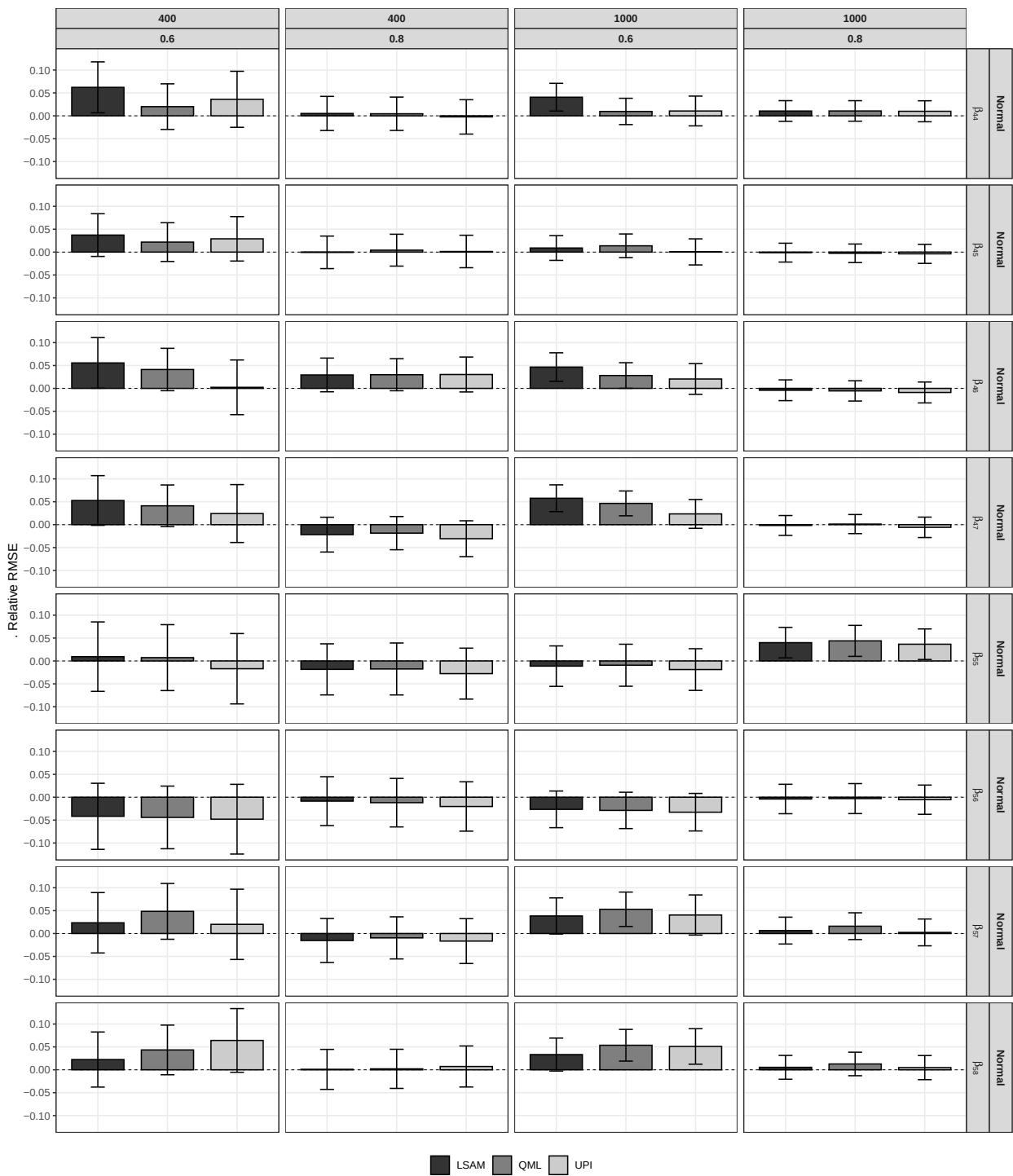


Figure S23

Additional Distributional Conditions Difference: Relative RMSE for Simulation 2 – Normal (Zeta)

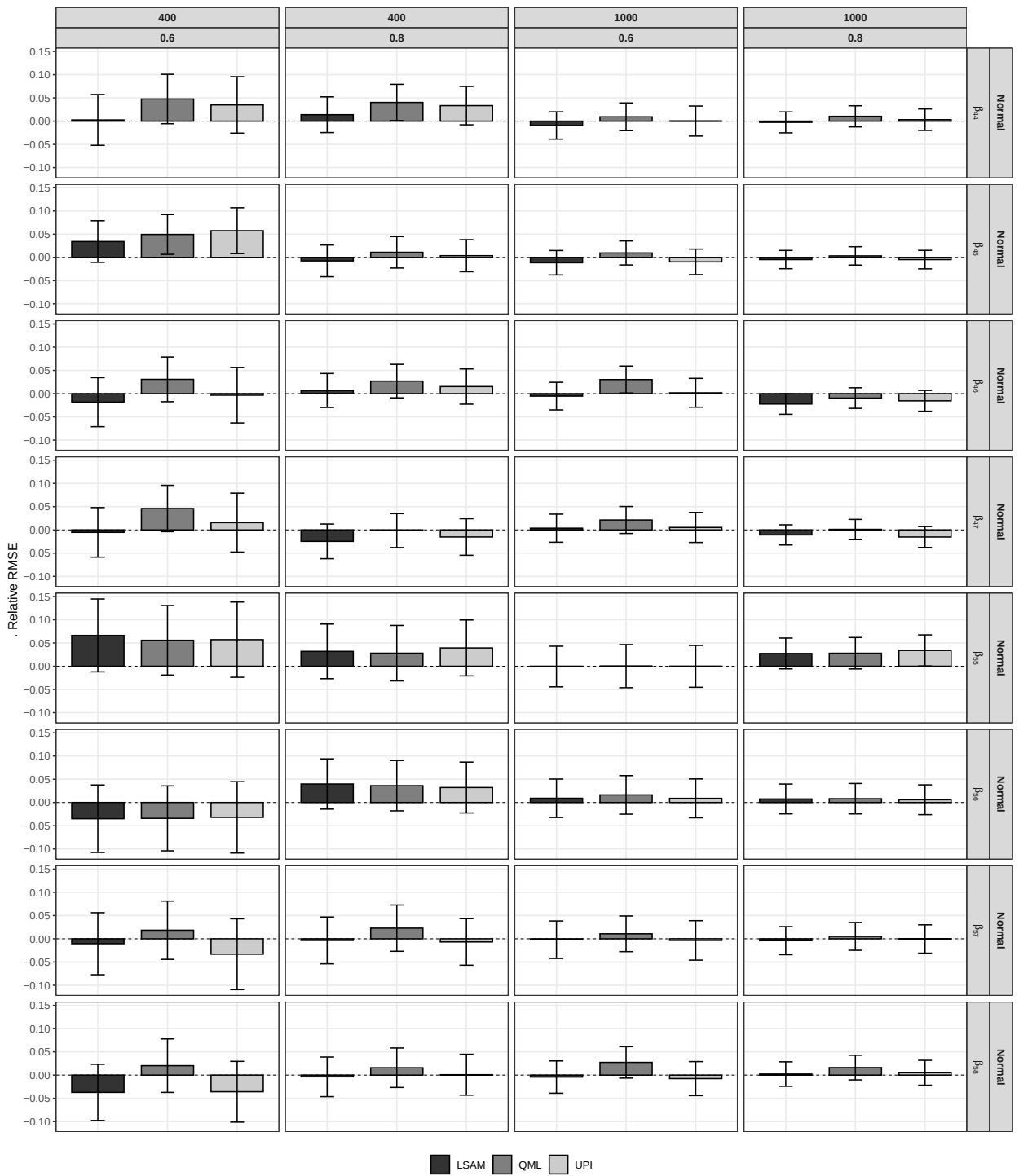


Figure S24

Additional Distributional Conditions Difference: Relative RMSE for Simulation 2 – Normal
(Epsilon & Zeta)

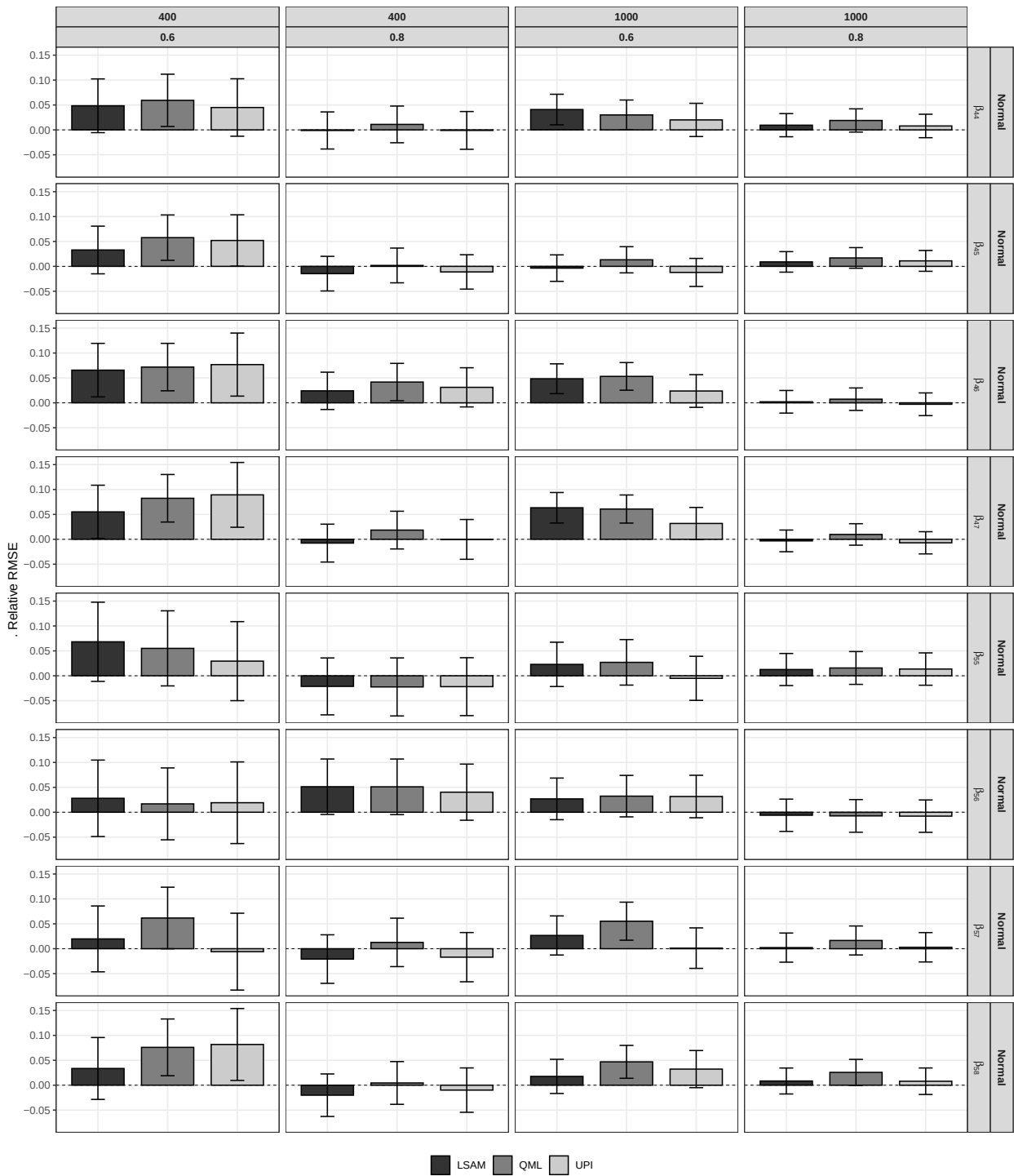


Figure S25

Additional Distributional Conditions Difference: Coverage Rate for Simulation 2 – Normal (Epsilon)

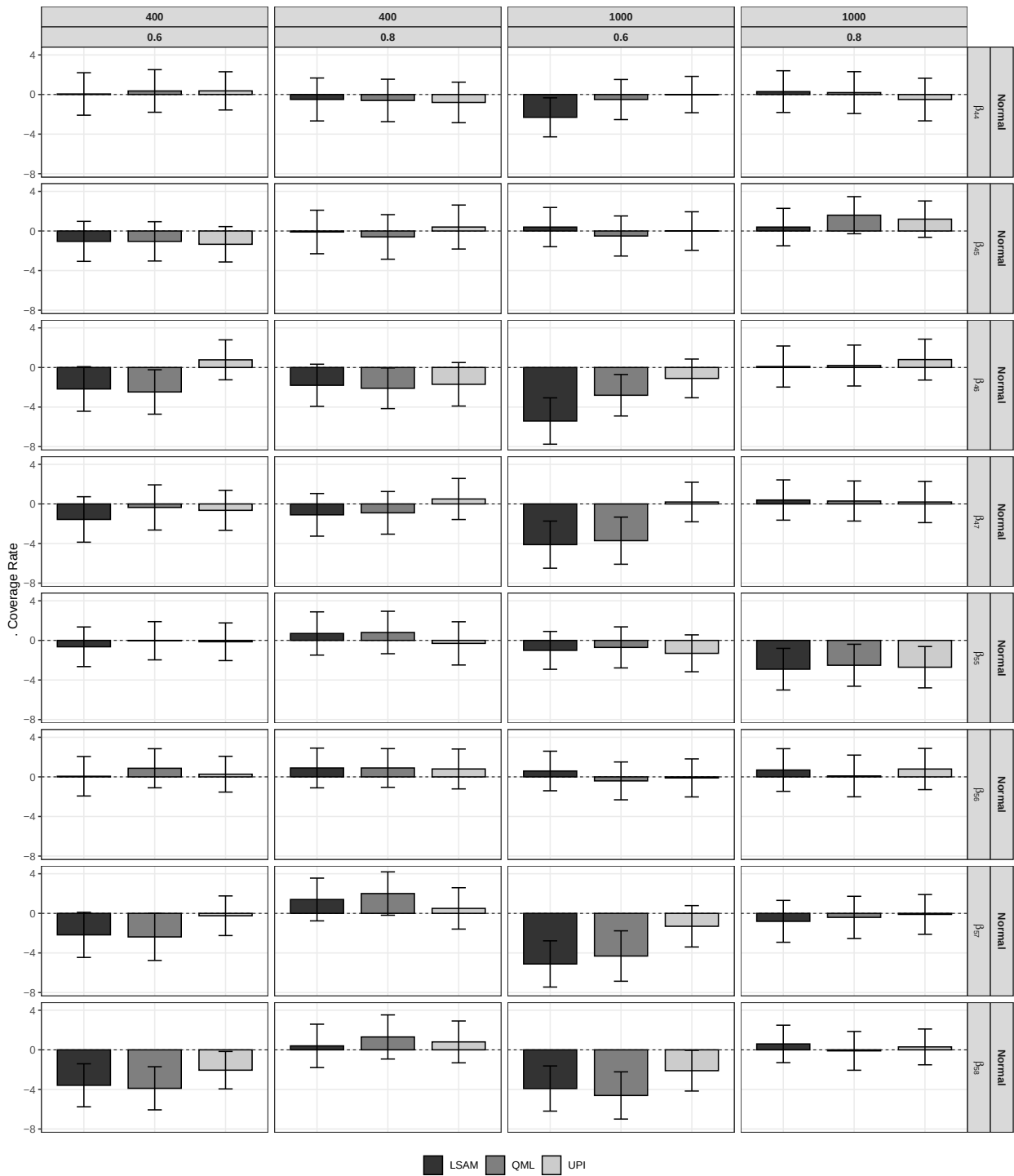


Figure S26

Additional Distributional Conditions Difference: Coverage Rate for Simulation 2 – Normal (Zeta)

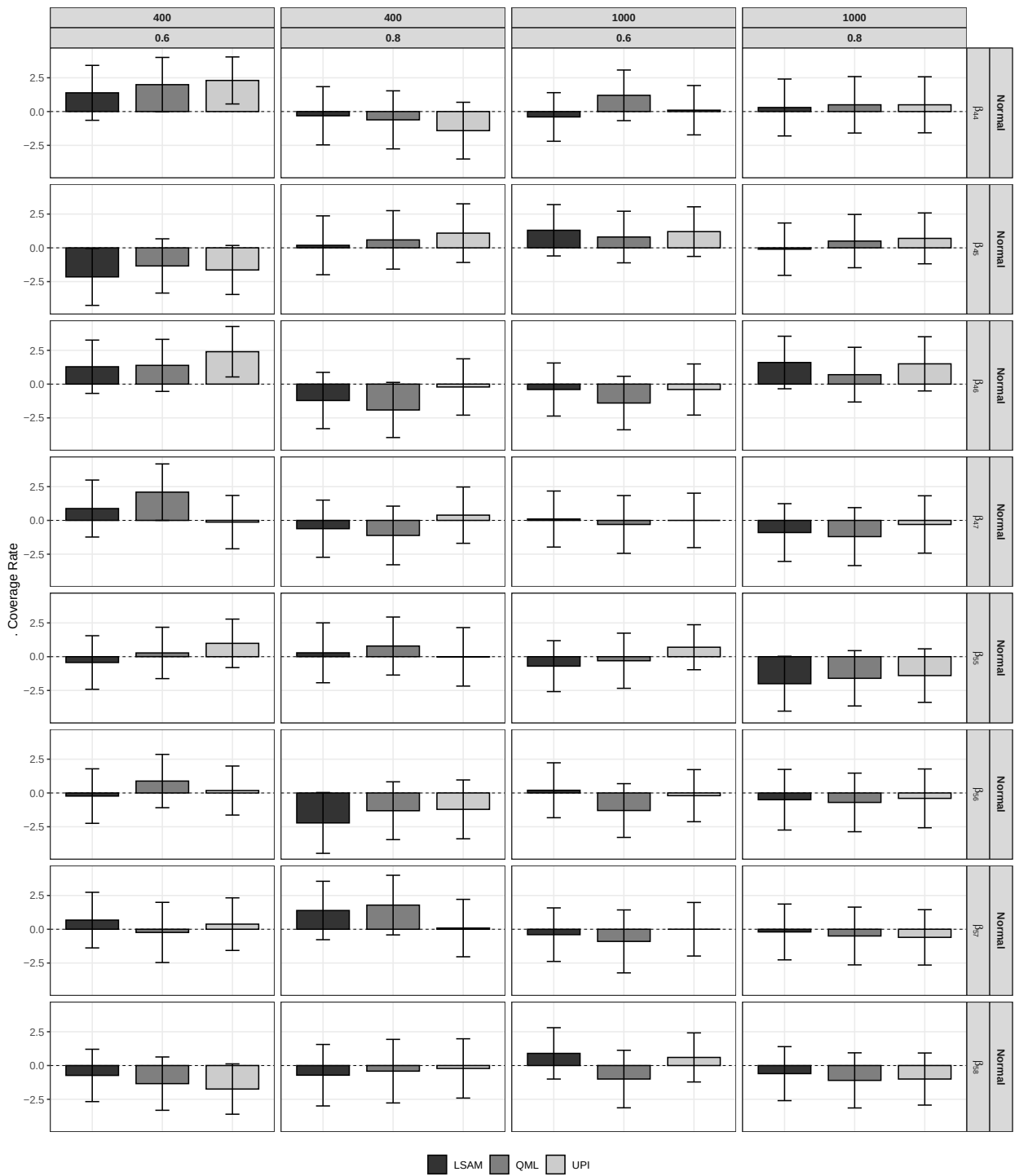


Figure S27

Additional Distributional Conditions Difference: Coverage Rate for Simulation 2 – Normal

(Epsilon & Zeta)

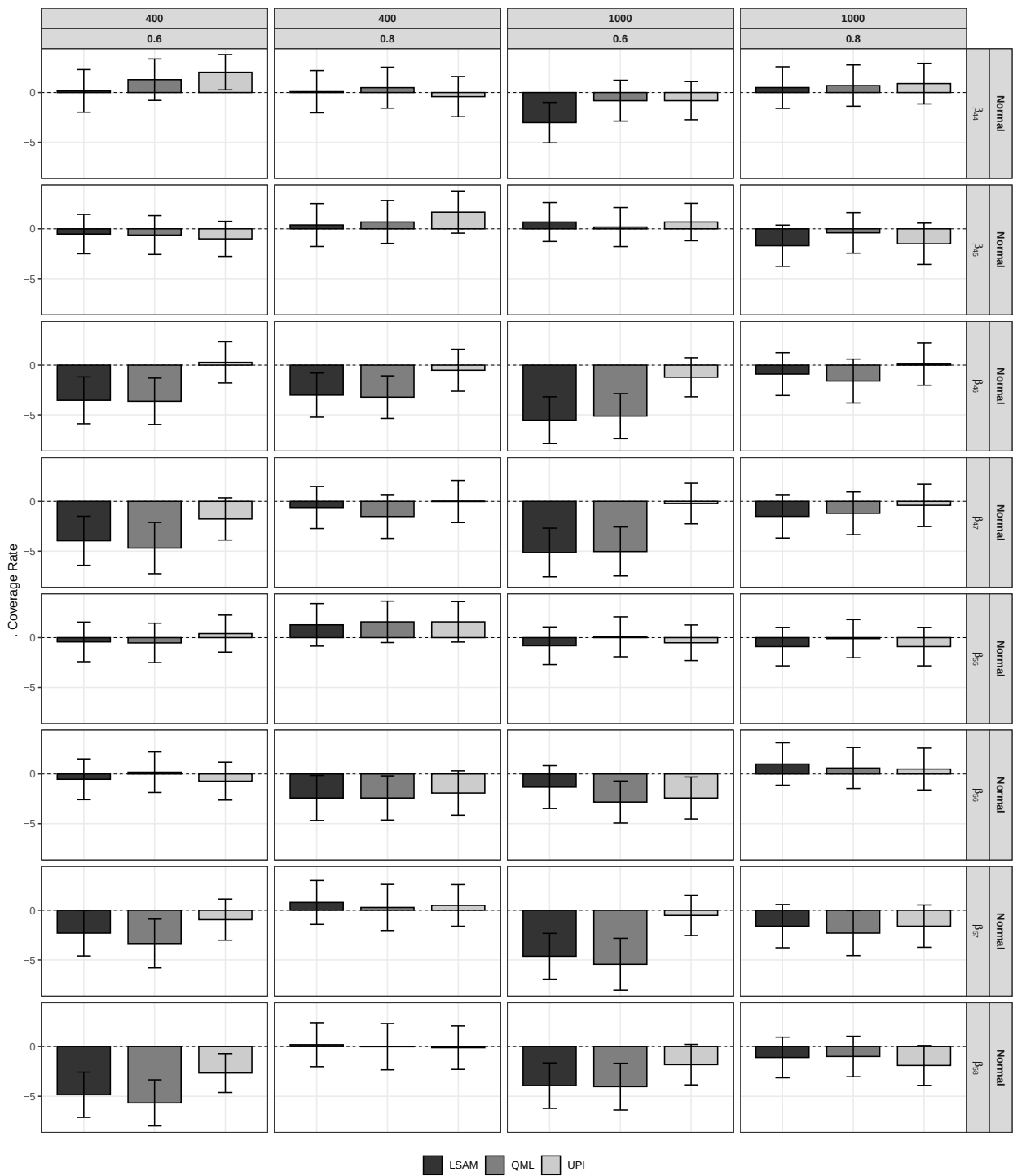


Figure S28

Additional Distributional Conditions Difference: SE/SD Ratio for Simulation 2 – Normal (Epsilon)

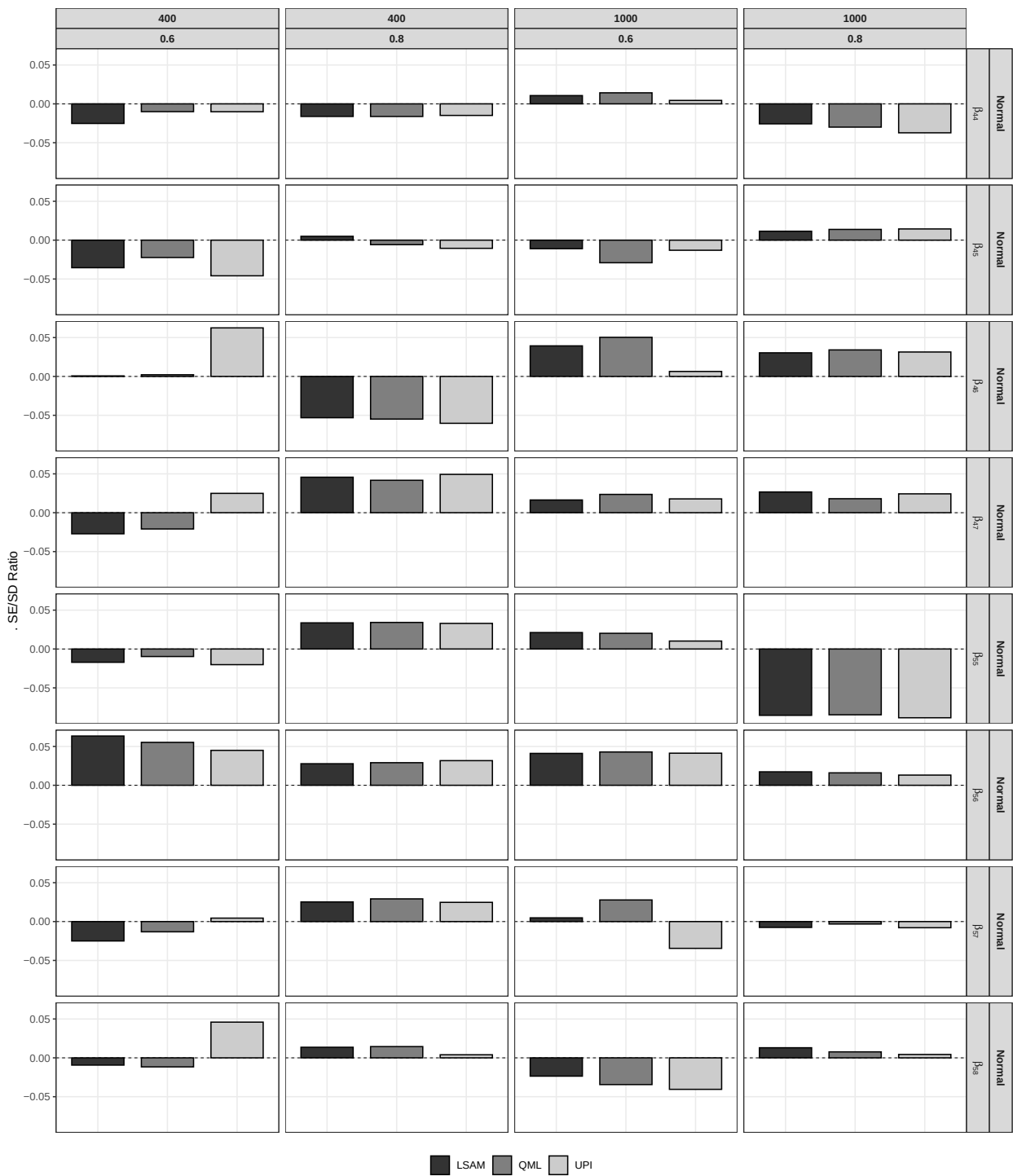


Figure S29

Additional Distributional Conditions Difference: SE/SD Ratio for Simulation 2 – Normal (Zeta)

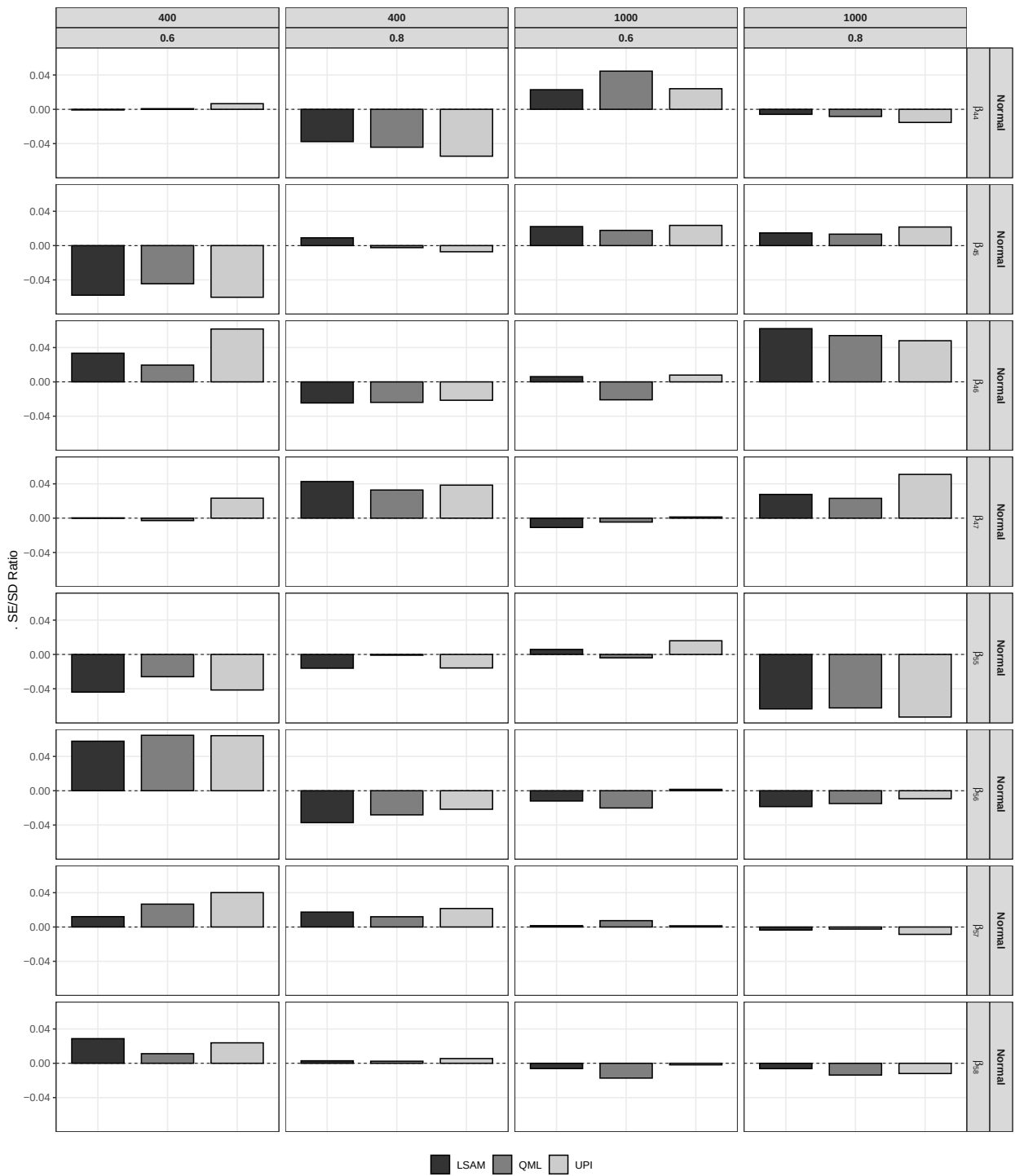


Figure S30

Additional Distributional Conditions Difference: SE/SD Ratio for Simulation 2 – Normal (Epsilon & Zeta)

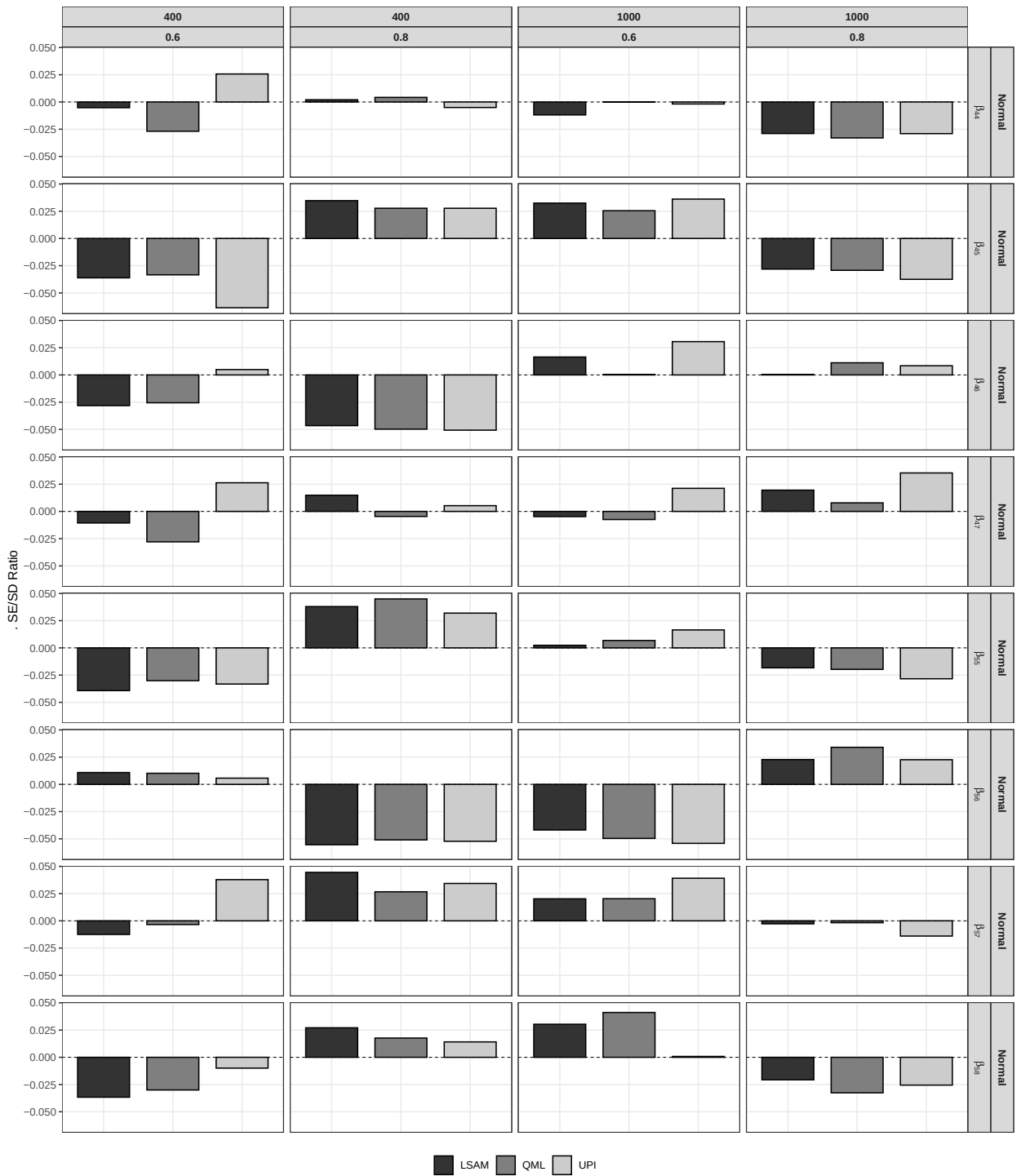


Figure S31

Additional Distributional Conditions Difference: Power for Simulation 2 – Normal (Epsilon)

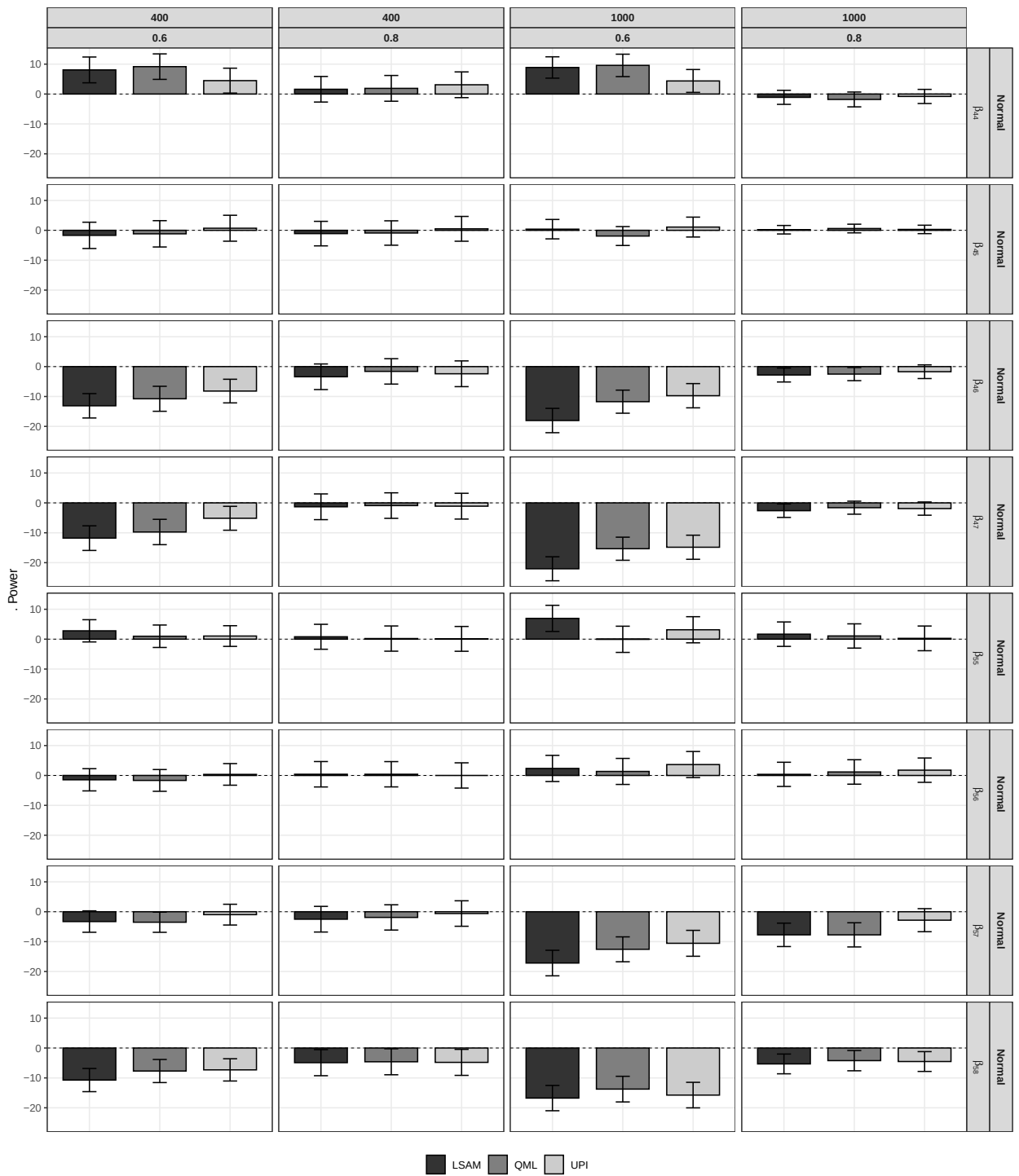


Figure S32

Additional Distributional Conditions Difference: Power for Simulation 2 – Normal (Zeta)

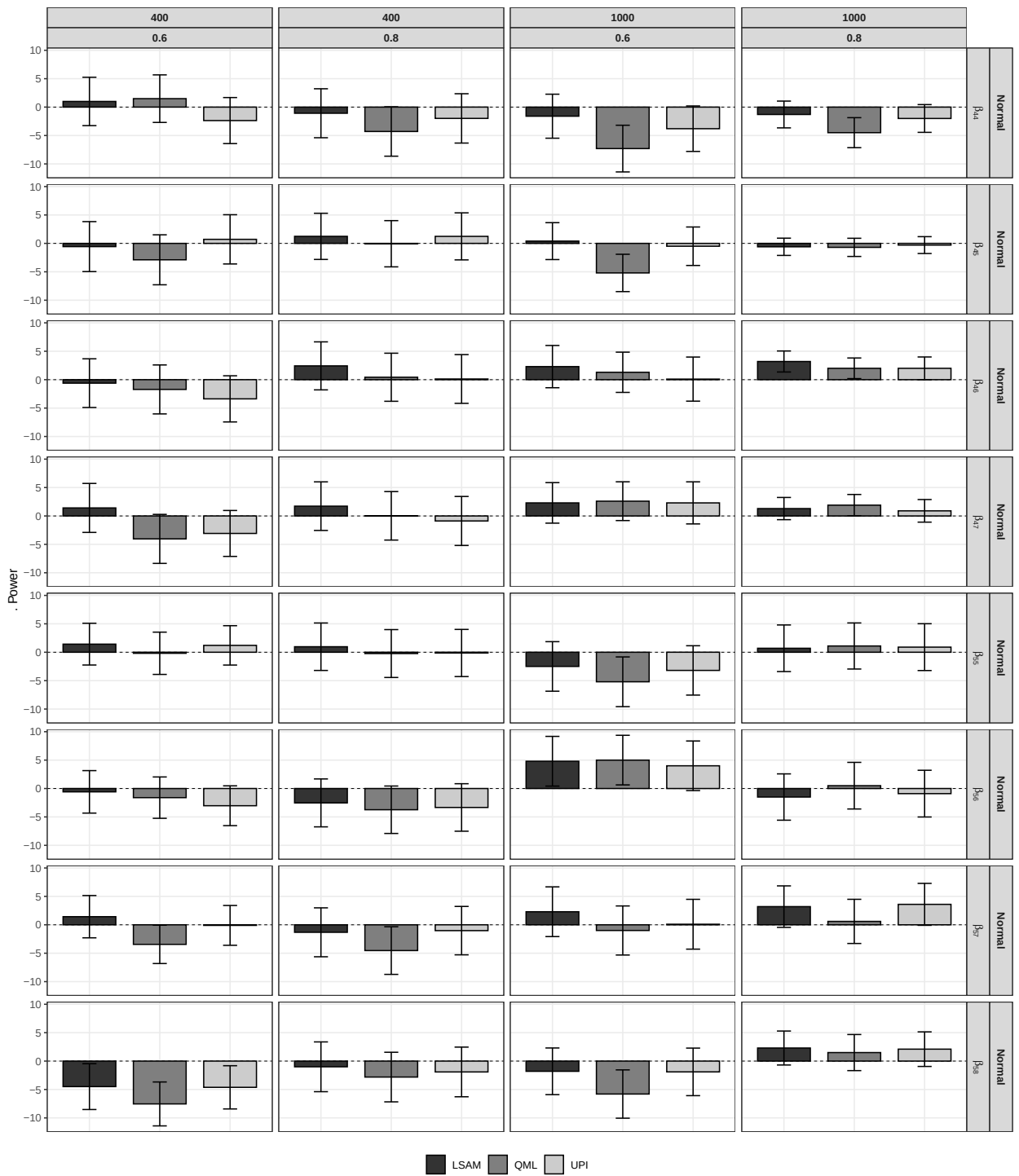


Figure S33

Additional Distributional Conditions Difference: Power for Simulation 2 – Normal (Epsilon & Zeta)

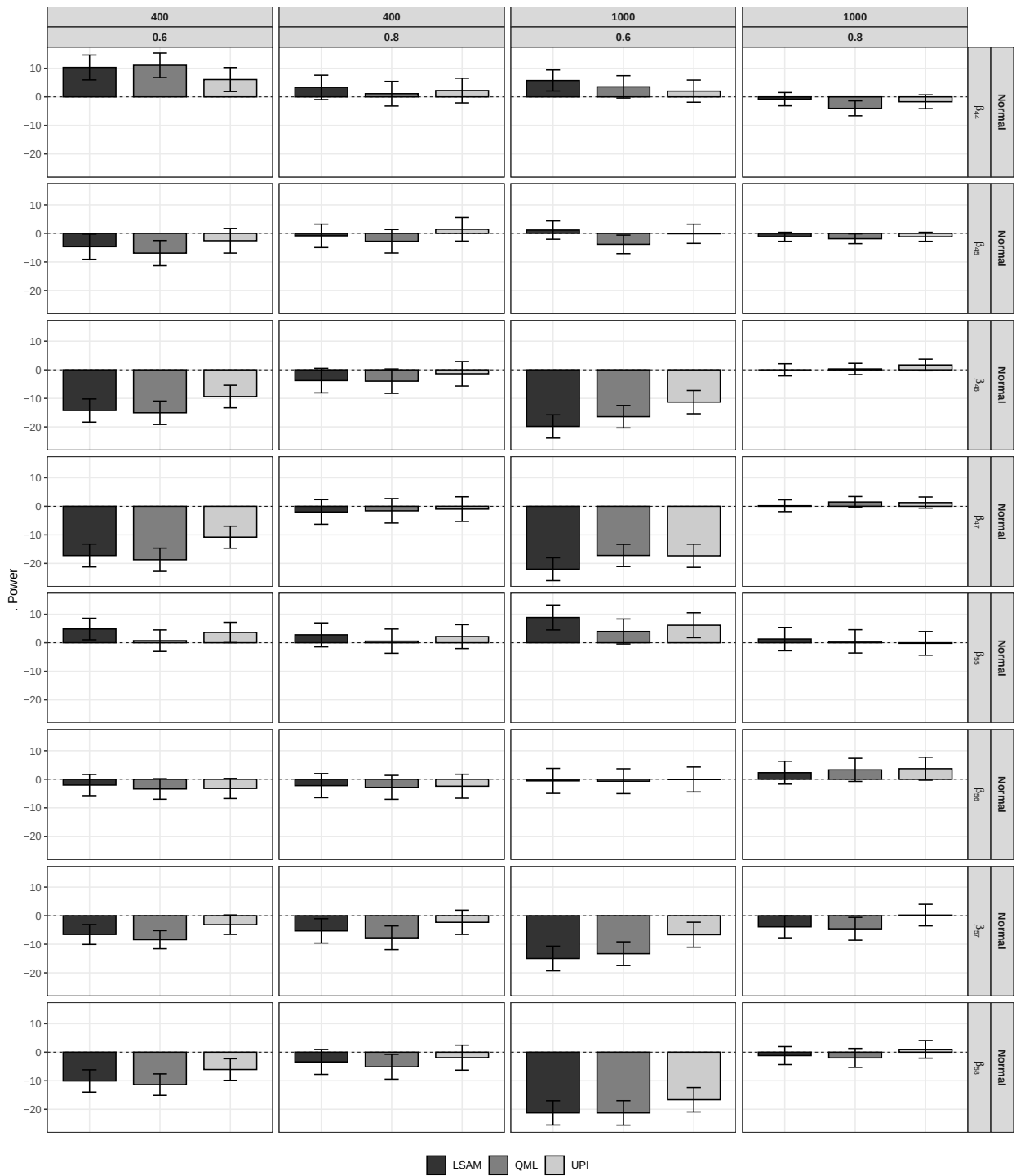


Figure S34

Additional Distributional Conditions Difference: Type I Error for Simulation 2 – Normal (Epsilon)

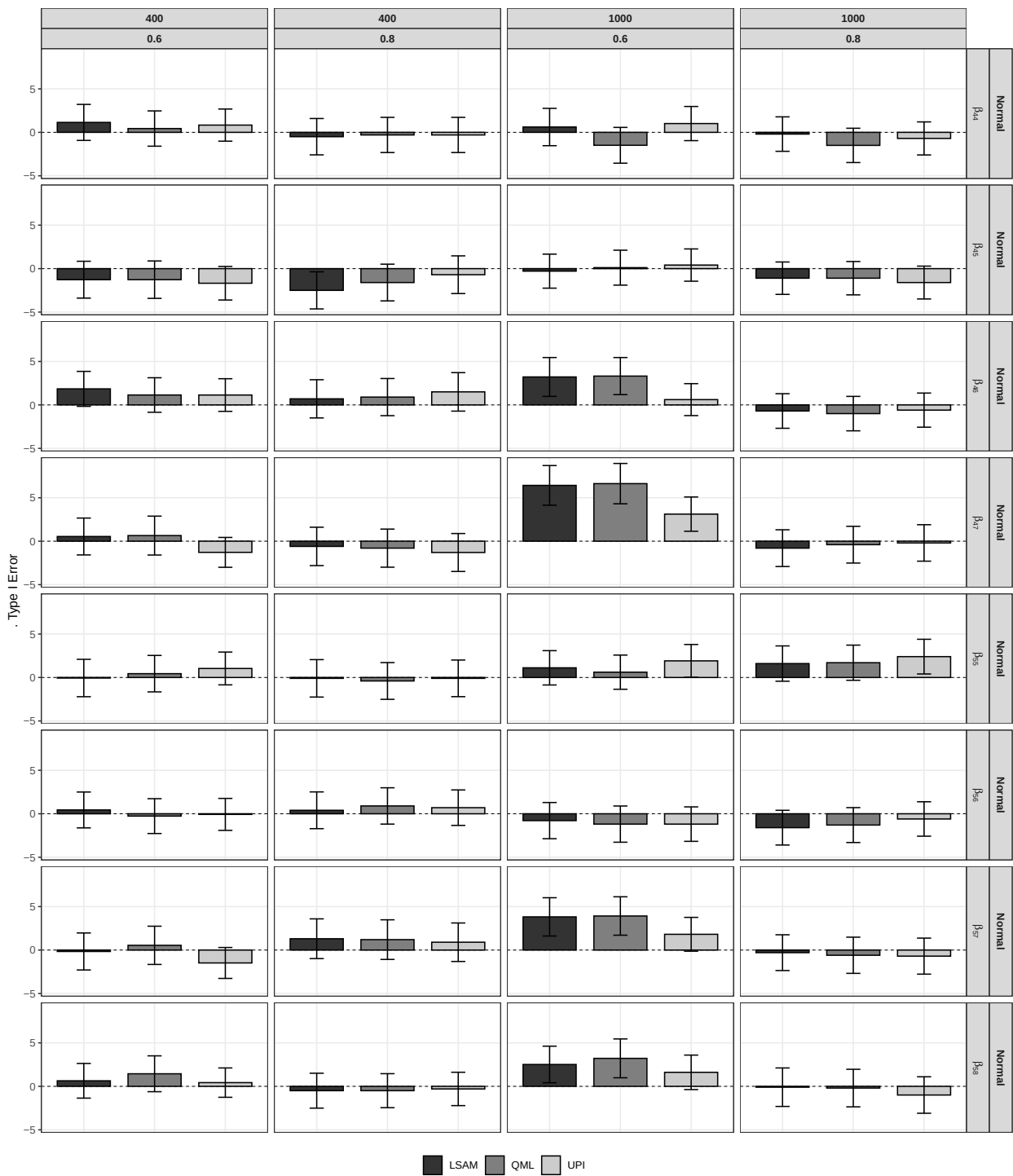


Figure S35

Additional Distributional Conditions Difference: Type I Error for Simulation 2 – Normal (Zeta)

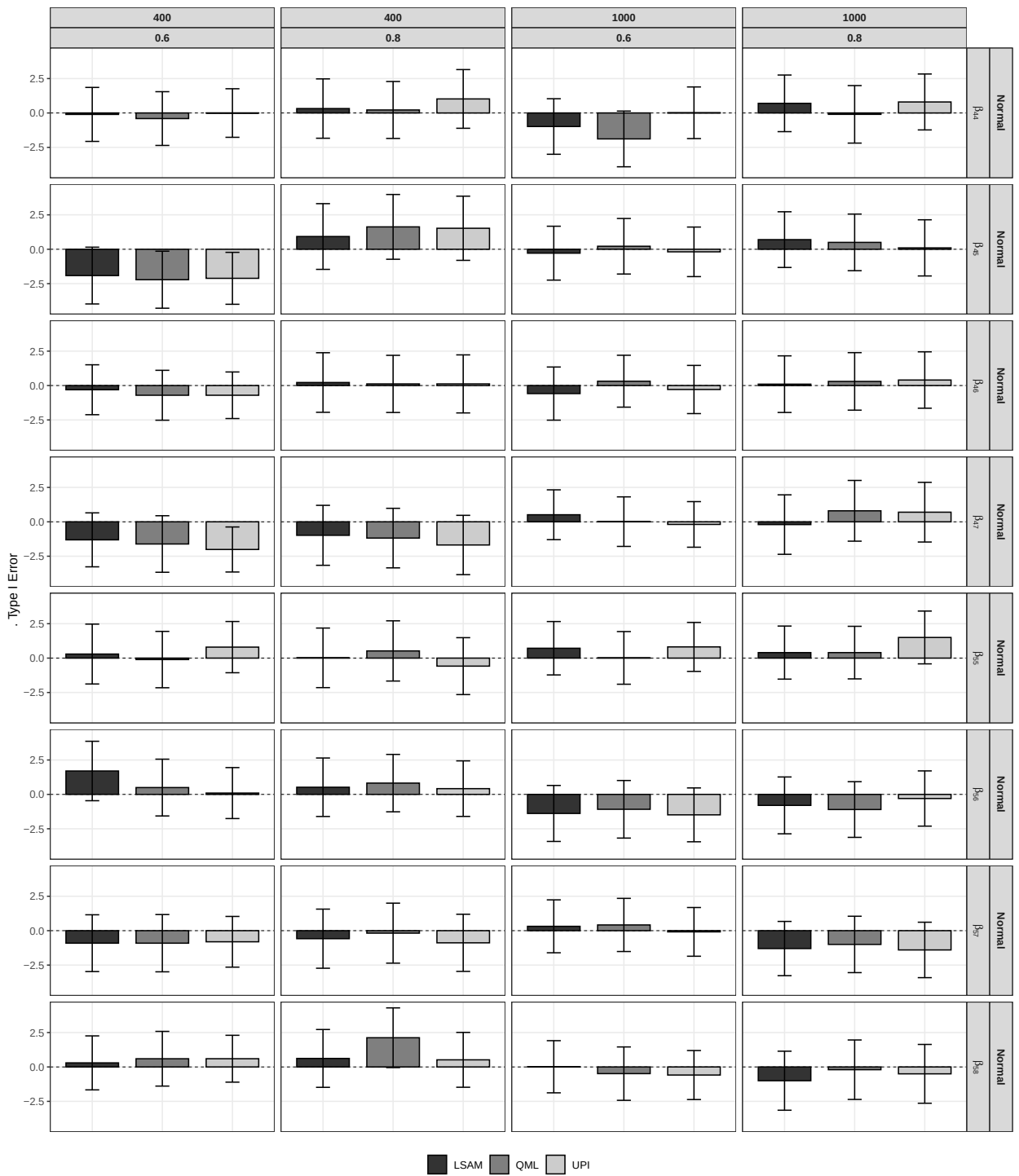


Figure S36

Additional Distributional Conditions Difference: Type I Error for Simulation 2 – Normal (Epsilon & Zeta)

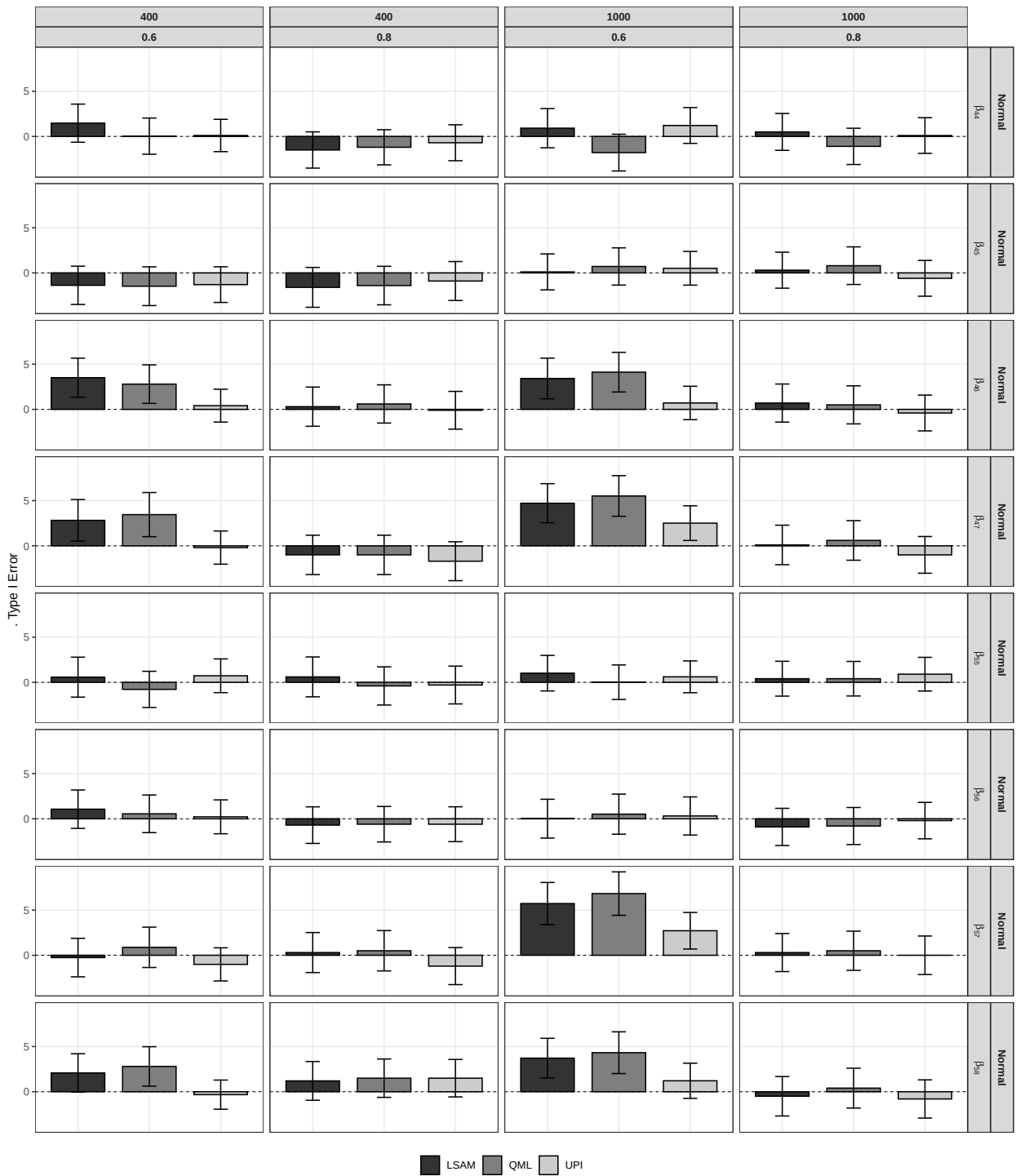


Figure S37

Additional Distributional Conditions Difference: Relative Bias for Simulation 2 – Right-skewed (Epsilon)

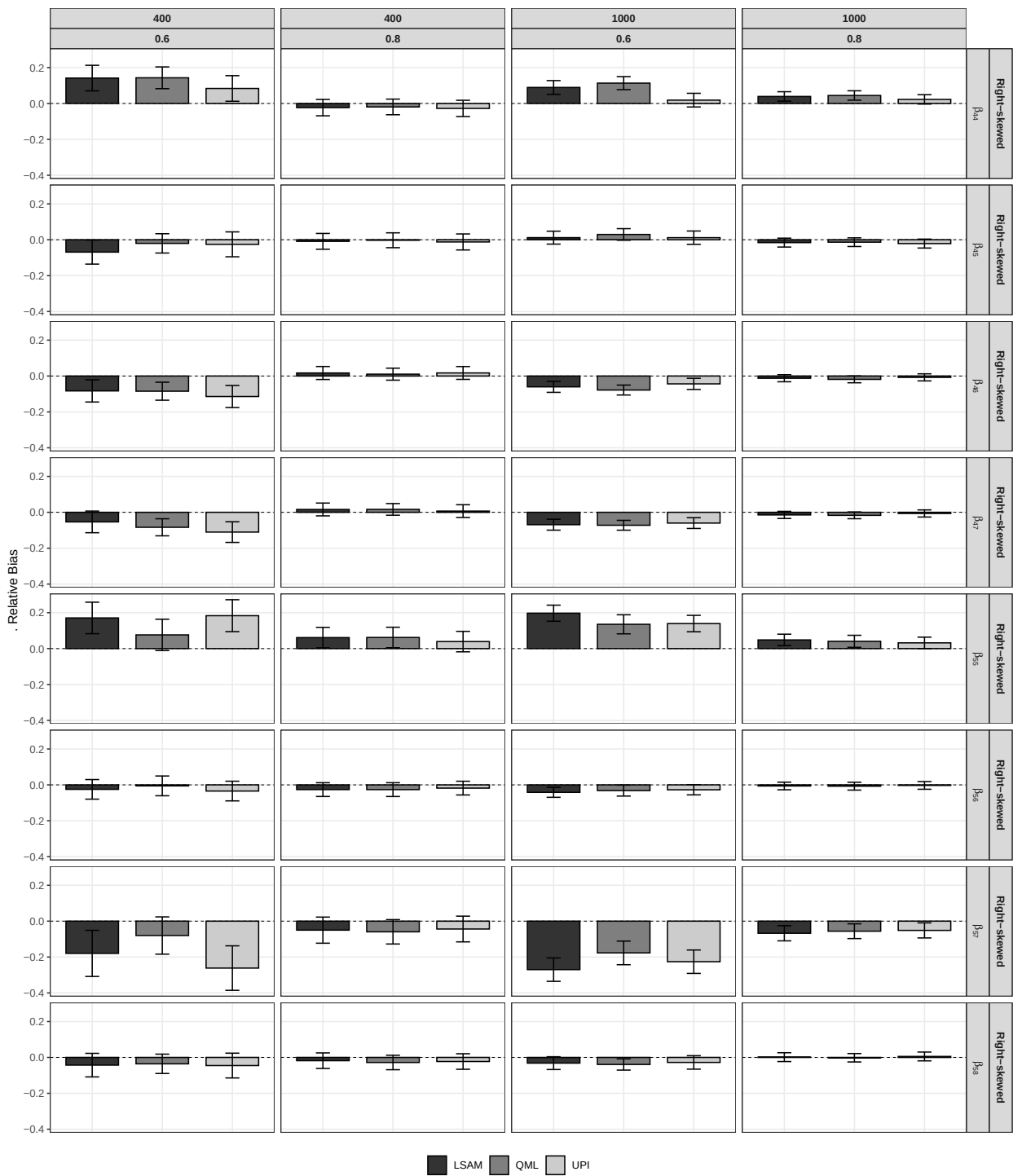


Figure S38

Additional Distributional Conditions Difference: Relative Bias for Simulation 2 – Right-skewed (Zeta)

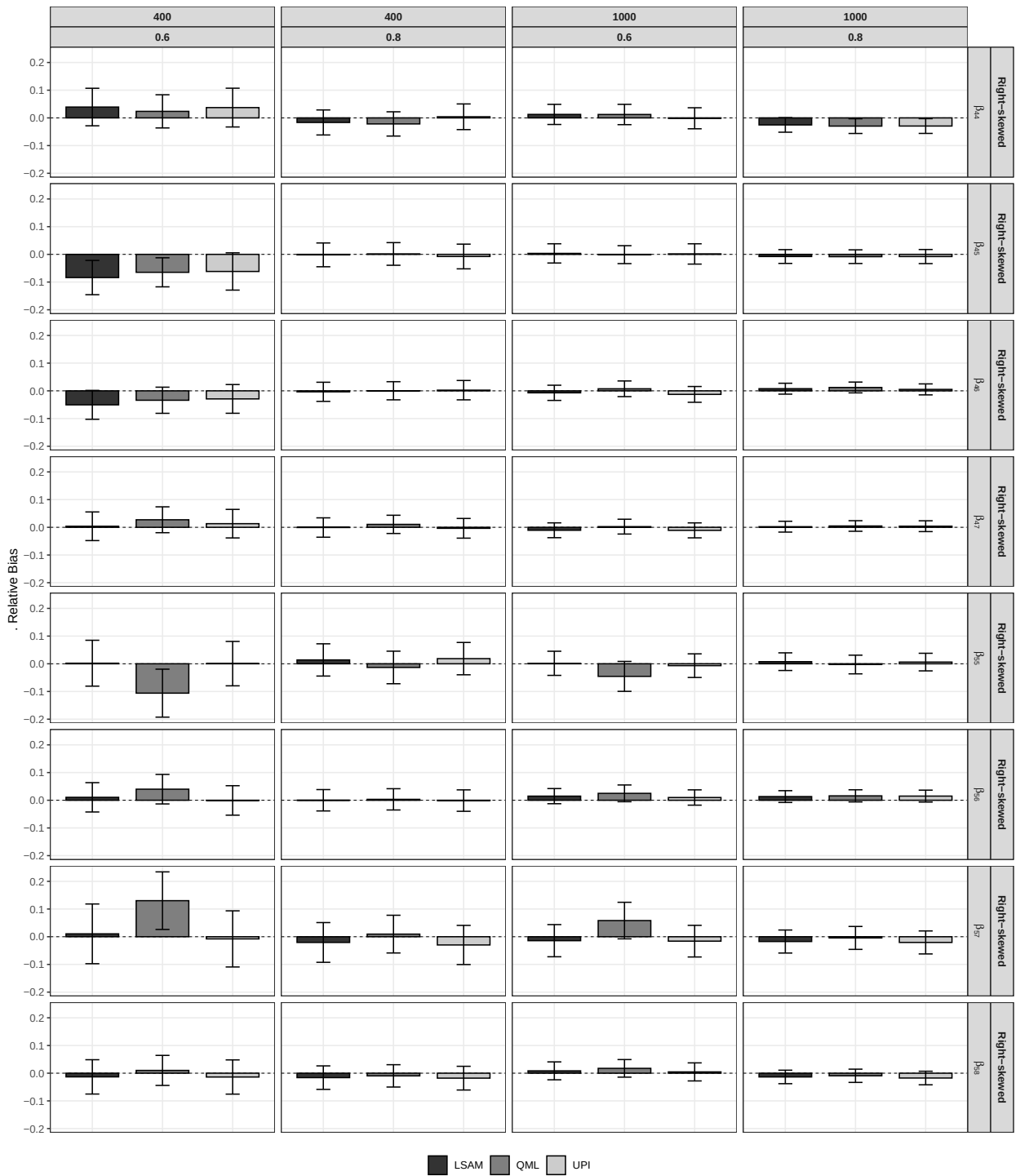


Figure S39

Additional Distributional Conditions Difference: Relative Bias for Simulation 2 – Right-skewed (Epsilon & Zeta)

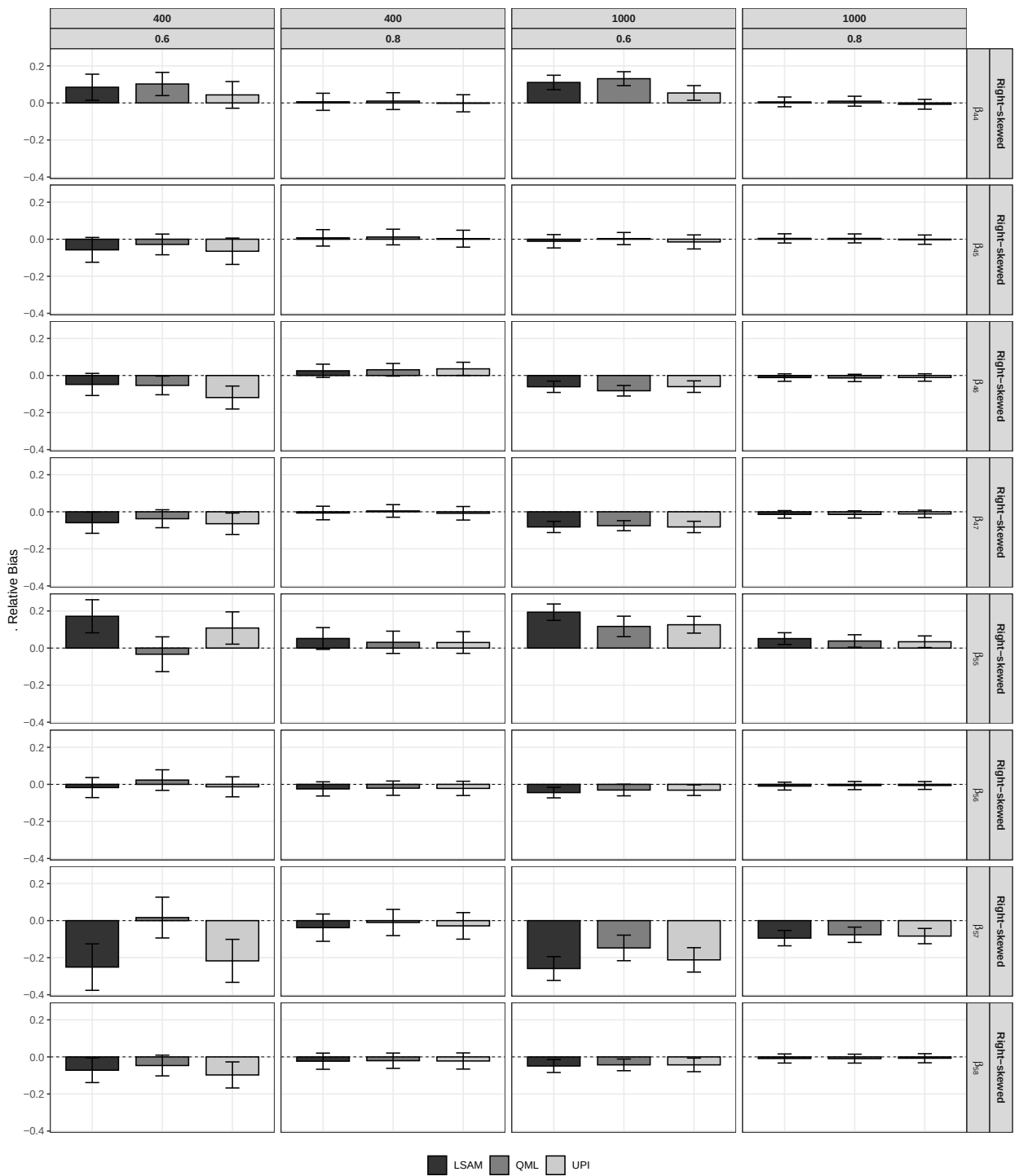


Figure S40

Additional Distributional Conditions Difference: Relative RMSE for Simulation 2 – Right-skewed (Epsilon)

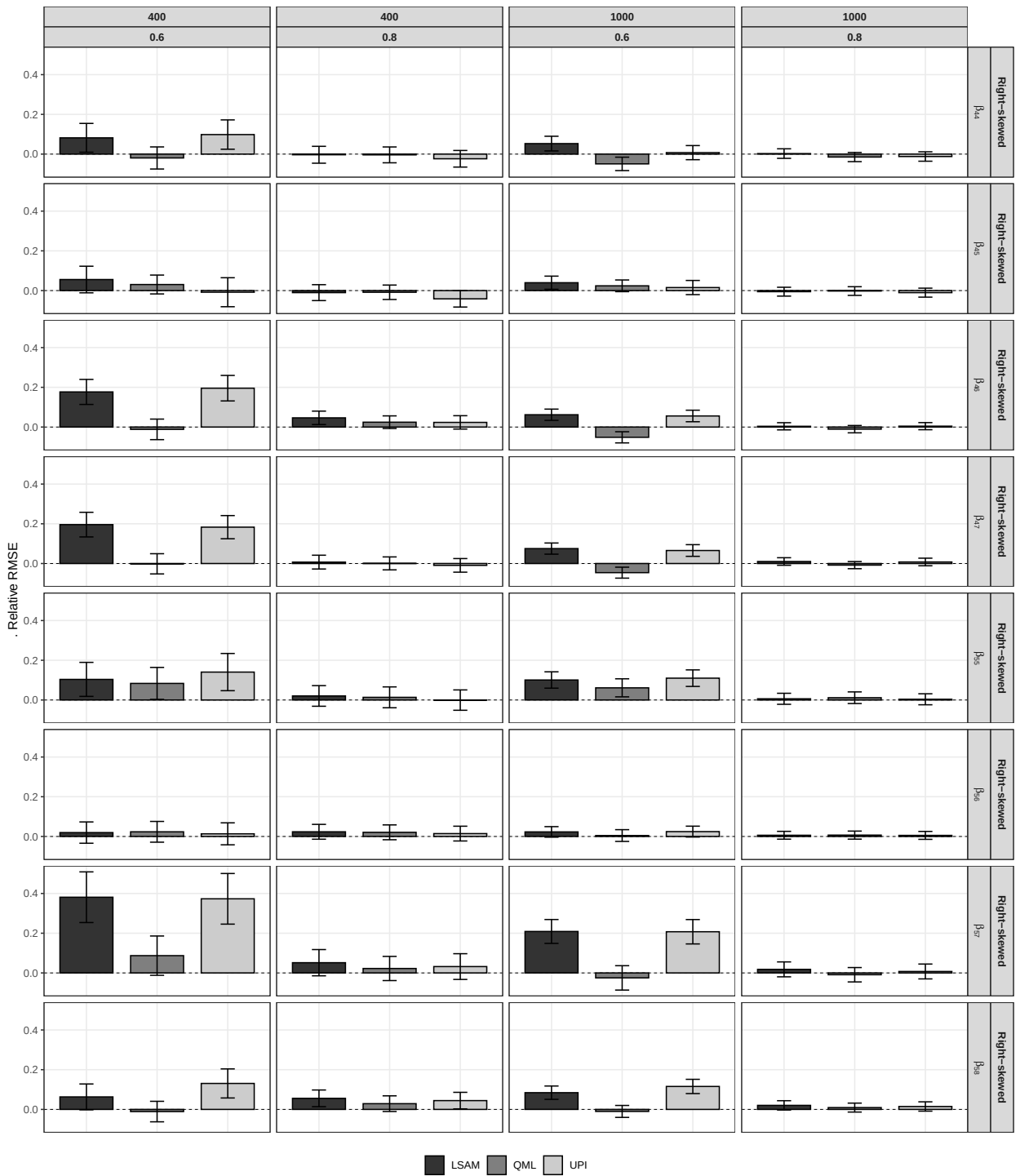


Figure S41

Additional Distributional Conditions Difference: Relative RMSE for Simulation 2 – Right-skewed (Zeta)

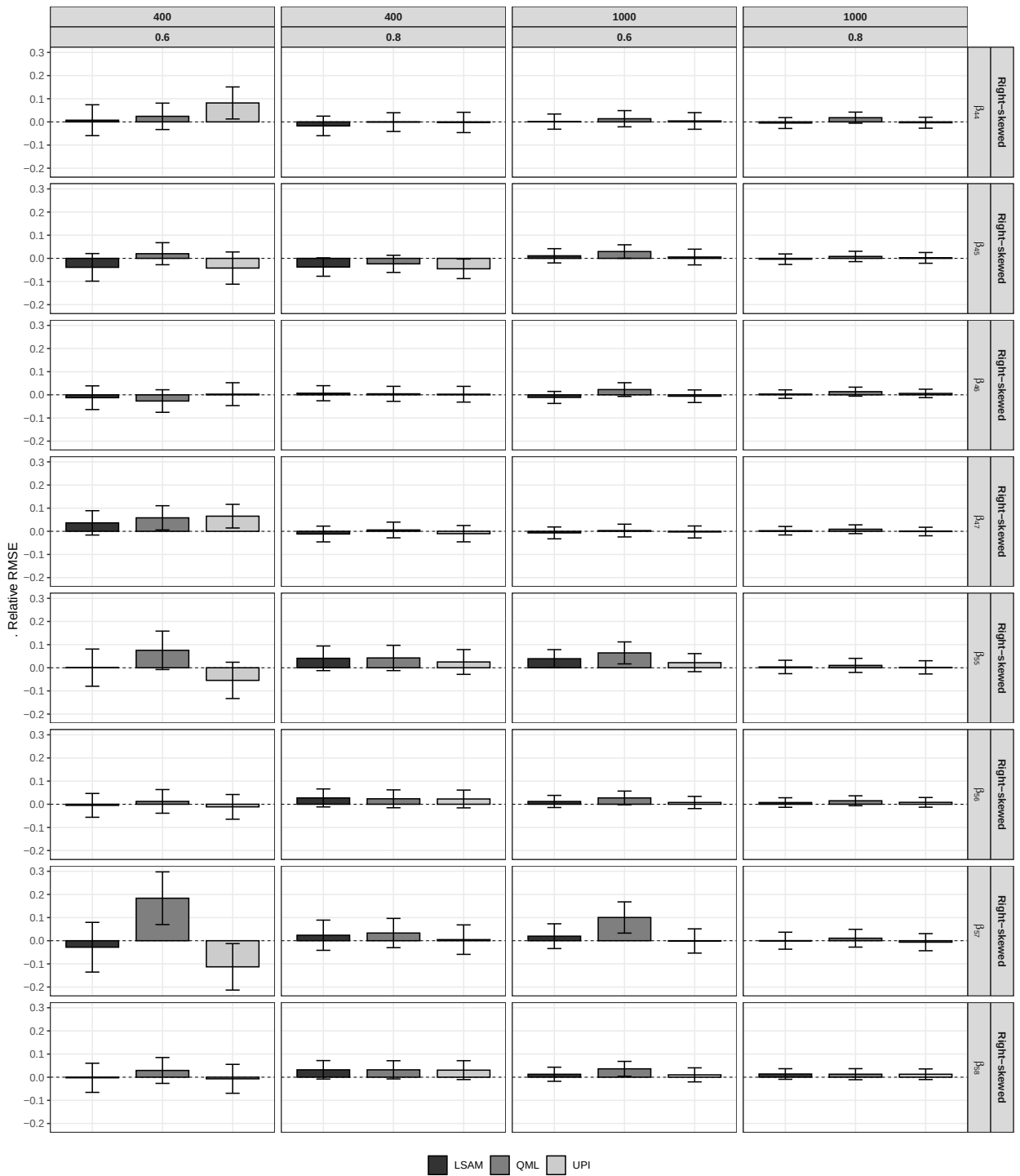


Figure S42

Additional Distributional Conditions Difference: Relative RMSE for Simulation 2 – Right-skewed (Epsilon & Zeta)

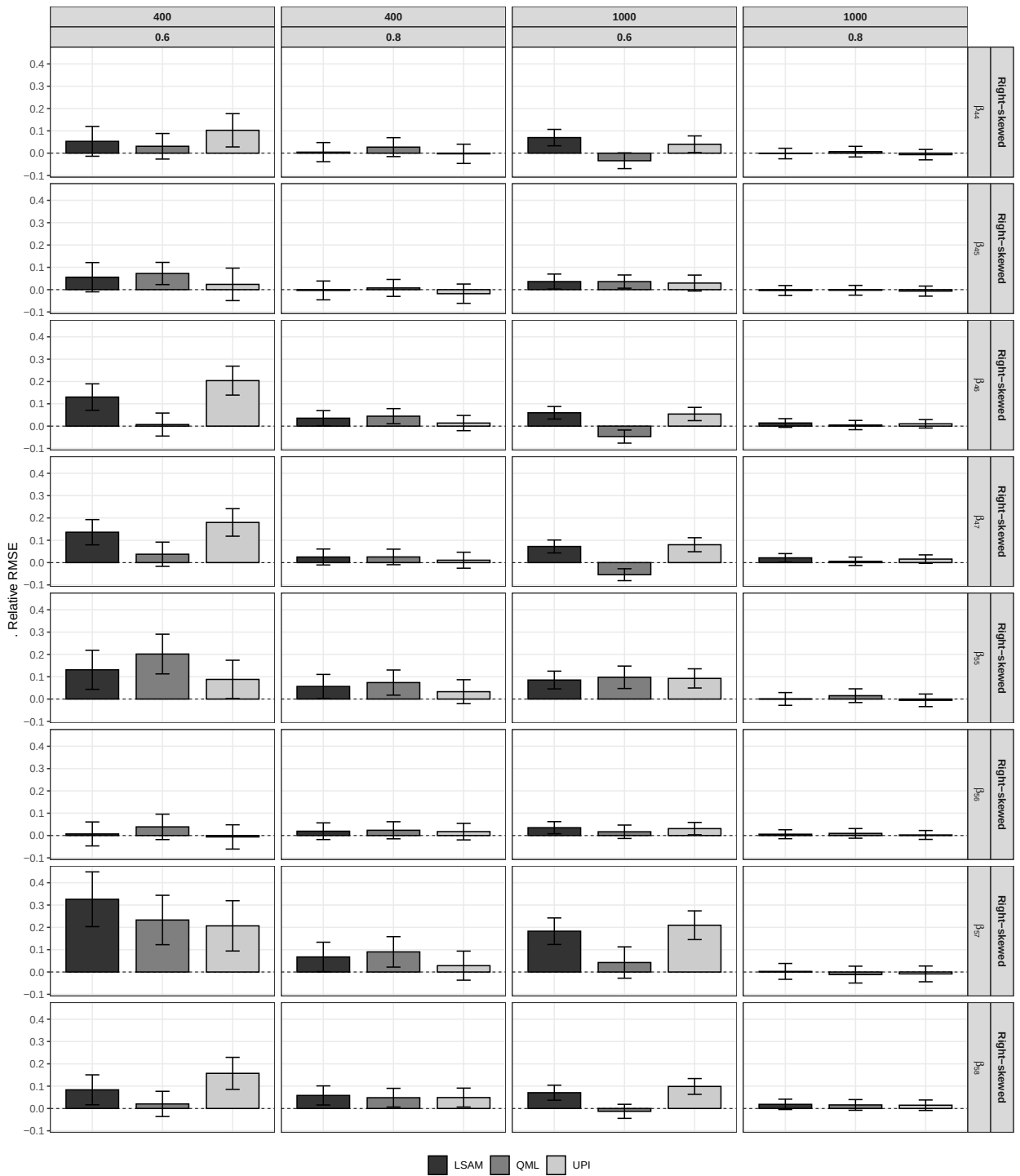


Figure S43

Additional Distributional Conditions Difference: Coverage Rate for Simulation 2 – Right-skewed (Epsilon)

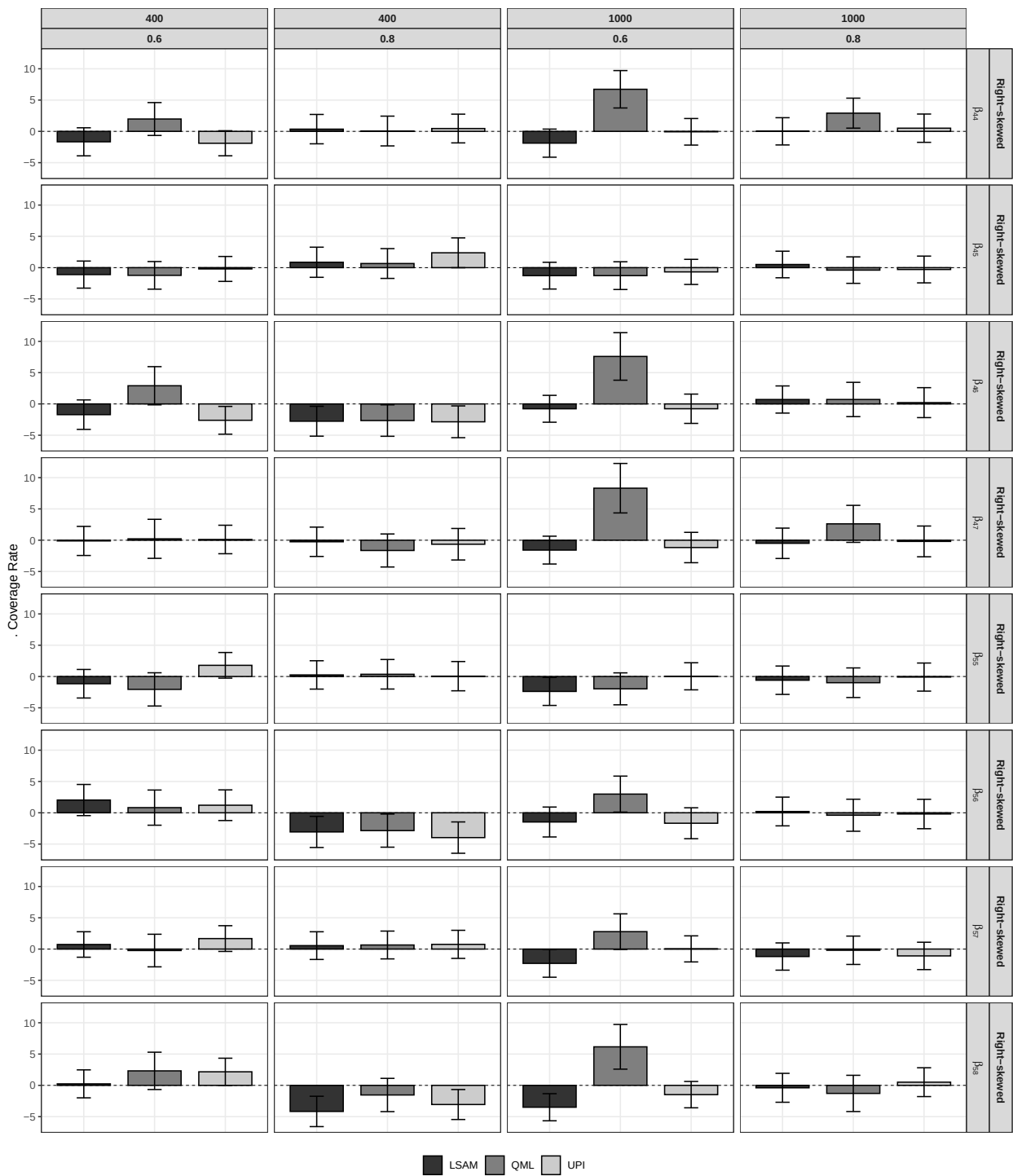


Figure S44

Additional Distributional Conditions Difference: Coverage Rate for Simulation 2 – Right-skewed (Zeta)

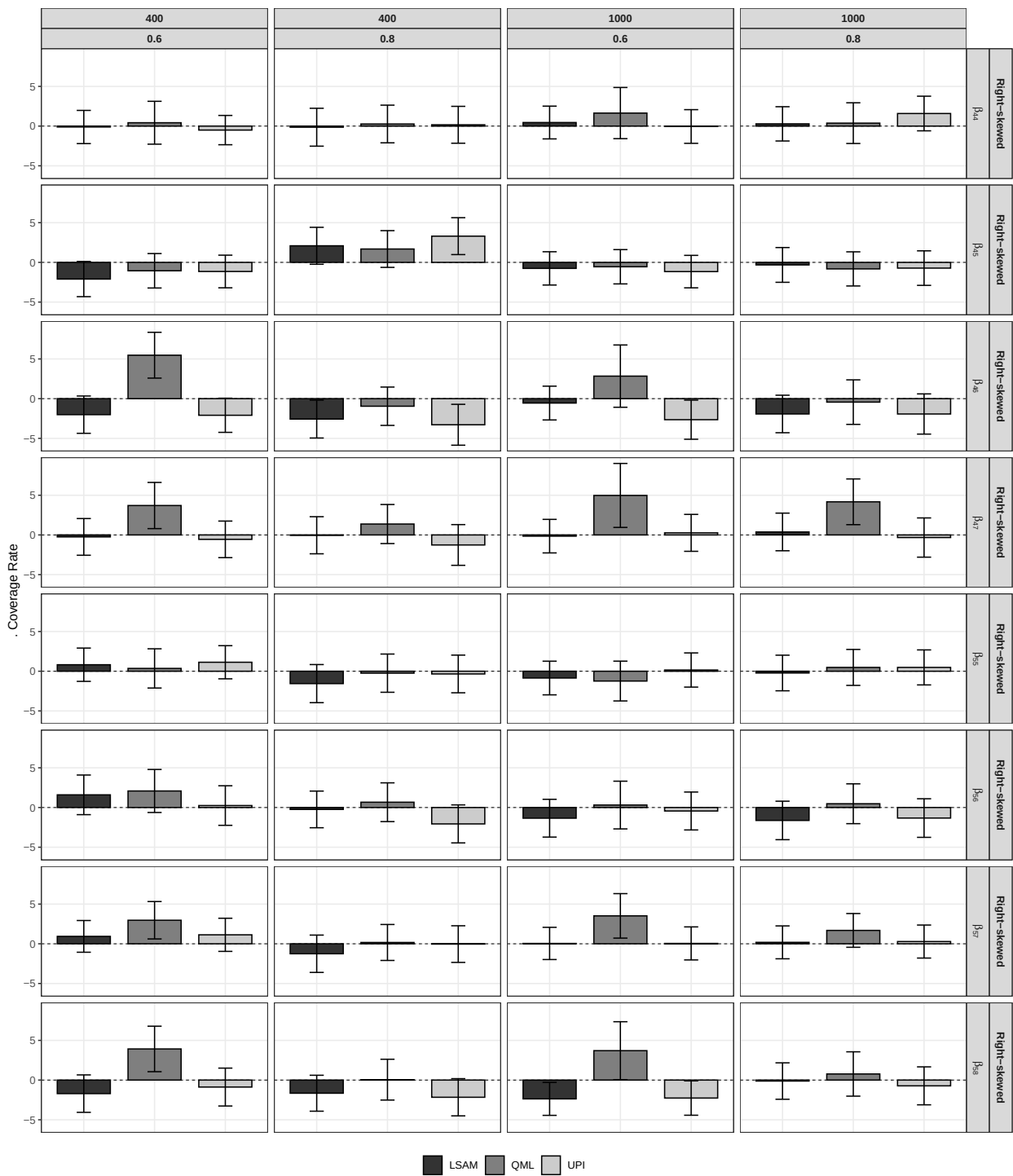


Figure S45

Additional Distributional Conditions Difference: Coverage Rate for Simulation 2 – Right-skewed (Epsilon & Zeta)

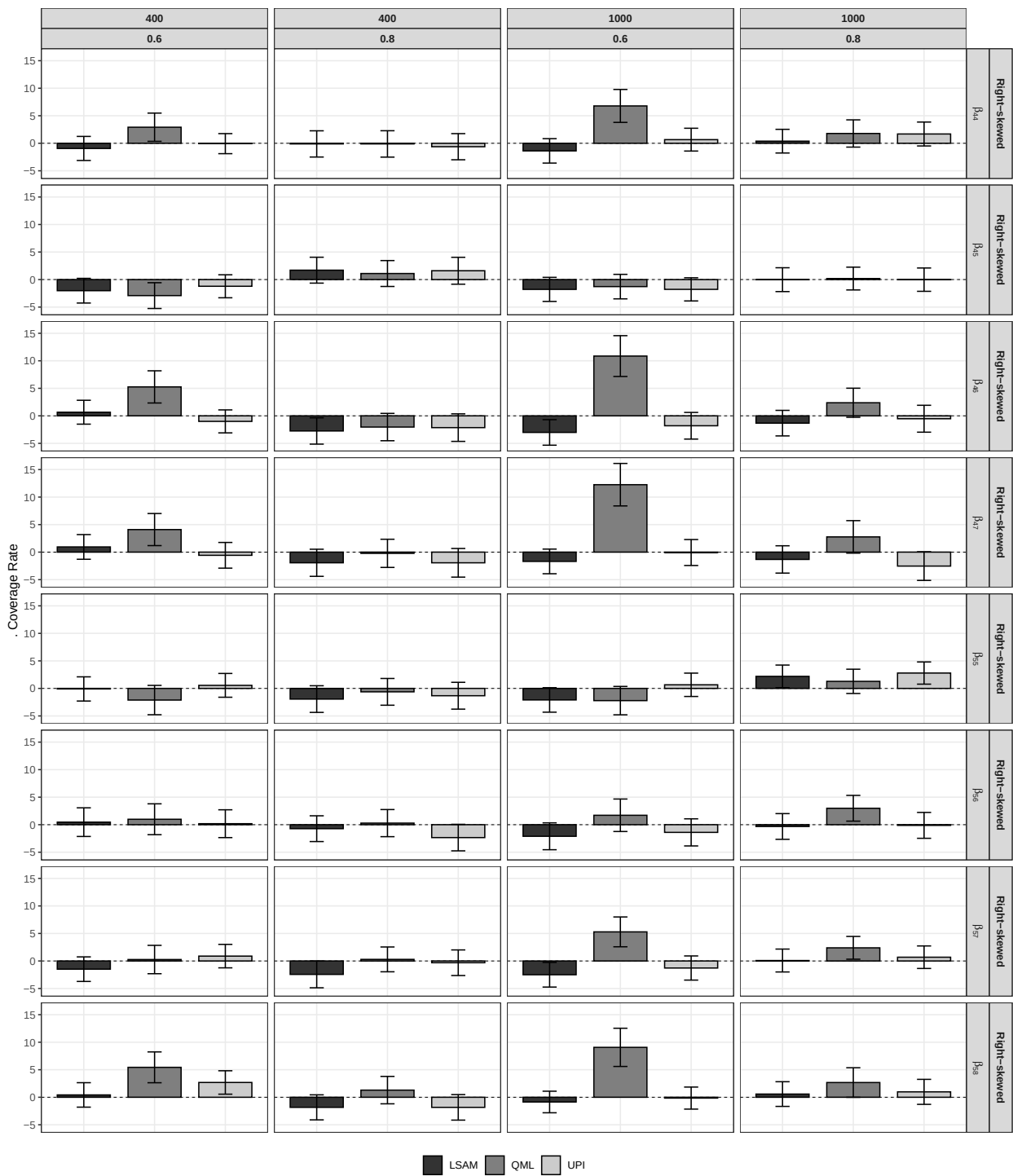


Figure S46

Additional Distributional Conditions Difference: SE/SD Ratio for Simulation 2 – Right-skewed (Epsilon)

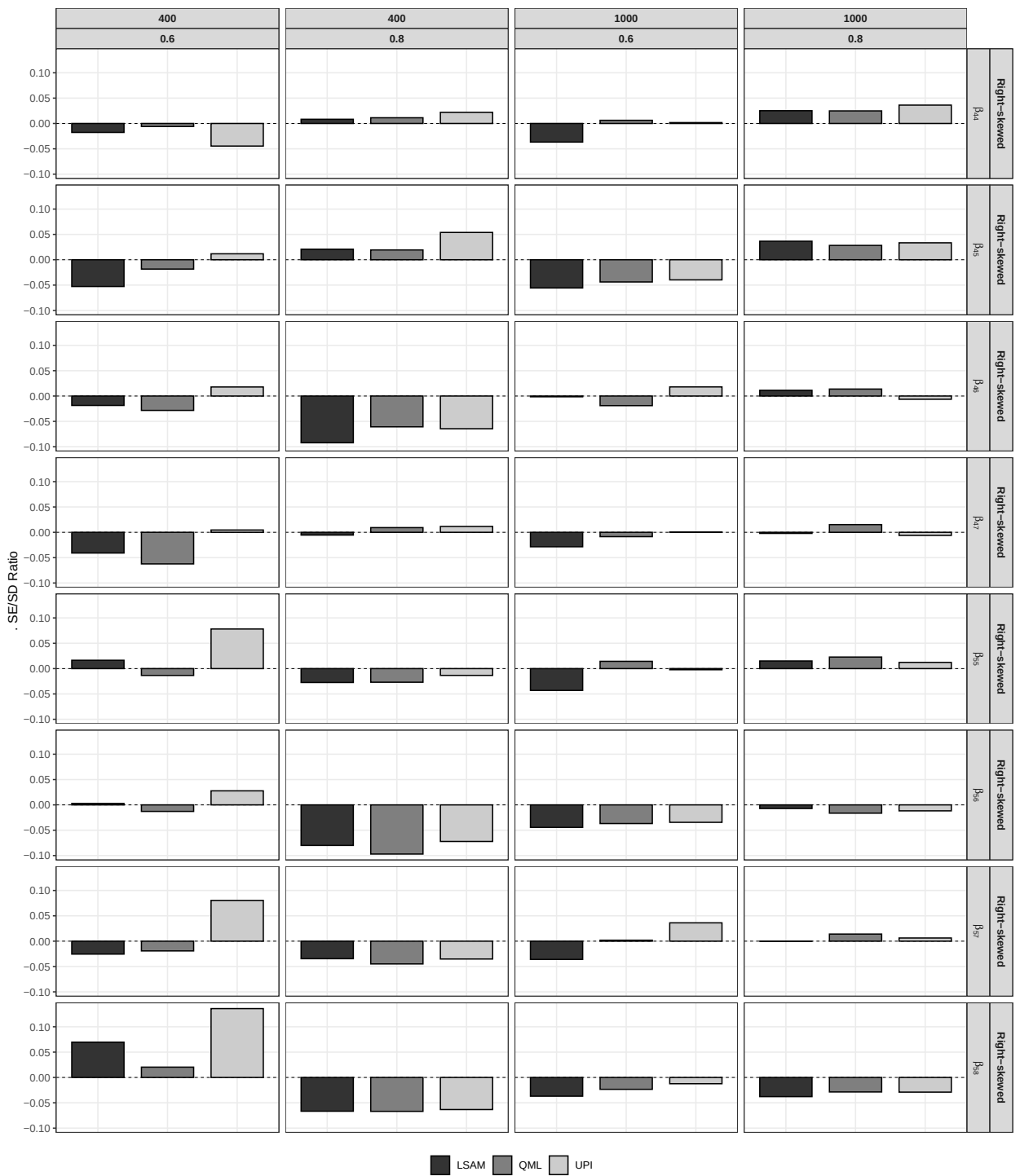


Figure S47

Additional Distributional Conditions Difference: SE/SD Ratio for Simulation 2 – Right-skewed (Zeta)

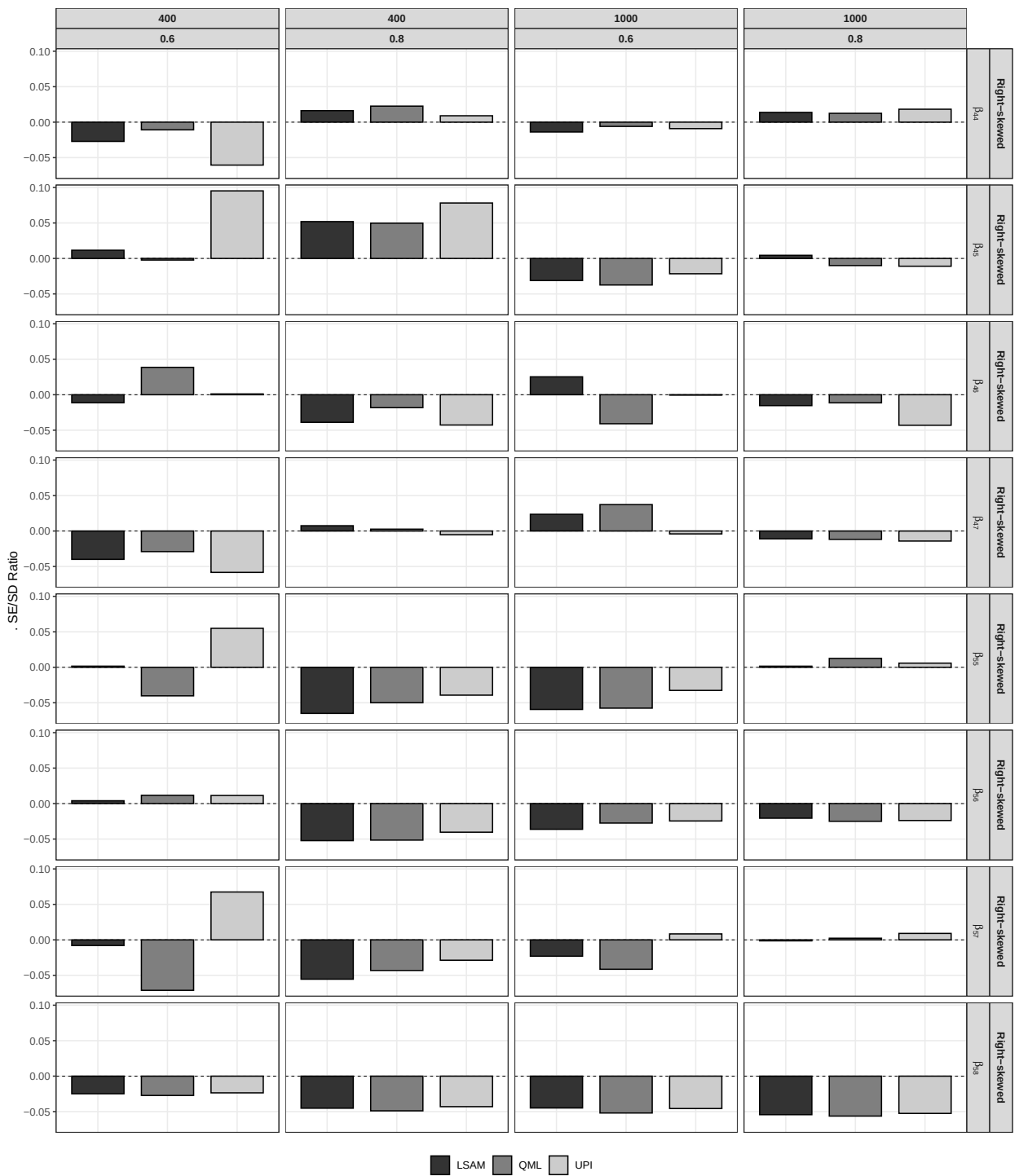


Figure S48

Additional Distributional Conditions Difference: SE/SD Ratio for Simulation 2 – Right-skewed (Epsilon & Zeta)

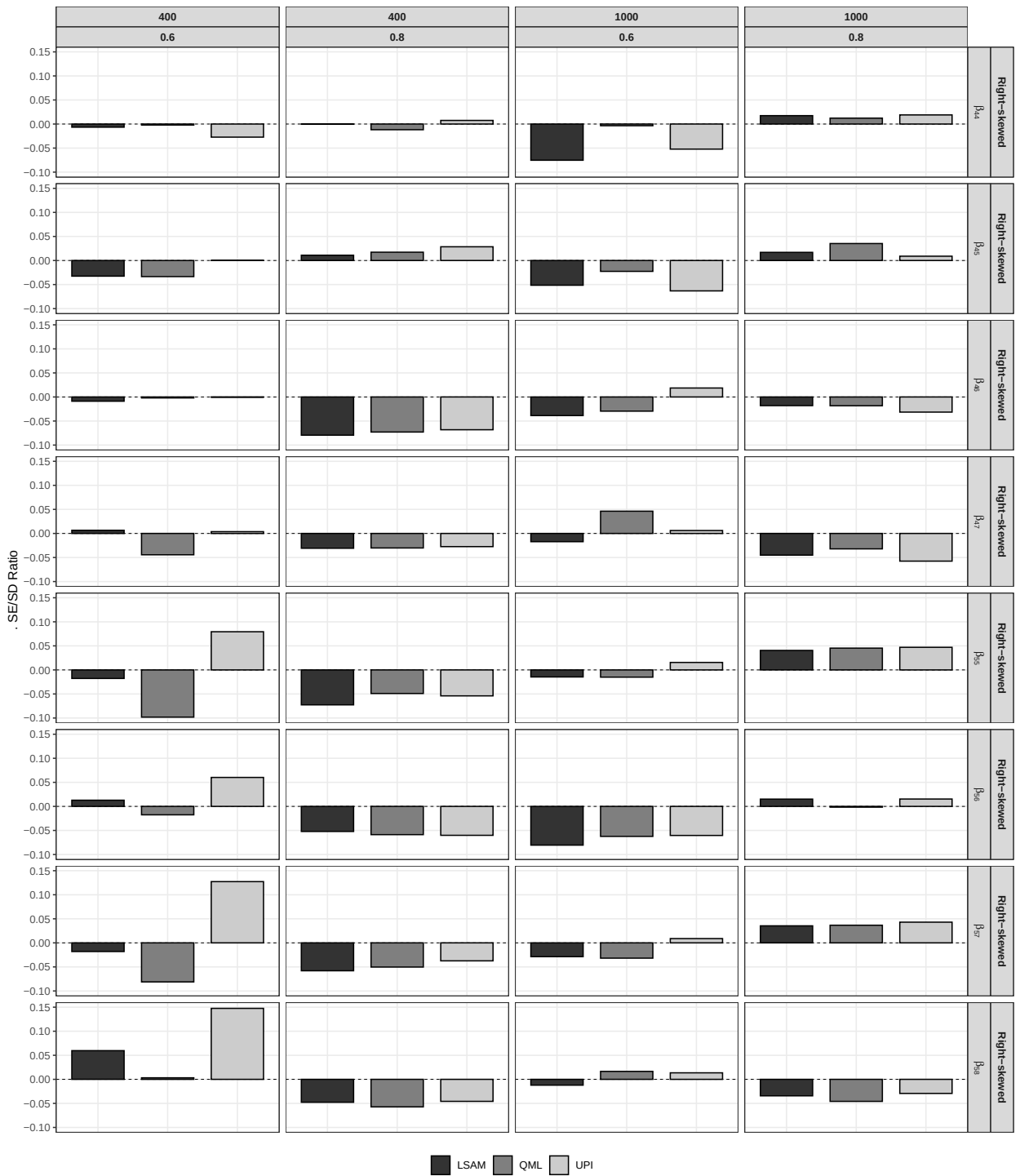


Figure S49

Additional Distributional Conditions Difference: Power for Simulation 2 – Right-skewed (Epsilon)

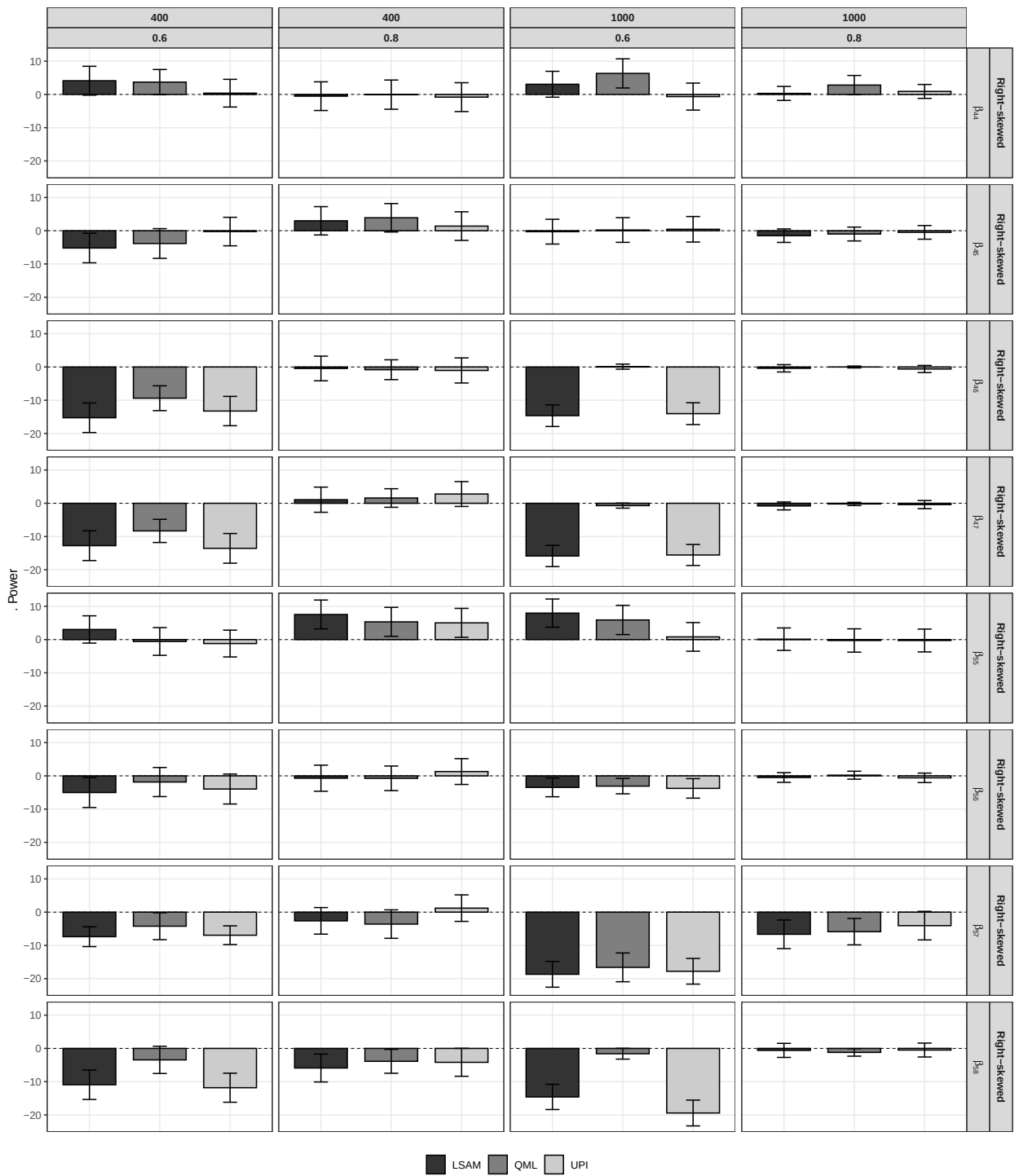


Figure S50

Additional Distributional Conditions Difference: Power for Simulation 2 – Right-skewed (Zeta)

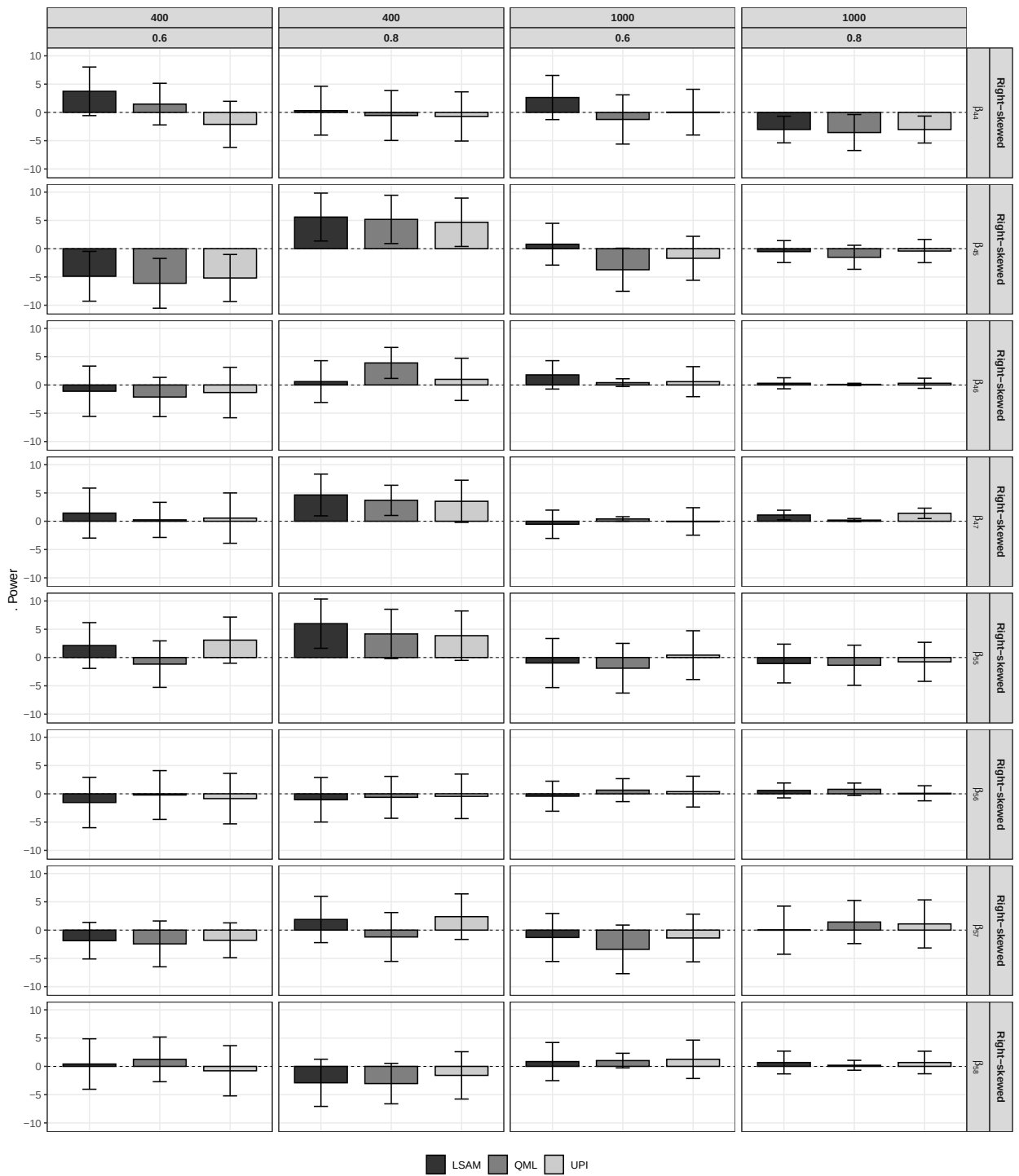


Figure S51

Additional Distributional Conditions Difference: Power for Simulation 2 – Right-skewed (Epsilon & Zeta)

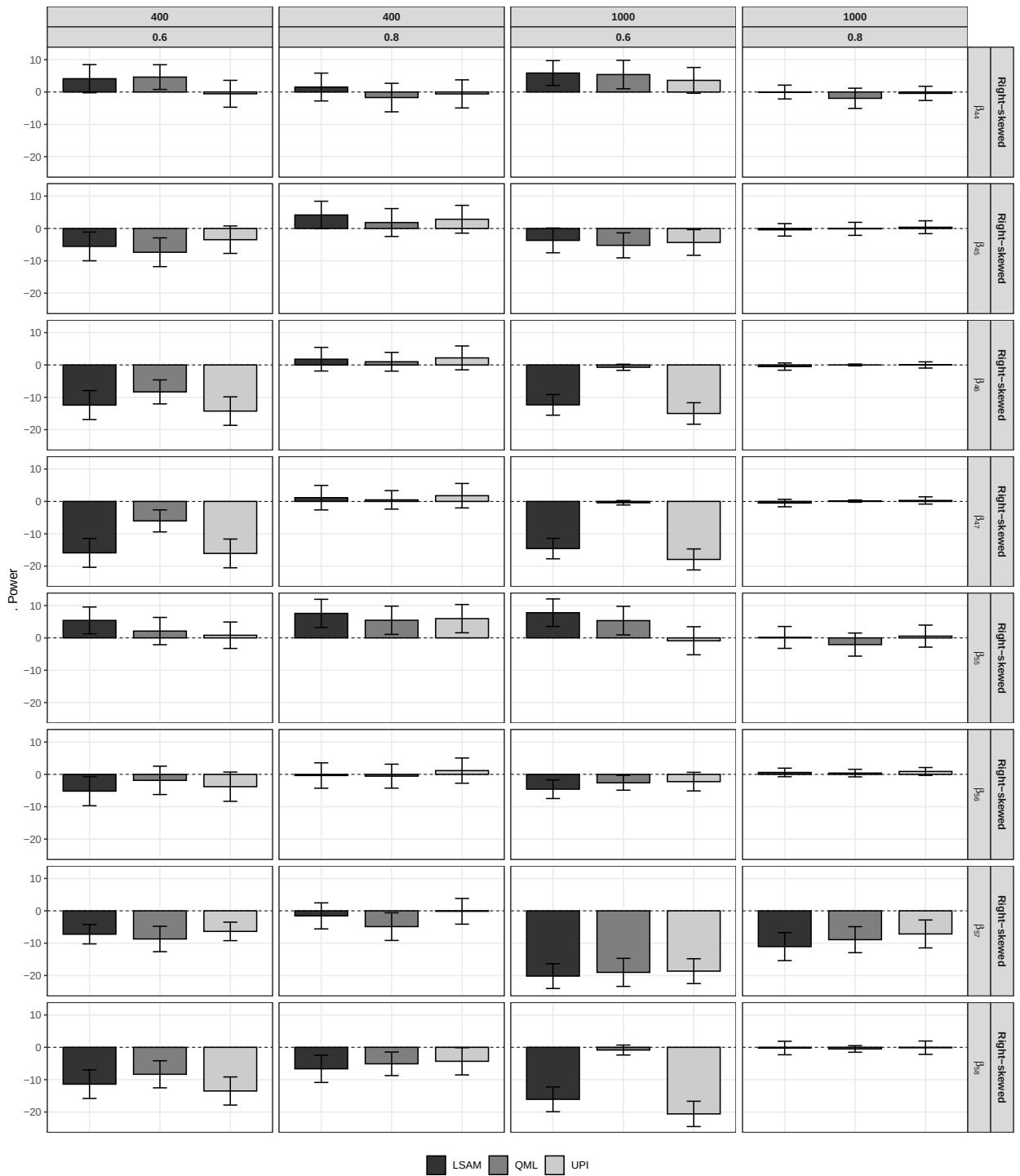


Figure S52

Additional Distributional Conditions Difference: Type I Error for Simulation 2 – Right-skewed (Epsilon)

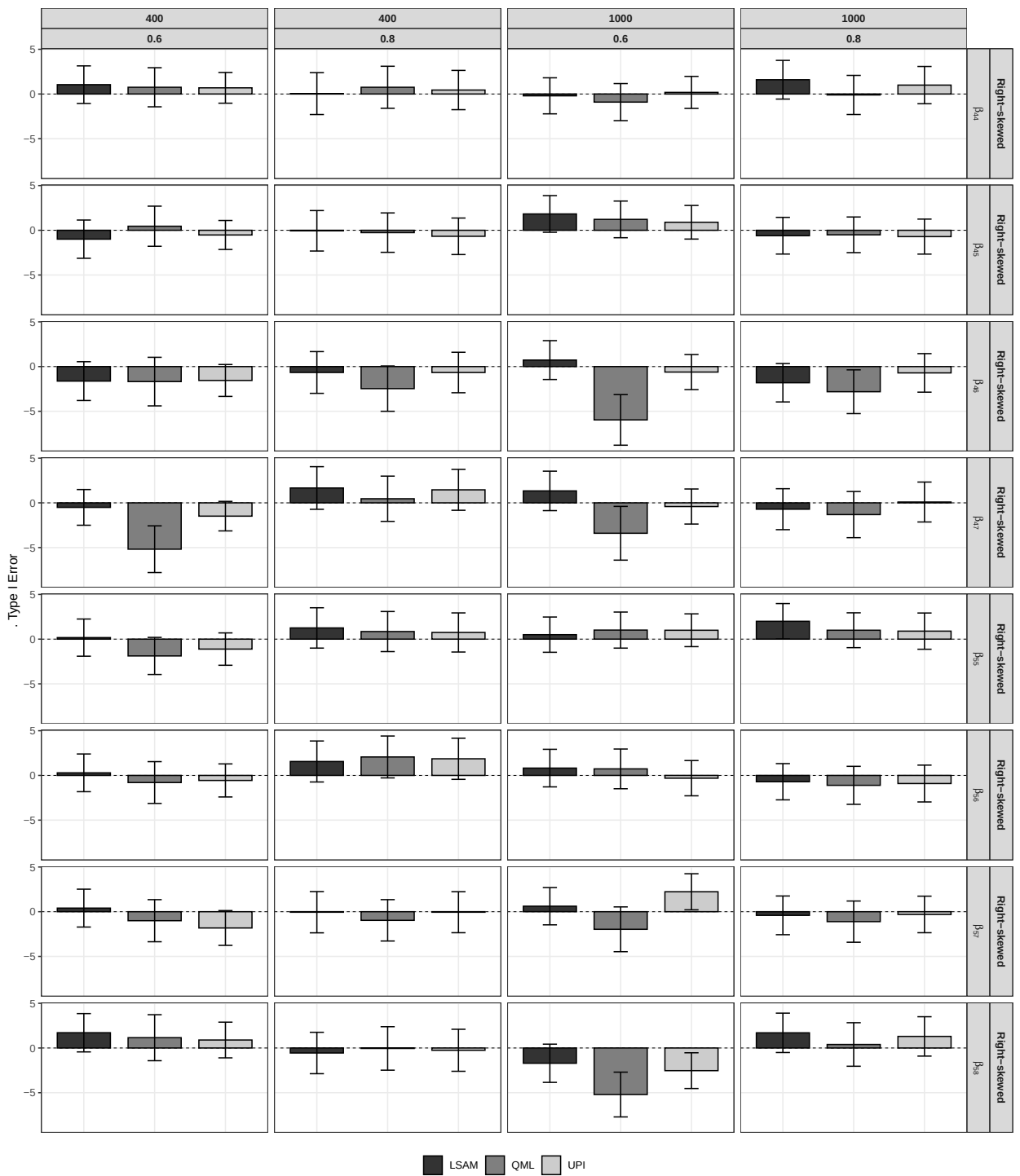


Figure S53

Additional Distributional Conditions Difference: Type I Error for Simulation 2 – Right-skewed (Zeta)

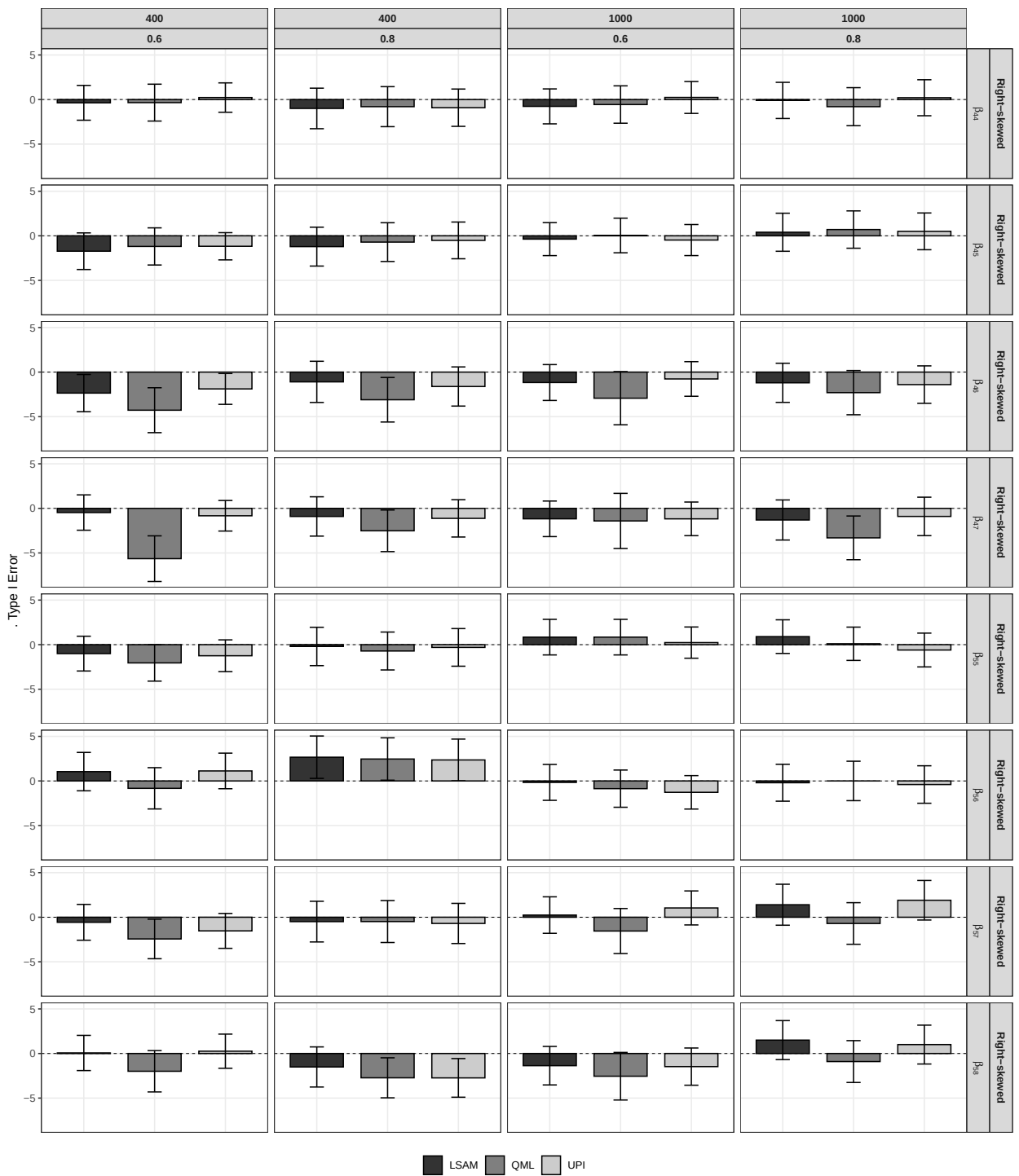


Figure S54

Additional Distributional Conditions Difference: Type I Error for Simulation 2 – Right-skewed (Epsilon & Zeta)

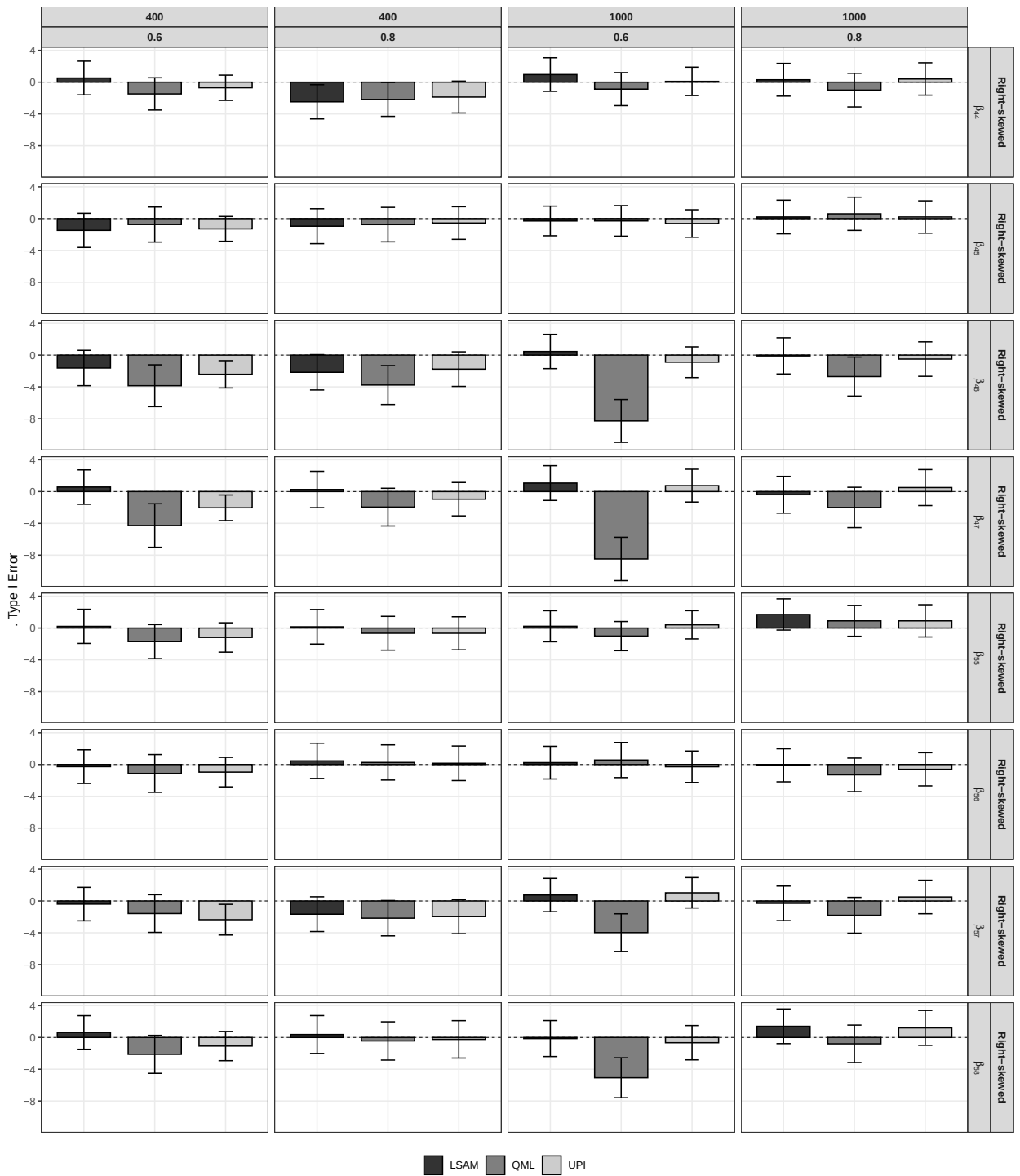


Figure S55

Additional Distributional Conditions Difference: Relative Bias for Simulation 2 – Uniform (Epsilon)

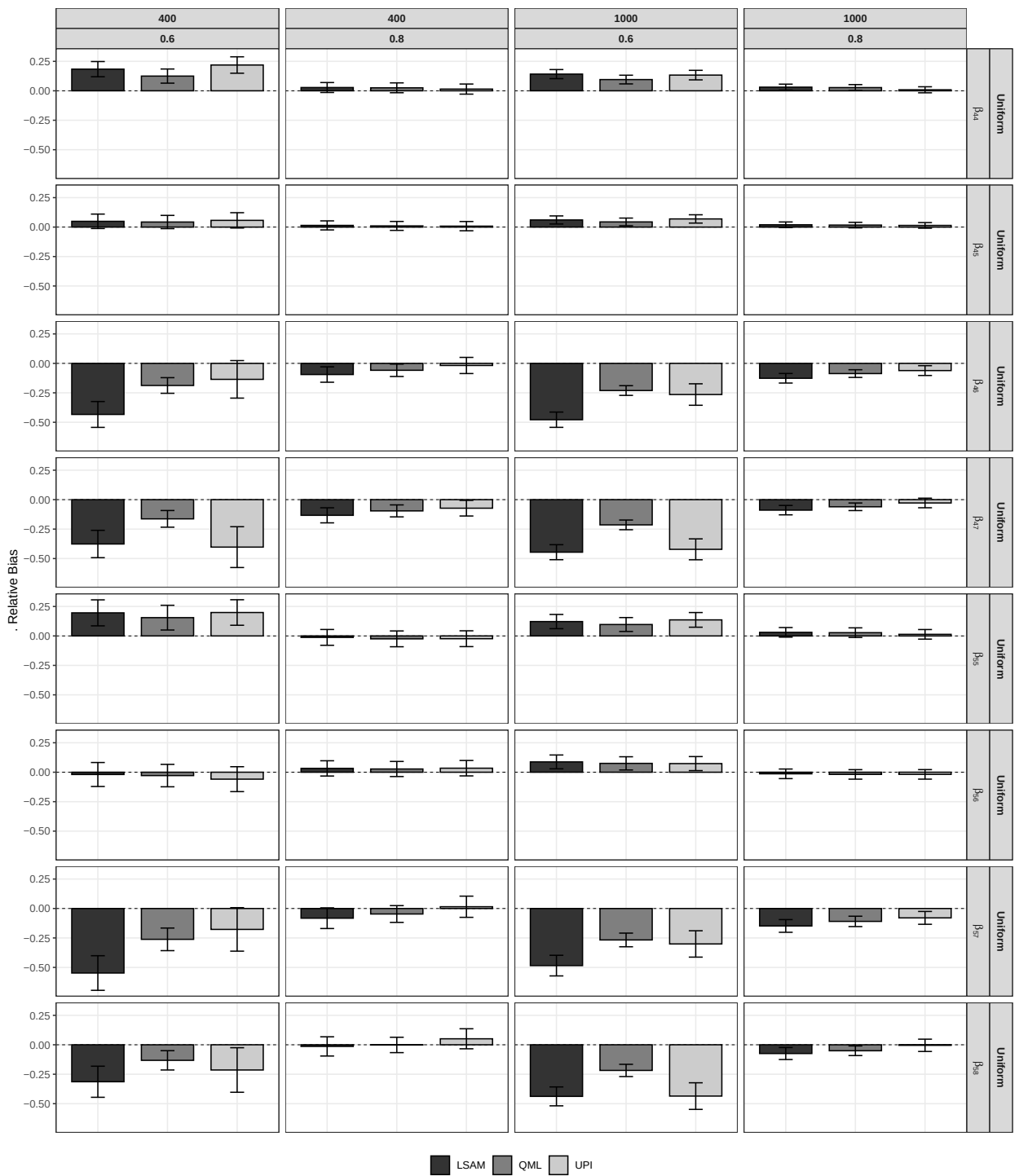


Figure S56

Additional Distributional Conditions Difference: Relative Bias for Simulation 2 – Uniform (Zeta)

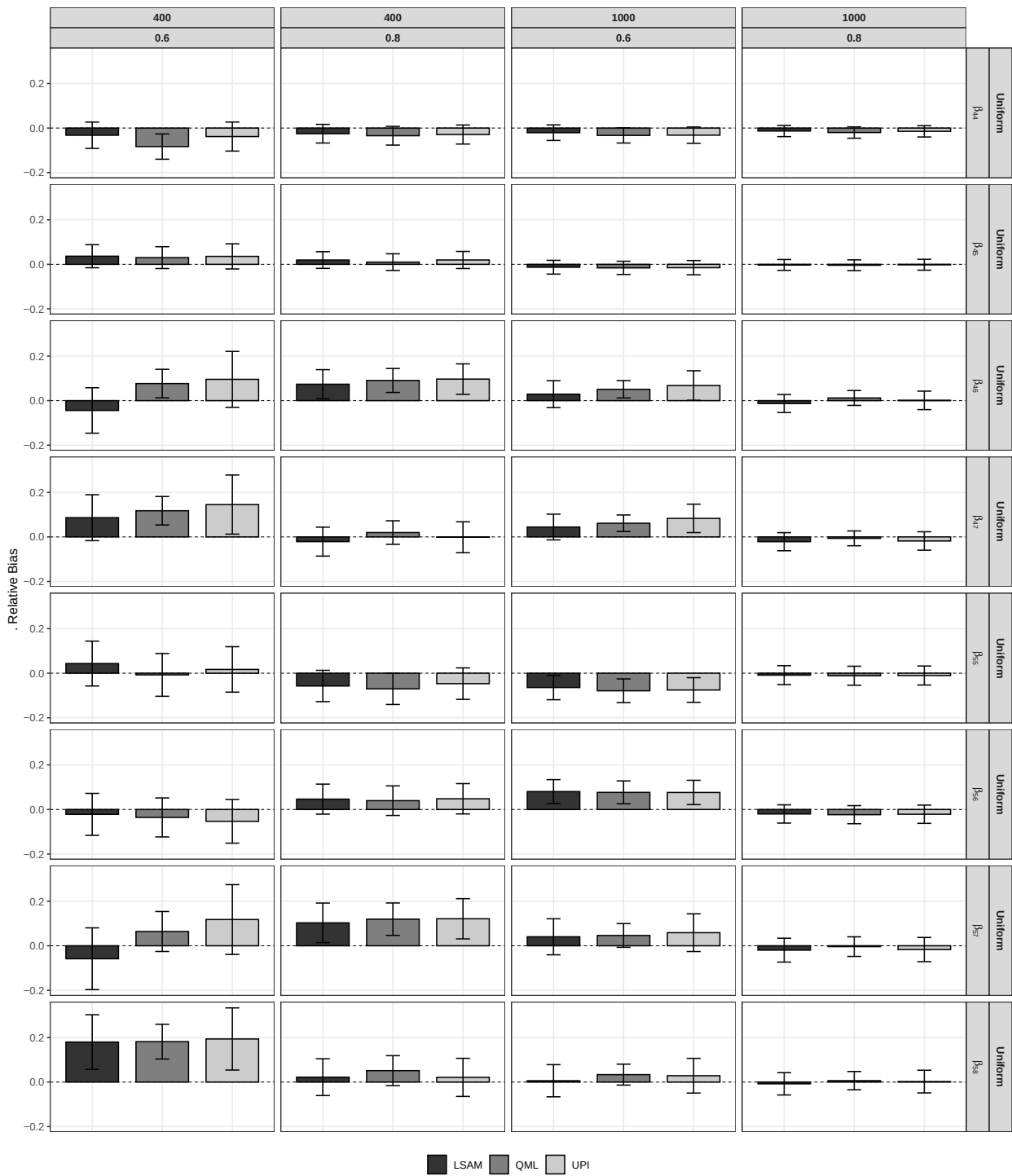


Figure S57

Additional Distributional Conditions Difference: Relative Bias for Simulation 2 – Uniform
(Epsilon & Zeta)

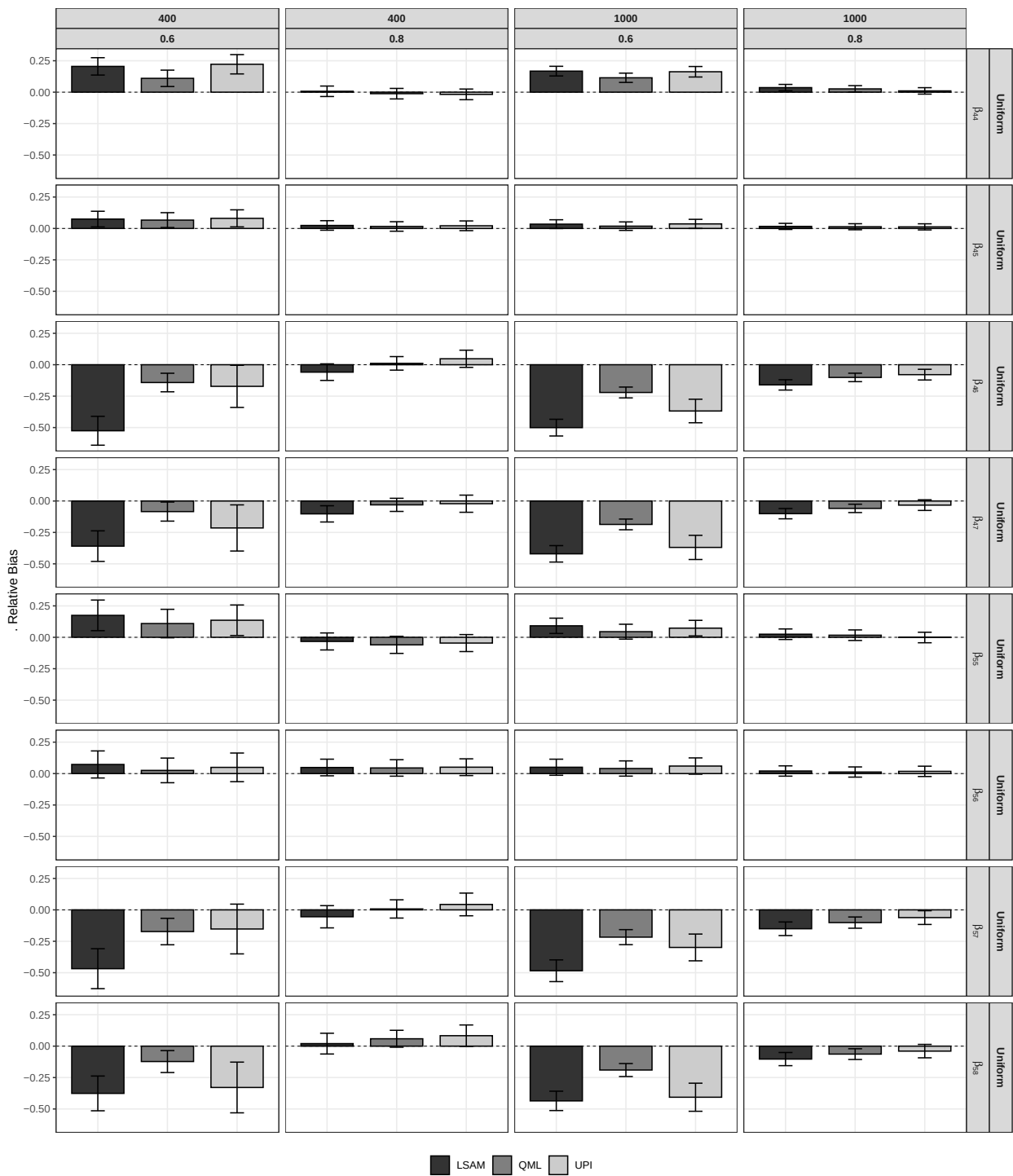


Figure S58

Additional Distributional Conditions Difference: Relative RMSE for Simulation 2 – Uniform (Epsilon)

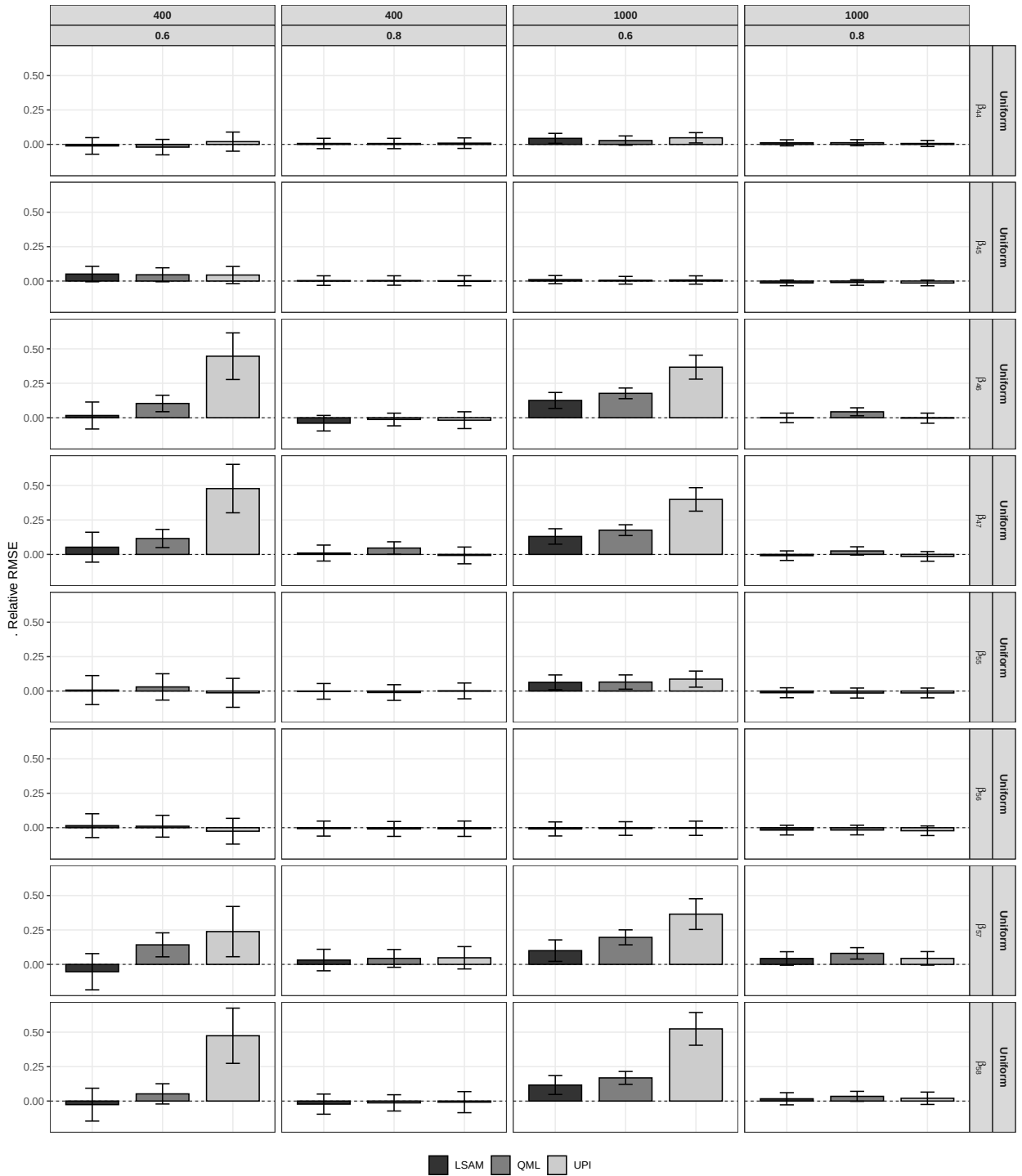


Figure S59

Additional Distributional Conditions Difference: Relative RMSE for Simulation 2 – Uniform (Zeta)

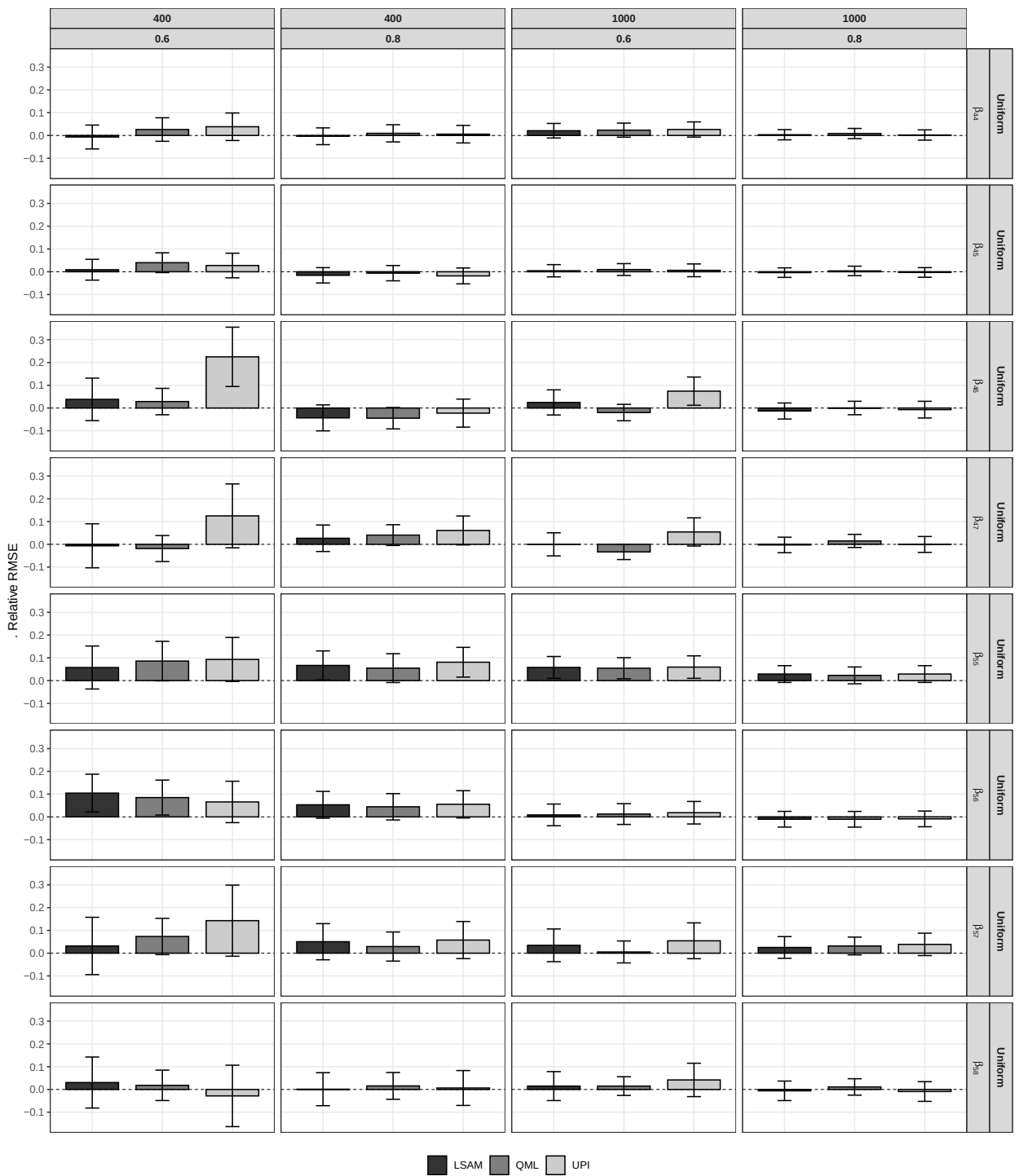


Figure S60

Additional Distributional Conditions Difference: Relative RMSE for Simulation 2 – Uniform (Epsilon & Zeta)

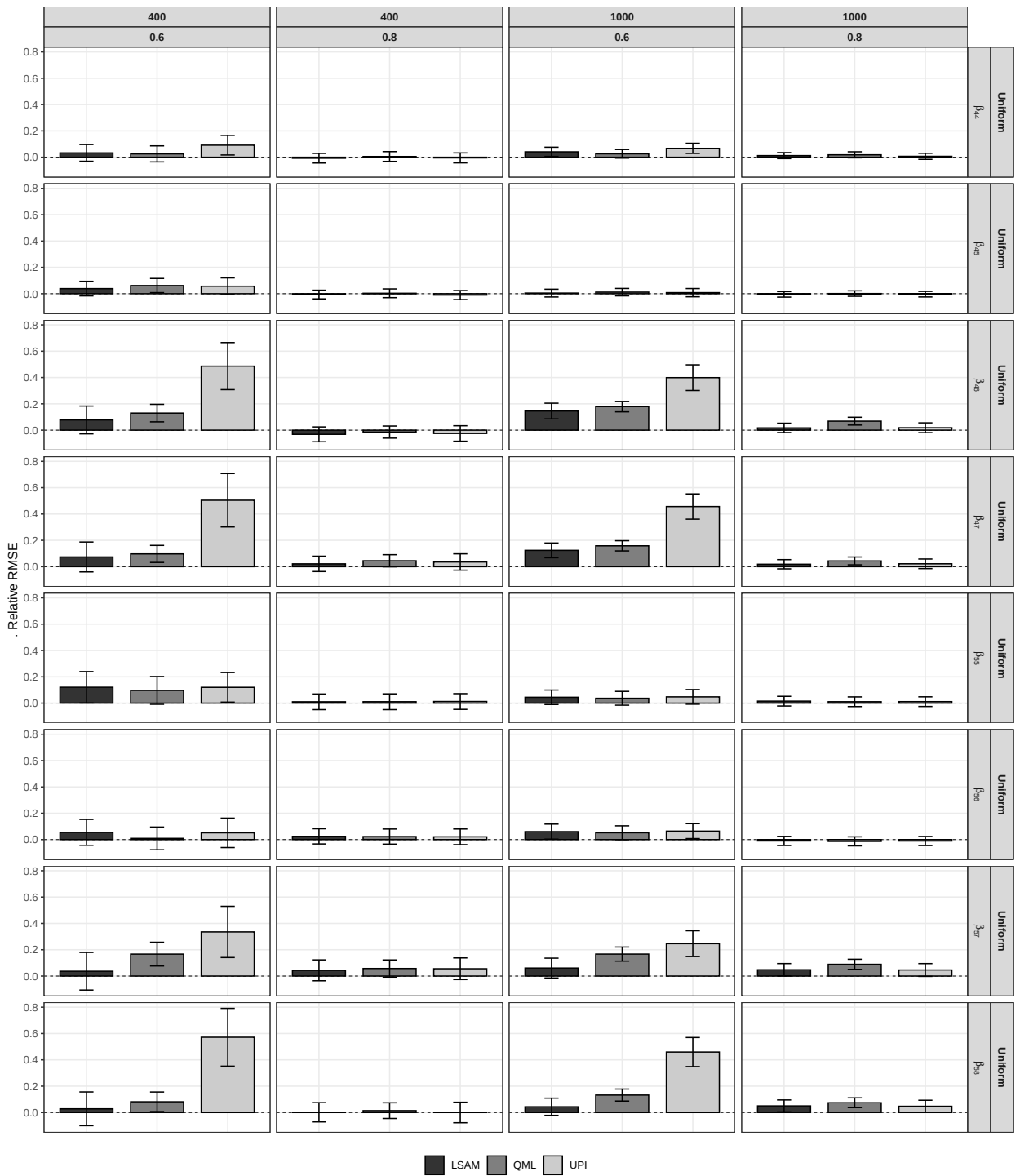


Figure S61

Additional Distributional Conditions Difference: Coverage Rate for Simulation 2 – Uniform (Epsilon)

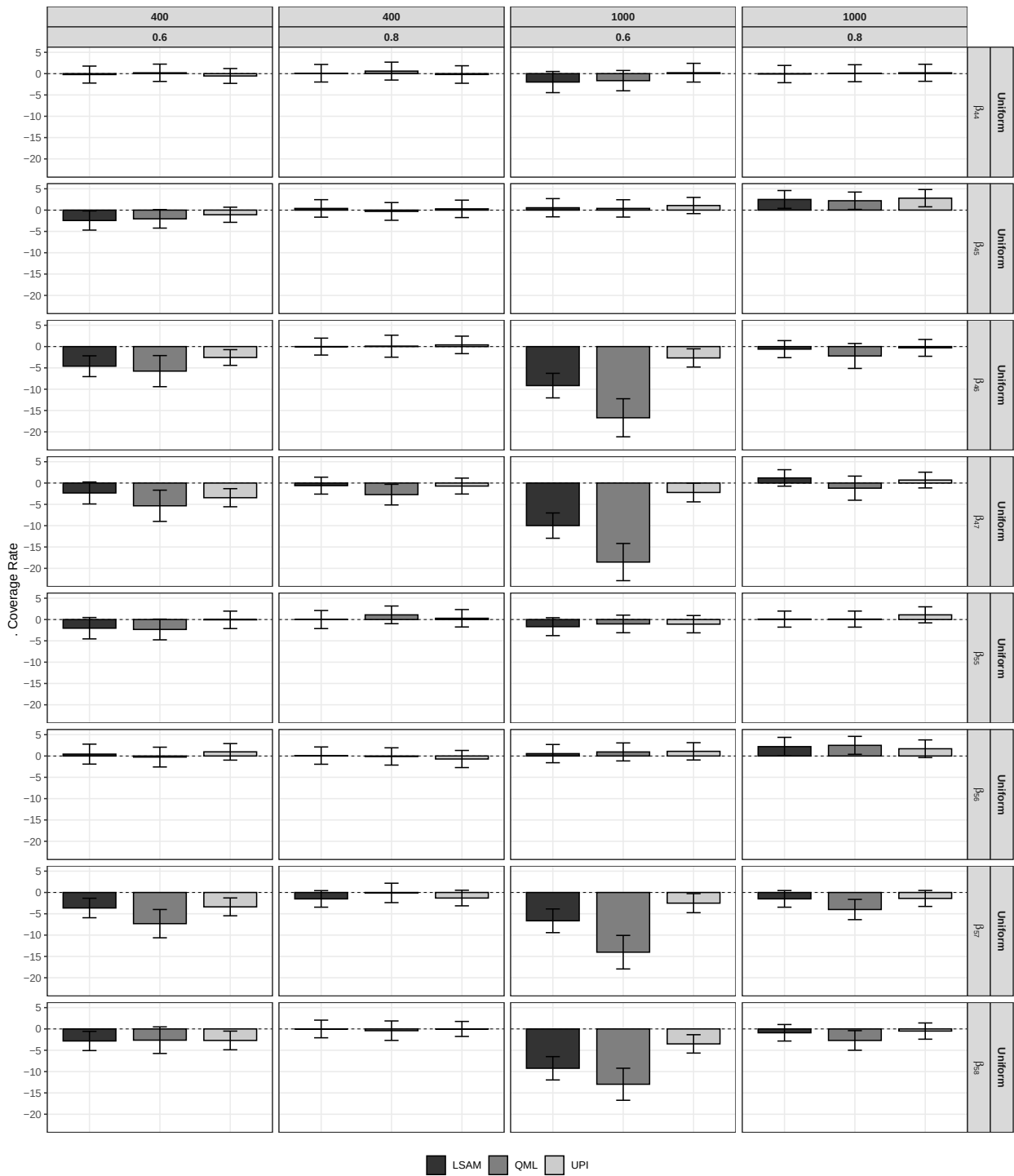


Figure S62

Additional Distributional Conditions Difference: Coverage Rate for Simulation 2 – Uniform (Zeta)

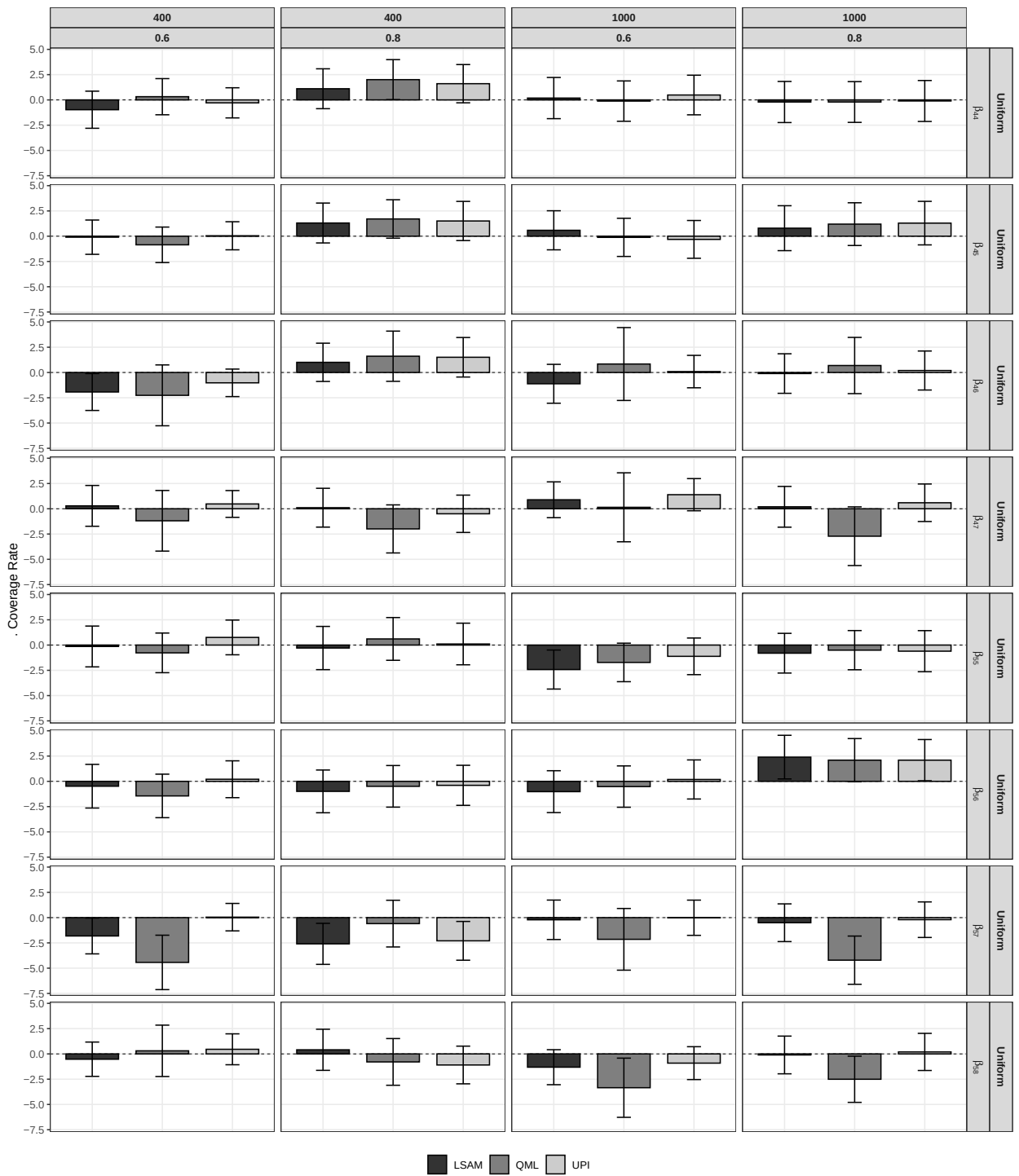


Figure S63

Additional Distributional Conditions Difference: Coverage Rate for Simulation 2 – Uniform (Epsilon & Zeta)

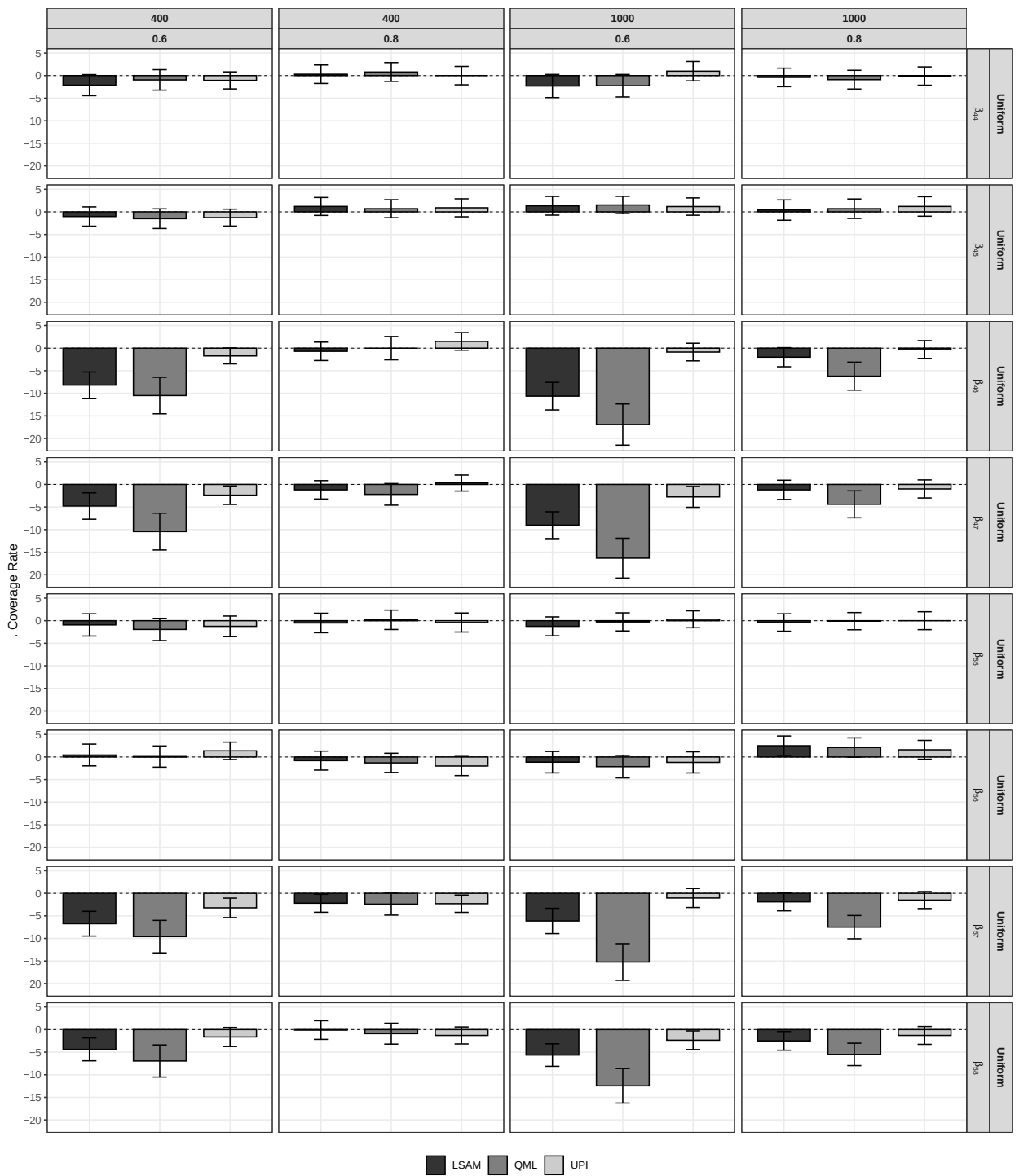


Figure S64

Additional Distributional Conditions Difference: SE/SD Ratio for Simulation 2 – Uniform (Epsilon)

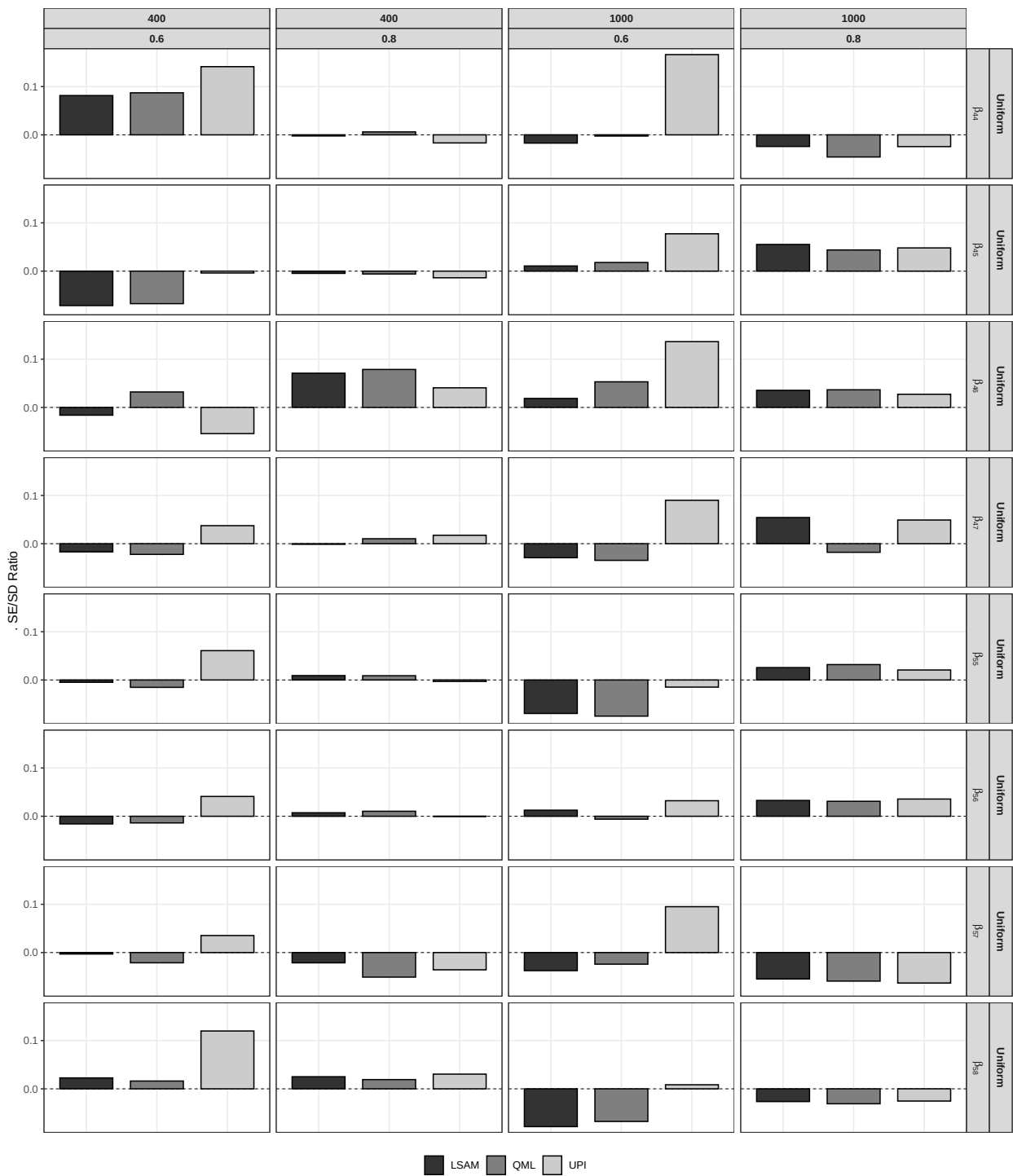


Figure S65

Additional Distributional Conditions Difference: SE/SD Ratio for Simulation 2 – Uniform (Zeta)

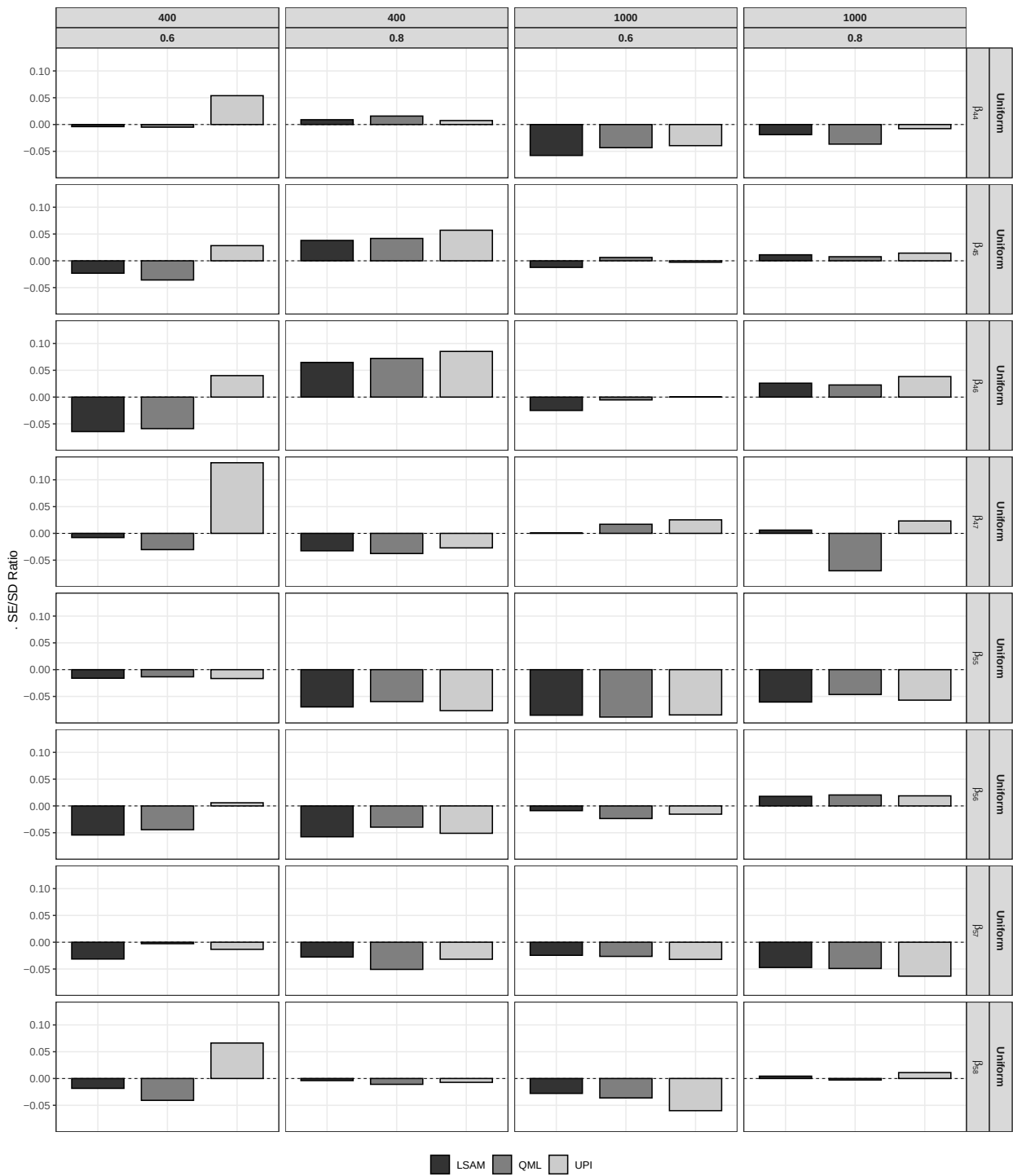


Figure S66

Additional Distributional Conditions Difference: SE/SD Ratio for Simulation 2 – Uniform

(Epsilon & Zeta)

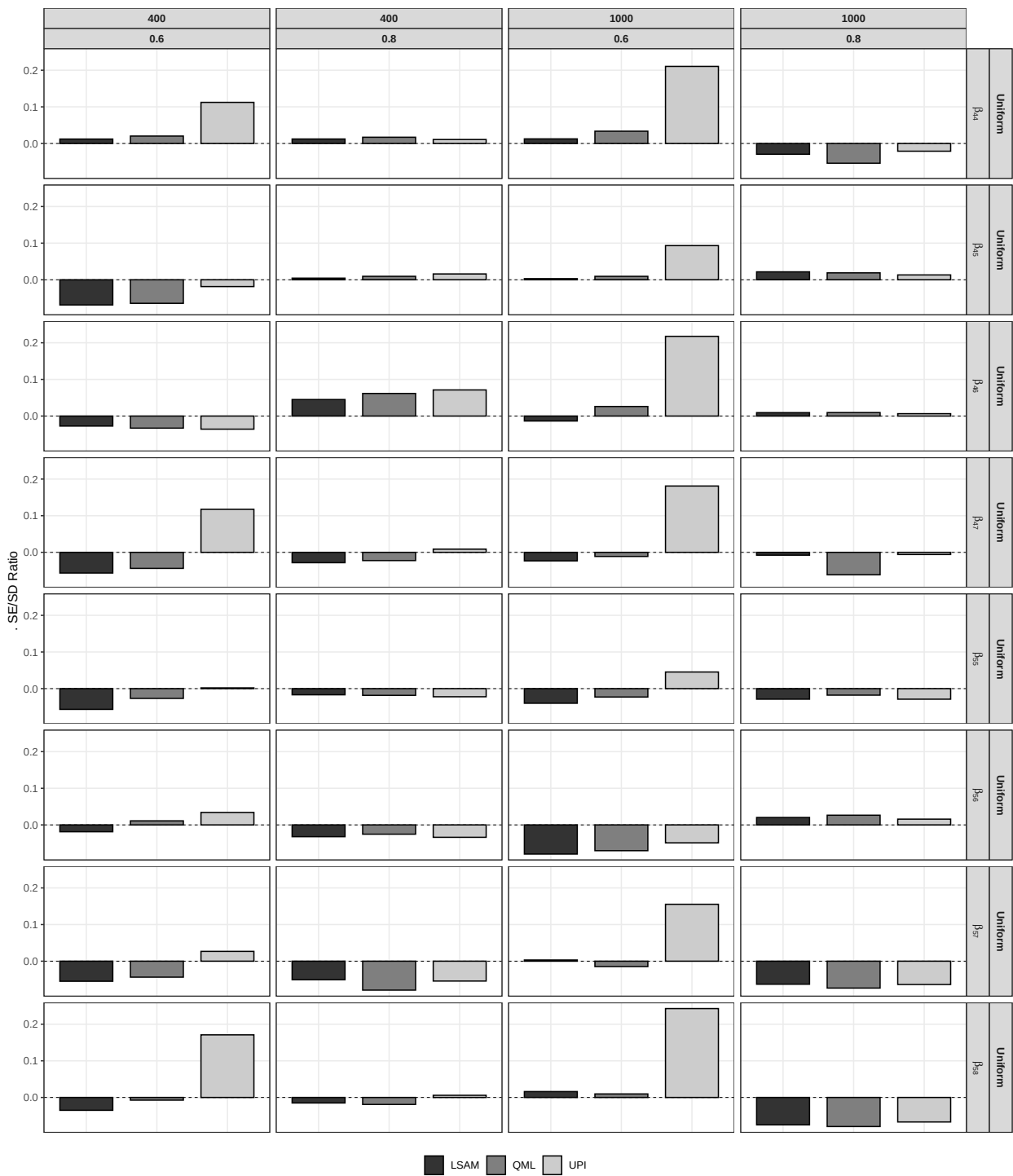


Figure S67

Additional Distributional Conditions Difference: Power for Simulation 2 – Uniform (Epsilon)

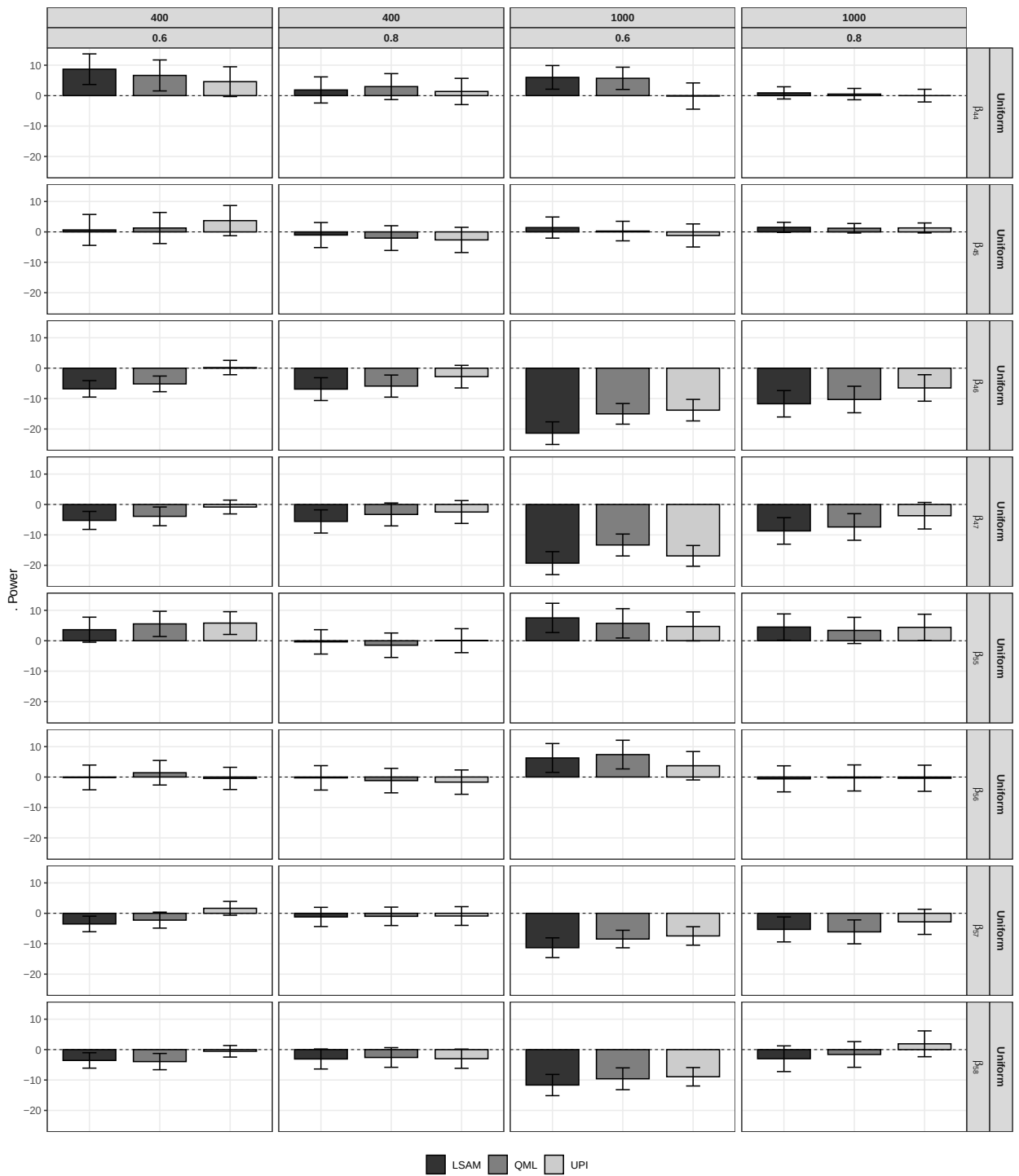


Figure S68

Additional Distributional Conditions Difference: Power for Simulation 2 – Uniform (Zeta)

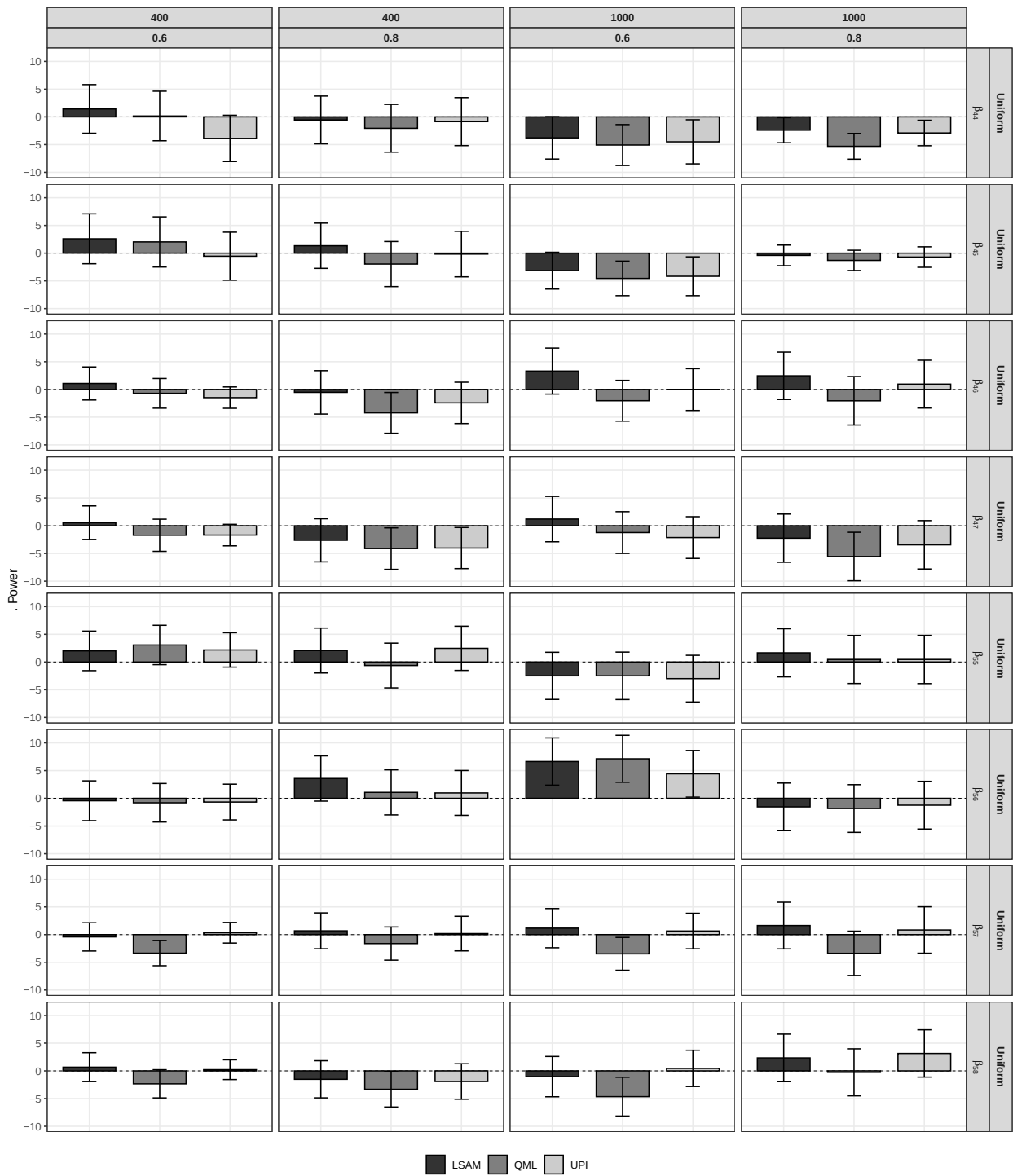


Figure S69

Additional Distributional Conditions Difference: Power for Simulation 2 – Uniform (Epsilon & Zeta)

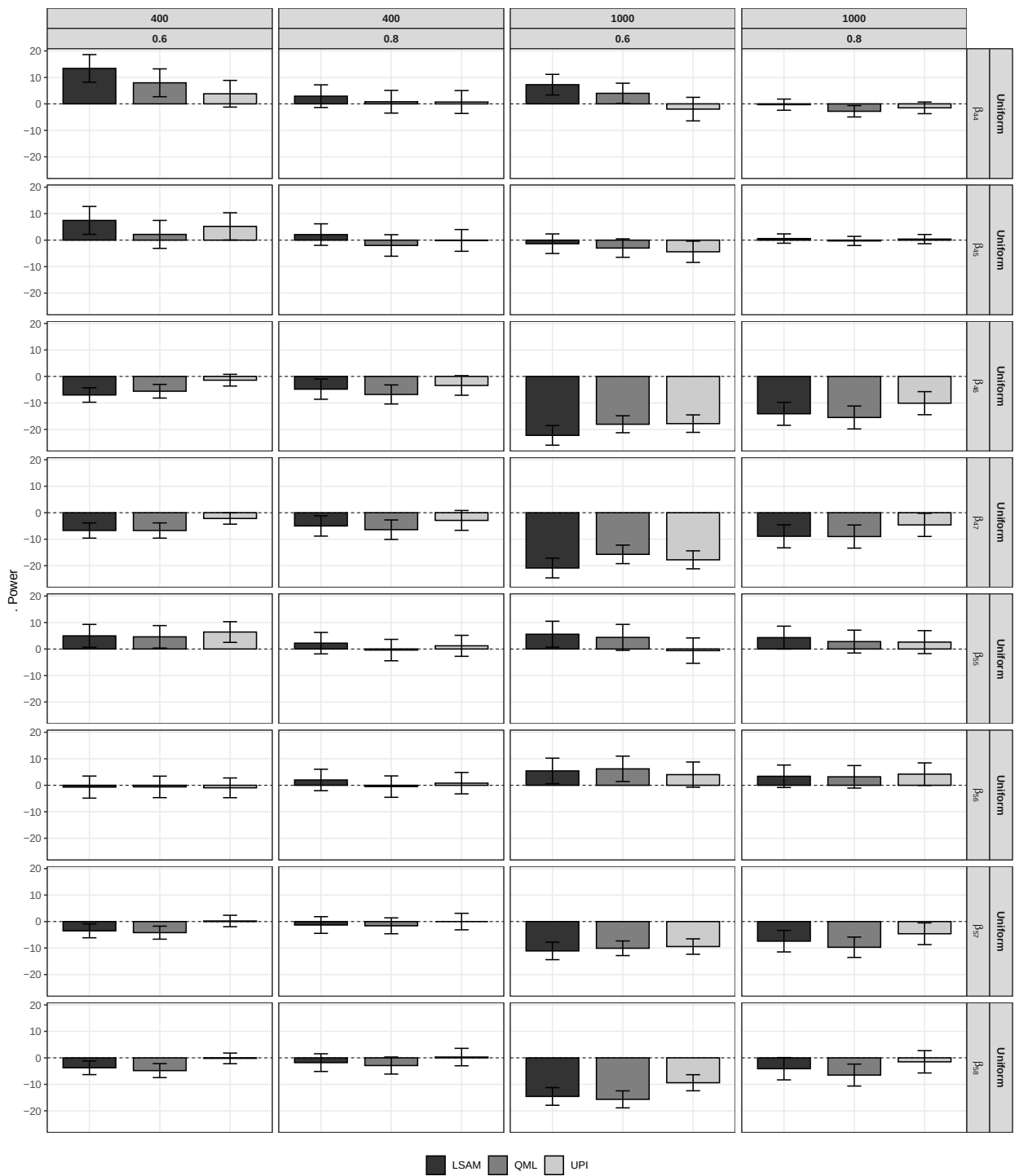


Figure S70

Additional Distributional Conditions Difference: Type I Error for Simulation 2 – Uniform (Epsilon)

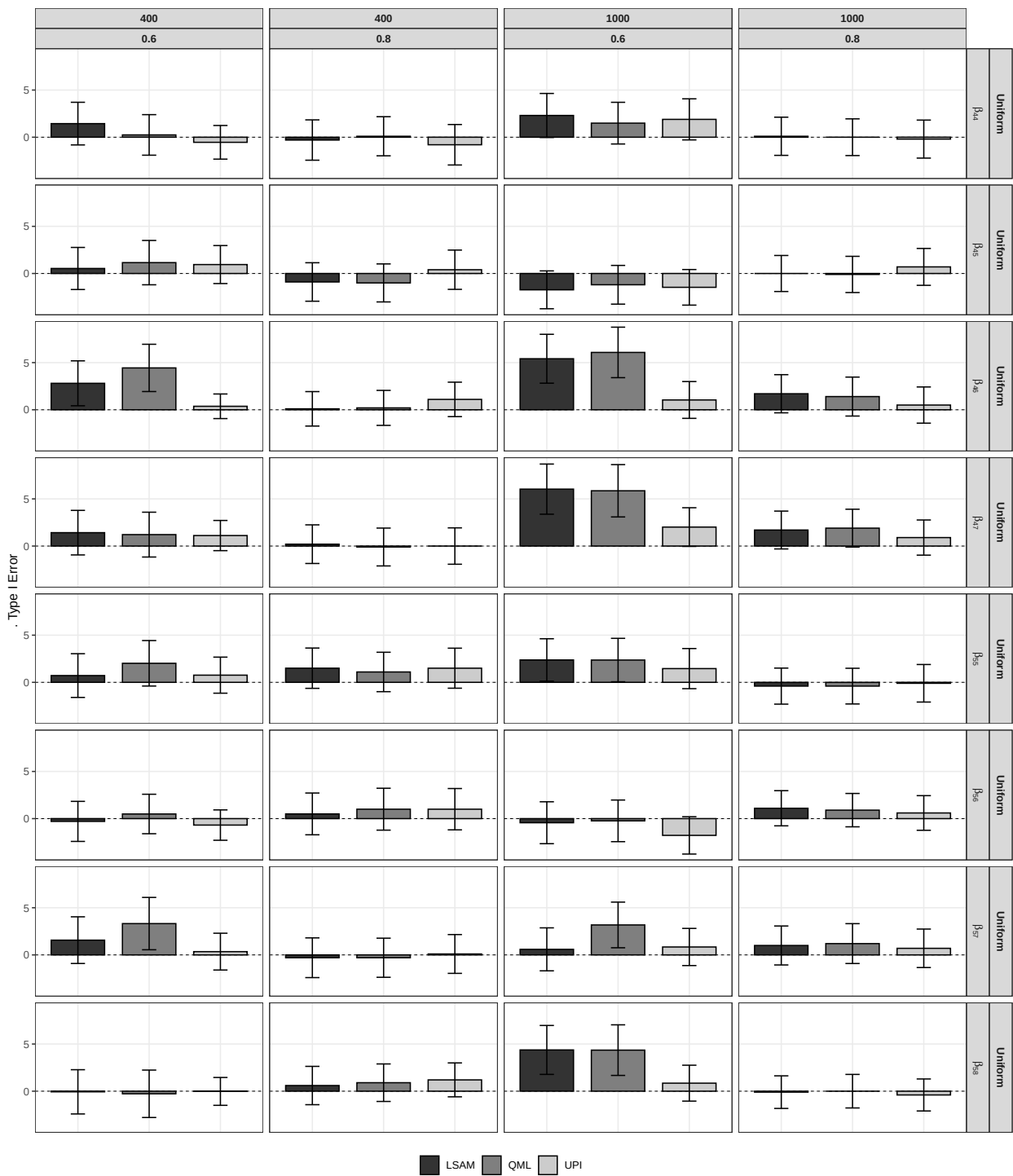


Figure S71

Additional Distributional Conditions Difference: Type I Error for Simulation 2 – Uniform (Zeta)

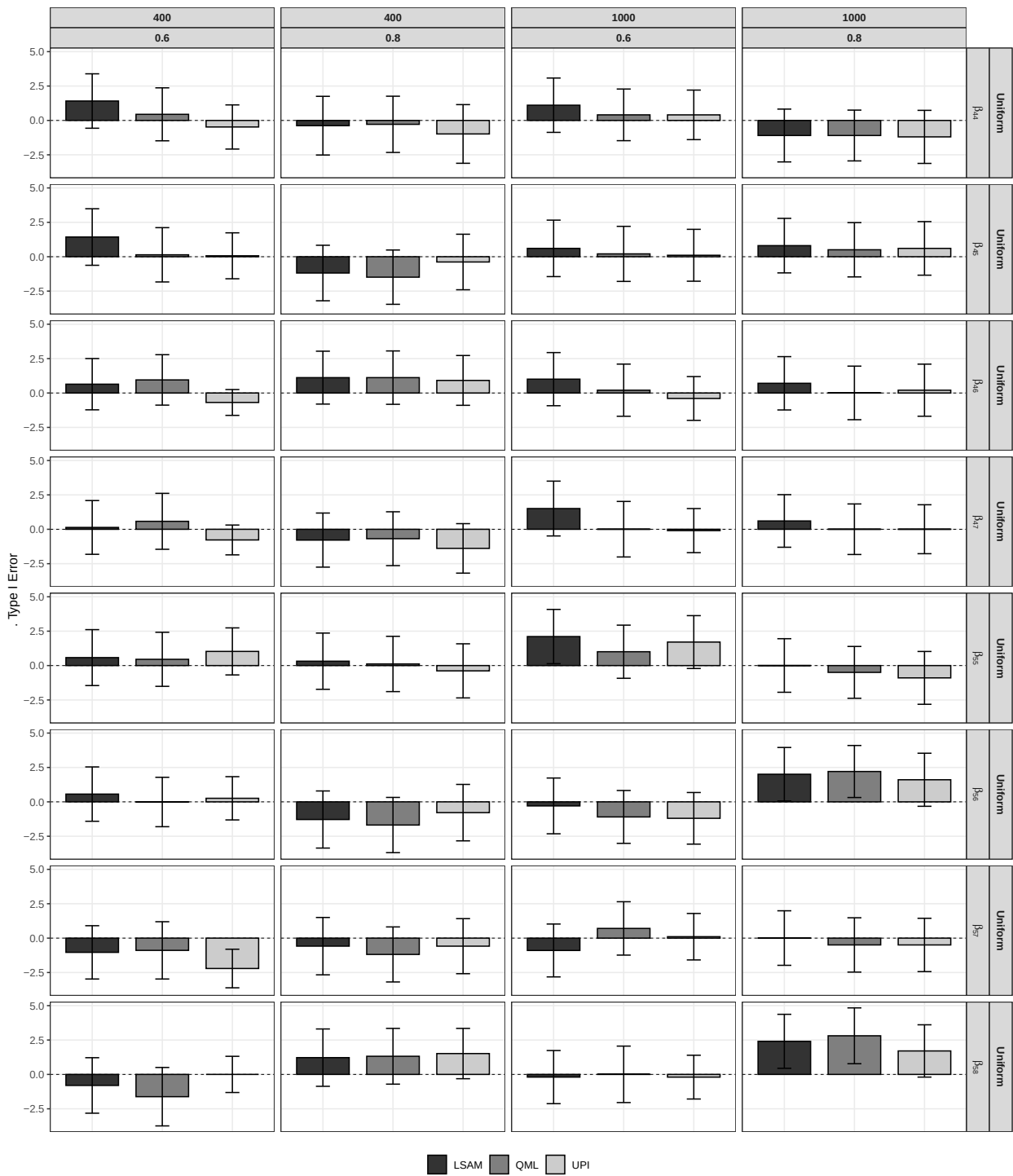


Figure S72

Additional Distributional Conditions Difference: Type I Error for Simulation 2 – Uniform
(Epsilon & Zeta)

